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Editors

Herbal Medicine Phytochemistry

Applications and Trends

Springer

Reference Series in Phytochemistry

This series provides a platform for essential information on plant metabolites and phytochemicals, their chemistry, properties, applications, and methods.

By the strictest definition, phytochemicals are chemicals derived from plants. However, the term is often also used to describe the large number of secondary metabolic compounds found in and derived from plants. These metabolites exhibit a number of nutritional and protective functions for human wellbeing and are used e.g. as colorants, fragrances and flavorings, amino acids, pharmaceuticals, hormones, vitamins and agrochemicals.

The series offers extensive information on various topics and aspects of phytochemicals, including their potential use in natural medicine, their ecological role, role as chemo-preventers and, in the context of plant defense, their importance for pathogen adaptation and disease resistance. The respective volumes also provide information on methods, e.g. for metabolomics, genetic engineering of pathways, molecular farming, and obtaining metabolites from lower organisms and marine organisms besides higher plants. Accordingly, they will be of great interest to readers in various fields, from chemistry, biology and biotechnology, to pharmacognosy, pharmacology, botany and medicine.

The Reference Series in Phytochemistry is indexed in Scopus.

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With 245 Figures and 197 Tables

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Preface

The advancement of evidence-based herbal medicine practice relies on understanding the chemical composition of medicinal plants and their bioactive components. Utilization techniques in herbal medicine draw from the traditional wisdom, knowledge, understanding, and practices of local healers. To ensure the safety and effectiveness of herbal medicines, ongoing quality assessment, and research are essential to help bridge the gap between traditional knowledge and modern scientific and societal methods. Herbal medicine phytochemistry identifies bioactive plant compounds, their mechanism of action, standardized herbal products, and the validation of traditional remedies. Challenges include the need for scientific validation, quality control, regulatory compliance, etc. Opportunities exist in the discovery of new medicines as well as the integration of herbal medicine into mainstream conventional healthcare. Quality assessment such as the precise identification of plant and phytochemicals and advanced chemical analysis can boost the cultural and ethical aspects and provide background for establishing relevant regulatory frameworks. Herbal medicine phytochemistry research is vital to integrate herbal medicine into modern healthcare effectively.

This book, entitled *Herbal Medicine Phytochemistry: Applications and Trends*, presents a roadmap through a broad interdisciplinary collection of reviews written by researchers, intellectuals, experts, practitioners, and professionals. The book focuses on various aspects of phytochemistry and herbal medicine. It delves into the historical context and the classification of phytochemicals in plants with herbal value as well as their utilization patterns. The book explores the diversity of medicinal plants used to treat viral, parasitic, bacterial, and viral diseases, emphasizing their nutritional profiles, bioactive components, and therapeutic potential. It also looks into specific plants like *Cola acuminata*, *Rosmarinus officinalis*, *Garcinia kola*, *Citrus aurantifolia*, *Citrus aurantium*, *Aframomum melegueta*, *Solanum torvum*, and others. It discusses herbal medicine and the use of plants in managing and treating various health conditions, including cardiovascular, respiratory, neurological, gastrointestinal, skin, cancer, arthritis, reproductive issues, eye, wound healing potential, and pregnancy. The utilization approach and practices in herbal medicine were also explored to highlight the significance of chemopreventive practices, trends in orthodox medicine practices, and the role of xenobiotics in traditional medicine. It further discusses quality control strategies, utilization methods in Africa and Asia, the value of herbal medicine in sustainable development, and its socioeconomic impact. The synergy of artificial intelligence in traditional medicine and its role in addressing antimicrobial resistance is also discussed. The book looks into herbal medicine's quality assessment and research needs, emphasizing topics like chemical and microbial contamination of herbal remedies, regulations, policies, and sustainability. It also covers the application of biotechnology in herbal medicine; the use of big data in herbal medicine; research needs related to medicinal plants; conservation and sustainable use of these plants; sustainable supply chain management in the herbal medicine industry; and the intersection of business, sustainability, and herbal medicine.

The book will be useful to students (undergraduates and postgraduates), academicians, researchers, experts, and practitioners in the fields of natural, life, health, clinical, and biomedical sciences, such as pharmaceuticals, herbal medicine, pharmacology, pharmacognosy, phytochemistry, industrial, food and public health scientists, human nutrition and dietetics, plant biology, and biotechnology/microbiology clinicians.

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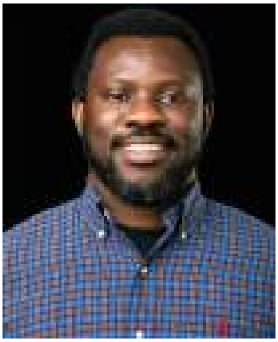
Dr. Sylvester Izah holds a Ph.D. in Public Health, and Applied Microbiology and Environmental Health, an M. Sc. in Applied Microbiology, and a B.Sc. in Biological Sciences from Niger Delta University, Nigeria. He is a licensed Environmental Health Specialist in Nigeria. He brings over ten years of experience from his work with an environmental consultancy firm specializing in soil, water, and atmospheric assessments. During the early stages of his career, Dr. Izah conducted studies on vegetation cover and biodiversity, focusing on species diversity, composition, and utilization. Presently, he serves as a lecturer at Bayelsa Medical University in Yenagoa, Nigeria, where he holds the position of Assistant Director of Academic Planning, Research, and Innovations. Dr. Izah's extensive research portfolio encompasses water, air, and soil quality, applied microbiology, biotechnology, public and environmental health, risk assessment, bioenergy, toxicology, medicinal plants, herbal medicine, and biodiversity. He has an impressive track record of over 300 peer-reviewed publications, including journal articles, book chapters, and edited books. Dr. Izah's Google Scholar total citation, h-index, and i10-Index are >6330, 44, and 169, respectively. His current research interests are centered around Sustainable Human-Environmental Health Interactions, covering areas such as environmental sustainability trends, toxicology, hygiene, sanitation, food science, public-environmental health assessment, medicinal plants, herbal medicine, bioactive and natural compounds, vegetation, and wildlife. Dr. Izah has also collaborated on research projects with colleagues worldwide and actively contributes as an editorial and review board member for several esteemed journals.



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An Introduction to Herbal Medicine Phytochemistry: Applications and Trends

Background

The advancement of evidence-based herbal medicine hinges on a comprehensive grasp of the chemical makeup of medicinal plants and their bioactive elements, with phytochemistry as an indispensable element in this endeavor. The utilization techniques and approaches in herbal medicine, often involving the preparation of plant extracts and remedies with a historical lineage, are deeply rooted in traditional wisdom and the expertise of local healers. To guarantee the safety, effectiveness, and uniformity of herbal medicines, continual quality assessment and research efforts are imperative, especially as integrating modern scientific methodologies gains increasing significance within this field.

This book has three main components: phytochemistry and herbal medicine, utilization approach and practices in herbal medicine, and quality assessment and research needs of herbal medicine (Fig. 1).

Phytochemistry and Herbal Medicine

Phytochemistry is the field of chemistry that looks into plant compounds' molecular structures. It involves the breakdown, separation, characterization, and identification of the numerous chemical constituents present in plants, encompassing a wide range of compounds like terpenes, alkaloids, flavonoids, phenolic compounds, etc. Phytochemistry plays a pivotal role in comprehending the chemical composition of plants and their possible applications, particularly in medicine. The major facets of phytochemistry are summarized in Fig. 2.

Botanical medicine, also known as herbal medicine or phytotherapy, typically involves utilizing plant-based substances for their therapeutic properties. This traditional healing approach has served as a primary treatment method across various cultures worldwide for millennia, encompassing a diverse array of civilizations. The main elements of herbal medicine are summarized in Fig. 3.

Herbal medicine and phytochemistry are closely intertwined fields. Phytochemistry plays a pivotal role in identifying and isolating bioactive plant compounds,

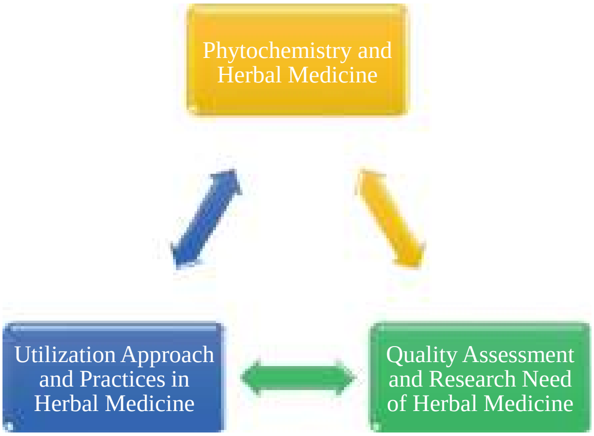


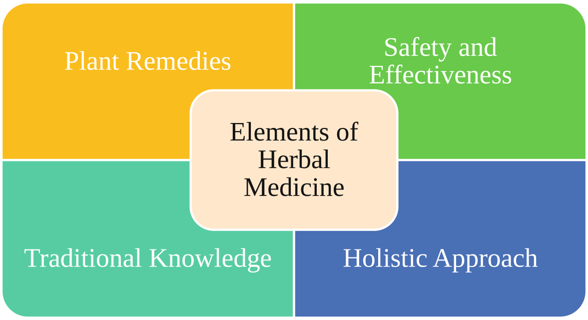
Fig. 1 Aspects of herbal medicine phytochemistry: trends and applications

Fig. 2 The major facets of phytochemistry

which are subsequently employed in herbal medicine. This scientific understanding not only promotes the development of standardized herbal products, ensuring consistency in dosage and quality, but also substantiates the legitimacy of traditional herbal remedies by unveiling their chemical composition and pharmacological effects.

Both phytochemistry and herbal medicine share common challenges and potential prospects. Obstacles include the need for rigorous scientific validation, quality control of the products, and addressing regulatory issues. Nonetheless, they also

Fig. 3 The main elements of herbal medicine



offer promising opportunities, such as the potential for discovering novel medications, more potent natural remedies, and the integration of complementary and alternative medicine into mainstream healthcare.

This part of the book has 33 chapters which are summarized as follows:

i. Historical Perspectives and Overview of the Value of Herbal Medicine: Herbal medicine has a rich historical context and continues to play a crucial role in healthcare, offering a holistic approach to healing by harnessing the therapeutic potential of plant-based remedies.
ii. Plant Food for Human Health: Case Study of Indigenous Vegetables in Akwa Ibom State, Nigeria: Indigenous vegetables in Akwa Ibom State, Nigeria, hold substantial nutritional value, emphasizing the importance of these local plant foods in promoting human health and well-being.
iii. Classification of Phytochemicals in Plants with Herbal Value: The classification of phytochemicals within plants with herbal value provides insights into the diverse compounds responsible for their medicinal properties, facilitating a deeper understanding of their therapeutic potential.
iv. Nutritional Profile, Bioactive Components, and Therapeutic Potential of Edible Flowers of Chhattisgarh, India: Edible flowers in Chhattisgarh, India, are not only nutritionally rich but also contain bioactive components that offer therapeutic benefits, making them a valuable resource for both food and health.
v. Phytochemical Constituents of *Rosmarinus officinalis* Linn. and Their Associated Role in the Management of Alzheimer's Disease: *R. officinalis* harbors phytochemical constituents that may contribute to the management of Alzheimer's disease, shedding light on its potential role in neurological health.
vi. Plants Used in the Management and Treatment of Male Reproductive Health Issues: Case Study of Benin People of Southern Nigeria: The Benin people in Southern Nigeria rely on specific plants for the management and treatment of male reproductive health issues, illustrating the significance of traditional herbal remedies in addressing such concerns.

vii. *Cola acuminata*: Phytochemical Constituents, Nutritional Characteristics, Scientific Validated Pharmacological Properties, Ethnomedicinal Uses, Safety Considerations, and Commercial Values: *Cola acuminata* exhibits a diverse range of phytochemical constituents, nutritional properties, validated pharmacological benefits, cultural significance, safety considerations, and commercial value, making it a multi-faceted plant of interest.

viii. Diversity of Medicinal Plants Used in the Treatment and Management of Viral Diseases Transmitted by Mosquitoes in the Tropics: Medicinal plants in tropical regions serve as a diverse resource for treating viral diseases transmitted by mosquitoes, offering potential solutions to public health challenges.

ix. Medicinal Plants in the Tropics Used in the Treatment and Management of Parasitic Diseases Transmitted by Mosquitoes: Administration, Challenges, and Strategic Options for Management: Medicinal plants in tropical areas are integral in combating parasitic diseases transmitted by mosquitoes, though challenges in administration require strategic management approaches.

x. Efficacy of Ethnoherbal Medicines with Anti-inflammatory and Wound Healing Potentiality: A Case of West Bengal, India: Ethnoherbal medicines in West Bengal, India, demonstrate efficacy in anti-inflammatory and wound healing applications, highlighting their valuable role in traditional healthcare.

xi. Classification Methods and Diversity of Medicinal Plants: The categorization of medicinal plants sheds light on the rich diversity of plant species used in traditional medicine worldwide. Also, the classification methods provide insights into the vast array of medicinal plants employed for diverse health purposes.

xii. Plants Used in the Management and Treatment of Cardiovascular Diseases: Case Study of the Benin People of Southern Nigeria: The Benin people in Southern Nigeria use herbs to manage and treat cardiovascular diseases, thereby offering valuable insights into traditional remedies for heart health. This practice also highlights the potential of indigenous knowledge in cardiovascular disease management, thereby bridging the gap between traditional and modern medicine.

xiii. Mechanistic Approaches of Herbal Medicine in the Treatment of Arthritis: The mechanism by which herbal medicine effectively treats arthritis, thereby contributing to the understanding of alternative therapies for debilitating conditions. The elucidation of the mechanisms may offer a scientific foundation for the use of herbal remedies in arthritis treatment.

xiv. Phytochemicals and Overview of the Evolving Landscape in Management of Osteoarthritis: The exploration of phytochemicals in osteoarthritis management provides an overview of how bioactive compounds found in plants are shaping the evolving landscape of treatment options.

xv. Assessment of the Phytochemical Constituents and Metabolites in the Medicinal Plants and Herbal Medicine Used in the Treatment and Management of Respiratory Diseases: The assessment scrutinizes the phytochemical constituents and metabolites in medicinal plants and herbal remedies, thereby focusing on their application in the treatment and management of respiratory diseases. This could contribute to the understanding of the therapeutic properties of herbal remedies for respiratory conditions.

xvi. Assessment of the Phytochemical Constituents and Metabolites in Medicinal Plants and Herbal Remedies Used in the Treatment and Management of Reproductive Diseases: Polycystic Ovary Syndrome: The phytochemical constituents and metabolites in medicinal plants and herbal remedies could be essential in the treatment of reproductive diseases such as polycystic ovary syndrome (PCOS). Findings could help address a critical area of herbal medicine application in women's healthcare.

xvii. Assessment of the Phytochemical Constituents and Metabolites in the Medicinal Plants and Herbal Medicine Used in the Treatment and Management of Skin Diseases: Phytochemical constituents and metabolites found in medicinal plants and herbal medicine emphasize their significance in the treatment and management of skin diseases. This chapter sheds light on the potential of herbal remedies in dermatological care.

xviii. Metabolites and Phytochemicals in Medicinal Plants Used in the Management and Treatment of Neurological Diseases: The role of metabolites and phytochemicals in medicinal plants and their application in managing and treating neurological diseases, providing insights into alternative therapeutic approaches for brain-related conditions. This contributes to the growing body of knowledge about herbal remedies for neurological health.

xix. Diabetes Treatment and Prevention Using Herbal Medicine: The chapter centers on using herbal medicine to treat and prevent diabetes, offering a holistic and natural approach to managing this prevalent metabolic disorder. This chapter underscores the potential of herbal remedies as a complementary strategy for diabetes care and prevention.

xx. Therapeutic Applications of Herbal Medicines for the Prevention and Management of Cancer: This chapter explores the potential of herbal remedies in preventing and managing cancer, focusing on their therapeutic applications and outcomes.

xxi. Herbs and Herbal Formulations for the Management and Prevention of Gastrointestinal Diseases: This chapter looks into the use of medicinal plants and herbal formulations to address gastrointestinal disorders, thereby shedding light on their role in disease prevention and management.

xxii. Herbal Medicine and Pregnancy: This chapter addresses herbal medicine during pregnancy, providing insights into its safety and efficacy, and emphasizing the well-being of expectant mothers.

xxiii. Herbal Medicine and Rheumatic Disorders Management and Prevention: This chapter discusses how herbal remedies can be employed to manage and prevent rheumatic disorders, highlighting their potential to improve the quality of life for affected individuals.

xxiv. Physiological and Biochemical Outcomes of Herbal Medicine Use in the Treatment of Hypertension: This chapter offers an overview of herbal medicine's physiological and biochemical effects on hypertension, shedding light on its role as an alternative treatment.

xxv. Evidence from the Use of Herbal Medicines in the Management and Prevention of Common Eye Diseases: This chapter provides evidence-based insights into utilizing herbal medicines for managing and preventing common eye diseases, emphasizing their potential benefits.

xxvi. Assessment of the Phytochemical Constituents and Metabolites of Some Medicinal Plants and Herbal Remedies Used in the Treatment and Management of Injuries: This chapter assesses the phytochemical constituents and metabolites of medicinal plants and herbal remedies used in injury treatment and management.

xxvii. Plants Used in the Management and Treatment of Female Reproductive Health Issues: Case Study from Southern Nigeria: This chapter offers a case study of plants used in the treatment of female reproductive health issues, focusing on the practices in Southern Nigeria.

xxviii. *Citrus aurantium*: Phytochemistry, Therapeutic Potential, Safety Considerations, and Research Needs: This chapter presents an overview of *Citrus aurantium*, discussing its phytochemical composition, therapeutic potential, safety considerations, and areas for further research.

xxix. Medicinal Spice, *Aframomum melegueta*: An Overview of the Phytochemical Constituents, Nutritional Characteristics, and Ethnomedicinal Values for Sustainability: This chapter provides an overview of *A. melegueta*, highlighting its phytochemical constituents, nutritional characteristics, and cultural significance for sustainability.

xxx. "Turkey berry (*Solanum torvum* Sw. [Solanaceae]): An Overview of the Phytochemical Constituents, Nutritional Characteristics, and Ethnomedicinal Values for Sustainability: This chapter discusses *S. torvum* and its phytochemical constituents, nutritional qualities, and ethnomedicinal significance in the context of sustainability.

xxxi. *Garcinia kola* Heckel. (Clusiaceae): An Overview of the Cultural, Medicinal, and Dietary Significance for Sustainability: This chapter explores the cultural, medicinal, and dietary importance of *G. kola*, emphasizing its role in sustainable practices.

xxxii. *Vernonia amygdalina* Delile (Asteraceae): An Overview of the Phytochemical Constituents, Nutritional Characteristics, and Ethnomedicinal Values for Sustainability: This chapter provides an overview of *V. amygdalina*, focusing on its phytochemical constituents, nutritional attributes, and ethnomedicinal value within the context of sustainability.

xxxiii. *Citrus aurantifolia*: Phytochemical Constituents, Food Preservative Potentials, and Pharmacological Value: This chapter focuses on *C. aurantifolia*, detailing its phytochemical components, potential for use as a food preservative, and its pharmacological value.

Utilization Approach and Practices in Herbal Medicine

Herbal medicine has a rich history of transcending cultures, emphasizing plant diversity, traditional knowledge, and holistic care. It has influenced medicine throughout history, with roots in traditional systems, European herbalism, and a twentieth-century resurgence. Today, herbal medicine finds uses in supplements, integrative healthcare, and personalized medicine. Despite its growth, it faces challenges including quality control, safety concerns, the need for scientific evidence, and ethical issues related to cultural appropriation (Fig. 4). Therefore, herbal medicine's holistic, plant-based approach remains significant, requiring continued research and responsible practices to ensure its safety and effectiveness in contemporary healthcare.

This part of the book has 16 chapters which are summarized as follows:

i. Chemopreventive Practices in Traditional Medicine: This chapter focuses on traditional medicine incorporated in a wide range of practices aimed at preventing disease, including dietary modifications and the use of specific medicinal plants, often rooted in historical knowledge and cultural traditions.
ii. Chemopreventive Strategies in Herbal Medicine Practice: Current Aspects, Challenges, Prospects, and Sustainable Future Outlook: This chapter focuses on herbal medicine practices especially chemopreventive strategies, with a focus on identifying specific medicinal plants and phytochemicals that exhibit cancer prevention properties, which can contribute to mainstream healthcare but face challenges in terms of standardization and regulatory compliance. Furthermore, the sustainable future outlook for herbal medicine in chemoprevention lies in strengthening scientific validation, addressing quality control

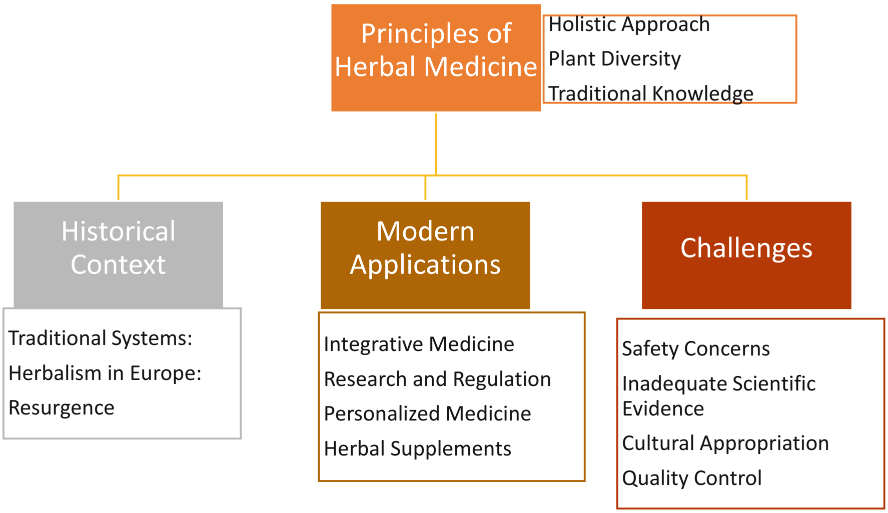


Fig. 4 Overview of the utilization approach and practices in herbal medicine

concerns, and fostering greater collaboration between traditional knowledge and modern research.

iii. Sustainable Current Trends and Future Directions in Orthodox Medicine Practice in Sierra Leone: In Sierra Leone, orthodox medicine practices are experiencing a shift toward more sustainable and community-oriented healthcare delivery, emphasizing accessibility, affordability, and cultural relevance. The future of orthodox medicine in Sierra Leone holds the promise of improved healthcare infrastructure, strengthened primary care, and integration with traditional healing practices to provide holistic health services.

iv. Herbal Medicine: A Case Study of Kherias, Lodhas, and Mundas Tribes in Southwest Bengal of India: The Kherias, Lodhas, and Mundas tribes in Southwest Bengal have preserved and continue to practice herbal medicine, employing a rich repertoire of plant-based remedies passed down through generations. This case study highlights the cultural significance and knowledge preservation of indigenous herbal medicine and underscores the need for their inclusion in discussions on healthcare sustainability.

v. Herbal Medicine—Exploring Its Scope Across Belief Systems of Indian Medicine: Herbal medicine in India transcends various belief systems, including Ayurveda, Siddha, and Unani, demonstrating a holistic approach to health and well-being that combines ancient knowledge with contemporary healthcare practices. This exploration sheds light on the versatility of herbal medicine across belief systems, fostering a more comprehensive understanding of its potential benefits.

vi. Xenobiotics in Traditional Medicine Practices and Quality Control Strategies: Traditional medicine practices sometimes involve the use of xenobiotics, foreign substances that require rigorous quality control measures to ensure safety and efficacy. Addressing the presence of xenobiotics in traditional medicine underscores the importance of quality control and regulation in these practices to safeguard public health.

vii. Herbal Medicine for Health Management and Disease Prevention: Herbal medicine offers an array of preventive and health management solutions, demonstrating its potential to reduce the burden of chronic diseases and improve overall well-being. The utilization of herbal medicine in health management presents an opportunity for a more holistic and cost-effective approach to healthcare.

viii. Utilization Methods and Practices of Herbal Medicine in Africa: Across Africa, diverse utilization methods and practices of herbal medicine are integral to healthcare, driven by cultural traditions and local knowledge. Understanding the rich tapestry of herbal medicine practices in Africa is vital for enhancing healthcare accessibility and addressing health disparities on the continent.

ix. Herbal Medicine Methods and Practices in Nigeria: In Nigeria, herbal medicine methods and practices have been deeply rooted in the culture and tradition, often involving the use of locally sourced plants and knowledge passed down through generations. These methods include the preparation of decoctions, infusions, and the use of specific plant parts for various ailments, showcasing the rich diversity of herbal remedies within the country.

x. Value of Herbal Medicine to Sustainable Development: Herbal medicine plays a vital role in sustainable development by offering affordable and locally available healthcare solutions, reducing the strain on modern healthcare systems and pharmaceutical industries. Additionally, the cultivation and sustainable harvesting of medicinal plants for herbal medicine contribute to biodiversity conservation and support the livelihoods of many rural communities.

xi. Herbal Medicine and Sustainable Development Challenges and Opportunities: Challenges related to herbal medicine and sustainable development include the need for quality control, standardization, and regulations to ensure safe and effective herbal remedies. However, opportunities lie in the potential for herbal medicine to provide alternative, ecofriendly healthcare solutions and promote sustainable practices in plant cultivation and harvesting.

xii. Current Trends on Phytochemicals Toward Herbal Medicine Development: Current trends in phytochemical research focus on identifying and isolating bioactive compounds from plants, understanding their mechanisms of action, and exploring their potential in developing new herbal medicines. The use of advanced analytical techniques and bioinformatics is enhancing our knowledge of phytochemicals, driving innovation in herbal medicine development.

xiii. Proximate Analysis of Herbal Drugs: Methods, Relevance, and Quality Control Aspects: Proximate analysis is essential for determining the nutritional and chemical composition of herbal medicine, ensuring its quality, safety, and efficacy. These methods are highly relevant in quality control, enabling the detection of contaminants and variations in herbal drug preparations, which is crucial for consumer safety and regulatory compliance.

xiv. Socioeconomic Values of Herbal Medicine: Herbal medicine contributes significantly to the socioeconomic well-being of communities involved in its production and trade, offering employment opportunities and income generation. Moreover, it serves as an accessible and cost-effective healthcare option for many people, particularly in regions with limited access to modern medicine.

xv. Harmonizing Tradition and Technology: The Synergy of Artificial Intelligence in Traditional Medicine: The integration of artificial intelligence with traditional medicine practices holds promise in enhancing the diagnosis, treatment, and management of health conditions based on extensive data analysis and predictive algorithms. This harmonization of tradition and technology showcases the potential for artificial intelligence to support the preservation and modernization of traditional medicine.

xvi. Antimicrobial Resistance and the Role of Herbal Medicine: Challenges, Opportunities, and Future Prospects: Herbal medicine presents opportunities to address antimicrobial resistance by providing alternative treatment options and potentially reducing the overuse of antibiotics. However, challenges include the need for scientific validation, quality control, and regulatory frameworks to harness the full potential of herbal medicine in combating this global health threat.

Quality Assessment and Research Needs of Herbal Medicine

Assessing the quality of herbal medicine is integral to ensuring the safety and reliability of these natural products. This process involves several key considerations, including precise plant identification to reduce the risk of adverse effects and contamination (Fig. 5). Advanced chemical analysis techniques, such as spectroscopy and chromatography, are essential for determining the chemical composition of herbal extracts and standardizing formulations. Moreover, establishing quality control protocols and guidelines is crucial to maintaining uniformity, safety, and potency in herbal products (Fig. 5).

There is a growing acceptance of herbal medicine in healthcare, but it requires further research to advance and integrate it into modern medical practices. This necessitates comprehensive studies to validate the therapeutic benefits and safety profiles of herbal remedies, addressing the concerns of both patients and healthcare professionals. Investigating the phytochemical composition of medicinal plants is vital to understanding the bioactive components underlying their therapeutic effects, facilitating the development of evidence-based herbal treatments. Furthermore, research must uncover potential interactions between herbal and pharmaceutical drugs to ensure patient safety and effective treatment approaches (Fig. 5). Acknowledging herbal medicine's cultural and ethical aspects is essential to preserve

Fig. 5 Quality assessment of herbal medicine practices

indigenous practices and prevent the exploitation of traditional knowledge. Strict regulatory frameworks, encompassing labelling laws and quality requirements, are needed to safeguard consumer confidence and public health. Additionally, more research is needed to determine how herbal medicine can be seamlessly integrated into conventional healthcare practices, considering its potential in complementary and alternative medicine approaches.

Research and quality assessment in herbal medicine (Fig. 6) are imperative for ensuring safety and effectiveness. First, accurate botanical identification is crucial, as using the wrong plant species can be harmful or ineffective; modern techniques like DNA barcoding and microscopy aid in precise identification. Moreover, the growing conditions of medicinal plants, influenced by factors like farming practices, climate, and soil quality, significantly affect their chemical composition and therapeutic potential, necessitating careful consideration for consistent quality. Good agricultural and collection practice provides a sustainable cultivation and harvesting framework, emphasizing quality control throughout production. Detection of authentication and adulteration is essential, as periodic substitutions with subpar or hazardous components can occur; analytical techniques like high-performance liquid chromatography and mass spectrometry help identify adulterants. Additionally, standardization of extraction, processing, and herbal product combination methods is essential for ensuring uniform quality and therapeutic efficacy (Fig. 6).

This part of the book has 19 chapters which are summarized as follows:

Fig. 6 Research needs in herbal medicine

i. Trace Metals Contamination of Herbal Remedies: Trace metals contamination in herbal remedies poses health risks due to potential toxic effects. Stringent quality control measures and standards are necessary to ensure consumer safety. Regulations and policies should address trace metal limits in herbal medicines, monitor their presence, and enforce compliance to safeguard public health.
ii. Regulations and Policies for Herbal Medicine and Practitioners: Establishing comprehensive regulations and licensing for herbal medicine practitioners is essential to ensure patient safety and standardize the practice. Striking a balance between traditional knowledge and modern regulations is key to promoting the integration of herbal medicine into mainstream healthcare systems.
iii. Toward Sustainability in the Source of Raw Materials for Herbal Remedies: Sustainable harvesting and cultivation practices are vital to conserve medicinal plant species and maintain a consistent supply of herbal remedies. Developing guidelines and incentives for sustainable sourcing can help preserve biodiversity and support local communities.
iv. Microbial Contaminants of Herbal Remedies: Health Risks and Sustainable Quality Control Strategies: Microbial contamination in herbal remedies can jeopardize consumer health. Quality control strategies, such as microbial testing and Good Manufacturing Practices, are essential for product safety. Sustainable sourcing, processing, and storage practices can mitigate microbial contamination risks in herbal medicines.
v. Adoption and Application of Biotechnology in Herbal Medicine Practices: The integration of biotechnology offers opportunities to enhance the quality, efficacy, and safety of herbal medicines. Regulation and ethical considerations must guide its responsible application. Biotechnological methods like tissue culture and genetic profiling can aid in the cultivation and conservation of valuable medicinal plant species.
vi. Challenges and Future of Nanotechnology in Global Herbal Medicine Practices: Nanotechnology holds promise for improving drug delivery and enhancing the therapeutic potential of herbal medicines, but challenges related to safety and regulation must be addressed. Collaborative efforts among scientists, regulators, and herbal practitioners are crucial to unlock the full potential of nanotechnology in herbal medicine.
vii. Eco-metabolomic Studies of Medicinal Plants and Herbal Medicine: Eco-metabolomic studies provide insights into the interactions between medicinal plants and their environments, aiding in sustainable cultivation and conservation efforts. This ecological perspective can guide the responsible use of medicinal plants and the development of herbal medicine industry practices that support environmental health.
viii. Physiological Ecology of Medicinal Plants: Implications for Phytochemical Constituents: Understanding the physiological ecology of medicinal plants helps elucidate how environmental factors influence the production of bioactive compounds, contributing to the standardization of herbal remedies.

Conservation strategies should consider the ecological requirements of medicinal plants to ensure a stable supply of quality raw materials.

ix. Big Data Application in Herbal Medicine – the Need for a Consolidated Database: The accumulation of big data can revolutionize herbal medicine research and practice, but a consolidated, accessible database is essential to harness this potential fully. Data sharing and standardized data collection methods can facilitate collaborative research and evidence-based decision-making in herbal medicine.

x. Challenges in the Storage of Herbal Medicine Products and Strategies for Sustainable Management: Proper storage of herbal medicine products is crucial to maintaining their efficacy and safety. Sustainable packaging and storage solutions can reduce environmental impacts. Strategies should address issues like humidity control and temperature stability, ensuring the long-term viability of herbal remedies.

xi. Herbal Medicine Formulation, Standardization, and Commercialization Challenges and Sustainable Strategies for Improvement: Herbal medicine formulation and standardization are necessary to ensure consistent product quality and efficacy. Sustainable practices can enhance market competitiveness. Collaboration between traditional healers, scientists, and industry stakeholders can lead to improved formulations and sustainable commercialization of herbal remedies.

xii. Research Needs of Medicinal Plants Used in the Management and Treatment of Some Diseases Caused by Microorganisms: Research is crucial to identify and develop herbal remedies for diseases caused by microorganisms. Collaboration between researchers and traditional healers can expedite discovery and validation. Addressing research needs can lead to the development of effective, evidence-based herbal treatments for infectious diseases.

xiii. Conservation and Sustainable Uses of Medicinal Plants Phytochemicals: Medicinal plants play a crucial role in traditional healthcare, with phytochemicals as their active compounds. Conservation efforts are essential to ensure the sustainable use of these plants, preserving their phytochemical richness and cultural significance.

xiv. Threats and Conservation Strategies of Common Edible Vegetables that Possess Pharmacological Potentials in Nigeria: Common edible vegetables in Nigeria possess valuable pharmacological properties, but they face threats from habitat loss and overharvesting. To safeguard these resources, conservation strategies must be developed to balance nutritional needs with sustainability.

xv. Sustainable Supply Chain Management in the Herbal Medicine Industry: Herbal medicine relies on various plant resources, making sustainable supply chain management crucial to ensure a consistent and ethical source of ingredients, while also supporting the livelihoods of local communities involved in cultivation and harvesting.

xvi. Consumer Perception and Demand for Sustainable Herbal Medicine Products and Market: Consumer demand for herbal medicine products with sustainability and ethical sourcing practices is on the rise. Understanding consumer perceptions and preferences is key to creating a thriving market for sustainable herbal remedies.

xvii. Indigenous Knowledge and Phytochemistry: Deciphering the Healing Power of Herbal Medicine: Indigenous knowledge plays a vital role in understanding the healing potential of herbal medicine. By combining traditional wisdom with modern phytochemistry, we can unlock the therapeutic properties of these natural remedies.

xviii. The Nexus of Business, Sustainability, and Herbal Medicine: The intersection of business, sustainability, and herbal medicine is an exciting frontier. It involves innovative approaches to commercializing herbal products while promoting environmentally friendly practices and ethical standards in the industry.

xix. Healing Trails: Integrating Medicinal Plant Walks into Recreational Development: Integrating medicinal plant walks into recreational development creates a unique experience for nature enthusiasts. These trails offer a blend of education, wellness, and conservation by introducing people to the world of medicinal plants while ensuring their sustainable use.

Conclusion

Herbal Medicine Phytochemistry: Applications and Trends offers a comprehensive exploration of the dynamic and ever-evolving world of herbal medicine and phytochemistry. Throughout this book, the rich tapestry of natural compounds found in plants, their intricate chemistry, and their diverse applications in healthcare, wellness, and beyond are presented. This is paramount considering the remarkable resurgence of interest in traditional herbal remedies, driven by the pursuit of natural alternatives, sustainability, and a profound understanding of the phytochemical treasures hidden within the plant kingdom, ancient cultures, and societies.

The book will take readers on a journey from the historical roots of herbal medicine to its modern renaissance, showcasing the valuable contributions of phytochemistry in unlocking the secrets of these medicinal plants. From the laboratories to the fields, it has revealed the diverse applications of herbal medicine, encompassing its use in traditional healing, mainstream healthcare, dietary supplements, and even cosmetics. It explores the transformative power of phytochemistry in identifying, isolating, and harnessing bioactive compounds that have the potential to combat diseases, alleviate symptoms, and enhance well-being. The innovative chapters featured in this book have illuminated the promising future of phytochemicals in modern medicine, paving the way for novel drug development, the creation of herbal formulations, and the enhancement of healthcare practices. It underscores the importance of responsible sourcing, sustainability, and quality control in the herbal medicine industry, highlighting the need to preserve nature's biodiversity and protect the heritage of indigenous knowledge. It has emphasized the imperative for rigorous research, safety evaluations, and standardized formulations to ensure that herbal medicine remains a safe and effective choice for individuals seeking natural solutions.

In a world where the demand for holistic and sustainable approaches to health is growing, *Herbal Medicine Phytochemistry: Applications and Trends* serves as an invaluable resource, shedding light on the boundless potential of phytochemistry and its impact on our lives. The book is a testament to the enduring significance of herbal medicine and the pivotal role of phytochemistry in the ongoing quest for wellness and the harmonious coexistence of humanity with the natural world. It is noteworthy that the world of herbal medicine and phytochemistry is a constantly evolving landscape. It is our hope that this book will provide readers, professional, practitioners, researchers, students, and enthusiasts with a deeper understanding of these fields and inspire curiosity, further research, policy establish, and responsible engagement. With this knowledge, we can look forward to a future where the healing powers of nature are harnessed, celebrated, and integrated into the healthcare systems of tomorrow.

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Part I

Introduction

Historical Perspectives and Overview of the Value of Herbal Medicine

1

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Abstract

Herbal medicine practices are as old as human history. Plants have been used to treat and manage different disease conditions in different parts of the world since time immemorial. Currently, about 80% of the global population uses herbal medicine directly or indirectly to manage or treat disease conditions. The use of plants to treat diseases depends on the local knowledge and cultural practices of the indigenous people of the area and requires a certain level of inborn skills or formal/informal training for the efficient and effective utilization of the herbal formulations. In herbal medicine, plants are used for spiritual processes including libation, sacrifice, appeasing of the gods, invocation, divination, food preparation, and as medicine. Whole plant or their parts are formulated into herbal medicine that is dependent on the bioactive constituents of the plant and their parts. Some diseases that have been successfully managed using herbal medicine include memory loss, cardiovascular issues, arthritis, osteoarthritis, digestive and respiratory diseases, reproductive problems, skin diseases, neurological issues, diabetes, diverse types of cancers, hypertension, and many others. Herbal medicines may be utilized in powder or liquid form and depending on the ailment may be ingested or externally applied on the affected parts of the body, but dosages are often unspecified or unclear. Herbal medicine is at a critical development stage, and there is a need for research and regulatory policies to support and standardize the processes involved in the exploration of raw materials, production of herbal remedies, branding, and marketing as well as sustainable utilization. Through research and regulatory policy, there will be greater clarity about herbal medicine processing, procedures, delivery routes, dosages, and compatibility, which will position the agelong process to be competitive, integrated into, or practiced alongside conventional medicine.

Keywords

Herbal medicine · Pharmacopeia · Traditional disease management · Physiological condition · Medicinal plants · Phytomedical research · Herbal medicine governance · Tradomedicine

Abbreviations

%	Percent or Percentage
AD	Anno Domini
AHM	Arab Herbal medicine
AM	Arabic medicine
AMM	Arabic materia medica
AMPK	5′ AMP-activated protein kinase
BC	Before Christ
BCE	Before the Christian Era
CAM	Complementary and alternative medicine
CAMHI	Conference of African Ministers of Health
COVID-19	Coronavirus disease 2019
FDA	Food and Drug Administration
HIV/AIDS	Human immunodeficiency virus acquired immunodeficiency syndrome
HPLC	High-performance liquid chromatography,
IIS	*C. elegans* insulin/IGF-1 signaling (IIS) pathway
JNK	c-Jun N-terminal kinases
MAPK	Mitogen-activated protein kinase
NFκB	Nuclear factor kappa-light-chain-enhancer
OAU	Organization of African Unity
ORID	Other related infectious diseases
SARS-CoV-2	Severe acute respiratory syndrome *coronavirus 2*
SATM	Standardized African traditional medicines
TB	Tuberculosis
TCM	Traditional Chinese Medicine
U.S.	United States
USA	United States of America
USD	United States Dollars
USG	United States Government
WHM	Western Herbal Medicine
WHO	World Health Organization

1 Introduction

Medicinal plants are the source and foundation for the effectiveness and efficacy of both traditional and modern medicine. In both industrialized and developing countries, medicinal plants are crucial in the management of diseases and are the foundation for many maladies that have been treated for thousands of years [85]. Traditional medicine is becoming more popular in both developed and developing nations due to its natural origins and perceived lack of negative side effects. Herbal medicine has gained increased attention at the policy level, as evidenced by the

Astana Declaration on Primary Healthcare [12], which promotes the impartial integration of both herbal medicine and scientific knowledge in basic healthcare. The declaration calls for greater political commitment to organizations and individuals that contribute to the provision of primary healthcare to everyone around the world. Herbal medicine is vital to primary healthcare because of the knowledge and capacity-building considerations. Nonetheless, it is impossible to not overstate the therapeutic significance of traditional medicine.

According to the World Health Organization (WHO), around 80% of the people in the world are reliant on herbal medicine for their basic healthcare needs ([85]; WHO, [208]). Based on the most recent estimates, two-thirds of the global populace are heavily reliant on unorthodox medical treatments because they are more widely accessible, more affordable, and more in line with patients' ideologies, placate the fears of negative effects from chemosynthetic medicines, satisfy the inclination for individualized healthcare, and enable widespread public access to personal and relevant health records [1, 37, 40]. As a result, several more individuals have embraced the use of indigenous plants. Chemically synthesized drugs are widely spread over the past century and have profoundly reformed medical services in several countries across the globe. Nevertheless, a larger fraction of the developing nation population continues to depend on natural remedies and traditional alternative medicine practitioners for their primary care. At least 90% of the African populace and 70% of the Indian descendants count on herbal medicine to aid and/or satisfy their healthcare needs [30, 49, 120]. Over 40% of all healthcare services provided in China are traditional medicine, whereas over 90% of general hospitals already established departments for traditional health medicine practice [54, 211, 216]. Furthermore, the use of traditional medicine is not restricted to third-world nations only; the last 200 years have witnessed an upsurge in the people's interest in natural remedies in developed nations, along with an expansion of the use of ethnobotanicals. Herbal medicine is at the heart of several healthcare systems across the globe – Western Herbal Medicine (WHM) is practiced in New Zealand, Canada, Australia, Great Britain, and Western Europe; Traditional Chinese Medicine (TCM) in China; Ayurveda in India; and Kampo in Japan. In addition, a variety of over-the-counter medicines approved by the European Union's Traditional Herbal Medicinal Product Regulation are widely available in pharmacies [62].

This chapter aims to provide a historical background to herbal medicine resources, practices and techniques, products, values, and challenges to the promotion of the agelong medical practice and highlight areas in need of attention to help increase the competitiveness of the practice and help integrate it into or ensure it is sustainably practiced alongside conventional medicine. It is organized into five parts – history of herbal medicine, utilization of herbal medicine, herbal medicine techniques, trends in herbal medicine, and herbal medicine research needs. This is the oldest form of medicine known to humans all over the world and is proven to be effective in the treatment and management of diverse diseases and ill health conditions. This work highlights the need for globally standardized procedures and research support to help clarify processing methods, routes of administration, doses, and compatibility issues.

2 Historical Perspectives of Herbal Medicine

It is difficult to clearly distinguish the history of herbal medicine into different eras because human societies evolved different herbal medicine practices and systems in isolation which contributed to the various forms of medicine available today. However, here we adopt a system that presents the views and historical practices of herbal medicine in some ancient human societies around different parts of the world, and these are arranged in no particular order.

2.1 Ancient Babylonians

The ancient Babylonians used plants as medical remedies as far back as 60,000 years ago. Written accounts of herbal medicine are at least 5000 years old in China and 2500 years old in Asia Minor and Greece [160]. There are several herbal medical systems, and each one's beliefs and practices are shaped by the area where it initially emerged [211]. Traditional Chinese medicine, which has been practiced for centuries, is the country's unique system [202]. The world's first known herbal book was published over 2000 years ago (*Devine Farmer's Classic of Herbalism*) and was written in China. Many herbal pharmacopeias and other monographs on certain plants are also available [200].

2.2 Ancient Hindustani Doctors

Ancient Hindu doctors and saints established Ayurveda, a medical system that has been practiced in India for over 5000 years. In its material medica, more than 1500 plants and 10,000 formulas are fully described. In contrast to Western medicine, the Indian government has acknowledged Ayurveda as a full healthcare system [53]. The Japanese herbal medication Kampo medicine has roughly 148 formulas and is over 1500 years old [205]. The majority of people continue to practice Ayurvedic medicine in India, Kampo medicine in Japan, Traditional Chinese Medicine, and Unani medicine in the Middle East and South Asia. The demand for herbal medicine and other botanicals by Western cultures has been continuously rising in the recent herbal renaissance period. Acupuncture and herbal treatment, which are both parts of TCM, are now more popular than ever in Western nations [21]. Together with herbal medications, other herbal items including cosmetics, perfumes, teas, health foods, and nutraceuticals are also widely used and make up a sizable component of the worldwide herbal market.

In actuality, it likely predates contemporary *Homo sapiens*. In Neanderthal graves in Iraq going back 60,000 years, archaeologists have discovered pollen and flower pieces from several different medicinal plants. Among them were species of *Ephedra*, *Centaurea*, *Senecio*, *Althea*, and *Achillea*. This can be evidence of the widespread usage of different herbal treatments at that very early time. According to legend, China has utilized *Cannabis* plant (*C. sativa* L.) for more than 8000 years [41].

The opium produced from the poppy (*Papaver somniferum* L.), which was first grown in Mesopotamia around 5400 years ago [19], has been used for medical purposes continuously ever since. Two fragments of the birch fungus *Piptoporus betulinus* (Bull.) Karst was found inside the mummified human body known as the "Iceman" that was found in the Italian Alps in 1991. According to scientists, this 5300-year-old person may have been utilizing the fungus as a medicine to cure intestinal parasites [24]. It is without dispute that herbal therapy has a long history.

2.3 Chinese Medicine

Chinese people have employed Traditional Chinese Medicine since ancient times. While components from animals and minerals have been employed, plants are the main source of cures. About 500 of the more than 12,000 items used by traditional healers are regularly used [103]. Botanical products are only used after being processed, which could include stir-frying or soaking in vinegar or wine, for example. In clinical practice, a complex and frequently individualized treatment may be prescribed after a conventional diagnosis. In China, conventional Chinese medicine is still widely practiced. Traditional medicine is regularly used by more than 50% of people, with rural areas having the highest rates of use. In China, there are over 5000 traditional medicines that makeup about one-fifth of the country's pharmaceutical sector [103]. Numerous herbal treatments made their way from China to the Japanese traditional medical systems. In the ninth century, the earliest pharmacopoeia of Japanese traditional medicine included a classification of native Japanese herbs [173].

2.4 Arab Herbalists

It is generally known that in the Middle Ages, Arab herbalists, pharmacologists, chemists, and doctors modified and enhanced classical Hippocratic-Greek medical knowledge. In addition, the vast majority of Arabs are Muslims, and Islamic philosophy and Arabic culture are intimately intertwined. As a result, Arabic medicine (AM), Arabic materia medica (AMM), and Arabic herbal medicine (AHM) are also known as Greco-Arab or Islamic medicine. Arabs in the Baghdad area were the first people in history to distinguish between medicine and pharmaceutical science in the eighth century. The Arab world is where the first pharmacies were opened (Baghdad, 754 CE). Certain medication formulations may still be found in pharmacopoeias today, and the forms utilized at the time are still used in treatment [116]. Mesopotamia produced the oldest records of herbs, which were recorded on clay tablets and written in cuneiform (dating back to 2600 BCE). The Ebers Papyrus, which dates back to 1500 BCE, is the most famous Egyptian medical record. It lists 700 herbal remedies (mostly derived from plants) and their dosage forms, including gargles, snuffs, poultices, infusions, pills, and ointments, as well as their carriers, including beer, milk, wine, and honey [171].

AHM has been practicing the use of organic and inorganic natural medicines for the prevention and cure of illnesses since the eighth century, including camel urine [123]. It is interesting to note that camel urine therapy had a significant cytotoxic impact on mouse bone marrow cells, according to pharmacological investigations [115]. Just 200–250 plant species are still used in traditional Arab medicine to treat different ailments, despite the Middle East being home to more than 2600 plant species, more than 700 of which are known for their usage as medical herbs or botanical insecticides [6].

From Alexandria to Sallum in Egypt, the western Mediterranean coastal area has 230 species of plants from 48 families; 89% of them were useful for medicine, 62% were common, around 24.9% were unusual, and 15% were rare [14]. The Mediterranean area and/or the worldwide market still sell or trades 236 plant species, 30 animal species, 29 organic compounds, and 9 materials of other or mixed sources that are still used to cure human ailments [6]. Around 250–290 plant species are still in use in the Mediterranean area, according to ethnopharmacologists' surveys of the region's plant species [101, 172]. A total of 129 plant species are used in AM in Israel to treat a range of illnesses. About 40 species of these plants are used to cure skin conditions, 27 species for digestive issues, 22 species for liver issues, 16 species for respiratory conditions and coughing, 22 species for different types of cancer, and 9 species for weight reduction and cholesterol lowering [15]. Yet throughout the Arab Empire, Islamic doctors employed more than 1400 different types of herbal remedies (632–1258). The late nineteenth and early twentieth centuries, up to the implementation of the Food and Drugs Act of 1906, were likely the height of herbal medicine use in the USA. Hundreds of patent herbal preparations, many of which included a significant amount of alcohol, were easily accessible on the US market before this regulation. Some of these, such as Swift's Syphilitic Specific, later shortened to SSS Tonic [191], and Lydia E. Pinkham's Vegetable Compound [192], continued to be sold and are still well remembered by many people today. Yet, the majority of these items and the absurd advertising that supported them were put out of business by the 1906 Act.

2.5 Aztec Medicine

This is a system of herbal medicine that evolved within Aztec societies where human health is managed holistically using spiritual and material bodies (plants and animal products and by-products). It began among the Nahuatl-speaking indigenous people of Mexico. Today, Mexico is the nation with the second-highest number of registered medicinal plants after China and has a long history of using medicinal plants [7, 50]. [36]. Indigenous traditional medicine in Mesoamerican societies began to emerge about 1500 B.C., beginning with the Olmec, the mother culture, and continuing with the Toltecs, Teotihuacans, and Mayans [170]. This came to an end, however, with the collapse and invasion of the Aztec empire in 1521 by Spanish colonizers, who lost a great deal of knowledge about the use of plants as medicines. After the conquest, however, it was possible to reconstruct the Aztecs' medical history. The *de la Cruz-Badiano Codex* (Código de la Cruz-Badiano), which was produced in 1552 by Nahuatl native Martin de la Cruz and translated by Juan Badiano, is the earliest book on medicinal plants used by Native Americans on the American continent [68]. Religion, astronomy, divination, and the polarity of cold and warm served as the foundation for Aztec medical doctrine. The estimated number of Indigenous people in Mexico now is 25.5 million (or 21.5%) [69]. The majority of herbal medicine used to treat common ailments is used by these Indigenous groups [29].

2.6 Indigenous or Native American Societies

Presently, 4.9% of Canada's population, or 1.7 million people, and 1.7%, or 5.6 million people in the USA, identify as Indigenous [110, 184]. The Indigenous people who resided on the land and were assimilated into several nations in Canada before the advent of European invaders, such as the Inuit and Metis, are known as First Nations [184]. Native Canadians' traditional medical practices are in danger of disappearing since they were historically replaced by the modern medical system and lost the trust of the populace as a result of conquering. As a result, governments, medical professionals, and European Canadian missionaries gradually banned indigenous healing practices [39]. For instance, with the Indian Act of 1876, Federal Indian policies increased and codified the monitoring and control of Native Americans' lives [52, 98].

Children from these villages were taken away by force in the 1960s and put in foster care to be adopted by non-Native families in Canada. Intergenerational trauma resulted from the assumption that the parents of these children did not have enough residences and educational opportunities to care for them [39, 88]. As a result, it has been challenging for the Indigenous community to transmit their ancestors' traditional medical knowledge. Nonetheless, this persecution and cultural marginalization have been reversing in recent decades. Indigenous peoples of Canada have used their traditional knowledge of medicinal plants for thousands of years, and it has been handed down verbally down the generations. Particularly significant in this sense for Indigenous peoples like the Metis, Cree, Dene, Sekani, Innu, Ojibwa, Chippewa, Abekani, and others are the boreal forest regions of Canada [195].

According to the most recent population census, the majority of the Indigenous people in the USA are American Indians, Alaskan Natives, and Hawaiian Natives, with a combined population estimated at five million (1.5%) [194]. Before the coming of European colonizers, Native Americans in the USA engaged in health practices that, during therapeutic sessions supported by goods like herbs, emphasized the connections between individuals (family and social groups) and the environment [215]. The loss of lands and other resources experienced by the American Indians and Alaskan Natives in particular during colonialism influenced their way of life, particularly the repression of their spiritual and healing practices, which is still happening today [215]. With the loss of their rights came the relocation of American Indians and Alaskan natives to reservations, separation from their holy places and medically significant flora and fauna. Congress was given extensive constitutional authority to negotiate with these groups in 1787. The American Indian Religious Freedom Act was enacted by the US Congress in 1978. However, many American Indians and Alaskan Natives are not recognized by the federal government [215].

Before the coming of the European conquistadors, traditional Native American medicine was used for generations. Yet these techniques were outlawed, and this information was given a lower value. Because of this, little is known about how these activities affect human health, and it is challenging to interpret both archaeological and human remains. The 1990 Native American Graves Protection and Repatriation Act, in particular, restricts research on the latter [193]. However, the contemporary tribal tribes are attempting to revive these historical rituals [215]. In the USA, the study of traditional indigenous medicine made from wild plants dates back more than a century [156]. The mesophytic woods of Appalachia are thought to include 1100 plant species of therapeutic use [25]. Eastern Tennessee, Western North Carolina, and Southeastern Virginia make up these woods; this area has a wide range of temperatures that favor the development of a wide diversity of plants. Although, traditional medicine and the use of medicinal plants can be traced back to Mesopotamia and Egypt, the foundation of European medicine was established by physicians and philosophers during Greek antiquity and the time of the Roman Empire, who began establishing a written consensus about what was considered effective medical knowledge. The oldest known fragment of a Greek herbal is preserved in the ninth book of Theophrastus of Eresos' (c. 370–287 BC) *Enquiry into Plants*, which is about the juices and therapeutic characteristics of plants [65]. Hippocrates of Kos (c. 460–370 BC), Galen (129–c. 199/200 or 216–217 AD), and Pedanios Dioscorides (first century AD) were the most illustrious and significant ancient writers who successfully merged botanical and medicinal knowledge. The most significant herbals ever published are considered to be Dioscorides' *De Materia Medica* and Galen's first alphabetical compilation of simple medications (*De simplicium medicamentorum facultatibus libri XI*). These writings had an impact on Mediterranean and European medicinal plant usage up to the eighteenth century via the constant copying and distribution of their content [11, 196].

Dioscorides discussed the therapeutic use of almost 900 herbal species, 600 medicinal plant species, 35 animals, and 90 mineral remedies [166]. A century later, Galen recorded roughly 850 simple medicines, most of which were similar to those named by Dioscorides but had far fewer medical use [100]. Each plant taxon was frequently referred to by a "code" made up of a list of the vernacular names and was known by various languages. This rigorous cross-referencing of names and uses served as a protocol for academic rigor to ensure accurate plant identification between regions, allowing herbal texts to be applicable in a wider geographical and cultural context and fostering the growth and consolidation of scientific knowledge.

2.7 Greek-Roman-European Herbalism

Greek herbals have a rather well-documented history, including their mutual effect and phylogeny. Lost copies may even be found by looking for missing connections [181]. Dioscorides, for example, cites and alludes to a variety of various writers and books, some of which have survived (such as works attributed to Theophrastus and Hippocrates) and others of which are now lost or have only partial copies left. Translations into Syriac served as a crucial intermediary step that permitted the transfer of Greek and Byzantine medical knowledge into Arabic and to the medieval Islamic world [169]. In addition to preserving classical medical knowledge, the Arab culture advanced it and made it possible for it to be transmitted back to the Occident via al-Andalus and the School of Salerno (about 1000–1300 AD) (711–1492; [196]). The analysis of the so-called Syriac Galen Palimpsest is likely to provide new and important insights on the transfer of Greco-Roman medical knowledge to the Arabic world. The earliest copy of Galen's writings on herbal remedies is now this manuscript, which is significant because it offers other readings from the more recent Greek versions that may be used to spot interpolations and other textual interpretations [18].

The middle and Northern European herbals were also affected by the traditional Mediterranean materia medica in terms of their organization, literary style, and substance. Back in the Medieval Ages according to monastic law, Benedictine monks were required to maintain therapeutic herb gardens, such as those at the abbeys at Montecassino, Italy, and St. Gall, Switzerland [190]. Classic Greco-Roman herbals and medicinal books underwent rigorous examination, commentary, and translation into contemporary languages throughout the Renaissance, garnering significant print runs. Renaissance herbals, however, also included the consensus of northern and central European medical folklore and were accompanied by woodcut drawings enforcing the commentators' botanical identifications.

The *Gart der Gesundheit* is recognized as the first thorough German herbal, authored by Johann Wonnecke von Kaub (ca. 1430–1504), physician to the city of Frankfurt and illustrated by Dutch artist Erhard Reuwich (1445–1505). [121]. The 435 monographs in the *Gart der Gesundheit* are based on texts from the Mediterranean region, including Pliny's *Natural History* from the first century AD, *The Canon of Medicine* by Avicenna (Ibn Sina, ca. 980–1037), the *Pseudo-Serapion* (Aggregator) by Ibn Wafid from the eleventh century in Toledo, and the *Circa Instans* by Matthaeus Platearius from the twelfth century [114].

The printing press' development led to a surge in the creation of European herbals and a race among writers, who weren't always free of nationalist emotions. The documentation of traditional folk knowledge on central and northern European materia medica was combined with the translation, commentary, and integration of ancient Greco-Roman knowledge by Peter Schoffer (1425–1503), Jan Stanko (1430–1493), Jean Ruelle (1474–1537), Otto Brunfels (1488–1534), Adam Lonitzer (1528–1586), Leonhard Fuchs (1501–1566), Hieronymus Bock (1498–1554), and William Turner (1508–1568). Nevertheless, a thorough examination of the problems of which plant species and applications were first recorded and in which herbal and how much materials were used is still absent.

The Greco-Roman and Arabian medical treatises were a major source of inspiration for the first formal pharmacopoeias. In addition to Hippocrates, Pliny, Dioscorides, and Galen, Matthaeus Platearius (twelfth century), Rhazes (865–925), and Avicenna were all significant figures (ca. 980–1037; [196]). Three texts written after 1000 AD at the School of Salerno, namely, the Antidotarium Nicolai (ca. 1100), the Antidotarium Nicolai Myrepsi (thirteenth century), and the Antidotarium or Grabadin of Pseudo-Mesue (ca. thirteenth century), are considered to be the forerunners of the official city pharmacopoeias due to the format and presentation used [196]. Constantinus Africanus (eleventh century), a Carthage herbal trader who collected Arabic medical works in North Africa and translated them into Latin, contributed a significant amount to the corpus of work created at the School of Salerno [121].

Roman era records of early imports of spices and herbal medicines from the Orient well-documented both the New World and Southeast Asia, showing the significance of exotic goods for European pharmacopoeias [198]. Prior to the creation of the European Pharmacopoeia, there were several factors that prevented attempts to synchronize pharmacopoeias, official or not, including the notion that locally sourced medicines would have a disproportionately high level of positive effects and the economic case for supporting locally produced goods [196]. The quest for effective treatments, often known as the trial and error method, during the outbreak of new epidemics (such as syphilis) and the testing of imported “strange” plant species supplied the laboratory for experimentation at the same time. After the Conquest, when Christopher Columbus and his ship returned from the Caribbean with their cargo, the syphilis pandemic was transferred from the Americas to Europe. The core wood of *Guaiacum officinale* L., a remedy used by the native people of the West Indies to treat syphilis, was imported to Europe and advertised against the disease, but later proved to be ineffective. This was because it was believed that the origin of specific diseases would be connected with the source of effective medicines [111]. Matthioli was the first to identify mercury’s particular usefulness against syphilis, even though Arabian pharmacists had brought the substance to Europe as a therapy for skin conditions [111].

The Swiss physician Paracelsus (1493–1541) attempted to depart from the traditional medical theories of the time, particularly those of Galen and Avicenna, and promoted high hygienic standards for surgery. Paracelsus is best known for developing the fundamental toxicology principle that “the dose makes the poison” [111]. Yet even Paracelsus, an alchemist, was susceptible to esoteric and symbolic thinking. The “doctrine of signatures” was put out by Paracelsus in De natura rerum in 1537 as a holistic and harmonic theory that could foretell the medicinal properties of plants based on characteristics like form, color, smell, and taste [121]. While the theory of signatures’ fundamental premise is founded in folk medical practices all over the globe, Paracelsus systematized it in conceptual contrast to Galen’s humoral pathology [121]. The German physician Samuel Hahnemann (1755–1843) developed homoeopathy, which still adheres to the idea of signatures today, although conventional herbal therapy in Europe is likely losing ground in this regard.

Since there were fewer early herbals and because they had more time to exert their causal influence on the spread of medical knowledge and popular herbal knowledge, the earlier herbals were generally more important. Due to their large print runs, which are estimated to have exceeded 30,000 for the older editions of *I Discorsi* alone, others, like Matthioli's Renaissance commentary on Dioscorides' Materia Medica (*I Discorsi*), originally published in 1544, also had a significant influence [99]. The work of Dioscorides was included into Matthioli's herbal, which also included 600 new medicinal species. It was also published in Italian and translated into more vernacular languages, including French, German, Czech, and Arabic, with a total of almost 60 distinct editions (Barberi, 1967–1970). *I Discorsi* was founded on Matthioli's experience as a medical doctor and botanist who was acquainted with the Mediterranean flora, making Matthioli's translation simple to understand [185].

2.8 Alkebu-Lan or African Herbalism

Alkebu-lan medicine may be the oldest form of herbal medicine practice and system from which every other system evolved considering the history and origin of humans. The lack of scripts and high secrecy associated with the practice in most parts of Africa makes it difficult to trace most of these practices, but remnants still exist to date in the form of African traditional medicine (ATM) on which a large proportion of the population depends on for their primary health care because it is cheap, easily accessible, and highly relatable [132, 134, 138, 139, 147, 152]. In large part, the system is connected to how plants are used both as food and medicine and their conservation [128–131, 133, 135–137, 140–142, 149–151]. According to Ozioma and Chimwe [153], ATM involves spiritualism, divination, and herbalism and permeates the culture, religious beliefs, attitudes, knowledge, and understanding systems of African communities. Practices include the use of materials (plants and or animal parts) with incantations, sacrifice, and exorcisms. The system was challenged and almost pushed out by the European and Arab colonizers who seek to increase the spread of their religion and modern medicine practice while the locals where challenged with economic (poverty) and other developmental issues. Enslaved Africans who were taken to different parts of the world may have contributed to the growth and development of the practice in those parts. Although enslaved they maintained small plots containing medicinal plants that are used in the preparation of herbal tinctures, decoctions, teas, incense, etc. for the treatment and management of diverse ailments. According to Okaiyeto and Oguntibeju [143], some factors contributing to the preservation of ATM include:

1. Accessibility and cost-effectiveness.
2. The perception that it is natural and safe.
3. Connection to their gods, dead relatives, and nature.
4. Superior efficacy to modern medicine.

5. Ability to self-medicate.
6. High levels of confidentiality associated with ATM.
7. Fear of erroneous diagnosis in the hand of modern medicine practitioners.
8. Long waiting periods are often associated with modern medicine in Africa.
9. Advertisement of herbal products, practices, and practitioners.

3 Utilization of Herbal Medicine

Globally, the interest and use of medicinal herbs have seen exponential growth with an estimated market value of USD 152 billion and a forecast of USD 350 billion by 2030 [47]. In Europe, Germany has the largest market for herbal medications with medical claims and those that are typically prescribed in pharmacies [62]. According to German manufacturer prices, the market costs €1.33 billion annually, which is proportionate to 20.7% of the entire European market. Other significant nations in conformity with the provision and sales of herbal medicines include Spain (6.3%), Poland (7.2%), Italy (17.1%), Russia (9.2%), and France (13.0%) [62]. A survey by Rashrash et al. [164] revealed that more than a third of adults in the USA utilize alternative medicine for the management of their health. The most frequently utilized alternative medicine, except for vitamins and minerals, was herbal products [16]. In most developed countries, over-the-counter sales of herbal medicine products can take in the form of plant oils, herbal supplements, or teas (such as chamomile, ginger tea, peppermint tea, green tea, etc.) sold in drug stores alongside conventional medicine. Traditional medical practices vary considerably from one nation to another and area to region due to influences from such things as culture, history, philosophy, and individual attitudes. Usually, their theory and practical application differ markedly from those of mainstream medicine [210].

Traditional medicine has been utilized in Nigeria to treat a variety of illnesses. In a study by Abubakar et al. [3], it was found that 85% of the respondents used traditional medicine to treat a variety of health issues, including diabetes, typhoid fever, pneumonia, pile, cancer, fever, measles, diarrhea, and cough. Accessibility, affordability, perceived safety, and therapeutic recommendation potential for treating numerous ailments are believed to be the major driving factors for the surge in the utilization of herbal medicine [124]. As such, the upkeep of human health has become the primary focus of the widespread usage of herbal medicine [4, 5, 70]. Past studies have revealed that Nigerians of various socioeconomic status engage in the use of herbal medicine [45]. Botanical treatments persisted in the form of tinctures, fluid extracts, and concentrated extracts. One might argue that such galenical medicines served as the cornerstones of period medicine. Products like the fluid extract of Ergot U.S.P. and the tincture of *Digitalis* U.S.P. were quite popular. With the advent of sulfanilamide in the middle of the 1930s, the first suppression of plant treatments started. The race was on as synthetic organic chemists created hundreds of novel compounds with potential therapeutic qualities in response to this new German antibacterial medication. Old plant medications were quickly superseded by new, often very helpful, synthetic pharmaceuticals for two reasons [167]. They could be marketed at a premium price to pay the expenses of the research needed in generating them while still allowing the maker to make a sizable profit since they were unique chemical entities covered by patent protection. Also, since they were unique chemical entities, it was simple to standardize their action and achieve predictable potency. In contrast, many of the traditional botanicals were likewise efficient but were abandoned since they could not be patented and did not guarantee profit margins. They were challenging to standardize since they often ascribed their action to many constituents, many of which remained unknown. Almost impossible was action uniformity. The issue is typified by Digitalis (*Digitalis purpurea*). The herb is prized as a treatment for congestive heart failure because it contains both short-acting and long-acting glycosides with rapid and slow onset, making it basically superior to any single glycoside isolated from it. However, it was difficult to accurately standardize the leaf in any way. Researchers eventually gave up after trying a variety of animals with little success, including guinea pigs, goldfish, chick hearts, frogs, cats, water fleas, and pigeons. As a result, digitalis leaf is no longer used in America. Eventually, almost all herbal remedies vanished from pharmacy shelves, and by the 1960s, American medications, unlike those used in the majority of other nations, were almost entirely synthetic. According to local custom, herbal remedies are utilized in several EU nations. For instance, according to the Allensbach Study from 2017, 70% of Germans have tried "natural medicines," with herbal treatments being the most popular kind. Without a prescription, by mail order, or in pharmacies, herbal medications worth 1.36 billion market data were sold in Germany alone in 2015 [67]. In addition to properly approved medications, other goods have been registered as traditional medicines due to their extensive history of usage. This presents some significant issues for ethnopharmacology, particularly in terms of how to establish a solid evidence foundation for the use of such items and define what exactly qualifies as a tradition of usage as a medicine. In the 1970s and 1980s, sophisticated European nations, particularly Germany, started to publish scientific and clinical papers demonstrating the medicinal and financial value of herbal treatments, which had never been completely abandoned in that region [175]. The oldest and maybe most diverse treatment approach is African Traditional Medicine. With its great biological variety and cultural diversity, which is reflected in regional variations in healing methods, Africa is regarded as the birthplace of humanity [10, 56]. Sadly, the systems of medications are still not well documented. Traditional African Medicine encompasses the body and mind holistically in all of its manifestations. Before administering medications, especially medicinal plants to address the symptoms, the traditional healer often determines the psychological cause of a disease and treats it [57].

The main natural resources ever employed by humanity for maintaining preventive, curative, and rehabilitative health in Africa include plants, minerals, and animals. These materials have been employed by traditional healers for more than 10,000 years, much like any other continent. In traditional African medicine, herbalists, midwives, and diviners are all involved. The diagnosis of illnesses, which in certain circumstances are said to be brought on by ancestor spirits and other forces, is the responsibility of diviners. Native plants are often used by traditional midwives to facilitate delivery. A herb trading market in Durban reportedly draws between 700,000 and 900,000 merchants annually from South Africa, Zimbabwe, and Mozambique since herbalists are so well-liked on the continent. There are smaller herb markets in almost every town [58].

4 Herbal Medicine Applications and Techniques

Plants contain a wide range of compounds, including minerals (molybdenum, potassium, calcium, sodium, zinc, magnesium, and so on) and phytonutrients which include sugars (cellulose, starch, rhamnose, sucrose, ribose, fructose, maltose, etc.), carotenoids (flavonoids, lignans, lutein, carotene, plant sterols, isothiocyanates, lycopene, and others), and polyphenols (cinnamic acid, vanillic acid, curcumin, ellagic acid, tannins, etc.) all of which play integral roles in the prevention of diseases and health management [59, 86, 95]. A majority of these compounds possess antioxidant properties, enhance immunity, prevent cancer, suppress inflammation, protect the brain, and effectively reduce the progression of aging [66, 95, 182, 199]. Plant components are used directly as chemotherapeutics; thus, ethnobotanicals are crucial in pharmacognosy research, drug discovery, and development. They can also be used as building blocks in the design and production of drugs or serve as models for pharmacodynamic substances [104]. Morphine (named after Morpheus – the Greek god of dreams) was the first physiologically effective pure compound, isolated from Papaver somniferum (opium) by Friedrich Serturner at the start of the nineteenth century [13]. This stimulated an avalanche of research on medicinal plants and led to the discovery of numerous lifesaving bioactive compounds of natural origin. Some known examples include artemisinin (*Artemisia annua*), aspirin (*Salix* spp.), atropine (*Atropa belladonna*), cannabinoids (*Cannabis sativa*), cocaine (*Erythroxylum coca*), codeine (*Papaver somniferum*), digoxin (*Digitalis purpurea*), galanthamine (*Galanthus nivalis*), quinine (*Cinchona officinalis*), and reserpine (*Rauwolfia* spp.), among others [13]. This epoch of drug discovery was further broadened to include microorganisms by Fleming's discovery of penicillin which established the scientific and economic platform for modern medicine and the biopharmaceutical industry [38].

According to recent aging studies, bioactive compounds have exhibited enormous promise for managing even the most severe ailments and may one day replace conventional medications [197]. Nevertheless, the effects of many plant products on humans are largely a mystery, despite the multitudes of research in this area. The best active ingredients and doses can be chosen to produce the optimal beneficial effects from herbal medicines by carefully analyzing the mechanisms by which they affect the aging process. It is better to examine in vivo the efficacy of biologically active substances and plant extracts with geroprotective characteristics. The animal model systems employed for this purpose include but are not limited to zebra fish (*Danio rerio*), mice (*Mus musculus*), fruit fly (Drosophila melanogaster), nematode

(*Caenorhabditis elegans*), and rats (Rattus norvegicus) [64, 118]. These models make it easier to thoroughly research how newly created medications affect the aging process. However, the model organism *C. elegans* is gaining increasing popularity in these studies because of its numerous advantages. They include a short life span; easy lab maintenance; transparent body for real-time imaging; advanced genetic, genomic, and molecular tools and manipulations; high genetic homology with humans; etc. [189]. Havermann et al. [61] revealed that the bioactive compound baicalein extended life span and improved stress resistance via the activation of Keap-1/Nrf-2 signaling pathway using C. elegans. In addition, luteolin and chrysin demonstrated life span extension via the activation of AMPK pathway in both *D. melanogaster* and *C. elegans* [97]. Rosmarinic acid, a natural polyphenol, has been shown to increase the average life span of *C. elegans* by upregulating the S pathway via *ins-18* and *daf-16*, the MAPK pathway via *skn-1* and *sek-1*, and stress resistance and antioxidant genes like *ctl-1*, *sod-3*, and *sod-5* [106]. Curcumin has also been reported to increase life span in *C. elegans*, which is dependent on the functions of *age-1*, *skn-1*, *sir-2.1*, *sek-1*, *unc-43*, *osr-1*, and *mek-1*, which are related to the S, MAPK, and JNK signaling pathways [105].

Furthermore, natural products remain the fulcrum of more than 60% of cancer therapeutics that are available or undergoing investigation. Over 70% of the globally approved 177 anticancer drugs are derivatives of natural products, and several of them have been optimized using combinatorial chemistry. Camptothecin (*Camptotheca acuminata*), a precursor for irinotecan and topotecan; combretastatin derived from *Combretum erythrophyllum*; paclitaxel, isolated from *Taxus brevifolia*; vincristine from *Catharanthus roseus*; noscapine from *Papaver somniferum*; and flavopiridol, a structural isomer of rohitukine derived from *Dysoxylum binectariferum* are some examples of plant-derived medications that have been employed for cancer therapeutics [31, 32, 158, 178]. In addition, alpinumi isoflavone and 4′-methoxylicoflavanone isolated from the stem bark of *Erythrina suberosa* exhibited cytotoxicity against human leukemia (HL-60 cells) through the suppression of the membrane potential in the mitochondria and activation of apoptotic proteins [96]. The chemoprotective and anticancer activity of curcumin has also been well established both in vivo and in vitro. Some of its known mechanisms include apoptotic stimulation via the inhibition of cyclin D1, cyclo-oxygenase-2, and other relevant NFκB target genes [188]; autophagy [27], mitotic arrest [55], and antioxidant and anti-inflammatory activities [48].

In addition, the inherent capacity of herbal medicine to fight COVID-19 became an important research focus during the global pandemic. Due to their potent suppression of SARS-CoV-2, traditional Chinese medications garnered a lot of interest. For instance, Qingfei Paidu decoction demonstrated an outstanding clinical efficacy of >90% in treating COVID-19 patients at all phases [119, 163, 201]. Shuanghuanglian, also popular in TCM, suppresses SARS-CoV-2 Mpro replication in a dose-dependent manner [187]. Natural products such as graveospene A, deguelin, and erianin have been utilized mono-therapeutically for the management of lung cancer in vitro, while baicalein, resveratrol, and ginkgolic acid were employed during the fight against SARS-CoV-2 [217]. Moreover, the combination of some Food and Drug Administration (FDA)-approved drugs with biologically active compounds has been reported to mitigate the pernicious effect of SARS-CoV-2 [217]. For instance, the combination of nelfinavir and cepharanthine, remdesivir and linoleic acid, cisplatin, and curcumin have all been reported [217]. It is noteworthy that more than 40% of pharmaceuticals approved by the FDA have come from natural sources or their derivatives [157].

Possibly, due to the phytochemical content of plants, many of them have been reported to be used as an alternative agent for the treatment of several diseases. Hence, they can be used to treat diseases caused by microbes possibly due to their antimicrobial potentials [42–44, 71–73, 77–83, 89–93]. It can be used to control vectors or insects such as mosquitoes [17, 74–76, 176, 177, 218]. They can be used as preservatives. Other diseases and conditions that can be managed with plants include memory loss, cardiovascular diseases, arthritis, osteoarthritis, digestive, respiratory, reproductive, skin, neurological diseases, diabetes, cancer, hypertension, gastrointestinal, and conditions such as pregnancy, etc. Two main factors that support the continued interest in traditional medicine in the African healthcare system include affordability and accessibility. Contemporary medical care is often too expensive, or not available. Secondly, although having an almost worldwide distribution, certain diseases like malaria and HIV/AIDS, disproportionately affect Africans more, and the reliability of traditional medicine makes them try it out [126, 127]. Medicinal plants are often the community's primary readily available source of healthcare in many regions of Africa. Also, they are often the patients' first choice. For the vast majority of these individuals, traditional healers provide information, counseling, and therapy to patients and their families in a personal way while also being aware of their patient's surroundings [10, 56, 58]. Africa is endowed with abundant biodiversity [144, 145, 161, 162, 174]; it is thought to have between 40 and 45,000 plant species with the potential to become developed, of which 5000 species are utilized medicinally. This is not unexpected given that Africa has a tropical and subtropical climate and that plants naturally collect significant secondary metabolites as a method of surviving in a harsh environment [109].

Written records of herbal medicine go back 5000 years in Egypt. The Edwin Smith, Ebers, and Kahun papyri include a wealth of information on Ancient Egyptian medicine. The Edwin Smith Papyrus and the Ebers Papyrus, both published in 1930 by Breathed and Bryan, respectively, are from the seventeenth and sixteenth centuries BCE. It is thought that these writings came from previous sources. These include spells and recipes for a wide range of illnesses and symptoms. They go through illness diagnosis and provide anatomical facts. They go into great depth into the anatomy, physiology, and medicine of the Ancient Egyptians. A gynecological literature called the Kahun Papyrus [51] discusses issues such the reproductive system, conception, pregnancy testing, childbirth, and contraception. Crocodile dung, honey, and sour milk are some of the substances that are recommended for contraception [168]. Pygeum (*Prunus africana*), a traditional African remedy for mild-to-moderate benign *prostatic hyperplasia*, has gained widespread acceptance outside of Africa. It has been offered in Europe since the 1970s. Pygeum barks are gathered annually in Madagascar and Cameroon in quantities of 600 metric tons and 2000 metric tons, respectively. The bark is brewed into tea in Africa [125].

It is offered everywhere in the globe in the forms of powders, tinctures, and tablets; often, it is mixed with other herbs thought to be beneficial for prostate issues. Customers experience less inflammation and cholesterol buildup, as well as easier urination. The significance of this therapeutic method in Africa may be shown by comparing the numbers of traditional healers and physicians. There is one traditional healer for every 700–1200 individuals in the Venda region of South Africa, compared to 1 doctor for every 17,400 people. For every 110 individuals, Swaziland has one traditional healer. The percentage is the same in Benin City, Nigeria. In urban Kenya, there is 1 traditional healer for every 833 people.

The chiefs of state and government of the former Organization of African Unity (OAU) in Africa acknowledged that 85% of the continent's people rely on it for their healthcare requirements. The OAU proclaimed 2001 to be the Decade of Traditional Medicine. Following this historic pledge by African leaders, the Plan of Action and implementation mechanism that were endorsed by the AU summit heads of state and government in Maputo in 2003 were adopted by the First AU Session of the Conference of African Ministers of Health (CAMH1), held in April 2003 in Tripoli, Libya. The Plan of Action's primary goal is for all Member States to recognize, embrace, advance, and institutionalize traditional medicine within the region's public healthcare system by 2010. [58]. Additionally, the Maputo Declaration on Malaria, HIV/AIDS and Other Related Infectious Diseases (ORID) of July 2003 resolved to keep supporting the Plan of Action for the AU Decade of African Traditional Medicine (2001–2010), particularly research in the area of treatment for HIV/AIDS, tuberculosis (TB), malaria, and ORID. The Lusaka Summit proclaimed the years 2001–2010 to be the OAU Decade for African Traditional Medicine in July of the same year. The 11 priority areas, which have been developed into strategic activities, are institutional arrangements, information, education, and communication, resource mobilization, research and training, cultivation and conservation of medicinal plants, protection of traditional medical knowledge, local production of standardized African traditional medicines (SATMs), partnerships, and evaluation. The African Union (AU) Member States have been carrying out the plan of action of the AU Decade of African Traditional Medicine and the priority interventions of the WHO regional strategy since 2001, namely, policy formulation, capacity building, research promotion, development of local production, including cultivation of medicinal plants, and protection of traditional medical knowledge and intellectual property rights [168].

5 Trends in Herbal Medicine

Now, several nations throughout the globe are still reacting to the Beijing statement from November 8, 2008. The World Health Organization assembly urged nations to take charge of their citizens' health by creating national policies, rules, and standards that would guarantee the proper, safe, and efficient use of traditional medicine [212]. Since traditional medicine has helped millions of people throughout the globe get cheap treatment, it has shown the ability to meet the WHO's goal for universal healthcare [26]. Studies demonstrate that patients who see a general practitioner with extra training in alternative medicine have lower healthcare expenses and death rates than those who do not support this claim [94]. Less use of prescription medications and hospital stays was said to have reduced costs. Hence, every government in the world has made the significance of complementary and alternative medicine (CAM), which mostly employs plant items as pharmaceuticals, a top priority. These medications are prescribed over the counter, utilized as over-the-counter home treatments, and serve as pharmaceutical industry raw ingredients. They make up a significant share of the global medication industry and help millions of people throughout the globe get basic healthcare. This section will examine the use, regulation, integration, and quality control trends for herbal medicine on a worldwide scale. Almost 100 million individuals in Europe use traditional medicine-related goods and procedures, with 5% utilizing it as an alternative to mainstream treatment and the same amount as a supplement to it [148]. According to reports from the UK, 40% of doctors reportedly recommend patients to alternative healthcare providers. Complementary and Alternative Medicine (CAM) is used by the populations of France, Canada, and Australia at rates of 49, 76, and 46%, respectively. In Asia, China is reported to have produced 83.1 billion US dollars worth of Chinese herbal medicine in 2012, while the Republic of Korea increased its yearly spending on traditional medicines from 4.4 billion US dollars in 2004 to 7.4 billion US dollars in 2009 [148].

Regarding regulation, conventional Western medicine is completely controlled by government agencies, as is the use of Traditional Chinese Medicine. Almost 1146 species had been listed by 2005 in the well-established Chinese Pharmacopoeia, which was created to assure the quality of Chinese material medicine. These species may be examined using over 479 different HPLC techniques, 45 TLC methods, and 47 GC methods [33, 34]. The Ministry of Standardization and Methodology in Japan has accepted and standardized over 148 formulas from various manufacturers [60]. In addition, over 210 formulations are available over the counter and are utilized in medical institutions. In Hong Kong, traditional healers are consulted by around 60% of the population. For instance, local researchers in Singapore employ software to examine multi-herb mixtures [203]. The usage of straightforward preparations (cut and dried) and those in other dosage forms is regulated by law (an example is existing statutes in Singapore). Only items that adhere to the necessary safety and quality criteria may be listed and authorized for production, importation, supply, or sale. In the USA, it was predicted that 14.8 billion dollars were spent directly on natural goods in 2008. According to further South American figures, traditional medicines are used by 71% and 48% of the populations of Chile and Colombia, respectively. In several of these nations, medical schools educate about traditional remedies. To assure the identification, quality, purity, potency, and consistency of herbal medications, the US FDA suggests a variety of tests. Chemical assays, distinctive markers, and chromatographic fingerprints are used in many testing for drug substances and products. Raw material process controls and drug substance validation are also conducted. Moreover, in the USA, the creation of herbal monographs has benefited greatly from microscopy [9].

Several African nations have not yet included these types of medications in their primary healthcare systems. According to Ghana, traditional medicine has been incorporated into South Africa's and Nigeria's current healthcare systems [209]. Others have partly incorporated it, such as Tanzania, Ethiopia, and Rwanda. More nations, such as Uganda, Chad, and Gabon, are creating integration policies. According to the World Health Organization, traditional medicines are the primary source of healthcare for 70–90% of Africa's rural population [210]. In a 10-year assessment, the World Health Organization found that the majority of African communities continue to rely on herbal therapy [213]. According to research by Chatora et al. [28], the majority of basic healthcare on the African continent is still provided by traditional medical practices and products. To substantiate Chatora's research report from 2003, Abdullahi [2] references the increased usage of traditional medicines in Africa despite the problems it faces. This indicates that, despite the changing circumstances, the need for herbal treatments has grown over time due to the growing population. The growth in chronic and noncommunicable illnesses, as well as their availability and cost, have all been linked to an increase in the use of traditional medicine, particularly herbal remedies, across the world [26, 102]. One reason for the widespread usage, for instance, is that middle- and low-income nations in Africa lack access to contemporary pharmaceuticals. According to a survey conducted in 36 low- and middle-income countries by Cameron et al. [23], a significant portion of the population cannot afford traditional medications. The evidence from nine African nations may support this.

The World Health Organization has created a traditional medicine plan for the years 2014–2023 to improve the integration of herbal medicines into member nations' primary healthcare systems. The main objectives of the approach are to assess, control, incorporate, and maximize the potential of traditional medicine for people's health and welfare [214]. By 2023, the plan will have identified some of the problems harming traditional medicines and strategies to work with the relevant member states and other stakeholders to resolve them. Its continuous use, expanding economic significance on a worldwide scale, advancements in research and development, intellectual property rights, and integration with healthcare systems are among the problems [214]. Evaluation of both safety and efficacy, quality control, regulatory and safety monitoring, education, and training of health authorities and control agencies are some of the areas in which research is required [211]. Australia is an island continent that has been inhabited by Aboriginal people for thousands of years. The many ecofloristic zones, including wet tropics, savannahs, evergreen forests, shrublands, grasslands, and wetlands, have taught over 500 different clan groupings or countries to coexist peacefully [87, 117]. While there are many distinct tribal groups in these very diverse places, what unites them all is their close and deep connection to nature and their utilization of its resources [107]. Native plants are an important part of their culture and are used to manufacture a variety of products, including food, medicine, narcotics, stimulants, adornments, ceremonial artifacts, weapons, clothing, shelter, tools, and artwork [35, 186]. Using local flora, animals, and abiotic resources over thousands of years, this native wisdom has developed [155]. Aboriginal lore serves as a means of knowledge preservation and transmission in the absence of written language. Songs, storytelling, dance, and other forms of art have been used to pass down practices and stories from one generation to the next. There are still hundreds of Aboriginal people who speak their native language and maintain the knowledge, songs, and rituals of their ancestors despite the disruption of oral traditions and lore practices by colonial contact, with English being the second or third language [113, 159]. Several Aboriginal clan groupings continue to practice Aboriginal ethnomedicine, although the amount to which it is employed varies greatly across rural and urban areas of Australia and between communities [146].

Aboriginal cultures still use numerous plants such as bush food and bush medicine, particularly in isolated locations. This makes Aboriginal plant knowledge the oldest surviving pharmacopoeia [154]. Australia has a total of 1511 plant species that have been documented as being used medicinally [154, 204]. The Aboriginal groups in the Northern Territory, New South Wales, South Australia, and Western Australia are responsible for the majority of the medicinal plant knowledge that has been documented in the literature [179]. Just a few publications on Aboriginal medicinal plants of Cape York Peninsula have been made, and there hasn't been much written on Native medicinal plants of Queensland [122, 180]. Several Native medicinal plants continue to be either unreported in the literature or unavailable to the general population [179]. Like many indigenous knowledge throughout the globe, Australian Aboriginal knowledge has been underexplored in a "Western scientific" sense and is at grave risk of disappearing entirely [84]. According to Stack [183], because the European immigrants brought their ailments and utilized their traditional treatments, Aboriginal traditional medicine did not play a role in their lives 200 years ago. Nonetheless, there exist documents that demonstrate that certain doctors and botanists collaborated with Native healers and conducted experiments with local plants for therapeutic reasons throughout the early years of European arrival. For instance, Denis Considen, the first assistant surgeon to Surgeon-General John White in the first fleet (1788–1794), claimed to have discovered indigenous medicinal plants before any other European physicians; however, his methods of discovery are unknown, and it is unclear whether he enlisted the help of indigenous informants [108]. Macpherson and Considen [108] noted the effectiveness of various indigenous medicinal plants, such as native sarsaparilla (*Smilax glyciphylla*) as an antiscorbutic and myrtle (*Eugenia australis*) and yellow gum (*Xanthorrhoea hastilis*) for dysentery [108]. In addition, local sarsaparilla was reportedly thought to be more pleasant than Jamaican or Central American varieties, according to Macpherson and Considen [108]. He said that Sydney herbalists had used it often as a trade item before 1927. Macadamia nuts were one of the plants that both the Aboriginal people and the European settlers traded regularly (*Macadamia* sp.). Notwithstanding the disruption of Native medical traditions by Europeans, Aboriginal people have made some significant contributions to global medical knowledge. For instance, Pearn [159] claims that Native childcare ethnomedicine is among "the world's oldest pediatric traditions" [159]. Native ethnomedical traditions are still a vibrant part of many Aboriginal communities today. Many groups in Australia employ various medicinal herbs depending on their local flora and habitat [146].

For instance, Southern Australia uses the fruits of the native plant *Solanum laciniatum*, whereas Eastern Australia once utilized *Solanum aviculare* (Kangaroo Apple) [183]. For joint swelling, both species were used as poultices [183].

The alkaloid solasodine, found in both of these *Solanum* species, serves as a precursor to cortisone and other steroids used in the creation of oral contraceptives (often known as "the pill") [112]. Due to their ability to biosynthesize this important phytochemical, these plants have been introduced to Russia and Eastern Europe, where they are now grown extensively [183]. A similar study was conducted on the congeneric *Duboisia myoporoides* when it was discovered that the native Australian Aboriginal narcotic plant pituri (*Duboisia hopwoodii*) was particularly useful. Aboriginals were chewing "pituri," a plant, in a manner like that of tobacco or East Indian betel, according to Joseph Banks, the first European botanist to visit the East Coast of Australia (in 1770) [46]. Ferdinand von Mueller, a botanist for the Victorian Government, recognized the plant as *Duboisia hopwoodii* a century later (in 1872) [165]. The related *Duboisia myoporoides* were discovered to generate hyoscine, now known as scopolamine, an alkaloid that is a very efficient remedy for motion sickness as reported by Von Mueller [63].

6 Research Needs in Herbal Medicine

There is a significant amount of work that needs to be done in the field of herbal medicines; however, this could be counterbalanced by the potential advantages of using these products, which are demonstrated by their consumption by significant populations, particularly in nations that are still developing. Be that as it may, it is necessary to research to ensure the identification of the specific compounds and bioactive elements contained within these herbal formulations [78, 79]. This will allow for the formulations to be purified and put to use treating specific diseases, as opposed to treating the disease with the whole plant [72, 73]. In addition, research is required to determine whether or not the use of extracted bioactive substances is preferable to the utilization of entire herbs. There is a significant need for additional studies to be conducted on the standardization procedures used to prepare herbal products. The evaluation of herbal medicine does not have a study approach that is considered adequate or generally acknowledged. It is necessary to have precise and repeatable procedures for the preparation of these medications to guarantee the distribution of the same herbal products despite differences in brand names and manufacturers. It is necessary to research to guarantee the safety of herbal goods in general. Further research is necessary to determine the processing methods, routes of administration, dosages, and compatibility with other herbal remedies or conventional treatments.

7 Conclusion

Herbal remedies have been used traditionally as medicine from the beginning of recorded human history. Plants have been used by humans in many parts of the world as a form of treatment and management for a wide variety of diseases and conditions. According to the World Health Organization (WHO) [206, 207], more than 80% of people all over the world utilize herbs as a kind of treatment when they are sick. Education is a crucial component in the usage of herbs since the capacity to utilize herbs as a therapy for diseases is based on the amount of information possessed by the native people of the area. Plants play a vital part in a wide variety of traditional religious rites, such as libation, sacrifice, the appeasing of the gods, invocation, and divination. In addition, plants are used in the production of food and the delivery of medicine. Plant parts are typically used in the manufacture of herbal treatments since these plant parts are the active elements in herbal medications. Some of the diseases and conditions that have been successfully managed by using herbal medicine include memory loss, cardiovascular disease, arthritis, osteoarthritis, digestive illness, respiratory illness, reproductive illness, skin illness, neurological illness, diabetes, cancer, hypertension, gastrointestinal illness, conditions such as pregnancy, and prenatal and postnatal procedures. Herbal therapy has also been shown to be effective in the treatment of a variety of other diseases and disorders. Herbal drugs are often administered either in powdered or liquid form, with the powdered form being the more prevalent of the two. Some of these products are intended to be used internally, while others can be put on the skin or rubbed into the part of the body that is problematic. There is still a significant amount of work that has to be done in the field of herbal medicines; however, this can be counterbalanced by the potential advantages that can be gained through the utilization of these items. Despite this, it is very necessary to conduct a study to guarantee the accurate identification of the particular chemical components as well as the physiologically active components that are found in these herbal treatments. There is a need for research to be conducted so that it can be determined whether or not the use of herbs in their unprocessed form, as opposed to the use of bioactive substances that have been extracted from the herbs, is more beneficial. There is a major need for additional studies to be conducted on the standardization procedures that are utilized during the manufacturing of herbal goods. This research should be conducted as soon as possible. To ascertain the processing methods, routes of administration, doses, and compatibility with other herbal medicines or conventional treatments, additional research is required.

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Part II

Phytochemistry and Herbal Medicine

Plant Food for Human Health: Case Study of Indigenous Vegetables in Akwa Ibom State, Nigeria

2

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Contents

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Abstract

The people of a particular culture and locale utilize plant resources within their environment for diverse purposes, e.g., as food. Plants that are utilized as food also contribute to the health and well-being of the populace. This chapter assesses the relationship between the people of Akwa Ibom State, Nigeria, and the crop plants found in their locality that contribute to health security. People from Akwa Ibom State rely mostly on vegetables for their diet and as medicines, and these plants play numerous roles in their culture. A total of 88 plants from 15 higher plant families were found to be used as food and medicine by the Akwa Ibom people. This chapter will describe these plants as well as their cultural practices, conservation status, and traditional knowledge and utilization patterns. These plants are either used solely or in combination with other plant resources to prepare different local relishes that possess medicinal value for the treatment and management of diverse ailments. These plants include *Cucurbita maxima, Gongronema latifolium, Abelmoschus esculentus*, *Piper guineense*, *Amaranthus hybridus, Justicia schimperi, Ocimum gratissimum, Telfairia occidentalis, Gnetum africana, Crassocephalum crepidioides, Heinsia crinita, Talinum triangulare, Vernonia amygdalina, Lasianthera africana, Colocasia esculentus, Cucumis sativus,* and *Ipomoea batatas*. Plants have been used as food and medicine for decades by all age groups in Akwa Ibom State, Nigeria, although some of them are considered underutilized. However, their utilization as food and medicine is on the increase in the present day, and they will most likely continue to be relevant in the future. Nonetheless, practical measures like value addition, better packaging, and conservation strategies are necessary.

Keywords

Ethnobotany · Indigenous vegetables · Akwa Ibom · Nigeria · Plant Conservation · Plant medicine · Food as medicine

1 Introduction

Ethnobotany is concerned with the dynamic relationship between people of a particular culture and their interactions with the diverse plant resources in their locality in terms of how they are utilized, what they are used for, when they are used, as well as their general relevance to life within the community or society. The field of ethnobotany evolved from the study of plants used by primitive societies [6, 42, 74, 76]. Subsequently, many other workers [43, 93] improved the concepts to include an interdisciplinary and holistic approach that focuses on the realm of the human-plant relationship. This realm of interaction between plants and human defined by economic, sociocultures, and environmental relationships is not limited to the use of plants for food, clothing, and shelter but also include their use for religious ceremonies, ornamentation, and health care [72, 80]. For instance, a previous survey revealed some vegetables eaten by Akwa Ibom people residing in Benin City, Nigeria, which highlighted ten vegetables including Ikong-ubong (*Telfairia occidentalis*), Mmong-mmongikong (*Talinum triangulare*), Atama (*Heinsia crinita*), Afang (*Gnetum africanum*), and Editan (*Lasianthera africana*), either consumed alone or in combination with other plant-based food resources or used in traditional medicines [99]. An ethnobotanical survey of plants is useful in identifying plants that serve in the alleviation of ailments [79, 80]. These plants work either in combination with one or more plants or are used solely. These plants have served and are still serving as leads in the production of orthodox medicines.

Indigenous vegetables are edible plant species peculiar to a particular locality or region with their diversity and distribution depending on geographical location and edaphic and climatic conditions [44, 46, 73, 75, 80]. Some of these plants are naturalized species and are well adapted to specific local conditions. According to Ogwu et al. [76, 77], these plants are grown for their leaves, succulent stems, young shoots, fruits, and/or combination of these plant parts. They may not require formal cultivation because they grow easily in the wild [69]. As protective foods, these indigenous vegetables are widely consumed in tropical regions including Nigeria as important sources of proteins, vitamins, and essential minerals [99]. The knowledge of indigenous plants used as food and medicines is often embedded in indigenous cultures but is slowly eroding with modernization. The decline of cultural diversity and the gradual erosion of human knowledge of medicinal plant species as well as their distribution, management, and methods of extracting medicinal chemicals is a contemporary challenge that is likely to increase in the future unless they are linked to their use as food [23, 24, 81].

This chapter aims to highlight some important vegetables with medicinal properties that are used by the people of Akwa Ibom State, Nigeria, and share their culinary uses and value in the treatment and management of diverse ailments. The chapters begin with an introduction to indigenous vegetables, Nigeria and Akwa Ibom State, and climax by presenting a list of indigenous vegetables associated with people from Akwa Ibom State and discussing their botany, traditional knowledge, and practices as well as their ethnomedicinal usage. The list of indigenous vegetables associated with Akwa Ibom people was sourced from personal knowledge of these vegetables, the work of Etukudo [31], and a few scholarly articles [95, 99]. The essence of the work is to give proper documentation of some of these species for easy access and to make recommendations on the need to conserve these vegetables as they are seen as underutilized crops.

2 Akwa Ibom State, Southern Nigeria

Akwa Ibom State (latitudes 4°32 and 5°53 North and longitudes 7°25 and 8°25 E) is flanked on the east, west, and north by Cross River, Abia, and Rivers states, respectively in the south-southern axis of Nigeria on the sandy deltaic coastal plain of the Guinea Coast (Fig. 1). To the south of the state is the Atlantic Ocean.

Akwa Ibom has a total land area of 8412 km^2 with altitudes of 45–70 m.a.s.l. and includes the Qua Iboe River basin and the lower Imo River basin [28, 99, 103]. The state is located within Nigeria's forest zone and has a tropical rainy climate. It has 31 local government areas composed mainly of the Ibibio, Eket, Annang, Oron, and Okobo ethnic groups. Ibibio is the largest ethnic group in Akwa Ibom and speaks the Ibibio language with diverse dialects [117]. The majority of the populace engages in farming, fishing, trading, hunting, wood-carving, raffia works, blacksmithing, pottery, ironworks, tailoring, and crafts creation. Native diets are mainly plant-based and are extolled for their medicinal values including efere afang, Edikang ikong,

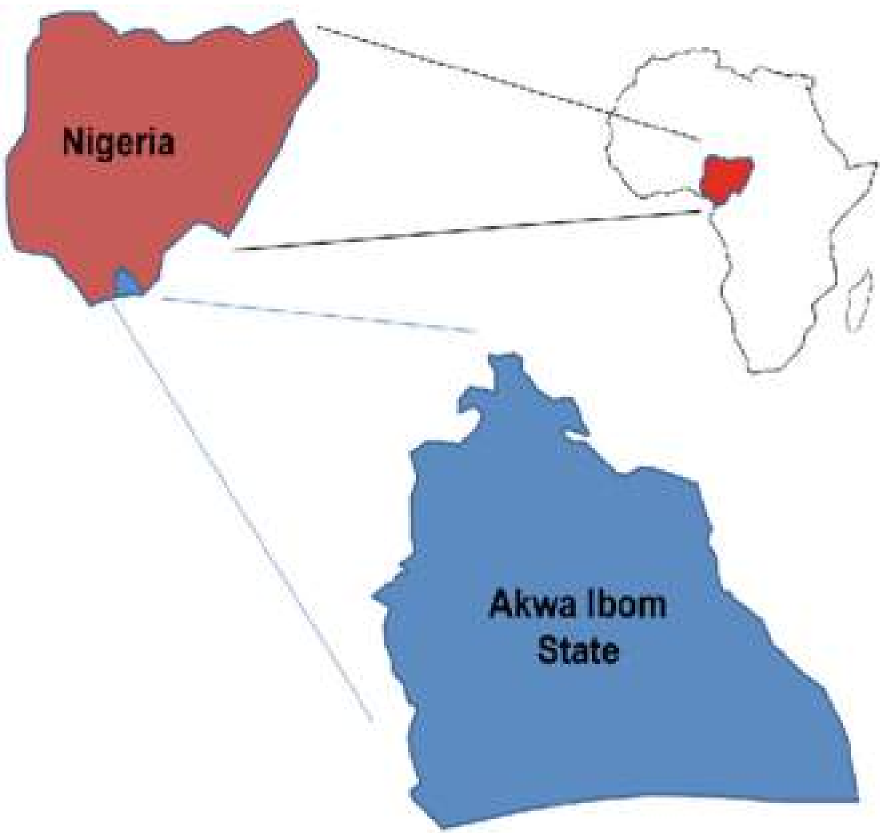

Fig. 1 Map of Africa showing Nigeria with Akwa Ibom highlighted in blue. (Source: Ref. [9])

efere etidot (bitter leaf soup), efere atama, efere editan, efere ikon, efere otong, afiaefere, eferendekiyak, efere etighi, efere ndukpauyo, efere mmongmmongikong, and eferenyama. Most of these soups go with coarse starchy food such as garri, pounded yam, wheat flour, and fufu.

3 Indigenous Vegetables Used as Food and Medicine in Akwa Ibom

The term vegetable is not strictly botanical but arbitrary and largely based on the mode of plant utilization rather than on their plant morphology [37]. For instance, Vainio and Bianchini defined vegetables as all edible plant parts including stem, stalk, root, tuber, bulb, leaves, flowers, fruits, or seeds, whereas Remison [106] suggested that plants are grown for their useable parts like leaves or young shoots, fruits, or a combination of usable plant parts like leaves and fruits. Vegetables can be fruity or leafy and may be soft and fleshy plant parts eaten raw or cooked, mature or immature, or both. Some plants used as vegetables also have horticultural values. Asaolu et al. [12] reported that vegetables are widely consumed in the processed, semi-processed, or raw form as part of a main dish or salad because of the taste they add to food. Many plants used as vegetables in Akwa Ibom are native to the Flora of Nigeria and considered resilient, adaptive, and tolerant to adverse climates. Knowledge of these plants has been transmitted from one generation to the next orally and without adequate documentation. The practice is still prevalent among rural and tribal communities within the state albeit unsustainable because of the potential loss of information about important species and best practices such as the method of propagation, mode of preparation, and preservation. These indigenous vegetables are cheap, flavory, and colorful and add aesthetic appeal to local diets [62]. They make up an important part of daily diets because of their carbohydrate, protein, mineral, vitamins, fiber, and other nutritional contents [32, 65]. In Akwa Ibom, despite their low calorific value, they are eaten either in the form of leaves, seeds, fruits, flowers or pods, roots, stems, and tubers and are a source of many nutrients including potassium, folic acid, vitamin A, and vitamin C [85, 91]. The low caloric content of these local vegetables is because plants produce food in the leaves but many do not store food in the leaves [32, 45, 73, 83]. Fibers in vegetables are known to promote digestion and prevent constipation [46, 64, 96–98]. The nutrients in vegetables can be absorbed as regulatory and protective nutrients, as well as for bodybuilding [12]. During the rainy season, vegetables are abundant in the wild, market, and home gardens and are relatively cheaper unlike in the dry season when there is general scarcity leading to higher prices as demands cannot be met [94]. A few of these vegetables are domesticated and marketed, thus serving as a source of household income.

Akwa Ibom State indigenes are well known for their local delicacies with high food and medicinal values. These delicacies are prepared using traditional methods. The scientific and common names of some of these vegetables are presented in Table 1, while their plant family distribution is presented

Table 1 Indigenous vegetables common to Akwa Ibom people

Family	Species	Names in Akwa Ibom
Gnetaceae	*Gnetum Africana*	Afang
Rubiaceae	*Heinsia crinita*	Atama
Portulaceae	*Talinum triangulare*	Mmongm mongikong
Cucurbitaceae	*Telfairia occidentalis*	Nkong
Icacinaceae	*Lasianthera Africana*	Editan
Asteraceae	*Vernonia amygdalina*	Etidod
Asclepiadaceae	*Gongronema latifolium*	Utasi
Malvaceae	*Abelmoschus esculentus*	Etighi
Lamiaceae	*Ocimum gratissimum*	Nton
Piperaceae	*Piper guineense*	Odusa
Amaranthaceae	*Amaranthus hybridus*	Iyanafia
Acanthaceae	*Justicia schimperi*	Mmemme
Araceae	*Colocasia esculentus*	Eka ikpori
Asteraceae	*Crassocephalum crepidioides*	Mkpafid
Convolvulaceae	*Ipomoea batatas*	Ediam makara
Cucurbitaceae	*Cucumis sativus*	Okokon
Cucurbitaceae	*Cucurbita maxima*	Ndise
Euphorbiaceae	*Microdesmis puberula*	Ntanebid, Ntabid
Zingiberaceae	*Aframomum melegueta*	Ntuen-ibok
Amaryllidaceae	*Allium sativum*	Etebe-owo inua
Liliaceae	*Aloe vera* (and *A. barbadensis*).	Akokafid
Gentianaceae	*Anthocleista djalonensis*	Ibu
Acanthaceae	*Acanthus montanus*	Mbara ekpe
Amaranthaceae	*Achyranthes aspera*	Udok mbiok, Udok mbiet
Amaranthaceae	*Alternanthera bettzickiana*	Nkpok isip essien
Amaranthaceae	*Cyathula prostrata*	Nkibe ubuk
Amaryllidaceae	*Allium cepa*	Ayim
Caricaceae	*Carica papaya*	Okpod, popo
Polygalaceae	*Carpolobia lutea*	Ikpafum
Anacardiaceae	*Lannea acida*	Ayara nsukakara
Anacardiaceae	*Lannea nigritana*	Odok eto
Anacardiaceae	*Anacardium occidentale*	Cashew
Anacardiaceae	*Mangifera indica*	Mango
Annonaceae	*Uvaria chamae*	Nkarika ekpo, atama nkana
Annonaceae	*Monodora myristica*	Enwun
Araceae	*Caladium bicolor*	Ikpon ekpo, udia edi
Araceae	*Xanthosoma sagittifolium*	Ikpon mbakara
Asteraceae	*Tridax procumbens*	Ayara utimense
Asteraceae	*Emilia sonchifolia*	Utime nse, usio mmon
Asteraceae	*Crassocephalum crepidioides*	Mkpafit
Boraginaceae	*Heliotropium indicum*	Otukeyin eka, esin ono
Bignoniaceae	*Spathodea campanulata*	Esenim
Bignoniaceae	*Newbouldia laevis*	Itumo, oboti

(continued)

Table 1 (continued)

Family	Species	Names in Akwa Ibom
Bignoniaceae	*Kigelia Africana*	Ntabinim
Bromeliaceae	*Ananas comosus*	Eyop mbakara
Burseraceae	*Dacryodes klaineana*	Eben ikot
Clusiaceae	*Allanblackia floribunda*	Udiaebion, ekporo-enin
Combretaceae	*Combretum micranthum*	Asaka
Combretaceae	*Terminalia ivorensis*	Nkot ebene
Convulvulaceae	*Ipomoea quamoclit*	Ediam ikanikot
Convulvulaceae	*Ipomoea pileata*	Mkpafiafian
Costaceae	*Costus afer*	Mbritem
Crassulaceae	*Bryophyllum pinnatum*	Ndodop afiaiy
Cucurbitaceae	*Citrullus colocynthis*	Ikon, ikpan
Cucurbitaceae	*Momordica balsamina*	Mbiadon edon
Dioscoreaceae	*Dioscorea dumetorum*	Enem, edidia iwa
Euphorbiaceae	*Euphorbia hirta*	Etinkene ekpo
Euphorbiaceae	*Jatropha curcas*	Ukim eyio
Euphorbiaceae	*Manihot esculenta*	Iwa
Euphorbiaceae	*Manniophyton fulvum*	Ekonikon
Euphorbiaceae	*Mallotus oppositifolius*	Uman nwariwa
Fabaceae	*Afzelia Africana*	Eyin mbukpo
Fabaceae	*Afzelia bella*	Eyin mbukpo
Fabaceae	*Albizia lebbeck*	Ubam
Fabaceae	*Baphia nitida*	Afuo
Fabaceae	*Cajanus cajan*	Nkoti
Fabaceae	*Cassia alata*	Adaya okon
Fabaceae	*Glycine max*	Nkoti eto
Fabaceae	*Lonchocarpus cyanescens*	Awa
Fabaceae	*Parkia biglobosa*	Ukon uyayak
Fabaceae	*Pentaclethra macrophylla*	Ukana
Fabaceae	*Pterocarpus erinaceus*	Ukpa
Fabaceae	*Pterocarpus santalinoides*	Nkpa-inyan
Fabaceae	*Tetrapleura tetraptera*	Uyayak
Hypericaceae	*Harungana madagascariensis*	Oton
Irvinginiaceae	*Irvingia gabonensis*	Uyo

in Fig. 2. Almost 80 plants from 15 higher plant families are presented here based on oral historical records and the work of Etukudo [31]. Cucurbitaceae and Asteraceae were the plant families with the highest representations. Earlier, Osawaru et al. [99] conducted an ethnobotanical survey of vegetables eaten by Akwa Ibom residents in Benin City, Nigeria, and reported a total of ten vegetables. Similarly, Ajibesin et al. [9] surveyed plants in Akwa Ibom State used for skin disease and other related ailments and documented 183 medicinal plant species representing 153 genera and 59 families.

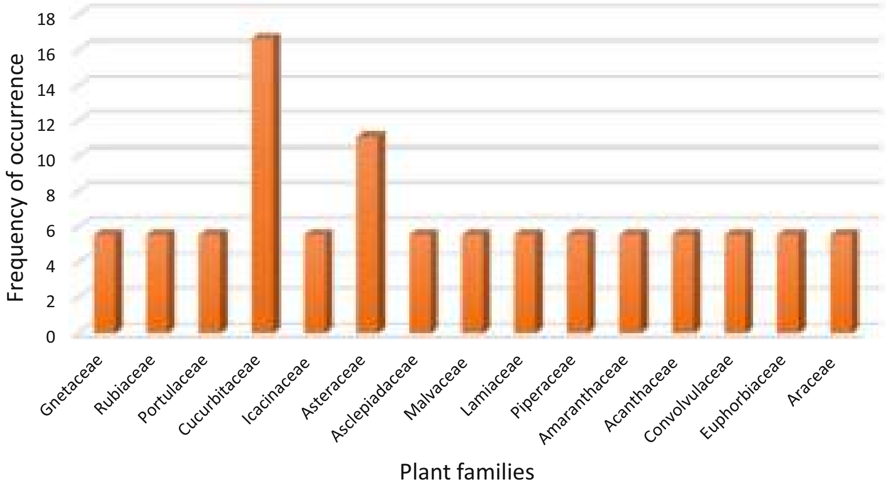

Fig. 2 Plant family distribution of indigenous plants used as food and medicine by Akwa Ibom people, Nigeria

The utilization mode and pattern, parts used, and cultivation status and sources of these indigenous vegetables used for food and health by Akwa Ibom people are presented in Table 2.

4 Botanical Description, Knowledge, and Practices Associated with the Vegetables Indigenous Used as Food and Medicine by Akwa Ibom People

The description and local utilization and knowledge and practices associated with some indigenous vegetables used as food and medicine in Akwa Ibom are presented below.

4.1 *Gnetum africanum*

Description, utilization, knowledge, and practices. It is a creeping vine. The leaves are collected as forest products. The leaves are used in the preparation of the local delicacy "Efere afang." The different recipes on how the dish is prepared mark the cultural identity of the Akwa Ibom people. Soup is one of the main meals for any kind of ceremony. The stem is used in preparations for medical concoctions to ease childbirth. The seeds and leaves are used in the treatment of piles, sore throat, and high blood pressure. It also serves as a purgative and the supple stem is sometimes used as rope. The mineral element composition of *G. africanum* is as follows: Ca, 130 ppm; Mg, 89.0 ppm; Fe, 76.1 ppm; and Zn, 1.3 ppm, while cadmium and lead were not detected [48].

Table 2 Status, utilization, and conservation status of the indigenous vegetables

Species	Status	Utilization (food and/or medicine)	Part used
Gnetum Africana	Wild; cultivated	Food, medicine	Leaves, stem, and seeds
Heinsia crinita	Wild; cultivated	Food, medicine	Leaves
Talinum triangulare	Wild; cultivated	Food, medicine	Leaves, stems
Telfairia occidentalis	Cultivated	Food, medicine	Leaves, seeds
Lasianthera Africana	Cultivated; wild	Food, medicine	Leaves
Vernonia amygdalina	Wild; cultivated	Food, medicine	Leaves
Gongronema latifolium	Wild; cultivated	Food, medicine	Leaves
Abelmoschus esculentus	Cultivated	Food, medicine	Leaves, seeds, pods
Ocimum gratissimum	Cultivated	Food, medicine	Leaves
Piper guineense	Cultivated	Food, medicine	Leaves
Amaranthus hybridus	Cultivated	Food, medicine	Leaves
Justicia schimperi	Cultivated	Food, medicine	Leaves
Colocasia esculentus	Cultivated, wild	Food	Whole plant
Crassocephalum crepidioides	Cultivated	Food, medicine	Leaves
Ipomoea batatas	Cultivated	Food, medicine	Leaves, roots
Cucumis sativus	Cultivated	Food, medicine	Whole plants
Cucurbita maxima	Cultivated	Food, medicine	Leaves, seeds, roots
Microdesmis puberula	Cultivated	Food, medicine	Whole plant
Aframomum melegueta	Cultivated	Medicine	Leaves, seeds
Allium sativum	Cultivated	Food, medicine	Bulb
Aloe vera (and *A. barbadensis*)	Cultivated	Medicine	Leaves, juice
Anthocleista djalonensis	Wild	Medicine	Bark, leaves
Acanthus montanus	Wild	Medicine	Stem, leaves
Achyranthes aspera	Wild	Medicine	Seeds, leaves
Alternanthera bettzickiana	Wild, cultivated	Medicine	Leaves, young shoot
Cyathula prostrata	Cultivated	Medicine	Leaves
Allium cepa	Cultivated	Food, medicine	Bulb, leaves
Carica papaya	Cultivated	Food, medicine	Leaves, seeds, fruits
Carpolobia lutea	Cultivated	Food, medicine	Leaves, fruits, stembark
Lannea acida	Semi-cultivated, wild	Medicine	Bark, leaves
Lannea nigritana	Cultivated, wild	Food, medicine	Seeds, fruit, roots
Anacardium occidentale	Cultivated, wild	Food, medicine	Bark, leaves, fruits
Mangifera indica	Cultivated	Food, medicine	Fruits, leaves, stem

(continued)

Table 2 (continued)

Species	Status	Utilization (food and/or medicine)	Part used
Uvaria chamae	Wild	Medicine	Root
Monodora myristica	Cultivated	Food, Medicine	Seeds
Caladium bicolor	Wild	Medicine	Leaves, rhizome, corm
Tridax procumbens	Wild	Medicine	Leaves
Emilia sonchifolia	Wild	Medicine	Leaves, young shoot
Crassocephalum crepidioides	Wild	Medicine	Leaves, roots
Heliotropium indicum	Wild	Medicine	Leaves
Spathodea campanulata	Cultivated	Medicine	Seeds, roots, bark
Newbouldia laevis	Cultivated	Medicine	Stem- bark
Kigelia africana	Cultivated	Medicine	Seeds, fruits
Ananas comosus	Cultivated	Food, medicine	Fruit
Dacryodes klaineana	Cultivated	Food, medicine	Fruit, leaves
Allanblackia floribunda	Semi-cultivated, wild	Medicine	Seed, bark
Combretum micranthum G	Cultivated	Medicine	Seed, leaves, roots
Terminalia ivorensis	Cultivated, wild	Medicinal	Bark, root
Ipomoea quamoclit	Cultivated	Medicine	Leaves
Ipomoea pileata	Cultivated	Medicine	Leaves
Costus afer	Wild, cultivated	Medicine	Leaves, fruits
Bryophyllum pinnatum	Cultivated	Medicine	Leaves
Citrullus colocynthis	Cultivated	Food, medicine	Seed, fruit
Momordica charantia	Cultivated	Food, medicine	Fruit, young shoots, leaves
Dioscorea dumetorum	Cultivated	Food, medicine	Root, tuber
Euphorbia hirta	Wild	Medicine	Young leaves and shoot
Jatropha curcas	Cultivated	Food	Seeds
Manihot esculenta	Cultivated	Food, medicine	Tuberous roots, young leaves
Manniophyton fulvum	Wild, semi-cultivated	Medicine	Seed, root, leaves, stem
Mallotus oppositifolius	Wild, semi-cultivated	Medicine	Leaves, stem bark
Afzelia africana	Semi-cultivated	Medicine	Leaves
Afzelia bella	Cultivated	Medicine	Leaves, bark
Baphia nitida	Cultivated	Medicine	Seeds
Cajanus cajan	Cultivated	Food, medicine	Seeds, shoot, leaves
Cassia alata	Wild, cultivated	Medicine	Leaves, seedpods
Glycine max	Cultivated	Food, medicine	Seeds

(continued)

Table 2 (continued)

Species	Status	Utilization (food and/or medicine)	Part used
Lonchocarpus cyanescens	Cultivated	Medicine	Leaves, roots
Parkia biglobosa	Cultivated	Food, medicine	Seed, Leaves
Pentaclethra macrophylla	Cultivated	Food, medicine	Stem, fruit, seed
Pterocarpus erinaceus	Cultivated	Medicine	Stem bark
Pterocarpus santalinoides	Cultivated	Medicine	Seed, leaves, root
Tetrapleura tetraptera	Cultivated	Food, medicine	Seed
Harungana madagascariensis	Cultivated	Food, medicine	Fruit, bark
Irvingia gabonensis	Wild, cultivated	Food, Medicine	Seed

Cultivation requirements and management within Akwa Ibom: The vegetables are often grown in the wild although some are cultivated in the home gardens. The stem cuttings are usually planted in the soil and properly watered till the plant germinates. As they grow, the plant requires some support such as a tree to grow into.

4.2 *Heinsia crinita*

Description, utilization, knowledge, and practices: It is usually cultivated or wild. The vegetable is used in the preparation of the local dish "Efere Abak." It is also used in pediatric care for the treatment of measles. The mineral element composition of *H. crinita* is reported as follows: K, 84.41 mg/g; Ca, 24.20 mg/g; Mg, 11.31 mg/g; and P, 9.14 mg/g [89].

Cultivation requirements and management within Akwa Ibom: The vegetable is propagated by seed and stem cuttings on loamy soil. The cultivated crops grow into a shrub or trees and are available all year round.

4.3 *Talinum triangulare*

Description, utilization, knowledge, and practices: The leaves and stems are used as a softener when cooking fibrous vegetables such as Afang (*G. africanum*), Atama (*H. crinita*), and fluted pumpkin (*T. occidentalis*). It is also used for its high medicinal value, usually in the treatment of diuretics and stomach problems [67].

Cultivation requirements and management within Akwa Ibom: Waterleaf is propagated by seed but also vegetatively by cutting 15–20 cm of the matured stem. It is a fast-growing plant that once established can easily reseed itself because it is selfpollinating and produces flowers early year-round. It is an ephemeral crop that is due for harvest between 35 and 45 days after planting (Udoh and Etim 2008). Some are wild or weedy.

4.4 *Telfairia occidentalis*

Description, utilization, knowledge, and practices. It is a vine plant often grown in a homestead garden. Due to its creeping nature, it is usually staked into a framework of bamboo, or a shelf is made for it to climb on when twining [88]. *Telfairia occidentalis* shoots and leaves are used in preparing Edikang ikong soup. Much importance is placed on the crop for its edible seeds which are rich in protein and fat and can be eaten as a whole by boiling. The sliced leaves can be mixed with coconut water and salt used for the treatment of convulsion. The leaves are also used in the management of anemia because of the high iron content, while the roots are used as rodenticides [35, 68].

Cultivation requirements and management within Akwa Ibom: Seeds are planted directly in the soil, typically in groups of three to increase output in a case of a failed germination. Although dependent on soil type, the fluted gourd can grow and subsequently produce many flushes of fruit over long periods [7]. The seeds cannot be stored for more than 3 days once they are extracted from the fruit.

4.5 *Lasianthera africana*

Description, utilization, knowledge, and practices. It is a perennial, glabrous shrub. The leaves are highly valued for their medicinal properties including the ability to relieve stomach complaints with or without diarrhea, as vermifugal, and for cold as a wash against headache. Leaf sap is considered anti-psoric and is crushed in warm water, which raises an abundant froth that is used to bathe feverish infants. Pulped leaf or pulped bark is made up into a dressing to bind over fractures. Tests for alkaloids have given mild response for the leaf, moderate for the bark, and strong for the root, and tannin is present in all these parts. Notwithstanding the presence of alkaloids and tannin, certain tribes like the Akwa Ibom people put the leaf in soup. The leaves are used as antacid, analgesic, antispasmodic, laxative, antipyretic, antiulcerogenic, antidiabetic, and antimalarial because of their bacteriostatic, anti-ulcer, fungicidal, antidiabetic, and antiplasmodial properties [29, 49, 50, 86, 87]. The plant also has social roles in ceremonies as well as in food preparation. The mineral element composition of *L. Africana* is as follows: potassium, 56.24 mg/g; calcium, 31.43 mg/g; magnesium 31.99 mg/g; and phosphorus, 5.00 mg/g [89].

Cultivation requirements and management within Akwa Ibom: It is cultivated through the propagation of adult stem cuttings [84].

4.6 *Vernonia amygdalina*

Description, utilization, knowledge, and practices. It is a shrub and can grow into a small tree. The leaves have a characteristically bitter taste and the extracts are used medicinally in the treatment of cough, fever, and pile and as a laxative. The vegetable is used in the preparation of the local relish such as efere etidod, and efere ikon. The leaves are washed before use to get rid of their bitter taste. The stems are used as a chewing stick for oral hygiene as well as the management of dental problems. They can be found as weeds in the wild and cultivated in the home garden. The active ingredient (vernonioside B) possesses antiparasitic, antitumoral, and antibacterial agents [114].

Cultivation requirements and management within Akwa Ibom: It is cultivated through the propagation of adult stem cuttings.

4.7 *Gongronema latifolium*

Description, utilization, knowledge, and practices: It is a climbing shrub usually used as a spice and vegetable [119]. In soups, it imparts its beautiful and agreeable bitter taste to the food. It is also eaten alone raw or added to other meals like plantain porridge, yam porridge, etc. The leaves are also incorporated into salads. The twigs and branches are used as chewing sticks [31]. Abbiv [1] records that the leaves are rubbed on joints to enable children to walk early. It is a cough, diabetes, malaria, and hypertension remedy [30]. Leaf extract or cold infusion of pounded leaf mixed with lime or pineapple juice is used as a blood tonic and is also known to expel intestinal worms and help with upset stomach and for the treatment of typhoid fever [30].

Cultivation requirements and management within Akwa Ibom: *G. latifolium* is propagated by soft stem or hard stem cuttings.

4.8 *Abelmoschus esculentus*

Description, utilization, knowledge, and practices. It is a tall erect annual plant with edible leaves, pods, seeds, and flowers. The leaves, flowers, and young shoots are eaten as vegetables. It contains mucilage which gives the soup a slimy, sticky texture. It can be sliced, dried, powdered, and preserved for future use. This serves as a flavoring and as a soup thickener. The seeds are eaten as snacks after roasting.

Cultivation requirements and management within Akwa Ibom: *A. esculentus* are cultivated through seedlings. Usually, the seeds are soaked overnight to have uniform germination. Two to three seeds are sowed per hole at 3 cm depth. The crop is usually watered regularly.

4.9 *Ocimum gratissimum*

Description, utilization, knowledge, and practices: It is a perennial shrub. The leaves have a strong fragrance with a strong aroma. The leaves are used as a seasoning in the preparation of pepper soup. Medicinally, it is used in the treatment of high fever and piles. The leaves may be taken for catarrh medication. A paste of the leaves is applied topically against ringworm and skin diseases. Crushed leaf extract together with leaf extracts is used to treat diabetes and pile. More so, most of the respondents claim that the scent from the leaves of the plant is also used to protect against snakes.

Cultivation requirements and management within Akwa Ibom: *O. gratissimum* is propagated by seeds and soft stem cutting and is usually available all year round.

4.10 *Piper guineense*

Description, utilization, knowledge, and practices: It is a climbing shrub commonly used as a spice in the preparation of meals especially soups like efereetek, edikangikong, afiaefere. This crop is used to facilitate the swallowing of coarse-textured starchy food. The vegetable is also used by Akwa Ibom people to add flavours to their local dishes. Ethnomedicinally, it is used in the regulation of the menstrual cycle. The fruits and leaves are used as a spice for preparing soup for postpartum women.

Cultivation requirements and management within Akwa Ibom: The crop is propagated by seeds.

4.11 *Amaranthus hybridus*

Description, utilization, knowledge, and practices: It is an erect annual plant with leaves used as vegetables in cooking porridge and vegetable sauce and also in the local delicacy ekpang nkukwo [125]. The crushed leaves are useful as an antidote for snake and scorpion bites. The plants are also used by the tribal people to feed animals.

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds and they germinate readily if the soil is warm.

4.12 *Justicia schimperi*

Description, utilization, knowledge, and practices: It is an annual or perennial herb. The leaves are edible and are used in cooking soups, plantain porridge, and many other delicacies. The leaves are cooked in a palm fruit filtrate and fed to babies.

It is used as an enema to treat umbilical problems in infants. The leaves are also used in treating chest and heart problems (Inyang [31]).

Cultivation requirements and management within Akwa Ibom: It is usually propagated by seeds and germinates readily with the availability of rain.

4.13 *Colocasia esculentum*

Description, utilization, knowledge, and practices: It is a perennial plant commonly grown for its starchy edible underground tuber that is consumed after boiling, frying, or roasting with soup or stew. It can be cooked as cocoyam porridge or grated and used to prepare ayan ekpang which is eaten with a mucilaginous soup rich in meat, periwinkle, and fish [31]. The corm and leaves are used for the preparation of the local delicacy ekpang nkukwo. They are mostly cultivated in home gardens around the homestead, while some can be found as weeds. The tender petioles are leaves that are also eaten when young. The matured leaves are used as wrapping materials. Consumption of the plant (corm and leaves) is believed to have effect on constipation, diarrhea, skin rash, injuries and neurological disorders.

Cultivation requirements and management within Akwa Ibom: The top of the corm or the whole cormel is used for propagation usually on ridges when the rains are reliable [106].

4.14 *Crassocephalum crepidioides*

Description, utilization, knowledge, and practices: It is an annual herb with soft stems. The leaves and tender shoots are edible as vegetables or spinach, especially by elderly people on very auspicious occasions. The leaves when used as soup cure indigestion and give general comfort and well-being to the gastrointestinal system [31].

Cultivation requirements and management within Akwa Ibom: It is propagated by seeds or through stem cuttings.

4.15 *Ipomoea batatas*

Description, utilization, knowledge, and practices: It is a herbaceous perennial creeping vine. The tender leaves and young shoots are eaten as vegetables or spinach. They are also used to wrap grated cocoyam mash during the preparation of ekpang nkukwo, especially in the dry season when the usual wrapping leaves are scarce and costly. Leaves are antidiabetic and antiscorbutic.

Cultivation requirements and management within Akwa Ibom: Cultivation is done by stew cuttings about 30 cm long or portions of tubers. Planting is usually done in ridges or mold. The crops are planted when the rains are well established [106].

4.16 *Cucumis sativus*

Description, utilization, knowledge, and practices: It is a trailing or climbing monoecious annual herb with an extensive and largely superficial root system [106]. The green fruits are eaten fresh in salad and are rich in vitamin B, iron, and calcium and believed to have diuretic and purgative effects. The leaves are eaten as vegetables either raw in a salad or cooked [20]. The fruit contains 95% water. Proteins, fat, carbohydrates, fiber, and ash make up the chemical composition of *Cucumis sativus*.

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds. The seeds are sown directly in ridges or molds or prepared planting holes. Plants seeded directly may be thinned to two or three plants per stand after establishment. The plant is sometimes stalked especially for some creeping cultivars [106].

4.17 *Cucurbita maxima*

Description, utilization, knowledge, and practices. It is an annual herb with thick climbing or creeping stems [107]. The leaves, tender shoots, and flowers of the plants are used as vegetables or pot-herb cooked in soups and is believed to help relive intestinal infections, stomach discomfort, and kidney issues. They are also used to prepare yams, cocoyam, sweet yam, and water porridge. The pulp is eaten after the whole fruit had been cooked. It can be eaten salted with stew or soup or as a vegetable.

Cultivation requirements and management within Akwa Ibom: It is propagated through direct seedlings in homestead gardens and grows well on organic matter [107].

4.18 *Microdesmis puberula*

Description, utilization, knowledge, and practices: *Microdesmis puberula* is a shrub or miniature tree. Edible leaves are used to prepare afang soup. The leaves may be pounded and mixed with alligator pepper seeds or ginger and applied on sprains, burns, and bruises for healing [31]. The twigs are used as chewing sticks to clean teeth and strengthen gums. It is sometimes taken orally or applied as an enema to treat diarrhea and is prescribed for pregnant women and young children [128].

Cultivation requirements and management within Akwa Ibom: It is propagated by seeds and can germinate massively in fallow land.

4.19 *Aframomum melegueta*

Description, utilization, knowledge, and practices: It is a perennial herb. The seeds are used as a traditional spice (ground or whole) for native ceremonies to welcome guests and ward off evil spirits. The seeds have a hot, pungent black-peppery flavor with a hint of citrus [113]. It is used traditionally in threes or sevens. It is used in almost if not all medicaments. The seeds are revulsive and carminative. They are used to strengthen drinks among other uses. The seeds are edible and are a stimulant.

Cultivation requirements and management within Akwa Ibom: It is propagated by seeds.

4.20 *Allium sativum*

Description, utilization, knowledge, and practices: It is a perennial herb. It is used to flavor soups, salads, and sausages. It is a classic ingredient of many cuisines and has a pungent, spicy flavor. The leaves are also edible. It is used in the treatment of several health-related issues like dysuria, and bronchitis [31]. They can also be used as antibiotics and antiseptics [102].

Cultivation requirements and management within Akwa Ibom: Propagation is by asexual reproduction. The cloves are planted in the soil and probably watered.

4.21 Aloe vera (*A. barbadensis*)

Description, utilization, knowledge, and practices: It is a perennial herb up to 160 cm tall, usually with or without a stem. It is a well-known medicinal plant with dermatological and other health benefits in Akwa Ibom. It is used in skin care, to alleviate stress, for heartburns, indigestion, etc. among other uses.

Cultivation requirements and management within Akwa Ibom: It is propagated vegetatively.

4.22 *Anthocleista djalonensis*

Description, utilization, knowledge, and practices: It is a tree growing in the wild. It is widely used as a strong purgative and diuretic. The bark is usually used to cure painful menstruation and gonorrhea. It is used as an abortifacient [26]. It is used in the treatment of several other ailments.

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds.

4.23 *Acanthus montanus*

Description, utilization, knowledge, and practices: It is a prickly semi-woody herb, nearly 2 m high. The plant is used as an ornamental because of its beautiful flowers and peculiar leaves. The leaves are taken by pregnant women to stop internal lower abdominal heat. Similarly, leafy twigs are used to cure several problems such as upset stomach, painful menstruation, cough, and whooping cough [31].

Cultivation requirements and management within Akwa Ibom: It is vegetatively propagated.

4.24 *Achyranthes aspera*

Description, utilization, knowledge, and practices: *Achyranthes aspera* is a perennial, erect herb used in the treatment of dropsy, piles, and boils in children as well as of asthma, bleeding, bronchitis, cold, cough, colic, debility, dog bite, dysentery, headache, ear complications, leukoderma, pneumonia, renal complications, scorpion bite, snake bite, and skin diseases [51]. It is also used as herbal medicine, especially in obstetrics and gynecology to treat abortion, induction of labor, and postpartum bleeding [56].

Cultivation requirements and management within Akwa Ibom: It grows in the wild as an invasive species.

4.25 *Alternanthera bettzickiana*

Description, utilization, knowledge, and practices: *Alternanthera bettzickiana* is an herbaceous perennial herb. It is used in amenity planting [31]. The leaves are cooked as a vegetable and used in the management of anemia in children, as well as hepatitis, tight chest, and asthma, and to improve general health and well-being [60].

Cultivation requirements and management within Akwa Ibom: It is propagated by seeds or cuttings.

4.26 *Cyathula prostrata*

Description, utilization, knowledge, and practices: It is an annual to perennial branched herb/shrub often harvested for local use, especially as a medicine but also as food. It is used to treat many ailments including dysentery, wounds, and eye trouble [20].

Cultivation requirements and management within Akwa Ibom: It is propagated by seeds and stem cuttings.

4.27 *Allium cepa*

Description, utilization, knowledge, and practices: It is a perennial plant grown as an annual. It is used as a spice, flavoring agent, food plant, and vegetable. The immature and matured bulbs may be eaten fresh or cooked and eaten as a vegetable. *Allium cepa* is added to soups and sauces as a seasoning and flavoring agent. Onions are eaten raw to ease cough [31]. It is used as hypotensive, diuretics, and depurative.

Cultivation requirements and management within Akwa Ibom: It is propagated by sowing the bulb directly into the soil.

4.28 *Carica papaya*

Description, utilization, knowledge, and practices: Large herbaceous perennial plant with a soft single stem and sparely arranged leaves at the top of the trunk, whereas the lower trunk is scarred where leaves and fruits are born [112]. The fruits are edible. It is also used medically for weak digestion. Pawpaw juice is used in the treatment of eczema and ulcer [31]. Traditionally, the leaves are used in the treatment of malaria as well as an abortifacient and purgative agent or smoked to help remedy asthma (Titanji et al. 2008).

Cultivation requirements and management within Akwa Ibom: This plant is cultivated by seed.

4.29 *Carpolobia lutea*

Description, utilization, knowledge, and practices: It is a shrub or small tree with sweet edible fruits. The soft stem is used as a chewing stick. It is used to cure stomachaches and bone fractures and to boost sexual performance [70]. The root is used in the treatment and management of reproductive issues including as an aphrodisiac, the facilitation of childbirth, and treatment of sterility as well as other ailments like headache and worm infestation [63].

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.30 *Lannea acida*

Description, utilization, knowledge, and practices: It is a multipurpose shrub cultivated for its medicinal and edible values. Its berry-like fruits occur in large clusters and may be consumed fresh or dried [66]. The plant parts are used in the treatment of various ailments including skin injuries and inflammations, general body pain, gastrointestinal problems, fever and malaria, gynecological and pregnancy disorders, hemorrhoids, skin diseases, and infections [39].

Cultivation requirements and management within Akwa Ibom: Propagation is done by seeds.

4.31 *Lannea nigritana*

Description, utilization, knowledge, and practices: It is a small shrub with brownish flowers and fruits. It is used for amenity planting. The leaves are used to treat burns hence its Ibibio name (Uma-ikan). The bark is used by Akwa Ibom people to treat intestinal pains and dysentery. The leaves are administered as an enema for treating abdominal complaints [19].

Cultivation requirements and management within Akwa Ibom: Propagation is done by seeds.

4.32 *Anacardium occidentale*

Description, utilization, knowledge, and practices: It is a shrub cultivated for diverse uses including as medicine and as a source of commodities (raw material). The fruits are eaten raw or cooked and usually have an astringency that leaves the mouth feeling furry. The leaves are used as a febrifuge and in the treatment of malaria [10]. The fruit is anti-scorbutic, astringent, and diuretic [109]. The nutshell is used locally to draw tattoos. The sap or bark is considered to be a contraceptive [15]. The root is used as a purgative.

Cultivation requirements and management within Akwa Ibom: Propagation is done by seeds and usually requires very little management practices.

4.33 *Mangifera indica*

Description, utilization, knowledge, and practices: *M. indica* is a large, evergreen tree grown for its edible fruits. It is one of the oldest cultivated plants. The fruit is consumed for its fresh spicy flavored taste. The bark is used either through decoction or infusion is used to treat jaundice, rheumatism, diarrhea, and dysentery. The flowers are used to repel mosquitoes [15]. The leaves have astringent properties and are infused to reduce blood pressure and treat asthma, cough, and diabetes [15].

Cultivation requirements and management within Akwa Ibom: Propagation is done by seeds and usually requires very little management practices.

4.34 *Uvaria chamae*

Description, utilization, knowledge, and practices: It is a climbing shrub or small tree. The plant is harvested from the wild for edible fruits, while the medicinally important roots are sold in local markets. The root purgative, respiratory, and antipyretic benefits are used by Akwa Ibom people to treat dysentery, nasal congestions, and catarrh [31].

Cultivation requirements and management within Akwa Ibom: It grows in the wild.

4.35 *Monodora myristica*

Description, utilization, knowledge, and practices: it is a perennial edible plant. The seeds are used as a spice because of the nutmeg-like odor and taste. The seed has anti-sickling properties and is consumed to maintain good health [122]. The stem and bark are used in the treatment of stomachaches, fever pains, and eye disease [57].

Cultivation requirements and management within Akwa Ibom: It is cultivated by seeds.

4.36 *Caladium bicolor*

Description, utilization, knowledge, and practices: *Caladium bicolor* also called fancy-leafed caladium, elephant's ear, and heart of Jesus because of its leaves and ornamental value. In Akwa Ibom, it grows in the wild as tubers and produces berry fruits with several small ovoid seeds. Fresh leaves are used in the treatment of angina, while dried powdery leaves are placed on sores for them to heal and as facial treatments [17].

Cultivation requirements and management within Akwa Ibom: It grows in the wild.

4.37 *Tridax procumbens*

Description, utilization, knowledge, and practices: *Tridax procumbens* is a prostrate herbaceous plant that is covered with erect stiff hairs [120]. The leaves are opposite, simple, and thick and with dense hairs. The extracts of *Tridax procumbens* leaves have antimicrobial activity and stimulate wound healing [115]. In Akwa Ibom, the leaves are crushed and used in the treatment of skin spots [31].

Cultivation requirements and management within Akwa Ibom: It grows in the wild.

4.38 *Emilia sonchifolia*

Description, utilization, knowledge, and practices: *Emilia sonchifolia* is a branching, annual herb with lyrate-pinnatilobed leaves [52, 66]. In Akwa Ibom, the plant leaves are crushed to release the juice which is then applied to wound, measles, and rashes.

Cultivation requirements and management within Akwa Ibom: It is propagated wild by seed.

4.39 *Crassocephalum crepidioides*

Description, utilization, knowledge, and practices: It is an erect annual, slightly succulent herb. The fleshy, mucilaginous leaves and stems may be eaten despite the presence of plant toxins [5, 25, 38]. The leaves and stems help with indigestion and act as a laxative, purgative, and remedy for lever problems [13]. In Akwa Ibom it is also used to treat wounds, boils, and burns.

Cultivation requirements and management within Akwa Ibom: It is propagated wild by seed.

4.40 *Heliotropium indicum*

Description, utilization, knowledge, and practices: *Heliotropium indicum* is a robust annual herb that occurs as a weed [101, 105]. Traditionally, the plant leaves are used for wound healing, antidote, bone fracture, boils, febrifuge, eye infections, menstrual disorders, nerve disorders, kidney problems, and antiseptic purposes [116].

Cultivation requirements and management within Akwa Ibom: It is propagated wild by seed.

4.41 *Spathodea campanulata*

Description, utilization, knowledge, and practices: It is a tree that grows between 7 m and 25 m (23–82 ft) tall and is valued for its timber used as firewood, fodder, and fence [33, 41, 120]. The sap sometimes stains yellow on fingers and clothes. The seeds are used as a poison to kill pests [21]. The stem, bark, and leaves of *Spathodea campanulata* may be crushed and soaked in water or gin and used for the treatment of skin eruption, skin lesions, burns, ulcers, and bruises [8].

Cultivation requirements and management within Akwa Ibom: It is propagated wild by seed.

4.42 *Newbouldia laevis*

Description, utilization, knowledge, and practices: Newbouldia laevis is a fast-growing evergreen shrub or small tree [19]. In Akwa Ibom, the stem, bark, and root are used to prepare concoctions for treating boils, skin lesions, and spots.

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.43 *Kigelia africana*

Description, utilization, knowledge, and practices: *Kigelia africana* is a tree grows that bears poisonous fruits and can grow up to 60 cm. The wood of *Kigelia africana* is pale brown or yellowish, undifferentiated, and not prone to cracking. [108]. The stem and bark are used in Akwa Ibom in the treatment of wounds and sore.

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.44 *Ananas comosus*

Description, utilization, knowledge, and practices: Ananas comosus is a herbaceous, monocot biennial, or perennial tropical plant that produces fleshy, edible fruits. In Akwa Ibom, it is commonly known as Eyop mbakara. Mature pineapple is used in traditional medicine by mixing the fruit peel with other materials to treat rashes [8] or stomachache.

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.45 *Dacryodes klaineana*

Description, utilization, knowledge, and practices: *Dacryodes klaineana* is an evergreen perennial tree known as Eben Ikot in Akwa Ibom. The leaves and roots are used in the treatment of skin spots and rashes.

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.46 *Allanblackia floribunda*

Description, utilization, knowledge, and practices: *Allanblackia floribunda* is an evergreen tree, which grows up to 30 m [55]. The seeds are eaten fresh or made into jams and jellies. Stem decoction and leaves are used to treat skin infections.

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.47 *Combretum micranthum*

Description, utilization, knowledge, and practices: *Combretum micranthum* is a shrub is known as Asaka in Akwa Ibom. The seeds are edible. Infusions of the leaves are used in the treatment of skin lesions and spots [8].

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.48 *Terminalia ivorensis*

Description, utilization, knowledge, and practices: *Terminalia ivorensis* is known as Nkot ebene in Akwa Ibom. The plant is useful in the treatment of ulcers, cuts, wound sores, body pains, high temperatures, hemorrhoids, and diuresis, as well as malaria and yellow fever [20, 59].

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.49 *Ipomoea quamoclit*

Description, utilization, knowledge, and practices: Ipomoea quamoclit is a herbaceous vine from the family Convolvulaceae known as Ediam ikanikot in Akwa Ibom. The leaves are Poultice and are used traditionally for the treatment of boils and small open injuries.

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.50 *Ipomoea pileata*

Description, utilization, knowledge, and practices: *Ipomoea pileate* is an annual or perennial twining herb that grows from a thick taproot. In Akwa Ibom, it is commonly known as Mkpafiafian. The leaves are crushed in water and applied to skin spots.

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.51 *Costus afer*

Description, utilization, knowledge, and practices: *Costus afer* is a tall perennial semi-woody herb. In Akwa Ibom, it is generally known as Mbritem and is mostly used for several ailments including arthritis, rheumatism, eye treatments, nasopharyngeal affections, stomach pains and laxatives, sexual stimulants, antidepressants, venereal diseases, febrifuges, leprosy, dropsy, swellings, oedema, and gout stem.

Cultivation requirements and management within Akwa Ibom: It is propagated by seed.

4.52 *Bryophyllum pinnatum*

Description, utilization, knowledge, and practices: *B. pinnatum* is commonly called the resurrection plant or cathedral bells. In Akwa Ibom, it is generally known as Ndodop Afiaiyo. The leaf extract is used for the treatment of kidney stones, ulcers, bacterial and viral infections, asthma, and ulcers. It has analgesic properties, and recent research shows that it has antimutagenic and anticancer properties [127].

Cultivation requirements and management within Akwa Ibom: Propagation is done by seeds. In addition, plantlets arise from the edges of their leaves and take root when it drops on the ground. It requires little care, but during the hot seasons, it should be shielded from direct sunlight or else the leaves will be burned.

4.53 *Citrullus colocynthis*

Description, utilization, knowledge, and practices: *C. colocynthis* is a perennial creeping herbaceous vine commonly found in desert areas. It looks like a watermelon, and common names include bitter apple, desert gourd, bitter apple, vine of Sodom, etc. In Akwa Ibom, it is called Ikon. The seed oil has antioxidant, anticancer, antidiabetic, and insecticidal properties. The fruit is bitter and pungent and used as a purgative, anthelminthic, and carminative cure for tumors, ulcers, asthma, leukoderma, etc. The seeds are crushed and used to clear skin spots and abscesses [2].

Cultivation requirements and management within Akwa Ibom: Cultivation is by seeds. It requires good light and good soil-based compost. It requires care only at the initial stage because it thrives in pots and nursery beds at the initial stage.

4.54 *Momordica charantia*

Description, utilization, knowledge, and practices: *M. balsamina* is commonly called an African pumpkin. In Akwa Ibom, it is called Mbiadon Edon. It is a tendril-bearing, wild climber with small lobed leaves, monoecious flowers, and ovoid ellipsoid fruits which are softly warted and fleshy. The leaves, fruits, seeds, and bark contain resins, alkaloids, flavonoids, glycosides, steroids, terpenes, cardiac glycosides, and saponins that are known to have medicinal properties including antiviral, antiplasmodial, antidiarrheal, antiseptic, antibacterial, antiviral, antimicrobial, analgesic, and hepatoprotective properties [40]

Cultivation requirements and management within Akwa Ibom: Cultivation is by seeds. Care is required at the initial stage because it thrives better in a nursery.

4.55 *Dioscorea dumetorum*

Description, utilization, knowledge, and practices: *D. dumetorum* is commonly called bitter yam. In Akwa Ibom, it is called Anim. It has distinct trifoliate leaves each having 3–7 veins per leaflet. The vine has ridged ascending prickles and a characteristic twining to the left. The vine grows annually from the underground tubers which are deeply lobed and occur in clusters just below the surface of the soil.

Bitter yam lowers blood sugar level, boosts fertility, reduces inflammation, enhances brain function, and prevents cancer [4].

Cultivation requirements and management within Akwa Ibom: Propagation is by seed or cutting off the tuber, setts, or bulbils. It requires staking for maximum yield.

4.56 *Euphorbia hirta*

Description, utilization, knowledge, and practices: *E. hirta* is commonly referred to as an asthma plant and is a hairy herb In Akwa Ibom, it is called Etinkene ekpo. It exudes white latex as soon as it is cut. It is used as a traditional medicine to treat asthma, skin ailments, and hypertension [110].

Cultivation requirements and management within Akwa Ibom: It goes in the wild.

4.57 *Jatropha curcas*

Description, utilization, knowledge, and practices: *J. curcas* is a semi-evergreen shrub or small tree. In Akwa Ibom, it is called Ukim Eyio. It is used for the treatment of rheumatism, toothache, jaundice, dysentery, and gum bleeding and inhibits HIV by inducing cytopathic effects with low cytotoxicity, etc. [100].

Cultivation requirements and management within Akwa Ibom: It can be cultivated by both seeds or stem cuttings and it needs little management to survive.

4.58 *Manihot esculenta*

Description, utilization, knowledge, and practices: *M. esculenta* is commonly called cassava.

In Akwa Ibom, it is called Iwa.

It is a perennial shrub plant with conspicuous, almost palmate alternate leaves which are deeply parted into 5–9 lobes. It produces elongated tubers. The stems can grow up to 5 m high. It is used to induce labor, treats dehydration in people with diarrhea, and is effective for tiredness and sepsis [11].

Cultivation requirements and management within Akwa Ibom: This is mostly propagated using stem cuttings. To enhance productivity, the stem cuttings must be planted in the right position on flat ground for soft soil and on mounds for hard soil.

4.59 *Manniophyton fulvum*

Description, utilization, knowledge, and practices: *M. fulvum* is a straggling bush or liane and in Akwa Ibom is called Ekonikon. Its stem can be about 30 m long to

10 cm thick. It is used for the treatment of whooping cough, leprosy, dysentery, piles, and painful menses [20].

Cultivation requirements and management within Akwa Ibom: It grows in the wild, although it can be propagated by seed and requires little management.

4.60 *Mallotus oppositifolius*

Description, utilization, knowledge, and practices: *M. oppositifolius* is a shrub or small tree with pale greenish-gray bark which is slightly rough and scaly and covered with mixed simple and stellate hairs [104]. In Akwa Ibom, it is commonly called Uman nwariwa. The leaves are useful for the treatment of epilepsy, diarrhea, inflammation, dysentery, convulsion, pain, etc. [3].

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds, although vegetative propagation may be possible given its easy growth. They have a self-supporting growth form and hence do not need much management for survival. Similarly, it grows in the wild.

4.61 *Afzelia africana*

Description, utilization, knowledge, and practices: *A. africana* is a tree in Akwa Ibom called Eyin Mbukpo. The leaves are alternate, petiolated, or paripinnate, up to 30 cm long with 7–17 pairs of leaflets; it has sweet-smelling flowers and produces oblong straight flattened dehiscent pod fruits which are black-brown [27, 34, 92]. The leaves are used for pain relief and treating digestive problems such as constipation and vomiting and internal bleeding.

Cultivation requirements and management within Akwa Ibom: It is propagated by seeds and vegetative techniques through budding. Since the seeds are readily preyed upon by animals, care should be taken for the right depth while planting.

4.62 *Acacia ataxacantha*

Description, utilization, knowledge, and practices: *A. ataxacantha* is a woody, shrub or small tree growing up to 10 m in height. In Akwa Ibom, it is called Mbara Okpok and is used in the treatment of abscesses, backache, cough, dental caries, toothache, pneumonia, malaria, sores, wounds, stomach problems, etc. [61].

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds as well as stem cuttings.

4.63 *Afzelia bella*

Description, utilization, knowledge, and practices: *A. bella* is a tall tree. The leaves produce sweet-scented flowers. The ground bark is used for the treatment of topical skin infections. Bark decoctions and macerations are taken to treat intestinal parasites, diarrhea, menstruation problems, rheumatism, and hemorrhoids and can be used as a tonic. Leaves are administered against constipation [71]. It is commonly called pod mahogany and referred to as Enyin Mbukpo in Akwa Ibom.

Cultivation requirements and management within Akwa Ibom: Propagation is via seeds which must be sown no deeper than 2 cm with the hilum facing downward. Little management is required.

4.64 *Baphia nitida*

Description, utilization, knowledge, and practices: *B. nitida* is commonly called camwood and Afuo in Akwa Ibom. It is a shrubby, leguminous, hardwooded tree with evergreen leaves. It can grow to about 4–5 m tall. It has characteristic smooth, green leaves which are oval shaped measuring about 10–15 cm long. It produces flowers that are white and pea-like and have a yellow center. The leaves have inflammatory, antidiarrheal, and analgesic effects. The extract of camwood can be formed into a soft soap-like material that promotes healthy skin [111].

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds and cuttings (from young parts). It does not require special management [22].

4.65 *Cajanus cajan*

Description, utilization, knowledge, and practices: *Cajanus cajan* is called Nkoti in Akwa Ibom and is an erect, branched, hairy shrub. The leaves are used for treating diabetes, sores, skin irritations, hepatitis, measles, jaundice, dysentery, stabilizing menstrual period, expelling bladder stones, etc. [126].

Cultivation requirements and management within Akwa Ibom: Propagation is via seeds in holes 2 m apart. The management required constant weeding for the first 2 months.

4.66 *Cassia alata*

Description, utilization, knowledge, and practices: *Cassia alata* is called Adaya Okon in Akwa Ibom. It is an annual shrub or occasionally a biannual herb with zygomorphic flowers. Seed extracts are used to cure skin diseases, while the leaves are used to treat constipation and fresh leaves are applied to the skin to treat fungus [90].

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds.

4.67 *Glycine max*

Description, utilization, knowledge, and practices: *Glycine max* is called Nkoti eto in Akwa Ibom. It is an erect, bushy, hairy annual legume. Seeds are used for the prevention and treatment of cardiovascular diseases, diabetes, cancer, obesity, cholesterol, etc. [14].

Cultivation requirements and management within Akwa Ibom: Cultivation is by seeds and requires little management.

4.68 *Lonchocarpus cyanescens*

Description, utilization, knowledge, and practices: *Lonchocarpus cyanescens* is a shrub. In Akwa Ibom, it is called Awa. All the plant parts are medicinal and can be used in the treatment of diarrhea, leprosy, yaws, arthritis, rheumatism, genital stimulants, dysentery, etc. [36].

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds and it requires little management to thrive.

4.69 *Parkia biglobosa*

Description, utilization, knowledge, and practices: *Parkia biglobosa* is called Ukon Uyayak in Akwa Ibom and is a perennial deciduous tree. It is useful for the treatment of hypertension, wound healing, antimalaria, management of bacterial infection, etc. [53].

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds and it doesn't require special management.

4.70 *Pentaclethra macrophylla*

Description, utilization, knowledge, and practices: *Pentaclethra macrophylla* is called Ukana in Akwa Ibom. It is a large size tree with log bipinnate compound leaves. It has antibacterial and antimicrobial properties, and the seeds are used to treat infertility.

Cultivation requirements and management within Akwa Ibom: Cultivation is by seeds and little management is required to make it grow.

4.71 *Pterocarpus erinaceus*

Description, utilization, knowledge, and practices: *Pterocarpus erinaceus* is commonly called bar wood and also known as Ukpa in Akwa Ibom. The leaves are 3–7-cm-long petioles with hairy pinnae. It has a golden yellow flower that grows up to 13 mm long. The fruit is straw colored, circular, flattened, and indehiscent. It has a seeded pod of 4–7 cm in diameter. These pods contain seeds. It is used to treat inflammatory disease, malaria, anemia, ulcer, rheumatism, dermatitis, etc.

Cultivation requirements and management within Akwa Ibom: It is propagated by planting nursery-raised seedlings or rooted cuttings. Careful management is required for its survival before sowing which requires dipping the seeds in water or sulfuric acid for some time. After germination, frequent pruning is required [123].

4.72 *Pterocarpus santalinoides*

Description, utilization, knowledge, and practices: *Pterocarpus santalinoides* is a tree with slash yellowish-white exuding drops of red gum when slashed. It produces fruits that are orbicular with the flattened indehiscent pod which is hairy and pal brown in color. The pods have seeds [58]. It is commonly called Mututi in English and Nkpa-inyan in Akwa Ibom. It is used for the treatment of gastroenteritis.

Cultivation requirements and management within Akwa Ibom: Cultivation is by seeds.

4.73 *Tetrapleura tetraptera*

Description, utilization, knowledge, and practices: *Tetrapleura tetraptera* is commonly called aridan or gum tree in English and Uyayak in Akwa Ibom. It is a single-stemmed, robust, perennial tree [54]. It is used for the treatment of diabetes, hypertension, asthma, epilepsy, schistosomiasis, arthritis, etc.

Cultivation requirements and management within Akwa Ibom: Cultivation is by seeds that require scarification for germination due to their hard coats.

4.74 *Harungana madagascariensis*

Description, utilization, knowledge, and practices: *Harungana madagascariensis* is called Oton in Akwa Ibom and is a small bushy tree. It produces almond-scented white or cream flowers and globular berry-like small fruits [16]. It is used in the treatment of chest pain, urogenital infections, ringworm, eczema, malaria, etc.

Cultivation requirements and management within Akwa Ibom: Propagation is by seeds and it requires thinning after planting to enhance growth.

4.75 *Irvingia gabonensis*

Description, utilization, knowledge, and practices: *Irvingia gabonensis* is commonly called Uyo in Akwa Ibom. It is useful for weight loss and treatment of high cholesterol and diabetes [18].

Cultivation requirements and management within Akwa Ibom: Cultivation is by seeds.

5 Conclusion

The plants documented here are regularly used by Akwa Ibom people as food and medicine despite the lack of any elaborate institutional effort to manage, protect, and conserve the genetic resources. The food and medicinal benefits of these plants are linked to the phytochemical composition and non-secondary metabolites that are bioactive at different levels because their ingestion in diverse forms influence biological processes. This is unsustainable because of the ongoing biodiversity loss, climate change, and sociocultural transitions away from traditional past and calls for formal management strategies. Knowledge about the food and medicinal values of these plants are mostly held by community elders, village heads, chiefs, head of households, academics, researchers, and traditional medical practitioners. There is a need to undertake the collection, characterization, and conservation of the plant resources recorded in here to protect them from ongoing changes and ensure future generations can benefit from these resources. By documenting the relevant information connected to the utilization and cultivation of these plant resources as well as encouraging their cultivation on large scales, employing the use of plant breeding programs for better qualities and employing the use of conservation methods for all varieties, these resources and their inherent values can be saved

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Classification Methods and Diversity of Medicinal Plants

3

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Abstract

This chapter discusses the diversity and classification of medicinal plants. Medicinal plants occupy a place of prime importance due to their complementary applications in modern medicines, and their use as food supplements, nutraceuticals, pharmaceutical intermediates, folk medicines, and secondary metabolites they provide for novel drugs. Therefore, an understanding of the diversity and classification of medicinal plants is useful for practitioners of ethnomedicine, botanists, and the common man for home remedies. Medicinal plants are mostly either angiosperms or gymnosperms with about 4171 and 706 plant species, respectively. Among angiospermic medicinal plants, dicotyledons have the most common representation of medicinal plants. Medicinal plants may be classified based on the following; parts used, habitat, habit, therapeutic, and ayurvedic formulations. The majority of medicinal plants are herbaceous, and about 51.3% grow mainly in the wild. Tropical rain forest provides the highest percentage of species (54.6%), followed by home gardens (25.4%) and unused land (19.5%). The medicinal use of about 72 (38.5%) plants has not before been reported. Many medicinal plants and their related indigenous values have not been adequately documented despite numerous investigations on medicinal plant products. The bulk of plant parts utilized in phytomedicine are the leaves (145 species), followed by the roots (119 species), and the stem (53 species). Herbs make up the highest percentage of species in terms of habit (44%), followed by trees (26%), shrubs (16%), lianas (5%), and others. Also, the sustainable use of medicinal plants should be ensured, and thorough documentation of the traditional knowledge linked with their usage should also be made and passed down across generations.

Keywords

Ayurvedic · Bioactive · Classification · Diversity · Herbs · Medicinal plants

Abbreviations

BGCI	Botanic Garden Conservation International
IUCN	International Union for Conservation of Nature
AKSU	Akwa Ibom State University
Ag	Agriculture
Aq	Aquaculture
FV	Food value
CO	Construction
UR	Urbanization
NT	Near threatened
NYBA	Not yet assessed
Fd	Food
Md	Medicine

Om	Ornamental
Tb	Timber
UV	Use-value

1 Introduction

A medicinal plant can be any plant in which one or more of its components contains bioactive substances, which can be utilized therapeutically or as precursors to the manufacture of valuable pharmaceuticals. Since time immemorial, natural plant products have occupied a place of prime importance in the field of medicine. They continue to be vital to many who do not have access to modern medicines or cannot afford synthetic drugs. Medicinal plants, for example, herbs and trees contain chemical compounds known to modern and ancient civilizations for their bioactive properties. Until the development of chemistry, particularly, the synthesis of secondary metabolites in the nineteenth century, medicinal plants and herbs have been the sole source of bioactive compounds capable of healing human ailments. Researchers believe that if medicinal herbs are taken in the appropriate doses and forms, they will be more effective than modern pharmaceutical drugs (synthetic drugs).

Plant products contain secondary metabolites that heal a variety of ailments and conditions in humans and animals. Over 75% of the world's population relies mainly on plants and their products for primary healthcare. Approximately 30% of all plant species were used for medical purposes at some point in history [1].

In developed nations like the United States, it has been estimated that up to 25% of all medications are made from plants. Fast-developing nations like China and Singapore make up as much as 80% of the total. As a result, countries like China use medicinal herbs for economic purposes considerably more frequently than the rest of the globe. Indigenous systems of medicine are employed to treat rural people, and these nations contribute two-thirds of the botanicals used in contemporary medical practices. Over 80,000 of the 250,000 higher plant species that exist on Earth are used for medicinal purposes [2].

Medicinal plants are plants that have the potency in the treatment of illnesses and the prevention of specific problems and diseases that impact both humans and animals. There are other varieties of herbal treatments that may differ from region to region, producing a similar pattern of "size" and "shapes." Natural plants have excellent secondary metabolites or chemical properties from their roots down to leaves even in their saps. Leaves of some herbs, for example, Karpooravalli (*Coleus ambonicus*), Bitter leaf (*Vernonia* species), Podina (*Mentha arvensis*), finger root (*Uvaria chamae*), Neem (*Azadirachta indica*), Thudhuvalai (*Solanum trilobatum*), Fern (*Diplazium sammatii*), Basil (*Ocimum sanctum*) and some edible species of fungi (mushroom), etc., are used by humans today. Some leaves have secondary metabolites that treat skin conditions, colds, headaches, constipation, and indigestion, among other conditions. Therefore, plants used for their specific health benefits to both human and animal health are considered medical plants [3, 4].

A significant number of plants are rarely used whole; at least one of its parts (leaf, stem, root, rhizome, bark, seeds, buds and leaves, etc.) can be utilized medicinally. The value of other parts of the same plant can vary. Plants that produce secondary metabolites can also be utilized as food, condiments, or even for purifying drinks. Since the beginning of time, the theory of signatures, which was systematized in the sixteenth century, has been essential and significant in the distinction by analogy of plants useful for human healing. From the seventeenth century, however, it has been expanded, contested, and canvassed, and by the eighteenth century, the elite community has completely abandoned it. According to data on plant use around the world, 14–28% of plants are included for their therapeutic properties. According to twenty-first-century surveys, 3–5% of patients in Western nations, 80% of local inhabitants in poor nations, and 85% of populations in the Southern Sahara chose medicinal plants as their primary form of treatment [2, 5].

There are still many people who practice traditional medical approaches. According to estimates, 80% of the world's population, or about four million people, cannot afford the goods produced by the Western pharmaceutical sector. They have to rely on the use of conventional medications, many of which have botanical origins. The inventory of medicinal plants, which includes over 20,000 kinds of plants, provides extensive documentation of this fact. Despite the overwhelming influence, reliance, and technological advancements in synthetic medications, a sizable portion of the global population still favours drugs derived from plants due to their accessibility and lower cost.

This chapter seeks to discuss the methods of classification of medicinal plants. We will also give details on their diversity, usage, formulation, habits, nutrition, habitats, nature, size, life span, as well as their therapeutic values.

1.1 Medicinal Plant Species and Their Therapeutic Value

Different plant groups have different medicinal plants that are synonymous to these groups depending on the number of species that belong to the said group. A breakdown of the different plant groups as well as the total number of species is given in Table 1. These plants and their bioactive components have been exploited

Table 1 Plant species with therapeutic value under different plant groups

Different plant groups	Number of plant species
Angiosperms:	
Monocotyledons	676
Dicotyledons	3495
Gymnosperms:	
Thallophytes	250
Bryophytes	54
Pteridophytes	402
Total	**4877**

Source: [6]

Table 2 Plant species, drugs, and their uses

Plant Species	Drugs	Uses
Catharanthus roseus	Vinblastine	Anti-cancer
Tradescantia spathacea	Taurine	Anti-inflammatory
Catharanthus roseus	Ajmalacine	Anti-cancer, hypotensive
Erythroxylum coca	Cocaine	Local anaesthetic
Rauvolfia serpentina	Rescinnamine	Tranquilizer
Rauvolfia serpentine	Reserpine	Tranquilizer
Papaver somniferum	Codeine	Anti-tussive
Cinchona species	Quinine	Antimalarial, amoebic dysentery
Erythroxylum coca	Cocaine	Tropical anaesthetic
Papaver somniferum	Morphine	Painkiller
Colchicum autumnale	Colchiceine, Amide	Anti-tumour
Xanthium strumarium	Tomentocin	Anti-sedative
Papaver somniferum	Codeine	Anti-cough
Digitalis species	Cardiac glycosides	Congestive heart failure
Artemesia annua	Artemisinin	Anti-malarial
Plumbago indica	Plumbagin	Anti-bacterial, antifungal
Gossypium species	Gossypol	Anti-spermatogenic
Allium sativum	Allicin	Anti-fungal
Xanthium strumarium	Xanthinin	Hair growth
Ricinus communis	Ricin	Anti-fungal
Digitalis thevetia	Digitoxin, Digoxin	Cardiotonic
Thevetia species	Nerrifolin	Cardiotonic
Podophyllum emodi	Podophyllin	Anti-cancer
Aspilia Africana	Aspicine	Wound healing
Musa parasidiaca	Musadine	Cuts and sores
Sesamum indicum	Sesamin	Anti-anaemic
Tithonia diversifolia	Tithonine	Skin diseases
Senna occidentalis	Carfeine	Pain relief
Tinospora cordifolia	Tinosporone	Anti-rheumatism
Eryngium foetidium	Arialgeine	Analgesic and inflammation
Colchicum autumnale	Colchicine	Anti-tumour, anti-gout

Sources: [7–10]

by pharmaceutical companies for many years now to produce drugs with several benefits, some of which are listed in Table 2.

2 Methods of Classifying Medicinal Plants

It has been estimated that there are 250,000 species of higher plants on earth; more than 80,000 species have at least some known medicinal uses. There are around 5000 species with specific medicinal benefits. They are classified according to: parts used, habit, habitat, therapeutic value, and Ayurvedic formulation [3, 9, 10].

2.1 Some Medicinal Plants Based on Parts Used

The following are some medicinal plants used based on their parts:

Whole plant: *Boerhaavia diffusa, Phyllanthus neruri, P. amarus, Euphorbia hirta*
Rhizome: *Curcuma longa*, *Zingiber officinale*
Roots: *Uvaria chamae, Daucus carota, Nuaclea latifolia*
Stem: *Tinospora cordifolia, Acorus calamus*
Bark: *Saraca asoca, Manifera indica, Azadiractita indica, Persea americana*
Leaf: *Indigofera tinctoria, Lawsonia inermis, Aloe vera, Sesamum indicum, Centalla asiatica*
Flower: *Biophytum sensitvum, Mimusops elenji*
Fruit: *Aframumum melequeta*, *Solanum* species, *Psidum gaujava, Cucumis melo*
Seed: *Datura stramonium, Cola nitida, Cola acuminata, Persea americana*

2.2 Some Medicinal Plant Based on Habit

The following are some medicinal plants used based on their habits:

Grass: *Cynodon dactylon*
Sedge: *Cyperus rotundus*
Herbs: *Vernonia cineria, Emilia sonchifolia*
Shrubs: *Solanum* species, *Capsicum* species
Climbers: *Asparagus racemosus, Dioscorea* species, *Phildendron domesticum, Cleome ciliate, Smilax anceps*
Creeper: *Commelina erecta, Commelina begalensis, Ipomoea batatas, Ipomoea cordatotriloba*
Trees: *Azadiractita indica, Mangifera indica, Dacroydes edulis, Croton* species.
Lianas: *Heinsia bussei, Tetracarpidium conophorum*

2.3 Some Medicinal Plant Based on Habitat

The following are some medicinal plants used based on their habitats:

Tropical: *Andrographis paniculata*
Sub-tropical: *Mentha arvensis*
Temperate: *Atropa belladona*

2.4 Some Medicinal Plant Based on Therapeutic Values

The following are some medicinal plants used based on their therapeutic values:

Anti-malaria: *Cinchona officinalis, Artemisia annua*
Anti-cancer: *Catharanthus roseas, Taxus baccata*
Anti-ulcer: *Azadirachta indica, Viscum album*
Anti-diabetic: *Catharanthus roseus, Musa acuminata, Dioscorea bulbifera*
Anti-cholesterol: *Allium sativum*
Anti-inflammatory: *Curcuma domestica, Desmodium gangeticum, Eryngium foetidum*
Anti-viral: *Acacia* species
Anti-bacterial: *Plumbajo indica*
Anti-fungal: *Allium sativum*
Anti-diaharrhoeal: *Psidium guajava, Curcuma domestica*
Anti-hypertensive: *Allium sativum, Persea appericara*
Anti-cardiotonic: *Digitalis* species*, Telfairia occidentalis, Peristrophe bicalyculata*

2.5 Medicinal Plant Based on Ayurvedic Formulation

This type of medicine is practiced in some parts of the world specifically in India, for example, *Desmodium gangeticum*, *Solanum indicum, Gmelina arborea, Premna spinosus, Ipomoea maxima, Emilia sonchifolia, Cardiospermum halicacabum, Ficus racemose, Ficus microcarpa, Phyllanthus emblica, Terminalia bellerica*.

3 Diversity in Medicinal Plants

Plantae as a Kingdom has existed for about 410 million years as green algae evolve from water to land, medicinal plants included. The land is a rich resource of medicinal plants and was comparatively not colonized. However, terrestrial habitats give more light and carbon dioxide needed by plant for growth and survival.

Medicinal plants are multicellular and mostly photosynthetic and may be terrestrial or aquatic and some are epiphytic or ephemeral. Medicinal plants can be found almost in all habitats on Earth.

According to research carried out at the Botanical Garden in the main campus of Akwa Ibom State University at Ikot Akpaden, Mkpat Enin Local Government Area of Akwa Ibom State – Nigeria by Okon et al. [11]. The Garden has a rich floral diversity of medicinal plants that have been conserved for the past ten years both in situ and ex situ mainly for research purposes. According to Sikolia and Omondi [12], the preservation of plants in botanical gardens dates back to the beginning of time. According to Botanic Garden Conservation International (BGCI), a botanic garden is an establishment that is accessible to the general public for assessment and evaluation, has a respectable level of permanence, and conducts scientific or technical research on plants (including species of wild origin) in its collections that are used for medicinal, ornamental, cultural, or other purposes. The Botanical Garden of Akwa Ibom State University (AKSU), Nigeria, was the site of a study by Okon et al. [11] that indicated the percentage habits of medicinal plants found in the highlighted

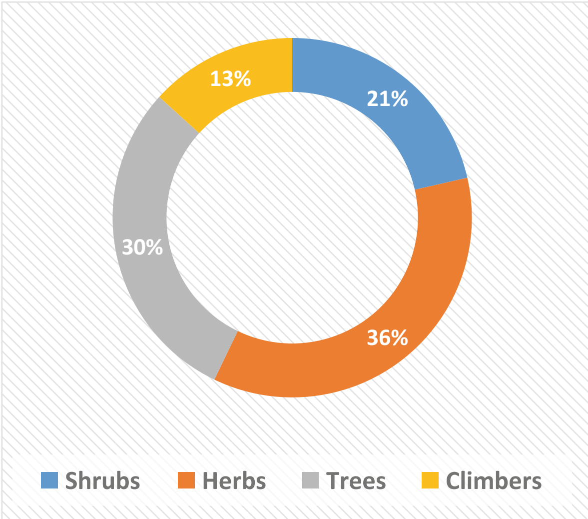

Fig. 1 Percentage distribution of medicinal plants habits in AKSU Botanical Garden, Nigeria [11]

area of the AKSU botanical garden examined documented diversity of medicinal plants based on their habits as follows: herbs as highest (35.71%), trees follow with 29.59%, then shrubs with 21.43%, while climbers were the least with 13.27% (Fig. 1).

Medicinal plant diversity can also be attributed to how well protected they are. To determine the condition of their conservation, the total number of plant species discovered in the AKSU Botanical Garden were cross-checked on the websites of two conservation organizations: the International Union for Conservation of Nature (IUCN) and Natural Resources/Nature Serve. Research indicates that 42.86% of the medicinal plants identified at the AKSU Botanical Garden are assessed as least concern/secure and face minimal risk, while 52.04% are rated as data deficient/ unranked. Near threatened species were recorded at 4.08% and 1.02% susceptible [11].

3.1 Medicinal Plant Diversity Based on Their Habitat

Based on the different characteristics and morphology of plants, there are diverse types of plants. Medicinal plants are grouped into the following categories according to their habitat [13, 14].

3.1.1 Hydrophytes

The word hydrophytes means plants that grow near water or in. These plants have weak vascular tissue (xylem and phloem), soft stems, and poor root systems. Here the majority of the tissue has air spaces and is spongy.

Hydrophytic Plants Include the Following Characteristic Features

a. Submerged plants, for example, *Elodea*, *Hydrilla*, *Sphgnum*, *Zostera*, *Vallisneria,* and *Potamogeton*, etc.
b. Fixed-floating and free-floating, for example, *Eichhornia, Utricularia*, *Wolffia, Salvinia, Egeria, Lemna, Ceratophyllum, Pistia*, *Trapa,* and *Eichornia,* etc.
c. Amphibious, that is, partly submerged, for example, *Sagittaria*, *Ranunculus aquatilis,* and *Alisma plantago*, etc.). Two angiosperms that are also marine, for example, *Thalassia* and *Zostera*.

3.1.2 Hygrophytes

Hygrophytic plants require wet, dark, and moist surroundings to develop and grow. Their fragile, spongy stems and root systems limited growth. Their leaves are completely formed and have stomata. Examples of common plants are several types of grasses, ferns, and begonias [6].

3.1.3 Halophytes

Halophytes are plants that can thrive in salty environments like soil or water. They can adapt or withstand high salt concentrations, such as those of NaCl, MgCl2, and MgSO4, quite well. They possess peculiar, negatively geotropic breathing roots known as pneumatophores, such as *Rhizophora racemosa, Rhizophora mangle Ceriops,* and *Avicennia* etc., which are some common examples.

3.1.4 Mesophytes

Mesophytes are mostly angiosperms that thrive in environments with little or moderate water supply. They always grow bigger and faster. They also have a strong root and leaf system. Both woody and herbaceous stems are possible. Some of them grow mesophytically in the summer and xerophytically in the winter, much like deciduous trees.

3.1.5 Xerophytes

Most xerophytic plants are medicinal. Examples include *Ziziphus*, *Nerium, Acacia, Euphorbia, Amaranthus, Argemone,* etc. These plants do well in xeric, arid, or places with insufficient water availability. Due of their ability to store water in their stems, many xerophytes are succulents like *Opuntia*, leaves like *Bryophyllum* species, and *Agave* or roots, for example, *Asparagus* species [7].

3.1.6 Epiphytic Plants

These plants can be found growing as an epiphyte on trees or on the trunks or branches of other plants, such as orchids or lichen like *Mangifera indica, Persia americana, Citrus* species, *Kola* species, and *Pterocarpus* species, etc. Epiphytic plants are thought of as the parasitic space occupants on other plants. The interaction between an orchid, a commensal, and a tree, the host, is an illustration of commensalism where the host is unharmed [13].

3.1.7 Parasitic Plants

Parasitic plants such as *Cuscuta, Nuytsia floribuda, Monotropa uniflora, Castilleja,* and *Striga* are parasites that grow on other plants on the roots of *Sorghum bicolor* (jowar). They produce a specialized structure mainly for feeding called haustoria that form a functional cap between the two plants.

3.2 Medicinal Plant Diversity Based on Habit

Medicinal plants in the group of angiosperms are categorized into four different groups depending on their form, shape, and size [14, 15].

3.2.1 Herbaceous Plants

These are plants with short, green, and soft stems. Usually with a brief or short life span like wheat or gram. Many herbaceous plants have a much-reduced underground stem portion, but the aerial branch with flowers at the apex develops from the underground part during reproduction. Such a stem is recognized as a scape, for example, *Allium cepa* (onion) and *Allium sativum* (garlic) in the family Amaryllidaceae [7].

3.2.2 Shrubby Plants

Shrubby plants are more advanced and significantly woody with branches than herbaceous plants. They lack an axis but generally have multiple stems, common examples are Ixora dwarf, black kodia, *Schefflera*, henna, roses, and China roses [16].

3.2.3 Arborescent (Trees)

They are usually taller than bushes with thick, rigid, and woody. Arborescent have a noticeable trunk, for example, *Irvingia* species, *Khaya* species, *Hura crepitans*, *Garcinia* species, and *Musanga cecropioides*.

3.2.4 Culms

The nodes and internodes in culms are very apparent. Most plants in these groups have hollow internodes. A typical example is *Bambusa vulgaris* (bamboo), which are grasses but cannot be classified as herbs, shrubs, or trees, etc.

3.2.5 Flowering Vine

These are very weak stem plants, normally supported by another plant. Common examples are Wisteria, Kiwifruit, ivy, and morning glory, etc.

3.3 Medicinal Angiosperm Diversity Based on the Nature of Stem

Some medicinal plants under angiosperm can be grouped based on the type of stem they possess [14, 15].

3.3.1 Erect Stem

Plants in this group develop upright/upward with hard or strong stems. Based on their strong stems, most trees, shrubs, and many herbaceous plants can stand upright on the ground without any support from another plant. Such plants include *Terminalia catappa*, *Terminalia ivorensis*, *terminelia superba*, *Irvingia* species, *Khaya* species, *Hura crepitans*, *Garcinia* species and *Musanga cecropioides, Hibiscus rosa-sinensis, Zea mays,* and *Lysimacchia vulgaris,* etc.

3.3.2 Creepers

Here, the plant's stem dangles and creeps on the ground, and the roots grow from the nodes (i.e., leaves come from nodes) of the culm from which the branches emerge. Consider strawberries and grasses (Cynodon and Oxalis). Adventitious roots are created by the nodes and the length of the stem.

3.3.3 Trailers

Trailers resemble creepers, they do not grow adventitious roots at nodes. There are two different kinds of trailers: Prostrate/procumbent – the stem creeps on the ground, as in the case of *Evolvulus, Tribulus,* or Decumbent – when the prostrate stem extends its tip, as in the case of *Portulaca* and *Linderbergia.*

3.3.4 Climbers

These plants have long, weak stems and are equipped with roots, petioles, spines, and tendrils that allow them to climb or attach themselves to other plants or objects for support. They are of three different types: **Rootlet Climbers** – Roots are produced at the nodes to assist in climbing the object, for example, *Tecoma*, *Pothos,* and *Piper betal* (pan). **Hook Climbers** – *Bougainvillea*, *Duranta,* and *Carrisa* species where the thorns are the modification of an axillary vegetative bud which helps in climbing. Unlike in *Bignonia*, the terminal leaflet is modified into a hook which helps in climbing, **Tendril Climbers** – The tendrils here are thread-like in structure which helps the plant in climbing, for example, *Lathyrus sativus*, *Pisum sativus*, *Clematis*, *Nepenthes*, *Smilex*, *Gloriosa*, *Antigonon*, *Vitis,* and *Passiflora,* etc. [16].

3.4 Medicinal Plant Diversity Based on Size

The medicinal angiosperm plants come in a wide range of sizes. The smallest angiosperm medicinal plant is the rootless aquatic plant *Wolffia*. It has a diameter of roughly 0.1 mm. Aquatic *Lemna* has a diameter of roughly 0.1 cm. With a height of more than 100 metres, the *Eucalyptus regnans* tree is the tallest angiosperm medicinal plant. Some eucalyptus trees can reach heights of 130 metres. The largest plant species is the banyan tree (*Ficus bengalensis*). It can cover a space of between two and five acres and contains more than 200 prop roots.

3.5 Medicinal Plant Diversity Based on Life Span

Based on life span, angiosperms are categorized into the following four groups [14].

3.5.1 Ephemerals

These types of medicinal plants reach the end of their life span relatively quickly before the start of the true dry season. These plants, such as *Argemon Mexicana, Cassia tora*, and *Solanum xanthocarpum*, among others, are not genuine xeric and are frequently referred to as drought evaders, escapers from drought, or drought avoiders.

3.5.2 Annuals

As soon as they have finished their life cycle within a year, medicinal plants like *Oryza sativa* (rice), *Triticum aestivum* (wheat), and Cicer arietinum (gramme), among others, die.

3.5.3 Biennials (Biannuals)

These classes of medicinal plants finished their entire life cycle in two years. Only the first year after planting do they show signs of vegetative development; the following year, they begin to produce flowers, fruits, and seeds as well as attain full maturity. These plants are typically herbs, such as carrot, turnip, and radish plants.

3.5.4 Perennials

Perennial medicinal plants can live for more than 200 years. The Kolkata Botanical Garden is home to a massive banyan tree (*Ficus bengalensis*). A *Ficus religiosa* that can be found in Gaya is about 2500 years old. After reaching full maturity, a large number of medicinal perennials bloom and bear fruit during a particular time of the year. They are polycarpic and include *Mangifera indica*, *Acacia* species, and *Cocos nucifera*. Some perennial medicinal plants, such as *Agave* (Agarvaceae) and *Bambusa vulgaris* (Poaceae), are monocarpic, meaning they only bear fruit once in their lives. All annuals and biennials are monocarpic [16].

3.6 Medicinal Plant Diversity Based on Nutrition

Based on their method of nutrition, medicinal plants in this category are separated into the following groups [14].

3.6.1 Autophytes/Autotrophs

They produce their own food. They are divided into two groups: chemotrophs, which use chemical energy to manufacture their food, and phototrophs, which use sunlight to perform photosynthesis to generate their food.

3.6.2 Heterotrophs

Here, plants are unable to produce their own food and must rely on external sources. Heterotrophs can be insectivorous, saprophytic, parasitic, or symbiotic. Other plant varieties include Polygamous plants, for example, *Mangifera indica*, Stolon plants, for example, *Ajuga*, *Stachys,* and *Mentha*, seedless vascular plants, sucker plants, for example, red raspberry, lilac, and *Forsythia*, air layering plants, for example, *Forsythia*, jasmine, *Hamamelis,* and *Philodendron,* cutting plants. There are different kinds of medicinal plants, and each one needs a certain setting in order to grow and propagate. The essential elements are sunlight, nutrients, water, air, soil, and temperature that plants rely on for their life, germination, growth, development, and full maturity [14–16].

3.7 Diversity in Frequency of Occurrence and Relative Density of Medicinal Plants

The frequency of occurrence and relative density of some medicinal plants found/ grown in Uyo Local Government Area, Nigeria; *Cleome ciliata, Diplazium sammatii, Hibiscus rosa-sinensis, Justicia insularis,* and *Sesamum indicum* each had the highest frequency of occurrence of 60% each while medicinal plants such as *Bombax buonopozense, Glyphaea brevis, Heinsia bussei, Talinum triangulare,* and *Vernonia polysphaera* had the least frequency value of 20% each. In density value, *Cleome ciliata* dominated (8000 plants/ha) while the lowest value was associated with *B. buonopozense* (40 plants/ha) as reported in Table 3 by Okon [17].

In Ikot Ekpene Local Government Area, Nigeria (Table 4), *Justicia insularis, Lasienthera africanum, Microdesmis puberula,* and *T. triangulare* had the highest frequency of occurrence of 60% each while species like *B. buonopozense, Curcubita maxima, D. samamatii, Pterocarpus mildbraedii*, *Vernonia amygdalina,* and *V. polysphaera* had the lowest frequencies of occurrence (20%). *D. sammatii* and *Pterocarpus mildbraedii* had the highest and lowest density values of 1200 plants/ha and 40 plants/ha, respectively. The frequency occurrence of medicinal plants in Eket Local Government Area as shown in Table 5 revealed that *Glyphaea brevis* had the highest frequency of occurrence (80%) while the least frequency was associated with species like *Amaranthus hybridus* (20%), *C. ciliata* (20%), *D. sammatii* (20%), and *M. puberula* (20%). *S. indicum* had the highest density value of 4200 plants/ha while *B. buonopozense* had the least density value of 40 plants/ha. Also, the results revealed that *Gnetum africanum* was near threatened. Threats that affected the medicinal plants were as a result of agricultural activities, aquaculture, food value, construction, and urbanization (Tables 3, 4 and 5) [17].

3.8 Diversity in Use-Value Index of Medicinal Plants in Akwa Ibom State, Nigeria

The use-value index was determined based on the different categories for a particular medicinal plant. Results obtained showed a high index for the following medicinal

Table 3 Frequency of occurrence and relative density of medicinal plants in Uyo, Nigeria

Vegetables	Habitats	Frequency (%)	Density (Plts/ha.)	Rel. Freq.	Rel. Density	Conservation Status	Threats
Amaranthus hybridus	Herb	40	220	6.67	1.37	NYBA	Fv, Ag
Bombax buonopozense	Tree	20	40	3.33	0.25	NYBA	Ag, UR, CO
Cleome ciliata	Herb	60	8000	10.00	49.81	NYBA	CO
Cucurbita maxima	Herb	0	0	0	0	NYBA	–
Diplazium sammatii	Herb	60	1600	10.00	9.96	NYBA	–
Glyphaea brevis	Shrub	20	80	3.33	0.49	NYBA	–
Gnetum africanum	Herb	0	0	0	0	NT	Fv, Ag, Aq
Heinsia bussei	Herb	20	60	3.33	0.37	NYBA	–
Hibiscus rosa-sinensis	Shrub	60	180	10.00	1.12	NYBA	UR, CO
Justicia insularis	Herb	60	1000	10.00	6.23	NYBA	UR, CO
Lansianthera africamum	Shrub	40	240	6.67	1.49	NYBA	Fv, UR, CO
Microdesmis puberula	Shrub	40	240	6.67	1.49	NYBA	CO, UR
Piper guineense	Herb	40	60	6.67	0.37	NYBA	Fv, Ag
Pterocarpus mildbraedii	Tree	0	0	0	0	NYBA	–
Sesamum indicum	Herb	60	2200	10.00	13.69	NYBA	Fv, CO, UR
Sterculia tragacantha	Tree	40	80	6.67	0.49	NYBA	CO, UR
Talinum triangulare	Herb	20	800	3.33	4.98	NYBA	Fv, Ag, UR
Telfairia occidentalis	Herb	0	0	0	0	NYBA	Fv, UR, CO
Vernonia amygdalina	Shrub	0	0	0	0	NYBA	Fv, UR, CO
Vernonia polysphaera	Herb	20	1260	3.33	7.85	NYBA	CO, UR

Ag agriculture, *Aq* aquaculture, *FV* food value, *CO* construction, *UR* urbanization, *NT* near threatened, *NYBA* not yet assessed
Source: [17]

Table 4 Frequency of occurrence and relative density of medicinal plants in Ikot Ekpene, Nigeria

Vegetables	Habitats	Frequency (%)	Density (Plts/ha.)	Rel. Freq.	Rel. Density	Conservation Status	Threats
A. hybridus	Herb	40	280	7.14	4.93	NYBA	Fv, Ag
B. buonopozense	Tree	20	60	3.57	1.06	NYBA	Ag, UR, CO
C. ciliata	Herb	40	400	7.14	7.04	NYBA	CO
C. maxima	Herb	20	300	3.57	5.28	NYBA	–
D. sammatii	Herb	20	1200	3.57	21.13	NYBA	–
G. brevis	Shrub	40	300	7.14	5.28	NYBA	–
G. africanum	Herb	0	0	0	0	NT	Fv, Ag, Aq
H. bussei	Herb	0	0	0	0	NYBA	–
H. rosa-sinensis	Shrub	40	300	7.14	5.28	NYBA	UR, CO
J. insularis	Herb	60	1000	10.71	17.61	NYBA	UR, CO
L. africamum	Shrub	60	360	10.71	6.34	NYBA	Fv, UR, CO
M. puberula	Shrub	60	460	10.71	8.09	NYBA	CO, UR
P. guineense	Herb	40	60	7.14	1.06	NYBA	Fv, Ag
P. mildbraedii	Tree	20	40	3.57	0.70	NYBA	–
S. indicum	Herb	0	0	0	0	NYBA	Fv, CO, UR
S. tragacantha	Tree	0	0	0	0	NYBA	CO, UR
T. triangulare	Herb	60	600	10.71	10.56	NYBA	Fv, Ag, UR
T. occidentalis	Herb	0	0	0	0	NYBA	Fv, UR, CO
V. amygdalina	Shrub	20	220	3.57	3.87	NYBA	Fv, UR, CO
V. polysphaera	Herb	20	100	3.57	1.76	NYBA	CO, UR

Ag agriculture, *Aq* aquaculture, *FV* food value, *CO* construction, *UR* urbanization, *NT* near threatened, *NYBA* not yet assessed

Source: [17]

Table 5 Frequency of occurrence and relative density of medicinal plants in Eket, Nigeria

Vegetables	Habitats	Frequency (%)	Density (Plts/ha.)	Rel. Freq.	Rel. Density	Conservation Status	Threats
A. hybridus	Herb	20	140	3.33	1.52	NYBA	Fv, Ag
B. buonopozense	Tree	40	40	6.67	0.43	NYBA	Ag, UR, CO
C. ciliata	Herb	20	420	3.33	4.55	NYBA	CO
C. maxima	Herb	40	260	6.67	2.81	NYBA	–
D. sammatii	Herb	20	1200	3.33	12.98	NYBA	–
G. brevis	Shrub	80	480	13.33	5.19	NYBA	–
G. africanum	Herb	0	0	0	0	NT	Fv, Ag, Aq
H. bussei	Herb	60	140	10.00	1.52	NYBA	–
H. rosa-sinensis	Shrub	40	180	6.67	1.95	NYBA	UR, CO
J. insularis	Herb	40	600	6.67	6.49	NYBA	UR, CO
L. africamum	Shrub	0	0	0	0	NYBA	Fv, UR, CO
M. puberula	Shrub	20	200	3.33	2.16	NYBA	CO, UR
P. guineense	Herb	0	0	0	0	NYBA	Fv, Ag
P. mildbraedii	Tree	40	120	6.67	1.29	NYBA	–
S. indicum	Herb	40	4200	6.67	45.45	NYBA	Fv, CO, UR
S. tragacantha	Tree	0	0	0	0	NYBA	CO, UR
T. triangulare	Herb	40	400	6.67	4.33	NYBA	Fv, Ag, UR
T. occidentalis	Herb	0	0	0	0	NYBA	Fv, UR, CO
V. amygdalina	Shrub	40	220	6.67	2.38	NYBA	Fv, UR, CO
V. polysphaera	Herb	60	640	10.00	6.93	NYBA	CO, UR

Ag agriculture, *Aq* aquaculture, *FV* food value, *CO* construction, *UR* urbanization, *NT* near threatened, *NYBA* not yet assessed
Source: [17]

plants: *T. triangulare, V. amygdalina, P. guineense, P. mildbraedii, L. africamum, J. insularis, G. fricanum, C. maxima, A. hybridus,* and *T. occidentalis* while the least use-value was recorded in *M. puberula, B. buonopozense, S. tragacantha, G. brevis, H. bussei* and *V. polysphaera*, *C. ciliata,* and *D. sammatii.* Also, all the ten medicinal plants selected for analysis were observed to be available in the wild except *H. rosa-sinensis* (Table 6) [17].

3.9 Medicinal Plants Diversity Among Tree Canopies

Most temperate moist forests in different regions are home to a variety of flora that are pivotal in sustaining the livelihoods of the local plant communities. The abundance and distribution of plant life in these forests, particularly medicinal species, is comparatively underreported (with scant or no supporting documentation). Zubair et al. [18] identified a total of 45 species from 34 different groups in their study site in North-Eastern Pakistan. The majority of plants were identified as being medicinal (45%), followed by food (26%) species that also included some medicinal components. Based on richness and abundance, tree canopy cover has an impact on the general growth of ethnomedicinal plants. In comparison to open spaces and closed canopy, the site with partial canopy showed the highest diversity, richness, and dominance. These results are crucial in revealing the abundance of medicinal flora variety in the Balakot temperate forest in the north-east and the potential for preserving the native plant communities' means of subsistence through public–private collaborations in several sectors.

It has been observed that the diversity of vegetation has a direct impact on the biogeochemical cycles, ecosystem processes, and vegetation structure of a particular ecosystem [19, 20]. Statistics on plant biodiversity in mountainous areas offer vital information and insight on the suitability of the particular habitat, ecosystem productivity, and successional pathway prediction [21–23].

The engine room and hotspots of plant diversity, metabolic processes, ecological zones, and contacts between creatures and the atmosphere are thought to be found under forest canopies [24]. The maintenance of biodiversity and meeting multiple environmental needs are both dependent on the canopy cover of the forest stand [25, 26]. Nearly 40% of the world's current plant and animal life is supported by various forest canopies, of which 10% are recognized as canopy authority [27]. Different strata of medicinal plant species rely on forest canopies for sunlight, nutrients, and assistance with reproduction. According to research, the density and quantity of understory vegetation are directly impacted by disturbance in the various layers. In a forest's understory vegetation, medicinal plants play a crucial function [28, 29]. Holscher and Adnan [30] evaluated the variety and abundance of medicinal plants in old growth forest and degraded forest, reporting that the density of medicinal plants in old growth forest was significantly higher than that in the latter.

Furthermore, it was noted that the quantity of quantitative vegetation surveys in the Himalayas is extremely little and insufficient to draw a precise picture of the diversity of understory vegetation [30, 31]. For the purpose of developing strategies

Table 6 Plant diversity/categories and use-value index of medicinal plants

Scientific name	Family	Common name	Local name	Harvest type	Uses	No. of categories Used	No. of respondents	∑	Rank/%	UV
M. puberula	Euphorbiaceae	–	Ntabid	Wild	Md	1	5	5	12/10	0.02
B. buonopozense	Bombacaceae	Red silt cotton tree	Ukim	Wild/ cultivated	Tb	1	2	2	15/4	0.02
S. tragacantha	Sterculiaceae	African tragacanth	Udod Eto	Wild/ cultivated	Md	1	4	4	13/8	0.02
S. indicum	Pedaliaceae	Sesame	Etehedeh	Wild/ cultivated	Fd, Md	2	14	28	9/28	0.04
G. brevis	Tiliaceae	–	Ndodido	Wild/ cultivated	Md	1	6	6	11/12	0.02
H. rosa-sinensis	Malvaceae	Red hibiscus	Frawa	Cultivated	Md, Om	2	14	28	9/28	0.04
H. bussei	Rubiaceae	–	Atama Idim	Wild	Md	1	2	2	15/4	0.02
V. polysphaera	Compositae/ Asteraceae	Bitter leaf	Asio-isong	Wild/ cultivated	Md	1	10	10	10/20	0.02
C. Ciliata	Capparidaceae	Consumption weed	Mkpat unen	Wild	Md	1	3	3	14/6	0.02
D. sammatii	Athyriaceae	Fern	Nyama Idim	Wild	Om	1	1	1	16/2	0.02
T. triangulare	Portulaceae	Waterleaf	Mmon-mmon ikon	Cultivated	Fd, Md, Om	3	42	126	6/84	0.06

V. amygdalina	Compositae/ Asteraceae	Bitter leaf	Etidod	Cultivated	Fd, Md	2	49	98	2/98	0.04
P. guineense	Piperaceae	Guinea black pepper	Odusa	Wild/ cultivated	Fd Md	2	50	100	1/100	0.04
P. mildbraedii	Papilionaceae	White Camwood	Mkpa	Wild/ cultivated	Fd, Md, Tb	3	48	144	3/98	0.06
L. africamum	Icacinaceae	–	Editan	Wild/ cultivated	Fd, Md	2	50	100	1/100	0.04
J. insularis	Acanthaceae	Hunter's weed	Mmeme	Wild/ cultivated	Fd, Md	2	50	100	1/100	0.04
G. africanum	Gnetaceae	African salad	Afang	Wild/ cultivated	Fd, Md, Om	2	46	92	4/92	0.04
C. maxima	Cucurbitaceae	Melon pumpkin	Ndise	Wild/ cultivated	Fd, md	2	37	74	7/74	0.04
A. hybridus	Amarantheceae	Amaranth	Inyan-afia	Wild/ cultivated	Fd, Md	2	43	86	5/86	0.04
T. occidentalis	Cucurbitaceae	Fluted pumpkin	Ikon ubon	Cultivated	Fd, Md	2	49	98	2/98	0.04

Notes: *Fd* food, *Md* medicine, *Om* ornamental, *Tb* timber, *UV* use-value
Source: [6, 17].

and employing sustainable management techniques, it is critical to document the variety of vegetation, particularly the situation of medicinal plants in the area. It is important that these untamed and priceless resources are well-maintained because a large section of the forest communities rely on them for economics, food, construction, and medical uses, as well as the restoration of ecosystem functions and biodiversity. Similar to this, there are not many studies on how various forest canopies affect the variety of medicinal plants. In numerous studies, the biodiversity of diverse medicinal plants in the chosen forest is portrayed, as well as the impact of different canopies on the variety and abundance of the medicinal plants. Despite the fact that there are many studies emphasizing the importance and diversity of medicinal plants in many places, this research offers a new dimension to our understanding of how differing forest canopies affect the variety, distribution, and abundance of medicinal plants.

3.10 Medicinal Plant Cover and Abundance Between the Various Forest Canopies

Research revealed that the comparison of the different forest canopies reported that the partial canopy had 65% medicinal plants as the maximum number, followed by the closed canopy with 56% and 41% for open spaces.

3.11 Types of Medicinal Plant Canopy

3.11.1 Closed Canopy

The closed canopy recorded about 56% of medicinal plant species identified and classified as extremely important medicinal plants, in which the most dominant one was *Valeriana wallichii* (Frequency (F) = 100%; Density (D) = 173). *Perilla frutescens* was the second medicinal plant species that was significantly present (F = 70%; D = 113). *Geranium wallichianum* (F = 60%; D = 56), a threatened and endemic medicinal plant (Zubair et al. 2021). Infrequent medicinal plants species were also recorded, the most treasured of which was *Podophyllum hexandrum* Royle (F = 10%; D = 1). Some edible plants of about 25% were noted and the most prominent was *Potentilla indica* (Andrews) T. Wolf having the maximum F and D (F = 100%; D = 320). Other medicinal plants such as *Viburnum grandiflorum* Wall. (F = 10% D = 2) and *Dryopteris filix-mas* (L.) Schott (F = 20% D = 18) were not quite as abundant (Table 7).

3.11.2 Open Spaces

Zubair et al. [18] reported the frequency (F) and Density (D) of the vegetation in the open space to be little to no different when compared to the closed canopy. The difference was that the open spaces were the representation of grasses and forage with 47% in the majority that were utilized by the grazing community. The most abundant of medicinal plants were *Chrysopogon gryllus* (F = 70%; D = 171), *Trifolium repens* L. (F = 60%; D = 185), *Gladiolus communis* (F = 50%;

Table 7 Species and classification in closed canopy in Naga forest

Species	Classification
Perilla frutescens	Medicinal
Adiantum stenochlamys	Medicinal
Viburnum grandiflorum	Fruit
Chrysopogon montanus	Animal forage
Lavatera cachemiriana	Medicinal
Podophyllum hexandrum	Medicinal
Geranium wallichianum	Medicinal
Arisaema triphyllum	Medicinal
Dryopteris filix-mas	Vegetable
Sonchus asper	Animal forage
Chrysopogon gryllus	Animal forage
Potentilla indica	Fruit
Gentiana decumbens	Medicinal
Geranium nepalense	Medicinal
Dryopteris austriaca	Vegetable
Valeriana wallichii	Medicinal

Source: [18]

D = 37), *Primula denticulata* (F = 30%; D = 11), *Cyperus rotundus* (F = 30; D = 46). Plants species showing medicinal characteristics (some chemical constituents) were also represented in this paper, having about 41% of the species, *Hypericum perforatum* (F = 30%; D = 41) being the most prominent and important followed by *Plantago ovata* (F = 20% D = 21) and *Rumex nepalensis* (F = 20%; D = 13), others were quite rare, having minimum frequencies and densities. Edible vegetables were not very dominant and hence were sparsely scattered around different canopies. Among the edibles, only *Potentilla indica* (F = 40% D = 56) had considerable representation (Table 8).

3.11.3 Partial Canopy

Partial canopy cover depicted a different trend away from open and closed spaces and recorded the greatest number of medicinal plant species. The maximum number of medicinal plants was 65%, the most dominant species being the *Mentha spicata* (F = 50%; D = 41) followed by *Polygonum amplexicule* (F = 30%; D = 55) and *Skimmia laureola* (F = 20%; D = 58), while the other 52% of the medicinal plant species were quite rare and were not uniformly distributed in the canopy [18]. Among the rare species, the most important medicinal plant species present in the canopy were *Paeonaia emodi* (F = 10%; D = 35), *Berberis lycium Royle* (F = 10%; D = 1), *Asparagus racemosus* (F = 10%; D = 14), and *Geranium wallichianumi* (F = 10%; D = 40). As compared to the closed canopy-covered site, this area had quite a few varieties of grass species, the most dominant being the *Chrysopogon gryllus* (F = 100%; D = 150) and *Trifolium repens* (F = 40%; D = 244). In this site, a few edible species were represented. A quite popular edible plant, *Potentilla indica* (F = 80%; D = 299), was in abundance when compared to other edible vegetables while other important edible plants such as *Dryopteris filix-mas* (F = 10%; D = 18), *Phyllanthus niruri* (F = 10%; D = 14), and *Solanum nigrum* (F = 10%; D = 16) were quite rare as assessed in the canopy (Table 9).

Table 8 Species and classification in open spaces in Naga forest

Species	Classification
Hyoscyamus niger	Medicinal
Skimmia laureola	Medicinal
Cyperus rotundus	Animal forage
Cynodon dactylon	Animal forage
Rubus niveus	Fruit
Adiantum stenochlamys	Medicinal
Primula denticulate	Animal forage
Artemisia absinthium	Medicinal
Potentilla collettiana	Fruit
Rumex nepalensis	Animal forage
Gladiolus communis	Aromatic
Chrysopogon gryllus	Animal forage
Plantago ovata	Medicinal
Hypericum perforatum	Medicinal
Trifolium repens	Animal forage
Potentilla indica	Fruit

Source: [18]

Table 9 Species and classification in partial Canopy in Naga forest

Species	Classification
Asparagus racemosus	Vegetable
Adiantum stenochlamys	Medicinal
Polygonum amplexicaule	Medicinal
Dryopteris austriaca	Vegetable
Paeonaia emodi	Medicinal
Geranium wallichianum	Medicinal
Arisaema triphyllum	Medicinal
Miscanthus nepalensis	Animal forage
Rumex nepalensis	Medicinal
Berberis lyceum	Medicinal
Plantago major	Medicinal
Potentilla indica	Fruit
Chrysopogon gryllus	Animal forage
Skimmia laureola	Medicinal
Dryopteris filix-mas	Vegetable
Trifolium repens	Animal forage
Mentha spicata	Medicinal

Source: [18]

4 Conclusion

This chapter has extensively discussed the diversity and different methods of medicinal plant classification. Medicinal plants are plants that have the potency in the treatment of ailments and prevention of certain diseases and conditions which affect humans and animals alike. It is estimated that out of the 350,000 higher plant species on earth, more than 80,000 have medicinal properties. They are classified according to: part use, habit, habitat, and therapeutic value as well as Ayurvedic formulation. Plantae as a Kingdom has existed for about 410 million years as green algae evolve from water to land, medicinal plants included. The land is a rich resource of medicinal plants and was comparatively not colonized. This chapter expounds on the diversity and classification of these medicinal plants to aid understanding and usage of these plants. This chapter has also offered a new dimension to our understanding of how differing forest canopies affect the variety, distribution, and abundance of medicinal plants. Research has revealed that from the comparison of the different forest canopies, the forest ecosystem with the partial canopy had 65% medicinal plants as the maximum number, followed by the closed canopy with 56% and 41% for open spaces. Thus, for the purpose of developing strategies and employing sustainable management techniques in the conservation of medicinal plants, it is critical to document the variety of vegetation, particularly the situation or status of medicinal plants in the area. It is important that these untamed and priceless resources are well-maintained and conserved because a large section of humans living in forest communities rely on them for economics, food, construction and medical uses, as well as the restoration of ecosystem functions and biodiversity.

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Classification of Phytochemicals in Plants with Herbal Value

4

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Abstract

In the past decade, there has been an increased concern about the effects of medicinal plants. Traditional medicinal herbs from diverse habitats and locations can be evaluated as novel treatment and prevention methods for injuries and diseases. Natural products, especially secondary metabolites in medicinal herbs, including those utilized in conventional and ethnic health care systems, provide prospective components for developing novel drug candidates. Phytochemicals have many potential roles as they can protect plants from enemies and act as antimicrobial, anti-inflammatory, antidiabetic, and chemopreventive agents. Their identification and classification are usually according to the chemical formula, such as flavonoids, terpenoids, alkaloids, saponins, phytosterols, carotenoids, essential oils, nonessential amino acids, and aromatic and aliphatic acids. Each group has characteristics, including anticancer, anthelmintic, and antigenotoxic. Additionally, they can offer direct/indirect protection against pathogens or hazardous conditions. Due to their potency and cost-effectiveness, phytochemicals have recently received considerable interest in this area. The impact of medicinal plant utilization is international and has been developing in many countries. Notably, as a potential source of alternative treatments, traditional medicine has attracted attention worldwide. This chapter will focus on classifying phytochemicals (primary and secondary metabolites), identifying some active secondary metabolites (such as flavonoids, alkaloids, terpenoids, phytosterols, and phenolic compounds), studying their potentiality in the treatment of some disorders, and the modern research advances in herbal medicine field.

Keywords

Phytochemicals · Herbal medicine · Flavonoids · Alkaloids · Cancer treatment · Antimicrobial properties · Artificial phytochemicals production

Abbreviations

5-FU	5-fluorouracil
ABTS	2,2′-azinobis-(3-ethylbenzothiazoline-6-sulfonic acid)
ACE	Angiotensin-converting enzyme
ATP	Adenosine triphosphate
COX	Cyclooxygenase
DPPH	α, α-diphenyl-β-picrylhydrazyl
ECa-09	Esophageal squamous carcinoma cell in the human
ESCC	Esophageal squamous cell carcinoma
HL-60	Promyelocytic leukemia cell line
HNSCC	Head and neck squamous cell carcinoma

IC_{50}	Half-maximal inhibitory concentration
JAK/STAT	The Janus kinase/signal transducers and activators of the transcription pathway
NF-B	Nuclear factor-B
Nrf-2	Nuclear factor-erythroid factor 2-related factor 2

1 Introduction

Phytochemicals (secondary metabolites) are the chemical substances synthesized by the plant through various chemical mechanisms. Most phytochemicals support plants' ability to compete with other plants and provide defense against diseases, herbivores, or abiotic challenges, such as high UV radiation doses. Usually, humans and animals can detect or smell and sometimes taste the secreted phytochemicals. Moreover, phytochemicals contain various chemical constituents, including alkaloids, flavonoids, saponins, steroids, terpenoids, and many other active groups. Plant metabolites can be classified depending on their physiological function in the plant system as follows: (1) Primary metabolites: They are biomolecules that directly affect the metabolism and the development of different species, and they are broadly distributed in plants, thus being appropriate for use as an edible human source. In addition, these primary components (carbohydrates, proteins, lipids, vitamins, and nucleic acid constituents) are produced through a complex system of metabolic processes to fulfill energy demands and serve as precursors for some physiological processes and biochemical synthesis of vital components (secondary metabolites) [1]. (2) Secondary metabolites are derived from the primary metabolites, in particular developmental stages of the plant. These sophisticated compounds have diverse purposes in the plant system (defense against enemies) and potential biological activity, such as medicinal components [2].

The identification and nomenclature of plant species and understanding their relationships to other species are fundamental for many researchers. Moreover, the botanical identification and taxonomic positions of studied plants, especially medicinal herbs, are crucial in the scientific investigation of their therapeutic usage or the foundation of crude pharmaceuticals. Numerous medicinal plants belonging to diverse plant families include Fabaceae, Apiaceae, Apocynaceae, Lamiaceae, Malvaceae, Mimosaceae, Papaveraceae, Phytolaccaceae, Asteraceae, Boraginaceae, Brassicaceae, Caryophyllaceae, Cesalpinaceae, and Cucurbitaceae [3].

Attributed to their significant biological activity, phytochemicals have been used for decades in traditional medicine, reflecting the critical therapeutic benefits of these chemical components [4]. Additionally, different tissues and organs of medicinal herbs may have unique therapeutic characteristics at distinct stages of development [5]. They are involved in valuable industries like pharmaceuticals, cosmetics, and fine chemicals [6].

In this chapter, the classification of some phytochemicals, including main active groups (terpenoids, flavonoids, alkaloids, phytosterols, and phenolic components) are highlighted and their general biosynthesis pathway, their vital activity in the treatment of some diseases, artificial production of secondary metabolites, and some recent advances in their research are presented.

2 Classification of Phytochemicals

The secondary metabolites are very specialized and can be prevalent in several plant species. The complicated combinations of plant secondary metabolites provide distinctive chemical characteristics among the plant classes, which is an essential classification tool for taxonomists [7]. A summary of some active compounds of different medicinal plant species and their potential treatment effects are presented in Table 1.

2.1 Flavonoids

Flavonoids belong to the polyphenol family of plant secondary metabolites, with more than 6,000 structures, widely distributed in various parts of the plant due to the multiple potential benefits provided by the experts that have received significant attention [56]. Like most metabolites described, flavonoids are essential for adapting plants to their environment, helping to cope with biotic and abiotic stresses, and having critical pharmacological activities. The biosynthesis of flavonoids was studied by using many plant species, including *Arabidopsis thaliana*, *Petunia hybrida*, and *Zea mays*. The synthesis of flavonoids begins with the phenylalanine pathway; multiple enzymes are involved, which makes flavonoids one of the rich families in the plant kingdom [57–59]. The basic structure of most flavonoids contains two phenyl rings (A and B), connected by a heterocyclic pyrene ring with an oxygen atom (C-ring) with the general formula of C6-C3-C6 (Fig. 1). Usually, flavonoids are water-soluble metabolites and classified following the position of ring B into isoflavone and other types, while according to the degree of saturation of the central heterocyclic C-ring can be subdivided into seven categories, including flavanone, flavone, flavanonol, flavanol, flavonol, anthocyanidins, and chalcones (Fig. 2) [60].

The most significant flavonoids are quercetin, quercitrin, and kaempferol, which are expressed in about 70% of all plant species. Several flavonoids play a vital role in anticancer and improving cardiovascular health. For example, hesperidin (Hsp) is a vital flavonoid with high anticancer activity. Also, quercetin is an antioxidant flavonoid, improving blood vessel health and reducing cardiovascular disease risk [61, 62].

2.1.1 Isoflavone

The distribution of isoflavone in plants is limited. It is mainly existed in soybean and legumes and known as phytoestrogens. It has a unique structure to other flavonoids; the position of the B ring for isoflavone is in the third place of the C ring, while the B ring is in the second place of the C ring for other flavonoids. Isoflavone has potential

Table 1 Overview of some bioactive secondary metabolites in herbal medicine

Scientific name	Family of the plant	Active chemical compound	Medical importance	Reference
Cajanus cajan	Fabaceae	Hydroalcoholic, cyanidin-3-monoglucoside, longistylin C	Antioxidant, anti-inflammatory activities, anticancer	[8, 9]
Coriandrum sativum	Apiaceae	Linalool, γ-terpinene, and α-pinene	Hepatoprotective	[10]
Calotropis procera	Apocynaceae	Group of cardiac glycosides (calotropin, calactin, and calotoxin)	Anti-ATPase, antidiarrheal, anticonvulsant, and antiviral	[11–13]
Psychotria nervosa Sw.	Rubiaceae	Emetine (emetine dihydrochloride) and cephaeline	Anti-dengue virus infection	[14–17]
Hippeastrum puniceum (Lam.) Kuntz	Amaryllidaceae	isoquinoline alkaloids (9-O-demethyllycoramine, 9-demethyl-2α-hydroxyhomolycorine, lycorine and tazettine)	Antioxidant, antifungal, antiparasitic, and acetylcholinesterase inhibitory activity	[18]
Hymenocallis caribaea (L.) Herb	Amaryllidaceae	Narciclasine and prancristatin	Antineoplastic compounds	[15, 19, 20]
Thespesia populnea (L.) Sol. ex Corrêa	Malvaceae	Thespesenone and dehydrooxoperezinone-6-methyl ether	Antihepatotoxic compounds	[21–23]
Lagerstroemia speciose (L.) Pers.	Lythraceae	Corosolic acid and ellagitannins	Diabetes treatment, antihyperlipidemic, and antioxidant activities	[24]
Persea americana Mill.	Lauraceae	Peptone, b-galactoside, and glycosylated abscisic acid	Hypertension, stomachache, bronchitis, diarrhea, and diabetes	[25]
Cocos nucifera L.	Arecaceae	Folate (vitamin B_9)	Reduce the risk of breast cancer and anemia during pregnancy	[26, 27]
Acorus calamus L.	Acoraceae	β-asarone, eugenol, asaronaldehyde, and acorin	Antidiabetic, anti-obesity, antihypertensive, anti-inflammatory, and anticonvulsant activities	[28–31]
Alstonia scholaris L. R. Br.	Apocynaceae	Ditamine, echitamine, echitenine, alschomine, isoalschomine, tubotaiwine, and N^b-oxide	Anti-ulcer, antirheumatic, carminative, aphrodisiac, and antiperiodic	[32]

(continued)

Table 1 (continued)

Scientific name	Family of the plant	Active chemical compound	Medical importance	Reference
Phyllanthus epiphyllanthus	Phyllanthaceae	Michellamine B (novel alkaloid), gallotannins, and triterpenes	Inhibition of human immunodeficiency virus (HIV), hepatitis B virus, and antibacterial activity against typhoid fever bacteria	[15, 33–35]
Citrus limon L.	Rutaceae	Eriodictyol	Prevention of diabetic retinopathy, antioxidant, anti-inflammatory, and analgesic effects	[36, 37]
Euphorbia hypericifolia L.	Euphorbiaceae	Isomotiol, espinendiol A, ursolic acid, juglangenin A, and teuviscin A	Anticancer activity	[38–41]
Valeriana officinalis L.	Caprifoliaceae	Isovaleric acid, gamma-aminobutyric acid, valerenic acid, and valerine	Relieves mild nervous tension and helps in sleep	[42–44]
Eucalyptus obliqua L'Hér.	Myrtaceae	1,8-cineole, catechins, isorhamnetin, luteolin, kaempferol, phloretin, and quercetin	Anticancer, antioxidant, antiseptic, stimulant, antimalarial, anthelmintic action, anti-inflammatory, and antihistaminic	[45–48]
Panax ginseng C. A.Mey.	Araliaceae	Ginsenosides	Improve phagocytosis, interferon production, and vasodilation, and affects the hypoglycemic activity	[49–51]
Stachytarpheta jamaicensis (L) Vahl	Verbenaceae	Gamma-aminobutyric acid and dopamine	Inhibition of the chicken pox virus	[15, 52]
Moringa oleifera Lam.	Moringaceae	Quercetin, kaempferol glycosides, myricetin, and epicatechin	Hypolipidemic, hypoglycemic, antioxidant, anti-inflammatory, and anticancer properties	[53–55]

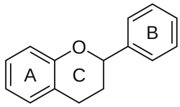

Fig. 1 Basic structure of flavonoids

antibacterial and antioxidant activities [64, 65]. Besides, it is protective against acute lung injuries, cardiovascular diseases, and breast cancer [66–68].

2.1.2 Flavanone

Flavanone is the direct precursor of most flavonoid synthesis. It has a saturated and oxidized C ring. For instance, naringenin and hesperetin are analogue compounds for flavanones and excipients richer in citrus, lemon, grapefruit, and fruit peels. Naringenin is potentially antiviral and antitumoral, preventing cardiovascular disease [69–71]. Hesperetin and its derivatives positively improve acute lung injury, diabetes, and Alzheimer's disease [72–75].

2.1.3 Flavone

Flavone is one of the largest subgroups of flavonoids. Compared to flavanones, besides having a keto group at position four on the C ring, there is also a double bond between the C-2 and C-3 of the C ring. Flavone is present in practically all plant tissue. It has commercial and pharmaceutical value as neuroprotective [76, 77]. For example, apigenin protects neuronal cells from injury by inhibiting microglia cells [78]. In addition, some studies showed that apigenin is safe for humans, which makes it a suitable target for future drug development [79, 80].

2.1.4 Flavanol

Flavanol has a saturated and oxidized C ring and a hydroxyl group in C-3 or C-4 of the C ring. It is mainly found in cocoa, fruits, cereals, and vegetables [81]. Flavanols are important bioactive compounds in *Theobroma cacao*. Researchers have proven its efficacy in anti-inflammatory, neuroprotective, and cardiovascular diseases [82–85]. Recently, Hidalgo et al. [86] investigated the therapeutic activity of epicatechin and epigallocatechin-3-gallate (the main active compound in green tea) flavanols toward nonalcoholic fatty liver disease. Their findings reflected several beneficial properties besides its ability to prevent or treat nonalcoholic fatty liver disease, such as antihyperglycemic, antihypertensive, antithrombotic, anti-inflammatory, and antifibrotic effects. Moreover, following the US Food and Drug Administration classification, they reported its safety on humans as harmless.

2.1.5 Flavonol

Flavonol has a hydroxyl group in C-3 of the C ring, like flavanol. However, there is also a double bond between the second and third carbons in the C ring. Kaempferol and quercetin are representative compounds of flavonols. Both have anxiolytic, antiviral, and anti-inflammatory effects [62, 87]. Tian et al. [88] studied the antioxidant (through the activities of phagocytosis, DPPH, and ABTS radical scavenging)

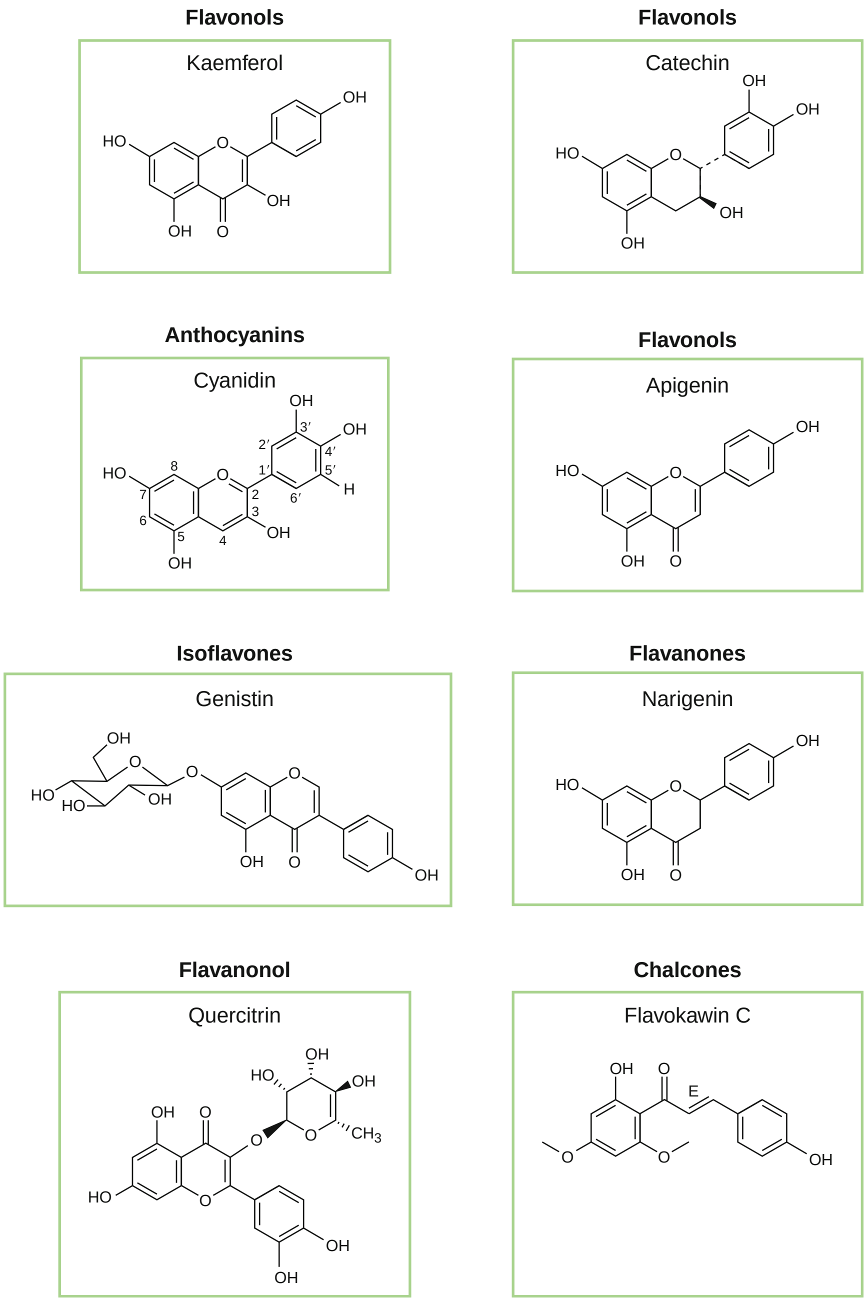

Fig. 2 Examples of different flavonoid categories. (Modified after Bešlo et al. [63])

and anti-inflammatory properties of some flavonols (kaempferol, luteolin, quercetin, and apigenin). The results showed that quercetin is a promising anti-inflammatory and antioxidant substance potentially as a therapeutic adjunct for inflammatory disorders and oxidative stress. Furthermore, preliminary findings from this study indicated that antioxidant activity is directly correlated with the number of phenolic hydroxyl groups. A comparison of anti-inflammatory and antioxidant activities revealed that compounds with enol groups were better than those without.

2.1.6 Flavanonol

Flavanonol has a hydroxyl group at position C-3 and a keto group at C-4. Its compounds can exist in free or combined form and are mainly abundant in citrus. Additionally, flavanonols are found in various medicinal herbs families, like Rutaceae, Leguminosae, and Rosaceae [89]. Flavanonol has significant medical value in vascular protection; it has been found that taxifolin has some therapeutic properties, including preventing the damage of vascular structure in diabetic patients, improving capillary microcirculation, and improving blood flow in the retina [90].

2.1.7 Anthocyanidins

Anthocyanidins are water-soluble pigments widely distributed in flowers and can also be detected in leaves, roots, and fruits (strawberry, chokeberry, and elderberry) [91–95]. The soluble pigment helps plants show different colors (red, orange, blue, and green). Consequently, approximately all angiosperm species contain anthocyanins [96, 97]. The structure is characterized by the absence of ketone in the C-4 position of the central heterocyclic ring and the positive charge of the first oxygen atom in the C ring. In flora, anthocyanins are the most prevalent anthocyanidins (containing glycosylated flavylium ion). Up to 300 anthocyanidins have been identified [98], whereas there are approximately 8,000 possible anthocyanins, such as the various anthocyanidin subtypes and the glycosylated [99].

Polyphenolics, especially anthocyanins, have recently become more relevant in treating chronic diseases like cardiovascular diseases [100]. Pergola et al. [101] investigated that the blackberry extract (about 88% of the cyanidin-3-glucoside out of the total anthocyanin content) inhibited the deleterious cardiovascular activity of nitric acid. Moreover, Graf et al. [102] studied the antiatherogenic activity of anthocyanins in the grape-bilberry extract. The results demonstrated that a dose of 1.55 mg/L decreased the total content of cholesterol and triglyceride levels in treated models. Additionally, anthocyanins reflected a physiological disruption of many groups of fatty acids, including a reduction in saturated fatty acids and an elevation in the long-chain fatty acids in plasma. Consequently, these findings showed the antiatherosclerosis activity of anthocyanins-rich grape-bilberry juice.

2.1.8 Chalcones

Chalcones are open-chain flavonoids (alpha, beta-unsaturated ketones) lacking the structural formula's complete C ring. The basic structure of chalcones includes two conformations: cis and trans isomers [103]. It is widely distributed in medicinal herbs and can be modified into diversified forms (prenylated chalcones (3, 4, 7), licochalcones, and dihydrochalcones). Chalcones and derivatives have been reported for pharmacological activities in the aspect of antiviral, cardiovascular disease prevention, antimicrobial, antioxidant, analgesic, and antidiabetic functions [104–108]. Many researchers reported the pharmaceutical properties of chalcones. Tang et al. [109], Acharjee et al. [110], and Attarde et al. [111] showed the antidiabetic activity of chalcone, piperonal chalcones derivative, and 3-(4-hydroxyphenyl)-1-phenylprop-2-en-1-one (chemically synthesized chalcone) exhibited significant suppression of the activity of α-glucosidase and α-amylase.

2.2 Alkaloids

Alkaloids are a prominent structurally diverse family of heterocyclic substances containing nitrogen in the heterocyclic ring and obtained from amino acids [112]. They exhibit a variety of significant physiological impacts on humans and other mammals. Morphine, strychnine, quinine, ephedrine, and nicotine are prominent alkaloids. Alkaloids are mainly found in plants and are particularly prevalent in certain flowering plant families [113]. Around 20% of higher plant species are estimated to contain alkaloids, with several thousand distinct varieties, besides some other alkaloid-rich families, including Ranuculaceae, Amaryllidaceae, and Solanaceae [114]. They are primarily associated with plant defense against herbivores (feeding deterrence) and pathogens (antibacterial and antifungal activities). Traditional and recent applications of alkaloids range from 25–75% in pharmaceuticals, demonstrating their enormous medicinal potential [115, 116]. The fundamental quality of alkaloids is no longer the requirement for an alkaloid, and the chemical reactivity of nitrogen atoms permits at least four classes of nitrogenous substances.

Moreover, several synthesized substances with equivalent structures are also referred to as alkaloids [117]; however, others can produce salts with organic acids like oxalic and acetic acids. Certain botanical alkaloids, such as solanine in the Solanaceae family, exist in a glycosidic form. The biosynthetic pathway of the alkaloid is involved, especially in the decarboxylation of substances [118]. For example, aspirin is the most common antiplatelet (pain treatment) medicine, prepared mainly from salicin alkaloids originating from the willow plant (*Salix alba* L.) [119].

According to their biochemical precursor and heterocyclic ring arrangement, alkaloids have been categorized into several groups, including indole, piperidine, tropane, purine, pyrrolizidine, imidazole, quinolizidine, isoquinoline, and pyrrolidine [112, 120]. Fig. 3 shows the structural formula of some potent alkaloid substances.

Some pure isolated alkaloids and synthetic analogues (berberine, serotonin, dopamine, and gamma-aminobutyric acid) are employed as fundamental therapeutic agents globally due to their antispasmodic, antimalarial, antibacterial, and analgesic activities [121, 122]. Alkaloids can prevent the beginning of several degenerative disorders by scavenging free radicals or interacting with the oxidation reaction catalyst. Numerous studies have evaluated the vast spectrum of pharmacological properties of alkaloids

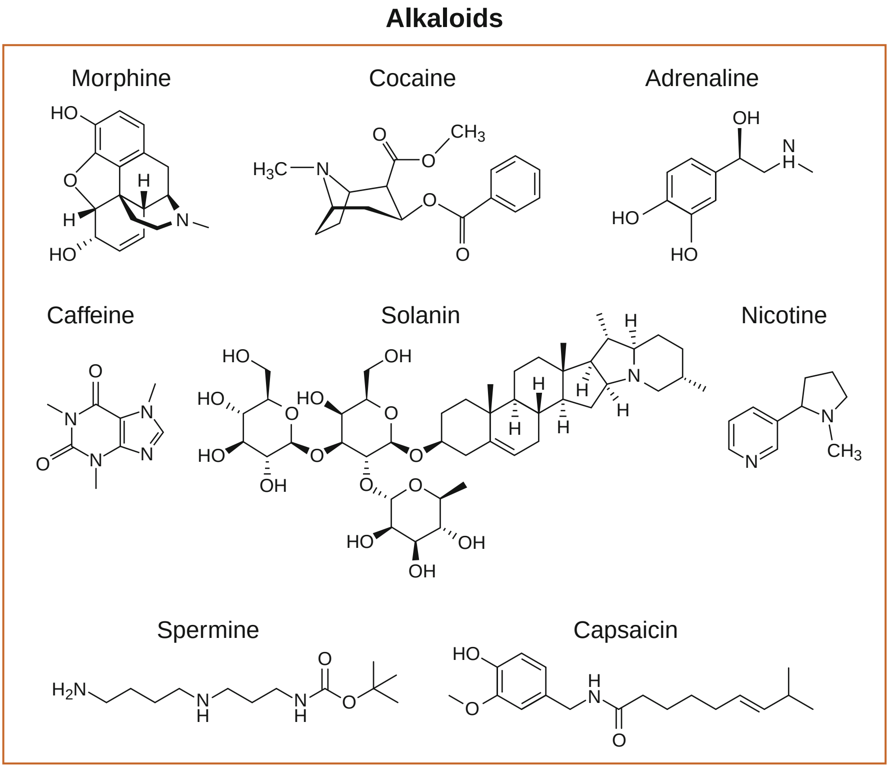

Fig. 3 Examples of some vital alkaloid compounds

from diverse plants [120]. For instance, the active ingredients of *Papaver somniferum* L., the opium poppy (morphine), and its methyl ether derivatives (codeine) are comparatively nonaddictive analgesics [123]. Moreover, some alkaloids stimulate the cardiovascular or respiratory systems. Uzor [124] reported that quinine, chloroquine, amodiaquine, mefloquine, artemisinin, and artemether are the most vital antimalarial alkaloidal drugs. In addition, Kouam et al. [125] isolated three novel antimalarial indolosesquiterpene alkaloids from the bark extract of *Polyalthia oliveri*. Several studies [112, 126–131] investigated the activity of some active alkaloid constituents, such as atropine (from *Atropa belladonna*) acts as an antidote; ephedrine (from *Ephedra sinica*) can be used as antiasthmatics; and emetine (from *Carapichea ipecacuanha*) as antiprotozoal. Furthermore, noscapine (from *Papaver somniferum*) is used as an antitussive and used as a product in the following names Bequitusin, Degoran, and Tussisedal. In addition to pharmaceutical drugs, some alkaloids have been used as antiplatelet substances. Rutaecarpine, an alkaloid derived from *Evodia rutaecarpa*, was demonstrated as an antiplatelet, and its derivatives (3-methylenedioxyrutaecarpine, 3-chlororutaecarpine, and 3-hydroxyrutaecarpine) reflecting a significant enhancement of its role by interacting with several mediators of coagulation factors [132, 133].

2.3 Terpenoids

Terpenoids are the largest class of phytochemicals (plant-specialized metabolites) with several significant pharmacological effects [134]. They are isoprene-based (5-carbon) metabolites commonly known as isoprenoids. Terpenes can be divided into six categories based on the number of isoprene units in their chemical structure: monoterpene, sesquiterpene, diterpenes, triterpenes, tetraterpenes, and polyterpenoids (Fig. 4) [135]. Terpenoids exist mainly in flowering plants and are responsible for the distinct aroma, flavors, scents, and colors of many species, such as eucalyptus, cinnamon, tomatoes, sunflower, ginger, and clove [136].

Hundreds of terpenoid compounds possess pharmaceutical characteristics; for example, most terpenes in *cannabis* sp. (tetrahydrocannabinol) have diverse pharmacologic effects, including anti-inflammatory, sedative, immunomodulatory, and neuromodulatory properties [137].

2.3.1 Monoterpenoids

The chemical structure of monoterpenes and their derivatives contains two isoprene units. They incorporate three subclasses, acyclic monoterpenes, monocyclic monoterpenes, and bicyclic monoterpenes. For example, the *Cymbopogon citratus*

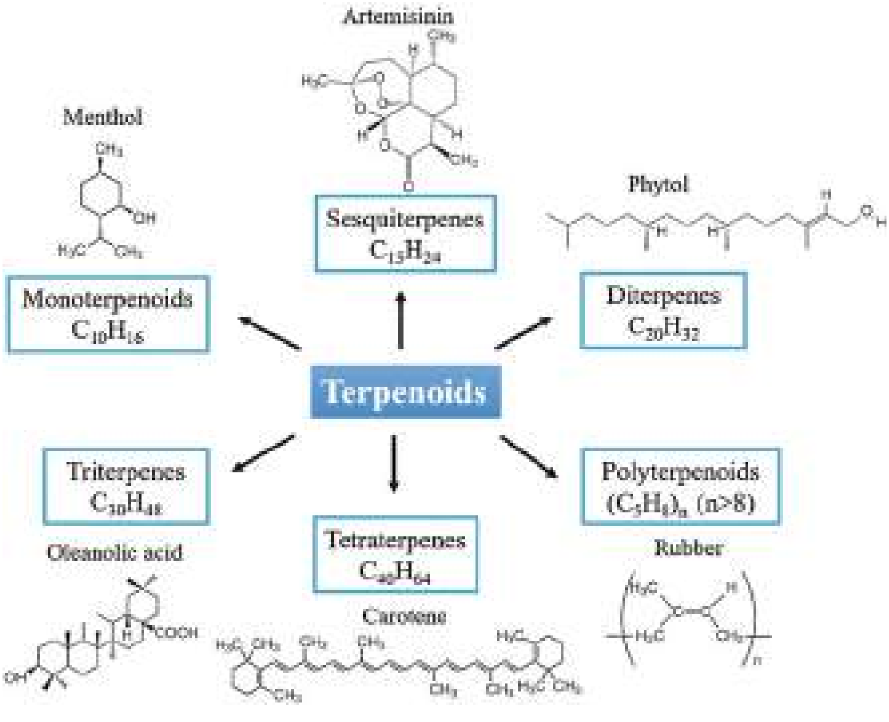

Fig. 4 Chemical structure of some representative terpenoid substances

(lemongrass) produces a large quantity of citronellol (acyclic monoterpene). Santos et al. [138] investigated that citronellol significantly affects bacterial, fungal, allergic, spastic, and diabetic resistance. More acyclic monoterpenes showed antibacterial, anti-inflammatory, and antianxiety properties, such as linalool and its derivatives [139, 140]. In addition, menthol is a monocyclic monoterpene with a prevalent analgesic activity in pain control [141]. α-and β-pinene are representative bicyclic monoterpenoid substances; however, they share the same molecular formula of C10H16, their chemical structures are distinct, and both exhibit anticancer and antimicrobial bioactivity [142–144].

2.3.2 Sesquiterpenes

Sesquiterpene compounds are the most common kind of terpene with three isoprene units. It is subdivided into straight sesquiterpenes (e.g., farnesol and nerolidol) and annular sesquiterpenes based on the stability of the carbon rings (e.g., artemisinin and guaiol). Almost 50% of sesquiterpene compounds contain lactones [145]. It has significant biological and physiological functions. Recent research demonstrated that sesquiterpene lactones could contribute to inflammation and cancer therapies [146, 147].

2.3.3 Diterpenes

Diterpene compounds comprise four isoprene units with one or more carbon rings within their molecular structure. Saha et al. [148] reported that diterpenes had a broad spectrum of microbial inhibition, including bacteria, viruses, and fungi. In addition, several studies reflected that it possesses anticancer, antidiabetic, and antitumor properties [149–153]. Diterpene compounds can not only inhibit the growth and spread of cancer cells but also promote the apoptosis of these cells. Thus, they are widely used to treat liver, lung, breast cancers, and other malignant tumors [151, 154, 155].

2.3.4 Triterpenes

Triterpene chemicals refer to six isoprene units–containing terpenes. The most common triterpene molecules are tetracyclic and pentacyclic, which have four- and five-carbon rings, respectively. Triterpenes showed potential medical activities; for example, oleanolic acid might combat diabetes by suppressing the function of α-glucosidase [156]. Moreover, Sohag et al. [157] revealed that lupeol (pentacyclic triterpene) safeguard against cardiovascular, renal, liver, and skin disorders. Sun et al. [158] summarized the effect of *Centella asiatica* triterpenes in treating common diseases and the underlying mechanisms, such as Alzheimer's disease, acne, chronic and recurrent liver injuries, pelvic inflammatory disease, and rheumatoid arthritis.

2.3.5 Tetraterpenes

Tetraterpenes are one of the isoprenoids containing eight isoprene units with molecular formula C40H64 – they involve many pigmentary substances (carotenoids) [159] and nonpigmentary compounds (poduran) [160]. Carotene is the first isolated tetraterpene discovered in carrots. Furthermore, beta-carotene has significant antioxidant effects that help in the treatment of many chronic diseases, such as optical diseases and particular cancers [161]. Zerres and stahl [162] reported that carotenoids are characterized by UV-absorbing properties and can be used as photoprotectants. They revealed that carotenoids provide comprehensive photoprotection after supplementation or consuming carotenoid-rich meals. Additionally, lycopene offers therapeutic potential to treat human malignancies [163].

2.3.6 Polyterpenoids

Polyterpenoid molecules have more than eight isoprene units and have a greater molecular weight than other terpenes. These can be extracted from some citrus oils or tree sap, and their yields fluctuate according to the growing seasons (e.g., rosin and its derivatives) [164]. Few investigations have been carried out on the pharmacological characteristics of polyterpenoids. Rubber is a natural polyterpenoid compound with between 1500 and 15000 isopentenyl units. de Barros et al. [165] and Mendonca et al. [166] showed that it influences wound healing and oxytocin-sustained release.

2.4 Phenolic Components

Phenolic acids are a large family of secondary aromatic metabolites. They are extensively distributed in their seeds, roots, stems, and leaves and are associated with xylans, pectin, and lignin. Similarly, to flavonoids, phenolic acids are essential members of the phenolic family. They comprise the most frequently distributed nonflavonoid phenolic molecules in plants and exist in various forms, like free, conjugated soluble, and insoluble bound. Phenolic compounds are the derivatives of benzoic and cinnamic acids [167]. The most common benzoic acid derivatives are p-hydroxybenzoic acid, salicylic acid, gallic acid, and ellagic acid, whereas the most prevalent derivatives of cinnamic acid are p-coumaric acid, caffeic acid, and ferulic acid [168]. Phenolic acids compounds can be classified into two fundamental subclasses hydroxybenzoic and hydroxycinnamic acids (Fig. 5).

In hydroxybenzoic acid, one of the hydrogens on the benzene ring is substituted by the carboxyl group (—COOH). The majority of hydroxybenzoic acid compounds exist as free acids, while a few occur as esters or glycosides. Hydroxycinnamic acids comprise a three-carbon chain connected to the benzene ring (C6H5CHCHCOOH) with at least one hydrogen atom that an OH group can replace [169]. In drug development technology, phenolic components have a significant interest worldwide due to their diverse chemical structures and biological activities. They play a crucial function in antihypertensive, antidiarrheal [170], neuroprotective, antidepressant, anti-inflammatory, anticancer, and antihyperglycemic drugs [171]. In addition, phenolic acids can suppress the activity of acetylcholinesterase and butyrylcholinesterase, which play a crucial role in treating Alzheimer's disease [172].

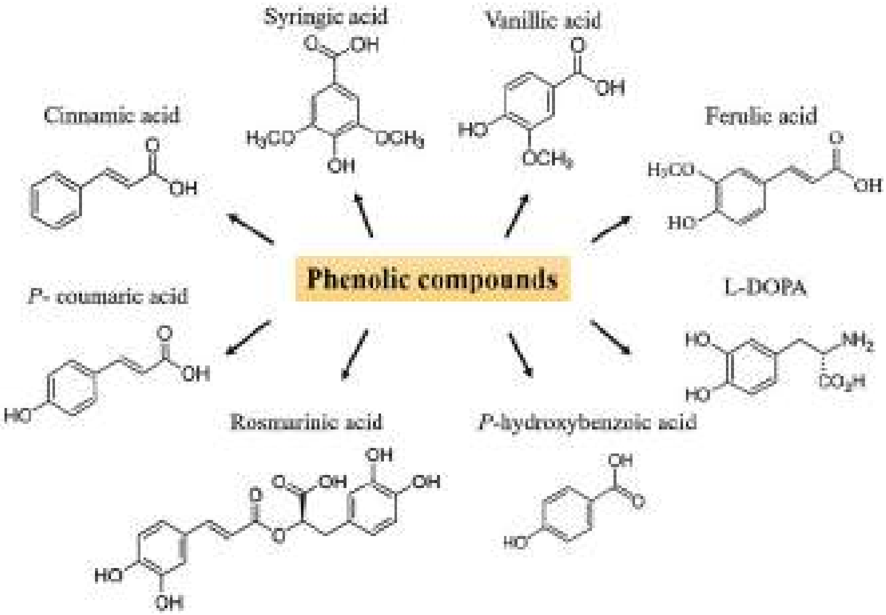

Fig. 5 Examples of some phenolic compounds

2.4.1 Hydroxybenzoic Acids

Hydroxybenzoic acid comprises a benzene ring, a carboxyl group, and additional hydroxyls, methyl (CHn), or methoxy (CHnO) groups attached to the benzene ring. The primary components of hydroxybenzoic acids include gallic, p-hydroxybenzoic, protocatechuic, and vanillic-hydroxybenzoic acid components produced as a resultant product of aromatic amino acids through the shikimic acid pathway [173]. Here, chloroformate intermediate is used to convert shikimic acid to L-phenylalanine. Consequently, L-phenylalanine is translated into p-hydroxybenzoic acid (a precursor to other phenolic acid metabolites) [174]. Moreover, the hydroxylation and methylation of its benzene ring produces other hydroxycinnamic acids (including ferulic and caffeic acids) or hydroxybenzoic acids (such as protocatechuic and p-hydroxybenzoic acids). It is hypothesized that hydroxybenzoic acid is synthesized by hydroxycinnamic acid with a similar structure via CoA-dependent (oxidative) or CoA-independent (nonoxidative) or a mixture of both mechanisms [169].

P-hydroxybenzoic acid, its derivatives, and vanillic acid are the major phenolic acids in many plant species, including cereals, flaxseeds, chia, and sunflower seeds [175–177]. Most of the hydroxycinnamic acid derivatives in chokeberry are chlorogenic and neochlorogenic acids. At the same time, strawberries and red raspberries involve predominant types of hydroxybenzoic acids, such as p-coumaric acid, p-hydroxybenzoic acid, and ellagic acid, respectively [168]. Hydroxybenzoic acid has contributed to significant medical research advances. Adefegha et al. [178] reported that some hydroxybenzoic acids have antidiabetic properties. For instance, gallic and protocatechuic acids inhibit the activity of type 2 diabetes–related enzymes (amylase and glucosidase). Furthermore, Yi et al. [179] investigated that the last mentioned acids (extracted from muscadine grapes) exhibited anticancer activities against cervical cancer.

2.4.2 Hydroxycinnamic Acids

From a structural perspective, hydroxycinnamic acids contain functional groups that can contribute to the coordination, like phenolic hydroxyl and carboxyl groups [180]. In addition, some compounds have phenolic hydroxyl groups and carboxyl groups of hydroxycinnamic acid and have one or two methoxy groups adjacent to phenolic hydroxyl groups on the benzene ring that have multiple coordination sites (which is crucial to the formation of complex and diverse compounds) [181]. Hydroxycinnamic acid is produced by lignin's metabolic pathway, which provides mechanical support for the plant cell wall [182]. It is catalyzed by phenylalanine ammonia-lyase, which deaminases L-phenylalanine to generate (E)-cinnamic acid. Then it undergoes further enzymatic transformations to create some related metabolites [183].

Recently, hydroxycinnamic acids' antioxidant, anti-inflammatory, and antibacterial properties have been investigated and employed in various pharmaceutical domains [184]. Many studies focused on vasodilation, antioxidant activity, and inflammation characteristics of hydroxycinnamic acid. Recent research showed that hydroxycinnamic acid compounds could reduce the severity of cardiovascular diseases [185, 186], hypertension [187, 188], depression [189], and diabetes [190]. Moreover, Tresserra-Rimbau et al. [179] and Adriouch et al. [180] studied that the high-intake percentages of hydroxycinnamic acids in daily meals significantly reduced the risks of cardiovascular disease in two countries (Spain and France).

2.5 Phytosterols

Phytosterols are cholesterol-like molecules found in various plants, with high concentrations occurring in the cell membranes of vegetables, fruits, and other plants. Phytosterol is a 3-hydroxy compound of cyclopentaphthene. The structure of phytosterols is similar to cholesterol; the main difference is that sterols are usually connected with a methyl or ethyl group at C-24, and common sterols have unsaturated double bonds between C-22 and C-23. Their structure comprises a steroid skeleton characterized by the saturated bond between C-5 and C-6 of sterol moiety [191]. Their aliphatic side chains are connected to C-17, while the hydroxyl groups are attached to the C-3 atom. They can be divided into sterols and stanols, representing unsaturated and saturated molecules. Phytosterols exist in four forms in plants, in the form of fatty acid esters, free alcohols, sterol glycosides, and acylated sterol glycosides [192]. They mainly exist in free and esterified forms (in edible oils), are found in plants, and cannot be synthesized in the human body. People mostly get phytosterols by ingesting dietary fiber, such as vegetable oil, beans, nuts, vegetables, and fruits [193]. Phytosterols show the characteristics of directly inhibiting tumor growth, including reducing cell cycle progression, inhibiting tumor metastasis, and inducing apoptosis, such as reducing the risk of oesophagal and ovarian cancers [194]. Campesterol, β-sitosterol, ergosterol, and stigmasterol are four representative compounds widely studied in treating diseases (Fig. 6).

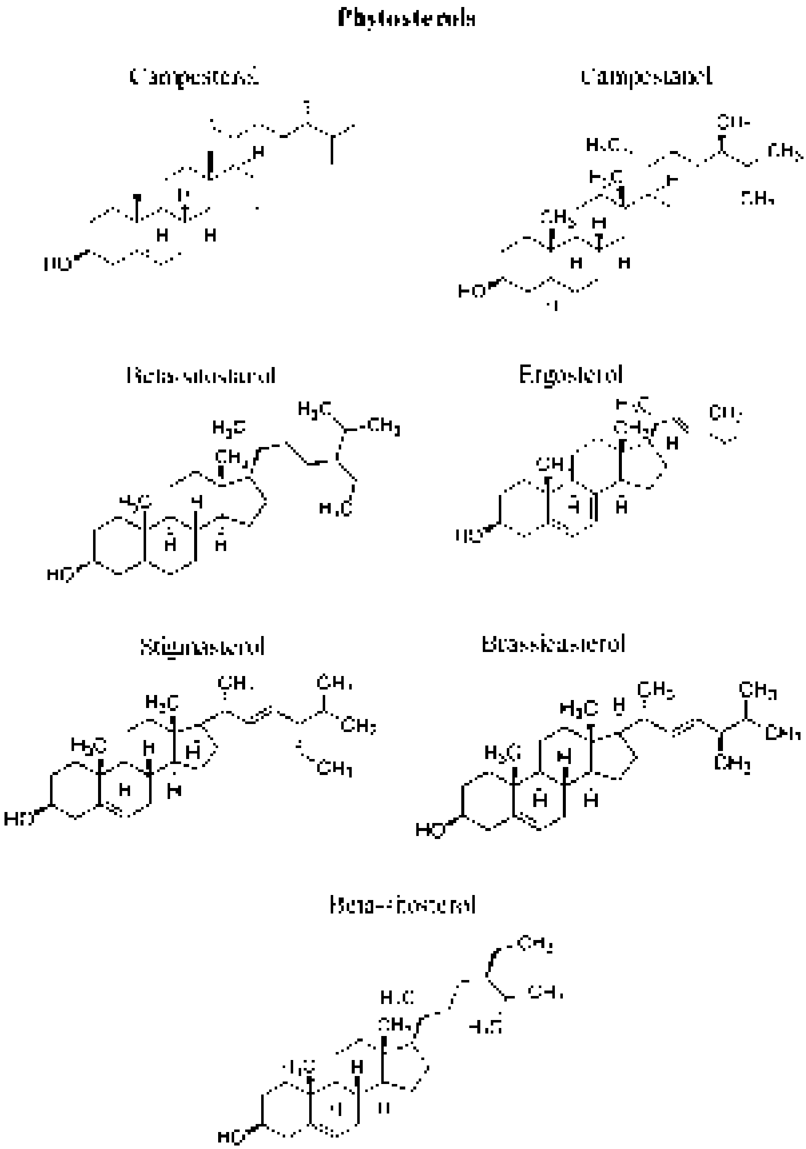

Fig. 6 Examples of some phytosterol compounds

Campesterol is abundant in canola and corn oils [195]. Campesterol has played an important role in inhibiting the proliferation of ovarian cancer cells and has the potential as a new antiovarian cancer drug. It was isolated from *Strychnos innocua* (Delile) and has been found to potentially affect antibacterial activity [196].

β-sitosterol has an unsaturated double bond on C5-C6, easily oxidized by active oxygen. Because of this structural characteristic, oxyphytosterols are one of the most common phytosterols in plants. Oxyphytosterols can play an important role as anti-inflammatory and antivirus agents. β-sitosterol can interfere with various cell signaling pathways, including cell cycle, proliferation, anti-inflammatory, hepatoprotective, and antioxidation [197]. Moreover, von Holtez et al. [198] discovered that low concentrations of β-sitosterol in cancer treatment dimensioned the development of cancer cells and even caused cancer cell death.

Ergosterol is the major steroidal component of the cell membrane of filamentous fungi and is not present or a minor component in most higher plants. It also exists in yeast cell walls and the mitochondrial membrane [193]. Its content has been widely used as an estimate of fungal biomass in soil and the aquatic environment because it was found that there was a strong correlation between ergosterol content and dry fungal biomass [194].

Stigmasterol is mainly found in soybeans. In addition to reducing cholesterol activity, stigmasterol has anti-inflammatory effects. Gabay et al. [199] showed that stigmasterol extracted from the bark of *Butea monosperma* (Lam.) Taub. has hypoglycemic, antiperoxidative, and thyroid-inhibiting effects.

3 Role of Secondary Metabolites in Diseases Treatment

Phytochemicals have been increasingly studied in disease treatment globally. In this section, the role of some phytochemicals in treating some disorders, including hepatic, renal, and cardiovascular disorders, cancer treatment, antimicrobial, anti-inflammatory activities, and neurological disorders, are discussed.

3.1 Hepatic Disorders

Nowadays, alternative therapy methods for several ailments are being embraced that use phytochemicals derived from natural resources. Despite significant developments in contemporary medicine, a persistent issue has been the lack of suitable and effective hepatoprotective medication [200]. Recent research has focused mostly on phytochemical screening to identify distinct types of biological activity. Recent years have seen a tremendous increase in interest in phytotherapy research as researchers look for new pharmacological cures. Numerous phytochemical substances, especially from medicinal herbs, are incorporated into the treatment of liver problems, including *Hedyotis corymbos*, *Rosa damascene*, *Casuarina equisetifolia*, *Cajanus cajan*, *Trichosanthes dioica*, *Glycosmis pentaphylla*, *Justicia gendarussa*, *Tinospora crispa*, *Bixa orellana*, *Moringa oleifera*, *Premna esculenta*,

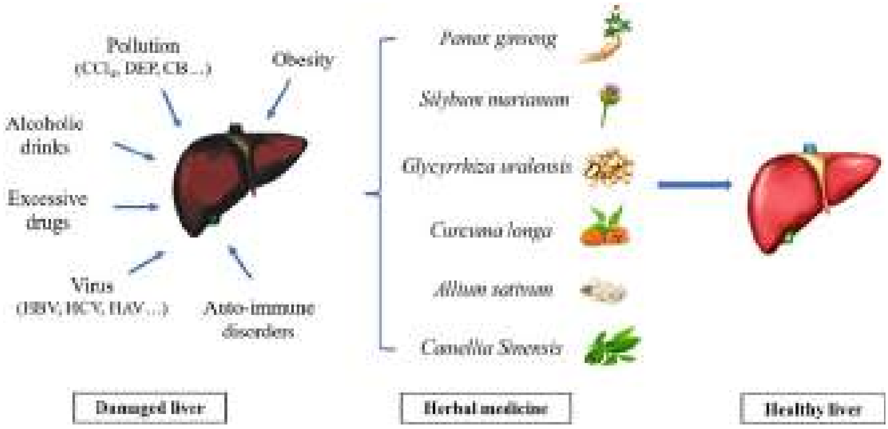

Fig. 7 Some vital phytochemicals support the hepatic disorders treatment. (modified after Das et al. [202])

Dendrophthoe pentandra, *Argemone mexicana*, *Leea macrophylla*, *Physalis minima*, *Synedrella nodiflora*, *Hylocereus polyrhizus*, *Caesalpinia bonduc*, *Piper chaba*, and *Bombax ceiba* (Fig. 7) [201].

Aguirre et al. [203] investigated some polyphenol components, like resveratrol (trans-3,4′,5-trihydroxystilbene) and stilbenoid involved in the enhancement of glycemic control and glucose tolerance, decrease in lipid levels, as well as anti-lipogenic, anti-inflammatory, and antioxidant properties. In addition, thymohydroquinone exhibited hepatoprotective characteristics toward drug-induced hepatotoxicity [204].

3.2 Renal Disorders

Renal damage and chronic kidney disorders are severe medical cases globally; the average number of patients increased by 8–16%. Besides Chinese herbal medicine metabolites, bioflavonoids, resveratrol, quercetin, and curcumin were demonstrated to be potential in chronic kidney disorders [205]. Green tea contains various polyphenolic substances, among them is epigallocatechin-3-gallate. Many reports showed the reactivity of epigallocatechin-3-gallate in renal damage treatment by delaying lupus nephritis via boosting the antioxidant Nrf2 pathway and diminishing the NLRP3 inflammasome activation. Additionally, ursolic acid (pentacyclic triterpenoid) is frequently found in fruit peels, herbs, and spices. Kunkel et al. [206] investigated that ursolic acid has a protective activity against chronic kidney disorders by suppressing the activity of STAT3 and the NF-B mechanism, hence decreasing the inflammatory activity. Moreover, allicin (organosulfur compound) enhances renal function by modifying the AT1, Nrf2/Keap1, and eNOS pathways and lowering CKD-mediated systemic hypertension [207].

3.3 Cardiac Disorders

According to current epidemiological forecasts, the globe is on course to experience a vascular typhoon of cardiovascular disease burden. In emerging nations, coronary heart disease raised by 120% in women and 137% in men in 2020 [208]. Therefore, researchers searched for natural plant-derived cardiovascular medicines. Soya bean isoflavones reported a powerful ability to manage the risk factors for cardiovascular disorders. Key isoflavones found in soya beans are present as glycosides (e.g., genistein, daidzein, and glycitin) and support cardiac function through various methods [209]. Many epidemiological, clinical, and experimental investigations revealed a positive correlation between green tea consumption and cardiovascular health, as green tea contains a high yield of catechins [210]. In addition, an epidemiological study suggests dietary carotenoids may lower the incidence of ischemic stroke, myocardial infarction, and coronary heart disease. Several phytochemical components, such as glucoraphanin, cucurbitacins, diosgenin, sulforaphane, and tocopherols, provide cardiovascular protection, reduce oxidative stress, enhance lipid profiles, and lower the blood pressure [211, 212].

3.4 Cancer Treatment

The number of cancer patients is still rising yearly, despite research into many ways to prevent and treat the disease. Reregulating cellular processes has been the focus of cancer treatment. Many clinical trials have examined cancer treatments using radiation, chemotherapy, antibody therapy, and immunotherapy. Because they cause deleterious effects on healthy cells, radiation and chemotherapy have serious adverse effects. Immunotherapy and antibody therapy exhibit exact cancer-targeting abilities, but they have a small therapeutic window and are sometimes expensive [213]. About 25% of medications utilized in cancer therapeutic settings come from plants (anticancer activity). For example, Lestari et al. [214] reported the anticancer activity of curcumin (1,7-bis(4-hydroxy-3-methoxyphenyl)-1,6-heptadiene-3,5-dione) (isolated from *Curcuma longa*) in colorectal cancer treatment.

Additionally, curcumin can be investigated as an antiangiogenic, anti-inflammatory, and antioxidant drug. Polyphenol flavonoid resveratrol (3,4,50-hydroxystilbene) can stop the growth of *H. pylori* and the division of gastric NSCLC cells preventing/treating gastric cancer [215]. Recent research revealed that the secondary metabolite hypericin from the plant *Hypericum* L. inhibited the overexpression of the ABC transporter in the HL-60 subclone leukemia cells via accelerating mitoxantrone-induced cell death [216]. Furthermore, β-carotene has been demonstrated to enhance the suppression activity of the anticancer drug (5-FU) on the development of tumors by oesophagal carcinoma (Eca109) [217].

3.5 Antimicrobial Activity

In the past 50 years, the development of vaccines and antibiotics has successfully eliminated infectious diseases such as *Mycobacterium tuberculosis* and smallpox, saving many lives. Plant secondary metabolites, such as flavonoids, phenolic, alkaloids, organic acids, and essential oils, have a significant role in antimicrobial properties. *Saussurea gossypiphora* contains multiple secondary metabolites, including apigenin and luteolin, which have antimicrobial activity against *Escherichia coli* and *Staphylococcus aureus* [218]. Other examples of heterologous expression of phytochemical pathways have shown it is possible to develop plants into pharmaceutical production platforms. *Crocosmia* spp. is an ornamental plant found to produce montbretin A, a potent inhibitor of human pancreatic amylase, which is a promising treatment for type II diabetes pending clinical validation.

Moreover, rosmarinic acid extracted from *Origanum vulgare* L. has significant antibacterial activity against *Helicobacter pylori* associated with an ulcer [219]. Flavonoids have multitarget characteristics, which can target bacterial plasma membranes to inhibit drug-resistant bacteria through different sterilization mechanisms. Isoprenylation of flavonoids plays a crucial role in antibacterial activity. Both α-mangostin and isobavachalcone target phospholipids, which destroy the membrane homeostasis and combine with the outer membrane permeabilizer against gram-positive bacteria. In addition, the antibacterial activity of phenolic secondary metabolites is attributed to the acidic characteristics of hydroxyl groups. The compounds change the permeability of cells, interfere with the enzymes involved in productivity, interrupt protein synthesis, and eventually lead to cell death [220]. Alkaloids affect the integrity of the cell wall of *Candida albicans*, leading to mitochondrial dysfunction, which in turn leads to the upregulation of oxidative stress. Similarly, essential oils have antimicrobial mechanisms that affect the ATP concentration and peptidoglycan of *Escherichia coli* and *Staphylococcus aureus*, resulting in cell wall damage [221].

3.6 Anti-Inflammatory Activity

Inflammation is the common process in the body's defensive response to hazardous stimuli, such as biological factors (bacteria, fungi, parasites, and viruses), physical factors (ultraviolet rays, extreme temperature, radioactive substances, and mechanical damage), chemical elements (a poisonous gas), and others. The uncontrolled inflammatory reaction is one of the main causes of allergies, cardiovascular diseases, and autoimmune diseases. Inflammation can cause various conditions, such as asthma, arteriosclerosis, and arthritis, which greatly threaten health [222]. Treating some inflammation-related diseases depends on steroidal and nonsteroidal anti-inflammatory drugs but treating such drugs will have several side effects. Therefore, using natural herbal medicines in treatment is particularly important. How to develop drugs with anti-inflammatory activity by using natural herbal medicine has become the hotspot of research. Natural herbal medicines can synthesize various anti-inflammatory compounds, including flavonoids, terpenoids, alkaloids, and essential oils.

Turmeric has anti-inflammatory and anti-arteriosclerosis effects. In India, turmeric is a traditional medicine used to treat rheumatic diseases [223]. It is reported that the genus *Ipomoea* and *Alstonia* species in Egypt have anti-inflammatory activity with little side effects. Furthermore, *Ipomoea pescaprae* extract has a therapeutic impact on dermatitis caused by jellyfish sting and oedema caused by ethyl phenylpropionate in experimental animals [224]. Studies illustrated this activity is due to the separation of some lipid and phenolic compounds.

In addition, licorice is the most frequently used traditional Chinese medicine; it can produce a variety of bioactive secondary metabolites, such as flavonoids, polysaccharides, and triterpenes. They have anti-inflammatory activity and are important as antibacterial, antiviral, and hepatoprotective substances [225]. Stilbenoids are found in tea, grapes, nuts, and berries. Resveratrol, pterostilbene, and piceatannol are well-known stilbenoids with indisputable anti-inflammatory activity in vitro and in vivo. Stilbene compounds are a group of plant phytoalexin polyphenols that can protect plants against pathogenic bacteria [226]. In folk medicine, these compounds are extensively used to treat skin inflammation, hepatitis, and stomachache. At the same time, they also have anticancer, neuroprotective, and antiviral properties. Recent studies reported that phenylpropanoids in essential oils have anti-inflammatory activities. Phenylpropionic acids are organic substances produced by plants that play an important role in preventing and treating trauma and infection. They are widely used in the medical field due to their corresponding pharmacological characteristics [227]. For instance, *Cinnamomum cassia* is commonly used to treat gastritis, anti-inflammation, and dyspepsia [228].

Moreover, alkaloids are widely distributed in *Menispermum dauricum* DC., *Tripterygium wilfordii*, and *Begonia kunmingshanensis* [229]. Wei et al. [230] stated that alkaloids have strong anti-inflammatory activity and are commonly used in treating ankylosing spondylitis, systemic lupus erythematosus, and other diseases. Berberine, sinomenine, dauricine, tetrandrine, stephanine, and lycorine in isoquinoline alkaloids have good anti-inflammatory effects. Piperidine alkaloids have anti-inflammatory, anti-arrhythmia, and anticancer effects. In addition to anti-inflammation, terpenoid alkaloids have antipyretic, analgesic, and antihypertensive effects [231].

3.7 Neurological Disorders

Neurological disorders affect the central nervous system and its peripheral nerves. The symptoms of neurological disorders include headaches, memory loss, slowness of movement, speech disorders, limb weakness, or pain. The causes of the disease are complicated; some neurological diseases have obvious heritability. Furthermore, environmental factors and age also affect nervous system health. The active ingredients in plants are very important in preventing and treating neurological diseases. Herbal medicines can delay or reduce the symptoms of neurological disorders. For example, tea polyphenols are a well-known active ingredient in tea. It can inhibit acetylcholinesterase and butyrylcholinesterase in improving Alzheimer's disease. In addition, tea polyphenols can modulate and shape gut microbiota to boost immunity and enhance sleep quality by the gut-brain axis. Phenolic substances extracted from grapes have a good antioxidant effect, which can reduce the formation of free radicals and prevent damage to protective cells, helping to alleviate neurological disorders and brain aging. Moreover, numerous research reported that cannabinoids (isolated from *Cannabis Sativa*) have sedative, anti-inflammatory, and analgesic effects. Thus, it can potentially relieve various neurological diseases [232, 233].

4 Artificial Phytochemicals Production

Herbal medicines have been utilized since antiquity, and several therapeutic plant secondary metabolites are employed directly as medications and raw resources for semisynthetic alterations. Their structural diversity, which frequently limits the cost-effective chemical synthesis and relatively low extracted content of many plant species, mandates the field-cultivated materials. The novel biotechnological fabrication techniques of these chemicals offer a variety of benefits, including predictable, steady, and longtime sustainable production, adaptability, and more straightforward isolation and purification [234]. Many researchers use genetic engineering to manipulate plant metabolism to manufacture artificial phytochemicals. This entails the introduction of genes that encode enzymes involved in synthesizing specific bioactive components in plants. For instance, scientists have successfully modified plants to generate secondary Taxol® (paclitaxel) metabolite. It is an isoprenoid extracted mainly from *Taxus brevifolia* plant and widely used in cancer treatment [235–237]. Moreover, the artificial synthesis of phytochemicals involves using genetically engineered microbes or "cell factories" engineered to manufacture certain compounds, including carotenoids, flavonoids, and terpenoids. The extracted chemicals are subsequently utilized as dietary supplements or food additives to enhance their nutritional value.

Plant synthetic biology attempts to avoid several challenges by overexpressing bioactive chemicals for large-scale synthesis and purification. The complete characterization of the biosynthetic process for the target molecule is the critical step in synthetic biology. Recognizing the interactions between produced intermediates and identifying the necessary components of a biosynthetic route based on conventional classes of enzymes can assist in completing several fragmented molecular mechanisms [238]. Moreover, contemporary sequencing technology has progressed, so whole genome sequencing and phylogenetic analyses of nonmodel species are now feasible [239]. For example, Natt et al. [240] identified a biosynthetic mechanism for producing and extracting colchicine (alkaloid) from *Gloriosa superba*. The study was conducted through previously derived RNA sequencing data of many plant species following metagenomics to ascertain eight critical genes in the pathway and to reassemble a 16-gene scheme in *Nicotiana benthamiana* with overexpression to generate N-formyldemecolcine (the colchicine precursor).

5 Recent Advances in Herbal Medicine

To understand the most recent advances in herbal medicine achievements, in this part, groups of studies conducted on the potentiality of plant secondary metabolites activity in disorders treatment are discussed. Several natural products derived from Chinese herbal medicine display anticancer properties, such as antiproliferative, proapoptotic, antimetastatic, and antiangiogenic effects, as well as the ability to regulate autophagy, modify multidrug resistance, balance immunity, and augment chemotherapy in vitro and in vivo. For instance, the therapeutic activity of curcumin, epigallocatechin gallate, berberine, artemisinins, ginsenosides, ursolic acid, silibinin, emodin, triptolide, cucurbitacins, tanshinones, ordonin, shikonin, gambogic acid, artesunate, wogonin, β-elemene, and cepharanthine has been discussed in more than 100 published research articles [241]. Curcumin has been found to possess diverse pharmacological effects, such as anticancer, anti-inflammatory, and anti-xidative activities, as supported by clinical data and comprehensive research [242, 243].

Curcumin and its derivatives have demonstrated promising potential as therapeutic agents for various malignant conditions, including cancer. Several research studies have demonstrated that curcumin and its derivatives can impede the growth of tumors in different human body regions, such as the head, neck, skin, ovarian, and gastric cancers[244–246]. Liu et al. [243] reported that the natural polyphenol curcumin (extracted from turmeric rhizome) suppressed the Janus kinase/signal transducers and activators of the transcription (JAK/STAT) pathway in cultured oesophagal squamous cell carcinoma (ESCC) cells. The attachment of cytokines to receptors activates JAKs, which phosphorylates STATs. STATs are translocated into the nucleus after being dimerized and phosphorylated to regulate gene expression. These genes, including cyclins and antiapoptosis proteins, are crucial for cell proliferation and survival [242]. The results showed that curcumin inhibits JAK2 activation, downregulating STAT3 signaling, suppressing cell proliferation and colony formation, cell cycle arrest, and apoptosis. In addition, curcumin substantially reduced tumor growth in xenografts derived from ESCC patients. These findings indicated that curcumin is a powerful agent for preventing ESCCs harboring constitutively active STAT3 proteins. In addition, curcumin substantially reduced tumor growth in xenografts derived from ESCC patients. These findings indicated that curcumin is a powerful agent for preventing ESCCs harboring constitutively active STAT3 proteins. A study by Sivanantham et al. [247] reflected the anticancer activity of curcumin and its role in treating head and neck squamous cell carcinoma (HNSCC). They combined curcumin with three anticancer drugs, including docetaxel, doxorubicin, 5-fluorouracil, and diammine dichloroplatinum (II) (cisplatin), and evaluated their joint influence on the HNSCC cell line NT8e. The outcomes showed that the combined administration of 5- fluorouracil or doxorubicin with curcumin significantly inhibited NT8e cancer cell proliferation and enhanced apoptosis. NT8e cells treated with 5-fluorouracil or doxorubicin in combination with curcumin exhibited apoptosis via inhibition of Bcl-2 and elevation of Bax, caspase-3, and poly-ADP ribose polymerase. The researchers undergo some confirmation experiments to ensure the results are obtained through DAPI staining and decreased red/green fluorescence by JC-1 observations of apoptotic cell features, such as membrane blebbing, nuclear condensation, and cell contraction.

Some phenolic substances collaborate with nonsteroidal anti-inflammatory drugs to block pro-inflammatory mediators' functioning or gene expression, such as cyclooxygenase (COX). Diverse phenolic components can also act on transcription factors, like nuclear factor-B (NF-B) and nuclear factor-erythroid factor 2-related factor 2 (Nrf-2), to upregulate or downregulate antioxidant response pathway constituents. Moreover, they may suppress enzymes involved in developing human diseases and have been utilized for treating various common human disorders, such as hypertension, metabolic issues, incendiary infections, and neurological disorders [247]. Hypertension is a prevalent and frequently progressive condition associated with a high risk of heart failure and complications [248]. According to estimates, up to 25% of adults struggle with hypertension globally [249]. Hypertension is a serious and becoming more prevalent global health issue. Multiple investigations have demonstrated that polyphenol-rich foods effectively prevent and treat hypertension, particularly through angiotensin-converting enzyme (ACE) inhibition [250]. Patten et al. [235] recently discovered about 74 plant families with substantial ACE inhibitory activity. Similarly, Han et al. [251] proved that some cocoa polyphenols, including catechins, flavonol glycosides, anthocyanins, and procyanidins, are bioavailable substances with antihypertensive properties by inhibiting ACE [252].

In addition, an in vitro study carried out by Bhandari et al. [253] investigated the antidiabetic efficacy of two soluble active compounds(−)-3-O-galloylepicatechin and (−)-3-O-galloylcatechin found in the ethyl acetate Bergenia ciliate extract served as catalysts for the dose-dependent suppression of porcine pancreatic amylase and rat intestinal maltase activation. The half-maximal inhibitory concentration (IC_{50}) value is the compound concentration required for inhibiting enzyme activity by 50%. The in vivo and in vitro suppressive enzyme activity of these compounds against α-glucosidase and α-amylase indicate that they have outstanding prospects for development as a treatment for type 2 diabetes [254].

Aging can be described as an accumulation of multiple detrimental changes in cells and tissues, which increases the risk of disease and mortality with advancing age. The free radical and oxidative stress theory [255] represents one of the most commonly accepted hypotheses for the aging mechanism. Despite normal conditions, oxidative damage occurs; however, as antioxidative and restoration processes become less effective with age, the rate of oxidative destruction rises [256, 257]. Plasma antioxidant capacity is correlated with antioxidant dietary consumption; it has been observed that a diet rich in antioxidants can mitigate the negative effects of aging and behavior. Multiple investigations suggest that a combination of antioxidant/anti-inflammatory polyphenolic substances found in fruits and vegetables may be effective antiaging compounds [258]. Anthocyanins, a category of flavonoids, are exceptionally abundant in brightly colored fruits, like berries, Concord grapes, and grape seeds. Anthocyanins have potent anti-inflammatory and antioxidant effects and inhibit lipid peroxidation and COX-1 and -2, which are inflammatory mediators. The antioxidant activity of formulations of flavonoid-rich fruits and vegetables, including spinach, strawberries, and blueberries, is very high. According to the results of Shukitt-Hale et al. [259], supplemental nutrition with spinach, strawberry, or blueberry extracts in a control diet was similarly effective in restoring age-related deficits in the brain and behavioral function of geriatric rats. Moreover, tea catechins have potent antiaging characteristics, and drinking green tea abundant in these catechins could potentially delay the advent of aging [260].

6 Conclusion

Natural plant secondary metabolites (phytochemicals) have unique chemical structures with a wide range of diversity in medicinal and biological characteristics. They significantly treated some diseases and chronic disorders (cancers, cardiovascular, hyperglycemic, and type 2 diabetes). Moreover, they are involved in many pharmaceutical industries, like cosmetics and food supplements. The classification and the taxonomic position of each group and plant species are crucial in understanding each phytochemical's distinct function and characteristics (physical and chemical properties, synthesis pathways, and extraction methods). Furthermore, nature may have more based on the number of secondary metabolites identified and extracted. Considering the developments in synthesis technology as well as the invention of more advanced isolation and analysis approaches, it should be possible to identify a significant amount of these additional phytochemicals. Moreover, artificial phytochemical manufactoring also offers the potential to increase the effectiveness and safety of natural phytochemicals. Scientists can confirm the absence of impurities and pollutants by synthesizing molecules in a sterile atmosphere. In addition, they can alter the structure of a substance to increase its efficacy or decrease its toxicity. Recently, many researchers reflected on the potency of many isolated active metabolites in treating some vital disorders and may replace commercial and chemically synthesized drugs, especially those offered for tumor treatment.

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Cola accuminata: Phytochemical Constituents, Nutritional Characteristics, Scientific Validated Pharmacological Properties, Ethnomedicinal Uses, Safety Considerations, and Commercial Values

5

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Contents

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Abstract

The tropical plant *Cola accuminata*, more often known as Cola nut, has a long history of traditional and ethno-medical applications. Our discussions have covered many topics related to *Cola accuminata*, such as its phytochemical components, nutritional qualities, scientifically proven pharmacological effects, ethnomedicinal usage, safety concerns, and commercial values. Caffeine, theobromine, tannins, flavonoids, and phenolic compounds are only some of the many phytochemicals found in *Cola accuminata*. Its antioxidant, anti-inflammatory, anti-microbial, anti-glucose, neuroprotective, and cancer-fighting activities can be attributed to these components. Scientific studies proving the plant's usefulness in a variety of in vitro and in vivo models give data supporting these pharmacological potentials. *Cola accuminata* has a wide variety of historical and contemporary applications in traditional and folk medicine. It has been used in traditional medicine for a variety of purposes, including as a stimulant, to improve memory and focus, as an aphrodisiac, to treat respiratory diseases, and to control gastrointestinal issues. In addition to its diuretic properties, *Cola accuminata* has been used in wound healing and ulcer care. Nevertheless, when working with *Cola accuminata*, it is essential to prioritize safety. *Cola accuminata* contains caffeine, which in large doses can cause jitteriness, anxiety, and a rapid heartbeat in those who consume it. Care should also be required during pregnancy, lactation, and with certain medical conditions to avoid adverse reactions, drug interactions, gastrointestinal distress, and other complications. In addition, *Cola accuminata* has marketable qualities. Beverages, flavours, and conventional herbal medicines all benefit from their use. Moreover, the pharmaceutical sector is also interested in its features for the creation of new medicines and therapeutic agents.

Keywords

Cola accuminata · Bioactive compounds · Therapeutic potentials · Ethnomedical utilizations · Safety respects · Medicinal plants · Herbal medicine

1 Introduction

The cola nut (*Cola accuminata*), which is the produce of the Cola tree, is considered to be a fruit of paramount significance in West Africa, where it is used for a variety of uses by numerous ethnic backgrounds [1–4]. The fruit tree is not just native to West Africa, where it is cultivated, but it also has its origins in Africa as a whole. According to Unya [5], cola nut (sometimes spelt as Kola nut) belongs to the plant family Sterculiaceae, which has approximately 125 species of trees native to the tropical rainforests of Africa. The most common species in Nigeria are *Cola nitida* and *Cola accuminata* and it takes up an uncommon position among West Africans where it is extensively eaten [5]. The nuts of *Cola nitida* can be identified with its two cotyledons, while *Cola accuminata* can be differentiated with its two to seven cotyledons [6]. Moreover, the colour of *Cola nitida* varies between pink, white, and red, whereas *Cola accuminata* has a colour variation of white and pink [6].

The tree holds special significance in the people's social lives as well as their religious practices throughout the tropical regions of West Africa. Cola nut is highly valued across the entirety of West Africa by people of all socioeconomic backgrounds and religious persuasions, including but not limited to Muslims, Christians, and animists [7]. It is usually offered as a revered sacrifice and as a sign of reverence for the recipient among major tribes in Nigeria. It is also an essential component of getting people together in the community, in addition to playing an important role in a wide variety of rituals of advancement and events, including those that serve to bind legal agreements and treaties [8].

In the report of Sprague and Adega [9, 10], who comment on the vitality and significance of cola nut among the West African people. Portuguese explorer who travelled to the region in the year 1587 reported that a great deal of the inhabitant of the area he came across in his expeditions chewed on the cola fruit to quench their thirst as well as enhance the culinary delights of water [9]. Similar medicinal properties of the cola fruits observed by the explorer were also noted in other publications written by voyageurs at the same time. Other publications have also captured African practices, such as the use of the nut to fortify the digestive tract as well as battle cirrhosis of the liver [11, 12].

In Nigeria, the nuts are not only cultivated in a significant amount, but they are also considered to be an agricultural product that is of fundamental value in the daily routine of those who reside there due to the numerous functions that they play. As a result, the country located in the West African region is the highest production of cola nuts. According to Danquash [13], a significant part of the cola nuts produced in Nigeria are eaten within the borders of the country, while the remaining 10% is shipped abroad. According to Asogwa et al. [14], there is a vast amount of soils with high, moderate, and minimal reproductive rates in Nigeria that have the potential to be capitalized on economically for the production of cola nuts as part of an efficient land use management policy. These kinds of fertile soils have long since been discovered across various regions of the country.

Generally plants have shown that it is a new candidate for the production of drugs. From time immemorial plants have been used to treat and manage disease conditions as well as the preservation [15] and control of insects [16–21]. Therefore this chapter focuses on phytochemical constituents, nutritional characteristics, scientific validated pharmacological properties, ethnomedicinal uses, safety considerations, and commercial values of *Cola accuminata.*

2 History and Cultural Significance of *Cola accuminata*

Various Nigerian ethnic groups, such as the Igbo, Hausa, and Yoruba ethnic groups, make extensive use of *Cola accuminata* in the practice of traditional medical treatments as well as in cultural customs [22]. The plant has been used for medicinal purposes for a very long time, and at one point in history, it had been employed as an instrument of currency. A number of research studies, such as one that was conducted by Odugbemi [23], have provided evidence that it has been utilized in traditional medicine to treat conditions such as headaches and migraines a high fever and gastrointestinal problems.

According to Kammamposal and Laar [24], in the Nigerian culture, cola nuts are frequently given as a token of reverence and openness to visitors, and they are also used within traditional celebrations such as marriage ceremonies and naming rituals throughout a variety of cultural backgrounds. Accordingly, Kanu [25] asserts that the kola nut ceremony is a crucial communal rite in Nigerian culture. The celebration entails those participating in the breaking of kola nuts to share them with one another. The ceremony frequently goes hand in hand with dancing, musical entertainment, and the telling of stories.

The utilization of *Cola accuminata* in Nigerian traditional beliefs is further evidence of the historical significance of this plant in Nigerian society. The nuts are frequently presented as sacrifices to the gods because it is widely held that they possess mystical and defensive qualities. They are additionally employed in clairvoyance celebrations, during which they are tossed onto a special enchantment surface in order to decipher correspondence from the celestial beings [24]. This practice dates back to ancient Africa.

In Igbo land, *Cola accuminata* plays a significant role in their cultural and religious festivities. Cola nuts are broken open and shared during a significant communal rite in Igbo culture known as the kola nut ceremony [26]. Songs, dancing, and oral narration are common parts of the event, which is valued for its ability to bring people together and promote an atmosphere of neighbourhood [27]. Nwankwo et al. [28] noted that the kola nut is also an embodiment of tranquillity and has been utilized to mediate conflicts among various social groups. The fruits are also reported to possess mystical and defensive attributes, and they are frequently employed to make a tribute to the deities in customary faith. Tossed onto a divinity podium, they are employed to decipher information from the celestial beings in religious rituals [27].

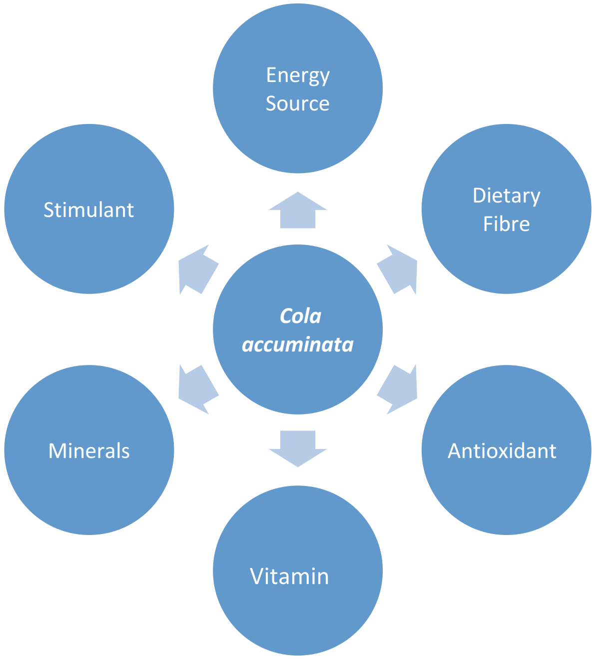
The Yoruba have traditionally associated *Cola accuminata* with warmth and close relationships. It is a common gesture of hospitality and reverence. The giving and receiving of Cola nuts are a vital component of social events and customary festivities [29]. The close connection between *Cola accuminata* and Yoruba religious and spiritual practices also demonstrates the plant's societal importance. During practices and rites led by traditional priests and priestesses, the nut is often used as an offering to the gods and ancestral deities. It is commonly used as a channel for interacting with the ethereal world due to its revered metaphysical status. In addition, the Cola nut is used extensively in their indigenous faith for divination purposes. Ifa divination, a multifaceted system that aids in decision-making, enlightenment, and reuniting with past generations and the divine, counts this as one of its key elements. Thus, the nuts are a symbol of the sanctity and significance of divination rituals [30].

Consequently, the historical and cultural significance of *Cola accuminata* is attested by the fact that it is still used and recognized in local communities today. The nut represents many different things, including national pride, friendliness, religious faith, and traditional medicine.

3 Nutritional Properties of *Cola accuminata*

Cola accuminata fruit, also referred to as cola nut, can be used in a number of different ways to improve one's diet because it contains a rich variety of vitamins and minerals. Approximately 2% of catechin, caffeine, theobromine, and kolatin can be found in the fruits [31]. The nuts can be baked, crushed, or gnawed at as well as be used as an additive to beverages like tea or dairy products, as well as breakfast cereal

Fig. 1 Nutritional properties of *Cola accuminata*



like porridge [32]. The flavour of the entire nuts is unpleasant whenever they are gnawed at whereas they leave an excellent aftertaste that brings out the flavours and deliciousness of additional food items in the dish [33]. Figure 1 shows a list of a few of the dietary uses of the *Cola accuminata* nuts.

3.1 Stimulant

Cola accuminata is famous for its stimulation qualities, which are mainly attributable to the existence of caffeine and theobromine in the nut. It is commonly eaten for the purpose of enhancing heightened awareness, bettering a person's energy demand, and fighting feelings of fatigue [34]. The caffeine content in *Cola accuminata* varies between 1% and 3%, dependent on the diversity and processing techniques deployed [35]. Theobromine is another stimulant compound that belongs to the xanthine family. It has a similar stimulatory effect as caffeine, but it is less potent. Theobromine levels in *Cola accuminata* varies between 0.5% and 1.5% [36].

3.2 Energy Source

Cola accuminata is an excellent source of carbohydrates, which can be utilized by the body to produce energy [37]. The carbohydrates found in *Cola accuminata* deliver a rapid boost to one's energy levels, which can be helpful during times of extended physical activity or instances of strenuous working out. According to Beck [38], the majority of the carbohydrates in *Cola accuminata* are found primarily in a range of sugars. These sugars include glucose, fructose, and sucrose. *Cola accuminata* also has fatty acids, which are an additional significant source of energy and are present in the plant [39]. In addition to being an excellent source of fatty acids, it is also necessary for the body to be able to absorb fat-soluble vitamins. A wide variety of essential amino acids, which are known as the fundamental constituents of proteins, are also found in *Cola accuminata.* When the body requires more energy, it is capable of converting amino acids into a usable form. According to Dah-Nouvlessounonh et al. [37], the high content of amino acids in *Cola accuminata* adds to the nutritional value of the plant when used as an important form of energy.

3.3 Dietary Fibre

The nut is an excellent source of dietary fibre, which is the portion of plant-based foods that cannot be digested. It is important for maintaining gastrointestinal tract functioning and is also an important factor in overall well-being. Consuming foods rich in dietary fibre can assist in the regulation of stool production, the avoidance of diarrhoea, and the promotion of a digestive system that is in good condition [40]. According to Okwu and Morah [41], it may also play a role in an impression of feeling satiated and contribute to staying on top of one's weight.

The nut has also been reported to contain soluble fibre, which is a type of fibre that disintegrates when immersed in water and produces a substance with the consistency of gel when it reaches the gastrointestinal tract [42]. According to Fernandez-Murga et al.'s [43] research, soluble fibre not only assists in controlling the amount of sugar in the blood but also minimizes blood cholesterol levels and fosters an optimal gut microbiome. Insoluble fibre is another fibre constituent in *Cola accuminata* which assists in adding mass to the diet and delays bloating by keeping bowel movements regular. It favours consistency and accounts for the general well-being of the gastrointestinal tract [43].

3.4 Minerals

The cola nut is an excellent source of a variety of minerals, including potassium, calcium, magnesium, and phosphorus, among others. Minerals such as these are required for the human body to carry out a wide variety of biological functions. Calcium in *Cola accuminata* is essential to keep healthy teeth and bones, as well as for adequate functioning of the muscles and nerves signalling [44]. The nuts of *Cola accuminata* are a good source of magnesium, which is a vital mineral that plays a role in countless biological processes in human beings, as well as the generation of vitality, the functioning of muscles, and ensuring the restoration of a healthy cardiovascular activity [45].

Cola accuminata is also an excellent source of the mineral potassium, which is essential to the regulation of fluid balance, the functioning of nerves, and the twitching of muscles [46]. Adequate potassium consumption has been linked with reduced blood pressure [45, 47]. According to Adeoye et al. [45], the mineral phosphorus can be found in *Cola accuminata*. This mineral plays an important role in maintaining healthy bones, creating energy, and synthesizing deoxyribonucleic acid (DNA). Zinc which is found in the nuts of *Cola accuminata* is very essential for the body's immune system functioning, the healing of wounds, and the production of DNA. According to Adeoye et al. [45], zinc is also important for preserving the health of the skin, hair, and nails. In addition, iron in the nuts is necessary for the production of red blood cells as well as the distribution of oxygen all over the body. Accordingly, Georgieff [48] affirmed that iron is essential for the generation of energy as well as for brain development.

3.5 Antioxidant

These substances are found in cola nuts, and these assist in protecting the body from the oxidative stress that is triggered by free radicals [49, 50]. Some examples of antioxidants found in *Cola accuminata* include phenolic chemical substances. Antioxidants have been linked to a decreased likelihood of developing long-term illnesses in addition to a general improvement in wellness and overall health [51, 52]. According to Adedapo et al. [53], the phenolic compounds found in *Cola accuminata*, which include flavonoids and tannins, are responsible for the plant's powerful antioxidant activity. The nuts of *Cola accuminata* is rich in catechins, which are a type of flavonoid that have powerful antioxidant properties. According to Adeyemi et al. [54], catechins have the ability to eliminate free radicals while safeguarding cells from damage caused by oxidative stress.

Cola accuminata is also an organic source of vitamin C, which is a potent antioxidant that assists in neutralizing free radicals as well as protecting the body cells from it [55]. It contains a sizeable amount of vitamin E, which is an antioxidant that is able to dissolve in fat. The substance helps in protecting the cell membranes of the body from damage caused by oxidative stress [56]. Moreover, *Cola accuminata* is loaded with proanthocyanidins, which are powerful antioxidants recognized to have a variety of medical advantages, such as acting as a shield for the cardiovascular system and soothing effects on inflammation [54].

3.6 Vitamins

These nutrients can be obtained naturally from the nuts. Thus, *Cola accuminata* is a natural source of vitamins, including vitamin C and a number of B vitamins. According to Adedapo et al. [53], *Cola accuminata* contains vitamin C, which is also referred to as ascorbic acid. Vitamin C works as an antioxidant and plays a role in the formation of collagen as well as the primary objective of the immune system and the uptake of iron. The nuts of *Cola accuminata* is also an organic source of B vitamins, including thiamine (B1), riboflavin (B2), niacin (B3), pantothenic acid (B5), pyridoxine (B6), and folate (B9). According to Oboh et al. [55] and Adedapo et al. [53], these vitamins are essential for the formation of red blood cells, the proper functioning of the body's nervous system, and the breakdown of calories. The nuts also contain Vitamin E which is a fat-soluble vitamin with antioxidant abilities that aid in safeguarding the cells against damage from oxidative stress [56]. According to Adaramogye et al. [56], *Cola accuminata* is a good source of vitamin K, which plays a role in the clotting of blood and maintaining bones in good condition. When included in a diet that is otherwise well-balanced, the vitamins that are found in *Cola accuminata* add to the total nutritional worth of the species and can give a variety of nutritional advantages to those who eat it. It is also essential to be aware that the number of vitamins of *Cola accuminata* may differ from one variety of the plant to another, due to its environmental factors as well as handling methods used.

4 Phytochemical Constituents of *Cola accuminata*

4.1 Proximate Composition of *Cola accuminata*

The proximate composition of *Cola accuminata* as reported by several authors varies in percentage composition among the constituent variables assessed as a result of its environmental factors and methods of samples processing. Table 1 gives a summary of the breakdown of the constituent elements as reported by some scholars.

4.1.1 Moisture Content

The moisture content of *Cola accuminata* is reported to vary between 5.80% and 54.40% of its dry matter content [–, 6, 35, 37, 39, 57, 58, 59]. This deviation of moisture content in the various investigations would definitely be associated with the duration the samples were dried before they were used for evaluation. This is in accordance with the report of a study conducted by Lowor et al. [60], who asserted in

Table 1 Proximate composition of *Cola accuminata*

Parameters	% Dry matter	References
Moisture content	3.39–54.4	[6, 35, 37, 39, 57, 58, 59]
Dry matter	45.6–96.62	[6, 35, 37, 39, 57, 58, 59]
Total ash	1.46–4.96	[6, 37, 39, 57]
Crude fat	1.20–15.72	[6, 37, 39, 57, 59]
Crude protein	8.80–11.95	[6, 37, 39, 57]
Total sugars	5.57–6.27	[37, 64]
Reducing sugars	15.22–71.16	[37, 64, 65]
Energy value (Kcal/g)	88.98–98.98	[37, 64]
Crude fibre	7.35–15.80	[6, 37, 39, 59]
Carbohydrate	33.84–67.22	[6, 37, 39, 59]

their report that the moisture content of a test sample varies according to their drying time. Thus, the relatively higher moisture content in *Cola accuminata* nuts in some samples would serve as an indicator that the product was going bad as any food sample that contains an excessive amount of moisture has the potential to support the growth of microbes. This is responsible for the vast majority of the biological, chemical, and physiological activities that take place in a plant [61]. Nevertheless, a low amount of moisture content in the nuts, will make it possible for them to be stored for an extended period of time without them becoming rancid.

4.1.2 Protein Content

Protein content in *Cola accuminata* ranges from 8.80% to 10.64% [6, 37, 39, 57]. The disparity between the protein content of *Cola accuminata* could be attributed to soil and environmental circumstances arising from the fact that the lack of water in the soil would contribute to inadequate amount of nitrogen availability in the plant's tissues [22]. In addition, physiological features of a plant can be innately affected by its genetic composition [62] as it imposes a higher degree of resistance to the nitrogen supply and water deficiency in a plant. Nevertheless, the relatively high protein content of *Cola accuminata* nuts could be used to supplement the amount of protein needed in the body for growth and development [63]. In addition, [6] opined that the nuts of *Cola accuminata* would serve as a suitable alternative for the production of curd or as an additive material to other vital minerals for the formation of a gel in food products especially protein gel which serves as a structural matrix for holding water, flavour, sugars, and food ingredients.

4.1.3 Fat Content

Cola accuminata has a crude fat content that is anywhere from 1.20% to 15.72%. This data indicates that the crude fat content of *Cola accuminata* is highly variable with different studies having reported widely varying amounts. According to Adeniyi et al. [39], analysis of the nutritional profile of *Cola accuminata*, the nut had a crude fat content of 2.87%. The fat content of *Cola accuminata* was calculated to be 1.20% by [57], while the fat content of *Cola accuminata* was reported to be higher at 10.55% by [6].

Similarly, a study by Dah-Nouvlessounon et al. [37] found that *Cola accuminata* has a crude fat content of 15.72% and *Cola accuminata*'s fat content was 5.33%, which was reported according to an analysis by Adeyeye and Ayejuyo [59]. These studies shed light on the fact that the crude fat content in *Cola accuminata* varies, depending on a number of variables such as the variety, the growing conditions, and the processing methods used. These numbers are averages, and there may be some variation in fat content between different samples.

4.1.4 Ash Content

Cola accuminata's total ash content can be anywhere from 1.46% to 4.96%. The presence of mineral components in *Cola accuminata* has been suggested by the fact that various studies have reported varying values for the plant's ash content. In their analysis of *Cola accuminata*'s nutritional profile, Adeniji et al. [39] found that the plant had a total ash content of 2.35%, while according to research by Mustapha [57], *Cola accuminata* has an ash content of 1.46%. Nevertheless, *Cola accuminata*'s total ash content was reported to be 4.96% in a study conducted by Dah-Nouvlessounon et al. [37]. Thus, the mineral composition of *Cola accuminata* has been elucidated by these studies. After plant matter has been completely burned, inorganic mineral residue is left, which is represented by the ash content. It is an indicator that *Cola accuminata* contains beneficial minerals like calcium, potassium, magnesium, and phosphorus and is, therefore, crucial to the body's normal functioning and preservation of health.

4.1.5 Total and Reducing Sugar Content

Both Kingsley et al. [64] and Dah-Nouvlessounon et al. [37] report that the total sugar content of *Cola accuminata* is relatively high compared to their nuts, ranging from 5.57% to 6.27%, while its reducing sugar content varies between 15.22% and 71.16% [37, 64, 65]. These variations in total and reducing sugar content of the nut is attributed to factors such as the cultivar, growing conditions, and maturity of the plant.

4.1.6 Carbohydrate and Energy Value

Carbohydrates, found in nuts like *Cola accuminata*, are an important macronutrient because they provide the body with energy. Carbohydrates, mainly in the forms of dietary fibre and sugars, are present in nuts of *Cola accuminata* in varying amounts of 33.84–67.22 as reported by the studies of Adeniji et al. [39], Mustapha [57], Abulude [6], and Adeyeye and Ayejuyo [59]. In addition, dietary fibre, a complex carbohydrate, is important for digestive function and bowel regularity. It helps with digestion and stool bulk, and it may be useful in warding off constipation. Sugars from carbohydrates are quickly absorbed by the body and used for fuel. They are quickly metabolized and absorbed, supplying energy for workouts and mental processes. Thus, *Cola accuminata* nuts may be a better choice for managing blood sugar levels than other snack options because they contain some natural sugars but are generally low in sugar compared to other snack options [66].

4.2 Antioxidants Properties

4.2.1 Free Radical Scavenging Activity

The high concentration of phenolic compounds and flavonoids in *Cola accuminata* is responsible for its potent free radical scavenging activity. These antioxidants eliminate dangerous free radicals, protecting cells and tissues from oxidative damage. Antioxidant phenolic chemicals present in *Cola accuminata* have been widely acknowledged for their value. Catechins, flavonols, and phenolic acids are just a few examples of these molecules that can boost the plant's antioxidant capacity. The antioxidant and anti-oxidative properties of phenolic compounds such as flavonoids have long been recognized. They are essential in lowering oxidative stress due to their powerful antioxidant capabilities. *Cola accuminata* has a high antioxidant capacity thanks to the presence of flavonoids, which help to scavenge free radicals and lessen oxidative damage. The antioxidant and free radical scavenging properties of tannins are well recognized, and *Cola accuminata* has a lot of them. It has been established that tannins have potent antioxidant properties, protecting cells from oxidative stress-induced damage. This capacity of *Cola* species extract to scavenge free radicals and protect against oxidative stress was reported in a study by Fabunmi and Arotupin [67].

4.2.2 Hydrogen Peroxide Scavenging

Hydrogen peroxide (H_2O_2) is a reactive oxygen species that can damage cells. The scavenging ability of *Cola accuminata* against hydrogen peroxide was reported by Adedapo et al. [53]. According to the results of the study, the oxidative damage caused by H_2O_2 can be reduced by using a methanol extract of *Cola accuminata*'s leaves and stems. The extremely reactive chemicals known as free radicals have been linked to oxidative damage in cells and the progression of illness. *Cola accuminata*'s ability to scavenge free radicals aids in the neutralization of these potentially damaging radicals and protects cells from oxidative stress [68].

4.2.3 Metal Chelating Capacity

Cola accuminata has been shown to have metal chelating capabilities, making it useful for reducing metal-induced oxidative stress. Iron and copper, for example, can be catalysts in reactions that generate free radicals. By binding and sequestering certain metals, *Cola accuminata*'s chelating activity reduces their involvement in oxidative processes. Adedapo et al. [53] found that a methanol extract of *Cola accuminata* leaves and stems had a high metal chelating ability.

4.2.4 Lipid Peroxidation Inhibition

Blocking the chain reaction of lipid peroxidation results in cellular damage when reactive oxygen species are present. The research by Adedapo et al. [53] shows that *Cola accuminata* has anti-oxidant properties that protect against lipid peroxidation. Strong reduction of lipid peroxidation was observed in the methanol extract of *Cola accuminata* leaves and stems, suggesting its potential in protecting cells and tissues from oxidative damage. To prevent cellular damage caused by lipid peroxidation, which takes place when free radicals oxidize lipids in cell membranes, antioxidants are used. Inhibition of lipid peroxidation by *Cola accuminata* protects cell membranes against oxidative stress, as reported by Oboh et al. [68].

4.2.5 DNA Protection

Genomic material is protected from oxidative damage thanks to the DNA-protective qualities included in *Cola accuminata*. The DNA-protective activities of the aqueous extract of *Cola accuminata* seeds against oxidative stress-induced damage were studied by Adeyemi et al. [54]. This outcome suggests that *Cola accuminata* had the potential to aid in DNA maintenance and protect the body against mutations caused by oxidative stress. Consequently, these results emphasize *Cola accuminata*'s anti-oxidant capability, which makes it a significant natural resource for warding off the health risks associated with oxidative stress. *Cola accuminata* is able to shield cells and tissues from oxidative damage because of the wide variety of antioxidant qualities it possesses.

5 Traditional and Ethno-Medicinal Uses of *Cola accuminata*

The following traditional and ethno-medicinal uses of *Cola accuminata* have been documented in various studies and supported in its historical and cultural significance. However, it is important to note that while these traditional uses provide valuable insights, further scientific research is necessary to validate the efficacy, safety, and appropriate dosage of *Cola accuminata* for each specific use. Some of the traditional and ethno-medicinal uses of *Cola accuminata* are indicated in Fig. 2.

5.1 Digestive Aid

Cola accuminata has traditionally been used as a digestive aid. It has been hypothesized that it stimulates digestion by increasing gastric motility and secretion of

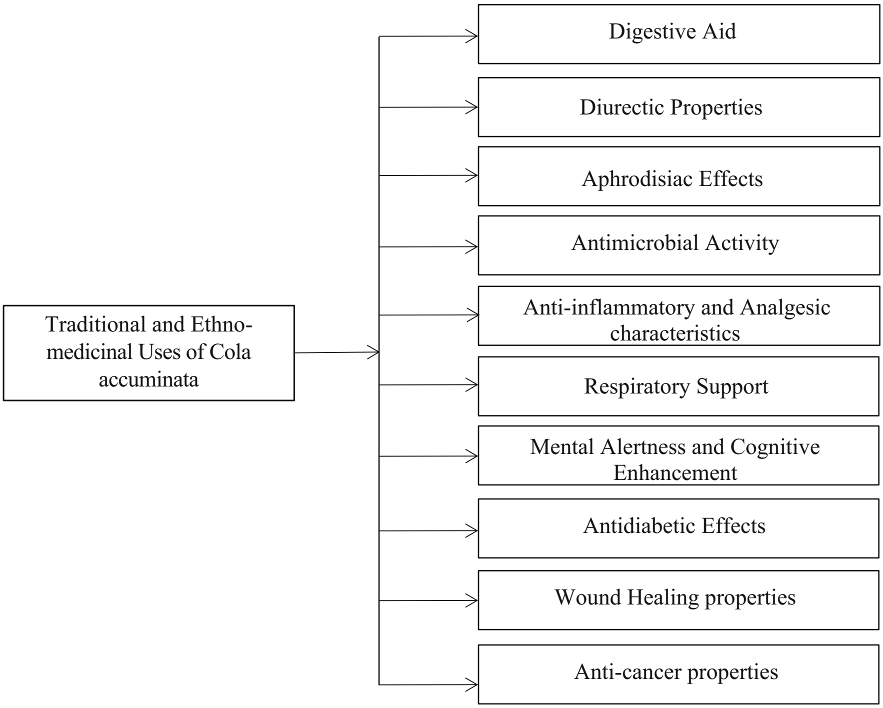


Fig. 2 Traditional and ethno-medicinal uses of *Cola accuminata*

digestive enzymes. Symptoms like bloating and flatulence can be reduced, and digestion can be enhanced [69–71]. *Cola accuminata* has been used traditionally for the treatment of gastric ulcers, and recent research suggests that this use may be justified because of the plant's anti-ulcer efficacy [72].

5.2 Diuretic Properties

Cola accuminata is commonly used as a diuretic because it stimulates the body to produce more pee. This can help the body get rid of accumulated fluids and waste products. *Cola accuminata* has long been used as a diuretic in the treatment of oedema and hypertension [73]. Udegbunam et al. [74] found that taking extracts from the plant *Cola accuminata* led to an increase in urine volume and an increase in the excretion of salt and potassium.

5.3 Aphrodisiac Effects

Cola accuminata has long been known for its aphrodisiacal properties. It may improve virility, libido, and performance in the bedroom. Traditional aphrodisiac use of *Cola accuminata* may be attributable to the plant's stimulating qualities, such as its caffeine content [75]. Nwonuma et al. [76] demonstrate that *Cola accuminata* may have pro-fertility benefits by increasing sperm count and motility.

5.4 Antimicrobial Activity

Significant antibacterial activity against a wide range of infections has been demonstrated by *Cola accuminata*. *Cola accuminata*'s antibacterial characteristics explain why it has been used historically to treat infections and wounds. Omwirhiren et al. [77], Wink [78], and Kamatenesi-Mugisha et al. [79] claim that it inhibits the growth of bacteria, fungi, and viruses, even certain strains that have developed resistance to antibiotics. Authors have variously reported that the bioactive substances such tannins, alkaloids, flavonoids, etc., are responsible for the antimicrobial activity of plants [80–101].

5.5 Anti-inflammatory and Analgesic Characteristics

Cola accuminata has traditionally been used for relief from pain and management of inflammatory diseases due to its anti-inflammatory and painkiller characteristics. Arthritis, rheumatism, and other inflammatory disorders may benefit from its anti-inflammatory and pain-relieving properties [56]. Flavonoids, phenols, and other bioactive chemicals are responsible for its anti-inflammatory and analgesic actions [68].

5.6 Respiratory Support

In the treatment of respiratory disorders like asthma, bronchitis, and cough, *Cola accuminata* has long been traditionally used to aid the body's natural defences. It is reported to have bronchodilatory and expectorant qualities that make it easier to breathe by breaking up mucus and opening airways [102]. *Cola accuminata* has shown promise as an anti-asthmatic agent, with preliminary research indicating it may relax bronchial smooth muscle and lessen inflammation [103].

5.7 Mental Alertness and Cognitive Enhancement

Cola accuminata is well-known for its stimulating features, such as its high caffeine level, which helps keep the mind alert and improves cognitive performance. It has a long history of use to boost memory, focus, and other mental abilities. Reduced weariness, increased alertness, and enhanced cognitive function have all been linked to the stimulating impacts of *Cola accuminata* on the central nervous system [104, 105]. Multiple studies have shown that *Cola accuminata* improves cognitive function, therefore it may be useful in treating dementia and Alzheimer's disease [106].

5.8 Antidiabetic Effects

Cola accuminata has shown remarkable antidiabetic capabilities, which brings us to our seventh point. Studies back up the folk applications of *Cola accuminata* for diabetic management. Blood glucose levels are lowered and glucose tolerance is enhanced, indicating hypoglycaemic activity [107, 108]. *Cola accuminata*'s anti-diabetic effects may be due to the presence of bioactive substances such alkaloids, flavonoids, and tannins [68].

5.9 Wound Healing Properties

Cola accuminata has long been used to aid in the recovery from wounds. It has been hypothesized to have tissue-regenerating and wound-protecting capabilities. Extracts of the plant *Cola accuminata* have been found to have healing and regenerative properties in studies [109]. Antimicrobial, anti-inflammatory, and antioxidant characteristics may account for its wound-healing actions [110].

5.10 Anti-cancer Properties

Cola accuminata has gained interest for its possible anti-cancer qualities. *Cola accuminata* extracts have shown anticancer efficacy in preliminary experiments against a number of cancer cell lines [106]. These cancers include breast, colon, prostate, and lung. *Cola accuminata* may have anti-cancer properties due to the presence of bioactive substances such as flavonoids and phenolic acids, which can suppress tumour growth and induce death [65].

6 Modern Scientific Research on Pharmacological Potentials of *Cola accuminata*

The pharmacological effects of *Cola accuminata*, often known as Cola nut, have been the focus of several recent scientific studies. These investigations have been conducted to determine whether or not *Cola accuminata* contains any bioactive chemicals for medicinal use. In Fig. 3, we highlight some of the most exciting recent developments in the study of *Cola accuminata*'s therapeutic potential.

6.1 Antimicrobial Activity

Adebayo et al. [111] tested *Cola accuminata* extracts for antibacterial activity against bacteria and fungus. The extracts inhibited drug-sensitive and drug-resistant

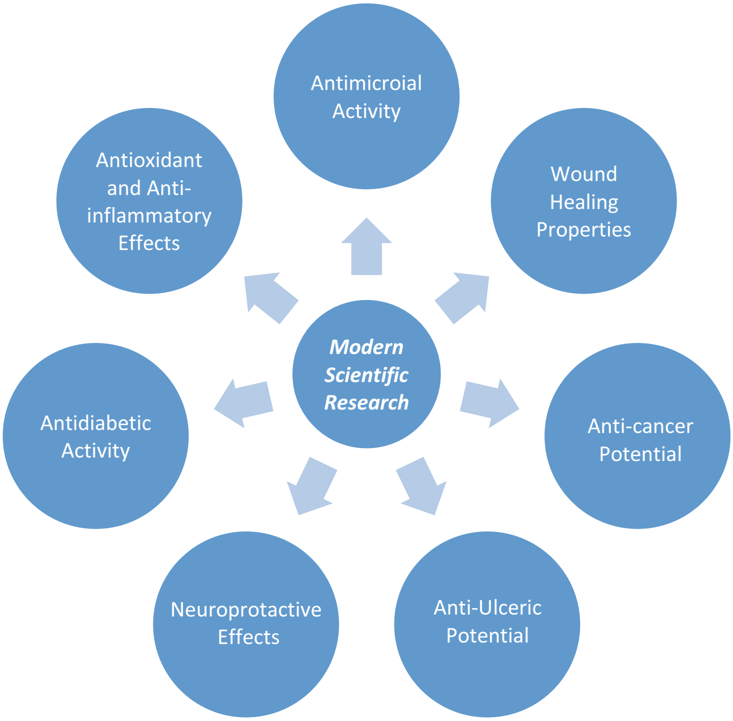


Fig. 3 Modern scientific research on pharmacological potentials of *Cola accuminata*

bacteria in the study. Bioactive substances such as tannins, alkaloids, and flavonoids had broad-spectrum antibacterial properties, according to the study. Also, in a study by Aiyegoro et al. [112], the authors tested *Cola accuminata*'s antibacterial efficacy against a variety of microorganisms. The extracts inhibited Gram-positive, Gram-negative, and drug-resistant microorganisms indicating that tannins, alkaloids, and flavonoids in the extracts had powerful antibacterial effects, according to the study. These studies therefore show that *Cola accuminata* possesses antibacterial capability. Tannins, alkaloids, and flavonoids in the plant provide broad-spectrum antibacterial activity. These chemicals inhibit drug-sensitive and drug-resistant bacteria and fungi, suggesting they could be used to generate new antimicrobials. These findings show promise, but more research is needed to identify and isolate *Cola accuminata*'s antibacterial bioactive components.

6.2 Antioxidant and Anti-inflammatory Effects

Amadi and Nwachukwu [113] and Akinmoladun et al. [114] examined *Cola accuminata* extract antioxidant properties. Phenolic chemicals and flavonoids in the extracts gave them antioxidant properties. These antioxidants neutralize free radicals, reduce oxidative stress, and protect cells from oxidative stress. These effects were linked to *Cola accuminata*'s antioxidant phenolic chemicals and flavonoids. Accordingly, Oboh et al. [115] reported *Cola accuminata* extracts reduced inflammation in animal studies. The extracts greatly suppressed pro-inflammatory mediators and enzymes, showing their anti-inflammatory potential. The extracts modify inflammatory pathways and inhibit pro-inflammatory molecule production, according to the study. Similarly, Olurishe et al. [116] tested *Cola accuminata* extracts for anti-inflammatory properties in a cell model. The extracts inhibited pro-inflammatory mediators dose-dependently, suggesting anti-inflammatory properties. Bioactive chemicals in the extracts disrupt inflammation signalling pathways, according to the study. Therefore, the phenolic chemicals and flavonoids in the plant neutralize free radicals and reduce oxidative stress. The extracts also suppress pro-inflammatory mediators and enzymes suggesting that *Cola accuminata* may alleviate oxidative damage and inflammation.

6.3 Antidiabetic Activity

Victoria et al. [42] and Osadee [117] tested *Cola accuminata* extracts for anti-diabetic effects in animals. The extracts have strong hypoglycaemic effects, reducing blood glucose. Tannins and flavonoids, which increase the generation of insulin and inhibit carbohydrate-digesting enzymes, may explain *Cola accuminata*'s anti-diabetic effects. Also, Ezejiofor et al. [118] examined *Cola accuminata* extracts' anti-diabetic properties in animal models. The extracts dramatically lowered blood glucose and enhanced insulin sensitivity. Bioactive chemicals in *Cola accuminata* improve glucose absorption and metabolism, according to the study. In a similar study, Boudiba et al. [119] examined *Cola accuminata* extracts' anti-diabetic effects in cellular models, and the results showed that the extracts significantly inhibited the metabolism of carbohydrates and glucose absorption enzymes. These findings imply that *Cola accuminata*'s antidiabetic action may be due to its capacity to inhibit carbohydrate-digesting enzymes, lowering blood glucose release. Thus, the *Cola accuminata* possesses anti-diabetic properties. The tannins and flavonoids in the plant increase insulin synthesis, inhibit carbohydrate-digesting enzymes, and increase glucose absorption, making it hypoglycaemic. These findings suggest *Cola accuminata* could be used to produce antidiabetic medicines.

6.4 Neuroprotective Effects

Imam-Fulani et al. [120] examined *Cola accuminata* extract's neuroprotective properties in animals. The extract improved memory and learning. *Cola accuminata*'s antioxidants minimize oxidative stress and preserve neurons, according to the study. Similarly, Oboh et al. [68] investigated *Cola accuminata* bioactive components' neuroprotective properties. The study examined how these components affect memory and cognition. *Cola accuminata* extract bioactive components protected against loss of memory and dementia. Bioactive chemicals alter neurotransmitter systems and increase neural signalling, according to the researchers. Furthermore, Oboh et al. [121] and Ishola et al. [122] examined how *Cola accuminata* extract protects against brain damage. In animal experiments, the extract reduced oxidative stress and protected neurons. The antioxidant qualities of *Cola accuminata* may help treat and prevent neurological illnesses, according to researchers. Thus, these research demonstrate *Cola accuminata*'s neuroprotection. The plant's antioxidant qualities may protect neurons from oxidative stress and death. *Cola accuminata* improves memory, learning, and cognitive function. These findings show that *Cola accuminata* may provide neuroprotective chemicals for treating and preventing neurodegenerative disorders. However, there is still need to identify and isolate these bioactive chemicals and understand their neuroprotective processes in further researches.

6.5 Anti-cancer Potential

Nugraha et al. [123] examined *Cola accuminata* extract's cytotoxicity on breast cancer cell lines. The extract suppressed breast cancer cell multiplication and induced apoptosis. Bioactive chemicals in *Cola accuminata* may fight cancer, according to the study. *Cola accuminata* was also tested in lung cancer cell lines by Alsaeedi et al. [124]. The extract inhibited lung cancer cell growth and induced apoptosis. *Cola accuminata*'s bioactive ingredients selectively kill cancer cells, according to the study. *Cola accuminata* extract was tested on colon cancer cell lines by Percival et al. [125]. The extract killed colon cancer cells and inhibited cell multiplication. The extract's capacity to alter cell growth and survival signalling pathways may explain its anti-cancer action. In Nugraha et al. [123], the authors examined *Cola accuminata*'s anti-cancer effects in prostate cancer cell lines. The extract inhibited prostate cancer cell multiplication and induced apoptosis. The bioactive chemicals in *Cola accuminata* may suppress prostate cancer cell proliferation by interfering with important biological processes.

6.6 Anti-Ulceric Properties

Cola accuminata extract was tested in rats with aspirin-induced stomach ulcers by Holcombe et al. [126]. The extract reduced ulcer size and improved stomach mucosa histology. The study authors hypothesized that the extract's gastroprotective effects came from increased mucus production, antioxidant defences, and oxidative stress reduction. Nkanu [127] tested *Cola accuminata*'s anti-ulcer effects in different rat ulcer models. The extract reduced gastrointestinal lesions, stomach acid output, and pepsin activity, preventing ulcers. Antioxidant enzymes and gastric mucus increased, indicating better stomach lining protection. Suntar et al. [128] evaluated *Cola accuminata* extract's effects on rats' ethanol-induced stomach ulcers. The extract reduced ulcer index, stomach acid output, and lipid peroxidation to treat ulcers. The extract increased gastric mucin and antioxidant enzyme activity, strengthening the mucosal barrier. These findings imply *Cola accuminata* may treat ulcers. The extract may reduce gastric acid output, protect gastrointestinal mucosa, boost mucus formation, and act as an antioxidant. These ways help the drug preserve the gastric mucosa, heal ulcers, and prevent them. These findings support the traditional usage of *Cola accuminata* to treat ulcers.

6.7 Wound Healing Properties

Ishola et al. [122] and Edwards et al. [129] assert that *Cola accuminata* extract helped to heal and improved wound contraction, size, and re-epithelialization. The extract accelerated wound-healing mechanisms like collagen deposition, angiogenesis, and antioxidant activity, according to the researchers. Agbaje and Charles [130] tested *Cola accuminata*'s wound-healing ability in rats using excision and incision wound models. The extract improved wound closure, tensile strength, and collagen formation. Researchers believed the extract's antioxidant and anti-inflammatory properties helped mend wounds. Diallo et al. [131] tested *Cola accuminata* extract for burn wound treatment in rats. The extract improved wound healing by shrinking wounds, re-epithelializing, and depositing collagen. The extract stimulated fibroblast proliferation and migration, improving wound healing. Scientists believed the extract's angiogenesis and anti-inflammatory effects helped mend wounds. These trials confirm *Cola accuminata*'s wound dressing efficacy. This extract promotes wound healing, re-epithelialization, collagen production, antioxidants, and anti-inflammation. Its wound-healing capabilities make it a viable natural therapy.

7 Some Herbal Formulations Involving *Cola accuminata* as Active Ingredients

Traditional and folk medicine uses herbal preparations using *Cola accuminata* as an active ingredient. These compositions mix *Cola accuminata* with other plant-based substances for synergistic effects and improved medicinal results. Some example of *Cola accuminata* herbal formulations in Nigeria are shown below.

7.1 Agbo Formulation

Nigerian herbalists use agbo to cure a variety of diseases. *Cola accuminata* is mixed with medicinal herbs. Akande et al. [132] tested an agbo formulation with *Cola accuminata, Morinda lucida*, and *Alstonia boonei* for antibacterial activity. This herbal formulation inhibited Gram-positive and Gram-negative bacteria, confirming its traditional use.

7.2 Ewuro-Ajekobale Formulation

Yoruba folk medicine practitioners in Nigeria treats diabetes using Ewuro-Ajekobale formulation. This formulation contains *Cola accuminata*, bitter leaf (*Vernonia amygdalina*), and wild tomato (*Solanum torvum*). Musabayane et al. [133] examined the hypoglycaemic effects of the formulation in diabetes animal models and this herbal composition showed considerable blood glucose decreases, indicating its diabetes management potential.

7.3 Ciklavit Herbal Preparation

Traditional Yoruba medicine practitioners treats menstruation issues with this herbal preparation. It has *Cola accuminata, Citrus aurantifolia*, and other therapeutic herbs. Ciklavit was tested on experimentally induced menstruation problems in female rats by Cyril-Olutayo et al. [134], and its anti-inflammatory and analgesic properties supported its use in menstrual problems.

7.4 Agbo-Iba

Yoruba traditional medicine uses a herbal concoction called Agbo-Iba to treat fever and malaria. *Cola accuminata* is combined with A*zadirachta indica* and *Mangifera indica*, two more therapeutic plants. Nwabuisi [135] examined the efficacy of the Agbo-Iba formulation against plasmodium. The findings provided strong evidence for its traditional use as an antimalarial treatment, showing considerable reduction of *Plasmodium falciparum* development.

7.5 Orogbo Formulation

This herbal mixture is used in traditional medicine to treat gastrointestinal problems like diarrhoea and dysentery. It includes *Cola accuminata* together with *Garcinia kola* and *Vernonia amygdalina*, both of which are plants. Odukanmi et al. [136] evaluated the effectiveness of the *Cola accuminata* in preventing diarrhoea in animal models. Supporting its traditional use, the extract significantly reduced the frequency and severity of diarrhoea.

7.6 Opa Eyin Formulation

Traditional healers rely on this herbal concoction to treat skin diseases and wounds. Included are *Newbouldia laevis* and *Cola accuminata*, both of which have medicinal uses. The Opa Eyin formulation's antibacterial efficacy against several bacteria and fungi was studied by Bolawaa et al. [137]. Significant inhibitory effects were seen against the investigated microorganisms, suggesting that the formulation may be useful in the treatment of skin infections.

7.7 Agbo-Jedi Formulation

The formulation is a traditional herbal concoction with aphrodisiac effects, widely used in traditional medicine. *Cola accuminata* is combined with *Mondia whitei* and *Citrullus colocynthis*, two other therapeutic plants. Akinboro et al. [138] and Adeyemi et al. [139] examined the aphrodisiacal efficacy of the Agbo-Jedi formulation. The traditional use of the mixture as an aphrodisiac was supported by the data showing a considerable increase in sexual behaviour and reproductive indices.

7.8 Agbo-Atosi Formulation

This is a herbal concoction used in Nigerian folk medicine to treat hypertension. *Cola accuminata* is combined with other therapeutic herbs like aloe barbadensis and onion. Eiya and Igharo [140] and Akande et al. [132] looked into the antihypertensive effects of the Agbo-Atosi formulation and reported that the formulation had promising antihypertensive effects, with blood pressure levels significantly reduced.

7.9 Oroki Formulation

This herbal formulation with hepatoprotective qualities is widely utilized in traditional medicine in Nigeria. It includes therapeutic herbs like *Cola accuminata* and Garcinia kola. Emeleku [141] evaluated the Oroki formulation's hepatoprotective efficacy in animal models. The formulation showed promising results in protecting against hepatotoxicity, indicating its potential in managing liver health.

7.10 Osomo Formula

Osomo is a local herbal aphrodisiac. It contains *Cola accuminata* and *Allium sativum/cepa*. Adeyemi and Owoseni [142] examined the polyphenolic content and biochemical evaluation of the formulation and reported the formulation showed significant composition of the parameters assessed, thereby supporting the traditional use of this formulation as an aphrodisiac.

7.11 Epa-Ijebu Formulation

This is a traditional herbal concoction with aphrodisiac characteristics that is utilized in most traditional medicine. *Cola accuminata* is combined with *Mondia whitei* and *Carpolobia lutea*, two other therapeutic herbs. The aphrodisiac effects of the Epa-Ijebu combination were tested on animals in a study by Enitan [143]. The combination was shown to significantly improve sexual performance and libido, lending credence to its historical role as a natural aphrodisiac [143]. In summary, *Cola accuminata* has a wide variety of uses in traditional herbal formulations, as evidenced by these examples. It is commonly used for its aphrodisiac, gastrointestinal health, anti-malarial, and anti-inflammatory properties.

8 Safety Considerations and Side Effects of *Cola accuminata*

Some of the safety considerations and side effects of consuming *Cola accuminata* to be discussed under this subhead are shown in Fig. 4.

8.1 Caffeine Content

The plant *Cola accuminata*, among others, has the naturally occurring chemical caffeine. It has been scientifically proven to stimulate the central nervous system. Most people can manage modest amounts of caffeine with no problems, but too much can have negative effects. *Cola accuminata*'s caffeine level is a major cause for concern because of its link to irritability and nervousness. Caffeine in excess can overstimulate the nervous system, causing irritability, nervousness, and even anxiety. Caffeine use has been linked to both difficulty falling asleep and maintaining a restful slumber [144]. One of the negative effects of consuming too much caffeine is a rapid heartbeat. Caffeine causes a transient increase in heart rate due to its stimulatory action on the cardiovascular system. This effect may be more

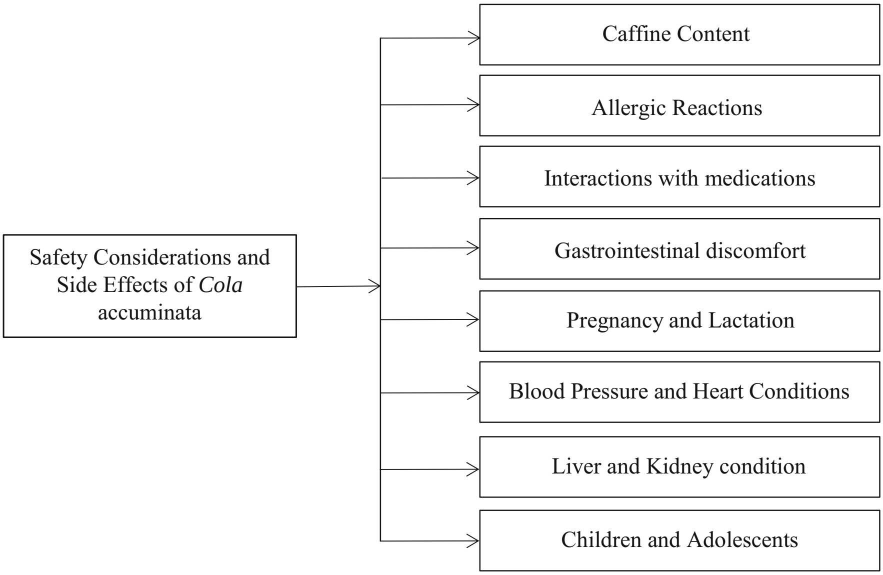


Fig. 4 Safety considerations and side effects of *Cola accuminata*

pronounced in caffeine-sensitive people or people with pre-existing heart issues and may cause palpitations or abnormal heart rhythms [144].

Another common negative reaction to caffeine overdose is gastrointestinal distress. Caffeine increases stomach acid production, which can lead to indigestion, heartburn, and other gastrointestinal distress. It may also have minor diuretic effects, resulting to increased urine production and frequency in certain people [145, 146]. Thus, caffeine tolerance varies greatly from person to person. People with pre-existing problems like heart disease, high blood pressure, or gastrointestinal issues may be particularly vulnerable to the negative effects of caffeine. Nehlig [144] stressed the importance of moderation in caffeine consumption, including *Cola accuminata*, in order to reduce the risk of adverse consequences.

Nevertheless, *Cola accuminata* can be consumed safely provided that consumers are aware of their unique caffeine sensitivity and take into account their total caffeine intake from sources like coffee, tea, and other caffeinated beverages or meals. People who are more susceptible to caffeine's effects or who have health issues that can be made worse by caffeine should cut back on their intake [144].

8.2 Allergic Reactions

Although it is uncommon for people to have an adverse reaction to *Cola accuminata*, it is nonetheless vital to be aware of the possibility. An allergic reaction occurs when the immune system overreacts to a normally harmless chemical. Symptoms of an allergy can present in a number of ways, including on the skin, in the lungs, or throughout the body. Some people have severe allergic reactions to *Cola accuminata*, including hives, swelling of the face and throat, and trouble breathing, which can occur after eating or touching the plant. These reactions to *Cola accuminata*'s allergenic components might manifest themselves quickly or build up over time [147]. Hence, there is a need to stop using *Cola accuminata* and get medical help right away if an allergic reaction is imminent. As such, products containing *Cola accuminata* should be used with caution by people with allergies or a history of adverse reactions. In these circumstances, it is best to talk to a doctor or allergist about getting tested for allergies and getting advice on how to consume *Cola accuminata* safely. *Cola accuminata* allergies appear to be uncommon because people rarely report experiencing negative reactions to the plant. However, when drinking or coming into contact with *Cola accuminata*, it is crucial to remain cautious and respond appropriately to any indicators of allergic reactions.

8.3 Interactions with Medications

Cola accuminata's bioactive ingredients, such as caffeine, may interfere with the effectiveness of several pharmaceuticals. These interactions might lessen or heighten the likelihood of desired results, depending on the drugs involved. If you are currently taking any drugs, you should discuss using *Cola accuminata* with your doctor because of the possibility of drug interactions. One of *Cola accuminata*'s most significant components, caffeine, can interfere with the effectiveness of drugs used to treat heart disease, mental illness, and breathing problems. Because of the potential impact on the safety of patients and treatment outcomes, Baur et al. [148] emphasized the need of taking into account such interactions.

Beta-blockers, used to treat hypertension and other cardiovascular disorders, may have an adverse reaction to caffeine. The blood pressure-lowering benefits of beta-blockers may be diminished by caffeine consumption [149]. Caffeine can also have an influence on the metabolism of, or increase the risk of side effects associated with, certain psychiatric drugs, including selective serotonin reuptake inhibitors (SSRIs) and lithium [150]. In addition, caffeine can interact with bronchodilators like theophylline, which are used to treat respiratory disorders. Theophylline's negative effects can be amplified by caffeine, leading to a faster heart rate and tremors [151]. Given the potential for negative interactions with other medications, it is best to check with a doctor before using *Cola accuminata*.

8.4 Gastrointestinal Discomfort

Intestinal distress has been reported in a small number of people who have used *Cola accuminata* or its extracts. Caffeine and other components of the plant may contribute to uncomfortable side effects such gastrointestinal distress, heartburn, and diarrhoea. One of the primary ingredients of *Cola accuminata* is caffeine, which has been shown to stimulate the digestive system, leading to more gastric acid output and possibly stomach discomfort [145]. In addition, some people may experience diarrhoea as a result of the laxative effects of caffeine [152]. Some people may be more susceptible to caffeine's gastrointestinal side effects than others.

Cola accuminata has a number of chemicals that can affect gastrointestinal function, including caffeine, tannins, and polyphenols. Particular tannins have been linked to astringent characteristics and the potential to cause gastrointestinal discomfort in some people [153]. *Cola accuminata* preparations include several chemicals, which may cause gastrointestinal distress in those who are sensitive to them. It is important to note that the amount of *Cola accuminata* consumed, individual tolerance, and pre-existing gastrointestinal problems may all have a role in the occurrence and severity of gastrointestinal discomfort. As a result, you should pay attention to how your body reacts and make dietary adjustments accordingly.

8.5 Pregnancy and Lactation

Caffeine, a popular stimulant found in *Cola accuminata*, can enter the foetal circulation after crossing the placenta [154]. A higher likelihood of miscarriage, premature birth, low birth weight, and issues with growth have all been linked to maternal caffeine consumption during pregnancy [155]. Infants of moms who drink too much coffee may develop behavioural and sleep problems and experience abdominal distress as a result [144]. Given the current state of knowledge, expectant and nursing mothers should proceed with caution while consuming *Cola accuminata* or items containing it. It is best to talk to a doctor about your specific situation and any hazards involved so you can make an educated choice.

The dearth of particular studies on this plant in pregnant and breastfeeding women is largely to blame for the scarcity of data on the safety of *Cola accuminata* at these stages of a woman's life. In order to protect the health of both mother and child, healthcare providers normally recommend that pregnant and nursing women limit their caffeine intake from all sources, including *Cola accuminata*.

8.6 Blood Pressure and Heart Conditions

Cola accuminata contains caffeine, which has been shown to temporarily increase heart rate and blood pressure. People who have hypertension or cardiovascular disease should use caution when consuming *Cola accuminata* or products containing it, and should keep an eye on their blood pressure and heart rate. In these situations, you should talk to a doctor before using *Cola accuminata*. The stimulant effects of caffeine on the central nervous system are accompanied by a transient elevation in blood pressure and heart rate [156]. This reaction is temporary and is felt by the vast majority of people. However, those who already have hypertension or heart problems may be more susceptible to caffeine's negative effects.

Caffeine can induce a transient increase in blood pressure, thus those with hypertension should be careful about how much they drink [157]. Caffeine, particularly *Cola accuminata* products, should be limited by people with hypertension in order to maintain healthy blood pressure levels. Maintaining a blood pressure reading within a healthy range requires regular monitoring. Caffeine should also be consumed with caution by those who have heart disorders such as arrhythmias or coronary artery disease. Some cardiac arrhythmias may be triggered or made worse by caffeine, and caffeine may also increase the heart's workload [156]. Before taking *Cola accuminata* or products containing it, those with heart issues should talk to a doctor about the possible effects on their condition.

8.7 Liver and Kidney Condition

Cola accuminata or goods containing it should be consumed with caution by people with liver or kidney disorders. *Cola accuminata*'s presence of caffeine and other chemicals may have an impact on several body systems. In these situations, you should talk to a doctor before using *Cola accuminata*. The liver and kidneys are essential organs for detoxification and waste removal. Caffeine, a stimulant, can affect several systems because of its metabolism and excretion. Caffeine and other substances may not be processed properly by people with liver or kidney disorders, which could have negative consequences. Changes in indicators of liver and kidney function were observed in rats given *Cola accuminata* extract in a study by Olatunji et al. [158]. The study authors cautioned against using *Cola accuminata* for anyone who already has liver or kidney disease.

Some people may have a diminished ability to metabolize caffeine due to liver diseases such as liver disease or poor liver function. Extended contact with caffeine, or its build-up, has been linked to hepatic stress [150]. Caffeine and its metabolites may also have an effect on renal function in those with kidney disorders such as chronic kidney disease or impaired kidney function [150]. Individuals with pre-existing liver or kidney issues should not consume *Cola accuminata* or products containing it without first consulting a healthcare expert. A medical specialist is most equipped to analyse an individual's health situation, weigh the pros and downsides, and recommend a safe dosage or alternate treatment.

8.8 Children and Adolescents

Due to its caffeine level and potential effects on developing bodies, children and adolescents should consume *Cola accuminata* moderately. Caffeine can impact sleep, growth, and health in this population. *Cola accuminata* should not be used in children and teenagers without medical advice. Caffeine, a stimulant, affects children and adolescents more than adults. Children and teenagers may be particularly sensitive to caffeine due to their developing bodies. Caffeine can alter sleep patterns and make it hard to fall and remain asleep [159]. Sleep disturbances can harm children and teenagers' growth, cognition, and well-being. In addition to sleep disruptions, high coffee intake in this population may impact growth and development. Caffeine can lower adolescent bone mineral density, which can lead to osteoporosis later in life [159]. Caffeine can also impair nutritional absorption, which could impede growth and development.

Caffeine may potentially harm kids' cardiovascular health. In people with cardiovascular problems, excessive caffeine can raise heart rate and blood pressure [159]. Some people may be more sensitive to caffeine's stimulating effects. Given these concerns, *Cola accuminata* should be limited in children and adolescents and used under medical supervision. A doctor can advise on the right amount of caffeine for this population and weigh the dangers and benefits.

9 Commercial Properties of *Cola accuminata*

Cola accuminata has the potential to be economically valuable and used in a wide range of businesses thanks to its commercial features. This includes its uses in the nourishment and beverage sector, the pharmaceutical sector, and the cosmetics sector to mention but a few (Fig. 5). Let us go into greater depth about these investment properties for businesses, using appropriate sources.

9.1 Food and Beverage Industry

Cola accuminata has been used for decades in the food and beverage industry as a flavouring agent in the creation of carbonated beverages such as cola drinks. The addition of caffeine and other bioactive components gives these drinks their own unique flavour. Candies, chocolates, and desserts all benefit from the addition of *Cola accuminata* extracts and infusions [160]. *Cola accuminata*'s natural components make it a valuable commodity in the food and drink sector.

9.2 Pharmaceutical Industry

The pharmaceutical sector is interested in *Cola accuminata* because of its possible therapeutic characteristics. The pharmacological effects of the bioactive chemicals found in *Cola accuminata* have been extensively researched. As previously mentioned, these effects can be broken down into four categories: antioxidant, anti-inflammatory, antimicrobial, and anti-cancer [31, 68, 122, 161]. Because of these

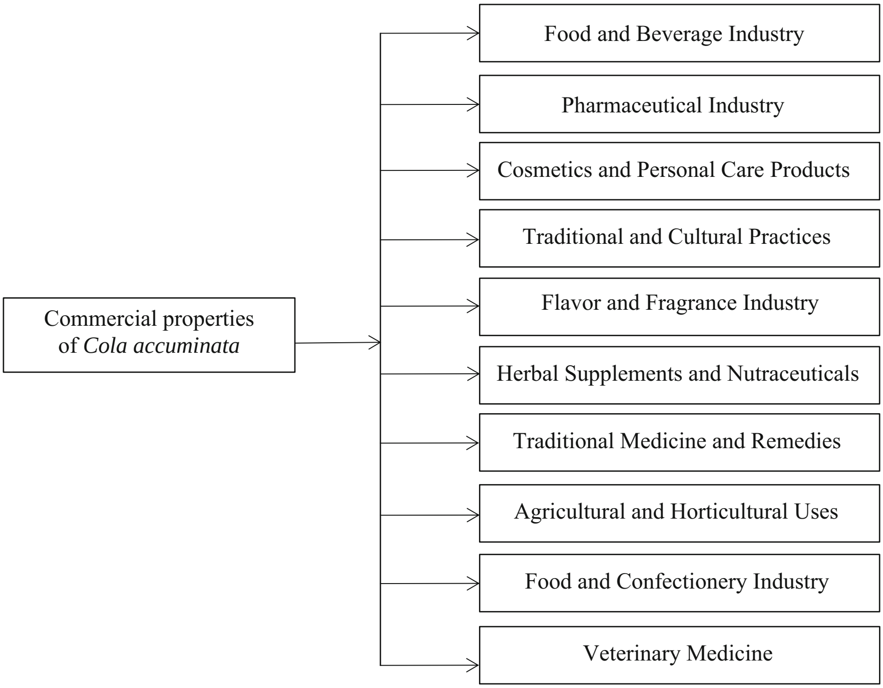


Fig. 5 Commercial properties of *Cola accuminata*

qualities, *Cola accuminata* is a promising ingredient in the creation of natural pharmaceuticals and functional meals.

9.3 Cosmetics and Personal Care Products

Extracts from the *Cola accuminata* plant are used in the cosmetics business. *Cola accuminata*'s antioxidant properties make it a promising addition to cosmetics. The skin benefits from antioxidants because they shield it from oxidative stress, which is linked to aging and environmental damage. To give antioxidant advantages and support healthy skin, *Cola accuminata* extracts can be added into formulations including creams, lotions, etc. The trend towards organic and all-natural cosmetics and toiletries necessitates the usage of natural substances like *Cola accuminata*. Extracts from the *Cola accuminata* plant are used in the production of a wide variety of cosmetics and toiletries. Because of its high antioxidant content, this plant is a welcome addition to many topical skincare formulations. Extracts from the *Cola accuminata* plant have been shown to reduce the effects of oxidative stress on the skin and to boost its appearance in youth [162, 163].

9.4 Traditional and Cultural Practices

Cola accuminata has cultural and traditional value in several locations in addition to the economic purposes described above. *Cola accuminata* has spiritual and ceremonial significance in various societies. *Cola accuminata*, from which the cola nut is harvested, is utilized in religious ceremonies and as a sign of hospitality and brotherhood [164, 165]. *Cola accuminata*'s social and symbolic worth is enhanced by these cultural practices.

9.5 Flavour and Fragrance Industry

Extracts and oils from the *Cola accuminata* plant are used to give products a cola-like aroma and flavour in the flavour and fragrance business. Perfumes, colognes, and other scented goods make use of *Cola accuminata* for its distinctive aroma. *Cola accuminata*'s singular aroma profile enhances the complexity and depth of perfumes in many forms [14, 104].

9.6 Herbal Supplements and Nutraceuticals

Nutraceuticals and herbal supplements made with *Cola accuminata* are becoming increasingly popular. Caffeine and other bioactive chemicals in it suggest it could be used in energizing supplements and medicines. Stimulant effects, cognitive enhancement, and general health benefits are all possible selling points for these items [106]. The incorporation of *Cola accuminata* in such products meets the rising demand for organic and plant-based options in the supplement market.

9.7 Traditional Medicine and Remedies

Cola accuminata plays a significant role in some regions' traditional medicine systems. *Cola accuminata* is used by traditional healers and practitioners to treat a wide range of problems, including gastrointestinal issues, weariness, and lack of concentration [42]. It has aphrodisiac powers, at least in the eyes of some societies. *Cola accuminata* has commercial value since it is used in traditional medicine, which is still common in some cultures.

9.8 Agricultural and Horticultural Uses

The *Cola accuminata* tree has numerous practical applications in agriculture and horticulture. Insects and other pests can be kept away from crops thanks to substances found in the plant's seeds and husks [166, 167]. Because of its historic value and visually appealing foliage, *Cola accuminata* is also occasionally cultivated as an ornamental tree.

9.9 Food and Confectionery Industry

Cola accuminata is used to improve the flavour and scent of many foods and sweets in the food and confectionary industries. To give baked products, candies, chocolates, ice creams, and other sweets with the familiar cola flavour, cola flavouring is often added. Energy bars and other functional food products can benefit from *Cola accuminata*'s naturally occurring caffeine [168].

9.10 Veterinary Medicine

Potential uses of *Cola accuminata* extracts in veterinary medicine have been investigated. Their potential to fight against various animal diseases and antimicrobials are being researched. *Cola accuminata*'s bioactive components have therapeutic potential for animal wellness and good health [158].

10 Conclusion

Cola accuminata, or the Cola nut, is a plant that has been used traditionally and holds great cultural significance in many parts of the world. *Cola accuminata*'s phytochemical components, nutritional profile, scientifically confirmed pharmacological qualities, ethnomedicinal applications, safety concerns, and commercial value have all been discussed at length. These results demonstrate the many uses and benefits of this plant. *Cola accuminata* has been shown to contain a number of bioactive substances, such as alkaloids, tannins, flavonoids, phenolic compounds, and caffeine. The plant's antioxidant, anti-inflammatory, antibacterial, anti-cancer, and neuroprotective benefits can be attributed in part to these components. *Cola accuminata* may have medicinal uses in the treatment and prevention of disease due to the presence of certain phytochemicals.

Carbohydrates, proteins, lipids, vitamins, and minerals are just some of the many nutrients that *Cola accuminata* is a good source of. The energizing effects of the caffeine in it might help you stay awake and focused. Caffeine is a stimulant, but consuming too much of it can cause negative side effects like irritability, nervousness, and a faster heart rate. So, it is best to take it easy sometimes.

The use of *Cola accuminata* in traditional and folk medicine dates back many centuries. It has been used for its stimulating effects, and it may also help with things like memory and focus, digestion, and wound healing, among other things. Several of these traditional applications have been validated by scientific studies, demonstrating the plant's usefulness in a variety of contexts. However, there are a number of factors that must be taken into account while determining the safety of *Cola* *accuminata*. *Cola accuminata*, like other caffeine-containing drinks, can cause a variety of unwanted side effects if consumed in large quantities. *Cola accuminata* should be used with caution and in consultation with medical specialists by people with certain health issues, including cardiovascular illnesses, psychological conditions, liver or renal conditions, and pregnancy or lactation.

Cola accuminata has substantial commercial value. It is a staple in the beverage industry, where it is used to impart the signature cola flavour and natural caffeine content into classic cola drinks. The plant is used in a wide variety of other contexts, such as aromatherapy, veterinary medicine, the cosmetics and personal care industries, and the food and confectionary industries. Its distinctive aroma, flavour, and bioactive qualities all add to its commercial value and popularity as a formulating component. Nevertheless, more study is needed to determine its full potential and how best to apply it to enhance human health and well-being.

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Citrus aurantium: Phytochemistry, Therapeutic Potential, Safety Considerations, and Research Needs

6

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Abstract

Citrus aurantium, commonly referred to as sour orange or bitter orange, holds significant importance both in biological and economic terms. Throughout history, humans have turned to nature in their pursuit of health and wellness. Among the plants that have historically played a role in enhancing fitness, *Citrus aurantium* stands out. A diverse array of phytochemical constituents present in *Citrus aurantium* have been closely tied to its various biological activities, encompassing areas such as gastrointestinal disorders, insomnia, headaches, cancer treatment, antiseptic properties, antioxidant effects, and antispasmodic effects. Beyond its pharmacological relevance, *Citrus aurantium* also boasts numerous non-pharmacological applications that make it particularly intriguing. It serves as a food preservative agent, contributes to aromatherapy practices, acts as a pesticide, provides raw materials for the pulp and paper industry, lends its aromatic qualities to the food processing and cosmetics sectors, and offers potential as an anti-aging agent. Despite its impressive array of properties, the utilization of *Citrus aurantium* and its derivatives has been linked to certain unwanted side effects. While some studies have largely cleared the plant of safety concerns, others have cast doubt on its safety. This chapter delves into the complex aspects surrounding the safety of *Citrus aurantium*'s phytochemistry and its derived products. Ultimately, the chapter advocates for the ongoing use of this plant, but with a careful awareness of its limitations to prevent any undesirable effects.

Keywords

Citrus aurantium · Bitter orange · Phytochemicals · Non-pharmacological uses · Economic relevance

Abbreviations

ALT	Alanine transminase
AMPKα	Activated protein kinase alpha
AMPKα	AMP-activated protein kinase alpha
AST	Aspartate aminotransferase
CAVAPs	Crude polysaccharides of *C. aurantium* L. var. Amara Engls
DPPH	2,2-Diphenyl-1-picrylhydrazyl
ERK	Extracellular signal- regulated kinase
EtOAc	Ethyl acetate
FAS	Fatty acid synthase
FDA	Food and Drug Administration
FRAP	Ferric reducing antioxidant power
GGT	Gamma-glutamyl transferase
GSK3β	Glycogen Synthase Kinase 3 Beta
IL-1β	Interleukin-1β
IL-6	Interleukin 6
iNOS	Inducible nitric oxide synthase
JNK	c-Jun N-terminal kinase
MAPK	Mitogen-activated protein kinase
MTD	Maximum tolerated dose
NAFLD	Non-alcohol fatty liver disease
NASH	Non-alcoholic steatohepatitis
NF-kB	Nuclear factor kappa B
NOAEL	No-observed-adverse-effect-level
NOEL	No-observed-effect-level
Nrf2	Nuclear factor erythroid 2-related factor 2
ORAC	Oxygen radical absorbance capacity
PGC-1α	PPARγ co-activator 1α
PTFC	Pure total flavonoids from citrus
SCD1	Stearoyl-CoA desaturase 1
TAC	Total antioxidant capacity
TBARS	Thiobarbituric acid reactive substances
TNF- α	Tumor necrosis factor α
UCP1	Uncoupling protein-1
WADA	World Anti-Doping Agency

1 Introduction

Citrus fruits stand apart due to their unique structural composition, which includes seeds, pulp, oil cells, and a white membrane layer [1]. These fruits are remarkable sources of essential nutrients such as vitamin C, bioactive flavonoids, and folic acid. These components play diverse roles in reducing inflammation and enhancing the body's immune system. Native to Australia, Malaysia, New Caledonia, and a few other countries, *Citrus aurantium*, commonly known as sour orange, has a significant geographical presence [2]. Over time, it was cultivated in regions including Spain, France, North and South Africa, and various other parts of the tropical and temperate world, owing to its robustness compared to other Citrus species [2]. Among the cultivars of *Citrus aurantium*, '*Citrus aurantium* Changshanhuyou' (CACH) stands out. This cultivar is the result of a hybridization process involving *Citrus grandis* and *C. sinensis* [1].

In recent years, there has been a notable increase in citrus production, leading to a greater focus within the fruit processing industries on creating various products such as juices, essential oils, flavorings, antioxidants, and acidifying ingredients for food applications [2]. However, these advancements generate a substantial amount of solid waste from citrus, including peels, seeds, and membrane residue, during the processing stage. This solid waste from citrus is not without value, as it can be recycled and put to various uses, primarily serving as a source of beneficial phytochemicals, vitamins, and minerals that have potential applications in treating various health conditions [2].

Furthermore, the fruits of *Citrus aurantium* (*C. aurantium*) offer more than just essential oils; they also contain compounds of the flavonoid type, which exhibit a range of biological effects. These compounds contribute to the fruit's health-promoting qualities, including antioxidative, antibacterial, antiallergic, anticancer, and antidiabetic properties [1–3]. *C. aurantium* has been utilized in traditional Chinese medicine to address various health issues such as anxiety, cancer (particularly lung and prostate), digestive problems, and obesity. Additionally, its uses extend to serving as a stimulant, appetite suppressant, and remedy for motion sickness, indigestion, and constipation [2, 4]. Moreover, it has been explored for treating cancer and cardiovascular diseases [2]. Notably, the aroma of its essential oils has been harnessed to induce sleep in individuals prone to insomnia [5–7], and these oils also exhibit potent pesticidal properties [3, 8–10].

Beyond its compelling biological significance, *C. aurantium* possesses intriguing non-pharmacological attributes. Its essential oils, for instance, display antimicrobial and antioxidant characteristics, rendering them valuable as natural food preservatives [11]. Furthermore, due to its exceptional qualities, *C. aurantium* serves as a suitable raw material for producing pulp and paper [12]. Industries such as food processing and cosmetics also harness its aromatic nature [13–15]. Moreover, *C. aurantium* has found relevance in skin treatments, particularly as an anti-aging agent [16].

The prominence of *Citrus aurantium* has increased, notably as a replacement for weight-loss products previously utilizing *Ephedra sinica* Stapf. in Farw., as it contains p-synephrine, an alkaloid of the phenylethanolamine type. This compound has hunger-suppressing effects and shares a chemical relationship with adrenergic medications [3]. The protoalkaloid p-synephrine has garnered attention for its use in weight loss and athletic supplements, as well as potential risks associated with its consumption [17–19]. Although certain studies have largely absolved *Citrus aurantium* and p-synephrine of reported risks, safety concerns remain [18, 20–22]. Recent scientific research has extensively explored the potential effects of various parts of *Citrus aurantium*, including its blossoms, fruits, and essential oils [3].

This chapter provides a comprehensive discussion of the phytochemistry, pharmacological and non-pharmacological potentials, safety considerations, and research gaps related to *Citrus aurantium*. The aim is to evaluate the current scientific endeavors focused on toxicological assessments of the plant, while also setting the groundwork for future research initiatives.

2 Overview of the Non-pharmacological Uses of *Citrus aurantium*

2.1 The Role of *Citrus aurantium* in Food Preservation

The interest in natural preservatives has increased tremendously due to the undesirable tendencies of synthetic preservatives [23]. Owing to its antimicrobial and antioxidant properties, the essential oil of *Citrus aurantium* has become an attractive alternative to the use of synthetic preservatives [11]. This section explores the antimicrobial and antioxidant properties of *Citrus aurantium* in a view to appraising its food preservative potentials.

2.1.1 Antibacterial Properties of *Citrus aurantium* and Its Food Preservative Potential

Citrus aurantium essential oil is a terpenoid-rich mixture with antibacterial effect, claim Anwar et al. [11]. The plant's phytochemicals' close interaction with the target cell and subsequent solubilization of the cell membrane give rise to the plant's antibacterial characteristics. In a study conducted by Ben Hsouna et al. [24], it was discovered that *Citrus aurantium* contains the following monoterpenes: 27.5% limonene, 14% terpineol, 4.3% E-ocimene, and 2.4% delta-3-carene. Sesquiterpenes are said to make up 30.2% of the essential oil, with (E)-nerolidol having the highest concentration (17.5%). Several bacteria and fungi, including *Candida albicans*, *Staphylococcus epidermis*, and *Escherichia coli*, have been demonstrated to be suppressed by terpineol [11]. The essential oils of *Citrus aurantium* were investigated for their antibacterial activity against a panel of 13 bacterial strains using agar diffusion and broth microdilution procedures. According to the findings [24], the plant's essential oils exhibit a moderate-to-strong antibacterial activity. The antibacterial efficacy of *Citrus aurantium* essential oil against the two foodborne pathogens *Escherichia coli* and *Salmonella enterica* in apple juices was studied by Friedman et al. [25] in 2004. The capacity of the essential oil to preserve food was demonstrated by their discovery that it was effective against the two bacteria developing in the apple juice. Additionally, *Pseudomonas fluorescens* and *Aeromonas hydrophila*, two bacteria that can ruin fish, were discovered to be resistant to the antibacterial effects of *Citrus aurantium* essential oils. The antibacterial activity was determined by measuring the MIC using the disc diffusion method at two temperatures, including the bacteria's optimal temperature (30 °C or 37 °C) and the refrigerated temperature (4 °C). As a food preservative, the essential oil was demonstrated to be effective at both temperatures against the intended bacteria [26].

2.1.2 Antifungal Properties of *Citrus aurantium* and Its Food Preservative Potential

Many fungi, including *Aspergillus niger*, *Aspergillus flavus*, *Aspergillus nidulans*, *Aspergillus fumigants, Fusarium graminearum*, *Fusarium oxysporum*, *Fusarium culmorum,* and *Alternaria alternate,* are known to be responsible for food spoilage [24]. Therefore, an agent that can inhibit the growth of these fungi will do well as a food preservative. Fortunately, the essential oils of *Citrus aurantium* have shown profound antifungal activity by inhibiting the growth zone of the above-mentioned fungi. The antifungal capacity of *Citrus aurantium* has been attributed to the limonene and (E)-nerolidol content of the essential oils [27]. Therefore, the essential oils of *Citrus aurantium,* particularly the limonene and (E)-nerolidol isolates, would serve well as a food preservative agent.

2.1.3 Antioxidant Properties of *Citrus aurantium* and Its Food Preservative Potential

Antioxidants are known to limit food spoilage by preventing the autoxidation of pigments, flavors, lipids, and vitamins [28]. The essential oils of *Citrus aurantium* have shown profound antioxidant properties that can be harnessed as a food preservation [24, 29, 30]. Furthermore, the peel and fruit juice of *Citrus aurantium* have been demonstrated to possess antioxidant properties [31, 32]. The antioxidant properties of *Citrus aurantium* were confirmed via the 2, 2-diphenyl-1-picrylhydrazyl (DPPH) assay, β-carotene bleaching test, total antioxidant capacity (TAC) assay, and thiobarbituric acid reactive substance (TBARS) assay [24, 29–32]. Indeed, the collective antioxidant properties of *Citrus aurantium* suggest its use as a food preservative agent in the food industry.

2.2 The Role of *Citrus aurantium* in Aromatherapy

The non-pharmacological therapeutic use of essential oils derived from aromatic plants is known as aromatherapy. Aromatherapy has been used to treat a variety of ailments, including body aches, nausea, vomiting, anxiety, melancholy, stress, and insomnia [33]. Due to their high volatility, citrus essential oils can alter the body's hemodynamic parameters or blood flow via controlling circulation via the autonomic nervous system [33]. In particular, the inhalation of *Citrus aurantium* has been observed to possess sedative, hypnotic, and anti-anxiety properties [34, 35]. Furthermore, *Citrus aurantium* essential oils have been used to improve sleep quality in type 2 diabetes mellitus patients, pregnant women, and postmenopausal women [5–7]. Given its ability to improve sleep quality in the above-stated conditions, its effects on several other conditions that distort sleep quality in humans can be assessed.

2.3 The Role of *Citrus aurantium* in Pulp and Paper Production

One major drawback in pulp and paper production is the insufficiency of raw materials [12]. Two main attributes distinguish the *Citrus aurantium* tree as a suitable raw material for pulp and paper production viz its fast growth and chemical composition. In particular, the high cellulose and holocellulose content and low lignin content, which is essential for paper production, stands *Citrus aurantium* out in paper production [12]. Therefore, the *Citrus aurantium* tree stands as a good raw material for pulp and paper production.

2.4 The Role of *Citrus aurantium* in Food Processing and Cosmetics Industries

Citrus aurantium has mostly been investigated as a flavoring additive by the food processing sectors. In addition to alcoholic and non-alcoholic beverages, marmalades, ice cream, sweets and candies, soft drinks, gelatins, and cakes, it has also been used to impart aroma to culinary products [14, 15]. As well as being used to make infusions, *Citrus aurantium* blossoms are also utilized to make perfumes and cosmetics thanks to the essential oil they yield [13]. Additionally, *Citrus aurantium* fruit peel is used to make marmalade, and the fruit's juice is added to salads to provide a tart flavor [36]. Finally, *Citrus aurantium* is a component of some South American and Asian recipes, including yoghurt-based side dishes and hot soups with salt and pepper [37].

2.5 The Role of *Citrus aurantium* in Pesticides Production

Various plant parts have demonstrated pesticidal potentials particularly for the control of mosquito larva [38–42]. Furthermore, essential oils and an isolated part of *Citrus aurantium* have been found to have highly potent pesticidal effects. *Citrus aurantium* and limonene, an isolated component, were found to have fumigant activity against *Bemisia tabaci* (silverleaf whiteflies) in an in vitro experiment and to cause insect mortality between 41.00 and 47.67% within 24 h of exposure at 2.5 and 20.0 l/L air concentration [9, 43, 44]. Additionally, it was found that after 24 h of exposure, *Citrus aurantium* essential oil had a larvicidal effect with LC50 and LC90 values of 22.64 and 83.77 mg/L, respectively [8, 45]. The active component was found to be limonene, and *Citrus aurantium* fresh peeled ripe fruit had potent larvicidal effect against the mosquito vector *Anopheles stephensi* at an LC50 value of 31.20 ppm [3]. *Citrus aurantium* essential oils have also been effective in controlling rice weevils and saw-toothed grain insects [10].

3 Phytochemistry and Bioactive Compounds of *Citrus aurantium*

The well-known pharmacological or therapeutic potentials of plants such as antimicrobial, anti-cancer, antiproliferative, hypolipidemic, and cardio-protective effects are associated with the secondary metabolites, or phytochemicals they possess [42, 46–52]. There are a ton of different chemicals in the *C. aurantium* plant. The

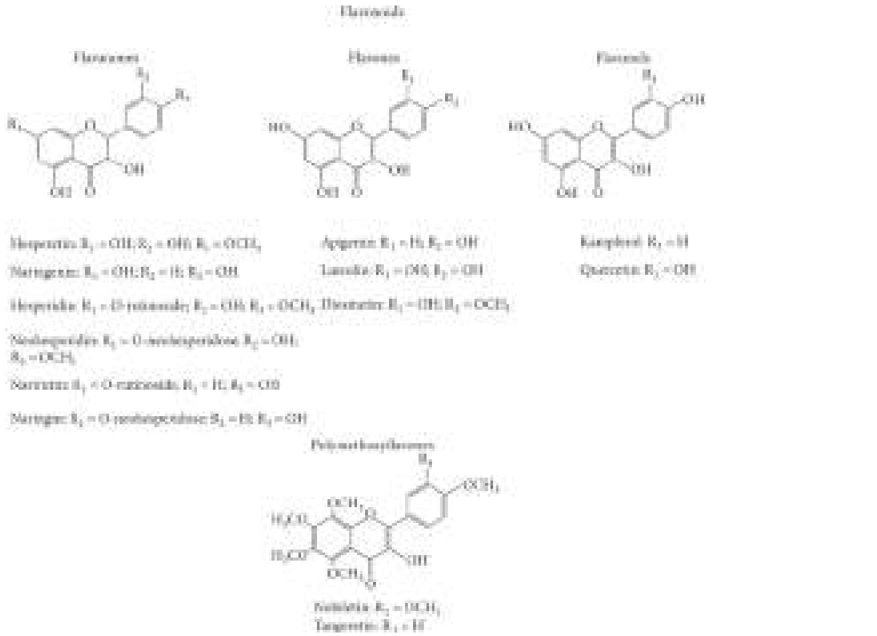
Fig. 1 *Citrus aurantium* with their bioactive compounds. (Source: Maksoud et al. [53] (CC BY 4.0 https://creativecommons.org/licenses/by/4.0/))

primary bioactive substances found in *C. aurantium* are flavonoids, which are further divided into flavanones, flavones, and flavonols. There are also alkaloids such as p-synephrine and other limonoids like limonin and nomilin [53]. Figure 1 depiction of *Citrus aurantium*'s juice, blossoms, seeds, leaves, and peels is followed by a detailed analysis of each part's bioactive chemical concentration. It is important to remember that the location, growing season, and harvesting time all significantly impact the chemical composition or quantity of biomolecules [53].

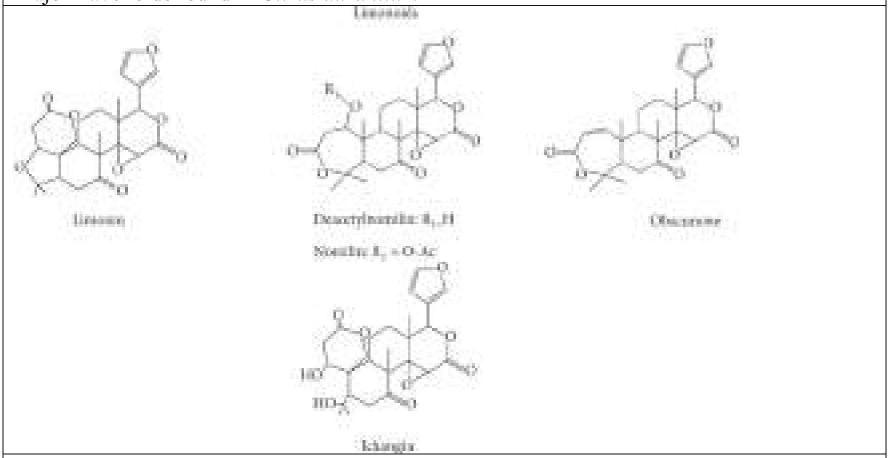
One of the several chemical components of *C. aurantium*, flavonoids from the phenolic family, have been recognized as significant due to their physiological and pharmacological relevance as well as their health benefits. Flavones, flavanones, flavonols, and anthocyanins are the derivatives of flavonoids commonly found in *C. aurantium* (Fig. 2) [3].

Flavanones are the main flavonoids present in *C. aurantium* and the two most common free flavanones found so far are hesperetin (4-methoxy-3,5,7-tri-hydroxyflavanone) and naringenin (4,5,7-trihydroxyflavanone) [3, 54]. The structural similarities between hesperetin and naringenin include the presence of aglycones or glycosides and two hydroxyl groups at positions C-5 and C-7, respectively [3, 55]. Hesperidin and neohesperdin, which are conjugates with rutinose and neohesperdose, respectively, are the two most common glycosides of hesperetin [3]. Naringenin (naringenin-7-neohesperidoside) and narirutin (naringenin-7-rutinoside) are the two most widely used naringenin glycosyl derivatives [3, 55].

The flavones in their aglycon or/and glycosidic form are the second most prevalent type of flavonoids found in *C. aurantium*. Apigenin, luteolin, and diosmetin are the three free flavones that are most frequently detected in *C. aurantium* [3]. O-glycosides and C-glycosides are the two primary forms of flavone glycosides [54].



Major flavonoids found in *Citrus aurantium*



Major limmonoids found in *Citrus aurantium*

Fig. 2 Structure of major chemical composition of *Citrus aurantium*. (Source: Suntar et al. [3] (CC BY 4.0 https://creativecommons.org/licenses/by/4.0/))

Examples of methoxylated flavones found in *C. aurantium* include nobiletin and tangeretin, meaning that all or almost all their hydroxyls have been capped by methylation [3]. *C. aurantium* may also contain very small quantities of flavonols like quercetin and kaempferol, especially in glycosidic form [3].

The second class of secondary metabolites identified in *C. aurantium* are limonoids (Fig. 2). They are oxygenated triterpenoids because of the exceptionally large quantities of oxygen atoms (7–11) present in their structures [3]. There are glucosidic (bitter taste) and aglyconic (water-insoluble but flavorless) types of limonoids. The most significant limonoid is limonin, which has been recognized as a citrus component [3].

P-synephrine is the chemical that is most common among the phenylethylamine alkaloids found in *C. aurantium* [3]. This substance has a hydroxyl group linked to the para position of the benzene ring and structural similarities with ephedrine [3] (Suntar et al. 2018). The peel of unripe fruits is the part of the plant that has the largest amount of p-synephrine [3].

The chemical constitution of *C. aurantium* is a complex one even though the peel is made up of unique and familiar essential oil. The essential oil of citrus fruit especially *C. aurantium* affords them the strong flavor and smell that have been linked to several medicinal activities. Apart from the essential oil found in the peel of *C. aurantium*, there are also synephrine, flavones, alkaloid, *N*-methyltyramine, and carotenoids, and octopamine. The principal bioactive components of *C. aurantium* are synephrine and *N*-methyltyramine with 0.24–1.45% (g/g) and 0.19–0.83% (g/g) respectively [35].

Other phytoconstituents of *C. aurantium* include neohesperidin, noradrenaline, tyramine quinoline, rhoifoli, 5-odesmethyl nobiletin, narcotin, tryptamine, lonicerin, nobiletin, and naringin.

The essential oil constituent of *C. aurantium* peels contains majorly limonene (about 90%) and then vitamin C, coumarins, triterpenes, pectin, carotene, and flavonoids. The flavonoids constituents are notable for their efficacy against fungi, inflammation, and bacterial infections. There is however variation in the essential oil constitution in the leaves, peel, and flowers of *C. aurantium*. For instance, the essential oil of *C. aurantium* leaves is dominated by linalyl acetate which is about 50% composition while linalool (constituting 35% of the essential oil) is the major makeup of the flowers.

Unripe fruits of *C. aurantium* contains cirantin (which is generally believed to be a contraceptive) including: acetic-acid, decylpelargonate, alphahumulene, gamma-elemene, decanal, alphapinene, alpha-terpineol, geranyl acetate, alpha-terpinyl-acetate, alphaylangene, 5-hydroxyauranetin, ascorbic-acid, isolimonic acid, palmitic acid, aurapten, isoscutellarein, beta-elemene, valencene, betaocimene, beta-pinene, cadinene, benzoic acid, decylaldehyde, geranic acid, caryophyllene, delta-cadinene, dodecen-2-al-(1), dipentene, 4-terpineol, hesperidin, caryophyllene, isotetramethylether, neryl acetate, caprinaldehyde, sabinene, llinalylacetate, nonanol, stachydrine, limonin, nootkatone, tannic acid, octyl acetate, terpinen-4-ol, naringenin, p-cymol, cryptoxanthin, furfurol, tetra-o-methyl-scutellarein, acetaldehyde, pectin, farnesol, linalyl acetate, transhexen-2-al-1, pentanol, umbelliferone, phenol, nerolidol, pyrrol, delta-3-carene, lauric aldehyde, citral, rhoifolin, naringin, aurantiamene, d-citronellic acid, alpha-phellandrene, hexanol, d-limonene, geraniol, citronellic acid, gamma-terpinene, isosinensetin, citronellol, (+)-auraptenal, dodecanal, alphapinene, butanol, dl-linalool, nomilin, methanol, cinnamic acid, l-stachydrine, sinensetin, nonylaldehyde, linalool, tangeretin, octanol, malic acid, terpenyl acetate, citronellal, nerol, pcymene, terpinolene, linalool, cis-ocimene, dlterpineol, thymol, pelargonic acid, cis-ocimene, geranyl oxide, trans-ocimene, phellandrene, carvone, nobiletin, undecanal, phenylacetic acid, duodecylaldehyde, mannose, violaxanthin, limonene, pyrrole, delta-3-carene, neral, d-nerolidol, l-linalool, camphene, gamma-terpinene, indole, formic acid, beta-copaene, alpha-ionone, ethanol [35] (Table 1).

As previously discussed, the phytochemical constituents of *C. aurantium* are majorly the glycosides, flavonoids, carbohydrates, alkaloids, tannins, saponins, terpenoids, steroids, and phenolic compounds [75]. Phytochemical analysis of the leaf extract of *C. aurantium* revealed tannins, phytosterols, terpenoids, proteins, saponins, flavonoids, essential oils, and carbohydrates [58]. Several authors have also reported their findings on the phytochemical constituents of various parts of *C. aurantium*. Variations in percentage of bioactive compounds and chemical composition in their reports have been linked to the period of harvest, type of solvent, extraction method, geographical area, or growing season. For instance, the phytochemical screening by Ishaq et al. [75] was in tandem with that of Rauf et al. [76] where phenols, flavonoids, steroids, and tannins were reported but contravenes the study by Gunwantrao et al. [77] where no terpenoids and flavonoids were reported in ethanol peel extract of *C. aurantium*.

In a recent study by Babajide et al. [78] whereby different solvents (n-hexane, ethanol, and aqueous) were used for extraction, the leaf extract of *C. aurantium* indicated the presence of terpenoid, flavonoid, saponin, tannin, and glycoside but for saponin that was not present in the n-hexane extract. This report was also in agreement with the result from the works of Khudhair et al. [79] where saponins, flavonoids, and tannins were reported in the aqueous leaf extract of *C. auratium* and that of Rao et al. [80] reported the absence of absence of phlobatannins and saponins but the confirmation of flavonoids, tannins, and terpenoids in the ethanol and aqueous extracts of *C. aurantium*. Also, confirmation of flavonoids in this study agrees with the report of He et al. [81] and that of Khudhair et al. [79] where the seeds and peel extracts of *C. auratium* revealed abundant cardiac glycosides, alkaloids, steroids, tannins, saponins, and flavonoids.

Because of the versatility of *C. aurantium,* the phytochemical constituents have been linked to its biological and physiological activities against different ailments. For instance, phenolic compounds in *C. aurantium* have been reported from several studies to have inhibitory potential [82]. In the same way, the flavonoid constituent of *C. aurantium* has been linked to antioxidant, antiproliferative, anti-inflammatory, antimicrobial, anti-diabetic, anticarcinogenic, cardio-protective, and anticholesterolemic activities in promoting the general well-being in humans [35, 83]. The main components in *C. aurantium* seeds have been reported to be flavonoid (56%) at maturity and phenolic acids constituent is 22% [68]. The efficacy of *C. aurantium* as an antioxidant agent is associated with their natural capacity in quenching free radicals and disrupting the radical chains [83].

The extraction of bioactive agents from *C. aurantium* is also very important. As previously said, various factors could be responsible for different chemical components and percentage of the active agents obtained. By and large, extraction of

Table 1 Bioactive compounds and extraction methods of *C. aurantium* plant

S/N	Plant part	Compounds	Extraction means	Class of compound	Solvent	Reference
1	Leaves	α-terpineol, limonene, linalyl acetate, and linalool,	–	Essential oil	–	[56, 57]
		O-cymene, eucalyptol, 4-terpineol, β-pinene, α-pinene, D-limonene, α-terpineol, sabinene, β-linalool, and β-myrcene	Hydro-distillation	Essential oil	–	[58]
		Total phenols, flavonoids, flavonols, proantho-cyanidins, polymerized phenols, hydrolyzable tannins, and soluble phenols	Solvent extraction	Phenolic compounds	Methanol: water (80: 20)	[59]
2	Peels	Sesquiterpenes, monoterpenes	–	Essential oil		[60, 61]
		n-nonadecanoic, abinene, cyclohexene, α-pinene, n-decanal, octyl aldehyde, myrcene, dimethoxybenzylidene, α-terpineol, δ-guaiene, limonene, n-octanol, β-fenchyl alcohol, hexasiloxane, β-pinene, γ-gurjunene, cyclotrisiloxane, octyl aldehyde, and thyocynic acid	Microwave steam distillation	Essential oil	Aqueous	[62]
		Synephrine, N-methylthyramine, octopamine, and carotenoids	–	Flavones and alkaloids	–	[60, 61]
		Geranyl acetate, β-pinene, limonene, linalyl acetate, myrcene, geranial, sabinene, linalool, β-caryophyllene, γ-terpinene, and neral	Peel rinds squeezing	Monoterpene hydrocarbons and oxygenated monoterpenes	n-hexane	
		p-coumaric and ferulic acids	Hydro-distillation	Phenolic compounds	Diethyl ether	[31]

		Galacturonic acid	Microwave-assisted extraction	Pectin	Citric acid aqueous solution	[63, 64]
		Limonene, α-terpinene, octanal, camphor, α-terpinyl acetate, and 1, 8-cineole	Hydro-distillation	–	–	[65]
3	Seeds	Hesperidin, naringin, neohesperidin, and narirutin	Soxhlet extraction	Flavonoids	Acetone and petroleum ether	[66, 67]
		Hesperidin, rutin, naringin, naphtorecinol, quercetin, resorcinol, epigallocatechin, kaempherol, apigenin, neohesperidin, and catechin	Solvent extraction	Flavonoids	Methanol	[68]
		p-coumaric acid, syringic acid, rosmarinic acid, vanillic acid, gallic acid, and trans-2-hydroxycinnamic acid	Solvent extraction	Phenolic acids	Methanol	[68]
		Vanillic acid – Hydroxybenzoic acids; and caffeic acid, *p*-coumaric acid, trans-ferulic acid - Hydroxycinnamic acids	Solvent extraction	Phenolic acid	Methanol	[69]
		Free stigmasterol, esterified β-sitosterol, free β-sitosterol, and campesterol	Soxhlet extraction	Phytosterols	n-hexane	[66]
		Arachidic acid, stearic acid, linoleic acid, cerotic acid, oleic acid, and palmitic acid, and palmitoleic acid	Soxhlet extraction	Fatty acid	n-hexane	[66]
4	Flowers	Rutin, pyrogallol, caffeic acid, gallic acid, quercetin, syringic acid, and naringin	Solvent extraction	Phenolic and flavonoids	Ethanol, methanol, and aqueous	[70]

(continued)

Table 1 (continued)

S/N	Plant part	Compounds	Extraction means	Class of compound	Solvent	Reference
		Hotrienol, linalool, nerol, and terpineol	Soxhlet extraction	Oxygenated monoterpenes	Ethanol	[71]
		Tetrapentacontane and dotriacontan	Soxhlet extraction	Aliphatic hydrocarbons	Ethanol	[71]
		β-pinene, β-myrcene limonene, and β-ocimene	Soxhlet extraction	Monoterpene hydrocarbons	Ethanol	[71]
		Neryl acetate and linalyl acetate	Soxhlet extraction	Esters	Ethanol	[71]
		Hexadecanoic acid, linalool, D-glucuronic acid, daphnetin, octadecenoic acid, D-limonene, phthalic acid, and pyrrolidinone	Solvent extraction	–	80% ethanol	[72]
		Farnesol, linalool, geranyl acetate, linalool acetate, and nerolidol	Microwave-assisted hydro-distillation, Microwave-assisted hydro-distillation solvent microwave extraction and solvent-free microwave extraction	–	Aqueous	[73]
5	Fruit	Synephrine, neohesperidin, and naringin	Ultrasound-assisted aqueous two-phase extraction	Flavonoid	Ethanol	[74]
6	Juice	Limonene, α-thujene, and α-phellandrene	Solvent extraction	Monoterpene hydrocarbons	Ether-pentane (1:1)	[31]
		Caryophyllene oxide	Solvent extraction	Oxygenated sesquiterpenes	Ether-pentane (1:1)	[31]
		Ferulic acids and *p*-coumaric	Solvent extraction	Phenolic acids	Ether-pentane (1:1)	[31]
		Quercetin and rutin	Solvent extraction	Flavonoid	Ether-pentane (1:1)	[31]

bioactive compounds from different parts of *C. aurantium* or their wastes has also been of benefits to the agro-industrial activity while the metabolites serve beneficial roles in various sectors like pharmaceutical, food, cosmetic, beverage, and other industries [53]. Since no single extraction method is regarded as the principal reference point for extraction of active compounds, the following are the established and unconventional extraction means on *C. aurantium* plant parts: Soxhlet extraction, microwave-assisted extraction, supercritical fluid extraction, solvent extraction, hydro-distillation, and ultrasound-assisted extraction [53].

Specifically, bioactive molecules in *C. aurantium* parts like leaves, juice, flowers, seeds, and peels are presented in Table 1.

4 Scientific Evidence-Based Therapeutic Potentials of *Citrus aurantium*

C. aurantium like other species of the citrus family has been attributed to have therapeutic properties in deterring several life-threatening diseases [84] which in recent times has been in high demand due to the shift in the use of natural products for therapeutic purposes as opposed to conventional orthodox medicines. From the early days of human civilization, citrus fruits have been used not only as food but also as medicine, according to Gao et al. [1], there are documented use of Citrus in ancient traditional Chinese medicine hence the need to extensively study and utilize its potential in modern medicine, its use has been reported in Japan and Korea [53] and in Africa [85]. Several species in the Citrus genus like *C. aurantifolia, C. limon, C. reticulata,* and *C. paradisi* are favored for use as medicinal constituents are of their chemical constituents, pleasant aromas, and availability. *C. aurantium* unlike other species of Citrus has just been recognized for its therapeutic properties, the chemical constituents of *C. aurantium* are known to have several benefits to human health, and more benefits have been discovered in recent times.

The ethnomedical uses of *C. aurantium* and CACH are of interest to the scientific community. It promotes the regular, in-depth exploration of the various pharmacological routes that control CACH. Currently known extracts and isolated compounds exhibit a range of pharmacological actions, including anti-inflammatory, antioxidant, anticancer, hypolipidemic, and organ protection. The distinct pharmacological action is explained in this section and then summarized in the following list (Fig. 3).

4.1 Antibacterial Activity

Extracts and essential oils from citrus fruits are known to be natural antimicrobials particularly bacteria used in the food industry as well as in cosmetic industry [86]. Several authors have reported the antibacterial properties of *C. aurantium* extracts against certain species of bacteria which are mostly biostatic for the first 2 days and then biocidal after 7–8 days. Aladekoyi et al. [86] reported a 0.25 mm maximum zone of inhibition against *Pseudomonas aeruginosa* but showed no zone of inhibition against

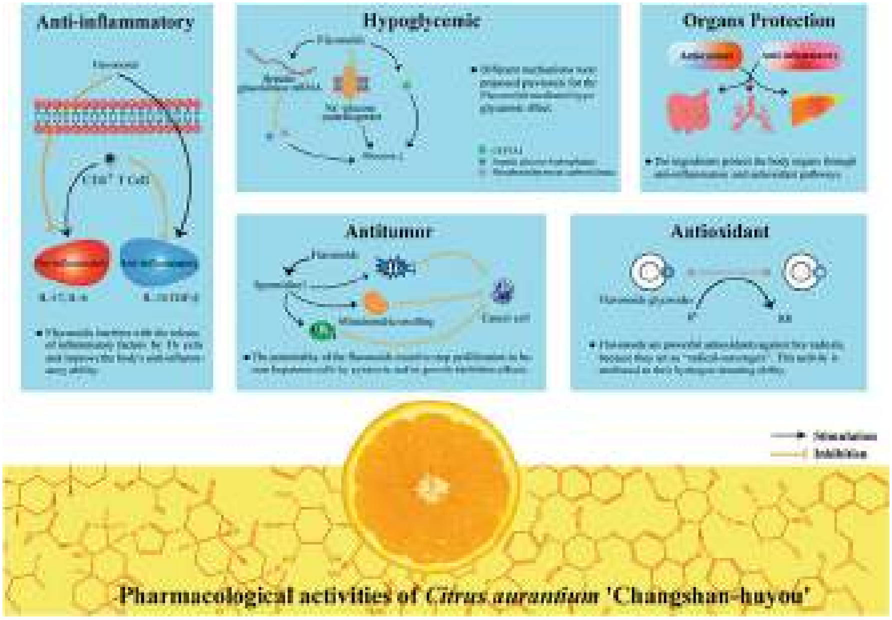


Fig. 3 Distinct therapeutic potentials of *Citrus aurantium* and CACH. (Source: Gao et al. [1] (CC BY 4.0 https://creativecommons.org/licenses/by/4.0/))

Escherichia coli and *Klebsiella* from an antimicrobial susceptibility assay done using oil extracted from the seeds of *C. aurantium*. Essential oils extracted from *C. aurantium* showed a high inhibition zone from 8 to 18 mm when used as an antibiofilm inhibitor against *Bacillus subtilis* and *Stenotrophomonas maltaphilia* [87]. In a study done by Benzaid et al. [88], it was observed that essential oil from the flowers of *C. aurantium* showed a reduction in the growth of *Streptococcus mutans* and the expression of its virulent genes. According to Okla et al. [89], when essential oils from the bark, leaves, and branches of *C. aurantium* were tested against *Erwinia amylovora, Dikeya solani,* and *Agrobacterium tumefaciens*, the notable diameters of inhibition zone came with the increase in the measure of the oil used from the initial 10–25 μl. Karabiyikli et al. [90] studied the effect of *C. aurantium* juice on *Listeria monocytogenes* and *Salmonella typhimurium* and it showed these microbes were able to survive for 2 days in pH neutralized juice and after a seven-day incubation period no growth was found suggesting that the low pH, length of incubation and temperature are enabling factors for the antimicrobial property of *C. aurantium*. Gopal [91] reported significant inhibition zones against Gram-positive *Bacillus subtilis* and *Staphylococcus* and Gram-negative *Klebsiella pneumonia* and *Escherichia coli* using extracts from *C. aurantium* leaves. A study done in Algeria by Haraoui et al. [92] on the antibacterial profile of Citrus plants showed that *C. aurantium* showed a higher antimicrobial activity comparable to *C. limon*; from their observations *C. aurantium* contains certain phytochemicals and the sensitivity and the tolerance level of some bacteria made them more prone to being denatured by these chemicals.

Organisms tested against *C. aurantium* juice that showed high antibacterial activity in their study include Gram positive: *Staphylococcus aureus, Bacillus subtilis, Staphylococcus epidermidis, Micrococcus luteus*, and *Enterococcus faecalis* and Gram negative: *Klebsiella, Pseudomonas aeruginosa,* and *Escherichia coli.* According to Teneva et al. [93] from their study on the antimicrobial activity of essential oils, the antimicrobial activity of the oils is based on their chemical composition; bioactive alcohols like linalool and carveol enhance the antimicrobial activities. From their investigations, Gram-negative bacteria were more resistant to the essential oils because of their outer cell membrane which obstructs the dispersion of the oils to the cytoplasm of the cell through the cell membrane showing zones of inhibition of 9–10 mm while the Gram-positive bacteria were more sensitive to the essential oils with a measured zone of inhibition of 12.5 mm.

4.2 Antifungal Properties

Citrus oils have been harnessed for use in dermatological formulations for topical applications on the skin and scalp to prevent and cure diseases due to their perceived antifungal properties. Several research have shown the efficacy of *C. aurantifolia* but not much has been done for *C. aurantium* over time; citrus oil was studied by Clavaud et al. [94] against the fungal species Malassezia furfur in vitro, it was observed that the oil had a biostatic effect on the organism with zone of exhibition of 2.6 mm, since most *Citrus* spp. has similar phytochemicals though in varying proportions it is assumed that the oil from *C. aurantium* may also have antifungal properties. Kačániová et al. [87] however reported using *C. aurantium* oil in an antibiofilm study resulting in high zones of inhibition against *Penicilium crustosum, P. expansum,* and *P. citrinum.* Ellouze et al. [95] observed a moderate effect of *C. aurantium* leaves essential oil on *Saccharomyces cerevisiae* and *Mucor ramannianus* with the 9.2 and 5 mg/ml minimum inhibition concentration respectively when compared to its antibacterial properties.

4.3 Antioxidant Properties

Antioxidants are stable molecules that donate electrons to free radicals and neutralize them and reducing their capacity to damage cells in the body [96]. According to Maksoud et al. [53], *C. aurantium* is a great source of antioxidants, it has a high level of flavonoid among other phytochemicals [1] whose hydrogen donating ability makes it a strong antioxidant against free radicals. The antioxidant properties of *C. aurantium* may vary according to the plant part used and the antioxidant levels may be dependent on the extraction method. From research by Shi et al. [97], they observed that the dry and immature fruits of *C. aurantium* contain a high flavonoid content of 76.22% and 4 types of flavonoids were observed namely: hesperidin, naringin, narirutin, and neohesperidin. Bendaha et al. [98] worked on the antioxidant activity of essential oil from the peels of *C. aurantium* fruit grown in Eastern

Morocco using 2,2-Diphenyl-1-picrylhydrazyl (DPPH) assay and observed that the essential oil did not show any substantial antioxidant activities as the DPPH capacity ranged from 7–15% displaying a weak DPPH scavenging capability. The essential oil was found to consist of myrcene and more than 90% limonene; myrcene and limonene are not considered to play a major in determining the scavenging activity. Teneva et al. [93] studied the antioxidant activity of *C. aurantium* peel oils using DPPH assay and results show that limonene, β-myrcene, and α-pinene were the main components resulting in 85.22%, 4.3%, and 1.29% respectively. When compared to other antioxidants *C. aurantium* L essential oil showed a higher antioxidant activity at 88.1% in the DPPH assay and this can be explained by the different composition of *C. aurantium* species.

4.4 Anticancer/Cytotoxic Activity

Treatment for cancers through conventional medications can be effective but with many side effects which may lead to damage to other body cells as well as the targeted cancer cells. To this end, new research aimed at using diet for cancer treatment has been on the rise in recent times; by regulating the metabolic niche of the cancer cells and using antitumor (specific active food molecules) that can target the metabolic pathways of the cancer cells [1]. This is aimed at disrupting the increase and spread of the cancer cells after identifying the pathogenesis of the tumor cells. Bian et al. [99] have reported the use of the Chinese herbal drug Weifuchen tablet which contains *Fructus aurantii*; the dried unripe fruits of *C. aurantium* in combination with Red Gingeng (Renshen) and Isoden amethystoides for relief from precancerous lesions in people with gastric cancer through the regulation of intestinal microbial balance. This implies that *C. aurantium* is a good herbal alternative for cancer treatment.

Yao et al. [100] experimented on the use of the active ingredients contained in *C. aurantium* in the treatment of lung cancer, the study which was validated by both molecular and cellular experiments proved that nobiletin in *C. aurantium* inhibits the development of non-small cell lung cancer (NSCLC) as a key ingredient through targets and other related pathways. Though further clinical studies are needed, it was established that nobiletin inhibited the growth of NSCLC and the expression of certain cancer cells was affected.

According to Gao et al. [1], neohesperidin a type of flavonoid has a negative growth impact on human hepatoma cells and *Citrus aurantium* “Changshan-huyou” (CACH) also has anti-tumor activity by intervening with the cell cycle of the cancer cells and impeding their growth.

Shen et al. [101] evaluated the extracts of crude polysaccharides of *C. aurantium* L. var. Amara Engls (CAVAPs) and showed that they enhance the immune system. When RAW264.7 cells were exposed to CAVAPs, the CAVAPS stimulated the production of tumor necrosis factor-α (TNF-α) and interleukin-1β. Further evaluation showed that CAVAPs may possibly activate macrophage signaling pathways through mitogen-activated protein kinase (MAPK) and nuclear factor kappa B (NF-kB).

Research carried out by Han et al. [102] on the apoptotic activities of the peels of *C. aurantium* (CME) on U937 human leukemia cells showed that CME inhibited the growth of U937 cells. CME-induced apoptosis was dose-dependent and triggered by curbing Bcl-2 and IAPs related to the activation of Akt through an intrinsic pathway. Akt is an upstream signal that controls Bcl-2 and IAPs and regulates cell multiplication and programmed death and the study confirmed that CME suppressed Akt phosphorylation, and this is associated with apoptosis.

4.5 Effects on Cardiovascular Activity

Most research on *C. aurantium* are said to be beneficial but some research have looked at the possible toxicity of it. Like other Citrus species, *C. aurantium* contains the alkaloid p-synephrine; known for its anti-obesity properties and treatment of digestive problems because of its fat-decreasing properties in traditional medicines, however according to Arbo et al. [103] it has been flagged as a toxic alkaloid potential risk to the human cardiovascular system by the National Collegiate Athletic Association (NCAA) hence the need for in-depth research. Arbo et al. [103] observed that *C. aurantium* extracts and p-synephrine showed no significant effect on the cardiovascular systems of mice and suggested that the toxicity may arise from the combination of p-synephrine with other stimulants such as caffeine, salicin, and amphetamines. This is in tandem with the observations of other authors including [104] who reported that the association of high doses of p-synephrine together with caffeine and ephedrine caused death due to cardiac complications such as hypertension, strokes, arrhythmias, and heart attack. Hansen et al. [45] also observed that the use of p-synephrine either in its pure state or in *C. aurantium* extract on test animals showed no significant increase in blood pressure and heart rate even at doses much higher than that used in humans. Test animals treated with 95% p-synephrine showed lower increases in blood pressure and heart rate when compared with animals treated with *C. aurantium* extract which suggests that other components of the extracts can be attributed for these changes, there were also significant increases in the cardiovascular parameters when caffeine given together with p-synephrine implicating the addition of the caffeine for the increased heart rate and blood pressure. The reports of the negative effects of p-synephrine have not been confirmed in other studies.

Stohs [105] and Penzak [106] in their independent studies both observed that on its own p-synephrine has low toxicity but when in combination with other components like caffeine, salicin and ephedrine in products can induce cardiovascular effects. Kang et al. [107] (2016) studied the effects of neroli oil on the cardiovascular system as a vasodilator in mice (neroli oil extracted from the flowers of *C. aurantium* var. amara which is rich in limonene and linalool is used to manage anxiety and stress and also to treat high blood pressure and other cardiovascular symptoms) was studied due to the claims that it causes hypotension and bradycardia in adolescents and observed that in the aorta rings of the mice that are already precontracted with prostaglandin f2α, vasodilation was induced, but for rings precontracted with nitric oxide synthase inhibitor the relaxation effects decreased. This induced muscle relaxation was partially reversed by 1H-[1, 2, 4] oxadiazolo [4,3-a] quinoxalin-1-1, a soluble guanylyl cyclase inhibitor. Also, neroli oil suppressed the effect of extracellular Ca^{2+}-dependent contraction in a dose-dependent manner. Ultimately the research indicated that the relaxation effects of neroli affect both the vascular smooth muscle and the vascular endothelium, but no negative effect was observed.

4.6 Anti-obesity Activity

C. aurantium is considered a sport enhancer and swift weight loss ingredient and as an alternative therapeutic dietary supplement as it contains compounds such as flavonoids and polyphenols both of which can reduce the accumulation of lipids [108]. Stohs et al. [109] in their review of human clinical trials involving the use of *C. aurantium* and its alkaloid p-synephrine observed that from over 20 studies carried out with over 50% of the 360 test subjects were obese. P-synephrine was given in combination with caffeine, guarana, green tea, gingko, ginseng, and yerba mate making it difficult to ascertain if weight loss was as a result of p-synephrine or the other components; be that as it may, it was deduced that *C. aurantium* is the most likely causative agent for weight loss because of its effects on metabolic processes including increase in metabolic rate, appetite suppression, and lipolysis. More so 50% of the subjects were caffeine consumers before the commencement of the studies but were still obese. *C. aurantium* was found to show increase in the metabolic rate and energy expenditure of the subjects and when consumed for up to 12 weeks may result in eventual weight loss, hence it can be consumed as dietary supplement at defined doses.

According to research by Li et al. [110], blossoms of *C. aurantium L* var. amara Engl have been reported to inhibit lipid deposits, to better understand the constituents from the blossom extracts responsible for the bioactivity and the mechanism for the lipid breakdown were studied. Using chloroform to harvest the extracts, 14 compounds were identified and nobiletin, trigonelline hydrochloride, and 7-demethysuberosin were more. The chloroform extracts significantly decreased lipid accumulation and improved the plasma biochemical profile in high-fat diet-fed mice by enhancing the gut microbiota. At higher doses, the chloroform extracts increased the abundance of Lachnospiraceae while at lower doses increased the abundance of Erysipelotrichaceae and reduced the ratio of Firmicutes to Bacteroidetes.

Park et al. [108] investigated the anti-obesity mechanism of *C. aurantium* by administering it in high-fat diet-induced obese mice for 8 weeks. Eighty percent ethanol extracts of immature dried *C. aurantium* fruits were analyzed and found to contain naringin 20.1% and neohesperidin 14.4% and other phytochemicals in lesser quantities. The effect of *C. aurantium* on promoting adipogenesis and thermogenesis is dependent on activated protein kinase alpha (AMPKα) activation (the energy sensor and regulator of metabolic homeostasis). From their observations, there were elevated phosphorylation levels of AMPKα and ACC in the 3T3-L1 adipocytes and culture brown adipocytes due to the *C. aurantium*. The anti-obese effects of *C. aurantium* were dependent on the activation of the AMPKα pathway due to the elimination of the anti-adipogenic and pro-thermogenic effects of the *C. aurantium* impeding the AMPKα pathway through the introduction of cell culture pre-treatment. The result was a significant decrease of body weight, serum cholesterol, and adipose tissue weight in the mice suggesting that *C. aurantium* has the potential of being used as an anti-obese agent as it inhibits white adipogenesis and induces brown adipocyte thermogenesis through AMPKα activation.

Verpeut et al. [111] worked on the anti-obesity effects of *C. aurantium* (6% synephrine) and *Rhodiola rosea* L. (3% rosavins and 1% salidroside) in combination on diet-induced obese rats. From their investigations, *C. aurantium* or *R. rosea* alone did not suppress appetite in normal weight rats, but *C. auratium* and *R. rosea* in combination when administered registered a 10.5% appetite suppression by elevating the hypothalamic norepinephrine and the frontal cortex dopamine. It can however be deduced that the combination of *C. aurantium* and *R. rosea* are effective in altering appetite suppression and obesity because of the interaction between the two extracts and their bioactive components.

4.7 Hypoglycemic Activity

According to Gao et al. [1], the use of *C. aurantium* in treating and preventing diabetes and hyperlipidemia is receiving widespread because its mechanism has not been fully understood yet. The assumption is that the hypoglycemic effects of *C. aurantium* are achieved by improving lipid metabolism, regulation of the intestinal flora, and the increased consumption of glucose. Jia et al. [112] researched on the use of neohesperidin isolated from *C. aurantium* in in vivo treatment of hypoglycemia and hypolipidemia in KK-ay diabetic mice. The results showed a significant reduction in fasting glucose, serum glucose, total cholesterol, triglycerides, leptin level, and glycosylated serum protein in the treatment mice when compared to the control at the end of the experiment. At the end of the six-week period Neohesperidin treatment showed a high potential as a hypoglycemic agent in vivo.

From the investigations of Zhang et al. [113], naringin and neohesperidin isolated from *C. aurantium* var. *changshanensis* showed significant effects on glucose consumption by Hep G2 cells through the process of increased phosphorylation of AMPK expression in the cells. Neohesperidin and naringin showed enhancement of glucose consumption in a manner like metformin. Campbell et al. [114] investigated the lipid-lowering effects of an infusion of *Rawolfia vomitoria* leaves and *C. aurantium* fruits, a Nigerian anti-diabetic infusion consumed locally and attributed to decrease blood glucose level when consumed with the patient expected to stick to a low carbohydrate and fat diet with strict no alcohol consumption. The study was conducted on 6–11-week-old lean mice including inbred and outbred mice. There was no significance in their appearance but there was a transitory effect on the motor activity of the mice on the first day of administration. After 6 weeks, there was an observed significant weight loss when the dosage corresponding to ten-fold human daily dosage was administered to genetic diabetic mice which had been maintained on a carbohydrate-deficient diet. The control group did not have a significantly different diet from the treatment group; however, the treatment group had a significantly higher serum triglyceride level proposing that the lipid might be from internal stores. The treatment mice also showed a reduced stearoyl CoA desaturase activity.

4.8 Anti-inflammatory Activity

C. aurantium like other citrus fruits is highly rich in flavonoids and other bioactive natural products. Examples of citrus flavonoids include narirutin, neohesperidine, and naringin and they display strong biological activities like anti-inflammation and are likely therapeutic choices for treatment of metabolic dysregulation [115]. Wang et al. [116], from their research on the fruits of *C. paradisi*; a hybrid of *C. aurantium* as an anti-inflammatory for the inhibition of asthma allergens observed that the dry immature of *C. aurantium* contains narirutin, neohesperidin, and naringin which are the primary active components, and they exhibit powerful inflammatory activities and inhibit airway remodeling in asthma allergens by regulating the Smad2/3 and MAPKs signaling pathways. The overexpression of inflammatory factors is related to the symptoms of diseases like chronic gastritis and peptic ulcer and *C. aurantium* can be used in treating them [1].

Jiang et al. [115] worked on the anti-inflammatory effects of total flavonoids extracted from *C. aurantium* hybrid QZK (TFCH) on non-alcoholic fatty liver disease in vivo and it was observed that TFCH suppresses both systemic and intrahepatic inflammation and attenuates hepatic lesions in obesity-induced rats. The effects were marked by the suppression of NF-Kb and MAPKs. From this result, it can be deduced that the extracted flavonoid component in TFCH is the main activator of anti-non-alcoholic fatty liver disease property of the extract and is a new source for the development of therapeutic medicine for the treatment of anti-non-alcoholic fatty liver disease.

Pallavi et al. [84], from their investigations on the anti-inflammatory and anti-nociceptive properties of the peels of Citrus fruits common in India, observed that all citrus fruits contain naturally high levels of phytochemicals like flavonoids, steroids, terpenoids, glycosides, carotenoids, phenolic compounds, and alkaloids; and showed that *C. medica* fruit peel extract exhibited significant anti-inflammatory and analgesic property in a dose-dependent manner which is similar to the standard indomethacin drug, this was followed by *C. reticulate*, *C. aurantium*, *C. aurantifolia,* and lastly *C. grandis* and generally the anti-inflammatory property was more evident in the later phases of time interval.

CACH can be used to treat both chronic gastritis and peptic ulcers, which are predominantly brought on by the overexpression of inflammatory factors. Jiang et al. [115] on the in vivo anti-inflammatory properties of TFCH showed that TFCH at doses of 50 to 100 mg kg^{-1} significantly lowered TNF-α, IL-6, IL-1, and other inflammatory

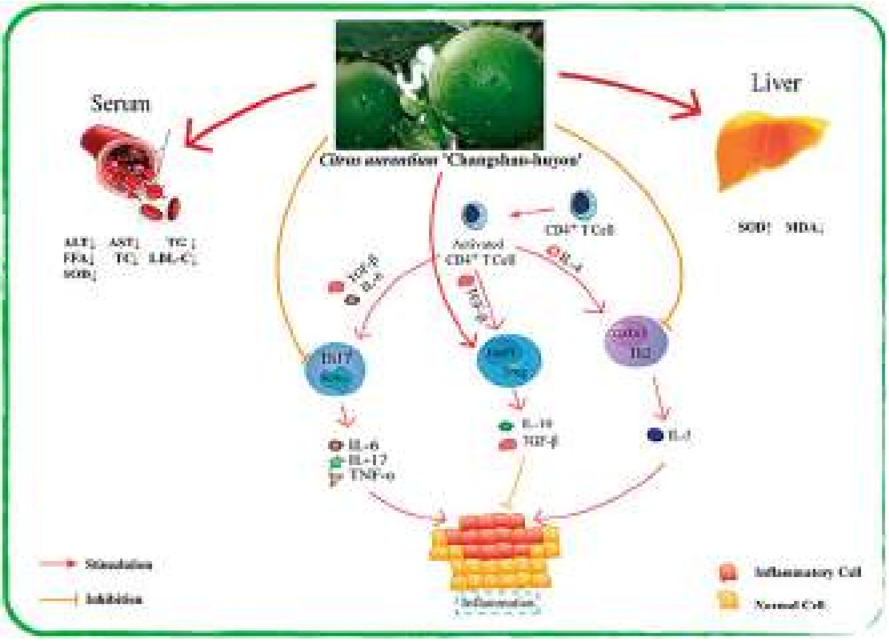


Fig. 4 Anti-inflammatory mechanism of CACH. (Source: Gao et al. [1] (CC BY 4.0 https://creativecommons.org/licenses/by/4.0/))

factors. According to the study, NF-B/MAPK is the main mechanism via which TFCH manifests its anti-inflammatory characteristics [1]. Figure 4 illustrates the most recent research on the anti-inflammatory mechanism of CACH based on the findings [1].

4.9 Hepato-Protective Activity

Jiang et al. [115] stated the protective properties of flavonoids found in *C. aurantium* extracts using the disease model of non-alcohol fatty liver disease (NAFLD) as discussed earlier. According to Yu et al. [117] the pathological features of NAFLD are simple fatty liver, hepatic sclerosis, and steatohepatitis and in some cases liver cancer, the microenvironment of the liver has been implicated for the development and rapid deterioration of NAFLD when it is inflamed and the oxidative stress. To prevent and treat liver diseases using *C. aurantium* is achieved through antioxidant, anti-inflammatory, and regulation of the intestinal microflora. Shi et al. [97] examined and evaluated the effects of TFCH on non-alcoholic steatohepatitis (NASH), a liver disorder involving hepatocellular injury over the course of NAFLD, using dried immature fruits of the *C. aurantium* cultivar known as TFCH. Using the antioxidant response element route of nuclear factor erythroid 2-related factor 2 (Nrf2), the NASH model was assessed in vivo and in vitro for 26 weeks and 8 weeks, respectively. The CCK-8 experiment demonstrated the anti-apoptotic effects of TFCH on

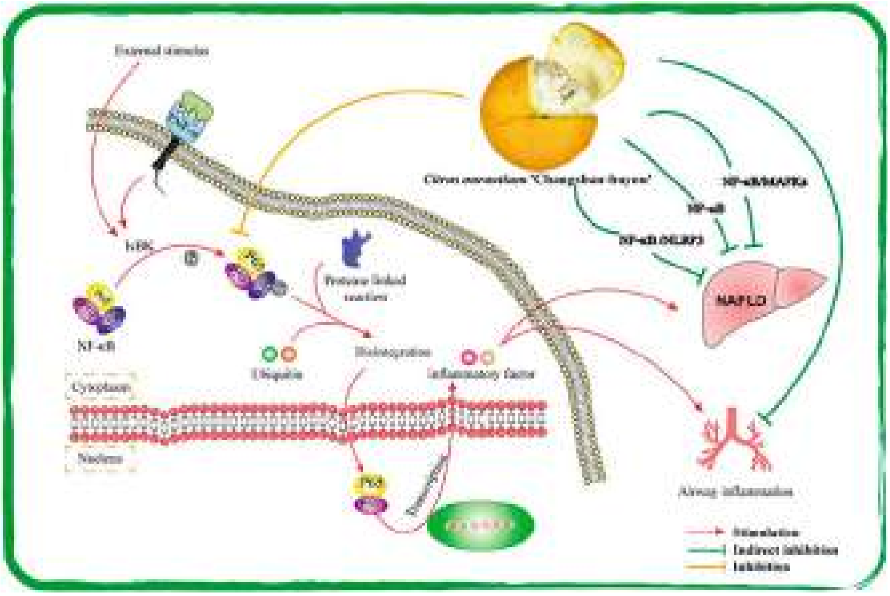


Fig. 5 Mechanism of CACH protection on organs through NF-κB pathway. (Source: Gao et al. [1] (CC BY 4.0 https://creativecommons.org/licenses/by/4.0/))

LX-2 cells grown with FFA. In vivo and in vitro measurements of lipid parameters and oxidative stress markers revealed that TFCH lowers oxidative damage in the NASH model and has a dose-dependent effect on antioxidant capacity. Analysis of Nrf2 and target gene protein expression in mouse liver and human LX-2 cells revealed that TFCH up-regulates these genes' protein expression and significantly alters the Nrf2-ARE signaling pathway. This suggests that TFCH has hepatoprotective and anti-oxidative effects on NASH and that TFCH therapy may eventually offer a unique therapeutic alternative for the treatment of NASH.

For instance, extensive research has been done on NF-B, a crucial protein in the pathways regulating liver inflammation in NAFLD. Jiang et al. [115] (2019) used male SD rats to develop an HFD-induced NAFLD model and isolated and produced TFCH. By limiting the phosphorylation of IB to stop the breakdown of NF-B and so preventing the creation and release of inflammatory components, TFCH reduces the inflammatory milieu in the liver. The results were also corroborated by the research's most successful drug group (polyene phosphatidylcholine capsule group, 196.3 mg/kg) (Fig. 5) [1].

4.10 Intestinal Adjustment

Drug usage should be regulated through treatments in the intestinal milieu since unfavorable drug reactions frequently occur in the intestines [1]. According to Chen et al. [118], consideration must be given to NSAID side effects that cause small intestine injury to take NSAIDs safely. Pure total flavonoids (PTFC) from the *C. aurantium* var. have been used as a form of treatment for digestive disorders, while the benefits and mechanisms of action are yet unknown. To show the protective advantages of PTFC, the study investigated the potential role of autophagy in the process of diclofenac and MSAIDs caused small intestinal injury. In rats with NSAID-induced small intestinal injury and IEC-6 cells treated with diclofenac, the expression levels of the tight junction (TJ) proteins ZO-1, claudin-1, and occludin light chain 3 (LC3)-II, as well as autophagy-related 5 (Atg5), were shown to be lower than in the control group. In the intestinal disease-induced rat group, p-PI3K and p-Akt expression levels increased in comparison to the control group but decreased in the PTFC group. Pretreatment with PTFC preserved the intestinal mucosal barriers and decreased NSAID-induced small intestine damage by increasing autophagy via the PI3K/Akt signaling pathway.

He et al. [119] analyzed the gut microbiota composition in NAFLD mice by pure total flavonoids from citrus (PTFC) treated high-fat diet using PTFC and the outcome of this was that PTFC adjusted the gut microbial structure, the abundance of the intestinal flora and improved the proliferation of Bacteriodeae, Akkermansia, and Christensenellaceae indicating that the advancement of NAFLD is linked to the gut microbiota and PTFC can modulate this target to therapeutic role.

4.11 The Role of *Citrus aurantium* in Skin Treatment

Linalool, linalyl acetate, and terpineol have been shown to have anti-aging properties when present in isolated form [120]. The antioxidant properties of many skin care products are linked to their anti-aging benefits, and these ingredients have been shown to have alluring antioxidant properties. Additionally, exposure to sunshine makes aging-related enzymes like collagenase and elastase work harder. These enzymes particularly degrade the collagen and elastin in the skin, resulting in wrinkles and skin sagging that are associated with getting older [121, 122]. However, *Citrus aurantium* essential oils have been demonstrated to have anti-aging properties since they inhibit the activity of the enzymes collagenase and elastase and are rich in linalool, linalyl acetate, and terpineol [16].

5 Safety Considerations from the Use of *Citrus aurantium* and Its Products

5.1 *Citrus aurantium* and Cardiovascular Toxicity

As asserted earlier, *Citrus aurantium* came to the limelight following the drawback encountered with the use of *Ephedra sinica* as a weight loss and athletic performance supplement, and its subsequent ban by the Food and Drug Administration (FDA) in April 2004 [123–127]. The suggestion of *Citrus aurantium* as a replacement for

Ephedra sinica is due to the structural similarities its main phytochemical, p-synephrine, shares with the Ephedra isolate, ephedrine. Ephedrine was observed to be associated with myocardial infarction, hypertension, and stroke [124, 128]. p-synephrine is a protoalkaloid extracted from the immature fruit or peel of *Citrus aurantium* with widespread use as a weight loss and athletic performance supplement [18]. It is an isomer of synephrine with a hydroxyl group in the para position, while other isomers have a hydroxyl group in the meta and ortho positions even though it is widely accepted that p-synephrine is safer than ephedrine, its toxicity index and range of safety have been controversial [126, 129].

5.2 The Controversies

In 2008, a reported case of a young man who presented with a myocardial infarction after the use of a synephrine-containing substance raised a huge medical concern [19]. The authors of the report hypothesized that myocardial infarction might have been due to coronary spasm followed by thrombosis which developed due to synephrine tablets abuse [19]. Aside the toxicity adduced to synephrine, the authors called the attention of readers to another study that asserted that *Citrus aurantium* contains a number of flavonoids, including 6′,7′-dihydroxybergamottin and furocoumarin, which selectively block the intestinal cytochrome P450 enzyme, CYP3A4, in bioavailability study [124, 130]. Consequently, increasing the serum level of the drugs metabolized by CYP3A4, and resulting in undesirable pharmacological interactions [124]. Indeed, several articles have further discussed the cardiovascular toxicity of *Citrus aurantium* and its product [106, 130–144]. More intriguingly, in 2010, *Citrus aurantium* was listed in the Customer's report as being possibly unsafe and a dietary supplement “to avoid” [145]. However, in many cases of reported adverse effects, the individuals were either exercising or had recently exercised [126, 135, 139, 142]. On the other hand, Stohs et al. [18] dissented from many toxicity reports for a variety of reasons, such as the use of complex mixtures of ingredients in products containing bitter orange extract, the disseminating of false information by governmental organizations, the misunderstandings surrounding the isomeric forms of synephrine and their various pharmacological properties, and the misrepresentations of p-synephrine, particularly its mistaken identity with other substances.

5.3 The Hansen et al. and Dr. Stohs' Debate on *Citrus aurantium* Safety: A Major Controversy

In view of the avalanche of articles attributing major hazardous potentials to *Citrus aurantium* and its product, Hansen et al. [17] intended to investigate the cardiovascular toxicity potentials of *Citrus aurantium* in exercised rats. Sprague-Dawley rats were administered a daily dose of 10 or 50 mg of synephrine/kg body weight (in both extract and pure form) for a total of 28 days, either alone or in combination with

25 mg/kg body weight of caffeine. Additionally, the rats were made to run for 30 min a day, 3 days a week, on a treadmill. Additionally, a group of rats was made to run on a treadmill for the same period while receiving a caffeine dose of 25 mg/kg body weight [17]. The body temperatures were carefully monitored, QT intervals, blood pressures, and heart rates of the rats. It was demonstrated that both dosages of synephrine considerably increased systolic and diastolic blood pressure for up to 8 h following treatment, even though the effects of caffeine seemed to have the largest impact on heart rate and body temperature. The scientists concluded that exercise, coffee, and synephrine all significantly affect blood pressure and heart rate. However, in a Letter to the Editor of the Journal of Cardiovascular Toxicology, Stohs [22] questioned the procedures and outcomes of the trial carried out by Hansen et al. [17]. The authors, according to Stohs [22], did not sufficiently discuss the clinical importance and relevance of *Citrus aurantium* and p-synephrine for humans, did not review the most recent scientific research on these two substances, and did not accurately relate their doses in rats to typically used doses in humans. Regarding the dose, Stohs [22] argued that the 50 mg/kg body weight of synephrine administered to the rats – equivalent to 8.1 mg/kg body weight in a man and roughly 649 mg for a typical 80 kg person – was simply too high for an adverse effect to go undetected [17, 22]. Stohs [22] noted that the authors' dose was 16 times the typical human dose as a result. Even yet, Stohs [22] noted that the trial's increases in systolic and diastolic blood pressure, which were 9.7% and 9.0%, respectively, were insufficient to cause concern. Hansen et al. [17] argued in defense of their methodology by saying that they used 10 mg/kg body weight of synephrine rather than 50 mg/kg body weight. According to the authors, the maximum dose of synephrine that can be given to a person is 1.62 mg/kg body weight. The authors reported that at this dose, systolic and diastolic blood pressure in the rats increased statistically significantly and remained for up to 8 h, indicating a pathogenic situation. Furthermore, Stohs [22] identified certain works that appear to defend synephrine, although Hansen et al. [17] claimed that they were not aware of such. *Citrus aurantium* caused some degree of cardiovascular damage in strenuous-exercising rats, according to the scientists' findings [17].

5.4 Other Toxicity Considerations

In an experiment by Deshmukh et al. [21] to assess the acute and 14-day oral toxicity of *Citrus aurantium* in rats, the LD50 of the plant extract standardized to 50% p-synephrine exceeded 5000 mg/kg body weight in female rats. Additionally, it was reported that no toxicity occurred when the plant extract was given orally to female rats for 4 days straight at a dose of 2000 mg/kg body weight [21]. A 14-day repeated dose of 50% p-synephrine-containing *Citrus aurantium* extract at the doses of 250, 500, 1000, and 2000 mg/kg/day had no appreciable impact on the body weights, relative and absolute organ weights, clinical chemistry and hematological parameters, gross pathological findings, or food intake of the experimental animals [21]. Two male rats with gastrointestinal impaction had perished after taking 2000 mg/kg, it was determined during necropsy. Additionally, during the second week of treatment with 1000 mg/kg and 2000 mg/kg/day for 15 and 45 min, respectively, rats were reported to exhibit brief indications of repeated head-burrowing in the bedding material (hypoactivity) [21]. Despite the short-term and transient clinical symptoms that were noticed at the dose, it was decided that the maximum tolerated dose (MTD) of *Citrus aurantium* extract standardized to 50% p-synephrine was 1000 mg/kg/day [21]. 500 mg/kg body weight was the no-observed-effect-level (NOEL) for *Citrus aurantium* extract in the study. The researchers contended that *Citrus aurantium* extract had a sufficiently large margin of safety to satisfy any serious concerns. *Citrus aurantium* was tested for safety in rats over a sub-chronic 90-day period by Deshmukh et al. [20], who discovered that doses of 100, 300, and 1000 mg/kg/day of the 50% p-synephrine-containing extract exhibited no unfavorable clinical effects. Furthermore, the extract's NOEL was 300 mg/kg, and its no-observed-adverse-effect-level (NOAEL) was 1000 mg/kg, according to the authors' assertions [20]. The authors highlighted that *Citrus aurantium* extract is quite safe as a result.

Despite having useful characteristics, *Citrus aurantium* essential oil has been demonstrated to contain a small amount of toxicity [146]. *Citrus aurantium* essential oil was neither an irritant nor a sensitizer to 25 volunteers when tested at 10%, but it did produce sensitivity in 1.5% of all dermatitis patients when tested at 2% [147–149]. *Citrus aurantium* essential oil has also been linked to an increase in light sensitivity, especially in those with fair skin. When an essential oil is eaten, photosensitivity occurs more frequently than when it is administered topically to the skin and exposed to bright light [35]. As a result, it is advisable to avoid staying out in the sun for too long after applying the oil.

6 Research Needs for *Citrus aurantium* as a Therapeutics

Different authors have suggested that in order to fully understand the therapeutic potentials of plants it will be important to identify, isolate, and purify the bioactive compounds that make the plant confer such therapeutic potentials [48, 49, 51, 52, 150–156].

To gain a comprehensive understanding of the therapeutic advantages of *Citrus aurantium*, it is imperative to bridge the information gap among the available studies. Various aspects of toxicity, including acute, sub-acute, and chronic toxicities, alongside the pharmacological safety profiles of the plant, necessitate thorough investigation. Before embarking on clinical trials, it is essential to research the short-term and long-term toxic effects of the plant's compounds on animals. To facilitate this endeavor, a comprehensive grasp of the precise chemical composition of these molecules is crucial. This understanding will facilitate a more profound insight into the metabolic processes that give rise to these beneficial compounds. Moreover, an enhanced comprehension of metabolic engineering can significantly improve the synthesis and accumulation of these compounds. *Citrus aurantium* holds significant importance in sustainable biodiversity preservation and utilization for both current society and future generations. The bitter orange plant not only offers novel opportunities to develop innovative medications to address a spectrum of debilitating ailments [157], but it also remains an extensively untapped source for isolating and characterizing specific chemical components.

Studies investigating the bioactivity potential of *Citrus aurantium* have revealed a diverse array of biological effects attributed to its flowers, fruits, essential oils, and phytoconstituents. These effects encompass antimicrobial, antioxidant, cytotoxic, anxiolytic, antidiabetic, anti-obesity, and anti-inflammatory properties [3]. Nevertheless, further comprehensive and prolonged research is imperative to ensure the plant's safety, particularly with regard to understanding interactions between the plant's components and medications, as well as determining appropriate dosage levels. P-synephrine, characterized as a relatively inefficient adrenergic agonist, is not expected to have notable effects on the cardiovascular system. It's worth noting that despite the absence of concrete evidence pointing to potential toxicity or cardiovascular risks, the NCAA has prohibited its use. In contrast, the World Anti-Doping Agency (WADA), an organization collaborating with Olympic athletes, permits its usage [3].

Multiple scientific studies have underscored the distinct advantages that *Citrus aurantium* holds over other citrus species. Comparative analyses have consistently highlighted bigarade leaves, juice, or flowers as possessing the highest levels of total phenolic content (TPC), total flavonoid content (TFC), and antiradical activity. A significant category of bioactive compounds, including phenols, flavonoids, alkaloids, vitamin C, and essential oils, furnishes compelling evidence. These secondary metabolites confer a range of beneficial biological and therapeutic properties, such as antioxidants, antibacterial agents, anti-cancer agents, anti-diabetic agents, anti-obesity agents, and anxiolytics. Moreover, their utility extends to the cosmetic and functional food sectors, where these compounds find diverse applications. While most of these bioactive components are distributed throughout the entirety of the *Citrus aurantium* plant, distinct identity markers for phytochemical content and proportions have been identified in various plant sections. Remarkably, certain byproducts resulting from the processing of *Citrus aurantium*, such as the peels left behind after juice extraction, emerge as significant reservoirs of valuable biomolecules with promising potential applications. This underscores the importance of repurposing residual biomass, traditionally viewed as waste, within the context of the evolving bioeconomy. Literature also outlines the extraction techniques employed to obtain the bioactive constituents from *Citrus aurantium*. However, to comprehensively grasp their mechanisms of action and ascertain safety via interactions with biological models, advanced clinical research mandates the incorporation of further analytical methods to fractionate and purify these active components [53].

The identification criteria for *Citrus aurantium* present ample opportunities for enhancement. Variations in origins and cultivation practices can lead to significant variations in components due to the distinctive nature of Chinese herbals and diverse cultivation methods. The concurrent presence of multiple chemical elements and compounds often results in varying outcomes in pharmacological activity tests. Because of these conflicting factors, certain extracts utilized in research fail to provide comprehensive details about their extraction procedures or processing conditions. Moreover, some studies have not successfully established accurate identification of *Citrus aurantium* or precise descriptions of extraction sites. These issues contribute to a decline in the overall quality of studies, thereby diminishing their utility as credible references.

Citrus aurantium has yielded over a hundred types of chemicals, many of which exhibit bioactivity. While investigations verifying pharmacological activity have mainly concentrated on specific constituents, such as flavonoids from *Citrus aurantium*'s peel, limited attention has been given to other bioactive components like limonins, organic acids, and additional phenols. Consequently, more reliable, and substantiated data are essential to support the concept of *Citrus aurantium's* "multi-target, comprehensive intervention." Further clarification and comprehensive exploration of the bioactive constituents and pharmacological processes of *Citrus aurantium* are imperative. Notably, researchers frequently prioritize outcomes and overlook the comprehensive mechanisms through which the active components in *Citrus aurantium* contribute to disease prevention and treatment.

The future of medical research on *Citrus aurantium* will revolve around developing activity screening models rooted in pharmacological actions, unearthing novel bioactive ingredients, and dissecting its pharmacological mechanisms. Key areas of focus include establishing robust sources and diversity identification criteria for *Citrus aurantium*, incorporating innovative activity screening models into pharmacological evaluations, and leveraging advanced technologies such as digital light processing and 3D printing to align research with clinical application criteria. Furthermore, despite traditional uses and preliminary evidence from animal models suggesting *Citrus aurantium*'s potential in treating lung and bronchial issues, the current findings lack solid scientific substantiation. The importance of accumulating toxicological and pharmacokinetic data cannot be overstated, as it lays the groundwork for clinical research and the development of *Citrus aurantium*-related products [1].

Citrus peel, leaves, and blossoms harbor essential oils of remarkable value for various reasons. Citrus fruits represent one of the most cultivated fruit crops globally, resulting in the ready availability of these oils as byproducts of citrus processing. The prized attributes of citrus essential oils, which encompass therapeutic benefits related to scent, along with fungicidal, virucidal, and bactericidal activities, render them valuable constituents in cosmetics, fragrances, and potentially eco-friendly substitutes for chemical-based antimicrobials in diverse applications. These characteristics have the potential to drive "green consumerism," promoting the utilization and cultivation of plant products. However, achieving standardization in commercial citrus essential oil composition and conducting further studies to unravel the mechanisms of action of active chemicals remain vital endeavors [158]. There remains ample room for refining their utilization and application.

7 Conclusion

The *Citrus aurantium*, often called bitter or sour orange, is necessary for various utilizations, both in the pharmacological and non-pharmacological world. People have, throughout history, looked to nature as a source of health and well-being for themselves. *Citrus aurantium* is a plant that stands out among traditionally beneficial in increasing physical fitness. Numerous phytochemical components that may be found in *Citrus aurantium* have been shown to have close relationships to the fruit's wide variety of beneficial biological effects, such as the treatment of cancer, gastrointestinal issues, inability to sleep, headaches, and spasticity. *Citrus aurantium* is a very intriguing compound because, in addition to its significant role in the field of pharmacology, it also has a wide variety of applications outside of the pharmaceutical industry. It is used as a pesticide, a source of raw materials for the pulp and paper business, and aromatic properties for the food preservation and processing and cosmetics industries, and it has the potential to work as an anti-aging agent. *Citrus aurantium* and the products derived from it have an incredible number of benefits. Yet, the usage of *Citrus aurantium* has also been connected to a variety of adverse side effects. Even though specific investigations have mostly exonerated the plant of any safety infractions, other research has questioned the matter.

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Medicinal Spice, *Aframomum melegueta*: An Overview of the Phytochemical Constituents, Nutritional Characteristics, and Ethnomedicinal Values for Sustainability

7

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Abstract

This chapter explores the botanical characteristics, traditional uses, culinary applications, and medicinal properties of the medicinal spice, *Afromomum melegueta* [Roscoe.] K. Schum., Zingiberaceae (Guinea pepper or Grain of Paradise). The herbaceous plant is part of the cultural identity and heritage of West Africa where it is considered to have originated and has been used and cultivated for centuries and plays vital roles in traditional medicine, rituals, and ceremonies of various Indigenous cultures in the region and other parts of the world. *A. melegueta* is also cultivated in parts of Central and South America, the Caribbean, Southeast Asia, and some regions of Oceania. It was one of the valuable commodities exchanged along with other spices, such as black pepper and cardamom, through the trans-Saharan trade routes. Some of the culinary uses and applications of *A. melegueta* include spice, flavor enhancer, distilled spirit, baking, seasoning blend, craft brewing, and condiments. The herbal medicinal value, distinctive aroma, and fragrance of the plant are attributed to phytochemical constituents such as essential oils (like myristicin, limonene, beta-caryophyllene, and pinene), fatty acids, terpenoids, tannins, glycosides, steroids, carotenoids, phenols, alkaloids, flavonoids, vitamins, and minerals. Some ethnomedicinal uses of the plant seed (oil and extract) include digestive aid, anti-inflammatory, antioxidant properties, weight reduction and management, pain relief, respiratory health, antimicrobial, stomach upset, aphrodisiac, and fever reduction. However, further pharmacognostic and clinical research is required to verify the ethnomedical benefits of the plants. Business opportunities abound in the cultivation, processing and packaging, export and import, culinary ventures, health and wellness products, and fair trade initiative aspects of the plant. As global interest in spice continues to grow, it is essential to prioritize sustainable cultivation practices, fair trade, and phytomedical exploration of vital medicinal spices like *A. melegueta* to ensure its long-term availability, conservation, sustainable utilization, and benefit to local communities.

Keywords

Guinea pepper · Medicinal spice · Herbal products · Herbal medicine · Grain of paradise · Phytochemicals

1 Introduction

Aframomum melegueta [Roscoe.] K. Schum, commonly known as Guinea pepper or Grains of Paradise, is a spice and medicinal plant native to West Africa. It belongs to the Zingiberaceae family, which also includes ginger and turmeric. Guinea pepper has a long history of culinary and medicinal use and plays a significant role in the culture and cuisine of the regions where it grows. *A. melegueta* is believed to have originated in the rainforests of West Africa, specifically in the region that includes present-day Nigeria, Ghana, and Ivory Coast [50, 78]. This perennial plant thrives in the hot and humid tropical climate of the African rainforests, where it can be found growing both in the wild and under cultivation. The name "Grains of Paradise" is often used to refer to the seeds of *A. melegueta*. The term "paradise" in this context alludes to the spice's exotic and prized nature, which was highly valued in medieval Europe [87, 95]. *A. melegueta* is a robust herbaceous plant characterized by its large leaves, tall stalks, and unique flowering structures. The plant's leaves are large, oblong, and typically dark green. They grow alternately along the stems. The stems of Guinea pepper can reach a height of up to 2 meters (6.6 feet). They are sturdy and bear the plant's leaves and flowers. Guinea pepper produces beautiful, red-veined flowers that grow in spike-like clusters. The flowers are tubular and have a distinctive appearance [8, 9]. The fruit of *A. melegueta* is a capsule containing numerous seeds. These seeds, known as Guinea grains or Grains of Paradise, are the primary part of the plant used for culinary and medicinal purposes. Although the seeds are the most famous part of the plant, the roots and rhizomes also contain aromatic compounds and have been used in traditional medicine [6, 29, 30].

In recent years, *A. melegueta* has gained recognition in the global culinary scene. Chefs and food enthusiasts have rediscovered its unique flavor and have started to incorporate it into their dishes [5]. This revival has led to increased demand for Grains of Paradise in international markets. The spice trade, which historically included Grains of Paradise among its prized commodities, has seen a resurgence in interest. Small-scale farmers in West Africa are cultivating Guinea pepper to meet the growing demand from gourmet chefs and artisanal food producers worldwide. Grains of Paradise has a rich history in culinary traditions, both in Africa and beyond. They are known for their complex flavor profile, which includes notes of citrus, pepper, and warmth. Some of the culinary uses and applications of *A. melegueta* include the following:

Spice: Grains of Paradise is primarily used as a spice to season a wide range of dishes. They are often used in small quantities due to their potent flavor. In West African cuisine, they are a key ingredient in spice blends like pepper soup spice and suya spice [69].

Flavor Enhancer: Guinea pepper can enhance the flavor of meats, stews, soups, and sauces. It is particularly well-suited for game meats and hearty, savory dishes [71].

Distilled Spirits: Grains of Paradise is a botanical ingredient in some gin recipes, contributing to the complexity of the spirit's flavor [67–69].

Baking: In European cuisines, Grains of Paradise was historically used in baking, especially in gingerbread and spiced bread recipes [82].

Seasoning Blends: They are sometimes added to spice blends, such as garam masala, to provide a unique and exotic flavor [69, 71].

Craft Brewing: In recent years, craft brewers have experimented with adding Grains of Paradise to beer recipes, creating unique and flavorful brews [90].

Condiments: Guinea pepper can be used to make flavored oils and vinegar, adding a spicy kick to dressings and marinades [90].

In African cuisines, Guinea pepper is more than a spice; it is part of the cultural identity and heritage of the region. However, in addition to its culinary applications, *A. melegueta* has a rich history of traditional uses in various cultures. It has been valued for its medicinal properties and has played a role in religious and cultural practices. In traditional African medicine, Guinea pepper has been used to treat a wide range of ailments. It is believed to have digestive, anti-inflammatory, antimicrobial, and analgesic properties [11, 20, 51]. Grains of Paradise has a reputation as an aphrodisiac in some cultures. They have been used to enhance libido and sexual performance [7]. In some African cultures, Guinea pepper has been used as a protective charm or talisman to ward off evil spirits and negative energy. The seeds have been used in religious rituals and ceremonies in West African traditional religions. Guinea pepper has been associated with prosperity, abundance, and good fortune in some African cultures. While traditional medicine has long recognized the medicinal properties of *A. melegueta*, modern research has begun to shed light on the potential health benefits of this spice [14]. Some areas of interest include the following:

Anti-inflammatory: Grains of Paradise contains compounds with anti-inflammatory properties, which may have applications in reducing inflammation and related conditions.
Antioxidant: The spice is a source of antioxidants, which can help protect cells from oxidative stress and damage.
Digestive Health: In traditional medicine, Guinea pepper has been used to aid digestion. Some research suggests that it may have gastroprotective effects.
Metabolic Health: Preliminary studies have explored the potential impact of Guinea pepper on metabolic health, including its role in regulating blood sugar levels and promoting weight management.
Analgesic: Some traditional uses of Guinea pepper include pain relief. Research into its analgesic properties is ongoing.

This chapter aims to discuss the phytomedicinal value of *A. melegueta*. It presents the origin, distribution, and botany of *A. melegueta* wherein it is shown that the widespread distribution of the plant has led to its incorporation into a variety of culinary and medicinal traditions. This includes culinary and traditional medicine uses. The next sections address the origin, distribution, botanical, nutritional, and phytochemical properties as well as the food uses of *A. melegueta*. Other parts of the chapter are on the ethnobotanical, pharmacognosy, and clinical works, herbal products, toxicity, and commercial values of *A. melegueta*. It is believed to have various medicinal properties, including digestive, anti-inflammatory, and antimicrobial effects [20, 66]. *A. melegueta* is a crop with great potential for sustainable cultivation. It thrives in the agroforestry systems of West Africa, where it can be grown alongside other crops, such as cocoa and oil palms. This intercropping approach contributes to biodiversity and provides additional income to farmers. Sustainability practices, including organic cultivation and fair-trade principles, are being promoted to ensure that the production of Guinea pepper benefits both farmers and the environment.

2 Origin and Distribution of *Aframomum melegueta*

Aframomum melegueta, commonly known as Guinea pepper or Grains of Paradise, is native to West Africa. It is believed to have originated in the tropical rainforests of this region. Specifically, the plant's natural habitat spans several West African countries, including Nigeria, Ghana, Ivory Coast, Liberia, Sierra Leone, and Togo [36]. However, the exact place of origin within this region is difficult to pinpoint precisely, as Guinea pepper has been used and cultivated in West Africa for centuries, making it an integral part of the region's culinary and cultural heritage. Cultivation efforts can be found in parts of Central and South America, the Caribbean, Southeast Asia, and some regions of Oceania [12]. In these areas, it is grown to meet the global demand for this unique spice. Guinea pepper's name (pride of paradise) alludes to the historical trade routes that brought this spice to Europe during the medieval period. It was one of the valuable commodities exchanged along with other spices, such as black pepper and cardamom through trans-Saharan trade routes [33, 34]. These trade networks connected West Africa to Europe, the Middle East, and Asia, introducing the world to the unique flavors of Guinea pepper.

In its native West Africa, Guinea pepper is not only a spice but also a symbol of cultural identity and heritage. It has played a significant role in traditional medicine, rituals, and ceremonies of various Indigenous cultures in the region. Today, Guinea pepper remains an essential part of West African cuisines and is also gaining recognition in the global culinary scene for its distinct and complex flavor profile. While Guinea pepper is now cultivated in other parts of the world due to its increasing popularity, its true origin and cultural significance are deeply rooted in the vibrant and diverse landscapes of West Africa. Guinea pepper is primarily native to West Africa [17, 93]. This region is where the plant has its natural habitat and has been traditionally grown and used for centuries. However, due to its popularity in culinary and medicinal applications, it has garnered interest worldwide, leading to its cultivation in various regions beyond its native habitat. Today, *A. melegueta* is cultivated in other tropical regions around the world. These regions typically share a similar climate to its native West Africa, characterized by a hot and humid tropical climate. Countries in Central and South America, the Caribbean, Southeast Asia, and parts of Oceania have seen cultivation efforts to meet the growing global demand for Guinea pepper. Guinea pepper has gained recognition as a unique and flavorful spice in international cuisines. As a result, it is traded and used by chefs, culinary enthusiasts, and the food industry in many parts of the world. While it may not be grown commercially in all these regions, it is imported and incorporated into dishes and spice blends. In some countries with suitable climates, *A. melegueta* is cultivated as a horticultural or ornamental plant. These efforts may not necessarily be for commercial spice production but for the plant's esthetic and aromatic qualities. In regions with nontropical climates, such as parts of Europe and North America, Guinea pepper is occasionally grown in controlled environments like greenhouses. This allows for cultivation in climates that would otherwise be unsuitable. The spice is valued for its unique flavor profile, reminiscent of citrus and pepper, and has made it a sought-after ingredient in global cuisine.

3 Botanical Characteristics of *Aframomum melegueta*

The ecology of *A. melegueta* is closely tied to its native habitat in West Africa. This perennial plant thrives in tropical regions with well-drained, fertile soils and adequate rainfall. It typically grows in the understory of rainforests, benefiting from the partial shade provided by taller trees. Physiologically, *A. melegueta* is characterized by its rhizomatous growth, with shoots emerging from underground rhizomes. It produces lance-shaped leaves and distinctive red or orange fruits that contain aromatic seeds. The plant's physiology enables it to adapt to its native rainforest environment, where it plays a role in the ecosystem and serves as a valuable spice in human cultures.

According to Amponsah et al. [9], botanically, *A. melegueta* is a herbaceous perennial plant. Herbaceous plants lack woody stems and persist for several growing seasons. The plant typically reaches a height of 1–2 meters (3.3–6.6 feet) when fully mature. The leaves of Guinea pepper are large, oblong, and dark green. They are arranged alternately along the stems. The leaves contribute to the plant's lush appearance. The stems of *A. melegueta* are sturdy and upright. They support the plant's leaves, flowers, and fruits. The stems are green and become more lignified (woody) as the plant matures. Guinea pepper produces striking, tubular flowers. The flowers are often described as being red-veined and have a distinctive appearance. They grow in clusters on spike-like inflorescences. The flowering period can vary depending on growing conditions. The fruit of *A. melegueta* is a capsule that contains numerous small seeds. These seeds are the part of the plant most commonly used as a spice and are known as Grains of Paradise. The capsules are typically brown or reddish-brown when mature. While the seeds are the most famous part of the plant, the roots and rhizomes of Guinea pepper also contain aromatic compounds. In traditional medicine and some culinary applications, the roots and rhizomes are used along with the seeds. Guinea pepper has a clumping growth habit, meaning that it forms clusters of stems arising from the same underground rhizome. This growth habit helps the plant spread in its natural habitat. The seeds of *A. melegueta* are known for their aromatic and complex fragrance, which includes notes of citrus, pepper, and warmth [3]. This unique fragrance is a key factor in their culinary and medicinal uses. The plant is native to the tropical rainforests of West Africa, where it thrives in the hot and humid climate. It is often found growing in the understory of the rainforest, receiving filtered sunlight with other tropical crops [57, 64, 65]. These morphological characteristics, including the unique appearance of the flowers and the aromatic seeds, make *A. melegueta* easily recognizable. The seeds, in particular, are highly prized for their flavor and fragrance, which have contributed to their historical and contemporary culinary and medicinal uses.

In addition to its ecology within its native range, Guinea pepper is cultivated in tropical regions worldwide. Cultivation practices aim to replicate the plant's native growing conditions to ensure healthy growth and the production of high-quality seeds. The plant has a distinctive appearance, and aromatic seeds have made it a sought-after spice in culinary traditions worldwide. It is known for its complex flavor profile, which combines elements of citrus, pepper, and warmth, making it a valuable addition to a wide range of dishes and spice blends.

4 Nutritional Value and Uses of *Aframomum melegueta*

Aframomum melegueta is not primarily consumed for its nutritional content but for its unique flavor and aromatic properties [40]. Nonetheless, *A. melegueta* contains many vital nutritional components but is not a significant source of macronutrients like carbohydrates, proteins, or fats. However, it does contain various bioactive compounds and phytochemicals that contribute to its flavor and potential health benefits. Some of the nutritional components and bioactive compounds found in Guinea pepper are as follows:

Essential Oils: Guinea pepper seeds contain essential oils, which are responsible for their distinctive aroma and flavor. These oils include compounds like myristicin, limonene, and pinene, which contribute to its complex fragrance [19].

Phenolic Compounds: Phenolic compounds are antioxidants found in Guinea pepper. These compounds have potential health benefits due to their ability to neutralize harmful free radicals in the body [1, 24].

Alkaloids: Some alkaloids, such as guineensine and aframonine, are present in Guinea pepper. These compounds may have bioactive properties and could contribute to their medicinal uses [18].

Flavonoids: Flavonoids are a group of polyphenolic compounds known for their antioxidant properties. While Guinea pepper contains some flavonoids, they are not a primary source [16].

Vitamins and Minerals: Guinea pepper is not a significant source of vitamins and minerals [53]. However, it may contain trace amounts of vitamins and minerals depending on the soil and growing conditions.

The primary use of Guinea pepper is as a spice. Its seeds are used to flavor a wide range of dishes, including soups, stews, sauces, and meat-based dishes. It adds a complex, citrusy, and peppery flavor with a hint of warmth. Grains of Paradise is often included in spice blends, such as garam masala, curry blends, and various regional spice mixtures. It enhances the overall flavor profile of these blends. In historical European cuisines, Guinea pepper was used in baking, particularly in gingerbread and spiced bread recipes. It added depth and complexity to baked goods. Some gin recipes include Guinea pepper as a botanical ingredient. It contributes to the flavor complexity of the spirit. Guinea pepper can be used to make flavored oils, vinegar, and condiments. These are used to add a spicy kick to dressings, marinades, and dipping sauces.

In West African traditional medicine, Guinea pepper has been used to treat various ailments. It is believed to have digestive, anti-inflammatory, and analgesic properties. Some traditional uses of Guinea pepper include its role in promoting digestive health. It has been used to alleviate digestive discomfort and stimulate appetite. Research into the anti-inflammatory properties of Grains of Paradise is ongoing, and it may have potential applications in reducing inflammation. The phenolic compounds in Guinea pepper give it antioxidant properties, which can help protect cells from oxidative stress and damage. In traditional medicine, Guinea pepper has been used for pain relief. While more research is needed, some studies suggest it may have analgesic effects. In some cultures, Guinea pepper has been considered an aphrodisiac, believed to enhance libido and sexual performance. In certain African cultures, Guinea pepper has been used as a protective charm or talisman to ward off evil spirits and negative energy. The seeds of Guinea pepper have been used in religious rituals and ceremonies in West African traditional religions. The spice is associated with prosperity, abundance, and good fortune in some African cultures and has cultural significance in various rituals and celebrations [31, 58–60]. In recent years, Guinea pepper has experienced a culinary revival. Chefs and food enthusiasts have rediscovered its unique flavor and have started to incorporate it into their dishes. This has led to an increased demand for Grains of Paradise in international markets. The global spice trade has also seen a resurgence of interest in Guinea pepper. Small-scale farmers in West Africa are cultivating Guinea pepper to meet the growing demand from gourmet chefs and artisanal food producers worldwide.

5 Global *Aframomum melegueta* Business, Market, and Supply Chain

Aframomum melegueta, commonly known as Guinea pepper or Grains of Paradise, has gained recognition in the global market due to its unique flavor profile and potential health benefits. To understand the global business, market, and supply chain of *A. melegueta*, there is a need to explore various aspects, including production regions, market trends, supply chain dynamics, and business opportunities.

5.1 Production Regions

West Africa is the primary production region of *A. melegueta*: Guinea pepper is primarily cultivated and harvested in West African countries, including Nigeria, Ghana, Ivory Coast, Liberia, Sierra Leone, and Togo. These countries have the ideal tropical rainforest climate for its growth. However, expansion to other tropical regions is ongoing due to increasing global demand. Hence, cultivation efforts have expanded to other tropical regions worldwide. Central and South American countries, the Caribbean, Southeast Asia, and parts of Oceania have seen cultivation initiatives to meet the growing demand for this unique spice.

5.2 Market Trends

Guinea pepper has experienced a culinary revival in recent years, with chefs and food enthusiasts appreciating its complex flavor. It is increasingly used in gourmet cuisine, craft beverages, and artisanal food products [54–56, 76]. The spice's potential health benefits, including anti-inflammatory and antioxidant properties, align with consumer trends toward healthier eating. This has driven interest in incorporating Guinea pepper into wellness-focused products. Some gin producers include Guinea pepper as a botanical ingredient in their recipes. The craft beverage industry's growth has contributed to increased demand. Guinea pepper is integral to various ethnic and regional cuisines, particularly in West African and North African dishes. The growing popularity of these cuisines has expanded its market. Guinea pepper is a key component of spice blends such as garam masala, curry blends, and regional seasoning mixtures. These blends are popular for their convenience and flavor enhancement.

5.3 Supply Chain Dynamics of *A. melegueta*

Cultivation: The supply chain begins with the cultivation of *A. melegueta*. Small-scale farmers in West Africa often grow it alongside other crops like cocoa and oil palms.

Harvesting: Guinea pepper is harvested when the capsules containing the seeds mature and turn brown or reddish-brown. Harvesting methods can vary but typically involve manually plucking the capsules.

Processing: After harvesting, the seeds are separated from the capsules. Depending on the scale of production, this can be done manually or with the help of machinery.

Packaging: Processed seeds are typically packaged in airtight containers to preserve their flavor and aroma. Packaging may vary based on market destinations, including bulk packaging for wholesale and retail-ready packaging for consumer markets.

Distribution: Distribution channels include local markets, international spice traders, and specialty food distributors. Guinea pepper is often imported by countries worldwide.

Retail and Consumer: Grains of Paradise is available in various forms, including whole seeds and ground spice. Consumers can purchase them in stores or online for use in culinary applications and home cooking.

5.4 Business Opportunities with *A. melegueta*

Business opportunities exist in the following areas concerning *A. melegueta:*

Cultivation: Entrepreneurs and agricultural investors can explore opportunities in cultivating Guinea pepper, especially in regions with suitable climates.

Processing and Packaging: Establishing processing and packaging facilities for Guinea pepper seeds can create value-added products for export and local markets.

Export and Import: International traders can participate in the export-import business of Guinea pepper, connecting producers in West Africa with global markets.

Culinary Ventures: Chefs, culinary entrepreneurs, and food manufacturers can incorporate Guinea pepper into their products to meet the demand for unique and exotic flavors.

Health and Wellness Products: Developing health and wellness products that incorporate Guinea pepper's potential health benefits can tap into consumer interest in functional foods.

Sustainable and Fair-Trade Initiatives: Supporting sustainable and fair-trade practices in the supply chain can create business opportunities focused on ethical and environmentally responsible sourcing.

6 Ethnomedicinal Value of *Aframomum melegueta*

Aframomum melegueta contains a variety of primary and secondary phytochemicals, which contribute to its flavor, aroma, and potential health benefits. Some of the primary and secondary phytochemicals found in *Aframomum melegueta* are the following:

Essential Oils: Guinea pepper seeds are rich in essential oils, which are responsible for their distinct aroma and flavor [49, 80]. Some of the primary components of these essential oils include myristicin, limonene, pinene, and beta-caryophyllene [23, 43, 85]. Some of the properties of these essential oils include the following:

- Myristicin: known for its aromatic and spicy flavor
- Limonene: imparts citrusy notes to the spice
- Pinene: adds pine-like and woody aromas
- Beta-caryophyllene: contributes to the spice's peppery and earthy character

Fatty Acids: While not a primary source, Guinea pepper seeds contain small amounts of fatty acids, including oleic acid and linoleic acid [27, 47].

Phenolic Compounds: Phenolic compounds are antioxidants found in Guinea pepper [3, 45, 75]. These secondary metabolites have potential health benefits due to their ability to neutralize harmful free radicals. Examples of phenolic compounds in Guinea pepper include vanillic acid, caffeic acid, syringic acid, quercetin, and kaempferol. Some of the properties of these compounds include the following:

- Vanillic acid: known for its sweet and vanilla-like flavor
- Caffeic acid: has antioxidant properties and contributes to the spice's flavor
- Syringic acid: an antioxidant with potential health benefits

- Quercetin: a flavonoid with anti-inflammatory properties
- Kaempferol: another flavonoid with antioxidant and anti-inflammatory effects

Alkaloids: Some alkaloids are present in Guinea pepper seeds, and while they are not primary sources, they may have bioactive properties [28]. Examples of alkaloids found in *Aframomum melegueta* include guineensine, malic acid, and aframonine [38, 77, 81].

Flavonoids: Flavonoids are a group of polyphenolic compounds with antioxidant properties [73]. While Guinea pepper contains some flavonoids, they are not the primary source of these compounds.

Terpenoids: Terpenoids are secondary metabolites found in Guinea pepper, contributing to its aromatic properties [6]. Some terpenoids include limonene and pinene.

Sesquiterpene Lactones: While not as common as other compounds, some sesquiterpene lactones have been identified in Guinea pepper [26, 41, 91]. These compounds are known for their diverse biological activities.

Tannins: Tannins are polyphenolic compounds that may be present in Guinea pepper, albeit in relatively small quantities [10, 39]. They can contribute to the astringency of the spice.

Glycosides: Some glycosides may be found in Guinea pepper seeds, though their presence and types can vary ([46]; 2017).

Steroids: Steroids are compounds that may be present in small quantities in Guinea pepper, contributing to its chemical diversity [79].

Carotenoids: While not as prominent as in some other spices, Guinea pepper may contain trace amounts of carotenoids, which are responsible for the coloration of some fruits and vegetables [61, 62, 86].

It is important to note that the exact composition of phytochemicals in Guinea pepper can vary based on factors such as the plant's growing conditions, location, and maturation. These phytochemicals collectively contribute to the unique flavor, aroma, and potential health benefits associated with *A. melegueta*. Additionally, ongoing research continues to uncover the diverse chemical constituents present in this spice, further deepening our understanding of its properties and potential uses. While further research is needed to fully understand the bioactive properties of Guinea pepper, it continues to be a popular and intriguing spice in the culinary world with potential applications in functional foods and traditional medicine.

Some of the ethnomedical uses of the bioactive compound in *A. melegueta* include the following:

- **Digestive Aid**: In traditional African medicine, Guinea pepper has been used to aid digestion. It is believed to stimulate the digestive system, relieve indigestion, and alleviate gastrointestinal discomfort [22, 29].
- **Anti-inflammatory**: Guinea pepper has been traditionally employed as an anti-inflammatory agent. It is used to reduce inflammation and soothe inflammatory conditions such as arthritis and joint pain [52].
- **Antioxidant Properties**: The phenolic compounds found in Guinea pepper are antioxidants. Antioxidants help protect cells from oxidative stress and damage

caused by free radicals. Traditional uses may include harnessing these antioxidant properties for overall health [63, 74].

- **Pain Relief**: In some traditional healing practices, Guinea pepper has been used for relief from pain and rheumatism [13, 92]. It is believed to have analgesic properties and may be used topically or consumed to alleviate pain. Aqueous seed extract of grain of paradise has been shown to possess peripheral analgesic activity through actinociceptive activity of the intraperitoneal dose [88, 92].
- **Respiratory Health**: Guinea pepper has been used in traditional remedies for respiratory issues such as coughs and congestion [21]. It may be used as an ingredient in herbal formulations to help ease respiratory discomfort. The study by El Dine et al. [21] showed that the methanolic extract of the plant has antiadhesive concentration-dependent effects on microorganisms in the lung carcinoma line.
- **Antimicrobial and Antihelminthic**: Some cultures have used Guinea pepper as an antimicrobial and antihelminthic agent [37]. It may be employed to help combat infections, although further research is needed to confirm its efficacy [44, 70].
- **Aphrodisiac**: In certain African cultures, Guinea pepper has been considered an aphrodisiac. It is believed to enhance libido and sexual performance, and it is sometimes included in preparations meant to boost sexual vitality [4, 15, 35].
- **Ritual and Spiritual Uses**: Guinea pepper has cultural and spiritual significance in some African traditions. It is used in rituals, ceremonies, and protective charms to ward off evil spirits and promote positive energy [25, 32, 83].
- **Weight Reduction and Management**: The plant seeds and seed extracts are consumed to reduce and manage weight in many Indigenous communities across the world including diabetic patients [48, 84, 89]. The work of Yoneshiro et al. [96] suggest the plant extract adopts adoptive thermogenesis to reduce body fat in humans.
- **Stomach Upset**: Guinea pepper is sometimes used to alleviate stomach upset, including nausea and vomiting, in traditional medicine [42]. López-Ríos et al. [42] reported that the plant seed is taken by postmenopausal women as a remedy for hot flashes, anxiety, and depressive symptoms.
- **Cardiovascular Protection**: The seed and seed extract of the plant is taken to protect and increase cardiovascular activities. It is also used to treat hypertension in Indigenous communities in the Global South [2].
- **Fever Reduction**: In some African cultures, Guinea pepper has been used to help reduce fever and alleviate symptoms of illnesses associated with fever.
- **Nephroprotection:** The seed and seed extract of the plant is taken to reduce kidney injury and protect the organ from adverse damage [2].

Table 1 presents some of the ethnomedicinal properties of *A. melegueta* as well as the parts used and bioactive compounds responsible for the effects.

Table 1 Ethnomedicinal properties of *A. melegueta* as well as the parts used and bioactive compounds responsible for the effects

Ethnomedical uses	Part used	How it is used	Bioactive compound responsible
Digestive aids	Seeds	Crushed or powdered seeds are often added to food or herbal formulations to aid digestion and relieve indigestion	Myristicin, limonene, pinene, and phenolic compounds
Anti-inflammatory	Seeds, oil	Seeds may be ingested, or oil extracts are applied topically to reduce inflammation and alleviate joint pain	Phenolic compounds, and terpenoids
Antioxidant properties	Seeds, extracts	Guinea pepper seeds or extracts are consumed to provide antioxidant support for overall health	Phenolic compounds, flavonoids, and terpenoids
Pain relief	Seeds, oil	Seeds may be crushed and applied topically as poultices or mixed with carrier oils for massage to alleviate pain	Phenolic compounds, terpenoids, and alkaloids
Respiratory health	Seeds, preparations	Seeds are sometimes included in herbal preparations for respiratory issues, promoting clear airways	Terpenoids and essential oils
Antimicrobial agent	Seeds, extracts	Extracts or powdered seeds are used in traditional remedies to combat infections, both topically and internally	Terpenoids, phenolic compounds, and alkaloids
Aphrodisiac	Seeds, preparations	Ground seeds may be included in aphrodisiac preparations believed to enhance libido and sexual vitality	Alkaloids and terpenoids
Ritual and spiritual uses	Seeds, whole	Whole seeds are used in rituals, ceremonies, and protective charms to ward off negative energy and evil spirits	Cultural and spiritual significance
Stomach upset	Seeds	Seeds are occasionally used to soothe stomach upset, including nausea and vomiting	Terpenoids and phenolic compounds
Fever management and reduction	Seeds	Seeds are sometimes used to reduce fever and alleviate symptoms associated with fever	Terpenoids and alkaloid

7 Pharmacognosy and Clinical Research on *Afromomum melegueta*

7.1 Pharmacognosy of *Aframomum melegueta*

Pharmacognosy is the study of natural products from plants and other natural sources to understand their properties, composition, and potential pharmacological uses. *Aframomum melegueta* has been of interest to pharmacognosists and researchers due to its unique bioactive compounds [73]. Some aspects of the pharmacognosy of *Aframomum melegueta*:

Morphological Identification: Pharmacognosists study the morphological characteristics of the plant, including its leaves, flowers, and seeds, to ensure proper identification. Guinea pepper is known for its distinctive seeds enclosed in capsules.

Chemical Composition: Researchers analyze the chemical composition of Guinea pepper, including its essential oil content and specific bioactive compounds. Gas chromatography-mass spectrometry (GC-MS) is often used to identify the chemical constituents.

Quality Control: Pharmacognosists are involved in establishing quality control standards for Guinea pepper products. This helps ensure that commercial products meet certain quality criteria.

Extraction and Isolation: Techniques for extracting and isolating bioactive compounds from Guinea pepper are explored, allowing for further studies on their properties and potential applications.

Phytochemical Screening: Researchers conduct phytochemical screening to identify the presence of various compounds, such as alkaloids, flavonoids, phenolic compounds, and terpenoids, which contribute to the spice's properties.

7.2 Clinical Research on *Aframomum melegueta*

Clinical research involves studying the effects of Guinea pepper on human health and well-being through controlled experiments and trials [72, 94]. While Guinea pepper has a history of traditional uses, clinical research is ongoing to validate these uses and explore potential therapeutic applications. Some areas of clinical research on *Aframomum melegueta*:

- **Anti-inflammatory Properties**: Studies aim to investigate the anti-inflammatory effects of Guinea pepper and its potential in managing conditions characterized by inflammation, such as arthritis.
- **Antioxidant Effects**: Clinical trials may assess the antioxidant properties of Guinea pepper and its ability to protect cells from oxidative damage. This research is relevant to overall health and may have implications for chronic diseases.
- **Digestive Health**: Guinea pepper's traditional use as a digestive aid is a subject of interest. Clinical studies may explore its effects on gastrointestinal health and its role in improving digestion.
- **Pain Management**: Research may investigate the analgesic properties of Guinea pepper and its potential for pain relief, especially in conditions associated with pain and discomfort.
- **Respiratory Health**: Clinical trials may explore the use of Guinea pepper or its extracts in respiratory health, particularly in managing conditions like coughs and congestion.

- **Antimicrobial Activity**: Guinea pepper's potential antimicrobial properties may be examined in clinical studies to assess its effectiveness against infections.
- **Aphrodisiac Effects**: Research may delve into the traditional use of Guinea pepper as an aphrodisiac, exploring its impact on libido and sexual health.
- **Safety and Dosage**: Clinical studies also aim to establish safe dosages and assess any potential side effects or contraindications associated with the use of Guinea pepper.

8 Some Herbal Products and Formulations of *Afromomum melegueta*

Aframomum melegueta has unique flavor profile that has led to its incorporation into various herbal products and formulations. Some examples of herbal products and formulations that may include *A. melegueta* are the following:

Spice Blends: Guinea pepper is a common ingredient in spice blends and seasoning mixtures. It adds depth and complexity to spice mixes used in various culinary applications. For example, it is a component of traditional spice blends like garam masala and curry powder.

Digestive Bitters: Some herbal digestive bitters and digestive aids may include *A. melegueta* among their ingredients. These formulations are designed to support healthy digestion and alleviate digestive discomfort.

Herbal Teas: Guinea pepper may be included in herbal tea blends, often in combination with other herbs and spices known for their digestive or warming properties. These teas are consumed for their flavor and potential digestive benefits.

Aphrodisiac Formulations: Due to its traditional reputation as an aphrodisiac, Guinea pepper may be incorporated into herbal formulations designed to enhance libido and sexual vitality. These products may be marketed as dietary supplements or natural aphrodisiacs.

Aromatherapy Blends: The essential oil derived from *A. melegueta* may be used in aromatherapy blends for its unique aroma. It can be diffused or added to massage oils for its aromatic properties.

Traditional Medicine: In some traditional healing systems, *A. melegueta* is used in herbal formulations to address various health concerns, such as digestive discomfort, joint pain, or respiratory issues.

Flavoring for Alcoholic Beverages: Guinea pepper is sometimes used to flavor alcoholic beverages, particularly craft beers and gins. It contributes to the complex and aromatic profiles of these drinks.

Nutraceuticals: *A. melegueta* extracts or powders may be used as ingredients in nutraceutical products, such as dietary supplements or functional foods. These products may claim various health benefits attributed to the spice's bioactive compounds.

Culinary Oils and Vinegar: Some specialty oils and vinegar may be infused with Guinea pepper to impart their unique flavor and aroma to dressings, marinades, or cooking oils.

Herbal Liqueurs: In some regions, herbal liqueurs or aperitifs are made using Guinea pepper, among other botanicals. These beverages may have a spicy and aromatic character.

9 Toxic Compounds in *Afromomum melegueta*

Aframomum melegueta is generally considered safe for culinary use when consumed in moderation. However, like many spices and plants, it contains certain compounds that, in excessive amounts, can potentially have adverse effects on health [80]. Some of the compounds in Guinea pepper that may have toxic effects if consumed in large quantities are as follows:

Essential Oils: Guinea pepper is rich in essential oils, which contribute to its aroma and flavor. While these oils are generally safe for consumption, excessive consumption of essential oils, especially when not properly diluted, can lead to gastrointestinal discomfort, skin irritation, or allergic reactions in some individuals.

Alkaloids: Some alkaloids have been identified in Guinea pepper. In moderate amounts, alkaloids may have physiological effects, but excessive consumption can lead to toxicity. Common alkaloids found in plants include caffeine and nicotine.

Phenolic Compounds: While phenolic compounds are known for their antioxidant properties, excessive intake of phenolic-rich foods or supplements may lead to gastrointestinal irritation or other adverse effects in sensitive individuals.

Tannins: Guinea peppers may contain tannins, which are polyphenolic compounds. In large amounts, tannins can have an astringent effect and may interfere with the absorption of certain nutrients.

Spice Allergy: Some individuals may be allergic to specific spices, including Guinea pepper. Allergic reactions can range from mild skin irritation to severe respiratory symptoms. Individuals with spice allergies should avoid consuming Guinea pepper.

It is important to emphasize that the potentially toxic effects of these compounds are typically associated with excessive consumption or misuse. When used in culinary applications as a spice, Guinea pepper is considered safe for most people. However, as with any food or spice, moderation is key, and individual tolerance may vary. Additionally, it is essential to be cautious when using essential oils derived from Guinea pepper or any other plant. Essential oils are highly concentrated and should be used sparingly and appropriately diluted when used for aromatherapy, massage, or other applications.

10 Conclusion

Grains of Paradise is a spice with a rich history, diverse culinary uses, and potential medicinal properties. It is native to the rainforests of West Africa and has made its mark in cuisines around the world, adding complexity and depth to a wide range of dishes. Its unique flavor, reminiscent of citrus and pepper, has earned it a special place in the hearts of chefs and food enthusiasts. Beyond the culinary realm, Guinea pepper has deep cultural and traditional significance, serving as a medicinal herb, an aphrodisiac, and a protective charm in various cultures. Modern research is beginning to uncover its potential health benefits, including anti-inflammatory and antioxidant properties. As the global spice trade experiences a resurgence of interest in this unique spice, it is crucial to prioritize sustainable cultivation practices and fair-trade principles to ensure that the production of *A. melegueta* benefits local communities and ecosystems. In a world where culinary diversity is celebrated and the search for new flavors never ends, *A. melegueta* offers a tantalizing journey into the heart of West African cuisine and culture, reminding us of the rich tapestry of flavors and traditions that make our global culinary heritage so vibrant and exciting.

In conclusion, Guinea pepper is a spice with a rich history, diverse culinary uses, and potential medicinal properties. While not a significant source of macronutrients, it contains various bioactive compounds and phytochemicals that contribute to its flavor and potential health benefits. Its unique flavor profile, reminiscent of citrus and pepper, has earned it a special place in the world of culinary arts and traditional medicine. As global interest in this spice continues to grow, it is essential to prioritize sustainable cultivation practices and fair trade to ensure its long-term availability and benefit to local communities. *A. melegueta* has a growing presence in the global culinary and spice market. Its unique flavor, culinary versatility, and potential health benefits have contributed to its popularity. Opportunities abound for those interested in various aspects of the Guinea pepper supply chain, from cultivation and processing to distribution and culinary ventures. Sustainability and fair trade principles are also essential considerations for businesses in this industry.

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Turkey Berry (*Solanum torvum* Sw. [Solanaceae]): An Overview of the Phytochemical Constituents, Nutritional Characteristics, and Ethnomedicinal Values for Sustainability

8

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Abstract

This chapter aims to discuss the phytomedicinal value of turkey berry (*Solanum torvum* Sw.). The shrub or small tree, turkey berry, is native to tropical and subtropical regions of Africa, Asia, and the Americas and has a long history of use in various culinary and traditional medicinal practices across different cultures. However, it is highly adaptable to different ecological niches, which allows it to grow in a variety of habitats, from coastal areas to highlands. The eggplant or tomato-like fruits of *S. torvum* have a unique aroma, sweet taste, and high phenolic contents that are essential to the dietary and medicinal properties of the plant. The plant is rich in vitamins, minerals, dietary fiber, protein, and secondary phytochemicals like flavonoids, alkaloids, tannins, saponins, essential oils, glycosides, and steroids. It is used to prepare curry and gravy dishes, stir-fries, pickles and chutneys, soups and stews, sauces and sambals, fried snacks, stuffed dishes, rice-based dishes, and other traditional dishes. The ethnobotanical values of *S. torvum* include rituals and symbolism, culinary traditions, economic uses, and traditional medicine. The plant is used to aid digestion as well as an anti-inflammatory, antioxidant, antimicrobial, immune modulation, cardiovascular health maintenance, wound healing, antidiabetic, weight management, and fever reduction. Clinical research on turkey berry is ongoing, and there is still much to learn and explore regarding the potential health benefits and therapeutic applications of the plant. Some herbal products containing turkey berry include herbal teas, herbal powders, herbal tinctures and extracts, capsules, herbal blends, and infusions. Turkey berry is generally considered safe for consumption when used in moderation as a food or traditional herbal remedy. However, like many botanicals, excessive or improper use can potentially lead to adverse effects.

Keywords

Ethnobotany · Clinical research · Herbal products · Phytochemicals · Toxicity · Antimicrobials

1 Introduction

Turkey berry (*Solanum torvum* Sw.), also known as wild eggplant, devil's fig, or pea eggplant, Sundakkai in Tamil Nadu, India, Makheua Phuang in Thai, and Izote in Mexico, is a plant that belongs to the Solanaceae family, which includes other well-known members such as tomatoes, potatoes, and bell peppers [5, 42]. This plant is native to tropical and subtropical regions of Africa, Asia, and the Americas. Turkey berry has a long history of use in various culinary and traditional medicinal practices across different cultures [3, 25, 68].

Understanding the botanical characteristics of turkey berry is essential for recognizing, cultivating, utilizing, and conserving the plant. Turkey berry is a perennial plant that can grow as a shrub or small tree, typically reaching heights of 1 to 3 m (3 to 10 ft). It features woody stems and branches with numerous leaves and fruit clusters [25]. The leaves of turkey berry are simple, alternate, and dark green. They are generally elliptical or ovate, with a slightly serrated margin [66]. The flowers of *S. torvum* grow in clusters on the plant and are typically small, star-shaped, and white to pale lavender. The fruit of turkey berry is a small, round, or oval berry, about 1–2 cm (0.4–0.8 in.) in diameter [110]. When mature, the fruits turn from green to yellow or orange and contain numerous seeds. The fruit resembles miniature eggplants or tomatoes. They have a unique aroma, sweet taste, and high phenolic contents that are essential to the dietary and medicinal properties of the plant [126]. It is worth noting that different cultivars of turkey berry exist with variations in fruit size and color. Some have green or white fruits, while others have red or purple ones. The plant is well adapted to tropical and subtropical climates and can thrive in a variety of soil types, including sandy, loamy, and clay soils [116]. It is often found in disturbed habitats, such as roadsides and fallow fields, and it often grows as a weed in agricultural fields [64, 130].

Turkey berry is a widely distributed plant, with native populations found in various regions around the world. In Africa, turkey berry is native to several African countries, including Nigeria, Ghana, Cameroon, Madagascar, and other tropical and subtropical regions across the continent [57, 132]. In Asia, *S. torvum* is prevalent in countries such as India, Sri Lanka, Malaysia, Thailand, and Indonesia. Turkey berry has also been reported in the Americas, particularly in tropical regions of the Caribbean, Central America, and South America [8, 42]. Due to its adaptability and hardiness, turkey berry has become naturalized in many other parts of the world, including parts of the United States, Australia, and various Pacific islands [46].

Turkey berry has a rich history of ethnobotanical uses in traditional medicine and culinary traditions across different cultures. Its diverse applications include:

Culinary Uses: Jena et al. [63] reported that turkey berry is commonly used in cooking various culinary dishes, particularly in Asian (South Indian) and African cuisines. It is added to soups, stews, curries, and stir-fries. The bitter and slightly tangy taste of turkey berry adds a unique flavor dimension to dishes. It is often used to balance the flavors of other ingredients.

Traditional Medicine: This includes

- Digestive Aid: In traditional medicine, turkey berry is used to aid digestion and alleviate gastrointestinal discomfort [19].
- Anti-inflammatory and Antioxidant: Some cultures use turkey berry as a natural remedy for inflammatory conditions and the plant's antioxidant properties have led to its use as a remedy for various health issues [25, 86].
- Fever Reduction: In some regions, turkey berry is employed to help reduce fever and symptoms of illnesses [5].

Pest Control: The leaves and fruits of turkey berry contain compounds that can act as natural insect repellents [33, 66]. For this role, they are sometimes placed in storage containers to protect grains and pulses from pests [94–96, 100].

Livestock Feed: In some areas, turkey berry leaves and fruits are used as supplemental fodder for livestock [12, 38, 90, 104].

This chapter aims to discuss the phytomedicinal value of turkey berry. It will present the origin, distribution, and botany of *S. torvum* wherein it is shown that the widespread distribution of the plant has led to its incorporation into a variety of culinary and medicinal traditions. This includes culinary and traditional medicine uses. The next sections address the nutritional and phytochemical properties as well as the food uses of turkey berry. Turkey berry is a versatile ingredient in many cuisines. It is used in curries, soups, stews, and stir-fries. Its slightly bitter and tangy flavor adds depth to dishes. Turkey berry has a history of use in traditional medicine across its distribution areas. Other parts of the chapter are on the ethnobotanical, pharmacognosy, and clinical works, herbal products, toxicity, and commercial values of turkey berry. It is believed to have various medicinal properties, including digestive, anti-inflammatory, and antimicrobial effects. The adaptability of turkey berry to various regions and environmental conditions has allowed them to become an integral part of diverse culinary traditions and traditional healing practices. The plant's versatility, from its multiple uses in regional cuisines to its potential health benefits, underscores its significance in global biodiversity and human culture.

2 Origin and Distribution of Turkey Berry (*Solanum torvum*)

The exact origin of *S. torvum* is unknown but researchers believe that the plant likely originated in tropical and subtropical regions of Africa, Asia, and South America where it can be found in the wild [48, 132]. In addition to being found in the wild in these regions, it has cultural and agricultural significance through a long history of cultivation and use in traditional cuisines and medicinal practices. Turkey berry is well-adapted to tropical and subtropical climates, where it thrives in warm and humid conditions. It is known for its ability to grow in a variety of soil types, including sandy, loamy, and clay soils, as long as there is good drainage. The plant can be found in disturbed habitats such as roadsides, fallow fields, and areas with disrupted vegetation. Turkey berry can be cultivated in gardens and farms, but it also grows as a wild plant in many regions. Turkey berry is commonly found in tropical and subtropical regions of Africa, Asia, the Americas, and other parts of the world. Its adaptability to different ecological niches allows it to grow in a variety of habitats, from coastal areas to highlands. Genetic studies have indicated high genetic diversity in *S. torvum* populations in Africa and Asia, reinforcing the idea of its long-term presence in these areas [134, 135]. Also, there have been archaeological findings of *S. torvum* seeds in ancient sites in Africa, Asia, and South America providing further evidence of its historical cultivation and consumption. Within these regions, turkey berry goes by a multitude of local names in different regions, reflecting its cultural significance and versatility in traditional dishes. Some of these names include Igbo or Oha (Igbo language) in Nigeria where it is a key ingredient in soups like Oha soup, Sundakkai and Bhatkatiya in Tamil and Hindi (India) where it is a popular addition to South Indian cuisine [10, 124]. It is called Makheua Phuang in Thailand and is used in various dishes, including curries and stir-fries. It is commonly called wild eggplant or bitter ball in the Caribbean and is used in traditional recipes. In Brazil, turkey berry is called Berinjela-do-mato and is used in many Brazilian local cuisines [79].

3 Distribution and Naturalized Regions of Turkey Berry (*Solanum torvum*)

Turkey berry has an extensive distribution that spans multiple continents and regions because it thrives in a variety of climates and habitats, making it a resilient and adaptable plant. Due to its adaptability and ability to grow in a wide range of conditions, turkey berry has become naturalized in various other regions around the world. Some examples include the United States (found in states like Florida and Hawaii, where it grows as a weed), Australia, and the Pacific Islands [47, 50]. Turkey berry exhibits adaptability to diverse habitats and climatic conditions, which contributes to its widespread distribution. It can thrive in:

(a) **Africa:**

West Africa: Turkey berry is native to several West African countries, including Nigeria, Ghana, Cameroon, and Ivory Coast. It is a staple in many West African cuisines, where it is used in dishes such as soups and stews [49, 99].

East Africa: The plant is also found in East African countries like Kenya, Tanzania, and Uganda, where it is used both in cooking and traditional medicine.

Southern Africa: Turkey berry is present in Southern African countries like Zambia and Zimbabwe, where it is known by various local names.

(b) **Asia:**

India: Turkey berry is widely cultivated and consumed in India, where it is a common ingredient in various regional cuisines.

Southeast Asia: In countries like Thailand, Malaysia, and Indonesia, turkey berry is used in curries, stir-fries, and traditional dishes.

Sri Lanka: Known as Waran, it is a popular vegetable in Sri Lankan cuisine.

Philippines: In the Philippines, it is called tabatsoy and is used in local dishes.

China: Turkey berry is found in parts of southern China.

(c) **Americas:**

Caribbean: Turkey berry is native to the Caribbean region, where it is used in local dishes and traditional remedies. It is known as a wild eggplant in some Caribbean countries.

Central America: In countries like Costa Rica and Nicaragua, it is called berenjena cimarrona and is used in traditional dishes.

South America: Turkey berry is present in South American countries such as Brazil, Colombia, and Venezuela, where it is known as berenjena cimarrona or jilo.

(d) **Tropical and Subtropical Regions**: Turkey berry prefers warm and humid tropical and subtropical climates, making it well-suited for regions with consistent temperatures and rainfall [34, 125].

The distribution and naturalization of Turkey berry in these regions is connected to

– **Various Soil Types**: The plant can grow in a variety of soil types, including sandy, loamy, and clay soils, as long as there is good drainage.
– **Disturbed Habitats**: It often thrives in disturbed habitats, such as roadsides, fallow fields, and areas with disrupted vegetation.

– **Cultivated Settings**: While it can grow as a wild plant, turkey berry is also cultivated in gardens and farms in many regions for its culinary and medicinal uses.

4 Botany and Cultivation of Turkey Berry (*Solanum torvum*)

Solanum torvum is a perennial plant that can grow as either a shrub or a small tree. Its growth habit depends on environmental conditions and cultivation practices [7, 121]. Depending on the growth conditions and regional variations, turkey berry plants can appear as compact shrubs with dense foliage or as small trees with a more open canopy. The plant's overall appearance can be bushy or somewhat spindly, with multiple branches and leaves (Fig. 1). The plant has a fibrous root system. The size of turkey berry plants can vary widely. They typically range from 1 to 3 m (3 to 10 ft) in height, but some individuals can grow taller under favorable conditions [120]. The plant features woody stems and branches. These stems are often green when young, turning brown or gray as they mature. Turkey berry has simple, alternate leaves that are dark green and slightly glossy. The leaves are typically elliptical or ovate with a slightly serrated margin [70]. The leaves are typically dark green, but leaf color can vary depending on factors like age and environmental conditions. The leaf blades are usually elliptical or ovate, with a smooth or slightly serrated margin. The leaves are generally smooth and slightly glossy [9, 51]. The leaves of turkey berry are simple, meaning they consist of a single leaf blade. They are arranged alternately along the stems. The flowers of *S. torvum* are small, star-shaped, and white to pale lavender. They are typically arranged in clusters or inflorescences [52]. Turkey berry produces small, star-shaped flowers that are typically white to pale lavender. The flowers are arranged in clusters or inflorescences at the tips of branches. These flowers are hermaphroditic, meaning they have both male and female reproductive structures within the same flower. The fruit of turkey berry is a small, round, or oval berry, typically measuring

Fig. 1 Leaves of turkey berry (*S. torvum*)

Fig. 2 Flowers and seeds of turkey berry (*S. torvum*)

about 1 to 2 cm (0.4 to 0.8 in.) in diameter [18, 80, 84]. When ripe, the fruit changes from green to yellow or orange and contains numerous small seeds (Fig. 2). The fruit bears a resemblance to miniature eggplants or tomatoes. The fruit of *S. torvum* is a small berry that starts green and matures to yellow or orange. When the fruit is ripe, it is typically 1 to 2 cm (0.4 to 0.8 in.) in diameter. Each fruit contains numerous tiny seeds [61, 74]. The fruit has a slightly bitter and tangy taste, which contributes to its culinary and culinary appeal. It is worth noting that *S. torvum* variants and cultivars exist with differences in fruit size, color, and taste, depending on the region and local preferences. These features are linked to factors such as soil type, climate, and regional genetic variations [59, 133]. These characteristics contribute to the plant's adaptability and versatility in different environments and culinary traditions.

Cultivating turkey berry can be a rewarding endeavor due to the increasing culinary and medical application of the crop. It is also not challenging because the crop is highly versatile [76, 105]. Turkey berry is well-suited to tropical and subtropical climates, and it can thrive in a variety of soil types. It thrives in warm and humid tropical and subtropical climates but prefers temperatures between 25 °C and 35 °C (77 °F to 95 °F) and cannot tolerate frost. The planting location must also receive plenty of sunlight, as turkey berry requires full sun for optimal growth [118]. Turkey berry can grow in various soil types, but it prefers well-drained soil with a pH level between 6.0 and 7.5 [122]. Soil fertility may also be improved by adding organic matter, such as compost or well-rotted manure, to enhance nutrient availability [106]. Turkey berry can be grown from seeds or propagated from stem cuttings. If planting from seeds, soak the seeds in water for 24 h before planting to improve germination rates. Sow the seeds directly into the prepared soil, spacing them about 1 to 2 ft apart in rows or a well-prepared garden bed [117].

Water the seeds or cuttings thoroughly after planting. Turkey berry requires consistent moisture for optimal growth. Keep the soil evenly moist but not waterlogged. Water deeply when the soil surface begins to dry and ensure good drainage to prevent waterlogging, which can lead to root rot. Applying a layer of mulch around the base of the plants helps retain moisture, suppress weeds, and regulate soil temperature [129]. Organic mulch like straw or compost is ideal. Turkey berry benefits from regular fertilization with balanced, slow-release fertilizers [123, 127]. Farmers can use a general-purpose fertilizer or one specifically formulated for vegetables. Follow the recommended dosage on the fertilizer label and avoid over-fertilizing, as excessive nitrogen can result in excessive vegetative growth at the expense of fruit production. Pruning can help manage the growth of turkey berry plants and encourage better fruit production. Pinch back the growing tips of young plants to encourage bushier growth [30, 31]. Provide support or trellises for the plants, especially if they are grown as tall shrubs or small trees, to prevent sprawling and improve air circulation. Turkey berry is relatively resistant to pests and diseases, but occasional issues may arise. Monitor your plants regularly and take appropriate measures if pests or diseases are detected [32, 62]. Common pests of turkey berry include aphids, whiteflies, and fruit flies. Use organic pest control methods or insecticidal soap if necessary. Inspect the plants regularly for signs of diseases like powdery mildew or leaf spot, and treat promptly with appropriate fungicides if needed [20, 58]. Turkey berry typically begins to produce fruits within a few months after planting [119, 128]. Harvest the fruits when they are firm, green, and about 1 to 2 cm (0.4 to 0.8 in.) in diameter. Ripe fruits turn yellow or orange but should be harvested before they become too ripe to maintain optimal flavor. Harvesting regularly encourages continued fruit production. It is a versatile and flavorful addition to many culinary dishes, and its adaptability to various soil types and climates makes it an attractive option for home gardeners and farmers in tropical and subtropical regions.

The diversity of *S. torvum* encompasses a range of aspects, including its botanical characteristics, cultivars, regional variations, and uses in different cultures [1]. This diversity highlights the plant's adaptability and versatility in various environments and its rich history of culinary and medicinal applications.

Botanical Diversity of *Solanum torvum*:

- Varieties: *Solanum torvum* variants and cultivars have differences in fruit size, color, and taste. These variations are found in different regions and have led to local preferences for specific types [6, 29].
- Plant Morphology: The size and shape of turkey berry plants can vary. They can grow as small shrubs or small trees, depending on environmental conditions and local cultivation practices.

Regional Diversity of *Solanum torvum*:

- Culinary Uses: Turkey berry is used differently in various culinary traditions around the world. It is incorporated into diverse dishes, including soups, stews, curries, and stir-fries, each with its unique flavors and cooking methods.
- Local Names: Turkey berry is known by numerous local names in different regions, reflecting its cultural significance. For example, it is called Sundakkai in Tamil Nadu, India, Makheua Phuang in Thailand, and njama njama in Cameroon. These names emphasize its role in local cuisines.
- Traditional Medicine: The uses of turkey berry in traditional medicine also vary by region. Different cultures have developed remedies based on their

understanding of the plant's properties, resulting in diverse applications for ailments ranging from digestive issues to inflammation [87].

Culinary Diversity of *Solanum torvum*:

- Preparation Methods: The culinary diversity of turkey berry is evident in various ways it is prepared. It can be fried, roasted, boiled, or used fresh, depending on the dish and cultural practices.
- Complementary Ingredients: The choice of ingredients that accompany turkey berry varies by region. Some cultures pair it with spices, coconut, or other vegetables to create unique flavor profiles.
- Dish Types: Turkey berry is used in a wide array of dishes, such as Indian sambar, Thai green curry, Filipino pinakbet, and Ghanaian kontomire stew. The plant's adaptability to different cuisines underscores its versatility.

Variability in Traditional Medicine:

- Remedial Practices: Traditional medicinal uses of turkey berry are diverse and can include treatments for digestive problems, fever, inflammation, and more. Local herbalists and healers may have distinct methods of preparation and application.
- Preparation Techniques: The methods for preparing turkey berry-based remedies can vary, such as infusions, decoctions, poultices, or direct application of crushed leaves or fruits.

Environmental Adaptability:

- Habitat Range: Turkey berry exhibits adaptability to various ecological niches. It can thrive in tropical and subtropical regions, ranging from coastal areas to highlands, and in different soil types [22].
- Cultivation Practices: Depending on local agricultural practices, turkey berry may be cultivated in gardens, farms, or allowed to grow naturally in fields and disturbed habitats. Its adaptability makes it suitable for both subsistence and commercial cultivation.

Nutritional Diversity of *Solanum torvum*:

- Nutrient Content: The nutritional composition of turkey berry can vary based on factors like soil quality, cultivation methods, and maturity at harvest. Variations in vitamins, minerals, and phytochemicals are observed [14].
- Dietary Significance: Turkey berry provides essential nutrients and dietary fiber, contributing to the nutritional diversity of diets in regions where it is consumed regularly.

Medicinal Versatility of *Solanum torvum*:

- Health Conditions: The potential health benefits of turkey berry are diverse, including antioxidant, anti-inflammatory, and digestive properties. Different cultures may emphasize specific health aspects based on their traditional knowledge.

- Application Methods: Traditional healers and practitioners may employ various parts of the plant, including leaves, fruits, or roots, for medicinal purposes, leading to a diverse range of applications.

Variability in Preparation and Presentation:
- Culinary Creativity: The culinary diversity of turkey berry is marked by the creativity in its preparation. It can be a prominent ingredient or a complementary one, lending unique flavors and textures to dishes.
- Regional Specialties: Specific dishes featuring turkey berry become regional specialties, celebrated for their unique taste and cultural significance.

5 Nutritional Properties and Food Uses of Turkey Berry (*Solanum torvum*)

Turkey berry is not only valued for its culinary and medicinal uses but also for its nutritional content. It is important to note that turkey berry has a slightly bitter and tangy taste, which can be an acquired taste for some. Its culinary use varies by region, and it is prized for its ability to add depth of flavor to dishes [2, 4, 91–93]. Turkey berry is not only valued for its culinary and medicinal uses but also for its nutritional content [11]. While the nutrient composition can vary depending on factors such as maturity and cultivation methods, these are some key nutritional properties:

1. **Vitamins:** Turkey berry is a source of various vitamins, including vitamin C, vitamin A, and vitamin B-complex vitamins like niacin and riboflavin.
2. **Minerals:** It contains essential minerals such as potassium, calcium, and iron, which are important for maintaining overall health.
3. **Dietary Fiber:** Turkey berry is a good source of dietary fiber, which aids in digestion and supports gastrointestinal health [60].
4. **Protein:** While not a primary source of protein, turkey berry contains a moderate amount, making it a valuable addition to vegetarian diets.
5. **Secondary Phytochemicals:** The plant is rich in phytochemicals, including flavonoids, alkaloids, and phenolic compounds, which contribute to its potential health benefits [13].

Solanum torvum is widely used in culinary traditions across different cultures. It is appreciated for its unique flavor and versatility in a variety of dishes. Some common food uses of turkey berry include:

1. **Curry and Gravy Dishes:** Turkey berry is a popular ingredient in many curry and gravy dishes, particularly in South Asian and Southeast Asian cuisines. It adds a slightly bitter and tangy flavor to the dishes, complementing the richness of the sauce.

2. **Stir-Fries:** In stir-fry dishes, turkey berry is often used alongside other vegetables and protein sources. Its firm texture holds up well to stir-frying, and it adds a distinct taste to the overall dish.
3. **Pickles and Chutneys:** In some regions, turkey berry is used to prepare pickles and chutneys. The berries are typically marinated with spices, oil, and other seasonings. These pickles and chutneys can be served as condiments or side dishes.
4. **Soups and Stews:** Turkey berry can be added to soups and stews for extra flavor and nutrition. It is often used in traditional dishes like sambar (a South Indian lentil stew) and sinigang (a Filipino sour soup).
5. **Sauces and Sambals:** Turkey berry is sometimes included in sauces and sambals (spicy condiments) to enhance their flavor profiles. It can be used in combination with other ingredients like chilies, garlic, and herbs.
6. **Fried Snacks:** In some cultures, turkey berry is coated in batter and deep-fried to make crispy snacks. The fried turkey berries are enjoyed for their crunchy texture and savory taste.
7. **Stuffed Dishes:** Turkey berry can be stuffed with various fillings, such as spices, ground meat, or lentils, and then cooked. Stuffed turkey berries are common in Indian cuisine and other regional cuisines.
8. **Rice and Rice Dishes:** In certain dishes, turkey berry is added to rice during cooking to infuse its flavor into the grains. It is also used in rice-based dishes like biryani and pilaf.
9. **Traditional Dishes:** Turkey berry is a key ingredient in traditional dishes like njama njama in Ghana, Terong Belanda in Indonesia, and Koottu in South India. These dishes are celebrated for their unique combination of flavors.
10. **Medicinal Preparations:** While not a food use, turkey berry is sometimes used in medicinal preparations, including herbal teas and infusions.

6 Phytochemical Constituents and Potential Health Benefits of Turkey Berry (*Solanum torvum*)

Turkey berry (*S. torvum*) contains a variety of phytochemical constituents, which are natural compounds found in plants. These phytochemicals contribute to the plant's flavor, aroma, and potential health benefits. Some of the notable phytochemical constituents found in turkey berry include:

1. **Alkaloids:** Alkaloids are naturally occurring organic compounds that often have physiological effects on humans and animals [97, 103]. Turkey berry contains various alkaloids, including solanidine and solasonine. Alkaloids may have diverse effects, ranging from antimicrobial to anti-inflammatory properties [15, 101].
2. **Flavonoids:** Flavonoids are a group of polyphenolic compounds found in many fruits and vegetables, including turkey berry. These compounds are known for

their antioxidant properties and potential health benefits, such as reducing oxidative stress and inflammation.

3. **Glycosides:** Glycosides are compounds composed of a sugar molecule (glycone) and a non-sugar moiety (aglycone). Turkey berry contains various glycosides, which can have different biological activities. Some glycosides may play a role in the plant's medicinal properties [23].
4. **Tannins:** Tannins are polyphenolic compounds found in turkey berry and other plants. They can contribute to the astringent taste of certain fruits and are often used in traditional medicine. Tannins have antioxidant properties and may have applications in wound healing and digestive health [35].
5. **Phenolic Compounds:** Phenolic compounds, including phenolic acids and phenolic glycosides, are present in turkey berry [36, 102]. These compounds are known for their antioxidant and anti-inflammatory activities. They may contribute to the potential health benefits of the plant.
6. **Steroids and Triterpenoids:** Turkey berry contains steroidal compounds and triterpenoids. These compounds are part of the plant's chemical makeup. Some steroidal compounds may have anti-inflammatory properties [28, 75].
7. **Saponins:** Saponins are glycosides with a characteristic foaming property. They are found in various plant species, including turkey berry. Saponins may have immune-modulating and cholesterol-lowering effects [21, 78].
8. **Essential Oils:** Some varieties of turkey berry may contain essential oils, which are volatile aromatic compounds responsible for the plant's fragrance. These oils can vary in composition and may contribute to the plant's flavor.

It is important to note that the phytochemical composition of turkey berry can vary depending on factors such as plant variety, growing conditions, and maturity at harvest [37]. While these phytochemicals contribute to the plant's potential health benefits, further research is needed to better understand their specific effects and mechanisms of action. Additionally, the presence of these compounds in turkey berry has made it a subject of interest in both traditional and modern herbal medicines, as well as in culinary applications. The consumption of turkey berry has been associated with several potential health benefits, although further research is needed to fully understand its mechanisms and efficacy. Some of these benefits include:

1. **Antioxidant Properties:** The presence of phytochemicals in turkey berry gives it the typical antioxidant properties, which can help combat oxidative stress and reduce the risk of chronic diseases.
2. **Anti-inflammatory Effects:** Turkey berry may possess anti-inflammatory properties, which could be beneficial for individuals dealing with inflammatory conditions.
3. **Digestive Health:** Traditionally, turkey berry has been used to support digestive health and alleviate gastrointestinal discomfort [53].
4. **Antimicrobial Activity:** Some studies suggest that turkey berry may have antimicrobial properties, which can help protect against various infections [39].

5. **Antidiabetic Potential:** Preliminary research indicates that turkey berry may play a role in blood sugar regulation and could be explored as an adjunct in diabetes management.
6. **Weight Management:** The dietary fiber in turkey berry can contribute to feelings of fullness, potentially aiding in weight management.
7. **Nutrient Boost:** Incorporating turkey berry into the diet can provide essential vitamins and minerals that support overall health [77].

7 Ethnobotanical Value of Turkey Berry (*Solanum torvum*)

The ethnobotany of turkey berry (*S. torvum*) reveals its rich history of traditional uses across various cultures and regions. It underscores the multifaceted role of the plant in various cultures and regions. From culinary traditions to traditional medicine and cultural symbolism, this plant has a long history of diverse uses. Its adaptability and versatility make it an integral part of many culinary and healing traditions, contributing to its enduring importance in the ethnobotanical landscape.

This versatile plant has been valued for centuries for its culinary, medicinal, and cultural significance.

Rituals and Symbolism: In certain cultures, turkey berry holds symbolic significance and is used in rituals, ceremonies, and offerings. It may be included in traditional healing rituals or as a protective charm.

Culinary Traditions: Turkey berry is a staple ingredient in many traditional dishes and contributes to the cultural identity of the cuisine in which it is used [54].

Economic Uses: Turkey berry is cultivated and harvested for both domestic consumption and commercial purposes, contributing to the livelihoods of farmers and local economies in regions where it is grown [16, 56, 95].

Traditional Medicine: Turkey berry has a long history of medicinal uses in traditional herbal medicine systems across different cultures and regions. While scientific research is ongoing to validate some of these traditional uses, some of the common medicinal uses attributed to turkey berry include:

- **Digestive Health**: This is one of the most widely recognized traditional uses of turkey berry is its role in promoting digestive health. It is believed to have mild laxative properties and is used to alleviate constipation and improve bowel regularity. Turkey berry is also used to relieve indigestion, bloating, and gas [55].
- **Anti-inflammatory Properties**: In traditional medicine, turkey berry is used to reduce inflammation and alleviate pain associated with conditions like arthritis and joint pain. Poultices made from crushed leaves or heated leaves are sometimes applied topically to reduce inflammation.
- **Antipyretic (Fever-Reducing) Effects**: In some cultures, turkey berry is used to lower fever and manage symptoms of fever-related illnesses. It is believed to have antipyretic properties that help reduce body temperature during fever [17, 131].

- **Diuretic and Kidney Health**: Turkey berry has diuretic properties and is used in some regions to promote urination and support kidney health. It is believed to help in the elimination of toxins and excess fluids from the body [88].
- **Wound Healing**: In certain traditional medicine systems, turkey berry is used topically as a poultice or ointment to promote wound healing. It is believed to have antimicrobial properties that may help prevent infection.
- **Respiratory Health**: Some traditional remedies include turkey berry for managing respiratory conditions such as coughs and colds. It is believed to have properties that can help alleviate symptoms of respiratory ailments [45].
- **Antioxidant Effects**: Turkey berry contains antioxidants, including flavonoids and polyphenols, which may help combat oxidative stress and reduce the risk of chronic diseases. These antioxidants are thought to play a role in overall health and wellness.
- **Potential Antidiabetic Effects**: Preliminary studies have suggested that turkey berry may have potential antidiabetic effects. Some research indicates that it may help lower blood sugar levels, making it of interest in the management of diabetes.

It is important to note that while turkey berry has a history of traditional medicinal use, scientific research is ongoing to better understand its active compounds and their potential therapeutic benefits [40]. As with any herbal remedy, it is advisable to consult with a healthcare professional before using turkey berry or its extracts for medicinal purposes, especially if you have underlying health conditions or are taking other medications. Additionally, the quality and safety of herbal preparations can vary, so it is essential to use them with caution and under appropriate guidance.

8 Pharmacognosy and Clinical Research on Turkey Berry (*Solanum torvum*)

One way by which modern scientific research is exploring the potential health benefits of turkey berry and its bioactive compounds for better understanding and to lead to new applications and products is pharmacognosy and clinical research trials. Pharmacognosy is the study of natural products obtained from plants and their applications in medicine. *S. torvum* has been a subject of interest in pharmacognosy due to its traditional medicinal uses and the presence of various phytochemical constituents. Pharmacognostic works on turkey berry typically involve one of the following as outlined in Nurit-Silva et al. [89]:

1. **Morphological Identification:** Pharmacognosy begins with the accurate identification and authentication of the plant material. In the case of turkey berry, this involves recognizing its distinct morphological characteristics, including the growth habit, leaves, flowers, and fruits.
2. **Phytochemical Analysis:** Pharmacognostic studies of turkey berry involve the analysis of its phytochemical constituents. This includes identifying and

quantifying alkaloids, flavonoids, glycosides, tannins, phenolic compounds, steroids, triterpenoids, saponins, and other secondary metabolites present in the plant. Various analytical techniques, such as chromatography and spectroscopy, may be employed to determine the composition of phytochemicals.

3. **Quality Control:** Pharmacognosy plays a vital role in the quality control and standardization of herbal medicines and dietary supplements containing turkey berry extracts. Establishing quality parameters ensures the consistency and safety of herbal products.
4. **Medicinal Uses:** Pharmacognostic studies provide insights into the traditional medicinal uses of turkey berry. This includes documenting its historical applications in treating digestive disorders, inflammatory conditions, fever, and more. The identification of specific bioactive compounds helps understand the plant's pharmacological properties.
5. **Pharmacological Studies:** Pharmacognosy contributes to the exploration of turkey berry's pharmacological activities. Researchers investigate its potential as an anti-inflammatory, antioxidant, analgesic, diuretic, and more. Animal and in vitro studies are conducted to evaluate the plant's bioactivity.
6. **Formulation Development:** Turkey berry may be incorporated into various herbal formulations, traditional medicines, or dietary supplements. Pharmacognostic data assist in selecting the appropriate parts of the plant, extracting methods, and dosage forms for formulation.
7. **Toxicological Studies:** Assessing the safety of turkey berry is an essential aspect of pharmacognosy. Toxicological studies help determine the plant's potential adverse effects and safe usage levels.
8. **Standardization and Regulation:** Pharmacognostic data contribute to the development of monographs and standards for turkey berry in pharmacopeias and regulatory frameworks.
   – Standardization ensures the consistency and quality of herbal products containing turkey berry.
9. **Future Research:** Ongoing pharmacognostic research may uncover new phytochemical constituents and potential medicinal uses of turkey berry. Modern pharmacological and clinical studies can validate its traditional applications and explore new therapeutic avenues.

Clinical research on *S. torvum* is ongoing, and while there is still much to learn, several studies have explored its potential health benefits and therapeutic applications. Some areas of clinical research related to turkey berry include:

1. **Antioxidant and Anti-inflammatory Properties**: Some studies have investigated the antioxidant and anti-inflammatory effects of turkey berry. Antioxidants can help protect cells from oxidative stress, while anti-inflammatory properties may have implications for conditions like arthritis [65].
2. **Antidiabetic Effects**: Research has examined the potential antidiabetic properties of turkey berry. It has been suggested that it may help lower blood sugar levels, making it of interest in diabetes management [107].

3. **Gastrointestinal Health**: Turkey berry's traditional use as a digestive aid has prompted studies to assess its effects on gastrointestinal health. Research has explored its potential role in alleviating indigestion, promoting regular bowel movements, and reducing symptoms of gastrointestinal disorders [67].
4. **Antimicrobial Activity**: Some studies have investigated the antimicrobial properties of turkey berry. It has been tested against various bacteria and fungi, suggesting potential applications in controlling microbial infections [24, 26].
5. **Hepatoprotective Effects**: Turkey berry has been studied for its hepatoprotective effects, indicating its potential to protect the liver from damage caused by toxins or certain diseases [69].
6. **Wound Healing**: Research has explored the wound-healing properties of turkey berry. It has been tested in both animal and clinical studies to assess its potential to promote the healing of wounds and injuries [41].
7. **Immune Modulation**: Some studies have looked at the immunomodulatory effects of turkey berry [27, 111]. This research aims to understand its impact on the immune system and its potential applications in immune-related conditions [43].
8. **Anticancer Properties**: Preliminary studies have investigated turkey berry's potential anticancer properties, particularly its effects on cancer cell proliferation and apoptosis (cell death). However, more research is needed in this area [108].
9. **Cardiovascular Health**: Research has explored the effects of turkey berry on cardiovascular health markers, such as blood pressure and lipid profiles. These studies aim to determine its potential role in heart health [71].
10. **Toxicological Studies**: Clinical research includes safety and toxicity assessments to evaluate the potential adverse effects of turkey berry and to establish safe usage levels [44, 109].

These studies provide valuable insights into the potential health benefits of turkey berry, more research is needed to confirm and fully understand its therapeutic applications [112]. Additionally, the results of clinical studies can vary, and individual responses to herbal remedies may differ. As a result, it is advisable to consult with a healthcare professional before using turkey berry or its extracts for medicinal purposes, especially if you have underlying health conditions or are taking other medications.

9 Herbal Products of Turkey Berry (*Solanum torvum*)

Turkey berry is utilized in various herbal products and traditional preparations across different cultures. Its diverse phytochemical composition and potential health benefits have made it a valuable ingredient in herbal remedies and dietary supplements. Some herbal products and traditional preparations that incorporate turkey berry:

- **Herbal Teas:** Turkey berry leaves and fruits can be dried and used to prepare herbal teas. These teas are often consumed for their potential digestive and antioxidant properties [113].

- **Herbal Powders:** Dried and powdered turkey berry can be used as a dietary supplement. It is available in capsule or powder form and is sometimes marketed for its potential health benefits [113].
- **Tinctures and Extracts:** Turkey berry extracts, in the form of tinctures or liquid extracts, may be used as herbal remedies. These concentrated forms allow for easy dosage and administration.
- **Traditional Remedies:** In traditional medicine systems, turkey berry is used in various herbal remedies for digestive disorders, fever, inflammation, and other ailments. For example, in some regions of Africa, it is used in concoctions and infusions to treat stomachaches and indigestion.
- **Culinary Herbs and Spices:** Turkey berry is a popular culinary herb in some regions. It is used to add flavor to various dishes, particularly in South Indian and Southeast Asian cuisines. It is often used in curries, stews, and stir-fries to impart a slightly bitter and tangy taste.
- **Herbal Ointments and Topical Preparations:** Turkey berry leaves may be used in the preparation of herbal ointments or poultices for topical applications. They are believed to have anti-inflammatory and wound-healing properties.
- **Herbal Capsules and Tablets:** Some herbal supplement manufacturers offer turkey berry capsules and tablets as part of their product line. These products are often marketed for their potential health benefits.
- **Herbal Blends:** Turkey berry may be combined with other herbs and botanicals to create specialized herbal blends targeting specific health concerns, such as digestive health or immune support.
- **Herbal Infusions:** Turkey berry leaves and fruits can be steeped in hot water to make herbal infusions. These infusions are often consumed for their potential medicinal properties and pleasant flavor.
- **Traditional Tonics:** In some cultures, turkey berry is used as an ingredient in traditional health tonics or elixirs believed to boost overall wellness [73, 80].

Turkey berry is a part of various herbal products and traditional preparations, but scientific research is ongoing to better understand their specific health benefits and mechanisms of action. Also, as with any herbal product, it is advisable to consult with a healthcare professional before using turkey berry or its extracts for medicinal purposes, especially if you have underlying health conditions or are taking other medications. Additionally, the quality and safety of herbal products can vary, so it is essential to choose reputable sources and products.

10 Toxicity of Turkey Berry (*Solanum torvum*)

Turkey berry is generally considered safe for consumption when used in moderation as a food or traditional herbal remedy [72]. However, like many botanicals, excessive or improper use can potentially lead to adverse effects. Some considerations regarding the toxicity and safety of turkey berry include:

Solanine Content: Turkey berry, like other plants in the Solanaceae family (nightshades), contains certain alkaloids, including solanine and solasonine. Solanine is known for its potential toxicity when consumed in large quantities. The solanine content in turkey berry can vary depending on factors such as the plant's age and maturity. To minimize the risk of solanine toxicity, it is advisable to cook turkey berry before consumption, as cooking can help reduce the alkaloid content. Overripe or unripe fruits may have higher solanine levels [82, 114].

Allergic Reactions: Some individuals may be allergic to certain compounds present in turkey berry. Allergic reactions, though rare, can include symptoms like skin rashes, itching, and gastrointestinal discomfort. If you have known allergies to plants in the Solanaceae family, exercise caution when trying turkey berry for the first time.

Gastrointestinal Effects: When consumed in large amounts, turkey berry may cause gastrointestinal upset, including nausea, vomiting, and diarrhea. This is more likely to occur with excessive intake [115].

Interaction with Medications: Turkey berry may interact with certain medications or medical conditions. For example, it may affect blood sugar levels, which could be problematic for individuals with diabetes. If you are taking medications or have underlying health conditions, it is important to consult with a healthcare professional before incorporating turkey berry into your diet or as a supplement.

Raw Consumption: As a precaution, it is recommended to avoid consuming raw turkey berry, especially if you are unsure about its ripeness. Cooking the berries can help neutralize potentially harmful compounds and improve their palatability.

Safe Dosage: There is no established standard dosage for turkey berry as an herbal remedy. If you plan to use it for medicinal purposes, start with small amounts and monitor your body's response. Avoid excessive or prolonged use.

Consultation with Healthcare Professionals: Before using turkey berry or any herbal remedy for medicinal purposes, consult with a qualified healthcare professional, especially if you have pre-existing health conditions, are pregnant or nursing, or are taking medications.

Although turkey berry is generally considered safe for culinary use and traditional herbal remedies when consumed in moderation, it is essential to exercise caution, especially when using it for therapeutic purposes. If any adverse effects or concerns about its safety arise it is recommended to seek medical advice promptly. Individuals with known allergies, sensitivities, or specific medical conditions should be particularly vigilant and consult healthcare professionals.

11 Commercial Value of Turkey Berry (*Solanum torvum*)

Solanum torvum holds commercial value in various ways, particularly in regions where it is cultivated and used for culinary and medicinal purposes. Some aspects of its commercial value include:

Culinary Use: Turkey berry is an integral part of the cuisine in many regions, particularly in South Asia, Southeast Asia, and Africa. It is used in a wide range of dishes, adding a unique flavor and texture to curries, stews, and stir-fries. In regions where it is in high demand, the sale of fresh turkey berries in local markets can contribute to the income of small-scale farmers.

Export and Trade: In countries where turkey berry is grown commercially, it may be exported to international markets to cater to the culinary preferences of immigrant populations and consumers interested in global cuisines. Turkey berries are sometimes dried and packaged for export.

Traditional Medicine: Turkey berry is used in traditional medicine systems in various cultures for its potential health benefits [84]. The sale of turkey berry-based herbal remedies and dietary supplements can contribute to the herbal products industry.

Herbal Products: Turkey berry is incorporated into herbal products, including capsules, tablets, herbal teas, and tinctures, which are sold in health food stores and online markets. These products are marketed for their potential digestive, antioxidant, and anti-inflammatory properties.

Agroforestry and Farming: Turkey berry cultivation can be part of agroforestry systems, contributing to diversified farming practices. It can be grown alongside other crops, providing additional income and resources to farmers [81].

Traditional Tonics and Elixirs: In some cultures, turkey berry is used as an ingredient in traditional health tonics and elixirs. These tonics are sold in local markets and may contribute to the livelihoods of traditional healers [83].

Pharmaceutical and Cosmetics Industry: Some phytochemicals found in turkey berry have potential applications in the pharmaceutical and cosmetics industries. Extracts or compounds from turkey berry may be used in the development of new products.

Sustainable Agriculture: Turkey berry cultivation can be part of sustainable agriculture practices, as it is a hardy plant that can thrive in various conditions. Its cultivation may contribute to soil fertility and crop rotation strategies [85].

Food Processing Industry: Processed turkey berry products, such as pickles and chutneys, are available in some markets and cater to consumer preferences for convenience.

Research and Development: Ongoing research on the phytochemicals and potential health benefits of turkey berry may lead to the development of new commercial products and applications.

The commercial value of turkey berry can vary by region and market demand. Additionally, the quality and safety of turkey berry products are essential considerations, especially in the herbal products and dietary supplement industries. Regulations and quality control measures play a role in ensuring the integrity of turkey berry-based products.

12 Conclusion

Solanum torvum is a versatile plant with a rich history of use in culinary and traditional medicinal practices. Its botanical characteristics, wide distribution, ethnobotanical uses, nutritional properties, potential health benefits, and culinary applications make it a valuable and intriguing plant. As interest in natural and traditional remedies continues to grow, turkey berry remains an important resource with the potential for further scientific exploration and utilization in modern healthcare and cuisine. Turkey berry (*S. torvum*) is a remarkable plant with a rich history of cultivation, culinary use, and traditional medicine across diverse cultures and regions. Its origin in tropical and subtropical regions of Africa and Asia has given rise to its widespread distribution and adaptability to different climates and habitats. Its versatility in culinary applications, potential health benefits, and cultural significance make it a valuable and intriguing plant that continues to thrive and contribute to various aspects of human life around the world. Understanding its origin and distribution provides insight into the global significance of this unique plant. The botany of turkey berry (*S. torvum*) encompasses its growth habit, leaves, flowers, fruits, and adaptability to various environments. Its simple, yet distinctive, botanical features make it easily recognizable, and its versatility in terms of growth conditions and culinary uses has contributed to its significance in different cultures around the world. Pharmacognosy plays a crucial role in the comprehensive study of turkey berry, encompassing its morphological identification, phytochemical analysis, medicinal uses, pharmacological properties, and quality control. This multidisciplinary approach contributes to the understanding and utilization of turkey berry in traditional and modern medicine.

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Garcinia kola Heckel. (Clusiaceae): An Overview of the Cultural, Medicinal, and Dietary Significance for Sustainability

9

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Abstract

This chapter aims to discuss the cultural, medicinal, and dietary significance of *Garcinia kola* Heckel (Clusiaceae) from a sustainability standpoint. The small- to medium-sized evergreen tree is native to the dense humid tropical rainforest ecosystems of West and Central Africa and grows alongside other traditionally valued forest crops. Also known as bitter kola or kola, the plant is deeply rooted in the cultures and traditions of West and Central Africa and has played a significant role in the lives of indigenous communities for centuries by offering a rich tapestry of uses and cultural significance. It bears bitter-tasting kola seeds. These seeds, with their intense bitterness, have become iconic and emblematic of the region's cultural heritage. Some food uses of bitter kola include snacks, flavoring agents, preparation of traditional soup, beverage infusion, medicinal concoctions added to food, and condiments and seasoning. Some of the cultural significance of *G. kola* include as a symbol of hospitality, traditional rituals and ceremonies, traditional medicine, food, and flavoring (as seasoning), stimulant and energy boost, social and ceremonial uses (like weddings and religious gatherings), offering and prayers, cosmetic and skin care, trade and economic value, and cultural conservation. Bitter kola is renowned for its various medicinal properties, and it has been used in traditional African medicine for centuries. It contains garcinol which has shown promise in research for its anticancer effects as well as kolaviron which has therapeutic effects. Some medicinal uses and potential health benefits of bitter kola include antimicrobial properties, antioxidant activity, anti-inflammatory effects, hepatoprotective properties, antidiabetic, antimalarial activity, cardiovascular health, analgesic and anti-inflammatory effects, anticancer potential, digestive aid and gut health improvement, aphrodisiac, and weight loss and management. The use of *Garcinia kola* for health management is generally considered safe when consumed and used in moderation. However, like many natural products, bitter kola should be used with caution, and excessive consumption can lead to potential adverse effects due to the bitter taste and sensitivity, caffeine content, potential allergic reactions, gastrointestinal distress, interactions with medications, and liver distress from overconsumption. Some herbal formulations prepared based on traditional knowledge that contain bitter kola include bitter kola extracts or tinctures, capsules, supplements, traditional herbal remedies, chewing sticks (snacks), traditional drinks and tonics (bitters), and honey infusions. This chapter suggests the need to incorporate sustainable practices and further research into the phytomedicinal and traditional benefits of the plant.

Keywords

Bitter kola · Kola · Nuts · Sustainability · Ethnomedicine · Cultural value · Molecular docking · Herbal formulations · Herbal medicine

1 Introduction

Garcinia kola Heckel, commonly known as bitter kola, is a tropical flowering plant from the plant family, Clusiaceae. It is native to West and Central Africa and is particularly found in countries like Nigeria, Cameroon, Ghana, and Cote d'Ivoire. Bitter kola is a small, evergreen tree that produces fruit pods containing seeds known as kola nuts or simply kola. The intensely bitter-tasting kola may not be to everyone's liking and consumption is often limited to small quantities or specific preparations. These seeds are revered for their cultural, medicinal, and dietary significance in the regions where they grow and beyond [81]. Other diverse uses are linked to the phytochemical constituents of the fruits and tree bark which include several bioactive compounds [67, 69, 70]. They may be incorporated into some food products because of their medicinal and cultural value but not used as a staple food source in most diets within the native range of the crop [47]. Instead, bitter kola is valued for its role in traditional practices, customary events, festivities, and family celebrations as well as health benefits. Therefore, the tree is celebrated for its cultural, medicinal, and dietary significance in the regions where it grows. The seeds of *G. kola* are the most celebrated part of the plant and are commonly referred to as kola nuts. These seeds are known for their cultural significance, traditional uses, and potential medicinal properties [48]. Bitter kola seeds are used as a symbol of hospitality, a traditional remedy for various ailments, and an ingredient in traditional rituals and ceremonies [56]. Bitter kola holds a special place in the cultural and social traditions of many West and Central African communities. It is often used as a symbol of hospitality and is offered to guests as a sign of respect and welcome. In some cultures, presenting a kola nut during gatherings or ceremonies is a customary gesture.

The phytochemical compounds found in *G. kola* are not only beneficial for human health but also contribute to the defense mechanisms, ecophysiology, and adaptive capabilities of the plants, helping them ward off pests, diseases, and environmental stressors [95]. Bitter kola is rich in phytochemical constituents like flavonoids, polyphenols, alkaloids, saponins, tannins, terpenes, and many others, and these compounds have actual and potential health benefits against chronic diseases such as cancer, heart disease, and diabetes [71, 81]. Bitter kola is typically consumed in small quantities due to its intense bitterness. Its primary use is for its cultural and traditional significance, as well as its potential role in traditional medicine for various ailments [64, 80]. As such, bitter kola is not considered a significant dietary source of essential nutrients, but it may offer certain health benefits in the context of traditional medicine and folklore [65, 82]. However, scientific research on its specific nutritional content and health effects is limited, and further studies are needed to better understand its potential contributions to human health. Additionally, the ethnobotany of the bitter kola tree is a testament to the intricate relationship between people and plants in West and Central Africa [97–99, 114, 118]. The plant has deep roots in the traditions and folklore of West and Central Africa, which include:

Symbol of Hospitality: Bitter kola is considered a symbol of hospitality in many African cultures. It is often presented to guests as a gesture of respect and welcome. In some regions, offering a kola nut during social gatherings or ceremonies is a customary practice.

Traditional Medicine: Bitter kola has a rich history in traditional African medicine. It has been used for centuries to treat various ailments and is renowned for its medicinal properties [63, 106]. The seeds are believed to possess numerous health benefits, and they are used in various remedies.

Rituals and Ceremonies: Bitter kola is incorporated into various traditional rituals and ceremonies. The presence of kola is often associated with spiritual and cultural practices, including naming ceremonies, weddings, and religious rites.

Economic Value: The kola nut trade has economic significance in some regions, with bitter kola being an important commodity for trade and exchange [89–91]. Beyond its botanical characteristics, *G. kola* is culturally, medicinally, and economically significant. The seeds of the plant, known as kola nuts, have been used traditionally for various purposes, including:

Cultural and Social Traditions: Kola nuts are often presented as a symbol of hospitality and respect in many African cultures [62]. They are commonly offered to guests during social gatherings and ceremonies.

Traditional Medicine: Bitter kola has a rich history in traditional African medicine. It is believed to have various medicinal properties and has been used to treat ailments such as digestive issues, infections, and coughs [25]. Some people also associate it with aphrodisiac properties and improved energy levels [12].

Clearly, from its role as a symbol of hospitality to its use in traditional medicine, bitter kola weaves itself into the fabric of daily life and cultural practices. Its bitter taste is not just a flavor but a reflection of its deep-rooted significance. The use of bitter kola extends far beyond its nutritional value; it embodies a cultural identity, a source of livelihood, and a bridge between the physical and spiritual realms. As communities continue to cherish and celebrate bitter kola, they also recognize the importance of conserving this invaluable botanical treasure for future generations. In addition to its medicinal and cultural significance, bitter kola is consumed as a dietary supplement in some regions. The kola nuts are chewed or ground into a powder to make a paste that can be mixed with water or other beverages. Bitter kola is known for its intensely bitter taste, which is attributed to the presence of compounds like caffeine and theobromine [105].

G. kola, like many other rainforest plants, plays a significant role in its native ecosystem. It contributes to the ecological balance and provides essential services to the environment and wildlife [83]. The kola nut seeds are consumed by various animals in the rainforest, including rodents, primates, and birds. These animals aid in the dispersal of the seeds, helping the plant propagate and colonize new areas. Bitter kola contributes to the biodiversity of the rainforest by providing habitat and sustenance for a variety of wildlife species. It is part of the intricate web of life that characterizes these lush ecosystems. Some animals in the rainforest may also benefit from the medicinal properties of bitter kola. For instance, primates that consume the seeds may experience some of the plant's therapeutic effects [81]. While *Garcinia kola* thrives in its natural habitat, it faces several conservation challenges, primarily due to deforestation and habitat destruction in African rainforests. As human populations and economic activities expand, the rainforests are increasingly threatened by logging, agriculture, and urbanization. This poses a risk to the biodiversity and survival of plant species like bitter kola. Efforts are being made by conservation organizations and governments in African countries to address these challenges [92, 96, 115, 117]. These initiatives include promoting sustainable forestry practices, establishing protected areas, and raising awareness about the importance of preserving the rainforest ecosystem.

This chapter aims to discuss the cultural, medicinal, and dietary significance of *G. kola* from a sustainability standpoint. The chapter presents the origin, distribution, and botany of *G. kola* as well as the diverse food and traditional medicine uses. Other parts of the chapter focus on the ethnobotanical, pharmacognosy, and clinical works, herbal products, toxicity, and commercial values of *G. kola*. It is believed that bitter kola has various medicinal properties, including anti-inflammatory, antioxidant, anti-inflammatory, and antimicrobial effects [53]. *G. kola* is a crop with great potential for sustainable cultivation. The small- to medium-sized evergreen tree is native to the dense humid rainforest ecosystems of West and Central Africa and grows alongside other traditionally valued forest crops. This chapter suggests the need to incorporate sustainable practices, including organic cultivation, agroforestry, and further research into the phytomedicinal and traditional benefits of the plant such as the use of molecular docking techniques with compounds from *G. kola* using the 3D structures of these compounds and target proteins, as well as specialized software and expertise in computational chemistry.

2 Origin and Distribution of *Garcinia kola*: The Bitter Kola Tree

The center of origin of *G. kola* is believed to be in the rainforests of West and Central Africa where the plant has enjoyed a long history of cultivation and widespread distribution throughout the region (Fig. 1). However, it is generally accepted that bitter kola's natural habitat is within the lush, tropical rainforests of Africa, which includes several countries like Nigeria, Cameroon, Ghana, Cote d'Ivoire (Ivory Coast), Sierra Leone, Liberia, Equatorial Guinea, Gabon, Congo, and the Democratic Republic of Congo in lush, humid environments [107]. These areas are also known for their rich biodiversity and diverse flora within which *G. kola* has naturally thrived for centuries [83]. In these rainforests, bitter kola has deep cultural, medicinal, and ecological significance. It has been used in traditional medicine, is a symbol of hospitality, and plays a role in the local ecosystems as a source of sustenance for various wildlife species. Wild stands of *G. kola* are also supported by present and ongoing cultivation efforts.

In Nigeria, the plant is extensively grown in various southern states such as Ogun, Ondo, Edo, and Cross River. Nigerian farmers have played a significant role in the production of bitter kola, making it a substantial part of the country's agricultural economy [61]. In Cameroon, *G. kola* is also prevalent in the southern

Fig. 1 Distribution of *Garcinia kola* in Africa. (Source: Maňourová et al. [81] (CC BY 4.0))

and central regions of the country. The plant is found in the dense rainforests that characterize these areas. Bitter kola is cultivated and harvested in regions with suitable rainforest conditions in Ghana and has a long history of traditional use especially in traditional ceremonies. In Cote d'Ivoire, kola can be found in the rainforests of the southwestern part of the country. Nigeria, Cameroon, Ghana, and Cote d'Ivoire are among the primary producers of bitter kola but the plant is also present in other West and Central African nations with suitable rainforest habitats including Sierra Leone, Liberia, Equatorial Guinea, Gabon, Congo, and the Democratic Republic of Congo. It thrives in the humid, equatorial climate and is well adapted to the dense, evergreen rainforests where it receives abundant rainfall and high levels of humidity.

The life span of *G. kola* can vary depending on a range of factors, including environmental conditions, disease, and other stressors [108]. Generally, bitter kola is a long-lived tree with a potential for longevity under favorable circumstances. In the wild, *G. kola* life span can extend from several decades to potentially over a century. However, in cultivated settings or agroforestry systems, the life span may be influenced by factors such as management practices and environmental conditions. The longevity of *G. kola* is closely tied to the environmental conditions in which they grow. Adequate rainfall, suitable soil conditions, and a lack of significant stressors can contribute to longer life spans. Conversely, factors like drought, disease, or damage from pests may shorten a tree's life. The susceptibility of bitter kola trees to diseases and pests can influence their lifespan. Healthy trees that are well-maintained and protected from threats are more likely to live longer. Human activities, such as logging and habitat destruction, can significantly impact the life span of bitter kola trees in the wild. Deforestation and logging can lead to prematurely removing these trees from their natural habitat. The presence of young trees and successful regeneration through seed dispersal, as well as the presence of both male and female trees for pollination (as bitter kola is dioecious), can contribute to the perpetuation of the species. It is worth noting that *G. kola* has no specific fixed life span, individual trees can live for several decades or more under favorable conditions. Bitter kola can thrive in its natural rainforest habitat due to the long reproductive cycle and adaptive capabilities of the plant. However, like many species in tropical rainforests, the ongoing challenges of habitat loss and environmental changes due to human activities can pose threats to its continued existence in the wild. Conservation efforts are essential to protect and sustain populations of *G. kola* and other valuable rainforest species.

3 Botanical Overview of *Garcinia kola*

G. kola belongs to the plant family Clusiaceae (formerly known as Guttiferae) within the broader classification of angiosperms, eudicots, and Malpighiales. The *Garcinia* genus comprises over 200 species of flowering plants. The genus is known for containing numerous species, many of which are tropical evergreen trees and shrubs. *Garcinia kola* shares its genus with other *Garcinia* species, some of which have also been of cultural and economic importance. Some relatives of *G. kola* are *Garcinia mangostana* (mangosteen), *Garcinia cambogia* which has small, pumpkin-shaped fruit and is native to Southeast Asia and India, *Garcinia gummi-gutta*, which is widely used in Southeast Asia in traditional cooking due to the strong sour flavor to dishes, *Garcinia indica* (kokum or kokam), which is native to the Western Ghats region of India and valued for its fruit that is used in culinary applications and traditional medicine, *Garcinia atroviridis* (asam gelugur), a tropical fruit tree native to Southeast Asia with sour fruits used in traditional cooking, especially in Malaysian and Thai cuisine and *Garcinia livingstonei* (imbe) that is native to Southern Africa and has edible fruits with a sweet and tangy flavor. *G. kola* is a small- to medium-sized evergreen tree with a straight trunk and dense, glossy leaves [45]. The tree produces fruit pods containing seeds, which are commonly referred to as kola nuts. These seeds are the most renowned part of the plant and are known for their intensely bitter taste.

3.1 Morphological Characteristics of *Garcinia kola*

The plant is a small- to medium-sized evergreen tree that typically reaches heights ranging from 7 to 15 m, although some specimens can grow up to 30 m under favorable conditions. The leaves of *G. kola* are simple, glossy, and dark green [45]. Leaves are arranged oppositely on the branches and are elliptical or oblong. The leaves have a prominent midrib and lateral veins. Bitter kola trees produce small, greenish-yellow to pale-green flowers [24]. The flowers are unisexual, meaning they occur as either male or female on the same tree. They are typically solitary or arranged in clusters at the leaf axils. The fruit of *G. kola* is a berry-like capsule. It is approximately 5–7 cm in diameter and contains several seeds. When ripe, the fruit turns brown or reddish-brown. Inside the fruit are the seeds, which are commonly referred to as kola nuts. These seeds are the most celebrated part of the plant and are known for their intensely bitter taste. Seeds of *G. kola* are brown, oblong, and roughly 2–4 cm in length. They have a hard outer shell and are divided into segments, each containing a cotyledon. The bitterness of the seeds is due to the presence of compounds such as caffeine and theobromine [56, 102].

3.2 Reproductive Features of *Garcinia kola*

Garcinia kola is a dioecious plant, meaning it has separate male and female flowers on different trees. To produce fruit, both male and female trees need to be present in close proximity for pollination to occur. The reproductive features of *G. kola* are essential for understanding how this plant reproduces and produces seeds for propagation. Kola is a dioecious plant, which means it has separate male and female trees. In dioecious species, individual trees are either male or female, and both types are required for successful reproduction. This reproductive strategy is common in many tree species and ensures genetic diversity in the population. The flowers of *G. kola* are unisexual, meaning that individual flowers are either male or female. This characteristic is typical in dioecious plants. The male flowers produce pollen, which contains the male reproductive cells (sperm). These flowers typically have stamens with anthers that release pollen. Female flowers have structures called carpels that contain the ovules, which eventually develop into seeds if fertilized. The female flowers receive pollen from the male flowers to initiate fertilization. The flowers are typically pollinated by insects, particularly bees and butterflies. These insects transfer pollen from the male flowers to the female flowers while foraging for nectar. After successful pollination, the female flowers develop into fruit. The fruit of *G. kola* is a berry-like capsule. Inside the fruit, the fertilized ovules develop into seeds. These seeds are commonly referred to as kola nuts. The number of seeds produced can vary depending on various factors, including the health of the tree and the effectiveness of pollination. Bitter kola seeds are primarily dispersed by animals. Various wildlife species, including rodents, primates, and birds, consume the seeds and then disperse them in different locations [88]. This process aids in seed dispersal and allows new plants to be established in different areas. Once a bitter kola seed is dispersed and finds a suitable environment, it germinates. Germination is the process by which a seed sprouts and begins to grow into a new plant. As the seedling grows, it develops into a juvenile tree, eventually reaching maturity and producing flowers and fruits. This reproductive cycle continues as long as there are both male and female trees in proximity to facilitate pollination. This reproductive strategy enhances genetic diversity within the species and contributes to the sustainability of bitter kola populations in their natural habitat.

4 Phytochemical Constituents of *Garcinia kola*

Bitter kola is rich in biologically active compounds and some of them are responsible for the color, flavor, and disease resistance of the plants. These chemical compounds are not necessarily considered essential nutrients like vitamins and minerals, but they play a crucial role in human health [6, 34, 35]. Consuming a diet rich in phytochemicals has been associated with numerous health benefits, including antioxidant and anti-inflammatory effects, as well as potential protection against chronic diseases such as cancer, heart disease, and diabetes [43, 71]. The phytochemicals in *G. kola* include flavonoids, polyphenols, alkaloids, saponins, tannins, terpenes, and many others [46].

Some of the alkaloids found in *G. kola* are caffeine and theobromine and they are known to possess pharmacological effects on humans [58, 59]. For instance, caffeine has a stimulating effect on the central nervous system and can increase alertness and energy levels when consumed while theobromine has mild stimulant properties and can dilate blood vessels. Although the levels of caffeine and theobromine in bitter kola may not be as high as in coffee or chocolate, their presence contributes to the plant's stimulant properties [87]. Tannins found in *G. kola are* polyphenolic compounds that are known for their astringent taste [84]. They can bind to proteins and minerals, and they have antioxidant properties. Polyphenols are a group of naturally occurring compounds known for their antioxidant properties. The plant also contains ellagic acids, which are a kind of polyphenol with antioxidant properties [85]. Ellagic acid has been studied for its potential health benefits, including its role in protecting cells from oxidative damage. Examples of flavonoids found in *G. kola* are quercetin and kaempferol [18]. These compounds have antioxidant, anti-inflammatory, and potential anticancer properties [109]. Quercetin has been studied for its potential role in reducing the risk of chronic diseases while kaempferol has been associated with various health benefits, including cardiovascular health [55]. Saponins are natural compounds also found in kola with diverse biological activities and are often responsible for the foaming properties of certain plant extracts. Garcinol is a type of polyisoprenylated benzophenone found in *Garcinia* species, including bitter kola [27, 51, 52]. It has been studied for its potential anti-inflammatory and antioxidant properties. Garcinol has also shown promise in research for its anticancer effects. Kolaviron is a complex bioactive compound found in bitter kola [86]. It is a mixture of flavonoids and other bioactive compounds. Kolaviron has been extensively studied for its potential therapeutic effects, including its antioxidant, anti-inflammatory, and hepatoprotective properties [28]. It is considered one of the key

Table 1 Health benefits of phytochemicals found in *Garcinia kola*

Phytochemicals in *G. kola*	Health benefits and uses
Alkaloids	
Caffeine	Natural stimulant, increases alertness and energy. Potential to improve mental focus and concentration. May alleviate fatigue and drowsiness. Used in some traditional remedies for its stimulating effects.
Theobromine	Potential vasodilatory properties may support cardiovascular health. Mild stimulants can have a relaxing effect on smooth muscles.
Tannins	
Ellagic	Potential role in cancer prevention and treatment. May have anti-inflammatory effects. Strong antioxidant, helps protect cells from oxidative damage.
Flavonoids	
Quercetin	Potential for reducing the risk of chronic diseases. May support heart health by improving blood vessel function. Anti-inflammatory properties may reduce inflammation. Potent antioxidant, protects cells from free radical damage.
Kaempferol	Potential anticancer properties. Cardiovascular benefits, support heart health. Antioxidant helps neutralize harmful free radicals. Anti-inflammatory effects may reduce inflammation.
Saponins	
Garcinol	Investigated for its potential in cancer prevention and treatment. Antioxidant effects, help combat oxidative stress. Potential anti-inflammatory properties.
Kolaviron	Strong antioxidant, that protects cells and tissues from oxidative damage. May have antimicrobial and antiviral activities. Hepatoprotective properties support liver health. Anti-inflammatory effects, may alleviate inflammation.
Polyphenols	
Gallic acid	Potent antioxidant, that protects cells from oxidative stress. Potential for cancer prevention and treatment. May support heart health by improving blood vessel function. Anti-inflammatory effects, reduce inflammation.

compounds responsible for some of the medicinal properties attributed to bitter kola. Bitter kola contains gallic acid, a polyphenolic compound with antioxidant properties. Gallic acid has been studied for its potential health benefits, including its role in protecting cells from reactive oxygen species. Some of the health benefits of secondary phytochemicals found in bitter kola are presented in Table 1.

5 Nutritional Properties of *Garcinia kola*

Bitter kola is used as a food and flavoring ingredient in various cultural contexts and is primarily known for its intense bitterness, which is not always suitable for all tastes [17, 93, 103, 104]. Additionally, it is consumed in moderation due to its bitter taste and potential effects on digestion. Common food uses of *Garcinia kola* are presented in Table 2.

Table 2 Common food uses of bitter kola

Food and related uses	Description
Snack (chewing snack)	Bitter kola seeds are often chewed as a snack. They have a distinctly bitter taste and are enjoyed for their flavor and potential health benefits.
Flavoring agent	Ground or powdered bitter kola is sometimes used as a natural flavoring agent in traditional African cuisine, adding a unique bitterness to dishes.
Traditional soup	Bitter kola may be added to traditional soups and stews, especially in West African cuisine, to enhance the flavor and impart its distinctive bitterness.
Beverage infusion	Bitter kola can be infused in water or other liquids to create a bitter, aromatic beverage. This infusion is sometimes used for its potential medicinal properties.
Medicinal concoction	In some cultures, bitter kola is used to prepare medicinal concoctions or teas to address various health issues, such as digestive discomfort or coughs.
Condiments and seasoning	Ground or crushed bitter kola can be used as a condiment or spice to season dishes, similar to how other spices like pepper or chili are used.

Garcinia kola is consumed for its cultural, traditional, and potential medicinal properties rather than its nutritional content. However, bitter kola is not typically consumed as a primary source of nutrition, it does contain some nutrients and bioactive compounds. Some nutritional properties of *G. kola* include:

Bitterness: The most prominent characteristic of bitter kola is its intense bitterness, which is attributed to the presence of compounds like caffeine and theobromine. This bitterness is not related to its nutritional value but rather to its flavor profile [110].

Phytochemicals: Bitter kola contains bioactive phytochemicals many of which have known health benefits like stimulatory effects on the central nervous system and antioxidant properties. Bitter kola contains antioxidants, which can help protect cells from oxidative damage. Antioxidants play a role in overall health and may contribute to the plant's potential health benefits [111–113]. Kolaviron which is also present in bitter kola has therapeutic effects. These compounds are more related to their traditional and medicinal uses rather than their nutritional content [119].

Vitamins and Minerals: Bitter kola contains trace amounts of certain vitamins and minerals, although not in significant quantities to meet daily dietary requirements. These may include small amounts of vitamin C, vitamin A, and calcium [6, 34, 35].

Fiber: Bitter kola may contain dietary fiber, although the fiber content is relatively low. Fiber is important for digestive health and can help regulate bowel movements.

6 Traditional Uses of *Garcinia kola*

The ethnobotany of *Garcinia kola* is deeply rooted in the cultures and traditions of West and Central Africa. This remarkable plant has played a significant role in the lives of indigenous communities for centuries, offering a rich tapestry of uses and cultural significance. The small- to medium-sized evergreen tree that is native to the

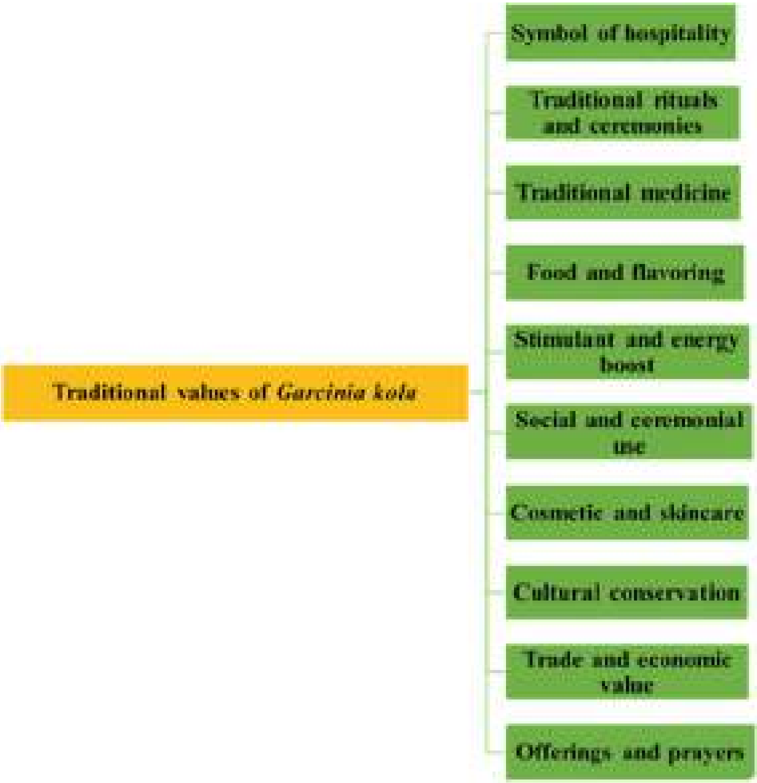

Fig. 2 Cultural values of *Garcinia kola*

dense humid rainforests of West and Central Africa bears bitter-tasting kola seeds. These seeds, with their intense bitterness, have become iconic and emblematic of the region's cultural heritage. Some of the cultural significance of *G. kola* include (Fig. 2):

- **Symbol of Hospitality**: Bitter kola is synonymous with hospitality in many West African cultures. It is customary to offer kola nuts to guests as a gesture of warm welcome and respect. The act of sharing bitter kola during social gatherings and ceremonies signifies unity and goodwill.
- **Traditional Rituals and Ceremonies**: Bitter kola plays a pivotal role in various traditional rituals and ceremonies across West and Central Africa. Its presence can be seen in rites of passage, ancestral worship, and spiritual offerings. The nuts are often included in sacrificial offerings to deities, ancestors, or spirits, depending on the cultural context.
- **Traditional Medicine**: One of the most profound ethnobotanical aspects of bitter kola is its esteemed place in traditional African medicine [22]. The plant has been harnessed for centuries to address a wide array of health issues. It is traditionally

believed to have therapeutic properties for ailments such as coughs, colds, malaria, and stomach aches [38, 44]. Others are:

  - **Digestive Aid**: The intense bitterness of bitter kola is thought to be beneficial for digestion [25]. It is often consumed after meals to alleviate digestive discomfort and promote healthy digestion.
  - **Antimicrobial Properties**: In traditional practices, bitter kola has been used for its potential antimicrobial properties [42, 54]. It is regarded as a natural remedy to combat certain infections.
  - **Anti-inflammatory Effects**: Some traditional uses involve the application of bitter kola for its anti-inflammatory effects [1]. It is used externally to reduce inflammation and promote healing.
- **Food and Flavoring**
  - **Chewing Snack**: Bitter kola seeds are occasionally chewed as a snack. The unique and intense bitterness of the seeds is appreciated by those who have acquired a taste for them. It is a common sight to see people chewing bitter kola, especially during social gatherings and leisure time.
  - **Seasoning**: Bitter kola can be ground or crushed and used as a seasoning or condiment in traditional African dishes. It imparts a distinctive bitterness to the food, enhancing its flavor. Bitter kola-infused dishes are celebrated for their unique taste.
- **Stimulant and Energy Boost**
  - **Caffeine Content**: Bitter kola contains caffeine, a natural stimulant. This caffeine content contributes to its traditional use as a source of increased alertness and energy. It is often consumed to combat fatigue and drowsiness, especially during long journeys or night vigils.
- **Social and Ceremonial Use**
  - **Weddings and Celebrations**: Bitter kola occupies a prominent place in weddings and celebratory ceremonies. It is shared among participants as part of the cultural festivities, symbolizing joy and unity.
  - **Religious Gatherings**: In both cultural and religious gatherings, bitter kola is distributed among attendees as a symbol of unity and shared values. It is believed to foster harmony and goodwill among the participants.
  - Traditional ritual: Bitter kola is used in various traditional rituals and ceremonies in Africa, often associated with spiritual and cultural practices.
- **Offerings and Prayers**: Bitter kola is considered a sacred plant of spiritual significance by some communities. It is used in offerings and prayers to seek protection, blessings, or favor from the spiritual realm [60]. The intense bitterness of the seeds is believed to make them particularly potent in spiritual contexts.
- **Cosmetics and Skincare**: Bitter kola has been used topically for skin care and cosmetic purposes. It is believed to have properties that benefit the skin, such as reducing blemishes and promoting a healthy complexion.
- **Trade and Economic Value**: Bitter kola holds commercial economic significance beyond its cultural and traditional roles. It is traded and sold in local markets, serving as a source of income for communities engaged in its cultivation and trade [116]. This economic value has contributed to its continued cultivation and importance.

- **Cultural Conservation**: The cultural and traditional importance of bitter kola has sparked efforts to conserve and protect this species and its natural habitat. Communities recognize the need to preserve the tree that holds such a central place in their heritage [94, 100, 101].

7 Modern Scientific Research on Ethnomedical Uses and Pharmacological Potentials of *Garcinia kola*

Garcinia kola has also been studied for its potential pharmacological properties. These properties are largely attributed to the various phytochemical constituents found in the plant. Pharmacological properties of *Garcinia kola* include (Fig. 3):

Fig. 3 Health benefits of *Garcinia kola*

Antimicrobial Properties: Bitter kola has been investigated for its antibacterial properties [68]. Some studies suggest that extracts from bitter kola may have inhibitory effects against certain pathogenic bacteria, potentially making it useful in combating bacterial infections [23, 42]. Research has shown that bitter kola contains compounds with antimicrobial properties, which may help combat certain bacteria and viruses [57]. It has been used traditionally to treat infections, coughs, and colds. Research has also indicated that bitter kola extracts may exhibit antifungal activity [66]. This property could have applications in addressing fungal infections.

Antioxidant Activity: Bitter kola contains various phytochemicals, including flavonoids, polyphenols, and tannins, that possess antioxidant properties [68]. Antioxidants help neutralize harmful free radicals, reducing oxidative stress and potentially contributing to overall health. The presence of compounds like ellagic acid, quercetin, and gallic acid in bitter kola contributes to its antioxidant effects [72]. Bitter kola can help protect the body's cells from oxidative damage, potentially contributing to overall health and well-being [4].

Anti-Inflammatory Effects: Some studies have explored the anti-inflammatory potential of bitter kola. Inflammation is a key factor in various chronic diseases, and the anti-inflammatory properties of bitter kola compounds like garcinol and kolaviron may have therapeutic implications [2, 3, 5, 73]. Some compounds found in bitter kola have anti-inflammatory properties, which may make it beneficial for alleviating pain and inflammation in conditions like arthritis.

Hepatoprotective Properties: Kolaviron, a bioactive compound found in bitter kola, has been studied for its hepatoprotective (liver-protective) effects [2, 3]. It may help protect the liver from damage caused by toxins and oxidative stress ([8, 26]; Adaramoye and Adeyemi 2012). This hepatoprotective property is particularly significant in regions where liver diseases are prevalent.

Antidiabetic Potential: Some research has suggested that bitter kola may have antidiabetic properties. Extracts from bitter kola have been studied for their potential to lower blood glucose levels and improve insulin sensitivity [7, 19, 63]. These effects could be beneficial for individuals with diabetes or those at risk of developing the condition.

Antimalarial Activity: Traditional medicine practices have long used bitter kola for its potential antimalarial properties [38]. While more research is needed, some studies have explored the plant's efficacy against the malaria parasite. Bitter kola's antimalarial potential may be attributed to its phytochemical constituents.

Cardiovascular Health: The antioxidant and anti-inflammatory properties of certain bitter kola compounds may have positive effects on cardiovascular health [20, 22, 49]. These properties can help reduce the risk of cardiovascular diseases by protecting blood vessels and reducing inflammation.

Analgesic and Anti-Inflammatory Effects: Some traditional uses of bitter kola involve its application as an analgesic (pain reliever) and anti-inflammatory agent, particularly for addressing musculoskeletal pain and inflammation [79]. These effects may be attributed to its anti-inflammatory compounds, which can provide relief from various painful conditions.

Anticancer Potential: While research is in its early stages, some studies have explored the potential anticancer properties of bitter kola compounds, including garcinol and kolaviron [9, 10, 21]. These compounds have demonstrated anti-cancer effects in laboratory studies, but further research is needed to determine their clinical applications [29, 74].

Digestive Aid and Gut Health Improvement: Bitter kola is often consumed to relieve digestive discomfort, such as bloating, indigestion, and constipation [75]. It is believed to stimulate digestion and promote the flow of gastric juices.

Aphrodisiac: Bitter kola is sometimes regarded as an aphrodisiac and is believed to enhance sexual performance and libido in some cultures [14–16]. However, scientific evidence supporting these claims is limited.

Weight Loss: Some individuals have explored the potential of bitter kola for weight loss due to its reputation as an appetite suppressant [72, 75, 76]. However, more research is needed to confirm its effectiveness for this purpose.

Although bitter kola shows promise in various pharmacological areas, much of the research is preliminary, and further studies are required to fully understand its mechanisms of action and therapeutic potential [82]. Additionally, the traditional uses of bitter kola should not replace conventional medical treatments, and individuals with specific medical conditions should consult healthcare professionals for appropriate guidance and treatment.

8 Toxicity of *Garcinia kola*: Safety Considerations and Side Effects

The use of *Garcinia kola* for health management is generally considered safe when consumed and used in moderation. It has a long history of traditional use in West and Central Africa for various purposes, including as a cultural symbol, a traditional remedy, and a dietary component. However, like many natural products, bitter kola should be used with caution, and excessive consumption can lead to potential adverse effects. Some potential toxic considerations associated with bitter kola include:

- **Bitter Taste and Sensitivity**: The most notable characteristic of bitter kola is its intensely bitter taste. Some individuals may find the taste unpleasant or intolerable, which can limit their consumption. Additionally, individuals with heightened sensitivity to bitter flavors may experience discomfort when consuming bitter kola.
- **Caffeine Content**: Bitter kola contains caffeine, a natural stimulant. While the caffeine content in bitter kola is not as high as in coffee or certain other sources, excessive consumption of caffeine can lead to side effects such as nervousness, restlessness, insomnia, and increased heart rate. People who are sensitive to caffeine should consume bitter kola in moderation.
- **Potential Allergic Reactions**: As with any food or natural product, some individuals may be allergic to specific components of bitter kola. Allergic reactions

can range from mild symptoms like itching or hives to more severe reactions that may require medical attention. If you have known food allergies or experience allergic symptoms after consuming bitter kola, you should avoid it.

- **Gastrointestinal Distress**: Excessive consumption of bitter kola seeds can sometimes lead to gastrointestinal discomfort, including stomachaches, indigestion, or diarrhea. The bitter taste of the seeds may also trigger nausea in some individuals.
- **Interactions with Medications**: Bitter kola contains bioactive compounds that could potentially interact with certain medications. If you are taking prescription medications or have underlying health conditions, it is advisable to consult with a healthcare professional before adding bitter kola to your diet to avoid potential interactions.
- **Liver Health**: While bitter kola has been studied for its potential hepatoprotective (liver-protective) properties, consuming extremely large quantities of bitter kola or its extracts may have unintended effects on liver function [11, 13]. Moderation in consumption is key.
- **Pregnancy and Lactation**: Pregnant and breastfeeding women should be cautious when consuming bitter kola. Limited information is available on the safety of bitter kola during pregnancy and lactation, so it is advisable to consult a healthcare provider before using it during these periods.
- **Children and Seniors**: Due to differences in sensitivity and metabolism, children and seniors may be more vulnerable to the effects of bitter kola. Care should be taken to limit consumption and ensure that it does not adversely affect their health.
- **Individual Variability (Quality and Source)**: The quality and safety of bitter kola products can vary. It's important to obtain bitter kola from reputable sources to ensure it has been handled and processed appropriately. Individual tolerance and sensitivity to bitter kola can vary widely. What is well-tolerated by one person may cause discomfort in another. Start with small quantities to assess your response.

As with any dietary supplement or natural product, it is important to use bitter kola in moderation and be aware of your tolerance. However, individual tolerance and sensitivity can vary, so it's wise to exercise caution and practice moderation. If you have specific health concerns, allergies, or are taking medications, it is advisable to consult with a healthcare professional before incorporating bitter kola into your diet or using it for medicinal purposes. If you have specific health concerns or are taking medications, consult with a healthcare professional before incorporating bitter kola into your diet. Additionally, if you experience any adverse reactions after consuming bitter kola, discontinue use and seek medical advice.

9 Recommendation for Further Research: Molecular Docking Studies with Compound from *Garcinia kola*

Molecular docking is a computational technique used in drug discovery and bioinformatics to predict the binding interactions between small molecules (ligands) and target proteins (receptors) [36]. Conducting specific molecular docking studies with compounds from *G. kola* using the 3D structures of these compounds and target proteins, as well as specialized software and expertise in computational chemistry. General steps involved in molecular docking as suggested in Adewole et al. [18] include:

1. **Data Preparation**: Obtain the 3D structures of the compounds from Garcinia kola (e.g., using chemical structure databases or software tools). Acquire the 3D structure of the target protein or receptor of interest (e.g., from protein structure databases or experimental methods like X-ray crystallography).
2. **Ligand Preparation**: Prepare the ligands (compounds from Garcinia kola) by assigning partial charges and optimizing their 3D conformations. This step ensures that the ligands are in a suitable format for docking.
3. **Receptor Preparation**: Prepare the receptor (target protein) by assigning partial charges, adding hydrogen atoms, and optimizing its 3D conformation. This step ensures that the receptor is in a suitable state for docking.
4. **Grid Generation**: Create a grid or docking box around the active site of the target protein. The grid defines the region where ligands will be docked and scored for binding affinity.
5. **Docking**: Use molecular docking software (e.g., AutoDock, AutoDock Vina, or other specialized tools) to dock the prepared ligands into the binding site of the target protein. The software explores various conformations and orientations of the ligands within the binding site and evaluates their interactions with the receptor based on scoring functions that account for van der Waals forces, electrostatic interactions, and other factors. Docking simulations may include energy minimization and molecular dynamics simulations to refine the ligand-receptor complexes.
6. **Scoring and Analysis**: After docking, the software calculates binding scores and ranks the ligands based on their predicted binding affinities. Lower scores typically indicate stronger binding interactions. Researchers analyze the docking results to identify potential ligands from Garcinia kola that have favorable binding interactions with the target protein. Visualization tools are often used to examine the ligand-receptor complexes and understand the nature of the interactions, such as hydrogen bonds or hydrophobic contacts.
7. **Validation and Interpretation**: Molecular docking results should be validated through experimental studies, such as binding assays or structural biology techniques, to confirm the predicted binding interactions. Interpret the docking results in the context of the specific biological or pharmacological question being investigated.

It is important to emphasize that molecular docking is a computational tool for predicting binding interactions, and its accuracy depends on the quality of the input data and the reliability of the scoring functions used. Experimental validation is crucial to confirm the predictions made through docking studies. Additionally, conducting molecular docking studies with compounds from *G. kola* would require access to the 3D structures of these compounds, which may not be readily available in public databases. Therefore, specialized research efforts would be needed to obtain or generate these structures for docking studies.

10 Some Herbal Formulations Involving *Garcinia kola* and their Active Ingredients

Garcinia kola is often consumed as a whole seed or used in traditional remedies. However, it is less common to find standardized herbal formulations involving bitter kola, especially within commercial herbal products in Western markets [116]. However, some traditional and contemporary formulations do exist, often leveraging the bioactive compounds found in bitter kola for specific health benefits. Some examples include:

1. **Bitter Kola Extracts or Tinctures**: Some herbal product manufacturers offer bitter kola extracts or tinctures. These liquid formulations typically involve the extraction of bioactive compounds from bitter kola seeds using solvents. They may be marketed for their potential health benefits, including their antioxidant properties and use in traditional medicine.
2. **Bitter Kola Capsules or Supplements**: In some regions, you can find bitter kola capsules or dietary supplements. These products are designed to provide a standardized dose of bitter kola's bioactive compounds, often with a focus on its potential health benefits.
3. **Traditional Herbal Remedies**: In many traditional healing practices across West and Central Africa, bitter kola is used as an ingredient in various herbal remedies. These formulations may include other locally available medicinal plants and are often used to address specific health conditions. For example, bitter kola may be combined with other herbs to create a remedy for coughs or digestive issues.
4. **Bitter Kola Chewing Sticks**: Bitter kola is sometimes incorporated into chewing sticks or "oral health sticks." These sticks are used for oral hygiene and teeth cleaning. The natural bitterness of bitter kola is believed to help with oral health.
5. **Bitter Kola in Traditional Drinks and Tonics**: In some traditional cultures, bitter kola is added to herbal drinks, tonics, or concoctions. These preparations are believed to have various health benefits, such as energy enhancement, digestive support, and immune system boosting.
6. **Bitter Kola and Honey Infusions**: Bitter kola can be infused in honey to create a sweet and bitter combination. This infusion is sometimes used for its potential health benefits, including its supposed immune-boosting and energy-enhancing properties.

It is important to note that while these formulations may be based on traditional knowledge and practices, scientific research on the specific health effects of bitter kola and its active compounds is ongoing. If you're interested in using herbal formulations involving Garcinia kola, it's advisable to consult with a healthcare professional or herbalist who is knowledgeable about traditional remedies. They can guide the appropriate use and dosage of such products, as well as any potential interactions or contraindications with other medications or health conditions.

11 Commercial Value of *Garcinia kola*

Garcinia kola holds commercial value primarily due to its cultural significance and traditional uses, as well as its potential medicinal properties. Although it may not be a major cash crop like some other agricultural products, bitter kola plays a vital role in the economies of certain regions in West and Central Africa and some aspects of the commercial value of *G. kola* include:

Cultural Significance: Bitter kola is deeply embedded in the cultures and traditions of West and Central Africa. It is often used as a symbol of hospitality and goodwill. The exchange of bitter kola during social gatherings, ceremonies, and rituals is a common practice, contributing to its cultural value.

Traditional Medicine: Bitter kola has a long history of use in traditional African medicine. Many indigenous communities believe in its therapeutic properties for addressing various health issues, including coughs, colds, malaria, and digestive problems. This traditional use adds to its commercial value as a natural remedy.

Trade and Local Markets: Bitter kola is traded and sold in local markets across West and Central Africa. It serves as a source of income for local communities engaged in its cultivation and trade. The sale of bitter kola in local markets contributes to the livelihoods of many people.

Export Market: Bitter kola has gained some recognition in international markets due to its potential medicinal properties and as a unique botanical product. In recent years, there has been a growing interest in the export of bitter kola to other countries, including Europe and the United States, where it is sometimes used in traditional medicine practices or as a dietary supplement.

Pharmaceutical and Nutraceutical Industries: Some pharmaceutical and nutraceutical companies have shown interest in the potential medicinal properties of bitter kola's bioactive compounds. They may use bitter kola extracts or compounds in the development of herbal supplements or traditional medicine products, adding to its commercial value.

Food and Beverage Industry: Bitter kola is occasionally used as a flavoring or seasoning in traditional African dishes, contributing to its value in the culinary industry. It imparts a distinctive bitterness to food, enhancing its flavor.

Oral Health Products: Bitter kola is sometimes incorporated into oral hygiene products, such as toothpaste and chewing sticks. These products may be marketed for their potential oral health benefits.

Cosmetics and Skincare: In some regions, bitter kola is used topically for skincare and cosmetic purposes. It is believed to have properties that benefit the skin, such as reducing blemishes and promoting a healthy complexion.

Research and Development: Scientific research on bitter kola's potential medicinal properties and bioactive compounds has spurred interest in its commercial applications. This includes the development of herbal formulations and products.

The commercial value of bitter kola is not as prominent as that of major cash crops, its importance lies in its cultural heritage, traditional uses, and potential in various industries. As interest in natural remedies and traditional medicine grows, bitter kola continues to find a place in both local and international markets, contributing to the economies of the regions where it is cultivated and traded.

12 Future Prospects of Therapeutic Roles of Garcinia kola

The therapeutic roles of *Garcinia kola* have garnered significant interest due to its traditional uses and the presence of bioactive compounds with potential health benefits. Yet much research is still needed to fully understand its mechanisms of action and therapeutic potential because bitter kola holds promising prospects in various areas of healthcare and wellness. Some potential future therapeutic roles of bitter kola include:

- **Antioxidant and Anti-Inflammatory Properties**: Bitter kola is rich in bioactive compounds like kolaviron, quercetin, and ellagic acid, which possess antioxidant and anti-inflammatory properties [30, 31]. These properties make bitter kola a potential candidate for managing oxidative stress-related diseases and chronic inflammatory conditions [39].
- **Antimicrobial and Antiviral Activity**: Preliminary studies have suggested that extracts from bitter kola may exhibit antibacterial, antifungal, and antiviral properties [50]. These findings raise the possibility of using bitter kola as a natural remedy for infectious diseases.
- **Antidiabetic Potential**: Research has indicated that bitter kola extracts may help lower blood glucose levels and improve insulin sensitivity. With the increasing prevalence of diabetes worldwide, bitter kola may have a role in managing this chronic condition.
- **Hepatoprotective Effects**: Kolaviron, a bioactive compound in bitter kola, has been studied for its hepatoprotective (liver-protective) properties [32, 40]. As liver diseases remain a global health concern, bitter kola could have applications in supporting liver health.
- **Cardiovascular Health**: Bitter kola's antioxidant and anti-inflammatory properties may have positive effects on cardiovascular health. It could potentially contribute to reducing the risk of heart disease by protecting blood vessels and reducing inflammation [41, 51].
- **Antimalarial Activity**: Traditional medicine practices have long used bitter kola for its potential antimalarial properties [33]. Further research may lead to the development of natural antimalarial remedies.
- **Analgesic Uses**: Bitter kola has been traditionally used as an analgesic and anti-inflammatory agent for addressing musculoskeletal pain and inflammation. Its potential role in pain management warrants further exploration.
- **Cancer Research**: Some studies have investigated the anticancer properties of bitter kola compounds, such as garcinol and kolaviron [37, 51]. While research is in its early stages, these compounds have shown promise in laboratory studies as potential agents for cancer prevention and treatment [74].

- **Nutraceuticals and Dietary Supplements**: Bitter kola extracts and compounds may find applications in the development of nutraceuticals and dietary supplements aimed at promoting overall health and well-being [82].
- **Cosmeceuticals and Skincare Products**: Bitter kola's potential benefits for skin health and its antioxidant properties may lead to its incorporation into cosmeceuticals and skincare products.
- **Functional Foods and Beverages**: Bitter kola may be used as an ingredient in functional foods and beverages, offering consumers a natural source of bioactive compounds with potential health benefits.
- **Continued Research and Clinical Trials**: Future research endeavors, including well-designed clinical trials, will help validate the therapeutic roles of bitter kola and establish its safety and efficacy in specific medical conditions [77, 78].

Although bitter kola shows promise in various therapeutic areas, much of the research is still in the early stages, and further studies are required to establish its clinical applications. As research progresses, bitter kola may find its place in modern medicine and complementary healthcare, offering potential alternatives or adjuncts to existing treatments.

13 Conclusion

The evergreen foliage, dioecious reproductive system, and distinctive kola nuts make *G. kola* a unique and valuable plant species in the tropical rainforest ecosystem. Beyond its ecological role, the plant's cultural and medicinal importance has solidified its place in the hearts and traditions of the people of the region. Bitter kola is a culturally significant plant in West and Central Africa with a history of traditional medicinal use. It offers potential health benefits, but it is important to approach its consumption with caution, especially if you have underlying medical conditions or are taking medications. Always consult with a healthcare professional before incorporating bitter kola or any herbal remedy into your diet or wellness routine. Kola is a culturally, medicinally, and ecologically significant plant that has been integral to the local traditions and ecosystems in West and Central Africa and beyond for centuries. However, the plant is facing numerous sustainability and conservation challenges due to habitat destruction, efforts are underway to protect this valuable plant and the rich biodiversity of African rainforests. The cultural and historical importance of *G. kola*, combined with its potential health benefits, make it a captivating subject of study and a testament to the intricate relationship between people, plants, and the environment in which they thrive. Preserving the natural habitats of plants like bitter kola is not only essential for the well-being of local communities but also for the conservation of the Earth's biodiversity and the continued exploration of the secrets hidden within the rainforests of Africa.

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Vernonia amygdalina Delile (Asteraceae): An Overview of the Phytochemical Constituents, Nutritional Characteristics, and Ethnomedicinal Values for Sustainability

10

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Contents

Abstract

This chapter focuses on the ethnomedicinal values of bitter leaf (*Vernonia amygdalina* Delile (Asteraceae)) as well as the nutritional and phytochemical properties of the plant. The chapter discusses ongoing pharmacognostic and clinical research works on the plant as well as herbal products containing bitter leaf or its active ingredients, toxicity and safety considerations, and commercial

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values. *V. amygdalina* is a tropical perennial plant native to West Africa. The plant is widely valued for its culinary, medicinal, and nutritional uses and has a distinctive bitter taste. Its bitter taste, once moderated, adds a unique flavor profile to traditional dishes and soups. Bitter leaf is often exchanged as a symbol of hospitality and goodwill during traditional events in West Africa. In traditional African medicine, bitter leaf has a long history of use for its health benefits. It is used as a remedy for various ailments including malaria, fever, gastrointestinal issues, pain relief, immune system boosting, antimicrobial and anti-parasitic properties, nervous system regulation, cardiovascular health maintenance, digestive aid, fever reduction, diabetes management, stomachache, and others. These health benefits are linked to the phytochemical constituents of the plant. Prominent phytochemicals in bitter leaf include vernodalin, vernolepin, vernonioside, vernomygdin, and others. Further research to ensure the conservation of the plant and the sustainability of practices associated with it is paramount.

Keywords

Bitter leaf · Ethnomedicine · Phytochemistry · Bioactive compounds · Antioxidants · Antimicrobial · Culinary uses · Traditional medicine

1 Introduction

Vernonia amygdalina Delile (Asteraceae), also known as bitter leaf or African bitter leaf, is a tropical plant native to West Africa. Bitter leaf is cultivated in many West African countries, including Nigeria, Ghana, Cameroon, and others. It is often grown in home gardens and small farms. The plant is widely valued for its culinary, medicinal, and nutritional uses [5, 6]. Bitter leaf is characterized by its distinctive bitter taste and is a staple in many West African cuisines and traditional medicine practices. Different cultures prepare it in various ways, incorporating it into a wide range of dishes and soups [76, 79, 80]. However, bitter leaf is the key ingredient in the popular native soup in Southern Eastern Nigeria called ofe onugbu (bitter leaf Soup). The soup is typically made with a mixture of vegetables, meat or fish, and seasonings. The bitterness of the leaves adds a unique flavor to the soup. Bitter leaf can also be used in various vegetable dishes, stir-fries, and side dishes [9, 10, 64, 65]. It is often cooked with other vegetables and spices to balance its bitterness. To make the leaves less bitter, they are typically washed and sometimes soaked in water before use in culinary preparations. This process helps reduce the intensity of the bitter taste.

In traditional African medicine, bitter leaf has a long history of use for its potential health benefits. It is believed to have medicinal properties and is used in various remedies to address ailments such as malaria, fever, gastrointestinal issues, and more [18]. Bitter leaf contains antioxidants, which can help protect cells from oxidative damage [11]. These properties are attributed to the presence of bioactive compounds like flavonoids, polyphenols, and others [16, 17]. Bitter leaf is consumed to alleviate digestive discomfort and promote healthy digestion. It is believed to have mild laxative properties. Also, research has indicated that extracts from bitter leaf may possess antimicrobial properties, which could be valuable in combating certain infections. Bitter leaf has been studied for its anti-inflammatory effects, which may have implications for managing inflammatory conditions. Further, bitter leaf is a source of essential vitamins and minerals, including vitamins A, C, and K, as well as minerals (like calcium, magnesium, and iron) and dietary fiber [23]. Bitter leaf contains dietary fiber, which is important for digestive health and can help regulate blood sugar levels. While not a primary source of protein, bitter leaf contains some protein content [19, 22]. Bitter leaf is a versatile plant with cultural and culinary significance, and it continues to be studied for its potential health benefits. Its bitter taste, once moderated, adds a unique flavor profile to traditional dishes and soups. Additionally, as interest in natural remedies and traditional medicine grows, bitter leaf may find broader applications in wellness and healthcare [12]. Bitter leaf has cultural significance in social gatherings, ceremonies, and rituals across West Africa. It is often exchanged as a symbol of hospitality and goodwill during these events.

The diversity of *Vernonia amygdalina* reflects its adaptability to a range of ecological conditions and its cultural significance in various West African communities. This diversity is a valuable resource for both culinary traditions and potential medicinal applications, as different plant characteristics may offer unique flavor profiles and bioactive properties. Scientists and researchers continue to explore and document this diversity to better understand and utilize the potential of bitter leaf in agriculture, nutrition, and healthcare [63, 77]. *V. amygdalina* exhibits considerable diversity in terms of its morphology, habitat, and genetic makeup across its native range in West Africa. This diversity is shaped by various environmental factors, such as climate, soil conditions, and altitude, as well as genetic variations within the species. The leaves of bitter leaf plants display variations in the size and shape of their leaves. Leaves are typically broad and serrated, but variations exist. The plants can vary in growth habit, with some individuals growing as small shrubs and others as taller, more tree-like plants. The flowers of *V. amygdalina* can vary in color, size, and petal arrangement, although they are generally small and clustered. It is highly adaptable to a range of ecological niches and can be found in both forested and non-forested areas. It grows in lowland rainforests, savannas, and cultivated fields and can be found at varying altitudes, from lowland areas to higher elevations in some regions. Genetic diversity within the species contributes to variations in traits such as leaf chemistry, disease resistance, and adaptability to different environmental conditions [52]. Genetic studies have shown that different populations of bitter leaf may exhibit variations in their genetic makeup, which can influence the plant's characteristics and adaptability to local conditions. Traditional medicinal practices involving bitter leaf can vary from one region to another. Different communities may use it to address specific health concerns, resulting in variations in the formulations and remedies. Bitterness levels in bitter leaf can vary among individual plants and populations. Some plants may have more intensely bitter leaves, while others may be less bitter. Techniques to reduce bitterness are also variable.

This chapter aims to discuss the ethnomedicinal value of bitter leaf. The chapter presents the origin, distribution, and botany of *V. amygdalina*. The next sections address the nutritional and phytochemical properties as well as the food uses of bitter leaf. *V. amygdalina* has a long history of use in traditional medicine across West Africa. Other parts of the chapter are on the ethnobotanical, pharmacognosy, and clinical works, herbal products, toxicity, and commercial values of *V. amygdalina*. It is believed to have various medicinal properties, including digestive, anti-inflammatory, and antimicrobial effects. The environmental remediation and high adaptability of bitter leaf contribute to their growing widespread distribution whereas the efficacy associated with herbal remedies prepared from the plant makes it an integral part of traditional healing practices. However, further research to ensure the conservation of the plant and the sustainability of practices associated with it is paramount.

2 Origin and Distribution of *Vernonia amygdalina*

Vernonia amygdalina is native to tropical and subtropical regions of West Africa, but the exact origin within this region is not definitively known. It is largely hypothesized to have originated in the forested areas of West Africa, which encompass several countries along the western coast of the continent and a range of ecological zones, including lowland rainforests, savannas, and cultivated fields, where *V. amygdalina* has adapted and thrives [28]. The plant has a long history of traditional use in various West African cultures, and its use predates recorded history. It has been cultivated and harvested for its leaves, which are valued for their distinctive bitter taste and potential health benefits. Today, *V. amygdalina* is widely cultivated and grown in home gardens, small farms, and agricultural fields across West Africa. It is a significant part of the culinary and medicinal traditions in the region and continues to be an important cultural and economic resource. The plant's adaptability to various ecological conditions and its cultural significance have contributed to its spread and cultivation in different West African countries. It is used in a variety of traditional dishes and is an integral part of many cuisines in the region. Additionally, its medicinal properties have led to its continued use in traditional African medicine for various health concerns [69].

The distribution of bitter leaf is closely tied to the indigenous peoples and culture of West Africa, who have utilized it for generations as a valuable resource for both sustenance and traditional medicine [34]. Bitter leaf has played a vital role in the culinary and medicinal traditions of these communities, and its history is deeply embedded in their cultures. While *V. amygdalina* is native to West Africa, it has been cultivated and distributed beyond its native range due to its cultural and economic importance. Bitter leaf is cultivated in many West African countries, including Nigeria, Ghana, Cameroon, Ivory Coast, and others. It is often grown in home gardens, small farms, and agricultural fields. The plant is adapted to a range of ecological conditions, from lowland rainforests to savannas, and it is known for its hardiness and ability to thrive in diverse environments. *V. amygdalina* is distributed across multiple countries in West and Central Africa, with Nigeria being one of the primary producers and consumers of bitter leaf. In addition to its widespread distribution within West Africa, bitter leaf has garnered some recognition in international markets, particularly among African diaspora communities and those interested in African cuisine and traditional medicine [16, 17]. The distribution of *V. amygdalina* across different regions and ecological zones has led to variations in the plant's characteristics. Different populations of bitter leaf may exhibit variations in leaf size, bitterness levels, and adaptability to local conditions. These variations have contributed to the plant's adaptability and resilience, allowing it to flourish in a range of settings. As *V. amygdalina* is of significant cultural and economic importance, efforts have been made to promote its sustainable cultivation and harvesting practices. Organizations and initiatives have emerged to support small-scale farmers and promote sustainable agricultural practices for bitter leaf production.

African diaspora communities around the world, including those in Europe, North America, and other continents, have brought their culinary traditions with them. As a result, *V. amygdalina* is grown and used in some of these communities to recreate traditional dishes. In recent years, there has been an increasing global awareness of African cuisine and traditional ingredients. This has led to the availability of *V. amygdalina* in international markets, particularly in areas with a significant African diaspora. The plant's potential medicinal properties have attracted the interest of herbalists and practitioners of traditional medicine beyond Africa. As a result, bitter leaf extracts and preparations have been explored and used in various parts of the world for their potential health benefits. Scientific research on *V. amygdalina* has gained recognition globally. Researchers and scientists from different parts of the world have studied the plant's bioactive compounds, potential health benefits, and medicinal applications. While the distribution of *V. amygdalina* has expanded beyond its native range, its cultural and culinary significance remains strongest in West and Central Africa. In these regions, the plant continues to be an essential ingredient in traditional dishes, a symbol of hospitality and goodwill, and a valuable resource in traditional medicine. The global distribution of *V. amygdalina* reflects its versatility and adaptability as well as its potential to contribute to diverse culinary traditions and potential health benefits worldwide.

3 Botanical Characteristics of *Vernonia amygdalina*

Vernonia amygdalina belongs to the genus *Vernonia*, which is part of the large Asteraceae family. The genus *Vernonia* includes over 1000 species distributed across various continents, but *Vernonia amygdalina* is one of the most well-known and widely used species within this genus. Its vernacular names, such as bitter leaf, aptly describe its defining characteristic: an intensely bitter taste. The plant has a rich history deeply intertwined with the cultures of West Africa, where it is native. Its leaves, known for their bitterness, are used in a wide range of culinary dishes, traditional remedies, and cultural practices.

V. amygdalina is a perennial plant that can vary in size from a small shrub to a taller, more tree-like plant. Its growth habit can be erect or spreading. The stem of *V. amygdalina* is typically green, herbaceous, and somewhat woody at the base, especially in older plants. The leaves are the most distinctive feature of the plant [31]. They are broad, elliptical to ovate, and typically measure about 5–12 centimeters in length. The leaves have serrated or toothed margins. The upper surface of the leaves is usually dark green, while the lower surface may be lighter in color, often with a slightly pubescent (hairy) texture [32]. The leaves are arranged alternately along the stems of the plant [31]. The plant produces small, tubular flowers that are typically lavender to purple. These flowers are clustered in inflorescences, forming composite flower heads. The inflorescences vary in size and shape and can be found at the ends of branches. The plant typically blooms during its flowering season, which may vary depending on local climatic conditions [35]. The fruits of *V. amygdalina* are small achenes, often with a tuft of hairs attached to aid in wind dispersal. The defining characteristic of bitter leaf is its intense bitterness. This bitterness is attributed to the presence of certain compounds in the leaves. The roots and stem of the plant are less commonly used than the leaves. The stem is typically green and shrubby. Bitter leaf exhibits a variable growth habit. It can range from a small shrub with a bushy appearance to a taller, more tree-like plant. It can grow as a single stem or have multiple stems, depending on the conditions [47]. *V. amygdalina* possesses reproductive features typical of flowering plants. These features enable the plant to reproduce and propagate itself. It is primarily pollinated by insects, particularly bees and butterflies. These pollinators visit the flowers in search of nectar and, in the process, transfer pollen from one flower to another, facilitating fertilization. These reproductive features ensure the continuation of *V. amygdalina*'s life cycle and its ability to produce new generations of plants. While the plant is primarily valued for its culinary and medicinal uses, understanding its reproductive biology is essential for its conservation and sustainable cultivation [62] (Fig. 1).

Some botanical relatives of *V. amygdalina* include:

Vernonia calvoana (African Ironweed): *V. calvoana* is a closely related species to *V. amygdalina and* is native to Central and West Africa. Both plant species share some botanical characteristics, and like bitter leaf, *V. calvoana* has been used in traditional medicine practices.

Vernonia cinerea (Little Ironweed): *V. cinerea* is a species found in various parts of the world, including Asia, Africa, and Australia. While it is not native to Africa, it is part of the *Vernonia* genus and shares a common genus with *V. amygdalina*.

Vernonia guineensis (Guinea Ironweed): *V. guineensis* is another species within the *Vernonia* genus that is found in West and Central Africa. It is closely related to *V. amygdalina* and is used in similar ways in traditional medicine and sometimes in culinary preparations.

Vernonia galamensis (African Ironweed): *V. galamensis* is native to West Africa and is cultivated for its seeds, which are used for oil extraction.

Vernonia anthelmintica (Bitter Weed or Wormseed): While not native to Africa, *V. anthelmintica* is another species within the *Vernonia* genus that is known for its seeds, which have been used traditionally for their anthelmintic (anti-parasitic) properties.

Fig. 1 *Vernonia amygdalina* plant

Vernonia diversifolia (Variable-leaved Ironweed): *V. diversifolia* is a species found in various parts of the world, including Africa with diverse characteristics and traditional medicinal uses.

Vernonia adoensis (Ethiopian Ironweed): *V. adoensis* is native to East Africa, including Ethiopia, and has potential ethnomedicinal uses.

4 Environmental Value of *Vernonia Amygdalina*

Vernonia amygdalina possesses several environmental values and benefits, contributing to the ecosystems where it grows [3, 9]. While it is primarily known for its culinary and medicinal uses, it also plays a role in the environment in various ways. According to Ikhajiagbe et al. [33] some of these roles include:

Soil Stabilization: Like many other plants, *V. amygdalina* helps stabilize soil through its root system. The roots of the plant bind the soil particles, reducing erosion caused by rainfall and runoff.

Biodiversity Support: Bitter leaf, as a native plant in West and Central Africa, provides habitat and forage for local wildlife. Insects, including pollinators like bees and butterflies, visit its flowers for nectar, contributing to local biodiversity.

Carbon Sequestration: *V. amygdalina* absorbs carbon dioxide from the atmosphere through photosynthesis. This process helps mitigate climate change by sequestering carbon and converting it into organic matter.

Nitrogen Fixation: Some plants, including certain species within the Asteraceae family (to which *Vernonia amygdalina* belongs), have associations with nitrogen-fixing bacteria in their root nodules [19]. These bacteria convert atmospheric nitrogen into a form that plants can use as a nutrient. While not all members of the family have this ability, some may contribute to nitrogen enrichment in the soil.

Erosion Control: In its native region (West and Central Africa), flood and erosion are a growing challenge but the growth of *Vernonia amygdalina* can help prevent soil loss by stabilizing slopes and riverbanks.

Medicinal Plant Conservation: *V. amygdalina*'s value as a medicinal plant has led to conservation efforts in some regions to ensure its continued existence [61]. Preserving its natural habitat helps protect the plant and other species that depend on it.

Traditional Agroforestry Systems: In some traditional agroforestry systems, bitter leaf may be grown alongside other crops and trees, contributing to diversified and sustainable agricultural practices that are beneficial for both the environment and local communities.

The presence of *V. amygdalina* in local ecosystems contributes to maintaining ecological balance and supporting biodiversity. Recognizing the environmental values of this plant can contribute to its conservation and sustainable use in its native habitat and land management. The plant also exhibits some weedy characteristics through their

Rapid Growth: *V. amygdalina* is a fast-growing plant. Under favorable conditions, it can establish itself quickly and spread in an area.

Seed Production: The plant produces seeds in abundance. These seeds have adaptations, such as tufts of hairs that aid in wind dispersal. This can facilitate the plant's spread in the wild.

Adaptability: Bitter leaf is adaptable to a range of ecological conditions, from lowland rainforests to savannas. This adaptability allows it to thrive in diverse environments, including disturbed or cultivated areas.

Vegetative Propagation: In addition to sexual reproduction through seeds, *V. amygdalina* can propagate vegetatively. This means that new plants can sprout from stem cuttings or even root fragments under certain conditions.

Resilience: The plant is hardy and can tolerate some degree of environmental stress, including drought conditions.

Naturalized Populations: In areas where it is cultivated, *V. amygdalina* may escape cultivation and become naturalized. This can lead to the plant growing in unintended locations.

Competitive Ability: In some situations, *V. amygdalina* may outcompete other plant species for resources such as light, water, and nutrients.

In some cases, *V. amygdalina* may be intentionally cultivated as a crop, and its rapid growth and adaptability can be advantageous. In other cases, particularly in natural ecosystems, it may be considered invasive if it displaces native vegetation. Efforts to manage the spread of *V. amygdalina* in natural ecosystems should be guided by principles of conservation and biodiversity preservation. In cultivated areas, appropriate agricultural practices can help control its growth and prevent it from becoming a weed.

4.1 Bioremediation Value of *Vernonia amygdalina*

Vernonia amygdalina has demonstrated potential bioremediation value in certain environmental contexts. Bioremediation is the use of living organisms, including plants, to mitigate or clean up environmental contaminants, such as heavy metals and organic pollutants. Some bioremediation roles attributed to *V. amygdalina* in Shittu and Ikhajiagbe [81] include:

Heavy Metal Accumulation: Bitter leaf can accumulate heavy metals from the soil. Some studies have shown that it can accumulate metals such as lead, cadmium, and nickel from contaminated soil. This characteristic makes it a candidate for phytoremediation, a specific form of bioremediation that uses plants to remove, stabilize, or detoxify contaminants in soil and water.

Hyperaccumulation Potential: Some plants, including certain species within the Vernonia genus, are known as hyperaccumulators. These plants can accumulate exceptionally high concentrations of heavy metals in their tissues without displaying toxicity symptoms. *V. amygdalina* has shown the potential for hyperaccumulation of heavy metals, particularly in its leaves. This ability is advantageous in removing and concentrating contaminants for subsequent remediation.

Soil Improvement: Bitter leaf can improve soil quality through its root system and organic matter deposition. As it grows and matures, the plant's roots help enhance soil structure and nutrient content. Improved soil conditions can indirectly support bioremediation efforts by promoting the growth of other remediation plants and microorganisms.

Sustainable Remediation: The use of *V. amygdalina* for bioremediation aligns with sustainable and environmentally friendly remediation practices. It reduces the need for harsh chemicals and mechanical interventions, minimizing potential environmental harm.

Although *V. amygdalina* has shown promise in certain phytoremediation studies, its use in practical remediation applications depends on several factors, including the specific contaminants involved, site conditions, and regulatory considerations. Bioremediation is a complex process that requires careful planning and monitoring. Additionally, the effectiveness of bioremediation using *V. amygdalina* may vary depending on the extent of contamination and the suitability of the plant for the specific site. Therefore, it is essential to conduct site-specific assessments and consider the broader ecological implications when using plants for bioremediation purposes.

5 Cultural Significance of *Vernonia amygdalina*

Vernonia amygdalina holds significant historical and cultural importance in West and Central Africa. Its multifaceted significance is deeply woven into the cultural, culinary, and traditional medicinal practices of the region. Some cultural significance of bitter leaf include (Fig. 2):

- **Traditional Medicine**: Bitter leaf has a long history of use in traditional African medicine systems. Indigenous communities have employed it as a remedy for various health conditions, including malaria, fever, gastrointestinal issues, and more [18]. The plant's phytochemical composition, including bitter compounds and bioactive constituents, is believed to contribute to its potential medicinal properties.
- **Culinary Traditions**: Bitter leaf is a fundamental ingredient in West and Central African cuisine. It imparts a unique and cherished flavor to numerous dishes and soups. One of the most famous dishes featuring bitter leaf is bitter leaf soup (ofe onugbu), a delicacy enjoyed across the region. The plant is also used in vegetable dishes, stews, and side dishes.
- **Cultural Symbolism**: Bitter leaf has symbolic significance in various cultural practices and rituals. It is exchanged as a symbol of hospitality, goodwill, and friendship during social gatherings, ceremonies, and celebrations. Its presence in

Fig. 2 Cultural values of *Vernonia amygdalina*

traditional offerings and exchanges strengthens social bonds and fosters community ties.

- **Medicinal Knowledge Transfer**: The use of *V. amygdalina* in traditional medicine has been passed down through generations within African communities. Elders and traditional healers play a crucial role in transferring knowledge about the plant's medicinal properties and applications to the younger generation.
- **Cultural Identity**: Bitter leaf contributes to the cultural identity of West and Central African communities. Its inclusion in traditional dishes and remedies reflects the region's culinary heritage and medical practices. It distinguishes West and Central African cuisine from other global culinary traditions, adding a distinctive flavor profile.
- **Ethnobotanical Heritage**: *V. amygdalina* is an integral part of the ethnobotanical heritage of West and Central Africa. It exemplifies the region's rich botanical knowledge and the relationship between people and plants. Ethnobotanical studies have further illuminated the plant's uses and cultural significance within indigenous communities.
- **Cultural Conservation**: The cultural and historical importance of *V. amygdalina* has spurred efforts to conserve and protect the plant's natural habitat. Conservation initiatives aim to safeguard this valuable botanical resource.

1. **Phytochemical Constituents of *Vernonia amygdalina***

   Vernonia amygdalina contains a diverse array of phytochemical constituents, which contribute to its medicinal and nutritional properties. These phytochemicals are responsible for the plant's characteristic bitterness and its potential health benefits.

   Sesquiterpene Lactones: These are bitter compounds that are primarily responsible for the plant's intense bitterness. Sesquiterpene lactones, such as vernodalin, vernolepin, and vernomygdin, are characteristic phytochemicals of bitter leaf [24, 43].

   Alkaloids: Bitter leaf contains alkaloids, including vernodalin alkaloid, which may contribute to its medicinal properties [18, 43].

   Flavonoids: Flavonoids are a group of polyphenolic compounds found in bitter leaf. They have antioxidant properties and may play a role in protecting cells from oxidative damage [11, 29, 30].

   Steroids and Triterpenoids: Some steroids and triterpenoids, such as vernonioside A1 and vernonioside B1, have been isolated from *V. amygdalina*. These compounds may have potential pharmacological activities [20, 43].

   Phenolic Compounds: Phenolic compounds, including caffeic acid, chlorogenic acid, and quercetin, are present in bitter leaf. These compounds have antioxidant and anti-inflammatory properties [11].

   Saponins: Bitter leaf contains saponins, which are glycosides with surfactant properties. Saponins have been studied for their potential health benefits, including cholesterol-lowering effects [24].

Glycosides: Glycosides are compounds that contain a sugar molecule bonded to another molecule. Bitter leaf contains various glycosides, and some may have biological activities.

Tannins: Tannins are a group of polyphenolic compounds found in bitter leaf. They have astringent properties and may have antimicrobial effects [25, 26].

Minerals and Vitamins: Bitter leaf is a source of essential minerals such as calcium, magnesium, and iron. It also contains vitamins like vitamin A, vitamin C, and vitamin K.

Essential Oils: Bitter leaf may contain essential oils that contribute to its aroma and potential therapeutic properties [41].

The phytochemical composition of *V. amygdalina* can vary depending on factors such as the plant's age, environmental conditions, and geographic location. The combination of these phytochemical constituents gives bitter leaf its characteristic bitterness and makes it a valuable resource in traditional medicine and culinary practices in West and Central Africa. Additionally, ongoing research is uncovering the potential health benefits and pharmacological activities of these phytochemicals. Some of the essential phytochemical constituents of bitter leaf, parts found, and their usefulness are presented in Table 1.

V. amygdalina is a leafy green vegetable that is valued for its potential medicinal properties and nutritional content. While bitter leaf is primarily recognized for its bitterness and medicinal uses, it also contains various essential nutrients that contribute to its nutritional value. The approximate concentrations of mineral nutrients in bitter leaf are presented in Table 2. Bitter leaf is a source of several vitamins, including vitamin A, vitamin C, and vitamin K. These vitamins are essential for various bodily functions, including immune support, vision, and blood clotting. Bitter leaf contains important minerals such as calcium, magnesium, and iron. Calcium is crucial for bone health, magnesium is involved in various enzymatic processes, and iron is essential for red blood cell formation and oxygen transport. Bitter leaf is a good source of dietary fiber, which is important for digestive health. Fiber helps regulate bowel movements, prevent constipation, and may assist in managing weight loss [12]. Bitter leaf contains various antioxidants, including flavonoids and phenolic compounds. These antioxidants help protect cells from oxidative stress and may have anti-inflammatory properties [75]. Bitter leaf is rich in phytochemicals, such as sesquiterpene lactones and alkaloids, which are responsible for its characteristic bitterness. Some of these phytochemicals have been studied for their potential health benefits. Bitter leaf is a low-calorie vegetable, making it a suitable choice for those looking to manage their calorie intake [48]. While not a high-protein source, bitter leaf does contain some protein, which is essential for tissue repair and overall growth and development. Bitter leaf contains carbohydrates, primarily in the form of dietary fiber and small amounts of sugars. Carbohydrates provide energy for the body. Bitter leaf is naturally low in fat, making it a heart-healthy choice when included in a balanced diet. It is important to note that the nutritional content of bitter leaf can vary based on factors such as plant variety, growing conditions, and how it is prepared and cooked. Bitter leaf is often

Table 1 Some phytochemical constituents of bitter leaf, parts found, and usefulness

Phytochemical constituent	Types	Medicinal uses and potential health benefits	Part of the plant found
Sesquiterpene lactones	Vernodalin, Vernolepin, and Vernomygdin	Bitterness, anti-inflammatory, antioxidant, pain relief, and may support the immune system	Leaves
Alkaloids	Vernodalin, Vernomenin, Vernolepin, Epivernodaline, cyanogenic alkaloids	Pain relief, anti-parasitic properties, and regulations of the nervous system	Various parts
Flavonoids	Kaempferol, Quercetin, Luteolin, Rutin, Myricetin, Catechins	Antioxidant, anti-inflammatory, cardiovascular health, and support of the immune system	Leaves and stems
Steroids and triterpenoids	β-Sitosterol, Campesterol, Stigmasterol, Lupeol, β-Amyrin, Ursolic acid, Oleanolic acid, β-Amyrin, Friedelin, Betulinic acid	Anti-inflammatory, immunomodulatory, anti-cancer effects, and support skin health	Leaves
Phenolic compounds	Caffeic acid, Chlorogenic acid, Epicatechin	Antioxidant, anti-inflammatory, antioxidant, anti-inflammatory, and cardiovascular health as well as aid digestion	Leaves
Saponins	Vernoniosides, bitter saponins, steroidal saponins	Cholesterol reduction, antioxidant, and immune support	Leaves
Glycosides	Cardiac glycosides; Iridoid glycosides and Anthraquinone glycosides	Cardiac health, diuretic properties, and antioxidant effects	Leaves
Tannins	Gallotannins, proanthocyanidins, and ellagitannins	Astringent, antimicrobial effects, astringent, wound healing, antioxidant, and antimicrobial properties	Leaves
Minerals and vitamins	Calcium, potassium, sodium, phosphorus, iron, etc.	Nutritional and health value. Calcium for bone health, iron for anemia prevention, and vitamins for overall health	Leaves
Essential oils	Terpenes, eugenol	Aroma, and potential therapeutic properties. Aroma therapy has potential for skin care as well as calming effects	Leaves

used in traditional dishes in West and Central Africa, where it is typically cooked or added to soups and stews.

Note that bitter leaf's bitterness may require certain culinary techniques to make it more palatable, its potential nutritional and medicinal properties make it a valuable addition to traditional diets in many regions. However, individuals with certain medical conditions or sensitivities may need to consume bitter leaf in moderation,

Table 2 Approximate concentrations of mineral nutrients in bitter leaf

Mineral nutrient	Approximate concentration (per 100 g of fresh leaves) in mg
Calcium	309
Magnesium	109
Potassium	470
Sodium	11
Phosphorus	66
Iron	1.9
Zinc	1.6
Copper	0.14
Manganese	1.3

and it is advisable to consult with a healthcare professional or nutritionist for personalized dietary recommendations.

6 Ethnomedicinal Uses of *Vernonia amygdalina*

Vernonia amygdalina is a versatile and culturally significant plant with a rich history of ethnomedicinal uses in various regions of West and Central Africa. This plant is celebrated not only for its culinary contributions but also for its potential therapeutic properties. Bitter leaf has been a staple in the diets of various West and Central African communities for centuries. Its bitter taste, which gives the plant its name, is a defining characteristic that has shaped its culinary and medicinal roles. In many African cultures, bitter leaf is not merely an edible green but also a symbol of resilience and adaptability. The plant thrives in diverse ecological conditions, and this resilience is often associated with qualities that are admired and respected by people. Additionally, the bitterness of the leaves is seen as a metaphor for life's challenges, suggesting that strength and healing can be found even in adversity. The ethnomedicinal uses of bitter leaf are deeply intertwined with cultural practices and belief systems. Traditional healers, often referred to as herbalists or traditional medicine practitioners, play a vital role in preserving and passing down the knowledge of using bitter leaf and other medicinal plants. These healers are highly respected in their communities and are considered custodians of invaluable wisdom. Some of the prominent ethnomedicinal uses of *V. amygdalina* include (Fig. 3):

- **Malaria Treatment and Prevention**: Perhaps one of the most well-known traditional uses of bitter leaf is its role in the management of malaria. Various parts of the plant, including the leaves and stem, are used to prepare decoctions and infusions believed to have anti-malarial properties [2, 18]. The bitter leaf's role in malaria treatment aligns with its historical significance, as malaria is a prevalent and debilitating disease in many parts of Africa.
- **Fever Reduction**: Bitter leaf is often employed as a natural remedy to reduce fever. The leaves are typically brewed into a medicinal tea or used in poultices for

Fig. 3 Ethnomedicinal values of *Vernonia amygdalina*

external application. The plant's ability to lower body temperature is valued in regions where fever is a common symptom of various illnesses.

- **Digestive Health**: Bitter leaf is recognized for its digestive benefits. It is believed to aid in digestion, alleviate indigestion, and provide relief from gastrointestinal discomfort. It is sometimes consumed as a tea or added to soups and stews to promote digestive well-being.
- **Diabetes Management**: Traditional medicine practitioners in some regions use bitter leaf as an adjunct treatment for diabetes [12]. While modern scientific research on this aspect is ongoing, bitter leaf is believed to have properties that help regulate blood sugar levels.
- **Skin Care**: Bitter leaf's anti-inflammatory and antimicrobial properties make it a popular choice for treating various skin conditions. It is used topically to alleviate itching, soothe rashes, and support wound healing.

- **Cough and Respiratory Health**: Bitter leaf is often included in herbal preparations to alleviate coughs, colds, and respiratory infections. Its potential anti-inflammatory and immune-boosting properties make it a valuable component in traditional cough remedies.
- **Pain Relief**: Bitter leaf is considered an analgesic, and it is used to alleviate various types of pain, including headaches and body aches [54]. Some traditional preparations may be applied externally to painful areas.
- **Stomach Ulcer Soothing**: In some traditional systems, bitter leaf is used to soothe and potentially promote healing in cases of stomach ulcers [45]. Its reported anti-inflammatory properties may contribute to this application [66].
- **Hepatitis Support**: Traditional healers in certain regions use bitter leaf as a supportive therapy for hepatitis. It is believed to promote liver health and assist in recovery.
- **Immune System Support**: Bitter leaf is seen as a natural immune booster. Its regular consumption is thought to strengthen the immune system and enhance overall health.
- **Weight Management**: In some cultures, bitter leaf is incorporated into weight management strategies. It is believed to play a role in appetite regulation and metabolism [12].
- **Anemia Management**: Bitter leaf contains iron and other nutrients, making it a potential remedy for anemia. It may be used to support individuals with iron-deficiency anemia.
- **Hypertension**: Some traditional practitioners use bitter leaf to help manage hypertension. While research is ongoing in this area, it is believed that the plant's compounds may contribute to blood pressure regulation.

It is important to note that the ethnomedicinal uses of bitter leaf are often passed down through generations and are deeply rooted in cultural traditions. These traditional remedies are valued for their accessibility, affordability, and perceived effectiveness, particularly in regions where access to modern healthcare may be limited. Modern scientific research are ongoing into the ethnomedicinal uses of *V. amygdalina* and are well-documented in traditional systems. Modern scientific research has begun to explore the plant's potential therapeutic properties. Studies have investigated the bioactive compounds present in bitter leaf and their mechanisms of action. Some of the key findings and ongoing research areas include:

Antioxidant Properties: Bitter leaf contains various phytochemicals, such as flavonoids and phenolic compounds, which exhibit antioxidant activity. These antioxidants are believed to protect cells from oxidative stress, which is linked to various chronic diseases.

Anti-inflammatory Effects: Bitter leaf's anti-inflammatory properties have been studied, and it is considered a potential natural remedy for inflammatory conditions.

Antimicrobial Activity: Research has indicated that bitter leaf extracts may have antimicrobial properties, which can be useful in combating infections.

Hepatoprotective Potential: Studies suggest that bitter leaf may have hepatoprotective effects, potentially supporting liver health.

Immunomodulatory Effects: Bitter leaf's impact on the immune system is a subject of interest, and research is ongoing to better understand its immunomodulatory properties.

Antidiabetic Potential: Bitter leaf's role in blood sugar regulation has garnered attention, and investigations into its potential as an antidiabetic agent continue [12, 49].

It is worth emphasizing that while traditional knowledge is valuable, scientific validation is critical to fully understand the safety and efficacy of bitter leaf in modern healthcare. Researchers are working to bridge the gap between traditional and evidence-based medicine by conducting rigorous studies to assess the plant's potential benefits. Despite the promise of *V. amygdalina* as a source of natural remedies, several challenges and considerations exist:

Standardization: Traditional preparations of bitter leaf vary in terms of plant part used, preparation methods, and dosages. Standardization is crucial for ensuring consistent and safe therapeutic outcomes.

- **Safety Concerns**: While bitter leaf is generally considered safe for most individuals, there can be adverse effects if consumed in excessive quantities. Specific populations, such as pregnant women and individuals with certain medical conditions, should exercise caution.
- **Integration with Modern Medicine**: Integrating traditional remedies like bitter leaf into modern healthcare systems requires careful evaluation, regulation, and collaboration between traditional healers and healthcare professionals.
- **Preservation of Traditional Knowledge**: Preserving and passing down traditional knowledge of medicinal plants, like bitter leaf, is vital. Efforts to document and protect this knowledge are ongoing.
- **Sustainability**: The popularity of bitter leaf and other medicinal plants can lead to overharvesting and habitat degradation. Sustainable harvesting practices must be promoted.

The future of *Vernonia amygdalina* in modern healthcare lies in multidisciplinary research that combines traditional wisdom with scientific rigor. Collaborations between ethnobotanists, pharmacologists, healthcare practitioners, and local communities are essential to harness the plant's potential for the benefit of public health.

7 Pharmacognostic and Clinical Research Involving *Vernonia amygdalina*

Clinical research on *Vernonia amygdalina* will culminate in an understanding of the potential therapeutic properties and safety for human use. While traditional and ethnobotanical knowledge has long recognized the plant's medicinal value, modern clinical studies aim to provide scientific evidence to support its traditional uses and explore new applications.

- **Antimalarial Properties**: Some clinical studies have investigated the antimalarial properties of *Vernonia amygdalina*. These studies aim to assess its efficacy in the treatment and prevention of malaria, which is a prevalent and serious disease in many regions where the plant is used traditionally [2].
- **Diabetes Management**: Research has explored the effects of bitter leaf extracts or preparations on blood sugar regulation [12]. Clinical trials may investigate the potential of bitter leaf to help manage diabetes and its complications.
- **Liver Health**: Clinical studies may assess the impact of bitter leaf on liver health, including its hepatoprotective properties. Researchers aim to understand its role in supporting liver function and mitigating liver-related conditions.
- **Antioxidant and Anti-inflammatory Effects**: Investigations into the antioxidant and anti-inflammatory properties of bitter leaf are ongoing. Clinical trials may explore its potential in reducing oxidative stress and inflammation, which are associated with various chronic diseases.
- **Immune Modulation**: Some research focuses on the immunomodulatory effects of *Vernonia amygdalina* [50, 71]. Clinical studies aim to understand how it may enhance immune responses and strengthen the body's defense mechanisms.
- **Antimicrobial Activity**: Clinical research may assess the efficacy of bitter leaf extracts against specific pathogens, including bacteria, viruses, and parasites [28]. This research could have implications for infectious disease management.
- **Cancer Research**: Studies have explored the potential anticancer properties of bitter leaf compounds [38]. Clinical trials may investigate its effects on specific types of cancer and its mechanisms of action.
- **Blood Pressure Regulation**: Clinical investigations may assess the impact of bitter leaf on blood pressure regulation, potentially leading to therapeutic applications for individuals with hypertension [1, 46].
- **Safety and Tolerance**: Clinical studies also focus on evaluating the safety and tolerability of bitter leaf preparations in human subjects [73]. This includes assessing potential side effects and interactions with medications.
- **Dosage and Formulation**: Research may determine the optimal dosage and formulation of bitter leaf extracts or supplements for therapeutic use [73]. This information is important for developing standardized products.
- **Nutritional Benefits**: Clinical trials can examine the nutritional properties of bitter leaf, including its nutrient content and potential as a dietary supplement [55].
- **Traditional Medicine Validation**: Clinical research plays a role in validating the traditional uses of bitter leaf in various traditional medicine systems [7]. This can help bridge the gap between traditional and evidence-based medicine.

Note that the findings from this research work may vary depending on factors such as the specific preparation of bitter leaf, dosage, and patient populations. Collaborations between researchers, healthcare professionals, traditional healers, and local communities are crucial for conducting meaningful and culturally sensitive clinical studies on *V. amygdalina*. Additionally, regulatory considerations and ethical guidelines play a role in the design and implementation of clinical trials involving medicinal plants.

8 Toxicity and Safety Considerations of *Vernonia amygdalina*

Vernonia amygdalina is generally considered safe for consumption when used in culinary and traditional medicinal practices. It has a long history of use as a food ingredient and in traditional medicine across West and Central Africa [27, 60]. However, like many plants, bitter leaf may have certain toxic or adverse effects under certain circumstances or when consumed in excessive amounts. According to Eze et al. [27] some of the toxic considerations of bitter leaf include:

- **Bitterness and Palatability**: Bitter leaf gets its name from its bitter taste, which is a characteristic feature. While the bitterness is generally tolerated in culinary preparations, some individuals may find it unpalatable in large quantities. Bitterness can also vary among different plant varieties.
- **Cyanogenic Glycosides**: Like many plants, bitter leaf may contain small amounts of cyanogenic glycosides, which can release toxic cyanide when metabolized [16, 17]. However, the levels of these compounds are usually very low and not considered a significant health risk when consuming bitter leaf as part of a balanced diet. Proper cooking and preparation methods can further reduce the potential for cyanide release.
- **Pregnancy and Lactation**: Pregnant and lactating women are often advised to consume bitter leaf in moderation. Excessive consumption of certain phytochemicals found in bitter leaf, such as sesquiterpene lactones, during pregnancy may have adverse effects [39, 40]. It is advisable for pregnant and nursing women to consult with healthcare professionals before incorporating bitter leaf into their diets.
- **Allergies**: Some individuals may be allergic to specific compounds found in bitter leaf, leading to allergic reactions. Symptoms can include skin rashes, itching, or gastrointestinal discomfort. If allergic reactions occur, discontinuing the consumption of bitter leaf is recommended.
- **Digestive Discomfort**: In some cases, excessive consumption of bitter leaf may lead to digestive discomfort, including diarrhea or stomach upset [42, 45]. It is advisable to consume bitter leaf in moderation and observe how your body responds.
- **Interactions with Medications**: Bitter leaf may interact with certain medications due to its phytochemical content. It is important to inform healthcare providers about the use of bitter leaf, especially when taking prescription medications, to avoid potential drug interactions.
- **Individual Sensitivity**: People may vary in their sensitivity to the bitterness of bitter leaf and its potential effects on digestion. Some individuals may tolerate it well, while others may find it less agreeable.

It is important to note that the vast majority of people in regions where bitter leaf is traditionally consumed incorporate it into their diets without experiencing adverse effects. Bitter leaf is valued for its potential health benefits and nutritional value, and it is considered a safe and nutritious food when used appropriately. To mitigate potential risks and ensure safe consumption:

Cook Bitter Leaf Properly: Cooking, boiling, or blanching bitter leaf can help reduce bitterness and minimize the potential release of cyanide compounds [17]. Traditional culinary practices often involve cooking bitter leaf in soups, stews, or as a side dish.

Moderation: As with any food or herbal remedy, consuming bitter leaf in moderation is advisable. Excessive consumption of any substance can lead to adverse effects.

Consult Healthcare Professionals: It is recommended to consult certified and practicing healthcare professionals before using bitter leaf to treat or manage illness if you have underlying medical conditions, are pregnant or nursing, or are taking medications.

Although *V. amygdalina* is generally safe for most people when used in culinary and traditional medicinal practices, it is essential to be mindful of individual sensitivities and potential interactions with medications. As with any dietary ingredient, balance and moderation are key to safe and enjoyable consumption.

9 Some Herbal Formulations Involving *Vernonia amygdalina* and Their Active Ingredients

Vernonia amygdalina is a versatile plant that is often incorporated into various herbal formulations in traditional medicine across West and Central Africa. These formulations may combine bitter leaf with other herbs, roots, or natural ingredients to address specific health concerns.

Malaria Remedies: Bitter leaf is frequently used in herbal formulations to combat malaria [13]. It is often combined with other antimalarial herbs, such as *Artemisia annua* (sweet wormwood) or *Morinda lucida* (brimstone tree). These formulations may be consumed as decoctions or infusions to reduce fever and alleviate malaria symptoms [2, 15].

Digestive Tonic: Bitter leaf is sometimes included in digestive tonics or bitters. These formulations often contain a blend of bitter herbs and spices to promote digestion, alleviate indigestion, and stimulate appetite [8, 37]. Bitter leaf's natural bitterness contributes to its role in these remedies.

Cough and Cold Remedies: Bitter leaf is used in herbal preparations aimed at relieving coughs, colds, and respiratory infections [70]. It may be combined with other respiratory herbs like ginger, garlic, and honey to create soothing and immune-boosting remedies.

Anti-Inflammatory Mixtures: Bitter leaf's potential anti-inflammatory properties make it a valuable component of herbal formulations aimed at reducing inflammation [13, 53, 83]. These mixtures may include other anti-inflammatory herbs like turmeric, ginger, and neem.

Liver Support Formulas: Bitter leaf is used in herbal formulations to support liver health and promote detoxification [13, 36]. These remedies may include herbs, such as dandelion root, milk thistle, and burdock root.

Wound Healing Salves: Bitter leaf extracts or poultices are sometimes applied topically to wounds, cuts, and skin irritations to promote healing and prevent infection [44]. The plant's antimicrobial properties are believed to contribute to its effectiveness in this regard.

Diabetes Management: In some traditional systems, herbal formulations are prepared to help manage diabetes [14, 74]. Bitter leaf may be a key ingredient, as it is believed to have blood sugar-regulating properties. Other diabetes-supportive herbs such as fenugreek or cinnamon may be included.

Immune-Boosting Tonics: Bitter leaf is often incorporated into immune-boosting tonics to enhance overall health and strengthen the immune system [50, 72]. These formulations may contain a combination of immune-enhancing herbs like garlic, ginger, and echinacea.

Anti-hypertensive Mixtures: Some herbal preparations aim to help manage hypertension. Bitter leaf's potential role in blood pressure regulation is considered when formulating these remedies, and it may be combined with other blood pressure-supportive herbs like hawthorn or olive leaf [51].

Anti-parasitic Remedies: Bitter leaf is used in herbal formulations designed to combat intestinal parasites [21]. These remedies may include other anti-parasitic herbs like wormwood, black walnut, or cloves.

Note that specific herbal formulations and recipes can vary widely across different cultures and regions. Traditional medicine practitioners, often referred to as herbalists or traditional healers in West and Central Africa, possess unique knowledge and expertise in crafting these remedies based on local traditions and available resources. While traditional herbal formulations can offer potential health benefits, it is essential to exercise caution and consult with healthcare professionals, especially when using herbal remedies alongside prescription medications or for the treatment of serious medical conditions. Additionally, the safety and efficacy of these formulations should be considered in light of scientific research and evidence-based practices.

10 Commercial Value of *Vernonia amygdalina*

Vernonia amygdalina holds commercial value in several ways, primarily driven by its culinary and potential medicinal uses. Its commercial significance varies across different regions and markets [59, 78].

Culinary Use: Bitter leaf is a popular and essential ingredient in the cuisines of many West and Central African countries. It is used in a wide range of traditional dishes, soups, stews, and sauces. Due to its unique bitter flavor, it adds depth and complexity to the taste of these dishes, making them culturally significant. In some regions, processed or packaged bitter leaf products are available for convenience.

Traditional Medicine: Bitter leaf has a long history of use in traditional medicine across Africa [67, 68]. As interest in herbal and natural remedies grows globally, there is potential commercial value in the development and marketing of herbal products containing bitter leaf extracts or formulations. These products may be promoted for their potential health benefits, including immune support, anti-inflammatory properties, and digestive health.

Pharmaceutical and Nutraceutical Industries: The phytochemical compounds found in bitter leaf, such as flavonoids, phenolic compounds, and sesquiterpene lactones, have attracted the attention of researchers and the pharmaceutical industry. There is ongoing exploration of the potential use of these compounds in drug development and nutraceutical products.

Export Market: Bitter leaf has gained recognition in international markets, especially among African diaspora communities. It is exported to countries with large African populations or those interested in African cuisine. In some regions, dried or frozen bitter leaf is exported to meet the demand of African expatriates and consumers interested in African culinary traditions.

Herbal and Dietary Supplements: Bitter leaf is sometimes used as an ingredient in herbal supplements and dietary products. These supplements are marketed for various purposes, such as immune support, detoxification, and digestive health.

Cultivation and Farming: As the demand for bitter leaf increases, there may be opportunities for commercial cultivation and farming of *Vernonia amygdalina*. Controlled cultivation can ensure a consistent supply of high-quality bitter leaf for domestic and international markets.

Traditional Food Processing: Traditional food processors who specialize in preparing and packaging bitter leaf for sale may contribute to its commercial value. These businesses may operate in local markets and play a role in preserving culinary traditions.

Research and Development: Research institutions and organizations may invest in the research and development of bitter leaf-based products and formulations. This can lead to innovations in the food, pharmaceutical, and nutraceutical industries.

Herbal Tea and Beverage Industry: Bitter leaf can be used as an ingredient in herbal teas and beverages. Its unique flavor and potential health benefits make it a candidate for inclusion in herbal tea blends targeting health-conscious consumers.

Export Regulations: Regulatory bodies in some countries may establish standards and regulations for the export and import of bitter leaf and related products. Compliance with these regulations is essential for market access and commercial success.

The commercial value of *Vernonia amygdalina* makes it evident that it is essential to consider sustainable cultivation, harvesting practices, quality control, and adherence to safety standards to ensure the long-term viability of the bitter leaf industry. Additionally, collaborations between researchers, traditional healers, farmers, and entrepreneurs can contribute to the responsible and profitable commercialization of this valuable plant.

11 Future Prospects of Therapeutic Roles of *Vernonia amygdalina*

The prospects of *V. amygdalina* in therapeutic roles are promising and multifaceted. As interest in natural and plant-based remedies continues to grow, coupled with ongoing scientific research, *Vernonia amygdalina* holds great potential for contributing to various aspects of modern healthcare. Research has shown that *Vernonia amygdalina* contains a variety of phytochemicals, including flavonoids and phenolic compounds, which exhibit antioxidant and anti-inflammatory properties. These properties are important in combating oxidative stress and chronic inflammation, which are underlying factors in many chronic diseases [41]. The potential immunomodulatory effects of bitter leaf suggest its use in enhancing immune function. As immune support becomes increasingly important, bitter leaf formulations may find a place in promoting overall health and resilience. Bitter leaf has been traditionally used for its antimicrobial properties [4]. Research is ongoing to understand its effectiveness against various microorganisms, including bacteria, viruses, and parasites. This could have implications for infectious disease management. Preliminary studies have explored the antidiabetic potential of *Vernonia amygdalina*. As diabetes remains a global health concern, there is interest in developing bitter leaf-based products or therapies for blood sugar regulation. Bitter leaf is used traditionally to support liver health and detoxification. Further research into its hepatoprotective properties may lead to the development of liver support products or therapies. Some studies suggest that bitter leaf may play a role in blood pressure regulation. This could be valuable for individuals with hypertension or those at risk of high blood pressure. Bitter leaf has attracted attention for its potential anticancer properties. Investigations into the mechanisms of action of its bioactive compounds may lead to the development of novel cancer therapies or preventive strategies. Bitter leaf may be incorporated into nutraceuticals and dietary supplements targeting specific health concerns. These products could offer convenient and standardized ways to access its therapeutic benefits. The isolation and characterization of bioactive compounds from bitter leaf may pave the way for the development of pharmaceutical drugs derived from its natural compounds [82]. This could result in innovative treatments for various health conditions. Beyond its medicinal properties, *V. amygdalina* continues to be a culinary staple. Innovative chefs and food scientists may explore new ways to incorporate bitter leaf into culinary creations, promoting both its unique flavor and potential health benefits [56–58]. As the demand for bitter leaf increases, there will be a growing need for sustainable cultivation practices to ensure a consistent supply. Conservation efforts to protect wild populations of *Vernonia amygdalina* may also become essential. Collaborations between traditional healers and modern researchers can help preserve and document the rich traditional knowledge associated with bitter leaf. This knowledge transfer can inform future therapeutic developments. It is important to emphasize that the therapeutic roles of *Vernonia amygdalina* are an evolving field of study, and further research is needed to validate its efficacy and safety for specific health conditions. Collaboration between scientists, healthcare professionals, traditional healers, and local communities will be crucial in harnessing the full potential of this remarkable plant. Additionally, regulatory considerations, quality control, and standardized formulations will play a role in its future therapeutic applications.

12 Conclusion

The use of bitter leaf in ethnomedicine expresses the deep connection between nature, culture, and healthcare. The growing ethnomedicinal use reflects the wisdom passed down through generations in West and Central Africa. While its bitterness defines its taste, it is the plant's potential therapeutic properties that make it truly remarkable. From malaria treatment to immune system support and beyond, bitter leaf continues to play a vital role in the lives of many people. As modern science sheds light on its bioactive compounds and mechanisms of action, there is optimism about its potential contributions to modern healthcare. However, the complex intersection of traditional knowledge, scientific inquiry, and cultural heritage makes it essential to respect and value the lessons of the past while embracing the possibilities of the future. Therefore, the story of *V. amygdalina* is a testament to the enduring power of nature to heal, nurture, and inspire. *V. amygdalina* is a remarkable plant with a rich history deeply rooted in West Africa. Its origin within the region can be traced back to the tropical and subtropical zones where it has thrived for centuries. Bitter leaf's significance transcends its botanical characteristics, encompassing cultural, culinary, and medicinal dimensions that have been integral to the lives of West African communities. From the renowned bitter leaf soup (ofe onugbu) to its use in traditional medicine, bitter leaf has left an indelible mark on the cultural landscape of the region. Its cultivation and distribution have expanded beyond its native range, making it a valuable resource not only within West Africa but also in international markets. As awareness of the actual and potential health benefits grows, *Vernonia amygdalina* continues to be a subject of scientific inquiry, further unraveling its diverse uses and potential applications in modern medicine and nutrition.

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Citrus aurantifolia: Phytochemical Constituents, Food Preservative Potentials, and Pharmacological Values

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Abstract

This chapter focuses on the phytochemical constituents, food preservative potentials, pharmacological values, safety considerations, and research needs of *Citrus aurantifolia*. The phytochemical components of *Citrus aurantifolia* offer diverse health benefits, including strengthening the immune system, supporting collagen synthesis, enhancing iron absorption, reducing inflammation (vitamin C), providing anti-inflammatory and antioxidant effects, and antiviral activities (Flavonoids). Also, the other phytochemicals, including limonoids, essential oils, alkaloids, coumarins, terpenes, and phenolics, have specific health effects. *Citrus aurantifolia* is an antioxidant and anti-inflammatory compound that may alleviate conditions like arthritis. The essential oils from the fruit can inhibit pathogenic microbes, while limonoids and coumarins possess anticancer potentials. *Citrus aurantifolia* extracts can also be used to reduce stress and manage diabetes. They also function as an analgesic, anti-cholinesterase agent, immune modulator, antidiarrheal, antiulcer, anthelmintic, and, surprisingly, a contraceptive method for some women. *Citrus aurantifolia* offers natural antimicrobial properties effective against foodborne pathogens, preserving freshness and flavor while reducing reliance on synthetic preservatives. On safety considerations, *Citrus aurantifolia* can trigger allergies, irritate the stomach, erode dental health, heighten photosensitivity, and interact with medications. Overconsumption may lead to dehydration and peel pesticide residues may transfer to the juice. Proper handling and storage are crucial. Potential research needs include comprehensive phytochemical profiling and isolation, investigation of antioxidant, antibacterial, and antifungal properties, research on bioavailability and pharmacokinetics, clinical trials for health applications, safety and toxicology studies, synergy with other compounds, and sustainable cultivation and processing methods.

Keywords

Disease management · Medicinal plants · Phytochemistry · Human health · Safety considerations · Plant-based therapeutics

1 Introduction

Ethnobotany plays a pivotal role in unraveling the traditional applications of plants for medicinal and therapeutic purposes. This field, which investigates the dynamic interplay between humans and plants, enables us to tap into the treasure trove of ancestral knowledge. Herbal medicine, a specialized branch of ethnobotany, takes full advantage of this wisdom, leveraging it to harness the therapeutic potential of various plant species [33]. By uniting the insights of traditional practices with the rigor of scientific exploration, ethnobotany safeguards the invaluable knowledge of indigenous cultures.

Indigenous communities often reside in remote environments and profoundly connect with their natural surroundings. Through generations living in harmony with the land, they have cultivated a deep understanding of the indigenous flora and the myriad ways these plants can be used. Ethnobotanical studies serve as a means of preserving this collective wisdom [22].

Herbal medicine, rooted in ethnobotanical knowledge, involves applying compounds sourced from plants for therapeutic purposes [58]. *Citrus aurantifolia* emerges as a prime exemplar of the vast potential inherent in herbal therapy. The fruit, replete with various health-enhancing phytochemical constituents, takes center stage in numerous scientific investigations.

Researchers are particularly drawn to the juice extracts of *Citrus aurantifolia* (Izah and Odubo). The juice is rich in phytochemicals that provide numerous health benefits and has become the central focus of extensive research [47]. In essence, this citrus fruit embodies the marriage of traditional herbal wisdom with contemporary scientific exploration, offering a compelling case for the potential of herbal medicine to enhance human well-being.

Citrus aurantifolia, commonly known as lime, is studied for its delightful flavor and rich composition of bioactive compounds. The antimicrobial properties of *Citrus aurantifolia*, primarily attributed to essential oils like citral and limonene, position it as a viable natural option for food preservation. These compounds exhibit efficacy in inhibiting the growth of harmful bacteria and fungi, extending the shelf life of food products without the need for synthetic preservatives [47]. Incorporating *Citrus aurantifolia* into food preservation methods aligns with the growing consumer demand for unprocessed and safer food choices.

Beyond its role in food preservation and culinary applications, *Citrus aurantifolia* offers significant pharmacological properties. It possesses Vitamin C, with its anti-inflammatory and antioxidant attributes, is valuable in addressing chronic diseases and combating oxidative stress. *Citrus aurantifolia* contains limonoids, flavonoids, and essential oils such as citral and limonene, each with a spectrum of health advantages, including anti-inflammatory, antibacterial, and even anticancer effects [39]. Limonoids, mainly, show promise in cancer prevention and treatment, making *Citrus aurantifolia* a valuable addition to a well-rounded diet. Furthermore, essential oils of *Citrus aurantifolia* have even been used in aromatherapy to alleviate stress and anxiety.

This chapter looks into the capacity of natural products to enhance human well-being, exemplifying the harmonious blend of traditional wisdom and modern science in harnessing plant-based remedies for the betterment of health.

2 Phytochemical Constituents of *Citrus aurantifolia*

Citrus aurantifolia fruit has a relatively high citric acid content in comparison to *Citrus limon* (lemon) [59, 71]. *Citrus aurantifolia* has remarkable phytochemical constituents. These natural compounds are found in various plant parts, including the fruit, leaves, and essential oils [47].

Secondary metabolites are chemical compounds produced by plants, and they play a crucial role in defending against microbial and insect attacks [46, 55–57]. These secondary metabolites encompass various categories, including flavonoids, terpenoids, phenolics, limonoids, and alkaloids, which can be further categorized into volatile and non-volatile components. In the case of *Citrus aurantifolia*, the various parts, such as stems, leaves, flowers, seeds, and fruit, serve as sources for bioactive components [45, 47]. In a review study by Indriyani et al. [45], it was revealed that flavonoids and alkaloids are found in all parts of the plant (leaf, fruit, rind, seed, stem, root, and bark), while essential oil is found in the leaf and peel; terpenoids in the leaf, peel, seed, root, and bark; limonoids in the seed; and phenolics in the leaf, fruit, rind, seed, stem, and bark. The authors further reported that the peels of citrus fruits that are not commonly used contain secondary metabolites with significant antioxidant properties compared to other plant parts.

2.1 Alkaloids

Alkaloids constitute a broad category of naturally occurring organic compounds primarily found within plant materials [13, 24, 82]. They often trigger pharmacological responses in both humans and various other organisms. The vastness of this compound class is evident from the isolation of more than 10,000 distinct alkaloids in their natural contexts, underscoring their immense diversity. Alkaloids are present in a wide array of plant species, and it is believed that plants have evolved this diversity of alkaloids as a defense mechanism against herbivores and various threats.

These alkaloids can manifest pharmacological effects when administered internally or applied externally for medicinal purposes. Among the alkaloids commonly associated with plants from the *Citrus* genus, synephrine, tyramine, and octopamine stand out as the most prevalent [98]. Alkaloids have been reported in different parts of the *Cirtus auratifolia*, including peels [40], leaves [1], and fruit juice [47]. Due to their remarkable biological properties, citrus alkaloids and glycosides exhibit significant potential as pharmacological supplements.

2.2 Phenolic Compounds

Research indicates that the presence of phenolic compounds plays a significant role in how plants respond to various pathogens, including fungi, viruses, and bacteria, thereby enhancing their ability to fend off infections [90]. In the case of *Citrus aurantifolia* fruit, the level of phenolic compounds can be influenced by the fruit's degree of maturity [45]. It's noteworthy that *Citrus aurantifolia* fruit exhibits the highest phenolic content. This content varies as the fruit matures, and the extract solvent could have an impact (positive or negative) on the phenolic compounds [45]. Singh et al. [90] reported the presence of four phenolic acids: gallic acid (26.85 g/g), tannic acid (14.27 g/g), and traces of ferulic and coumaric acids in the pulp of *Citrus aurantifolia*. The authors reported that tannic acid has the highest concentration at 26.85 g/g, while gallic acid is the predominant phenolic acid in the pulp.

2.3 Limonoids

Limonoids, a group of chemical constituents, are prevalent in citrus fruits. Citrus fruits serve as rich sources of limonoids, with relatively high concentrations of these compounds. Two distinct forms of limonoids can be identified in citrus fruits: glycosides and aglycones. Aglycones can break down into acidic mono-lactones, neutral dilactones, and neutral dicarboxylic acids [45]. Limonoid aglycones are known for their bitter flavor and insolubility in water, contributing to their nomenclature [45, 66]. Oranges, in particular, are associated with a specific chemical responsible for their bitter taste. Both limonin and nomilin are glycosides, highlighting their significance [6, 45].

Limonoids belong to the subgroup of oxygenated tetracyclic triterpenoids within the broader category of triterpenoids. These compounds possess diverse biological effects, such as anticarcinogenic, antioxidative, antibacterial, larvicidal, antimalarial, and antiviral properties, as well as hypoallergenic, anti-inflammatory, antiproliferative, antimutagenic, anticarcinogenic, antitumor, and anti-obesity and hyperglycemic effects [45]. In a study by Castillo-Herrera et al. [19], methanol and acetone were used to extract limonoids from *Citrus aurantifolia* seeds, and findings showed that methanol yielded a limonoid concentration of 1.65 mg/g, while acetone resulted in a lower concentration of 0.779 mg/g.

2.4 Terpenoids

Terpenoids are a class of chemical compounds known for their low molecular weight and volatility. They are commonly present in essential oils and play significant roles, exhibiting antibacterial properties [45]. Terpenoids exert their antibacterial effects by interacting with the cell membrane, characterized by its lipophilic nature, resulting in membrane rupture. Their hydrophobic nature makes the cytoplasmic membrane a primary target for these compounds. Authors have reported that *Citrus aurantifolia* parts contain α-pinene and d-limonene [10, 15, 18]. These two compounds are characteristic components of monoterpenes, a subgroup of terpenoids, and are known for their versatile biological functions, such as antibacterial, antiseptic, and anticancer properties [45]. *Citrus aurantifolia* is rich in various types of terpenoids, including monoterpenes, alcoholic terpenes, aldehyde terpenes, ketone terpenes, and ester terpenes [45].

2.5 Flavonoids

Flavonoids, a typical class of secondary metabolites in many plant species, are prevalent among various plants, including *Citrus aurantifolia*. Specifically, *Citrus aurantifolia* is known for its rich production of flavonoids, such as 2-phenylbenzyl-pyrone. Citrus flavonoids exhibit a wide range of biological activities, acting as antioxidants, altering enzyme functions, reducing cell proliferation, and serving as antibiotics, hypoallergenic, antiulcer, antidiarrheal, and anti-inflammatory agents [45, 67]. Flavonoids such as apigenin, rutin, kaempferol, quercetin, and nobiletin have been reported in *Citrus aurantifolia* extract [45, 67]. Additionally, flavonoids like eriocitrin and hesperidin (which possess antioxidant properties) have been reported in *Citrus aurantifolia* extract [60]. Different solvents have been used to extract flavonoids from other parts of *Citrus aurantifolia* including ethanol on the leaves [42] and methanol [67, 94].

2.6 Essential Oils

Citrus aurantifolia oil, exclusively sourced from *Citrus aurantifolia* plants, constitutes a vital essential oil. The peel of *Citrus aurantifolia* yields an essential oil composed mainly of terpenes (75%), oxygenated compounds (12%), and sesquiterpenes (3%), with steam distillation being the primary extraction method [45]. The volatile constituents of *Citrus aurantifolia* include, monoterpenes such as d-limonene and terpinene, sesquiterpenes, hydrocarbons, and their oxygenated derivatives like geranial, nonanal, neryl, and linalool, as well as aldehydes, ketones, acids, alcohols, and esters [45]. Terpenes, a subgroup within this category, play a pivotal role in the essential oil's composition [45].

Furthermore, non-volatile components responsible for various qualities include fatty acids, long-chain hydrocarbons, sterols, waxes, and limonoids. Findings indicate that the primary constituents of *Citrus aurantifolia* essential oil are d-limonene (35.98%), a-pinene (9.02%), a-terpineol (8.12%), and citral (7.49%) [45, 83, 93]. Furthermore, linalyl acetate, geraniol, citral, a-pinene, a-terpineol, felandrene, sesquiphellandrene, citronellol, neryl acetate, fencone, geranyl acetate, and farnesene could also be present in the essential oil [45]. The two primary components of *Citrus aurantifolia* essential oil are citral (which gives the distinctive citrus flavor and aroma) and limonene (which can potentially inhibit the growth of harmful bacteria and fungi, owing to its potent antibacterial properties).

2.7 Vitamin C (Ascorbic Acid)

Vitamin C, also known as ascorbic acid, is crucial in supporting the body's ability to defend against illnesses and diseases [9]. Vitamin C is pivotal in helping various physiological processes, including bolstering the immune system, stimulating collagen production to maintain healthy skin and connective tissues, facilitating the absorption of dietary iron, and acting as an effective scavenger of free radicals to mitigate inflammation and oxidative stress.

Citrus aurantifolia is esteemed for its rich vitamin C content, also referred to as ascorbic acid [88]. The quality of acid *Citrus aurantifolia* fruit is primarily determined by its vitamin C content, which varies depending on the side of the tree from which the fruit is harvested [88].

In the high and mid hills zone of Nepal, fruits from the south side had the highest levels of ascorbic acid (79.6 mg and 69.9 mg), while fruits from the central part had the lowest levels (62.8 mg and 55.1 mg) [88]. Conversely, in the Terai region, fruits from the north side had the highest concentrations (58.7 mg), and fruits from the center had the lowest concentrations (41.8 mg) [88]. Al-Aamri et al. [9] reported variations in ascorbic acid content in *Citrus aurantifolia* can be attributed to factors like the nature of the fruit, geographical origin, or the extraction solvent used. The study further showed that the amount of ascorbic acid in fruit juice was considerably lower than that in the extract. The ascorbic acid content in the juice was higher in citrus species collected from Lamjung at a higher altitude (760 m) than in those from Nawalparasi at a lower height (300–400 m) [9]. This variation may be influenced by fruit position on the tree, environmental conditions, ripeness, citrus species, variety, temperature, and other factors, as ascorbic acid levels in citrus fruits are not stable and fluctuate over time [9] (Table 1).

3 Food Preservative Potentials of *Citrus aurantifolia*

Citrus aurantifolia is a desirable option as a food preservative due to its natural antibacterial and antioxidant properties. It excels in inhibiting the growth of bacteria, guarding against foodborne illnesses, and maintaining the flavor and freshness of food products. *Citrus aurantifolia* has the ability to reduce reliance on artificial preservatives aligns with the growing consumer demand for healthier and more natural food options, driven by increasing awareness of the risks associated with synthetic preservatives. *Citrus aurantifolia* is a compelling alternative to chemical preservatives with significant potential in ensuring the safety, quality, and longevity of a diverse range of food products as we explore eco-friendly and clean-label food

Table 1 Key phytochemicals found in *Citrus aurantifolia*

Phytochemical component	Description and health benefits
Vitamin C (ascorbic acid)	Strengthens the immune system Promotes collagen synthesis for healthy skin and connective tissues Enhances dietary iron absorption It acts as a free radical scavenger, reducing inflammation and oxidative stress
Flavonoids	It can impact the characteristic color of the fruit. Offers anti-inflammatory and antioxidant effects. Exhibits antiviral activities May reduce the risk of chronic diseases like cancer and heart disease
Limonoids	Abundant in the peel and seeds Under investigation for potential anticancer effects It could impede the growth of specific cancer cells
Essential oils	It is derived from the peel and leaves and is highly fragrant Primary components include citral and limonene Citral contributes to the citrus flavor and is used in aromatherapy for stress reduction Limonene has antibacterial properties and can inhibit harmful bacteria and fungi
Alkaloids	Alkaloids can have various effects on the body depending on type and concentration
Coumarins	Coumarins are associated with anti-inflammatory and anticoagulant properties
Terpenes	Terpenes are responsible for the distinctive aroma and may have potential anticancer and anti-inflammatory actions
Phenolic	Phenolic compounds in plants offer potent antioxidant properties, reducing inflammation and lowering the risk of chronic diseases, such as heart disease

Table 2 Food preservative potentials of *Citrus aurantifolia*

Aspect	Food preservative potentials of *Citrus aurantifolia*
Natural antimicrobial properties	*Citrus aurantifolia* has inherent antimicrobial properties. This makes it valuable for food preservation and extending the shelf life of various food products
Inhibition of foodborne pathogens	*Citrus aurantifolia* has antimicrobial properties that are effective against common foodborne pathogens, including *Staphylococcus aureus*, *Escherichia coli*, *Pseudomonas aeruginosa*, *Enterobacter aerogenes*, *Proteus*, and *Salmonella* species
Preservation of freshness and flavor	*Citrus aurantifolia* is rich in essential oils and phytochemicals, which preserve the sensory quality of food by inhibiting microbial growth and oxidative processes
Reduced dependence on synthetic preservatives	*Citrus aurantifolia* offers a natural alternative to synthetic preservatives, aligning with the consumer trend towards minimally processed foods with clear labels
Potential applications in different food products	*Citrus aurantifolia* is versatile and adaptable, suitable for a wide range of food products

preservation solutions. Table 2 summarizes the food preservative potentials of *Citrus aurantifolia.*

3.1 Natural Antimicrobial Properties

Citrus aurantifolia possesses inherent antimicrobial properties, making it a valuable component in food preservation [47]. Extensive research has focused on the anti-bacterial capabilities of essential oils derived from this *Citrus aurantifolia* species, such as citral and limonene [45]. These essential oils exhibit potent antimicrobial effects by disrupting various organisms' cellular membranes and metabolic processes. This attribute renders *Citrus aurantifolia* an ideal choice for extending the shelf life of diverse food products, thereby lowering the risk of contamination and spoilage.

A study by Olaniran et al. [75] yielded significant findings regarding using *Citrus aurantifolia* juice as a preservative. The study demonstrated that when *Citrus aurantifolia* juice was applied at a concentration of 4%, it exhibited the highest bacteriostatic and fungistatic efficacy. The authors concluded that *Citrus aurantifolia* juice was an effective preservative and demonstrated antimicrobial properties. This suggests that *Citrus aurantifolia* juice has the potential to be used as an alternative to chemical preservatives, offering a natural and effective means of preserving food products.

3.2 Inhibition of Foodborne Pathogens

Foodborne pathogens, bacteria, parasites, and viruses that can induce illness when ingested via contaminated food substantially threaten food safety and public health. Fortunately, several species of plants and plant-derived compounds harbor essential antimicrobial properties that can be harnessed to inhibit the proliferation of these diseases. *Citrus aurantifolia*'s antimicrobial properties make it effective in inhibiting the growth of common foodborne pathogens, including *Staphylococcus aureus, Escherichia coli, Pseudomonas aeruginosa, Enterobacter aerogenes, Proteus,* and *Salmonella* species [47]. Therefore, incorporating *Citrus aurantifolia* extracts or essential oils into food products can help reduce the risk of foodborne illnesses, ensuring food safety for consumers.

3.3 Preservation of Freshness and Flavor

Citrus aurantifolia exhibits the potential to serve as a natural preservative, capable of preserving the flavor and freshness of food. This citrus fruit is rich in essential oils and phytochemicals, both of which contribute to protecting the sensory quality of food products by inhibiting microbial growth and oxidative processes. This quality is particularly advantageous in maintaining the aroma, taste, and texture of a diverse range of culinary items, from sauces and dressings to baked goods and dairy products [49].

3.4 Reduced Dependence on Synthetic Preservatives

The widespread use of artificial chemical preservatives in the food industry has raised concerns about potential adverse health effects. In response, *Citrus aurantifolia* offers a natural alternative to synthetic preservatives, aligning with the increasing consumer preference for minimally processed food products with clear labels. By reducing our dependence on artificial preservatives, *Citrus aurantifolia* contributes to creating healthier and more natural food choices, reflecting the broader trend of favoring unprocessed options.

3.5 Potential Applications in Different Food Products

Citrus aurantifolia holds promise as a versatile food preservative suitable for a wide range of food products. Its adaptability allows for diverse applications, including beverages such as zobo [49], condiments, sauces, baked goods, dairy items, and even the preservation of fruits and vegetables in their fresh state. This adaptability makes it a valuable asset, benefiting businesses seeking natural and safe food preservation methods and consumers seeking such options.

4 Pharmacological Values of *Citrus aurantifolia*

Plants have played an indispensable role in managing and treating an array of conditions, encompassing peptic/gastric ulcers, malaria, anemia, diarrhea, diabetes, obesity, cancer, inflammation, hypertension, rheumatism, microbial infections, and a plethora of other ailments. One such plant that stands out is *Citrus aurantifolia*.

Kawaii et al. [61] have reported that all parts of *Citrus aurantifolia*, including the stem, leaves, roots, flowers, and fruit, can be employed to treat microbial infections. Different reviews on the traditional uses of *Citrus aurantifolia* highlight its antibacterial, antidiabetic, antifungal, antihypertensive, anti-inflammatory, anti-lipidemia, antioxidant, anti-parasitic, and antiplatelet activities [71]. Additionally, it has been documented for its efficacy in treating cardiovascular, hepatic, osteoporosis, and urolithiasis diseases, as well as its potential as a fertility promoter [36, 74, 86, 96, 100].

The juice and essential oil of *Citrus aurantifolia* have found widespread application in the pharmaceutical, food, and cosmetic industries due to their medicinal properties and fragrant characteristics. Aibinu et al. [6] have reported that when the juice is combined with sugar and palm oil or honey, it forms a potent cough-relieving mixture, further underscoring the versatile utility of this plant in healthcare and well-being. This section focuses on the pharmacological values of *Citrus aurantifolia* and its potential contributions to human health (Table 3).

Table 3 Pharmacological properties and benefits of *Citrus aurantifolia*

Pharmacological property	Properties and uses of *Citrus aurantifolia*
Antioxidant	*Citrus aurantifolia* is a rich source of vitamin C, offering antioxidant protection against oxidative stress and free radical damage, reducing the risk of chronic diseases
Anti-inflammatory	Phytochemicals, including flavonoids, in *Citrus aurantifolia* exhibit anti-inflammatory effects, potentially alleviating symptoms in conditions like arthritis and inflammatory bowel disease
Antimicrobial	Essential oils in *Citrus aurantifolia*, such as citral and limonene, have strong antimicrobial properties and are effective against many pathogenic bacteria and fungi
Anticancer	The limonoids and coumarins in *Citrus aurantifolia* have demonstrated anticancer potential by inhibiting the growth of various cancer cells
Stress and anxiety	Essential oils from *Citrus aurantifolia* are used in aromatherapy to reduce stress and anxiety
Antidiabetic	*Citrus aurantifolia* extracts can be used in managing diabetes by modulating glycogen metabolism
Anti-plasmodial	*Citrus aurantifolia* leaf extracts have anti-plasmodial activity, potentially contributing to the treatment of malaria
Analgesic	*Citrus aurantifolia* leaves have been found to alleviate pain without adverse effects, offering a natural analgesic option
Anti-cholinesterase	*Citrus aurantifolia* methanol extracts exhibit anti-cholinesterase activity by inhibiting acetylcholinesterase and butyrylcholinesterase (BChe), potentially impacting cognitive health
Anti-modulatory	*Citrus aurantifolia* juice has been shown to modulate the immune response
Anti-diarrhea	*Citrus aurantifolia* has been traditionally used to alleviate diarrhea, with bioactive compounds potentially reducing symptoms of indigestion.
Antiulcer	Specific components of *C. aurantifolia*, such as citral and 4-hexen-3-one, have demonstrated inhibitory effects on *Helicobacter pylori* growth, which could be valuable in addressing ulcer-related conditions
Anthelmintic	*Citrus aurantifolia* peel constituents have shown the ability to inhibit the reproduction of parasites such as certain helminths
Antifertility	Some women have used *Citrus aurantifolia* juice as a contraceptive method due to its acidity and phytochemical constituents that can destroy human sperm cells

4.1 Antioxidant Properties

One of the critical pharmacological values of *Citrus aurantifolia* lies in its antioxidant properties. The fruit is an abundant source of vitamin C, a well-known antioxidant that helps combat oxidative stress and free radical damage. These harmful molecules are implicated in various chronic diseases, including cardiovascular disease, cancer, and aging. Regular consumption of *Citrus aurantifolia* can help protect cells and tissues from oxidative damage, reducing the risk of these diseases.

Numerous studies have highlighted using essential oils derived from *Citrus aurantifolia* leaves as natural antioxidants [71], particularly in the pharmaceutical and food industries, owing to their relatively safe toxicological profiles. In a study by Al-Aamri et al. [9], it was found that 31 essential oils sourced from citrus fruits exhibited superior antioxidant properties compared to Trolox. Authors have documented that the presence of D-limonene in both the leaves and rind of *Citrus aurantifolia* accounts for its antioxidant properties and the ability to combat cell damage induced by free radicals [39, 65, 81]. Furthermore, the flavonoids found in *Citrus aurantifolia* have demonstrated antioxidant and anticancer attributes by inhibiting cell division in cancerous cells, as noted in the research conducted by Abubakar et al. [3].

4.2 Anti-inflammatory Effects

Inflammation is a natural defense mechanism in the human body designed to eliminate or restrict the spread of harmful agents. While it is a typical response to tissue injury, it can become uncontrolled in chronic autoimmune conditions like rheumatoid arthritis, Crohn's disease, and allergic reactions such as asthma and anaphylactic shock [8, 84]. The phytochemical constituents of *Citrus aurantifolia*, such as flavonoids, have demonstrated anti-inflammatory properties [71]. Inflammation is a central component of many chronic conditions, including arthritis and inflammatory bowel disease. The natural compounds in *Citrus aurantifolia* can help mitigate inflammation, relieving individuals suffering from these ailments. *Citrus aurantifolia* leaves and rind contain D-limonene as a significant constituent, which has been reported to reduce inflammatory markers associated with conditions characterized by chronic inflammation, such as osteoarthritis.

4.3 Antimicrobial Activity

Plant has emerged as a source of new antimicrobial probability due to their ability to inhibit gram-positive and harmful microbes [50–54]. The essential oils found in *Citrus aurantifolia*, particularly citral and limonene, exhibit potent antimicrobial properties. These compounds have traditionally been used to inhibit the growth of pathogenic bacteria and fungi. As a result, *Citrus aurantifolia* has applications in the field of pharmacology as a potential source of natural antimicrobial agents. This can be particularly valuable when antibiotic resistance is a growing concern.

Numerous reports have highlighted the potent antimicrobial properties of *Citrus aurantifolia* juice against a broad spectrum of pathogenic microorganisms (bacterial and fungal) [7, 31, 32, 63, 77, 87, 95, 97], including *V. cholerae*, *S. aureus*, *K. pneumonia*, *S. pyogenes*, *Actinobacillus* sp., *Citrobacter* sp., *Shigella* sp., as documented by Jayana et al. [59] and Adebayo-Tayo et al. [4]. The effectiveness of this antimicrobial activity can be attributed mainly to the complex chemical composition of the juice, which encompasses various chemical classes such as terpenes, aldehydes, alcohols, esters, phenols, ethers, and ketones, as highlighted by Al-Aamri et al. [9]. Moreover, extracts derived from *Citrus aurantifolia* fruit have been found to exhibit antimicrobial properties against microorganisms like *Candida albicans*, *Aspergillus niger*, and *Aspergillus fumigatus*, as reported by Aibinu et al. [6] and Abubakar et al. [2].

In addition to the fruit extracts, the ethanolic extracts obtained from *Citrus aurantifolia* leaves have demonstrated bactericidal effects against a range of bacterial species, including *Bacillus subtilis*, *Escherichia coli*, *Salmonella*, and *Streptococcus feacalis*, as observed by Oboh and Abulu [73].

These antimicrobial properties are attributed to bioactive phytochemicals in different parts of *Citrus aurantifolia*. Flavonoids, in particular, have been recognized for their influence on arachidonic acid metabolism, enabling them to combat a wide array of bacteria by forming complexes with extracellular and soluble proteins within bacterial cell walls, as explained by Adebayo-Tayo et al. [3].

4.4 Anticancer Potential

Cancer is a primary global public health concern, the second leading cause of death. In the United States in 2015, there were approximately 1.6 million new cancer cases, resulting in roughly 500 thousand cancer-related deaths [89]. Perhaps one of the most exciting pharmacological values of *Citrus aurantifolia* is its potential for cancer prevention and treatment [71]. The limonoids in this citrus fruit have shown anticancer properties by inhibiting the growth of cancer cells. These compounds may help reduce the risk of cancer development and support treating various cancer types. An intriguing study by Patil et al. [78, 79, 80] revealed the anticancer potential of *Citrus aurantifolia* fruit extract. They found that at a concentration of 100 μg/ml, this extract inhibited the growth of human colon SW-480 cancer cells by an impressive 78% after 48 h of exposure. This remarkable effect can be attributed to the fruit extract's substantial amounts of limonene and dihydrocarvone. Additionally, when a 25 μM concentration of *Citrus aurantifolia* peel extract was used, it displayed the capability to inhibit the growth of colon SW-480 cancer cells in humans by 67% after 72 h of exposure. This inhibition is attributed to three key coumarins: 5-geranyloxy-7-methoxycoumarin, *Citrus aurantifolia* tin, and isopimpinellin found in the peel extract.

Patil et al. [78, 79] reported that 100 μg/ml of *Citrus aurantifolia* juice extract effectively halted approximately 73–89% of pancreatic Panc-28 cancer cell growth in humans after 96 h of exposure. This remarkable effect can be attributed to limonoid substances, such as limonexic acid, isolimonexic acid, and limonin in the juice extract. Meanwhile, *Citrus aurantifolia* seeds extract exhibited the ability to stop the growth of pancreatic Panc-28 cancer cells in humans with an inhibitory concentration of 50% (IC50) ranging from 18 to 42 μM after 72 h of exposure. This inhibition was achieved through compounds like limonin, limonexic acid, isolimonexic acid, β-sitosterol glucoside, and limonin glucoside.

In another study, Gharagozloo et al. [35] reported that *Citrus aurantifolia* fruit juice extract from Iran, at concentrations ranging from 125 to 500 μg/ml, effectively inhibited the growth of breast MDA-MB-453 cancer cells after 24 h of exposure. Furthermore, Adina et al. [4] found that a concentration of 6 and 15 μg/ml of *Citrus aurantifolia* peel extract from Indonesia was capable of inhibiting the growth of breast MCF-7 cancer cells, targeting different phases of the cell cycle, specifically G1 and G2/M, respectively, after 48 h of exposure.

Additionally, Castillo-Herrera et al. [19] reported that limonin extract derived from *Citrus aurantifolia* seeds in Mexico displayed remarkable inhibitory effects on the growth of L5178Y lymphoma cells, with an IC50 ranging from 8.5 to 9.0 μg/ml. These findings underscore the promising potential of *Citrus aurantifolia* extracts in the fight against various types of cancer.

4.5 Stress and Anxiety Relief

Anxiety, a common mental health condition often associated with comorbidity, poses a significant psychological, social, and economic burden and an elevated risk of physical illness [62, 76]. The pharmacological treatment of anxiety and mood disorders includes drugs like beta blockers, benzodiazepines, monoamine oxidase inhibitors, tricyclic antidepressants, and selective serotonin reuptake inhibitors. Still, they have notable limitations, such as comorbid psychiatric disorders, acute poisoning, and addiction [91]. The essential oils from *Citrus aurantifolia* are often used in aromatherapy for their calming and stress-reducing effects. These essential oils can be explored in pharmacology as a natural alternative to pharmaceutical anxiolytics. The aroma of *Citrus aurantifolia* can help alleviate stress and anxiety, promoting mental well-being. Furthermore, Guzmán et al. [37] reported that in China, Brazil, and Mexico, the fruit peel and juice of *Citrus aurantifolia* have been employed as tranquilizers and anxiolytic medications.

4.6 Antidiabetic Properties

Diabetes mellitus, a metabolic disorder characterized by elevated blood sugar levels, stems from issues with insulin secretion or action, affecting the metabolism of carbohydrates, fats, and proteins [23]. Type 1 diabetes results from the autoimmune destruction of pancreatic β-cells, necessitating external insulin supplementation for patients to maintain their well-being [20]. Conversely, Type 2 diabetes is associated with dysfunctional insulin levels that impair glucose uptake.

Type 2 diabetes, which can manifest as a chronic or mild condition, can eventually lead to vascular complications, dysfunction, and an increased risk of coronary artery and peripheral vascular diseases. This ailment is a pressing global public health concern, as the World Health Organization reports that approximately 422 million individuals have diabetes, with around 90% of these cases being Type 2 diabetes [44].

Ethnobotanical research has revealed a notable preference for extracts from the Rutaceae family in diabetes management. For instance, the essential oil of *Citrus aurantifolia* has shown potential in inducing apoptosis in pancreatic cell lines and reducing hyperglycemia in alloxan-induced rats. Furthermore, it may possess antidiabetic properties through its ability to modulate glycogen metabolism [44]. Narang and Jiraungkoorskul [71] have reported the antidiabetic potential of *Citrus aurantifolia*.

4.7 Antidepressant

In the current era, mental and neurological disorders stand out as one of the foremost global health concerns. Among these, anxiety and mood disorders are particularly prevalent and carry a heavy psychological and socio-economic burden, increasing the risk of physical ailments. Conventional medications for anxiety and mood disorders include beta-blockers, benzodiazepines, monoamine oxidase inhibitors, tricyclic antidepressants, and selective serotonin reuptake inhibitors. However, these medications have limitations, such as the potential for co-existing psychiatric disorders, acute toxicity, drug interactions, and the development of physical dependence, leading to intolerable side effects [91]. As a result, there has been a quest for alternative and innovative treatments, and promising results have emerged from using plants. For instance, Chinese, Brazilians, and Mexicans have employed the fruit peel and juice extract of *Citrus aurantifolia* as tranquilizers and anxiolytic therapies. At the same time, the leaves can be used as tranquilizers. Additionally, the flowers of *Citrus aurantifolia* are used in Mexico to alleviate anxiety and nervousness. In some parts of Latin America and Africa, an infusion of dried *Citrus aurantifolia* flowers is a nerve-soothing sedative [91].

4.8 Anti-plasmodial

Turning to anti-plasmodial treatments, malaria, primarily transmitted through the bite of female Anopheles mosquitoes carrying *Plasmodium falciparum*, poses a significant global health threat. The parasite is introduced into human red blood cells during the mosquito's bite, leading to a malarial infection once it multiplies in the bloodstream. Malaria claims the lives of a staggering live in many region of the word with Africa being the most endemic region of the diseases [17, 26, 28, 48, 72, 101]. Furthermore, malaria kills 235,000 to 639,000 individuals worldwide annually [43, 70, 99]. Furthermore, resistance to conventional antimalarial drugs has been documented, a significant breakthrough emerged from Chinese researchers who discovered the efficacy of Artemisinin (Qinghao), a traditional Chinese medicine, in combatting drug-resistant forms of malaria infection [30]. In Africa, *Citrus aurantifolia* has been employed to treat various illnesses, with the fruit juice being recognized for its ability to alleviate swelling resulting from mosquito bites. Furthermore, Ettebong et al. [30] have reported that *Citrus aurantifolia* leaf extract exhibits substantial anti-plasmodial activity in vivo.

4.9 Analgesic

The management of analgesic conditions often necessitates synthetic molecules, which, unfortunately, can lead to various adverse effects on the body. These effects may vary in severity and impact critical systems like the kidneys and digestive tract. Plant-based treatments have gained attention as an alternative approach to addressing analgesic ailments [21]. In a recent study by Al-Snafi [12], it was found that administering an extract of *Citrus* species to mice effectively alleviated their analgesic condition. Significantly, this natural remedy exhibited no adverse effects and did not result in mortality among the test subjects.

4.10 Anticholinesterase Activity

Alzheimer's disease is a progressive neurological disorder characterized by cognitive deficits and behavioral abnormalities. It has been well-documented that the progression of Alzheimer's disease is linked to the presence of reactive oxygen species, emphasizing the importance of using antioxidants to reduce disease advancement and minimize neuronal degeneration. Additionally, a cholinergic deficit resulting from basal forebrain degeneration calls for strategies to enhance acetylcholine production – an essential neurotransmitter for cognitive functions – to restore cognitive deficits. This approach is known as the cholinergic hypothesis [14, 85, 92].

Notably, *Citrus aurantifolia* methanol extracts in n-hexane fractions have significant anti-cholinesterase effects. This effect is achieved through the inhibition of acetylcholinesterase (AChe) and butyrylcholinesterase (BChe), as demonstrated in plasma assays, as reported by Loizzo et al. [67].

4.11 Anti-modulatory Activity

Citrus aurantifolia, commonly known as lime, has been found to exhibit immune-modulatory activity, as demonstrated in a trial utilizing mitogen-activated culture mononuclear cells. In this study conducted by Gharagozloo and Ghaderi [34], the juice of *Citrus aurantifolia* was assessed for its impact on the immune response.

The results indicated that the application of 250 and 500 micrograms of *Citrus aurantifolia* juice led to a significant inhibition of the proliferation of phytohemagglutinin-activated mononuclear cells. This suggests that *Citrus aurantifolia* juice can modulate the immune response, potentially by regulating the activation and proliferation of immune cells.

This finding underscores the potential immunomodulatory properties of *Citrus aurantifolia*, which may have implications for its use in various contexts, such as supporting immune health or developing therapies to regulate immune responses. Further research and exploration of these properties could provide valuable insights into the potential applications of *Citrus aurantifolia* juice in immunology and health.

4.12 Anti-diarrhea Activity

Diarrhea is a condition that can pose a significant threat to patients, potentially leading to severe pain and, in extreme cases, even death if it remains untreated or is inadequately managed [25, 52]. While diarrhea can affect individuals of all age groups, it is especially concerning in infants and children, necessitating special attention. Several factors are known to be direct or indirect triggers for diarrhea, including bacterial and viral infections, allergies, malabsorption issues, and food poisoning [25].

Citrus aurantifolia has been traditionally used to support digestive health. Its bioactive compounds may aid in reducing symptoms of indigestion and promoting overall gastrointestinal well-being. This pharmacological value is particularly relevant in digestive disorders and conditions. He et al. [41] conducted a study revealing that pre-treatment with dried fruits of *Citrus aurantifolia* significantly alleviated 2,4,6-trinitrobenzene sulfonic acid-induced diarrhea in rats. This finding suggests a potential anti-diarrhea effect associated with *Citrus aurantifolia*, offering promise for managing this distressing condition.

4.13 Antiulcer

Helicobacter pylori, a prominent pathogen associated with developing various ulcer-related conditions such as gastritis, peptic ulcers, gastric adenocarcinoma, and mucosa-associated lymphoid tissue lymphoma, has become widely prevalent in the human population [11, 27]. This is primarily due to its high level of antibiotic resistance, which poses a significant challenge to the effective treatment and eradication of *Helicobacter pylori*. Consequently, there is a growing need for new alternatives to address this issue.

In regions with a high prevalence of *Helicobacter pylori*, citrus fruit has been reported to reduce the occurrence of gastric cancer [16, 64]. Lee et al. [64] further demonstrated that specific components of *Citrus aurantifolia*, such as citral and 4-hexen-3-one, can inhibit the growth of triple drug resistant *Helicobacter pylori* strains. Notably, the 4-hexen-3-one constituent exhibited a more substantial inhibitory effect on the urease activity of triple drug resistant *Helicobacter pylori* strains as its concentration increased, indicating the potential of these citrus components as a promising approach in addressing *Helicobacter pylori*-related conditions.

4.14 Anthelmintic

Helminthosis is a parasitic infection known for its detrimental impact on animal health, causing growth impairment, reducing survival rates, hindering reproductive performance, and eventually leading to the death of infected animals. This poses a substantial economic burden on humans, especially livestock farmers. Currently, chemotherapy is the most effective means of treating helminthosis, but it has significant cost implications and raises concerns regarding developing resistant helminth parasites.

While the fruit peels of *Citrus aurantifolia* have gained recognition for their medicinal properties in managing various illnesses and infections, they are rarely included in the diet of livestock [29, 91]. Enejoh et al. [29] conducted a study revealing that the constituents found in *Citrus aurantifolia* peel exhibit a noteworthy ability to inhibit the hatching of *Heligmosomoides bakeri* eggs, a trichostrongyloid parasite naturally occurring in wild mice. Additionally, these constituents were effective in killing the helminth larvae in vitro. This suggests a potential role for *Citrus aurantifolia* peel in combating helminth infections and highlights the significance of exploring this natural remedy in livestock health and management.

4.15 Antifertility

The ability to reproduce offspring is a fundamental and vital aspect of life. However, various health conditions can interfere with this function. When individuals face challenges in their ability to conceive and reproduce, it can give rise to multiple concerns and potentially lead to additional health-related issues, such as anxiety and depression. Interestingly, reports have shown that infertility can also be induced intentionally by using various plant extracts by individuals who do not wish to have children. For example, the juice of *Citrus aurantifolia*, commonly known as *Citrus aurantifolia*, has been employed by African women as an alternative contraceptive method, along with douching, to prevent pregnancy and sexually transmitted diseases [74].

Furthermore, in vitro studies conducted by Okon and Etim [74], Hair et al. [38], Masud et al. [69], and Longo et al. [68] have unveiled antifertility findings regarding the juice of *Citrus aurantifolia*. This citrus juice can destroy human immunodeficiency virus (HIV) and human sperm cells. This effect is attributed to the juice's high acidity and phytochemical constituents. These studies shed light on the potential applications and consequences of using *Citrus aurantifolia* juice in the context of antifertility and protection against certain health risks.

In summary, *Citrus aurantifolia* offers a range of pharmacological values due to its rich composition of bioactive compounds. Its antioxidant, anti-inflammatory, antimicrobial, and anticancer properties make it a valuable addition to pharmacological research in preventing and treating chronic diseases. Additionally, its stress-relieving and digestive health benefits contribute to its potential role in enhancing overall well-being. As we continue to explore the health benefits of natural products, *Citrus aurantifolia* stands as a promising candidate in the field of pharmacology for its diverse and potent pharmacological values.

5 Safety Considerations of *Citrus aurantifolia* Juice Extracts

Even though juice extracts from *Citrus aurantifolia* can enhance culinary capabilities and offer potential health benefits, it is essential to be aware of the associated risks. Some of the related risks of *Citrus aurantifolia* are as follows:

allergies, stomach sensitivity, dental health, photosensitivity, medication interactions, hydration, etc. (Table 4). Therefore, achieving a safe and enjoyable experience hinges on consuming these extracts in moderation and making well-informed choices.

6 Research Needs on the Juice Extracts of *Citrus aurantifolia*

Citrus aurantifolia juice extracts hold significant promise as a valuable natural resource with various potential health benefits. To harness research, it is essential to encompass the identification and isolation of bioactive compounds, followed by comprehensive clinical trials and safety assessments. Researching this will advance the understanding of the natural compounds in plants and pave the way for innovative preventive and therapeutic applications in nutrition, medicine, and food science. Table 5 shows the potential research needs on the juice extracts of *Citrus aurantifolia.*

Table 4 Safety considerations of *Citrus aurantifolia* juice extracts

Consideration	Safety considerations of *Citrus aurantifolia* juice extracts
Allergies	*Citrus aurantifolia* can trigger allergic reactions, ranging from mild skin rashes to severe anaphylaxis. Therefore, individuals with citrus allergies should use extreme caution when consuming *Citrus aurantifolia* juice
Stomach sensitivity	*Citrus aurantifolia* juice contains high acidity, which can irritate the stomach and digestive tract, potentially causing discomfort, heartburn, or acid reflux. Therefore, those with digestive disorders should consume *Citrus aurantifolia* juice cautiously
Dental health	The acidity in *Citrus aurantifolia* juice can erode tooth enamel, increasing the risk of dental issues like tooth sensitivity and cavities. Therefore, rinsing with water after consumption is advised, and avoid brushing teeth immediately afterward
Photosensitivity	*Citrus aurantifolia* juice may contain compounds that make the skin more sensitive to sunlight, potentially leading to phytophotodermatitis and skin damage when exposed to the sun. Therefore, caution must be exercised when applying *Citrus aurantifolia* juice to the skin or ingesting it before sun exposure
Medication interactions	*Citrus aurantifolia* juice can interact unfavorably with certain medications, altering their effectiveness or causing side effects
Hydration and electrolytes	*Citrus aurantifolia* juice has a natural diuretic effect due to its potassium content, which can lead to dehydration and electrolyte imbalances when consumed excessively
Pesticide residues	*Citrus aurantifolia* peels may carry pesticide residues that can transfer into the juice from the soil
Safe handling and storage	Proper handling and storage practices are essential for food safety

Table 5 Research needs on the juice extracts of *Citrus aurantifolia*

Research area	Potential research needs on the juice extracts of *Citrus aurantifolia*
Phytochemical profiling and isolation	Comprehensive profiling of phytochemicals in *Citrus aurantifolia* juice extracts Isolation and purification of these compounds to understand their specific health contributions
Antioxidant effects	Research on how *Citrus aurantifolia* reduces oxidative damage and inflammation
Antibacterial and antifungal activities	Studying the extent to which *Citrus aurantifolia* juice extracts inhibit the growth of pathogenic microorganisms – potential applications as natural food preservation and antibiotic alternatives
Bioavailability and pharmacokinetics	Research on how the body absorbs, distributes, metabolizes, and eliminates the bioactive constituents of the juice. Essential for optimal dosage and administration
Clinical trials and health uses	Conducting well-designed clinical trials to assess the effectiveness of *Citrus aurantifolia* juice extracts in preventing and treating conditions such as cancer and chronic inflammatory disorders
Studies on the product safety and toxicology	Comprehensive toxicology studies to establish safe consumption dose levels for *Citrus aurantifolia* juice extracts
Synergistic effects with other components	Study how *Citrus aurantifolia* juice extracts interact with other natural compounds and medications
Research into sustainable cultivation practices and processing technologies	Research into sustainable cultivation methods and processing technologies to preserve the integrity of bioactive components and ensure a consistent supply of high-quality raw materials

7 Conclusion

Investigating *Citrus aurantifolia* and its juice extracts reveals a multifaceted natural resource with significant potential in nutrition, medicine, and food preservation. The phytochemical constituents present in *Citrus aurantifolia* contribute to its distinct properties, and research in several areas is vital to unlock its full potential. Firstly, the phytochemical profiling and isolation of specific compounds, such as vitamin C, flavonoids, and essential oils, form a fundamental research need. Understanding the bioactive compounds in *Citrus aurantifolia* enables us to harness their potential in various applications. Moreover, the antioxidant properties of *Citrus aurantifolia* juice extracts are a noteworthy avenue of research. Exploring these properties in the context of oxidative stress-related diseases has promising implications for preventive and therapeutic measures. These extracts' antimicrobial and antifungal activities are another significant area of study. These natural extracts show potential for use as food preservatives and alternatives to synthetic antibiotics. Bioavailability and pharmacokinetics studies are essential to optimize the health benefits of *Citrus aurantifolia*. Understanding how the body processes these compounds ensures effective administration. To validate the practical health applications of *Citrus aurantifolia* juice extracts, transitioning from in vitro and animal studies to well-designed clinical trials is necessary. This research can help us unlock their potential in preventing and managing various health conditions. Additionally, safety and toxicology studies are crucial to establishing the safe dosage levels of these extracts, assuring consumers and healthcare providers of their safety. Considering potential synergistic effects with other compounds, particularly in combination therapy, offers exciting prospects for enhancing therapeutic outcomes. Finally, sustainable cultivation and efficient processing methods should be researched to ensure a consistent supply of high-quality raw materials while maintaining the integrity of the bioactive compounds.

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Phytochemical Constituents of *Rosmarinus officinalis Linn.* and Their Associated Role in the Management of Alzheimer's Disease

12

Marcella Tari Joshua

Contents

Abstract

Neurodegenerative disorders, such as Alzheimer's disease, lead to a gradual and irreversible loss of neurons and brain function, resulting in progressive dementia characterized by disoriented plaques and neurofibrillary tangles composed of beta-amyloid and hyperphosphorylated tau proteins, respectively. Currently, there is no permanent treatment for these disorders. However, herbal plants and formulations, along with certain research findings, have reported the potential therapeutic use of rosemary leaves to mitigate these conditions more pleasantly. *Rosmarinus*

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officinalis Linn., commonly known as rosemary, holds significant economic importance and is often used as a flavor enhancer in soups. In the field of phytomedicine, this herb is gaining recognition as researchers investigate not only its unique taste and aroma but also the various health benefits of its constituents when extracted. The phytotherapeutic components of rosemary, such as rosmarinic acid and other derivatives found in its essential oils, including caffeic acid, p-coumaric acid, rosmanol, carnosol, carsonic acid, 4-hydroxybenzoic acid, beta-amyrin, oleanic acid, and urosolic acid, offer numerous advantages. These constituents possess anti-memory loss properties and various other health benefits, primarily due to their inhibitory effects on factors contributing to memory problems. This chapter sheds light on the impact of *Rosmarinus officinalis Linn.*, specifically rosemary leaves and its essential constituents, on health benefits. These benefits include its potential as an anti-memory loss agent, possibly attributed to the presence of flavonoids, triterpenoids, diterpenoids, and antioxidants. The review chapter critically examines the protective systematic effect this natural medicinal plant produces, such as being a stimulus as well as a protective agent that promotes healthier brain functionality.

Keywords

Evidence- Rosemary Leaves · *Rosmarinus officinalis Linn*. Rosmarinic Acid · Anti-memory loss · Flavonoids · Amyloid Beta-Protein · Dementia

Abbreviations

AChE	Acetylcholinesterase
AD	Alzheimer's Disease
AGE	Advanced Glycation End Product
AGE-RAGE	Advanced Glycation End Product-Receptors of Advanced Glycation End Product
ALS	Amyotrophic Lateral Sclerosis
BChE	Butyrylcholinesterase
CA	Carsonoic Acid
CJD	Creutzfeldt-Jakob Disease
CORT	Corticosterone
FDA	Food and Drug Administration
FTD	Frontotemporal Dementia
KEAP1	Kelch-Like Associated Protein 1
LBD	Lewy Body Dementia
MS	Multiple Sclerosis
NRF2	Nuclear Factor Erythroid-2-Related Factor
p-CA	P-Coumaric Acid
PSP	Progressive Supranuclear Palsy
RA	Rosmarinic Acid
RAW 264.7	Murine Macrophage Cell Line
UA	Ursolic Acid

1 Introduction

Phytochemistry is the branch of chemistry that focuses on the study of the chemical compounds found in plants, particularly those with medicinal properties. It involves the identification, isolation, characterization, and analysis of the various chemical constituents present in plant species. Phytochemistry plays a crucial role in understanding the therapeutic potential of medicinal plants and in developing herbal medicines.

Herbal medicine, also known as phytotherapy or botanical medicine, is a traditional or alternative healthcare system that utilizes medicinal plants and their extracts to prevent, alleviate, or treat various health conditions [3, 4, 24, 48]. It relies on the knowledge of the phytochemicals present in plants and their pharmacological effects on the human body. Herbal medicine has been practiced for centuries in different cultures and is still widely used today as a complementary or alternative approach to conventional medicine.

Medicinal plants are plants that contain bioactive compounds with therapeutic properties [45–47, 50–52]. These plants have been used for centuries in traditional medicine systems around the world to treat a wide range of ailments and diseases. Medicinal plants serve as the primary source of ingredients for herbal medicines. They are valued for their natural compounds, which can have anti-inflammatory, antioxidant, analgesic, antimicrobial, and other beneficial effects on health [40–44].

Therefore, phytochemistry delves into the chemical composition of plants, herbal medicine employs medicinal plants and their extracts for health purposes, and medicinal plants are the natural sources of bioactive compounds used in herbal remedies and traditional medicine systems. These fields are interconnected and contribute to our understanding of the healing potential of the natural world.

Rosemary, scientifically known as "Rosmarinus officinalis Linn.," is an evergreen shrub known for its aromatic and medicinal properties. It has received approval from the European Union [8] and is commonly found growing around the Mediterranean region. This plant is classified under the Lamiaceae plant family and is characterized by slender, needle-like leaves on erect woody stems. The leaves are green on the upper side and white underneath. Rosemary branches are firm with a cracked bark, and the woody stems are brown and square-shaped. It produces small light blue to white flowers in a distinctive inflorescence, typically blooming from late spring to early summer.

Various parts of the rosemary plant, including its leaves, flowers, and twigs, are processed to yield phytonutraceuticals [75]. Rosemary is cultivated in different regions worldwide and holds multiple uses. It is a staple in Mediterranean cuisine, used to make tea, incorporated into cosmetology products, and has a history in traditional and contemporary medicine for treating issues such as colds, coughs, and rheumatoid problems. Given its diverse applications, exploring the neuron-protective properties of rosemary leaves as a potential treatment for memory loss associated with neurodegenerative disorders is essential.

Numerous epidemiological studies have shown that a diet rich in polyphenols from rosemary leaves (*Rosmarinus officinalis* L.) can reduce the risk of Alzheimer's disease, a prevalent form of dementia in elderly individuals [7, 60, 86]. Among the various phyto-constituents found in *Rosmarinus officinalis*, compounds like caffeic acid, rosmarinic acid, oleanolic acid, urosolic acid, rosmanol, carnosol, carsonic acid, piccrosalvin, beta-amyrin, and rosmarinic acid stand out as promising anti-memory agents. These compounds appear to mitigate the progressive development of dementia by inhibiting alpha-beta aggregation and reducing synaptic toxicity in vitro [64].

According to Farmanfarma et al. [27], the accumulation of certain proteins, particularly in developed countries, has been linked to the development of brain cancer, accounting for approximately 3% of all cancer types, especially within the category of nervous system malignancies. Among these, 48% are classified as primary tumors in the central nervous system, with about 57% falling into the category of glioblastoma, a type of glioma that predominantly affects males between the ages of 75 and 84 and is also potentially linked to Alzheimer's disease. Extracts from *Rosmarinus officinalis* L. have been utilized to counteract these effects.

Reports by Sayorwan et al. [76] and Czajkowski and Nazaruk [19] regarding essential oil extracts from rosemary herb highlight their therapeutic benefits in slowing memory loss and preventing Alzheimer's disease, supported by evidence-based research. Sayorwan et al. [76] demonstrated that inhaling essential oils extracted from the Rosmarinic acid found in rosemary leaves via the olfactory nerve had a significant impact on brainwave activity. This stimulation influenced mood states and cognitive processes, enhancing alertness and mental activity, ultimately leading to reduced sleepiness, increased productivity, and a sense of refreshment.

Furthermore, Czajkowski and Nazaruk [19] reported that Rosmarinic acid from rosemary leaves had protective properties against neurotoxicity induced by the deposition of tau proteins and beta-deposits in the brain. This therapeutic potential of Rosmarinic acid had a gradual effect on the development of dementia, a prevalent form of neurodegenerative disorder characterized by the aggregation of tau proteins and beta-amyloid in the brain.

Regarding gliomas, which are highly toxic brain tumors, the use of rosmarinic acid as a treatment therapy for glioblastoma has been reported but has shown limited success in preventing neurological dysfunction and death [27]. Frontotemporal disorders, resulting from damage to neurons in the frontal and temporal regions of the brain, manifest as emotional difficulties, communication problems, and impaired working and walking abilities.

In a study conducted by Rasoolijazi et al. [71], the anti-memory effects of rosemary leaves were assessed using a behavioral evaluation involving eight rats in a Morris water maze test. The experiment involved dividing the rats into two groups: a positive group induced with neurotoxins (scopolamine) and a control group provided with regular rat chow and water ad libitum. The positive group exhibited signs of depression and reduced mobility due to the induced neurotoxins. Subsequently, the induced rats were orally treated with extracts from *Rosmarinus officinalis* L. Hidden cameras were strategically placed in different sections of the animal housing facility to record the rats' swimming patterns and behavioral activities. The rats underwent learning and spatial memory retention phases.

The rats received varying doses of rosemary leaf extracts dissolved in distilled water, with doses of 50, 100, and 200 mg/kg administered through oral gavage

(1 mL). After treatment, the laboratory mice were induced with a single dose of scopolamine intraperitoneally. The rats were then observed for their ability to locate the platform within 60 s, considering the process of navigating to the correct environment and the reduction of dormancy in each mouse. The results of the experiment highlighted the potential neuroprotective effect of the rosemary leaf extracts on the proper functioning of the hippocampus, particularly emphasizing the effectiveness of rosmarinic acid, a key constituent of rosemary leaves.

Currently, there is a growing need for further research on the therapeutic potential of *Rosmarinus officinalis Linn.* in the treatment of cognitive impairment. Existing studies in this area are limited, and there is a growing interest among researchers to conduct more comprehensive research in the future to gain deeper insights into the various aspects of the rosemary plant [5]. The objective of this chapter is to explore evidence-based research that demonstrates the therapeutic properties of Rosmarinic acid and other phytochemical constituents found in rosemary leaves as potential candidates for mitigating memory loss in individuals.

2 Overview of Neurodegenerative Diseases

Neurodegenerative diseases are a group of conditions characterized by a gradual decline in the function and structure of the nervous system, particularly affecting neurons in the brain and spinal cord. These illnesses can significantly diminish a person's quality of life and often lead to the progressive loss of cognitive and physical abilities. Neurodegenerative diseases typically involve the accumulation of abnormal proteins, oxidative stress, and inflammation within the brain, although their precise causes are not always fully understood. Several well-known examples of neurodegenerative diseases are summarized in Table 1.

1. Alzheimer's disease: This is the most common neurodegenerative disorder, primarily impacting memory and cognitive functions. It is characterized by the buildup of tau protein tangles and beta-amyloid plaques in the brain.
2. Parkinson's disease: Parkinson's disease affects motor coordination and movement and is associated with the loss of dopamine-producing neurons in the brain. Symptoms often include tremors, stiffness, and slow movements.
3. Huntington's disease: This genetic disorder leads to the degeneration of neurons in the basal ganglia and cortex, resulting in cognitive decline, psychiatric symptoms, and motor impairment.
4. Amyotrophic lateral sclerosis (ALS), also known as Lou Gehrig's disease: ALS affects motor neurons in the brain and spinal cord, causing issues with breathing, speaking, swallowing, muscle weakness, and paralysis.
5. Multiple sclerosis (MS): In this autoimmune disease, the immune system targets the myelin sheath surrounding nerve fibers in the central nervous system, leading to various neurological symptoms like fatigue, coordination problems, and sensory disturbances.

Table 1 Common neurodegenerative diseases

Neurodegenerative disease	Description	Key features/symptoms
Alzheimer's disease	Affects memory and cognition, characterized by beta-amyloid plaques and tau protein tangles in the brain.	Memory loss, cognitive decline.
Parkinson's disease	Impairs movement, linked to dopamine neuron loss.	Tremors, rigidity, bradykinesia (slow movement).
Huntington's disease	Genetic disorder, causes basal ganglia and cortex neuron degeneration.	Motor dysfunction, psychiatric symptoms.
Amyotrophic Lateral Sclerosis	ALS or Lou Gehrig's disease, affects motor neurons in brain and spinal cord.	Muscle weakness, paralysis, speaking/ swallowing difficulties.
Multiple Sclerosis (MS)	Autoimmune disease attacks myelin sheath, central nervous system.	Fatigue, coordination problems, sensory disturbances.
Frontotemporal Dementia (FTD)	Degeneration of frontal and temporal brain lobes.	Behavior/personality changes, language deficits.
Amyloidosis	Group of rare diseases, amyloid protein deposits in organs, including the nervous system.	Organ dysfunction, neurological symptoms.
Creutzfeldt-Jakob Disease (CJD)	Rare, fatal prion disease affecting the brain.	Rapid cognitive decline, muscle stiffness/twitching.
Progressive Supranuclear Palsy	Rare disorder affecting movement and balance.	Eye movement issues, speech/swallowing problems.
Lewy Body Dementia (LBD)	Characterized by Lewy body protein deposits in the brain, similar to Parkinson's symptoms.	Cognitive decline, visual hallucin

6. Frontotemporal Dementia (FTD): FTD involves the degeneration of the frontal and temporal lobes of the brain, resulting in changes in behavior, personality, and language skills.
7. Amyloidosis: Amyloidosis encompasses a group of rare disorders where abnormal proteins called amyloids accumulate in various organs, including the nervous system, causing neurological symptoms and organ dysfunction.
8. Creutzfeldt-Jakob disease (CJD): CJD is a fatal prion disease that damages the brain, causing muscle twitching, stiffness, and rapid cognitive decline.
9. Progressive supranuclear palsy (PSP): PSP is a rare neurological condition that impairs balance and movement, often leading to difficulties in swallowing, speaking, and eye movements.
10. Lewy body dementia (LBD): LBD is a type of dementia characterized by the presence of abnormal protein deposits called Lewy bodies. It leads to cognitive impairment, visual hallucinations, and movement symptoms resembling Parkinson's disease.

While the causes, symptoms, and progression rates of these neurodegenerative diseases vary, they all share a common feature: the deterioration of the nervous system, resulting in significant disability, often necessitating specialized care and treatment.

The diagnosis of neurodegenerative diseases is characterized by the prevalence of ongoing memory loss, which is most commonly associated with dementia. This condition often manifests as a transition from the normal aging process, affecting many older individuals, but it is increasingly being observed in younger people as well. Dementia is influenced by various factors, including depression, anxiety, physical health, sleep patterns, lifestyle choices, aging, gender, and the cognitive capacity of individuals, particularly in terms of reasoning, as assessed by the function of the Hippocampus [39].

The amygdala, a brain region, plays a vital role in cognition, learning, memory, emotional regulation, and response to stress. However, many aging individuals experience issues related to thinking, changes in behavior, reasoning, and memory, resulting in reduced productivity in their daily lives.

Neurodegenerative diseases like Alzheimer's disease, Parkinson's disease, Huntington's disease, and motor neuron diseases are considered multifactorial age-related conditions that affect the intricate central nervous system of the brain, typically occurring with aging. Alzheimer's disease, in particular, is the most common form of dementia and is closely associated with a decline in cholinergic signaling to the brain. It currently affects approximately 50 million people worldwide, with expectations of this number tripling by 2050.

Several modifiable risk factors are linked to the development of Alzheimer's disease and dementia, including depression, obesity, diabetes, low levels of physical and mental activity, and hypertension. Additionally, factors such as sleep disturbances and anxiety have been reported to have connections to memory loss. Alzheimer's disease is characterized by irreversible neuron loss and impaired brain function, potentially due to neurotoxicity induced by beta-amyloid proteins and etau and beta-amyloid protein plaques buildup.

In summary, neurodegenerative diseases are associated with memory loss and cognitive decline, affecting individuals across various age groups. Alzheimer's disease is a prevalent form of dementia with modifiable risk factors, and its impact on global health is expected to increase significantly in the coming decades.

3 Overview of Alzheimer's Disease

Alzheimer's disease is a progressive and degenerative condition that primarily affects memory, thinking, and behavior. It falls under the umbrella of dementia, which describes significant cognitive decline interfering with daily life. While it's most commonly associated with older individuals, Alzheimer's, also known as early-onset Alzheimer's, can impact people in their 40 and 50 s. The key aspects of Alzheimer's disease are summarized in Table 2.

Table 2 Concise summary of key aspects of Alzheimer's disease

Aspect	Description
Definition	Progressive and degenerative brain disorder
Common symptoms	Memory loss, cognitive decline, behavior changes
Pathology	Accumulation of beta-amyloid plaques and tau tangles in the brain
Risk factors	Age, genetics, family history, cardiovascular health, head injuries, lifestyle
Stages	Early stage, middle stage, late stage
Diagnosis	Comprehensive evaluation, ruling out other causes
Treatment	No cure, medications to manage symptoms, non-drug interventions
Caregiving	Significant burden on caregivers, support needed
Research	Ongoing efforts to understand causes and develop treatments

1. **Symptoms:** Alzheimer's often begins with mild memory problems and difficulty performing cognitive tasks. Over time, these symptoms worsen, leading to confusion, disorientation, language difficulties, impaired judgment, and personality changes. In advanced stages, individuals may struggle to recognize loved ones or perform everyday tasks.
2. **Pathology:** A hallmark of Alzheimer's is the accumulation of abnormal protein deposits in the brain. These include tau tangles within cells and beta-amyloid plaques between nerve cells. These abnormalities disrupt communication among brain cells and lead to their death.
3. **Risk Factors:** While the exact cause of Alzheimer's remains uncertain, several risk factors have been identified. Age is a primary factor, with risk increasing significantly after the age of 65. Family history and genetics also play a role. Other risk factors include cardiovascular health, head injuries, and lifestyle factors like diet and exercise.
4. **Stages:** Alzheimer's typically progresses through three stages: early, middle, and late. Early-stage symptoms include mild memory issues, progressing to greater confusion and memory loss in the middle stage. In the late stage, significant cognitive decline, loss of communication skills, and round-the-clock care are often required.
5. **Diagnosis:** Diagnosis involves a comprehensive evaluation, including medical history, cognitive tests, neurological assessments, and sometimes brain imaging. While there's no definitive test for Alzheimer's, diagnosis is typically reached by ruling out other potential causes of cognitive impairment.
6. **Treatment:** Currently, there's no cure for Alzheimer's disease. However, some medications can help manage symptoms and slow disease progression, particularly in the early stages. Non-pharmacological interventions such as cognitive training and lifestyle changes can also enhance the quality of life.
7. **Caregiving:** Alzheimer's places a significant burden on caregivers, often family members. Providing physical and emotional care for someone with Alzheimer's

can be challenging. Support groups, respite care, and professional assistance can help reduce caregiver stress.

8. **Research:** Ongoing research aims to better understand the causes of Alzheimer's, develop effective treatments, and ultimately find a cure or preventive measures. Studies focus on genetics, inflammation, and lifestyle choices contributing to the disease.

4 Biology, Cultivation, Nutritional and Proximate Composition of Rosemary Leaves

Rosemary leaves, belonging to the Lamiaceae family, are characteristic of the Mediterranean region and are widely recognized for their diverse biological properties, making them valuable in both traditional medicine and culinary applications [36]. While rosemary cultivation is not as abundant in places like Nigeria compared to more developed tropical regions, efforts are underway to encourage farmers to cultivate this versatile herb in larger quantities.

Rosemary thrives in warm climates with moderate humidity levels and exhibits good heat tolerance. It prefers temperatures between 55° and 80° Fahrenheit and requires well-draining soil to flourish [10]. Therefore, successful cultivation should take place in locations with favorable climatic conditions.

Cultivating rosemary involves sexual reproduction and requires loamy-rich soil that is well-moistened and offers excellent drainage to enhance the plant's aromatic intensity [49]. Alternatively, rosemary can be propagated from non-woody sprigs. To achieve this, softwood cuttings are taken in spring, preferably after the plant has flowered, without including the flowers themselves. These cuttings, measuring around 3–5 in. and taken just below a branching point or leaf node, are placed in a compost mix with a significant portion of vermiculite. The container is then kept in a suitable environment, such as a propagator, cold frame, or windowsill, ensuring the soil remains consistently moist but not overly wet. After approximately 4 weeks, when the plant offers firm resistance upon a gentle tug, it is considered mature and ready for harvest [58].

Cultivating rosemary from seeds is typically done in the spring, although it requires patience due to the slow germination process and relatively low success rate. To start, the seeds should be placed in a well-drained compost pot indoors, ideally at least 3 months before the growing season begins. It's recommended to use a mix of loam compost with plenty of vermiculite or perlite. After moistening the seeds, the pot can be placed in a propagator. Seedlings typically emerge within 14–28 days and can then be removed from the propagator. Once they reach a height of around 3 in., they should be transferred to a larger pot to continue maturing. Harvesting can occur when the stems become strong [58].

Rosemary leaves are highly valued as a culinary herb, adding flavor to various dishes, particularly meats like lamb and fish, lemon sauces, stews, and beans. Additionally, they possess numerous medicinal properties, including antioxidant and antibacterial qualities. The key chemical constituents responsible for these properties are Rosmarinic acid, Carnosol, Carnosic acid, and essential oils rich in terpenes and terpenoids, including alpha-pinene, 1,8-Cineole, alpha-terpineol, camphor, borneol, and myrcene [16].

In terms of nutritional content, fresh rosemary leaves (*Rosmarinus officinalis Linn.*) contain essential minerals such as manganese, potassium, magnesium, copper, calcium, iron, along with vitamins A, C, B6, thiamine, riboflavin, and folate. Rosemary is cholesterol-free and provides around 131 calories per gram serving when consumed. In addition to these nutrients, it contains pyridoxine, folate, ursolic acid, carnosol, carnosic acid, rosmanol, 4-hydroxybenzoic acid, and riboflavin [59].

Furthermore, research by Ezza and Malia [25] suggests that rosemary extract may have memory-enhancing properties. This effect is likely attributed to its primary constituent, rosmarinic acid, a potent flavonoid known for its potential to reduce the risk of memory loss. Rosmarinic acid works in a complex manner within the brain by modulating the neurotransmission of acetylcholinesterase, an enzyme that plays a crucial role in the production of the neurotransmitter acetylcholine. An imbalance in acetylcholinesterase levels can lead to cognitive issues. Table 3 summarizes the biology, cultivation, nutritional, and some uses of rosemary leaves.

Table 3 Examples of some common neurodegenerative disorders and their symptoms

Aspect	Description
Biology	*Rosmarinus officinalis* – Rosemary is an aromatic, evergreen herb native to the Mediterranean region. It belongs to the Lamiaceae family.
Cultivation	Rosemary is grown as a perennial shrub. It prefers well-drained soil and full sun exposure. It can be cultivated in gardens, pots, or containers.
Nutritional composition	Rosemary leaves are low in calories. They contain dietary fiber, vitamins, and minerals. Notable vitamins include vitamin A, vitamin C, and vitamin B6. Important minerals include calcium, iron, and magnesium.
Bioactive compounds	Rosemary contains several bioactive compounds, including rosmarinic acid and antioxidants like flavonoids and phenolic acids.
Flavor and aroma	Rosemary has a strong, woody aroma and a slightly bitter, pine-like flavor. It is commonly used as a seasoning in various culinary dishes.
Medicinal and culinary uses	Rosemary is used in traditional medicine for its potential health benefits, including improved digestion and cognitive function. It is a popular herb in cooking, enhancing the flavor of dishes like roasted meats, soups, and stews.

5 Active Chemical Composition of Rosemary (Rosmarinic Acid)

In recent times, there has been a noticeable increase in the recognition of the value of using natural medicinal plant products to address various health concerns. This growing awareness has led to a heightened interest in understanding the chemical compounds present in these plants, particularly those with properties that can mimic the effects of certain diseases, such as Alzheimer's disease. This discussion focuses on the rosemary plant, which contains essential oils known as Rosmarinic acid and is recognized for its potential in mitigating the effects of Alzheimer's disease.

Rosmarinic acid is a group of phytochemicals found in rosemary leaves, and they encompass flavonoids and other phenolic compounds, including terpenes. These compounds can be extracted from rosemary using solvents like ethanol, acetone, and hexane [1]. The extraction process often involves the use of supercritical carbon dioxide, highlighting the bioactive nature of these substances [59, 62].

Research has shown that these isolated compounds play a role in inhibiting structural neurodegenerative changes [15]. Furthermore, the removal of beta and tau proteins has been associated with improved Alzheimer's disease management in elderly individuals [53]. Some specific compounds identified in rosemary include rosmaridiphenol, alpha-amyrin, botulin, luteolin, diterpenes, and beta-amyrin.

It's worth noting that the composition and concentration of these isolates can vary based on factors such as the plant's country of origin, drying and storage methods, weather conditions, cultivation practices, genetic variation, and harvest timing [23].

Examining each of these constituents in detail, it's noteworthy that phenolics and flavonoids have been reported to mimic the action of insulin in the treatment of diabetes mellitus [49]. The phytochemical active constituents, including rosmarinic acid, have unique mechanisms of action that contribute to their therapeutic functions as anti-memory agents. Understanding these individual properties is crucial for appreciating their potential health benefits.

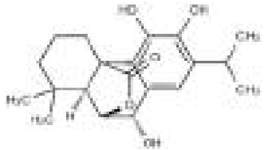
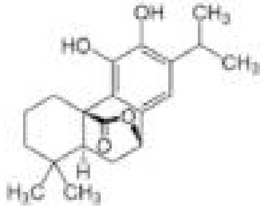
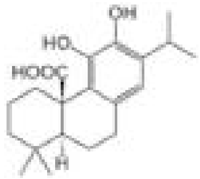
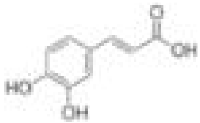
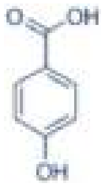
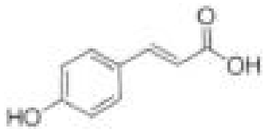
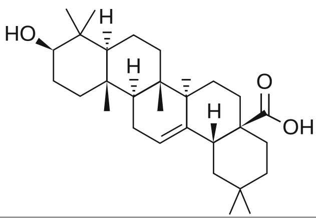
6 Some Phytochemical Constituents of Rosemary Leaf (*Rosmarinus officinalis Linn.*) for the Alleviation of Alzheimer's Disease

Rosemary leaves contain a range of active bioactive chemical constituents, each with its own unique properties and potential health benefits, particularly in alleviating Alzheimer's disease. Some of these constituents, including rosmarinic acid, caffeic acid, p-coumaric acid, carnosol, ursolic acid, 4-Hydroxybenzoic acid, and carnosic acid, belong to various chemical classes, such as monoterpenes, diterpenes, and polyphenolic compounds. The chemical names, molecular formulas, molecular weights, and chemical classifications of these constituents are detailed in Table 4. Additionally, the chemical structures of these phytochemical constituents are illustrated in Table 5.

Table 4 Summary of some identified phytochemical constituents of *Rosmarinus Officinalis Linn*

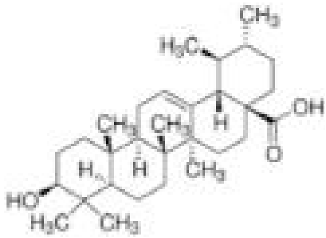
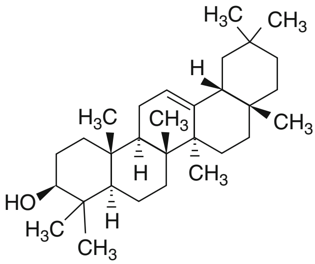
Phytochemical constituents	Compound name	Molecular formula	Molecular weight (g/mol)	Chemical classification	References
Rosmanol	20-Deoxocarnosol	$C_{20}H_{26}O_5$	346.4	Phenolic diterpenes	[15, 74]
Carnosol	4aR-(epoxymethano)phenanthrene-2-one	$C_{20}H_{26}O_4$	330.4	Ortho-phenolic diterpene	[15]
Carnosic acid	Abieta-8,11,13-triene	$C_{20}H_{28}O_4$	332.4	Benzenediol abietane diterpene	[15]
Rosmarinic acid	3,4,Dihydroxyphenylactic acid	$C_{18}H_{16}O_8$	360.1	Flavonoids	[15, 79]
Caffeic acid	3-(3,4,Dihydroxyphenyl)-2-propenoic acid	$C_9H_8O_4$	180.2	Phenols	[9]
4-Hydroxybenzoic acid	Para-Hydroxybenzoic acid	$C_7H_6O_3$	260.2	Phenols	[34, 68, 83]
P-Coumaric acid	(2E)-3-(4-Hydroxyphenyl)prop-2-enoic acid	$C_9H_8O_3$	164.2	Phenols	[73]
Oleanic acid	4a(2H)-carboxylic acid	$C_{18}H_{34}O_2$	456.7	Pentacyclic triterpenoid	[26, 35, 78]
Ursolic acid	Urs-12-en-28-oic-acid	$C_{30}H_{48}O_3$	456	Pentacyclic triterpenoid	[91]
Beta-Amyrin	(3-beta)-oliean-12-en-3-ol	$C_{33}H_{58}O_5$	498.9	Pentacyclic triterpenoid	[12, 85]

Table 5 Chemical structures of the phytochemical constituents of *Rosmarinus Officinalis Linn*.

Phytochemical constituents	Chemical structures	References
Rosmanol		[67]
Carnosol		[87]
Carnosic acid		[87]
Rosmarinic acid		[22]
Caffeic acid		[89]
4-Hydroxybenzoic acid		[56]
P-Coumaric acid		[28]
Oleanic acid		[69]

(continued)

Table 5 (continued)

Phytochemical constituents	Chemical structures	References
Ursolic acid		[20]
Beta-Amyrin		[17]

To obtain these valuable constituents, rosemary leaves sourced from local farms are harvested and carefully extracted. Various extraction methods can be employed, depending on the polarity of the specific compounds being targeted. These methods may include traditional hydro-distillation, maceration, decoction, and solvent extraction using solvents like ethanol, methanol, or n-hexane. Alternatively, more modern techniques, such as “Supercritical Fluid Extraction” (SFE), can be utilized. The choice of extraction method depends on the specific compounds of interest and the desired clinical application as a phytomedicine product [8, 36].

6.1 Caffeic Acid

Caffeic Acid (CA) has been identified as a metabolite derived from phenolic acid and has shown promise in reducing oxidative stress and brain damage by inhibiting 5-Lipoxygenase, according to reports by Cai et al. [14]. Additionally, Andrade et al. [9] conducted research indicating that CA has significant potential as it inhibits the fibrillation of amyloid alpha-beta peptides within neuronal membranes, as observed in an in-vitro analysis using an experimental rat model. It’s worth noting that while CA did disaggregate amyloid-beta peptides, its activity exhibited a step-wise directional increase over time due to the presence of lipid membranes in the rat brain. This characteristic makes caffeic acid a potential constituent in addressing memory loss, particularly in diseases like Alzheimer’s.

6.2 4-Hydroxybenzoic Acid

Also known as p-hydroxybenzoic acid, 4-Hydroxybenzoic acid is a monohydroxy benzoic crystalline salt that is moderately soluble in chloroform and highly soluble in alcohols and acetone, classified as a "polar organic solvent." It serves as a precursor to aspirin and is naturally found in substances like rosmarinic acid and coconuts [21]. This compound possesses several biological properties, including neuroprotective and antioxidative effects, which contribute to combating reactive oxygen species in the brain [32].

6.3 P-Coumaric Acid

P-coumaric acid, a natural compound with the chemical formula $C_9H_8O_3$, is a hydroxycinnamic acid [82] that forms an esterified system with tartaric acid known as tartaric p-coumaroyl ester [73]. This compound has immunoregulatory properties and is soluble in water and other solvents like diethyl ether and ethanol. P-coumaric acid can cross the blood-brain barrier and has a neuroprotective role, as evidenced by its ability to reduce neurotoxicity induced by alpha-beta amyloid proteins [33] in the hippocampal neuronal region of the brain, mitigating cognitive impairment [72]. It is also found in vegetables like rosemary leaves and is biosynthesized from cinnamic acid. Recent research by Yu et al. [88] demonstrated that p-coumaric acid has a neuroprotective effect against memory impairment induced by chronic corticosterone [28]. This effect was attributed to the reduction of oxidative stressors in the brain, along with interactions with neuronal receptors in the hippocampus [65]. Specifically, p-coumaric acid appeared to interfere with the interaction between corticosterone-induced advanced glycation end products and their receptors, restoring balance and protecting against depression and memory impairment [88].

6.4 Rosmarinic Acid

Rosmarinic Acid is a naturally occurring compound found in rosemary leaves. It has a chemical structure represented as $C_{18}H_{16}O_8$ and belongs to the Lamiaceae family. This compound has distinct characteristics, including a molecular weight of 360.1 g/mol, solubility in various solvents like water and formaldehyde, a melting point of 171–175 °C, and a boiling point of 694.7 °C [79].

Rosmarinic acid is known for its complex actions on inflammatory processes, including inhibiting cytokine release from activated T-cells and preventing T-cell activation. These properties make it relevant in addressing memory loss and allergic inflammatory reactions [63]. As the field of immunology and research techniques advances, the mechanisms of action of rosmarinic acid are becoming increasingly intricate.

In terms of memory loss, studies suggest that rosmarinic acid has anti-inflammatory and neuroprotective properties. It may protect against cognitive damage and aid in recovery [35]. The mechanism behind its action involves suppressing the accumulation of beta-amyloid plaques in the brain, which is associated with Alzheimer's disease. Rosmarinic acid accomplishes this by modulating the neurotransmission of acetylcholinesterase, an enzyme that affects the production of the neurotransmitter acetylcholine. By normalizing acetylcholinesterase and acetylcholine levels in the brain, rosmarinic acid contributes to improved cognitive function [25].

Furthermore, Rosmarinic Acid is considered the most active component of rosemary leaves and one of the most important polyphenols. It acts on receptors in the hippocampus, enhancing memory and combating memory loss caused by the accumulation of beta-amyloid proteins. It works by cleaving the amyloid precursor protein (APP) in the presence of the enzyme alpha-secretase, preventing the formation of amyloid plaques [77].

Rosmarinic acid is classified as an aqueous phenolic compound [34] and is known for its anti-inflammatory properties, primarily due to the presence of the RosA moiety. Given the rising concerns about memory impairment, incorporating rosemary leaves into our diet may enhance brain activity. Rosmarinic acid plays a vital role in addressing memory-related issues and neuroprotection ([83]; Guo et al. [31]) offering potential benefits in treating and preventing memory loss disorders.

6.5 Rosmanol

Rosmanol, with a chemical structure represented as $C_{20}H_{26}O_5$, is a compound found naturally in rosemary leaves. Traditionally, it was extracted using organic solvents like n-hexane, and the bioactive compounds were purified [55]. However, modern extraction methods such as Pasteurized Liquid Extraction (PLE) and Supercritical Fluid Extraction (SFE), using carbon dioxide (CO_2) and co-solvents like ethanol or methanol, have replaced these older methods. SFE offers improved selectivity, better fractionation capabilities, and higher extraction yields [74].

For enhanced absorption and potential anti-memory effects, combining Rosmanol and carnosol is recommended. Rosmanol is commonly found in the Lamiaceae family, such as in rosemary leaves, and is a polyphenol of 3,4-Dihydroxyphyllique, synthesized from carnosol in the human metabolic pathway [18]. Previous studies have identified Rosmanol as a potent anti-inflammatory compound that can inhibit the activation of nuclear factor pathways responsible for cell damage [54].

In terms of its mechanism of action in memory loss, research by Ayodeji et al. [11] on Wistar rats examined its inhibitory effects on iron-induced lipid peroxidation and key enzymes like Acetylcholinesterase (AChE) and Butyrylcholinesterase (BChE), which play crucial roles in brain neurotransmission. The study evaluated the antioxidant activity of Rosmanol as compared to galantamine, a typical cholinesterase inhibitor. The results showed that Rosmanol significantly increased scavenging activity in the brain cells of the rats. Molecular docking evaluations revealed a binding activity with AChE and BChE, indicating its potential affinity for inhibiting cholinergic enzymes and lipid peroxidation in brain neurons [2].

The scavenging activity of Rosmanol, acting as a cholinesterase inhibitor for improved memory function, is attributed to secondary metabolites known as flavonoids. These flavonoids are considered safe and have been shown to have neuroprotective effects. They prevent the action of acetylcholinesterase and enhance memory [13]. In the experiments, Rosmanol exhibited a binding energy of −11.1 kcal/mol, surpassing that of the galantamine inhibitor, which had a binding energy of −10.8 kcal/mol [80].

Rosmanol, with its neuroprotective and cholinesterase-inhibiting properties, holds promise in addressing memory loss and cognitive function, potentially offering therapeutic benefits for individuals experiencing memory-related disorders.

6.6 Carsonic Acid

Carsonic acid, an abietane diterpenoid extracted from rosemary leaves, has been associated with anti-memory effects [38]. It is obtained through an evaporation extraction method using hexane and subsequent dissolution in methanol, followed by complete evaporation. This compound is a carboxytricyclic compound categorized as a member of catechols and a monocarboxylic acid. It features a hydroxyl and carboxylic group at positions 11 and 12, and it is also substituted at position 20. As a member of the carboxytricyclic group, it falls under the category of catechols and is a conjugated acid of carnosate.

In terms of its mechanism of action in addressing memory loss problems, Carsonic Acid (CA) has demonstrated neuroprotective properties (Guo et al. [31]) in in-vivo experiments conducted by Takumi et al. [84] and Lipton et al. [57]. In these studies, mice were subjected to a downregulation of microglia and increased serum levels of proinflammatory cytokines, including tumor necrotic factor alpha. This proinflammatory cytokine activated the KEAP1/NRF2 pathway, leading to an increase in anti-inflammatory factors. The mice were then treated with 10 mg/kg body weight of CA extracts administered via the transnasal route for periods ranging from 14 days to 3 months.

The results showed significant behavioral and histological improvements in the treated mice, suggesting a translational progress. Histological examination of the brain indicated strong evidence that the formation of reactive oxygen species, responsible for oxidative stress and contributing to neuronal and synaptic damage, plays a role in the pathogenesis of memory impairment [57].

6.7 Oleanic Acid

Oleanic acid, also known as oleanolic acid ($C_{30}OH_{48}O_3$) [78], also possesses a chemical modification involving triple bonds, specifically the C-3 hydroxy, C-12, C-13, and the addition of a fourth one, C-28 carboxylic acids. This leads to the creation of a sequence of new synthetic oleanane triterpenoids [37, 81]. This compound is found naturally as 3, beta-hydroxyolean-12-en-28-oic acid and is biologically active in approximately 1620 plant species, including *Rosmarinus officinalis Linn.* [26, 30], making it widely distributed in the plant kingdom.

Oleanic acid is a phytochemical constituent of saponins, which act as anti-memory agents, particularly in Alzheimer's disease, by blocking the action of the enzyme nitric oxide synthase involved in brain inflammation. Its affordability and availability make it a suitable candidate for further synthetic modification and production through in-vitro techniques such as Heterological Biogenesis or Artificial Biology [70, 90].

6.8 Ursolic Acid

Ursolic acid has been reported to alleviate early subarachnoid hemorrhagic brain injury in experimental rat models [91]. When rats were fed with 25–50 mg per body weight per day (25/50 mg/kg/day) of ursolic acid extract, significant reductions in brain inflammation were observed after a 48-h observation period. This effect was attributed to anti-inflammatory constituents, including intercellular adhesion molecule-1, inducible nitric oxide synthase, and matrix metalloproteinase-9, along with a decrease in apoptosis scores, showcasing the neuroprotective properties of ursolic acid.

6.9 Beta-Amyrin

Beta-amyrin is a natural chemical compound widely distributed in plants, including rosemary leaves. It is classified as a pentacyclic triterpenoid and is considered a component of glycyrrhizin, used as a sweetener and having pharmacological activity. Beta-amyrin originates from the precursor oleanolic acid and is positioned at the 3 beta-position between positions 12 and 13 [12, 85].

Beta-amyrin exhibits neurological effects, including memory regulation [66]. Research has demonstrated its neuroprotective properties, particularly in an Alzheimer's mouse model with synaptic plasticity in the hippocampus. In clinical experiments, mice with Alzheimer's disorder were treated with beta-amyrin extract from rosemary leaves, showing positive effects. This treatment was compared with a positive control group that received minocycline, which has been reported to have both positive and negative outcomes in Alzheimer's rat models [6, 29, 61].

7 Conclusion

Alzheimer's disease, a form of memory loss known as Vascular Dementia, is becoming a growing concern, not only among the elderly but also in younger individuals. Various factors, including physical inactivity, aging, stress, and depression, contribute to this alarming trend. The conventional approach of using drugs that interact with multiple receptors, resulting in low bioavailability, has led to a shift toward exploring phytomedicine. Rising healthcare costs and a shortage of specialized healthcare professionals have prompted scientists to consider phytomedicine as an alternative therapeutic approach for individuals with conditions like Alzheimer's.

Phytomedicine, which involves investigating the therapeutic properties of traditional plants, offers a promising avenue for enhancing drug efficacy and therapy. Scientists are increasingly recognizing its potential, as nature often provides abundant solutions to the challenges faced in pharmacological interactions. Within the spectrum of phytomedicine, the constituents of *Rosmarinus officinalis Linn.*, including caffeic acid, rosmarinic acid, rosmanol, p-coumaric acid, beta-amyrin, among others, have been highlighted for their potential in alleviating memory loss associated with Alzheimer's and dementia.

Numerous research studies have reported the anti-memory effects of rosmarinic acid on Alzheimer's disease and dementia. These studies have elucidated the mechanisms by which these constituents can aid in the treatment of memory impairment, particularly in the hippocampus and other affected regions. In light of this extensive body of scientific evidence, it is imperative to emphasize the need for further exploration of the therapeutic potential of *Rosmarinus officinalis Linn.*, commonly known as rosemary leaf, within the context of Ayurveda medicine.

Furthermore, advancements in analytical methods, including molecular and genetic techniques, enable the precise isolation and quantification of rosmarinic acid and other constituents of *Rosmarinus officinalis Linn.* This knowledge is critical for developing dosage formulations for drug therapy. Given the favorable conditions for cultivating rosemary leaves, it is advisable to encourage widespread cultivation of this plant. Following harvest, regular consumption of rosemary leaves in accordance with recommended dosages by regulatory authorities such as the Food and Drug Administration (FDA) can significantly contribute to maintaining optimal brain function and cognitive abilities.

Moreover, there is a need for further clinical trials to establish detailed guidelines for incorporating rosemary leaf as a form of drug therapy, possibly in the form of nutraceuticals. Rosmarinic acid's inhibition of acetylcholinesterase in the brain positions it as an herb with anti-memory loss properties in adult therapeutic medicine. If these principles are well integrated into the clinical realm of drug therapy with FDA authorization, older individuals can effectively adopt these practices to ensure they receive the right nutritional benefits from this herb in their daily lives. This could potentially have a substantial impact on preserving good memory and cognitive function.

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Diversity of Medicinal Plants Used in the Treatment and Management of Viral Diseases Transmitted by Mosquitoes in the Tropics

13

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Abstract

Mosquitoes are insects with slender bodies, three pairs of jointed legs, a pair of wings, and an elongated mouthpart for sucking the blood of vertebrate animals including humans. The saliva of mosquitoes causes itchy rashes when deposited on the skin of humans during blood feeding. Many female mosquito species are capable of ingesting and/or transmitting infectious pathogens such as viruses that cause diseases like chikungunya fever, dengue fever, Japanese encephalitis, West Nile fever, yellow fever, Zika fever, and others to humans. These viral diseases are primarily transmitted through the bites of infected female mosquito (*Aedes*, *Anopheles*, and *Culex*) species in the Culicidae family. The listed viruses are endemic in the tropics and subtropics of Africa and Southeast Asia, where they cause the emergence and re-emergence of mosquito-borne viral diseases. These diseases are responsible for thousands of deaths reported annually within the affected regions. Treatment and management of mosquito-borne viral diseases with orthodox medicines are common in several parts of the world. But due to the toxicity of synthetic drugs, their alternatives (mainly from medicinal plants) are beginning to receive wider attention. Therefore, the age-long practice of using plants and herbs by herbalists in the tropics to treat different ailments with believable reports from patients and herbalists is being empirically evaluated presently. Based on verifiable reports of antiviral properties of some medicinal plants, this chapter intends to discuss medicinal plants diversity in the tropics, highlight recent updates on their use in the treatment and management of mosquito-borne viral diseases, and succinctly espouse some isolated and identified phytochemicals that are potentially active against the listed mosquito-borne viral diseases.

Keywords

Mosquito · Viruses · Diseases · Medicinal plants · Recent updates · Isolated phytochemicals

Abbreviations

CDCP	Centers for Disease Control and Prevention
DENV-1	Dengue virus 1
DENV-2	Dengue virus 2
DENV-3	Dengue virus 3
DENV-4	Dengue virus 4
HIV	Human immunodeficiency virus
IDPs	Internally displaced persons
RdRp	RNA dependable RNA polymerase

RNA	Ribonucleic acid
UV	Ultra violet
WHO	World Health Organization

1 Introduction

Mosquito-borne viral diseases are spread through bites of female mosquitoes that are infected with viruses. Mosquito-borne viral diseases in the tropics include chikungunya, dengue, Japanese encephalitis, West Nile fever, yellow fever, Zika, and others. Mosquitoes that carry these viruses (vectors) belong to a few genera – *Aedes*, *Anopheles*, and *Culex.* These mosquito genera are well distributed in all the countries of Africa, Asia, South America, and parts of Europe. Over seven hundred million people globally suffer from mosquito-borne diseases with more than one million deaths annually [1–3]. Among the diseases, chikungunya is caused by an alphavirus that is primarily transmitted through the bites of infected female *Aedes* mosquitoes (*Aedes aegypti*, *Aedes albopictus*, and *Aedes furcifer*) [4]. The virus is presently endemic in the tropics and subtropics of Africa and Southeast Asia. The transmission cycles in these regions are between non-human primates and *Aedes* mosquito species. For dengue, the disease is rather caused by flavivirus which is transmitted through the bites of infected female *Aedes* mosquitoes (*Aedes aegypti* and *Aedes albopictus*). Dengue virus is a positive-stranded ribonucleic acid in the family Flaviviridae that causes one of the major morbidity and mortality in the tropics and subtropics. The disease is causing approximately ten thousand deaths with about 100 million symptomatic infections every year in more than 125 countries of the world and is being transmitted by *Aedes* mosquitoes [5, 6]. Japanese encephalitis is also another flavivirus disease that is spread and transmitted by *Culex* mosquito species. The virus is the leading cause of encephalitis in many tropical Asian countries with about 68,000 cases of symptomatic infections annually. The disease is endemic in about 24 WHO countries in Southeast Asia and Western Pacific regions with about three billion people at risk of being infected [7].

For West Nile, the disease is caused by a positive-stranded ribonucleic acid flavivirus in the family Flaviviridae. The virus is transmitted through the bites of infected *Culex* mosquito species (*Culex quinquefasciatus*, *Culex stigmatosoma*, *Culex thriambus*, *Culex pipiens*, and *Culex nigripalpus*). These *Culex* species are responsible for infecting birds such as blue jays and crows with West Nile virus and can occasionally infect other mammals such as dogs, horses, and humans. Unfortunately, West Nile virus can as well be transmitted by other species of birds [8–11]. Yellow fever is also another disease caused by flavivirus. The virus is transmitted through the bites of infected female *Aedes* mosquitoes (*Aedes aegypti* and *Aedes albopictus*). The virus is endemic in both North and South America and in the tropics of Africa. For clearer distinction, yellow fever is capable of causing epizootic infections in non-human primates and yellow fever outbreaks in humans [12]. Thus, a section referred to as the sylvatic reservoir system is in the primates; hence, yellow fever transmission encompasses urban (human and urban *Aedes* mosquitoes) and sylvatic (non-human primates and forest-restricted *Aedes* mosquitoes) cycles. The sylvatic cycle is responsible for the major cases reported in South America while the transmission of the virus from monkeys to humans or from human to human through bites of infected mosquitoes is currently reported in Africa alone [13–17]. In the case of Zika, the disease is also caused by a flavivirus that is transmitted through the bites of infected female *Aedes* mosquito species (*Aedes aegypti*, *Aedes africanus*, *Aedes albopictus*, and *Aedes hensilli*). *Aedes albopictus* has been noted to be the likely major vector due to its link to the Zika virus outbreak in Gabon in 2007. Unfortunately, the virus has been reported to be transmitted from mother to foetus, through blood transmission, organ transplantation, laboratory exposure, and sexual transmission [18–20]. Outside the 2007 outbreak of Zika virus in Gabon, other outbreaks have occurred in Micronesia and Brazil [21]. Furthermore, over 6000 cases of symptomatic Zika virus infections have been reported between 2015 and 2017 in the USA [22].

These mosquito-borne viral diseases have caused a reasonable economic loss for both families and government and thus aggravate poverty in developing countries. For instance, a dengue episode causes an ambulatory patient to lose 14.8 working days and $514 while non-fatal hospitalized patients lose 18.9 working days and $1491 in eight burdened countries [23], hence making mosquitoes the most arthropod of medical and veterinary importance. Mosquitoes are insects in the family of Culicidae that are grouped with the common flies. They have slender bodies, three pairs of jointed legs, a pair of wings, and an elongated mouthpart for sucking the blood of vertebrate animals including humans. The saliva of a mosquito when transferred to humans during blood sucking can cause itchy rashes. Many female mosquito species are capable of ingesting and/or transmitting infectious diseases between humans or from animals to humans. Thus, different female mosquito species are transmitters of alphavirus and flavivirus pathogens, (Fig. 1) that cause mosquito-borne viral diseases such as chikungunya fever, dengue fever, Japanese encephalitis, West Nile fever, yellow fever, and Zika fever to humans [24].

Treatment and management of mosquito-borne viral diseases with orthodox medicines are very common. But due to the toxicity of orthodox-based medicine, their alternatives have gained attention. Based on the traditional use of plants and herbs by herbalists in the tropics to treat different ailments with believable reports from patients and herbalists, attention is being shifted to using medicinal plants, as a novel approach to treating viral diseases. Medicinal plants are highly diversified in the tropics and hence, have been used to treat and manage symptomatic viral infections by traditional healers. Medicinal plants in the tropics with complex bioactive phytochemicals that are rich for pharmaceutical purposes have been carefully selected and used in their crude forms by past generations, and the practice was handed over to the present generation based on folklores. There are many plant extracts and isolated compounds that have been reported to be antiviral agents and are less harmful, and less expensive as compared to orthodox medicines [25–28]. Based on verifiable reports of the antiviral properties of medicinal plants against viral diseases in the literature, this chapter shall discuss and tabulate medicinal plant

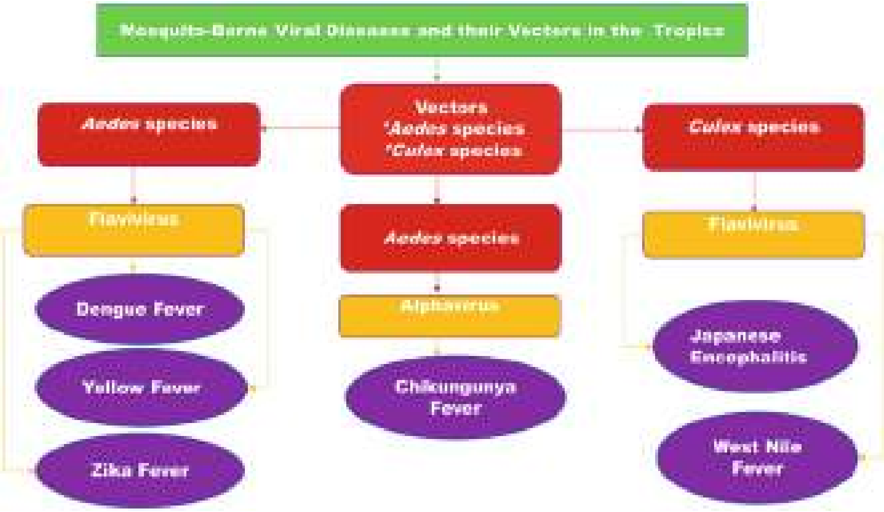

Fig. 1 Overview of viral diseases transmitted by mosquitoes in the tropics

diversity that is used in the treatment of mosquito-borne viral diseases in the tropics. This chapter will also highlight recent updates on the use of medicinal plants in the treatment and management of mosquito-borne viral diseases and as well espouse the isolated and identified phytochemicals that are potentially active against mosquito-borne viral diseases.

2 Overview of Distribution of Mosquito-Borne Viral Diseases in the Tropics

Environmental changes as a significant factor have induced viral diseases such as dengue fever which is an emerging serious public health importance and ranked the most concerned mosquito-borne viral disease with the potential of causing an epidemic. Dengue cases have recorded a multifold increase within the last 50 years with a staggering record of impacts on human and economic costs. One of the major fears about dengue is the fast spread of its vector: *Aedes aegypti* mosquito which has spread to more than 20 countries of Europe and to the Caribbean Islands where it was depicted to also vector chikungunya [24]. Mosquitoes are continually causing the emergence and re-emergence of mosquito-borne viral diseases that are threatening the world's public health systems and status. For instance, dengue fever estimated at 2.5 billion globally [29] affected about 75,454 people, and chikungunya fever affected 18,639 in India [30]. Other mosquito-borne viral diseases such as West Nile, Yellow fever, Zika, and others have also impacted human health and their socioeconomic importance is severely felt.

2.1 Chikungunya Fever

Chikungunya is a viral disease and its vectors are *Aedes aegypti* and *Aedes albopictus* mosquitoes. The virus which these *Aedes* mosquito species carry belongs to the Alphavirus genus in the family Togaviridae [31]. Chikungunya fever is thus a mosquito-borne disease that is now a worldwide public health threat. In the tropics, there are reports of outbreaks due to a high prevalence of *Aedes* mosquito species that transmit the virus to humans, circulation of the virus, and high population density in the urban cities of tropical countries [32]. *Aedes aegypti* is the most prevalent species in urban centres and is the major vector of the virus but as of 2006, *Aedes albopictus* was depicted to contribute to the spread of the virus to Europe and the Americas [33–36]. *Aedes aegypti* and *Aedes albopictus* are the major vectors of chikungunya in Asia and Indian Ocean region but other species of *Aedes*, *Culex annulirostris*, *Mansonia uniformis*, and *Anopheles* species are denoted to vector chikungunya virus [24].

There are three poly groups of the chikungunya virus (West African poly group; Eastern, Central, and Southern Africa poly groups; and the Asian poly group) that are distinctly separated geographically [37]. The virus causes chikungunya fever in man and was first diagnosed in Tanzania in the early 1950s. Since then, there have been subsequent outbreaks of the disease in sub-Saharan African countries and Asia [38]. Chikungunya epidemics have occurred in South-East Asia, India, Islands of the Pacific, and the Indian Ocean respectively in the last two decades leading to more than six million infections [4]. The virus has also been recorded in the Western Hemisphere with over two million cases of infection in about 50 American countries and 114 countries globally [39]. Chikungunya outbreaks were restricted in Africa (Democratic Republic of the Congo, Uganda, Central Africa Republic, Guinea, Angola, South Africa, Malawi, Nigeria, and other sub-Saharan countries) and Asia (Thailand, Cambodia, India, Myanmar, Philippines, Laos, Vietnam, Indonesia, Malaysia, and Sri Lanka) between the 1960s and 1980s [40]. There was an outbreak in a French territory in the Indian Ocean between 2005 and 2006, and about 266,000 inhabitants were affected. In 2007, the outbreak got to countries within the WHO South-East Asia Region and India with about 1.4 million cases. The disease was reported for the first time in Europe in 2007 in Northeast Italy. Similarly, the disease transmission was reported for the first time in 2013 at the WHO Region of the Americas, the Caribbean Island of Saint Martin, and other islands of the region [40].

There was a report of an average loss of more than 106,000 disability-adjusted life yearly, between 2010 and 2019, and in the year 2020, about 170,000 cases of chikungunya virus fever were reported globally, causing a higher rate of infant morbidity [41–43]. From 2004, chikungunya virus epidemics occurred in Kenya with 70% of her population being infected. Since then, it has spread to other countries like the Reunion Island, Thailand, Malaysia, India, Hong Kong, Taiwan, Sri Lanka, and other European and American countries [44–48]. Symptoms associated with acute infection include arthralgia/polyarthralgia, high fever, headache, myalgia, skin rashes, joint swelling, and nausea [49]. Chikungunya virus though causes low mortality with 0.07% death but patients with polyarthralgia can nurse it for months or even years after the acute phase of infection has been resolved and with joint pains that can last for months [41, 50].

2.2 Dengue Fever

Dengue fever is one of the viral infections imposing enormous economic burdens on the people of the tropics and subtropical countries. It is also referred to as backbone fever because it causes severe body pain mainly in the joints. The disease is caused by four flavivirus serotypes (dengue virus 1, dengue virus 2, dengue virus 3, and dengue virus 4) that belong to Flaviviridae family with dengue virus 2 being the most deadly virus. The virus is being harboured by female *Aedes aegypti*, *Aedes albopictus*, and *Aedes polynesiensis* mosquitoes which are known to thrive more in urban and semi-urban areas [51, 52]. Though the infection is asymptomatic and mild, the number of carriers is increasing considerably at the global level. Moreover, the actual number of infected cases is not comprehensively reported, or wrongly diagnosed with other febrile infections. The infected female mosquito bites in the day time and infect humans with the virus. More than 300 million people are reported to be infected every year, and about 22,000 of them are likely not to survive the infection [53].

Global infection of dengue fever is estimated at 2.5 billion with yearly recurrent or new cases at 50 to 100 million [51, 54, 55]. More than 40% of the world's population (amounting to about 2.5 billion people) are currently at risk of dengue infection. Going by WHO estimate, over 100 million people globally are infected annually, and about 500,000 of them especially children must be treated in the hospital while 2.5% of the infected persons cannot survive the disease. Dengue was earlier localized in just 9 countries of the world before 1970 but has now risen to endemic status in more than 100 countries (Africa, the Americas, South-East Asia, Western Pacific, and the Eastern Mediterranean). A report in 2010 showed that about 2.3 million in the Americas, Western Pacific regions, and South-East Asia are the ones mostly affected [24]. The disease is endemic in over 100 countries of Africa, Asia, America, Eastern Mediterranean, and Western Pacific regions. About 20 countries in Africa recorded dengue epidemics between 1960 and 2017 [56].

Due to the international trade of used tyres, timber, bamboo, and other similar stuff that provide breeding space for *Aedes*, another secondary dengue vector (*Aedes albopictus*) seen in Asia has spread to North America and Europe. *Aedes albopictus* is noted to have a wider geographical distribution because of its resilience and ability to survive in urban and rural environments. The eggs are resistant to harsh weather and therefore, can be viable all through the length of the dry season [24]. Female *Aedes aegypti* is the main vector of the dengue virus which is transmitted to humans when the female *Aedes* mosquito sucks blood from human beings. Once a female *Aedes* is infected, it can transmit the virus throughout its life cycle. Interestingly, *A. aegypti* are urban-restricted mosquitoes that breed in most man-made containers such as used tyres, beverage cans, plastic bottles, and the like. *Aedes* usually bite in the early hours of the morning and evening before dusk. The female can bite many people during its feeding phase. Infected individuals show signs of headache, fever, severe muscle and joint pains, and hemorrhagic fever which may lead to dengue shock syndrome which is a resultant effect of severe cases of dengue fever [57, 58].

Dengue fever is a severe flu-like illness that manifests with high fever, severe headache, muscle and joint pains, nausea, vomiting, swollen glands, or rash. Dengue is rarely fatal but severe ones come with possible serious complications, with symptoms that include low temperature, severe abdominal pains, and rapid breathing. Dengue virus has four serotypes (DENV-1, DENV-2, DENV-3, and DENV-4). Recovery from any of the serotypes confers a lasting immunity to the serotype. However, if infected by another serotype, it will still cause a severe dengue fever if not managed on time. Dengue infection presents symptoms that are related to malaria and other febrile conditions; hence, it is difficult to diagnose or predict if a patient is at risk or not at the time of medical examination [59]. Hemorrhagic fever was first reported in the 1950s during the dengue epidemics in Thailand and the Philippines. Presently, it is one of the primary roots of illnesses and mortality in many countries in Asia, India, and Latin America [60]. Untreated dengue cases may lead to complications such as respiratory distress, fluid accumulation, severe bleeding, organ damage, and others [61].

2.3 Japanese Encephalitis

Japanese encephalitis is a zoonotic disease (infectious disease transmitted from animals to humans) that contributes to one of the cases of morbidity and mortality in Southeast Asia and the Western Pacific area [62]. Japanese encephalitis is caused by a virus in the flavivirus (Flaviviridae) family. The virus is carried and transmitted to man by a *Culex* mosquito species and could cause permanent neuropsychiatric disorder in over 40% of cases and 20% death [63]. In some cases, the virus is transmitted by a tick. The disease affects mainly children and elderly people with compromised immunity [64].

Japanese encephalitis prevalence is highest in the tropics (China, Southeast Asia, Philippines, India, Japan, and Korea) where some major outbreaks have been reported [65]. Cases of Japanese encephalitis (mostly children below 5 years) are estimated at 50,000 leading to 10,000 deaths in a year. The disease exists in Asia at the islands of the Western Pacific in the East and extends to the Pakistanis border in the West and from the Republic of Korea in the North to Papua New Guinea in the South. Countries like Japan, the Korea Republic, Taiwan, and China are currently having considerably reduced cases but the disease has extended to India, Napa, and Sri Lanka where the disease is an important public health problem. Japanese encephalitis is mainly transmitted in rural agricultural provinces, especially among those living in flooded rice farms. It can as well occur near urban centres, and in temperate zones of Asia, as the virus transmission is seasonal [24].

Culex tritaeniorhynchus is the primary vector and transmits the virus; however, animals such as pigs and wading birds are potential transmitters of the virus. Humans are minor hosts as they may not be able to develop considerable concentrations of the virus in their bloodstreams that could infect female mosquitoes that are feeding. Japanese encephalitis virus when transmitted by an infected female *Culex* species to humans will usually show mild or no symptoms. Only a small fraction of humans infected develop inflammation of the brain that is associated with high fever, sudden onset of headache, disorientation, coma, tremors, and convulsions. About 25% of severe cases can be fatal while 30% of survivors would have lasting central nervous system damage [24].

2.4 West Nile Virus Disease

West Nile virus is a flavivirus-causing disease in the family Flaviviridae; the virus is transmitted by a *Culex* mosquito. West Nile virus disease presents fever, muscle weakness, encephalitis, and meningitis as symptoms without a known or medically acceptable treatment and no available vaccine yet. West Nile virus extends its distribution to Africa, Asia, Australia, and Europe and has spread to the USA, Mexico, Canada, and the Caribbean after being detected in New York City in 1999 [66]. It is maintained naturally in a *Culex* mosquito-bird-mosquito transmission cycle while man, horses, and other mammals serve as incidental hosts [67]. The majority of the individuals infected with the virus present no symptoms while about 20% of infected people experience some level of clinical symptoms such as West Nile fever but only a few will have severe West Nile neuro-invasive diseases, potentially fatal cases such as encephalitis, meningitis, acute fluid paralysis, fatigue, muscle weakness, and persistent tremors [68–70].

2.5 Yellow Fever

This disease is a known viral haemorrhagic fever that was before now regarded as a lethal disease until its vaccine was effectively developed. The disease is known to have an acute phase with symptoms such as fever, muscle pain, headache, shivers, loss of appetite, nausea, and vomiting. Most patients usually get relieved after 3–4 days but 15% of them enter a toxic phase making the patient to have jaundice and may be bleeding which could appear in the vomit. Factors such as less vaccinated population, urbanization, deforestation, climate change, and population movements have led to an increase in yellow fever cases; hence, about 200,000 infections and 30,000 cases of death are recorded yearly [24].

Yellow fever virus is transmitted by species of *Haemagogus* and *Aedes* mosquitoes to monkeys and humans. In African countries, yellow fever outbreaks have occurred in recent years as seasonal workers, nomadic, and displaced people were affected. North and Central America, the Caribbean, and Asia have had their share of the disease in the last 30 years due to an increase in air travel and sporadic increases in the distribution of *Aedes aegypti* in urban areas [24].

2.6 Zika Virus Disease

Zika virus is another flavivirus disease transmitted by *Aedes* species, which was identified in 1952 and circulated in Brazil in 2015 but was discovered in 1947 in Uganda with a transmission cycle between primates and wild *Aedes aegypti* which later infected humans [71]. The viral infection is now a serious threat to public health because of its suspected effect on the development of foetuses in pregnant women and malformations of congenital organs. It can as well cause meningitis and Guillain-Barre syndrome in severe cases [72–75].

Symptoms associated with Zika virus disease are similar to other febrile illnesses earlier stated, thus making its early diagnosis difficult. Zika virus infection is as well reported to be transmitted sexually; hence its global emerging epidemic threat has been warned by World Health Organization [76–78]. Proposed plans to fight the virus are to develop vaccines and screen molecules to inhibit different phases of the viral life-cycle [79–81].

3 Diversity of Medicinal Plants Reported to Be Used in the Treatment and Management of Mosquito-Borne Diseases in the Tropics

Based on the biology and life history of mosquitoes, it won't be out of place to say that mosquitoes are the most deadly insects globally. The assertion is because of many debilitating diseases transmitted to over one billion people every year [24]. Mosquitoes are vectors of many diseases such as malaria, dengue, yellow fever, lymphatic filariasis, chikungunya, Japanese encephalitis, West Nile fever, and Zika which collectively kill over one million people globally every year. Larvicidal and ovicidal activity of leaf and seed extract of plants against mosquitoes (*Aedes aegypti* and *Anopheles stephensi*) have been reported in some countries (India and Nigeria) [80–85].

Based on the World Health Organization's estimate, about 80% of the global population resort to medicinal plants as the basic source of their medical/healthcares [86]. Plants are known to produce phytochemicals such as alkaloids, flavonoids, essential oils, terpenoids, and polyphenols with different chemical structures. These chemicals are what plants use to defend themselves against microbial infection, herbivorous and insects attack, and environmental stress [87]. Phytochemicals have several organic activities that are antiviral, such as influenza, herpes simplex, hepatitis, and dengue, hence benefiting man, in fighting the above diseases [88–94].

Africa and countries in the tropics have a diverse range of medicinal/herbal plants that are used by traditional medicine herbalists for different ailments. However, there is a paucity of documented data on medicinal plants of the tropics in literature. In this section of the chapter, different reports of traditional medicines against mosquito-borne diseases in Africa and other countries in the tropics will be discussed. A concise review of scientific evaluation and validation of medicinal plants that have therapeutic and prophylactic properties against viral-causing diseases such as chikungunya, dengue, Japanese encephalitis, West Nile fever, yellow fever, and Zika fever. By extension, the records have shown that many plants in the tropics are medicinal and therefore, could be extended to the treatment and management of mosquito-borne viral diseases. This section, therefore, will highlight reports of different medicinal plants used in the treatment and management of mosquito-borne viral diseases that are documented in scientific literature.

3.1 Medicinal Plants for Treating and Managing Chikungunya Virus Fever

There are plant extracts that are reported to have anti-chikungunya virus properties (Table 1) with an interesting report that epigallocatechin gallate, derived from *Camellia sinensis* and *Curcuma longa* plants, can prevent the chikungunya virus from attaching to the host's cells and harringtonine from *Cephalotaxus harringtonia* blocking replication of the chikungunya virus in vitro study [95–97]. A study noted that extraction of phytochemicals for the treatment of diseases is dependent on the polarity of solvents; hence, hexane and chloroform are less polar solvents and therefore, could only extract alkaloids, coumarins, fatty acids, and terpenoids while more polar solvents such as ethyl acetate, ethanol, methanol, and water will extract saponins, tannins, flavones, polyphenols, terpenoids, anthocyanins, polypeptides, and lectins from plants [98].

Plants that have been reported to inhibit chikungunya virus at different levels of concentration include *Picrorhiza kurroa*, *Ocimum tenuiflorum*, *Terminalia chebula*, *Commiphora wightii*, *Cedrus deodara*, and *Zingiber officinale.* Among these plant

Table 1 Isolated phytochemicals reported to be used in the treatment and management of chikungunya

Isolated compound	Plant species	Plant's part	Medicinal activity	Reference
Coumarin (A and B)	*Mammea americana*	Seeds	*Active in Vero cells*	[37]
Curcumin	*Curcuma longa*	Bulb	Anti-chikungunya virus	[96, 99]
Berberine	*Berberis vulgaris*	Roots, stem-bark	Anti-chikungunya virus	[100–102]
Andrographolide	*Andrographis paniculata*	NS	Anti-chikungunya virus	[99]
Silymarin	*Silybum marianum*	Fruits, seeds	Anti-chikungunya virus	[103]
Baicalein	*Scutellaria baicalensis*	Roots	Anti-chikungunya virus	[104]
Silvestrol	*Aglaia foveolata*	NS	Anti-chikungunya virus	

Note: Phenols, saponins, alkaloids, flavonoids, and tannins are bioactive compounds contained in *Oroxylum indicum* plant which are anti-chikungunya virus [105, 106]; *NS* not stated

Table 2 Medicinal plants reported in India for the treatment and management of chikungunya fever

Plant family	Plant species
Acanthaceae	*Andrographis paniculata*
Asteraceae	*Pluchea lanceolata*
Brassicaceae	*Lepidium sativum*
Burseraceae	*Commiphora wightii*
Burseraceae	*Boswellia serrata*
Combretaceae	*Terminalia chebula*
Combretaceae	*Terminalia bellerica*
Cyperaceae	*Cyperus rotundus*
Euphorbiaceae	*Emblica officinalis*
Fabaceae	*Glycyrrhiza glabra*
Lamiaceae	*Ocimum sanctum*
Menispermaceae	*Tinospora cordifolia*
Ranunculaceae	*Nigella sativa*
Rubiaceae	*Rubia cordifolia*
Scrophulariaceae	*Picrorhiza kurroa*
Solanaceae	*Withania somnifera*
Verbenaceae	*Vitex negundo*
Zingiberaceae	*Zingiber officinale*
Zingiberaceae	*Curcuma longa*

Source: [107]

species, *Ocimum tenuiflorum* and *Terminalia chebula* inhibited 100% chikungunya virus at a concentration of 200 μg/ml [32]. Plants such as *Gynura bicolor*, *Hydrocotyle sibthorpioides*, *Ocimum americanum*, and others listed in Table 2 have anti-chikungunya virus; hence, their phytochemicals could be isolated and characterized as lead compounds in developing anti-chikungunya medicine. Based on extractants, water-extracted phytochemicals are weak against the chikungunya virus compared to other extractants [31]. Resistance and cost are some of the factors necessitating alternative means of anti-chikungunya virus. Thus, plant sources become a target due to their availability, cheapness, and diverse phytochemicals.

3.2 Medicinal Plants for Treating and Managing Dengue Virus Fever

Due to resistance, cost, and other medical logistics, the need to search for novel anti-dengue medications that are organically produced is voting for the use of medicinal plants which have been used widely by traditional medicine herbalists to treat febrile ailments. This option is anchored on safety, non-toxicity, and less harmful reports of the use of medicinal plants in the tropics. About 80% of Africans and Asians depend on traditional medicine as the basic healthcare for febrile diseases due to cost and geographical constraints [108]. At the moment, there is no specific treatment for dengue infection that is globally accepted among the medical professions. However, strategies to eliminate the disease are currently going on such as immunization, modulation of the host's immunity, and interfering with the host's ribonucleic acid [109]. Until these scientific innovations and breakthrough are presented, the traditional herbalists in the tropics have started using herbal formulations of plant-based materials to treat dengue infection which has been applauded by modern scientific reports. An example is the use of *Carica papaya*, a plant that has been used widely by traditional medicine herbalists to treat many febrile illnesses including dengue fever [59]. Hence, medicinal plants have been employed by traditional medicine herbalists, especially as one of the therapeutic approaches to treating and managing the disease. The practice of using medicinal plants in India to treat dengue viral infection is copious and has shown to be efficacious and better than using a single isolated compound. For instance, azadirachtin from *Azadirachta indica* was inefficient against dengue infection but the whole extracts strongly inhibited the dengue infection [60] which may be attributed to the synergistic effect of the compounds in the plant.

Medicinal plants have been used in developing countries to treat all known diseases and are as well becoming popular in developed countries [110]. Hence, 80% of the global population attends to their healthcare requirements by using medicinal plants [111]. Medicinal plants (Table 3) such as *Euphorbia hirta* and *Carica papaya* among other 31 species of plants are among the folkloric reports of medicinal plants against dengue infection in many African communities [112, 113]. The diversity of medicinal plants used in treating human-associated febrile infections such as dengue is estimated at 52,000 species [114]. These plants are reported to reverse and prevent dengue infection from reaching a critical stage by possessing different potentials to inactivate, inhibit, and increase blood platelet count, white blood cells, and neutrophils. Some can act as anti-dengue, prevent bleeding, and downregulate dengue infection [115].

Medicinal plants have diverse chemicals that are biologically capable to inhibit the virus replication cycle. As noted earlier, no medically approved drug is in the public domain for targeting the dengue virus. Other drugs that are approved to be positive for other viruses have been reported not to be effective as dengue virus is resistant to them, thus necessitating the use of medicinal plants to combat dengue disease. A study has presented isolated phytochemicals and their molecular structures that are depicted as anti-dengue (Table 4). The way these structured compounds work includes inhibition of the dengue virus at the post-infection stage, virus budding and secretion, binding of the peptide to the virus, and inhibition of the virus at an early stage of infection. Some will inhibit the dengue virus or have immunomodulatory activity, inhibit protease activity and secretion of virus particles, and inhibit entry. Some will block the activity of non-structural proteins (NS1, NS2, and NS3) especially NS3 protease, inhibit virus adsorption and host cell internalization, early stage of the virus cycle, and from binding to receptors. They can as well

Table 3 Medicinal plants reported for the treatment and management of dengue in the tropics

Plant family	Plant species	Reference
Acanthaceae	*Andrographis paniculata*	[115–117]
Acoraceae	*Acorus calamus*	[116]
Amaranthaceae	*Alternanthera philoxeroides*	[116, 117]
Caprifoliaceae	*Lonicera japonica*	[116]
Caricaceae	*Carica papaya*	[115–117]
Chordariaceae	*Cladosiphon okamuranus*	[113, 114]
Clusiaceae	*Garcinia mangostana*	[116]
Cucurbitaceae	*Momordica charantia*	[115]
Elaeagnaceae	*Hippophae rhamnoides*	[116, 117]
Euphorbiaceae	*Euphorbia hirta*	[116]
Euphorbiaceae	*Cladogynos orientalis*	[115]
Euphorbiaceae	*Euphorbia hirta*	[116, 117]
Fabaceae	*Acacia catechu*	[116]
Fabaceae	*Glycyrrhiza glabra*	
Fabaceae	*Leucaena leucocephala*	[115]
Fabaceae	*Mimosa scabrella*	
Fabaceae	*Tephrosia madrensis*	
Fabaceae	*Tephrosia crassifolia*	
Fabaceae	*Tephrosia viridiflora*	
Fabaceae	*Castanospermum australe*	[117]
Fabaceae	*Mimosa scabrella*	
Fagaceae	*Quercus lusitanica*	[116, 117]
Flagellariaceae	*Flagellaria indica*	[115]
Gramineae	*Cymbopogon citratus*	[116]
Halymeniaceae	*Cryptonemia crenulata*	[115]
Labiatae	*Ocimum sanctum*	
Lamiaceae	*Basilicum polystachyon*	[116]
Lamiaceae	*Ocimum tenuiflorum*	
Liliaceae	*Allium sativum*	
Marcgraviaceae	*Schwartzia brasiliensis*	
Meliaceae	*Azadirachta indica*	[116, 117]
Menispermaceae	*Cissampelos pareira*	[117]
Moraceae	*Ficus septica*	[116]
Myrtaceae	*Psidium guajava*	[115, 116]
Nyctaginaceae	*Boerhavia diffusa*	[116]
Orchidaceae	*Gastrodia elata*	[117]
Phyllanthaceae	*Phyllanthus urinaria*	
Phyllophoraceae	*Gymnogongrus torulosus*	[115]
Phyllophoraceae	*Gymnogongrus griffithsiae*	
Piperaceae	*Piper retrofractum*	
Poaceae	*Cymbopogon citratus*	
Rhizophoraceae	*Rhizophora apiculata*	[116, 117]
Rubiaceae	*Oldenlandia uniflora*	[116]

(continued)

Table 3 (continued)

Plant family	Plant species	Reference
Rubiaceae	*Pavetta canescens*	
Rubiaceae	*Tarenna asiatica*	
Rubiaceae	*Uncaria tomentosa*	[115]
Sapindaceae	*Doratoxylon apetalum*	[116]
Sapindaceae	*Nephelium lappaceum*	
Saururaceae	*Houttuynia cordata*	[115]
Schisandraceae	*Schisandra chinensis*	[116]
Solieriaceae	*Meristiella gelidium*	[115]
Verbenaceae	*Lippia alba*	
Verbenaceae	*Lippia citriodora*	
Verbenaceae	*Lippia citriodora*	[117]
Zingiberaceae	*Curcuma longa*	[116]
Zingiberaceae	*Kaempferia parviflora*	
Zingiberaceae	*Boesenbergia rotunda*	[116, 117]
Zosteraceae	*Zostera marina*	[115–117]

decrease viral RNA production, act as virucidal such as catechin, and inhibition of viral particle assemblage [110].

3.3 Medicinal Plants for Treating and Managing Japanese Encephalitis

In a quest to also treat patients infected with the Japanese encephalitis virus with herbal formulations, a study [157] evaluated the antiviral activity of *Isatis indigotica* plant. The author reported low cytotoxicity of the plant but potentially inhibited Japanese encephalitis virus replication in vitro study. The plant's extract reduced the virus yield, block the virus attachment to the host, and showed virucidal activity. *Isatis indigotica* extracts were reported to contain active compounds such as indigotin, indirubin, indican, and isatin. Among these isolated phytochemical compounds of *I. indigotica*, indirubin was appraised to have strong protective potential against the Japanese encephalitis virus and proved less toxic to the host's cells.

Because of the need to develop safe and cheaper antiviral medicines, Roy et al. [158] also evaluated *Trachyspermum ammi* plant against the Japanese encephalitis virus. The authors reported that the essential oil obtained from the plant through distillation was able to inhibit the virus in vitro study, thus showing the anti-Japanese encephalitis virus activity. The essential oil of *Trachyspermum ammi* is reported to contain thymol, α-pipene, p-cymen, and limonene, which are its major curative agents against many diseases caused by fungi, bacteria, and viruses. The essential oil inhibited the Japanese encephalitis virus with low cytotoxicity and without the formation of plagues [159].

Table 4 Isolated phytochemicals reported to be used in the treatment and management of dengue

Isolated compound	Plant species	Plant's part	Medicinal activity	Reference
Castanospermine	*Castanospermum australe*	NS	Against dengue activity	[118]
Quercetin, quercetinin, chlorogenic acid, hyperoside	**Houttuynia cordata*	Leaves	Against dengue activity	[119, 120]
Rutin	*Spondias mombin*		Inhibition, anti-replication of DEV-2	[121]
Quercetin, ellagic acid	*Spondias tuberosa*	NS	Inhibition, anti-replication of DEV-2	
Fucoidan	*Cladosiphon okamuranus*	NS	Inhibition of dengue virus 2	[122]
Galactans	*Cryptonemia crenulata*, *Gymnogongrus torulosus*	NS	Inhibition of DEV-1,3, anti-replication in Vero cell	[123]
Kappa-carrageenan	*Gymnogongrus griffithsiae*, *Meristiella gelidium*	NS	Inhibition of dengue virus 2 and 3	
Trigocherrins A, trigocherriolides B and C	*Trigonostemon cherrieri*	Bark, wood	Dengue RdRp	[124]
Tatanan A	*Acorus calamus*	Roots	Anti-dengue virus	[125]
Andrographolide	*Andrographis paniculata*	Leaves	Inhibition of dengue virus 2	[126–128]
Nimbin, desacetyl nimbin, salanin	*Azadirachta indica*	Leaves	Inhibition of dengue virus 2	[129, 130]
Kaempferol 3-O-rutinoside, epicatechin	*Azadirachta indica*	Leaves	Inhibition of dengue virus 2	[131]
Gedunin, pigmol	*Azadirachta indica*	Leaves	Anti-dengue virus	[132]
Pinostrobin, pinocembrin, alpinetin, cardamonin, panduratin A	*Boesenbergia rotunda*	Rhizomes	Anti-dengue virus, dengue virus 2	[133–135]
Pareirarine, cissamine, magnoflorine	*Cissampelos pareira*	Flowers	Inhibition of dengue virus 2, 3	[93, 133–136]
Rutin, scopoletin	*Cladogynos orientalis*	Leaves, roots/stem	Anti-dengue virus, dengue virus 2	[137, 138]
Galatan	*Cryptonemia crenulata*	Alga	Anti-dengue virus, dengue virus 2, 3, 4	[135]
Curcumin	*Curcuma longa*	Rhizomes	Inhibition of dengue virus	[135, 140]
Pectolinarin, acacetin-7-O-rutinoside	*Distictella elongata*	Leaves, fruits	Anti-dengue virus 2	[93, 139]
Quercetin	*Hippophae rhamnoides*	Leaves	Anti-dengue virus 2	[93, 115, 135, 140]

(continued)

Table 4 (continued)

Isolated compound	Plant species	Plant's part	Medicinal activity	Reference
Linoleum, limonene	*Lippia alba*	NS	Inhibition of dengue virus in Vero cells	[141]
Catechin, gallic acid, naringin, quercetin	*Psidium guajava*	Roots, leaves, bark	Anti-dengue	[142]
Baicalein	*Scutellaria baicalensis*	Root	Inhibition of dengue virus 1, 2, 3, 4	[143, 144]
Fucoidan, glucuronic, fucose	*Cladosiphon okamuranus*	Alga	Impede dengue virus 2; anti-dengue effect	[122, 145]
Azadirachtin	*Azadirachta indica*	Leaves	Anti-dengue virus 2	[130]
Castanospermine	*Castanospermum austral*	Seed	Anti-dengue virus 1,2,3,4	[118]
Chartaceones A-F	*Cryptocarya chartacea*	Bark	Anti-dengue virus 2	[146]
Betulinic acid 3 β-caffeate	*Flacourtia ramontchi*	Stem/bark	Anti-dengue virus	[147]
DL-galatan	*Gymnogongrus torulosus*	Red seaweed	Anti-dengue virus 2	[148]
α mangostin	*Garcinia mangostana*	Fruit	Anti-dengue virus 1,2,3, 4	[149]
Galactomannans	*Mimosa scabrella*	Seeds	Anti-dengue virus 1	[150]
Geranin	*Nephelium lappaceum*	NS	Anti-dengue virus 2	[151]
Catechin	*Psidium guajava*	Bark	Anti-dengue virus 2	[152]
Schisandrin A	*Schisandra chinensis*	NS	Anti-dengue virus 1,2,3, 4	[153]
Trigocherrin A1, A, B	*Trigonostemon cherrieri*	Bark, wood	NS	[124]
Hirsutine	*Uncaria rhynchophylla*	NS	Anti-dengue virus 1,2,3,4	[154]
Zosteric acid	*Zoster marina*	NS	Anti-dengue virus 1,2,3,4	[155]
Diallyl disulphide	*Allium sativum*	NS	Anti-dengue virus 2	[156]

Note: * when the concentration of *Houttuynia cordata* compound is used individually, the inhibition rate would be moderate but when combined with others, it will result in a greater rate of inhibition activity; *NS* not stated, *RdRp* RNA dependable RNA polymerase

3.4 Medicinal Plants for Treating and Managing West Nile Virus Disease

Medicinal plants with their isolated and purified organic products are rich resources of anti-virus which have been shown to target viral life cycles either as inhibitors of virus entry, prevention of replication, gathering, or release of virus [160]. West Nile virus is less lethal but its wild spread and re-emergence is posing a global health

Table 5 Isolated phytochemicals reported to be used in the treatment and management of West Nile virus

Isolated compound	Plant species	Plant's part	Medicinal activity	Reference
Delphinidin	NS	Flowers, fruits	Alteration of WNV virus replication, virucidal	[162]
Epigallocatechin	*Camellia sinensis*	Leaves	Alteration of WNV virus replication, virucidal	
Palmatine	*Coptis chinensis*	NS	Inhibition of NS2B-NS3	

Note: *NS* not stated, *WNV* West Nile virus, *NS2B* non-structural protease 2B, *NS3* non-structural protease 3

threat due to its asymptomatic characteristics; hence, Goh et al. [161] reported few phytochemical compounds that are capable of altering, preventing, and inhibiting West Nile virus (Table 5).

3.5 Medicinal Plants for Treating and Managing Yellow Fever

A few surveys of medicinal plants used by traditional medicinal herbalists used in the treatment and management of yellow fever showed about 95 species of plants in 52 families (Table 6). The bark and leaves of these plants are the most frequently used. The plants are reported not to only be cheap and affordable but prevent, and treat, yellow fever patients with acceptable tolerance and no serious side effects [163]. The majority of the African population are sceptical about receiving vaccines, especially in rural communities; hence, they prefer to treat and manage yellow fever with herbal medicine. For instance, *Cochlospermum tinctorium*, *Eucalyptus globulus*, *Mangifera indica*, and *Musa sapientum* are reported as medicinal plants for treating and managing yellow fever in some parts of Central and Northern parts of Nigeria. *Musa sapientum* is the most reported plant in the treatment of yellow fever. Other plants used in the treatment of yellow fever include *Azadirachta indica*, *Alstonia boonei*, *Anacardium occidentale*, *Carica papaya*, *Citrus aurantifolia*, *Senna occidentalis*, *Zingiber officinale*, and others [163].

3.6 Medicinal Plants for Treating and Managing Zika Virus Disease

A few medicinal plants evaluated against Zika virus are *Berberis vulgaris* which contains berberine compound and *Rheum palmatum* and *Coptis* species, which contain emodin compound. These compounds have been shown to be virucidal, reducing about 80% of virus infectivity in Vero cells. Emodin, for instance, can block the entry of viruses and inhibit the processes associated with viral particle release from cells [166] and also has a modest effect as a pre-treatment medicine by

Table 6 Medicinal plants reported for the treatment and management of yellow fever in the tropics

Family	Species	Reference
Acanthaceae	*Hygrophila auriculata*	[163]
Aizoaceae	*Trianthema pentandra*	
Alliaceae	*Allium cepa*	[164]
	Allium sativum	
Anacardiaceae	*Mangifera indica*	[163, 164]
	Anacardium occidentale	[163]
	Sclerocarya birrea	
	Spondias mombin	
Annonaceae	*Enantia chlorantha*	[163, 164]
	Annona senegalensis	[163]
Apocynaceae	*Strophanthus kombe*	[164]
	Leptadenia hastata	[163]
	Alstonia boonei	[165]
Arecaceae	*Cocos nucifera*	[164]
	Elaeis guineensis	[163]
Aristolochiaceae	*Aristolochia ringens*	
Asteraceae	*Chromolaena odorata*	[164]
Bignoniaceae	*Stereospermum kunthianum*	[163]
	Kigelia africana	
Bixaceae	*Cochlospermum tinctorium*	
Bombacaceae	*Adansonia digitata*	
Boraginaceae	*Cordia africana*	
Bromeliaceae	*Ananas comosus*	[163, 164]
Caesalpiniaceae	*Senna alata*	[164]
	Senna occidentalis	
Caricaceae	*Carica papaya*	[163, 164]
Celastraceae	*Hippocratea indica*	[164]
Celastraceae	*Celastrus indica*	[163]
Clusiaceae	*Garcinia kola*	
Combretaceae	*Guiera senegalensis*	
	Terminalia avicennioides	
	Anogeissus leiocarpus	
	Combretum micranthum	
Compositae	*Vernonia amygdalina*	
Connaraceae	*Byrsocarpus coccineus*	[163, 164]
Cucurbitaceae	*Citrullus lanatus*	[163]
	Momordica charantia	[164]
Ebenaceae	*Diospyros mespiliformis*	[163]
Euphorbiaceae	*Jatropha curcas*	[164]
	Sapium grahamii	[163]
	Euphorbia hirta	
	Euphorbia unispina	
	Croton lobatus	[164]

(continued)

Table 6 (continued)

Family	Species	Reference
Euphorbiaceae	*Ricinus communis*	[163]
Fabaceae	*Mucuna solan*	[164]
	Parkia biglobosa	[163]
	Prosopis africana	
	Cassia occidentalis	
	Detarium senegalense	
	Tamarindus indica	[163]
	Abrus precatorius	[164]
	Cassia tora	
Gentianaceae	*Anthocleista djalonensis*	[164]
Gramineae	*Cymbopogon citratus*	[165]
Lamiaceae	*Ocimum gratissimum*	[164]
Lauraceae	*Cinnamomum verum*	[163]
Lythraceae	*Lawsonia inermis*	[164]
Malvaceae	*Sida acuta*	[165]
Meliaceae	*Khaya senegalensis*	[164]
Meliaceae	*Azadirachta indica*	[163]
	Entandrophragma utile	
Mimosaceae	*Parkia biglobosa*	
	Acacia nilotica	
Moraceae	*Ficus asperifolia*	[164]
	Ficus sycomorus	[163]
	Ficus platyphylla	
	Ficus polita	
Moringaceae	*Moringa oleifera*	[163]
Musaceae	*Musa sapientum*	
Myristicaceae	*Pycnanthus angolensis*	
Myrtaceae	*Eucalyptus globulus*	
	Psidium guajava	
	Syzygium aromaticum	
Nymphaeaceae	*Nymphaea lotus*	
Oleaceae	*Olea europaea*	
Phyllanthaceae	*Securinega virosa*	
Piperaceae	*Piper guineense*	
Plumbaginaceae	*Plumbago zeylanica*	
Poaceae	*Sorghum caudatum*	[164]
	Cymbopogon citratus	[163]
	Saccharum officinarum	
Rhamnaceae	*Ziziphus mauritiana*	
Rubiaceae	*Morinda lucida*	
	Nauclea diderrichii	
Rutaceae	*Citrus aurantifolia*	[163, 164]
	Citrus aurantium	[164]

(continued)

Table 6 (continued)

Family	Species	Reference
Sapotaceae	*Nicotiana tabacum*	
	Solanum torvum	[165]
Tiliaceae	*Corchorus olitorius*	[164]
	Triumfetta cordifolia	
Xanthorrhoeaceae	*Aloe barteri*	[163]
Zingiberaceae	*Aframomum melegueta*	
	Curcuma longa	[164]
	Zingiber officinale	

interfering with virus entry [167]. At low concentrations, berberine and emodin are effective prophylaxis agents as berberine has the potential to spread to organs (liver, kidneys, brain), and muscles, and can remain in those places [168], which is a preferred distinction needed in the treatment and management of Zika virus [167]. Berberine is also reported to have low toxicity as no serious side effects have been reported in humans [169–171]. Thus, berberine and emodin phytochemical compounds are considered for their virucidal effect and impair Zika virus particles in vitro study. In endemic regions, both berberine and emodin are acceptable recommendations for the prevention, treatment, and management of patients infected with the Zika virus [167].

Apart from the general reports that plant products (polyphenols, flavonoids, alkaloids, terpenes, and others) have strong potential to inhibit the replication of the Zika virus, some isolated phytochemical compounds (Table 7) have been reported to work against Zika virus in humans. For instance, curcumin which is a phytochemical compound from *Zingiber officinale*, has been shown to have a strong potential against the Zika virus by possessing inhibiting potential on the viral envelope and can be very useful, especially during outbreaks of Zika infection [162, 172]. Similarly, phytochemical extracts and compounds isolated from seaweed plants are evaluated for anti-Zika virus prospect [173] while another study [174] reported that *Aphloia theiformis* extracts affected the organization of Zika virus particles severely, based on electron microscope examination.

4 Recent Updates on the Use of Medicinal Plants in the Treatment and Management of Mosquito-Borne Viral Diseases

Plants have emerged as credible candidate for replacing synthetic drugs. Hence, their discovery is a major boost to the fields of pharmaceutical chemistry and microbiology, herbal sciences, and humanity at large. Several plants have been reported to be active in the treatment of several diseases, especially those caused by microbes [176–195]. Viruses are seen as microbes in the field of microbiology, and the use of

Table 7 Isolated phytochemicals reported for the treatment and management of Zika virus in the tropics

Isolated compound	Plant species	Plant's part	Medicinal activity	Reference
Berberine	*Berberis vulgaris*	NS	Prevent virus entry; replication	[167]
Emodin	*Rheum palmatum*	NS	Anti-Zika virus, virucidal	
	Polygonum multiflorum	NS		
	Cassia obtusifolia			
Curcumin	*Zingiber officinale*	Root	Inhibit Zika virus	[166]
Lycorine	*Hippeastrum puniceum*	Flower	Anti-Zika virus	[175]
Pretazettine		Root		
Narciclasine		Bulb		
Narciclasine-4-O		NS		
β-D-xylopyranoside		NS		

Note: *NS* not stated

plants for the treatment of diseases caused by viruses has been a mere claim until now.

Basic scientific inquiries on the use of medicinal plants in the treatment and management of mosquito-borne viral diseases have been accentuated within the last few years due to viral infections such as COVID-19 and Ebola virus diseases leading to global pandemics and epidemics in some West African countries, respectively. Hence, many medicinal plants are depicted to have reasonable secondary metabolites against viral diseases. Empirical reports about isolated phytochemical compounds from medicinal plants which have performed optimally in many medical experiments have been reported. Thus, the formation of novel medicine from phytochemical constituents and plant-isolated compounds is beginning to be a major part of treating mosquito-borne viral diseases, especially in developing countries.

Based on the research reports succinctly discoursed above, it is glaring that a greater percentage of the population in the tropics use plant-based medicine as a basic form of their healthcare. Reasons for this practice are but not limited to the cost and safety of plant-based medicines which have been used by the generations past with little or no negative reports, culture and religious beliefs, inaccessibility of orthodox medicine due to higher cost, lack of primary healthcare centres, and trained professionals especially in rural communities [196]. Fortunately, a greater percentage of infected victims survive upsurges of viral infections (epidemics and pandemics) by relying on plant-based medicines which have drawn the attention of researchers to understudy the potentials of these plants that are used to treat and manage mosquito-borne viral diseases. In West Africa alone, about 124 species in 50 families of medicinal plants are reported to have antiviral properties including viral diseases transmitted by mosquito species, hence concurring with a report of higher medicinal plant diversity in the tropics [197–199].

Recent updates on the use of medicinal plants in the control [200–204], treatment, and management of mosquito-borne viral diseases are showing that about 52% which translate to 65 out of the 124 species are the ones that have been scientifically ascertained to have therapeutic potentials against mosquito-borne viral diseases especially dengue and other non-mosquito-borne viral diseases [196]. Progressively, 4 of the 65 plant species have their essential phytochemicals such as flavonoids, quercetin, morin, fisetin, naringenin, and hesperidin from *Citrus aurantifolia* and *Citrus paradise*. Alkaloids such as salidroside 2(4-hydroxyphenyl) from *Cucumis metuliferus* and ethyl β-D-glucopyranoside from *Loranthus micranthus*. Flavonoids (3,5-dicaffeoylquinic acid, acteoside, kaempferol 7-O-glucoside, bastadin-11) and stilbenes (vedelianin, schweinfurthin G, mappain) from *Macaranga barteri* are identified and isolated compounds responsible to potentially having countering activity against viral diseases. Other natural antiviral compounds isolated but not from plants such as mushroom (*Hypoxylon fuscum*) and lichen (*Ramalina farinacea*) include dihydropenicillic acid and sekikaic acid, respectively, with all the structures presented for further investigation and possible replication and production for treatment and management of viral diseases, including mosquito-borne viral diseases [196].

The use of medicinal plants in the treatment of mosquito-borne viral diseases is, therefore, becoming famous due to the bioactive compounds which are a very rich source of pharmaceuticals that contain reasonable antiviral properties. Hence, the World Health Organisation (WHO) and other traditional-based medicinal professionals have proposed that using medicinal plant extracts and their isolated compounds have been very useful in dealing with diseases especially dengue and other related viral diseases [205]. The WHO has gone ahead to consider isolated compounds from medicinal plants to be largely safe, non-toxic, and by comparison, assumed to be less harmful, and cheaper than orthodox medicines against viral diseases [205]. Currently, many researchers are scientifically exploring medicinal plants in an attempt to identify the active compounds that have antiviral properties since many of the secondary metabolites of the medicinal plants have been widely reported to have been used to treat and manage many viral ailments including mosquito-borne diseases such as chikungunya, dengue, Japanese encephalitis, West Nile fever, yellow, Zika, and others [206, 207].

5 Conclusion

Mosquito-borne diseases are contributing to a greater percentage of deaths in the world especially in the tropics, with their emerging and re-emerging infections being an untiring peril to the human population. Mosquito-borne diseases especially viral infections are seriously challenging the survival of the human population, especially children, gravid mothers, and immune-compromised adults. Mosquito-borne viral diseases remain one of the major causes of deaths globally with leading records of outbreaks in the human population and epizootic in non-human vertebrates. Mosquito-borne viral infections have caused severe economic problems in the tropics and subtropical regions of the world but are more endemic in Africa, America, Asia, Eastern Mediterranean Province, and Western Pacific. Regrettably, some of these diseases are prevalent in urban and semi-urban cities and are spread and transmitted by infected female *Aedes*, *Anopheles*, and *Culex* species. Symptomatic cases are generally presented with febrile characteristics, and their treatments vary depending on the orthodox therapeutic agents available. However, orthodox medicines are beginning to fail to meet the required expectations.

Presently, there are reports that medicinal plants and their bioactive compounds are effective against viral infections by either reducing viral loads or inhibiting viral multiplication. The diversity of the reported medicinal plants is very high in the tropics; thus, they have been used in the ancient days till now by herbalists who inherited or learned the art of using such plants to treat different ailments. Hence, a greater percentage of people in the tropics rely mainly on medicinal plants to treat and prevent diseases including mosquito-borne viral infections. Based on folklore and testimonies of patients and herbal medicinal practitioners, medicinal plants in the tropics could be a major source of producing natural antiviral medicines. Many medicinal extracts from various parts of plants such as leaves, flowers, bark, roots, stems, seeds, and whole plant and their isolated phytochemicals have been extensively reported to have reasonable antiviral properties based on; in vitro and in vivo studies. These empirical reports are corroborative of folkloric reports of medicinal plant potentials, which are used in dealing with viral infections and thus form the basis of recent reports of the use of medicinal plants to treat and manage mosquito-borne viral diseases in the tropics.

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Medicinal Plants in the Tropics Used in the Treatment and Management of Parasitic Diseases Transmitted by Mosquitoes: Administration, Challenges, and Strategic Options for Management

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Abstract

Mosquitoes are highly diversified insect group, globally distributed, and carry and transmit parasitic infections that are recognized threats to the world public health. Infections such as lymphatic filariasis and malaria are caused by filarial nematodes (*Wuchereria bancrofti*, *Brugia malayi*, and *Brugia timori*) and protozoans (*Plasmodium falciparum*, *P. vivax*, *P. ovale*, *P. malariae*, and *P. knowlesi*), respectively, which are transmitted to humans by mosquitoes. Filarial parasites are transmitted to humans and are spread by infected female mosquitoes (*Anopheles*, *Culex*, *Aedes*, *Mansonia*) while malaria parasites are spread and transmitted by infected female

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Anopheles mosquitoes to humans. Over 1.4 billion people in about 73 countries are at risk of lymphatic filarial infection while about 3.4 billion people in about 97 countries are at risk of malaria, contributing to about 90% of malaria deaths in developing countries. Orthodox means of treating the diseases have encountered some setbacks due to drug resistance, cost, accessibility, and other logistics, hence becoming necessary to employ sustainable and affordable alternatives. In this chapter, we shall emphasize entomological approach to managing mosquito-borne diseases which is rooted in controlling mosquito species at different stages of development. For the disease-causing parasites, various extracts of medicinal plants which are abound in the tropics and reported to have anti-filarial and anti-plasmodial properties shall be succinctly discussed. Biodiversity loss occasioned by continuous harvesting of plants and root structures, inefficient scientific production tools, paucity of documented data on medicinal plants, and lack of subsidy from government and other stakeholders shall be highlighted as some of the challenges against the use of medicinal plants for the treatment and management of mosquito-borne parasitic diseases. We shall also espouse strategic options such as creation of platforms that will link the health sectors in the ministry of health with traditional medicine practitioners in a view to establishing a reliable and responsive national agency that will protect plant-based medicines and her members. Also, to champion a common front for acceptance of plant-based medicines, product patency, and advocate for sustainable gathering of medicinal plants in the tropics is highly needed.

Keywords

Mosquito · Parasites · Diseases · Medicinal plants · Administration · Challenges · Strategic options

Abbreviations

ACT	Artemisinin-combined therapy
CDCP	Centers for Disease Control and Prevention
HIV	Human immunodeficiency virus
IDPs	Internally displaced persons
UV	Ultraviolet

1 Introduction

Mosquito-borne parasitic diseases are illnesses that are caused by parasites found in human populations which are transmitted from animal to humans or from human to humans by mosquito species. More than one billion people globally contract mosquito-borne parasitic diseases that cause about one million deaths every year. Mosquito-borne and other vector-borne diseases contribute about 16.66% of illnesses and disabilities suffered by children and pregnant women especially the poorest people in the society living in the developing countries [1]. Mosquito-borne parasitic diseases affect both urban and rural villages but most prevalent in villages that lack adequate living conditions such as housing and sanitation and more devastating for those with compromised immunity.

Mosquito is an insect with jointed legs, a pair of wing, and a modified mouthpart for sucking of blood of vertebrate animals including humans. Many female mosquito species are capable of transmitting infectious diseases between humans or from animals to humans. Thus, female mosquitoes are the most dreadful insects that transmit debilitating diseases to man and animals. They harbour infectious parasites such as protozoa, and nematodes in their midgut, which are transmitted from their salivary glands to their hosts (human beings and other vertebrate animals) when sucking their blood. The transmitted parasites (filariae and plasmodia) as outlined in (Fig. 1) cause filarial and malaria diseases, respectively, to man [1] and are known as mosquito-borne parasitic diseases.

Based on disease control (prevention, treatment, and management), medicinal plants are carefully selected from groups of plants which have been utilized in the tropics for years, for treatment and management of diseases from the ancient time till now. In traditional medicine, herbs and plant products are still being used mainly in Africa, Asia, and Latin America, and their use is currently extending to Australia and the USA. Inference from traditional medicine practitioners shows that the use of medicinal plants against diseases such as mosquito-borne parasitic diseases is economical, simple to use, easily accessible, and cheap while synthetic drugs require a lot of time, resources, and efforts (design and validation) from the onset. Plants are reported to have natural compounds in them which are anti: viruses, bacteria, protozoa, nematodes, fungi, oxidant, inflammatory, and carcinogenic [2–4]. Chemical compounds found in plants are generally known as phytochemicals and are secondary metabolites that protect plants against UV radiation, microbial infection, and insect attack [5], hence their use in the treatment and management of insect-borne parasitic diseases.

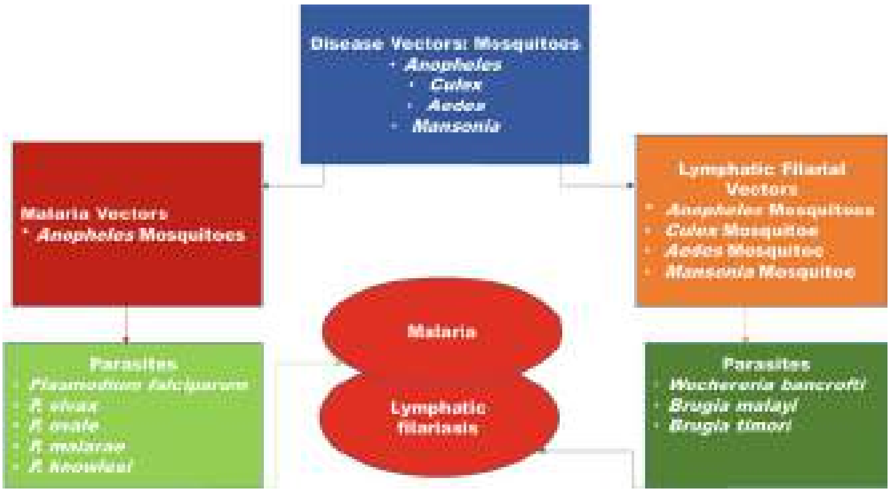

Fig. 1 Overview of mosquito-borne parasitic diseases and their vectors in the tropics

Some approaches employed in dealing with mosquito-borne diseases involve killing or repelling the mosquitoes with synthetic insecticides. Insecticides may be used to lace bed nets or spray the habitable homes. However, the use of insecticides impact negatively on the environment while the mosquitoes develop resistance to the killing powers of the insecticides. Mosquitoes' resistance to insecticides is a major factor driving against mosquito-borne diseases control strategies. Therefore, efforts are on for reasonable alternative remedies to deal with the population of mosquitoes at all stages of their life development. The most voted alternative is to use plant materials to control mosquitoes. Many researches have reported that some plants have the potentials to kill the larval or the adult stages of mosquitoes. For instance, natural products isolated from Chinese herbs and synthetic analogs of *Curcumin* killed the larval stages of *Aedes aegypti* [26]. Similarly, plants such as *Aframomum melegueta*, *Alstonia boonei*, *Croton zambesicus*, and *Newbouldia laevis* are reported in Nigeria as mosquito larvicidal agents [16] which are to be championed in researches related to management of mosquito-borne disease vectors. By extension, the management of mosquito-borne diseases may have been inspired by evidential reports of plasmodial resistance to malaria drugs from the quinines, chloroquinines, and now artemisinin combination therapy, thus requiring that new and affordable alternatives should be sought for. Thus, plant-based medicines stand a high chance of consideration in pursuant of novel mosquito-borne disease management that are rooted to natural resources.

Africa and countries in the tropics have a diverse range of medicinal/herbal plants which are used by traditional medicine practitioners or herbalists. However, there is a paucity of documented data of such medicinal plants from the tropics in open access literatures. In this chapter, we shall look at different reports of traditional medicines against mosquito-borne diseases in Africa and other countries in the tropics. A concise review of scientific evaluation and validation of medicinal plants that have therapeutic and prophylactic properties against parasite-causing diseases such as lymphatic filariasis and malaria shall be presented. Though few reports have shown that many plants in the tropics have various medicinal properties, this section will, therefore, limit discussion to mosquito-borne parasitic diseases such as lymphatic filariasis and malaria. The use of medicinal plants that are abound in the tropics to treat mosquito-borne parasitic diseases shall form the basis of discussion. Elaborate emphasis on phytochemicals and isolated compounds from the plants which have therapeutic and prophylactic potentials against mosquito-borne parasitic diseases (lymphatic filariasis and malaria) shall be highlighted as well. The subsequent section will highlight different preparation methods and administration of medicinal plants in the treatment and management of lymphatic filariasis and malaria, respectively.

2 Overview of Distribution of Mosquito-Borne Parasitic Diseases in the Tropics

Mosquito-borne parasitic diseases are previously envisaged as a peculiar problem in the tropical countries, but today, they are recognized threats to global public health because of mosquito diversity and their geographical spread, hence culminating to different human and animal diseases. Mosquito-borne parasitic diseases are globally spreading due to climate change, pattern of land use, and the continued increase in number of people moving from one country to the other, hence threatening almost half of the global population. Mosquitoes are responsible for majority of the vector-borne parasitic diseases in about 125 countries of the world especially in the tropics [6]. For instance, malaria prevalence alone in the world is estimated at 3.3 billion, and in India as a reference point, it causes about 1.85 million disability adjusted life per year, affected 1.95 million people between 2012 and 2013, while lymphatic filariasis affected 6 million people, thus impacting the socio-economic development of developing countries [7].

2.1 Lymphatic Filariasis

Lymphatic filariasis is commonly known as elephantiasis as a result of a man's skin being thickened and hardened because of excessive swelling due to lymphedema (lymph accumulation). This characteristic lymphedema is commonly obvious in the lower limbs and may affect the breasts, arms, and scrotum, hence making the disease one of the most painful and regrettably disfiguring disease. The disease is caused by round parasitic worms known as filariae (phylum Nematoda). There are three types of filariae (*Wuchereria bancrofti*, *Brugia malayi*, and *Brugia timori*) but *W. bancrofti* is responsible for about 90% of the disease [8]. Filarial parasites are likely to be the only multicellular organisms capable of infecting man with a debilitating disease. Filarial parasites are transmitted and spread by mosquitoes (*Anopheles*, *Culex*, *Aedes*, *Mansonia*) when they take up the microfilariae (the first larval stage of the filariae) in the bloodstream of man during blood feeding. These microfilariae will moult twice to become adult larvae (infective third larval stage) in the gut of mosquitoes which will then be transmitted to man during blood meal by an infected mosquito. The infective adult larvae wriggle into the tissues and enter the lymphatic vessels where they undergo two moults to become reproductive adults (males and females). The reproductive adults exchange their gametes during mating to produce microfilarial larvae which enter the bloodstream of man and are taken up by other uninfected mosquitoes for development [9].

Over 1.4 billion people in about 73 countries are at risk of lymphatic filariasis while more than 120 million of lymphatic filariasis have been reported, with about 40 million of them showing clinical symptoms [10, 11]. Among the total infected people, 40 million live in Africa and India, respectively, while South East Asia, Pacific, New Guinea and America share the remaining [12]. Endemicity of the disease is highest in Africa with about 39 countries having an estimated 390 million people being at risk of infection [13]. Lymphatic filariasis causes a devastating health challenge in addition to its socio-economic burden on infected population. It also causes acute and chronic morbidity among the people living in tropical and sub-tropical countries in Africa, Asia, Western Pacific, and parts of America, with its awful social significance [14]. The disease occurs when filarial parasites (nematodes) are transmitted to humans through a bite of infected mosquito species (*Aedes* spp., *Anopheles* spp., *Culex* spp., and *Mansonia* sp.). The parasites can live in the lymphatic system of an infected human between 6 and 8 years, produce millions of microfilariae that spread in the bloodstream, and significantly disrupt/weaken the immune system.

Lymphatic filariasis mostly presents no immediate identified symptoms but mutely damages the lymphatic system, kidneys, and immune system. Acute cases of lymphatic filariasis cause local inflammation on the skin, lymph nodes, and lymphatic vessels usually associated with tissue swelling known as lymphedema. Lymphatic filariasis occurs mainly in childhood but the pitiable symptom or condition is noted later or manifests in adulthood especially in males. The disease disorder is associated with damage of the lymphatic system, arms, legs, and genitals. These pitiable conditions cause severe pains, loss of optimum productivity, and social exclusion/stigma. Thus, 40 million out of the 120 million infected people with lymphatic filariasis are already defaced, disabled, and socially excluded.

The disease is known to upset over 25 million men mainly with genital disease, and over 15 million are with lymphedema. The South-East Asia region has the highest share of the disease at 65% while the African region has 30%, and other tropical areas share the remaining 5%. Different mosquito groups such as *Culex*, *Anopheles*, and *Aedes* are responsible for transmitting the filarial worms. *Culex* mosquitoes are the main culprit with a widespread distribution across urban and semi-urban areas. *Anopheles* are capable but mainly in rural areas while *Aedes* are responsible in the Pacific Islands and some parts of the Philippines [1].

2.2 Malaria

Malaria is a disease caused by protozoan parasites in the genus *Plasmodium* which accentuates fever, chills, a flu-like illness, and other symptoms. The symptoms are usually manifested in individuals, at least 7 days after exposure to bites of infected mosquitoes. A particular report showed that about 207 million people were down with malaria in 2012 alone, thus leading to unfortunate death of 627,000 of them. However, death rate is said to be decreasing by 42% in the world and 49% in the WHO region of Africa due to increase in prevention and control strategies by government and non-government organization. Nevertheless, about 3.4 billion people in about 97 countries of the world are at risk of malaria infection. Unfortunately, its burden is extremely peaked in sub-Saharan Africa and contributing to about 90% of malaria deaths. For instance, the Democratic Republic of the Congo and Nigeria are the countries with the highest burden of malaria, where four among ten malaria-infected patients die of the disease [1]. Moreover, young children, gravid women, HIV patients, internally displaced persons (IDPs), and non-immune migrants are mostly at risk of malaria infection. The poorest people living in remote villages where access to healthcare facilities is lacking are the most vulnerable to malaria infection.

Hitherto to the report above, about 40% of the world population is at risk of malaria, with half of the estimated 1.2 billion people that are at high risk of transmission found in Africa. They are concentrated in 13 countries with Nigeria, Congo, Ethiopia, Tanzania, and Kenya having over 50% of the cases [15] with Nigeria alone parading 25% of the cases. Malaria occurs all through the year in Southern Nigeria but seasonal in the northern region. *Plasmodium falciparum* (one of the protozoan parasites) is depicted as the major cause of malaria in Nigeria and responsible for malaria death cases. Thus, the estimated malaria cases in Nigeria as at 2006 in every 1000 individuals was 397 leading to 16 deaths and 3.9 fatality [16]. Reports have shown that malaria as a case study is still a global burden with over 3.4 billion people estimated to be at risk. The approximately 207 million cases of malaria recorded in 2012 with 80% of them occurring in Africa is an indication that the disease is though a global burden but more prevalent in the sub-Saharan countries. Hence, about 90% of infected persons in Africa don't survive the disease and 70% of them are children who are less than 5 years [17]. In 2019, about 229 million new cases were reported globally, out of which 409,000 of the patients died with about 94% occurring in the WHO African region. Demographically, the prevalence is more in pregnant women with 16.3% and children 0.6% [18–20]. Countries like Botswana, Namibia, Swaziland, Zimbabwe, and South Africa have approximately 15 million people that are at risk but 10 million of them are at a higher risk with 437 deaths [21].

Mosquito species such as *Anopheles arabiensis* is the main vector of *Plasmodium* responsible for transmission of malaria in the region. Female anopheline mosquito species are the vectors of five *Plasmodium* species (*Plasmodium falciparum*, *P. vivax*, *P. ovale*, *P. malariae*, and *P. knowlesi*) that are causing malaria in humans. *Plasmodium falciparum* and *P. vivax* are the most common but *P. falciparum* is more problematic with the highest rates of complications and mortality especially in sub-Saharan Africa [1]. *Anopheles gambiae* and *A. funestus* are the main vectors of malaria plasmodia species (*Plasmodium falciparum* and *P. vivax*) which are the deadliest parasites in sub-Saharan Africa [22].

3 Diversity of Medicinal Plants Used in the Treatment and Management of Mosquito-Borne Parasitic Diseases in the Tropics

Based on mosquito biology and ecology, the authors of this chapter who are inclined in entomology and parasitology, respectively, are propelled to describe mosquitoes as the most deadly insects existing on earth. The description is in connection with many debilitating diseases they transmit globally to over one billion people every year. Mosquitoes are vectors of many diseases such as malaria, dengue, yellow fever, lymphatic filariasis, chikungunya, Japanese encephalitis, West Nile fever, zika, and others which are collectively killing over one million people globally on a yearly basis. Management of mosquito-borne diseases is geared towards its vector (mosquito species) control and elimination at different stages of the vectors' development. Thus, larvicidal activity of six Indian plants against *Aedes aegypti* and *Anopheles stephensi* has been reported [23]. Others include larvicidal and ovicidal potentials of *Eclipta alba* against *Aedes aegypti* and leaf and seed extract of *Delonix elata* against *Anopheles stephensi* and *Aedes aegypti* [24, 25].

3.1 Lymphatic Filariasis and Medicinal Plants Used for Its Treatment

The diversity of medicinal plants against lymphatic filariasis, in particular, is estimated at 46 plant species, and many of them have the potentials to work against filarial infection in man [27]. Just like other herbal remedies against mosquito-borne diseases, the parts of plants used mostly to treat lymphatic filariasis are leaves, barks, roots, stems, bark and bulb but herbalists claim that active phytochemicals of the plants are concentrated in the roots underground [28, 29]. Extracts of medicinal plants (*Azadirachta indica*, *Polyalthia suaveolens*, *Andrographis paniculata*, *Bauhinia racemosa*, and *Haliclona oculata*) and others (Table 1) have shown to exert bio-efficacy through immunomodulatory elicitation response, either as alone treatment or as adjuvant [30]. The demand for medicinal plant-based formulations is increasing, hence culminating to studies that have ascertained the molecular structures and isolated phytochemicals (Table 2), which are effective against lymphatic filariasis. This has given rise to a scintillating impulse at which isolation of active compounds are top priority to meeting the pharmaceutical requirements to alternate or compensate the synthetic formulations against lymphatic filariasis.

3.2 Malaria and the Medicinal Plants Used for Its Treatment in the Tropics

Many tropical plants are endowed with phytochemical compounds which are bioactive agents that are effective against many ailments such as malaria. The phytochemicals (alkaloids, flavonoids, and terpenoids) are schizonticides. Anti-malaria activity of medicinal plants in the tropics especially the alcoholic-based extracts have been consistent in the reports of traditional medicinal herbalists [56]. It is noted that alkaloids target the apicoplast in the plasmodial parasites and inhibit their protein synthesis, while flavonoids obstruct the invasion of myoinositol and L-glutamine produced in an infected erythrocytes [57]. They can as well accelerate the level of erythrocytic oxidation, prevent the *Plasmodium* from synthesizing proteins, and prevent fatty acid biosynthesis [58]. In vivo screening for anti-malaria of 15 plant species against plasmodial parasites (*Plasmodium berghei*, *P. yoelii*) are reported to be active and successful. However, in in vivo studies, lower doses are reportedly not efficient as they can only suppress *Plasmodium* load at a higher dosage [16]. *Azadirachta indica* which is one of the widely used medicinal plants for different diseases was reported to deal with all the stages of maturation of plasmodium development by releasing oxidative stress on plasmodial gametocytes. Similarly, *Picralima nitida* plant has a reasonable in vivo anti-malaria action against early and old malaria infections while hexane extract of *Cocos nucifera* dealt with

Table 1 Medicinal plants reported to be used in the treatment of lymphatic filariasis in the tropics

Plant family	Plant species	References
Acanthaceae	*Andrographis paniculata*	[30, 31]
Alliaceae	*Tulbaghia alliacea*	[27, 32]
Amaranthaceae	*Achyropsis avicularis*	[27, 32]
Apiaceae	*Alepidea amatymbica*	[27]
	Trachyspermum ammi	[30, 31]
Aquifoliaceae	*Ilex mitis*	[27]
Araceae	*Acorus calamus*	
Araliaceae	*Cussonia paniculata*	
Asteraceae	*Aster bakerianus*	
	Berkheya setifera	
	Dicoma anomala	
	Eriocephalus L. sp.	
	Platycarpha glomerata	
	Senecio speciosus	
	Centratherum anthelminticum	[31]
	Neurolaena lobata	
	Sphaeranthus indicus	
	Aster bakerianus	[32]
	Dicoma anomala	
	Felicia erigeroides	[27]
Balanophoraceae	*Sarcophyte sanguinea*	
Betulaceae	*Alnus nepalensis*	[31]
Bignoniaceae	*Kigelia africana*	[27, 32]
Caesalpiniaceae	*Bauhinia racemosa*	[31]
	Caesalpinia bonducella	
Capparaceae	*Capparis tomentosa*	[27, 32]
Caricaceae	*Carica papaya*	[32]
Chalinidae*	*Haliclona oculata*	[30]
	Haliclona exigua	
Chenopodiaceae	*Chenopodium ambrosioides*	[27]
Combretaceae	*Terminalia chebula*	[31]
Convolvulaceae	*Ipomoea oblongata*	[27]
	Argyreia speciosa	[31]
Dioscoreaceae	*Dioscorea sylvatica*	[27]
Euphorbiaceae	*Croton sylvaticus*	
	Euphorbia clavarioides	
	Ricinus communis	[31]
	Mallotus philippensis	
	Excoecaria agallocha	
	Croton sylvaticus	[32]
	Euphorbia gorgonis	[27]
	Ricinus communis	[27, 30, 32]

(continued)

Table 1 (continued)

Plant family	Plant species	References
Fabaceae	*Elephantorrhiza elephantina*	[27]
	Albizia anthelmintica	[27]
	Acacia auriculiformis	[31]
	Butea monosperma	
	Pongamia pinnata	
	Psoralea corylifolia	
Ganodermataceae	*Ganoderma* sp.	[27]
Gunneraceae	*Gunnera perpensa*	
Hyacinthaceae	*Drimia depressa*	
	Eucomis autumnalis	[27, 32]
	Eucomis comosa	[27
	Ledebouria sp.	
	Eucomis bicolor	
	Urginea delagoensis	
Hypericaceae	*Psorospermum baumii*	[32]
Hypoxidaceae	*Hypoxis latifolia*	[27]
Lamiaceae	*Leucas cephalotes*	[31]
	Leucas aspera	
Malvaceae	*Hibiscus sabdariffa*	[30, 31]
	Hibiscus mutabilis	[31]
	Hermannia geniculata	[27]
Meliaceae	*Azadirachta indica*	[30]
	Xylocarpus granatum	[30, 31]
	Azadirachta indica	[31, 32]
Menispermaceae	*Tinospora crispa*	[31]
Mesembryanthemaceae	*Aptenia cordifolia*	[27]
Moraceae	*Streblus asper*	[31]
	Ficus racemosa	
Moringaceae	*Moringa oleifera*	[31, 32]
Pinaceae	*Cedrus deodara*	[31]
Piperaceae	*Piper betle*	
Plumbaginaceae	*Plumbago indica*	
Polygonaceae	*Rumex obtusifolius*	[27]
Portulacaceae	*Portulaca oleracea*	
Rhizophoraceae	*Cassipourea flanaganii*	
Rosaceae	*Cliffortia linearifolia*	
Rubiaceae	*Pentanisia prunelloides*	
	Morinda citrifolia	[30]
Rutaceae	*Aegle marmelos*	[31]
Sapindaceae	*Cardiospermum halicacabum*	[31, 32]
Saxifragaceae	*Saxifraga stracheyion*	[31]

(continued)

Table 1 (continued)

Plant family	Plant species	References
Solanaceae	*Datura stramonium*	[27, 31]
	Solanum aculeastrum	
	Withania somnifera	
	Solanum khastanum	
Verbenaceae	*Lantana camara*	[31, 32]
	Vitex negundo	[31]
Vitaceae	*Rhoicissus tridentata*	[27]
	Rhoicissus tomentosa	
Zingiberaceae	*Zingiber officinale*	[32]

Note: * = not plant but sponges

P. falciparum and *P. berghei* in both in vivo and vitro studies [59]. Empirically, these medicinal plants have reputable anti-malaria properties in in vitro experiment but not too strong in in vivo study except when the dosages are higher, which therapeutically are not meaningful [16].

To cut down on resources and time wastages on assumed medicinal plants, traditional medicinal herbalists and the old native people are usually consulted for anti-malarial plants. In one such consultations, 16 species of plants used by traditional medicine herbalists and the natives were beneficial as their voted anti-malaria plants, were chemically evaluated and proven to contain different phytochemicals (fagoronine, palmatine, jatrorrhizine, picraline, astonine, gendunin, meldenin, and others) that are anti-plasmodial; hence, they have been isolated and characterized [16]. Table 3 showcases some of the isolated and characterized phytochemicals contained in different medicinal plants, which are reported as effective remedies against malaria-causing parasites.

Phytochemicals contained in the plants include alkaloids, flavonoids, quassinoids, limonoids, terpenes, chalcones, coumarines, and others [76]. The concentration of these compounds is dependent on the solvents used to extract them. For instance, methanol extract yields more chemicals than water extract because it contains lipophilic compounds [77]. Hence, concentration of the chemicals determine how effective they will be against plasmodial parasites. Secondary metabolites of these chemicals are synergistically more effective against plasmodial parasites. The synergy of the compounds contained in a plant confers them with anti-malaria property, pointing that any plant without anti-plasmodial activity lack active metabolic compounds [78]. Medicinal plants used in the treatment of malaria in Uganda, for instance, are effective against malaria parasites and possess low level of toxicity such as leaves of *Artemisia annua*, *A. africana*, and all plant parts of *Carica papaya* and *F. virosa*. Those with high toxicity are that of petroleum ether leaf extract of *V. amygdalina* and dichloromethane leaf extract of *Microglossa pyrifolia* [79, 80].

In Africa, medicinal plants used in the treatment of malaria and other mosquito-borne diseases are not regularly documented and regulated by appropriate

Table 2 Isolated phytochemicals reported to be used in the treatment of lymphatic filariasis

Isolated compound	Plant species	Plant's part	Medicinal activity	References
Azadirachtin	*Azadirachta indica*	All parts	Inhibition of microfilariae	[33]
Andrographolide	*Andrographis paniculata*	NS	Anti-filarial, microfilariae	[34]
Minosamycin, xestospongin C, D, araguspongin C	*Haliclona oculata*	NS	Anti-filarial	
Gedunin, photogedunin	*Xylocarpus granatum*	Fruits	Anti-*B. malayi*, *B. pahangi*, *microfilarial activity*	[35–38]
Gossypetin, hibiscetine, sabdaretine, delphinidin	*Hibiscus sabdariffa*	Seeds	Anti-*B. malayi*, *female sterility*	[39, 40]
Platyphyllenone, alusenone, hirustenone, hirsutanonol	*Alnus nepalensis*	Leaves	Anti-*B. malayi*, anti-*microfilarial*	[41]
Galactolipid 1–3	*Bauhinia racemosa*	Leaves	Anti-*B. malayi*, anti-*microfilarial*	[42]
Cinnamic acid	*Cinnamomum* sp.	NS	Inhibition of microfilarial	[43]
Withanolide	*Withania somnifera*	Roots	Prevent *B. malayi*	[38, 44, 45]
Acaciaside- A, B	*Acacia auriculiformis*	Funicle	Anti-microfilarial, larvicidal	[46]
Coumarins	*Aegle marmelos*	Leaves	Inhibition of *B. malayi*	[47]
Diarylheptanoid	*Alnus nepalensis*	Leaves	Anti-*B. malayi*, anti-filarial	
Ferulic acid	*Hibiscus mutabilis*	Leaves	Anti-microfilarial	[48]
Glycyrrhetinic acid	*Glycyrrhiza glabra*	Roots	Anti-*B. malayi*	[49]
Oleanonic acid	*Lantana camara*	Stem	Anti-*B. malayi*, female sterility	[50]
Plumbagin	*Plumbago indica*	Roots	Worm immobility	[51]
B-sitosterol, +catechin-3-gallate, bergenin	*Saxifraga stracheyion*	Roots	Inhibition and movement of whole worms	[52]
Solamargine	*Solanum khastanum*	Fruits	Anti-filarial and microfilarial	[53]
Asperoside, strebloside	*Streblus asper*	Stem, bark	Anti-*B. malayi*, worm mortality, inhibition, immobility	[54]

(continued)

Table 2 (continued)

Isolated compound	Plant species	Plant's part	Medicinal activity	References
Withaferin A	*Withania somnifera*	NS	Anti-*B. malayi*, larvicidal	[44]
Eranoid, limonenes, β clemene, zingiberol, linalool	*Zingiber officinale*	Rhizome	Microfilarial reduction	[55]

Note: *NS* not stated

professional body or governmental agencies but still, traditional herbs and herbalists are being patronized for primary healthcare in relation to treating malaria symptoms. Providing and making documentations for medicinal plants that are used for prevention and treatment of malaria and other mosquito-borne diseases in the tropics would create a reasonable step to preserving and improving access to the use of medicinal plants. The goal of documenting medicinal plants used in the treatment of malaria disease will expedite prospective studies on the security and usefulness of medicinal plants and can provide a base for classifying particular chemical characters with anti-malaria activity which could be central in the development of standardized phytomedicine [21]. It has become imperative to search for organically alternative approach for the treatment of malaria since the causative agents (*Plasmodium* species) and the vectors (*Anopheles* species) have developed resistance to many treatment drugs and insecticides, respectively. Chloroquine resistance, for instance, is due to misuse while artemisinin combination therapy is a good and available option for treating malaria but quinolones (quinine, chloroquine, and mefloquinine) are known to cause cardiotoxicity. *Plasmodium* has also been reported to develop resistance to artemisinin combination therapy [81, 82].

In Nigeria, *Icacina senegalensis* Juss which undergone methanolic extraction competed favourably against chloroquine by suppressing about 80% of *P. berghei* while chloroquine suppressed about 92% of *P. berghei* after 5 days of mice infection with *P. berghei.* Though *I. senegalensis* for now has not been characterized, its anti-malaria potency was attributed to having alkaloids, flavonoids, and terpenes which are phytochemicals depicted to have anti-plasmodial agents [83]. Again, ethanolic leaf extract of *Pseudocedrela kotschyi* has been evaluated in; in vivo experiment for anti-malaria potential. Hence, the plant extracts were able to suppress and cure mice of *Plasmodium berghei berghei* infection at different dosages as the lethal dose of the extract was ascertained at 5000 mg/kg without acute toxicity. By comparison, the leaf extract at 400 mg/kg suppressed *P. berghei berghei* parasitaemia at 91% against chloroquine at 10 mg/kg which suppressed 94% of the *P. berghei berghei* parasitaemia density after 5 days of mice exposure [56]. In Western Nigeria, some authors [84] reported 37 species of plants in 25 families, which traditional herbalists have used to treat insect-transmitted diseases. Based on plant parts used, leaves constitute the most utilized parts at 43.2%, while bark is 32.4%, a whole plant is 10.8%, fruits and seeds are 5.4%, while root and whole plant are 7%. Both fresh and dry plants are used except occasionally where fresh is preferred [84]. In Southern

Table 3 Isolated phytochemicals reported to be used in the treatment of malaria in the tropics

Isolated compound	Plant species	Plant's part	Medicinal activity	References
Fagaronine	*Fagara zanthoxyloides*	Roots	Anti-*P. falciparum*	[60]
Akuammiline, akuammidine, akuammine, akuammine, akuammicine, picraline, alstonine, and β limonoids	*Picralina nitida*	NS	Anti-*P. falciparum*	[61]
Gendunin	*Khaya grandifoliola*, *Azadirachta indica*	NS, wood	Less active anti-*P. berghei*	[62, 63]
Meldenin	*Azadirachta indica*	Leaves	Anti-*P. falciparum*	[64]
Fissinolide	*Khaya grandifoliola*	Bark, roots, seeds	Anti-*P. falciparum*	[65]
Ursolic acid	*Spathodea campanulata*	Bark	Anti-*P. berghei*	[66]
Anthraquinones	*Morinda lucida*	NS	Anti-*P. falciparum*	[67, 68]
Sesquiterpene lactone, Tagitinin C	*Tithonia diversifolia*	Leaves	Anti-*P. falciparum*	[69]
Ajoene	*Allium sativum*	Bulb	Anti-*P. berghei*	[70]
Simalikalactone D	*Quassia Amara*	Leaves	NS	[71]
Diterpene	*Microglossa pyrifolia*	NS	Anti-*P. falciparum*	[20]
Nitidine	*Zanthoxylum chalybeum*	NS	Anti-*P. falciparum*	
Pristimarin	*Maytenus senegalensis*	NS	Anti-*P. falciparum*	[72]
Pinocembrin, flavonoid santin, clerodane diterpene	*Dodonaea angustifolia*	Leaves	Anti-*P. berghei*	[73]
Aloinoside	*Aloe macrocarpa*	Leaves	Anti-*P. berghei*	[74]
Aloin	*Aloe debrana*	Leaves	Anti-*P. berghei*	[75]
Otostegindiol	*Otostegia integrifolia*	NS	Anti-*P. berghei*	

Note: *NS* not stated

Nigeria, malaria transmission is experienced all through the year because it is a rain forest region that is characterized with a humid tropical climate. Thus, it supports greater number of plant diversity including medicinal plants that the natives and traditional medicinal herbalists use for different ailments. According to a review report [16], 24 plant species have been tested in in vitro studies and were found to have anti-plasmodial properties and low cytotoxicity values. *Nauclea latifolia*, for instance, was reported to inhibit *Plasmodium falciparum* erythrocytic cycle towards the last phase of schizogony between 32 and 48 h. To break some logistic barriers in the experimentation of medicinal plants against malaria, developing countries like Nigeria occasionally collaborate with foreign laboratories to conduct some in vitro and vivo studies to close the gap of lack of requisite laboratories in their native countries [16].

In Zimbabwe, 28 species of plants were reported to have anti-malaria properties but only 26 were taxonomically identified in 16 families [21]. Detailed analysis show that out of the 26 species, 38.5%, 30.8%, 23.0%, and 7.7% are trees, shrubs, climbers, and herbs, respectively. Twenty of the 26 species have anti-malaria (therapeutic) properties, 4 have preventive (prophylactic) properties, while the remaining 2 species serve the dual purpose. Among these species, only *Momordica foetida* and *Capsicum annuum* can be cultivated while others can only be harvested from the wild, a scenario that could lead to loss of biodiversity. Interesting report directly from the traditional medicine healers stated that some plants such as *Euclea natalensis* (Ebenaceae) may not have therapeutic or prophylactic values but may be added to boost strength and increase appetite in malaria patients [21].

In Ethiopia, 51 species of medicinal plants in 28 families against malaria have been reported with the family Fabaceae, Asteraceae, Euphorbiaceae, Aloaceae, Alliaceae, and Meliaceae having 6, 5, 4, 3, 3, 2, and 2 species, respectively, while other family members have 1 species each. Among these medicinal plants against malaria, *Allium sativum*, *Croton macrostachyus*, *Carica papaya*, and *Lepidium sativum* are the most frequently used plants to treat malaria [85]. Phytochemical compounds such as alkaloids, terpenes, limonoids, chromones, xanthones, flavonoids, and anthraquinones have been reported to be successful in the treatment of malaria. Anthraquinones and naphthalene derivatives are the most reported compound against malaria. These compounds are effectively reported in in vivo and vitro studies at 61% and 39%, respectively, against *Plasmodium falciparum* [85].

In Uganda, about 182 species of medicinal plants in 63 families, which have anti-malaria properties, but only 17 species are frequently used in malaria healthcare. The leaves constitute the highest parts that are used at 54.4%, and followed by roots at 17.4%, bark at 16%, while whole plant and other parts constitute 12.2% [86]. Among the 182 species of medicinal plants, 112 have been evaluated for treatment of malaria, but 108 were ascertained to be effective for malaria treatment.

In Ghana, 422 species of medicinal plants in 25 families that are highly diversified are reportedly used in malaria treatment, with Euphorbiaceae constituting 14.3%, Asteraceae 9.5%, and Poaceae and Fabaceae having 7.1% each as the most used families [87]. Trees are mostly used at 42.9%, herbs 28.6%, shrubs 16.7%, grasses 7.1%, and climbers 2.4%. About 75% of the plant materials are collected from the neighbourhood surroundings, while 25% are harvested from the wild and elsewhere. Based on plant parts used, leaves constitute 80%, fruits 13.3%, stem-bark 4.4%, and flowers 2.2%. Threats to medicinal plants include drought, farming, over harvesting, bush fire, and others [87].

In Kenya, 286 plant species in 75 families are reported to be useful in the treatment and management of malaria with Asteraceae, Fabaceae, Lamiaceae, Euphorbiaceae, Rutaceae, and Rubiaceae families. The percentage use of these listed plant families are 36.5%, 29.7%, 24.3%, 21.6%, and 17.6%, respectively [22]. Plant parts used are shrubs, trees, herbs, roots, bark, root-bark, and stem-bark at 33.2%, 30.1%, 29.7%, 19.4%, 10.8%, 10.5%, and 6.9%, respectively, hence similar to reports from other African countries. No serious report of threat associated with leaves usage but root and root structures (tubers and rhizomes) are potentially driving the plants to extinction not minding reports that plant parts below the ground contain more phytochemicals.

From the time immemorial, medicinal herbs have been used to treat malaria in Africa and other tropical countries. Medicinal plants are abound in the tropics with higher concentration of organic chemicals that have been used for treatment of malaria (Table 4). For instance, quinine and artemisinin are isolates of *Cinchona* and *Artemisia* plants that are natives to sub-Saharan Africa. Thus, the first plant-based anti-malaria drug was extracted from *Cinchona* plant as quinine in the form of alkaloid as reported by Baird et al. [88] and was later isolated and characterized (Saxene et al. 2003). Later in the nineteenth century, another medicinal plant *Artemisia annua* became another source of malaria treatment known as artemisinin which was combined with other therapies and referred to as artemisinin-combined therapy (ACT) [89]. This approach of malaria treatment has been the medically approved treatment in Nigeria since 2005 [90]. However, a lot of factors of production militate against artemisinin combination therapy production and therefore, the drug remains expensive that many people in the developing countries cannot afford it. It is therefore imperative that plants from the tropics especially in sub-Saharan African regions should be screened for anti-malaria properties since greater percentages of malaria death cases occur in the regions.

4 Preparations and Administration of Herbal Medicine Used in the Treatment of Mosquito-Borne Parasitic Diseases in the Tropics

Some of the parts of the medicinal plants used in the treatment of malaria disease are processed into powder and preserved in closed bottles. When in need, the powder is either soaked in cold or hot water to extract the active ingredients. After a while, the water is decanted or sieved and given to a malaria patient to drink. To make concoctions from the powders, the recommendations are usually a full, half, or quarter of either teaspoon or tablespoon or may be a pinch. The concoction is either prescribed to be taken twice or thrice a day as from 3 to 7 days or till the patient is certified healed. Where some fruits are to be used, they are swallowed whole. For instance, the fruit of *Capsicum annuum* (*Capsicum frutenscens*) is given to a patient to swallow without chewing because of its bitterness. In some case, especially for preventive measures, some leaves are eaten as vegetables such as *Momordica balsamina* and *M. foetida* [21].

Table 4 Medicinal plants reported in Africa for the treatment of malaria

Plant family	Plant species	References
Acanthaceae	*Justicia betonica*	[86, 91]
	Justicia anselliana	[86]
	Monechma subsessile	
	Thunbergia alata	
Alliaceae	*Allium sativum*	[16]
	Allium cepa	[86]
	Aloe kedongensis	
	Aloe volkensii	
	Aloe ferox	
	Aloe lateritia	
Aloeaceae	*Aloe dawei*	
Amaranthaceae	*Amaranthus hybridus*	
Anacardiaceae	*Anacardium occidentale*	[16, 91]
	Mangifera indica	[16, 86, 91]
	Rhus natalensis	[86]
	Rhus vulgaris	
Annonaceae	*Uvaria chamae*	[16]
	Xylopia aethiopica	[91]
	Annona senegalensis	[16]
	Enantia chlorantha	[16, 91]
Apiaceae	*Heteromorpha trifoliata*	[86]
	Centella asiatica	
	Carissa edulis	
	Carissa spinarum	
Apocynaceae	*Catharanthus roseus*	
	Funtumia africana	[16, 91]
	Picralima nitida	[16]
	Rauvolfia vomitoria	[91]
	Alstonia boonei	[16, 86, 91]
	Rauvolfia vomitoria	[16]
Araceae	*Culcasia faleifolia*	[86]
Aristolochiaceae	*Aristolochia elegans*	
	Aristolochia tomentosa	
	Cryptolepis sanguinolenta	[16]
Asclepiadaceae	*Gomphocarpus physocarpus*	[86]
	Pergularia daemia	[91]
	Aloe vera	[86]
Asteraceae	*Aspilia africana*	[16]
	Ageratum conyzoides	[86]
	Artemisia annua	
	Artemisia afra	
	Aspilia africana	
	Baccharoides adoensis	

(continued)

Table 4 (continued)

Plant family	Plant species	References
	Bidens grantii	
	Bothriocline longipes	
	Conyza bonariensis	
	Conyza sumatrensis	
	Crassocephalum vitellinum	
	Emilia javanica	
	Guizotia scabra	
	Gynura scandens	
	Melanthera scandens	
	Pluchea ovalis	
	Microglossa pyrifolia	
	Schkuhria pinnata	
	Sigesbeckia orientalis	
	Solanecio mannii	
	Sonchus oleraceus	
	Tagetes minuta	
	Tithonia diversifolia	
	Vernonia adoensis	
	Vernonia amygdalina	
	Vernonia cinerea	
	Vernonia lasiopus	
Bignoniaceae	*Spathodea campanulata*	[16]
	Stereospermum kunthianum	
	Markhamia lutea	[86]
	Spathodea campanulata	
	Newbouldia laevis	[16]
Bombacaceae	*Adansonia digitata*	
	Ceiba pentandra	[91]
Boraginaceae	*Heliotropium indicum*	
	Ananas comosus	
Caesalpiniaceae	*Cassia didymobotrya*	[86]
	Chamaecrista nigricans	
	Erythrophleum pyrifolia	
	Senna spectabilis	
	Senna siamea	[91]
	Senna podocarpa	
	Cassia hirsuta	[86]
Canellaceae	*Warburgia ugandensis*	
Cannaceae	*Canna indica*	[16, 91]
Caricaceae	*Carica papaya*	[16, 86, 91]
Celastraceae	*Hippocratea africana*	[16]
	Maytenus senegalensis	[86]
Chenopodiaceae	*Chenopodium ambrosioides*	
	Chenopodium opulifolium	

(continued)

Table 4 (continued)

Plant family	Plant species	References
Combretaceae	*Anogeissus leiocarpus*	[16]
	Guiera senegalensis	
	Terminalia avicennioides	
	Terminalia latifolia	
	Combretum molle	[86]
Compositae	*Chromolaena odorata*	[16, 91]
	Tithonia diversifolia	
	Ageratum conyzoides	[16]
	Eupatorium odoratum	
	Vernonia amygdalina	[16, 91]
	Vernonia cinerea	[16]
	Acanthospermum hispidum	[91]
Crassulaceae	*Kalanchoe densiflora*	[86]
Cucurbitaceae	*Cucurbita maxima*	
	Momordica foetida	
	Momordica balsamina	[16]
Dracaenaceae	*Dracaena steudneri*	[86]
Ebenaceae	*Euclea latideus*	
	Bridelia micrantha	
	Clutia abyssinica	
	Croton macrostachyus	
	Flueggea virosa	
	Diospyros mespiliformis	[91]
Euphorbiaceae	*Alchornea cordifolia*	[16]
	Bridelia ferruginea	
	Bridelia micrantha	
	Croton zambesicus	
	Phyllanthus amarus	
	Alchornea cordifolia	[86]
	Jatropha curcas	
	Macaranga schweinfurthii	
	Phyllanthus niruri	
	Shirakiopsis elliptica	
	Tetrorchidium didymostemon	
	Cajanus cajan	
	Crotalaria agatiflora	
	Crotalaria ochroleuca	
	Entada abyssinica	
	Entada africana	
	Erythrina abyssinica	
	Erythrina excelsa	
	Bridelia ferruginea	[91]

(continued)

Table 4 (continued)

Plant family	Plant species	References
Fabaceae	*Arachis hypogea*	[86]
	Indigofera arrecta	
	Indigofera congesta	
	Indigofera emerginella	
	Macrotyloma axillare	
	Pseudarthria hookeri	
	Rhynchosia viscosa	
	Senna absus	
	Senna didymobotrya	
	Senna siamea	
	Tamarindus indica	
Flacourtiaceae	*Oncoba spinosa*	
	Trimeria bakeri	
	Homalium letestui	[16]
Guttiferae	*Harungana madagascariensis*	[16, 86, 91]
	Allanblackia floribunda	[16]
Labiatae	*Ocimum gratissimum*	[16]
	Solenostemon monostachyus	[16]
	Hyptis suaveolens	[91]
	Ocimum gratissimum	
	Hyptis pectinata	[86]
Lamiaceae	*Aeollanthus repens*	
	Ajuga remota	
	Clerodendrum myricoides	
	Clerodendrum rotundifolium	
	Hoslundia opposita	
	Leonotis nepetifolia	
	Ocimum basilicum	
	Ocimum gratissimum	
	Ocimum lamiifolium	
	Plectranthus barbatus	
	Plectranthus caninus	
	Plectranthus cf. *forskohlii*	
	Rosmarinus officinalis	
	Tetradenia riparia	
Lauraceae	*Cassytha filiformis*	[16]
	Persea americana	[86]
Leguminosae-Caesalpinioideae	*Abrus precatorius*	[16]
	Afzelia africana	
	Cajanus cajan	
	Cassia occidentalis	
	Cassia siamea	
	Cassia sieberiana	

(continued)

Table 4 (continued)

Plant family	Plant species	References
	Cassia singueana	
	Daniellia ogea	
	Daniellia oliveri	
	Piliostigma thonningii	
Leguminosae-Mimosoideae	*Cylicodiscus gabunensis*	
	Parkia biglobosa	
	Prosopis africana	
	Tetrapleura tetraptera	
Leguminosae-Papilionoideae	*Erythrina senegalensis*	
	Indigofera pulchra	
	Lonchocarpus cyanescens	
	Pericopsis elata	
Loganiaceae	*Anthocleista djalonensis*	
	Anthocleista vogelii	
Loranthaceae	*Tapinanthus sessilifolius*	
	Tapinanthus constrictiflorus	[86]
Malvaceae	*Hibiscus surattensis*	
	Gossypium barbadense	[91]
	Gossypium arboreum	[16]
	Gossypium barbadense	
	Gossypium hirsutum	[16, 91]
	Sida acuta	[16]
Meliaceae	*Azadirachta indica*	[16, 86, 91]
	Khaya grandifoliola	[16]
	Khaya senegalensis	
	Khaya grandifoliola	[91]
	Carapa grandiflora	[86]
	Melia azedarach	
Menispermaceae	*Sphenocentrum jollyanum*	[16, 91]
	Cissampelos mucronata	[86]
	Acacia nilotica	
	Acacia sieberiana	
Mimosaceae	*Acacia hockii*	
	Albizia coriaria	
	Albizia grandibracteata	
	Albizia zygia	
	Newtonia buchananii	
Moraceae	*Ficus thonningii*	[16]
	Milicia excelsa	[16, 86]
	Antiaris toxicaria	[86]
	Ficus natalensis	
	Ficus saussureana	
	Ficus platyphylla	[16]

(continued)

Table 4 (continued)

Plant family	Plant species	References
Moringaceae	*Moringa oleifera*	[16, 86]
Musaceae	*Musa paradisiaca*	[86]
	Musa sapientum	[91]
Myricaceae	*Myrica kandtiana*	[86]
Myristicaceae	*Pycnanthus angolensis*	[86, 91]
Myrsinaceae	*Maesa lanceolata*	[86]
Myrtaceae	*Eucalyptus grandis*	
	Syzygium cordatum	
	Syzygium cumini	
	Syzygium guineense	
	Ormocarpum trachycarpum	
	Psidium guajava	[16, 86, 91]
Ochnaceae	*Lophira alata*	[16]
	Lophira lanceolata	
Palmae	*Elaeis guineensis*	
Papilionaceae	*Butyrospermum paradoxum*	[86]
Passifloraceae	*Passiflora edulis*	
	Mondia whitei	[91]
	Parquetina nigrescens	
	Pittosporum brachcalya	[86]
	Pittosporum mannii	
Poaceae	*Cymbopogon giganteus*	[16]
	Setaria megaphylla	
	Digitaria scalarum	[86]
	Imperata cylindrica	
	Zea mays	
	Cymbopogon citratus	[16, 86, 91]
	Saccharum officinarum	[16]
Polygalaceae	*Securidaca longipedunculata*	[86]
	Maesopsis eminii	
Portulacaceae	*Talinum portulacifolium*	
Rosaceae	*Prunus africana*	
	Rubus steudneri	
	Hallea rubrostipulata	
	Crossopteryx febrifuga	[16]
	Mitragyna inermis	
Rubiaceae	*Nauclea latifolia*	[16, 91]
	Coffea canephora	[86]
	Pentas longiflora	
	Vangueria apiculata	
	Citrus sinensis	
	Morinda lucida	[91]

(continued)

Table 4 (continued)

Plant family	Plant species	References
Rutaceae	*Citrus reticulata*	[86]
	Citrus aurantium	[91]
	Citrus sinensis	[16]
	Fagara zanthoxyloides	
	Citrus aurantifolia	[91]
	Citrus paradisi	
Salicaceae	*Trimeria grandifolia*	[86]
Sapindaceae	*Blighia unijugata*	
	Blighia sapida	[16]
	Lecaniodiscus cupanioides	[91]
Sapotaceae	*Manilkara obovata*	[86]
	Chrysophyllum albidum	[91]
Scrophulariaceae	*Striga hermonthica*	[16]
	Sopubia ramosa	[86]
	Scoparia dulcis	[16]
Simaroubaceae	*Quassia undulata*	
	Harrisonia abyssinica	[86]
	Quassia amara	[16]
Solanaceae	*Solanum erianthum*	
	Datura stramonium	[86]
	Physalis angulata	[91]
	Physalis peruviana	[86]
	Solanum nigrum	
	Capsicum frutescens	[91]
	Solanum nigrum	
Sterculiaceae	*Sterculia setigera*	[16]
Tiliaceae	*Triumfetta rhomboidea*	[86]
Ulmaceae	*Celtis durandii*	[16]
	Celtis africana	[86]
	Trema orientalis	[91]
Umbelliferae	*Steganotaenia araliacea*	[86]
Rutaceae	*Teclea nobilis*	
	Toddalia asiatica	
	Zanthoxylum chalybeum	
	Zanthoxylum leprieurii	
Verbenaceae	*Lippia multiflora*	[16]
	Stachytarpheta cayennensis	[16]
	Stachytarpheta indica	
	Lantana camara	[86]
	Lantana trifolia	
Vitaceae	*Cissus populnea*	[16]
Zingiberaceae	*Zingiber officinale*	[16, 91]
	Curcuma longa	[16, 86, 91]

Decoction which is the act of boiling of plants, and/or plant parts, in water is the commonest way of making the medicine out of the plants available for drinking. Concoction, on the other hand, is the processing of medicinal plants into powder and/or ingestion of fresh extract from plants. For decoction, and infusion (steam bath), preparations depend on plant type, parts, and choice which is mainly related to geographical communities. Decoction goes with water measurement and is drank orally. For concoction, some plants are dried at room temperature, pulverized, and transformed to powder. When in need, 2–3 tablespoon is added to water and boiled to obtain a decoction. Boiling of plants or plant parts has no reference time limit but can be estimated with the initial volume of water reducing to half of its original volume. Some are prepared by squeezing fresh leaves of a chosen plant to obtain 2–3 teaspoon of a juice-like liquid daily. In some cases, different plant leaves can be used to make an infusion with fresh roots that are pulverized and taken orally. Some squeezed juice of medicinal plant can be mixed with cold or warm water for bathing while some are eaten as vegetable for preventive measures [92]. For lymphatic filariasis, desired plants or parts are grinded into powder to increase the surface area of the plant so that a chosen extractant can easily extract reasonable quantity of active ingredient in the plant. Decoction and infusion methods of preparation are mostly employed while orally drinking of the prepared medicine for malaria treatment while topical application of concoction on the affected limbs and bathing with decoction are common methods of treating lymphatic filariasis.

In Ethiopia, crushing, powdering, maceration, decoction, and others are the commonest means of formulating medicinal plants. The medicines are orally administered as a route of delivery into the body. Water and some ingredients (honey, butter, salt, coffee, tea, and milk) are added while making the preparation to reduce the bitter taste of the plants, hence spicing the medicine and reducing toxicity [85].

In Ghana, leaves are more frequently used, mainly for decoction, hence ranking 95.35%, and decoction for steam bath and rubbing or massaging plant tissue on the body constitute 2.3% each [87]. Decoction of medicinal plants constitute 70.5%, infusion 5.4%, oil treatment and steaming 1.3%, and roasting 0.3%. Plant materials may be chewed, pounded, ground, or crushed while dry materials are burnt, smoked, and inhaled. Extracting solvents include methanol, water, ether, ethanol, dichloromethane, chloroform, ethyl acetate, and petroleum ether [22].

In Nigeria, a popular concoction known as Agbo-Iba is being reported to protect mice against *Plasmodium yoelii nigeriensis.* The concoction is formulated with leaves of *Cajanus cajan* (pigeon pea), *Latifolia* sp., *Cymbopogon giganteus*, *Nauclea latifolia*, *Euphorbia lateriflora*, *Mangifera indica*, and barks of *Cassia alata* and *Uvaria chamae* [93]. The experimentation of the Agbo-Iba formulation is decocted for 3 h and administered to mice orally. The result showed protection of the mice against *P. yoelii nigeriensis*. However, a report has it that Agbo-Iba was more of prophylactic instead of therapeutic for malaria treatment [93]. Similarly, a multi-herbal combination mixture containing the leaves of *Carica papaya*, *Azadirachta indica*, *Cymbopogon citratus*, and *Anacardium occidentale* are used as a steam therapy against malaria. The steam from the concoction in a cooking pot is guided for a patient to inhale a hot steam from the pot by covering both the patient and the hot pot.

Administration dosages of medicinal decoctions and concoctions vary, and no stringent precision exist among traditional medicine practitioners for the same ailment but are very certain about their recipes. None conformity and compliance to dosage by traditional medicine practitioners and patients, respectively, may hamper the goal of malaria treatment with medicinal plants. For instance, some patients may stop taking a *Cissampelos mucronata* roots concoction half way because of its bitterness. This concoction is usually prescribed to be taken thrice a day for 7 days [21]. However, dosages and strength of the medicine are peculiarly age dependent. In some cases, the volume of consumption range is between 100 and 500 ml, 250, and 100 ml for adults, older children above 5 years, and children less than 5 years, respectively, or from 1 to 3 tablespoon in reverse order. The chosen quantity could be taken one to three times daily for 7 days or until full recovery is ascertained [94, 78, 92].

In discussing medicinal remedies of plants about malaria, for instance, the terms anti-malaria and anti-plasmodia are mentioned but does not mean the same thing. Anti-plasmodium is related to in vitro study of medicinal plants while anti-malaria is related to in vivo study of medicinal plants. For instance, anti-plasmodial action of plant extracts in in vitro study will be rated 'very good' when the IC_{50} is less than 5 μg/ml, good when IC_{50} is greater than 5 μg/ml but less than 10 μg/ml, and moderate when the IC_{50} is 10ug/ml or less than 20 μg/ml. For anti-malaria, action of plant extracts in in vivo study will be rated 'very good' when the extracts suppress about or more than 50% at 100 mg/kg of a body weight in a day. It will be good when the extracts suppress malaria by more than 50% at 250 mg/kg of a body weight in a day and moderate when more than 50% of malaria is suppressed at 500 mg/kg of a body weight in a day [95]. Basic febrile symptom indicators for malaria, which traditional medicine practitioners depict to commence treatment, include feeling of cold/goose pimples, headache, fever, sweating, loss of appetite, body weakness/feeling sleepy, dizziness, vomiting, and body pains. Medicinal plant-based preparations are orally drunk, steam inhaled, topically applied, and rubbing of essential oils on the skin [21, 22].

5 Challenges in the Use of Medicinal Plants in the Treatment of Mosquito-Borne Parasitic Diseases in the Tropics

The use of medicinal plants to treat different diseases especially infectious ones [96–116] and control of vectors of some parasitic diseases [117–121] has transcended from folkloric phase to a novel practice that involves heuristic exploration and taking cognizance of all the processes (formulations and dosages) to be documented for posterity and reproducibility. However, like every other sector of human endeavour, the use of medicinal plants especially the act of harvesting, formulations, standardized dosages, and others present some challenges in their use. For instance, harvesting whole plant and making use of root structures are not sustainable practices since many of the medicinal plants are sourced directly from the wild and hence, could lead to biodiversity loss. Seasonal variation, environmental geography, and methods of extraction of active compounds are challenging factors that can influence the concentration of phytochemical extracts which are essential in the treatment of mosquito-borne parasitic diseases [122].

The process of extracting active compounds from medicinal plants for further examination and characterization poses some difficulties for researchers. For example, poor aqueous solubility, and rapid breakdown of compounds such as berberine and emodin (characterized plant compounds for treatment of mosquito-borne diseases) have been a challenge in pharmaceutical development of such compounds into readily available medicine [123–125]. Others include having enough quantity of the extracts to justify production cost and having the compounds that can work in both in vitro and vivo because some compounds are active in vitro at a small quantity which could not be enough for isolation, hence limiting their use in in vivo studies. Again, some parasites lack animal models for their empirical evaluation. For instance, filarial parasite such as *Wuchereria bancrofti* cannot survive or be maintained in any known experimental animal for in vitro and vivo studies, thus no screening model to be used to evaluate the potency of medicinal plants on the parasite. In some cases where phytochemical compounds are active in in vivo study, it occurs at higher dosages which are therapeutically not meaningful.

In most cases, the toxicity of most of the plant compounds are not yet evaluated to determine their toxicity level, hence limiting the acceptance of their medicinal potentials. The data necessary for selective index of the compounds are not worked out or not readily available at the disposal of professionals, hence their cytotoxicity importance is still questionable. More still, there is paucity of clinical data about safety and efficacy occasioned by lack of requisite laboratories especially in the developing countries, hence making practitioners not to have a common ground for standard preparations and dosages and contraindications of plant-based medicines.

6 Strategies for the Overcoming the Hurdles in the Use of Herbal Medicine in the Treatment of Parasitic Diseases Transmitted by Mosquito in the Tropics

Motivational incentives are spices that drive the wheels of production; hence, herbal medicine practitioners should be encouraged and promoted by organs of governments and multinationals. They should also be enlightened on the sustainable means of conserving biodiversity while providing healthcare services. It is therefore, imperative to cooperate with herbal medicine practitioners when carrying out scientific researches on medicinal plants. Their inclusion in the researches will reduce wastages in using many acclaimed medicinal plants, hence restricting scientific evaluation on the selected plants that are efficacious against parasitic diseases. For a sincere cooperation and willingness, the herbal medicine practitioners who are likely to hoard medicinal information of different plants should be enticed with some level of patency of the outcome of the researches.

There should be need to have policies on the use of traditional or herbal medicines to regulate traditional medicine practitioners with a goal to having standards and fine-tune researches that are leaned to their traditional claims. The policies shall convey the objectives of propagation, protection, and sustainable use of medicinal plants in the treatment of various ailments in their communities. The health sector should be organized through ministry of health to champion a bill for collaboration between healthcare sectors and traditional medicine practitioners to have a national council of traditional medicine practitioners and other medical practitioners to act as a body that would protect herbal products of her members. The isolated or purified compounds that emanate from scientific evaluations should undergo further investigation to cut down the limitations in the quest to develop novel plant-based medicines that are sustainable, cheap, and affordable. For parasites without reliable animal models, new animal models ought to be sought for so as to understand the host/parasite interactions for effective testing of medicinal plants in both in vitro and vivo studies.

7 Conclusion

Diseases associated with mosquitoes are previously seen as a peculiar problem in the tropical countries, but today, they are accepted threat to the world public health because of mosquito species diversity and their geographical spread. Mosquitoes are spreading globally due to climate change, pattern of land use, and the continued increase in number of people moving from one country to the other, hence threatening almost half of the global population. Mosquito-borne parasitic diseases affect both people in the urban and rural villages but most prevalent in villages without adequate living conditions such as housing and sanitation and more disturbing to those with low immunity. Many female mosquito species are culprits in transmitting infectious parasitic diseases such as lymphatic filariasis and malaria between humans and/or from animals to humans.

The use of medicinal plants to treat and manage mosquito-borne parasitic diseases in the tropics mainly in Africa is linked to cultural acceptability, efficacy, accessibility, and affordability. Greater percentage of people living in the rural communities in the tropical countries treat almost every ailment with traditional medicines that originate from plants which are prescribed by traditional medicine practitioners or herbalists. Medicinal plants are cautiously selected plants which have been used in the tropics for years, for treatment and management of diseases from the ancient time till now. In traditional medicine, herbs and plant products are still being used among the African, Asian, and Latin American people, and their use is currently extending to Australia and the USA.

Information from traditional medicine practitioners show that the use of medicinal plants against diseases such as mosquito-borne diseases is economical, simple to use, easily accessible, and cheap while synthetic drugs require a lot of time, resources, and efforts from the onset. Medicinal plants are reported to have natural compounds in them which are anti: viruses, bacteria, protozoa, nematodes, fungi, oxidant, inflammatory, and carcinogenic. Their use in traditional medicine is beginning to attract consideration among the healthcare professionals in the international community because of its current documentation in print and other forms of media.

Tropical countries are biodiversity hotspot for medicinal plants which traditional medicine practitioners and/or herbalists have used to treat mosquito-borne parasitic diseases with reliable successes. However, biodiversity loss occasioned by continuously harvesting of whole plants and root structure, lack of support and motivation from appropriate quarters, adequate scientific production tools, and paucity of documented data on medicinal plants are among the challenges against the use of medicinal plants used in the treatment and management of mosquito-borne parasitic diseases. Defined policies for collaboration between the organized health sectors through the ministry of health to have a consolidated national council of traditional medicine practitioners and other medicine practitioners ought to act as a body to protect medicinal products, and council members shall form alliance for acceptance, patronage, and sustainability of use of medicinal plants for treatment of mosquito-borne parasitic diseases in the tropics.

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Mechanistic Approaches of Herbal Medicine in the Treatment of Arthritis

15

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Abstract

Arthritis is a global concern, affecting the global population, especially older people. The allopathic treatments proved their efficacy but toxicities too, so the need is to establish the treatment which should be productive and safe too. Many herbal-based medicines are used in the traditional medicine system, proving their effectiveness in the disease. The need of the hour is to establish their scientific, mechanical approach to arthritis treatment, which can improve the quality of life of these patients. In recent years, scientific research has shown that plant bioactive compounds are efficacious in disease treatment. This chapter aims to establish the knowledge of herbal medicines in disease treatment through the available scientific research data and their applicability. The plant-related drugs and disease knowledge are essential for treating disease. Ayurveda is the best and gold standard in terms of efficacy. Ancient literature like Ayurvedic Pharmacopeia and Chark Samhita have not only classified but also provided evidence-based knowledge on many diseases. In this chapter, we have conceptualized the various bioactive compounds from herbal drugs in their role in disease management treatment. The plant's primary and secondary metabolites have massive potential for treating arthritis. We have discussed the targeted sites for the disease management and applicability of herbal drugs in arthritis through available scientific-based evidence.

Keywords

Arthritis · Herbal medicines · Bioactive compounds · Mechanism of action

Abbreviations

AIA	Adjuvant induced arthritis
Akt	Ak strain transforming
CIA	Collagen-induced arthritis
ERK	Extracellular signal-regulated kinase
FLS	Fibroblast-like synovial cells
MAPK	Mitogen activated protein kinases
NSAIDs	Non-steroidal anti-inflammatory drugs
PI3K	Phosphatidylinositol 3-kinase
RA	Rheumatoid arthritis

1 Introduction

Arthritis is a most worrying problem across the globe. Arthritis has prehistoric existence in the world medical background. According to ancient literature, Charka Susruta explained the kind of arthritis in well-known standard books like Charka and Susruta Samhitas. The symptoms of the disease include pain obstruction and stiffness of joint, bone, and muscle pain, which sometimes leads to paralysis in some cases. The term for arthritis is rheumatism, a joint disorder that reflects the existing blend of humoral and biomedical control globally.

Herbal drugs are defined as drugs having active constituents from natural plant parts like leaves, roots, flowers, and fruits. Herbal drugs have been used for the treatment of a variety of diseases from ancient times, and it is not an embellishment in the direction of the utilization of herbal drugs in the condition. Herbal medicines are amalgamated through the therapeutic knowledge of a generation of working physicians on an ancient system of medicine for more than long years. These days, investigators showed their enormous curiosity in therapeutic compounds that are consequent from plant origin since these drugs have the least toxicities and are not very expensive compared to available treatments. Herbal drugs have significant action in many disease treatments. Many researchers have explained the applicability of herbal medicine and their active constituents in managing arthritis treatment.

In this chapter, we have conceptualized the applicability of bioactive compounds in the treatment lane. The disorders start with joints, ligaments, muscles, tendons, bones, and nerves. Available literature suggests that the conventional medicine system incorporates complete body components. According to Charka, the rainy season, its moist, cold storms, and polluted water supply produce disturbances in wind, the primary reason for rheumatic problems. Diet, age, heredity, and lifestyle are important factors in disease development. The primary symptom is pain, which leads further to inflammation.

Pain is the sensation that leads to so many problems in day-to-day life. Pain can be tolerated or not depending on the person when it affects aged people, especially older ones, creating discomfort in daily activities. The primary problem in the case of pain is it is not measurable. Clinicians use a scale known as VAS (visual analogue scale) to treat the patients, which works on the perception of the patients. The recent evidence of chronic toxicities with the available allopathic treatment provided a path toward new research.

1.1 Inflammation

Inflammation is a foremost problem in arthritis, distinguished by redness, heat, inflammation, and failure of function. Arthritis symptoms are stiffness, pain, inflammation, and loss of joint tasks. The diverse type of arthritis contains distinguishing indication, diagnosis, and disease management [1]. The types of arthritis are osteoarthritis and rheumatoid arthritis.

A chronic inflammation-related disease that mainly involves the periarticular tissue, cartilage leads to deformity, functional disabilities, and significant disability as a result of many inflammatory transmitters secreted through macrophages, such as eicosanoids like prostaglandins, cytokines, leukotrienes, and reactive oxygen species [2].

RA is the diverse process involved in synovial cell proliferation, fibrosis with cartilage degeneration, and bone attrition progression reconciled by an inter-reliant complex of cytokines, proteolytic enzymes, and prostanoids.

The cytokines, precisely tumor necrosis factor and interleukin-1, have prominent functions in disease. The increased level of IL-1 in the synovial fluid can be seen in disease patients; the concentrations of IL-1 in the plasma reported by researchers correlate with the disease functioning [3].

1.2 Rheumatoid Arthritis

The different forms of the disease are as follows:

Palindromic rheumatoid arthritis
Rheumatoid spondylitis
Juvenile rheumatoid arthritis

1.3 Other Kinds of Arthritis

- Osteoarthritis
- Primary osteoarthritis – Affects older age population
- Secondary osteoarthritis – Affects all ages of populations
- Ankylosing spondylitis
- Infectious arthritis
- Gout

1.4 Epidemiology

The occurrence of the disease is 0.3–1% across the globe, majorly in developed nations [4]. The disease prominently affects women more than men (3:1), and the progression of the disease is mainly seen in ages 30–55 years. It affects 0.5–1.0% of adults, and 0.75% of disease cases are reported approximately in India [5]. Arthritis is highest among adults due to lack of physical activity compared to those who are inadequately active or have biological activity. The prevalence of disease may be due to concomitant diseases like cardiovascular, obesity, and diabetes. It is expected that the number of cases of the disease may increase with the highest percentage in adults due to current food habits and other factors.

1.5 Disease Etiology

The disease etiology is not known. Genetic propensity to environmental factors is probably responsible for the disease progression.

1. **Environmental factors**: The disease has been linked to mineral oils, silica exposure, diet factors, and blood transfusion.
2. **Role of sex hormones**: According to the prevalence of disease, it has been seen that disease prominently occurs in women. In female, it is observed that it generally starts at the perimenopausal, perinatal, and gestation period.
3. **Genetic**: The genetic mechanism of this disease is not precise; the genetic level's appearance compared to healthy patients strongly suggests that genetic inheritance is the factor that resembles the key for future-based discoveries for diagnosis and treatment. Some kinds of literature reported that persons with selective human leukocyte antigen genes have a greater risk for disease prevalence than those in which human leukocyte is not available.

1.6 Pathophysiology

The pathophysiology can be understood through Fig. 1. According to the disease pathophysiology of rheumatoid arthritis, it starts from the synovium. The level of cytokines plays a role in disease progression through cartilage destruction. The destruction of collagen leads to the narrowing of joint space and the destruction of the bone. The pannus formation occurs due to the exudation of fluid in the synovium. Activation of CD4 T cells releases the cytokines that cause the formation of fibroblasts by MMP and RANK ligand, leading to the activation of osteoclast and destroying the tissues, promoting the destruction of joints.

There are two chief mechanisms for the inflammation process: B cells and T cells activation by the immune system, which have significant functions linked with the disease. T cells are essential in distinguishing the antigen as "non-self," which produces cytokines. After that, the part B cells start to secrete and reproduce the

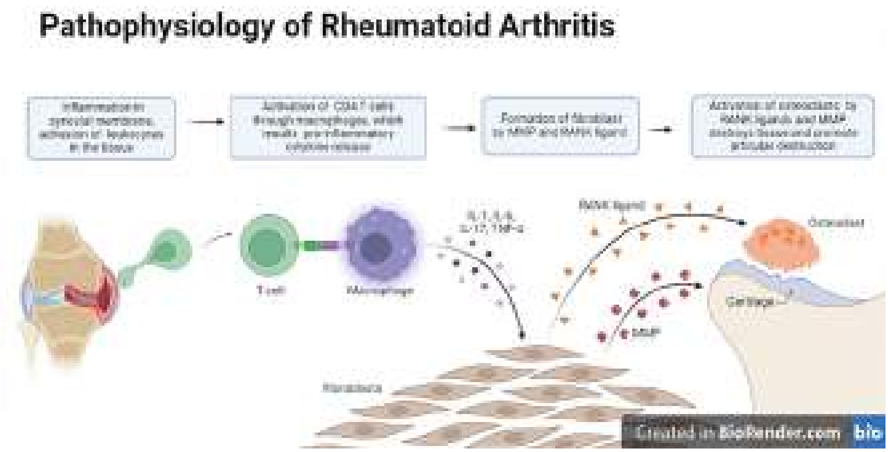

Fig. 1 Pathophysiology of rheumatoid arthritis

antibodies into the systemic circulation to identify antigens and activate the inflammation. Then, the various signaling pathways' role comes, which releases further pro-inflammatory mediators and causes the progression of the disease.

1.7 Responsibility of NF-κB/IκB-α in Inflammation Associated with RA

The disease pathophysiology entails the continual appearance of pro-inflammatory cytokines, chemokines that contain various sites like the presence of transcription factor inside their promoter, signifying the NF-κB involvement in the progression of the disease. The activation of tumor necrosis factor-alpha triggers NF-kB, which results in an increased level in the joints of rheumatoid arthritis patients. Furthermore, hyper (NF-κB) activity can also be seen in the macrophages of RA patients.

Classical and alternative pathways are the two essential signaling pathways that play a role in stimulating NF-kB. In these pathways, one common thing is IkB kinase complex activation that has the catalytic kinase subunits with a regulatory non-enzymatic protein named IKK. These NF-kB dimers are activated due to IKK-mediated phosphorylation-induced protein degradation of the IκB-α inhibitor, which liberates the active NF-κB transcription factor subunits to translocate to the nucleus and induce target gene expression. Without activating the signal, IkB sequesters NF-kB subunits and inhibits its translocation, consequently terminating transcriptional activity. In the classical signaling pathways, ligand binding happens on cell surface receptors such as a member of the Toll-like receptor superfamily, leading to the recruitment of adaptors to the receptor's cytoplasmic domain. In turn, these adaptors recruit the IKK complex, leading to phosphorylation and IB inhibitor degradation.

1.8 Role of Immunostimulants or Immunomodulators

The immune system assists in homeostasis through the active network that engages the applicability of numerous molecular basis cells capable of differentiation and reorganizing foreign particles from the self-cells. There are two types of immunity: innate and adaptive. Innate immunity works as the primary defense involving roles of cellular and molecular mechanisms that instantly respond to infection when any antigen attacks [6]. The natural immune system contains physical, chemical, and other cellular barriers. The group of cells like macrophages, neutrophils, and many more having foreign cells from self-act as cellular barriers [7]. If any pathogen or foreign cell can escape the first line of defense, immune system cells become the second line of defense or adaptive immunity [8].

1.9 NSAIDs: Other Inhibitors of Pro-inflammatory Mediators

The non-steroidal anti-inflammatory drugs treatment has been used for a long time to assuage arthritic pain and inflammation. Aspirin and Ibuprofen are drugs used extensively by clinicians to treat disease. Many NSAID products have become available for treating illness, but their toxicities are also high and concern global people. In the past, many painkillers were withdrawn from the pharmaceutical market, like rofecoxib and valdecoxib, due to cardiac toxicities. So, the safety of patients is a prime concern with efficacy.

2 Herbal Drugs and Applicability of Their Bioactive Compounds in the Treatment of Arthritis

Ayurveda has proven its efficacy in the disease since old times. Ayurveda originated from the Sanskrit words ayus and veda, which means denotation of life and knowledge. Ayurveda is a discipline of life focused on man and cures for diseases through the comprehension of medicinal plants [9]. The primary metabolites include that material that strengthens plant organization, in addition to the energy metabolism of the plant, e.g., proteins, carbohydrates, and fats as primary bioactive compounds. Secondary metabolites are the chemicals not utilized by plants for their organization or purpose. They are moderately available in small quantities with diverse application to protect plants from various microbes and insects, which also exhibits growth-regulating action in the plants. The scientific-based evidence proved the therapeutic applicability of these phytochemicals in disease management. Herbal plants used for the treatment of arthritis are summarized in Table 1. The review of available data suggests that herbal plants with bioactive compounds like flavonoids, triterpenoids, and phenolic compounds can be used in the treatment of arthritis. We have performed studies related to the plant-based drug for arthritis and found that researchers reported their work through the use of various animal models for

Table 1 Bioactive compounds of plant-based medicines for arthritis

S. no	Botanical name of plant	Family	Plant bioactive compounds	Reference
1.	*Abrus precatorius*	Fabaceae	Triterpenoid	[10, 11]
2.	*Acacia confusa*	Leguminosae	Flavonoids, melanoxetin	[12, 13]
3.	*Acacia hydaspica*	Leguminosae	Gallic acid, caffeic acid, rutin, catechin	[14, 15]
4.	*Aconitum vilmorinianum*	Ranunculacae	Diterpinoids, vilimorine A-D	[16, 17]
5.	*Albizia procera*	Fabaceae	Biochanin-A	[18, 19]
6.	*Allium sativum*	Liliaceae	Diallyl disulfide	[20, 21]
7.	*Alstonia boonei*	Apocynaceae	Phenolic acids, rutin, isoquercetin, flavonoliganes	[22, 23]
8.	*Alternanthera bettzickiana*	Amaranthaceae	Gallic acid, catechin	[24, 25]
9.	*Asystasia dalzelliana*	*Acanthaceae*	Steroids, flavonoids, alkaloids, tannins	[26]
10.	*Azadirachta indica*	Meliaceae	Nimbolide	[27, 28]
11.	*Berberis orthobotrys*	Berberidaceae	Flavonoids, phenolic compounds	[29]
12.	*Cissus quadrangularis*	Vitaceae	Russelioside B	[30, 31]
13.	*Capparis spinosa* L.	*Capparidaceae*	Flazin, guanosine, capparine	[32, 33]
14.	*Caesalpinia sappan*	Caesalpiniaceae	Sappanol, episappanol	[34, 35]
15.	*Calophyllum inophyllum*	Calophyllaceae	Triterpenoids	[36, 37]
16.	*Calotropis gigantea*	Apocynaceae	Lupeol	[38, 39]
17.	*Caltha palustris*	Ranunculaceae	Polysaccharide B	[40, 41]
18.	*Cayaponia tayuya*	Cucurbitaceae	Dihydrocucurbitacin B	[42, 43]
19.	*Celastrus*	Celastraceae	Celastrol	[44, 45]
20.	*Capparis erythrocarpos*	*Capparaceae*	Proteins, polyphenols	[46, 47]
21.	*Curcuma longa* L.	Zingiberaceae	Curcuminoids	[48, 49]
22.	*Chloranthus serratus*	Chloranthaceae	Terpenoids	[50, 51]
23.	*Clematis orientalis*	Ranunculaceae	Flavonoids and glycosides	[52, 53]
24.	*Clerodendrum serratum*	Verbenaceae	Terpenoids, steroids, flavonoids, and phenolics	[54, 55]
25.	*Cuscuta reflexa*	Convolvulaceae	Beta-sitosterol	[56, 57]
26.	*Drynaria quercifolia*	Polypodiaceae	Squalene, gamma tocopherol, N-hexadecanoic acid	[58, 59]
27.	*Eriobotrya japonica*	Rosaceae	Corosolic acid, oleanolic acid, and ursolic acid	[60, 61]
28.	*Ephedra gerardiana*	Ephedraceae	Alkaloids, flavonoids	[62, 63]
29.	*Euphorbia neriifolia*	Euphorbiaceae	Steroid and terpenoid	[64, 65]
30.	*Ficus bengalensis*	*Moraceae*	Flavonoids, terpenoid, sterols	[66, 67]
31.	*Fagopyrum cymosum*	Polygonaceae	Eugenol	[68, 69]
32.	*Ginkgo biloba*	Ginkgoaceae	Terpene	[70, 71]
33.	*Glycosmis pentaphylla*	*Rutaceae*	Sulfur-containing amides	[72, 73]

(continued)

Table 1 (continued)

S. no	Botanical name of plant	Family	Plant bioactive compounds	Reference
34.	*Glycyrrhiza glabra*	*Leguminosae*	Licochalcone A (flavonoid)	[74, 75]
35.	*Hemidesmus indicus*	Apocynaceae	Triterpenoids	[76, 77]
36.	*Ipomoea batatas*	Convolvulaceae	Polyphenols, 3-epifriedelinol	[78, 79]
37.	*Jatropha isabellei*	Euphorbiaceae	Alkaloid	[80, 81]
38.	*Justicia gendarussa Burm F*	*Acanthaceae*	Sterols, flavonoids	[82, 83]
39.	*Kadsura heteroclita*	Schisandraceae	Lignans and triterpenoids	[84, 85]
40.	*Lantana camara*	Verbenaceae	Triterpenoids	[86, 87]
41.	*Laportea bulbifera*	Urticaceae	Neochlorogenic acid, cryptochlorogenic acid, and chlorogenic acid	[88, 89]
42.	*Lawsonia innermis*	*Lythraceae*	Lawsochylin A and lawsonaphthoate, luteolin, apigenin	[90, 91]
43	*Leucas aspera*	*Lamiaceae*	Epicatechin, beta epicatechin, procyanidin, beta-sitosterol	[92, 93]
44.	*Linum usitatissimum*	Linaceae	Alpha linolenic acid	[94, 95]
45.	*Litsea cubeba*	Lauraceae	Litsecols	[96, 97]
46.	*Lonicerae japonica*	Caprifoliaceae	Lonicerin	[98, 99]
47.	*Melastoma malabathricum*	Melastomataceae	*B*-Sitosterol, melastomic acid	[100, 101]
48.	*Monotheca buxifolia*	Sapotaceae	Flavonoids, triterpenoids, vitamin E, phytol,	[102, 103]
49.	*Moringa rivae*	Moringaceae	Fatty acids, vitamin E	[104]
50.	*Olea europaea*	Oleaceae	Oleuropein, ligstroside	[105–107]
51.	*Panax ginseng*	Araliaceae	Ginsenosides Rg3, Rk1, and Rg5	[108, 109]
52.	*Phyllanthus amarus*	Phyllanthaceae	Phyllanthin and hypophyllanthin	[110, 111]
53.	*Pinus maritime*	Pinaceae	Flavangenol	[112, 113]
54.	*Piper betle*	Piperaceae	Hydroxychavicol	[114, 115]
55.	*Pistia stratiotes*	Araceae	Triterpenes, flavonoids	[116, 117]
56.	*Premna serratifolia*	Lamiaceae	Phenolic compounds and flavonoids	[118, 119]
57.	*Punica granatum*	Punicaceae	Tannins and anthocyanins	[120, 121]
58.	*Rhus verniciflua*	Anacardiaceae	Flavonol, fisetin	[122, 123]
59.	*Ribes alpestre*	Grossulariaceae	Phenolic and flavonoid	[124]
60.	*Ribes orientale*	Grossulariaceae	Polyphenolic compounds, flavonoid	[125]
61.	*Ruta graveolens*	Rutaceae	Alkaloids, polyphenols	[126, 127]
62.	*Salacia reticulata*	Celastraceae	Polyphenols,	[128, 129]
63.	*Salix nigra*	Salicaceae	Sterols, terpenes, flavonoids	[130, 131]

(continued)

Table 1 (continued)

S. no	Botanical name of plant	Family	Plant bioactive compounds	Reference
64.	*Saraca asoca*	Fabaceae	Polyphenols	[132, 133]
65.	*Sargassum wightii*	Sargassaceae	Alginic acid	[134, 135]
66.	*Saussurea lappa*	Asteraceae	Polyphenols, flavonoids	[136, 137]
67.	*Semecarpus anacardium*	Anacardiaceae	Flavonoids	[138, 139]
68.	*Sida rhombifolia*	Malvaceae	Rhombifoliamide, ß-sitosterol	[140–142]
69.	*Sinomenium acutum*	Menispermaceae	Sinomenine	[143, 144]
70.	*Solanum nigrum*	Solanaceae	Polyphenols, flavonoids	[145, 146]
71.	*Sophora flavescens*	Fabaceae	Flavonoids and alkaloids	[147, 148]
72.	*Strobilanthus callosus*	Acanthaceae	Lupeol, 19 alpha-h-lupeol	[149, 150]
73.	*Strychnos potatorum*	Loganiaceae	Alkaloids, ß-sitosterol	[151, 152]
74.	*Syzygium aromaticum*	Myrtaceae	Eugenol, terpenes	[153, 154]
75.	*Torilis japonica*	Apiaceae	Torilin	[155, 156]
76.	*Toxicodendron pubescens*	Anacardiaceae	Quercetin and rutin	[157–159]
77.	*Tragia involucrata*	*Euphorbiaceae*	Rutin, quercetin	[160, 161]
78.	*Trewia polycarpa*	*Euphorbiaceae*	Terpenoids, flavonoids	[162, 163]
79.	*Tridax procumbens*	Asteraceae	Flavonoids	[164, 165]
80.	*Trigonella foenum graecum*	Fabaceae	Steroids, alkaloids, polyphenols, flavonoids	[166, 167]
81.	*Urtica pilulifera*	Urticaceae	Phenolic compounds	[168, 169]
82.	*Vernonia cinerea*	Asteraceae	Steroids, flavonoids	[170, 171]
83.	*Vitex negundo*	Verbenaceae	Phenylnaphthalene-type lignans	[172, 173]
84.	*Wendlandia heynei*	Rubiaceae	Terpenoids, B-carotene	[174, 175]
85.	*Withania somnifera*	Solanaceae	Phenolic compounds, withanolides	[176–178]
86.	*Xanthium strumarium*	Asteraceae	Sesquiterpenoids, phenylpropenoids, lignanoids	[179, 180]
87.	*Yucca schidigera*	Asparagaceae	Resveratrol, spirostanol	[181, 182]

arthritis, which gives a direction in the treatment of disease and also attracts the phytoconstituents' role.

3 Bioactive Compounds and Their Targets in Disease

Bioactive compounds are the secondary metabolites isolated from herbal medicinal plants, animals, fungus, and microorganisms. Moreover, they are having pharmacological or toxicological effects on organisms, foremost to utilization in food and pharmaceutical industries. In the drug discovery process, the new property of these compounds gave the direction in the treatment of disease.

Treatment of any disease requires the knowledge of the targets for the drug delivery. Bioactive compounds of plant-based medicines have played a vital role in diseases, so the need of the hour is to explore these compounds with the moa that can give direction in disease management. As per the disease, etiology immune system plays a vital role, so targeting the cytokines in arthritis can give us the direction for treatment. In this section, we have highlighted that the various pathways like PI3K and Akt signaling, MAPK signaling, NF-κB, STAT signaling, and NRF2 signaling responsible for disease progression with the applicability of primary and secondary metabolites from herbal drugs in the lane of treatment reported by researchers.

3.1 Flavonoids

Flavonoids are successive compounds, and the structure of these compounds suggests that they are made up of phenyl rings and phenolic hydroxyl groups throughout their carbon chains [183]. Flavonoids have significant efficacy in treating arthritis and have been reported by many researchers that can be correlated with Table 1. Flavonoids also significantly influence therapy through inhibition of PI3K and Akt signaling pathways. This pathway plays a vital role in disease progression through fibroblast-like synovial cells that may secret an abundance of cytokines that incessantly excite FLS cells by unrestrained propagation of cells. PI3K and Akt signaling path is believed to be a connection between the propagation and programmed cell death of FLS cells. The studies on arthritis suggest that these signaling are highly articulated and reflect the consequence of the extreme movement of fibroblast-like synovial cells. Many pro-inflammatory mediators like IL-17 and IL-21 can support the inflammatory propagation of fibroblast-like synovial cells by stimulating and activating PI3K [184].

Some researchers reported that LY294002, a PI3K inhibitor, can improve over-growth and inflammation in synovial joints in animal studies [185].

Another critical action of flavonoids is the inhibition of the MAPK signaling pathway, which is also responsible for disease progression through stimulation of various kinase systems like ERK, p38, c-Jun N-terminal kinase, and ERK5, which control many enzymes cytokines and chemokines causes persistent inflammation and abnormal hyperplasia [186].

Some researchers revealed that a flavonoid isolated from *Glycyrrhiza uralensis* can considerably block the IL-1β-induced rheumatoid arthritis fibroblast-like synovial cell propagation by slow-down through the kinase system [187].

Flavonoids also inhibit the NF-κB STAT signaling pathways in disease. The NF-κB and STAT lead to shifting in the nucleus that causes inflammation, cell propagation, and apoptosis. Some studies suggested that flavonoid Genkwanin extracted from *Daphne genkwa* showed anti-arthritic activity by slowing down the p-NF-κB p-STAT3 in AIA mice [188].

Several researchers reported through their research on NRF2 signaling that it plays an essential role in arthritis through oxidative stress concerned with etiological and diagnostic parameters. The activation of the Nrf2 signaling leads a range of enzymes like heme oxygenase-1 HO-1 to be released, having antioxidant properties that regulate the oxidative stress condition in disease [189]. Some studies implicated that epigallocatechin 3-gallate has anti-arthritic potential up-regulation of Nrf2 and HO-1 in joints of CIA rats [190].

Moreover, a flavonoid calycosin isolated from the *Astragali radix* can inhibit the pro-inflammatory mediators in RA-FLS by up-regulation of Nrf2 and HO1, as reported by some researchers [191].

3.2 Lignans

Some researchers reported that lignans with antioxidant potential can inhibit the ROS through the NF-κB path, eventually decreasing the appearance of inflammatory cytokines and pro-inflammatory enzymes. Activating the AMPK and Nrf2 path increases antioxidant-related genes and promotes the release of anti-inflammatory cytokines [192].

3.3 Anthraquinone

The reported work by authors suggests that rhein is an anthraquinone, which demonstrated cell viability and differentiation without toxicities at humans' physiological levels but considerably decreased cytokine levels in urate crystal-activated macrophages [193].

3.4 ß-Sitosterol

ß-sitosterol structure is similar to the cholesterol group. The research work reported by authors revealed that ß-sitosterol showed marked inhibition on synovial angiogenesis through suppression and proliferation of endothelial cells that result in alleviated joint inflammation and bone erosion in CIA mice [194].

3.5 Terpenoids

Terpenes are an important class of organic compounds produced mainly by herbal plants. The researchers reported that terpenes moderate the immunologic response and cartilage destruction by inhibiting various cytokines responsible for disease progression [195].

4 Challenges with Herbal-Based Medicines

The major challenge with herbal-based medicine is standardization and quality control. Globally, various guidelines have been approved for standard, quality-related parameters of herbal drugs. WHO approved the guidelines for taking care of the standardization parameters of herbal medicines [196, 197]. This process involves the phytochemical investigation of raw material of plant-based treatment related to features like collection and management of plant material, therapeutic and safety profile measurement of the processed products with proper documentation based on knowledge, and proviso of produce products to consumers [198]. Growth and accomplishment of the guidelines of traditional or herbal drugs within diverse divisions of the globe are frequently tackled by quite a few confrontations [199, 200]. The problems often encountered in many countries are those connected to regulatory standings, evaluation of effectiveness with safety, and insufficient information concerning traditional and plant-related drugs in nationwide drug regulations. Many herbal medicines have inadequate information on their mechanism of action in disease treatment, and the adverse drug reactions are not known anymore. The interaction with other drugs data is not available.

The various parameters related to standardization and quality control are as follows.

4.1 Collection and Authentication

People use many herbal drugs on their traditional knowledge, causing problems in the collection, adversely affecting the yield and quality of chemical constituents, and errors in experiments. Authentication is the basic process before the experimentation of any herbal drugs.

4.2 Presence of Other Organic Materials

It's essential to collect, identify, and remove the other foreign organic materials during the collection because they can lead to errors in the experimental process and the purity of drugs too.

4.3 Ash Values Determination

Ash values are an essential method for the purity of crude drugs. The specific test is performed for the identification of the total ash value. These values also have significance with quality principles. The ash value also gives an idea of finding the suitable part for the extraction process.

4.4 Moisture in Plant-Related Drugs

The moisture content determination helps reduce the mistake in the judgment of the actual weight of drug material, i.e., little moisture recommends improved stability adjacent to product deprivation.

4.5 Extractive Yield

These are the investigative weights of the extractive yield of bioactive compounds of herbal drugs in different solvents. These give the knowledge of the relation between the solvent and plant metabolites, which can be utilized for isolation and impact the cost of herbal medicine.

4.6 Qualitative Profile Evaluation

Identifying and categorizing crude drugs with reverence to phytochemicals is essential in photo-constituent processing. It provides work for diverse investigative methods to recognize and separate the active ingredient. Bioactive compounds showing plans provide botanical characteristics, solvent selection, purification, and categorization of the active constituents, which are essential parameters for pharmaceutical utilization.

4.7 Chromatography

The detection of raw drugs is based on using major chemical constituents as markers. Chromatographic methods also play an essential role in identifying active plant-based constituents, which helps describe the relation with treatment. The process should be accurate and standardized.

4.8 Quantitative Evaluation of Chemicals

The plant phytoconstituents are one of the critical parameters for evaluating its efficacy in disease management and the applicability of major constituents, such as in dosage determination parameters.

4.9 Toxicological Screening

The toxicological screening is essential for determining toxic residues present in the preparation, which can lead to toxicities. Hence, it is essential to perform these studies per the available guidelines of the OECD. The processes are based on international guidelines based on scientific evidence.

Standardization is one of the critical parameters and needs for herbal drugs. Much literature is available on the uses of herbal medicines, but the problem is to prove the same through standardization parameters. Some issues may arise if someone is unaware of these drugs' therapeutic dosage. More work is to be needed in the standardization part of this medication.

Apart from these, additional factors affect the standardization and quality control of herbal drug's efficacy parameters for processing herbal medications.

4.10 Adulteration in Drugs

Adulteration affects efficacy as well as causes toxicities all over the world. There is a need to develop regulatory guidelines for adulteration of herbal drugs since quality testing is an important parameter.

4.11 Faulty Preparations

The irrational combination of drugs is a leading cause of drug-to-drug interaction. There is a need for strict guidelines for herbal preparations since the herbal market is increasing in terms of volume and consumption day-by-day. Faulty herbal preparation leads to adverse effects and reduced efficacy.

4.12 Storage

The ancient and current literature implicated the storage conditions of herbal drugs. Some factors like temperature, light, moisture, and air are essential parameters for primary and secondary metabolites as they directly affect the metabolites.

4.13 Dosage and Time

Many countries still use their traditional medicine system to treat various diseases. The Indian people use many herbal medicines and speak about dosage and time through their traditional knowledge. The ancient standard literature also implicates the therapeutic applicability of the same, so it is necessary to relate the traditional knowledge and scientific evidence to overcome the toxicities.

5 Prospects of Herbal Medicines in the Disease

The challenge for herbal medicine in arthritis management is the applicability of these formulations in disease management. In recent years, the surge in curiosity concerning the endurance of Ayurvedic formulations increased. From a comprehensive point of view, there is a transfer toward plant-based medicine, as the threats and the inadequacy of contemporary medicine have ongoing getting more perceptible, and the preponderance of Ayurvedic drugs are processed from herbal sources. In many countries, many herbal formulations are available for arthritis treatment, and people use these medicines without any prescriptions. Many small towns have some orthodox people who do not have proper qualifications and knowledge about these drugs but are selling them drugs. The need is to provide the treatment with the support of healthcare professionals, which is essential to create awareness among them for prescribing herbal medicine. It is the liability of the countries' regulators to ensure that the consumers get good quality medicine that is pure, effective, and safe for treatment. The regulatory setups are required for the clinical trials related to herbal drugs. The repositories should be available online for patient compliance, which can give a clear-cut idea for reporting adverse drug reactions with herbal medicines. The quality standards of raw materials related to plant sources of herbal medicines are a supreme consequence in mitigating their satisfactoriness in the current structure of the medical system. Most of the time, problems encountered by herbal drug manufacturers are the non-availability of raw materials for the quality control profile of herbal material and their formulations. Herbal product usage is increasing due to noise made by manufacturing companies through advertisements. Systems like Ayurveda still require awareness among the people and experiential support from contemporary medical sciences to make them plausible and tolerable for all [201]. The inventive investigation endeavors to describe the benefit of conventional systems of medicine with reverence to their safety profiles, which will possibly affect improved exploitation of these systems of medicine. Globally, several pharmacopoeias have offered herbal drug information through monographs describing the constraints and standards of many herbal medicines and several products completed by these standards, which are safe and productive. The value of herbal medicines in treating disease is increasing due to less toxicity when compared with allopathic medication. The herbal drugs market is expanding in terms of the volume of people who use these drugs as per their traditional knowledge. The acceptance of these drugs is increasing due to their safety margins. Many Pharma organizations have started manufacturing facilities for herbal medicines globally. Technological advancement is needed as primary and secondary metabolites of the drug have a prominent role in disease management. Herbal medicines can be a good choice compared to available drug treatments and can improve the daily life of these patients. As arthritis disease mainly affects older patients and has some concomitant treatment that increases chances of drug-to-drug interaction, Herbal medicines can be a good option for disease management as their safety margin is satisfactory. Many herbal formulations are available in the market for the treatment of arthritis and for improving the quality of life of these patients. Ethno-pharmacological studies are increasing regularly. The need is to relate traditional medicine knowledge with scientific evidence for appropriate formulations for patients' safety. Primary and secondary metabolites can be utilized for the future-based treatment of arthritis, as these metabolites can reduce the dosage due to their efficacy at minimum concentration. The time is to establish the ethanol-pharmacology concept on a broader base for the herbal medicines and changes needed in available manufacturing processes. The time is to show the ethanol-pharmacology idea on a wider base for the herbal medicines and modifications needed in available manufacturing processes. Labeling and product information for patients is also required to change from time to time for the acceptance purpose in disease treatment.

Almost three-fourths of the herbal medicines used wide-reaching were expected to be exposed after local treatment. As per the available data from WHO, the minimum percentage of contemporary drugs come from herbal sources used traditionally [202]. In the present scenario, almost all countries focus on the herbal medicine market due to its safety and efficacy. The growing demand for herbal medicine requires attention in terms of quality and methods for improving product processing and thought processes to get the maximum active bioactive compounds from the plant's raw material. The cost of processing bioactive compounds is high, and the need to develop methods like solvent selection is required. Many studies also focused on how that combination of water and alcohol can achieve better extractive yield. The modernization in processing methods not only affects the work but will also reduce the cost of medicine for the patients. All countries should focus on their native herbal drugs and explore their applicability in disease management. Every country should have an awareness program for their citizens to establish knowledge of herbal medicine. The various programs should be the plan for arthritis management for the stakeholders of the disease management. The time is to find the drugs that have not been explored well yet; many plants are not identified and stored in a database worldwide. Aquatic plants are another excellent option for treating arthritis in the future drug discovery process. The arthritic patients need to develop a formulation that can relieve pain and should have anti-inflammatory action. Patient tolerability to dosage is also a significant concern because many patients have some concomitant treatment.

6 Conclusion

Arthritis is increasing consistently, so it requires attention in diagnosing and treating the disease. The available allopathic treatment toxicities cannot be ignored. The need is to establish a treatment that is safe and effective. In this chapter, we have explained the mechanistic approaches of plant-related drugs with scientific evidence that can be utilized for future drug discovery processes. Herbal medicine can be an appropriate drug candidate for the management of arthritis. Perhaps more studies are needed to explore their potential. As per the etiology of the disease, cytokines play a crucial role in disease progression. We have highlighted the plant-based bioactive compounds such as flavonoids, lignans, anthraquinone, ß-sitosterol, and terpenoids that can be an option in future drug discovery as it has therapeutic values in the disease. The need of the hour is more research focusing on metabolites for getting a drug candidate and molecular basis for drug targeting.

The worldwide recognition and utilization of herbal medicines plus associated products continue to presume and increase. The problems involving adverse drug reactions at the current time are too much more dramatic, increasing in occurrence and no longer controversial since the previous delusions of concerning or else classifying herbal medicinal products as safer as they are extracted from natural origin. The truth is that saying safer and origin from nature are not the same; therefore, regulatory guidelines for herbal medicines must standardize with fortified approaches on the international level. The applicable regulatory requirements in diverse nations of the globe necessitate being proactive and enduring in the direction of suitable actions toward looking after public health by ensuring that every single herbal medicine permitted for sale is safe and of appropriate quality. Quality education specifically related to herbal medicine is needed for a better understanding. Globally, the guidelines should meet equal parameters. Through this, quality herbal medicine will be available for the patients. The pharmacist should give all the information related to drugs, like drug-to-drug interactions, for the betterment of the disease patients.

The contributors of plant-based drugs, like doctors, nurses, and pharmacists, have minute training in understanding the effects of herbal medicines on the health of their patients. Several are inadequately informed about these products, and how to use them is a big question. Sufficient training and teaching are required because patients use different types of prescription and counter-product drugs. The fact is that healthcare professionals have to play a vital role in expressions of their precious assistance to safety assessments of medications with the significance of the same. All contributors to herbal medicines should be adequately allowed to observe the safety of herbal medicines. It can be achieved through partnerships with conventional healthcare professionals. The need of the hour is to create such an environment for smooth functioning and adequate knowledge about the use and safety of herbal medicines. The qualification of every doctor, nurse, pharmacist, and patient linked to herbal drugs is essential for treating and understanding the misuse of herbal medicines. To improve the quality of life of arthritis patients, it is required to develop the formulation based on natural sources.

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Assessment of the Phytochemical Constituents and Metabolites in the Medicinal Plants and Herbal Medicine Used in the Treatment and Management of Respiratory Diseases

16

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Abstract

Worldwide, chronic obstructive pulmonary disease, asthma, lung cancer, and cystic fibrosis are some of the top causes of death and morbidity. As a result, these illnesses place a significant burden on healthcare systems, economies, and societies in many nations. Chronic respiratory illnesses impact the respiratory airways, lung parenchyma, and pulmonary vasculature, among other parts of the respiratory system. Respiratory disorders are a fairly frequent diagnosis in children, adolescents, and adults in the era of air pollution. Both upper and lower respiratory system disorders can have a significant negative impact on overall health as well as economic and psychological burdens. Because they are eco-friendly and have few side effects, plant-based solutions are receiving a lot of attention as a means of treating and preventing health issues. It is believed that active plant contact and exposure to nature are good for both physical and mental health. The immunological and cardiovascular systems are the main targets of plant-based medications. The medicinal properties of both terrestrial and marine botanicals are an effective control of a variety of disorders. Biologically active compounds with distinct values may be isolated from both. Therefore, the main aim of this chapter is to know the potential of pulmonary diseases and to provide the knowledge of plant therapy or some medicinal herbs for curing.

Keywords

Chronic pulmonary disease · Lung disease · Respiration · Immune · Psychological · Airborne · Botanicals · Medicinal herbs · Anti-inflammation

Abbreviations

AYUSH	Ayurveda, Yoga and Naturopathy, Unani, Siddha, and Homeopathy
ACE2	*Angiotensin-converting enzyme*
AKT	Ak strain transforming
BAX	Bcl-2-associated X protein
CD	*Cluster of differentiation*

CDK	Cyclin-dependent kinases
COPD	Chronic obstructive pulmonary disease
COVID-19	*Corona virus disease – 2019*
CXCR4	C-X-C motif chemokine receptor 4
DMDS	Dimethyl disulphide
ET	Endotracheal Tube
FEV	Forced expiratory volume
GSH	Glutathione
HIV	Human immunodeficiency virus
HUVEC	*Human umbilical vein endothelial cells*
LOX	Lipoxygenase
MCT	Medium chain triglycerides
MDA	*Malondialdehyde*
NOS	Nitric oxide synthase
PAP	Pulmonary artery pressure
PASMC	*Pulmonary artery smooth muscle cells*
PDE	Phosphodiesterase enzyme
RNA	Ribonucleic acid
ROS	Reactive oxygen species
RVH	Right ventricular hypertrophy
RVSP	*Right Ventricular Systolic Pressure*
SARS CoV-2	Severe acute respiratory syndrome coronavirus 2
SOD	Superoxide dismutases
TGF	Transforming growth factor
TMS	Transcranial magnetic stimulation
VEGF	*Vascular endothelial growth factor*
WBC	White blood cells

1 Introduction

Medicinal herbs plays a critical role in the well-being of rural, remote, and indigenous people. Plants with medicinal properties are used to cure a wide range of illnesses in both humans and animals. Scientists are interested in the ethnomedical and nutritional uses of locally sourced plant products, prompting a search for bioactive chemicals. Carbohydrates, protein, and fat are all found in medicinal plants. Physiological, metabolic, and morphological processes rely on these substances, making them essential for human survival [1]. Medicines, dietary supplements, and other healthcare items often contain ingredients originating from plants. Antioxidants, hypoglycaemics, and hypolipidaemics are just a few examples of the types of phytochemicals found in plants that play a significant role in the search for new and improved therapeutic components. Many pharmaceuticals have their origins in plants, either as direct or indirect sources [2]. Humans rely on plants for a wide variety of things, including food, shelter, fibre, and medicine. Herbal remedies are the primary choice for health care among rural residents. Even in places where conventional medicine is readily available, there has been a recent uptick in curiosity about herbal remedies. Minerals and plant-based chemicals found in medicinal plants are two examples of the types of physiologically active components that have been shown to have an array of impacts on human physiology [2]. As they are an abundant source of several bioactive chemicals, essential oils are becoming more and more popular. They are preferred over chemical antioxidants due to their antioxidant and antibacterial capabilities as well as their GRAS (generally recognised as safe) status [3].

Botanicals are used as ingredients in medications, cosmetics, dietary supplements, and herbal teas, among other healthful products, and have seen a surge in popularity in recent years [3]. Plants used for medical purposes have been found to have bioactive substances that may reduce the risk of developing serious illnesses like cancer, cardiovascular disease, and diabetes. Medicinal herbs are essential in the treatment of oral health problems such halitosis, gums bleeding, dental caries, ulcers of mouth, and gingivitis due to their high effectiveness and low risk of adverse effects. Many plant species have unique mechanisms for producing secondary metabolites [3]. A lack of certain vitamins and minerals in the diet has been linked to a variety of potentially serious health problems, including malnutrition and a host of diseases caused by micronutrient deficiencies. Historically, all medicines were derived from plants, either in the form of a crude extract mixture or a simpler sample of plant parts. The primary advantage of using plant-based drugs is that they are non-toxic, far safer than their synthetic counterparts, while also providing significant therapeutic benefits and being much cheaper.

The last several decades have seen a surge of curiosity about medicinal plants and their traditional applications across the Indian subcontinent. Many young people nowadays are leaving rural communities to pursue higher education and work in urban centres, and with them, there has been a decline in the value placed on traditional medicines. The rich medicinal plant knowledge of the country is in danger due to the present trends of deforestation, environmental degradation, and modernization in the area. Hence, herbal knowledge is only accessible by the elderly, and only a select few have access to conventional therapies. Hence, by recording indigenous people's traditional plant medicine, we can close the generational gap in herbal knowledge. So, the goal of this chapter is to examine the big picture of the various herbal plants, formulations, and plant constituents that are currently being utilized to treat respiratory ailments.

2 Phytoconstituents Contributed in Pulmonary Disease Treatment

Plants contain molecules called phytochemicals, which have protective and disease-fighting qualities. Phytochemical components such as triterpenoids, quercetin, resveratrol, etc., have shown significant clinical efficacy against cardiovascular and respiratory problems [4]. In Table 1, we have outlined just a handful of the plant-based chemicals that are effective in treating respiratory disorders.

Table 1 Plant-based chemicals in treating respiratory disorders

Phytochemicals	Disease
Androsin	Bronchial asthma
Apple polyphenol	High blood pressure in the lungs
Curcumin	Bronchitis, emphysema, cystic fibrosis
Ligustrazine	High blood pressure in the lungs
Quercetin	Asthma, COPD, pneumonia, pulmonary hypertension
Resveratrol	Asthma, COPD, pneumonia, pulmonary hypertension, pneumonia, sarcoidosis
Salidroside	High blood pressure in the lungs
Triterpenoids	Pulmonary hypertension, chronic bronchitis

3 Medicinal Plants as Beneficial Agents

The reduction in blood pressure diastolic and activity of sympathetic nervous system and the induction of more relaxed, calmed, and natural sentiments are only a few of how nature and active plant engagement are thought to benefit physical and mental health [5]. Plants are the most common form of life and are widely believed to be the most emblematic component of a natural setting. It is important to remember that indoor plants have their unique challenges that cannot be faced in the outdoors. As a result, houseplants can be read in two ways [6] either as a representation of nature or as a product of human engagement with the natural world. The presence of even a small number of houseplants has been linked to fewer sick days taken at work and a greater sense of satisfaction, comfort, and well-being. The visual density of urban tree coverage has been found to positively correlate with both stress recovery [7] and landscape choice [8] in various studies. There should be more interest and use of energy-efficient indoor plants that can improve air quality and moderate local climate. There are plant species and cultivars that are particularly effective at removing volatile organic chemicals [9]. Diffusion of volatile organic molecules into a plant's leaf depends on several factors, including the leaf's stomatal features, wax layer, and hair growth, all of which vary from plant to plant [10]. Members of the Araliaceae family tended to have intermediate to high removal rates of organic compounds which are highly volatile, as compared to the family of Araceae which have demonstrated low [11] rates of removal. Ferns, followed by herbs, had the highest formaldehyde elimination rates, according to research by [10]. Dry deposition and atmospheric dispersion are two mechanisms that vegetation can use to its advantage to improve air quality [12]. The study of dry deposition, which analyses the procedure by which pollutants are at least momentarily removed from the ambient air, takes into account interception, sedimentation, capture, biological processes, and other factors. Pollution levels can be lowered locally by the use of green infrastructure. Many studies have found that persons who reside in more peaceful environments (rural areas, big green spaces) report higher levels of happiness [13]. Vegetation in urban areas might mitigate some of the adverse effects of noise pollution caused by human activity [14]. Certain green areas can provide thermally comfortable circumstances because they permit deeper water penetration into the ground, increasing the amount of water available for evaporation and so aiding in the further cooling of the surrounding air [15]. The Ministry of AYUSH (Government of India) has occasionally issued distinct guidelines for the treatment of respiratory issues dependent on different medicine systems of India. A respiratory infection called bronchitis is marked by a rise in mucus production and airway inflammation [10]. It is challenging to separate from other illnesses, such as influenza, the common cold, etc., that frequently cause cough. In the upper respiratory, mild viral infection system is referred to as the "common cold." Each year, separate influenza outbreaks, a viral infection of the respiratory system, occur. According to Rawal et al. [16], acute respiratory distress syndrome is a fatal inflammatory lung injury that develops along with the acute failure of the respiratory as a consequence of clearly identified non-pulmonary and pulmonary insults [17]. Emphysema is characterized by persistent breathlessness and an inability to tolerate physical exercise due to damage to the respiratory membrane. There is evidence of a rise in neutrophils, macrophages, T-lymphocytes, and eosinophils in the terminal air spaces and emphysematous tissue of the pulmonary parenchyma. A deteriorating lung condition called COPD is marked by airway narrowing. The inflammatory reaction causes emphysema in the small air cells but chronic bronchitis in the larger airways [18]. These various methods, which the hospitals use following their areas of expertise and modern medicine primarily as adjuvants and may be pertinent for boosting immunity, treating fevers, or providing other respiratory treatments, are as follows:

3.1 Aush Kwath

The AYUSH Ministry encourages AYUSH kwath used as a pre-made mixture for promoting the general public's health. *Ocimum sanctum* leaves, *Cinnamomum verum* stem barks, *Zingiber officinale* rhizomes, and Piper nigrum fruits make up the mixture. Ayush Kwath, Ayush Kudineer, and Ayush Joshanda are a few of the titles under which the formulation is marketed in the market. The market offers it in tablet and powder shapes. These herbs are effective treatments for different viral diseases and are said to boost immunity [19, 20].

3.2 Samshamani Vati

Ayurvedic medicine, that is, Guduchi ghana vati (Samshamani vati), is used to treat all kinds of fevers. It is also used as an anti-inflammatory and painkiller [21]. Samshamani vati, an aqueous extract of Tinospora cordifolia, is thought to have immunomodulatory activities as a result of the synergistic effects of the many chemicals present [22]. Additionally, it works well for many infectious illnesses.

Hareetaki Agasthaya, a popular "Avaleha kalpana," Agastya Haritaki Rasayana contains more than 15 herbal components and is used to treat a variety of respiratory infections. The majority of its components demonstrated antiviral, anti-asthmatic, anti-inflammatory, and immunomodulatory properties [23–25].

3.3 Anuthaila

Arn. and Retz. (Leptadenia reticulate) Wight has been used in the therapy of throat, bronchitis, and asthma issues. Anuthaila Anuthaila contains about twenty ingredients. Similarly, *Ocimum sanctum* and *Sesamum indicum* oil are suggested for a variety of ailments, including dry cough, asthma, migraines, and respiratory illnesses [26]. There are accounts of *S. indicum* seeds combined with *Tachyspermum ammi* seeds that are being used to treat colds, dry coughs, and lung conditions [27].

3.4 Roghan-e-Baboona

A traditional Unani medicine called roghan-e-baboona is used to manage inflammation and asthma. The primary component of Roghan-e-Baboona is Matricaria chamomilla L. flowers. It is made from *M. chamomilla* flowers, which have been proven to be effective for sore throats and acute viral nasopharyngitis [28].

3.5 Sharbat-e-Sadar

It is a formulation polyherbal syrup of unani region plants and is extensively used for the common cold, cough, and respiratory diseases. A key component, *Trachyspermum ammi*, was found to promote the growth of B-cells and to block the neutralization of antibodies to the Japanese encephalitis virus [29]. According to Singh et al. [30], *Adhatoda vasica* inhibits the HIV protease, and *Bombyx mori* was found to boost immune reactions against viral infection. Other components with antiviral and immunomodulatory properties include *Ziziphus jujuba*, *Glycyrrhiza glabra*, *Onosma bracteatum*, and *Ficus carica*.

3.6 Asgandh Safoof

Withania somnifera Dunal, also known as Asgandh Safoof Asgand, is a widely used medicinal herb in India. In the Unani school of medicine, the root powder is used as an immunomodulator. According to reports, the extract of roots substantially raises counts of CD8+ and CD4+ as well as blood profile measurements, particularly WBC and platelet counts (Bani et al., 2006). By disrupting the connections between the host ACE2 receptor and the viral S-protein receptor binding domain, aqueous suspension prevented SARS-CoV-2 entrance and demonstrated strong inhibitory action against the mitogen-induced proliferative response of T lymphocytes [31].

4 PH Treatment with Medicinal Plants

4.1 Allium Sativum

The plant species Garlic (*Allium sativum* L.) has been utilized in traditional medicine for centuries. Cysteine sulfoxides and other sulphur-containing compounds are primarily responsible for the therapeutic effects of *A. sativum*. The category includes allicin, which has the potential to be converted as thiosulfinates [32]. It has been shown that *A. sativum* has therapeutic effects, including as lowering the level of glucose in the blood, decreasing oxidative stress, fighting cancer, as well as reducing inflammation, boosting the immune system, warding off infections, and protecting the heart [33]. *A. sativum* may have the ability to hyperpolarize the membrane of smooth muscle cells, eNOS activity increases, and thereby relax vascular smooth muscles. According to research by Fallon et al. (1998), the vasorelaxant action of a fine extract in the powder form of *A. sativum* can slow the pH development in rats exposed to hypoxia. These are Bunge Allium macrostemon. Traditional Chinese medicine makes use of Bunge Allium macrostemon, a member of the family having Amaryllidaceae, for its many pharmacological activities, including those of being antitumourous, antiasthmatic, antioxidant, and immune-enhancing immunity [34]. The primary component of *A. macrostemon* is dimethyl disulphide (DMDS), which has several biological functions including melanin synthesis regulation, anti-hypertensive and anti-inflammatory, and is thus utilized for myocardial ischemia treatment [35].

4.2 Allium Ursinum

Also known as bear's garlic or wild garlic, it is a plant that is endemic to Eurasia. It is an Amaryllidaceae family member. It has a long history of application in alternative medical practices [36]. *A. ursinum* is capable of producing a miscellaneous set of chemical structures, which are ultimately responsible for the organism's biological activity. The aroma, which is reminiscent of garlic, is caused by flavonoid glycosides and sulphur components, which are two of the most important chemicals. Several of the chemicals produced by *A. ursinum*, such as phytosterols and galactolipids, are unique to the species [37]. On human platelets it may be antiaggregatory effects, inhibition effect on 5-LOX (5-lipoxygenase), and synthase of prostaglandin-endoperoxide (PTGS) enzymes in conditions of in vitro, high antioxidant activity and biosynthesis of cholesterol inhibition in vitro are few essential pharmacological activities that are exhibited by *A. ursinum*. Other important pharmacological activities include the following. In addition to this, its activity has been linked to blood pressure reduction as well as an inhibition of the PDE 5A (phosphodiesterase enzyme). *A. ursinum* contains flavonoids and saponins, both of which have the potential to suppress the activity of PDE5A [38]. There have been reports of an upregulation of PDE5 in PH [39]. Remodelling of vascular pulmonary and RVH can be described through the rise of RV/(LV + S) ratio and thickness of pulmonary arterial wall, respectively. In contrast to untreated hypertensive rats which are untreated, rats having pulmonary hypertension treated with *A. ursinum* are shown to be remodelling the pulmonary vascular and RVH [40] (Table 2).

4.3 Mimosa Pigra

Throughout Africa, the Americas, and Asia, the *Mimosa pigra* L. is widely used as a medicine due to its status as a member of the Fabaceae family. This plant is used against a variety of conditions, including heart problems, gastrointestinal issues, and infections [41]. In a research on the pharmaceutical potential of *Mimosa pigra* against hypoxia-induced PH in rats, Rakotomalala et al. (2013) discovered that *Mimosa pigra* can upregulate NO generation, downregulate PAP and RV/(LV + S) and p38 MAPK expression and phosphorylation in lung tissue. In rats with pulmonary hypertension, P38 MAPK is activated [42]. This research suggests that *M. pigra* may have a beneficial effect on PH outcomes.

4.4 Trifolium Pratense

Trifolium pratense, or Red clover, is a member of the Fabaceae family of plants. Its original range includes West Asia, Europe, and Northwest Africa. *T. pratense* relies heavily on several different flavonoids [43] for its health benefits. *T. pratense* contains isoflavones, which a recent study on animal models found to inhibit delicate muscle contractions in the uterus, ileum, and bladder [44]. Jiang and Yang (2016) showed that feeding broiler chickens with PH an extract of T. *pratense* containing isoflavones decreased ET-1 in the hens' serum and lungs while simultaneously increasing NOS secretion. To a greater extent, ET-1 can stimulate MMP-2 expression, which is associated with respiratory vascular remodelling [45]. *T. pratense* can raise serum levels of NOS and NO due to its phytoestrogen activity [46]. These results imply that Trifolium pratense isoflavones may be useful for treating PH in avian models, where it has been observed to increase NO and decrease ET-1 expression.

4.5 Tulsi (Ocimum Sanctum)

A sacred Hindu herb that has a very large history of medical usage and is now being studied for its potential to alleviate symptoms of bronchial asthma and bronchitis. Tulsi's main ingredient, Eugenol, has a variety of therapeutic effects, including antimicrobial, analgesic, antispasmodic, and adaptogenic. Symptoms of bronchitis and the duration of the illness were reduced in randomized, double-blind, placebo-controlled clinical trials with the combination of primrose root tincture and thyme

Table 2 Medicinal plants used for various diseases

Name of the plant	Scientific name and family	Methods used	Parts used	Purpose of use
Spiny amaranth	*Amaranthus spinosus* Amaranthaceae	Decoction	Foliage	Used for treatment of cough and asthma
Mango	*Mangifera indica* Anacardiaceae	Freshly consumed and decoction	Foliage and fruits	Wound cuts and diabetes can be cured
Soursop	*Annona muricata* Annonaceae	Freshly consumed and decoction	Foliage and fruits	Kidney infection, diabetes, hypertension, and wound cuts
Custard apple	*Annona squamosa* Annonaceae	Poultice	Foliage	Stomachache and headache
Indian snakeroot/ Sarpagandha	*Rauwolfia serpentina* Apocynaceae	Decoction	Foliage	Cure cold and cough
Bagawak	*Philodendron lacerum* Araceae	Poultice	Foliage	Sprain and stomachache
Aloe	*Aloe vera* Asphodelaceae	Extraction	Foliage	Burning of the skin
Common mugwort	*Artemisia vulgaris* Asteraceae	Decoction and poultice	Foliage	Diseases of the lung, blood vomiting, ringworm, and cough
Sambung	*Blumea balsamifera* Asteraceae	Decoction	Foliage	Cold, headache, flu, fever, urinary tract infection, and stomachache
African marigold	*Tagetes erecta* Asteraceae	Pounding and poultice	Leaves	Wound and cuts
Fukien tea tree	*Carmona retusa* Boraginaceae	Poultice	Leaves	Cough and body pain
Indian cherry	*Carmona dichotoma* Boraginaceae	Decoction	Leaves	Post-delivery fever
Papaya	*Carica papaya* Caricaceae	Pounding and extraction	Leaves	Malaria and dengue
Rangoon creeper	*Quisqualis indica* Combretaceae	Decoction	Leaves	Kidney infection
Sweet potato	*Ipomea batatas* Convolvulaceae	Decoction	Leaves	Diabetes and cut and wound infection

(continued)

Table 2 (continued)

Name of the plant	Scientific name and family	Methods used	Parts used	Purpose of use
Bitter-melon	*Momordica charantia* Cucurbitaceae	Cooking and extraction	Leaves and fruits	Diabetes and cut and infection of the wound
Alingaro	*Elaeagnus alingaro* Elaeagnaceae	Decoction	Roots	Headache, stomachache, Kidney infection, muscle pain, gall bladder stone and liver disease
Chinese laurel	*Antidesma bunius* Phyllanthaceae	Decoction	Foliage	Arthritis and kidney infection
Croton	*Codiaeum variegatum* Euphorbiaceae	Poultice	Foliage	Body pain
Snake weed	*Euphorbia hirta* Euphorbiaceae	Decoction	Stem, leaves and roots	Malaria and dengue
Physic nut	*Jatropha curcas* Euphorbiaceae	Poultice	Foliage	Sprain and stomachache
Bitter cassava	*Manihot esculenta* Euphorbiaceae	Decoction	Foliage	Diabetes, wound and cuts
Quick stick	*Gliricidia sepium* Euphorbiaceae	Pounding	Foliage	Vomiting, diarrhoea, and kidney infection
Lima bean	*Phaseolus lunatus* Euphorbiaceae	Cooking	Seeds	Diabetes and wound infection
Glorybower	*Clerodendrum intermedium* Fabaceae/ Lamiaceae	Poultice	Foliage	Sprain and stomachache
Spearmint	*Mentha spicata* Lamiaceae	Decoction	Foliage	Cold, cough, and headache
Common basil	*Ocimum basilicum* Lamiaceae	Decoction	Foliage	Cough and cold
Oregano	*Origanum vulgare* Lamiaceae	Decoction	Foliage	Cold, cough, headache,
Cat's whiskers	*Orthosiphon aristatus* Lamiaceae	Decoction	Foliage and roots	Kidney infection
Painted nettle	*Plectranthus scutellarioides* Lamiaceae	Poultice	Foliage	Kidney infection, cold, and cough

(continued)

Table 2 (continued)

Name of the plant	Scientific name and family	Methods used	Parts used	Purpose of use
Fragrant premna	*Premna adorata* Lamiaceae	Extraction and poultice	Foliage and stem	Skin disease and allergies
Chinese chase tree	*Vitex negundo* Lamiaceae	Decoction	Foliage	Cough and cold
Cinnamon	*Cinnamomum mercadoi* Lauraceae	Decoction	Foliage	Vomiting, nausea, and hypertension
Avocado	*Persea Americana* Lauraceae	Decoction	Foliage	Kidney infection, stomachache, hypertension, and diabetes
Queen's flower	*Lagerstroemia speciosa* Lythraceae	Decoction	Foliage	Kidney infection
Drumstick tree	*Moringa oleifera* Moringaceae	Eating in a fresh state and decoction	Fruits and foliage	Diabetes and wound Infection
Guava	*Psidium guajava* Myrtaceae	Decoction	Foliage	Diarrhoea, wounds and cuts Infection
Java plum	*Syzygium cumini* Myrtaceae	Eating in a fresh state and decoction	Fruits and foliage	Diabetes and wound infection
Bilimbi	*Averrhoa bilimbi* Oxalidaceae	Decoction	Foliage	Fever
Screw pine	*Pandanus tectorius* Lythraceae	Decoction	Foliage	Fever and cold, stomachache, flu, kidney infection, wounds cuts infection, and diabetes
Pepper elder	*Peperomia pellucida* Moringaceae	Decoction	Foliage	Body pain and hypertension
Betel pepper	*Piper betle* Myrtaceae	Poultice	Foliage	Sprain
Love grass	*Chrysopogon aciculatus* Poaceae	Decoction	Roots	Stomachache, headache
Cogongrass	*Imperata cylindrica* Poaceae	Decoction	Roots	Muscle pain, headache, stomachache, liver disease, kidney infection, and gall bladder stone
Calamondin	*Citrofortunella microcarpa* Piperaceae	Extraction	Fruits	Cold and cough

(continued)

Table 2 (continued)

Name of the plant	Scientific name and family	Methods used	Parts used	Purpose of use
Star apple	*Chrysophyllum cainito* Sapotaceae	Eating in a fresh state and decoction	Fruits and foliage	Diabetes and wound infection
Bell pepper	*Capsicum annuum* Poaceae	Poultice	Foliage	Boil and wounds
Burr bush	*Triumfetta bartramia* Tiliaceae	Decoction	Roots	Muscle pain, headache, stomachache, kidney infection, gall bladder stone and liver disease
Turmeric	*Curcuma longa* Rutaceae	Decoction	Rhizome	Stomachache

fluid extract [47]. Some research [48] also shows that ivy (*Hedera helix*) leaf extract helps reduce chronic bronchitis symptoms. In contrast to thymoquinone, a different phytochemical, nigellone, derived from black cumin was reported to display anti-spasmodic and facilitative effects on pulmonary clearance [49].

4.6 Argemone Mexicana (Papaveraceae)

Brahmadandi, the Indian plant *A. mexicana*, is a regular sight along highways there. The most active compounds identified from *A. mexicana* for the curing of asthma are berberine and protopine [50]. Native Andhra Pradesh residents have found success in treating asthma by taking a powder made from the seeds two times a day for 2 weeks [51]. At a concentration of 50 mg/kg, aqueous extracts from the stem of *A. mexicana* have been shown to reduce milk-induced leukocytosis and eosinophilia, two hallmarks of the allergic response [52].

4.7 Piper Longum (Piperaceae)

Piperine extracted from *P. longum* (long pepper) fruit has shown antiasthmatic and anti-bronchitis action (Hussain et al., 2011), and the plant is native to India. Alkaloids, steroids, glycosides, and flavonoids are all present in *Piper longum* fruit. The guinea pig's bronchospasm caused by histamine was successfully reduced by the fruit [47].

4.8 Tamarindus Indica (Fabaceae)

Breathing disorders, vaginal and uterine symptoms, and inflammation are just a few of the many conditions that *T. indica* (Tamarind), a traditional tropical medicine, is renowned for treating in folk medicine. Researchers have found that an extract of *T. indica* leaves has potent anti-asthmatic activity due to its bronchodilating, anti-histaminic (H1-antagonist), and anti-inflammatory properties [53].

4.9 Ageratum Conyzoides

The New World's tropical regions are home to the annual herb *Ageratum conyzoides*, abbreviated as *A. conyzoides*, which is a member of the Asteraceae family (previously the Compositae). Yet, it is also found throughout the rest of the world's tropical and subtropical climates. This plant's hydro-alcoholic leaf extract has been shown to have significant antihistaminic activity and to prevent clonidine-induced catalepsy in rats when given orally at doses of 250–1000 mg/kg [54].

5 Phytochemicals Used for PH Treatment

The Orchidaceae genus includes the *Eulophia macrobulbon* (Corduroy orchid) species. In Asia, Europe, and America, conventional medicine frequently makes use of orchids. Narkhede et al. (2016) state that they also treat cancer, diabetes, autoimmune, and inflammatory illnesses, as well as pulmonary and other diseases. PDE5 inhibitors, flavonoids, and terpenoids are present in the ethanolic extract of *E. macrobulbon*, indicating that it may be a pulmonary vasodilator [55]. A treatment with an ethanolic extract of *E. macrobulbon* decreases calcium chloride 2–induced in vitro contraction of PA rings and lowers the ratio (RV/LV + S), according to research on MCT-induced PH in rats by Wisutthathum et al. [56]. Pigs with high cholesterol have their coronary arteries relaxed by *E. macrobulbon* extracts because of stilbenes and their compounds. As a result, it has been proposed that *E. macrobulbon* extract may regulate pH and cause PAs (like coronary arteries) to relax.

5.1 Apple Polyphenol

Especially treat conditions like gastric mucosal injury, cancer, inflammation, and cardiovascular disease, doctors often prescribe apple polyphenol, a chemical extracted from several apple fruits (*Malus* spp.) [57]. Apple polyphenol can influence hemodynamic indicators such as mPAP and PVR, which may reduce the contraction of pulmonary vascular rings, as discovered by Hua et al. [58] in their investigation of hypoxia-induced PH in rats. It reduces cytosolic Ca2+ in PASMCs in vitro. Apple polyphenol increases eNOS expression, boosts NO levels, and reduces caspase-3 and iNOS expression in PAEC at the genetic level. Our results demonstrate that apple polyphenol has therapeutic potential as a treatment for PH, as it reduces contraction and increases NO levels to bring about its therapeutic impact.

5.2 Asiaticoside

The saponin asianoside, isolated from the plant *Centella asiatica* (L.) Urb., has a wide variety of biological activities that have led to its widespread use in traditional medicine. These include effects on temporary cerebral ischemia and reperfusion that are antioxidant, anti-inflammatory, antihepatofibrotic [59], and neuroprotective. In research of hypoxia-induced PH, [60] discovered that asiaticoside therapy decreased RVH and mPAP while simultaneously increasing cGMP and NO concentrations and decreasing ET-1 concentrations in the blood. Not only does this chemical prevent apoptosis in endothelial cells, but it also stimulates nitric oxide synthase (NOS) and activates AKT. Evidence suggests that asiaticoside can influence NO production, which slows the development of PH via phosphorylating and activating the serine/threonine protein kinase/eNOS pathway. According to research by Wang X. B. et al. [61] on hypoxia-induced PH in rats and PASMCs, treatment with asiaticoside raises RVH and mPAP, lowers the production of TGF-bR2, TGF-b1, and TGF-bR1, and inhibits the phosphorylation of Smad2/3 in lung tissue of animals. Smad2/3 phosphorylation is shown to increase when PH develops. Asiaticoside decreases both the quantity and migration of PASMCs, as well as the expression of TGF-b1. Thus, asiaticoside inhibits cell growth and encourages cell death through apoptosis to deliver its therapeutic benefits [61] (Fig. 1).

5.3 Astragalus Polysaccharides

The polysaccharides in *Astragalus membranaceus* Fisch. ex Bunge is the actual chemical responsible for the plant's many pharmacological effects, including its anti-inflammation and antioxidant properties [62]. According to Yuan et al. (2017), therapy with astragalus polysaccharides improves hemodynamic indicators

Fig. 1 Medicinal properties of phytochemicals

including PVR, RVH, mPAP, and medial wall thickness in rats with MCT-induced PH. Increased levels of NO and overexpression of eNOS were also noted in the therapy groups. Astragalus polysaccharides were effective in reducing pro-inflammatory mediator levels. The pho-IkBa expression level was also shown to be decreasing. Astragalus polysaccharides, from the research, can be an effective treatment for PH by inhibiting inflammation and by boosting NO levels, which improves hemodynamic indicators. Pho-IkBa is a marker for the evaluation of inflammation.

5.4 Baicalin

Flavonoid baicalin is extracted from the roots of the Scutellaria baicalensis Georgi plant and has several pharmacological effects, including antioxidant, anticancer, anti-inflammatory, and apoptosis-inducing [59]. Because of its anti-fibrotic effect, baicalin has been shown to reduce collagen I expression. It was shown by Wang et al. [61] that hypoxic conditions play a role in PH via pulmonary arterial collagen. In their investigation of hypoxia-induced PH, Liu et al. (2015) found that baicalin treatment improved hemodynamic parameters such as mean arterial pressure (mPAP), mean systemic arterial pressure (mSAP), and right ventricular (RV)/(LV + S) percent. Collagen I is a marker of PH development in lung tissue, and baicalin inhibits its protein and mRNA expression. Baicalin has a much larger effect on protein expression, collagen I mRNA than it does on collagen III, which is worth noting. Baicalin reduces collagen I expression by increasing ADAMTS1 expression. As baicalein inhibits collagen I synthesis and improves hemodynamic parameters, it may be useful as a therapy for PH [63].

5.5 Rhodiola Tangutica

Golden root, also known as *Rhodiola tangutica*, is a significant plant used in Qinghai and Tibetan ancient medicine. This plant is used to cure a range of diseases, most notably to avoid colds brought on by high altitude [64]. Phenylethanols, flavonoids, and terpenoids are a few of the *R. tangutica* enrich fraction's physiologically active compounds [65]. According to Yuan et al. (2017), therapy with astragalus polysaccharides improves hemodynamic indicators including PVR, RVH, mPAP, and medial wall thickness in rats with MCT-induced PH. These three proteins are essential for the transition of the cell cycle from G0/G1 phase to S phase. A regulator of CDKs is P27kip1. As a result, controlling these variables can reduce cell proliferation, which in turn reduces medial vessel thickness and contributes to structural alterations in pulmonary hypertension. In a lab study on isometric tension changes brought on by a force transducer in rat's pulmonary artery verified the vasorelaxant activity of *R. tangutica* [63]. As a result, these investigations suggest that *R. tangutica* may be used as a successful pH treatment.

5.6 Andrographis Somnifera

In particular in India, ashwagandha, a plant belonging to the Solanaceae family, is utilised as a herbal remedy to treat a number of diseases [66]. HPLC study identifies withanolide A and withaferin A as the most important active components in *W. somnifera* root powder extract. *W. somnifera* is used medicinally to treat a number of ailments, such as cancer, Alzheimer's, ageing, inflammation, and bone-building [67]. It is also used to lower blood pressure and cholesterol levels [66]. Kaur et al. (2015) discovered that therapy with *W. somnifera* can lessen RVP and RVH as well as PCNA expression in a study on MCT-induced pulmonary hypertensive mice. Procaspase-3 expression rises as a result of the *W. somnifera* therapy, inducing apoptosis in the pulmonary vessels. Additionally, *W. somnifera* therapy might lower the ROS concentrations in pulmonary tissue. Its anti-inflammatory impact is related to the ability of it to raise IL-10 levels while lowering NFkB and TNF-a levels. Withanolides discovered that *W. somnifera* has an anti-inflammatory impact by inhibiting TNF-a, which in turn inhibits the activation of NFkB [68]. *W. somnifera* also alters the expression of HIF-1a and eNOS in lung tissue (HIF-1a increases in hypoxia conditions). Increased eNOS action may result in higher NO levels, which relax the blood vessels. *W. somnifera* antioxidant, vasorelaxant, and anti-inflammatory properties make it a promising herb for the treatment of PH.

5.7 Luteolin, Apin, and Rhoifolin

Luteolin, Apiin, and Rhoifolin are three kinds of flavonoids used in traditional medicine. Studies have shown that the bulk of plants and veggies contain flavonoids. These flavonoids have biochemical and pharmaceutical effects that include anti-inflammatory, anti-allergic, and spasmolytic properties. The slow cardiac fibres of dogs may be inhibited by luteolin, apiin, and rhoifolin, which can also lower aortic pressure and capillary respiratory pressure in healthy animals. In a research on hypoxia-induced hypertension in dogs, Occhiuto and Limardi [69] showed that structural changes in the flavonoids (apiin, rhoifolin, and luteolin) significantly alter their antihypertensive activity. They showed that rhoifolin decreases aortic pressure but has no effect on pulmonary vascular resistance, whereas apiin and luteolin exhibit antihypertensive characteristics.

5.8 Resveratrol

Resveratrol has a polyphenolic composition and is a potent antioxidant. Due to its numerous biological effects, it has a variety of medicinal uses, including anti-inflammatory, antioxidant, and anti-aging properties [70]. Because it prevents ACT, which promotes apoptosis in human endometrial cancer cells [71], resveratrol possesses anticancer characteristics. Guan et al.'s [72] study on hypoxic pulmonary artery smooth muscle cells (PASMCs) shows that resveratrol administration may inhibit PASMC migration and proliferation, which in turn might reduce p-AKT and AKT expression in PAMSCs. Aside from that, MMP-2 and MMP-9 production is decreased by resveratrol. Due to MMPs' activation of the PI3K/AKT signalling pathway, PASMC migration and proliferation are enhanced [73]. Resveratrol may boost NO production, reduce oxidative stress, inflammation, and cell proliferation, according to some research. As a result, it might enhance PH and pulmonary artery endothelium performance. Resveratrol can also stop the contraction and inflammation that lead to PAH in the rat pulmonary artery dysfunctions and cardiac myocyte hypertrophy [74]. In a study on hypoxia-induced PH in vitro and in vivo, Xu et al. (2016) showed that resveratrol has the ability to improve PH and reduce inflammatory mediators in the lungs of rats. Additionally, resveratrol treatments cause a decrease in H2O2 in the lungs of rats. According to Prabhakar and Semenza [75], ROS and NO boost HIF-1a expression whereas resveratrol's antioxidant activity decreases HIF-1a expression. HIF-1a is a crucial gene in the evolution of PH. Resveratrol boosted the expression of the proteins Trx-1 and Nrf-2. Antioxidant proteins like Trx-1 are balanced by the antioxidant sensor Nrf2. Resveratrol boosts SOD and GSH production, Trx-1 and Nrf-2 protein levels, and PASMC growth. It also reduces ROS, H2O2 production, and HIF-1a expression. These indicators demonstrate the potency of resveratrol as an antioxidant. In contrast to resveratrol, a synthetic substance called trimethoxystilbene (TMS) is produced by methylating resveratrol and has more activities [31]. Increasing the involvement of apoptosis and reducing cell proliferation, TMS is more effective than resveratrol in treating pH, according to a 2012 research by Gao et al. on PASMCs. Resveratrol is therefore proposed to be a potent treatment for PH because of its antioxidant and anti-proliferative properties.

5.9 Carvacrol

The leaves of the herbs thyme (*Thymus* spp.) and oregano (*Origanum* spp.) are the source of the phenolic compound carvacrol. Pharmacologically, carvacrol has been shown to have antiviral, antibacterial, anticancer, and antioxidant effects. In a study of hypoxia-induced PH in rats, Zhang et al. (2016) found that carvacrol decreased RVH and pulmonary artery thickness. By boosting GSH and SOD activity and lowering MDA levels in pulmonary artery vascular smooth muscle cells, it can also mitigate oxidative stress. Carvacrol treatment decreases Bcl-2 (anti-apoptotic protein) protein expression and mRNA, enhances caspase-3 expression, and decreases Procaspase-3 expression. Apoptosis is inhibited and cell survival is promoted by ERK1/2 and Akt. The ERK1/2 and PI3K/Akt pathways are also down-regulated by carvacrol therapy. These results show that carvacrol can reduce oxidative stress and increase apoptosis in PASMCs, both of which are important in reducing PH.

5.10 Nobiletin

Flavonoid nobiletin originates in the rinds of citrus fruits. In light of its wide range of biological actions, including antioxidant, anti-inflammatory, and anti-tumour effects, this molecule has several potential uses [76]. According to Cheng et al. (2017) in a research of MCT-induced pulmonary hypertensive rats, treatment with nobiletin enhances hemodynamic markers such as RVSP and RVH, suggesting that it has a positive impact on hypoxia-induced respiratory vascular remodelling. Nobiletin prevents the phosphorylation of Src, STAT3, and PDGF-BB in vitro, which prevents the expression of Pim1 and NFATc2, two Src and STAT3 target genes. In the end, this chain of genes acts to halt further cell division. Therefore, nobiletin's ability to block the proliferation-promoting Src/STAT3 axis means it can slow the development of pH.

5.11 Panax Notoginseng Saponins

The biological and therapeutic effects of *Panax notoginseng* Saponins (PNS) are used in Therapeutic Potential and Cellular Mechanisms (TCM). They include antioxidant, vasorelaxant, and inhibitory impact on vascular smooth muscle receptor of the voltage-gated Ca2+ channels [35]. Injecting PNS into rats with hypoxia-induced pulmonary hypertension improved hemodynamic indicators like mean pulmonary arterial pressure (mPAP), mean capillary permeability (mCAP), and right ventricular to left systolic ratio (RV/(LV + S)), and thus decreased p38MAPK mRNA expression in lung tissue of these rats. The expression of P38MAPK, a key indicator of physiological stress, rises as PH worsens. Research suggests that PNS may be an option for treating PH.

5.12 Quercetin

In addition to its anti-inflammatory, antioxidant, anti-tumour, anti-proliferative, inducing apoptosis, anti-metastatic, vasodilator, and blood pressure-lowering properties, the flavonoid quercetin is found in a wide variety of plants and fruits. Quercetin may also slow the development of pH in rats, according to certain studies. Quercetin treatment improves hemodynamic markers such PAH and RVH, as shown in research of hypoxia-induced PH in rats by He et al. (2015). By blocking the TrkA/AKT signalling axis, quercetin suppresses PASMC growth and triggers apoptosis in the cells. Increased expression of BAX and decreased expression of Bcl-2 were both found to occur after quercetin administration. Protein expression of cyclin D1 is upregulated by quercetin, while cyclin B1 and Cdc2 are downregulated. Cell migration regulators including MMP2, MMP9, CXCR4, integrin b1, and integrin a5 are under their control because of their ability to suppress their expression. All medicinal effects of herbs and phytochemicals are triggered by one of six mechanisms:

(a) **Antiproliferation:** Antiproliferative effects can be achieved through the use of plant extracts and phytochemicals such as *W. somnifera*, *R. tangutica*, genistein, berberine, magnesium lithospemate B, isohynchophylline, nobiletin, and quercetin all of which work by inhibiting the expression of genes or suppressing pathways involved in cell proliferation. Reducing the expression of CDK 4 and 6, cyclin D1, PDGF-BB, PP2AC, and PCNA while simultaneously increasing the expression of the p27kip1 (cyclin inhibitor) marker, which suppresses the ACT/GSK3b and Trk A/ACT pathways, leads to decreased cell proliferation [65].

(b) **Antioxidant:** The antioxidant effects of *B. vulgaris*, *C. rhipidophylla*, *K. odoratissima*, *M. oleifera*, *T. pratense*, carvacrol, *T. arjuna*, oxymatrine, magnesium lithospemate B, and resveratrol in the treatment of PH occur via increases in GPx, SOD, CAT, and GSH and decreases in ROS, H2O2, and MDA. These metrics point to less oxidative stress and less tissue damage overall [72].

(c) **Antivascular remodelling:** It has previously been explained that vascular remodelling is a crucial event in the development of PH. Several plant extracts and phytochemicals have their effects by inhibiting the process of vascular remodelling by first reducing ET-1 expression and then MMP-9 and MMP-2 production. As an example, *K. odoratissima*, *C. rhipidophylla*, *T. pratense*, *S. miltiorrhiza*, baicalin, asiaticoside, punicalagin, ginsenoside Rb1, and resveratrol are all examples of plant extracts and phytochemicals that have a vascular anti-remodelling effect and thus inhibit the progression of PH [60, 72].

(d) **Vasodilator:** Increased blood vessel constriction and subsequent PH is a major contributor to PH. The major vascular relaxation inducers reduce TXA2 and cytosolic Ca2+ levels while boosting NO levels by increasing eNOS expression. *A. sativum, A macrostemon, K. odoratissima, E. macrobulbum, S. miltiorrhiza, M. pigra, T. pratense, S. securidaca, W. somnifera*, Apple polyphenols, polydatin, and resveratrol are all examples of plant extracts and phytochemicals with these characteristics [54].

(e) **Apoptosis inducer:** Wsomnifera, T. arjuna, baicalin, apple polyphenols, isohynchophylline, carvacrol, quercetin, punicalagin, salidroside, and resveratrol are only a few phytoextracts and phytochemicals that have been demonstrated to enhance apoptosis and hence improve PH. They carry out their function by suppressing the activity of ERK1/2/STAT3 and PI3K/ACT, limiting the action of growth factors like HIF-1a, TGF-b1, and AhR, or increasing the expression of apoptosis-inducing genes like bax and caspase-3 while concurrently decreasing the expression of BCL2 (an anti-apoptosis factor). By causing lung tissue cells, PAECs, and PASMCs to undergo apoptosis, these chemicals stop the progression of PH [58].

(f) **Anti-inflammation:** Increased inflammation is a significant mechanism in PH. It has been demonstrated that a variety of phytochemicals and plant extracts, such as Astragalus polysaccharides, *Withania somnifera*, punicalagin, oxymatrine, and resveratrol, can reduce levels of inflammatory mediators such IL-6, IL-10, TNF-a, IL-1b, NFkB, VEGF, and pho-IkB, which in turn can reduce chronic inflammation. As a result, they can halt the inflammatory process, which in turn slows the development of PH [70].

6 Pulmonary Disease Treatment Using Indian Herbal Medicine

6.1 Asthma

The candlenut tree (Aleurites moluccana) may be found all over the United States and has long been used as a folk remedy for a variety of ailments, including bronchial asthma, joint discomfort, fever, and migraines. This is because it has several beneficial qualities, including anti-viral, anti-nociceptive, anti-hypersensitivity, and anti-microbial [77]. Studies on a semisolid herbal medication made from *A. moluccana* leaf extract show promise as a phytomedicine for the treatment of asthma [78]. This extract showed anti-inflammatory, analgesic, and wound-healing properties in pre-clinical experiments. The therapeutic benefits of *Nigella sativa* (Black cumin) extracts are extensive and include but are not limited to the following: anti-inflammatory, anti-oxidant, antihistaminic, anti-allergic, immunomodulatory, antitussive, and bronchodilatory effects. Patients' asthma symptoms were reduced when they were given *N. sativa* seed, boiling extract, or oil, according to clinical trials [79]. Mountain knotgrass, or *Aerva lanata*, is a widespread weed that thrives in the country's warmer plains. The ethanol extract of this plant has anti-asthmatic properties due to its ability to cause catalepsy and degranulation of mast cells [80]. One additional medicinal herb that can stabilize mast cells is *Bacopa monnieri* (Brahmi). Extracts from medicinal plants like *Bauhinia variegate* (Rakta Kanchnar), *Cassia sophera* (Kasaunda), *Casuarina equisetifolia* (Whistling Pine), and *Clerodendrum serratum* (Bharangi) have the potential to treat asthma by inhibiting immune responses like bronchial and tracheal constriction brought on by histamine and clonidine.

6.2 Chronic Obstructive Pulmonary Disease

In addition to its usage as a Siddha medicine, the leaves and fruits of the *Solanum nigrum* (Manathakkali) plant are commonly utilized as a culinary item throughout most of India. Inflammation-induced oedema can be reduced by using extracts from its leaves. By blocking the ability of AP-1 and NF-kB to bind to DNA, a glycoprotein isolated from *S. nigrum* was discovered to reduce the production of inflammatory chemicals. Reduced levels of superoxide and nitric oxide, both of which have a great value in the disease physiology of chronic obstructive pulmonary disease (COPD), were observed in those who took an ethanolic extract of *Boerhavia diffusa* (Punarnava). In addition to being able to relax muscles and prevent infection, it also protects cells from damage [81]. Vasadi syrup and Shwasaghna dhuma, two Indian herbal medicines, have been shown to significantly improve COPD symptoms and forced expiratory volume in 1 second (FEV1%) in clinical trials involving patients with COPD. Equal parts of the following ingredients make up the Vasadi syrup: *Curcuma longa* (Haridra), *Justicia adhatoda* (Vasa), *Clerodendrum serratum* (Bharangi), *Coriandrum sativum* (Dhanyaka), *Zingiber officinale* (Shunthi), *Tinospora* *cordifolia* (Guduchi), *Piper longum* (Pippali), and *Solanum virginianum* (Kantakari). Contents of Shwasaghna dhuma are dry leaves of *Datura stramonium* (Dhatura), powder of seeds of *Solanum virginianum* (Kantakari), seeds of *Hyoscyamus niger* (Khurasani ajwain), *Trachyspermum ammi* (Ajwain), *Curcuma longa* (Haridra), Potassium Nitrate (Kalmi shora), and *Cannabis sativa* (Bhanga) in equal parts [82].

6.3 Bronchial Asthma

Ocimum sanctum (Tulsi), an ancient medicinal herb, is effective against both acute bronchitis and chronic bronchial asthma. Tulsi's main ingredient, Eugenol, has a wide range of therapeutic effects, including anti-microbial, analgesic, antispasmodic, and adaptogenic [53]. In double-blind, randomised, placebo-controlled clinical studies, the combination of thyme fluid extract and primrose root tincture decreased bronchitis symptoms and the length of the disease [47]. Some research [48] also shows that ivy (*Hedera helix*) leaf extract helps reduce chronic bronchitis symptoms. In contrast to thymoquinone, a different phytochemical, nigellone, derived from *Nigella sativa* (Black cumin) was reported to display antispasmodic and facilitative effects on respiratory clearance [49].

6.4 Lung Cancer

In Indian cuisine, *Curcuma longa* (Turmeric) is a staple ingredient. In vitro studies using the A549 lung cancer cell line revealed that *C. longa* extract inhibited telomerase activity and exhibited lethal effects in a dose-dependent manner. The phytochemical conferone, which was isolated from the Ferula species, was found to have weak cytotoxic effects against the A549 lung cancer cell line [76]. The cytotoxic action of *Annona muricata* (Mamaphal) can be seen in the A549 lung carcinoma cell line, and this can be traced back to the plant's constituent chemicals, annomuricin A and B [83]. Using a mouse model, researchers showed that alcoholic extracts of *Andrographis paniculata* (Kiryat) had chemopreventive benefits by increasing lung levels of superoxide dismutase, catalase, and DT-diaphorase [30]. An extract of *Phyllanthus urinaria* (Jaramla) demonstrated anti-angiogenic efficacy in a mouse model in which Lewis lung carcinoma cells were implanted by inhibiting neovascularization in the tumour cells as well as the migration of HUVEC [51]. Other medicinal plants that are used to treat lung cancer symptoms include *Glycyrrhiza glabra* (Liquorice), *Zingiber officinale* (Ginger), *Ocimum sanctum* (Tulsi), *Adhatoda vasica* (Malabar nut), and *Terminalia chebula* (Myrobalan) [52].

6.5 Pneumonia

The antibacterial properties of the *Verbascum* species, better known as Mullein, have led to its usage in the treatment of pneumonia as part of traditional herbal medicine.

Verbascum fruticulosum extract showed strong antibacterial activity against the multidrug-resistant *Streptococcus pneumoniae* strain, suggesting its potential for application in the treatment of pneumonia. Similar to the results seen with *V. fruticulosum* [84], the extract of *Urtica urens* (Dwarf nettle) also showed anti-microbial action against *S. pneumoniae* in the same investigation. Beetroots, or *Beta vulgaris*, are a popular vegetable. Antibacterial activity, as measured by a concentration-dependent expansion of the zone of inhibition against Klebsiella pneumonia, was present in alcoholic leaf extracts of *B. vulgaris* fractionated by n-hexane and chloroform [85]. *Nepeta glutinosa* (Benth), *Ficus racemosa* (Gular fig), *Terminalia chebula* (Myrobalan), *Ricinus communis* (Castor bean), and *Vitex negundo* (Chaste tree) are some other medicinal plants that have been used to cure pneumonia [86].

6.6 COVID-19

The medicinal plants' essence, such as *Astragalus membranaceous* (Katira), *Cullen corylifolium* (Scurfy Pea), *Agastache rugosa* (Indian mint), *Cassia alata* (Candlebush), *Mollugo cervine* (Carpetweed), *Gymnema Sylvestre* (Gurmar), *Tinospora cordifolia* (Gurjo), and *Quercus infectoria* (Manjakani), is capable of inhibiting the action of coronavirus (Benarba and Pandiella (2020). Due to their ability to inhibit the SARS-CoV-2 mechanism of action by obstructing the activity of spike protein, the primary protease, ACE-2, and their receptor, phytochemicals like curcumin, quercetin, luteolin, withaferin A, apigenin, amaranthin, and gallic acid have the potential to be used as drug candidates [34]. An Ayurveda treatment plan was successful in curing COVID-19 symptoms in a 43-year-old guy, according to a case study. Sudarsana Churna was used to reduce fever, Talisadi Churna was used to restore a sense of taste, and Dhanwantara Gutika was used to help with breathing difficulties [87].

7 Pharmaceutical Industry and Herbal Plants

The plant world naturally contains phytochemicals, which have demonstrated their therapeutic value as a source of a broad range of biological activity [88]. Various active components found in plants allow us to speculate about their possible therapeutic value. Herbivores, microbes, and plants use the structures of secondary metabolites as protection systems within the cell by interfering with molecular targets [89]. For the creation of plant-based medicines and pharmaceuticals, it is extremely beneficial to identify as many chemical units and groups as possible, such as phenol and organic acids, carbohydrates, and amino acids [90]. Terrestrial or marine botanicals can yield biologically active substances, and the value of these substances relies on their characteristics [91]. Plants' nutritional and therapeutic properties lead to effective disease management for a variety of conditions. Plants demonstrate the importance of their flexible ecomorphology for environments. Animal, human, insect, and overall ecosystem well-being are all directly influenced by plants. Farming and herbal gardening are increasingly being studied in depth.

Climate, soil, and geographic variables all have an indirect or direct impact on plant development [92]. Unfortunately, many medicinal plants are in danger of going extinct due to overharvesting, habitat destruction, a lack of sustainable agriculture, and conservation practices. However, by controlling their accumulated secondary compounds, plants can get around many obstacles (physical, biological, and chemical) [93, 94]. Only 6% of the estimated 215,000–500,000 higher plant species in the globe have had their biological activity screened [94]. Economic commerce in more than 2000 different medicinal and aromatic plant species threatens 10% of those species in at least one European country [95]. Over 82% of drug substances are directly derived from natural products, and over 50% of drug substances come from chemicals separated from plants or herbs [96]. According to Markets 2019, the plant extracts industry is expected to grow from its current estimated value of USD 23.7 billion to USD 59.4 billion by 2025. The use of medications to relieve some of the most prevalent respiratory disease symptoms, such as a sore tongue, cough, nasal congestion, etc., is supported by extensive data [14]. In the accessible literature, reports of individual or combined herbal sources have been made. Homeopathy is widely used, and using it has been linked with positive outcomes.

8 Conclusion

In conclusion, medicinal plants have been used for centuries to cure respiratory diseases and provide relief from respiratory symptoms. They contain bioactive compounds that have antiviral, antibacterial, anti-inflammatory, and immune-modulatory properties, which make them effective in treating respiratory infections and diseases. Studies have shown that various medicinal plants, such as ginger, turmeric, eucalyptus, garlic, liquorice, and thyme, can be used to treat respiratory infections and diseases like asthma, bronchitis, and pneumonia. However, it is essential to note that the use of medicinal plants as a treatment for respiratory diseases should not replace conventional medicine but should be used as a complementary therapy. In addition, more research is needed to determine the safety and efficacy of these medicinal plants in the treatment of respiratory diseases. Nonetheless, their use can offer a natural and cost-effective way to manage respiratory diseases and improve overall respiratory health. Therefore, the integration of medicinal plants into respiratory disease management programs can be a promising approach to improving respiratory health outcomes. Medicinal plants have been used for centuries as a natural remedy for respiratory diseases. They offer a variety of therapeutic benefits, including anti-inflammatory, antioxidant, and expectorant properties, which make them an effective treatment option. Many of these plants have been studied extensively, and their efficacy in treating respiratory conditions has been proven by scientific research. Some of the commonly used medicinal plants for respiratory diseases include liquorice root, ginger, eucalyptus, thyme, and peppermint. These plants can be used in different forms such as teas, tinctures, and extracts and they can be easily incorporated into daily routines. While medicinal plants offer a natural and safe alternative to conventional medicine, it is essential to seek medical advice before using them, especially if you have pre-existing medical conditions or are taking medication. Additionally, it is crucial to purchase these plants from a reputable source to ensure their quality and purity. Overall, medicinal plants remain an important component of traditional medicine, and their role in the treatment of respiratory diseases cannot be ignored. With further research and exploration, these plants may provide new avenues for the development of effective and sustainable treatments for respiratory diseases.

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17 Assessment of the Phytochemical Constituents and Metabolites in Medicinal Plants and Herbal Remedies Used in the Treatment and Management of Reproductive Diseases: Polycystic Ovary Syndrome

Prasann Kumar, Subham Saurabh, and Khushbu Sharma

Contents

Abstract

The present study explores the phytochemical constituents and metabolites found in medicinal plants and herbal remedies traditionally used for treating and managing polycystic ovary syndrome (PCOS). This study's primary aims involve identifying, isolating, and characterizing bioactive compounds derived from natural sources. Subsequently, an assessment of their bioactivity will be conducted. Comparative analyses are undertaken to identify the most favorable approaches for managing polycystic ovary syndrome (PCOS), encompassing a thorough evaluation of safety and toxicity considerations. The study aims to establish standardized formulations and conduct clinical trials to assess these natural remedies' practical effectiveness. The primary objective is to develop evidence-based recommendations for

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responsibly utilizing medicinal plants and herbal remedies within the polycystic ovary syndrome (PCOS) treatment framework.

Keywords

Antioxidants · Herbal · Infertility · PCOS · Lumen · Reproductive system · No poverty · Zero hunger

Abbreviations

ANT	Anti-Mullerian hormone
DHEA	Dehydroepiandrosterone
ERs	Estrogen receptors
EV	Estradiol valerate
FSH	Follicle-stimulating hormone
IL	Interleukin
IVF	In vitro fertilization
LDLC	Low-density lipoprotein cholesterol
LH	Luteinizing hormone
LPA	lysophosphatidic acid
MDA	Malondialdehyde
NO	Nitric oxide
OC	Ovarian cancer
PCOS	Polycystic ovarian syndrome
TAC	Total antioxidant capacity
TC	Total cholesterol
TG	Triglyceride

1 Introduction

Traditional Asian medicine and plants have been used to treat various illnesses since ancient times. South Korea, Japan, and the tribal and agricultural populations of Bangladesh, China, India, Nepal, and Vietnam [22] for their profitable and user-friendly nature. Due to socioeconomic and geographic circumstances, ethnic people in many Asian nations rely heavily on medicinal plants for their primary healthcare in rural areas. Seventy to eighty percent of the population in developing nations depend on medicinal plants for primary healthcare, and ethnomedicine is becoming more popular in developed nations because it has almost no side effects. Ethnic people rely on the medicinal plants nearby to learn about the economic values and medicinal qualities of many plants and traditional medicine based on their needs, observations, and prior experiences [30]. Research on ethnomedicinal herbal plants led to the discovery of 75% of herbal drugs, and about 25% of modern medicines are derived from sources of widely used medicinal plants. Worldwide, the World Health Organization (WHO) has identified over 21,000 plant types for medicinal purposes.

Ayurvedic medicine, also known as Ayurveda, is a form of complementary and alternative treatment originating in India [6]. Ancient medicine used to treat issues with male reproduction: The herbal remedy made from *Centella asiatica, Hemidesmus indicus,* and *Hibiscus rosa* is used to cure spermaturia*. Helianthus fraternus, Dracaena terniflora*, *Sinensis*, and *Cuminum cyminum* are employed [14]. *Ocimum sanctum* and *Evolvulus alsinoides* are used to boost sperm concentration. To cure impotence, whole *Withania somnifera* is used (Table 1). Traditional medicine used to treat issues with feminine reproduction: *Celastrus paniculatus* and *Hibiscus rosa-sinensis* were used to cure leukorrhea. *Clerodendrum viscosum* and *Premna latifolia* have both been used to treat menorrhagia. Several plants were found to be effective in treating abortion, including *Ensete superbum*, *Mirabilis jalapa*, *Securinega leucopyrus*, *Celastrus paniculatus*, *Gardenia gummifera*, *Ziziphus oenoplia*, *Erythrina indica*, *Ixora coccinea*, and *Ziziphus rugosa*. To address dysmenorrhea*, Ocimum basilicum* and *Tabernaemontana divaricata* were used [16, 20, 21]. It was asserted that *Wrightia tinctoria* and *Diospyros montana* were usually helpful for treating different menstrual disorders and irregularities

Table 1 Plant species involved in disease prevention

Disease	*Plant species*	Part used	Use	Admin
1.	*Manihot esculenta*	Tuber, fresh	Vaginal infection, Vaginal discharge	Oral
2.	*Caesalpinia spinosa* (Molina) Kuntze	Seeds pods	Inflammation of ovaries	Oral
3.	*Ingo edulis* C. Martius	Huaba, Pacae, Guava, Pacai Flowers, fresh	Hair growth	Oral
4.	*Pelargonium roseum* Willd	Geranio	Flowers Uterus pain	Topical
5.	*Illicium verum* Hook.	Seeds, dried	Expel residues of feces in the stomach of newborn babies	Oral
6.	*Origanum vulgare* L.	Leave	Menstrual cramps	Oral
7.	*Salvia officinalis*	Whole plant	Control and regulate the menstrual cycle	Topical
8.	*Malva sylvestris* L.	Leave, stem	Pain	Topical
9.	*Mirabilis jalapa* L.	Root, fresh	Prostate, Pre-prostate cancer	Oral
10.	*Cynodon dactylon* L.	Stems, dried	Cysts of the ovary, Cysts of the uterus	Oral
11.	*Laccopetalum giganteum*	Leaves, fresh	Fertilization (Heat Ovaries)	Oral
12.	*Cinchona officinalis* L	Bark, dried	Fertility, Sexual potency	Oral
13.	*Cestrum strigilatum* R	Flowers, stem	Control and regulate the menstrual cycle	Oral
14.	*Typha angustifolia* L	Stems, dried	Prostate	Topical
15.	*Pilea microphylla* (L.)	Whole plant,	Prostate, Cysts	Oral
16.	*Lantana scabiosiflora*	Leaves, stem	Cold of the ovaries, Menstruation	Oral

(References: [14], [16], [20], [21])

(Table 1). The botanical names of the plants used to treat male and female reproductive disorders are listed alphabetically by disease. Additionally given are the components used in the drug preparation process, the dosage, and the recommended treatment time. Where accessible (Table 1), information about the other ingredients, if any, is also provided. The use of ethnomedicine as complementary and alternative medicine is growing in popularity despite the absence of scientific validation of the efficacy and safety of medicinal plants. It is also because it is cost-effective and has few side effects [34]. Pharmaceutical firms and scholars have recently concentrated on medicinal plants as a lucrative field. Polycystic ovary syndrome (PCOS), a prevalent and multifaceted endocrinological and metabolic disorder in women, is associated with infertility or subfertility. Initially referred to as the Stein-Leventhal syndrome, PCOS manifests as a severe condition characterized by ovarian enlargement and numerous small, undeveloped follicles [37]. It is a common female disorder with a prevalence rate of 2.2–26% globally and 11.96% among Indian teenagers [13]. Clinical manifestations of PCOS include menstrual irregularities or absence, abdominal obesity, acanthosis nigricans, and signs of androgen excess, such as acne or seborrhea, and insulin intolerance. Over the long term, PCOS increases the risk of endometrial cancer, type 2 diabetes, dyslipidemia, hypertension, and cardiovascular diseases. Notably, endometrial malignancy, cardiovascular disease, dyslipidemia, and type 2 diabetes mellitus are more prevalent in women with PCOS. The pathophysiology of PCOS primarily stems from abnormalities in the hypothalamic-pituitary axis, insulin secretion and action, and ovarian function. Obesity and insulin intolerance are closely linked to PCOS. Elevated insulin levels prompt the ovaries to produce testosterone, leading to anovulation, a significant cause of infertility. An altered luteinizing hormone/follicle-stimulating hormone (LH/FSH) relationship in patients with PCOS, with higher LH levels relative to FSH secretion, leads to increased androgen production by theca cells, resulting in irregular hormonal regulation, higher estrogen levels, erratic menstruation, and infertility [7]. Individuals with PCOS also exhibit an abnormal lipoprotein profile, characterized by elevated plasma triglyceride (TG) levels, slightly increased low-density lipoprotein cholesterol (LDL-C), and reduced high-density lipoprotein cholesterol (HDLC) levels. This lipid profile is often associated with heightened hepatic lipase activity, increased triglyceride content in circulation, insulin resistance, and elevated concentrations of smaller LDL particles. Current treatments for PCOS predominantly involve lifestyle modifications and pharmaceutical interventions. Lifestyle changes include dietary adjustments and supervised exercise to facilitate weight reduction. Pharmaceutical options like oral clomiphene citrate and metformin are employed for PCOS management; however, they may be accompanied by side effects such as nausea, vomiting, and gastrointestinal disturbances [5]. To mitigate these side effects, herbal medications are combined with pharmaceutical treatments to address PCOS symptoms effectively. This study aims to conduct a comprehensive analysis of the phytochemical constituents and bioactivity of medicinal plants and herbal remedies traditionally employed for managing reproductive diseases. The objectives encompass the identification and characterization of these compounds, the assessment of their bioactivity, the execution of comparative analyses to ascertain the most efficacious alternatives, the evaluation of safety and toxicity, the development of standardized formulations, the initiation of clinical trials, and the establishment of usage guidelines. The primary objective of this study is to provide evidence-based healthcare interventions that can improve the overall well-being of individuals impacted by reproductive health conditions.

2 Exploring Medicinal Plants and Herbal Remedies for Managing Reproductive Diseases with a Focus on Polycystic Ovary Syndrome (PCOS)

Polycystic ovarian syndrome (PCOS) is a complex metabolic disorder that significantly impacts the lives of numerous women during their reproductive years. This intricate condition poses challenges and potential implications for long-term health [44, 45]. Polycystic ovary syndrome (PCOS) can present a significant obstacle to a woman's desire to become a mother, as infertility frequently emerges as a prominent clinical manifestation of the condition. Nevertheless, the extensive ramifications of polycystic ovary syndrome (PCOS) transcend reproductive health, significantly impacting the psychosocial welfare of individuals affected by this condition. Polycystic ovary syndrome (PCOS) has attracted considerable attention due to its complex interconnections with other medical conditions. Polycystic ovary syndrome (PCOS) is often linked to diabetes, a metabolic disorder characterized by irregular blood glucose levels. Women diagnosed with PCOS face an increased risk of developing type 2 diabetes, which adds an extra health concern [37, 38].

Furthermore, the prevalence of obesity poses a significant concern in the lives of numerous women affected by polycystic ovary syndrome (PCOS). The presence of hormonal imbalances in this condition frequently leads to increased body weight and challenges in effectively managing weight [44, 45]. Consequently, this phenomenon exacerbates the susceptibility to diabetes and other metabolic complications. Polycystic ovary syndrome (PCOS) has been associated with hypertension, also known as high blood pressure. The co-occurrence of this condition seems to be prevalent, posing difficulties in the holistic management of individuals affected by this syndrome. Moreover, it is worth noting that polycystic ovary syndrome (PCOS) frequently leads to the grave outcome of heart disease, which is recognized as the primary cause of mortality in women.

Polycystic ovary syndrome (PCOS) is characterized by dysregulation of hormones, insulin resistance, and excessive body weight, increasing the likelihood of cardiovascular complications [26]. Consequently, women with PCOS should exercise prudence and vigilance in monitoring their cardiovascular health, emphasizing proactive interventions and adjustments to their daily habits. The range of lipid abnormalities presents an increasingly diverse range of health hazards linked to PCOS. Dyslipidemia, a medical condition characterized by aberrant levels of lipids in the bloodstream, is a common comorbidity in individuals with PCOS [28–30]. Elevated levels of low-density lipoprotein (LDL) cholesterol and triglycerides, in conjunction with diminished levels of high-density lipoprotein (HDL) cholesterol, have been implicated in the pathogenesis of atherosclerosis and the progression of cardiovascular disease. The increased prevalence of lipid abnormalities in individuals with PCOS adds a layer of complexity to the already intricate health profile observed in women affected by this condition. Autoimmune thyroiditis, characterized by chronic thyroid gland inflammation, represents an additional element within the intricate framework of PCOS [31, 32–34]. PCOS demonstrates an increased susceptibility to thyroid disorders, specifically autoimmune thyroiditis, a significant concern. This phenomenon adds to the array of difficulties that need to be tackled in the holistic administration of the syndrome. Maintaining optimal thyroid function is a crucial priority in the pursuit of overall health and well-being. The correlation between PCOS and malignancies exacerbates the overall impact of the condition. While PCOS has no direct causal relationship with cancer, existing evidence indicates a correlation between PCOS and an elevated long-term susceptibility to certain malignancies. Breast cancer, the most common malignancy among women, and endometrial cancer in the uterine lining are two noteworthy and worrisome health hazards [35–38]. The relationship between hormonal imbalances and PCOS is characterized by increased estrogen levels, which appears to contribute to an increased risk of developing cancer. Cancer is a poignant reminder of the importance of timely detection, cancer screenings, and preventive measures for women managing PCOS. The precise etiology of PCOS remains uncertain, highlighting the intricate interaction between genetic and environmental elements. The phenomenon is hypothesized to arise from a complex combination of innate genetic characteristics and external environmental influences. A genetic component in PCOS suggests a familial predisposition, as indicated by the high prevalence of the syndrome among close relatives. However, the activation of latent potential in PCOS is significantly influenced by environmental factors, ultimately resulting in its manifestation. The complex interplay between genetic and environmental factors gives rise to a syndrome that is a multifaceted puzzle that is still not fully comprehended. Extensive research has been undertaken to tackle PCOS's intricate challenges and comprehend, manage, and mitigate its effects. Various management strategies have been developed to improve the quality of life for individuals who self-identify as women and experience the effects of PCOS [39–41]. These strategies encompass various approaches, including adjustments to an individual's lifestyle, pharmaceutical interventions, and alternative therapies. Using medicinal plants has garnered considerable attention as a potential therapeutic approach, offering a holistic and nature-based strategy for managing PCOS. Throughout history, medicinal herbs have played a crucial role in traditional medicine systems, highly esteemed for their therapeutic properties and ability to effectively treat diverse health conditions. Botanical remedies have been employed within PCOS to alleviate the specific symptoms associated with this condition. PCOS has two significant symptoms in women: oligo/amenorrhea and hyperandrogenism. These symptoms have considerable implications for reproductive health and affect individuals' overall physical and mental well-being. Exploring medicinal plants as a potentially effective alternative therapy for PCOS represents a promising avenue for scholarly inquiry. Plants often demonstrate the presence of bioactive compounds that possess therapeutic properties, capable of addressing the hormonal imbalances and metabolic disruptions inherent to PCOS. By incorporating these natural remedies, which have strong roots in historical healing practices, individuals can adopt an integrative methodology to manage the syndrome proficiently. While it is not within the scope of this paragraph to conduct an extensive analysis of individual medicinal herbs and their corresponding properties, it is important to highlight the prevailing inclination toward natural and complementary treatments in the management of PCOS. By embracing these traditional botanical treatments, women diagnosed with PCOS can access various resources to rebalance hormone levels, regulate menstrual cycles, and alleviate the distressing symptoms commonly linked to this condition [42, 43]. To summarize, PCOS is a multifaceted metabolic disorder that substantially impacts women's health in the reproductive-age bracket. The wide-ranging health challenges associated with this phenomenon include diabetes, obesity, hypertension, heart disease, lipid abnormalities, and autoimmune thyroiditis. The ripple effect of this phenomenon extends to these various health conditions. The existence of malignant neoplasms, such as breast and endometrial carcinomas, underscores the severity of this syndrome [44, 45].

3 Unraveling the Enigmatic Nature of PCOS: Traditional Herbal Remedies as a Ray of Hope for Management and Well-Being

The enigmatic etiology of polycystic ovary syndrome (PCOS), intricately influenced by genetic and environmental factors, is a poignant testament to the unresolved enigmas within medical science [14]. Nevertheless, amidst these challenges, there is a glimmer of hope with the emergence of research on managing polycystic ovary syndrome (PCOS) and using medicinal herbs [16]. These traditional remedies, rooted in longstanding customs, offer an alternative approach to managing the complex array of symptoms associated with polycystic ovary syndrome (PCOS), thereby augmenting the overall well-being of individuals affected by this condition [20, 21]. As we navigate the dynamic terrain of PCOS research and treatment, using medicinal plants is a testament to the enduring nature and flexibility of healthcare methodologies across diverse societies and historical periods (Table 1).

4 Compounds from Plants for PCOS

Ecklonia cava: In addition, the administration of *Ecklonia cava* extract resulted in the normalization of various hormone levels, such as testosterone, estrogen, luteinizing hormone (LH), and follicle-stimulating hormone. Moreover, it demonstrated the ability to alleviate anti-Mullerian hormone (AMH) levels. The results of this study indicate that the extract in question has the potential to modulate multiple factors related to the development of ovarian follicles. The histological analysis provided additional evidence to support the assertion that rats administered with

Ecklonia cava extract demonstrated a notable decrease in symptoms associated with polycystic ovary syndrome (PCOS), resembling the characteristics observed in the ovaries of healthy individuals. This well-established natural remedy mitigates the effects of allergies and inflammation [21].

***Glycyrrhiza glabra*:** Glycyrrhiza glabra, commonly utilized in Korean herbal formulas, is recognized for its efficacy in addressing various health conditions such as inflammation, immune system modulation, hepatic failure, spasms, and metabolic disorders, specifically in women. A study conducted in 2018 by Lee et al. examined the impact of an ethanol extract of *Glycyrrhiza glabra* on female Sprague Dawley rat models that displayed symptoms similar to polycystic ovary syndrome (PCOS) induced by letrozole. The experimental group that received a combination of licorice and letrozole demonstrated a marked increase in follicle-stimulating hormone (FSH) recovery and a significantly reduced ratio of luteinizing hormone (LH) to FSH. According to Lee et al. [26], licorice extract demonstrated significant efficacy in mitigating polycystic ovary syndrome (PCOS) symptoms. These symptoms encompassed the thickening of the theca layer, the thinning of the granulosa layer of antral follicles, a decrease in the quantity of antral follicles, and the development of follicular cysts.

***Aegle marmelos*:** The present study investigated the potential antiandrogenic effects of a hydroalcoholic preparation obtained from *Aegle marmelos* in the management of polycystic ovary syndrome (PCOS) induced by letrozole in a female Wistar rat model. The present study investigated the impact of the hydroalcoholic extract derived from *Aegle marmelos* on ovarian histopathology and a range of biochemical and physical parameters. The research findings indicated significant elevations in blood luteinizing hormone (LH) levels and reductions in serum follicle-stimulating hormone (FSH) levels. An increased occurrence of corpus luteum, developing follicles, and oocytes enveloped by granulose cells [17].

***Bougainvillea spectabilis*:** In a 2018 study examining a rat model of polycystic ovary syndrome (PCOS) induced by estradiol valerate (EV), Ebrahim N.A. and colleagues conducted an assessment of the protective potential of *Bougainvillea spectabilis* leaves (BSL) extract on ovarian folliculogenesis. The research revealed that the group treated with estradiol valerate and *Bougainvillea spectabilis* extract exhibited significantly reduced blood levels of LH, estrogen, and glucose while also displaying elevated serum levels of FSH and antioxidants. Furthermore, this group's ovarian cortex featured corpora lutea, preantral, and antral follicles [18].

***Matricaria chamomilla*:** Various studies have explored the potential of natural remedies in alleviating symptoms and complications associated with polycystic ovary syndrome (PCOS), a complex metabolic disorder affecting women. For instance, research by Heidary et al. in 2018 examined the effects of chamomile on lipid and hormonal parameters in reproductive-age PCOS women. The study found oral chamomile capsule administration significantly decreased overall testosterone levels in PCOS women. However, it did not significantly affect dehydroepiandrosterone sulfate levels, LH/FSH ratio, or lipid factors [20].

***Cinnamomum zeylanicum*:** In 2018, Maryam Rafraf and her team conducted a study investigating how cinnamon supplements could affect women with PCOS.

They examined various factors, including blood total antioxidant capacity (TAC), malondialdehyde (MDA), and serum lipids. The women who took cinnamon supplements experienced significant improvements in their serum levels of total cholesterol (TC), low-density cholesterol, and high-density cholesterol. Additionally, the antioxidant capacity of their blood increased, indicating better protection against oxidative stress. They also noticed decreased malondialdehyde levels, a positive sign for their overall health.

***Galega officinalis*:** The potential protective effects of *Galega officinalis* in a rodent model of polycystic ovary syndrome (PCOS) induced by estradiol valerate. The study unveiled several encouraging results. The researchers noted noteworthy decreases in fasting blood sugar, insulin, testosterone, luteinizing hormone (LH), and follicle-stimulating hormone (FSH) concentrations, alongside substantial elevations in serum estrogen and aromatase levels. Additionally, the experimental group that received *Galega officinalis* demonstrated a significant augmentation in both pre-antral and antral follicles, accompanied by a notable reduction in cystic follicles. The results of this study indicate that *Galega officinalis* possesses potential advantages attributed to its antioxidant and insulin-related characteristics, as evidenced in a historical context tracing back to 1190 A.D.

***Moringa oleifera*:** This study investigates the impact of different doses of *Moringa oleifera* leaf extract on a polycystic ovary syndrome (PCOS) model that exhibits insulin resistance. The researchers made noteworthy observations during their investigation. Initially, a conspicuous decline in the accumulation of body weight was observed – a significant augmentation in folliculogenesis in the polycystic ovary syndrome (PCOS) ovary model. Additionally, androgen levels were declining in the cohort that underwent treatment with *Moringa oleifera*.

***Nigella sativa*:** The study encompassed a sample size of ten female participants clinically diagnosed with polycystic ovary syndrome (PCOS) and oligomenorrhea. Following the administration of *Nigella sativa*, noteworthy enhancements were observed in multiple health parameters, encompassing a considerable decrease in serum cholesterol levels, triglycerides, fasting blood sugar, insulin, aspartate aminotransferase (AST), and LH levels.

Vitis: This study investigates the impact of grape seed extract (GSE) on lipid and inflammatory markers in Wistar rats with polycystic ovary syndrome (PCOS) induced by estradiol valerate. The study primarily examined triglyceride (TG), total cholesterol (TC), high-density lipoproteins cholesterol (HDL-C), low-density lipoproteins cholesterol (LDL-C), and interleukin 6 (IL-6). Significantly, the cohort that was administered grape seed extract demonstrated noteworthy decreases in LDL-C, TC, and IL-6 concentrations. Administering specific doses of GSE had a beneficial effect on rats with polycystic ovary syndrome (PCOS). This effect was observed through the mitigation of dyslipidemia and reduction of inflammation, improving the overall systemic symptoms of the rats.

***Bambusa bambos*:** In recent studies, scholars have investigated a range of natural interventions to mitigate the difficulties associated with polycystic ovary syndrome (PCOS). This multifaceted metabolic disorder primarily affects women. The investigations above have produced encouraging results in mitigating particular symptoms and complications linked to polycystic ovary syndrome (PCOS). In a study conducted in 2018, the effects of processed *Nigella sativa* on menstrual irregularities associated with polycystic ovary syndrome (PCOS) were investigated. The findings of this study demonstrated noteworthy enhancements in cholesterol, triglycerides, blood sugar, insulin, AST, and LH levels after the treatment. The potential of grape seed extract (GSE) was examined in order to improve lipid profiles and mitigate inflammation in rats with polycystic ovary syndrome (PCOS). The findings of this investigation revealed favorable outcomes. Furthermore, the study conducted by examined the effects of bamboo seed oil on metabolic symptoms commonly associated with polycystic ovary syndrome (PCOS). This research revealed significantly reduced glucose, cholesterol, and triglyceride levels. The findings above highlight the potential of utilizing natural remedies to alleviate complications associated with polycystic ovary syndrome (PCOS) and improve the overall quality of life for individuals affected by this condition.

***Commiphora wightii*:** A study conducted in 2016 examined the effects of *Commiphora wightii* on hyperandrogenism in a polycystic ovary syndrome (PCOS) model. The findings of this study demonstrated a decrease in morphological abnormalities of ovarian follicles and a restoration of hormone levels to normal. Furthermore, a study conducted in 2015 examined the efficacy of hazelnut oil as a treatment for polycystic ovary syndrome (PCOS) in rats. The findings of this study indicated favorable results, including weight loss, restoration of hormonal equilibrium, and an augmentation in uterine weight. Moreover, a study conducted in 2016 revealed the potential of curcumin in regulating testosterone levels, mitigating insulin resistance, and preventing diabetic complications in rats with polycystic ovary syndrome (PCOS). These studies highlight the potential efficacy of natural remedies, including *Commiphora wightii*, hazelnut oil, and curcumin, in mitigating hormonal imbalances and metabolic disturbances commonly observed in individuals with polycystic ovary syndrome (PCOS) ([16, 24])

***Corylus avellana*:** The study's results indicated a significant reduction in body weight, restoring LH and FSH levels to normal, and increasing uterine weight. Furthermore, the researchers observed a substantial elevation in estrogen and progesterone levels and a decrease in testosterone. These findings highlighted the antioxidant properties of *Corylus avellana* oil and its efficacy in treating PCOS by regulating gonadotropins, steroids, and serum lipid parameters [16].

***Curcumin*:** Curcumin effectively restored serum testosterone levels to normal and successfully elevated progesterone levels, addressing hormonal imbalances associated with PCOS. Notably, curcumin can prevent insulin resistance and diabetic complications when administered orally, as evidenced by its capacity to halt the rise in HbA1c levels. Furthermore, the research revealed that curcumin was successful in eliminating cysts. These findings underscore the potential of curcumin as a promising medication for the treatment of both pathological and clinical abnormalities in PCOS.

Palm pollen: Palm pollen extract reveals reduced LH, estrogen levels, and cystic follicles, coupled with increased FSH, progesterone, and corpus luteum [23]. Similarly, research by Shetty et al. in 2015 found that *Vitex negundo* effectively lowered excessive androgen levels in rats with PCOS, accompanied by decreased glucose and testosterone levels. Additionally, in 2014, Sadrefozalayi et al. explored the renoprotective effects of the aqueous extract of *Foeniculum vulgare* in experimental PCOS rats, observing improvements in serum urea levels and kidney function. These findings underscore the potential of natural remedies, such as palm pollen, *Vitex negundo*, and *Foeniculum vulgare*, in alleviating symptoms and complications associated with PCOS.

Vitex negundo**:** The effects of *Vitex negundo* on female Sprague Dawley rats that had experienced hyperandrogenism due to PCOS. Through daily administration of the hydroalcoholic extract of *Vitex negundo*, they observed a significant reduction in elevated androgen levels. Additionally, the nirgundi extract led to a substantial decrease in serum glucose and testosterone levels. Notably, the group that received the nirgundi extract displayed the lowest incidence of follicular cysts and lesions.

Foeniculum vulgare**:** The research revealed that PCOS-afflicted rats exhibited lower serum urea levels. Notably, the administration of *Foeniculum vulgare* led to significant improvements in Bowman's space and acute tubular necrosis, restoring these parameters to a more normal state.

Allium cepa**:** The influence of an ethanolic extract derived from *Allium cepa* seeds on apoptosis modulation in rats with experimentally induced PCOS. The findings pointed to elevated levels of antioxidant capacity, a reduction in cyst formation, and a decrease in apoptotic granulosa cells, signifying the potential advantages of *Allium cepa* in managing PCOS. In addition, in 2010, Laxmipriya Nampoothiri and her colleagues investigated the efficacy of an Aloe vera gel formulation in a PCOS rat model induced by letrozole. The administration of Aloe vera gel resulted in enhanced ovarian health, the restoration of hormonal balance, and the maintenance of estrus cyclicity, underscoring its protective role against a PCOS phenotype [19].

Aloe barbadensis Mill**:** Aloe vera, scientifically classified as Liliaceae, was the focal point of research conducted by Laxmipriya Nampoothiri et al. in 2010. This study aimed to assess the effectiveness of a specific formulation of Aloe vera gel in a rat model of polycystic ovary syndrome (PCOS) induced by letrozole. Significant results were noted after applying Aloe vera gel, including a lack of weight increase and a decrease in ovarian atretic cysts. In addition, the oral administration of a 1 ml daily dosage of Aloe vera gel formulation for 45 days was crucial in restoring the subjects' estrus cyclicity, improving glucose sensitivity, and altering important steroidogenic activity. As a result, it served as a protective factor against the PCOS phenotype [33].

5 Exploring Natural Remedies for Enhancing Reproductive Health: A Focus on PCOS and Fertility

Polycystic ovary syndrome (PCOS) is a commonly encountered, heterogeneous, endocrinological, and metabolic disorder affecting women of reproductive age, often resulting in infertility or subfertility [14, 26]. Due to their diverse phytoconstituents, medicinal plants have been explored for their potential to manage various ailments, including PCOS [30]. Various herbal extracts following the onset of PCOS have been associated with reduced levels of blood testosterone and LH, alongside elevated blood progesterone and FSH levels [16]. Notably, these treatments have been observed to induce the development of various follicles at different growth phases, encompassing primary, antral, preantral, large oocytes, and corpora lutea in the treatment groups. Research suggests that many herbal extracts exhibit efficacy in treating PCOS, normalizing blood and sex hormone levels, resolving cysts, reducing insulin levels, and ameliorating other PCOS symptoms [21, 22]. Furthermore, some studies have emphasized the safety of these PCOS-targeted plants, attributing their favorable effects to the presence of phytoestrogens [23]. These compounds, characterized as weak estrogen antagonists, tend to exhibit more pronounced estrogenic effects when the body's estrogen levels are low in PCOS patients, rendering them safe for widespread utilization [24]. The medicinal plants discussed in this chapter can potentially enhance fertility through their interactions with the hypothalamic-pituitary-gonadal (HPG) axis and modulation of estrogen receptors (ERs). Additionally, these plants can provide an environment conducive to regulating ovulation, implantation, uterine embryo tolerance, and fetal development. They also show promise in averting reproductive transmitted bacterial, viral, and fungal infections, mitigating inflammatory reactions, hypersensitivity, and autoimmune disorders [2].

Estrogen receptors (ERs) mediate physiological responses to estrogen within specific tissues. While ERs are present in ovarian tissues, bone, and blood vessels, ERs are expressed in breast and uterine tissues. These ERs act as ligand-activated nuclear transcription factors, falling under the category of nuclear hormone receptors. Several plants contain phytoestrogens, which exhibit estrogenic effects in mammals. Notable examples of these substances with a strong affinity for estrogen receptors include formononetin, genistein, daidzein, and biochanin A. It is important to note that the concentration required for induction by these derivatives is significantly higher than anticipated despite their stronger binding to ER compared to ER [8]. These phytoestrogens contribute to the reduction in bone loss by lowering the levels of uric acid deoxypyridinoline. However, they do not impact osteogenic markers such as alkaline phosphatase and osteocalcin. Additionally, their vasoconstrictive properties make them beneficial in treating menopausal symptoms, such as reducing hot flashes, vaginismus, and dyspareunia [12]. Furthermore, bioactive compounds present in medicinal plants, including flavonoids and isoflavones, exhibit the potential to improve lipid profiles by reducing triglycerides and LDL cholesterol while increasing HDL cholesterol, which is crucial for reproductive health. Moreover, numerous studies have demonstrated that plants rich in polyphenols can impede the development of breast tumors by blocking pathways like IGF-1 (insulin-like growth factor 1)/P13K/Akt and ERK1/2 MAPK-Bax. Research into the effects of plant polyphenols, such as isoflavones, on the skin has shown that these compounds enhance the synthesis of hydroxyproline, a marker of collagen and elastic fiber synthesis, leading to vasodilation vascular endothelial growth factor (VEGF) in cutaneous blood vessels, regulation of sweat and sebaceous gland secretion, and control of collagen synthesis in the growth of hair layers, particularly in the context of menopause and infertility-related symptoms.

Punica granatum **(Pomegranate):** Pomegranates, identified by their scientific name *Punica granatum*, have garnered significant attention and are cultivated across various geographical areas, including Asia, the Middle East, and the Mediterranean [31]. Pomegranates possess abundant vital constituents, encompassing vitamin C, water, and flavonoids, namely anthocyanins, punicalagin, ellagic acid, and gallic acid. The pomegranate seeds contain phytoestrogens of significance, such as genistein, daidzein, coumestrol, and glutamic and aspartic acids [9]. Significantly, the utilization of pomegranate extract, containing these valuable phytoestrogens, has demonstrated the ability to regulate and alleviate symptoms commonly associated with polycystic ovary syndrome (PCOS) in animal studies involving rats affected by PCOS. The botanical extract in question plays a role in the augmentation of uterine wall thickness and the stimulation of mucus secretion by facilitating increased uterine blood flow through vasodilation. The increased production of mucosal secretions, facilitated by anti-inflammatory mechanisms, has been linked to a higher successful implantation rate. Mohammadzadeh et al. [32] conducted a rigorous clinical trial employing a triple-blind, randomized, controlled design, which involved a sample of 110 women who were in good health. The study yielded a noteworthy finding. The researchers discovered that the presence of calcium and tannins in pomegranate peel was associated with an enhancement in sexual satisfaction among women and a simultaneous reduction in the symptoms of inflammation and infection in their reproductive canal. Furthermore, a comprehensive examination of diverse human cell lines, such as ovarian (SKOV3), cervical (SiHa, HeLa), endometrial (HEC-1A), breast (MCF-7, MDA MB-231), and abnormal breast fibroblast (MCF-10A) cells, has unveiled the noteworthy characteristics of pomegranate extract as a selective estrogen receptor modulator (SERM). The observed effect of this activity can be attributed to its capacity to interact with estrogen receptors (ERs), suppressing cell line proliferation in laboratory and animal models, specifically in mice undergoing ovariectomy. Additionally, the study conducted a randomized controlled, triple-blind parallel trial to investigate the effects of pomegranate fruit extract on testosterone levels in the serum of 23 women diagnosed with polycystic ovary syndrome (PCOS). The findings revealed that the administration of pomegranate fruit extract reduced testosterone levels, improving the participants' lipid profile. Furthermore, a research study conducted on rats with polycystic ovary syndrome (PCOS) found that administering pomegranate fruit extract increased serum estrogen levels and a simultaneous improvement in related symptoms. The study reported significant findings after 81 days [22].

Matricaria chamomilla **(Chamomile):** *Matricaria chamomilla*, commonly known as chamomile, is a member of the chicory family and possesses a range of beneficial compounds, including flavonoids and antioxidants such as apigenin, choline, camelina, farnesene, matricin, and gallic acid. Numerous studies have demonstrated that chamomile extracts can substantially impact various physiological processes. The investigation of the growth and maturation of isolated mouse ovarian follicles within a three-dimensional culture system revealed the significant contribution of chamomile extracts. The extracts above exhibited the ability to enhance progesterone, 17-estradiol, and dehydroepiandrosterone levels in the culture medium while mitigating oxidative stress and reducing indicators of follicular diameter and antrum formation. Furthermore, the influence of chamomile resulted in an extended duration of oocyte survival. Applying chamomile extract has also exhibited the capacity to augment estrogen-dependent sexual characteristics in mice undergoing gonadectomy. The phenomenon above can be observed in facilitating modifications such as enhanced hair growth, body temperature fluctuations, and managing the menstrual cycle. Conducted a double-blinded clinical trial with a sample size of 80 postpartum pregnant women, all of whom had a gestational age of 40 weeks or more. The researchers observed that the administration of chamomile extract-containing capsules for 1 week resulted in a facilitation of labor onset. Furthermore, it is worth noting that this particular intervention resulted in a decreased duration of labor compared to the control group. The effects of chamomile plant extract on lactation have been elucidated through human research. According to it was observed that the administration of chamomile plant extract exhibited the potential to induce lactogenesis in women who were lactating. The observed mechanism can be attributed to the interaction between chamomile and dopamine receptors, which promotes galacta production. Furthermore, conducted a pilot randomized trial with a sample size of 56 women diagnosed with idiopathic hyperprolactinemia. The study findings suggest that chamomile might potentially play a role in regulating prolactin secretion in women. The findings of this study indicate that female participants who received a dosage of 5 ml of chamomile syrup twice daily experienced a reduction in prolactin levels after 4 weeks, as opposed to the group that received a placebo. In addition, the aroma of chamomile has been linked to increased intensity of uterine contractions during labor, as demonstrated in a controlled trial that included a sample of 130 female participants. Chamomile extract has the potential to provide supplementary advantages in comparison to chemical medications, such as mefenamic acid and nonsteroidal anti-inflammatory drugs (NSAIDs), in the prevention of postpartum pain among women. The inhibitory effect on COX-2 has been attributed to this phenomenon.

Vitex agnus-castus (Verbenaceae): Historically, *Vitex agnus-castus*, a plant belonging to the Verbenaceae family, has been employed for the management of acne and various menstrual disorders associated with corpus luteum insufficiency, including spasmodic dysmenorrhea, premenstrual symptoms, certain menopausal conditions, and insufficient lactation. The analysis of Verbenaceae plant extracts using liquid chromatography-electrospray ionization/mass spectrometry (LC-ESI/MS) confirmed the presence of various compounds. These compounds include orientin, catechin, rutin, rosmarinic acid glycoside, quercetagetin trimethyl ether, biochanin A, genistein, syringetin C glycoside, agnuside, kaempferol-7-*O*-glucuronide, luteolin-7-*O*-glucoside, homorientin, and isovitexin. Most of these compounds are isoflavones and flavonoids, which benefit women's reproductive health due to their strong binding affinity for estrogen ERs. The plant's flavones have been found to exert an inhibitory effect on the release of prolactin and FSH hormones by modulating the hypothalamic-pituitary-gonadal (HPG) axis. Additionally, its flavonoids have been shown to enhance the release of nitric oxide (NO) and cyclic guanosine monophosphate (cGMP) from vascular endothelial cells, thereby promoting increased blood flow to the endometrium ([4]. Approximately one in six couples is affected by the condition of infertility. When a couple cannot achieve pregnancy after 1 year, they are clinically diagnosed with infertility. The term "female infertility" describes situations in which the female partner is identified as the primary cause of infertility. Approximately 50% of infertility cases can be attributed to factors associated with female infertility, with approximately one-third of all cases solely attributed to female infertility. The term "unexplained sterility" is commonly employed in the context of reproductive processes. The term "unexplained infertility" describes cases where no identifiable cause for the inability to conceive has been determined. In general, the outcomes become apparent following the completion of all available tests by a couple. According to Beal [10], approximately 25% of couples experiencing infertility are diagnosed with unexplained infertility. The topic of discussion pertains to the issue of female infertility. The topic under discussion pertains to signs. Alterations in the menstrual cycle and ovulatory patterns among females can potentially indicate a medical condition associated with impaired fertility. One of the signs that can be observed is the presence of unusual menstrual cycles. Variations in menstrual patterns: bleeding that deviates from the typical flow in increased or decreased intensity. The duration between consecutive menstrual cycles fluctuates every month. The individual experiences recurring episodes of pain. The occurrence of cramps, pelvic discomfort, and back pain is possible. In certain instances, hormonal imbalances may contribute to infertility among women. In this particular scenario, it is plausible that supplementary indicators may be present. In addition to alterations in sexual drive and longing, according to Smith et al. [44], individuals may experience various skin alterations such as heightened acne, weight gain, hair loss or thinning, and the emergence of dark hair on the chin, torso, and lips. The diagnosis of female infertility refers to identifying and assessing factors contributing to a woman's inability to conceive or carry a pregnancy to term. The female individual and her significant other must undergo diagnostic examinations as a component of the inquiry into potential infertility. The woman has the potential to undergo the subsequent examinations: comprehensive physical examination, encompassing a thorough review of the patient's medical history. Laparoscopy, a minimally invasive surgical procedure involving inserting a specialized instrument through a small incision in the abdominal region, is employed to examine the reproductive organs. Additionally, blood tests are conducted to assess the levels of hormones associated with ovulation, while ultrasound scans are employed to detect the presence of fibroids. According to Lloyd and Hornsby [33], infertility can be attributed to various factors, including an underlying medical condition that adversely affects the fallopian tubes, inhibits ovulation, or disrupts hormonal balance. The medical conditions include endometriosis, polycystic ovarian syndrome, premature ovarian failure, pelvic inflammatory disease, uterine tumors, and polycystic ovary syndrome. Addressing any preexisting medical conditions that may contribute to reproductive challenges is imperative before initiating infertility treatment. Various treatment options can be considered if fertility restoration does not occur. The topic of interest pertains to the relationship between plants and the process of conception. Herbal medicines have historically been employed in the treatment of fertility issues. Historical records indicate that using herbal remedies to enhance male and female fertility dates back to approximately 200 A.D. The herbal fertility treatments encompass distinct botanical species and their corresponding extracts, which are believed to have advantageous effects on the reproductive system, hormonal equilibrium, and sexual drive. Both individuals seeking to enhance their chances of conception and couples experiencing fertility challenges utilize these medications. According to Agha-Hosseini et al. [1], herbal reproductive treatments have demonstrated potential in addressing hormonal imbalances, irregular menstrual cycles, erectile dysfunction, and impaired sperm motility. Throughout human history, individuals across various cultures and regions have explored and used herbal medicine for countless centuries. Infertility treatment is becoming more prevalent in Western societies among males and females. Nevertheless, the American Society for Reproductive Medicine asserts that there is limited scientific substantiation for the notion that herbal remedies can enhance fertility. Nevertheless, it is possible to provide credible evidence by utilizing evidence-based herbal medicine, specifically through clinical trials [11]. The emotional impact of infertility is well understood by women who undergo this condition. The conditions above encompass anxiety, isolation, prolonged depressive states, and difficulties concentrating on routine activities. Many infertility women may initiate their treatment by considering in vitro fertilization (IVF) techniques. Nevertheless, implementing these techniques incurs significant costs, rendering them financially inaccessible for certain women. Various natural and herbal remedies have been identified as potential interventions that may contribute to addressing the root causes of infertility and enhancing the likelihood of successful conception. According to Akhondzadeh et al. [3], incorporating a nutritious diet and engaging in consistent physical activity are lifestyle factors that can yield positive outcomes.

Ashwagandha: Ashwagandha, scientifically referred to as *Withania somnifera*, is an herbal remedy recognized as Indian ginseng. It has been observed to offer notable advantages for females encountering difficulties conceiving. This botanical remedy effectively promotes optimal functioning of the reproductive organs and helps maintain hormonal balance. Pomegranate has been found to possess properties that contribute to the toning of the endometrium in individuals who experience recurrent miscarriages. Pomegranate has been found to enhance female fecundity. This intervention facilitates the enhancement of uterine blood circulation and the thickening of the uterine lining, thereby reducing the likelihood of experiencing a miscarriage. Furthermore, it supports the optimal development of the fetus [15].

Cinnamon: Cinnamon has been found to promote healthy ovarian function, thereby potentially mitigating sterility. This therapeutic effect is associated with the potential to derive five distinct benefits from cinnamon consumption. Female infertility is a prominent factor contributing to infertility in individuals diagnosed with polycystic ovary syndrome (PCOS). According to a study published in the *Fertility and Sterility* journal in 2007, including cinnamon in the diet has been shown to potentially improve the regularity of menstrual cycles in individuals with polycystic ovary syndrome (PCOS). Additionally, it encompasses various factors that can hinder a woman's reproductive capacity, such as endometriosis, uterine fibroids, and amenorrhea, characterized by the absence of menstrual periods. Furthermore, it has been suggested that cinnamon possesses potential benefits in preventing yeast infections.

Chasteberry: Chasteberry, scientifically referred to as Vitex has been identified as an effective remedy for addressing infertility caused by hormonal imbalances within the body. The herb's ability to enhance ovulation is attributed to its elevated prolactin levels. Additionally, it combats polycystic ovary syndrome (PCOS).

Dates: The consumption of dates has been associated with potential enhancements in fertility due to the presence of various beneficial nutrients. According to these substances possess a rich concentration of essential nutrients such as vitamins A, E, D, and B, as well as iron and other minerals that are crucial for facilitating a woman's ability to conceive and successfully carry a pregnancy to full term. The maca root is a highly effective botanical remedy that has demonstrated efficacy in addressing both male and female infertility. This botanical specimen facilitates the synthesis of hormonally balanced substances and confers particular benefits to females afflicted with hypothyroidism due to its ability to enhance thyroid functionality. Moreover, it serves as a noteworthy reservoir of diverse nutrients that enhance fertility.

6 Ovarian Cancer and Its Prevention Through Herbal Medicines

Ovarian cancer (OC) is a prevalent form of cancer that affects a significant number of women annually, accounting for approximately 244,000 cases worldwide. This represents approximately 2% of all reported cancer cases globally. Nevertheless, obsessive-compulsive disorder (OC) exhibits notable variations across different geographical areas. In 2017, Fiji exhibited the highest prevalence of OC, with an estimated rate of 15 cases per 100,000 females. In contrast, China and specific regions of Africa reported a considerably lower incidence, with only four cases per 100,000 females. The incidence rate of ovarian cancer in women is approximately 1 in 70. Regrettably, when OC is concomitant with other gynecological malignancies, it emerges as the primary contributor to cancer-related mortality, leading to approximately 150,000 deaths globally annually. The elevated mortality rate can be ascribed to delayed detection, as symptoms frequently remain undetectable during the early stages and only become apparent during stages III and IV. Various therapeutic modalities have been developed to impede the dissemination of cancer cells, encompassing chemotherapy, laser therapy, radiation, gene therapy, hyperthermia, and surgical interventions. Medical interventions encompass a multifaceted approach, incorporating radiation therapy, chemical treatments, and surgical procedures. Although these methods possess certain benefits, it is essential to acknowledge their inherent limitations and drawbacks. Therefore, the scientific community is currently working to develop a more efficient treatment to tackle the obstacles presented by ovarian cancer.

Previous generations have used medicinal plants to address various human afflictions, including cancer treatment. Numerous botanical species have demonstrated various pharmacological attributes, encompassing antioxidative, antimicrobial, anticancer, and antidiabetic properties. One notable advantage associated with the utilization of plants in the field of medicine lies in their extensive collection of diverse phytochemicals, which possess the ability to specifically address a wide range of medical conditions. Searching for novel pharmaceuticals to combat cancer is contingent mainly upon identifying biologically significant phytochemicals [41]. Using biomolecules derived from plants has become a well-regarded and promising approach with considerable potential. Significantly, there has been a notable increase in the incorporation of pharmacologically beneficial botanicals in the process of pharmaceutical advancement during recent decades. Considerable research efforts have been undertaken over the past two decades to investigate the potential of natural products in enhancing the efficacy of cancer treatment, with a particular focus on numerous countries.

7 Other Therapeutic Plants' Ability to Prevent Ovarian Cancer

Africa is home to various indigenous medicinal plants traditionally employed for therapeutic purposes. Some of these plants include *Aframomum arundinaceum*, *Aframomum alboviolaceum*, *Aframomum kayserianum, Aframomum polyanthum*, *Echinops giganteus, Xylopia aethiopica, Piper capense*, and *Imperata cylindrica. Additionally, Gladiolus quartinianus, Vepris soyauxii [*25*]*, *Polygonum limbatum, Polyscias fulva, Beilschmiedia acuta, Crinum zeylanicum*, *Dioscorea bulbifera, Elaoephorbia drupifera, Solanum aculeastrum, Albizia schimperiana, Zanthoxylum gilletii*, and *Strychnos usambarensis* have demonstrated their efficacy in the treatment and management of various malignancies, including cancer [16]. The observed plants have demonstrated a notable cytotoxic impact, particularly in regions with a high cancer incidence, and have exhibited resistance to pharmaceutical interventions. As an illustration, in the context of in vitro investigations involving Korean medicinal plants to examine their potential anticancer properties, it was observed that the ethyl acetate fraction derived from the methanolic extract of *Lespedeza cuneata* exhibited a noteworthy cytotoxic impact on A2780 human ovarian carcinoma cells. The IC50 value, indicating the concentration required to inhibit 50% cell growth, was determined to be 77.25 ± 2.05 μg/mL. Similarly, the compound known as (+)-9′-*O*-(α-l-rhamnopyranosyl) lyoniresinol, derived from this particular plant, demonstrated in vitro antiproliferative activity against A2780 cells, with an IC50 value of 77.24 ± 2.05 μM. Significantly, a study involving athymic mice and human SKOV3 ovarian cancer xenografts observed that the tea seeds (*Camellia sinensis*), known to contain saponins, exhibited cancer chemopreventive effects. In addition, it has been observed that lysophosphatidic acid (LPA), a biologically active lipid, can enhance the invasive properties of cancer cells and promote their migration. This effect is accompanied by the phosphorylation of STAT3, which triggers the secretion of interleukin-6 (IL-6) and interleukin-8 (IL-8). Curcumin, a polyphenolic compound derived from the plant species *Curcuma longa*, possesses a range of curcuminoids and exhibits potential therapeutic efficacy in ovarian cancer. It is worth noting that traditional practices in various regions of Nigeria have employed various plant species, such as *Kigelia africana*, *Pistia stratiotes*, *Chenopodium africanum*, *Nymphaea lotus*, *Parquetina nigrescens*, *Nicotiana tabacum*, Nigrescentongensis, *Elaeis guineensis*, *Piper guineense*, *Aframomum melegueta*, and *Petiveria alliacea*, for the treatment of cancer, including ovarian cancer. In particular, *Securidaca longipedunculata* has gained recognition as an effective remedy for ovarian cancer [42]. Although there have been dedicated studies on the Ijebu people, an ethnic group within the Yoruba community, similar practices have been observed in various regions, including Ogun State in northwest Nigeria.

8 Conclusion

This evaluation examines the phytochemical constituents and metabolites in medicinal plants and herbal remedies utilized to treat and control reproductive ailments, specifically emphasizing conditions such as polycystic ovary syndrome (PCOS) and ovarian cancer. This analysis underscores the considerable potential of plant-based therapies in advancing women's reproductive health. This study examines various medicinal plants and compounds, such as *Ecklonia cava*, *Glycyrrhiza glabra*, and *Punica granatum*, and their potential to alleviate reproductive diseases and prevent ovarian cancer. These substances demonstrate promising properties in this regard. The complex nature of polycystic ovary syndrome (PCOS) and ovarian cancer necessitates a comprehensive approach to both treatment and prevention. These herbal remedies can alleviate the symptoms associated with these conditions and tackle the root causes, thereby providing a natural and complementary alternative to conventional medical interventions. Furthermore, this study elucidates the potential therapeutic efficacy of less prevalent botanicals such as *Aegle marmelos*, *Bambusa bambos*, and *Corylus avellana*, which have historically been employed in various regions across the globe. The assessment above highlights the significance of ongoing research on the effectiveness and safety of these botanical treatments. Scientific validation and clinical trials are crucial in establishing the optimal dosages, potential adverse effects, and potential interactions with other medications. This will assist healthcare professionals in making well-informed decisions and providing recommendations to patients who are seeking alternative treatment options. Moreover, it is imperative to underscore that herbal remedies should not be regarded as substitutes for conventional medical interventions but as adjunctive modalities. Patients should seek guidance from healthcare professionals and herbalists to develop personalized treatment strategies, considering their unique circumstances and requirements. Within women's reproductive health, investigating botanical remedies and their constituents holds considerable potential for forthcoming scholarly inquiry and practical implementation. As further knowledge is acquired regarding these therapeutic interventions, their incorporation into holistic healthcare approaches can enhance treatment efficacy and improve the overall well-being of individuals grappling with reproductive disorders and ovarian malignancies. This assessment incentivizes the broader medical and scientific community to engage in collaborative efforts, conduct further research on these botanical agents, and advocate for their responsible utilization in treating and preventing reproductive diseases.

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Assessment of the Phytochemical Constituents and Metabolites in the Medicinal Plants and Herbal Medicine Used in the Treatment and Management of Skin Diseases

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Abstract

The most significant organ present in the human body is the skin. It helps in protecting the body from different microbial organisms. Skin also helps regulate body temperature and enables one to sense touch, heat, and cold. A variety of skin diseases are widespread and frequently occurring health problems that affect people of all ages, from newborns to elders. These diseases can cause harm in several different ways. It is essential to maintain healthy skin to maintain a healthy body. People can suffer from various skin diseases that affect their skin, causing skin cancer, herpes, cellulitis, acne, and eczema. Many wild plants and their parts having medicinal properties are widely available. Those plants are frequently applied to treat these diseases and their symptoms. Plants have been used by humanity since the beginning of time. In addition to being cheap, natural treatments are also claimed to be safe. The raw material can also be used to produce novel synthetic agents, as it is suitable for this purpose. In this chapter, you will find a review of some plants that can be used to treat skin diseases and different symptoms produced due to these diseases and a summary of the recent technological advancements in this area during the past few years.

Keywords

No poverty · Zero hunger · Skin disease · Eczema · Herpes · Cancer · Health · Symptoms

Abbreviations

COLIPA European Cosmetic, Toiletry, and Perfumery Association (Now Cosmetic Europe)
CPE Catechin polyphenols

DPPH	2, 2-Diphenyl-1-picrylhydrazyl
HICV-RT	Human Immunodefeciency Virus-Reverse Transcriptase
HuDe	Human Dermal
IL	Interleukin
LCMS	Liquid Chromatography-Mass Spectrometry
NO	Nitrous oxide
PCPSO	Pomegranate Cold-Pressed Seed Oil
PFJ	Pomegranate Fermented Juice
PGE2	Prostaglandin E2
UHPLC	Ultra-High-Performance-Liquid Chromatography
UV	Ultraviolet
UVA	Ultraviolet Aging
UVB	Ultraviolet burning

1 Introduction

Since immemorial times, a wide range of plants of medicinal value have been utilized as a medicine to treat different diseases. More attention has been gained due to their remedial effects [1]. Other traditional herbs from various habitats can be a new strategy to protect against skin infections. The antimicrobial properties of those plants are now in demand due to their remedial properties. Due to a lack of awareness regarding phytomedicines and deforestation, the value of medicinal herbs is declining. These modern medicines come with many challenges, creating side impacts and toxicity [2]. Mother Nature has provided us with many herbal plants contributing to the health care system. These plant-based medicines are beneficial against various diseases and disorders. Due to lower side effects, there has been an increase in the adaptability and acceptability of these phytomedicines in the recent era [3]. The plants having herbal remedies are efficacious against various skin disorders, skin cancers, eczema, fungal and bacterial infections, acne, etc. [4]

Human skin is considered the largest organ of the human body and is the first defense line to protect the body from many diseases and disorders. Skin disorders occur worldwide, affect all age groups, and present a significant health burden in developed and developing countries. Healthy skin can only be achieved through the use of herbal medicines. The demand for herbal medicines is increasing progressively every day due to their low price and the hazardous effects of modern medications. These medicines have gained a vital contribution to health care [5] due to the presence of various bioactive components like glycosides, tannins, alcohols, aldehydes, etc. Different therapeutic agents can be discovered from these bioactive components and be a future general asset. Cosmetics have also been prepared to look beautiful. In the olden days, many homemade remedies were prepared with different plants and their parts, like banana, neem, sugar beet, *aloe vera*, saffron, papaya, carrot, neem, tulsi, marigold, etc. All these plants are used as medicinal herbs and various homemade remedies to manage different skin diseases. Polyherbal formulations, plant extracts, powders, oils, face packs, soaps, etc., can be prepared with these herbs. People use various synthetic products as cosmetics, but these products may lead to side effects such as skin reactions, inflammations, swellings, irritations, etc., use herbal skin products that are safe for the skin to avoid various side effects and allergies. Various synthetic products are used for oily and dry skin, but those products are not up to the mark. However, the homemade herbal remedies used in different combinations had a marked effect when applied on the skin with few side effects. Dental care products and perfumes also come under cosmetics to prevent tooth decay and body odor. There are different types of skin: oily skin and dry skin. Applying facewash, lotion, and sunscreen for protection from UV radiation and using moisturizing creams for cold climates. All these skin care products should be of herbal origin for better care. Among young stars, acne is the biggest problem that spoils most people's look, appearance, and beauty. It happens due to access sebum secretion, which increases oils in the skin. This can be treated with acne-treating products using gels and creams.

Due to stress and anxiety, many youngsters face aging experiences, which can be prevented by applying herbal skin care products and taking good care of their mental health. Anti-aging helps skin firmness and smoothness, improving brightening and a young appearance.

With the help of various herbal preparations containing fruit pulp extracts, blackheads and black spots can be cleared as these products supply a good amount of antioxidants, which helps give a toned skin. Herbal plants also help brighten skin and provide a good look, enhancing the complexities of the skin and giving it a shiny appearance. Many people face skin dehydration, which ultimately dries up, and the skin is subject to peeling. Various herbal cosmetics preparations can be applied to the skin to increase its moisturizing capacity.

Hair fall has become a significant problem for many people due to dandruff. Most men are facing such issues. So, protecting the hair from dandruff is very important to prevent baldness. The application of herbal anti-dandruff shampoo can do this. Various hair oils can also be provided to control hair fall. Growing hair follicles on the face and all over the body is a significant problem many women face. This can be overcome by applying various herbal preparations to prevent hair growth [6]. The focus of this chapter is to assess the role of phytomedicine in skin infection management.

2 Skin Layers and Their Functions

Skin is an organ of tissues that work together to perform unique and critical functions that provide overall protection. The skin consists of multiple layers, of which epidermal tissue is the most superficial layer attached to the dermis and external to the hypodermis. The epidermal layer is keratinized and protects the skin cells. The dermis is the vascular layer that provides flexible tissues. Additionally, it comprises several connective tissues, sweat glands, hair follicles, and connective tissues and helps provide support and flexibility to the tissues. It also helps to regulate body temperature and provides physical protection against injury. It is a barrier against the external environment, preventing the entry of bacteria and other harmful substances. The innermost layer of the hypodermis is one of the body's most vascularized layers, consisting of loose connective tissues that serve as energy reserves and maintain the body's temperature (Fig. 1). It also contains sweat glands, hair follicles, and oil glands. The hypodermis also plays a vital role in the body's immune system by providing a physical barrier to keep out infectious agents.

Absorption of the drug using the skin is a route through which the drug enters the skin. Drug absorption depends on factors like concentration, time duration, medicine stability, skin condition, and body part exposure. Skin absorption is the transport of chemicals using the skin's outer surface as well as into the circulation. While transporting herbal medication into the dermal region for the treatment of skin diseases, a series of steps are involved in releasing herbal medicines in the dosage form. These steps include formulating the dosage forms, selecting the appropriate delivery system, and ensuring the release of the medicine at the desired rate. Additionally, the stability and safety of the drug must be considered to provide the highest quality of treatment. After a therapeutic photothermal constituent penetrates through the epidermal layer of the skin, it diffuses into the dermal region of the body. The light energy then activates the photothermal branch and produces photothermal effects, leading to the desired therapeutic outcome. The photothermal effects can be monitored and controlled by modulation of light energy. As soon as the herbal constituent reaches the circulation system, it is absorbed by the body. Figure 2 below depicts a graph illustrating how drugs penetrate the body. The drug is then

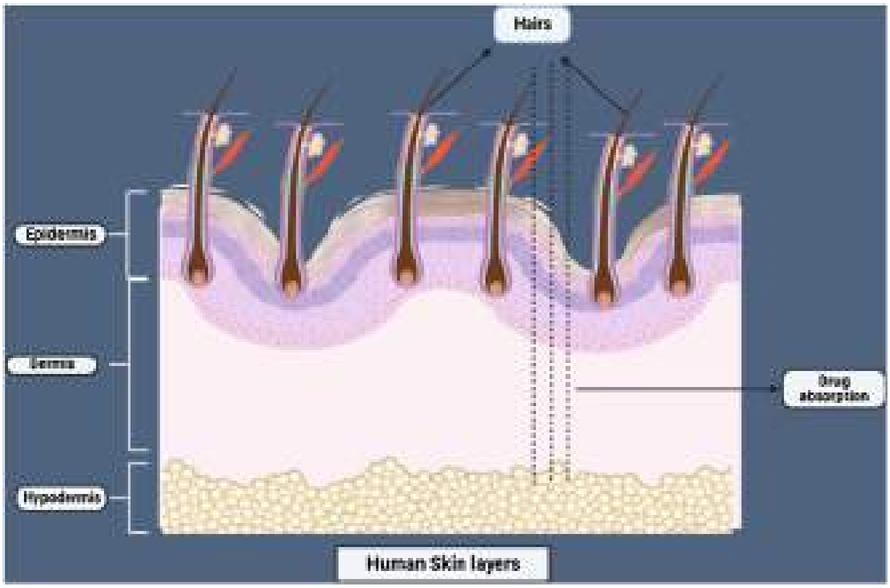

Fig. 1 Drug absorption through the skin that transport the chemicals through outer surface of the skin and through blood circulation

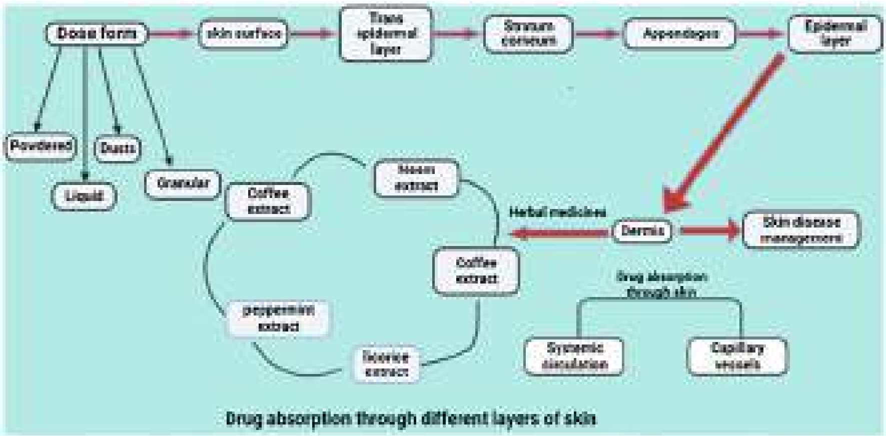

Fig. 2 Drug absorption through different layers of skin, by means of systemic circulation and capillary vessels

distributed through the body, achieving the desired therapeutic effect. The light energy can then be turned off, ending the photothermal effect. The drug is then metabolized and eliminated from the body.

3 Diverse Skin Conditions and Diseases: Characteristics, Effects, and Pathogen Involvement

3.1 Skin Disease

It includes various types of conditions that create irritation or inflame in the skin. Skin diseases also cause rashes or other changes in skin appearance. All the skin diseases mentioned below are not due to pathogens. The list of pathogens occurring in skin diseases is mentioned in Table 1. Various skin conditions can affect individuals, each with distinct characteristics and effects. These conditions include acne, characterized by oil clogging in skin follicles and the formation of bacteria and dead skin in pores; alopecia areata, which leads to hair loss in small patches; atopic dermatitis (eczema), causing itchy skin, dryness, swelling, and cracking; psoriasis, resulting in scaly, swollen, and hot skin; Raynaud's phenomenon, involving periodic reduced blood flow in fingers, toes, and other body parts with associated changes in skin color and numbness; rosacea, typically presenting with thickened skin and pimples on the face; skin cancer, marked by uncontrolled abnormal development of skin cells; vitiligo, leading to pigment loss in skin patches; prurigo, causing itchy skin when exposed to the sun; argyria, involving changes in skin color as the body accumulates a silvery buildup; chromhidrosis, a pigmentation disorder; epidermolysis bullosa, a connective tissue-related disorder resulting in fragile skin prone to blisters and tears; Harlequin ichthyosis, characterized by the presence of thick, hard patches on the skin at birth;

Table 1 List of pathogens causing skin infections

Pathoges	Skin infections	References
Bacterial pathogens		
Propionibacterium acnes	Acne, dendruss, psoriasis, and blepharitis	[9]
Microsporum audouinii., Trichophytion spp.	Tinea capitis	[87]
Staphylococcus aureus	Atopic dermatitis and skin and soft tissue infections	[9]
Fungi		
Malassezia dermatis, M. furfur, M. globose, M. japonica, M. obtuse, M. restricta, M. slooffiae, M. sympodialis, and M. yamatoensis	Tinea versicolor, psoriasis, and atopic dermatitis	[88]

lamellar ichthyosis, which leads to the development of a waxy skin layer, revealing scaly and red skin; and necrobiosis lipoidica, a condition where rashes on the lower legs may progress to ulcers.

3.2 Cause of Skin Disease

Certain lifestyle factors for developing skin disease include (a) trapping of bacteria in the hair follicles or pores, (b) sun rays, (c) viruses, (d) parasites on the skin, e.g., any conditions affecting a person's immune system, (e) certain medicines, bowel infections, (f) kidney diseases, (g) thyroid problems, (h) diabetes, etc.

3.3 Diagnosis of Skin Disease

Medical professionals employ various diagnostic methods to assess skin conditions. These methods include performing a biopsy to examine a small-skin sample under a microscope, conducting a culture to identify the presence of fungi, bacteria, or viruses on the skin, administering a skin patch test to detect allergic reactions to chemicals, utilizing a Wood's light test involving UV light to examine the skin, employing diascopy to assess skin patches by applying pressure with a microscope, using dermoscopy for inspecting skin lesions and conducting a Tzanck test to examine fluid from a blister (Fig. 3).

3.4 Various Skin Diseases Caused by Pathogens

Table 1 presents the list of pathogens causing skin infections

3.4.1 Bacterial Infections

The skin's external surface is mainly affected by bacteria. Skin acts as a medium for bacterial growth. It also allows commensal bacteria to grow on the skin to protect it

Fig. 3 Methods for diagnosing skin disease

from disease-causing bacteria. Some gram-positive bacteria like corynebacterium, staphylococcus, micrococcus, and streptococcus are notorious bacterial pathogens for the skin. A pathogenicbacteria allows bacteria to adhere, grow, and invade the skin's surface. With an increase in bacteria, disruption of the integumentary barrier, invasion by these colonizing bacteria emanates, and skin and soft tissue infections develop.

3.4.2 Viral Infections

It is believed that viruses can penetrate the stratum corneum, which is the top layer of the skin and is susceptible to infection. The stratum corneum is a natural barrier to illness, but viruses can pass through the skin, leading to disease. Viruses can also be transmitted through contact with contaminated surfaces or objects. Therefore, it is essential to take precautions such as washing your hands often and avoiding contact with people who are sick. As well as infecting the epidermis, it also infects the subcutaneous layer. To prevent infection, avoiding touching your face is essential, as this is an easy way for viruses to enter your body. Additionally, it is necessary to regularly clean and disinfect surfaces and objects that are frequently touched. There are a variety of viral skin infections, such as measles, warts, chickenpox, etc., which are not healable by antibiotics [7]. Therefore, practicing good hygiene and wearing protective clothing is essential to avoid contracting the virus. Vaccination is another effective way to reduce the risk of viral skin infections.

3.4.3 Fungal Infections

Fungal infection occurs in the depth of the skin tissue. But sometimes, the conditions may occur superficially, like skin, nails, or hair, called athlete's foot infection or ringworm infection, where trial antibiotics are needed to be taken, which in the long-term cause severe disorders [7].

3.4.4 Tumors and Skin Cancers

The lack of melanin causes skin cancer, which ensues in vitiligo, an auto-immune disorder. The scarcity of melanin cells causes albinism, which may cause skin cancer when exposed to UV radiation. If detected, there is a chance for the survival of patients. [7].

3.4.5 Skin Pigmentation

With the increase in the melanin content of the body, hyperpigmentation occurs, which causes skin irritation and premature aging [7].

Trauma: The burns, cuts, or skin injury leads to a pathogenic attack known as trauma [7].

3.4.6 Rashes

Inflamed-skin conditions can cause pain, itching, and discomfort, leading to scarring and psychological distress. Treatment of such conditions can include topical medications, antibiotics, and lifestyle changes. It is essential to seek professional medical advice when dealing with skin inflammation.

3.4.7 Psoriasis

Infections such as psoriasis that affect the skin can harm the patient's quality of life and negatively affect the patient's outlook on life. Having psoriasis can be physically and emotionally draining. Symptoms can be uncomfortable and embarrassing; flare-ups can cause pain, itchiness, and inflammation. These symptoms can interfere with everyday activities and make it difficult for the patient to enjoy their life. Excessive proliferation of the skin can result from either hereditary factors or hyperproliferation of the skin. Psoriasis is an autoimmune disorder that causes the body to mistakenly attack healthy skin cells, resulting in a rapid overproduction of skin cells that accumulate on the skin's surface. This leads to the characteristic red, scaly patches of psoriasis. In addition, psoriasis can lead to psychological distress, as patients may feel embarrassed or self-conscious about their appearance. Three types of psoriasis are known to exist: nail psoriasis, which causes pitting and yellowing of the nails along with severe hyperkeratosis; skin psoriasis, which causes yellow pus to appear on the skin and blisters to appear near the joints; and psoriatic arthritis, in which bone erosion occurs near the joints [8]. Psychological distress can be exacerbated by the physical discomfort caused by the condition and the fact that psoriasis can be visible to the public. Covering nail or skin psoriasis with clothing is often tricky, and

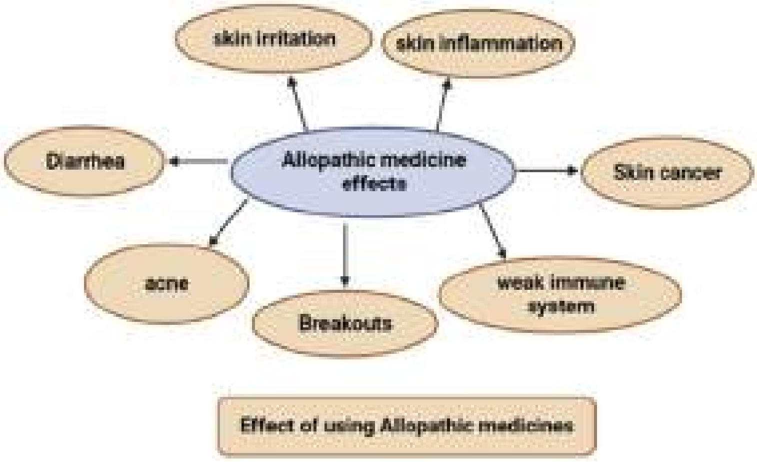

Fig. 4 Side effects of allelopathic medicines

psoriatic arthritis can cause severe pain and joint stiffness. All of these can lead to social isolation and a decrease in self-esteem.

3.5 Allopathic Approaches for Skin Disorders

Skin disorders treatment with allopathic medicines causes severe ripple effects and can be diminished by phytomedicines. Numerous hazardous effects occur due to these allopathic medicines, like skin inflammation, irritation, breakouts, and diarrhea. Immunosuppressants lead to skin cancer and a weak immune system [5] (Fig. 4).

4 Role of Phytomedicines for Skin Infection

4.1 *Azadirachta indica*

Neem, a member of the family Meliaceae, plays a significant role in the promotion of better health due to the presence of a high content of antioxidants, azadirachtin, glycerides, polyphenols, triterpenes, limonoids, beta-sitosterol, quercetin, catechins, carotenes, and vitamin C that contribute to its health-promoting effects. It has been used in Chinese, Unani, and Ayurvedic medicine for thousands of years. It has been used for centuries as a traditional medicine due to its anti-inflammatory, antibiotic, antifungal, and antiviral properties. It is also a key ingredient in many natural skin care and hair care products. The paste prepared from leaves helps treat acne, eczema, scabies, skin allergies, and psoriasis. Neem is also known to have antibacterial and anti-ulcer effects, making it an excellent tool for boosting the overall health of individuals. Additionally, it has been shown to help reduce cholesterol levels, improve digestion, and regulate blood sugar levels. Neem is also known for its antioxidant, anti-inflammatory, and anti-cancer properties that help boost immunity and relieve stress and anxiety. Neem is known to have antiviral and antifungal properties. The findings from earlier reports have confirmed that the role of neem and its constituents play an essential role in the pathogenesis of many diseases [86]. It treats various illnesses, such as malaria and intestinal parasites. In addition, neem has also been used to treat eye infections, gastrointestinal disorders, and fever. The bark of this plant helps maintain healthy gums, increases oral hygiene, and can also be used as an antiviral against chicken pox due to its antiviral properties. It can also reduce swelling, promote healing, and has contraceptive effects that can help manage neurodegenerative disorders and diabetes. As far as treating eczema, scabies, and other skin disorders is concerned, neem paste has a variety of beneficial effects. The product contains essential fatty acids that provide moisture and texture to the skin during the healing process by inhibiting the growth of inflammatory cells. Several studies have shown that the oil extracted from neem can treat ringworm, psoriasis, and scrofula diseases [10]. Oil extracted from neem is also one of the best herbal remedies for treating various skin disorders, allergies, infections, inflammations, and other skin disorders [11, 12]. The oil extracted from neem also reduces wrinkles, dark spots, and discoloration, excellently making it a tremendous natuan for minor skin irritations and infections. Applying neem and turmeric as lotion during night hours helps prevent scthe abies, eczema, and various logic problems. Neem can also be taken orally, either as a tea, in caps as a tea, or in capsule form to boost the immune system. Neem is also a natural insect repellent, making it useful for insects from gardens and homes away.

4.2 *Glycyrrhiza glabra*

It is a sweet and aromatic herb that comes from the licorice family. This herb's active metabolite is glycyrrhizin, an antiviral, anti-diabetic, antifungal, expectorant, demulcent, mild laxative, antioxidant, and skin-whitening substance. It is important to note that liquiritin, an antioxidant, is the anotis component of licorice that contributes to skin lightening. It also contains glycyrrhizin, which is known to have antibacterial properties that can help keep skin acne-free. It is believed to have many health benefits and has been used in traditional herbal medicine for centuries. Licorice has anti-inflammatory, antibacterial, and antioxidant properties that improve overall health and skin health. It also has a mild sedative effect, making it a popular choice for people looking for natural sleep aids. Despite its ethnopharmacological value, licorice has been explored worldwide due to its ethnopharmacological properties. Research has also suggested that it may positively affect cardiovascular health. Glycyrrhizin also shows potential in treating liver disorders, chronic bronchitis, and ulcers. In addition, research suggests that licorice may be beneficial in supporting cognitive function and mental health. Additionally, it is also a good source of valuable flavonoids and saponins as well. It may also help to reduce the risk of cancer and heart disease. The benefits of this plant provide a place for research and discovery of other benefits and potent effects that the plant has to offer. Licorice has also been used in traditional medicine for centuries, making it a trusted and reliable source of health and wellness. In addition, it is widely available and affordable, making it an attractive option for those seeking natural remedies. Glabridin is another constituent of *G. glabra,* which can lighten the skin's pigment due to its ability to inhibit the activity of tyrosine B16 in melanoma cells. Due to its anti-inflammatory and antioxidant properties, it reduces the appearance of wrinkles and improves skin texture. It has also been found to have beneficial effects on the skin, such as reducing redness and irritation. Additionally, licorice can also help protect skin from UV radiation and reduce melanin production, which can help with skin whitening. As a cosmetic, the herbal extracts of this plant can be used to depigment the skin and protect it from UV rays [13]. This can benefit those who want to lighten their skin tone or reduce the appearance of dark spots. According to various reports, root extracts of *G. glabra* have been proven effective in treating acne, eczema, and psoriasis. Furthermore, its anti-inflammatory properties can help reduce redness, pain, and swelling caused by skin conditions [14]. Licorice gel can control atopic dermatitis in patients with asthma and hay fever in their family history. Additionally, licorice helps to protect against UV radiation. It can also be used to treat cold sores and heal wounds. In combination with polyethene glycol, this gel can be applied at night along with polyethene glycol to reduce itching on the face [14]. Licorice gel can thus be used to reduce the severity of atopic dermatitis and for anti-aging purposes. It is important to note that licorice gel should be used in moderation and with the advice of a healthcare professional. In addition, some studies have shown that licorice extract also reduced pigmentation in the skin of black mollyfish by inhibiting tyrosine synthesis due to the dispersal of melanin (which contributes to skin tanning) from its location in the body. However, caution must be taken when using licorice gel as it can cause irritation and other skin conditions. It is, therefore, recommended to do a patch test before using the gel. As a result of the nutritional properties of licorice extract, new possibilities exist for the development of beautifying preparations that are safe, effective, and provide effective treatment for allergic dermatitis [15]. Such practices should be developed using carefully selected, high-quality raw materials and tested adequately in pre-clinical and clinical trials. This way, licorice gel can be used safely and effectively for its intended purpose.

4.3 *Ocimum sanctum*

It has become apparent that the "queen of herbs," *Ocimum sanctum* [16], has become a drug of choice for many people. Studies have found that this herb has anti-inflammatory, antibacterial, antifungal, and antiviral properties. Some essential constituents of basil are eugenol, ursolic acid, linalool, carvacrol, and rosmarinic acid, which are believed to have anti-inflammatory and antioxidant properties [16]. It has been used to treat various health conditions, ranging from colds and coughs to skin diseases and digestive issues. It is also known to boost immunity and fight oxidative stress. As it has been reported, it is used in several countries to treat various skin diseases and disorders, including allergies, rashes, insect bites, snake bites, scars, leucoderma, blisters, chickenpox, and roundworm infections. In addition, it has been used in traditional medicine to treat cancer, diabetes, and cardiovascular diseases. It is thought to reduce inflammation and boost the immune system. There is also evidence that the phyto extracts of basil are helpful in the reduction of itchy skin that occurs due to acne, pimples, and scars [17]. Studies have shown that basil effectively treats various conditions, including anxiety, inflammation, and skin irritation. Therefore, *O. sanctum* would be a striking plant for developing cosmetics. Its strong antibacterial and antifungal properties make it a viable option for producing natural skin care products. With its numerous therapeutic benefits, *O. sanctum* has the potential to be a valuable ingredient in the cosmetic industry. The first report of anti-aging activity has been investigated using collagenase, elastase, and hyaluronidase activity inhibition under in vitro conditions. These results suggest that basil could help reduce the visible signs of aging. Thus, basil could be a powerful and natural ingredient for skin care products. Some studies also confirmed the presence of rosmarinic acid in basil, which possesses potent antioxidant and anti-inflammatory activity that can protect the skin from damage caused by free radicals. Moreover, it can stimulate collagen production, which in turn can help reduce wrinkles and fine lines. *Extracts of basil* have inhibitory activities against oxidation, inflammation, collagenase, elastase, and hyaluronidase [18]. This indicates that *O. sanctum* extracts have the potential for use in cosmetics and dermatological applications due to their anti-aging properties. It can also be used to boost the immune system and reduce stress levels. Another report from a study includes the beneficial effects of herbal extracts as safe, effective, potent antibacterial, antiviral, and antifungal agents in controlling skin blisters, treatment of wound infections, chickenpox, and roundworm infections by modifying lifestyle [19]. Many skin disorders can be managed using tulsi plants, which have been found to have good herbal potential. Basil extract formulations and base (containing no extracts) were applied to the cheeks of 11 healthy human volunteers to determine the presence of erythema and melanin. Significant results were observed in the base product, while erythema decreased in the formulation. The study concluded that tulsi plants have the potential to manage skin disorders. Even though the melanin content of the skin increased significantly with the application of the base, it decreased with the application of the formulation [14]. The results indicate that the formulation containing the basil extract had a significant anti-inflammatory effect on the skin and a skin-lightening outcome. This suggests that basil extract could be helpful in skin care products. In addition, the essential oil from basil leaves has been used to treat headaches and other ailments. The compounds present in tulsi are believed to contribute to the therapeutic effects of basil, although further research is needed to understand the mechanisms of action fully.

4.4 *Andrographis paniculata*

This plant has been used for decades in the subcontinents of India to treat various diseases. It contains diterpenes, lactones and flavonoids, alkanes, keto aldehydes, and andrographolide. It helps treat pyrexia, dyspepsia, malaria, colic pains, dysentery, and diarrhea with aerial parts of the plants, roots, leaves, and whole plants of *Andrographis paniculata*. It has also been used to treat common colds and sore throats and improve the immune system. The plant is known for its antioxidant, anti-inflammatory, and anti-cancer properties. In this plant, there is a bioactive compound called andrographolide that is highly bioavailable and has anti-inflammatory, antioxidant, and other activities that can be used as a therapeutic agent to treat inflammatory skin diseases [20]. Andrographolide also exhibits various pharmacological activities, such as anti-tumor, anti-arthritis, and anti-diabetic. Furthermore, it can also be used as an analgesic and anti-diabetic agent. *A. paniculata* helps cure primary skin infections like eczema and leukoderma. Moreover, it can be mixed with oil and applied to treat wounds as a paste on the skin [21]. This compound also helps reduce inflammation and itching while promoting the healing of wounds. It is also known to have antibacterial, antifungal, and antiviral properties.

4.5 *Emblica officinalis*

*As a medicinal herb, Emblica officinalis, also known as Indian gooseberry or Amla, is used for the treatment of various skin disorders and diseases. A*mong the chemical constituents, it contains tannins, polyphenols, gallic acid, alkaloids, emblicanin A and B, phyllembein, quercetin, ascorbic and ellagic acids, vitamins, and minerals. It is also believed to have anti-inflammatory and antioxidant properties, which can help the body fight against diseases. Amla can be eaten raw or taken as a supplement and has been used in traditional medicine for centuries. It has been found that the fruits of this plant have immense pharmacological and medicinal applications. It also boosts immunity, improves digestion, and reduces cholesterol levels. Amla has been used to treat diabetes, high blood pressure, and other health conditions. Amla is highly valued in Ayurveda for its therapeutic properties and is used in various herbal preparations. It is also known to be a rich source of antioxidants, essential for fighting free radicals and maintaining overall good health. In addition to its antioxidant properties, it also has anti-inflammatory and wound-healing properties. Amla is also rich in vitamin C, which helps boost the immune system and protect against infectious diseases. Furthermore, it is known to have anti-aging benefits, helping to keep skin looking youthful and healthy. Several conventional therapeutic agents can be replaced with remarkable herbs due to their outstanding efficacy and lack of side effects [22]. Amla is one such herb that provides a wide range of health benefits. Ayurveda, an ancient Indian medical practice, highly recommends amla and rasayan due to their antioxidant properties as part of its daily healthcare regimen that can help protect the body from the damaging effects of free radicals. They can also help to promote healthy skin, hair, and nails. According to various reports, amla contains high amounts of vitamin C, making it an effective hair tonic and treatment for warts and skin infections [23]. Amla is also rich in minerals like calcium, phosphorus, iron, and carotene, essential for healthy bones, teeth, and muscles. Rasayan provides various health benefits, including promoting better digestion, boosting immunity, and relieving stress [22]. In addition, amla is known to have anti-inflammatory properties, which can help reduce inflammation and pain associated with arthritis. It has been suggested to help lower cholesterol levels. Oral consumption of amla helps improve the condition in females, and drinking the extract improves the skin's elasticity and reduces the appearance of wrinkles [24]. Additionally, this fruit is known to be full of vitamins and minerals that are essential for overall health and well-being. It inhibits microbial adhesion, inactivates enzymes, suppresses influenza A virus, prevents viral absorption, and treats HIV by inhibiting HIV-RT [22]. Furthermore, it has been used to treat various skin conditions such as acne, eczema, and psoriasis.

4.6 *Aloe vera*

It is one of the most valuable and oldest herbs belonging to the family Liliaceae, indigenous to India, and belongs to the genus Aloe vera*. Aloe vera* has traditionally been used to treat various ailments, including skin conditions, digestive issues, and inflammation. It contains many nutrients, including folic acid and vitamins A, C, E, and B12. It also contains calcium, manganese, zinc, and magnesium minerals. The most investigated active compounds are aloe-emodin, aloin, aloesin, emodin, and acemannan [28]. Several species of Aloe have been used to prevent and treat skin-related problems for a long time. *Aloe vera* also contains a range of anti-inflammatory compounds, which have been found to have antibacterial, antifungal, and antiviral properties. It is a natural ingredient in many skin care, hair care, and health products. Depending on the individual application, aloe materials can be used for cosmetic, radiation, cancer, wound healing, and antimicrobial purposes [25]. It is also known to aid digestion, reduce constipation, and help regulate blood sugar levels. *Aloe vera* can also be taken orally as a juice or in capsule form. Additionally, *aloe vera* can be applied directly to the skin as a topical remedy. There is a derivative of *aloe vera* called aloesin, which helps to lighten dark patches, age spots, and acne marks. *Aloe vera* is also known to possess anti-inflammatory and antioxidant properties, which can help to reduce inflammation and protect the skin from environmental damage. It can even help to reduce the signs of aging, making skin look smoother and more youthful. As one of the most potent skin healers on the planet, this herb is used to treat sunburns, insect bites, wound infections, cuts, itching, and swelling in people worldwide. *Aloe vera* is also beneficial for treating skin conditions such as psoriasis, eczema, and acne. It is also used to reduce the appearance of scars, stretch marks, and dark spots. As a result of its antioxidant, antiaging, and anti-wrinkle properties, it purifies and detoxifies the skin. Aqueous extract from this plant is crucial in repairing damaged skin cells, reducing wrinkles, curing acne, reducing pimples, treating eczema, and working as a herbal gel for glowing skin [26]. Kumar et al. [27] published an eminent report regarding the application of *aloe vera* to wounds and burns. *Aloe vera* also helps treat psoriasis and other skin problems due to its anti-inflammatory and antibacterial properties [28]. It is also used in the treatment of skin cancer [29].

4.7 *Carica papaya*

Carica papaya is a herbaceous succulent plant, popularly known as pawpaw, belongs to the Caricaceae family and is considered the powerhouse of all vital nutrients for medicinal use. The roots, pulp, bark, seeds, and peels can be utilized for various ailments due to abundant vitamins and enzymes. The active components in papaya are alkaloids, glycosides, tannins, saponins, and flavonoids, responsible for medicinal activity. Platelet count can be increased if leaf juices are given to dengue fever patients [30]. The principal constituent of papaya is papain, which has antioxidant, antifungal, and antiviral effects. It is one of the safest and most widely promoted nutraceuticals against diseases like eczema, warts, and skin infections, preventing premature aging and scabies and reducing itching and wrinkles. Due to its inflammatory activity, papaya extract has wound-healing properties. Papaya peels can be used as potent sunscreen and, when mixed with honey, can be used as a skin-lightening and moisturizing agent. Adding lemon juice to papaya pulp can be used as anti-dandruff, and blending essential oils with papaya can be a good muscle relaxant [31].

4.8 *Jatropha curcus*

Jatropha is a drought-resistant, perennial plant belonging to the family Euphorbiaceae, famous for the production of biodiesel along with several medical applications. It contains terpenoids, coumarins, alkaloids, phenols, steroids, flavonoids, tannins, lignoides, and coumarins. Latex of Jatropha contains Jatrophine, Jatropha, and curcumin with anti-cancerous properties that help heal central skin lesions with wound healing properties and have purgative, anti-helminthic, and abortifacient effects. The oil produced from seeds help manage parasitic skin infections, which validates the plant in minimizing skin infections by promoting the healing of wounds with potent anti-inflammatory and antimicrobial processes [32]`.

4.9 *Zingiber officinalis*

Ginger belongs to the Zingiberaceae family, which can be used as a medicine and flavoring agent. Among the many essential components of ginger, gingerol *helps form new blood vessels at the site of damaged skin, promoting the repairing of skin cells and reducing the pain the patient feels.* Gingerol has potent medicinal properties like anti-inflammatory, antioxidant, and anti-tumor properties and can help protect against bacteria and viruses. It also contains carbohydrates, lipids, terpenes, and phenolic compounds, a popular ingredient in many Asian cuisines. It is also known for herbal ointments that can treat primary skin lesions. These ointments usually contain natural ingredients such as *aloe vera*, tea tree oil, and lavender oil, which can help reduce inflammation and promote healing. They are gentle and typically well-tolerated by people with sensitive skin. It has been shown that the oil obtained from seeds can be used to treat parasitic skin infections through its ability to promote wound healing and wound care and its potent anti-inflammatory and antimicrobial properties [33]. It also has antifungal and antiviral properties, making it an ideal choice for treating skin infections. Furthermore, it can help to protect the skin from further damage and irritation. Ginger can be consumed in various forms, including fresh, powdered, dried, and as an oil or extract. A ginger extract can help improve the structure and function of the skin, as well as reduce the formation of wounds that do not heal and make skin susceptible to disease. It also helps to improve the production of collagen, which is essential for skin health and increases the body's antioxidant capacity. Ginger may also help reduce wrinkles, age spots, and other signs of aging. It helps reduce inflammation, which is a significant cause of skin damage and can help reduce the risk of skin cancer. In addition, ginger can help to regulate oil production in the skin, helping to keep it balanced and healthy. In a study conducted on corticosteroid-treated rats, pre-treatment with topical ginger and curcumin extract improved the healing of induced abrasion skin wounds [34]. The study concluded that ginger and curcumin extract have a protective effect on the skin and can help wound healing and reduce inflammation. It also helps to reduce pain and can improve digestion. Gingerol can also help improve circulation, lower cholesterol levels, and improve immunity. It has also effectively combat bacterial, fungal, and viral infections. *The root extracts of ginger have been explored for their activity against various skin problems.* Ginger can also be a natural remedy for headaches, nausea, arthritis, and menstrual cramps. It can also treat multiple respiratory issues such as cough, cold, and sore throat. *The herbal extracts of ginger help in wound healing by improving skin texture and structure.* Additionally, ginger can relieve digestive issues such as indigestion and bloating. *The combination of ginger and turmeric helps to speed up the healing process of wounds.* It has been observed that when ginger extract, which is also an anti-inflammatory agent, is applied to mice ears, croton oil-induced edema can be inhibited by inhibition of cytokines cyclooxygenase and nitric acid production. In addition, ginger extract has been found to reduce inflammation and pain in cases of rheumatoid arthritis. Due to its antioxidant properties, ginger helps to protect against oxidative stress and tissue damage. There has also been evidence that wound-healing mechanisms have been observed in humans when these results have been simulated [34]. These findings suggest that ginger and turmeric can help to reduce inflammation and support the healing of wounds. In addition, applying these natural ingredients can reduce the need for other medications, such as antibiotics.

4.10 *Eucalyptus globulus*

It is estimated that there are more than 600 species of eucalyptus in the large genus Eucalyptus within the family Myrtaceae. Eucalyptus is also a famous ornamental tree, grown for its exciting bark and the shade it provides. It contains sideroxylonal C, leptospermone, grandinol, cypellogins A, B, and C, (+)-oleuropeic acid, cypellocarpins A, B, and C, and isoleptospermone. Many of these species are native to Australia and are used for their medicinal and aromatic properties in many cultures worldwide. Eucalyptus provides a wide variety of herbal by-products, food additives, antioxidants, healing properties of wounds, and protection against many fungal infections. They are also known to have anti-inflammatory, antiseptic, and antiviral properties. It is also used as an insect repellent, and its leaves and bark can be used to make tea. There are reports of different species of eucalyptus, like *Eucalyptus globulus, E. maculate*, and *E. viminalis,* in managing acne, scabies, and skin infections. Also, it offers a cooling sensation to significant cuts and wounds at the site of application, thereby reducing inflammation at the site of application [35].

4.11 *Euphorbia hirta*

The Euphorbia hirta *plant is a pantropical weed and is part of the Euphorbiaceae family.* It contains alkaloids, flavonoids, terpenes, chromenes, and sterols from all the plant parts. Quercetin is the active metabolite in euphorbia with potent antioxidant and anti-inflammatory properties that can be used to treat severe wounds, boils, and swellings. It is one of the oldest herbal medicines used today and is an effective remedy for acne, warts, and cancer. *E. hirta* is also known for its anti-inflammatory, antiseptic, and antifungal properties, making it a popular choice for treating skin ailments. It can also treat colds, fevers, and stomach pain. If you mix the extract with warm coconut oil and turmeric, it can be used to relieve the itching of boils and sores that are caused by bacteria. Roots of *E. hirta* are also adequate food nutrients for infants and breast-feeding mothers and are effective against snakebites. It has sedative properties and treats urinary tract infections [36]. The extract of *E. hirta* is also used to treat fever, diarrhea, and dysentery. It also improves digestion and treats respiratory problems such as asthma. It has analgesic properties and is used to relieve pain. Not only that, but it is also used to treat fever, cough, and other respiratory ailments. In addition, it is an effective anti-inflammatory and antiseptic. It can be administered topically to relieve pain and reduce swelling.

4.12 *Ficus carica*

In temperate climates, the fig tree (*Ficus carica* L.) is one of the world's oldest cultivated trees and one of the most popular edibles. It produces a sweet, nutritious, and versatile fruit enjoyed in fresh and dried forms. Figs are a high source of fiber, vitamins, and minerals, making them a great addition to any diet. It also contains phenolic acids and flavonoids. They are also known for their anti-inflammatory and antioxidant properties. There are two types of figs: fresh and dried. Fresh figs are available in the summer and early fall, while dried figs are open year-round. Figs can be enjoyed in various dishes, from salads to desserts. They can also be eaten as a snack on their own. Fresh figs are widely eaten near production areas because they are highly perishable. Figs have several nutrients, including fiber, potassium, calcium, and iron. Dried figs are a good source of dietary fiber and calcium. They are also a good source of minerals like magnesium, manganese, and copper. Figs are a good source of antioxidants, which can help protect the body from damage caused by free radicals. Several health benefits can be gained from the fruits, including treating skin ulcers, eczema, acne, skin dryness, and wounds, and preventing the aging process. It has also been found to have cholesterol-lowering properties and can aid in weight loss. Figs may also help reduce the risk of certain types of cancer. It has been shown that when applied to human cheeks for 8 weeks to investigate melanin, sebum, and trans-epidermal edema, the fruits reduced hyperpigmentation, acne, and wrinkles after 8 weeks of application. Figs may also have anti-inflammatory properties and can help protect the skin from environmental damage. Furthermore, they are a good source of dietary fiber and other essential nutrients. It has been suggested that the herbal extract of ficus can be used as a source for future skin care products [37]. Figs are a rich source of antioxidants, which can help to repair and protect the skin from oxidative stress. They also contain polyphenols, which can help to reduce the appearance of wrinkles and fine lines. Overall, figs can be a powerful source of nutrition for skin health.

4.13 *Allium cepa*

Since ancient times, onions have been used in the traditional Ayurvedic system of medicine. It contains Kaempferol, Myricetin, Quercetin, Quercitrin, and Allyl sulfide. For managing skin allergies and anti-aging, fruits and bulbs contain essential constituents such as pyruvic acid and quercetin. Onion helps as a remedy for various infections, including healing wounds, scarring, and cancers. Onions are rich in antioxidants, which help to reduce inflammation and fight infection. They also contain sulfur compounds that help to kill bacteria, fungi, and viruses. Onion juice is often used as a topical treatment for skin infections, and the poultice helps reduce scars. Raw or cooked onions can also be consumed for various health benefits, including improved digestion and cardiovascular health. In addition, onion extract has also been found to help prepare new herbal skin care products due to its anti-aging properties [38]. Onion extract is also rich in antioxidants, which help protect the skin from environmental damage, eczema, and psoriasis and reduce inflammation. It can also help reduce wrinkles and make the skin more elastic. Onion extract can also lighten the skin, reduce acne, and tighten facial pores. It can even help reduce dark circles under the eyes. The onion can be a skin irritant because it stimulates blood circulation, thus causing the skin to itch. Onion extract is also an effective antiseptic, helping to reduce the risk of infection and soothe sunburns and other minor skin irritations. Finally, it can also act as a natural skin toner. The poultice can also treat boils and warts and produce heat to speed up healing. Due to its strong scent, the onion can also be used as a natural insect repellent. It can also be used as an antiseptic to help protect wounds from infection. It inhibits ultraviolet B (UV-B) Inhibits free radical scavengers.

4.14 *Abrus percatorius*

It is also known as a jequirity bean or rosary pea containing a potent poison known as abrin. It also has luteolin, orientin, isoorientin, glycyrrhizin, abrectorin, abrusoside A to D, abrusogenin desmethoxycentaviridin-7-o-rutinoside, and abruquinones D, E, and F. The leaves with lime can be applied to scars, sores, blisters, and skin allergies. It helps stimulate blood circulation, prevents gray hairs, and heal wounds and leukoderma [40]. It also helps in controlling inflammation and cure rabies. The seeds also have insecticidal properties and can control worm infections [39].

4.15 *Daucus carota*

Carrots are the oldest cultivated vegetables and produce different colored varieties with different nutrients and flavors. It contains beta carotene, fiber, vitamin K1, potassium, antioxidants, and lutein. The presence of rich beta carotene maintains immunity, vision, and skin health by converting it into vitamin A in the body. Carrot oil is considered the bootstrapper of rich available vitamins A and E, which is beneficial for skin and hair and prevent wrinkles and aging [41]. It also produces ample vitamin C, producing collagen in the body. It is an essential protein for making the skin elastic and preventing wrinkles. It is also an excellent moisturizer with antiseptic and calming effects. It helps in detoxification to treat different skin problems. When mixed with sunflower, the roots of this plant can be a good anti-tanning agent. A study was conducted to determine the SPF of carrot oil as it contains high B-carotene concentration. Four oils, i.e., carrot seed oil, wheat germ oil, jojoba oil, and olive oil, are selected for B-carotene, a vitamin A precursor. It also contains vitamin B1, B2, B5, B6, and B9. Another study was executed with *Aloe vera*, *Cucumis sativus* and *Daucus carota* to prepare a polyherbal cosmetic cream. Oil in water preparations is incorporated with stearic acid and cetyl alcohol. The results of the studies showed no redness, edema, irritation, or inflammation on the skin of individuals, indicating that the formulation is stable and safe for the skin [42].

4.16 *Menta piperata*

Peppermint, an aromatic plant hybrid mint, is a combination of the flavors of spearmint and wild mint. It contains essential oils that contain menthone and carboxyl esters, mainly menthyl acetate, and dried peppermint contains volatile oil containing 1,8-cineol, menthone, menthol, menthofuran, and methyl acetate. Since ancient times, this herbal remedy has been used for various ailments, such as digestive problems, headaches, colds, and skin conditions. It helps reduce dandruff and the growth of hair lice due to its antimicrobial and antiseptic effects. It increases blood circulation in the scalp and can be used in the preparation of creams, shampoos, and various hair and skin products and also helps in preventing hair loss [43]

4.17 *Beta vulgaris*

Beetroot, the root vegetable, is deep red. It contains flavonoids, betalains (betaxanthins and betacyanins), saponins, polyphenols, and inorganic nitrate. It also contains potassium, zinc, sodium, copper, phosphorus, calcium, magnesium, iron, and manganese.

It contains flavonoids, betalains (e.g., betacyanins and betaxanthins), saponins, polyphenols, and inorganic nitrate. It also contains potassium, zinc, sodium, copper, phosphorous, calcium, magnesium, iron, and manganese. It has many health benefits, such as boosting stamina, reducing inflammation, and protecting cells from damage. Beetroot can be eaten raw, roasted, boiled, or juiced. Beetroot juice is rich in antioxidants and nutrients which improve skin health. When the phytoconstituents were investigated from their aqueous and methanolic extracts, they were rich, especially in quercetin, kaempferol, and phenolic compounds. Ointments were prepared at different concentrations and treated against acne and psoriasis, which showed a healing effect for acne. It was also observed that dried leaf extracts were more predominant than fresh leaf extracts. The dose prepared in the solution was more effective than ointment. The same result was obtained in the case of psoriasis, but in this case, the psoriasis ointment dose was more efficacious [44].

4.18 *Musa paradisiaca*

Beauty treatments, especially for skin and hair, are most effective with bananas. It contains Leucocyanidin (flavonoids). A small piece of banana skin placed on the wart with the yellow side out can treat warts. Banana peel can also help treat poison ivy rashes by rubbing the peel over the rashes. Rubbing peel on sore skin can increase discomfort. It helps reduce skin allergies, warts, poison ivy rashes, irritation, and inflammation. In such cases, a thick paste can be applied, reducing irritation and inflammation. Drinking smashed bananas in milk 2–3 times a day can treat skin allergies [23].

4.19 *Persea americana*

The avocado is an evergreen tree called an alligator or avocado pear. Due to the presence of vitamin C, vitamin E, carotenoids, and polyphenols, these compounds have an antioxidant effect, protecting cells from free-radical damage. It contains flavonoids, tannins, phenolic compounds, saponins and alkaloids, hydroxybenzoic acid, ferulic acid, catechin, caffeic acid, coumaric acid, chlorogenic acid, and triterpenoid glycosides. It also helps in the treatment of psoriasis, wrinkles, and stretch marks [45, 46].

4.20 *Calendula officinalis*

An annual and perennial herb with medicinal properties. Calendula CO2 extract acts as a well-standard cosmetic preparation, especially for the preparation of ointments for different skin diseases as well as for the preparation of cosmetics. It contains adinol, carotenoids, isorhamnetic, saponins, triterpenes, sesquiterpenoids, scopoletin, flavonoids, quercetin, and kaempferol. Anti-inflammatory principles of marigold by CO2 extraction are due to pentacyclic triterpene alcohols such as faradiol-3-monoesters and taxa sterols, which are the essential ingredients [47]. The extract was proven to be an effluent scavenger of hydrogen peroxide radicals under in vitro studies within the mitochondria of rat cardiac muscles. The section shows excellent antioxidant activity due to carotenoids, flavonoids, and polyphenols [48]. Water-alcohol extracts of marigolds were checked for their antimicrobial, antifungal, and antiviral properties against acne and dermal problems among young stars at the beauty and edicationsl centre "Top Beauty" in Sofia [49].

4.21 *Coffea arabica*

Coffee has been cultivated and consumed for centuries and contains caffeine, a stimulant that boosts energy and chlorogenic acid, which reduces redness when exposed to UV. It also has antioxidants and other beneficial compounds. Coffee can produce collagen and elastin and protect the skin from moisture loss. Coffee can protect against photo aging. According to the study by researchers, when hairless mice were exposed to UVB rays, and then caffeine applied to them 3 times/week for 11 weeks, they showed limited photodamage. This may be due to apoptosis (programmed cell death) of UVB, which removes damaged skin cells before they cause aging and skin cancer. An innovative facial cream has been designed that includes ingredients that help in improving the appearance of wrinkles, firmness, redness, and texture. According to different researchers, coffee bean extract rejuvenates skin aging. There were significant improvements in the formation of wrinkles, firmness, redness, and consistency of facial skin, and it also helps to prevent the development of aging and inflammatory skin disorders [50–53].

4.22 *Cucumis sativus*

It is a creeping plant widely cultivated for its edible fruit, which contains 95% water, 4% carbohydrates, and 1% protein with negligible fat content. It contains alkaloids, flavonoids, and a cyanogenic glycoside. *Cucumis sativus* extract mixed with *Glycyrrhiza glabra* extract and almond oil was evaluated for redness, edema, inflammation, and irritation. It was observed to be safe to use for skin [54]. The extracts of *Cucumis sativus* were investigated for their antimicrobial and antioxidant properties, and their cosmetic value in the treatment of acne shows the presence of phenolic and flavonoid compounds in fresh and dried plant extracts. An herbal cream prepared with fresh cucumber extract mixed with tea tree oil and linseed oil showed good spreadability, consistency, ease of removal, and no evidence of phase separation [55]. Herbal cream made of cucumber fruits, *Cyperus rotundus* roots, and almond oil were tested, showing no edema, redness, inflammation, or irritation during irritancy studies. These plant formulations are safe for use on the skin [56]. An emulsion of 3% cucumber extract and lemon oil was prepared to adjust the odor. The formulation was then evaluated to see the effects on sebum, erythema, melanin, moisture, and water loss on the skin of healthy human volunteers for 4 weeks. The odor was adjusted with a few drops of lemon oil. Over time, the smell disappeared due to the volatilization of lemon oil. The formulation shows eloquent effects on skin sebum secretion in which there was an increase in erythema while skin dehydration level and melanin in the skin decreased [57].

4.23 *Allium sativum*

It is an aromatic herb known for its biological properties and is a good source of antioxidants. It contains allicin, alliin, S-allyl cysteine, and methiin. When administered orally, it affects ultraviolet protection, cancer treatment, and immunologic properties.

Topically, garlic extract is effective against viral and fungal infections, alopecia areata, skin aging, psoriasis, keloid scar, leishmaniasis, and cutaneous horn. It also helps in skin rejuvenation and wound healing. However, the effectiveness of oral and topical garlic extract application is not yet appropriately explored [58]. An herbal shampoo was prepared with aqueous extract of *Allium sativum,* coconut oil, castor oil, and olive oil. The results of this combination help produce stable herbal shampoo with good foam formation, foam quality, and proper retention [59].

4.24 *Vitis vinifera*

Grapes are a rich source of antioxidants due to the presence of valuable phenolic compounds. It also contains flavonols, proanthocyanidins, flavon-3-ols, myricetin, flavonoids, peonidin, resveratrol, tannins, quercetin, anthocyanins, kaempferol, ellagic acid, and cyaniding. These phenolic compounds have anti-aging, anti-inflammatory, and antimicrobial actions and can permeate through the skin's barriers. The phenolic compounds present in grapes are the source of natural ingredients for cosmetics [60], fighting against acne [61], skin tightening and healing [62], reduction of dark circles surrounding eyes, hydration, anti-aging [63, 64], and UVB radiation protection [65, 66]. The active ingredients extracted from grapes can be used as cosmetics like skin-conditioning, antioxidants, flavoring agents, and colorants. The safety of concentration and use of these active ingredients from grapes has been confirmed by The Cosmetic Ingredient Review Expert Panel [67].

4.25 *Punica granatum*

The pomegranate is a fruit with a great deal of nutritional value, and its juice is an excellent source of antioxidants, vitamin C, and polyphenols, which may have health benefits in the future. It also contains gallic acid, ellagic acid, punicalagin's, and punicalins. Studies have shown that drinking pomegranate juice can improve heart health, reduce inflammation, and even boost memory. It is also a rich source of fiber, potassium, and vitamin K. A pomegranate extract, which is made from the peel, seeds, or flowers of the fruit, has been used in traditional medicine systems for a long to treat inflammations, infections, and cancers, among other conditions. An evaluation was carried out in a study in which creams were formulated with neem oil, jamul powder, and carrot powder at different concentrations based on the antioxidant properties of the phytoextracts and evaluated [68]. Upon evaluation of the extract and seed oil, it contained high levels of ellagic acid (peel, juice, seed extract) and punic acid (seed oil). Aside from these components, pomegranate derivatives contain other polyphenols that make them practical and valuable as anti-inflammatory and anti-aging agents for topical application. There are several advantages for using pomegranate seed oil, including its moisturizing and nourishing properties. Additionally, pomegranate seed oil can help reduce wrinkles and signs of aging and promote skin elasticity and firmness. It can also help to protect the skin from sun damage and environmental stressors. As a result, it is excellent for mature and aging skin, dry and cracked skin, and irritated and sunburnt skin. Pomegranate seed oil is also non-greasy and has anti-inflammatory properties, making it suitable for all skin types. It can also be a natural remedy to reduce acne, diminish blemishes and scars, and balance oily skin. This oil is used for treating acne, eczema, psoriasis, and rosacea and assisting in several specialist facials and body treatments. Additionally, pomegranate seed oil is a potent antioxidant, rich in fatty acids and vitamin C, which helps reduce inflammation and protect against free radical damage. It also helps to improve the skin's elasticity and can help to reduce wrinkles and fine lines. The combination of pomegranate and oil extracted from seed is highly effective in reducing and preventing fine wrinkles and firming up the epidermis [69]. It can also provide intense hydration and help to reduce inflammation while providing antioxidant protection. Pomegranate seed oil is also known to be a natural sun protectant and can help reduce age spots' appearance. Fermented juice and seed oil flavonoids of pomegranate were studied for antioxidant and eicosanoid enzyme inhibition properties. Pomegranate fermented juice (PFJ) and cold-pressed seed oil (PCPSO) showed vigorous antioxidant activity. A study of total phenols in PCPSO showed the presence of punicic acid (65.3%) along with palmitic acid, stearic acid, oleic acid, linoleic acid, and three unidentified peaks from which two peaks are probably isomers of punicic acid [70]. A study was conducted to check the sunscreen activity of combined *Pongamia pinnata* leaf extract and *Punica granatum* peel extract. To determine the sun protection factor, absorption spectroscopy, transmission spectroscopy, and the COLIPA standard method were used. Absorption spectroscopy measures the amount of UV radiation absorbed by the sunscreen, while transmission spectroscopy measures the amount of UV radiation transmitted through the sunscreen. The COLIPA standard method measures the amount of UV radiation reflected by the sunscreen. Combining the results of these three methods allows the sun protection factor to be accurately determined. Evidence suggests that sunscreens do a better job of protecting against UVA and UVB rays following the study by Patil and Fegade [71]. This is because the COLIPA standard method is designed to measure the radiation reflected by the sunscreen rather than the amount absorbed. This allows for a more accurate measurement of the sun protection factor, vital in protecting skin from the harmful effects of UVA and UVB rays.

4.26 *Cucurbita maxima*

The pumpkin is a fruit rich in nutrients, antioxidants, and fiber and has various health benefits, especially for the eyes, heart, skin, and immune system. Beta-carotene, an antioxidant in pumpkin, helps protect the cells from damage from free radicals and can reduce inflammation. It also contains vitamin A, which helps maintain healthy vision, and vitamin C, which helps strengthen the immune system. It contains peptides, fixed oils, carotenoids, polysaccharides, para-aminobenzoic acid, proteins, γ-aminobutyric acid, and sterols. Additionally, due to the presence of fiber in pumpkin, it helps to keep the gut healthy and can reduce the risk of certain diseases. Pumpkin also contains zinc, phosphorus, iron, magnesium, calcium, ferulic acid, alpha-linolenic acid, vitamin A, and vitamin B. The vitamins and minerals in pumpkins are essential for maintaining healthy vision, skin, cardiovascular, and immune health. The vitamins A and B are necessary for healthy skin and eyes, while zinc, magnesium, iron, phosphorus, and calcium are essential for bone, muscle, and heart health. Ferulic acid and alpha-linolenic acid's antioxidant and anti-inflammatory properties also benefit overall health. In addition to these nutrients, it contains other components such as copper, niacin, thiamin, riboflavin, salicylic acid, ferulic acid, unsaturated oils, retinol, antioxidants, and beta-carotene. As a powerful antioxidant and mild alternative to retinoic acid, pumpkin is an exfoliation accelerator. Zinc in pumpkin helps in wound healing by boosting the immune system and promoting the production of collagen in the skin, which helps keep the skin elastic and reduces the appearance of wrinkles. Beta-carotene and retinol are both antioxidants that help to protect the skin from damage caused by free radicals. The salicylic acid and unsaturated oils help to exfoliate the skin and reduce the appearance of pores. In addition to being rich in vitamin A derivatives, pumpkin also contains retinoids that hold a high affinity for the retinoic acid receptors in the skin, activating the cellular turnover process. In addition to its keratolytic properties, pumpkin enzymes are an excellent non-abrasive preparation for much stronger exfoliations while at the same time reducing the oxidative and free radical stress associated with chemical exfoliants [72]. Retinoids are a form of vitamin A that helps to reduce wrinkles, improve skin firmness, and even out skin tone. Pumpkin's retinoids target the retinoic acid receptors in the skin, which helps to promote skin cell turnover and reduce oxidative and free radical damage. This makes pumpkin enzymes an ideal choice for gentle exfoliation that can produce results similar to more potent chemical exfoliants.

4.27 *Camellia sinensis*

Green tea is one of the richest sources of antioxidants and catechins. Other compounds like theaflavins, Quercetin, L-Theanine, GABA, statin, and aromatic volatiles are also present. Furthermore, it has astringent properties as well. The raw material of green tea has a wide range of applications as a raw material. The catechins in green tea are known to have anti-inflammatory and antibacterial properties and are great for cardiovascular health. The astringent properties help to reduce inflammation, soothe skin, and even help to reduce wrinkles. Additionally, the raw material of green tea can be used in various ways, from skin care products to herbal teas. It also contains anti-inflammatory, antibacterial, and antiviral properties capable of boosting the immune system and protecting the body from infections [73]. The astringent properties of green tea work to tighten and tone the skin, which can reduce the appearance of wrinkles and other signs of aging. The antibacterial and anti-inflammatory properties can help reduce redness and inflammation, while the antiviral properties can help protect against viruses and other pathogens. As a result of the presence of polyphenols and other ingredients in the product, it may also provide some skin benefits. The polyphenols in green tea can neutralize free radicals, which are molecules that can cause cellular damage. This can lead to both anti-aging and skin protective benefits. Additionally, green tea can help to reduce inflammation and redness from skin conditions such as acne or rosacea. The antibacterial and antiviral properties of green tea can also help to reduce the risk of bacterial or viral infections on the skin. It has been reported that the formulations of green tea have been evaluated regarding the corneum transepidermal water loss status: water content, viscoelastic-to-elastic ratio (UV/UE), and skin microrelief. According to the results revealed from applying green tea extract for an extended period, this extract has a prolonged moisturizing effect on the skin. This can be attributed to the polyphenol and proanthocyanidin compounds present in green tea, which have antioxidant properties that help to protect the skin from environmental damage. Additionally, the polysaccharides in green tea help increase the skin's water content and create a protective barrier that helps lock in moisture. The skin microrelief improved significantly due to the topical application of the formulation [74]. The moisturizing effect of the green tea extract was due to its ability to increase the water content of the stratum corneum and reduce the transepidermal water loss. This resulted in a higher viscoelastic-to-elastic ratio (UV/UE) and improved skin microrelief. These effects were long-lasting, indicating the potential of green tea extract to be used as a moisturizing ingredient.

4.28 *Curcuma longa*

Curcumin is one of the main components of turmeric, a common spice known for its antioxidant and anti-inflammatory properties, which can help with various ailments such as arthritis and digestive issues. It also contains saponin, alkaloid, sterol, tannin, flavonoid, phytic acid, and phenol. It is also believed to have cancer-fighting properties. An evaluation of the effect of a cream prepared from turmeric extract on skin sebum secretion in humans was conducted using a sebumeter to measure the impact of the cream on human volunteers. The results showed that the cream had an anti-sebum effect and improved the skin's elasticity, which is beneficial in reducing wrinkles and other signs of aging. The anti-inflammatory and antioxidant properties of curcumin were also demonstrated, as the cream improved skin hydration and decreased the appearance of wrinkles. The DPPH method was used to determine the extract's antioxidant activity. In the 6th week after applying the cream, there was a significant increase in the observed sebum values. This suggests that the curcumin extract was able to penetrate deep into the skin and have a positive effect on the sebum levels, which helps to maintain healthy skin and reduce wrinkles. Additionally, the DPPH method is often used to measure antioxidant activity, and the findings suggest that the curcumin extract had a significant positive effect [75].

4.29 *Crocus sativus*

The spice of saffron is a precious spice, an expensive and rare spice that has been used in medicine for centuries. There are three main constituents of saffron, namely, croccin, picrocrocin, and crocetin, as well as safranal. It is harvested from the crocus flower and has many medicinal properties. It treats various illnesses, such as depression, anxiety, and insomnia. Saffron is also used in cooking to flavor and color dishes. Saffron is also known to have inflammatory and antioxidant properties, making it a voguish component in traditional medicines. It is also used in perfumes and cosmetics and is widely used in the Middle East and India. It is used to evaluate the antioxidant properties of saffron by using the DPPH (2, 2-Diphenyl-1-picrylhydrazyl) method (2, 2-Diphenyl-1-picrylhydrazyl). Using a hexameter, a water-in-oil (w/o) formulation of this plant extract was evaluated in human volunteers for the amount of melanin and erythema in their skins. A significant depigmentation and anti-erythematic effect were observed on human skin due to these results. In addition, applying the saffron formulation may also be beneficial in treating melanoma in humans. Further studies are necessary to assess the effect of *C. sativus* extract on melanoma. Animal studies can also be conducted to understand the mechanism of action better. Additionally, the safety of long-term use of the extract should be determined [76].

4.30 *Fragaria vesca*

Strawberries are not actual berries but are the richest source of vitamin C, manganese, folate, potassium, antioxidants, polyphenols, anthocyanins, and vitamins. Anthocyanins, ellagic acid, flavonols. The potential photoprotective capacity of different strawberry-based formulations was assessed in human dermal fibroblasts (HuDe) exposed to UV radiation, which showed noticeable photoprotection in HuDe, increasing cell viability due to the presence of Coenzyme Q10 (antioxidant) [77]

4.31 *Passiflora incarnata*

It is a flowering tropical vine which gained a lot of attention as it is a powerful source of antioxidants. Seed Oil is a superior salve, rich in vitamin C, calcium, riboflavin, niacin, iron, and phosphorus. Due to the high amount of fatty acids and liquid texture, a wide array of personal care products to nourish oily and dry skin are prepared from the seed oil of passion fruit. It increases vitality and improves hair growth. It can also be included in bath care products as it has a calming, sedating effect, and promotes relaxation. The oil is included in massage applications due to its anti-inflammatory, anti-spasmodic, and sedative properties [78].

4.32 *Petroselinum crispum*

This plant is generally considered a versatile herb that provides a concentrated source of nutrients. Rice in vitamins, minerals, and antioxidants helps boost the immune system and protect against illness. It also contains Flavonols (kaempferol and quercetin) and flavones (apigenin and luteolin). Additionally, it has anti-inflammatory properties that help in reducing puffiness and swelling. Many cosmetic industries use parsley to freshen the face, minimize discoloration, wrinkles, dark age spots, and freckles, prevent red sites, and strengthen hair. Several in vitro studies have shown that the herb has oil-controlling properties [79]. It can also be used as an antiseptic and can be used to treat acne. Parsley also has antimicrobial properties, which can help to protect the skin from infection.

4.33 *Theobroma cacao*

Cocoa beans are well-known, as they are rich in polyphenols. It contains carboxylic acid, phenolic acid, fatty acid, flavonoids (flavonol and flavones), stilbenoids, and terpenoids. Cocoa beans are a rich source of antioxidants and phytonutrients, linked to reducing inflammation, improving heart health, and promoting healthy blood sugar levels. They are also a good source of fiber and minerals such as magnesium, manganese, and zinc. To measure the antioxidant capacity of cocoa pods, a UHPLC extraction method was used to extract from cocoa pods that shows the quantification of the total polyphenols and flavonoids present in the cocoa pods. This method allowed researchers to determine cocoa beans' antioxidant capacity and their beneficial effects on the human body. To assess the impact of pod extract on skin whitening, elastase assays, collagenase assays, and sun screening tests were conducted to determine if they would inhibit the action of skin-degrading enzymes, and mushroom tyrosinase assays were performed to determine if they would impede the movement of skin-degrading enzymes. The elastase, collagenase, and sun screening assays showed that the pod extract had an inhibitory effect on skin-degrading enzymes, indicating that it could have a whitening effect. The results of the mushroom tyrosinase assay showed that the pod extract inhibited the activity of the mushroom tyrosinase, which further supported the idea that it could potentially have a whitening effect. The study results revealed that cocoa extract had antioxidant properties and showed a more decisive action against enzymes such as elastase and collagenase than other extracts. This is important because these enzymes can break down collagen and elastin, essential proteins that keep skin looking firm and youthful. By inhibiting the activity of these enzymes, the cocoa extract could help to protect the skin and reduce the signs of aging. Despite its limited performance as a UVA sunscreen agent, CPE also showed potential as a UVB sunscreen. The cocoa extract contains a type of polyphenol known as catechin polyphenols (CPE). Polyphenols have been shown to inhibit certain enzymes that break down collagen and elastin. By inhibiting these enzymes, CPE helps protect the skin from damage and reduce the signs of aging. In addition, CPE also has UVB protection properties, although its performance in this area is limited. CPE was found to contain carboxylic acid, phenolic acid, fatty acid, flavonoids (flavonol and flavones), stilbenoids, and terpenoids when LCMS analysis was conducted [80]. There are scientific validations for using cocoa-derived phytoconstituents as a practical approach for skin protection [81]. Cocoa butter is a common ingredient in different skin moisturizers, but cocoa's beneficial effects on the skin may extend beyond its topical agent. According to some studies, it has been found that cocoa flavonoids protect the skin from UV damage caused by the sun's UV rays [82]. Cocoa flavonoids contain antioxidants that have been found to reduce inflammation and improve skin elasticity. The appearance of dark spots and wrinkles is also reduced. In addition, these flavonoids can help protect against further skin damage caused by UV rays.

4.34 *Psidium guajava*

Guava is a tropical fruit full of nutrients and contains fibers, vitamin C, and folic acid. The leaves have inflammatory action and potent antibacterial activity. Acne and black spots can be eliminated with guava leaf extract. Guava leaves blended with water can be used as a scrub to remove blackheads. Due to its antioxidant properties, it protects the skin from damage and improves texture, skin tone, and skin tightening. The leaves can also give an instant cure for itchiness as they contain allergy-blocking compounds. An evaluation was done with the essential oil of guava against four dermatophytes, namely, *Microsporium canis, Trichophyton rubrum*, *T. tonsurans,* and *T. verusossum,* that were isolated from the infected part of the nail, skin, scalp, and genital organs of patients from the district hospital of Bareilly. Excellent inhibitory effects were given by the oil against all the pathogenic dermatophytes [83]. It has also been reported that aqueous extracts of guava leaves inhibit the growth of pathogenic microbes like *Proteus mirabilis,* *Streptococcus pyogenes, Escherichia coli, Staphylococcus aureus,* and *Pseudomonas aureginosa,* which is more effective than the organic extracts. These results also suggest that aqueous extracts can be used as a natural antibacterial agent, providing a potential alternative to synthetic antibiotics. More research is necessary to confirm the efficacy of this method. Furthermore, the extract acne can also be used to formulate oral antibacterial drugs that are used to treat surgical and skin infections [84]. Extracts from medicinal plants can treat many bacterial infections and may provide an effective alternative to synthetic antibiotics. Further research is needed to confirm the efficacy of this method. Various herbal plants containing different metabolites and their uses are mentioned below in Tables 2, 3, and 4.

5 Conclusion

Phytoherbal remedy is an art of healing with plant-based medicine, penetrating deeply into Indian culture. With new emerging technology and modern medicinal practices, people from every part of the world depend on herbal healing. It is increasing daily due to its admiration, lower cost, less toxicity, few side effects, and widespreadness in our natural ecosystem. Moreover, to fulfill the needs of the growing human population and their deteriorating health conditions, nature is contributing an immense amount of herbs for the survival of human beings and other organisms on the planet. Phytomedicines provide tremendous options for curing various skin infections. Different herbal medicines have been tested against skin infections like rashes, acne, eczema, psoriasis, pigmentation, fungal, viral and bacterial infections, skin cancers, inflammation, cuts, wounds, etc. In these presented chapters, there is a mention of different plant-based products, from tulsi, neem, jatropha, ginger, garlic, cucumber, grapes, pomegranate, cucumber, pumpkin, turmeric, tea, passion fruit, papaya, etc., to treat various skin problems. There are mentions of multiple phytoconstituents from those plants that help treat different skin remedies. Consequently, the demand for herbal cosmetics has been increasing tremendously. Healthy and natural skin maintenance has become vital as it may lead to many severe conditions. Cosmetics can be prepared from a single plant or combinations of plant parts and polyherbal formulations. These formulations can be further explored against harmful microorganisms that cause severe health hazards. A comprehensive screening of these natural herbs can be a new commanding and valuable target for developing herbal-based industries that tie up with pharmaceutical companies. The positive effects of herbal medicines may help researchers reduce skin infections and screen out newer leading molecules, appropriate approaches and new processes in drug discovery.

Table 2 Applications of herbal phytomedicines in skin disorders

Sl no	Common name	Botanical name and family	Useful against skin infections	Parts used
1	Neem	*Azadirachta indica*, Meliaceae	Leaves paste in treating acne, eczema, scabies, skin allergies, psoriasis	Leaves, barks
2	Licorice	*Glycyrrhiza glabra*, Leguminosae	Skin lightening, anti-inflammatory, depigmentation, atopic dermatitis, protect skin from UV radiation, and reduce melanin production, allergic dermatitis	Roots
3	Tulsi	*Ocimum santhum,* Labiatae	Cure rashes, scars, leucoderma, allergy, blisters, reduce inflammation, and boost immune system, cure chickenpox, insect bites, round worm infections	Leaves
4	Bitterweed	*Andrographis paniculata*, Acanthaceae	Eczema, leucoderma, cuts, and wounds	Aerial parts, roots, and leaves
5	Amla	*Embillica officinalis,* Euphorbiaceae	Treatment of warts, skin infections, and prevents premature aging, reduces itching, wrinkles, and scabies	Fruit
6	*Aloe vera*	*Aloe barbadensis,* Lilaceae	Helps in healing wounds, burns, sunburns, insect bites, cuts, itching swelling, scabies, and dandruff	Leaves
7	Papaya	*Carica papaya,* Caricaceae	Eczema, warts	Roots, pulp, bark, seeds, and peels
8	American purging nut	*Jathropha curcus,* Euphorbiaceae	Major skin lesions, wound healing	Fruits, seeds, and roots
9	Ginger	*Zingiber officinalis,* Zingiberaceae	Skin problems	Root and rhizomes
10	Eucalyptus	*Eucalyptus globulus,* Myrtaceae	Heals fungal infection, skin problems, and healing of wounds	Leaves
11	Asthma plant	*Euphorbia hirta,* Euphorbiaceae	Major wounds, boils, swelling, and acne	Root
12	Common fig	*Ficus carica,* Moraceae	Hyperpigmentation, wrinkles, itchy skin and skin irritation, eczema, skin ulcers, and acne	Fruits
13	Onion	*Allium cepa,* Amaryllidaceae	Skin allergy, anti-aging, scars, and stimulates blood circulation	Bulbs
14	Rosary Pea	*Abrus precatorius,* Fabaceae	Leucoderma, gray hair prevention, and wound healing property	Leaves
15	Carrot	*Daudus carota,* Apiaceae	Hair and skin moisturizer, eczema, and warts	seeds

(continued)

Table 2 (continued)

Sl no	Common name	Botanical name and family	Useful against skin infections	Parts used
16	Peppermint	*Menta piperata,* Labitae	Reduces dandruff and lice	Seeds
17	Beetroot	*Beta vulgaris,* Chinopodiaceae	Acne, psoriasis	Roots and leaves
18	Banana	*Musa paradisiaca,* Musaceae	Skin allergies, warts, poison ivy rashes, irritation, and inflammation	Peel and fruit
19	Avocado	*Persea americana,* Lauraceae	Psoriasis, wrinkles, and stretch marks	Fruit
20	Pot marigold	*Calendula officinalis,* Asteraceae	Anti-inflammatory	Flowers and leaves
21	Coffee	*Coffee arabica, C. canephora*	Wrinkles, firmness, and redness	Leaves
22	Cucumber	*Cucumis sativus,* Cucurbetaceae	Photoaging, skin disorders	Fruit
23	Garlic	*Allium sativum,* liliaceae	psoriasis, alopecia areata, keloid scar, and wound healing	Bulb
24	Grape	*Vinis vitifera,* vitaceae	Removal of fungal and viarl infections, leishmaniasis, control aging of skin, and rejuvenates skin	Berry
25	Pomegranate	*Punica granatum,* Lythraceae	Antioxidant	Seeds
26	Pumpkin	*Cucurbita maxima,* Cucurbetaceae	Aid in the healing process and exfoliators	Fruit
27	Tea	*Camellia sinensis,* Theaceae	Antiaging, antioxidant and astringent activity, and skin roughness (62)	Leaves
28	Turmeric	*Curcuma longa,* Zingiberaceae	Skin sebum secretion	Rhizomes
29	Saffron	*Crocus sativus,* Iridaceae	Melanin and erythema, management of melanoma	Flower
30	Strawberry	*Fragaria ananassa,* Rosaceae	Antioxidant, photoprotection	Fruit
31	Passion fruit	*Passiflora edulis,* Passifloraceae	Hair growth, treating dry skin, treating muscular aches, and swelling	Fruit seed
32	Parsley	*Petroselinum crispum,* Apiaceae	Removes dark age spots, face discoloration, freckles, wrinkles, and prevents red spot occurence, growing and strengthening hair, control oils	Leaves

(continued)

Table 2 (continued)

Sl no	Common name	Botanical name and family	Useful against skin infections	Parts used
33	Cocoa Bean	*Theobroma cacao,* Malvaceae	Sunscreen, antioxidant properties,	Pods
34	Guava	*Psidium guajava* Myrtaceae	Acne, dark spots, and blackheads	Leaves

Source: [5, 6]

Table 3 Constituents and mechanism of action of herbal plants [5, 6]

Sl no	Common name	Chief constituents	Mechanism of action
1	Neem	Azadirachtin, limonoides, vitamin C, glycerides, polyphenols, triterpenes, quercetin, chtechins, carotenes, and beta-sitesterol	The product contains essential fatty acids and during healing add moisture and texture to the akin thereby inhibiting the inflammatory cell growth
2	Licorice	Glycerrihizin, liquiritin	Tyrosin B16 inhibition of melanoma cells by dispersed tyrosin B16
3	Tulsi	Contanis ursolic acid, rosmarinic acid, oleanolic acid, linalool, carvacrol, β-caryophyllene, and eugenol	Blocks cylcoxygenase lipoxygenase pathways of arachidonic acid metabolism and immune response enhancement
4	Bitterweed	Diterpenes, lactones flavonoids, alkenes, ketonaldehydes, and andrographolide	It inhibits the inflammatory mediators nitrous oxide release (NO), interleukin (IL), and prostaglandin E2 (PGE2)
5	Amla	Ascorbic acid, polyphenols (ellagic acid, chebulinic acid, gallic acid, chebulagic acid, apeigenin, quercetin, corilagin, leutolin, etc.)	Inhibits adhesion of microbes, causes inactivation of enzymes. Influenza A virus suppression and prevention of viral absorption, treats HUV by inhibiting HIV-RT
6	*Aloe vera*	Anthraquinones, phenolics, flavonoids, chromones, anthrones, steroids, alkaloids and tannins anthraquinones, phenolics, chromones, anthrones, flavonoids, steroids, alkaloids, and tannins	The proliferation cells of skin, blood cells production, enhancement of elastin and collagen production, suppresses the immunity of skin, self-hydration of skin, and removal of dead skin cells due to penetration
7	Papaya	Papain, phenolic acids, chymopapain, cyanogenic glucosides, tocopherol, cystatin, glucosinolates, and vitamin C	Increases prothrombin, removes necrotic tissues
8	American purging nut	Tannins, flavonoids, phenols, saponins, lignoides, steroids, coumarins, terpenoides, and alkaloids	Phagocytosis of bacterial cells, during proliferative phase release of cytokine can cause the migration and division of cells

(continued)

Table 3 (continued)

Sl no	Common name	Chief constituents	Mechanism of action
9	Ginger	Carbohydrates, lipids, terpenes, and phenolic compounds	Cytokinin cyclooxygenase and nitric acid is inhibited, inhibits nuclear factor kappa B (NF-Kb) cells
10	Eucalyptus	Sideroxylonal C, (+)-oleuropeic acid, cypellocarpins A, B, and C, cypellogins A, B, and C, leptospermone, isoleptospermone, grandinol	Inhibits gram-negative and gram-positive bacteria by interaction of trichosolan with enzymes of the fatty acid pathway
11	Asthma plant	Alkaloides, flavonoids, terpenes, chromenes, and sterols from all the plant parts	Serum cholesterol triglycerides and urea reduction in alloxan-induced m ice, due to antioxidant and free radical scavenger
12	Common fig	Phenolic acid and flavonoides	Inhibits tyrosin, reduces melanin content
13	Onion	Kaempferol, muricetin, quercetin, quercitrin, and allyl sulfide	Inhibits ultraviolet B (UV-B), inhibits free radical scavenger.
14	Rosary Pea	Luteolin O-rutinoside, orientin, abrectorin, isoorientin, desmethoxycentaviridin-7-, abrusoside A to D, abrusogenin, glycyrrhizin, and abruquinones D, E, and F	Collagen synthesis enhancement through elevated hydroxyproline
15	Carrot	Beta carotene, fibrem vitamin K1, potassium, antioxidants, and lutein	Acts on epidermal cells
16	Peppermint	Essential oils of peppermint contains menthone and carbaryl esters, mainly menthyl acetate while dried peppermint contains volatile oil containing menthyl acetate, menthone, menthofuran, menthol, and 1,8-cineol	Improvement in the circulation of bold in scalp
17	Beetroot	Contains betalains (e.g., betaxanthins and betacyanins), saponins, flavonoids, inorganic nitrate, and polyphenols. It also contains sodium, potassium, phosphorus, calcium, magnesium, copper, iron, zinc, and manganese	–
18	Banana	Leucocyanidin (flavonoids)	Reduce skin irritation and inflammation
19	Avocado	Flavonoides, saponins, phenolics, tannins, alkaloids, hydroxybenzoic acid, catechin, chlorogenic acid, ferulic acid, coumaric acid, caffeic acid, and triterpenoid glycosides	Protect cells from free radical damage
20	Pot marigold	Adinol, carotenoids, isorhamnetin, saponins, triterpenes, sesquiterpenoids, scopoletin, flavonoids, quercetin, and kaempferol.	Effective scavenger of H2O2 radicals, change in malondialdehyde level, superoxide dismutase glutathione, ascorbic acid, and catalase

(continued)

Table 3 (continued)

Sl no	Common name	Chief constituents	Mechanism of action
21	Coffee	Chlorogenic acid	Reduce redness on skin due to ultraviolet exposure, increase apoptosis (programmed cell death)
22	Cucumber	Alkaloids, flavonoids, and a cyanogenic glycoside	Skin sebum secretion
23	Garlic	Alliin, methiin and S-allylcysteine, and allicin	Immunologic properties, cutaneous microcirculation, and protection against UVB
24	Grape	Presence of phenolic acids, flavon-3-ols, myrcetin, flavonols, flavonoids, resveratrol, quercetin, tannins, ellagic acid, kaempferol, anthocyanins, cyaniding, and proanthocyanidins peonidin	Skin protection against UVB radiation, antioxidant, antiaging, and skin tightening
25	Pomegranate	Gallic acid, ellagic acid, punicalagins, and punicalins	Antioxidant potential
26	Pumpkin	Carotenoides, polysaccharides, paraaminobenzoic acid, fixed oils, peptides, proteins, sterols, and γ- aminobutyric acid.	Exfoliation accelerator, activation of cell turnover
27	Tea	Catechins, theaflavins, quercetin, L-Theanine, GABA, statin, and aromatic volatiles	Antioxidant and astringent activity
28	Turmeric	Saponin, alkaloid, sterol, tannin, flavonoid, phytic acid, and phenol.	Skin sebum secretion
29	Saffron	Crocin, Picrocrocin, and safranal	Change in the levels of skin erythema and melanin, management of melanoma
30	Strawberry	Vitamin C, manganese, folate, potassium, antioxidants, polyphenols, anthocyanins, and vitamins. Anthocyanins, ellagic acid, and flavonols	Photoprotective capacity, increasing cell viability
31	Passion fruit	Riboflavin, niacin, iron, and phosphorus	Anti-inflammatory, anti-spasmodic, and sedative properties
32	Parsley	Flavonols (kaempferol and quercetin), flavones (apigenin and luteolin)	Control oil secretion on the skin
33	Cocoa bean	Carboxylic acid, flavonoids (flavonol and flavones), stilbenoids, fatty acid, phenolic acid, and terpenoides	Higher inhibition toward tyrosinase enzyme, shows better action against collagenase and elastase enzymes
34	Guava	Fibers and vitamin C, Folic acid	Destroy the free radicals that damage skin

Source: [5, 6]

Table 4 List of some other phytomedicines

Sl no	Skin care products	Scientific name and family	Phytoconstituents	uses	Reference
1	Coconut oil	*Cocos nucifera*, Arecaceae	Glycerides and lower chain fatty acids	Baking and cooking, used as hair oil	[89]
2	Sunflower oil	*Helianthus annuus*, Asteraceae	Lecithin, tocopherols, carotenoids, and waxes	Products prepared for face and body	[90]
3	Jojoba oil	*Simmondsia chinenesis*, Simmondsiaceae	Linear liquid wax esters, alcohols, fatty acids, sterols, and vitamins	Used as moisturizer exotic fragrances carrier oil, removes odor	[91]
4	Olive oil	*Oleaeur opaea*, Oleaceae	Triolein,tripalmitin, trilinolein, tristearate, monosterate, triarachidin,squalene, β- sitosterol, and tocopherol	Skin and hair conditioners like lotions, shampoos, etc.	[85]
5	Rhodiola rosea	*Lignum rhodium,* Crassulaceae		Enhance physical endurance, increase in work productivity, longevity, resistance to high altitude sickness, treatment of fatigue, anxiety, anemia, impotence, gastrointestinal problems, skin disorders and infections, and nervous system disorders	[92]
6	Ginkgo	*Ginkgo biloba*, Ginkgoaceae	Glycosides, mostly quercetin and kaempferol derivatives, and terpenes	Antioxidant, anti-inflammatory, reduce UBV sunrays	[93]
7	Henna	*Lawsonia inermis*, Lythraceae	Gallic acid, resin, mannitol, fats, mucilage, glucose, and alkaloid traces	Henna powders are produced for hair application and mehendi designs on hands and legs	[94]
8	Shikakai	*Acacia concinna*, Leguminosae	Saponins, flavonoids, tannin, alkaloid, sugar, and anthraquinone glycosides	Hair cleaning and washing, hair growth improvement, purgative, expectorant, and emetic	[95]
9	Tamarind	*Tamarindu syndical*, Fabaceae	Rich source of sugars, contain minerals and excellent source of vitamin B, contains high antioxidant phenolic compounds	Improves skin health	[96]

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Metabolites and Phytochemicals in Medicinal Plants Used in the Management and Treatment of Neurological Diseases

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Contents

Abstract

In medicine, neurological disorders can be well defined as disorders which affect the brain and nerves found all through the body of humans as well as the spinal cord. Structural, biochemical or electrical abnormalities in terms of biochemistry,

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electrical or structure in the brain, spinal cord or other nerves can lead to a range of symptoms. Some good examples of such symptoms include poor or loss of coordination and sensation, seizures, muscle weakness, paralysis, pain, confusion and an altered sense of consciousness. There are many causative factors when it comes to neurological disorders, this may include lifestyle, genetics, congenital abnormalities, brain injury, infections and nerve and spinal cord injuries. Neurological disorders include but are not limited to the following: Alzheimer's disease, brain aneurysm, epilepsy and seizures, meningitis, multiple sclerosis, muscular dystrophy, Parkinson's disease, stroke, migraine, headaches and encephalitis. Plants used in the treatment and management of neurological disorders include *Andrographis paniculata*, *Areca catechu*, *Centella asiatica*, *Sesamum indicum*, *Cinnamomum cassia*, *Euphorbia tirucalli*, *Piper methysticum*, *Areca catechu*, *Angelica sinensis*, *Centella asiatica*, *Aegle marmelos*, *Gardenia jasminoides*, *Tripterygium wilfordi*, *Xylopia aethiopi*, *Curcuma longa*, *Panax ginseng*, *Garcinia kola*, *Glycine max*, *Schizandrae chinensis*, *Ginkgo biloba*, *Piper nigrum*, *Withania somnifera*, *Cyperus rotundus*, *Suctellaria baicalensis*, *Silybinisus laborinum*, etc. these plants possess some key phytochemical groups such as Phenolics, alkaloids, saponins, flavonoids, triterpene, glyco withanolides, sesquiterpenes, lignans, neolignans, etc. While the active constituents/metabolites include curcumin, wogonin, flavonoid, silibinin, stilbenoid/resveratrol, hyperforin, rutin, ferulic acid, paeoniflorin, safranal, crocin, naringenin, berberine, chrysin, thymoquinone, nardostachysin, jatamansic acid, jatamansone, etc. These phytochemicals have been reported by many researchers to possess and exhibit neuroprotective properties. The plants discussed in this chapter as well as their active constituents provide great insight into affordable means of treating and managing neurological disorders as well as a basis for pharmaceutical companies searching for novel sources for the production of neuroprotective pharmacotherapy drugs.

Keywords

Brain · Curcuma longa · Curcumin · Metabolites · Naringenin · Neurological disorder · Panax ginseng · Phytochemicals

1 Introduction

Neurological diseases or disorders can be described as diseases/disorders which particularly affect or disturb the brain, spinal cord as well as nervous coordination which run throughout the human body. Distortions or abnormalities which can be biochemical, structural or electrical as the case may be in the brain, nerves or spinal cord can create an assortment of symptoms. These symptoms include seizures, poor coordination, paralysis, confusion, muscle weakness, altered levels of consciousness, loss of sensation and pain [1].

There are a great number range of specific causes of neurological diseases/ disorders. The prominent of these causes are genetic disorders, contagions, deformities or disorders, malnutrition which could be linked to the health of one's environment, lifestyle, serious injuries to the brain, nerves as well as the spinal cord are also amongst the common causes of neurological diseases.

Let's also note that there's a big difference between neurological disorders and mental disorders (which could also be referred to as psychiatric illnesses, mainly deformities of ones way of thinking, behaviour as well as feeling ultimately resulting in anguish or diminishing of function). On the other hand, there are numerous neurological diseases, very few are rare ones while others are very common [1]. Figures from the US National Library of Medicine put the number of neurological disorders at 600.

Disabilities associated with neurological diseases/disorders are numerous, including cerebral palsy, brain tumours, epilepsy, autism, learning disabilities, ADD and neuromuscular disorders. A few of these disorders appear from childbirth thus making them congenital. Aside neurological diseases being congenital, some are actually caused by trauma, physical defects, tumours, degeneration and contagions all of which causes injury to the nervous system which will ultimately determine the magnitude of vision, communication, cognition, movement and hearing impairment [1].

The use of several synthetic agents and medication with neuroprotective properties has been properly documented. However, there are some discomforting side effects associated with these synthetic neuroprotective medications which include fretfulness or nervousness, fatigue, dry mouth, tiredness, drowsiness, struggle with balance, etc. However, the use of plants and herb with neuroprotective properties in recent times has gained prominence globally. This has led many plant scientists globally to focus on and research more into the metabolites and phytochemicals present in the said plants and their mechanism of neurological moderation and neuroprotective conferment against neurodegeneration.

Plants used in the treatment and management of neurological diseases and disorders include but are not limited to *Andrographis paniculata*, *Areca catechu*, *Centella asiatica*, *Sesamum indicum*, *Cinnamomum cassia*, *Euphorbia tirucalli*, *Piper methysticum*, *Areca catechu*, *Angelica sinensis*, *Centella asiatica*, *Aegle marmelos*, *Gardenia jasminoides*, *Tripterygium wilfordi*, *Xylopia aethiopi*, *Curcuma longa*, *Panax ginseng*, *Garcinia kola*, *Glycine max*, *Schizandrae chinensis*, *Ginkgo biloba*, *Piper nigrum*, etc. Some key phytochemical groups in the plants listed above include Phenolics, alkaloids, saponins, flavonoids, triterpene, glyco withanolides, sesquiterpenes, lignans, neolignans, etc. While the active constituents/metabolites include curcumin, wogonin, flavonoid, silibinin, stilbenoid/resveratrol, hyperforin, rutin, ferulic acid, paeoniflorin, safranal, crocin, naringenin, berberine, chrysin, thymoquinone, nardostachysin, jatamansic acid, jatamansone, etc.

According to Phani et al. [2], saponins, phenolic acids, flavonoids, polyphenols, terpenoids, etc. are regarded as the most essential phytochemicals from dietary sources. These phytochemicals work by decreasing or by retrogressive cellular injury and by decelerating advancement of neuronal cell loss.

This chapter with explicitly discuss the numerous medicinal plants that have been used in the management of neurological diseases and disorders globally already. It will also highlight the phytochemical groups and bioactive constituents/metabolites and their mechanism of neurological moderation and neuroprotective conferment against neurodegeneration as well as providing the basis for pharmaceutical companies searching for novel sources for the potential production of neuroprotective pharmacotherapy drugs.

2 Types of Neurological Diseases

Several disorders of the nervous system exist, which require professional healthcare physicians. Some of these disorders include: Alzheimer's disease, brain aneurysm, epilepsy and seizures, meningitis, multiple sclerosis, muscular dystrophy, Parkinson's disease, stroke, migraine, headaches, encephalitis, etc.

2.1 Alzheimer's Disease

According to John Hopkins Medicine [3], over 5.2 million people in America are affected with Alzheimer's disease particularly those above 65 years of age as well as thousands under the said age as well. Alzheimer's disease which is also regarded as the most common form of dementia affects about two-thirds of women as well.

The motor function of Alzheimer's patients is not affected when compared to other forms of dementia until in the later stages. Common behaviours associated with Alzheimer's disease include [3]: language descent, misperception, diminished memory, behaviour and thinking, agitation, compromised judgement, diminished ability to communicate, changes in behaviour and personality, impaired awareness, incapability to think of following directions and emotional indifference.

2.2 Brain Aneurysm

This (brain, intracranial or cerebral aneurysm) can be defined as a ballooning effect or swelling which occurs as a result of the weakening in a particular area specifically in the wall of the brain's blood vessel. Brain aneurysms usually do not show any symptoms and are insignificant in size (10 mm) and are usually not noticed until rupturing occurs. The smaller the aneurysm, the lower the risk and chances of rupturing [3].

More so, sentinel haemorrhage which occurs sparingly may show symptoms before rupture. This is as a result of a small quantity of blood leakage into the brain. These aneurysms symptoms arise sometimes because of the pressing on other adjourning structures like the eye nerves which results in a loss of vision and suspension of eye movement [3]. Some of the symptoms of an unruptured brain aneurysm include change in vision, reduced eye movement, headaches and pains in the eyes.

2.3 Epilepsy and Seizures

Epilepsy happens to be the utmost common neurological disorder which affects patients regardless of background, race or age. According to John Hopkins Medicine [3], over

2.2 million people in America live with epilepsy. Epilepsy is a nervous system disorder that involves the brain, thus making the patients vulnerable to repeated seizures.

One is described as epileptic or having epilepsy if he/she has two or more seizures. However, seizures may be caused by several factors that may disturb the networks between the brain and nerve cells such as concussion of the brain, alcohol, drug withdrawal, high/low blood sugar, high fever, etc. a few of the major causes of epilepsy include strokes, tumours, brain damage caused by either injury or illness, and disproportion in neurotransmitters.

2.4 Meningitis

An inflammation or swelling of the meninges results in the disease meningitis. Meninges here are the three layers that surround and protect the brain and spinal cord. It is usually, an infection that results in the swelling of the meninges [3].

Meningitis caused by viruses is more severe and more of a threat to life than those caused by less common bacterial. Meningitis caused by viruses is usually transmitted via different forms such as pitiable hygiene, sneezing, coughing and sparingly through common insects like ticks and mosquitoes [3]. Some symptoms of meningitis include misperception, seizures, stiffness of the neck, sleepiness, extreme temperature, headache, vomiting, photophobia and pains in body joints.

Symptoms observed in children include: fever, difficulty waking, backache, vomiting, loss of appetite, agitated, loud cry and change in skin colour to pale.

2.5 Parkinson's Disease

Parkinson's disease is a very serious neurodegenerative disorder that is progressive and shreds away all motor capabilities consequently, reducing the balance in individuals usually above 65 years of age. Parkinson's disease is sometimes regarded as hereditary or occurs extemporaneously [3]. However, it is known that this disease may occur as a result of dead brain cells in the "substantia nigra" resulting in the termination of dopamine production, hence loss of muscle movement or control [3].

Recent researches are currently targeted at investigating genes responsible for Parkinson's disease as well as discovering new complex biochemical pathways which is involved in the disease for targeted therapy.

2.6 Stroke

Stroke happens as a result of the stoppage of blood flow to the brain which put the patient in an emergency state. Brain functions appropriately when there is adequate nutrient and oxygen supply, hence, stoppage of blood supply notwithstanding duration may create a lot of complications [3].

Loss of brain functions due to dead brain cells may result in the patient's inability to carry out the following functions [3]: movement, eating, making speech, remember or think, regulate bowel and bladder movement and regulate additional dynamic body functions.

Strokes occur unannounced without prior notice. Strokes are of two types: haemorrhagic and ischaemic stroke.

Risk factors of stroke include: 140/90 or higher blood pressure, chronic smokers, usage of contraceptives, patients with heart diseases, diabetes, elevated blood cholesterol, drug abuse, obesity, excessive alcohol intake, not exercising the body, old age (from 55 years), genetics and males are more at risk of having a stroke than women.

2.7 Migraine

The intensity of migraines can range from modest intensity to tremendously severe ones which are usually characterized by an excruciating beating feel. Often time migraines occur on one part but can also occur on a whole or either part of the head, face or neck [3]. Photophobia and sensitivity to smell and noise are associated with migraines while vomiting is one of the most common symptoms. Migraines can occur as a result of the following: excessive alcohol consumption, insomnia, loss of water due to dehydration, menstruation, grinding of tooth, extreme weather changes and starvation.

2.8 Encephalitis

This is caused by swelling of the brain's active tissues either caused by an autoimmune reaction or infection leading to serious swelling in the brain with resultant effects like photophobia, rigid neck, headaches, seizures as well as mental misperception [3].

Observed symptoms of encephalitis include photophobia, sound sensitivity, headaches, high body, temperature, fever, unconsciousness and stiffness of the neck.

3 Medicinal Plants Used in the Treatment of Neurological Disorders

Medicinal plant can be any plant in which one or more of its parts contains bioactive substances which can be used for therapeutic purposes or which are antecedents for the synthesis of useful drugs. Right from time immemorial, natural plant products have occupied a place of prime importance in the field of medicine. They continue to be vital to many who do not have access to modern medicines or cannot afford synthetic drugs. Medicinal plants, e.g., herbs, and trees contain chemical compounds known to modern and ancient civilizations for their bioactive properties. Until the development of chemistry, particularly, the synthesis of secondary metabolites in the nineteenth century, medicinal plants and herbs have been the sole source of bioactive compounds capable of healing human ailments. Researchers believe that if medicinal herbs are taken in the appropriate doses and forms, they will be more effective than modern pharmaceutical drugs (synthetic drugs).

Plant products contain secondary metabolites that heal a variety of ailments and conditions in humans and animals. Over three-quarters of the world's population rely mainly on plants and their products for primary healthcare. About 30% of the entire plant species worldwide, at one time or the other was used for medicinal values [4].

It has been estimated that in developed countries like the USA, plant drugs constitute as much as 25% of the total drugs. Fast developing countries such as China and Singapore, their contributions are as much as 80%. Therefore, the economic uses of medicinal plants are much more in countries such as China than the rest of the world. These countries provide two-third of the plants used in modern systems of medicine and the health care delivery of rural population depends on indigenous systems of medicine. Of the 250,000 higher plant species on earth, more than 80,000 have medicinal purposes [5].

Current reviews and research have shown that there are a lot of medicinal plants with several phytochemicals which have been reported to decline neuropathy in animal models of CIPN. Curcumin, rutin, quercetin, matrine, euphol, thioctic acid, rosmarinic acid and cannabinoids are the most relevant natural products with therapeutic effects in CIPN [6] (Table 1).

4 Medicinal Plants, Bioactive Compounds/Phytochemicals and Mechanisms of Action

4.1 *Panax ginseng*

This medicinal plant is well known for its numerous healing properties and it is indigenous to the Chinese/Koreans. The plant has been credited with the ability to aid patients resist sicknesses by boosting the patient's immune system and has been used in the treatment of several illnesses including neurodegenerative disorder, diabetes, cancer, hypertension, etc. According to a research by Nah et al. [51], the active compound found in *Panax ginseng* ginsenoside, which is a saponin, has the potential to impede voltage-dependent Ca^{2+} channels linked by a G protein receptor sensitive to the Ca^{2+} which modulates neuronal Ca^{2+}. The neuroprotective effect of ginsenoside occurs when it acts on dopaminergic neurons by impeding the promotion of the nigral iron level, thus dropping expression of divalent metal transporter (DMT1), elevating ferroportin (FP1) expression in Parkinson's disease's patient [52].

4.2 *Ginkgo biloba*

Ginkgo biloba is widely used as a plant supplement with great build-up of bioactive compounds such as bilobalide, ginkgolides A, B, and C, quercetin,

Table 1 List of medicinal plants in the treatment of neurological disorders

Botanical names	Common Names	Family	Bioactive compounds/ phytochemicals	Mechanism of action/effects	References
Andrographis paniculata	King of bitters	Acanthaceae	Diterpene lactone and Andrographolide	Decreases of aggregation of platelet. Increases endothelial nitric oxide synthase (eNOS) activity	[7]
Areca catechu	Areca, Betel-nut Palm, Pinang	Arecaceae	Alkaloids, Arecoline, Guvacine, Arecaidine	Involvement in the binding activity of muscarinic (M2)	[8, 9]
Sesamum indicum Linn	Sesame	Pedaliaceae	Sesamol	By improving cognitive functions and the reduction of oxidative stress	[10, 11]
Cinnamomum cassia	Chinese cinnamon or cassia	Lauraceae	Coumarin	Reduces cold allodynia	[12]
Euphorbia tirucalli	Milk bush	Euphorbiaceae	Euphol	Reduces persistent and mechanical hypersensitivity	[13]
Piper methysticum	Kava	Piperaceae	Kavain, Yangonin, Kavalactones, Methysticin	Modulation of γ-aminobutyric acid type A (GABAA) receptors	[14]
Angelica sinensis	Garden angelica	Apiaceae	Ferulic acid	Involvement in antioxidant action	[15]
Centella asiatica	Indian pennywort	Apiaceae	Asiatic acid/triterpene	Protects against dysfunction caused by 3-nitropropionic acid (3-NP). Decreases acetylcholinesterase (AChE) and involvement in antioxidant activity	[16, 17]
Aegle marmelos	Bilwa or bael	Rutaceae	Alkaloid-amide and Aegeline	Reduces reactive oxygen species (ROS)	[18]
Gardenia jasminoides	Common gardenia or cape jasmine	Rubiaceae	Crocin/carotenoid	Decreases acetylcholinesterase (AChE) and oxidative stress	[19]

Tripterygium wilfordii	Thunder god vine	Celastraceae	Triptolide	Increases cognitive function, complemented by reduction in neuroinflammation and Aβ deposition as well as the improvement of memory	[20, 21]
Xylopia aethiopica	Spice tree	Annonaceae	Xylopic acid	Reduces inert motorized hyperalgesia	[22]
Curcumin longa	Turmeric	Zingiberaceae	Curcumin	Upsurge manifestation of brain neurotropic factor by the reversal of hippocampal neurogenesis caused by stress	[23, 24]
Panax ginseng	Ginseng	Araliaceae	Ginsenoside and triterpene glycosides	Promotion of hippocampal gamma-aminobutyric acid (GABA) level in the brain through the increased enzyme activities of glutamate decarboxylase. Reduces Ca^{2+}	[25]
Garcinia kola	Bitter kola	Clusiaceae	Kolaviron	Reduces necrotic cell death and myeloperoxidase (MPO)	[26]
Glycine max	Soybean	Fabaceae	Genistein and flavonoid	Reduces monoamine oxidase (MAO) and inflammation	[27, 28]
Schisandrae chinensis	Chinese magnolia vine	Magnoliaceae	Schizandrin	Plays a part in acetylcholinesterase (AChE) and γ-Aminobutyric acid type A (GABAA) receptors	[29]
Ginkgo biloba	Maidenhair trees	Ginkgoaceae	Ginkgolides, Gastrodin, Vanillin, Bilobalide	By promotion of hippocampal gamma-aminobutyric acid (GABA) levels.	[30]
Piper nigrum	Black pepper	Piperaceae	Piperine	Suppresses the actions of N-methyl-D-aspartate (NMDA) receptor	[31]
Withania somnifera	Indian Winter cherry	Solanaceae	Withanolides/Withanols/ Withamides/Withanone/Amido compounds	Stabilization of mood	[32]
Cyperus rotundus	Nut grass	Cyperaceae	α-Cyperone	Decreases of aggregation of platelet	[33]

(continued)

Table 1 (continued)

Botanical names	Common Names	Family	Bioactive compounds/ phytochemicals	Mechanism of action/effects	References
Scutellaria baicalensis	Huang Qin or Chinese skullcap	Lamiaceae	Wogonin/flavonoid	Decrease in monoamine oxidase A (MOA-A) and synthesis of thrombin	[34]
Silybinisus laborinum		Fabaceae	Silibinin	Decreases acetylcholinesterase (AChE) and reactive oxygen species (ROS)	[34, 35]
Vitis vinifera	Common grape	Vitaceae	Stilbenoid/Resveratrol	Decreases beta-amyloid plaques	[36]
Hypericum perforatum	St. John's wort	Hypericaceae	Hyperforin and Rutin	Decrease in monoamine oxidase A (MOA-A) and monoamine oxidase B	[37]
Paeonia lactiflora	Common garden peony	Paeoniaceae	Paeoniflorin	Controlling adrenergic systems	[38]
Crocus sativus	Saffron crocus	Liliaceae	Safranal, Crocin	By promotion of hippocampal gamma-aminobutyric acid (GABA) levels	[39]
Citrus paradise	Grapefruit	Rutaceae	Naringenin	Reduces the inflammation of cytokines	[40]
Citrus sinensis	Orange	Rutaceae	Naringenin	Reduces the inflammation of cytokines	[40]
Berberis genus	Barberry	Berberidaceae	Berberine	Decreases acetylcholinesterase (AChE)	[41]
Hypericum afrum	–	Hypericaceae	Flavonoid	Decrease in monoamine oxidase A (MOA-A)	[27]
Cytisus villosus	Brooms	Fabaceae	Chrysin	Decrease in monoamine oxidase A (MOA-A)	[27]
Nigella sativa	Black seed	Ranunculaceae	Thymoquinone	Upsurge in GABAergic tone via opioid receptor	[42]
Ficus platyphylla	Broad leaf fig	Moraceae	Saponin	Overturns excitatory and inhibitory synaptic traffic. Works by the suppression of excited synaptic traffic	[43]
Nardostachys jatamansi	Balachara	Valirenaceae	Nardostachysin, Jatamansic acid, Jatamansone	Enhances dopamine	[44]

Biota orientalis	Oriental arborvitae	Cupressaceae	Pinusolides	Confers neuroprotection	[45]
Camellia sinensis	Green tea	Theaceae	Epigallocatechin gallate/catechin	Decreases acetylcholinesterase (AChE) and reduces oxidative stress	[19, 46]
Valeriana officinalis	Garden heliotrope	Caprifoliaceae	Hesperidin/flavonoid	Reduces oxidative stress	[35]
Angelica gigas	Korean angelica	Apiaceae/ Umbelliferae	Decursin/pyranocoumarin	Decrease in monoamine oxidase A (MOA-A)	[47]
Galanthus nivalis	Snowdrop	Amaryllidaceae	Galanthamine	Inhibits acetylcholinesterase AChE	[48]
Gastrodia elata	Tian ma	Orchidaceae	Gastrodin	GABA transaminase inhibition	[30]
Ziziphus jujube	Common jujube	Rhamnaceae	Spinosin, Jujubosides	Reduction of hippocampal hyper-action	[49]
Allium sepa	Onions, shallots	Alliaceae	Allium and Allicin	Activation of sulphides containing allyl conveys neuroprotective properties by the initiation of mitochondrial uncoupling proteins	[50]
Allium sativum	Garlic	Alliaceae	Allium and Allicin	Activation of sulphides containing allyl conveys neuroprotective properties by the initiation of mitochondrial uncoupling proteins	[50]
Ocimum sanctum	Tulsi	Lamiaceae	Essential oils	Involvement in dopamine functions	[2]
Rubia cordifolia	Common madder	Rubiaceae	Terpenoids	Increases gamma-aminobutyric acid (GABA) levels	[2]
Cymbopogon citratus	Lemongrass	Gramineae	Essential oils	Increases gamma-aminobutyric acid (GABA) neurotransmission	[2]
Salvia lavandulaefolia	Spanish sage	Lamiaceae	α-Pinene, 1,8-cineole β-Pinene	Inhibits acetylcholinesterase AChE	[2]

isorhamnetin, kaempferol, etc. *Ginkgo biloba* has been reported to have positive effects on neuropathy (vincristine-induced) [53]. Research also shows the impeding properties of *Ginkgo biloba* through the nitrosative- and oxidative-induced neural damage on diabetic neuropathy in animal model [54]. Reports by Taliyan and Sharma [55] also show ginkgolide B regulation of neural damage via cellular pro-apoptotic pathways.

4.3 *Curcuma longa*

Curcuma longa also known as turmeric is known for its numerous benefits and medicinal properties. In relation to neurodegenerative disorder, curcumin found in *Curcuma longa* has been reported to be beneficial in management and treatment of the disorder. Xu et al. [56] reported curcumin benefits which includes betterment of behavioural deficits and protects the neurons against ischemic cell death as observed in animal models. Other researches have also reported that curcumin has been used in in vitro treating of age-related neurodegenerative diseases. Lingering stress-induced damage of hippocampal neurogenesis has been reportedly upturned by curcumin as well as the potential to increase the manifestation of brain-derived neurotrophic factor (BDNF) [56, 57].

4.4 *Allium sativum* and *A. sepia*

Allium sativum (garlic) and *A. sepia* (onions) belongs to the family Alliaceae. They possess high quantity of allium and allicin (organosulphur compounds) which have been reported to have positive effects on neurodegenerative disorders and free radical foraging actions. Activation of sulphides containing allyl conveys neuroprotective properties by the initiation of mitochondrial uncoupling proteins [50].

4.5 Camellia sinensis

Camellia sinensis also referred to as tea, in which the popular black and green teas are produced from depending on their degree of fermentation. This plant is a great source of polyphenolics. Phytochemicals and other bioactive compounds found in *Camellia sinensis* include flavonoids (aflavins) and catechins (epicatechin, gallocatechin, epigallocatechin-3-gallate (EGCG), epigallocatechin) all of which drops chemotherapy-associated dose-preventive side effects in the early stage of therapy [58]. ECCG has been reported in neurological disturbances studies in many models as a result of its antioxidant and anti-inflammatory properties [59, 60].

5 Chemical Structures of Bioactive Compounds/Phytochemicals Found in Plants Used in the Treatment and Management of Neurological Disorders

See Figs. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, and 28.

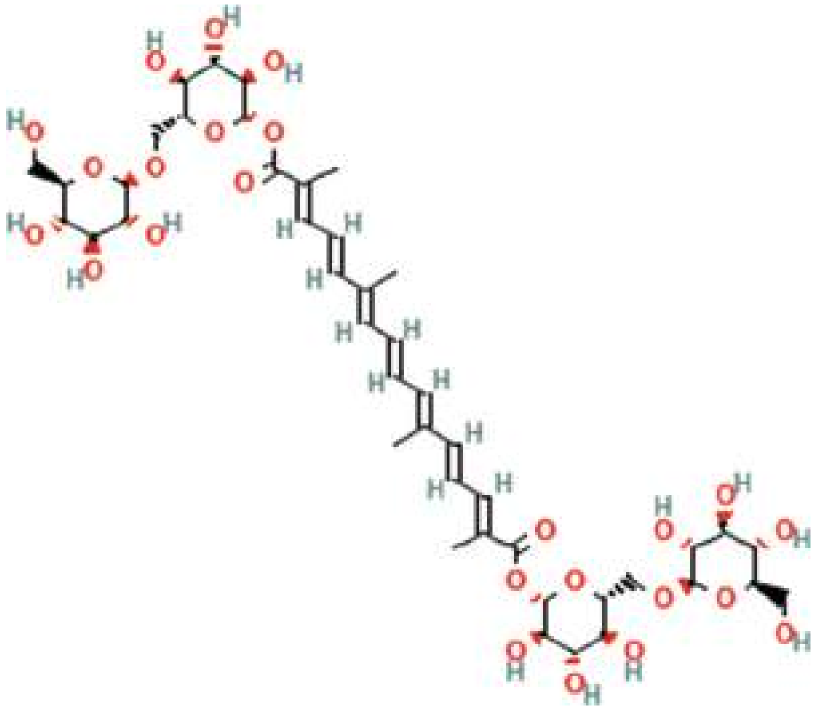


Fig. 1 Crocin (*Crocus sativus*)

Fig. 2 Naringenin (*Citrus sinensis*)

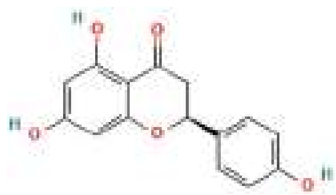


Fig. 3 Silibinin (*Silybinisus laborinum*)

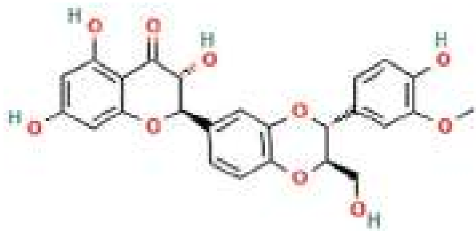


Fig. 4 Curcumin (*Curcumin longa*)

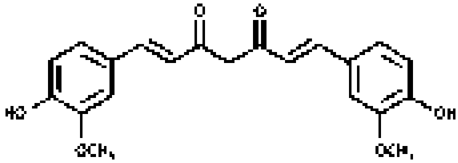


Fig. 5 Spinosin (*Ziziphus jujube*)

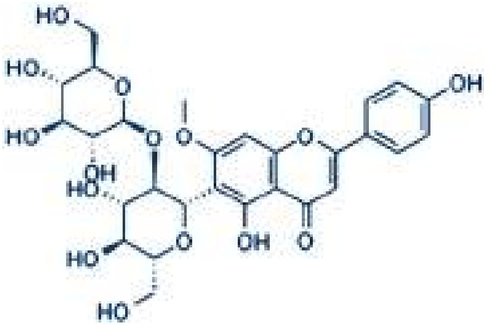


Fig. 6 Arecaidine (*Areca catechu*)

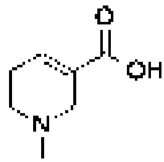
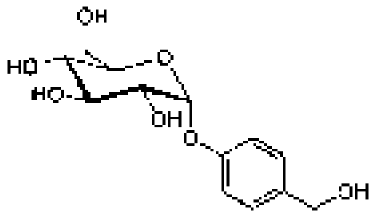


Fig. 7 Gastrodin (*Gastrodia elata*)



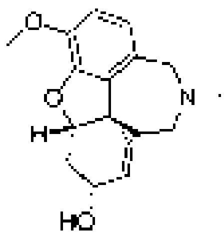


Fig. 8 Galanthamine (*Galanthus nivalis*)

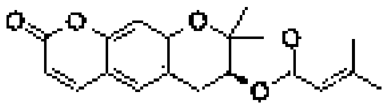


Fig. 9 Decursin (*Scutellaria baicalensis*)

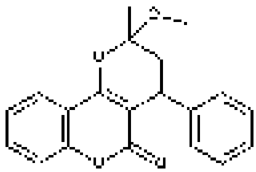


Fig. 10 Pyranocoumarin (*Scutellaria baicalensis*)

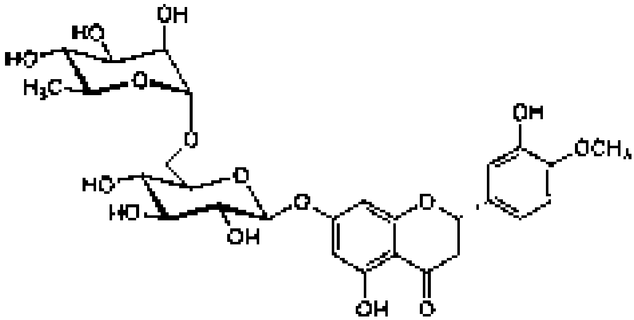


Fig. 11 Hesperidin (*Valeriana officinalis*)

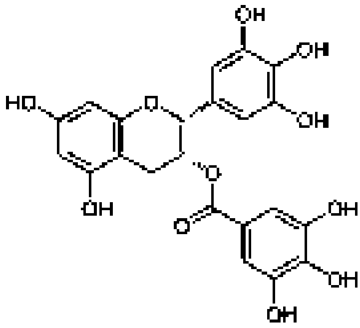


Fig. 12 Epigallocatechin gallate (*Camellia sinensis*)

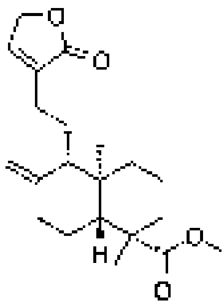


Fig. 13 Pinusolides (*Biota orientalis*)

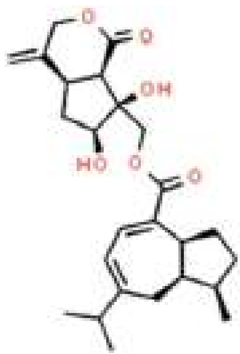


Fig. 14 Nardostachysin (*Nardostachys jatamansi*)

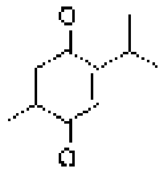


Fig. 15 Thymoquinone (*Nigella sativa*)

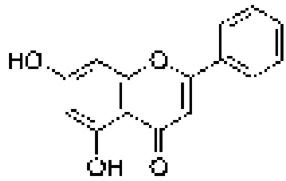


Fig. 16 Chrysin (*Cytisus villosus*)

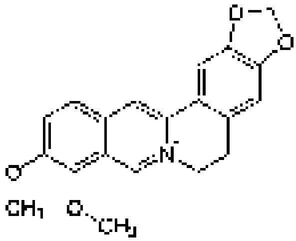


Fig. 17 Berberine (*Berberis* genus)

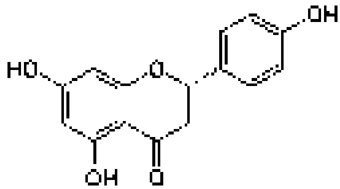


Fig. 18 Naringenin (*Citrus paradise*)

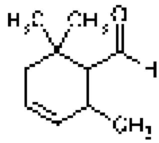


Fig. 19 Safranal (*Crocus sativus*)

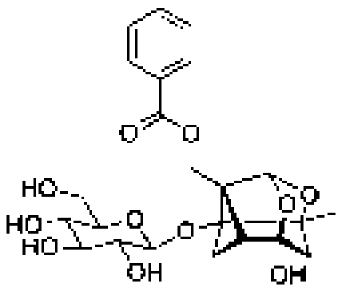


Fig. 20 Paeoniflorin (*Paeonia lactiflora*)

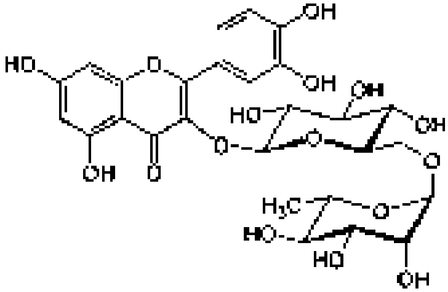


Fig. 21 Rutin (*Hypericum perforatum*)

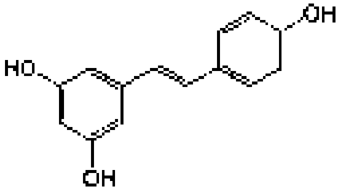


Fig. 22 Stilbenoid (*Vitis vinifera*)

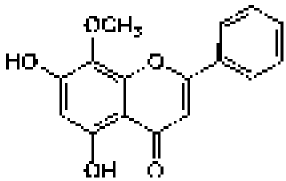


Fig. 23 Wogonin (*Suctellaria baicalensis*)

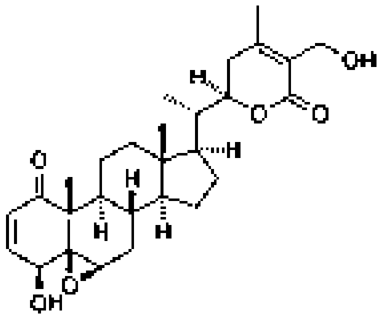


Fig. 24 Withanolide (*Withania somnifera*)

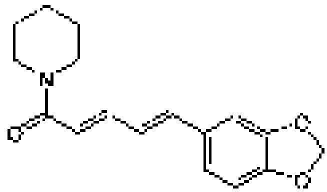


Fig. 25 Piperine (*Piper nigrum*)

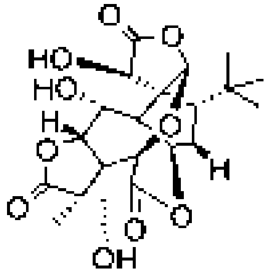


Fig. 26 Ginkgolides (*Ginkgo biloba*)

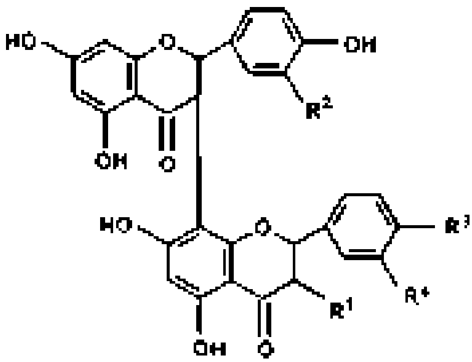
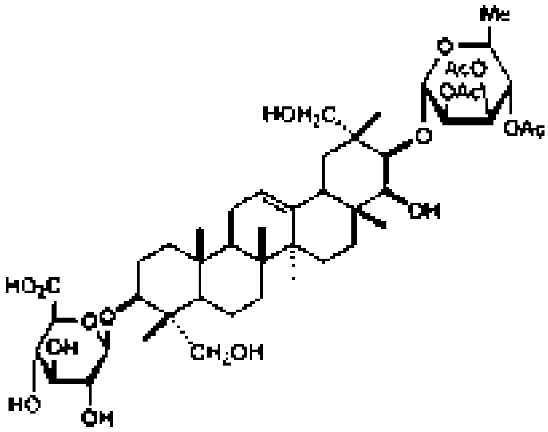


Fig. 27 Kolaviron (*Garcinia kola*)

6 Conclusion

Neurological diseases or disorders can be described as diseases/disorders which particularly affect or disturb the brain, spinal cord as well as nervous coordination which run throughout the human body. There are a great number of specific causes of neurological diseases/disorders. The prominent of these causes are genetic disorder, contagions, deformities or disorders, malnutrition which could be linked to the health of one's environment, lifestyle, serious injuries to the brain, nerves as well as the spinal cord are also amongst the common causes of neurological diseases. The use of several synthetic agents and medication with neuroprotective properties has

Fig. 28 Triterpene glycosides (*Panax ginseng*)



been properly documented. But there are some discomforting side effects associated with these synthetic neuroprotective medications which include fretfulness or nervousness, fatigue, dry mouth, tiredness, drowsiness, struggle with balance, etc. This chapter has reviewed a lot of medicinal plants with several phytochemicals and their mechanism of actions which have been reported to decline neuropathy. The plants discussed in this chapter as well as their active constituents provide a great insight into affordable means of treating and managing neurological disorders as well as a basis for pharmaceutical companies searching for novel sources for the production of neuroprotective pharmacotherapy drugs.

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Diabetes Treatment and Prevention Using Herbal Medicine

20

Babalola Ola Yusuf, Rukayat Abiola Abdulsalam, and Saheed Sabiu

Contents

Abstract

Diabetes is a cluster of prevalent hormonal conditions that display abnormally high blood glucose level. Diabetes is caused by the improper utilization of insulin by body cells or absence of pancreatic synthesis of insulin. Diabetes management involves measures used in maintaining diabetic condition to be close to normoglycaemia. This includes the use of medications such as insulin therapy and oral antidiabetic agents, exercise, weight loss, diet, and natural remedies. 65–80% of people worldwide use herbal medicine for their main medical care, which entails using medicinal plants to treat, diagnose, and prevent illnesses. Due

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to their accessibility and alleged absence of adverse effects, most people, especially those who live in developing nations, prefer to treat their diabetes with medicinal plants including fenugreek, cinnamon, bitter lemon, and *Gymnema sylvestre*. Since many of these plants inhibit amylase, encourage insulin secretion, and aid in glucose uptake, they have similar antidiabetic effects to those of common oral antidiabetic drugs. There have been efforts to isolate bioactive components in antidiabetic plants as well as comprehending how they work due to the recent increase in interest in medicinal herbs and technological advances.

Keywords

Diabetes · Diabetes management · Herbal treatment · Traditional medicine · Medicinal herbs

1 Introduction

Diabetes is an enduring metabolic ailment branded by high blood glucose levels [1]. There are approximately 422 million diabetics in the world, and this figure is steadily rising [1]. Either insufficient pancreatic insulin synthesis or inappropriate insulin use by body cell is to blame for the disorder [2]. It is regarded as the foremost causes of illness and mortality [3] and can cause several consequences, such as cardiovascular disease, renal failure, blindness, as well as amputation [2]. Dietary vicissitudes, insulin treatment, and oral hypoglycaemic drugs are typical management options for diabetes [1]. Herbal medicine has been utilized for ages to treat diabetes and has lately grown in favour of supplemental therapy [1]. The utilization of phytotherapy in the management and prevention of diabetes is examined in this chapter. Diabetes has long been treated with herbal medicines in conventional healthcare systems of Chinese (TCM) and Indian (Ayurveda) [4]. Herbal remedies work in a variety of ways, including by improving insulin sensitivity and glucose absorption [4].

2 Herbal Medicine

2.1 The Herbal/Traditional Medicine Structure

Traditional medicine, commonly known as complementary and alternative medicine (CAM), includes an array of procedures and approaches, including the use of therapeutic exercises, spirituality, and medicinal herbs for treatment, prevention, and maintenance of health [5]. Due to regional variations in historical, economic, and lifestyle circumstances, these behaviours, which are grounded on ideas and knowledge, fluctuate significantly from one place to another [6]. Traditional folk medicine has been practised and improved upon since the Stone Age [7]. The theory and practice of traditional medicine are said to work together to diagnose, treat, and prevent ailments. It is based on earlier information and discoveries that were frequently shared verbally through oral tradition, oral lessons from ancestors, or more recently, written down [7, 8]. The most prevalent kind of treatment in contemporary medical settings is the usage of medicinal plants or herbal medications [9].

Herbal remedies are frequently used in conventional medicine, which emphasizes their important place in ethnomedicine [10]. The investigation of pharmacognosy and the use of herbal plants is the foundation of conventional medicine [11, 12], which is known as herbal medicine also referred to as phytomedicine or phytotherapy [13]. Herbal medicine, phytotherapy, and phytomedicine are additional names for the research on pharmacognosy and the use of herbal plants, which is the cornerstone of contemporary medicine [14]. The WHO claims that herbal remedies are pharmaceuticals that have undergone processing and labelling and are utilized to cure illness or enhance life's quality. They include active compounds found in above- (e.g. seed, flower, leaf, and bark) or below-ground plant components (e.g. root, rhizome) or certain combinations of plant parts [15, 16]. Traditional Chinese herbal medicine and Ayurveda are the most well-known and widely used traditional healthcare systems [17, 18]. Among these systems are the following:

- System of Traditional African Medicine

  Traditional African medicine encompasses a range of conventional medical practices, such as native herbalism and African theology, and often involves the involvement of shamans, midwives, and herbalists [19]. Shamans, midwives, and herbalists are frequently involved in traditional African medicine, which includes a variety of Western medicinal techniques such native herbalism and African religion [19]. Ancient healing magic may be more common than Western medicine [7] and has a longer history and wider prevalence on the African continent than other traditional medical practices [20]. Different specialties of African traditional medicine, which is a complete healthcare system, include divination, spiritualism, and herbalism. However, in some circumstances there might be some overlap [7, 21].

  An individual who practises medical care in their tribe by utilizing methods such as herbs, minerals, animal parts, and incantations, among others, in accordance with their community's traditions and beliefs is known to be a traditional therapist/healer [7]. They should be considered as skilled, experienced, adaptable, and trustworthy [22]. Depending on the culture, there may be various classifications of traditional healers, including high priests, herbalists, midwives, priestesses, diviners, seers/spiritualists, and witch doctors [7]. In certain societies in Nigeria, these individuals are identified by their native names, such as *Babalawo*, *Buka*, or *Dibia* [23]. It is believed in ancient African medicine that certain aspects of the practice are only shared with initiates. Therefore, to gain access to these components, initiation into a cult is usually required [7]. According to this form of medicine, illnesses are caused by spiritual forces rather than random chance. As a result, the prescribed remedy typically involves a plant-based treatment that is believed to possess therapeutic qualities as well as symbolic and spiritual significance [7].

- Traditional Chinese Medical System

  Traditional Chinese medicine (TCM) is a complementary medical methodology that originates from Chinese traditional medicine [24]. TCM encompasses a wide range of health and therapeutic practices that are typically viewed as incompatible. These practices include traditional beliefs, intellectual theories, Confucian ideology, herbal treatments, dietary interventions, physical activities, various medical specialties, and different schools of thought [25]. Traditional Chinese medicine (TCM) is based on the philosophy of yinyangism, which combines the *Yin Yang* theory and the Five Phases hypothesis [26]. This philosophy applies the concept of *yin* and *yang* to the human body, where the top and back are considered as *yang*, and the bottom is viewed as *yin* [27]. *Yin* and *yang* are also used to describe various biological processes and disease symptoms. For example, sensations of cold and heat are considered *yin* and *yang* symptoms [27]. Traditional Chinese medicine (TCM) includes a list of medications that are believed to address specific symptom clusters by enhancing either *yin* or *yang* [29]. Additionally, strict guidelines have been established for the order of actions, countermeasures, and other interactions among the Five Phases theory [28]. The *zàng-fû* concept is closely linked to each of these Five Phases tenets and significantly affects the TCM body model [28]. The Five Phases concept is frequently utilized in both diagnosis and treatment within TCM [28] (Fig. 1, Table 1).

  In traditional Chinese medicine, the vital energy of the body, known as *ch'i* or *qi*, is believed to flow through meridians that are linked to multiple organ systems and physical functions through forks [31]. Rather than focusing on structural

Fig. 1 Relations between *Wu-xing* (the Five Phases of TCM) [29]

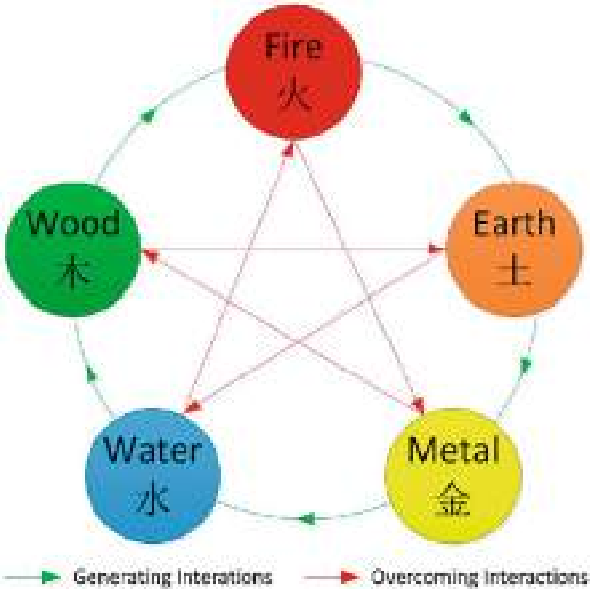

Table 1 The arrangement of things according to the Five Phases character and *záng-fû* organ as well as distribution

Phenomenon	Wood	Fire	Earth	Metal	Water	References
Direction	East	South	Centre	West	North	[29]
Colour	Green or violet	Red or purple	Yellow or pink	White	Black	[30]
Climate	Wind	Heat	Damp	Dryness	Cold	[28]
Taste	Sour	Bitter	Sweet	Acrid	Salty	[28]
Záng organ	Liver	Heart	Spleen	Lung	Kidney	[29]
Fû organ	Gall bladder	Small intestine	Stomach	Large intestine	Bladder	[29]
Sense organ	Eye	Tongue	Mouth	Nose	Ears	[30]
Facial part	Above bridge of nose	Between eyes and lower part of the eyes	Bridge of nose	Between eyes and middle part	Below cheekbone	[30]
Eye part	Iris	Inner/outer corner of eye	Upper and lower lid	Sclera	Pupil	[30]

details, traditional Chinese medicine places greater emphasis on physiological processes such as digestion, respiration, and temperature regulation [32]. The perception of illness in this system is that it reflects an imbalance or disharmony in the operations or relationships of various elements, including *yin*, *yang*, *qi*, *xuê*, *zàng-fû*, meridians, and their interplay with the surroundings and the human body [27]. Traditional Chinese medicine (TCM) considers recognizing a "pattern of disharmony" to be a critical step in the diagnostic process, as it forms the basis of therapy [28, 33]. However, identifying various patterns, which is also known as "pattern discrimination," is considered to be the most challenging aspect to master within TCM [34].

- Traditional Indian Medical System

  Ayurveda is a type of complementary therapy, being a 2000-year-old tradition originated from India [35]. Although it has changed over time, its beliefs and methods, which include herbal medicines, dietary programmes, meditation, yoga, massage, enemas, laxatives, and medicinal oils, are frequently regarded as pseudoscientific [36, 37]. Samkhya and Vaisheshika are two philosophical ideas that served as the foundation for Ayurveda, as well as Buddhist and Jain principles [38]. Maintaining balance is the major goal of Ayurveda, and it is thought that stifling natural desires might make one sick [39]. In fact, according to Ayurveda, even suppressing a sneeze may result in shoulder ache [40]. Ayurveda places a strong emphasis on a number of elements, including healthy eating practices, enough sleep, and appropriate sexual behaviour [39]. The human body, according to Ayurveda, is made up of three components: *doshas*, which are humoral substances; *malas*, which are waste products; and *dhatus*, which are the body's tissues [40]. Ayurveda recognizes three primary physiological humours or *doshas*, namely, *Pitta*, *Kapha*, and *Vata* [41]. These *doshas*, collectively known as *tridosha*, are often associated with the nervous system by contemporary writers [41]. An individual's mental state can be influenced by the *doshas* in Ayurveda [42]. The dominance of any *dosha* impacts an individual's bodily composition (prakriti) and character, as each dosha possesses distinct qualities and performs specific tasks in the body and mind [42, 43] (Fig. 2).

Ayurveda considers an imbalance in both mental and physical *doshas* as a significant cause of illnesses [45]. To diagnose illnesses, Ayurveda employs eight diagnostic methods, which include examining the urine, stool, speech, touch, vision, tongue, appearance, and pulse [45]. In Ayurveda, physicians rely on the five senses to diagnose patients, and for example, they use hearing to observe breathing and speech [46]. Ayurveda emphasizes the study of the *marman* or *marma* points, which are points in the body that could be potentially lethal [46]. Contemporary Ayurveda primarily focuses on improving the body's metabolism, promoting effective elimination of substances, and enhancing longevity, while two of the original eight branches of Ayurveda concentrate on surgical procedures [46]. There is also a significant emphasis on exercise, yoga, and meditation in this approach [47]. To maintain good health, Ayurveda subscribes to the Dinacharya school of thought, which emphasizes following the natural rhythms of the brain for activities such as waking, sleeping, and working [48]. The Sattvic diet is recommended as a part of the dietary regimen. In addition, important hygiene practices include regular showers, teeth brushing, tongue scrubbing, oil pulling, caring for the skin, and cleaning the eyes [48].

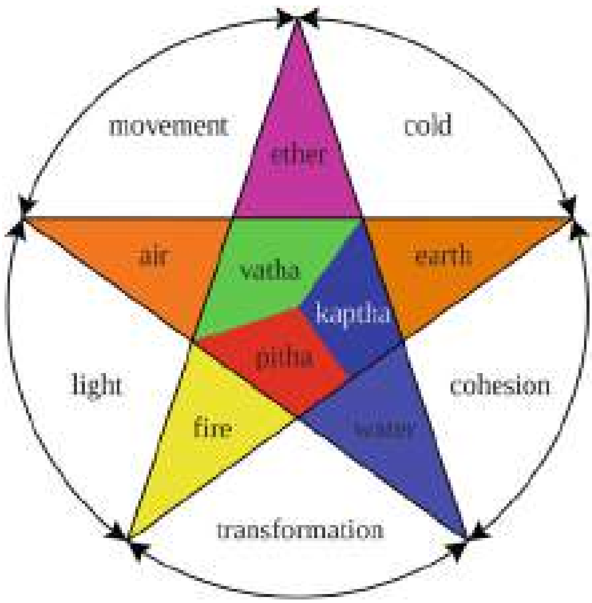

Fig. 2 The three *doshas* in Ayurveda are made up of five elements [44]

2.2 Herbal Preparation

There are different methods for preparing herbal remedies that vary based on regional and cultural practices. It is permissible to use both fresh and dried parts of plants, and a specific method is chosen based on experience to maximize effectiveness and minimize potential harm [7]. In general, there are several techniques for preparing herbal remedies, including:

(a) Extraction: The extraction method involves using a solvent in a specific weight-to-volume ratio. Sometimes, the solvent may evaporate and transform into a jelly-like substance [7].
(b) Infusion: To create an infusion, the plant materials are soaked in hot or cold water for a little while. Honey or another addition can be added to stop spoilage [7, 49].
(c) Decoction: It is created by making woody part of wood boil for a fixed period and then filtering out the solids. Potash can be used both as a preservative and to aid in the extraction process [49].
(d) Tincture: Tinctures, which are alcohol infusions, can be diluted before use if they are too concentrated [50].
(e) Ashing: Ashing involves burning the dry ingredients until they turn to ash, filtering the ashed samples, and then mixing them with water or food for consumption [7].

    There are several other types of herbal preparations. Lotions are thin treatments applied to the body, while liniments are surface treatments that come in watery, semi-liquid, or oily forms and contain bioactive ingredients [7]. Poultices are made from freshly macerated plant parts that retain their liquid and are applied to the skin [49]. Snuffs, which are made from powdered dried herbs, are burned to create charcoal and inhaled through the nose [7]. Gruels are grain-based cereals or porridges that are mixed with pulverized dried vegetables or their ash and consumed. Sometimes, mixtures of multiple plants are used to enhance the combined effects of the substances [49].

Aside from the usual methods of taking herbal remedies, such as orally, rectally, topically, or nasally, there are other ways to consume them. These include passive inhalation or smoking a poorly constructed cigar made of dried plant materials [7]. Another method involves inhaling the volatile oils released from heating or burning the plant matter [7]. These techniques can be helpful in relieving congestion, headaches, and respiratory issues. Sitz baths, however, are specifically used for treating piles [7, 51].

2.3 Prevalence of Use

Herbal treatments are a popular form of primary medical treatment worldwide, with estimates indicating that 65–80% of people use them [52]. Conventional medicine is responsible for about 40 percentage of medical services in China [53], and about 80% of African utilize conventional remedies to assist with healthcare difficulties [54]. As of 2018, the cases of herbal medicine usage varied across several cultural groups [55], and people's perceptions of herbal remedies are shaped by ancient traditions and lifestyle [56, 57]. For instance, the significant utilization of spices and herbs in Australian culture is recognized as a way to preserve food [57, 58]. According to data from the WHO, the use of phytomedicine was highest in European countries and Southeast Asia in 2019, at a rate of 91%, followed by Africa at 87%, and the lowest usage was in the America at 80% [6].

People who suffer from chronic illnesses such as HIV/AIDS [59], diabetes, asthma, end-stage kidney disease [60], and cancer [61] are more likely to use herbal remedies. Furthermore, several factors, including sex, age, race, level of education, and socioeconomic status, have been linked to the frequency of natural remedy usage [62]. Additional factors include arthritis, depression, anxiety, hypertension [63], and various chronic conditions [64, 65].

3 Diabetes Management

The principal aim of managing diabetes is to regulate sugar level as close to normoglycaemia, without causing it to drop too low [66]. To achieve this goal, individuals may need to make changes to their diet, engage in regular exercise, lose weight, and use medications such as insulin, oral medications, or herbal remedies [67].

Being well-informed about diabetes and vigorously partaking in treatment are imperative, as people who manage their blood sugar well have low risk of complication [68, 69]. Managing diabetes typically involves modifying one's diet, engaging in physical activity, losing weight, and using appropriate medications (insulin, oral, and herbal) [67].

The American College of Physicians recommends maintaining a target glycated haemoglobin (HbA1c) level of 7–8% for diabetes treatment [70]. Health conditions including high blood pressure, obesity, metabolic syndrome, as well as physical idleness can exacerbate the negative effects of diabetes [71].

The same principles for controlling diabetes can be applied to the general population, but interventions may need to be tailored to specific populations. Further research is necessary to determine if self-monitoring is effective for individuals with serious mental illnesses as well as type 2 diabetes and if the outcomes are comparable to those seen in the general population [72].

3.1 Medication

To be effective, most treatments for diabetes work by reducing glucose levels in different ways [73]. It is widely accepted that maintaining tight control over blood glucose levels helps people with diabetes to avoid additional health problems such as kidney or eye issues [74, 75]. The kind of medications used here include the following:

- Oral antidiabetic medications: Type 2 diabetes is frequently managed with oral antidiabetic medicines [76]. Metformin, the first-line treatment, is so due to its proven effectiveness in reducing mortality [76]. Metformin reduces glucose production by the liver [78]. There are several classes of pharmaceuticals, mostly administered orally, that can aid in managing hyperglycaemia in individual having type 2 diabetes such as insulin-stimulating production (like sulfonylureas), reduce glucose uptake from the gastrointestinal system (like acarbose), block the metabolic function of the enzyme DPP-4 which inhibits incretins like GLP-1 and GIP (like sitagliptin), improve the body's insulin responsiveness (like thiazolidinediones), and increase glucose excretion from the body (like SGLT2 inhibitors) [78] (Table 2).
- Insulin Therapy: Insulin is an anti-glycaemic agent drug that lowers high sugar levels, especially in people with type 1 diabetes and, to a lesser extent, in people with type 2 diabetes as well as associated consequences [86]. The World Health Organization recognizes insulin as an essential medication that should be widely available and accessible to people globally [6, 87]. Type 1 diabetes results from a deficiency of insulin, and insulin is commonly used to manage this condition [88]. Subcutaneous injections or insulin pumps are the typical methods of administering insulin, but some types can also be given through intravenous or intramuscular injections [86, 88]. The speed at which insulin is processed in the body differs, resulting in various peak times and lengths of effect [88]. Insulins that act more quickly often have extended peak durations and remain in the body for longer periods [89]. In the management of type 2 diabetes, a sustained-release form of insulin is usually introduced initially, while continuing with oral medications [77]. Insulin dosages are then increased until the desired glucose levels are achieved [90].

  It is important to monitor patients regularly and provide comprehensive education on insulin therapy, as improper administration can have serious consequences. For example, if food intake is reduced, the insulin dosage may need to be reduced as well [91]. Physical exercise affects the absorption of glucose by cells in the body, which is regulated by insulin with varying onset times and durations. This means that exercise can reduce the need for insulin [92]. However, insulin therapy also carries risks due to the inability to consistently monitor an individual's blood glucose level and adjust insulin infusion accordingly [93].
- Herbal Medication: For centuries, people have used herbal medicine to treat several illnesses, including diabetes [94]. Diabetes is a chronic condition that results in high blood sugar levels, which can lead to severe complications if not managed effectively. Despite the availability of conventional treatments like oral hypoglycaemic drugs and insulin therapy, numerous individuals with diabetes opt for complementary and alternative therapies, such as herbal medicines [94]. Fenugreek, cinnamon, bitter melon, and *Gymnema sylvestre* are some of the herbs that have been employed in the past to treat diabetes. These herbs are believed to help

Table 2 The antidiabetic drugs that can be taken orally have varying modes of action and potential side effects

Oral agent	Mechanism of action	Side effect	References
Sulfonylurea, e.g. gliclazide, glibenclamide, glipizide, glimepiride	Sulfonylureas act on pancreatic beta cells by binding to and blocking the KATP channels, which are sensitive to ATP and located in the cellular membrane. This action prevents potassium from leaving the cell, leading to depolarization. As a result, voltage-gated channels for Ca^{2+} open, causing an increase in cytosolic Ca^{2+}. This increase in Ca^{2+} levels results in a greater integration of insulin granules with the phospholipid bilayer and consequently, an increase in the production of mature insulin	Hypoglycaemia, weight gain, gastrointestinal upset, headache, and hypersensitivity reactions	[79, 80]
Postprandial glucose regulators, e.g. repaglinidine, nateglinide	Postprandial glucose control medications work by interacting with KATP channels in the cellular membrane of pancreatic beta cells, which are sensitive to ATP. These medications block the postprandial glucose control mechanism by preventing potassium from leaving the cell, leading to depolarization. This depolarization causes the opening of the Ca^{2+} channels which is voltage-gated. This allows an influx of Ca^{2+} ion into the cell increasing the concentration. The increased Ca^{2+} concentration facilitates the interaction of the membrane of the cells and insulin granules, increasing the synthesis of mature insulin. However, the effects of these medications are short-	Hypoglycaemia and weight gain	[81]

(continued)

Table 2 (continued)

Oral agent	Mechanism of action	Side effect	References
	lived with a peak of two to 4 h after administration and a duration of less than 6 h, as they are rapidly absorbed and eliminated from the body		
Biguanide, e.g. metformin	Biguanides work by activating the AMP-activated protein kinase (AMPK). The breakdown and synthesis of fats and carbohydrates, insulin signalling pathways, as well as the maintenance of overall ATP homeostasis heavily depend on this protein. When AMPK is activated, expression of the small heterodimer partner is increased, which leads to the reduction in the synthesis of genes partaking in glucose production in the liver, particularly those responsible for phosphoenolpyruvate carboxykinase as well as glucose-6-phosphatase	Gastrointestinal upset and lactic acidosis	[82, 83]
Thiazolidinedione, e.g. pioglitazone, rosiglitazone	Thiazolidinediones (TZDs) are a class of drugs that work by activating PPARs, which are specific nuclear receptors also known as PPAR-gamma or PPARG. They are categorized as PPARG agonists. Free fatty acids (FFAs) and eicosanoids are the main ligands for these receptors. When the PPAR receptor is activated in combination with the retinoid X receptor (RXR), it can affect gene expression, suppressing some genes and increasing the transcription of others. The transcription of certain genes mainly causes adipose cells to store more fatty	Weight gain and oedema	[84, 85]

(continued)

Table 2 (continued)

Oral agent	Mechanism of action	Side effect	References
	acids, leading to a decreased transported of fatty acids. Consequently, energy is generated through the breakdown of sugars, especially glucose, to produce energy for various biological processes		
Alpha-glucosidase inhibitors, e.g. acarbose	Alpha-glucosidase inhibitors work by hindering the conversion of oligosaccharides into monosaccharides, which is carried out by alpha-glucosidase enzymes in the small intestine. Acarbose, a type of alpha-glucosidase inhibitor, lowers postprandial blood glucose levels by competing with dietary polysaccharides for alpha-glucosidase enzymes, which reduces the absorption of glucose	Gastrointestinal upset	[81]

regulate blood sugar levels by boosting insulin production, reducing sugar cravings, enhancing insulin sensitivity, and slowing down carbohydrate absorption [95]. Various herbal remedies, including *Acacia arabica*, *Allium sativum*, *Aegle marmelos* L., *Zingiber officinale*, *Aloe barbadensis*, and *Eucalyptus globulus*, are naturally insulin-mimetic or insulin secretagogues [93]. Some herbs, such as *Juniperus communis*, *Salix planifolia*, and *Pinus banksiana*, help in the efficient absorption of glucose [95], while others serve as α-amylase inhibitors, such as *Eugenia cumini* seeds (632 ρg/ml) [96], *Aloe barbadensis* (30 ρg/ml) [96], *Morus alba* (1440 ρg/ml) [97], *Pterocarpus marsupium* (0.9 ρg/ml) [98], and *Costus pictus* (2510 ρg/ml) [99], among others.

3.2 Current Research on Chinese and Ayurvedic Medicines

One of the first methods of treating and managing illness dates back to the Chinese and Ayurveda medicine [100]. Chinese and Ayurvedic medicine make up more than 65% of all herbal on the market today [101]. It is crucial to analyse the conclusions the most recent research has produced.

Emblica officinalis's antidiabetic properties were evaluated by [102]. According to their findings, 200 mg/kg b.w. of the fruit extract significantly lessened sugar level in the blood. In their study on *Salacia reticulata*, [103] found that compounds isolated from the plant, such as Salcinol and kotalanol, had the capacity to inhibit -glucosidase. Combinations of *S. reticulata* and *Catharanthus roseus* L. showed reduced level of fat and glucose in rats induced to diabetes. Furthermore, they say that in comparison to controls, herbs lowered level of glucose in the blood of diabetic-induced rats. Inhibition of the enzymes aldose, glucosidase, as well as pancreatic lipase is what mostly lowers blood glucose levels. Research carried out in vitro employing a rat pancreatic cell line RINm5F have revealed that the fractions of *Tinospora cordifolia* stem rich in isoquinoline alkaloids exhibit properties of insulin release and mimicry. Similarly, in vivo experiments have also confirmed these effects. These fractions include palmatine, jatrorrhizine, and magnoflorine [104]. Another isoquinoline alkaloid, called berberine, has been the subject of research and has been proven effective in both experimental and human diabetes. According to reports, berberine can reduce high blood sugar levels just as effectively as metformin [105]. Furthermore, the stem and root of *Tinospora cordifolia* contain neosporin, isocolumbin, palmatine, tinocordiside, cordioside, and sitosterol compounds, which are believed to possess properties such as antidiabetic, anti-hyperlipidemic, and antioxidant properties [103]. *Aegle marmelos* (L.) Corrêa demonstrated a considerable drop in HbA1c, PPBG, and BMI, while *Allium sativum* L. showed a significant decrease in PPBG and DBP; *Aloe vera* L. was found to significantly reduce BMI, and *Anethum graveolens* L. was found to suggestively lower fasting insulin, insulin resistance, LDL, and other parameters, according to [106] systematic review and meta-analysis from 2022. They arrive at the conclusion that Ayurveda botanicals have favourable effects on outcomes connected to type 2 diabetes [107]. conducted a research on people who have been diagnosed of type 2 diabetes, where they evaluated the impact of a hydroalcoholic extract of *Crocus sativus* L. on various factors such as HbA1c, lipid profile, fasting blood glucose, and renal and liver function tests. According to the study findings, the extract was potent in reducing HbA1c, fasting blood glucose, serum cholesterol concentration, and LDL-c levels, by improving insulin sensitivity and enhancing antioxidant enzyme activity. However, there was no noticeable difference observed in the liver and renal function tests. The study conducted by [108] on *Azadirachta indica* leaves and twigs revealed that the aqueous extract had a significant effect in reducing fasting blood sugar, post-prandial glucose level, insulin resistance, as well as HbA1c when compared to a placebo. Moreover, neem was found to be more effective than the placebo in terms of improving endothelial function, reducing oxidative stress, and systemic inflammation. The extract was effective across all doses, but it had no impact on lipid profile or platelet aggregation. As per [109], supplementing with *Anethum graveolens* (dill) powder led to significant reductions in total cholesterol, malondialdehyde, and low-density lipoprotein cholesterol levels in the group treated. The homeostatic typical valuation of insulin resistance and mean serum levels of insulin also showed significant improvement. The intervention group had considerably greater average of serum total antioxidant capacity and high-density lipoprotein compared to baseline assessment. The frequency of digestive complaints related to colonic motility was the only one that significantly decreased with supplementation.

The treated groups had considerably lower mean scores than the control group when malondialdehyde, low-density lipoprotein cholesterol, insulin, and total cholesterol levels were considered. Additionally, the treatment group's mean improvements in high-density lipoprotein were substantially higher in comparison to that of the control group. In an experiment conducted by [110], rats are being nourished diet heavy in fat and receiving streptozotocin (STZ/HFD) treatment, which resulted in distorted insulin resistance (IR) pointers across the three IR valuation methods used in the study. Their insulin and sugar levels were much higher than normal throughout the experiment. The study used EmbliQur, a treatment that comprises *Emblica officinalis* and *Curcuma longa*, and it is verified to be effective in reducing the elevated BGL caused by HFD/STZ and improving IR in all animals subjected to the IR test. According to a study conducted by [111] on the results of using separated compounds and unrefined components from *Striga orobanchioides* Benth on diabetic rats induced by streptozotocin, pentacyclic triterpenoids were found to be present (81.5% w/w) in the ethylacetate-methanolic fraction of the ethanol extract. Betulin was discovered through spectroscopic analysis of the isolated chemical. The bioactive fraction and betulin demonstrated substantial inhibition of digestive enzymes in vitro, and the group receiving betulin at a dose of 40 mg/kg showed substantial improvement in their level of diabetes. Gene expression studies disclosed that betulin at a dosage of 40 mg kg^{-1} improves liver glucose usage, sinks hepatic inflammation, and surges insulin making. In silico tests revealed that the plant compound has a strong ability to impede the enzymes DPP-IV, α-glucosidase, GK, and α-amylase.

According to [112], JQ-R, a Chinese herbal medicine preparation, consists of three active components, namely, the polyhydric alcohols of *L. japonica*, *A. membranaceus* saponin, and the whole alkaloid fraction of *C. chinensis*. These fractions have been verified to possess antidiabetic effects in diabetic KK^{Ay} mice and in vitro. The aforementioned components efficiently control the metabolism of glucose, reduce oxidative stress, and regulate related inflammatory responses. By promoting glucose absorption through the insulin-dependent PI3K-AKT signalling pathway, JQ-R was effective in improving glucose metabolism partially in diabetic KK^{Ay} mice. According to a study by [113], mulberry leaves were found to have potential in treating diabetes through in vivo and in vitro tests. Aqueous extract of the leaves which revealed alkaloids and flavonoids as the main hypoglycaemic components was given to diabetic mice, producing rates of inhibition of $87.29 \pm 1.32\%$ and $86.12 \pm 1.79\%$, respectively, on glucosidase activity. When given to diabetic rats, aqueous mulberry leaf extract improved glucose tolerance by 19.02% and lowered blood sugar levels by 28.17%. The extract also had positive effects on gut flora, reduced kidney damage, and lowered blood levels of insulin, tumour necrosis factor-α (TNF-α), glycosylated serum protein, and free fatty acid. These results imply that the beneficial interactions between the different components of the leaves of mulberry may be responsible for their hypoglycaemic actions. According to [114], the study found that the polyphenols of mulberry leaf, which consist of various compounds such as rutin, benzoic acid, hyperoside, and chlorogenic acid, effectively suppressed glucose transport. The degree of inhibition increased over time and with higher concentrations of MLPs. The blood glucose lowering impact of *Dendrobium officinale* and the related mechanisms were demonstrated in a study by [115]. The study found that *Dendrobium officinale* had an impact on the cAMP-PKA and Akt/FoxO1 signalling pathways conciliated by glucagon. This resulted in the encouraged synthesis of glycogen as well as inhibition of glycogen breakdown and gluconeogenesis in the liver. In a study conducted by [116], it was discovered that polyphenols derived from *Azadirachta indica* have the ability to enhance glucose absorption in muscle tissue and decrease the functioning of the proteins responsible for breaking down polysaccharides. As per [117], *Cordyceps militaris* extract exhibits a blood sugar-lowering effect. In addition, it has the ability to improve and repair damaged pancreatic islet cells, increase the thymus index, reduce pro-inflammatory cytokines release, and limit the release of cytokines. This indicates that the immunomodulatory and inflammatory-reducing qualities of *Cordyceps militaris* are closely associated. The study also showed that quercetin has a curative impact on diabetic retinopathy in rats with the disease. Furthermore, the study found a correlation between the treatment outcome and the level of expression of HO-1. In a study conducted by [118], a diabetic rat was treated with a microcapsule encapsulating quercetin, and the results on enteric nerves were assessed. The outcomes showed that by reducing oxidative stress, this therapy may mitigate the nerve damage brought on by diabetes [119]. conducted a study in 2022 that found ZSP to significantly increase the PI3K/AKT pathway's activity through increasing the expression of Akt1 and Akt2 proteins and the copying of the 4 catalytic PI3K subunit genes pik3ca, pik3cb, pik3cd, and pik3cg. The phosphorylated forms of these proteins were also considered due to their role in PI3K and AKT activation. The outcomes disclosed that ZSP enhanced PI3K/AKT signalling at both translational and transcriptional levels when likened to the model group. However, there was a discrepancy in the normal group's PI3K and AKT mRNA and protein expression, which could be due to possible differences in post-transcriptional regulation between wt/wt and db/db animals. Additionally, the high insulin levels in db/db mice may donate to the elevated PI3K and AKT expression as a result of ongoing insulin stimulation.

3.3 Several Traditional Herbal Medicines' Modes of Action

Plants exhibit antidiabetic properties by reactivating pancreatic cells to increase insulin production, inhibiting glucose absorption in the gastrointestinal tract, or promoting metabolites involved in insulin-dependent activities. The existence of certain compounds, such as polyphenols, flavonoids, terpenoids, and coumarins, in medicinal plants is believed to be responsible for their antidiabetic effects [120]. Different herbal remedies have diverse methods of demonstrating their antidiabetic function. Some of these include the following:

- Ginseng (*Panax* spp.): The greatest study has been done on this medicinal herb's hypoglycaemic effects. Ginseng's geographic origin, amount, dispensation, and diabetes types have the main effects on its potency and effectiveness [121]. *Panax ginseng*, sometimes recognized as Chinese or Korean ginseng, possess the uppermost medicinal potential [121]. Ginseng diminishes blood sugar by regulating several different mechanisms (Fig. 3). *Panax ginseng* at a concentration of 0.1–1.0 g/mL was found to significantly increase the amount of insulin released from pancreatic islets of rats which was isolated when glucose is present at an amount of 3.3 mmol/L [122]. Oral ingestion for 20 days of 100 mg/kg b.w. of heat-processed American ginseng was taken and was found to decrease serum glucose levels, glycosylated proteins, and glycated haemoglobin (HbA1c) in rats with diabetes caused by streptozotocin (STZ). Additionally, the treatment increased values of creatinine clearance and reduced the storage of N (ϵ)-(carboxymethyl) lysine and its receptors in the kidneys [123]. Ginsenosides are one of the numerous constituents that have been isolated from ginseng and are regarded to be the active ingredient. Polysaccharides, peptides, and poly acetylenic alcohols are additional components [124]. The water-soluble extract of ginseng root has the ability to reduce diabetes when compared to its fat-soluble components. The ginsenosides present in both American and Panax ginseng are thought to improve insulin resistance and its sensitivity at the target tissues [121]. There are some foods that don't appear to assist lower blood sugar levels since they have low ginsenoside contents [125].

- Bitter lemon (*Momordica charantia*): Research has indicated that bitter melon exhibits hypoglycaemic effects in cell culture, animal experiments, as well as clinical trials involving humans [126, 127]. The bitter melon's antidiabetic ingredients include alkaloids, vicine, charantin, polypeptide-P, and other

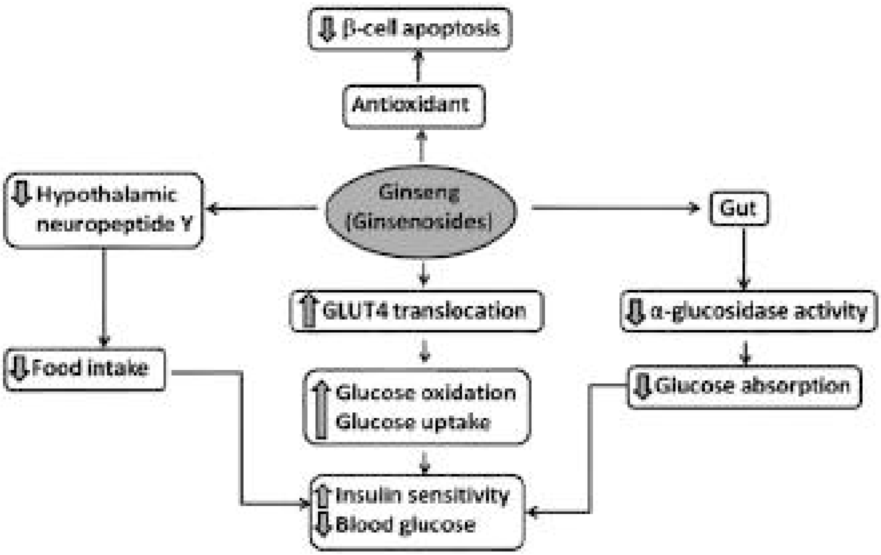

Fig. 3 The mechanism by which ginseng increases glucose metabolism [121]

non-specific bioactive antioxidants [121]. Bitter melon's oleanolate-glycoside was found to enhance glucose tolerance in individuals with type 2 diabetes by inhibiting the absorption of sugar in the intestines [121]. Bitter melon has been verified to reduce levels of glucose in the blood in mice with diabetes caused by STZ by preventing cell peroxidation and apoptosis, increasing the amount of liver and muscle glycogen, and stimulation of the enzymes in the liver such as phosphofructokinase, glucokinase, and hexokinase activity [128]. In diabetic rats caused by STZ, the inactivity of liver fructose-1,6-bisphosphatase and glucose-6-phosphatase activities resulted in decreased gluconeogenesis. However, overexpression of glucose-6-phosphate dehydrogenase (G6PDH) in red blood cells and hepatocytes enhanced glucose oxidation [129]. In summary, research conducted on both humans and animals has demonstrated that bitter melon plays a role in regulating glucose and lipid metabolism (as depicted in Fig. 4). Although the exact mode of action is still being investigated, bitter melon can be utilized as a dietary supplement or herbal medicine for managing diabetes and/or metabolic conditions [130].

- Fenugreek (*Trigonella foenum-graecum*): A type of medicinal product made from a plant's dried seed that is indigenous to North Africa, China, and India [121]. Fenugreek was used in Ayurveda for millennia as both a demulcent (an agent that soothes inflammation of the throat, nose, mouth, or skin) and a laxative [121]. Fenugreek seed powder users apparently experience decreased postprandial glucose levels [132]. This can be due to a slowdown in gastrointestinal glucose absorption and a bulk laxative action. Fenugreek, however, also appears to boost insulin release since it contains 4-isoleucine [131, 133]. Research has demonstrated that fenugreek seeds are effective in improving glycaemic regulator and reducing resistance of insulin in people with type 2 diabetes that is moderate. Furthermore, fenugreek seed supplementation has a positive effect on hypertriglycaeridemia and increases HDL-C levels. Supplementation with

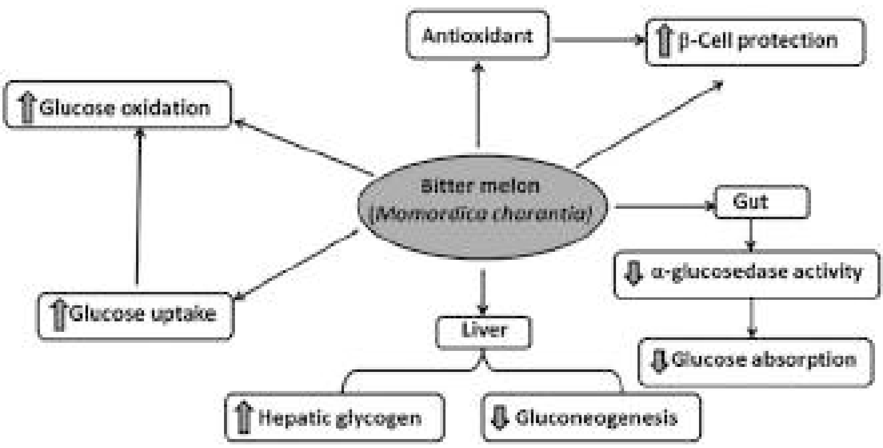

Fig. 4 Bitter melon's mechanism of action in lowering blood sugar [121]

fenugreek leaves improves body weight and liver glycogen levels, and its impact on glucose metabolism is significant, similar to the effects of glibenclamide [134]. Studies conducted on experimental diabetes indicate that fenugreek leaf powder has the ability to reduce oxidative stress. In rats with diabetes, administration of fenugreek has been found to decrease lipid peroxidation and significantly enhance the antioxidant system [135].

- *Coptis chinensis*: In China, *Coptis chinensis* is widely utilized for the management of diabetes. The active component of the *Coptis chinensis* plant is an isoquinoline alkaloid known as berberine. The plant stems, bark, roots, and rhizomes contain it [121]. Giving diabetic rats berberine through intragastric administration at doses of 100 and 200 mg kg^{-1} resulted in a reduction of fasting blood sugar levels, total cholesterol, triglycerides, and low-density lipoprotein cholesterol (LDL-C) in the blood. At the same time, there was an increase in high-density lipoprotein cholesterol (HDL-C), nitric oxide (NO) levels, and prevention of the elevation of glutathione peroxidase (GSH-px) and superoxide dismutase (SOD) [136]. Berberine can bring about a reduction in body weight and an increase in insulin response, which may be due to various mechanisms. In MIN6 cells and rat islets, it decreases glucose-stimulated insulin secretion (GSIS), boosts AMPK activity, and enhances the translocation of GLUT4 in adipocytes and myotubes [137, 138]. In addition, it raises the synthesis of PPAR α/δ/γ/ proteins in the liver and enhances the expression of insulin receptors in the liver and skeletal muscle cells, which stimulates the use of glucose by cells in the presence of insulin [139]. Furthermore, in rats with diabetes induced by STZ, berberine significantly lowers the activity of disaccharidases and α-glucuronidase in the intestine [140]. A berberine derivative called dihydroberberine (dhBBR) displayed advantageous benefits in vivo in animals fed a high-fat diet [141]. Figure 5 provides an overview of the several ways that berberine lowers blood sugar levels.
- Banaba (*Lagerstroemia speciosa*): Banaba belongs to the same family as the banaba plant. Banaba leaf extracts are highly favoured in the Philippines and Southeast Asia, and they are frequently employed by diabetics in North America. Banaba comprises two significant components that have two significant impacts on individuals with diabetes [121]. Corosolic acid and ellagitannins are two components found in banaba extracts that appear to function similarly to insulin and stimulate the activation of insulin receptors [142]. The main reasons for the above-mentioned effect are thought to be either the inhibition of tyrosine phosphatase or the stimulation of tyrosine kinase insulin receptor. Initial clinical investigations indicate that individuals suffering from type 2 diabetes who take a special banaba extract (named glucosol) for 2 weeks experience a notable 10% drop in their the levels of sugar in the blood as opposed to those who receive a placebo [143]. Research has shown that banaba possesses a hypoglycaemic impact; however, additional investigation is required to assess its potential long-term effects [121].
- *Agaricus* mushroom (*Agaricus* spp.): Despite having its roots in Brazil, *Agaricus* is now readily accessible in Japan, China, and other Asian countries. It has carbohydrates like betaglucans, which, like those in other mushrooms, seem to boost immune function markers and serve as an immunostimulant [121]. *Agaricus*

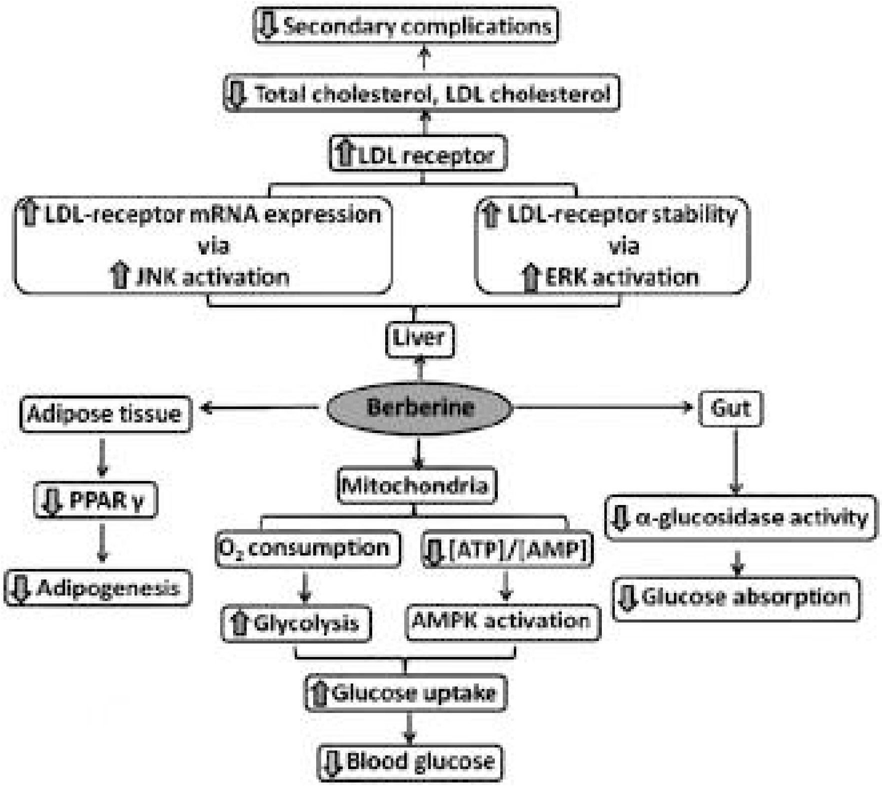

Fig. 5 The mechanism of action of berberine in controlling blood sugar [121]

mushrooms contain certain compounds that can enhance sensitivity of insulin as well as lowering resistance of insulin in individuals with type 2 diabetes. Additionally, these compounds may elevate adiponectin levels, which could potentially decrease insulin resistance [121]. Clinical studies have shown that individuals with type 2 diabetes who augmented their regular treatment with half gram of *Agaricus* mushroom extract thrice daily had decreased overnight levels of insulin in comparison to others who abstained from eating the mushroom [144]. For the majority of people, the *Agaricus* mushroom extract appears to be harmless when used for a period of up to 12 weeks. However, it can cause potentially harmful drops in blood sugar levels (hypoglycaemia) [144].

3.4 Surgery

If type 1 diabetics experience severe complications from their illness, such as end-stage kidney disease that requires a kidney transplant, a pancreas transplant may be considered as an uncommon alternative [145, 146].

Weight loss surgery, which is often used to treat obesity, can also be a successful therapy for obese patients that are diabetic [147]. Following surgery, many patients are competent to maintain normoglycaemia without the need for medication, which can reduce the risk of long-term mortality [148, 149]. However, it is unclear what BMI thresholds should be used to determine if surgery is appropriate [149]. It is recommended that individuals who are unable to control their blood sugar levels and weight should consider this option [150].

3.5 Self-Management and Support

In countries like the United Kingdom that have a general practitioner system, diabetes treatment is typically provided outside of hospitals. Hospital-based care is usually only necessary in cases of complications, difficulty in controlling blood sugar levels, or for research purposes. General practitioners and specialists may work together on occasion to provide care. Additionally, managing diabetes through telehealth from the comfort of one's own home can be a successful strategy [151].

Computer-based self-management interventions are utilized to gather data for educational programmes that aim to aid self-care in type 2 diabetics [152]. Nonetheless, there is insufficient proof to back the effects on various physiological, psychological, and emotional outcomes such as blood pressure, cholesterol, weight, dietary changes, depression, depression-related quality of life, and other related factors [152, 153].

- Glycaemic surveillance and management: Glycaemic monitoring and control refers to the process of measuring and managing blood glucose levels in individuals with diabetes [154]. Blood glucose monitoring using a glucometer is a common method employed for this purpose. If necessary, adjustments are made to insulin dosages, dietary habits, and exercise routines to ensure that blood glucose levels remain within a predetermined target range [155, 156]. Regular monitoring is critical in reducing short-term and long-term issues associated with diabetes, including hypoglycaemia, renal damage, and cardiovascular disease. It is recommended to consult with a healthcare professional to determine the most appropriate target range and monitoring frequency for your individual needs [153]. Additionally, measuring haemoglobin A1c levels every three to 6 months offers an average of blood glucose levels over the past two to 3 months and is vital for effective monitoring [157]. Managing blood glucose levels effectively in addition to monitoring blood glucose also requires a healthy diet, regular exercise, and appropriate medication use, including insulin and oral hypoglycaemic agents [158]. Individuals with diabetes should work with a healthcare professional to develop a personalized plan for monitoring and regulating their blood glucose levels.
- Diabetes education: Diabetes education pertains to educating people on the essential expertise as well as ability to treat their disease, which involves understanding the causes; recognizing warning signs, symptoms, and possible

complications; as well as learning preventative and treatment strategies [159]. This education encompasses several topics, such as comprehending the effects of diabetes on the body, monitoring blood glucose levels, making necessary insulin adjustments, creating healthy meal plans, engaging in physical activity and exercise, understanding medication usage, identifying and avoiding potential complications, managing emotional well-being and stress, and many more [159]. Diabetes education can be offered in various settings, including hospitals, clinics, community centres, and online [159]. Self-management lessons, workshops, or one-on-one counselling sessions may be included, and it can be provided either in a group or individual context [87, 160]. Proper diabetes education and receiving appropriate care can lead to improved overall quality of life, decreased chances of complications, and more fulfilling lives for individuals with diabetes. By collaborating with healthcare professionals to design a customized education programme and attending diabetes education sessions on a regular basis, individuals with diabetes can stay informed about new developments in science and treatment methods [159].

4 Prevention of Diabetes

Type 1 diabetes has no known mitigating measures [161]. However, healthy lifestyle choices, such as diet and exercise modifications, can often prevent type 2 diabetes from developing or postpone it [162], and it constitutes approximately 85–90% of all global instances [163].

4.1 Lifestyle

Numerous healthy lifestyle practices have been established to assist in the prevention of diabetes. It is believed that smoking cigarettes significantly increases the risk of developing diabetes [164]. Stopping smoking is a critical preventive measure as it increases the chances of developing diabetes and its associated consequences [165]. According to a 2015 study, there is a clear link between the amount of exposure to cigarette smoke along with the chance of developing type 2 diabetes, with the risk increasing as the amount of exposure to cigarette smoke increases [164]. The 2014 Surgeon General's Report highlights the position of promoting smoking termination as a crucial public health strategy to combat the global diabetes epidemic, stating that people who smoke frequently have 30–40% higher likelihood of developing type 2 diabetes than individuals who do not [166].

4.2 Alcohol

In 2010, approximately 41% of adults worldwide consumed alcohol based on available data [167]. The WHO reported that on average, individuals above

15 years of age consume 6.2 L of unadulterated alcohol per year, corresponding to 13.5 g [168]. A 2015 report indicated that 86.4% of individuals 18 years or older in the United States reported consuming alcohol [169]. The consumption of alcohol has the potential to increase the likelihood of developing diabetes. Furthermore, extreme alcohol ingestion portends various chronic health conditions, including diabetes mellitus and its associated problems [169]. Multiple studies have revealed different associations between alcohol ingestion and diabetes risk, including positive, null, U-shaped, or J-shaped relationships [170]. While liquor ingestion may have various effects on blood sugar levels, studies have found that prolonged and regular drinking can cause upsurge insulin resistance [171]. To prevent the negative effects of alcohol on insulin resistance and blood sugar, it is recommended to minimize alcohol intake to a maximum of two drinks per day. Additionally, it is advised to consume alcohol with food or a snack, rather than on an empty stomach [172]. Average alcohol consumption is linked to a minimized chance of developing type 2 diabetes mellitus (T2DM) compared to those who drink minimally or not at all, while consuming excessive amounts of alcohol carries the same danger as not drinking [169]. Assessment of cohort research as a whole conducted in various countries has provided convincing evidence that consuming fewer than 30 g of alcohol in a day is linked with reduced risk of developing T2DM in the populace [173]. Another study suggests that the highest risk reduction for developing T2DM is 18% for individuals who consume between 10 and 14 g of alcohol per day [174]. Research has indicated that alcohol consumption may prevent diabetes from occurring in men and women by delaying its onset [175, 176]. According to research, the consumption of approximately 24 g and 22 g of alcohol per day for women and men, respectively, is linked to the highest level of protection against developing diabetes. However, doses exceeding 50 g for women and 60 g for men daily are deemed harmful [177]. The frequency of alcohol use can also affect the level of risk [174]. Binge drinking, which involves consuming large amounts of alcohol in one go, has been linked with amplified risk of heart disease and other negative health consequences when compared to individuals who consume the same amount of alcohol but with a more frequent consumption pattern spread out across three to 4 days per week [174, 177].

According to a study that followed participants over time, drinking alcohol may advance the chances of getting type 2 diabetes mellitus [178]. Appraisal of multiple studies found that women who drink a lot of alcohol daily are more susceptible to type 2 diabetes development, but this correlation isn't seen as strongly in men [177]. Another study shows that excessive alcohol intake can prevent the body from controlling glucose levels, putting both men and women at higher risk of developing type 2 diabetes [179]. A summary of numerous studies that looked at participants over time found no considerable contrast in the likelihood of getting type 2 diabetes mellitus between those who drink heavily and those who consume less than 48 g of alcohol per day [180]. However, consuming large amounts of alcohol can worsen issues with managing blood glucose levels, which can promote the likelihood of having type 2 diabetes and its related complications [178]. There is evidence that drinking alcohol can lower insulin resistance [181]. Alcohol intake can affect how the body processes glucose, whether someone has diabetes or not. Drinking alcohol on an empty stomach can surge the risk of hypoglycaemia by reducing the making of glucose and breakdown of stored glucose, especially when combined with certain medications and low glycogen levels [182]. Explanations for how alcohol can add to the progression of diabetes includes decreased activity in beta cells, increased insulin resistance, and a reduction in glucose tolerance. Studies have found a connection between higher blood glucose levels and decreased beta cell activity, as well as increased alcohol consumption [183]. Frequent and excessive alcohol intake can make it harder for the body to maintain healthy blood glucose levels, which can cause problems with glucose tolerance and increase insulin resistance [184]. Alcohol consumption can lead to weight increase, an important trigger for type 2 diabetes, according to data [184, 185].

4.3 Diet and Exercise Modification

Combining a healthy diet with regular exercise can reduce your chances of developing diabetes, help you lose weight, lower your cholesterol levels, and reduce your blood pressure [160]. The American Diabetes Association (ADA) endorses maintaining a healthy weight, engaging in at least two and a half hours of physical activity per week (even just taking long walks can be enough), consuming foods with moderate amounts of fat (about 80% of caloric intake should come from healthy fats), and getting enough fibre from sources like whole grains [186]. Getting more exercise can help lower type 2 diabetes risk, particularly after eating a high-carbohydrate meal that causes a spike in blood sugar levels [187, 188].

Increase your intake of whole grains, fibre, and the polyunsaturated fats in fish, nuts, and vegetable oils [172, 188]. Red meat and other foods high in saturated fat should be avoided so as to lessen the likelihood of diabetes development [189].

5 Conclusion

Diabetes has long been treated and prevented with herbal therapy. Herbs work in a variety of ways, including by boosting glucose absorption, lowering oxidative stress, and improving insulin sensitivity. It is encouraging that herbal medicine can be used to treat and prevent diabetes. However, despite these functions, there need to be more research on identifying bioactive constituents of plants and elucidating their mechanism of action.

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Therapeutic Applications of Herbal Medicines for the Prevention and Management of Cancer

21

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Contents

Abstract

Plants serve as a revered base for the natural products that support human health. There is growing evidence that Ayurvedic medicinal herbs and the secondary metabolites they produce can be used to treat cancer. Numerous research have documented the value of herbal plants in immune system modulation, cancer

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patient survival, and quality of life. For the treatment and prevention of cancer, Ayurvedic herbal remedies are included into mainstream therapy. Humans are becoming more and more reliant on herbal remedies because they are safer and more potent than manufactured pharmaceuticals. The value of medicinal plants is rising due to its potential and uses. The use of Ayurvedic medicinal herbs for the prevention and treatment of various malignancies is examined in this chapter's review work. This chapter also discusses a report on the potential use of phytoconstituents in the treatment of cancer.

Keywords

Ayurvedic medicinal plants · Cancer · Phytoconstituents

Abbreviations

DNA	Deoxyribonucleic acid
S.Nr	Serial number
WHO	World Health Organization

1 Introduction

Herbalism is a term that literally refers to the use of plants to treat a variety of human illnesses. The word "herb" derives from a Latin word that designates a plant's vegetative and reproductive parts that are used for therapeutic purposes. The use of medicinal plants as a source of drugs to treat or prevent various diseases may be dated back about five millennia in recorded accounts from India's early civilisations. The natural products that enhance human health are derived from plants. The Plantae might contain a range of therapeutically useful physiologically active compounds, as well as unique templates for synthesising analogues and stimulating tools for better understanding biological processes [1–8].

According to the WHO in 2011, approximately 88% of the world's population relies on herbal medications that have therapeutic efficacy. Long before the prehistoric era, plants were employed for medical purposes. Chinese, Egyptian, and Unani authors of old or ancient writings all agreed that items derived from medicinal plants are utilised as areas of medicine. Over a thousand years ago, Vaids and Hakim from various civilisations used medicinal herbs for therapeutic purposes [9–20].

1.1 Traditional Medicine System

Traditional systems of drugs still are being widely practised in various ways. The use of synthetic drugs and its negative impact and high cost leads to faith of human beings on medicinal plants. Utilisation of plant materials in the form of herbal medicines progressed because of improper supply of medicines and increase in population. The diversified and verdant plant kingdom is renowned for their therapeutic value [21–32]. More utilisation of plants for medicinal purposes enhanced because of development of varied herbal therapies. According to the Turkish Journal, significant value of medicinal plants or herbal plants increased importance over allopathic medicines because of presence of many narcotic compounds. Medicinal plants bestowed with varied compounds having biotic origin and showing medicinal and therapeutic values. Medicinal plants are exceptionally a magnificent possibility for curing human diseases because of presence of analgesic, anti-inflammatory, anti-tumour, and antiviral properties. Medicinal plants have various better properties over chemical medicines because plant-derived compounds are more tolerant, non-toxic, and beneficial for the human body [33–48].

India is the birth place and vast repository of various plants having medicinal value. Among many ancient and primitive civilisations, Indian civilisation has been evidenced for medicines assessed from herbs. Flora of India provides path for processing of herbal drugs. Nearly eight thousand herbal medicines are documented in India by AYUSH. Various types of medications are accepted by Indian culture which was assessed by Unani, Siddha, and Ayurvedic systems [49–62].

80% of the world's population, according to the WHO (2012), relies on herbal medicines for health reasons. Around 21,000 plant species with therapeutic properties that are used globally for basic health were also documented by the WHO (2012). More than 30% of a plant species' total biomass was employed for therapeutic reasons. The contribution from quickly developing nations like China and India is at most 80%. India and China contributed more towards the herbal medication. Rural population of these countries mainly rely on indigenous medication system. Treatment is associated with natural system having maximum positive impact and minimal negative impact. Remedies based on nature are good for human beings and medicinal plants play very vital role in herbal treatment. Medicinal plants are restrained as ironic possessions of components which can be utilised for drug formation. Therapeutic value of plants mainly depends on medicinal properties of plants [63–85].

The biologically active chemicals found in the plant kingdom constitute a largely untapped source of potential medicines as well as useful tools and special templates for studying biological processes. In India, the use of plant composites for therapeutic purposes has grown over time [80–82, 84–88]. Traditional medicines have played a significant part in addressing the world's healthcare requirements in the past, are still doing so today, and will continue to do so in the future. The Ayurvedic structure in India has described an enormous number of those medicines buttressed plants or plant material and therefore the determination of their morphological and pharmacological or pharmacognostical characters. It can offer a better considerate of their active principles and mode of accomplishment. Young people nowadays are completely dependent on allopathy, therefore using medicinal plant mixtures to halt the disease's sneaky effects is an alternative to Western medications, which often have terrible side effects. Secondary metabolites found in medicinal plants, such as flavonoids, flavones, anthocyanins, lignans, and coumarins, have been shown to have anticancer activities. Items are made from herbs, especially in Ayurveda. The use of traditional remedies in India was emphasised in the ancient Vedas and other writings [89–101]. Ayurveda has two primary goals: first, to promote health and second, to treat disease. Ayus and Veda, two Sanskrit terms that translate to "life" and "knowledge," respectively, are the origin of Ayurveda. Literally, it signifies understanding of and research into life. Ayurveda, the science of life, may be a part of India's illustrious Vedic legacy. Ayurveda, the oldest system of medicine in India, uses plant extracts to combat cancer-causing chemicals and eradicate tumours associated with cancer. Three major expositions on Ayurveda – the Charaka Samhita (a work on ancient Indian medicine), the Sushruta Samhita (a text on ancient Indian surgery), and the Kashyapa Samhita – are well known (text on gynaecology and child health). Ayurveda which prescribes a variety of medications for preserving health, vitality, youthfulness, and longevity. Ayurveda considers health as a equilibrium of delight or happiness of the soul, sense, and mind. Medicines prepared by Ayurvedic mechanisms well proved for cancer treatment having anti-cancerous properties [102–114]. Ayurvedic medicines have an ability to fight against the cancerous tumours. Preparations of medicines by Ayurvedic method act as an ancillary or a concomitant treatment alongside radiotherapy or chemotherapy. Negative impact of chemotherapy and radiotherapy can be reducing by usage of Ayurvedic medicines. Ayurvedic medicines assist to detain the cancerous cells where radiotherapy and chemotherapy have significant side effects. The ancient therapeutic art and science of Ayurveda focuses on enhancing, sustaining, and balancing mind-body health. It also takes a holistic approach to health and wellness. Meditation is the cornerstone of Ayurvedic medicine's approach to mind-body healing. Ayurvedic medicines that are sold in many parts of the world have undergone substantial research about their anti-cancer properties. Both Charaka and Sushruta Samhita referred to granthi and arbuda as the equivalents of cancer (malignant or major neoplasm). Based on the doshas involved, both are frequently inflammatory or non-inflammatory. The Ayurveda definition of health is the harmonious with coordination of vata, pitta, and kapha system within the body [115–131].

1.2 Cancer and Its Molecular Basis

Cancer is characterised by uncontrolled and unrestrained cell proliferation. It appears as a growth in a clinical context. A malformed mass of tissue known as neoplasm grows erratically and keeps growing long after the factors that first triggered its growth have been removed. A tumour is described as benign if its characteristics are believed to be essentially harmless, which implies that it wouldn't spread to neighbouring or distant regions, could be surgically removed without difficulty, and didn't constitute a severe threat to the patient's life [132–141]. Malignant tumours are collectively referred to as cancers, which are derived from the Latin word for crab, because they have a crab-like effect on the tissues. A malignant tumour has the ability to spread to distant areas (metastasize), harm nearby structures, and ultimately lead to death. Not all cancers result in death; with early detection and appropriate care, certain cancers can be cured [142–178].

Cells that proliferate expand quickly.

- Nucleotide sequence changes due to genetic mutation
- Agents that cause cancer, or carcinogens

Such genetically modified crops, rising cigarette use, fast food, alteration, or mutation may be inherited through the germ line or acquired through the action of environmental forces. Environmental agents Rapid development, neighborhood invasion, and the ability to metastasize far away are characteristics of malignant neoplasms. The ability of a cancer cell to proliferate in the absence of growth signals, resistance to signals that would normally stop a cell from dying, ability to form new blood vessels, ability to spread to other tissues, ability to metastasize to distant organs, and failure to repair damaged DNA are just a few of the key changes that occur in a cancer cell [179–202].

1.3 Common Cancers

Men are most likely to have lung, prostate, colorectal, stomach, and liver cancers worldwide. The most often diagnosed cancers in women are those of the breast, cervix, colorectum, lung, and stomach [203–217]. Oral, lung, stomach, and colorectal cancers are becoming prevalent among men in India. The most prevalent cancers in women are breast, cervix, colorectal, ovarian, and oral cancers [213, 216, 218–230]. According to the geographic region, prevalent social customs, and socioeconomic strata, different malignancies have different incidence rates. For instance, oral malignancies are more common in the Indian subcontinent than in Western nations. This is a result of increased use of chewable tobacco products including gutkha, paan, paan masala, khaini, and supari, among others. Due to inadequate genital cleanliness, cervical cancer is more prevalent in women from lower socioeconomic groups. Those who consume more fatty foods and fewer fibre in their diets are more likely to develop colorectal cancer [231–242].

1.4 Common Causes of Cancer

There are many agents who causes cancer or increases the risk of cancer. Some major agents are as follows (Fig. 1). Also some risk factors of cancer are shown in Table 1.

Fig. 1 Common causes of cancer [100]

Table 1 Some common conditions and risk factors of certain cancer

Tobacco	The single most significant preventable risk factor for cancer death worldwide is tobacco use. It is thought to be responsible for 22% of cancer deaths annually, according to the WHO. The vast majority of lung cancers are linked to smoking. Those who don't smoke have also been linked to malignancies by passive smoking. Danger rises as smoking frequency increases. The usage of smokeless tobacco is more common in the Indian subcontinent. They include using raw tobacco, betel quid, gutkha, pan masala, masheri, and other substances
Alcohol	Drinking alcohol increases your risk of acquiring a number of cancers that are related to alcohol. It has a synergistic impact when taken with smoking that significantly raises the risk of cancer development compared to when either chemical is consumed alone
Obesity	People with BMIs over 30 are considered obese. They are more likely to get cancer, diabetes, and heart disease. Regular exercise and a healthy, balanced diet are crucial for battling the obesity pandemic
Areca nut	It is also referred to as supari in India. It is called as Pan or betel quid and can be chewed either alone or in combination with betel leaf, catechu, and slaked lime. It is linked to a condition called submucous fibrosis, which gradually narrows the chewer's mouth aperture and is a sign of malignancy
Pollution	This refers to the environmental pollution of the air, water, and soil caused by the usage of carcinogenic substances. Both ingestion and air exposure to these compounds, which are known as carcinogens, have the potential to cause cancer

(continued)

Table 1 (continued)

Occupational exposure	Numerous compounds, such as asbestos, cadmium, ethylene oxide, benzopyrene, silica, ionising radiation, including radon, tanning booths, aluminium and coal manufacturing, and the manufacture of iron and steel, are known to cause various cancers. Understanding these is essential because the majority of these cancers are preventable with the correct knowledge and safety precautions
Radiation	Exposure to ionising radiation has been associated with a greater risk of acquiring a number of cancers. UV radiation has been related to skin cancers such basal cell carcinoma, squamous cell carcinoma, and melanoma
Biological agents	A wide range of germs, parasites, and viruses that raise people's chance of developing cancer. Oropharyngeal and cervical cancers have been linked to the human papilloma virus (HPV). Hepatitis B and C are associated with liver cancer. Schistosomiasis, a parasite infection, is associated with an increased risk of bladder cancer. *H. pylori* infection may result in stomach cancer
Lifestyle factors	High body mass index is associated with cancers of the oesophagus, colorectum, breast, endometrium, etc. Consuming red meat increases your chances of colon and rectal cancer. A nutritious diet that includes fruits, salads, and vegetables has been found to reduce the risk
Unprotected sex	Unprotected sex also causes cancer in males or females

2 Importance of Traditional Plant-Based Drugs in Treatment of Cancer

Plant-based drugs are becoming more and more common all around the world. Recent studies on herbal plants or medicine have made substantial progress in the pharmacological study of several plants used in traditional medical systems. The secondary metabolites or compounds found in therapeutic plants include tannins, terpenoids, alkaloids, and flavonoids. These compounds determine the therapeutic effectiveness of the plants, especially their antimicrobial activity. The use of plant-derived remedies as an indigenous treatment in traditional systems of medicine has been related to the advent of plant-derived medications in contemporary medicine. It has been discovered that several of the plants have potent anticancer properties. Cancer is treated with medicines made from plants [211, 212, 243–248]. The traditional herbal plants used to treat cancer are listed in the table below.

2.1 List of Traditional Plants Used to Treat Cancer

There are so many plants that possess anticancer properties. Some of these plants and their chemical constituents are shown in Table 2.

Table 2 Plants with anticancer potentials

Sr. No.	Botanical name	Family	Chemical constituents	Types of cancer
1.	*Ferula assafoetida*	Apiaceae	Coumarin compounds	Liver cancer [9]
2.	*Thymus vulgaris*	Lamiaceae	Thymol	Prostate cancer [10], head cancer [11]
3.	*Thymbra spicata*	Lamiaceae	Thymol	Lung cancer [12]
4.	*Taverniera spartea*	Fabaceae	Isoflavonoid compounds	Breast and prostate cancer [13]
5.	*Peganum harmala*	Nitrariaceae	Alkaloids	Breast cancer [14], cervical cancer [13, 22]
6.	*Viola tricolor*	Violaceae	Flavonoids	Cervical cancer [23]
7.	*Achillea wilhelmsii*	Asteraceae	Phenolic compounds	Colon cancer [24]
8.	*Mentha pulegium*	Lamiaceae	Piperitone	Blood cell cancer [25]
9.	*Ammi visnaga*	Apiaceae	Visnadine	Breast cancer [26]
10.	*Camellia sinensis*	Theaceae	Epicatechin	Breast cancer [27]
11.	*Avicennia marina*	Acanthaceae	Flavonoids	Breast and larynx cancer [15, 16]
12.	*Silybum marianum*	Asteraceae	Flavonoids	Colorectal cancer [17], breast cancer [18]
13.	*Artemisia absinthium*	Asteraceae	Artemisinin, isorhamnetin	Colon cancer [19], blood cancer [20]
14.	*Curcuma longa*	Zingiberaceae	Curcumin	Breast [21], prostate [28], cervix [3], larynx cancer [29]
15.	*Crocus sativus*	Iridaceae	Phenolic compounds	Sarcoma, oral cancer [30, 31]
16.	*Zingiber officinale*	Zingiberaceae	Flavonoids	Urinary cancer [32], prostate cancer [33]
17.	*Olea europaea*	Oleaceae	Oleic acid	Breast cancer [34], colon cancer[35]
18.	*Taxus baccata*	Taxaceae	Taxol	Prostate cancer [36], pancreatic cancer [37], blood cell cancer [38]
19.	*Nigella sativa*	Ranunculaceae	Thymoquinone	Prostate [39]
20.	*Allium sativum*	Amaryllidaceae	Allicin	Cervical cancer [40], breast [41, 42] colon [43, 44], larynx [45]
21.	*Lepidium sativum*	Brassicaceae	Vitamins (A, B, C, and E)	Breast cancer [46]

(continued)

Table 2 (continued)

Sr. No.	Botanical name	Family	Chemical constituents	Types of cancer
22.	*Trigonella foenum-graecum*	Fabaceae	Alkaloids	Breast cancer [47, 48]
23.	*Glycyrrhiza glabra*	Fabaceae	Glycyrrhizin	Kidney cancer [48]
24.	*Physalis alkekengi*	Solanaceae	Physalins	Cervical cancer [50, 51]
25.	*Lagenaria siceraria* Standl	Cucurbitaceae	Vitamins B and C	Lung cancer [52], breast cancer [53]
26.	*Ferula gummosa*	Apiaceae	Sesquiterpenes	Lung cancer [54], skin cancer, stomach cancer [55]
27.	*Urtica dioica* L.	Urticaceae	Phenolic compounds	Prostate cancer [56–58]
28.	*Ammi majus*	Apiaceae	Coumarin compounds	Breast cancer [59]
29.	*Rosa damascene*	Rosaceae	Phenolic compounds	Lung cancer [60], breast cancer, cervix [61]
30.	*Astragalus cystosus*	Fabaceae	Lectins, flavonoids	Lung cancer [62]
31.	*Myrtus communis*	Myrtaceae	Myrtucommulone	Breast cancer [63–65]
32.	*Vinca rosea*	Apocynaceae	Vindoline	Breast cancer, larynx cancer [64]
33.	*Citrullus colocynthis*	Cucurbitaceae	Quercetin	Liver cancer [66]
34.	*Polygonum aviculare*	Polygonaceae	Flavonoids and alkaloids	Breast cancer [67], cervical cancer [68, 69]
35.	*Astrodaucus orientalis*	Apiaceae	A-thujene and a-copaene	Breast cancer [70]
36.	*Actinidia chinensis*	Actinidiaceae	Polysaccharide	Hepatic cancer [71]
37.	*Aegle marmelos*	Rutaceae	Lupeol	Breast cancer and leukaemia
38.	*Agave americana*	Agavaceae	Steroidal and alkaloid	Cancerous tumour
39.	*Aloe vera*	Asphodelaceae	Emodin	Leukaemia, stomach cancer [72]
40.	*Alpinia galanga*	Zingiberaceae	Pinocembrin	Leukaemia
41.	*Amoora rohituka*	Meliaceae	Amooranin	Lymphocytic [73]
42.	*Andrographis paniculata*	Acanthaceae	Flavonoids	Colon cancer [74]

(continued)

Table 2 (continued)

Sr. No.	Botanical name	Family	Chemical constituents	Types of cancer
43.	*Annona muricata*	Annonaceae	Acetogenins	Lung cancer and liver cancer [75]
44.	*Apis mellifera*	Apidae	Protein	Lymphoid cancer [76]
45.	*Ananas comosus*	Bromeliaceae	Bromelain	Leukaemia [77], gastrointestinal carcinoma [78]
46.	*Angelica sinensis*	Apiaceae	Polysaccharide	Cervical cancer [79], brain tumour [80], glioblastoma multiforme [81]
47.	*Annona species*	Annonaceae	Acetogenins	Nasopharyngeal cancer [82]
48.	*Arctium lappa*	Asteraceae	Arctigenin [85]	Prostate cancer [83, 84]
49.	*Artemisia asiatica*	Asteraceae	Iso-liquiritigenin	Liver tumour
50.	*Astragalus membranaceus*	Fabaceae	Swainsonine	Gastrointestinal cancers and liver cancer [85]
51.	*Azadirachta indica*	Meliaceae	Limonoids and nimbolide	Leukaemia [78]
52.	*Bauhinia variegata*	Caesalpiniaceae	Cyanidin glucoside	Breast cancer [79]
53.	*Berberis vulgaris*	Berberidaceae	Berbamine and berberine	Prostate cancer, oral cancer [80]
54.	*Betula alba*	Betulaceae	Betulinic acid	Prostate cancer [82], thyroid [86], breast cancer [84], colon cancer [85]
55.	*Betula utilis*	Betulaceae	Ursolic acid and betulin	Liver cancer, breast cancer [78]
56.	*Bolbostemma paniculatum*	Cucurbitaceae	Tubeimoside-5	Glioblastoma cancer [79]
57.	*Cannabis sativa*	Cannabaceae	Cannabinoids	Breast cancer, colorectal cancer [80, 81]
58.	*Catharanthus roseus*	Apocynaceae	Vincristine and vinblastine	Non-Hodgkin's lymphoma and Hodgkin's disease
59.	*Cinnamomum cassia*	Lauraceae	Coumarin	Promyelocytic leukaemia and basal cell carcinoma [84]
60.	*Colchicum luteum*	Liliaceae	Colchicines	Hodgkin lymphoma

(continued)

Table 2 (continued)

Sr. No.	Botanical name	Family	Chemical constituents	Types of cancer
61.	*Combretum caffrum*	Combretaceae	Combretastatin	Colon cancer and lung cancer [85]
62.	*Coriandrum sativum*	Apiaceae	Quercetin	Adenocarcinoma [87, 88]
63.	*Daphne mezereum*	Thymelaeaceae	Mezerein	Lung cancer and lymphocytic leukaemia [89]
64.	*Echinacea angustifolia*	Asteraceae	Arabinogalactan	Oesophagus cancer [90]
65.	*Emblica officinalis*	Phyllanthaceae	Emblicanin a and b	Liver cancer
66.	*Fagopyrum esculentum*	Polygonaceae	Inhibitor-1 protein,	Lymphoblastic leukaemia [91]
67.	*Ginkgo biloba*	Ginkgoaceae	Ginkgolides	Invasive oestrogen-receptor breast cancer
68.	*Glycine max*	Fabaceae	Genistein and daidzein	Cancer in oral
69.	*Gossypium hirsutum*	Malvaceae	Gossypol	Breast cancer and melanoma [92]
70.	*Indigofera tinctoria*	Fabaceae	Tannins and phenols	Lung cancer [84]
71.	*Lentinus edodes*	Polyporaceae	Terpenoids and lentinan	Colon cancer and lung carcinoma [93]
72.	*Linum usitatissimum*	Linaceae	Lignans	Breast cancer [80, 94]
73.	*Nothapodytes foetida*	Icacinaceae	Scopoletin, camptothecin	Colon cancer [85, 95]
74.	*Ochrosia elliptica*	Apocynaceae	9-methoxy ellipticine	Breast, kidney cancer [85, 96]
75.	*Ocimum sanctum*	Lamiaceae	Ursolic acid	Fibrosarcoma
76.	*Oldenlandia diffusa*	Rubiaceae	Stigmasterol	Stomach cancer [88, 97]
77.	*Origanum vulgare*	Lamiaceae	Rosmarinic acid	Lung cancer [85]
78.	*Panax ginseng*	Araliaceae	Polyacetylene	Stomach cancer [85]
79.	*Pfaffia paniculata*	Amaranthaceae	Cytotoxic substances	Oestrogen positive breast cancer [88]
80.	*Picrorhiza kurroa*	Plantaginaceae	Picrosides-i, ii, iii	Liver cancer [90]
81.	*Plumbago zeylanica*	Plumbaginaceae	Plumbagin	Fibrosarcoma cancers [90]
82.	*Podophyllum hexandrum*	Berberidaceae	Podophyllotoxin	Leukaemia [87]

(continued)

Table 2 (continued)

Sr. No.	Botanical name	Family	Chemical constituents	Types of cancer
83.	*Prunella vulgaris*	Lamiaceae	Oleanolic acid	Oral cavity [88]
84.	*Psoralea corylifolia*	Fabaceae	Psoralidincorylfolinin	Osteosarcoma [88]
85.	*Viscum album*	Santalaceae	Phenolic compounds	Testis cancer [89]

3 Conclusion

Cancer is the unchecked growth of malignant cells or tissues within the body. A cancerous growth is referred to as a malignant tumour or malignancy. A growth that is not cancerous is referred to as a benign tumour. Each of the sequential, connected processes that restrict the rate of cancer spread is sequential and interconnected. The traditional medical systems that are currently in use each have their own linked concepts and historical settings. While some of them, like Ayurveda traditional medicines, are spreading over the globe, some of these, like Tibetan traditional medicine, are still mostly exclusive to the country of origin. Many compounds found in plants have been demonstrated in clinical studies to have chemoprotective effects. Inhibiting angiogenesis is a novel approach to cancer treatment. The proper selection and application of this plant could very well aid in the management of cancer through antiangiogenic therapy. Plant-derived anticancer medicines are in high demand because they effectively inhibit cancer cell lines. In order to meet demand and be sustainable, the exploitation of these agents needs to be controlled. The attempt has been made to compile a list of the many different plant species found throughout the world that are either regularly utilised or being researched for their potential anticancer properties.

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Herbs and Herbal Formulations for the Management and Prevention of Gastrointestinal Diseases

22

Nitu Singh, Urvashi Sharma, Bharat Mishra, Ashish M. Kandalkar, and Sachin Kumar Jain

Contents

Abstract

Diseases affecting the digestive tract's activities are referred to as GIT disorders. Stomach or abdominal pain, dyspepsia, bloating, diarrhea, constipation, dysentery, vomiting, and gastroenteritis are typical gastrointestinal illnesses. Patients with gastrointestinal issues are routinely treated with herbal or alternative treatments. In several regions of the world, herbs have historically been trusted for the management of numerous illnesses. They are considered vital bases of pharmaceutical items, especially herbal remedies, and their use considerably impacts the delivery of basic health care. Because they have minimal side effects when used

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appropriately, herbal medications are becoming more and more popular with Westerners. In recent years, a widespread trend has been there toward a resurgence of interest in a conventional system of medical care. According to the WHO, traditional medicine has a place in the primary healthcare system. Medicinal plants remain to be a significant source of treatment in developing nations. The majority of people in developing nations, or about 88% of them, are thought to rely on a traditional system of medicine for their medical care. When treating GI problems, evidence-based, rational phytotherapy which employs herbs and herbal preparations to provide the desired therapeutic effect plays a significant role. Given the variety of these disorders, it stands to reason that a wide range of herbal products can be utilized in both prevention and treatment. This chapter aims to review the use of various herbs and herbal preparations for managing and preventing gastrointestinal diseases as well as their safety and efficacy.

Keywords

Gastrointestinal disorders · Herbal medicines · Herbs · Herbal formulation · Safety

Abbreviations

5-HT	5-hydroxytryptamine
BMI	Body mass indices
DFHE	Dill Fruit Hydroalcoholic Extract
DGL	Deglycyrrhizinated licorice
DNA	Deoxyribonucleic acid
ESCOP	European Scientific Cooperative on Phytotherapy
FD	Functional Dyspepsia
FGIDs	Functional gastrointestinal disorders
GERD	Gastroesophageal Reflux Disease
GI	Gastrointestinal
GLP-1	Glucagon-like peptide 1
H2	Histamine Type 2
IBS	Irritable Bowel Syndrome
IL 10	Interleukin 10
LES	Lower esophageal sphincter
L-NAME	L-N^{G}-Nitro arginine methyl ester
MMP	Matrix metalloproteinase
NERD	Nonerosive reflux disease
PDS	Postprandial distress syndrome
PKS	Polyketide synthases
PMO	Peppermint Oil
POAL	Probiotics originating from Aloe leaf)
PPIs	Proton pump inhibitors
QOL	Quality of Life
RCT	Randomized control trial

TCM	Traditional Chinese medicine
TID	Thrice a day
VEGF	Vascular endothelial growth factor
WHO	World Health Organization
XO	Xanthine oxidase

1 Introduction

The gastrointestinal tract (GIT), also known as the human digestive system, is made up of a hollow, twisting muscular tube that starts from the oral cavity, where food is taken in the mouth and travels through the throat, esophagus, stomach, and intestines before ending at the rectum and anus, from where digested food is discharged. Solid organs of the digestive system cover the gallbladder, pancreas, and liver. GI tract bacteria, also called gut flora or microbiome, aid digestion. Bacteria are the most prevalent and diverse group of microbial eukaryotes (fungi, protozoa, archaea, and viruses) that make up the human gastrointestinal microbiota. Its bacterial ecosystem is extensive and complex, with up to 1014 bacteria present – more than ten times as numerous as somatic human cells [1]. The GI tract's primary functions include digestion, absorption, excretion, and protection. Such tasks are carried out by a few specialized organs extending from the mouth to the anus. The stomach and small intestine are majorly responsible for digestion and absorption due to physical mechanisms like retropulsion along with chemical factors such as the presence of bile and enzymes in the small intestine. Desiccating and compacting waste, which is then kept in the rectum and sigmoid colon until it is evacuated, are the main jobs of the large intestine [2]. Ailments affecting the digestive tract's activities, like absorption of food and liquid, digestion, or excretion, are referred to as gastrointestinal disorders [3]. These conditions are brought on by infections with different bacterial, viral, and parasite species [4, 5].

1.1 Types of Gastrointestinal Disorders

There are two types: functional and structural.

Functional gastrointestinal disorders (FGIDs) are ailments where the digestive system appears normal but does not operate as it should. Functional disorders can affect anyone at any age, including adults, teens, and children. Yet, rather than being the outcome of an illness or infection, FGIDs cause sensitivity and a variety of GI symptoms due to improper functioning. The three main traits of FGIDs are dysfunction in the brain-gut axis, sensation, and motility.

Motility is the term for the muscular activity of the GI tract. The sensation is produced by the GI tract's nerves in response to stimuli (e.g., digestion of food). Functional GI illnesses may have nerves that are so responsive that even normal contractions can be painful or uncomfortable. A failure in communication between the brain and the GIT is known as a brain-gut dysfunction. FGIDs may impair the regulatory pathway between brain and gut function. Some of the indications of FGIDs are nausea, stomach pain, constipation, bloating, burping, flatulence, diarrhea, vomiting, and indigestion [6].

Structural gastrointestinal disorder refers to conditions where the colon seems aberrant and is not functioning properly. Sometimes the structural anomaly must be surgically removed. GI structural illnesses include inflammatory bowel disease, colon cancer, stenosis, colon polyps, hemorrhoids, and diverticular disease. Untreated structural GI problems usually aggravate symptoms and lead to life-threatening consequences [6].

1.2 General Signs of Digestive Disorders

Clearly, the conditions and symptoms of digestive illnesses are unique and vary from individual to individual. Nonetheless, most digestive problems exhibit certain indications and symptoms. Figure 1 illustrates typical indications [7].

The few commonest gastrointestinal conditions are:

1. Celiac Disease
2. Irritable Bowel Syndrome (IBS)
3. Lactose Intolerance

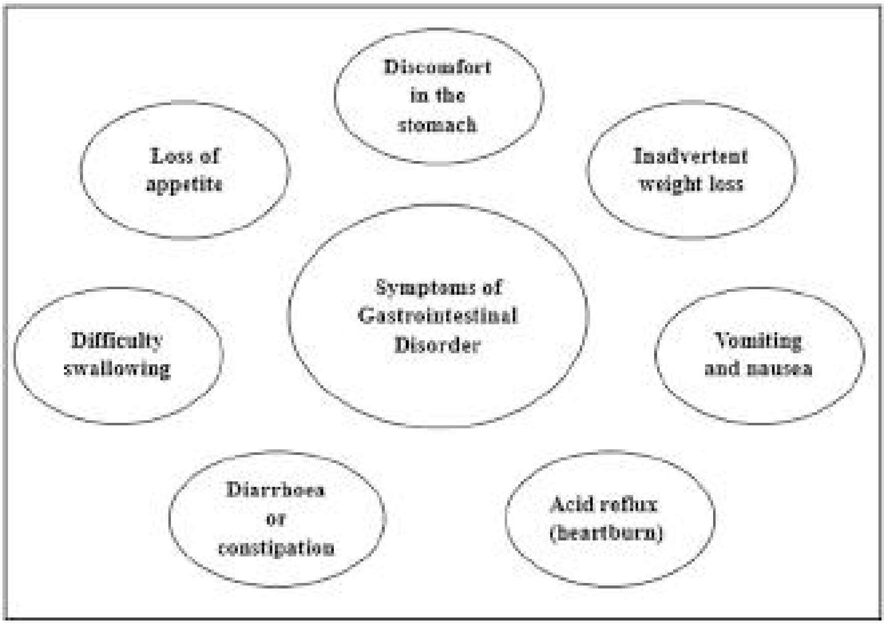

Fig. 1 Common symptoms of gastrointestinal disorders

4. Chronic Diarrhea
5. Constipation
6. Gastroesophageal Reflux Disease (GERD)
7. Peptic Ulcer Disease
8. Crohn's Disease
9. Ulcerative Colitis
10. Gallstones
11. Chronic and Acute Pancreatitis
12. Liver Disease
13. Diverticulitis [7]

1.3 Synthetic Medicines for Gastrointestinal Disorders and Their Adverse Effects

Gastrointestinal disorders are typically treated with the following class of medications.

- **Antacids**

  An antacid is a blend of many components having various salts of calcium, magnesium, and aluminum as their active substances. The antacid functions by neutralizing the acid present in the stomach by inhibiting the action of the proteolytic enzyme pepsin. The elderly and newborn populations are particularly vulnerable to negative consequences. It is not advised to take antacids frequently in this group for reasons of safety. Table 1 lists the therapeutic applications and negative effects of antacids [8].
- **Histamine H2-receptor antagonists (Histamine Blockers)**

  For a range of gastrointestinal disorders, including those listed in Table 1, stomach acid-suppressing drugs also called H2 receptor blockers or H2 receptor antagonists (H2RAs) are routinely prescribed. H2RAs prevent the endogenous ligand histamine from binding and showing its action by reversibly attaching themselves to receptors available on stomach parietal cells, hence reducing stomach acid output. Antagonists of the H2 receptor are often well tolerated. Headache, weariness, drowsiness, fatigue, abdominal pain, constipation, or diarrhea are examples of mild side effects. There is a correlation between the usage of H2RAs and some side effects, as shown in Table 1, in individuals who are over 65 or have hepatic or renal impairment [9].
- **Proton pump inhibitors**

  Proton pump inhibitors (PPIs) prevent the final step of gastric acid secretion by blocking the hydrogen/potassium adenosine triphosphatase (H+/K+ ATPase) enzyme, also known as the "proton pump," in the parietal cells of the stomach. The following conditions associated with stomach acid secretion are advised for the prevention and treatment of PPIs. In general, doctors do not worry about PPIs having serious side effects when used over a brief amount of time say 2 weeks,

Table 1 Reported adverse effects of synthetic medications

S.No.	Category of medications	Therapeutic use	Adverse effects	References
1	Antacids	Heartburn symptoms of GERD Gastric and duodenal ulcers Gastritis due to stress Non-ulcer dyspepsia Bile acid caused diarrhea Biliary reflux Constipation	Constipation Hypophosphatemia Hypercalcemia Osteopenia Microcytic anemia Fecal impaction Nausea Vomit Stomach cramps	[8]
2	Histamine H2-receptor antagonists (Histamine Blockers)	GERD Ulcer of duodenum or GIT, hypersecretion of gastric juices, and indigestion and heartburn	Delirium, confusion, hallucinations, or slurred speech	[9]
3	Proton pump inhibitors	Gastro-esophageal ulcers, peptic ulcer disease, non-steroidal anti-inflammatory drug-associated ulcers, Zollinger-Ellison syndrome, and eradication of *Helicobacter pylori*	Dementia, fracture risk, hepatocellular carcinoma, nosocomial pneumonia, chronic kidney disease, nutritional deficiencies, vitamin B12, iron and calcium deficiency, bacterial overgrowth, gastric polyps	[10]
4	Prokinetics	Functional dyspepsia	Parkinson-like symptoms, Dystonia, Akathisia, Galactorrhoea, and breast engorgement	[11]

but reports of adverse effects arise with the growing use of these drugs on long-term use. According to current studies, PPIs can be consumed only for a shorter duration of time in the lowest possible dose which is effective enough because using them for a longer time may pose adverse effects like reduced nutritional absorption, dementia, infections, kidney disease, and hypergastrinemia. The therapeutic uses and side effects of chronic PPI usage are listed in Table 1 [10].

- **Prokinetics**

  Dopamine antagonists (metoclopramide, domperidone, levosulpiride, and itopride) and serotonin (5-HT) receptor agonists for example cisapride and mosapride make up the majority of the prokinetic medications used in clinical settings. Although the effectiveness of all prokinetic medications for the treatment of gastrointestinal hypomotility disorders is well recognized, these medications are also linked to a number of negative side effects, which are mentioned in Table 1 [11].

2 Herbal Therapy for the Management and Prevention of Gastrointestinal Diseases

Herbal therapy has significantly improved human health because of its capacity to promote health, heal illnesses, and treat and rehab patients. Herbal medicine, also known as phytomedicine, or phytotherapy, comprises herbs, herbal components, or formulations, which have plant-based active chemicals or other compounds [12]. Numerous studies indicate that traditional people around the world frequently employ plant-based therapies or herbal medications to address problems in the digestive system. People can treat several digestive-related illnesses simply by regularly incorporating herbal substances into their diet for improving the performance of the digestive system. These herbal treatments function in a variety of ways, either by easing the bowel movements, enhancing the digestive process, raising the bowel movement frequency, detoxifying the body, eliminating toxins, minimizing gas, settling an upset stomach, and decreasing the bloating and digestive discomfort. All societies have employed medicinal plants throughout history as a type of medicine for the treatment of various human maladies, and they still have a crucial role in our current technology-based civilization. Approximately half of all medications currently used in clinical settings are plant-derived components or natural drug products or their derivatives [13]. For their health care, almost 80% of individuals in underdeveloped nations mostly use traditional herbal remedies that contain plant extracts or their active ingredients [14]. Secondary metabolites, also known as phytochemicals, are abundant in medicinal plants, and their derivatives display a wide range of biological characteristics, including anti-plasmodial, anti-inflammatory, anticancer, and antioxidant effects [13, 15, 16].

This review's objective is to provide some illustrations of studies that have been successful in establishing the efficacy of various gastrointestinal illnesses treated with plants.

2.1 Herbs Used in Gastrointestinal Diseases

2.1.1 *Aloe barbadensis* Miller

Aloe barbadensis Miller is one of more than 400 species of aloe that are native to arid tropical and subtropical weathers, including the southern United States, also referred to as *Aloe vera* (Fig. 2a). They are all members of the Liliaceae family and originated in South Africa [17]. Only a few aloe species have recently been thought to be important commercially, with *A. vera* being of utmost strength and thus the maximum well-liked herb in the research community [18].

Secondary metabolites produced by *A. vera* are recognized to be beneficial [17, 19]. The two most important secondary metabolites are anthraquinones and tricyclic aromatic quinines, both of which are widely distributed. Chrysophanol and Aloe-emodin are the two main anthraquinone derivatives that are found in nature [20]. It has been suggested that the type III polyketide biosynthesis pathway is used

Fig. 2 (**a**) Leaves of *Aloe barbadensis Miller* (**b**) Dried ripe fruits of *Anethum graveolens L* (**c**) Dried ripe fruits of Carum carvi L. (**d**) Fruits of Emblica officinalis (**e**) Dried ripe fruits of *Foeniculum vulgare* Mill. (**f**) Dried roots of Glycyrrhiza glabra (**g**) Leaves of Mentha piperita (**h**) Seeds of Plantago ovata Forsk (**i**) Fruits of Trachyspermum ammi L. (**j**) Seeds of Trigonella foenum-graecum L (**k**) Rhizome of Zingiber officinale

to generate the tricyclic aromatic quinines found in aloe [21]. The important secondary metabolites in *A. vera* gel are aloesin, aloin, and aloe-emodin (an oxidative derivative of aloin) [22].

For the treatment of stomach infections due to *H. pylori*, *aloe vera* gel acts as a special and effective natural remedy. It exhibits antibacterial action against both susceptible and resistant *H. pylori* strains. The cytoprotective properties of *Aloe vera* leaf extracts are explored for their benefits in digestion and peptic ulcer treatment [23]. *Aloe vera* latex is frequently used to treat constipation because it contains anthraquinone glycosides, which are known to have laxative effects [24, 25]. In a double-blind, randomized controlled trial (RCT) with 28 healthy individuals, it is discovered that aloin had a greater laxative effect than the stimulant laxative phenolphthalein [26]. The *L. brevis* strains were isolated from naturally fermented *A. vera* gel and called POAL (probiotics from aloe leaf) strains because they displayed exclusionary resistance to a variety of antibiotics while preventing the growth of various dangerous enteropathogens in the gut [27]. The colonic flora converts the aloin included in the gel into reactive aloe-emodin, which has a purgative effect. Aloe-emodin, a compound derived from *A. vera*, via decreasing MMP-2/9 reduces the migration of colon cancer cells and suppresses vascular endothelial growth factor (VEGF) and Ras homolog gene family member B through lowering the nuclear translocation and binding of DNA to NF-KB, assuring the effectiveness of aloe-emodin tumor via multiple target sites [28].

2.1.2 *Anethum graveolens* Linn

Dill, *Anethum graveolens* L. (Apiaceae family), is a well-known herb that is frequently used as a spice and also produces essential oil (Fig. 2b). It has been utilized in Ayurveda treatment since ancient times. It is a flavorful annual herb. In Ayurvedic medicine, dill is well known in the treatment of gastrointestinal pain, colic, and to aid with digestion. In Unani medicine, it is utilized for colic, digestive issues, and even for diarrhea [29].

Anethum is a component found in gripe water and used for the treatment of colic pain in infants and flatulence in young children [30]. The seed is stomachic, somewhat diuretic, stimulant, and galactagogue with a pleasant aroma [31, 32]. Dill oil from grapes contains the monoterpenes carvone and limonene as their primary ingredients [33]. Three different compounds phellandrene, dill ether, and myristicin combine to give the dill herb its distinctive aroma [34, 35]. The essential oil from the seeds decreases intestinal cramps and spasms, thereby calming colic [36, 37]. The volatile oil acts as an appetite booster and relieves gas aiding in digestion. Chewing the seeds helps to reduce bad breath [38]. The mucosa is strongly protected by *A. graveolens* L. seed extracts, that possess antisecretory, anti-ulcer, and anti-HCl and ethanol-induced stomach ulcers in mice [39]. The seeds of *A. graveolens* L. were used to extract two flavonoids, quercetin and isorhamnetin, which have antioxidant qualities and may be utilized to fight free radicals. This outcome might help avoid peptic ulcers [40, 41]. The HCl extract of dill fruit strongly relaxed the contractions caused by a range of spasmogens in rat ileum, supporting its importance in traditional medicine for digestive problems [42].

In another study, the effects of methanolic extract of dill on both spontaneous contractions and even those induced by acetylcholine were explored. Cumulative dosages of dill methanolic extract significantly reduced ($p < 0.01$) the spontaneous rat ileum contractions. The extract also reduced the contraction induced by acetylcholine in a dose-dependent manner ($p < 001$). The findings confirmed that the methanolic dill extract had a relaxing effect on the isolated rat gut. Dill extract prevented both naturally occurring and acetylcholine-induced ileum contractions [43].

2.1.3 *Carum carvi* Linn

C. carvi also known as Caraway is an essential medicinal plant belonging to the Apiaceae family (Fig. 2c). It has been grown for a very long time in Egypt, Australia, China, and Iran, as well as in the north and center of Europe [44]. It has been used traditionally as a spice in food and drinks as well as an herbal alternative for the treatment of GI diseases such as dyspepsia, diarrhea, various spasmodic symptoms, bloating, and flatulent colic [45].

Two important natural chemicals are limonene and carvone, having mucoprotective and antiulcerogenic actions on gastroduodenitis and duodenal peptic ulcers [46]. Caffeic acids, quercetin, kaempferol [47, 48], fatty acids (10–18%) (linoleic and oleic acids, petroselinic), essential oil (3–7%), protein (20%), and carbohydrates (15%) are the main constituents of caraway fruits [49]. As per European Pharmacopoeia, caraway fruit possesses essential oil up to 3%, the primary component is D-carvone which is up to 50–65%, (+)-limonene (45%), and carveol and dihydrocarveol (<1.5%). The odor is due to its main constituent D-carvone [50, 51].

An alcoholic extract of caraway has been found to have antispasmodic properties that block the spasmogens acetylcholine and histamine-induced smooth muscle spasms [52, 53]. Caraway oil demonstrated significant levels of selectivity in a study on 12 intestinal bacteria, suppressing the growth of possible pathogens to levels that did not affect the study's beneficial bacteria. This outcome was related to the efficacy and value of caraway oil in traditional medicine for the management of dysbiosis, which is associated with a variety of gastrointestinal and systemic disorders [54]. Caraway extracts reduced acid and leukotriene production, increased mucin secretion, and released prostaglandin E2 preventing indomethacin-induced stomach ulcers in a dose-dependent manner. Histological testing also supported the antiulcerogenic action, which was linked to the flavonoid content and free radical scavenging abilities of the food [55]. In one clinical trial, patients with functional dyspepsia (FD), according to Rome III criteria who were on a placebo with their regular medications, such as PPIs, anticonvulsants, antihistamines, antidepressants / TCAs, beta-blockers, pain modulators, H2RAs, and antacids, were compared with those who were taking enteric-coated capsules encapsulating caraway oil and menthol. The results of the study showed that the symptomatic reduction in the intervention group was quantitatively greater than that in the placebo group. Clinical Global Impressions were improved by intervention and placebo treatments in 61% and 49% of patients, respectively ($p = 0.23$). The treated group experienced no significant side effects [56]. The effectiveness of hot or cold olive oil poultices in curing IBS for 3 weeks, followed by 2-week "wash-out" periods, was compared to the efficiency of hot poultices created with caraway poultices in a cross-over RCT. Two-thirds of patients who used a caraway oil poultice on their skin rated the treatment as satisfactory or very good. The overall IBS-QOL (Quality of Life) score showed a statistically significant difference between caraway oil and the other oils. Comparatively, Caraway oil reported adequate alleviation at 51.8%, compared to hot olive oil 23.5% and cold olive oil 25.8% [57].

2.1.4 Emblica Officinalis

Emblica officinalis holds a revered place in Ayurveda, an ancient medical system practiced in India (Fig. 2d). It is a member of the Euphorbiaceae family. The Indian gooseberry or *Phyllanthus emblica* is another name [58, 59]. The juice of this fruit contains Vitamin C, i.e. 4.78 mg mL^{-1} [60, 61] in larger quantities. Several different acids that are present include gallic acid, ellagic acid, chebulinic acid, chebulagic acid, 1-O-galloyl-beta-D-glucose, quercetin, 3,6-di-O-galloyl-D-glucose, 1, 6-di-O-galloyl beta-D glucose, corilagin, 3-ethylgallic acid (3-ethoxy-4 [62]. Kaempferol-3-O-alpha-L-(6″- ethyl)-rhamnopyranoside and kaempferol-3-O-alpha-L-(6″-methyl)-rhamnopyranoside are flavonoids found in *P. emblica* [63].

P. emblica can improve meal digestion, absorption, and assimilation when used regularly. People that use it discover they prefer the taste of food more. Additionally, it enhances iron absorption for healthy blood. Stomach acids are balanced. For reducing mild to severe hyperacidity, *P. emblica* is the best choice. It is administered medicinally to alleviate diarrhea. In cases of dysentery, it is provided by the locals as a fruit decoction blended with sour milk. Fenugreek seeds infused with leaves are used to treat chronic diarrhea [64].

Polyphenols from *P. emblica* L. are widely used for gastrointestinal organ protection. Since *Helicobacter pylori* are responsible for stomach ulcers, the bioactive substances of amla act as one of the potent inhibitors of these strains in-vitro [65].

The *Phyllanthus emblica* extract was investigated for antiulcer and anti-secretory activities using several models to prompt GI ulcers in mice such as pylorus ligation, administration of indomethacin or necrotizing agent (25% sodium chloride, 0.2 M sodium hydroxide, and 80% ethanol), and hypothermia induction are all used to cause gastrointestinal ulcers [66]. In all the test models employed, oral treatment of extract at doses of 250 mg/kg and 500 mg/kg significantly reduced the development of stomach lesions. In a different experiment, mice with pancreatitis caused by L-arginine had their blood levels of lipase and IL-10 reduced when *P. emblica* L. (rich in tannins and gallic acid) in 200 mg/100 g was administered [67]. The research also showed that the group of rats in *P. emblica* had greater amounts of nucleic acids, pancreatic protein, DNA synthesis rate, and pancreatic amylase. The histological changes in the colon of mice with colitis brought on by acetic acid were also lessened by the methanol fruit extract of *P. emblica* [68]. The daily intake of 500 mg/tablet of *P. emblica* extract daily twice decreased the frequency and intensity of vomiting and indigestion when compared with the placebo group [69].

2.1.5 *Foeniculum vulgare* Mill.

A biennial plant from the Apiaceae family, *Foeniculum vulgare* Mill. (Fig. 2e) has both aroma and therapeutic properties. Due to its flavor, it is a medicinal herb that humans have been aware of and used since ancient times. It was grown in practically every nation [70]. Due to the valuable nutritional makeup of fennel, which includes the presence of important fatty acids, a diet containing the right amount of fennel may have potential health benefits [71].

Fennel water also possesses qualities just like anise and dill water. And when combined with syrup and sodium bicarbonate, this collection forms the domestic "gripe water," that is given to treat newborn flatulence. By adding boiling water to a teaspoonful of smashed fennel seeds, fennel tea can be made which is also used as a carminative [72]. From *F. vulgare*, flavonoids such quercetin-3-rutinoside, rosmarinic acid, and eriodictyol-7-rutinoside have been extracted [73]. The most prevalent flavonoids in *F. vulgare* include quercetin-3-glucuronide, isoquercitrin, kaempferol-3-glucuronide, quercetin-3-arabinoside, isorhamnetin glucoside, and kaempferol-3-arabinoside [74]. There have also been reports of aqueous extract of *F. vulgare* found to include quercetin-3-O-galactoside, kaempferol-3-O-rutinoside, quercetin-3-O-galactoside, and kaempferol-3-O-glucoside [75]. While the methanolic extract was used to isolate quercetin 3-O-rutinoside, quercetin 3-O-glucoside, and kaempferol 3-O-rutinoside, the ethyl acetate extract also yielded quercetin, kaempferol, and isorhamnetin 3-O-rhamnoside. These flavonoids have outstanding anti-inflammatory and antinociceptive properties [76]. Additionally, it was claimed that the antioxidants quercetin, rutin, and isoquercitrin had immunomodulatory properties [77]. Chlorogenic acids and rosmarinic acid (6.8% and 14.9% respectively) are the main phenolic components obtained in the methanolic extract of fennel seeds, while the main flavonoids are quercetin and apigenin (17.1% and 12.5%, respectively) [78].

Fennel essential oil reduces intestinal gas while also controlling the intestine's smooth muscle movement. *Foeniculum vulgare* is approved for the treatment of several symptoms of dyspepsias such as gastrointestinal atony, a feeling of heaviness in the stomach, and many others. It can be taken alone or in conjunction with other herbal remedies for the treatment of these conditions. Fennel can be added to treatments that contain anthraquinone ingredients to lessen the frequency of stomach pain that is frequently related to this kind of laxative [79]. The significant anti-ulcerogenic effect was shown by an aqueous extract of *F. vulgare* on a rat stomach on which lesions were brought on by ethanol. It was discovered that pre-treating with an aqueous extract considerably lessened the harm caused by ethanol to the stomach. When compared to the control group of animals, the effect of the aqueous extract was greatest and statistically significant ($P < 0.001$) in the 300 mg kg^{-1} group of mice. Aqueous extract of *F. vulgare* also markedly decreased levels of malondialdehyde in whole blood while markedly raising levels of Vitamin C, nitrite, Vitamin A, nitrate, and beta-carotene. As a result, rats' stomach mucosal lesions caused by ethanol were prevented by using aqueous fruit extract of *F. vulgare* [80].

2.1.6 *Glycyrrhiza glabra* Linn.

Commonly known as **Licorice**, it belongs to the Leguminosae family and has the scientific name *Glycyrrhiza glabra* (Fig. 2f). Both in parts of Europe and Asia, this therapeutic herb can be found [81].

The triterpenoid saponin glycyrrhizin is the primary constituent of licorice root, also referred to as glycyrrhizinic acid or glycyrrhizic acid, and is typically present in between 6% and 10%. The aglycone molecule (glycyrrhetinic acid) is produced by the hydrolysis of glycyrrhizin due to intestinal flora and a sugar moiety, causing both to be absorbed [82].

By removing the glycyrrhizin molecule, deglycyrrhizinated licorice (DGL) a processed licorice extract is produced that is used to treat peptic and aphthous ulcers. Flavonoids are those parts of DGL that are active. In animal trials, these compounds showed remarkable resistance to the development of chemical-induced ulcers [83]. A few more components of licorice include isoflavonoids (like, licoricone, isoflavonol, glabrol, and kumatakenin), coumarins like herniarin and umbelliferone, chalcones, triterpenoids, lignins, amino acids, amines, sterols, gums, and volatile oils [84].

Even though, glycyrrhetinic acid was the first medication to be shown to speed up the healing process in duodenal and gastric ulcers [85], many doctors who used to use licorice to treat peptic ulcers have started using DGL. As it has been proven to be superior and efficient to glycyrrhetinic acid with no side effects [86]. Over the years, numerous clinical investigations have revealed DGL to be a potent drug for ulcers. It has proven to be incredibly successful to treat stomach ulcers [87–91]. In one trial, DGL (760 mg, thrice a day) or a placebo was given to 33 individuals with stomach ulcers for 1 month. The DGL group (78%) experienced a considerably larger decrease in ulcer size than the placebo group (34%) [89].

DGL was found to be superior to cimetidine (Tagamet), ranitidine (Zantac), and antacids in multiple head-to-head comparison studies, both for the short-term treatment of peptic ulcers and for maintenance therapy [87, 88, 91]. For example, a hundred volunteers were given DGL (760 mg, between the meals, TID) or Tagamet in a head-to-head comparison with 200 mg of Tagamet thrice a day, with a bedtime dose of 400 mg [88]. Despite the fact that Tagamet has several serious adverse effects, using DGL is very safe. Since it was demonstrated in human tests to minimize the gastrointestinal bleeding brought on by aspirin, DGL is strongly advised in patients needing treatment for longer duration with ulcerogenic medications including aspirin, NSAIDs, and corticosteroids, to prevent gastric ulcers [90]. Duodenal ulcers can benefit from DGL as well. DGL was used to treat 40 patients who had more than six relapses the year prior and had chronic duodenal ulcers that had been present for four to 12 years. For 8 weeks, a daily dose of 4.5 g of DGL was given to half of the patients, while the other half had 3 g daily for the same period. Typically, within 5 to 7 days, all 40 patients had significant improvement, and no one needed surgery at the time of the one-year follow-up [92].

2.1.7 *Mentha piperita* Linn.

The earliest medicinal herb utilized in both Eastern and Western traditional medical systems is *Mentha piperita* (Fig. 2g) belonging to the family Lamiaceae. Herbalists have employed *M. piperita* as an emmenagogue, antiseptic, antispasmodic, astringent, carminative, antipruritic, antiemetic, diaphoretic, antibacterial, rubefacient, anticatarrhal, and analgesic [93]. It has historical usage as a spasmolytic for a variety of digestive issues, including colic in newborns, flatulence, diarrhea, dyspepsia, nausea, and vomiting, as well as to lessen gas and cramps [94]. In addition to menthol, peppermint oil also contains limonene, menthone, isomethone, cineole, isopulegol, pulegone, menthofuran, caron, and menthyl acetate [95]. Menthol is the main component of peppermint oil. According to the British Herbal Compendium [96], the leaves of piperita were used to cure biliary problems, indigestion, and intestinal colic. Its usage as a cholagogue and carminative in conditions affecting the gastrointestinal system, gallbladder, and bile ducts was advised by the German Commission, European Scientific Cooperative on Phytotherapy (ESCOP).

The major component of Dabur's "Pudina Hara," an Ayurvedic medicine that is used for the treatment of several stomach-related issues like gas, indigestion, acidity, etc., is *M. Piperita* oil. Mostly the oil is employed for treating the conditions like IBS, liver issues, ulcerative colitis, biliary tract ailments, and gallbladder problems. The gastric emptying rate is increased following *M. piperita* oil administration in a blinded, controlled research in patients with dyspepsia and normal gastrointestinal function [97]. It has a statistically significant impact on gynecological surgery patients who experience nausea [98]. Research indicates that *M. piperita* causes intestinal smooth muscle relaxation by lowering the calcium flux to the jejunum and large intestine [99]. Menthol and similar terpenes exhibit choleretic effects in in-vivo investigations and are useful in the treatment of a patient with bile duct and gallbladder cholesterol stones [100]. Although, piperita is acknowledged to be safe and is approved by Food and Drug Administration (FDA), still people with GI reflexes must use it cautiously because it might cause gallbladder discomfort and bile duct blockage [94].

2.1.8 *Plantago ovata* Forsk

The psyllium plant, *Plantago ovate* (Fig. 2h) of the family Plantaginaceae, is currently grown all over the world, including India [101]. It is originally from Persia. Several members of the *Plantago* genus are collectively referred to as "psyllium," and their seeds are a fantastic source of both soluble and insoluble fiber and are used commercially to manufacture mucilage [102]. The term "mucilage"refers to a class of transparent, colorless gelling substances that are produced by the seed coat and are frequently referred to as the husk or psyllium husky. For ailments such as hypercholesterolemia, colitis, diabetes, colon cancer, and inflammatory bowel disease, *P. ovate* is being utilized in many cultures as a home remedy [103]. Mucilaginous polysaccharide, mostly constitutes rhamnose, xylose, galactose, galacturonic acid, and arabinose, is the main component of the seeds and husk. Fatty acids are also present in the seeds, namely linoleic, oleic, and linolenic acids in concentrations of about 40.6%, 39.1%, and 6.9% respectively making up the majority. Dietary fiber of *P. ovata* may be extracted and has medical benefits, making it a useful source for making low-calorie foods [102].

Chronic constipation, digestion, hemorrhoids, moderate diarrhea, relief of duodenal ulcer, colon cleaning, IBS, demulcent, GERD, and other disorders are regularly treated with psyllium husk as a therapeutic agent [104]. Being an herbal, non-irritating laxative, which helps promote digestion, psyllium husk has been suggested. It transforms into a gelatinous, viscous substance that absorbs water to carry out a specific function when submerged in water. Fibers of husk are usually used to treat constipation and are frequently taken as a supplement. Clinical studies have consistently indicated the raised moisture level in stool, causing it to become heavier and softer while facilitating bowel movements [105]. On the other side, water uptake by psyllium has been demonstrated to impede colon transit and gastric emptying, which is advantageous for people who have diarrhea or stools that are incompatible with liquid stools. Both the number of bowel motions and the strain on the gastrointestinal walls are increased. Hemorrhoids and diverticulitis can be treated with psyllium husk [106]. Because of the accompanying health benefits, the FDA has authorized using psyllium husk in eatable items [107].

According to numerous research, using psyllium husk for medicinal purposes is risk-free and has few negative effects [103]. However, to validate its pharmacological qualities and postulate mechanisms of action for improvements to human health, further research is essentially required.

2.1.9 *Trachyspermum ammi* Linn.

Trachyspermum ammi L. (Fig. 2i), a drug of the Apiaceae family, is a highly prized seed spice with significant therapeutic value [108]. Ajwain fruits consist of brownish essential oil of 2–4%, of which thymol makes up the majority (35%–60%) [109].

Carvacrol, dipentene, α and β-pinenes, α-terpinene, and γ-terpenine as well as para-cymene are all components of the nonthymol fraction thymene [110]. The plant also contains trace quantities of myrcene, caphene, and α-3-carene. Alcoholic extracts possess a saponin that is extremely hygroscopic. A steroid-like compound and a yellow, crystalline flavone have been extracted from the fruits, along with a glucoside, 6-O-β-glucopyranosyloxythymol [111], and 25% of oleoresin that contains volatile oil [112]. One of the main ingredients of *T. ammi* oil is carvone which is 46% [113].

The fruits of *T. ammi* have been used medicinally for amoebiasis, expectorant, stomachic, carminative, and antiseptic purposes. The investigation of calcium channel blockage, which is revealed to facilitate the spasmolytic activities of herbs, supports the conventional usage of Ajwain in gastrointestinal ailments, like colic and diarrhea, as well as in hypertension [114]. Various ulcer models were used to demonstrate the antiulcer activity of *Trachyspermum ammi* fruit. The ulcer index and the proportion of ulcer prevention in mice pre-treated with ethanolic extract considerably lowered in all models. According to the findings, the extract significantly reduced ulcerative lesions, indicating protection ($p < 0.001$) in comparison to the animals in the control group [115]. When *T. ammi* was added in infusion, the gastric acid amount rose because it would cause the secretion of gastric acid to increase. *T. ammi* enhanced gastric acid output by about four times [114].

T. ammi decreased meal transit time in in-vivo experiments with rats, hence boosting the production of bile acids, and/or increasing digestive enzymes activity [116]. One of the studies found that the essence made from the seeds of *T. ammi* considerably decreased ($P < 0.05$) the contractions caused by the neurotransmitter acetylcholine (10^{-4} M). Upon adding the essence of *T. ammi* (0.01%) to the organ bath, it quickly began to have an inhibitory impact on acetylcholine-induced contraction; it reached its peak within 0.5 min and remained for at least 5 min. On the other hand, the same doses of *T. ammi* demonstrated a substantial antispasmodic activity [$P < 0.05$] with cumulative doses of 10^{-5}–10^{-2} M acetylcholine. The highest and lowest inhibitory effect on contraction of about 96.16% and 88.17% were elicited by concentrations of 0.01% and 0.002% *T. ammi*, respectively, in presence of 10^{-3} M acetylcholine with the statistically significant [$P < 0.05$] variations in the anti-spasmodic activity of both the concentrations [117].

2.1.10 *Trigonella foenum-graecum* Linn.

Trigonella foenum-graecum L. (Fabaceae) is also known as fenugreek (Fig. 2j). It has been a valuable spice for generations [118, 119]. One of the many chemical components of fenugreek is steroid sapogenins. It has been established that the fenugreek embryo, which is oily, contains diosgenin. In the stem alkaloids such as Nicotinic acid, trigocoumarin, trigonelline, and trimethyl coumarin are present. Seeds contain mucilage that stands out in specific [29]. The stem contains volatile oil, mucilage up to 28%, fixed oil of about 5% with a great aroma and a bitter taste, alkaloids namely choline and trigonelline, 22% proteins, and a substance that gives the stem its yellow color [120]. Fenugreek has about 58% carbohydrates, 23–26% protein, 6% fat, and 25% dietary fiber [121].

Fenugreek seeds have been found to lessen the GI adverse effects of tablets of metformin in people having diabetes type 2. Fenugreek seeds enhanced and controlled the intestinal microbiota, increased mucus secretion, and selectively stimulated intestinal bacteria. Fenugreek seeds work by decreasing the GI adverse effects of metformin tablets and increasing acceptance of them. This allows for sustained use of the pills as a treatment [122]. Fenugreek seed extract, oil, and powder have been found to hasten the healing of mucosal wounds in mice. In their research, mucilage-containing fenugreek seeds reduced intestinal and stomach swelling by creating a thin, supple covering on the fortifications of the affected organs [123, 124].

Similar to ranitidine, fenugreek seeds are used to treat gastroesophageal reflux [125]. It lessens bloating as well. The number of bacteria was reported to increase by polysaccharides isolated from fenugreek seeds, just like or even superior to inulin in behavior [126]. Fenugreek seeds have antiulcer properties. Seeds of fenugreek have a comparable effect as a popular proton pump inhibitor omeprazole, used to treat digestive problems such as gastroesophageal reflux disease, gastric and duodenal ulcers, and gastritis [127]. The hydro-extract and a fraction of gel extracted from seeds of fenugreek demonstrated notable ulcer-protective properties in a rat model of alcohol-provoked stomach ulcers, which are due to its actions on mucosal glycoproteins and anti-secretory activity. Additionally, fenugreek seed extract prevents alcohol-caused peroxidation of lipids and following mucosa damage likely through increasing the antioxidant capability of the stomach mucosa [127]. An aqueous trigonella seed extract has also been proven by researchers to have an antioxidant capacity [128], and it is well-known that giving antioxidants to rats can prevent ethanol-induced stomach damage [129]. Additionally, it has been discovered that soluble gel from fenugreek seeds prevents gastric lesion formation more effectively than omeprazole. The gastro-protective and anti-secretory properties of seeds of fenugreek may be caused by the polysaccharide content of the gel or the flavonoids [127]. This theory may be true because fenugreek seeds are acknowledged to have great quantities of flavonoids and it was proved that flavonoids guard the mucosa by reducing the growth of certain necrotic agent-induced lesions [130].

2.1.11 Zingiber officinale

Ginger (Family: Zingiberaceae) is a rhizome (Fig. 2k) that is utilized as a spice all over the world. Depending on the growth, harvesting, and processing conditions, this rhizome includes an extensive range of volatile and non-volatile chemicals in varying amounts [131]. Ginger is a crucial dietary component that has a carminative action, eases intestinal cramping, reduces strain on the lower esophageal sphincter [LES], and prevents bloating, gas, and dyspepsia [132–134]. It is a key ingredient in Asian medicine and has been utilized for a long time to treat a number of ailments, including digestive disorders like heartburn, gas, colic, vomiting, and diarrhea including pain in the abdomen, nausea, and cramping [135, 136]. The majority of the components of ginger that have been extracted by various analytical techniques are gingerols [137].

It improved the excretion of digestive enzymes. Clinical trials have demonstrated that ginger is just as active as dimenhydrinate [138] as well as being more efficient than vitamin B6 [139] in alleviating morning sickness and nausea during pregnancy. In addition, it has been suggested in the management of chemotherapy-related vomiting [140], gynecological laparoscopy [141], and postoperative preventive antiemetics [142]. Additionally, ginger is also suggested as a remedy for motion and sea sickness [143]. Ginger's exact mode of action is uncertain. It may, however, block receptors of serotonin and have antiemetic effects on the central nervous and gastrointestinal tract [144]. In a clinical trial, 100 mg ginger extract or 2 g of rhizome for 2 days was given and their effect on GI motility was examined [145]. The findings showed that the intervention group's gastrointestinal motility significantly improved when compared to the control group. Ginger has been demonstrated to quicken stomach emptying and increase antral contractions in healthy people [146]. In research, no variations were observed in the length of the fundus, GI symptoms, or the levels of the gut peptides like ghrelin, motilin, and GLP-1 of serum in individuals with functional dyspepsia [147].

2.2 The Herbal Formulations Used in Gastrointestinal Disorders

2.2.1 Triphala (India)

An Ayurvedic herbal formulation Triphala is made up of three herbs as shown in Table 2 [148, 149]. Tannins, ellagic acid, gallic acid, and chebulinic acid are their major constituents. These powerful antioxidants may be responsible for some of the formula's immunomodulatory effects [150–152]. Other bioactive substances found in Triphala include saponins, flavonoids like luteolin and quercetin, anthraquinones, fatty acids, amino acids, and several types of carbohydrates [151]. The human gut microbiome also converts triphala-derived polyphenols like chebulinic acid into active metabolites, possibly in vitro to reduce oxidative damage [153]. The usage of Triphala for gastrointestinal health, in general, is likely its most well-known benefit. Triphala extracts made from both aqueous and alcohol have been reported to stop diarrhea in animal trials [154]. Triphala also has enteroprotective properties, which are most likely brought on by its high antioxidant content, at least in part. In a mouse model, Triphala restored reduced protein, glutathione, and phospholipid amounts in the intestinal villi of the brush border while concurrently lowering myeloperoxidase and XO levels in the epithelium of the intestine [155]. Triphala had gastro-protective effects on stress-induced ulcers in rats [156]. Triphala was tested in a human clinical experiment on individuals with gastrointestinal diseases, and it was found to improve the occurrence, yield, and constancy of stools while reducing constipation, mucus, abdominal pain, hyperacidity, and flatulence [157]. The antioxidant properties and great quantities of flavonoids in Triphala were responsible for the healing effect, which also decreased colitis in mouse screening methods [158].

Triphala contains phytochemicals like quercetin and gallic acid that are known to encourage the evolution of Bifidobacteria and *L. bacillus* species by suppressing the

Table 2 Herbal formulations used in gastrointestinal disorders with their constituents

S.no	Formulation Name	Herbs used in the formulation	Indication	References
1	Triphala	Fruits of *Terminalia chebula* (family Combretaceae), *Terminalia bellerica* (family Combretaceae), and *Phyllanthus emblica* (Phyllanthaceae)	Appetite stimulation, gastric hyperacidity reduction, constipation, abdominal pain, and flatulence	[148, 149, 155]
2	STW-5 (IBEROGAST®)	*Iberis amara* (family Brassicaceae), Roots of *Angelica Sinensis* (family Apiaceae), flowers of *Matricaria chamomilla* L. (Family Asteraceae), fruit of *Carum carvi* L. (family Apiaceae), fruits of *Silybum marianum* (family Asteraceae), leaves of *Melissa officinalis* (family Lamiaceae), leaves of *Mentha x piperitae* (family Lamiaceae), *Chelidonium majus L.* (family Papaveraceae), and roots of *Glycyrrhiza glabra* (family Fabaceae)	Functional dyspepsia Irritable bowel syndrome	[166–171]
3	Peppermint oil (PMO)	*Mentha piperata* (family Lamiaceae)	Digestive tract smooth muscle relaxant Diarrhea Irritable bowel syndrome Antispasmodic	[172, 174, 177–179]
4	Zhizhu Kuanzhong	*Rhizome of Atractylodes macrocephala* (family Asteraceae), fruits of *Citrus aurantium* (family Rutaceae), roots of *Bupleurum* (family Apiaceae), and fruits of *Crataegus pinnatifida* (family Rosaceae)	Functional dyspepsia	[180, 181]
5	Rikkunshito RKT	Tuber of *Pinellia ternata* (family Araceae), roots of *Panax ginseng* (family Araliaceae), rhizome of *Atractylodis lanceae* (family Asteraceae) *Pachyma hoelen* (family Polyporaceae) *Aurantii nobilis pericarpium*	Dyspepsia Stress-induced gastric hypersensitivity Epigastric fullness, heartburn, burp, and nausea.	[184, 185, 194, 195]

(continued)

Table 2 (continued)

S.no	Formulation Name	Herbs used in the formulation	Indication	References
		(family Rutaceae), fruits of *Zizyphi fructus* (family Rhamnaceae)		
6	DA-9701	Seeds of *Pharbitis nil* (family Convolvulaceae) tubers of *Corydalis turtschaninovii* (family Papaveraceae)	Functional dyspepsia	[197, 198, 200–202]
7	Pudin Hara	Pudin Hara Pearls: leaves of *Mentha piperata* (family Lamiaceae) and *Mentha spicata* (family Lamiaceae) Pudin Hara Liquid: *Mentha piperata* (family Lamiaceae)	Indigestion, functional dyspepsia	[204]

formation of unwanted gut flora like *E. coli* [159–162]. Additionally, lactic acid bacteria have enzymes (such as tannase) that can break down plant tannins like the gallic acid in Triphala [163–165]. For instance, chebulinic acid, a polyphenol derived from Triphala, is metabolized by the human gut microbiota into products called urolithins that is able to fend off oxidative injury.

2.2.2 STW-5 (Iberogast®) (Germany)

A liquid concoction known as STW-5 is created by combining extracts from nine well-known herbs as indicated in Table 2, that were obtained using alcohol in a certain ratio. It is marketed in Europe as an over-the-counter drug and has been clinically used for many years in German-speaking nations [166, 167]. The concoction has distinctive ingredients, like extracts of well-known herbs produced by distinct extraction methods with stable quantities of components [167].

STW-5 demonstrated spasmolytic effects in an experiment using a guinea pig ileal muscle strip by decreasing acetylcholine and spasms brought on by histamine. Because of *Iberis amara* the basal resting tone is increased and the atonic segment is contracted. These double actions make it spasmolytic and tonic and were reliant on the basal tone of the gut. The duodenum, jejunum, and colon also experienced these effects [168].

315 individuals with functional dyspepsia as indicated using the Rome II criteria were included in a randomly, double-blind, placebo-controlled research. The human volunteers were randomly assigned to receive either a placebo or STW 5, 3×20 drops/day for an 8-week treatment after a washout period of a week. Six months were spent monitoring the patients. The gastrointestinal symptoms' (GIS's) evolution during the treatment period served as the primary endpoint. In both groups, the GIS improved. Following 4 and 8 weeks of treatment, the STW 5 group showed a

clinically meaningful improvement that was more than the placebo. Global evaluations by researchers and patients reaffirmed STW 5's appreciable superiority [169, 170].

The effectiveness and security of STW 5 in treating IBS were examined in one random, double-blind, and placebo-controlled research [171]. The test used a gastroenterological expert panel-developed abdominal symptom score, which contained eight specific IBS indications assessed on a 4-point Likert scale, as one of its major effectiveness endpoints. After a 1-week washout period, for four weeks, 300 patients were randomized to receive STW 5, two research medications, or a placebo. In comparison to the placebo, the IBS symptom score dropped by 1.5 points because of STW 5, which was clinically relevant and statistically significant ($p < 0.0004$). Similar results were found for the stomach pain, which was assessed using four-point Likert scales for each of the four abdominal quadrangles. STW 5 once more demonstrated a statistically significant advantage over placebo [171].

2.2.3 Peppermint Oil

Steam distillation is used to create peppermint oil (PMO) through plant leaves of *Mentha piperata* as indicated in Table 2 [172]. Its numerous constituents (>80) include 35–55% of menthol, 20–31% of menthone, menthyl acetate in 3–10%, cineol, and several other volatile oils [173–175]. Menthol, which naturally occurs as a pure stereoisomer, appears to be its main component and active element [174].

There are numerous lines of evidence that point to PMO's potential role as a digestive tract smooth muscle relaxant [174]. In guinea pig ileal smooth muscle, PMO and its ingredient menthol inhibit calcium channels, which aids in relaxing intestinal smooth muscle, according to in vitro studies [176]. A different investigation discovered that PMO greatly decreased the guinea pig coli's sensitivity to acetylcholine, 5-hydroxytryptamine, and histamine [99]. By limiting the uptake of extracellular calcium ions via voltage-dependent channels, they hypothesized that the PMO relaxes GI smooth muscle [99, 175]. PMO reduced stomach spasms when administered orally or topically [177, 178].

According to Rome III criteria, in a recent study, patients with mixed IBS or IBS with diarrhea participated in a randomized controlled experiment of 4 weeks to test the effectiveness and tolerability of a new PMO formulation intended to sustain the drug release within the small intestine. A solid-state customized enteric-coating matrix designed to deliver peppermint oil to the small intestine that is triple-coated with prolonged release and fewer potential side effects was used in their research. At the end of the experiment, the PMO group had a 40% reduction from baseline in the whole IBS scale scores when compared with the placebo group 24.3% ($p = 0.0246$). In addition, compared to the placebo, there was a superior perfection in numerous distinct GI symptoms and also in rigorous or intolerable symptoms [179].

2.2.4 Zhizhu Kuanzhong (China)

Since 2002, a proprietary fixed combination formulation known as Zhizhu Kuanzhong (ZZKZ) has been sold commercially in China [180]. Four herbs as indicated in Table 2 combine to make it. The primary ingredients Shan Zha and

Bai Zhu have been investigated in animal pharmacological investigations, and it has been found that they speed up intestinal propulsion and gastric emptying when compared to control treatments [181, 182]. Zhi Shi is frequently used as a TCM treatment for FD on its own. Zhi Shi demonstrated an inhibitory effect in in vitro tests on the impulsive retrenchment of the pyloric circular smooth muscle strip [183]. It has been demonstrated that chai hu has an anti-anxiety and antidepressant effect in addition to speeding up small intestinal transit time and gastric fluid emptying [180].

ZZKZ, a patented formula, enhanced intestinal propulsion and motility as well as stomach emptying in rats [180, 181]. ZZKZ was demonstrated to be highly efficacious than placebo for 392 patients meeting Rome III postprandial distress syndrome criteria when taken at a dose of 3×2 capsules daily for functional dyspepsia. This particularly enhanced satiety and fullness after meals. Since no serious side effects had been documented in any of the RCTs or post-marketing monitoring data, it suggests that this medication is safe [180, 181].

ZZKZ was shown to be non-inferior to cisapride in FD symptom control and gastric emptying in a limited Phase II, RCT with 403 FD patients [182]. In a study, St. John's wort and domperidone combined with ZZKZ at the same dosage of three capsules per day had comparable benefits on FD patients' scores for depression and anxiety [183].

2.2.5 Rikkunshito (RKT) (Japan)

As stated in Table 2 [184, 185], RKT is an extracted granule for oral use made up of a dried mixture of eight unprocessed herbs. RKT has been shown to improve GE, antral contractions, and fundic relaxation in addition to reducing GI symptoms like dyspepsia or reflux [186]. It has been proposed that each component of RKT exhibits synergistic effects through a complicated interacting route [187, 188].

By triggering the esophageal acid clearance mechanisms, RKT was able to minimize clinical symptoms and the amount of time that acid was present in the distal esophagus during 24-hour esophageal pH monitoring tests in a trial with eight children who had symptoms of gastroesophageal reflux [189]. In a pilot survey, RKT treatment for 8 weeks enhanced clearance of the esophagus by lessening LES residual pressure throughout swallowing and elevating the entire frequency of bolus transit and peristaltic contraction rate in esophageal multichannel impedance and manometry in 30 patients from PPI-refractory NERD [190]. Rikkunshito produced a relaxing effect on the stomach smooth muscles of a rat, which was brought about by the Ca2+-stimulated K+ channel being activated [191]. RKT is demonstrated to improve adaptive relaxation of the guinea pig stomach when it is isolated [192]. RKT demonstrated a high bile salt binding capacity and may guard against bile acid exposure harming the esophageal mucosa. Refractory GERD and duodenogastroesophageal reflux can both benefit from it [193]. RKT can improve stomach accommodation and reduce stress-induced gastric hypersensitivity [194]. In Japan, during a placebo-controlled, randomized trial with 42 patients having chronic idiopathic dyspepsia, the acetaminophen absorption technique revealed that RKT possesses prokinetic activity for enhancing gastric emptying, it also decreases upper gastrointestinal symptoms such as nausea, heartburn, epigastric fullness, and burp in comparison with placebo [195]. According to a randomized, parallel comparison experiment in Japan with 104 patients with GERD who were resistant to PPIs, the effectiveness of RKT with a standard dose of PPI on decreasing acid-related dysmotility symptom and reflux symptoms was comparable to that of a double dose of PPI [196].

2.2.6 DA-9701 (South Korea)

The Korean Food and Drug Administration approved the novel herbal medication DA-9701, which was created in South Korea, in May 2011 [197]. It contains ethanolic extracts of plants indicated in Table 2. These two herbs are extensively utilized as an herbal medication in Korea, China, and Japan since ancient times to treat gastrointestinal as well as gynecological disorders. Pharbitidis semen is employed as an intestinal peristalsis activator and an analgesic for the stomach [198]. Tuber of Corydalis is prescribed as an analgesic or an antispasmodic for the gastrointestinal tract [199]. The active ingredients in DA-9701 are corydaline and tetrahydropalmatine in Corydalis tuber and chlorogenic acid in Pharbitidis semen. The pharmacological investigation has shown that DA-9701 exhibits dopamine D2 antagonistic, 5-HT1A agonist, adrenergic a2 agonist, and 5-HT4 agonist action [197, 198].

According to the Rome II criterion, 462 Korean FD patients participated in a randomized, controlled, multi-center trial that revealed the effects of 30 mg DA9701 TID and 50 mg itopride hydrochloride TID on FD symptoms [200]. At 4 weeks, the itopride group had a responder rate of 36%, whereas the DA-9701 group had a responder rate of 37%. DA-9701 with itopride significantly decreased from baseline, and all consumer complaints related to gastrointestinal. In short research, FD patients according to Rome III definitions and the presence of H. pylori, the group having DA-9701 presented better indication recovery rates (73.3%) at 12 weeks as compared to the elimination group (60%). Due to the short sample size, there was no statistically significant difference [201]. In a subclass examination, DA-9701 greatly lowered the reflux symptom score in participants 65 years of age or older when compared to the placebo [202]. After taking 30 mg of DA-9701 TID for 24 days, 27 patients who satisfied Rome III's parameters for functional constipation showed improved involuntary movement of the bowel, stool form, and subjective symptoms of constipation [203].

2.2.7 Pudin Hara (India)

Two Ayurvedic preparations manufactured by Dabur India Ltd. called Pudin Hara Pearls and Pudin Hara Liquid include two Mentha species as shown in Table 2 and are intended for indigestion. Along with other ingredients, preservatives, and colors, each 180 mg of Pudin Hara Pearls contains 0.174 ml of Mentha oil obtained from the aerial part of *Mentha piperata*, and 0.034 ml of Spearmint oil (*Mentha spicata*, Aerial part, Oil). In addition to other ingredients, preservatives, and colors, each 1 ml or 20 drops of Liquid Pudin Hara contain 0.0337 ml of *Mentha piperata* oil.

One capsule of Pudin Hara Pearls (thrice a day) or ten drops of Liquid Pudin Hara was given to 79 individuals with functional dyspepsia after mealtimes for 5 days or until signs subsided, whichever came first. Based on improvements in indications such as nausea, acid regurgitation, heartburn, stomach discomfort, and upper abdominal bloating as well as the overall assessment of the investigator and patient therapeutic efficacy was determined from the baseline. Safety was evaluated based on the emergence of negative events and modifications to the hematological and biochemical profiles. Liquid Pudin Hara and Pearls of Pudin Hara dramatically reduced the symptoms of functional dyspepsia, including the above-mentioned conditions. Although no statistically significant variations were observed in the research products while comparing the effectiveness in treating the symptoms of dyspepsia Pudin Hara Pearls were more effective in treating bloating of the upper abdomen, heartburn, acid regurgitation, and stomach pain, while Liquid Pudin Hara was more effective at treating nausea and vomiting as well as the feeling of acidity. In any of the study subjects, no adverse events were recorded during the investigation. Pudin Hara Pearls and Pudin Hara Liquid could be used safely and effectively in treating functional dyspepsia [204].

3 Prospects and Challenges of Herbal Medicine Formulation

A single herbal remedy or medicinal plant may include hundreds of natural ingredients, and a polyherbal formulation has many times as many as one. Single active ingredient may practically be hard to identify in such a study, especially when a herbal product is a blend of two or more herbs [205]. Raw material quality is influenced by environmental factors, farming practices, and good collection techniques, such as plant selection and cultivation, in addition to intrinsic (genetic) aspects. It is challenging to conduct quality controls on the raw materials for herbal medicines due to a variety of factors. Correct identification of medicinal plant species, specific storage, and unique cleaning processes for diverse materials are significant requirements for the quality control of raw materials, according to good manufacturing practice (GMP). The main difficulties come in the quality control of final herbal medicines, particularly herbal mixtures [205]. As a result, compared to other pharmaceuticals, the overall specifications and procedures for quality control of final herbal products continue to be far more complicated. The World Health Organization (WHO) continues to support the implementation of quality assurance and control measures, such as National Quality Specifications and Standards for Herbal Materials, GMP, labeling, and licensing schemes for manufacturing, to assure the safety and efficacy of herbal medicines. The use of incorrect plant species, adulteration of herbal products, contamination, overdosage, misuse of herbal medications by consumers or healthcare professionals, and the combination of herbal medications with other medications are just a few of the causes of adverse effects resulting from the consumption of herbal medicines. Most manufacturers of herbal medicines are not properly informed about the value of taxonomic botany and documentation, and this presents specific obstacles during the identification and collection of medicinal plants used in herbal remedies. It is vital to use the most widely used binomial names for medicinal plants to avoid confusion caused by the common names. Botanists, phytochemists, pharmacologists, and other key players must work effectively together to monitor herbal medicine [205].

In human health care systems around the world, medicinal herbs have been crucial not only in treating illnesses but also as a potential resource for maintaining good health. The herbal industry has the potential to significantly impact the world economy. Future international labeling guidelines should effectively address quality issues given the rising use of herbal products. To comprehend the use of herbal medicines, standardization of methods and quality control data on safety and efficacy are required. The formation of businesses based on medicinal plants in undeveloped countries has been severely hampered by a lack of understanding of the potential social and economic benefits of doing so. Much research is being done on herbal medications to incorporate them into brand-new drug delivery systems [206]. Innovative approaches to reducing drug instability, like the use of suspension, biodegradable cellulose, therapeutic proteins, nanoparticles, and emulsifiers, can be especially helpful. Preventing drug deterioration brought on by environmental factors is a major issue when it comes to medication stability [207]. To maintain medication stability, certain procedures, including packing and container standards, must be followed. Pharmacokinetics and ADME research are essential for ensuring the efficacy, toxicity, and safety profile of herbal therapies because the side effects and toxicity of herbal treatments have been well-documented, including kidney damage, the development of stones, acute neuropathy, and newborn mortality. The validation of efficacy, safety, and toxicity levels occurs at the most crucial stage of developing herbal medications, which includes clinical trials and ethical considerations [207]. Such validation is necessary to compete with existing allopathic medicines and survive in the global medication market. For herbal medicine producers to maintain the quality of their products, competent authority regulations for quality and safety criteria are essential [208].

4 Conclusion

Numerous studies show that traditional people around the world frequently employ plant-based therapies or herbal medications to address digestive system problems. Different ethnic groups have employed a wide variety of herbs to cure different digestive issues from generation to generation. Herbal drugs are becoming more and more well-liked among Westerners because, when used properly, they have few negative effects. Recently, there has been a general trend toward a recovery of interest in a traditional healthcare system. People can treat a range of digestive-related ailments by regularly including herbal medicines in their diets, which can enhance how well the digestive system works. Numerous secondary metabolites or phytochemicals found in plants can have a part in the management and inhibition of certain gastrointestinal problems. These herbal treatments for digestion function in a variety of methods, counting encouraging effortless bowel movements, enhancing digestion, growing the incidence of bowel movements, purifying the body and eliminating contaminants, settling an upset stomach, and minimizing gas, bloating, and discomfort in digestion. Unfortunately, not enough research has been done on the vast majority of plant species utilized in indigenous traditional medicine in many underdeveloped nations. Triphala is a potent polyherbal blend with a wide range of medicinal benefits for preserving homeostasis as well as illness prevention and therapy. Numerous scientific investigations have documented evidence-based support for Triphala's many traditional uses. It may be possible to explain STW 5 (Iberogast) observed therapeutic effectiveness in treating hypotonic and spastic dysmotility symptoms of functional dyspepsia and irritable bowel syndrome, at least in part, by the pharmacological evidence that points to a dual-action principle. The practice of STW-5 as a prospective alternate medicine for functional gastrointestinal disorders, particularly functional dyspepsia and irritable bowel syndrome, will be made possible by further research on their involvement in the pathophysiology of FGIDs. Clinical experiments and in vitro investigations have shown that peppermint oil appears to reduce IBS indications, particularly stomach pain, by soothing smooth muscles in the stomach. It is necessary to have pertinent data derived from thorough studies to support the efficacy and security of PMO. Rats' intestinal propulsion and motility as well as stomach emptying were improved by the patented formula Zhizhu Kuanzhong. It was discovered to be more effective at treating FD when used alone or in combination with conventional Western therapy. RKT enhances stomach motor activity, together with gastric accommodation and emptying, and boosts esophageal clearance and motility, according to basic research and clinical investigations. DA-9701 increases stomach emptying and antral motility in healthy and sick situations, according to a number of preclinical and human investigations. Furthermore, it was asserted that DA-9701 might lessen constipation and GERD in the particular group. By using Pudin Hara Pearls and Pudin Hara Liquid, functional dyspepsia symptoms such as acid regurgitation, heartburn, a sense of acidity, stomach pain, and upper abdominal bloating were dramatically reduced. Both Pudin Hara Liquid and Pudin Hara Pearls may be used safely and successfully to treat functional dyspepsia. For the establishment of the scientific basis for activity and to more accurately assess the superiority, efficiency, and security of these formulations, ongoing examination of traditional plant medicines is necessary. Well-designed clinical trials will offer the required proof of efficacy to validate herbal medicines. In a small number of clinical trials, the safety and acceptability of conventional and herbal medicines used to treat digestive system illnesses have been examined. The findings generally demonstrate that there have been minimal side effects observed. Recent research on conventional plant-based therapeutics has produced evidence that supports additional studies in the hope that new treatments for digestive illnesses may be created.

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Herbal Medicine and Pregnancy

23

Priyanka Devi and Prasann Kumar

Contents

Abstract

The usage of herbal medications is increasingly commonplace worldwide, and this trend is also noticeable among pregnant women. Pregnant women may use herbal medicine to improve their general health. Although self-diagnosis and self-dosing are common in herbal medicine, they are unsafe and highly discouraged. It is advisable to use herbs for pregnancy management only under the guidance of a midwife, physician, herbalist, naturopathic, or homeopathic doctor because each pregnancy is unique. This chapter discusses the use of herbal remedies during pregnancy and the types of herbal remedies used. Herbal medicines include herbs, herbal preparations, and finished products made from herbs, and they contain active compounds from plants with known or suspected therapeutic advantages.

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To ensure the best outcome for pregnant women and their babies, herbal medicine products should only be recommended for use by individuals who are well-versed in the types of herbs, their parts (e.g., roots, leaves, etc.), dosage, and how they could be used (e.g., capsules, tonics, teas) before consuming herbs during pregnancy. Most pregnant women who use herbal medicine conceal their usage from their attending conventional medical professionals. However, most women were encouraged by their families and friends to utilize herbal medications because are considered safer, more effective, and had fewer adverse side effects than modern medicine. During pregnancy, herbs that are considered safe to use are usually those used for food and other related purposes. The most common form of these supplements is a tablet, a tea, or an infusion.

Keywords

Pregnancy · Tonics · Baby · Roots · Leaves · Herbs · Naturopathic · No poverty · Zero hunger

Abbreviations

ACOG	American College of Obstetricians and Gynecologists
AEDs	Antiepileptic drugs
FDA	Food and Drug Administration
HMs	Herbal medicine
UDP	Glucuronosyltransferase

1 Introduction

In many societies across the globe, the use of medicinal plants to promote a healthy pregnancy and facilitate a smooth delivery has been a long-standing tradition. In the past three decades, there has been a noticeable surge in the availability of comprehensive literature on plant-based therapies in Western countries. Despite the medical community lacking substantial evidence to deem homeopathic remedies unsafe during pregnancy, this perception still endures. It is worth noting, however, that most natural remedies employed during pregnancy and labor have not undergone rigorous scientific scrutiny. While anecdotal evidence may offer some support for their use, it is imperative to establish scientific validation regarding their safety and efficacy.

Nevertheless, women in North America and elsewhere continue to rely on botanical remedies they have come to know through word of mouth or natural birthing literature. This study draws from a range of sources, including HerbClips, a resource provided by the American Botanical Council, and various herbal and pregnancy literature. These sources shed light on the subject [16].

Pregnancy triggers a series of significant physiological changes in a woman's body, often leading to a variety of discomforts such as nausea, vomiting, indigestion, and diarrhea. It is a common response for expectant women, including the use of natural remedies, to self-medicate to alleviate these conditions [54]. This trend has contributed to the substantial increase in the global utilization of natural remedies by pregnant women [14, 54].

The prevalence of natural medication use during pregnancy is influenced by various factors, including geographic region, ethnicity, cultural practices, and socio-economic status. To investigate this phenomenon, researchers conducted a cross-sectional study involving 9459 pregnant women across 23 countries in Europe, encompassing Western Europe, Northern Europe, Eastern Europe, North and South Americas, and Australia. Among the participants, 28.9% reported the use of herbal medicines during their pregnancies. Notably, the countries with the highest reported usage of natural medications were Russia (60.0%), Poland (49.8%), and Australia (43.8%) [57].

Furthermore, Hwang et al. [37] found that countries like Iran (49.2%), Egypt (27.31%), Bangladesh (70%), Iraq (53.7%), Palestine (40%), and Taiwan (33.6%) also exhibited a substantial prevalence of natural medication use among pregnant women.

While previous studies indicate the widespread use of natural medicines during pregnancy in sub-Saharan Africa [6], there remains a scarcity of data on the prevalence of herbal medicine usage among expectant women in numerous sub-Saharan African nations. This study provides insights into the reasons, potential adverse effects, and effectiveness of the most commonly used herbal medicines during pregnancy in sub-Saharan Africa and the varying frequencies of herbal drug utilization across different countries in the region.

The utilization of herbal medicine (HM) during pregnancy is a common practice, with the majority of users being female, and many continue using HMs even during pregnancy [24]. However, the prevalence of HM use during pregnancy can vary widely depending on factors such as region, race, culture, and socioeconomic status, ranging from 7% to 55% [24, 114]. For example, in Australia, 34% of pregnant women use HMs [27], while in the European Union, the figure is as high as 50% [36, 62]. In contrast, in the United States and Canada, 6–9% of pregnant women use HMs [64].

Pregnant women often turn to herbal products to support their health and address non-life-threatening conditions, viewing them as a safe and natural alternative to conventional medications, particularly for issues like nausea and constipation [27, 38]. Notably, some HMs, such as black cohosh (*Actaea racemosa*) and blue cohosh (*Caulophyllum thalictroides*), have a well-documented history of use as uterine tonics and labor inducers [27]. However, despite reports of associated risks [109], there remains a lack of comprehensive information regarding their safety and efficacy during pregnancy.

One key issue is the widespread availability of HMs without needing a prescription in many countries. This can lead to challenges when pregnant women self-prescribe HMs alongside traditional medications recommended by their healthcare providers [99]. Patients may not always disclose their use of HMs to their healthcare providers, often because they do not perceive them as potentially hazardous substances [1].

This chapter serves as a comprehensive exploration of the multifaceted landscape of herbal medicine use during pregnancy. Its primary objective is providing valuable education to medical practitioners and expectant mothers. It seeks to shed light on the prevalence of herbal medicine use among pregnant women and, more crucially, to highlight the potential interactions between herbal medicines (HMs) and conventional medications employed to manage underlying medical conditions or pregnancy-related issues.

Physiological changes that occur during pregnancy can trigger or exacerbate various medical conditions. For instance, hypertensive disorders, encompassing chronic hypertension, fetal hypertension, and preeclampsia, contribute to approximately 10% of pregnancy-related complications in the United States. Some expectant mothers grapple with preexisting conditions like epilepsy or asthma, with approximately 9% reporting a history of asthma. Moreover, the use of antiepileptic drugs (AEDs) during pregnancy has been associated with potential risks, including adverse effects like miscarriage, antepartum, and postpartum hemorrhage.

Within this chapter, we delve into the demographics of pregnant herbal medicine users and explore the risks associated with herbal medicine usage during pregnancy. By highlighting the key medical conditions that herbal remedies aim to address, we aim to provide a comprehensive understanding of the nuances surrounding this practice.

In essence, the use of herbal medicine during pregnancy is a topic that requires thoughtful consideration, well-informed decision-making, and the active engagement of healthcare professionals. By examining prevalence, risks, and interactions with conventional medications, we endeavor to equip both medical practitioners and expectant mothers with the knowledge necessary to make informed choices regarding herbal medicine use during pregnancy. This knowledge ultimately plays a pivotal role in ensuring the best possible outcomes for both mother and child.

2 Navigating Pregnancy: A Guide to Herbal Medicine and Medicinal Plants

Pregnancy is a remarkable period marked by profound physical and emotional transformations as women's bodies adapt to nurture and develop a new life. Pregnancy is a profound and intricate life-altering journey, encompassing a multitude of physical, emotional, and social transformations. This process typically spans 9 months and culminates in the birth of a child. This journey often brings about a host of typical symptoms, such as morning sickness, sleep disturbances, and heartburn. In response to these challenges, many individuals turn to herbal remedies and medicinal plants as an alternative or complementary approach to conventional medical care.

The various facets of the pregnancy voyage are summarized in Table 1. It's imperative to acknowledge each pregnancy's uniqueness and its individualized experiences. Consultation with healthcare professionals is indispensable for

Table 1 Key stages and aspects in pregnancy

Topic	Key points
Conception	Pregnancy begins with sperm fertilizing an egg.
	Zygote forms in the fallopian tube.
	Zygote implants in the uterine wall.
Prenatal development	Zygote evolves into an embryo.
	Divided into three trimesters.
	Each trimester is marked by changes and milestones.
Pregnancy symptoms	Includes morning sickness, fatigue, mood swings, and appetite changes.
	Symptoms can vary among individuals.
Prenatal care	Essential for a healthy pregnancy.
	Involves routine check-ups, prenatal vitamins, and screening for complications.
Diet and nutrition	Balanced diet crucial for fetal development.
	Avoidance of harmful substances like alcohol and specific medications.
Physical changes	Pregnancy results in weight gain, breast modifications, and abdominal expansion.
	Hormonal changes can lead to skin alterations like stretch marks.
Emotional changes	Pregnancy-related emotional adjustments due to hormonal shifts and anticipation of parenthood.
	Effective communication and emotional support are vital.
Maternal health	Monitoring the health of the expectant mother is paramount.
	Conditions like preeclampsia and gestational diabetes may require specific care.
Labor and delivery	Culmination of pregnancy with uterine contractions leading to birth.
	Various delivery methods, including vaginal birth and cesarean section.
Postpartum period	A phase of physical and emotional recuperation post-childbirth.
	Involves caring for the newborn and adapting to parenthood.
Challenges	Some pregnancies may face complications like ectopic pregnancies, miscarriages, or preterm birth.
	Awareness and timely medical intervention are crucial.
Parenting and childcare	Pregnancy initiates the lifelong journey of parenthood.
	Considerations include childcare, parenting approaches, and the well-being of both child and parents.
Support system	Quality of pregnancy and postpartum period depends on a strong support system including partners, family, friends, and healthcare professionals.
Cultural and societal factors	Cultural and societal factors influence an individual's pregnancy experience.
	Wide variations exist across different geographic regions and communities.
Fertility and reproductive health	Vital concepts for those trying to conceive.
	Infertility concerns and assisted reproduction options warrant discussion.

(continued)

Table 1 (continued)

Topic	Key points
Family planning	Decisions about family size and birth spacing significantly impact pregnancy discussions.
	Timing and methods of contraception are part of family planning.
Legal and ethical issues	Complex topics like surrogacy, adoption, and abortion are discussed in the context of pregnancy.
	These matters involve moral, legal, and psychological considerations.

personalized guidance and addressing specific concerns or medical conditions during pregnancy. Additionally, emotional support and open communication are indispensable components for expectant parents to navigate this transformative journey effectively.

Herbal medicine, an ancient practice deeply ingrained in human culture, involves using plants and their derivatives for therapeutic purposes [7, 8, 48]. Medicinal plants have been significant in various cultural traditions, serving as remedies for various health conditions [40–47]. The use of medicinal herbs and plants during pregnancy has garnered considerable attention due to the potential relief they may offer for the common discomforts experienced by expectant mothers. However, it also raises substantial concerns regarding their effectiveness and safety.

The use of herbal remedies and medicinal plants during pregnancy necessitates a careful evaluation, as it entails a delicate balance between potential benefits and safety, always with the overarching goal of safeguarding the well-being of both the expectant mother and the developing fetus. This discussion endeavors to provide participants with a comprehensive understanding of the subject, enabling them to make well-informed choices concerning their health and well-being during this transformative period, benefiting themselves and their unborn children.

This chapter section lays the foundation for a comprehensive exploration of the use of herbal medicine and medicinal plants during pregnancy. The associated potential risks that underscore the importance of seeking expert medical advice, identifying known safe herbs, discussing appropriate dosage considerations, exploring potential benefits, and navigating the legal aspects are summarized in Table 2. Furthermore, we will consider the varying responses of individuals to herbal remedies and complementary therapies, emphasizing the necessity for personalized guidance and informed decision-making.

In summary, safety should be a paramount concern for pregnant individuals, and consultation with healthcare providers is essential. While some medicinal plants and herbal remedies may be used cautiously to address common discomforts, they should complement rather than replace conventional medical care during pregnancy. Informed and open discussions with healthcare professionals are vital for making well-informed decisions regarding the use of herbal remedies during pregnancy.

Table 2 Aspects of herbal medicine and medicinal plants in pregnancy

Topic	Key points
Safety concerns	Limited scientific data on medicinal plants' safety and fetal effects during pregnancy.
Consultation with healthcare professionals	Pregnant individuals should always seek advice from healthcare providers before using herbal remedies.
Known safe herbs	Some herbs are considered safe during pregnancy when used in moderation.
	Examples: ginger, peppermint, and red raspberry leaf (in the third trimester with supervision).
Exercise caution with some herbs	Avoid herbs like black cohosh, pennyroyal, and blue cohosh, which can induce preterm labor or miscarriage.
Regulation and quality	Quality of herbal products varies; use products from reputable sources to ensure safety and purity.
Dosage and duration	Adhere to recommended dosages and usage durations for safe herbal product use during pregnancy.
Potential benefits	Some herbs may offer advantages for common pregnancy discomforts, but use cautiously and under supervision.
Herbal treatments for common pregnancy symptoms	Examples of herbal treatments for common discomforts.
Individual variability	Response to herbs varies between individuals, emphasizing the need for personalized guidance.
Alternative therapies	In some cases, alternative therapies like acupuncture or aromatherapy can provide relief without potential risks.

3 The Herbal Approach to Pregnancy Care: Tradition, Safety, and Well-Being

The use of herbal plants in traditional medicine has a long-standing history, spanning centuries, and this practice extends to the realm of pregnancy care. Incorporating herbal remedies into pregnancy-related healthcare is a venerable tradition observed across diverse cultures worldwide. These remarkable botanicals, possessing a multitude of therapeutic properties, provide pregnant women with natural solutions to alleviate various discomforts and enhance overall well-being, all while minimizing potential risks associated with pharmaceutical medications. When employed judiciously and under the guidance of knowledgeable practitioners, herbal medicine during pregnancy can serve as a valuable and holistic approach to maintaining a healthy pregnancy and addressing common maternal and fetal health concerns.

Pregnancy is a unique and transformative phase in a woman's life, characterized by physiological, emotional, and hormonal changes. Many expectant mothers contend with a range of common discomforts, including morning sickness, heartburn, insomnia, anxiety, and swollen ankles. Herbal medicine offers a diverse array of herbal remedies that can be harnessed to safely and effectively manage these symptoms [81, 82]. One of the most renowned herbs for alleviating pregnancy-related nausea and vomiting is ginger (Zingiber officinale), with a history of use spanning centuries, and its efficacy in curbing morning sickness is supported by modern scientific research. Ginger can be consumed as tea, capsules, or candied ginger.

Another commonly used herb during pregnancy is chamomile (*Matricaria chamomilla*). Chamomile tea is a soothing and gentle remedy for anxiety, insomnia, and digestive discomfort, rendering it a valuable herb for expectant mothers. The calming effects of chamomile can help reduce stress and promote better sleep, which is especially beneficial during pregnancy when sleep disruptions are expected.

Beyond managing specific discomforts, herbal medicine during pregnancy encompasses herbs recognized for their uterine toning and nourishing properties. For example, raspberry leaf (*Rubus idaeus*) is a well-regarded herb for strengthening uterine muscles and preparing the body for labor when consumed as tea in the second and third trimesters. Nettle (*Urtica dioica*), celebrated for its rich nutritional profile, provides essential vitamins and minerals, including iron and folic acid, crucial for both the developing fetus and the expectant mother's health. Nettle tea is often recommended to combat anemia, improve circulation, and alleviate swelling, which are common concerns during pregnancy.

Herbal medicine also addresses the emotional and psychological aspects of pregnancy. Herbs like St. John's Wort (*Hypericum perforatum*) and lavender (*Lavandula angustifolia*) can offer relief from anxiety and mood swings. St. John's Wort, known for its antidepressant and anti-anxiety properties, can be safely used in moderation. Lavender essential oil, when diluted and applied topically or diffused, can help soothe nerves and promote relaxation.

However, it is essential to underscore the need for caution when considering herbal remedies during pregnancy. Safety is paramount, as certain herbs may be contraindicated during pregnancy due to potential adverse effects. Pregnant women should consult a qualified herbalist, naturopathic doctor, or midwife experienced in herbal medicine to select appropriate herbs and dosages tailored to their unique needs. Additionally, expectant mothers should be aware of herbs to avoid during pregnancy. Herbs such as black cohosh (*Actaea racemosa*), pennyroyal (*Mentha pulegium*), and tansy (*Tanacetum vulgare*) have been associated with adverse pregnancy outcomes and should be strictly avoided. Herbs that stimulate uterine contractions, like blue and black cohosh, should not be used without proper supervision during pregnancy.

Regulation and quality control of herbal products are crucial considerations when incorporating herbal medicine during pregnancy. Pregnant women should obtain herbs from reputable suppliers to ensure their purity, free from contaminants, and that they have been responsibly harvested and processed. Choosing organically grown herbs whenever possible is essential to minimize exposure to pesticides and other potentially harmful chemicals.

It is essential to emphasize that herbal medicine should not replace standard prenatal care, including regular check-ups with healthcare providers, ultrasound scans, and appropriate screenings. Instead, it should complement conventional medical care by addressing specific concerns and promoting overall well-being, providing a holistic approach to pregnancy care.

The importance of informed and cautious herbal medicine use during pregnancy cannot be overstated. Each pregnancy is unique, and what may be safe and beneficial for one expectant mother may not be suitable for another. This underlines the necessity of individualized care and guidance from a qualified herbalist or healthcare practitioner. Such an approach ensures that the chosen herbs are appropriate for the expectant mother's health and pregnancy. Herbal medicine offers a holistic and natural approach to supporting maternal and fetal health during pregnancy. When used with caution and under the guidance of experienced practitioners, herbal remedies can relieve common discomforts, nurture the mother's overall well-being, and even prepare the body for labor and delivery. However, pregnant women need to approach herbal medicine with a discerning eye, seeking professional guidance to ensure the safety and efficacy of their chosen herbs [98]. With the right approach, herbal medicine can be a valuable and harmonious addition to a healthy and happy pregnancy journey. Numerous studies have indicated that pregnant women have a range of options for consuming herbal medicines (HMs) during pregnancy. These options include unprocessed herbal mixtures, herbal extracts, finished and branded herbal medicines, as well as nutritional supplements containing specific HMs, vitamins, and minerals [57]. In Table 3, you can find a list of medicinal plants commonly used during pregnancy and their intended purposes.

Pregnant women across the globe are increasingly turning to herbal remedies as they seek natural solutions to common discomforts associated with pregnancy. Among the most widely embraced herbal options are ginger, chamomile, peppermint, Echinacea, cranberry, Vaccinium, and garlic. These herbal choices transcend geographical boundaries, recognized for their potential to alleviate typical pregnancy-related ailments.

For instance, ginger is highly regarded for its ability to combat morning sickness, while chamomile and peppermint are sought after for their calming properties. Cranberry and Echinacea are known for their respective benefits in promoting urinary health and bolstering the immune system. Garlic is appreciated for its versatile medicinal properties.

Regional preferences for specific herbal remedies may vary. In Eastern and Western Europe, uva ursi has traditionally been employed for its purported urinary tract benefits. Conversely, in Eastern European nations, herbal options like motherwort, centaurium erythraea, and lovage have taken precedence in the repertoire of pregnant women. These regional variations are deeply rooted in historical practices and cultural traditions.

Understanding the popularity and regional disparities in herbal choices among pregnant women is of paramount importance for both healthcare professionals and expectant mothers. It facilitates informed decision-making, underscoring the significance of effectiveness and safety. This global perspective serves as an invaluable resource for promoting the well-being of both mothers and their unborn children as herbal medicine continues to play a role in pregnancy care.

In Sweden, expectant mothers often incorporate dietary supplements such as Floradix® IRON + HERBS, ginseng, and valerian into their routines [35]. The decision to use herbal medicines during pregnancy can yield benefits for both the

Table 3 Different medicinal plants used by pregnant women in their different physiological conditions of pregnancy

Physiological condition	Medicinal plants	Scientific names
Cold and flu	Echinacea	*Echinacea purpurea*
	Elderberry	*Sambucus nigra*
Pain (Gastralgia and other types of pain)	Chamomile	*Matricaria chamomilla*
	Peppermint	*Mentha × piperita*
	Lavender	*Lavandula angustifolia*
Gastrointestinal disorders, constipation, flatulence	Fennel	*Foeniculum vulgare*
	Dandelion	*Taraxacum officinale*
Anxiety, stress	Valerian	*Valeriana officinalis*
	Lemon balm	*Melissa officinalis*
Urinary tract infection	Cranberry	*Vaccinium macrocarpon*
	Uva Ursi	*Arctostaphylos uva-ursi*
Fetal health promotion	Raspberry leaf	*Rubus idaeus*
	Nettle leaf	*Urtica dioica*
Labor preparation, facilitation, and induction	Black cohosh	*Actaea racemosa*
	Blue cohosh	*Caulophyllum thalictroides*
Milk production and secretion	Fenugreek	*Trigonella foenum-graecum*
	Blessed thistle	*Cnicus benedictus*
Nausea and vomiting	Ginger	*Zingiber officinale*
Stress and anxiety	*Matricaria chamomilla* (chamomile)	*Matricaria chamomilla*
Digestive discomfort	Peppermint	Mentha × piperita
Immune support	Echinacea	*Echinacea purpurea*
Uterine tonic for late pregnancy	*Rubus idaeus* (raspberry leaf)	*Rubus idaeus*
Sleep and relaxation	*Lavandula angustifolia* (lavender)	*Lavandula angustifolia*
Nutrient support	*Urtica dioica* (stinging nettle)	*Urtica dioica*
Preparing for labor	*Rubus idaeus* (red raspberry leaf)	*Rubus idaeus*

mother and the unborn child, and they are occasionally employed to manage specific health issues. Some of the most frequently mentioned uses of medicinal plants during pregnancy include addressing concerns like nausea and vomiting, urinary tract infections (UTIs), childbirth preparation and comfort, common colds and flu, digestive issues, pain management, improving fetal outcomes, preventing abortion, managing anxiety, maintaining general health, and alleviating swelling [33, 57].

4 Global Perspectives on Herbal Medicine Use During Pregnancy: Practices, Prevalence, and Safety Considerations

The utilization of HMs is not confined to specific nations; however, certain HM practices are more prevalent in particular regions. For instance, expectant mothers in India and Ghana display a higher tendency to use HMs for the prevention of abortions and to enhance the health of their unborn children, compared to their counterparts in more industrialized regions [5, 15].

These practices are often rooted in a variety of factors, some of which are directly related to the well-being of the fetus. For example, the prevention of newborn hyperbilirubinemia, protection of the unborn child from injury, promotion of increased prenatal weight, and support for optimal brain development are among the commonly cited reasons [23, 105]. Surprisingly, certain indications also strongly influence societal perceptions of pregnant women. In places like Lesotho and Zimbabwe, it is customary for pregnant women to use HMs, and adherence to cultural standards often mandates this practice [72, 94, 95, 103].

The use of regional medicinal plants for prenatal care is widespread in developing nations, especially in Africa and Asia [68, 96, 97, 112]. These medicinal herbs are consumed in various forms, including decoctions, medications, and soups. For example, in South Africa, Zulu women in their final trimester frequently consume a concoction known as "Isihlambezo." This mixture, which comprises several medicinal herbs, is believed to promote a favorable pregnancy and facilitate uncomplicated labor [17]. Some of the most commonly used ingredients in this mixture include *Agapanthus africanus* (L.) Hoffmanns, *Pentanisia prunelloides* (Klotzsch ex Eckl. & Zeyh.), *Clivia miniata* (Lindl.) Verschaff. Walp., and *Gunnera perpensa* L. [17].

The specific formulations employed often depend on several factors, including the preferences of the expectant mother, the traditional healer's recommendations, and the woman's current health status. It's worth noting that the uterine toning effects of these herbal mixtures are activated upon consumption, but excessive doses leading to heightened vaginal stimulation may result in adverse effects. For example, the "Isihlambezo" mixture has been linked to low newborn birth weights, embryonic meconium spotting (a potential sign of prenatal distress and oxygen deprivation), and uterine breach [17].

Malawi, situated in southeastern Africa, is renowned for its picturesque landscapes, diverse cultural heritage, and the long-standing practice of utilizing herbal medicine and medicinal plants. Traditional healers and local communities in Malawi have relied on the healing properties of indigenous plants to address a wide array of health concerns. This practice is deeply ingrained in Malawian culture and operates in tandem with modern medical approaches.

The realm of Malawian herbal medicine encompasses a plethora of plant species, each possessing distinctive therapeutic attributes. Notably, the Moringa tree (*Moringa oleifera*) stands as a symbol of nutrition and offers a potential solution to combat malnutrition, a prevailing challenge in the nation. Additionally, Malawi's abundant flora includes plants like *Artemisia annua*, known for their antimalarial properties, providing a natural alternative in the fight against this prevalent disease. In Malawi, the “Mwanamphepo” mixture is commonly used to induce labor and consists of various herbs, including *Ampelocissus obtusata* (Welw. ex Baker) Planch., *Cyphostemma hildebrandtii*, and *Cissus verticillata*, often consumed with oatmeal just before childbirth (Maliwichi-Nyirenda and Maliwichi 2010).

Medicinal plants are integral to the Malawian healthcare landscape. The country boasts a diverse array of plant species renowned for their medicinal applications. For example, Senna (*Cassia abbreviata*) is employed for its laxative qualities, while the African potato (*Hypoxis hemerocallidea*) is sought after for its potential to address a spectrum of health issues.

This exploration of herbal medicine and medicinal plants in Malawi seeks to shed light on the profound interplay between the nation’s rich biodiversity and its healthcare traditions. We delve into the cultural significance, traditional healing methodologies, and the potential integration of these herbal therapies into the broader healthcare framework. Furthermore, we address the vital issue of sustainability concerning medicinal plants and their role in delivering accessible and cost-effective healthcare solutions to the people of Malawi.

In Korea, the herbal combination “Anjeonicheon-tang,” which includes therapeutic plants like Atractylodes macrocephala Koidzumi white rhizoma, *Glycyrrhiza glabra* L., and Panax ginseng C.A.Mey, is used to address pregnancy-related symptoms, particularly hyperemesis gravidarum. In Taiwan, a tea known as “An-Tai-Yin,” composed of medicinal plants aimed at inducing labor, is commonly consumed by expectant mothers [20]. However, mothers who consumed An-Tai-Yin were found to have an increased risk of connective tissue, joint, and ocular abnormalities in their babies [20].

Research conducted by Pallivalapila et al. [102] suggests that the first and third trimesters witnessed the highest prenatal use of herbal medicines. During the first trimester, herbal remedies were often used to manage early pregnancy symptoms like morning sickness, nausea, and gastrointestinal issues. In contrast, the third trimester use aimed at aiding in childbirth and preparing the cervix for delivery [79, 97]. Nevertheless, it was not uncommon for women to use herbal medicines throughout their entire pregnancy with the goal of promoting both maternal and fetal health. Notably, no consistent trends have been observed by scholars [110], and some women continued using these herbal remedies until they felt better [77].

According to Kennedy et al. [58], married women with higher levels of education were more inclined to use complementary and alternative medicine (CAM) that included herbal medicines during pregnancy (Forster et al. 2006; [2, 57, 58]). This demographic of herbal medicine consumers was typically middle class, Caucasian, non-smoking males over the age of 30 who had prior experience with herbal remedies. On the other hand, in Eastern European countries, pregnant women under the age of 20 were more likely to use herbal medicines [57].

In contrast, in Africa and Asia, women who incorporated herbal medicines during childbirth tended to be under the age of 30, possess lower socioeconomic status, lack a high school diploma, and reside in remote areas with limited access to public healthcare facilities [54, 113]. These pregnant women often sought advice from friends, family, traditional birth attendants, local plant vendors, and herbalists before deciding to use herbal medicines [54, 57, 113].

Media coverage played a role in influencing the decision to use herbal medicines during pregnancy in industrialized nations, as reported by Hall et al. [33]. In Eastern Europe, pregnant women typically followed their doctor's recommendations regarding herbal medicine use [57].

The practice of using both herbal medicines and traditional remedies during pregnancy is common worldwide [36, 79, 110, 111]. It is worth noting that when herbal medicines are used in conjunction with conventional drugs, they can either enhance or diminish the effects or side effects of these medications. However, many pregnant women refrain from disclosing their herbal medicine usage to healthcare providers due to the stigma attached to this practice.

Overall, a significant proportion of pregnant women turn to herbal medicines and traditional remedies for support during pregnancy, labor, and postpartum, with the prevalence of herbal medicine use varying based on demographic factors such as geographical location, cultural background, and educational level. This highlights the urgent need for research into the efficacy and safety of herbal medicines and traditional plant remedies in pregnant individuals, and potentially, in humans. Table 4 summarizes the global Perspectives on Herbal Medicine Use During Pregnancy: Practices, Prevalence, and Safety Considerations.

5 Herbal Medicine in Pregnancy: Nurturing Maternal and Fetal Health

The utilization of herbal plants in traditional medicine has a rich historical tradition that spans centuries and extends its influence to prenatal care. Across diverse cultures globally, the incorporation of herbs into pregnancy-related healthcare has been a time-honored practice. These botanical wonders, with their diverse therapeutic attributes, offer expectant mothers natural remedies to alleviate a variety of discomforts and enhance overall well-being, all while mitigating potential risks associated with pharmaceutical medications.

When used judiciously and under the guidance of knowledgeable practitioners, herbal medicine during pregnancy can be a valuable and holistic approach to fostering a healthy gestation period and addressing common maternal and fetal health concerns. Pregnancy is a remarkable and transformative phase marked by physiological, emotional, and hormonal changes. Many expectant mothers grapple with a range of discomforts, including morning sickness, heartburn, insomnia, anxiety, and swollen ankles. Herbal medicine provides a spectrum of botanical solutions that can effectively manage these symptoms safely.

One of the most renowned herbs for alleviating pregnancy-related nausea and vomiting is ginger (*Zingiber officinale*). With a history of use spanning centuries, modern scientific research substantiates ginger's efficacy in curbing morning

Table 4 Global perspectives on herbal medicine use during pregnancy: practices, prevalence, and safety considerations

Topic	Key findings
Regional variation in HM practices	Herbal medicine (HM) practices during pregnancy vary across regions. In India and Ghana, there's a higher tendency to use HMs for pregnancy health compared to industrialized regions. Cultural factors often influence this choice.
Prevalent reasons for HM use	Reasons for using HMs during pregnancy include preventing newborn hyperbilirubinemia, protecting the unborn child from injury, promoting increased prenatal weight, and supporting optimal brain development.
HM use in developing nations	Developing nations, particularly in Africa and Asia, widely use regional medicinal plants for prenatal care. These medicinal herbs are consumed in various forms, but some formulations may have adverse effects.
Country-specific HM practices	Specific examples include South African Zulu women using "Isihlambezo," Korean "Anjeonicheon-tang" for hyperemesis gravidarum, and Taiwanese "An-Tai-Yin" to induce labor (with reported risks to babies).
Trimester-specific HM use	HM use peaks during the first and third trimesters. In the first trimester, it's used to manage early pregnancy symptoms, while the third trimester use aims to aid in childbirth and prepare the cervix for delivery.
Demographic variations	Demographic factors like education and age influence HM use. Higher education levels correlate with increased CAM usage, while in Eastern Europe, younger women are more likely to use HMs.
Socioeconomic and cultural factors	In Africa and Asia, women with lower socioeconomic status, lacking a high school diploma, and residing in remote areas tend to use HMs. Societal perceptions and cultural practices often influence HM use.
Media influence	Media coverage affects HM use in industrialized nations. In Eastern Europe, pregnant women typically follow their doctor's recommendations regarding herbal medicine use.
HM use alongside conventional medicine	Many pregnant women use both herbal medicines and conventional remedies worldwide. The combined use can enhance or diminish the effects of conventional drugs, but there's a stigma associated with HM use.
Need for further research	The prevalence and safety of herbal medicines and traditional plant remedies during pregnancy call for further research and exploration, taking into account regional, cultural, and demographic differences.

sickness. It can be incorporated into one's diet in various forms, such as ginger tea, capsules, or candied ginger.

Chamomile (*Matricaria chamomilla*) is another herb frequently employed during pregnancy. Chamomile tea is a soothing and gentle remedy for anxiety, insomnia, and digestive discomfort, making it a valuable herbal option for expectant mothers.

Its calming effects can help reduce stress and facilitate improved sleep, which is especially valuable during pregnancy when sleep disruptions are common.

However, it is crucial to exercise caution when considering herbal remedies during pregnancy, with safety being of utmost importance. Some herbs can be contraindicated due to their potential to induce adverse effects. Pregnant women should consult with qualified herbalists, naturopathic doctors, or experienced midwives to ensure the selection of appropriate herbs and determine suitable dosages tailored to their specific needs.

In addition to professional consultation, expectant mothers should familiarize themselves with herbs that must be avoided during pregnancy. Herbs like black cohosh (*Actaea racemosa*), pennyroyal (*Mentha pulegium*), and tansy (*Tanacetum vulgare*) have been associated with adverse pregnancy outcomes and should be categorically avoided. Furthermore, herbs known to stimulate uterine contractions, such as blue and black cohosh, should not be used without appropriate supervision during pregnancy.

The regulation and quality control of herbal products are paramount considerations when incorporating herbal medicine during pregnancy. Pregnant women should source their herbs from reputable suppliers to ensure that these botanicals are free from contaminants and have been harvested and processed responsibly. Opting for organically grown herbs whenever possible is essential to minimize pesticide exposure and potential exposure to other harmful chemicals.

It is essential to reiterate that herbal medicine should not replace standard prenatal care. Regular check-ups with healthcare providers, ultrasound scans, and appropriate screenings are non-negotiable components of comprehensive pregnancy care. Herbal medicine should complement conventional medical practices by addressing specific concerns and enhancing overall well-being.

When approached with prudence, herbal medicine during pregnancy offers a holistic and natural methodology for supporting maternal and fetal health throughout this unique journey. When carefully selected and monitored under the guidance of experienced practitioners, these herbal remedies can effectively alleviate common discomforts, nurture maternal well-being, and prepare the body for labor and delivery. Pregnant women, however, need to approach herbal medicine with a discerning eye, seeking professional guidance to ensure the safety and efficacy of their chosen herbs. With a thoughtful and informed approach, herbal medicine can be an invaluable and harmonious addition to a healthy and joyful pregnancy experience.

6 Raspberry Leaves in Pregnancy: A Natural Supportive Remedy

Various herbal plants play a role in pregnancy support, particularly tonic herbs that are deemed safe for regular consumption during pregnancy. These herbs serve a dual purpose: they are non-toxic and not highly therapeutic, making them suitable for expectant mothers, while also nourishing the mother and fortifying the cervix in preparation for a smooth childbirth. Among the herbs falling into this category, we find raspberry leaves (*Rubus idaeus*), partridge berries (*Mitchella repens*), and stinging nettles (*Urtica dioica*).

Raspberry leaves have a long history of use by pregnant women in China, Europe, and North America. Both fresh and dried raspberry leaves can be steeped in hot water to create a flavorful tea with high nutritional value, believed to promote the well-being of the pregnant woman's cervix. Additionally, women at risk of miscarriage have been encouraged for centuries to incorporate raspberry leaf tea into their daily routines throughout pregnancy, as it is believed to aid in carrying the pregnancy to full term. This recommendation underscores the significance of proper nourishment in reducing the likelihood of pregnancy and childbirth complications, such as miscarriage, postpartum bleeding, and premature or delayed labor [3].

Raspberry leaves are a rich source of essential nutrients, including vitamins A, B complex, C, E, as well as minerals like calcium, iron, phosphate, and potassium [4]. Furthermore, they contain magnesium and iron [10]. The herb's bitterness, invigorating properties, and soothing effects can be attributed to its tannins, polypeptides, and antioxidants. Of particular importance is the alkaloid fragarine, discovered in 1941, which is known to inhibit uterine contractions [12]. Some reports suggest that fragarine strengthens the muscles of the uterus and pelvis, potentially easing the process of childbirth. While animal studies have shown that raspberry leaf can both stimulate and relax the uterus, human research in this regard has been limited. However, a recent observational study involving raspberry leaf tea in women of reproductive age indicated a reduced risk of preterm or delayed labor and a decreased need for medical intervention during labor. Notably, the plant has not been linked to adverse pregnancy outcomes, and there is no evidence to suggest it is toxic or teratogenic. While raspberry leaf tea has been promoted for use throughout pregnancy, its significant impact on the endometrium is sometimes recommended only in the third trimester [25]. Additionally, chewing on small fragments of raspberry leaf tea ice cubes may provide a moist tongue, which, in turn, could facilitate uterine contractions.

7 Herbal Remedies for Pregnancy: Efficacy and Safety

Approximately 25% of pregnancies end prematurely, and the causes can vary, including the mother's immune system rejecting the embryo, having an incompetent cervix, or carrying a nonviable fetus. Miscarriage is a distressing issue, and women at risk of losing their pregnancies often face limited options. Early pregnancy symptoms, like spotting and cramps, typically lead to bed rest until the threat either subsides or results in a miscarriage [119].

However, two North American native plants, black haw (*Viburnum prunifolium*) and false unicorn (*Chamaelirium luteum*), offer potential for miscarriage prevention.

Black Haw: Historically, Eclectic Physicians used both black haw and its close relative, cramp bark (*Viburnum opulus*), for similar purposes, such as preventing and treating abortion, preparing for labor, and alleviating false labor and postpartum symptoms [118]. Black haw has been found effective when used in small, gradual amounts as a preventative measure or in larger doses for imminent situations, according to Eclectic practices [120]. As a result of widespread use in both alternative and conventional medicine, we now have a better understanding of the pharmacological effects of black haw. Its active compounds, salicin and scopoletin, act as efficient uterine antispasmodics, soothing and relaxing uterine muscles. In modern Chinese therapy, black haw is employed to safeguard pregnancies that might otherwise end in abortion due to cervical laxity or uterine instability. It remains unclear whether the herb's efficacy varies depending on its form, whether consumed as a pill, tea, or extract. Although clinical studies are limited, black haw and cramp bark are generally considered safe for use throughout pregnancy [116]. However, due to the potential risk of syncope (fainting), it is advisable for pregnant or nursing women to avoid high doses or regular usage of black haw. There is also anecdotal evidence suggesting that cramp root might be a viable alternative.

False Unicorn: False unicorn, commonly used in herbal preparations to enhance fertility, reduce the risk of abortion, and prepare the body for childbirth, is associated with various benefits. This herb is particularly known for its ability to address uterine incontinence and strengthen a cervix that has loosened due to recurrent pregnancies. It is believed that taking false unicorn before pregnancy may enhance a woman's fecundity [101]. In certain situations, women turn to false unicorn to either prevent miscarriage or alleviate the symptoms of an impending loss, such as bleeding and cramps [65].

While there is limited rigorous scientific testing on the safety of false unicorn during pregnancy, it is generally considered safe by most herbalists. Some women have used false unicorn even after the onset of abortion, taking frequent doses to successfully carry their pregnancies to full term. There is some indication that the plant may lead to increased levels of human chorionic gonadotropin in the blood, potentially averting abortion. Additionally, cases have been reported where false unicorn and other herbs effectively halted excessive bleeding associated with abortion.

Despite the absence of clinical studies regarding the pharmacological effects of false unicorn, it is commonly believed to possess qualities that restore ovarian and uterine health while acting as an emmenagogue. Contrary to popular belief, it does not stimulate the uterus during pregnancy, but it may contribute to hormone regulation and strengthen the liver and intestinal system [19, 78, 80]. False unicorn can be consumed in various forms, including as a drink, extract, or pill. The specific active components in these different forms are not well documented due to limited research on the plant's mechanisms. Given its acrid flavor, false unicorn is typically preferred in pill or extract form rather than being brewed into a beverage.

Labor Induction: In North America, women often turn to various herbs for labor induction, as highlighted by the research of Moshi and Mhame [71]. Some commonly used herbs for this purpose include blue cohosh (*Caulophyllum thalictroides*), black cohosh (*Cimicifuga racemosa*), and even beetroot (*Trillium erectum*).

Blue Cohosh, a plant native to North America, holds a long history of use by Native Americans and Eclectic Physicians for its medicinal properties, particularly the roots and tubers. It was employed to hasten labor in cases where it had stalled due to maternal frailty, exhaustion, or a lack of reproductive energy [56, 94, 95]. During the 1800s, it gained popularity in both Western and African-American traditional medicine as a means of inducing childbirth and increasing menstruation flow. In modern herbal medicine, blue cohosh is highly regarded for its uterus-stimulating, antispasmodic, and emmenagogue properties [39, 91, 100]. This plant owes its oxytocic effects to flavonoids such as caulosaponin and caulophyllosaponin, making it one of the most potent natural labor inducers. It is often blended with other herbs, particularly black cohosh, in a 1:1 ratio to expedite labor. However, it's important to note that some of blue cohosh's active components are not water soluble, making it more effective as a preparation rather than a tea.

Recent concerns regarding the safety of blue cohosh have emerged. There have been reports of increased fetal distress, as evidenced by meconium-stained amniotic fluid or fetal arrhythmia, and elevated fetal heart rates associated with its use. Some case reports highlight severe health issues in newborns whose mothers used blue cohosh, particularly when used alongside black cohosh. One study [49] described a case where a woman took an unknown quantity of blue and black cohosh to induce labor, resulting in the newborn exhibiting symptoms of illness, necessitating hospitalization and medical intervention for convulsions, renal injury, and artificial respiration. This has raised concerns about the safety of blue cohosh.

Blue cohosh contains compounds like caulosaponin, which researchers believe may contribute to these health problems by constricting cardiac blood vessels and inducing harmful myocardial responses. Some experts have argued that our understanding of these plants' dangers is based on studies conducted on individual components in animals, often using unrealistically high doses. The toxic impact of caulosaponin in animal studies was observed at levels far exceeding what a human would typically consume and could be fatal.

Furthermore, there have been documented cases of nausea, severe gastrointestinal discomfort, and other adverse reactions associated with blue cohosh use. Additionally, blue cohosh contains at least ten alkaloids, including the teratogenic-methylcytosine and extremely embryotoxic sparteine. There's also concern regarding anagyrine, a substance found in the plant's roots, which has been linked to congenital disabilities, such as twisted calf syndrome. Although there are no direct evidence of such issues in humans, a case report connected a similar genetic malformation in a human to the consumption of anagyrine-contaminated goat's milk. However, teratogenic effects of anagyrine have not been observed in studies involving rat embryos.

Given the growing body of evidence pointing to potential risks, it's advisable to exercise caution when considering the use of blue cohosh. The continued use as a labor induction method in North America suggests a need for further research and education on its safety [29, 92] (▶ Chap. 59, "Research Needs of Medicinal Plants Used in the Management and Treatment of Some Diseases Caused by Microorganisms"). Conversely, black cohosh, another plant with labor-inducing potential, has a lower reputation outside North America. While no poisonous, carcinogenic, or teratogenic traits have been established, some European authorities recommend against its use during pregnancy and breastfeeding [50, 107]. Extremely high doses of black cohosh may lead to unpleasant side effects, including intense pain, vertigo, nausea, and vomiting. It is more common to use blue cohosh, but midwives often recommend a combination of these plants to expedite labor. Black cohosh is believed to complement the effects of blue cohosh by regulating the frequency and intensity of uterine contractions. Its purported properties include being antispasmodic, calming, vasorelaxant, and hypotensive.

Beetroot, a plant historically used by Native Americans in North America, particularly its rhizomes, has been employed for centuries to aid in childbirth and reduce postpartum hemorrhage. The term “beth” in beetroot likely stems from “birth” [119]. This maroon-flowered plant was commonly used by early American colonies, Shakers, and Eclectic Physicians to treat conditions such as vaginal bleeding and heavy menstrual flow [13]. While scientific investigations into beetroot’s pharmacological effects are lacking, ample evidence suggests its efficacy in inducing uterine contractions. It is known for its properties as an antihemorrhagic and emmenagogue for the genitourinary system, and it also possesses expectorant, astringent, and antimicrobial qualities [87, 89, 90].

In modern herbal medicine, beetroot, often consumed as a drink or extract, is a common choice for stimulating labor that has stalled. Laboratory tests have shown that saponin glycosides present in beetroot and its close relative, *Trillium grandiflorum*, have significant antifungal properties. However, the glycosides in beetroot have not been thoroughly examined regarding their chemistry and toxicity. Clinical studies on beetroot are highly unlikely due to its limited recognition, even though they could provide valuable insights into its phytochemistry and pharmacology. While more research is needed, beetroot may be considered a potentially safer option for inducing labor compared to other alternatives [22].

8 The Efficacy and Safety of Herbal Medicine in Pregnant Women

Historically, people have traditionally relied on the wisdom and experience of their elders to ascertain the safety and effectiveness of specific medicinal plants. Over the past two to three decades, extensive preclinical and clinical research has been conducted to shed light on the efficacy and safety of various therapeutic plants and herbal combinations. Some of these studies have involved in vivo and in vitro assessments to evaluate their potential embryotoxicity.

For instance, ginsenosides, the active components in ginseng, had direct teratogenic effects on rat fetuses, as determined in an in vitro investigation [18]. This study further inferred that these herbal medicines, due to their oxytocic activity, might lead to uterine hyperstimulation, potentially resulting in fetal ischemia and premature birth, based on their findings using an isolated rat uterus model.

In another study by Mahmoudian et al. [66], the impact of valerian (*V. officinalis* L.) preparation on the development and functioning of mouse embryonic brain cells was explored. This research revealed that the use of valerian during the second trimester significantly reduced the levels of zinc in the embryonic brain, a critical component crucial for normal brain development [85, 93]. As a result, it is advisable for pregnant women to exercise caution when considering the use of valerian products.

Furthermore, research involving thermopsis lanceolata R.Br., an expectorant, indicated that pregnant rodents exposed to this herbal medicine had offspring with reduced body weights and craniocaudal dimensions by the time they reached 20 days of age [59]. While the pups did not experience an impact on their postnatal survival rate, their sensory, motor, and emotional-motor behaviors were altered. Consequently, the use of thermopsis or any product containing it for therapeutic purposes is discouraged due to these findings.

9 Interactions Between Medications and Herbal Supplements

9.1 Use of Homeopathic Medicines During Pregnancy and the Risk of Epilepsy

Women who experience frequent seizures may become pregnant, but the scientific research on herbal remedies for epilepsy is somewhat limited. Some herbal medicines (HMs) might increase the risk of seizures, cause unwanted side effects, or interact with conventional antiepileptic drugs (AEDs). An exception is a medication derived from the *Cannabis sativa* L. species that has received FDA approval for the treatment of certain types of epileptic seizures. Specifically, cannabidiol extracted from *Cannabis sativa* has shown promise in the management of certain forms of epilepsy. Interestingly, while these cannabis preparations have been available in several other countries for some time, the United States has only recently gained access to them. These products come in various forms, with some containing additional compounds like tetrahydrocannabinol.

However, it's worth noting that cannabis use during pregnancy has been associated with adverse effects on both the mother and the fetus, as pointed out by Gunn et al. [32]. Furthermore, in vitro studies have shown that cannabidiol inhibits several cytochrome enzymes, including CYP3A4, CYP3A5, and CYP2D6, while inducing CYP1A1 [52]. Additionally, these studies have demonstrated that cannabidiol activates the UDP-glucuronosyltransferases UGT1A9 and UGT2B7, as highlighted by Yamaori et al. [121–123]. These complex interactions could affect the processing of AEDs and may lead to treatment failures or adverse effects, altering the levels of these medications in the bloodstream. Notably, antiepileptic drugs such as clobazam, rufinamide, topiramate, zonisamide, and eslicarbazepine have been found to interact with cannabidiol, as reported by Gaston et al. [30].

While some herbal medicines are used during pregnancy, there is less scientific evidence to support their use compared to their application in treating epileptic patients. Additionally, the pharmacokinetics of various antiepileptic drugs are intricate, and they may interact with other drugs or herbal medicines in ways that can influence the course of treatment. Herbal medicines commonly used by individuals with epilepsy include St. John's Wort (*Hypericum perforatum* L.), *Ginkgo biloba* L., *Borago officinalis* L., *Valeriana officinalis* L., and *Cannabis sativa* L., as indicated by Liu et al. [63].

9.2 Pregnancy-Related Hypertension and the Use of Herbal Medicine

Pregnancy can lead to various forms of hypertension, including chronic, prenatal, preeclamptic, and chronic hypertension with concurrent preeclampsia, as noted by Lai et al. [61]. Pregnancy-related hypertension is one of the major causes of maternal and fetal illness and mortality during pregnancy, as highlighted by Mustafa et al. [74]. Preeclampsia, characterized by the onset of hypertension and proteinuria after 20 weeks of pregnancy, is on the rise, particularly among older women and those with underlying medical conditions [74, 115]. Townsend et al. [115] further emphasize that preeclampsia poses a significant risk for preterm birth and maternal mortality. Moreover, it can have detrimental effects on various bodily systems, including the renal, liver, neurological, cardiovascular, and nervous systems (e.g., eclampsia, cerebral blood clots, and bleeding, acute fatty liver of pregnancy, hepatic periportal inflammation, glomerular endotheliosis, and acute kidney injury).

Managing pregnancy-related hypertension is essential to reduce the risk of complications, including renal damage. One of the antihypertensive medications used for cases of mild to severe hypertension is labetalol [21]. It is important to avoid angiotensin-converting enzyme inhibitors during pregnancy, and alternative medications like methyldopa and nifedipine are recommended [21].

In 2017, the American College of Obstetricians and Gynecologists (ACOG) released updated recommendations for the management of sudden-onset significant hypertension during pregnancy based on the best available data [21]. Sublingual nifedipine with immediate release is the first-line treatment for pregnant women with abrupt-onset hypertension. In cases where immediate treatment is needed, hydralazine or labetalol administered intravenously are also options [21]. Most pregnant women are prescribed either nifedipine or labetalol.

The metabolic pathways of these medications differ. Nifedipine is metabolized in the body into pyridine and nifedipine carboxylic acid, primarily through the enzyme CYP3A4 [124]. Labetalol, on the other hand, undergoes significant metabolic changes in the human liver and intestines. According to Jeong et al. [51], glucuronidation, mainly mediated by enzymes UGT1A1 and UGT2B7, plays a central role in the elimination of labetalol. This process results in the formation of two inactive drug substances known as alcoholic glucuronide and o-phenyl glucuronide.

It's important to be aware that the use of herbal medicines by pregnant women, whether under medical supervision or not, can potentially alter the way these drugs are metabolized.

9.3 Using HM During Pregnancy and Asthma

Asthma is a respiratory condition that impacts the bronchial airways of the lungs, and it is estimated to affect approximately 4–8% of pregnant women [73, 76, 86]. Inadequate management of asthma during pregnancy has been associated with adverse outcomes for both the mother and the infant. These maternal outcomes may include conditions such as hypertension, gestational diabetes, preterm delivery, and even maternal mortality. Similarly, infants of mothers with poorly controlled asthma may experience fetal malformations, illness, and, in severe cases, even mortality. Multiple studies have provided evidence supporting these associations [11, 69].

Research conducted by Al Ghobain et al. [9] and McLaughlin et al. [69] underscores the importance of effectively managing asthma during the perinatal period, showing that this leads to better outcomes for both mothers and their children. Consequently, the US Department of Health and Human Services (HHS) recommends giving high priority to pregnancy-specific asthma treatment. The primary focus of such treatment is to keep the airways open, reduce the need for short-acting inhaled beta2-agonists, and minimize the risk of severe exacerbations, which helps protect both the mother and her infant from experiencing low oxygen levels.

Standard daily asthma management during pregnancy often involves a combination of medications. This includes the use of a long-acting beta-agonist like salmeterol or formoterol, a leukotriene receptor antagonist such as montelukast or zafirlukast, and a short-acting beta2-agonist like albuterol. In some cases, systemic corticosteroids like prednisone, prednisolone, or methylprednisolone may be prescribed. Additionally, healthcare providers may recommend the use of methylxanthines like theophylline or 5-lipoxygenase inhibitors like omalizumab or cromolyn as part of the treatment regimen [75, 76].

10 Conclusion

Pregnant women and their healthcare providers must make informed decisions regarding the use of pharmaceuticals or natural remedies. However, the safety and efficacy of many plant-based treatments during pregnancy remain poorly documented, often relying on anecdotal evidence. In some cases, herbal remedies might offer alternatives to invasive medical procedures, potentially reducing complications associated with malnutrition. Natural remedies like black haw and false unicorn could help save fetuses that might otherwise be lost. Compared to hospital inductions, herbal labor induction methods may provide pregnant women with a greater sense of control during childbirth.

The herbal remedies discussed here deserve increased attention for their potential to assist expectant mothers and their caregivers in achieving safe, healthy pregnancies and satisfying birthing experiences. Nevertheless, natural treatments should be approached with a degree of skepticism. Recommendations for any medication, whether natural or synthetic, should be founded on comprehensive laboratory research and clinical trials.

The combination of herbal medicines (HMs) or other natural products with conventional medications during pregnancy raises growing concerns, particularly due to our limited knowledge of potential interactions and associated risks. Many pregnant women use HMs without consulting their healthcare providers, despite the documented positive outcomes HMs can have on maternal and infant health before, during, and after pregnancy. The common assumption that HMs are inherently safe can lead to unintended consequences.

Both pregnant women and medical professionals would benefit from enhanced education and a shift away from the misconception that HMs are universally harmless. There is a real risk of interactions between HMs and traditional over-the-counter medications. For example, pregnant women taking aspirin and HMs with antiplatelet properties face an elevated risk of hemorrhage, which can, in severe cases, lead to fatal consequences. The lack of research in this area is a significant concern, as little is known about the effects of HMs on fetal development and the potential adverse reactions when combined with conventional medications.

Moreover, the potential for medication interactions can significantly compromise the management of certain medical conditions during pregnancy, such as hypertension, asthma, and seizures. The reluctance to conduct research during pregnancy makes it challenging to determine the optimal medication dosage for a woman in her third trimester, especially when she is concurrently using HMs that may interfere with the effectiveness of conventional treatments. Equally concerning is the lack of study into the embryotoxic effects of HMs. Clinical pharmacologists, scientists, and healthcare providers should strive to better understand the results of experimental and clinical research concerning the harmful effects of HMs on fetal development.

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Herbal Medicine and Rheumatic Disorders Management and Prevention 24

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Abstract

Rheumatoid disorder is a chronic inflammatory disorder of the joints that causes discomfort, inflexibility, swelling, deformity, and sometimes disability. Its cause remains a mystery. The severity of irreversible joint degradation may be mitigated with early diagnosis and therapy with disease-modifying anti-rheumatic drugs, as recommended by the American Academy of Rheumatology. Although effective treatments were quickly identified, harmful side effects often undermine their effectiveness. Herbal treatments are gaining in popularity and effectiveness for several forms of arthritis. According to a recent US poll, 90% of people with rheumatoid arthritis use complementary therapy, like herbal medications. The evidence for herbal treatments for four prevalent arthritic conditions back pain, fibromyalgia, osteoarthritis, and rheumatoid arthritis is briefly reviewed in this chapter. Some plants commonly used for the prevention or therapy of rheumatoid disorder are *Zingiber officinale*, *Piper methysticum*, *Ephedra* sp., *Oenothera biennis*, *Panax ginseng*, *Ginkgo biloba*, *Allium sativum*, *Ganoderma lucidum*, *Tripterygium wilfordii*, *Uncaria tomentosa*, *Boswellia serrata*, *Tinospora cordifolia*, *Tinospora cordifolia*, *Tribulus terrestris, Curcuma longa*, *Caesalpinia sappan*, *Clematis mandshurica*, *Rosmarinus officinalis, Elaeagnus angustifolia, Crocus sativus*, *Camellia sinensis*, *Matricaria chamomilla*, *Punica granatum*, *Cinnamomum cassia*, *Centella asiatica*, *Phyllanthus amarus*, *Psidium guajava*, *Moringa oleifera*, *Nigella sativa*, *Cannabis sativa*, *Andrographis paniculata*, *Artemisia annua*, and *Piper betle*. The possibility of treating rheumatic diseases using these plants as well as the chemical components that have this impact will be discussed in silico, in vitro, in vivo, and in clinical terms.

Keywords

Clinical study · Herbal medicine · In silico · In vitro · In vivo · Rheumatoid

Abbreviations

ADAMTS	Disintegrin and metalloproteinase with thrombospondin motifs
BSA	Bovine serum albumin
BW	Body weight
CAM	Complementary treatments
CFA	Complete Freund adjuvant
CIA	Collagen-induced arthritis
COX	Cyclooxygenase
CRP	C-reactive protein
DMARD	Disease-modifying anti-rheumatic drugs
DMS	Duration of morning stiffness
EAE	Experimental autoimmune encephalomyelitis
IL	Interleukin
IL-1B	Interleukin-1β
INOS	Inducible nitric oxide synthase
KG	Kilogram
LPS	Lipopolysaccharide
MAPK	Mitogen-activated protein kinase
MG	Milligram
MMP	Metalloproteinases
MSM	Methylsulfonylmethane
MTX	Methotrexate
NO	Nitric oxide
NSAID	Non-steroidal anti-Inflammatory drugs
PDB	Protein Data Bank
PGE2	Prostaglandin-E2
RA	Rheumatoid arthritis
RF	Rheumatoid factor
ROS	Reactive oxygen species
SJC	Number of swollen joints
SLE	Systemic lupus erythematosus
TJC	Number of painful joints
TLR	Toll-like receptor
TNF-α	Tumour necrosis factor-alpha
VAV	Visual analog scale
WHO	World Health Organization
WT	Wild type

1 Introduction

Rheumatic illnesses include fibromyalgia, osteoarthritis, and rheumatoid arthritis (Fig. 1). Rheumatic disorders are associated with decreased productivity, substantial disability, and diminished quality of life [1]. Conventional treatment for rheumatic

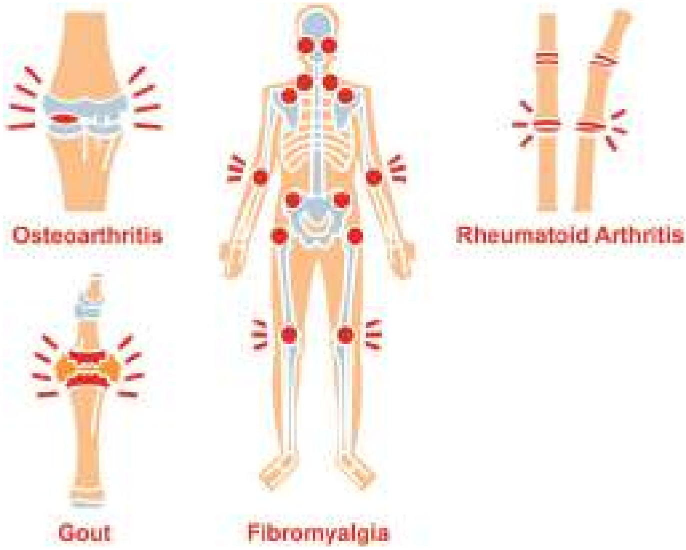

Fig. 1 Rheumatic disorders

disorders usually uses disease-modifying anti-rheumatic drugs (DMARDs), non-steroidal anti-inflammatory drugs (NSAIDs), and biologics. Some individuals may not react well to these conventional therapies. Thus, people with rheumatic illnesses usually seek alternative and complementary treatments (CAM) [2–4].

According to the World Health Organization, complementary and alternative medicine includes a wide variety of healthcare practices that are not integrated into the mainstream healthcare system and are not part of the country's tradition [5]. Under the guise of complementary and alternative medicine, a range of methods may be utilized to prevent or treat illnesses. Five groups can be used to categorize complementary and alternative medical practices, such as acupuncture, homoeopathy, herbal remedies, dietary ingredients, or additions derived from nature [6]. Around 60–90% of rheumatoid arthritis patients have employed complementary and alternative therapy, according to prior studies [7].

Prevention techniques for rheumatoid arthritis are classified as primary, secondary, or tertiary interventions. The objective of primary prevention is to prevent the onset of illness by removing particular risk factors and boosting an individual's resistance to the condition. The purpose of secondary prevention is to slow the development of a disease from its latent or asymptomatic phase to its symptomatic phase. Hence, a secondary preventive intervention aims to thwart the processes of disease development before they manifest as a disease. Cancer screening programmes such as mammography and colonoscopies are examples of this strategy. Tertiary prevention has the purpose to postpone or mitigate the effects of a disease that has already taken hold. Here is where the majority of rheumatic illnesses are being treated, with rheumatologists seeking to avoid the progression of the disease to a disability or early death when a patient presents with clinically evident disease (swollen joints in RA, or skin rash in SLE) [5].

The therapeutic objectives for rheumatoid arthritis are to lessen symptoms, stop the disease's progression, and improve quality of life. The conditions considered include analgesic relief, inflammation reduction, articular structure protection, function maintenance, and systemic involvement control before starting rheumatoid arthritis treatment [8].

This chapter examines various traditional herbs from China, India, and other Asian nations that are used as arthritis treatments. It also will go into greater detail about in silico, in vitro, in vivo, and clinical trial research on arthritis-curing plants. The data are gotten by searching the keyword (clinical study, herbal medicine, in silico, in vitro, in vivo, rheumatoid) at Google Scholar, PubMed, and other Internet sources.

2 Herbal Medicine

Herbal medicine is generally defined as a raw material or a preparation derived from plants with a therapeutic or medicinal effect that benefits human health. Herbal medicine is composed of raw materials or materials that have gone through further processing from one or more plant species. Below are some plants that have benefits as herbal medicine.

2.1 *Zingiber officinale*

Zingiber officinale (ginger) belongs to the Zingiberaceae family and has numerous advantages because of its compounds. In in silico studies, their bioactive compounds 8-gingerol, 6-gingerol, and zingerone had the potential to be inhibitors of several rheumatoid arthritis proteins [10]. These compounds carried out molecular binding on several rheumatoid arthritis protein targets, namely, cyclooxygenase-2 (COX-2), interleukin-1β (IL-1β), macrophage colony-stimulating factor (MCSF), matrix metalloproteinases-9 (MMP-9), and tumour necrosis factor-alpha (TNF-α) proteins (Fig. 2) [2]. An in vitro experiment utilizing the aqueous extract of ginger at 125 g/ml revealed that its anti-inflammatory activity was higher than diclofenac (58% vs. 52%) at the same concentration. In another study, prostaglandin-E2 (PGE2), IL-6, and IL-8 were inhibited by the 1 g/ml of ginger ethanol extract on the mRNA expression of inducible nitric oxide synthase (iNOS) and COX-2 in Caco-2 adenocarcinoma cells [11]. Rats were induced by I/R and fed once per day for 3 days in a row before the application of 6-gingerol (1, Fig. 5) from *Zingiber officinale* Roscoe extract. When administered orally at 25 and 50 mg/kg BW, the interleukin-1 and 6 and TNF-α were found to be at lower levels, as was the neutrophil count [12]. Expression of IL-12 and TGF-α in the central nervous system and serum was shown to decrease when ginger essential oil was given to experimental autoimmune encephalomyelitis (EAE) mice at 200 and 300 mg/kg [13]. In different clinical trial, it was discovered that ginger, either by itself or in combination with *Echinacea* extract (15 patients over 60 years old, 25 mg of ginger, and 5 mg of *Echinacea* extract for 30 days), considerably reduced knee discomfort [9].

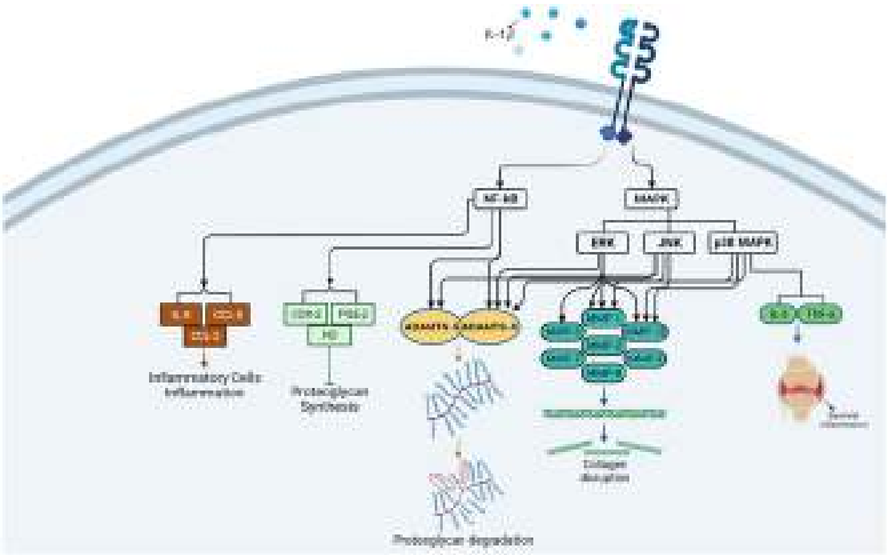

Fig. 2 Mechanism of rheumatic arthritis that is influenced by several protein targets [9]

2.2 *Piper methysticum*

Piper methysticum (kava) is a plant originating from Papua, Indonesia. Their bioactive compounds include alkaloids, flavonoids, tannins, saponins, anthraquinones, and essential oils. Based on the outcomes of molecular docking of the side chain with the morpholine ring occupying the polar domain and ring, one of the compounds from this plant showed action in suppressing iNOS protein is known as protein data bank (PDB) ID: 3E67 [10]. In in vitro investigation of nitric oxide (NO) inhibitory generation, this extract had an inhibition rate of 84.0% at a dose of 10 mM (IC_{50} 14 6.4 M) with the least amount of cytotoxicity (Fig. 3). In addition, the kavain (2, Fig. 5) in this extract significantly reduced TNF-α production induced by *E. coli* lipopolysaccharide (LPS) in primary macrophages of WT mice. In in vivo evaluation, this compound also has a significant anti-inflammatory effect on mice-induced CAIA (collagen antibody-induced arthritis) [14].

2.3 *Ephedra* Sp.

A herb known as *Ephedra* is frequently used in traditional medicine to treat fever, headaches, cough, and shortness of breath. *Ephedra alte* contains β-sitosterol (3, Fig. 5) and androstan-3-one efficiently inhibited inflammation and reduced oxidative stress by inhibiting five target proteins such as IL-6, toll-like receptor-4 (TLR4) hybrid, TNF-α, TLRs, and IL-1β. Fresh chicken egg albumin denaturation

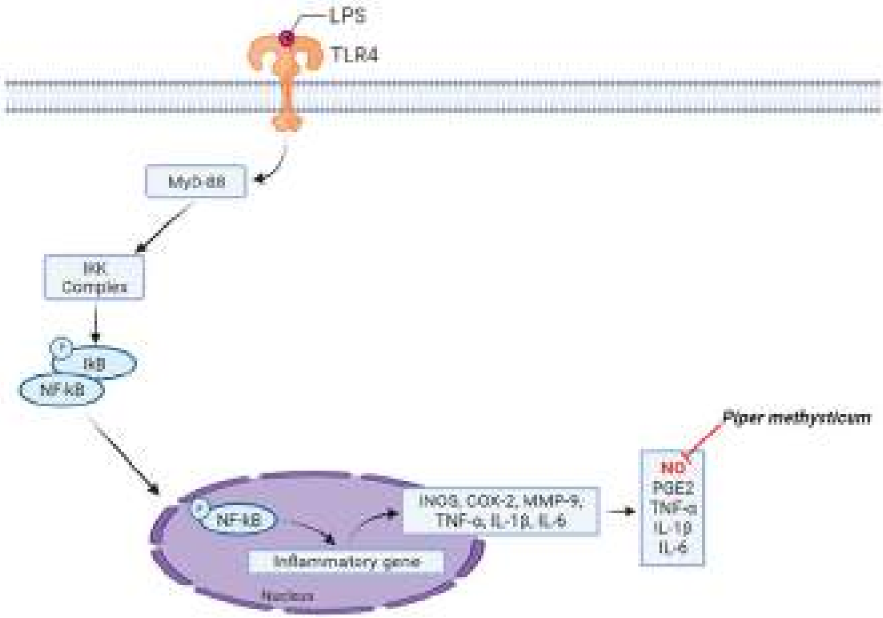

Fig. 3 Mechanism of *Piper methysticum* in inhibiting NO in RAW 264.7 cells

and membrane stabilization were performed on ethanol, ethyl acetate, n-butanol, and aqueous extracts of *Ephedra gerardiana* at 6400 μg/mL in vitro experiments involving bovine serum albumin (BSA) inhibition tests. Their aqueous extract exhibited the greatest inhibitory impact (98.5%). Then n-butanol and ethyl acetate inhibited 92.1 and 85.7%, respectively. Meanwhile, the ethanol extract with the same concentration showed an inhibitory effect of 99.1%. All extracts showed better inhibitory effect results compared to aspirin (80.5%) [15]. In other species, extracts of *Ephedra alata* and *Ephedra fragilis* inhibited NO in RAW cells 264.7 by 62% and 70% at 50 mg/mL, respectively [16]. ESP-B4 (an acidic hetero-polysaccharide) derived from *Ephedra sinica* also demonstrated reduced LPS-induced cytokine production. Moreover, pre-treatment with ESP-B4 greatly reduced the phosphorylation of mitogen-activated protein kinases (MAPKs) brought on by LPS [17]. *Ephedra gerardiana* at 200 mg/kg reduced leg volume and diameter in formaldehyde-induced rats [18]. To cure rheumatoid arthritis, it has been demonstrated that the ESP-B4 from *Ephedra sinica* inhibited the TLR4 signaling pathway, which reduces the production of inflammatory agents and cytokines [17]. This plant activity in rats given Freund's complete adjuvant was examined (FCA). Rat foot oedema can be reduced and haematological and biochemical alterations were greatly improved by this extract. Moreover, in all extracts-treated mice, the overproduction of pro-inflammatory cytokine was markedly reduced. Nonetheless, there was a significant rise in IL-4 and 10 [18].

2.4 *Oenothera biennis*

NO, TNF-α, IL-1, and thromboxane B2 (TXB2) production can be reduced by the sterol components found in *Oenothera biennis* oil or evening primrose oil (EPO). Sterols do not, however, lower PGE2 [19]. TNF-α, IL-6, and IL-1 were significantly inhibited by this extract in peritoneal macrophages (PMs), and extracellular signal-regulated ERK signaling, P38 of MAPK, and LPS-activated NF-kappaB (NF-kB) were also inhibited by *Oenothera biennis* L. [19]. A study of the arthritic group's response to EPO revealed a substantial decrease in body weight and an increase in the ankle, plasma angiopoietin-1, and TNF-α [20]. In arthritic rats, EPO increased platelet count, white blood cell count, and high ankle diameter. EPO therapy significantly reduced the levels of inflammatory biomarkers in arthritic animals, including rheumatoid factor, TNF-α, IL-1β, and IL-6. As measured by disease activity score-28 (DAS28), IL-17, tacrolimus (TAC), rheumatoid factor (RF), and C-reactive protein (CRP) values after 3 months in clinical studies on 90 rheumatoid arthritis patients who received EPO 500 mg/TDS, the results might lessen DAS28 score and severity of rheumatoid arthritis patients [21].

2.5 *Panax ginseng*

With binding affinities of −7.58 and −7.89 kcal/mol for NF-kB protein, the ginsenoside (4, Fig. 5) derived from *Panax ginseng* has an anti-inflammation medicinal agent. Moreover, it reduced the reactive oxygen species (ROS) in anti-inflammatory experiments utilizing an in vitro model of NF-kB in RAW 264.7 macrophages activated by LPS [22]. It boosted the expression of early and late-stage differentiation markers (alkaline phosphatase, collagen type-I, and osteocalcin) during osteoblast differentiation on MC3T3-E1 cells that were exposed to H_2O_2. It also decreased the overexpression of pro-inflammatory cytokines such as IL-1, NO expression, and ROS generation [23]. Ginsenoside also suppresses osteoclastogenesis in human CD14+ cells [24]. In IL-1-stimulated chondrocytes, ginsenoside Rg3 increased the expression of aggrecan and collagen type 2 (COL-II), two chondroprotective proteins, while decreasing MMP-1 and MMP-13 [25]. A mouse model of osteoarthritis was used to demonstrate that ginsenoside Rg5 dramatically ameliorated osteoarthritis symptoms by decreasing cartilage breakdown and cell death. Whereas other research demonstrated that 15 mg/kg of ginsenoside-Rg5 orally for 1 month lowered levels of IL-1, TNF-α, NO, and iNOS by 67, 54, 32, and 49%, respectively [26]. Ginsenoside Rg1 administration to osteoarthritis mice attenuated cartilage degradation and lowered the loss of type II collagen and MMP-13 levels [27]. In the joints of collagen-induced arthritis (CIA) mice treated with *Panax ginseng*, pro-inflammatory cytokines including IL-17, IL-6, TNF-α, and IL-1 were significantly reduced [28]. Fifty-two postmenopausal women with a diagnosis of degenerative arthritis of the hand and symptoms of oedema and pain in the hands had therapy with 3 g red ginseng daily for 12 weeks, then both blood oestradiol and endometrial thickness were unaffected by ginseng [28, 29]. It had considerably elevated serum OC concentrations and a dramatically decreased DPD/OC ratio, which represents bone resorption and bone formation. Twelve weeks of treatment greatly alleviated knee arthritis symptoms [30].

2.6 *Ginkgo biloba*

Ginkgo biloba is a plant from the Ginkgoaceae family and is classified as a phytopharmaceutical or medicine. It had a potential effect as an anti-inflammatory based on its ability to bind to each target protein (such as COX-2, iNOS, TNF-α, TLR-4, and IL-1β) by forming H-bonds and hydrophobic interactions [31]. In addition, *Ginkgo biloba* extract inhibited IL-1-induced chemokine synthesis, as evidenced by a decrease in THP-1 cell motility to the cell culture medium and a decrease in IL-1-stimulated iNOS expression and NO release [32]. The presence of bilobalide in *Ginkgo biloba* extract had a crucial role in suppressing the synthesis of iNOS, COX-2, and MMP-13 in ATDC5 chondrocytes stimulated by IL-1, decreasing the activity of ECM-degrading enzymes, and increasing the expression of cartilage anabolic protein [33–35]. The ginkgolide C inhibited the release of pro-apoptotic factors and stimulates the release of anti-apoptotic proteins that are produced by H_2O_2. This substance inhibited the production of MMP-3 and MMP-13, ADAMTS, and iNOS. ECM breakdowns via inhibiting Cox-2 and SOX-9 [36].

In mice models of osteoarthritis produced by ACLT, ginkgolide C had been shown to suppress joint discomfort, cartilage degradation, and aberrant subchondral bone remodeling [37]. Moreover, bilobalide inhibited the development of aberrant osteophytes in the subchondral bone and stopped the degeneration of cartilage after trauma [38]. The ginkgolide B reduced blood levels of IL-1, IL-6, MCP-1, TNF-a, MMP-3, and MMP-13 and increases IL-10 [34]. In clinical studies, *Ginkgo biloba* extract treatment for 6 weeks in 51 OA patients had much greater benefits than glucosamine/chondroitin [39, 40].

2.7 *Allium sativum*

Allium sativum is commonly used as a medicinal plant. Based on its bioactive constituents (S-allyl cysteine, polyphenols, and flavonoids), this plant has anti-inflammatory properties. In in silico testing, the presence of the pyridoxamine from *Allium sativum* inhibited all receptors except NT5C2-1497 against the four target proteins (XO, ADA, GDA, and NT5C2-1498). It indicated that the substance has the potential to be employed as an anti-gout agent [37]. Meanwhile, in a study using *Allium sativum* methanol extract, it was shown that the extract had the highest inhibition value at 100 μg/ml of 88.5% in denaturation protein egg albumin and 61.2% in bovine serum albumin [41]. This plant also showed regulation of the IL-17 gene which significantly did not influence cell proliferation and lymphocyte proliferation (T CD4+ and CD8+) [42]. In a mouse model of arthritis, treatment with

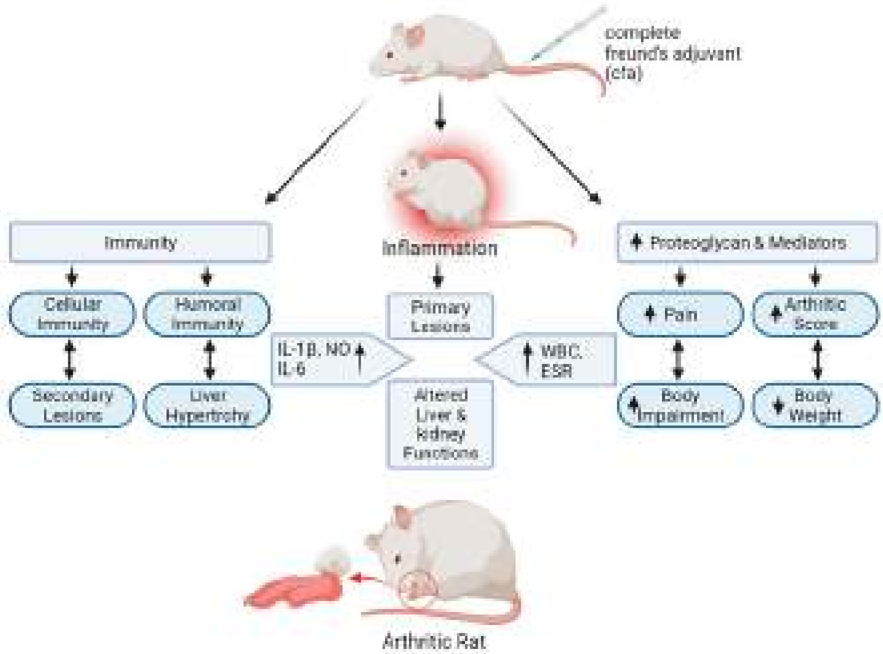

Fig. 4 Mechanism and pathway of CFA-induced arthritis [44]

100 M diallyl trisulfide suppressed the production of pro-inflammatory cytokines and regulated immunological activity by restoring the balance between Th17 and Treg cells [39]. In addition, administration of diallyl disulfide (20 and 50 mg/kg) in a CFA rat model of rheumatoid arthritis (Fig. 4) induced a significant decrease in leg volume and oedema formation; this compound also reduced inflammation, decreased white blood cell count, and increased RBC count in CFA-induced mice [40]. Seventy women with rheumatoid arthritis were separated into two groups for a clinical experiment, with one group receiving 1000 mg of garlic and the other receiving a placebo for 8 weeks. The results showed that serum C-reactive protein decreased significantly, pain intensity also decreased decreases, and the number of swollen joints also decreased [43].

2.8 *Ganoderma lucidum*

Ganoderma lucidum is a medical mushroom that contains triterpenoids, β-glucan, sterols, coumarins, mannitol, organic germanium, vitamins, flavonoids, and minerals. In silico sterol from this plant was directly bound actively to p38 and p65 to suppress their activation [45]. Furthermore, sterol has a crucial role in reducing inflammation in macrophages, as it may greatly reduce LPS-induced cell polarization and the production and expression of mRNA from pro-inflammatory mediators such as NO, TNF-α, IL-1, and 6 [46, 47]. *Ganoderma lucidum* extract's oil can prevent CIA-induced cartilage degradation in knee synovial membrane inflammation. The GLS oil group demonstrated a significant drop in knee eosinophils, and this oil lowered IL-6 mRNA expression. *Ganoderma lucidum* at 250 mg/kg BW was effective as an anti-inflammatory drug [48]. Clinically, this plant had an analgesic effect for patients with active rheumatoid arthritis, which is generally safe and well tolerated due to demonstrated activities, including significant antioxidant, anti-inflammatory, or immunomodulating properties [49].

2.9 *Tripterygium wilfordii*

Tripterygium wilfordii (thunder god vine) is commonly used in TCM to treat inflammatory conditions such as rheumatoid arthritis and autoimmune. Based on the structure of the CD2 protein, the triptolide derivative (5R)-5-hydroxy triptolide (LLDT-8) from this plant received the greatest docking score. This compound effectively blocked immune and inflammatory signaling pathways [46]. Another compound of tripterygium glycoside showed an anti-arthritis effect in rheumatoid arthritis therapy by initiating apoptosis in affected synovial lymphocytes or fibroblasts to prevent their proliferation, restrain Th17 cell differentiation, and inhibit angiogenesis [47]. Their alkaloids also had an anti-arthritis effect and significantly reduced leg swelling and suppress articular cartilage degeneration in CIA-induced mice. In addition, it was found that levels of IL-6, IL-8, NF-kB, and TNF-α in CIA serum mice decreased significantly [50]. Clinically, tripterygium glycosides significantly influenced the duration of morning stiffness (DMS), the number of painful joints (TJC), the number of swollen joints (SJC), the visual analog scale (VAS), the protein C-reactive (CRP), the erythrocyte sedimentation rate (ESR), and the rheumatoid factor (RF) in rheumatoid arthritis patients [51]. *Tripterygium wilfordii* extract at a single dose had a significant efficiency compared to MTX, while their combination gave a good significance than a single dose of MTX [52].

2.10 *Uncaria tomentosa*

The cat's claw plant (*Uncaria tomentosa*) is found in the peat forests of West Kalimantan, Indonesia. Some of the empirical benefits of this plant are used for ulcers, rheumatism, cirrhosis, gonorrhea, and female genital tract cancer. It contains alkaloids, flavonoids, polyphenols, triterpenoids, steroids, saponins, and tannins. This plant reduced osteoclast differentiation induced by RANKL without affecting cell viability [53]. Also, injection of this extract reduced ecto-nucleoside triphosphate diphosphohydrolase activity in mice models of CFA-induced arthritis but does not affect ecto-nucleoside triphosphate diphosphohydrolase and elevated erythrocyte adenosine deaminase activities in healthy animals [54]. In a separate study, administration of *Uncaria tomentosa* extracts decreased the RANKL/OPG ratio, an essential gene involved in osteoclast differentiation, reducing TRAP-positive cells lining alveolar bone and raising bone-specific alkaline phosphatase (BAP) and anabolic bone markers in plasma rat arthritic models [53]. Clinically, this plant was given to patients with advanced cancer for 8 weeks to improve their quality of life and reduce fatigue and inflammation that occurs [55]. During hemorrhagic cystitis (HC) in rats, quinovic acid glycoside was demonstrated to reduce oedema, bleeding, IL-1β levels, and neutrophil migration, exhibiting a protective impact on CH-induced urothelial damage, control of visceral discomfort, and reduction of IL-1 levels [56].

2.11 *Boswellia serrata*

Boswellia serrata (salai/salai guggul) is one of the components of the ancient and most famous herb in traditional Indian medicine, Ayurveda. This plant is beneficial for the treatment of arthritis, as well as diarrhoea, dysentery, ringworm, boils, and fever, according to empirical evidence (antipyretic). This plant is pharmacologically effective as an anti-atherosclerotic, analgesic, anti-inflammatory, and anti-arthritic. Boswellic acid (5, Fig. 5) and curcumin from this plant had the potential effect for anti-rheumatism as indicated by the high binding energy, which is between −8.66 and −9.67 [57]. The dry extract of this plant (0.1 μg/ml) showed pro-angiogenic activity, whereas the hydro enzymatic extract did not show any effect on the migratory capacity of endothelial cells [58]. Treatment of this extract for 21 days to CIA-induced Wistar rats altered numerous parameters, including MPO, LPO, GSH, and NO at doses 100 and 200 mg/kg BW. Furthermore, it resulted in a considerable reduction in numerous inflammatory mediators and a rise in IL-10 levels [59]. The injection of this extract at 180 mg/kg BW resulted in a rise in the rat's body weight and a decrease in knee diameter, leg thickness, ankle diameter, foot volume, and arthritis index, but the impact was less than that of the conventional cyclophosphamide medication. Clinically, a 12-day study of curcumin complex or its combination with boswellic acid improved pain-related symptoms in individuals with degenerative hypertrophic knee osteoarthritis. Curamin®, which contains *Curcuma longa* and *Boswellia serrata* extracts, improved the treatment of osteoarthritis due to the synergistic impact of curcumin and boswellic acid [60]. Similar to ibuprofen, the combination of *E. angustifolia* and *Boswellia thurifera* relieved pain and enhanced function in knee osteoarthritis patients [61]. Comparing methylsulfonylmethane (MSM) 5 g and boswellic acid (BA) 7.2 mg in 60 patients with knee arthritis revealed a statistically significant difference between the experimental group and the placebo group in the number of patients using anti-inflammatory medicines [62].

2.12 *Tinospora cordifolia*

Tinospora cordifolia (gurjo) is a traditional medicinal herb having anti-inflammatory properties that use to treat rheumatism, gout, bruising, fever, and stimulating hunger.

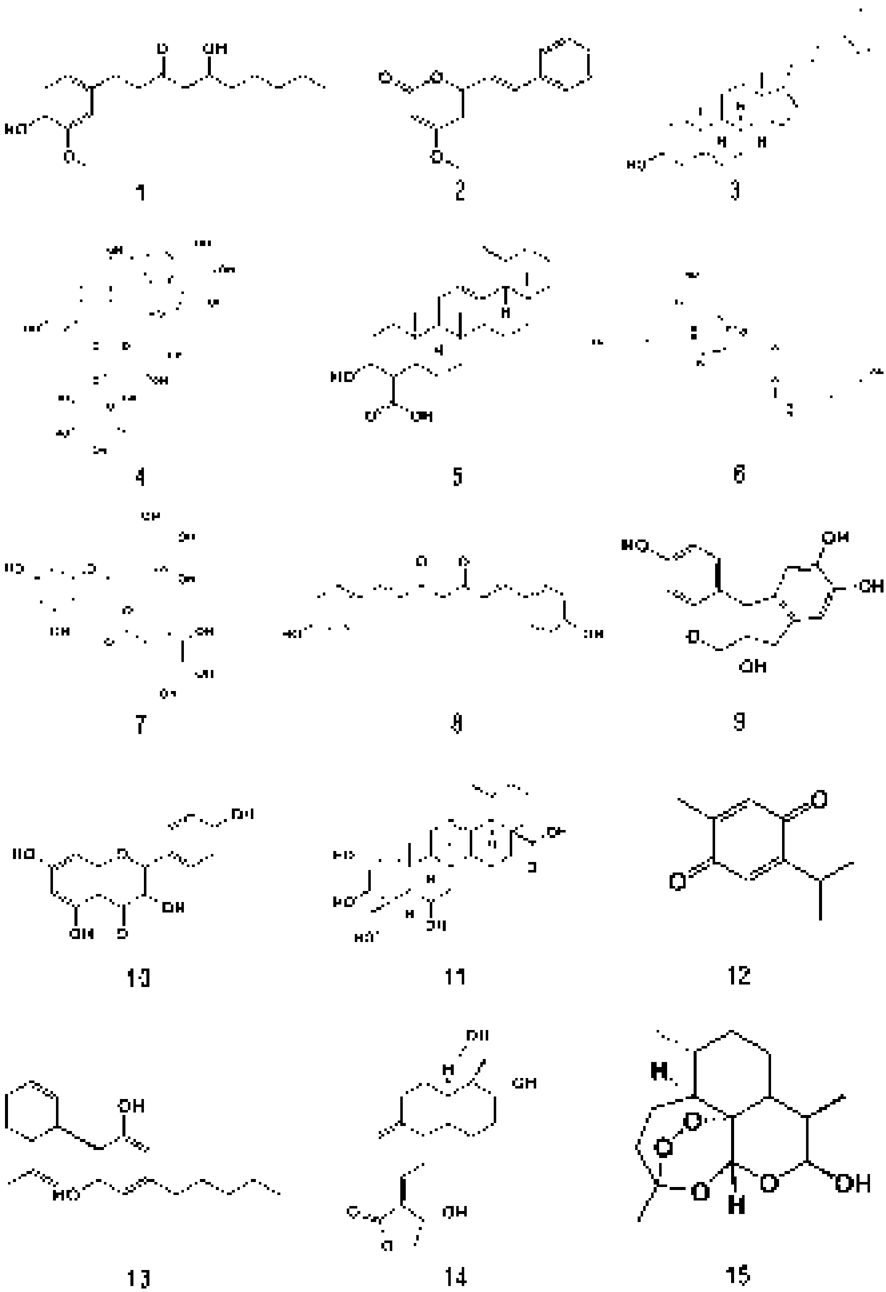

Fig. 5 Structure of chemical compounds that have the potential for anti-rheumatoid disorders therapy

It contains alkaloids, saponins, glycosides, tannins, polyphenols, flavonoids, picroretin, berberine, tinocrisposide, and columbin. This plant contains a tyramine-Fe molecule that forms a complicated binding with COX-2 (COX 2-tyramine Fe) [63]. Their extract increased the expression of genes that regulate osteoblastogenesis and mineralization, such as type I collagen, alkaline phosphatase, bone sialoprotein, osteopontin, osteonectin, osteomodulin, and osteocalcin, and has a positive effect on osteogenesis and bone formation [64]. Oral treatment of 50 mg/kg of *Tinospora cordifolia* extract to ovariectomized rats for 21 days prevented bone loss and suppressed osteoclast development [64]. This plant suppressed TNF-α and IL-1 and reduces NO generation in RAW cells 264.7 [65]. The chloroform extracts of this plant downregulated pro-inflammatory biomarkers in THP-1 macrophages by inhibiting NF-B nuclear translocation in THP cells [66]. Their extract considerably lowered levels of pro-inflammatory like IL-1, 6, and 17 in rat models of RA-induced arthritis. Hence, it may be inferred that the extract inhibited bone destruction by altering the ratio of mediators in bone remodeling (RANKL and MMP-9) that support anti-osteoclastic activity [67].

2.13 *Withania somnifera*

Withania somnifera (ashwagandha) is one of the endemic plants from India and is believed to improve physical and mental health as well as prevent various diseases, such as antimicrobial, immunostimulant, anti-inflammatory, anti-arthritis, antioxidant, and anti-diabetic. Their activity conducted cannot be separated from the role of the active compounds contained in this plant. Withanolide B present in the *Withania somnifera* had an excellent inhibition value of −7.8 kcal/mol for rheumatoid arthritis by targeting TNF-α through in silico analysis [68]. The ethanol extract of this plant inhibited 5-LOX (65%) in vitro with an IC_{50} of 0.92 mg/mL [69]. While their chloroform and water extracts were able to control the expression of TNF, IL-1, and IL-6 by reducing the production of iNOS and by downregulating NF and activator proteins. These extracts did not affect the expression of TNF-α. Moreover, the extract inhibited the migration of activated microglia by reducing MMP expression [70]. The alcoholic extract of this plant significantly decreased the diameter of the right leg's tarsal joint. In addition, the histology findings revealed that the joint membranes of mice administered this extract did not exhibit an inflammatory reaction [71]. Ashwagandha at a dose of 200 mg/kg has shown to minimize pannus production in rheumatoid arthritis joints [72]. Their extracts showed dose-dependent relief of disease severity based on reduced lipid peroxidation and ceruloplasmin, which lowered some serum levels of TNF-α, IL-1β, COX-1, and COX-2 [73]. Ashwagandha at 300 mg/kgBB increased the secretion of IL-10, lowered pro-inflammatory cytokines, and inhibited NF-kB activity [74]. The anti-inflammatory effect in adult zebrafish of comparable size and weight was determined by extracting fresh leaves of this plant with methanol and water. Due to phenolic acids and flavonoids, this extract can block TNF-α channels in zebrafish [75]. In collagen-induced rheumatic rats, administering root powder at 600 mg/kg body weight reduced the severity of arthritis by enhancing functional recovery of motor activity and radiological scores [76]. In a separate trial, this plant extract delayed morphine's analgesic impact. Moreover, it may inhibit hyperalgesia in the tail-flick test [77]. The anti-inflammatory effect of this root aqueous extract was investigated by measuring TNF-α, IL-1, 6, and 10 levels in rats with collagen-induced arthritis. Oral treatment of 300 mg/kg of their aqueous root extract reduced ROS and normalized MMP-8 levels in collagen-induced arthritic mice [74]. Clinically, the consumption of 500–1000 mg ashwagandha tablets by 77 patients for 8–12 weeks resulted in managing arthritis symptoms in these patients [78].

2.14 *Tribulus terrestris*

Tribulus terrestris is a member of the family Zygophyllaceae. This plant has been utilized in TCM and Ayurveda for centuries. Its pharmacological properties included anti-inflammatory, analgesic, hepatoprotective, and anti-cariogenic. Its action cannot be separated from the existence of active chemicals such as flavonoids, steroids, saponins, and alkaloids in this plant. Tribulusamide D and N trans-caffeoyl tyramine in the ethanol extract of *Tribulus terrestris* possessed anti-inflammatory activity, as indicated by the ability of the compounds to inhibit NO induced by LPS and PGE2 by reducing iNOS which can be induced and expression of COX-2. Tribulusamide D (6, Fig. 5) decreased the production of LPS-induced inflammatory cytokines, including IL-6, IL-10, and TNF-α, in RAW 264.7 macrophage-induced LPS [79]. These extracts were capable of repairing LPS and inducing excessive NO release and transcription of inflammatory cytokine genes. This plant inhibited LPS-induced increases in intracellular iNOS and NF-kB, which are responsible for cellular NO and cytokine production, respectively [80]. In vivo anti-arthritis action was proven by a substantial decrease in pro-inflammatory cytokines in an arthritic rat model treated with the extract at a dose of 200 and 300 mg/kg orally for 21 days [81]. In addition, this extract inhibited NO synthase 2, COX-2, TNF-α, and IL-6 in animals with MIA-induced OA [82].

2.15 *Curcuma longa*

Curcuma longa or turmeric is a plant that comes from the Zingiberaceae family. Traditionally, turmeric is often used as a treatment such as an antioxidant, antibacterial, anti-inflammatory, and antiviral. This plant includes important components such as curcumin, resin, curcuminoid, curcumin diglucoside, curcumin monoglucoside, sophoricoside, and epigallocatechin gallate (7, Fig. 5). Molecular docking scores and energy-energy ligand-protein interactions demonstrated that turmeric reduced TNF-α and IL-1 via strong molecular interactions [83]. The bisdemethoxycurcumin (8, Fig. 5) exhibited potent anti-inflammatory activity in conjunction with the suppression of NO, xanthine oxidase, and lipoxygenase. This chemical also inhibited the LPS-induced synthesis of COX-2, iNOS, IL-6, and TNF in RAW macrophages [84]. Turmeronol strongly inhibited the LPS-induced synthesis of PGE2 and NO, as well as the mRNA expression of the corresponding synthetic enzymes. Furthermore, this compound effectively reduced the LPS-induced mRNA and protein elevation of IL-1, IL-6, and TNF-α. Turmeronol derivates suppressed NF-kB nuclear translocation in a manner comparable to that of the NF-B inhibitor pyrrolidine dithiocarbamate, but not that of curcumin (another NF-kB inhibitor) [85]. Curcumin lowered inflammatory activity 6 h after zymosan-induced arthritis in rats. Curcumin at 1 mg/kg was more efficacious than prednisone in the first 6 h following arthritis onset. It did not have the same anti-inflammatory impact as prednisone after 12, 24, and 48 h. It was more effective than prednisone at decreasing inflammatory infiltration after 48 h, independent of the prednisone dosage employed [86]. Clinically, treatment of theracurmin containing curcumin 180 mg/day orally every day for 8 weeks considerably reduced reliance on celecoxib to the point where no adverse effects are noticed [87]. Compared to placebo, treatment with curcuminoids was linked with considerably higher decreases in WOMAC, VAS, and LPFI ratings. Curcuminoids are a safe and effective alternative therapy for OA [88]. *Curcuma longa* inhibited inflammation and had therapeutic benefits, such as a drop in VAS/WOMAC score level [89].

2.16 *Caesalpinia sappan*

Caesalpinia sappan known as secang is a member of the Caesalpiniaceae family. This plant is utilized in several nations, particularly in Southeast Asia. In Indonesia, secang is historically used as an immune-boosting and immune-maintenance herbal beverage. The part of this plant often used as herbal medicine is bark because it contains active compounds and is used in treating various diseases such as inflammation, diarrhoea, anti-osteoporosis, and other conditions. Phytochemical compounds in secang include brazilin, sappanone, cassane, and caesalpiniaphenol. Cassane cured rheumatoid arthritis via the TCR, TLR, VEGF, and osteoclast differentiation pathways, which are strongly associated with antigen recognition, inflammation, angiogenesis, and osteoclastogenesis [90, 91]. 3,7-dihydroxy-4H-chromen-4-one and 4-hydroxy-3,5-dimethoxybenzaldehyde exhibited potent inhibition of NO with IC_{50} values of 14.5 and 21.5 μM [92]. In CIA rats, brazilin (9, Fig. 5) at 10 mg/kg BW decreased the severity of acute foot oedema. Bone mineral density was considerably lower in CIA rats and appeared to rise proportionally with the administration of methotrexate and brazilin. The microstructural analysis demonstrates that brazilin reduced joint degradation, surface erosion, and enhanced bone growth. Brazilin significantly lowered blood levels of inflammatory cytokines in rats with CIA [93]. Sappanchalcone prevents chronic disease progression, leg oedema, arthritis severity, radiography, and histomorphometric alterations. Serum levels of pro-inflammatory cytokines were considerably lowered [94].

2.17 *Clematis mandshurica*

Clematis mandshurica roots are commonly used to treat inflammatory disorders such as gout, arthritis, and tetanus. This plant is traditionally used to relieve pain and fever in Korea. It contains hederagenin which inhibited the protein expression of iNOS, COX-2, and NF-kB at doses of 10, 30, and 100 μM triggered by LPS and generated NO, PGE2, TNF-α, IL-1β, and IL-6 on RAW 264.7 cells [95]. Feruloyl and iso-feruloyl derivatives at 50 mM exhibited a moderate inhibitory effect ranging from 29.3% to 38.7%, whereas the 3-carboxy-indole derivative exhibited a strong inhibitory effect with an IC_{50} of 7.7 to 12.4 mM [96]. The 5-O-isoferuloyl-2-deoxy-D-ribono lactone (5-IRL) stimulated inducible mRNA expression of COX-2 and iNOS [97]. This plant extract lowered arthritis scores relative to the control group in CIA-induced rats. Normal rats treated with this plant had no adverse effects on gut microbiota and SCFA metabolism [100]. This root extract at 0.25, 0.5, and 1 mg/mL inhibited protein expression LPS caused by iNOS, COX-2, NF-kB, and MAPKs (ERK, JNK, and p38) as well as the generation of TNF-α, IL-1β, IL-6, NO, and PGE2 [98]. The anti-oedema activity of hederagenin was demonstrated by the dependent reduction of LPS-induced mRNA levels of iNOS and COX-2 and the results of the carrageenan-induced rat hind limb oedema test. In addition, 30 mg/kg of hederagenin prevented carrageenan-induced increases in skin thickness, inflammatory cell infiltration, and mast cell degranulation [95]. Clematichinenoside significantly reduced leg swelling in AIA mice. It also increased levels of IL-10 and TGF-β1 released from ConA-activated PP lymphocytes, while decreased levels of IL-17 A and TNF-α [99].

2.18 *Rosmarinus officinalis*

Rosmarinus officinalis often known as rosemary is a member of the Lamiaceae family. This plant is utilized as traditional medicine like anti-diabetic, anti-inflammatory, and antioxidant. This plant contains carnosic acid, carnosol, and rosmarinic acid. Carnosic acid suppressed the expression of pro-inflammatory cytokines, including TNF-α, IL-1β, IL-6, IL-8, IL-17, and MMP-3, and downregulated the creation of RANKL. Moreover, it suppressed osteoclastogenesis and bone resorption [100]. The 70% ethanol extract of this plant exhibited oxidative damage and an enzymatic antioxidant defence system and anti-inflammatory effects [101]. In rats with arthritis, the aqueous extract of this plant reduced oxidative damage and enhanced antioxidant capacity by boosting GSH levels. Also, this extract decreased leg oedema, the number of leukocytes recruited in the femorotibial joint cavity, and the weight of lymph nodes and delayed the formation of subsequent diseases [102]. At 60 mg/kg, the methanol extract of this plant exerted a potentially effective anti-arthritic by controlling inflammation in an adjuvant-induced experimental model [103].

2.19 *Elaeagnus angustifolia*

Elaeagnus angustifolia or oleaster is a plant that comes from the Elaeagnaceae family. Kaempferol (10, Fig. 5) contained in this plant significantly inhibited IL-1-β-induced protein production from inflammatory mediators such as inducible iNOS and COX-2. It also inhibited general matrix-degrading enzymes (MMP-1, MMP-3, MMP-13, and a disintegrin), MMP with a thrombospondin-5 motif produced by IL-1β, hence preventing collagen II degradations [104]. The extract at a dose of 32 mg/kgBW significantly reduced serum concentrations of MMP-3 and MMP-13 [105] and at 1000 mg/kg effectively decreased leg oedema swelling [106]. In a randomized clinical trial conducted on 90 female osteoarthritis patients, this extract at 15 g/day for 8 weeks reduced serum levels of TNF-α and MMP-1 [107]. In addition, kaempferol reduced osteoarthritis symptoms with an efficacy comparable to that of ibuprofen [108].

2.20 *Crocus sativus*

Crocus sativus known as saffron is a member of the Iridaceae family and a traditional remedy for chronic disorders such as asthma, arthritis, and respiratory problems. Active chemicals found in saffron include phenolic and carotenoid compounds. The extract of this plant exhibited anti-inflammatory properties such as NO and IL-6 by murine RAW 264,7 cells. This extract possessed anti-osteoclastogenic properties by a full suppression of tartrate-resistant acid phosphatase (TRAP)-positive osteoclast production and a reduction in the expression of key osteoclast-related genes [109]. The extract administered orally at 25, 50, and 100 mg/kg BW to rats with arthritis for 47 days decreased TNF-a and IL-1 levels in CSE-2 and 3-group rats compared to rheumatic mice [110]. Dosages of 200–2000 mg/kg of saffron ethanol extract exhibited an anti-inflammatory effect of 18.2–27.8% against mice treated with 49.5% DEX (15 mg/kg) [111]. Histological scores of joint inflammation, bone erosion, chondrocyte mortality, cartilage surface erosion, and bone erosion in CIA mice treated with 40 mg/kg crocin were equal to those of normal mice. The protein expression levels of MMP-1, MMP-3, and MMP-13 in CIA mice treated with 40 mg/kg crocin were lowered to normal levels. In addition, crocin inhibited the expression of TNF-a, IL-17, IL-6, and CXCL8 in CIA rats' blood and ankle tissue [112, 113].

2.21 *Camellia sinensis*

Green tea also known as *Camellia sinensis* is one of the herbal plants frequently utilized in traditional medicine such as losing weight, lowering cholesterol, and overcoming an inflammation. This plant has several active compounds including flavones, flavonols, flavanones, catechins, and isoflavones. Epigallocatechin gallate in this plant had a bond to TNF-α of −90.90 and to the IL-1 target protein of

−120.75 [83]. This compound showed anti-rheumatoid arthritis efficacy by killing synovial fibroblast cells (66.1%), downregulating TNF-α (93.33%), IL-6 (87.97%), and inhibiting p-STAT3 (77.75%) [114]. A dose of 400 mg/kg/BB of *Camellia sinensis* extract had anti-arthritis effectiveness of 60% and boosted levels of superoxide dismutase, catalase, and glutathione by 34%, 59%, and 50% [115]. At 400 mg/kg/BB, this extract effectively reduced TNF-α, IL-1, and IL-6 levels in the joint tissue of rats. Compared to the arthritic and infected group, the therapy of this extract exhibited substantial reductions in cartilage degradation and pannus development, and there was no evidence of inflammation in the small intestine [116]. Clinically, administration of this extract for 4 weeks resulted in a significant difference in the average VAS pain score, total WOMAC, and WOMAC physical function between the green tea and control groups [117].

2.22 *Matricaria chamomilla*

Matricaria chamomilla (camomile) is a member of the Asteraceae family and used for treating stomach discomfort and sleeplessness, as well as for anti-inflammatory, antibacterial, and sedative effects. The compounds from this plant had comparable scores in the active site by the free binding energy ($\Delta G = 54.19$ kcal/mol) [118]. The extract at 400 mg/kg BW decreased blood calcium, phosphate ion, and magnesium levels by 54.01%, 27.73%, and 20%, respectively. In addition, this dose also decreased creatinine and alkaline phosphatase levels by 29.41% and 27.82%, respectively [118, 119]. Compared to diclofenac and placebo, chamomile oil greatly reduced the patient's requirement for acetaminophen. Nevertheless, there were no statistically significant changes in the WOMAC questionnaire's domains, and no adverse effects were reported by patients taking chamomile oil [120].

2.23 *Punica granatum*

Punica granatum is used as an anti-inflammatory, antibiotic, antibacterial, etc. It contains flavonoids, polyphenols, alkaloids, anthocyanins, tannins, etc. The ellagic acid derived from the *Punica granatum* could bind to NF-kB protein; however, the various binding efficiencies are examined based on the energy gained for binding and the number of hydrogen bonds generated. The calculated bond-free energy of the NF-kB-EA complex was 7.1 kcal/mol with two hydrogen bonds [121]. The 70% ethanol extract of this plant decreased IL-1-inducible, iNOS, COX-2, and MMP-13 protein production in primary mouse chondrocytes (PRC) [122]. At a dose of 50 mg/kg, the butanol fraction increased RBC and Hb while decreasing WBC and ESR [123]. The injection of 500 and 750 mg/kg of this ethanol extract for 4 weeks resulted in a reduction in paw volume; a rise in latency time and levels of ALT, AST, and ALP; and a substantial drop in RF and MDA activity compared to the rheumatism control group. The extract reduced the levels of IL-1 and TNF in rats with arthritis. Histological analysis revealed that therapy with the extract reversed greatly the histological alterations produced by arthritis [124]. Rosmarinic acid considerably decreased the severity of rheumatism, leg volume, joint diameter, white blood cell count, and erythrocyte sedimentation rate. Moreover, it dramatically boosted body mass, haemoglobin, and red blood cells [125]. In a mouse model of osteoarthritis caused by collagenase type II, POMx (150 mg/kg) reduced knee-associated inflammation and nociception and corrected alterations in the weight-bearing ratio of experimental mice [122]. In clinical studies, 250 mg of pomegranate enhanced disease activity and multiple blood indicators of inflammation and oxidative stress in individuals with rheumatoid arthritis [126]. Pomegranate juice enhanced physical function and stiffness, as well as it is antioxidant and MMP-13 inhibiting capabilities in osteoarthritis patients [127]. Women with knee OA who took 500 mg of pomegranate peel twice a day for 8 weeks saw a reduction in pain and an improvement in symptoms [128].

2.24 *Cinnamomum cassia*

Cinnamomum cassia is a plant that belongs to the Lauraceae family and is traditionally used to treat gastritis, blood circulation abnormalities, dyspepsia, and inflammatory conditions. This plant contains several active compounds such as cinnamic aldehyde, coumarin, tannin, methyl dihydromelilotoside derivative, hierochin B, balanophonin, quercetin, and luteolin that were able to interact with JAK2, mPEGS-1, COX-2, IL-1β, and PPARγ target proteins [129]. This plant possessed antipyretic, anti-inflammatory, antibacterial, anti-diabetic, and anticancer properties. The ethyl acetate fraction of this plant inhibited carrageenan-induced leg swelling in rheumatoid arthritis and LPS-induced NO generation in murine immune cells [130]. Cinnamaldehyde from this plant dramatically reduced synovial inflammation in rats with arthritis. It was related to the inhibition of IL-1β. This compound was also capable of reducing HIF-1α activity by preventing succinate build-up in the cytoplasm. This reduction inhibits IL-1β synthesis because HIF-1α promotes the expression of NLRP3, which is important in the assembly and processing of IL-1β by the inflammation. In addition, cinnamaldehyde suppressed the expression of the GPR91 succinate receptor, hence inhibiting HIF-1α activation even more [131]. Cinnamomi lactone inhibited MMP-3, IL-1β, and MMP-1 gene expression in FLS cells [132]. Stems of this plant considerably lowered MDA levels and joint swelling, accompanied by a rise in GSH levels. This extract dramatically decreased joint swelling, IL-1, and TNF levels in rats with arthritis caused by CFA. The expression of TNF-α receptors was reduced in mice treated with indomethacin or *Cinnamomum cassia* stem extract [133].

2.25 *Centella asiatica*

Centella asiatica is a plant that is often used in wound healing and contains saponins, asiaticoside, asiatic acid, and carotenoids. The 2000 g/ml of *C. asiatica* extract inhibited protein denaturation and membrane stabilization in the greatest percentages of 89.76% and 94.97%, respectively [134]. Madecassoside and madecassic acid (11, Fig. 5) directly inhibited the release of pro-inflammatory cytokines and IL-10 from CD4+ CD25+ T cells [135]. The extract dramatically reduced mechanical allodynia and hyperalgesia, spontaneous pain, and motor changes. Their efficacy was greater than or equivalent to that of the reference medication, triamcinolone acetonide (100 μg intramuscularly). The histological study of MIA+14G1862-treated rats revealed the improvement of several morphological parameters, with significantly positive effects on joint space and fibrin deposition [135]. The methanol fraction of *Centella asiatica* (150 and 250 mg/kg) considerably mitigated the severity of CIA and decreased synovial inflammation, cartilage erosion, and bone erosion, as demonstrated by histological and radiological data. The increase in plasma levels of TNF-α, IL-1, IL-6, IL-12, and NO in CIA mice was significantly attenuated by treatment with *Centella asiatica*. Serum levels of anti-collagen type-II antibodies were significantly lower in the treatment group than in the rheumatic group in rats given *Centella asiatica* (150 and 250 mg/kg) [136].

2.26 *Phyllanthus amarus*

Phyllanthus amarus is a medicinal plant that is often used to treat pain and includes the Phyllanthaceae family. Based on its pharmacological activity, this plant is often used to treat diabetes, malaria, analgesics, arthritis, and hypertension. This plant contains compounds such as filantin, hypophylantine, filantenol, nirantin, and Quercetin. In a cartilage explant model, *P. amarus* extract (PAE) and its major components, including phyllanthin and hypophyllanthin, were evaluated. Following 4 days of incubation, the DMMB binding assay and zymography were used to assess the release of sulfated glycosaminoglycans (sGAGs) and MMP-2 activity, respectively, in a culture medium. The chondroprotective potential of PAE and its major components against IL-1β-induced degradation of cartilage explants was demonstrated by decreased levels of sGAGs and MMP-2 activity in culture media, which were consistent with increased uronic acid and proteoglycan content in explants, compared to IL-1β treatment [137]. Phyllanthin considerably inhibited the compound's response to LPS-stimulated prostaglandin overproduction via a mechanism connected to the modification impact of COX-2 protein and gene expression. Moreover, phyllanthin dramatically prevented the release of mRNA and the production of IL-1ß and TNF-α. In a concentration-dependent way, phyllanthin also dramatically downregulated IκBα, NF-kB (p65), and IKKα/β phosphorylation and inhibited JNK, ERK, p38MAPK, and Akt activation [138]. In addition, the extract suppressed the synthesis of pro-inflammatory (TNF-α, IL-1β, PGE2) and the expression of COX-2 protein in LPS-stimulated U937 human macrophages. Pre-treatment of this plant also substantially inhibited the pro-inflammatory marker mRNA transcription in U937 macrophages [139].

Phyllanthus amarus extract was able to modulate the development of physical parameters (e.g. weight loss, increase in body temperature, reduction in hind limb volume, and severity of arthritis), bone mineral density, haematological and biochemical disturbances, serum cytokine production, and levels of MMPs together with their inhibitors in synovial fluid. Examination of the knee joint's histopathology demonstrated that the extract efficiently lowered synovitis, pannus development, bone resorption, and cartilage disintegration [140]. The extracts and leaves of two *Phyllanthus amarus* and *Phyllanthus fraternus* elevated the threshold for mechanical withdrawal (mechanical hyperalgesia) to return to the original reaction. The percentage of maximal achievable effectiveness and the percentage of suppression of hyperalgesia further demonstrate the efficiency of *Phyllanthus* extract [141]. In 40 OA patients, *Phyllanthus amarus* extract nanoparticles administered through phonophoresis showed a substantial rise in the visual analog scale (VAS) and 6-MWT following therapy in both groups. The phonophoresis (PP) group had a more substantial reduction in VAS pain levels and an increase in 6-MWT than the ultrasonic therapy (UT) group [142].

2.27 *Psidium guajava*

Psidium guajava is a plant that is known for its many benefits in Indonesia. The phytochemical analysis of guava leaves revealed the presence of flavonoids, saponins, tannins, phenols, carotenoids, and quercetin. In addition, epicatechin-3-O-gallate shows tremendous promise as a selective anti-inflammatory COX-2 inhibitor due to its strong bond strength and stability, as well as its low bond energy when bound to the target protein (6-COX) [143]. Guava leaf extract significantly inhibited LPS-induced production of NO and PGE2 in RAW264.7 macrophages by inhibiting the expression and activity of iNOS and COX-2 which was partially induced by the downregulation of ERK1/2 activation [144]. The ethanol extract of this leaf considerably suppressed leg oedema in the chronic phase, as measured by an increase in oedema beginning on day 20. The most beneficial dose of EEPG for lowering arthritis scores was 250 mg/kg. Histopathological analysis revealed that the knee's synovial membrane and cartilage have been repaired [145]. Extract of this leaf was also capable of reducing the number of lymphocytes and leukocytes. The group that received extract dosages of 250 and 750 mg/kg BW showed a substantial reduction in leukocytes both before and after treatment. Following administration of 250, 500, and 750 mg/kg BW of ethanol extract of *P. guajava* leaf, the number of lymphocytes was also reduced. This leaf extract at 750 mg/kg BW suppressed leukocytes and lymphocytes most effectively [146]. In the 12th week of the clinical experiment, knee pain and stiffness (a subgroup of the JKOM score) were considerably lower in the guava group than in the placebo group. Utilizing a visual analog scale (VAS) to measure knee pain, a significant correlation was seen between treatment effect and test length, with the guava group having a lower VAS score after 12 weeks than the placebo group. In conclusion, continual use of guava leaf extract can alleviate knee discomfort, indicating a possible preventative benefit against OA symptoms [147].

2.28 *Moringa oleifera*

Moringa oleifera or commonly known by the local name “moringa” is a type of plant that has various benefits. One of the potentials possessed by moringa based on phytochemical analysis (flavonoids, saponins, and polyphenols) is as an anti-inflammatory. The moringa extract inhibited the production of inflammatory cytokines (IL-1β, TNF-α, and IL-6) and the degradation of cytoplasmic IB- and nuclear translocation of p65 protein, resulting in reduced NF-kB transactivation. The moringa extract also inhibited NFk-B activation in RAW 264.7 cells, hence reducing the levels of pro-inflammatory mediators like NO, IL-1β, TNF-α, and IL-6 [148]. In vivo investigation of moringa leaf extract at 250 and 500 mg/kg in rats with arthritis showed higher inhibitory action against inflammatory leg oedema with fewer toxicity and adverse effects than the indomethacin-treated group [149]. Oral treatment of *Moringa oleifera* (25, 50, and 100 mg/kgBW) in rats induced with CFA significantly reduced joint inflammation as evidenced by reductions in joint diameter, rheumatic scores, and inflammatory cell infiltration [150]. Oral administration of *Moringa oleifera* at doses between 250 and 500 mg/kg lowered the rheumatic index and sole oedema. It is believed that moringa can reduce serum pro-inflammatory cytokines when administered [151].

2.29 *Nigella sativa*

Nigella sativa (black cumin) is one of the herbal plants that are frequently employed as a traditional medicine due to the presence of its active constituents. One of the pharmacological activities of this plant is as an anti-inflammatory. This plant produces thymoquinone (12, Fig. 5) which acts as an anti-inflammatory and anti-proliferative in osteoarthritis. This compound inhibited apoptosis-regulated signaling kinase 1 (ASK1) in response to TNF-induced activation of phospho-p38 and phospho-JNK. Moreover, TNF-α promotes ASK1 phosphorylation at the Thr845 site in RA-FLS selectively [152]. Application of this plant oil (1.82 mL/kg) to arthritic rats dramatically decreased the onset of arthritis by 56%. The oil possessed anti-inflammatory, anti-rheumatic, and analgesic properties [153]. Thymoquinone derivative considerably lowered the total leukocyte count, inflammatory clinical score, and blood urea and serum creatinine levels [154]. Clinically, therapy with *Nigella sativa* caused an increase in the number of swollen joints, as compared to the initial and placebo groups. This plant treatment led to a drop in CD8 + and a rise in the proportion of CD4+ CD25+ T cells and the CD4+/CD8+ ratio. In the NS group, a significant negative connection was identified between CD8+ change and CD4+ CD25+ T cell change [155]. *Nigella sativa* lowered the DAS28 score considerably compared to the placebo group. Serum levels of IL-10, TNF-, MDA, SOD, catalase, TAC, and NO did not alter significantly from the placebo group [156].

2.30 *Cannabis sativa*

Cannabis sativa is a plant that comes from the Cannabidaceae family and is often used as an antidepressant, for fatigue, and mental disorders. The active compounds of this plant include cannabinoids, tryptophan, and cannabidiol (13, Fig. 5). Tryptophan had an inhibition value of −5.6 against TNF-α protein compared to ascorbic acid and linoleic acid [157]. Cannabidiol contributed to the reduction of cell viability, proliferation, and RASF IL-6/IL-8 production. Moreover, pre-treatment with TNF enhanced intracellular calcium and absorption of cationic PoPo3 viability dyes in RASF [158]. *Cannabis sativa* oil decreases the survival rate of MH7A cells and induces apoptotic cell death in a dose- and time-dependent manner. In MH7A cells treated with HO, lipid accumulation and intracellular ROS increase. Then co-treatment with the antioxidant tyrone substantially negated the harmful impact of H_2O_2 on MH7A cells. The result was that ROS levels decreased, cell viability was restored, and apoptotic cell death was significantly decreased [159]. Cannabidiol increased the amount of early apoptotic B cells at the expense of viable cells and decreased the production of TNF and IL-10 when independently stimulating T cells. CBD boosted IL-10 synthesis in PBMC when B cells were triggered by T cells but decreased TNF levels when T cells were activated independently. CBD inhibited IL-10 production in PBMC/rheumatoid synovial fibroblast co-cultures when B cells were stimulated independently of T cells [160]. In a dose-dependent manner, administration of cannabidiol in a rat model of OA decreased joint oedema, limb posture score as a spontaneous pain rating, immune cell infiltration, and synovial membrane thickening. Immunohistochemical study of bone marrow (CGRP, OX42) and dorsal root ganglia (TNF-α) demonstrated a dose-dependent decrease of pro-inflammatory biomarkers, with 6.2 and 62 mg/day being effective levels [160]. Cannabidiol resulted in improvement in patient-reported outcome measures, including pain on the visual analog scale; arm, shoulder, and hand disabilities; and single-assessment numerical evaluation scores, compared to the control group in a clinical trial involving 10 patients with arthritis of the thumb. Physical metrics associated with a range of motion, grip, and pinch strength were found to be comparable [161, 162].

2.31 *Andrographis paniculata*

Andrographis paniculata popularly known as sambiloto is one of the nine most extensively used medicinal herbs in Indonesia, as well as other nations. This plant demonstrated antithrombotic, antiviral, immunosuppressive, and anti-inflammatory properties. Andrographolide is one of the primary active chemicals in this plant. This compound dose-dependently inhibited IL-1β and IL-6 release from TNF-α-stimulated RA-SFs [163]. It was effective at alleviating arthritis symptoms by reducing neutrophil infiltration and necrosis in the ankle joint, relieving systemic inflammation, accelerating LPS-activated neutrophil apoptosis, and inhibiting extracellular-dependent autophagy-dependent neutrophil formation [164]. Andrographolide (14, Fig. 5) was also capable of dose-dependently suppressing

MMP-1, 3, and 9 inhibited by these substances [165]. In in vivo experiments, andrographolide reduced blood levels of anti-CII, TNFα, IL-1β, and IL-6 to considerably reduce the severity of arthritis and joint destruction [163]. By directly targeting NF-κB, andrographolide inhibited the inducible activation of iNOS and COX-2 as indicated by immunoblot analysis [166]. Compared to mice administered dexamethasone, animals treated with andrographolide had a reduction in leg oedema, cell cytotoxicity, and NO generation [167]. Moreover, the administration of this molecule enhanced the anti-rheumatic efficacy of MTX. Combining AD and MTX decreased inflammatory symptoms in CFA mice additively. The hepatoprotective effect of AD and MTX combination treatment was characterized by an improvement in blood indicators, potentially related to antioxidant activity, and verified by histological alterations in the liver. Moreover, combination treatment substantially decreased blood levels of TNF-α, IL-6, and IL-1β [168]. On days 28, 56, and 84, individuals treated with 300 and 600 mg/day of paractin experienced significantly less discomfort than the control group. Compared to the placebo group, stiffness, physical function, and tiredness scores improved significantly in both paractin-treated groups [169].

2.32 *Artemisia annua*

Artemisia annua is a medicinal plant originating from sub-tropical regions such as China. This plant has several properties including being antimalarial, anti-inflammatory, and effective in treating autoimmune diseases. These benefits and properties cannot be separated from the role of their active compounds such as artemisinin, alkaloids, saponins, sterols, triterpenes, tannins, and coumarins. In RAW 264.7 cells, casticin and chrysosplenol D decreased the LPS-induced production of IL-1β, IL-6, and MCP-1, limited cell motility, and lowered the LPS-induced phosphorylation of IB and c-JUN. In addition, it was a JNK SP600125 inhibitor that prevents the inhibition of cytokine release by chrysosplenol D. Application of casticin and chrysosplenol D inhibited ear oedema produced by croton oil and dexamethasone in rats [170]. The dihydroartemisinin (15, Fig. 5) contained in *Artemisia annua* extract was able to inhibit bone loss in ovariectomized rats so that osteoclast formation will also be inhibited [157, 171, 172]. The clinical administration of *Artemisia annua* extract treatment at 150 and 300 mg twice a day orally for 12 weeks resulted in a substantial decrease in pain ratings and joint swelling [10].

2.33 *Piper betle*

Betel leaf (*Piper betle*) is one of the plants used in traditional medicine by the community. Betel leaves contain secondary metabolites including flavonoids, tannins, saponins, and alkaloids. Moreover, betel leaves have the potential for different therapeutic properties, including anti-inflammatory properties. The 4-allyl-1,2-diacetoxybenzene exhibited stronger interactions with less binding energy to RA targets than MMP-1(−6,4 kcal/mol), TGF(−6,9 kcal/mol), and IL-1(−5,4 kcal/mol) [173]. Betel leaf methanol extract (PBME) was able to suppress protein denaturation more efficiently than normal ibuprofen, with an IC_{50} concentration of 20.5 μg/ml compared to 17.5 μg/ml for ibuprofen [173]. Moreover, the recovery of rat joint injury demonstrates the anti-arthritic action of PBME (250 and 500 mg/kg BW). The reduction of leg oedema and rheumatic scores, backed by radiographic and histological alterations, demonstrates that combined treatment with allylpyrocatechol (APC) and MTX dramatically improved arthritic parameters. This combination inhibited the production of pro-inflammatory cytokines, such as TNF-α and IL-6. In addition, the APC-MTX combination reduced cachexia, splenomegaly, and oxidative stress associated with these conditions [174]. The use of lower (200 mg/kg) and higher (400 mg/kg) dosages of *Piper betle* extract showed a dose-dependent reduction in SGPT, SGOT, and ALP, leg volume, foot diameter, and increased body weight as compared to the rheumatism control group. Serum rheumatoid factor was significantly decreased in all treated animals. In addition, the anti-arthritic effect was validated by histology of the synovial joints of treated rats, which revealed decreased constriction of the inter-tarsal joint gaps. The acquired findings are comparable to the standard [175–179].

Medicinal plants have activity as anti-rheumatoid disorders through various mechanisms depending on their secondary metabolites. Table 1 summarizes the mechanism of anti-rheumatoid from the selected plants in this chapter of the book. Some are through the inhibition of PGE2, IL-6, IL-8, and iNOS, etc. In more detail the mechanism of action of medicinal plants as anti-rheumatoid disorders can be seen in Table 1.

3 Conclusion

Arthritis disorders are associated with reduced productivity, substantial disability, and decreased quality of life. Conventional therapies include disease-modifying anti-rheumatic drugs (DMARDs) and non-steroidal anti-inflammatory drugs (NSAIDs).

Table 1 The mechanisms of anti-rheumatoid disorders form the selected plants

Plant	Mechanism
Zingiber officinale	Aqueous extract of ginger at 125 g/ml revealed that its anti-inflammatory activity was higher than diclofenac Ethanol extract of ginger inhibit PGE2, IL-6, IL-8, and iNOS Essential oil of ginger decreasing expression of IL-12 and TGF-α
Piper methysticum	Extract of *Piper methysticum* inhibit nitric oxide (NO) Kavain compound reduce production of TNF-α
Ephedra sp.	β-Ssitosterol and androstan-3-one inhibit IL-6, TLR4, TNF-α, and IL-1β Extract of *Ephedra alata* and *Ephedra fragilis* inhibited NO *Ephedra sinica* inhibit production of cytokine

(continued)

Table 1 (continued)

Plant	Mechanism
Oenothera biennis	*Oenothera biennis* inhibit TNF-α, IL-6, and IL-1 Evening primrose oil inhibit TNF-α and reduce IL-1β, and IL-6
Panax ginseng	*Panax ginseng* reduce of reactive oxygen species (ROS), IL-1, NO expression, and ROS. Ginsenoside compound suppresses osteoclastogenesis and increased the expression of aggrecan and collagen type 2 (COL-II) Ginsenoside decreasing of MMP-1 and MMP-13
Ginkgo biloba	*Ginkgo biloba* extract inhibit IL-1, iNOS, NO, MMP-13 Ginkgolide B reduces IL-1, IL-6, MCP-1, TNF-a, MMP-3, and MMP-13 and increases IL-10
Allium sativum	*Allium sativum* regulate of the IL-17 gene
Ganoderma lucidum	Sterol compound reduce pro-inflammatory mediators such as NO, TNF-α, IL-1, and 6
Tripterygium wilfordii	Compound of tripterygium glycoside prevent their proliferation, restrain Th17 cell differentiation and inhibit angiogenesis Alkaloid compound from *Tripterygium wilfordii* decrease levels of IL-6, IL-8, NF-kB, and TNF-α
Uncaria tomentosa	*Uncaria tomentosa* extracts decreased the RANKL/OPG ratio Quinovic acid glycoside was demonstrated to reduce oedema, bleeding, IL-1β levels, and neutrophil migration
Boswellia serrata	*Boswellia serrata* extract reduce IL-10 levels *Boswellia serrata* extract altered numerous parameters, including MPO, LPO, GSH, and NO
Tinospora cordifolia	*Tinospora cordifolia* suppressed TNF-α and IL-1 and reduces NO generation Chloroform extract *Tinospora cordifolia* downregulated pro-inflammatory biomarkers in THP-1 *Tinospora cordifolia* extract lowered levels of pro-inflammatory like IL-1, 6, and 17
Withania somnifera	Chloroform and water extracts *Withania somnifera* were able to control the expression of TNF, IL-1, and IL-6, COX-1, and COX-2
Tribulus terrestris	Ethanol extract of *Tribulus terrestris* inhibit NO induced by LPS and PGE2 Tribulusamide D compound decreased the production of LPS-induced inflammatory cytokines, including IL-6, IL-10, and TNF-α, in RAW 264.7
Curcuma longa	Bisdemethoxycurcumin compound suppresses NO, xanthine oxidase, and lipoxygenase and inhibits COX-2, iNOS, IL-6, and TNF Turmeronol compound inhibited the LPS-induced synthesis of PGE2, NO, IL-1, IL-6, and TNF-α
Caesalpinia sappan	Cassane compound cured rheumatoid arthritis via the TCR, TLR, VEGF, and osteoclast differentiation pathway 3,7-dihydroxy-4H-chromen-4-one and 4-hydroxy-3,5-dimethoxybenzaldehyde exhibited potent inhibition of NO
Clematis mandshurica	Hederagenin inhibited the protein expression of iNOS, COX-2, and NF-kB Clematichinenoside compound significantly increase level of IL-10 and TGF-β1
Rosmarinus officinalis	Carnosic acid compound suppressed the expression of TNF-α, IL-1β, IL-6, IL-8, IL-17, and MMP-3.

(continued)

Table 1 (continued)

Plant	Mechanism
Elaeagnus angustifolia	Kaempferol compound significantly inhibited IL-1β, MMP-1, MMP-3, MMP-13
Crocus sativus	*Crocus sativus* extract inhibit NO and IL-6 *Crocus sativus* extract decreased TNF-a and IL-1 levels Crocin compound inhibited the expression of TNF-a, IL-17, IL-6, and CXCL8
Camellia sinensis	Epigallocatechin gallate compound killing synovial fibroblast, downregulating TNF-α, IL-6 and inhibiting p-STAT3
Matricaria chamomilla	*Matricaria chamomilla* extract decreased blood calcium, phosphate ion, and magnesium levels
Punica granatum	70% ethanol extract of *Punica granatum* decreased IL-1-inducible, iNOS, COX-2, and MMP-13 Ethanol extract *Punica granatum* reduced the levels of IL-1 and TNF
Cinnamomum cassia	Compound from *Cinnamomum cassia* were able to interact with JAK2, mPEGS-1, COX-2, IL-1β, and PPARγ target proteins Cinnamaldehyde inhibit IL-1β, reducing HIF-1α Cinnamomi lactone inhibited MMP-3, IL-1β, and MMP-1
Centella asiatica	Madecassoside and madecassic acid directly inhibited the release of pro-inflammatory cytokines and IL-10 *Centella asiatica* extract reduce TNF-α, IL-1, IL-6, IL-12, and NO
Phyllanthus amarus	*Phyllanthus amarus* extract decreased levels of sGAGs and MMP-2 Phyllanthin compound reduce IL-1ß and TNF-α
Psidium guajava	Guava leaf extract inhibited LPS-induced production of NO and PGE2
Moringa oleifera	The moringa extract inhibited the production of inflammatory cytokines (IL-1β, TNF-α, and IL-6) The moringa extract also inhibited NFk-B and reducing NO, IL-1β, TNF-α, and IL-6
Nigella sativa	Thymoquinone compound inhibited apoptosis-regulated signaling kinase 1 (ASK1) in response to TNF-induced activation of phospho-p38 and phospho-JNK *Nigella sativa* led to a drop in CD8+ and a rise in the proportion of CD4+ CD25+ T cells and the CD4+/CD8+ ratio
Cannabis sativa	Cannabidiol compound contributed to the reduction of cell viability, proliferation, and RASF IL-6/IL-8 production Cannabidiol decreased the production of TNF and IL-10 Cannabidiol boosted IL-10 synthesis
Andrographis paniculata	Andrographolide compound inhibited IL-1β and IL-6 Andrographolide suppressing MMP-1, 3, and 9 Combination of andrographolide and methotrexate substantially decreased blood levels of TNF-α, IL-6, and IL-1β
Artemisia annua	Casticin and chrysosplenol D compound decrease IL-1β, IL-6, and MCP-1
Piper betle	4-Allyl-1,2-diacetoxybenzene exhibited stronger interactions with MMP-1, TGF, and IL-1 Allylpyrocatechol (APC) and methotrexate (MTX) inhibit the production of pro-inflammatory cytokines, such as TNF-α and IL-6.

However, this therapy has several disadvantages related to gastrointestinal, cardiovascular, and nephrotoxic effects. Therefore, alternative medicine, such as the consumption of herbal medicine, is highly beneficial for complementary therapy. However, more information is required about plants or compounds that potentially be used to treat rheumatic disorders. In this book chapter, we identified 33 plants containing more than 15 compounds for anti-rheumatoid arthritis activity, osteoarthritis, gout, etc. The articles mentioned in the chapter book apply several study designs, including in silico, in vitro, and in vivo techniques. The findings of this review indicate that these plants have an effective mechanism for inhibiting multiple protein targets in rheumatic disorders, including TNF-α, interleukins, NO, iNOS, ROS, MMP, NF-kB, COX-1, COX-2, PGE2, MCSF, and others. Regarding the activity of several plants and compounds on arthritis protein targets, however, additional research is still required so that plants with potential anti-arthritis disorder activity can be identified.

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Evidence from the Use of Herbal Medicines in the Management and Prevention of Common Eye Diseases

25

Prasann Kumar, Aman Khokhar, and Shipa Rani Dey

Contents

Abstract

The fact that botanical compounds have been extensively used to cure various diseases and ailments throughout history is well known. Several of these compounds have been reported to be powerful antioxidants, anti-inflammatory agents, and antiapoptotic agents. Furthermore, these mechanisms have also been involved in ocular diseases, such as age-related macular degeneration (AMD), glaucoma, diabetic retinopathy, cataracts, and retinitis pigmentosa. Recent studies have demonstrated that several epidemiological and clinical studies have been performed that have shown the benefit of plant-derived compounds on these ocular pathologies, such as curcumin, lutein, zeaxanthin, ginseng, and many others in recent years. Cell cultures and animal models have shown promising results when these drugs have been used to treat eye diseases in the laboratory. It is important to note that although there appear to be many significant correlations between these botanical compounds, further investigation is necessary to determine their mechanistic pathways. This will facilitate widespread pharmaceutical application and the development of non-invasive alternatives for significant eye diseases to provide non-invasive solutions to those diseases.

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Keywords

Eye · Glaucoma · Cataracts · Diabetic retinopathy · Macular degeneration · Lutein

Abbreviations

AMD	Age-related macular degeneration
AMPK	AMP-activated protein kinase
AREDS	Age-related eye disease study
CB1	Cannabinoid receptor type 1
CBD	Cannabidiol
CGRP	Calcitonin gene-related peptide
COX-2	Cyclooxygenase-2
DR	Diabetic retinopathy
EGCG	Epigallocatechin gallate
fERG	Flash electroretinogram
GBE	*Ginkgo biloba* extract
GO	Graves' ophthalmopathy
HSP72	Heat shock proteins
HSV	Herpes simplex virus
HUVECs	Human umbilical vein endothelial cells
IGF-I	Insulin growth factor-1
IOP	Intraocular pressure
JNK	c-Jun N-terminal kinase
KRG	Korean red ginseng
LDLs	Low-density lipoproteins
LIRD	Light-induced retinal degeneration
MCP	Monocyte chemoattractant protein
MDA	Malondialdehyde
NF-kappa B	Nuclear factor kappa B
NO	Nitric oxide
NOS	Nitric oxide synthase
NTG	Normal tension glaucoma
PACAP	Pituitary adenylate cyclase-activating peptide
PCG-1alpha	Peroxisome proliferator-activated receptor gamma coactivator 1alpha
POAG	Primary open-angle glaucoma
Rb1	Retinoblastoma protein 1
RCTs	Randomized controlled trials
Rg3	Retinoblastoma protein 3
RGC-5	Retinal ganglion cell
ROS	Reactive oxygen species
RP	Retinitis pigmentosa
RPE	Retinal pigment epithelium

THC	Delta-9-tetrahydrocannabinol
TNF-alpha	Tumor necrosis factor-alpha
UV	Ultraviolet
VF	Visual field
WHO	World Health Organization

1 Introduction

The World Health Organization (WHO) estimates that approximately 34 million blind people live worldwide [10]. The eye has several natural defense mechanisms that it can use to protect itself from injury or infection. Tears, for instance, maintain a moist environment around the eye and physically remove any foreign particles, such as dust or microorganisms, that may have settled there. They also keep the eye from drying out. Moreover, tears have been found to contain several anti-infective substances, such as lysozymes and interferon. The eyelashes and eyelids keep the eye's surface moist while serving as a barrier between the ocular surface and the outside world. On the other hand, these protective mechanisms can become compromised occasionally, leading to eye inflammation. Eye infections are frequently and commonly reported in the South African province known as the Eastern Cape. These infections are brought on by exposure to bacteria, fungi, viruses, and other types of microorganisms. Traditional healers and herbalists recommend plant-based remedies for treating conditions like these [34, 56]. According to the various ancient and modern Chinese pharmacopoeia, numerous herbal treatments have been utilized for a significant amount of time to treat different eye conditions. They effectively treat night blindness, cataracts, floaters, and glaucoma, among other eye conditions. The usefulness of these treatments is deduced from anecdotal evidence and specific clinical results rather than going through the rigorous process of modern systematic scientific research, which would involve testing on animals and participating in clinical trials. In addition, the efficacies have a reputation for having a large amount of variation due to herbal remedies needing to be standardized in terms of their sources, qualities, combinations, or preparation methods [59, 60, 62, 63, 65, 66]. This is because these aspects of herbal remedies can affect their effectiveness. Despite this, the historical treatments documented for a long time are a goldmine for discovering natural products. Herbal extracts and herbal molecules that have been defined offer a superior alternative for the continued research and development of herbal medicines, making it possible to circumvent the challenges posed by variations. Because it has such a high metabolic rate, the eye is constantly damaged by the light's photooxidation. Damage caused by oxidative agents can quickly occur in the tissue cells of the eye. Even though there are several pathological factors associated with ocular diseases, such as age, inflammation, oxidative stress, genetic predisposition, tumorigenesis, diabetes, angiogenesis, and ischemia, it appears that oxidative stress plays a pivotal role in the pathogenesis of ocular diseases. Any of these contributors can cause ocular diseases, but they all work together.

The Aim and Objective of the Chapter

- To analyze the effectiveness of herbal medicines in managing glaucoma symptoms and preventing progression.
- To investigate the potential of herbal remedies in reducing intraocular pressure for glaucoma patients.
- To assess the impact of herbal treatments on retinal health in diabetic retinopathy cases.
- To examine the role of herbal supplements in preventing cataract formation and progression.
- To explore the antioxidant properties of herbal extracts for protecting against age-related macular degeneration.
- To evaluate the safety and efficacy of herbal therapies in alleviating dry eye symptoms.
- To investigate the mechanisms by which herbal medicines can improve visual function in eye diseases.
- To identify potential herbal interventions for reducing ocular inflammation in various eye conditions.
- To examine the impact of herbal supplements on improving ocular blood flow in glaucoma.
- To assess the clinical outcomes of using herbal treatments in conjunction with conventional therapies for eye diseases.
- To determine the neuroprotective effects of specific herbal compounds in glaucoma management.
- To analyze the anti-inflammatory properties of herbal medicines for reducing eye discomfort and redness.
- To investigate the role of herbal remedies in preventing or delaying the onset of cataracts.
- To explore the potential of herbal therapies in addressing oxidative stress in diabetic retinopathy.
- To evaluate the safety and tolerability of herbal supplements as adjunct treatments in ophthalmology.
- To examine the impact of herbal interventions on improving contrast sensitivity in eye diseases.
- To assess the effectiveness of herbal treatments in preventing vision loss associated with age-related eye conditions.
- To investigate the mechanisms of action of herbal compounds in preserving retinal health.
- To explore the synergistic effects of combining herbal and traditional glaucoma medications.

Several different risk factors have been associated with age-related macular degeneration, also known as AMD, including getting older, smoking cigarettes, being exposed to sunlight, and consuming a diet that is high in fat and low in antioxidants, in addition to having specific genetic variants (such as the polymorphism of complement factor H), has been shown to increase the risk of developing cardiovascular disease. There is a correlation between these potential dangers and oxidative pressure, ultimately damaging retinal tissue. This includes damage to retinal pigment epithelium, photoreceptor cells, and choroidal capillaries. Glaucoma causes progressive and irreversible death of retinal ganglion cells, damage to the optic nerve and loss of vision, even without increased intraocular pressure. This is because glaucoma causes pressure to build up inside the eye. Oxidative stress is responsible for cytotoxic effects, a degeneration of the retinal ganglion cells, and damage to the trabecular meshwork, all of which contribute to an obstruction of the flow of an increase in intraocular pressure as well as the production of aqueous humor [44, 58, 68, 69]. The gradual clouding of the lens that occurs because of prolonged exposure to light is the underlying cause of cataracts. This photooxidation is responsible for the breakdown of biological components of the lens, most notably crystallin. Molecules that possess crystalline structures and contain oxidized thiol groups tend to establish disulfide bonds, resulting in the agglomeration of crystals and the consequent onset of cataract formation. Oxidative stress is increased due to diabetes, which leads to damage to capillary cells and the microvasculature of the retina. An impairment in glyceraldehyde-3-phosphate dehydrogenase is another factor that contributes to the pathogenesis of diabetic retinopathy brought on by diabetes. This factor incorrectly activates significant biochemical pathways [46]. The medical condition known as Graves ophthalmopathy, or GO, is characterized by an abnormal eyeball bulging and is partially brought on by hyperthyroidism. It affects anywhere between 25% and 50% of Graves' patients. Roughly 5% of patients being treated have a condition that could harm their vision. Numerous pieces of evidence indicate that the pathogenesis of GO may involve oxidative stress [2]. The acceleration of essential metabolic processes and cellular oxidative metabolism in mitochondria results from excessive thyroid hormone production, resulting in an elevation in the generation of reactive oxygen species (ROS). The presence of an oxidative environment has the potential to induce cellular proliferation in fibroblasts located within the eye, which can lead to eye protrusion. Some research showed that active GO patients had higher levels of 8-hydroxy-20-deoxyguanosine (8-OHdG) in their urine and that GO orbital fibroblasts had higher levels of 8-OHdG, superoxide anions, malondialdehyde (MDA), and hydrogen peroxides. Among the elderly population, blindness ranks as the second most significant fear, following death. Common causes of blindness include ocular diseases like glaucoma, cataracts, and age-related macular degeneration (AMD). The development of cataracts is a significant challenge faced by the elderly population and has a considerable bearing on the cost of medical care. The lens is the component of the eye that is particularly vulnerable to oxidative damage. Cataracts are caused by the exposure of underlying epithelial cells to both exogenous and endogenous reactive oxygen species, in addition to cellular death and degeneration. The phenomenon above results in the cross-linking and aggregation of the crystallin proteins present in the lens, leading to the development of cataracts. The formation of cataracts is one of the age-related processes that is significantly impacted by oxidative stress. Because the type of cataract and its pigmentation are determined by the intensity and form of oxidative stress, this factor makes preventing cataracts more difficult. Dry eye syndrome is a prevalent ocular condition. It can cause eye pain, vision problems, and possible corneal surface damage. The fact that more and more people are going to ophthalmology clinics with symptoms of dry eye syndrome shows how important etiology is. It also shows how important it is to find out what causes dry eyes and stop doing those things. Surgeons should identify and assess preoperative risk factors while taking measures to prevent factors that may exacerbate this condition. Given its established status as a recognized syndrome in periorbital surgery, it is imperative to avoid this condition. The patient will be able to have a more straightforward recovery process if all potential risk factors associated with dry eye syndrome have been identified and considered. This will reduce the likelihood of the patient developing this complication. In 2007, according to the International Dry Eye Work Shop, dry eye disease was a condition that impacted both the tear film and the ocular surface. Symptoms of dry eye disease include pain, blurred vision, unstable tear film, and eye surface damage. Symptoms include a foreign body sensation, burning, itching, blurred vision, pain, and photophobia; scleral injection; ocular discomfort; and blurred vision. Non-healing epithelial defects, corneal thinning, infections, corneal perforation, and permanent vision impairment are some visual complications that can result from all these symptoms.

2 Different Types of Eye Infections

Bacterial eye infections: Microorganisms such as *Haemophilus influenza*, *Staphylococcus aureus*, *Streptococcus pneumonia*, *Pseudomonas aeruginosa*, *Escherichia coli*, *Bacillus cereus*, *Neisseria gonorrhoeae*, *Staphylococcus epidermidis*, *and Chlamydia trachomatis* are responsible for the eye infection [29]. The bacteria *Staphylococcus aureus* and *Staphylococcus epidermidis* are the most common causes of external eye infections. *C. trachomatis* is the pathogen that leads to trachoma. This condition is the primary contributor to ocular morbidity and is the foremost infectious etiology of blindness globally [83]. According to the World Health Organization's estimation, the global population of individuals affected by trachoma is approximately 146 million. Some symptoms associated with bacterial eye infections include irritation, burning, tearing, and typically a mucopurulent or purulent discharge. It is possible to wake up with your eyelids stuck together. This is especially common in the morning. Even though bacterial eye infections are typically considered self-limiting, if they are allowed to go untreated, they can progress into more severe conditions threatening the patient's vision.

Fungal eye infections: *Fusarium oxysporum*, *Aspergillus niger*, *Fusarium solani*, *Candida albicans*, *Aspergillus flavus*, and *Penicillium notatum* are some of the fungal species that can cause infections in the eye [27]. The treatment of these infections poses a significant challenge and has the potential to lead to blindness. Redness, blurred vision, and sensitivity to light are some of the symptoms. Amphotericin B and Natamycin are two antifungal agents that can be applied topically to the eye to treat fungal infections [11, 19].

Viral eye infections: Coxsackieviruses, Herpes simplex virus-1, and adenoviruses are the three most common culprits behind viral infections of the eye [40]. In developed countries, ocular infection caused by HSV-1 is the primary etiology of visual impairment. Herpes simplex type 1 accounts for more than 95% of ocular herpes infections. Currently, 51 human adenovirus serotypes are known, which can be broken down into 6 species (A–F). In particular, the eyes can become infected with species D. Eye infections caused by viruses are highly contagious and are passed on through direct contact, most commonly through the sharing of items that have been in connection with the eye secretions of an infected person. For instance, the virus can be passed on when someone who already has individuals come into contact with contaminated surfaces, such as door handles, after touching their eyes. Additionally, sharing objects in connection with an individual's eyes, such as towels or pillowcases, between two people can also facilitate transmission. Antiviral medications like acyclovir, famciclovir, and valacyclovir can shorten the infection's duration and lessen the severity of the symptoms. Glaucoma is a common form of ocular disease that affects people all over the world. The demise of retinal ganglion cells characterizes glaucoma, the formation of a hollow at the optic nerve head, and, typically, a rise in intraocular pressure (IOP). Although elevated, the etiology of ganglion cell degeneration and optic disc cupping has long been attributed to intraocular pressure (IOP). However, research on low-tension glaucoma has provided evidence that challenges this notion, indicating that IOP may not be a sufficient or necessary factor in developing the disease. The cause of this condition is still unknown. Other mechanisms involving the neurosensory retina have been proposed. Recent research has brought attention to the possible neurotoxic function of glutamate. It has indicated that apoptotic mechanisms could be implicated in the deterioration of ganglion cells [38]. Glaucomatous eyes of dogs, monkeys, and humans have been found to have abnormally high levels of glutamate in the vitreous. Many of its morphological features resemble those of human congenital glaucoma, including an expanded retinal surface, an atypical corneal endothelium exhibiting cellular degeneration, insufficiently differentiated cells with a collapsed trabecular meshwork, and an adhesion of the anterior aspect of the iris to the posterior cornea. It is believed that several stimuli can start or start the process of apoptosis in glaucoma. Therefore, elevated intraocular pressure (IOP) and the accumulation of organelles at the optic nerve head may be able to obstruct axoplasmic flow, thereby impeding the circulation of trophic factors [30]. Mitosis, conjunctival hyperemia, and the inflammatory response in the eye are comprised of the breakdown of the blood-aqueous barrier, resulting in the subsequent leakage of protein into the aqueous humor. The nature, duration, and strength of the noxious stimulus determine the extent of these reactions. Substance P and calcitonin gene-related peptide (CGRP) are two examples of C-fiber neurotransmitters that have been demonstrated to play an essential role in the ocular response to injury. Both neurotransmitters are involved in healing [26]. Recent research has identified pituitary adenylate cyclase-activating peptide (PACAP), a C-fiber neuropeptide that has been shown to play a role in ocular inflammation in rabbits. Any transmitter removed from local nerve fibers will diffuse into the anterior chamber because there is no barrier between the iris, the ciliary body, and the anterior chamber. This is because the anterior chamber is directly connected to the ciliary body. Because of this, research on the release of transmitters benefits significantly from using the eye as a model. Nitric oxide (NO) is a molecule with a short half-life that participates in various biological processes. Recent observations suggest that NO may significantly regulate ocular function in its physiological or pathophysiological aspects. Therefore, the presence of NOS activity in the anterior uvea of rabbits has been demonstrated. At the same time, NOS immunoreactivity has been observed by immunizing nerve fibers in the uvea of rat eyes. Both findings can be found in the literature. It was discovered that administering L-NAME to the rabbit via intravenous injection caused a reduction in the regional blood flow in the uvea. It is interesting to note that the activation of ocular C-fibers in response to a minor injury (infrared irradiation of the iris) appears to involve the participation of nitric oxide (NO).

Ocular injuries – Exposure to industrial chemicals like acid and alkali, radiation energy like ultraviolet light, and physical trauma all contribute to ocular injuries in the workplace. Although injuries to the eye caused by plant saps are uncommon, they can have serious consequences [42].

Conjunctivitis – Conjunctivitis is a medical condition characterized by mucous membrane inflammation that lines the eyelid's inner surface. Various things can cause conjunctivitis, such as viruses, bacteria, allergies, injuries, or other foreign substances. The following are some symptoms: itchiness, watery eyes, bloodshot eyes, pain, and occasionally blurred vision. Depending on what caused conjunctivitis in the first place, there are a few different treatments available. In most cases, antibiotics are prescribed for the treatment of bacterial infections. Eyewashes can also be made from several herbal combinations [9].

3 Major Eye Diseases Share Common Mechanistic Pathways

The most common causes of blindness can be broken down into four categories: age-related macular degeneration (AMD), cataracts, glaucoma, and other retinal diseases such as retinitis pigmentosa (RP) and diabetic retinopathy (DR) [74]. According to the findings of an epidemiologic study that the Eye Diseases Prevalence Research Group carried out, it is estimated that by the year 2020, there will be 30.1 million people in the United States with cataracts. Furthermore, it was estimated that 4.1 million individuals aged 40 and above in the United States would have diabetic retinopathy [15]. Approximately 2.95 million individuals in the United States would receive a diagnosis of age-related macular degeneration (AMD), and about 2.2 million individuals would receive a diagnosis of glaucoma. It is interesting to note that all these diseases are linked to aging and that the mechanisms that underlie their respective etiologies and pathophysiologies are similar in several respects. The pathways above encompass apoptotic factors, inflammation, and oxidative stress, offering valuable insights into areas that may be amenable to targeted interventions. Oxidative stress is induced by reactive oxygen or nitrogen species, and lipid peroxidation is responsible for the death of ocular cells in many eye diseases. Various pathological pathways are associated with inflammatory factors, including nuclear factor kappa B (NF-kappa B) and tumor necrosis factor-alpha (TNF-alpha). It is interesting to note that the mechanisms of these pathways often intersect with those of numerous botanical compounds. Oxidative stress causes the generation of reactive oxygen species, leading to their interaction with mitochondria, subsequently triggering the activation of the JNK pathway, ultimately resulting in the cell's death. This review will concentrate on AMD, DR, RP, and glaucoma as pathologies and the possible benefits of botanical compounds in preventing and treating these eye diseases. This is because all these conditions have a significant impact on populations all over the world.

Age-related macular degeneration (AMD): Age-related macular degeneration is a persistent retinal condition commonly observed in individuals aged 50 years or above. The degeneration of photoreceptor and RPE cells in the macula results in a loss of central vision, both of which are necessary to provide sharp and clear vision. Smoking, one's racial or ethnic background, and one's family medical history are various factors that have been identified as potential contributors to the development of age-related macular degeneration, a prevalent cause of blindness among elderly individuals on a global scale, according to numerous epidemiologic studies. Oxidative stress and inflammation are essential mechanistic pathways to the pathology [31]. As per the results of the National Eye Institute's Age-Related Eye Disease Study (AREDS), the consumption of dietary supplements that comprise antioxidants and zinc has the potential to diminish the likelihood of advanced AMD onset and consequent vision impairment to a considerable extent [4]. This research opens the door for discovering potentially beneficial natural compounds for managing and prophylaxis of age-related macular degeneration (AMD). The investigation of the active ingredients of various plants has revealed their potential as antioxidants and anti-inflammatory agents. These characteristics are crucial in managing and avoiding age-related macular degeneration (AMD) and other ocular ailments which have already been present.

Glaucoma: Glaucoma is a term that refers to a set of ocular disorders that may interrupt the transmission of visual stimuli from the eye to the brain [3]. Most glaucoma cases are attributed to optic nerve damage resulting from the apoptosis of retinal ganglion cells [78]. The medical term for the heightened force within the eye is intraocular pressure (IOP), frequently used in clinical settings. Glaucoma patients typically have their intraocular pressure (IOP) lowered using oral medications, eye drops, or surgery [47]. This is the primary treatment method for the disease. Even though the elevation of intraocular pressure (IOP) is considered a leading etiological factor in the development of glaucoma, many cases continue to worsen even after the eye pressure has been brought down to the normal range. Developing novel and creative approaches to damage prevention and mitigation and reducing intraocular pressure is essential in such situations. The role of apoptosis in the pathogenesis of glaucoma has been well established. Therefore, investigations into the efficacy of neuroprotective agents may yield promising results. Various chemical components are found in plants that exhibit neuroprotective qualities and have the potential to be helpful in the treatment and prevention of glaucoma.

4 Diabetic Retinopathy (DR) and Retinitis Pigmentosa (RP)

Individuals diagnosed with either type 1 or type 2 diabetes are susceptible to the development of diabetic retinopathy. Diabetic retinopathy (DR) is considered one of the primary causes of blindness worldwide, in addition to age-related macular degeneration and glaucoma. It is estimated that DR affects 140 million people [85]. Changes in the retinal blood vessels bring on the pathology in this case. There is a possibility that blood vessels will swell or leak, and the emergence of anomalous blood vessels on the retina's surface is a possibility. Four stages comprise the DR pathology: severe no proliferative retinopathy, moderate no proliferative retinopathy, mild no proliferative retinopathy, and proliferative retinopathy. Proliferative retinopathy is the final stage. No treatments are necessary for the first three stages of the disease; even so, patients in these phases are required to regulate their blood cholesterol, blood pressure, and blood sugar [21]. This is done to prevent the disease from progressing further in these stages. Laser surgery is required to stem fluid flow out of the eye when proliferative retinopathy is present. However, in severe bleeding, a vitrectomy may be necessary to remove blood from the macula (the central part of the retina) [85]. The laser treatment may help reduce the size of the abnormal blood vessels, but a vitrectomy is necessary in these circumstances. Another disease that affects the retina is called retinitis pigmentosa, or RP for short. This condition causes rod and cone cells in the retina to become damaged, resulting in impaired vision and, in severe cases, total blindness. Retinitis pigmentosa is a rare disease affecting only 1 in 4000 people in the United States [75]. Genetic predisposition is the primary factor that puts a person at risk for developing this condition. Because there are currently no treatments available for retinitis pigmentosa, the state is a significant source of concern, even though it is not as common as other eye diseases. However, results from earlier experiments conducted on mice suggested that high doses of antioxidants, such as vitamin A palmitate, may slow the progression of the disease. Numerous plant extracts have the potential to have significant effects on the treatment as well as the prevention of both RP and DR. However, because no studies have been conducted in a clinical setting, there are plenty of questions regarding the potential advantages of taking such supplements.

Cataract: A cataract is an eye disease that commonly affects older people and typically begins to manifest in people's eyes around the age of 60. The lens becomes cloudy due to this pathology because of the breakdown of proteins, which causes the lens to be affected. Damage to the lens causes vision to become blurry, as well as a reduction in the sensitivity to colors and shapes; the lens is an essential component for achieving focus on objects situated at varying distances, whether they are nearby or distant. It is known that several risk factors, including smoking, exposure to ultraviolet (UV) light, diabetes, eye injury, and family history, can increase the likelihood of developing cataracts [91]. Patients who suffer from cataracts can easily be cured by undergoing surgery. The cloudy lens causing the condition will be surgically removed during the procedure, and an intraocular lens will be inserted. It is estimated that cataracts cause 51% of the world's blindness. However, surgery can restore sight in most patients who undergo it; several nations lack access to sophisticated eye care services. Fortunately, new research suggests that limiting protein degradation in the eye by eating botanical compounds rich in antioxidants may help reduce the risk of cataracts.

5 Therapeutic and Adverse Effects of Herbal Remedies

5.1 Ginkgo *(Ginkgo biloba)*

GBE, an extract derived from the *Ginkgo biloba* plant, has been extensively documented in various sources and is made by processing the leaves of a tree estimated to have originated in China more than 250 million years ago. For centuries, *Ginkgo biloba* extract has been utilized in traditional Chinese medicine and is thought to help improve cognitive function. It is widely available in supplement form and is popular among people looking to improve their memory and focus. As we all know, *Ginkgo biloba* extract (GBE) is predominantly comprised of flavonoids and terpenoids, with a total of more than 60 distinct bioactive compounds, of which 30 are exclusive to GBE and cannot be found anywhere else in nature [16]. GBE is a safe and effective natural supplement with a variety of applications. It has been employed to treat various medical conditions, including depression, memory loss, anxiety, and fatigue. Research suggests that GBE can also help improve mental clarity and focus. In addition, it is an herbal supplement most frequently used by individuals who have reached the age of 65 or above [13]. GBE exhibits a high degree of tolerability and presents a limited incidence of adverse effects. Therefore, it can be safely used as a supplement for elderly patients looking for natural remedies for their health issues. It has been found that the pathogenesis of glaucomatous optic nerve degeneration remains largely unknown. However, evidence suggests that neuroinflammation, oxidative stress, and optic nerve ischemia may be involved in its development. GBE can act as an antioxidant and reduce these oxidative stress levels, thus potentially slowing down the progression of glaucoma. It also has anti-inflammatory properties, which can help reduce the inflammation in the optic nerve and improve patients' vision. According to research, gamma-butyrobetaine, or GBE, is a potent antioxidant that can protect tissues against damage caused by free radicals, just like other antioxidants, like vitamins C and E, do. These properties of GBE make it a promising option for treating glaucoma. GBE supplementation is safe and effective in early studies. Additional investigation is required to validate its effectiveness in the treatment of glaucoma. By contrast, GBE operates at the organelle level by providing molecular stabilization to the mitochondria. The fact that it has this property sets it apart from the other agents. During oxidative stress, it has been observed that alterations in mitochondrial function can render retinal ganglion cells more susceptible to the deleterious impact of oxidative stress [23, 49]. GBE has been found to increase the stability of the mitochondria and reduce the damage caused by oxidative stress, suggesting that it may be effective in treating glaucoma. Further research is needed to confirm its efficacy.

According to the findings of one study, the application of GBE has the potential to decrease the quantity of reactive oxygen species within cultured neuronal cells and protect the mitochondrial membrane. The vasodilatory properties of GBE have been discovered to have potential benefits for enhancing coronary and peripheral circulation. Additionally, GBE has been found to have rheological effects that could improve blood viscosity [33]. Both findings were made possible by the fact that GBE has vasodilatory properties. Additionally, GBE can lessen the number of active cells (such as glial cells) in low-grade inflammation [12]. GBE is regarded as a neuroprotective agent because of its antioxidant, vast regulatory, and anti-inflammatory properties. As a result, there has been a suggestion that it could serve as a viable treatment option for individuals with glaucoma. Researchers have conducted four randomized controlled trials (RCT) to investigate GBE's impact on glaucoma. The first study was a double-blind, placebo-controlled crossover experiment on patients with normal tension glaucoma (NTG) [73]. The participants were allocated randomly to either of the two intervention groups: 4 weeks of GBE administration followed by 8 weeks of placebo or the opposite. Even though GBE supplementation led to statistically better visual field (VF) indices compared to baseline, the observed VF indices did not exhibit sustained improvement after the washout period, and no statistically significant alterations in intraocular pressure (IOP) were noted. In contrast, [25], who conducted a similar cross-over study, the study did not observe any statistically significant variations in the visual field (VF) indices or intraocular pressure (IOP). The treatment order and duration were identical in both studies; however, the second study only included patients recently diagnosed with NTG and taking topical hypotensive agents. The fact that the first study was carried out on a population from Europe and the second study was carried out on a population from Asia both suggest that differences in patient populations' ethnicities might be another factor that contributed to the findings. Another randomized controlled trial has demonstrated that using GBE has been observed to elicit a favorable impact on blood circulation in individuals diagnosed with NTG. This proposition implies that GBE can potentially serve as a therapeutic intervention for NTG. Further research is needed to confirm the findings and assess the safety and efficacy of GBE as a treatment. Specifically, following 4 weeks of *Ginkgo biloba* extract (GBE) supplementation compared to a placebo, there were notable enhancements in ocular blood volume, blood flow, and blood velocity due to GBE intake. The results of this study indicate that *Ginkgo biloba* extract (GBE) could serve as a viable therapeutic option for individuals with normal tension glaucoma (NTG). However, further clinical trials are needed to confirm these results. Further studies should also be conducted to understand the long-term impact of GBE supplementation on NTG patients. In addition 2016, Dewi Sari and colleagues conducted a study on individuals diagnosed with primary open-angle glaucoma (POAG). The study involved the administration of GBE to the participants, and the researchers observed noteworthy enhancements in various parameters, including VF indexes, superior and inferior retinal nerve fiber layer thickness, malondialdehyde (a plasma-derived oxidative stress marker), and glutathione peroxidase (an antioxidant enzyme). These improvements were observed after 6 months of GBE administration compared to the placebo group. The study concluded that GBE could be a potential treatment option for POAG. It is noteworthy to mention that there were no reported negative consequences associated with the administration of GBE. Nevertheless, it is unknown whether, statistically speaking, there was a significant disparity in the level of improvement observed between the treatment group and the baseline group. Additional investigation is required to validate the efficacy of *Ginkgo biloba* extract (GBE) as a therapeutic intervention for primary open-angle glaucoma (POAG). Different dosage levels and treatment durations should be investigated to determine the optimal treatment protocol. Clinical trials should be conducted on the safety and effectiveness of *Ginkgo biloba* extract (GBE) in managing primary open-angle glaucoma (POAG). There are numerous benefits associated with Ginkgo. Studies have demonstrated the potential of Ginkgo to enhance memory, mitigate the likelihood of developing dementia, and ameliorate circulation. The safety and tolerability of this substance are widely acknowledged among the general population. Furthermore, Ginkgo has few known side effects and is widely available. GBE is typically well-tolerated, although its adverse effects should be considered, which are the most severe related to the fact that GBE acts as an antithrombotic. This means, it can increase the risk of bleeding, especially in those taking other medications that thin the blood. Therefore, it is important to consult a physician before taking Ginkgo to ensure there are no potential interactions with any other medications. There have been several case reports in which ocular complications have been identified, such as retinal hemorrhage and hyphema. In contrast, systemic effects have been identified, like subdural hematoma and subarachnoid hemorrhage have been reported in other case reports. Ginkgo should not be taken without consulting a doctor, as it may have adverse effects. These can range from ocular complications to systemic effects like subarachnoid hemorrhage. Following the doctor's advice on dosage and usage when taking Ginkgo is important. Nevertheless, it has been found that the incidence of bleeding is not significantly different between patients who take GBE and those who take a placebo (based on the results of the two randomized controlled trials (RCTs) that examined the effect of GBE on elderly patients [17].

Bilberry *(Vaccinium myrtillus)*: Since the sixteenth century, people have used bilberry fruit for its medicinal properties. About half of the glaucoma patients who responded to a survey about using herbal remedies for treatment said they took bilberry [77]. Anthocyanin, a type of flavonoid, is the bilberry component responsible for its medicinal properties [76]. Like GBE, bilberry's proposed therapeutic effect is based on the evidence that a neurodegenerative process characterizes glaucoma, and this substance has the potential to act as a neuroprotective agent and combat that process. This statement pertains to the augmentation of retinal ganglion cell density, the fortification of optic nerve architecture, the amelioration of retinal ganglion cell resilience against mechanical or ischemic perturbations, and the mitigation of neuroinflammatory processes. Several proposed mechanisms of action exist for bilberry, including 1) antioxidative properties, 2) decreased capillary fragility, 3) collagen stabilization and promotion of collagen production, and 4) prevention of the production and release of proinflammatory compounds. Research into the use of bilberry as a treatment for glaucoma has been conducted due to the positive effects that bilberry has on antioxidation and blood circulation. This study is a retrospective investigation examining the controlled impact of bilberry supplementation and GBE for 6 to 59 months. It found that those who took the supplements improved their visual acuity, while those in the control group saw a decline in their visual acuity [80]. Additionally, VF improved, but the statistical analysis revealed no significant differences in VF between the treatment and control groups. There was no discernible impact on IOP from the use of bilberry supplements. A randomized controlled trial lasted 24 months, and individuals who were administered anthocyanin supplements exhibited a decrease in VF mean deviation. Nevertheless, the reduction in mean deviation was notably inferior in the group receiving treatment compared to the group receiving placebo. In the studies discussed above, the use of bilberry was not associated with any adverse effects. Conversely, an overdose of bilberry can result in symptoms such as icterus, anemia, and cachexia.

Marijuana *(Cannabis sativa)*: Marijuana, also referred to as cannabis, has a long history of use for therapeutic purposes dating back thousands of years; however, its acceptance in Western medicine has a tumultuous past [1]. In 1937, the United States passed a law that made marijuana use illegal. The use of marijuana for medicinal purposes was made legal in California in 1996, making it the first state to do so. Since then, a cumulative count of 32 states and the District of Columbia have enacted legislation similar to California's, legalizing the medicinal use of marijuana. The increasing legalization of marijuana and a shift in social attitudes toward its use have led to a greater likelihood of glaucoma patients utilizing marijuana as a therapeutic alternative. Cannabidiol (CBD) and delta-9-tetrahydrocannabinol (THC) are the chemical compounds responsible for marijuana's physiological effects. Marijuana is made up of over 400 different compounds (CBD). There is a lack of knowledge regarding the mechanism of action responsible for the ocular effect that marijuana produces. The endocannabinoid system relies on the crucial G-protein receptors, cannabinoid receptor type 1 (CB1) and cannabinoid receptor type 2 (CB2). The CB1 receptor is situated in the ciliary body, muscles, trabecular meshwork, and Schlemm's canal, indicating its potential influence on the production of aqueous humor and the outflow of trabecular and uveoscleral fluids. On the other hand, the CB2 receptor is known to regulate cytokine release in the immune system [6]. It is hypothesized that marijuana possesses a neuroprotective property, among other mechanisms, reducing free radicals and inhibiting apoptosis. The effects of marijuana smoking on a limited number of healthy participants were investigated by Hepler and Frank, revealing a reduction of 30% in intraocular pressure (IOP). On the other hand, only about 65% of people would feel this [24]. The pressure-lowering effect lasts for only a short time – between 3 and 4 hours on average. THC can be administered through various routes, including inhalation, oral ingestion, sublingual administration, intravenous injection, and topical application. The topical application of cannabinoid extracts is not an optimal route of administration for treating glaucoma due to the low aqueous solubility and highly lipophilic nature of these extracts, which result in poor ocular penetration. This is contrary to what one might assume to be the case. The study conducted did not reveal any noticeable variation in the intraocular pressure (IOP) among the subjects who were administered with topical 1% THC or light mineral oil (control) within a short duration [36]. However, WIN55212-2, a more recently developed topical synthetic cannabinoid, lowers intraocular pressure (IOP) by 20–30% in glaucoma patients, but the maximal effect takes place after an hour [72]. The study demonstrated that synthetic cannabinoids, namely, BW29Y and BW146Y, when orally administered to patients with glaucoma, reduced intraocular pressure (IOP). The findings indicated that BW146Y was equally effective in reducing IOP compared to the consumption of THC through smoking marijuana. However, BW29Y did not exhibit any significant impact on IOP. Notably, CBD increased intraocular pressure (IOP) in mouse eyes at intervals of 1 and 4 hours following topical administration, in contrast to THC [48]. In a study conducted on humans, analogous outcomes about elevated intraocular pressure (IOP) were noted after the sublingual administration of greater cannabidiol (CBD) after a lapse of 4 hours. Marijuana use has been associated with various negative outcomes, including psychotropic effects such as cognitive impairment, euphoria, dysphoria, decreased coordination, decreased short-term memory, time distortion, and sleepiness. Cardiovascular effects such as tachycardia, hypotension, conjunctival hyperemia, and decreased lacrimation have also been linked to the use of this substance. Moreover, the potential hypotensive impact of marijuana on intraocular pressure (IOP) may be counteracted by the concomitant reduction in blood pressure, resulting in inadequate blood supply to the optic nerve and consequent ischemic alteration [50]. If you smoke marijuana, you risk developing emphysema and possibly lung cancer. Topical THC can cause corneal injury and local irritation in some people. These are some of the side effects specific to the route of administration. There is a need for additional concern regarding the possible addictive properties and the level of tolerance. This is a particularly concerning finding because for patients to attain round-the-clock intraocular pressure (IOP) regulation through marijuana consumption, they must engage in smoking activities approximately six to eight times within 24 hours. However, this frequency of use is likely to lead to the development of cannabinoid use disorder. The monthly cost is anticipated to be approximately $690 to keep this dose, a sizeable increase from the currently prescribed medications.

Curcumin: Curcumin, also called *Curcuma longa*, commonly known as turmeric, is a widely utilized spice in South Asian cuisine. Curcumin is a naturally occurring compound obtained from the *Curcuma longa* plant. It has been extensively used for its anti-inflammatory properties in managing various inflammatory disorders [23]. Curcumin is a polyphenol with lipophilic properties, rendering it insoluble in aqueous solutions; however, it can maintain its integrity in environments with an acidic pH, such as the human stomach [49]. Interaction with various molecular targets for inflammation is required for curcumin's anti-inflammatory effects. For instance, this is achieved through the regulation of enzymatic activity, including but not limited to nitric oxide synthase (iNOS), cyclooxygenase-2 (COX-2), and lipoxygenase. This is how it works to regulate the inflammatory response. Inflammatory cytokines such as monocyte chemoattractant protein (MCP), TNF, interleukins, and migration inhibitory protein are all produced in lower quantities when curcumin is present [18]. This study investigates the impact of curcumin, an anti-inflammatory and antioxidative agent, on light-induced retinal degeneration (LIRD) in rat models and retina-derived cell lines. Studies have demonstrated the anti-inflammatory and antioxidative properties of curcumin. Retinal neuroprotection was noted in rats supplemented with 0.2% curcumin for 2 weeks. Through the inhibition of NF-kB activation and the downregulation of inflammatory gene expression, curcumin protected the retina from the effects of LIRD. When used in experiments as a pretreatment for the utilization of retina-derived cell lines, namely, 661W and ARPE-19, curcumin has been observed to confer cellular protection against hydrogen peroxide (H_2O_2)-induced cell death. This protective effect is attributed to activating cellular protective enzymes, including HO-1 and thioredoxin [33]. These results were found in studies. The study revealed that the cytoprotective properties of curcumin against oxidative stress induced by H_2O_2 were augmented when human retinal cells were incubated with 15 M curcumin. This effect was attributed to the upregulation of HO-1 expression, which decreased reactive oxygen species (ROS) levels [12]. Curcumin can also change the expression of NF-B, AKT, NRF2, and growth factors, which stop inflammation and protect cells. Studies with real-life rat models have demonstrated that curcumin directly impacts ocular disorders. Curcumin has been shown to have protective effects. The present study investigates the impact of oral administration on Wistar albino rats with streptozotocin-induced diabetic retinopathy in recent DR studies. Studies have demonstrated the potential of curcumin to inhibit the development of cataracts in rat models induced by selenite ingestion, galactose ingestion, naphthalene exposure, and diabetes [25]. The inclusion of curcumin in the dietary regimen of diabetic rats was found to confer protection to their lens against cataractogenesis. This was attributed to preserving the chaperone-like function of lens-crystallin [14]. Curcumin has been linked to several unwanted side effects since it was first studied in 1976, even though it has shown some promise as a possible natural treatment. Alterations to chromosomes and DNA are the primary types that can occur at higher doses of curcumin. Curcumin has been identified as the preferred compound for managing and prophylaxis of age-related macular degeneration (AMD), cataracts, and diabetic retinopathy (DR). However, it is noteworthy that curcumin possesses antioxidative and anti-inflammatory properties that may result in adverse effects at elevated dosages.

Lutein and zeaxanthin: Understanding that macular pigments comprise carotenoids such as lutein and zeaxanthin is important. They are found in the macula, the part of the retina responsible for the sharpest central vision, and are essential for healthy vision. Lutein and zeaxanthin are believed to act as “sunglasses for the macula,” protecting it from the negative effects of blue light. Generally speaking, these carotenoid pigments are found in higher concentrations in the macula and retina of humans. Carotenoids are present in various fruits and vegetables, including but not limited to kale, spinach, and maize. They are frequently consumed in the form of dietary supplements. Research has suggested that augmenting the consumption of lutein and zeaxanthin can potentially mitigate the likelihood of age-related macular degeneration. In addition, they can be found in a wide variety of fruits and vegetables, including red grapes, kale, spinach, corn, kiwi, and kiwifruit [37]. Lutein and zeaxanthin are also found in egg yolks, oranges, and mangoes. They may also have a protective effect on the eyes from UV light. In addition, carotenoids have the potential to exhibit anti-inflammatory and antioxidative characteristics. Multiple epidemiological studies have demonstrated an association between decreased likelihood of age-related macular degeneration and elevated levels of lutein and zeaxanthin in the bloodstream. Notably, this relationship has been observed in studies such as the Age-Related Eye Disease Study (AREDS) and the Case-Control Study Group for Eye Diseases in the United States. This suggests that consuming foods rich in lutein and zeaxanthin may help reduce the risk of AMD [52–54, 55, 57]. Furthermore, supplementation with lutein and zeaxanthin has also reduced the risk of AMD. There is still some debate about the exact mechanism by which lutein and zeaxanthin work, but it has been hypothesized that carotenoids have the potential to safeguard the macula and outer retinal segments of photoreceptors against oxidative stress through the activation of an antioxidant cascade, which subsequently deactivates reactive oxygen species [24]. Furthermore, it has been proposed that lutein and zeaxanthin play a significant role in the metabolic processes of photoreceptor cells, thereby contributing to the preservation of optimal visual health. Research has indicated that lutein and zeaxanthin possess the potential to safeguard the macula against harm caused by blue light. For this hypothesis to be accepted, it is necessary to consider that lutein and zeaxanthin possess antioxidant properties. These antioxidants help to scavenge free radicals and protect the macula from oxidative damage. Thus, lutein and zeaxanthin play a vital role in maintaining healthy vision. In addition, lutein and zeaxanthin also perform the function of light filters within the eye, absorbing blue light before it reaches the retina, thereby protecting the retina from damage. Lutein and zeaxanthin can be found in many fruits, vegetables, and supplements, making them easy to obtain. A balanced diet with these nutrients is essential for maintaining healthy vision. As a result, the retina is protected from the harmful effects of LIRD, which can be caused by acute light exposure and exposure to highlight levels. The consumption of lutein and zeaxanthin has been associated with a potential decrease in the likelihood of developing age-related macular degeneration and cataracts. Taking these nutrients regularly can help improve vision and eye health. A study on cultured ARPE-19 cells indicates that providing lutein and zeaxanthin led to decreased photooxidative damage and suppressed gene expression linked to inflammation in RPE cells. The results of this study indicate that lutein and zeaxanthin can safeguard the eyes against oxidative stress and inflammation commonly linked to the deterioration of vision due to aging. This can significantly improve individuals' eye health and vision as they age. Furthermore, several epidemiological studies have revealed an inverse correlation between the amount of macular pigment and the likelihood of age-related macular degeneration (AMD) among older individuals. Therefore, augmenting the dietary consumption of lutein and zeaxanthin can potentially mitigate the likelihood of acquiring age-related macular degeneration (AMD). Supplementation with these two carotenoids has also been beneficial for reducing the risk of AMD. As per a recent clinical study that was published in an American Academy of Ophthalmology journal, the administration of lutein and zeaxanthin supplements exhibited an improvement in visual function and deceleration in the advancement of AMD pathology among patients diagnosed with early AMD [48]. Additionally, the research above revealed a positive correlation between the intake of lutein and zeaxanthin supplements and increased macular pigment optical density. This suggests that the intake of lutein and zeaxanthin supplements has been suggested to potentially mitigate the likelihood of age-related macular degeneration (AMD) by augmenting the optical density of the macular pigment. Zeaxanthin has also been shown to enhance the visual function of older male patients with AMD in clinical studies that were similar to this one. This indicates that lutein and zeaxanthin supplementation may have a protective effect against AMD and may also help improve the visual function of people already suffering from AMD. The consumption of lutein and zeaxanthin supplements may be a useful choice for preventing macular degeneration in the future [50]. Additional research is required to ascertain the precise dosage and duration necessary to elicit the protective effects of lutein and zeaxanthin supplementation. It is also important to consider the potential side effects of supplementing these carotenoids. Finally, diet is a key factor in determining an individual's risk for AMD and should also be considered.

Saffron: Because of its ability to reduce toxicity, saffron is one of the spices frequently used in traditional medical practices since it reduces toxicity. Saffron is recognized for its antioxidative and anti-inflammatory characteristics. It has been employed to treat diverse medical conditions, from dermatological disorders to gastrointestinal ailments. Additionally, it is believed to have mood-boosting effects. To provide it with its antiapoptotic properties, crocin and crocetin are the active ingredients that are present in them. These two compounds work together to reduce oxidative stress, thus providing saffron with its health benefits. Studies have also suggested that saffron has been found to possess potential health benefits in reducing the incidence of specific types of cancer and cardiovascular disease. The antioxidant carotenoids crocin and crocetin are known to prevent the onset of apoptosis by protecting cells from the effects of reactive oxygen species. Saffron also has anti-inflammatory effects, and consumption of certain foods may have a beneficial effect on reducing inflammation and promoting overall health. Additionally, saffron can help to reduce the risk of depression and improve cognitive function, concentration, and memory. There is evidence that crocin inhibits apoptosis in serum-starved PC12 cells exposed to hypoxic conditions, peroxidation of membrane lipids, and caspase-3 activation in serum-starved PC12 cells. Furthermore, research has indicated that saffron may have potential therapeutic benefits in addressing oxidative stress and inflammation in brain tissue, thereby indicating its potential utility in treating Alzheimer's disease. In addition, it also increases the levels of glutathione (GSH) in the body. It prevents the activation of the JNK pathway, which plays a crucial role in the downstream signaling cascade initiated by ceramide [45]. These findings suggest that saffron could be a potential therapy for Alzheimer's and other neurological diseases. Further research is needed to validate these findings and explore saffron's full potential as a therapeutic agent. It has been demonstrated that low GSH levels in the body lead to enhanced sensitivity of the cells to agents that induce apoptosis; dietary supplementation with saffron may potentially safeguard cells against damage and death by preserving glutathione (GSH) levels within the body

[51]. This suggests that saffron may exhibit a safeguarding impact against cellular damage and mortality caused by oxidative stress. Further research will be necessary to confirm these findings and to elucidate the exact mechanisms by which saffron exerts its protective effects. The addition of crocin to the RGC-5 retinal ganglion cell line inhibited oxidative stress in the cells, as shown by a decrease in the production of caspase-3 and -9, thereby preventing the death of the cells [84]. The findings of this investigation propose that saffron exhibits promising prospects in the prophylaxis and management of neurodegenerative ailments. Additional investigation is required to establish the effectiveness and safety of saffron for the aforementioned medical conditions. These findings were gleaned from research conducted on the RGC-5 cell line. Numerous studies have examined the potential impact of saffron as a dietary supplement, including clinical trials conducted on both human and rodent subjects. The effects of prolonged exposure to bright light were significantly mitigated in rats given saffron supplements. Falsini et al. [20] found that when saffron was supplemented at a dose of 20 milligrams per day for 90 days, the study observed notable enhancements in the macular photopic flash electroretinogram (FERG) parameters, specifically in terms of amplitude and modulation threshold [20]. This supports the idea that saffron may be effective as a natural remedy for treating eye diseases caused by prolonged exposure to strong light. Further research is needed better to understand the effects of saffron on eye health. The findings above were obtained through clinical trials on human subjects diagnosed with early age-related macular degeneration (AMD). The administration of saffron through oral supplementation resulted in a noteworthy enhancement in macular function. However, the precise mechanisms underlying the advantageous impacts of saffron on photoreceptors and bipolar cells remain unclear. Studies on rats have shown evidence of saffron's ability to inhibit cell death in situations where the rats were exposed to intense light. Based on the findings of these preclinical investigations, it can be inferred that saffron exhibits neuroprotective properties. These findings suggest the possibility of saffron as a prospective therapeutic intervention for addressing retinal degenerative pathologies. Additional investigation is required to comprehensively comprehend the impacts of saffron on ocular health and its prospective therapeutic applications. According to the findings of the clinical trials, adding saffron to one's diet may temporarily improve one's retinal function; age-related macular degeneration can manifest in its early stages. As part of the research, it has been demonstrated that the inhibitory effect of saffron on the proteolysis of water-soluble protein fractions within the lens can prevent the formation of selenite-induced cataracts in Wistar rats. This is attributed to saffron's antiapoptotic and antioxidative properties. Additional investigation is required to comprehend the precise mechanisms underlying the impact of saffron on ocular well-being and whether its benefits can be sustained in the long term. Additionally, it is imperative to consider all possible adverse effects of consuming saffron supplements. This is accomplished through the saffron's ability to prevent cell death. Even though the studies do not provide conclusive evidence for the potential neuroprotective properties of saffron in age-related macular degeneration (AMD) and its potential role in the prevention of cataracts, despite the need for further investigation, the data above exhibit potential

for the advancement of prophylactic and remedial applications of nutritional supplements in combatting the ailments above. Further research is needed to understand the exact mechanisms of saffron's neuroprotective effects and validate the efficacy of saffron as a therapeutic or prophylactic agent in treating the aforementioned ocular conditions.

Catechin – Catechin is a polyphenolic antioxidant frequently discovered in green tea [28]. Epigallocatechin gallate (EGCG), found in the most significant quantity in green tea, is a catechin that has powerful antioxidative effects. Researchers have shown in the past that previous research has demonstrated that the administration of EGCG in conjunction with sodium nitroprusside via intraocular injections resulted in a safeguarding impact on the retinal photoreceptors, which suggests that EGCG may have a beneficial effect on patient individuals who experience ocular ailments characterized by oxidative stress [86]. This protective effect was further confirmed in a recent study that found that EGCG improved the visual acuity of individuals diagnosed with age-related macular degeneration. Thus, EGCG is a promising therapeutic agent for ocular diseases. Catechins have a few different mechanisms of action, including mitigating glutamate-induced toxicity via lipid peroxidation and protein modification, eliminating reactive oxygen species, and the oxidative changes of low-density lipoproteins. The oral administration of EGCG has been shown to the findings of the study indicate that the administration of the substance in rat models may have a potential therapeutic effect in mitigating light-induced retinal neuronal degeneration, thereby potentially serving as a prophylactic measure against photoreceptor cell death [87]. Similarly, when catechin was given to Sprague-Dawley rats with cataracts caused by N-methyl-N-nitrosourea, the cataract-caused apoptosis in the lens epithelium was stopped. This may help treat or prevent cataracts in people. Additionally, RPE cell migration and adhesion can be inhibited by EGCG, which offers a potential preventative action against AMD. As a result, catechin is a compound that should be considered for treating and preventing diseases like AMD.

Ginseng – The Panax ginseng plant's root, commonly called ginseng in traditional Chinese medicine, has been widely utilized in this field for an extended period. It is believed to help boost energy levels and improve overall well-being. It is also thought to possess antioxidant and anti-inflammatory properties, which may help to protect the body from disease. There are several active components in ginseng, but the most important are ginsenosides. Ginsenosides are thought to be responsible for many of the purported health benefits of ginseng, as they appear to have antioxidant, anti-inflammatory, and immune-modulating effects. Empirical evidence suggests that these substances positively impact cognitive abilities and physical aptitude. Ginsenosides, a class of steroidal saponins, have demonstrated the ability to impede glutamate-induced neurotoxicity, lipid peroxidation, and calcium influx into cells. These effects may contribute to preventing apoptosis [78]. The ginsenoside saponins Rb1 and Rg3 have been found to inhibit apoptotic cascades in cells. There has been evidence that retinoblastoma protein 1 (Rb1) and retinoblastoma protein 3 (Rg3) inhibit NMDA glutamate receptor activity, thus suppressing tumor necrosis factor-alpha production and providing neuroprotective effects to cultured cortical cells, which have been shown to have a wide range of pharmacological effects [7], targeting a wide range of tissues. This suggests that Rb1 and Rg3 could be potential drug targets for neurodegenerative diseases. Further research on their effects on other tissues is warranted. The medicinal use of ginseng dates back several millennia. In recent years, studies have found that ginseng may benefit the brain, including improved cognition and memory. It is thought that ginseng may work by increasing levels of Rb1 and Rg3, which could provide new insights into how it may be used as a treatment. As a result of oral administration of Korean Red Ginseng (KRG) to patients with glaucoma during clinical trials, there have been significant increases in the amount of blood flowing to the retina in the temporal peripapillary areas of the eye. The potential outcome of this could be a decrease in the intraocular pressure and a postponement in the advancement of the ailment. Further research is needed to confirm the efficacy of KRG in treating glaucoma. To prevent glaucoma, augmenting the blood flow to the retina in the initial phases of the ailment could potentially yield advantageous outcomes [39]. Further research should also focus on the effects of KRG on other ocular diseases. Additionally, further studies are needed to identify the long-term efficacy and safety of KRG for the treatment of glaucoma. It is commonly thought that glaucoma is characterized by the enlargement of blood vessels and diminished blood circulation, which are significant risk elements that may lead to optic nerve damage in individuals with glaucoma. The efficacy of KRG in lowering intraocular pressure has been demonstrated, potentially mitigating the likelihood of optic nerve impairment. Further research should provide more evidence to support this hypothesis. The evidence indicates that ginsenosides derived from ginseng may potentially contribute to preventing age-related macular degeneration (AMD) due to their ability to inhibit TNF-a. This could suggest that KRG can be used to reduce the risk of AMD. Further studies are needed to confirm this hypothesis. Undoubtedly, inflammation is a principal risk factor associated with age-related macular degeneration (AMD). Therefore, further research into the effects of KRG as an anti-inflammatory agent could be beneficial in reducing the risk of AMD. Additionally, research into the long-term safety of KRG would be beneficial for determining its potential as an AMD prevention strategy. It is believed that patients suffering from conditions such as age-related macular degeneration, glaucoma or cataracts may benefit from ginseng's antiapoptotic and antioxidant properties. This could potentially assist in the retardation of the advancement of the ailment, mitigation of symptoms, and enhancement of the general visual capacity. Further investigation is required to establish the effectiveness of ginseng as a treatment for age-related macular degeneration (AMD). It is noteworthy that extensive research has been conducted on the advantageous impacts of ginseng for individuals with diabetes, encompassing the mitigation of blood glucose levels, regulation of weight gain, and augmentation of insulin secretion. Therefore, ginseng may have similar benefits for those suffering from AMD. It is also important to consider the safety of ginseng when using it as an AMD treatment. Further research is needed to assess the potential risks of using ginseng for AMD treatment. In a study that was conducted very recently, it was shown that ginseng treatment significantly reduced the amount of retinal oxidative stress that was caused by diabetes in mouse models, which were associated with diabetes. Nonetheless, it remains ambiguous whether this phenomenon holds for human beings. More studies need to be conducted to understand the long-term effects of ginseng on AMD. Many of the findings in this study were based on the fact that ginseng has antioxidative properties. Ginseng also has anti-inflammatory effects, which could help reduce the symptoms of AMD. Additional investigation is required to comprehensively comprehend the potential efficacy of ginseng as a treatment for age-related macular degeneration. Even though researchers have given more attention to the effects of ginseng on AMD and diabetes than to the effects of ginseng on cataracts caused by selenite in rats, ginseng has also been shown to have the ability to reduce the severity of cataracts caused by selenite. This suggests that ginseng may have potential benefits for people with cataracts. However, more studies are needed to confirm if ginseng can be used as a potential cataract treatment. Furthermore, a team of Korean researchers led by Lee et al. [43] have identified the non-saponin component of ginseng as one of the key agents responsible for preventing cataracts. Their research found that the non-saponin component of ginseng could block the damage caused by oxidative stress on the proteins in the eyes and ultimately reduce the risk of cataracts. Further research is required to validate the effectiveness of ginseng as a plausible remedy for cataracts. As a supplement, ginseng is an extremely promising compound for further investigation into the treatment of macular degeneration, diabetic retinopathy, and even cataracts in the future. In addition, ginseng has also been linked to improved visual acuity and contrast sensitivity, indicating potential benefits for treating age-related vision loss. Further research is needed to determine the full potential of ginseng for eye care. This is because ginseng is renowned for its potent antioxidant properties. Ginseng also contains compounds that are thought to have therapeutic effects on the eyes and can help protect the eyes from oxidative stress. It is also believed to reduce inflammation and improve blood circulation, benefiting eye health.

Resveratrol – Researchers have suggested that red wine's protective effects on human health could explain why France has a lower mortality rate than other countries due to the lack of cardiovascular diseases in the country. Researchers suggest this is linked to the French consuming much more red wine than other countries. This has led to studies into the potential health benefits of moderate red wine consumption. The findings indicate that red wine may confer health advantages, including a decreased cardiovascular disease risk. In red wines, large amounts of polyphenols can be found in them. Polyphenols possess antioxidative and anti-inflammatory characteristics, potentially mitigating the susceptibility to specific ailments. However, it is important to remember that these benefits are only achievable if red wine is consumed in moderation. It is widely believed that polyphenols are a class of compounds that are capable of exhibiting a variety of properties, including this phenomenon that involves the suppression of platelet aggregation, the production of eicosanoids that promote inflammation and blood clotting, and the hindrance of endothelin synthesis which triggers vasoconstriction. These properties can reduce the risk of coronary heart disease and other cardiovascular diseases. However, drinking too much red wine can have unwanted side effects, including an increased risk of liver disease, high blood pressure, and obesity. Additionally, too much alcohol can lead to impaired judgment, poor impulse control, and long-term health issues. Therefore, it is important to consume red wine responsibly in moderation. It is well known that red wines have particular health benefits for the cardiovascular system. Therefore, it is important to consume red wine in moderation and combine it with a balanced and healthy diet. Regular physical activity and other lifestyle factors can also help to maximize the health benefits of drinking red wine. The administration of resveratrol in mice that were subjected to a high-fat diet resulted in a noteworthy improvement in insulin sensitivity, a reduction in insulin-like growth factor-1 (IGF-I) levels, and an elevation in the activity of AMP-activated protein kinase (AMPK) and peroxisome proliferator-activated receptor gamma coactivator 1alpha (PCG-1alpha), which in turn has been shown to result in significantly longer survival and health [8]. The results of this study indicate that resveratrol could potentially and significantly impact the preservation of metabolic equilibrium and the mitigation of metabolic disorders. Further studies are needed to understand resveratrol's effects on metabolic health better. Additionally, it is important to consider the potential side effects of the long-term use of resveratrol supplements. Furthermore, resveratrol has also been shown to reduce inflammation and improve mitochondrial health. Thus, it is possible that resveratrol could be beneficial for overall health and well-being. However, more research is needed before any definitive conclusions can be made. The researchers also found that supplementation with resveratrol reduced oxidative stress, vascular endothelial growth factor, and the development of early vascular lesions in rat and mouse models of diabetes [32]. However, there is still debate over the efficacy of resveratrol as a supplement. Additionally, more research is needed to determine the potential long-term effects of resveratrol supplementation. In patients with diabetic retinopathy, this quality may be helpful due to its ability to delay or even stop the death of cells. Despite these potential benefits, resveratrol must be studied further before it can be recommended as a treatment for any condition. Further research is needed to confirm the potential benefits of resveratrol supplementation. As a result of inhibiting CHOP and IRE1 expression, resveratrol prevented injury-induced capillary degeneration and endoplasmic reticulum stress in rats. However, further studies are needed to determine the long-term effects of resveratrol supplementation. Additionally, studies should be conducted to assess the efficacy of resveratrol in humans. It was demonstrated by the fact that it could protect the body against both conditions. Thus, it is likely that resveratrol could be used as a therapeutic agent to treat conditions associated with capillary degeneration and endoplasmic reticulum stress. Further research is needed to explore the potential of this promising compound further. There is some evidence that resveratrol may be able to be a novel drug in terms of treating vascular dysfunction in the retina [71]. However, animal studies have shown conflicting results. More research is needed to confirm the effectiveness of resveratrol in humans. Additionally, further investigation is required to assess the compound's safety over an extended period. This is because retinal ischemia plays a significant role in the development of closed-angle glaucoma and diabetic retinopathy, as well as other types of eye diseases. Thus, it is imperative to evaluate the effectiveness of resveratrol and appraise the possible hazards linked to its prolonged consumption. Further research is needed to clarify the effects of resveratrol on eye health. There is no doubt that resveratrol is an efficient antioxidant and that the potential benefit of this substance lies in its ability to mitigate the oxidation of low-density lipoproteins (LDLs), thereby conferring cellular protection against the deleterious effects of free radicals. However, more research is needed to fully understand the potential risks of the long-term use of resveratrol on eye health. Additionally, further studies should be conducted to determine the optimal dosage for humans. Additionally, the compound resveratrol has exhibited potential in promoting increased blood flow, which may be a preventative measure against vessel damage and apoptosis of optic nerve cells in individuals with glaucoma. The results of these studies could help guide further research into the potential therapeutic uses of resveratrol for eye health. Additionally, these findings could be used to develop new treatments for glaucoma and other eye diseases. It has been shown that resveratrol, the induction of heme-oxygenase-1 and the inhibition of pro-oxidant intracellular heme effects, neuronal cell cultures following strokes were observed to exhibit a mitigated response, provide neuroprotection and demonstrate a novel pathway for cellular neuroprotection [90]. These findings provide evidence of the potential for resveratrol to treat glaucoma by protecting the neural cells from oxidative damage. Further research is needed to investigate the cellular mechanisms of resveratrol-mediated neuroprotection further. The administration of resveratrol as a supplement to rats resulted in a reduction of malonyl dialdehyde levels and an increase in glutathione levels in the lens. This study also revealed that resveratrol supplementation effectively suppressed selenite-induced oxidative stress and the development of cataracts. Further studies are needed to uncover which pathways are regulated by resveratrol and how to optimize the dose and timing of resveratrol for maximal neuroprotection. Additionally, human safety and efficacy trials are needed to confirm these findings. Because oxidative stress plays a crucial role in the pathophysiology of significant ocular ailments, including cataracts, age-related macular degeneration, glaucoma, and diabetic retinopathy. The characteristics of resveratrol may hold significant potential in creating innovative therapeutic approaches and preventative measures for these conditions. Further research is needed to explore the full potential of resveratrol as a therapeutic and preventive agent for major ocular diseases. Studies should focus on the long-term administration of resveratrol and its effects on ocular health. Ultimately, these studies may lead to novel treatments and preventive interventions. *Salvia miltiorrhiza*, also referred to as Asian red sage or Danshen, comprises salvianolic acid B. This potent polyphenolic antioxidant is water-soluble and possesses anti-inflammatory properties. Danshen is also known as *Salvia miltiorrhiza*. A diabetic retinopathy mouse model was used in a recent investigation conducted to examine the impact of intravenous administration of Danshen. Ischemia of the blood capillaries is one of the most serious complications that can arise from DR. This condition is characterized by a change in structure; the thickening of the basement membrane of the capillaries induces the condition. Under such circumstances, the prompt elimination of oxygen radicals following ischemia is hindered, which destroys the permeability membrane and the formation of edema due to the destruction of nerve cells due to lipid peroxidation. This can permanently damage the capillaries, decreasing the organ's oxygen supply. Furthermore, this can cause organ failure and other serious health issues. The administration of Danshen to hypoxic and ischemic retinal tissues may facilitate the restoration of blood-oxygen transportation, enhance the absorption of retinal hemangiomas, and consequently avert visual impairment if it is injected into the tissues of the retina that are suffering from hypoxia and ischemia. There is also evidence that Danshen may help patients with diabetes maintain healthy blood sugar levels by scavenging free radicals [88]. There has also been evidence that Danshen is also beneficial for treating glaucoma, with intravenous use of Danshen after glaucoma treatment resulting in a reduction of retinal ganglion cell damage as a result of intravenous use of Danshen. Danshen has been suggested to have the potential to stabilize the visual field during the middle to late stages of glaucoma based on the outcomes of prior clinical trials. In other research, in rabbits, Danshen has demonstrated the ability to prevent the mortality of retinal ganglion cells and suppress the activation of nuclear factor kappa B (NF-kB) by the inflammatory protein TNF-[89]. Like ginseng, Danshen's neuroprotective mechanism may operate through the inhibition of NMDA receptor antagonist activity, similar to how ginseng protects neurons. This suggests that Danshen could be a prospective therapeutic intervention for neurodegenerative disorders, including but not limited to Alzheimer's and Parkinson's disease. Additional investigation is required to ascertain the effectiveness and safety of Danshen for human consumption. The results of preclinical studies conducted on patients suffering from ocular diseases involving oxidative stress have shown promising results for Danshen. Clinical trials conducted in humans are now needed to explore the therapeutic potential of Danshen further. If results prove successful, Danshen could provide an effective and safe treatment for neurodegenerative diseases. Among the most common diseases are diabetic retinopathy, age-related macular degeneration, and cataracts. Danshen has been shown to reduce inflammation in the eye's blood vessels. It also helps to improve vision and protect the retinal cells from further damage caused by oxidative stress. Furthermore, it could potentially mitigate the likelihood of ocular disease-induced visual impairment.

Quercetin **–** Researchers have paid the most attention to quercetin as one of the flavonoids, and a diverse array of plant-based sources, including but not limited to Brassica vegetables, black and green teas, and various types of berries, contain this particular nutrient, which are all great sources of this flavonoid. Quercetin is acknowledged for its anti-inflammatory characteristics and has been associated with a decreased susceptibility to specific categories of cancer and cardiovascular ailments. Additionally, it may also have the potential to improve cognitive performance. Researchers found that populations with diets rich in flavonoids, such as those located in the Mediterranean region, tend to have lower rates of cardiovascular disease and are more likely to live longer than those with diets low in flavonoids [22], which sparked an interest in flavonoids. Studies have suggested that flavonoids can improve memory, focus, and attention. Moreover, flavonoids have also been linked to improved mood, enhanced cognitive function, and reduced cognitive decline. Much research has been conducted on quercetin's anti-inflammatory and antioxidative properties [82]. However, there are currently no medications based on quercetin; these products have received approval from the regulatory body known as the Food and Drug Administration. Quercetin is a type of flavonoid that has been found to have many health benefits. More research is needed to understand the potential benefits of quercetin and how it can be used to treat various illnesses. Research on the protective effects of quercetin was demonstrated in the retinal cell line ARPE-19 by demonstrating its ability to inhibit proinflammatory molecules and directly inhibit the intrinsic apoptosis pathway of the cells. Further research is needed to understand the effects of quercetin on human cells and its potential use as a therapeutic agent. Currently, quercetin can be found in various supplements and dietary sources. The findings of an in vitro study involving RF/6A rhesus choroids-retinal endothelial cells indicate that the administration of quercetin led to a dose-dependent reduction in the migration and tube formation of the cells. These results suggest the potential of quercetin to be used as an anti-angiogenic agent in the treatment of retinal diseases. Further studies need to be conducted to evaluate the effects of quercetin on human cells in vivo. Those two processes are crucial to the phenomenon of retinal angiogenesis, which is a distinctive hallmark of age-related macular degeneration and is the subject of inquiry. Therefore, quercetin could be a promising therapeutic agent for treating age-related macular degeneration. Further research should also be done to evaluate the efficacy and safety of quercetin for other retinal diseases. The administration of quercetin after oxidative damage exhibited a dose-dependent reduction in cellular damage and senescence. The findings of this investigation indicate that quercetin exhibits promise as a viable therapeutic intervention for managing age-related macular degeneration and other retinal pathologies. Additional investigation is required to comprehensively comprehend the effectiveness and safety of quercetin for these medical conditions. Additional research involving human RPE cells grown in culture further demonstrated that this was the case. Quercetin also showed protective effects on the human RPE cells against oxidative stress, indicating its potential as a therapeutic agent for retinal diseases. Further research should be conducted to investigate the effects of quercetin on human RPE cells in vivo. It has also been shown that quercetin (along with other natural flavonoids) significantly decreased the production of reactive oxygen species (ROS) in retinal cell cultures that were subjected to oxidative stress as a result of ascorbate and $Fe+2$ [5]. This suggests that quercetin may be useful in treating and preventing retinal diseases caused by oxidative stress. Further research should explore the potential of quercetin to reduce ROS production in vivo. According to a study, human umbilical vein endothelial cells (HUVECs) were used to determine the effect of quercetin in reducing human umbilical vein endothelial cell viability following exposure to oxidants. The study's findings indicate that quercetin can mitigate oxidative stress and cellular apoptosis in HUVECs. The results of this study indicate that quercetin may possess the ability to act as a potent agent in mitigating oxidative stress and safeguarding the retina against harm. Researchers have also studied the effects of quercetin on laser-induced choroidal neovascularization in vivo, finding that quercetin significantly reduced the size of the choroidal neovascularization compared to untreated controls when compared to untreated controls. These results demonstrate the potential for quercetin to be an effective treatment for reducing the risk of vision loss caused by laser-induced choroidal neovascularization. Further research is needed to determine the long-term efficacy of quercetin in clinical settings. Quercetin was observed to impede the expression of heat shock proteins, particularly HSP72, in rat models of glaucoma, thereby hindering the neuroprotective properties of these proteins in the models above. This suggests that quercetin may have a detrimental effect on the vision of glaucoma patients. Therefore, further research is necessary to understand healthfully quercetin's implications on vision. Even though there is some evidence that quercetin may have anticataract effects, scientists have not yet understood the exact mechanisms by which this may be achieved [35, 61, 64, 65, 67, 70]. Hence, additional investigation is required to explore the potential of quercetin in safeguarding visual function. Additionally, more studies should be conducted to evaluate the safety of quercetin in long-term use. The compound also acts as an antioxidant, modulating glycation, epithelial cellular signaling, calpain protease activity, and sorbitol-aldose reductase activity [79]. Additional research should be conducted to explore the other potential benefits of quercetin. Furthermore, the impact of quercetin on different age groups should be studied to determine its efficacy in different populations. In fact, due to its ability to inhibit aldose reductase, quercetin served as a gold standard for cataract inhibition in 2011 when Gacche and Dhole tested the abilities of other flavonoids that also showed similar abilities. Further research on the efficacy of quercetin for cataract inhibition is needed to determine the full extent of its potential to improve vision. Long-term studies should be conducted to assess the impact of quercetin on different age groups over time. The precise mechanisms underlying the impact of quercetin on cataract formation and its effect on the eye's lens remain unclear, despite its status as the most prevalent flavonoid in the human diet and the existence of multiple plausible pathways for its physiological activity [41, 81]. More research is needed to understand how quercetin affects eye health and vision. In particular, more research is needed to understand its long-term effects on different age groups. Also, studies should be conducted to determine whether quercetin can prevent or treat cataracts. Although there is significant evidence associating quercetin and other flavonoids with medical benefits, particularly regarding ocular health, additional research is necessary to establish the suitability of this promising compound as a therapeutic intervention for ocular inflammatory diseases and cataracts. Additional research is warranted to explore the impact of quercetin on various ocular ailments. It is recommended that clinical trials be undertaken to assess the safety and efficacy of quercetin in the treatment of various ocular conditions.

6 Conclusion

Finding evidence to support the use of complementary and alternative medications is difficult. Still, nutritional supplements and herbal remedies are easily marketed in popular culture as promoting health benefits. While these medications may not have been proven effective, many people take them to improve their health. However, it is essential to understand the risks associated with taking these medications and the potential for interactions with other medicines. As a result of patients with eye diseases frequently using complementary and alternative medicine (CAM) products, ophthalmologists must be knowledgeable about the appropriate use of these products in common eye diseases. They should also be aware of potential adverse events or interactions associated with using CAM products. Furthermore, healthcare providers should be mindful of the potential for additive effects when CAM products are used concomitantly with traditional medications. Several studies have shown that specific antioxidant vitamins and zinc can significantly reduce the progression of age-related macular degeneration (AMD) in patients by slowing retina deterioration. Therefore, healthcare providers should be knowledgeable about the potential benefits of CAM products and should consider recommending them to patients who may benefit from them. Additionally, healthcare providers should educate patients on adequately using the potential risks associated with using complementary and alternative medicine (CAM) products. As a result of the reviewed studies, antioxidant vitamins and mineral supplements are not currently supported by evidence in the literature about their effectiveness in preventing or treating cataracts in healthy individuals. Therefore, healthcare providers should discourage their use, as they may be a waste of money and resources. Patients should be aware that using said products carries potential hazards, and there is a dearth of substantiating evidence regarding their effectiveness. In both diabetic retinopathy and glaucoma research, these products demonstrate no benefit. Patients should be encouraged to discuss with their healthcare provider any alternative treatments that are evidence-based and offer a better chance of success. Furthermore, healthcare providers should stay current on the latest research to make the best patient decisions. The challenges associated with later studies on the effects of antioxidants on cataract development and progression are attributed to several factors. These include the variability within and between study populations, inadequate definition of subjects' lifetime exposure to antioxidants, reliance on self-reported dietary intake, imprecise assessment of antioxidant nutrients, variations in the doses and formulations of vitamins used in different studies, and the duration of the follow-up period required to evaluate cataract development and progression. Therefore, it is difficult to draw definitive conclusions from studies on the association between antioxidant vitamins and cataract risk. Further research is needed to understand the potential role of antioxidants in cataract prevention. In terms of determining whether herbal remedies can be used in the treatment of eye diseases, currently, there exists a dearth of observational studies or prospective clinical trials that satisfy the prerequisites for level I or level II evidence. Thus, more research is necessary to assess the efficacy and safety of herbal remedies in treating eye diseases. Further studies should also evaluate the role of antioxidants in cataract prevention. Consequently, it presents a challenge to formulate any definitive inferences regarding the effectiveness of herbal remedies in addressing ocular ailments. Given the lack of reliable evidence, caution should be exercised when using herbal remedies to treat eye diseases. Further research should focus on the potential short- and long-term effects of herbal remedies on eye health.

Additionally, more research is needed to explore the potential of herbal remedies in preventing eye diseases.

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26 Assessment of the Phytochemical Constituents and Metabolites of Some Medicinal Plants and Herbal Remedies Used in the Treatment and Management of Injuries

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Abstract

This chapter discusses the phytochemical constituents and metabolites in some medicinal plants and herbal remedies used in the treatment and management of injuries. A meta-analysis was conducted to identify the studies that investigated the chemical composition and biological activity of these natural remedies. The medicinal plants and herbal medicine included in the review were chosen for their traditional use in wound healing and were found to contain a range of phytochemicals with known medicinal properties, such as flavonoids, terpenoids, alkaloids, and saponins. These phytochemicals have been proven to exhibit antioxidant, anti-inflammatory, and wound-healing activity in various in vitro and in vivo studies. In addition, the review identified several metabolites produced by these medicinal plants and herbal medicine that may contribute to their therapeutic effects. Overall, the results of this review suggest that medicinal plants and herbal medicine may be valuable sources of phytochemicals and metabolites with medicinal properties, and encourage their usage in the treatment and management of wounds. However, further research is needed to fully understand the mechanisms of action and potential therapeutic effects of these compounds. This involves research into the best doses and combinations of natural remedies for various sorts of wounds, along with research into their safety and effectiveness in clinical settings.

Keywords

Wound healing · Antioxidant activity · Traditional medicine · Therapeutic effects · In vitro · Flavonoids

Abbreviations

5-LOX	5-lipoxygenase
ALT	Alanine transferase
AMPK	Activated protein kinase
AST	Aspartate aminotransferase
BC	Before Christ
bFGF	Basic fibroblast growth factor

CADs	Cichoric acids
CC	Creative Commons
CFE	Calendula flower alcohol extracts
CFS	Chronic fatigue syndrome
COX-2	Cyclooxygenase-2
DCQA	Dicaffeoyl quinic acid
DPPH	2,2-Diphenyl-1-picrylhydrazyl
EGF	Epidermal growth Factor
ET-1	Endothelin-1
GABA	Gamma amino butyric acid
HB-EGF	Heparin-binding epidermal growth factor-like growth factor
HPA axis	Hypothalamic–pituitary–adrenal axis
IFN-γ	Interferon gamma
IGF-1	Insulin-like growth factor-1
IL-1β	Interleukin-1 beta
IL-6	Interleukin-6
KGF	Keratinocyte growth factor
MCF	Monocyte chemoattractant protein
MIP	Macrophage inflammatory protein
MMP	Matrix metalloproteinase
MPP	1-Methyl-4-phenylpyridinium
MTT	(3- [4,5-Dimethylthiazol-2-yl]-2,5 diphenyl tetrazolium bromide)
NF-yB	Nuclear factor y B
NF-κB	Nuclear factor kappa B
NK cells	Natural killer cells
NO	Nitric oxide
NrF2	Nuclear factor erythroid 2-related factor
O/W	Oil-in-water
PCL	Polycaprolactone
PDGF	Platelet-derived growth factor
PGE2	Prostaglandin E2
ROS	Reactive oxygen species
STAT3	Signal transducer and activator of transcription 3
TGFs	Transforming growth factors
TGF-α	Transforming growth factor alpha
TGF-β	Transforming growth factor beta
TIMPs	Transforming inhibitors of metalloproteinases
TNF-α	Tumour necrosis factor alpha
TRPA 1	Transient receptor potential ankyrin 1
TRPV 1	Transient receptor potential cation channel subfamily V member 1
UVB rays	Ultraviolet B rays
VEGF	Vascular endothelial growth factor
VPF	Vascular permeability factor

1 Introduction

The largest organ in the body is the skin. The protection against physical harm from the environment, control of body temperature, and sensitivity to stimuli are just a few of the important jobs that skin performs. It serves different functions, and any harm or injury makes the organism more vulnerable to other biological and physical dangers, leading to the wound. A wound is a type of injury to the human body which forces the skin to break and subsequently damages the affected underlying tissue [1].

Herbal medicines, also known as phytomedicine, include herbs (such as seeds, leaves, bark, and roots), herbal materials (such as essential oils, and fresh juices), herbal preparations (such as tinctures, extracts, and fatty oils), and completed herbal products. It is a chemical compound combination from multiple chemical classes, the bioactivity of which combine to produce effects as a result of combining activities of the individual components [2, 3]. Additionally, other inert compounds serve as ballast or fillers. The medicine in question is therefore a complex mixture, both chemically and in relation to how different bioactivities interact to generate a particular effect. Contemporary synthetic drugs are very expensive, sometimes dangerous, and have withdrawal symptoms that come back. Nonetheless, because they come from a natural source and already have been utilized from the beginning of time, herbs are typically claimed to be safe [2]. Herbal medicine has been utilized for centuries to treat an array of ailments [4, 5], and it remains a popular choice for people seeking natural remedies [6]. At the locations of injury, wounds are the rupture of the anatomical and functional continuity of cells and tissues. They may be brought on by mechanisms that damage the tissue physically, chemically, microbiologically, or immunologically [7]. Each year, 14 million persons are projected to suffer from burns and wounds, with more than 80% living in countries with low and moderate incomes [7]. Nevertheless, this might be underreported because wounds are an everyday event which can befall anybody any time, and burns and minor wounds are not always treated medically.

The skin protects our bodies against both chemical and physical harm, as well as loss of water, and assists in maintaining homeostasis in the body. In everyday life, skin damage occurs when the skin fails in its defensive role, resulting in the emergence of an injury on the surface of the skin. The skin defends our internal body system against the outside world, and thus regulates temperature and maintains homeostasis. To sustain its functions after damage, the skin's integrity must be restored [8]. A complex circulatory and neural network connects the outer epidermis, dermis, and lower subcutaneous layer of human skin. The epidermis, located on the outermost layer, is primarily made up of keratinocytes, which establish an airtight barrier for protection alongside Langerhans cells, melanocytes, and Merkel cells [9]. The dermis is located beneath the epidermis and is connected to it by the basement membrane, which is a slim layer of extracellular matrix (ECM) composed primarily of collagen IV, integrins, perlecan, laminins, and nidogen [10]. The dermis has a complex composition that differs quite significantly from the epidermis [10]. It is made up of a slim layer of extracellular matrix (ECM), which serves as a framework for hair follicles, fibroblasts, sweat glands, blood vessels, and other mesenchymal cells. Additionally, it contains chemicals that modulate homeostasis, such as enzymes and growth factors [11]. The dermis is further divided into multiple layers, with the papillary layer nearest to the basement membrane and made up of poorly organised thin collagen fibres that house a dense population of fibroblasts [12]. The skin's complexity makes it extremely challenging to duplicate in a laboratory environment.

The tear and exposure of the skin could end up in illnesses and lesions, which could give rise to severe diseases and possibly fatality. As a result, one of the main subjects in medical research on wound treatment is how to repair these wounds in as little time as possible and with the lowest possible level of effect on the wound in regards to appearance [13, 14].

The aim of this chapter is to explore the phytochemical constituents and metabolites present in medicinal plants and herbal remedies used in the treatment and management of injuries, with a focus on their traditional use in wound healing. The objectives of this chapter are to identify the various phytochemicals present in these natural remedies, examine their biological activity, and evaluate their potential therapeutic effects on wounds.

To achieve the aims and objectives of the chapter, the authors have conducted a meta-analysis of studies that have investigated the chemical composition and biological activity of medicinal plants and herbal remedies used in wound healing. The traditional use of these natural remedies and various phytochemicals and metabolites present in these remedies that may contribute to their therapeutic effects have also been identified and examined. In addition, the in vitro and in vivo studies that support the antioxidant, anti-inflammatory, and wound-healing activity of these phytochemicals have been discussed.

The chapter is organized into different sections that discuss the phytochemicals and metabolites present in medicinal plants and herbal remedies used in wound healing. Each section focuses on a specific class of phytochemicals, such as flavonoids, terpenoids, alkaloids, and saponins. The biological activity of these compounds and their potential therapeutic effects on wounds have also been evaluated.

The information used in this chapter was sourced from various peer-reviewed journals, scientific databases, and books. The information presented in this chapter is supported by a meta-analysis of studies that have investigated the chemical composition and biological activity of medicinal plants and herbal remedies used in wound healing. The authors have also relied on traditional knowledge and practices, as well as modern scientific research, to support the use of these natural remedies in wound healing.

1.1 Herbal Medicine Classifications

According to Balap and Gaikwad [2], there are three main classifications of herbal medicine:

- Chinese herbal medicine
- Western herbal medicine
- Traditional Indian medicine
- Chinese herbal medicine: The oldest traditional treatise known to exist on the topic was written by Huang-ti, the Yellow monarch (2697–2597 BC), who is responsible for coordinating and standardizing Chinese herbal remedies. Due to the health benefits of both, the majority of Chinese herbalists combine one principal herb with a diet. The Indian subcontinent employs herbalism in the same manner. In Chinese medicine, there are four main ways to combine herbs. When two or more herbs are used together to complement one another's effects on the body, complementary herbs are used. Herbs that are helpful have different uses and effects. When the other herb gives supportive catalysis, one herb takes the initiative. Herbs of fright lessen the risk of wound by reducing the ferocity and power of partner herbs in an admixture. Cancelling herbs, that is, a mixture of laxative and astringent herbs, are employed in reducing constipation caused by the latter. Herbs used to reduce constipation caused by such injury are utilized to remove unwanted side effects [2].
- Western herbal medicine: Before the twentieth century, Western herbal medicine predominated the medical landscape; however, conventional or inorganic medicine founded on scientific medical models has since supplanted it. The general public's attitude towards conventional medicine has changed recently, though. The bilious, sanguine, choleric, and melancholy temperaments, which determine the makeup of the patient, the level of cold, wet, heat, or dry administered to sickness conditions and herbs, and Hippocrates' sense of humour were the foundations of European herbalism until the seventeenth century.
- Traditional Indian medicine: Traditional Indian medicine, also regarded as Ayurvedic medicine, is an ancient Indian health system that focuses on general healthcare, medicine, dietary guidance, and health education. Ayurveda medicinal system is founded on the three distinct biotic humours of Vata, which are Pitta (fire), (air), and Kapha (water). These tridoshas are present in everyone, but one element is constitutionally dominant. The primary factor defines the constitutional strengths and limitations. Vata is known for its lack of power and stamina, as well as breathlessness, indigestion, bloating, and circulation problems [15].

1.2 Traditional Use of Herbal Remedies in Wound Healing

Herbs are frequently used in combination in traditional medicine to produce a synergetic effect. It implies that the effects of numerous plants combined together can be larger when compared to the effects of each individual herb alone. This is known as a 'herbal preparation' or 'herbal formula' and is a regular technique in traditional medicine.

Traditional healers in various cultures possess a plethora of knowledge regarding the application of medicinal herbs in wound healing. They have established specialized procedures and methods for producing and applying herbal treatments, which they frequently combine with other therapies like massage, dietary modifications, or acupuncture.

One advantage of employing herbal therapies for healing wounds is that they may be adjusted to the needs of the individual. Practitioners of traditional medicine frequently consider the patient's general state of health and welfare, as well as specific wound features such as severity, location, and type. This personalized approach can aid in the process of healing and encourage a faster recovery.

An additional benefit of medicinal herbs is that they are frequently less expensive and easier to obtain than prescription drugs or medical treatments. A lot of herbs can be cultivated in the garden or bought at agricultural markets or natural food shops. This could make them an especially enticing choice for people who prefer a natural healing approach or those who do not have access to medical services.

The field of science has begun to understand the phytochemical components and metabolites that give these plants their medicinal properties [16]. Studies have demonstrated that substances such as polysaccharides, flavonoids, triterpenoids, and curcumin have anti-inflammatory and wound-healing properties [16]. With this knowledge, medicinal plants can be used effectively to treat and manage wounds, which helps to improve the healing process.

2 Wound Classification

Wounds are often categorized based on the underlying aetiology of wound development. They include the following.

2.1 Acute Wounds

Acute wounds are those that recover completely within 5–10 days, and at most, within 30. Acute wounds typically happen during surgery or due to acute tissue injury. Trauma (such as burns or injuries), a subsequent disease condition (such as leg ulcers), or a mix of both can result in acute skin lesions [17]. These wounds heal on their own using a prompt and systematic healing process that returns the skin's structural and functional integrity [18]. In contrast to chronic wounds, which take months to fully recuperate, acute wounds take just a few weeks. A recent injury that has not gone through the wound-healing stages is characterized as an acute wound. These might be superficial wounds that involve both the dermis and the epidermis, or extend beyond these layers and cause damage to the skin that compromises the subcutaneous layer [16]. Tissue damage/injury in an acute wound typically takes place throughout a systematic, phase of time correction that successfully restores functional and anatomical integrity. Cuts or surgical incisions frequently result in acute wounds [19].

2.2 Closed Wounds

Closed wounds allow blood to exit the circulatory system, but it still remains within the human body and includes contusions or abrasions, hematomas or blood tumours, and crush injuries [20]. Bumps and bruises make it obvious [21].

2.3 Open Wounds

In this scenario, blood leaks out with obvious bleeding from the person. It is further classified as follows: incised wounds, laceration or tear wounds, puncture wounds, abrasions or superficial wounds, and penetration wounds.

2.3.1 Incised Wounds

This wound has minimal tissue damage and no tissue loss. Sharp instruments like scalpels and knives are the major culprits.

2.3.2 Laceration or Tear Wounds

This non-surgical injury, when combined with other types of stress, causes tissue damage and loss.

2.3.3 Puncture Wounds

They are brought about by an object puncturing the skin, like a needle or a nail. Infection is common in these wounds because dirt can seep deep into them.

2.3.4 Abrasions or Superficial Wounds

Abrasion results from slipping on a rough surface. The top layer of skin, or epidermis, is scraped off during this process, exposing nerve endings and causing a painful injury.

2.3.5 Penetration Wounds

Penetration wounds are generally the result of an item, for instance, a knife, passing through the skin.

2.3.6 Gunshot Wounds

They often result from a bullet or similar projectile that enters the body.

2.4 Chronic Wounds

These wounds do not pass via the typical healing phases and have instead progressed to a level of pathological inflammation. They require a longer time to heal or they become recurrent [22] (Fig. 1).

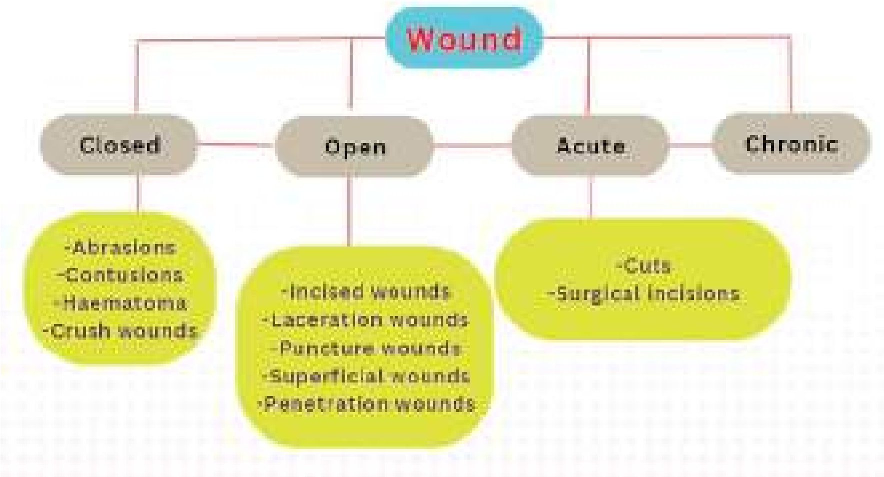

Fig. 1 Types of wound summarized

3 Wound Healing

Wound healing is a complicated process which comprises the interaction of various cell types, inflammatory mediators, and extracellular matrix components, and also several signalling pathways [23]. It is defined by a well-coordinated series of activities aimed to restore the skin's mechanical integrity and barrier function [24]. Tissue damage initiates healing, and this has four time-dependent phases: haemostasis, inflammatory, proliferative, and remodelling phases [25] (Fig. 2).

3.1 Haemostasis Phase

Coagulation and haemostasis occur immediately after injury to prevent bleeding [18, 19]. These mechanisms necessitate a series of interconnected procedures to secure blood vessels and establish matrixes for cell influx required in the latter stages of the healing process [20, 21]. Extrinsic and intrinsic routes trigger the coagulation cascade, resulting in platelet aggregation and the formation of clots [19]. During haemostasis, three steps occur in rapid succession. The first reaction is vascular contraction, which happens when blood vessels constrict to reduce the loss of blood. In the second step, platelets attach within a few seconds of a blood vessel's epithelial wall rupturing to establish a temporary plug over the vessel wall breach. Collagen, when exposed at the injury location, enhances platelet adhesion to the wound site. The third and last phase is blood clotting, often known as coagulation. The coagulation cascade, also regarded as secondary haemostasis, is defined as a set of stages

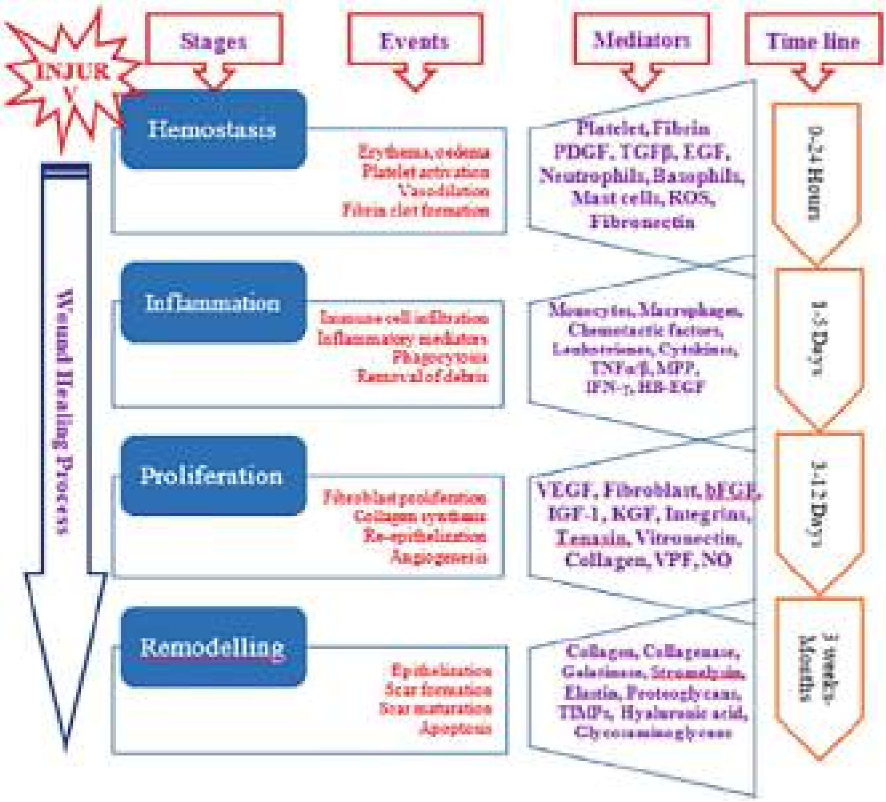

Fig. 2 Schematic representation of wound-healing phases and mediators involved [26]. (Adapted from Khaire et al. [26]. "An Insight into the Potential Mechanism of Bioactive Phytocompounds in the Wound Management" by Manisha Khaire is licensed underCCBY4.0)

that occurs as a result of tissue injury-induced bleeding, with each step activating the next and eventually producing blood clots. Coagulation strengthens the haemostatic plug by forming fibrin threads, which act as 'molecular glue'.

It takes around 60 s for the first set of fibrin strands to entangle with the wound. The fibrin-rich platelet block is completely established after a few minutes [19–22]. Growth factors derived from platelets, also known as platelet-derived growth factors, with cytokines also aid wound healing by activating/triggering and recruiting neutrophils, endothelial cells, fibroblasts, and macrophages [17, 21].

The phytochemicals and metabolites of medicinal herbs have been found to possess different effects on the different stages of wound healing. For example, flavonoids exhibit anti-inflammatory and antioxidant effects, which can promote the early wound-healing stages [23]. Triterpenoids, on the other hand, have been found to have qualities that reduce inflammation and promote healing [25]. Man can heal wounds in situ via continuous tissue regeneration and repair [27]. Curing chronic and acute wounds, along with burn wounds, follow the same basic steps: haemostasis, inflammation, propagation, fibroplasia, deposition of collagen, re-epithelialization, contraction, remodelling, and maturation [28]. The skin's ability to regenerate and restore the barrier function to the external environment after tissue damage is critical for survival. Healing of wounds is crucial following surgical incisions, burns, and chronic ulcers [29].

3.2 Inflammatory Phase

This is divided into two stages: the early inflammatory phase and the late inflammatory phase [30]. The early inflammatory phase/stage initiates molecular mechanisms which result in the infiltration of neutrophils at the site of the wound, whose primary objective is to avert infection [31]. Several chemoattractive drugs begin to attract neutrophils to the site of the wound within 24–36 h of damage. Neutrophils are responsible for the removal of foreign substances, microbes, and injured tissue from wounds by producing oxygen-derived free radicals and proteolytic enzymes [32].

The neutrophils are evacuated from the area once the task is finished. Then macrophages arrive at the injury site and carry on with the phagocytosis before moving on to the next healing phase in the course of the late inflammatory phase. Macrophages are crucial regulatory cells that store a considerable number of powerful growth factors in subsequent stages of the inflammatory response [31]. Lymphocytes are the last cells to arrive at the wound area (in the course of the late inflammatory phase), and they perform a vital role in the control of collagenase they are later on necessary for remodelling of collagen, extracellular matrix component formation, and breakdown [30].

3.3 Proliferative Phase

The proliferative phase includes the key healing mechanisms and begins three days following the injury, and goes on for about a fortnight. After the injury, myofibroblasts and fibroblasts multiply for the first three days in the surrounding tissue [33] prolyl moving into the wound, drawn there by substances like platelet-derived growth factors (PDGFs) and transforming growth (TGFs), which are generated by platelets and inflammatory cells [34]. When fibroblasts enter the wound, they proliferate rapidly and begin to deposit the new extracellular matrix by generating matrix proteins which include type 1 and 3 procollagen, proteoglycans, fibronectin, and hyaluronan. The synthesis of collagen is a critical event. Moreover, collagen is a critical part of all wound-healing stages because it helps to maintain wound integrity [16].

3.4 Remodelling Phase

Scar tissue production occurs through wound remodelling (Fig. 3d, above) and is finished in roughly one year or more [35]. Once the lesion heals, capillary growth ceases, fibroblast and macrophage density is decreased by apoptosis, and blood supply and metabolic process diminish [35]. The result is a developed scar with fewer blood vessels, cells, and a strong tensile strength [30]. Though the majority of injuries are the consequence of basic wounds, local and systemic factors can change and slow down the finely regulated process of repair, resulting in injuries whose healing is not orderly and timely, and grow into non-healing, chronic wounds. Consequently, they are classed as either chronic or acute wounds subject to the tissue's capacity to mend the injury [16].

4 Some Facts About Chronic and Acute Wounds

Though chronic wounds take several months to heal, acute wounds require just a few weeks to recover entirely. A fresh wound that has not yet developed via successive wound-healing stages is classified as an acute wound. These might be superficial wounds that involve both dermis and epidermis or full-thickness damage to the skin that compromises the hypodermal layer. Surgical incisions, abrasions, lacerations, and thermal wounds are all examples of acute wounds. They heal promptly via the regular inflammatory process, new tissue formation, and remodelling. Growth factors, together with cytokines that are released close to the injury site, regulate the healing of the acute wound [36].

On the other hand, chronic wounds do not heal normally and are not treated in a timely and organized manner [37]. The healing process is incomplete and disrupted by different circumstances that cause haemostasis, inflammation, propagation, or remodelling to be delayed by one or more stages [16]. Tissue hypoxia, infection, exudate, necrosis, and high inflammatory cytokine levels are instances of such causes [38]. The injury's continuous inflammatory state leads to a response by the tissue in a cascade format which, when combined, keeps the injury from healing. These wounds usually relapse because the procedure of healing is disorganized and the anatomical and functional outcomes are poor [37]. Different factors, like neuropathic disorders, venous and arterial insufficiency, high blood pressure, vasculitis, and/or burns, cause chronic wounds [39] (Fig. 4).

5 Herbs Utilized in Herbal Medicine for the Treatment of Wounds

Plants have been used for therapeutic purposes by humans throughout the course of time. Herbal medicine, also referred to as phytotherapy, is an ancient practice which treats numerous health ailments with plants and their extracts. Accidents, burns, and surgeries can all result in wounds.

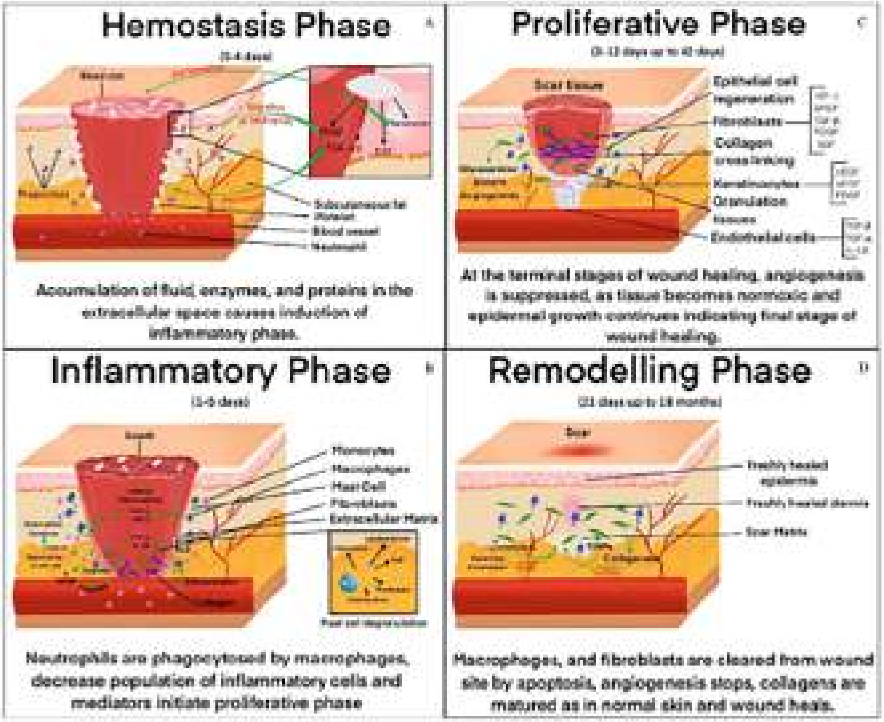

Fig. 3 Healing of wounds: pathophysiology and the mechanisms of different mediators involved. (Adapted from Khaire et al. [26]. "An Insight into the Potential Mechanism of Bioactive Phytocompounds in the Wound Management" by Manisha Khaire is licensed underCCBY4.0.) (**a**) Haemostasis phase, (**b**) inflammatory phase, (**c**) proliferative phase, and (**d**) remodelling phase. bFGF: basic fibroblast growth factor; EGF: epidermal growth factor; IGF-1: insulin-like growth factor-1; IL-1β: interleukin-1 beta; KGF: keratinocyte growth factor; MCF: monocyte chemoattractant protein; MIP: macrophage inflammatory protein; MMP: matrix metalloproteinase; PDGF: platelet-derived growth factor; NF-κB: nuclear factor kappa B; TGF-α: transforming growth factor alpha; TGF-β: transforming growth factor beta; TIMPs: inhibitors of metalloproteinases; VEGF: vascular endothelial growth factor. Haemostasis (**a**): a clot forms, creating a momentary obstacle to loss of fluid and the entrance of pathogens, which act as a storehouse for antimicrobials and bioactive substances, producing a functional extracellular matrix that supports the infiltration and movement of immunological cells, and initiating tissue healing pathway. (**b**) Inflammatory phase: An initial step that entails activating damage-associated molecular patterns, followed by the generation of free radicals, and the creation of reactive biomolecule-recruiting immune cells; then antimicrobial species are released, followed by immune cells which infiltrate and produce amplifying alarmin signals (which are intrinsic, expressed constitutively, chemotactic, and proteins/peptides that are immune activation which are released as a consequence of degranulation, cell injury or even death, or in reaction to immunological stimulation), plus keratinocyte and fibroblast activation. (**c**) Proliferation phase: this includes fibroblast, keratinocyte, and endothelial migration plus multiplication; the resolution of inflammation; extracellular matrix and collagen production, as well as a reduction in vascular permeability; the neovascularization of new lymphatic vessels and capillaries; re-epithelialization; and the new development of granular tissue. (**d**) Extracellular matrix/collagen turnover (production and breakdown); remodelling of the extracellular matrix and readjustment; contraction of the extracellular matrix; apoptosis of the fibroblast and endothelia; repigmentation [16]

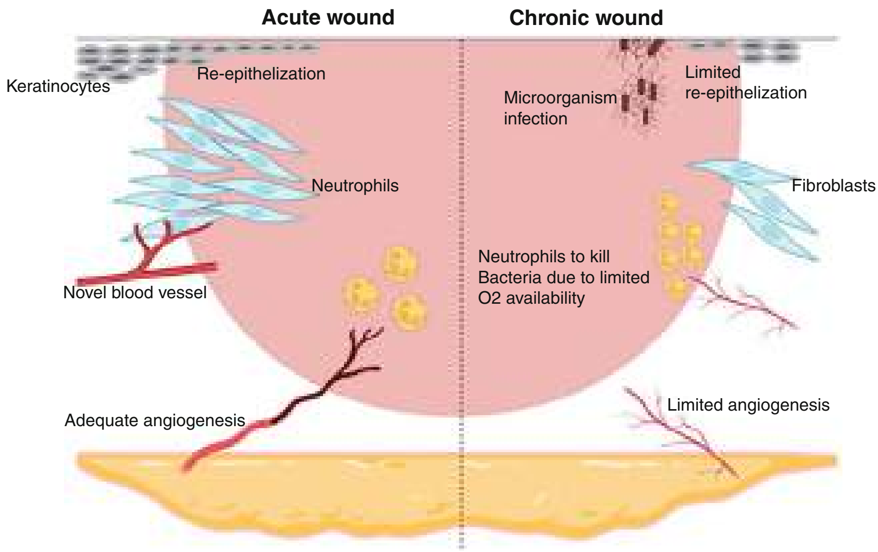

Fig. 4 Processes that are engaged in both acute and chronic wounds. Appropriate angiogenesis accompanied by re-epithelialization promotion, fibroblast propagation, and then neutrophil anti-infection activity within acute wounds (left side). Chronic wounds as seen on the right: featuring persistent local bacterial contagions, and failure of new blood vessel development. The proliferation of fibroblasts is decreased, and neutrophil anti-infection capabilities are limited by inadequate angiogenesis. (Adapted from Vitale et al. [16]. "Phytochemistry and Biological Activity of Medicinal Plants in Wound Healing: An Overview of Current Research" by Vitale et al. is licensed underCCBY4.0)

Although some wounds can be managed through simple first-aid procedures like disinfecting and dressing, others necessitate more specialized medical attention. Herbal medicine is a natural method of wound treatment that can facilitate healing while also lowering the likelihood of infection. Wounds, whether mild or serious, are common medical conditions that can benefit immensely from herbal treatment. Many herbs have been discovered to possess anti-inflammatory, antimicrobial, antioxidant, and wound-healing properties, making them effective in wound treatment. Herbs with antimicrobial properties can help decrease the chances of infection by preventing viruses, bacteria, and fungi from growing and reproducing. This is especially vital for open wounds, which can become infected. Herbs with anti-inflammatory properties can assist in reducing inflammation and swelling, enabling the body's built-in healing mechanisms to take place and so improving wound healing. Herbs with antioxidant properties help protect the body from harmful free radical and oxidative stress. Herbs with wound-healing properties can aid in the growth and renewal of healthy tissue, necessary for the repair of injured skin and tissue. In this chapter, we will look at a few of the herbs that are often used in herbal therapy to treat wounds. It is crucial to emphasize, however, that herbal medication should not be used in place of professional medical treatment.

One area where herbal medicine has been demonstrated to be particularly effective is in wound management and treatment. A diverse range of therapeutic plants and herbs are utilized in standard wound treatment, and scientists are just beginning to comprehend the phytochemical components and metabolites that give these plants their therapeutic effects.

Plants that are traditionally used for wound treatment include *Aloe vera* [7], *Calendula officinalis* [40], *Centella asiatica* [41], and turmeric [42].

6 Phytochemical Composition of Selected Plants Used in the Treatment and Management of Wounds and Infections Resulting from Wounds

Phytochemical analysis, the investigation of the chemical makeup of plants, has helped identify the specific compounds which give these plants their medicinal properties. All plants naturally produce chemical molecules as by-products of their primary metabolites carbohydrates and fats, and their supplementary metabolites, sugars, and sterols [43], which are only present in a few plant genera or species. The pigments, as well as secondary metabolites, could be converted into drugs that have therapeutic effects on people. As a result, this has led to the growth of standardized extracts and isolated compounds, which can be used, that is, wound treatment with increased safety and efficacy.

6.1 *Aloe vera*

A plant from the tropics with a wide geographic range that thrives in hot, arid climates is *Aloe vera*. The gel can be extracted from an *A. vera* leaf's mucilaginous centre. For several thousand years, the plant has been utilized and remains a key component in numerous commercial burn and skin injury care products [44]. Along with polysaccharides, enzymes, amino acids, minerals, and carbohydrates, the gel of *Aloe vera* also possesses vitamins B, A, E, and C and anthraquinone.

According to research, the glycoprotein fraction taken from *A. vera* encourages the growth and migration of cells [45].

6.2 *Calendula officinalis*

Calendula officinalis originates from the Asteraceae family and is a widely cultivated herb used for medicinal purposes in India, the United States, China, and Europe. It is known by a variety of common names, like pot marigold and marigold. *C. officinalis* has historically been applied externally to heal minor burns, other skin issues, and wounds [46, 47]. Herpes, wounds, scars, and products for the skin and hair can all be treated with *C. officinalis* infusions, tinctures, liquid extracts, creams, or ointments [48]. Chemical and pharmacological studies conducted over the past ten years have revealed that *C. officinalis* contains a vast array of additional metabolites with several pharmaceutical activities that attest to its medicinal use [16]. Triterpenoids, both esterified and free, flavonoids, quinones, carotenoids, coumarins, volatile oil, polyunsaturated fatty acids, along with amino acids, are the most active substances [40]. Studies using in vitro fibroblast and keratinocyte cells along with in vivo rodent animal models made up the majority of the studies on the roles of *C. officinalis* in the healing of acute wounds [49–52]. Most clinical research has primarily focused on chronic wounds [40], while only one clinical trial examined acute wounds [53]. In vitro, tests of calendula flower alcohol extracts (CFE) showed the presence of anti-inflammatory and anti-oedematous properties [25]. They also promoted angiogenesis in the chorioallantois membrane model, increased human keratinocyte and fibroblast migration and proliferation, and decreased collagenase activity. Transcription factor NF-kB was activated by CFE, which then affected the inflammatory stage in keratinocytes, whereas when it had a negligible impact on keratinocyte migration, it enhanced interleukin-8 chemokine levels at both the translational and transcriptional levels.

6.3 *Centella asiatica*

Centella asiatica (Gotu Kola, or Asian pennywort) is utilized to accelerate wound healing. *Centella asiatica* aerial section extracts are reported to help with healing chronic ulcers about their size, depth, and location. A punch-type wound has been demonstrated to benefit from the impact of asiaticoside, a compound isolated from *Centella asiatica*. Isolated triterpenes from *Centella asiatica* enhance glycosaminoglycan synthesis and collagen remodelling. Additionally, oral medication administration of the madecassoside in Gotu kola enhances angiogenesis and collagen synthesis at the wound site [54]. It contains several compounds like Asiatic acid, madecassoside, madecassic acid, triterpenoids, flavonoids, and asiaticoside which are responsible for its wound-healing abilities by stimulating collagen formation [41].

6.4 *Hypericum perforatum*

The herb *Hypericum perforatum*, also referred to as St. John's Wort, for millennia, has been utilized in traditional European herbal medicine and is widely distributed in Asia, Europe, and Africa [55]. It has also become naturalized in America. Because of this species' burn/wound healing and anti-inflammatory abilities, its medicinal qualities have been known since Hippocrates' time [56]. A compilation of prescriptions from a Cypriot monastery mentions the herb's use in treating cuts. Tinctures and oils from the herb are used to cure a variety of diseases, including myalgia, minor burns, bruises, abrasions, contusions, and ulcers [57]. Pharmacological research has shown that the presence of a variety of biologically active compounds is what gives *H. perforatum* its medicinal benefits. These compounds include flavonoids (quercetin, hyperoside, rutin, and quercitrin), naphtodianthrones (hypericin and pseudohypericin), xanthones (Norathyriol), prenylated acylphloroglucinols (hyperforin and hyperforin), and sesquiterpenes-rich essential oil [57]. According to research, these substances exhibit anti-inflammatory, antiseptic, antiviral, and wound-healing properties [58].

6.5 Turmeric

Turmeric, or *Curcuma longa*, a common spice, has been discovered to possess the active compound curcumin which possesses wound-healing and anti-inflammatory qualities. Since it is an active component present in the ginger family and in the *Curcuma longa* root, curcumin has been used for both medicinal and culinary purposes for many years. Ayurvedic traditional medicine practitioners utilize curcumin in treating diabetes, liver disorders, airway disorders, skin injuries, and asthma [59, 60]. Curcumin is utilized in conventional Chinese medicine as a common treatment for stomach pain. It is a widely researched nutraceutical, with frequent usage for decades by various ethnic groups. An extremely pleiotropic chemical has now been shown to correspond with important cellular pathways at the DNA replication, translation, and post-translational levels. Pro-inflammatory cytokines, adhesion cell molecules, NF-yB, apoptosis, prostaglandin E2, cyclooxygenase 2, STAT3, 5-LOX, and β-transforming growth factor, phosphorylase kinase, triglycerides, heme oxygenase-1, ET-1, AST, creatinine, and ALT which are observed in the goal pathways. Several of the helpful effects of curcumin are produced by changing the pericellular and extracellular matrix, according to experimental results from numerous in vitro and in vivo studies [21]. Therefore, it may not come as a surprise that curcumin promotes fibroblast growth, granulation tissue development, and collagen build-up during the healing of cutaneous wounds [61].

6.6 Comfrey

Comfrey is a herb that has been used for centuries to heal broken bones and wounds. It contains compounds like Allantoin, which promotes cell growth and repair.

It helps to stimulate the growth of new skin cells, which can hasten the recovery process. Rosmarinic acid has anti-inflammatory properties, Mucilage creates a protective barrier above a wound, helping to keep it moist and promoting healing, Saponins have antimicrobial properties, inulin, a polysaccharide located in the comfrey root, is considered to promote the growth of new cells and helps in healing [62], and *Symphytum officinale* is a pyrrolizidine alkaloid believed to be responsible for comfrey's capacity to heal wounds, but which has been linked to liver damage and has been banned in several countries [63]. In an experiment, comfrey leaf extract was employed to determine whether it could aid in wound healing. The topical compositions of glycerol alcohol solution, carbomer gel, and oil-in-water (O/W) emulsion were investigated. O/W emulsions have proven to be the most effective at accelerating the process of recovery. From the 3rd to the 28th day, inflammatory cell infiltration decreased from 3% to 46% while collagen deposition rose from 40% to 240%. However, the best outcomes were obtained with an 8% extract prepared as an emulsion [64].

6.7 *Echinacea*

Echinacea is a herb known for its immune system-boosting properties [65]. This was shown by its ability to stimulate the monocytes and natural killer cells, which are often deployed in innate immunity against infections [66]. This antimicrobial property makes it suitable for use to treat burns, minor cuts, and skin irritations. The plant *Echinacea* and its component chemical echinacoside look to have beneficial anti-inflammatory and tissue-healing properties [43]. Scientific research has proven the anti-hyaluronidase properties of echinacoside, an *Echinacea* coffee oil conjugate. The authors in [67] studied the significance of *Echinacea* on the hyaluronidase enzyme in pig vocal cords along with the herb's relevance in wound healing. *Echinacea* dosages of 300, 600, and 1200 mg were applied topically to the affected side at random. The findings demonstrated that anti-hyaluronidase therapy was successful in accelerating vocal cord recovery.

6.8 *Arnica montana*

Arnica montana is used topically to treat some ailments such as joint pain, bruises, wound healing, muscle aches, sprains, superficial phlebitis, insect bite swelling, and inflammation from bone breaks. Recent research indicates that it might be beneficial in treating scalds [68, 69]. Increased total phenolic acid levels in *Arnica montana* L. roots were also correlated with mycorrhization [70]. *A. montana* contains 150 therapeutically active substances, which include sesquiterpene lactones such as helenalin and 11 alpha, 13-dihydohelenalin, together with their carbonic acid ester short chains (0.1–0.5% of dry matter in leaves, 0.3–1% in flower shoots), flavonoids (0.6–1.7%) using capillary electrophoresis [71] in the form of flavonoid glucuronides, flavonoid glycosides, and flavonoid aglycones.

Other components of *A. montana* are arnidiol; carotenoids; diterpenes; coumarins (umbelliferone and scopoletin); polyacetylenes; pyrrolizidine alkaloids (tussilagine and isotussilagine) [72]; phenolic acids (Heriguard, limonin, 1.0–2.2%) and caffeic acid [26]; dicaffeoyl quinic acid (DCQA) derivatives (cynarin, ZINC13451209, and 4,5 dicaffeoyl quinic acid); oligosaccharides; and lignans [73]. It consists of sesquiterpene lactone derivatives like tigloyl, methacryloyl, metacryl, isobutyryl, and isovaleryl helenalin [74]. The composition and quantity of phytochemicals such as helenalin esters, phenolics, caffeic acid derivatives, and dihydrohelenalin esters which exist in the flower shoots are different, depending on climate and latitudinal variations, according to a phytobiological study of *A. montana.* Several researchers have discovered that the plant's flowers are mostly rich in active components [73, 74]. Sesquiterpene lactone quantity and nature vary with latitude. Flowers gathered from high-latitude grasslands primarily possess helenalin esters, whereas flowers collected from lower-latitude pastures primarily contain dihydrohelenalin esters. Another study looked at the significance of environmental conditions on the degree of xanthotoxin lactones in 10 German grasslands. Xanthotoxin lactone concentration was higher (0.59–1.10%) [75].

6.9 Chickweed

Chickweed or *Stellaria media* Linn. is a herb that has already been traditionally utilized in treating skin irritations and wounds. It contains compounds that can decrease inflammation and promote cell growth [76]. Chickweed has a high concentration of lipids [77], triterpenoid [78], C-glycosyl flavonoids [79], phenolic compounds and flavonoids [76], sitosterol, phlobatannins, pentasaccharide and alkaloids [80], and saponins [81]. Furthermore, (S)-2-aminohexanedioic acid, 6,7 dimethylheptacosane, oxalic acid, 5-acetoxydotetracont-3-en-1-ol, saponarin, and saccharopine are bioactive metabolites identified from chickweed aerial parts [82]. Another phytochemical investigation discovered active biological metabolites in several regions of chickweed and its preparations [76]. Flavonoids are mostly saponarin, apigenin 6-C-glucoside (isovitexin), apigenin glucoside arabinoside, isovitexin 7,2-di-O-glucoside, isovitexin 7-O-galactosyl-2-O-glucoside, apigenin 6-C-α- apigenin 6-C-β-D- galactosyl-8-C-β-L-arabinoside, quercetin 3-O-rutinoside (rutin), apigenin 6,8-di-C-α- L-arabinoside, L-arabinosyl-8-C-β-D-galactoside, apigenin 6-C-β-D-glycosyl-8-C-β-D- galactoside, coumarins, likewise ascorbic acid [83]. Luteolin, kaempferol, quercetin, rutin, sinapic acid, protocatechuic acid, vanillic acid, catechin hydrate, and resorcylic acid are phenolic chemicals found in chickweed leaf extracts [84]. Other chemicals extracted through the aerial section of the chickweed include 6-methylheptyl-3′-hydroxy-2′-methylpropanoate, 2, 2, 4-trimethyloctan-3-one and 2, 4, 5, 7-tetramethyloctane [76]. These vital metabolites exhibit anti-inflammatory properties and aid in weight loss [85].

Hydroxycinnamic acids, catechins [86], saponosides, ascorbic acid, mucilage, polyphenols, silicon, and carotene are important antioxidant active components [87]. Chickweed possesses a high concentration of phenolic acid, flavonoids, phlobatannins, alkaloids, tocopherol, and Vitamin C, all of which play important antioxidant roles in the human body [76].

6.10 Lavender

Lavender is a herb that has been traditionally utilized in treating minor cuts, scalds, and other derma irritations. It has antiseptic and anti-inflammatory properties, which can assist in lowering the risk of infection and promote healing [88]. Before flowering, lavender flowers (*Lavandulae flos*) are gathered to make the medicinal raw material. Lavender's principal biologically active agents are phenolic compounds, triterpenes, essential oil components, and sterols [89]. Lavender is mostly recognized for its essential oil, which is available in concentrations ranging from 2% to 3% [43]. By steam distillation or hydrodistillation, it is extracted from the petals. Linalyl acetate and linalool are the essential oil's two main components, with concentrations ranging from 1.2% to 59.4% and 9.3% to 68.8% respectively [43]. Both the high concentration of linalyl acetate and linalool and their relative proportions affect the quality of lavender essential oil [90]. Terpenes such as limonene, borneol, eucalyptol, camphene, camphor, −ocimene, 1,8-cineol, fenchone, acetate, α-terpineol, lavandulol, β-caryophyllene, α-pinen and geraniol, and also non-isoprenoid cyclic components such as octenol, octanon, octanol, and octenyl acetate are the major chemicals [89]. Polyphenols are another significant class of chemical elements found in lavender flowers. These are plant secondary metabolites that possess diverse biological characteristics. Flowers, leaves, stalks, fruits, and seeds are just a few of the plant's components where they are found [91]. Almost 8000 polyphenolic chemicals have so far been discovered [92]. They are identified through single or multiple cyclic rings and various numbers of hydroxy groups in a molecule about chemical structure [93]. The shikimate pathway is used in the biosynthesis of polyphenols [94]. They could be separated into numerous different classes, including lignans, coumarins, stilbenes, flavonoids, and phenolic acids [94]. The majority of phenol compounds are discovered in mixtures with organic acids, sugars, and esters [93]. The hydroxybenzoic acid derivatives and the 3-coumaric acid derivatives are the two categories of phenolic acids [95]. The most prevalent hydroxycinnamic acid in the *Lavandula* genus is rosmarinic acid [96]. The principal participants in this group are 3-hydroxybenzoic acid, benzoic acid, syringic acid, 4-hydroxybenzoic acid, vanillic acid, gallic acid, protocatechuic acid, homovanillic acid, and homoprotocatechuic acid [97]. The research of [98] in a rat model showed that topical lavender oil treatment dramatically improved the synthesis of collagen by fibroblasts, which was associated with elevated TGF- β expression in wound abrasions and increases the production of granulation tissue by synthesizing collagen, tissue remodelling by switching from type III to type I collagen, and wound shrinking.

6.11 *Panax ginseng*

This is a popular medicinal plant in Japan, China, Eastern Siberia, and Korea [21]. Recollection is also thought to boost immunity, improve physical agility, and lessen weariness [21]. Hence, the herb is utilized in treating anxiety, chronic fatigue syndrome (CFS), and depression. It has been demonstrated that *Panax ginseng* has vasodilatory, blood lipid-controlling, anticancer, anti-inflammatory, antibacterial, anti-allergic, anti-ageing, and immunomodulatory properties [99]. Several bioactive compounds make up *Panax ginseng*, but a type of saponin is its most effective active component (also known as ginsenosides by Asian scientists and panaxosides by Russian researchers) [21]. Preparations of the root of *Panax ginseng* are attested to shield the skin from severe UVB rays and significantly accelerate the healing process after laser burns and excision wound damage [21]. According to studies, *Panax ginseng* extracts boost human dermal fibroblasts' ability to proliferate, promote keratinocyte migration, and produce collagen ex vivo. However, ginsenoside Rb2, which is separated from *Panax ginseng*, has been revealed to stimulate the growth of the epidermis in raft kelp culture by boosting fibronectin and the receptor, epidermal growth factor, and receptor expression, as well as collagenase I and keratin, which are all essential/critical for wound healing [100]. According to Park and Park [101], *Panax ginseng* may promote the repair of full-thickness skin lesions and also has a great effect on skin regeneration, suggesting its usage as a wound dressing and development as a food product.

6.12 Neem

Neem (*Azadirachta indica*) is renowned for its antifungal, anti-ulcer, anticancer, antiviral, antibacterial, and antioxidant properties in wound dressing. The anti-oxidation of nitric oxide in RAW 264.7 cell lines was evaluated by [102]. Inside wells coated with integrated collagen and 1000 g/mL of *Azadirachta indica* extract, the nitric oxide level was determined to be 10 g/mL. The biocomposite film possesses effective nitric oxide antioxidant and anti-inflammatory properties. In additional studies, the scientists used the *Azadirachta indica* -incorporated collagen film cell lines to perform the antioxidant activities and the biocompatibility test. The combined collagen films of its extract (400 g/mL) demonstrated an 80% improvement in DPPH scavenging ability and had at least 80% cell viability using the MTT assay. Research shows that the herb contains azadirone, azadirachtin, nimbin, nimbidin, and nimbinin [103].

6.13 German Chamomile (*Chamomilla recutita*)

For German chamomile, scientists investigated the effectiveness of PCL/PS electrospinning nanofiber membranes as active wound dressings that contain chamomile. Quercetin, apigenin, luteolin, patuletin, and their glucosides are some of the specific phenolics and flavonoids that provide the Asteraceae plant, C. *recutita* (L.) Rauschert, its therapeutic properties [21]. The most uncommon flavonoid discovered in chamomile plants, apigenin, has a notable impact on how quickly wounds heal [104].

Nanofibers were demonstrated to be efficient against microorganisms, *C. albicans* (fungi) and *S. aureus*, showing inhibitory zones measuring around 7.6 mm in width, according to studies of antifungal and antibacterial in vitro [104]. The mesenchymal stem cells' survival and ex vivo cell adherence to the nanofibers were demonstrated by the MTT assay. The authors' research revealed that the nanofibers, mixed with about 15% of *Chamomilla recutita* extract, could cure a large proportion of the injury 14 days following therapy, which was verified using a rat injury model. This injury evaluation revealed that collagen accumulated in the tissue's dermis together with re-epithelization as well as the absence of necrosis [104] (Table 1 and Fig. 5).

7 Conclusion

This review is a significant addition to natural product research and wound therapy. As such, it has the potential to influence both research and development and clinical treatment in this field. Going forward, the outcomes of this review could be utilized to direct future studies into the individual phytochemicals and metabolites that are most beneficial in wound treatment and management. This might facilitate the invention of novel chemicals with wound-healing characteristics, which could then be turned into new patient medicines. Furthermore, the study could have practical implications for the use of herbal medicine in wound treatment. Clinicians and patients could benefit from a more evidence-based approach to selecting herbal treatments for the care of wounds if specific plants and metabolites that are beneficial in treating wounds are defined. Overall, the literature evaluation has the potential to assist and guide future research and clinical practice in natural products research and wound healing.

The evaluation of phytochemical components and metabolites in medicinal plants and herbal medicine used in wound therapy and management is an important area of research. Traditional medicine usage for wound healing has been applied for centuries and continues to be significant in modern times. Many medicinal plants contain phytochemicals such as phenolic compounds, flavonoids, terpenoids, and alkaloids, which have been demonstrated to have wound-healing properties. Furthermore, by the growing need for natural and safe alternatives to traditional therapy, using herbal medicine to treat wounds has attracted increased attention in recent times. The assessment of phytochemical elements and metabolites in medicinal plants used for wound remedy has shown encouraging results, and the discovery of active chemicals in these plants may lead to the development of novel and effective wound-healing medications.

Finally, the evaluation of phytochemical components and metabolites in herbal medicine and medicinal plants used in the treatment and management of wounds has the potential to provide a better understanding of these plants' modes of action, in addition to helping to create novel and effective wound treatments.

Table 1 Plants and their constituents used in wound repair

Scientific name	Common name	Plant part and formulation	Phytochemical compositions	Biological function	Mechanism of action	References
Aloe vera* (L.) *Burm. f. anagallis	Aloe	Corpulent leaves, gel	Flavonoids- (aleosin, aloin, rhein, emodin) Polysaccharides- (glucomannan, acemannan, and acetylated polymannan)	Antibacterial, anti-inflammatory, epithelialization process	Regulates inflammatory response; regulates the autophosphorylation of signalling proteins; enhances collagen deposition and angiogenesis; boosts the propagation of fibroblasts and mildly stimulates the migration of keratinocytes.	[105–107]
Calendula officinalis	Pot marigold	Flowers	Triterpenoids, Flavonoids- (5-O-caffeoylquinic acid, caffeic acid, isorhamnetin-isorhamnetin-3-O-rutinoside, rosmarinic acid, medicocarpin, nicotiflorin, venoruton, and isoquercetin) Quinones – coumarins	Re-epithelialization process, anti-inflammatory	Increases the appearance of inflammatory mediators; enhances keratinocyte and fibroblast proliferation; boosts collagen biosynthesis and angiogenesis; suppresses lipoxygenase activity; decreases glutathione levels.	[26, 52, 108, 109]
Centella asiatica	Gotu kola	Leaf	Triterpenoids/ triterpenes Flavonoids (Asiatic acid, madecassoside, madecassic acid, and asiaticoside	Antioxidant, antimicrobial, anti-inflammatory, neuroprotective, and wound healing	Collagen depositing and re-epithelialization in a punch-type wound Collagen repair and biosynthesis of glycosaminoglycans Synthesis of collagen and angiogenesis at the wound site	[110, 111]
Hypericum perforatum	St. John's wort	Flower	Hyperforin, hyperoside, hypericin, quercetin,	Antioxidant, antiatherogenic activity, antidepressant, oestrogen-	Inhibition of lipoxygenase activity and prostaglandin E reduction; shorten the interval of	[112–115]

(continued)

Table 1 (continued)

Scientific name	Common name	Plant part and formulation	Phytochemical compositions	Biological function	Mechanism of action	References
			flavonoids, tannins, and saponins	mimetic activity anti-inflammatory, analgesic, gastroprotective	inflammation, stop fibroblast migration during the process of wound healing	
Curcuma longa	Turmeric	Rhizomes	Diarylheptanoids/ curcuminoids Curcumin, demethoxycurcumin, bisdemethoxycurcumin, 5′-methoxycurcumin, dihydrocurcumin, Sesquiterpenes turmerone, zingiberene, arturmerone, fatty acids, sesquiterpenoids, monoterpenoids, proteins	Antioxidant, gastroprotective, anti-inflammatory, antiseptic, antibacterial, antimicrobial, antimutagenic, anticancer, antifungal, hepatoprotective, cardioprotective, hypoglycaemic, antiarthritic, antidepressant	Prevent local activation and appearance of NF-κB genes which mediate joint inflammation and destruction, reduce histamine production and prolong the impact of cortisol against inflammation, reduce in 9,2-benzanthrazen-induced DNA adducts and consequent reduction of carcinogenesis; granulation and tissue formation, increase fibroblast migration, increase TGF-β production; collagen deposition; increase fibroblast proliferation; inhibition of membrane phospholipid peroxidation and surge in liver lipid metabolism; inhibition of the increase in thiobarbituric acid-reactive substance and protein carbonyl and reversing changed activities of antioxidant enzymes	[25, 116–121]
***Symphytum officinale* L.**	Comfrey	Leaves	Tannins, terpenoids, triterpenoids, amino acids,	Anti-inflammatory, analgesic, antioxidant,	Decreasing inflammation, apoptosis, and oxidant stress,	[122–126]

			flavonoids, reducing agents, flavones, saponins, alkaloids, allantoin, polysaccharides, phenolic acids (caffeic, rosmarinic, and caffeoylquinic acids), glycosides, pyrrolizidine phytosterols, and triterpene saponins	antitumour, antiapoptotic, neuroprotective, antibacterial, hepatoprotective, antihistamine	through the NF-kB, Nrf2, and caspase-3 pathways; correcting cholinergic disorders, correcting the antioxidant-oxidative imbalance, delaying the occurrence of the pro-inflammatory event in primary human epithelial tissue	
Echinacea	Purple coneflower	Roots, flowers, leaves	Phenolics – echinacoside, caftaric acid, chlorogenic acid and cichoric acids (CADs), cynarin Alkylamides – polysaccharides/ glycoproteins Arabinogalactans	Antioxidant, antibacterial, anxiolytic, anti-inflammatory, anti-hyaluronidase, analgesic, tissue healing, antimicrobial, antifungal, cyclooxygenase, and lipoxygenase enzyme activities, immunostimulant, vitamin C	Stimulate phagocytosis, T-cell production, lymphocytic activity, cellular respiration (anti-oxidation) action against tumour cells, inhibit hyaluronidase enzyme secretion	[127–131]
Arnica montana	Mountain tobacco	Flowers	Sesquiterpene Flavonoids – flavonoid glucuronides, flavonoid glycosides, and flavonoid aglycones Lactones – Helenalin,11a,13-dihydohelenalin Essential oils – monoterpenes, fatty acids, thymol derivatives, and sesquiterpene; carotenoids;	Antiphlogistic, inotropic, anti-inflammatory, antibiotic, uterotonic, immunomodulatory, antiplatelet, analgesic and anti-rheumatic, antibacterial, antifungal, antioxidant	Increase blood circulation to the affected area, which can improve the delivery of nutrients and oxygen to the injury site, aiding in the healing process; reduce inflammation, stimulate the immune system, promote tissue regeneration, and scale down development of scar tissue	[71, 73, 74, 132]

(continued)

Table 1 (continued)

Scientific name	Common name	Plant part and formulation	Phytochemical compositions	Biological function	Mechanism of action	References
			a triterpene – arnidiol; diterpenes; necine bases (isotussilagine and tussilagine); coumarins (scopoletin and umbelliferone); polyacetylenes; phenolic acids (caffeic acid, chlorogenic acid, and cynarin, 1.0–2.2%); dicaffeoyl quinic derivatives (cynarin, ZINC13451209, and 4,5 dicaffeoyl quinic acid); lignans; and oligosaccharides			
***Stellaria media* (Linn.)**	Chickweed	Whole plant, leaves, flowers, roots, stems	Flavonoids, saponins, isoflavonoids, tannins, alkaloids, triterpenoids, phenolic acids, anthraquinone, and phenolic compounds	Anti-anxiety, antiobesity, antifungal, antidiabetic, antibacterial, anti-inflammatory, anti-leishmanial, and antitoxicity	Promote the propagation and movement of fibroblasts; produce collagen and other extracellular matrix components; stimulate the immune system, support angiogenesis, and accelerate the removing the wound's debris and dead tissue	[76–78, 80, 81]
Lavandula angustifolia	Lavender	Flowers	Flavonoids, coumarins, terpinen-4-ol, camphor,	Anti-inflammatory, analgesic, anxiolytic,	Inhibit pro-inflammatory cytokines and enzyme production, e.g.,	[90, 133, 134]

			cineole, linalyl acetate, linalool	antioxidant, antimicrobial, sedative	cyclooxygenase-2 (COX-2), TNF-α, (IL-6), and prostaglandin E2 (PGE2); interact with pain receptors in the body, such as the transient receptor potential cation channel subfamily A member 1 (TRPA1) and subfamily V member 1 (TRPV1); regulate the activity of gamma-aminobutyric acid (GABA) neurotransmitter in the brain; disrupt the cell membrane of microbes; scavenge free radicals and reduce oxidative stress; activate GABA receptors in the brain and reduce the activity of the system for the histamine neurochemical	
Panax ginseng	Korean ginseng	Roots	Ginsenosides, phenolic compounds (flavonoids and phenolic acids); polysaccharides; essential oils; polyacetylenes; peptides	Anti-inflammatory, antioxidant, adaptogenic, immunomodulatory, antidiabetic, antitumour, cognitive enhancing	Inhibit the manufacture of enzymes and proinflammatory cytokines, such as interleukin-6 (IL-6), TNF-α, COX-2, and PGE2; regulate the sympathetic nervous system and hypothalamic–pituitary–adrenal (HPA) axis; scavenge free radicals and reduce oxidative stress; stimulate the production of natural killer (NK) cells and T cells; induce apoptosis, deposit collagen, enhanced proliferation and migration of epidermal cells	[101, 135–138]

(continued)

Table 1 (continued)

Scientific name	Common name	Plant part and formulation	Phytochemical compositions	Biological function	Mechanism of action	References
					as well as keratinocytes; enhance angiogenesis, increase proliferation and trafficking of fibroblasts and boost the protein expression of collagen	
Azadirachta indica	Neem	Leaves, bark, seeds	Azadirachtin, nimbin, nimbinin, quercetin, salannin, beta-sitosterol, kaempferol, azadirone, meliacin, nimbidin, tiglic acids, fatty acids	Anti-pyretic, anti-inflammatory; antiviral; antioxidant; immunomodulatory	Inhibit the manufacture of pro-inflammatory cytokines; activated protein kinase (AMPK) activation, an enzyme required in regulating the metabolism of glucose; inhibit the replication of viral particles by blocking the viral protease activity; scavenges free radicals and reduces oxidative stress; inhibit the assimilation of cholesterol in the intestines	[102, 103, 139]
Matricaria chamomilla	German chamomile	Flowers	Chamazulene, apigenin, bisabolol; matricin; coumarins (herniarin, umbelliferone); quercetin; luteolin; patuletin	Anti-inflammatory; antispasmodic; antioxidant; sedative	Inhibit the making of pro-inflammatory cytokines and enzymes such as COX-2; free radicals scavenging and reduction of oxidant stress; bind to benzodiazepine receptors in the brain; relax smooth muscles; promote an immune response	[104, 111]

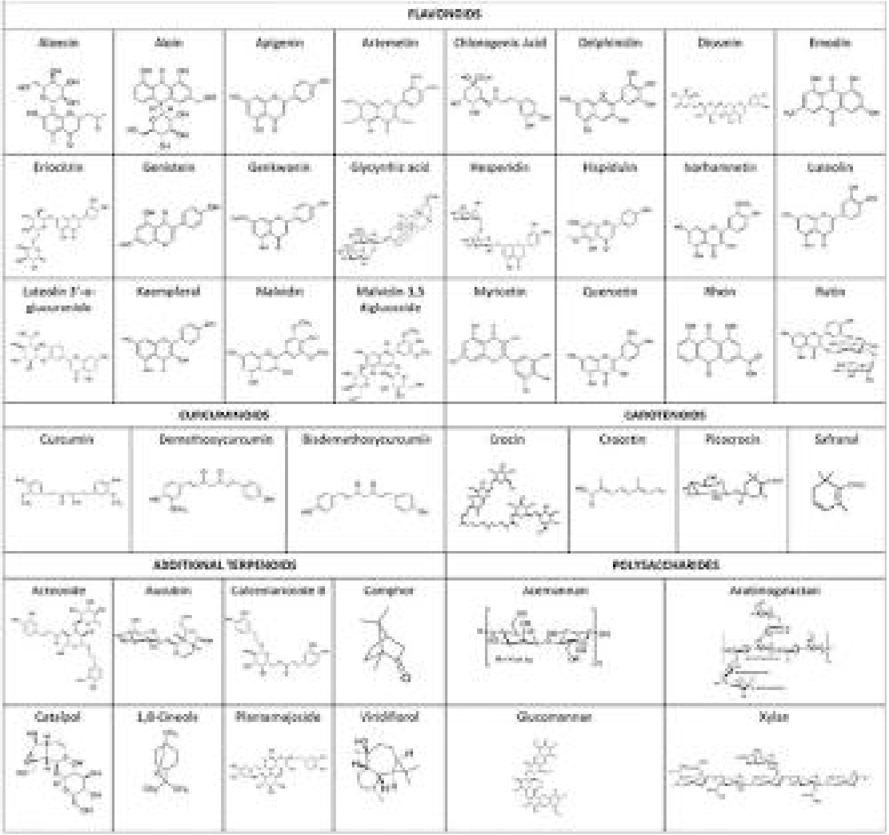

Fig. 5 The structures of selected primary phytochemical metabolites involved in wound healing. (Adapted from Vitale et al. [16]. "**Phytochemistry and Biological Activity of Medicinal Plants in Wound Healing: An Overview of Current Research**" by Vitale et al. is licensed under CC BY 4.0)

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Phytochemicals and Overview of the Evolving Landscape in Management of Osteoarthritis

27

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Abstract

Osteoarthritis (OA), characterized by degeneration of the joints, is forecast to be the major cause of disability for older people by 2030. The primary pathology of osteoarthritis are cartilage and bone rebuilding. Osteoarthritis management consists of pharmacotherapy, physiotherapy, and surgery. OA should traditionally be treated using herbal medications in various medical systems around the world in addition to steroid use. These provide symptomatic relief from pain and inflammation, but minimally reduce joint destruction. Structural damage needs to be effectively prevented and should be the main goal of new therapeutic regimens. Phytoconstituents especially belonging to phenolics, flavonoids, steroids, terpenoids, sterols, sesquiterpenes, triterpenes, and phytoalexins class exhibit antioxidant, anti-inflammatory, anti-arthritic effects and prove beneficial in OA. These phytochemicals modulate different targets like cyclooxygenases, lipoxygenases, and other signaling pathways involved in the pathophysiology of OA, thereby providing immense benefits. These also serve as potential leads for the development of new chemical entities as potential anti-inflammatory molecules of diverse utility. A number of clinical trials have found that herbal remedies and their components are essential to mitigate osteoarthritis. Reflex plus™ is a patented combination of collagen hydrolysate and rosehip extract. Ingredients in the blend have been thoroughly researched for their efficacy in arthritis. Collagen hydrolysate tends to accumulate within cartilage and assists in repairing the damage caused by osteoarthritis and other cartilage alterations. The rose hip works on the anti-inflammatory pathway to reduce joint discomfort and inflammation. The most important pro-inflammatory cytokines implicated in the pathogenesis of osteoarthritis are TNF-α and IL-1β, which act on synoviocytes and chondrocytes by specific interactions with cytokine receptors at the cell surface, assisting in the construction of cartilage for the effective functioning of the joints and the relief of pain associated with osteoarthritis and also help in the protection of cartilage. The purpose of this chapter is to incorporate the impact of phytochemicals on various molecular targets of inflammation into the overall management of osteoarthritis.

Keywords

Osteoarthritis · Anti-inflammatory · Herbal medicine · Phytochemicals · Rose hip

Abbreviations

BMD	Bone mineral density
CH	Collagen hydrolysate
COX-1	Cyclooxygenase -1
COX-2	Cyclooxygenase -2
FDA	Food and Drug Administration

Fib3-1	Fibrin-3-1
Fib3-2	Fibrin-3-2
FSTL1	Fibulin-3 peptide and follistatin-like protein-1
GMPs	Good Manufacturing Practices
GOPO	Galactolipid
GS	Glucosamine sulfate
IL-1	Interleukin-1
IL-6	Interleukin-6
IL-8	Interleukin-8
LDL	Low-density lipoproteins
MMP-13	Matrix metalloproteinase-13
NSAIDs	Non-steroidal anti-inflammatory drugs
OA	Osteoarthritis
PMN	Polymorphonuclear
PUFAs	Polyunsaturated fatty acids
RONS	Reactive oxygen and nitrogen species
ROS	Reactive oxygen species
TNF- α	Tumor necrosis factor alpha
USDA	United States Department of Agriculture
UV-B	Ultraviolet B

1 Introduction

Osteoarthritis (OA) is the most prevalent joint disease. It is the most common type of arthritis, with a higher chance of reduced mobility (defined as assistance in walking or climbing stairs) in individuals over 65 years old who have affected knees [1]. OA is a multifactorial condition with disparities in structure and function throughout the joint. The changes that occur in articular cartilage as a result of aging can lead to the degeneration of tissue and a pathological process characterized by a gradual and typically progressive deterioration of tissue structure and function [2]. In a typical knee joint, the cartilage remains intact and undamaged, preserving the joint's health and functionality. Similarly, the meniscus, a crucial component for shock absorption and stability, remains healthy and fully functional. There are no signs of osteophytes, which are bony outgrowths that can affect joint movement [3]. Moreover, the subchondral bone and ligaments exhibit no abnormalities, maintaining their normal state. Conversely, in a knee joint affected by osteoarthritis, there is a breakdown of the cartilage, leading to compromised joint integrity. The meniscus also suffers damage, reducing its ability to effectively cushion and stabilize the joint [4]. Osteophytes, which are indicators of degeneration, become present, potentially hindering smooth joint movement. Additionally, the subchondral bone undergoes remodeling and sclerosis, contributing to the altered joint structure. Ligaments may also be affected by these changes (Table 1). All the diseases associated with aging, osteoarthritis is perhaps the most strongly correlated and causes significant mobility impairment. The aging process results in significant changes to the structure, composition of the matrix, and mechanical properties of articular cartilage. Most

Table 1 Difference in normal knee and osteoarthritis knee

Normal knee	Osteoarthritis knee
No cartilage damage	Cartilage breaking down
Healthy meniscus	Damaged meniscal
No osteophytes	Presence of osteophytes
Normal subchondral bone and ligaments	Bone remodeling and sclerosis

individuals with fibrillation of the articular surface do not suffer from joint pain or dysfunction. Therefore, the age-related fibrillation of articular cartilage does not necessarily lead to the progressive degeneration of articular cartilage that results in osteoarthritis [5, 6].

Pain and stiffness in the affected joints are the most common signs of osteoarthritis. Discomfort usually intensifies when the joint is moved or toward the end of the day. Joints may become stiff after a period of rest, but this typically subsides relatively quickly once movement resumes. Symptoms of the condition often fluctuate without any obvious cause, with periods of discomfort lasting for several weeks or months, followed by periods of relief. On occasion, the affected joint may become swollen. Swelling in the joints, particularly in the fingers, can take on a knobby and hardened appearance due to the formation of additional bone tissue, or a soft and puffy appearance caused by the thickening of the joint lining and the accumulation of extra fluid in the joint capsule (Fig. 1).

The joint may produce grinding or crackling noises during regular movement, which is known as crepitation. At times, the muscles around the joint may appear weak or atrophied. Engaging in exercises that strengthen the muscles supporting the joints can prevent this from happening [7]. The exact cause of osteoarthritis remains uncertain. It is more than just a result of regular "wear and tear," and the likelihood of developing osteoarthritis is influenced by various factors [8], such as:

1. Age: Osteoarthritis typically emerges in the late forties, possibly due to age-related bodily changes like muscle weakness, weight gain, and reduced efficiency in the body's healing ability.
2. Gender: Osteoarthritis is typically more prevalent and severe in women in most joints.
3. Obesity: Being overweight, particularly in weight-bearing joints like the hip and knee, is a common cause of osteoarthritis [9].
4. Joint Injury: Arthritis can develop in a joint later in life due to a severe injury or surgery. Osteoarthritis is not caused by regular activity or exercise; however, the risk may increase with very strenuous or repetitive activities or physically demanding occupations.
5. Abnormalities in the joints: If person were born with abnormalities, OA may begin early and be more severe than usual [10].
6. Weather and Diet: Several individuals with osteoarthritis may notice that changes in weather worsen their pain, especially when the atmospheric pressure drops before it rains. However, weather changes do not cause osteoarthritis; they can

Fig. 1 Normal and osteoarthritis knee

only influence the symptoms. However, your risk of developing osteoarthritis is influenced by your weight somewhat than other specific dietary factors [11].

Accurate diagnosis of arthritis is fundamental to its treatment. There are numerous types of arthritis, and each may require different treatments. Typically, diagnosing osteoarthritis involves evaluating an individual's symptoms, medical history, the progression of their symptoms, and the physical examination of the affected joint [12]. Osteoarthritis impacts various joints across the body, leading to discomfort and reduced mobility. The affected joints encompass the neck, shoulders, lower back, hips, elbows, base of the thumbs, fingertips, knees, ankles, and the base of the toes. In these areas, the cartilage gradually deteriorates, causing pain, stiffness, and limited joint movement [13]. Osteoarthritis commonly targets these regions, contributing to challenges in daily activities and prompting the need for management strategies to alleviate its effects. Physical examination of joints involves screening for tenderness over the joint, Crepitus, also known as joint grating or creaking, bone swelling, excessive fluid, difficulty in movement, instability of joint, weakness of muscles or thinness that supports the joint.

Blood tests such as CBC, ESR, and rheumatoid factor are usually normal in OA. X-rays may not always be useful in diagnosing osteoarthritis or determining the appropriate treatment options. However, they can detect the presence of calcium deposits in the affected joint.

In rare cases, an MRI scan of the knee may prove to be useful. High-frequency radio waves can also be used to create images of soft tissue and bone changes that would not be evident on an X-ray. Its primary function is to uncover other potential joint and bone abnormalities that may be causing the symptoms [14].

2 Pathophysiology of OA

OA is deemed an organ disease that affects all joint structures. Subchondral osteosclerosis, articular marginal osteophytes, and moderate, chronic, nonspecific synovial inflammation are all associated with progressive loss of articular cartilage in synovial joints [15, 16].

2.1 Articular Cartilage

Despite the involvement of multiple joint tissues, osteoarthritis has long been characterized mainly by interruption of the repair method in damaged cartilage as a consequence of biochemical and biomechanical changes in the joint. The normal maintenance of physiological homeostasis in articular cartilage is controlled by chondrocytes, which generate a structural matrix consisting mainly of type II collagen and proteoglycans [17]. Chondrocytes can produce less effective collagen (type I collagen), smaller and less dense proteoglycans, more degradable enzymes, and nitric oxide, which produces different inflammatory mediators, including IL-1, when subjected to mechanical stress and interleukin (IL)-1. The imbalance between breakdown and synthesis of the extracellular matrix leads to a vicious cycle, resulting in the loss of articular cartilage. This process does not cause clinical symptoms until the innervated tissue is affected, as articular cartilage is avascular [18].

2.2 Osteophytes

Osteophytes near the end of the joint can expand at this point. Osteophytes are cartilage-covered bony growths. Osteophytes are classified into three types: traction spines, where ligaments and tendons adhere to bone, inflammatory spines from vertebral bodies, and osteochondrocytes, which generate cartilage by synovial metaplasia. They can cause discomfort in spinal osteoarthritis but are advantageous in lower extremity osteoarthritis because they support the joints [19, 20].

2.3 Synovial Inflammation

The synovial membrane is essential for appropriate functioning of joints by providing nourishment to chondrocytes via synovial fluid and joint space, as well as removing waste products and breakdown products from the matrix. Synovial mucosal cells produce hyaluronic acid and lubricin, which help to protect and maintain articular cartilage [21]. In osteoarthritis, synovial inflammation contributes to various clinical symptoms, such as pain, and correlates with the progression of the disease. In addition, synovitis is a main element in the pathophysiology of osteoarthritis through the action of several soluble mediators. The association between arthroscopically assessed synovitis and the fraction of functional impairment or pain insights continues to be debatable [22].

3 Molecular Targets of OA

Cytokines, proteinases, lipid mediators, and reactive oxygen species stimulate chondrocytes to produce enzymes that break down cartilage. In patients with osteoarthritis, levels of IL-6, IL-8, and MMP-13 are significantly higher in osteophyte-derived osteoblasts [23]. The expression of IL-6 and IL-8 genes in osteoblasts was found to increase in response to non-physical mechanical stress, and the degree of stress was found to be directly proportional to the magnitude of this increase. This suggests that inflammatory factors play an even greater role in the formation of osteophytes.

Furthermore, IL-6 stimulates the production and expression of MMP-13 in osteoblasts derived from osteophytes and those located in the subchondral region affected by osteoarthritis. Increased IL-8 and MMP-13 expression can cause cartilage degradation through chondrocyte enlargement [24]. Growth factors, which are responsible for the synthesis of the normal matrix, including insulin-like growth factor-1, bone morphogenetic protein, platelet-derived growth factor, and transforming growth factor, play a crucial role in inhibiting the activity of pro-inflammatory cytokines and supporting repair. Osteoarthritis can cause cartilage damage. They enhance anabolic activity and proteoglycan production in chondrocytes while inhibiting catabolic activity [25].

Currently, there are no biomarkers that can be considered reliable tools for diagnosing and predicting the progression of osteoarthritis in routine clinical practice. Fibulin-3 peptide and follistatin-like protein-1 (FSTL1), on the other hand, are extracellular proteins with promise as osteoarthritis biomarkers. Fibrin-3 is frequently distributed in several kinds of tissues and blood vessels of different sizes, and can prevent vascular encroachment and angiogenesis [26]. In addition, it also improves the cartilage in osteoarthritis. An osteoarthritis patient's urine and serum have a larger proportion of their two fibrin-3 fragments (Fib3-1 and Fib3-2) than controls. Increased levels of Fib3-1 were found to be associated with aging and hormone levels, while Fib3-2 levels were not related to gender, age, or menopause. FSTL1, which is activated by ischemic stress and inflammatory mediators, is expressed in human tissues. It is important to note that this information is paraphrased and not intended to be used as medical advice. Serum FSTL1 levels have been found to correlate with inflammatory status, making it a potential biomarker for rheumatoid arthritis and other autoimmune diseases. FSTL1 is thought to play a role in the etiology of arthritis and is activated by ischemic stress and inflammatory mediators in human tissues. The levels of serum FSTL1 were significantly higher in patients with osteoarthritis than in healthy individuals, and in women, they were correlated with the severity of the disease and the expansion of joint space [27].

4 Management of OA

Although there is still no cure for osteoarthritis, there are treatments that can relieve symptoms shown in Fig. 2. These include: 1. Changes in lifestyle, such as increased physical activity, muscle-building exercise, and weight loss (where relevant); 2. Painkillers, e.g., pain-relieving creams or gels or steroid injections; 3. Physical therapies, e.g., application of heat, splints, or other aids; 4. Surgery; 5. Complementary treatment and supplements [28].

4.1 Physical Activity

Many patients are concerned that exercising would aggravate their discomfort and create further joint injury. Physical activity can help protect your joints by strengthening the muscles that support them, as well as control pain, relieve stress, and decrease excess weight. If the discomfort makes it difficult to move, take a pain medication like acetaminophen and warm up the bothersome joint beforehand, or apply an ice pack [29]. There are three types of workouts:

Movement exercises: Exercises that involve movement are beneficial to posture, strength, and flexibility. The exercises softly and gently loosen up the joints to a range of motion that feels pleasant to the touch [30]. This coupled with low intake and low serum levels of vitamin D are linked to increased risk of OA progression [31].

Strengthening workouts: These are exercises that use water, light weights, or resistance bands to provide resistance and strengthen the muscles that support and move your joints. Because of a natural dread of pain, many individuals become less

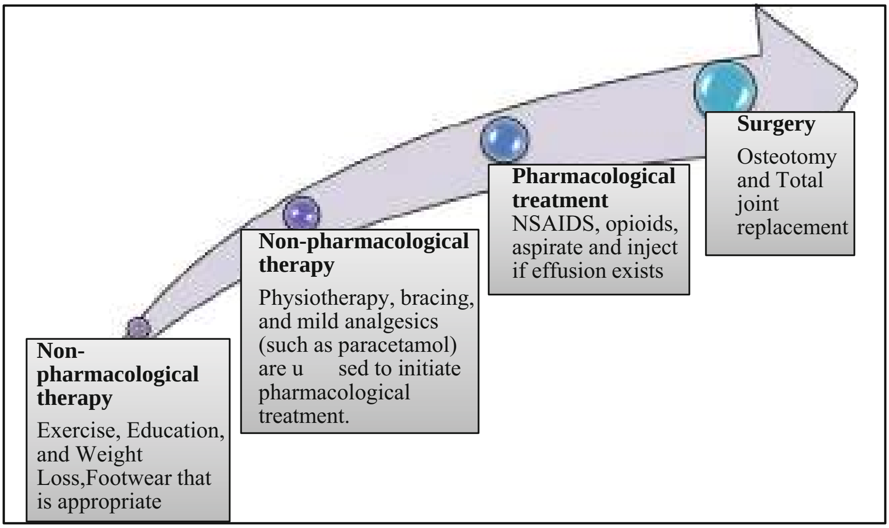

Fig. 2 Management approaches of osteoarthritis

active when they get arthritis, which can lead to increased damage causing muscles to weaken that can cause muscles to weaken and even become noticeably thinner, making it even more difficult for joints to function normally to retain [32].

Aerobic exercise: For individuals experiencing joint pain, walking, cycling, and swimming are all great forms of exercise. You could also consider using a stationary bike or elliptical machine. Walking in the shallow end of a pool is an effective way to improve leg muscle strength. Pools designed for hydrotherapy or aquatic therapy are typically warmer than regular swimming pools. Water provides a comforting and pain-relieving effect, and the buoyancy helps in weight support, making it easier to perform some stretching exercises while waiting for daily activities like cooking or doing household chores [33, 34].

Weight loss: Even a small reduction in weight can have a significant impact on symptoms, particularly for those who are overweight. A well-balanced diet is the most effective way to achieve weight loss. To reduce calorie intake, one should avoid consuming foods that are high in fat and sugar. At the same time, it's important to have a balanced diet that includes all major food groups. Gradual increment in physical activity can also assist with weight loss. There are various ideas regarding what is good and harmful to eat when you have arthritis, but no single diet has been shown to assist. Consulting with a nutritionist is recommended to determine the most appropriate dietary plan. Some studies suggest that including fatty fish or fish-derived oils in the diet may be beneficial in alleviating symptoms of certain types of arthritis [35, 36].

4.2 Medication

The medications that are broadly taken for OA do not influence the actual condition. However, they can help reduce the side effects of inflammation and tension. NSAID gels and creams: Topical creams, gels, and patches containing nonsteroidal anti-inflammatory drugs (NSAIDs) can be applied directly to the skin for pain relief. Drug stores and stores sell ibuprofen and diclofenac gels over the counter. Others, such as ketoprofen, are only available with a doctor's prescription [37]. Creams, gels, and fixes can be effective for specific joints, most notably the knees and hands, but they cannot treat joints that are deeper under the skin, such as the ankles and hips. They are quite safe and are typically used as the first line of defense in drug therapy [38, 39].

Capsaicin cream: Capsaicin cream is a powerful pain reliever derived from the pepper plant (capsicum). It is very beneficial for knee and hand osteoarthritis, as are NSAID creams and gels. It should be used on a daily basis. When most people first take capsaicin, they feel a heat or burning sensation, which usually goes away within a few days. Capsaicin is extremely safe, just like NSAID creams, gels, or fixes [40, 41].

Paracetamol: The safest form of pain treatment to attempt first is paracetamol. It is best to take them before the pain becomes unbearable. Spending extra money on expensive paracetamol brands, which can be obtained without a prescription at drugstores and supermarkets, does not provide any advantages [42, 43].

Nonsteroidal anti-inflammatory drugs (NSAIDs): Nonsteroidal anti-inflammatory drugs (NSAIDs) are a more effective pain relief option compared to acetaminophen. Ibuprofen is the most commonly used NSAID and can be purchased in low to medium doses at pharmacies and supermarkets. If necessary, the doctor may advise high doses of ibuprofen, such as naproxen or diclofenac. Celecoxib and Etoricoxib, two newer forms of NSAIDs, are said to reduce the incidence of stomach ulcers and discharges [44, 45].

Steroid injections: Long-acting steroid injections are occasionally delivered straight into a sore joint. The injection usually begins to work within a day or two and can relieve pain for several weeks or even months, particularly in the knee or thumb. Steroid injections are frequently utilized for managing painful osteoarthritis or abrupt, intense pain triggered by crystals in the joint. The doctor may also provide a steroid injection if the arthritis could interfere with an important event [46, 47].

4.3 Physical Therapies

Heat and cold: Heat applied to a painful joint might help ease pain. To prevent skin damage, use a hot water bottle covered in a towel or a microwaveable wheat bag. Another option is to soak your hands or feet in a bowl of warm water for a short period. In case of joint inflammation, an ice pack covered in a towel can help relieve swelling and discomfort. Apply ice for up to 20 minutes every few hours [48].

Splints and other supports: Orthoses, braces, and other supports are available to assist with problematic joints. When osteoarthritis has changed the orientation of a joint, these can be incredibly helpful. It is best to seek expert counsel from a competent specialist or physiotherapist before making your decision [49, 50].

Shoes: Choosing comfortable, durable shoes influences not just your feet, but also your knees, hips, and spine. The perfect shoe would have a thick but delicate bottom, a delicate upper, and plenty of space at the toe and arch. To relieve pressure on areas of the foot not affected by osteoarthritis, the doctor or podiatrist may offer orthotics for various foot problems [51, 52].

Crutches: Some people use a cane. Holding it in the other hand can also mitigate knee or hip pain. It is best to obtain the counsel of a knowledgeable medical professional because your reasons for utilizing a stick will dictate which side it should be used on. It is crucial that a walking stick is adjusted properly according to the body size, a good provider should do this. Check that the handle feels comfortable in the hand if there is arthritis in the hands [53, 54].

Posture: Posture affects joint alignment, among other things. This in turn can lead to joints or joint parts being subjected to greater stress than usual [55].

4.4 Surgery

The majority of patients with osteoarthritis do not require surgery. However, if the arthritis is severe and the symptoms have a significant influence on quality of life, the doctor may recommend surgery. Every year, thousands of hip and knee replacement procedures are performed for osteoarthritis, and other joint replacement surgeries are becoming more popular [56]. There are additional surgical options than joint replacement. Washing out loose bone fragments and other tissue from the knee can sometimes be accomplished using keyhole surgery. This is known as arthroscopic lavage. Fusion can be quite effective for some tiny joints. This means that the bones in a joint are surgically fixed, preventing movement and, as a result, pain. Although the motion loss may appear weird at first, fusion frequently produces favorable outcomes [57].

4.5 Complementary Treatment and Supplements

Complementary therapies can help ease some of the symptoms of arthritis, such as pain and stiffness, as well as some of the adverse effects of medicine. Relaxation practices, for example, can improve overall safety. Some of the most well-known treatments:

1. Acupuncture involves the insertion of thin needles into specific points on the body, which is said to help restore the natural balance of health by correcting imbalances in energy flow. There is evidence that it can help with osteoarthritis symptoms. NICE does not currently recommend this treatment for osteoarthritis.
2. The Alexander Technique entails becoming more conscious of your posture and moving more efficiently. It has been proven to be very efficient for back pain but is less marked for osteoarthritis.
3. Aromatherapy uses vegetable oils to support health and well-being. The oil can be vaporized, inhaled, exploited in showers and burners, or used as portion of a scented healing back massage. Aromatherapy Is Effective for Osteoarthritis Symptoms, according to Research No, yet some people enjoy the following advantages, such as relaxation support.
4. Although there is limited evidence that it is beneficial in alleviating osteoarthritis symptoms, a good massage can be relaxing and rejuvenating.
5. Osteopathy and chiropractic involve manipulating the body's alignment manually and applying pressure to soft tissues. The objective is to correct structural and mechanical irregularities, improve mobility, reduce pain, and facilitate the body's natural healing process. Chiropractice may provide relief in spine or extremity OA associated pain as reported in aretrospective study conducted to understand the impact of chiropractic management in patients with OA and spine pain. There is no specific scientific evidence that osteopathy is effective in the treatment of osteoarthritis [63].

Use of supplements: Arthritic patients frequently experiment with a range of supplements, including herbal medicines, vitamins, minerals, and supplements. There is minimal evidence that they improve arthritis or its negative effects in many patients. Before taking any supplements: 1. Research the supplements you intend to use; not all of them are suitable for everyone. 2. Consult your doctor or pharmacist to ensure if it is safe to combine with prescription or over-the-counter treatments. After taking medication, keep an impact diary to see if the supplement (s) are having an effect [58].

Glucosamine: Glucosamine is a substance that is present in the body, mainly in tissues like cartilage, tendons, and ligaments. It can also be obtained from the shells of crustaceans such as shrimp, crab, and lobster. Some studies have suggested that taking dietary supplements of glucosamine, particularly in the knee, may be useful in treating osteoarthritis. There seems to be proof that there is no such thing as instant pain relief. It will take several months to visualize if it works. If it doesn't work after 2 months, it possibly won't work [59, 60].

Chondroitin: Chondroitin is normally present in our body and is said to help make ligaments more versatile. Animal studies indicate that it may help slow cartilage disintegration. To a certain extent. Expect no progress for at least 2 months. Assuming the ligaments are badly damaged, chondroitin cannot be used [61].

Fish oils: Fish oil and fish liver oil are generally regarded good for the joints. There isn't sufficient information to say if it's effective. Fish liver supplements are often high in vitamin A, so it's important not to overeat. Fish oil supplements tend to be lower in vitamin A, so it's safer if you really desire to take benefit of the high fish oil content [62].

5 Herbal Medicines in OA

Herbal medicine refers to the purpose of different sections of medicinal plants. Herbology has a long practice of use outside of conventional medicine. It is the collection of information, skills, and practices that are used to preserve health as well as to prevent, diagnose, improve, or treat health. Many physical and mental diseases. The earliest written validation of the help of medicinal plants to formulate medicines was observed on his 5000-year-old Sumerian clay tablets at Nagpur. It includes 12 recipes and references over 250 plant ingredients, including alkaloids like poppy, henbane, and mandrake. The sacred Vedic publications of India pertain to healing with the help of abundant plants in the country. Pen TSao, a Chinese treatise on the utilization of roots and plants, was authored in 2500 BC by Emperor Shen Nong. Author of 365 remedies (dried bits of medicinal herbs). Many of them are still in existence today [64, 65].

The risks connected with herbal medications are introduced onto the market without a mandated safety or toxicological evaluation of their efficacy. Many of these nations also lack efficient procedures to control natural medicine manufacturing and quality laws [65]. Many countries face problems with regulatory status, safety and efficacy evaluation, quality control, safety monitoring, and insufficient or erroneous information about traditional supplementary items [66].

The difficulties in assessing safety and efficiency stem from the fact that the requirements, inquiry procedures, standards, and methodologies necessary to evaluate the safety and efficacy of natural medicines are more involved than those required for traditional medications and pharmaceuticals [67, 68]. The point is, no one can deny that it's a lot more perplexing. Routine medical care. A single home-grown pharmaceutical or medicinal plant may have hundreds of natural elements, and natural therapeutic items combined may include many more. This is essentially unimaginable when evaluating individual dynamic components in this way, especially if the natural product is a mix of at least two spices [69].

5.1 Rose Hip

Rose hip oil is extracted from dried *Rosa canina (R. canina*) fruits acquired from Hyben Vital in Langeland, Denmark. The fruit of the Rosaceae (rosebush) rose family *Rose hip* ranges in hue from red to orange. This species is indigenous to Europe and is widely dispersed there [70]. In fact, it is the most abundant rose species in Central Europe. Furthermore, the dog rose is endemic to portions of North Africa and Central Asia. European invaders transported *R. canina* to the Americas, South Australia, New Zealand, and southern Africa hundreds of years ago, where it escaped cultivation, naturalized, and now thrives in the wild [71].

R. canina is a big shrub or small tree with an arched trunk that grows to a height of 2.74 meters. The five-petaled blossoms, which range in color from white to pink and bloom alone or in tiny clusters in June and July, bloom alone or in tiny clusters. In September and October, the blooms are followed by persistent brilliant red berries, which are often referred to as fake berries (hips). The true fruit is a tiny hairy mass containing seeds in the rump. Dog roses are propagated through spreading, sucking, layering, and sowing, which is most commonly done by loin-eating birds. *R. canina* is tough to control due to its reproductive habits and massive size. Some countries have put in place control measures to prevent their spread and categorize them as invasive species [72].

Rose hip has long been utilized for its astringent flavor. It was claimed to relieve diarrhea, dysentery, thirst, coughing, and "blood vomiting." Nicholas Culpeper, an English herbalist, botanist, physician, and astrologer from the seventeenth century, observed that ripe hips were rendered temperate (usually a mixture made with sugar or honey), which was described as "gently binding the belly and diverting from humor and fluid flow." He also said that the dry, powdery loins dissolve stones, make urinating easier, and alleviate colic.

Rose hips are utilized in various formulations to treat a wide range of ailments such as colds, diabetes, diarrhea, edema, fever, gastritis, gout, stiffness, sciatica, renal and lower urinary tract diseases. They are also used as a diuretic and laxative, and for blood cleansing purposes. Additionally, they are used to enhance the flavor of several food items including jams, jellies, teas, soups, infusions, syrups, beverages, pies, bread, and wine, owing to their vitamin C content [73, 74].

In the northern region of Portugal, people consume the fruit in its raw form, and extracts or brandy are made by steeping it for its diuretic, anti-diarrheal, anti-rheumatic, anti-inflammatory, respiratory, decongestant, energizing, and tonic properties, as well as for treating skin infections. The young shoots and fruit are eaten uncooked, and the fruit is utilized to make spirits and cakes. Similarly, it is also produced in rural Cantabria, which is located in northern Spain [75, 76].

Although the FDA has not officially recognized *R. canina* as generally recognized as safe (GRAS) for use in food, extracts and oils obtained from other species of *Rose hips* such as *R. alba*, *R. centifolia*, *R. damascena*, *R. gallica*, and their variants have been approved for use in cosmetics, foods, and medicines. However, the USDA considers *Rose hips* from *R. canina* or *R. rugosa* as a food [77, 78]. The use of *Rose hip* or *Rose hip* extract is allowed as an ingredient in dietary supplements, provided that the manufacturer complies with current Good Manufacturing Practices (cGMPs) and notifies the FDA within 30 days of marketing if they make a structure-function claim. In the FDA's cGMPs for regulating dietary supplements, the term "*rosehips*" is mentioned multiple times in the preamble [79, 80]. The FDA provides detailed specifications and labeling requirements for different types of dietary supplements that contain rosehips, such as "*rosehips* dietary supplement" and "vitamin C from *rosehips* dietary supplement." [81].

5.2 Phytochemistry of *R. canina*

With the aid of tandem mass spectrometry, gas chromatography, high-performance liquid chromatography, thin-layer chromatography, diode array detection, and thin-layer chromatography, a new study was started to discover the chemicals found in *R. canina* fruits. These studies allowed for the differentiation of various fruit combinations from *R. canina* [82]. Future investigation should focus on finding additional substances that might be responsible for some of the plant's bioactive qualities that have been found in recent medical studies. Quantitative component studies, which are also lacking, might point to the creation of legal *R. canina* breeding and post-harvest control methods as well as support for therapeutic agents [83].

Triterpene

The terpene family is one of the most diverse and variable groups of natural phytochemicals. Over 4000 different complex molecules, many of which are typically outside the realm of chemical synthesis, are among them. Simple triterpenes can operate as signaling molecules, whereas complex glycosylated triterpenes or saponins provide pest and microbe protection. It can be put to a lot of different uses. Triterpenes, which are precursors to sterols, are made by both plants and animals. Sterols are crucial membrane structural underpinnings and, as steroid hormones, they also contribute to cell signaling. Triterpenes, however, are not thought to be essential for healthy development and growth. Among the components of rosehip, both simple and conjugated triterpenes are highly regarded [84].

Flavonoids

Phytochemicals called flavonoids are part of the phenylpropanoid family of compounds. Two aromatic rings are connected by three carbon bridges to form the fundamental backbone of a flavonoid. This skeletal structure is produced by two different biosynthetic pathways. One aromatic ring and a three-carbon extension are obtained via the malonic acid pathway via phenylalanine, and the other is obtained through the condensation of three acetic acid-derived units. The most prevalent substituents for flavonoids, which typically exist as glycosides, are sugars. However, flavonoids can have a variety of substituents [85].

Anthocyanins, flavones, flavanols, isoflavones, flavanols, and flavanones are the first six subclasses of flavonoids. The anthocyanins, which are beta-glycosides with a sugar in the third position, are the largest group of colored flavonoids. Anthocyanins work as attractants, enticing animals (pollinators) to flowers and fruits through visual cues. Another essential attribute of flavonoids is their capacity to protect cells from ultraviolet B (UV-B) exposure. While permitting visible wavelengths to pass through, they gather in the epidermal layers of the skin and capture harmful UV-B radiation [86, 87]. Compared to anthocyanins and carotenoids, flavones and flavanols trap light at fewer frequencies. Flavones and flavanols cannot be seen by the human eye because of this. Because they are linked to the UV patterns of flowers known as "Nectar Guides," they can appear to insects that perceive UV tones in a range of light. It has been demonstrated that isoflavones have a variety of biological functions, such as antibacterial and insecticidal qualities [88].

The in vitro findings from flavonoid bioactivity screens were supported by in vivo research on the resistance of plasma and lipoproteins to ex vivo oxidation after ingestion of flavonoid-rich meals. A lower incidence of infectious diseases, cancer, inflammation, heart disease, ischemic stroke, atherosclerosis, and other chronic diseases, is linked to consuming foods high in flavonoids, particularly fruits and vegetables. Quercetin, one of the few flavonoids, has shown fascinating bioactive properties in both in vitro and some in vivo investigations [89, 90]. Rutin and its glycosides (rutin and quercitrin, for example) have shown anti-inflammatory activity in gastrointestinal inflammation models, perhaps through downregulation of nuclear factor kappa-beta signaling. Quercetin may also influence eicosanoid biosynthesis, protect low-density lipoproteins (LDL) from oxidation, reduce clotting, and relax smooth muscle in the heart. Interestingly, the flavonoid tiliroside (kaempferol 3-O-d-(6-p-coumaryl)-glycopyranoside) discovered in *R. canina* dramatically reduced the oxidation of human LDL in vitro. It has been shown in humans to have anti-obesity properties as well as antioxidant, cytotoxic, and anti-complementary properties [91, 92].

Carotenoids

Tetraterpenoids called carotenoids absorb light with wavelengths of 400–500 nm. Carotenoids, which give fruits and flowers their colors, are therefore found in plants as red, orange, and yellow pigments. Prevents photochemical processes from occurring, particularly oxidation (radical scavenger) [93, 94]. Animals must obtain their carotenoids from their diet because they are unable to synthesize them. A 40-carbon structure with an isoprene base unit makes up the majority of carotenoids. There are typically two subgroups of carotenoids. (1) Oxygen-containing xanthophylls (lutein, zeaxanthin, cryptoxanthin, etc.); and (2) non-hydroxylated hydrocarbons (i.e., −carotene, −carotene, and lycopene). The structure of a carotenoid (the number of conjugated double bonds and the presence or absence of oxygen) has a direct impact on the compound's color [95, 96].

Dietary carotenoids have been linked to the prevention of angina pectoris, apoptosis induction, and mammary cell proliferation. It has also been proposed that carotenoids may aid in human prostate cancer prevention [97, 98]. The positive effects of carotenoids that have been discovered may be the result of this discovery, either because carotenoids are better suited to serve as indicators of high consumption of fruits and vegetables or because the effects of epidemiological studies have been attributed to substances unrelated to carotenoids [99, 100]. However, some carotenoids have reportedly been found to have intriguing physiological properties. According to human research, lutein and zeaxanthin, which are found in high concentrations in the macula of the human retina, may protect the outer segment of the macula and the retinal photoreceptors from oxidative stress are seen [101].

Diets high in lutein and zeaxanthin are associated with a lower prevalence of nuclear cataracts in older women and a lower risk of age-related macular degeneration. Additionally, it has been suggested that lycopene, when given in the form of tomato paste or extract, can treat non-eosinophilic airway inflammation and prostate cancer. Because it is a result of administering a combination of compounds, it is problematic [102, 103].

Vitamins

Vitamins are organic chemicals that are produced by plants and some lower animals. Vitamins play a variety of activities in the body, including coenzyme activity, precursor activity, antioxidant activity, calcium and phosphorus intake regulation, and coagulation regulation (blood coagulation). Numerous illnesses and diseases in humans are caused by vitamin deficiencies [104].

Vitamins that dissolve in water, like vitamin C and the B vitamins, cannot be stored by the body and must be consumed continuously through food. Water-soluble vitamins that aren't used are excreted. Recent studies have concentrated on the antioxidant activity of vitamins C and E [105].

Apart from scurvy prevention, which is attributed to vitamin C's role in collagen synthesis, vitamin C is involved in a number of significant enzymatic syntheses. It is crucial for the production of bile acids and the conversion of cholesterol to these acids. Vitamin E, which is assumed to work largely as an antioxidant, is thought to protect PUFAs of plasma membrane phospholipids and plasma lipoproteins. According to recent studies, vitamin E reduces protein kinase C activity, although the physiological significance of this impact is unknown [106].

Osteoblasts, which promote bone formation, and osteoclasts, which encourage the breakdown of mineralized tissue (breakdown of bone structure), control bone formation. Osteoclasts produce ROS (reactive oxygen species), and ROS encourage the breakdown of collagen chains, which is crucial for bone remodeling. Proteolytic enzymes like elastase and metalloproteinases can be activated by elevated oxidants, further harming the bone extracellular matrix. Antioxidant supplements may therefore aid in restoring bone mineral density (BMD) [107].

Ascorbic acid, a component of vitamin C, is crucial for procollagen and stable collagen secretion, both of which are necessary for the development of connective and bone tissue. *Rose hips* beneficial effects as an osteoporosis treatment are further supported by their high polyphenol content. Oxidative stress-related bone loss is known to be lessened by it. Strong evidence supports the idea that combining the seeds and skins of the *R. canina* subspecies helps people feel less pain and function better every day. It is evident that both the peel and the seed are required because a study on pure peel powder found no impact on patients' reported pain and discomfort [85, 108].

The beneficial effects of *R. canina* have not yet been fully explained in terms of their biochemical underpinnings, though. According to this perspective, the only molecule, a galactolipid by the name of GOPO that has anti-inflammatory properties, appears to be significant. Collagen plays a significant role in the healing process and the high natural vitamin C content in *R. canina* is readily absorbed. Linoleic acid which contributes to the COX-1 and COX-2 inhibitory effects found in rose hips may also offer an explanation in addition to triterpene acids [109, 110].

At the molecular level, OA involves an inflammatory process that produces excessive levels of RONS (reactive oxygen and nitrogen species). RONS generated by chondrocytes and associated with OA include superoxide, hydrogen peroxide, nitric oxide, and peroxynitrite [111]. Despite the fact that RONS is required as a second messenger in the control of normal physiological processes and as a destructive agent in the immune system's defense against pathogens, aberrant chondrocyte metabolism reduces its physiological buffering capacity [112]. An overabundance of anything causes oxidative stress. Proteins, lipids, nucleic acids, and extracellular matrix constituents can all be damaged by RONS overproduction. Induced cell damage and RONS signaling both amplify the inflammatory response at the same time. These comprise transcription factors, matrix-degrading enzymes (such as metalloproteinases MMP-3 andMMP-13), inflammatory cytokines (interleukins IL-1 and IL-6), and this process is closely linked to permanent cell cycle exit and chondrocyte senescence (also known as the "senescence-associated secretory phenotype") [113, 114].

TNF-α, IL-1, and IL-6 are pro-lipolytic inflammatory cytokines produced by adipocytes. Moreover, it prevents the synthesis of lipids. Lipids function as a negative feedback mechanism by suppressing the synthesis of type II collagen and proteoglycan in addition to promoting the production of prostaglandins, matrix metalloproteinases (MMPs), and other cytokines. In this method, pro-inflammatory cytokines cause bone resorption and cartilage matrix disintegration [115, 116].

5.3 Clinical Trial Study on Rose Hip Extract

This study aims to evaluate Reflex Plus's impact on people with osteoarthritis (OA). The study will last 12 or 24 weeks, not including the roughly 1-week screening period. Reflex Plus, a health supplement made of collagen hydrolysate and rosehip extract, is the test product under investigation. The supplement is believed to assist in the development of cartilage, which is essential for proper joint function and reducing arthritic pain. It is also helpful in managing joint discomfort and safeguarding cartilage. The primary aim of the study is to assess the difference in knee pain between the start and end of the trial, as determined by the WOMAC osteoarthritis index sum score, when comparing the supplement and placebo.

Study Phase: Phase 4

Condition: Osteoarthritis

Intervention: The dietary supplement contains 5 g of Reflex Plus Collagen Hydrolysate and 0.55 g of Rose hip Aqueous Extract. The placebo includes Fructose, Orange Flavor, and Sucralose.

Study: During the study duration, the Active Comparator Reflex Plus should be taken once a day before breakfast, using sachets.

Interventions: During the course of the study, the Reflex Plus placebo comparator dietary supplements placebo sachets should be consumed once daily before breakfast. Dietary Supplement is Placebo.

Status: Completed.

Rose hip extract contains galactolipids.-1,2-di-O-[(9Z,12Z,15Z)-octadeca-9,12,15trienoyl]-3-O-b-D-galactopyranosylglycerol, betulinic acid, oleanolic acid, ursolic acid, vitamin C, vitamin E, beta-carotene, lycopene and linoleic acid are all found in this supplement. Also contains monogalactosyldiglyceride and digalactosyldiglyceride. These active components control the inflammatory response and stop cartilage from being destroyed. Additionally, *Rose hip* extract has been shown to lower levels of the acute phase protein CRP and serum creatinine while also inhibiting peripheral blood polymorphonuclear leukocytes (PMN). Additionally, it was noted that *Rose hip* use lessens pain, stiffness, and disease severity while COX inhibits −2 and contributes anti-inflammatory properties. Rose hip powder was proven to be an effective dietary supplement for her OA knee patient. However, more research is needed to determine the therapeutic relevance of the reported good experimental outcomes.

6 Conclusion

OA is a frequent condition among the elderly. The first line of management consists of lifestyle changes and physical therapy, which is followed by analgesics and NSAIDs. However, these medications only treat symptoms and have no impact on how the disease develops naturally. Nutraceuticals refer to dietary supplements that are believed to modify the inflammatory process and, as a result, impact the natural progression of osteoarthritis. Collagen peptides are similarly effective in treating OA, although there are various formulations available on the market, each with its unique pharmacological effects. As a result, standardizing collagen peptide formulation is required before utilizing it to treat OA. Supplements containing collagen hydrolysate (CH) and glucosamine sulfate (GS) have been shown to be safe, however reports on their effects have been inconsistent. Despite lacking sufficient research, the European Society for Clinical and Economic Aspects of Osteoporosis and Osteoarthritis has recommended the use of nutraceuticals as a first-line treatment option for managing OA. Rose hip powder is considered a valuable nutraceutical in the treatment of OA patients due to its anti-inflammatory, chondroprotective, and immune-modulatory properties. However, more extensive research is necessary to establish its efficacy in the future.

R. canina, also known as rosehip, is a remarkable and gorgeous medicinal plant. It is a very potent source of antioxidants and anti-inflammatory compounds and can grow in very rocky soils. Additionally, it contains a lot of vitamins, particularly vitamin C. *Rose hips* are very effective at treating osteoarthritis in humans and some animal models, according to a number of clinical studies. It is crucial to note that a dose response, one of the pillars of the evaluation of prescription drugs, has been confirmed for *Rose hip* subspecies. There is proof that *Rose hip* treatment may be beneficial for conditions like rheumatoid arthritis, cardiovascular disease, and skin conditions like wrinkles. However, the amount of active ingredients in *Rose hips* varies. Different types of rose seeds exist. To identify the species that offer the greatest potential, a careful comparative analysis is all that is necessary. It is interesting to note that the *R. canina* variant of osteoarthritis was able to reduce the use of emergency medications by up to 50%, particularly when discussing anti-inflammatory diets and anti-inflammatory plants. As a result, you can alter your diet and possibly reduce the cost of your prescription drugs.

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Physiological and Biochemical Outcomes of Herbal Medicine Use in the Treatment of Hypertension

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Annaletchumy Loganathan and Natalia Shania Francis

Contents

Abstract

Complementing herbs with anti-hypertensive drugs is a standard health treatment for those with uncontrolled high blood pressure. This review compares the effects of completing different types of herbs with anti-hypertensive drugs on the physiological and biochemical outcomes such as blood pressure, body mass index

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(BMI), waist-hip circumference, glucose levels, hemoglobin level, and lipid profiles. Ambulatory blood pressure measurements (ABPMs) help detect blood pressure variations and serve as a cardiovascular risk marker for hypertension patients. In the clinical practice guidelines, ABPMs are included in diagnosing and managing isolated office hypertension among pregnant women with masked or labile hypertension, as the figure denotes blood pressure readings over 24 h. Point-of-care (PoC) testing devices provide rapid results and information at the location and time of care, the measuring procedure of which can even be done at home. This practice eliminates negative biases, and PoC equipment is capable of chalking up measurements as accurate as those of laboratory-based biochemical analyzers. PoC reduces cost, waiting time, and the necessity for multiple visits improves quality of life (QoL) and increases accessibility of people living in remote areas to medical testing. However, taking ABPMs in actual clinical settings and PoC technology are grossly underutilized in the Asian regions. Hence, this chapter will review evidence of complementing herbal medicine with anti-hypertension therapies and the physiological and biochemical outcomes using ABPMs and PoC devices.

Keywords

Herbal medicine · Traditional herbal medicine · Hypertension · Ambulatory blood pressure monitor · POC · Blood pressure · Blood lipids · Blood glucose · Hemoglobin

Abbreviations

ABPM	ambulatory blood pressure measurements
ACE	American Council on Exercise
BMI	body mass index
DBP	diastolic blood pressure
HDL-C	high-density lipoprotein cholesterol
LDL-C	low-density lipoprotein cholesterol
NO	nitric oxide
PoC	point of care
QoL	quality of life
SBP	systolic blood pressure
T&CM	traditional and complementary medicine
TC	total cholesterol
TCM	traditional Chinese medicine
TG	triglycerides

1 Introduction

Hypertension is a known global health issue that affects people regardless of age, race, and socio-economic background; it is a leading cause of premature death. Hypertension is defined as persistent elevations of systolic blood pressure (SBP) of

140 mm/Hg or greater and diastolic blood pressure (DBP) of 90 mm/Hg or more significant, taken at least twice on two separate occasions [1]. The prevalence of hypertension varies worldwide but is more common in low- and middle-income countries, and ironically, the number of affected people being undertreated is proportionally higher [2]. In Malaysia, the number of people diagnosed with hypertension from 2015 to 2021 increased by 14.1% among individuals above the age of 30 years [3].

The anti-hypertensive medication is often the first-line treatment; however, many people complement medication with herbal medicine to treat hypertension. Herbal medicine could be raw plants that are dried, minimally processed, mildly boiled, infused or herbal materials, herbal preparations, and finished herbal products that contain active ingredients, parts of plants, or other plant materials, or combinations [4, 5]. However, the safety and efficacy of concurrently taking herbal medicine and anti-hypertensive drugs have yet to be validated; the same goes for the potential risks and benefits. A study conducted by Asgary et al. [6] reported that anti-hypertensive drugs prescribed with *Vaccinium myrtillus* efficiently controlled hypertension. Besides blood pressure, beneficial evidence was also observed on parameters like BMI, glucose, and lipid when herbs such as *Vaccinium arctostaphylos* (Whortleberries) were complemented with anti-hypertensive drugs in treating blood pressure [7].

Subsequently, previous studies have also reported adverse effects associated with complementing herbal medicine and anti-hypertensive drugs. For instance, when *Nigella sativa* was taken with ACE inhibitors and diuretics in a study carried out in Indonesia, side effects such as dyspepsia, nausea, and constipation were observed among the participants [8]. Moreover, musculoskeletal pain, fatigue, breathlessness, itching, and even epigastric discomfort were exhibited among the patients who concomitantly consumed amlodipine with *Phyllanthus emblica* extracts [9]. Therefore, monitoring the physiological and biochemistry outcomes of using herbal medicine concurrently with any anti-hypertensive drug is imperative.

Globally, point-of-care (PoC) devices are increasingly popular monitoring instruments for measuring physiological and biochemistry outcomes near patients' bedside. PoC devices are handy in several settings, including remote areas. PoC-measured values obtained immediately within 5 min were comparable to those obtained in laboratories [10]. One of the prime PoC devices that measures physiological outcomes is the ambulatory blood pressure monitor (ABPM), a non-invasive tool that monitors repeated and continuous blood pressure over 24 h. This is useful as blood pressure varies throughout the day, and a single measurement may not be accurate enough to diagnose a person with hypertension [11]. Previous studies have reported that ABPM is the gold standard as it provides consistency and is an excellent diagnostic tool for hypertension [12].

Moreover, portable PoC devices can measure biochemistry parameters of lipid profiles such as cholesterol, triglycerides, glucose, and hemoglobin levels. PoC devices can provide producible physiological and biochemistry outcomes at a reduced cost, provide timely information regarding patients' health status, and reduce the necessity for multiple visits [12]. A home blood glucose monitoring device is essential for diabetes patients to monitor their glucose levels. In accordance, the ABPM and other PoC devices are extremely useful to patients taking herbal medicine anti-hypertensive drugs concurrently; they can make better decisions by observing the changes in blood pressure as well as the biochemistry level after ingesting both in different scenarios at a time, or at an additional time, in the short term or the long term, respectively. In addition, the monitoring devices must be user-friendly and suitable for individuals of all age groups suffering from chronic ailments.

A limitation was reported after using an ABPM in a study: less tolerability as patients may face difficulties self-fixing the machine. However, the ABPM blood pressure measurements have many advantages as they can be saved, downloaded, or printed and shown to medical personnel at clinical visits [13]. Moreover, a study conducted in Korea also elucidated similar beneficial outcomes of a PoC device known as the GCare lipid analyzer, which can measure lipid and glucose levels in a single machine. Other limitations of PoC devices reported are unreliable, unusable, and biased measurements compared to laboratory analyzers. Nevertheless, the PoC-produced measurements using capillary blood were in the acceptable range [14].

There are divergent outcomes of using PoC devices to gauge the concurrent effects of taking herbal medicine and conventional anti-hypertensive drugs. Thus, a thorough investigation among patients with hypertension is warranted. This chapter reviews the physiological and biochemistry outcomes of concurrently taking herbal medicine and anti-hypertension drugs among hypertensive adults.

2 Epidemiology of Herbal Medicine Use with Anti-hypertensive Drugs

Globally, the concurrent use of herbal and anti-hypertensive drugs to treat hypertension is a prevalent trend. In a study carried out among the South African communities, 21% of the study population was reported to have used traditional herbal medicine with anti-hypertensive drugs to treat their hypertension [15]. On the other hand, Jordan said an astounding 51.4% of the population uses local herbal medicine with anti-hypertensive medications, a trend similar to those in other countries like China and Iraq [16]. This could be attributed to the affordability and availability of these various local medicinal herbs and the strong beliefs of the communities in these countries to consume herbs as an effective means of curbing hypertension [16].

In Tanzania, 22.1% of the study population on anti-hypertensive drugs to manage hypertension used herbal medicine such as garlic, ginger, beetroots, papaya, moringa seeds, and Chinese medicine [17]. In a study carried out by Joachimdass [18] in a sub-urban setting of Malaysia, 30.6% of the study population was found to complement their medical treatment with self-prepared herbs such as *Centella Asiatica*, *Momordica charantia*, *Carica papaya*, *Cucumis sativus*, and *Parkia speciosa* with the highest usage. The previous study reported these factors were associated with complementing herbal medicine with conventional anti-hypertensive drugs: being females, having higher education levels, having high socio-economic backgrounds, and dealing with multiple co-morbidities [19].

Even though increasingly hypertensive patients are taking herbal medicine and conventional anti-hypertensive drugs concurrently, the physiological and biochemistry outcomes vary from patient to patient. In a previous study conducted in Tehran, Iran, a local herb, *V. arctostaphylos* berry, was complemented at different doses among subjects who were on prescribed anti-hypertensive drugs, the combinations of which were reported to control both systolic and diastolic readings [20]. The phytochemical compounds contained in the herbal plant, such as chlorogenic acid, anthocyanin, and others, produce no side effects in the study subjects [4, 7]. Similar findings were also observed in another study in Egypt; ginger slices were taken by the study population on ACE inhibitors [21]. However, in a previous study conducted among hypertension patients who used ginger powder alone did complain of gastrointestinal discomfort and diarrhea [22].

3 Evidence-Based Herbal Medicine Use in the Treatment of Hypertension

Even though many pharmaceutical drugs are readily available to treat hypertension, some people may prefer herbal medicine as an alternative and complementary therapy. Below are some evidence-based herbal medicines studied for their potential benefits and effectiveness in combating hypertension.

The first example is garlic (*Allium sativum)*, a renowned herb used since ancient times for its blood-pressure-lowering effects. In a clinical trial in Australia, the participants consumed two garlic capsules rich in a biochemical constituent named allicin over a 12-week intervention period [23]. The results showed that 11.8 mm/Hg significantly reduced the study participants' mean systolic blood pressure; specific blood lipid parameters like LDL and total cholesterol also registered improvements.

Besides, *hibiscus sabdariffa* is another herb that potently affects blood pressure readings. Based on a study published by Mckay and colleagues [24], it was discovered that drinking hibiscus tea for 6 weeks lowered systolic and diastolic readings by an average of 7.2 mm/Hg and 3.1 mm/Hg, respectively. In addition, the supplementation of hibiscus also brings about hypocholesterolemic effects because of its antioxidant properties, regardless of the age and gender of the consumers.

Furthermore, *Olea europaea* L. (olive leaf) extracts also exhibit blood-pressure-lowering effects in a clinical trial. It was reported that positive outcomes could be seen in blood pressure and lipid profiles due to the active phytochemistry constituent, namely, oleuropein, found in the olive leaf extracts. This is because oleuropein, a polyphenolic compound, inhibits angiotensin-converting enzyme (ACE) through its hypotensive actions [25]. *Momordica charantia* L. (bitter gourd) is commonly used to prevent hypertension. A meta-analysis study reported no significant difference after consumption of bitter gourd; however, reductions in systolic and diastolic measurements were observed among younger adults when used on a short-term basis [26].

Hypertensive adults should always be cautious when consuming herbal medicine and be more transparent about their usage to healthcare providers, as there could be some adverse effects, which may lead to poor hypertension management.

4 Evidence on Herbal Medicine Interaction with Anti-hypertensive Drugs

Nowadays, herbal medicine as an alternative or complementary therapy is rising. However, this herbal complementation to medical treatment may lead to potential interactions between the two, producing adverse effects on the body. Moreover, the effectiveness of conventional medication could also be impacted negatively because of the herbs' complementation. Hence, the following information sheds light on the interactions between herbal remedies and anti-hypertensive drugs.

On the other hand, Hawthorn is an excellent example of an herbal remedy that may improve blood pressure readings. On the other hand, there have been reports of explicit interactions between hawthorn and two classes of conventional first-line drugs that are typically used to treat high blood pressure. The drugs include beta-blockers, for instance, atenolol and propranolol, as well as calcium channel blockers like amlodipine. Both drugs work by dilating the blood vessels; concurrent intake of hawthorn may notably lead to a spike in the effectiveness of the drugs, which in turn causes repercussions like excessive lowering of blood pressure [27].

Panax ginseng (ginseng), which possesses ginsenoside-Rg3, is a steroidal saponin compound that provides anti-hypertensive effects. These outcomes are likely mediated by myogenic response inhibitions on blood vessels [28]. Nevertheless, interactions between ginseng and anti-hypertensive medications such as calcium channel blockers (amlodipine) have been noticed. A study by Jeon et al. [29] elucidated that this interaction may lead to elevated blood pressure readings and dire consequences as the drug's potency may be diminished. Similar outcomes of high blood pressure readings due to decreased effectiveness of drugs were also discovered when *Hypericum perforatum* (St. John's wort) interacted with calcium channel blockers, digoxin, and warfarin [27].

Furthermore, *Glycyrrhiza glabra* (liquorice root) is commonly used in beverages and as a herbal antidote. Research done by Deutch et al. [30] sheds light on the interfering effects of liquorice with hypertensive medication, such as ACE inhibitors. The active phytochemistry constituent, namely, glycyrrhizin, which is found in liquorice, causes interactions with anti-hypertensive drugs that are metabolized by Cytochrome P450 3A4 (CYP3A4), an enzyme found in the liver and intestine, which increases blood pressure. Conversely, *Salvia miltiorrhiza* (Danshen) can interact with diuretics and lead to uncontrolled hypotension outcomes [27].

Table 1 contains the types of herbs and drugs that lead to improvements when used concomitantly, as well as the phytochemical constituents and mechanism of action that lead to specific outcomes.

Table 1 Examples of herbal medicine with anti-hypertensive properties used in clinical studies

Author, year	Number of subjects	Type of herbs	Results and implications	Country
[20]	100	400 mg *V. arctostaphylos* berry leaf extract capsule	• The *V. arctostaphylos* berry leaf extract capsule significantly reduced systolic and diastolic blood pressure due to the presence of gallic acid, chlorogenic acid, and anthocyanin • Another study further justified that the anti-hypertensive effect of this leaf extract was indeed due to the presence of anthocyanin, chlorogenic acid, and quercetin • These constituents mediated actions such as endothelial nitric oxide synthase activation, ACE inhibition, as well as calcium antagonistic action [7]	Tehran, Iran
[21]	40	Ginger slices	• Blood pressure readings were controlled and found to be less than 120/80 mm/Hg over 1 month when ginger slices were complemented Improvements were attributed to the phytochemical constituents found in ginger such as gingerols when taken concurrently with conventional medicine When 1.2 g of powdered ginger was used instead of slices, reductions in blood pressure were still observed although not significantly. Overall, ginger is still deemed to possess hypotensive effects as it can inhibit ACE and work as an antagonist [22]	Menoufia University Hospital, Egypt

5 Evidence on the Integration of Herbal Medicine into the Conventional Medical Practices

Some types of complementary and herbal medicine practices directly integrated into the conventional medical methods in Malaysia, Shanghai, and Beijing are traditional Chinese medicine, acupuncture, and traditional massages. Herbal medicine practices are integrated with the mainstream of conventional medicine for various ailments, namely, chronic pain, post-stroke and cancer treatments, hypertension, and even for postpartum mothers who are in recovery. In Malaysia, 15 hospitals have started integrating traditional and complementary medicine (T&CM) into modern healthcare systems, including Malay, Chinese, and Indian traditional treatment practices, homeopathy, chiropractic, osteopathy, and Islamic medicine [31]. The T&CM system integration is similar to the application in Shanghai and Beijing, respectively; to receive herbal medicine, the patients must first seek help from Western medicine practitioners and obtain a referral letter with indications of their health problems.

In Malaysia, with a referral letter from a general practitioner, the patients can be treated by doctors who have majored in traditional and complementary medicine and assisted by trained nurses [31]. However, in Shanghai and Beijing, certified Traditional Chinese clinicians will treat patients with a referral [32]. It was reported that one-third of the patients who visited Yuetan Chinese Community Health Service (Yuetan CHS) in Beijing were referred to traditional Chinese medicine practitioners (TCM), as the conventional treatments were unsuccessful. Table 2 contains information on the types of T&CM provided and the specific types of diseases treated in these hospitals [31–34].

Integrating T&CM into modern medical systems is still an area of ongoing research, particularly in treating hypertension and other diseases at hospitals. Limited concrete evidence is available on the types of herbs that can be complemented with drugs and their potency in treating various disorders. Thus, it is of utmost importance to continually and extensively evaluate the efficacy of integrating herbal medicine into conventional medicine practices; it would be helpful for the relevant authorities to develop comprehensive regulatory frameworks for the usage and practice. Moreover, more quality research should be done to overcome the methodological limitations of integration in hospitals; adequate training and education should be provided to medical personnel on safely implementing the integration [35].

6 Anthropometry Measurements

Anthropometry is the quantitative measurement of the physical body using non-invasive tools such as height-measuring devices and weighing scales. The measures comprise height, weight, body mass index (BMI), head, waist, and hip circumferences, skin fold thickness, and waist/hip ratios, all of which can be utilized to assess the risk of cardiovascular diseases such as hypertension [36]. The measurements of waist/hip and the resultant ratios determine body fat distribution, which is correlated

Table 2 Examples of conventional medical practices with the integration of herbal treatments worldwide

Location	T&CM provided	Ailments treated	References
Cheras Rehabilitation Hospital, Kuala Lumpur, Malaysia	Basti therapy, Shirodhara, traditional massage and acupuncture	Chronic pain and post-stroke	[31]
National Cancer Institute, Putrajaya, Malaysia	Herbal therapy for cancer patients and acupuncture	Chronic pain, post-stroke, chemotherapy-induced side effects	[33]
Sungai Buloh Hospital, Selangor	Traditional Indian medicine (varmam therapy)	Pain management in knee osteoarthritis, shoulder, cervical and lumbar	[31]
Shanghai Yueyang Integrated TCM and Western Medicine Hospital	Traditional Chinese and Western medicine, massage disciplines such as tuina and acupuncture as well as dermatology	Spine, stroke, heart rehabilitation, and several disorders	[34]
Yuetan Chinese Community Health Service (CHS), Beijing	Acupuncture and herbal remedies for (Management of common and chronic diseases and physical rehabilitation)	Knee and back pain, headaches, menopausal symptoms, infertility, hypertension, gastrointestinal disorders, urinary problems, and emotional/mental problems	[32]

with hypertension and insulin resistance. In contrast, changes in fat mass contribute to raised blood pressure readings [37]. Therefore, anthropometry measurements taken at health facilities will facilitate identifying patients at risk for hypertension.

6.1 Body Mass Index (BMI)

Higher BMI levels are often associated with co-morbidities such as hypertension, diabetes, and elevated cholesterol readings. The following are some of the studies that examined the BMI outcomes of hypertensive individuals who consumed herbal products and conventional medicine at the same time.

A randomized controlled trial investigated the effects of *Cynara scolymus* (Artichoke) combined with the anti-hypertensive drug captopril on hypertensive patients' blood pressure and BMI readings. After an eight-week intervention period, the patients with co-morbidities such as obesity and diabetes were observed to show improvements in blood pressure and BMI levels [38]. Artichoke is rich in inulin and oligofructose phytochemistry bioactive compounds, which help maintain a balance in gut microbiota. This decreases the amount of inflammation in the body and promotes the modulation of hormones in the gastrointestinal tract, leading to BMI improvements [39].

Another study reported that green tea contains catechin polyphenols, which increase the hormone leptin, which maintains average body weight in the long term [40]. However, the reported results are inconsistent; it is uncertain whether the consumption of green tea improves weight loss and BMI levels. A study by Bogdanski and colleagues [41] reported no significant changes in the BMI of obese hypertensive participants when they complemented green tea extracts with anti-hypertensive drugs for 3 months. Hence, green tea consumption improves weight loss, inflammatory markers, and glucose homeostasis, but its effects on BMI require more research, mainly when used with anti-hypertensive drugs [40].

6.2 Fat Mass

The correlation between body composition and hypertension risk is inconsistent in the previous studies. Ye et al. [42] revealed a positive association between excess fat mass and elevated blood pressure among adults. Direct fat mass measurements require sophisticated equipment, such as machines with dual-energy X-ray imaging technologies and densitometry. Fat mass measurements are indirectly taken using the skinfold method and a bioimpedance machine [43]. The American Council on Exercise (ACE) has developed a guideline for men and women; men with fat mass percentages of above 25% and women above 32% are classified as obese. The following paragraphs explain the outcome of fat mass measurements for concomitant herbal medicine usage with anti-hypertensive drugs among hypertensive adults.

In a study by Boix-Castej'on et al. [44], hypotensive effects and the outcomes of fat mass percentage were observed for consuming two capsules of herbal medicine containing *Lippia citriodora* and *Hibiscus sabdariffa* extract, the participants of which were adults with stage-1 hypertension and were obese. Significant reductions were noted in fat mass percentage and systolic blood pressure readings. Moreover, the improvements in weight loss and BMI were also attributed to the participant's compliance with a regimen of proper exercise and diet. The combination of polyphenolic compounds such as verbascoside and hibiscetin from *Lippia citriodora* and *Hibiscus sabdariffa* was another probable reason for the reduction in fat mass percentage.

Another study reported effects on fat mass percentage among hypertensive adults who consumed flaxseed. The flaxseed, which contains lignans of secoisolariciresinol diglucoside, reportedly can lessen hypertension and diabetes; it can control obesity and bring about weight reductions due to the anti-inflammatory properties and presence of antioxidants [45]. The study participants with hypertension consumed the flaxseed for over 12 weeks, showing a significant reduction in BMI levels and overall fat mass percentage. A plausible reason could be the high amounts of soluble and insoluble fibers in the flaxseed, which help lose weight. Moreover, lifestyle factors such as regular exercise and a healthy diet contribute to the improvements [46]. However, more research is needed to properly implement self-management of body fat mass using herbal medicine and prescribed drugs.

6.3 Waist/Hip Circumferences

The waist/hip circumference (w/h) ratio is high among people with raised blood pressure and abdominal obesity [47]. However, there are plausible studies on the w/h ratio outcomes of hypertensive adults, particularly those who consume herbal medicine and anti-hypertensive drugs concurrently. The following studies detail the evidence obtained on the w/h ratios:

In a study by Dicks et al. [48], adults with hypertension and type 2 diabetes consumed flavanol-rich cocoa powder with multiple anti-hypertensive drugs for 12 weeks; the results showed a significant reduction of w/h ratios. The consumption of cocoa powder and anti-hypertensive drugs concomitantly reduced endothelin-1 system activation and inhibition of the ACE enzyme [49].

Reportedly, hypertensive adults who consumed herbal supplements containing hibiscus, olive leaf, and green tea with conventional medication experienced reductions in the w/h ratio and BMI levels, in addition to improvements in blood pressure readings [50]. Similarly, in another study, saffron consumed concurrently with anti-hypertensive drugs was reported to improve the w/h ratio and other anthropometry parameters due to the presence of carotenoids [51]. Carotenoids are a known antioxidant compound that prevents the damaging effects of singlet oxygen on human lymphocytes, reducing the risk of hypertension and obesity [52].

All in all, extensive research is still needed to investigate the outcomes of concurrent consumption of herbs and anti-hypertensive drugs in depth. Nevertheless, based on past studies of adults with co-morbidities such as hypertension or obesity, concomitant usage of herbal remedies and medicines seems to yield positive changes in w/h ratios.

7 Blood Pressure

Blood pressure is the pressure exerted on the walls of the arteries due to the force of the blood pushing against the walls when the heart pumps blood to all parts of the human body. It is measured in millimeters of mercury (mm/Hg) by an office blood pressure machine or ambulatory blood pressure monitor (ABPM); it is written using two figures determining a person's blood pressure levels. Systolic blood pressure is the first value measuring the pressure in the arteries when the heart beats. Subsequently, the second figure denotes the diastolic blood pressure. It is indicative of the pressure when the heart rests between beats. An ideal blood pressure reading would be 120/80 mm/Hg; when an individual has readings above 140/90 mm/Hg, that person would be classified as suffering from hypertension. It is vital to realize that the tasks may vary throughout the day depending on a person's schedule, stress levels, and several other factors that may affect the quality of life and the likelihood of being predisposed to other co-morbidities. Thus, monitoring blood pressure readings is essential and should be undertaken regularly at a health facility or at home [1].

7.1 Ambulatory Blood Pressure Monitor (ABPM)

Blood pressure measurements via an ambulatory blood pressure monitor have elucidated the effectiveness of detecting blood pressure variations and the prevalence of cardiovascular risk markers in hypertensive patients. The device can be used as a treatment guide for detecting treated but uncontrolled hypertension, as it can measure the blood pressure readings throughout 24 h. On the other hand, in the clinical practice guidelines, ABPM is included to diagnose and manage isolated office hypertension among pregnant women, masked and labile hypertension [11]. ABPM is the gold standard as it provides consistency; it is a better predictor for hypertension as repeated measurements can be done, leading to more excellent reproducibility of average blood pressure taken with increased precision [12].

In a clinical trial in Poland, the hypertensive patients consumed standardized tomato extract with diuretics and acetylsalicylic acid concomitantly over a 4-week duration; the average blood pressure reading measured using ABPM was 127/78 mm/Hg. The readings were repeated every 15–30 min over 24 h. When combined with the drugs, the presence of lycopene from the extract brought about a synergistic hypotensive effect that significantly improved both systolic and diastolic blood pressure readings [53].

Similarly, another study in China reported that the average 24-h ABPM reading was 131/84 mm/Hg, and the measurements were repeated every 20–30 min at the baseline appointment and after 8 weeks. The Chinese herbal formula, Songling Xuemaikang capsule (SXC), was used with losartan, which improved overall blood pressure readings. The changes could be attributed to puerarin, the active ingredient of Pueraria lobate found in SXC, as it can upregulate mRNA expression of ACE-2 and reduce plasma angiotensin II levels [54].

7.2 Office Blood Pressure Monitor

Since the olden days, sphygmomanometers, or office blood pressure monitors (OBP), have been the fundamental tool for measuring blood pressure. The measurements obtained via the OBP make diagnosing and managing hypertension easier as patients could be referred for treatment to prevent raised blood pressure readings [55]. However, readings obtained using OBP machines may come with errors due to masked hypertension. Thus, to avoid such mistakes and increase the accuracy of tasks, the machine should be calibrated by comparing the readings with those obtained manually, and the measurements should be triplicated to acquire an average assignment [56].

Moreover, some other steps could be taken to prevent errors: ensure the patients are seated comfortably without talking and have rested for a few minutes before beginning with the measurements [56]. In a study carried out in Thailand among hypertensive adults who consumed an array of herbs with anti-hypertensive drugs, the average blood pressure reading was 147/81 mm/Hg as measured by an office blood pressure machine. Two readings were taken at 15 intervals and then averaged

[57]. Thus, compared to readings obtained via ABPM across varying studies, the OBP measurement seems to give elevated blood pressure outcomes.

Lai and colleagues [54] explain two significant reasons for the higher blood pressure readings obtained when measured using an office blood pressure machine, compared with those obtained using the ABPM in this study and other previous studies. The first reason could be attributed to the phenomenon known as white-coat hypertension. When an office blood pressure machine measures an individual's blood pressure, the person tends to be less relaxed due to the presence of the medical personnel or equipment. However, several steps have been taken to mitigate the situation. The following reason could be this: office blood pressure measurements are commonly accepted in the morning before the patients consume their prescribed medicine, and readings tend to be higher during the day. In contrast, ABPM readings are taken many times at regular intervals over 24 h, even while a person is asleep at night; besides, there are no external factors that may cause a spike in the readings.

8 Point-of-Care Technology Evaluation and Biochemical Outcomes

Point-of-care (PoC) devices and their utility for patients' self-management of chronic disease states have been well known since the onset of the COVID-19 pandemic. In clinical settings, PoC devices provide rapid laboratory diagnostic results; they are handy monitoring tools for managing patients and titrating medication doses [58]. Several situations have prompted the transition to using PoC devices: the convenience of self-monitoring at home, further improving disease management and healthcare cost-effectiveness.

Many devices are available for home use to manage chronic disorders such as diabetes mellitus, hypertension, congestive heart failure, and anticoagulation. Many of these devices have built-in software capabilities, enabling patients to share important health information with healthcare providers via a computer. Official visits to health facilities revealed long turnaround times for patients to receive results and attend follow-up consultations; in comparison, results are immediately generated using a PoC device, and appropriate actions can be taken as soon as possible. The frequency of HbA1c produced by the PoC device is higher than in laboratories. However, another study on PoC devices reported varied sensitivities for detecting covid-19 antiCOVID-19 lower for IgM, and higher for IgG compared with those of the gold standard, automated chemiluminescent immunoassay [59]. In another study, the newly developed SARS-CoV-2 PoC device was claimed to provide higher sensitivities of neutralizing antibody production. Even though the acquisition of immunoglobulin and neutralizing antibodies varies according to vaccine type, age, days after vaccination, pain degree after immunization, and immunization diseases, this PoC device can be used to give clinical recommendations; it is a decision tool to advise a person whether to receive additional therapy through the immediate rapid testing [58]. A multi-centered cross-sectional study conducted at nine hospitals in Guangzhou reported the risk of having dyslipidemia by examining blood lipids using a PoC device [60]. An analysis of the PoC survey was conducted remotely in Australia; the findings showed timely access to pathology results and subsequent follow-up of patients for treatment, which was practicable despite the long distances to the nearest laboratory [61].

Utilizing PoC devices could be a quick and helpful way to assess health outcomes when herbal medicine is used as a means of self-management, which can then evaluate the statement that "herbal medicines are all safe." It is feasible to introduce PoC devices in China; the widespread use might raise dyslipidemia awareness, improve treatment, and facilitate better control.

9 Blood Glucose Level

Patients with diabetes and hypertension may be high-risk candidates for heart and brain disorders. Blood glucose levels must be adequately controlled along with other cardiovascular risk factors.

9.1 Fasting Blood Glucose Level

For hypertensive patients with metabolic syndrome, consuming Yiqi Huaju Formula, a traditional Chinese herbal medicine mixture, in addition to standard care, significantly improved fasting blood glucose and other assessed parameters [62]. In a different trial, patients with metabolic syndrome who took Pycnogenol® maritime pine bark extract for 6 months had a reduction in fasting blood sugar levels, going from a baseline of 123 mm/Hg to 106.4 mm/Hg after 3 months and eventually to 105.3 mm/Hg at the end of the study [63].

A unique green aquatic plant called Wolffia globosa mankai is a dietary source of high-quality protein. In a randomized crossover study with participants who were abdominally obese, the intervention group was given a shake containing Wolffia globosa to replace dinner. In contrast, the control group was assigned a shake containing yogurt. The study participants showed lower fasting blood sugar levels the following morning (83.2 mg/dL to 86.6 mg/dL) [64].

In India, diabetes and hypertension are common. Fenugreek seeds are frequently taken to treat diabetes and hypertension. Patients with diabetes got 10 gm of fenugreek seeds in hot water daily in a randomized, controlled study, while the control group did not. The patients had type 2 diabetes mellitus, managed by diet, exercise, and insulin or oral hypoglycemic drugs. According to statistical analysis, the study group's fasting blood glucose levels significantly decreased in the fifth month, whereas the study group's HbA1C significantly reduced in the sixth month [65]. On the other hand, treatment with 162 mg/d of onion skin extract quercetin decreased ABP in hypertension patients [66]. Prehypertensive males who consume an olive leaf extract high in phenols do not significantly affect fasting blood glucose levels or other measures [67].

9.2 2-Hour Post-Prandial Blood Glucose Level

Clinical investigations have indicated various advantages of herbal medications in lowering blood glucose. A herbal product containing nettle leaf 20% (w/w), berry leaf 10% (w/w), onion and garlic 20% (w/w), fenugreek seed 20% (w/w), walnut leaf 20% (w/w), and cinnamon bark 10% (w/w) was fed to study participants who failed to control blood glucose with two oral medications and were unwilling to inject insulin for 12 weeks in a double-blind, controlled study by Parham et al. [68]. In tests conducted during fasting and two hours after eating, the glucose level was dramatically reduced after consuming the herbal medicine mixture. Additionally, glycated hemoglobin A1c (HbA1c) dropped from 0.33% to 0.20%. Finally, after using the herbal treatment, the degree of insulin resistance dramatically decreased from 1.9 to 1.4.

An oral glucose tolerance test has improved due to the effects of olive leaf extracts. According to a study, borderline diabetics' blood glucose levels dramatically lowered at 30 and 60 minutes when 1 g of olive leaf was supplied with 300 g of white rice. In both cases, it was believed that the polyphenols in olive leaf extracts inhibited the activity of intestinal and salivary amylases. However, it is also possible that the aglycones in olive leaf extracts compete with the glucose released from food in the gut for glucose receptors, reducing absorption. According to the study, the use of olive leaf extracts by prehypertensive men revealed no significant effects on fasting glucose, insulin, or other indices [69].

10 Lipid Profile

Total cholesterol (TC), triglycerides (TG), high-density lipoprotein cholesterol (HDL-C), and low-density lipoprotein cholesterol (LDL-C) are lipid profile measures in the blood that are frequently evaluated. Increased TC, TG, LDL-C, and decreased HDL-C are signs of dyslipidemia. Dyslipidemia is a known modifiable risk factor for hypertension, ischemic heart disease, and stroke. Participants with dyslipidemia received treatment with the Unani polyherbal medicine Jawarish Falafili (JF) in a prospective randomized research. When JF was ingested, significant decreases in mean blood TC, TG, and LDL-C were seen at baseline compared to the control group who took atorvastatin for dyslipidemia. However, neither group's mean serum HDL-C levels significantly changed after treatment [65]. Compared to the control group in a different trial, consumption of olive leaf extract was also linked to physiologically significant decreases in TC, LDL-C, and TAG, with no change in the effect on HDL-C [70]. The TC and LDL-C reductions seen using olive leaf extracts and JF could be equivalent to decreased total CVD risk when considering earlier statin trials [69]. It is unknown what processes underlie the lipid-lowering effects of herbal therapy.

The consumption of phenolic components of herbal medicine, on the other hand, appears to reduce the activities of critical cholesterol-regulatory enzymes, 3-hydroxy-3-methylglutaryl-CoA (HMG-CoA) reductase (the main target of statins), and acetyl-CoA cholesterol acyltransferase (ACAT), leading to a reduction in cholesterol biosynthesis [71]. Contrarily, neither in the entire trial group nor in the subgroup of hypertension patients did quercetin supplementation for 6 weeks impact fasting serum concentrations of total cholesterol, LDL cholesterol, or HDL cholesterol. As opposed to this, Lee et al. [72] discovered that supplementing male smokers with quercetin (100 mg/d) for 10 weeks considerably decreased fasting serum concentrations of total cholesterol and LDL cholesterol and significantly elevated serum HDL-C. The length of quercetin administration, the study participants' baseline characteristics, and the dosage of quercetin taken internally may be to blame for these conflicting outcomes.

Traditional medicine uses *Phyllanthus emblica*, a plant high in antioxidants, tannins, and vitamin C, to cure various illnesses. Patients with essential hypertension were randomly assigned to receive a *Phyllanthus emblica* capsule (500 mg) or a placebo twice daily, in addition to their usual medications, over 12 weeks in a randomized controlled experiment. In the beginning, the treatment and placebo groups were comparable. Patients with essential hypertension have a satisfactory safety profile when taking *Phyllanthus emblica*. However, using *Phyllanthus emblica* failed to improve oxidant status, antioxidant capacity, or lipid profile, and it did not result in any further drops in systolic and diastolic blood pressure levels [9].

In Egypt, the management of hypertension has traditionally involved the use of *Hibiscus sabdariffa* and *Olea europaea*. When captopril 25 mg, *Hibiscus sabdariffa*, and *Olea europaea* were administered to Egyptian patients with grade 1 essential hypertension for 8 weeks, the subjects' mean triglyceride levels significantly decreased [73].

11 Hemoglobin Level

Some signs of rising hemoglobin (Hb) levels may cause a rise in systolic and diastolic blood pressure. A HemoCue Hb 201+ analyzer (HemoCue AB, Ngelholm, Sweden) routinely analyses Hb levels in finger stick capillary samples. In research by Atsma et al. [74], both male and female adults with hypertension showed a positive rise of Hb with both SBP and DBP. Several scientific explanations for the connection between Hb and blood pressure have been proposed. First, pulse wave velocity, a measure of arterial stiffness, is closely correlated with Hb, which raises SBP and DBP [75]. Furthermore, Hb may relax smooth muscle cells, change peripheral resistance, and thus modify blood pressure by scavenging nitric oxide (NO) generated in the endothelial cells that line blood arteries. Increased free Hb might bind to NO, which tightens the blood vessels and raises blood pressure [76]. Third, elevated Hb concentrations would result in elevated blood viscosity. Previous studies have shown that increasing hematocrit and hemoglobin levels cause blood to become more viscous, possibly damaging cardiovascular function by affecting blood pressure. Iron content in herbal medicine and its ability to treat anemia make it a popular drug. Hematopoiesis is significantly influenced by iron. However, since other elements such as alkaloids, flavonoids, saponins, tannins, calcium, zinc, and vitamins C and K contribute to the body's ability to absorb iron, the therapeutic potential of the herbs cannot be determined only based on their availability of iron. For instance, vitamin C helps the body absorb iron [77]. Some herbal medicines were beneficial for treating iron deficiency anemia, but studies that are more likely to be accurate have primarily focused on this influence on blood pressure control.

12 Conclusions

Blood pressure, blood glucose levels, blood lipids, and hemoglobin levels can all be affected, albeit to differing degrees, by the various herbal medicine protocols. Other confounding factors, most likely conventional medicines, smoking, exercise, co-morbidities, and food, have been the primary contributors to the changes in hypertension treatment. Certain targeted herbal medicine elements have the potential to complement hypertension treatment. However, it is essential to evaluate each patient's pharmacological therapy individually.

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Efficacy of Ethno-herbal Medicines with Anti-inflammatory and Wound Healing Potentiality: A Case of West Bengal, India

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Abstract

Inflammation is the hallmark of several immune defence mechanisms against pathogenic as well as non-pathogenic challenges. However, uncontrolled inflammation can often lead to chronic pathological conditions such as diabetes, autoimmunity, cancer, and other cardiovascular diseases. Present anti-inflammatory drugs pose serious physiological threats in the long run, showing severe side effects and thus, there is a huge demand for alternative drugs. Several medicinal plants used by different ethnic, indigenous, and tribal communities across the world show anti-inflammatory and wound-healing capabilities. One such indigenous community of Sub-Himalayan Bengal of India is the Rajbanshi community, who have established their own folk medicine system. This chapter discusses in detail the traditional Rajbanshi healers, the medicinal plants they use, their preparation of the medicinal formulation, as well as route and dose of administration. Plants used to treat various inflammatory ailments such as allergy, common cold, diabetes, skin diseases, cut, burn, gastrointestinal problems, and wound healing properties are enlisted in this chapter. The study also evaluates seven of the plants surveyed for judging their capability to scavenge free radicals, inhibit delayed-type hypersensitivity, and stimulate antibody production in the murine model. This chapter provides an opportunity to unravel new anti-inflammatory drugs with the potential for immunotherapeutic applicability.

Keywords

Herbal Medicine · Rajbanshi · Anti-inflammatory · Anti-oxidant · Wound healing · Immunostimulatory · DTH

Abbreviations

Ab	Antibody
ACE	*Ageratum conyzoides*
Ag	Antigen
ANOVA	Analysis of variance
CAE	*Centella asiatica*
CCE	*Cajanus cajan*
CGE	*Coccinia grandis*
COPD	Chronic obstructive pulmonary disease
CPCSEA	Committee for the Purpose of Control and Supervision of Experiments on Animals
CSE	*Cleome spinosa*
DDE	*Drymaria diandra*
DNFB	2,4 dinitrofluorobenzene
DPPH	2,2-Diphenyl-1-picrylhydrazyl Radical Scavenging assay
DTH	Delayed type hypersensitivity
EDTA	Ethylenediaminetetraacetic acid
HA	Hemagglutination assay
MA	Mannitol

MDA	Malondialdehyde
MME	*Mikania micrantha*
NSAIDs	Non-steroidal anti-inflammatory drugs
ROS	Reactive oxygen species
SCID	Severe combined immunodeficiency disease
SRBC	Sheep Red Blood Cells
TBA	Thiobarbituric acid
TCA	Trichloroacetic acid

1 Introduction

Inflammation is a host immune response to protect the body against pathogenic and non-pathogenic infection and injury. As a general mechanism, inflammation gradually initiates the healing process to recover from the local tissue injury and restores the body's homeostasis. However, the case of chronic inflammation and severe hypersensitivity, where the healing process is not restored properly, can lead to chronic inflammatory conditions affecting several organs such as the heart, liver, lungs, kidney, pancreas, gastrointestinal tract, or even brain. These situations can lead to serious inflammatory-associated disorders such as diabetes, pancreatitis, atherosclerosis, Alzheimer's, rheumatoid arthritis, osteoarthritis, tuberculosis, asthma, pulmonary fibrosis, COVID-19, and cancer [7, 16–18, 20, 51, 52, 54]. Synthetic anti-inflammatory drugs are beneficial in treating inflammatory diseases but with prolonged use, they can cause severe life-threatening side effects. Non-steroidal anti-inflammatory drugs/ NSAIDs, such as aspirin, ibuprofen, fenoprofen, indomethacin, diclofenac, flurbiprofen, naproxen, piroxicam, tenoxicam, and oxaprozin which are commonly used, cause serious deleterious effects leading to cardiovascular risks, gastrointestinal and hepatotoxicity, hypertension, and even neurotoxic effects [3]. Thus, newer and alternative approaches, showing comparable effectivity with less or no deleterious side effects need to be taken into consideration. Medicinal plants contain many phytoconstituents which show comparable efficacy as synthetic drugs [57]. There are several traditional methods practised by indigenous communities to treat inflammatory illness using herbal formulations [19] that need to be explored for new and safer anti-inflammatory drugs.

Plants are one of the most important source of medicine in many parts of the world [13–15, 21–31, 35–37, 49, 50]. In spite of enormous progress in modern medicine and pharmaceutical research, the use of medicinal plants still remains an important part of daily life. With the arrival of modern chemistry in the mid-eighteenth century, the ability to isolate, and purify specific compounds and synthesize them in large quantities became the foundations of the current pharmaceutical industry. Natural plant extracts of therapeutic relevance are of paramount importance as reservoirs of structural and chemical diversity [1, 40] and are used commercially across the globe. India represents a country of diversified ethnic groups with rich knowledge of biological resources and traditional medicines. Here, Ayurveda still remains dominant, compared to modern medicine. The basis of the ancient wisdom of Ayurveda medicine was that a system as complicated as the human body could not easily be cured by single compounds, but rather to reset harmony of the spirit and body by the administration of combinations of medicines. Different indigenous communities in India treat their daily ailments based on the knowledge that they have been sharing among their community for generations [12, 33]. Rajbanshi community is one such indigenous group of people, who have been living for thousands of years in the Terai and Brahmaputra valleys of Eastern Himalaya, which is rich in biodiversity. Rajbanshis are traditionally agriculturists and at the same time being deeply rooted in mother nature they have acquired great knowledge of plants with medicinal values in due course of time [44]. Rajbanshi traditional healers, called Kabiraj, have vast knowledge regarding the use of plant species for the treatment of various ailments. The new millennium with its developed technology has provided many opportunities for validation of the uses of natural products. The present work would like to unravel the vast knowledge of the Rajbanshi people regarding their ancient traditional medicinal practices, mainly in curing inflammation and healing wounds. At the same time, we have also put efforts to validate the efficacy of some of these medicinal plants through laboratory experiments which could provide preliminary data to explore their potentiality in a larger perspective as anti-inflammatory agents. We believe this work will pave the path for further work in the field of immunotherapy against inflammatory diseases.

2 Overview of Herbal Medicine Practices in the West Bengal Area of India

West Bengal is a state of India, situated between 27°13′15″ to 21°25′24″N latitude and 85°48′20″E 89°53′04″E longitude and the total area that the state covers is 88,752 Km^2. The state has diversified landforms, with mountains, plateaus, hills, plains, and coastal regions. In the northern region, it has the Himalayas and in the south, it has the Bay of Bengal and the Sundarbans coastal area. Due to this varied geographical pattern, West Bengal is rich in biodiversity with respect to both its flora and fauna. There are several ethnic and tribal communities in West Bengal, in the southern region there are mainly the Santhals, Koras, Mundas, Mahali, Kora, Meches, Lodhas, Bhumijs, Oraon, Kherias, and in the northern districts there are mainly the Rajbanshis, Koches, Meches, Rabha, Lepcha, Gorkhas, and Adivasis [48]. All of these communities stay very close to nature, mostly nearby forests, and have enormous knowledge regarding the use of medicinal plants, which has been passed to them from one generation to another. Folk and ethnic medicine has become an integral part of their culture and tradition. Almost every house in the community has medicinal plants that are used as a regular source of remedy to deal with day-to-day health problems such as cuts, injuries, burns, fractures, fever, headache, diarrhoea, skin rash, allergy, etc. Usually, at first, they try to cure the health problem happening in the family with their own indigenous knowledge of household herbal remedies. And in case it is not being cured by them, they approach the traditional healers. The traditional healers are commonly known as Ojha, Kabiraj, and Mahan, on whom the local people depend for their treatment either due to their age-old faith or also often due to their financial constraints [2].

Every medicinal plant that is being used for treatment has been given a local name in the vernacular language by the community people. Among the varied parts of the plants being used, the aerial parts are mostly preferred in preparing the medicine [46]. A study by Sarkar and Mandal [47] mentioned that the Santal, Munda, Lodha, and Mahali tribes have their own traditional way to treat different ailments, where they use different plants such as *Shorea robusta, Andrographis paniculata, Adhatoda vasica, Caesalpinia pulcherrima, Calotropis gigantea, Butea monosperma, Cajanus cajan, Tagetes patula, Ocimum sanctum, Psidium guajava,* and *Moringa oleifera*. For curing diarrhoea, the Santal and Lodha tribe often use roots of *Shorea robusta*, leaves of *Psidium guajava,* and also bark and leaves of *Butea monosperma*. Leaves and roots of *Datura metel* and *Calotropis gigantea* are used by Santal, Munda, and Lodha community for treating dog bites. Various parts of *Cajanus cajan* is used by the Munda and Lodha community to treat Jaundice. The Santal and Lodha community use roots and leaves of *Andrographis paniculata* during snake bite. *Adhatoda vasica* is used for treating cough and cold, *Datura metel* for treating asthma by Santal, Lodha, and Munda communities. However, these tribes now believe that the practice of treating diseases with ethno-medicines is diminishing gradually due to the decline in the number of plant species in their localities and the adjoining areas. Another work was carried out by Paul [42] in the district of Purulia, which is located in the western part of West Bengal and is predominated by Toto and Birhor tribes. The study says that these tribes as well as local communities use several plants for treating fever, such as stems of *Tinospora cordifolia*, roots and leaves of *Calotropis gigantea*, roots of *Hemidesmus indicus* and *Thysanolaena agrostis* Nees, the entire plant of *Andrographis paniculata* wall, and the bark of *Randia dumetorum.* Leaves and roots of *Breynia retusa* Alston and leaves of *Ocimum canum* for treating pneumonia and cough, the entire plant of *Alangium salviifolium* for snake bite, bark and leaves of *Streblus asper* for bronchitis, and roots of *Clitoria ternatea* and *Mimosa pudica* are used for treating infertility. Indigenous people of this district also used leaves/bulbs of *Allium cepa* Linn, the entire plant of *Andrographis paniculata* wall, and the entire plant of *Polygonum plebeium* R.Br. for treating diarrhoea and dysentery.

Raj and co-workers [43] did an extensive study by interviewing indigenous people in the villages nearby Chilapatta Reserve Forest, which is situated in the sub-Himalayan region of West Bengal. The indigenous people of this area mostly suffer from stomach-related disorders such as gastroenteritis, dysentery, stomach pain, indigestion, stomach worm, diarrhoea, and ulcers, and thus, there are about 40 different plant species that are being used for stomach-related ailments. Next to that are the plant species that are used for healing different types of cuts and wounds such as *Justicia adhatoda* L., *Justicia gendarussa* Burm. f., *Alternanthera brasiliana* (L.) Kuntze, *Rauvolfia serpentina* (L.), *Thespesia populnea* (L.), *Asparagus racemosus* Willd., *Ageratina adenophora* (Spreng.), *Ageratum conyzoides* L., *Eupatorium odoratum* L., *Oroxylum indicum* (L.) Benth, *Bryophyllum pinnatum* Kurz., *Shorea robusta* Gaerth f., *Acacia catechu* (L. f.) Willd., *Cassia sophera* L, *Leucas aspera* (Willd.) Spreng, *Melastoma malabathricum* L, *Melia azedarach* L, *Moringa oleifera* L, *Datura metel* L, *Solanum nigrum* L, *Christella dentata*

(Forssk.), *Cissus repanda* Vahl, *Alpinia malaccensis* (Burm.f.) Roscoe, *Curcuma caesia* Roxb., and *Curcuma longa* L. For treating cough and cold they mainly used *Andrographis paniculata* (Burm.f.) Wall. *ex* Nees, *Justicia adhatoda*, *Centella asiatica* (L.) Urb, *Alstonia scholaris* (L.) R. Br., *Phoenix sylvestris* (L.) Roxb, *Terminalia bellirica* (Gaertn.) Roxb, *Elaeocarpus ganitrus* Roxb. ex G. Don, *Ricinus communis* L, *Ocimum sanctum* L, *Melia azedarach* L, Moringa oleifera L., *Myristica longifolia* Wall, *Syzygium cumini* (L.) Skeels, *Piper nigrum* L., and *Lantana camara* L. Another study by Dutta et al. showed that the tribal population of Cooch Behar used different herbal formulations mainly to treat abdominal disorders, jaundice, liver problems, and intestinal worms, and also to heal cuts and wounds [9].

This traditional knowledge has become the basis of primary healthcare being practised by different ethnic and indigenous communities of West Bengal. The use of traditional medicine is important for providing effective treatments that are prepared for the individual's needs. Additionally, Ayurveda, a traditional Indian system of medicine, has been shown to be effective in treating a variety of conditions, including diabetes, heart disease, and cancer. Furthermore, traditional medicine has been found to be an effective form of preventative care, in this region. Herbal formulations prepared by making powder or paste of different parts of plants are mainly used for curing diseases such as ulcers, diabetes, skin diseases, vomiting, diarrhoea, liver problems, fever, neurological disorder and gynaecological disorders.

3 Herbal Medicine Practices in West Bengal

3.1 Study Area and the Survey Work

The study area comprises north of West Bengal, India, located at 26°22′N latitude and 89°29′E longitude, covering the Cooch Behar district (Fig.1a). Local practitioners/traditional healers were interviewed through structured questionnaires. A total of 20 informants, where 13 of them were traditional healers or Kabiraj, were interviewed to obtain details about their methods of treatment. Written consent was taken from each interviewed person, who was willing to share their knowledge. The information was recorded on the basis of the names of the plants, their medicinal use, the formulation of the herbal preparation, and the methods of administration. Interviews and discussions were conducted in the local Rajbanshi language.

3.2 Leaf Extract Preparation

Most of the plant specimens were collected from the medicinal garden of traditional healers and some from nearby natural vegetation. The collected plant specimens were preserved with 2% mercuric chloride in ethanol and were deposited at the Department of Botany, University of North Bengal for identification. A unique accession number was assigned to each herbarium specimen by the Department of Botany. For extract preparation, leaves of the plants were collected, cleaned, shade

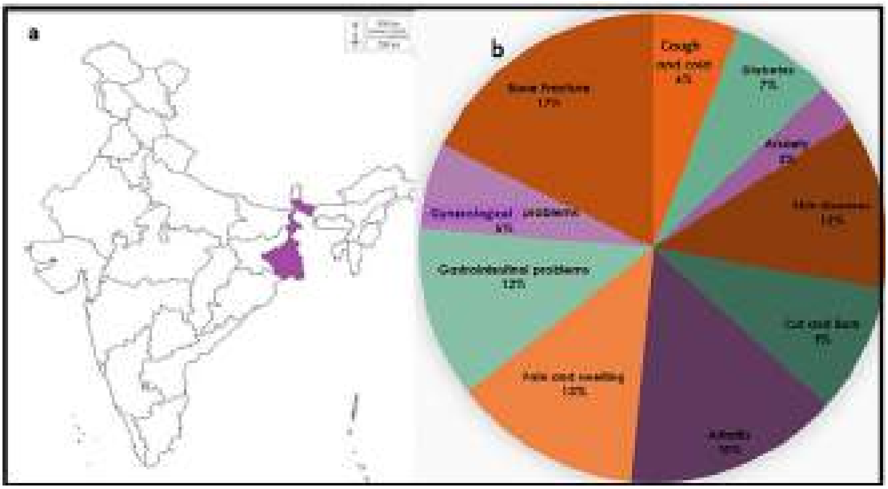

Fig. 1 (**a**) Location of West Bengal in the country map of India; (**b**) Different types of diseases cured by the Rajbanshi community with their indigenous herbal preparations

dried, and ground into powder form with a mechanical grinder. For each type of plant, 10 gm of ground leaves were then soaked in 100 ml of ethanol (80% v/v) in a 1:10 ratio and placed for 3 days in a magnetic stirrer. After 3 days, the extract was filtered first with Whatman paper (2) followed by a Millipore syringe filter (0.22 mm). The dry weight of the extracts was determined by evaporating to dryness through Rotary Vaccum (Buchi). Seven plants were chosen for further investigations, namely, *Ageratum conyzoides* (ACE), *Centella asiatica* (CAE), *Cajanus cajan* (CCE), *Mikania micrantha* (MME), *Coccinia grandis* (CGE), *Cleome spinosa* (CSE), and *Drymaria diandra* (DDE).

3.3 Free Radical Scavenging Assay

Inflammation leads to the generation of reactive oxygen species (ROS), which in due course of time causes oxidative stress and damage to the tissue. Natural compounds which are capable of quenching/scavenging ROS could be considered a potential anti-oxidant and anti-inflammatory agent. Thus, free radical scavenging assays were performed.

3.3.1 2,2-Diphenyl-1-picrylhydrazyl Radical Scavenging (DPPH) Assay

The protocol followed by Mahdi-Pour et al. [39] and Patel et al. [41] was considered for DPPH scavenging assay with some modifications. Different concentrations of 2 μl of extracts containing 0.0292 mg/ml, 0.0584 mg/ml, 0.0876 mg/ml were mixed with 800 μl of DPPH solution in methanol. The solution mixture was incubated for 30 min in the dark at room temperature. Gallic acid (GA) was used as a positive control. At 517 nm, the optical density (OD) was estimated through a spectrophotometer (HITACHI, U-2900). The free radical scavenging was calculated as a percentage of inhibition according to the following formula:

$$\text{Percentage of DPPH radical \% scavenging} = \{(A0 - A1)/A0\} \times 100$$

Where absorbance of the control reaction is denoted as A0 and A1 is of the extracts/ethanol.

3.3.2 Hydroxyl Radical Scavenging Assay Through Malondialdehyde (MDA) Generation

In vitro hydroxyl radical scavenging properties of different plant extracts were investigated using Fe^{2+}- ascorbate- EDTA (Ethylenediaminetetraacetic acid)- H_2O_2 system. The reaction mixture of 1 ml final volume contained L-ascorbic acid (1 Mm), 2-deoxy D-ribose (2.8 mM), ferric chloride (2 mM), hydrogen peroxide (H_2O_2) (1 mM), EDTA (1 mM) and phosphate buffer (4 mM) was incubated at 37 °C for 1 h with different concentration of leaf extract (0.146 mg/ml and 0.0584 mg/ml). TBA-TCA reagent containing 2.8% trichloroacetic acid and 1% aqueous thiobarbituric acid (TBA) was administered into the mixture. The reaction mixture was then boiled (15mins) for generating pink-coloured MDA-TBA chromogen. Once the reaction mixture had cooled the OD was estimated at 532 nm in UV- spectrophotometer (Make-Hitachi, U2900) [5, 45]. Mannitol (MA) a well-known anti-oxidant was used as a positive control to compare the level of efficacy of the plant extract. The efficacy of different plant extracts and ethanol as vehicle control was calculated assuming the level of MDA generation to be 100% in tubes deprived of extracts or ethanol and the inhibition percentage was calculated with the following formula:

$$\text{Percentage of inhibition} = \frac{\text{MDA generated in untreated tube} - \text{MDA generated in extract/ethanol treated tube}}{\text{MDA generated in untreated tube}} \times 100\%$$

3.4 Experiments on Animal Model

3.4.1 Animal Model

Swiss albino mice of 10–12 weeks (both male and female) were considered for in vivo experiments. Each experiment was repeated at least thrice taking 10 mice in each set. All experimental procedures were followed as guided by the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), Govt. of India. The Institutional Animal Ethics Committee of Cooch Behar Panchanan Barma University approved the protocols (Ref. No. CBPBU/IAEC/ZOO/010/2018).

3.4.2 Delayed Type Hypersensitivity (DTH) Assay

Delayed-type hypersensitivity (DTH) reactions represent a clinically important class of immune responses that has wide-ranging significance for various health and disease conditions [39]. Several pathogens such as mycobacteria, different proteins, as well as haptens can induce DTH reaction [34, 55]. Thus, to judge the anti-inflammatory property of the plant extract, delayed-type hypersensitivity model induced by a strong hapten, 2,4 dinitrofluorobenzene (DNFB) was considered. Swiss albino mice were injected subcutaneously with DNFB made in acetone. For primary sensitization, 0.000001% DNFB was injected in the subcutaneous region of the right foot pad of the mouse. The reinjection was done after 8 days on the left foot pad with 0.0001% DNFB. Different leaf extracts prepared in ethanol were administrated (25 μl dose) intravenously 1 h before resensitization. For each experiment, an equivalent amount of ethanol (alcohol) was administered in a separate set of animals 1 h prior to resensitization [6]. The size of the inflammation of the foot pad (diameter) was measured through a slide calliper and has been indicated as the degree of inflammation.

3.4.3 Hemagglutination Assay (HA Titre)

To understand whether the plant extract can stimulate antibody production with its treatment, HA titre was performed, where the antigen (Ag) and antibody (Ab) reaction can be observed as a visible agglutination ring at the bottom of the titre plate. Sheep erythrocyte (SRBC) was used as an antigen for immunization. Different leaf extracts/ethanol (25 μl dose) were injected intravenously 1 day before immunization. 0.1 ml of the 25% of SRBC (Ag) was injected intravenously in the lateral tail vein of the mice for immunization. The mice were sacrificed on the seventh day of immunization and serum was collected and stored at −20 °C. On the day of the experiment, the serum was heat inactivated at 56 °C for 45 min in the water bath. The heat-inactivated serum (Ab) was sterilized by passing through Millipore filters (0.45 μm, Sartorius) before experimentation. The experiment was carried out in a 96 well micro-titre plate where the first well contained 1:10 dilution of the heat-inactivated serum and the second well onward two-fold serial dilution (1:20 to 1:20,480) was made with PBS. Then in each well, 0.1% of antigen (Ag) was added before the incubation for 4–6 h at 37 °C in a humidified atmosphere. The reciprocal value of serum dilution in the last well giving positive haemagglutination reaction was considered as agglutination titre value. There will be the least agglutination in one of the wells down the row, beyond which no agglutination is visible, is considered as the end point of agglutination [4, 53].

4 Study Outcome

4.1 Outcomes of the Survey

A total of 20 informers were interviewed during the survey work. In this study we found that most of the healers are aged over 50 years old. The common diseases that were treated with different medicinal plants were cough and cold, diabetes, anxiety, skin diseases, cut and burn, pain and swelling, bone fracture, gynaecological problems, as well as gastrointestinal problems (Fig. 1b).

During this review we have recorded 37 specimens of medicinal plants used by the traditional healers, which are *Mikania micrantha, Stephania japonica,* Ageratum conyzoides, *Cleome spinosa, Cajanus cajan, Sida acuta, Eclipta prostrata,* Centella asiatica, *Ludwigia perennis, Drymaria diandra, Peperomia pellucida, Coccinia grandis, Vitex negundo, Plumbago zeylanica, Terminalia arjuna, Andrographis paniculata, Leucas zeylanica, Cissus quadrangularis, Heliotropium indicum, Calotropis procera, Scoparia dulcis, Trapa bispinosa, Ocimum sanctum, Achyranthes aspera, Euphorbia hirta, Eucalyptus globulus, Ocimum gratissimum, Abroma augusta, Mirabilis jalapa, Trichosanthes dioica, Piper nigrum, Mangifera indica, Tinospora cordifolia, Momordica charantia, Moringa oleifera, Azadirachta indica, and Curcuma longa* (Fig. 2, Tables 1 and 2).

Fig. 2 Medicinal plant used by Rajbanshi community for treating inflammatory diseases and for wound repairing and healing: *1. Mikania micrantha* 2. *Stephania japonica 3.* Ageratum conyzoides *4. Cleome spinosa 5. Cajanus cajan 6. Sida acuta 7. Eclipta prostrata 8. Centella asiatica 9*. *Ludwigia perennis* 10. *Drymaria diandra* 11. *Peperomia pellucida 12. Coccinia grandis 13. Vitex negundo 14. Plumbago zeylanica 15. Terminalia arjuna 16. Andrographis paniculata 17. Leucas zeylanica 18. Cissus quadrangularis 19. Heliotropium indicum 20. Calotropis procera* 21. *Scoparia dulcis 22. Trapa bispinosa 23. Ocimum sanctum 24. Achyranthes aspera 25. Euphorbia hirta 26. Eucalyptus globulus 27. Ocimum gratissimum 28. Abroma augusta 29. Mirabilis jalapa 30. Trichosanthes dioica 31. Piper nigrum 32. Mangifera indica 33. Tinospora cordifolia 34. Momordica charantia 35. Moringa oleifera 36. Azadirachta indica 37. Curcuma longa*

Table 1 Medicinal plant used by Rajbanshi community for treating various inflammatory diseases and for wound healing and repairing. The scientific name, common name, parts used, its preparation, and treatment are mentioned

Sl no.	Scientific name	Local name	Parts used	How it is being used	Ailments cured by Kabiraj
1	*Mikania micrantha*	Ashamlata	Leaves, stem	Boiled with water and decoction	Eczema, fungal infection, wound healing, stomach ache
2	*Stephania japonica*	Muchilata	Leaves, root	Mixed with other plants and boiled with water	Astringent, healing bone fracture, diarrhoea, fever, headache, urinary diseases
3	*Ageratum conyzoides*	Uchundi/ Bonpudina	Leaves, whole plant	Paste of the leaves mixed with other plants	Arthritis (joint inflammation), antifungal, bleeding control
4	*Cleome spinosa*	Hurhure	Leaves	Decoction	Bacterial infection, swelling, wound healing
5	*Cajanus cajan*	Arahar	Leaves, fruit	Leaf decoction	Jaundice; dysentery, wound healing
6	*Sida acuta*	Banmethi/ kureta	Leaves, whole plant	Decoction, juice of the leaves	Skin diseases, fever, common cold, wound healing
7	*Eclipta prostrata*	Keshraj/ kalakeshari	Whole plant	Paste with coconut oil	Promote hair growth, insomnia, dermatitis
8	*Centella asiatica*	Thankuni	Leaves	Two to three leaves were chewed daily during early morning, paste the leaves and apply 1–2 drops to the eye	Stomach problem, dysentery, conjunctivitis, wound healing
9	*Ludwigia perennis*	Bon lobongo	Leaves, roots	Boiled leaves or paste	Used to lower down fever and also for skin cut and wound healing
10	*Drymaria diandra*	Hargila	Whole plant	Plant juice	Laxative, inflammation
11	*Peperomia pellucida*	Luchipata	Whole plant	Plant juice/leaf paste	Stomach problem, cut and wounds

(continued)

Table 1 (continued)

Sl no.	Scientific name	Local name	Parts used	How it is being used	Ailments cured by Kabiraj
12	*Coccinia grandis*	Telakochu	Leaves and fruit	Fruit used as vegetable	Diabetes, osteoarthritis
13	*Vitex negundo*	Nishinda	Leaves	Mixed with different plants with mustard oil, honey, and water and directly applied to the affected area	Arthritis (joint inflammation)
14	*Plumbago zeylanica*	Agechita/ Chitrak	Leaves, roots	Powder of the leaves, paste of the leaves and roots	Piles, common cold cough, improve digestive system
15	*Terminalia arjuna*	Arjun	Bark	The bark of the tree was dried, ground, and then mixed with water/ milk	Heart-related problems, bone fracture
16	*Andrographis paniculata*	Kalmegh	Leaves	One or two washed leaves soaked in water overnight were chewed	Control diabetes, health tonic, treat intestinal worm
17	*Leucas zeylanica*	Dondokolosh	Whole plant	Used as a leafy vegetable (sauté with mustard oil), decoction	Fever, body dche, treat stomach parasite
18	*Cissus quadrangularis*	Harjora	Whole plant	Mixed with other plants with different solvents	Arthritis (joint inflammation), pain, healing of broken bone
19	*Heliotropium indicum*	Hatisur	Whole plant	Mixed with other plants with mustard oil and honey	Arthritis (joint inflammation), healing of broken bone
20	*Calotropis procera*	Akanda	Leaves	Mixed with other plants with mustard oil and honey, roasted leaves applied directly in inflamed area due to arthritis	Arthritis (joint inflammation), pain
21	*Scoparia dulcis*	Chinigur	Leaves	Juice of leaves with water taken orally	Diabetes, jaundice, skin diseases
22	*Trapa bispinosa*	Jolsingara/ paniphol	Fruit	Fresh fruits, or dry fruits made into powder	Helps to clean bowel system, treating ulcers

(continued)

Table 1 (continued)

Sl no.	Scientific name	Local name	Parts used	How it is being used	Ailments cured by Kabiraj
					and wound, flour of fruit is nutritious for health
23	*Ocimum sanctum*	Tulsi	Leaves	Juice of the leaves mixed with honey	Helps to treat cough, cold, and throat infection
24	*Achyranthes aspera*	Apang	Leaves/ roots	In the form of juice for oral consumption or for local application on skin	Consumption of juice helps to treat asthma, cough, pneumonia Local application of the juice helps to treat skin rash and itchiness Brushing tooth with the stem of helps to cure toothache
25	*Euphorbia hirta*	Dudhiya	Leaves	Juice made up of tender leaves	Used to cure respiratory diseases like asthma and bronchitis and irregular menstruation
26	*Eucalyptus globulus*	Eucalyptus	Leaves	Leaves were mixed with other plants to prepare paste	Used to treat arthritis; smoke from the burnt dried leaves helps to reduce cough
27	*Ocimum gratissimum*	Ramtulsi	Leaves	Leaves are crushed and mixed with honey	Helps to treat cough and cold
28	*Abroma augusta*	Ulotkombol	Root, bark, stem	Extracts from root, stem, and bark	Helps to cure gynaecological problems
29	*Mirabilis Jalapa*	Sandhyamalati	Leaves	Juice of the leaves	Leaf juice applied on the wound for healing

Almost all parts of medicinal plants were used for the preparation of medicinal formulations. However, leaves were used mostly compared to roots, stems, or flowers. Depending upon the disease two or more parts of the plants were used to prepare the formulation for the treatment. The mode of application of medicines was topical, orally, or by inhaling the smoke. Herbal preparations were often mixed with mustard oil, coconut oil, honey, or milk. It was observed that many of the traditional

Table 2 List of herbal formulation prepared by Rajbanshi Kabiraj for the treatment of different ailments

Sl. No.	Herbal Ingredients	Solvent	Procedure	Result
1	Leaves of *Trichosanthes dioica* *Curcuma longa* Roots of *Asparagus racemosus*	Water	Leaves were mixed with *Curcuma longa* and roots of *Asparagus racemosus*, crushed and mixed with water	The filtered portion is taken daily to cure allergic problems
3	Seeds of *Syzigium cumini* Internal white part of *Mangifera indica* seed	Honey	All of the ingredients were dried and ground in powder form. After that mixed with solvent for preparation of pills and sun dried.	This combination is very much effective for cough and cold and diabetes
4	*Andrographis paniculata* *Vitex negundo* G*lycyrrhiza glabra* *Azadirachta Indica*	Water	All the ingredients were sun dried, ground into powder form and prepared as pills	One pill every day in empty stomach will help to cure worms of abdomen and also important for healthy stomach
5	V*itex negundo* *Azadirachta indica* *Neolamarckia cadamba* *Datura metel* *Allium sativum* *Curcuma longa*	Mustard oil	All the ingredients were boiled in mustard oil until the oil becomes black	Its helps in pain of arthritis
6	*Vetiveria zizanioides* Rice (*Oryza sativa*)	Water	Roots of the grass were mixed with rice which was soaked overnight and made into paste. The paste was taken after meal	Helps to cure Haematuria and hematemesis
7	*Centella asiatica,* *Leucas zeylanica*	Water	All the ingredients were boiled with water and then dried as pills	Pills are taken after meals and its very much effective for jaundice, body ache, abdominal dysfunction
8	*Datura metel* *Calotropis procera.* *Curcuma longa*	Mustard oil	All the ingredients were made into paste and then mixed with mustard oil and boiled for 20–30 mins. The content was then filtered and the oil collected was used.	This oil is very helpful for reducing pain and inflammation caused by fracture of bone
9	*Withania somnifera*	Water, ghee	A thick paste of all the ingredients was prepared	Its helps in arthritis and also help in joining of

(continued)

Table 2 (continued)

Sl. No.	Herbal Ingredients	Solvent	Procedure	Result
	Cissus quadrangularis *Leucas zeylanica* *Bambusa polymorpha* *Moringa oleifera* *Terminalia arjuna* *Tragia involucrata* *Piper nigrum* *Zingiber officinale* *Vitex negundo*	(clarified butter)	and applied on the area of fracture or on the area of dislocated bone	bone and also helps to reduce the pain of bone dislocation
10	Leaves of *Trichosanthes dioica* *Asparagus racemosus* *Phyllanthus emblica* *Terminalia chebula* *Curcuma longa* *Tinospora cordifolia*	Water	All the ingredients were ground and boiled with 250 ml of water for 30 min. The filtered water was taken every day on an empty stomach	Its helps to cure allergy

healers were interested in treating bone-related problems, like fractures of bone, dislocation of bones, and arthritis. For treating bone fracture they prepare a paste of fresh leaves of *Cissus quadrangularis, Vitex nigundo, Calotropis procera*, and *Heliotropicum indicum* mixed with different solvents and the paste is applied directly on the affected area. According to the local healers *Mikania micrantha, Eclipta prostrata*, and *Sida acuta* are the medicinal plants used for skin, hair-related problems, and also for the treatment of allergies. Application of paste of *Eclipta prostrata* also known as Kalakeshari/Keshraj in vernacular language, helps in the regrowth of hair, dermatitis, and curing insomnia if the leaf paste was mixed with coconut oil. For treating cough and cold, a decoction of fresh leaves of *Ocimum sanctum* (Holy basil or Tulsi) was used. *Ocimum gratissimum*, *Piper nigrum*, and *Leucas zeylanica* were also used to treat cough and cold. *Piper nigrum*, known as golmorich, and its fruits are traditionally used for flavouring food recipes. Ranjbanshi community used it for treating asthma, cough, headache, menstrual pain, gastric problems, and urinary problems too. The roots of *Vetiveria zizanioides* ground with scented-soaked rice and the paste was consumed to cure Haematuria

Fig. 3 Some herbal medicinal formulation in the form of syrup and pills prepared by Rajbanshi community

and Hematemesis. According to traditional medicinal practitioners *Heliotropicum indicum* was very useful in treating conjunctivitis. *Tinospora cordifolia* also known as Guloncho in vernacular language is a deciduous climbing shrub that is traditionally used to cure diabetes, reduce hypertension, and also enhance memory. The methods of application of herbal formulations are different depending on the ailment being treated and are mentioned in Tables 1 and 2. In Fig. 3, different types of medicines prepared by Kabiraj's has been shown. Seven different plants which were mainly used to treat inflammation and for healing wounds by the Rajbanshi community were chosen for comparative analysis of their anti-oxidant, anti-inflammatory, and immunostimulatory activities (Fig. 3). The dry weights of the ethanolic leaf extract and their accession number are represented in Fig. 4.

4.2 Antioxidant Property of the Different Plant Extracts

4.2.1 DPPH Scavenging

In the present investigation, three different doses (0.0292 mg/ml, 0.0584 mg/ml, and 0.876 mg/ml) of different ethanolic leaf extracts have been considered. The percentage of inhibition of DPPH generation by different extracts has been compared to the blank set, where the purple colour gradually decolorizes into pale yellow indicating DPPH scavenging. Out of seven plant extracts , DDE, CAE, CSE, and CGE showed a high percentage of inhibition in all three doses compared to other extracts. Starting from the lowest dose to the highest dose, CSE exhibited the maximum inhibition, comprising 48.80 $\pm$ 3.38 (0.0292 mg/ml), 63.82 $\pm$ 7.39 (0.0584 mg/ml), and

Fig. 4 The identification accession number and the dry weight of the ethanolic leaf extract of seven different plants used for assessing their anti-oxidant, anti-inflammatory, and immunostimulatory properties

72.26 ± 3.23 (0.876 mg/ml), whereas with DDE and CGE the percentage of inhibition ranged between 41% and 50% from lower to higher doses. The percentage of inhibition was low with the lower doses of CAE, around 34% to 36%, however, with the highest dose (0.876 mg/ml) it showed 51.38 ± 3.37% of DPPH inhibition (Fig. 5). The result of CSE at 0.876 mg/ml dose was comparable to Gallic acid, a well-known anti-oxidant which exhibited inhibition of DPPH between the range of 74% and 86% from lower to a higher dose.

4.2.2 Hydroxyl Radical Scavenging

Hydroxyl radical is one of the major reactive oxygen species, formed continuously during the process of reduction of oxygen to water in living organisms. This hydroxyl radical that is being generated can cause a breakdown of disulphide bonds in several proteins, especially fibrinogen, which can lead to abnormal protein configuration as observed during cancer, atherosclerosis, and in neurological disorders. Thus, the present study investigates the protective effect of plant extracts against damage due to hydroxyl radical generation. Two different concentrations of ethanolic leaf extract were considered in the present investigation, 0.0584 mg/ml and 0.146 mg/ml to investigate the scavenging of hydroxyl ion generation. The concentration of malonaldehyde (MDA) generated in the reaction mixture is indicative of the degree of hydroxyl radical generation. The inhibition of hydroxyl generation has been compared with the blank set (Fig. 5).

MDA generation was found significantly lower in almost all the extracts in both the concentrations (0.146 mg/ml and 0.0584 mg/ml) except CCE, where the inhibition of hydroxyl generation was not significant compared to the vehicle control (Fig. 6). The level of inhibition of hydroxyl generation was highly significant for

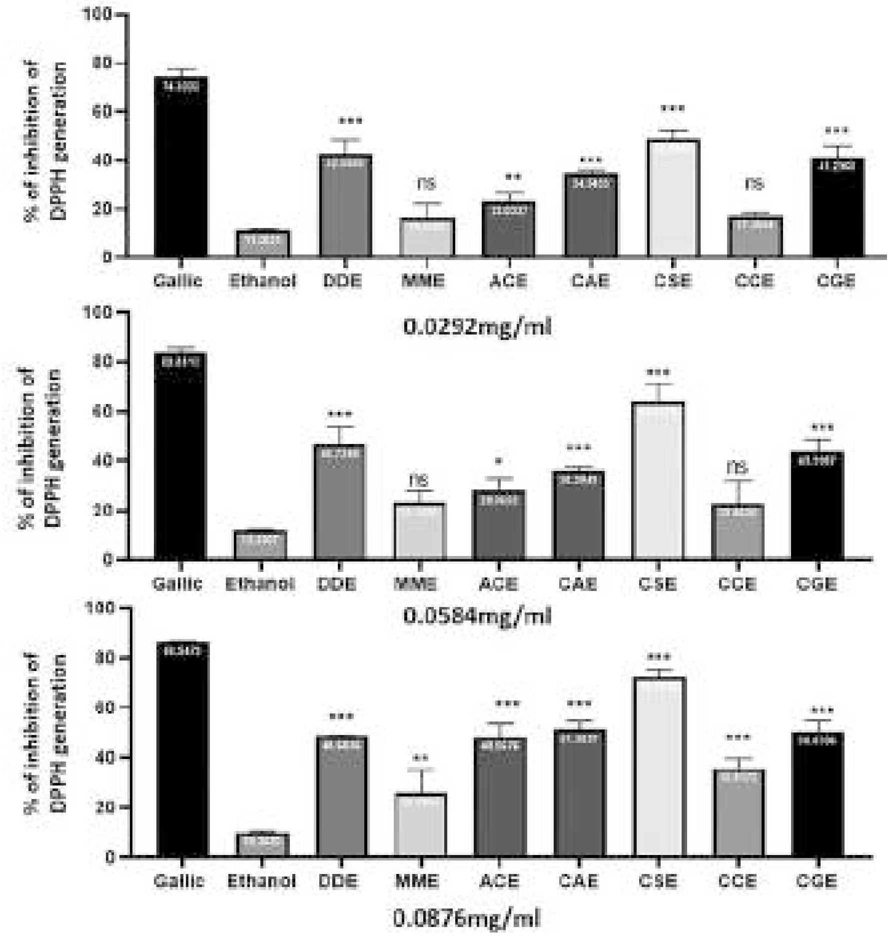

Fig. 5 Inhibition of DPPH generation by different plant extracts at 0.0.292 mg/ml, 0.0584 mg/ml, 0.0879 mg/ml. Gallic acid (Gallic) has been used as a positive control. Results were expressed as Mean ± SD ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, *ns* non significant. The significance of the antioxidant property has been estimated comparing the value of the vehicle control (ethanol)

CSE and ACE in both concentrations compared to others. However, CAE and DDE also exhibited high antioxidant properties compared to CCE and CGE. As observed in DPPH scavenging (Fig. 5), here also CSE could significantly inhibit MDA generation up to 52.84 ± 1.84% at 0.0584 mg/ml dose and 67.47 ± 3.94 at 0.146 mg/ml, dose indicating high potentiality in scavenging hydroxyl radical generation. ACE (58.21 ± 6.97%), CAE (57.55 ± 1.29%), CGE (55.44 ± 4.73%), and DDE (48.61 ± 7.24) at 0.146 mg/ml dose also showed better efficacy in inhibiting hydroxyl radical generation (Fig. 6). Mannitol (MA) has been used as a positive control to understand the potentiality of the antioxidant property of the extracts, where CSE showed to some extent similar effects with mannitol.

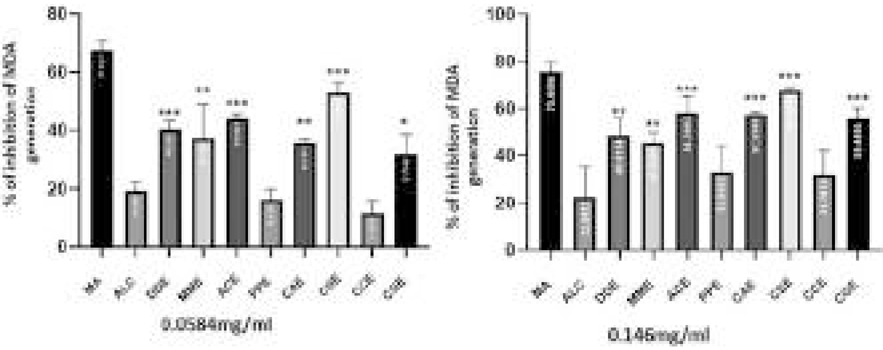

Fig. 6 Inhibition of hydr oxyl ion generation by different ethanolic leaf extracts at 0.0584 mg/ml and 0.146 mg/ml dose. Mannitol (MA) has been used as a positive control. Result expressed as Mean ± SD ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, non-significant has been denoted by ns. The significance of the antioxidant property has been estimated comparing the value of vehicle control (ethanol)

4.3 Anti-inflammatory Property of Different Plant Extracts

Delayed type hypersensitivity (DTH) is a vital defence against several pathogens such as *Mycobacterium tuberculosis, Mycobacterium leprae*, *Leishmania sp*, *Histoplasma*, *Coccidioides* fungi and is also involved in tissue rejection and cancer immunity [10, 32, 38, 55, 56]. Thus, to understand the potentiality of the different ethanolic leaf extracts for curing deleterious DTH response, DNFB, a strong hapten has been used to induce DTH reaction. In case of DTH reaction in the skin, there is an early infiltration of non-specific cells during primary sensitization with the antigen, which is followed by a specific response upon resensitization with the same antigen, which is mainly T cell-mediated response. This antigen-specific hypersensitive response leads to erythema (redness of skin) and oedema at the site of the antigenic challenge in the sensitized animal [4]. The hypersensitivity response is usually delayed, showing effects after 48–72 h of resensitization. In our animal model, the DTH response exhibited within 48 h and gradually got severe. Ethanolic leaf extracts were injected intravenously 1 h before resensitization. Among all the ethanolic leaf extracts, CSE showed the best result in alleviating the DTH inflammation faster. On the 12th day of resensitization with CSE treatment the paw size was 0.295 ± 0.023, which is almost the size of a 0 h normal paw (0.276 ± 0.014). The other extracts such as CAE (0.301 ± 0.005) and ACE (0.313 ± 0.012) also showed better inhibition compared to other extracts, however, the foot pad healing process was still in continuation (Fig. 7a). In the untreated and vehicle (ethanol) control groups, the foot pad muscles as well as paw digits were also lost in the course of inflammation. However, in the plant extract-treated groups, the foot pad loss gradually recovered and the inflammation also subsided much earlier than in the control groups. In the treatment groups, none of the mice exhibited digit loss (Fig. 7b). In the case of the control groups it took more than 3 weeks to subside

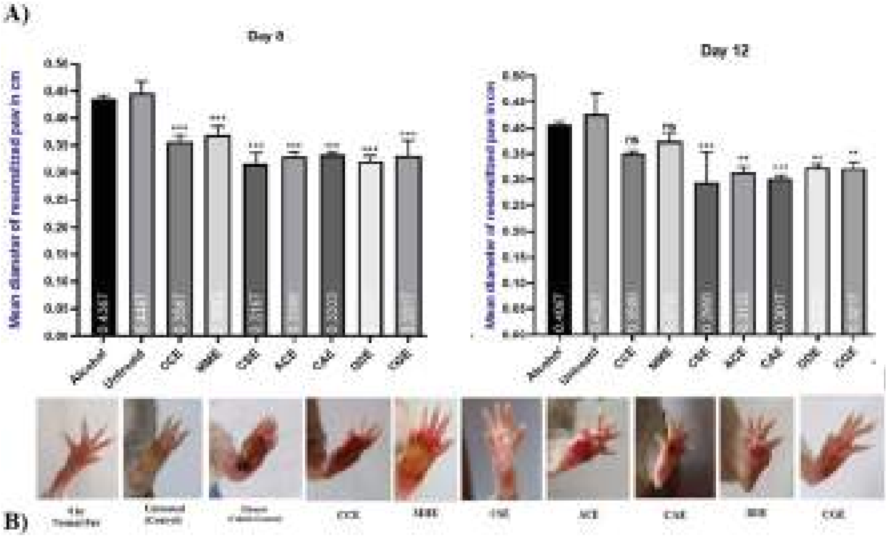

Fig. 7 (**a**) DTH (*delayed type of hypersensitivity*) reaction of resensitized paw in treated, untreated (control), and ethanol (vehicle control) at 8th and 12th days of DTH inflammation, represented by mean diameter in cm, considering the day of resensitization as day 0. Results are expressed as mean ± SD, ***$p < 0.001$, **$p < 0.01$ and *$p < 0.05$, non-significant value has been denoted by ns. (**b**) Photograph shown are of the resensitized paw on the 12th day. The far left photograph shows the normal paw (0 h, before the resensitization)

the DTH inflammation, although the vehicle control groups (ethanol) recovered a bit earlier than the untreated one. CSE-treated mice never exhibited digit loss and the inflamed foot pad eventually healed back to its normal size within 12 days of resensitization, suggesting its strong anti-inflammatory and wound-healing properties. The DTH paw treated with CAE and ACE extract resumed back to normal paw diameter in 13–14 days.

4.4 Immunostimulatory Property of the Different Plant Extracts

The antibody or humoral-mediated immune response during primary immunization was investigated through a haemagglutination titre (HA) assay, which measures the kinetics of primary immunity. The mode of plant extract treatment is similar to the DTH assay, where the mice were injected intravenously with different leaf extracts, 1 h before the immunization with sheep RBC (SRBC). It was interesting to observe that almost all the plant extracts were capable enough to boost antibody (Ab) production higher than the ethanol-treated and untreated mice, however, the level of stimulation varied (Fig. 8). In the serum collected from the primary immunized mice, the highest antibody (IgM) titre value was shown with CSE (*Cleome spinosa*) treatment which was 1:2560, followed by CCE (*Cajanus cajan*) 1:1280 and then MME (*Mikania micrantha*), ACE (*Ageratum conyzoides*), CAE (*Centella

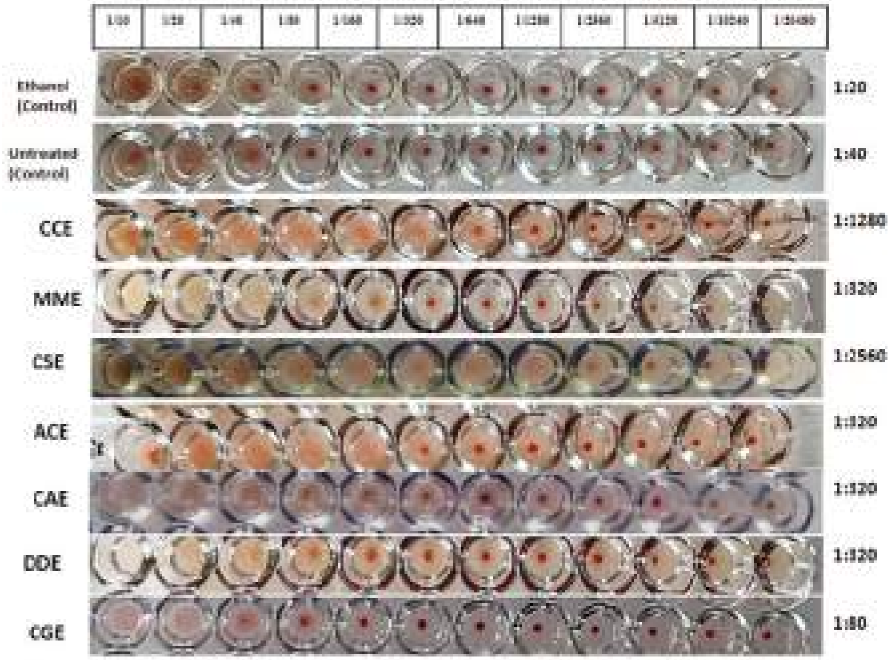

Fig. 8 Antibody response after the treatment of different plant extract. Hemagglutinin titre (HA) value shows the level of antibody production during primary immune response, mostly IgM (immunoglobulin M). The titre value mentioned on the extreme right indicates the last dilution of serum capable of agglutinating antigen

asiatica), DDE (*Drymaria diandra*) where all showed 1:320 titre value, and CGE (*Coccinia grandis*) showed the least with 1:80. Whereas, the Ab titre value was 1:20 with ethyl alcohol and 1:40 in the untreated group.

5 Discussion of Findings

Among the several ethnic communities in West Bengal, Rajbanshi community are the ancient tribe of Koch kingdom, belonging to the Indo-Mongoloid race. This chapter documents the ethnomedicinal knowledge shared by the Rajbanshi community of the Cooch Behar district of West Bengal, India. Cooch Behar is a historic location rich in culture and heritage, marked as a princely state since the sixteenth century, and is known to be the last ruling Hindu dynasty of Bengal before British rule. The Rajbanshi community inhabits mostly the northern part of West Bengal which comes under the sub-Himalayan region. The river system here plays an important role in its civilization by providing a source of water for irrigation and for drinking purposes. Torsa, Tista, Mahananda, Karatowa, and their tributaries are important rivers of this region. This Terai region of Bengal is covered with dense forest, inhabiting a huge number of exotic and wild species of flora and fauna. This area is often subjected to heavy rainfall and flash floods every year. These two important geographical features having rivers and forests makes them suitable for agricultural practices and at the same time makes them vulnerable to disease outbreak, due to drinking contaminated water during the rainy season and the dense forest that harbours parasitic diseases. The diseases that have always been prevailing in this region are dysentery, diarrhoea, cholera, malarial fever, smallpox, goitre, whooping cough, common cold, and skin diseases. In spite of the fact that the people were affected with serious diseases, the healthcare facility was never up to the mark. Due to a lack of proper healthcare facilities, the community had to be dependent on their indigenous traditional herbal medicine. In almost every household in this community, a herbal medicinal garden exists, this way they also contribute in conserving the species. Traditional medicine has been utilized by indigenous communities for centuries and the practice is still deeply rooted in the Rajbanshi community to treat a variety of ailments. Though with due course of time, the availability of modern healthcare services improved, the community still continues to rely on traditional medicine, highlighting the importance of plants in the treatment of illnesses. The results of the survey illustrated the continued significance of traditional medicine in the lives of these indigenous people in the study area. The present study documented the ancient practice of traditional medicine by the Rajbanshi community of India. About 37 medicinal plants were documented that are used to treat inflammation and for healing different types of wounds by Rajbanshis. This study highlights the plants which are focused mainly to cure inflammation, inflammation-related disorders, and healing wounds. Though some of the plants were also used for treating different types of fevers, such as *Stephania japonica, Sida acuta, Ludwigia perennis,* and *Leucas zeylanica*. For treating diarrhoea, dysentery, and for stomach-related disorders *Stephania japonica*, *Cajanus cajan*, *Mikania micrantha*, *Centella asiatica*, and *Leucas zeylanica* are used. The community used several parts of plants for preparing the herbal formulations, starting from the entire plant to its leaves, stem, roots, buds, flowers, seeds, fruits, and bark. Leaves were used the most for preparing the herbal formulation. Different modes of application were preferred for treatment such as consuming the filtrate of the leaf, applying the paste on the wound/inflamed area, inhaling the smoke of burnt leaves, or simply consuming it as cooked vegetables. The herbal preparations were utilized mostly to heal wounds and sores, abdominal pain/ache, cure jaundice, liver problems, as well as for anti-parasitic activities. However, the traditional healers of Cooch Behar district have now inclined more towards treating bone-related problems, such as breaking or dislocation of bones, and arthritis.

Almost all living beings are subject to attack by foreign pathogenic agents. Our immune system plays the most important role in combating those foreign particles and acts as a barrier in establishing the infection induced by pathogens as well as non-pathogenic agents. The non-specific immune system acts as the primary barrier by releasing several mediators and recruiting immune cells at the target site for prompt response through inflammation. Whereas, the specific immune response is target specific which responds through antibody production or through cell-mediated immune response. Immunostimulatory agents from the plant world have the potential to activate the immunological responsiveness of an organism directly at the cell level or by inducing the production of mediators. Plants and plant-derived products are safer, eco-friendly, low-cost, fast, and less toxic as compared with conventional treatment methods and act specifically on target cells without affecting normal cells when administered in proper doses. It strengthens the overall activity of the immune system including those elements that are most capable of controlling inflammatory conditions. Thus, the present study has selected seven different plant species (*Ageratum conyzoides*, *Cenetella asiatica*, *Cajanus cajan*, *Mikania micrantha*, *Coccinia grandis*, *Cleome spinosa*, and *Drymaria diandra*) which were mainly used to cure inflammatory diseases and for healing different cuts and wounds by the Rajbanshi people. Ethanolic extracts of these seven different plants were evaluated for their anti-inflammatory and immunostimulatory properties. The purpose was to look for plants with immunopharmacological properties to cure inflammatory diseases in humans. Among the seven plants investigated, *Cleome spinosa* (CSE), *Cajanus cajan* (CAE), and *Drymaria diandra* (DDE) showed high immunopharmacological properties. They exhibited anti-oxidant and anti-inflammatory properties by scavenging free radical generation, alleviating deleterious DTH response, and healing the DTH wound faster. Knowing the fact that ROS generation is one of the crucial components in aggravating inflammation and causing tissue damage, a plant product that can scavenge ROS and at the same time minimize tissue damage induced by deleterious DTH reaction and heal the wound faster without any digit loss in the murine model could turn out to be a wonderful drug.

The HA titre results were quite fascinating as with CSE and CCE treatment, the antibody titre in the host serum increased several folds higher than the control groups upon antigenic challenge. This immunostimulatory property of *Cleome spinosa* and *Cajanus cajan* has been reported here for the first time. However, other plants such as *Ageratum conyzoides* (ACE), Centella asiatica *(CAE), Mikania micrantha* (MME), and *Dymaria diandra* (DDE) also exhibited similar properties but with lower effectivity. There are certain advantages of using plants to boost the immune system as they can be utilized directly as feed and can stimulate both systemic and mucosal immunity. Our study shows that ethanolic leaf extract *Cleome spinosa* and *Cajanus cajan* plant extract can stimulate the antibody-mediated immune system in a murine model when challenged with sheep RBC. There are several pathological conditions where the immunocompromized host shows more susceptibility towards infections due to certain environmental or genetic factors [11]. One such condition is severe combined immunodeficiency (SCID) disease where there is a deficit of both humoral and adaptive immune response leading to uncontrolled damage [8]. Immunodeficiency can also arise during treatment with various agents/drugs during malignancy or for allograft rejection, leading to secondary immunodeficiency, compromising the host defence mechanism. During immunosuppression induced by pathogens, a herbal compound that has the potential to activate the humoral arm of the immune system can prove to be a wonderful agent to induce antibody production to neutralize the pathogenic effect or binds with the antigen facilitating its destruction by other cells. Thus, it will be interesting to study in detail the immunopharmacological properties of *Cleome spinosa* and *Cajanus cajan* in the future with pathogenic challenges. At the same time, we also suggest that further research on *Cleome spinosa* (CSE), *Cajanus cajan* (CAE), and *Drymaria diandra* (DDE) might provide some important leads to the search for new drugs to cure pathogenic as well as non-pathogenic inflammatory diseases.

6 Conclusion

Globally, the use of medicinal plants is widespread in the developing world, providing primary health care, livelihood improvement, and even a source of income for many people. As a result, the importance of preserving and protecting these plants is of paramount importance for the continued health of both local communities and the global population. Studies have demonstrated that medicinal plants can offer a wide range of therapeutic benefits, from treating chronic and acute illnesses to improving overall wellness. Furthermore, medicinal plants can be used to enhance nutrition, reduce the use of synthetic drugs, and provide economic opportunities for communities. As a result, the conservation of medicinal plants is imperative to ensure that their medicinal properties are not lost. Moreover, research into the potential medicinal uses of plants is an important part of the health care industry. This research can help to identify new medicines, develop safer drugs, and improve the efficiency of existing medicines. Furthermore, the use of medicinal plants can also help to reduce the cost of health care, as many of these plants are readily available and do not require the use of expensive pharmaceuticals. In this study we have documented different ethnomedicinal plants which are mainly used by the Rajbanshi community. Through ethnobotanical studies, it is important to document the indigenous knowledge associated with the plants of a certain region. This knowledge can be used to help conserve and utilize the biological resources of the region, as well as to uncover potential bio-cultural benefits. In addition, the collection of local names and traditional uses of the plants can provide important insights into the culture of the indigenous people of the area.

Strategies related to drug discovery are taking a new turn by including the knowledge gained from traditional and alternative system of medicines. The world still holds enormous medicinal plants and still there are many indigenous ethnic communities that use plants as the important source of treatment. Both need to be explored extensively, for the development of new anti-inflammatory drugs, as a majority of the plant species have not been investigated for their potentiality to cure inflammatory disorders which are non-cytotoxic. Thus, learning from the indigenous communities about their age-old traditional treatment methods and implementing it for the betterment of the mankind is the need of the hour.

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Plants Used in the Management and Treatment of Cardiovascular Diseases: Case Study of the Benin People of Southern Nigeria

30

Matthew Chidozie Ogwu, Happiness Isioma Ogwu, and Enoch Akwasi Kosoe

Contents

Abstract

There is a global increase in cardiovascular diseases (like stroke, heart attack, and heart failure), especially in developing countries. Healthcare organizations in these low- and middle-income countries are unable to handle the current and projected economic and social effects of CDVDs and are struggling to deal with other diseases and developmental challenges that disproportionately impact the region. This chapter aims to produce an inventory of plants and plant-based remedies used by Benin people, Nigeria, for the treatment and management of CDVDs. Fourteen locally identifiable key informants including professional

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herbalists, traditional healers, and herbal remedy sellers were administered questionnaires to document plants and plant-based remedies used in the management and treatment of CDVD by Benin people of Southern Nigeria. From the survey, a total of 17 medicinal plants belonging to 14 plant families were identified, documented, and collected as used by Benin people for the treatment and management of CDVD. The plants included *Theobroma cacao, Aloe barbadensis, Momordica charantia, Laurus nobilis, Morinda citrifolia, Carica papaya, Allium cepa, Ficus exasperate, Elaeis guineensis, Olea europaea, Persea Americana, Moringa oleifera, Beta vulgaris, Allium sativa, Hunteria umbellate, Zingiber officinale,* and *Curcuma* species. The dominant plant families were Lauraceae, Amarylidaceae, and Zingiberaceae with two representatives each. The plants were mostly collected from home gardens, markets, and local forests and within either cultivated or wild systems. Although plants and plant-based remedies were mostly taken to address hypertension and high blood pressure, they are also used to manage stroke and to address CDVD-related chest pains without any side effects. Medicinal plants used in the management and treatment of CDVD contain natural bioactive compounds that accumulate in the plants as secondary metabolites, such as alkaloids, sterols, terpenes, flavonoids, saponins, glycosides, and tannins. The fundamental problem that most herbal medications address is the lack of standardization that may be used to classify them even though these effects usually overlap.

Keywords

Cardiovascular diseases · Plant medicine · Benin people · Ethnobotany · Heart attack · Non-communicable diseases · Healthcare systems

Abbreviations

ACE	Angiotensin-Converting Enzyme
AVG	Aloe Vera Gel
CDVD	Cardiovascular Disease(s)
NO	Nitric Oxide
NR	Not Reported
OLE	Olive Leaf Extract
OPP	Oil Palm Phenolic
SSA	Sub-Saharan African
UN	United Nations
WHO	World Health Organization

1 Introduction

There has been a considerable rise in cases of cardiovascular diseases globally [222]. CDVD is now considered among the principal cause of mortality globally. The incidences of diseases are rising at an alarming rate, especially in developing countries [102, 138, 149]. CDVD refers to any disease of the heart and blood vessels [102]. Also, cardiovascular disease was defined by Abunnaja and Sanchez [3] as "the pathologic process (usually atherosclerosis) affecting the entire arterial circulation, not just the coronary arteries" [3]. The most common CDVDs are cardiomyopathy, aortic disease, hypertension, strokes, coronary heart disease, high blood pressure, heart attack and failure, and peripheral arterial disease [102, 108, 125, 244]. CDVD causes economic, sociocultural, and health burdens to societies globally and accounts for around 31% of deaths every year [199, 236]. Globally, the World Health Organization (WHO) estimates that cardiovascular disease claims about 17.9 million lives [19, 42, 65, 122, 138, 199].

Healthcare organizations in low- and middle-income countries are unable to handle the current and projected economic and social effects of CDVDs, and the economic burden on high-income countries is unsustainable [116]. The CDVD crisis has increased the strain on the healthcare systems of sub-Saharan African (SSA) nations, which are already trying to deal with other diseases and developmental challenges that disproportionately impact the region [78, 85, 135, 139, 141, 145, 166, 183]. Also, information about region and age-specific death from CDVDs suggests SSA is disproportionately affected and so are people above 45 years [59, 227, 239]. Some causes of CDVD include unhealthy habits like smoking, and others like high cholesterol, hypertension, obesity, excess inactivity, and diabetes mellitus [149]. By 2025, the United Nations (UN) wants to reduce fatalities from noncommunicable diseases including cardiovascular disease by 25%. The current therapeutic approaches' economic costs to lower the three main CDVD risk factors – hyperlipidaemia, atherosclerosis, and hypertension – cannot be sustained on a worldwide scale [116].

Regarding the treatment of CDVD in SSA, herbal therapy is the most preferred healthcare for those affected but is also often complemented with modern medicines. It is arguably so because of sociocultural and economic motivations like affordability, availability, self-medication, and history of successful use in traditional communities [87, 128]. Herbal therapy for CDVD in SSA mostly revolves around the use of medicinal plants and spirituality because of their fewer side effects, reliability, accessibility, availability, and ability to self-medicate. Moreover, modern medicine is considered by some native people to be ineffective in the treatment and management of certain disease conditions [175]. Herbal medicines refer to the use of plant parts, materials, extracts, preparations, and products either alone or their combinations to address diverse medical uses [81, 128, 140, 143, 144, 146, 147]. Traditional medicine is a common healthcare means in developing countries of the Global South, whereas complementary or alternative medicine refers to modern medicine used in developed nations [33]. The use of traditional medicine in SSA and other parts of the Global South is often attributed to their accessibility, effectiveness, familiarity, perceived safety, naturalness, cost friendliness, and culture [29, 49, 127, 134, 178, 209].

This chapter aims to produce an inventory of the plant and plant-based remedies used by Benin people, Nigeria, for the treatment and management of CDVDs. A focused approach was adopted to select informants along with a non-experimental validation of the plants used through a literature review of the phytochemical and pharmacological information supporting the medicinal activities of the plants and plant-based remedies used in the treatment and management of CDVDs. Also, this chapter documents the traditional medicinal practices for the treatment and management of cardiovascular diseases and enumerated how these plants are used (i.e., either alone or in combination).

2 Plants and Plant-Based Remedies for the Treatment and Management of Cardiovascular Diseases in Benin City, Southern Nigeria

A survey was carried out in the six Local Government Areas that make up Benin City, Edo State (6.34° N and 5.60° E; Fig. 1) and occupied by Benin people of Southern Nigeria to document medicinal plants and traditional plant-based remedies used in the treatment and management of CDVDs. Benin City lies within the humid tropical rainforest zone of Nigeria but its vegetation has been heavily altered by

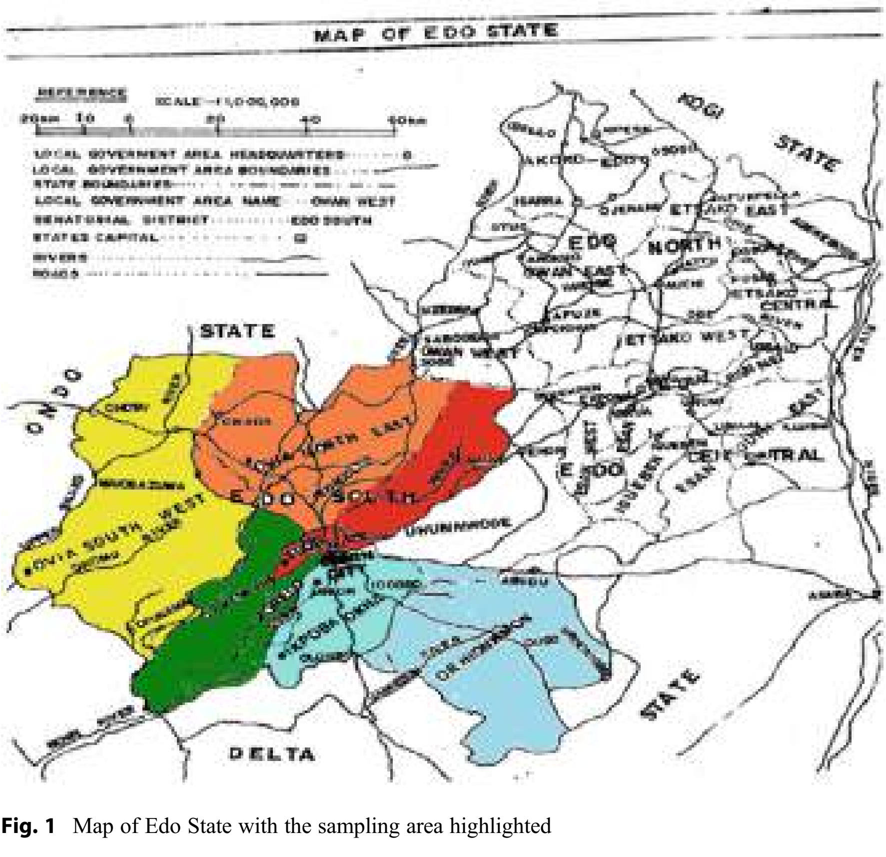

Fig. 1 Map of Edo State with the sampling area highlighted

anthropogenic activities and human population growth and is currently a mosaic of secondary forest [161–163]. The climate is monsoonal with two distinct seasons (dry and rainy season) with high rainfall (2000–3000 mm), temperature (20–40 °C), and average atmospheric humidity of 28% and with radiation of 1600 h per year [150, 152]. A detailed description of the geography, biodiversity, and soil features of the study area is present in Osawaru et al. [155–160], Osawaru and Ogwu [153, 154], Ogwu and Osawaru [142].

The survey to document traditional medicine practices and higher plants used for managing and treating CDVD by Benin people was conducted between January and June 2021 using survey sheets for information collection, a digital camera, an audio recording device for interviews, and a mobile phone-enabled position marker. The survey focused on randomly selected and locally identifiable professional herbalists, traditional healers, and herbal remedy sellers using semi-structured, open-ended questionnaires in English and/or preferred native languages. The 14 randomly selected respondents and interviewees are duly registered with all the relevant authorities (Table 1). The questionnaires were administered to the key informants (full-time traditional healers, part-time practitioners, elderly people who know the

Table 1 Demographic information of respondents

Characteristics	Frequency	Relative Percentage (%)
Marital Status		
Single	2	14.29
Married	12	85.71
Divorced	0	0.00
Unmarried but living with a partner	0	0.00
Educational Level		
Primary school	1	7.14
Secondary school	4	28.57
Technical degree	2	14.29
College/University	7	50.00
Employment Status		
Housewife	0	0.00
Student	0	0.00
Unemployed	3	21.43
Full-time employee	0	0.00
Part-time employee	11	78.57
Gender		
Male	4	28.57
Female	10	71.43
Occupation		
Trader	6	32.86
Student	3	21.43
Natural therapist	2	14.29
Pharmacist	2	14.29
Teacher	1	7.14
Tribe		
Benin	14	100.00

traditional values of the plants, relatives, and acquaintances who at one time or the other used some of these plants for the treatment and management of CDVD). The respondents animated a field survey to identify and collect plants and plant parts used for the treatment and management of CDVD. Specific information about the plants like their botanical and local names, part of the plant used, method of preparation and dosage, application, and treatment duration were asked and recorded. Collections were made from homestead farms, distant farms, and different forests present in the study area.

Data collected include the socio-demographic features of the survey respondents (e.g., marital status, educational level, employment status, gender, occupation, and tribe/ethnicity). Other pertinent data collected included plants used for CDVD treatment and/or management and preparation of the herbal remedies. Data collected were processed and analysed using Windows Microsoft Excel 2016 version software to understand inherent patterns. The therapeutic relevance of each identified species was also recorded. Based on the knowledge of the informants, a total of 17 medicinal higher plant species that are used for the treatment and management of CDVD were identified by a taxonomist at the University of Benin, Nigeria, and documented.

2.1 Medicinal Plants and Plant-Based Remedies Used by Benin People for the Treatment and Management of Cardiovascular Diseases

From the survey, a total of 17 medicinal plants belonging to 14 plant families were identified, documented, and collected as used by Benin people for the treatment and management of CDVD (Tables 2 and 3). The plants included *Theobroma cacao, Aloe barbadensis, Momordica charantia, Laurus nobilis, Morinda citrifolia, Carica papaya, Allium cepa, Ficus exasperate, Elaeis guineensis, Olea europaea, Persea Americana, Moringa oleifera, Beta vulgaris, Allium sativa, Hunteria umbellate, Zingiber officinale,* and *Curcuma* species. Some of these plants including *Allium sativum*, *Persea americana*, *Hunteria umbellate, Moringa oleifera*, and *Allium cepa* have been previously documented as useful for managing and treating CDVD [138]. The dominant plant families are Lauraceae, Amaryllidaceae, and Zingiberaceae with two representatives each. The other plant families are Malvaceae, Liliaceae, Cucurbitaceae, Rubiaceae, Caricaceae, Moraceae, Apocynaceae, Arecaceae, Moringaceae, Oleaceae, and Amaranthaceae. The habitat of these plants ranges from evergreen rainforests to warm temperate zone (Table 2). The plants were mostly collected from home gardens, markets, and local forests and within either cultivated or wild systems (Table 3). Earlier, Osawaru and Ogwu [154] confirmed the presence of plants with herbal medicine value as well as plant-based herbal remedies in local markets in Benin City and its environs. The plants and plant-based remedies were taken to address the following CDVDs – stroke, high blood pressure, hypertension, and chest pains without any side effects. It is known that plants are used to treat people with CDVDs like heart attacks, hypertension, cerebrovascular disorder, and heart failure [95, 126, 138]. Although these plants and

Table 2 Ethnobotanical documentation of plants and plant-based remedies used in the treatment and management of cardiovascular disorders

Plant and Plant-Based Remedy						
Plant name	Scientific name	Usefulness	Plant Family	Plant Habitat and Origin	Efficacy	Known Side Effects
Cocoa and Aloe Vera	*Theobroma cacao* *Aloe barbadensis*	Stroke and high blood pressure	Malvaceae Liliaceae	Evergreen tropical rainforest Tropical/subtropical forest (xerophytic environment)	Successful	None
Bitter melon/ bitter squash	*Momordica charantia*	High blood pressure	Cucurbitaceae	Terrestrial	Successful	None
Bay leaf	*Laurus nobilis*	High blood pressure	Lauraceae	Mediterranean area	Successful	None
Indian mulberry or hog apple	*Morinda citrifolia*	High blood pressure	Rubiaceae	Shady forests	Successful	None
Pawpaw	*Carica papaya*	High blood pressure	Caricaceae	Tropics of America	Successful	None
Onions	*Allium cepa*	Hypertension	Amaryllidaceae	Temperate zone	Successful	None
Sandpaper plant	*Ficus exasperata*	Hypertension	Moraceae	Moist deciduous Fringe Forest	Successful	None
Garlic, ginger, and turmeric	*Allium sativa* *Zingiber officinale* and *Curcuma* species	Hypertension	Amaryllidaceae Zingiberaceae	Terrestrial	Successful	None
Abeere	*Hunteria umbellata*	Hypertension	Apocynaceae	Terrestrial	Successful	None
Palm kernel nut Moringa seed	*Elaeis guineensis* *Moringa oleifera*	Stroke	Arecaceae Moringaceae	Tropical or subtropical region	Successful	None
Olive seed	*Olea europaea*	High blood pressure	Oleaceae	Mediterranean basin	Successful	None
Avocado pear	*Persea americana*	Chest pain	Lauraceae	Tropical/temperate region	Successful	None
Beetroot	*Beta vulgaris*	Hypertension	Amaranthaceae	Warm temperate zone	Successful	Yes, not to be taken with malaria medication

Table 3 Primary usage or methods of using these plants for treatment and management of cardiovascular disorders

Plant Remedy	Parts Used	Methods of Administration	Duration of Use	When Used	Dosage	Storage Method	Source of Plant	Wild or Cultivated	Collection Time
Cocoa Aloe Vera	Leaves and seeds	Infusion	Until recovery	Morning and evening	2 teaspoons	Canning	Garden and forest	Cultivated	Anytime
Bitter melon	Fruits and seeds	Aqueous steeping	Daily	Morning and evening	A glass	Freezing	Market and garden	Cultivated	Anytime
Bay leaf	Leaves	Aqueous steeping	Daily	Morning and evening	A glass	Bare floor or dry place	Market	Cultivated	Anytime
Indian Mulberry	Fruits and seeds	Aqueous steeping	Daily		25 ounces	Freezing	Garden	Cultivated	Anytime
Pawpaw	Fruits	Aqueous steeping	Daily	All the time	2–3 glasses	Freezing	Garden and market	Cultivated	Anytime
Onions	Whole bulb and bulb peels	Aqueous steeping	Daily	All the time	As many as possible	Freezing	Market	Cultivated	Anytime
Sandpaper plant	Leaves	Aqueous steeping	Daily	3 times daily	NR	Freezing	Garden	Wild	Anytime

Garlic Ginger Turmeric	Roots	Raw/aqueous steeping	3 weeks	Morning and evening	NR	NR	Market	Cultivated	Anytime
Abeere	Fruits and seeds	Raw/aqueous steeping	Daily	Anytime	2–3	NR	Market/ Garden	Cultivated	Anytime
Palm kernel nut Moringa seed	Seeds	Oral	Daily	Morning and evening	2 tablespoons	Canning	Market/ garden	Cultivated	Anytime
Olive seed	Seed	Infusion	Daily	3 times daily	Half tumbler	NR	Market	Cultivated	Anytime
Avocado pear	Seed	Raw	1 week	Once daily	2 tablespoons	NR	Market	Cultivated	Anytime
Beetroot	Fruit and seed	Oral	Until recovery	Morning	1 bottle	Freezing	Market	Imported	Morning

NR Not reported

their herbal remedies have active substances with pharmacological and prophylactic properties, they often have little to no side effects when taken in moderation. These bioactive compounds include phenolics, flavonoids, nitrogen, anti-inflammatory, terpenoids, antioxidants, and other secondary metabolic compounds. Other researchers [26, 101, 199] have documented and confirmed the in vitro and in vivo efficacy of *Ginseng*, *Ginkgo biloba*, *Gastrodia elata,* Nerium *oleander*, Salvia *miltiorrhiza*, *Dracocephalum moldavica, Daucus carota*, *Ganoderma lucidum*, *Tinospora cordifolia*, *Hydrocotyle asiatica, Bombax ceiba, Terminalia arjuna*, *Amaranthus viridis, Andrographis paniculate, Mucuna pruriens*, *Picrorhiza kurroa*, and *Gynostemma pentaphyllum in the treatment and management of CDVD.* The consumption or use of these plants reduces the risk of CDVDs like congestive heart failure, hypertension, ischaemic heart disease, and arrhythmias [96, 99, 127, 138, 179, 186].

2.2 Enumeration of Plants Used in the Management and Treatment of Cardiovascular Disease by Benin People in Southern Nigeria

2.2.1 Cocoa (*Theobroma cacao*)

The use of cocoa in traditional medicine is linked to the phytochemical constituents in the plants like flavanol, polyphenol, and procyanidin in significant concentrations [25, 41, 197]. These compounds have nutritional and pharmacological benefits. The polyphenols in cocoa have cardiovascular-protective properties by modulating different inflammatory markers connected to atherosclerosis [99]. The flavanol in cocoa also has a regulatory capacity and helps with total dietary requirements [99, 109, 229]. Cocoa consumption has been linked to flow-mediated vascular dilatation, blood pressure, total lipid profiles, and insulin resistance [74, 75, 88, 184].

Also, methylxanthines have been reported from cocoa extracts and like theobromine and caffeine help enhance the vascular and central nervous system and work in tandem with flavonols and other secondary metabolic compounds [151, 194, 197]. Both procyanidins and flavanols have been connected to diverse cardiometabolic and cardiovascular advantages in humans through the improvements of endothelium-dependent vasodilation, blood pressure regulation, inflammation, and platelet activation [40, 46, 63, 73, 75, 103, 115, 119, 165, 176, 181, 185, 188, 193, 195, 197, 214]. Also, it provides insight into the distribution, absorption, metabolism, and excretion of flavonols [5, 164, 187].

2.2.2 Aloe Vera (*Aloe barbadensis*)

Aloe vera is an important tradomedicinal plant and its innermost leaf layer contains diverse minerals and secondary metabolites [190, 200]. Aloe is used as medicine by diverse cultures and societies globally worldwide since time immemorial [105]. Aloe vera leaves contain high amounts of anthraquinone [200]. The inner layer leaf of Aloe vera is called Aloe vera gel (AVG) or aloe gel and it does not contain those essential anthraquinone compounds [190]. The plant has several beneficial properties including those lipid-lowering, immunomodulatory effects, antioxidant, anti-inflammatory, and hepatoprotective effects [190, 200]. Aloe vera contains polysaccharides like acetylated polymannan, glucomannan, acemannan, and mannose-6-phosphate that have therapeutic and pharmacological effects. AVG has direct and indirect cardioprotective impacts through hyperglycaemia, hyperlipidaemia, hypertension, and obesity effects [190]. The mineral nutrients in AVG (like iron, zinc, copper, and selenium) also participate in enzymatic, cell metabolic, and antioxidant functions with net positive effects on CDVD. Also, AVG contains a lot of antioxidant vitamins which have been linked to decreased serum levels of and risk of coronary heart disease. Both lophenol and cycloartenol are present in aloe and they can regulate blood glucose [200].

2.2.3 Bitter Melon (*Momordica charantia*)

Bitter melon offers diverse health advantages for reducing illnesses and enhancing life [191]. It is grown in most countries with tropical or sub-tropical climates and popular medicinal food in many developing countries [219]. Bitter melon's whole fruit, seeds, and leaves control oxidative stress and prevent fat build-up, and contain a variety of bioactive substances, including alkaloids, polypeptides, vitamins, and minerals. It aids in controlling blood cholesterol levels, and defending the body against cardiovascular diseases like atherosclerosis [22, 191, 203]. Also, bitter melon improves glucose and lipid metabolism [35, 208, 218]. The medicinal benefits of bitter melon are linked to its high antioxidant qualities, which are partly owing to the presence of secondary phytochemicals like isoflavones, phenols, flavonoids, anthroquinones, terpenes, and glucosinolates that give the fruit its bitter flavour [17, 47, 64]. Loss of potential health benefits may occur when active bitter components are removed by a variety of debittering techniques and selective breeding [207]. Bitter melon contains vitamins, minerals, and flavonoids [17, 117]. Because it contains a variety of bioactive components, bitter melon exhibits pharmacological effects that have been linked to improved health, including the ability to scavenge free radicals and have hypoglycaemic and hypolipidemic effects [191, 230].

2.2.4 Bay Leaf (*Laurus nobilis*)

Bay leaf can be used in alternative medicine and is effective as a blood pressure-lowering medication [14, 54, 71]. Bay leaf consumption decreases CDVD risks [98]. This plant has a track record for effectively treating disease, has few adverse effects, and is accessible. Bay leaves contain tannins, flavonoids, citrate, eugenol, and essential oils [98, 215]. Antioxidant chemicals, which include tannins and flavonoids, are present in bay leaves and have health benefits regarding blood pressure [54, 98]. The availability and prevalence of bay leaf are anticipated to aid in spreading awareness and promoting it as a herbal alternative for health. Bay leaf is a plant that is often used in the neighbourhood as a complementary medicine. It is a diuretic and has been used to treat eructation, epigastric bloating, poor digestion, and flatulence [70]. In addition to its hypoglycaemic benefits, bay leaf can improve diabetic patients' capillary function, lipid metabolism, liver, and kidney function, as well as their antioxidant status [51, 69]. The glucosidase enzyme reduces blood sugar levels, and the phenolic chemicals in bay leaves could block it [94]. The bay leaf extract may cause pancreatic cells to secrete more insulin or that it acts similarly to insulin [221]. The fact that these extracts diminish gluconeogenesis and glycogenolysis, decrease glucose absorption, and improve peripheral glucose utilization also suggests that they may have hypoglycaemic effects [51, 170].

2.2.5 Indian Mulberry (*Morinda citrifolia*)

Indian mulberry is one of the main components in several traditional formulas that are sold all over the world [53]. Mulberries and their extracts are used to treat both acute and chronic illnesses because they have powerful antimicrobial, anti-hyperlipidaemic, anti-inflammatory, and anti-cancer properties [238]. The mulberry roots, bark, leaves, fruits, and stem twigs are rich sources of vital bioactive compounds [53, 93, 167, 238]. Mulberries are a great source of phosphorus, iron, riboflavin, calcium, vitamin (C and K), potassium, and other essential minerals for our bodies [13, 93]. Additionally, they contain a sizable amount of dietary fibre as well as a variety of organic substances, such as phytonutrients, zeaxanthin, resveratrol, anthocyanins, lutein, and different polyphenolic compounds [93]. This plant's extraordinary benefits in decreasing blood cholesterol and serum glucose levels make it possible to utilize them. The presence of numerous bioactive components in this plant, including flavonoids, polyphenols, alkaloids, terpenoids, and steroids, is what gives it these qualities [93].

2.2.6 Pawpaw (*Carica papaya*)

Pawpaw has high dietary fibre, protein, carbohydrates, vitamins, and mineral nutrients. The plant is used to address diverse ailments including for treating indigestion and gastric ulcer. The consumption of pawpaw fruit helps reduce blood pressure [52, 169]. It offers numerous health advantages, including lowering blood pressure, promoting wound healing, assisting with digestion, lowering the risk of heart disease, diabetes, and cancer, and enhancing blood glucose control in diabetics [4]. Pawpaw seeds and fruits have potent antihelminthic and antiamoebic properties [11, 148]. Ascorbic acid, flavonoids, cyanogenic glucosides, and glucosinolates are just a few of the active substances found in pawpaw leaves that have been shown to lower lipid peroxidation levels and increase total antioxidant power in the blood, thus they can be used either for the prevention or treatment of CDVD [11, 196, 202].

2.2.7 Onions (*Allium cepa*)

Onions are regarded as essential vegetables [20, 30, 76, 83, 220]. Many cultures have valued onions for their culinary and therapeutic uses because they are rich sources of flavonols and organosulfur compounds [66, 131]. Also, it contains flavonoids and phenolics which have antioxidant, anti-cholesterol, anti--inflammatory, and anticancer properties. The plant is rich in protein, water, selenium, and different vitamins (B1, B2, and C) as well as potassium, diverse polysaccharides, and essential oil [225]. These phytochemicals have been linked to several health advantages and improvement in diseased conditions [20, 30, 76, 83, 86, 112, 220]. The variety of phytonutrients in onions is known to have important and extensive biological activity. Onions exhibit potent anti-atherogenic effects that are related to a variety of bioactivities [86, 112, 220]. The high amounts of flavonoids and organosulfur compounds are linked to these biological actions, which are mostly antioxidant and anti-inflammatory in nature [30, 220, 225, 235].

2.2.8 Sandpaper Plant (*Ficus exasperate*)

Sandpaper plant is a deciduous shrub or tree and extracts from the leaf have demonstrated high concentrations of secondary phytochemicals like alkaloids, tannins, flavonoids, and saponins [237]. It has been used to treat difficult childbirth, bleeding, and diarrhoea. Sandpaper leaf has hypoglycaemic potential and is utilized to manage, regulate, and/or treat CDVDs like hypertension. The leaf extract is used to treat cardiac problems and reduce blood pressure [10, 38]. Also, in the treatment of numerous degenerative diseases, the bark, leaves, fruits, and latex of sandpaper plants contain astringent, anti-oxidative, carminative, anti-inflammatory, and anticancer agents that can be used to treat diabetes, skin conditions, ulcers, dysentery, diarrhoea, stomach aches, haemorrhoids, and hypertension [12, 38, 91]. The plant's ability to treat or manage several ailments is supported by the presence of phytochemicals. Its high quantities of flavonoids and saponins, however, imply that taking the plant in excessive doses over a sustained period can cause toxicity [237].

2.2.9 Garlic (*Allium sativa*)

Garlic, the most popular herbal dietary supplement on the market, is famous for its vast health benefits, especially in the treatment and prevention of CDVDs [16, 80]. It has a lot of sulphur-based compounds and is a good source of carbohydrates as well as proteins [2, 80, 233]. More than 200 unique compounds that can shield the human body from a wide range of ailments can be found in garlic alone. Garlic's sulphur-containing components provide the body with protection by promoting the development of specific advantageous enzymes [60, 120]. Garlic contains sulphur compounds, and diverse enzymes, as well as essential minerals (like calcium, germanium, iron, copper, magnesium, potassium, zinc, and selenium), vitamins (A, B1, and C), fibre, and water [60]. Also, garlic extracts exhibit anti-inflammatory, antioxidant, and anti-microbial properties [110, 204, 234]. The consumption of garlic helps lower blood pressure, total cholesterol, body mass index, triglycerides, and inflammatory indicators [28, 34, 60, 80, 210]. Additionally, it can raise HDL-c levels and enhance several cardiovascular parameters, including carotid intima-media thickness, microcirculation, post-occlusive reactive hyperaemia, epicardial and periaortic adipose tissue, and low attenuation plaque [60, 80].

2.2.10 Ginger (*Zingiber officinale*)

Ginger is rich in antioxidant and anti-inflammatory compounds [124, 243]. Some of these compounds are effective against CDVD like hypertension and other diseases [18, 189]. The phenolic compounds in ginger include gingerols, paradol, and shogaol that have therapeutic and pharmacological potentials and are helpful in the management of inflammatory and oxidative stress diseases [114, 132, 189, 212]. In addition to its significant nutritional contributions, ginger has several medicinal and immune-boosting benefits. Additionally, they are crucial in the control of blood pressure and the treatment of cardiovascular diseases. The utilization of ginger-based products will help to advance the human health system [48, 82, 104, 114, 130].

2.2.11 Turmeric (*Curcuma* species)

Turmeric has spiritual significance and uses to treat various CDVDs and other diseases such as digestive, respiratory, and hepatic illnesses or disorders [9, 100, 106, 129, 171, 172, 192, 228]. Having a low bioavailability and pleiotropic activity, turmeric is a naturally occurring phenolic compound. Turmeric has been used as a cooking spice and a traditional treatment to heal skin conditions, cuts, and wounds since the dawn of time [92, 198]. It has been recognized as an essential dietary supplement and nutraceutical [72, 206]. There is evidence that the primary yellow bioactive component of turmeric has a variety of biological effects. These consist of its antioxidant, anti-inflammatory, anti-mutagenic, anti-carcinogenic, anti-coagulant, antifertility, antidiabetic, antibacterial, anti-fungal, anti-viral, anti-fibrotic, anti-venom, anti-ulcer, hypotensive, and hypocholesterolaemic activities [92, 192, 198, 228].

2.2.12 Moringa (*Moringa oleifera*)

Moringa tree is common in Asia and Africa and is appreciated for its therapeutic properties as the "miracle tree" [8, 19, 97, 111]. It has traditionally been utilized as both a food source and a medication to treat several illnesses like anaemia, diabetes, and infectious or cardiovascular ailments [1, 19, 62, 216]. The biggest medicinal and nutritional benefits are found in the leaves and seeds of the moringa plant. Protein, minerals, and antioxidant chemicals are plentiful in the leaves [19, 27, 77, 111]. In addition to being nutrient-rich, moringa also has anti-nutrients such as flavonoids that function as antioxidants [211, 223]. The primary phytochemical components of moringa are phenolic acids, flavonoids, saponin, tannins, alkaloids, glycosides, and glucosinolates [31, 32, 168, 226]. The bioactive components in moringa have synergistic therapeutic effects that include lowering blood sugar levels, anti-inflammatory properties, cardioprotective, anticancer, antimicrobial, neuroprotective, and immune system modulation [118, 223]. The phytochemicals N,L-rhamnopyranosyl vincosamide, isoquercetin, quercetin, quercitrin, and isothiocyanate that have been linked to functional activities related to cardiovascular disorders have been found in Moringa. The phytochemicals, such as quercetin and N,L-rhamnopyranosyl vincosamide, have molecular antioxidant, anti-inflammatory, and anti-apoptotic properties. These result in increased cardiac contractility and damage prevention for the heart's structural integrity. Additionally, these substances function as endothelium protectors and natural vasorelaxants [19, 113, 213].

2.2.13 Olive (*Olea europaea*)

Olive tree is considered one of the oldest tree species, and historically, indigenous inhabitants relied on its fruits and by-products, such as olive oil, as their primary source of nutrition [56, 67, 241]. Olives are a good source of vitamins, minerals, and carbohydrates. Numerous studies have found links between the "Mediterranean diet's" health advantages and longer life spans and lower occurrences of chronic degenerative diseases [67]. Olive leaf polyphenols and flavonoids have been shown to have anti-inflammatory, anti-carcinogenic, anti-hypertension, and antibacterial characteristics; they are therefore crucial to the outcomes that have made olive leaves significant. Olive leaves contain secoiridoid phenols, which are unique to the plant [217].

Olive seed protein hydrolysates have been shown to exhibit significant antioxidant and angiotensin-converting enzyme (ACE) inhibitor capabilities and to lower micellar cholesterol solubility [57]. Also, olive leaf extract (OLE) has lipid-lowering properties and contains significant amounts of the phenolic antioxidant known as oleuropein, far more than what is found in olive fruit or olive oil [21, 58, 68, 173, 205]. Olive oil contains at least 30 phenolic compounds, the majority of which are simple phenols (tyrosol and hydroxytyrosol), secoroids (oleuropein and ligstroside), and lignans (1-acetoxypinoresinol and pinoresinol) [42, 224]. The phenols in olive oil include straightforward compounds like vanillic, gallic, coumaric, caffeic acids, tyrosol, and hydroxytyrosol [224]. In several populations, phenolic compounds from virgin olive oil have had a positive impact on systolic and diastolic blood pressure [42, 121, 201]. A large amount of monounsaturated fat and a low amount of saturated fat are two of olive oil's key qualities. The primary monounsaturated fat, oleic acid, has several health-improving qualities, including a decrease in CDVD, neurological disorders, and cancer. Virgin olive oil is also rich in several beneficial compounds and polyphenols are likely the most important of these bioactive chemicals [42, 232]. Olive oil's phenolic components are extensively researched for their antioxidant capacity [67, 241]. It was discovered that several phenols from olive oil (like hydroxytyrosol and oleuropein) operate as radical scavengers and diminish the production of superoxide anions, neutrophil respiratory bursts, and hypochlorous acid. This may account for the decreased incidence of cancer and coronary heart disease that has been linked to the Mediterranean diet [42, 67, 231, 241]. Olive oil is resistant to heat oxidation and possesses high antioxidant potential within biological systems [42, 201, 241, 242]. Tocopherols and phenols are the principal substances enhancing stability during heat oxidation. The stability is also aided by the oil's high oleic acid content [67, 241].

2.2.14 Avocado (*Persea americana*)

The avocado pear is a widely grown and popular tropical fruit that is eaten all over the world [23, 36]. It also has significant levels of protein, potassium, and unsaturated fatty acids, which are uncommon in other fruits [23]. Its oil is high in monounsaturated fatty acids and is utilized to boost high- and low-density lipoprotein levels [137]. The chemicals included in the lipidic component of this fruit, including omega fatty acids, phytosterols, tocopherols, and squalene, have been credited with its health advantages. The health benefits of avocados include lowering cholesterol and avoiding cardiovascular illnesses [23, 137, 177]. This tree has been known to possess several therapeutic characteristics, including emmenagogue (fruit) and cicatrizant (fruit pulp). Its leaves have also been used to treat digestive disorders and skin infections and irritation. Additionally, seed oil has long been used as an ointment to soothe pain and soften the skin around wounds, as well as for the treatment of dry hair and other diseases [39, 137]. The primary bioactive substances found in avocado include tocopherols, carotenoids, polyphenols, and phytosterols. While carotenoids and tocopherols are primarily present in the avocado pulp, polyphenols can be found in the fruit's pulp, peel, seed, and leaves. Waste extracts, therefore, have a variety of biological properties, such as antimicrobial, anticancer, anti-inflammatory, antidiabetic, and antihypertensive properties [15, 89, 107, 137, 177].

2.2.15 Beetroot (*Beta vulgaris*)

Beetroot is grown in many nations throughout the world. It is consumed as part of a normal diet and is frequently used in food processing as a food colouring ingredient [37, 61, 245]. Proteins, sugar, carbs, vitamins (B complex and vitamin C), minerals, and fibre are all found in abundance in beetroot. The phenolic chemicals and antioxidants in beet include sesquiterpenoids, coumarins, triterpenes, carotenoids, and flavonoids (rhamnocitrin, tiliroside, kaempferol, astragalin, and rhamnetin) and are also present in significant amounts [37, 174]. Beetroot contains antioxidant and anti-inflammatory effects, anti-carcinogenic and anti-diabetic activities, as well as hepato-protective, hypotensive, and wound healing capabilities [37, 43, 61, 123]. Bioactive substances like dietary NO_3, betanin, antioxidants, and phenolic compounds are found in beetroot. Beetroot contains a variety of additional bioactive chemicals that may have health advantages, especially for conditions marked by chronic inflammation [37]. By enhancing nitric oxide synthesis, controlling gene expression, or modulating the activities of proteins and enzymes involved in these cellular processes, beetroot rich in nitrate and bioactive compound contents improves cardiovascular and metabolic functions [37, 45, 240]. The body uses nitric oxide (NO) for vasodilation to lower blood pressure and improve nutrients and oxygen supply within the body. According to these outcomes, beetroots may be useful in the prevention and treatment of cardiovascular disease [24, 90, 123, 180, 182, 240]. The beetroot-cereal bar exhibits high levels of antioxidant, phenolic, and nitrate components. A beetroot-cereal bar might be a good way to help individuals who have risk factors for cardiovascular disease by improving their cardiovascular parameters [44].

2.2.16 Palm Kernel Nut (*Elaeis guineensis*)

Contrary to common belief, the excess consumption of palm kernel oil does not increase the risk of having CDVD. The work of Ismail et al. [84] could not find any association but recommend their consumption in moderation to ensure good cardiometabolic health. Oil palm is abundant in West Africa and is the chief source of cooking oil [79]. Although it contains about 50% saturated fatty acid and high amounts of unsaturated fatty acids, it is not known to promote either arterial thrombosis or atherosclerosis [50]. Rather it has been documented that oil palm phenolic (OPP) found in oil palm has cardioprotective effects through mechanisms and pathways such as cholesterol biosynthesis pathway, anti-inflammatory properties, and antioxidants. OPP is recoverable from oil palm milling aqueous waste products. The CDVD effects of oil palm extolled by Benin people might likely be linked to OPP and other properties of the plant fruits.

2.2.17 Abeere (*Hunteria umbellata*)

Abeere is widely used in traditional medicine in SSA for CDVD as well as managing and treating sexually transmitted diseases, anaemia, obesity, diabetes, and stomach ulcers. The plant can be found in tropical forests all over SSA. The medical benefits of the plant are linked to its phytochemical compositions which include tannins, alkaloids, flavonoids, phenolics, and saponins, which have been implicated in lipid peroxidation, antioxidant, and free-radical scavenging activities [7, 55]. Recently, Odukoya et al. [138] implicated the use of the seed, stem, and bark of the plant in the treatment and management of CDVD. These parts of the plants contain diverse water-soluble alkaloids. It is likely that the nitric oxide synthase and arginase inhibitory effects on endothelial function contribute to the CDVD roles of the plant [136]. Another consideration is that the alkaloid fraction of *H. umbellata* helps to reduce weight gain, which otherwise may increase the risk of CDVD [6]. Hence, *H. umbellata* has a cardioprotective effect by contributing to weight loss.

3 Conclusion

Since time immemorial, plants have been used in the treatment of diverse CDVDs and recent epidemiological research has shown that eating foods derived from certain plants lowers the chance of developing cardiovascular disease. These plants will continue to play significant roles in sustaining human societal and individual well-being. Medicinal plants used in the management and treatment of CDVD contain natural bioactive compounds that accumulate in the plants as secondary metabolites, such as terpenes, flavonoids, alkaloids, sterols, glycosides, saponins, and tannins. These compounds have both therapeutic properties and pharmacological effects on humans. Traditional remedies have been utilized to treat cardiovascular illnesses for a very long time but have only started to attract the interests of researchers, government agencies, and large co-operations as they seek to explore alternative medicine, promote their usage, and search for natural raw materials and remedies for illnesses of global concern like CDVD. The fundamental problem that most herbal medications address is the lack of standardization that may be used to classify them even though these effects usually overlap. The chemical constituents of medicinal plants used for managing and treating CDVD should be further explored as potential raw materials for modern medicine as well as to understand their mode of action to improve the quality of life for individuals and society in general. Strong cardioactive glycosides are present in several plants, and they have favourable inotropic effects on the heart. The advantages of plant-based diets for the prevention of CDVD need to be investigated further. The phytochemical components found in natural goods are being researched for their potential to shield the cardiovascular and vascular systems from further harm.

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Abstract

The intake of flowers as food is documented in a variety of cultures as part of customary dishes and alternative therapies. But numerous types of edible flowers are more than just a treat because they include important nutrients like protein, vitamins, carbohydrates, fats, and therapeutic potential. Flowers, with their favourable sensory and nutritional properties and the existence of bioactive components advantageous to human health, form a key sector for developing the food industry. It is essential to encourage the local use of flowers in order to maintain traditional norms, and many studies have been conducted to better understand the social and cultural contexts in which edible flowers are consumed. With the global market for natural and healthy diets, the nutritional characteristics, therapeutic advantages, active chemicals, and cooking techniques of edible species have been explored. Current techniques for extracting natural bioactive chemicals from flowers are helping to discover their constituents and produce food industry-beneficial additives. Correct taxonomy, toxicology, and a comprehensive practice guide for commercially grown flowers, processing, and preservation are still needed to promote edible flower usage. The scientific and technical information about edible flowers' nutritional profile, therapeutic effects, and phytochemical composition is evaluated and addressed to improve knowledge, dietary practices, and dietary investigation.

Keywords

Edible flowers · Chhattisgarh · Food · Therapeutic potential · Nutritional profile · Bioactive components

Abbreviations

ABTS	2,2′-azino-bis (3-ethylbenzothiazoline-6-sulfonic acid)
AGEs	Acute gastroenteritis
AMD	Age-related macular degeneration
BSA	Bovine serum albumin
BSA/fructose	Bovine serum albumin/fructose
BSA/glucose	Bovine serum albumin/glucose
CCl_4	Carbon tetra chloride
CVD	Cardio vascular disease
DMB	Dimethyl benzene
DNA	Deoxyribonucleic acid
DPPH	2,2-diphenylpicrylhydrazyl
HePG2	Human hepatoma cell
HIV	Human immunodeficiency virus
HPLC	High-performance liquid chromatography
HPLC-DAD-ESIMS	High-performance liquid chromatography equipped with photo-diode-array detection-Mass

HPLC-PDA	High-performance liquid chromatography-photo-diod-array
LPS	Lipopolysaccharide
MAP kinase	Mitogen-activated protein kinase
MCF-7	Human breast cancer cell line
MPPs	Mitochondrial processing peptidase
ORAC	Oxygen radical absorbance capacity
Pl3K	Phosphatidylinositol-3 kinase
T47D	Human breast cancer cell line

1 Introduction

One of the most important ecosystems in Chhattisgarh may be situated on the Indian subcontinent, which is home to a vast array of unique plant and animal species [1]. These areas in Chhattisgarh, also known as the central region of India, conserve a wide variety of plant life and provide a wealth of wood and non-timber materials, such as gums, with significant economic values [2–4]. The first step in protecting these unique natural biological components is to measure and record plant variation. We can always count on healthy ecosystems, which are beneficial to humans in a variety of ways, to be characterized by high biodiversity [5]. Food webs, food chains, bioremediation, nutrient availability, and human subsistence all rely on biodiversity to function in harmony [6]. There are approximately 6922 km^2 of forest, including deep forest and open forest, as well as mangroves. It covers around 21.5% of the nations at the geographical region level. It has several applications, including community forestry, agroforestry management, replanting, and the restoration of damaged or demolished commercial and industrial sites [7, 8]. The term "agroforestry"simply refers to a combination between agriculture and forestry that has been employed for centuries [9–11]. Agroforestry practices are also becoming a realistic means of decreasing the adverse environmental impacts of the current state of affairs [12]. Chhattisgarh has several useful and nutritional plants since its forests comprise 42% of the state's total territory and are composed of tropical moist and tropical dry deciduous trees. From a conservationist's perspective, it is crucial to keep all of these temperate forests alive and well, as they are native to a wide variety of plant varieties and serve as a safe haven for many different kinds of wildlife [6]. Hence, securing these biotic assets is critical to maintaining a healthy ecosystem. The nutritional value of food and its accessibility have long shaped the eating habits of forest-dwelling communities, and people in these areas typically harvest their own food. Roots, rhizomes, tubers, fruits, seeds, flowers, and leaves are all consumed by those who live in the forest as a means of subsistence. As a result, these edible plants are crucial to the long-term viability and food security of people who make their homes in the nation's forests [13].

The state of Chhattisgarh is located at 80°15′ to 84°24′ E longitude and 17°46′ to 24° N latitude. The state consists of both hilly and flat areas throughout its entirety.

On average, 60 in. of precipitation fall each year. Rice is the primary crop produced in the state. Because of the state's exceptionally diverse plant and animal life, Chhattisgarh is often referred to as the "herbal state." Around 42% of the land area of the state is covered by woods. People who live in tribal communities are wholly reliant on the forest for their food and a variety of other needs [14]. Although farming is the chief source of income for indigenous communities, the forest and its resources are equally vital to the survival of indigenous and folk communities by providing for a wide range of needs, including food, medicine, fibres, and shelter. They rely heavily on farming to meet their food needs, but they also forage for wild roots, tubers, leaves, flowers, and fruits. In terms of ethnobotany, the state is not well known; the only important studies that have been written down can be found there. In this chapter, only edible flowers and their healing properties, as well as a few other ethnobotanical traits, are discussed. On the other hand, there is a lack of details concerning edible plants. This chapter discusses the wild plant species that they harvest for their edible qualities and then devour. Flowers have been eaten for a long time, but recently there has been a resurgence of interest in doing so, both for their visual beauty and any potential health benefits they may provide. New interest in edible flowers has led to a lot of research on their nutritional value, medicinal value, and phytochemical components [15]. There are many different phytonutrients in flowers, and some of them may help protect against chronic diseases. This chapter discusses the nutritional and pharmacological activities of edible flowers from Chhattisgarh, India. Whether edible flowers are to be considered functional foods or a component of our diets, however, more findings referring to their security and application are required [16, 17].

2 Edible Flowers of Chhattisgarh

Flowers are natural substances that are not only beautiful but also known for their nutritional profile and therapeutic perspective [18, 19]. People have been able to benefit in many ways from natural resources like wild edible plants. These benefits reflect long-standing interactions that have existed between plants and humans throughout different periods of history. Because of this, there is no question that people eat edible flowers that grow in the wild because the term farming has never been clearly defined. The edible flowers of Chhattisgarh originate from an extensive range of plant groups, genera, and species; however, the specific number of flowering plants that can be consumed varies from region to region. There is a wide variety of plants, but only a few are selected as edible, making accurate recognition of edible flowers crucial. A variety of edible flowers can be purchased commercially, and their popularity is growing rapidly. These blooms are most popular when consumed in moderation, but they also have many other uses in the kitchen, including in baked products, drinks, marmalade, salads, smoothies, and even as a replacement for vegetables in savoury dishes. There are so many edible flowers that it would be impossible to study them all [20, 21].

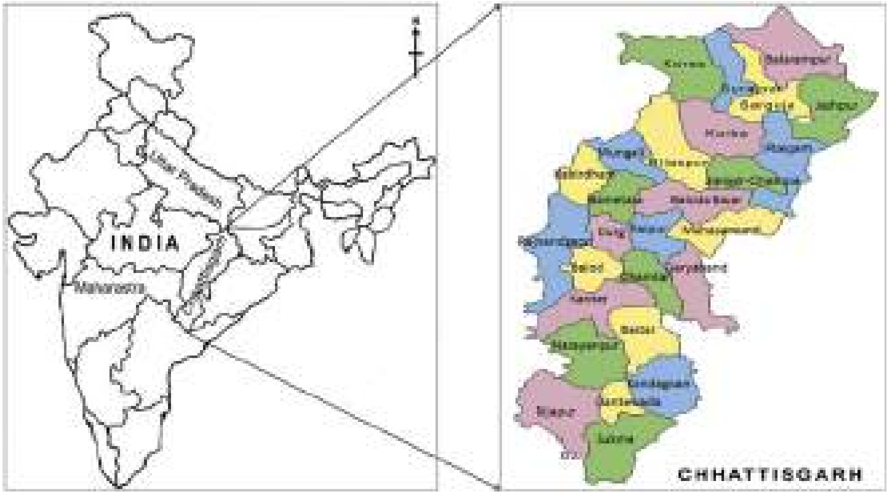

Fig. 1 Distribution of edible flowers in Chhattisgarh

3 Distribution of Edible Flowers in Chhattisgarh

An edible flower is defined as a safe, harmless, and non-toxic flower with health advantages that is taken as a component of human food. Since ancient times, they have consistently been a significant factor in supplying living organisms with various nutrients. It has been demonstrated that consuming flowers is a cultural tradition in a huge number of locations, including Raipur, Bilaspur, Raigarh, Janjgir-Champa, Baloda-Bazar, Bastar, Sarguja, Koriya, Gariyaband, Sakti, Sarangarh-Bilaigarh, Korba, Durg, and other locations of Chhattisgarh, India (Fig. 1). Edible flowers have become extremely popular around the world recently due to their attractive beauty and the therapeutic advantage associated with their specific aroma, flavour, and colouring [15]. Edible flowers have recently been at the centre of numerous investigations in the fields of food science, technology, and even pharmaceuticals. The goal of this chapter was to enhance the attractiveness of edible flowers as prospective food products by discussing their phytochemical components, health benefits, and toxicity. There is evidence that humans have been eating flowers for at least 2000 years. Several regions, including India, Pakistan, Sri Lanka, Vietnam, the USA, Australia, Kenya, and South Africa, have reported the consumption of edible flowers. Nowadays, you may eat flowers in salads, in light curries, or even by themselves as a vegetable. Certain edible flowers are suitable for stuffing or using in stir-fries [16, 22].

4 Traditional Use of Edible Flowers

The flowers are more than just decorative additions to the savoury dishes and sweet confections; rather, they offer a novel variety of tastes and boost the nutritional content of the dishes. They can be used as fresh salads, savoury dishes, soups, beverages, candies, jellies, spices, and colours. For example, *Tagetets erecta, Azadirachta indica, Rosa gallica, Cardia dichotomo, Brasica oleracea,* and *Tamarindus indicus* flowers are employed in a variety of culinary applications in dried form, powdered form, crystallized form, and foam form [23, 24].

There are many plants that have conventionally been used as edible flowers, such as *Madhuca latifolia* Roxb.*, Bombax malabaricum, Butea monosperma, Holarrhena antidysenterica, Indigofera casioides, Tamarindus indicus, Holostemma rheedianum Spreng, Crotalaria juncea L., Crotalaria incana Rottl., Celastrus paniculata Willd, C. renigera (Wall) Gagnep., C. fistula L., C. arborea* Roxb.*, Bauhinia racemosa Lam., Alangium salvifolium, Brassica oleracea var. botr. Linn, Couroupita guianensis Aubl, Brassica oleracea, Cordia dichotoma, Cucurbita L., Moringa oleifera, Hibiscus sabdariffa L., Azadirachta indica, Solanum xanthocarpum, Solanum xanthocarpum, Hibiscus rosa-sinensis, Helianthus annuus L., Rosa gallica, and Tagetet erecta.* The comprehensive information, however, was not provided.

The following key point describes the traditional uses of edible flowers:

- Provides a major portion of the calories and nutrients needed on a yearly basis [25].
- It is possible to use it as a main dietary replacement, especially during times of seasonal food crisis.
- For use as a salad topping, appetizer, main dish, beverage, or sweet course.
- Edible flowers have been used in a wide variety of recipes for tea, baked goods, sauces, jellies, syrups, flavoured liquors, vinegars, honey, and oils.
- Certain edible flowers that are safe to eat can be incorporated into stir-fried dishes.
- Offering an option for the application of natural dyes and colourants in the food industry.
- Salads, appetizers, soups, main dishes, confectionery, and beverages all benefit from their colourful, aromatic flavours, and refreshing aromas [26].
- Even though edible flowers are popular in their current forms, they can also be baked, frozen (as ice cubes), bottled in sweets, or stored with volatiles [15].
- Edible flower petals can be steeped in oil or vinegar to provide a floral fragrance.
- Sugar and egg whites can be employed to crystallize flowers for candy [27].
- Flowers’ essential oils are applicable in aromatherapy treatments.
- Edible flowers are included in therapeutic diets.
- Edible flowers were employed as garnishes on dishes served to the nobles, particularly during formal events like feasts and festivals [15, 28]. The uses of many edible flowers in Chhattisgarh are listed in Table 1.

5 Edible Flowers as Cultural and Regional Foods

A state that is located in the centre of India is called Chhattisgarh, because the term “Chhattisgarh“is made up of two words: first “Chhattis” and second “Garh,” which means 36 forts. The name Chhattisgarh is meant to represent the large group of forts

Table 1 Uses and family description of edible flower of Chhattisgarh, India

S. N.	Plant Local Name	Scientific Name	Edible parts	Family	Uses	References
1	Mahua	*Madhuca latifolia* Roxb.	Flower	Sapotaceae	The fresh as well as dry flowers are eaten along with seed	[29]
2	Semal	*Bombax malabaricum*	Flower	Bombacaceae	The young flower buds, mixed with Mahua flowers, are eaten during food scarcity	[29]
3	Palash	*Butea monosperma*	Flower	Fabaceae	The young flower buds are used as vegetable along with pickled oil	[29]
4	Koraya	*Holarrhena antidysenterica*	Flower	Apocynaceae	The flower bud is cooked as vegetable. Boiled flowers are cooked and eaten as vegetable	[29, 30]
5	Girhul	*Indigofera casioides*	Flower	Fabaceae	Flower bud is cooked as vegetable and curry	[29]
6	Imali	*Tamarindus indicus*	Flower	Fabaceae	Flowers are eaten as vegetable	[29]
7	Konga	*Holostemma rheedianum Spreng*	Flower	Asclepiadaceae	Flowers are eaten	[30]
9	Sun	*Crotalaria juncea L.*	Flower	Fabaceae	Flowers are eaten as vegetable	[30]
10	Jangli sun	*Crotalaria incana Rottl.*	Flower	Fabaceae	Flowers are cooked as vegetable	[30]
12	Kujur	*Celastrus paniculata Willd*	Flower	Celastraceae	Flowers are used as vegetable	[30]
13	Khilbiri	*C. renigera (Wall) Gagnep.*	Flower	Caesalpiniaceae	Flowers are eaten as vegetable	[30]
14	Amaltas	*C. fistula L.*	Flower	Caesalpiniaceae	Flowers are eaten as vegetable	[30]
15	Kumhi	*C. arborea* Roxb.	Flower	Lecythidacae	Flowers are eaten as vegetable	[30, 31]
16	Kachnar	*Bauhinia racemosa Lam.*	Flower	Caesalpiniaceae	Young flowering buds are used as vegetable	[30]
17	Ankol	*Alangium salvifolium*	Flower	Alangiaceae	Flowers are eaten as vegetable	[30]
18	Phoolgobhi	*Brassica oleracea var botr. Linn*	Flower	Brassicaceae	Flowers are eaten as vegetable	[14]

(continued)

Table 1 (continued)

S. N.	Plant Local Name	Scientific Name	Edible parts	Family	Uses	References
19	Kalaspati	*Couroupita guianensis Aubl*	Flower	Lecythidaceae	Flowers are eaten as vegetable	[32]
20	Broccali	*Brassica oleracea*	Flower	Boraginaceae	Flowers are eaten as vegetable	[32]
21	Bohar	*Cordia dichotoma*	Flower	Boraginaceae	Flowers are eaten as vegetable	[32]
22	Makhana	*Cucurbita pepo*	Flower	Cucurbitaceae	Flowers are eaten as vegetable	[32]
23	Munga	*Moringa oleifera*	Flower	Moringaceae	Flowers are eaten as vegetable	[32]
24	Amari patuwa	*Hibiscus sabdariffa L.*	Flower	Malvaceae	Flowers are eaten as vegetable	[32]
25	Neem	*Azadirachta indica*	Flower	Meliaceae	Flowers are eaten as vegetable	[32]
26	Bhaskatiya	*Solanum xanthocarpum*	Flower	Solanaceae	Flowers are eaten as vegetable	[32]
27	Kela	*Solanum xanthocarpum*	Flower	Musaceae	Flowers are eaten as vegetable	[32]
28	Madar	*Hibiscus rosa-sinensis*	Flower	Malvaceae	Flowers are eaten as vegetable	[32]
29	Surajmukhi	*Helianthus annuus L.*	Flower	Aesteraceae	Used as edible oil	[32, 33]
30	Gulab	*Rosa gallica*	Flower	Rosaceae	Flowers are eaten as vegetable	[32]
31	Merigold	*Tagetets erecta*	Flower	Aesteraceae	Flower are eaten as salad	[23]

that are located in the area. Almost 70% of the community is employed in agriculture, giving rise to the region's other name, the "Rice Bowl" of Central India. Rice is the major crop in the province, and it also makes up the majority of the food of the people who live here. It is abundantly clear that as the rate of urbanization rises, people's conventional dietary patterns are shifting in the direction of a culture around unhealthy food. Nevertheless, nearly 80% of the population of Chhattisgarh lives in rural areas. As a consequence of this, the state's traditional culture is still being practised, and the majority of the people's eating habits have not altered [34]. Foods from Chhattisgarh are both tasty and nutritious. Because of the state's extensive forest area, the rural and tribal people who live there supplement their diets with forest-grown roots, tubers, leaves, flowers, and fruits. They add a distinct flavour and beneficial properties of nature to traditional Chhattisgarh dishes. There are various types of leaves, tubers, roots, and flowers eaten locally here [35, 36]. This chapter delves into the typical flower-based food of Chhattisgarh and gives information on the flowers, leaves, tubers, rhizomes, and roots used in these dishes. Vegetarianism is highly prevalent in Chhattisgarh, and the state's cuisine makes excellent use of locally grown vegetables and other nutritious ingredients. Every nation's traditional cuisine has survived and is the healthiest option for its citizens. There is a worldwide trend towards continually low fruit and vegetable consumption [37]. Flowers, rhizomes, fruits, leaves, and other plant-based foods make up a significant food group that is rich in necessary minerals and vitamins; ensuring adequate consumption of this group is especially crucial in Chhattisgarh, which suffers from chronic nutritional deficits [38]. The food cultures of Chhattisgarh are the path forward towards adopting sustainable practiscs and making use of the native materials at our disposal to enhance our regular intake. Not only does the processing and preparation of ethnic foods recognize the inventiveness of the local inhabitants and their wealth of cultural and historical heritage, but it also validates their tremendous competence in supporting the existence of the environment in its entirety [39].

Dietary practices are a significant factor in determining the state of people's healthcare. The growth of the human population has been paralleled by the steady evolution of the culture surrounding local foods. It is determined by geographical position, environmental conditions, seasonal fluctuations, soil composition, water supply, forest area, agricultural practices, immigration, and the impact of conquerors, as well as the labour routines of the inhabitants in the area. The culture around traditional foods is based on an experiential context that develops over the course of generations. Food's function in human existence is not confined to simply satisfying appetite; rather, it has a far larger meaning and is an essential component in practically every facet of human existence, including the family, social relationships, festivals, and spiritual rites. The residents of a region have access to the healthiest, most nutritious food options available thanks to the traditional cuisine that has been passed down through the generations [40, 41]. Accepting unhealthy, out-of-region eating habits in the name of globalization is not only unethical but also potentially damaging to people's health. In 2015, the Global Burden of Disease study found that bad food was the largest cause of immorality worldwide, largely due to the shift from a conventional to a modern eating pattern. Each nation's traditional cuisine has stood the test of time and is the healthiest option for its citizens. While rice and vegetables are the most common foods in Chhattisgarh, pulses, green leafy vegetables (GLV), tubers, flowers, and roots are also part of a typical Chhattisgarh lunch. The rural and tribal residents of the state have acquired the ability to do with what nature provides because of the extensive forest cover in their area. The ethnic food of Chhattisgarh is thought to have beneficial health effects, which are supported by both traditional medical practices and more contemporary scientific studies. The food culture of Chhattisgarh is both abundant and extremely varied. The state's numerous dietary patterns are intricately connected to aspects of social authenticity, religious practice, and other sociocultural factors, in addition to local farming techniques and a broad variety of readily available food. Well over 50% of the state's inhabitants can acquire their meals and vegetables from the crops they grow, and edible flowers are a significant component of those crops. It is farmed or cultivated in almost every region, and about half of Chhattisgarh's population uses it as their primary source of nutrition [42].

6 Nutritional Profiles of Edible Flowers

Although the nutritional value of many forest meals is unclear, there seems to be sufficient evidence to suggest that forest foods are beneficial to one's health in terms of their nutritional content. It is of the utmost priority to support experiments on the nutritional values of foods found in forests, since this will inspire people to consume a larger amount of food and offer them a more optimal nutritional profile [14, 43]. Edible flowers have been a component of the human diet for thousands of years. Their therapeutic characteristics and subsequent health benefits have led to their classification as a type of plant cuisine. *Brassica oleracea var botr. Linn, Brassica oleracea, Cordia dichotoma, Cordia dichotoma, Moringa oleifera, Hibiscus sabdariffa L., Solanum xanthocarpum, Helianthus annuus L., Rosa gallica, Tagetes*, *Madhuca latifolia Roxb., Crotalaria juncea L., Bombax malabaricum, Butea monosperma, Tamarindus indicus, C. fistula L., and Bauhinia racemosa Lam.* are the many types of edible flowers that have been eaten since ancient periods. Flowers that are harmful or otherwise unsafe for humans or have poor nutritional benefits are not typically included in human food [44, 45]. Certain flower varieties include poisonous compounds that may reduce their nutritional value, such as alkaloids, oxalate, cyanogenic glycosides, nitrate, or phytate, or cause serious harm to users [15, 44, 46]. These edible flowers are sold commercially or used in human food. It is important to confirm the nutritional profile and other properties associated with human nourishment for the foods that are generally accepted as edible. Edible flowers vary widely in terms of total carbs, soluble fibre, and micronutrients, depending on the type of edible flower [15, 26, 44, 46, 47]. The majority of edible flowers are composed entirely of water molecules (more than 72%) and contain the least concentration of protein and vitamins. Edible flowers also include varying quantities of carbohydrates, crude fibre, fat, lipids, and minerals. Edible flowers include a variety of bioactive components, like pigments, carotenoids, phenolic compounds, terpenoids, flavonoids, and volatile oils, which offer a broad range of beneficial qualities. The proximate composition and mineral content concentration of many edible flowers from Chhattisgarh, India, are displayed in Tables 2 and 3, respectively.

7 Bioactive Components of Edible Flowers

Bioactive components are clinically evident, naturally produced organic molecules that do not constitute nutrients but have therapeutic properties that promote public health. They enhance the plant's appearance, aroma, and taste in addition to protecting it from illness and environmental damage [72]. Most of the bioactive chemicals in edible flowers are polyphenol compounds, carotenes, flavonoids, and anthocyanins [15, 71]. Carotenoids and flavonoids give flowers their colour and antioxidant capacity [73]. Lutein and flavonoids have been examined for their potential to treat cytotoxic cancers [73–76]. Several types of polyphenol components, including phenolic acids, flavonoids, flavonols, and anthocyanins, are found in edible flowers, which have strong antioxidant potential, act as a shield against free radicals, and have been linked to improved human metabolism [77–80]. Polyphenols are known to have antioxidant properties in in vitro and in vivo examinations [81]. Many therapeutic properties may be contributed by phenolic components, carotenes (colourants) [82], isothiocyanates, volatile oils (the primary source of the flowers' aroma) [83, 84], and cyclotides, which are circular plant peptides. The ingestion of edible flowers has substantially expanded recently due to their multiple health advantages, which are contributed in part by the abundance of phytonutrients found in edible flowers. Although many investigations have been conducted, there has not been a comprehensive literature review prepared on the nutritional profile of edible flowers and the phytonutrients found in them. Prior to recently, exploration of edible flowers focused on their nutrition, aroma, and volatile oils [85, 86]. However, phytonutrients, the primary beneficial constituents of edible flowers, have received increased attention in investigations. Phytochemicals are the active molecules found in vegetables and plant foods that significantly lower the chances of serious illnesses like tumours, fever, cancer, CVD, inflammation, and cytotoxicity [87, 88]. Figure 2 displays the various bioactive components of edible flowers from Chhattisgarh.

7.1 Anthocyanins

Anthocyanins are water-soluble phytochemicals that belong to the flavonoid group. These are the natural colour pigments that are responsible for the colouring of flowers. The pH value, metal ions, and co-pigments all play a role in determining their unique coloration. Anthocyanins have been demonstrated to have a higher potential for eliminating free radicals [89]. These molecules also play a significant role in the prevention of several illnesses, including malignancy, cardiac disease, overweight, hypertension, and diabetes. Anthocyanin can be found in high

Table 2 Proximate composition of edible flowers of Chhattisgarh India

S. N.	Plant Scientific Name	Moisture Content (g/100 g)	Total Ash (g/100 g)	Total Protein (g/100 g)	Total Fat (g/100 g)	Fibre (g/100 g)	Carbohydrate Content (g/100 g)	Total Energy (kcal)	References
1	*M. latifolia* Roxb.	79.58	1.5	6.35	1.6	10.8	–	–	[48]
2	*Bombax malabaricum*	7.4	6.55	5.3	1.21	14.4	32.47	290.31	[49, 50]
3	*Butea monosperma*	19.4	11.2	6.5	8.1	6.1	43.9	–	[51]
4	*Holarrhena antidysenterica*	79.61	4.27	7.31	7.17	6.01	53.38	561.9	[52]
5	*Indigofera casioides*	47.6	5.2	2.63	1.13	19.65	23.92	484.26	[53]
6	*Tamarindus indicus*	42.7	3.32	2.4	0.17	1.79	56.72	216.86	[54]
7	*Crotalaria juncea L.*	10.85	3.31	36.43	20.42	15.32	45.19	227.34	[55]
8	*C. fistula L.*	5.27	12.17	12.43	1.72	19.82	41.32	298.34	[56]
9	*Bauhinia racemosa lam.*	8.2	7.28	6.92	6.57	1.98	76.17	238.29	[57, 58]
10	*Alangium salvifolium*	83.9	1.61	2.71	3.8	2.95	111.67	209.13	[59]

11	*Brassica oleracea var botr. Linn*	85.58	8.02	36.15	7.23	10.17	39.5	304.42	[60]
12	*Couroupita guianensis Aubl*	–	8.94	11.23	–	6.78	–	–	[61]
13	*Cordia dichotoma*	79.29	–	2.33	–	–	31.66	–	[62]
14	*Cucurbita pepo*	69.91	9.61	11.9	2.09	3.51	36.8	226	[63]
15	*Moringa oleifera*	–	78.7	178.7	39.89	–	–	–	[64]
16	*Hibiscus sabdariffa L.*	7.13	11.09	17.29	3.73	24.51	38.59	920.21	[39, 65, 66]
17	*Solanum xanthocarpum*	91.17	1.56	1.79	0.44	–	–	32.6	[67]
18	*Hibiscus rosa-sinensis*	82.92	1.41	1.55	1.35	1.53	2.74	228.47	[68, 69]
19	*Rosa gallica*	11.18	1.83	7.94	16.09	3.71	52.23	4.22	[70]
20	*Tagetets erecta*	83.38	0.89	1.35	0.31	9.18	14.14	28. 03	[71]

Table 3 Mineral content concentration of edible flowers

S. N.	Plant Scientific Name	Ca (mg/100 g)	P (mg/100 g)	Fe (mg/100 g)	Zn (mg/100)	Cu (mg/100 g)	Co (mg/100 g)	Mg (mg/100 g)	K (mg/100 g)	References
1	*Madhuca latifolia* Roxb.	45	22	–	–	–	–	–	–	[48]
2	*Bombax malabaricum*	–	–	–	–	–	–	–	–	
3	*Butea monosperma*	44.5	–	44.4	–	–	–	–	–	[51]
4	*Holarrhena antidysenterica*	–	–	–	–	–	–	–	–	
5	*Indigofera casioides*	0.93	0.025	0.032	0.013	0.003	0.001	0.32	0.16	[53]
6	*Tamarindus indicus*	74	0.17	2.8	0.1	0.87	0.012	0.15	1.14	[54]
7	*Alangium salvifolium*	12	–	1.37	–	0.32	–	19	168	[59]
8	*Brassica oleracea var botr. Linn*	4.37	5.91	0.92	0.55	0.051	0.021	1.95	33.91	[60]
9	*Cordia dichotoma*	0.46	–	0.51	0.35	–	–	–	7.83	[62]
10	*Cucurbita pepo*	2.61	–	0.21	0.06	–	–	2.39	19.97	[63]
11	*Hibiscus sabdariffa L.*	155.57	39.5	27.6	3.85	0.71	–	38.52	23.04	[65] [40, 66]
12	*Solanum xanthocarpum*	24.2	18.7	0.79	0.67	0.35	0.42	14.47	86.9	[67]
13	*Hibiscus rosa-sinensis*	4.33	–	1.49	0.83	–	–	–	23.66	[68, 69]

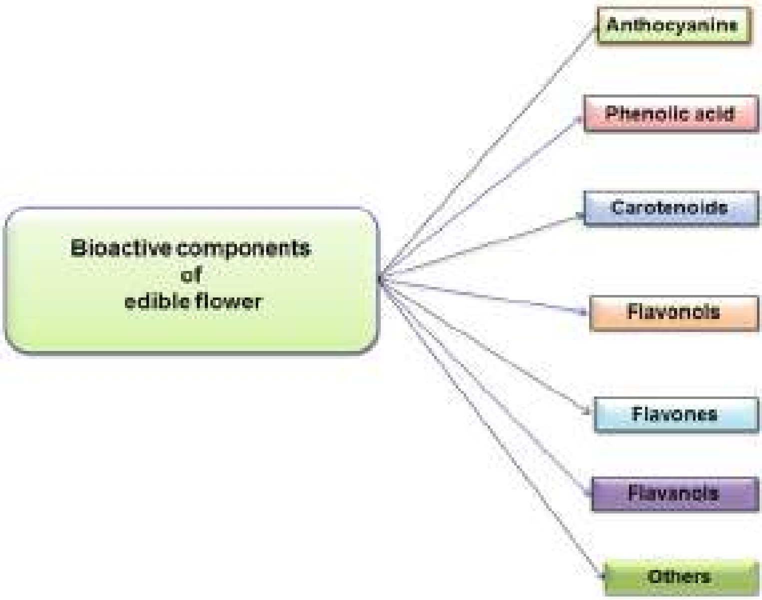

Fig. 2 Bioactive components of edible flowers

concentrations in *Rosa gallica, Solanum xanthocarpum, Hibiscus rosasinensis, Hibiscus sabdariffa L., Moringa oleifera, Cordia dichotoma, Cordia dichotoma, and Tagetets*, among other edible flowers of the plants. Kumari et al. (2017) [90] examined the anthocyanin of various Indian cultivars of *Rosa gallica*. Employing HPLC-PDA, she identified cyanidin 3,5-di-O-glucoside and pelargonidin 3,5-di-O-glucoside. The cyanidin 3,5-di-O-glucoside has been the chief anthocyanin responsible for the reddish and pinkish colour of the edible flower. Pelargonidin 3,5-di-O-glucoside has also been found as an anthocyanin in the carrot-red type of colour of the edible flower [91].

7.2 Carotenoids

Carotenoids are natural pigments found in plants. They give flowers their colours, which range from yellow to orange to red. Carotenoids contain many isoprenoid molecules [92]. The proportion of carotenoids in blossoms varies substantially across floral taxa. Carotenoids are a vital part of both human and animal nutrition and are absorbed from healthy meals and supplementation. According to the findings of numerous studies, carotenoids might decrease the development of vitamin A deficiency, age-related macular degeneration (AMD), blindness, UV exposure, malignancy, cytotoxicity, and CVD. Several studies have shown that Tagetes flowers could be a good source of carotenoids because they contain lutin and its derivatives. At a low amount, zeaxanthin, antheraxanthin, β-cryptoxanthin, β-carotene, α-carotene, zeinoxanthin, and violaxanthin have also been detected in many edible flowers like *Indigofera casioides*, *Crotalaria juncea L., Bauhinia racemosa L., Brassica oleracea,* etc. *Madhuca latifolia, Tamarindus indicus, Moringa oliefera, Rosa gallica, and Cardia dichotoma* flowers have been found to contain many carotenoids like lutein, flavoxanthin, β-carotene, α-carotene, lycopene, rubixanthin, zeaxanthin, α-cryptoxanthin, 13-*cis*-carotene, α-carotene, *trans*-carotene, 9-*cis*-carotene, luteoxanthin, violaxanthin, zeaxanthin, neoxanthin, and lutein epoxide [89, 92–95].

7.3 Flavonols

The group of flavonoids is referred to as flavonols. It is one of the most successful sources of flavonoids. The flavonols contain many sub-groups, such as quercetin, kaempferol, isorhamnetin, and myricetin, along with their derivatives. Flavonols were widely found in edible flowers. They were found in *Rosa gallica* [96], *Moringa oliefera, C. fistula L., Cardia dichotoma, Solonum xanthocarpum, Hibiscus rosa sinensis*, and cactus in a variety of morphologies [97]. *Tegetete*, *Rosa gallica, and Couroupita guianensis Aubl*. also noted the variety of primary flavonols that are present in phenolic flavonoids. Researchers have found that the most common flavonol aglycones in *Rosa gallica* petals are kaempferol and quercetin [91, 92, 98–100]. The majority of edible flowers contain large concentrations of flavonols in comparison to other flavonoids. Shrivastav et al. (2016) [29] reported that the edible flowers of *Butea monosperma, Bombox mealricum,* and *Madhuka latifolia* Roxb. could have isorhamnetin, quercetin, and their glycosides. Ekka and Ekka (2016) [30] also found that *Crotalaria juncea* flowers had several types of flavonols, like kaempferol, quercetin, isorhamnetin, and their glycosides. The above findings provided more evidence that flavonols could be found in many common edible flowers.

7.4 Flavones

Flavones are a subclass of flavonoid whose skeleton is 2-phenylchromen-4-one (2-phenyl-1-benzopyran-4-one). Flavones are a derivative of the Latin word "flavus," which describes a yellowish colour pigment. Flavones are widely available in edible flowers of many kinds. Acacetin, chrysoeriol, apigenin, luteolin, and their corresponding glucosides are the main flavones. There is a rich supply of flavones in a variety of edible flowers. There was a high concentration of flavones in all types of edible flowers that originated from the Asteraceae family, such as *Helianthus annuus* and *Tegetes* [16]. There is a rich supply of flavones in a variety of edible flowers. *Tamaricus indicus* flowers were discovered to contain a significant amount of luteolin-7-glucoside [29]. *Chrysanthemum* and *Tegetet* flowers contain greater amounts of flavones, including acacetin, acacetin-7-O-glucoside, apigenin,

apigenin-7-O-glucoside, luteolin, and luteolin-7-O-glucoside. These flavones have been claimed to have anti-inflammatory action. *Rosa gallica* and *Moringa olifera* both contain luteolin in their respective preparations. Apigenin glycosides were the predominant flavones identified in *Tegetet* [101].

7.5 Flavanols

Flavanols have been found in edible flowers that are made up of catechin, epicatechin, epicatechin gallate, and epigallocatechin gallate. Many researchers reported that *Rosa gallica* had a greater concentration of flavanols in edible flowers [16, 96]. Catechin, epicatechin, epicatechin gallate, and epigallocatechin gallate are the chief constituents of *Rosa gallica*. Many flavanols have been reported in *Brassica oleracea, Madhuca latifolia*, *Butea monosperma, and Bombox malabaricum* flowers [96, 98, 102, 103]. Many researchers reported flavanols (catechins) in *Chrysanthemum* flowers in the highest concentration. The majority of the crude extract of *Rosa gallica* flowers is composed of flavanols, specifically catechin and epigallocatechin gallate [104].

7.6 Phenolic Acids

To simplify, "phenolic acids" refer to all phenolic compounds with a single carboxylic acid group. Phenol-carboxylic acid, mostly called polyphenols, is an abundant phenolic acid found in many plants and edible flowers. They can be found in many different types of edible flowers. Another important class of secondary metabolites found in edible flowers are phenolic acids. There have been many reports of the presence of phenolic acids in *Rosa gallica* [105], *Cardia dichotoma, Alangium solvifolum, Chrysanthemum*, and *Brassica oleracea* [106]. Other phenolic acids comprise protocatechuic acid, caffeic acid, and gallic acid. An important category of phytochemicals that are identified in edible flowers is called phenolic acids. *Solanum xanthocarpum* flowers contain many phenolic compounds like coumaric acid, quercetin, rutin, ferulic acid, hydroxybenzoic acid, caffeic acid, and cinnamic acid in higher concentrations. *Indigofera cosioides* blossoms contain significant amounts of polyphenolic compounds [107].

7.7 Others

Various types of phytochemicals like carotenoid (β-carotene), phytosterols (β-sitosterol) [108], alkaloids, lignans [109–111], neolignans [112], coumarins, and bisabolo are also found in various edible flowers. The quantities of these phytonutrients were significantly low, and their functional properties were not as potent [113, 114].

8 Therapeutic Potential of Edible Flowers

The purpose of this chapter has been to examine the conventional uses, phytonutrients, therapeutic properties, and nutritional profiles of various edible flowers in Chhattisgarh, India. Flavonols, flavones, flavonoids, anthocyanins, phenolic acids, and their derivatives were discovered to be ubiquitous phytonutrients in a variety of locally eaten flowering plant species, all of which were linked to the edible flowers' reported therapeutic properties. Edible flowers have various health advantages, including antioxidant, anti-inflammatory, anticancer, anti-diabetic, anti-obesity, neuroprotective, anti-carcinogenic, anti-microbial, hepatoprotective, analgesic, and antiulcer characteristics [115–117]. These effects are due to flower phytonutrients and secondary metabolites. Advantages to health from edible flowers are also discussed [118]. The therapeutic potential in edible flowers of Chhattisgarh, India, are displayed in Fig. 3.

8.1 Anti-oxidant Activity

Oxidative stress, ageing, neurological illnesses, cerebrovascular disease, metabolic disorders, cardiovascular disease, tumours, and cytotoxemia are all caused by an excess production of free oxidative radicals. The edible flowers possess strong antioxidant properties. The antioxidant activities were investigated using both *in-vitro* (DPPH, ABTS, and ORAC) and *in-vivo* (lipid peroxidation assay)

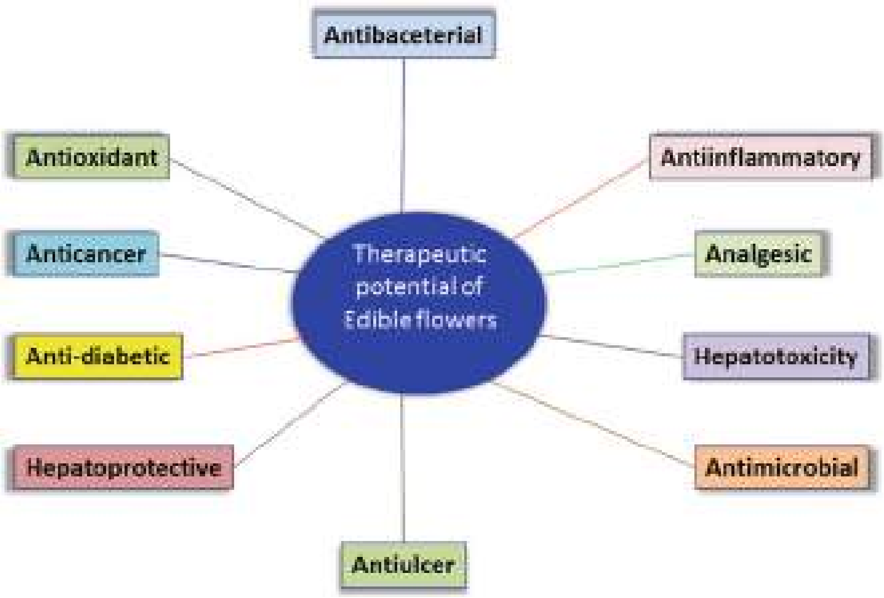

Fig. 3 Therapeutic potentials of edible flower

techniques [119]. Many researchers reported that edible flowers displayed higher antioxidant potential [120, 121]. Some edible flower extracts, like *Madhuca latifolia* Roxb., *Butea monosperma*, *Tamarindus indicus*, and *Holarrhena antidysenterica* have shown greater capacity to scavenge free radicals [122, 123]. Edible flowers have been shown to have considerable anti-oxidant potential owing to the presence of anthocyanins, flavonoids, phenolic acids, alkaloids, terpenoids, and glycosides. Edible flowers have a stronger antioxidant potential than other plant parts like leaves, roots, rhizomes, and barks, which may be due to the higher number of phenolic compounds, particularly flavonoids [124]. The phytochemical flavonoids, alkaloids, terpenoids, steroids, carbohydrates, and glycosides in edible flowers might help them fight free radicals. The biological processes that occur within the human organism generate free radicals as an unwanted consequence. Due to reactive compounds, they cause the oxidative destruction of lipids, proteins, and ultimately DNA. A well-balanced anti-oxidative inhibition system can help counteract damage. Edible flowers are known to be rich providers of a number of different natural bioactive substances, such as vitamins. Healthy food sources, including polyphenols, are of major importance because they provide better antioxidant sources. Phenolic compounds have been shown to have a good impact on chronic health issues such as CVD, diabetes, hypertension, heart disease, hepatoprotective, and neurological illnesses [125–127].

8.2 Anti-inflammatory

The inflammatory response is a necessary and helpful part of the body's physiology. Its main function is to protect the host from both poisons made by the body and microorganisms that come from outside the body. Inflammatory conditions can start off as a short-term acute condition, but they can also turn into a long-term chronic condition, which can cause irreparable harm to the host's health. The anti-inflammatory properties of *Madhuca latifolia* Roxb., *Bombax malabaricum, Butea monosperma, Holarrhena antidysenterica, Indigofera casioides, Tamarindus indicus, Crotalaria juncea L., C. fistula L., C. arborea* Roxb., *Couroupita guianensis Aubl, Cordia dichotoma, Moringa oleifera, Hibiscus sabdariffa L., and Solanum xanthocarpum* demonstrated strong inflammation action. Evaluation of the inflammation effect of edible flower extracts on chronic inflammatory diseases was conducted using cell line assays such as lipopolysaccharide (LPS)-induced RAW 264.7 macrophage activation and in vivo assays such as dimethylbenzene (DMB)-induced ear vasodilatation, acetic acid-induced fluid accumulation, and carrageenan-induced paw oedema. The results indicated that edible flower extracts reduced the severity of acute inflammation. Inflammation is a natural process that causes the immune system to work harder [128]. It is an essential part of the body's physiologic function and acts as a protective mechanism against bacterial infections, viruses, and poisons. The cell simulation method was used to investigate the characteristics of inflammation in the extract of a variety of edible flowers. Many edible flowers possess strong inflammation potential [89, 129].

8.3 Anti-cancer

Investigations in epidemiology have demonstrated that a diet rich in edible flowers, fruits, rhizomes, and vegetables is closely connected with a lower risk of developing long-term diseases [130]. Some edible flowers involve *Chrotalarea juncea* L., *Chrotalarea incana* Rottl., *Rosa gallica*, *Hibiscus sabdariffa*, and *Casea fistula* L. [131–133]. Edible flowers exhibit powerful actions against cancers of the liver, bladder, prostate, breast, and colon. The ability of edible flowers to prevent cancer is mostly attributable to phytonutrients like gallic acid, protocatechuic acids, flavonoids, and rugethanoids B [133, 134]. These edible flowers had the ability to provide cancer-fighting medications. Edible flowers have shown significant activity against a variety of different cancers. The growth and development of malignant breast tumours can be inhibited by consuming modest doses of lutein from marigold extract in their diet. In recent studies, the MTT test was used to show that the extracts of *Bauhinia tomentosa* flowers and *Tagetes erecta* petals were effective against the liver cancer cell lines HePG2 and MCF-7. Skin disorders such as skin damage and skin cancer have emerged as a topic of attention for people all over the world in recent years. The anticancer properties of *Hibiscus sabdariffa*, which is an abundant source of flavonoids, have been the subject of extensive research [135–137]. Researchers tested its cytotoxicity on T47D breast cancer cells and obtained positive results [138]. In a manner comparable to that which was addressed previously, a few studies demonstrated the anticancer action of edible rose blooms. The findings suggested, taken together, that edible flowers might be an even more effective source of anticancer drugs [139].

8.4 Anti-obesity

The build-up of excess fat in the body, the process of fatty acid synthesis, and an imbalance in energy levels all contribute to obesity, also known as overweight, which is associated with adipocyte differentiation and a higher risk of health-related issues. Substances with natural bioactive properties can be utilized to prevent obesity. Obesity or being overweight happens when a person takes in more calories than they use up, and it always leads to a number of metabolic and long-term conditions, such as trouble sleeping, high blood pressure, hypertension, high cholesterol, and diabetes [140]. Several edible flowers, including *Madhuca latifolia* Roxb.*, Bombax malabaricum, Butea monosperma, Holarrhena antidysenterica, Indigofera casioides, Tamarindus indicus, Crotalaria juncea L., C. fistula L., C. arborea* Roxb.*, Couroupita guianensis Aubl, Cordia dichotoma, Moringa oleifera, Hibiscus sabdariffa L., and Solanum xanthocarpum*, demonstrated an inhibiting effect on obesity [141, 142]. Various phytochemicals acted as mediators between the enzyme and biochemical pathways in lipid burning. *Rosa gallica* extract either controlled calorie consumption and expenditure by decreasing α-amylase activities and preventing carbohydrate or starch uptake or prevented adipocyte development by influencing the PI3K and MAP-kinase pathways.

8.5 Neuroprotective Activity

Alzheimer's disease and Parkinson's disease are both common forms of dementia that cause memory loss. Neurodegeneration is closely related to age, and both of these disorders are connected with ageing. Many researchers discovered, through the use of a cerebral I/R model, it has the potential to significantly enhance the neurological deficiency levels as well as the infract volume [143]. In the *Caenorhabditis elegans* model, flower extracts of *Tagetes erecta* showed potential antioxidant and neuroprotective characteristics. Polyphenols, mainly laricitin and its glycosides, were responsible for these activities and were shown to be responsible for them [144]. Degradation of dopaminergic neurons is also the root cause of ischemia, which is one of the main causes of death and chronic disability. The potent neuroprotective effect of edible flowers, such as *Chrysanthemum, Rosa gallica*, and *Honeysuckle*, was assessed in an in vivo model using glutamate, arachidonic acid, and 6-hydroxydopamine (6-OHDA)-induced serious injuries, posterior cerebral artery occlusion-induced focal cerebral ischemia, and the neurotoxin 1-methyl-4-phenylpyridinium. MPP+'s ability to cause apoptosis in human neuroblastoma cells could be stopped by the phenolic acid found in edible flowers [89].

8.6 Anti-diabetic

Metabolic disorder diabetes mellitus is associated with reduced or insufficient insulin secretion or function. Diabetes can also be referred to as "adult-onset diabetes." As a consequence, the body is unable to metabolize glucose, which ultimately leads to increased quantities of glucose in the blood. Many edible flowers have the capacity to reduce high levels of glucose in the blood, which was evaluated in comparison to insulin [145]. It was found that *Catharanthus roseus* flower ethanol extracts were better at lowering blood sugar levels than the main diagnostic medicine. Due to the liver using more glucose, hypoglycaemia has been seen [146]. Because *Butea monosperma* flowers cause a significant increase in the amount of glucose that is taken up by *Saccharomyces cerevisiae*, there is greater potential for their use in the treatment of diabetes [147]. Various phytochemicals, particularly flavonoids and terpenoids, have a stimulating impact on the body's ability to take in glucose. The anti-diabetic benefits of these flowers are caused by the suppression of enzymes involved in the metabolism of carbohydrates, the control of glucose transporters, the renewal of beta cells, and the enhancement of insulin-generating function [148].

8.7 Hepatoprotective Activity

Edible flowers and their natural active components have received an abundance of interest due to their remarkable ability to treat hepatic issues. The aqueous extract of *Madhuca latifolia* was examined, and the result indicated that it has a strong hepatoprotective reaction against carbon tetrachloride. This was a direct consequence of the serum levels of transaminases, bilirubin, alkaline phosphatase, and triglycerides regenerating themselves. Both the water and methanolic extracts of the edible flowers of *Butea monosperma* demonstrated significant hepatoprotective effects. Another aqueous extract of *Bombax ceiba* flowers showed that these flowers have hepatoprotective activity against CCl_4-induced hepatic injury [29, 149]. Abdelhafez et al. (2018) [150] investigated the possible hepatoprotective activity of *Hibiscus malvaviscus* against the hepatic injury caused by CCl_4 in mice, and findings indicated that dichloromethane and ethyl acetate fractions considerably reduced the seriousness of hepatic damage in rodents.

8.8 Antimicrobial Activity

A plant extract of *Azadirichta indica* is used in the production of natural high-antimicrobial agents. The flowers of *Brassica oleracea* and *Cardia dichotoma* plants were effective against both gram-negative and gram-positive bacteria [14]. Also, antibacterial activity has been found in both ethanolic and aqueous extracts of *Hibiscus rosa-sinensis* against a number of foodborne bacterial infections [151]. *Rosa gallica* methanolic extracts have an antimicrobial effect that is effective against a variety of bacterial and yeast strains [89].

8.9 Others

Numerous plant families and groups across Chhattisgarh, India, rely on edible flowers as a primary means of ensuring their continued access to food and means of survival. It is believed that around 800 different plant species are used as food in India, the majority of which are eaten by the indigenous people. Consuming edible flowers is vital for the eating habits of humans in several communities and is intricately tied to practically all areas of the sociocultural, spiritual, and social welfare of these humans and communities. There is a critical role for the consumption of edible flowers in satisfying the nutritional requirements of rural populations [147, 152]. There are many varieties of edible flowers, and these flowers have been used in a wide variety of ways across a variety of socioeconomic and geographic settings. In addition, there have been substantial reports that suggest that eating wild foods has been a common activity for a very long time [153]. Many of the therapeutic advantages associated with consuming edible flowers are due to the high concentrations of phytonutrients. Both *Rosa gallica* and *Hibiscus*, which are members of the Malvaceae family, have a number of medicinal properties. *Rosa gallica* was shown to be anti-cholesterol, anti-hypertensive, and anti-diabetic [16]. Both delphinidin-3-O-sambubioside and cyanidin-3-O-sambubioside were important, but the biological causes are still unknown. Hibiscus effects of on epileptic seizures and fertility suppression. Also, it helped with healthy hair, tissue repair, and immunology. Two chrysanthemum species were found to significantly reduce the formation of multiple AGEs in the bovine serum albumin (BSA)/glucose and BSA/fructose systems [154, 155]. This finding provides the possibility for the formulation of new diagnostic approaches for the prevention of diseases linked to glycosylation. In in vivo studies, the use of *Rosa gallica* extract resulted in a reduction in blood pressure in both acute and chronic situations. Anti-HIV-1 properties have been identified in rose species that could be consumed [16]. Despite substantial research into the possible health advantages of natural active components in edible flowers, surprisingly little is known about these plants.

9 Overview of Commercial Edible Flowers

Flowers have always played an important role in human culture as well as in the business and decorating worlds. Edible flowers are popular all over the world because they are pretty and healthy. Flowers can be used in many ways. They can be used to add scent, flavour, and colour, and they can also be used to decorate and make food and drinks. Before people can eat flowers, though, their biochemistry needs to be carefully studied. Roses, marigolds, hibiscus, calendula, and other flowers like these were often used in different kinds of food. The floriculture market gets a boost when flowers are grown specifically for their culinary qualities. Floriculture in India is considered an economy with tremendous growth potential. The importance of commercial floriculture with regard to international trade is growing. The liberalization of economic and trade policy created the foundation for the growth of flower production with an emphasis on exportation. Flowers are cultivated not just for their aesthetic value but also because of the nutritional and therapeutic benefits that they offer. Edible flowers have been utilized in a variety of ways throughout a wide range of international locales, including for their nutritional worth, medicinal function, flavour, form, and aesthetic appeal [15, 26]. Many flowers consumed are referred to as edible flowers. Until they are put to use, many parts of flowers, like sepals, petals, carpels, and filaments, are often plucked. Generally, edible flowers are taken in their entirety; however, depending on the kind of flower, certain sections of the flower should be ingested. In addition, it is necessary to remove some sections of certain flowers because of their bitterness. They are saved for use at a later time by employing preservation methods such as drying, freezing, or steeping in oil [156].

Regional and national preferences in food preparation and consumption are shifting. More and more consumers are learning the importance of eating well and choosing foods that meet their nutritional needs. Studies have shown that edible flowers are extremely beneficial to human health due to their abundance of beneficial chemicals, such as antioxidants, anti-carcinogens, vitamins, and many others [157]. Considering the interconnectedness of elements like effective input utilization, cost benefits, improved product quality, and ecological processes is also crucial to the enterprises' success. More people are aware of the serious decline in natural resources and the deterioration of the environment, and this has increased the pressure to make any creative activity as eco-friendly as possible [158]. Flowers have long been used as key components in a wide variety of dishes. Edible flowers are employed for both their aesthetic value and their sensory qualities [159]. The visual appeal of a meal can be affected by both the ingredients used and the presentation. Quality and aesthetics affect the appearance of meals. Edible flowers can boost a dish's attractiveness.

Edible flowers have not been grown by the floriculture industry in the past, but doing so could open up a new market and give the industry a new perspective. Certain flowers are deadly; therefore, it is important to know which ones to avoid [160]. Flowers that are often used in cooking include the *Chrysanthemum*, *Tegetets erecta*, *Madhuca latifolia* Roxb*., Bombax malabaricum, Butea monosperma, Holarrhena antidysenterica, Indigofera casioides, Tamarindus indicus, Crotalaria juncea L., C. fistula L., C. arborea* Roxb*., Couroupita guianensis Aubl., Cordia dichotoma, Moringa oleifera, Hibiscus sabdariffa L., and Solanum xanthocarpum*.

Edible flowers were a big part of traditional ways of life that are coming back thanks to globalization and better consumer knowledge. Now, more than ever, edible flowers are all the rage as a trendy new food trend in both well-known and new countries, making competition fierce in this previously untapped, unique area. Because of the COVID-19 situation right now, the flower business is looking into new options. While the number of people who grow and market edible flowers (in places like supermarkets, flea markets, and online) grows, the expectations of consumers and chefs alike continue to rise. They will be the cutting-edge components of next-generation food varieties [28]. When a flower is grown for its edible petals, it boosts the economy in a way that is hard to measure. In addition to their aesthetic value, flowers serve an essential function in agriculture: pollination. The primary factor driving the expansion of the global edible flower market is the increasing application of edible flowers as a supplement in the nutritional, food, and pharmaceutical industries. This has a beneficial effect on edible flower yield. The demand for them is rising all around the world since chefs and regular people alike see them as a healthy and secure meal option. You can also find them in a variety of dishes, drinks, decorations, baked goods, and salads. When compared to the costs of growing other crops, these ones are prohibitively expensive. The purchase of improved varieties for organic agriculture is a major cost driver. Edible flowers have a short shelf life, so they need to be eaten right away after being sold at the market. There are many geographic areas in the market for edible flowers, like Raipur, Bilaspur, Durg, Jagadalpur, and Raigarh city of Chhattisgarh, India. At the time of the evaluation, it was thought that the unique flavour and taste of edible flowers were the main reason for the growth of the regional market. Nevertheless, currently, there is no reliable evidence for the Indian edible flower market. This is due to the fact that the majority of edible flowers in India are harvested from the wild. As a result, there is a requirement to strengthen the market in economies that are experiencing economic growth [89, 161].

10 Conclusions

Edible flowers are becoming more popular in modern nutrition, mostly because of improvements in the nutritional value of a variety of foods and meals. Generally, this has a potentially therapeutic influence on public health. Edible flowers are a natural product that can be found all over the world. The vast majority of them have multiple phytochemicals that are good for your health, which is becoming more and more interesting. Numerous edible flowers have been used as a healthy diet for a long time. Recent research has shown that they are good for your health and found the active molecules and how they work to treat illness. Also, studies have been done to find out if common edible flowers are safe to eat, the best way to use them, and how much to use. Yet, due to the vast number of edible flowers that may be found across the globe, only a small portion of these flowers have been researched. We think that more research should be done to find the best ways to use edible flowers. This would make edible flowers more likely to be used as food ingredients and would also help to avoid any possible risks. Edible flowers have renewed interest in them, both because they look nice and because they may have health benefits. As their popularity has grown, more attention has been paid to their nutritional profiles, therapeutic potential, and natural bioactive components. Edible flowers have a wide range of phytonutrients. Some of these have been shown to reduce inflammation and act as antioxidants, which are two of the main reasons why we age and get sick over time. The composition and bioactivity of edible flowers have been studied. However, further data on the security and use of edible flowers is required before they can be termed functional foods or a part of our diet. Edible flowers are getting more attention in the present day as a result of their remarkable nutritional potential. Nutritionally and phytochemically, edible flowers are a great resource. The prominent phytochemicals found in edible flowers include phenolic compounds, carotenoids, terpenoids, flavonoids, and anthocyanins. Effective therapeutic potential such as anti-diabetic, anti-cancer, cytotoxicity, anti-inflammatory, antibacterial, hepatoprotective, analgesic, antimicrobial, antiulcer, and neuroprotective is present in edible flowers. The worldwide culinary and pharmaceutical industries are increasingly turning to edible flowers as a source of nutritionally rich ingredients, creating huge potential for the flower-based edibles market. Hence, the discovery of edible flowers has led to new opportunities for combating hunger, expanding agricultural production, creating new streams of revenue, and saving endangered species of wild flowers. Researchers focusing on edible flowers and their applications may find the current study a useful resource, as it covers a thorough understanding of all these issues.

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Plants Used in the Management and Treatment of Male Reproductive Health Issues: Case Study of Benin People of Southern Nigeria

32

Matthew Chidozie Ogwu and Moses Edwin Osawaru

Contents

Abstract

Through an ethnomedicinal survey, this case study presents a list of medicinal plants used for male reproductive health care by the Benin people of Southern Nigeria and their associated ethnobotanical knowledge and practices. Information was obtained from randomly selected traditional healers, tradomedical practitioners, and native midwives from the main tribal groups within the cities using a semistructured open-ended questionnaire. A total of 30 medicinal plants belonging to 22 plant families were identified, documented including *Alchornea cordifolia, Aloe vera*, *Bambusa vulgaris, Calophyllum inophyllum, Cananga odorata, Carica papaya, Cassia alata*, *Cassia mimosoides*, *Chasmanthera dependens, Cissampelos mucronata, Cissus populnea, Citrullus vulgaris, Citrus*

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aurantifolia, *Combretum racemosum, Cyathula prostrata*, *Cynodon dactylon, Diodia scandens, Musa paradisiaca*, *Nauclea latifolia, Newbouldia laevis*, *Parkia biglobosa*, *Plumeria rubra, Rauwolfia vomitoria, Sansevieria trifasciata, Secamone afzelii, Sida acuta, Solanum nigrum, Tectona grandis, Trianthema portulacastrum,* and *Zea mays*. These plants contribute to the cultural well-being of indigenous people and are used in the treatment of venereal diseases like gonorrhea and syphilis, impotency, genito-urethra disorders, cleansing of the genital tracks, boosting sexual appetite and performance (as an aphrodisiac), boosting sperm counts to address pre-testicular, lifestyle and post-testicular causes and factors of male infertility. The dominant plant families are Apocynaceae and Poaceae with three plant representatives each and followed by Rubiaceae, Menispermaceae, and Fabaceae with two representatives each. Together these dominant plant families account for 41.4% of plants encountered in the study. The parts of the plant used in the improvement of male fertility by the Benin people of Southern Nigeria include leaves, fruits, shoot and root, flowers, leaves and root, root and flowers, seeds, stem, whole plant, root and bark, and bark. The medicinal plants documented in this chapter would benefit from research and technological innovations. The vast knowledge and practices associated with medicinal plants need to be acknowledged and conserved not only through sustained local but also global efforts. The chapter contributes toward understanding, promotion, documentation, and safeguarding of indigenous knowledge, practices, and plant resources used in addressing male reproductive issues.

Keywords

Infertile men · Sperm count · Venereal disease · Traditional medicine · Benin people

1 Introduction

Traditional medicine is the earliest form and the only health care means available to many Nigerians. This medical approach is used to treat various communicable and noncommunicable human diseases/ailments such as cough, dysentery, insomnia, cardiovascular diseases, mental illness, and infertility. Male infertility is the inability of a man to impregnate a fertile female over a minimum period of 6 months to 1 year [39, 46]. Some reversible and irreversible underlying conditions that may cause male infertility include reduced sperm motility, semen abnormalities, genital tract infection, high scrotal temperature (varicocele) and testicular disorder, excess alcohol and drug use (like anabolic steroids, ranitidine, antidepressant, and cimetidine) and abuse of illegal drugs, disturbance to the endocrine, systemic disease, genetic abnormalities, obesity, immunological factors, etc. [27, 31–33, 75]. Recently, Turner et al. [78] described male infertility as a women's and societal health issue because of stigma, unequal burden-sharing, and lack of assisted reproductive technology for diagnosis.

Male infertility has resulted in so many breakdowns in marriages across the globe. This reproductive health issue is increasing globally with little to no concerted effort [27, 72]. This may not be unconnected to the poorly understood mechanisms that are linked to male infertility like environmental, psychological, genetical, and biochemical mechanisms [8, 9]. The major means for diagnosing male infertility is through semen characterization to detect abnormality in quantity, appearance, quality, vigor or motility, and concentration. Over the last 50 years, evidence has shown that sperm count has been on the decrease as a consequence of increases in male infertility. Infertile men typically fail to impregnate fertile females due to a lack of enough or deformed sperm with excess abnormal morphology. Also, other factors such as psychological factors, which include stress, depression, guilt, and low self-esteem generally, contribute to male infertility.

This chapter aims to produce an inventory of the plants used by the residents of Benin City Nigeria for addressing male reproductive and sexual health conditions. A focused approach was adopted and carried out along with a nonexperimental validation of the plants used through a literature review of the phytochemicals and pharmacological information supporting the medicinal activities. This chapter documents the traditional medicinal practices for improving male fertility as well as in the treatment and management of male fertility issues and enumeration of how these plants are used (i.e., either alone or in combination).

2 Global Infertility and Infertility in Men

A global pandemic caused by infertility will have socioeconomic and developmental consequences besides individual and public health impacts. Infertility has been linked to individual and communal psychological, physical, spiritual, economic, and sociocultural decline mainly due to the distress, depression, and anxiety of being infertile [73, 80]. Infertility is a global health, socioeconomic, and development issue with no standardized reporting format. The problem threatens not only physical health but can lead to a serious impact on the mental and social well-being of infertile couples. At the moment, developing countries are disproportionately affected by infertility and cases continue to show an upward trend due to insufficient fertility care, caregivers, and facilities in low-income countries as well as effectiveness and affordability where available [54, 55]. On the other hand, Borumandnia et al. [12] reported a downward trajectory in infertility in high-income and developed countries due to a low tendency of reproduction and a high number of infertility care facilities. Generally, about 40–45% of infertility in men is due to the “male factor” i.e., alteration or decline of sperm concentration, motility, or morphology [31]. The concept of male factor is hinged on sperm and sperm quality, which is typically used as a surrogate measure of male fecundity. It is estimated to affect about 15% of couples or around 48.5 million couples worldwide with males solely responsible for 20–30% and 50% of cases, respectively [2]. Infertility is considered lower in men than in women

Infertility may be primary (when pregnancy has never been achieved) or secondary (when at least one prior pregnancy has been achieved). Secondary infertility is the most prevalent form in women due to reproductive tract infections whereas primary infertility is more common in men depending on several complex factors [18, 37, 79]. Male infertility may be linked to:

1. Pre-testicular causes and factors
2. Lifestyle factors and causes
3. Post-testicular causes and factors

Pre-Testicular Causes and Factors. These are conditions that impede the production and transportation of sperm and include

- Hypogonadotropin hypogonadism (low testosterone).
- Undiagnosed and untreated coeliac disease.
- Medications (particularly chemotherapies).
- Severe injury or diseases like diabetes, HIV AIDS, thyroid-linked diseases, Cushing syndrome, heart attack, liver or kidney failure, impaired sperm production, and chronic anemia.
- Genetic abnormalities such as Robertsonian translocation.
- Sexually transmitted diseases that scar the male reproductive system, impair functions, and block sperm passage like repeated *chlamydia trachomatis,* and gonorrhea infections. Human papillomavirus (genital warts).
- Poor nutrition affects sperm quality and quantity.
- Environmental factors like occupational or long-term, intensive exposure to chemicals and toxins (e.g., agrochemicals like herbicides and pesticides, bisphenol A, phthalates, and organochlorines and heavy metals like lead, cadmium, or arsenic) that reduce sperm count and quality, testicular functions, or alters the endocrine systems.
- Varicoceles (i.e., the enlargement of testicular veins which transports oxygen-depleted veins out of the scrotum resulting in the inefficient circulation of blood out of the organ), which occurs in about 15% of men and around 40% of infertile men and results in increased testicular temperature, which affects sperm production, movement, and shape [32, 33].
- Aging can reduce the count, motility, and genetic quality of sperm. This is a gradual process of male fertility decline.

Lifestyle Factors and Causes. These are factors that cause physical and emotional stress, which have the potential to temporarily reduce male fertility especially sperm count and quality and erectile dysfunction. Some issues that have been linked to low sperm count and quality like:

- Testicular overheating due to high fevers, saunas, and hot tubs.
- Excessive use of drugs like cocaine, heroin, and marijuana.
- Heavy alcohol consumption.

- Excessive cigarette smoking.
- Prolonged bicycling exposes the perineum (i.e., the region between the scrotum and the anus) to extreme shocks, vibrations, and scrotal injuries.

Post-Testicular Causes and Factors. These are conditions that decrease male fertility after testicular sperm production like

- Genital tract and ejaculation defects (premature, delayed, or retrograde ejaculation).
- Obstruction of the seminal ducts or ejaculatory duct obstruction.
- Vas deferens obstruction or lack of vas deferens.
- Infection like prostatitis.
- Retrograde ejaculation.
- Ejaculatory duct obstruction conditions like aspermia, prostatic pain, oligo asthenospermia and azoospermia, painful ejaculation, and hematospermia [34].
- Hypospadias – urethra not located at the tip of the penis.
- Erectile dysfunction or impotence.

3 Plants Used in Male Reproductive Health Care: Case Study of Benin City, Southern Nigeria

A survey was carried out in the six Local Government Areas that make up Benin City, Edo State (6.34° N and 5.60° E; Fig. 1) and is occupied by Benin people of Southern Nigeria to document medicinal plants used in the treatment and management of male fertility issues. Benin City lies within the humid tropical rainforest zone of Nigeria but its vegetation has been heavily altered by anthropogenic activities and human population growth and is currently a mosaic of secondary forest [17]. The climate is monsoonal with two distinct seasons (dry and rainy season) with high rainfall (2000–3000 mm), temperature (20–40 °C), and average atmospheric humidity of 28% and with radiation of 1600 h per year [56, 57]. A detailed description of the study area including soil characteristics can be found in Osawaru et al. [60–66], Osawaru and Ogwu [58, 59], Ogwu and Osawaru [43].

The survey to document traditional medicine practices and higher plants used for male reproductive care by Benin people was conducted between July and November 2021 using survey sheets for information collection, a digital camera, an audio recording device for interviews, and a mobile phone-enabled position marker. The survey focused on randomly selected and locally identifiable professional herbalists, traditional healers, native midwives, and herbal remedy sellers using semistructured, open-ended questionnaires in English and/or preferred native languages. The randomly selected respondents and interviewees are duly registered with all the relevant authorities. The questionnaires were administered to the key informants (full-time traditional healers, part-time practitioners, elderly people who know the traditional values of the plants, relatives, and acquaintances who at one time or the other used some of these plants for male reproductive health and fertility). The respondents

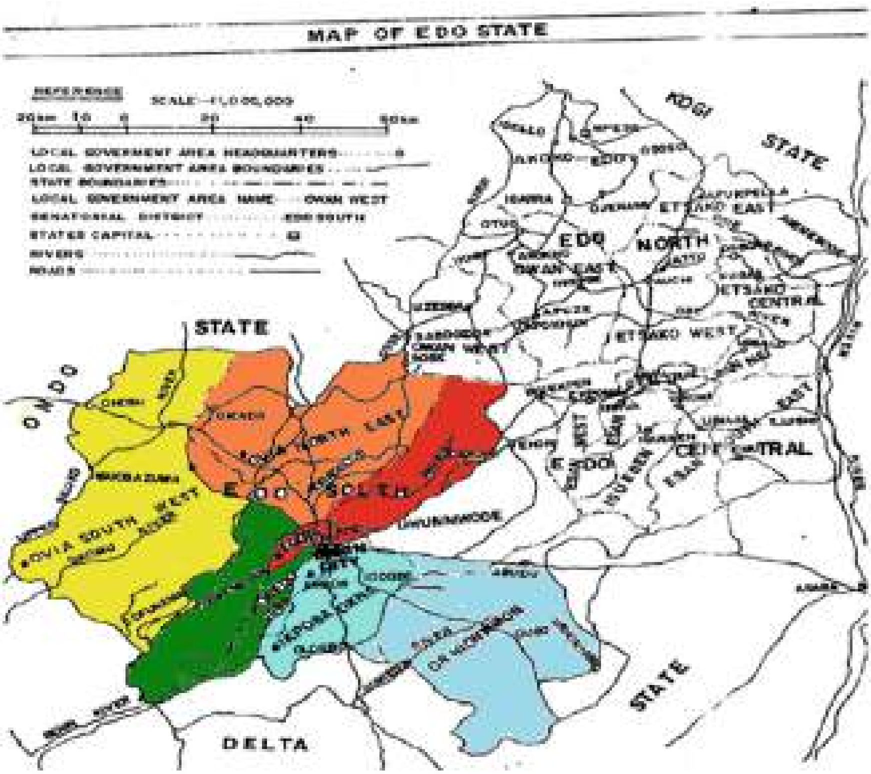


Fig. 1 Map of Edo State with the sampling area highlighted

animated a field survey to identify and collect plants and plant parts used for the treatment and management of male infertility and reproductive care. Specific questions like the botanical names, local names, plant parts used, dosage, method of preparation, application, and duration of treatment were asked during the oral interview, and the information supplied by these people was recorded. Collections were made from homestead farms, distant farms, and different forests present in the study area.

Data collected include the sociodemographic characteristics of the respondents (gender, age, marital status, educational qualification, religion, tribe/ethnic group, occupation, name of village, and languages spoken), and information on the plants used for the treatment and management of male infertility and preparation of herbal remedies for reproductive care. Data collected were processed and analyzed using Windows Microsoft Excel 2016 version software, which was also used to draw charts to understand inherent patterns. The therapeutic importance of each identified species was also recorded. Based on the knowledge of the informants, a total of 30 medicinal plants used for the treatment and management of male infertility were identified by a taxonomist at the University of Benin.

3.1 Medicinal Plants Used by Benin People for the Treatment and Management of Male Infertility and Reproductive Healthcare

From the survey, a total of 30 medicinal plants belonging to 22 plant families were identified, documented, and collected as used by Benin people for the treatment and improvement of male fertility (Tables 1 and 2). The plants included *Alchornea cordifolia, Aloe vera*, *Bambusa vulgaris, Calophyllum inophyllum, Cananga odorata, Carica papaya, Cassia alata*, *Cassia mimosoides*, *Chasmanthera dependens, Cissampelos mucronata, Cissus populnea, Citrullus vulgaris, Citrus aurantifolia*, *Combretum racemosum, Cyathula prostrata*, *Cynodon dactylon, Diodia scandens, Musa paradisiaca*, *Nauclea latifolia, Newbouldia laevis*, *Parkia biglobosa*, *Plumeria rubra, Rauwolfia vomitoria, Sansevieria trifasciata, Secamone afzelii, Sida acuta, Solanum nigrum, Tectona grandis, Trianthema portulacastrum,* and *Zea mays* (Table 1). They are used for treatment of venereal diseases like gonorrhea and syphillis, impotency, genito-urethra disorders, cleansing of the genital tracks, boosting sexual appetite and performance (as an aphrodisiac), and boosting sperm counts (Table 1). Hence, the plants can be used to address pre-testicular, lifestyle, and post-testicular causes and factors of male infertility in line with the reports of Roozbeh et al. [74] and Boroujeni et al. [11]. For greater contribution in addressing global male infertility issues, the active agents in these plants require a collaborative effort to isolate them as a prodrug candidate and elucidation at the molecular, cellular, and clinical levels to understand their mechanism of action [1, 28]. The plant families encountered in this study included Amaranthaceae, Annonaceae, Apocynaceae, Asparagaceae, Bignoniaceae, Caesalpinioideae, Calophyllaceae, Caricaceae, Combretaceae, Cucurbitaceae, Euphorbiaceae, Fabaceae, Lamiaceae, Liliaceae, Malvaceae, Menispermaceae, Musaceae, Poaceae, Rubiaceae, Rutaceae, Solanaceae, and Vitaceae (Table 2). The dominant plant families are Apocynaceae and Poaceae with three plant representatives each and followed by Rubiaceae, Menispermaceae, and Fabaceae with two representatives each (Table 2). Together these dominant plant families account for 41.4% of plants encountered in the study.

Plants possess medicinal value in addition to their nutritional importance [40–42]. From time immemorial, plants and plant extracts have been used medicinally in all parts of Nigeria especially as fertility agents without producing apparent toxic effects [39, 53]. This is backed by the report of D'Cruz et al. [16] wherein they opined that plant medicine is safe for addressing male reproductive dysfunction and other illnesses such as cancer and diabetes. Generally, there is a paucity of data and information on the safety of herbal medicine products and practices compared to modern medicine [22, 38, 69]. Where data exist, variations exist in terms of the significant toxic effect of medicinal plants on the reproductive system. The work of Nozhat et al. [38] using spearmints (Mentha spicata) suggests the absence of significant toxic effects on the reproductive system. However, toxic levels of heavy metals like cadmium, lead, arsenic, mercury, and copper have been reported in herbal medicines with significant geographical correlations in their concentrations

Table 1 Commonly used plants in the improvement of male fertility by the Benin people of Southern Nigeria

Scientific name	Family name	Common/Benin name	Parts used	Usefulness
Alchornea cordifolia	Euphorbiaceae	Christmas bush/Uwonwen	Leaves	To treat gonorrhea and other venereal diseases
Aloe vera	*Asphodelaceae*	Aloe vera/Aloe	Leaves	Treatment and management of impotency
Bambusa vulgaris	Poaceae	Bamboo/Ekpokoro	Young shoot	Treatment and management of gonorrhea and other venereal diseases
Calophyllum inophyllum	Calophyllaceae	Alexandrian laurel ball tree/ Ebe-ori	The kernel	Used in the treatment and management of gonorrhea and genito-urethra disorders
Cananga odorata	Annonaceae	Cananga tree\Erhan-uwa	Flower	Used for the treatment and management of impotency
Carica papaya	Caricaceae	Paw paw/Ughoro	Leaves and roots (used together)	For the treatment and management of venereal diseases and cleansing of the genital tracks
Cassia alata	Caesalpiniodeae	Candle bush/Asunwon	Root and flower	For the treatment and management of venereal diseases (specifically syphilis)
Cassia mimosoides	Fabaceae		Seed	It is used as an and to boost sexual performance
Chasmanthera dependens	Menispermaceae	Climbing plant/Aghoghon-ewaki	Root	A decoction made from it is taken to treat and manage gonorrhea.
Cissampelos mucronata	Menispermaceae	Abuta/Ewaki-nokhua	Leaves and roots	Leaves and roots are made into herbal juice and used as an aphrodisiac to boost sexual performance
Cissus populnea	Vitaceae	Bush mango/Iri-ogbono	Stem	Used for the treatment and management of venereal diseases
Citrullus vulgaris	Cucurbitaceae	Watermelon/Watermelon	Fruit	Fruits are eaten in large quantities as a remedy for urinary and genital tract problems arising from gravel and stone in the bladder
Citrus aurantifolia	Rutaceae	Lime/*Alimo-negiere*	Leaf	It is used to treat gonorrhea and urine retention
Combretum racemosum	Combretaceae	Christmas rose/Akoso-namwen	Root	It is used as an aphrodisiac

Cyathula prostrata	Amaranthaceae	Prostrate pasture weed/Ebe-ekperhon	Whole plant	Used for the treatment of gonorrhea
Cynodon dactylon	Poaceae	Stubborn grass/Iruvvba-ebo	Root	It is used in the treatment of secondary syphilis and irritated bladder
Diodia scandens	Rubiaceae	Not registered/Iyekegul	Leaf	It is used in treating venereal diseases
Musa paradisiaca	Musaceae	Plantain/Oghede	Root	Root juice is used to treat gonorrhea and urine retention; also, it is used in treating some other venereal diseases
Nauclea latifolia	Rubiaceae	African peach/Ero	Root and leaf	Roots and leaves are taken to improve male fertility
Newbouldia laevis	Bignoniaceae	Boundary tree/Ikhinmwin	Root with the leaves	It is used as an aphrodisiac and to boost sexual activities
Parkia biglobosa	Fabaceae	African locust bean/Ugbore	Seed	The seeds are used as an aphrodisiac
Plumeria rubra	Apocenaceae	Temple tree/Obadan-atori	The latex	Used for the treatment and management of venereal diseases
Rauwolfia vomitoria	Apocynaceae	The poison devil's pepper/Akata	The root bark	It is used as an aphrodisiac and to improve sexual performance
Sansevieria trifasciata	Asparagaceae	*Sansevieriatrifasciata*Prain/Asparagaceae	Leaf	It is used in the treatment of sexual weaknesses and to improve sexual performance
Secamone afzelii	Apocynaceae	Secamone/Iri -egile	Whole plant	It is used to cure venereal diseases like gonorrhea
Sida acuta	*Malvaceae*	Wire weed/Aranrenvbi	Seed	Used for the treatment and management of gonorrhea
Solanum nigrum	*Solanaceae*	European black nightshade/Ebeakpe	Leaf	The leaf juice is used for the treatment of gonorrhea
Tectona grandis	Lamiaceae	Teak/Ovbiakke	Bark	It is used as an aphrodisiac agent
Trianthema portulacastrum	*Aizoaceae*	Desert horse purslane/Eyen	Whole plant	Used for the treatment and management of gonorrhea and other sexual infections
Zea mays	Poaceae	Corn/oka	Seed	Roasted or boiled seeds are taken as an aphrodisiac to improve sexual performance and boost sperm count

Table 2 Distribution of plant species encountered among higher plant family

Higher plant family	Number of plant species	Relative percentage
Amaranthaceae	1	3.45
Annonaceae	1	3.45
Apocynaceae	3	10.35
Asparagaceae	1	3.45
Bignoniaceae	1	3.45
Caesalpinioideae	1	3.45
Calophyllaceae	1	3.45
Caricaceae	1	3.45
Combretaceae	1	3.45
Cucurbitaceae	1	3.45
Euphorbiaceae	1	3.45
Fabaceae	2	6.90
Lamiaceae	1	3.45
Liliaceae	1	3.45
Malvaceae	1	3.45
Menispermaceae	2	6.90
Musaceae	1	3.45
Poaceae	3	10.35
Rubiaceae	2	6.90
Rutaceae	1	3.45
Solanaceae	1	3.45
Vitaceae	1	3.45

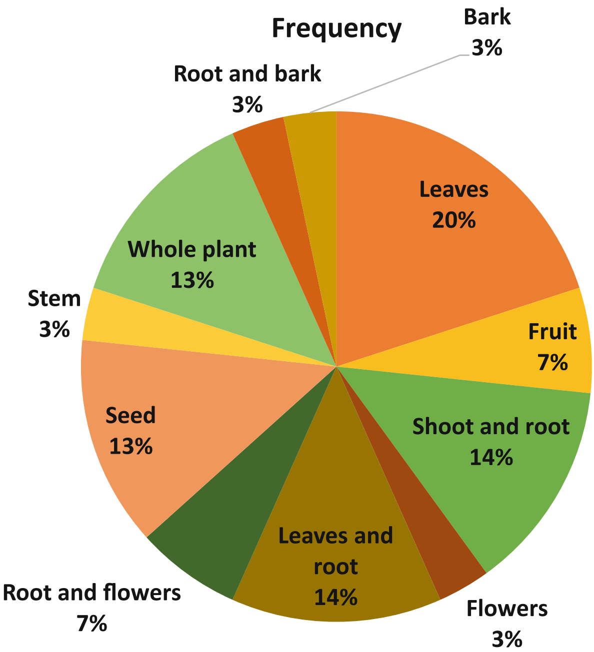
[7, 25, 36]. The use of these plants and herbal medicine products that contain their extract poses long-term health (cancer) risks [53].

In recent times, the use of plant and plant extracts for male fertility healthcare is on the increase because of the shifting attention from synthetic drugs to natural plant products [44, 45, 47–49]. Some of these plants are used to address sexual hormonal imbalances in males and females. Phytochemical screening of medicinal plants supports the presence of bioactive materials that can affect the regulation of conception and reproduction [20, 81]. Idu and Onyibe [26] documented a total of 300 plants belonging to 274 genera and from 77 families for the treatment of various illnesses, and among those plants listed includes those used in the promotion of male fertility among the people of Edo state. Trial and error play a huge role in the initial discovery and use of these medicinal plants for the treatment and management of male fertility issues. For instance, one or more plants are first used in the treatment or management of a single ailment (e.g., gonorrhea and some other venereal diseases) and then if the patient does not get relief from the plant within a specified period, another plant is tried, secondly, if the desired plant cannot be found on a specified time, another plant is taken as a substitute. The parts of the plant used in the improvement of male fertility by the Benin people of Southern

Table 3 Frequency of plant parts used in the improvement of male fertility by the Benin people of Southern Nigeria

Plant part	Frequency	Relative percentage
Leaves	6	20
Fruits	2	6.67
Shoot and root	4	13.33
Flowers	1	3.33
Leaves and root	4	13.33
Root and flowers	2	6.67
Seeds	4	13.33
Stem	1	3.33
Whole plant	4	13.33
Root and bark	1	3.33
Bark	1	3.33

Fig. 2 Percentage of plant parts used



Nigeria include leaves, fruits, shoot and root, flowers, leaves and root, root and flowers, seeds, stem, whole plant, root and bark, and bark but the most commonly used parts are the leaves (20%), seeds (13.33%), whole plant (13.33%), leaves and root (13.33%), and shoot and root (13.33%) (Table 3 and Fig. 2). This finding is supported by the reports of Jaradat et al. [28] wherein seeds, roots, and leaves were used for infertility treatment.

3.2 Enumeration of Medicinal Plants Used by Benin People for the Treatment and Management of Male Infertility and Reproductive Healthcare

3.2.1 *Alchornia cordifolia* (Common Name – Christmas Bush and Native Name – Uwonwen)

Alchornea cordifolia, is an evergreen dioecious shrub or small tree found almost throughout tropical Africa and has fleshy seeds (Fig. 3).

PLANT PART USED: Leaves.

FOLK USE: To treat gonorrhea and other venereal diseases and is supported by the work of Boniface et al. [10]. The phytochemical constituent of the plant has significant bioactivity and a high therapeutic value.

PREPARATION OF REMEDY: It is either macerated or chewed.

DOSAGE: Not specified.

3.2.2 *Aloe vera* (*A. barbadense*) (Common Name Aloe Vera)

The plant is a cactus-like succulent perennial with strong and fibrous roots and numerous, persistent, fleshy leaves, proceeding from the upper part of the root, tapering, thick, and usually beset at the edges with spiny teeth. They can be easily cultivated especially in tropical and subtropical regions of the world. Many of the species are branching and can grow up to about 30–60 feet in height with stems as much as 10 feet in circumference (Fig. 4).

Fig. 3 *Alchornea cordifolia*

Fig. 4 *Aloe vera*

PLANT PART USED: Leaf.

FOLK USE: It is used to treat and manage impotency and to improve sperm count, motility, and other characteristics. The plant can increase testosterone levels and sperm quality and quantity [23, 52].

PREPARATION: The leaf is made into juice and taken with water.

DOSAGE: To be taken twice daily.

3.2.3 *Bambusa vulgaris* (Common Name – Bamboo, and Native Name – Ekpokoro)

The plant forms moderately loose clumps and has lemon-yellow culms (stem) with dark green leaves. The culms are not entirely straight, not easy to split, inflexible, thick-walled, and strong. The densely tufted stems can grow up to 20 m high and 10 cm thick across the base. They can be cultivated and found in the wild (Fig. 5).

PLANT PART USED: Young tender shoot.

FOLK USE: Taken to cure gonorrhea and other sexually transmitted diseases. It is used to improve reproductive vigor.

PREPARATION OF REMEDY: Shoot extract is soaked in alcohol for several days.

DOSAGE: Two shots twice a day.

3.2.4 *Calophyllum inophyllum* (Common Name – Alexandrian Laurel Ball Tree, and Native Name – Ebe-Ori)

Calophyllum inophyllum is a large evergreen plant that can grow up to 60 m tall and found in tropical regions of the world. It is a low-branching and slow-growing tree

Fig. 5 *Bambusa vulgaris*

Fig. 6 *Calophyllum inophyllum*

with a broad and irregular crown glossy oval-shaped leaves, scented flowers, and round green fruits. It usually reaches 8–20 m in height (Fig. 6).

PLANT PART USED: The fruit/kernel. Oil is produced from the kernel for use in male reproductive health.

FOLK USE: Used in the treatment of gonorrhea and genito-urethra disorders like swelling and wounds. The kernels are naturally antiseptic and anti-inflammatory [13, 70].

PREPARATION FOR USE: The kernel oil is gotten.
DOSAGE: One spoon full three times daily.

3.2.5 *Cananga odorata* (Common Name – Canaga Tree, Native Name – Erhan-Uwa)

Cananga is a fast-growing tropical tree that originates in Southeast Asia (Indonesia, Malaysia, and the Philippines) and can grow above 5 m per year. The evergreen leaves are smooth and glossy, oval, pointed, and with wavy margins and 13–20 cm long. The scented flower is drooping, long-stalked with six narrow, greenish-yellow petals (Fig. 7).

PLANT PART USED: Flowers.
FOLK USE: Cures impotency. It may be used as a male contraceptive because of its anti-spermatogenic effects [19].
PREPARATION OF REMEDY: Decoction prepared from the flowers.
DOSAGE: Two shots twice daily (morning and evening).

3.2.6 *Carica papaya* (Common Name – Paw Paw or Papaya and Native Name – Ughoro)

Papaya has a single stem growing and can reach 10 m tall, with spirally arranged leaves at the top of the trunk. The lower trunk is conspicuously scarred to mark where previous leaves and fruits were borne. The large leaves may reach 70 cm in diameter and contain a significant amount of antioxidants (Fig. 8).

Fig. 7 *Cananga odorata*

Fig. 8 *Carica papaya*

PLANT PART USED: The leaves and roots are used together.

FOLK USE: Treatment of gonorrhea and other venereal diseases. The leaves can be used as an aphrodisiac to boost libido as well as functional sterility as a contraceptive [15, 30, 35].

PREPARATION OF REMEDY: Infusion of the green leaves and roots.

DOSAGE: Three times daily (half a glass cup).

3.2.7 *Cassia alata* (Common Name – Candle Bush And Native Name – Asunwon)

Cassia alata is a shrub that can reach 4 m in height with pinnate leaves that are between 50 and 80 cm long. The leaves close in the dark while the yellow inflorescence looks like a candle. The fruits are borne in a straight pod that is approximately 25 cm long while the seeds are dispersed by water and animals (Fig. 9).

PLANT PART USED: The root and flower.

FOLK USE: For the treatment and management of venereal diseases (like syphilis). It also has hormone-boosting functions [50, 51, 77].

PREPARATION POF REMEDY: The leaves and roots are made into powder and mixed with lime.

DOSAGE: Two shots are taken twice daily.

3.2.8 *Cassia mimosoides* (Common Name – Senna Tea)

It is a low-lying to an upright diffuse shrub that can grow up to 1.5 m or more in height. The pinnate leaves are 7–10 cm long, with 40–60 pairs of narrow leaflets, and a solitary, sessile glind on the ranchis below the leaflets. Flowers grow one or two

Fig. 9 *Cassia alata*

Fig. 10 *Cassia mimosoides*

together in the axils of the leaves. Shining, small, and yellow. Pods are strap-shaped, flat, and about 5 cm long, containing rhomboid, dark-brown seeds (Fig. 10).

PLANT PART USED: Seed.

FOLK USE: Aphrodisiac to boost sexual appetite and performance and increases fertility by improving sperm characteristics.

PREPARATION OF REMEDY: Seed extract is made.
DOSAGE: Taken twice daily (half of a glass cup).

3.2.9 *Chasmanthera dependens* (Common Name – Climbing Plant And Native Name – Aghoghon-Ewaki)

The plant is a woody climber (liana) in dense evergreen or semi-deciduous lowland forest and riparian woodland of the drier Guinean zone from Sierra Leone to Nigeria, and Cameroun across Ethiopia and Somalia. They may be propagated through their seeds. The roots and leaves are considered edible (Fig. 11).

PLANT PART USED: Root.
PREPARATION OF REMEDY: Decoction prepared with root or root extracts and soaked in alcohol.
FOLK USE: The decoction is taken to treat venereal diseases like gonorrhea. It has fertility-enhancing functions and androgenic activities [21, 71].
DOSAGE: One shot is taken three times per day.

3.2.10 *Cissampelos mucronata* (Common Name – Abuta and Native Name – Ewaki-Nokhua)

Cissampelo smacronata is a liana with woody rootstock in tropical Africa. The leaves are subpeltate, ovate-cordate, and covered in grayish indumentum, mucronate at the apex, flower axillary, or in false racemes (Fig. 12).

PLANT PART USED: The leaves and roots are used.
PREPARATION OF REMEDY: The leaves and roots are made into juice for the preparation of bitter tonic.

Fig. 11 *Chasmanthera dependens*

Fig. 12 *Cissampelos macronata*

FOLK USE: For the treatment of genital-urinary diseases.
DOSAGE: Half of a glass cup twice daily.

3.2.11 *Cissus populnea* (Common Name – Bush Mango and Native Name – Iri-Ogbono)

Cissus populnea is a strong woody liana, 8–10 m long, 7 cm in diameter, dispersed generally throughout the region from the coast to the Soudanian and Sahelian woodland, of Nigeria (Fig. 13).

PLANT PART USED: Stem.
FOLK USE: Treatment of venereal diseases.
PREPARATION OF REMEDY: The decoction of the stem is mixed with the decoction prepared with *Alchornea cordifolia* root. It improves sperm quality and quantity and may be used as an aphrodisiac to enhance sexual performance [5, 50].
DOSAGE: One shot is taken twice daily (morning and evening).

3.2.12 *Citrullus vulgaris* (Common Name – Water Melon)

Citrullus vulgaris is a vine-like flowering plant originally from the Southern parts of Africa but is commonly cultivated in Central and Northern Nigeria. it is a large, sprawling oval-shaped annual plant with coarse, hairy pinnately lobed leaves and white to yellow flowers that are grown mainly for the edible fruit, which is considered a berry, botanically. The fruit is a smooth hard rind, usually green with light to dark green stripes or yellow spots, and a juicy, sweet interior flesh, usually deep red to pink, but sometimes orange, yellow, or white, with many seeds (Fig. 14).

Fig. 13 *Cissus populnea*

Fig. 14 *Citrullus vulgaris*

PLANT PART USED: Fruit.

FOLK USE: The fruit is eaten in large quantities because the pulp is considered a remedy for the urinogenital problems arising from gravel and stone in the bladder. It is considered a genital cleanser. It is believed to enhance sperm motility and kinetics [29].

PREPARATION OF REMEDY: No definite preparation. It depends on the user and is often preferred fresh or made into a juice.
DOSAGE: Not specified.

3.2.13 *Citrus aurantifolia* (Common Name – Lime and Native Name – Alimo-Negiere)

Citrus aurantifolia is a shrubby tree with many thorns. It can be grown indoors and the leaves are harvested regularly for use in the treatment and management of male reproductive issues. The trunk rarely grows straight and has many branches. The leaves are ovate and resemble orange leaves. The flowers are about 2.5 cm in diameter and are yellow-white with a light purple tinge on the margins. Flowers and fruit may be seen on the plant throughout the year, but are mostly abundant in the early rainy season in Nigeria (Fig. 15).

PLANT PART USED: Leaves.
FOLK USE: Taken to treat gonorrhea and urine retention problems. The high amount of antioxidants in the plant has a positive effect on male fertility [3].
PREPARATION OF REMEDY: Hot decoction of the leaves is prepared with water.
DOSAGE: Half a glass cup is taken three times daily.

3.2.14 *Combretum racemosum* (Common Name – Christmas Rose and Native Name – Akoso-Namwen)

It is a shrubby liane with opposite or subopposite leaves. It is common in West Africa and other wet tropical biomes (Fig. 16).

Fig. 15 *Citrus aurantifolia*

Fig. 16 *Combretum racemosum*

PLANT PART USED: The root.

FOLK USE: It is used as an aphrodisiac to improve sexual performance and appetite. Its root is known to help with infections.

PREPARATION OF REMEDY: The root is soaked in alcohol and the extract is obtained.

DOSAGE: One shot is taken three times per day.

3.2.15 *Cyathula prostata* (Common Name – Prostrate Pasture Weed and Native Name – Ebe-Ekperhon)

It is an annual to perennial erect or ascending herb that may grow up to 50 cm in height. The roots appear at the nodes while the stem is obtusely quadrangular, thickened above the nodes, often tinged with red, and covered with patent and fine hairs. The leaves are oppositely arranged, simple, and rhomboid obovate to rhomboid-oblong form and vary in size between. The inflorescence is an erect elongated raceme, terminal and in highest leaf-axis and dull pale green, hairless within and eternally clothed with patent long and white hairs. The fruit is an ellipsoid utricle, thin-walled, hairless, one-seeded, and surrounded by stiff perianth (Fig. 17).

PLANT PART USED: Whole plant.

FOLK USE: Used for the treatment of gonorrhea and other venereal diseases. It is also taken to improve sexual performance and sperm production [4]. Male sexual problems connected to obesity are treated or managed using the plant [24].

PREPARATION OF REMEDY: Decoction of the whole plant is made using water.

DOSAGE: Half a glass cup is taken three times daily.

Fig. 17 *Cyathula prostata*

3.2.16 *Cynodon dactylon* (Common Name – Stubborn Grass, Bermuda Grass, and Native Name – Iruvvba-Ebo)

The blades are gray-green and short, usually 2–5 cm long with rough edges. It is a low-statured and creeping perennial found in almost all tropical and subtropical regions. The erect stems can grow up to 30 cm tall and are slightly flattened, and often tinged purple. The seeds are produced in a cluster of six spikes at the top of the stem. It has a deep root system. The grass creeps along the ground and roots wherever a node touches the ground, forming a dense mat (Fig. 18).

PLANT PART USED: Root.

FOLK USE: It is used in the treatment of secondary syphilis and irritated bladder. It is effective in remediating stress-induced male sexual dysfunction and is believed to be a potent aphrodisiac [14].

PREPARATION OF REMEDY: Root decoction is made and soaked in water or alcohol.

DOSAGE: Two tea cups are taken per day.

3.2.17 *Diodia scandens* (Common Name – Sarmentosaswart and Native Name – Iyekegul)

Straggling, scrambling, or procumbent, often with many lateral branches from the main stem and compound leaves. The leaf blades are yellowish green, elliptic, apex, narrow to the base, and scabrid above with dense very short to longer tubercle-based hairs. Leaves are scabrid and jungle-thicket-like (Fig. 19).

Fig. 18 *Cynodon dactylon*

Fig. 19 *Diodia scandens*

PLANT PART USED: Leaf.
FOLK USE: it is used in treating venereal and urinogenital diseases.
PREPARATION OF REMEDY: Leaf decoction is prepared.
DOSAGE: Not specified.

3.2.18 *Musa paradisiaca* (Common Name – Plantain and Native Name – Oghede)

Musa paradisiaca is a large and fast-growing evergreen perennial monocotyledonous plant that can reach 9 m tall when mature. The above-ground part of the plant is a false stem or pseudostem and consists of leaves that are huge and paddle-like. Each pseudostem can produce a single flowering stem. After fruiting, the pseudostem dies, but an offshoot may develop from the base of the plant (Fig. 20).

PLANT PART USED: Root.

FOLK USE: The root juice is used to treat gonorrhea and urine retention; also, it is used in treating some other venereal diseases. It has an enhancing effect on male reproductive functions and semen quality [6].

PREPARATION OF REMEDY: Juice is made from the root.

DOSAGE: Three shots should be taken daily (morning, afternoon, and evening).

3.2.19 *Nauclea latifolia* (Common Name – African Peach and Native Name – Ero)

Nauclea latifolea is a deciduous shrub or tree with an open canopy, usually branching from low down the bole, and can reach 30 m in height depending on soil and moisture conditions. The edible fruit is gathered from the wild for local use (Fig. 21).

FOLK USE: For general male fertility and improving sexual performance.

PLANT PART USED: Root and leaves.

Fig. 20 *Musa paradisiaca*

Fig. 21 *Nauclea latifolia*

PREPARATION OF REMEDY: It is prepared by making a decoction from the root and the leaf and is typically soaked in alcohol.

DOSAGE: One glass cup is taken twice daily.

3.2.20 *Newbouldia laevis* (Common Name – Boundary Tree and Native Name – Ikhinmwin)

Newbouldia laevis is a fast-growing evergreen shrub or small tree. It only reaches a height of 8 m in the west within its range but can attain a height of up to 20 m in the east. The bole can be up to 90 cm in diameter, but it is usually less (Fig. 22).

PLANT PART USED: Root with the leaves.

FOLK USE: It is an aphrodisiac in action. It is also used to address a hormonal imbalance in men for fertility enhancement.

PREPARATION OF REMEDY: Root with leaves soaked in either palm wine or dry gin.

DOSAGE: About half a glass cup three times daily if soaked in alcohol (less than 40% alcohol) or one shot three times daily if soaked in dry gin (about 40% alcohol).

3.2.21 *Parkia biglobosa* (Common Name – African Locust Bean and Native Name – Ugbore)

Parkia biglobosa is a perennial deciduous semi-cultivated tree that can reach 20 m in height or some cases, up to 30 m. It provides a canopy for several plants that grow within its shade. The tree is considered to be fire resistant (heliophyte) and is characterized by a thick dark gray-brown bark. It has a dense spreading umbrella-

Fig. 22 *Newbouldia laevis*

Fig. 23 *Parkia biglobosa*

shaped crown and a cylindrical trunk. The pods are referred to as locust beans and are pinkish in the beginning but later turn dark brown when fully mature. Each pod can contain up to 30 seeds (Fig. 23).

PLANT PART USED: Seeds.
FOLK USE: The seeds are aphrodisiacs and improved sperm quality and quantity.
PREPARATION OF REMEDY: To be chewed.
DOSAGE: Not specified.

Fig. 24 *Plumera rubra*

3.2.22 *Plumeria rubra* (Common Name – Temple Tree and Native Name – Obadan-Atori)

Plumera rubra is a deciduous shrub or small tree that grows up to a height of 2–8 m. It has a thick succulent trunk and sausage-like branches that are covered with a thin gray bark. The branches are brittle and when broken, ooze a white latex that can be irritating to the skin and mucous membranes. The large green leaves can reach 50 cm long and are arranged alternately and clustered at the end of branches. The flowers are terminal, appearing at the end of the branches over the summer (Fig. 24).

PLANT PART USED: The latex from the bark.
FOLK USE: For the treatment of venereal diseases like syphilis and gonorrhea. The plant is believed to have a regulatory function on fertility [76].
PREPARATION OF REMEDY: The latex tapped from the bark.
DOSAGE: A very small dosage of about 0.2 g to 0.3 g is taken and not more than that is recommended for use.

3.2.23 *Rauwolfia vomitoria* (Common Name – Devil's Pepper and Native Name – Akata)

Rauwolfia vomitoria is a shrub or small tree that grows up to 8 m in height all over West Africa. The older part of the plant contains no latex, unlike the younger parts. The branches are whorled and the nodes are enlarged and lumpy. Leaves are in threes, elliptic-acuminate to broadly lanceolate. Flowers are minute and sweet-scented, and branches of the inflorescence are distinctively puberulous with hardy-free corolla lobes. Fruits are fleshy and red (Fig. 25).

Fig. 25 *Rauwolfia vomitoria*

PLANT PART USED: Root bark.

FOLK USE: It is used as an aphrodisiac. It is used to address stress-induced infertility issues in men and to improve fertility.

PREPARATION OF REMEDY: The root back is soaked in gin.

DOSAGE: One shot is taken twice daily.

3.2.24 *Sansevieria trifasciata* (Common Name – Viper's Bowstring Hemp and Native Name – Erhurhue-Ekpen)

The plant is an evergreen perennial plant and forms dense stands that are spread by way of its creeping rhizomes, which are sometimes above ground. Its stiff leaves grow vertically from a basal rosette. Mature leaves are dark green with light gray-green cross-banding and usually range between 70 cm and 90 cm long and 5 cm and 6 cm wide (Fig. 26).

PLANT PART USED: Leaf.

FOLK USE: It is used in the treatment of sexual weakness and is considered an aphrodisiac. Used to maintain sexual rigor and improve performance.

PREPARATION OF REMEDY: Decoction of the leaf with *Piper guineensis.*

DOSAGE: About half a glass cup is taken twice daily.

3.2.25 *Secamone afzelii* (Common Name – Secamone and Native Name – Iri -Egile)

It is a scandent creeping woody shrub or climber, of secondary jungle and savanna thickets, common on unkempt farmland and in boundaries. It has compound leaves (Fig. 27).

Fig. 26 *Sansevieria trifasciata*

Fig. 27 *Secamone afzelii*

PLANT PART USED: Whole plant.
FOLK USE: It is used to cure venereal diseases like gonorrhea and improve fertility.
PREPARATION OF REMEDY: Decoction is prepared using the whole plant.
DOSAGE: The whole plant is used to cook light-boiled soup and is eaten together with the soup.

Fig. 28 *Sida acuta*

3.2.26 *Sida Acuta* (Common Name – Wire Weed and Native Name – Aranrenvbi)

It is an undershrub, with mucilaginous juice, aerial, erect, cylindrical, branched, solid, and green. The leaves are alternate, simple, lanceolate to linear, rarely ovate to oblong, obtuse at the base, acute at the apex, coarsely and remotely serrate, and the petiole is shorter than the blade. The inflorescence is a racemose (Fig. 28).

PLANT PART USED: The seeds.
FOLK USE: It is used for treating venereal diseases like gonorrhea.
PREPARATION OF REMEDY: Infusion of the seed is made.
DOSAGE: Not specified.

3.2.27 *Solanum nigrum* (Common Name – European Black Nightshade and Native Name – Ebeakpe)

It is a common herb or short-lived shrub, found in many wooden areas, as well as disturbed habitats. It can reach a height of 30–120 cm while the leaves average between 4 and 7.5 cm long and about 2 and 5 cm wide and are ovate to heart-shaped, with wavy or large-toothed edges, both surface hairy or hairless. The flowers have petals greenish to whitish (Fig. 29).

PLANT PART USED: Leaves.
FOLK USE: The leaf juice is used for the treatment of gonorrhea and other venereal diseases.
PREPARATION OF REMEDY: Juice prepared with or obtained from the leaves.
DOSAGE: Half of a glass cup is taken twice daily.

Fig. 29 *Solanum nigrum*

Fig. 30 *Tectona grandis*

3.2.28 *Tectona grandis* (Common Name – Teak and Native Name – Ovbiakke)

Teak is a large deciduous tree that measures up to 40 m tall with gray to grayish brown branches. Leaves are ovate-elliptic to ovate, 15–45 cm long by 8–23 cm wide, and are held on robust petioles that are 2–4 cm long. Leaf margins are entire (Fig. 30).

PLANT PART USED: The bark.

FOLK USE: It is used as an aphrodisiac agent to improve sexual appetite and performance.

PREPARATION OF REMEDY: The bark juice is commonly extracted with and soaked in alcohol.

DOSAGE: About half a glass cup is taken daily if extracted with palm wine but one shot three times daily if extracted with gin (about 40% alcohol).

3.2.29 *Trianthema portulacastrum* (Common Name – Desert Horsepurslane and Native Name – Eyen)

It is an annual herb forming a prostrate map or clump with stems up to a meter long. It is green to red, hairless except for small lines of hairs near the leaves, and fleshy. The leaves have small round or oval blades up to 4 cm long borne on short petioles. Solitary flowers occur in leaf axils (Fig. 31).

PLANT PART USED: The whole plant.
FOLK USE: Used for the treatment of gonorrhea.
PREPARATION OF REMEDY: The whole plant is dried and pulverized for use.
DOSAGE: Not specified but it is taken with cereals like pap (ogi).

3.2.30 *Zea Mays* (Common Name – Corn, Maize, and Native Name – Oka)

It is a vigorous annual grass, varying greatly in size according to race and growth conditions, culms, stout, often with prop roots from the lower nodes, many-noded, terminated by inflorescences of the male spikelet, and with one or more female spikelet in the axils of leaves below the tassel, leaf blade expanded, usually drooping, usually green (Fig. 32).

Fig. 31 *Trianthema portulacastrum*

Fig. 32 *Zea mays*

PLANT PART USED: The seeds.
FOLK USE: Taken to boost sperm count and as an aphrodisiac to boost sexual performance and appetite.
PREPARATION OF REMEDY: Grounded seeds are taken as pap or the seeds are roasted and consumed.
DOSAGE: Not specified.

4 Conclusion

The survey reported in this case study revealed 30 medicinal plants that are used by the Benin people of Southern Nigeria for promoting male fertility. The scientific, common, and indigenous names of the plants along with the parts used, dosage (where they exist), and the methods of preparation for the treatment and management of diverse male infertility issues were documented. The plants that were identified belong to 23 families. The survey revealed that Poaceae, Apocynaceae, Rubiaceae, Menispermaceae, and Fabaceae are the dominant plant families as they have the most plant representations with at least two plants within the family used to promote male fertility among Benin people. The mode or method of administration of the plant is entirely based on oral administration either by making a juice out of the plant, by making a decoction of the plant or by infusion of the plant. The various parts of the plant used are the plant leaf, the plant stem, the plant root, the plant seed, stem bark, root bark young shoot, and fruit juice. It is worth noting that the genetic diversity of medicinal plants is continuously in decline and is increasingly threatened by extinction due to human population exploitation, unsustainable harvesting and utilization techniques, loss of habitats, and unmonitored and unregulated trade in medicinal plants. These plants contribute to the culture and well-being of indigenous people together. Medicinal plants used to treat illnesses including fertility issues would benefit from research and technological innovations. The vast knowledge and practices associated with medicinal plants need to be acknowledged and conserved not only through sustained local but also global efforts. It is important to recognize the roles played by remote communities and indigenous people as custodians of the world's medicinal plant genetic heritage.

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Plants Used in the Management and Treatment of Female Reproductive Health Issues: Case Study from Southern Nigeria

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Moses Edwin Osawaru and Matthew Chidozie Ogwu

Contents

Abstract

Through an ethnomedicinal survey, this case study presents a list of medicinal plants associated with female fertility, pre- and postnatal care among Benin and Warri residents in Southern Nigeria, and their associated ethnobotanical knowledge and practices. Information was obtained from randomly selected traditional healers, tradomedical practitioners, and native midwives from the main tribal groups within the cities using a semistructured open-ended questionnaire. A total of 24 medicinal plant species belonging to 21 families were of ethnomedicinal importance for female fertility, pre- and postnatal care of Benin and Warri

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residents in Southern Nigeria. Some of the plants include *Psidium guajava, Pennisetum purpureum, Oenothera biennis, Aloe barbadensis, Asparagus racemosus, Gossypium hirsutum, Newbouldia laevis, Phyllanthus amarus*, and *Afrormosia laxiflora*. These plants were said to have activities against menopausal symptoms, hormonal effects, fertility, and anti-fertility effects as well as abortifacients, contraceptive efficacy, aphrodisiac properties, treating reproductive tract infections, postnatal complications, relieving uterine bleeding, etc. Herbs and trees are the most used growth forms whereas leaves and seeds are the most cited plant part used in most of the tradomedical remedy preparations. Most of the remedies were prepared from a single and/or multiple plant source with other ingredients. The chapter contributes toward understanding, promotion, documentation, and safeguarding of indigenous knowledge, practices, and resources used in addressing female reproductive issues.

Keywords

Reproductive health · Female fertility · Postnatal care · Traditional healers · Menopause

1 Introduction

From time immemorial, man's life has always been dependent on plants around him, from which he draws all his basic needs [1–7]. Despite the effectiveness of chemically synthesized medicines, plants have been used medicinally in all civilizations [8, 9]. Medicinal plants contribute to the development of new pharmaceuticals to address new or existing medical issues including reproductive issues [10]. A large proportion of the Global South populace in rural and suburban areas including Nigerians rely on traditional or ethnomedicine for their primary health care due to the accessibility, availability, effectiveness, affordability, and inherent trust associated with it. This agrees with the view of CGSPS [11] that an estimated 85% of developing nations mainly use traditional health care systems. About 33% of Nigeria's population is classified as poor with women accounting for a substantial amount of the group and living in remote rural communities [12, 13]. According to WHO (2010) an estimated 25% of maternal deaths occur during pregnancy. Almost 20% of global maternal death happen in Nigeria with about 600,000 death and 900,000 near death mainly due to bleeding in pregnancy, infection, preeclampsia, eclampsia, and anemia in pregnancy [11, 14, 15]. Other issues include low access, inadequate, inefficient, and inappropriate reproductive health care system in Nigeria for the large youthful and reproductively active population [12, 14]. There is a need for high-quality care to maintain reproductive and sexual vigor as well before, during, and after pregnancy and childbirth.

Appropriate medical use of Nigeria's rich and diverse flora can contribute to addressing female reproductive health issues. For instance, *Alstonia boonei* stem bark and root and *Cymbopogon citrates* leaves that are used to treat fever and itching during pregnancy and epilepsy and *Bryophyllum pinnatum* leaves that is used to treat cough [9, 16–18]. *Ficus exasperata* is important for treating venereal diseases [16]. *Ocimum gratissimum* and *Vernonia amygdalina* are used by Usen people in Edo State Nigeria to boost immunity for mother and child after delivery and to regulate breast flow, respectively [5]. However, these medicinal plants are not isolated from the ongoing global biodiversity loss due to environmental change and anthropogenic activities as well as erosion of ethnomedicinal knowledge and practices that characterize the diverse ethnic groups of Nigeria [19–22]. The major issue connected to the erosion of ethnomedicinal knowledge and practices is the aging and death of the custodians of this knowledge, migration, and a lack of interest by the younger generation [8, 23, 24]. Put together, traditional medicine may be under threat. Confronted with illness and disease, early men discovered a wealth of useful therapeutic agents in the plant and this ageless system is still valuable today. The empirical knowledge and systems associated with medicinal substances in plants were and is still being passed on by oral tradition [5, 6, 25, 26].

This chapter aims to produce an inventory of the plants used by the residents of Benin and Warri for addressing female reproductive and sexual health conditions. A focused approach was adopted and carried out along with a nonexperimental validation of the plants used through a literature review of the phytochemical/pharmacological information supporting the medicinal activities. Here, we defined reproductive conditions and infertility according to Ogbe et al. [9] and Olooto et al. [27] as those that affect reproductive and sexual success as well as the prevention of conception and infertility as the inability to achieve pregnancy over 6–12 months despite adequate, intentional, and regular (i.e., about 3–4 times per week), unprotected sexual intercourse, respectively. Some causes of female infertility include leukorrhea, menopause, sexually transmitted infections (like chlamydia and gonorrhea), preexisting conditions (like malaria, HIV/AIDS, anemia, and malnutrition) abnormal menstrual cycles, faulty uterus and ovaries, menstrual disorder, and inappropriate lifestyles like excessive smoking and drinking that alter the hormonal pattern and ovulation [28–31]. Miscarriage (i.e., vaginal bleeding before 20 gestational weeks resulting in fetal abortion) is a common complication in pregnancies [32–34]. Prenatal health care help prepare for safe childbirth and parenthood and is beneficial for the survival of the mother, and prevents, detects, alleviates, and manages actual or potential health problems of the fetus and expectant mother. It is important to note that female genital mutilation increases the likelihood of complications during childbirth [35–37].

At present, there is little to no documentation of higher plants used for female fertility, pre- and postnatal care by Benin and Warri residents in Southern Nigeria. This chapter will document the traditional medicinal practices for female fertility, prenatal and postnatal of Benin residents and identify and evaluate how these plants are used (i.e., either alone or in combination).

2 Plants Used in Female Reproductive Health Care: Case Study of Benin and Warri Cities, Southern Nigeria

2.1 Methodology

Benin City is in Edo State (6.34° N and 5.60° E) and to the South is Delta State where Warri City is located (5.55° N and 5.77° E). Benin and Warri Cities lie within the humid tropical rainforest zone of Nigeria but their vegetation has been heavily altered by anthropogenic activities and human population growth and is currently a mosaic of secondary forest [9]. The climate is monsoonal with two distinct seasons (dry and rainy season) with high rainfall (2000 mm–3000 mm), temperature (20–40 °C) and atmospheric humidity and with radiation of 1600 h per year [38, 39]. A detailed description of the study area including soil characteristics can be found in Osawaru et al. [40]; Osawaru and Ogwu [41, 42]; Ogwu and Osawaru [43].

Survey to document traditional medicine practices and higher plants used for female fertility, prenatal and postnatal care by Benin and Warri residents was conducted between July and November 2021 using survey sheets for information collection, a digital camera, an audio recording device for interviews, and a mobile phone enabled position marker. The survey focused on randomly selected and locally identifiable professional herbalists, traditional healers, native midwives, and herbal remedy sellers using semistructured, open-ended questionnaires in English and/or preferred native languages. The randomly selected respondents and interviewees are duly registered with all the relevant authorities. The questionnaires were administered to the key informants (full-time traditional healers, part-time practitioners, elderly people who know the traditional values of the plants, relatives, and acquaintances who at one time or another used some of these plants for female fertility). The respondents animated a field survey to identify and collect plants and plant parts used for female fertility, prenatal and postnatal care. Specific questions like the botanical names, local names, plant parts used, dosage, method of preparation, application, and duration of treatment were asked during the oral interview, and the information supplied by these people was recorded. Collections were made from homesteads, distant farms, and different forests present in the study area.

Data collected include the sociodemographic characteristics of the respondents (gender, age, marital status, educational qualification, religion, tribe/ethnic group, occupation, name of village, and languages spoken), information on the plants used for female fertility, prenatal and postnatal care, and preparation of herbal remedies. Data collected were processed and analyzed using Windows Microsoft Excel 2016 version software, which was also used to draw charts and graphs to understand inherent patterns. The variables were presented as percentages. The therapeutic importance of each identified species was also recorded. Based on the knowledge of the informants, a total of 24 plants were identified and properly identified by a taxonomist at the University of Benin and through literature review.

2.2 Demographic Characteristics of Respondents

The demographic characteristic of the respondents and animators is presented in Table 1. There were more female (80%) compared to male (20%) respondents. The most dominant age category was 36–45 years (40%) and most respondents were married (66.67%) and had at least secondary school education (53.33%). Marital status, gender, culture, occupation, affordability, accessibility, and education level are major drivers for the practice, acceptance, and use of traditional medicine [44, 45]. Economic status is no longer a major determinant factor for the use of traditional medicine but techniques (like mind-body methods), spirituality, dietary practices, and comfort [46]. African traditional religion was the dominant religion of the professional herbalists, traditional healers, native midwives, and herbal remedy

Table 1 Demographic characteristics of respondents

Characteristics	Frequency	%
Gender		
Male	3	20
Female	12	80
Age range		
18–20	0	0
21–25	3	20
26–35	2	13.33
36–45	6	40
46 and above	4	26.67
Marital status		
Single	3	20
Engaged	0	0
Married	10	66.67
Divorce	2	13.33
Highest educational qualification		
Primary	3	20
Secondary	8	53.33
University	4	26.67
Religion		
Christian	8	53.33
Muslim	0	0
African traditional religion	7	46.67
Others	0	0
Occupation		
Tradomedical practitioner (healer, native midwives, nurses, etc.)	8	53.33
Government worker	1	
Trader	6	40
Private company worker	0	0
Unemployed	0	0

sellers surveyed in Benin and Warri Cities (Table 1). The religious affiliation documented in the survey is different from those reported in Curlin et al. [47] wherein it was reported that the majority of alternative medicine practitioners are most likely not to have any religious affiliation. However, they noted that these practitioners describe themselves as very spiritual and indirectly use religious beliefs in their dealings. On the other hand, higher spirituality and religious affiliation among contemporary medicine practitioners positively correlate with their acceptance to integrate alternative medicine into their practice [47]. This may be exploited in the Global South where religious affiliation is high among contemporary medicine practitioners to integrate and increase the relevance of traditional medicine practices for female fertility, pre- and postnatal care. A greater number of the preferential and concurrent use of herbal and contemporary medicine has been documented in parts of the Global South like Tamale, Northern Ghana [48]. According to Ngere et al. [49], this may be connected to the perceived sociocultural relevance of belief systems in disease etiology and diagnosis as well as caregiving and care reception in intervention formulation. In more developed countries, age and chronic diseases are the most important factors for accepting and using herbal medicines [50].

2.3 Medicinal Plants Used by Benin and Warri Residents for Female Reproductive Health Care by Benin and Warri Residents in Southern Nigeria

A total of 24 medicinal plants are used by Benin and Warri residents for the treatment and management of female reproductive issues. The 24 plants recorded from this survey are distributed into 21 families including Myrtaceae, Poaceae, Lamiaceae, Fabaceae, Apiaceae, and Lauraceae (Tables 2 and 3). The 24 plants included *Psidium guajava, Pennisetum purpureum, Oenothera biennis, Aloe barbadensis, Asparagus racemosus, Gossypium hirsutum, Newbouldia laevis, Phyllanthus amarus, Vitex agnus-castus, Portulaca oleracea, Afrormosia laxiflora, Agelaea obliqua, Alstonia boonei, Aneilema hockii, Apium graveolens, Brunfelsia splendida, Cucumeropsis mannii, Euphorbia deightonii, Heeria insignis, Macaranga barteri, Penianthus zenkeri, Persea americana, Azadirachta indica,* and *Carica papaya*. The diversity of the flora suggests the strength and diversity of reproductive health issues that may be treated with them and the wide variety of medicinal recipes [51]. Fasola [52] documented 61 plants from 32 families used in addressing female reproductive problems in Ibadan, Oyo State, Nigeria whereas Ogwu et al. [5] recorded 36 plant species belonging to 25 plant families in Usen Edo State, Nigeria. Joudi and Ghasem [53] and Ogwu et al. [5] reported the diverse uses of plants from three families – Asteraceae, Fabaceae, and Malvaceae for reproductive ailments. In the current study, Euphorbiaceae had the highest plant species representation followed by Myrtaceae. Tsobou et al. [54] recorded 70 plants from 37 families used in the treatment of reproductive ailments in Cameroon. In a similar study, Nduche et al. [55] documented 62 plant species from 41 families that are used in Ebonyi State, Nigeria to address fertility-related

Table 2 Plants used by Benin and Warri residents for the treatment of female fertility, prenatal and postnatal issues

S/No.	Collection source	Plant name	Family	Common name	Status	Cultivation status	Origin
1	Home garden	*Psidium guajava*	Myrtaceae	Guava	Not assessed	Cultivar	Exotic
2	Roadside	*Pennisetum purpureum*	Poaceae	Napier grass	Not assessed	Weed	Indigenous
3	Roadside	*Oenothera biennis*	Myrtaceae	Evening primrose	Not assessed	Weed	Exotic
4	Home garden	*Aloe barbadensis*	Asphodelaceae	Aloe vera	Endangered	Cultivar	Exotic
5	Roadside	*Asparagus racemosus*	Asparagaceae	Asparagus grass	Endangered	Weed	Exotic
6	Home garden	*Gossypium hirsutum*	Malvaceae	Cotton	Threatened	Cultivar	Exotic
7	Distant farmlands/forest	*Newbouldia laevis*	Bignoniaceae	Akoko	Not assessed	Wild	Indigenous
8	Roadside	*Phyllanthus amarus*	Euphorbiaceae	Indian gooseberry	Not assessed	Weed	Exotic
9	Distant farmlands/forest	*Vitex agnus-castus*	Lamiaceae	Chaste tree	Not assessed	Wild	Exotic
10	Home garden	*Portulaca oleracea*	Portulacaceae	Purslane plant	Not assessed	Cultivar	Exotic
11	Distant farmlands/forest	*Afrormosia laxiflora*	Fabaceae	Afrormosia	Endangered	Wild	Indigenous
12	Home garden	*Agelaea obliqua*	Connaraceae	Esura	Least concern	Cultivar	Indigenous
13	Distant farmlands/forest	*Alstonia boonei*	Apocynaceae	Stool wood	Least concern	Wild	Indigenous
14	Market	*Aneilema hockii*	Commelinaceae	Aneilema	Least concern	Weed	Indigenous
15	Home garden	*Apium graveolens*	Apiaceae	Celery	Least concern	Cultivar	Exotic
16	Home garden	*Brunfelsia splendida*	Solanaceae	Kiss me quick	Vulnerable	Cultivar	Exotic
17	Home garden	*Cucumeropsis mannii*	Cucurbitaceae	Egusi melon	Least concern	Cultivar	Indigenous
18	Roadside	*Euphorbia deightonii*	Euphorbiaceae	Ejagham	Not assessed	Weed	Indigenous
19	Distant farmlands/forest	*Heeria insignis*	Anacardiaceae	Heer plant	Not assessed	Wild	Indigenous
20	Distant farmlands/forest	*Macaranga barteri*	Euphorbiaceae	Agbasa	Not assessed	Wild	Indigenous
21	Distant farmlands/forest	*Penianthus zenkeri*	Menispermaceae	Ewe	Not assessed	Wild	Indigenous
22	Home garden	*Persea americana*	Lauraceae	Avocado	Least concern	Cultivar	Exotic
23	Home garden	*Azadirachta indica*	Meliaceae	Neem	Least concern	Cultivar	Exotic
24	Home garden	*Carica papaya*	Caricaceae	Papaya	Least concern	Cultivar	Exotic

Table 3 Distribution of the plant families of plants used for female reproductive health care by Benin and Warri residents in Southern Nigeria

Families	Number of species
Myrtaceae	2
Poaceae	1
Asphodelaceae	1
Asparagaceae	1
Malvaceae	1
Bignoniaceae	1
Euphorbiaceae	3
Lamiaceae	1
Fabaceae	1
Cucurbitaceae	1
Connaraceae	1
Apocynaceae	1
Commelinaceae	1
Apiaceae	1
Solanaceae	1
Anacardiaceae	1
Menispermaceae	1
Lauraceae	1
Meliaceae	1
Caricaceae	1
Portulacaceae	1

issues. These reports agree with the submission by Gill [56] that Nigerian flora is invaluable to the health system care of Nigerians. Some medicinal plants have multi-curative abilities where they are used in the preparation of reproductive health remedies for multiple ailments and taken as infusions, decoctions, and powdering [52, 57]. In the present study, most of the plants documented were reported to be associated with more than one female reproductive health issue as a remedy. Nonetheless, these native plants are often neglected or underutilized albeit they possess a substantial amount of undescribed benefits [58, 59]. A significant proportion of the African populace uses traditional plants and traditional healers and elders to manage health issues including the treatment of reproductive health problems [54]. The knowledge and practice are deep-rooted in the sociocultural background of the locale [60]. In some cases, a blend of indigenous and modern scientific knowledge is adopted to address certain reproductive health problems [5].

When these plants are used for female reproductive care, they are prepared and used in the form of oral tinctures, macerations, concoctions, or infusions and applied topically in rare cases [61, 62]. The survey of Benin and Warri residents revealed that the plants were mostly used alone and are commonly soaked in hot water or gin or boiled in water to extract the bioactive constituents (Table 4). In some cases, the plant seeds are dried and then grinded. These plant parts contain bioactive

Table 4 Plants used by Benin residents for female reproductive care, the parts used (whether alone or in combination with other herbal remedies), and the mode of administration is presented

S/N.	Botanical name	Part used	Decoction means	Alone or in combination with	Mode of administration
1	*Psidium guajava*	Leaves and stems	Soaking in hot water or gin	Alone	Swallowed
2	*Pennisetum purpureum*	Leaf extract	Soaking in hot water or gin	Alone	Swallowed
3	*Oenothera biennis*	Seed oil extract	Soaking in hot water or gin	Alone or in combination with *Cocos nucifera*	External application and/or swallowed
4	*Aloe barbadensis*	Leaf extract	Used fresh	Alone	Swallowed or applied externally
5	*Asparagus racemosus*	Root extract	Soaking in hot water or gin	Alone or in combination with other plant extracts	Swallowed
6	*Gossypium hirsutum*	Leaf and seed extract	Soaking in hot water or carbonated soft drink	Alone	Swallowed
7	*Newbouldia laevis*	Leaf and stem bark	Soaking in hot water or gin	Alone	Swallowed
8	*Phyllanthus amarus*	Leaf extract	Soaking in hot water or gin	Alone or in combination with yam extract	Swallowed
9	*Vitex agnus-castus*	Leaf and stem	Soaking in hot water or gin	Alone	Swallowed
10	*Portulaca oleracea*	Leaf and stem	Soaked in gin	Alone	Swallowed
11	*Afrormosia laxiflora*	Leaf extract	Soaking in hot water or gin	Alone	Swallowed
12	*Agelaea obliqua*	Leaf extract	Soaking in hot water or gin	Alone	Swallowed
13	*Alstonia boonei*	Leaf extract	Soaking in hot water or gin	Alone	Swallowed
14	*Aneilema hockii*	Leaf extract	Soaking in hot water or gin	Alone	Swallowed
15	*Apium graveolens*	Leaf and seed extract	Soaking in hot water or gin	Alone or in combination with other herbal remedies	Swallowed
16	*Brunfelsia splendida*	Leaf extract	Soaking in hot water or gin	Alone	Chewed and swallowed
17	*Cucumeropsis mannii*	Seed extract	Dried and grinded	Alone and/or in combination with other herbal remedies	Swallowed

(continued)

Table 4 (continued)

S/N.	Botanical name	Part used	Decoction means	Alone or in combination with	Mode of administration
18	*Euphorbia deightonii*	Succulent stem	Used fresh	Alone	Swallowed
19	*Heeria insignis*	Leaf extract	Soaking in hot water or gin	Alone	Swallowed
20	*Macaranga barteri*	Leaf and seed extract	Soaking in hot water or gin. Seeds are grinded.	Alone and/or in combination with other herbal remedies	Swallowed
21	*Penianthus zenkeri*	Root, stems, and leaf extract	Soaking in hot water or gin	Alone and/or in combination with other herbal remedies	Swallowed
22	*Persea americana*	Seeds	Dried and grinded	Added to meals	Swallowed
23	*Azadirachta indica*	Leaves	Boiled in water and soaked	Alone	Swallowed
24	*Carica papaya*	Seeds	Dried and grinded	Added to meals	Swallowed

polyphenolic compounds like isoflavones and flavonoids that confer beneficial reproductive health effects on women [63].

The plants used by Benin and Warri residents for female reproductive care and the parts used, whether alone or in combination with other herbal remedies, and the mode of administration is presented in Table 4. The most common plant part used is the leaf and they can be used alone or in combination with other local herbal remedies. The decoction made from the plants is more likely to be ingested than externally applied.

The commonly used plant part in the preparation of traditional medical remedies by Benin and Warri residents for the treatment and management of female reproductive health issues is presented in Fig. 1. Besides the whole plant, various plant parts are commonly collected and used in the preparation of herbal remedies including stem, bark, root, leaves, whole plant fruits, and seeds [6, 52, 64–66]. However, in the present study, leaves followed by leaves and seeds and leaves and stem bark were the most used parts, while root, stems and leaves, root, oil, and stem had the least usage. Other workers have made a similar observation regarding the plant parts that are most commonly used including Idowu et al. [67]; Thirupathy [68]; and Shosan et al. [69] wherein leaves were reported as the most used plant part for the treatment and management of reproductive issues. These leaves may be used alone or in combination with other plant parts in the preparation of traditional remedies [70]. In agreement with the finding of Tsobou et al. [54] most herbal remedies are prepared as a decoction. We also observed that all the plants were used as a decoction. The method of preparing the herbal remedy is diverse but in most cases are soaked in dry gin (approximately 40% alcohol) or boiled in hot water

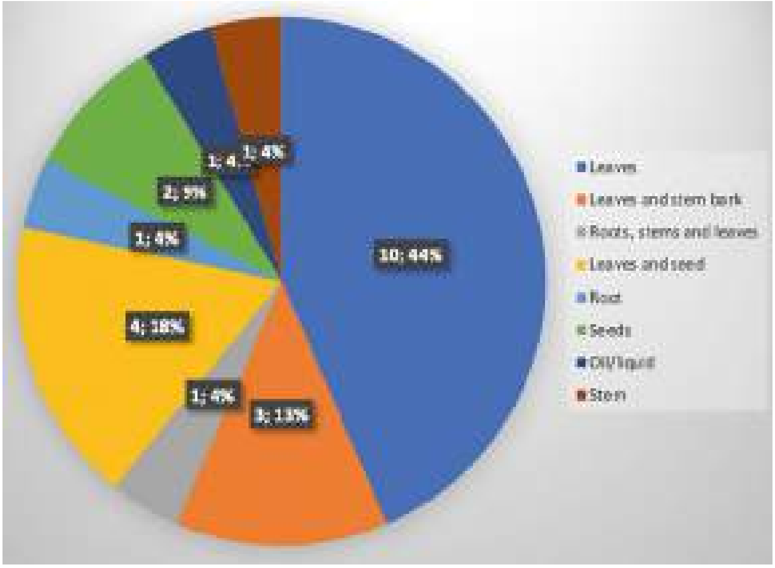

Fig. 1 Frequency of plant parts used for reproductive health care by Benin and Warri residents in Southern Nigeria

[6, 52, 71]. This work has highlighted the importance of plant-based medicine in female reproductive care, which is in line with the findings of Tsobou et al. [54] that medicinal plants are invaluable in the management of reproductive healthcare. Some of these plants used for female reproductive care have activities against menopausal symptoms, hormonal effects, fertility, and anti-fertility effects as well as abortifacients, contraceptive efficacy, aphrodisiac activity, reproductive tract infections, childbirth complications, and efficacy in relieving uterine bleeding, etc. Medicinal plants contain secondary metabolites that have important in vivo and in vitro functions to both ovarian folliculogenesis and steroidogenesis in many animal species [72].

2.4 Enumeration of Medicinal Plants Used for Female Reproductive Health Care by Benin and Warri Residents in Southern Nigeria

2.4.1 *Psidium Guajava* (Common Name – Guava)

It is a perennial evergreen tree. Flowering starts within the first 2 years and trees typically reach maturity after 5–8 years. Leaves are oppositely arranged while the fruit exudes a strong, sweet, musky odor when ripe and may be round, ovoid, or pear shape. The central pulp of the fruit is concolorous. Guava leaves are known to have attenuating effects on hypertension, hepatic steatosis, and other oxidative stress diseases because of their high oxidation activities [73–75].

Fig. 2 *Psidium guajava* leaves and fruits

Parts used: Leaves and stem bark. The leaves contain rutin, kaempferol, epicatechin, naringenin, isoflavonoids, gallic acid, and flavonoids [76].

Mode of administration: Tea made from leaves or bark is taken to regulate menstrual flow and pre-pregnancy stomach cleansing (Fig. 2).

2.4.2 *Pennisetum Purpureum* (Common Name – Elephant Grass)

A perennial grass that easily forms dense thickets with ecosystem-impacting characteristics functions especially on hydrology cycles, biophysical dynamics, and community composition. The plant has hormone-regulatory, antioxidant, and immune capacities as well as postpartum blood pressure management [77–79]. The decoction made from the plant is also used to manage menstrual pains and facilitate childbirth.

Parts used: Leaves.

Mode of administration: Young leaves are boiled and used in the treatment of venereal diseases preventing pregnancy and for superovulatory treatment as a hormonal control (Fig. 3).

2.4.3 *Oenothera Biennis* (Common Name – Evening Primrose)

A biennial plant that grows up to 1.2 m. It is not frost tender and flowers from June to September. The plant is hermaphrodite and is often polluted by Lepidoptera. The stem is hairy and often purple-tinged. Bright yellow, four-petaled flowers that open at night. The seed remains in the soil and germinates when the soil is disturbed. It is used in the treatment of mastalgia (breast pains) and hot flashes [80]. Hutcherson

Fig. 3 Mature *Pennisetum purpureum* plant

et al. [81] reported that using the plant decreases the risk of cesarean section, macrosomia, preeclampsia, and stillbirth. The plant is relevant for the cervical preparation of women about to undergo hysteroscopy [82].

Parts used: Oil extract from the plant.

Mode of administration: Used to improve sexual appetite, treat menopause symptoms, and regulate estrogen (Fig. 4).

2.4.4 *Aloe Barbadensis* (Common Name – Aloe Vera)

A succulent evergreen perennial that is stemless or short-stemmed and grows up to 60–100 cm. Leaves are thick and fleshy, green to gray-green. Leaf margins are serrated with small white teeth. The plant is used to treat internal and external injuries during pregnancy and after childbirth [83]. External application of aloe vera helps increase the sensitivity, elasticity, and functioning of reproductive organs and increases estrogen formation. Also, Maharjan et al. [84] connected the use of aloe vera gel to ovarian steroid restoration and protection.

Parts used: Leaves and leaf extract (juice).

Mode of administration: Leave and leaf extract are used to prepare juice and taken to improve ovary development, and follicle-stimulating hormones and to rejuvenate the uterus (Fig. 5).

2.4.5 *Asparagus Racemosus* (Common Name – Asparagus Grass)

Possess small pine-needle-like phylloclades that are mostly uniform and shiny green. The white flowers are short and spiky. Fruits are blackish-purple, globular berries. It has a tuberous adventitious root system that is about one meter in length and tapering at both ends. The plant is used to increase fertility and to address stress-related issues

Fig. 4 Mature *Oenothera biennis* plant with flowers

in pregnant women [85, 86]. It also has hormonal balancing effects in women, which increases the chances of pregnancy as well as lactation-boosting functions.

Parts used: Root extract.

Mode of administration: Extracts from the roots are used to balance female reproductive hormones (Fig. 6).

2.4.6 *Gossypium Hirsutum* (Common Name – Cotton)

The plant is an annual subshrub with alternate leaves, petiolate, palmately 3–5 lobed, hirsute, and cordate blades. Fruit is a dehiscent capsule (4–6 cm long), spherical smooth, light green, and bearing hairs on the epidermis strongly attached to the seed coat. Well-developed taproot with numerous lateral roots. A herbal remedy prepared from cotton regulates labor-inducing proteins and supports uterine quiescence [87]. The plant is also used to increase fertility and address health issues like fever, headaches, gastrointestinal problems, and nausea during pregnancy.

Fig. 5 Mature *Aloe barbadensis*

Fig. 6 *Asparagus racemosus*

Parts used: Leaves, seeds, and roots.

Mode of administration: Extracts from leaves and seeds are used in the treatment of menstrual disorders, symptoms of menopause, inducing labor, and afterbirth cleansing, and improve breast milk production. It is applied vaginally and on the breast surface (Fig. 7).

Fig. 7 *Gossypium hirsutum* leaves and flowers. (Source: Morris [88])

2.4.7 *Newbouldia Laevis* (Common Name – Boundary Tree, Akoko)

A perennial tree that grows up to 15 m high. The stem is woody. Each leaflet is oblanceolate 10–20 cm long and 5–10 cm broad, serrated, sessile, or with short petioles. The fruit is a long, pendulous, dehiscent capsule about 30 cm long dotted with glands. The plant leaves have uterine contractile effects and antihemorrhagic functions [89, 90]. It is also used to treat fever during pregnancy.

Parts used: Leaves and bark.

Mode of administration: Leaves are squeezed in water and applied in the virginal to stop bleeding in threatened miscarriage. The leaves are cooked and eaten by pregnant women to facilitate safe and easy delivery and milk production after birth. The barks are used to prepare a solution and used to treat stomach pains and cleanse the stomach in preparation for childbirth (Fig. 8).

Fig. 8 *Newbouldia laevis* leaves. (Source: Nwokolo [91])

2.4.8 *Phyllanthus Amarus* (Common Name – Sleeping Plant, Indian Gooseberry Plant, Seed under Leaf)

A slender annual herb with an erect stem and leafy angular branches. The leaves are very small and the minute pedicelled flowers are very numerous. The plant is used to regulate hormones and for hepatoprotection during pregnancy because of its immunomodulatory capacity and for the treatment of jaundice, diabetes, dysentery, urogenital diseases, and malaria before pregnancy [92–95]. It is also used to treat gonorrhea and other wart-like genital issues.

Parts used: The whole plant.

Mode of administration: The plant is soaked in alcohol (dry gin) and allowed to stand. The extract is taken orally to treat dysentery and stomach aches both during pregnancy and after delivery (Fig. 9).

2.4.9 *Vitex Agnus-Castus* (Common Name – Sleeping Plant, Indian Gooseberry Plant)

A deciduous hermaphrodite shrub or herb. It is pollinated by Insects and grows well in sandy and loamy soils that are well-drained even in nutritionally poor ones. Suitable pH: acid, neutral, and basic (alkaline) soils. It does not grow well in the shade. The plant is used mainly to manage and treat uterine bleeding disorders and mastodynia [96]. It is also used to improve sexual dysfunction in women undergoing menopause [97]. The plant helps with stress during pregnancy by reducing prolactin and has a hormone-stabilizing effect on prepregnant women.

Parts used: Leaves and stems.

Fig. 9 *Phyllanthus amarus* leaves. (Source: Ghosh et al. [92])

Mode of administration: The leaves and stems are soaked in dry gin and the extract is taken orally to treat premenstrual syndrome and menstrual disorder. It increases menstrual flow (Fig. 10).

2.4.10 *Portulaca Oleracea* (Common Name – Purslane Plant)

An annual or perennial prostrate succulent herb. Stem may spread on the ground or be erect. They are smooth and fleshy, more or less reddish, and could reach 40 cm. It produces mucilage in high amounts that enhance the flexibility of female reproductive organs. The plant has antiaging effects on female genitalia as well as numerous antioxidants and antiatherogenic activities. The plant is also used to fight infections and complications during and after childbirth [62].

Parts used: Leaf and stem extract.

Mode of administration: Extract is taken to improve women's fertility and libido (Fig. 11).

2.4.11 *Afrormosia Laxiflora* (Common Name – Aformosia, African Teak)

A small to medium-sized straight tree bearing crooked, drooping branches forming a disheveled crown. It is common in woodland savanna and fringing to dry dense

Fig. 10 Mature *Vitex agnus-castus* with flowers

Fig. 11 *Portulaca oleracea* leaves and flowers. (Source: SEINet Portal Network [98])

Fig. 12 *Afromosia laxiflora* leaves and flowers. (Source: Green Institute [101])

forests. It is used to boost sexual appetite and performance, cure venereal diseases, and stimulate higher milk production [99, 100].

Parts used: Leaves.

Mode of administration: Leaves are soaked and used as a laxative and stomach cleanser in preparation for pregnancy (Fig. 12).

2.4.12 *Agelaea Obliqua* (Common Name – Esura)

Scrabbling shrub with shining leaves. Flowers pure white or tinged yellow, green or pink, fragrant; fruits scarlet, in forest and secondary growth. The plant is taken to support and boost sexual functions [102].

Part used: Leaves.

Mode of administration: Leaves are consumed raw or in processed form due to their aphrodisiac properties and to treat mild vaginal infections (Fig. 13).

2.4.13 *Alstonia Boonei* (Common Name – Stool Wood)

A tree with grayish bark that exudes a bitter-tasting latex. The leaves may be elliptic or oblanceolate in shape. Whorled and angular branches covered with white lenticels. Inflorescence is repeatedly branched in whorl, with flowers in threes. Herbal decoction prepared with the plants are taken to address fever, insomnia, joint and body pains, and diarrhea. It is used to address abdominal pains, pelvic congestions, ovarian cysts, uterine fibroid, menstrual pains, and microbial infections before, during, and after pregnancy [105].

Part used: Leaves.

Fig. 13 Agelaea species. (Source: Willis [103]; Martin and Harvey [104])

Mode of administration: Leaves in combination with alligator pepper and other medicinal plants are used to prepare soup for pregnant women that are believed to allow the fetus to develop well and promote safe delivery (Fig. 14).

2.4.14 *Aneilema Hockii* (Common Name – Aneilema)

Tufted perennial with erect or straggling shoots. Roots are thick and fleshy. Leaves are spirally arranged. The large, bluish flowers readily distinguish this species.

Fig. 14 *Alstonia boonei* leaves and flowers. (Source: Ileke [106])

Capsules are generally longer, less pubescent, and have seeds in capsules that are emarginated and lack a terminal apicule. The plant is taken to increase fertility and fight venereal diseases. It is also used to treat malaria during pregnancy [107].

Part used: Leaf and root extract.

Mode of administration: Leaf and root extract are taken to prevent miscarriage and maintain the health of the mother and fetus (Fig. 15).

2.4.15 *Apium Graveolens* (Common Name – Celery)

Biennial to perennial herb. The leaves are lobed on each side of the central axis and has a row of two or more lobes on each side of the central axis. The leaves are used to increase fertility and balance hormone production. Herbal decoction prepared from the seed is used to stabilize the blood pressure of pregnant women [109]. The plant is also known to have antioxidant properties relevant for addressing oxidative stress issues in pregnant women.

Part used: Leaves and seed extract.

Mode of administration: Herbal extract prepared with *Apium* species to improve reproductive vigor in women. It is also taken to correct any sexual dysfunction in women (Fig. 16).

2.4.16 *Brunfelsia Uniflora* (Common Name Kiss Me Quick)

A medium-sized, evergreen shrub or small tree with diffuse, spreading branches. It usually grows 0.5–3 m tall. The plant is known to have antioxidant properties relevant for addressing oxidative stress issues in pregnant women.

Part used: Leaf.

Fig. 15 *Aneilema hockii.* (Source: Hyde et al. [108])

Mode of administration: The leaf is believed to increase the chances of getting pregnant in young women (Fig. 17).

2.4.17 *Cucumeropsis Mannii* (Common Name – Melon)

A climber with stems that are sparse crisped-hairy. Leaf-blade are broad ovate-cordate or reniform-cordate. Fruits are cylindrical-ellipsoid, and rounded. Seeds ovate in outline, compressed, smooth, and white. The plant is used to protect and maintain a healthy female genital system and to boost fertility [112].

Part used: Seeds.

Mode of administration: Crushed seeds are blended and mixed with other herbal remedies to improve female fertility and conception (Fig. 18).

Fig. 16 *Apium graveolens* leaves. (Source: Dennis et al. [110])

2.4.18 *Euphorbia Deightonii* (Common Name – Ejagham)

A shrub, branching from the base without a distinct trunk, to 6 m high. The plant appears to be always cultivated. It is commonly grown as a hedge-plant, the lowest branches becoming buried in the ground thus providing a thick barrier to the base. The plant contains copious white latex, which is said to be poisonous. It is an irritant in wounds. An herbal remedy prepared with the plant is used to improve endocrine pathways and treat polycystic ovary syndrome [63].

Part used: Succulent stem.

Mode of administration: Chewed and swallowed (Fig. 19).

2.4.19 *Heeria Insignis* (Common Name – Heer Plant)

A branched shrub or small tree. Leaf undersurface is silvery-white and flowers small, and creamy; fruits are glossy and black. Inflorescences of terminal and axillary, much-branched panicles. The leaves are boiled and used in the treatment of viral and microbial infections before and during pregnancy [114]. It contains important secondary metabolites. It is used to prevent and treat malaria before and during pregnancy.

Fig. 17 *Brunfelsia uniflora.* (Source: Makoto [111])

Part used: Leaves.

Mode of administration: The plant is taken to improve lactation after childbirth (Fig. 20).

2.4.20 *Macaranga Barteri* (Common Name – Macaranga, Akan)

An evergreen climbing shrub or a tree. The spiny bole have spreading and spiny branches. It is a dioecious species. Leaves and seed extracts are used to improve fertility, treat stomach ache, and fight venereal diseases like gonorrhea.

Part used: Leaf and seed extracts.

Fig. 18 Opened *Cucumeropsis mannii* fruits. (Source: Morris [113])

Fig. 19 *Euphorbia deightonii*

Mode of administration: Leaf and seed extract are taken as an abortifacient, to treat sexually transmitted infections, and stomach cleanser in preparation to get pregnant (Fig. 21).

Fig. 20 *Heeria insignis.* (Source: Tropical Plants Database [115])

2.4.21 *Penianthus Zenkeri* (Common Name – Ewe)

A dioecious, evergreen shrub or small tree. Leaves are arranged spirally, simple, and entire; with stipules absent. Seed ellipsoid, seedling with epigeal germination.

Part used: Root, stem, and leaf.

Mode of administration: The roots, stem, and leaf extract are used as an aphrodisiac, and branches, bark, and roots are so used. They are also used to treat impotence (Fig. 22).

2.4.22 *Persea Americana* (Common Name – Avocado, Pear)

It is a medium to large evergreen tree (although some varieties lose their leaves before flowering starts). The tree canopy ranges from low, dense, and symmetrical to upright and asymmetrical. Leaves are variable in shape (elliptic, oval, lanceolate). They are often pubescent and reddish when young, becoming smooth, leathery, and dark green when mature. Flowers are yellowish green, and 1–1.3 cm in diameter. The plant seed is used for hormonal management and to improve lactation [117, 118].

Part used: Leaf and seed extract.

Mode of administration: Extract from the leaf and seed are taken to improve female reproductive hormone and fertility control as well as improve maternal health (Fig. 23).

2.4.23 *Azadirachta Indica* (Common Name – Neem, Dongoyaro)

A fast-growing evergreen tree with wide and spreading branches. The leaves are opposite and pinnate with missing terminal leaflets. Petioles are mostly short with

Fig. 21 *Macaranga barteri* leaves. (Source: Willis [103]; Martin and Harvey [104])

Fig. 22 *Penianthus zenkeri* leaves and fruits. (Source: Davidson and Christoph [116])

Fig. 23 *Persea americana* leaves and fruits. (Source: Krist [119])

axillary panicle drooping flowers. The fruit endocarp contains 1–3 seeds with a brown coat. The plant is used to manage menstrual flow, improve fertility, and treat malaria during pregnancy [120, 121].

Parts used: Leaves and flowers.

Mode of administration: A decoction of the leaves is taken orally to treat malaria during pregnancy (Fig. 24).

2.4.24 *Carica Papaya* (Common Name – Pawpaw, Uhoro)

It is a tree with a stem that yields copious white latex. It has a weak, soft wooden stem crowned by a terminal cluster of large edible fruits. It is used to boost sexual performance and fertility in women through hormonal balancing and increase in estrogen production.

Parts used: Seeds and Leaves.

Mode of administration: The seeds are used to cause abortion in pregnant women, especially young girls. The leaves are used to treat malaria during pregnancy as a prenatal and postnatal measure (Fig. 25).

3 Conclusion

Though traditional reproductive health service is considered affordable and accessible, the younger generation is more interested in modern medicine, which is characteristically more expensive and less available in certain parts of Benin and Warri cities. The issue is driving the decline and neglect in the practice and resources used

Fig. 24 Azadirachta indica leaves and flowers

Fig. 25 *Carica papaya*

for female reproductive health care, respectively. This is further compounded by the process of urbanization and Western culture, which adds an extra layer to the complex and multidimensionality of female reproductive dysfunction. The threats to this indigenous practice and knowledge system require an urgent revitalization strategy to protect indigenous medical knowledge. Also, only little scientific evidence exists to prove the efficacy of the remedies and plant species employed to treat or manage reproductive disorders. Addressing these issues could lead to the confirmation of their efficacy and the discovery and development of new drugs for modern medicine. Moreover, the medicinal plants documented in this study are considered invaluable for addressing female reproduction dysfunction through traditional approaches and need conservation attention. Their conservation will help preserve the heritage and plant diversity.

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Part III

Utilization Approach and Practices in Herbal Medicine

Current Trends on Phytochemicals Toward Herbal Medicine Development 34

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Abstract

Traditional plants are the most promising source of phytochemicals for creating herbal medicines. Numerous biosynthetic metabolic networks are important in the biogenesis of phytochemicals and their accumulation in plants under the strong influence of environmental factors. Primary and secondary metabolites generated from plants, or phytochemicals, have great promise as chemotherapeutics for preventing or treating potential public health problems. Pharmaceutical enterprises prepare plant-derived phytochemicals into herbal medicines using various conventional methods while considering the properties of the phytochemicals. These conventional methods of producing phytochemicals as herbal medicines have several advantages and drawbacks. Additionally, the pharmaceutical industry has been forced by drug resistance and population growth to speed up the development and processing of phytochemicals from plants used as herbal medicines to address today's human health problems. Functionalization of plant metabolic networks and its regulatory bioparts has proposed a paradigm change for phytochemicals production in model microbial platforms. Based on this existing situation, the first part of the current summary provides an overview of several plant-derived phytochemicals, their medicinal properties, the biosynthetic network in plants, and conventional methods for processing them. The development of herbal medicines as a sustainable route forward is the subject of the second half, which is centered on modern omics, systems metabolic engineering, and synthetic biology techniques to improve phytochemical synthesis using plant bioparticulates inside microbial platforms.

Keywords

Phytochemicals · Herbal medicine · Plant extracts · Environmental stress factors · System metabolic engineering · Synthetic biology · Sustainable microbial cell factory

Abbreviations

AD	Alzheimer's disease
ADME/T	Absorption, distribution, metabolism, excretion, and toxicity
ASFV	Asian swine flu virus
ATHB	*Arabidopsis thaliana* Homeobox-leucine zipper protein
BAAV	Bovine adeno-associated virus
BC	Breast cancer
CoA	Coenzyme A

CT	Condensed tannins
DMAPP	Dimethylallyl diphosphate
DNA	Deoxy ribonucleic acid
DPPH	2,2-diphenylpicrylhydrazyl
DXP	1-deoxy-d-xylolose-5-phosphate
EC	Esophageal carcinoma
Eca109	Esophageal cancer cell line 109
EGCG	(2)-epigallocatechin-3-gallate
EOs	Essential oils
FA	Ferulic acid
FeLV	Feline leukemia virus
FPP	Furanosyl pyrophosphate
GMOs	Genetically modified organisms
GPP	Geranyl diphosphate
HBV	Hepatitis B virus
HCC	Hepatocellular carcinoma
HCT-15	Human colon cancer type 15
HCV	Hepatitis C virus
HeLa	Henrietta Lacks cell line
HGC	Human gastric carcinoma
HIV	Human immunodeficiency virus
HSPs	Heat shock proteins
HSV	Herpes simplex virus
HT	Human colorectal adenocarcinoma type
IC50	Half-maximal inhibitory concentration
IL	Inerleukine
IMF	Intramuscular fat
IPP	Isopentenyl diphosphate
IRES	Internal ribosome entry site
KYSE30	Human Asian squamous cell carcinoma 30
LTL	Longissimus thoracis et lumborum
MEP	Methylerythritol phosphate
MH7A	TNF-α-stimulated synovial fibroblasts
MICs	Minimal inhibitory concentrations
miRs	Micro Ribonucleic acids
MMP-9	Matrix Metalloproteinase-9
MVA	Mevalonic acid
Nfe2l2	Nuclear factor erythroid 2-related factor 2
NS	Nuclear signaling
OVCAR	Ovarian cancer
PAL	Phenylalanine ammonia lyase
Pcs	Platelet concentrates
PD	Parkinson's disease
Ppargc1a	Peroxisome proliferative activated receptor, gamma, coactivator 1 alpha

PPC	Purple prairie clover
PPT	Podo phyllo toxin
pso	Postsymptom onset
PTHR1	Parathyroid hormone/parathyroid hormone-related protein receptor 1
RA therapy	Rheumatoid Arthritic
RdRp	RNA-dependent RNA polymerase
RLV	Rauscher murine leukemia virus
RNA	Ribonucleic acid
ROS	Reactive oxygen species
RT	Reverse transcriptase
Saos2	Sarcoma osteogenic2
SGC	Sarcoma gastric cercinoma
SKOV3	Epithelial human ovarian carcinoma 3
SPL	Squamosa promoter binding protein-like
StAR genes	STAR steroidogenic acute regulatory genes
T2DM	Type 2 diabetic mellitus
TB	Tuberculosis
TNF	Tumor necrotic factor
Topo II	Topoisomerase II
TPS	Terpenoids
U2OS Line	Human Bone Osteosarcoma Epithelial Cells
UV	Ultraviolet
WHO	World Health Organization

1 Introduction

Plants have long been a crucial source for developing novel medications [1]. Numerous phytocompounds found in them aid in treating and preventing various ailments. Herbal medicine uses a variety of medicinal plants, herbs, or their parts to treat illnesses while boosting overall health and the recovery of disorders [2]. Herbal medicines are the primeval form of healthcare known to us. Over time, a growing trend in the development of herbal medicines has been observed. From different studies, we have seen about 70–85% of the rural population, if we exclude the western countries of the world, have used traditional medicine as their only healthcare system and still continue to do so. The reason lies not only in the poverty of the region to buy expensive drugs but also that they are more culturally acceptable and have continued this practice since ancient era [3]. Plants have the ability to secrete secondary metabolites. Phytochemicals present in herbal plants are bioactive chemicals that provide us certain desirable health benefits. These phytochemicals also have some commercial importance like its usage as a pharmacological tool to study various biochemical processes, tannins acting as astringent, and quinones being used as antimicrobial agent. As it relates to immunity, certain phytochemicals strengthen the immune system, preventing the body from being harmed by illnesses [4]. There is also the notion that every ailment has a suitable treatment method built into it by nature. The invention of novel herbal medications, which in many aspects have proven to be superior to conventional pharmaceuticals [5], has been facilitated by the growing research and improvements in the extraction and characterization of phytochemicals.

According to many recent World Health Organization (WHO) investigations, these phytochemicals are present in more than 75% of persons who only rely on herbal medications. The phytochemicals saponins, tannins, anthraquinones, terpenoids, and many others are all found in herbal plants. It has been established that each of these phytochemicals has biomedical applications that may be further developed into herbal medications to treat diseases more effectively [6]. Herbal medicines are cost effective and have been proved advantageous over allopathic medicines which have adverse side effects and toxicity in many approaches. These have over a long period of time resulted in tremendous growth for producers of herbal medicines. People without prescriptions have also been shown to utilize more herbal medications [7]. Leaves, stems, barks, seeds, roots, and extracts from these are used in herbal medicine. Though herbal medicines are not able to cure rapid sickness and possess complexity of standardization, yet they have been proved beneficial in many ways like being cost-effective, recyclable, and with comparatively less side effects as they have been derived from natural source – plants – providing more protection with enhanced tolerance and much more. Through decades, these medicines have been tested and widely used to cure various fatal diseases like cardiovascular diseases, cancer, type 2 diabetes, and a lot to be added to the list [8]. Researchers have further found that phytochemicals possess antioxidant abilities. According to studies, the overproduction of oxidants in the human body causes serious damage to big macromolecules like DNA, proteins, and lipids. This pathogenesis of fatal diseases like cancer, cardiovascular disorders, and many others is due to this damage. The phytochemicals' antioxidant property has been extremely helpful in the treatment and prevention of chronic diseases. Researchers have revealed that many of the antioxidant phytochemicals have been found to possess anti-inflammatory actions [9, 10]. They reduce inflammation by inhibition of prostaglandin production and nuclear factor activity. They also inhibit enzyme production and increase the production cytokines which adds on a better defense mechanism to the human body. These antioxidant phytochemicals other than possessing anti-inflammatory actions and strong oxidant also have free radical scavenging activities which further forms the basis of other bioactivities and provides several other health benefits. Other than the chronic diseases, obesity is rising in alarming rates all around the world. Not only obesity comes with other health-degrading factors like respiratory problems, infertility, sleeping problems, and osteoarthritis, but also it is becoming a major health burden with social and economic costs. Phytochemicals present in herbal medicine have been able to cure obesity eradicating other health issues coming with it [11].

Phytochemicals also provide us a lot of other health benefits when consumed as they consist essential nutrients like vitamins, minerals, and proteins for optimal health. As all products come with pros and cons, here too with herbal medicines a few things are to be kept in mind. At times, all herbal medicines are considered safe as they do not use any synthetic products and are derived from natural source, but sometimes they also produce certain side effects and allergic reactions [12]. They may also result in nauseousness, vomiting, diarrhea, exhaustion, seizures, and other symptoms. This frequently happens as a result of self-dosing, when the patient does not fully understand the medication or is unaware of any potential allergens in the medication, leading to health problems later. Because herbal medicines may be purchased without a prescription, many individuals do so without learning everything there is to know about them. Despite this, herbal medicines are an excellent alternative to costly medications. Despite these obstacles, herbal medications have been shown effective through years of human research. Some medications have been combined with other herbs to successfully balance the side effects, making them safe for human use with the advantage of having no negative side effects. However, some medications have been withdrawn owing to toxicity. In Asian nations, herbal medications are often employed, and they have been effectively used for decades to synchronize immune system activities. This category includes at least 35 plants, and other herbs are utilized to create pharmacological medicines. As a result, research on herbal treatments has increased throughout the years. An effort has been made to highlight numerous phytochemical developments in the evolution of herbal therapy in this issue [13].

2 Outlines on Phytochemicals from Plants as Potent Herbal Medicine

Phytochemicals are called "Secondary metabolites" as they are produced due to the reaction of primary metabolites as amino acid, sugars, etc. Several phytochemicals that are derived from plants and are potent herbal medicines have been classified as follows:

2.1 Terpenoids

With five carbon isoprene units that can be combined in various ways, terpenoids make up the biggest class of secondary metabolites. A subgroup of terpenes called terpenoids has undergone alteration, moving or removing various functional groups and oxidized methyl groups at various locations. They are further divided into monoterpenes and have ten carbon atoms and two isoprene units; sesquieterpenes have three isoprene units; diterpenes have four; sesterpenes have five; and triterpenes have six. Terpenoids contained in herbal treatments are used to make anticancer medications, which are used to treat terminal conditions like cancer. These are additionally used in the development of antimalarial drugs [14].

2.2 Polyphenols as Herbal Medicine

According to studies, more than 8000 polyphenolic compounds, which are derived from the amino acid phenylalanine or its closely related component shikimic acid, are found in various plants. These molecules generally exist in conjugated form, where hydroxyl groups are attached to one or more sugar residues. These are further divided into phenolic acid, flavonoids, lignans, and stilbenes, which also have antifungal properties and two phenyl moieties connected by a two-carbon methylene bridge and two aromatic rings connected by three carbon atoms. Polyphenols aid in the treatment of chronic illnesses, such as reducing the prevalence of coronary heart disease. Additionally, they boost plasma antioxidant capabilities, which lower the chance of developing numerous degenerative illnesses [15].

2.3 Phytosterols as Herbal Medicine

More than 300 families of phytosterols have been found by researchers, some of which include beta-sitosterol, beta-sitostanol, campestanol, campesterol, and stigmasterol. In addition to lowering the high risk of cardiac arrest, phytosterols are particularly effective at treating excessive cholesterol. Due to their structural resemblance to cholesterol, they compete with cholesterol for the digestive system's attention. Now that sterol is digested in place of cholesterol in our bodies, part of the cholesterol is excreted as waste. Phytosterols are also an excellent treatment for diabetes and obesity [16, 17].

2.4 Alkaloids as Herbal Medicine

Alkaloids are composed of at least one nitrogen atom and are a form of amine, and their structures are incredibly varied. The different subparts of alkaloids include morphine, quinine, nicotine, ephidrine, and strychnine. These phytochemicals offer a number of therapeutic benefits, including the ability to cure arythmias and to lessen uterine hemorrhaging, hay fever, and bronchial asthma [18].

3 Therapeutics Features Plant-Derived Phytochemicals as Herbal Medicine

3.1 Saponins

Saponin has anti-inflammatory qualities, making it a viable RA therapy. A component of *Aralia taibaiensis* root extract called saponin can suppress the migration and proliferation of synoviocytes, specifically MH7A cells (IL-1, IL-6, IL-8, and TNF), and can lower the production of proinflammatory cytokines in vitro. A triterpenoid saponin called Tubeimosid I that was isolated from *Bolbotemma paniculatum* has the capacity to reduce MMP-9 expression in vitro. Recent evidence suggests that a variety of saponins, including those found in soy, ginseng, dioscins, and avicins, have played a variety of roles in the treatment of tumors. Few antitumorigenic medication combinations with saponins result in synergistic effects with possible growth suppression [19]. Angiopoietic and endothelial tyrosine kinase receptor expression can be induced by saponins from *Rhizoma dioscoreae nipponicae* [20].

3.2 Phenols

Large amounts of ferulic acid (FA), a naturally occurring phenolic component, are found in herbal plants. Numerous therapeutic advantages of FA have been documented, including ones that are anti-inflammatory, antioxidant, anticancer, and antidiabetic. It is important to remember that FA has prooxidant and antioxidant properties in addition to a range of antibacterial activities against all infections. In addition to preventing arsenic-induced toxicity in the male genital system, FA has a protective effect through reducing oxidative stress and testicular damage. Regulating the expression of the Nfe2l2, Ppargc1a, and StAR genes also increased the quality of the sperm [21]. The rhizome of the turmeric plant yields curcumin (*Curcuma longa*), one of the most important bioactive polyphenolic compounds. This chemical has well-documented anti-inflammatory, antioxidant, and anticancer properties. Researchers have discovered that curcumin can protect against fatal diseases in animal models of ailments like Alzheimer's disease (AD), Parkinson's disease (PD), depression, multiple sclerosis, ischemic stroke, etc. Curcumin's role in preventing stomach cancer has been documented in numerous studies. When curcumin was given orally to animal models of gastric cancer, carcinogenesis and tumor growth were significantly reduced. The anticancer effects of curcumin were demonstrated in HeLa cervical cancer cells. The results show that curcumin's IC50 value of 34.23 M/ml was effective in slowing the development of cancer cells. Curcumin usage inhibits cervical cancer cell invasion, proliferation, and enhanced cell death [22]. Salvianic acid B, a significant water-soluble polyphenolic acid, is produced by Radix Salviae Miltiorrhizae, a traditional herbal remedy that has been utilized as an antioxidant in Chinese medicine for thousands of years. There are numerous medicinal plants that contain gallic acid, including *Barringtonia racemosa*, *Cornus officinalis*, *Cassia auriculata*, *Polygonum aviculare*, *Punica granatum*, *Rheum officinale*, *Sanguisorba officinalis*, *Terminalia chebula*, and *S. chinensis*, as well as dietary spices like thyme and clove. According to certain reports, gallic acid has an inhibitory effect on bacterial dihydrofolate reductase and an excitation effect on topoisomerase IV-mediated DNA breakage in various bacteria. Alkyl gallates can also breach the bacterial cell membrane and obstruct cellular respiration and the electron transport chain [23].

3.3 Flavonoids as Herbal Medicine

Apples, grapes, red raspberries, shallots, cabbage, vegetables, and almonds are just a few examples of foods that contain quercetin, a bioactive flavonoid component. The antioxidant and anticancer effects of quercetin are well-known, as well as its ability to prevent metastasis and trigger apoptosis. Because of its biological and pharmacological effects on disorders such as atopic dermatitis, renal fibrosis, myocarditis, and neurodegenerative diseases, previous research have suggested that quercetin may decrease the advancement of degenerative diseases. It has also been applied to the treatment of cardiovascular disease and cancer. Li et al. [24] revealed quercetin's anticancer efficacy on cell growth and inhibition rates of U2OS and Saos2 human metastatic osteosarcoma cells through lowering PTHR1 activity. Myricetin is a bioactive flavonoid that is present in a wide range of edible plants, such as grapes, oranges, berries, and teas. Myricetin exhibits significant biological properties such as anti-inflammatory, antioxidant, and anticancer action. Numerous cancer cell lines, including those from the stomach (HGC-27 and SGC-7901), esophagus (EC9706 and KYSE30), ovarian (OVCAR-3, A2780/CP70, and SKOV3), colon (HCT-15, and HT-29), and cervix (HeLa), have verified the function of myricetin in preventing the proliferation of cancer cells and encouraging apoptosis. It also inhibited glucosidase and amylase, which helped prevent type 2 diabetic mellitus (T2DM) [25]. Luteolin, an active flavonoid component present in a range of plants, including broccoli, celery, pepper, and thyme, activates p53 and induces autophagy in hepatocellular carcinoma (HCC) cells. Numerous pharmacological and biological properties of luteolin have been shown, including anti-inflammatory, anticancer, antioxidant, and antidiabetic actions for the treatment of diabetic nephropathy. Eca109 esophageal cancer cells undergo apoptosis as a result of luteolin [26]. The results show that in esophageal cancer Eca109 cells, luteolin, a natural chemical, promotes caspase3, caspase9 mRNA, and protein expressions during cell death [27].

3.4 Tannins as Herbal Medication

Recently, many investigations have been carried out to determine the pertinent antioxidant activity of tannins. Much attention has been given to the antioxidant capabilities of tannins, which include protecting against osteoporosis, cancer, and cardiovascular disease. From the leaves of *F. altissima*, condensed tannins (CT) have also shown to have better antioxidant activity. According to a study, tannins at doses of 0.13%, 2.25%, and 0.45% significantly increased feed conversion rates, decreased cecum levels of ammonia, isobutyric, and isovaleric acid, decreased the depth of ileal crypts, and decreased intestinal bacterial proteolysis [28]. The mechanism of tannins was investigated by researchers using a bacteriostatic model. *E. coli* O157:H7 strain 3081 was tested on condensed tannins (CT) from purple prairie clover (*Dalea purpurea* Vent; PPC), which have anti-*E. coli* O157:H7 action. After 24 h, optical density at 600 nm was measured to assess bacterial growth. CT prolonged the lag period and slowed *E. coli* O157: development. The ability of a hydrolyzable tannin derived from chestnut to control *Clostridium perfringens* growth in the gut has been studied [29].

According to certain studies, tannins can shield the body from viruses like norovirus, bovine adeno-associated virus (BAAV), and HIV. When tannins' inhibitory effects on HIV-1 were evaluated, the results showed that tannins prevented the development of syncytia, lytic effects, the synthesis of the viral p24 antigen, and HIV-1 reverse transcriptase (RT) enzyme activity. Some plant extracts exhibit substantial inhibition of HIV-I integrase activity, according to results of a different investigation on this subject. Results showed that (2)-epigallocatechin-3-gallate (EGCG), an active component of green tea, directly interacted with the hepatitis C virus (HCV) in vitro and prevented the virus from adhering to the cell surface [29]. Condensed tannins' ability to kill parasitic adults and free-living larvae provided evidence for their potential use in the management of various stages of parasite development. In conclusion, the direct mechanisms include restricting larval development to lessen the establishment of infected third-stage larvae in the host and reducing spawning to inhibit the motion performance of the parasite [28, 29]. The indirect mechanisms include boosting the human immune system and infection resistance.

3.5 Alkaloids and Its Medicinal Impacts

Alkaloids are well-known for their ability to relax muscles. One such example is D-tubocurarine, which has antiparasitic properties because it can impede the acetycholine receptor sites that allow muscle relaxation at neuromuscular junctions. Using DPPH and metal chelating tests, it was discovered that the linearilobin, linearilin, and lycotonine browniine norditerpene alkaloids, which were isolated from the roots of *Delphinium linearilobum*, have antioxidant activity. The alkaloid 6-methoxydihydrosanguinarine was shown by Yin et al. [29] to be capable of controlling the synthesis of reactive oxygen species (ROS) and inducing apoptosis. Sanguilutine and chelilutine, two quaternary benzo[c] phenanthridine alkaloids, have comparable types of action processes. The cytokine TNF is widely acknowledged to be crucial for managing inflammation. Conophylline is said to prevent TNF-induced NF-B activation by lowering the expression of TNF-receptors on the cell surface. The Amaryllidaceae alkaloids lycorine and lycoricidinol either blocked protein synthesis or altered how cysteine/methionine was absorbed into macrophages, hence lowering TNF-production. Wirasathien et al. have demonstrated the antituberculosis, antimalarial, and cytotoxic effects of aporphine alkaloids from the aerial part of *Pseuduvaria setosa* on *Mycobacterium tuberculosis*, breast cancer (BC), and small cell lung cancer (NCIH187) cell lines. The antiviral activities of Eudistomin, a novel oxathiazepine ring containing alkaloids isolated from *Eudistoma olivaceum* against RNA viruses like *Coxsachie A*-21 and equine rhinovirus and against DNA viruses like HSV-1, HSV2, and Vaccinia virus, have been mentioned by Gul and Hamann in their review on indole alkaloids. While [30] established the anti-HIV action of Coscinamide alkaloids derived from *Coscinoderma* sp., [31] demonstrated that Dragmacidin alkaloids obtained from *Spongosorites* sp. were reported to suppress in vitro replication of feline leukemia virus (FeLV).

3.6 Essential Oils (EOs) as Herbal Medicine

Lavender oil has MICs that range from 0.16 to 11.90. Numerous microorganisms, including the majority of bacteria, yeasts, and filamentous fungus, are susceptible to the antibacterial effects of lavender oil. Additionally, it exhibits antisepsis action. Additionally, EO aids in the treatment of pain, profound anxiety, sleep disorders, and stress. On *Aspergillus*, *Candida*, and *Dermatophyte* strains, extract from *Lavandula luisieri* shows a fungicidal action. According to recent investigations, a number of the constituents of lavender essential oil have been revealed to have anticancer and antimutagenic properties. *Staphylococcus aureus*, *Staphylococcus pyogenes*, and *Haemophilus influenza* are just a few of the gram-positive and gram-negative microorganisms that are resistant to eucalyptus oil's powerful cytotoxic and antibacterial effects. Eucalyptus oil can be used as an adjuvant in immunosuppression and for the treatment of tumors. Peppermint oil, which has fungistatic and fungicidal activities with minimal inhibitory concentrations (MICs) of 0.5–8 ul/ml, inhibits the development of *Staphylococci*. *Candida albicans* cannot produce biofilms when given a dose of 2 ul/ml of PO. Oral infections and oral biofilm are strongly inhibited by Melaleuca gel, which includes an extraction from the *Melaleuca alternifolia* plant [32]. Three *Candida* species – *C. albicans, C. glabrata*, and *C. tropicalic* – are susceptible to the therapeutic effects of lemon oil. Dentistry makes extensive use of the eugenol-containing clove oil's therapeutic capabilities. It also prevents *C. albicans* from forming germ tubes. Additionally, it exhibits significant inhibitory effect against *Staphylococcus* spp. that are multidrug resistant. The plant from which cinnamon oil is extracted, *Cinnamomum zeylanicum*, possesses qualities that include antibacterial, fungicidal, antimutagenic, antioxidant, and antiparasitic [32].

4 Biosynthetic Networks for Phytochemicals Biogenesis in Plants

Over time, there has been a lot of interest in the role of miRNAs in regulating plant-secondary metabolites and metabolic processes. This work improves our knowledge of the functions of miRNAs in the control of miRNA-targeted genes and the manufacture of essential oils, terpenoids, alkaloids, and saponins. The class of phenyl-propanoids includes the majority of secondary metabolites found in plants. These compounds are derived from the aromatic amino acid phenylalanine. Flavonoids, phenolic acids, stilbenes, coumarins, and monolignols make up the majority of what they contain. The secondary metabolites indicate its important roles, and it is abundant in the majority of plant kingdoms. It is interesting to note that active miR-8154 and miR-5298b also increase the phenyl-propanoid content, whereas the target genes for the miRNAs (miR-9662, miR-894, miR-172, and miR-166) are essential for controlling the metabolic biosynthetic process of phenyl-propanoid saponin in plants [33].

The cytosolic mevalonic acid (MVA) system and the plastidial methylerythritol phosphate (MEP) pathway are the two separate processes that produce terpenoids. In contrast to abiotic stress, biotic stress results in a significant production of terpenoid compounds. According to transcriptional study, the terpenoid backbone biosynthesis in the leaves of *Ananas comosus* var. *Bracteatus* is carried out by roughly 43 identified unigenes. Terpene production is also regulated by the *Ferula gummosis* miRNAs miR-2919, miR-5251, miR-838, miR-5021, and miR-5658. However, it is hypothesized that miR-5021 and miR-5658 regulate the *Arabidopsis* transcription factors SPL7, SPL11, and ATHB13, which are known to be involved in terpenoid production.

Indole, piperidine, tropane, purine, pyrrolizidine, imidazole, quinolizidine, isoquinoline, and pyrrolidine are among the substances that make up alkaloids. These substances are produced by the metabolism of many alkaloids. Many miRNAs, including pso-miR-13, pso-miR-2161, and pso-miR-408, were found to regulate the alkaloid synthesis pathway in the Opium poppy plant (species of flowering plant in the *Papaveraceae* family), according to research published in 2014. The pso-miRNA-2161 functions as an intermediary molecule in the synthesis of benzylisoquinoline alkaloids and is regulated by the mRNAs for the enzymes S-adenosyl-L-methionine, 30-hydroxy-*N*-methylcoclaurine, and 40-O methyltransferase [34].

Essential oils are important secondary metabolites of plants that have been used in a wide range of ethnobotanical medicines since the beginning of time. Essential oils are created via two intricate natural biochemical processes that involve several enzyme activities. Isopentenyl diphosphate (IPP) and its isomer dimethylallyl diphosphate (DMAPP), the universal precursors of essential oil biosynthesis, are produced by either the cytosolic enzymatic MVA (mevalonic acid), also known as the 2-C-methylerythritol-4-phosphate pathway or the plastidic and enzymatic 1-deoxy-d-xylolose-5-phosphate (DXP) pathway. To create prenyl-diphosphates, which act as substrates for either geranyl diphosphate (GPP; C10) or fernesyl diphosphate in the specific plant cell component (FPP; C15), prenyl diphosphate synthases further combine isopentenyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP). Terpene synthases are a vast category of enzymes that produce essential oils, which are the ultimate terpenoid molecules (TPS) [35, 36].

5 Environmental Factors' Effects on the Production of Phytochemicals in Plants

Weak knowledge exists regarding how the environment affects the quantity of secondary metabolites found in fruits and vegetables. Although descriptions of metabolic pathways are improving, it is still unclear and difficult to quantify how environmental factors affect such pathways. With the exception of light quality, the main hypothesis concerning how environmental conditions affect the production and accumulation of carotenoids and polyphenolic chemicals in fruits and vegetables will be reviewed in this talk. First, there is a connection between the primary and secondary metabolisms. The biosynthesis of carotenoids and polyphenolic compounds is influenced by the carbon status. Second, the importance of secondary metabolites in helping plants adapt to their environment is now generally acknowledged. Reactive oxygen species (ROS), specifically, may be involved in how biotic and abiotic constraints affect the synthesis of carotenoids, according to substantial evidence.

In order to help farmers in developed and developing countries produce fruits and vegetables with higher phytochemical concentrations without using genetically modified organisms (GMOs), all of these hypotheses and modeling prospects are presented and discussed from a controlling point of view. According to a theory that is being debated, fruits and vegetables grown organically have higher amounts of polyphenols and perhaps carotenoids, which may have health advantages. The importance of secondary metabolites in plants' ability to adapt to their environment is now well-known, which brings us to our second point. Reactive oxygen species (ROS) may operate as a mediator between biotic and abiotic limitations and the synthesis of carotenoids, according to data, which strongly support this theory. All of these concepts and the possibility of modeling are presented and analyzed from a controlling standpoint. It has been hypothesized that fruits and vegetables grown using organic methods contain greater levels of polyphenols and perhaps carotenoids, which in turn contribute to their health advantages [37].

5.1 Secondary Metabolites in Plants as a Function of Light Intensity

A well-known indole alkaloid called camptothecin reacts to environmental stresses, and changes in light irradiation conditions can affect how quickly it accumulates. Overshadowing, for instance, has been shown to cause biochemical alterations in plants, particularly leaves. For instance, *Camptotheca acuminata*'s leaves showed a higher concentration of camptothecin under severe shadowing than the lateral roots did during 100% full sunshine (as compared to 27% full sunlight) [38].

5.2 Bioavailability of Phytochemicals

In vitro and animal studies have demonstrated that several phytochemicals offer a great deal of promise for application in therapeutic settings. When these drugs are employed clinically, however, this is frequently not the case; a lot of them are limited because of their xenobiotic activity (the presence of foreign molecules) before they reach their intended organ or tissue. Phytochemicals frequently encounter difficulties due to their quick metabolism and indigestibility. Additionally, they may be impacted by other phytochemicals included in the consumed material as well as interactions between various plant components [39, 40].

5.3 Biogenesis of Phenolic Compounds

Phenolic compound biogenesis Plant PC molecule architectures often differ greatly from one another. Contrarily, simple phenols have just one aromatic ring, oxy-benzoic and oxyric acids have two rings plus a tri-carbon fragment, flavonoids have two rings, and oxy-coumarins contain benzene and pyrone rings. Simple phenols also have just one aromatic ring. PCs differ in where they are located inside the molecule and have distinct oxy-group numbers. Based on it, there are mono-, di-, and polyphenols. In PCs, the hydrogen of the oxy group may occasionally be replaced by methyl or another functional radical. It was discovered that PCs are produced through the polyketide pathways acetate-malonate and shikimate. Along the shikimate route, phenyl-propanoid substances such as hydroxy-cinnamic acids and coumarins are produced. The major enzyme of the phenylpropanoid pathway, phenylalanine ammonia-lyase (PAL), uses the phenylalanine aromatic amino acid, which is produced by the shikimate route. This activated p-coumarate reacts with malonyl-CoA and the essential enzyme chalconsintase to form chalcones. The latter deaminates aminocinnamic acid, resulting in the production of p-coumarin CoA. Naringenin, a precursor to other flavonoids, is produced as a result of the product's isomerization. The PAL enzyme, which changes phenylalanine from deamination to transcinnamic acid, was initially described in the chapter [41].

Information on the function of the PAL was obtained by examining how the enzyme activity was impacted by biotic and abiotic stressors (light, wound, phytopathogens, and alterations in mineral balance). A de novo link was discovered between the expression of the relevant gene, PCs accumulation, and PAL biosynthesis. Other well-established regulation mechanisms were founded on the feedback principle in transgenic tobacco plants and posttranslational phosphorylation changes of the enzyme. The acetate-malonate route for PC production is common in lichens, fungi, and microorganisms. Through its participation, flowering plants produce flavonoids, with the acetate-malonate route creating one ring and the shikimic pathway creating the phenylpropanoid ring. When the molecules condense, coumarin-CoA and three molecules of malonylCoA result in the formation of a chalcone. The process is catalyzed by chalcon synthase, the main enzyme in the polyphenol flavonoid biosynthesis block. The generated tetraoxychalcon is converted into the flavanone naringenin, which begins nearly all other families of flavonoids with the exception of chalcones and dihydrochalcones. Flavonoids, which are derivatives of 3-phenylchroman, are also capable of producing a large number of isoflavonoids, which are derivatives of 2-phenylchroman. It was discovered that the intermediates and final products of the flavonoid pathway accumulated where they were synthesized. In spite of the tremendous structural variety of phenolic compounds, there are only two primary biosynthetic routes for them: shikimic and acetate-malonate. In higher plants, the shikimic pathway is a crucial part of cellular metabolism [41].

5.4 Time-Scale Variability

Most frequently, each population of plants has a unique and wild type secondary metabolite profile. Plant secondary metabolite profiles and contents can be linked to a number of variables, including the season, hour of the day, tissue age, and temperature because this particular attribute is time-dependent. Sesquiterpene lactones and flavonoids accumulate in the *Vernonieae* species (*Asteraceae*), which are Indigenous to central Brazil. *Lychnophora* spp. and *Eremanthus* spp. are two examples of plants from this Compositae that display a range of metabolite profiles and bioactivity. The volatile terpenes from *Virola surinamensis* also accumulate in an unusual manner; they modify their rates of accumulation in response to the seasonal patterns of the Amazon jungle. For instance, evening monoterpene levels are lower during a rainy winter. Seasonal patterns of secondary metabolite synthesis in plants can be explained in terms of UV radiation, humidity, and high temperatures. For instance, the elder leaves of *Medicago sativa* accumulate summer cuts, medicagenic acid, and saponins [42].

6 Traditional Approaches for Production and Processing of Plant-Derived Phytochemicals

Podophyllotoxin Aryltetralin Lignin

Given that *Podophyllum peltatum*, *P. hexandrum*, *Linum reader*, and *L. narbonese* develop themselves in the rhizomes and roots of some plants, it is possible that these substances contain aryltetralins. The inventory of PPT from natural sources have been constrained as a result of overharvesting and environmental risks. In addition to being an effective anticancer agent by itself, it also acts as a precursor for the creation of a number of semisynthetic derivatives, including etoposide and teniposide, both of which have significant commercial value because of their effectiveness against acute leukemia, testicular/small cell lung cancer, Hodgkin and non-Hodgkin's carcinoma, and, respectively, acute leukemia and non-Hodgkin's carcinoma. PPT functions primarily by inhibiting Topo II, which blocks protein tyrosine kinase and causes DNA damage and cell death by stopping the cell cycle in the G2 phase. The organic combination of this emulsion is sensitive because of the complex chemical makeup and great cost-effectiveness [43].

6.1 The Health Benefits of Phytochemicals Derived from *Catharanthus roseus:* Possibility of Use as an Anticancer Drug

Vinblastine, vincristine, vindoline, vindolidine, vindolicine, vindolinine, and vindogentianine are a few of the anticancer alkaloids found in *C. roseus*. They alter the dynamics of microtubules to cease cell division and start apoptosis. Vinblastine and vincristine were the first of these plant-derived anticancer medications to be used in clinical settings. Hodgkin's disease, lymphosarcoma, choriocarcinoma, neuroblastoma, breast carcinoma, and various organs in acute and chronic leukemia have all been treated with vinblastine sulfate. Vincristine sulfate, an oxidized form of vinblastine, is effective in treating reticulum cell sarcoma, lymphocytic leukemia, Hodgkin's disease, Wilkins' tumor, and acute leukemia (in children). It also stops mitosis during the metaphase.

Other indole alkaloids from *C. roseus* have been demonstrated to be sensitive to a variety of cancer cell lines. It was reported that the human promyelocytic leukemia HL-60 cell line was repressed by catharoseumine, a novel monoterpenoid indole alkaloid that was isolated from the entire plant of *C. roseus*. The IC50 value for this alkaloid is 6.28 M, making it a potent inhibitor. In addition, three newly isolated dimeric indole alkaloids – 14′,15′-didehydrocyclovinblastine, 17-deacetoxycyclovinblastine, and 17-deacetoxyvinamidine – and five previously identified substances – vinamidine, leurosine, catharine, cycloleurosine, and leurosidine – exhibited an IC50 range of 0.73–10.67 M in vitro inhibition of cell viability against the human breast. It is significant that the recently identified bisindole alkaloid cathachunine from *C. roseus* showed antitumor effects on human leukemia cells but significantly less cytotoxicity than on healthy human endothelium cells, demonstrating that it only reduced activity on leukemia cells. Along with examining each individual *C. roseus* alkaloid's anticancer properties, the effects of the crude extract of *C. roseus* as a whole were investigated on a number of cancer cell lines. Recent investigations have demonstrated the *C. roseus* root and stem extract's strong in vitro cytotoxic activity against a number of cancer cell lines. Similar to this, Fernández-Pérez et al.'s work showed that the potent anticancer activity of the indole alkaloid-enriched extract from *C. roseus* cell cultures was not a result of a single molecule acting alone but rather a result of the combined action of multiple bioactive substances. These findings demonstrated how the bioactive elements in *C. roseus*, which have already been discovered in other plant materials and are being explored as a therapy option, cooperate to positively combat cancer cells. By increasing bioavailability, lowering metabolism and excretion, complementing modes of action, reversing resistance, and mitigating adverse effects, substances with little to no activity compared to the primary active components assist the active component in reaching the target. This support makes this phenomenon conceivable [44].

6.2 Phytochemicals and Therapeutic Plants That Fight Viruses

Many plants have been utilized in medicine from the beginning of time because of their powerful healing abilities. Many of these herbs have been used in conventional medicine to treat viral illnesses. The main findings regarding plant extracts' antiviral properties are compiled in Table 1. Only a few of the listed extracts underwent in vivo testing; other extracts were examined in cell culture. Proteins, terpenes, flavonoids, alkaloids, and other glycosides were crude extracts from which several phytochemicals were found, extracted from, and purified. Several plants contain antiviral compounds. For instance, rutin, a flavonoid glycoside found in numerous plants, is effective against the avian influenza virus as well as the HSV-1, HSV-2, and parainfluenza-3 viruses. Quercetin, abundant in plants aglycone of rutin, may prevent a variety of viruses that might cause sickness. The highly pathogenic influenza virus, dengue type 2 virus, HSV-1, poliovirus, adenovirus, Epstein-Barr virus, Mayaro virus, Japanese encephalitis virus, respiratory syncytial virus, and HCV are among these viruses. Its antiviral action mode was looked into in a few instances. Its capacity to inhibit the activity of a subset of heat shock proteins (HSPs) involved in viral IRES (internal ribosome entry site) translation is one well-known means of suppressing HCV. In response to stress, cells create proteins known as HSPs. Inhibiting HCV replication and the HCV NS3 protease in a subgenomic HCV RNA replicon cell culture is another approach. Inhibitions of rhinovirus pathogenic processes like endocytosis, viral genome recording, and protein synthesis are also caused by quercetin. According to a different study, quercetin acts in a more targeted manner, slowing down dengue virus type-2 replication but not viral attachment and entry processes. In addition, quercetin and three additional flavonoids – the 3,3′,4′,5,5′,7-hexahydroxyflavone in myricetin, the 3,3′,4′,5,6,7-hexahydroxyflavone in quercetagetin, and the 5,6,7-trihydroxyflavone in baicalein – effectively inhibited the reverse transcriptases of the HIV and Rauscher murine leukemia virus (RLV). Myricetin is a flavonoid that is found in a variety of fruits, berries, nuts, vegetables, and wild plants [45].

Table 1 Phytochemical activities of herbal plants with antiviral features

Herbal plants	Nature of extract	Disease	Phytochemical activity	References
Cyperus rotundus	Hydro-alcohol-by-pass extract	HBV	Cyperene-3, 8-dione, 14-hydroxycyperotundone, 14acetoxycyperotundone, and 3βhydroxycyperenoic	[69]
Viola diffusa	Ethanolic extract	HBV	2β-hydroxy-3,4secofriedelolactone-27-oic acid, 2β,28βdihydroxy3,4secofriedelolactone-27oic acid, 2β, 30β-dihydroxy-3,4-secofriedelolactone-27-lactone and stigmastane, and stigmast-25-ene-3β,5α,6β-triol	[70]
Vitisla brusca	Methanol extract	SA-11 and human (HCR3) rotaviruses	Resveratrol, piceatannol, transarachidin-1, and transarachidin-3	[71]
Ficus benjamina	Ethanolic extract	HSV-1, HSV-2	Rutin, 3-O-rutinoside, and 3-O-robinobioside of kaempferol	[72]
Curcuma longa L.	Curcumin plant extract	HTLV-1	Decreased regulation of Jun D protein in HTLV-1-infected T-cell lines	[73]
Tulbaghia violacea	Plant extract	Cardiovascular disease (CVD)	Inhibition of angiotensin-converting enzyme (ACE)	[74]
Helichrysum ceres	Plant extract	Cardiovascular disease (CVD)	Lowers blood pressure	[75]
Cucurbita pepo	Aqueous extract	Nonalcoholic fatty liver disease	α-glucosidase and amylase inhibition lowers hyperglycemia	[76]
Camellia sinensis	Methanolic extract	Diabetes mellitus	Epigallocathechin-3-gallate	[77]
Capparis spinose	Hydroalcoholic extract	HIV-1	Protein	[78]

(continued)

Table 1 (continued)

Herbal plants	Nature of extract	Disease	Phytochemical activity	References
Cyperus rotundus	Hydroalcoholic extract	HBV	Cyperene-3, 8-dione, sugetriol-3, 9-diacetate, 14-hydroxycyperotundone, 14-acetoxycyperotundone, and 3-hydroxycyperenoic acid	[69]
Diospyros kaki	Aqueous extracts	Human rotavirus	Lutein, vitexin, apigenin-7-O-glucoside, licoflavonol, glyasperin D, 18-glycyrrhetinic acid, and licocoumarone	[71]
Cassia alata	Crude extracts	Antioxidant protective effects	Hepatotoxin-induced liver damage in rats	[79]
Leucojum vernum	Methanolic extract	HIV-1	2-O-acetyllycorine and homolycorine	[80]

Elagic acid and myricetin from the aronia fruit were found to be effective in vivo and in cell cultures against a range of influenza subtypes, including an oseltamivir-resistant form. Many plants include the flavone aglycone apigenin (4′,5,7-trihydroxyflavone), which has a wide range of antiviral properties against the influenza A virus, enterovirus 71, foot-and-mouth disease virus, HCV, and ASFV. Notably, a number of plant-derived flavonoids are known to possess antiviral effects. Inhibiting the synthesis of nucleoproteins is one mechanism through which six phytochemicals (apigenin, baicalein, biochanin A, kaempferol, luteolin, and naringenin) out of 22 different flavonoids were effective against the avian influenza H5N1 virus in human lung epithelial (A549) cells. Baicalin, the glucuronide of baicalein, was also effective against enterovirus, dengue virus, respiratory syncytial virus, Newcastle disease virus, human immunodeficiency virus, and hepatitis B virus in addition to being effective against a variety of viruses. There are numerous ideas as to how baicalin prevents the spread of viruses. According to a study on the avian influenza H5N1 virus, baicalin, for instance, decreases the generation of HBV, the templates for viral proteins, and HBV-DNA synthesis and reduces IL-6 and IL-8 production without altering IP-10 levels. Oleanolic acid and ursolic acid, two triterpenoids that are abundant in plants, have been shown to inhibit enterovirus 71 replication and reduce the pathogenicity of HCV NS5B RdRp, respectively. Standardized elderberry extract, which also has *Sambucus nigra* L. as an active component, is a successful treatment for fever, colds, and influenza A and B [45].

7 Phytochemicals and Benefits of Selected Agricultural Residues in Livestock Nutritive Diet or Dietary Needs

7.1 *Malus domestica* Wastes Produced Nutritious Dietary Resources

Apple trash contains pectin as well as a wide range of other advantageous components, including carbohydrates, vitamin C, dietary fiber, and minerals. It has been demonstrated that adding apple pulp to chickens' diets considerably improves their productivity and fertility rates. Due to the high pectin content of apple pulp, it is essential to add multienzymes to the apple pulp ration to ensure that the pulp is more effective for chickens. Additionally, it has been claimed that feeding poultry apple pulp decreases their uric acid levels dramatically while increasing their blood glucose levels [46].

7.2 Nutritious Dietary Resources Derived from Avocado Skin and Seeds

Animal feed has occasionally included avocados. Animal waste feed made from avocados is derived from the fruit that is meant for human consumption. Because certain commercial requirements and industrial procedures are not followed, the seeds and skin are, however, thrown away. Unsaturated fatty acids, tocopherols, and other significant phytochemical components are abundant in avocados and aid animals – including pigs – in triggering healthy biological responses. According to Grageola et al., avocado eating helps to maintain the nitrogen and energy balances of young pigs. Furthermore, according to Wood et al., the taste and juiciness of avocados are regarded as sensory attributes that could influence the intramuscular fat (IMF) and fatty acid composition of meat. In their study, Hernandez-Lopez et al. contrasted the muscle of the treated pigs' experimental group with the muscle of a different group of pigs fed a control diet devoid of avocado waste. We investigated the oxidative stress, color stability, and fat content of pigs. Avocado consumption significantly reduced the amount of lipids in the LTL muscle, increased the content of unsaturated fats, and reduced intramuscular fat (IMF) [46].

7.3 Biorefinery of Waste from Plants

Each year, agricultural activities and subsequent food processing result in large amounts of plant-based waste. The majority of plant-based trash comes from agricultural residues other than grains, such as maize stover, wheat straw, and rice straw. Due to their wide availability and relatively uniform composition, these agricultural byproducts are appealing feedstocks for large-scale industrial biorefineries. The beverage industry, which produces a lot of pomace such as grape pomace and apple pomace, is the source of the second-largest amount of food waste. The pomace also includes significant amounts of high-value functional components, including polyphenols, essential oils, and vitamins, in addition to plant fibers. Waste made from plants mostly contains cellulose, hemicellulose, and lignin as its main structural components. Biofuels, organic acids, and nano-cellulose materials can all be made from cellulose. Hemicellulose can be broken down into the important chemical compounds xylose, xylite, and furfural. Lignin is a natural binder and glue that can also be utilized to create useful chemicals like phenols. Other deposits rich in plant-derived waste, such as sugars, proteins, and oils, as well as phytochemicals, should also be collected because they have the potential for a variety of uses in addition to structural compounds. Well-balanced essential amino acid proteins made from plant waste can be added to food to enhance its sensory and functional properties. For instance, soy protein has been used in imitation cheese, soy milk, and other goods. Two phytochemicals, polyphenols and carotenoids, have been related to health-promoting effects like lowering cholesterol and lipid oxidation. In addition to having health-promoting properties, the inclusion of these antioxidant chemicals in food matrices can increase the shelf life of those items and prevent the development of bad tastes. Vegetable oils present in plant-derived waste can be used to make sugar-based surfactants (such alkyl polyglucosides) with low toxicity and good detergent properties, in contrast to typical surfactants manufactured from fossil fuels. The extraction, purification, and manufacture of valuable chemicals, minerals, and biofuels from plant-derived wastes are therefore the subject of increased research [47].

8 Modern Approaches to Ameliorate Phytochemicals' Biogenesis to Mitigate Human Health Demands

Due to new knowledge of the hazardous ingredients and reactions contained in those synthetic medicines, many users of synthetic drugs and cosmetics are now becoming more interested in natural items. Due to their safety, ease of synthesis, and widespread availability, phytomedicines are among the natural therapeutic compounds derived from microorganisms, insects, and mammals that are acknowledged to have significant pharmaceutical value. The use of plant-based natural products in pharmaceutical, nutraceutical, and dietary supplement manufacturing has increased recently [48]. However, suggested plant molecular factories for the production of isoprenoids, including tocopherols and carotenoids, an important component of the human diet, have been made. Numerous plant-based natural products are utilized as medications to treat severe illnesses including HIV and newly emerging diseases that have recently gained popularity [49, 50]. The fact that natural phytoproducts are being used to treat neurological and cardiovascular disorders, such as Alzheimer's [51], diabetic neuropathy [52], and cardiomyopathy [53], is remarkable. Natural plant-based medication products are generated using contemporary safety precautions and isolation techniques, and they are intended to treat drug-sensitive and multidrug-resistant infections [54, 55]. Yellow fever [56], Zika fever [57], TB [58], and other protozoal diseases are all wiped out thanks to the improved mosquitocidal properties of natural phytomedicines.

The most recent figures for the years 1981–2014 show that around 42% of the 1562 newly approved medications are from natural medicinal plants [59]. Meanwhile, more than half of the 1073 novel types of small molecule medications that were licensed between 1981 and 2010 were natural bioactive chemicals [60]. It has proven difficult due to the complexity of the chemical compositions of natural substances, the wide variation in their structures, nonspecific adsorptions, false positive results, and the undetectable trace levels of active components [61]. These obstacles represent the actual challenges in the identification of bioactive components and prediction of their possible mechanisms of action. The good choice of in vitro and in vivo assays for screening of the plant bioactivity is a very important step in drug discovery from plants. The chosen bioassays should be characterized by simplicity together with good sensitivity and reproducibility. Numerous in vitro models using purified proteins, cell-based target-oriented or phenotypic assays, can be utilized for evaluation of the biological activities of natural products, in addition to the isolated tissues or organ models, and also in vivo preclinical animal models can be used [60]. Previously, plant-derived extracts and compounds were screened using the forward pharmacological strategy, which starts with in vivo animal tests, organ, tissue models, or bacterial preparations before moving on to in vitro testing to determine the mechanism of action. Prior to characterizing the molecular target of the active compounds, forward pharmacology, also known as phenotypic drug discovery, begins with the identification of phenotypic changes in complex biological systems to assess functional activity. Until the development of modern molecular biology technologies and the Human Genome Project, this traditional approach to drug discovery was the main approach [60]. A reverse pharmacology approach is used when the screening process begins with the testing of plant-derived compounds (or "libraries") against precharacterized disease-relevant protein targets in order to identify "hits," the biologically active compounds that are then studied using in vivo animal models with the goal of their validation. The only difference between the forward and reverse pharmacology techniques is the point at which the tests are used. Reverse pharmacology starts with the identification of the promising pharmacological target in order to produce the promising hit compounds, which are subsequently verified in vivo and regarded as target-directed drug discovery [62]. Molecular structures that may be recognized compounds produced from plants can be suggested to have a protein ligand binding property using in silico simulations. Compounds that perform well in in silico simulations can make excellent candidates for additional experimental testing. The success rates of applying virtual screening for activity forecasts are rising [63]. In silico simulations can be a useful filtration tool for both identifying ADME/T features and forecasting novel behaviors for a known natural substance [64]. New binding sites on proteins with known structures can also be found using these computational methods. To locate solvent-accessible protein surface cavities that serve as significant ligand binding sites and may afterward be computationally analyzed, pocket finders are used [65]. The biological effects of the active substances might be seen by controlling how signals are transmitted throughout the body and preserving a healthy metabolism. Alternately, they might work through interacting with enzymes and receptors, which are common disease-related medication targets [66]. The discovery of new therapeutic targets could be aided by the screening of biologically active natural compounds that target these enzymes and receptors, as well as by understanding the molecular activity of bioactive small natural molecules [67].

9 Conclusion and Future Outlook

Traditional herbal medicine practitioners have been verbally transmitting instructions on how to make herbal medicines for decades. Normally, they do not keep records, but the WHO has recently made public efforts to chronicle the use of medicinal herbs by conventional healers across the world. As a result, initiatives to record ethnomedical information on plants have risen in many developing nations. It has made it simpler to prove their pharmacological efficacy through science. People will be more knowledgeable and happy with effective medication treatment and improved health status after these regional ethnomedical formulations have undergone thorough scientific evaluation and dissemination [67]. Plants continue to hold promise as sources of medicinal medicines and as a foundation for the extraction of semisynthetic chemical compounds used in the food, cosmetic, and fragrance industries [68]. Additionally, the acceptability and use of plant-based health care products in the cosmetics business is growing. Therefore, medicinal plants have a significant role in the growth of the whole economy as a source of both commercial money and healthcare. The sales of herbal supplements and treatments are expected to increase in the next years, and meeting this need will be a booming market for herbals [66].

This implies that the pharmaceutical industry, medical professionals, and researchers will go to emerging nations like China, India, and others for their supply. Because these nations are the top exporters of raw herbal materials and have the widest variety of medicinal plant species. The use of plants in their whole or as pharmaceuticals will continue in the future due to their rising appeal as a secure and less expensive alternative to traditional medicinal agents.

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Proximate Analysis of Herbal Drugs: Methods, Relevance, and Quality Control Aspects

35

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Abstract

Since the beginning of human history, people all over the world have relied on the healing properties of plants to treat a variety of illnesses. To sustain life, carbohydrates, lipids, fibre, and protein are all necessary nutrients. Studying plants and their immediate environments is essential for establishing the nutritional significance of plants. The proximate composition of the food is made up of a number of different constituents, including the moisture content, ash content, fat content, protein content, and carbohydrate content. The food industry may be concerned with these food materials for product improvement, quality control (QC), or monitoring purposes. Rapid QC methods or more precise but time-consuming official processes may be applied. Assessment of a consistent and representative sample, as well as accurate results, requires meticulous attention to detail during the sample collection and preparation phases. The valuation methods for moisture content, ash value, crude lipid, total carbohydrates, starch, total free amino acids, and total proteins are presented in a clear manner. The chemical composition of the feed determines its potential nutritive value; hence, the proximate principles are established first in the evaluation of feed quality. Considering their nutritional relevance can help determine their value because many herbal plant species are used for food in addition to their medical properties.

Keywords

Plant materials · Herbal drug · Proximate analysis · Methods · Relevance · Quality control

Abbreviations

AC	Ash content
AIA	Acid-insoluble ash
AOAC	Association of official analytical chemists
API	Ayurvedic pharmacopeia of India
CF	Crude fat
CFC	Crude fibre content
CFi	Crude fibre
CYP	Cytochromes p450
DW	Distilled water
EPA	Environmental protection agency

EV	Extractive values
FAO	Food and agriculture organization
FDA	Food and drug administration
FI	Foaming index
GGE	Gram glucose equivalent
HCl	Hydrochloric acid
ISO	International organization for standardization
KF	Karl Fischer
LOD	Loss on drying
MC	Moisture content
NF	National formulary
OTC	Over-the-counter
PA	Proximate analysis
QC	Quality control
QCP	Quality control parameters
RT	Room temperature
SA	Sulphated ash
SI	Swelling index
TA	Total ash
TAC	Total ash content
TCC	Total carbohydrate content
TCM	Traditional Chinese medicine
TPC	Total protein content
USP	United States pharmacopeia
UV	Ultra-violet
WB	Water bath
WHO	World Health Organization
WSA	Water soluble ash

1 Introduction

Humans often use natural bioactive compounds derived from plants and soil to cure a variety of ailments [1]. Because of the physiological roles of these natural bioactive compounds, which are made up of phytonutrients and mineral elements [2]. Humans need to get enough nutrition, either in the form of a wide range of organic components or in a specific combination, in order for their physiological metabolism to work properly [3]. Energy-producing macronutrients such as carbohydrates, protein, and fat are necessary in large quantities, whereas critical micronutrients, including minerals and vitamins, are needed in small quantities [4]. Macronutrients are substances that are necessary in large quantities to provide energy. Microelements are added to many plant species to make them more therapeutic and less likely to get sick [5].

Herbal medications are in high demand around the world for basic healthcare because of the improved margins of safety and low cost that come along with using them [6]. The quality control (QC) of herbal medications results in a significant number of headaches. Therefore, the first and most important step is to select the appropriate kind of plant material, which includes substances that are effective in terms of treatment [7]. Herbal medicine manufacturing is done on a large scale, and the manufacturers face many challenges, including the use of low-quality raw materials, the absence of authenticating raw materials, the lack of availability of standards, the absence of quality control parameters (QCP), and the lack of adequate standardization procedures for individual medications [8, 9]. Consumers, on the other hand, are more likely to go for brands that adhere to established criteria. Therefore, standardizing Ayurvedic medications is an absolute necessity if one wants to ensure the efficacy, potency, safety, and purity of those preparations [10]. The use of combinations of herbal substances in traditional medicine further complicates the already complex issue of ensuring the quality of herbal products, which already present a number of their own particular challenges in this regard [11]. Therefore, when it comes to herbal treatments and goods, the standardization process needs to include the entirety of the relevant field of study, beginning with the growth of medicinal plants and continuing all the way to their clinical applications [12]. The World Health Organization (WHO) is actively involved in the standardization and quality control of herbal crude pharmaceuticals in order to oversee the physicochemical and proximate evaluation of these drugs [13]. Safety, efficiency, and sustainability assessments of completed products; documentation of safety and risk based on experience; provision of user product descriptions; and commercialization are all included in this evaluation [14]. In spite of the availability of current technology, pharmacognostic investigations, physicochemical studies, and proximate studies are both more reliable methods for the identification of plant-based medications. The proximate parameters are required for verifying the medications' identities as well as establishing their quality and level of purity [15].

Since many plant species valued for their therapeutic properties are also consumed for sustenance, assessing their nutritional relevance might shed light on their true value [16]. The majority of the diet is composed of carbohydrates, proteins, and fats, while minerals and vitamins play a relatively modest role. The nutritional content of plants is crucial because they make up the bulk of our diet [17]. In addition to these biochemicals, it has been noted that the moisture, fibre, ash, and energy values of different vegetables are significant for human health and soil quality [18]. In addition to containing pharmacologically significant phytochemicals, each species of medicinal plant also has its own unique nutritional content. We cannot live without these nutrients, as they are crucial to our bodily processes [19]. Carbohydrates, lipids, and proteins are examples of essential nutrients and biochemicals that help humans meet their energy and metabolic needs. Several substances having medical significance, such as vitamins, minerals, antioxidants, and anticarcinogens, are present in plants due to their diverse chemical composition [20]. Fibres, omega-3 fatty acids, vitamins, and minerals are all necessary for human wellness, and plants have enough of them [21].

The measurement of moisture content (MC), ash content (AC), crude fibre (CFi), total protein (TP), total energy value, volatile matter, and nutritional value has been widely used for over 160 years. Proximate analysis (PA) is a significant technique for the determination of the nutritional qualities of plant material and diets [22]. The quantitative measurement of bioactive molecules in plant materials and food products is referred to as "proximate analysis." Total protein content, crude fat, crude fibre, moisture content, ash content, total carbohydrates, total available energy, and nutritive value are mainly components of proximate analysis and may all be measured using a variety of methods, including extraction, Kjeldahl, Lowery, Biurets, Bradford, etc. [23]. It is possible to measure the relative quantities of protein, fat, moisture, fibre, ash, carbohydrate, nutritive value, and total available energy present in plant material using proximate analysis. These nutritional factors may be of importance to the food-service industry for a variety of reasons, including the production process, quality assurance, and regulatory compliance [24]. Total protein, total carbohydrates, crude fat, and crude fibre are all parts of plant materials that add up to the total amount of energy they provide. Moisture content and ash content are the only parts that add up to the plant material's mass. For determining the nutritional relevance of medicinal herbs, the proximate analysis of the plants plays an important role. The proximate analysis is a crucial index for determining how to categorize the nutritional value of a food material [25].

2 Moisture Content (MC)

The amount of water available in a plant component is referred to as its moisture content. In a broad variety of scientific and technical fields, it is employed as a tool. The moisture content can be determined by taking the dry weight and subtracting it from the initial weight. It is frequently expressed as the dew point temperature in grams of water per cubic metre of air (g/m^3) [26]. There are numerous procedures for measuring moisture content. Loss on drying (LOD) and Karl Fischer (KF) titration are the major procedures employed. Using conventional methods, the moisture content is measured in a hot air oven, as shown in the following scheme (Fig. 1) [27].

2.1 Moisture Content Protocol

1. Instantly weigh a pure sample and note the weight. This weight is called the "weight of the sample."

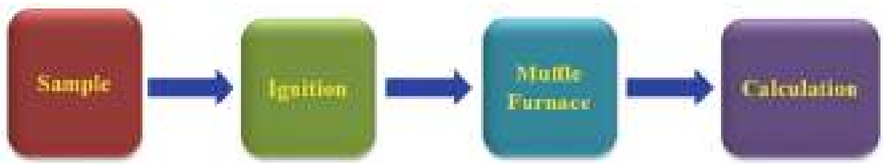


Fig. 1 Scheme of determination of moisture content

2. Dry the samples to a consistent weight using appropriate drying equipment.
4. Before using the sample, allow it to cool.
5. After allowing the sample to cool, reweigh it and write down the new weight. This weight is called the "dry weight of the samples."

2.2 Calculation

The following formula is used to estimate the percentage of moisture content of the sample plant material [26]:

$$\%\mathrm{Moisturecontent} = 1 - \frac{\mathrm{Weight}_{(\mathrm{dry\,sample})}}{\mathrm{Weight}_{(\mathrm{wet\,sample})}} \times 100$$

2.3 Relevance of Moisture Content

Moisture is an inherent constituent of crude drugs, and as much of it as is practically possible should be removed from the substance. One of the processes involved in transforming harvested plant material into crude pharmaceuticals is drying, which has a major impact on both quality and purity. The plant medicine is first garbled or washed to remove any remaining soil or debris, and then it is dried [28]. The drying of fresh material serves multiple purposes, the most important of which are to support the preservation process, "fix" the components (which means to stop the enzymatic or hydrolytic reactions from happening, which could alter the chemical make-up of the medicine), reduce their weight and mass, and allow further comminution, which is the process of crushing anything into a powder [29].

Inadequate drying makes it more likely that moulds and bacteria will cause spoilage, and it also opens the door for the enzymatic breakdown of active components. The typical moulds and bacteria that can grow on or in pharmaceuticals have relatively low moisture needs for their active growth; therefore, it is possible to find them on or in the drugs [30]. As a result, the process of drying the medicine needs to get the amount of moisture it contains down to a level that is below this crucial or threshold level. It is challenging to establish an accurate upper limit for the amount of moisture that can be tolerated in crude drugs [31]. This is due to the fact that enzymatic activity necessitates a certain level of moisture, and microbial demands for water content also vary depending on species and other environmental conditions (e.g., temperature, O_2 and CO_2 pressure, and light). Both the USP and the NF rarely make promises in this area [32]. If the humidity content of the medicine is decreased to 6% or less, however, it is possible to preserve the vast majority of medications without risk. One major exclusion is agar, which can have up to 20% moisture according to USP standards. Not only is the drug's final degree of dryness crucial, but also the rate of moisture removal and the circumstances under which it is eliminated are of equal importance [33]. If the rate is excessively slow, there is a high risk that a significant amount of spoilage may develop before the drying process is finished. As a result, drying should be completed as quickly as is practically practicable by the application of sound techniques. The amount of water content and other characteristics of the medications determine the amount of time needed for the drying process, which can range anywhere from a few hours to many weeks. The length of time that an elevated drying temperature is maintained is a crucial aspect to keep in mind because increasing the temperature accelerates destructive enzymatic reactions [34]. Most of these reactions, however, which are frequently encountered during the manufacture of crude medicines, only show a net effect up to about 45 °C. Yet, greater temperatures reduce the amount of time necessary for the drying process, and as a result, the amount of time that harmful reactions can take place is reduced as well [35]. Drying is accomplished through a variety of processes, the specifics of which might vary widely, but broadly speaking, drying processes can be natural or artificial. Physical ways include applying higher temperatures and/or lower pressures (vacuum), as well as infrared and radio-frequency radiation; chemical approaches include applying desiccants; or they may be biological, which involves the use of organisms that are genetically engineered to carry out the artificial method [36].

In most cases, the raw plant materials are subjected to drying and storage in order to serve as specimens for vouchers, in addition to serving extraction and isolation functions [37]. Although this is not the standard method of storage, the Ayurvedic Pharmacopoeia of India (API) does state that many quality assurance standards have been computed using the fresh plant matter without first drying it [38]. This was done without the plant material being subjected to heat. When it comes to stability, the percentage of moisture that a plant material has is quite essential. The following parameters are influenced by the amount of moisture present:

- Microbial growth, which is potentially harmful to the medicine and can take the form of either bacterial or fungal development.
- Enzymatic activity occurring within the plant tissues may pose a risk to the phytochemicals' capacity to maintain their structure.
- The process of drying the material results in a reduction in its overall volume, which is beneficial for the storage of the plant material.
- The process of further comminution and size reduction of the medicament is simplified after it has been dried [39].

3 Total Ash Content

Ash is defined as the inorganic by-product (residue) that remains after organic material in a sample has been ignited or completely oxidized, whichever occurs first. Measuring the ash level of a product is a component of its proximate composition for nutritional assessment, and it is a quality-related feature for specific food products owing to its enormous nutritional importance [40]. Furthermore, ashing is the initial stage in the preparation of a sample for particular elemental analysis, and it is also the most time-consuming. This laboratory activity uses the dry ash method in combination with an electric furnace in order to evaluate the ash content of a range of food items in a controlled environment [41]. The dry ash method is used to calculate the percentage of ash in the sample. The sample materials were burned at a high temperature in a furnace. The leftover inorganic substance is cooled, weighed, and employed to determine the mineral composition of the final product. In order to create an ash solution, 100 mL of 1 M HCl is used to dissolve the ash, and the resulting solution is centrifuged. It is common for people to look at the ash level of foods as part of quality assurance when they are trying to figure out how food works for their bodies. The ash content, measured in terms of a percentage of the dried sample's weight, describes how much of the sample is composed of ash [42].

3.1 Protocol

1. Place a certain amount of the crushed sample material, carefully weighed, in a suitably tarred dish (e.g., silicate or platinum), which has been earlier burned, cooled, and then weighed.
2. Combust the sample by progressively raising the temperature—not over 500 °C for 30 min in a muffle furnace—until it is completely devoid of carbon and ash.
3. Allow the sample to cool before weighing it.
4. Determine the ash content in mg/g [43].

3.2 Calculation

To calculate an estimate of the total ash content of a particular plant material, the following formula can be used [44]:

$$\text{Total Ash Content}\ \% = \mathrm{A}/\mathrm{B} \times 100$$

where A is the weight of ash and B is the weight of the dried plant.

The amount of ash in a crude medication is the amount of residue left over after burning. Typically, it stands for the inorganic salts that are already present in and adhere to the medicine, but it can also include inorganic material that has been added on purpose. In the case of certain specific medications, there is a sizable variation within rather tight bounds [45]. An ash analysis therefore provides information regarding the possible inorganic adulteration of a drug and serves as a basis for determining its identity and cleanliness. Tolerance levels for ash have been set for some legal medications. These standards typically stipulate a maximum allowable level of total ash (TA) or acid-insoluble ash (AIA). Total ash is what is left behind after burning; it consists of two types of ash: "physiological ash," produced when plant matter is burned, and "nonphysiological ash," which comes from ambient contaminants like sand and soil [46]. Physiological ash can be found in the sclereids of the pericarp of colocynth and the cork of liquorice, for instance. Nonphysiological ash includes things like sand or dirt that have been used as contaminants [47]. The main components of ash are carbonates, phosphates, silicates, and silica. Total ash content is not always a reliable indicator of the quality of herbal treatments because many plant components include large amounts of physiological ash, calcium oxalate in particular [48]. Hence, the amount of acid-insoluble ash in Chinese herbal medicine is still another indicator of its efficacy. Acid-insoluble ash refers to the percentage of total ash that remains undissolved after being exposed to aqueous HCl. In general, the amount of impure inorganic stuff can be gauged by looking at the ash or residue produced by an organic substance complex. In most cases, only a small amount of inorganic material is present that is not unpleasant and is difficult to eliminate throughout the purifying process [24]. Ashes help evaluate the potency and safety of powdered crude medicines. Ash that dissolves in water can be used as a proxy for the total amount of inorganic chemicals in a medication. The presence of the acid-insoluble ash, which is predominantly composed of silica, is indicative of contamination with granitic material [49]. Generally speaking, the ash level of fresh herbs is under 5%. In general, ash is not present in pure oils and fats, although it can be found in products like cured bacon (at 6%) and dry beef (at 11.6%) (wet weight basis). There can be zero to 4.1% ash in fats, oils, and lards, and between 0.5% and 5.1% ash in dairy foodstuffs. Ash content ranges from 0.2% to 0.6% in fresh fruit, fruit juice, and melons, and from 2.4% to 3.5% in dried fruits. Ash content is low (0.3% for pure starch and 4.3% for wheat germ) [50].

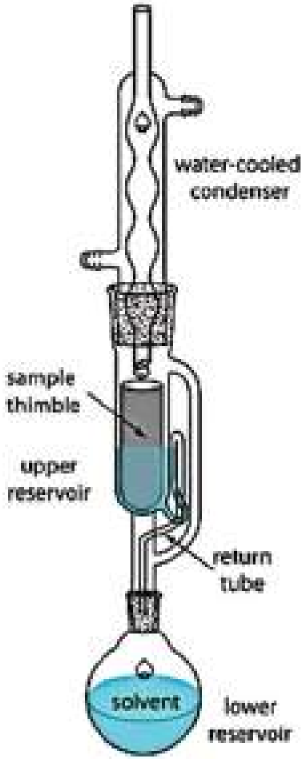
4 Crude Fat

The term “crude fat“refers to the raw or unprocessed combination of fat-soluble substances contained in plant material. The classic measurement of fat in fresh food is crude fat, often called total fatty acid or lipid content. Generally, the word “fat” refers to a group of compounds that are dissolvable in liquids like ether and chloroform but mostly immiscible in water. Fats are unsaturated lipids. Fats are normally solid at room temperature, whereas lipids are fluid [51]. The Food and Drug Administration (FDA) has officially described crude fats as molecules that have a lot of fatty acids in them, as well as triglycerides, phospholipids, non-esterified lipids, and salts [52].

4.1 Principle

Weigh out a specific quantity of the plant powder material. The thimble should be placed within the Soxhlet apparatus (Fig. 2). Put a dried, pre-weighed solvent flask underneath the apparatus, into which you will pour the needed amount of solvent before connecting it to the evaporator. Set the heating mantle to achieve a condensation speed of 4–5 droplets per minute, and then extract for 24 h at RT. After finishing, replace the thimble and use the apparatus to recover petroleum ether. Complete the extraction of petroleum ether in a water bath, followed by 30 min in a dried vacuum at 100 °C. Allow it to cool in the desiccator before weighing it [53].

Fig. 2 Soxhlet apparatus



4.2 Calculation

The following formula is used to determine the crude fat content of the particular sample [54]:

$$\text{Crude fat } \% = \frac{(W_2 - W_1)}{\text{Weight of sample}} \times 100$$

where W_1 is before weight of sample and W_2 is the weight of after weight of sample.

5 Crude Fibre (CF)

Crude fibre is a kind of insoluble fibre that may be observed in the eating component of the plant body. It is essentially cellulosic matter that has been produced as a by-product of the biochemical examination of vegetable ingredients. Crude fibre, on the other hand, is the non-soluble residue left behind after an acidic hydrolysis followed by an alkaline hydrolysis. This remnant comprises genuine cellulosic as well as insoluble lignin, among other things [55]. The amount of undigested lignocellulosic materials, saccharides, cellulose, and other components of this sort found in current diets is measured as “crude fibre.” These constituents have limited nutritional value but contribute the mass required for effective peristaltic movement in the intestine. The cellulosic component is mostly found as a by-product of the biochemical analysis of food [56].

5.1 Principle

A dried plant material that has been extracted with petroleum ether is first treated with a weak acid solution and then with an equally weak base solution. Then the residue is put in a petri dish after it is filtered [39].

5.2 Protocol

To prepare shade-dried plant material, weigh approximately 5 g into a flask. 200 mL concentrated H_2SO_4 is added to the flask and placed it under the condensation (Fig. 3) [57]. The mixture is brought to a boil in 2 min. Purified water is added to maintain volume and wipe away particulates sticking to the edges. Over low heat, it is heated for approximately half an hour, using antifoam if required. Using a vacuum filter, the water is filtered with filter paper in a funnel, and then it is rinsed with hot water to get rid of all the dirt and germs. Transferring the residues to the beaker, 200 mL of warm NaOH solution to the mixture is introduced. The container is replaced under the evaporator and returned to a simmer for 1 min. After 30 min of simmering, drain through a porous crucible. Rinse with hot water containing 1% HCl, then repeat with hot water. Using ethanol or methanol, crush the plant material

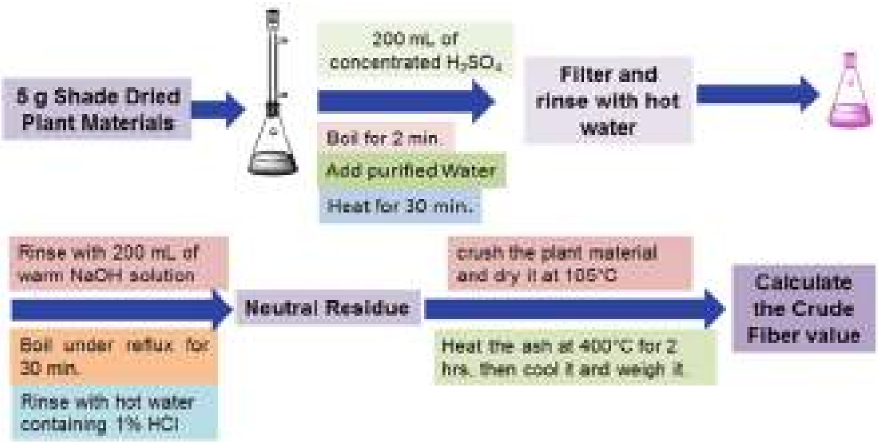


Fig. 3 Scheme for determination of Crude fibre

and dry it at 105 °C before cooling and weighing it. Heat the ash at 400 °C for 2 h, then cool it and weigh it. Calculate the crude fibre value using the following formula [43, 58]:

5.3 Calculation

$$\%\ \text{Crude fiber} = \frac{(\text{Wt. of Crucible} + \text{Dried residue}) - (\text{Wt. Crucible} + \text{Ash residue})}{\text{Weight of Sample}} \times 100$$

It is important to conduct an analysis of the diet that is consumed by humans or animals. More specifically, it is necessary to determine the amount of food that is ingested as well as the extent to which its dietary content is absorbed and maintained in the body. A digestibility study can be completed quickly using a procedure called quantification of faecal output. The determination of a food's crude fibre content is primarily done in order to get an idea of how much of food is indigestible by both animals and humans [59]. The organic compounds that are found in food or fodder that, while being subjected to treatment with chemicals at a certain temperature and concentration, maintain their insoluble state are typically referred to as crude fibres. They are believed to be adulterants due to the characteristics that they possess. A determination of crude fibre is carried out in order to identify any undesirable components that may be present in food or feed [54]. On the other hand, it is more practical to adulterate the medication powder as opposed to the herb or plant in its entirety. Synthetic compounds with the same chemical properties as those present in natural materials are often added for adulteration because they are less expensive. Oils of gaultheria and bitter almond, for example, may have methyl salicylate or benzaldehyde mixed into them. Both of these examples are examples of adulteration [43].

The crude fibre content of a plant can be estimated by boiling the plant tissue with a mild acid to remove the soluble material, then washing, drying, and weighing the remaining material to determine its cellulose, lignin, and cork cell content. The assessment of crude fibre is helpful in determining the amount of adulterants that are present in any given herbal sample, food sample, or fodder sample. The assessment of the crude fibre contributes to the process of distinguishing between the soluble and insoluble components, both of which are obtained from the various plant parts and the plant cells [47]. Crude fibres, unlike ash, have been found to be insoluble in water and resistant to digestion when boiled in H_2SO_4 and NaOH, indicating that they represent the more resilient sub-cellular structures within plants, together with certain less resistant components of cell walls, such as cellulose and pectin. Ash is the only component of the crude fibres that is not resistant to digestion. Chemicals like H_2SO_4 and NaOH are used in the primary technique for determining the amount of crude fibre present in a sample [48]. This technique requires that the material be processed in a specific order. The chemicals described above should be brought to a boil after being diluted to the appropriate concentrations. When the residue has been separated using filtration on a sintered glass filter, it should be washed, weighed, and ashed within a temperature range of 475–500 °C. The amount of crude fibre that was found in the sample is equivalent to the amount of weight lost during the ashing process [60].

6 Total Protein

Proteins are comprised of amino acids, which are organic substances formed of C, H, N, O, or S. Amino acids are the building blocks of protein. As per the FAO, proteins, like amino acids, are the basic components of muscle strength [61]. The quantity of proteins found in plant materials is referred to as the total protein. The Kjeldahl technique, the Dumas procedure, the Lawry method, and quantitative measurement techniques employing UV spectroscopy and refractive index methodologies are the most widely applied techniques to measure the protein content in plant materials and food products [62]. There are benefits and drawbacks to every approach. Direct amino acid testing, the Bradford technique, the Lowry approach, and the Kjeldahl technique were all used to quantify the proteins after they were separated using various procedures [63]. The Kjeldahl technique entails the use of a strong acid to digest food, which results in the production of nitrogen, which is then quantified using titration. As a result, a conversion factor is applied to the food's nitrogen content to derive its protein content [64, 65].

6.1 Kjeldahl Approach

Johann Kjeldahl designed the Kjeldahl technique in 1883. Basically, this is based on the idea that strong acids help food break down, which in turn releases nitrogen, which can be detected with the appropriate titration method [66]. Lab tests for Kjeldahl nitrogen or protein content may be done on a wide range of materials, including food and plant materials. Several scientific organizations, including the AOAC, EPA, and ISO, have approved and standardized the Kjeldahl method. Each stage of the Kjeldahl technique could be divided into three major components: digestion, distillation, and titration (Fig. 4) [26].

6.2 Digestion

By heating the sample in concentrated H_2SO_4, a homogeneous sample can be digested. The ultimate product is a solution of ammonium sulphate. The basic equation for the digestion of organic material is given as follows [39]:

$$\text{Organic}\,N_2 + H_2SO_4 \rightarrow (NH4)_2SO_4 + H_2O + CO_2 + \text{Byproducts}$$

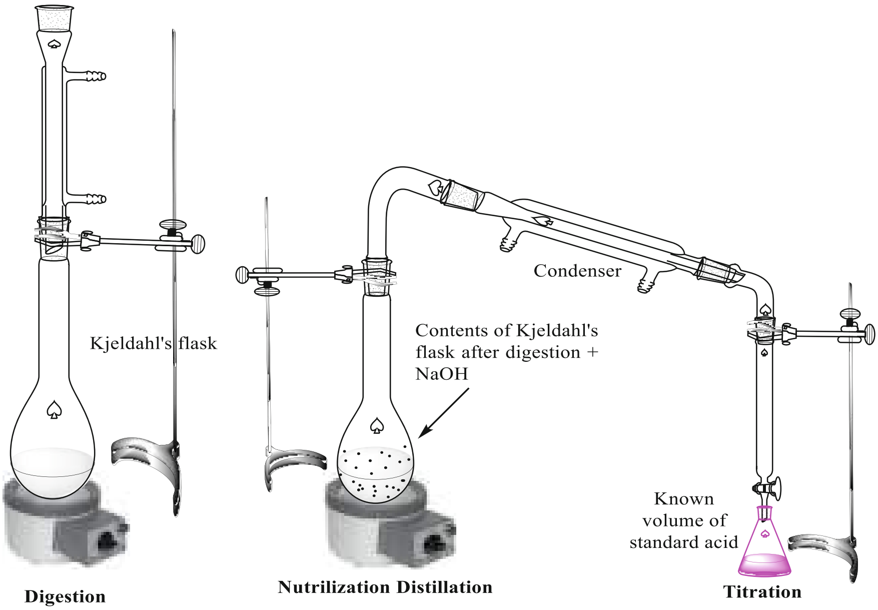


Fig. 4 Kjeldahl apparatus

6.3 Distillation

An excessive amount of base is mixed with the digested result, which causes the ammonium to be converted to ammonia, as shown in the equation given. Ammonia is recovered from the reaction product using distillation [26]:

$$(NH_4)_2SO_4 + 2NaOH \overset{\text{Heat}}{\rightarrow} 2NH_3 + Na_2SO_4 + 2H_2O$$

6.4 Titration

Titration is used to determine the quantity of NH_3 available in a receiving solution. The quantity of ammonia ions in the receiving solution may be used to estimate the nitrogen content of a sample material [64, 66].

Titration is classified into two categories:

1. Back titration
2. Direct titration

In both procedures, if ammonia is present in the distillate, the colour of the liquid will change.

6.5 Back Titration

A collection beaker containing a standard acidic solution that is calibrated is used to collect the ammonia. The received solution's excess amount of acid maintains its pH value, and back titration with bases is required to cause a change in the indicator [39]:

$$2NH_3 + 2H_2SO_4 \rightarrow (NH_4)_2SO_4 + H_2SO_4 \ \text{(No colorchange)}$$

$$\begin{aligned}(NH_4)SO_4 + 2H_2SO_4 + 2NaOH \rightarrow \ & Na_2SO_4 + (NH_4)_2SO_4 \\ & + 2H_2O \ \text{(Colorchange)}\end{aligned}$$

6.6 Direct Titration

When boric acid is used in place of a standard mineral acid in a titration, the ensuing chemical reaction is as follows [39, 43]:

$$\begin{aligned}NH_3 + 2H_3BO_3 \rightarrow \ & NH_4{}^+ + H_2BO_3{}^- + H_3BO_3 \\ & \text{(Colorchange)}\end{aligned}$$

When boric acid absorbs ammonia gas, ammonium-borate complexes are produced. Receiving solutions turn colours when ammonia accumulates:

$$\begin{aligned}2NH_4{}^+ + 2H_2BO^-{}_3 \rightarrow \ & H_2SO_4 + (NH_4)_2SO_4 + 2H_3BO_3 \\ & \text{(Color change in reverse)}\end{aligned}$$

7 Total Carbohydrate Content

Total carbohydrate content is made up of a variety of micronutrients and macromolecules such as dietary fibre, glucose, fructose, starches, proteins, total lipids or fat, and many others. All of these carbohydrate types are important for getting enough calories each day and living a healthy life, but eating too many of them, like any other macromolecule, can have negative effects [67]. Carbohydrates' primary role is to provide our body with energy so that it can perform critical functions such as muscle action, cognitive functioning, respiration, and many other processes [68]. Carbohydrates are mainly composed of saccharides, which are simple sugars. The majority of carbohydrate diets include polysaccharides, which are a group of many saccharides that are connected together [69]. The various procedures were employed to calculate the percentage of total carbohydrate content in the plant material assayed. Energy content is assessed by multiplying the percentages of protein, carbohydrates, and total lipids by 4 kcal/g and 9 kcal/g, respectively. There is an important role for carbohydrates in the upkeep and construction of the plant's materials. They are available as sugars as well as polysaccharides [70]. Diluted HCl is the first step in the process of breaking down carbohydrates into simple sugars. The dehydration of glucose or fructose to hydroxymethyl furfural occurs in a hot, acidic condition. Anthrone and this compound worked together to make a green-coloured product that could absorb at 640 nm [71].

7.1 Protocol

7.1.1 Extraction of Total Carbohydrate Content from the Plant Materials

(i) Weigh 1 g of the powdered plant material sample and transfer it to a bottom flask for testing.
(ii) Hydrolyse the plant material sample for 2 h in a boiling WB with 10 mL of concentrated HCl.
(iii) Then the bottom flask should be cooled down to 30 °C and neutralized with Na_2CO_3 until the bubbling stops.
(iv) Add distilled water to the boiled plant material to get the volume to 100 mL.
(v) Centrifuge the homogenization plant material at 2500 rpm for 5–10 min at RT until it has been moved to a centrifuge tube. Do this until it has been moved to the tube.
(vi) After that, remove the precipitate and employ it for estimating [39, 71].

7.1.2 Estimation of Total Carbohydrate Content in Plant Material Samples

(i) In the first, make a series of 1 mL to 5 mL of plant material and a standard solution in various test tubes.
(ii) Thereafter, transfer 0.5 and 1 mL of the carbohydrate extract of the plant material to 2 additional glass vials labelled.
(iii) Fill each test tube with DW to a volume of 1 mL.
(iv) Add 1 mL of DW to another test tube and label it as a blank solution.
(v) Next, add 4 mL of anthrone solution to each test tube along with the blank solution and mix well.
(vi) After thoroughly vortexing the test tubes, place them in a water bath for 10 min.
(vii) Each test tube should be quickly cooled down. Then, at 640 nm, measure the amount of greenish colour absorbed by the blank solution using a UV-Vis spectrometer.
(viii) This is how the percentage of total carbohydrate content in the herbal matter is found at the end of the process. It is expressed as GGE/100 g (gram glucose equivalent per 100 g) of the plant material sample [44, 72].

8 Total Energy Content

The total energy content of the diet and plant material is a significant characteristic to consider. The food we consume provides the energy our bodies require for exercise, communication, and work. Total energy content is the quantity of energy generated by the combustion or oxidation of one gram of plant material and is calculated in Kcal/g [73]. Total energy content provides the total available energy in plant sample material. Actually, total energy content is the sum of the values of total carbohydrate content, crude fat, and crude protein and is calculated using the factor of 16.736 for total carbohydrate content, 37.656 for crude fat, and 16.736 for crude protein [74].

8.1 Calculation of Total Energy Content

To quantify the total energy content of a food or plant material, multiply the number of grams of total carbohydrate content (TCC), total protein content (TPC), and crude fat (CF), respectively. Finally, combine the findings. Total energy content calculations are completed using a given formula [75]:

$$\text{Total Energy Content} = (\%\text{ of Crude protein} \times 16.736) + (\%\text{ of Crude fat} \times 37.656) + (\%\text{ of Total Carbohydrate content} \times 16.736)$$

9 Nutritive Value

As you eat and drink, you need certain nutrients to stay healthy. These nutrients are called “nutritional value.” Nutritional value, also known as “nutritive value,” is a measurement of how well these nutrients are balanced in the foods and beverages you consume and an indicator of a food’s influence on the nutritional content of the food [52]. This value is based on how quickly a meal can be digested as well as how many important components (protein, fat, carbohydrate, minerals, and vitamins) it has. The nutritive value of a food or plant material is the total amount of those nutrients that can be absorbed by the body and utilized for growth and maintenance [47]. Nutrient value analysis is used to evaluate adherence to standards for caloric intake, unhealthy fats, and salt, as well as to check the quantities of these macronutrients in meals. The nutritive value of plant material and foods relates to their composition and the effect of their ingredients on the health of the consumer. It talks about carbohydrates, lipids, proteins, vitamins, minerals, and salt consumption; as well as how much salt, fat, and sugar you eat [55].

10 Importance of Proximate Analysis

There are many diverse components in herbal medicines, and proximate analysis shines a light on numerous areas of detecting their physical and physicochemical features [76]. This kind of examination is useful for the initial identification of herbs, but it also provides insight into the quality of the plants employed in phytopharmaceutical formulations [45]. Fresh plants or plant components are typically the starting point, and these are then processed further through steps like drying and preservation. In several technical publications and monographs, pharmaceutical phytology details the steps necessary to convert a fresh plant or its constituent parts into a dried, potent remedy [47]. Phytomolecules are produced in the protoplast, moved across the plasma membrane to the tonoplast, and finally found in the vacuoles, where they are available to the cell's inhabitants in the form of cell fluid. Certain phytomolecules (like cellulose) with lipophilic components (such oils, balsams, and resins) are forced out of the cell via the plasmalemma [48]. Rose oil is produced when oil cells from the Lauraceae and Zingiberaceae families are dispersed in the cell fluid and chemically altered to produce a volatile oil that then fills the remaining space in the cell [77]. The Rutaceae family secretes lysogenic vacuoles, which dissolve with the cell wall, while the Umbelliferae family secretes schizogenic vacuoles, which expand intercellular gaps and produce etheric oil. The aqueous cell fluid is mostly used to store hydrophilic components, while lipophilic components are rarely found there [39]. When preserving plant materials, it is crucial to administer postharvest treatment in order to mitigate any negative outcomes (e.g., tubers, seeds, and certain dried fruit). Since the chemical components of medicinal plants can be altered by the quick or slow loss of water during processing (during the transformation of harvested plants into medications by drying and related features), understanding the postharvest physiology of crude pharmaceuticals is crucial [49]. So far, the low moisture content in the composition of the skin of the fruit or the seed case may impede the modifying processes. Extreme changes occur in the internal physical and physicochemical state of the cell as a result of water loss. Enzymes, which are usually present in the cell membrane of living plant tissues, interact with the active ingredients contained in the cell fluid when the plant cell dries. Many reactions, including hydrolysis, oxidation, and polymerization, might result from the interaction between a particular enzyme and its substrate [78]. While harvesting certain plants, the enzymes and active ingredients can react due to tissue disturbance because they are located in various (separate) cells. The idea behind drying raw pharmaceuticals is that they will be more stable in storage if they are dried out and no longer contain any moisture. Several decomposition-causing enzymes are stable during drying and can be reactivated by exposure to water [56]. Thus, the humidity level should be decreased to below the pharmacopoeia's minimum and maximum values. Many enzymes, including oxidases and peroxidases, are accountable for the formation of secondary metabolites in plants. The oxidation of phenols, unsaturated fatty acids, and terpenes is mostly attributable to these. Hydrolases can break down esters, glycosides, and polysaccharides. Morphine, quinine, and isomerases are essential for the manufacture of enantiomerically pure secondary phytochemicals such as the ergot alkaloids [79]. Medicines need to be dried as soon as feasible, and the enzymes must be denatured, to prevent them from being broken down by the body's natural processes. Drug stabilization refers to this process, which is rarely used in clinical settings. The results of an enzyme's work are not necessarily bad. Such procedures are often necessary for the release of active, aromatic, and other beneficial compounds [80]. After harvesting, there is no standard rule for how something should be dried or stored. To make standard-quality phytopharmaceuticals, low-quality raw materials provide a significant obstacle for botanical manufacturers. Although there are numerous botanical resources, official monographs rarely indicate any requirements for their use [70].

- Variations in the content of metabolites are seen due to geographical differences; these changes occur even when the standard is established and do not match those as stated in the pharmacopoeia.
- As herbal raw materials are so widely distributed, meeting demand is impossible.
- There is a risk that the extracted crude medicine is contaminated with insects or microorganisms, rendering it useless. Added preservatives or pesticides are left behind in unsafe quantities.
- It is challenging to sustain the excellence and clarity of herbal pharmaceuticals of pharmacopoeial quality due to the high demand for these drugs [39].

The quality of drugs is measured by how much of the active chemicals they really contain, whereas purity is determined by the absence of any foreign materials (organic or inorganic). Based on these elements, the product's functional and market value can be calculated. It is possible for a crude medication to meet all the statutory standards for purity and quality, depending on the concentration and nature of the ingredients [81]. In the process of gathering illicit substances from the wild, some inorganic or organic contamination is practically inevitable. Sand or other inorganic soil components are the primary inorganic materials in crude medicines. Metal, glass, or stone fragments may also appear, albeit less frequently. Very often, the organic material is from a species of plant that is not listed in the drug's official description, or it may be from a neighbouring plant of a different species [82]. Insects, moulds, and animal waste are some additional sources of contamination. These qualitative and quantitative parameters provide a clear idea about the distinctive features of crude pharmaceuticals. While these features are useful for informing the analyst on the nature and properties of crude medicines, it is only through further study of other parameters that it can be determined whether or not these substances are suitable [83]. The determination of several criteria, which have been discussed in more detail above, is a common practice in acquiring qualitative and quantitative data regarding the purity and quality of a crude medicament [59].

There is no conventional medicine equivalent to some plant-based pharmaceuticals, such as cardiac glycosides, which have been used for centuries. That is why studying medicinal plants and the bioactive chemicals they contain is so important. As a result, there has been a recent uptick in interest in herbal remedies [39]. Herbs continue to be valuable not just as treatments for a wide variety of human and animal illnesses but also as research tools for identifying novel bioactive chemicals that could be used to create novel pharmaceuticals. Several innovative treatments for cancer, high blood pressure, diabetes, infections, etc., have been produced thanks to the scientific exploitation of herbs used in traditional medicine [84]. Physical and chemical properties of plant medicines that may be used therapeutically include TA, AIA, WSA, SA, LOD, EV, FI, SI, CFC, bitterness value, haemolytic activity, and foreign organic matter [85].

Herbal formulations used in any medical system must first undergo the processes of authentication and standardization [86]. The class of base substances used in the formulation of herbal products is essential for the development of drug standardization. Herbal medication ingredients are typically traded as intangibles such as roots, bark, twigs, flowers, leaves, fruits, and seeds, making it difficult to visually authenticate the material utilized and leading to widespread adulteration [87]. Due to their varied composition—whether they are whole plants, sections of plants, or extracts from these sources—natural product standardization is a challenging endeavour. Quality reproduction of herbal products relies on strict quality control of raw materials. This has led to a surge in recent years in the standardization of many herbal medicines thought to have therapeutic effects [60]. False labelling is the first step in the dangerous use of any natural or herbal remedy. It is not uncommon for people to use the same colloquial name to describe two or more species that are, in fact, very distinct from one another [88]. Research into the pharmacognosy of medicinal plants, especially their physicochemical composition, holds the key to resolving all of these issues. Standardization and quality control of herbal products are defined by the WHO [89] as the procedures involved in the physicochemical assessment of crude pharmaceuticals, the safety, effectiveness, and sustainability evaluation of the completed product, the promotion of a product, the dissemination of information about that product to consumers, and the recording of the product's risks and benefits based on actual use [90]. All factors that affect the quality of herbal medicines should be considered in the standardization process [91].

11 Quality Control Aspects

In almost all conventional medical systems, quality control has been provided by the vigilant observation of trained professionals. They need to rethink their approach to quality assurance and the development of innovative techniques in light of the contemporary context. Hence, evaluating the efficacy of traditional herbal remedies is a hot topic right now [92]. It is necessary to explore the criteria associated with standardization in different batches in order to set a boundary for the reference standards used in quality control and quality assurance of herbal drugs. As a result, the established micro- and macro-standards can be used as indicators of the medicine's authenticity [60]. This is because several types of bacteria and mould, many of which originate in the soil, can be found on medicinal plant components, so determining microorganisms of great importance will be useful in identifying and authenticating the plant material, in addition to the microscopic method, which is the most cost-effective for authenticating the original content by tracing its origins [39]. When establishing criteria for selecting standard substances and carrying out quality control, it is vital to keep in mind that various ingredients in herbal medicines may have differing degrees of impact on the overall quality, safety, and effectiveness of the product. Contents such as moisture, ash, volatile matter, ash, fixed carbon, etc., are revealed. Water and organic materials are burned off during cooking, leaving behind an inorganic residue called ash that can be used as a proxy for the total mineral content of the food [24]. When compared to other components of food, minerals are not only stable but also resistant to heat. True drug samples may have total ash values that fluctuate over a wide range due to the presence of varying amounts of natural or physiological ash. A plant's ashes can reveal the types and amounts of minerals it once contained. Thus, the proximate study of herbal medications is a crucial topic in the field of natural product quality control [56].

11.1 Quality Assurance

The wild populations of several crucial plant species used in medicine have been depleted due to overharvesting, making it impossible to ensure a reliable supply for the foreseeable future. Health problems (whether caused by changes in lifestyle or not) are a major concern in the current setting of increasing urbanization and the resulting changes in lifestyle [84]. If and when demand for herbal remedies increases, it is anticipated that phytochemicals with supposed therapeutic capabilities for lifestyle-allied disorders will experience strong demand. Most of the raw materials used in manufacturing are delivered in bulk, which poses significant security risks. The distribution of raw materials in large quantities involves numerous middlemen. Little to no toxicological or carcinogenic testing is done at any point in the manufacturing process, and there is little oversight over the quality of the botanicals used [93]. There are a number of quality control measures in place to ensure that only the highest quality botanicals are exported, but concerns are especially high with regards to countries still in the development stage. Understanding the stringent quality standards of the herbal drug industry is crucial to the preservation of these rare plants for medicinal purposes [94]. There are two main ways to acquire medicinal plants: wild gathering or farming. The cultivated material is better suited for industrial applications like drug manufacturing. In the beginning, there are producers and primary collectors; then, there is an intermediary supply to local contractors; then, there is a supply to regional and large wholesale markets; finally, there is a supply to manufacturers via specialized suppliers; and finally, there is a supply to the end user. Yet, the supply chain must be optimized for effective output [94]. An examination of the current system's flaws reveals key areas for more regulatory oversight: First, the use of standardized cultivars; second, the use of standardized cultivation techniques; third, the superiority of raw ingredients at the source (as provided to manufacturers); fourth, the management of post-harvest treatment and processing; and fifth, the use of standardized formulations in accordance with regulatory mandates [39]. The plants include several phytoconstituents that contribute to their bioactivity, which presents a significant problem in the domain of standardization. It is not enough for a product to consistently have the right amount of active ingredients; they also need to be produced in a way that allows the body to absorb them. For the cultivators to make a profit, they need access to standardized genotypes that produce high levels of the active component. As a result, “niche” farmers who specialize in growing plants of concern for corresponding and unusual treatments can thank the procedures necessary for the ideal production of the active compounds for their success. Several older medicine formulations do not adhere to modern standards of practice [95, 96]. Those who do not have insufficient and ineffective quality checks that would provide the consumer with confidence in the drug’s efficacy. Some important steps that can be taken to help developing nations establish more profitable trading practices for herbal medicine are: careful processing, value-adding, and product display after collection; farmers and businesses working together to maximize farmers’ incomes; studying and developing new types of therapeutic plants; and encouraging regional or local farmer cooperation to ensure a reliable supply of inputs [97].

As many different substances may go into making herbal medicines, it might be difficult to standardize quality assurance standards and devise a foolproof validation process. The quality and amount of medicinal plants’ active ingredients can vary from harvest to harvest, even if the plants were harvested from the same location [98]. Moreover, various factors such as plant age, region of origin, harvest time, drying method, storage environment, production process, packing, etc., influence standards. Because the entire process of making a medication from a plant is considered an active entity, it is difficult to create appropriate guidelines [99]. The WHO procedures created in 1991 assist as a quality control mechanism for herbal medications. The most important parts of the WHO guidelines are: (1) ensuring excellence by checking raw plant material, plant preparation, and the final product; (2) ensuring durability by establishing a shelf life; (3) authenticating safety primarily through practice and toxicology studies; and (4) providing a recognized indication of productivity through activity assessment (in animals and humans). Although the WHO has offered guidance on how to evaluate herbal remedies, it is not within its scope to do so [93, 100]. Instead, the responsibility for ensuring compliance with WHO standards falls on the manufacturer, compendia review, and regulatory bodies. The production process must be monitored and assessed by regulatory agencies in accordance with the current regulatory regime [101].

12 Safety Concerns for Herbs

The chemical structures of herbal remedies are extremely diverse and complex. The use of natural compounds to alter organic capacity has also garnered much interest. They have been useful in the fields of chemistry, biology, and healthcare, where they have been essential in the identification of new therapeutic targets [102]. Natural products continue to play an important role in a wide range of therapeutic improvement and research initiatives because of their synthetically diverse variety and innovative mechanisms of activity. Some of the most important pharmaceuticals in a social insurance system are found among these natural goods, which have undergone fascinating and important changes in their ability to interface with numerous modified natural targets [103]. For instance, tiny compounds, which are produced by plants, microbes, and animals, have played a significant role in medication development. In addition to their widespread use in developing nations, plant-based healthcare solutions are also widely employed in affluent nations, either as a primary treatment method or in conjunction with traditional medicine [104]. The use of medicinal plants has long been a part of both conventional medicine and folk medicine, with several herbal prescriptions available for their use in the prevention and treatment of various disorders. It covers everything from widely used home remedies to professionally prepared herbal concentrations [105]. New biologic breakthroughs, including pharmacogenomic, metabolomic, and microarray techniques, can be used to conduct the in-depth examination of the pharmacological properties and safety of herbal-inferred therapies that is necessary [99, 106]. To add insult to injury, many people use herbal supplements along with their regular medication without telling their doctor, which can cause dangerous herb–drug interactions [107]. Herbs are becoming increasingly popular as dietary supplements and OTC medicines, leading to a greater need for scientific and clinical information for product assessment [108]. Several of the herbal medications currently in use have been around for a long time and have proven effective. Specific scientific backing is required to address the safety and effectiveness of the complex blend of herbal preparations and properly document trial data [109].

Phytoconstituent safety assessments and interactions with conventional drugs are subject to varying regulatory standards across nations [103]. Herbal medicines with known interactions with CYP isoforms must submit documentation to regulatory bodies prior to commercialization. Herb–drug interactions occur because various phytoconstituents have been found to either inhibit or activate cytochrome. Several phytoconstituents have been reported to interact negatively with standard pharmaceuticals. Many cases of herb–drug interactions have been documented [110]. This occurs most frequently when a patient conceals their herb use from their doctor. In some instances, the lack of sufficient scientific documentation and public knowledge may be to blame. Patients have been documented to experience a wide variety of side effects when exposed to phytochemicals such as allicin, quercetin, and silymarin [92, 111]. Patients may be using additional herbal remedies or nutritional supplements in addition to their prescribed medications without realizing it. Hence, we need to have an in-depth understanding of the pharmacology, mechanistic deed, drug relations, safety, and usefulness of herbal remedies to achieve the greatest possible benefit while minimizing the undesirable side effects [105]. The public's understanding of the potential for adverse reactions to drugs has expanded to include secondary metabolites seen in herbal remedies. It is obvious that combining natural items can interact and change the therapeutic advantages of these regularly used medicines [112]. Herbal medicines and OTC drugs may interact with drug-metabolizing enzymes in dangerous ways, leading to potentially fatal side effects if used together [113]. In order to better comprehend the prospective for herb–drug relations, studies on drug-metabolizing enzymes have been conducted [6].

12.1 Quality-Related Safety Issues

Plants may or may not include poisonous chemicals in addition to pharmacologically active molecules. The wrong herb may have been used, among other possible causes of toxicity in herbal remedies. The presence of other medications, hormones, or heavy metals in the soil can lead to this effect in plants. Providing safe herbal medicines and healthcare herbal products is vital to quality assurance [113]. The literature was scoured for accounts of side effects and possible drug interactions, and this list was developed. These side effects are plausible in patients taking herbal medicines, especially when used for an extended period of time or at large doses. The quality of raw herbs is crucial to the efficiency and safety of herbal medications. In contrast, medicinal plant raw materials and their definitions of quality vary greatly in terms of both the quality and quantity of their phytochemical profiles. That is why it is crucial to nail down the quality metrics that are inextricably related to reliability [114, 115]. Achieving uniform pharmacological action and facilitating significant contrast and meta-analysis of clinical records requires strict adherence to the harmonized quality standards. Traditional and modern herbal remedies sold commercially now have their constituents tested for quality and safety more than ever before. However, the safety and effectiveness of botanical medicines must be considered in any quality assurance paradigm for botanicals [109]. Depending on the context in which herbal products are being used and the health claims being made for them, a variety of terminologies, definitions, and quality factors are in use around the world. Complementary, alternative, and nonconventional medications, such as herbal products, are generally centred on traditional remedies, such as TCMs, Ayurveda, Siddha medicine from India, and Kampo medicine from Japan [6]. The dried, untreated, entire, or fragmented plants or plant matter are the starting point for the identifying descriptors in pharmacopeial herbal monographs, which are still relied on today by the therapeutic herb business and herbal component makers. Macroscopic, microscopic, and organoleptic analyses of arranged and dehydrated plant pieces revealed the identity of the constituents [116]. Yet, when material is fractured or sorted, evidence on leaf shape, extent, and variety, phyllotaxy, floral characteristics, and placement in the inflorescence may irreversibly vanish. This is especially true for specimens that lack the flowering and fruiting sections of the plant. These monographs serve as an identification guide, but they are useful only after the entire plant has been verified as genuine [39].

13 Conclusion

Proximate analysis of plant materials provides information about their total ash, moisture, protein, crude fibre, crude fat, carbohydrate, nutritional value, and energy content. It is vital to check the plant material's moisture content before eating it. The physiochemical components of plants and foods are influenced by the presence or absence of water, which in turn affects their flavour and texture when stored. Moisture content impacts food makers' processing and analysis. A plant material's properties, storability, utility, and quality depend on its moisture level. In agriculture, medicine, and insecticides, accurate moisture content evaluation is essential to product quality. The overall ash content of plant material helps estimate mineral reserves. Energy comes from crude fat in numerous ways. Crude fibre is the ratio of people's current diets' cellulosic, pectic, lignin, and other undigested elements to how much they eat. Crushing and separating grain from its vegetative condition is tested using crude fibre. Ripe plant-based foods have more crude fibre, which helps determine biological taste. Healthy eating promotes a healthy lifestyle, appropriate growth, development, and maturation, body mass maintenance, and a reduced chance of major illness. All of these improve health. The proximate study of medicinal plants is crucial for establishing their nutritional value. Proximate analysis is a useful indicator for categorizing the nutritional value of dietary ingredients. Herbal medicine proximate analysis is thus an important area of inquiry in the study of herbal material quality assurance.

Conflict of Interest We declare no conflict of interest.

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Abstract

The use of herbal medicines and herbal products has significantly increased in recent times and contributes to the socioeconomic sustainability of the rural and suburban populace. The socioeconomic importance of herbal medicine includes addressing health issues, revenue generation, cultural development, insights for novel medication development, and indigenous knowledge sharing and exchanges. The commercialization of medicinal plants enables traders in the locality to offer their products and expertise at a fair and reasonable price, thereby, making returns to support their family and community. Every aspect of herbal medicine formulation and use value chain (i.e., cultivation harvesting, processing, formulation, packaging, sale, use, and waste disposal) has diverse socioeconomic relevance from the local to global scale. However, the value chain sequence varies at different scales. For instance, at the local scale, it may be from smallholder markets to national and international trade with varying levels of actors along the sequence. Apart from their economic values, medicinal plants also serve as a source of employment and sustainable sourcing and stimulate quality control of the medicinal product. An innovative approach to subvert the increasing length of the value chain, the blockchain approach, can be employed. The concept of blockchain replaces the analog ledger approach, and it is a system that is built on decentralized data storage on several users, each forming a network node. This review, therefore, elaborates on the economic importance, value chain, recent approaches, and quality control of herbal medicine plants and products for sustainable production.

Keywords

Herbal medicine development · Socioeconomic sustainability · Disease management · Medicinal plants

Abbreviations

BC	Before Christ
CDC	Centers for Disease Control and Prevention
ECRA	Ethical Consumer Research Association
FDA	Food and Drug Administration

ICMPHD	International Center on Medicinal Plants and Herbal Drugs
USD	US Dollar
WHO	World Health Organization

1 Introduction

Herbal medicine refers to plant-derived substances that exist naturally and are used to cure ailments in local or regional healing practices. They are a combination of complex organic compounds that may be derived from plants either in raw or processed portions of plants [1]. Herbal medicine can be traced back to ancient societies. Herbal medicine entails the use of plant parts or whole for the formulation or as medicine to treat, manage disease conditions, and improve overall health and wellbeing [2–5]. Herbal medicines are described by the World Health Organization (WHO) as medical products which are labeled and contain active substances that may contain parts of plants that are seen aerially or under the ground, or other plant mixtures or materials [6–8]. Herbal medicines can be utilized as self-care solutions for everything from insomnia, stress, and mood problems to digestive issues, colds, and flu. Herbal therapy can be used to prevent and cure several ailments in humans, such as asthma, eczema, premenstrual syndrome, migraine, chronic fatigue, rheumatoid arthritis, cancer, menopausal symptoms, and irritable bowel syndrome. Many disorders are treated using herbal medicine. For example, the bark of willow plants contains a high concentration of salicylic acid, an active metabolite of aspirin [9]. It is estimated that over 80% of people worldwide use herbal formulations as a main health care system [10–12], which is rapidly increasing due to the toxicity and unpleasant responses associated with modern allopathic medications, which has also contributed to a quick rise in herbal drug makers [13]. Natural herbs were widely utilized in ancient times to cure and prevent a variety of illnesses. Based on the pros and cons in this discipline, new herbal medicines that are useful to health with no or little adverse effects are developed [14]. Since its inception around 8500 BC, medicinal plants have given humans curative medicines [15].

A large number of conventional medications currently on the market, used to treat both acute and chronic disorders, have been inspired by medicinal plants [15]. Substances derived from medicinal plants are frequently synthesized into synthetic derivatives utilized in clinical pharmaceuticals, hence broadening the variety of conventional medicines [16]. Many plants have been reported to possess pharmacological properties, particularly inhibition of microbial growth [17–33] and control of insects such as mosquitoes [34–40]. Herbal medicine has various advantages, including being a candidate for new antiviral, having fewer adverse effects than synthetic medications, and having a remarkable ability to inhibit the reproduction or synthesis of specific virus genomes [41]. Herbal medicines contribute to social, economic, and environmental sustainability in a variety of ways. Herbal medicine is important in promoting bioeconomy, which encompasses various value chains and economic activities which are based on biodiversity [42, 43]. Herbal medicines have always accounted for a significant portion of the medicine market. The global trade in medications derived from medicinal plants was projected to be between USD 32 and USD 43 billion [44]. These plants and the drugs they produce are prescribed in both developing and developed nations. To increase their reliance on their pharmaceuticals and build their economies, this should also be prioritized by other emerging nations. The growing nations must comprehend the worth of their native medicinal plants and their economic significance to achieve this [45, 46]. The promotion of their valued plants can thus draw in investors for the creation of pharmaceuticals and aid in the development of their own herbal medicine [44]. Therefore, using medicinal plants will go a long way in increasing the income of a country and enable the populace to have access to medicines, enhancing the nation's health services [44]. To this end, the value of herbal plants and their importance in the economy cannot be neglected.

This chapter addresses the socioeconomic values of herbal medicine, its concepts, value chains, challenges, and heavy metal contamination, and the recent approach to the value chain approach which involves the blockchain system. The information was synthesized from an intensive literature search and presented in a simplified approach for better understanding and easy assimilation. The information herein will go a long way in promoting sustainable socioeconomic activities and conscious practices for the distribution of herbal products in a technologically driven manner. Issues about prompt delivery of herbal medicine and reduction in the length of the value chain will also be discussed.

2 Social Values of Herbal Medicine

The practice of medicine is very important to the sustenance of sociocultural norms that hold many societies together around the world and has been successful in protecting and maintaining the health and wellbeing of many societies. The practice also has many socioeconomic aspects which are beneficial for the advancement of societies. The values cannot be overemphasized as over 90% of the medicines originate from plants. In ancient times, plants were used to address several diseases and ailments, most especially in areas that are thickly forested and hilly. Such areas can house diverse species which can be used for herbal purposes as well as generation of income and livelihood sustenance for the poor [44]. Those in the rural areas that have a vast knowledge of medicinal plants and this indigenous knowledge about the economically important plants are applied to treat various ailments that may occur to poor villagers.

For socioeconomic development, it is important to understand the scientific cultivation and uses of medicinal plants in a sustainable manner. Conversely, the scientifically cultivated herbal plants for commercial purposes are very few [38]. The medicinal extraction is very low and the abundance of herbal plants is reducing daily as a result of deforestation activities, urbanization, industrialization, overgrazing of lands, and intensive harvesting of plant species that are rare and endangered and have great medicinal purposes. The lack of support from the government, environmental ministries, inadequate management strategies, and institutional framework has greatly affected the management of valuable herbal plants. Also due to a lack of information about the genetic resources and the values of herbal plant species, countries have failed to enforce sustainable management strategies [26–29]. Presently, a large number of herbal plants are now extinct while a large number are considered threatened.

2.1 Economic Importance of Herbal Medicine

Herbal medicine, also called botanical medicines or phytomedicines, is a term used to describe herbs, their materials, the procedure of preparation of herbal materials, and already completed herbal products. These materials/products usually contain active components that are derived from the parts of plants or the plant materials from which the raw materials are obtained [7]. Examples include plant seeds, roots, bark, leaves, berries, or flowers [47]. Table 1 presents some plant materials and the active ingredients they contain.

Medicinal plants are known to be the source of many medications used in modern and conventional medicine with huge economic value and the driver of huge sectors. For instance, an herb, the foxglove plant (*Digitalis purpurea*) has been used since 1775 to cure cardiac glycoside as it is used to produce digitalis. Pure honey is a good product for the treatment of infected wounds and can act more quickly than usual. Similar to this, traditional birth practices use plant extracts from South Africa that have calming effects on the muscles. Table 2 presents the list of some herbal materials used in Nigeria and their economic importance. The commercial development of herbal medicinal plants has geared the increase in several situations around the value chain production, such as the scale of manufacture, convenience in terms of storage, and prompt delivery of cost-effective pharmaceuticals to low-income

Table 1 Herbal plants and active ingredients

S/N	Herbal plant(s)	Active ingredients	Uses	References
1	Meadowsweet plant (*Filipendula ulmaria*) and white willow bark	Salicylic acid	For the production of aspirin	[47]
2	Sweet wormwood (*Artemisia annua*)	Quinine	In the manufacture of antimalarial medications	[48]
3	Quinine tree (*Cinchona pubescens*)	Artemisinin	In the manufacture of antimalarial medications	[48]
4	Periwinkle (*Catharanthus roseus*)	Vinblastine and vincristine	In the production of an anticancer medication called vincristine	[49]
5	Opium poppy (*Papaver somniferum*)	It produces morphine, codeine, and paregoric	To cure diarrhea and pain-relieving medications	[49]

Table 2 Some herbal plants and their socioeconomic importance

Herbal plant	Functions	References
Basil (*Ocimum gratissimum*)	Cures diarrheal disorders	[50]
Poison devil's pepper (*Rauwolfia vomitoria*)	For hypertension, stroke, sleeplessness, and convulsions	[50]
Dry seeds of *Carica papaya*	For addressing intestinal parasitosis	[51]
Bitter kola (*Garcinia kola*)	Used in treating osteoarthritis due to its analgesic and inflammatory effects	[51]
The gel of aloe vera (*Aloe barbabensis*)	Used as an effective benzyl benzoate	[52]
Gingko (*Gingko biloba*)	It can treat patients with mild dementia and Alzheimer's disease, bone healing, anxiety, depression, eye health, and inflammation	[52]
Turmeric (*Curcuma longa*)	Preventing cancer and skin diseases, halting DNA mutations, and preventing pains caused by inflammatory diseases	[51]
Evening primrose oil (*Oenothera biennis*)	Used for mild skin conditions, breast pain, menopause, inflammation, diabetic neuropathy, and multiple sclerosis	[51]
Tea tree oil (*Melaleuca alternifolia*)	It is used to prevent athlete's foot, cuts, dandruff, and insect bites	[53]
Purple coneflower (*Echinacea purpurea*)	Used for colds, bronchitis, immunity, and upper respiratory infections	[53]
Grapeseed extract (*Vitis vinifera*)	It is used for treating cancer, edema, leg vein circulation, blood pressure, and lowering cholesterol	[52]
Lavender (*Lavadula angustifolia*)	It is used as an anti-stress, in the treatment of anxiety, lower blood pressure, and migraine	[52]
Chamomile (*Matricaria chamomilla*)	It is used to address stress, anxiety, insomnia, and cancer	[51]
Coriander (*Coriandrum sativum*)	It is rich in antioxidants and can cause urine retention	[51]
Fennel or saunf (*Foeniculum vulgare*)	It is used to treat cough, prevents bad breath, improve milk supply in women that are lactating, improves eyesight, and control cholesterol levels	[51]
Fenugreek (*Trigonella foenum-graecum*)	It controls cholesterol levels, purifies blood, lowers blood pressure, increases appetite, and cures diabetes and joint pains	[50]
Mint (*Mentha piperita*)	Boosts immunity, improves respiratory health, keeps mosquitoes away, and enhances mood	[50]
Tulsi/holy basil (*Ocimum tenuiflorum*)	It is an anticancer, treats indigestion, good for hair loss, diabetes, and heart diseases	[50]

nations. With all of these promising attributes, the commercial scale of production of herbal medicine can be promoted [54]. Food and Agricultural Organization has published the conditions and trends surrounding the production and commercialization of medicinal plants. This publication elaborates on the production and market value of medicinal plants and presents the diverse opportunities and challenges around the trade of medicinal plants to the world at large [55].

WHO [56] stated in its report that 30% of the entire brand of pharmaceuticals sold in the world today contained ingredients that were sourced from different parts of plant materials. About 600 million USD was estimated to have been incurred on the global production of herbal products in 2002, and the health care needs of about 80% of the population in developing nations depend solely on pharmaceuticals produced from plants. Therefore, the interests of countries with low income can be motivated by the use of medicinal plants, and this can form an important component in the developmental strategy of the economy and sustainable development. An estimation of 4 billion people, which amounts to about 80% of the population of the world in most developing countries, solely relies on the products from herbal medicine as a fundamental source of health care. Consequently, the practice of traditional medicine using herbs has been seen as a major part of their culture and traditions in rural communities [53]. Over 90% of the 1300 medicinal plants obtained from the wild are utilized in Europe, while 118 out of the 150 major prescriptions originate from natural sources in the USA [57].

Similarly, it is believed that pharmaceutical products in most industrialized countries also rely indirectly on medicinal plants [58]. In Asia, their medicinal plants account for a total export volume of around 50%, and the earnings from traditional medicine globally is about 45% [59]. Nigeria is a country that is richly endowed with diverse species of indigenous plants, which can be used in herbal medicine to address diverse diseases and heal different injuries. The issue of poverty can affect education, freedom, and health, in addition to the economic aspect [60], and is very evident in developing nations where the infrastructure and delivery of service is very poor. Rural poverty can be a major cause of inequality and intense degradation of the environment [61]. The livelihoods and eradication of poverty can be achieved through the field of agroforestry and the incorporation of herbal medicines in all its forms. The world can therefore create systems that are resilient to the worrying concerns about the environment and prevent disasters of all sorts [60].

The commercialization of medicinal plants has significant economic importance because it enables traders in the locality to offer their well-known products and expertise at a reasonable price, thereby making more returns to sustain their families and contribute to the social and economic development of their communities. Markets may also be established, which may have a national or global outreach where products that are of importance and newly identified can be advertised and sold to people. Nowadays, big herbal drug manufacturers are increasing their demand for medicinal plants as crucial raw materials. As a result, there is a great potential for creating jobs for those who are unemployed because the collecting, processing, and delivery of medicinal plant products demand a large input of the labor force [62–67]. Thus, by improving conventional processing by industries, cash earnings for locals can be raised [68]. Though only a few species of medicinal plants are manufactured on a large scale, cultivation sometimes may include the raising of medicinal plants with low demand for them globally [69]. China, the world's largest cultivator of medical plants, only plants on a bigger scale of between 100–250

Fig. 1 Economic importance of medicinal plants

species, and the wild environments produce about 80% of medicinal plants [70]. Europe cultivates medicinal plants of only 130–140 species out of the 1200–1300 native plant species found in the continent [71]. The importance of medicinal plants is presented in Fig. 1.

2.1.1 Health Purposes

Medicinal plants are plants that are used in the treatment and prevention of various health issues and are not typically believed to be dangerous to humans [72]. These medicinal plants may be either "wild plant species," which grow on their own in self-sustaining populations in ecosystems that may be natural or seminatural and can exist without direct human intervention, or "domesticated plant species," which can be as a result of the intervention of humans such as improved breeding and selection or management [73]. Medicinal plants are utilized as raw materials when active ingredients of medicinal plants are extracted, also pure ingredients (e.g., alkaloids like quinine and quinidine from cinchona bark, emetine from ipecacuanha root, glycosides from digitalis leaves, and sennosides from senna leaves) for synthetic vitamins or steroids, as well as natural preparations and indigenous medicines. Natural raw materials are provided by medicinal plants for the majority of traditional oral and nonoral drugs [74]. Many species of plants have several medical purposes, and probably there is rarely any plant known to have several medicinal uses [75]. It is also claimed that the majority of medications used in the pharmaceutical business are obtained from one or more products from plants, with different plant components having varied functions when used in the treatment of maladies [75]. The core of the pharmaceutical sector is perhaps best described as medicinal plants.

According to WHO [56], about 30% of pharmaceuticals sold in the world today contain compounds that were gotten from plant materials, with a global sales estimate from herbal products to be about 600 million USD in 2002. About 80%

of the entire population in developing countries relies majorly on pharmaceuticals that are produced from plants for their medical and health needs. Humans have traditionally employed medicinal plants to heal and treat ailments [16, 76]. Human cultures' evolution has heavily relied on the use of biodiversity, with plant species, particularly those with medicinal characteristics, being critical for human existence [15]. It occurs in several ways throughout the world and may be traced back to ancient history, with the oldest written records of herbal remedies found in India, China, Greece, Rome, Syria, and Egypt dating back approximately 5000 years. It was estimated in 2010 that the international trade of medicinal plants and products was USD 60 billion, and expected by 2050 to reach a total estimate of USD 5 trillion.

2.1.2 Revenue Generation

Medicinal plant production can be utilized as a technique to generate money, protect biodiversity, and promote local economic growth while preserving forests [77]. In low-income countries, there is a vital and strategic relationship between the values of medicinal plants accrues, as medicines and their production are very critical and essential in maintaining a healthy human population [17, 78, 79]. This relationship also has the potential to drive and sustain the economy where it is cultivated and marketed. Pharmaceuticals derived from plants are projected to be produced commercially on a large scale, and the approach of biotechnology will drive its production. This is due to its advantages in terms of the scale of production and economic growth, the safety of the product, easy storage and distribution network, and an opportunity to supply pharmaceuticals at a low cost to developing countries due to their purchasing power which may not be matched by any commercial chain system. Medicinal plants can contribute significantly to the security of livelihood, health care that is cheap and affordable, and economic wellbeing of the populace, making them the most valued non-timber forest products.

2.1.3 Novel Medications

Medicinal plants are a valuable source of novel medications all around the world [80, 81]. Over three-quarters of the medications used worldwide were plant derived and were discovered after serious leads from indigenous medicine [82]. Globally, the market value of medicinal plants and their products is projected to be more than $100 billion per year [83]. Nigeria is a developing country, and medicinal plants are commonly used for various treatments [84, 85]. Developing countries have valuable resources such as medicinal plants; this makes them very wealthy as they have cheap sources of raw materials for the production of medicines sold at high prices [55, 86–88]. Despite being tiny individual contributors, these medicinal plants contributed to the global trade of the country in 2006 by surpassing USD 60 billion [89]. Europe has been reported to import over USD 1 billion in value of medicinal and aromatic plants from Asia and Africa [67, 90, 91]. This trade is expected to grow greatly by the year 2050 since herbal treatments are becoming increasingly popular [92, 93]. Despite accounting for a lesser proportion of agricultural output, these therapeutic plants have the highest value in terms of weight in any traded plant. These are the pharmaceutical crops of high value, which have a great potential for their countries based on agricultural techniques which contribute significantly to the economy of a region [94]. Despite accounting for a lesser proportion of agricultural output, the value from therapeutic plants is highest in terms of weight when compared with any other plant that is traded.

2.1.4 Health of Native Communities

Medicinal plants that are endemic to a specific area serve an important role in maintaining and sustaining the health and wellbeing of native inhabitants. They are also posed to provide economic returns for the localities. As a result, they offer a living for a huge number of people in the area. Since a large percentage of plants are being sourced from forests in an unsustainable manner, the pressure on these natural resources has increased. Thus, the cultivation of desired medicinal plant species can be subjected to provide an alternative source of income for farmers in such a locality. People who earn a living by selling therapeutic plants, which they either cultivate or gather, benefit from medicinal plants [95–97]. Due to the ever-increasing need for medicinal herbs, various Asian countries have undertaken large-scale cultivation of specific species.

2.1.5 Subsistence Economies

Medicinal plants also play an essential role in rural people's subsistence economies, particularly for women, in an environmentally friendly way while preserving the biodiversity of natural medicinal plant products [62]. Traditionally, the poor, particularly women, usually source, collect, and dry the wild medicinal plants before transporting them to market. Based on this, the production and processing of medicinal plants in a systematic manner provide promising new revenue and opportunities for employment in others to improve the rural poor's livelihoods in an environmentally sustainable manner. These medicinal plant uses have the potential to expand the number of rural jobs significantly. The growing and management of plants with medicinal attributes are easily associated with the employment of women in rural areas [62, 64]. Women are typically recognized for successfully administering the duties required since activities connected to medicinal plant production match women's everyday necessities and work schedules. Women are involved in the entire production process, from gathering raw materials to drying and shipping them to the market. They also receive instruction from the natural drug sector to carry out the jobs. Thus, medicinal plants provide a source of livelihood in rural areas and aid in meeting basic health care needs [98]. Medicinal plants have played an essential role in the health and economy of rural villages since ancient times; rural settlements are located outside of economic development zones and generally remain in a condition of poverty, in some cases acute poverty. This circumstance forces individuals to rely on local resources for survival, such as food, medication, clothing, footwear, and shelter [63, 98, 99].

3 The Concept of Herbal Medicine Value Chain

The word value chain simply refers to a connecting line of different aspects of business activities in the creation of a product or service, from the point of purchasing raw materials through to transportation to the point of delivery to the market or consumers. The chain flow was described as a flow of activities that are mostly needed to create a finished product from an initial raw material such as field crops [100]. For most products, the process of the value chain has five major activities, which include inbound operations, operations, outbound logistics, sales and marketing, service, and four minor activities, which are procurement and purchasing, human resource management, technological development, and company infrastructure. The concept of the value chain was first described in a book titled *Competitive Advantage: Creating and Sustaining Superior Performance* by Professor Michael Porter, who at the time was at Harvard Business School in 1985. The analysis of the value chain, therefore, is done when a business or a company identifies the major, minor, and subactivities and goes further to evaluate the efficiency of each of those activities. An important element that should be taken into cognizance while discussing the value chain has different factors involved in the production process that are likely going to have different degrees of power, which may increase or decrease and would be a function of their income level [101].

Value chain analysis focuses on two major aspects. The first aspect is to know how several types of value chains take advantage of competition by altering the way a particular product is being processed and sold [101]. Also, value chain analysis is being deployed as a device for understanding the socioeconomic benefits, demerits, and dangers involved for the several members of the chain. Herbal medicine value chains simply refer to the various stages involved in the production, processing, distribution, and marketing of herbal medicines. It starts from the cultivation of the herbs used in preparing a medicine up to the final sale of the medicine to the end consumer. A value chain typically includes all the actors that are involved in the process, such as the farmers that are actively involved in cultivation, processors, manufacturers, distributors, wholesalers, and retailers. In herbal medicine, the value chain may involve the collection of wild herbs, the cultivation of medicinal plants, the drying and processing of the raw materials, the extraction of active ingredients that does the healing, the formulation and packaging of the final product, and the distribution and marketing of the product to customers [102]. Value chains play a critical role in determining the quality, safety, and efficacy of herbal medicines. By ensuring that all stages in the value chain process are properly managed, from the sourcing of raw materials to the marketing of the final product, it is possible to produce high-quality herbal medicines that meet the needs of consumers while also supporting the livelihoods of those involved in the value chain. It is essential to have a good understanding of the internal and external linkages among the manufacturing, processing, and marketing network; knowledge of this will give a clear view of why there is variation in the value of herbal medicines in various markets [101].

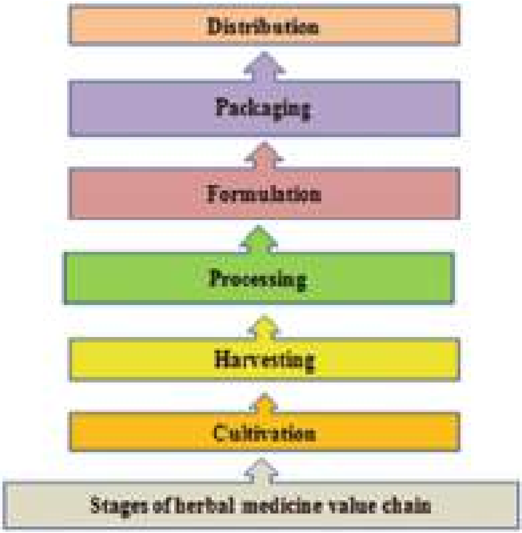

Fig. 2 Stages of herbal medicine value chain

3.1 Stages of Value Chain

The value chain of medicinal plants involves the production, processing, and distribution of plants or plant-derived products that are used for medicinal purposes. The chain in medicinal plants is a complex process that requires careful attention to detail at every stage to ensure that the final product is safe, effective, and of high quality. Figure 2 presents the flow stages of the herbal medicine value chain.

- **Cultivation:** The first stage involves the cultivation of plants deemed to have medicinal benefits. This can be done through farming, wild harvesting, or by growing plants in a greenhouse. Factors such as soil type, climate, and water availability can influence the quality and yield of the plants.
- **Harvesting:** Once the plants have reached maturity, they are harvested. This can be done manually or with the help of machines. It is important to harvest the plants at the right time to ensure that they have the maximum amount of active ingredients.
- **Processing:** The harvested plants are then processed to extract the active ingredients. This can involve a variety of techniques, such as drying, extraction, distillation, and fractionation. The extracted compounds are then purified and standardized to ensure consistent potency and quality.
- **Formulation:** The extracted compounds are then combined with other ingredients to create a final product, such as tablets, capsules, or liquid extracts. The formulation process can involve a range of techniques, such as blending, granulation, and compression.

- **Packaging:** Once the final product has been created, it is packaged and labeled. Packaging plays a key role in preserving the potency and stability of the good.
- **Distribution:** These involve the distribution of the product to retailers, wholesalers, or directly to consumers. Distribution can involve a range of channels, such as online sales, specialty stores, pharmacies, and health care providers.

3.1.1 Small/Medium-Scale Value Chain

Small- or medium-scale value chain involves the collection of primary products with a regional distribution where the healer is often also the retailer. These value chains rely on networks that are small and local or direct (one-on-one) relationships to secure the material they are processing or prescribing [101]. This has impacts on the sustainability and availability of the plants used as products. This type of value chain could have effects on the quality of the product, the benefits that are derived from it economically, the potential costs to the final consumer/patient, and it could also impact how quality can be monitored and the economic value that it generates. It was reported that in Ghana, most of the people selling herbal medicines were just full-time traders who had no idea about the source of the plant material because they did not gather the herbs themselves but just bought them from wholesalers [103]. This poses a big issue as if these retailers are unaware of the source of the plant material, they would be unaware of the compositions when it has been adulterated, and the general quality of the plant. It is very imperative to find ways of integrating the various elements of value chains. In Tanzania, the chain in value of herbal medicine consists of five actors; the harvesters who collect the plant materials; the wholesalers who process these plant materials by drying, milling, and packaging in bulk; the formulators who mix the different medicinal plants in different proportions to create a ready-to-consume product targeted at different diseases; distributors who are also sometimes retailers and healers; and then the final consumers/patients who consume these herbal medicines. Understanding this value chain is important to guide investments in the manufacture of herbal medicine [104]. In Ghana, some small- and medium-scale retailers have tried to formalize their products by using some modern techniques like bottling, but there are still the challenges of competing with foreign and imported brands in the market, which most times seem to be preferred by the consumers/patients [103].

3.1.2 The National Market Value Chain

This value chain involves the grower, collector, primary, middle manor processor, secondary middleman, retailer, and then the consumer/patient (Fig. 3). This a more complex value chain involving middlemen who serve as bridges between the producers and the retailers. They buy the plant materials in bulk from different farmers, process them, and then sell them to the retailers or consumers directly [105]. At this level, there are greater risks and more challenges to overcome as quality, purity, authenticity, and minimal level of contamination are all factors to be considered. Here only small amounts of the products are manufactured because of the rigorous pharmaceutical quality control measures put in place. The players in this chain,

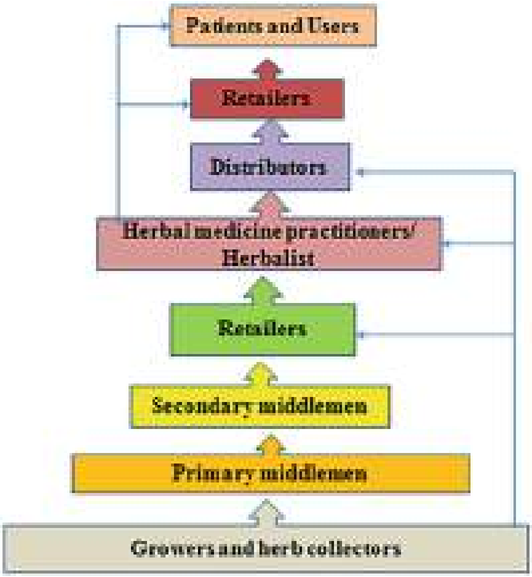

Fig. 3 National value chain of herbal medicine in many countries

especially the middlemen, have goals of scaling up the business and, therefore, will be ready to comply with regulatory institutions like the Food and Drug Administration (FDA) to pass mandatory tests and consequently secure a larger market [106]. This value chain is driven by innovations and has adopted a variety of modern equipment to increase production and guarantee quality. For example, instead of boiling the herbs to release active ingredients, alcohol may be used to extract the active ingredients [107]. Semiautomated filling machines, scanners, mixing machines, tube filling machines, and labeling machines are some of the new innovative technologies being used by the players in this value chain. The players in this chain strive to produce herbal products like tablets, pills, creams, capsules, and bottled mixtures with added preservatives to ensure a longer shelf life. This is so they can compete in the national market with other manufacturers but also with foreign herbal products [105, 107].

3.1.3 The International/Industrial Value Chain

This value chain consists of the grower, collector, middlemen/processor, exporter, consumer, country importer/distributor, retailer or practitioner, and the final consumer/patient (Fig. 4). This is a very complex value chain and involves more steps and more middlemen. Sometimes it is all controlled by a huge pharmaceutical company that controls all the different sectors till the final products and then even sales, advertising, and marketing [101]. This type of value chain has major impacts on cultural and ecological factors, as good practices are essential for this chain to work. Here the producers or retailers have little to no control over what happens with

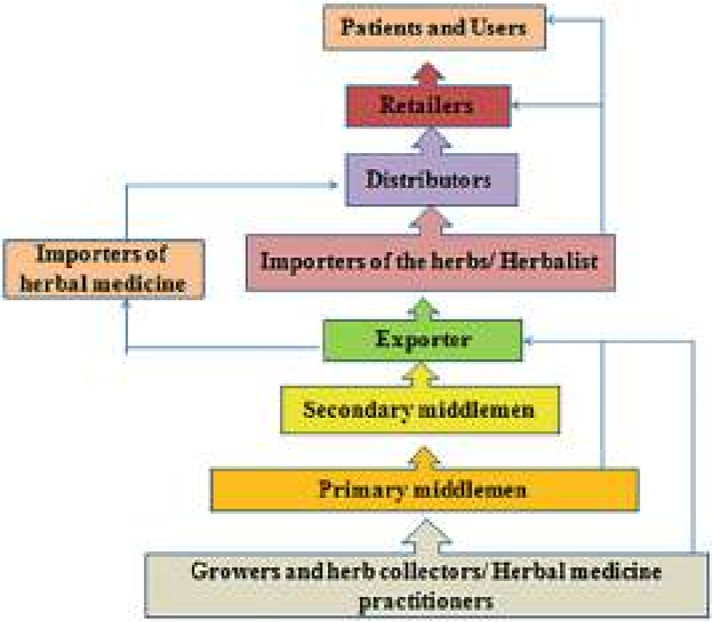

Fig. 4 International value chain of herbal medicine in many countries

the final product. Since products are being exported to other countries, lots of regulation and sophistication are required with an emphasis on commercialization, expanding the market reach in different countries [107]. The exportation of herbal products from India has increased from 69 million USD in 2005 to 128 million USD in 2009 [108]. Some Chinese farmers are now beginning to form a collaboration with sourcing pharmaceutical companies to gain access to a market that is more lucrative, thus changing the value chain. This approach can ensure the farmers get an agreed price for their products which will not be affected by conditions [105].

3.2 Value Chains of Herbal Medicines: Evidence

There is a significant amount of evidence to support the value chains of medicinal plants (Fig. 5). Some of the key findings from research studies and reports include:

- **Economic value**: Medicinal plant value chains can provide significant economic benefits to both producers and consumers. A study published in the *Journal of Ethnopharmacology* found that the global trade in medicinal plants is worth an estimated $60 billion annually.
- **Job creation**: Medicinal plant value chains can create employment opportunities, particularly in rural areas where other job opportunities may be limited.

Fig. 5 Values of herbal medicine

A report from the International Trade Center found that the medicinal plant sector is a significant source of employment in many developing countries.

- **Sustainable sourcing**: Medicinal plant value chains can be designed to promote sustainable sourcing practices that protect plant biodiversity and support conservation efforts. A report from the Food and Agriculture Organization of the United Nations found that sustainable harvesting practices can help to ensure the long-term viability of medicinal plant resources.
- **Quality control**: Medicinal plant value chains can include quality control measures to ensure that products are safe and effective. This can involve testing for contaminants, verifying the identity and potency of active ingredients, and ensuring compliance with regulatory standards.
- **Access to health care**: Medicinal plant value chains can help to improve access to health care, particularly in underserved communities. A study published in the *Journal of Public Health and Epidemiology* found that medicinal plants are widely used as a source of health care in many developing countries.

The evidence suggests that medicinal plant value chains can provide significant economic, social, and environmental benefits. However, it is important to ensure that these value chains are designed and implemented in a way that promotes sustainability, quality, and equitable access to health care. There have only been limited studies on medicinal plants' value chains. Researchers have reported that they are ethnographically based on value chains within the South American region [109, 110]. Also, it was reported that there are ecological pressures on plant species that have medicinal powers and rapid depletion in abundance and composition as a result of overexploitation [111]. Developing countries, an example of which is Nepal, rely majorly on the harvesting and marketing of plants with medicinal values which are sourced from the wild as their main source of income. Therefore, their research pointed out a key factor which was to enforce restrictions on the indiscriminate harvesting of medicinal plants and also advocating for the large-scale cultivation of these plants.

3.3 What Infers the Need for Herbal Medicine Value Chain Analysis?

Herbal medicine value chain analysis is a useful tool for evaluating and understanding the entire process involved in the production, distribution, and consumption of herbal medicines. Herbal medicine value chain analysis can help stakeholders in the industry to better understand the entire process of production, distribution, and consumption of herbal medicines and to identify areas for improvement and growth. The need for this analysis arises due to the following reasons:

- **Identifying opportunities for value addition:** Value chain analysis helps to identify opportunities for adding value to herbal medicines. It can help to identify areas where improvements can be made in production, packaging, transportation, and marketing, thereby increasing the value of the final product.
- **Improving efficiency:** Value chain analysis can help to identify inefficiencies in the production and distribution of herbal medicines, such as bottlenecks in the supply chain or areas where productivity can be improved. By addressing these inefficiencies, the overall efficiency of the value chain can be improved, leading to cost savings and increased profitability.
- **Identifying potential risks:** Value chain analysis can help to identify potential risks in the herbal medicine value chain, such as regulatory or market risks. This can help to mitigate these risks and ensure the sustainability of the industry.
- **Enhancing competitiveness:** Value chain analysis can also help herbal medicine producers and distributors identify areas where they can improve their competitiveness. By understanding their strengths and weaknesses relative to their competitors, they can develop strategies to improve their market position.

3.4 Benefits of Herbal Medicine Value Chains

The value chain of herbal medicine can provide a range of benefits, from economic growth and job creation to improved access to health care and the preservation of traditional knowledge. It is important to ensure that these value chains are designed and implemented in a way that promotes sustainability, quality, and equitable access to health care. There are many potential benefits associated with the value chains of herbal medicines. Here are some of the key benefits:

- **Economic benefits:** The herbal medicine value chain can contribute to economic growth and job creation. The production, processing, and distribution of herbal medicines can create employment opportunities in both rural and urban areas. In addition, the herbal medicine industry can generate revenue through exports.

- **Sustainability:** The herbal medicine value chain can promote sustainable practices such as the conservation of medicinal plant species and the use of organic farming methods. This can help to ensure that herbal medicines are produced in an environmentally sustainable way.
- **Cultural heritage:** The value chain of herbal medicine can help to preserve and promote traditional knowledge and cultural practices. Many herbal medicines have been used for centuries by indigenous communities, and the herbal medicine value chain can help to ensure that this knowledge is not lost.
- **Health benefits:** Herbal medicines can provide a range of health benefits and can be used to treat a variety of conditions. The value chain of herbal medicine can help to ensure that these products are of high quality and are effective in treating specific health conditions.

3.5 How to Conduct a Value Chain Analysis

Conducting a herbal medicine value chain analysis can help identify opportunities for improvement and support the development of a sustainable, high-quality herbal medicine industry. It is important to involve all important stakeholders in the analysis and to develop an action plan that is feasible and realistic. The analysis involves a systematic assessment of the various stages involved in the production, processing, distribution, and consumption of herbal medicines (Fig. 6). Here are some of the key steps involved in conducting a value chain analysis for herbal medicines:

Fig. 6 Steps in conducting value chain analysis

- **Identify the key stakeholders:** The first step is to identify the key stakeholders involved in the herbal medicine value chain. This can include producers, processors, distributors, retailers, regulators, and consumers.
- **Map the value chain:** The next step is to draw out a plan indicating the various stages of the value chain, from the production of medicinal plants to the distribution and consumption of herbal medicines. This can be done using a flowchart or other visual representation.
- **Assess the performance of each stage:** Once the value chain has been mapped out, the next step is to assess the performance of each stage. This can involve analyzing factors such as efficiency, quality, sustainability, and profitability.
- **Identify opportunities for improvement:** Based on the assessment of each stage, opportunities for improvement can be identified. This might involve improving production techniques, enhancing quality control measures, or developing new distribution channels.
- **Develop an action plan:** Finally, an action plan can be developed to implement the recommended improvements. This might involve developing new partnerships, investing in new technology or infrastructure, or providing training and capacity building to stakeholders.

4 Challenges Within Herbal Medicine Value Chains

The attention paid to medicinal plants, and their value chain is not significant, as it is an area faced with diverse difficulties, most especially in the supply chain. Difficulties are experienced at every step, with a major one being exploitation by middlemen, overharvesting of herbal plants in the wild, contamination, and substandardization of herbal products and activities along the value chain, and a lack of quality control and policies of activities along the value chain [10]. Herbal medicine has been widely used in the diagnosis, prevention, and treatment of various diseases, and regulation of the medicine is a big issue. The use has created a lot of challenges relating to the safety, quality, and efficacy of the herbs. To address this, validation as well as scientific and technological standardization is required for advancement in herbal medicine [106]. The use of herbal medicine by individuals in the world today has increased, thereby making the issues that are related to their safety a crucial aspect in the value chain. Most of the adverse effects caused by herbal medicines are connected with different problems [112]. The issue of quality control is a major issue in the herbal medicine value chain and can be classified into internal and external challenges.

- **Internal challenges:** It focuses on the complexity and nonuniformity of herbal medicine ingredients which can affect the quality and distribution of herbal medicines along the value chain.
- **External challenges:** It circles contamination from microbes, toxic chemicals, pesticide residues, adulteration, and misidentification of herbal medicines, which may be peculiar to the medicine value chain. These challenges can be reduced by the implementation of rigorous good agricultural collection practices (GACP) and

good manufacturing practices (GMP) in the value chain. Modern methods and techniques in analytical and pharmaceutical strategies can be employed.

There are diverse value chains in herbal medicine with various challenges peculiar to their activities. Herbal plant cultivation could improve the livelihoods of the people who are poor and own small pieces of land [113]. To address the major challenge in the value chain, which is the issue of sustained growth of medicinal plants, a fair distribution to the primary producers must become necessary. Also, the consumers and manufacturers indirectly pay a larger proportion of the money for middlemen's value because the chain is usually weak and inherent [10]. To address this, a value chain that is integrated with only producers and processors as commercial actors and NGOs which serve as promoters can create a value chain network that is economically robust and can benefit a lot of people rather than a few [114].

4.1 Challenges in Supply Networks

The increased trade and high value of medicinal products have brought about various challenges which are concerned with the supply of the product along the value chain. The concerns have either been on addressing a specific problem as regard quality, such as poor production systems and adulteration of medicinal products, or solving a specific concern in terms of social or economic context. The supply chain can also generate concerns such as sustainable supply, responsible sourcing, and sharing of benefits. Supply has been a major issue but increased awareness has been generated since 2012 in terms of the herbal materials value chain in a globalized market [101]. The core characteristics of the supply chain sector of the herbal medicine value chain have various links, such as:

- The herbal material may be sourced from the wild or cultured and a mix of the two sourced products can be supplied.
- There may be the presence of numerous primary producers (required as a necessity).
- The role of middlemen may be required to link complex value chains with international buyers.
- The lack of knowledge and poor practices toward herbal species that are overharvested or endangered.
- There may be an upscale increase in price when the medicinal product gets to the end users from the producers.

4.2 Challenges of Quality Considerations in Herbal Medicine Value Chains

Assurance of the quality of products from herbal medicinal sources is more complex than the procedure of quality assurance of pharmaceutical products. There is high variability in the composition of natural herbal medicinal products. Therefore, the assessment of quality must be considered in terms of the diversity and the value chains from which medicinal products may originate. The safety and quality of an herbal product on a short value chain is easy to control effectively and manage.

4.2.1 Challenges of Heavy Metal Contamination

There are two main reasons why herbal products are prone to contamination by heavy metals; these are pollution and natural accumulation. In developing countries, pollution from heavy industries is a major problem, and the issue has been addressed in some countries. Although the route pollutants find their way to cultivated lands through water channels will take a long time to remediate [65]. Some plants may accumulate heavy metals naturally, e.g., cadmium can be accumulated by *Perillia* sp or *Hypericum perforatum,* and some are found in areas where the subsoil contains heavy metals, e.g., cinnabar in the formation of rocks which can result to the high occurrence of mercury in some species. These problems can be reduced by careful identification and selection of sites for cultivation or apt monitoring and control in wild herbal plant species collection. There is less need for the use of pesticides in the short supply value chain, but due to expectations by customers and global supply a high level of pesticide usage is required to produce increased yield at a reduced cost. Training and careful management of the value chain regarding pesticide use and quantities can keep the number of pesticide residues in herbal plants within the allowable limits.

4.2.2 Challenges Associated with Antimicrobial Resistance

The antimicrobial resistance in herbal medicine usually arises when bacteria and fungi which are termed germs may overpower the herbal drugs that were fashioned to attack them. This implies that the germs that were meant to be killed by the herbal drugs are still living and they continue to grow. In some cases, such resistant germs or infections are usually difficult and almost impossible to eliminate. The Centers for Disease Control and Protection (CDC) reported such resistance as a public threat and has been linked to about 5 million deaths and killed a minimum of 1.27 million people across the world [115]. More than 35,000 deaths occur in the USA based on drug resistance. The extent of danger is not only observed when bacteria or fungi develop resistance to herbal drugs but when it is resistant to just one is a serious issue. For instance, some ailments/infections may require further treatments after herbal medications can cause serious side effects to patients and prolong recovery periods. With all of these events, it can be said that the issue of antimicrobial resistance can affect people at any life stage, therefore, making it a major health concern for the public.

The World Health Organization also reported antimicrobial resistance as one of humanity's major public threats [116], which is projected to claim over 10 million lives by 2050 and an economic cost of US $100 trillion globally [117]. The presence of resistant bacteria and fungi is becoming resistant to present antibiotics, and with few products in the pipeline, it is important to urgently look into the development of new strains by taking a cue from herbal medicine. The issues of resistance are very complicated and is multifaceted and may contain the following variables as reported [118, 119]:

- Lack of hygiene in hospitals, thereby increasing bacteria resistance
- The overuse of antibiotics in agricultural activities
- The spread of resistant varieties through international travels
- Excessive use by humans
- Lack of diagnosis on the appropriate antibiotic to use

Apart from the health sector, resistance can also occur in the food industry. Aflatoxins produced by *Aspergillus* spp. are a major mycotoxin in the food industry that is carcinogenic and highly toxic [120]. The contamination varies with the type of products; it is high in wheat, maize, nuts, and barley [121, 122]. The risks of contamination can be reduced by storage, processing, or transportation as the fungi grows when the conditions are right for it to thrive.

4.2.3 Challenges of Sustainable Sourcing of Herbal Products

There are many problems regarding the sustainability of herbal products and the value chain, one of which caused the withdrawal of some services in the markets [123]. Based on this, there are changes in the supply chain of goods, an increase in awareness, and increased demand by customers for herbal products. The reports of the Ethical Consumer Markets 2017 stated that the sustainability certification of herbal materials increased consumer spending [124]. The blockchain approach can go a long way in addressing the issue of sustainability as it can aid the collection and processing of herbal materials, thereby providing information that is the baseline for the products. Such information will be readily available for all the actors along the value chain. In a case where there is already a developed sustainable production system, the blockchain system can assist in certifying the production systems. Overexploitation of herbal materials can be prevented by tracking and information about the initial processing methods, thereby producing a better system of production.

4.2.4 Challenges of Intellectual Property

Intellectual property rights are an area of contention in herbal products and their value chain, and it is essential to abide by international agreements and national laws to forestall the development of herbal products rationally. Herbal medicines are mostly used in traditional settings in most developing countries and have become often adapted for use in developed countries, thereby creating a high commercial value for these products. These has created avenues for diverse conflicts in terms of patenting without the consent of or compensation to the original custodian of these herbal medicines. Originally, it is believed that the rights to this indigenous knowledge are bestowed on the indigenous people because they are close to the biological resources, have the knowledge, and know the appropriate items to use at a particular time. They are directly linked with the biodiversity conservation of these herbal plant species. In the present state, dilemmas, challenges, and several ethical questions have arisen based on the operations of the indigenous people toward conservation. Timmermans [125] reported that ways by which modifications can be made and new forms of ownership based on intellectual property that can settle the rift between indigenous people and modern science should be created. The author continued that the protection of herbal medicine intellectual property has diverse objectives and proposed strategies may affect one another as well as affect the goals and prioritization of proposed objectives. It was concluded that policies should address the multicomplexed questions and issues and try to balance them by developing a range of solutions to address each question and issue. Similarly, Mokgobi [126] opined that the integration of the traditional healing system and Western biomedicine can go a long way in addressing these challenges. As in South Africa, the number of traditional practitioners are huge and the government also supported the integration of the two systems of health care.

5 Recent Development in Herbal Medicine Value Chains and the Concept of Blockchain

There are various opportunities arising from the medicine value chains using a transdisciplinary approach [101]. A more recent approach that involves blockchain systems may be used to address some issues arising from the dynamic socioeconomic value chain frameworks such as the limitations in the number of middlemen in the value chain. This approach is quite useful in updated information dissemination and activities across the value chain. It is quite useful in supply chains with many stakeholders, and the benefits that accrue to producers, production channels, materials sourcing, and quality final herbal products can be achieved. The knowledge of an herbal product along the value chain is key in the exchange of goods and information, as well as information relating to the composition of the product, type, source, quality, and other information of importance about the product, which are stored in ledgers.

The concept of blockchain replaces the analog ledger approach, and it is a system that is built on decentralized data storage by several users, each forming a network node [124]. Information are usually stored in blocks, and all form a chain on the network which is only modifiable one block at a time. The technology is a very transparent database that is advanced and presents information sharing in a very transparent form within a business setup. It was said that there are enormous opportunities for using blockchain technology in food industries and herbal medicine distribution [126, 127]. This approach can be vital in complex transaction networks that lack trust in their value chains. The blockchain will provide a safe interphase for interaction between various actors along the value chain without necessarily building individual trust. The trust is usually in the blockchain system and optimized for better delivery [128, 129]. The negotiations can be validated using smart contracts, which are fused into the blockchain framework. These contracts are effective ways of validating contracts in the digital currency world [128]. When conditions on the contracts are met, payments will be automatically released to the parties, ensuring a fraud-free transaction.

The benefits of the use of blockchain technology in the herbal medicine value chain are:

- It will increase trust and reliability in parties or buyers that do not trust each other.
- It will remove third-party involvement, increase benefits economically, and decrease costs incurred in transportation and logistics.
- All records and transactions are saved and can be recalled for transparency and promotion of good practices.
- Small-scale producers can receive immediate benefits from commercialization.

6 Conclusion

The use of medicinal plants has become a popular activity all over the world due to their unique advantages and medicinal value. These medicinal herbs are becoming more and more important on a global scale as a result of their growing traditional use and cultural acceptance. They are also well recognized and have minimal side effects. These medicinal plants used for herbal medicine can be made more well known and used to boost the economies of low-income nations and provide jobs for their citizens. The majority of African nations rely heavily on the primary health care approach, hence it is crucial to integrate the use of medicinal plants into every aspect of primary health care in these nations. The rural population of nations needs high-quality raw materials for the development of drugs from medicinal plants to meet their fundamental health care demands. By including them in the processing and collecting of medicinal plants, the vast majority of people living in rural regions can also be given employment opportunities. The popularization of medicinal plants is crucial for the preservation of traditional knowledge. The state government should take action to educate the populace about the importance of these medicinal plants. Additionally, more resources should be allocated toward enhancing health-care-based systems. This will spur the development of the concerned industry. Farmers should be provided with top-notch marketing tools. It is important to use cutting-edge marketing strategies so that farmers experience fewer challenges and earn more money.

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Chemopreventive Practices in Traditional Medicine

37

J. Ekowati, Retno Widyowati, Norhayati, and Sachin Kumar Jain

Contents

Abstract

Chemotherapy is routinely used for the treatment of cancer. However, chemotherapy treatments, for example, 5-fluorouracil, doxorubicin, bleomycin, and cyclophosphamide, give various kinds of side effects and toxicity. To hinder the

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development of carcinogenic tumours, some patients in developing countries take traditional medicine as chemoprevention, an alternative, complementary, and supportive therapy, in spite of following the standard therapy protocol. Several traditional medicines from China, India, and Egypt are known for this purpose. There are some plants used as chemopreventive agents with several formulas such as *Annona muricata*, *Curcuma longa*, *Curcuma xanthorrhiza*, *Curcuma zedoaria*, *Euphorbia abyssinica, Garcinia mangostana*, *Gynura procumbens*, *Kaempferia galanga, Morinda citrifolia*, *Moringa oleifera*, *Nigella sativa*, *Phyllanthus urinaria,* and *Zingiber officinale*. Various numbers of *Annona muricata* extracts demonstrated anticancer activity in vitro and in vivo. Acting as a p53 regulator and pro-oxidant in MCF-7 cells, and also as a fatty acid synthase inhibitor in MDA-MB-231 cells, curcumin is collected from *Curcuma longa* and *Curcuma xanthorrhiza*. Through modulation of Bcl-2, p53, and PARP-1 protein levels, xanthorrhizol from *Curcuma xanthorrhiza* has pro-apoptotic activity. Curcumenone, curcumenol, and curdion, which are from *Curcuma zedoaria,* display pro-apoptotic activity in various cancer-induced cell lines and mouse models. The apoptotic effect induced by corilagin is mediated by extrinsic and mitochondrial pathways, whereas the apoptotic effect induced by geraniin is mediated by ROS-mediated stimulation, both in MCF-7 cells. However, the pro-apoptotic effects of corilagin and geraniin from *Phyllanthus urinaria* are distinct. *Nigella sativa* extract contains some active ingredients, including thymoquinone, which can extend the time of DMBA-induced tumour formation in mice. Active compounds from *Kaempferia galanga* can inhibit carcinogenesis by inhibiting proliferation of cell lines and also angiogenesis. Based on the above data, the use of chemoprevention in inhibiting carcinogenesis has a very good prospect for further development of sustainable food sources.

Keywords

Traditional medicine · Chemopreventive · Antioxidant · Anti-inflammatory · Antiproliferation · Apoptosis · Angiogenesis

Abbreviations

AGE	Acetogenin
ALK	Alkaloid
BDMC	Bisdemethoxycurcumin
CP	Cyclopeptide
CUR	Curcumin
DMC	Demethoxycurcumin
ERK	Extracellular signal-regulated protein kinase
FTG	Flavonol triglyceride
GCMS	Gas chromatography-Mass spectrometry
HBV	Hepatitis B virus
HPLC	High-performance liquid chromatography
MAPK	Mitogen-activated protein kinase

MG	Megastigmane
OVA	Ovoalbumin
PCNA	Proliferating cell nuclear antigen
ROS	Reactive oxygen species
XTL	Xanthorrhizol
YES	Yeast Extract Sucrose

1 Introduction

To date about 88% of all countries in the world use traditional medicine, such as herbal medicine, acupuncture, yoga, indigenous therapy, and others. Even 170 WHO member countries reported the use of traditional medicine; now it has become an important resource for health for centuries in communities throughout the world and is still a mainstay for some people who have uneven access to conventional treatment. Sociocultural practices and the existence of biodiversity from traditional medicine are invaluable resources for sustainable and diverse sustainable development for curing the diseases.

Defined as the unusual growth of cells in body tissue, cancer is a frequent condition in which destructive change occurs without apparent cause [1]. Factors such as sedentary lifestyle, poor eating habits, smoking, etc., can contribute to the growth of tumours [2]. According to the WHO, the number of cancer patients in the world increases by about seven million people every year, two-thirds of them in the developing countries. Chemotherapy, radiation exposure, and complicated medical procedures are required during the costly and side-effect-filled treatment of cancer.

Despite using chemical drugs to treat cancer, another way to cure or prevent cancer is by using phytotherapeutic agents or traditional medicine consisting of one active ingredient or mixture complexes of various origins of plant parts. Those herbs are commonly grown and known in some countries, like Asia and Africa. For example, over the last decade, the use of herbal medicines has increased in Indonesia based on the fact that Indonesia has many biodiversity contents of natural substances that can be utilized.

The aim and goal of this chapter are to investigate various traditional herbs from China, India, Indonesia, and other Asian countries that are used as chemoprevention practices. It also will go into more specifically about the active compounds of each herbal medicine, in vitro and in vivo assay, and chemoprevention mechanism research on some herbal medicine. The data were obtained by exploring the keywords (chemoprevention, herbal medicine, antioxidant, anti-inflammatory, apoptosis, angiogenesis, anticancer) in Google Scholar, PubMed, and other Internet sources.

2 Chemopreventive Practices in Traditional Medicines

Chemoprevention is the use of certain compounds, whether in the form of synthetic drugs or natural ingredients, to help reduce the risk of getting cancer, organ complications, or prevent the occurrence of these diseases. It can also be used for inhibition delay or to reveal carcinogenesis before invasion. Agents that can be used are classified into four categories, including hormonal, medications, diet-related agents, and vaccines. Tamoxifen and raloxifene are used in averting several types of breast cancer in women who are at high risk of developing the disease, for instance [3].

Several traditional medicines from Indonesia, China, India, and Egypt were recognized to be used for this purpose as chemopreventive agents with several formulas, such as *Curcuma longa*, *Annona muricata*, *Curcuma xanthorrhiza*, *Phyllanthus urinaria*, *Curcuma zedoaria*, *Gynura procumbens*, *Morinda citrifolia*, *Garcinia mangostana, Nigella sativa*, *Zingiber officinale*, *Moringa oleifera*, *Kaempferia galanga,* and *Euphorbia abyssinica.*

2.1 *Annona muricata*

2.1.1 Taxonomy

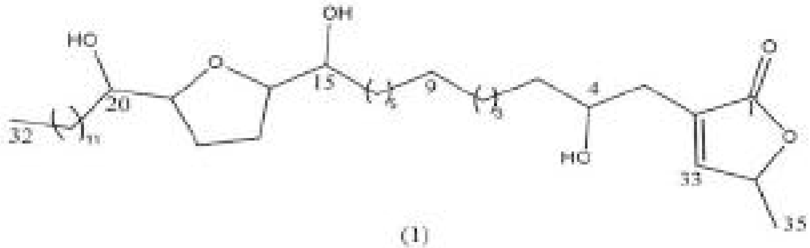
Native to South and North America, *Annona muricata,* or soursop, is a plant in the family Annonaceae. As shown in Fig. 1 [4, 5], it has an open circular canopy with large, glossy, dark, and green leaves characteristically.

The tree produces large, individual yellow flowers on woody stalks, and its edible fruits are able to weight above 4 kg. Each fruit segment contains a single, oval, smooth, hard, black seed. The leathery skin of the fruit is covered in short spines [6].

2.1.2 Phytochemistry

A. muricata contains numerous compounds as well as megastigmanes (MGs), alkaloids (ALKs), flavonol triglycerides (FTGs), cyclopeptides (CPs), essential oils, and phenolics (PLs). This plant also contains annonaceous acetogenin compounds (AGEs). *A. muricata* fruit contains nutrients and essential elements for humans because it contains elements such as Ca, Cu, Fe, K, Na, and Mg [4]. Additionally, as illustrated

Fig. 1 The fruits of *Annona muricata* L.



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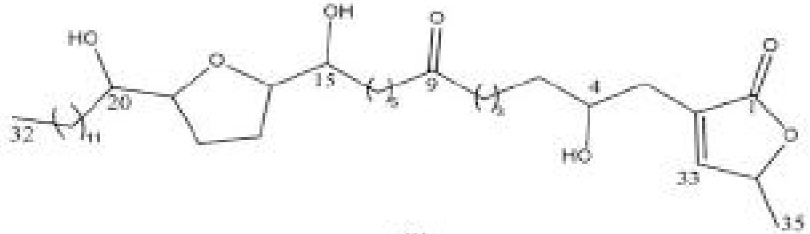

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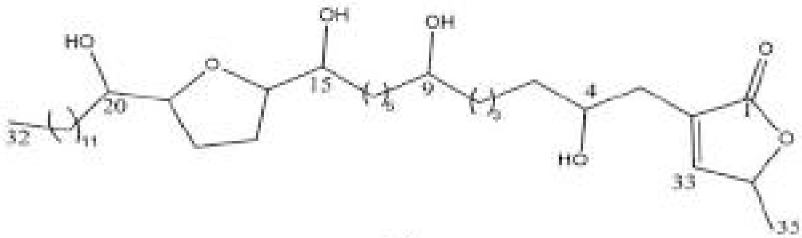

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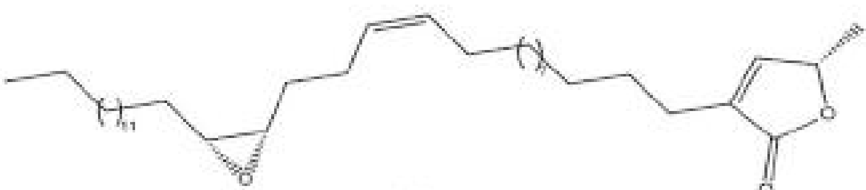


(4)

Fig. 2 Acetogenins from *Annona muricata* seeds and fruits: murisolin (1), annoreticuin-9-one (2), *cis*-annoreticuin (3), and sabadelin (4). (Ragasa et al. [7])

in Fig. 2, Ragasa & Shen [7] reported compounds such as acetogenins: murisolin (1), annoreticuin-9-one (2), *cis*-annoreticuin (3), and sabadelin (4).

Annona muricata (soursop) is a plant that contains several isoquinoline alkaloids, including anomuricine, anomurine, atherosperminine, coclaurine, coreximine, reticuline, and stepharine in its leaves, root, and stem barks. Additionally, annomuricatin is present in the plant's seeds. Besides, the fresh fruit pulp of *A. muricata* also contains the essential oil which consists of ethyl 2-hexenoate (8.6%), methyl 2-butanoate (2.4%), methyl 2-hexenoate (23.9%), methyl 2-octanoate (5.4%), α-terpineol (2.8%), β-caryophyllene (12.7%), 1,8-cineole (9.9%), linalool (7.8%), linalyl propionate (2.2%), and calarene (2.2%). The leaves of *Annona muricata* also contain annohexocin and mono-THF annonaceous acetogenin that display substantial repressive effects against six human cancer cell lines [8].

2.1.3 Usage as Chemoprevention

A. muricata has been the subject of numerous anticancer studies, including both in vitro and in vivo studies. While investigating a particular case, an elderly woman diagnosed with metastatic breast cancer consumed boiled *A. muricata* leaves along with Xeloda, resulting in the stabilization of the disease [9]. It is substantial in preventing cancer and tumour growth that was observed in *A. muricata* leaves that have been utilized in the development of tablet formulations containing the ethyl acetate-soluble fraction of the leaves. The fraction of AGEs can be used in adjuvant therapy [10]. On the other hand, as shown in Table 1, the primary mechanism of action has been investigated in several studies.

The effects of an aqueous extract made from the leaves of *A. muricata* on the benign prostatic hyperplasia (BPH-1) cell line and the prostates of rats were the subject of recent in vitro and in vivo experiments. The study showed that BPH-1 cell activity was reduced by the treatment. Following treatment for 72 h, the mRNA levels of Bax were upregulated and Bcl-2 were downregulated, along with the IC50 value of 1.36 mg/mL. Moreover, 2 months of treatment with the extract (at doses of 300 and 30 mg/mL) resulted in a reduction in the size of prostates of rats, likely due to apoptosis induction [11].

Promising antitumour effects of *A. muricata* also revealed in vivo investigation using mice as a subject with 7,12-dimethylbenzene anthracene (DMBA)-induced cell proliferation in the breast tissues. According to the investigation, *A. muricata* leaves administered orally were effective in preventing the damage of DNA induced by DMBA, which may help prevent the growth of breast carcinogenesis [12]. Furthermore, even at a reduced dosage of 30 mg/kg, the leaves suppressed the promotion and initiation stages of skin papillomagenesis in mice induced by DMBA and croton oil, respectively [13].

Annona muricata Linn. leaves have been shown to decrease the expression of the p53 protein in cancer patients infected with HPV, resulting in increased apoptosis. When a virus infected a cell, there is a stress increment and the p53 protein expression rises as a response. However, a reduction in the p53 protein expression can disturb the cell's ability to undergo apoptosis. Being the main role in apoptosis regulation, proteins such as p53, Bcl-2, and caspase-3 act as inhibitors of cancer. A decrease in apoptotic cells can cause an imbalance between apoptosis and cell proliferation, triggering the growth of cancer. Mutations in the p53 gene are responsible for the loss of p53 function in 50% of human cancer cells. The E6 protein of

Table 1 Anticancer studies of *A. muricata*

Plant part	Subject of study	Chemoprevention mechanism
Ethyl acetate extract of the leaves	Lung A549 cancer cells	Mitochondrial-mediated apoptosis, cell cycle arrest at G1 phase
Ethyl acetate extract of the leaves	Colon HT-29 and HCT-116 cancer cells	Mitochondrial-mediated apoptosis, cell cycle arrest at G1 phase, suppression of migration and invasion
Water extract of the leaves	Rat's prostate	Reduction of prostate size
Ethanolic extract of the leaves	Breast tissues of mice	Prevention of DMBA-induced DNA damage
Ethanolic extract of the leaves	DMBA/croton oil-induced mice skin papillomagenesis	Suppression of tumour initiation and promotion
Ethanolic extract of the leaves	DMH-induced colon cancer	Reduction of ACF formation
Ethanolic extract of the leaves	K562 chronic myeloid leukaemia cells	Induction of apoptosis
Leaves boiled in water	Metastatic breast cancer	Stabilization of disease
Ethyl acetate extract of the leaves	Azoxymethane-induced colon cancer	Reduction of ACF formation
Ethyl acetate extract of the leaves	Colon HT-29 cancer cells the leaves	Bioassay-guided isolation of annomuricin E and its apoptosis-inducing effect

Mugadhamtousi et al. [4]

HPV-16 and 18 can inactivate the p53 gene product through the ubiquitin-dependent proteolytic pathway (E6AP), leading to a decrease in p53 protein levels. Restoring the function of mutated p53 could trigger mass apoptosis, effectively killing cancer cells [14].

Astuti et al. [15] showed that the ethanol extract of *A. muricata* leaves contains antiproliferative effects on WiDr cancer cells. The range of IC50 values of the extract was from 6 to 12 μg/mL (after 48 h). Another study on ethanol extract of *A. muricata* from Tawangmangu also showed antiproliferation on WiDr cancer cells with an IC50 of 450.05 μg/mL, which was much larger than the previous study due to the differences in location and quality of the leaf essence. In addition, incubation with *A. muricata* on HepG2 liver cancer cells for 24 and 48 h showed LD50 values of approximately 80 and 180 μg/mL, respectively. Entering the G0 phase of G1 is a crucial choice in cell development. Additionally, in situations where the cellular environment is insufficient to cross the restriction point, it will return to the G0 phase.

Fig. 3 *C. longa* rhizomes with yellowish to orange colour

2.2 *Curcuma longa*

2.2.1 Taxonomy

Turmeric, which is a perennial shrub in the family Zingiberaceae and also known as *Curcuma longa* Linn, originates from India as depicted in Fig. 3 [16]. The plant can grow up to a height of 3–5 feet and is now cultivated in other tropical countries such as China and Sri Lanka. The roots of the plant are highly valued and are utilized for religious, culinary, and medicinal purposes in India [17].

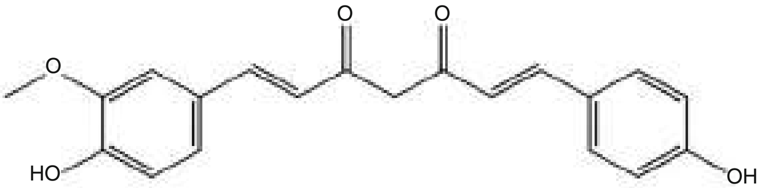
After harvesting, the turmeric rhizomes are washed and boiled in mildly alkaline water to soften them. They are then dried either in the sun or using an electric dryer. Turmeric is usually used as a colouring agent in diverse industries such as pharmacy, confectionery, and the food industry. It is also used for dyeing materials such as wool, silk, and cotton, and can be combined with other natural dyes to create a range of different shades [18].

2.2.2 Phytochemistry

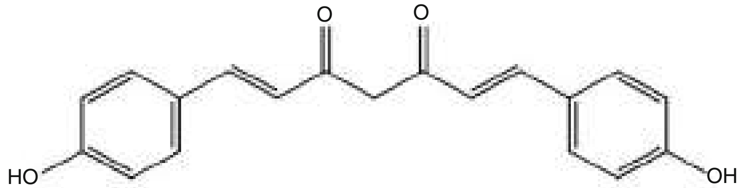
The three main chemical constituents of turmeric – curcumin (CUR), demethoxycurcumin (DMC), and bisdemethoxycurcumin (BDMC) – are illustrated in Fig. 4. These compounds, which belong to a group of chemicals called curcuminoids, make up approximately 77%, 17%, and 3% of turmeric, respectively. Each of these constituents has distinct biological activities [19–21].

Two active components that are found in turmeric are curcuminoids and volatile oil, usually found in the extraction of oleoresin from turmeric root. The essential oils in turmeric consist mostly of sesquiterpenes. Pungent scent of the spice is primarily acquired from α- and β-turmerones and aromatic turmerone (Ar-turmerone) [22]. Curcuminoids are not very soluble in water when the pH is acidic or neutral due to their chemical structure, but they are soluble in acetone, dimethyl, ethanol, methanol, and sulfoxide [21].

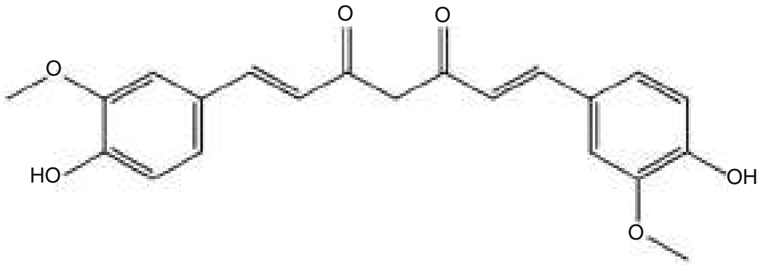
Several active compounds have been discovered in *C. longa*, including diarylheptanoids such as curcumin, demethoxycurcumin, and bisdemethoxycurcumin,



Demethoxycircumin (DMC)



Bisdemethoxycurcumin (BDMC)



Curcumin (CUR)

Fig. 4 The chemical structure of curcumin (CUR), demethoxycurcumin (DMC), and bisdemethoxycurcumin (BDMC). (Huang et al. [19])

as well as sesquiterpenoids such as α-curcumene, β-at-lantone, curcumol, diterpenoids, dehydrocurdione, phenolic acids, and polysaccharides [23, 24]. In previous investigations, some sesquiterpenoids have also been identified. In an effort to identify new biological elements present in *C. longa* and enhancing its chemical components, the rest of the fractions of the plant were further fractionated, resulting in the discovery of a new bisabolane-type sesquiterpenoid, (6S)-2-hydroxy-6-[(1′S,5′R)-(4′-methene-5′-hydroxyl-2′-cyclohexen)-2-methylheptan-4-one], and turmerone Q (1). Six known compounds, including three sesquiterpene derivatives (2–4), two diarylheptanoids (5–6), and one diphenyl alkane (7), are illustrated in Fig. 5 which were also identified [25–28].

The essential oil of *C. longa* rhizome and leaf contains several compounds, with the most abundant being α-phellandrene (9.1%), α-turmerone (24.4%), ar-turmerone

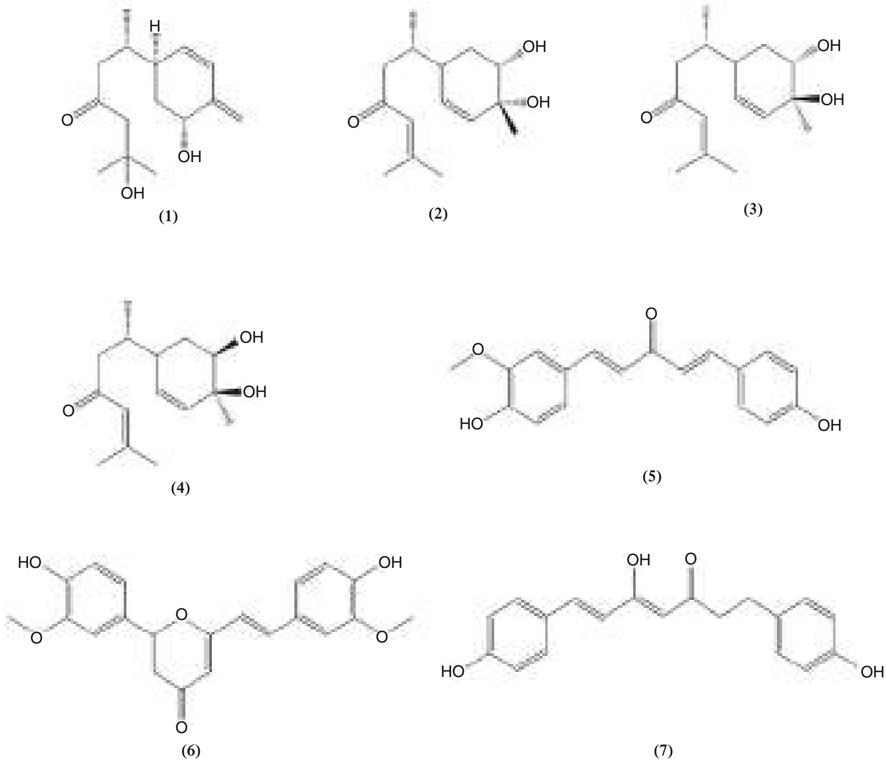


Fig. 5 Structures of compounds 1–7. (Yuan et al. [25])

(25.5%), β-sesquiphellandrene (5.1%), β-turmerone (14.0%), terpinolene (7.2%), (Z)-β-ocimene (5.5%), undecanol (7.1%), p-cymene (5.5%), α-zingiberene (4.8%), ar-curcumene (1.6%), β-caryophyllene (2.9%), and 1,8-cineole (1.3%) [29, 30].

2.2.3 Usage as Chemoprevention

C. longa possesses multiple pharmacological activities, including anti-inflammatory, antioxidant, hepatoprotective, anticarcinogenic, antidiabetic, antimicrobial, and anti-depressant effects. It has also been studied for its potential use in treating cardio-vascular disease, gastrointestinal disorders, and neurological disorders [31].

In rabbits, the consumption of turmeric powder has been shown to increase the secretion of mucin, which may help protect against irritants in the gastrointestinal tract. While turmeric exhibits both anti-ulcer and ulcerogenic properties, research has found that curcuma, a compound found in turmeric, is as effective as ranitidine in protecting the gastric mucosal layer. When administered orally, turmeric has been found to inhibit gastric acid and gastric juice secretion, as well as prevent ulcer formation, similar to the properties of ranitidine. Turmeric has also been shown to have a protective effect on the cardiovascular system by lowering the levels of cholesterol and triglyceride, lowering the susceptibility of low-density lipoprotein (LDL) to lipid peroxidation, and preventing platelet aggregation. Animal experiments suggest that the presence of turmeric extract and curcumin in turmeric can potentially reduce the occurrence of tumour formation in test animals. This research shows that turmeric might be an effective anticancer ingredient [32].

Kim et al. [33] demonstrated chemopreventive effects in an in vitro model related to hepatitis B virus (HBV). A study was conducted on HBV X protein (HBx) transgenic mice which aimed to analyse the effects of chemopreventive mechanism in vivo. The mixture of a *Curcuma longa* Linn (CLL) was concentrated with dextrose water by boiling and then lyophilized, and the resulting extracts were administered to the mice for 2–4 weeks starting at 4 weeks of age or for 3 months starting at 6 months of age. The liver tissue was analysed for histological changes and the expression of genes associated with HBx. Additionally, CLL treatment led to an increase in p-p53 expression. These findings indicate that CLL could be advantageous in preventing and postponing liver cancer by affecting the initial and later phases of liver disease. Therefore, CLL may be a potential candidate for chemopreventive agents against liver cancer [34].

In traditional Indian Ayurvedic medicine, turmeric (*Curcuma longa* L.) has long been used for a variety of medical conditions. In addition, turmeric has also been reported against DENA-induced cancer of the skin, stomach, colon, oral cavity, and liver in rats [35]. Turmeric provides a chemopreventive effect against HCC induced in Wistar rats and relatively improves biochemical parameters so that it can control and delay the initiation of carcinogenesis [36]. Curcumin's antioxidant effects, which can significantly lower the level of oxidative stress that contributes to the growth of cancer, maybe the cause of its chemopreventive properties [37].

The various levels of regulation in cellular growth and apoptosis that are targeted by curcumin to exert its anticancer effects are depicted in Fig. 6. Curcumin not only affects transcription factors, oncogenes, and signalling proteins in a vertical manner, but it also acts in different stages of carcinogenesis, starting from the initial DNA mutations to the development of tumours, their growth, and metastasis. Due to its broad and multifaceted impact on the regulatory processes of cell growth, curcumin is a promising candidate for chemotherapy in several types of human cancers [38].

Several studies have demonstrated that curcumin can effectively block the stimulation of NF-κβ by blocking the phosphorylation of IkBα through IKK. This hindrance of NF-κβ results in a reduction in the expression of several pro-inflammatory cytokines, nitric oxide synthase, MMP-1, MMP-3, MMP-9, COX-2, and cyclin D1. As a consequence, curcumin has the potential to decrease cell proliferation, growth, and tumour angiogenesis [39]. Furthermore, curcumin can directly inhibit COX-2 and 5-lipoxygenase, indicating its direct anti-inflammatory activity, which plays an active role in reducing inflammation in the tumour microenvironment [40].

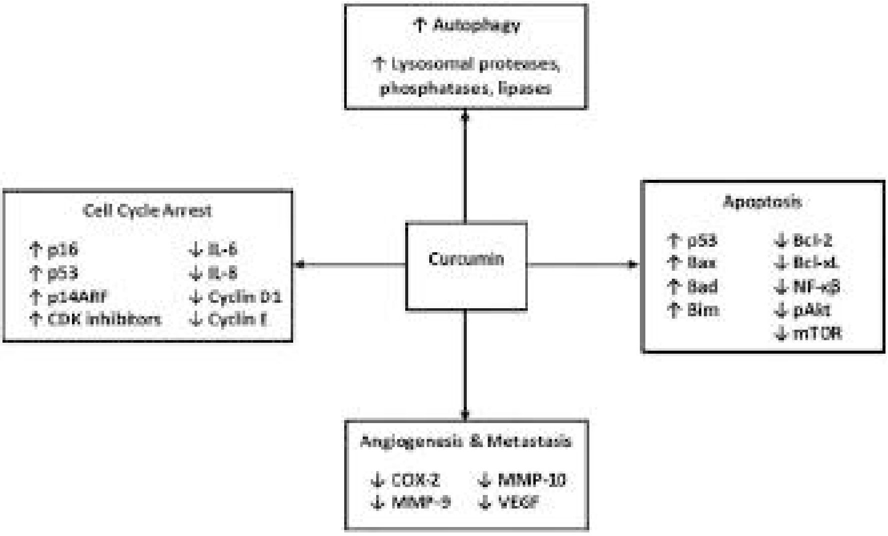


Fig. 6 Overview of the anticancer effects of curcumin. (Wilken et al. [38])

2.3 *Curcuma xanthorrhiza*

2.3.1 Taxonomy

Curcuma xanthorrhiza Roxb., also known as Java turmeric, is a plant in the Zingiberaceae family and is commonly grown in the Southeast Asian region [41]. Although it originates from Indonesia, it is also grown both in the wild and in cultivation in countries such as the Philippines, Thailand, Malaysia, and Sri Lanka [42]. The tuber of the plant has a circular shape and a yellowish skin on the outside as depicted in Fig. 7. The rhizomes have a fragrant aroma and a bitter taste. Historically, the rhizomes of *Curcuma* species have been used as a source of yellow dye, along with a spice and food preservative, flavouring agent, and remedy for treating various diseases, owing to their chemical compounds.

2.3.2 Phytochemistry

C. xanthorrhiza contains several major components as well as starch (48–59%), volatile oils (12%), including phellandrene, camphor, tumoural, cineol, borneol, and xanthorrhizol, sesquiterpenes (ar-curcumene, β-curcumene, bisabolane, germacone, lactone), flavonoids (apigenin, catechin, epicatechin, quercetin, kaempferol, luteolin, myricetin, naringenin), and curcuminoids [43].

The majority of these chemicals are curcuminoids and terpenoids, which possess essential biological properties [44]. Additionally, xanthorrhizol is the predominant secondary metabolite gained from the essential oil of its rhizome based on various studies [45]. However, other parts of the plant, such as the flower bract, are likely to contain various essential secondary metabolites. For instance, the flower bract

Fig. 7 The rhizome powder of *Curcuma xanthorrhiza* Roxb

consists of xanthorrhizol, α-curcumene, β-elements, *trans*-caryophyllene, β-farnesene, camphor, and isoborneol [46].

2.3.3 Usage as Chemoprevention

C. xanthorrhiza possesses multiple pharmacological activities, including anti-inflammatory, antioxidant, and anticancer effects by inhibition of glutathione transferase [47–49, 51].

Devaraj et al. [50] reported that the antioxidant properties of *C. xanthorrhiza* were evaluated in vivo on male Sprague–Dawley rats weighing between 150 and 200 g that were remedied with carbon tetrachloride (CCl_4) as a hepatotoxin. The hexane fraction of *C. xanthorrhiza* demonstrated its ability to act as a hepato-protector through its antioxidant properties [52]. In a separate study by Widyastuti et al. [53], the antioxidant characteristics of *C. xanthorrhiza* were investigated across diverse places. The antioxidant activity of *C. xanthorrhiza*'s methanol extract obtained from different locations was evaluated, as well as its overall phenolic and total flavonoid content.

Itokawa et al. [54] reported during the initial antitumour evaluation multiple compounds (ar-turmerone, β-atlantone, α-curcumene, and xanthorrhizol) that were extracted from *C. xanthorrhiza*'s rhizome and tested on rats with sarcoma 180 ascites. Notably, xanthorrhizol was found to significantly suppress the formation of tumour spots in lung tissue and the growth of tumour masses in the abdomen. Molecular analysis further demonstrated that xanthorrhizol exhibited inhibition of COX-2 expression, phosphorylated extracellular signal-regulated kinase (ERK), and matrix metalloproteinase-9 (MMP-9) in metastatic mice, providing additional support for its antitumour activity.

XTL is a highly active and abundant component found in the essential oil of the *Curcuma xanthorrhiza*'s rhizomes [55]. In addition to its nephroprotective and hepatoprotective effects demonstrated in male ICR mice remedied with cisplatin, XTL has also shown potential as a suppressor of carcinogenesis. The cytotoxic impact of XTL might be due to the presence of its phenol group, which is known to possess cytotoxic activity. XTL's anticancer mechanisms are varied, affecting various aspects of cellular growth and apoptosis. The morphology associated with apoptosis is typically described by chromatin condensation, DNA fragmentation, cell shrinkage, and elongated lamellipodia. The mechanisms underlying XTL's anticancer properties are strongly linked to its ability to function as anti-inflammatory agent and an antioxidant, induce apoptosis, and arrest the cell cycle [43].

Lee et al. [56] reported that XTL, the marker compound found in *C. xanthorrhiza*, contributes to its anti-inflammatory ability. Research has shown that XTL exhibits in vitro anti-inflammatory activity, as demonstrated by its ability to inhibit the production of nitric oxide (NO) and prostaglandin E2 (PGE2) in lipopolysaccharide-induced mice leukemic monocyte-macrophage cell RAW 264.7. This inhibition leads to a reduction in the expression of cyclooxygenase-2 (COX-2) and inducible nitric oxide synthase (iNOS). Also, XTL can inhibit the expression of COX-2, iNOS, TNF-α, and IL-6 in induced microglial cells. Moreover, the effects of XTL in vivo using mice with severe inflammation induced by 12-O-tetradecanoylphorbol-13-acetate (TPA) are known and found that XTL could inhibit the expression of ODC, COX-2, and iNOS in the mice dermis. These results suggest that XTL may play a role in reducing the levels of IL-6 and TNF-α, and inhibiting the activation of COX-2 and iNOS via the NF-kB pathway, leading to decreased production of PGE2 and NO. Additionally, preliminary findings suggest that the hydroxyl group of XTL is crucial for its pharmacological effectiveness as acetylation of this group leads to a loss of activity [57].

Furthermore, earlier studies have also demonstrated that xanthorrhizol can promote apoptosis by inducing the mitochondrial pathway (dependent on p53) in liver cancer cells (HepG2) and cervical cancer cells (HeLa) [58].

2.4 *Curcuma zedoaria*

2.4.1 Taxonomy

Curcuma zedoaria Rosc. is a plant species native to India that has also been widely cultivated throughout Southeast Asia, as well as Indonesia. The appearance of *Curcuma zedoaria* is similar to ginger on the outside and turmeric on the inside. This perennial plant can grow up to 1 m in height, and its main rhizome is ovoid in shape. The leaf blade of *Curcuma zedoaria* typically measures 80 cm in length and often features purple on both surfaces, blotches running along the midrib. This plant is illustrated in Fig. 8 [59–61].

C. zedoaria is a rhizomatous perennial plant with herbaceous, erect stems that develop into pseudo-stems, bulbs, and branches. It also has subterranean cylindrical rhizomes and succulent roots, with some roots producing terminal storage structures that can take the form of round or elongated tuber-like roots known as t-roots [62].

Fig. 8 The rhizome of *Curcuma zedoaria* (Christm.) Roscoe

2.4.2 Phytochemistry

Plants produce secondary metabolites that vary in type and concentration across different species. These metabolites serve as a defence mechanism against threats and are also used in plant identification. *Curcuma zedoaria* primarily contains terpenoids, predominantly sesquiterpenoids, phenolics, tannins, alkaloids, terpenoids, saponins, and steroids [59]. The rhizome oil of *Curcuma zedoaria* consists primarily of monoterpenoids (15–20%) and sesquiterpenoids (80–85%). The primary compounds of the essential oil of *C. zedoaria*'s rhizome are curzerene, epicurzerene, curzerenone, curdione, debromofiliforminol, β-sesquiphellandrene, 1,8-cineole, curcumenene, p-cymene, and α-phellandrene. On the other hand, the primary compounds of the leaf oil are dehydrocurdione, isoborneol, α-Terpinyl acetate, and selina-4, 7(11)-dien-8-one [63].

The content of secondary metabolites of essential oil from *Curcuma zedoaria* rhizome is influenced by various factors, including the environment in which the plant grows. A study of Indonesian *Curcuma zedoaria* oil found it to be comprised of 21 constituents, with 13 sesquiterpenoids (55.6%) and 7 monoterpenoids (40.1%) [64]. The main compounds of *Curcuma zedoaria* rhizome oil vary across different phytogeographical locations in Asia. *Curcuma zedoaria* from India is known to contain curzerenone, epi-curzerenone, or 1,8-cineole as major components, whereas epi-curzerenone or furanogermenone are the chemical markers of the original species for *Curcuma zedoaria* from China. On the other hand, the typical components of *Curcuma zedoaria* originating from Taiwan and Japan are curcumenol and dehydrocurdion [65].

Curcuma zedoaria rhizome extract using hexane and dichloromethane contains several compounds, including 16 dial dehydrocurcumin, curcumol, procurcumin, curcumone, labda-8,12diena-15, komoson II, zerumbone epoxide, zederon, 9- isopropylidene-2,6-dimethyl-11octylo[6. 2.1.01, 5]undec-6-en-8-ol, curcumanolide A, curcuzedoalide, germacrone, and gweicurculactone, furanodiene, germacrone-4,5-epoxide, calcaratarin A, isoprocurcuminol, germacrone-1,10-epoxide, and zerumin A [66]. In addition, 27 compounds were successfully isolated from *Curcuma zedoaria,* including curcumin, curcuminoids, curcumenone, dihydrocurdione, zedoaronediol, curcumanolide, furanodiene, furanodienone, zedorone, curzeone, curzerenone, germacrone, germacrone, 13-hydroxygermacroneethylpara methoxycinnamate, β-eudesmol, 18 β-turmerone, epicurzerenone, curzerene, 1,8-cineole, dihydrocurcumin, zingiberene, neocurdione, curdione, and α-phellandrene [67].

2.4.3 Usage as Chemoprevention

Curcuma zedoaria has been used as both a food and drug additive for a long time. In traditional cuisine, *C. zedoaria* possesses several activities, including anti-tumour/cancer, anti-inflammatory, pain relief, cholesterol-lowering, and antimicrobial effects [68]. Zedoary rhizome extracts exhibit anticancer, analgesic, antiallergic, anti-inflammatory, antibacterial, antifungal activities, and antiparasitic against *Entamoeba histolytica* [63]. Components obtained from *Curcuma zedoaria* extract such as curcumin, curzerenone, and zedoarone are effective as antifungals [69]. Zedoaria oil absorbed by sand granules in larvicidal tests against *Aedes aegypti* compared to abate has a potential effect with a lethal dose of 50% and 99% [70].

As analgesic, steroid and curcuminol components obtained through hydroalcoholic extract of the rhizome of this plant have stronger analgesic activity than the control drug because their mode of action does not involve the opioid system [71]. Methanol extract of *C. zedoaria* also has anti-inflammatory effects because it inhibits the activity of the COX-2 pathway and prostaglandin biosynthesis [72]. *Curcuma zedoaria* has a natural antioxidant content, namely diferuloylmethane, which comes from the essential oil of its rhizome. This oil can be used in preventing and slowing down the ageing process associated with degenerative diseases at a dose of 20 mg/ml [73].

Curdione has been presented to hinder the growth of breast cancer cell in vivo and in vitro. The mechanism behind this effect is believed to involve curdione-induced apoptosis, which is characterized by enhancing the expression of caspase-3, caspase-9, and cleaved Bax. Furthermore, it inhibits the proliferation of both MCF-7 cells and MDA MB-231 [74, 75]. Both curcuminoid and curcuminone compounds derived from *Curcuma zedoaria* demonstrated potent cytotoxic activity resistant to WEHI-3, HL60 leukaemia, and HUVEC cells, with IC_{50} values of 25.6, 106.8 μM, and 69.6 μM, respectively [76].

Malaysian and Indonesian *Curcuma zedoaria* essential oils, rich in camphor, zerumbone, and curzerenone, were found to exhibit strong cytotoxic activity resistant to lung, breast, and cervical cancer cells, with IC_{50} values ranging from 6.4 to 22 μg/mL in cell viability tests [64]. *Curcuma zedoaria* rhizome extract 0.1–0.2% given orally to experimental mice induced by murine B16F10 melanoma cells for 30–45 days increased the number of lymphocytes and NO production by macrophages, thereby inhibiting cytotoxic mediators of B16F10 melanoma cells and has an antimigratory effects on cancer cells that can inhibit metastasis [77].

The content of the polysaccharide fraction in the daily dose of *Curcuma zedoaria* should be 6.25 mg/kg in mice induced by S-180 sarcoma cells that can inhibit the growth of solid tumours by 50%, thereby reducing tumour size and preventing chromosomal mutations [78]. In addition, the content of ethyl p-methoxycinnamate, curcuminoids, bisdemothxycurcumin, isocurcumenol, and demothxycurcumin in *Curcuma zedoaria* can inhibit OVCAR-3 cells proliferation (human ovarian cancer), leukaemia (HL-60), NIH 3 T3 fibroblast cell metaplasia, HCT-15 and HT-2930 colon cancer cells, kidney embryonal cell malignancy (HEK293)14, and beta tumerone content in *Curcuma zedoaria* ethanol extract show an effect in inhibiting Hep-2 cells and hepatocellular carcinoma cells [79].

2.5 *Phyllanthus urinaria*

2.5.1 Taxonomy

Phyllanthus urinaria is classified as a competitive weed due to its substantial amount of seeds, tolerates shade well, and has a far-reaching system of root. Along the underside of its stems can be found numerous small green-red fruits as depicted in Fig. 9 [80]. The Euphorbiaceae family includes *Phyllanthus urinaria* L., a plant species that has been known for its medicinal properties. This species can be discovered in tropical areas across the globe, such as in India, Malaysia, Indonesia, Japan, and the United States [81].

2.5.2 Phytochemistry

Several functional compounds have been successfully isolated from *Phyllanthus urinaria* such as coumarins, carboxylic acids, flavonoids, tannins, and lignans as shown in Table 2. Polyphenols are the major bioactive components in *Phyllanthus urinaria,* including 9 flavonoids and 17 tannins. In addition, phenylpropanoids, including coumarins, lignans, and phenols, have been identified in *Phyllanthus urinaria*. Terpenoid class compounds such as 2 triterpenoids (oleanolic acid and glochidiol), 2 diterpenoids (spruceanol and cleistanthol), 1 sesquiterpene (cloven-2β, 9α-diol), and 1 monoterpene [(6*R*)-menthiafolic acid] were also successfully isolated from all *P. urinaria* plant extracts. Several other components have been isolated, consisting of glycosides, volatile oils, and steroids [81].

Fig. 9 *Phyllanthus urinaria* L.

Table 2 Chemical constituent of *Phyllanthus urinaria*

Category	Type	Compounds
Polyphenols	Tannins	Excoecarianin
		Trimethyl-3,4-dehydrochebulate
		Phyllanthusiin C
		Epigallocatechin, epicatechin-3-gallate, epicatechin, epigallocatechin-3-gallate
		Gallocatechin-3-gallate
		Repandinin B, furosin, repandusinic acid A, mallotinin, geraniin, acetonylgeraniin D, corilagin
		Phyllanthusin F, phyllanthusin G
	Flavonoids	Rhamnocitrin
		Rutin, quercetin
		Naringin
		Quercetin 3-O-α-L-(2,4-di-O-acetyl) rhamnopyranoside-7-O-α-L-rhamnopyranoside, quercetin 3-O-α-Lrhamnopyranoside
		Quercetin 3-O-b-D-glucoside, kaempferol 7-methyl ether
		Kampferol
Terpenoid	Sesquiterpene	Cloven-2β, 9 α-diol
	Monoterpene	(6R)-menthiafolic acid
	Diterpenoids	Cleistanthol, spruceanol
	Triterpenoids	28-norlup-20(29)-ene-3, 17β-diol, betulin, β-betulinic acid, 3-oxo-friedelan-28-oic acid, oleanolic acid, 3R-E-coumaroyltaraxerol, 3R-Z- coumaroyltaraxerol, glochidiol, oleanolic acid
Phenylpropanoid	Coumarins	Ellagic acid, brevifolincarboxylic acid
		Methyl brevifolincarboxylate
		Ethyl brevifolincarboxylate
	Lignans	Phyllanthin, hypophyllanthin, nirtetralin, niranthin, phyltetralin, (+)-dihydrocubebin, (+)-lyoniresiol, (7R,7’R,8S,8’S)-icariol A2, evofolin B, 4-oxopinoresinol, ()-syringaresinol,(−)-episyringaresinol
		Virgatusin, lintetralin, 5-demethoxyniranthin, urinatetralin
	Phenols	Ferulic acid, p-hydroxybenzaldehyde, 3,5-dihydroxy-4-methoxybenzoic acid, dehydrochebulic acid trimethyl ester
		Gallic acid, methyl gallate
Glycosides	Glycosides	β-Sitosterol-3-O-β-d-glucopyranoside
		Gentisic acid 4-O-b-d-glucopyranoside, syringin
Volatile oils	Jasmonate derivatives	(+)-Cucurbic acid, (+)-methyl cucurbate, methyl [lR,2R,2’Z]-2-(5′-hydroxy-pent-2′ -enyl)-3-oxocyclopentaneacetate
	Aldehyde derivatives	5-Hydroxymethyl-2-furaldehyde
Steroids	Steroids	β-Sitosterol, [3β,22E]-stigmasta-5,22-diene-3,25-diol
		Daucosterol

Du et al. [81]

2.5.3 Usage as Chemoprevention

Both extracts of aqueous and methanol of *Phyllanthus urinaria* have been shown to hinder the metastasis of A549 cells by suppressing their migration and invasion through the hypoxia and ERK1/2 signalling pathways, while the Fas receptor expression pathway is involved in the induction of apoptosis in human osteosarcoma 143B cells by a hot water extract [82, 83].

Phyllanthus urinaria is known to induce apoptosis along with antimetastatic action by targeting ERK and hypoxia pathways [84]. Corilagin is a polyphenolic tannic acid gained from *Phyllanthus urinaria* that is commonly known to selectively induce autophagy and apoptosis in cancer cells but not in healthy cells [85].

Several studies have scientifically proven the apoptosis-inducing activity of *P. urinaria* in tumour inhibition. Being a bioactive compound of *Phyllanthus urinaria,* corilagin induces apoptosis and has a mechanism through intrinsic and extrinsic pathways by downregulating procaspase-3, procaspase-8, PARP, procaspase-9, Bcl-2, and upregulating caspase-8, cleaved PARP, caspase-9, and Bax mediated by the production of ROS in the repression of breast cancer cells [85, 87]. The anticancer potential of *P. urinaria* may be attributed to several mechanisms shown in Table 3 [86].

Table 3 Apoptotic effects of extracts of *Phyllanthus urinaria* in cancer

Extract/ constituent	Cancer type	Study type	Chemopreventive mechanism
Corilagin	Breast carcinoma cells (MCF-7), MDA-MB-231 and xenograft	In vitro and in vivo	Downregulation of procaspase-8, procaspase-3, PARP, Bcl-2, and procaspase-9 and upregulation of caspase-8, cleaved PARP, caspase-9, and Bax
Aqueous	Human cancer cells	In vitro	DNA fragmentation, membrane blebbing, and formation of apoptotic bodies
Aqueous	Lewis lung carcinoma cell line	In vitro	Induce apoptosis via downregulation of Bcl-2 and upregulation of caspase-3
Aqueous	Human nasopharyngeal carcinoma cell line	In vitro	DNA fragmentation upregulated caspase-3 and decreased Bcl-2
Aqueous	Human osteosarcoma 143B cell	In vitro	Activation of Fas receptor/ligand expression, upregulated of bid, tBid, and Bax, and decreased Bcl-2
Corilagin	Ovarian cancer cell SKOV3	In vitro	Upregulated caspase-3 activity and loss of mitochondrial membrane potential
Alcohol	Stomach cancer cell line	In vitro	Apoptosis of MKN28 cell
Ethylene glycol, ethyl acetate, methanol, and water	Hep-2 (alveolar epithelial carcinoma cell line), MCF-7, HeLa (cervical cancer line)	In vitro	Activate caspase-3, caspase-8, and suppress Bcl-2

Saahene et al. [86]

Table 4 Antiangiogenic effects of extracts of *Phyllanthus urinaria*

Extract/ constituent	Cancer type	Study type	Chemopreventive mechanism
Aqueous	Mice bearing Lewis lung carcinoma cells, HUVECs	In vitro and in vivo	Suppressed neovascularization in mice
Aqueous and methanolic	MeWo and PC-3HUVECs	In vitro	Suppressed migration, invasion, and microcapillary like-tube structure formation in HUVECs
Aqueous	Human osteosarcoma xenograft mice	In vivo	Inhibition of cluster of differentiation 31
Ellagic acid	Chorioallantoic membrane in chicken embryo	In vivo	Inhibition of MMP-2
Methanolic	Male Sprague–Dawley rats, HUVECs	In vitro and in vivo	Suppression of vascular growth and tube formation
Aqueous	Hepatocellular carcinoma	In vitro and in vivo	Inhibition of migration

Saahene et al. [86]

The bioactive component of *Phyllanthus urinaria*, ellagic acid, can be in anti-angiogenic in vivo studies by hindering MMP-2 [88]. Methanol extract of *Phyllanthus urinaria* at a concentration of 100 μg/mL can inhibit blood vessel growth by 56.7% and tube formation by 35.6% [89]. Thus, angiogenesis regulation has been shown to suppress tumour progression. *Phyllanthus urinaria* can inhibit angiogenesis to suppress some cancers as shown in Table 4 [86].

In an in vivo study using mice bearing Lewis lung carcinoma, the development of *Phyllanthus urinaria* angiogenic inhibitors resulted in a significant reduction in tumour progression and size, as well as inhibition of tumour neovascularization and induction of apoptosis. Notably, these effects were achieved with limited cytotoxicity from the extract [90]. *Phyllanthus urinaria* possesses the strongest dose-dependent antimetastatic and antiangiogenic activity via reduction of matrix metalloproteinase-2 [91]. The mechanism underlying the suppression of human osteosarcoma xenograft growth by *Phyllanthus urinaria* is the induction of apoptosis characterized by increased Bax/Bcl-2 ratio and inhibition of differentiation cluster 31 through regulation of mitochondrial function and fusion proteins [92].

2.6 *Gynura procumbens*

2.6.1 Taxonomy

Gynura procumbens is a plant species native to Southeast Asia, commonly found in Indonesia, Malaysia, the Philippines, and Thailand. It is an annual evergreen shrub that typically grows to a height of 1–3 m and has distinctive purple and fleshy stems.

Fig. 10 *Gynura procumbens*

The leaves are ovate or lanceolate, measuring between 3.5 and 8 cm in length and 0.8 and 3.5 cm in width. The plant produces panicle-shaped yellow flower heads, each measuring approximately 1–1.5 cm in length as shown in Fig. 10 [93, 94].

2.6.2 Phytochemistry

The leaves of *Gynura procumbens* are often consumed as food, with no toxic effects. *Gynura procumbens* is known to contain many active chemical constituents, including tannins, flavonoids, terpenoids, sterol glycosides, and saponins. Additionally, it is rich in some constituents, such as kaempferol-3-O-β, kaempferol, and 3,5-dicaffeoylquinic acid methyl ester, quercetin, terpenoids, alkaloids, saponins, tannins, and astragalin [94, 95]. Caffeoylquinic acid and its derivatives are the most important constituents isolated from the leaves or aerial parts of *Gynura procumbens* [96].

2.6.3 Usage as Chemoprevention

Gynura procumbens is a necessary tropical medicinal plant. The medicinal properties of this substance have been the subject of numerous studies, particularly in Southeast Asian countries. *Gynura procumbens* has been used as medicine for household remedies against various human diseases such as fever, rashes, kidney, migraine, constipation, hypertension, diabetes mellitus, and cancer. In addition, *Gynura procumbens* leaves or leaf extracts have activities as anti-herpes simplex virus, antihyperglycaemic, antihyperglycaemic and antihyperlipidemia, anti-inflammatory, reducing hypertension, antiproliferation in human mesangial cells, antioxidant, and anticarcinogenic. Moreover, the presence of defence proteins has been identified in *Gynura* leaves such as miraculin, peroxidase, and thaumatin-like protein. The potent protein fraction SN-F11/12 is known to hinder breast cancer proliferation in vitro [97, 98].

Meiyanto et al. [99] examined that for the DMBA-induced breast cancer development in rats, ethanolic extract of *G. procumbens* has potential chemopreventive-resistant properties. The investigation discovered that overseeing the concentrate at dosages of 250, 500, and 750 mg/kgBW decreased growth frequency by 60%, 30%, and 20% individually. The effective doses in suppressing tumour multiplicity are 500 and 750 mg/kgBW, while the dose of 250 was less effective in this case.

Hamid [100] investigated that inhibiting changes in the epithelial cells of the mammary gland, as demonstrated through histopathological changes, resulted in the potential of ethanolic extract of *Gynura procumbens* leaves in preventing breast cancer. Two distinct doses were administered to Sprague–Dawley rats for treatment. These were extracted (300 and 750 mg/kg bw) for 7 weeks. DMBA was used to induce breast cancer, and tumour progression was monitored until 12 weeks after the last DMBA initiation. Histopathological changes were quantified using the HE method. Results showed that both doses of the extract significantly inhibited mammary carcinogenesis.

Nurulita et al. [101] reported the cytotoxic properties of the ethyl acetate fraction of *Gynura procumbens* leaves (FEG) on colon and breast cancer cells, as well as its selectivity. Quercetin was identified in FEG using both HPTLC and HPLC, but it was not found to be the main active compound. FEG inhibited the growth of WiDr, MCF-7, and T47D cells and demonstrated lower effects on T47D cells than on WiDr and MCF-7 cells. The growth inhibitory effect of quercetin was higher on WiDr and MCF-7 cells compared to FEG. FEG was found to be less inhibitory towards NIH3T3 cells than all other cell lines, demonstrating a selective effect against cancer cells. The selectivity index of FEG on WiDr, MCF-7, and T47D cells was 4.97, 2.77, and 7.79, respectively.

Shwter et al. [102] reported that the administration of *Gynura procumbens* to the rats in this study resulted in a significant increase in colon GST activity. GST is an important enzyme in the detoxification of carcinogens and is involved in the conjugation of electrophilic compounds with glutathione, making them more water-soluble and easier to excrete [103]. The induction of GST activity by *Gynura procumbens* suggests that it may have chemopreventive potential by enhancing the detoxification of carcinogens [104, 105].

Furthermore, rats that were fed with *Gynura procumbens* had considerably lower numbers of aberrant crypt foci (ACF) compared to the carcinogen group ($P < 0.001$). It also appeared that the frequency of ACF in the distal colon was substantially greater than that in the proximal colon in all experimental groups ($P < 0.001$). Nonetheless, the quantity of ACF observed in both the distal and proximal colon was significantly decreased in 500 mg/kg *Gynura procumbens*, 250 mg/kg *Gynura procumbens*, and reference groups (21.40 ± 7.09; 23.20 ± 4.7, and 32.0 ± 2.2, respectively) in comparison to the carcinogen group (130.60 ± 21.93) ($P < 0.001$). The experimental groups that received 500 mg/kg and 250 mg/kg *Gynura procumbens* demonstrated a reduction of approximately 83.6% and 82.2%, respectively, in the total number of crypts compared to the carcinogen group [102]. Earlier research has validated the utility of ACF as a biomarker for colon cancer [106].

In addition, Shwter et al. [102] showed that the superoxide dismutase activity (U/mL) in rats fed with *Gynura procumbens* 250 mg/kg was found to be the highest at 11.55 ± 0.331, followed by rats fed with *Gynura procumbens* 500 mg/kg at 8.41 ± 0.941, while the carcinogen group rats fed with AOM showed the lowest activity at 5.87 ± 0.311. The results showed that treatment diets utilized in feeding the rats had significantly higher SOD activity compared to the carcinogen group. SOD is an enzyme that plays a crucial role in preventing the formation of ACF by converting superoxide anion radicals, which are potent carcinogens, into hydrogen peroxide and molecular oxygen. The H_2O_2 produced by SOD is additionally diminished to water by catalase. These findings are consistent with previous studies demonstrating lower SOD activity in various tumour cell lines [107, 108].

2.7 *Garcinia mangostana*

2.7.1 Taxonomy

Garcinia mangostana is a tropical plant that lives and is cultivated in Southeast Asia. This plant can grow up to 6–25 m in height and has a dark purple to reddish fruit colour [109]. The main feature of this fruit, which is known as 'mangosteen', is that at the end there is a simple inflorescence as depicted in Fig. 11 [110].

2.7.2 Phytochemistry

Polyphenolic compounds were identified through phytochemical screening of *Garcinia mangostana* as well as anthocyanins, flavonoids, xanthones, and benzophenones as illustrated in Fig. 12 [111]. But the main bioactive compounds in *Garcinia mangostana* are xanthone derivatives. Being members of the polyphenol group, xanthones are secondary metabolites that are recognized by their tricyclic aromatic ring structure [112]. This compound was obtained from the extraction results, especially in the pericarp of the mangosteen fruit.

Garcinia mangostana has yielded more than 50 isolated xanthones. Phytochemical studies conducted on the DCM fraction from *Garcinia mangostana* rind resulted in five xanthones known as 9-hydroxycalabaxanthone, 8-deoxygartanin, gartanin, γ-mangostin, and α-mangostin [113]. Meanwhile, the ethanol extract produced 14 isolates, including mangosteen, 8-deoxygartanin, garninon, gartanin, 1,5,8-

Fig. 11 The mangosteen fruit or *Garcinia mangostana*

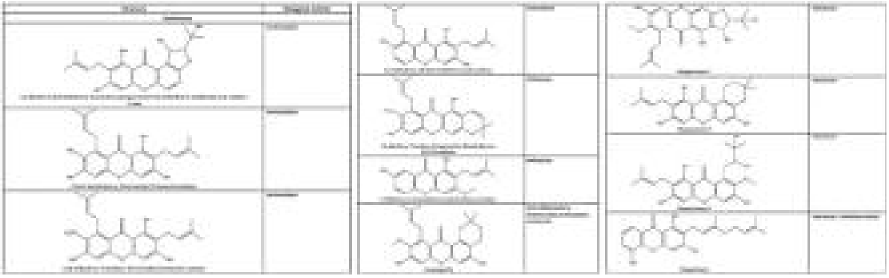


Fig. 12 Chemical compound and its biological activity. (Murthy et al. [111])

trihydroxy-3-methoxy-2[3-methyl- 2-butenyl] xanthone, trapezifolixanthone, padiaxanton, tovophyllin A, garnison B, 1,3,7-trihydroxy-2,8-di-(3-methylbut-2-enyl)xanthone, mangosteen D, 2-geranyl-1,3,5-trihydroxysanthone, 1,7-dihydroxy-2-(3-methylbut-2-enyl)- 3-methoxyxanthone, and 7-O-Demethyl mangostanine. The most abundant and frequently isolated xanthone for research is α-mangostin [114].

2.7.3 Usage as Chemoprevention

There are various biological activities shown by *Garcinia mangostana* such as antioxidant, anti-inflammatory, and anticancer effects. In vivo test results demonstrated which extracts and isolates from *Garcinia mangostana* can increase antioxidant enzymes, like SOD, CAT, GPx, and GSH, or by reducing markers of oxidative stress, such as MDA levels [115]. Compounds that contribute to the antioxidant activity of *Garcinia mangostana* include α-mangostin, γ-mangostin, gartanin, 8-deoxygartanin, and mangostanin [111].

In vivo tests have demonstrated that *Garcinia mangostana* can serve as an effective anti-inflammatory agent through several mechanisms, including repression of eosinophil recruitment via PI3K inhibition, as well as decreasing levels of OVA-induced IgE, cytokines TGF-β1 and (IL-4, IL-5, IL-13), and downregulating NFκB p65 levels in nuclear protein. These activities are contributed by α-mangostin and γ-mangostin [116]. Mangostin compounds also can inhibit bacterial growth [117].

The pericarp of *Garcinia mangostana* fruit contains xanthones, which have been found to exhibit antitumour effects. Many of the compounds in pericarp extract have been shown to be active in cancer models in various organs, including the colon, breast, lung, cervix, prostate, etc. The use of mangosteen extracts has been associated with several mechanisms, such as apoptosis induction, metastasis prevention, and angiogenesis prevention.

In an in vitro study, α-mangostin can induce apoptosis in oral cancer cells. The effects of xanthone derivatives include increased Bax expression and higher levels of caspase-3 and cleaved PARP, which leads to G1 phase cell cycle arrest. α-Mangostin also reduces CDK4, CDK2, cyclin D, and cyclin E levels, while increasing levels of p21 and p27 [118]. Other studies reported that α-mangostin induced apoptosis in animal models with oral cancer. The effect that can be observed is declined tumour volume and weight [119].

Carcinogenesis can be halted by inhibiting angiogenesis, which is a viable strategy due to the role of VEGF in stimulating endothelial cell proliferation, migration, and differentiation to form new blood vessels. VEGF receptor binding results in conformational changes and activation of the MAPK and PI3K/Akt signalling pathways [120]. In vitro studies have shown that α-mangostin can reduce VEGF expression in T47D breast cancer cells and inhibit its biological effects by interfering with its signalling pathway [121].

Moreover, α-mangostin has been found to restrain VEGF-induced hyper-permeability, increase migration, proliferation, and tube formation in retinal endothelial cells and limit vascular growth through the suppression of VEGF-induced VEGFR2 phosphorylation, as reported by Jittiporn et al. [122]. In addition, a distinct study exhibited the suppressive impact of α-mangostin on VEGF-triggered migration, proliferation, and formation of tubules from human umbilical vein endothelial cells [123].

2.8 *Morinda citrifolia*

2.8.1 Taxonomy

Morinda citrifolia is a small tree or an evergreen shrub that is native to Southeast Asia. It can grow up to 10 m in height and has a deep taproot and a far-reaching lateral root system that makes it hard to remove once established. The bark of the tree is grey or light brown, while the sapwood is soft and yellow-brown in colour. The leaves are opposite, glossy, membranous, and deeply veined as depicted in Fig. 13. The fruit changes from green to yellow to white as it ripens and has a potent smell that is sometimes described as unpleasant [124]. *Morinda citrifolia* has been consumed for more than 2000 years for various therapeutic purposes [125].

2.8.2 Phytochemistry

Morinda citrifolia has anthraquinone compound found in seeds, heartwood, root, and flowers. Ricinoleic acid was found in the seed. On the other hand, the root contains morenone 1, morenone 2, 7-hydroxy-8-methoxy-2-methylanthraquinone, and damnacanthal. From the heartwood, three anthraquinones were isolated: morindone, physcion, and anthraquinone glycoside. The flowers of *M. citrifolia* were identified as having one anthraquinone glycoside and two flavone glycosides. Besides, β-sitosterol and ursolic acid were found in the leaves. Numerous studies have found that *M. citrifolia* fruits contain a variety of chemical components, including nonvolatile compounds such as acetyl derivatives of asperuloside, caproic acid, glucose, and caprylic acid [126]. In addition, the juice extract of these fruits has been reported to contain over 50 volatile compounds, including terpenes, aldehydes, ketones, and sulphur compounds [127].

Fig. 13 The fruit of *Morinda citrifolia*

2.8.3 Usage as Chemoprevention

Morinda citrifolia has been traditionally used as a medicinal plant for several ailments, such as diabetes, antioxidant, and cancer. All parts of the plant have been utilized for their therapeutic properties. Numerous studies have highlighted the pharmacological activity of the fruit, which contains alkaloid compounds that have been found to inhibit α-glucosidase [128].

Batista et al. [129] reported that polysaccharide extracted from *Morinda citrifolia* significantly reduces oxidative damage by increasing GSH levels and reducing the content of MDA and NO_3/NO_2. In addition, polysaccharides can inhibit IL-1β, COX-2, and TNF-α expression. These results indicate that polysaccharide exhibits an anti-inflammatory effect on colitis by reducing inflammation response and oxidative stress in the inflamed colon. Hence, *Morinda citrifolia* extract has therapeutic potential against inflammation diseases, such as inflammatory bowel disease [130].

In a study conducted on rats, researchers investigated the potential neuroprotective effects of *M. citrifolia* juice against ischaemic stress-induced brain damage [131]. Specifically, they focused on the impact of fruit juice in preventing the growth of post-ischaemic glucose intolerance as this is thought to be a mechanism of brain protection. The results showed that *M. citrifolia* juice was effective in protecting the brain from damage and preventing the development of glucose intolerance. Besides, the neuroprotective effect of *M. citrifolia* juice was evaluated in scopolamine-induced rats [132]. The results demonstrated that the juice had a beneficial effect in preventing memory loss and protecting against oxidative damage caused by scopolamine. This was achieved by maintaining superoxide dismutase (SOD) activity. The findings suggest that *M. citrifolia* juice could potentially serve as an alternative therapy in the prevention of Alzheimer's disease.

Purwanto and Martono [133] reported that *M. citrifolia* fruit's ethanolic extract exhibited varying effects on class mu GST activity, with concentrations of 1% and 5% (w/v) inducing the activity while 10% (w/v) inhibiting it. In the case of DMBA treatment, all concentrations of the extract significantly suppressed class mu GST activity in rats' lungs. These findings show that it is potentially becoming a supportive agent for chemotherapy drugs. Numerous in vivo and in vitro experiments have investigated the potential anticancer effects of *Morinda citrifolia* extracts, as depicted in Table 5. These studies have highlighted the plant's ability to induce apoptosis, inhibit angiogenesis, prevent migration, and suppress tumour growth and proliferation, leading to the elimination of cancer cells shown in Table 6 [134].

Morinda citrifolia is capable of hindering the proliferation of cancer cells by suppressing the AKT/NF-κB signalling pathway, leading to apoptosis [135]. Additionally, *Morinda citrifolia* downregulates the expression of Ki67 cell proliferation and PCNA protein, inhibits the antiapoptotic protein Bcl-2, and upregulates apoptotic caspase-3 protein to increase the apoptotic pathway and eliminate tumour cells. Moreover, this plant has demonstrated the ability to interfere with cell migration, inhibiting metastasis and halting tumour cell development [136].

2.9 *Nigella sativa*

2.9.1 Taxonomy

Nigella sativa flower has 5–10 petals and is light blue and white in colour. *Nigella sativa* seeds, also known as 'Alhabba Al-sauda' in Arabic and 'Kalvanji' in Urdu and Hindi, are often used especially in the Middle Eastern countries as depicted in Fig. 14 [137, 138].

2.9.2 Phytochemistry

Nigella sativa has various active ingredients as well as thymoquinone and monoterpenes [139]. Thymoquinone isolate is composed of several compounds, including p-cymene nigeglanine, dithymoquinone, thymohydroquinone, thymol, nigellicine, nigellidine, nigellimine, t-anethol, and 4-terpineol as illustrated in Fig. 15 [140, 141]. *Nigella sativa* also can be a source of vegetable oil because it contains various fatty acids, including linoleic acid and oleic acid [142].

Nigella sativa seeds are usually used as a herbal medicine for various ailments. *Nigella sativa* or black cumin has a variety of nutrients. More than 20% of carbohydrates are contained in the seeds. In addition, black cumin seeds have a protein content of 3.7–4.7% and 3.4–4.7% fibres. Elements of vitamins and minerals also include various nutrients that plants have [141].

2.9.3 Usage as Chemoprevention

Nigella sativa has been found to exhibit significant antiproliferative effects resistant to several cancers, including liver, colon, breast, cervix, lung, pancreas, prostate, etc. Multiple in vitro and in vivo studies have reported that the plant's anticancer activity is treated by modulating various biological processes and pathways [142].

The subject that was used to study the antitumour effect of thymoquinone, an isolate from black cumin, was a mouse tumour xenograft model of breast cancer. Treatment by inducing p38 phosphorylation and ROS production in breast cancer cells results in significant pro-apoptotic and antiproliferative effects. Furthermore,

Table 5 Pre-clinical evidence obtained from plant extract

Cancer type	Cell model	Intervention description	Comparator/ combination	Mechanism of action
Breast	Human breast tumour explant	Noni juice	Not started	Inhibition of capillary growth and degenerate newly formed vessels
	CYP 2C19 enzymes isolated from human recombinant Sf9 insect cells	Fruit juice	Not started	Dose-dependently inhibited CYP 2C19 aromatase enzyme which decreases oestrogen production
	MDA-MB-231 human breast cancer carcinoma	Unclear (butanol extract)	LPS	Induced apoptosis through interaction with TLR4
	MCF-7 human breast cancer cell; MDA-MB-231 human breast cancer carcinoma	Fractionation of 90% ethanolic fruit extract	Paclitaxel (cytotoxicity assay)	Arrested cell proliferation and cell cycle; induced apoptosis; increased caspase activities; decreased ROS production and mitochondrial membrane potentials
	MCF-7 human breast cancer cell	95% ethanolic leaf extract	Not started	Inhibited cell migration; suppressed migration of cancer cells
Lung	A549 human lung adenocarcinoma cells; LL2 mouse Lewis lung carcinoma cells	50% ethanolic leaf extract	Erlotinib 50 mg/kg	Extract inhibited cell proliferation and induced apoptosis; induced cell cycle arrest and activated caspase apoptotic activity
	A549 human non-small cell lung cancer (NSCLC)	Fermented fruit juice	Not started	Induced apoptosis, inhibited proliferation, invasion, and migration of cells; downregulated phosphorylation of AKT, p50, and STAT3 protein
	A549 human lung carcinoma cells	Fruit juice	Not started	Inhibited the manganese-induced HIF-lα protein expression; interfered with manganese's signalling to activate PKB, ERK-1/2, JNK-1, and S6; repressed the induction of HIF-Iα protein by desferoxamine or IL-Iβ

(continued)

Table 5 (continued)

Cancer type	Cell model	Intervention description	Comparator/ combination	Mechanism of action
	A549 human lung cancer	Dried seeds	Not started	Induced intracellular nuclear damage; induced apoptosis and cell cycle arrest; accumulated ROS production and disruption of mitochondrial potential
Eye	Y79 retinoblastoma cell	Fruit extract	Not started	Apoptotic morphology observed; modulated caspase activation
Leukaemia	Jurkat cells (human T lymphocytes)	50% ethanol leaf extract	Not started	Induce apoptosis and cell cycle arrest; enhanced caspase activities

Chanthira Kumar et al. [134]

thymoquinone was found to inhibit the expression of antiapoptotic gene proteins, including survivin, Bcl-xL, XIAP, and Bcl-2 in breast cancer cells and breast tumour xenografts. Additionally, thymoquinone was observed to increase the levels of superoxide dismutase, catalase, and glutathione in rat liver tissue [138].

The proliferation of human LoVo colon cancer cells can be significantly reduced by administering 20 μmol/L thymoquinone. This treatment can also lower the levels of p-Akt, p-PI3K, β-catenin, and p-GSK3β, thereby inhibiting the downstream COX-2 expression. As a result, COX-2 expression is reduced, leading to lower PGE2 levels and suppressed EP2 and EP4 activation [143]. In another study, thymoquinone was found to increase the total cell death index and activate apoptosis at 2 μM, with the effect decreasing as the dose increased. Thymoquinone can also cause mitochondrial outer membrane permeability (MOMP) and activate autophagic cell death with the stimulation of p38 and JNK in CPT-11-R LoVo colon cancer cells [144].

Thymoquinone has been studied for its potential role in inhibiting the migration, proliferation, and invasion of lung cancer cells. In particular, administering thymoquinone at a dose of 10, 20, or 40 μmol/L has been found to suppress PCNA, MMP9, cyclin D1, and MMP2 mRNAs. It can also inhibit MMP2 gelatinase activity, reduce the expression of MMP9 and cell cycle inhibitor P16, and inhibit the migration, invasion, and proliferation of A549 cells by reducing the ERK1/2 pathway [145, 148]. In other studies, thymoquinone has shown effectiveness against lung cancer cell proliferation by increasing Bax expression, downregulating Bcl2 protein, increasing the Bax/Bcl2 ratio, increasing p21 expression, decreasing cyclin D expression, downregulating IKK1 and NF-kappa B levels, and increasing TRAIL 1 and 2 receptor expression [146, 147].

Table 6 Pre-clinical evidence from *M. citrifolia*'s phytochemistry

Cancer type	Cell model	Phytochemical	Comparator/ combination	Mechanism of action
Breast	4TI mouse breast cancer cell; MCF-7 human breast cancer cell; MDA- MB-231 human breast cancer carcinoma	Nordamnacanthal	Not started	Increased population of early apoptotic and late apoptotic cells; induced cell cycle arrest
	MCF-7 human breast cancer cell	Damnacanthal	Not started	Induced apoptosis and stimulate cell cycle arrest; activation of caspase activity; enhanced expression of pro-apoptotic genes and proteins
	MCF-7 human breast cancer cell	Damnacanthal	Doxorubicin (0.2–0.55 μg/ mL)	Induced cell cycle arrest and apoptosis in combination treatment; upregulation of pro-apoptotic proteins and downregulation of antiapoptotic proteins in combination treatment
	MCF-7 human breast cancer cell	Polysaccharides	Not started	Induced DNA damage by apoptosis through activation of p53 and caspase-3 proteins
Liver	Hep G2 human hepatocellular carcinoma	Damnacanthal	Not started	Affected cell survival; inhibited growth and clonogenic potential; induced apoptosis; inhibited cell migration
Skin	MUM-28 human melanoma cell	Damnacanthal	Not started	Inhibited proliferation and increased apoptotic; increased caspase activities; upregulated pro-apoptotic protein
Colon, prostate, and	HCT-116 human colon cancer cell; HT-29 human	Damnacanthal	Not started	Suppression of cell growth; affected cell cycle regulation

(continued)

Table 6 (continued)

Cancer type	Cell model	Phytochemical	Comparator/ combination	Mechanism of action
breast cancer	colorectal adenocarcinoma cell; PC-3 human prostate cancer cell; MCF-7 human breast cancer cell			

Chanthira Kumar et al. [134]

Fig. 14 The seeds and the flowers of *Nigella sativa* Linn

2.10 *Moringa oleifera*

2.10.1 Taxonomy

Moringa oleifera is a member of the Moringaceae family and the most widely grown species [149]. *Moringa oleifera*, which is referred to as the drumstick tree and a versatile plant that is considered a miracle tree, is depicted in Fig. 16. Every part of the plant is utilized for various purposes and is deemed to have medicinal significance. Moreover, the plant is an exceptional source of both essential nutrients and bioactive compounds. It is a very nutrient-dense vegetable tree with a wide range of applications. The tree itself is relatively slender, with drooping branches. Nevertheless, it is typically cut back annually to a height of 1 m or less and then left to regrow to maintain pods and leaves within reach [150].

2.10.2 Phytochemistry

A wealth of bioactive substances, including phenols, glucosinolates, tocopherols, carotenoids, and ascorbic acid, have also been found in *Moringa oleifera* plants [149]. Vitamin A is abundant in fresh leaves from *M. oleifera*. Carotenoids with pro-vitamin A potential are abundant in *M. oleifera* leaves. In addition, *M. oleifera* leaves have a higher content of vitamin C than oranges, at 200 mg/100 g. *M. oleifera*

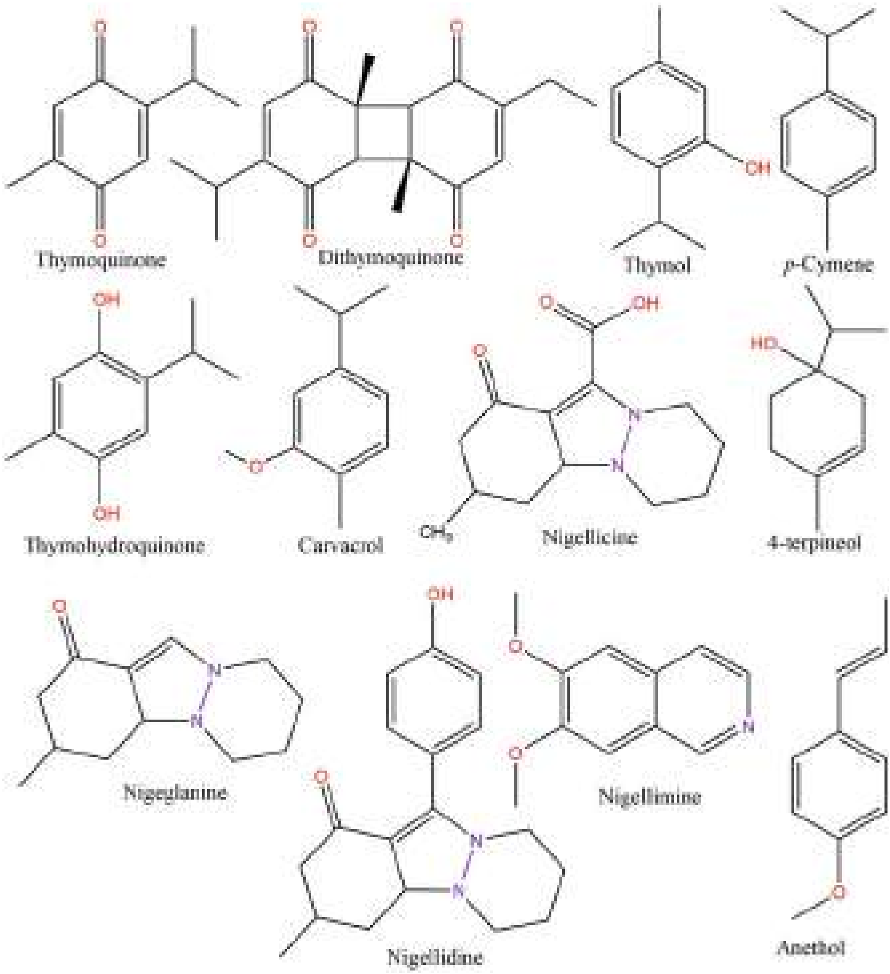


Fig. 15 Some compounds have been isolated from *Nigella sativa.* (Ansary et al. [141])

fresh leaves provide high levels of vitamin E that are comparable to those in nuts [151]. This is significant because vitamin E has been found to suppress cell proliferation in addition to acting as an antioxidant [152, 153].

Moringa oleifera has several phytochemicals that are found in the plant's leaf, branch, seed, pod, and root. According to Flora and Pachauri [154], the main phytochemicals of *M. oleifera* include glucosinolates, phenolics, flavonoids, crude fats, fatty acids, key nutrients, minerals, and total protein. In addition, glucosinolates are secondary metabolites that identify Brassicales plants, including *M. oleifera,* and are only found in the Brassicales order except for the root; most of the *M. oleifera* plant is covered in glucosinolates. *M. oleifera* has a high concentration of aromatic glucosinolates, including benzyl glucosinolates (glucotropaeolin), 2-phenylethyl

Fig. 16 Images of *M. oleifera* leaves and seeds

glucosinolates (gluconasturtiin), and p-hydroxybenzyl glucosinolates (sinalbin). Furthermore, the predominant substance found in *M. oleifera*'s leaves is benzyl glucosinolates [155].

Many compounds of eight *M. oleifera* cultivars from India were found when the carotenoid examined yielded luteoxanthin, lutein, zeaxanthin, and β-carotene. Lutein was found to be the main component of the leaves and fruits, making about 53.6 and 52.0%, respectively, of the total carotenoids. In addition, 90 mg/g, 8 g/mL, and 27 g/mL of total phenolic, flavonoid, and β-sitosterol components were found in the leaves of *M. oleifera* [156, 157]. Also, an abundance of different bioactive substances like glycerol-1-9-octadecanoate, glucosinolates, isothiocyanates, and glycoside compounds are additional anticancerous elements [158].

2.10.3 Usage as Chemoprevention

In Indonesia, *Moringa* is a plant that grows as shrubs or trees and is extensively distributed. The locals utilize this herb to treat a variety of sickness concerns. The indigenous people of Timor Island use the *Moringa* plant to cure anaemia, hypertension, respiratory issues, ear and mouth infections, and skin and respiratory illnesses [157]. There have been various biological activities in *M. oleifera*'s plant, including protection against gastric ulcers, antidiabetic, anti-inflammatory, and anticancer [159].

M. oleifera recently showed the promising cytotoxic and antitumour properties. The anticancer activity of *M. oleifera* is not limited to its pure compounds. In a study conducted by a research team, they examined that the effects of an aqueous extract of *M. oleifera* pods on colon cancer showed a huge decrease in the nuclear antigen indices of growing cells ($P < 0.05$) and suppressed the expression of inducible iNOS and COX-2, which are associated with cancer development [160]. This suggests that the extract has potential as a natural anticancer agent. The inhibition of leaves extract (200 μg/ml) rate reached 80%. HepG2 liver cancer cell survival may be decreased by an alcoholic extract of *M. oleifera*'s fruit [161]. The antitumour efficacy against colon cancer cells was proven in 2015. According to an in vitro study, colon and breast cancer cells were cytotoxic to *M. oleifera*'s alcoholic leaf and bark extracts (above 500 g/ml). It decreased the proliferation of HCT-8 and MDA-MB-231 cells and blocked in G2/M phase [162].

Several studies suggest that *M. oleifera* is safe and effective in treating various conditions. For instance, a study reported *Moringa* leaf's extract improved the efficacy of chemotherapy in reducing human pancreatic cancer cells growth, such as p34, Colo 357, and Panc-1, by hindering the NF-kB signalling pathway [163]. Similarly, the use of *Moringa* leaf extract reduced the growth of B16F10 melanoma cells and induced the death of 22% of cancer cells [164].

M. oleifera has been found to obstruct several signalling pathways, including the nuclear factor-kappa B (NF-κB) and STAT5 signalling pathways, as well as to a lesser extent the STAT1/STAT2 signalling pathways. GMG-ITC (4-(α-l-rhamnopyranosyloxy)-benzyl ITC) is an active inhibitor of these pathways, which are all involved in cancer [165]. *Moringa* has also been shown to produce reactive oxygen species (ROS) in cells, which can lead to apoptosis by upregulating caspase 3 and caspase 9, both of which are part of the apoptotic pathway. This ROS production is thought to be a potential anticancer agent produced by *Moringa*. Furthermore, research has found that concentrates from *Moringa* can increase glutathione-S-transferase expression, which helps defend cells against oxidative damage [166]. The table of anticancer efficacy of *M. oleifera* is shown in Table 7 [167].

Bhadresha et al. [168] reported evaluating the in vitro anticancer effects of *M. oleifera* leaf extracts. The human cancer cell line A549 experienced a dose-dependent, substantial reduction in cell growth when treated with *M. oleifera* leaf extracts. *M. oleifera* leaves aqueous extracts were examined for cytotoxicity against lung cancer A549 cell lines at a range of various concentrations of 100–500 g/ml by using MTT assay. The percentage of A549 cells is still viable after 24 h of incubation with treatments compared to controls. The results showed that sensitive A549 cells lost viability at doses of 400 and 500 g/ml, causing 30% and 15% more cell death than the control, respectively ($P < 0.05$).

2.11 *Zingiber officinale*

2.11.1 Taxonomy

Zingiber officinale, or ginger, is used as a spice, flavouring agent, food, and herbal medicine because of its pungent taste. Aromatic rhizomes (underground stem) are depicted in Fig. 17. Zingiberaceae is the family of flowering plants, and *Zingiber* is the genus. *Zingiber officinale* (ginger) is a species of perennial plant with yellow flowers. The rhizomes, or tuberous roots, of the ginger plant offer a flavourful and aromatic scent [169, 170].

Furthermore, the ginger order of flowering plants are Zingiberales, which have about 53 genera, 8 families, and more than 1300 species. The tropics are habitats to many Zingiberales species, many of which are used as shade plants. Both species with bilateral flowers and those with radially symmetrical flowers can be found in the monocot order Zingiberales [171].

Table 7 Mechanism of action from *M. oleifera* in cancer

Type of effect	Plant of *M. oleifera*	Type of extract	Treatment doses/ extract	Animal model/cell lines	Outcome
Anticancer	Leaves	Extract	3125%	Wistar rat oral cancer cells	Reduced VEGF expression in oral cancer cells in Wistar rats
	Leaves	Ethanolic extract	3125%	Male Wistar rats	Decreased the expression of HSF1 in oral cancer induced by benzo[a] pyrene
	Leaves	Aqueous extract	1 gr/10 mL boiling water	Human pancreatic epithelioid carcinoma cell line PANC-1	Pancreatic cancer cell survival and metastatic activity were significantly decreased It further inhibited tumour growth
	Leaves	Active MO-derived fractions	3-Hydroxy-β-ionone	SCC15 cell line	MO extracts and its 3-hydroxy-β-ionone fraction provided the anticancer activity by inhibiting development and inducing apoptosis of SCC15 cell lines
	Seeds	Water, 80% ethanol extract and ethanolic extracted powder further fractionated in other solvents	0, 50, 100, 200, 300, and 500 μg/mL	MCF7 breast cancer cell	Crude water extract, hexane, and dichloromethane fractions of crude ethanolic extract of seeds inhibited the proliferation of MCF7 with IC50 values 280, 130, and 26 μg/mL, respectively

Arora and Arora [167]

Fig. 17 The fresh rhizome of *Zingiber officinale* plant

2.11.2 Phytochemistry

Zingiber officinale has been chemically analysed and found to have more than 400 constituents. *Zingiber officinale* contains 3–8% lipids, 38–70% carbohydrates, phenolic chemicals, and terpenes as the main components of ginger. The bioactive components of ginger, notably phenolic chemicals and terpenes, are responsible for its nutritional benefits. Some of these substances, including those from the groups of shogaols, gingerols, paradols, zingerone, and zingiberenes, have been reported to have the ability to modify some biological activity. Compared to other chemicals, shogaol (18–25%) and gingerols (23–25%) are more prevalent. Besides, there are also proteins, minerals, water, phytosterols, raw fibre, ash, and some vitamins. The bioactive compound *Zingiber officinale* is illustrated in Fig. 18 [172].

Zingiber officinale contains gingerenone-A, quercetin, 6-dehydrogingerdione, and zingerone. Additionally, ginger contains a number of terpene components, including a-curcumene, a-farnesene, B-bisabolene, B-sesquiphellandrene, and zingiberene, which are considered to be the main components of ginger essential oil. The other several substances found in *Zingiber officinale* (ginger) are diarylheptanoids and the phenol gingerol. It has been used to treat nausea and vomiting caused by motion sickness and other conditions, and it might be somewhat effective [169].

There are several metabolites in ginger. However, the essential components are volatile; therefore, this plant is volatile when cooked or dried. Shogaol, gingerol, paradol, and zingerone are some of the non-volatile bioactive substances that are associated with their pharmacological effects. Regarding the presence of the B-hydroxy keto group, the gingerols (4-gingerol, 6-gingerol, 8-gingerol, and 10-gingerol) are thermolabile bioactive chemicals that are transformed to shogaols during the degradation of ginger, and shogaols can be hydrogenated into paradols. The pharmacological activities, pharmacokinetics, and bioavailability of both gingerol and shogaol compounds have many differences, with gingerol being more bioactive [169, 173].

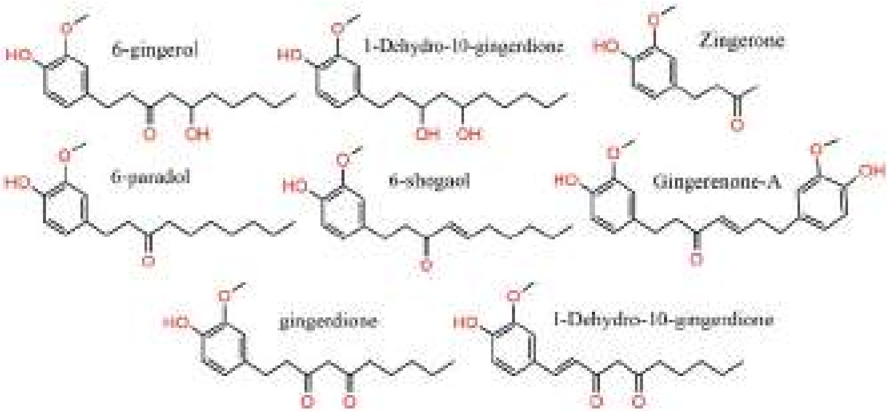


Fig. 18 Several bioactive compounds of ginger's rhizome. (Ishfaq et al. [172])

2.11.3 Usage as Chemoprevention

As one of the spices, and used for medical purposes, *Zingiber officinale* has been known for thousands of years for its usage [174]. Ginger has attracted a lot of attention from scientists due to its medicinal ability to control the activity of several enzymes. The herb has also been used to treat hypertension, dementia, helminthiasis, constipation, diarrhoea, toothaches, gingivitis, bronchitis, and asthmatic respiratory diseases, infectious diseases, anti-inflammatory, antioxidant, and cancer [175, 176]. Ginger has been shown through several previous studies to both block and treat a number of cancers as shown in Table 8. Also, the common mechanisms for anticancer involve the induction of apoptosis and the suppression of the proliferation of cancer cells [169].

Rhode et al. [177] reported that all of the cell lines examined experienced significant growth inhibition after being treated with ginger in cultured ovarian cancer cells. The study discovered that, among the individual ginger components examined in vitro, 6-shogaol is the most active. NF-kB activation was inhibited by ginger therapy, and VEGF and IL-8 secretion was also reduced. We conducted an experiment to evaluate the bioactivity of 6-, 8-, and 10-gingerol, as well as 6-shogaol, on A2780 ovarian cancer cells. The results showed that none of the three gingerols (6-, 8-, and 10-gingerol) had any impact on the viability of the ovarian cancer cells, while 6-shogaol can have a significant growth inhibition as seen in cells treated with whole ginger extract.

Ginger has demonstrated anticancer properties against colorectal cancer. It can prevent the growth and multiplication of colorectal cancer cells. In one study, 6-gingerol was found to halt the proliferation of colon cancer HCT116 cells. Additionally, it has been discovered that the reduction in tumour growth is associated with the activity of leukotriene A4 hydrolase, which was further supported by an in silico method [178].

Ginger extract with its components has been proven in preclinical tests to have chemopreventive and anticancer effects on gastric cancer. An in vitro study revealed

Table 8 The major pathways for anticancer activity of *Zingiber officinale*

Phytochemistry	Investigation type	Subjects	Treatment doses	Chemoprevention mechanism
6-Shogaol	In vitro	LNCaP, DU145, and PC-3 human prostate cancer cells	10, 20, and 40 μM	Inducing apoptosis; inhibiting STAT3 NF-kB signalling; downregulating the expression of cyclin D1, survivin, c-Myc, and Bcl2
6-Gingerol	In vitro	HeLa human cervical adenocarcinoma cells	60, 100, and 140 μM	Inducing cell cycle arrest in the G0/G1 phase; decreasing the levels of cyclin A, cyclin D1, and cyclin E1; increasing the expression of caspase; inhibiting the mTOR signalling pathway
10-Gingerol	In vitro	Human and mouse breast carcinoma cells	50, 100, and 200 μM	Inhibiting cell growth; reducing cell division; inducing S phase cell cycle arrest and apoptosis
6-Gingerol, 10 gingerol, 6-shogaol, and 10-shogaol	In vitro	PC-3 human prostate cancer cells	1, 10, and 100 μM	Inhibiting prostate cancer cell proliferation; downregulating the expression of MRP1 and GSTπ
Ginger extract	In vitro	HT29 human colorectal adenocarcinoma cells	2–10 mg/mL	Promoting apoptosis; upregulating the caspase 9 gene; downregulating KRAS; ERK; Akt; and Bcl-xL
	In vivo	Female Swiss albino mice	100 mg/kg	Activating AMPK; decreasing the expression of cyclin D1 and the level of NF-kB; increasing the expression of p53

Mao et al. [169]

that 6-gingerol induces apoptosis of gastric cancer cells. By boosting caspase-3/7 activation, it aids TNF-related apoptosis-inducing ligand (TRAIL)-induced apoptosis. By suppressing TRAIL-induced nuclear factor-kappaB (NF-κB) activity and downregulating the cytosolic inhibitor of apoptosis (cIAP)-1, 6-gingerol was able to cause apoptosis. In addition, 6-gingerol and 6-shogaol can decrease the growth of gastric cancer cells by harming microtubules [179].

Zerumbone is one of the ingredients in Asian ginger and also prevents the proliferation of pancreatic cancer through various mechanisms. According to the study, zerumbone induces apoptosis of PANC-1 cells. Reactive oxygen species (ROS) production and overexpression of the p53 and p21 proteins were linked to the induction of apoptosis in zerumbone-remedied PANC-1 cells [180–182]. This result suggests its induced apoptosis of PANC-1 cells via the p53 signalling pathway.

Studies conducted in vitro show that *Z. officinale*'s active ingredients help prevent liver cancer. According to a study, 6-shogaol has been revealed to induce apoptotic cell death of Mahlavu hepatoma cells through an oxidative stress-mediated caspase-dependent mechanism. 6-Shogaol-induced apoptosis of Mahlavu cells can be mediated by glutathione (GSH) depletion [183].

In addition, a number of additional pathways have been implicated in the reduction of cell growth by induced 6-gingerol and apoptosis in human colorectal cancer cells. These include downregulation of the NAG-1 beta-catenin, cyclin D1, PKCepsilon, and GSK-3β pathways, as well as protein degradation [184]. According to Radhakrishnan et al. [185], 6-gingerol's anticancer properties may be attributed to its ability to inhibit the ERK1/2/JNK/AP-1 pathway.

2.12 *Euphorbia abyssinica*

2.12.1 Taxonomy

Euphorbia abyssinica is the species of flowering plants which is well known for its therapeutic properties [186]. *Euphorbia abyssinica*, often known as desert candle, is a monoecious geophyte that can reach a height of 9 m as shown in Fig. 19 [187, 188].

Euphorbia abyssinica is a succulent, spiky tree with a height of up to 10 m. Its milky sap is poisonous. If the sap came into contact with the body, it would result in blistering, irritating dermatitis, keratoconjunctivitis, blindness, nausea, and vomiting. Euphorbon is a dangerous poison in plants that is found in many members of the Euphorbiaceae family [189].

Fig. 19 *Euphorbia abyssinica* plants

2.12.2 Phytochemistry

Euphorbia abyssinica was found to contain only low levels of diterpenes during chemical screening. In addition to euphorbol, euphol, oleanolic acid, lupeol, β-sitosterol, and β-sitosterol-3-O-glucoside, minor components such as ingenol esters and lathyrane derivatives have been reported in its latex. Moreover, the plant is composed of key ingredients such as rubber, waxes, and resins, and has demonstrated strong antifungal efficacy against *Aspergillus niger*, *Aspergillus flavus*, and *Candida albicans*, in addition to containing 8(R)-hydroxy-dec-3(E)-en-oic acid [190].

EL-Fiky et al. [190] have reported a new hydroxy unsaturated fatty acid called 8 (R)-hydroxy-dec-3(E)-en-oic acid isolated from biologically active components of *E. abyssinica,* and its chemical structure was fully determined. Four compounds were isolated and identified: lupeol, β-sitosterol, oleanolic acid, and β-sitosterol-3-O-glucoside.

In particular, ingenol and ingenol esters, which are tigliane-type diterpenoids, have shown promising activity against various types of cancer cells, making them a target of interest in drug discovery. Therefore, the identification and isolation of these diterpenes in Euphorbia species is important for their potential therapeutic applications primarily responsible for their antitumour activity [191].

2.12.3 Usage as Chemoprevention

Euphorbia abyssinica is actually believed to have traditional medicinal uses in Ethiopia for treating a variety of diseases, including ascariasis, gonorrhoea, leprosy, rabies, allergy (known locally as 'mitch'), toothache, syphilis, tinea infestation, TB, and skin cancer. These uses have been documented in ethnobotanical studies and suggest that the plant may have potential as a source of natural remedies. Plant latex and ground-up roots are pulverized as pastes or ointments to the affected area as a treatment for skin cancer. Meanwhile, there are no scientific findings related to the efficacy of this treatment for this particular species. The observations of two further similar species of *Euphorbia*, *E. tirucalli* and *E. prolifera*, have reported their usage as anticancer and antitumour treatments in traditional medicine outside of Ethiopia [192].

Furthermore, there have been numerous studies demonstrating various pharmacological activities of *Euphorbia* species. These activities include antiproliferative, cytotoxic effects, and antipyretic-analgesic effects [193]. El-Hawary et al. [194] reported that *Euphorbia abyssinica* has been shown to have cytotoxic activity against CACO2, which are human colon adenocarcinoma cells (IC50 11.3 μg/mL).

According to Ahmed et al. [195], both fruit and aerial parts are extracted from *Euphorbia abyssinica* by using ultra-performance liquid chromatography-mass of both extracts. In the DPPH experiment, the *E. abyssinica* spectrometry (UPLC-MS) revealed a rich source of a variety of ingredients. The identification and assessment of the antioxidant activity of 39 compounds were found in both aerial parts and fruit extracts of *E. abyssinica*. The fruit extract exhibited stronger antioxidant activity than the aerial part extract, as determined by three in vitro experiments with IC50 values of 85.1 ± 1.07 and 562.3 ± 1.01 g/mL, respectively. In addition, the study found that increasing concentrations of the *E. abyssinica* fruit extract triggered an increase in the protein expression of NQO1, a chemopreventive marker, in Hepa1c1c7 cells.

Emam et al. [196] revealed that 80% ethanolic extract of *E. abyssinica* was also cytotoxic and hindered the growth of cervical cancer cells (HeLa) by 55.2%. The ethanolic crude extracts of *E. abyssinica* exhibited 50% more growth or inhibition on HeLa cancer cells at a concentration of 30 μg/ml. In HeLa cancer cells, the IC_{50} value for crude extracts of *E. abyssinica* was 18.6 μg/ml. A strong cytotoxic activity was shown by *Euphorbia abyssinica*, but it was less potent than doxorubicin. *E. abyssinica* has a cytotoxic impact with an IC_{50} value of 18.6 μg/ml. The National Cancer Institute (NCI) defines an extract as potentially active against cancer cells if its IC_{50} value is $\leq$30 μg/mL. The IC_{50} value is the concentration of a substance needed to inhibit 50% of cell growth in vitro and is commonly used as a measure of cytotoxicity. Therefore, if a crude extract has an IC_{50} value of 30 μg/mL or lower, it is considered a promising candidate for further purification and investigation for anticancer activity.

2.13 *Kaempferia galanga*

2.13.1 Taxonomy

Kaempferia galanga is a pungent medicinal herb from a dried rhizome belonging to the family Zingiberaceae. The Zingiberaceae family consists of the stem-less herb, which belongs to the *Kaempferia* genus. In some places, it is also known as sand ginger or aromatic ginger [197]. *Kaempferia galanga* is a monocotyledonous plant in the Zingiberaceae family as shown in Fig. 20, which has roots and stems that are embedded in the ground [198, 199].

2.13.2 Phytochemistry

The chemical diversity of *K. galanga* is due to its ability to produce various secondary metabolites that are recognized to have a broad spectrum of biological functions. Terpenoids such as diterpenes, sesquiterpenes, and monoterpenes are the major components of *K. galanga*. Some of the identified terpenoids are sandaracopimaradiene-9α-ol, kaempulchraol I, and kaempulchraol L. Phenolic compounds such as galangin, kaempferol, and quercetin have also been identified in *K. galanga*. Anti-inflammatory, antioxidant, and anticancer properties are shared by these compounds. Moreover, cyclic dipeptides, diarylheptanoids, and polysaccharides have also been isolated from *K. galanga* and are known to possess various biological activities [200].

The essential oil extracted from the rhizomes of *Kaempferia galanga* L. was analysed using gas chromatography-mass spectrometry (GC-MS) method, and the major components of the oil were identified as borneol, ethyl cinnamate, ethyl-p-

Fig. 20 The rhizomes of *Kaempferia galanga* L.

methoxycinnamate, 8-cineole, γ-cadinene, 1, δ-carene, linoleoyl chloride, ethyl-m-methoxycinnamate, camphene, and α-pinene [201].

2.13.3 Usage as Chemoprevention

The main regions in Southern China where *K. galanga* is cultivated are Guangxi, Guangdong, and Yunnan. Because of its advantageous therapeutic effects on conditions such as inflammations and tumours, it was utilized as a folk remedy in China, which is a reservoir of valuable bioactive components. *K. galanga* was utilized in India to treat intestinal wounds and urticaria. Furthermore, it might be used as a food condiment [202, 203].

The rhizome of *Kaempferia galanga* is abundant in essential oils and a traditional remedy for indigestion, colds, chest and stomach pain, headaches, as well as an expectorant, diuretic, and carminative. Additionally, *K. galanga* extracts have anti-inflammatory, cytotoxic, antihypertensive, and hypolipidemic effects on the body, and are also used for the treatment of swelling [204].

Srivastava et al. [203] reported that the rhizome of *K. galanga* exhibits anti-proliferative properties, with EPMC being the primary phytochemical present in the rhizome. To investigate this further, the MTT assay was used to assess the anti-proliferative effects of the methanol extract, hydro-distillate, and EPMC on cancer cell lines specific to different organs. Cell growth has been seen to slow down in a dose-dependent manner. The highest concentration of EPMC, hydrodistillate, and extract (50 μg/mL) inhibited the proliferation of MDA-MB-231 cancer cells by 48%, 43%, and 36%, respectively. Similarly, the proliferation of WRL-68 cells was inhibited by EPMC, hydro-distillate, and extract at 50 μg/mL, with respective inhibitions of 37%, 42%, and 46%.

Hamid et al. [205] investigated the inhibitory effect of the extract of *Kaempferia galanga* L. on the expression of VEGF and Cox-2 in the chorioallantois membrane of embryonated chicken eggs (ECE). It was using 24 ECE and divided into six groups: C+, which was inoculated with solvent; C-, which was not inoculated; T1, T2, T3, and T4, which were inoculated with bFGF 60 ng and treated with extract dosages of 30, 60, 90, and celecoxib 60 mg, respectively. Besides, it was detected that the expression of VEGF and Cox-2 used the immunohistochemical method. The extract of *Kaempferia galanga* L. inhibited the expression of VEGF and Cox-2 in the chorioallantoic membrane as a result.

Swapana et al. [202] revealed that *Kaempferia galanga* exhibited good cytotoxicity against HeLa cervical cancer cell and HSC-2 mouth squamous cell carcinoma. The cytotoxicity was assessed using a modified MTT assay, and the IC50 values for sandaracopimaradiene-9α-ol, kaempulchraol I, and kaempulchraol L were found to be 75.1 μM, 74.2 μM, and 76.5 μM, respectively, against HeLa cervical cancer cell, and 69.9 μM, 53.3 μM, and 58.2 μM, respectively, against HSC-2 mouth squamous cell carcinoma.

According to the previous study, the essential oils extracted from *K. galanga* have shown some antitumour activity. To investigate their effects on MKN-45 cells, we evaluated the volatile oil's impact on cell cycle and apoptosis using flow cytometry (FCM). We treated the cells with different doses of volatile oil (1.56 g/d, 0.78 g/d, and 0.39 g/d) and found that the growth suppression rates of gastric cancer were 57.2%, 28.0%, and 5.0%, respectively. The high-dose volatile oil-treated group was the most effective, arresting the MKN-45 cells at G0/G1 phase and inhibiting their growth [206].

3 Conclusions

It is reported that natural compounds from traditional medicines, such as *Annona muricata, Curcuma longa*, *Curcuma xanthorrhiza, Curcuma zedoaria, Phyllanthus urinaria, Gynura procumbens, Garcinia mangostana, Morinda citrifolia*, *Nigella sativa*, *Moringa oleifera, Zingiber officinale*, *Euphorbia abyssinica,* and *Kaempferia galanga,* are useful for their potential chemopreventive properties. The research showed that they can act to inhibit cancer through various mechanisms to obstruct the proliferation and growth of cancer cells. Moreover, they can also help reduce oxidative stress and inflammation, which are the key factors in the development and the spread of cancer, induce apoptosis, or inhibit angiogenesis.

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Chemopreventive Strategies in Herbal Medicine Practice: Current Aspects, Challenges, Prospects, and Sustainable Future Outlook

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Abstract

Cancer is one of the high-profile diseases and is responsible for millions of deaths worldwide every year. Even though there are many therapeutic strategies (natural and chemical), many issue have been still unresolved in order to advance cancer therapy. For this, a lot of efforts are being put forwarded by oncology research to find new and efficient anticancer therapeutical strategies that can overcome the discomfort caused by conventional methods. Many of the chemically derived drugs and anticancer treatments already exist, but are known to have toxic effects. Hence, the demand for naturally derived anticancer agents is used as an alternative for which nature remains the main supplier, especially plants. Many plant-derived secondary metabolites, like alkaloids, brassinosteroids, flavonoids, polyphenols, and phytosterols, have potential as anticancer agents, unaided or in formulations, and have antioxidant activity, apoptosis induction, cancer cell line inhibition and cytotoxicity, multitarget specificity, etc. Many of these plant-derived drugs preexist in research and have been cleared for clinical trials, mostly in phases I–III. This chapter provides a thorough overview of the therapeutic drugs derived from plants and also advances a clinical assessment of recently investigated natural compounds in terms of their anticancer modes of action. Besides, some new technologies like nanococheleates, nanoparticles, and nanoliposomes are being used in the administration of anticancer therapies for target specificity. These plant-derived anticancer agents are in high demand as they are efficient inhibitors of cancer cells lines; hence, their exploration needs to be managed to be sustainable. These have the potential to minimize the harmful effects typically associated with conventional medicines. Unlike certain synthetic drugs, herbal medicines being derived from plant sources and may possess compounds that have a lower risk of toxicity or adverse reactions. Finally, the current review describes an in-depth analysis of the most innovative advances of several efficient therapeutic natural agents for the encapsulation and target specificity in basic and applied cancer research.

Keywords

Nanoparticles · Sustainability · Chemopreventive · Environment

Abbreviations

BRs	Brassinosteroids
CDK	Cyclin-dependent kinase
HDAC	Histone deacetylases
NP	natural products
WHO	World Health Organization

1 Introduction

Any region of the body can develop cancer, which is characterized by the unchecked growth and dissemination of abnormal cells. Cancer has the ability to infect nearby tissues and spread to the other organs. It is a complex disease influenced by various factors, including genetics and environmental exposures, certain lifestyle choices, etc., and is among the leading sources of death and mortality in humans. In spite of many advanced technologies, cancer deterrence is extremely superior to treatment and causes death of many patients during treatments. For treatment, early diagnosis (first stage) and removal of tumor by surgical resection and radiation therapy are among the most common methods. The positive chemopreventive results have increased the awareness among the people globally. Further, for chemoprevention, natural or synthetic compounds are used to prevent, suppress, or even reverse carcinogenesis as an alternative to simplify this frightening burden on public health [1]. Among these chemopreventive natural compounds, phytoconstituents are the most common as they show properties of preventive and suppressing agents [2]. Three major types of natural chemopreventive drugs, that is, natural products, semi-synthetic, and synthetic drugs, are derived from natural products [3]. The perusal of previous studies revealed that more that 25% of medical prescriptions for cancer patients dispensed contain active ingredients from plants [4, 5]. Nature harbors an immense repository of chemical compounds that have the potential to serve as valuable therapeutic agents. Many plants have a long history of traditional use in various medicinal systems, indicating their potential medicinal value. However, it is crucial to conduct systematic scientific investigations to identify and isolate bioactive compounds from these plants and evaluate their efficacy and safety. With advancements in technology and research methodologies, there are increasing opportunities for the discovery of novel anticancer drugs from natural sources. Exploration can involve screening a wider range of plant species, including those from diverse ecosystems and unexplored regions. Additionally, focusing on traditional medicinal systems and indigenous knowledge can provide valuable leads for identifying plants with anticancer potential [6]. However, unlike Western medicine system, traditional herbal medicine a mixture of many secondary metabolites and works on multiple signaling pathways and molecular targets [7, 8]. Therefore, using these medications may have a variety of benefits that may reduce toxicity, while others may boost bioavailability, maximizing their efficacy in improving health and well-being.

Since ancient times, natural products (NPs) and their derivatives have played a key role in pharmacotherapy for cancer and other infectious diseases owing to their phytoconstituents, which are changes with geographical and environmental distributions. Throughout history, various civilizations have relied on natural products as a source of medicine. Plants, fungi, marine organisms, and other natural sources have been extensively explored for their therapeutic potential. NPs are being employed to treat a great variety of ailments, including cancer and infectious diseases, due to the presence of bioactive compounds in them. NPs and their derivatives have been used in pharmacotherapy for cancer and other diseases for centuries [9]. The effectiveness of NPs in pharmacotherapy is attributed to their diverse phytoconstituents. Phytoconstituents are chemical compounds found in plants and other natural sources that possess medicinal properties. These compounds can include alkaloids, flavonoids, terpenoids, polyphenols, and many others. Different types of NPs contain unique combinations and concentrations of these phytoconstituents, which contribute to their therapeutic effects [9]. Importantly, the geographical and environmental factors under which NPs are grown or obtained can significantly impact the composition and concentration of their phytoconstituents. Plants and organisms in different regions exhibit variations in climate, soil composition, and other environmental factors, leading to distinct phytochemical profiles. This phenomenon is known as the chemical fingerprint of a natural product and contributes to the diversity and uniqueness of NPs obtained from different geographical locations. More than 60% of existing anticancer drugs have been derived from natural resources, which cover a significant portion of the world's population [10]. By harnessing the phytoconstituents present in NPs, researchers and medical professionals have been able to develop drugs and treatments for cancer and infectious diseases. Many successful anticancer drugs, such as paclitaxel derived from the Pacific yew tree, and antibiotics, such as penicillin derived from the *Penicillium* fungus, have been derived from natural sources. With increasing interest in herbal medicines globally, scientists are researching for functional or bioactive metabolites [11]. Recently, researchers are actively studying functional or bioactive metabolites found in NPs, such as curcumin, emodin, genistein, gingerol, kaempferol, quercetin, resveratrol, and tryptanthrin [12, 13]. These natural products (NPs) acquire early detection of the cancerous cells and also have many benefits, viz., easy availability, cost-effectiveness, no toxicity, multitargeting efficacy, drug resistance, and high potential for chemosensitization. Many studies have determined the importance of NPs and their effect reduces many chronic diseases. However, their use is inadequate because of their limited life span and low solubility [1]. So, this is the call of the hour, to look for the new curative strategies to overcome the current therapy resistance with more efficient, less toxic strategies, and more selectivity. Therefore, the NPs obtained from the natural resources and their uses in the development of novel molecular leads need to be broadly reviewed [14].

Despite the abundance of natural resources, only a small fraction of plants have been investigated for bioactive compounds. This necessitates further exploration of natural sources to discover novel anticancer drugs. Traditional herbal medicine, unlike the Western medicine system, consists of mixtures of secondary metabolites that work on multiple signaling pathways and molecular targets. This complexity may contribute to decreased toxicity and increased bioavailability of these medicines. Numerous clinical trials are currently underway to evaluate the efficacy of NPs as chemopreventive agents. Overall, the passage emphasizes the potential of natural compounds in cancer prevention and the need for further research and exploration to harness their benefits effectively. This chapter aims to collect data on active NPs as chemopreventive agents, including their current aspects, challenges, and future prospects for sustainable development. NPs have been used traditionally in various cultures for gastrointestinal diseases, analgesia, and anti-inflammatory purposes, and to promote overall health and vitality. Additionally, NPs are being used in numerous clinical trials as chemopreventive medications, which are covered in more detail later in this chapter.

2 Cancer: Etiology and Management Approaches

Most of the chronic diseases, cancer, cardiovascular, diabetes, etc., are most prevalent and costly to treat; however, these can be avoided through lifestyle changes. Among these, cancer is a growing public health problem globally, and in spite of costly chemotherapy treatments, it is estimated that cancer death rates could increase by 45% by 2030 [15]. The main reason for the increased cancer death may be low effectiveness of the present chemotherapeutic drugs as these are costly and also has some adverse side effects [15]. Hence, it is the call of the time to look for alternative nontoxic approaches and cost-effective methods, which also show some therapeutic resistance, effectiveness in preventing cancer resurgence, and improvement of life standards. Cancer shows a great deal of heterogeneity; despite this, transformation of a normal cell into precancerous lesion to malignant tumors by alternation in many signally pathways is among the common factor [16, 17]. The reason for such neoplastic transformation includes (i) exposure to UV, gamma, and X-ray radiations; (ii) infectious microorganisms (*Helicobacter pylori*, papilloma); and (iii) many carcinogenic agents [18, 19]. However, physiological anomalies like oxidant–antioxidants homeostasis, which, if get disrupted, results in more oxidants, and further leads to damaged DNA [20, 21] by imposing oxidative stress [22, 23]. Such oxidative damage to these macromolecules can lead to cancer development by inducing inflammatory signaling [24, 25].

3 Plant-Derived Drugs in Cancer Treatment

Cancer is among the leading sources of death and mortality in humans, and cases are constantly increasing, with changes in lifestyle or environment. Irrespective of the recent advance technologies, deterrence of cancer is extremely superior to treatment and causes the deaths of many patients during treatments. For treatment, early diagnosis (first stage) and removal of tumor by surgical resection and radiation therapy are among the most common methods. The positive chemopreventive results have increased the awareness among the people globally. Further, for chemoprevention, natural or synthetic compounds are used to prevent, suppress, or even reverse carcinogenesis as an alternative to simplify this frightening burden on public health [1]. Among these chemopreventive natural compounds, phytoconstituents are the most common as they show properties of preventive and suppressing agents [2]. These natural products (NPs) acquire early detection of the cancerous cells and also have many benefits, viz., easy availability, cost-effectiveness, no toxicity, multitargeting efficacy, drug resistance, and high potential for chemosensitization. The importance of NPs and their effect reduces many chronic diseases, which have been reported by many workers. Despite these benefits, the use of NPs is inadequate because of their limited life span and low solubility [1]. So, this is the call of the hour to look for the new curative strategies to overcome the current therapy resistance with more efficient, less toxic strategies, and more selectivity.

NPs and their derivatives have played a key role in pharmacotherapy for cancer and other infectious diseases owing to their phytoconstituents [9] for many years now, which are changes with geographical and environmental distributions. As reported previously, more than 60% of the existing chemopreventive drugs were obtained from natural resources [26], which covers over 80% of world's population [27]. With increasing interest in herbal medicines globally, scientists are researching for functional or bioactive metabolites [26, 27]. These include curcumin, emodin, genistein, gingerol, kaempferol, quercetin, resveratrol, tryptanthrin, etc. [1]. Therefore, the NPs obtained from the natural resources and their uses in the development of novel molecular leads have been/or need to be broadly reviewed [14]. So, the aim of this chapter is to collect data on active chemopreventive agents (NPs), current aspects, and challenges, and provide future prospects for sustainable development.

NPs are being used to cure many gastrointestinal diseases, such as analgesia, anti-inflammatory, increasing vitality, and sustain good health since ancient times in China, Egypt, India, and Mesopotamia [12, 13]. Natural chemopreventive drugs can be classified into three major types: natural products, semi-synthetic drugs, and synthetic drugs derived from natural products. Natural products are compounds directly derived from natural sources, such as plants, animals, or microorganisms. These compounds have a long history of use in traditional medicine and often exhibit diverse biological activities. Semi-synthetic drugs are derived from natural products but undergo chemical modifications to enhance their therapeutic properties. These modifications can improve drug stability, potency, or pharmacokinetics. Synthetic drugs derived from natural products are entirely synthetic compounds designed based on the chemical structure or pharmacophore of a natural product. These drugs aim to mimic or improve upon the bioactive properties of the natural compound while providing advantages such as increased potency or selectivity. By harnessing the wealth of natural compounds and employing chemical modifications or synthesis, scientists can develop a range of chemopreventive drugs with diverse structures and properties, offering promising avenues for cancer prevention and treatment [28]. Between 1980 and 2002, over 69% of the licensed anticancer medicines came from natural sources [29]. The perusal of previous studies revealed that more than 25% of medical prescriptions for cancer patients dispensed contain active ingredients derived from plants [4, 5] Despite the treasure of natural bounty, about 15% of the total plants have been reported with bioactive compounds so far [30], which leads to the need for more exploration of natural wealth in search of novel anticancer drugs [6, 7]. However, unlike Western medicine system, regarding their composition and mechanism of action targets, traditional herbal medicine is often a mixture of various secondary metabolites and can work on multiple signaling pathways and molecular targets [7, 8]. In contrast to natural chemopreventive drugs, which often utilize complex mixtures of compounds derived from natural sources, Western medicine tends to focus on isolated active compounds that target specific molecular pathways. Western medicine emphasizes the identification and isolation of individual active constituents from natural sources to understand their mechanisms of action and optimize their therapeutic effects. This reductionist approach allows for a more precise understanding of the specific components responsible for a drug's activity. The contrast between Western medicine's focus on isolated active compounds and the utilization of natural mixtures in natural chemopreventive drugs highlights different approaches to drug development. While Western medicine emphasizes targeted therapies and precise control, natural chemopreventive drugs often leverage the complexity and potential synergistic effects of natural sources. Both approaches have their merits and continue to contribute to the development of effective treatments for various health conditions. This characteristic of herbal medicine can potentially provide several functions, such as reducing toxicity and increasing bioavailability of active compounds. Several clinical trials using NPs as chemopreventive medicines are currently ongoing, which have been discussed later in this chapter.

4 Current Prospective

4.1 Plant Compounds with Medicinal Properties

Plants or plant-based medicines have been utilized for thousands of years by populations in Asia, Africa, and even developed countries as folk remedies or for health advantages. The WHO reports that many countries still rely on plant-based remedies as their primary source of medicine, which have shown to be effective for treating medical conditions [31, 32]. Many secondary metabolites/compounds have been identified and extracted from plants for their anticancer properties, which include alkaloids, phenolic compounds, and terpenoids (Table 1).

4.2 Phytosterol

These are the plant sterols frequently used in the human diet with unique properties of reducing cholesterol absorption in the gut. Medicinally, these are reported to weaken the inflammatory response of neutrophils and macrophages [32]. It has been linked to a number of positive health effects, including being an antioxidant and antipyretic, anti-inflammatory [107], treating benign prostatic hypertrophy, preventing breast cancer and colon cancer, reducing inflammation, protecting the liver, and treating rheumatoid arthritis [108]. The substance has previously been found in numerous plants, including *Vanda roxburghii* [109] and *Withania coagulans* [110]. Due to their potent therapeutic properties, they are thus a useful research topic. Some of these phytosterols like sulforaphane, obtained from broccoli, and Withaferin A, obtained from *Withania somnifera,* are summarized in Table 1.

Table 1 Natural chemopreventive compounds, their sources, and physiological targets

Preclinical uses of compound	Natural resource	Affected targets	References
Acetylbritaniilactone	*Inula britannica*	Decreases cyclin A, D1, E, and CDK2	[33]
Aclarubicin	*Streptomyces* sp.	Affects topo-isomerase-I and II	[34, 35]
Allyl sulfides	*Allium sativum*	Cell cycle arrest via mitochondrial signaling	[13, 36, 37]
Apigenin	Celery plant	Decreases cyclin A, D1, E and CDK2	[38]
Berberine	*Berberis* sp.	Increases CDKN1A and 1B	[39]
Betulinic acid	*Betula pubescens*	Inhibition of topo-isomerase, production of ROS	[40]
Bufotalin	*Venenum bufonis*	Degradation of CDC25A, CDK1	[41]
Butyrolactone −1	*Aspergillus terreus*	Inhibition of CCDK1/cyclin B complex	[42]
Capsaicin	Chili peppers	Cell cycle arrest and growth inhibition	[43]
Chebulagic acid	*Terminalia chebula*	Increases CDKN1B and decrease NF-kB	[44]
Combrestatin	*Combretum caffrum*	Inhibition of angiogenesis and microtubule depolymerization	[45, 46]
Curacin A and D	*Lyngbya majuscule*	Inhibition of tubulin polymerization	[47]
Curcumin	*Curcuma longa*	Increases ING4, CDKN1A, 1B	[48, 49]
Daunomycin	*Streptomyces peucetius*	Acts as intercalating agents	[26]
Diindolyl methane	Crussiferae	Prevention of tumor development (in vivo)	[50, 51]
Dineolignan	*Saurus chinensis*	Decreases CDK-6 and cyclin D	[52]
Discodermolide	*Discodermia dissoluta*	Acts as microtubules-stabilizing agents	[53]
Dolastanin 10 and 15	*Dolabella auricularia*	Prevents mitotic assembly	[54, 55]
Doxorubicitn	*Streptomyces peucetius*	Topoisomerase-II inhibitor	[26]
Epigallocatechin-3-gallate	Green tea	Increases CDK1A, 1B, 2A, and 2C	[56, 57]
Epothilone A, B, C D	*Sorangium cellulosum*	Mitotic arrest by stabilizing tubulins	[58, 59]
Fangchinoline	*Stephaniae moore*	Decreases cyclin A, D1, E	[60]
Flavopiridol	*Dysoxylum binectariferum*	Tyrosine kinase inhibitor, cell cycle arrest; apoptotic induction	[61, 62]
Indole-3-carbinol	Brassicaeae family	Decreases CDK-2, cyclins D1	[63]

(continued)

Table 1 (continued)

Preclinical uses of compound	Natural resource	Affected targets	References
Isorhamnetin	*Hippophae rhamnoides*	Increases CDKN1A	[64]
Limonoids	Oranges and lemons	Apoptotic induction`	[65, 66]
Myricetin	*Myrica cerifera*	Inhibition of matrix metalloproteinase	[67, 68]
Pacilitaxel	*Taxus brevifolia*	Stabilizes tubulin polymers	[69, 70]
Palmatine	*Berberis lyceum*	Increases CHEK2 and decreases CDC25	[71]
Phloridzin	Apple	Increased expression of tyrosinase genes	[72]
Podophyllotoxin	*Podophyllum peltatum*	Causes ss-DNA and ds-DNA breaks	[73, 74]
Proanthocyanidins	*Vitis vinifera*	Decreases CDK-1, 4, and 6	[75, 76]
Pseudolaric acid B	*Pseudolarix kaempferi*	Decreases CDC2 and 25	[77]
Quercetin	*Quercus* spp.	Reduced proliferation and angiogenesis	[78, 79]
Retinoic acid	*Dacus carrota*	Decreases telomerase activity	[80, 81]
Rhizoxin	*Rhizopus chinensis*	β-Tubulin binding	[82, 83]
Rocaglaminde-A	Aglaia	Degradation of CDC25A, induces ATM pathway	[84]
Roscovitine	Radish	Inhibition of cyclin-dependent kinases (CDKs)	[85, 86]
Rosamultic acid	*Rosa multiflora*	Decreases CDK4, 6 and cyclin D1	[87]
Sanguinarine	*Papaver somniferum*	Apoptotic induction and caspase activation	[88, 89]
Silibinin	*Silybum marianum*	Regulation of tumor necrosis factors, decreases CDK2`	[90–92]
Spongistanin 1	*Hyrtios erecta*	Binds to the site on β-tubulin	[93]
Sqamocin	*Annonaceous* sp.	Increases CDKN1A and 1B	[94]
Sulforaphane	Broccoli	Inhibition of viability and in vitro proliferation	[95, 96]
Tamarixetin	*Tamarix ramosissima*	Increases cyclin B1, BUB1, and CDKN1A	[97]
Tanshinone IIA	*Salvia miltiorrizae*	Decreases CDC-2	[98]
Theaflavin	Black tea	Apoptotic induction via Fas/ caspase-8 and Akt/pBadu	[99, 100]
Tryptoquivaline	*Aspergillus fumigatus*	Inhibits MAP2, Tau-induced tubulins	[101]
Vinblastine	*Catharanthus roseus*	Inhibits tubulin polymerization	[102, 103]

(continued)

Table 1 (continued)

Preclinical uses of compound	Natural resource	Affected targets	References
Vincristine	*Catharanthus roseus*	Inhibits tubulin polymerization	[102, 103]
Withaferin A	*Withania somnifera*	Targeting on Hsp90, cell cycle arrest	[104]
Wogonin	*Scutellaria radix*	Decreases cyclin A, D1, E	[105]
Zerumbone	*Zingiber zerumbet*	Decreases cyclin B1 and CDKs	[106]

4.3 Phenolic Compounds

The phenolic compounds are the most commonly occurring secondary metabolites, which are the products of pentose phosphate, phenylpropanoid, and shikimate pathways in plants [111]. Besides their medicinal importance and protection against pathogens, these compounds positively affect growth and reproduction of plants also [112]. Additionally, like most of the allied compounds, it also shows antioxidant, anti-inflammatory, and anticancerous properties [113]. The caffeic acid is a well-known phenolic compound and is popularly known for reducing the level of cholesterol, free fatty acids, phospholipids, and triglycerides in rats [114], and also can reduce the chances of skin cancer. The compound is also known to be dispersed in many fruits and vegetables. This compound is also detected from various other plants, such as *Origanum vulgare* and *Ocimum gratissimum* [115]. Chlorogenic acid, which is created through the esterification of cinnamic acids like caffeic acid and quinic acid, is another member of the phenolic chemical family. Some of the phenonic compounds, such as apigenin, combrestanin, flopiradiol, indole-3-carbinol, isorhamnetin, myricetin, phloridizin, quercitin, retinoic acid, rhizoxin, tamarixetin, and wogonin, along with their physiological role, are summarized in Table 1.

4.4 Alkaloids

Alkaloids, a group of naturally occurring organic compounds, are characterized by their alkaline properties and are typically found in various parts of the plants, such as leaves, stems, roots, and seeds. Alkaloids exhibit a wide range of chemical structures and biological activities and are medicinally important as effects on the human body, ranging from pain relief and stimulant properties to toxicity. Some well-known alkaloids include caffeine, cocaine, morphine, nicotine, quinine, and strychnine. Few famous alkaloids, like berberine, capsaicin, curacin A and fangchinoline, paclitaxel, palmatine, rosamultic acid, sanguinarine, vinblastine, and vincristine, are of plant origin and known for their anticancerous properties. The details of these compounds, along with the natural resources and their physiological targets, are discussed in Table 1.

4.5 Triterpenoids

These are also a group of secondary metabolites produced by pentacyclic triterpenoids (betulinic acid), which show direct or indirect antioxidant properties that are beneficiary to the cardiovascular disease [116]. Betulinic acid is a naturally occurring triterpenoid and widely dispersed throughout the plant kingdom, especially in *Betula* spp. [117], where it was reported for the first time in the species *B. alba* [117]. It exhibits a variety of medicinal properties and biological activities, such as inhibition of HIV [118], antibacterial [119], and also anticancer activities [120]. Another naturally occurring pentacyclic triterpenoid, named oleanolic acid, is prevalent in the members of Oleaceae family, especially in the species *Olea europaea*, after which the compound was named, and is still a main source (commercial) of the compound [121]. After this, the compound has been isolated in about 1620 plants, which are highly known for their food and medicinal properties [122]. In the plants, oleanolic acid is reported in cuticles with waxes, helps plants in excessive water loss, and also as defense line against the pathogens, for example, fungal pathogens [123] or other allelopathic agents [123]. Another naturally occurring triterpenoid, ursolic acid, is a secondary metabolite of the plants and usually occurs in the stem bark, fruit peels, and even in leaves [124]. The perusal of the literature reveals a number of medicinal and pharmacological properties. The compound is also reported with a number of anticancerous activities [124], antimalarial activity, and antibacterial activities [125] as it affects the peptidoglycan metabolism. Many other triterpenoids, for example, zerumbone, a monocyclic sesquiterpene, are obtained from *Zingiber officinale* and are known for their anticancerous properties as they decrease cyclin B1 and CDKs (Table 1). Similarly, pseudolaric acid B, obtained from *Pseudolarix kaempferi*, is known to decrease cdc2 and cdc25 (Table 1).

4.5.1 Brassinosteroids (BRs)

Brassinosteroids (BRs) are another group of plant hormones that have been investigated for their potential chemopreventive properties. Chemoprevention refers to the use of natural or synthetic substances to prevent, inhibit, or reverse the development of cancer [126]. Here are some ways in which brassinosteroids may play a role as chemopreventive herbal medicine [127]:

1. Antioxidant and anti-inflammatory effects: Brassinosteroids possess antioxidant properties and can reduce inflammation. These effects are important in chemoprevention since chronic inflammation and oxidative stress are known to contribute to the development and progression of cancer. By reducing inflammation and neutralizing free radicals, brassinosteroids may help protect cells from DNA damage and inhibit the formation of cancerous cells.
2. Cell cycle regulation: Brassinosteroids have been found to influence the cell cycle progression and cell division in various cell types. Abnormal cell cycle regulation is a hallmark of cancer, and compounds that can restore or modulate the cell cycle have potential as chemopreventive agents. Brassinosteroids may help maintain proper cell cycle control, preventing the uncontrolled proliferation of cancer cells.

3. Apoptosis induction: Apoptosis, or programmed cell death, is a natural process that eliminates damaged or potentially cancerous cells. Dysregulation of apoptosis can contribute to tumor formation and growth. Some studies have suggested that brassinosteroids can induce apoptosis in cancer cells, promoting their programmed death. This property is crucial in eliminating cancer cells before they become clinically significant.
4. Immune modulation: The immune system plays a critical role in recognizing and eliminating cancer cells. Brassinosteroids have been shown to modulate immune responses by regulating the activity of immune cells, such as T cells, natural killer (NK) cells, and macrophages. By enhancing immune surveillance and promoting immune cell activity, brassinosteroids may aid in the prevention and control of cancer.
5. Antiangiogenic effects: Tumor growth relies on the formation of new blood vessels, a process known as angiogenesis. Inhibiting angiogenesis can limit tumor growth and metastasis. Some studies have suggested that brassinosteroids can suppress angiogenesis by inhibiting the production of pro-angiogenic factors. This antiangiogenic property may help prevent the development and spread of cancer.

It is important to note that the research on brassinosteroids as chemopreventive agents is still in its early stages, and further studies are needed to validate their effectiveness and safety. Additionally, the optimal dosage, delivery methods, and potential interactions with other medications need to be determined. Two natural BRs, 28-homocastasterone and 24-epibrassinolide, have been observed with anticancer effects [127].

4.6 Plant-Derived Anticancer Drugs

Drugs derived from plants are preferred in cancer treatment owing their natural origin, easy availability, and the potential for administration as part of dietary intake [128, 129]. Normally, these plant-derived drugs are generally nontoxic to human cells [130], although there are some such as cyanogenetic glycosides, lectins, lignans, saponins, and some taxanes [130, 131]. Plant-derived drugs can be classified into four classes: methyltransferase inhibitors, DNA damage preventive drugs or antioxidants, histone deacetylases (HDAC) inhibitors, and mitotic disruptors [128]. These classes target specific molecules or pathways involved in cancer cell growth and proliferation. These compounds have been discussed in Table 1 with names of the compound, natural resource, and affected targets. The plant secondary metabolites, such as isoflavones, isothiocyanates, pomiferin, and sulforaphane, are considered to be histone deacetylases inhibitors that inhibit carcinogenic proteins [127]. Further, derivatives of vinca alkaloids, vinblastine, vincristine, vindesine, vinflunine, and vinorelbine also inhibit the binding to β-tubulin in microtubules [132]. Similarly, taxanes (paclitaxel and its analog docetaxel) act as microtubule disruptors and inhibit cell-causing cell cycle arrest and apoptosis [133, 134]. Paclitaxel was one of the first drugs to be observed with massive implication for cancer treatment, while vinblastine and vincristine were among the first medications to be separated [132]. Table 1 provides a list of various compounds derived from natural resources and their preclinical uses in cancer research. These compounds target specific molecules or pathways involved in cancer cell growth and proliferation.

5 Challenges of Anticancer Phytochemicals

Anticancer phytochemicals face several challenges in their development and application. The developments of plant-derived anticancer agents for clinical trials require significant evidence of efficacy before approval for use in patients. In spite of the significant anticancer properties of NPs, it needs to be checked prior to clinical application. So, absorption, poor aqueous solubility, inadequate therapeutic potential, poor penetration in target cells, and toxicity are some of the characteristics that still are of major concern before applying these compounds in the clinical treatment of cancer [135, 136]. For example, compounds like camptothecin, colchicine, and podophyllotoxin, and their derivatives are in limited use due to side effects. Some others, like vinca alkaloids, are being used in formulations as they show limited effect singly. Another challenge is their limited bioavailability as they may be poorly absorbed and distributed in the body, reducing their effectiveness of reaching cancer cells. The complex mixture and variability of phytochemicals from different plant sources make standardization difficult, hindering consistent efficacy and safety. Moreover, the lack of specific targeting may result in off-target effects and potential toxicity. Interactions with other drugs and therapies, as well as the presence of side effects, need to be carefully managed. Additionally, the limited clinical evidence and the absence of intellectual property protection pose hurdles for commercialization and investment in research. Overcoming these challenges requires extensive research, robust clinical trials, interdisciplinary collaboration, and regulatory support to harness the full potential of phytochemicals in the fight against cancer. Beside these challenges, clinically tested phytochemical compounds also face challenges like characterization, extraction, optimization, and synthesis of anticancer compounds. Thus, new advances are required in analytical and computational methodologies to ease the process of identification, optimization, synthesis, and modification of new phytochemicals.

5.1 Low Solubility and Absorption

As discussed in the previous section, the phytochemical compounds are limited in availability and also have lower potential in cancer treatment. To solve this issue, lipid-based formulations and nanotechnology have been used to enhance bioavailability, solubility, and target specificity of these compounds [136, 137]. It mentions specific examples, like the encapsulation of vinca alkaloids and paclitaxel in liposomes and other nanoformulations to improve their therapeutic efficacy. However, it also noted that while many nanomedicines have been investigated, only a few have progressed to clinical trials, and even fewer have been cleared for clinical application [138]. This encapsulation enhances performance by controlling release and also providing protection for the bioactivity against external environment. For example, liposomes, such as Marqibo®, have been cleared by the FDA for the treatment of adult leukemia [139]. Similar efforts have also been done for paclitaxel, resulting in the development of various nanoformulations, including polymeric nanoparticles, polymeric conjugates, polymeric micelles, and liposomes [138]. However, despite the extensive investigation of nanomedicines, only a few formulations have progressed to clinical trials, and even fewer have been approved for clinical application. For instance, a hydroxypropyl-methacrylamide copolymer–paclitaxel formulation developed by Pfizer was discontinued in phase I due to high neurotoxicity observed in rats. Poly (L-glutamic acid)-paclitaxel is one of the few formulations to reach phase III of clinical trials but was ultimately discontinued in 2016 because it did not demonstrate significant results over the current standard [140]. In case of irinotecan, its hydrophobic nature, low stability at physiological pH, and side effects limited its therapeutic use. To address these limitations, new drug formulations incorporating nanoparticles with carbohydrates, dendrimers, peptides, and polymer conjugates have been investigated [141].

Similar challenges have been posed by low solubility and availability of certain compounds in clinical application, which limits their potential for cancer treatment. Resveratrol, for example, has a short circulation life of a few minutes, while quercetin works at low concentrations in the blood, which is insufficient for effective anticancer activity [142]. To address these limitations, many studies have explored innovative strategies to increase the availability of these compounds. These strategies include the development of liposomes, nanoformulations, nanoparticles, and other delivery systems. For instance, gingerol in combination with proliposomes increases in vitro inhibition of HepG2 cancer cells and also improves oral bioavailability in vivo [143]. In another study, nanoparticles of gold-coated iron oxide significantly improve the activity and stability of SFN (sulforaphane) against human breast cancer cells. Furthermore, mesoporous silica nanoparticles formulated with hyaluronic acid have been used for the delivery of curcumin [144]. The anticancer activity of curcumin was evaluated in both in vitro and in vivo experiments, and the results revealed enhanced the anticancer activity, potentially due to improved bioavailability. These examples highlight the potential of innovative strategies and nanotechnology-based formulations in enhancing the availability and effectiveness of compounds in clinical applications. By improving solubility, stability, and bioavailability, these approaches aim to overcome the limitations that hinder the therapeutic use of such phytoconstituents in cancer treatment [145].

Nowadays, many clinical trials are exploring modern strategies to improve the bioavailability of these chemotherapeutic drugs, but the number of these studies is relatively limited compared to the extensive research published on novel strategies [137, 140]. This indicates that there is still a gap between preclinical research and clinical implementation in this field. As the understanding of drug delivery systems, nanoformulations, and other innovative strategies continues to advance, there is a growing interest in translating these findings into clinical applications. The need to improve the bioavailability and efficacy of these chemotherapeutic drugs remains a critical goal in cancer treatment.

5.2 Phytochemicals in Combination with Conventional Chemotherapeutics

The previous studies of phytochemical drugs revealed that single-target chemical agents are inefficient in cancer treatment owing to rapid molecular adaptations of cancer cells, and their limited effects, when used alone [145]. As a result, a combinatorial approach involving amalgamation of phytochemicals with conventional chemotherapeutic drugs proposes to achieve synergistic effects, increase toxicity in cancer cells, reduce therapeutic doses, and address toxicity concerns. Numerous studies have been conducted to evaluate the efficacy of these combinatorial approaches [146–148]. Resveratrol is a successful example of these drugs and used in vitro and in animal models as an adjuvant with other drugs, like doxorubicin, paclitaxel, and temozolomide. Similarly, curcumin has been employed as a chemosensitizing agent in combination with cisplatin, docetaxel, irinotecan, paclitaxel, and vincristine, leading to decreased drug resistance and enhanced efficacy compared to single treatments [133, 149, 150]. Other phytochemicals such as apigenin, berberine, betulinic acid, and gingerol have also been evaluated in combinatory assays to enhance the sensitivity of cancer cells to chemotherapeutic drugs. Clinical trials have also explored the combination strategy, as indicated in Table 1. Although more studies are still needed, the use of phytochemicals as chemosensitizers shows promise in cancer therapy. By combining them with conventional chemotherapeutics, it may be possible to overcome drug resistance, enhance therapeutic efficacy, and reduce the overall toxicity associated with higher drug doses.

6 Future Sustainable Development

Chemopreventive herbal medicine is poised to play a crucial role in future sustainable development. As the world seeks more sustainable solutions in healthcare, herbal medicine offers a natural and eco-friendly approach. By harnessing the healing properties of plants, it reduces reliance on synthetic drugs, minimizing their environmental and social impacts. Furthermore, the cultivation and sustainable harvesting of medicinal plants promote biodiversity conservation and support local economies, particularly in rural communities. Integrating traditional knowledge systems and personalized preventive healthcare approaches can lead to a more holistic and culturally sensitive healthcare system. Embracing chemopreventive herbal medicine can pave the way for a sustainable future where human health, environmental preservation, and socioeconomic well-being are all interconnected and mutually beneficial.

6.1 Enhancing Drug Administration

With the advancement and emergence of naturally derived drugs, new technologies are/or being developed using these anticancer compounds. The combination of these plant-based drugs, viz. combination of alkaloids, diterpenes, and lignans in plant extracts, is reported with some enhanced anticancer properties and also reported with improved therapeutic efficacy [132, 134]. For example, extracts of *Artemesia monosperma*, *Origanum dayi*, and *Urtica membranaceae*, in combination, were tested on cancer cell lines [134], and the results revealed killing effect on cancer cells; however, normal human lymphocytes and fibroblasts were not affected. Further, these drugs were known to induce apoptosis by increasing sub-G_1 phase populations with lower content of DNA and condensed chromatin, and also increase caspase-3 activation [134]. As a result, plant products are more widely used as medicinal agents than those made of hazardous chemicals, and more clinical trials should take these factors into account.

Further, for the effective administration of new drugs (natural compounds), they need to be a perfect alteration to the modern treatments, such as chemotherapy. This can be attained by developing nanoparticles, which act as delivery system in drugs to reach the target sites. The anticancer natural compound is limited in the clinical because of the high dosages if applied separately, but shows great result if applied in combination with nanoparticles [133]. Bromleain, a proteolytic compound, isolated from *Aanas comosus*, is more effective in cancer treatment in combination with nanoparticles than free [133]. These bromelain-loaded NPs trigger apoptosis in benign cells by regulating the proapoptotic expression and antiapoptotic proteins. Similarly, gold nanoparticles in formulation with *Antigono leptopus* extract (powdered) and copper oxide nanoparticles of *Acalypha indica* showed cytotoxic effect against MCF-7 cell lines of breast cancer [149]. Paclitaxel, a natural compound from *Taxus* spp. (gymnosperms), has been cleared for clinical trials and early treatment settings. However, modern researchers are aiming to use magnetic mesoporous silica nanoparticles to control the drug release and enhance target specificity [150], so that paclitaxel can be controlled externally. Success was also reported in another drug, quercitin, where formulation with superparamagnetic magnetite nanoparticles was tested against MCF-7 cell lines of breast cancer [151]. This procedure has been successful in increasing the drug's capacity to stop the spread of cancer as well as lower undesirable effects [150]. Additionally, nanocochleates and nanoliposomes are being used by researchers in anticancer activities [151] by oral or inhalable ingestion, which is more comfortable and economical. Additionally, clinical trials with noscapine, a benzylisoquinoline alkaloid derived from the poppy plant, were constrained by its intractable characteristics [152]. The formulation of nanoliposomes with 9-bromo-noscapine and its subsequent increase in cytotoxicity and apoptosis in lung cancer cell lines compared to the free compound is an exciting development. The use of nanoparticles in cancer treatment has gained significant interest and is showing promising results as a natural alternative to conventional chemotherapy. Nanoparticles, such as liposomes, can improve the delivery of therapeutic agents to cancer cells, enhancing their efficacy while minimizing side effects.

The small size and unique properties of nanoparticles allow them to selectively target tumor cells, penetrate cellular barriers, and release the drug payload efficiently. This targeted approach increases the concentration of the therapeutic agent at the site of the tumor, leading to enhanced cytotoxicity and apoptosis. The application of nanoliposomes as a delivery system for 9-bromo-noscapine demonstrates the potential of nanoparticles in improving cancer treatment outcomes, offering a more targeted and effective approach that may reduce the toxicities associated with traditional chemotherapy.

6.2 Medicinal Plant Demands

Due to the cytotoxic impact they have on cancer cells and lack of toxicity on healthy cells, plant-derived medicines have undergone a number of effective clinical trials and are now widely used and in great demand for clinical development. A lot of species have been investigated globally (especially in developing countries) where medicinal plants are used and herbal therapies are practiced for primary treatment [153–155]. A WHO report estimated that the trade value of plant-derived drugs was US$100 billion in 2007, which was anticipated to reach US$5 trillion by 2050 [127]. This huge demand for plant-based drugs in developing countries is putting pressure on the floral diversity and increase the threat of extinction of these plants of this overexploitation of floral diversity continues. Normally, when these plants are harvested, only a part of the plant (bark, leaf, bulbils, tuber, etc.) is used, which may threat its survival to extinction [156]. So, to increase the sustainability of medicinal plants these countries, conservation of these plants is vital and also the use of all plant parts including bark, leaf, stem, and root should also be included in the exactions of bioactive compounds and also in treatment (if applied directly). Other conservation methods are storing viable seeds, cryopreservation, germplasm conservation, tissue culture, etc. [157, 158], which allows industrial utilization the medicinal plants [158]. Besides the developing nations, cultivation practices of these plants are undergoing even in developed nations to meet with increasing demands of natural drugs [156]. These cultivation practices may release pressure on floral diversity. Recently, most attention is drawn toward the fruit berries and crucifer vegetables [159]. Nowadays, industrial raw products with anticancer properties like grapes and grapes seed extract are being added in food stuffs due to the high polyphenolic content. In wine industry, grape stems are used as raw by-product of wine making; however, waste may be used for anticancer drug development due to its high polyphenolic content and antioxidant properties. The utilization of industrial waste from medicinal plants can indeed present a profitable scheme to address environmental issues while harnessing their medicinal properties. Industrial waste from medicinal plants often contains valuable bioactive compounds that can be extracted and utilized for various purposes. By repurposing and recycling this waste, we can reduce environmental pollution and contribute to a more sustainable and circular economy. Furthermore, the bioactive compounds present in these plant waste materials may have therapeutic potential. For example, the utilization of waste-derived compounds in the prevention of DNA damage by alleviating oxidative stress is a promising application [160]. Oxidative stress is known to contribute to various health conditions, including cancer. By utilizing these waste-derived compounds, we can potentially develop natural antioxidants that can help protect DNA from damage and promote overall cellular health. Additionally, the anticarcinogenic potential of waste-derived compounds against specific types of cancer, such as cervical cancer and thyroid cancer, is encouraging. Research studies exploring the anticancer properties of these compounds can lead to the development of specific treatments or preventative measures for these cancers. This not only contributes to the field of cancer research but also highlights the value of utilizing waste materials from medicinal plants in addressing significant health concerns. In summary, the utilization of industrial waste from medicinal plants offers a dual benefit of addressing environmental issues while capitalizing on their medicinal properties. By repurposing waste materials, we can contribute to a more sustainable and environmentally friendly approach, while simultaneously exploring their therapeutic potential for various health conditions, including cancer prevention and treatment [160, 161].

7 Conclusion

Cancer is one of the high-profile diseases globally and millions of people died from cancer or related diseases in low-income countries [160]. As a result, there is a huge market for its prevention and treatment. Numerous chemically generated medications and anticancer therapies currently exist [162], but they are known to have deleterious effects on tissues that are not their intended targets, leading to a variety of health problems. Therefore, there is a demand for naturally derived anticancer agents as an alternative. Many plant secondary metabolites, such as alkaloids, brassinosteroids, flavonoids, polyphenols, and phytosterols, have been studied for their potential use as anticancer agents (collectively) with antioxidant activity, apoptosis induction, cancer cell line inhibition and cytotoxicity, multitarget specificity, etc. (Table 1). Many of these plant-derived medications have entered clinical trials after being utilized in research. (Table 1), mostly in phase I to phase III (vinca alkaloids and paclitaxel). These compounds are easily available naturally and possess cytotoxicity to cancer cells, while relatively no effect on normal cell. Besides, some new technologies like nanocochleates, nanoparticles, and nanoliposomes are being used in the administration of anticancer therapies for target specificity. These formulations can be used to control prolonged drug release and support efforts to develop novel, target-specific medications with fewer severe side effects. However, the floral biodiversity is in danger, especially in emerging nations, due to the rising demand for these plant-derived medicines made from valuable, endangered, or uncommon medicinal plants [156]. This load on floral biodiversity and shortage of utilizing raw by-products in industries can be overcome by adopting conservation strategies [157, 161]. The high demand for plant-derived anticancer agents due to their efficiency in inhibiting cancer cell lines highlights the need for sustainable exploration and management of these resources. Sustainable management involves various aspects, such as conservation of medicinal plants, responsible harvesting practices, and promoting cultivation or farming of these plants. It is important to identify and prioritize plant species that have demonstrated potent anticancer properties and work toward their sustainable production and utilization. This includes implementing regulations and guidelines for wild harvesting, promoting organic farming practices, and supporting local communities engaged in the cultivation of medicinal plants. Additionally, research efforts should focus on identifying alternative sources or sustainable synthesis methods for these anticancer agents to reduce reliance on wild harvesting. By managing the exploration of plant-derived anticancer agents in a sustainable manner, we can ensure their availability for future generations and contribute to the long-term viability of these valuable natural resources.

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Sustainable Current Trends and Future Directions in Orthodox Medicine Practice in Sierra Leone

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Abstract

The socioeconomic, spiritual, and cultural importance of traditional medicine resonates well with majority of Sierra Leonean who use traditional medicine for healing and income purposes. This chapter discusses the traditional medicines status, opportunities in Sierra Leone and proffer remedies for traditional medicine dosage, challenges, misconception, and the sustainability of traditional medicines for the socioeconomic development of Sierra Leone. It is also believed that traditional healing powers inherited by native Doctors are mostly associated with spiritual powers from ancestral spirits. The sources of traditional medicine range from plant roots, leaves, fruits, animal parts, insects, rodents, birds, among others. Although some plants are prepared by macerating plant leaves in either cold or hot water, others plant parts like the leaves and barks are pounded, sieved, and water added to prepared the infusion for ingesting. Plant parts are in some instance burned to ashes and applied to cure rare illness. Study revealed that traditional medicines are used to cure sicknesses such as malaria, fever, bone fracture, demons, stomach ache, dysentery, piles, among other sicknesses, in Sierra Leone. Two recognized categories of traditional healers are known in Sierra Leone; they are traditional birth attendant and traditional healers commonly known as herbalist. However, the processing of most traditional medicines is complex and complicated. The study concludes that traditional medicines have been contributing to the socioeconomic development of hard to reach communities across Sierra Leone since the precolonial era. This has enabled most herbalist to gained self-employment and recognition through traditional medication and treatment. Nonetheless, traditional knowledge on forests medicines is being threatened by technological advancement and modern civilization in Sierra Leone. In such, the future direction of traditional medicine and its sustainability in Sierra Leone is uncertain due mainly to technological advancement, population growth, and climate change.

Keywords

Traditional medicine · Sierra Leone · Threats · Forests · Healthcare system · Treatment · Sicknesses · Plants

Abbreviations

FAO	Food and Agricultural Organization
GOSL	Government of Sierra Leone

IUCN	International Union for Conservation of Nature
SLENATHA	Sierra Leone National Traditional Healer Association
TM	Traditional Medicine
USD	United States Dollars
WHO	World Health Organization

1 Introduction

Traditional medicine's socioeconomic, spiritual, and cultural importance resonates well with majority of Sierra Leonean especially the elderly who are residing in rural communities. The spiralling significance of traditional medicine in tackling various illnesses in rural and hard to reach societies of developing countries especially in Africa cannot be over emphasized [1]. From ancient times, herbal medicines derived from plants in the wild has been useful in treating diverse types of diseases and illnesses hence alleviating poverty and pain in rural societies. It is undeniable over the past centuries that, traditional medicines in Africa has played a pivotal role in the maintenance of basic health system especially in locally deprived communities [2, 3]. For centuries now, natural plant resources have been a critical resource for the maintenance of basic healthcare in mostly deprived communities on the planet since time immemorial [4]. Globally, Traditional Medicine (TM) has been referred to as native medicine; ethnomedicine; local medicine; folk medicine among other names and is the oldest health therapy in Africa [5]. Traditional medicine has been widely used across Africa for the treatment of various types of illnesses long before the arrival of orthodox western medicines [6]. Traditional medicine is the combined knowledge and skill used by traditional people based on their beliefs, norms used to cure and maintain good health in local communities. Before the colonial era and the arrival of cosmopolitan medicine into Africa, traditional medicine was the sole source of medical treatment accessible in urban and rural societies across Sierra Leone [5].

These medicines are also used to cure physical and mental illnesses as well as diagnose and treat spiritual encounters [7, 8]. Traditional medicines have served as an alternative health remedy for majority of African populations especially in rural communities. Traditional medicines comprise diverse chemical substances capable of healing various diseases [9]. According to WHO estimate, 80% of the global population uses traditional medicines as indigenous therapies to cure many sicknesses [10]. It is reported that over 170 countries used traditional medicine as a supplement to expensive orthodox medicine. From time immemorial, traditional medicine is considered an integral resource used to cure diseases in communities without conventional medical facilities. Traditional medicine has been and is still considered a mainstay of traditional people's health [10] especially in developing countries like Sierra Leone.

Across Africa, traditional medicines are not considered a substandard medicine to conventional drugs, but instead used as an alternative for locals without access to hospitals and other health facilities [11]. Traditional medicines especially in developing countries is used to treat fungal and other skin diseases that have proven to be complex for Western drugs [12, 13]. Moreover, traditional medicines are relatively cheap as compared to Western medicines that are expensive and mostly unreachable in some developing countries like Sierra Leone. Globally, traditional medicine has gained momentum and is considered a trillion-dollar health remedies, healthy living and industries for cheap pharmaceuticals products [10]. Traditional medicines are sources of artemisinin and aspirin drugs but the actual importance of these medicines is understated [10]. These drugs extracted from traditional medicines are used to cure various sicknesses such as diarrhea, skin rash, bone fracture, wounds, etc. [14, 15].

Traditional medicines are a critical source of health therapies in deprived rural communities across Africa and Sierra Leone in particular. In most societies of developing countries, traditional healers are labeled as native doctors due to the fact that they play an important spiritual, cultural, and ancestral mouthpiece role for communities without good health facilities. According to Clarke, [16] and Semenya and Maroyi, [14], traditional healers are considered as the first option in treating sick people and are readily available any time their service is needed in most rural communities in Africa.

Over the past decades, many studies have been investigating the role of traditional medicine in supporting the healthcare system of developing countries especially in Africa. However, information on traditional medicine trend and future direction of traditional medicine in Sierra Leone is scarce if not unavailable. This chapter intends to bridge this gap and throw light on the importance and sustainable future of traditional medicine practices in Sierra Leone.

2 Overview of Traditional Medicine and Practice

In Sierra Leone, literature on the medicinal role forests plants in supporting the healthcare system is limited. Traditional healers are known to provide diverse medical care especially in rural or remote communities where they live using plant herbs, animal's parts, incantations, among other methods based on traditional belief and norms [17, 18]. Across Sierra Leone, there are two recognized categories of traditional healers; (1) Traditional healers (herbalist) and (2) Traditional birth attendants. The former is generalist in the treatment of all kinds of sicknesses using traditional herbs collected from forests [19] while the latter use traditional herbs to deliver pregnant women. The Sierra Leone National Traditional Healer Association (SLENATHA) was established in 1992 with a mandates to monitor traditional practitioner's activities in the country. In Africa and Sierra Leone in particular, distinguished scholar's publications have encompassed a wide scope of traditional medicines, herbalist, and sicknesses cure by traditional medicines [20]. Eisah et al. [21] conducted a study on the various uses of medicinal plants by indigenes of forest edge communities in Moyamba District Sierra Leone. Similarly, Johnny et al. [22] carried out a survey to ascertain the various uses of medicinal plants in southern Sierra Leone. Alternately, Jusu and Sanchez, [23] investigated the threats and opportunities associated with medicinal trade in Sierra Leone. The ethnobotanical importance and conservation status of sacred groves associated with the Kpaa Mende in Sierra Leone was investigated by Lebbie AR, Guries RP [24]. The medicinal plants of Pujehun District, Sierra Leone and their uses were recorded by Macfoy CA, Sama AM [25]. Ranasingle et al. [26] investigated the combined use of plants herbs in treating malaria within Bo district of Sierra Leone. Lebbie and Turay, [27] examined plant use as antivenom against snake bites in Southern Sierra Leone. Furthermore, Lebbie et al. [28] assessed the ethnomedicinal knowledge of plant among herbalists in Liberia. In South Africa, Okaiyeto et al. [29] investigated the hostile effects and cytotoxic ability of traditional herbs in Africa. In addition, Hughes et al. [30] studied the relevance of traditional medicine in curing chronic non-communicable diseases in South Africa. Ngarivhume et al. [31] explored the various traditional medicines used by herbalist to cure malaria illnesses in Zimbabwe.

Globally, studies on traditional medicine's well-being and health importance have gained momentum in recent times thereby triggering a global debate on the subject matter among the scientific community [32–34]. These publications have shed great light on the socioeconomic and spiritual importance of traditional medicines to healthcare system. A recent publication of FAO [35], assessed "*forest pathways for green recovery and building inclusive, resilient and sustainable economies.*" Munyaradzi [36] did a critical analysis on the ethical dilemmas surrounding herbal medicine and spirituality. Karrjalainen et al. [37] examines the link between forests and human well-being and identified the challenges affecting this relationship. Mirzaie et al. [38] assesses the possibility of using traditional medicines to treat Covid-19. Noronha et al. [39] reviewed traditional medicines' ability to treat malaria in India and discovered that traditional medicine plays a central role in supporting the healthcare system of India. Similarly, Suswardany et al. [32] critically reviewed the role of traditional medicine in treating malaria ailment in Asia-pacific region. Marqures et al. [34] tried understanding the traditional healing morals and beliefs in ensuring good health and well-being of society. Eshete and Molla [20] studied the cultural importance attached to medicinal plants used to cure sicknesses among the indigenous people of Suro Barguda District Ethiopia. Khatib et al. [40] inspected the various common uses of traditional medicines in Syria.

2.1 Traditional Medicine Plant Richness in Africa

Africa accounts for 25% of the global plant species with an estimate of 45,000 native plant species used as traditional medicine. Off this estimate, about 5000 plant species are used as traditional medicine to treat malarial, pain, cancer, bone fracture, and spiritual encounters [41]. In addition, plants have been reported to possess several pharmacological attributes including antimicrobial [42–51], insecticidal [52–54], analgesic, anti-depressant, anti-malaria, antianxiety, antidiabetes, antimicrobial, antioxidants, anti-helminthes, anti-ulcer, anti-inflammatory, hyperlipidemia, anti-hypercholesterolemic, antiatherogenic, antitrypanosomal, anti-modulatory, antispasmodic, antidiarrheal, and antifertility activities, among others [45]. Across developing countries like Sierra Leone, Liberia, Ghana, Nigeria, etc., traditional medicines are called by different names such as native medicine, herbal medicine, local medicine, ethnomedicine, folk medicine, bush medicine, primitive medicines, alternative medicine, among other names [5]. Traditional medicine is the oldest form of healthcare system that has stood the test of time in low- and middle-income countries.

In Sub Sahara Africa, scientific research has been carried out to authenticate plant extract use for medicinal purposes over the past decades. Literature suggests that, traditional medicine utilization knowledge is being transferred to generation mostly via tribal tradition link [2, 55]. In Sierra Leone, for instance, plant resources have been known to cure different sicknesses and diseases especially in hard to rich communities where Western medicinal services are limited. Drugs such as chemotherapy, artemisinin, and morphine are all derived from African plant species. These drugs and other traditional medicine derived from plants have directly and indirectly contributed to the socioeconomic development of African countries since time immemorial. However, different countries and communities differ in plant resources used as traditional medicine due to their culture and religious beliefs. According to WHO report, about 20,000 plant species have been proven to have medicinal attributes and is used by 80% of the remote global population to cure various sicknesses and diseases. Remote communities in Africa and Sierra Leone in particular relies greatly on traditional medicines for their health problems. Others rely on traditional medicines to solve spiritual encounters and appease their ancestors. Some plants can cure multiple sicknesses or diseases and are used for other purposes as well. For instance, some plants are used for clothing, fodder, food, fuel, medicines, shelter, and medicine [56].

2.2 Plant Part's Use to Prepare Traditional Medicine

In most parts of the world, traditional medicines are extracted from different parts of a plant by herbalist and spiritualists. Some medicinal properties are extracted from the roots while other are extracted from the leaves, stem, fruits, and bark. Similarly, the seeds of some plants serve as medicinal component in most trees. The various techniques used to prepare medicinal properties from plants are: boiling, soaking, heating, chewing, and rubbing [24, 57]. In addition, some traditional medicines are consumed as hot tea on a daily basis. Leaves, roots, and stem parts of medicinal plants are sometimes sun-dried, ground, and mixed with clay soil for external application to cure bone fracture, or reduce swelling of body parts [24]. The grinded plant supplements are mostly molded into lumps and applied on the affected body part. In Sierra Leone, traditional medicines from plant play a significant role in the livelihood activities of remote rural communities [22]. These plant parts used as medicine are processed via various means. Some plant parts such as leaves, fruits, barks, and twigs are boiled to extract the medicinal component from them. Other plant parts like the bark, leaves, and stem are ground and sometimes sun-dried to enhance it healing power. Plant parts that are mostly ground and dried are used externally while those boiled are used for drinking or heating. Some plants produce oil components, which are mostly rubbed to treat skin diseases or swelling. In some cases, two or more plants are mixed to form an infusion decoction that is consumed to cure various sicknesses [24].

Sierra Leone is endowed with abundant natural resources in the form of plants, wild animal's minerals, water, among others [58]. The use of herbal plant medicines in Sierra Leone dates back to the return of free slaves in the eighteenth century. Across Sierra Leone, diverse variety of traditional herbs are used as traditional medicines to cure countless diseases and rare illnesses not treated in conventional healthcare facilities [22, 59]. The absence of alternative Western medicines during those eras forced native Sierra Leonean to explore the medicinal properties of trees found around them. In Sierra Leone plants are the primary traditional medicines that is relatively available and cheap to acquire in any community [22]. The continuous utilization of traditional plant medicine knows no boundary and is practiced in many parts of the world. However, different regions and ethnic groups in different countries use different plant component to cure different illnesses troubling the local population. Although a fraction of the population uses animal component to cure illnesses, however, majority rely on plant supplement to curb various sicknesses especially in remote rural communities in Africa. Plants are not only used by human to cure sicknesses, but are also consumed by animals for health reasons especially in the northern part of Sierra Leone [22, 60].

Animal parts are also used for herbal or medicinal purposes in Sierra Leone [61]. The bones and skin of certain animals are used to cure many sicknesses in Sierra Leone. For example, the bones of Bamboo are believed to possess great power and are used to cure physical weakness in both male and female. While some drink the boiled water of these animals, others tie the bones around their waist or neck for muscular power and spiritual protection. In developing countries with poor infrastructure, these healers are located in nearly every rural village setting as well as in the busy urban areas. While underappreciated, they may play a key role in solving the continent's diverse health problems that governments have not been able to handle since time immemorial. Traditional medicine involving traditional healers and the use of herbs, animal parts, spiritual therapies, manual therapies, and exercises remains common in sub-Saharan Africa, where more than 80% of the people use herbs to treat their illnesses [10].

2.3 Traditional Medicine Dosage and Quantification Controversies in Sierra Leone

Although herbal medicines play a key role in solving the health crisis in Sierra Leone over the past centuries, critics argue that the lack of accuracy in prescription in terms of how much to consume; when to consume, and interval to consume render these medicines inferior to Western medicines [62]. Also, the absence of adequate warning of their side effects or reaction by traditional healers further cast doubt on the safety of these drugs as compared to Western medicines. Some traditional healers across

Africa and Sierra Leone in particular, only instruct patients to rub or drink local herbs given to them but the quantity to drink or amount to rub externally is never specified. The absence of traditional prescription directives has led to the loss of thousands of lives in rural communities especially rural people across Sierra Leone. Local patients who survive over-dosage sometimes go insane. In addition, most traditional medicines from local healers come with strict instruction to maintain the healing power of the herb. Most traditional medicine prepared by herbalist have rules and regulation to follow if the therapy is to act accordingly. For instance, some traditional healers will instruct patients not to lay the herbs on the ground for any reason. Failure to comply by these rules most times render the medicines useless. Also, some traditional medicines are forbidden to be touched by women and exchanged between people. For instance, some traditionally made infusions can't be handed over directly from one client to another, instead, the medicine is placed on the ground for the other client to pick from the ground. Other instructions include applying only in the morning, or at midnight or preventing the opposite sex from touching it during application.

2.4 Traditional Medicine Preparation and Processes in Sierra Leone

The preparation of herbs for medicinal purposes differ with herbalist, regions, and traditional beliefs and religious practices across regions and countries. Certainly, there is no one-size-fits-all approach in preparing or processing traditional medicines. Some medicines are prepared by macerating plant leaves in either cold or hot water to prepare the infusion for drinking [24]. Some plant parts like the leaves and barks are pounded, sieved, and water added to generate the required infusion needed for drinking. In other cases, plant parts are burned to ashes in order to prepare the required remedies. However, in certain tribal or ethnic regions in Sierra Leone, herbal remedies are either boiled, heated, or sun-dried. For instance, medicinal herbs that require pounding in the northern part of Sierra Leone before infusion may not need pounding in the southern part of the country. Similarly, some plants are used to cure different illnesses along different ethnic communities in Sierra Leone. In some cases, two or more plants are collected and mixed together to form a drinkable infusion. The process could either be through pounding, cuttings, boiling, sun-drying, burning, or heating of various herbs. These processes however have strict rules to follow while preparing theses herbs for infusion. The negligence of these rules sometimes render the prepared herbs useless or powerless.

2.5 Traditional Medicine Contribution to the Spiritual and Social Well-Being of Rural Communities in Sierra Leone

In rural communities today, traditional healers are the borne to managing disease and other sicknesses that affect their livelihood and wellbeing. Most of these herbalists use spiritual enchantment to remedy various ailment and the same time drive away evil spirits that are believed to be the cause of these ailments [20, 33, 36, 63]. In Sierra Leone, traditional medicines still remain the most easily affordable and accessible treatment source of various illnesses in the healthcare system of the country. This could be attributed to the inability to access modern healthcare system and where available the cost is beyond the earning capability of most rural people. Traditional medicines serve as a nexus of interaction among herbalist, indigenous people, and the formal healthcare system in Sierra Leone. Traditional herbs collected from nearby forests in Sierra Leone is mostly utilized for the treatment of diverse illnesses suffered by the indigenous people of these communities across Sierra Leone. Besides treating ailments, traditional medicines have a more holistic tactic in enhancing rural livelihood and fulfilling the beliefs and norms of indigenous people in Sierra Leone. For instance, herbalist with their traditional medicines play a key role in societal activities considered as sacred ancestral tradition that also serve as social platforms in local communities in Sierra Leone. In Sierra Leone and beyond, herbalists with their traditional medicines are embedded within the healthcare system and are respected for their spiritual healing of sicknesses that modern healthcare can't detect or diagnose. These traditional medicines provide perfect societal imbalance through physical and spiritual healing of all class of people in society [63–66]. Traditional medicines have provided massive economic, socio-cultural and spiritual benefits for practitioners engage in the act since the independence of most countries in Africa.

However, critics have associated herbalist and traditional medicine with witchcraft, evil spirits, and magic, which are morally considered harmful to the well-being of society [67, 68]. Therefore, must people in urban and rural communities in Africa and Sierra Leone in particular view traditional medicine with negative lenses. Also, the use of traditional herbs to harm or kill people from afar or to manipulate people's destiny [69] are reasons why traditional medicines importance is undervalued and less reported in Africa and Sierra Leone in particular.

3 Orthodox Medicine Sustainability

3.1 Role of Forests in Sustaining Traditional Medicine

Forests are of national and global importance especially in developing countries like Sierra Leone [35]. Literature has proven that there is a direct relationship between forests, human health, and well-being [70]. Forests have been enhancing adequate ecosystem functions and services for mankind since time immemorial [71]. Plants or trees provide ecosystem services especially for poor and sustained communities around the world. In addition, forests serve as a direct or indirect reservoir for plant and their products in Sierra Leone. Forests in Sierra Leone and around the world are known to provide adequate ecosystem services that support human health and well-being [70]. In addition, forests help in regulating contagious disease by providing critical drugs needed to heal these sicknesses. Forests are further considered as "plant safe." Besides the role the forests play in conserving plants or trees, the forests provide spiritual, recreational, aesthetic, and cultural services deemed relevant to society. Medicinal plants collected from the forests play a critical role in enhancing healthy living conditions for societies [72]. Forests serve a crucial function in building resilient, inclusive, and sustainable economies across the globe [35]. Literally, majority of medicinal drugs used to cure various sicknesses today are derived from the forest ecosystems [35]. In Sierra Leone, 90% of the traditional medicines are derived from both primary and secondary forests located close to dwelling communities. The presence of large track of forests (secondary forests) in Sierra Leone is an indication that traditional medicines practices is not under existential threat but will flourish overtime if properly managed.

3.2 Traditional Medicines' Usage Perception Among the Twenty-First Century Generation in Sierra Leone

Across Africa, the indigenous knowledge about traditional medicine is widely believed to be transferred from elders or ancestors with great knowledge and belief in the practice to their younger generation. Despite the reliance of the health sector on traditional medicines to cure communicable, noncommunicable, and spiritual illnesses across Sierra Leone, the usage and knowledge of traditional herbs among the young generation is low and inadequate. According to the World Health Organization [8], 80% of African's population uses traditional medicine to meet their basic healthcare needs but majority of these are from the adult population. Besides, the medicinal values of traditional herbs, civilization has added more values to traditional herbs for cosmetics, perfume, dye, and flavors [72, 73]. The improvement in living conditions especially in urban settings of Sierra Leone has deterred the transfer of traditional medicine knowledge from elders of the current generation to the future generations.

In Sierra Leone, traditional plant medicines' knowledge is lodged with elders in society. For instance, a herbalist lamented the following: *"I inherited the skills and knowledge to be a traditional healer from my grandfather,"* [74]. Moreover, civilization and advancement in healthcare system technology is making traditional medicines use unpopular among the current generation of young Sierra Leonean. The unawareness of the importance of traditional medicine is associated with the inadequate availability of innovative scientific evidence on the beneficial importance of traditional medicines to society over the years [73]. Ahmad et al. [75] and Cámara-Leret et al. [76] cautioned that local knowledge on traditional medicines and its practices is negatively impacted by civilization and modernization. The authors warned that if nothing is done in time, great and valuable knowledge about traditional medicine might be lost and not be transferred to the future generation [77]. Across Sierra Leone, most youths are reluctant to take traditional medicine and have since refused to practice the profession. Their excuse for refusing to take traditional medicines is aligned with hearsay side effects and the physical appearance of prepared infusion. The knowledge about traditional medicine is important for the continuity of the practice as drugs can be extracted from fresh plant components, leaves, roots, barks, flowers, buds, stems, and twigs. However, the refusal of the young generation to practice the profession paint a bleak future for the sustainability of the practice.

3.3 Traditional Medicine Contribution to the Socio-Economic and Health System in Sierra Leone

There are approximately more than 45,000 traditional healers across Sierra Leone [78]. Traditional medicine trade in Sierra Leone is estimated at USD 64,000 annually [79]. Interestingly, 30–40% of traditional medicine trade value or income in Sierra Leone is undocumented due to lack of appropriate data base. *Traditional healers have long held a prominent position in Sierra Leone, particularly in rural areas where the majority of the population live and where access to formal health care services is limited* [74]. These traditional healers support the health care system that is deficient in trained and qualified doctors in Sierra Leone and at the same time contribute to the socioeconomic development of rural communities. Medicinal herbs collected by traditional healers contribute to the social well-being of local indigenes by treating sicknesses like diabetes, dysentery, gonorrhea, bone fracture, impotency, and river blindness. People designated as herbalist or traditional healers generate their income from the profession and are considered as key stakeholders in most communities across Sierra Leone [78]. Traditional medicines have over the years not only contributed to solving general health problems but has also been used to prevent, diagnose, and treat both mental and physical ailment [80]. For many traditional medicine practitioners, healing people is their main source of financial income and livelihood. Most of these traditional healers have large families that depend on them for sustenance and survival. More so, about 45,000 or more traditional healers are self-employed thereby reducing the number of unemployment in Sierra Leone. In Sierra Leone, traditional healers are nicknamed "Bush Doctors" or "JuJu Man" and are known to stay in rural settings. Most of the mental health cases in Sierra Leone are treated by traditional healers based on the belief that the ailment is linked with spiritual interference caused by demons and evil spirit [81]. The quest of traditional healers to ensure well-being and healing in society is sometimes rewarding to them and their families through signs of financial appreciation given by healed patients [81]. The fewer number of Western trained medical practitioners created a gap that traditional healers are filling through the treatment of mental disorder, bone fracture, diagnosis, and prevention of problems that Western trained practitioners can't detect [81, 82]. In rural settings across Sierra Leone, traditional healers play an important role in promoting and sustaining affordable health for all irrespective of socioeconomic status. For majority of Sierra Leonean living in rural communities, traditional herbs are not considered inferior to Western medicines. Instead, they are considered as their first aid healthcare treatment capable of treating and detecting illnesses that Western medicine cannot treat or cure with minimal expense. The support given by traditional healers to the health sector in Sierra Leone cannot be overemphasized.

4 Traditional Medicine Threats Face in Sierra Leone

4.1 Threats Facing Traditional Medicine Conservation in Sierra Leone

The increased use of traditional medicine in Africa and Sierra Leone in particular over the past century has attracted huge global focus by the public, policy makers, health practitioners, and researchers, among other key stakeholders in Africa [62, 83, 84]. Irrespective of the global recognition and healthcare importance of traditional medicine, the healing therapy is not without danger or harm. Traditional medicine has been reported to have adverse effects especially when taken incorrectly. Besides the incorrect administering threats facing traditional medicine's usage in Sierra Leone, decline in forests areas has emerged as a major factor threatening the availability of herbs in the near future. Some of the challenges are deforestation, negative perception, bias against Western drugs, practice secrecy, absence of regulatory body for traditional medicines consumption, wildfire, climate change, shifting cultivation, fuel wood collection and charcoal burning, illegal logging and mining, and urbanization [58, 85–90]. The various threats facing traditional medicine sustainability in Sierra Leone are presented in Table 1.

Also, knowledge of traditional herbs is mainly primitive and experience-based therapy, whereas Western medicines are evidence-based. This poses a threat to traditional medicine's knowledge sharing and expertise for the young and inexperienced generation. To be able to diagnose, treat, and prevent spiritual attacks using traditional medicine, past experience knowledge is a prerequisite for the practitioner. In addition, initiation into the secret society is also a prerequisite to practice traditional healing using herbs from the forests. These prerequisites are of little interest to the current generation hence threatening the continuity of the traditional medicine practice. In Sierra Leone, traditional medicine's practice is mostly done by elders and experienced individuals who live among the people they treat. These illnesses curing ability is mostly received from ancestors either physically or in dreams and passed on from father to son or mother to daughter or any other interested member in society before the elder (herbalist) passed on to another life beyond. Nonetheless, this transition process from elderly ancestors to younger siblings is at stake due to technological advancement in medicinal science, rural to urban migration, increase in literacy rate, and public perception about traditional medicines among the elite of society. These challenges pave the way for local scammers who wish to make fast money from fake traditional medicine practices at the expense of vulnerable society members. Unfortunately, there is no active organization monitoring traditional healer's competencies and therapies administered to the public in Sierra Leone. This latest trend of development in traditional

Table 1 Traditional medicine challenges, sources and remedy

Traditional medicine challenge	Literature sources	Proposed remedy
Deforestation	[58, 87, 91]	Ban deforestation, provide alternative livelihood and enforce environment Act
Negative perception	[58, 62, 85, 92]	Increase awareness about the importance of traditional medicine
Shifting cultivation	[58, 86, 93]	Introduce monoculture practices and encourage swamp farming
Climate change	[58, 94, 95]	Adhere to climate change mitigation and adaptation strategies
Fuelwood collection and charcoal burning	[23, 96, 97]	Establish woodlot plantations and provide alternative energy sources
Improper diagnosis	[98]	Improve on diagnosis precision
Total acceptability	[98]	More awareness should be raised
Conservation	[99]	Training on sustainable traditional medicines harvest
Wildfire	[87, 100]	Institute bylaws on wildfire
Lack of functioning policy and legislation on traditional medicine	[23, 85, 89]	Design and develop policies and legislations on traditional medicines use and regulation
Fake traditional healers	[5, 85]	Provide legal framework and license to be issued to herbalists
Religious beliefs	[5]	Increase awareness raising
Plant extinction threats Technological advancement	[87] [101, 102]	Employ good forest management TM should be transformed from experience to evidence based

medicine practice has the tendency to discourage experience and genuine practitioners in societies (Table 2).

5 The Future of Traditional Medicine in Sierra Leone

Although traditional medicine knowledge transfer is under great threat, its uses and dependence to supplement affordable and cheap healthcare system makes it still the most trusted healing therapy in Sierra Leone. The use of traditional medicine is common among the rural population in Sierra Leone and the trend is expected to continue into to the next generation. Nonetheless, the acceptability of traditional medicine and its inclusion into the mainstream healthcare system in Sierra Leone is still a daunting task especially by the elite in urban areas. Over the next decades, the number of traditional medicine practitioners in Sierra Leone is expected to decrease but its practice will continue in remote and hard to reach communities where modern healthcare system is unreachable. In a resource poor country like Sierra Leone, access to Western or modern healthcare system is marred by challenges such as distance, cost, limited medical practitioners, unequal distribution of health clinic,

Table 2 Tree species and their medicinal significance in Sierra Leone

	Name of tree species	Illness treated	Tree parts use	Preparation	Sources
1	*Anisophyllea laurina*	Eye problem, dysentery	Bark	Grinding/ drying	[21, 103]
2	*Cajanus cajan*	Measles	Leaves	Boiling	[22, 27, 25]
3	*Cassia sieberiana*	Stomach ache, malaria	Leaves and fruits	Edible/ boiling	[22, 25, 28]
4	*Ficus capensis*	Closure of the frontal suture in babies	Leaves	Grinding/ heating	[25]
5	*Moringa oleifera*	Malaria	Leaves, roots, fruit, and stem	Boiling, grinding, drying	[22, 61]
6	*Morinda chrysorhiza*	Malaria	Leaves	Boiling	[61]
7	*Phyllanthua Discoideus*	Bone fraction	Leaves	Grinding	Field study
8	*Terminalia ivorensis*	Malaria, joint and back pain. Yellow fever	Stem, bark	Grinding and boiling	Field data 2022
9	*Parinari excelsa*	Toothache, dysentery, stomach pain	Root, stem, bark	Scrapping bark, grinding	Field Study 2022
10	*Spondias mombin*	Menstrual period pain	Leaves	Boiling/ grinding	Field study 2022
11	*Termarindus indica*	Diarrhea, abdominal pain, and dysentery	Bark, seeds, leaves	Heating, boiling, and grinding	Field study 2022
12	*Mangifera indica*	Head and stomach ache, malaria	Leaves, back, and root	Boiling	Field study 2022
13	*Anacardium occidentale*	Stomach ache, skin rash, and diabetes	Bark, leaves Nut, and fruit	Boiling and drying	
14	*Carapa procera*	Fever, dysentery	Seed	Dry	[79, 104]
15	*Hevea brasiliensis*	Heal wounds and sores	Bark and latex	Grinding	[104]
16	*Upapaca Guinanisis*	Treat male impotence and head ache	Bark, leaves	Boiling and heating	[104]
17	*Kola Nitida (kola nut)*	For easy child delivery, stomach pain, prostate cancer	Leaves, bark, nut,	Boiling and nursing chewing	[22, 24]
18	*Garcinia Kola*	Dysentery, diarrhea, back pain impotency, malaria, headache	Leaves and roots	Drying and heating	[79, 104]

(continued)

Table 2 (continued)

	Name of tree species	Illness treated	Tree parts use	Preparation	Sources
19	*Psidium guajava*	Dysentery, diarrhea, cough, stomach pain, ulcers	Leaves, fruits, and bark	Grinding, chewing, and boiling	[21, 104]
20	*Paullinia pinnata*	Ulcer and pile	Leaves, bark, and seed	Grinding	[21]
21	*Diospyros heudelotti*	Children disease	Leaves	Heating	[25]
22	*Parkia biglobosa*	Ulcer and pain	Leaves and bark	Grinding	[89]
23	*Azadirachta indica*	Dysentery, diarrhea, ulcer, diabetes	Leaves, fruits, and barks	Grinding and boiling	[89]
24	*Ceiba pentandra*	Paralysis, diabetes, leprosy	Bark, roots, and thorns	Boiling and grinding	[22, 89]
25	*Daniellia thurifera*	Skin and parasitic infection, headache, and cough	Roots, leaves, and bark	Grinding, chewing, and heating	[89, 104]
26	*Funtumia africana*	Urinary incontinence, cancer fever	Seeds, leaves, bark, and roots	Heating, boiling, and drying	[24, 89]
27	*Acacia pennata*	Tooth ache	Leaves	Heating	[24]
28	*Anthonotha macrophylla*	Dysentery and snake bites	Leaves	Beating, grinding	[24]
29	*Irvingia gabonensis*	Kidney, asthma, and liver	Leaves, roots, and barks	Grinding and boiling	[103]
30	*Cassia sieberiana*	Pain and headache	Bark and roots	Drying	[79]
31	*Alstonia boonei*	Pain, cancer, and dysentery	Bark and latex drying		[79]

[a]This data was collected in the field by the researcher to complete this aspect of the chapter. Field data 2022

and the absence of required drugs in health centers. These challenge stimulate the dependability of indigenous people on traditional medicine, which is cheap, affordable, closer to patients, and easy to acquire. Indigenous people residing in rural communities across Sierra Leone are knowledgeable and will continue the use of traditional medicine to cure ailment at the expense of Western drugs. The resilience nature of traditional medicine in villages and town is due to the acceptability of the practice among citizens of these communities. If traditional medicine knowledge is to be preserved for the next generation, then stakeholders should lead the participation process to entice community involvement in the practice, especially the younger generation. The possibility that traditional medicine will be incorporated or accepted into the mainstream health system in Sierra Leone is unrealistic over the next decade.

Furthermore, the trend of natural disasters and anthropogenic activities is also posing the greatest threats against biodiversity conservation and preservation in the face of growing population in the near future. The increase in shifting cultivation activities, wildfire outbreak, intensive land use change activities, and other land degradation activities pose a great threat to the existence of some endangered medicinal plants in Sierra Leone. The International Union for Conservation of Nature (IUCN) have listed the following plants among other species as threatened in Sierra Leone [105]. The plant species include: *Terminalia ivorensis, Gilbertodendron splendidum, Pterocarpus erinaceus*, *and Gilbertodendron bilineatus.* Most of these species have significant medicinal values that will be lost if these species go extinct [91, 105].

5.1 Conclusion

Traditional medicine is the combination of knowledge and skill by traditional people based on their beliefs and norms to cure and maintain good health in the local communities in Sierra Leone and other parts of the world. Traditional medicine usage in Africa and Sierra Leone in particular has been in existence since time immemorial. Before colonization and independence in Africa, traditional medicine was considered to be the only remedy to cure illnesses physically and spiritually. While traditional medicine has many positive impacts in maintaining the sound healthcare system in remote communities, the practice is not without harmful effects due to incorrect dosage and poor unhygienic preparation of infusion. In spite of the reliance of the health sector on traditional medicines to cure communicable, non-communicable, and spiritual illnesses across Sierra Leone, the usage and knowledge of traditional herbs among the young generation is low and discouraging. The socioeconomic, spiritual, and cultural importance of traditional medicine resonates well with majority of Sierra Leonean who use traditional medicine for healing and income purposes. This could be attributed to the easy access, low cost, and the presence of the herbalist in these communities. In Sierra Leone, there is no one-size-fits-all approach in preparing or processing traditional herbs. While some plants are prepared by macerating plant leaves in either cold or hot water, others plant parts like the leaves and barks are pounded, sieved, and water added to prepare the infusion for consumption. In some instances, plant parts are burned and the ash is used to cure the illness. Traditional medicines are used to cure sicknesses such as malaria, fever, bone fracture, demons, stomach ache, dysentery, piles, among other sicknesses, in Sierra Leone. The review discovered that there are two recognized categories of traditional healers; they are traditional healers and traditional birth attendant.

It is concluded that Sierra Leone is endowed with abundant natural resources in the form of plants, wild animal's minerals, water, among others, which support traditional medicine practitioners. However, threats like deforestation, shifting cultivation, climate change, wildfire, negative perception, religious beliefs among others, are making the progress and sustainability of the practice difficult especially in the twenty-first century. It is therefore, recommended that more awareness raising be done on the importance of traditional medicine to the health status of rural communities in order to motivate the younger generation to join the practice for sustainability.

5.2 The Way Forward

In order to preserve plants with medicinal attributes, measures should be taken to conserve threatened plants. In addition, awareness raising on the importance of medicinal plants should be intensified. The use of traditional medicine to cure rare illnesses should be encouraged by motivating the younger generation through incentives and moral support. Traditional medicine usage is mostly associated with traditional norms and beliefs, therefore, its application and usage should be encouraged to preserve the culture and norms in society.

To prevent fake herbalist from endangering the life of poor rural people, traditional healer's associations and related organizations should review and enact relevant policies that will ensure the sustainability of the herbal medicines in Sierra Leone. Moreover, traditional healers must be given license to practice their professional after going through rigorous assessment by experience herbalist. Any practitioner practicing the profession should be an experienced, competent, resourceful, passionate, and trusted spiritualist or herbalist with a moral attitude. Also, extractions, infusions, decoctions, and other forms of herbal medicine preparation should be done with extra care to reduce side effects or spoilage. Prepared infusions should be tested to detect any harmful reaction that could be envisaged after using the healing substance. Therefore, sound awareness raising should be started to inform the general public about the negative side effects of traditional medicines. In addition, detailed ethnobotanical surveys should be done in Sierra Leone to ascertain the wider functional role of plants in society in relation to the societal beliefs, norms, and traditions of the people. Information gathered should be published to spread plant knowledge to the young generation to show the interaction among plant, environment, and people. Traditional medicine practitioners should be cautious when administering diagnosis to avoid giving misleading infusions to patient. Traditional medicine preparation should follow standard herbal procedures to maintain hygiene of the prepared infusion or decoction. Herbal products like leaves, roots, fruits, barks, among others, should be prepared in hygienic ways to minimize side effects, and dosage must be prescribed correctly.

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Herbal Medicine: A Case Study of Kherias, Lodhas, and Mundas Tribes in Southwest Bengal of India

40

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Contents

Abstract

Ethnobotany is a well-established branch of science today. It includes in its totality the idea and concept of people about plants as living organisms including beliefs and actions about survival, attitudes of people toward the destruction of vegetation, its careful conservation through faith (taboos, totems, clan, etc.), or practical understanding of plants and their continued improvement by selection or manipulation. Ethnobotany has great relevance in a region whose flora is associated with various

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uses. The present ethnobotanical study deals with tribes *like* Kheria (Sabar), Lodha, and Munda of southwest Bengal. These tribes are somewhat primitive to this region and have lived for a long period in the natural settings of the region. During this period, a total of 249 plant species have been collected from the study area of lateritic West Bengal. These plants belong to 84 families and 205 genera of the plant kingdom. According to the uses made by Kherias, Lodhas, and Mundas, the plants can be classified as foods and drinks, fodder, medicine (human), veterinary medicine, value-added items, magical beliefs, cultural practices, etc. Besides this, periments have been done to ascertain the phytochemicals present in the claimed five medicinal plants like *Curculigo orchioides* (Amaryllidaceae), *Asparagus racemosus* (Liliaceae), *Smilax ovalifoila* (Liliaceae), *Elephantopus scaber* (Asteraceae), and *Hemidesmus indicus* (Asclepiadaceae). To make it imperative to society, the traditional tribal system of medicine and wisdom must be documented not only because it has always helped to make a breakthrough in the treatment of virulent diseases but also to acknowledge the tribal heritage of using plant resources.

Keywords

Ethnobotany · Tribes · Lateritic West Bengal · Kheria · Lodha · Munda

1 Introduction

Ethnobotany is the scientific exploration of how diverse societies and cultures utilize plants. It involves investigating the intricate relationship between people and plants, delving into their historical utilization in areas such as sustenance, medicine, attire, habitation, tools, and religious or cultural ceremonies. Ethnobotanists are intrigued by the acquisition of plant-related knowledge and practices across different cultures and how these practices have evolved. Ethnobotany is primarily concerned with the documentation and preservation of the traditional knowledge held by local or indigenous communities regarding the management, utilization, and cultivation of plants. Ethnobotanists often delve into the use of plants for medicinal purposes, exploring the active compounds within plants and their applications in treating various illnesses. Many plants hold significant roles in the cultural and spiritual practices of various societies. Ethnobotanists study how plants are integrated into rituals, ceremonies, and belief systems. Ethnobotany investigates the use of plants as food sources, encompassing traditional farming methods and culinary practices. Ethnobotanists also contribute to the conservation of plant resources and the promotion of sustainable practices, recognizing that the knowledge of indigenous communities can be valuable for natural resource management.

By bridging the gap between traditional ecological wisdom and contemporary scientific research, ethnobotany plays a pivotal role in safeguarding cultural diversity, biodiversity, and the sustainable utilization of plant resources. Its significance extends beyond academic realms to practical applications in fields like pharmacology, agriculture, and conservation. Moreover, it offers profound insights into the intricate dynamics of human-environment interactions.

Ethnobotany is a multidisciplinary science, involving in its domain, Anthropology, Sociology, Archeology, Geography, Pharmacognosy, Medical, Economic Botany, and several other allied branches of Botany. On December 4, 1896, Harshberger first applied the term "ethnobotany" to the study of plants used by primitive and aboriginal people. Most of the authors [6–8, 16, 32, 34, 36, 37, 75–79] have interpreted ethnobotany as the total relationship between primitive people and their plant surroundings in the widest sense. But Feulks [17] and Richard [72], Martin [47], Osawaru and Ogwu [63], Ogwu et al. (2016, [56, 58]) have included the entire realm of economic botany, which even includes the modern uses of plants. Ethnobotany deals with not only the useful relationship between plants and men (the man of past and present) but also with the impact of man on his environment and vegetation. The subject ethnobotany includes in its totality the idea and concept of people about plants as living organisms including beliefs and actions about survival, attitudes of people toward the destruction of vegetation, its careful conservation through faith (taboos, totems, clan, etc.), or practical understanding of plants and their continued improvement by selection or manipulation [20, 52–54]. Ethnobotany has great relevance in a region whose flora is associated with various uses such as foods, fodder, medicines (human and veterinary), fuel house building materials, oilseeds, narcotics, beverages, dye, mordant masticatories, gums, resins, agricultural implements, field implements, material culture, magico-religious belief, religious rites, beliefs and rituals from time immemorial. In India, considerable literature exists on medicinal and economic plants (such as Watt 1989–96; [12–15, 35, 42, 45]). Ethnobotany is the discipline that explores how diverse cultures and societies engage with and harness the potential of plants. This field delves into the intricate interplay between humans and plants, encompassing the traditions, rituals, and day-to-day applications of various plant species among indigenous and local communities. Ethnobotanists delve into the multifaceted roles of plants in specific cultural settings, investigating their contributions to sustenance, medicine, shelter, ceremonial practices, and other facets of daily life. By scrutinizing the cultural, ecological, economic, and medicinal significance of plants, ethnobotanists help safeguard biodiversity through the preservation and documentation of traditional plant knowledge.

This chapter aims to provide data on plant wealth of primitive areas and regions where plant and plant parts are widely used for various purposes of human beings, livestock, and others and can add knowledge toward the attitude of the inhabitants to the vegetation of a region and its impact on the environment. The study area for this case study is the lateritic tract of West Bengal covering the parts of civil districts of Bardhaman, Birbhum, Bankura, Paschim Medinipur, Purulia, and Jhargram between latitudes 21° 75′ and 24° 33′ N.

2 Integrative Healthcare: Bridging the Gap between Herbal and Modern Medicine

In the contemporary world, there is a growing trend toward integrative healthcare, where individuals combine elements of both herbal and modern medicine to leverage the benefits of each approach. It is crucial to seek guidance from healthcare professionals to make well-informed decisions regarding the most appropriate treatment options based on individual health needs and circumstances [48–50]. The approach to disease treatment can vary significantly, depending on the nature and severity of the illness and individual factors. Various standard disease treatment methods are outlined in Table 1. The choice of treatment is contingent upon the disease type and stage and the patient's overall health. Consulting medical experts is essential to determine the most suitable treatment plan, with the potential need for a combination of therapies to address different facets of the disease.

Herbal medicine and modern medicine represent two distinct healthcare approaches, each possessing its advantages and limitations. A comparison of these two treatment methods in the contemporary world is summarized in Table 2. Herbal medicine, also known as phytotherapy or botanical medicine, is an approach to healing that harnesses the therapeutic properties of plants and plant-derived compounds for the prevention, treatment, or alleviation of various medical conditions [1, 2, 31, 60, 61].

Table 1 Standard methods of disease treatment

Treatment methods	Standard methods of disease treatment
Pharmaceutical medications	Prescribed or over-the-counter drugs manage diseases by addressing their symptoms or underlying causes.
Surgery	Surgical procedures remove tumors, repair organ damage, or correct structural abnormalities.
Physical therapy	Physical therapy helps treat musculoskeletal disorders that affect physical function and mobility.
Radiation therapy	Targeted radiation therapy is used to eliminate cancer cells in diseases like cancer.
Chemotherapy	Medications are utilized to combat cancer cells, either eradicating them or slowing their growth, often in combination with surgery or radiation.
Diet and lifestyle modifications	Healthy dietary and lifestyle changes, including exercise and stress management, can manage or reverse conditions like heart disease or type 2 diabetes.
Immunotherapy	The immune system is harnessed to fight diseases, especially cancer and other life-threatening infectious and non-infectious diseases, using methods such as monoclonal antibodies and vaccines.
Gene therapy	Experimental gene therapy aims to replace or repair faulty genes in the treatment of certain hereditary diseases.
Alternative and complementary	Some individuals integrate traditional medical treatments with alternative therapies like acupuncture, herbal remedies, or meditation.
Preventive measures	Disease prevention involves a healthy lifestyle, regular check-ups, and vaccinations to reduce the risk of contracting diseases.
Mental health treatment	Psychotherapy and medications are used to manage symptoms and improve well-being, especially in cases of conditions like depression and anxiety.
Rehabilitation	To regain physical strength and function after injuries or surgeries, rehabilitation is essential.
Hospice and palliative care	In advanced or terminal illnesses, palliative care focuses on symptom relief and enhancing the quality of life. Hospice care provides support during the final stages.

Table 2 Comparison of modern and herbal medicine

Aspect	Herbal Medicine	Modern Medicine
Source of treatment	Derived from natural plants, herbs, and botanicals.	Typically synthesized chemicals or biotechnological products.
Historical roots	Rooted in centuries-old traditions and indigenous knowledge.	Evolved through rigorous scientific research and clinical trials.
Regulatory oversight	Often less regulated, leading to variability in product quality.	Heavily regulated by government agencies, ensuring safety and efficacy.
Treatment approach	Holistic, addressing underlying causes and focusing on prevention.	Often symptom-specific, aiming to target and treat specific medical conditions.
Scientific evidence	With limited scientific validation, some herbs have established benefits.	Evidence-based, backed by extensive research, clinical trials, and approval by appropriate agencies.
Consistency in dosage	Dosage may vary based on plant source and preparation methods.	Precise and standardized dosages in pharmaceuticals.
Side effects	Generally considered safer with fewer side effects, but potential for interactions and adverse reactions.	Well-documented side effects and drug interactions, but risk of more severe adverse effects.
Speed of action	May take longer to show results, focusing on gradual and long-term effects.	Typically faster acting, especially in emergencies.
Chronic conditions	Often used for chronic conditions and overall wellness support.	The primary choice for acute and complex medical conditions.
Integration with modern medicine	Used in complementary and alternative medicine alongside modern treatments.	Mainstream medical approach for diagnosis and treatment in many cases.

This traditional healing system has deep roots in various cultures across the globe, persisting as a popular practice in the modern era. The various aspects of herbal medicine are summarized in Table 3. Herbal medicine continues to evolve and integrate with contemporary medical systems, playing a significant role in healthcare worldwide, whether as a complementary or primary mode of treatment.

Plants' therapeutic potential is attributed to their phytochemicals [28–30]. Phytochemistry, the study of plant-based compounds and their chemical composition, is a cornerstone of herbal medicine. Phytochemistry serves as a fundamental element in herbal medicine by elucidating the chemical composition of medicinal plants, enabling the recognition and isolation of active compounds, ensuring product quality and safety, and enhancing our comprehension of herbal remedies' mechanisms of action. It effectively bridges the gap between traditional plant-based healing and modern scientific methodologies, facilitating the development of evidence-based herbal medicine. The pivotal role phytochemistry plays in herbal medicine includes compound identification, isolation of active compounds [22, 23–27, 38–41], quality control, safety assessment, mechanism of action understanding, standardization, drug development, herbal combinations, safety and efficacy, and bioprospecting (Table 4).

Table 3 Aspects of herbal medicine

Aspect	Aspects of herbal medicine
Plant-based remedies	Herbal medicine can utilize various plant parts (leaves, roots, bark, flowers, seeds) in medicinal forms like teas, tinctures, capsules, and topicals.
Traditional knowledge	The knowledge of medicinal plant remedies is passed down through generations, relying on oral traditions and written records.
Holistic approach	Herbal medicine adopts a holistic approach to healthcare, addressing underlying causes of illnesses, not just symptoms.
Cultural and regional diversity	Herbal medicine practices vary across cultures and regions, creating diverse plant-based therapies over time.
Natural healing	Herbal medicines are favored for their natural origins, believed to have fewer side effects than synthetic drugs.
Scientific validation	Recent scientific efforts evaluate herbal treatments, identifying active compounds and establishing evidence-based practices.
Regulatory variation	Regulations governing herbal medicine differ among countries, ranging from stringent to permissive.
Common uses	Herbal medicine is employed for various health concerns, from common ailments to chronic illnesses and wellness support.
Potential risks	Concerns include contamination, mislabeling, adverse interactions, and side effects.
Consultation with professionals	Healthcare or herbalist consultation is advised, particularly for individuals with medical conditions or on other medications.

3 Method

3.1 Source of Information

Various investigators (namely, [3, 9–11, 18, 32–34, 68, 81–83, 84]) have suggested various tools for ethnobotanical research.

Method recognized as adequate for this purpose may be divided into three broad categories:

(a) Ethnobotany of the present: through extensive field studies, carried out among the tribal people. The studies are most effective and informative because the information is recorded on the spot.
(b) Ethnobotany of the past: the glimpses that can be obtained in several ways such as the study of Herbarium and Museum materials, Old literature and archeological remains, etc.
(c) Calculation of use value of the species studied: The use value [70] of each species is calculated by the total number of uses as recorded by interrogating the medicine men and the total number of medicine men interrogated according to the formula given below.

Table 4 Roles and aspects of phytochemistry in herbal medicine

Information	Aspects of phytochemistry in herbal medicine
Compound identification	Phytochemistry identifies and characterizes chemical compounds in medicinal plants, such as alkaloids, flavonoids, saponins, and terpenoids. An understanding of plant chemistry could reveal its potential therapeutic properties.
Isolation of active compounds	Phytochemistry aids in isolating active constituents responsible for a plant's healing properties. These compounds are used in standardized herbal remedies for consistent efficacy.
Quality control	Phytochemical analysis ensures the quality and uniformity of herbal products, helping manufacturers meet regulatory and safety standards by confirming specific constituent presence and concentration.
Safety assessment	Phytochemistry evaluates the safety of herbal remedies by identifying potentially toxic components guiding the establishment of safe dosage guidelines.
Mechanism of action understanding	Knowledge of plant chemical composition deepens the understanding of how herbal medicines work on a molecular level, aiding in identifying specific biological pathways and action mechanisms.
Standardization	Phytochemistry contributes to standardizing herbal products by quantifying active compound concentrations, ensuring consistent therapeutic benefits across various remedies.
Drug development	Research in phytochemistry can lead to developing pharmaceutical drugs derived from plant compounds, with many modern pharmaceuticals having plant origins.
Herbal combinations	Understanding the chemical composition of different plants facilitates the creation of herbal combinations that enhance therapeutic effects or minimize adverse effects.
Safety and efficacy	Phytochemistry supports an evidence-based understanding of the safety and efficacy of herbal medicines, vital for recognition within the medical community and by regulatory agencies.
Bioprospecting	Phytochemical research can lead to the discovery of new chemicals with potential therapeutic applications, contributing to ongoing bioprospecting efforts for natural remedies.

Use value of a species **(UVis) = $\sum$Uis / nis**

Where UVis = Use value of a species
Uis = Total number of uses of the species
nis = Total number of medicine men interrogated

3.2 Fieldwork

The present study is the outcome of careful and planned fieldwork in several selected tribals, particularly Kherias, Lodhas, and Mundas during the last 4–5 years. The areas taken up for study were those where the concentration of these ethnic groups was believed to be intense and the habitats were to some extent isolated from urban populations so far as communication and culture were concerned.

3.3 Informants

As in many other societies, the Mukhya is the hereditary head among the tribals. They are considered good informants in the tribal community. *Vidya*, the medicine men of said tribal communities, are also the informants. These persons are generally the most respected and rather indispensable members of the society. In some places, it is found that old women are the practitioners mainly of female diseases.

3.4 Method of Procuring Information

The procedure of gathering data, in general, was the same as mentioned by [32–34]. It included either interviews of the informants or witnesses of the uses during the period of work in the field.

3.5 Method of Recording Information

A pertinent question may be asked as to what reliance can be put on the authenticity of the statement of the informants. Efforts were made especially in this direction and the same plant materials were made the subject of discussion with various informants on different days and at different habitats.

3.6 Evaluation of Data

Data obtained from different sources as mentioned above have been categorically scrutinized and compared with earlier important publications. In this case medicinal plants – "Glossary of Indian Medicinal Plants" and its supplement [13, 14]; "Wealth of India: Raw Material" ([4]); and "Medicinal Plants" [35] have been taken as standard references.

4 Results and Discussion

4.1 Plant Distribution

During the field studies, a total of 249 plant species were collected from the study area of lateritic West Bengal (Southwest Bengal), which are used by the tribals such as Kherias, Lodhas, and Mundas. Out of 249 plant species, 25 are Monocots, 222 are Dicots, and 2 are Pteridophytes (Fig. 1).

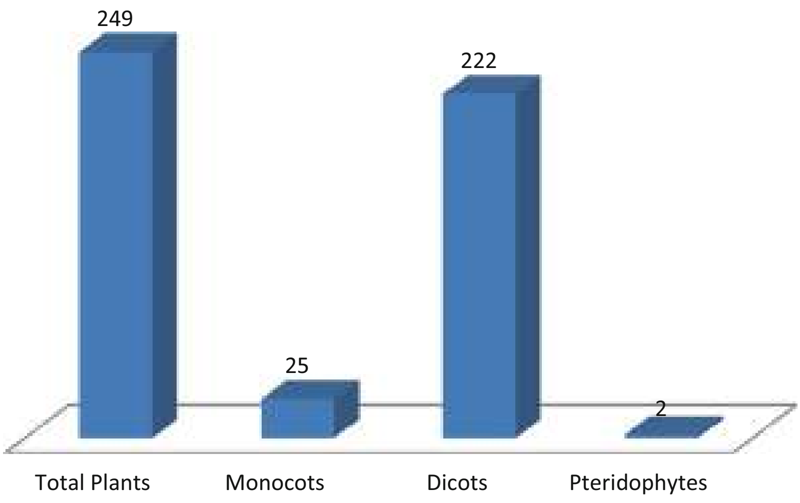

Fig. 1 Distribution of plant species collected during the study

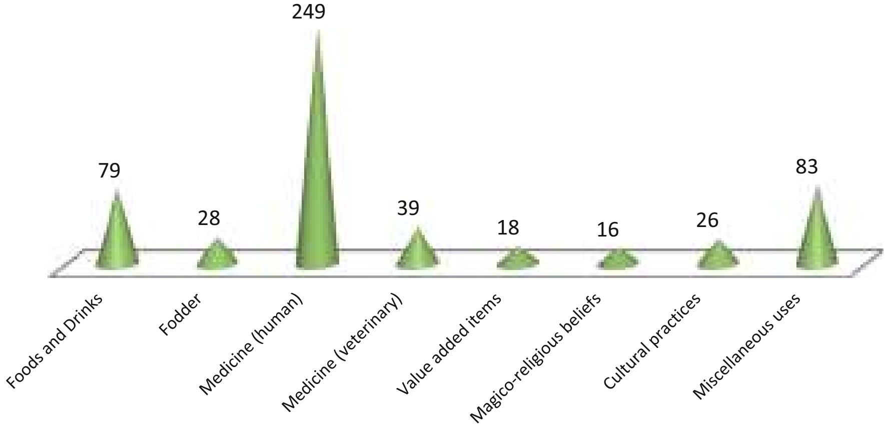

Fig. 2 Distribution of plant species, according to usages by different tribal communities

These plants belong to 84 families and 205 genera. According to the usages made by Kherias, Lodhas, and Mundas, the plant species have been classified as follows (Fig. 2):

1. Foods and Drinks: 79 species.
2. Fodder: 28 species.
3. Medicine (human): 249 species.
4. Medicine (veterinary): 39 species.
5. Value-added items: 18 species.
6. Magico-religious beliefs: 16 species.

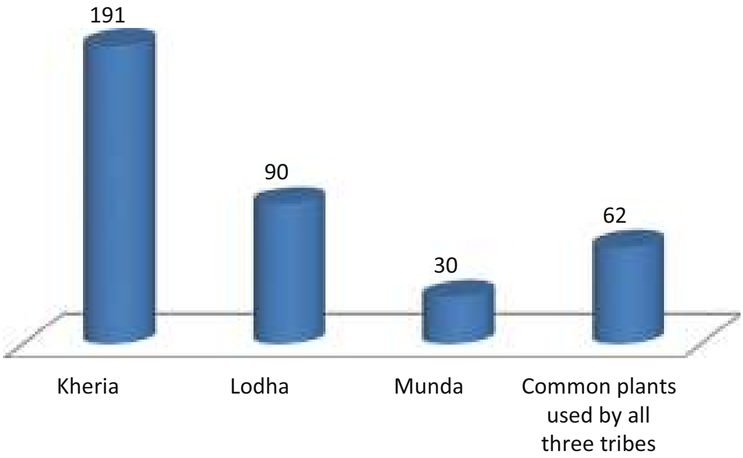

Fig. 3 Number of plant species are used by different tribal communities of the study area

7. Cultural practices: 26 species.
8. Miscellaneous uses: 83 species.

The number of plant species being used by different ethnic groups is recorded below (Fig. 3):

Kheria: 191.
Lodha: 90.
Munda: 30.

Common plants used by all three tribes: 62.

From the analysis, it has been observed that the Kheria is much more knowledgeable about the plant wealth of their surroundings followed by Lodha and Munda. It is interesting to mention that there are plants, e.g. *Abrus precatorius, Abutilon indicum, Haldinia cordifolia, Alstonia scholaris, Asparagus racemosus, Combretum roxburghii, Gymnema sylvestris, Holarrhena pubescence, Semicarpus anacardium,* etc. are commonly used by all the tribal groups. So it can be easily assumed that there is the cross-cultural relationship between the tribal groups of a particular geographical region.

4.2 Ethnobotanical Aspects

4.2.1 Food and Drinks

Though rice is the staple food for all the ethnic communities of this region, a large number of other grains like '*Marua*' (*Elusinecora coma*), *Penisetrum typhoides, Paspalum scrobiculatum, Paspalum orbiculare, Panicum maximum, Echinoclow* *acrusponis,* 'Wild rice' (*Oryza ruphipogon*), *Zea mays, Chinopodium albm,* etc. and wild plants *Dioscorea* plant species*, Asparagus racemosus, Madhuca indica, Diospyrus montana,* etc. are used as supplementary food. In general, the gathering of food plants, their processing, and preparation for consumption are the major routine work of women and children. Male members sometimes collect the fuel wood and help in harvesting the crop if necessary. A total number of 75 plant species are used as food and only 4 plant species are used to prepare drinks and beverages.

It is also observed that plants belonging to the families Poaceae, Fabaceae, Amaranthaceae, Asteraceae, Euphorbiaceae, Araceae, Lilliaceae, Basellaceae, Bracicaceae, Cucurbitaceae, Fabaceae, Musaceae, and Zingiberaceae are in major usage as staple or supplementary food by them.

4.2.2 Fodder

Out of 28 plant species being used as fodder, about 9 species, namely, *Haldiniab cordifolia, Bauhinia purpurea, Canavalia gladiata, Chenopodium album, Chloris virgata, Gardenia gummifera*, *Terminalia perviflora,* and *Rivea hypocretariformis* are less known or unknown as fodder plants.

4.2.3 Medicine (Human)

All the 249 plant species described in one way or the other are used as medicinal plants of which plant parts like roots – 94, stems – 13, barks – 27, leaves – 81, fruits – 17, latex – 18, seeds – 21, bulbs – 8, and tubers (Table 5) are mostly important parts to be used. Whole plants are used in 29 cases only. Besides these, plant parts like flowers, apical buds, spines, etc. are also in use.

The prevailing diseases of the tribals in this region are diarrhea, dysentery, dyspepsia, hepatic diseases, malarial fever, pneumonia, skin diseases like eczema, scabies, ulcer, venereal diseases, conjunctivitis, colic pain, and others [43, 59, 62]. Some of these diseases such as hepatic and skin diseases, etc. are possibly due to consumption of country liquor namely, rice-beer, *mahua* liquor which are consumed without proper hygienic care. Besides these, some diseases are reported as specific to men or women or children in the tribal communities. Men occasionally suffer from headache, eczema, bronchitis, venereal diseases etc., but constipation and appendicitis are uncommon among them. It is because of consumption of large amount of leafy vegetables, which contain sufficient cellulose fibers, and makes the bowl clear. On the other hand, it may be the cause of colic pain, dysentery, indigestion, etc. Debility of sex is recorded among some persons of the communities though it is not very frequent among Kherias and Lodhas. Common female diseases are anemia, white discharge with mucus, dysmenrrhoea (painful menstruation), hysteria, leucorrhea, etc. The practice of abortion is not uncommon. But mostly tetanus is associated with it and is the cause of mortality of carrying mothers. Ignorance and unhygienic condition practiced during delivery of babies is the major cause of this disease.

Similarly some new and important reports on use of a combination of several plants or their parts are also given here. The stem barks of five plants, namely, *Mangifera indica*, *Shorea robusta*, *Spondias cytherea*, *Madhuca indica*, and

Table 5 List of plant parts used by *Kherias, Lodhas,* and *Mundas* of this region

Sl. No	Scientific name	Family	WP	R	T	S	B	Le	Ap	La	Fr	J	Fr. P	Se	Ol
1	*Abrus precatorius*	Fabaceae		+		+									
2	*Abutilon indicum*	Malvaceae		+		+									
3	*Acacia arabica*	Mimosaceae						+	+						
4	*Acacia auriculiformis*	Mimosaceae		+		+		+			+				
5	*Acalypha indica*	Euphorbiaceae		+											
6	*Achras sapota*	Sapotaceae					+				+				
7	*Achyranthes aspera*	Amaranthaceae	+	+		+		+							
8	*Acorus calamus*	Araceae		+				+							
9	*Adina cordifolia*	Rubiaceae		+					+	+					
10	*Aegle marmelos*	Rutaceae		+				+			+				
11	*Aerva lanata*	Amaranthaceae		+				+							
12	*Ageratum conyzoides*	Asteraceae	+					+							
13	*Ailanthus excelsa*	Simarubiaceae		+				+							
14	*Alangium lamarckii*	Cornaceae		+			+	+							
15	*Albizia procera*	Mimosaceae					+								
16	*Allium cepa*	Liliaceae			+										
17	*Allium sativum*	Liliacea			+										
18	*Aloe vera*	Liliaceae						+		+					
19	*Alstonia scholaris*	Apocynaceae		+			+			+					
20	*Alternanthera ficoides*	Amaranthaceae	+	+											
21	*Alternanthera sessilis*	Amaranthaceae	+					+							
22	*Alysicarpus bupleuifolius*	Fabaceae	+	+											
23	*Alysicarpus monilifer*	Fabaceae		+											
24	*Alysicarpus vaginalis*	Fabaceae	+	+											
25	*Amaranthus spinosus*	Amaranthaceae	+	+											
26	*Amaranthus viridis*	Amaranthus		+											

28	*Ammannia baccifera*	Lythraceae		+											
27	*Ammannia indica*	Lythraceae		+											
29	*Amorphophallus paeoniifolius*	Araceae			+										
30	*Ananas sativa*	Bromeliaceae						+	+						
31	*Andrographis paniculata*	Acanthaceae	+			+		+							
32	*Anisomeles ovata*	Lamiaceae						+							
33	*Anona squamosa*	Anonaceae						+			+			+	
34	*Anthocephalus chinensis*	Rubiaceae		+			+	+	+						
35	*Aristida adscenscionis*	Poaceae		+											
36	*Aristolochia indica*	Aristolociaceae		+											
37	*Artocarpus lakoocha*	Urticaceae					+				+				
38	*Asparagus racemosus*	Liliaceae		+											
39	*Azadirachta indica*	Meliaceae					+	+						+	
43	*Barleria prionitis*	Acanthaceae						+	+						
44	*Barringtonia acutangula*	Lecythidaceae					+								
45	*Basella alba*	Basellaceae						+							
46	*Bassia latifolia*	Ebenaceae					+							+	
40	*Bauhinia purpurea*	Caesalpininaceae		+			+								
41	*Bauhinia vahlli*	Caesalpiniaceae						+	+						
42	*Bauhinia variegata*	Caesalpiniaceae		+					+						
47	*Biophytun reinwardti*	Oxalidaceae		+											
48	*Blumea flava*	Asteraceae						+							
49	*Blumea mollis*	Asteraceae													
50	*Boerhaavia diffusa*	Nyctaginaceae	+												
51	*Bombax ceiba*	Bombaceae		+			+								
52	*Borassus flabellifer*	Aracaceae									+	+			
53	*Brassica napus*	Brassicaceae				+		+							
54	*Butea monosperma*	Papilionaceae		+										+	

(continued)

Table 5 (continued)

Sl. No	Scientific name	Family	WP	R	T	S	B	Le	Ap	La	Fr	J	Fr. P	Se	Ol
55	*Butea superba*	Papilionaceae					+	+							
56	*Cajanus cajan*	Papilionaceae						+							
57	*Calotropis gigantia*	Asclepiadaceae		+						+					
58	*Calotropis procera*	Asclepiadaceae						+		+					
59	*Canavalia gladiata*	Fabaceae						+							
60	*Canscora decussata*	Gentianaceae	+					+							
61	*Capsicum annuum*	Solanaceae						+							
62	*Cardiospermum halicacabum*	Sapindaceae	+					+							
65	*Careya arborea*	Lecythidaceae		+			+			+	+				
63	*Carica papaya*	Caricaceae													
64	*Carissa spinarum*	Apocynaceae		+											
66	*Cascabula thevetia*	Apocynaceae								+	+				
67	*Cassia fistula*	Caesalpiniaceae									+	+	+		
69	*Cassia sophera*	Caesalpiniaceae						+						+	
68	*Cassia tora*	Caesalpiniaceae		+				+							
70	*Catharanthus roseus*	Apocynaceae		+				+							
71	*Celosia argentea*	Amaranthaceae		+				+							
72	*Centella asiatica*	Apiaceae	+					+							
73	*Chenopodium album*	Chenopodiaceae		+				+							
74	*Chloris varigata*	Poaceae		+		+									
75	*Cissus quadrangularis*	Vitaceae				+									
76	*Citrus medica*	Rutaceae						+					+		
77	*Cleistanthus collinus*	Euphorbiaceae													
78	*Cleome viscosum*	Capparidaceae		+											
79	*Clerodendron indicum*	Verbinaceae		+											
80	*Clerodendron viscosa*	Verbinaceae		+											

81	*Clitoria ternatea*	Papilionaceae		+										+	
82	*Coccinia cordifolia*	Cucurbitaceae		+				+							
83	*Cocos nucifera*	Aricaceae						+							+
84	*Combretum roxburghii*	Combretaceae				+		+							
85	*Commelina peludosa*	Commelinaceae	+					+							
86	*Costus speciosus*	Zinziberaceae		+	+										
87	*Croton bonplandianum*	Euphorbiaceae						+		+					
88	*Croton roxburghii*	Euphorbiaceae		+				+							
89	*Curculigo orchioides*	Hypoxidaceae			+										
90	*Curcuma aromatica*	Zingiberaceae			+										
91	*Curcuma longa*	Zingiberaceae			+										
92	*Cuscuta reflexa*	Convolvulaceae				+									
93	*Cynodon dactylon*	Poaceae	+												
94	*Cyperus rotundus*	Cyperaceae		+											
95	*Datura metal*	Solanaceae						+				+			
96	*Dendraphtoe falcata*	Loranthaceae					+								
97	*Deeringia merospermae*	Amaranthaceae													
98	*Derris scandens*	Fabaceae								+					
99	*Desmodium gangeticum*	Fabaceae		+											
100	*Desmodium motorium*	Fabaceae	+												
101	*Dioscorea pentaphylla*	Dioscoriaceae			+										
102	*Diospyrus Montana*	Ebenaceae					+				+				
103	*Drosera burmanii*	Droceraceae	+							+					
104	*Eclipta alba*	Verbinaceae	+												
105	*Emblica officinales*	Euphorbiaceae										+		+	
106	*Enhydra fluctuans*	Asteraceae	+												
107	*Erythrina varigata*	Lamiaceae						+							
108	*Eugenia caryophyllifolia*	Myrtaceae												+	

(continued)

Table 5 (continued)

Sl. No	Scientific name	Family	WP	R	T	S	B	Le	Ap	La	Fr	J	Fr. P	Se	Ol
109	*Eupatorium ayapana*	Asteraceae		+				+							
110	*Eupatorium odoratum*	Asteraceae						+							
111	*Euphorbia hirta*	Euphorbiaceae		+											
112	*Euphorbia nerifolia*	Euphorbiaceae		+				+							
113	*Euphorbia tirucalli*	Euphorbiaceae								+					
114	*Evolvulus alsinoides*	Convolvulaceae		+											
115	*Evolvulus nummularius*	Convolvulaceae		+											
116	*Feronia elephantum*	Rutaceae		+											
117	*Ferula asafoetida*	Apiaceae								+					
118	*Ficus benghalensis*	Moraceae		+						+					
119	*Ficus racemosa*	Moraceae									+				
120	*Ficus reliogiosa*	Moraceae								+					
121	*Ficus virens*	Moraceae					+								
122	*Flacourtia indica*	Flacourtiaceae	+												
123	*Flemingia cahappar*	Fabaceae		+				+							
124	*Foeniculum vulgare*	Apiaceae						+			+				
125	*Gardenia gummifera*	Rubiaceae							+						
126	*Gliricida maculata*	Fabaceae		+			+								
127	*Gloriosa superba*	Liliaceae		+	+										
128	*Gmelina arborea*	Verbenaceae		+											
129	*Gnaphalium indicum*	Asteraceae	+	+											
130	*Grangea maderaspatana*	Asteraceae		+											
131	*Gymnema sylveatris*	Asclepiadacee		+				+							
132	*Hamarthria compressus*	Poaceae													
133	*Helicteres isora*	Sterculaceae									+				
134	*Heliotropium indicum*	Boraginaceae	+												

135	*Hemidesmus indicus*	Asclepiadaceae		+				+							
136	*Hibiscus cannabinus*	Malvaceae													
137	*Hibiscus rosa-sinensis*	Malvacae						+							
138	*Holarrhena antidysenterica*	Apocynaceae		+			+								
139	*Holoptelia integrifolia*	Ulmaceae		+			+								
140	*Hygrophila auriculata*	Acanthaceae													
141	*Icnocarpus frutescens*	Apocynaceae						+		+					
142	*Ionidium suffruticossum*	Violaceae		+											
143	*Ipomoea reptans*	Convolvulaceae				+									
144	*Jatropha curcas*	Euphorbiaceae								+					
145	*Jatropha gossypifolia*	Euphorbiaceae						+		+					
146	*Justicia adhatoda*	Acanthaceae						+							
147	*Justicia jendarussa*	Acanthaceae		+				+							
148	*Kalanchoe pinnata*	Crassulaceae						+							
149	*Lagenaria vulgaris*	Cucurbitaceae						+							
150	*Lagerstroemia flos-reginae*	Lytraceae		+											
151	*Lantana camara*	Verbinaceae		+											
152	*Lawsonia alba*	Lythraceae						+							
153	*Lemna trisulca*	Lemnaceae	+												
154	*Leonotis nepetifolia*	Lamiaceae		+				+							
155	*Leonurus japonicum*	Lamiaceae		+											
156	*Leucas aspera*	Lamiaceae						+							
157	*Lindenbergia indica*	Scophulariaceae				+									
158	*Lippia nodiflora*	Verbenaceae						+							
159	*Lycopersicum esculentum*	Solanaceae									+				
160	*Lygodium circinatum*	Schizaceae		+				+							
161	*Mangifera indica*	Anacardiaceae					+								
162	*Marselia quadrifolia*	Marseliaceae	+												

(continued)

Table 5 (continued)

Sl. No	Scientific name	Family	WP	R	T	S	B	Le	Ap	La	Fr	J	Fr. P	Se	Ol
163	*Martynia annua*	Pedaliaceae		+										+	
164	*Melilotus phillipensis*	Fabaceae				+									
165	*Merimea tridentata*	Convolvulaceae		+				+							
166	*Mimosa pudica*	Mimosaceae		+											
167	*Mimusops elengi*	Ebnaceae												+	
168	*Mitragyna perviflora*	Rubiaceae					+								
169	*Momordica charantia*	Cucurbitaceae						+						+	
170	*Morinda tinctoria*	Rubiaceae		+				+							
171	*Moringa oleifera*	Moringaceae		+				+							
172	*Mukuna pruriens*	Papilionaceae							+						
173	*Musa balsamia*	Musaceae									+				
174	*Musa paradisiaca*	Musaceae						+			+				
175	*Nicotiana tabacum*	Solanaceae	+												
176	*Nigella sativa*	Rananculaceae									+				
177	*Nyctanthes arbor-tristis*	Oleaceae						+						+	
178	*Ocimum basilicum*	Lamiaceae						+							
179	*Ocimum sanctum*	Lamiaceae		+				+							
180	*Opuntia stricta*	Cactaceae				+									
181	*Oryza rufipogon*	Poaceae				+									
182	*Oxalis corniculata*	Oxalidaceae	+					+							
183	*Paederia foetida*	Rubiaceae	+					+							
184	*Pedilanthus tithymaloides*	Euphorbiaceae								+					
185	*Pergularia daemia*	Asclepiadaceae		+											
186	*Peristrophe bicalyculata*	Acanthaceae				+		+							
187	*Phoenix acualis*	Aracaceae		+								+			
188	*Phyla nodiflora*	Verbinaceae		+											

189	*Piper betle*	Piperacee						+							
190	*Piper longum*	Piperaceae		+											
191	*Piper nigrum*	Piperaceae									+				
192	*Plumbago zeylanica*	Plumbaginaceae		+		+									
193	*Plumeria acutifolia*	Apocynaceae						+							
194	*Polygonum hydropiper*	Poligonaceae	+	+											
195	*Pongamia pinnata*	Fabaceae					+							+	
196	*Premna herbacea*	Verbinaceae		+											
197	*Psidum guajava*	Myrtaceae						+	+						
198	*Pterocarpus marsupium*	Fabaceae									+				
199	*Punica granatum*	Punicaceae						+							
200	*Rauvolfia serpentina*	Apocynaceae		+				+		+					
201	*Rauvolfia tetraphylla*	Apocynaceae													
202	*Ricinus communis*	Euphorbiaceae												+	
203	*Rivea hypocreteriformis*	Convolvulaceae		+				+							
207	*Ruellia tuberosa*	Acanthaceae		+											
204	*Saccharum officinarum*	Poaceae				+						+			
205	*Saraca asoca*	Lamiaceae					+								
206	*Schleicherea oleosa*	Oliaceae					+							+	
207	*Scoparia dulsis*	Scophulariaceae	+												
208	*Semecarpus anacardium*	Anacardiaceae		+										+	
209	*Sesamum orientale*	Papilionaceae												+	
210	*Sesbania gladiflora*	Papilionaceae						+							
211	*Shorea robusta*	Dipterocarpaceae					+	+							
212	*Sida acuta*	Malvaceae						+							
213	*Sida cordifolia*	Malvaceae		+											
214	*Smilax macrophylla*	Liliaceae		+											
215	*Smilax ovalifolia*	Liliaceae		+											

(continued)

Table 5 (continued)

Sl. No	Scientific name	Family	WP	R	T	S	B	Le	Ap	La	Fr	J	Fr. P	Se	Ol
216	*Solanum melongena*	Solanaceae		+											
217	*Solanum torvum*	Solanaceae	+												
218	*Solanum tuberosum*	Solanaceae			+										
119	*Solanum xanthocarpon*	Solanaceae		+											
221	*Sonchus brachyotus*	Asteraceae	+					+							
222	*Sphaeranthus indicus*	Asteraceae						+						+	
223	*Spondias pinnata*	Anacardiaceae					+							+	
224	*Stephania harnandifolia*	Menispermaceae		+											
225	*Streblus asper*	Moraceae					+								
226	*Strychnos nux-vomica*	Loganiaceae					+							+	
227	*Syzygium cumini*	Myrtaceae					+							+	
228	*Tagetes patula*	Asteraceae						+							
229	*Tamarindua indica*	Tamarindaceae						+							
230	*Terminalia arjuna*	Combritaceae		+			+								
231	*Terminalia bellerica*	Combretaceae									+			+	
232	*Terminalia chebula*	Combretaceae									+				
233	*Thespesia lampas*	Malvaceae		+											
234	*Thunbergia erecta*														

235	*Tinosphora cordifolia*	Menispermaceae		+		+									
236	*Trapa natans*	Trapaceae													
237	*Trema orientalis*	Urticaceae					+								
238	*Trewia nudiflora*	Euphorbiaceae					+								
239	*Tribulus terestris*	Zygophylaceae		+										+	
240	*Trichosanthes dioica*	Cucurbitaceae						+							
241	*Urena lobata*	Malvaceae		+											
242	*Vanda tessellata*	Orchidaceae						+							
243	*Vitex negundo*	Verbinaceae						+							
244	*Withania somnifera*	Solanaceae		+											
245	*Xanthium indicum*	Asteraceae		+											
246	*Zingiber officinale*	Zinziberaceae			+										
247	*Ziziphus jujuba*	Rhamnaceae		+			+								
248	*Ziziphus oenopliea*	Rhamnaceae													
249	*Zornia gibbosa*	Fabaceae	+					+							

Where WP – Whole Plant, R – Root, T – Tuber, S – Stem, B – Bark, Le – Leaves, AP – Apical bud, La – Latex, Fr – Fruit, J – Juice, Fr. P - Fruit Pulp, Se – Seed, and Ol – Oil

Terminalia arjuna are found to be used in dysentery and diarrhea. Similarly a combination of 4 plants is used against any type of stomach pain, 3 plants as an antidote of snake venom, 4 plants against piles, 5 plants against sexual problems, 5 plants against hepatic troubles, 3 plants against hair falling, 2 plants as anti-pregnancy drug, 3 plants against malaria fever, 3 plants against anemia, 3 plants against tuberculosis, 3 plants against baldness, 3 plants against paralysis, 2 plants against rickets, 5 plants against high blood pressure (Table 6).

4.2.4 Medicine (Veterinary)

The present study reveals that 39 plant species are used as veterinary medicine. The major diseases covered are dysentery, diarrhea, intestinal worm, sores, broken horns, bone fracture, dislocation of bones, dyspepsia, fever, delivery problems, milking problem, foot and mouth disease, swelling of throat, etc. Kheria tribe uses the leaf decoction of *Blumea lacera* orally for dentition. The fruit powder of *Cassia fistula* is orally given to cows during pregnancy for good milk production after the delivery. *Amaranthus spinosus* is used widely for milching cows to increase milk flow. To cure sores *Abuitilon indicum* seed paste is used externally with coconut oil. To get rid of ticks of pets the leaf extract of *Anthocephalus chinensis* is mixed with camphor and smeared on the skin. The fume created by burning of green leaves of *Azadirachta indica* is used as mosquito repellant in the cowsheds.

It is observed that the usages of plants like *Abuitilon indicum, Anthocephalus chinensis, Blumea lacera, Caloptropis procera, Combretum roxburghii, Tamarindus indicus,* and *Vitex negundo* are practically unknown in the veterinary world.

4.2.5 Value-Added Items

The technology used by the tribals in their livelihood depends on the plant wealth. They use a good number of plants for building and shelter purposes, making different implements (hunting, fishing, agricultural, etc.), preparing poisons (used for prey of birds, animals, and fishes), making various material cultural items like utensils, oil extracting equipment, musical instruments (drums, *Kendra*-a string instrument), etc., for extraction of tannin, preparation of country liquor, various artifacts, etc. Tribal people use wooden combs made out of the young timber of specific plants like *Alstonia scholars* and *Gmelina arborea.* Washing powder especially as shampoo is prepared from the ash made after burning leaf of *Acacia auriculiformis.* A mucilaginous shampoo like preparation is made from the flower of *Hibiscus rosa-sinensis.* The colorful seeds of *Abrus precatorious* are used to make ornaments like necklace, earring, etc. The hard shell of the fruit of *Feronia limonia* is used to make different decorative materials. Broom made out of grasses like *Aristida adscenscionis, Vetiveria zizanoides, Thysolena maxima, Sida acuta, Impereta cylindrical* are widely used. They are also marked items in this region. *Sal (Shorea robusta*) leaves are used for making leaf plates and cups. It is also a marketable item. The leaves are stitched with the stem of *Aristida adscenscionis* and leaf petiole of *Azadirachta indica.* The spines of *Barleria prionitis, Acacia nilotica* and thorn of *Aegle mermelos* are used for designing tattoo marks on the body.

Table 6 Combination of plant species used against different ailments

Sl. No.	Name of diseases	No. of plant species used
1	Abdominal pain	4
2	Abnormal discharge of semen	19
3	Against dry cough	5
4	Anemia	5
5	Anti-allergic	3
6	Anti-bile’s	3
7	Anti-drug addiction	3
8	Anti-tumor	1
9	Baldness	8
10	Birth control	4
11	Bleeding due to injury	6
12	Blood purification	2
13	Body eruption	5
14	Bronchitis	4
15	Burn injury	2
16	Carbuncle	9
17	Cold cough	10
18	Conjunctivitis	2
19	Dandruff	7
20	Delivery pain	2
21	Dental pain	13
22	Diabetes	10
23	Diarrhea	2
24	Dysentery	35
25	Dyspepsia	20
26	Ear drop	2
27	Ear pus and pain	12
28	Ergot of uterus	1
29	Eye drop	2
30	Fever	5
31	Fractured bone	7
32	Gastric ulcer	4
33	Glycosurea	2
34	Gonorrhea	2
35	Headache	9
36	Hemorrhage	26
37	Hernia	1
38	High blood pressure	6
39	Hypnotism	1
40	Impotency	1
41	Increase Immunity power	7
42	Irregular menses	22

(continued)

Table 6 (continued)

Sl. No.	Name of diseases	No. of plant species used
43	Itches and scabies	10
44	Jaundice	10
45	Leucorrhea	5
46	Lice	1
47	Malaria	10
48	Mental disorder	3
49	Nasal bleeding	1
50	Paralysis	8
51	Physical weakness	3
52	Piles	7
53	Premature fall of hair	9
54	Repellant	1
55	Rheumatism	17
56	Rickets	2
57	Ring worm	8
58	Scorpion bite	10
59	Small pox	7
60	Snakebite	23
61	Stomach pain	15
62	Swelling of body	6
63	Tarka of child	1
64	Teeth pain	5
65	To increase mother milk	5
66	Tonsillitis	7
67	Tuberculosis	2
68	Unwanted pregnancy	4
69	Urine Clarence	1
70	Vomiting	1
71	Weak heart	2

4.2.6 Magico-Religious Belief

The distinction between religious faith on certain plants and belief in magical power of certain plants is not always sharply demarcated. So here these beliefs are classed under magico-religious beliefs.

1. Annulet: The plants associated with it are mostly for preventing and guarding evil spirits and incurable diseases.
2. Plants for magical cures are associated with curing ailments through charms and others. Such as Lodhas believe that the touch of green leaves of *Azadirachta indica* cures pox.
3. Plant like *Alangium salvifolium* used by Kheria for curing of all sorts of ailments. They believe that the plant is an abode of ghost.

4.2.7 Cultural Practice

Good numbers of plants are associated with the tribal cultures of the study area. They are artificially categorized as follows:

(i) Worship – plants associated with worship and offerings to God, Goddess, and evil spirits.
(ii) Community worship – plants associated with social customs and community worship.
(iii) Religious beliefs and practices – plants considered as sacred plants or associated with religious taboos, totems, myth, legends, etc. The plants are conserved and protected [21].
(iv) Tattooing and body ornamentation – plants used for making tattoo marks on body for body ornamentation for detecting the clan, etc.

It is interesting to mention that the plants used for tattooing purpose are mostly dye-containing plants and practically unknown or less known to the outsiders.

4.2.8 Miscellaneous Use

Plants associated with miscellaneous uses in the lives of the tribal people can be classified into the following:

(i) Masticatories.
(ii) Hallucination.
(iii) Tooth brush.
(iv) Insect repellent, etc.

4.3 Use Value

The use values of different plants dealt here with range from 1 to 6. Out of 249 species, 135 species have their use value between 1 and 2, 32 species have use values within 2.1-3.0, 64 plants have use values within 3.1-4.0, 7 species have use values within 4.1-5.0, and only 11 species have use values within 4.1-6.0 (Fig. 4).

4.4 Phytochemical Study of Some Selected Samples

Phytochemical screening for active constituents present in the alcoholic extracts of the following plants was carried out using chromatography and spectral techniques.

1. *Curculigo orchioides.*
2. *Asparagus racemosus.*
3. *Smilax macrophylla.*
4. *Elephantopus scaber.*
5. *Hemidesmus indicus.*

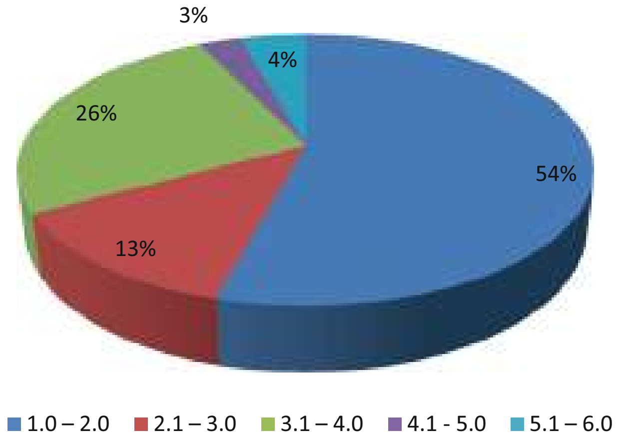

Fig. 4 Percentage of species with different use values

All these samples were tested for specific active constituents and then subjected to thin layer chromatography (TLC) along with UV spcctrophotometry, which are described below:

4.4.1 Phytochemical Screening for Active Constituents

All the samples were chemically tested to ascertain the presence of active constituents, namely, alkaloids, flavonoids, steroids, terpenoids, tannins, and saponins in the extracts. The various chemical tests were conducted and the results obtained in respect of alcoholic extracts of above five plants (1–5) are given in Table 7. Chemical tests indicated the clear presence of steroids, triterpenoids, and flavonoids in the alcoholic extracts of five plants.

The migration of different components of the alcoholic extracts of five different plant samples relative to the solvent front on thin layer chromatogram is expressed by the R_f values. The R_f values of the spots of the alcoholic extracts of five different samples are given in Table 8.

R_f = Distance moved by the spot from starting point / Distance of solvent front from starting point [R_f = Retardation force].

From Table 8 it is clear that sample 1 is the mixture of ten different components, sample 2 is the mixture of six components, sample 3 is the mixture of five components, sample 4 is the mixture of seven components, and sample 5 is the mixture of six components.

The plants studied for their phytochemicals are also used in Indian traditional medicinal system of medicine and have been studied earlier. *Curculigo orchioides* is used in Ayurvedic medicine. It has been observed that glycosides are present in the rhizomes. The tribes referred to in this study also use the rhizome part of the plant. Free sugars, mucilage, hemicellulose, and other polysaccharides are found to be present in this study. The plant yields a flavonone glycoside, which is a powerful

Table 7 Phytochemical screening for active constituent in the alcoholic extracts of five plants

Sample	Alkaloid	Steroid	Terpenoid	Flavanoid	Saponin	Tannin
Curculigo orchioides	–	+	+	+	–	–
Asparagus racemosus	–	+	+	+	+	–
Smilax ovalifoila	–	+	+	+	+	–
Elephantopus scaber	–	+	+	+	–	–
Hemidesmus indicus	–	+	+	+	+	–

(+) = Present, (–) = Absent

Table 8 R_f values of different spots of alcoholic extracts of five different plant samples

	R_f Values									
Sample	1	2	3	4	5	6	7	8	9	10
Curculigo Orchioides	0.06	0.22	0.30	0.35	0.37	0.65	0.72	0.76	0.80	0.85
Asparagus Racemosus	0.35	0.39	0.51	0.56	0.77	0.85				
Smilax ovalifoila	0.35	0.38	0.42	0.67	0.87					
Elephantopus scaber	0.38	0.41	0.69	0.74	0.79	0.83	0.89			
Hemidesmus indicus	0.29	0.36	0.71	0.78	0.84	0.90				

The digits 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 represent the R_f values

uterine stimulant in guinea pigs, rats, and rabbits notwithstanding the gravid or non-gravid state [80]. *C. orchioides* is used extensively in Ayurvedic formulation for a wide variety of ailments, specially as a general tonic and as an aphrodisiac. In the Philippines, it is used for skin diseases [46]. The rhizome is also used to make edible flour by many tribal people. Curculigoside, a phenolic glycoside, has been characterized in the rhizome [44]. In the present study, Kherias are found to use the rhizome against snakebite and in sexual debility of women. Lodhas use the plant against scorpion stinging. Mundas use the flower as hallucinating agent. Earlier work done (Bharatiya Vidya Bhavan [5]) in *Asparagus racemosus* shows that saponins are present. Gaitonde and Jethmalani [19] have observed antioxytocic and ADH activity in the saponin fractions isolated from the roots of this species. The plant has been used in Ayurveda as a galactogogue (*Stanya*) to increase milk secretion during lactation. It is also used as a general tonic and as an aphrodisiac. The plant's action as a galactogogue has been reported by Patel et al. [69]. In a study by Sabnis et al. [73], it was shown that in the estrogen-primed rats *A. racemosus* could cause both increase in the weight of mammary lobulo-alveolar tissue and the milk yield. It was attributed to the action of release of cortico-steroids or an increase in prolactine. All the three tribes, namely, Kheria, Lodha, and Munda, use the plant as a general tonic and a promoter of growth and good health. Kherias and Mundas use the plant as a sex stimulant for male. Other than the said uses, Kherias and Lodhas use the plant against Gonorrhea. It is also used as food supplement. Mundas use the root against dysentery and also as a food supplement.

Smilax ovalifoila is a widely used plant in Indian traditional systems of medicine. The genus produces a well-known drug known as *Sarsaparilla,* which is also found in the root of ***Hemidesmus indicus*** (Indian *Sarsaparilla*). Root of the species is used in venereal diseases, rheumatism, urinary complaints, and dysentery. Kherias and Lodhas use the root of this plant to avoid unwanted pregnancy, whereas Mundas use its leaf paste to control the abnormal discharge of semen of male.

There is no record of usage of ***Elephantopus scaber*** in traditional medicine of India like *Ayurvedic* system. Photochemicals like Germacranolide dialactones, amino acids, triterpenoides, sterols, and long-chain fatty acids have been reported in this plant in earlier studies. The use of the root of the plant as emollient (making skin soft) and is given in dysuria, diarrhea, dysentery, and swellings or pain in the stomach has been reported in other traditional medicinal system. The root is given to arrest vomiting, powdered with pepper it is used in toothache. Bruised leaves boiled in coconut oil are applied to ulcers and eczemas. Kherias use the root paste of the plant against elephantiasis. Lodhas use its leaf paste in dysentery. Mundas use the leaf paste to prevent warms in children and also in elephantiasis.

Hemidesmus indicus (Indian *Sarsaparilla*) is a very commonly used plant in indigenous medicinal system. Bharatiya Vidya Bhavan [5] studied its phytochemistry in detail. Essential oil and saponins have been isolated from root extract. The aqueous extract of its root induced a slight increase in urine output in rats; in dogs none of the fractions had any diuretic action [74]. Blood pressure was reported raised in dogs by the drug. The antibacterial activity of the root extract and the essential oil [71] has been reported. It is used as a febrifuge, diuretic, antidiarrheal, and to stimulate lactation. Clinical trials have shown a benefit in ringworm infection and for malnutrition. Kherias use the plant as a febrifuge and also as a stimulant of milk secretion in mothers. They also use it against old ulcers, eczema, and other skin diseases. Lodhas use the plant in snakebite and also as a febrifuge. Mundas use the plant in controlling dysentery.

4.5 Attitude Toward Diseases and Their Diagnostics and Treatment

It is said that nature has created some plant in the world for every ailment, all we have to do is to find it. There is a cure for every disease, men has to find it out.

Tribals of the district are followers of animism. It leads to the belief that disease and certain spirits or supernatural powers cause deaths [51]. Sometimes bigot faith of dead spirits who live with living bodies is in existence. These beliefs have had a significant influence on their attitude and psychology about the ailments. They have to take steps to counteract the inimical influences of the hidden powers, to pacify, propitiate, or satisfy some and to fight and punish others like witches. This is the origin of some of the holy or sacred incantations, invocations, recitations, offerings, sacrifice, amulets or talisman, which are regarded as efficacious in the tribal faith.

5 Conclusion

In conclusion, the study has documented plants and plant parts used by Kherias, Lodhas, and Mundas from an ethnobotanical standpoint. This study highlighted gaps between traditional ecological wisdom and contemporary scientific research. The significance of this report extends beyond academic realms to practical applications in fields like pharmacology, agriculture, and conservation by providing insights into the intricate dynamics of human-environment interactions [55, 57, 64–67]. The choice of plant and plant parts used in the treatment of ailments by people *in* Kherias, Lodhas, and Mundas was observed to be contingent upon the disease type and stage and the patient's overall health. However, consulting medical experts remains essential to determining the most suitable treatment plan, with the potential need for a combination of therapies to address different facets of the disease. Most importantly, the pivotal role phytochemistry plays in herbal medicine and societal development includes compound identification, isolation of active compounds, quality control, safety assessment, mechanism of action understanding, standardization, drug development, herbal combinations, safety and efficacy, and bioprospecting were highlighted. Ethnic groups within Kherias, Lodhas, and Mundas believe that their environment plays key roles in the type of plants found around them, and in turn, the plant and plant parts used in addressing their health issues. Both the Mukhya i.e., the hereditary tribal head, and Vidya, the medicine men within these communities, are very knowledgeable about plants and plant parts that should be used for specific ailments. They are also generally respected and rather indispensable members of the society. Most of the plants used by the communities are Dicots and mostly as foods and Drinks, Fodder, Medicine (human and animals), value-added items, magico-religious beliefs, cultural practices, and miscellaneous uses. Rice was the main staple food for all the ethnic communities in addition to other grains like '*Marua*' (*Elusinecora coma*), *Penisetrum typhoides, Paspalum scrobiculatum, Paspalum orbiculare, Panicum maximum, Echinoclow acrusponis,* 'Wild rice' (*Oryza ruphipogon*), *Zea mays, Chinopodium albm,* etc. and wild plants *Dioscorea* plant species*, Asparagus racemosus, Madhuca indica, Diospyrus montana,* etc. are used as supplementary food. In general, the gathering of food plants, their processing, and preparation for consumption are the major routine work of women and children. Male members sometimes collect the fuel wood and help in harvesting the crop if necessary. The plants were from some of the following higher plant families Poaceae, Fabaceae, Amaranthaceae, Asteraceae, Euphorbiaceae, Araceae, Lilliaceae, Basellaceae, Bracicaceae, Cucurbitaceae, Fabaceae, Musaceae, and Zingiberaceae. Finally, all the 249 plant species described in one way or the other are used as medicinal plants of which plants.

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Herbal Medicine: Exploring Its Scope Across Belief Systems of the Indian Medicine 41

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Abstract

The acceptance and use of any system of medicine is influenced by belief of their practitioners, founders, teachers, and most importantly the general public. According to the World Health Organization, 80% of herbal medicines are utilized worldwide for better heath. The strong belief systems that underpin herbal medicine are crucial to comprehending the concepts of health, disease, disease physiology, disease pathology, and treatment modalities. One of the oldest medical systems is Unani, which primarily uses plants as a form of treatment. The fundamental knowledge of Unani medicine was collected by famous scholars of ancient time like *Buqrat* (Hippocrates 460 BC, Greece), *Jalinoos* (Galen 129 CE, Anatolia), *Zakariya Razi* (Rhazes 854, Persia), *Ibn Sina* (Avicenna 980, Persia), and later on *Hakim Ajmal Khan* (1868, India) emerging to be a significant contribution to its spread in India. The foundation of the Unani system is based on the core theory of the seven basic physiological principles, temperament, self-regulating power called "*Tabiat*," and six essential factors. Unani scholars employ these theories as therapeutic and diagnostic tools in their clinical practices. Patients and general population's belief on herbal medicine particularly Unani medicine varies. Some common believes are the following: It is restrained to Muslims only, is harmless, can be taken without advice of a physician directly over the counter, is only effective in chronic diseases, and can be taken in addition along with conventional medicine. The physician should be a religious or spiritual preacher, according to the general public's perception of Unani practitioners. These viewpoints are similar to those of other branches of traditional medicine, which all employ holistic approaches to treat illnesses rooted in a close connection to the associated religion. As a result, different groups of people favor these branches of medicine for their medical needs.

Keywords

Belief system · Unani · Herbal medicine · Ayurveda · Homeopathy

Abbreviations

A & UTC	Ayurvedic & Unani Tibbia College
AKTC	Ajmal Khan Tibia College
AYUSH	Ayurveda, Yoga, Unani, Siddha, Homeopathy
CCIM	Central Council of Indian Medicine
CCRIM & H	Central Council for Research in Indian Medicine and Homoeopathy
CCRUM	Central Council for Research in Unani Medicine
CT	Cupping Therapy
IMPCL	Indian Medicines Pharmaceutical Corporation Limited
ISM & H	Indian Systems of Medicine & Homoeopathy
MMT	The Munzij w Mushil Therapy
NIUM	National Institute of Unani Medicine

NIUM	National Medicinal Plants Board
NRHM	National Rural Health Mission
NTC	Nizamia Tibbia College
PCIM	Pharmacopoeia Commission for Indian Medicine
PLIM	Pharmacopoeial Laboratory for Indian Medicine
RT	Regimenal Therapy
SEF	Six Essential Factors
TKDL	Traditional Knowledge Digital Library
TM	Traditional Medicine
UM	Unani Medicine
USM	Unani System of Medicine
WCT	Wet cupping Therapy

1 Introduction

While there is much to be learned from looking at the drawings, records, skeletal remains, and surgical equipment of early humans, it is thought-provoking to extrapolate their sense of death and sickness. History not recorded or written is difficult to understand. It is possible that as they acquired cognitive abilities, they discovered by trial and error which plants could be consumed as food, which were poisonous, and which offered some therapeutic benefits [1]. Folk medicine, also known as domestic medicine or traditional medicine, which is primarily centered on the utilization of plants or herbs, minerals, and animals as sources of health, had its start in this way and is still practiced today. However, early humans did not view illness and death as natural events [2]. There were common illnesses like colds, coughs, and fevers that were treated with the available herbal remedies and were considered to be a normal part of life, but there were serious and incapacitating illnesses that came from a supernatural source, the result of a curse that the victim's enemy had placed on them, a visit from a malicious expert, or the handiwork of a wrathful god who projected an arrow, stone, or worm into the victim's body. The next phase was to attract the wandering spirit back to its rightful abode within the body, where it could then be driven away either by a dart or a demon [3, 4].

In Britain, France, Peru, and other parts of Europe, trepanned skulls from the prehistoric past have been discovered. The fact that many of them exhibit signs of healing shows that those who practiced trepanning, also known as trephining, thought the disease would leave the victim's body by drilling a hole in their head. Although it is rapidly disappearing, certain Indigenous people still practice it in specific areas of Algeria, Melanesia, and possibly elsewhere [5].

Early human society's medical practices were heavily influenced by magic and religion. Blasphemies, dancing, scowling, and all the tricks of the magician were used along with medication or treatment. Magicians were the only doctors that used amulets, broken bones, and ornaments, which is still common today and has ancient roots [6].

Early healers used a holistic approach (mental, physical, and spiritual) to demonstrate their abilities, and because both healer and patient had faith in the efficacy of treatments, even those that had no evident physical impact on the body would have made a patient feel better. This alleged placebo effect is still relevant in modern systems [7].

The emergence of writing and the calendar foreshadowed the beginning of written history. The earliest records of knowledge are only found on clay tablets with cuneiform seals and indicators used by ancient Mesopotamian physicians. The Code of Hammurabi, which was composed in the eighteenth century BCE by a Babylonian king, is kept on a stone pillar in the Louvre Museum in France [8]. A more precise image can be created by looking at Egyptian medicine. The first physician to appear was Imhotep, chief minister to King Djoser in the third millennium BCE. He constructed one of the first pyramids, the Step Pyramid at 'aqqrah, and was thereafter worshipped as the Egyptian deity of medicine and associated with the Greek god Asclepius [9]. A more certain understanding can be gained from studying Egyptian papyri, especially those discovered in the nineteenth century by Ebers and Edwin Smith. The former is a list of cures with the appropriate spells or incantations, while the latter is a surgical treatise on the treatment of wounds and other injuries [10–12].

2 Belief Systems of the Indian Medicine

The ancient Indian system of medicine is regarded as the most systematic and holistic system, both in terms of its theories and its therapeutic approaches, among all the ancient civilizations of the world [13]. There is no shortage of writing on the medical systems and literature of ancient India, according to an examination of Indian medical historiography [14]. However, the individuals in charge of the healing have not gotten much recognition. The Sanskrit word "*Ayushman-bhava*" which means long life is the source of the term "AYUSH" [15]. Since the time of the Mahabharata, this phrase has been used frequently as a blessing for a healthy, long life. Currently, the Commission for Scientific and Technical Terminology has chosen the term "AYUSH" to refer to all "traditional and non-conventional systems of health care and healing, including Ayurveda, Yoga, Unani, Naturopathy, Siddha, Sowa Rigpa, Homoeopathy." The entire medical system covered by AYUSH is based on the time-tested holistic method of treating illness and promoting healing. Even in the contemporary period, AYUSH plays a crucial role in fostering Indigenous healthcare practices and maintaining traditional medical knowledge. It also offers populations better, more accessible, and affordable healthcare.

2.1 Belief System of Unani Medicine

It is fundamentally different to compare holistic Unani medicine to reductionist Western medicine. In order to provide better healthcare, Unani physician should be well-versed in the fundamental principles of Unani medicine, including disease diagnosis, drug selection, and treatment methodology [16].

The two states of the human body according to Unani philosophy are *Sehat* (state of health) and *Marz* (state of disease). A person should have a healthy constitution and morphology in accordance with their needs to be in a state of health, which is a naturally occurring physiological condition. As a result, the body functions normally and the individual has a tranquil and healthy life. Harmony of the seven basic components (*Umoor Tabiya*), *Arkan* (Element), *Mizaj* (Temperament), *Akhlat* (Humors), *Azaa* (Organs), *Arwah* (Pneuma), *Quwa* (Faculties or Powers), and *Afal* (Functions) are responsible for healthy and peaceful life while changes in any one of these elements can lead to illnesses or even death [17]. Both the cause and prevention of disease employ *Ilmul Asbab*. In Unani philosophy, Asbab (causes), *Alamat* (symptoms), and *Ilaj* (treatments) should all be very precise and tailored to the individual. Therefore, a treatment plan that enables the eradication of disease-causing factors and the restoration of health must include an understanding of *Ilmul Asbab*. *Asbab Sitta Zaruriyya*, the six essential factors for a better and healthier lifestyle, are one such preventive strategy that the treating physician should be well-versed in [18].

Classification of *Asbab* (Causative factors) is illustrated in Fig. 1.

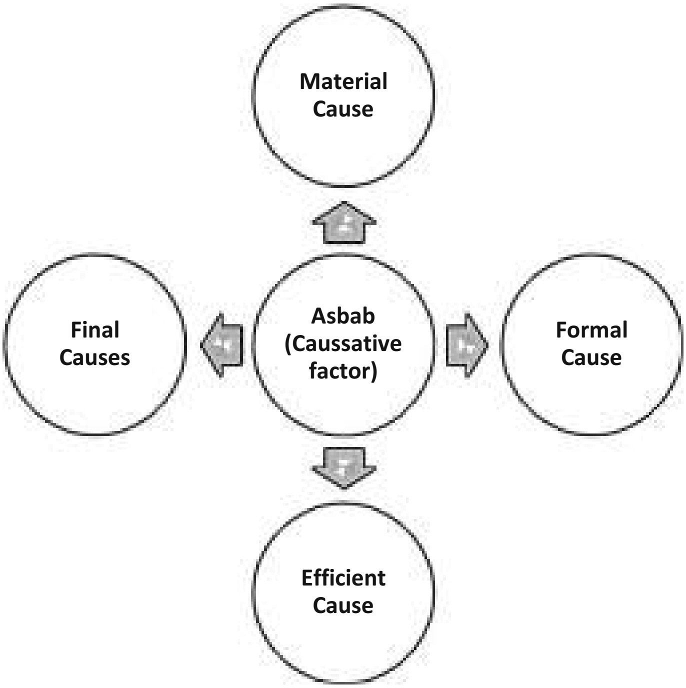

Fig. 1 Classification of *Asbab* (Causative factors)

Above are examples of material causes being *Arkan* (Primordial Essence), *Akhlat* (Humors), *Aza* (Organs), and *Arwaah* (Pneuma), whereas examples of formal causes include *Mizaj* (Temperament), *Quwa* (Faculties), and *Tarkeeb* (Structure). The final cause contains *Afal* (Functions), which are the desired natural result. An organ's dysfunction reveals an organ's disorder. The efficient cause is made up of six essential factors (Air and water, physical and mental movement, physical and mental rest, sleep and wakefulness, body retention, and evacuation), as well as nonessential factors [19].

One of the enduring contributions to the science of medicine is the Hippocratic humoral theory. Based on the current scientific and philosophical views of the time, ancient physicians developed a revolutionary system for deciphering and treating disease as well as maintaining health [20]. The humoral hypothesis of disease was named after this framework. The philosophies of the ancient Greeks were blended into moral theory in order to explain health and sickness as well as provide recommendations for treatment. The foundation of the humoral hypothesis is the firsthand observation of natural phenomena. Hippocrates in particular attentively studied the onset of disease in his patients and drew conclusions from his findings, as did other ancient Greco-Arabic doctors. The origin of disease's material causes was the humoral pathogenesis of illness [21].

2.2 Concept of *Tabiat* (Supreme Power/Immune System) in Unani Medicine

The preservation of one's health and the normal constitution of the human body are the major objectives of Tabiat as per Unani fundamentals [22].

A strong *Tabiat* maintains the physiological function of each organ and supports and generates a healthy body. It also shields the body against unhealthy situations and pathological conditions. Maintaining the integrity of the body requires the association of the seven fundamental components (*Umoor Tabiya*), which have opposing and similar characteristics and make up corporeal creatures. Additionally, due to their oppositional nature, these components have a propensity to separate from corporeal objects. To keep them organized, a force is therefore required, and *Tabiat* is the power that unifies all elements. It is the supreme planner since it creates a healthy environment inside the body and prepares it to fight disease. It is also known as a person's individual hidden power or administrative system known as *Quwate-Mudabbar-e-badan*. When it is strong, a person is not prone to illness of any kind; when it is weak, a person is more likely to become ill [23].

Diseases are thought to be brought on by atmospheric microorganisms, according to the theory of allopathic medicine. The environment in which people live contains a variety of elements and causes that alter the body's natural homeostasis. The health of the body is affected by these elements, but not all of the population is equally affected. This suggests that everyone possesses a secret ability that, depending on the individual, can defend them when it is strong. *"Tabiat-i-Insaniya"* is the name of this secret power. It is the force that unknowingly and involuntary controls and manages the human body. The concept of existence of *Tabiat* can be preserved by interpreting it in accordance with other medical systems within the bounds, as doing so would uphold Unani Tibb's honor and legitimacy [24].

It is compared to immunity by some academics. Although immunity is a part of *Tabiat*, it is not the same as *Tabiat*. Cameron Gruner, who has a firm belief in the idea of *Tabiat*, is the individual who has this belief [25]. According to him, Tabiat is a more complete idea in Unani Tibb than the germ theory or any other theory. He adds that Ibn Sina's embrace of this idea is the main reason he likes him. Loyal Sherald and Dr. Weil Fraid Narth Field were also ardent proponents of the idea or theory of *Tabiat*. According to Dr. Weil Fraid Narth, it makes no difference whether someone agrees with this idea or not. These academics held a firm belief in the idea of Tabiat despite the fact that they were not Unani Tibb scholars [26].

2.3 Epidemic and Pandemic in Unani Medicine

Nearly all of the historical civilizations had an impact on the development of Unani medicine. Hippocrates wrote a treatise about pandemics and epidemics (460–380 BC). Ancient medics regularly used the term "epidemic" in the same way as it is used now, referring to any illness that affects a large number of people in one place at once [27]. The classic Unani literature also touch on the fundamental ideas of epidemics and pandemics, including their causes and preventative strategies. According to the genesis approach, imbalances in the humors or production of abnormal excessive humors within the body and invasions by foreign bodies that target the humors are the two fundamental causes of epidemics [28]. The second group of diseases includes infectious and epidemic conditions. Ancient literature mentions a number of precautions against pandemic and epidemic situations. In his Canon of Medicine, Avicenna described the epidemic diseases as *Amraz-e-Wabai* and advised isolation at home when one was present. *Al - Rhazes* had the concept of quarantine [29].

In addition to isolation and quarantine, three additional measures – (i) environmental purification utilizing specialist herbal pharmaceuticals as fumigants or sprays, (ii) immunomodulation and health promotion, and (iii) usage of preventive and symptom-specific medications – are of utmost importance [30]. According to recent scientific studies, several herbal single and compound formulations that contain antipyretic, anti-infectious, and numerous pharmacologically active substances are indicated in classical Unani scriptures. These formulations may provide new insights into the prevention and management of epidemics and pandemic [31, 32].

2.4 Diagnostic Tools in Unani Medicine

Physical examination and observation are used to make diagnoses in Unani medicine. More focused on the signs and symptoms of illnesses rather than laboratories

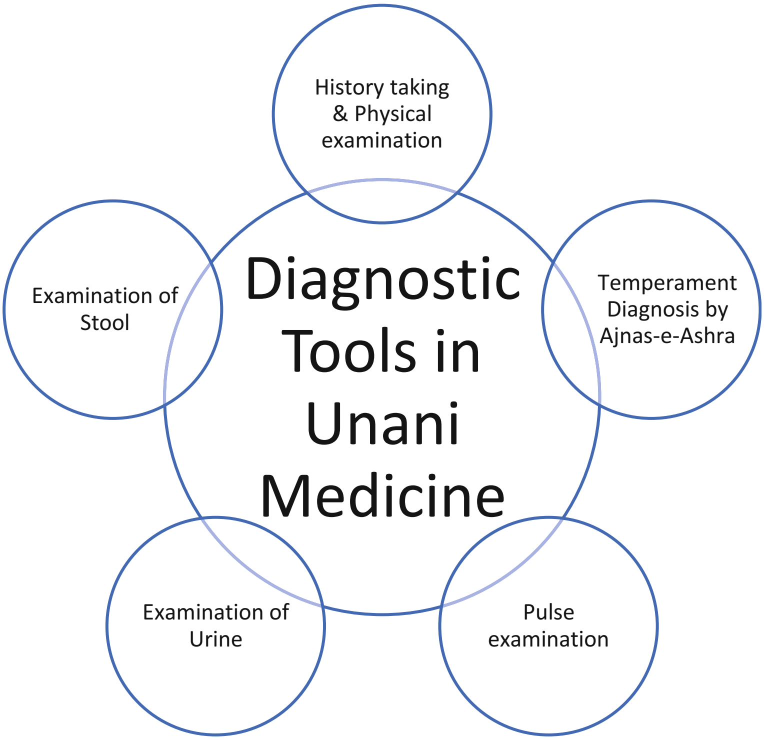

Fig. 2 Diagnostic tools in Unani medicine

values. By taking a detailed history and performing a physical exam, subjective parameters are accessed [33].

In some instances where they are helpful on a secondary level, it also has the intellectual foundation to employ contemporary medical diagnostic techniques. For instance, it does not hesitate to use endoscopy to determine the amount of structural discontinuity because it views it as a fundamental pathology. The history, physical examination including temperament (*Mizaj*), pulse examination (*Nabz*), and inspection of excreta (*Bol w Baraz*) are all included in the unmatched comprehensive diagnosis of Unani medicine. Figure 2 is depicting the diverse diagnostic tools used in Unani medicine to arrive at a comprehensive understanding of the patient's history [34].

2.4.1 The History Taking and Physical Examination

It is carried out in accordance with both general guidelines and the specific guidelines required by the patient's individual situation, primarily under the guidance of the "Ten Fundamental Categories" (*Ajna's Ashara*) of *Mizaj* [35].

2.4.2 Examination of *Nabz* (Pulse)

According to Unani medicine, the human pulse results from the heart and arteries' alternating contraction and expansion, which regulates breathing through inspired air. This idea states that every pulse consists of two motions and two pauses. The ten parameters that comprise the pulse's foundation are size, strength, speed, consistency (elasticity), fullness, temperature, rate, frequency (constancy), regularity, and rhythm. Complex pulses have also been thoroughly documented in Unani medicine. Based on general criteria, specific pulse types linked to each ailment are also identified [36]. Classical Unani writings indicate several conditions to check the pulse for both patient and physician. When checking the patient's pulse, for instance, the thumb should be held upward. At the time of the evaluation, both the patient and doctor should neither be suffering from any emotional or mental disorders, nor be hungry or have a full stomach. It is critical to compare the patient's pulse to that of a healthy, normal person. By placing the pulp of all the patient's finger tips on the patient's radial artery, the physician should move the patient's little finger toward his elbow and his index finger toward his thumb. When checking the patient's pulse, the physician should use his right hand; when supporting the patient's elbow, he should use his left [37]. The examiner's finger pulps should be smooth and soft, not harsh and rigid. In addition, the physician should not be experiencing any medical conditions that would interfere with his ability to concentrate during the examination. It is important for the patient and the physician to get adequate sleep and minimize outside distractions. In addition to closely monitoring the patient's pulse, the doctor should be observing the patient's eyes and facial expressions [38].

2.4.3 Examination of *Bawl* (Urine)

Since ancient times, doctors have employed urinalysis (or uroscopy) as a diagnostic technique [39]. According to ancient literature, urine is the end product of all digestive processes that take place inside the liver and blood vessels. The physiological functions and metabolic status of the liver, kidney, and conditions of other vital organs can all be immediately determined by carefully examining the urine. Additionally, urine also reveals information regarding other systemic diseases [40]; physical examination of urine is important which includes quantity, color, odor, consistency, foam or froth, clearness and turbidity, and sediments [41].

2.4.4 Examination of *Baraz* (Stool)

Additionally useful in the diagnosis of many disorders is the physical examination of stool. During a physical examination, the existence of foreign bodies is looked for as well as their color, amount, consistency, and size [42].

2.5 Line of Treatment in Unani Medicine

The management of diseases in the Unani system depends on the pathology involved in the disease process. The basics line of treatment is shown in Fig. 3:

2.5.1 Removal of the Causes

The pathophysiology of the illness or exact causes must be identified and removed by following a suitable regimen for optimal disease management [43].

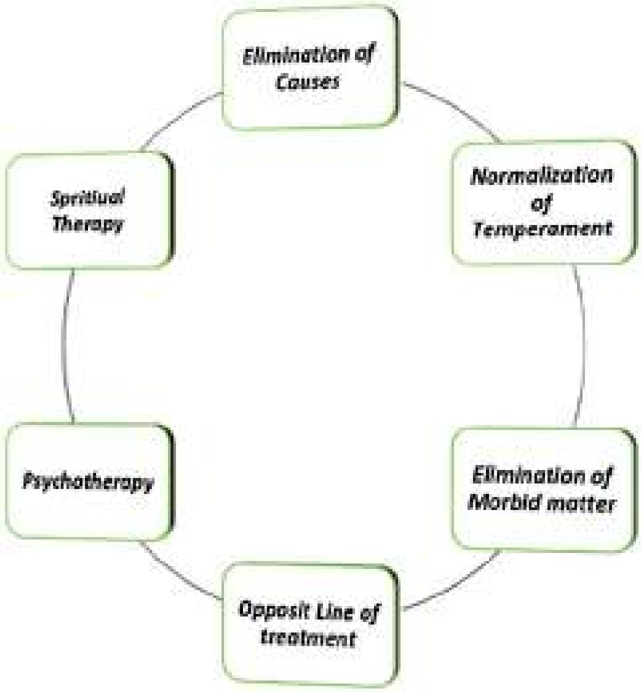

Fig. 3 The principles of treatment in Unani medicine

2.5.2 Normalizing of Alter Temperament

When an illness alters a person's normal temperament without affecting quantity or quality of humor, all that is really needed is for the temperament to return to normal. It may be corrected by only altering a person's lifestyle, or by using an appropriate regimen or medication to treat the sickness [43].

2.5.3 Elimination of Morbid Material

A variety of therapies are used to treat the diseases when a change in temperament results from the quantity and quality of morbid materials accumulating in the body, including concoction, purgative, vomiting, cupping, Venesection, and leeching therapies [44].

2.5.4 Opposite Line of Treatment

The administration of a regimen or medicine whose temperament is opposed to the illness is the fundamental therapeutic principle of the Unani medicine in order to cure morbid temperament and treat illness [45].

2.5.5 Holistic Approach

Specific habits, the surrounding environment, and the patient's physical, emotional, temperamental, and humoral state, as well as the health of the vital organ, are in consideration while treating the diseases. Also, it is important to consider the body constitution, the correct diagnosis, and course of therapy for better treatment.

2.5.6 Psychotherapy

Unani medicine uses herbal remedies, counseling, and changes to body-mind functions, such as sleep and eating habits, to treat mental illnesses. By maintaining a healthy lifestyle by incorporating the six essential components, one can promote the recovery of both physical and mental disorders [46].

2.5.7 Spiritual Treatment

Through its knowledge of the *Ruh* (spirit), which is regarded as the ultimate regulator of the human body, the Unani System of Medicine has evolved to recognize the relevance of spiritual health and therapy. One of the basic tenets of Greco-Arabic medicine is the notion of the reciprocal relationship between the soul and the body. Greek-o-Arabic medical experts gave a full explanation of how these two aspects of the human being interact, and they subsequently used this information to improve patient care. The patient's mental and spiritual well-being should always be given top priority by a doctor because the soul and the body are reciprocal. According to *Al-Razi*, "the doctor, even though he has his doubts, must always make the patient believe that he will recover, because the state of the body is linked to the state of the mind." Inspiring the patients to avoid negative emotions like grief and fear and getting them involved in joyful pursuits like chest games and hunting can all help them recover more quickly. Psychosomatic therapies, according to Unani doctors, are commonly used as actual treatments [47].

2.6 Therapeutic Approach in Unani Medicine

The initial method to treatment comprises the creation of a routine to normalize and balance the external or environmental variables (such as air, water, and food) involved in ailments and diseases. Other methods, such as treatment with natural remedies, may be suggested if this turns out to be in morbid matter. For pharmacotherapy, a different course of therapy or drugs with a specific temperament is used to treat particular ailments [48].

Figure 4 depicts modes of treatment in Unani System of Medicine:

2.6.1 Regimenal Therapy

It also called *Ilaj Bit Tadbeer* (Ilaj means "therapy" or "treatment," and *Tadbeer* signifies "regimen" or "systematic plan"). Consequently, *Ilaj Bit Tadbeer* refers to a course of therapy that attends to the ill person while maintaining general health. Regimenal therapy mostly consists of nonpharmacological strategies that regulate how people's lifestyles are changed for the purpose of disease prevention and treatment [49]. Ancient Unani physicians described numerous regimens for the care of illnesses, either by itself or in conjunction with other remedies. Classical

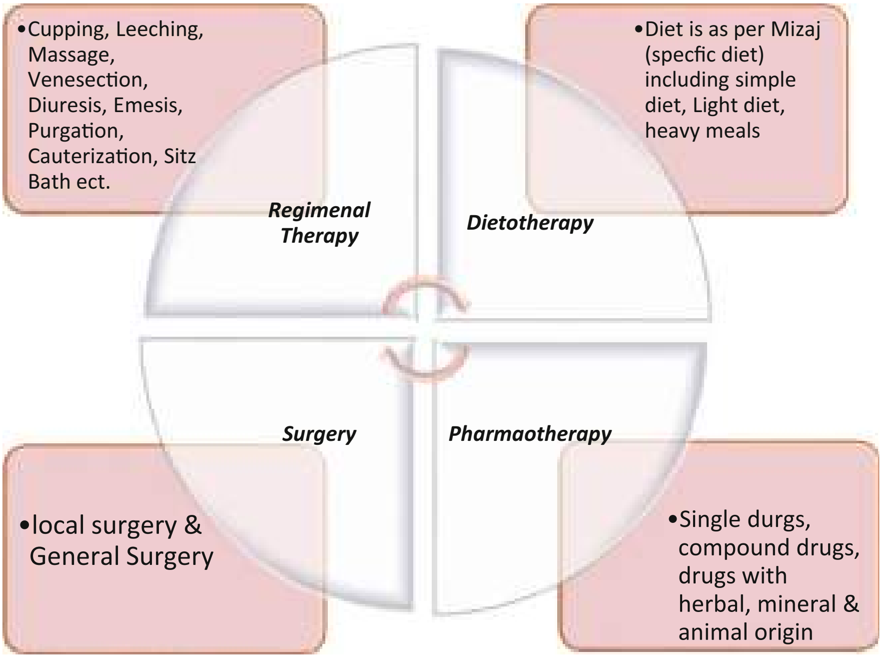

Fig. 4 Modes of treatment in Unani System of Medicine

writings for the eradication of the body's diseased humors or their diversion are referenced in the following regimens.

(a) ***Munzij w Mushil* (Concoction) therapy:** This method of healing is highly effective in managing chronic illnesses such as arthritis, fevers, obesity, diabetes, etc. For all chronic illnesses and illnesses more than 40 days, or acute illnesses more than 7 days, the Unani scholars *Jalinoos, Raban Tabri*, and *Razi* highly prescribe the concoction followed by purgation. Duration of concoction therapy depends on the humor involved followed by *Tabreed* (cold regimen). *Tabreed* is compulsory to avoid any unwanted adverse effect of purgatives and also to bring normalcy in temperament [50].

(b) **Cupping therapy**: Due to thousands of years of clinical experience in Unani medicine, the use of cupping therapy (*Hijama*) and detailed procedure with proper indications and contraindications are very extensively documented. Classical texts categorize CT into three main categories [51]:
   - One is Wet cupping Therapy (WCT) or cupping with bloodletting called Hijama bil-shurt in Unani medicine.
   - Second is Dry cupping Therapy or cupping without bloodletting called Hijama bila-shurt.
   - Third is Fire cupping also called Hijama-e-Nariya.

(c) **Massage:** The two primary mechanisms of action of massage indicated in the traditional Unani literature are, respectively, the evacuation of morbid matter and the diversion of morbid humors [52].

(d) **Leech therapy:** It is also called *Irsale Alaq* (Leech or Hirudo therapy); it is a bloodletting technique that entails the removal of a significant amount of blood from the body with the help of medicinal leeches for the prevention and treatment of numerous ailments [53].

(e) **Venesection:** In the Unani medical system, *Fasd*, or venesection, is a traditional and antiquated form of treatment. The Arabic term for "to open" is called "*Fasd*." Complete evacuation occurs throughout this phase, draining blood and dominant humors from veins. Most bloodletters reach a vein in an arm, leg, neck, or other area of the body using a small, delicate tool called a lancet. Then, after securing the region with a tourniquet and gently grasping the lancet between their thumb and fingers, they pierce the vein either lengthwise or diagonally. Through a perpendicular cut, the blood vessel is severed. They put the blood in measuring basins made of excellent Venetian glass that is expertly crafted [54].

(f) **Emesis:** Vomiting is referred to as "*Qay*" in the Unani medical system, and historically, Unani physicians have employed this procedure to heal many diseases. *Qay* is an effective and simple treatment. The medications used for these purposes are known as emetics, they either stimulate the stomach and cause reverse peristalsis or stimulate the location where vomiting occurs, and certain emetic prescriptions are required in order to treat a given sickness and provide the intended results. It is one of the best regimens to evacuate bilious and phlegmatic morbid matter, particularly from the liver and stomach. Along with its preventive advantages, it also has a number of therapeutic uses, including in cases of melancholia, vertigo, hemiplegia, jaundice, sciatica, and gout, among others [55].

(g) **Diuresis:** It is a regimen for eliminating waste from the body through urination. In a variety of medical conditions, this route is used to evacuate morbid matter or waste product. When diuretic drugs are added to a treatment plan to treat conditions like ascites, vertigo, back pain, joint pain, etc., it works best [56].

(h) **Enema:** The best way to treat colicky abdominal pain, diseases of the kidneys or bladder, and other inflammatory conditions of the abdominal cavity is to administer an enema through the anus, which is the best way to remove feces and other morbid matter from the intestines [57].

(i) **Diaphoresis:** Sweating serves as a form of bodily waste product elimination. This is accomplished through various techniques like hot fomentation, hot baths, hot poultices, exercise, massage, and also by introducing some specific herbs. This routine is also beneficial for boosting skin nutrition and lowering body temperature in cases of fever [58, 59].

(j) **Diversion of matter:** The Unani system defines *Imala* as the diversion of diseased matter, which may be present in vital organs and may harm the body's basic functions if it accumulated. There are two different types of diversion, one of which drives morbid humors toward the distant organ and the other toward the nearby organ. In this instance, when the morbid humors have just gathered and are not agitated in one organ, they are diverted to the next organ, and vice versa [58, 59].

(k) **Sitz bath:** One method of bathing involves immersing the lower body, particularly the hips and buttocks, in water with or without medication. This regimen is useful in the treatment of diseases of the gastrointestinal, rectal, anal, prostate, kidney, urinary bladder, urethral, testicular, vaginal, and uterine systems. Four various forms of sitz baths, including hot, cold, moderate, and compound (alternating hot and cold), are widely used for treatment [60].

(l) **Turkish bath:** The Unani system has used the *hammam* for millennia. In a typical *hammam*, there should be four separate compartments, each with a distinct temperature. For instance, the first, second, third, and fourth chambers should all be at various degrees, ranging from warm and hot to cold and normal. The hammam's hot water and air are beneficial to the body since they warm and moisturize the skin, respectively. The body is refreshed by this regimen, which also enhances hunger and allows waste to be expelled through the skin [61].

(m) **Irrigation:** In this procedure, the disease portion is irrigated with a medicated decoction or oil from a height above. A small amount of the medicine may pass through the skin and help in evacuating morbid matters. According to *Ibn Sina*, this regimen is particularly effective in treating a wide range of disease conditions, including head illnesses [62].

(n) **Fomentation:** As part of this protocol, the medicine powder is heated while wrapped in a cotton cloth before being applied locally as a fomentation to reduce inflammation and discomfort of a particular part. Unani pharmacopoeias include a number of formularies for the treatment of various illnesses using hot and cold fomentation [63].

2.6.2 Dietotherapy

The word "diet" comes from the Greek word "diaita," which denotes a way of life. It is described as both liquid and solid food items that are regularly ingested during daily life or as a prescribed amount of food that has been specially prepared for a certain illness or condition of health [64]. It is specifically used to maintain or restore strength, vigor, and vitality when they have been compromised by illness. To recognize the relative usefulness of foods in treating disease, one must first understand the quality and quantity of diet. For example, a wide variety of food products are known to have diaphoretic, laxative, or diuretic effects [65]. Different dietary classifications have been established to classify these foods according to their nutritional value, which includes light and soft diets, extremely nutritious diets, less nutritious diets or highly nutritious diets in moderation, and good diets that promote the production of chyme. Many of the nutritional products that have been described as medicines in classical texts, such as *Gulqand*, *Halwa*, *Murabba*, whey, vinegar, barley water, and mutton soup, are frequently administered either by themselves or in combination with other medicinal substances. The importance of nutrition in the Unani System of Medicine is highlighted by this all-encompassing approach to diet [66].

2.6.3 Pharmacotherapy

It is called *Ilaj Bil Dawa* in Unani medicine, in which single or compound formulation is used for treatment of various diseases. In view of Unai doctors, this therapy approach is one of the most successful because it is natural, environmentally safe, and less invasive. Pharmacotherapy may also produce good results when it is combined with various regimens or specific diet. The choice of medications for any given treatment is based on three fundamental ideas. First is drug quality in terms of temperament, second is drug quantity in terms of weight and potency, and third is administration time. The type and nature of the condition determine which medication is best. The type of affected organ, the severity of the illness, and other relevant variables including sex, age, weight, habit, habitat, season, prior therapy received, and disease stages all affect the weight and efficacy of the medication [67]. The Unani pharmacopoeia has about 2000 medicines derived from many sources, including plants, minerals, and animals. The classical writings of Unani medicine contain specialized methods for processing herbs, minerals, animals, and poisonous drugs. A range of dosage forms, including powders, decoctions, infusions, pills, and semisolid preparations like Majoon and Jawarish, are also covered in detail, along with the dosage and timing of consumption [68].

2.6.4 Surgery

Since ancient times, the Unani System of Medicine has always included surgery as a kind of treatment. A book named *Kitāb al-Taṣrīf li-man ʻajiza ʻani'l-Taʻlīf*, written by Arab Unani physician *Abul Qasim Zahrawi*, was composed of 30 volumes and covered issues in medicine, surgery, pharmacy, and other health sciences. The final volume, which has 300 pages, is devoted to surgery [78]. For the first time in medical history, he treated surgery as a separate subject. He discussed a number of treatments, innovations, and techniques, including tonsillectomy, tracheotomy, craniotomy, caesarean section, dentistry, and the removal of kidney stones and cataracts. Antiseptic drugs, styptic drugs, and other categories of medications are utilized in cases requiring surgical interventions according to the Unani System of Medicine [69].

2.7 Drugs and Dosage in Unani System of Medicine

2.7.1 Drug Source

There are three types of natural medicines: those derived from plants, minerals, and animals. According to its origin, Unani medications are categorized into three categories [70]:

(a) Plant-based medications use ingredients like roots, stems, bark, leaves, flowers, seeds, fruit, gum, resin, and extract, among others.
(b) Different metals, metal ores, and nonmetals in their natural forms are used in mineral origin medications.
(c) Animal origin medicines involve animal glands, tissues, and occasionally animal poisons.

2.7.2 Dosage Forms

The dosage forms were developed with the intention of satisfying all of the core objectives of pharmaceutical processing, including palatability, assimilability, stability, and an increase in safety and efficacy. In the Unani System of Medicine, drugs are typically used in four different forms: solid, semisolid, liquid, and vapor [71].

Pill (*Hab*), tablet (*Qurs*), powder (*Safuf*), and other solid dosage forms are available. Semisolid dosage forms include *Jawarish, Majoon, Khamira, Lauq, Itrifal*, and others. Decoction (*Joshanda*), infusion (*Khisanda*), distillate (*Arq*), syrup (*Sharbat*), drops (*Qatur*), and others are liquid medication forms. Vapor dose types include fragrances, *Lakhlakha*, steam inhalation, and fumigation. Unani doctors can also provide medication through pessary, suppository, liniment, Sitz bath, etc.

Kushta are also utilized to treat a variety of diseases. *Kushta* are fine powdered medical preparations made by calcining pharmaceuticals of metal, mineral, and animal origin with enhancing and correcting herbs, which transform them into organic forms that are biologically assimilable. The word *Kushta* derived from the Persian word *Kushtan*, which means "to kill," refers to all of the aforementioned changes. It should be mentioned that adding herbs recommended by the Unani System of Medicine results in extra benefits. Some of the therapeutic uses of *Kushta* are given as follows [72]:

- *Kushta-i Jast* showed hepatoprotective activity in CCl_4-induced liver damage.
- *Kushta-i Shibb* elicited protective action against myocardial infarction in rats.
- *Kushta-i Hart̠āl* is found effective in diabetes.
- *Kushta-i Nuqra* exhibited analgesic activity against chemical, thermal, and electrical noxious stimuli.
- *Kushta-i Ṭilā Kalān̠* (Unani gold preparation) augmented cell-mediated and humoral immunity.

2.7.3 Specific Processing of Drugs

The toxicity and danger of medications derived from plants, minerals (metals), and animals were known to Unani scholars. However, they can be sufficiently treated to make them safe for therapeutic usage [72]. Comparative pharmacological examinations of dangerous medicines such as arsenic, strychnos nux-vomica, aconitum napellus, etc. in their natural and processed forms have shown that the processed forms almost completely lose their toxicity when compared to the unprocessed, natural form. In the Unani System of Medicine, processing has goals that go beyond detoxification, such as improving efficacy and pharmacokinetic optimization, which involves achieving the desired rate of absorption and transport to the target tissue, among other things. The Unani System of Medicine has produced a tremendous amount of novel processing procedures. Several of the key methods are Burning (*Ihraq*), Bathing (*Ghasl*), Roasting (*Tahmis*), Frying (*Taqliya*), (*Tashwiya*), Purification (*Tasfiya*), etc. [73].

2.7.4 Corrective Drugs

The doctors of the Unani System of Medicine were aware of a variety of potential medication side effects (ADR) [72]. As a result, when explaining the properties of pharmaceuticals, they included their potential side effects as well as their corrective drugs. For instance, the commonly used laxatives *Operculina turpethum* L and *Citrullus colocynthis,* which cause tenesmus, are remedied by adding *Zingiber officinalis*. Similarly, the antiaging drug *Aloe barbadensis miller* induces intestinal abrasions and tenesmus, which are treated by *Sterculia urens* [74].

2.8 Prevention and Management of Diseases by Six Essential Factors in Unani Medicine

The six essential factors, whose suitable use in a person's life is decided by their temperament, provide one of the outstanding examples of the observational and philosophical genius of the Unani System of Medicine. As a matter of fact, the Unani System of Medicine identifies all the factors that affect health and sickness, classifying some as essential and others as nonessential [75].

The scope of the essential factors is exceedingly broad, and their inventive uses or rectifications are much more numerous. For instance, clean, abundant air is crucial for healing; therefore, homes should be roomy and cities should have adequate security. Contrarily, unavoidable issues can be dealt with by employing corrective factors [76].

The second aspect is "Movement and Rest," which are essential for good health and should be done in accordance with one's temperament. Movement can be active or passive; even someone who is paralyzed can engage in swinging, a practice that serves as exercise. In addition, exercise is necessary for all organs, including the senses, including the eyes, which should be used to look at lovely items, scenes, and complicated patterns, and the ears, which should be used to listen to music and songs. This demonstrates the brilliance of Unani scholars who discovered several elements and straightforward classification techniques that could group them into a manageable number of categories for easy application. Figure 5 illustrates the six essential factors impacting the overall management of disease state in patients [77].

2.9 Historical Evaluation of Unani Medicine and Contribution of Greek, Arab, and Indian Physicians

2.9.1 Unani Tibb in Greek Origin

The Unani System of Medicine, as indicated by its name, has deep-rooted Greek origins, with a historical journey that traverses through Egypt and Mesopotamia, ultimately embraced and enhanced by the Arabs [78]. In India, Unani medicine gained prominence, thanks in large part to influential physicians like *Hakim Ajmal*

Fig. 5 Six essential factors in Unani medicine

Khan, who skillfully adapted its principles to the local geographical and human context, making it one of the most significant traditional medical systems in the country. Unani medicine also demonstrated its commitment to progress by delving into molecular research in the 1920s to understand the molecular impacts of its treatments. Additionally, it successfully institutionalized medical practices through the development of curricula, organizations, and professional bodies [76].

The botanical foundation of Unani therapy finds its origins in ancient Egypt, where the use of plants in medical treatment was highly esteemed. The Egyptian culture's proficiency in surgery and medicine is evident in the historical papyri studies. Renowned Egyptian physicians like Amenhotep (1550 BC) and Imhotep (2800 BC) left their mark in the annals of medical history. Mesopotamia, too, made significant contributions, as evidenced by their diagnostic instruments for testing urine and stool samples. The Greek era of Unani medicine was initiated by Asclepius (*Asqalibus*, 1200 BC), a prominent physician and scientist who laid the foundation for the Greek approach to Unani medicine. During this era, the Greeks synthesized medical knowledge, drawing from Egyptian and Babylonian medical expertise, and began shaping the art of medicine [79].

(a) ***Buqrat*** **(Hippocrates):** The leading physician throughout the classical era of Unani medical history was Hippocrates (460–370 BC). He chronicled the current medical knowledge and placed an emphasis on the natural causes of disease in order to provide the groundwork for the development of medicine as a disciplined science. Observation, experience, and logical concepts were the three pillars of Hippocratic medicine, and they are still relevant in the fields of science and medicine today [80].
(b) ***Jalinoos*** **(Galen):** Galen was a Roman Greek physician, surgeon, and philosopher. Galen had an impact on the development of various scientific domains, including anatomy, physiology, pathology, pharmacology, and neurology as well as philosophy and logic. Galen is recognized as one of the most accomplished of all ancient medical specialists. Galen, a physician, was the first person to use the pulse as a sickness symptom. Representative research fields include urology, neurology, myology, respiration, reproductive medicine, and neurology [79, 80]. He advanced drug research and clinical use. He discussed the heart's valves, named the seven unique pairs of cranial nerves, and pointed out the structural differences between arteries and veins.
(c) **Dioscorides:** The first author of the illustrated book De Materia Medica (*Kitab al-Hasha'ish*), which contains 600 herbal remedies, was a man with a vast understanding of medicinal plants. With the exception of Dioscorides original additions, the book included all prior knowledge of pharmacology [81].

2.9.2 Arab Physician Contribution in Unani Tib

Following the fall of the Greco-Roman civilization, the expansive *Arabic*-speaking region of the Middle Ages made significant contributions to human knowledge, particularly in the realms of science and medicine. The translation of Unani medical texts into Arabic began in the *Umayyad* era but gained widespread popularity during the early years of the *Abbasid* century. Between 750 and 850 AD, a period of intensive translation efforts was succeeded by the publication of original Arabic works authored by scholars in the region [82].

Arabian physicians conducted extensive philosophical and scientific research, critically examining existing medical knowledge while contributing new insights to the field. They achieved remarkable advancements in various disciplines, including philosophy, general science, technology, and, notably, medicine, which they revolutionized [83]. Prominent figures during this era included *Ibn Sina, Abu Sahal Masihi, Zakariya Razi,* and *Rabban Tabri,* all of whom were exceptional physicians [79, 80].

One notable achievement was the creation of the "*Materia Medica*," a comprehensive compilation of 1400 drugs, greatly influenced by the work of *Ibne Baitar*. In the field of chemistry, *Jabir bin Hayyan* made significant contributions. Additionally, a pioneering figure in the realm of eye surgery, *Hunain bin Ishaq*, advanced the field with cutting-edge treatment methods, laying a solid foundation for ophthalmic surgery [84].

The progress in surgery can be better appreciated by examining the work of the Spanish Unani scholar *Abu-al-Qasim Zahrawi*, particularly his treatise *"Kitab* *al-Tasrif*." Furthermore, the Arabs' contributions extended to the development of obstetrics and pediatrics, which became distinct fields of study. Arab scholars conducted research and authored books on these subjects, further enriching the corpus of medical knowledge during this transformative period [85].

Ibn Zuhr and *Ibn Rushd*, two other Spanish scholars, helped establish Unani medicine in Europe. Numerous books were written by *Ibn Zuhr*, also known as Avenzoar, including "*Kitab al-Iqtisad*" (*Book of Moderation*), "*Kitab al-Aghdhiya*" *(Book of Aliments),* and "*Kitab al-Taysir fi al-Mudawat wa al-Tadbir*" *(Book of Facilitation on Therapeutics and Dietetics).* "*Kitab al-Taysir fi al-Mudawat wa al-Tadbir*" *(Book of Facilitation on Therapeutics and Dietetics),* one of these compilations, is to evaluate his contributions to the development of surgery. He did postmortems on sheep as part of his clinical study on the management of ulcerating disorders of the critical organs and underlined the importance of providing surgical trainees with a full and practical anatomy instruction [85]. He proposed the idea of a training program or internship for medical students to address this. Additionally, he indicated when a doctor should quit treating a surgical problem and drew a line between a general surgeon and a general physician. By proposing a variety of novel diseases and treatments, he contributed to the improvement of surgical and medical knowledge by developing the concept of pre- and postsurgery diet and management. Following this, the Unani system emerged in India [86].

2.9.3 Unani Scholars from India

The Unani System of Medicine, introduced in India in the seventh century, has undergone significant development and evolution spanning over 1200 years. Throughout this lengthy history, countless researchers and scholars have contributed to its development, leading to its widespread acceptance among the population. The support of Delhi *Sultans* and *Mughal* emperors further solidified its position in India [79]. Adapted to the unique geo-human context of the nation, the Unani system has become one of the most effective and beloved Indigenous medical systems. Notable figures like *Muhammad ibn Yusuf Harwi*, in his work on geriatric care and antiaging therapy, explored various techniques and pharmaceuticals aimed at delaying the effects of aging. Influential proponents of the system, including *Ali Gilani, Hakim Alvi Khan*, and *Akbar Arzani*, made significant contributions through teaching, clinical treatment, and prolific book authorship. Ali Gilani's famous commentary on all five volumes of A*l-Qanun*, written during *Akbar's* rule, was frequently referenced by Unani scholars in Iran and the Middle East [87].

The underlying ideas and key practices of Unani medicine were created by the *Arabs* and Persians, but Indian scholars explained and applied them to a degree that was unsurpassed in the countries where it originated. In contrast to *Muhammad Najm-al-Ghani's* greatest Indian compendium, which was released in 1930 and contains descriptions of 2500 natural commodities, *Ibn Baitar's* ancient pharmacopoeia offers descriptions of 1400 medicinal plants and minerals [88]. By examining their therapeutic effects in line with Unani principles, new drugs created in India or modified from the Indian system of medicine, for instance, were introduced into the Unani System of Medicine. *Takmile Hindī*, a company founded by *Shāh Ahl Allāh*, serves as an excellent illustration of this venture. Existing medications' new effects were also identified*; Muḥammad Ḥusayn Shīrāzī* was a notable contribution to this [86]. In the realm of pharmacy, there have been several developments and novel drug formulations created, such as the delicately edible *Khamira* developed during the Mughal era [85]. Unani medicine, deeply rooted in historical tradition, has a rich history of addressing a diverse array of medical conditions, including immunosuppressive disorders such as HIV, syphilis, and gonorrhea. In India, numerous scholars, such as *Abd al-Fatah and Khwaja Rizwan Ahmad*, have authored approximately 100 treatises in Arabic, Persian, and Urdu, offering insightful interpretations of *Ibn Sina's* works [80]. Notably, the nineteenth century marked a period of remarkable innovation, exemplified by the pioneering contributions of *Hakim Azam Khan*, who made groundbreaking advancements in medicine and pharmacology within the Unani system [78].

During this time, a government organization in Hyderabad State played a crucial role in disseminating Unani references and textbooks. Influential educators like *Hakim Kabeeruddin* significantly contributed to the dissemination of Unani knowledge [79]. He worked diligently to translate old Unani texts from Persian to Urdu, harmonizing Unani philosophy with contemporary scientific concepts. This collaborative effort resulted in the creation of textbooks suited to the modern era. He also held important positions at institutions such as the *Nizamia Tibia College in Hyderabad* and the *Ayurvedic and Unani Tibia College in New Delhi* [81].

In the early twentieth century, Unani medicine embraced modern insights from molecular medicine while preserving its traditional philosophical foundations. Scholars like *Hakim Muhammad Feroz al-Din* incorporated contemporary prescriptions, shared their findings with the wider medical community, and encouraged Unani doctors to critically assess them. Their efforts extended to researching the molecular mechanisms underlying alternative therapies for later stages of healing and conducting comprehensive evaluations of Unani treatments, encompassing both positive and negative outcomes [86]. The late nineteenth and early twentieth centuries witnessed significant contributions from families such as the *Hyderabadi Nizam*, *Lucknowi Azizi*, and *Dilli Sharifi* family, further demonstrating the enduring relevance and progress of the Unani medical system over time [82].

The renowned educator and physician, *Hakim Ajmal Khan*, made significant contributions to the advancement of the Unani System of Medicine in India. He initiated a comprehensive effort to modernize the Unani system, bringing about significant changes in its academic, clinical, and research aspects. To create a prototype college for oriental systems of medicine, he extensively studied European medical institutions. This initiative resulted in the establishment of the *Ayurvedic & Unani Tibia College in Karol Bagh*, New Delhi, founded by Lord Hardinge in 1916 and inaugurated by *Mahatma Gandhi* in 1921. This unique institution remains the only one in India offering education in both *Ayurvedic* and *Unani* medical systems concurrently [87].

Following in *Hakim Ajmal Khan's* footsteps, the renowned physician *Hakim 'Abd al-Hamid* modernized Unani medicine production on a large scale [86]. He founded *Hamdard Davakhana*, a company dedicated to manufacturing high-quality Unani medicines for domestic and international markets. Additionally, he was instrumental in establishing the Institute of History of Medicine & Medical Research in New Delhi, later known as *Jamia Hamdard*. This institution has evolved over time into a multifaceted center with a distinguished faculty and a specialized Unani treatment hospital [88].

In the latter half of the twentieth century, *Hakim M.A. Razzaque* continued *Hakim Ajmal Khan's* vision, placing a strong emphasis on research and scientific endeavors. He played a pivotal role in shaping the structure of the Central Council for Research in Unani Medicine (CCRUM) and in establishing the National Institute of Unani Medicine in Bangalore [83]. Serving as the first director of the organization and as the Indian government's deputy Unani advisor from 1979 to 1991, he laid the foundation for the research in Unani System of Medicine in the country [87].

3 Conclusion

The use of traditional medicine in healthcare and disease management dates back to thousands of years since the earliest of civilizations. With the advent of new religions and beliefs, it has been both exalted and condemned. More than secular factors influence clinical presentation, diagnosis, course of therapy, formulations employed, and even the type of specialist seen in Unani medicine, not just in India but also globally. We anticipate that modernizing the documentation process for generating evidence while taking into account the epistemology and guiding principles of the system will undoubtedly help Unani medicine reclaim its rightful position among the contemporary world's healthcare systems and pave the way for a vibrant, forward-thinking, and promising future. Reverse pharmacology, pharmacoepidemiology, retrospective treatment-outcome surveys, and whole-system approaches are investigative techniques. Connecting the best aspects of Unani medicine to modern technological interventions, substantiated by in-depth studies and robust clinical trials, will help validate the therapeutic utility of Unani system. This chapter hopes to fuel research and documentation of scientific efforts done across the globe to propel this system as one of the mainstream medicinal systems. It also envisages attraction of new practitioners to develop an evolutionary approach in prescribing Unani medicine in routine clinical practice and recording observations of expected and unexpected clinical responses in a scientific manner backed by appropriate statistical analytics.

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Xenobiotics in Traditional Medicine Practices and Quality Control Strategies

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Contents

Abstract

Traditional medicine has been recognized globally as an effective tool for managing an array of ailments that confront humanity. Before the advances in sciences that revolutionized the pharmaceutical industry, ancient civilizations were dependent on herbal preparations for their therapeutic needs and

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occasionally even prophylaxis demands. The applications of Chinese medicine, for instance till date speak volumes to the therapeutic potentials of carefully crafted traditional medical practices. However, the pollution of agricultural fields owing to indiscriminate applications of agrochemicals has led to contamination of soils, water, air, and wildlife creating a vicious cycle of xenobiotics in the environment. Since herbal plants do not grow in isolation, they are often polluted with heavy loads of trace organic and inorganic pollutants circulating in the environment. The concentration of which may be dependent on the source of the plant, geographical location, proximity to source contaminants release point, topography of the area, flooding, industrial activities, and discharges, among other factors. Xenobiotics of unknown toxicity may be taken in especially by rural dwellers who depend largely on herbal preparations for their pharmaceutical needs. Since these preparations are not administered at scientifically determined doses, they constitute a considerable source of exposure to pollutants with associated adverse health effects and even death. This chapter reviews the state of pollutants in herbal preparations, their sources and potential threats on consumers as recently reported in the literature. It highlights quality control measures as a necessary approach to minimize the menace of xenobiotics contamination of herbal medicinal products. It concludes by proffering viable measures to arrest the presence of contaminants in herbal products. These measures, when taken, would hopefully increase the quality of herbal preparations and increase optimal utility while minimizing the hazardous effects arising from the presence of contaminants. From pleasure-seekers in sexual enhancers to weight-loss products for the obese to chronic medical conditions and microbial infections, botanicals have remained a regular companion in several homes on both sides of the rural-urban divides.

Keywords

Xenobiotics · Herbal medicinal preparations · Quality control · Global south · Adverse side effects · Analytical techniques

Abbreviations

CSOs	Civil Society Organizations
DDT	Dichlorodiphenylethanes
E-wastes	Electronic wastes
FAO	Food and Agricultural Organization
FDA	Food and Drugs Agency
FTIR	Fourier Transform Infrared Spectroscopy
GC-MS	Gas Chromatography-Mass Spectrometry
HCH	Hexachlorocyclohexanes
HPLC-MS	High-Performance Liquid Chromatography-Mass Spectrometry
HPTLC-MS	High-Performance Thin Layer Chromatography-Mass Spectrometry
HMPs	Herbal Medicinal Products

ICP-OES	Inductively Coupled Plasma-Optical Emission Spectroscopy
LC-NMR	Liquid Chromatography-Nuclear Magnetic Resonance
LRAT	Long-Range Atmospheric Transport
MRL	Maximum Residual Levels
OOCPs	Organochlorine Pesticides
OPs	Organophosphorus Pesticides
PPCP	Pharmaceuticals and Personal Care Products
PCNB	Pentachloronitrobenzene
PCDDs	Polychlorinated Dibenzo-paradioxins
PCDFs	Polychlorinated Dibenzofurans
PDE-5	Phosphodiesterase type-5
PAs	Pyrrolizidine Alkaloids
POPs	Persistent Organic Pollutants
PM	Particulate Matter
TCM	Traditional Chinese Medicine
THM	Traditional Herbal Medicine
UNEP	United Nations Environmental Program
WHO	World Health Organization

1 Introduction

Botanicals are a rich source of novel phytochemicals which carry pharmacological activities in biological systems [1, 2]. These have often been employed as the only source of medicals for rural dwellers. The modern man occasionally tends toward botanicals to meet his needs, owing to the rich presence of phytochemicals and nutraceuticals with desirable bioactivities [3, 4]. Humans' quest for botanicals varies with their needs. Pleasure seekers may go after sex-enhancers, while those desiring to lose weight patronize slimming herbal medicinal preparations [5–7]. Sufferers from microbial infections seeking relief against an array of pathogenic microbes of varying genetic composition, inflicting different medical conditions with a range of pathophysiological presentations, also depend on botanicals [8–11].

The ancient man depended upon herbs to cure himself of his ailments. The active ingredients in herbs are often the starting materials of refined pharmaceuticals, or aid as a guide in the chemical synthesis of similar active ingredients or more potent compounds through modifications of original isolated bioactive molecules [12]. The acceptance and patronage of herbal preparations among urban dwellers is growing [13] even among medical practitioners in the developed nations [14] as well as medical students [15] thereby leveling the steep slope that hitherto existed between consumers of rural and urban dwelling. Some of the factors attributed to increased herbal use in the global south include low socioeconomic status, long travel distance between patients' houses and locations of health care facilities, level of education, and inadequate health insurance coverage [13, 16]. On a comparative basis, the rural dwellers show more propensity of depending on Traditional Herbal Medicine

(THM) than the urban dwellers. Hughes et al. [17] reported the odds of dependence on THM are 12-to-1 against the rural dwellers. Among the several predisposing factors, sex, age, and religious affiliation present inconsistent pattern to account for herbal patronage or otherwise [18].

The global consumption of herbal medicinal preparations is high [3] thus, it is important for it to be monitored for public health concerns. These can be achieved through stringent quality control measures and increasing education and endorsement of medical practitioners [14] around the importance of Traditional Herbal Medicines (THMs) and relevance of quality of the products being sold. Sadly, these products are contaminated with xenobiotics of varying nature, sources, and concentrations [19–21]. With most people estimated at 80% from emerging economies meeting their health needs through THM [3, 12], it is imperative to ensure the safety of these consumers through ensuring highest quality THMs, thereby preventing further health complications which would worsen pressures on already fragile and overstretched health care systems and maximizing the benefits from THM. Thus, it is important to understand this nexus-presenting situation, to be armed in the choice and use of herbal products while avoiding the contaminants and their attendant adverse side effects. This is a multifaceted task encompassing producers, regulatory agencies, consumers, and advocates such as academia and civil society organizations. The objectives of this chapter are to demonstrate the global use of THMs, elucidate the presence of xenobiotics in these products and trace their sources, and then discuss the quality control measures that could be employed to ensure healthy THMs are available for all consumers.

2 Global Patronage of Herbal Preparations

The application of medicinal plants for therapeutic purposes is as old as human history, and spreads across the globe. It has remained an important part of trado-medical practice till today. Data shows that about 80% of the world's population rely on herbal preparations for their primary health care demands [2, 3, 22]. The medicinal values of medicinal plants are attributed to their phytochemical content. The most important of these phytochemicals include alkaloids, flavonoids, tannins, and phenolic compounds [23]. In the last decades, the surge in demand for THMs industrial products has produced a corresponding increase in the circulation of a large variety of products in the markets useful to both complementary and alternative medicine [3]. According to a WHO report, herbal medicines represent approximately 20% of the total drugs marketed globally. The total global market worth of herbal products is estimated at over US $72 billion as at 2020 [24]. Additionally, it is projected that the percentage of plant-derived prescription drugs is a least 25% of total prescription drugs in the market [21]. Medicinal plants are a good source of various natural antioxidants notably used for the treatment of diverse diseases worldwide [25]. Botanicals, or their products, have been employed in the prevention and treatment of several chronic diseases such as arthritis, cardiovascular diseases, diabetes, inflammatory diseases, and other ailments [26]. They have also been reported to possess antimicrobial potential [27]. Furthermore, the problems of the extreme cost of synthetic drugs, besides the residual effects on livestock products, adverse effects, and evolution of drug resistance genes in pathogenic organisms have led researchers to continue searching for powerful, safe, unconventional, and economically friendly natural drugs of plants and animals origin.

2.1 Economic Potentials of Herbal Preparations: The Case of Traditional Chinese Medicine

Traditional Chinese medicine (TCM) is reported as an important characteristic of Chinese culture and medical tradition for thousands of years now [28, 29]. Nowadays, Traditional Chinese medicine has gone beyond the borders of Asia and has become internationalized. It is now often combined with orthodox single-target drugs to attain complete cure of complex diseases [30]. For instance, its being a long time now since the WHO recommended artemisinin combination therapy as an important management procedure for malaria [31]. The proceeds of traditional herbal medicine (THM) have been projected to reach trillions of dollars before 2028 [32]. The growing popularity of traditional and herbal medicine raises the important question to its likelihood of contamination and the prospects that long-term exposure to such contaminants could result in threatening health concerns [33, 34].

2.2 Risks and Challenges Associated with Herbal Preparations

The increasing demand and the pressure to meet with sufficient supply has influenced THM productions. This is accompanied by either unintended or even intentional contaminations to increase quantity hence maximizing profit [1, 35]. Contamination of herbal products by potentially toxic elements could arise from pollution of food supply chain processes. These include the growing conditions, the harvesting procedures, and the storage techniques [28, 36]. Moreso, the increasing popularity of herbal medicine throughout the world has led to increasing fraudulent practices along with possible negative effects due to contamination, adulteration, or misidentification of plant species [1, 35] and those linked to the natural toxicity of the plants [37, 38]. Because of the increased demand from consumers, there is enhanced urgency to assess the potency of the products to ensure they are safe for the consumers [19]. In addition, the risk of adverse effects due to the presence of xenobiotics such as pesticide residues, heavy metals, etc., or microbiological contaminants mandates close toxicological assessment to eliminate potential safety concerns [19].

3 Xenobiotics: Definition and Scope

Xenobiotics refers to "chemicals found but not produced in organisms or the environment." These are compounds of synthetic origin which maybe present in trace concentrations and/or high concentrations, often used to service domestic, industrial, pharmaceutical, and agricultural demands whose residues may remain in the environment. The technological revolution observed by the twentieth century resulted in the manufacturing of several compounds targeted at improving daily life such as antibiotics, agrochemicals (fertilizers, pesticides, and all the other agrocides), dyes, pharmaceuticals, and personal care products (PPCPs), additives, etc. These compounds may not be found naturally occurring in the environment or occur naturally at concentrations markedly peculiar from those resulting from anthropogenic activities [39]). Xenobiotics are characterized by their physicochemical characteristics, such as ionizability, small molecular size, hydrophilicity, lipophilicity, volatility, and polarity, which makes their identification, quantification, and removal a tedious task [39]. The term xenobiotics also describes some naturally occurring chemicals present in the environment at an excessively high concentration [40]. Xenobiotics may be obtained in water, soil, plants, air, animals, and in even humans owing to direct contact or transfer through different medium and food chain. They are classified as biocides/pesticides, pharmaceuticals, illicit drugs, personal care products, and wastes [41] (Fig. 1).

Xenobiotics may also be categorized according to an array of factors. These may include physical state, uses, nature, and physiological modulatory effects. Moreover, their impacts on the environment and by extension humans' health are of great concern since protracted contact with even minute concentrations may give rise to toxicities evident by mutagenic, carcinogenic, or teratogenic effects [39]. Due to the increase in population growth and the rise in urbanization, several human activities have led to widespread environmental contamination by xenobiotics of different kinds. Thus, the high demand for pharmaceuticals cum the increasing population growth has placed the public at risk of exposure to xenobiotics. In addition, the development of illicit drugs has further led to the release of harmful pollutants into the environment. The outcome of this environmental pollution posed numerous short- and long-term consequences on the natural ecosystem.

3.1 Reported Xenobiotics Found in Herbal Preparations

Xenobiotic toxicity is a great threat to traditional medical practice and the use of herbal plants. It greatly impacts the quality of botanicals as starting materials, herbal extracts, and the safety, and consequently the marketability of THMs. Thus, effective quality assessment and control of foreign chemicals (xenobiotics) present in herbal plants used in food industries, pharmaceuticals, and traditional medical practice have become indispensable [42]. Although the use of THMs has witnessed a marked increase in developed countries over the last decades, many health dangers could be

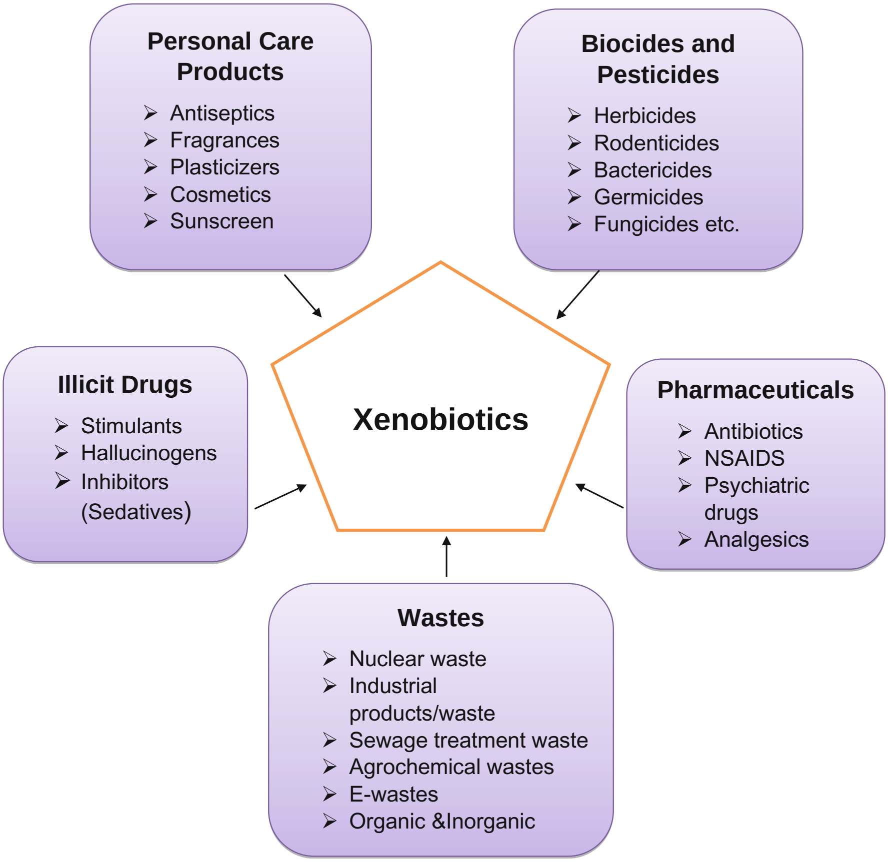

Fig. 1 Sources of xenobiotics

attributed to the THMs and their mixtures due to the contamination of agricultural soil, underground water, and even atmosphere [43–45].

Heavy metals including Nickel (Ni), Aluminum (Al), Lead (Pb), Cadmium (Cd), and Arsenic (As) which occur naturally in the environment have the potential to accumulate in both plants and human organs. According to the Environment Protection Agency, these metals are the commonest heavy metals associated with environmental pollution [46]. It has been well-documented that soil may be contaminated with these toxic metals due to processes such as automobile and industrial emissions, polluted irrigation water, atmospheric dust, and the use of agrochemicals (pesticides and fertilizers). The pollution of the environment ultimately results in the contamination of botanicals of herbal value [47]. Accordingly, the WHO has long emphasized the necessity of the quality control assessment of herbal products, such as heavy metals analysis [48]. In a recent study to examine the extent of herbal contamination by heavy metals using Inductively Coupled Plasma-Optical Emission Spectroscopy (ICP-OES), 25 pharmaceutical herbal products for infants and

15 traditional herbs used in Jordan were analyzed. Al, Pb, and Ni were detected in 76, 88, and 4% of the analyzed samples of pharmaceutical herbal products and in 87, 93, and 13% of the analyzed samples of THMs, respectively [49]. The data obtained ranged within the acceptable limits of toxic heavy metals based on the World Health Organization (WHO) specifications. The results also proved the extreme use of medicinal plants as an alternative medicine should be embraced cautiously while considering the safety levels [49]. Some heavy metals have been detected above the maximum permissible daily limits with the heavy metals' concentrations found between 9.61 and 52.74 μg/g for Pb while Cd was from 0.19 to 1.75 μg/g in herbal products examined in Iran [50]. Another research by Kohzadi et al. analyzed the concentration of heavy metals in eight diverse types of THMs and eight dissimilar kinds of herbal distillates marketed in Sanandaj, Kurdistan, Iran. The author found that the THMs contained a significantly higher level of Fe, Al, Pb, Cr, As, Cd, Mn, Zn, and Ni ($p < 0.02$) more than the herbal distillates [20]. A summary of several other case reports in the past can be found in the article cited [49, 51]. Undoubtedly, a high level of exposure to these metals has detrimental impacts on human wellbeing, as several of these metals are associated with abnormalities such as cancers and damage to certain organs [36]. Lead for instance besides being recognized as a potential carcinogen may also accumulate in the gray matter of the brain thereby destroying red blood cells and inducing neuronal damage and loss of synaptic connections [3]. Some other metals such as Mn, Cu, Zn, Fe, and Cr are essential nutrients for the normal biological functions and supporting physiological processes of the human body. However, higher concentrations above certain permissible limits might lead to toxicity of vital organs in the body [52] Table 1.

Another route of contamination of medicinal herb relies on the use of agrochemicals. Excessive microbial growth, the presence of pesticides, and orthodox drugs have been widely reportedly found in some herbal products [55]. Asian traditional herbal medicines are the highest studied types of herbal product. To ensure consumer safety, maximum residual levels (MRLs) have been established for pesticides [56].

Table 1 List of some heavy metals, their sources, potential toxic effects, and their maximum permissible limits in foods [53, 54]

Toxic metals	Uses/sources	Toxic effects	Permissible limits (mg/l)
As	Pesticides	Skin diseases and lung cancer	0.1–0.2
Cd	Plastics, pigments, batteries, plating	Kidney damage, lung cancer, and bone disorder	0.05–2.0
Cr	Dyes, alloys, tanning	Liver and kidney damage allergic dermatitis, respiratory effects	0.05–0.15
Pb	Batteries, wire, and cable, alloy	Neurological effects, hematopoietic system damage, and reproductive effects	0.01–3.0
Hg	Chloro alkali industry, pesticides, thermometer, batteries	Neurological effects, kidney damage	0.05–1.0

According to research by Wang et al. to determine the presence of about 168 pesticides from analyzing 1017 samples of Chinese herbal medicines, 76.0% of the investigated samples contained multiple pesticides' residues, contrasting the small fraction of 10.8% which was free of the targeted pesticides' residues [57]. In another development, a recent publication summarizing the findings from routine inspection targeting organochlorine pesticides in Chinese herbal medicine, spanning over 10 years, 1665 samples for 37 kinds of Chinese Herbal Medicine were analyzed employing the Quick, Easy, Cheap, Effective, Rugged, and Safe (QuEChERS) method, coupled with gas chromatography (GC) featuring electron capture detection (ECD). Based on Asian pharmacopeias acceptable maximal residue levels for organochlorine pesticides, pentachloronitrobenzene (PCNB) contamination of *Ginseng radix,* and the total dichlorodiphenylethanes (DDT) and PCNB contamination of *Panacis quinquefolii radix* presents critical issues to worry over. Chien et al. [58] observed that based on Korean Pharmacopeia residue limit, DDT level exceeded the Asian acceptable maximum residue levels by nearly 60% in *Panacis quinquefolii radix*. The exceeding rate based on Japan Pharmacopoeia was about 50% while based on the Hong Kong Material Medical Standard, the residue exceeded the limit by nearly 20%. In addition, the PCNB level according to the Pharmacopeia of the Peoples' Republic of China exceeded the Asian acceptable maximum residue levels by 40% for *Panacis quinquefolii radix,* and 30% for *Ginseng radix* [58].

Being one of the most patronized Chinese herbal medicines, chrysanthemum (*Dendranthema grandilora*) is widely engaged as a heart-cleansing herb and blood detoxifying agent, owing to its potent anti-inflammatory, anti-oxidative stress, as well as anti-tumor effects [59, 60]. Pesticide residues in comparatively high levels have been found in commercial chrysanthemum flowers and chrysanthemums [61, 62]. For instance, according to research by Feng et al., a count of 25 pesticides were observed with incidences of detection exceeding 10% in 75 chrysanthemum tea samples [63]. Commonly observed pesticides were carbendazim, bifenthrin, and imidacloprid having a detection rate of 91%, 68%, and 63% being the top three pesticides respectively [63]. Furthermore, in a quantitative analysis of multi-pesticide residues involving some 50 commercial chrysanthemum-tea samples, at least 5–13 pesticides' signatures were identified in 56% of samples [59]. The risk of pesticide contamination of herbal medicine presents a worrying situation all over the world, and there is need for a more focused attention toward quality assessment and control measures for herbal drugs.

Similarly, mycotoxins in plant material represent another potential health risk on the use of herbal drugs. Mycotoxins which are secondary metabolic products may be secreted onto or into medicinal plants by some species of fungi. There are non-volatile compounds with a relatively low molecular weight. The most reported mycotoxins in herbal products are those produced by species of fungi of the genus *Fusarium, Aspergillus,* and *Penicillium*. Aflatoxins, ochratoxins, fumonisins, and tricothecenes, are the four major groups of mycotoxins. There are highly neurotoxic, mutagenic, hepatotoxic, carcinogenic, teratogenic, immunotoxic, and nephrotoxic. Hence, their presence in medicinal plants and food products represents an interesting area of research. The International Agency for Research on Cancer classified

Aflatoxin B1 as a carcinogen of Group-1 standing and is the most toxic mycotoxin [55, 64]. Mycotoxin contamination of medicinal herbal products has been investigated by various research groups which have reported elevated content of mycotoxins beyond the maximum acceptable limits [65, 66].

3.2 Sources of Xenobiotics Found in Herbal Preparations: Focus on the Global South

Several anthropogenic activities account for the release of toxic xenobiotics into the environment, including Pharmaceutical and Personal Care Products (PPCPs) and their metabolites, cleaning products, surfactants, food additives, metalloids, microplastics, pathogens, livestock treatment and excretion, wastewater discharges from sewage treatment plants and industrial effluents, industries and production plants and associated industrial formulations, nanomaterials, rare earth elements, and of course a wide range of agro-industry related biocides [41]. Other sources of herbal contaminants of biological origin include bacteria and bacterial spores, viruses, yeasts, molds, protozoa (amoebae, helminths) nematodes, insects, and such other organisms as earthworms, acarus [67, 68]. The prevalence of microbial pathogens in herbal preparations could be high, posing as a further complication to public health concern especially for elderly ones. De Sousa Lima et al. [69] investigated 132 herbal preparations donated by users for microbial analysis. They reported 51.5% and 35.6% had bacterial and fungal growth, respectively while analyzed water samples showed 77.85 and 66.7% were positive for total coliforms and *E. coli*, respectively [69]. The predominantly observed microbial contaminants were *Salmonella* spp., *S. aureus*, *P. aeruginosa,* and *E. coli* with percentage isolates in the order of 49.2%, 34.8%, 25.8%, and 14.4%, respectively. Let's take a closer look at the sources of xenobiotics in herbal preparations.

3.2.1 Agrochemicals, Pharmaceutical, and Personal Care Products

Herbal drugs are known for containing pesticide residues, which may have accumulated due to exposure from agricultural practices, contact with contaminated soils during farming, such as spraying, and administering fumigants during storage [3]. The broad term Pesticides comprises chemicals used to eliminate pests and are classified as nematocides, fungicides, insecticides, herbicides, rodenticides, and others depending on their specific activities [3, 70]. Structurally, pesticides are categorized under organochlorine pesticides (OCPs) such as hexachlorocyclohexanes (HCH) and dichlorodiphenylethanes (DDT)]. Others include organophosphorus pesticides (OPs) such as malathion, dichlorvos, and parathion; nitrogen-containing pesticides such as atrazine and propazine; pesticides derived from plants such as rotenoids and pyrethroids, among others. Between 1990 and 2016, the surge in the application of pesticide-active ingredients per unit of cropland attained 75% increase according to UNEP Report on Pesticides and Fertilizers [71]. The database for European Union pesticides consists of about 1426 pesticide active ingredients out of which only 476 are currently approved for use, 901 of the ingredients are banned, and 49 are still pending approval [72]. Pesticides whose application is directly to the soil may be carried by rainwater run-offs into water bodies such as rivers while others may yet percolate into the groundwater. Pharmaceutical and Personal Care Products consumed by humans are metabolized into different metabolites, among which there may be derivatives whose toxicities may exceed that of the parent molecule(s) and, upon release from the body, they get into environment [41]. After excretion, toxic metabolite ends up in sewage/wastewater treatment plants from where there are partially removed through treatment and then eventually dumped in receiving water bodies such as lakes, rivers, oceans, as well as other environmental matrices such soil, groundwater, etc. Agricultural soils can also be polluted by the unsuitable disposal of unwanted or expired pesticide packaging, pesticides, and during cleaning of application equipment [43]. Alternatively, it could be by accidental discharge of pesticides from damaged containers, leaky pipes, underground storage tank spills, or intentional deposits at waste dump sites [43, 73]. Both Pharmaceutical and Personal Care Products and pesticides enter the food chain through absorption into the food chains either by plants, aquatic organisms, or animals. These residual pesticides and their metabolites, or degradation intermediates could remain in plants, or in the soil eventually becoming a source of contamination for THMs [29].

Organochlorines stimulate the central nervous system which could result in tremors, hyperexcitability, and seizures. Although organochlorine pesticides generally do not elicit immediate toxic effects compared to organophosphates or carbamates pesticides, they are extremely hazardous due to their environmental persistence and owing to their higher lipophilicity, they tend to accumulate in animals' tissue as they move through the food chain [43]. There have been worldwide reports of residual organochlorine pesticide and their degradation products detected in soil, plants and animals' tissues and human breast milk [55]. The foremost harmful effects linked to overexposure to organophosphates are symptoms of the nervous system, such as paresthesia, discoordination, dizziness, headache, convulsions, or tremor. Organophosphates inhibit acetylcholinesterase resulting in the accumulation of acetylcholine in the nerve tissue and at the effector's organ, and culminating in the sustained stimulation of cholinergic synapses. Exposure to organophosphate results has also been associated with delayed neuropathy [3, 70]. The European Pharmacopeia and United States Pharmacopoeia acceptable limits for some common pesticides are summarized in Table 2.

3.2.2 Industrial Release

Xenobiotics such as toxic chemical substances may be discharged during a production process or as a final product. They may be released into the environment intentionally or accidentally, from automobile or industrial emissions which may cause widespread contamination of the ecosystem. Gas emissions associated with the operation of automobile and coal-fired power generating stations release toxic chemicals into the environment which can impact human health. The acid gases and greenhouse gases emitted during the combustion of coal contribute substantially to acid rain and have been linked with global warming [74–77]. The generation of

Table 2 Examples of some common pesticides and their residual limits in Medicinal Plants [55]

Substances	Ph. Eur. And USP
Dichlorodiphenyltrichloroethane (DDT)	1.0
Malathion	1.0
Diazinon	0.5
Parathion	0.5
Alachlor	0.02
Aldrin and dieldrin	0.05
Dichlorvos	1.0
Dithiocarbamate	2.0
Fonofos	0.05
Chlordane	0.05
Hexachlorocyclohexane	0.3

Ph. Eur. And USP (European Pharmacopeia and United State Pharmacopeia)

particulate matter and such gases as sulfur dioxide and nitrous oxides is associated with coal-fired power plants employed in electrical generation [78]. These substances may combine through chemical reactions to form fine particles in the atmosphere [78]. The reactive chemicals may spread over thousands of kilometers from the source of emission and result in the widespread distribution of xenobiotics. Besides the environmental harm orchestrated by greenhouse gases, acid rain, and other emissions, the contaminated air emitted from coal-powered electricity generating stations harbors a significant level of toxic compounds that negatively impact on both plants and human health [78, 79].

Medicinal plants are easily contaminated through absorption of heavy metals released into the water, soil, and air. Mostly, the soil is contaminated with toxic metals through atmospheric deposition from point sources such as an array of industrial processes, rainfall, atmospheric dust, and the application of plant protection agents [78]. Consumption of heavy metal-contaminated herbal drugs allows the entrance of the toxicants accumulated in herbs into the human body which can upset the normal functions of various organs in the body as earlier discussed. Among the heavy metals widely studied, Pb, As Hg, and Cd are very toxic and mutagenic even at minute concentrations. Instances abound where malformation of organs, physiological abnormalities, and ill health in humans have been traced to heavy metals toxicity have been reported [3, 20, 49, 80]. Besides humans, animals and plants also suffer upon exposure to toxic levels of heavy metals.

3.2.3 Wastewater and Irrigation

Contaminant constituents and concentrations in wastewater depend on the process producing the effluents and the technology employed in their treatment [43]. In developing countries, wastewater has been reportedly used for irrigation purposes in agriculture, especially in farmlands located within proximity of cities where they are generated. The practice of recycling wastewater benefits agripreneurs who harness the abundant wastewater loaded with organic matter cheaply serving dual purposes, namely irrigation and as organic fertilizers [81]. However, this practice exposes the plants to trace elements such as heavy metals and organic contaminants notably

POPs and PFAS, which abound in wastewater as well as represent a source of harmful germs, pharmaceuticals, and personal care products which are potentially toxic to plants [82, 83]. Toxic elements present in wastewater can pollute water supplies such as surface waters from rivers, streams, and impact agricultural soils, the environment, and humans either directly or through the food chain. The crops and medicinal plants become exposed to contaminants and may accumulate unwanted quantities of potentially toxic elements. Consumption of plants irrigated with untreated wastewater poses a high risk for human health due to the potential exposure to pathogens, and uptake and bioaccumulation of potentially toxic chemicals in edible parts of the plants [43]. Increased levels of heavy metal accumulation in plants have been reported from areas with a protracted history of growing on untreated and/or treated wastewater, previous dumpsites as well as plants growing along heavy traffic ways [3, 78].

3.2.4 Wastes Disposal

Agricultural and industrial wastes must be collected and managed in an environmentally safe manner. Poor waste management represents a potential route for xenobiotics entering the environment. For example, non-biodegradable irrigation tubing, mulching films, small tunnel films, and protective fleeces used in agricultural farms need extra care in their post-use management to ensure they are not left behind on site, thereby causing environmental pollution. Polystyrene seedling trays must be collected and removed from the fields after planting season [43]. In the event of poor waste management, the possibility of them being dumped indiscriminately such as in the open countryside, or disposal in open space is high which must be minimized. The wastes may subsequently be burnt in the open space to reduce volume. Toxic emissions and partially combusted particles containing contaminants such as the polychlorinated dibenzo-para-dioxins (PCDDs) and polychlorinated dibenzofurans (PCDFs) generated from such uncontrolled fire can then pollute agricultural soils and bioaccumulate in the food chain at both short and long-range [43].

3.2.5 Herbal Contamination by Adulterants

Adulteration of herbal preparations could be regarded as the deliberate or unintended introduction of other materials possessing different chemical properties hence inferior or differing therapeutic effects other than the original material [1]. The challenge of herbal medicinal products adulteration [35, 84] and associated cases of adverse drug reactions [85, 86] has become a problem with global spread [87]. Of 5957 herbal products surveyed using DNA-based methods, Ichim [87] reported the continental prevalence of herbal products adulteration was in the descending order of 79%, 67%, 47%, 33%, 27%, and 23% for Australia, South America, Europe, North America, Africa, and Asia, respectively. The Federal Drug Agency (FDA) of the US Tainted Supplements Report covering between 2007 and 2012 mentioned 332 dietary supplement products were adulterated; 95% of which were classified as sexual performance enhancers [84]. Sexual performance-enhancing products are commonly linked with contamination. For instance, these products are reported to contain synthetic phosphodiesterase type-5 (PDE-5) such as tadalafil, sildenafil, and vardenafil. These have up to 50 analogs with few structural modifications capable of eliciting similar pharmacological properties as the original products [88, 89]. Their contaminating presence raises questions, about whether the acclaimed therapeutic effects of such products could be attributed to the constituent botanicals or the contaminants. Similarly, Skalicka-Wozniak et al. [6] opined the associated adulterants expose consumers to adverse side effects, other than the desired pleasure outcomes. Herbal products for slimming purposes are another group of blockbusters owing to the global distribution of obesity. Often, marketed products for slimming down are commonly contaminated with sibutramine hydrochloride monohydrate (an anti-obesity drug that functions through the inhibition of serotonergic and noradrenergic reuptake) with its unapproved analogs [6]. Another study in Brazil reported a similar trend of sibutramine contamination from 106 sampled herbal preparations [90]. The Dutch national pharmacovigilance center Lareb reported about 15% of total reported serious adverse drug reactions traceable to adulterating contaminants. The authors [86] reviewed twenty-six herbal medicinal preparations and reported the presence of adulterants such as pollens, rodents, dust, insects, fungi, microbes, parasites, toxins, mold, pesticides prescription drugs and/or toxic heavy metals. These contaminants and adulterants are associated with severe clinical conditions such as multi-organ failure, metabolic acidosis, renal or liver failure, and perinatal stroke, among others [86]. Sadly, other contamination may be deliberately caused by vendors to increase the commercial value of their herbal products. Some of the deliberate means achieving such goals include substitution with subordinate commercial varieties, use of artificially manufactured substitutes, replacement with expired drugs, incorporation of worthless heavy materials of similar color or looks, introduction of inferior drugs [1].

3.2.6 Naturally Occurring Xenobiotics in Plants Used for Herbal Preparations

We have explored anthropogenic activities which lead to environmental pollution and other natural factors, which culminate in the pollution of plants and hence herbal medicine preparations sourced from such contaminated plants. However, there is another dimension to herbal products contamination. These could be associated with natural processes of the plants or associated plants in the field. The plant families of *Fabaceae, Boraginaceae*, and *Asteraceae*, produce pyrrolizidine alkaloids (PAs) as a chemical means of self-defense mechanism against herbivores [37]. While growing on the field, PAs were shown to be transferred from a producer *Senecio jacobaea* to acceptor plants grown within the vicinity of the producer, hence indicating a situation of producer-acceptor exchange hence cross-contamination of other plants with PAs, in a manner previously unknown, but thought to occur through the roots of the producer plants thereby contaminating surrounding soils [91]. The presence of PAs in plants other than producing plants could have far-reaching consequences on consumers of such plants derived products. The PAs have been detected in herbal preparations or formulations such as medicinal teas, phytopharmaceutical formulations, or other plants-based drug products [37] which imparts negatively on the medicinal value of the contaminated botanicals. The presence of weeds has been associated with the development of pyrrolizidine alkaloids (PAs) which has become a major source of contamination and have been discussed among health authorities for several years, culminating in the establishment of limiting values for PA content in herbal preparations [38]. They elicit toxic effects with the lungs being the primary target organ as the site of metabolic transformation [92]. PAs-induced toxicity does not spare the liver and other tissues of humans and livestock [92]. This class of natural xenobiotics has become important in quality control since their composition is affected by species of plant, location of growth, age of the plant with content getting to its peak preceding the flowering stage or early bud [93]. Such phenomenon poses constraints on the path of harmonizing the active ingredients content of herbal preparations.

3.3 Impact of Transportation of Xenobiotics on the Environmental Distribution of Pollutants

Transportation of xenobiotics is facilitated by both abiotic and biotic media. Among the two, abiotic processes hold a greater sway on the environmental distribution of pollutants. This entails the displacement of contaminants present in water, air, as particulate matter and transported by wind, or by the action of water, or soil solution, or even soil masses [94]. Once introduced into the environment, part of the xenobiotics (e.g., pesticide) get adsorbed to soil debris, and yet others become bound with the liquid and/or vapor phases. The extent to which a potentially toxic chemical is distributed between the liquid, adsorbed, and vapor phases greatly influences its subsequent distribution in the environment. Because xenobiotics may exist in all three phases, several transport routes influence their environmental distribution [95]. Potentially toxic chemicals such as agrochemicals applied on the surface of soil may degrade in situ or be transported from the target area through processes such as leaching into groundwater, runoff into nearby streams, and/or volatilization into the atmosphere.

3.3.1 Winds

Pollutants present in or adsorbed to soil can be transported around and offsite as aerosols or particles blown by the winds. Winds may transport only small particles, except in rare occasions where nonselective particles as much as a grain size can be easily transported as observed during hurricanes or cyclones. Thus, wind facilitates the movement and dispersion of xenobiotics to distant sites far away from point sources [94]. Wind transportation of chemicals – persistent organic pollutants, for instance – could be either long-range or short-range. Long-range transport potential (LRTP) for persistent organic pollutants (POPs) – most of which are xenobiotics – is determined by the strength of the wind, and weaker turbulence [96]. Thus, at higher atmospheric levels, xenobiotics could be transported over a thousand kilometers. Several factors are significant in facilitating air movement especially in long-range atmospheric transport (LRAT) involving persistent organic pollutants (POPs). This LRAT is reportedly responsible for the detection of POPs in extreme environments.

Tibetan Plateau in China is the largest and highest plateau on earth, a location which might be thought to be free of POPs. However, Wang et al. [97] reported the presence of a wide range of POPs on Tabetan Plateau in quantities that far exceeded those measured at other mountainous areas. Predominant POPs were identified as dichlorodiphenyltrichloroethane (DDT) and hexachlorocyclohexane (HCH) besides recently emerging compounds including polyfluoroalkyl substances and hexabromocyclododecanes, which have been widely distributed in fauna on the plateau [97]. They attributed the POPs' presence on the plateau to the LRAT impacts of The Indian Monsoon, East Asian Monsoon, and Westerly winds with polluting sources identified as being from the surrounding counties. Particulate matter (PM) size is instrumental in the transportation of POPs. Odabasi et al. [98] investigated particulate matter distribution in air from an industrial and urban settlement in Izmir, Turkey [98]. They reported proximity to the source of emission heightens the air content of POPs and holds the prospects of further contributing to the potential toxicity to surrounding plants when employed in herbal medicinal applications. Size distribution of particulate matter shows preferential deposition (>50%) in the fine particulate matter ($d_p < 2.5$ μm) because fine particles have higher sorption capacity than coarse particles (2.5 μm $< d_p <$ 10 μm) due to the relatively higher organic content component of fine PM compared to the coarse PM [98]. Other factors such as temperature and the octanol-air partition coefficient (K_{OA}) of POPs increase fine PM composition content and POPs composition, while increasing temperature decreases fine PM with significant correlations for up to 51% of studied compounds [98]. Other factors linked with LRAT are increasing height associated with wind speed, temperature changes, and reduction in decaying effects owing to surface friction [96].

3.3.2 Surface Runoff

Chemicals in surface runoff occur by mechanisms involving the erosion of sediment with adsorbed chemicals and the dissolution of chemicals into the runoff water [95]. Erosion of solid soil particles along with adsorbed xenobiotics occur through the independent or combined action of air, water, and temperature. It is facilitated by increase in the exposed surface area of the soil. In areas with a steep gradient, mass-wasting processes such as creep, slides, rock flows, debris flows, rockfalls, slumps, and block glides of the eroded materials are observed due to the influence of gravitational force. Environmental distribution of pollutants is also possible via transport in water in the form of particulate matter or in dissolved form; in ground or surface waters [94, 95]. In the case of surface waters, soil fragments are introduced into water bodies and flocculate as particles downstream by sliding, rolling, and saltation and are deposited further downstream. The rate or distance of flow depends largely on turbulence, flow velocity, grain size, shape, and density.

3.3.3 Leaching into Groundwater

Xenobiotics also frequently pollute groundwater by the process of leaching. Leaching is a process by which contaminants are freed from the solid phase into the aqueous phase orchestrated by the dissolution and desorption from their previously supported phases. Leaching of pollutants into groundwater is a vital process since it is closely linked with most pollutants of eminent environmental concern. The dissolved transport in waters allows pollutants to easily migrate to other sites or environmental compartments far from the source [94, 95].

3.3.4 Volatilization

The process of volatilization entails the release of a chemical (contaminant) from a solid or a liquid medium into the atmosphere or into the gaseous phase. Once in the atmosphere, potentially toxic xenobiotics may undergo degradation or become deposited in nontargeted areas through wet or dry deposition [99, 100]. Often, part of the chemicals volatilized into the atmosphere are transported and subsequently deposited in rivers, streams, and other important water bodies [95, 101, 102].

3.3.5 Drifts

Spray drifts represent another means of migration of xenobiotics into distant part of the environment away from source points. For instance, pesticide emissions can occur as spray drift in concentrated droplets or as pesticides attached to dust particles [103]. Xenobiotics spread from drift and volatilization can be transported through the atmosphere to forested areas and deposited directly into streams via tree wash-off.

Toxic elements and compounds emitted from industrial plants and the operation of automobile and coal-fired power stations release toxic chemicals into the environment.

Taken together, transportation of xenobiotics contributes greatly to the wide-spread distribution of xenobiotics and consequently results in risk to agricultural process located even far away from the source point.

4 Universal Variations in Botanicals: Impact on Quality Control of Herbal Preparations

The therapeutic impacts of botanicals are ascribed to the phytochemical variety, content, and composition present in the plant. Thus, fresh leaves tend to present better therapeutic effects and hence most employed in traditional medical practices. However, the presence of phytochemicals in plants varies with differences in environment where it is grown, climate, age of the plant/plant part, composition, soil components, and geographical region where the plant is grown [104]. Since these factors vary from one plant to another, and soil composition and components vary largely with anthropogenic activities; these factors as expected would contribute largely to the problems of standardization. Moreover, there are preferential seasons for harvesting different plant parts for use in herbal preparations. For instance, roots and rhizomes are better collected at the end of vegetation period, leaves and herbs at bloom, bark in the spring, flowers preferably at anthesis or shortly after opening, while fruits and seeds are collected after maturity or ripe [105]. During herbal preparations, aqueous formulations are administered.

The water used in preparing the solutions may be contaminated, as they tend not be a universally similar quality of water to use from wherever the formulation is being made. However, one of the pressing public health concerns of the twenty-first century remains clean water for human use [106]. Nigeria is a prime example of the situation in the global south. While water resources are abundant in most parts of the country [107, 108] especially during rainy seasons which occur between April and November in most parts of the country; drinking water remains an out-of-reach commodity for the majority. Some of the key sources of contamination of drinking water resources in Nigeria arise from such causes as agricultural activities, abattoir effluents, industrial effluents, leachates, oil spill, scrap yards, and municipal waste [109] thereby contaminating even drinking water resources. For instance, only about 11% of Nigerians have access to pipe borne water, and only 4% premises are supplied with pipe-borne water [110]. The same source noted massive contamination of drinking water with *E. coli* at 68% and 70% at the source and point of consumption, respectively. The presence of pollutants in water implies the polluting contaminants are added into the herbal preparation, further contributing to the problem of loss of standardization.

5 Improvements in Quality Control Strategies: A Panacea for Herbal Products Contamination

Different kinds of contaminants have been associated with herbal preparations. Commonly mentioned ones include microorganisms, fungi and associated mycotoxins, heavy metals and other elemental impurities, trace organics such as pesticide residues, pyrrolizidine alkaloids through weeds [8, 11, 38, 111]. As the kinds of contaminants vary widely, so also their sources. Therefore, an effective quality control strategy will involve an all-encompassing approach, covering the different kinds of contaminants identified. All these are applicable in the global south, hence the need to address all of them. The entire discussion would be beyond the scope of a single book chapter. The kinds of approaches employed would include chemical, biological, microscopic, physical, and biological evaluations [10]. Several analytical techniques are being engaged in checking the integrity of herbal preparations. The ever-increasing analytical sensitivity associated with high throughput DNA sequencing enables its use in determining untargeted contaminants and simultaneous multi-taxa identification [87] has presented special prospects in improving the quality of herbal preparations. While chemical analytical techniques can identify and quantify these contaminants and toxins, they usually suffer a limitation in their ability to the source of the contaminant, a gap DNA barcoding leverages upon [112, 113]. Besides, using amplicon metabarcoding, labeled ingredients are verified and substituted, adulterated, and added species are identified, thus further proving to be an indispensable tool in quality control assessment. Several other techniques have been applied in the detection of contaminants in herbal preparations. Elemental analysis has employed the techniques of graphite furnace atomic absorption, flame absorption spectroscopy, cold vapor atomic absorption [111], and inductively coupled plasma emission optical emission spectroscopy, ICP-OES. The hyphenated analytical techniques of Gas Chromatography-Mass Spectrometry (GC-MS), High-Performance Liquid Chromatography-Mass Spectrometry (HPLC-MS), Liquid Chromatography-Nuclear Magnetic Resonance (LC-NMR), High-Performance Thin Layer Chromatography-Mass Spectrometry (HPTLC-MS) [114], Fourier transform infrared spectroscopy (FTIR) as well as simpler analytical techniques such as column chromatography [5] have particularly played key roles in the chemical evaluation of HMPs. Thus, the applications of these techniques in regulatory laboratories, manufacturing companies, academic research spaces can tremendously ensure the appropriate quality of herbal products are available for sales and consumption. Though the costs be large, there is no short cut to take to ensure health and safety of the hundreds of millions of inhabitants found in the national boundaries of the global south communities.

As often observed, the global south is composed of evolving economies lacking the financial capacity to implement the array of analytical techniques outlined above. While this is a disadvantage, it is also a huge advantage to the global community in disguise. Here in this region lies large numbers of unexploited natural products of promising immense therapeutic importance. Collaborations between academics, governments, and Civil Society Organizations (CSOs) will bridge this gap and release the avalanche of untapped resources for the benefit of humanity.

6 Summary

Xenobiotics are commonly identified in herbal medicinal products across the globe. As contaminants, they vary in composition and content. This variation is attributable to environmental factors, geographical location where the plant was sourced, age of the plant and even intraspecies variations are reported, among several other factors. Some of the major identified contaminants may be classified under different classes viz.: pharmaceuticals, pesticides (agrochemicals), other industrial products/heavy metals, nuclear wastes, illicit drugs, and personal care products. These contaminants could be contained in industrial discharges, untreated or partially treated wastewater effluents, and irrigation water, solid and e-waste disposals, among others. The contaminants could be transported from point release to further distances and distributed to other locations by winds, surface run-off, leaching, volatilization, and drifts. Different analytical techniques have been employed in identifying and sometimes quantifying these contaminants. Often employed analytical techniques are inductively coupled plasma emission optical emission spectroscopy, ICP-OES, HPLC, HPLC-MS, LC-MS, GC-MS, capillary electrophoresis, and high-throughput DNA-based sequencing techniques. Naturally occurring xenobiotics are another class of compounds, such as pyrrolizidine alkaloids, that have been identified in herbal preparations with established fatalities. Although the universal composition of botanicals varies largely, with the application of stringent quality control procedures, the problem of xenobiotics contamination can be minimized. This will instill efficacy hence confidence in such products, boost economic activities around

the trade in herbal products, lessen public health burden, and achieve the desired goal of using traditional medicinal products. Most importantly, it will cater for the rural dwellers of the global south, whose readily available medical relief products are the traditional herbal preparations, hence increasing economic viability and general well-being.

7 Conclusion

Herbal medicinal products (HMPs) are a viable means of global trade and beneficial in sustaining fragile health care systems around the globe, especially in the developing global south. Plagued by contaminants, the good intent for their consumption may become ill-fated, owing to the presence of contaminants. Contaminants which are of different kind and arises from different sources such as microbial, elemental, pesticides' residues, and natural sources pose serious threats resulting in adverse drug side effects and sometimes death. These must be controlled through carefully crafted, and economically feasible quality control measures, spearheaded by governmental regulatory bodies, researched upon and regularly checked by the academic researchers. CSOs may lead the awareness campaigns while ensuring appropriate information is disseminated to all stakeholders in the value chain of HMPs. The global south must form collaborations with developed economies to identify and monitor for contaminants of public concern as a measure toward monitoring for the control of their presence in herbal products.

8 Recommendations

- Following national surveillance over a 30-year period, van Hunsel et al. [85] concluded that pharmacovigilance of herbal products in necessary to guarantee citizens' safety following consumption. While the Dutch national pharmacovigilance lab and database at Lareb offers such services, in many countries of the global south such as Nigeria, a detailed analysis of herbal medicinal products may be lacking, hence exposing the citizens to further medical complications.
- Comprehensive value chain overhaul of the HMPs is necessary, especially for registered products. Since these are often sourced from botanicals, comprehensive soil analysis and air monitoring should be periodically carried out to establish best-growing conditions for cultivated botanicals targeted for HMPs production. The siting of such farms should also consider environmental factors such as topography, presence of point source pollution releases.
- Registration of HMPs with statutory regulatory bodies should be carried out to ensure standardization and quality control measures are achieved and maintained. This will enable the producing companies to be accountable to the governments and consumers for the products they sell. This is challenging in the global south, where most herbal preparations come from local traditional medical practitioners. On the flip side, such attend to fewer persons hence limiting the scope of

consumers and consequently the extent of public health implications that may be recorded. Even though small, the impacts must still be minimized through locally sustainable approaches,

- Local dwellers should be enlightened on the side effects and potential toxicities associated with herbal medicinal products, to minimize wanton consumption and limit it to only in cases where necessary.
- Traditional medical practitioners should be enlightened on the best practices to employ in harvesting, preparations, storage, and administration of botanicals and resultant herbal preparations to minimize the associated adverse effects. This could be part of a national or regional quality control approach in ensuring public safety and improving public health, as was also recommended previously [11].
- Registered HMPs must be monitored to ensure adherence to production processes just like conventional pharmaceuticals production processes are under the radar of regulatory agencies. Also, labels must be checked to ensure conformity with actual ingredients content.
- Furthermore, community advocacy by learned persons, institutions, concerned civil society organizations, national bodies associated with HMPs and traditional medical practices should be done. This will enlighten stakeholders and through these efforts improve public health and safety while maximizing the anticipated health, economic, and traditional gains associated with the herbal products.
- Regulatory agencies must step up their activities to ensure the safety of the populace. This is beneficial especially for licensed products. This should be surveyed regularly to ensure consistency with labels and uniformity of constituents. This is important since the active ingredient content of botanicals may vary with season, site of cultivation, and several other factors that are highly variable.

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Herbal Medicine for Health Management and Disease Prevention

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Abstract

Herbal remedies have been developed and used since the dawn of human civilization. Herbal medicines preponderantly deal with the medicinal application of a broad spectrum of plant regimes for health management and disease prevention. Wider varieties of algae, fungus, and lichen exudates have also been considered for herbal medicine using traditional methods such as infusion, decoction, maceration, distillation, expression, fractionation, and purification. However, conventional channels are insufficiently successful to generate significant demand for industrial-scale manufacture of herbal medicines. Systems synthetic biology and metabolic engineering have proved important in boosting productivity and cutting costs in the production of herbal drugs in this area. Additionally, there are regulatory and bioethical problems for health management and illness prevention at the current state of the art that make it difficult to mass-produce herbal medications. Based on this current scenario, current chapter deals with general outline on herbal medicines; its impact on health management and disease prevention; impact of system synthetic biology and metabolic engineering to accelerate herbal medicine development; and brief outline on regulatory-bioethical aspects upon usage of herbal medicine for health management and disease control.

Keywords

Herbal medicine · Health management · Disease prevention · Systems synthetic biology · Metabolic engineering

Abbreviations

4CL	4-coenzyme ligase
ADRs	Adverse drug reactions
AYUSH	Ayurveda, Yoga and Naturopathy, Unani, Siddha, and Homeopathy
BEO	Bergamot essential oil
BUN	Blood urea nitrogen
CHS	Chalcone synthase
CodR	Codeinone reductase
DAC	Department of Drugs and Cosmetics
DR	Drug resistance

EGFR	Estimated glomerular filtration rate
FDA	Food and Drug Administration
HIV	Human immunodeficiency virus
HSS	Homospermidine synthase
ILK1β	Suppresses interleukin-1β
IPA	Interpretative phenomenological analysis
LCY	Lycopene cyclase
LIS	Linalool synthase
MPNPs	Medicinal plant natural products
NHPR	Natural Health Product Regulations
NHPs	Natural health products
PKs	Polyketides
RIAD	RIα-binding peptide anchoring disruptor
RIDD	RIα-binding peptide detector disruptor
RNAi	RNA interference
SARP	Streptomyces antibiotic regulatory protein
SDG	Sustainable Development Goal
SIRT3	NAD-dependent deacetylase sirtuin-3
TGA	Therapeutic Goods Administration
UAE	Urinary albumin excretion
UTIs	Urinary tract infections
VEGF	Vascular endothelial growth factor
WHO	World Health Organization

1 Introduction

Herbal medicine is the oldest and traditional way of pharmaceutical industry. In the arena of medical products, it has been crucial. Natural products always have come from plant roots, flowers, fruits, seeds, leaves, bark, fruits peels, even entire plant, etc. They consist therapeutic activities and primitive human world, and most of the old healthcare have depended on natural ingredients of medicine rather than synthetic medicine. In India, the forest produces over 90% of the herbs and medicinal plants. Between 1950 and 1970, almost 100 distinct plant-based medications were released to the American market, including the higher plant-derived medicines vincristine, deserpidine, and rescinnamine [1]. When chemical analysis first became possible in the early nineteenth century, scientists started to extract and alter the active components from plants. Later, scientists started creating their own versions of plant chemicals, which led to a gradual fall in the usage of herbal remedies in favor of pharmaceuticals. Recent demands for drug production and quantity have increased due to increased resistance to pathogenic microorganisms, and traditional methods of the production of natural medicines are unable to meet these demands. Sustainability of production and green development has become increasingly prominent demands. Life science has a modern interrelated part synthetic biology that has an impact on traditional metabolic engineering and concept of systems biology [2].

2 General Outline and Classification of Herbal Medicine

Herbal medicine, according to World Health Organisation (WHO) guidelines, encompasses herbs, herbal materials, active components with therapeutic actions, and even negative consequences. Herbs are the roots, flowers, leaves, bark, fruit peel, seeds, gums, and so on. Herbs are classified into distinct categories based on their applications and activities. There are different types of herbs as they have different uses and activities. Herbal materials refer to some extra processes on herbs to get a typical extract. These extra processes are may be boiling, blending, grinding, and even by soaking the herbs. By these processes, extract product may be fruit juice, powder of seeds, peels, and barks, even dry petals of flowers, leaves, etc. Active ingredients mean why the herbs are active or effective? Herbs must include some element that allows them to function as active or effective products, which can be any enzyme, vitamin, antioxidant, antibacterial, therapeutic activity, and so on. The entire herbal remedy may be regarded as an active ingredient when it is impossible to identify the active component. Successfully preventing, diagnosing, and treating bodily and mental illness is referred to as therapeutic action. Side effects are any unintended consequences of using a drug at its recommended dosage for treating humans [3]. "Preparations manufactured industrially consisting of active ingredient(s) that are/are purely and naturally original, not chemically altered plant substance(s), and are/are responsible for the overall therapeutic effect of the product," is another description of a herbal medicine. Consumers are turning to plant-derived preparations because they believe that since HMs are natural products, they are safe – if not safer – than conventionalized pharmaceuticals [4]. The quality of a natural product is also determined by user feedback. Categorization of a full herbal product occurs in a variety of methods.

2.1 Classification of Herbal Remedies According to Their Uses

The majority of culinary herbs are spices that are used to flavor food and have been utilized in traditional medicine for thousands of years due to their presumed and known health benefits. For instance, mint, curry leaves, fennel seeds, fenugreek seeds, black pepper, turmeric, ginger, garlic, parsley, cinnamon, clove, cardamoms, honey, etc. Ornamental herbs mean they are used for decoration and fragrance of surrounding air because of their colorful flowers, leaves, and good smell. Some of these are lavender, rose petals, myrrh lemon leaves, *Clitoria ternatea* (common name bluebellvine), rosemary, chives, oregano, aloe vera, etc. Medicinal herbs consist any of one protectiveness such as antimicrobial, antioxidant activities, etc., by which they can prevent disease by internally and externally. Most of the medicinal herbs are common as culinary herbs such as mint, fennel seeds, fenugreek seeds, neem, turmeric, ginger, garlic, cinnamon, clove, cardamom, honey, curry leaves, etc. Besides more herbs are vasaka, hibiscus leaves, green chiretta, *Hygrophila auriculata* (common name kokilaksha), *Centella asiatica* (thankuni leaves), aloe vera, Indian snakeroot, *Bacopa monnieri* (common name water hyssop), etc.

2.2 Classification of Herbal Medicine Based on the Period of Life

Annual herbs, such as anise, basil, borage, parsley, saffron, and others, blossom for only one season and then die. Biennial herbs are plants which live more than one or up to two seasons like caraway seeds, prime rose, etc. Perennial herbs grow for more than two seasons. Some of these are aloe vera, lavender, bay leaves, rosemary, lemon balm, etc.

2.3 Classification of Herbal Medicine Based on the Properties

Aromatic herb, as the name suggests, gives a pleasant odor, for example, lavender, rose petals, or a strong odor to reduce bad odor. For example, ginger, myrrh lemon leaves, clove, etc. Bitter herbs contain phenols, phenolic glycoside, alkaloids, and saponins. They commonly act as blood purifier like neem; besides, they contain anticatarrhal, antipyretic, vermifuge activities, e.g., aloe vera, cascara, pumpkin, etc. Diuretic herbs cause the body to lose fluid through the urine system. Asparagus, blessed thistle, corn silk, and parsley are a few foods that are healthy for the liver, kidneys, and circulatory system. Polysaccharides found in mucilaginous herbs give them a slick, mild flavor that tastes pleasant when consumed with water. The majority of mucilages, including glucomannan from konjac root, Irish moss, althea, aloe, burdock, and fenugreek, absorb toxins from the stool instead of being broken down by the human digestive system. Herbs that are nutritious impart their nutrition to the consumer. They are primarily available as food products, but they also have some medical properties like fiber, mucilage, and diuretic activity. As necessary nourishment, they also include carbohydrates, proteins, healthy fats, vitamins, and minerals. For example, apple, asparagus, banana, barley grass, bilberry, broccoli, cabbage, carrot, grapefruit, hibiscus, lemon, onions, orange, papaya, pineapple, wheat germ, etc. [3].

2.4 Classification of Herbal Medicine Based on Extraction Processes

Some herbs are used after drying and grinding, like orange peel, fenugreek seeds, and hibiscus leaves. Some herbs are used after soaking, like curry leaves, fennel, ginger, mint, etc. Some herbs are used after blending, like vasaka, thankuni leaves, kokilaksha leaves, Indian snakeroot, etc. Two essential criteria or qualities that must be taken into consideration for an HM to be assessed and evaluated under the proper registration procedure are the presentation of the product and the purpose for which it is provided. Dietary supplements are only permitted in oral preparations under the USA regulatory authorities; all other preparation types fall under the category of botanical pharmaceuticals, which have the toughest registration requirements.

3 Traditional and Current Methods for Developing Herbal Medicines

As ancient as human civilization itself is the history of herbal medicine. Herbal remedies refer to a natural substance that must also evolve together with our lifestyle. Without growth characteristics, purity, efficacy, and manufacturing speed, herbal medication development is useless. Each phase of technique extraction and production is accountable. It is also necessary to investigate how they affect the body. A large portion of the Indian population is depending on the Indian system of Medicine – Ayurveda, an ancient science of lives till now [5]. But natural medicine has been excluded by regular uses of synthetic drugs due to disqualification of traditional production of natural medicine. This use of synthetic medicine acts as a threat to the environment and biodiversity as well. In order to acquire adequate enhancing capacity, it is essential to promote and integrate herbal medicines into primary healthcare. This should be done in a sustainable manner. It could lead to the development of new drugs, increased access to healthcare, the preservation of forest resources, and assistance in closing socioeconomic gaps and meet the Sustainable Development Goal (SDG) as well as the World Health Organizations' (WHO) promotion of herbal medicines. Several factors can influence traditional methods like (1) study of effectiveness of plant remedies, (2) an unhappiness with the effects of synthetic medications and a consumer perception that herbal treatment is superior, (3) the high cost and side effects of the majority of contemporary medications, (4) patients' beliefs that their doctors have not accurately diagnosed the issues, leading them to believe that herbal therapies are a superior alternative, and finally (5) herbal remedies need some development approaches that can influence their efficacy, quality, safety and budget friendly cost and conference to move towards herbal remedies like increasing quality of extraction of flavors or chemicals from plants by soaking in water, called infusion, and extraction of water-soluble products by boiling plant, roots, etc. A synthetic cell is transformed into a suspension of intact cell to get a better quality of medicine. There are further quality control procedures, such as distillation, which is used to distil all types of liquid goods, fractionation, which is used to separate subcellular components, and purification, which is used to check purity, among others. Before studying the approaches of development of herbal medicine, it requires to have proper knowledge on trends for traditional method or natural method of herbal medicine developments.

3.1 Conventional System's Effects on Basic Healthcare

Herbal medicine has benefited greatly from India's reputable and well-coordinated healthcare system. Before the growth and dissemination of contemporary science, traditional systems of herbal treatment were the reasonable and essential component of several cultures. Additionally, it suggests a number of efficient medications for the management of hepatic problems [26, 28].

3.2 Efficiency of Traditional System

The roots, leaves, bark, and other plant parts are used to make herbal products. They undoubtedly contain a variety of biologically and chemically active substances, as described in the point that follows. Plants are deemed medicinal if they exhibit pharmacological properties with potential therapeutic use. As a result of millennia of trial and error, these practices are typically well-known, but they need to be carefully explored if we want to develop fresh treatments that meet the demands of modern therapy.

3.3 The Objectives of This Field's Research

The active components of medicinal plants must be identified, and the safety, efficacy, and ongoing action of the extracts must be confirmed. It is possible to synthesize, structurally modify, or simply extract these active components with greater efficiency by the isolation of these substances and the characterization of their structure. Research on medicinal plants must follow a strict protocol. Simple technical mistakes frequently diminish the value of studies on natural goods. By ensuring the process of raw material cultivation, identification, recognition, collection, and preservation, a natural medication will become a well-liked herbal medicine among consumers. To ensure that the desired outcome of the products is achieved, it is necessary to process, store, and use raw materials and finished products in accordance with current Good Manufacturing Practices (cGMP). To hold this eco-friendly management, it must require development in each portion. Some strategies or approaches must be followed by this management. Manufacturing and formulation-based strategies: Making a safe and effective herbal medication requires several important procedures, including ensuring the correct number of recipients and revalidating the composition. Utilizing contemporary technologies and creating patient-centric innovative herbal formulations will increase consumer acceptance of the medication for both treatable and untreatable conditions. The right dosage for the right disease is a crucial component of disease prevention. Therefore, randomize clinical trials in multicenter, multinational, and multicultural settings are an essential strategy for spreading herbal medicine around the world.

3.4 Host-Based Strategies

Drug resistance (DR), which can be either inherent or acquired, refers to the failure of a drug's capacity to effectively kill pathogens or inhibit their metabolism. For instance, *Staphylococcus aureus* and *Neisseria gonorrhoeae* almost have immunity to survive in benzyl penicillin. In addition to developing naturally, drug resistance is also unnecessarily accelerated by sloppy practices in drug supply and use. Most of the microorganisms that cause these disorders have a resistance to first-line medications that ranges from zero to approximately 100%. Drug inactivation or modification, target site modification, metabolic pathway modification, and decreased drug accumulation are the four primary mechanisms for DR. Race and poverty are socioeconomic characteristics that have an impact on pharmacological therapy accessibility and adherence. Racial variety, host-defense differences, genetic shift and drift, and other non-modifiable limiting factors are cited to support the safety and efficacy of herbal treatment. Therefore, trials that are multicenter, multicultural, and multiracial may help to solve the issue of spreading herbal medicine globally.

3.5 Consumer Satisfaction-Based Strategies

Natural ingredients are constantly used to make herbal medicines. However, they also have certain detrimental effects on the human body. Typically, consumers expect herbal medicines to be pure, effective, safe, and affordable. If these demands are fulfilled, then the herbal medicine will gain popularity. Adverse drug reactions (ADRs) of an each and every drug must be followed by repeatedly taking a single dose for a specific period. Therefore, it is crucial to prescribe herbal medicines to consumers after conducting thorough clinical research and gathering vast amounts of data.

3.6 Laboratory-Based Strategies

Evaluation of drug absorption, distribution, chemical changes in the body, effects and routes of drug metabolite excretion, biochemical and physiological effects of drugs on the body or on microorganisms or parasites inside or on the body, mechanisms of drug action, and the relationship between drug concentration and effects (pharmacodynamics) are all crucial factors in the development of natural medicine. Drug-drug interactions, drug-food interactions, drug-plant interactions, and even drug-enzyme interactions have been seen after a certain drug has been applied, such as the human body, any microorganism, or any plant. Carelessness, ignorance of the active ingredients in the relevant medications, or incorrect handling could all be contributing factors to these interactions. The proper personnel must be hired at the right time and location, a biosafety level must be established, standard containments must be used, and high-tech equipment for in-vivo and in-vitro investigations must be used in order to achieve the goal of herbal medicine globally [28].

3.7 Quality or Fertility or Maturity of Raw Material-Based Strategies

All of the previously described raw material-based tactics cannot be successful without the quality of herbal medicine, fertility, or maturity. We know that the natural medicine is produced from living organisms such as plants, microorganisms, and animals, but when a natural bioactive product is coming from plants, it needs to be mature and pure within plant cells. Therefore, the entire plant needs to be grown and mature properly, which is doubtlessly a time-dependent process; otherwise, researcher or collector will not be able to collect proper stage of bioactive component. In that instance, if natural products can be extracted from microorganisms, the method will be simpler and faster. A microbial cell may mature and produce bioactive chemicals thousands of times faster than a complete plant. The role of microorganisms in the natural medicine have been discussed in many articles. It is best to start by noting that the maritime environment has a great variety of ecosystems that are adapted to a wide range of environmental factors, including temperature, nutrient availability, and lit/unlit areas. So, in the marine ecosystem, most dominant are sponges, some fungi, then sea whips, sea fans, soft corals, tunicates, opisthobranch mollusks like nudibranchs, sea hares, echinoderms like starfish, sea cucumbers, and bryozoans like marine algae, including green algae, brown algae, red algae, golden algae, and diatoms, as well as marine bacteria are also involved in producing effective bioactive compounds.

From 1928, Alexander Fleming's discovery of lysis and inhibition of the growth of *Staphylococcus* on Petri dishes colonized by the penicillin from fungus *Penicillium rubens* (named at the time as *P. notatum*, and until recently as *P. chrysogenum*) which was isolated from the fungus *Penicillium notatum*, makes believe that microorganisms are a rich source of unique bioactive compounds [29, 30]. There are many fungi, which have the capability to produce antibacterial bioactive compounds like statins and mevastatin from *Penicillium citrinum*, lovastatin from *Aspergillus terreus*, strobilurins from *Strobilurus* species, and so on. Not only fungus, there are many *actinomycete* bacterial species those can produce antifungal, antitumor activity, and so on. Some bioactive components and producing microorganisms are mentioned in Table 1.

4 Health Effects of Herbal Medication

From the beginning of human civilization, herbal medicines have lot of uses to cure disease. All herbal medicines are typically derived from six main sources, including plants, animals, minerals/earth, man-made and semi-man-made materials, microorganisms, and recombinant DNA technologies [3]. There should be a focus on health rather than sickness, according to well-known herbal medicine systems such as traditional Chinese medicine and Ayurvedic medicine. About 100 years ago, natural herbs were the mainstay of human ailment treatment. Eighty percent of people globally, or around $60 billion yearly, rely on herbal medicines for some part of their basic healthcare, according to a recent estimate by the World Health Organization. According to estimates, 25% of current medications – including aspirin, artemisinin, ephedrine, and paclitaxel – are derived from plants that were originally used traditionally. The use of plant-based pharmaceuticals and dietary supplements has recently increased. Researchers in the fields of pharmacology, microbiology,

Table 1 Applications of various herbal remedies as well as the bioactive chemicals they contain and the microorganisms that produce such compounds

Brand name of medicine	Precursor bioactive compound	Producing microorganism	Applications	References
DAFLON 500	Flavonoids	*Annulohypoxylon elevatidiscus, Xylaria papulis BCRC 09F0222, Tricoderma* species	It has antibacterial, antifungal, antitumor, anti-inflammatory, and antiaging activities	[8]
Mevastatin	Compactin	*Penicillium compactum*	Helps to maintain cholesterol	[9]
Amoxil, Bactocil, Nacfil, etc.	Penicillin G	*Penicillium rubens*	Prevent bacteria from synthesizing peptidoglycan and used to treat throat infections, syphilis, meningitis, etc.	[10]
Pciltaxel (PTX)	Taxol	*Sorangium cellulosum*	It has antitumor and antifungal activities	[11]
Tygacil	Tigecycline	*Streptomyces aureofaciens*	It can effect on tetracycline resistance bacteria	[12]
Afinitor Disperz	Everolimus	*Streptomyces hygroscopicus*	It is an immunosuppressive agent	[13]
Invanz	Ertapenem	*Streptomyces cattleya*	It is also an antibacterial agent	[14, 15]
Elidel	Pimecrolimus	*Streptomyces hygroscopicus*	It is anti-inflammatory for skin disease	[16]
Myfortic, CellCept	Mycophenolic acid	*Penicillium stoloniferum*	Immunosuppressive agent in organ transplantation	[17]
Brostallicin	Distamycin A	*Streptomyces distallicus*	It has anticancer activity	[18]
Gilenya	Fingolimod	*Isaria sinclairii*	It can treat a very rare disease, multiple sclerosis	[19]
CellCept, Myfortic	Mycophenolic acid (MPA)	*Penicillium brevicompactum*	It has immunosuppressive activity	[20]

botany, and natural product chemistry are scouring the planet for phytochemicals and other potential drug "leads" that might be used to treat a variety of disorders. Some of the plants have been found to have antibacterial, antifungal, anticancer, antidiuretic, anti-inflammatory, and antidiabetic properties. Mental diseases are treated with plant-based medications. Among the most prevalent are skin disorders, cancer, diabetes, jaundice, hypertension, and tuberculosis [6].

In line with WHO, the current pharmacopoeia contains at least 7000 medical substances that are derived from plants. Significant changes in plant features have been linked to different soil types in several medicinal and aromatic plants.

Examples include morphine from the opium poppy, digoxin from foxglove, quinine from the bark of cinchona trees, and aspirin from willow bark. Herbs are high in nutrients since they are derived from plants in the same way. However, occasionally they have had unforeseen results. They could cause asthma, headaches, dizziness, agitation, dry mouth, seizures, exhaustion, tachycardia, nausea, vomiting, and diarrhea. They might also cause allergic reactions. There have also been reports of severe adverse effects. Kava-kava has been linked to cases of hepatotoxicity (liver damage), and the majority of herbal medicines have been linked to anaphylactic reactions that can be fatal [7]. Due to their hazardous chemical or heavy metal content or adverse drug interactions, even herbal remedies can induce kidney failure and liver damage in some users. Before releasing herbal medications into the market, the Therapeutic Goods Administration (TGA), according to Professor Roger Byard, should subject them to independent testing. Byard also suggested that when a product did not adhere to legal requirements, legal action should be explored. According to TGA statistics on complementary and alternative medicines from 2015 to 2016, there were even greater rates of quality issues, according to Dr. Ken Harvey, a medicinal drug policy specialist with Medreach. "Toxic side effects of herbal medicines used in traditional societies have typically not been reported, and this is often cited in favour of their safety," Byard added. Therefore, any substance considered a medicine, whether natural or not, can have both positive and negative impact on consumers. Some herbs and their positive and negative effects on human health are discussed in Table 2.

5 Impact of Herbal Medicine on Disease Prevention

It has long been known that plants have healing qualities. Indigenous societies have employed traditional herbal therapy to treat a wide range of illnesses for ages [43]. The potential of botanical agents for illness prevention can be found through historic uses of therapeutic plants. Due to the decreasing efficacy of some synthetic drugs and the rising contraindications to their use, the question of using natural drugs comes up [44]. The therapeutic efficacy of medicinal plants, particularly their antibacterial properties, is determined by secondary metabolites or compounds contained in the plants, such as tannins, terpenoids, alkaloids, and flavonoids [45]. Different medicinal plants can be used to prevent illnesses such chronic illnesses, metabolic syndrome, malaria, diarrhea, tuberculosis, asthma, and urinary tract infections.

5.1 Chronic Diseases

Numerous chronic diseases, including diabetes, cancer, cardiovascular disease, inflammatory bowel disease, and neurologic disorders, are on the rise as a result of considerable changes in human nutrition and lifestyle [44]. One of the most

Table 2 Different herbs, their active components, and their positive and negative effects on human health

Herbs	Texture of extracted product and extracted portion	Active ingredients	Positive impact	Negative impact	References
Azadirachta indica (neem)	Juice and oil from leaves and bark	Azadirachtin, nimbidin, nimbidol, ascorbic acid, quercetin, and polyphenolic flavonoids	Prevents diabetes, fungal and bacterial infections, and rashes It fights against tumor, cancer, etc. Even regulates apoptotic regulation	Taking neem seeds and bark by mouth is unsafe for child and during pregnancy It may be responsible for vomiting, diarrhea, etc.	[22]
Aloe barbadensis miller (aloe vera)	Gel from leaves	Vitamins A, C, E, and B12, folic acid, salicylic acid, enzymes, minerals, sugars, lignin, saponin, and amino acids	Lowers blood sugar and cholesterol Fights against acne, skin disease, dandruff, anal fissure, etc.	It can cause cramping, diarrhea, and imbalance of electrolyte in blood. It can disrupt other drugs (diabetes and heart disease)	[23, 24]
Mentha piperita L. (mint)	Liquid juice or oil from leaves and whole plant	Menthol, menthyl, acetate, menthone, rosmarinic acid, isomenthol, vitamins C and B6, iron, sodium, calcium, etc.	Peppermint fights against diarrhea, indigestion, cough, headache, migraines, menstrual cramp, chicken pox, etc. It reduces morning sickness, vomiting during pregnancy, even toxicity induced by arsenic, etc.	Heartburn, gallbladder, inflammation, sometimes liver damage is also caused by peppermint Menthol causes jaundice in new born babies, allergic reaction, etc.	[25]
Curcuma longa L. (turmeric)	Juice from raw roots and powder from dry roots	Main leader of the advantages of *Curcuma longa* is curcumin (yellow polyphenolic pigment) Curcumin consists of carbohydrates, protein, fat, and another important component is curcuminoid that contains curcumin, demethoxycurcumin (DMC), and bidemethoxycurcumin (BDMC)	It fights against diabetes, inflammation, cancer, bacterial and viral diseases, etc. It lowers joint pain and chances of heart disease. It helps to lose weight, promotes good sleep, prevents production of the melanin, etc.	After all goodness, it is responsible for a few negative effects such as dermatitis, allergies, digestive disorders (abdominal pain and diarrhea), synergistic effects, etc. Sometimes it acts as an inhibitory agent for other drugs and human sperm mortality to alter fertility	[26, 27]

Zingiber officinale (ginger)	Juice from underground stem	The major compounds are gingerol, shogaol, zingiberene, zingerone, terpene, vitamins C and B6, minerals, etc.	It reduces headache, cough, frequency of vomiting, and nausea significantly. In a little amount, this herb can recover muscle strength. Even it is useful regarding improving biochemical obesity indicator and weight loss	It has no such harmful effect, but among the adverse effect, gastronomical related symptoms. Sometimes it is responsible for causing diarrhea during heavy menstrual bleeding	[28, 29]
Allium sativum L. (garlic)	Juice from lower part of shoot system	Allicin is the sole active compound of *Allium sativum.* (allinase enzyme activate and produce allicin from alliin when chopped or crushed) Other important compounds are 1-propenyl thiosulfonate, (E,Z)-4,5,9-trihidodeca-1,6,11-triene 9-oxide (ajoene), and y-L-glutamyl-S-alkyl-L-cysteine. Steam distilled garlic oil consists of diallyl, allyl methyl, and dimethyl mono to hexasulfides	To recover from cardiovascular disease, garlic has significancy since ancient time. It has antitumor, antidiabetic, antimicrobial, antiprotozoal, antifungal, and antidiabetic effects. Even it succeeded to prevent chemical-induced hepatotoxicity	It increases the risk of bleeding, heartburn, and digestive problems Garlic contains a variety of sulfur compounds that may cause bad breath especially when eaten in large amount	[30]
Murraya koenigii (curry leaves)	Juice from leaves	It contains copper, fiber, phosphorus, carbohydrates, magnesium, vitamins A, C, and B6, some amino acids, etc.	It helps to lower cholesterol, maintain eye and liver health, treat damaged hair, lose weight, improve blood circulation (menstrual problems and gonorrhea), prevent or treat diabetes, anemia, treat dysentery, diarrhea, morning sickness, and nausea, even control other drugs' side effects, and also flush out toxins and body's fat content	It creates allergic reactions (rashes, itching, and swelling) It may cause stomach upset due to high fiber and low blood pressure	[31, 32]

(continued)

Table 2 (continued)

Herbs	Texture of extracted product and extracted portion	Active ingredients	Positive impact	Negative impact	References
Cinnamomum verum (cinnamon)	Powder from dry bark	Most important components are cinnamaldehyde and trans-cinnamaldehyde Besides, it contains sodium, potassium, vitamins C and B6, protein, iron, calcium, magnesium, etc.	It helps to low diabetes, cholesterol, heart disease, infection, allergies, tooth decay, etc. It is also used to treat HIV, cancer, Alzheimer's disease, and it is good for our intestine	It normally has no negative effects, but excessive use is bad for the lips and mouth, can induce allergic responses, can cause liver damage, disturb other medications, and is not appropriate to take as a treatment during pregnancy or post-pregnancy (breastfeeding)	[33]
Syzygium aromaticum (clove)	Powder from dry flower buds	Main bioactive compounds are eugenol, eugenyl acetate, and caryophyllene Besides, iron, calcium, sodium, magnesium, vitamins A and B6, etc., are present in this herb	It is good for diabetes, oral and dental problems, stomach upset problems such as vomiting, bloating, indigestion, diarrhea, etc. It acts as immunity booster, bone strength developer, etc.	It can cause severe side effects such as seizures, liver damage, and fluid imbalances. In the large amount, it is not safe for pregnant or breastfeeding women	[34]
Trigonella foenum-graecum (fenugreek seeds)	Powder from seeds and juice from leaves	It contains alkaloids like trimethylamine, neurin, choline, etc.; amino acids like leucine, isoleucine, histidine arginine, etc. Besides, vitamins C and B6, calcium, sodium, potassium, iron, and proteins are present in this herb	It is used to cure vitamin A deficiency disease called beriberi, mouth ulcers, boils, bronchitis, infection of the tissues beneath the surface of the skin (cellulitis), tuberculosis, chronic coughs, chapped lips, baldness, cancer and kidney ailments, etc. It is good for intestine, digestive system, gastritis, hair growth, post-pregnancy breast milk production, etc.	It is responsible for heartburn; extreme use of this can cause diarrhea, nausea, headaches, liver toxicity, harmful drop in blood sugar, etc. It is not safe during pregnancy, and it can produce birth defect in both animal and human	[35]

Foeniculum vulgare (fennel)	Powder from dry seeds, leaves, and flower	The main ingredients are trans-anethole, 2-pentanone, fenchone, and benzaldehyde-4-methoxy	Women use it for menstrual cramp. It is good for intestine and digestive system	It increases skin sensitivity to sunlight and makes sunburn more likely. It is not safe in large quantities during pregnancy. It can cause allergies, bleeding difficulties, and disruptions in hormone secretion, which can lead to breast cancer, uterine cancer, ovarian cancer, uterine fibroids, and other problems	[36]
Hibiscus rosa-sinensis (hibiscus)	Powder from dry petals and leaves	It contain tannins, anthraquinones, quinines, phenols, flavonoids, alkaloids, saponins, cyclopropanoids, methyl sterculate, 2-hydroxyserculate, malvalate, and beta-sitosterol; even minerals, proteins are also present	It helps to lower cholesterol, manage blood pressure, relieving menstrual pain, improve digestion, lose weight, even prevent cancer It has antibacterial, antidepressant, and antiaging properties. As it is good for hair growth, it has been used in hair care products, etc.	After all goodness, it has some side effects also. Like the people who has low blood pressure or hypotension, hibiscus is strictly prohibited It may cause dizziness, fainting, heart damage, etc. It also effects the estrogen levels due to which the hibiscus product is not applicable for women those are on birth control treatment or hormone replacement therapy	[37, 38]
Centella asiatica (gotu kola)	Juice from leaves	The primary active constituents of *Centella asiatica* are saponins or triterpenoids which include asiaticosides, in which a trisaccharide moiety is linked to the aglycone asiatic acid, madecassoside, and madasiatic acid	These triterpene saponins and their sapogenins are mainly responsible for the wound healing and vascular effects by inhibiting the production of collagen at the wound site. It has antidepressant properties, sedative and anxiolytic properties, antiepileptic properties, antioxidant properties, etc. It is	It has no such adverse effects, but it may cause skin allergy, burning sensations, headache, stomach upset, dizziness, and extreme drowsiness when consumed in large amount	[39]

(continued)

Table 2 (continued)

Herbs	Texture of extracted product and extracted portion	Active ingredients	Positive impact	Negative impact	References
			good for gastric ulcer, inflammation, etc.		
Hygrophila auriculata (kokilaksha)	Juice from leaves or entire plants	The whole plant contains lupeol, stigmasterol, isoflavone, glycoside, and alkaloid. Seeds contain asterol 1,2,3,4, astercanthine, histidine, lysine, phenylalanine, etc. Fresh flower contains apigenin-7-0-glucoside	The ash of dry plant is used to treat renal calculi, roots are used in the treatment of jaundice, and seeds powder is used to treat impotency, fewer sperm counts, and general debility Even it is known to increase hemoglobin in blood or purifier of blood	Some negligible side effects are found in this herb. Seeds are not suitable during pregnancy, and for regular use, consumers must consult with doctor	[40]
Andrographis paniculata (green chiretta)	Juice and paste of leaves	The leaves of *Andrographis paniculata* contain many bioactive compounds including diterpene, lactones, diterpene glucoside, flavonoids, etc. Some microparticles are polylactic-glycolic acid, alginic acid, and glucan derivatives. Several nanoparticles are polymeric nanoparticles, solid lipid nanoparticles, gold nanoparticles, nanocrystal, and micro and nanoemulsion Its bioavailability was obtained by a lipid combination of lecithin, glyceryl behenate, glycerol monostearate, and surfactant solution 3% Tween-80	As it is a bitter herb, it is good for diabetes and blood circulation or purification. It is good for liver disease, influenza, common cold, appetite stimulant, stomach ulcers, and inflammation. It has antioxidant properties due to which it reduces toxicity of hydrogen peroxide, hydroxyl radical, and singlet oxygen those are responsible for oxidative stress It is also used for leprosy, pneumonia, tuberculosis, malaria, cholera, rabies, and even HIV/AIDS	It is safe up to 340 mg per day; otherwise, it can cause side effects such as diarrhea, vomiting, rashes, headache, runny nose, and fatigue. It is unsafe during pregnancy when taken by mouth, as it cause miscarriage Andrographis might cause slow blood clotting and reduce blood pressure	[41, 42]

well-liked plants, *Moringa oleifera*, has a number of important health advantages. The *Moringa oleifera* leaf extracts are helpful in preventing the main symptoms of diabetes mellitus, including effective control of blood sugar or insulin levels, improvement of insulin tissue sensitivity, and protection against organ damage under long-term hyperglycemia. *M. oleifera* leaf and seed extracts show enhanced effects in strengthening intracellular antioxidant defense like catalase, superoxide dismutase, and glutathione to lower lipid peroxidation products and repress pro-inflammatory markers like tumor necrosis factor-α, interleukin (IL)-β, and monocyte chemoattractant protein-1 [46]. Skimming, one of the key pharmacological active compounds found in *Hydrangea paniculata*, lowers blood urea nitrogen (BUN) and urine albumin excretion (UAE),urine albumin excretion (UAE) and suppresses interleukin-1 (IL-1).Reduces the expression of interleukin-1 (ILK1) and IL-6. This can significantly meliorate renal function and suppress the immune complex deposition [47]. Inhibiting lipid accumulation in adipocytes and boosting lipolysis are effects of consecutive water and methanol extracts of *Nelumbo nucifera* petals. Furthermore, preparations of *Nelumbo nucifera* petals clearly exhibit agonist and antagonist actions against cannabinoid and serotonin receptors, respectively [48]. For both type 2 diabetes and hyperuricemia, the extract from the leaves of the *Morus alba* plant may be a great alternative [49]. Andrographolide, major diterpene lactones found in *Andrographis paniculata*, exhibits notable inhibitory effects on various cancers. By decreasing the transcriptional and translational levels of VEGF, EGFR, cyclin A, and cyclin B expression, it inhibits the growth of tumors. Additionally, *Andrographis paniculata* extract has anti-infective, anti-inflammatory, antioxidant, immunostimulant, and antidiabetic properties [50, 51]. The fruits of *Pterodon pubescens* are frequently used to treat respiratory disorders, sore throats, and rheumatic diseases. By lowering blood sugar, cholesterol, and triglyceride levels, the ethanolic extract of *P. pubescens* also exhibits anti-inflammatory and antinociceptive properties and prevents mechanical and thermal hyperalgesia in the postoperative pain [52–54]. Different constituents like vitamins, flavonoids, minerals, etc., are found in the different parts of *Carica papaya* that develop different functions such as antidiabetic activity, wound healing activity, antimalarial activity, etc., for preventing disease. *C. papaya* seeds water extract also acts as a potent free radical scavenger, providing protection to Detroit 550 fibroblasts that underwent H_2O_2 oxidative stress by its antioxidant effect [55, 56]. Jian-Pi-Shen formula has [composed of *Astragalus mongholicus* Bunge, *Atractylodes macrocephala* Koidz., *Dioscorea oppositifolia* L., *Cistanche deserticola*, *Wurfbainia compacta* (Sol. Ex. Maton) Skornick and A.D. Poulsen, *Salvia miltiorrhiza* Bunge, *Rheum palmatum* L., and *Glycyrrhiza glabra* L.] enhanced perindopril suppression of chronic kidney disease progression by increasing the expression of SIRT3, modulation of mitochondrial dynamics, and antioxidant effects [57]. Ca^{2+} is mobilized by bergamot essential oil (BEO) from *Citrus bergamia* Risso from extracellular and intracellular sources in endothelial cells through IP3-mediated Ca^{2+} release and affect the promotion of Ca^{2+} influx via an SOC mechanism [58].

5.2 Alzheimer's Disease

The most common neurodegenerative ailment, Alzheimer's disease (AD), produces progressive dementia by limiting mental capacity development and obstructing neurocognitive function [59]. *Ginkgo biloba* extract has positive effects on issues with insufficient blood flow, loss of consciousness, depression, and headaches. The attenuation of apoptosis, an anti-inflammatory effect, an inhibition of membrane lipid peroxidation, and an inhibition of the development of amyloid aggregates are only a few examples of the molecular and cellular mechanisms that are shown [60]. Due to the presence of curcumin, which inhibits plaque formation in the brain, oxidative stress, and amyloid pathology, and regulates a number of transcription factors connected to inflammation, *Curcuma longa* has anti-inflammatory effect against Alzheimer's disease [60, 61]. *Melissa officinalis* (also called lemon balm) extract contains rutin as the main flavonoid compound, and the primary phenolic ingredient, rosmarinic acid, enhances brain-derived neurotrophic factor and nitric oxide synthase gene expression to improve memory in Alzheimer's patients. The ability to reduce agitation is another benefit of *Melissa officinalis* extract [62, 63]. *Panax ginseng* extract contains ginsenosides and gintonin that contribute to AD amelioration by inhibiting Aβ-induced neurotoxicity, activating nonamyloidogenic pathway to reduce Aβ formation and enhancing acetylcholine and choline acetyltransferase expression in brain. *Ficus racemosa* is known to have the same function to prevent this disorder [64, 65]. Therapeutic benefits of the class of chemicals known as bacoside-A, which are derived from the *Bacopa monniera* plant, include memory and cognitive function enhancement. Bacoside-A significantly reduces cytotoxicity, fibrillation, and in particular the interaction of amyloid-beta (1-42) (Aβ42) with membranes, a peptide important in the progression and toxicity of Alzheimer's disease [66, 67]. Besides, plants such as *Salvia officinalis, Punica granatum, Rosmarinus officinalis*, etc., have significant effects in Alzheimer's disease prevention by AchE and BchE inhibition, antioxidant activity, neuroprotective activity, anticholinesterase activity, etc. [59].

5.3 Metabolic Syndromes

The metabolic syndrome is a collection of factors that increase the risk of developing cardiovascular disease, such as central obesity, insulin resistance, glucose intolerance, dyslipidemia, and hypertension. People who have the metabolic syndrome are more likely to suffer from peripheral vascular disease, stroke, and coronary heart disease, as well as other conditions caused by plaque formation in artery walls.

A Chinese herbal extract SK0506, a mixture of active ingredients of *Gynostemma pentaphyllum, Coptis chinensis*, and *Salvia miltiorrhiza,* is known to have endpoint effects on impaired glucose tolerance, hyperinsulinemia, and hyperlipidemia, and on visceral obesity. This oral agent may have the antidiabetic property to prevent type 2 diabetes [68, 69].

5.4 Urinary Tract Infection

Urinary tract infections (UTIs) are very common infection. In a study, cranberry products were found to be effective than antibiotics for the prevention of recurrent UTIs in women. Furthermore, *Rosa canina* has the potency to prevent UTI in women undergoing a caesarean section [70].

5.5 Malaria

One of the most serious parasite diseases in the world and a major killer, especially in underdeveloped nations, is malaria [71]. Hexanic extract of *Xylopia aromatica* root wood, *Xylopia emarginata* root bark, *Casearia sylvestris* var. *lingua* leaves, stem wood, and stem bark, and *Cupania vernalis* leaves was found to have the strongest antiplasmodial activity [72]. K1 strain of *Plasmodium falciparum* can be foreclosed by the antimalarial activity of 80% EtOH extract of *Goniothalamus marcanii* [73]. Leaf ethyl acetate and methanol extract of *Phyllanthus emblica* and flower bud extract of *Syzygium aromaticum* also may serve as antimalarial agents even in their crude form [74].

5.6 Cholera

Saraca indica bark extract exhibits various pharmacological effects such as antioxidant activity, anti-breast cancer activity, and vibriocidal activity [75, 76]. Along with *S. indica,* ethanolic extract of *Lawsonia inermis*, *Syzygium cumini*, *Terminalia belerica*, and *Allium sativum* are also known to have anti-vibrio potential to prevent cholera [76].

5.7 Pneumonia

In a study, Zamzam water extract of *Echinops adenocaulos* and aqueous extract of *Urtica urens* was detected to have antibacterial activity against multidrug-resistant strain of *S. pneumonia* [77]. The acetone, hexane, and methanol fractions of *Parietaria judaica* exhibit antibacterial activity, while the potent antifungal activity against *Candida albicans* has been found in the acetone fraction. Besides, the fractions have strong cytotoxic and antiproliferative activity against HeLa cancer cell [78]. Additionally, due to its antibacterial properties, the crude extract of *Beta vulgaris* leaves can suppress the growth of *Klebsiella pneumoniae* [79].

5.8 Typhoid

MA 001, manufactured from *Citrus aurantifolia*, *Spondias mombin*, *Lantana camara*, *Bidens pilosa, Trema occidentalis, Psidium guajava, Morinda lucida, Vernonia amygdalina, Persea americana, Paullinia pinnata, Momordica charantia*, and *Cnestis ferruginea* medicinal herbs, has been utilized for the treatment of typhoid fever [80].

5.9 HIV

Prostratin, an antiviral substance made by *Homalanthus nutans*, shows substantial antiviral activity against multiple strains of HIV-1 and HIV-2 and has been utilized in traditional Samoan herbal medicine to treat "yellow fever" (hepatitis) and other viral illnesses. Prostratin interacts with a cellular target required for viral entry in order to predominantly impede the entry step of the HIV replication cycle [81, 82].

6 Acceleration on Herbal Medicine Development Involving Metabolic Engineering, Synthetic Biology, and Systems Biology

Many naturally occurring substances that are derived from plants have numerous pharmacological and medicinal uses. The creation of medicinal plant natural products (MPNPs) results from specialized plant metabolism. This is accomplished by a variety of enzymes from diverse classes that catalytically convert core metabolites into secondary metabolite molecules like terpenoids like artemisinin and paclitaxel, alkaloids like camptothecin, and polyketides (PKs) like flavonoids. However, because of their slow growth, limited yield, prolonged time requirements, and intricate structural design, these compounds are relatively difficult to acquire. Consequently, there has been a rise in the demand for their industrial output [83–85]. Traditional metabolic engineering and ideas from systems biology are combined in synthetic biology, a brand-new branch of contemporary multidisciplinary life science and systems science. Due to recent developments in synthetic biology and supporting technologies like DNA synthesis, sequencing, and analytical techniques, many natural products of plant origin are not synthesized by microorganisms. As a result, metabolic and protein engineering have advanced to the point where both can be used to engineer the biosynthesis of a specific medicinal molecule by developing engineered microorganisms. Optimized and transformed organisms are characterized by environmental friendliness, low carbon emission, and economic effectiveness. They can continuously and successfully synthesize specified target molecules with high yield. Instead of depending simply on the traditional discovery and separation or difficult synthetic chemistry to obtain a restricted number of herbal pharmaceuticals, many natural drugs and analogues can now be manufactured fairly inexpensively by creating new biosynthetic pathways [84, 86–89]. The biosynthesis of herbal medicines like artemisinin, paclitaxel, tanshinone, noscapine, etc., is used as an example to demonstrate how scientific advances in synthetic biology can be used for the synthesis of natural plant compounds.

6.1 Selection of Host Species for a Heterologous Plant Pathway to Overproduce MPNP Precursors

The choice of the ideal host species for the pathway to be engineered is the first step in the heterologous production of MPNPs. Based on characteristics including ease of cloning, culture, and appropriateness for the new enzymes and chemicals, host species are chosen. The first host species of choice for the production of MPNPs may be plant cells. For instance, *Nicotiana benthamiana* is a promising heterologous host to manufacture cannabinoids in planta because of its high transformation rate, natural capacity to accommodate the supply of precursor GPP, and assistance in redirecting the current pathway towards cannabinoid synthesis. However, the fundamental disadvantage of employing plant species as hosts is that they are very sluggish in comparison to microorganisms; for this reason, microbial hosts are frequently preferred [84, 89, 90]. The most popular tools for genetic modification are *Escherichia coli* and *Saccharomyces cerevisiae* because they are simple to cultivate, have a variety of platform strains available, and have the ability to produce too much of the desired substance [91, 92]. Additionally, the growth of *Corynebacterium glutamicum* strains results in the commercial production of amino acids such as L-arginine and L-proline [93]. Any bacterial or plant species can serve as a model organism. Their metabolic pathways have been extensively studied and can be utilized to direct the overexpression and knockout modification of certain enzymes and genes for the overproduction of important metabolite precursors. Due to the complexity of cellular metabolism, the synthesis portion of metabolic engineering of *S. cerevisiae* is comparably excessive while the analysis portion is limited. In order to mimic the subcellular localization used in MPNPs biosynthesis in plants, cellular microcompartments like mitochondria and peroxisomes can be used [92, 94, 95]. The doubling time of *E. coli* is 3–4 times shorter than that of *S. cerevisiae*, suitable for the very high expression of enzymes, and has a different profile of native metabolites available compared to *S. cerevisiae*. Therefore, it is thought that both strains of *E. coli* and *S. cerevisiae* as well as numerous other plant species are possible hosts that, following genetic manipulation, produce an excess of alkaloids, terpenoids, flavonoids, and other therapeutic chemicals.

6.2 Increasing Precursor Supply

Enhancing precursor supply is a successful method for the high synthesis of important medical natural compounds that may be used to native and heterologous generating strains. Serpentine, strictosidine, ajmalicine, and other monomeric indole alkaloids, as well as dimeric indole alkaloids like vinblastine and vincristine, are all produced by *Catharanthus roseus* [96]. Significantly, more distinct alkaloids are synthesized when upstream rate-limiting enzymes like geraniol synthase, secologanin synthase, and MAP kinases are overexpressed [97]. The overexpression of hyoscyamine 6β-hydroxylase (H6H) (key enzyme for the conversion of hyoscyamine to scopolamine) increases the scopolamine production in *Datura innoxia* hairy roots [98]. The overexpression of codeinone reductase (CodR) significantly increased the amount of codeine and morphine in transgenic hairy roots lines of *Papaver bracteatum* [99]. Foreign enzymes from other plant species may also aid in the production of specific metabolites. For example, the overexpression of *hyoscyamine* 6β-*hydroxylase* genes of *Hyoscyamus niger* in hairy root cultures of *Scopolia lurida* and the expression of heterologous *Vitreoscilla* hemoglobin in the hairy roots of *Hyoscyamus niger* enhance the accumulation of scopolamine [100, 101]. In *Panax ginseng* root culture, the overexpression of ginsenosides biosynthetic pathway key gene PgSQS1, responsible for upregulating the expression of SE, β-amyrin synthase, and cycloartenol synthase, results in increase of phytosterols and ginsenosides [102, 103]. The overexpression of linalool synthase (LIS) gene from *Clarkia breweri* in *Lavandula latifolia* plants increased the linalool content up to tenfold [104]. The accumulation of anthocyanins has boosted output and yield as a result of the overexpression of MYB transcriptional factors. Hop plants were genetically modified by Agrobacterium utilizing MYB transcription factors from *A. thaliana*, which resulted in a definite increase in flavonoid synthesis [105]. The use of microorganisms for the manufacturing of medical products has drawn interest as a solution to the challenging downstream plant extraction process and poor production. A strategy was used by Fowler et al. to find and validate fresh genetic alterations that would increase flow to malonyl-CoA and boost the production of flavonoids in *E. coli* [106]. Coenzyme ligase (4CL) from *Medicago truncatula* and chalcone synthase (CHS) from *Vitis vinifera* has been expressed in *S. cerevisiae* that resulted 28-fold increased production of naringenin [107]. Martin et al. increased the generation of isoprenoid precursors in *E. coli* by introducing the MVA route from *S. cerevisiae* [108]. Amorphadiene synthesis was boosted in *S. cerevisiae* by increasing the flow from pyruvate to acetyl-CoA in the MVA route through overexpression of acetaldehyde dehydrogenase and the insertion of a *Salmonella enterica* acetyl-CoA synthetase variation [109]. The metabolic engineering of photosynthetic organisms has also increased the production of terpenoids [110].

6.3 Engineering Pathway-Specific Components

Numerous factors, including the modification of regulatory elements, the management of pathway elements, protein engineering, etc., are taken into account while constructing the most efficient pathway for the production of therapeutic natural products. For instance, the positive regulator of antibiotic synthesis has been identified as the gene clusters producing the Streptomyces antibiotic regulatory protein

(SARP). The production of antibiotics by *Streptomyces griseus* has increased by about six times when the SARP from the *fdmR1* has been overexpressed [111, 112]. The transcriptional and translational control of individual enzyme may be a promising approach to fine-tune metabolic pathways but determining the proper enzyme level can be challenging [88]. For example, an inducible promoter system was utilized to determine optimal expression levels for each gene in *S. cerevisiae* for the production of benzylisoquinoline alkaloids [113]. Additionally, it may be possible to modify enzymes through evolutionary engineering to boost the pathway's effectiveness.

6.4 Suppression of Competitive Pathways and Gene Editing

The elimination of competitive pathways that can obstruct the native biosynthetic pathway for the manufacture of specific desired products is another tactic, as is the downregulation or deletion of specific genes. For instance, *Artemisia annua* produced significantly more artemisinin when the expression of SQS, a crucial gene of the sterol route (competitive process of artemisinin biosynthesis), was suppressed by RNA interference (RNAi) [114]. By using RNA interference (RNAi) technology, the lycopene ϵ-cyclase (LCY-ϵ) gene was downregulated, which resulted in a 21-fold increase in the amount of β-carotene in sweet potato transgenic calli [115]. To improve NADPH availability (the limiting factor of high flavonoid production) and boost flavonoid production in *E. coli*, a combination of gene knockouts was identified [116]. The output of (+)-valencene in *N. benthamiana* plants increased somewhat after the endogenous 5-epi-aristolochene synthase and squalene synthase, which compete for the FPP pool, were silenced [117]. Higher plants are incapable of homologous recombination; therefore, they cannot delete a particular gene. However, the CRISPR/Cas9 technology for gene editing has recently solved this problem [110, 118]. For example, using a CRISPR/Cas9 approach, Zakaria et al. introduced detrimental mutations into *hss* gene encoding *homospermidine synthase* (HSS) in *S. officinale* that successfully eliminated pyrrolizidine alkaloids which are toxic to humans [119].

6.5 Designing New Precursor Biosynthetic Pathways

There are numerous methods that can be used to build a novel biosynthetic pathway in a heterologous host. This entails disassembling a pathway into its component parts, each of whose unique enzymes is evaluated and optimized in a host before being combined when fully assembled. Horinouchi et al., for instance, divided the biosynthesis of flavonoids into three modules: substrate synthesis, PK synthesis, and post-PKS modification. In *E. coli*, known PKs were produced by altering these three plasmid components and combining them with precursor-directed biosynthesis

[84, 120]. An IPA method for terpenoids synthesis was created by Clomburg and colleagues [121] that could convert isoprenoid precursor with the fewest possible steps. The healing plant *C. roseus* produced numerous chlorinated alkaloids when bacterial tryptophan halogenases were added [122]. The quickest, ATP-independent pathway from formaldehyde to acetyl-CoA was created by Lu et al. by reusing glycolaldehyde synthase and acetyl-phosphate synthase [123].

6.6 Protein Engineering

An important tool for changing the substrate specificity of biosynthetic enzymes, protein engineering is a process by which a protein sequence is altered by substitution, insertion, or deletion of nucleotides in the encoding gene [124]. This process increases the effectiveness of enzymes for the development of biocatalysis. By creating a synthetic chimeric *bidomain P450* that could mimic the structure of the bacterial *P450BM-3*, Koffas et al. were able to improve the rate at which isoflavones were produced in *E. coli* [125]. By using a pair of short peptide tags (RIAD and RIDD) that increased the carotenoid synthesis in *E. coli* by 5.7 folds, Kang W and colleagues constructed a scaffold-free enzyme assembly [126]. It may be advantageous to use these powerful metabolic engineering tools to successfully generate the intended medicinal herbal medicines. Metabolic engineering of artificial but natural therapeutic compounds. By knowing the pathways and the related enzymes, this goal may be accomplished.

7 Regulations and Bioethical Concerns with the Use of Herbal Remedies for Illness Prevention and Health Management

7.1 Adverse Effects of Herbal Medicine on Health

In spite of having the beneficial aspects of medicinal herbal products, many medicinal plants exhibit undesirable reactions that can cause serious injuries and life-threatening conditions. Like all effective synthetic drugs, herbal medicines are also associated with different drug reactions including toxicity (nephrotoxicity, hepatotoxicity, cardiotoxicity, neurotoxicity, and skin toxicity) and overdose, drug-herb interaction, allergic reaction, etc. The toxicity can occur due to the excessive dosage intake, misuse of herbal medicine, self-treatment, and environmental factors [127, 128]. Although *Ginkgo biloba* plant extracts seem to be safe for humans' health, they may also be the cause of the herb's toxicity, genotoxicity, and carcinogenicity, as well as its unpleasant side effects include headaches, vertigo, restlessness, nausea, vomiting, diarrhea, and skin sensitivity [129, 130]. A study by Liu et al. demonstrated the aconite poisoning after the ingestion of traditional Chinese medicine containing *Aconitum carmichaelii*, as it contains poisonous alkaloids [131]. The overdose of combination of *Aconitum kusnezoffii* and *Aconitum carmichaelii* has shown the toxic effects as bradycardia and hypotension. But the correct dose after processing would not result in bradycardia as the toxicity of aconite reduces due to processing [132]. Administration of *Glycyrrhiza glabra* has been linked to adverse reactions such as hypertension and secondary disorders brought on by hypokalemia. Subacute exposure to *G. glabra* causes the adrenal-pituitary axis to become inhibited and lowers the amount of iron in the liver [133]. The nephrotoxicity of different plants have been shown. For example, *Tripterygium regelii* Sprague et Takeda poisoning heads to acute kidney injury and renal failure. *Rheum palmatum L.* and its anthraquinone compounds can lead to renal damage under certain condition. *Averrhoa carambola L.* contains oxalate in high quantity that causes kidney damage. *Guaiacum officinale L.* and *Arctostaphylos uva-ursi* increase formation of stone [128, 134]. As a type IV hypersensitivity reaction to urushiol, *Toxicodendron diversilobum* (poisonous oak) and *Toxicodendron rydbergii* (poison ivy) are the most often occurring plants to cause allergic contact dermatitis [135]. St. John's wort, or *Hypericum perforatum*, has side effects such as allergic reactions, headaches, dizziness, restlessness, nausea, vomiting, constipation, and more [21].

7.2 Overview of Regulatory Aspects and Approval of Herbal Medicine

Different countries have their own legal policy to regulate the usage of herbal drug. This is because of the adverse effects due to poor quality or improper use, cultural aspects, and the lacking scientific study on herbal medicines. In order to establish the fundamental standards for evaluating the quality, safety, and effectiveness of herbal medicines, the WHO has developed recommendations [136]. Below is a quick description of the regulatory frameworks of several, chosen nations. The Department of AYUSH is the regulating body, and the Department of Drugs and Cosmetics (DAC) Act of 1940 and the Cosmetic Laws of 1945 regulate herbal drugs in India. The Indian System of Medicine (ISM) is being expanded, and the Department of AYUSH is in charge of defining standards, regulating, promoting, and supervising it. The Department of Indian Medicine and Homoeopathy (ISM & H) was founded in 1995 with the goal of encouraging assistance to the public in regards to safe and effective services that satisfy pharmacopoeial requirements and AYUSH quality standard [137]. This department employs AYUSH to obtain wonderful health [138]. In Malaysia, any herbal product falls under the regulated product category. Anyone in the marketing industry who wants to sell a herbal product must first register it. Traditional items and health supplements are the two categories under which these items may be registered with the Malaysian Registrar of Businesses or Suruhanjaya Syarikat Malaysia [138].

In Australia, the Therapeutic Goods Act 1989, which governs therapeutic products, regulates both complementary and herbal medicines. A two-tiered method was created for the regulation of medicinal commodities based on risk. Because they may be made from pre-examined materials, complementary medications are thought to carry a lesser risk and are not reviewed before being released. However, there are individual assessments for the licensed medications' effectiveness, quality, and safety [139]. The US Food and Drug Administration categorized the botanical items as a medicine, food, or dietary supplement based on the claims and intended use. According to the FDA, the medication must be sold under an NDA that has been authorized. In accordance with the Dietary Supplement Health and Education Act of 1994 [140], the US FDA controls goods advertised as dietary supplements. Natural health products (NHPs) are what are referred to as "Classified as Medicines" in Canada, which is how the majority of natural goods are categorized. These fall under the 2004-enacted Natural Health Product Regulations (NHPR), which are overseen by Health Canada (HC). Additionally, a collection of NHP monographs has been created, to which an application may certify as their sole source of proof of efficacy and safety [141].

8 Conclusion and Future Perspectives

Due to its economic significance, traditional medical treatment employing herbal medicine is a fast expanding healthcare system that is now widely employed in many nations across the world. Numerous types of categorized herbs have been shown to contain numerous medicinal components, including alkaloids, flavonoids, terpenoids, saponins, etc., that have demonstrated antibacterial and antifungal actions. Research is being done on several plants that might be helpful in the prevention of many deadly diseases, and the idea of a medical natural substance having multiple uses is gaining popularity [142]. The majority of herbal natural goods are produced using conventional methods; however, these methods' main drawback is the product's poor yield. The large-scale industrial manufacturing of the required pharmaceutical goods has been made possible by advances in our understanding of the metabolic and regulatory networks at the genome level, as well as natural product biosynthetic pathways [143]. For the high output that is detailed in Fig. 1, several metabolic engineering components, such as protein engineering, pathway engineering, etc., as well as synthetic biology and systems biology, have been extremely helpful tools. Along with the advantages of herbal therapy over modern treatment, concerns about adverse reactions are also becoming more prominent [144]. Since "safe" and "natural" are not the same thing, herbal medicines should not always be taken for granted just because they come from sources that are deemed to be "safe." Therefore, the three most crucial factors in herbal therapy are safety, effectiveness, and quality control. As a result, it is imperative that worldwide regulatory standards for herbal medicine be established,

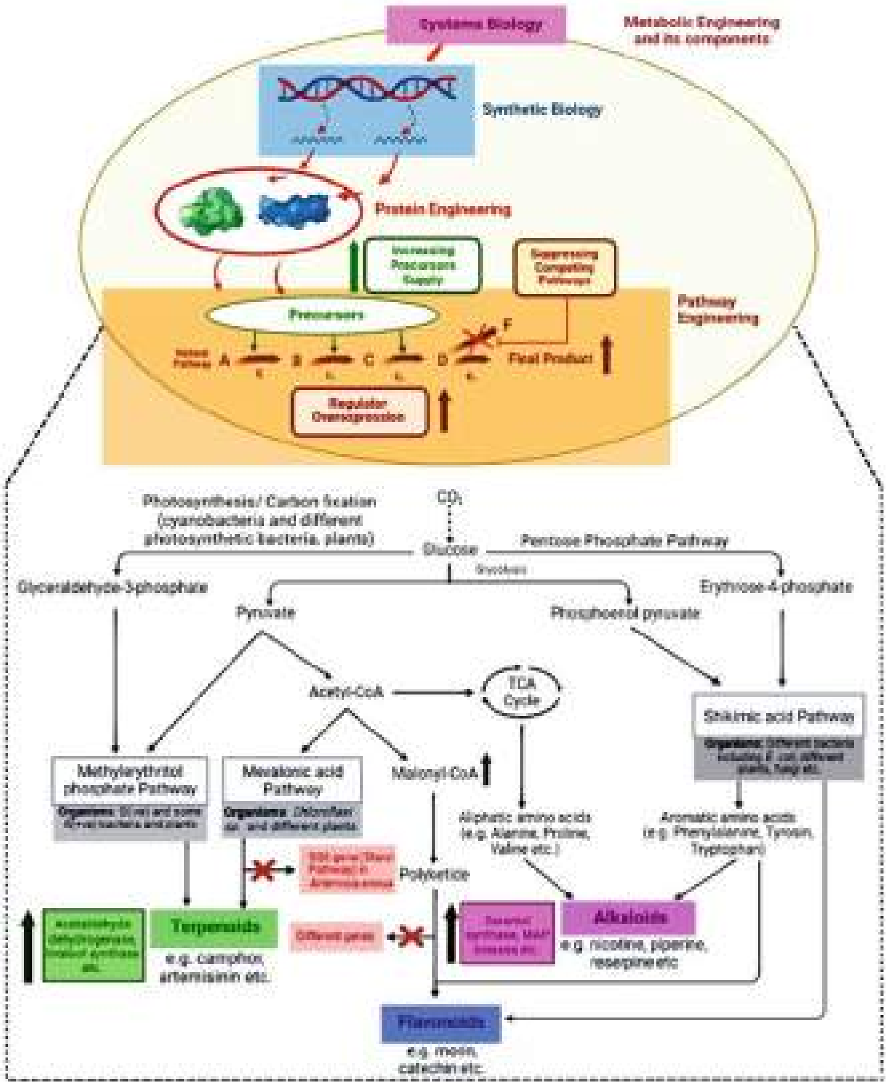

Fig. 1 Putative metabolic engineering and synthetic biology tools for microbial production of herbal-based medicines towards health management and disease prevention

and all national regulatory agencies must take aggressive measures to safeguard the public's health by guaranteeing the efficacy and safety of herbal remedies. The conclusion drawn from this chapter is that herbal medicine may prove to be a viable replacement for synthetic chemical-based medications.

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Utilization Methods and Practices of Herbal Medicine in Africa

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Abstract

A comprehensive healthcare system, African traditional medicine is divided into three degrees of specialization: divination, spiritualism, and herbalism. The traditional healer offers medical treatments in accordance with the predominant cultural and religious ideas, knowledge, and attitudes in his society. The belief that illness has both natural and supernatural origins necessitates its treatment via both physical and spiritual methods, including divination, incantations, animal sacrifice, exorcism, and the use of herbs and other natural remedies. The use of herbal medicines involves integrating a number of therapeutic techniques from indigenous medical systems that may date back many generations. This integration frequently yields helpful instructions for the choice, preparation, and application of herbal formulations for the treatment, control, and management of a wide range of illnesses. Herbal medicine has been used in Africa since ancient times and is being used now via divination, interviews, medical reports, and viewpoints from both the spiritual and physical realms. Despite the current improvements in mainstream medicine, the use of herbal medicines for basic healthcare needs in Africa is still on the rise significantly. The African Perspective of Illness and Disease, Herbal Medicine in Africa, Clinical Practice of African Traditional Medicine, Some Traits in Herbal Medicine Practices from Selected African Countries, Factors Responsible for Increased Usage of Herbal Medicine in Africa, Adverse Effects of Herbal Medicine, Influence of Regulatory Policies in Safety of Herbal Medicine, Challenges of Monitoring Safety of Herbal Medicines are discussed in this chapter.

Keywords

Herbal medicine · Africa · Spirituality · Divination · Herbalism · Healthcare

Abbreviations

AIDS	Acquired Immunodeficiency Syndrome
ASICUMPON	Association for Scientific, Identification, Conservation, and Utilization of Medicinal Plants of Nigeria
ATM	African Traditional Medicine
CNS	Central Nervous System
DSHEA	Dietary Supplement Health and Education Act
EU	European Union
FDA	Food and Drug Administration
GACP	Good Agricultural Collection Practical
GMP	Good manufacturing Practice
HIV	Human immune virus
HRH	Human resources in health

PHC	Primary health care
THM	Traditional herbal medicine
THP	Traditional health Practitioners
THR	Traditional Herbal Registration
TM	Traditional Medicine
TMK	Traditional Medical Knowledge
TMPs	Traditional Medical Practitioners
UK	United Kingdom
WHO	World Health Organization

1 Introduction

Herbal medicine, often known as herbalism, is the method used by a civilization to cure, identify, and prevent disease that are not based on scientific evidence. It is context-specific since it is determined by a culture's knowledge and values, much as social constructions and risk negotiations. These methods become "complementary, non-conventional or alternative medicine" [1] when contemporary civilizations embrace them outside of their original setting.

As a result, traditional medicine (TM) native to the many African civilizations is known as African traditional medicine (ATM). African traditional medicine is well renowned for providing essential healthcare to address the psychological, emotional, spiritual, cultural, and social requirements of the African people, and it is notably thought to be safer than manufactured medications due to its natural ingredients [2]. Traditional herbal medicine (THM) has been around since at least the Stone Age. It seems that traditional healing and magic are significantly more prevalent than Western medicine in Africa [3, 4] and are much older than certain other traditional medical professions. African traditional medicine is a comprehensive healthcare system with three levels of specialization: divination, spiritualism, and herbalism [2, 5]. In certain cases, these specializations may overlap.

Herbal medicine is the most often used and favored form of TM treatments out of all of them [6, 7]. Many individuals have started using herbal therapy as a supplement to or an alternative to allopathic treatment as the practice gains greater international acceptance [8, 9]. According to available data, 80% of Africans [10–12] depend on herbal medication for their basic healthcare (PHC) requirements. This use of herbal medicines is supported by a wide range of factors. These explanations vary depending on the culture and ethnicity. The availability and accessibility of herbal medications, their efficacy, safety, cost, and their cultural compatibility are often cited justifications for their usage [13].

In many emerging African nations, the availability of herbal medications far surpasses that of allopathic treatments. Therefore, the majority of individuals turn to herbal remedies when they are unwell or are having health issues. The extensive usage of herbal remedies in Africa and other underdeveloped nations may also be explained by the ease with which practitioners can get them [14]. There are hardly many allopathic doctors in rural settings; they are mostly found in metropolitan regions.

Therefore, according to Abdullahi [15], traditional healers, especially herbal medicine practitioners, continue to be the primary source of health icare for millions of people living in rural areas. The African Perspective of Illness and Disease, Herbal Medicine in Africa, Clinical Practice of African Traditional Medicine, Some Traits in Herbal Medicine Practices from Selected African Countries, Factors Responsible for Increased Usage of Herbal Medicine in Africa, Adverse Effects of Herbal Medicine, Influence of Regulatory Policies in Safety of Herbal Medicine, Challenges of Monitoring Safety of Herbal Medicines are discussed in this chapter.

2 African Perspectives of Illness and Disease

In an African culture, there is a justification for why a person is afflicted with a certain illness at a specific period. Ayodele [16] asserts that sicknesses are often caused by witchcraft or sorcery, deities or ancestors, as well as being inherited and occurring naturally. A sick person is seen differently by African culture than by Western allopathy. Natural, cultural, and societal factors are often cited as the sources of sickness [17]. Cultural or sociological illness is said to be linked to supernatural causes like angered spirits, witchcraft, or alien/evil spirits, even for conditions that are now recognized as well as understood in modern medicine, such as hypertension, sickle-cell anemia, cardiomyopathies, and diabetes. According to ancient African tenets, a person's identity is comprised of their body, soul, conscience, and society. If all three factors were well-balanced, it would be a sign of great health; otherwise, it would point to illness.

As a result, the physical health of a patient is just one part of treatment that may improve their emotional, moral, and social well-being as well. African traditional medicine is distinct from complementary and alternative medicine in a number of ways. For example, many African cultures practice faith healing (spiritual healing), therapeutic occultism, male and female circumcision, tribal marks, treatment for snake bites, treatment for whitlow, removal of tuberculous lymphadenitis in the neck, cutting of the umbilical cord, ear piercing, removal of the uvula, extraction of a carious tooth, abdominal surgery, and other infections [18]. African traditional medicine cannot be seen as an alternative since it is the continent's original system of healthcare.

3 Herbal Medicines

Herbal medicine (HM) is a part of African traditional medicine and is sometimes used interchangeably. It is the oldest and still widely used kind of medicine in the world [19]. HM is used by people of many cultures and socioeconomic backgrounds [20, 21]. Herbal medicines are defined by the World Health Organization (WHO) as those that contain whole plants, parts of plants, or other plant materials (such as leaves, bark, berries, flowers, roots, and bark), and/or their extracts, for therapeutic use or for other benefits in humans and sometimes animals.

The traditional healer, or herbalist, in the unique and well-known discipline of herbal medicine specializes in the use of plants to treat a wide variety of ailments. They play a pivotal role because of their comprehensive knowledge of the pharmaceutical procedures necessary to turn native plants into medicines. It seems that the use of herbal treatments is common in all cultures. However, even in cases when the same ailment is being treated, the plants employed and the therapeutic approach are different from one region to another [22, 23]. Medicinal plants are plants whose organs make up various compounds, which can be used for treatment of ailments [24–56]. Some herbs, for instance, have been used for decades in traditional medicine despite a lack of scientific proof supporting their efficacy. In this context, crude medicines of natural or biological origin refer to these plants in whole or in part due to their medicinal properties. Medicines may be classified as “organised” or “unorganised” based on whether they are made from plant parts that include cells, such as leaves, bark, roots, etc., or from acellular plant components [57].

Herbal medication is widely accessible and affordable compared to contemporary allopathic treatment [58, 59]. Because common medicinal plants are generally well known, particularly in rural regions, there is less need for consultation with traditional healers, except for the treatment of chronic disorders [58]. Even in cases when traditional healers consult, there is a lack of agreement among them on the methods for making herbal remedies and the appropriate dose [60]. However, in Africa, at least 80% of people depend on medicinal plants for their healthcare, according to WHO [61]. Herbal medications have a number of benefits, including cheap cost, accessibility, acceptance, and what seems to be minimal toxicity [62, 63].

In many parts of Africa, traditional healers still possess the knowledge of the plant species used and the techniques for making and giving the treatment, particularly for critical illnesses. The use of these drugs is still shrouded in secrecy and rivalry, and healers are often reluctant to impart their expertise to anybody other than close friends [64].

3.1 Methods of Preparation and Administration

Diffent techniques or methods have been employed in the preparations of traditional medications according to different cultures and their geographical locations. Various plant materials in dried and fresh states are been used. For the sake of this chapter a particular approach has been outlined, which can bring about the herbal medication efficacy and safety. A few common preparation techniques are shown in Fig. 1:

Lotions, which are diluted liquid treatments applied to the skin, and liniments, which are applied topically, come in liquid, semiliquid, and oily forms that all include the active components. Freshly macerated plant components are used to make poultices, which are then applied to the skin. Snuff is dry, powdered plant material that is inhaled via the nasal passages. Plant charcoal is produced when dried plants are burned. Sometimes mixtures made from many plants are developed to maximize the benefits of the composite plants [65, 66].

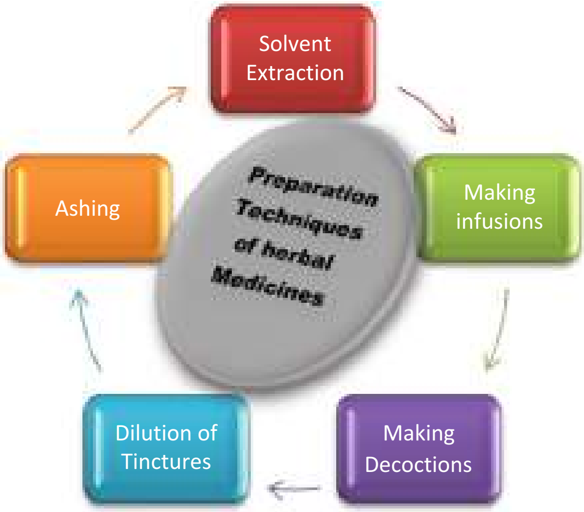


Fig. 1 Preparation techniques of herbal medicine

3.2 Ethnobotanical Surveys

Through ethnobotanical surveys, which look at plants in connection to human culture, information about plants is gathered. Although several plants are utilized in traditional African medicine, little is known about their active components. Ethnobotanical surveys engage with the local population and their surroundings; as a result, they are participatory methods that allow the community to share their knowledge of the plants' practical use [67]. This might entail identifying, preserving, documenting, and using therapeutic plants. The majority of ethnomedical data lacks adequate validation. Despite the fact that many writers have published a wealth of information on plants with medicinal properties in Africa [62, 66]. These provide a wide range of data for scientific validation and investigation [68].

Many chronic illnesses may be effectively treated by using plants as medication. For instance, according to information from Nigerian folkloric medicine, *Ocimum gratissimum* is used to cure diarrheal disorders whereas *Rauvolfia vomitoria* is used to treat hypertension and other nerve problems. Others include *Garcinia kola* seeds for pain and inflammation, *Aloe vera* for skin conditions, *Carica papaya* seeds for intestinal parasites, *Citrus paradise* seeds for resistant urinary tract infections, and pure honey for treating chronic wounds [69–73].

For plants from other African nations, the same is true [74]. The majority of these curative qualities have been known for a while thanks to observations supported by evidence. Scientific studies aim to comprehend the plant chemical components viz-a-viz their medicinal benefits. A plant's therapeutic action is a result of its intricate chemical makeup, with various plant sections having various therapeutic benefits. Alkaloids, glycosides, tannins, acids, coumarins, sterols, phenols, and other chemical compounds present in plants are what give rise to the diverse therapeutic actions

[75, 76]. Many current drugs have been based on these substances or were first produced from them. For instance, aspirin is made from salicylic acid, which is obtained from the bark of *Salix alba* and the meadowsweet plant, *Filipendula ulmaria* [77].

Antimalarial medications include artemisinin and quinine, which are derived from the bark of the *Artemisia annua* plant. Leukemia is treated with the anticancer medications vincristine and vinblastine, which were produced from the Madagascar periwinkle (*Catharanthus roseus*). While digitoxin is a cardiac glycoside produced from the foxglove plant (*Digitalis purpurea*), morphine and codeine, which are derived from the opium poppy (*Papaver somniferum*), are used to treat diarrhea and provide pain relief [74, 78].

Additionally significant resources for the cosmetic industry are medicinal plants. As synthetic chemistry developed toward the beginning of the twentieth century, the usage of herbal medicines decreased. Therefore, when synthetic pharmaceuticals lost their efficacy owing to high levels of resistance as well as increased toxicity and expense, there has been a revival of interest in plant-based medicines in more recent years. More than half of all synthetic medications now in use are thought to be originated from plants [79].

4 Clinical Practice of African Traditional Medicine

Clinical practices are the terms used to describe therapeutic, educational, promotional, and rehabilitative services in African traditional medicine. The process of assessing a person's medical conditions and treating them is also referred to as clinical practise. These traditional healthcare services are delivered via tradition and culture as dictated by a certain philosophy, in which the norms and taboos within are closely observed and serve as the foundation for traditional health practitioners' acceptance in the communities they serve [80].

According to WHO health is "a condition of total physical, mental and social well-being and not only the absence of sickness or infirmity" [61, 81] and believes that everyone has the right to be healthy as a basic human right. The term "health triangle" refers to the interrelationship between one's physical, mental/emotional, and social well-being. Every civilization in traditional Africa had to come up with ways to solve its problems due to the identification of sickness and diseases. Around the world, several communities have unique medicinal herbs and is developed in a considerable amount of period. African traditional communities used herbal medicine, surgery and nutritional treatment like contemporary Western treatment practices [82].

These medicinal advancements were made long before the arrival of the Europeans. Treatment successes recorded were codified, and occasionally the right preparation and dosage techniques were prescribed. The contents and method of preparation varied according to the illness but were also influenced by other variables including geography, sociology, and economics. The important thing to note, however, was that patients were often healed of their physical or psychological

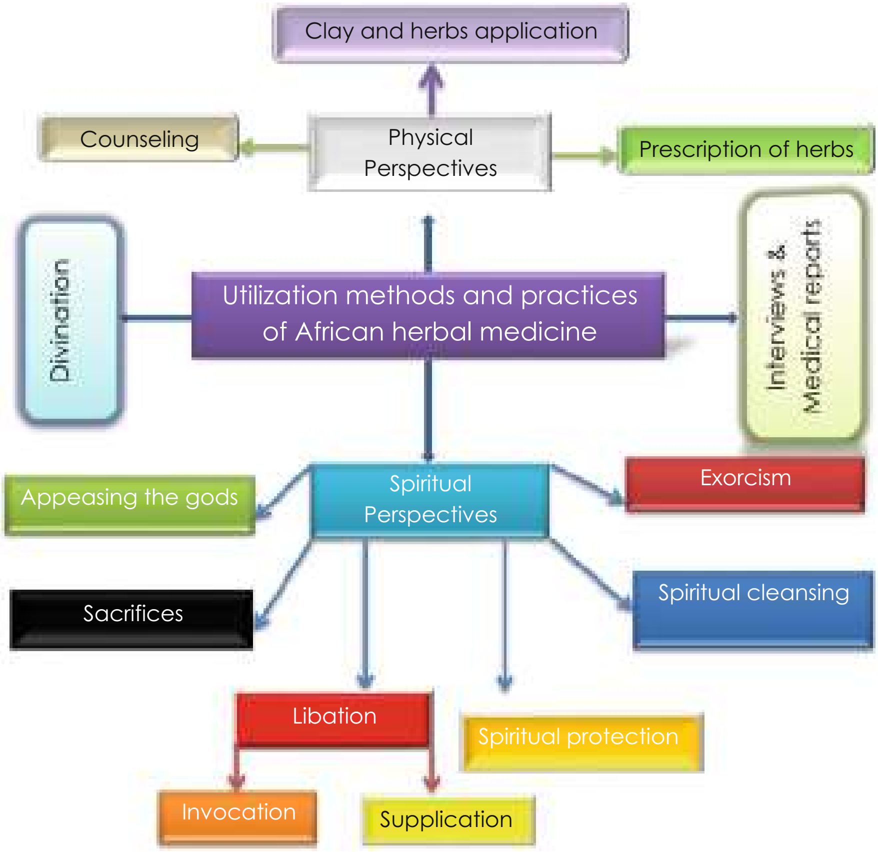


Fig. 2 Utilization methods and practices of African herbal medicine

diseases [83]. Traditional health practitioners (THPs) in African traditional medicine evaluate patients using the following approaches, as indicated in Fig. 2, in order to diagnose, cure, and prevent illness.

4.1 Divination

Divination entails communicating with the divine beings. It is a process through which information about a person or situation of disease is gained through the use of symbolism that is randomly placed to learn about curing diseases. It's also seen as a means of getting knowledge that is beyond human comprehension. Divination is a method in which seers get information by talking to spiritual entities including deities, ghosts, and ancestors [84]. Therefore, it is a crucial component in the traditional African method of illness diagnosis. Communicating with the non-human world is done while trying to establish the reason why the ailment or whether the sick person broke a rule. The bulk of the artifacts are divining bones, including bones from lions, hyenas, anteaters, baboons, crocodiles, wild pigs, goats, antelopes, etc. The bones stand for all the factors that may affect a person everywhere, regardless of their culture [85]. Divination is often the initial step in African traditional medicine and therapy because of its enlightening abilities [86].

4.2 Interviews and Medical Reports

Some traditional healers may do oral interviews to learn more about the patient's medical history, treatment history, and length of time in that condition. They are able to address the situation thanks to this method. In cases when a patient is unable to express their own wishes, the healer may ask a member of the patient's family to do so on their behalf. After the healing process is complete in the contemporary period, customers or patients are also encouraged to have medical diagnosis to make sure they are indeed recovered. The medical reports are often used as documentation for future reference and as a means of reassuring potential customers of their expertise and integrity. Because of the comprehensive approach used by the healers, no artificial boundaries are imposed between the healing process's natural and spiritual components [87]. Thus, the spiritual and physical spheres of health are seen from two key angles.

4.3 Spiritual Perspective

Cases with a spiritual component are treated as follows:

(i) Spiritual protection: If it is believed that an assault by bad spirits is the root of the illness, the individual will be safeguarded using amulet and talisman especially making craft bodies and spiritual bath to ward off the negative spirits. The rituals are intended to drive away bad and hazardous forces, spirits, or elements in order to get rid of any risks or evils that may have affected a family or society [88].

(ii) Sacrifices: Sometimes sacrifices are necessary to appease the spirits, gods, and ancestors. Animals like dogs and cats are occasionally sacrificed or buried alive at midnight to protect the soul of a dying person on the belief that their spirits are strong enough to replace life [84]. The fact that they are domesticated and raised among people raises the possibility that they might risk their own lives to protect their human family. This is particularly true in cases when the animal passes away mysteriously, leading people to assume that it gave up its life in exchange for that of its owner. Sometimes rituals are carried out to consecrate certain plants since they are essential to the healing process. Before, during, and after the manufacture and administration of medication, divine and ancestor sanctions are thought to be essential [89].

(iii) Spiritual purification: A sick person may be obliged to take a bath (water or animal blood) at particular times for a set period of time. Some communities in Ghana regularly engage in this activity [88].

(iv) Please the gods: If an illness is believed to be brought on by casting a curse or breaking taboos to please the gods or ancestors depending on how serious the situation is. A person is often expected to give certain objects for sacrifice and/or libation, including immaculate animals (doves, cats, dogs, goats, and birds), local gin, cola nuts, eggs, and plain white, red, or black fabric. Usually, the gods specify these things. Depending on the nature and severity of the situation, the used things may be dumped into the river, let to decay, or put at strategic locations, often at crossroads on the outside of the village [90].

(v) Exorcism is the practice of driving out evil spirits. Many traditional tribes have the view that bad spirits are mostly to blame for illnesses, particularly mental illnesses. Only a priest or other religious leader with the necessary credentials and authority may execute exorcisms. Sometimes the devil was represented with a wax or clay figure, which was finally destroyed. Exorcisms often include whipping the person or touching them with unusual items like animal tails and other objects to frighten the ghost away. They may also involve singing and dancing to the rhythm of drums. The person who is possessed will get a little upset before eventually calming down after the spirit has been expelled from the body. This practice is also performed for those who are mentally challenged. They believe that the possessed individual will not achieve freedom until they are released from the influence of the evil spirit. Exorcism is thus seen to be essential [91].

(vi) Libation: In a libation, liquid – typically local gin – is poured over the ground or other inanimate objects, and then words are chanted or recited over it. It is often considered a kind of prayer. Historically, beer was the only alcoholic beverage available, but wine, whisky, schnapps, and gin have all joined the mix in recent decades. Corn flour combined with water is a common alternative in certain countries, while in others people drink coconut water or make wine from palm fruit [92]. Some forms of libation pouring include an invocation, a prayer, and a final conclusion.

- **Invocation:** They begin by calling upon the ancestors, the soil, and the supreme God. Pouring libations is a manner of showing respect to the dead and appeasing the spirits, as believed by those who partake in the practice.
- **Supplication:** After invocation, supplication asks the spirits, gods, or ancestors to help the worshippers obtain spiritual consecration (cleaning) for themselves or their society due to transgressions like breaking taboos (s). The specifics of what is prayed for vary according to the circumstances [92].
- **At the end:** After the libation pouring, they express gratitude by calling upon spirits and ancestors. They end up cursing their enemies for wishing them harm, which suggests that praying for the success of one's adversary is a waste of time. In turn, their enemies, witches, and bad-skilled opponents

will all die [17]. In this procedure, the individual delivering the prayers would pour the drink into the ground and the witnesses would then respond to each prayer point.

4.4 Physical Perspectives

The following strategies are used if the sickness is physical in nature:

(a) Herbs are administered to the sick individual in accordance with the nature of their ailment. Each prescription comes with detailed instructions on the dosage, timing, and how to prepare the herb.
(b) Application of clay and herbs: Some therapeutic processes may benefit from the application of clay and plants. It is also sometimes employed in ceremonies to ward off the bad spirits that cause disease, along with certain particular plants and clay.
(c) Counseling: The sick person is occasionally advised on the dos and don'ts of treatment, the foods to eat or avoid, and to generally behave well in accordance with established social and cultural norms. If this is not done, the good spirits will withdraw their blessings and protection, which will leave the sick person vulnerable to illness, death, drought, and other misfortunes. When a taboo has been broken, this is often done [93].

In addition to offering effective and affordable treatments for the common illnesses that plague populations in the African region (such as malaria, stomach infections, respiratory issues, rheumatism, mental health issues, bone fractures, infertility, complications of childbirth, etc.), THPs also provide counseling, advice, and solutions to prevent future recurrence.

5 Unusual Aspects of the Use of Traditional Herbal Remedies in a Few African Nations

The physical world around us, including the cattle, trees, people, and towns, as well as the invisible, supernatural world of spirits, powers, and sicknesses, may all be understood in an African manner, just as there is an African way of comprehending God [94, 95]. Over the course of centuries, people created distinctive indigenous healing practices that were tailored to and characterized by their culture, beliefs, and environment [61]. These practices met the health requirements of their communities.

Different ethnic groups and cultures have created various healthcare systems and therapeutic approaches in order to address the many diseases, symptoms, and causes that they are aware of. Despite differences, there are significant parallels between how traditional medicine is practiced in many African nations. The World Health Organization (WHO) has encouraged the development of national policy and regulations as crucial indicators of the degree to which traditional medicine and complementary and alternative medicine are integrated within a country's national healthcare system in response to the growing popularity of these practices. The following is a description of several nations' unusual customs:

5.1 Ghana

Herbal medicine in Ghana is often used as the number one line of defense against disease, particularly in rural regions. Traditional healers are often used for a variety of reasons, including the challenge of inability to get access to healthcare institutions, inadequate infrastructures, and the cost of care. In addition, the patient to doctor ratio for conventional healers is 1:200 as opposed to the approximately 1: 20000 ratio for medical practitioners. This has a significant impact on how healthcare decisions are made. The choice of kind of healthcare is also influenced by other variables, including financial position, educational level, and recommendations from friends and family [96]. Traditional medicine has been practiced in Ghana for centuries. Most households have a basic familiarity with traditional medicine, which has been passed down through the generations through folklore, although this knowledge is usually held by spiritual healers**.**

Modern science-based medicine is widely accepted in Ghana, although traditional medicine is still highly regarded. They consider both the spiritual and physical sides of recovery. Herbal spiritualists, also known as "bokomowo," are widespread across the nation and engage in occult rituals, divinations, and prayers.

Traditional healers and practitioners in several Ghanaian groups, particularly in the Akan tribes, believe that breaking taboos may cause serious sickness in the offender(s) or community [97]. In the traditional religion of Africa, taboos play a significant role. A community or a group of people forbids them from doing or living certain ways.

The Ministry of Health of Ghana has established a Traditional Medical Directorate with the goal of creating a comprehensive, distinguishable, and standardized supplementary health system based on traditional and alternative medicine expertise. The current situation in Ghana involves the construction of institutions to combine scientific research into plant medicines with the inclusion of traditional medicine in university curricula [98]. Additionally, conventional medical colleges that provide degrees now prepare and certify herbal physicians.

5.2 Zambia

In Zambia, the number one rule is prognosis and using intricate therapeutic methods involving bush-based plants, several ceremonies, and the ultimate goal of curing sickness. Serious or persistent diseases call for "chizimba," which refers to permanently locking a sickness or illness away. To do this, a lizard must be killed, and its heart must be burned with charcoal and the roots of certain trees.

The mixture is applied to tiny wounds made on the affected region and left breast. You may utilize one plant alone or combine it with additional plants. The plant components are freshly collected, ground up, and allowed to dry before being immerged in H_2O or other solvents, such as alcohol. Major plant resources are utilized in powder form or burned to make charcoal. The majority of Zambia's 72 or so ethnic groups share 6 basic types of therapy: drinking, eating, drinking as porridge, making small wounds in the skin and applying, bathing with herbs, dancing to expel spirits, and steaming with boiling herbs. Nga:nga is the name of the traditional healer in Zambia [99].

5.3 Tanzania

Traditional medicine has been practiced separately from allopathic medicine in Tanzania since the time of the colonial period, but it is now in danger due to a lack of documentation, the loss of biodiversity in some areas as a result of resource discovery and excessive mining, climate change, urbanization, and agricultural modernization. Rural and urban dwellers in Tanzania utilize traditional medicine to treat both acute and chronic illnesses. There are four major categories of traditional healers: bone setters, herbalists, traditional birth attendants, and diviners. Indigenous medical knowledge began to disappear as the majority of traditional healers aged and died, the young people who were supposed to carry on the tradition shied away from it, and those in remote regions died of AIDS. Lack of information about highly threatened or endangered medicinal plant species in Tanzania was another barrier to the growth of traditional medicine [100]. The Ministry of Health now oversees the practice of traditional medicine. By raising knowledge of the value of traditional medicine and medicinal plants in healthcare and educating traditional health practitioners on best practises, conservation, and sustainable harvesting, efforts are being made to expand the use of traditional medicine [101].

5.4 South Africa

In South Africa, traditional medicine is a daily part of thousands of people who live therein. Actually, the usage of traditional medicines, often known as muti, is reported to be at 80% of the population. The term "muti," which is derived from the phrase "medicinal plant," designates drugs that are traditionally obtained from plants, minerals, and animals.

Traditional healers of the southern Bantu group are called Sangoma or Inyanga, and they are the keepers of healing power. Depending on the patient's condition, a female traditional practitioner may use one of many treatment modalities during a normal session. Antibiotics and immune system boosters are used to treat disease and weakness, while sedatives are used to treat depression, medications that induce vomiting are used to cleanse the digestive system, and bitter tonics are used to stimulate appetite. Before giving them the proper herbal remedies, she often gave them advice [102].

5.5 Kenya

Kenya, like many other nations in Sub-Saharan Africa, is struggling with a lack of health workers, especially in rural regions. According to anecdotal data, at least 80% of the rural dwellers are poor and only have access to traditional medical practitioners (TMPs) [19].

There is relatively little quantitative research or literature on traditional medicine, alternative healing methods, the demand for traditional medical practitioners, or the function that they play in delivering specific healthcare services to the rural poor in Kenya. TMPs are thus now underrepresented in HRH planning processes because of a lack of official government recognition, and their operations are still uncontrolled. Hospitals are chosen if they are accessible and economical, according to data gathered from the community [103].

In Kenya, alternative medicine practitioners are know as allopathic physicians by a large margin. They follow the same customs as other African nations. They often mix contemporary and natural medications, particularly if they have chronic illnesses like HIV/AIDS, high blood pressure, cancer, or diabetes [104].

5.6 Nigeria

Apart from their Western equivalents, who practice like other tribes, Nigeria's distinct ethnic groupings have separate medical professionals. They are referred to as “babalawos” by the Yoruba, “dibia” by the Igbo, and “boka” by the Northerners or Hausas [105]. People's lives have been touched by traditional and herbal treatments, particularly in rural regions where access to mainstream medicine is limited [105]. The high cost of Western treatment makes traditional medicine appealing, in addition to the challenge of proper approach and the concern of adultrated or phoney medications. Most modern herbalists and naturopaths are well-versed in the healing qualities of plants and the social construction of illness. They use this information in conjunction with state-of-the-art skills and procedures learned at any of the various schools dedicated to either herbal medicine or homoeopathy, which allows them to effectively treat and prevent illness. A list of Nigerian medicinal plants and their therapeutic benefits has been released by the Association for Scientific, Identification, Conservation and Utilization of Medicinal Plants of Nigeria (ASICUMPON) [65]. Information from other books has also been valuable [62, 68]. Lack of sufficient standardization and safety rules is still Nigeria's biggest issue with herbal medication [106]. However, the attention and participation of educated and scientifically oriented individuals in the practice of herbal medicine have greatly reduced the mystique and enhanced the acceptance of these treatments by a broader proportion of the population than might otherwise be suspicious.

6 Adverse Effects of Herbal Medicines

Herbal medicines are widely utilized because of their purported benefits, but they are not risk-free. Even while there is evidence that shows medicinal herbs may be beneficial, they can also have unintended consequences if used improperly or in excess. The likelihood of unintended outcomes rises when consumption is indiscriminate, careless, or uncontrolled, and when adequate standards are absent. These topics have been the subject of several international symposia on the study of medicinal plants [107]. Many African plants are poisonous, yet some of the continent's flora also has promising therapeutic applications [79].

In Nigeria, researchers observed that many people use herbal medicine despite the fact that many of them are uninformed of the potential toxicity of these medicines [74]. Unfortunately, several all-natural remedies have been linked to potentially fatal adverse effects [108].

The current measures are inadequate in negative effects monitoring of herbal medicine [61, 109]. Because of their all-natural and biodegradable makeup, herbal medications are often seen as being just as safe as their synthetic counterparts. A very small number of patients that make use of herbs and their supplements relate that information to their physicians. Many adverse pharmaceutical reactions may not be formally disclosed to various health regulatory authorities or organizations, since no pharmacovigilance studies are being undertaken and patients are not revealing information to health services. Poisoning from herbs may be difficult to diagnose. When damage is thought to have been caused by a herbal remedy, a conclusive prognosis might be challenging to draw an inference without a complete study of the herbal formulations and its components [110].

When taken in conjunction with conventional treatments, herbal remedies have a very low rate of negative effects [61]. Several investigations have shown that reported adverse drug reactions may frequently be attributed to incorrect preparation and usage of herbal therapies. It was discovered that liver issues were the most significant indicators of toxicity as a consequence of prolonged usage, which was shown in a study of WHO [61].

These plants' poisonous ingredients, such as saponin, tannins, alkaloid, oxalates, etc., are probably to blame for the toxicities that have been documented. Microbial infection as a result of unsanitary circumstances during preparation is another significant cause of toxicity in herbal medicines that is worth highlighting [111]. When herbal medications are used with certain conventional pharmaceuticals or supplements, toxicity may also emerge from a herb–drug interaction [57]. Toxic effects from misusing and misidentifying plants are also possible.

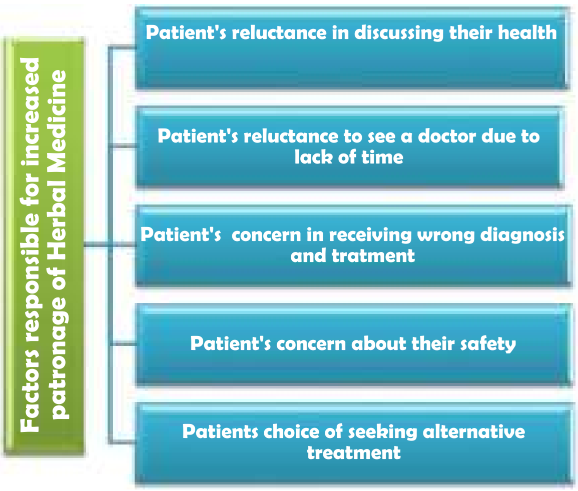
Therefore, it is important to convey accurate information of developing risk result as a crucial component of pharmacovigilance, which might really enhance patients' health and safety. This necessitates more coordination between conventional and contemporary medical professionals and pharmaceutical regulatory agencies. If known, the response time interval of a medicine and the incidence of an negative response may aid in determining the causative relationship in the management of pharmacovigilance [79]. Such details may be very helpful in interpreting medication safety signals and making judgments about further precautionary measures to be implemented in relation to future usage.

7 Factors Responsible for Increased Patronage and Self-Medication with Herbal Medicine

Many people believe that the various substances in herbal remedies work together synergistically, but this is not always the case. Herbal therapies are mainly made up of fragments of plants or unpurified plant extracts. The recent spike in public interest in herbal remedies has been attributed to a number of factors, including varying assertions about the efficacy or effectiveness of plant medicines. Unhappiness with the outcomes of synthetic drugs hoping that traditional remedies may bring about useful treatment of various ailments for which synthetic therapy with drugs proved unsuccessful or insufficient; reasons for this include (i) consumer preference for natural therapies and an increased interest in alternative medicines; (ii) an unshakeable conviction that herbal products are superior to manufactured products; and (iii) the high cost and unpleasant side effects of most conventional therapies.

There are a number of factors that contribute to patients and individuals turning to herbal remedies for treatment on their own (see Fig. 3), including (i) patients' reluctance to discuss their medical issues and concerns about the privacy of their health information, (ii) patients' reluctance to see a doctor due to a lack of time or (iii) patients' concerns about the possibility of receiving the wrong diagnosis and treatment due to vague or nonspecific symptoms, and (iv) patients' concerns about the safety of trying (v) patients' right to pick their own physician is inspiring them to try out alternative treatments like herbal remedies, with many making their decisions based on hearsay such "it helped a friend of mine" [112]. Furthermore, due to the

Fig. 3 Factors responsible for increased patronage of herbal medicine



influence of religion and the development in spiritual awareness [113, 114], many individuals are becoming more inclined to accept the therapeutic effectiveness of a course of treatment that is led by faith or intuition rather than by scientific logic.

When the body's innate capacity for healing is highlighted, the allure of herbal therapies increases [112]. These items have been significantly projected into larger spotlight by numerous herbal medicine producers' and their sales representatives' marketing tactics and efforts in addition to all of the aforementioned elements. Numerous media advertisement, especially those on television and radio, have greatly raised consumer awareness and given herbal goods an excessive amount of legitimacy and credibility [112, 115]. To appeal to the many age groups prevalent in the society, these advertisements are thoughtfully prepared. Herbal remedies are popular among the young because of the positive psychological benefits they have, the stress relief they provide, and the delay or prevention of the aging process. Herbs are recommended for their antiaging or rejuvenating benefits on the elderly, and for their fat-burning and beauty-enhancing properties in women [112].

8 Influence of Regulatory Policies on Safety of Herbal Medicines

It has been noted that the designation of many of these items as foods or dietary supplements in various countries is the primary cause of the majority of the issues related to the use of traditional and herbal medicines. As a result, before sale, no proof of the quality, effectiveness, or safety of these herbal medications is needed. In a similar vein, manufacturing standards and quality checks are often less stringent or monitored, and conventional healthcare providers may not always be qualified or certified. As a result, both national health authorities and the general public now have serious concerns about the safety of traditional and herbal medications [116]. Up until 2011, a customer in the UK may get a herbal product through one of three legal channels. The most prevalent method is the use of an unregistered herbal treatment, which is exempt from quality and safety requirements as well as consumer safety information [117]. The prepared tradional products must get license from the appropriate authorities before manufacturing can be tagged as "conventional" items or register as "traditional herbal medicinal products" [117, 118]. Similar to conventional medications, approved herbal medications are subject to quality, safety, and effectiveness standards. Therefore, for safe usage, they must be accompanied with detailed literatures which includes warnings, instructions, usage procedure, complications, storage instructions, and regulatory information. Typically, a booklet with this information is put inside the product packaging [117]. On the other hand, certain herbal medications cannot receive a license to market these items since there is inadequate proof of repeatable effectiveness to fulfil regulatory criteria [119].

In light of this, the UK established a "simplified registration procedure"for traditional herbal medicines known as the Traditional Herbal Medicines Registration Process. Herbal medicinal products sold under this system are expected to meet rigorous quality and safety standards, to recognize traditional uses as valid indications for their use, and to provide consumers with safety information in the form of a leaflet [117]. However, in different parts of the globe, particularly poorer nations, a wide variety of unregulated herbal products are sold without much in the way of oversight. The prevalent misconception that natural products are safe to use without risk of poisoning or adverse reactions may potentially lead to overexposure or even death. This misconception is not exclusive to underdeveloped countries. Even in advanced countries, people are drawn to "natural" products without fully understanding the risks associated with them, particularly with prolonged or excessive use [120].

9 Challenges of Monitoring Safety of Herbal Medicines

It is long overdue for both herbal products and pharmaceuticals to be included in pharmacovigilance systems, considering their widespread use. Since it is vital to evaluate the hazards connected with the use of herbal medicines primarily in terms of population exposure [121, 122], the efficacy and safety of the formulations has raised a cause for concern in public health. Protecting the public's health and promoting their safe use requires rigorous toxicity evaluation and vigilant pharmacovigilance [123]. This is because there has been a rise, in many regions of the globe, in the number of poisoning cases traced back to the use of herbal remedies. When laws concerning the use of traditional or herbal medicine are drafted and put into effect in different parts of the world, they often run into a number of problems. The regulatory status of traditional, complementary/alternative, and herbal medicines; the evaluation of their safety and efficacy; quality control; safety monitoring; and insufficient or poor knowledge about these medicines among national drug regulatory authorities are all problems faced and shared by many countries [122].

9.1 Challenges in the Regulatory Status of Herbal Medicines

Each country has its own criteria for classifying and defining herbal medicines. Depending on the rules controlling foods and medicines, the same medicinal plant may be categorized as a food, a functional food, a dietary supplement, or a herbal remedy in different countries. It also makes it hard to define the concept of herbal medicines for the purposes of national drug regulation [122], which may be confusing for patients and consumers. In the USA, natural products are regulated under the Dietary Supplement Health and Education Act (DSHEA) of 1994 [124].

A "dietary component" is a part of a product that is intended to be ingested as a dietary supplement. Vitamins, minerals, herbs, and other botanicals may be included in such products [124]. If a herb was on the market before 1994, the DSHEA may exclude it from further toxicity testing. In this scenario, the Food and Drug Administration (FDA) must provide evidence that a herbal medication or "dietary ingredient" poses a risk to human health. Another serious problem in many nations is that regulatory information on herbal drugs is frequently not exchanged between regulatory agencies and safety monitoring or pharmacovigilance centeres [121].

9.2 Challenges in the Assessment of Safety and Efficacy

It is an undeniable fact that there is a complexity of the requirements, research procedures, standards, and methodologies needed to assess the safety and effectiveness of herbal medicines in comparison to orthodox or conventional pharmaceuticals [122, 123]. Only one traditional remedy/plant part may include different hundreds of active ingredients, a combination traditional remedy may have many more of that amount. The amount of time and resources needed would be enormous if each active component in each plant used to design or create a herbal medication had to be separated. When a herbal product is a blend of two or more plants, such analysis may virtually be impossible [122].

9.3 Challenges in Quality Control of Herbal Medicines

The quality of the raw components used in their production has a significant impact on the safety and efficacy of these herbal medicines. Environmental variables and high-quality agriculture are all examples of extrinsic factors that influence source material quality in addition to intrinsic (genetic) features. Due to the complex interplay between these factors, it is difficult to perform quality controls on the herbal medicine raw materials [121, 122]. Species identification, proper storage, and material-specific sanitation and cleaning procedures are all vital for ensuring the quality of raw materials in the pharmaceutical industry. One of the most common challenges in ensuring the finished traditional herbal formulation quality, particularly the combination of traditional medicines, is validating the existence of all plants or starting components [122].

As a result, overall requirements and processes for quality control of final herbal products continue to be much more challenging [121, 122, 125] compared to those of conventional medications. In countries where herbal medicines are regulated, the WHO continues to support the development of quality assurance and control measures like national quality specifications and standards for herbal materials, Good Manufacturing Practices (GMP) for herbal medicines, labeling, and licensing schemes for manufacturing, import, and marketing [121].

9.4 Challenges in Safety Monitoring of Herbal Medicines

Increased use of herbal products in developed countries, reliance on plants by many in developing countries as a primary source of medicine, lack of regulation of herbal medicines in most countries, and the occurrence of high-profile adverse events have all contributed to a heightened awareness of the need to monitor safety and increase understanding of the potential harmful and beneficial effects associated with the use of herbal products in recent years.

Accidental ingestion of the wrong plant species, herbal products adulterated with other, unregistered medicines, contamination with toxic or hazardous substances, overdosing, misuse of herbal medications by consumers or healthcare professionals, and the concurrent use of herbal medications with other medications are all potential causes of negative outcomes following ingestion of herbal medicines. It is far more challenging to analyze adverse events connected with the use of herbal remedies than it is with conventional pharmaceuticals [122, 123], despite the growing interest among consumers, regulatory authorities, and healthcare professionals in doing so. Also, it is known that determining safety is complicated by factors such as the source of the plant, the variety of processing techniques used, the route of administration, and the possibility of drug interactions [126].

Identifying and gathering medicinal plants used in herbal therapies might be difficult since most herbal medicine manufacturers lack an expertise of taxonomy, botany, and documentation [127]. In order to eliminate the confusion caused by the common names, it is crucial to adopt the most generally used binomial names (including their binomial synonyms) for medicinal plants. For instance, the narcotic component included in *Artemisia absinthium* L., which goes by at least 11 different common names, has the potential to disrupt the central nervous system and lead to general mental impairment. Seven of the names used in common do not correspond to their botanical counterparts. *Valeriana officinalis* (garden heliotrope), which contains valepotriates with hypnotic and muscle relaxing effects, and *Heliotropium europaeum* (heliotrope), which contains potent hepatotoxic pyrrolidine alkaloids, are commonly mistaken with one another due to the popularity of common names. When reporting adverse drug reactions to herbal drugs, it is critical to provide the full scientific name of the plant, the component of the plant that was used, and the manufacturer's name. Therefore, the success of herbal medication safety monitoring relies on effective collaboration between botanists, phytochemists, pharmacologists, and other relevant players.

10 Importance of Traditional Medicine for Indigenous Peoples and Local Communities

Besides its obvious value in healing, traditional medicine also serves as a major economic driver in many regions. Traditional medicine, which must be emphasized, may have a significant role in shaping the character of a certain group. Pre-industrial tribes still participate in wild collecting, domesticating, cultivating, and managing medicinal plant resources since they are the ones who first identified most of the plants being used today [128]. Numerous indigenous peoples and local communities are helped by this economic activity, which in turn encourages the preservation of TM. While some therapeutic plants are grown for commercial purposes, the majority are still gathered from the wild [129].

Indigenous people and local groups may be knowledgeable about medical uses of plants as well as how to prepare and gather them. The indigenous people and local groups that have traditionally utilized herbal remedies may benefit greatly from this knowledge, as can anybody attempting to export and use medicine outside of its natural setting. TMK could also influence a society's spirituality and way of life. Traditional African medicine, for instance, is distinguished by a holistic worldview that sees all living things as part of a single, interconnected whole from which they get their life energy [130]. The practice of spiritual healing, which is regarded to be mediated by supernatural or spiritual forces, may be part of traditional African medicine. For psychological counseling and medical care, the majority of Africans with HIV/AIDS rely on traditional healers and herbal remedies [131].

11 Future Prospective

The advances recorded in the utilization of herbal medicine in Africa is imminent. The economic prospect of the utilization practice is enormous. However, the process can be intensified by promoting industrial and government plans and funding especially applying the biotechnological approaches in medicinal plant cultivation and processing. Also establishment of a scientific training and retraining programmes of traditional healthcare professionals should be encouraged. The implementation of positive strategies or blueprints for proper standards and regulation of herbal medicine practice is highly recommended. There is need for collaborative efforts to culminate in the beneficial effects of the practice of herbal medicine in Africa to avoid reprimand attitudes and perception of herbal medicine practices. The conservation and sustainable methods of harvesting medicinal plants need to be given top priority for habitat preservation. Proper documentation of medicinal plant types, their ecology, and method of distribution as well as various traditional extraction techniques for the preservation of cultural practices should be given attention. A partnership of practitioners of herbal medicine with scientist and researchers should be reassuring for proper collaboration and safeguarding the intellectual property rights of the herbal medicine practitioners.

12 Conclusion

The use of herbal medicines involves integrating a number of therapeutic techniques from indigenous medical systems that may date back many generations. This integration frequently yields helpful instructions for the choice, preparation, and application of herbal formulations for the treatment, control, and management of a wide range of illnesses. The availability and accessibility, efficacy, safety, cost, and cultural compatibility are the most often cited benefits of using herbal remedies in Africa. In African traditional medicine, practitioners evaluate patients utilizing their knowledge of the following techniques: divination, interviews and medical reports, spiritual and physical perspectives in order to diagnose, cure, and prevent illnesses.

In Africa, there has long been use of herbal medicine via various means and customs. The usage of herbal medicines for basic healthcare needs is still on the rise in Africa despite the advancements in Western medicine.

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Herbal Medicine Methods and Practices in Nigeria

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Abstract

The Nigerian ecosystem is most varied and contains diverse biological resources and biodiversity. The ecosystem serves as a source of medicines, foods, and diverse natural products that form the basis of traditional medicine in Nigeria. The use of traditional/herbal medicine has been in existence since time immemorial, and its practices are diverse. Its peculiarity and use are principally a function of the ethnic community in which the herbal practice is performed. It is used based on the active photochemical contained in the plants and therefore is a major component of treatment processes in over 300 ethnic communities in the country. The practice of herbal medicine is faced with a major challenge of documentation and a lack of clinical evidence to justify its use in most rural areas. This has posed a serious problem in meeting a requirement for herbal medicine that is standard and can be adopted in clinical practice. The country is rich in a collection of plants that can be used for medicinal purposes and herbal treatments, and their efficacy has been reported by research. Plants such as moringa (*Moringa oleifera*), neem (*Azadirachta indica*), and guava (*Psidium guajava*) to mention a few have been reported and found to exhibit good microbial inhibition. It is important to report the richness of Nigerian herbal medicine and its efficacy in treatment activities which will serve as evidence to support the incorporation of this approach into clinical medicine procedures. A number of plants in Nigeria have also been reported to contain terpenes, glycosides, alkaloids, and polyphenols which have therapeutic potential. Compounds such as orosunol, alloeudesmenol, and hanocokinoside have been identified but are yet to be researched in Nigeria. This chapter tends to highlight the medicinal plants in Nigeria, their application and use, and various therapeutic activities they can perform. Conversely, there is a dire need for information and research into the use of herbal plants to cure neglected tropical diseases (NTDs) that can be facing humans in Nigeria. It also highlights the prospects and future trends in the use of herbal medicine in the quest for acquiring a basis for herbal medicine procedures to be incorporated into Nigerian clinical care.

Keywords

Phytochemicals · Herbal use · Microbial agents · Medicinal plants · Conservation

Abbreviations

ACE	Angiotensin-1-converting enzymes
AIDS	Acquired Immune Deficiency Syndrome
ARSO	African Regional Standard Organization
AU	African Union
CNS	Central Nervous System
FTIR	Fourier-transform infrared spectroscopy
GC-MS	Gas Chromatography-Mass Spectrometry
GRAS	Generally Regarded As Safe

HIV	Human immunodeficiency virus
IUCN	International Union of Conservation and Nature
NMC	Nigeria Medical Council
NMR	Nuclear magnetic resonance
NNMDA	Nigerian Natural Medicine Development Agency
NTDs	Neglected Tropical Diseases
NTMA	Nigerian Traditional Medical Association
NUMHP	Nigerian Union of Medical Herbal Practitioners
SARS	Severe Acute Respiratory Syndrome
TNMP	Traditional Nigerian Medicine Practitioner
UV	Ultraviolet
WHO	World Health Organization

1 Introduction

Herbal medicine is the process employed in the use of herbal plants to treat diseases thereby enhancing the health and well-being of humans [1]. The potency of herbal plants differs and must be taken with caution. The use of herbal medicine is crucial in the provision of healthcare to the populace in rural and most developing and underdeveloped countries. Its use is also adopted in some developed countries in the treatment of some ailments. Herbal medicine may provide healing for ailments, the welfare of the community, growth, and the creation of wealth for individuals. The World Health Organization [2] reported the issue of global health is a major concern, as infectious diseases have been reported to be a cause of threat to humans and have accounted for 61.7% (5.9 million) out of the 9.6 million deaths occurred in the sub-Saharan regions of Africa. The use of plants with medicinal values in health care has been used since the olden years. Studies have shown evidence of the efficacy of medicinal plants and have provided information on the plant-based compounds that have therapeutic uses [3]. Based on this, researchers are developing research tools that are modern to develop herbal medicine processes and paying the way for its adoption into modern medicine [4]. At the concoction dosage, herbal medicines of plant origin are generally regarded as safe (GRAS) and this is based on their usage [5]. Based on this, the natural and abundant source of active drugs are plants which are very useful in the ethno-medicinal cure of various ailments [6].

Herbal medicine has been seen as an integral part of traditional medicine with a wide range of characteristics and components recognized by the World Health Organization (WHO). Their practice cuts across various sectors such as health practices, Indigenous knowledge, beliefs, spiritual therapies, and techniques which may be carried out in unison or in combination to address issues relating to diseases, and diagnose or prevent all forms of illness. In developed countries, the concept of herbal medicine has evolved into contemporary medicine. The use of herbal medicine is defined by the beliefs, environment, and culture and how it satisfies the needs and health concerns of the locality [3]. The increased and widespread use of herbal medicine has promoted its integration into the national healthcare systems in some countries by World Health Organization. This will go a long way in the development of policy and regulations nationally and easy incorporation into the national healthcare policy of the country.

Herbal medicines basically refer to all herbs, herbal preparations, and materials, or finished products from herbal products that may contain active ingredients from parts of plants or materials. It is also called botanical medicines or phytomedicines. The materials from plants to be used are the roots, seeds, berries, flowers, or barks and are components of many drugs in conventional medicine. The dependence on herbal medicine for health issues is a major activity in over 80% of the population in Asian and African countries [3]. It was estimated by World Health Organization that about 70–80% of the population depends on alternative medicine in developed countries of the world; it is the most popular form of practice and very lucrative in the international market. The major issue is the irresponsible or unregulated use of these herbal products which are capable of increasing the risk of toxicity in humans. The effectiveness and safety of herbal medicine products and practices lack scientific evidence making it difficult to evaluate their performance [4]. Different reactions have been reported when used or used in combination with conventional medicines, and despite its international adoption, there is no international standard for its evaluation. Dearth in the knowledge of plant population and its use as medicines is a major threat to the sustainability of herbal plants.

In Nigeria, the application of herbal medicine is more pronounced in the grassroots areas as a measure of clinical care. It employs approaches that hover around the use of holistic natural healing methods, and the growth and wealth creation of the locality. Due to this, recent research is being geared towards developing a framework that can be adapted to modern health practices using modern research tools and equipment [7]. The practices of herbal medicine are endemic to every society as they have developed methods for approaching this practice. The concept of illness and health involves the individual and totality of being which covers the social, emotional, spiritual and physiological, and immediate environment. The system of traditional medicine in Nigeria encompasses the following: traditional systems – which have been practiced by several Nigerians over decades; folk systems – which focus on the culture, faith, and beliefs of people; Indigenous system – which involves the knowledge level and resources that belong to the local people; and communal system – which involves resources collectively owned. In Nigeria, there are over 300 ethnic groups that are diverse and the practice of herbal medicine differs across these localities [8].

The acceptance and use of Indigenous herbal medicine must first be located within the global and historical concept of the locality. Despite the existence of herbal medicine over the years, a conflict of interest in the knowledge and use of these herbal medicines exists. This occurred especially when the European cultures crept into our health system in the fifteenth century and it distorted the traditional herbal medicine system and the local practices while ensuring the objectives of the European cultures are met [3]. Herbal medicines can be used by adults, pregnant women, and even children, and it has been evaluated, but their use in the population without health issues is yet to be evaluated empirically. With the immeasurable importance of herbal medicine, its use may produce several negative effects such as rashes, allergic reactions, asthma, headaches, nausea, vomiting, and diarrhoea [1]. Its use must be prescribed by qualified and registered personnel. When herbs are used in combination with pharmaceutical medications, interactions based on their activities may occur. Based on this, it is important to inform medical doctors about the herbal mixture taken at that point of diagnosis. The issue of unregulated use of herbal plants and mixtures is of great concern as it may affect organs of the body when in excess. This review, therefore, describes herbal plants, their use, various properties of herbal plants, and future prospects of herbal plants in Nigeria due to the large number of plant species that have not been analyzed for phytochemical constituents.

1.1 Concept of Herbal Medicine

Herbal medicine as described by the World Health Organization involves the total skills, knowledge, and practices which is based on various theories, experiences, and Indigenous beliefs of different cultures and used in the prevention, treatment, diagnosis, or improvement of health, both physical and mental. Its practice is common in other African countries and Asia, and an interesting area of research is the polypharmacological formulation of extracts from plants for use in healthcare [7]. The screening of herbal plants based on their phytochemical constituents is done to ascertain their antimicrobial activities against a broad spectrum of microorganisms. The resistance activities by most bacteria have caused a serious problem [8] and have stimulated continuous research in this area for the development of therapeutic agents that are safe. Plants with important components and promising attributes in inhibiting microorganisms that cause diseases and their use in ethnomedicine have been documented [9]. Research activities are in place to understand the functioning of herbal plants, and these results have formed a prototype for about 25–30% of marketed drugs [10].

1.2 Herbs, Traditional Knowledge, and Concept of Health in Nigeria

The health and well-being of individuals in every society differ as a result of various developmental approaches, knowledge, and practices. In most cultures of the world, the concept of good health and the condition of illness involve the physical, emotional, social, spiritual, and psychological well-being within its environment [8]. Herbal medicine in Nigeria has its roots in the traditional, folk, communal, and Indigenous systems thereby forming a link with culture, faith, and beliefs which has evolved over the years and is at the centre of the national economy [9]. The concept of herbal medicine is based on different experiences, beliefs, sociocultural diversity, and diverse ethnic groups [10], and this has limited the development of uniform herbal medicine practices.

WHO [11] described herbal/traditional medicine as "the totality of skills, knowledge, and practices that is based on belief, theories, and experiences which are endemic to different cultures." Herbal medicine is used in maintenance, prevention, improvement, diagnosis or treatment of illness, both mental and physical [12]. Based on this, it can be said that historic experiences dictate the form and contents of herbal medicine to be used for disease treatment and it is within a social and ecological context. Over the years, the perception of Nigerian herbal medicine usage has persisted among educated Nigerians most especially in the medical profession [13]. Based on this, the Traditional Nigerian Medicine Practitioner (TNMP) is constantly involved in ceremonies that are targeted towards restoring individuals to social, spiritual, and psychological harmony [14]. Since the 1960s, the Nigerian research community and some other post-colonial states in the developing world realized the inherited healthcare system and health needs of the people [15]. After these conscious efforts in the development of herbal medicine, WHO reported an increasing global trend to include herbal medicine in healthcare systems in developed and developing countries to address chronic ailments which conventional medicine could not address [2]. This has provided a leeway for the government to explore ways of connecting Indigenous healing practices with the medical systems.

2 Phytochemical Components of Herbal Plants

Phytochemicals are bioactive organic compounds that are contained in herbal plants and have medicinal purposes. This phytochemical can function in defensive roles to fight against chronic diseases and genetic and infectious diseases. They are found in grains, fruits, vegetables, and other forms of products from plants [16–18]. They can perform activities that are metabolic in nature, functioning as primary metabolites in all plants such as fats and sugars, and secondary metabolites providing specialized functions, and are found in plants that are of small range. Secondary metabolites are described as the biomolecules produced from sustainability or adaptation processes by plants in their native environments. These result from factors such as pollination, predation, and protection from drought and are not required for survival by the plant [19]. These metabolites and pigments can be processed into various drugs because of their healing effects, such as examples listed in Table 1.

Herbal plants are screened that differ in bioactive compounds; some of the compounds are tannins, alkaloids, and saponins [25]. The functional classes of phytochemicals that have therapeutic potential in them are:

- Anticancer agents
- Antioxidants
- Detoxifying agents
- Immunity-potentiating agents
- Neuropharmacological agents [19]

Table 1 Plants and the secondary metabolite they contain

S/N	Secondary metabolites	Plant source	Botanical names	References
1	Quinine	Cinchona plant	*Cinchona officinalis*	[20]
2	Digoxin	Foxglove plant	*Digitalis purpurea*	
3	Codeine and morphine	Poppy plant	*Papaver somniferum*	
4	Inulin	Dahlias plant	*Dahlia pinnata*	
5	Phytoalexins	Orchids	*Orchis militaris*	[21]
6	Teroenoids and steroids	Orchid plant	*Orchis militaris*	
7	Triterpenes and steroids	Foxglove plant	*Digitalis purpurea*	[22]
8	Stilbenoids and bibenzyls	Orchid plants	*Dendrobium chrysotoxum, Orchis militaris*	
9	Phenantherens	Orchids	*Gymadenia conopsea, Bletilla striata, Dendrobium moniliforme,* and *Cypripedium macranthos*	
10	Alkaloids	Orchids belonging to the genus *Dendrobium*	*Dendrobium nbile, D. chrysanthum, D. anosmum, D. crepidatum,* and *D. findlayanum*	[23]
11	Coumaris	Sweet woodruff and tonka bean	*Galium adorantum* and *DIpteryx odorata,* respectively	
12	Polyphenols	Ginger, guava, neem, and moringa	*Zingiber officinale, Psidium guajava, Azadirachta indica,* and *Moringa oleifera*	[23, 24]
13	Terenes	Moringa, guava, and neem ginger	*Moringa oleifera, Psidium guajava, Azadirachta indica,* and *Zingiber officinale*	
14	Glucosides	Guava, ginger, neem, and moringa	*Psidium guajava, Zingiber officinale, Azadirachta indica,* and *Moringa oleifera*	

Each functional class of phytochemical listed consists of various compounds which have multiple functions and varying levels of potency. Some plants can produce phytochemicals that have distinct functions and belong to the biochemical motifs [26]. Some of such bioactive compounds are listed in Table 2.

2.1 Phytochemical Screening of Medicinal Plants

In most plants used as medicines and aromatic plants, the major sources of active ingredients are found in the stems, barks, flowers, roots, or leaves [19]. The mode of action of these ingredients can be connected to the ecological or biochemical identities of plants and animals when functioning in the human central nervous

Table 2 Bioactive compounds with their distinct functions

S/N	Bioactive compound	Functions	References
1	Titerpenoids	Anti-inflammatory activity	[27]
2	Tannins	Anti-microbial, anti-inflammatory, and astringent properties	
3	Saponins	Blood cleansers, antibiotics, and expectorants	
4	Alkaloids	The effects are significant on the central nervous system (CNS)	
5	Glycosides	Increases the forces of systolic concentration	
6	Carotenoids	Anti-aging, antioxidant, anti-diabetic, anti-cancer, pro-vitamin, hepatoprotective properties, anti-obesity, and immune modulating	[28]
7	Phenolic compound	Antioxidants, attractants (carotenoids and flavonoids), structural polymers (lignin), defence response chemicals (phtyoalexins and tannina), and ultraviolet (UV) screens (flavonoids)	[29]
8	Terpenoids	Antioxidants, anti-cancer, anti-microbial, anti-inflammatory, and antallergic	[30]
9	Peptides	Anti-microbial, anti-cancer, antioxidative, anti-inflammatory, immunomodulatory, and anti-hypertensive	[31]
10	Fatty acids	Anti-cancer, anti-immune disorders, anti-vardiovacular diseases, and anti-osteoporosis	[32]
11	Terpens	Antioxidants, anti-cancer, antimicrobial, anti-inflammatory, food preservation, and antallergic	[30]

system [33]. The steps that are taken to adequately utilize the bioactive compounds from plants involve the following steps as described by [34]:

- Extraction
- Primary screening
- Isolation
- Purification
- Characterization
- Pharmacological and toxicological analyses

The principle of pharmaceutical extraction involves the division of microorganism tissues and plant constituents with the aid of a buffer solution or solvent that is selective and carried out according to protocols that are standardized [35, 36]. The quality of extract from plants is influenced by the part (root, bark, or leaf) of the sample, the extraction method, and the solvent used and concentration during extraction. The chemical constituent effects depend on the type of plant material, the source of the plant, the processing extent, the moisture level, and the size of particles [37]. The composition and quality of secondary metabolites are influenced by factors such as the type and timing of phytochemical extraction, the concentration and nature of the solvent, the temperature of processing, and the analyses of polarity [37]. As reported by [34, 38, 39], purification, identification, and characterization of

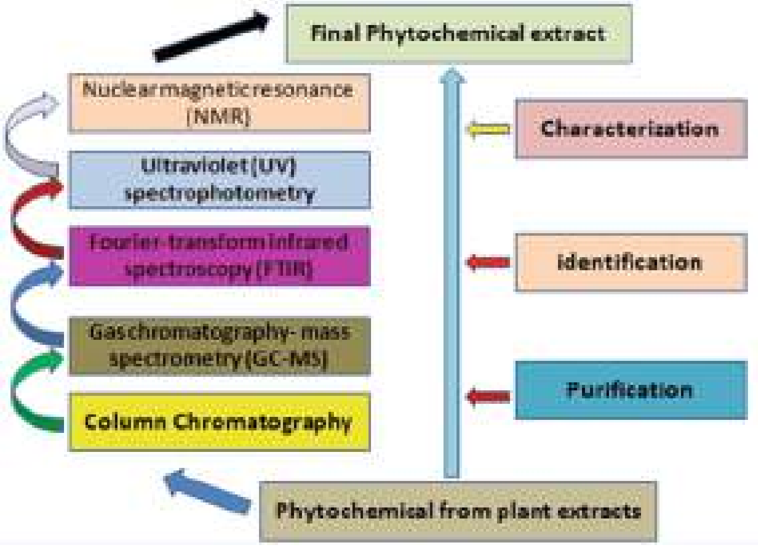


Fig. 1 Stepwise characterization of phytochemical groups. (Source: Refs. [34, 38])

phytochemicals from herbal extracts are carried out following the stepwise techniques of GC-MS (Gas Chromatography-Mass Spectrometry), FTIR (Fourier-transform infrared spectroscopy), UV (Ultraviolet), and finally NMR (Nuclear magnetic resonance) as illustrated in Fig. 1.

2.2 Processing Effects of Phytochemicals from Medicinal Plants

The antioxidant activities in herbal plant extracts can either increase or decrease as a result of the method of processing the herbal plants. Various methods can impact the availability of phytochemicals and their ability for use in pharmacology. An example is the sandpaper leaf (*Ficus exaperata*) which has the ability to inhibit arginase and angiotensin-1-converting enzymes (ACE). The best processing method is the soaking method, and it is far better than boiling, blending, and hand maceration [40]. The health benefits associated with herbal plants in mono or poly formulations result from a mixture of phytochemicals in a whole form and produce an overall action that may be due to the combined properties of antioxidants [36, 41]. The use of different processing methods, thermal or non-thermal, has been studied [42].

3 Medicinal Plants in Nigeria

Nigeria is blessed with diverse species of flora which may be used for medicinal purposes. Despite the discovery of antibiotics and their recorded successes, death caused by infectious diseases is ranked second in the world [43]. The quest for new

Table 3 Some Nigerian medicinal plants and their therapeutic functions

S/N	Plants with medicinal properties	Point of action	References
1	*Macaranga barteri* and *Ageratum conyzoides*	Antiviral	[45–48]
2	*Gossypium arboretum, Cajanus cajan, Heeria insignis, Momordica charantia,* and *Amaranthus spinosus*	Anti-parasitic	[49–52]
3	*Azadirachta indica, Justicia flava, Cassytha filiformis, Allamanda cathartica, Morinda lucida, Enantia chlorantha,* and *Landolphia owariensis*	Anti-parasitic, anti-fungal, and antibacterial	[50, 52–56]
4	*Ficus exasperate, Adansonia digitate, Cola nitida, Argemone mexicana, Annona senegalensis, Citrullus colocynthis,* and *Vernonia amygdalina*	Antibacterial	[50, 52, 56–59]
5	*Garcinia kola Buchholzia coriace*	Analgestic, antioxidant, anti-microbial, anti-inflammatory, anti-hypocholesterolemic, antiviral, and anti-diarrhoea	[60–62]
6	*Cola nitida and C. acuminata*	Anti-proliferative, anti-androgenic, and anti-coronary activities	[63, 64]
7	*Buchholzia coriacea*	Antioxidant, anti-microbial, and antitumor properties	[68, 69]
8	*Buchholzia coriacea*	Anti-allergies, anti-inflammation, platelet aggregation, free radicals, anti-ulcer, and hepatoxins potentials	
	Buchholzia coriacea	Anti-microbial, anti-diabetic, anti-helmintic, anti-ulcer, and anti-plasmodial activity	[65–67, 70–76]

compounds with therapeutic processes to treat diseases that are infectious has increased research in this area for the best plants that may contain massive drug compounds. Some of the herbal plants and their antimicrobial activities are presented in Table 3. The resistance by microbes to antibiotics has also stimulated the continuous research and need for a potent and safe therapeutic agent [44].

There have been several reports by researchers on plants that can be used for the treatment of ailments. Among the Esan in Edo state, 70 herbal plants were identified for the use of treating various sicknesses and ailments [77]. Among the Fulanis and Hausas, 72 plants were identified that can be used for treating cancer and disorders in Northern Nigeria [78]. In south-western Nigeria, about 73,143 individuals and 132 species of plants were identified as important in cancer treatment and management, diabetes, and haemorrhoids [79–81]. It was reported by Rufus [82] that the majority of tree species that grow in the Niger Delta have medicinal and pharmaceutical importance. Some plants used in Nigeria for herbal treatment and the specific ailments they addressed are presented in Table 4.

Plants regarded as weeds are very surprising to have phytochemicals that are useful for disease treatments and an important role in the fight against microorganisms [90]. Some weeds are consumed as drugs in orthodox medicine while some have been used locally for disease treatments [91, 92]. Most medicinal plants have become rare as a result of land clearing for structural development, and this may be a setback for the development of herbal medicine in Nigeria. The government can be blamed for its inefficient role in environment conservation issues and policy implementation around this. Another threat is the loss of culture which has made youths and younger generations to fill differently about herbal medicine. This is basically due to the recent development in technology and the rapid rate of urbanization. At this point, herbal medicine associations need to swing into action.

3.1 Antimicrobial Activities of Medicinal Plants

There are several activities herbal plants can address. The phytochemical in these plants can be extracted by using an aqueous solution, ethanol, dichloromethane, and methanol. The medium for extraction dictated the potency of the phytochemical and work rate [93]. Some antimicrobial activities addressed by different medicinal plants/extracts are highlighted:

- ***E. coli***

It can be addressed by the application of leaf and root extracts of *Terminalia glaucescens* [87, 94]. It is a gram-negative bacterium that can be addressed by the application of methanol extracts from medicinal plants [44]. The aqueous and ethanolic extracts of *Zingiber officinale* have shown multidrug resistance to food-borne pathogens. The methanol, water, chloroform, leaves, and petroleum ether extracts of *Senna alata* have antimicrobial properties at a concentration of 500 μg/mL and fight the bacteria isolates [95].

- ***S. typhi***

It can be addressed by the application of leaf and root extracts of *Terminalia glaucescens* [87, 94].

- ***Bacillus subtils***

The methanol extracts from some medicinal plants can fight against clinical isolates of these gram-positive bacteria. The methanol, water, chloroform, leaves, and petroleum ether extracts of *Senna alata* have antimicrobial properties at a concentration of 500 μg/mL and fight the bacteria isolates [95].

Table 4 Some Nigerian medicinal plants, parts, and their uses

S/N	Family	Scientific name	Habit	Part of plant used	Uses	References
1	Anacardinaceae	*Anacardium occidentale*	Tree	Fruits, stem, leaves, and bark	Treatment of malaria, typhoid fever	[81]
2	Asteraceae	*Tridax procumbens*	Herb	Leaves	For treatment of skin infections and sores on the skin	
3	Bignoniaceae	*Kigelia africana* Benth	Herb	The whole plant	Back ache and stomach ache	
4	Bombacaceae	*Bombax buonopozense*	Tree	Leaves and bark of stem	As blood tonic, for stomach ache, and skin infection treatments	
5	Caricacae	*Carica papaya*	Shrub	Stem bark, fruits, and leaves	Catarrh, cough, dysentery, malaria, diabetes, venereal diseases, and purgative	[80]
6	Cucurbitaceae	*Telfaria occidentalis*	Climber	Leaves	Treatment of convulsion and as a blood tonic	
7	Dioscoreaceae	*Dioscorea bulbifera*	Climber	Fruits	Treatment of fever and boils	
8	Dioscoreaceae	*D. dumentorum* (Kunth)	Climber	Tubers and leaves	Abdominal pains and malaria treatment	
9	Euphorbiaceae	*Euphorbia heterophylla*	Herb	Roots and leaves	Treatment of skin infections and as a purgative	
10	Euphorbiaceae	*Jatropha curcas*	Shrub	Leaves	Treatment of haemorrhoids	[79]
11	Fabaceae (Papilionioideae)	*Baphia nitida*	Small tree	Roots, bark, and leaves	Treatment of small pox, diabetes, constipation, and skin disease	
12	Fabaceae (Papilionioideae)	*Centrosema pubescens*	Climber	Leaves	Treatment of diseases occurring on the skin	
13	Guttiferae	*Garcinia kola*	Tree	Stem bark, foots, and fruits	Treatment of diabetes, cough, tooth ache, anti-cancer, and dysentery	
14	Icacinaceae	*Icacina trichanta*	Shrub	Roots and leaves	As a purgative, toothache, abortificient, aphrodisiac, and rheumatism	

15	Lilaceae	*Allium ascolanicum*		Whole plant	Treatment of dysentery, diabetes, anti-cancer, and haemorrhoids	[83]
16	Menispermaceae	*Rhigiocarya racemifera*	Climbing plant	Leaves and seeds	As an aphrodisiac and sedative	[78]
17	Moraceae	*Milicia excelsa*	Tree	Leaves, roots, and bark	For insomnia, malaria, nausea, and abdominal pain	
18	Olacaceae	*Olax subscorpioidea*	Tree	Twigs, bark, leaves, roots, and stem	Treatment of venereal diseases, guinea worm, yellow fever, and jaundice	[84]
19	Rutaceae	*Citrus aurantifolia*	Tree	Bark, roots, and stem	Gonorrhoea, diabetes, and haemorrhoids	[85]
20	Zingiberacea	*Aframomum melegueta*	Herb	Leaves and seeds	Stimulants and haemorrhoids	
21	Sapindaceae	*Paullinia pinnata*	Woody climber	Seeds, roots, and leaves	Leprosy, dysentery, jaundice, and dysentery	[79]
22	Solanaceae	*Nicotiana tabacum*	Shrub	Leaves	A stimulant, convulsion	
23	Steruliaceae	*Theobroma cacao*	Tree	Roots, seeds	Stimulant, toothache, and gingivitis	
24	Tiliaceae	*Corchorus olitorius*	Woody herb	Seeds, roots, and leaves	Fever, asthma, and diarrhoea	
25	Polygalaceae	*Carpolobia lutea*	Shrub	Bark, leaves	Aphrodisiac	
26	Annonaceae	*Annona senegalensis*	Tree	Leaves, fruits, and roots	Treatment of toothache, cancer, and various venereal diseases	[40]
27		*Monodora myristica*	Tree	The seeds	Treatment of headache, constipation, anaemia, and arthritis	
28		*Uvaria barteri*		The leaves	Toothache	
29		*Enantia chlorantha*	Tree	Tree bark	For treatment of jaundice and fever	
30	Apocynaceae	*Picralima nitida*	Shrubs		Treatment of dysentery, fever, and diabetes	[86]

(continued)

Table 4 (continued)

S/N	Family	Scientific name	Habit	Part of plant used	Uses	References
				The roots and tree bark		
31		*Alafia barteri*	Liana	The roots and leaves	Treatment of diabetes	
32		*Funtumia elastic*	Tree	The latex and stem	Treatment of jaundice and pile	
33		*Holarrhena floribunda*	Tree	The leaves, bark, and the whole plant	Treatment of jaundice, gonorrhoea, dysentery, and anti-malaria	
34		*Alsotonia boonei*	The tree	The leaves and roots	Treatment of fever, diabetes, and haemorrhoids	
35		*Raicolfia vomitoria*	Thee	The leaf sap, bark, and root	Treatment of measles, jaundice, convulsion, herpes, and internal disorders in man	
36	Asclepiadaceae	*Pergularia daemia*	Climber	The whole plant	Treatment of pile, diabetes, and fever	[41]
37		*Gongronema latifolium*		The stem and roots	Treatment of haemorrhoids and anthelmintic	
38		*Calotropis procera*	Shrub	The latex, roots, bark, and leaves	Treatment of diarrhoea, cough, dysentery, elephantiasis, ringworm, leprosy, and diabetes	
39	Asteraceae	*Chromolaena odorata*	Herns	Leaves	Treatment of dysentery, headache, and antimicrobial	[46]
40		*Ageratum conyzoides*	Herb	The whole plants	Treatment of ulcer, diabetes, menstrual pain and irregular menstruation, and diarrhoea	
41	Bombacaceae	*Adansonia digitata*	Tree	The bark, pulp, leaves, and fruits	Treatment of diabetes, malaria, and asthma	[87]
42		*Ceiba pentandra*	Tree	The oil and leaves	Treatment of sores, laxative, and rheumatism	
43	Moraceae	*Fisus sur*	Tree	Roots	Haemorrhoids	
44	Connaraceae	*Cnestis ferruginea*	Shrun	Roots and leaves	Treatment of fever, laxative, tooth ache, and haemorrhoids	
45		*Byrsocarpus coccineus*	Shrub	Roots and leaves	Treatment of cancer, jaundice, pile, dysentery, and haemorrhoids	

46	Concolvulaceae	*Merremia tridentate*	Climber	The whole plant	Treatment of gonorrhoea	
47	Cucurbitaceae	*Momordica charantia*	Climber	The fruit, leaves, and whole plant	Treatment of burns, ulcers, pile, diabetes, convulsion, and skin infections	[88]
48	Euphorbiaceae	*Alchornea laxiflora*	Shrub	The roots and leaves	For typhoid fever treatment and as antioxidant	
49		*Acalypha fimbriata*	Herb	Leaves	Treatment of asthma, ulcer, and rheumatism	
50		*Crton lobatus*	Herb	Roots and leaves	Rheumatism and skin diseases	
51	Fabaceae (Caesalpimioideae)	*Caesalpina bunduc*	Shrub	The roots, seeds, and leaves	Treatment of diabetes, fever, and anthelmintics	
52		*Albizia adianthifolia*	Tree	Bark of stem	Treatment of cough, gonorrhoea, night blindness, and haemorrhoids	
53	Fabaceae (Mimosoideae)	*Albizia zygla*	Tree	Bark and leaves	An astringent	[89]
54		*A. ferruginea*	Tree	The leaves, roots, and stem bark	Constipation and dysentery	
55	Fabaceae (Papilionooideae)	*Dalbergiella welwitschii*	Shrub	The leaves, roots, and stem	Menstrual order, diabetes, purgative, and anthelmintics	
56		*Ptrocarpus Osun*	Tree	Bark of the stem and root	Asthma, eczema, and infections on the skin	
57	Longaniaceae	*Anthoclesta djalonensis*	Tree	Stem bark and leaves	Abdominal pain, diabetes, eczema, and a purgative	
58	Marantaceae	*Thaumatococcus danielli*	Herb	Fruit	Treatment of diabetes	
59	Melastomataceae	*Heterotis rotundifolia*	Herb	The entire plant	Venereal diseases and antimicrobial	
60	Menispermaceae	*Rhigiocarya racemifera*	Climbing plant	Seeds and leaves	An aphrodisiac and sedative	

- ***Staphylococcus aureus***

The methanol extracts from some medicinal plants can fight against clinical isolates of these gram-positive bacteria.

- ***Pseudomonas aeruginosa***

It is a gram-negative bacterium that can be addressed by the application of methanol extracts from medicinal plants [44].

- ***Klebsiella pneumonia***

It is a gram-negative bacterium that can be addressed by the application of methanol extracts from medicinal plants [44].

- ***Salmonella typhi***

The tree bark of ethanol and acetone extracts of *Azadirachta indica* (*A. Juss*) has significant bacterial activity on the microbes at a concentration of 25–40 mg/mL [96]. The aqueous and ethanolic extracts of *Zingiber officinale* have shown multi-drug resistance to food-borne pathogens.

- ***Aspergillus niger***

The leaves of *Diospyros* spp. have the ability to resist this microbe [97].

- ***Proteus aeruginosa***

The methanol, water, chloroform, leaves, and petroleum ether extracts of *Senna alata* have antimicrobial properties at a concentration of 500 μg/mL and fight the bacteria isolates [95].

3.2 Anti-malarial Activities of Medicinal Plants

The incidence of malaria is a threat to global health and has killed millions of people worldwide in sub-Saharan Africa. Multi-resistant malaria parasites have become the order of the day, and this has resulted in the quest for developing combined formulations through research to address the parasite and develop a more efficient therapeutic drug [98]. Some of the combined drugs used are the artemether-lumefantrine and the sulfadoxine-pyrimethamine. *Allamanda cathartica* and *Bixa orellana* are plants that have anti-malarial properties [93], and *Alchornea laxiflora* (Breth.) is traditionally used for malaria treatment in Nigeria and belongs to the family Euphorbiaceae. The oral application of *Enantia chlorantha* has been used traditionally to treat ailments, most especially malaria, and can cause toxic effects

Table 5 Herbal plants, extracts, and results in treating malaria parasites

S/N	Malaria parasite	Herbal plant	Mode of Extraction	Results	References
1	*Plasmodium falciparum*	Sensitivity to *Diospyros monbuttensis* (Egun eja)		High anti-plasmodial activity ($IC_{50} = 3.2$ nM)	[98]
		Sensitivity to *Momordica charania* (Ejirin)		High anti-plasmodial activity	
		Sensitivity to *Morinda lucida* (Oruwo)	Ethanolic	Low anti-plasmodial activity ($IC_{50} = 25$ nM)	
		A. laxiflora		Significant anti-malarial activity in mice	[100]
		P. berghi	Ethyl acetate	Fights against chloroguine-sensitive (Pf3D7) and resistant (PfINDO) strains in mice	
2	*Eimeria* parasites	*M. lucida*		Drugs for poultry birds and veterinary purposes	[101]
3	*Plasmodium berghei*	*L. owariensis*	Methanol	Significant anti-plasmodial activity in albino mice due to presence of flavonoids, alkaloids, saponins, and tannins in the extracts	[102]

when administered in excessive dosage [99]. Table 5 presents dome herbal plants and their effects on malarial parasites.

Azadirachta indica is a common plant in Africa, and its extracts have exhibited potential for pharmacological anti-plasmodial activities [103]. The plant belongs to the Meliaceae family, its use as anti-malarial treatment is not yet ascertained, and this area calls for more research [103]. It was reported in an in silico study that nimbinone, numbone, and margolonone were identified as compounds that are functional and have the ability to modulate *P. falciparum* heat shock protein 90 (PfHsp90) activities.

3.3 Anti-fungi Activities of Medicinal Plants

They are infections caused by fungi and have the tendency of causing serious problems and life-threatening diseases [104, 105]. Fungal diseases that are severe may occur from other health issues such as asthma, cancer, organ transplant, and human immunodeficiency virus (HIV) [106]. These diseases vary based on the region as presented in Table 6. Different herbal plants have diverse anti-fungal activities some of which are presented in Table 7.

Table 6 Fungal infections in some countries and prevalence

S/N	Countries	Fungal infections	Prevalence	References
1	Nigeria	Cryptococcal antigenemia and subclinical histoplasmosis	HIV/AIDS patients, babies in neonatal intensive care	[107, 108]
2	Cameroun	Cryptococcal meningitis, oesophageal candidiasis, *Pneumocystis* pneumonia, disseminated histoplasmosis, and invasive aspergillosis	Adults	[109]
		Tinea capitis	School children	[109]
3	Mozambique	*Emergomyces* and recurrent *Candida* vulvo-vaginitis	HIV and hospitalized patients	[110]

Table 7 Some herbal plants, mode of action, and antifungal activities

S/N	Herbal plant	Mode of action	Mode of extraction	Treatment	References
1	*Calotropis procera*	Inhibition of *Microsporum* and *Trichophyton* species after inoculation for 10 days	Water	Eczema, leprosy, syphilis, malaria, and cutaneous infections	[111]
2	*Spondias mombin* (bark and leaves)	Contains alkaloids, terpenoids, glycosides, flavonoids, and saponins	Crude methanol	Anti-candidal effects	[112]
3	*Psidium guajava*			Active on *Candida albicans* from patients that are infected	[113]
4	*Alchornea laxiflora* leaf extract	Contains flavonoids, alkaloids, tannins, saponins, and reducing sugars		Antibacterial and anti-fungal activities	[59]

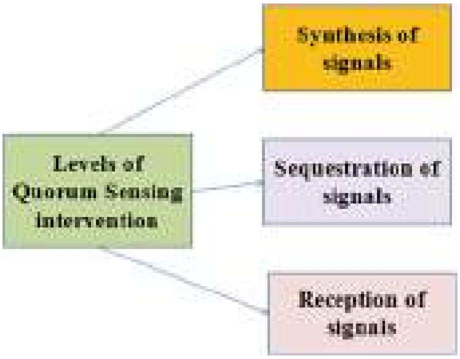
3.4 Antiviral Activities of Medicinal Plants

A fundamental approach towards producing an antiretroviral drug is reverse transcriptase encoding which acts in viral genome reverse transcription in human immunodeficiency virus type 1 (HIV-1) [22, 114]. Some plants were screened and showed antiretroviral activities (Table 8); examples of such plants are *Ancistrocladus korupensis* [103] and *Ancistrocladus congolensis* [115] produced michellamine A and B [21], and *A. congolensis* [115].

Table 8 Some medicinal plants with antiviral properties and their activity

S/N	Country	Virus	Form of extracts	Herbal plant	Part of plant extract	Activity	References
1	Nigeria	Echovirus prototypes 7 (E7), 13 (E13), and 19 (E19)	Methanolic	*Macaranga barteri*	Leaves	Highest anti-viral activity on E7 and E9	[116]
				Ageratum conyzoides	Leaves	Activity highest on E7 and E19	
				Mondia whitei	Leaves	Leaves extract on E7 and E19	
2	China	HIV-1 virus		*Rheum palmatum* and *R. officinale extracts*		Inhibition of HIV-1 reverse transcriptase-associated DNA polymerase (RDDP) and ribonuclease H activities	[116]

Fig. 2 The levels of quorum sensing interventions. (Source: Ref. [120])



3.5 Mechanism of Action of Medicinal Plants

Due to the rare occurrence of disease outbreaks in plants in the wild, it can be said that they have competent defence mechanisms to fight and protect against infectious diseases. Different action mechanisms can stimulate the action of antimicrobial activities such as:

- Enzymes and toxin activity inhibition
- Bacterial membrane damage
- Virulence factors suppression
- Biofilm formation
- Protein synthesis inhibition
- Quorum quenching [117]

Tannins can bind proteins, and this is their major mode of action thereby causing the synthesis of proteins in the cell [23, 118, 119]. The quorum sensing intervention during the mode of action synthesis is in three levels as presented in Fig. 2.

4 Methods of Herbal Medicine Preparations in Nigeria

Herbal medicine can be prepared using various methods, and principally, the medicinal plants are exposed to various treatments so as to extract the active ingredients and phytochemicals present. Some of the methods are:

- Extraction
- Distillation
- Expression
- Fractionation
- Purification

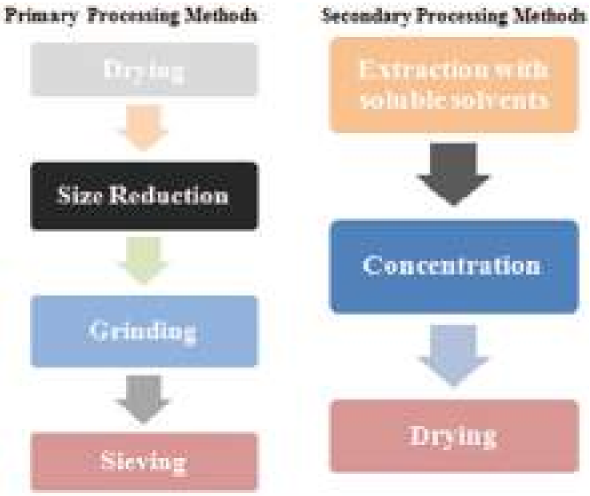


Fig. 3 Primary and secondary medicinal plant-processing methods

- Concentration
- Fermentation

The method of processing is very vital and is connected with the potency of the active ingredients in medicinal plants. The parts to be processed may be the rhizomes, roots, seeds, bark, fruits, leaves, stems, and flowers. It is therefore essential to determine the suitable technique of processing so as to maintain the optimum quality of the herbal product because the value of the herbal product is related to the active ingredients in the plants. The major processing steps are expressed in the flow chart in Fig. 3. There are two types of processing methods: primary and secondary processing methods. The products derived from secondary processing methods can be in the form of extracts which can be whole, concentrated, or powdered form. The form of the extract is then standardized based on herbal regulations [121]. The most common method of extraction is by boiling the plant to dissolve the chemicals. The roots, stems, bark, and rhizomes may be dissolved, and the process is called decoction. In this process, the plant is first dried then cut, so it can dissolute maximally, and finally boiled in water to extract oils, organic compounds, and other dissolved compounds [122]. Instead of water, glycerol or aqueous ethanol may be used for extraction.

4.1 Practices of Herbal Medicine

The origin of herbal medicine is ancient as it involves the use of medicinal plants to heal all forms of diseases and ensures the well-being of humanity. In practice, the healer can make use of substances such as money, seeds, bones, cowry shells, animal shells, use of herbs, and some other spiritual methods to cure ailments. The practice and its use have been studied extensively in Nigeria curing chronic illnesses among the adult and paediatric populations. It has been used to treat epilepsy [123], diabetes [124], hypertension [125], cancer [126], asthma, and sickle cell [127]. The studies on the use of herbal products among the general population are few [128, 129]. Oreagba et al. [130] assessed the herbal medicine use and prevalence among adults without illness in Lagos, Nigeria. Currently, the use of herbal medicine has various patterns used by respondents during the study, and its practices have been reported among adults with hypertension. The rate in the study was three times the value (23%) that was reported in children with epilepsy, sickle cell, and asthma. Some of the natural herbal medicine preparations used are agbo jedi-jedi, agbo iba, herbal toothpaste, jebu-ode mixture drink, splina, and omega roots.

4.2 Relevance of Herbal Medicine Use

Herbal plants naturally occurring are exposed to little industrial processing and are used to treat different ailments. They are known to be completely active ingredients, which may be concentrated in the leaves, roots, and stems, and are used for herbal preparations. The plants can produce materials such as oils, gums, juices, glycosides, and essential oils and may contain recipients. Herbal medicine played an important role in the treatment of Severe Acute Respiratory Syndrome (SARS) in China [130]. It helped to contain and treat severe issues of SARS. In Nigeria, the use of traditional herbal medicine is beginning to gain popularity. Its historical health benefits have shown that herbal medicine is an important and significant part of alternative medicine. It is a part and parcel of alternative treatment methods. Herbal medicine involves the use of different parts of the plant. At the moment, a high percentage of Nigerians, most especially in rural and remote areas, use herbal medicines as a principal source of healthcare [54]. It can also provide economic returns for people who depend on the use as a means of income. There are some factors affecting the acceptance of traditional medicine in Nigeria. They are:

- Some herbal remedies are affordable and very efficient.
- The use of herbs as food and medicine has been time immemorial therefore creating an avenue for acceptance.
- Communities may have a historic/cultural tie with some herbal medicines they use.
- Herbal medicine is perceived to have a strong emphasis on health than disease treatment.

5 Proposed Nigerian Medicinal Plants for Standardization

With the immeasurable benefits from various Nigerian medicinal plants and efforts to improve and harmonize the practice of traditional medicine in Africa, it has been proposed by the African Regional Standard Organization (ARSO) which is formed by the African Union (AU) to adopt ten plants that have medicinal properties in

Table 9 The proposed ten Nigerian plants for certification by Nigerian Medical Council (NMC)

S/N	Common name	Botanical name
1	Moringa	*Moringa oleifera*
2	Bitter kola	*Garcinia kola*
3	Scent leaf	*Vernonia amygdalina*
4	African bush mango	*Irvingia gabonensis*
5	Yellow yam	*Dioscorea bulbifera*
6	Red stinkwood tree	*Prunus africana*
7	Baobab	*Adansonia digitata*
8	Zobo	*Hibiscus*
9	Cashew	*Anarcadium occidentale*
10	Bitter leaf	*Vernonia amygdalina*

Source: Ref. [131]

Nigeria. The body is a nongovernmental organization that is aimed at standardization and regulation of clinical practices in Africa. The ten plants are categorized as Functional Food Plants of Africa and are listed in Table 9. The listed plants are captured in Nigeria as Food and Medicine.

The proposed plants were selected because they have been well-researched and have economic, financial, social, and health impacts. For instance, the African bush mango is a drug in Europe used for weight loss. It is locally called Ogbono and is of various types. Because of the lack of standards, people sell products that are not genuine as cases have been reported about people from Asia who sold drugs that they claim have contents that are not found in their drugs. *Prunus africana* is a plant on the red list of the International Union of Conservation of Nature (IUCN) as it cannot be traded or moved between country boundaries. The plant functions basically for inhibiting prostate enlargement and is commonly used in South Eastern Nigeria.

- ***Moringa oleifera***

It is commonly called the drumstick tree and is referred to with various names based on ethnicity in Nigeria. It has been reported that the plant strengthens the immune system of the body, is a natural energy booster, can cure migraines, asthma, and headaches, has an antibiotic potential, can reduce arthritis and pains, and inflammations, and can restrict the growth of tumours. The flower of the plant when boiled with soymilk is a good aphrodisiac. The fruit can boost sperm count and improves egg fertilization chances in men. It is enriched with iron and has more Vitamin C content than oranges. It has four times more calcium than milk and more Vitamin A than carrots. It contains double the protein as milk and triple the potassium as a banana. It can function in body detoxification and expelling harmful substances as wastes. It has the potential to address issues of malnutrition and can be taken by people with Human Immune-deficiency Virus (HIV)/Acquired Immune Deficiency Syndrome (AIDS). Nursing mothers who consume the plant have been reported to produce more healthy milk, and children gain more weight after leaves are added to their diets as supplements.

- ***Garcinia kola***

It is commonly called bitter cola, and it belongs to the Guttifereae family. It is called differently based on ethnicity in Nigeria. It has been reported by scientists in Nigeria that eye drops made from this plant have the ability to prevent blindness in patients suffering from glaucoma.

- ***Vernonia amygdalina***

It is called the bitter leaf and belongs to the family Compositae. It is called differently by various ethnic groups in Nigeria. It is a good panacea for cancer, diabetes, damage of the liver, infections that resist drugs, and safety in childbearing. The plant has been reported to pass human trials as an anti-diabetic medication and has received a United States Patent 6,531,461. The aqueous extract of the plant collected from Benin City has also been received by United States Patient 6,849,604 as a treatment for cancer. Researchers in Nigeria have reported the plant as a good protection against liver damage by acting as an antioxidant. The roots and leaves have been reported to increase the contraction of the uterus and motility thereby promoting safer childbirth. It is also useful in the management of the placenta that is retained, malaria, infertility, colic pains, and painful menstruation.

- ***Anacardium occidentale***

It is commonly called cashew and belongs to the family Anacardiaceae. It is called jambe in Hausa, and Kaju or Kantonoyo in Yoruba. The extract of seeds is an effective anti-diabetic agent. The tree and its products have been known to be an effective anti-inflammatory agent and reduce blood sugar and diabetes. Eating fruit, nut, and leaf extracts is good for tackling high blood pressure. The nuts can improve the sensitivity of a reflex called the baroreflex which is important in maintaining the level of blood pressure. A gram-positive bacterium that is responsible for the decay of teeth, tuberculosis, acne, and leprosy can be killed when cashew nuts, apples, and shell oil are consumed.

- ***Ocimum gratissimum***

It is commonly called scent leaf and is a member of the family Lameacea which is a mint family. It is referred to by the Yorubas as effirin and Ibos as arigbe or nchuanwu. It is a herb that can reduce physical and emotional stress, and improve anxiety and depression. Locally, the plant is used for treating diarrhoea, headache, and infection of the kidney and wart worms. Leaves contain thymol oil which is an antiseptic and can prevent bites by mosquitoes.

- ***Irvingia gabonensis***

It is a common fruit consumed in Nigeria and West Africa and is commonly called Wild mango or West African mango. It also has other names such as native mango, dika nut tree, bush mango, and dika bread tree. The name of the fruit varies with the ethnic group in Nigeria. Ogbono helps to reduce fat in the abdomen, reduce weight, and lower cholesterol and the chances of diabetes, cancer, stroke, heart disease, and kidney failure.

- ***Dioscorea bulbifera***

It is commonly called several names such as yellow yam, air yam, air potato, potato yam, cheeky yam, and bitter yam. It belongs to the family Diosoreaceae. It is known to cure dysentery, diarrhoea, and other ailments.

- ***Prunus africana***

It is called the red stinkwood tree and belongs to the family Rosaceae. The wood has an unpleasant odour which is the reason for the name stinkwood. The bark is used to treat health conditions by traditional healers such as enlargement of the prostate, erectile dysfunction, diseases of the kidney, infections of the urinary tract, stomach upset, chest pain, stomach ache, wound, baldness in males, and inflammation. The bark is also an aphrodisiac that enhances sexuality and vitality, especially in men. The plant has been reported to have bioactive substances with anti-cancer, anti-inflammatory, anti-microbial, and antiviral effects. Some of the compounds are sterols, coumarin, triterpenes, and flavonoids which have medicinal properties. The harvesting of the plant has been on the increase due to its properties and has the potential of saving the world from deadly prostate cancer.

- ***Adasonia digitata***

It is commonly called baobab by the English, Portuguese, and French. It is called several names based on ethnicity in Nigeria. The seeds, leaves, fruits, flowers, fruit pulp, roots, and bark baobab are edible and have been extensively researched. The fruit pulp is rich in calcium, vitamin C, phosphorus, carbohydrates, potassium, fibres, proteins, and lipids. It can be used in seasoning as an appetizer and in the production of juices. Seeds contain high levels of zinc, magnesium, phosphorus, iron, sodium, manganese, lysine, calcium, iron, and thiamine. It has biological properties such as antimicrobial, and anti-malaria, and can cure diabetes, anaemia, diarrhoea, and anti-inflammatory activities. The active ingredients are flavonoids, phytosterols as well as vitamins and minerals.

- ***Hibiscus sabdariffa***

It is commonly called zobo or roselle and is a member of the family Malvaceae. It is known to lower blood pressure and cholesterol and inhibits the invasion of human prostate cancer cells. The polyphenols that are contained in the leaves of hibiscus have the ability to inhibit the growth and destroy the growth of melanoma cancer [131].

5.1 Threats of Nigerian Herbal Medicine Sources

Some of the herbal associations are the Nigerian Traditional Medical Association (NTMA) and the Nigerian Union of Medical Herbal Practitioners (NUMHP), and their roles are:

- Ensuring quality control.
- Regulating pricing.
- Regulating the mode of herbal product marketing.
- The screening of members for authenticity: Fake members are sent out of the organization.
- Lobbying for recognition from the government, and this has been achieved by the organization.
- Interaction with an alternative practice of healing which are the orthodox practitioners of medicine, and this has so far not been achieved by the organization.
- Establishing gardens with herbal plants across the country.

The Nigerian Natural Medicine Development Agency (NNMDA) is also concerned with the regulation of herbal practices and has initiated activities and programs that are aimed at documenting medicinal plants and animals used in herbal medicine. They develop strategies that can be used to boost the nation's Indigenous non-medication, and techniques used in the practice of herbal medicine. The management of herbal plants is crucial because of the value they have and are potential drug sources for improved health care and economic promotion and biodiversity conservation. Many plant species have become scarce to collect in the wild due to the prevailing threats; likewise, the species available are prone to disappear due to the present condition in the wild. If they disappear, where can the species be sought? It was said by Soladoye and Olowokudejo [132] that if this continues, many industries and factories will become extinct. About 480 plant species in Nigeria are endangered as a result of increasing pressure from the population, forest logging, and clearing of lands for development and urbanization [133, 134]. It was reported by the International Union of Conservation and Nature (IUCN) [135] that about 560 Indigenous tree species are found in Nigeria, 16 critically endangered, 18 endangered, 138 vulnerable, 2 species are conservation dependent, 15 near threatened, 10 data deficient, and 173 least concerned. This translates that the endangered flora in the Nigerian ecosystem is fast increasing and the role of government has not been promising in the use of herbal plants. A vital approach may be to adopt approaches for the conservation of herbal plants thereby preventing the loss of total forest [136].

6 Future of Nigerian Herbal Medicine Practices

Considering the numerous herbal plants found in Nigeria, the future of herbal medicine seems very promising. There is quite a number of plants with medicinal potentials that are yet to be investigated for their compositions so as to be used as appropriate. The study of herbal plants and their therapeutic potentials have created a space for understanding the build-up of drugs, design, and development of the sector. The future of herbal plants can positively affect the practice of medicine as new diseases and pathogens are discovered daily and this calls for the use of complementary medicines to address them [10]. There is also an urgent need for new strains of drugs as microorganisms have begun to resist antibiotics; there is an immediate and urgent demand for discovering safe and natural compounds that can be used for this purpose. There is also an increase in the use of herbal plants as nutraceuticals and functional foods, thereby used as preventive medicine and searching for a solution to the issue of drug resistance by microorganisms [3].

7 Conclusion

From this review, it is evident that Nigeria has diverse medicinal plants which are identified by different names based on ethnicity. These plants have been seen to have potential as an alternative therapy to avert the spread of pathogens that are multidrug-resistant and the toxicity of some antibiotics. It has also been observed that these medicinal plants are on the verge of total disappearance as a result of the destructive practices by humans who do not value their use; therefore, sustainability and conservation are far from these medicinal plants. Nigeria lacks adequate information on some medicinal plants and most importantly their characterization and chemical composition towards their use. The investigation of bioactive components in Nigerian medicinal plants is few; therefore, the ethno-medicinal and phytochemical screening of these plants are yet to be validated. The knowledge which is Indigenous about medicinal plants is also diminishing as younger generations need to be enlightened on the importance of medicinal plants and the role they can play in human health. Due to the enormous therapeutic uses of medicinal plants, adequate research into novel plants and compounds can cure neglected tropical diseases (NTDs) that are endemic in Nigeria. There is also a dire need to strategize approaches that are reliable for the conservation of medicinal plant diversity and documentation of knowledge that are Indigenous for generations as native plant species are an important constituent of Nigerian and African traditional medicine.

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Value of Herbal Medicine to Sustainable Development

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Abstract

Herbal medicine utilization has the potential to stimulate sustainable development and influence livelihood of societies in diverse ways. This review will update and inform healthcare authorities and decision-makers about the value of herbal medicine in actualizing the sustainable development goals of adequate healthcare system for all by 2030. This chapter is poised to provide an overview about the role of herbal medicine in stimulating sustainable development in remote societies. Thus, the chapter reviewed and extracted important information's from a plethora of sources on sustainable development subject matter. The review highlights common types of herbal medicines used to treat illnesses, adverse effects, challenges, and the way forward in making herbal products acceptable globally. However, the growing demand for herbal medicine today is greatly contributing to the growth of developing countries' economy while at the same time improving livelihoods. The study finds that the increase in demand and value of herbal medicines today is closely connected with the rise in knowledge of the therapeutics, phytochemical properties, availability, affordability, and inexpensiveness of the herbal products as compared to modern drugs. The chapter further concludes that herbal medicine sustainability is impeded by various factors ranging from uncontrolled deforestation, climate change, absence of policies on herbal medicine usage, anthropogenic activities, and limited collaboration between modern and herbal medicine practitioners. It is concluded that herbal medicine plays a key and central role in the sustainable development of local communities through the strengthening of healthcare system. It is recommended that more collaboration be sought between modern healthcare specialist and local herb practitioners in developing countries.

Keywords

Herbal medicine · Sustainable development · Healthcare · Sicknesses · Phytochemical properties · Herbal products

Abbreviations

AIDS	Acquired immune deficiency syndrome
COVID-19	coronavirus disease 2019
CVO	Covid organic
FAO	Food and Agricultural Organization
FRA	Forest Resource Assessment
GEF	Global Environmental Facility
SARS	Severe acute respiratory syndrome
SDGs	Sustainable Development Goals
UN	United Nations
UNDP	United Nations Development Programme
USD	US Dollars
WHO	World Health Organization

1 Introduction

Herbal medicine is described as the combination of knowledge, skills, cultural experience, beliefs, and theories applied in the maintenance of human health through herbal materials prepared from active plant ingredients to diagnosis, prevent, and treat complex and complicated mental and physical illness incur by humans [1–3]. The increase in the demand of herbal medicine over the past decades is closely connected with its affordability, efficacy, cultural acceptability, inexpensiveness, and the treatment of diseases deemed incurable by modern health drugs and practitioners. These medicinal properties and qualities from herbal medicines have attracted the attention of modern health practitioners to incorporate herbal products in clinical research to unearth more healing properties in herbal products [1]. According to the WHO [4] estimate, about 50–70,000 plant species have medicinal attributes and component contained in them [5]. The plant components from which herbal medicines are derived are flowers, seeds, roots, stems, bark, fruits, and leaves [6]. These herbal components comprise of nutritional supplements that are extracted and prepared for intake [7]. In addition, herbal medicines are formulated into major forms: powdered and liquid [8–10]. Herbal medicines have in time immemorial ensured that local people live healthy lives and stay well all through their lives.

The role of herbal medicine in the maintenance of healthcare system is critical and differs by countries, regions, and countries. Scientific literature suggests that more than 80% of the world population still relies on herbal medicines for their healthcare needs [11–13]. For instance, Asia and Africa utilized more herbal medicine than other continents. That is to say, 80% of South Africa's population use herbal medicine to meet their healthcare needs as compared to 40% of Sierra Leonean relying on herbal medicines for medication [14, 15].

Globally, the value and the sustainable use of herbal medicine have been studied extensively over the past decades [16–18]. According to WHO [19], herbal medicines are described as the use of herbs consisting of therapeutic properties emanating from plants parts such as bark, leaves, seed, flowers, and roots. The increase in value and demand for herbal medicine is growing and contributing to the sustainable development of mankind overtime [20]. Nonetheless, the manner in which herbal medicine contributes to the sustainable development of society is poorly understood, and research especially in a developing country like Sierra Leone is unrealistic. The health of humans especially those living in rural communities greatly depend on ecosystem for the provision of goods and services that support healthy living in the form of medicine, food, shelter, clean water, and recreation among others [21]. Herbal medicines have been in existence for many decades with little or no major complication and implication [22]. In the era of pandemic outbreak, herbal medicines are witnessing a growing demand and awareness in terms of the benefits and importance of herbal medicines in tackling complex pandemic ailment especially in developing countries. These herbal medicines are critical in supporting the sustainable development goals in diverse ways. The sustainable development goals are the UN set of 17 goals with 169 targets to better the lives of mankind by 2030 [23]. In particular, SDG 3 focuses mainly on the sustainable health of human being.

1.1 Sustainable Development in the Context of Herbal Products in Developing Countries

Globally, herbal natural products have in time immemorial been used in many countries and industries to cure various ailments and have attracted global attention over the past centuries [24]. The sustainable development of herbal products is impinged by the high levels of dosage and misconception problems characterized by lack of clinical approval of herbal medicine drugs. However, the growing demand for herbal products today is greatly contributing to the growth of developing countries' economy and enhancing rural livelihoods. In such, the sustainable harvesting of herbal products has the potential to benefit herbal practitioners, improve their livelihood activity, contribute to environmental sustainability, social equity, and economic wealth, and improve healthcare system in remote communities.

Herbal medicine usage and utilization are rooted in indigenous knowledge and embedded in most local community lifestyle for years [25, 26]. These products have been used to provide natural solutions to healthcare issues in societies in a sustainable manner [24]. Natural herbal products are mostly categorized into food and medicinal products and are mostly derived from plant species. These herbal products have contributed to the sustainable development of society through livelihood security, affordable healthcare system, provision of income, creation of job for traditional healers, and provision of food supplement for local indigenes [25, 27]. While 80% of the developing world relies on herbal medicines for the healthcare needs, the developed world indirectly depends on herbal medicine for healthcare with 25% of modern pharmacy products derived from herbal plants [25, 28]. These herbal medicines are vital in the sustainable development goal agenda for the current and future generations. Herbal medicines play a critical role in achieving the sustainable development goal target of healthy lives in a healthy environment.

1.2 Herbal Medicine Sustainable Development Potential

As the sustainable development goal is geared toward living a healthy live for all, herbal medicines play an integral part in achieving this goal. Recent studies suggest that a wide range of therapies from herbs plays a critical role in the well-being of people regardless of race, color, and social status [29]. Specifically, herbal medicines stand the chance to meet target 3.8 of the Sustainable Development Goals (SDGs) that aim to ensure that there is a universal access to healthcare system for all that is free from financial risk, considered safe, and affordable for all class of people and is effective in fighting future pandemic [30].

Herbal medicine is a catalyst of sustainable development due to the fact that it enables new drugs to be discovered, readily available for all class of people, and cheap and support socioeconomic activities and livelihood. Herbal medicine further contributes to sustainable development through the provision of self-employment, household income, livelihood enhancement, support to healthcare system, and socioeconomic activities among others [29]. Today, majority of the global population consider plants as the primary source of healthcare that plays a critical role in the sustainable development of societies [31]. Plant species used to prepare herbal medicine has the potential to enhance the production of life-saving drugs that will contribute to sustainable development of the current and future generation [32]. Hence herbal medicine is an important tool for the sustainable development of healthcare system especially in developing countries [33]. In low-income countries, the sustainable economic development of local economies is greatly dependent on natural products like herbal medicines derived from plants. The sustainable use of herbal medicine provides a reliable source of livelihood and household income for remote communities of low- and middle-income countries like Sierra Leone. In such, the transfer of herbal medicine knowledge from generation to generation is a sustainable means of preserving drugs that will benefit society positively.

In Sierra Leone, herbal medicine plays a primary role in the day to day healthcare activities of people and is being considered an alternative source of employment, income generator, and environmental sustainability alternative. In Sierra Leone and other parts of the world, trade in herbal medicine has been a sustainable source of livelihood for traditional herb practitioners, middle men, and hawkers across the street and towns. Herbal medicines ensure the access to local medication easy for hard to reach communities within Sierra Leone while at the same time strengthening the fragile healthcare system that in turn enhances sustainable development [33]. Herbal medicines don't only enhance the realization of the sustainable development goals but serve as a source of income for local practitioners. Trade and practice in herbal medicine serve as a source of sustainable income and self-employment for local practitioners in developing countries.

1.3 Herbal Medicinal as a Sustainable Source of Income

Herbal medicine has been and is still a sustainable source of income in developing countries around the world. Herbal medicine practice and trade are important industry and business ventures that serve as the source of income for many herbal medicine practitioners around the world [34]. For instance, Sierra Leone has more than 45,000 herbal medicine practitioners nationwide [18]. Across Sierra Leone and in other part of Africa, many local indigenes relied on the additional income provided through herbal medicine sale and practice to supplement their basic needs on a daily basis. According to Zion Market Research, the global income generated by the sale of herbal medicine was reported to be estimated at USD 166 billion in 2021 and is expected to reach USD 348 billion by 2028 at an annual rate of 11.2% [22]. For instance, the value of herbal medicine trade in Sierra Leone over the past decades is estimated to be USD 64,000 annually [15]. The world herbal medicine market value and size in 2021 was valued at USD 151.91 billion and in 2022 was valued at USD 165.66 billion and is projected to worth 347.50 billion by 2029 respectively [35]. Similarly, the UN Comtrade [36] statistics reported that aromatic and medicinal plants are valued around USD 800 million per year at a growth rate of 15–25% [37].

Nonetheless, the trade value or income generated by this trade over the past decades in Sierra Leone is undocumented and under reported. In most part of Africa, the trade and income received from the sale of herbal medicines serves as an important source of livelihood and a major contributor to household income and the gross national products [34].

Across Africa, herbal medicine used for healing various illnesses is used as a profession to earn income [38]. Herbal medicine practitioners in middle- and low-income countries use this profession as trade upon which their sustenance and survival depend. Hence, they directly or indirectly demand a reward for their service in cash or in kind. Any given amount is mostly accepted, but the more you give, the more attention would be given to your situation. Compared to modern healthcare treatment cost, herbal medicines are perceived to be less expensive, more affordable, easily accessible, and readily acceptable in society today [39]. In most low- and middle-income countries where poverty and unemployment are rampant, herbal medicine practitioners' cost of medication is considered by many as cheap, affordable, and an important source of healthcare drugs for the low-income earners [40]. Across Africa and in other parts of the world, the herbal medicine trade is mostly done by the informal sector and is now traded in local markets by individual hawkers on the streets of urban and peri-urban communities. A study conducted by

Mander et al. [34] in South Africa indicated that about 74% of herbal practitioners, street traders, and herb harvesters are women mostly from rural settings. Herbal medicine products are sometimes sold as raw materials or process mixture of different plants by traditional health practitioners or traders for profit making. Recent estimate in herbal medicine trade by WHO is estimated around US$100 billion and still has a great potential of expansion in the next coming years. The WHO estimated that herbal medicine annual growth rate is around 15% and is derived from trade in herbal raw materials, medicinal plants, and other herbal products. The higher demand of herbal medicine is stimulated by shortage and high cost of western drugs as well as the introduction of counterfeit or fake drugs by criminal drug cartels.

2 Global Concept of Herbal Medicine Utilization

Herbal medicines include the combination of traditional knowledge on plant therapeutic experience spanning from many generations [41]. Herbal medicines are considered to offer a great deal of alternative treatment for sicknesses ranging from disease and other life-threatening ailments that positively affect the health of societies. Herbal medicines have over the past centuries evolved and became a strategic mainstream healthcare therapy with great drug development prospect for the human race [40]. There are different points of views on herbal medicines depending on localities and civilization. According to WHO [42], herbal medicine is characterized and includes any herbal material prepared by health practitioners from natural products containing active ingredient capable of treating complex ailment to those treating common cold and fever. Herbal medicine is perceived by traditional healers to have phytoconstituents that are well-matched with the human health system [43]. Herbal medicines comprise of many plant-extracted substances in the form of homemade infusion, dried products, seed, leaves, bark, etc. [44].

Herbal medicines have been in use for centuries as is considered a life-saving dietary that prevents and cures diseases and are considered by many as alternative medicines to western drugs. Even today, 60% of the global population and 80% of the population of developing countries rely on herbal medicines to supplement primary healthcare especially remote societies [45]. Herbal medicines have been reported to cure sicknesses like AIDS, skin disease, cancer, hypertension, diabetes, mental disorder, and tuberculosis among others [41]. On the global stage, herbal medicine is still supporting the health system of some countries like Egypt, China, Bhutan, India, South America, Mali, Sierra Leone, South Africa, and many more in tackling complex and complicated ailments that modern drugs can't treat efficiently [41]. The rise in demand for herbal medicines is closely associated with the herb's abundance, presumptive efficiency, easy access, inexpensiveness, quick reaction, and safety [41]. Nonetheless, critics have raised series of concerns about the prescription, consumption, standardization, and side effects posed by herbal medicines over the years [46]. For instance, the State Health Director of California in the USA once issued a warning in 2001 about the prescription and consumption of herbal product made from Anso Comfort capsules on the pretext of undeclared

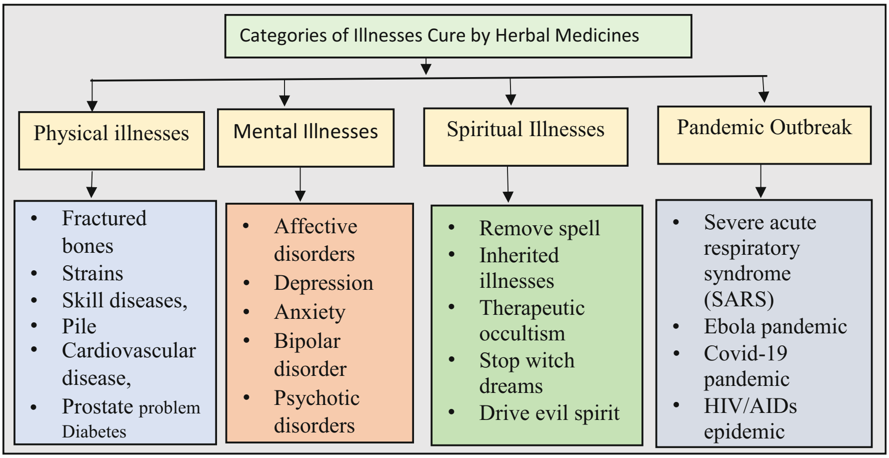

Fig. 1 Category of illnesses cured by herbal medicine

benzodiazepine chlordiazepoxide in the herb [46]. In most places around the world especially in developing countries, herbal medicines have been proven to cure physical, mental, spiritual, and pandemic illnesses (Fig. 1). The value of herbal medicine is hard to quantify due to the fact the practice does not require legal license in most cases.

2.1 The Value of Herbal Medicine to Society Today

The forest ecosystem is widely recognized as a reservoir with great importance in terms of job creation abilities, provision of food supplement, and other ecosystem services of great livelihood benefits and significance [47]. The value of herbal medicine can be expressed in healthcare service provided and income generated through trading. The increase in demand and value of herbal medicines today is closely connected with the rise in knowledge of the therapeutics, phytochemical properties, and the availability, affordability, and inexpensiveness of the herbal products as compared to their conventional medicine counterpart. This knowledge has aided the switch from 100% use of modern medicine to 50% use of herbal products [47]. This switch has boosted the market value especially during the COVID-19 pandemic when herbal medicines were used as a supplement to western vaccines. Nonetheless, there are a number of factors driving or propelling the use and value of herbal medicine especially during the pandemic and post-pandemic period. These factors are not limited to rising herbal therapy knowledge, affordability, growing public awareness, homeopathy, and dietary supplement treatment cost but by failure of modern medicine in treating complicated illness [7].

Herbal medicine has been in used for centuries and demonstrate great therapeutic phytochemical benefits, safety, and efficacy that has and will stand the test of time.

Herbal medicine value and contribution have not only been realized in the effective treatment of complicated illnesses but strive to meet the health needs of society for many centuries. For many in deprived communities around the world, herbal medicine is the only source of healthcare that is readily available, affordable, and accessible for all class of people. The phytochemical and therapeutic components of herbal medicines have contributed to the development of modern drugs sold in pharmacies around world today. Historically, more than 30% of modern drugs are directly or indirectly derived from plants. Recent data shows that lower income nations are the key actors and consumers of herbal medicine due to few healthcare infrastructure and weak purchasing power in those countries [19]. Hence, these societies value herbal medicine than other communities today. Besides the various illnesses cured by herbal medicines, the products have shown great potential in tackling emerging pandemic outbreak. For instance, countries like China and Madagascar employed herbal medicine in the fight against COVID-19 and other recent outbreaks. In such, if attention is given to herbal medicines and advance research is conducted, these products have the potential to tackle emerging pandemic outbreak with ease in the near future.

2.2 The Role of Herbal Medicines in Tackling Emerging Health Pandemic

The COVID-19 caused by SARS-CoV-2 and the Ebola formerly known as Ebola hemorrhagic fever are the two most recent pandemics that have devastated and killed millions around the world [48]. Scientific evidence has proved that herbal medicine helps to ease SARS-COVID-19 infectious disease [49]. During these pandemic periods, low- and middle-income countries like Sierra Leone, Guinea, Liberia, Togo, and Ghana relied on vaccine sharing arrangement from wealthy nations to vaccinate its population. Nonetheless, only around 10% of middle- and low-income countries received at least a single dose of vaccine [50]. Although COVID-19 and Ebola has negative impact on lives and the world order, it also has positive impact on the use and reliability of herbal medicine to tackle future pandemic outbreak. In most low- and middle-income countries, people have turned to herbal medicines to cure illness caused by the pandemic using well-prepared herbal infusions [51]. In some parts of the world, herbal medicines increase exponentially during the COVID-19 periods. For instance, in Malaysia, some ethnic groups use herbal medicines to tackle the Covid-19 pandemic during the peak period when vaccine access was difficult [51]. Similarly, in Madagascar, the Malagasy Institute of Applied Research developed a cure for coronavirus called "Covid Organics (CVO)," an organic herbal concoction. This organic herbal concoction cure was believed to cure COVID-19 patients suffering from the infection [52]. This organic herbal concoction was requested by Tanzania, Equatorial Guinea, the Republic of Congo, Guinea Bissau, and Sierra Leone among other countries in Africa [52]. According to a recent study conducted by Ekor [53], herbal medicines contained phytochemical components that have the potential of curing or deterring coronavirus infection accordingly. However, more studies are ongoing to prove the efficient of herbal medicines as an alternative complementary approach in tackling COVID-19 and other future pandemic outbreaks using herbal drugs to provide needed immunity [54]. In Asia and Africa, local populations are refocusing on herbal medicines as remedies to cure many emerging diseases at a low cost [55]. In Africa, countries like Madagascar have shown the importance of herbal medicine in fighting complicated and complex health crisis capable of wiping out generations. The emergence of global pandemic like COVID-19 and Ebola has brought recognition to herbal medicines potential in curing complex ailment that western drugs struggle to cure at a cheap cost. A recent study by Demeke, Woldeyohaninsg, and Kifle [49] revealed that herbal medicines have the capability of interfering with COVID-19 pathogenesis by hindering the SAR-COVID-19 replication process and infection. The study identifies *Mentha piperita*, *Allium sativum*, *Nigella sativa*, *Citrus* spp., *Allium cepa*, *Artemisia annua*, and *C. sinensis* plant species to contain phytochemical components that inhibit SARS-COVID-19 and is consumed through herbal drink [49, 56]. Similarly, studies conducted by Vellingiri and Rajagopalan [57], Luo [58], and Chan et al. [59] discovered that herbal medicines can reduce COVID-19 severity and in some cases prevent the virus from replication. For instance, India and China use herbal medicines alongside modern drugs to boost the immunity of patients infected by COVID-19 [60, 61]. Moreover, in China, herbal medicines demonstrate the ability to reduce mortality rate, inhibit replication, and improve symptoms signs [58]. Some plant species used as herbal medicine are presented in Table 1. The use of herbal medicines in tackling emerging global pandemic cannot be overemphasized. Herbal medicines have the potential to support various forms of livelihood activities especially in remote and hard to reach communities of low-income countries.

2.3 Herbal Medicine and Its Impact on Livelihood Activities

Globally, the significance of herbal medicine to mankind is basically infinite due to the roles played in enhancing livelihood through the sale of medicinal herbs and the principal source of income for herbal practitioners and their families. Over the past decades, herbal medicines have served mankind in treating complex ailment considered nontreatable and at the same time served as source of livelihood for practicing practitioners. Natural products like herbal medicine also have the potential to enhance livelihood via consumption, treating ailment, and sale of products to provide an economic safety net for herbal medicine practitioners [62, 63]. A study conducted by Timmermann and Smith-Hall [63] concluded that herbal medicine sale gives an opportunity to local herbal medicine practitioners to earn financial incomes capable of supporting their household and enhancing livelihood activities. A similar study conducted in the mountainous region of Nepal found that herbal plant plays an integral role in contributing to livelihood and annual income [64]. Rasul et al. [65] and Shrestha and Bawa [66] supported these claims in their study by concluding that 21% and 85% of household income came from the sale of herbal medicines in Nepal. Across the world, the sustainable use of herbal medicinal plant products is integral in

Table 1 Common herbal plants and their medicinal properties around the world

Scientific name	Family	Medicinal Properties	Reference
Mentha piperita	Lamiaceae	Treat fever, cold, oral mucosa, throat inflammation, and digestive issues and as antiviral and antifungal	[49, 76]
Allium sativum L	Liliaceae	Treat diseases like asthma, bronchitis, influenza, COVID-19, and others	[49, 77]
Allium cepa	Alliaceae	For cough and other infections of the air tubes in children	[49, 78]
Citrus spp.	Rutaceae	Contains hesperidin and rhoifolin properties: treat SARS-COVID-19 pro-inhibition in a dose-dependent manner	[49, 79]
Angelica decursiva	Apiaceae	Contains columbianadin and decreases airway inflammation	[49, 80]
Hibiscus sabdariffa	Malvaceae	Contains anthocyanin-converting enzyme, decreases plasma aldosterone and serum angiotensin	[81]
Salvia officinalis	Lamiaceae	Contains polysaccharide and emollient/ potent antitussive activity	[82]
Aloe barbadensis Mill	Asphodelaceae	Enhances potent antitussive/emollient activity and contains polysaccharide	[82]
Malva sylvestris L	Malvaceae	The Plants Contains Polysaccharide that enhances Gil/Vitaceae as well as Tetrastigma hemsleyanum Diels	[82]
Torreya nucifera	Taxaceae	Amentoflavone, authentic flavones. SARS-3CL pro-inhibition/62% at 100 μg/mL	[49, 83]
Artemisia annua	Asteraceae	Contains many biological metabolic, antitumor, antimicrobial, and immunomodulatory properties	[56, 84]
Ginkgo biloba	Ginkgoaceae	The herb improves circulation and cognition and the leaves aid in asthma and cardiovascular disorders	[85, 86]
Piper methysticum	Piperaceae	Plant used to cure nervous system illnesses, helps in relaxation, induces sleep, and helps in weight reduction	[86]
Echinacea purpurea	Asteraceae	Plant used to cure snakebites, syphilis, toothache, chronic arthritis, among others	[1, 86, 87]
Eleutherococcus senticosus	Araliaceae	Increase body's resistance against stressful situations, treat hypertension, diabetes, depression	[88]
Ephedra sinica	Ephedraceae	Plant used to treat cough, myasthenia, rheumatism, hay fever, enuresis, narcolepsy, among other sickness	[1, 86]

supporting livelihood activities in low- and middle-income countries. For instance, in the northern part of Sierra Leone, the livelihood of most communities depends on incomes derived from herbal medicine practice and healing. The town named "Kabala" is well known for the mystical power possessed by herbal practitioners in that part of the country and is the first choice for people to visit when inflicted with black magic sickness that cannot be cured by modern drugs. Besides, besides livelihood support provided, herbal medicine also supports healthcare system, boosts financial income, and enhances cultural diversity of people [67]. In countries like Guinea, Afghanistan, Ethiopia, Mali, Senegal and Sierra Leone, herbal medicine trade and practice boost household income and support livelihood activities while at the same time save lives. Besides herbal medicine practitioners, middle men dealing in herbal medicine products also rely on these herbs to provide livelihood support for themselves and their household. Herbal medicines do not only support livelihood activities but also contribute to the sustainable development of societies especially in developing countries. A study conducted by Abisoye et al. [68] found that income earned from herbal medicines sale or trade is used to support the livelihood of household by paying for children education, healthcare, feeding, shelter construction, and covering of other expenses. In some remote parts of Sierra Leone, herbal medicine practice and trade are full-time employment with huge financial income potential [62]. Herbal medicine practice has the potential to reduce poverty, enhance social equity, and provide integral resources that enhance sustainable livelihood. Moreover, herbal medicines play a critical role in sustaining local economies and rural livelihood for remote communities and societies. Traditional medicines can also ensure environmental sustainability through increased biodiversity and conservation of herbal plants [69]. Nevertheless, the contribution of herbal medicine to livelihood and household income in most countries around the world is not well captured and documented due to the absence of available database in these countries.

3 Common Herbal Plant Species Use for Medicinal Purposes Worldwide and in Sierra Leone

Plants with medicinal properties are globally a valuable source of healthcare medication from herbal products derived from the natural environment. Scientific literature has confirmed that plants are the primary source of herbal medicine around the globe and they are rich in phytochemical substances used for medication [16, 70]. Plant species differ with geographic location, climate, vegetation, and soil pattern (Table 1). Although some plants are found within a wide geographical area spanning through different continents, some are only found in specific countries or regions (Table 2). The extraction, preparation, processing, and administration of herbal medicine differ by country, culture, experience, and educational level [71]. Similarly, the number of illness or sicknesses that can be cured by herbal plants products also differs by region, country, culture, age, experience, spiritual powers, herb acceptability, and societal norms. Generally, herbal medicines derived from plant products treat various illnesses ranging from severe acute respiratory syndrome (SARS), HIV, prostrate problems, inflammation, cardiovascular disease, and depression, among other ailments [72–75]. Herbal drug products can be prepared, processed, and consumed in different forms and ways such as infusion, syrup, concoction, ointment, teas, rubs, powder,

Table 2 Some herbal plants, diseases cured, and mode of preparation in Sierra Leone

Scientific name	Family	Disease/sickness cured	Part use	Mode
Adenia lobata	Passifloracea	Abscesses, deworming	Leaf, stem	Decoction
Anisophyllea laurina	Anisophylleaceae	Eye problems, stomach pain	Leaf	Infusion, Squeezed juice
Anthostema senegalense	Euphorbiaceae	Abortion	Leaf	Infusion, squeezed juice
Carica papaya	Caricaceae	Constipation, dysentery, fetish, purgative	Leaf	Infusion, squeezed juice
Landolphia dulcis	Apocynaceae	Penis enlargement	Root	Infusion
Cassia alata	Leguminosae	Malaria, constipation	Leaf, bark	Infusion
Citrus aurantifolia	Rutaceae	Constipation, fever	Leaf, root	Infusion
Ficus exasperate	Moraceae	Abortion, malaria	Leaf, bark	Infusion
Funtumia africana	Apocynaceae	Impotency in male	Leaf, bark	Infusion
Gmelina arborea	Lamiaceae	Malaria	Bark, leaf	Infusion
Garcinia kola	Clusiaceae/ Guttiferae	Purgative, throat infection, diarrhea, long sexual intercourse	Seed, root, and bark	Infusion
Mangifera indica	Anacardiaceae	Malaria, constipation, headache	Bark, stem,	Infusion
Manihot esculenta	Euphorbiaceae	Constipation, body pain, fetish	Leaf	Infusion
Marsdenia latifolia	Asclepiadaceae	help child walk faster, convulsion, purgative	Stem, bark	Decoction, moldy paste
Mimosa pudica	Fabaceae	Pregnancy pain, skins boil,	Leaf	Moldy paste
Moringa oleifera	Rubiaceae	Heart attack, stomach ache, deworming, malaria	Leaf	Decoction,
Musa sapientum	Musaceae	Ulcer, vomiting, diabetes,	Leaves, fruit	Decoction,
Nauclea latifolia	Rubiaceae	Malaria, typhoid, stomachache, headache,	Bark, stem, leaf, root	Infusion
Parkia biglobosa	Leguminosae	malaria, diarrhea, diabetes, hemorrhoid	Leaf, bark	Decoction, Infusion
Spondia mombin	Anacardiaceae	Menstrual problem	Leaf	Decoction
Tecoma stans	Bignoniaceae	Fever, stomachache	Stem, leaf	Infusion
Terminalia ivorensis	Combretaceae	Dysentery, diarrhea	Leaf, bark	Infusion

(continued)

Table 2 (continued)

Scientific name	Family	Disease/sickness cured	Part use	Mode
Terminalia superba	Combretaceae	Heal sore, malaria, diarrhea, skin boil	Leaf, bark	Infusion, squeezed juice
Vernonia amygdalina	Asteraceae	Malaria, elephantiasis, constipation, fetish	Stem, root, bark	Infusion, squeezed juice
Ficus capensis	Moraceae	Stomachache	Bark	Decoction
Ficus laprienrii	Moraceae	External bleeding	Leaf	Decoction
Heinsia pulchella	Rubiaceae	Heat rash	Leaf	Decoction
Hibiscus esculentus	Malvaceae	Diarrhea	Bark	Decoction
Hymenocardia lyrata	Euphorbiaceae	Skin disease	Bark	Decoction
Hyptis suaveolens	Labiatae	Fever	Leaf	Decoction
Ipomoea involucrata	Convolvulaceae	Gonorrhea	Stem	Decoction
Jatropha curcas	Euphorbiaceae	Stomachache	Leaf	Decoction
Mareya micrantha	Euphorbiaceae	Skin disease	Leaf	Decoction
Microglossa afzelii	Compositae	Headache	Leaf	Decoction
Musanga cecropioides	Moraceae	Toothache	Flowers	Decoction
Napoleona vogelli	Lecythidaceae	Cough	Bark	Decoction
Newbouldia laevis	Bignoniaceae	Elephantiasis	Root	Decoction
Ocimum viride	Labiatae	Headache	Leaf	Decoction
Olyra latifolia	Gramineae	Stomach pain	Fruit	Decoction
Parinari excelsa	Rosaceae	Headache	Bark	Decoction
Parinari macrophylla	Rosaceae	Bone fracture	Leaf	Decoction
Phyllanthus discoideus	Euphorbiaceae	Fever	Bark	Decoction
Pterocarpus santalinoides	Papilionaceae	Body pain	Leaf, flower	Decoction

Source: 2022 field work

capsule, etc. [72]. However, critiques argue that herbal medicines lack standardization in terms of prescription, consumption, and application. Table (1) presents the common herbal plants around the world and their medical properties, while

Table (2) presents herbal plant used to cure diseases, mode of preparation, and parts of plants used to prepare medication in Sierra Leone.

4 Drivers of Herbal Medicine Use

In both developing and developed countries, the patronage of the populace toward herbal product utilization is on the increase based on details that include but are not limited to the following factors: insufficient knowledge about herb preparation process, complexity of phytochemical properties, dosage misconceptions, and religious and cultural beliefs, among others, which are the influencing factors shaping herbal medicine intake [89]. For some people, herbal medicine is part of their cultural and spiritual belief that can't be easily broken [90]. However, for others herbal medicine consumption and application are associated with witchcraft and evil spirit. Herbal medicine intake and acceptability in developing countries are mainly influenced by trust in plant, tribal belief, as well as the magical power of the practitioners. However, in developed countries, the intake of herbal medicine depends on proven scientific evidence and the accuracy of clinical trial results obtained from series of experiments [91]. Another prominent reason why herbal medicines are gaining popularity is the fact that the drugs are less expensive and easily available as compared to conventional medicines (Table 3). The integration of

Table 3 Factors influencing herbal medicine use and characteristics

Influencing factors	Characteristics	Source
Age and marital status	Elderly people believe more herbal medicine than younger people base on experience	[89]
Accessibility of health service	Long distance to clinics, medical fees	[37]
Accessibility of herbs	Natural herbs are mostly abundant within the patient locality	[92]
Spiritual significance	Supernatural power belief	[89]
Educational level	Less educated people rely on herbs more	[93]
Religious belief	Motivates or discourages herb intake	[94]
Healthcare staff attitude	Most conventional healthcare practitioners embarrass patients	[95]
Cultural beliefs	Influence herbal medicine intake	[96]
Affordability of healthcare services	Some remote places lack healthcare services; hence, herbal medicine is the only option	[97]
Primitive preparation method	The primitive techniques used to process herbal medicines discourage some people	[98]
Cost-effectiveness	Herbal medicines are freely harvested	[99]
Financial income	Does not require huge sum of money	[93]
Lack of awareness	Most people lack awareness about herbal medicine benefits	[93]

clinical trial for herbal drug safety and efficacy assessment of different herbs is critical for the global acceptance of herbal drugs in the near future.

5 Negative Effects of Herbal Medicines and Its Sustainable Development Implication

Although herbal medicines are derived from natural plants, it is worth noting that plant products don't simply mean safe and without side effects [100]. As such, herbal medicine safety has been a major problem in local healthcare system especially in low-income countries and has attracted huge global debate on the standardization of herbal products for people's well-being [101].

Scientific evidence has shown that some herbal medicines have serious side effects such as fatigue, headache, vomiting, and asthma among other negative reactions [100]. Like western or conventional medicines, herbal medicines are not risk-free, neither free from side effects [101]. Some herbs from plants can cause additional sicknesses and stimulate adverse reaction on patient if wrongly prescribed. The perception or idea that herbal medicines are completely safe without any adverse effects is untrue and deceptive [101]. Studies have revealed that a range of herbal medicines have produce undesirable side effects that are life threatening and can lead to early death [102]. Studies conducted by Ernest [103] and Ekor et al. [102] alleged that some herbal products turn into poison leading to the death of most patients. The consequences of wrong herbal medicine prescription and the inadequate knowledge about the potential adverse effects and reactions have sent many patients to their early graves in deprived communities around the world. In spite of the positive contribution of herbal medicines to the sustainable development of societies, herbal therapies remain a threat to the health of many patients due to wrong prescription, little or no clinical testing, or standardization of products [102]. Moreover, the causative factors of herbal medicines' adverse effects are closely associated with mislabeling and misidentification of some herbal products by illiterate or semi-illiterate herbal practitioners [102]. The absence of accurate dosage prescription, proven clinical test, and safety assurance information is complicating the healing potential of herbs related medication around the world.

In such, the negative attributes associated with some herbal medicines properties undermine the sustainable development goal 3 and its related targets. Most herbal products have adverse and irreversible life-threatening side effects that have been killing patients slowly in developing countries, hence impeding the sustainable development drive. The growing report on the adverse side effects of herbal medicines is also impeding the sustainable development of novel drug derived from plants [101]. The root cause of herbal medicines adverse effects is closely aligned with the primitive techniques used in preparing and extracting phytochemical properties and the knowledge level of concern practitioners [101].

6 Challenges Facing Herbal Medicines' Sustainability

Herbal medicine sustainability around the world is facing multiple challenges ranging from deforestation, climate change vulnerabilities, limited research in herbal medicines, conservation and management, among others (Fig. 2). The analysis of these challenges is highlighted below.

6.1 Deforestation

Although herbal therapies have proven to be beneficial to society, however, the therapies' sustainability is face with complex and grave challenges that threatened the continuous availability of herbal drugs. The high rate at which herbal medicine plants are disappearing today is alarming and needs urgent attention to be addressed to avoid extinction. The forest ecosystem is the main reservoir for herbal medicines but is rapidly shrinking despite actions being initiated to halt deforestation and by extension to restore degraded lands globally. The recent Forest Resources Assessment (FRA) estimates show that 420 million of forested areas were deforested between 1990 and 2020 [104]. The high rate of deforestation taking place around the globe presents an uncertain future for medicinal plant existence in few years to come [105]. This situation may then have greater impact on global health system especially for majority of the population in developing countries who find it difficult to access conventional medicines [105, 106]. The high rate of deforestation in most

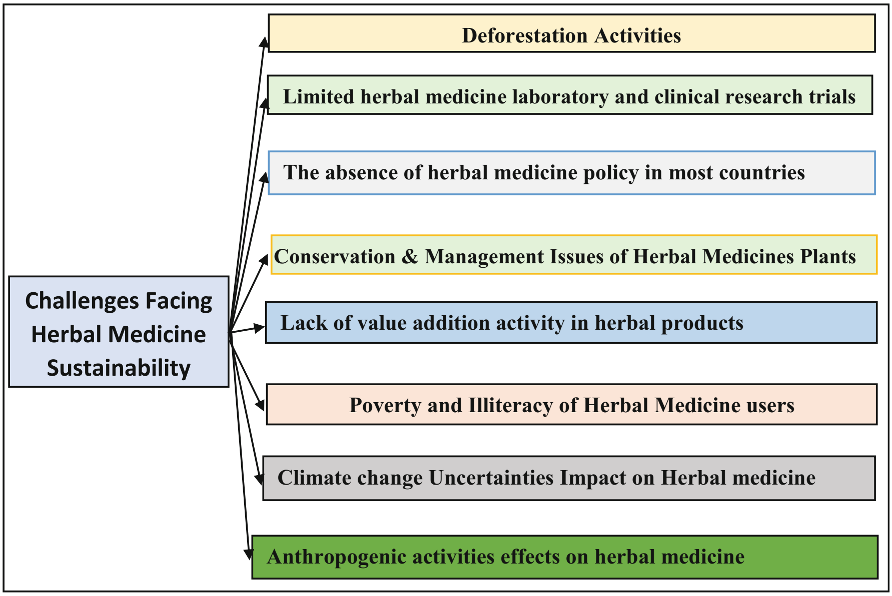

Fig. 2 Challenges facing herbal medicine sustainability

developing countries is connected with biomass energy production [107] and shifting cultivation.

6.2 Limited Herbal Medicine Laboratory and Clinical Research Trials

Herbal medicine has been used to treat various sicknesses and considered the backbone to local healthcare system for decades especially in hard to reach communities. However, only limited clinical or laboratory research work is being conducted globally on a step by step basis to prove the efficacy and safety of herbal medicines as compared to conventional medicine [108]. The absence of adequate clinical experience for herbal medicines affects user confidence and dependability of the product on the international stage. A handful of research conducted over the past years have shown that plant contains antibacterial, antiviral, anticancer, phytochemical, nutraceuticals, and Ayurvedic therapies that can help cure pandemic outbreaks like COVID-19 and Ebola [109]. The absence of adequate number of clinical or laboratory research in herbal medicine therapies has triggered the high rate of mortality resulting from side effects and wrong dosage prescription worldwide [109]. The tropical climate in most developing countries supports herbal plant growth, distribution, and diversity, but the absence of medical research facilities in those countries prevents the actual healing potential of herbal medicine from being realized. According to Alam et al. [109] and Ren et al. [110], some Chinese herbal medicine has been proven to prevent or have the ability to prevent certain critical illnesses from progressing to a deadly and irreversible state. However, the fewer research conducted on herbal medicines hinders the conservation and sustainability of herbal products across Africa and beyond [111, 112].

6.3 The Absence of Herbal Medicine Policy in Most Countries

The safety and efficacy connected with herbal medicines is the growing concern for health authorities and the general public today. The absence of sound and updated policies on herbal medicine development, regulation, monitoring, safety, quality control, processing, and harvesting is limiting the full healing potential of herbal medicines. In addition, the omission of herbal drugs in our healthcare and medical learning system and institutions is jeopardizing the future sustainability of herbal drug acceptability [90, 113]. For instance, the presence of national policy on herbal drugs can help shape and achieve the sustainable conservation and management of herbal plants. Sound policy on herbal medicines can provide the required policy direction and framework needed for sectors to follow for the long- and short-term sustainable utilization of herbal drugs. In some communities of low-income countries, there are by-laws to prevent wildfire outbreak, social unrest, and community forest degradation among others but certainly most don't have by-laws to protect the unsustainable exploitation of herbal plants [90].

6.4 Conservation and Management Issues of Herbal Medicine Plants

The greatest threats facing herbal medicine availability today are sustainable harvesting and management of herb reservoir found around dwelling communities [111]. The extraction method of herbal medicines differs from one region to another and is exploitative in most cases. For instance, most herbs are extracted from stem bark and roots, hence exposing the plant to local extinction and in some instances death [111, 114]. In addition, overharvesting of herbal medicines has the potential to make endangered plant extinct in a short period of time [111, 114]. The uncontrolled utilization and harvesting of herbal medicines by community members to meet the constant health demand are responsible for the death of many plants with huge medicinal properties. Extraction methods like stem debarking, cutting, and uprooting affect the conservation and management of herbal products on a sustainable basis. Stem debarking, for instance, blocks the flow of plant material from the root to the leaves, thereby increasing insect attack and shortening the life of the plant [115].

Furthermore, the rapid exploitation of herbal products through global warming, urban sprawl, decline in natural ecosystem [106], and population growth posed a serious threat to the future sustainability of herbal medicines around the world [116]. In such, the overexploitation of herbal products across the world has resulted in the decline of herbal medicines and the emergence of gene pool and the loss of biodiversity for the current and future generation [116].

6.5 Lack of Value Addition Activity in Herbal Products

The high demand of herbal medicines motivates herbal practitioners to produce more, but the absence of value addition activity renders herbal products inferior to Western drugs [117]. Although herbal medicines are known to be a good source of biomolecules and other phytochemical derivatives, the presence of minimal value addition activities in the processing and preparation of the herbs prevents breakthrough in complex problems supposed to be cured by herbal products. The absence of value addition in herbal medicines discourages investment, reduces income generated through herbal product sale, and stalls local livelihood and economic activities [117]. Value can be added to the raw herbal products harvested from the wild through adequate packaging, branding, drying, milling, labeling, and mixing different products to form a remedy that can cure more sicknesses. In some parts of the world, the harvesting of herbal products is done haphazardly, and as such, drying, grading, storage, and improper processing influence the value of the final herbal products [118].

6.6 Poverty and Illiteracy of Herbal Medicine Users

Poverty is a big problem in most regions or countries where herbal medicine reliance is high. Recent literature suggests that high poverty and illiteracy rates are the major sources of unsustainable herbal medicine harvesting, prescription, and exploitation [119]. Studies have recently detected a relationship between poverty and mass exploitation of natural resources including herbal products. Furthermore, illiteracy and unemployment in herbal resource-rich communities pose a threat to the sustainability and conservation of resources due to massive overexploitation [118]. Poverty-stricken and deprived communities consider herbal medicine as the only alternative source of healthcare and income and as such exploit the resources to meet their increasing daily demands [118] unsustainably.

6.7 Impacts of Climate Change Uncertainties on Herbal Medicine

According to Applequist [120], climate change has proven to have adverse effects on both human and plant species. For centuries, herbal products from plants have benefited humanity in diverse ways especially in communities where western medicine are difficult to reach or purchased. Nonetheless, climate change uncertainties and vulnerabilities threaten the abundance and sustainability of these herbal therapies around the globe [120]. The evolving trend in climate change uncertainties in the form of drought, landslides, flooding, erratic rainfall pattern, and desertification poses a serious challenge against plant extinction [119]. In drier regions around the world, climate change uncertainties are leading to mass extinction of valuable plant resources and wild animals. In addition, climate change has had a detrimental effect on herbal plants through decline in herbal plants and to some extent extinction of some herbs. Moreover, climate change affects plant survival, phytochemical properties, and production ability of most plants. In some ecosystems, climate change has altered the environmental conditions and ecological setup [120–123].

6.8 Effects of Anthropogenic Activities on Herbal Medicine Sustainability

Time and again, traditional medicine has proven to be a reliable source of medicines capable of curing and preventing various diseases especially in developing countries where locals can't afford to buy expensive western drugs. However, anthropogenic activities continue to hinder the growth and survival of herbs with important medicinal properties [124]. In such, herbal medicine abundance and availability have been altered by anthropogenic activities [125]. In recent times, shifting cultivation, charcoal and firewood collection, land use change activities, and livestock grazing have resulted to the decline in traditional herbs used as medicine to cure ailments [124].

7 The Way Forward in Promoting Herbal Medicine Usage and Sustainability

For herbal medicine production to be sustainable, some measures and policies need to be put in place in order to tackle challenges affecting herbal product availability and conservation (Fig. 3). Furthermore, herbal medicine usage should be strengthened at both national and regional levels through the inclusion of herbal medicine in the healthcare system of countries [126]. Furthermore, more search should be conducted to identify the potential phytochemical properties of some unexplored herbal plants. Furthermore, research collaboration between herbal and modern medicine practitioners should be encouraged and promoted at national and regional levels (Fig. 3). In addition, more value addition activities should be undertaken to upgrade herbal product appearance and intake in order to compete with western drugs. Stakeholder engagement and awareness raising drive on the importance of herbal medicine management and conservation should be encouraged at both national and regional levels.

8 Conclusions

Herbal medicine contribution to the healthcare system of most countries especially developing ones has been immersed over the years. It is concluded that plants with medicinal properties are globally a valuable source of healthcare medication more so

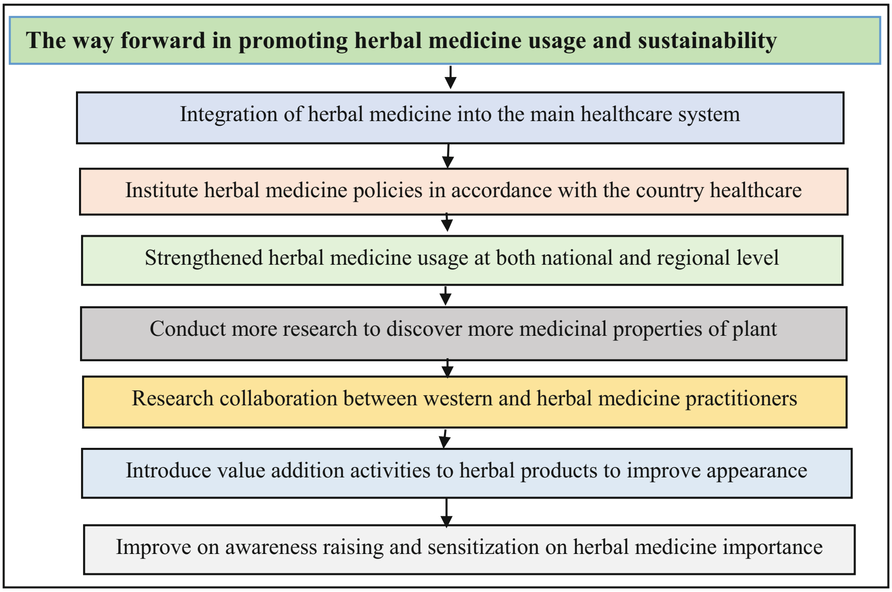

Fig. 3 Way forward in promoting the sustainable development of herbal medicine

for the less privileged in the society. The contribution of herbal medicines ensures the sustainable development of countries and their local economies. The upsurge in demand and value of herbal therapy today is closely linked with the rise in knowledge of the therapeutics, phytochemical properties, availability, and affordability of herbal medicine and the inexpensiveness of the herbal products as compared to modern drugs. The review accomplishes that herbal medicine sustainability is impeded by various factors ranging from policy absence, anthropogenic actions, uncontrolled deforestation, and limited collaboration between herbal and western healthcare practitioners. The review concludes that herbal drugs play a critical and central role in the sustainable development of local communities through the strengthening of healthcare system and the availability of drugs at any time among others. If adequate conservation and management measures are put in place, herbal products have the potential to trigger sustainable development revolution in the healthcare system of low-income countries. It is concluded that herbal medicine contribution to sustainable development is important in achieving the sustainable development goal 3 set for 2030. It is recommended that more research and collaboration be undertaken to integrate herbal medicine into the healthcare system of countries.

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Herbal Medicine and Sustainable Development Challenges and Opportunities

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Abstract

Eco-friendly, bio-friendly, reasonably priced, and usually safe herbal medicines have moved from the margins to the center of medicine as a result of expanding research in the field of traditional medicine over the past few decades. A variety of herbal medicines have been created over the past few decades for both the prevention and treatment of diseases that are safe for the environment, organically produced, and affordable due to extensive study in the conventional system of medicine. Natural herbs have been used for a long time to treat and prevent many illnesses. However, the major obstacles to the growth of herbal medicine include regulation, industrialization, biopiracy, inadequate infrastructure, overuse and irrational use of medicinal herbs, and loss of biodiversity. Promoting and incorporating herbal medicines into basic healthcare develops sufficient capacity enhancement. This should be viewed as a sustainable route. It possesses the potential to support the Sustainable Development Goals (SDG) and the promotion of herbal medicines by the World Health Organization (WHO), as well as to facilitate the development of new drugs, improved access to healthcare, the preservation of forest resources, and assistance in bridging socioeconomic gaps.

Keywords

Herbal medicines · Traditional medicine · Organically · Biodiversity · Sustainable development · Healthcare

Abbreviations

AYUSH	Ayurveda, Yoga and Naturopathy, Unani, Siddha, and Homeopathy
CNS	Central nervous system
GABA	Gama-aminobutyric acid
GACP	Good agricultural and collection practices
GLP	Good laboratory practice
GMP	Good manufacturing practices
ICDS	Integrated child development services
ISM	Indian System of Medicines
JSY	Janani Suraksha Yojana
MAO	Monoamine oxidase
MDGs	Millennium Development Goals
NMPBs	National Medicinal Plant Boards
NRCH	National Reproductive and Child Health
NRHM	National Rural Health Mission
RCT	Randomized clinical trials
SDG	Sustainable Development Goals
SOP	Standard operating procedures
TCM	Traditional Chinese medicine
UNICEF	United Nations International Children's Emergency Fund
WHO	World Health Organization

1 Introduction

The utilization of plant species, particularly those having medicinal characteristics, has been essential for human survival throughout the evolution of human cultures [1]. Since their earliest beginnings, which date back to 8500 BC [1], medicinal plants have offered humankind curative therapies. Traditional Chinese medicine and Ayurveda are two notable cultures that have used and continue to use medicinal plants [2]. However, during the colonial era, these medical practices were disregarded and even made illegal by officials because they were thought to be primitive and unscientific. To use the potent medicinal substances found in plants as components in allopathic medicines, scientists began to extract and alter them when chemical analysis was accessible in the first decade of the nineteenth century [4]. At least one active component originating from plants is found in at least 25% of allopathic prescription drugs [88]. According to a recent study, herbal medications are incredibly safe and equally efficient as conventional ones. Figures show that only 15% of the 250–400 thousand plant species in the globe have undergone phytochemical investigation, and only 6% have undergone systematic biological activity screening [89]. As essential medications, more than 120 chemical compounds derived from plant sources are used, and over 60% of drugs used in therapeutic uses involve natural substances or their derivatives [90].

The word "herb" is used explicitly in botanical literature to describe nonwoody vascular plants without enduring wood stems, such as annual, biennial, and specific perennial crops (including the majority of the monocot species). The term is used more widely in the study of pharmacology, where it is used as a synonym for "plant" to refer to all trees, shrubs, and herbs. Herbal medicine is defined by the WHO (World Health Organization) as a practice that uses herbs, herbal supplies, herbal preparations, and final herbal products that include active substances extracted from plants, other plant materials, or combinations of these [3]. The stems, leaves, blossoms, roots, and seeds of plants are used to make these herbs [4].

Additives like dilute solutions, solvents, or preservatives are frequently added to plant-based medications in addition to plant components, either in their raw or processed forms or active substances [4, 5]. These active substances protect the plant from harm and illness and improve its scent, flavor, and color. They are categorized by scientists as phytochemicals and fall under a variety of categories, such as tannins, alkaloids, flavonoids, glycosides, saponins, and terpenoids [6]. The potential for phytonutrients to improve the well-being of people has been established over time by science [7]. Herbal sedatives and stomachic combinations, for example, contain aromatic plant species with medicinal essential oils with antibacterial, stomach-soothing, and antispasmodic properties. Species of plants that have a significant tannin content are used in mixes for the treatment of stomach ulcers and diarrhea because they typically have antibacterial, astringent, and anti-inflammatory qualities [8, 9]. Herbal remedies can potentially improve access to healthcare and help realize Sustainable Development Goal (SDG) number 3 – wellness and excellent health – in addition to playing a part in the economic, social, and environmental sustainability spheres. The production of medicinal plants can be used to improve income generation or as a source of alternate income, safeguard biodiversity, and foster regional economic development while preserving forests [10]. By keeping forests healthy, we may avoid irrevocable climate change. For instance, the Amazon is a significant carbon sink and contributes significantly to the global and regional water cycle [11, 12]. Severe changes in land use, particularly those connected to extensive deforestation and forest degradation, have been indicated to significantly influence regional and worldwide precipitation patterns [13]. Recognizing herbal medicines and their ability to stop forest destruction is now more crucial than ever before. Additionally, it might cause people to quit their present, less sustainable jobs in the forestry and mining industries. As a result, a bioeconomy may result in an expanded enterprise that, in spite of generating income, has the power to support and protect natural environments from harmful land use practices. The value chain must be analyzed to ensure all activities are sustainable, as development carries the danger of increased strain on natural ecosystems [13]. Additional study on the subject is necessary to improve awareness of how herbal medicine development may help ensure healthcare access, biodiversity preservation, and socioeconomic growth in this environment.

2 The Historical Significance of Herbal Therapy

According to historical Babylonian reports, people first used plants as medicines 60,000 years ago. At least 5000 years have passed since the writing of herbal therapy in Egypt and China, and 2500 years have passed since Asia Minor and Greece [14]. Multiple herbal medical systems exist, and the foundations and procedures in each are affected by the part of the world in which it was developed [15]. China has utilized traditional Chinese medicine (TCM) for quite a while [16]. The Celestial Farmer's Classic of Herbalism, the world's earliest herbal text, was written in China approximately 2000 years ago. Herbal written material provides access to multiple herbal pharmacopoeias and numerous monographs on specific botanicals [17]. Ayurveda is a historical Hindu medical discipline practiced for more than five thousand years in India and was created by ancient Hindu practitioners and philosophers. More than 1500 floral species and around 10,000 mixtures are exhaustively described in its Materia Medica. Unlike traditional Western medicine, the Indian government recognizes Ayurveda as an extensive healthcare system [18]. Kampo medicine, a category of Japanese herbal medicine with a history extending back over 1500 years, consists of 148 unique formulations [19].

2.1 Sustainable Development and Global Commitments

The intrinsic worth of natural assets and their social advantages must be comprehended, particularly in light of the global prevalence of medicinal plant use. But when seen from the viewpoint of environmental history, it stands out for the ways that people have used nature to further their ends. Inequality results from industrialized countries' increased consumption levels without considering the requirements of developing countries [20]. A human-centered viewpoint has made many environmental problems worse and contributed to the present endangerment of natural resources and medicinal plants. Since the 1987 release of the Brundtland Report, in which sustainability is defined as "addressing the demands of the present without jeopardizing the capacity of future generations of people to meet their own needs," sustainable growth has been an essential strategy for development [21]. In addition, it ensures continual access to the resources necessary to establish a less disruptive and highly resource-intensive economy [22]. It is an approach designed on the idea that economic, ecological, and social factors must work together to assure economic growth while fulfilling community and ecological necessities [20]. Rio de Janeiro, Brazil, hosted the United Nations Conference in 1992 on the Environment and Development, resulting from a growing global environmental movement. Agenda 21 results from the Earth Summit, which focused on the emergent need for a worldwide sustainable strategy that requires national, regional, and municipal effort [20, 22]. Agenda 21 included 120 recommendations for action. Over time, Agenda 21 was incorporated into the United Nations Sustainability Goals and Millennium Development Goals (SDGs and MDGs). To ensure that the needs of people are catered to both now and in the future, they were established in 2015 and include 17 goals and 169 targets that cover a broad range of topics. Regarding the requirement to appreciate medicinal plants as an element of primary healthcare, successful promotion and integration of herbal medicines could lead to a surge in the use of therapeutic plants and the development of the herbal medicines supply chain. A story that directly seeks to attain SDG 1 (No poverty), SDG 2 (Good health and well-being), and SDG 15 (Life on land) can be considered a means of achieving sustainable development (United Nations 2021).

2.1.1 SDG 1: No Poverty

Poverty affects freedom, health, and education and adversely affects the economy. This is particularly evident in emerging nations where poor infrastructure and service delivery, rural inequality, and environmental degradation are the leading causes of rural poverty [23]. Agroforestry and the incorporation of medicinal plants can help enhance livelihoods and contribute to the eradication of poverty in all its forms. SDG 15 (Life on land) has two goals that must be achieved to address the world's severe environmental concerns (United Nations, 2021).

2.1.2 SDG 3: Good Health and Well-Being

Incorporating plants for medicine into healthcare facilities can boost the availability of healthcare in both urban and rural areas, which is crucial for sustainable development. This is especially crucial in light of the effects of the COVID-19 pandemic and its spread of diseases such as AIDS, malaria, and neglected tropical diseases [24]. Plants with medicinal properties must develop new medications to improve healthcare.

2.1.3 SDG 15: Life on Earth

Earthly existence of protection, restoration, and development of ecosystem services and biodiversity require sustainable land use. The human population of tropical regions, which relies on natural resources for survival, must possess two characteristics to preserve its means of subsistence and confront the forthcoming difficulties of climate change [25]. Growing utilization and demand for medicinal plants may lead to an appreciation of the extant forest and its biodiversity in this context. Promoting the preservation of ecosystems and species in addition require equitable allocation of genetic resources by establishing a direct connection between medicinal plants and biodiversity.

The World Health Organization emphasizes the use and importance of safeguarding plants for medicinal purposes as one of the main actors in attaining the Sustainable Development Goals, which call for universal availability of healthcare [26, 27]. At the 1978 International Conference on Primary Healthcare held at Alma Ata, Kazakhstan, the World Health Organization (WHO) and the United Nations International Children's Emergency Fund (UNICEF) emphasized the need for immediate global action [28]. Since then, the WHO has remained dedicated to recognizing and encouraging the application of medicinal plants, which led to the development of a program for conventional medicine whose current version encompasses the years 2014–2023 [29]. The program's objective is to assist participating nations in developing and implementing a strategic plan that considers their unique capacities, goals, and regulatory and legal frameworks. Promoting herbal remedies is crucial for developing an alternative to traditional practices [30].

2.2 Health, Poverty, and Livelihoods

High levels of cultural diversity affect people's health in developing countries, particularly in tropical regions which are additionally affected by scarcity and infectious diseases [31, 32]. Since 1940, more than 300 contagious illnesses have emerged, impacting approximately 15 million individuals every year in the globe. In addition, one billion people worldwide are afflicted by neglected tropical diseases that have become endemic to rural areas with scarce resources and threaten the livelihoods of the people affected [32]. In the same regions, chronic diseases are also on the rise. Mexico will be used as an example because it is one of the biggest economies in Latin America and is still grappling with poverty and rural population inclusion [33]. Three thousand healthcare posts and 71 rural hospitals were established despite their attempts and accelerated expansion of their healthcare systems. As a result of the change, their healthcare system must now treat diseases connected to poverty and contemporary lifestyle disorders [31].

In the words of **Mphande** [32] who explains the association between health, poverty, and means of survival, the economic and health conditions of the population in developing nations influence their livelihood choices (Fig. 1). People are becoming increasingly aware that poverty can rapidly develop when a family's earnings are

Fig. 1 The relationship between poverty, health, and way of life with markers for each

restricted. However, declining health can also indirectly contribute to poverty by reducing the ability to support a family. This condition generates an endless cycle that jeopardizes all three pillars necessary for a prosperous existence. Figure 1 depicts the association between health, poverty, and means of subsistence, as well as the corresponding metrics for each factor.

3 Present State of Herbal Medicine: Higher Plants Are the Source of Modern Medicine

Due in significant part to medicinal plants, new medications are developed. According to the WHO, approximately 25% of current pharmaceuticals are derived from botanicals utilized in traditional medicine. Plenty more are synthetic analogues created using plant-derived model substances as a starting point. In addition, the WHO now recognizes medicinal plants as an integral component of primary healthcare [33]. Medications derived from plants have fundamentally transformed contemporary treatments. Vinblastine, extracted from the *Catharanthus rosesus* plant, is effective against malignancies of the testis, pharynx, Hodgkin's lymphomas, choriocarcinoma, and non-Hodgkin's lymphomas, among others [34]. The phophyllotoxin extracted from *Phodophyllum emodi* possesses anti-lymphoma, anti-lung, and anti-testicular characteristics. Taxol, which originates from *Taxus brevifolius*, is used to treat lung and ovarian tumors that have spread to these tissues. In 1953, serpentine was identified in the root of *Rauwolfia serpentina*, and it has been utilized successfully to treat hypertension and lower blood pressure [35]. Between 1950 and 1970, the US pharmaceutical industry reportedly introduced approximately 100 novel plant-based medications, which include deserpidine, reseinnamine, reserpine, vinblastine, and vincristine. Ectoposide, guggulsterone, teniposide, nabilone, plaunotol, Z-guggulsterone, llentinan artemisinin, and ginkgolides comprised among the novel plant-derived medications isolated between 1971 and 1990. More plant-based medications, such as Paclitaxel, Toptecan, Gomishin, and Irinotecan, are released to the pharmaceutical market between 1991 and 1995 [36]. Several researchers have recently discovered the chemical substances like quinine, digoxin, aspirin, ephedrine, atropine, and colchicine extracted from plants [37, 38].

3.1 Trends in Herbal Medicine Use

Herbal medicine continues to be extensively utilized in developing nations, and in the industrialized world, utterly 70% of physicians in France and Germany routinely prescribe herbal medications, based on estimates. Additionally, an increasing number of individuals are using herbal remedies [39]. Predictions indicate that 80% of the global population will use botanicals; this number could reach 95% in developing nations [40]. Traditional herbal remedies account for 30–50% of all pharmaceuticals used in China. In Ghana, Mali, Nigeria, and Zambia, herbal remedies constitute 60% of the first-line treatment. It is estimated that more than 50% of individuals in industrialized nations such as Europe, North America, and others have tried herbal remedies at least once [41]. In San Francisco, London, and South Africa, 75% of HIV/AIDS patients utilize natural treatments. Seventy to ninety percent of Canadians and Germans have used herbal remedies at least once. It is estimated that 158 million individuals in the USA use natural remedies, which continues to rise. The market for herbal remedies is presently worth over $60 billion annually and continues to expand [41, 42]. It is also important to remember that adult populations are more likely to use traditional and herbal medications. Since chronic diseases are increasingly common in this population, the long-lasting adverse effects of conventional medication therapies frequently discourage patients from using them for an extended period. Nevertheless, consistent application of herbal remedies offers a treatment without adverse effects. Due to these needs, herbal medicine has attracted interest and is rapidly gaining global recognition.

4 Challenges in the Medicinal Plants Sector

Human cultures in developing nations rely heavily on forest goods for their economic systems and way of life, so the continued growth of humanity is one of the factors causing concern regarding our capacity to meet our everyday needs for food and medicine. Due to this inclination, causing the forest and the goods it produces to continue to erode, it is difficult to satiate demand and safeguard irreplaceable bioresources [43]. The number of species in the Materia Medica is rising, but their purity and precise identification requirements have not kept pace. In addition, it is crucial to protect endangered medicinal plants due to the global deforestation crisis and the rise in herbal and ethnomedical practices. An enormous amount of plant materials and extracts are needed for the manufacturing of herbal formulations as well as for related industries, including food, nutraceuticals, cosmetics, perfumes, and toiletries, and because India, Tibet, China, and many African and Latin American countries are renowned for their traditional medicine, the majority of which, including Chinese medicine and Ayurveda and Unani in India, is primarily based on herbs [44]. Since market prices for medicinal plants and their by-products do not accurately reflect profits, supply, or demand, they only provide a nebulous understanding of how the market functions. After thoroughly examining the available information regarding the medicinal plant industry, we have identified the following essential characteristics and challenges.

4.1 Rising Demand

The World Health Organization (WHO) estimates that the annual market for medicinal plants is presently worth \$14 billion [45]. According to a WHO forecast, by 2050, the global market for restorative plant-derived raw materials will have grown between 15 and 25% yearly and be worth \$5 trillion. The annual value of medicinal plant trade in India is estimated at USD 1 billion [46]. India exported nearly three times as many Ayurvedic products during the previous two fiscal years (2001–2002 and 2002–2003). Due to the expected rise in demand for medicinal plants, numerous species were overharvested from the wild, causing their populations to decline. For instance, substantial quantities of Himalayan yew (*Taxus baccata*) have been harvested from the wild since its extract or parts, Taxol, have been approved for treating ovarian cancer. *Aconitum heterophyllum*, *Nardostachys grandiflora*, *Dactylorhiza hatagirea*, *Polygonatum verticillatum*, *Gloriosa superba*, *Arnebia benthamii*, and *Megacarpoea polyandra* are varieties of North Indian medicinal plants that have been overused for medicinal purposes and are now rare or endangered. Multiple species of medicinal plants are imperilled by excessive harvesting in the wild as they are used to treat a wide range of diseases [47, 48]. For instance, the *Hemidesmus indicus, Gloriosa superba, Aegle marmelos*, and *Phyllanthus emblica* are used to treat 34 diseases. The excessive use and ongoing devastation of medicinal plants [49] have hurt on the genetic diversity, supply of food, and manner of life of indigenous people who reside on the forest's periphery.

4.2 A Rise in Rarity

Due to the ongoing exploitation of several native medicinal plant species [50] and significant habitat destruction over the past 15 years [51], the overall population of high-value medicinal plant species has diminished over time. Biodiversity that is utilized by humans poses the biggest threat to medicinal plants [52]. Among the factors contributing to the disappearance of medicinal plant varieties is the decline of traditions that have governed the use of natural resources [53, 54]. Contemporary socioeconomic factors have repeatedly weakened these established regulations [55]. Numerous additional variables, such as habitat particularity, a restricted distributional scope, land use changes, the introduction of non-native species, habitat modification, alterations in the climate, significant grazing by livestock, the explosive growth of populace, population dispersion and deterioration, and genetic drift are responsible for the rarity of medicinal plant species [56–59]. Viruses, herbivores, and seed predators are natural adversaries that can significantly reduce the presence of rare species of medicinal plants in a given area [60, 61]. Even though humans and other animals consume medicinal plants, humans also experience physical disorders that can be treated by various species within the same genus. Several species of *Swertia*, for example, are used to treat malarial fever. Similar to how various *Berberis* species are used as a source of berberine to treat specific eye conditions, *Berberis* species appear in a diversity of forms. Moreover, even though they are less threatened by extraction than varieties with no alternatives, various species within the same genus have varying chemical concentrations, so there is a demand for them. Due to the absence of data/information on the population size and prevalence of rare plant species in the field, the classification has been limited to a handful of species based on herbarium specimens and expert opinion. Current assessments call into doubt the precision with which threat groups are assigned to species, particularly the number of endangered taxa in a given region. The problematic topography, hostile climate, and short plant life cycles exacerbate the difficulties associated with species assessment in the mountainous regions regularly at high altitudes. The majority of the material that is currently accessible was obtained from the more accessible regions of these mountains. Indigenous groups and commercial herb hunters also explore the same regions for medicinal plants. Because it varies in areas in which these rare medicinal plant species have been collected infrequently or never, the predicted density of classified rare medicinal plants is inaccurate [62].

4.3 Medicinal Plants Cultivation

Only 1% of the world's known plant species had access to agrotechnology, and less than 10% of those do not know how to reproduce [63, 64]. This pattern suggests that the development of agrotechnology ought to be one of the main study areas. To satisfy the growing demand for medicinal herbs, these plant types need to be cultivated. In addition to meeting the present need, agriculture may help preserve the genetic variety of medicinal plants in the wild. Agriculture enables the production of identical elements from which reliable standardization of products can occur. In addition, cultivation increases the likelihood of genetic advancements, improved quality control, and enhanced species identification. In large-scale cultivation, selecting the planting medium is another important consideration. Therefore, the plant material must be high quality, packed with active components, resistant to diseases and pests, and tolerant of its environment. To successfully cultivate all medicinal plants extensively, one must determine whether monoculture or polyculture is preferable.

Because many species of medicinal plants are mainly found in forest cover, agroforestry provides a valuable method for their farming and sustenance by:

1. Using medicinal plants that can survive in the shade as understory plants in multilevel ecosystems
2. Integrating fast-growing medicinal plants into established tree plantations
3. Planting plants with therapeutic properties for shade purposes

Regardless, it is recognized that cultivating medicinal plants is a complex undertaking, as demonstrated by the history of therapeutic plant cultivation. Medicinal plants like *Saussurea costus* have a lengthy cultivation cycle. Therefore, many farmers in the trans-Himalayan region of Northern India have shifted to cultivating common commodities such as peas, potatoes, and hops (*Humulus lupulus* [65]). It isn't easy to grow herbs for medicinal purposes in India since they are present in large amounts and are less expensive than many seasonal crops [66].

4.4 Bioprospecting and Biopiracy: An Emerging Problem

The once-remote verdant forests are now integral to the demanding multicultural politics and fast-paced, profit-driven global economy. For policymakers and economists, the declining demand for forest products, particularly medicinal plants, poses a challenge. Local medicinal plant traditions are significant on a global scale. The Rio Convention on Biological Diversity established a framework for the responsible and regulated utilization of biological diversity. The current climate for bioprospecting is marked by increasing suspicion and animosity regarding biopiracy and the rights of developing and developed countries to reap the benefits sharing [67]. The vast majority of obstacles associated with the legal protection of indigenous wisdom and the payment of remuneration to indigenous herbal practitioners for their expertise are extraordinarily complex. One group identifies the present system of payment or sharing of advantages under intellectual property rights as a new legal form of biopiracy. In contrast, another identifies it as a means of protecting the rights of knowledge proprietors. The diverse methods and procedures used in India, the USA, Europe, Canada, and other nations to grant patents on medicinal plants have created confusion [68]. Typically, patent protection is unavailable for plants, plant-related developments, and plant products such as seeds, flowers, fibers, and resins. However, any living object created by humans in the USA, whether through reproduction or laboratory research, is patentable. The Indian Preservation of Plant Varieties and Farmer's Rights Act of 2001 rewards farmers who engage in breeding initiatives. This legislation also includes arrangements for distributing benefits that recognize local populations as contributors of plant material [69].

The patentability of products varies greatly across developed and developing countries, including India. There were 31,25,603 patent applications from 91 countries, but only 301,177 (9.6%) were registered in developing countries, which is very unfortunate as they are the habitat of numerous plants having medicinal properties. The rest were filed in developed countries; 2.3% of those enrolled are from developing nations and 0.2% are residents. In addition, 97.7% of all applications for patents filed to date have been submitted by nonresidents for the primary purpose of monopolizing markets for export in developing nations [70]. Developing countries and numerous scientists interested in using medicinal plants require specific laws regarding the recording of the nationalities of samples and the equitable distribution of the benefits within the countries of origin, the inventor, and business benefactors. Several developed nations are hesitant to embrace these clauses. These disagreements have irritated numerous scientists who argue that organic products will always be the finest resource for developing novel medications. In 2004, at the Kuala Lumpur Conference, the delegates of 188 nations decided to develop a structure acceptable to all signatories to reduce such disputes and prepare for the discovery of new drug sources. The proposed framework will thus be presented for discussion at the forthcoming meeting in 2006, which Brazil will host.

4.5 Enhancing the Market System Medicinal Plants

Herb gatherers, local middlemen, urban merchants, wholesale suppliers, producers, exporters, and herbal therapists are a few of the many involved players in the medicinal plant trade. The marketing structure for medicinal herbs is largely unregulated and unjust [55]. Typically, marginal cultivators and laborers are responsible for the harvest of medicinal plants. They earn money by selling therapeutic botanicals to cover everyday costs such as food, healthcare, and their children's education. They frequently ignore the genuine market prices for a vast array of medicinal plants. Even the vast majority of bottom-up stakeholders in the medicinal plant industry, which typically employs a top-down approach, don't realize there's a growing market for what they're selling. Farmers in the Chamoli district of Uttarakhand grew Kut (*Saussurea costus*) and Dolu (*Rheum emodi*). Yet they faced problems selling them because they were unfamiliar with the local distribution network. On the other hand, illegal trafficking of medicinal plants is widespread.

The following factors also pose challenges for the medicinal plant industry:

1. Many useful plants grow at an extremely slow rate.
2. Many medicinal plant species have a lengthy gestation time.
3. Inadequate methods of plant growth technologies for medicinal plant species.
4. Less production.

5. Unsuitable harvesting techniques.
6. Insufficient study on high-yield varieties.
7. Fluctuating demand and supply.
8. Poor manufacturers.
9. Insufficient marketing infrastructure – this term basically indicates that there aren't enough marketplaces selling these medicinal herbs. When we examine the market for accessing, storing, distributing, and adding value for different commodities (such as allopathy drugs), we can observe the significant difference.

Manufacturers frequently choose untamed plant species over cultivated medicinal herbs because wild plant species contain more chemicals. The harvesting seasons and various growth stages of a species also influence variations in chemical composition. The market for medicinal plants is largely underreported and inadequately regulated. Due to the extensive illicit commerce, the revenue generated by this industry is therefore inaccurate. The economic benefits and management costs of natural populations are frequently miscalculated [71].

4.6 Problems with Herbal Medications' Quality Control

Both intrinsic (genetic) and extrinsic factors, including the environment and good agricultural and collection practices (GACP), which include crop selection, cultivation, and harvest techniques, influence the overall quality of the basic materials used to produce botanical products. Combining these factors makes it difficult to conduct surveillance on basic ingredients. Implementing standard operating procedures (SOP) for authentication, quality, storage spaces, sanitation, and cleaning of raw materials is essential for achieving excellent laboratory and valuable manufacturing practices [72]. Concerns about the quality of the finished herbal product, especially mixed herbal goods, are common because of the many ways in which it is impossible to verify the presence of every single herbs or raw materials. Consequently, finished goods tend to be subject to stricter requirements and quality control procedures than other medications. By WHO recommendations, organizations for quality assurance and control measures have been established to oversee good manufacturing practices (GMP) for botanical products, licensing and labeling methods for their manufacture, commerce, and national marketing initiatives [73].

4.7 Harvesting and Processing Issues

The bad quality of herbal medicines results from various factors, including lack of processing technology, efficient farming practices, propagation methods, and a sloppy harvest, all contribute to poor quality produce [74].

Adulteration is the practice of using subpar replacements for the original drug substance. The replacement parts might come from the same or a different factory and be counterfeit, subpar, flawed, broken, or useless [75]. Adulteration can take place in two approaches:

- Intentional or direct adulteration
- Unintentional or indirect adulteration; for instance:
  1. Using artificially made ingredients, such as when nutmeg is combined with basswood or beeswax is tinted paraffin.
  2. Using subpar substances, such as *Ailanthus* as a belladonna adulterant or *Piper nigrum* as a papaya adulterant.
  3. With pharmaceuticals or poisonous substances, such as lead shot in morphine, limestone in asafetida, white oil presence in coconut oil, or cocoa butter combined with stearin or paraffin.
  4. Powdered substance adulteration, such as selling powdered liquor ice as cinchona or gentian combined with powdered olive stones [76].
  5. Dangerous levels of heavy metals have been found in many herbal formulations, including lead, arsenic, mercury, and others, according to research from several colleges [77].
  6. Adulteration of herbal formulas with readily available synthetic drugs: The US FDA found that some herbal supplements had sildenafil, warfarin, indomethacin, alprazolam, estrogen, etc. added to them.

4.8 Challenges Associated with Medicinal Plants' Clinical Research

Before describing a novel drug for conducting significant phase III trials, studying herbal drugs presents several problems that must be resolved. These challenges include ones connected to the study's plan, quality control, study budget, and regulatory requirements. The WHO released an operational strategy in 2005 covering the regulatory requirements that enable herbal product scientific trials [73].

Blinding is the most effective method for reducing bias and eradicating placebo effects in randomized clinical trials (RCT). Neither the two investigators nor the patient knows which therapy is being allocated, thus the name "double-blind." Maintaining double-blindness in herbal preparations can be challenging due to the multifaceted therapeutic approach employed in this healing process, which includes lifestyle, listening, counseling, explaining, and nutritional recommendations [78]. Other factors that must be taken into account and which may influence the results of clinical research include:

- The selection of controls: If the purpose of the investigation is to present evidence of a particular effect of the herbal medication, comparison comparability is required, therefore the rules are selected to be as similar to the group that will be receiving the therapeutic intervention as is practically possible. Finding a suitable power may be challenging for some odorous natural substances, such as ginger.

- Sample size calculations are necessary to carry out clinical studies.
- It has been recognized that there needs to be occupational harmonization, particularly in herbal product trials where the therapist is key [79].

4.9 Challenges Related to Pharmacovigilance of Herbal Medicines

Due to the growing use of herbal products including treatments worldwide, it is becoming increasingly essential to include herbal products and therapies in pharmacovigilance procedures. The major problem with implementation of pharmacovigilance in herbal medicines involves the following reasons [80] (Fig. 2).

Due to their extensive use worldwide, herbal medicines and products have been integrated into pharmacovigilance systems for quite some time. In this context, these goods' security has become a significant societal concern due to the necessity of basing risk assessments of herbal medications entirely on human interaction.

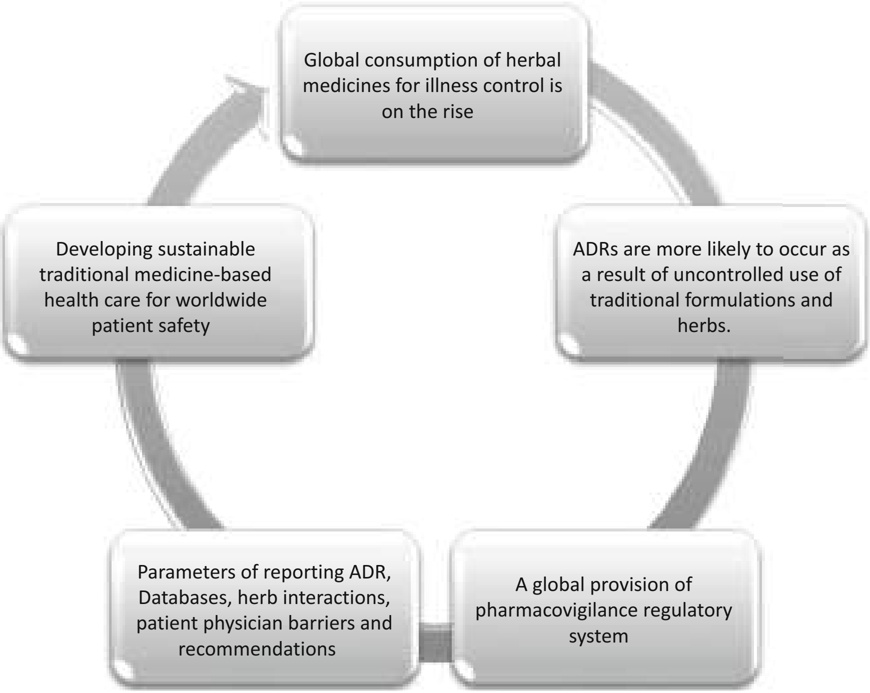

Fig. 2 Problem with implementation of pharmacovigilance

4.10 Problems with the Way Herbal and Conventional Drugs Interact

Many active phytoconstituents exist in different flora, each with a unique pharmacology, metabolism, and binding characteristic. These medications may also have pharmacokinetic or pharmacodynamic reactions with allopathic medicines. Garlic and ginger, which inhibit platelets and increase the risk of bleeding in warfarin-treated patients, increase bleeding in patients taking herbal drugs with a limited therapeutic window [81]. Other instances of how herbal medicines can combine

1. Ephedrine increases and prolongs the sympathomimetic impact by interacting with beta-blockers and MAO inhibitors (hypertensive crisis) [82]
2. Valerian and sedatives combination, causing more CNS depression [83]
3. Ginseng increases GABA metabolism and dopamine levels and reacts with MAO inhibitors [84]
4. Because of how KAVA (drug) and paracetamol combine, azole antifungals are more toxic to the liver [85]

5 Future Prospects of Herbal Medicine

Based on our understandings on herbal medicine, approximately 121 pharmaceutical drugs have been created in the last ten years [86]. At least 25% of currently available medications, including aspirin, picrotoxin, and many other substances with synthetic equivalents made from molecules extracted from plants, are thought to have their origins in plants [87]. Future expansion of the utilization of plants as sources of therapeutic agents will be accelerated by the increasing adoption of plant-based medications [86]. This has substantially increased the export of herbal remedies and attracted several pharmaceutical companies, which include multinationals [5]. The WHO's interest in chronicling ethnic communities' use of medicinal botanicals has resulted in increased scientific support for their application. Consequently, individuals will be better informed about the treatment's safety and effectiveness [14]. The oversight of herbs has contributed to the improvement of herbal goods, but additional adjustments are required to promote and advance high-quality research [4].

Conventional medical systems have gained popularity globally due to the public's fascination with herbal remedies and their astounding approval for their beneficial qualities with few to no adverse effects against various challenging health-related conditions. Sixty percent of the world's population currently relies on herbal or conventional treatments to treat malaria-related fever. Eighty percent of Africans, thirty to fifty percent of Chinese, forty-eight percent of Australians, seventy percent of Canadians, eighty percent of Germans, forty-two percent of Americans, thirty-nine percent of Belgians, and seventy-six percent of French individuals chose herbal or alternative treatments as their first line of defense against various diseases [88]. Seventy-five percent of people with HIV/AIDS in cities such as San Francisco, London, and the Republic of Africa use herbal medicines. Due to the availability of healthcare providers and affordability, countries such as Malaysia are spending more on costly prescription medications than on allopathic ones. Herbal medicines are gaining in popularity [89]. Projections show that many people in poor and developing countries get most of their healthcare from traditional drugs, and goods and remedies made from herbs are highly valued as essential parts of their cultures [90]. Herbal remedies are also widely used in many developed nations, while they are gaining increasing importance in the European Union, North America, Australia, and the UK [91].

5.1 Policies Proposed by AYUSH for the Healthcare Sector

- Ayush Ghutti, Bal Rasayana, Soubhagya Shunthi, Ajwain Ark, Pudina Ark, Punarnavadi Mandoor, and Tel Ksheerbala are some of the conventional medicines that will be added to the National Reproductive and Child Health (NRCH) Program.
- To come up with different ways for AYUSH remedies to work together with schemes like JSY-AYUSH, ICDS-AYUSH, breastfeeding early on, child development tracking, antenatal and postnatal care, etc.
- When used with the AYUSH medication Punarnavadi Mandoor to cure anemia during pregnancy.
- Verify if AYUSH treatments are accessible to major medical facilities.
- The use of AYUSH doctors in national programs for population stabilization and reproductive and child health.
- Utilizing the Indian methodology of traditional medicine to use in public health programs (NRHM), such as bringing in Ayurvedic medical professionals and paramedics.
- Spreading awareness of the Indian Traditional Medical System worldwide by giving seminars and conferences, working with foreign institutions, and initiating research there.
- The Institute for Research on Indian Systems of Medicine (CRISM) at the University of Mississippi in the USA was established by AYUSH as a joint Indo-American research facility.
- There are various universities in India teaching Ayurveda to international students from places like Japan, Italy, Russia, the USA, Australia, Holland, South Africa, Brazil, Canada, Hungary, Germany, France, Sri Lanka, Ukraine, and many more.
- Russia and India have entered an MoU that seeks collaboration in Ayurveda education, treatment, and study [92].

5.2 Industry

Over 10,000 industrialized traditional medicine facilities in India generate about $1 billion USD in net earnings from ISM and herbal systems. Over the past two decades, medications manufactured at AYUSH facilities have increased annually. Both ancient (such as pills, powdered form, medicinal oil, foods that have undergone fermentation, etc.) and modern-day medication forms can be used in the creation of Ayurvedic medicines (such as lotions, capsules, syrups, liniments, ointments, granules, and creams, among others). The Drugs and Cosmetics Act of 1940 governs this industry's production process and related regulations (1945). Entities involved with the production of both traditional and herbal medications must strictly adhere to GMP and GLP, which were developed by the Indian medical system's governing authorities [93].

5.3 India's Role and Future Perspectives

It is anticipated that herbal medicine will continue to be in high demand, and the prospects are optimistic. As a consequence of current ethnobotanical studies and the prolonged globalization of the trade markets for herbal remedies and medicinal plants [94, 95], the knowledge and use of innovative plant materials for domestic, medical, and industrial purposes are anticipated to increase. Moreover, India's economic growth depends on the prospective growth of botanical medicine. Priority should be given to ensuring sufficient quality assurance for herbal medication in order for consumers to purchase pure products. Herbal medicine's acceptance appeal is a significant advancement toward accomplishing the goal of delivering high-quality and affordable healthcare to people all around the world.

Demand for herbal medicines and supplements has skyrocketed with growing concern over the world's dwindling medical plant resources. India is in a privileged position regarding the variety of its climatic conditions and the profusion of its medicinal plant diversity. There is an extensive background to the Indian perspective on herbal medicines, which might be a source for future pharmaceutical innovations. According to reports, the Western Himalayan region is home to the medicinal plants that created 50% of British Pharmacopoeia's listed medications. The genetic variety of plants with therapeutic characteristics is just one example of the 8% percent of the entire world's biodiversity that may be found in India [96]. These facts demonstrate India's immense herbal medicine market potential. It is now critically important to combine modern and traditional knowledge. Following their discovery in plants, the pharmaceutical industry chemically synthesized several modern pharmaceuticals. Rare plants on the Indian peninsula are in danger of extinction due to excessive harvesting; saving them would benefit from the isolation, identification, and concomitant chemical response of therapeutically effective compounds derived from plant components. Growing economically valuable medicinal plants on wastelands can be an essential alternate income generator in rural areas. In Northeastern India, where numerous medicinal plants are produced, the socioeconomic status of the rural population has changed [97]. The government should motivate private gardeners to cultivate medicinal plants to ensure a constant supply. New jobs and prospects for employment would be created, contributing to economic sustainability in rural areas. For this sector to reach its full potential, we need feasible policies with a well-defined economic vision and sound planning techniques [98]. Today, encouraging the application of traditional herbal remedies is essential, as is investigating the integration of these treatments into context of public health. This potential has already been reached because India is home to abundant medicinal plants in great demand on the international market. If India wants the herbal medicine business to grow sustainably, it must fund studies into biological products, agrotechnology, standardized procedures, and quality control. To accomplish this, it is essential to comprehend the socioeconomic context and policies that promote research. Medicinal plant biodiversity must be safeguarded, and biopiracy must be stopped if this industry is to continue expanding. To demonstrate the scientific efficacy of our irreplaceable plant resources through exhaustive phytochemical, biological, and pharmacological research, collecting and documenting the extant traditional knowledge on them is necessary.

5.4 Bioeconomics

The green economy has evolved as a wholly novel idea in which industrial sectors have been forced to rethink their objectives, which are characterized by longevity and long-term viability, and focus growth on identifying new sustainable solutions [99]. In order to generate the components essential for human sustenance, this method entails using resources that can be renewed from the land and water, such as forests, crops, and microbes. It stands out as a suggestion for politicians and policymakers. Still, it has since garnered support from the general public, the research and development community, the financial community, and business leaders. The bioeconomy is more than just a branch of science or a particular industry; it is the framework for bringing together all of the significant businesses and endeavors built on bio-based products in a cutting-edge environment. Over 50 countries globally, including both developed and developing economies, now have bioeconomy strategies and initiatives in place. With the emergence of the bioeconomy, herbal medicine is becoming increasingly important to many forest-based occupations. In tropical regions, a bioeconomy transition has the potential to improve social engagement, protect and restore habitats, increase biodiversity knowledge, valorize livelihoods, and ultimately move away from the commodification of nature [99, 100]. Because it puts more pressure on natural resources and could result in overexploitation, the bioeconomy idea has been questioned until now. The emerging bioeconomy paradigm has also sparked the discussion over trade-offs between energy, food, and land. Furthermore, the bioeconomy idea has placed a strong emphasis on taking natural resources out of the environment, undermining significant benefits that ecosystems offer humans [100]. Such value chains must be assessed in order to make sure that over-exploitation is prevented and that local producers benefit from all activities, from cultivation to distribution.

5.5 Studies on Herbal Treatments

The plethora of scholarly articles that have recently been published strongly suggests that there is a growing concern among pharmacologists, microbiologists, and biochemists that botanists and natural product scientists are studying potential uses for medications for the management of multiple diseases [101, 102]. Most of these exercises focus on the isolation, purification, methodology for bioanalysis, and description of phytochemicals' pharmacological principles. In addition, research into herbal medicine aims to describe their molecular structures, mode of action, and potential toxicity [103]. In the twenty-first century, a paradigm shift is occurring in favor of standardizing herbal drugs for therapeutic purposes, resulting from search with corroborating data on herbal therapy, which numerous studies have supported and confirmed their effectiveness [104]. Concerns regarding their safety are: Ingestion of particular natural remedies has also been examined and approved using both in vivo and in vitro models [105].

Phytochemical and pharmacological studies of plants for therapeutic purposes were actively investigated throughout the past decade and conducted in universities and learning centers. Researchers are exerting considerable effort to identify and separate the bioactive chemical components, as well as to substantiate the efficacy and safety claims made about them [106]. Countless randomized clinical studies based on empirical proof have yielded favorable results in most natural remedies [107]. Understanding the mechanism of operation of plant bioactive ingredients has been facilitated by omics approaches, which has opened the way for the development and standardization of several herbal medications [108]. To contend successfully in biomedical science, it is crucial to remember that novel concepts and recent developments in the research field of herbal medicine have significantly impacted herbal remedies.

6 Recommendations for Expanding the Medicinal Plants Marketplace

An individual strategy for promoting commercial production, technological advancement, and a growth in medicinal herb exports is needed to take advantage of the current global interest in medicinal plants of Indian origin. Convergent efforts across all stages (including study findings, production, acquiring, processing, storage, manufacturing, and marketing) and an appropriate legislative framework are required to advance the medicinal plant industry. This evaluation identified several problems with the medicinal plant–based venture as well as potential solutions.

The choosing of cultivable medicinal plant species is an essential initial stage in the growth of the medicinal plant industry. The main reason for cultivating a particular medicinal plant variety is its economic viability. Additionally, according to National Planning Commission and National Medicinal Plant Boards (NMPBs), cultivation should be prioritized for rare species that cannot be eradicated from the wild because they tend to be expensive, in high demand, and are in scarce supply.

A medicinal plant may not be worth the cost to grow if it is abundant and easy to collect in the wild. Consequently, it is crucial to encourage the widespread cultivation of uncommon native varieties. Culturing medicinal plant species must be postponed until a reliable cultivation technique for the respective species is available. Numerous medicinal plant species can be grown in their natural niches in the varied climate and soil conditions of Northern India. To advance the medicinal plant industry, the following steps are required:

1. Documentation of traditional uses of medicinal plants
2. Certification of raw materials for quality control to be used in production
3. Build better agrotechnology for high-value medicinal plants
4. Indigenous systems of law should be formally recognized and protected
5. Develop a transparent procedure for issuing licenses to grow crops within a given time frame
6. Improve the methods of gathering and preparing medicinal herbs by doing continuing research and teaching

These initiatives could help rural and disadvantaged communities diversify their revenue sources and reduce their dependence on renewable resource bases [55, 59, 109, 110]. Agricultural funding organizations might help the medicinal plant industry thrive and research institutions support medicinal plant cultivators and producers by enhancing their comprehension of cultivation techniques [111]. Several variables, such as awareness among farmers and interest, beneficial policies by the government, secure and open markets, profitable prices of products, access to foundational and practical techniques, and the availability of skilled labor influence the successful cultivation of medicinal plants [112]. All of the current medical plant research must be distributed through a networked communication framework. Numerous herbs are used to treat diseases, but documentation is inadequate due to poor communication and infrequent application. Documenting the indigenous services of obscure and less-known medicinal plants is essential in order to communicate knowledge regarding their therapeutic value and relieve pressure on overfarmed species.

7 Conclusion

Herbal medicine is gaining popularity among the general population since it often does not produce any adverse reactions, has a lasting therapeutic impact on human wellness, and is frequently cost-effective even though the treatment typically lasts for a more extended period. If the information on clinical confirmation is prepared scientifically, it will not be difficult to coexist with modern medicine. The vast majority of the unprocessed ingredients (medicinal plants) utilized in the production of herbal therapeutic goods are taken from forests, which are the only environments in which these plants may thrive as a natural component of the flora. Producers of herbal medicines are required to develop the essential medicinal ingredients methodically so that they can guarantee the authenticity of the substances employed. Firms can additionally enter into buyback arrangements with the growers, in which they trade confirmed seeds and the relevant technology for a percentage of the production, and then purchase the full crop after it has been harvested. Before developing new products, the herbal business has to ensure that wild-harvested herbs are appropriately recognized and have those plants defined, maybe based on the amount of their active elements. If such provisions aren't in place, the description should be centered on the concentration level of the herbs' active ingredients. Using current biotechnological methods, the relevant government agencies ought to organize protection actions for known medicinal plants that can take place either in situ or ex situ.

During the production of medicine, "good manufacturing practices," also known as "GMP," must be utilized to guarantee the product's consistency and high quality. Regardless of the fact that these aspects have been defined in the past years via therapeutic experience, it is necessary to create clinical evidence regarding the security and efficacy of traditional herbal medicines to appeal to the contemporary mind. This must be done to win over the modern mind.

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Part IV

Quality Assessment and Research Need of Herbal Medicine

Trace Metals Contamination of Herbal Remedies

48

Health Implications and Quality Control Strategies

Sylvester Chibueze Izah, Odangowei Inetiminebi Ogidi, Adams Ovie Iyiola, and Chioma Peggy Obasi

Contents

Abstract

Trace elements are ubiquitous chemical agents. They are basically classified into two categories: vital and non-vital. The vital group has a beneficial role in biological systems at certain concentrations, while the non-vital group has no known direct biological function; hence, even a low concentration of it shows

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toxicity. Trace elements such as copper, zinc, magnesium, cobalt, lead, cadmium, chromium, arsenic, nickel, mercury, and iron have been detected in herbal medicine in different parts of the world. The trace metals enter the herbal medicine through the plant materials that are used in the formulation of the medicine; this could be due to the fact that plants have the tendency to bioaccumulate them in the soil. Furthermore, it could be from other materials, such as water used (for liquid herbal medicine), ingredients used in packaging them, etc. The concentration of these trace metals sometimes exceeds the tolerable level. Thereby posing a potential risk for the dysfunction of vital organs and tissues such as the skin, liver, kidney, and the various systems such as the cardiovascular, reproductive, digestive, respiratory, and nervous systems in the human body. If the disorder is not properly managed, it could lead to a chronic condition and even death. The trace contamination of herbal medicines can be avoided through quality assurance measures such as good manufacturing practices and good agricultural and collection practices for herbal medicines. Furthermore, there is a need for standardization and quality analysis at every stage of the value chain of herbal medicine, starting from the cultivation or collection of the herbs to the utilization of the formulated herbal medicine in the treatment of disease conditions.

Keywords

Trace element · Health implications · Quality control · Health hazards · Chemical contaminants

1 Introduction

Trace metals are metalloids with a particular density of greater than 5 g/cm^3 having a detrimental effect on both the environment and living beings [1, 2]. Trace elements are usually characterized by high electrical conductivity, malleability, and luster and have the ability to lose their electrons to form cations. They mostly naturally occur in the crust of the earth and differ in composition between different sites, resulting in spatial variances in concentrations within a vicinity.

Trace elements have been widely reported in different environmental matrices including air [3, 4], soil [5–10], sediment [11, 12], water [1, 13–17], foods such as fishes [18–25], palm oil [26], cow meat [27], palm wine [28], tomatoes [29], vegetables [30], *Saccharomyces cerevisiae* biomass [31], waste water [32–34], spices and culinary condiments [35], and herbal medicine [36–38].

The trace metals are basically divided into two categories: essential and nonessential. The essential trace metals play a significant role in enzyme synthesis in living cells, while the nonessential metals do not have any direct biological functions. Trace metals have a permissible level of effect on human health that will cause harm when exceeded. The high level of trace metals in human health is dangerous and could result in illness, abortion, preterm labor, and mental health retardation, especially in children. It can also cause high blood pressure, fatigue, kidney problems, and brain problems in adults [1]. The concentration of trace metals can be influenced by production factors such as fertilizer, pesticide, and the food processing industry, as well as the oil and gas sector [39], mining, industrial activities, urban runoff, etc. Also, natural activities like soil erosion and the natural weathering of the earth's crust are sources of trace metals in the environment.

The accumulation of trace metals in the body has many adverse health effects if allowed to continue for a long period of time. Trace metal exposure continues and is increasing in many parts of the world. Trace metals have significant environmental effects as pollutants and toxicity problems of increasing significance for ecological, nutritional reasons [40, 41]. Some trace metals commonly found in waste water include arsenic, cadmium, chromium, copper, lead, nickel, and zinc. These metals pose risks to the environment, which in turn affects human health.

Herbal medicine is an important remedy for diseases. Approximately 80% of the global population relies on herbal medicine for the treatment of diseases, especially among individuals residing in rural areas in many parts of the developing world [42, 43]. Herbal medicines are produced using plants as active ingredients [36–38]. Thus, the use of plant parts or the combination of different plant parts to form a medicine is increasing. This herbal medicine is basically formulated in powder or liquid form using parts of medicinal plants that haven proven to possess therapeutic potentials through research and development [44–67]. The focus of this chapter is to assess the level of trace metals in herbal medicine, the composition of herbal medicine used in the treatment of human diseases, and the human health implications of trace metals in herbal medicine. The chapter concludes by highlighting the quality control strategies for trace metals in herbal medicine.

2 Overview of Contaminants of Herbal Medicine

Herbal products are usually of plant origin. The use of medicinal plants is one of the oldest methods of treating illness and is mostly used in healthcare in low- and middle-income countries around the world. Its use has increased in developed countries because it is assumed to be harmless because it is derived from natural products. Presently, safety has become a major concern in the world. Most contaminants, such as metals and radionuclides, occur naturally in the atmosphere and on the ground. Some may occur from human activities in the environment, such as chemical factories, which later bioaccumulate in medicinal plants, which are active ingredients in the formulation of herbal medicine. Some contaminants may arise from the conditions in which the medicinal plant is cultivated, treatment during postharvest, or the manufacturing of products from the medicinal plants, such as residues from pesticides and fertilizers. Despite the perceived idea that medicinal herbs are safe, their compounds can interact with synthetic drugs, and some have side effects [68, 69].

With the increased usage of herbal medicine by humans and the expansion of the markets for herbal products globally, the issue of safety has become a major focus for both the health and public sectors. Countries around the world have begun regulating their herbal products based on various standards accrued to them. The World Health

Organization [70] developed some guidelines and strategies for herbal medicines focused on promoting the safety, efficacy, and quality of herbal products and medicines. This move is apt because the quality of herbal medicines directly impacts their efficacy and safety, as the most important step is controlling the quality of medicinal plants and the materials produced from herbs. This area is complex because it cuts across different areas ranging from the condition of the environment to agricultural practices during the propagation of herbal plants.

Many contaminants and residues that have the ability to cause serious harm to humans after the consumption of trace metals have been reported [71, 72]. The majority of these contaminants occur naturally in the environment, such as radionuclides, bacteria, or toxic metals. Some may be caused by the use of some materials in the past or present that leach hazardous substances that pollute the environment and medicinal plants. Examples of such are the factories' emissions and pesticide residues. It has been reported in recent research that herbal plants can absorb trace metals during growth [73]; this implies that there is a potential risk and danger to the health and well-being of human beings. These dangers can be reduced by ensuring the herbal medicines that contain such hazardous substances and contaminants do not get to the open market. The quality of the herbal plants and products must be checked optimally before reaching the public market.

The contamination of herbal medicine is a threat that affects humans globally, especially at threshold concentrations. The uptake by plants and animals and the accumulation in the food chain are threats to animal and human health [36, 74]. The contaminants in herbal medicine can be classified into biological, chemical, physical, and radioactive contaminants (Fig. 1). Many of these contaminants can cause several disorders, affecting the internal organs and causing various types of cancer

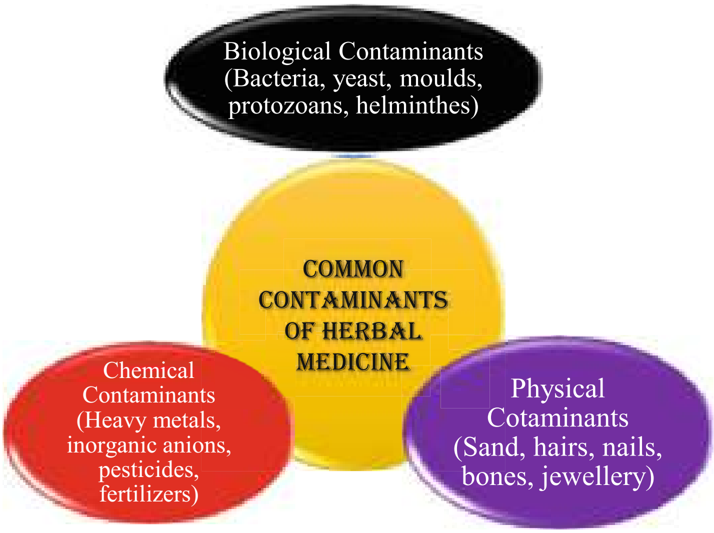

Fig. 1 Common types of contaminants of herbal medicine

depending on their toxicity level. WHO [70] reported some hazardous residues that may be present in herbal medicine, including organic solvent residues such as acetone, methanol, ethanol, and butanol, and residues from agrochemicals such as chemical fertilizers and pesticides. Some of the residues are unavoidable contaminants. For example, in most agricultural practices, the use of pesticides is increasing in order to kill, repel, or mitigate pests [75–87]. Some commonly used pesticides include fungicides, insecticides, and herbicides. Many chemical-based pesticides have the ability to persist in the environment for a long period of time. Therefore, some of the residues of the pesticides can enter the herbal medicine through the water, soil, air, and handling stages. Furthermore, traces of pesticide residues in medicine plants can be detected at different stages, including the plant itself, the production stage, and even finished products.

2.1 Common Types of Contaminants of Herbal Medicine

2.1.1 Biological Contamination

Biological contaminants are mainly life forms that can contaminate the quality of herbal medicine. The life forms can be classified into microorganisms (fungi, thus molds, yeasts, bacteria, and even viruses), parasites (protozoans and helminthes), plant residues, and animals (anthropods). Some of these contaminants, such as molds, can produce mycotoxins that are toxic to humans [88–90]. Some secondary metabolites are toxic to humans, and limits have been set for their levels in food and spices by European legislation. Limits have also been set for some medicinal herbs by the European Pharmacopoeia. The development of mycotoxin from mold depends on the following factors:

- The substrate on which the mold is growing
- The humidity of the environment
- The genetic predisposition of mold to produce mycotoxin
- The carbon dioxide to oxygen ratio
- The presence of other microbial species that are competitive and the presence of fungicides

2.1.2 Chemical Contamination

Chemical contaminants include metallic and nonmetallic toxicants such as nitrate, nitrite, arsenic, copper, mercury, cadmium, chromium, and lead; plant residues; fumigants; agrochemical residues; chemical-based pesticides; and fertilizers. This contamination may be caused by various sources; examples of such are the discharge of contaminated emissions from factories, contaminated water that runs off into rivers, lakes, and water bodies, the soil composition, and fertilizer composition.

2.1.3 Physical Contaminants

Physical contaminants are substances that can be seen with the naked eye, and they can contaminate herbal medicine. Some of the potential physical constituents of herbal medicine include sand, hair, nails, bones, jewelry, etc. The presence of foreign objects or physical contaminants may affect consumers.

3 Global Trace Metal Concentrations in Herbal Medicine

Many countries around the world, including India, Malaysia, Nigeria, the United Arab Emirates, Brazil, Ethiopia, Turkey, Ghana, China, Pakistan, Bangladesh, Iran, Poland, Zimbabwe, South Africa, Uganda, and the Sudan, have reported finding trace metals (copper, zinc, manganese, cobalt, lead, cadmium, chromium, nickel, mercury, and iron) in various herbal medicines (liquid and powder) (Table 1). The health, safety, and quality of the herbal product are among the factors that determine the global standard and regulatory criteria for the concentration of trace metals in herbal medicine. However, these legal specifications differ between nations [91]. The Italian Pharmacopoeia (FUI), the United States Pharmacopoeia (USP), and the European Pharmacopoeia (Ph. Eur.) are only a few examples of international laws on medicinal plants and products. Additionally, administrate frameworks have been developed to control the caliber of plant-based goods. Many countries have their own administrative procedures for herbal medicine or traditional medicine; however, China leads publications on medicinal plant analysis and number of inclusions in their Pharmacopoeia [92]. Before 1998, just 14 member nations, according to WHO [93], had laws governing herbal goods; by 2003, that number had risen to 53 (37%). Guidelines have been devised expressly to ensure that levels of trace metal concentration in medicinal and plant-based goods are within appropriate ranges in nations including Malaysia, Singapore, Canada, China, and Malaysia (Table 3). Although the maximum levels that are acceptable for Cadmium and Lead in raw materials are 0.3 mg/kg and 10 mg/kg, respectively [94], WHO has yet to publicize and establish these amounts.

The concentration of the reported trace metals in herbal medicine often exceeds the allowable level recommended by some countries such as Canada, China, Malaysia, Singapore, Thailand, the Republic of Korea, and the World Health Organization (Table 2). Basically, the presence of these trace metals in the herbal medicine is as a result of the traces of the metals in the plants used as active ingredients in the formulation of the herbal medicine, the water used in the formulation (in the case of liquid herbal medicine), processing equipment, and the environment.

4 Composition of Herbal Medicine Used in the Treatment of Human Diseases that Trace Metals Have Been Detected

Trace metals have been found in some herbal medicine used in the treatment of diseases conditions (Table 3), including those that cause piles, waist pains, sexual weakness, constipation, removal of excess phlegm, stomach pains, improvements in eyesight, menstrual pains, headache, infertility, body pains, high fever and toothache, menstrual pains, anemia, lumbago, dyspepsia, purifies blood, enhances

Table 1 Trace metals concentrations (ppm) in herbal medicine

Cu	Zn	Mn	Co	Pb	Cd	Cr	As	Ni	Hg	Fe	Types of herbal medicine (Liquid/Powdered)	Study locations and country	References
–	–	–	–	0.88	0.96	7.5	0.92	–	0.09	–	Powder	Dubai/UAE	[95]
9.1	–	3.9	–	0.13	0.08	1.67	1.3	1.21	0.002	–	Liquid	Kumasi/Ghana	[96]
–	–	–	–	75.0	45.8	–	–	–	–	–	Liquid	Benin-city/Nigeria	[97]
0.009	0.09	–	–	0.036	0.003	–	–	–	–	3.3	Liquid	Accra/Ghana	[98]
–	–	–	–	17.6	13.7	–	0.001	–	0.001	–	Powder	Kumasi/Ghana	[99]
7.7	21.2	–	–	55.2	–	–	–	0.80	–	–	Powder	Amhara/Ethiopia	[100]
16.2	24.5	27.5	8.5	3.25	2.01	5.3	–	3.5	–	–	Powder	Chittagong/Bangladesh	[101]
–	–	–	–	1.32	0.08	3.08	0.63	1.12	0.06	–	Powder	West Bengal/India	[102]
–	–	–	–	3.6	0.61	–	–	–	0.1	–	Powder	Sao Paulo/Brazil	[103]
41.6	146	–	–	23.5	1.11	–	–	–	–	821.0	Powder	Dubai/UAE	[104]
24	1.30	–	–	19.0	–	–	0.46	0.14	–	33.6	Powder	Mashonaland/Zimbabwe	[105]
15.8	81.4	208	–	481	187	3.6	–	–	40	2105	Powder	Khartoum/Sudan	[106]
1.1	–	0.7	–	–	0.003	–	–	–	–	–	Powder	Portharcout/Nigeria	[107]
0.03	–	0.35	–	–	–	–	–	–	–	42.7	Liquid	Ilorin/Nigeria	[108]
–	–	–	–	8.45	0.45	–	0.91	–	1.24	–	Liquid	Chennai/India	[109]
4.10	–	–	0.92	1.7	0.050	0.68	1.70	–	–	123	Powder	Campo/Brazil	[110]
–	7.1	6.9	–	18.2	–	–	16.7	–	76.2	13.6	Liquid	Abbottabad/Pakistan	[111]
0.60	0.41	–	–	–	–	–	–	–	–	–	Liquid	Lagos/Nigeria	[112]
66.9	101.6	17.6	–	–	–	–	–	–	–	–	Liquid	Ebonyi/Nigeria	[113]
42.5	281.9	849.5	–	14.6	0.32	41.6	–	9.96	–	1276.7	Powder	East Mediterranean/Turkey	[114]

(continued)

Table 1 (continued)

Cu	Zn	Mn	Co	Pb	Cd	Cr	As	Ni	Hg	Fe	Types of herbal medicine (Liquid/ Powdered)	Study locations and country	References
9.1	77.1	90.0	–	0.78	0.34	–	–	4.00	–	58.9	Powder	Karachi/Pakistan	[115]
–	–	73.5	–	–	–	–	–	–	–	339.3	Power	Eskisehir/Turkey	[116]
208.1	45.0	80.9	–	24.4	–	–	–	–	–	0.36	Powder	Enugu/Nigeria	[117]
11.1	–	–	–	3.2	0.44	–	1.0	–	0.1	–	Powder	Anhui/China	[118]
–	–	–	0.44	42.3	–	2.5	42.3	4.5	3.33	–	Powder	Tamil Nadu/India	[119]
–	–	–	–	52.7	1.75	–	–	–	–	–	Powder	Tehran/Iran	[120]
–	–	–	–	52.7	1.7	–	–	–	–	–	Liquid	Tehran/Iran	[120]
0.20	2.71	–	–	0.66	0.81	–	–	–	–	3.47	Liquid	Cape town/South Africa	[121]
–	0.22	–	–	0.1	0.03	0.01	–	0.01	0.06	–	Powder	Owerri/Nigeria	[122]
–	0.53	–	–	0.67	0.02	–	–	–	–	4.66	Powder	Kumasi/Ghana	[123]
–	19.6	–	3.79	15.5	–	12.69	–	10.9	–	104.5	Liquid	Wukari/Nigeria	[124]
9.2	15.2	–	–	–	5.65	–	–	–	–	18.2	Liquid	Abakaliki/Nigeria	[125]
–	–	–	–	0.92	0.34	–	–	–	–	–	Liquid	Enugu/Nigeria	[126]

4.14	38.9	–	–	33.8	–	–	–	54.0	–	257	Powder	Lagos/Nigeria	[127]
–	156.00	–	–	–	–	–	–	–	–	–	Liquid	Mumbai/India	[128]
–	–	–	–	2.26	–	–	–	–	–	–	Powder	Kerala/India	[129]
9.9	–	–	–	2.5	0.02	–	–	–	–	–	Liquid	Tehran/India	[130]
0.18	1.43	–	–	0.08	0.01	0.43	–	–	–	–	Powder	Assam/India	[131]
12.1	253.8	346.4	–	46.4	23.8	154.2	–	56.3	–	2731.8	Liquid	Swabi, Pakistan	[132]
7.0	30.0	61.3	3.6	–	–	–	–	12.9	–	248	Powder	Kano/Nigeria	[133]
–	–	–	–	12.6	1.66	29.5	0.31	15.1	0.75	–	Powder	Pimpri Pune/India	[134]
–	–	–	–	1.2	0.02	1.9	0.46	0.88	0.08	–	Powder	West Bengal/India	[135]
3.0	5.0	18.5	–	0.6	0.1	0.8	–	2.2	–	20.0	Liquid	Gdansk/Poland	[136]
0.71	1.5	15.0	–	1.0	0.39	–	–	–	–	3.1	Powder	Kualalumpu/Malaysia	[137]
4.3	0.43	2.3	–	0.009	0.05	–	0.074	0.005	0.011	25.1	Liquid	Kumasi/Ghana	[138]
0.27	3.6	44.9	–	1.1	0.10	0.15	–	0.34	–	18.4	Powder	Sokoto/Nigeria	[139]
0.29	0.09	0.18	–	0.09	–	7.16	0.004	1.0	0.002	8.47	Liquid	Ekiti/Nigeria	[140]
–	–	–	–	2.0	–	–	–	0.95	–	–	Liquid	Kyambogo/Uganda	[141]
–	–	–	–	13.8	0.42	8.7	–	7.5	2.46	–	Powder	Mumbai/India	[142]
22.2	78.1	–	–	–	–	–	–	54.5		1873	Powder	Peninsular/Malaysia	[143]

Table 2 Trace metal concentration limits in herbal medicinal products in some countries

S/N	Country	Nature of product	As, ppm	Pb, ppm	Cd, ppm	Cr, ppm	Hg, ppm	Cu, ppm	Total toxic metal as Pb, ppm
Concentration for herbal medicines									
1	Canada	Herbal materials (raw)	5.0	10.0	0.30	2.0	0.20	–	–
		Herbal products (finished)[a]	0.01	0.02	0.06	0.02	0.02	–	–
2	China	Herbal materials	2.0	10.0	1.0	–	0.05	–	20.0
3	Malaysia	Herbal products (finished)[b]	5.0	10.0	–	–	0.50	–	–
4	Singapore	Herbal products (finished)	5.0	20.0	–	–	0.50	150	–
5	Thailand	Herbal materials and finished products	4.0	10.0	0.30	–	–	–	–
6	Korea Republic	Herbal materials	–	–	–	–	–	–	30.0
7	WHO(2005) recommendations[b]	–	–	10.0	0.30	–	–	–	–
Concentration for herbal products									
1	National Sanitation Foundation draft proposal (Raw Dietary Supplement)		5.0	10.0	0.30	2.0	–	–	–
2	National Sanitation Foundation draft proposal (Finished Dietary Supplement)[a]		0.01	0.02	0.006	0.02	0.02	–	–

Source: Adapted from WHO [70]
Note: [a]Expressed as mg/day; [b]Expressed as mg/kg

Table 3 Composition of herbal medicine used in the treatment of human diseases that trace metals have been detected

Name of the herbal medicine	Composition	Indications	List the trace metals that have been reported	Types of herbal medicine (liquid/ powdered)	Study locations and country	References
Not disclosed	*Carica papaya*, Khaya	Piles, waist pains, sexual weakness, constipation, removal of excess phlegm, stomach pains, improves eyesight, menstrual pains, headache, infertility, body pains, high fever and toothache	Iron, zinc and lead	Liquid	Accra/Ghana	[98]
Not disclosed	Combination of several medicinal plant parts and honey	Piles, menstrual pains, anemia, abdominal pain, lumbago, dyspepsia, purifies blood, enhances circulation	Zinc and lead	Liquid	Accra/Ghana	[98]
Not disclosed	Combination of several medicinal plant parts	Relief of piles, constipation, waist pains, headache, fever, sexual weakness, fibroid, hernia	Iron, zinc, cadmium, and lead	Liquid	Accra/Ghana	[98]
Not disclosed	Combination of several medicinal plant parts	Ordinary piles, bleeding piles, eyes itching, sexual weaknesses, menstrual pains, constipation, rheumatism	Iron, zinc, cadmium, and lead	Liquid	Accra/Ghana	[98]
Not disclosed	Combination of several medicinal plant parts	Rheumatism, anemia, menstrual disorder, loss of appetite, tiredness, general disability, piles, fevers	Zinc and lead	Liquid	Accra/Ghana	[98]
Not disclosed	Combination of several medicinal plant parts	Rheumatism, anemia, menstrual disorder, loss of appetite, tiredness, general disability, piles, fevers	Iron, zinc, cadmium, and lead	Liquid	Accra/Ghana	[98]

(continued)

Table 3 (continued)

Name of the herbal medicine	Composition	Indications	List the trace metals that have been reported	Types of herbal medicine (liquid/ powdered)	Study locations and country	References
Not disclosed	Combination of several medicinal plant parts	Constipation, kidney-related problems, asthma, sexual weakness, menstrual imbalances, removes phlegm, appetizer	Iron, zinc, cadmium, and lead	Liquid	Accra/Ghana	[98]
Bobofen	Combination of several medicinal plant parts at varying concentration	Spastic gastric and intestinal ailments, colic, flatulence, adjuvant treatment of inflammatory states of the alimentary tract	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Bronchial	Combination of several medicinal plant parts at varying concentration	Adjuvant in upper respiratory tracts infections, particularly in cases of dry cough and in common colds with fever	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Cholagoga II	Combination of several medicinal plant parts at varying concentration	Metabolic disorders and insufficient secretion of bile, adjuvant in biliary dyskinesia	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Circulosan	Combination of several medicinal plant parts at varying concentration	Heart ailments resulting from neurosis and in climacterium period, adjuvant in mild heart rhythm disorders, and impaired heart efficiency	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Gastrosan	Combination of several medicinal plant parts at varying concentration	Alimentary tract inflammation, chronic gastric and duodenal ulcer disease, adjuvant in secretory and motor disorders of the alimentary tract	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]

Diabetosan	Combination of several medicinal plant parts at varying concentration	In prediabetes states, in latent diabetes, in an early stage of insulin-independent diabetes, adjuvant in diabetes treated with anti-diabetes drugs	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Infektoten	Combination of several medicinal plant parts at varying concentration	Common colds, adjuvant in influenza, and infections of upper respiratory tract with fever	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Nervosan	Combination of several medicinal plants parts at varying concentration	Nervous tension, anxiety, mild neurosis or insomnia, adjuvant in nervous disorders during puberty and climacterium periods	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Nervinum	Combination of several medicinal plant parts at varying concentration	Discomfort caused by nervousness, mild tension, and anxiety, difficulties in falling asleep	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Normosan	Combination of several medicinal plant parts at varying concentration	Constipations, adjuvant in metabolic disorders and meteorism	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Pyrosan	Combination of several medicinal plant parts at varying concentration	Diaphoretic in fever states, common colds, rheumatic ailments, muscle pains, adjuvant in a flu	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Septosan	Combination of several medicinal plant parts at varying concentration	Acute and chronic inflammations of the oral cavity and throat mucosa, adjuvant in gingivitis and tonsillitis, angina, and in purulent processes of the oral cavity	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]
Urosan	Combination of several medicinal plant parts at varying concentration	An agent increasing secretion of urine in renal calculosis and urinary sand, adjuvant in	Iron, zinc, cadmium, manganese, nickel, chromium, cadmium, and lead	Liquid	Gdansk/ Poland	[136]

(continued)

Table 3 (continued)

Name of the herbal medicine	Composition	Indications	List the trace metals that have been reported	Types of herbal medicine (liquid/ powdered)	Study locations and country	References
		inflammations of kidneys and urinary bladder				
Gokshuradi churna	Combination of several medicinal plant parts at varying concentration	Diuretic and cardiac, increases vitality and decrease body weakness, sexual performance	Lead, copper, arsenic and mercury	Powder	Manipal/India	[144]
Surya Sakthi Churna	Not disclosed	Not disclosed	Lead, copper, arsenic and mercury	Powder	Chennai/India	[109]
Anab	Not disclosed	Blood purifier	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Arq-e-badian	Not disclosed	GIT, UTI	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Arq-Mako	Not disclosed	Anti-inflammatory	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Asthmina	Not disclosed	Anti-asthmatic	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Banafsha	Not disclosed	RTI	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Bazori	Not disclosed	Diuretics	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Colgesic	Not disclosed	Anti-spasmodic	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Colic	Not disclosed	GIT disorders	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]

Demaghoon	Not disclosed	Enhance memory	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Digical	Not disclosed	GIT disorders	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Dinar	Not disclosed	Liver disorders	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Hazmeena tab	Not disclosed	Digestive	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Powder	Swabi, Pakistan	[132]
Injabar	Not disclosed	GIT disorders	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Mansofar tab	Not disclosed	Menstrual disorder	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Powder	Swabi, Pakistan	[132]
Masturin	Not disclosed	Menstrual disorder	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Pathri tore tab	Not disclosed	Kidney stone	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Powder	Swabi, Pakistan	[132]
Sposmodin	Not disclosed	Blood purifier	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Sharbate foulad	Not disclosed	Tonic	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Sharbate sadar	Not disclosed	RTI	Lead, cadmium, chromium, copper, nickel, zinc and iron	Liquid	Swabi, Pakistan	[132]
Suduri	Not disclosed	Cough	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Supari pak	Not disclosed	Leucorrhea	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Toot siah	Not disclosed	Cough	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]

(continued)

Table 3 (continued)

Name of the herbal medicine	Composition	Indications	List the trace metals that have been reported	Types of herbal medicine (liquid/ powdered)	Study locations and country	References
Urosinal	Not disclosed	UTI	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Y-lin	Not disclosed	Cough	Lead, cadmium, chromium, copper, nickel, zinc, and iron	Liquid	Swabi, Pakistan	[132]
Thymus vulgaris	Not disclosed	Antitussive, expectorant	Cadmium, lead, and copper	Liquid	Tehran/India	[130]
Matricaria recutita	Not disclosed	Peptic ulcer, gastritis	Lead and copper	Liquid	Tehran/India	[130]
Vitex agnus castus	Not disclosed	Menstrual disorder	Lead and copper	Liquid	Tehran/India	[130]
Crataegus oxyacanta	Not disclosed	Heartthrob, hypertension	Lead and copper	Liquid	Tehran/India	[130]
Hypericum perforatum	Not disclosed	Depression, migraine	Lead and copper	Liquid	Tehran/India	[130]
Urtica dioica	Not disclosed	Urinary problems	Lead and copper	Liquid	Tehran/India	[130]
Passiflora incarnate	Not disclosed	Anxiety, insomnia	Lead, cadmium, and copper	Liquid	Tehran/India	[130]
Dr Iguedo Goko cleanser®	Combination of several medicinal plants parts	Urinary disease, fertility inducer, stabilizing menstrual period	Cadmium, mercury, arsenic, lead, zinc, manganese, copper, nickel, chromium, and iron	Liquid	Ekiti/Nigeria	[140]
Not disclosed	Combination of several medicinal plant parts	Malaria, typhoid	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Irregular and painful menstruation	Zinc and copper	Liquid	Lagos/ Nigeria	[112]

Not disclosed	Combination of several medicinal plant parts	Energy booster, general overall well-being	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Diabetes, stomach disorder, pile	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Malaria and typhoid	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	All sexually transmitted diseases. Immune booster	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Irregular menstruation	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Vagina itching, burning and irritation, chronic pile and aids digestion	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Malaria	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Malaria	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Malaria, typhoid	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Pile, dysentery, constipation, diarrhea, irregular menstruation, men turgidity, withdraws protruding rectum, waist and stomachache	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Infertility, menstrual dysfunction, STD, mouth odor, acute stomachache, obesity	Zinc and copper	Liquid	Lagos/ Nigeria	[112]

(continued)

Table 3 (continued)

Name of the herbal medicine	Composition	Indications	List the trace metals that have been reported	Types of herbal medicine (liquid/ powdered)	Study locations and country	References
Not disclosed	Combination of several medicinal plant parts	Stomach disorder, indigestion, peptic/duodenal ulcer, improves blood circulation, eliminates pain, and discomfort	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Pile, hemorrhoid, waist pain, enhances effective digestion, reduces fatigue, improves libido, strengthens the heart and cardiac muscles, reduces blood sugar	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Glaucoma, eye ulcer, cataract	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
Not disclosed	Combination of several medicinal plant parts	Purifies the blood, eliminates toxic material from the blood vessels, reduces accumulated fat and cholesterol level, prevents edema, numbness, and reduces blood sugar	Zinc and copper	Liquid	Lagos/ Nigeria	[112]
C-lax	Combination of several medicinal plant parts	Laxative	Cadmium and lead	Powder	Tehran/Iran	[120]
Carvil	Combination of several medicinal plant parts	Weight loss	Cadmium and lead	Powder	Tehran/Iran	[120]
Garlet	Combination of several medicinal plant parts	Antihypertensive and lipid-lowering agents	Cadmium and lead	Powder	Tehran/Iran	[120]
Slim-quick	Combination of several medicinal plant parts	Appetite suppression Obesity weight loss	Cadmium and lead	Powder	Tehran/Iran	[120]

Galega	Combination of several medicinal plant parts	Anti-diabetic	Cadmium and lead	Powder	Tehran/Iran	[120]
Valiflore	Combination of several medicinal plant parts	Sedative hypnotic	Cadmium and lead	Powder	Tehran/Iran	[120]
Garcine	*Allium sativum*	Anti-hypertensive, anti-hyperlipidemic, antithrombotic, anti-atherogenicity	Cadmium and lead	Powder	Tehran/Iran	[120]
Ginko	*Ginkgo biloba*	Enhance memory	Cadmium and lead	Powder	Tehran/Iran	[120]
Green teadin	Combination of several medicinal plant parts	Lowering cholesterol, weight loss, preventing diabetes, and stroke	Cadmium and lead	Powder	Tehran/Iran	[120]
Aphrodit	Combination of several medicinal plant parts	In potency, premature ejaculation, sexual desire disorder, and low sperm count	Cadmium and lead	Powder	Tehran/Iran	[120]
Anethum	Combination of several medicinal plant parts	Lipid-lowering agents	Cadmium and lead	Powder	Tehran/Iran	[120]
Herbal formulation	Combination of several medicinal plant parts	Treatment of hyperlipidemia	Lead	Powder	Kerala/India	[129]
Herbal tea	Combination of several medicinal plant parts	Sore throat, headache and fatigue	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Ginseng pills	American ginseng	Fatigue and blood pressure	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Ban Kah Chai	Combination of several medicinal plant parts	Stomachache and diarrhea	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Li Chee powder	Combination of several medicinal plant parts	Mild diarrhea	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Green pills	Essential oil and radix	Fever	Copper, iron, nickel and zinc	Powder	Peninsular/ Malaysia	[143]

(continued)

Table 3 (continued)

Name of the herbal medicine	Composition	Indications	List the trace metals that have been reported	Types of herbal medicine (liquid/ powdered)	Study locations and country	References
Bo Ji Wan	*Cordyceps sinensis*	Excessive sputum, cough, and cold	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Chee Ke Wan	Essential oil and radix	Diarrhea and flu	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Chee Suat Tan	Radix and gypsum	Throat ulcer	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Seng Lian cough pills	Radix and essential oil	Cough	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Intamol	Essential oil and radix	Fever	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Guan Ying Chian	Bark of "Guan Ying Chian"	Fatigue and blood pressure	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Pang Nian Jin	Bark of "Pang Nian Jin"	Fatigue and relief cold	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Cold relief pills	Rhizome Belancandae and essential oil	Cold relief	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]

Flu pills	Essential oil	Flu	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Biau Leng San	*Artemisia*, malt, and coptis	Woman stomachache	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Oh Ouh Pau Chee San medicated powder	Mentha herb and *radix angelica*	Improving appetite	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Stomachache powder	Mentha herb and *radix angelica*	Stomachache and flatulence	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Soo Hup Wan tiny pills	Essential oil and rhizome cymbopogon	Indigestion and vomiting	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Tou Seah San (baby)	*Poriacocos* and semen *Plantago asiatica*	Stomachache	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Hung Lien Shang Ching Pien	Essential oil and ginseng	Reducing heat	Copper, iron, nickel, and zinc	Powder	Peninsular/ Malaysia	[143]
Deep roots	Combination of several medicinal plant parts	Male infertility	Zinc, copper, and manganese	Liquid	Abakaliki/ Nigeria	[113]

circulation Intestinal, respiratory, and metabolic disorders, as well as discomfort brought on by nervousness, mild tension, and anxiety, problems falling asleep, constipation, diabetes, ulcers, rheumatic illnesses, body inflammations, and renal issues, can all be treated and managed with the help of cominatin, according to Suchacz and Wesolowski [136]. These conditions include ulcers, diabetes, rheumatoid arthritis, body inflammations, and renal issues. Additionally, according to Kumar et al. [144], some herbal remedies can boost vigor, lessen physical weakness, and enhance sexual performance. The management and treatment of antitussive, expectorant, peptic ulcer, gastritis, menstruation disorder, hypertension, depression, migraine, urinary issues, anxiety, and insomnia are all conditions that certain herbal medication is useful in managing and treating, according to Ravanbakhsh et al. [136]. Malaria, typhoid, painful and irregular periods, diabetes, stomach issues, piles, vaginal itchiness, burning, and irritation, chronic piles that help with digestion, hemorrhoids, and waist pain, according to Idu et al. [112] strengthens the heart and cardiac muscles, boosts libido, lowers fatigue, and increases effective digestion. Herbal medication can be used to control or treat conditions like low blood sugar, infertility, menstruation dysfunction, STDs, mouth odor, acute stomachache, obesity, glaucoma, eye ulcers, and cataracts. The use of herbal medicine in the treatment of urinary illness, as a fertility enhancer, and in stabilizing menstrual cycles was also described by Udom et al. [140]. According to Mousavi et al. [120], certain herbal remedies have been shown to improve memory, lower cholesterol, prevent diabetes, prevent stroke, and be anti-hypertensive, anti-hypertensive, anti-hyperlipidemic, antithrombotic, anti-atherogenicity, and lipid-lowering agents. They also act as laxatives and aid in weight loss. According to Yap et al. [143], male infertility, diarrhea, copious sputum, cough, and cold.

5 Human Health Implications of Trace Metals

Trace metals are fundamentally divided into vital and non-vital groups based on their biological impacts. Iron, zinc, copper, chromium, cobalt, and manganese are typical examples of vital trace metals for human health and may be excellent for the body's metabolism [145, 146]. The non-vital trace metals, including cadmium, lead, and mercury, are regularly found in the environmental matrices and food sources of humans, just like the essential ones. Thus, the typical method of taking trace metals into the body is through the dermal route, inhalation, and, most importantly, ingestion of foods, drugs, or materials containing traces of these metals. However, the organs and tissues of the body may suffer if the concentration surpasses the permissible limit. Some of the typical roles of the essential metals are shown in Table 4.

Life support requires medicine, and herbal medicines contain trace metals that are detrimental to humans. Globally, herbal medicines have been reported to contain traces of toxic metals (Table 1). Trace metals often turn toxic when their concentration exceeds the permissible limit. In many developing nations, herbal medicine is one of the key resources for maintaining a healthy lifestyle. The majority of herbal medicines contain trace metal concentrations above allowable limits.

Table 4 Functions of vital trace metals to the human body

Essential trace metal	Roles	References
Iron	Involved in a number of biochemical and metabolic activities, including the transport of oxygen, the synthesis of deoxyribonucleic acid, the electron transport chain, and the control of cell development and differentiation	[147–149]
	Involved in some physiological processes, as well as the production of iron-containing enzymes and hemoglobin	[150]
Zinc	Vital for the generation of prostaglandins, bone mineralization, healthy thyroid function, taste acuity, blood coagulation, and maintaining sperm quality	[151]
	Vital for the immune system and wound healing	[146, 151, 152]
	It is necessary for connective tissue growth and maintenance, homeostasis, DNA synthesis, RNA transcription, cell division, and cell activation	[151, 152]
	Important structural, co-structural, catalytic, and regulatory capacities in enzyme molecules	[151–154]
	Essential for taste acuity, prostaglandin production, bone mineralization, proper thyroid function, blood clotting, cognitive functions, fetal growth, and sperm production	[146]
	Control of body fluid pH, production of collagen for hair and nail growth, improvement of cerebral development, and enhancement of sexual processes like testosterone secretion and prostate activity	[151, 155]
	Important for healthy organogenesis, neurotransmitter function, thymus development, taste sensibility epithelialization, pancreatic and gastric enzyme secretion, proper spermatogenesis and maturation, and sperm genomic integrity	[146, 156]
Copper	The production of red blood cells (hemoglobin), connective tissue metabolism, and bone growth are all examples of normal iron metabolism	[146, 152, 157]
	Numerous enzymes, including tyrosinase, uricase, and cytochrome oxidase, are involved in cell metabolism	[146]
	Important for several biological operations, such as hepatocyte and brain functioning, mitochondrial respiration, regulating hemoglobin levels, and embryonic growth	[158]
	Essential as a pro-oxidant, antioxidant, and for maintaining the strength of the skin, blood vessels, epithelium, myelin, and the thyroid gland	[152]
Manganese	Part of metalloenzymes and as a need for enzyme activation	[146]
	A need for the metabolism of mucopolysaccharides, fatty acids, and cholesterol	[146, 159]
Cobalt	It regulates the translocation of enzymes like homocysteine methyltransferase during the metabolism of methionine.	[146]
Chromium	Glucose tolerance factor biosynthesis	[146]

Adapted from Izah et al. [1]

The human body can create dangerous free radicals that can damage cells even at the cellular level when the amounts of trace metals in ingestible materials surpass optimal limits. These free radicals can cause diseases like cancer, harm to the immune system and central nervous system, damage to the intestinal tract, and harm to the lungs, kidneys, liver, and other vital organs. Additional acute human health symptoms include skin rashes, vomiting, dizziness, unexpected birth outcomes, and occasionally even death due to the intake of high concentrations of trace minerals by humans. Redox-active metals may also undergo redox cycling events because of their ability to produce reactive radicals like the superoxide anion radical and nitric oxide in biological systems [1]. According to Jomova and Valko [160], high levels of trace metals are linked to a wide range of different disorders, including cancer, cardiovascular disease, diabetes, atherosclerosis, neurological issues, and chronic inflammation. In addition, trace metal toxicity can cause muscle cramps, nausea, lethargy, poor nutrition, vomiting, pain, sweating, migraines, suffocation, speech difficulty, convulsions, anxiety, emotional instability, insomnia, and nausea. There is also a probability of digestive system problems as a result of exposure to trace metal-contaminated food, water [1], and herbal medicine. According to Rowel [161], exposure to trace metals such as lead, mercury, copper, cadmium, and chromium can cause cardiovascular disease, some types of cancer, and developmental issues in children, in addition to neurological effects like memory loss and behavioral changes.

Furthermore, trace metals are harmful to the human body because they can accumulate inside the body and result in deadly problems even in low quantities. These effects might be teratogenic, phytotoxic, carcinogenic, or synergistic. Trace metal concentrations can also harm cell membranes by interfering with cytoplasmic and nuclear functions [1]. According to Oguntona et al. [162], it could take months or even years for the chronic symptoms of trace metal exposure to manifest. Table 5 provides a list of the harmful consequences that trace metals in drinking water have on human health.

Arsenic is another nonessential metals that have been reported in food, water, and herbal medicine. High levels of arsenic exposure may result in cancer, dermatitis, neurological, pulmonary, gastrointestinal, cardiovascular, hematological, and neurological issues (Fig. 2) [163].

6 Quality Control Strategies of Trace Metals in Herbal Medicine

Quality control is the process involved in the standardization of an herbal product, drug, preparation, or substance. Quality control results in the standardization of herbal medicine and products. The authenticity of the herbal product is determined by morphological and microscopic identification, and the quality is determined by the physical and chemical characteristics based on the existing quality standards [187].

Table 5 Pathological effect of trace metal on human health

Metals	Pathological effects/symptoms	References
Lead	Liver and brain damage, infertility in women, and death.	[164, 165]
	Destruction to reproductive system	[165–167]
	Destruction to kidney	[164, 165, 167, 168, 169, 170]
	Destruction to nervous system	[164, 165, 166, 167, 170, 171]
	Birth weight loss and early birth	[172]
	High concentration results in anemia and brain edema	[168]
	Destruction or poisoning of cardiovascular system	[166, 173]
	Hair loss, lung fibrosis, skin allergies, an eczema flare-up, and varying kidney function	[172]
	Destruction to renal system	[166, 174]
	Anorexia, headache, malaise, diarrhea, lead-palsy, encephalopathy, and sleeplessness are among the symptoms of a gastrointestinal and neuromuscular illness	[174]
	Causes anemia, elevated blood pressure, and numbness in the fingers, wrists, or ankles	[164]
	Brain development is hampered, and the immune system is suppressed	[169]
Chromium	The oxidation of blood cells causes hemolysis and renal problems	[171]
	Injuries to the kidneys, liver	[164, 175]
	Circulatory, nervous system, and skin irritation issues	[175]
	Numerous health conditions, such as cancer, rapid breathing, irritation of the nasal mucosa, nose ulcers, runny noses, and breathing issues include asthma, coughing, and shortness of breath	[164]
	Cancer	[170]
Cadmium	Desmineralization of the bones, kidney, and liver damage	[164, 167, 170, 171, 174, 176, 177]
	Breathing issues and internal bleeding	[174]
Cobalt	Raises the likelihood of goiter	[168]
	Nauseousness and vomiting, cardiovascular, thyroid, and optical impairment	[178]
Manganese	Hinder dietary iron absorption, which can lead to iron deficient anemia	[179]
	Neurological condition	[170]
Copper	Abdominal pains, nausea, vomiting, diarrhea, headache, and dizziness	[180]
	Anemia, acne, adrenal insufficiency and hyperactivity, allergies, hair loss, rheumatoid arthritis, cancer, raised cholesterol, depression, diabetes, dyslexia, failure to thrive, exhaustion, fears, bone fractures, headaches, heart attacks, hyperactivity, hypertension, infections, inflammation, kidney and liver dysfunction, panic attacks, strokes, and decay in teeth	[164, 175]

(continued)

Table 5 (continued)

Metals	Pathological effects/symptoms	References
	Gastrointestinal malady	[170]
Zinc	Diarrhea	[181]
Iron	Metabolic and genetic illnesses	[182]
	Siderosis, a benign form of pneumoconiosis	[164]
Mercury	Death and respiratory failure harm the developing fetus and brain. The list of symptoms also includes nervousness, tremors, abnormalities in vision or hearing, memory issues, nausea, vomiting, and diarrhea, as well as elevated blood pressure, skin rashes, and itchy eyes	[163]
	Renal and central nervous system impairment	[164, 170]
	Abnormalities of the digestive system, irritation of the respiratory tract, collapse of the kidneys, and neurotoxicity	[169]
Nickel	Carcinogenic and toxic	[167, 170]
	Digestive issues, cardiovascular illness, pulmonary fibrosis, skin dermatitis, and kidney and lung dysfunction	[183–185]
	Agents that are carcinogenic, genotoxic, genotoxic, reproductively toxic, pulmonary toxic, nephrotoxic, and toxic to the kidneys and liver	[186]

Source: Izah et al. [1]

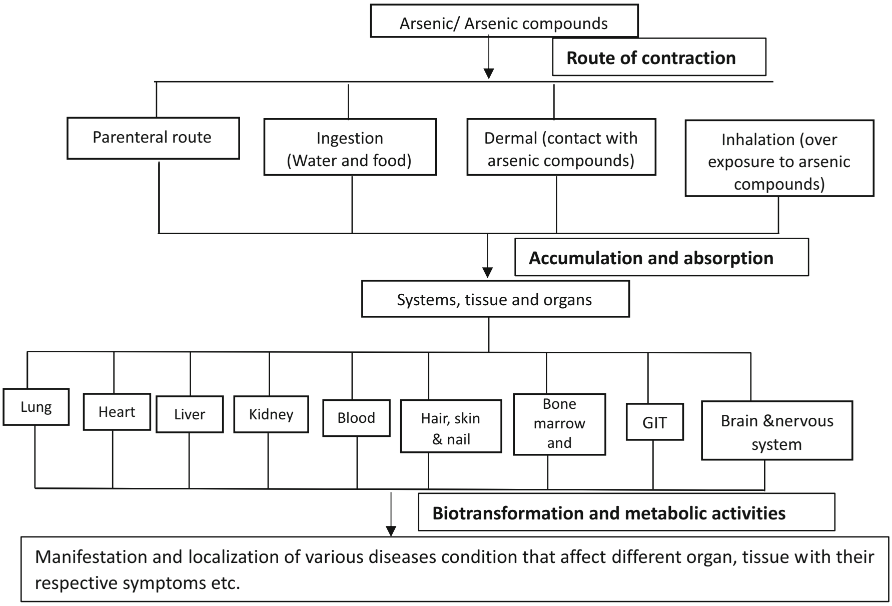

Fig. 2 Possible route of contracting the arsenic -related disease conditions (GIT = gastrointestinal tract). (Source: Izah and Srivastav [163])

The increase in adulteration, substitution, and the inadequacy of skilled personnel are the reasons for substandard herbal medicines. There is a need for advanced quality control processes and suitable standards for herbal medicine practices and use. Standardization is a major approach used in the identification of quality, purity, and confirmation of herbal products. As the primary method of identification, physical, chemical, and biological properties can contribute to the extent of purity of herbs. For the quality preservation of natural herbs and products, quality control is necessary.

Quality is of major concern to humans, and when herbal medicines are consumed, it is important that they address the well-being of humans. As there are various quality standards for pharmaceuticals, it is important that herbal medicine follow suit and be controlled as it is for chemically synthesized medicines [188]. Standardization and investigation of phytochemicals are usually done before the production of herbal medicine [189]. However, in many parts of the world this is not carried out.

The World Health Organization has been up to the task of fighting against contaminated herbal drugs and products and has developed documents and technical guidelines to ensure the quality and safety of herbal medicine, plants, and materials. Examples of such are:

- Guidelines on good agricultural and collection practices for medicinal plants.
- Quality control measures for medicinal plant materials.

The documents have provided explicit information on the determination of chemical, biological, pesticide residues, and radionuclear contaminants that may be present in herbal medicine and medicinal plants. They also provide information on appropriate test methods for contaminants and the regional quality specification standards for contaminants and residues that may be present in herbal plants and herbal medicine. It is also essential that these guidelines be reviewed from time to time in relation to the changing environment and human activities. Medicinal plants and materials that have medicinal potential can be consumed as a dietary supplement, as food, or as functional foods. A survey by World Health Organization [93] on the national policy and regulation of herbal medicines reported that the status, based on regulation that is ascribed mostly to herbal medicine, is as dietary supplements and health foods in over 100 countries. Therefore, more work is needed in the aspect of food safety, especially in connection with herbal medicines. This approach initiated a joint collaboration between traditional medicine and food safety teams with the Food and Agriculture Organization of the United Nations and the World Health Organization Joint Programme: Codex Alimentarius [70].

The processes involved in the registration and regulation of herbal medicine and its use vary from one country to another. In areas where they are regulated, herbal medicines are classified as prescription or nonprescription medicines. The World Health Assembly at its 56th session adopted a resolution on traditional medicine (WHA56.31), which urged member states to ensure safety, quality, and national standards for herbal medicines. It requested support from the Director-General of the World Health Organization in terms of technical information for monitoring and ensuring herbal medicine quality, safety, preparation, and information exchange.

Contamination in herbal medicine should be avoided as much as possible, and this can be achieved through quality assurance measures such as good manufacturing practices and good agricultural and collection practices for herbal medicines. The concentration of these contaminants may also reduce as a result of postharvest processing in herbal extracts and finished medicinal products during the manufacturing process.

To ensure guaranteed safety in the use of herbal medicines, several studies have been carried out in some countries of the world with the aim of safeguarding the concentration of trace metals. Most of these studies used a limited number of herbal samples and categories. Due to the hazardous effects trace metals can cause on the health of humans and the environment, some approaches were carried out by researchers to show that the concentration of trace metals in herbal medicines has a potential risk to human health and requires immediate attention [190]. Some of the approaches are:

- The exposure risk assessment [36].
- The hazard quotient [36].

Studies by Izah et al. [36] on the health risk assessment of trace metals due to the usage of certain herbal medicines could affect the health of humans negatively over a prolonged period of time. With this in mind, it is very important to carry out a risk assessment of trace metals in herbal medicine to ascertain their toxicity level. Many trace metals have no known biological function. Hence, even a low concentration of it suggests toxicity. Therefore, it is vital to check for any potential source of trace metals in the herbal medicine.

Compliance with guidelines on good agricultural and collection practices for medicinal plants and quality control measures for medicinal plant materials is essential in the production of herbal medicines that are of good quality. It is essential that the guidelines be strictly followed, from the cultivation of herbal plants to the sale of the herbal product. Foreign items found in medicinal plants and herbal drugs must be within the limits set by national or international regulations. Examples of such foreign items include insects, animal dungs, and other plant species.

The maximum concentration of trace metals in herbal medicine can be given based on the provisional tolerable intake values, which can vary from country to country. Herbal medicine products are not expected to contribute to the exposure of trace metal contaminants, and it is recommended that the contamination of trace metals be minimized. For global trade, it is essential to harmonize the limits of toxic metals and their standards in herbal medicine.

7 Conclusion

Traces of chemical substances are present everywhere. They fundamentally fall into two categories: vital and non-vital. The vital group has a beneficial impact on biological systems at specified concentrations, in contrast to the non-vital group, which has no recognized direct biological role and is toxic even at low quantities. Many regions of the world have reported finding trace levels of metals in herbal treatments, including copper, zinc, magnesium, cobalt, lead, cadmium, chromium, arsenic, nickel, mercury, and iron. Plant components used in the manufacture of herbal medicines may contain trace metals since plants have a tendency to bioaccumulate these substances in the soil. It could also originate from other substances, such as the chemicals used to package herbal medicines or the water required to create their liquid form. The concentration of certain trace elements sometimes goes over the permissible limit. As a result, there is a postnatal risk for dysfunction of the body's multiple systems, such as the skin, liver, and kidney, as well as the cardiovascular, reproductive, digestive, respiratory, and neurological systems. If the issue is not properly addressed, it could become a chronic sickness and cause death. Trace contamination of herbal drugs can be avoided through quality control measures such as adequate manufacturing practices and acceptable agriculture and gathering practices. Standardization and quality assessment are necessary at every stage of the value chain for herbal medicine, from the collection or cultivation of the herbs through the use of the prepared herbal medication to treat medical ailments.

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Regulations and Policies for Herbal Medicine and Practitioners

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Contents

Abstract

The absence of any form of a globally recognized and standardized regulation and policy for the coordination of the earliest form of medical care continues to hamper the development of this ageless and invaluable medical practice – herbal or traditional medicine. Considering that the availability, value, exploitation, exploration, and exportation of herbal medicine keep growing, this may be the

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best time to promote the safety, quality, and quantity evaluation as well as efficacy of herbal medicine products, practices, and practitioners through globally recognized regulations and integration into mainstream medicine. Existing community, national, and regional guidelines can serve as foundation materials for global policymaking and regulatory parameters as this continues to be the backbone of the calls from the World Health Organization. The advantage of this approach is connected to the prevention of communities, countries, and regions in the Global South (where herbal medicine use is widespread) and elsewhere from perceiving herbal medicine regulations and policies as entirely foreign, colonial, and disconnected from current realities. However, formal structures and institutions would be required to address and understand the transdisciplinary nature, complexities, and epistemology needed to dissect the cultural, clinical, and accessibility nexus of herbal medicine policy and regulation dimensions. The direct benefits of herbal medicine regulations and policy includes an increase in coordination and One Health security while preventing misappropriation. Current loopholes in community-, regional, and national-based herbal medicine regulations and policies are encouraging quack approaches like indiscriminate and unstandardized harvesting and processing of plants with herbal medicine potentials for herbal medicine formulation as well as herbal medicine production, storage, and use in the management and treatment of diverse ailments. Therefore, unified regulations and policies would serve as a reform tool with allowance for regular updates. However, such regulations and policies would first have to overcome the challenges associated with the diverse number of plants used in herbal medicine, their phytochemical constituent, production systems, and utilization patterns.

Keywords

Medical policies · Traditional medicine · Medical regulations · Herbal medicine safety · Herbal products

Abbreviations

%	Percentage
COFEPRIS	Federal Commission for the Protection against Health Risks
COVID-19	Coronavirus disease 2019
FDA	Food and Drug Administration
NHP	Natural Health Products
TCM	Traditional Chinese medicine
TM/CAM	Traditional medicine/complimentary and alternate medicine
USD	US Dollars
WHO	World Health Organization
WTO	World Trade Organization

1 Introduction

Herbal medicine has been used by humans for millennia for the treatment, management, and prevention of diseases as well as for health promotion and quality of life enhancement. The practice and products may be considered cosmopolitan despite only a little influence of globalization and the abandonment due to modern medicine. The use of herbal medicine in modern medicine has increased over time [51, 52, 58, 93]. For instance, around 80% of the global population now rely on herbal medicine for their primary healthcare, while 25% of drugs from contemporary medicine are made from plants [22, 90]. Herbal medicine is part of the heritage of almost all human communities, cultures, and traditions [75]. Herbal medicine has been practiced and utilized as the oldest treatment type in developed and developing nations [41, 60]. The World Health Organization (WHO) classified herbal medicine into three categories: raw plant materials, processed plant materials, and therapeutic herbal products [39, 88]. For many millions of people, herbal medications and traditional practitioners constitute the main source of healthcare and, in some cases, the only option [95]. This is because nature provides humans with food, clothing, housing, and medical needs. Herbal medicine practitioners could tell active plants from harmful ones and useful herbs with positive effects [41]. For nearly 80% of the world's population today, particularly in underdeveloped nations, herbal medicine remains the primary form of healthcare [38, 41, 46, 51, 63, 93]. Additionally, more people in rich countries like Germany, France, Italy, the United Kingdom, Spain, and the United States use herbs instead of regular drugs [19, 20, 90].

The procedures of herbal medicine are the culmination of centuries worth of therapeutic experience from previous generations of conventional doctors [51]. The concern of many scholars is that not all herbal medicines are safe; however, the many years of using herbal medicines provide enduring and time-tested modes of selection, preparing, and applying these herbs to address numerous health needs [4, 21, 22, 61, 62]. Due to their great safety compared to other therapeutic classes, herbal medications exhibit advantages in clinical therapies, particularly for cancer, hypertension, and many other incurable disorders [2, 28, 31, 45]. Several epidemic diseases have benefited greatly from the use of herbal remedies for both prevention and therapy [98]. For instance, in 2003, severe acute respiratory syndrome was treated using herbal medications, and the results were remarkably positive [47]. Additionally, the use of herbal remedies for the treatment of malaria finally led to the development of artemisinin, a herbal extract that is now a component of the worldwide standard treatment for *Plasmodium falciparum* welch malaria [82]. Furthermore, using herbal medicines therapy against coronavirus disease 2019 (COVID-19) is still a cost-effective solution in many nations and areas due to their efficacy, safety profile, and relatively reduced pharmaceutical costs [98].

Despite the perceived difficulties in regulating traditional, complementary/alternative, and herbal medicines on a global scale, nations have independently achieved significant progress in this area [4]. Since WHO adopted the Alma-Ata Declaration in 1978, to make primary care the key to healthcare and service for all, efforts have consistently been made to integrate traditional medicine into public health services [75]. Although existing herbal medicine policies and laws/regulations differ from country to country and region to region, they essentially continue to address the need for a global framework to ensure the safe use and practice of herbal medicines [4]. However, the differences in cultural orientation regarding the use and regulation of herbal medicines among different countries effectively reduced the opportunities to use herbal medicines to treat COVID-19 currently [98].

This chapter aims to address the need for standardized global regulations and policies for herbal medicine practice and practitioners. The chapter discusses the state of herbal medicine by highlighting available resources and the broad focus of the practice and practitioners. Thereafter, the chapter presents a viable guide for regulation and policy development through approaches, processes, and necessary tools. This is connected to the next section on the benefits of herbal medicine governance and an analysis of existing medical regulations and policies that may be relevant to herbal medicine practice and practitioners. The focus of the final part of the chapter is on the importance of sustainable regulations and policies for herbal medicine practice and practitioners.

2 State of Herbal Medicine: Resources, Practices, and Practitioners

Medicinal plants (i.e., plants that contain bioactive substances with therapeutic benefits [76]) are the core herbal medicine resources. These plants or their parts are used wholly or combined with other plants and chemicals (like water, alcohol, or soda) and used for disease prevention and health promotion through diverse strategies and are worth billions of USD per annum. Indigenous medicinal plants have long been used to cure varying diseases globally. The application of herbal medicinal plants as efficacious natural remedies in the treatment of diseases has gained prominence in recent decades as they have been recognized to pose relatively lower side effects as compared to chemical and synthetic medicines [53] and are also recognized to have widespread accessibility as compared to allopathic medicine [3, 56, 57, 66, 77]. Calixto [14] established that an estimated 40% of medicinal products used in recent times emanate from natural products (plants 25%, microorganisms (13%), and animals (3%)). Today, herbal medicine includes the use of herbs, herbal preparations, herbal materials, and finished herbal products containing active ingredients from plants, parts of plants, and other plant materials or combinations [94]. These materials are used in pursuit of Goal 3 of the United Nations Sustainable Development Goals to ensure healthy lives and promote the well-being of the world's populace.

There is widespread reportage on the use of herbal medicine globally. High patronage of herbal medicine uptake has been found in Southeast Asia. For instance, Cai et al. [12] discovered that over 1000 Chinese people use herbal drugs daily following prescriptions by Chinese medical practitioners with an annual uptake of herbal medicine clinching 200 million accounting for more than 40% of healthcare deliveries in the country [10]. Similarly, an estimated 70% of Japanese allopathic medical practitioners prescribe varying herbal and alternative treatments for their patients. Traditional Korean medicine usage is also pegged at 70% [11]. Interestingly, a nationwide survey by Rashrash et al. in 2015 in the United States found about 35% uptake of herbal supplements by adults on a routine basis [72]. The global market for herbal medicine was estimated at USD 71.19 billion in 2016 in a study conducted by Hexa Research [34], which has been projected to reach a whopping USD 5 trillion by the WHO [74]. In most developing economies, native healers are the focal health providers for the teeming populations. The implications for the preponderance of the native healers mean there is a better patient-to-healer ratio than the case of doctors-to-patient ratio in the case of allopathic medical practices.

There are peculiarities among practitioners of herbal medicine in Africa. In Ghana, herbal medicine is frequently used as the first line of defense against illness, particularly in rural areas. Traditional healers are widely used for several reasons, including the cost of care, lack of access to medical institutions, and poor roads and infrastructure. In addition, the patient-to-doctor ratio for conventional healers is 1: 200 as opposed to the approximately 1:20,000 ratio for medical practitioners. This has a significant impact on how healthcare decisions are made. Other determining elements, like economic position, level of education, and recommendations from relatives and peers, affect the type of health care a person chooses (Tabi et al. 2006; [59]). Herbal medicine is still widely recognized even though allopathic medicine has been embraced. It is a popular belief among some Akan communities in Ghana and other indigenous communities around Africa and Asia that individual(s) or an entire community may be at risk of severe illness or plague due to disobedience of native laws, customs, and culture that constitute a taboo [29, 32, 33], which is a fundamental architecture of African traditional religion. In Zambia, complicated treatment practices are often initiated following the diagnosis of a health condition by using herbal components accompanied by rituals in an attempt to cure the disease. Persistent illnesses call for *chizimba*, which means permanently locking a disease or illness away.

To do this, a lizard must be killed, and its heart must be burned with charcoal and the roots of particular trees. The combination is applied to small wounds made on the affected area. There are several ways in which herbal medicines are applied. For instance, a single plant can be used for herbal treatment or it may be used in conjunction with other plants. In whichever way the treatment is done, the acquisition, preparation, and utilization of the herbal medicines go through a similar procedure. The requisite parts of the plant are initially collected while fresh. These are then dried either at room temperature or in the open sun. Thereafter, depending on the mode of application or utilization, the dried parts of the plant are either soaked in water or other edible liquids, ground into powder form, or incompletely burned to obtain the residue for utilization. The modes of application also range from drinking the solution, adding the powder to meals, making incisions on specific parts of the human body and the powder then applied, bathing the extracts of the herbs, utilizing the steam from the boiling herbs, as well as dancing after applying the herbs to exorcise spirits. All these forms of treatment using herbal medicines are available to the ethnic groups in Zambia. Nga:nga is the name of the traditional healer in Zambia [79]. In South Africa, traditional medicine is a daily part of the lives of thousands of people. In actuality, the use of traditional medicines, often known as muti, is reported to be at 80% of the population. The term “muti,” which is derived from the words “medicinal plant,” designates drugs that are traditionally obtained from plants, minerals, and animals. Traditional medicine could also comprise minerals and animal parts in addition to plants.

A myriad of practices are also employed by traditional and herbal practitioners in the diagnosis and treatment of diseases. One dominant practice in herbal and traditional medical practice by indigenous people is divination. Divination is the practice of learning about healing via the use of symbolism that is placed at random to acquire knowledge about a person or sickness condition. In addition, it is viewed as a way to gain access to knowledge that is generally outside the scope of the rational intellect. It is a transpersonal technique wherein diviners converse with spiritual beings like gods, spirits, and ancestors to get knowledge [65]. Therefore, divination plays a crucial role in the traditional African method of disease diagnosis. The etiology of illnesses and diseases is said to be ascertainable through divination, particularly when one is suspected of breaking a taboo or contaminated by evil spirits as the source of the ailment. This is demonstrated by using cowry shells, throwing coins, seeds, dice, domino-shaped objects, or even actual dominoes, and other tools chosen by the diviner and the spirit to represent specific polarities on leather strips or flat pieces of wood. The majority of the items are divining bones, such as those from crocodiles, wild pigs, goats, antelopes, baboons, hyenas, anteaters, and lions. The bones are a representation of the forces that threaten any human being irrespective of his or her cultural orientation [16].

In herbal and traditional medical practice, interviewing and medical reporting are also a basis for the diagnosis and treatment of a health condition. Some traditional healers often rely on oral interviews to learn more about a patient’s medical history, where they have had treatment, and how long they have had the ailment which serves as the basis for finding the cure to a particular disease. Sometimes, to establish the history of the ailment, a family member could be interviewed if the patient is too ill to talk. After the interaction with the patient, the healers sometimes would advocate for the patient to access orthodox medical services for precise diagnosis. Patients are advised to return to the herbalist with the diagnosis for them to holistically heal them. The medical diagnosis of the patients is kept by the herbalist as evidence of their prowess in healing those ailments. Besides the cost of treatments which is often moderate and the easy access and availability of the herbalists, among other merits, the herbalists adopt a holistic approach to healing that involves the psychosomatic aspects of the person. Hence, there is no distinction between the natural and the spiritual or physical and the supernatural aspects of the person [80].

3 Regulation and Policy Development Approaches, Processes, and Tools

In parts of the world where the populace does not rely on herbal medicines and practitioners for their primary healthcare like in the European Union, United States, and Canada, herbal medicines are sold as food supplements albeit without or with little regulations [13]. Due to the contamination and adulteration of herbal medicinal products during their initial periods of discovery as having the capacity to treat various diseases [71], economies across the globe have enacted some guidelines and regulations to ensure the purity of these medicines to avoid complications after usage. The World Health Organization (WHO) in 2018 reported that about 90% of its member countries had some form of regulation or policy guideline on herbal medicine [94]. These regulations ensure herbal medicines are taken through some quality check and standardization processes such as physicochemical evaluation of medicinal products in their raw state through stability assessment to ensure the safety of processed products for consumption (Riedl et al. 2015).

Standardization of herbal medicines has been a complex phenomenon that involves a series of processes aimed at ensuring the quality of the final product. Given the different definitions of herbal medicines by various countries, standardization has for some time been at the individual country level. However, WHO has set guidelines lines for the standardization of herbal medicines [99]. The need for standardization has also been indicated by several studies [7, 23, 25, 30, 78, 84–86].

The first Herbal medicine strategy to be developed by the WHO was initiated between 2002 and 2005. However, due to the rapid expansion and the rapid utilization of traditional and herbal medicine, there was a shift in the reliance on herbal products which led to the incorporation of practices and practitioners in the traditional and herbal medicine fraternity stimulating the WHO to devise a comprehensive strategy which led to the development of the Traditional Medicine Strategy, 2014–2023, to regulate herbal medicine practices [95]. Three policy objectives underlie this strategic initiative. These include the building of a knowledge base that will allow for effective management of traditional and allopathic medicine within the national policy context. This is to help ensure the coordination and harmonization of the diversity of products, practitioners, and practices underlying herbal medicine. In addition, the 2014–2023 policy strategy seeks to strengthen the quality levels, safety, and right application or usage of both traditional and allopathic medicinal end products by regulating these as well as the activities of the practitioners. And finally, the policy strategy sought to make health coverage accessible to all through the integration of traditional and complementary medical service delivery as well as self-healthcare services.

There are however difficulties in the harmonization of global herbal medicine policies as different countries pursue varied regulatory strategies and implement varied policy legislations. This notwithstanding, there are several legislative frameworks on herbal products in many countries worldwide. For instance, in Mexico, legislation on herbal medicine has been incorporated into the Mexican General Health Law [73]. The Federal Commission for the Protection against Health Risks

(COFEPRIS) is a decentralized organization of the Ministry of Health charged with the responsibility of managing and promoting healthcare (COFEPRIS 2022) as enshrined in the General Health Law [73] and other legislative instruments. It sets the parameters for checking the safety and consumability of medical products including those generated from plant materials. In Canada, a whopping 1.7 million people identify themselves as indigenous and are often referred to as First Nation People [73]. Canada originated the policy of Natural Health Products (NHPs) in 1999 and incorporated herbal medicinal practices into its health system. This policy came into fruition in 2004 after the annexation of the NHPs in subsection 30(1) of the Food and Drugs Act [73]. These have legislative backing in the current national policy and national program on traditional and complementary medicine whose implementation is the reserve of provincial states and territories.

The Health Products and Health Branch of Food Canada have direct legislative authority over herbal products and as such ensure the licensing of all herbal medicine to ensure compliance with safety protocols before they are been commissioned into the market [37]. Herbal medicine products and raw materials are also sold in the market in developing countries [54, 55, 67]. In the case of the United States, there are institutions like the Food and Drug Administration (FDA), which, among other functions, is tasked with the responsibility of regulating the safety, effectiveness, as well as security of drugs, biological products, and food supplements. This implies that it has the mandate to regulate herbal medicinal products [83]. Amid these efforts at regulating herbal medicines, the United States is yet to develop a national plan that promotes the integration of herbal and complementary medicine into the healthcare system. There is, however, a National Center for Complementary and Integrative Health focused on supporting research as well as the provision of relevant information on complementary health products and practices [94].

African governments have also implemented policies and regulations aimed at regulating herbal and complementary medicines. For instance, in Kenya, a national policy document on traditional medicine and medicinal plants was drafted in 2005 which emphasizes the need to take account of all medicinal plants in the country while also establishing herbal gardens to ensure bio-conservation and also intensify research [64]. Similarly, in Ghana, many administrations have acknowledged the value of conventional medicine. This is demonstrated by the founding of the Ghana Psychic and Traditional Healers Association in 1961 and the Centre for Scientific Research into Plant Medicine in 1975. The establishment of the Food and Drugs Board in 1992, which among other things certified the sale of traditional medicine products to the general public, followed the establishment of the unit for the coordination of traditional medicine (which is now the Traditional and Alternative Medicine Directorate) in 1991 (Ministry of Health 2005). The TMPC Act, Act 575 was passed by the government in 2000 to create the Traditional Medicine Council, which is in charge of registering all Traditional Medical Practitioners in the nation. This, therefore, stands to reason that herbal medicine usage is now gaining moment, thus generating interest from international bodies and national governments to design legislative backing for traditional and complementary medicinal products.

4 Benefits of Herbal Medicine Governance

Public health and safety, job and income security, and biodiversity and resource management are major issues connected to herbal medicine. The focus of herbal medicine governance is to ensure that safe and quality herbal medicine products are available in the market to contribute to public health [18, 22]. The governance of herbal medicine will also ensure the sustainable harvesting and processing of materials used in formulating herbal medicine products and the job and income security of practitioners. Herbal medicine cannot be dissociated from allopathic medical delivery because a quality healthcare system and services are only achievable through informed policy and strategies. There is clinical trial data that abound stressing the efficacy of herbal medicine as well as herbal product components that are used in the manufacture of modern drugs. Herbal extracts are more clinically effective and accessible with fewer health risks [50, 81]. Sufficient examples have been published by clinical laboratory scientists demonstrating the potency of allopathic drugs that have been derived from herbal medicines (e.g., artemisinin) which saves millions of lives annually [26, 35]. Besides the modern governance approach, traditional governance approaches and systems may be instituted to regulate the herbal medicine sector because they shape the belief systems behind the practice and of practitioners [15]. On the other hand, modern governance systems within local spheres are considered top to be a form of downregulation and disconnected from realities. Negative experiences in the hands of modern medicine are one of the major reasons why many people prefer herbal medicine for their primary healthcare [91]. The major benefits associated with the governance of health practices as outlined in Busse et al. [8] include:

1. Providing clarity on the meaning of quality care and its connection to the health system
2. Creating a framework to assure and improve the quality of care provided by the health system
3. Evaluating care approaches based on current practices to align with cost-effectiveness and implementation
4. Addressing gaps by measuring and synthesizing existing quality healthcare knowledge and practice
5. Aligning national strategies with international guidance on quality healthcare provision
6. Ensuring accessible systems exist for an evidence-based assessment of quality healthcare strategies that address patient safety and culture
7. Clarifying the links and roles of stakeholders in assuring quality healthcare

Quality healthcare is central to the governance of every healthcare system because it links service delivery, healthcare workers, information, health products, financing, and governance to the goals, i.e., improved health, responsiveness, equity, and efficiency [9].

Fig. 1 Countries with national herbal medicine laws or regulations. (Source: adapted from WHO [94])

As of 2018, the number of countries with some form of a legal framework for herbal medicine is presented in Fig. 1. Surprisingly countries in the Global South with a long history of herbal medicine practice, production, administration, and use do not have any form of a legal framework for herbal medicine including Chad, Central Africa Republic, South Africa, Somalia, Eritrea, Congo, Gabon, Equatorial Guinea, Ivory Coast, Panama, Honduras, and Senegal.

Most of the existing national regulations are focused on prescription, nonprescription, and herbal medicines which include those that can only be purchased with a prescription from a physician, without a prescription for self-medication, and sold over-the-counter herbal drugs respectively (WHO [94]; Fig. 2).

According to WHO [94] out of the 249 member countries, 124 have some form of laws or regulation focused on herbal medicines, while 125 only have herbal medicine registration systems. Some of these are presented in Table 1.

These policies range from those that are partially or completely similar to conventional pharmaceuticals to those that are exclusive to herbal medicines. And in terms of their distribution by region, Europe and Africa dominate with 45 and 20 WHO member states with herbal medicine regulations, respectively, while Southeast Asia region had the lowest with only 10 WHO member states having herbal medicine policies as of 2018 (Figs. 3 and 4).

According to WHO [94], there is a clear increase in the number of countries with some form of herbal medicine laws and regulations from 65 in 1999 to 124 in 2018 (Figs. 3 and 5).

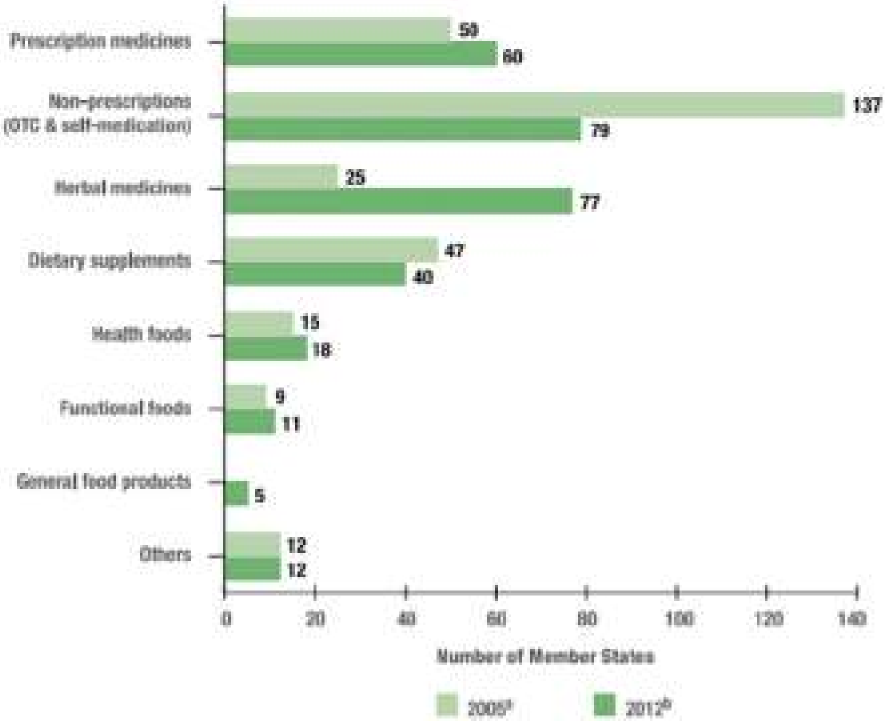

Fig. 2 Types and focus of national herbal medicine laws or regulations. (Source: adapted from WHO [94])

The importance of these herbal medicine governance frameworks in healthcare delivery is to align national structures with public healthcare foci of numerous international bodies and conventions. For instance, herbal medicine has gained recognition from the World Health Organization (WHO) in the Beijing Declaration as an important provider of primary healthcare [92]. The World Trade Organization (WTO) laid similar emphasis on the role of herbal medicine in strengthening healthcare in regions across the globe [97]. Even though the contribution of herbal medicine has been downplayed in many developed countries, these countries have alluded to the successes of this type of medicine in successfully controlling and managing chronic diseases associated with lifestyle [94]. The increasing usage of herbal medicine has been consolidated by rising healthcare costs and the rising cost of allopathic drugs. Herbal medicine aside from its curative and healing tendencies for medical conditions is also a representation of the skills, knowledge, and belief systems of indigenous people which are harnessed for the diagnosis and treatment of mental and physical ailments and the maintenance of general health conditions [94].

Table 1 Herbal medicine policies, regulations, and regulatory pathways

Country/ region	Policy and legislation	Year	Focus
United States	Dietary Supplement Health and Education Act	1994	Regulation of dietary supplements which are defined as any herb or part of a herb, botanical, and natural concentrates, metabolite, and constituent of extract
Canada	Natural Health Products Regulations (SOR/2003-196)	2003	Registration, licensing, labeling requirements, and recommended utilization approach of natural health products
European Union	Herbal Medicinal Products Committee European Directives 2001/83/EC and 2003/63/EC and amended versions like 2004/24/ EC (European Herbal Legislation)	2001, 2003, 2004	Provide guidelines for the authorization, evaluation, and safe use of herbal medicine products and resources (collection, characterization, manufacturing, testing, and quality control)
Brazil	RDC 48/2004	1967	Regulation of herbal drugs
India	Drug and Cosmetic Act and Rules	1940 and 1945	Regulation of herbal drugs and herbal medicine preparations
Ghana	TMPC Act and GMP rules	1992 and 2000	Registration of traditional medicine practitioner and their products
Nigeria	Decree Number 15/1993	1993	Regulation of finished herbal medicine products
United Kingdom	Medicines Act	1968	Registration, licensing, and regulation of herbal medicines
Korea Republic	Pharmaceutical Affairs Act	2015	Regulation of herbal medicinal preparations
Ethiopia	Proclamation # 176/99 and 1112/ 2019	1999 and 2019	Regulation of the preparation and finished herbal medicine products
China	Law of the People's Republic of China on Traditional Chinese Medicine (TCM law)	1986	Promotion and regulation of traditional Chinese medicine
Kenya	Pharmacy and Poison Act (cap 244)	2019	Control the trade and administration of drugs and poisons
Tanzania	Pharmaceutical and Poison Act	1978	Control the trade and administration of drugs and poisons
Hong Kong	Hong Kong Chinese Materia Medica Standards	2009	Ensure the safe use, trade, and quality of Chinese Materia Medica
Gabon	Ordinance # 001/95	1995	Recognition of herbal medicine
Australia	Therapeutic Goods Act	1989	Regulation of medicinal products containing such ingredients as

(continued)

Table 1 (continued)

Country/ region	Policy and legislation	Year	Focus
			herbs, vitamins, minerals, nutritional supplements, homeopathy, and certain aromatherapy preparations used in complementary medicine
Bolivia	Law # 1737	1996	Regulation of herbal medicine products and materials
Indonesia	Minister of Health Regulation # 007	2012	Provide criteria and permit for the distribution and sale of herbal medicine
Russia	Article 41 TM/CAM	1991	Regulation of herbal medicine prescription, use, sale, and production license
Democratic Republic of Congo	Organization of the Exercise of the Profession of Practitioner of Traditional Medicine	2002	Registration and regulation of herbal medicine formulation, content, and use
Argentina	Resolution 144/98	1998	Management of phytotherapeutic medicines and vegetable drugs
El Salvador	Decree # 55	1988	Regulation of herbal drugs as prescription and over-the-counter medicines
Peru	Decree # 584 and Supreme Decree # 002-92-SA	1990	Registration and regulation of herbal medicines

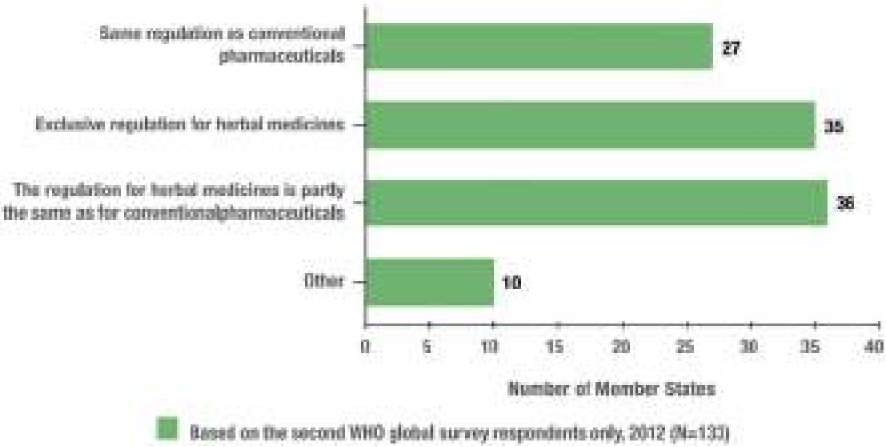

Fig. 3 Types of national herbal medicine policies reported by WHO member states. (Source: adapted from WHO [94])

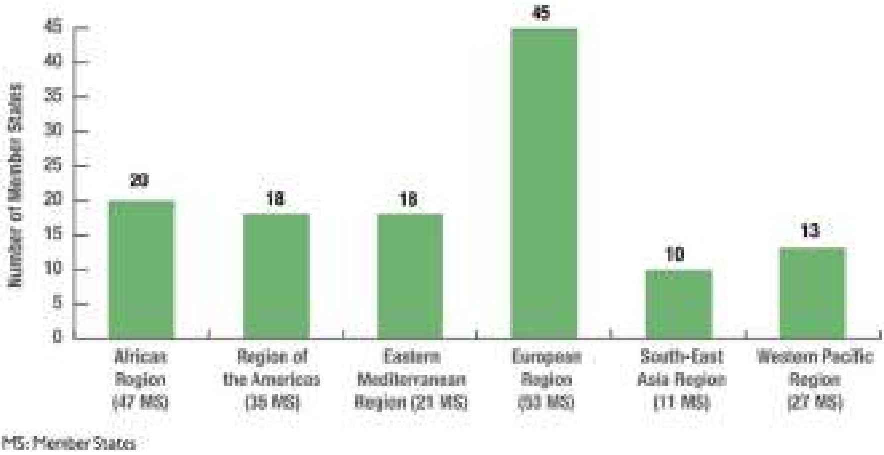

Fig. 4 Distribution of WHO member states with herbal medicine national policies by region. (Source: adapted from WHO [94])

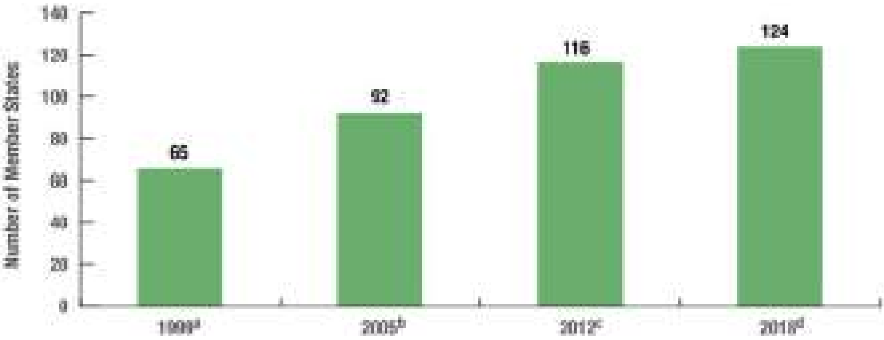

Fig. 5 Increase in the number of WHO member countries with some form of herbal medicine laws and regulations from 1999 to 2018. (Source: adapted from WHO [94])

Transforming existing national herbal medicine policies into a global policy strategy will help govern the education of practitioners and users, protect their right to practice and use herbal medicines, ensure safety and competence, and promote the reputation, quality, and efficacy of herbal medicines. A global policy will help address variations in care outcomes within and across countries in the world through an arsenal of strategies that are simple, cheap, and attentive to local needs (Busse et al. 2019). Existing evidence abounds to support the continued development of the practice, but initiatives need to be supported with appraisal mechanisms including research and policies [42]. For instance, this will regulate the admission of new practitioners and the validation of their practices, content, and accountability, balance the extraction and use of raw materials, and promote their continued education.

5 Medical Regulations and Policies that Are Relevant to Herbal Medicine and Practices

The pharmaceutical sector in recent times has shown a vested interest in studying various plant species because they can both be valuable and advantageous resources for the creation of novel medications and help preserve biodiversity [69, 87]. International agencies and national governments the world over have begun to establish regulatory frameworks for producing and distributing herbal medicines due to the increased interest in researching plants as therapeutic resources [89]. This is critical since many herbal products used may probably not have undergone any form of safety analysis to assess their effectiveness with negative consequences associated with some [70, 89]. According to Vyas et al. [89], there are difficulties in adapting herbal medications, which have been developed over many generations, to a contemporary regulatory system. Yet, China and India, two nations that have formalized training as a part of their national health systems to guarantee that traditional medicine attains high quality in the provision of healthcare, have prospered in this transformation [70].

To understand the usage of medicinal plants, it is essential to keep in mind that there are certain conceptual disparities while discussing herbal medicines. Medicinal plants can be used as a DIY remedy, as in tea, as a source of inspiration for new pharmaceuticals, or as an ingredient in herbal products that are sold commercially [49]. The market for herbal medicinal products has been split into herbal remedies and herbal medicinal products in Mexico, another country with a long history of employing medicinal plants [89]. According to the Mexican framework definition, herbal medicine is made up of components of a medicinal plant that have been combined or produced separately into a pharmaceutical form that is known to be effective for treating symptoms. It should not have any characteristics that could endanger human health, and all plant material must be cleaned following established standards. On the other hand, herbal medicines must adhere to stricter quality standards, which may include using standardized plant material or its derivatives that have been converted into various pharmacological forms [49]. The medicinal plant utilized in the herbal medical product needs to have a monograph in the national pharmacopeia proving it is effective and safe. The scope of regulation for herbal medicine in the event is a matter of a jurisdiction's regulatory frameworks due to differences in culture, history, and usage of herbal products [44]. For instance, the US surveillance agency Food and Drug Administration (FDA) has no regulatory framework for companies who manufacture dietary food supplements [44]; this notwithstanding, products containing medicinal plants with therapeutic capabilities require monitoring due to the possibility of negative side effects [43].

Given this complexity in pursuance of policy regulations presented as a result of the varying approaches that may be adopted by different governments, some underlying themes are required for them to be relevant. The World Health Organization has provided an important resolution by drafting the WHO *Traditional Medicine Strategy 2002–2005*. This breakthrough has three principal goals. First, it is aimed at helping to frame government policy. Second, it is meant to ensure safety, efficacy, and quality and to enhance access. Finally, it is aimed at promoting the right use of traditional medicine/complimentary and alternate medicine (TM/CAM) and herbal medicine [24]. These WHO Traditional Medicine Strategy objectives are crucial in bringing about increasing acceptability and relevance of herbal medicine and practices.

National governments need to specifically introduce regulatory systems on herbal medicine in the context of their local medical practices. These regulations are imperative for quality assurance and registration purposes and also to ensure that consumers used high-quality herbal products. Ensuring quality control for herbal medicine is a sophisticated venture as pointed out in several WHO documents including but not limited to Quality Control Methods for Medicinal Plant Materials and the General Guidelines for the Methodology on Research and Evaluation of Traditional Medicine [96]. Regulations on quality control of herbal medicine are thus important to ensuring compliance by small-scale manufacturers. Another important area in the field of regulation that will contribute to herbal medicine practice is the development of reliable information systems on the most common access to traditional and herbal medicine therapies which should be the result of intensive scientific and clinical studies. This will serve as a guide to TM/CAM and herbal medicine practitioners which will consequently help combat the irresponsive use of traditional medicine [17, 96].

There is also the need for medical regulations on the training and licensing of TM/CAM as well as herbal medicine practitioners to improve safety and build trust in herbal medicine therapies as well as promote the credibility of TM/CAM and herbal medicine practitioners in the eyes of consumers. Such rules may be established by either national or international authorities. Whether through statutory regulation or oversight by regional groups like professional associations in the shape of self-regulation, consumers will need to know where to get what information when it is needed. These regulations, like those suggested by the WHO, will go a long way to reduce or eliminate misinformation and bring sanity into the field of herbal and complementary medical practices and products. There are numerous guidelines for the training of TM/CAM and herbal medicine practitioners developed by WHO and other international agencies and national governments [17].

6 Importance of Sustainable Regulations and Policies for Herbal Medicine and Practitioners

For over millennia, herbal medicine has been relied upon as a form of treatment. However, due to the advent of allopathic medicine and expansion in the synthetic medicine industry, herbal medical practices and products have come under increasing threats that may undoubtedly threaten the continuous usage of herbal products if sustainable policy guidelines and regulations are not instituted. Mainstreaming herbal medicine into primary healthcare is one effective way of creating sustainable pathways to sustained utilization of herbal medicinal products and thus making it imperative to design policy frameworks and regulations to ensure continuous utilization of herbal medicine [49].

Sustainable regulations and policies for herbal medicine, herbal practitioners, as well as users of the product will yield fascinating results and rewards. For instance, the COVID-19 pandemic has reiterated the role of herbal medicine in the development of efficacious drugs and vaccines [49]. Jayk Bernal et al. [36] elicited several merits of herbal medicine including their effectiveness in blockading the multiplication of certain viral genomes, their quick usage to treat new antivirals, as well as producing products with fewer side effects as compared to synthetic medicine. A clinical study by Kumar et al. [40] consolidates this stance as his study revealed highly improved recovery rates from the COVID-19 virus in instances where allopathic medicine was supplemented with herbal medicine.

Herbal medicine also presents an avenue by providing access to explore herbal medicinal products and can contribute to the global commitment to realizing the Sustainable Development Goal 3 of attaining good health and well-being. Integrating herbal medicines into healthcare systems has the potential to increase access to health in both urban and rural settings, which is essential for sustainable development [27]. Herbal medicine also plays a critical role in different socioeconomic and cultural contexts. For instance, due to widening social and economic inequalities amid the devastating threats of climate change, especially on the production side, herbal medicine has become a topical agenda driving a lot of discourses [69]. Malhi et al. (2008) contend that the cultivation of medicinal plants can be exploited as a complementary option in the area of income generation, biodiversity conservation, and stimulating rapid local economic development while also stressing the importance of forest conservation. According to research [6, 48], the Brazilian Amazon plays a significant role in the regional and global water cycle as well as a crucial carbon sink in addition to its sociocultural significance. Chagnon and Bras (2005) explained that land use alternations, especially those linked to widespread land degradation and deforestation, influence precipitation patterns globally, thus stressing the need for herbal medicine sustainability through sustainable policy regulations and practices.

Herbal medicine has also been argued to tend to bioeconomy promotion as it recognizes economic activities and their associated value chains in the context of biodiversity [5, 69]. Complementing the above, in global attempts to attain climate change mitigating targets, keeping global temperatures at bay, and phasing out fossil resources, the bioeconomy has been highlighted as an important tool. Bioeconomy is noted not just as an academic disciplinary sphere or a business opportunity but serves as a conduit for the integration of all the relevant agencies and activities dependent on bioproducts under a single innovative hub [1, 87]. Besides, bioeconomy has direct benefits for the countries that transition to it, especially in the tropics where it could elicit the support of the locals, increase their livelihood pathways by conserving and restoring lost habitats, and bring people closer to nature, which changes their perspective in objectifying nature, with its attendant negative effects (Lima 2022). The bioeconomy idea has been contested up to this point since it puts more strain on natural resources and encourages overexploitation.

7 Conclusion

Existing national policies and regulations have a defining role in what may eventually become a global policy and regulation for herbal medicine practices, products, and practitioners. These existing national policies can provide the necessary regulatory and legal mechanisms necessary for ensuring global good practice, authentication guidelines, and safety and efficacy evaluation of herbal therapies. Also important is the need to maintain and promote herbal medicine practices as well as meet the socioeconomic needs of practitioners and the ecological and cultural security of resources through global policies and regulations. This will ensure that good manufacturing and administration practices are promoted, regulated, and adhered to by herbal medicine practitioners and users. Herbal medicine policies and regulations could also serve as means to harmonize the herbal medicine market and practice across different continents and bring professionals and consumers together. No doubt, almost every herbal-derived remedies require an assessment to ascertain their pharmacological qualities and safety, but this is not achievable with the current level of disinterest. Modern scientific technologies like next-generation sequencing as well as genomic testing and fingerprinting techniques can help address the pharmacogenomic and metabolomic benefits. Herbal medicine should not be evaluated at the same level as contemporary medicine when it is not receiving equivalent amount of budgetary and policy attention. However, on the practice and practical side, it is fair to compare their benefits using evidence-based medicine. Globalization efforts including goals like the United Nations Sustainable Development Goals should aim to integrate herbal medicine practice, practitioners, and resources. Herbal medicine is not only affected by goals and targets that are medico-centric but are also influenced by policies on biodiversity and environmental management. The threat to biodiversity is having a negative impact on the availability of plant resources that are used in herbal medicine by practitioners because most of them are harvested for use from the wild and less so from cultivated lands and poorly managed wild collections coupled with growing disinterest in farming may threaten the industry without relevant policies and regulations. Through policies and regulations, herbal medicine practice can be reinvented and the vital resources protected to meet and sustain the rapidly expanding herbal product market.

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Towards Sustainability in the Source of Raw Materials for Herbal Remedies

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Contents

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Abstract

Plants, especially herbs, are an essential part of the world's biodiversity; they are crucial to maintaining the environment. Natural ecosystems and the variety of plants that they support are crucial to human welfare and way of life because they offer the conditions for life to exist. For food, and other essentials for maintaining life, about 7000 plant species are used. A number of studies into the biological mechanisms and chemical make-up of herbal treatments have been made, and these investigations have proven to be quite helpful in identifying bioactive plants. The use of these therapeutic herbs is quickly spreading over the world due to the rising demand for herbal medications, herbal treatments, and their secondary metabolites. Unfortunately, the majority of herbal remedies found in natural settings are now increasingly at risk due to their unsustainable use and overexploitation. As a result, many highly valued species are fast becoming extinct. This situation is largely caused by a rise in human population, development, habitat alteration, and the increasing dependence of the vast majority of people on the planet's limited natural assets. This calls on management at all levels to pay attention to the usage and threat to these valuable resources to increase their sustainability.

Keywords

Medicinal plants · Human medicine · Anthropogenic impact · Overexploitation · Sustainable use · Conservation measures

Abbreviations

ROS Reactive oxygen specie
SAP Standard Agricultural Practises

1 Introduction

Plants are critical to sustaining the ecosystem since they are a vital component of the world's biodiversity. Because they provide the circumstances for life to exist, natural ecosystems and the variety of plants they sustain are essential to human well-being and way of life. Plant species are a valuable source of medications in addition to providing food, particularly when applied in conventional medicine to treat a variety of diseases [1]. Herbal remedies are a large natural resource and are essential for the provision of primary healthcare services, particularly in underdeveloped countries [2]. Since ancient times, herbal remedies and common herbal remedies have been used throughout the world to treat and alleviate a diverse array of human and animal diseases. Herbal remedies are also the most profitable sources of new medications in the world [3, 4]. About 118 of the top 150 prescription medications in the USA are produced using natural ingredients, as are 90% of the more than 1300 herbal remedies used in Europe [4]. In addition, 80% or more of residents in the developing world commonly used herbal remedies for their basic treatment, compared to around 25% of the recommended prescriptions in wealthy countries that are manufactured via the use of natural plant species [4]. The usage of medicinal herbs is growing quickly around the world due to the increased demand for natural health products, including secondary metabolites of medicinal plants [3, 5]. According to a very conservative estimate, the world loses at least one significant therapeutic candidate every 2 years, and the rate of plant extinction is currently between 100 and 1000 times quicker than expected by natural elimination [6]. According to the World Wildlife Fund and the International Union for Conservation of Nature, approximately close to 80,000 different types of flowering plant species are said to be utilised as medication globally [50]. Because of desecration of habitat and overuse, 15,000 of these species are at risk of going extinct [7], and 20% of their natural resources have already been depleted by an increasing number of animals and plants [8]. Although this problem has been known for decades, the extinction threat to herbal remedies has grown particularly in China [3, 9], India [9], Kenya [10], Nepal [10], Tanzania [11], and Uganda [11]. There is much research on the preservation and sustainable use of medicinal plants [1]. Numerous groups of suggestions have been made in order to safeguard them, such as the necessity for integrated conservation measures focused on both in situ and ex situ techniques, as well as the creation of protocols for species inventorying and assessment of status [12]. A viable conservation strategy for medicinal plants with declining populations is the wise use of natural resources. Due to the enormous demands of vast populations, the scenario is particularly terrible in China and South Africa.

2 Relevance of Herbal Remedies in Africa

The World Health Organization (WHO) has projected that 80% of Africans employ conventional herbal treatments. Plant extracts are used in traditional medicine in approximately 85% of cases [13]. This might indicate a strong reliance on natural medicines. Consider that there are around 500 traditional healers for every 40,000 individuals in Sub-Saharan Africa as opposed to only one doctor to grasp the extent of this dependency. Hence, the importance of herbal treatments in African culture cannot be overstated. A variety of issues, such as the growing cost and lack of orthodox treatments for individuals with average salaries, are contributing to the rise of interest in herbal therapies in Africa [14]. The fact that many diseases are becoming resistant to conventional treatments is another factor. For example, malaria parasites are becoming more and more resistant to chloroquine, one of the cheapest and most widely used medications in Nigeria to combat malaria [15]. Another well-known instance is the bacterial resistance to drugs. Another explanation could be that some diseases and infections, like HIV/AIDS, cannot be cured by Western orthodox treatment. As a result of the growth of the human immunodeficiency virus, extensive research has been done on plant derivatives that may be beneficial, especially for use in developing and developing nations.

The fact that employing herbal treatments has little to no drawbacks is another argument in favour of herbalism.

2.1 Raw Materials for Herbs Making/Remedies for Man

Herbal remedies have been used by cultures throughout history to treat a variety of diseases. A 5000-year-old clay slab from Sumer contains the oldest documented evidence of the use of herbal treatments [16]. Traditional medicinal plants continue to be the primary source of healing in many rural and developing countries today [17] (Fig. 1). The therapeutic plant components are prepared in a variety of ways (decoction, concoction, water homogenization, chewing, crushing, steam bath, smoke inhalation, etc.) and are administered through a variety of channels (oral, dermal, nasal, and other body parts) (Table 1). However, the most typical method of herbal preparation for human and cattle health problems is simply crushing and crushing a specific plant component or parts of a plant and homogenising them in water. When used by humans, the medicine is often first decanted. It was discovered that several of the medical plant species studied served many functions for the local population, including charcoal, firewood, building materials, feed, medicine, food, and household utensils.

2.2 Herbal Remedies for Domestic Animals

Many studies of the biological characteristics and chemical composition of ethnobotanical flora have been conducted, and ethnobotanical studies are very useful in the discovery of bioactive plants [18]. Interest in medicinal plants for veterinary health has grown as a result of the various side effects and the rising cost of pharmaceutical drugs [19]. Furthermore, since they represent a minimal risk of antibiotic resistance due to their complex chemical make-up and various biological activities, plants might replace antibiotics in the treatment of multifactorial diseases [20] (Fig. 1).

Erythrina senegalensis has a significant diuretic effect, in contrast to the purgative and convulsant properties of Cassia occidentalis leaves. *Andasonia digitala* is administered to cattle as an antidiarrhoeal [21]. According to reports, the plants *Guleria senegalensis*, *Anogassus leocarpus,* and *Selerocarya birrea* are effective in facilitating domestic animal pregnancies easier [22]. Madubunyi [23] asserts that the root extract of *Nauclea latifolia* has antihepatotoxic effects and prevents the growth of the *Trypanosoma brucei* infection. When used to treat anaemia in rabbits, a mixture of Sorghum bicolor and *Telfaria occidentalis* is more effective hematogenic than commercial hematogenic [24]. Studies have shown that papaya extracts are efficient against *Salmonella aureus* and *Escherichia coli* [25]. When a variety of northern Nigerian plant extracts were evaluated using the microtiter plate inhibition test against poliovirus, astrovirus, and parvovirus, an effective antiviral impact was discovered at values between 100 g/100 l and 400 mg/100 l [26]. It has been shown

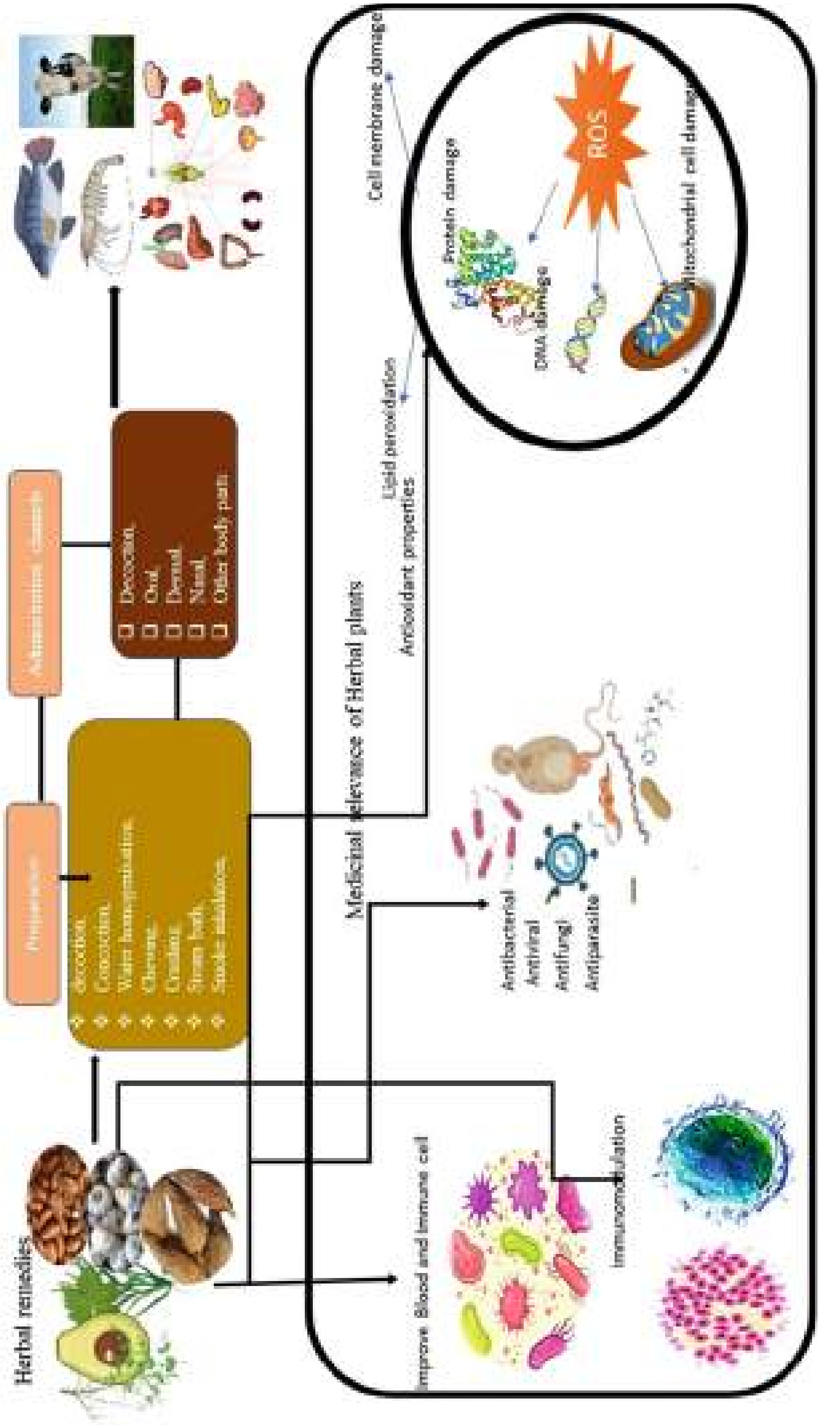

Fig. 1 Medicinal plants and their importance to man and animals

Table 1 Some herbal remedies of relevance to human health

Plant name	Image	Medicinal uses
Marigold		One type of perennial plant in the sunflower family is the marigold. It often has colours such as yellow, orange, red, and maroon and is a member of the Plantae kingdom. Marigold has several health advantages, including: Immune system booster Cures skin conditions Helps with period pain Cure oral sores Relieve joint discomfort Treats fungus-related illnesses Reduce conjunctivitis and eye infection Wards off insects and mosquitoes
Giant milkweed: *Calotropis gigantea*		Crown flowers are a great option for floral arrangements, as they include a group of waxy white or lavender blooms. As long as you plant crown flowers next to a location with direct sunlight, growing them in your yard should not be difficult. When it comes to Ayurvedic therapies for different diseases, it has a prominent position in treating toothaches and healing intestinal issues. May be useful for snakebites and treats a cough/ inhalation
Lemmon grass *Andropogon* spp.		A perennial grass belonging to the grass family, lemongrass is used both as a culinary and medicinal plant worldwide. The medicinal qualities and lemon-like smell are why people grow it. It may grow up to 10 ft tall with a crimson stem at the base. Many deodorants, soaps, teas, and cosmetics frequently use the grass genus as a component Excellent for aromatherapy, detoxifying the body, aiding in sleep difficulties, cure fever, preventing rheumatism, cleaning the skin, and protecting bodily cells
Aloe vera *Aloe barbadensis*		Evergreen succulent *Aloe vera* has very little or no stem. This plant, which has its origins in the Arabian Peninsula, thrives in tropical regions throughout the world. The plant is well-known among contemporary gardeners since it thrives both indoors and outdoors. The plant is used in making goods, including drinks, lotions, and ointments, to mention a few An effective antioxidant, frequently relieves constipation, lessens sunburn, prevents wrinkles, and lessens oral plaque

(continued)

Table 1 (continued)

Plant name	Image	Medicinal uses
Neem *Azadirachta indica*		A tall evergreen shrub known as neem or Indian lilac is grown for its lumber, resin, and fragrant seed oil. Although it is an evergreen plant, excessive dry circumstances can cause it to lose all of its leaves. In addition to being used medicinally, it may also be used as a pesticide to get rid of bugs indoors. It is used to prevent wounds, ulcers, pimples, and acne. May be used to cure a cough. It has antifungal and antibacterial characteristics, stimulates the kidney and liver, decreases oedema and inflammation, and reduces the risk of malaria
Soursop *Annona muricata*		A tiny evergreen tree from which medication is made in the pharmaceutical industry. It has recently received a lot of attention as a meal that fights cancer. It has a high concentration of antioxidant qualities, which is why many people like it. Antioxidant-rich prevents ulcers, treats herpes, boosts energy levels, and keeps joints and nerves in good condition
Moringa *Drumstick tree*		Drought-tolerant Moringa plants are widely grown for their medicinal stems, flowers, leaves, and pods. Although the whole plant is nutritious, the leaves provide the greatest nutrients. A healthy vegetable called moringa can help with issues like battling free radicals, maintaining the endocrine system, reducing inflammation, rejuvenating skin, safeguarding your cardiovascular system, and promoting wound healing
Turmeric *Curcuma longa*		Turmeric is a spice of the ginger family that contains bioactive ingredients. The plant is antibacterial and has many other health advantages due to these components. It is often brilliant orange and has a distinct earthy flavour and scent. It possesses potent antioxidant qualities. Capable of reducing the risk of cancer, lowering the risk of heart disease, improving heart health, balancing hormone levels, strengthening the immune system, and preventing hair loss
Peppermint *Mentha piperita*		The well-known member of the mint family is peppermint, obtained from the cross of watermint and spearmint. Today, peppermint is a popular flavouring for toothpaste, ice cream, soft drinks, and many more products. These are some of its therapeutic qualities. Improves the immune system, fights bad breath, eases gastrointestinal ailments, lowers stress, and eases muscular discomfort

(continued)

Table 1 (continued)

Plant name	Image	Medicinal uses
Garlic *Allium sativum*		*Allium sativum*, popularly known as garlic, is a plant that belongs to the Allium family and is widely used in cuisine around the world. It has a distinctive flavour that makes it a necessary component in many meals. Provides several additional health benefits in addition to easing those with blood and heart disease. It reduces disease, prevents Alzheimer's, lengthens life, improves athletic performance, stops hair loss, and removes pollutants from the skin
Pawpaw *Carica papaya*		Papaya aids in reducing inflammation and accelerating burn recovery. In addition, it includes lycopene, which enhances blood cell health and reduces the risk of blood clots. Papaya leaf juice is crucial in preventing stroke because it improves clotting, heals the liver, and destroys germs that obstruct digestion. It also helps to stimulate white blood cells and platelets
Mint plant *Mentha longifolia*		The 2–4-inch-long purple flower spikes on this plant are typical. When planted properly in the ground, the mint herb is a drought-tolerant plant. The leaves of this healing plant make a stimulating beverage and can provide a special flavour to the food. Very useful for diarrhoea treatment, removing fever and congestion, relieving nausea, and perhaps curing hepatitis
Thyme: Thymus vulgaris		One of the most used herbs of the mint family is thyme. In addition to several nutritional applications, it has been used to treat a variety of ailments. This decorative plant also has antibacterial and insecticidal qualities. The herb is available commercially in the form of tea and essential oils. Effective for stomach discomfort, diarrhoea, and sore throat. It may be used to treat arthritis, reduce blood pressure, treat coughs, and strengthen immunity
Milkweed *Asclepias syriaca*		A garden plant called a bloodflower provides nourishment for butterflies. It produces bright red blooms almost all year long. Among its health advantages are the following: prevents ringworm, acts to reduce discomfort, heals skin lesions, stimulates the nerve in the brain, combats jaundice, and increases blood-flow precision

(continued)

Table 1 (continued)

Plant name	Image	Medicinal uses
Bamboo *Bambusa vulgaris*		Historically, arteriosclerosis, cardiovascular disease, hypertension, and some types of cancer have all been treated using bamboo leaves
Cannabis sativa		*C. sativa* has also been used to treat rheumatism, epilepsy, asthma, skin burns, discomfort, and sexually transmitted diseases in addition to alleviating gastrointestinal symptoms
Pineapple *Ananas comosus*		The digestive enzyme found in pineapples, bromelain, has anti-inflammatory and analgesic qualities. This is beneficial if you have an injury, such as a burn or sprain, or an infection, such as sinusitis. Moreover, it lessens osteoarthritis-related joint discomfort. The vitamin C in pineapple juice helps keep inflammatory levels low
Albizia lebbeck		Several countries in Africa, Asia, and Australia have used *A. lebbeck* to prevent scabies, lung conditions, piles, bronchitis, stomach tumours, cough, eye issues, and other conditions
Mucuna pruriens		*Mucuna pruriens* (Fabaceae) is a well-known herbal remedy that is used to treat neurological problems, and male infertility, and as an aphrodisiac

that Ageratum conyzoides' crude extract has antibacterial activities [27]. Using the zone inhibition approach, the capability of the methanol leaf extract of *Balanites aegypti* to inhibit Shigella and Staphylococcus albus has been investigated. Staphylococcus aureus was used to confirm the antibacterial abilities of Aegypti [28].

According to Refs. [29] and [30], localized psoroptic mange infection in rabbits can be successfully treated by washing the skin lesions with fibrous palm kernel fruit waste mixed with lime, common salt, lime juice (*Citrus auraatium*), and palm oil for a few weeks [31]. Report that the mahogany seed oil (*Kyaya ivoriensis*) is useful in treating dermatitis brought on by mange and dermatophilosis. Powdered seeds from *Annona squamosa* and *Tephrosia vogelli* are efficient at preventing lice in ruminants and poultry, according to Ref. [29]. During the second day of treatment, tobacco leaf and stem extract is 100% effective against lice and continued to be so for around 56 days after the challenge in West African dwarf goats [32].

2.3 Relevance of Herbal Remedies in Aquaculture

Expanding aquaculture is typically linked to strengthening the culture, which results in congestion and poor water quality, which promotes breakouts of disease and demise [33].

To prevent and cure disease outbreaks and financial losses associated with sanitary deficiencies, veterinarian drugs are used routinely utilised in aquaculture [34]. Abuse of manufactured medications has many detrimental repercussions on both human health and the environment. The muscles of commercially produced animals develop antibiotic accumulation as a result of frequent use [35].

In addition, the use of antiparasitic drugs in bath treatments, such as trichlorfon or praziquantel, can cause resilience to arise and is bad for the environment and animals [36]. Most vaccinations are only effective against one type of disease since it is incredibly difficult to generate vaccines against several types of infections [37]. Although vaccinations are believed to be the most effective strategy to stop disease outbreaks in aquaculture, their high cost discourages fish producers from routinely using them. Alternative disease management tactics in aquaculture are becoming more and more necessary given the serious drawbacks of synthetic medications. Furthermore, because most infections are opportunistic and feed on weak or stressed animals, outbreaks of disease in aquaculture are typically related to the fitness and health of the animals. To protect fish from pathogen infections, alternative therapies should increase fish fitness and immunity [38]. As herbal remedies have been shown to offer a variety of bioactivities, including antistress, immunostimulant, and antiparasitic qualities (bacterial, fungal, viral, and ectoparasites), aquaculture may benefit from employing them as an alternative to chemotherapy [39]. Notwithstanding the recent surge in interest in the use of medicinal plants and plant extracts in aquaculture, rural fish producers have always used therapeutic herbs. The use of therapeutic herbs in aquaculture is supported by several scientific studies [40]. Plants' various active principles, including alkaloids, terpenoids, tannins, saponins, and flavonoids, have been shown to have a variety of benefits, including antistress, hunger stimulation, growth promotion, immune stimulation, and antipathogen capabilities in fish and shrimp aquaculture [41, 42]. When given to farmed fish, several plant extracts are said to increase intake and weight gain [43].

Fish treated with herbal remedies have improved immunological responses, according to [44]. A 42% increase in survival was seen in Nile tilapia (*Oreochromis niloticus*) that had been exposed to the bacterial pathogen *Aeromonas hydrophila* due to enhanced lysozyme, respiratory burst, alternative complement, and phagocytic activity [45].

The biological activities of herbal remedies with antibacterial properties are by far the most studied, according to numerous in vitro studies showing that certain plants have antibacterial action against both gram-positive and gram-negative marine bacteria [46, 47].

3 Operative Threats to Medicinal Flora

Several species of medicinal plants are threatened by both natural and human-induced reasons. However, threat levels differ from one location and one species to another.

3.1 Excessive Use for Local Purposes

A significant risk to herbal remedies, this species is human overexploitation for local use, which also jeopardises the habitats of some economically significant wild species of medicinal plants. In the wild, each of these plant species is subjected to both legal and illicit exploitation because of their therapeutic potential and are all sold locally, nationally, or internationally [48]. The greatest threat to these plant species is the abuse of some plants by locals for medical uses (such as *A. heterophyllum*, *S. costus*, and *P. kurroa*) and export to areas beyond the region [48]. A major threat to the biodiversity of rare herbal remedies is overexploitation for local use. Aconitum heterophyllum, a species with a high-risk index, needs immediate protection [49]. With these plant species, locals take a risk by collecting the entire plant or particular sections of it. The fundamental factor that causes the disappearance of economically viable herbal treatments is excessive and unchecked exploitation [50]. Dar and Naqshi [51] claim that decades of uncontrolled collecting of medical plants have caused numerous species to become extremely rare or even extinct in a region. Locals take a chance with these plant species by taking the entire plant or various sections of it. Excessive and unchecked exploitation is the main factor behind the demise of commercially lucrative medicinal plants in the Kashmir Himalayas [50]. Dar and Naqshi [51] claim that decades of unfettered collecting of medicinal plants have caused numerous species to become extremely rare, if not extinct. Due to overexploitation, the therapeutic plants in Kashmir Himalayas disappear with startling speed [48]. One of the main hazards to medicinal herbs is thought to be overuse [52]. The overuse of the estimated 15,000 different varieties of medicinal herbs puts them at risk [53].

3.2 Landslides/Eroding Soil

Landslides disrupt not only ecosystems but also a substantial section of the natural population [51]. Lapcha [54] asserts that landslides significantly contribute to the vulnerability of the wide variety of medicinal plants. Landslides frequently pose a threat to several species that are only found on mountain slopes. Two examples of plant species that perform well in muddy and damp environments that are prone to landslides and pose a serious threat to their survival are *Anemone rupicola* and *Corydalis cashmeriana* [50]. Due to the landslides' changes to the soil's physico-chemical composition, other ruderal species may proliferate more quickly, which might result in the extinction of endemic species whose natural settings were formerly favourable [50].

3.3 The Excessive Grazing

Most medicinal plant species live in places that are heavily grazed; grazing animals not only consume their leaves, but also destroy the reproductive portions of the species (inflorescence, flowers, and fruits), greatly limiting the size of their populations [48, 50]. When cattle graze the flowers extensively, the majority of individuals in diverse groups are unable to produce seeds (e.g. cows, sheep, and goats) (e.g. cow, sheep, and goats). Grazing and trampling, according to studies, are the factor that is decreasing the number of sought-after herbal remedies [55, 56]. Due to the removal and damage caused by overgrazing, plant populations are at risk of extinction [57, 58]. Overgrazing alters the biological niches in addition to directly devouring them, preventing the spread of new species [59]. Overgrazing hurts the environment, causing the formation of bogs, erosion of the soil, and loss of plant variety. Overgrazing has indirect consequences on the soil biota, damage to plant seedlings, and soil compaction, among others, as well as making the surface more vulnerable to erosion. These actions increase the susceptibility to erosion [48]. The vast majority of plants today appear rare in the vast majority of grazing locations because the grazed component of the species is frequently prone to fungal infection [60].

3.4 Forest Loss and Construction Process

The number of medicinal plant varieties that may still be found in the wild has decreased significantly dropped over the past 20 years as a result of the high rate of forest destruction. The natural habitats of these plants have been harmed by deforestation, which has decreased the variety of medicinal plant species in their original ecosystems. Deforestation poses a serious threat to Southeast Asia's biodiversity, according to Ref. [61]. Many detrimental consequences of deforestation on habitat exist, such as habitat degradation, fragmentation, shrinkage of patch and core areas, and isolation of suitable habitats [62]. Moreover, fragmentation makes it possible to allow weedy plants that require light to colonise new areas encroach on their native habitat on the borders of forests [63]. The array of herbal treatments has been impacted by the random development of highways and other structures in wild environments (forests, meadows, etc.). Some species of medicinal plant native environments have been severely affected by development activity [50]. Construction work was cited by [64] as the primary cause of the lack of diversity of the species. One of the main causes of species extinction that is often emphasised is habitat degradation and fragmentation caused by construction [65].

3.5 Illegal Commerce and Flash Floods

Various medicinal plants have been lost because of the hazard that flash floods offer to different sorts of plants around the world. The terrestrial flora's submergence considerably reduces the need for new individuals and reproduction; certain populations have even been seen to be uprooted. Among the greatest dangers to these species is the recent reports of the illegal trafficking of several therapeutic herbs. These species are at risk of going extinct as a result of the excessive collection of them for smuggling outside of their native habitats [51].

3.6 Invasion of Alien Species and Forage Overharvesting

Significant populations of medicinal plants have been greatly reduced as a result of alien plant species' extensive encroachment [66]. According to the observation of Pandey et al [49], the growth of native populations of *Sinopodophyllum hexandrum* has been severely impeded by the rapid proliferation in between forest gaps. The suppression or eradication of native species by invading species has a negative influence on biodiversity and the ecological system's ability to operate normally. They can gently influence the structure and species composition of the ecosystem by changing how nutrients are cycled through it [67]. Three important herbal remedies, *A. heterophyllum*, *Sinopodophyllum hexandrum*, and *A. absinthium*, are harvested along with other types of grass that the community consumes. The bulk of these plants are chopped down during the peak flowering season (July to August), as their seeds are not available for regrowth in the subsequent growing season. The *Acorus calamus* is chopped, which puts its existence in peril because it is a desirable plant species.

3.7 Uncontrolled Tourism

Due to a growth in tourism-related activities, the allowable number of tourist attractions within various ecological zones has been surpassed, jeopardising biodiversity. Certain species of medicinal plants are in danger of going extinct, and excessive visitor traffic in popular tourist destinations also wreaks havoc on their delicate environments. Trampling is one of the negative consequences that damage the medicinal flora in its natural environment [55, 56]. Unrestricted tourism activities are the main contributors to the species' scarcity, claim [57].

3.8 Conversion of Grasslands and Forests for Agricultural Purposes

Change in land use, which is the largest threat to biodiversity globally, is also a concern for medicinal plants [45]. Land use modifications remove the medicinal plant species' natural habitat, hence reducing their range and hastening their extirpation. The alteration of land use constitutes one of the main risks to biodiversity in many nations, which also results in the isolation of suitable ecosystems [62]. One of the main risks to medicinal plants is supposedly changes in land use brought on by increased agricultural output [68].

3.9 Quarrying and Mining Process

The medicinal flora is seriously endangered by unregulated mining. Mining operations currently pose a danger to the natural habitats of major herbal remedies. According to reports, the world's medicinal plants and their ecosystems are in peril due to mining, stone quarrying, and associated activities [69]. Three endangered fern species, as well as *Calotropis gigantea* and *Azadirachta indica* in Ewekoro, Nigeria, have lately been discovered to be threatened in Chhattisgarh, India [1, 54]. Additionally, industrial pollution from cement factories, notably cement dust, has put some plant species in danger. Disturbed habitats generated substantially fewer seeds than other natural areas where the plant thrives. This may be the case because a thick layer of cement dust has been spread across receptive stigmas, thereby preventing pollen from germinating there. It is important to investigate whether heavy exposure to cement dust has an impact on photosynthesis. Together with other risks, pollution was found to pose a serious threat to biodiversity [70]. Cement dust's chemical composition and pH have a significant impact on how plants utilise carbohydrates. Additionally, cement dust promotes leaf cell surface plasmolysis, which hinders starch production [71]. According to Ref. [72], alkaline cement particulate from electric filters purportedly prevented the production of reducing carbohydrates right away.

4 Sustainability in the Source of Raw Materials

In recent years, patients have mostly embraced herbal remedies, but the demand is increasing exponentially, and natural resources cannot keep up. Regrettably, the bulk of the herbal remedies found in undomesticated settings are currently going extinct as a result of their unsustainable use and overexploitation. Some highly valued species are rapidly becoming extinct as a result of causes such as population increase, urbanisation, loss of habitat, and an increased reliance of the vast majority of people on limited natural resources [73]. These human-made actions are thought to have significantly accelerated the pace of species extinction [74, 75]. In all, 96, 951 species have been assessed for danger categories, and 26,840 of them have been designated as threatened [51]; a few risks are habitat destruction or depletion, excessive utilisation of resources, invading species, industrial development, pollution, and rapid climate change [52]. Although it takes 3–5 years for certain wild herbs to organically regenerate, this period is excessive considering the increasing demand. As a result, it is becoming more and more difficult to responsibly use herbal resources. Long-term sustainable use of herbal resources requires striking a balance between social benefits, ecological stability, and raw material market demand [5]. To address issues of diminishing natural resources and the increasing market demand for wild herbal remedies, cultivation and natural nurturing were now the primary production methods [6]. Large-scale cultivation is a quicker and more efficient method than other methods of producing the required quantity of basic medicinal components. Although cultivation requires a large amount of farm space, the main obstacles to restricting its use to all medicinal plants are heavy metals, pesticide residue, plant disease, and pests. Moreover, the raw materials' output quality occasionally falls short of the required clinical requirements. Natural fostering, also known as a wild nursery or semi-imitational culture, is an innovative method of growing herbs. By integrating economic benefit with species conservation, the tension between the need for subsistence and the preservation of biodiversity can be effectively handled. By conserving and rebuilding healthy populations of wild species, China has profited from this technique [6]. It still has limitations, though, such as a drawn-out manufacturing cycle and subpar productivity, which prohibit it from swiftly satisfying the market's escalating demand. Although the production of raw medicinal components can benefit from natural care, wild harvesting, and cultivation, these methods alone cannot fully address the prevailing problems of sustainable exploitation of herbal resources.

4.1 Medicinal Herbs Under Traditional Management

Traditional legal systems for managing resources are typically more successful than enforced ones in stopping overharvesting and other harmful practises [76]. Traditional beliefs and practises include limiting how plant resources are used. The Indigenous knowledge linked with these cultural practises has been employed for many generations to fairly utilise the plant resources despite the fragile ecosystem in which they live. Knowledge passed down from generation to generation among Indigenous peoples is strongly related to the traditional method of land management method.

Herd expansion and crop rotation were the techniques adopted. These techniques provide the elders the ability to rest the grazing areas and feed their cattle a variety of browse and grass types, giving them enough ground cover for grazing throughout both the dry and rainy seasons. Furthermore, it is culturally unacceptable to cut trees, particularly for charcoal. Instead of being felled for animal feed, bushes or trees are lopped, and leaves, seeds, and pods are thrown with sticks. The preservation of biodiversity is aided by these traditional ways of life. Simonsen [77] made the same finding among pastoralists who resided in different locations. This may show how pastoral and agricultural groups might apply their combined management skills in situations where resources usage is similar. Given the grave circumstances, it is expected that urgent research attention would be given to biodiversity conservation priority in general and therapeutic plants in particular. Prioritising medicinal plant species whose primary distribution range includes the region under consideration, which means that they are almost ever known to exist elsewhere, is the most scientifically valid method for concentrating conservation efforts at the local level [78–80]. Regional rarity, followed by local rarity, and habitat vulnerability, would need to be assigned the greatest weights to use this method [81].

4.2 Conservation Approaches for Herbal Remedies

Regional rarity would need to be given the most weight, followed by local rarity and habitat vulnerability, to use this method [81]. More resources are being removed, mostly from wild populations. The worldwide demand for natural resources has expanded by 8–15% annually during the last few decades [7, 8]. There is a point at which a species' capacity to reproduce is irreparably compromised [82]. There are various alternate sets of guidelines for the sustainability of herbal treatments; offering both in situ and ex situ conservation is one example (Fig. 2).

Fig. 2 Conservation methods for herbal remedies

4.3 In Situ Conservation

Most of the therapeutic advantages of herbal treatments are derived from endemic species, which have molecules of secondary metabolism that react to environmental cues in their natural environments, while they might not appear in laboratory settings [83]. We can protect native species, natural communities, and the intricate web of interactions that connects them by maintaining complete communities in place [84]. Additionally, conservation in the wild increases the number of species that can be maintained while also strengthening the link between resource protection and sustainable use [85, 86]. The establishment of protected areas and the adoption of an an ecosystem-focused strategy as opposed to a species-focused one have been the primary goals of in situ conservation initiatives around the world [87].

4.3.1 Natural Reserves and Establishment of Wild Nurseries

One of the primary causes of the decrease in biodiversity is the state of the ecosystem [88]. Natural reserves protect important wild herb resources and were established to maintain and restore biodiversity [89]. There are now more than 12,700 protected areas worldwide, covering a total of 13.2 million km^2 (8.81% of the planet's geographical area) [90]. Analysing the contributions and ecological roles of certain habitats is necessary to conserve herbal remedies through by protection of important wild ecosystems [91].

Every natural wild plant habitat cannot be designated as a protected area due to financial limitations and competing land uses [92]. Species-focused cultivation and domestication of endangered herbal medicines are carried out in such a wild nursery, which is constructed in a protected area, natural habitat, or anywhere that is only a short distance from where the plants generally grow [93]. Wild nurseries can provide a useful strategy for the in situ preservation of medicinal plants that are rare, endangered, and in high demand, despite the fact that overexploitation, habitat degradation, and invasive species represent major dangers to the abundance of many wild species [94].

4.4 Ex Situ Conservation

For misidentified and endangered herbal treatments with retarded growth, limited supply, and a serious vulnerability to illnesses influencing replanting, ex situ conservation is a beneficial adjunct to in situ preservation [95]. Ex situ conservation, which is usually a swift response to preserve medicinal plant resources, tries to nurture and naturalise endangered species to secure their survival and occasionally to generate huge amounts of planting material required in medication synthesis [96]. When maintained in gardens remote from their natural environments, a variety of previously wild medicinal plant species can be chosen for and protected for reproduction in addition to maintaining potent activity.

4.4.1 Botanic Gardens and the Use of Seed Banks

According to Havens et al. [95], botanical gardens can protect ecosystems to aid in the survival of rare and threatened plant species. In addition, they are crucial for ex situ preservation [90]. Although live collections often only include a small number of specimens of each species and are consequently of limited utility for genetic conservation, botanical gardens offer numerous unique characteristics [2]. They often have a flora with a large variety of plant species that are grown locally under comparable conditions and are ecologically and taxonomically diverse [97]. Since seed banks offer a more efficient way than botanic gardens to preserve the genetic variety of many therapeutic plants ex situ, Li and Pritchard [98] advocate using them to preserve the biological and genetic diversity of wild plant species. The most well-known seed bank is the Millennium Seed Bank Programme at the Royal Botanic Gardens in the United Kingdom [99]. Seed banks make it easy to obtain plant samples for analysis of their characteristics and information that is useful for protecting the remaining natural populations [98]. Two difficult problems in seed banking are how to actively encourage the regrowth of wild populations and how to reintegrate plant species into the wild [98].

4.5 Cultivation Practice

Although wild medicinal plants are generally regarded as more potent than those found in domestic cultivation, the practice of domestic cultivation is common and well tolerated [100, 101]. Horticulture presents the opportunity to apply cutting-edge methods to address concerns with the cultivation of medicinal plants, like harmful components, pesticide pollution, low concentrations of active compounds, and incorrect identification of biological origin [102]. Production stability may be ensured, and the yields of active chemicals, which are virtually invariably secondary metabolites, can be improved when cultivating under-regulated growth conditions. The perfect conditions for water, fertilisers, optional additions, and climatic factors like temperature, light, and humidity are provided through growing methods.

4.5.1 Standard Agricultural Practises (SAP)

Standard agricultural practises (SAP) for herbal remedies have been developed in order to control production, guarantee quality, and facilitate the standardisation of herbal medicines [103]. Using an SAP method ensures that herbal medicines are of a high standard, are secure, and do not pollute [104]. A wide variety of subjects are covered by SAP, such as the ecological significance of production sites, germplasm, cultivation, gathering and safety aspects of pesticide assessment, macroscopic or microscopic authentication, chemical identification of bioactive compounds, and inspection of metal elements [105]. Several nations strongly promote the usage of SAP. For instance, in regions with these therapeutic plants, Chinese authorities have pushed for the manufacturing of widely used herbal medications. The use of organic fertilisers, which continuously supply nutrients to the soil and improve soil stability, has a considerable impact on the development of medicinal plants and the synthesis of important chemicals. For instance, *Chrysanthemum balsamita* produced more biomass and had a higher concentration of essential oils than plants that did not get organic fertilizers [106]. If we want medicinal plants to endure and develop over time, organic farming is becoming more and more crucial [107].

5 Sustainable Use

When therapeutic plants that grow slowly and in low abundance are harvested destructively, resources are frequently exhausted, and even species end up becoming extinct [108]. Therefore, it is critical to think about sustainable applications for medicinal plants and develop effective harvesting methods. For medicinal plants (such as herbs, shrubs, and trees), root and whole-plant harvesting does more harm than it does for leaves, flowers, or buds. Herbal remedies derived from whole plants or roots can be safely replaced by using the leaves of the plants as medicine. Wang et al. [109] discovered, for instance, that ginseng root and leaf-stem extracts have equal pharmacological activity, with the leaf-stem extract having the benefit of being a more sustainable resource.

6 Conclusion

The use of herbal remedies and phytonutrients, commonly referred to as nutraceuticals, continues to increase quickly around the world as more people turn to these products for the treatment of various health conditions in diverse national healthcare contexts.

According to estimates, up to four billion people, or 80% of the world's population, rely primarily on herbal remedies for their medical needs. In many societies, traditional medicine that uses herbs is valued as a fundamental component of culture. The adoption of herbal remedies has been a thing of global acceptance, its usage is not only limited to rural communities but also urban centres and across the developed nations. Also, their utilisation do not limit to man only, as highlighted; domestic and aquatic animals are also involved. Several factors that threaten the sustainable use of these herbal remedies must be investigated; both the scientist and other stakeholders must work together, although in different capacities to ensure sustainability in the use of natural gifts called herbal remedies.

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Microbial Contaminants of Herbal Remedies: Health Risks and Sustainable Quality Control Strategies

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Abstract

This chapter documents currently known pathogenic microorganisms that are associated with herbal remedies as well as their health impacts on consumers and discusses the need for standardized regulatory practices as a form of quality assurance of herbal remedies. Phytomedical practices are used since time immemorial but remain crude despite a growing number of users around the world. Importantly, there is a potential public health emergency as some herbal remedies contain microbial contaminants and heavy metals as a legacy of crude production techniques. Therefore, there is a need to put mechanisms in place to ensure that herbal remedies are safe for human use. Herbal remedies typically contain phytochemicals that exhibit antimicrobial activities, but issues linked to the toxicity of the crude extracts, method of extraction and purification, the dosage of administration, and microbial contaminants have negative implications on the practice, the general public, and users. Some microorganisms associated with herbal medicines have been known to cause serious harm, especially to children, the elderly, and immune-compromised people. Some microbes that have been isolated from herbal remedies include *Escherichia, Salmonella, Pseudomonas, Streptococcus, Staphylococcus, Klebsiella, Bacillus, Enterococcus, Proteus, Citrobacter, Moraxella, Serratia, Citrobacter, Enterobacter, Shigella Candida, Aspergillus, Cladosporium, Rhizopus, Mucor,* and *Alterneria*. Also, some harmful materials like heavy metals, dust, pesticidal residues, microbial contamination, and toxins have been isolated from adulterated herbal remedies. There is an urgent need to adopt more strategic quality control practices to reduce the density and diversity of microbes and other contaminants found in formulated and ready-to-use herbal medicines. Some important strategies would be knowledge sharing, formulating policies, and pharmacovigilance, which are capable of ensuring the quality, safety, and effectiveness of traditional and complementary medicine in the national health systems.

Keywords

Contaminants · Safety · Public health · Toxicity · Microorganisms · Quality control · Policy framework · Herbal remedies · Microbial contaminants

Abbreviations

%	Percentage
≤	Less than or equal to
CFU/ml	colony-forming unit per millilitre
FDA	US Food and Drug Administration
FTIR	Fourier-transform infrared spectroscopy
GC-MS	Gas chromatography and mass spectrometry
GMP	Good manufacturing practices
HPLC	High-performance liquid chromatography
HSCCC	high-speed counter-current chromatography
HS-CCC	high-speed counter-current chromatography
TLC	Thin layer chromatography
The USA	The United States
WHA	World Health Assembly
WHO	World Health Organization
μg/kg	Microgram per kilogram

1 Introduction

For centuries, plants and their derivatives have been used for food, treatment, and management of disease conditions and control of pests. Plant resources are a major source of food for humanity. Some of the common plants that are used as food have been comprehensively documented in the literature [1–20]. Some of the plants have been documented to possess pesticidal properties particularly effective in the control of mosquitoes [21–26]. Furthermore, certain plants have been reported to be effective in the management of disease conditions particularly the ones caused by infectious agents [27–43].

The use of plants as medicine has been documented before the era of modern medical practices most notably by the Chinese, Indians, Greeks, Southeast Asia, and Africa [44–48]. This unorthodox practice was the only means of treatment for various illnesses and diseases, but now it is most commonly referred to as "*alternate medicine*" or "Complementary and Alternative Medicine." Several advances have been made in modern medicine, but with the high cost of conventional medical care, poor or non-existent health insurance, and rise in human population, people, especially in low and middle-income countries, have fallen back to the use of herbal medicine for the prevention and cure of infectious and non-infectious diseases globally [49–51]. According to a survey carried out by World Health Organization [52], approximately 80% of people mostly in developing countries depend on medicines of plant origin for their basic healthcare needs. Street et al. [53] deduced from their study in South Africa that the use of herbal remedies by a vast majority of the population may be a result of several factors including ease of access to the plants, broad knowledge among Indigenous people, and affordability.

In developed and industrialized countries, people view herbal medicine as more wholesome, organic, safe, and promotes healthy living thereby less likely to have adverse effects as opposed to modern medicines which contain inorganic components and may contain certain constituents which may have been sourced through genetic modification, thus viewed as synthetic, and this has made herbal medicine quite popular in the recent decades [54–64]. With the increase in drug resistance of infectious disease-causing agents to conventional medicine, it has become imperative to consider other options for the treatment of infectious diseases [65]. As discussed by Hossaini et al. [66], Konieczynski et al. [59], and Calixto [67], pharmaceutical companies worldwide have begun the incorporation of plant and plant derivatives into conventional medicine for the treatment of various illnesses because of their high level of antioxidants and other phytochemicals; presently, an estimated 60% of antimicrobial and anticancer medications are derived from plant-derived and available for sale, with lots of others still at the clinical trials awaiting approval for use by the general public. According to WHO, plants are the best source for pharmaceutical and health industries in the coming years [68]. China and India are the biggest and second biggest exporters of herbal medicines, respectively, in the world [69], the main trade centres for herbal medicine include Germany, France, Italy, Spain, the United States, China, and Hong Kong [70].

At the moment in Africa, the use of herbal medicine is on an all-time increase, various plant parts (in their fresh and/or dried state), preparations, and finished products are in the market today with the marketers laying claims to their products being able to cure various ailments. There are no regulatory and standardized mechanisms in place to checkmate the authenticity of these products as regards their quality, safety, efficacy, and dosage [67, 71]. This is a major public health concern as medicinal plants even though they are considered organic can become the carrier of potential contaminants like bacteria, fungi, viruses, heavy metals, and pesticide residues, among others, through various points of handling from harvesting, storage, transportation, and preparation and the presence of microorganisms in such herbal remedies may have serious health implications to consumers [53, 71–76].

As observed by Nwankwo and Olime [71], Street et al. [53], Adeleye et al. [73], and Esimone et al. [74], pharmaceutical products of any kind must be sterile; herbal products in developing countries cannot be considered as sterile as most of them are processed and packaged without ascertaining the microbial status of the plants and this will ultimately decrease the therapeutic properties of the products. Regular monitoring of the processes involved in herbal products production, packaging, storage, and dispensing is of utmost importance. The framework or guideline for the manufacturers of these herbal preparations to adhere to and sanctions for defaulters are inadequate in many regions where the use of the herb for the management of diseases is high. The setting up of the aforementioned may spur them to improve their standards and processes. Microorganisms have been reported in every environment including physical, chemical, and biological. Microbes by their natures can withstand extreme environmental conditions, hence based on temperature they have been classified as psychrophiles, mesophiles, thermophiles, and hyperthermophiles. Human commensals and pathogens are mainly mesophilic. Several microbes have been reported in the air environment [24], drinking water sources [75–77], food sources of humans including ready-to-eat foods [12, 13, 16, 78, 79], palm wine [18], Palm Weevil (*Rhynchophorus phoenicis*), Palm oil [5, 8, 20, 21], herbal medicines, etc.

This chapter documents currently known pathogenic microorganisms associated with herbal remedies and their economic importance, health implications on consumers, and the need for standardized regulatory practices as a form of quality assurance of herbal remedies. It presents an overview of herbal medicine formulation and constituents, classifications of herbal medicines, sources of microbial contaminants found in herbal remedies and their microbial density, health implications of the microbial contaminants found in herbal medicine, and sustainable quality control strategies to prevent or minimize the microbial contaminants in herbal remedies.

2 Overview of Herbal Medicine Formulation and Constituents

Herbal medicine is popular worldwide, especially among people living in developing countries where there is limited research on the products' ability to scientifically prevent or cure the diseases and ailments attributed to them [80]. The World Health Organization describes herbal medicines as plant-derived preparations which contain raw or processed ingredients got from one or more plants and has therapeutic or other health benefits to humans [28, 29]. According to Osuntokun et al. [81] and Jamilu et al. [44], herbal medicine is also often used interchangeably with "phytomedicine" and refers to the use of one or more plant components and in some cases the whole plant including the seeds, flowers, leaves, roots, bark, fruits, rhizome, or stolon for medicinal purposes through extraction, concoctions, infusions, decoctions, and in finished products. Sometimes, herbal medicines contain active substances like animal and mineral materials that are not of plant origin and are still considered natural and organic in some countries. Pakdemirli et al. [70] stated that according to the WHO database, there are an estimated 20,000 medicinal plants commonly used in the world, and China, India, and the United States are the top three producers with 4941, 3000, and 2564, respectively, while others include Vietnam with 1800, Malaysia with 1200, Indonesia with 1000, and Turkey with 500.

A large number of medicinal plants have been identified for the essential oils they have which are useful because of their antimicrobial properties, and some have also been known to reduce cholesterol and lipids in the blood [82]. According to Osuntokun et al. [81], herbal medicine is used for not just curative purposes but also for libido enhancement mainly among sexually active males, improved fertility, immune system booster, body system detoxifier, and cleanser. Currently, there is a high rate of consumption of teas which are being used interchangeably with infusions derived from plant parts and are most commonly called herbal teas as they are not got from the *Carmellis sinensis* plant [83] because they are considered to be free of caffeine [84] and have several properties including antioxidants [84], anti-inflammatory [67, 84], anticancer [66, 85], antihypertensive [84], analgesic [66], and sedative [67].

Generally, herbal medicines have received tremendous attention with 80% of the world's population depending on herbal medicines as their primary healthcare needs [86–88]. Interest in herbal drugs has been a thrust area worldwide as they are safe, broadly effective, and with less or no adverse effects. Among the approved pharmaceutical products used today, 40% of them are derived from natural compounds. Natural compounds come in different herbal formulations containing various parts of the plants, be it a leaf, flower, root, fruit, seed, stem, wood, gum, rhizome, berry, twig, bark, lichen, fungi, algae, or the entire plant [89]. These formulations constitute one or more plant components or processed plant components at certain quantities of their bioactive compounds to provide the essential nutrient values to treat various ailments. Herbal formulations have been widely accepted as therapeutic agents exhibiting varied properties such as anti-inflammatory, antioxidant, anti-cancer, anti-diabetic, antihypertensive, anti-fertility, antibacterial, antifungal, antiviral, hepato-protective, anti-parasitic, anti-mutagenic, antidiarrheal, analgesic, and anti-aging as well as immunomodulatory [90–94]. Thus, the use of herbal medicines with complementary and alternative medicines for the treatment of several metabolic diseases as well as infectious and non-infectious diseases continues to grow worldwide [88]. In major populations of developing countries such as India and China, herbal medicine remains an integral part of their cultural practice and beliefs [95].

Plant compounds constitute phenols, flavonoids, phytosterols, terpenes, alkaloids, tannins, essential oils, vitamins, glycosides, fatty acids, and many other bioactive compounds which together put them as a very rich medicinal formulation. In almost every plant, phenolic compounds are the most abundant secondary metabolite and are chemically very heterogeneous. Derivatives of plant phenolics include phenolic acids, flavonoids, tannins stilbenes, and lignans, where flavonoids are the most abundant polyphenols. Among the flavonoids, catechin is present in tea, naringenin in grapes, daidzein, genistein, and glycitein, the main isoflavones in soybean, quercetin present in apples, and broccoli, and anthocyanin in almost all berry fruits. The functional hydroxyl group present in phenolic compounds acts as a hydrogen donor and, thus, can scavenge free radicals making them an excellent antioxidant [96] with a vast spectrum of biological activities. Herbal formulations containing fruits and vegetables in the form of juices or beverages can also accumulate high concentrations of phenolic acids, flavonoids, carotenoids, anthocyanins, coumarins, alkaloids, polyacetylenes, saponins, terpenoids, and tocopherols, and thus they become highly antioxidant, anti-inflammatory, anti-carcinogenic, and anti-aging [97].

Herbal formulations undergo different pre-extraction and extraction procedures to preserve the properties of the active biomolecules. Often the dried samples are preferred over the fresh samples as it helps in the removal of moisture and does not deteriorate faster than the fresh ones. Drying also prevents bacterial activity and restricts fungal growth; however, essential oils are often found in fresh samples. Plant samples can be air-dried, freeze-dried, or can be dried in a microwave or oven. Air drying is a lengthy process that can take 3–7 days to months or even a year depending on the type of plant. Drying in the microwave takes the shortest time; however, it may cause the degradation of essential phytochemicals [98]. Freeze drying is based on the principle of sublimation, a complex and expensive method of drying [98]. Once dried, the samples can be in powdered form by grinding to reduce the particle size. Powdered forms give more homogenized and uniform substances and will increase the rate of interaction between samples and solvents through the increased surface area. Smaller particle size <0.5 mm is considered ideal for efficient extraction [98]. For plants chosen based on their traditional practice, the same procedure of pre-extraction as mentioned by the traditional healers to get similar efficacy should be followed [99].

Extraction is one of the critical processes in separating and identifying the chemical components. Based on the polarity of solutes, the solvents chosen for extraction and fractions are obtained in a sequential process. The most commonly used solvents are water, ethanol, and methanol; however, chloroform, ether, acetone, and dichloromethanol are also used. The bioactive components are extracted depending on the solvent being used, for example, methanol and water can often elute tannins, terpenoids, anthocyanin, and saponins. Ethanol can extract tannins, terpenoids, polyphenols, flavonoids, and alkaloids. The choice of solvent mostly depends on what type of bioactive compound is desired, and then, bioactivity-guided fraction can lead to a large array of possible outcomes. To extract hydrophilic compounds, polar solvents such as methanol, ethanol, or ethyl acetate and for lipophilic compounds mixture of dichloromethane and methanol (1:1) or only dichloromethane are considered [100]. There are several conventional methods such as homogenization, maceration, infusion, digestion, decoction, percolation, and Soxhlet extraction [101–105]. Homogenization, maceration, infusion, digestion, and decoction are mostly based on the principle of soaking/dissolving the ground/powdered plant compound into the solvent and either agitating/shaking/heating or kept still for a certain duration and then filtering, whereas specific apparatus is being used in the case of percolation, hydro-distillation technique, and Soxhlet extraction, where the dried/ground/powdered plant material is added along with the solvent. Common solvents used in Soxhlet extraction are methanol, ethanol, or a mixture of alcohol and water [106, 107]. However, there are certain disadvantages of these conventional techniques as they usually require large amounts of organic solvents, and often high temperatures may lead to the decomposition of the active compound.

The non-conventional/modern techniques such as microwave-assisted extraction, ultrasound-assisted extraction, pulsed-electric field extraction, enzyme-assisted extraction, pressurized liquid extraction, supercritical fluid extraction, and surfactant-mediated techniques have several advantages over the conventional ones [102–106, 108]. These techniques improve extraction efficiency and reduce solvent consumption yields. Apart from these, there are several other chromatographic techniques such as column chromatography, thin layer chromatography (TLC,) and high-performance liquid chromatography (HPLC) which yields pure bioactive compound with the minimum solvent requirement. The final extracted plant compound can be in different formulations and administered in different doses depending on its bioavailability, potency, and toxicity. It can be in the form of the tincture (usually alcohol and water extracts), decoction (filtrate of boiled herbs in water), herbal glycerates (extraction in glycerine), capsules (solid dosage usually contained in gelatin container), tablets (a formulation containing glidants, lubricants, or binders), oxymels (sweet and sour herbal formulation with honey and vinegar), lozenges (sweet-flavoured solid preparation that dissolves slowly in the mouth), ointments/balms/creams (semi-solid preparation usually for external use), herbal suppositories (solid dosage meant for insertion into the rectum), herbal pessaries (meant for insertion into the vagina), herbal poultices/plasters/fomentations/compress (wrapping the mashed herbs or soaking a gauze/cloth in tincture or infusion over the skin), herbal ointments (warming massage medium), essential oils, juices, processed exudates, teas, syrups, soaps, powders, and pastes [109–117]. The finished herbal products need to undergo several tests to exclude microbial contamination, and toxicity, and to maintain the preservative efficiency, stability, and shelf life [111, 118–120]. The sections outlined herein will shed light on some of those parameters related to microbial contamination in herbal products.

3 Classification of Herbal Medicines

Herbal medicine is a broad term used to cover herbs, herbal materials, and finished herbal products [47, 121–123]. Herbal medicine can be classified into herbs, herbal materials, traditional herbal products, and standardized herbal products. However, according to Ghislenia et al. [122] a plant or a part of a plant like a stem or leaves can be considered herbal material and a finished product if it is well packaged and sold.

3.1 Herbs

Herbs refer to plants or plant parts that are used for medicinal purposes for healing [124, 125], supplementation of nutrients and prevention of illnesses and diseases, and as flavour enhancers either savoury or aromatic in food [126–128]. According to Ezekwesili-Ofili and Okaka [121] and Nikam et al. [47], herbs include roots, bark – mango, stems, fruits, seeds, and flowers – *Moringa*, leaves – neem, rhizomes – ginger and turmeric, bulbs – onions and garlic.

3.2 Herbal Materials

These are mainly raw materials from plants like their juices, gums, sap, essential oils, and resins considered to be medicinal which may be used immediately and/or dried, boiled, roasted, or milled into powders and used singly or in combination [129, 130]. According to Nikam et al. [47], these materials are processed by various local procedures and vary according to the country and the ailment they are targeted to treat, which may be steamed, roasted, or mixed with liquids as carriers.

3.3 Traditional Herbal Preparations

Herbal preparations include tablets or pills, capsules, syrup, drops, and waters made from crude herbal extracts in which the active properties have not been chemically separated [47, 131].

3.4 Standardized/Finished Herbal Products

These are products that contain extracts or other substances that have been purified; they can be combinations of more than one plant and could contain oils, and mineral compounds, and are systematically processed [69, 131]. They are termed mixed herbal products if more than one herb or herbal preparation is used. Any product that contains active substances that have been chemically profiled including isolates from other herbal materials and synthetic compounds cannot be termed herbal medicine [47]. According to Chandra et al. [69], and Indrayanto [45], the processes to achieving a standard herbal product must be thorough and its quality assurance guaranteed; products that contain genetically modified plants, chemically modified and purified compounds, and by-products of yeast, bacteria, and other organisms are not considered as herbal medicine. The processes involved from the harvest, storage, extraction, and purification/concentration for herbal products to be accepted in the global market according to Astutik, et al. [132] include three important sets of quality standards viz.: Good Agricultural Practices, International Standard for Sustainable Wild Collection of Medicinal and Aromatic Plants, and Good Manufacturing processes.

There are notable risks associated with the use of herbal remedies, mainly from the quality of the herbal preparations and finished products [123]. Astutik, et al. [132] observed from studies that in some Asian countries, standards for medicinal plants have been firmly established especially in Japan and China using Good Agricultural and Collection Practices, and six countries have Pharmacopeia on medicinal drugs including Indonesia, India, Bangladesh, Sri Lanka, Vietnam, and Thailand while nine countries have well-documented monographs on herbal drugs, namely, India, China, Korea, Indonesia, Malaysia, Vietnam, Myanmar, Thailand, and Sri Lanka. From investigations by Osuntokun et al. [81], in Nigeria traders and herbalists are not keen on revealing the formulae for their remedies; hence, they cannot be thoroughly assessed to ascertain their quality and safety for consumption. In Europe, there is the European Pharmacopeia which is used as the standard for all herbal products.

4 Sources of Microbial Contaminants Found in Herbal Remedies

Many factors can contribute to the poor quality of herbal remedies, from the point of harvest to the preparation of the products. Good manufacturing practices (GMP) are required for the production of herbal remedies to reduce the possible risks and damage that may arise from these products to guarantee not just the quality but also the safety and efficacy of the products. They include identifying the species of medicinal plants correctly, good hygiene, and sanitation methods for the materials used during the process but especially for the starting materials [88]. These contaminants can be further categorized into external and internal factors (Fig. 1).

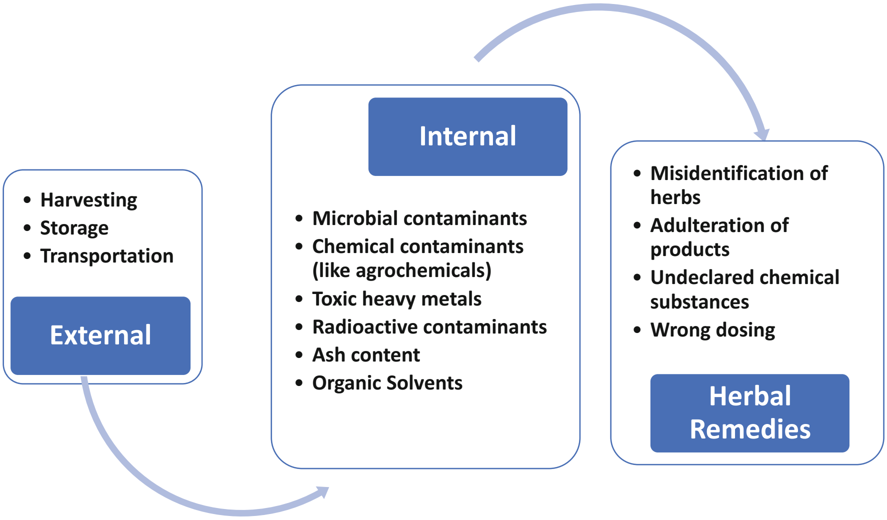

Fig. 1 Contaminants affecting the quality of herbal remedies. (Adapted from [124, 133, 134])

Microorganisms are ubiquitous, and it is nearly impossible to have products that are fully free of them. Regulatory bodies all over the world have set standards by which microorganisms can be tolerated. Microbial contamination of herbal remedies is a cause for concern as they create more illnesses and diseases for consumers rather than healing them. Many medicinal plants harbour naturally occurring microorganisms that are innocuous to the plants but can become pathogenic to humans by reducing the quality of the plant and causing spoilage. The microorganisms can also come from air and soil which carry millions of them [134]. According to Lutoti et al. [131], a broad range of microbial contaminants from bacteria, fungi, and viruses can be linked to medicinal plants and the extent to which these organisms multiply on the plants depend on the environmental dynamics which may impact the quality of the herbal preparation or product they are used to make. It is therefore imperative that quality control measures are put in place to assess products and reduce the risks for consumers. The presence and proliferation of microorganisms are determined by a range of factors from the plants themselves (i.e. their innate properties) and from extraneous sources.

Some plants may have less microbial load than others due to their morphological characteristics; plants with more distance from the soil may likely have fewer microorganisms than one that is close to the soil, and plants with tougher exteriors like hardened barks and exocarps may not be ideal for microbial growth. Some plants also have antimicrobial properties that will inhibit the growth of certain microorganisms on them; examples include fennel, and basil [134]; and several plants have been known for this which has led to the harnessing of these plants for use as therapeutics. Some plants have antioxidant properties which may restrict the growth of some microorganisms. Climatic factors such as humidity, precipitation, and temperature play a major role in the multiplication of microorganisms during the pre-harvesting and post-harvesting periods through poor handling practices and poor storage of both crude and processed plant materials. In tropical climates, mesophilic microorganisms multiply more as the temperature is generally optimum for their growth [71]. Street et al. [53] observed from their study in South Africa that most farmers, collectors, and traders of medicinal plants do not properly dry the plants which is the most used means of preservation; thus, high records of mould and yeast growth were observed.

5 Microbial Classes Found in Herbal Remedies

Microbial contaminants in any pharmaceutical product may reduce the potency of the product and can potentially cause adverse effects on the consumer; thus, it is expected that all medicinal products be assessed for their microbial quality before authorizing for consumption [71]. According to Kunle et al. [124], materials of plant origin have higher microbial contamination levels than those that have been synthesized, it is expected that herbal medicines carry some load of microorganisms mostly bacteria and moulds arising from the soil and plants where harvesting was done or from poor methods of handling, processing, and storage, and also natural microflora from the air predominantly aerobic spore-forming bacteria are predominant. Pharmacopoeias including the European Pharmacopeia, US Pharmacopeia, and British pharmacopeia have established the limit values for several microbial species that can be tolerated in herbal remedies.

5.1 Total Viable Counts

The total viable counts indicate the level of mesophilic aerobic bacteria in a given sample, according to the route of administration and the class of the herbal product. WHO Guidelines (2007) state that for herbal materials like teas and infusions that have been pretreated (e.g. with heat) or other materials like pastes and salves to be administered topically only, the acceptable limit for total aerobic bacteria count must be a maximum of 10^7 CFU/g or ml of the medicine, while for herbal materials that are ingested, the maximum limit is 10^5 per gram, and for herbal medicines that will require the use of boiling water for steeping before use, the limit is 10^7 CFU/g or ml. According to the European Pharmacopeia [135], the acceptable limit of total bacteria is as follows: 10^3 CFU/g or ml for non-aqueous oral preparations, 10^2 CFU/g or ml for aqueous oral preparations, 10^2 CFU/g or ml for remedies used for cutaneous treatments and vaginal treatments, and 10^4 CFU/g or ml and 10^7 CFU/g or ml for oral preparations containing raw material which is not pretreated and herbal products in which boiled water is added before use, respectively. The British Pharmacopeia [136] gave a general limit of 10^5 CFU/g or ml for all herbal medicines.

5.2 *Salmonella* Counts

The *Salmonella* species are pathogenic bacteria that arise from poor methods of processing and handling because they are widespread; the most well-known diseases caused by *Salmonella* species are diarrhoea-like illnesses like salmonellosis, enteritis, and typhoid fever. However, due to the high degree of virulence, the European Pharmacopeia [135] and the British Pharmacopeia [136] recommended that *Salmonella* sp. must be absent in all herbal medicines at any stage intended for ingestion; i.e. all herbal materials, preparations, and finished products must not contain any trace of *Salmonella* in them. Herbal preparations that are used topically or pretreated must also not have any colony of *Salmonella* as they can become opportunistic pathogens to the users.

5.3 *Shigella* Counts

As with *Salmonella*, *Shigella* is also a pathogenic bacterium and should not be found even in trace colonies in any given herbal remedy. According to WHO [137], also responsible for diarrhoea – illness – shigellosis is the second leading cause of childhood morbidity and mortality in Sub-Saharan Africa and South Asia; in recent times, there has been a notable increase in antimicrobial resistance in enteric bacteria particularly *Shigella*; hence, it is not expected to be identified in any form of herbal medicine, neither herbs, herbal materials, and herbal preparations, nor finished herbal products intended for use internally at any stage [52, 136].

5.4 *Staphylococcus* Counts

Staphylococcus species are normal skin flora in humans but are also a common cause of infections [138]. It accounts for minor ailments like boils and impetigo and can also cause serious diseases like septicaemia, staphylococcal gastroenteritis, pneumonia, folliculitis, and staphylococcal scalded-skin syndrome. It has also been reported to account for 49.2% of bacterial isolates in herbal medicines [139, 140]. Through poor manufacturing and handling practices, herbal medicines can become contaminated with *Staphylococcus* spp from humans [141]. According to The European Pharmacopeia [135] and British Pharmacopeia [136], *Staphylococcus* must not be present in herbal medicines for cutaneous, vaginal, and/or oral use i.e. *Staphylococcus aureus* must not exceed 1 CFU/g or ml.

5.5 Total Coliform

According to the WHO guidelines [52], when some Gram-negative bacteria are damaged, they produce endotoxins in their outer cell membranes which are released, and these endotoxins prompt antigenic response which causes resistance to infections of bacterial origin. Coliforms are intestinal bacteria typically associated with human faecal contamination; according to Kporou et al. [142], the presence of coliform indicates poor hygienic conditions during the preparation and storage of herbal medicines. Coliform bacteria include *Escherichia coli, Enterobacter,* and *Klebsiella;* according to Ameri et al. [143], they are found in the human gut and could be pathogenic especially certain strains of *Escherichia coli* which causes food poisoning; and in Iran's national standard, they are allowed to exceed 10^2 CFU/g or ml. WHO Guidelines [52] require a 10^4 CFU/g or ml limit of *Escherichia coli* for raw medicinal plants and herbal materials that require additional processing; for pretreated herbal materials, the limit is 10^2 CFU/g or ml (*E. coli*) and 10^4 CFU/g or ml (other enterobacteria), for herbal materials to be used internally and for herbal medicines that require boiling water to be added 10 CFU/g or ml (*E. coli*) and 10^3 CFU/g or ml (other enterobacteria). The European Pharmacopeia and British Pharmacopeia require a total absence of *E. coli* and other coliforms from all herbal medicines.

5.6 Fungi Counts

Yeasts and moulds are the most commonly found fungal species in herbal medicines, and they proliferate extensively in materials where moisture content and temperature are not properly controlled. The WHO Guidelines [52] expect that for herbal materials and raw medicinal plants intended for processing, mould yields must not exceed 10^5 CFU/g or ml, for pretreated herbal materials and for those that boiling water can be added before use 10^4 CFU/g or ml, 10^3 CFU/g or ml, respectively, for remedies intended for internal use and other related herbal medicines. British Pharmacopeia [136] states that the maximum limit for yeasts and moulds must not exceed 10^4 CFU/g or ml. According to WHO Guidelines [52] and Kunle et al. [124], some fungal species are known to produce mycotoxins which pose acute or chronic public health risks when ingested through food and/or medicines. Herbal remedies due to poor Good Manufacturing Practices (GMP) may contain mycotoxin-producing fungal species; these mycotoxins are usually produced as secondary non-volatile metabolites with low molecular weight on the crude herbal materials. Mycotoxins act in two ways by eliminating competing microbial species and aiding the invasion of the host tissues by parasitic fungal species. Fungal species most commonly associated with mycotoxins include *Penicillium, Aspergillus,* and *Fusarium,* and their toxins include aflatoxins (the most studied and recorded as a group 1 carcinogen in humans), fumonisins, trichothecenes, and ochratoxins. In infected humans, the mycotoxin severity may depend on the degree of exposure to the fungal species, the age and nutritional status of the individual, and the possible combination of other chemical agents the individual is exposed to (de Sousa Lima et al 2020). According to WHO Guidelines [52], aflatoxin levels must be closely monitored for quality control; several countries have different permissible limits, like Argentina; and herbs, herbal materials, and preparations used for teas must not exceed 5 μg/kg and must be absent for finished herbal medicines and those intended for internal use. In Germany, all materials used for the manufacture of herbal products must not exceed 45 μg/kg.

5.7 Other Microbial Species

Other bacterial species associated with herbal remedies include spore-forming bacteria, chiefly *Clostridium*, *Bacillus cereus,* and *Pseudomonas aeruginosa*. *Clostridium* species can be found in the soil, and they synthesize poisonous exotoxins which can either be in the form of enterotoxins which attack enteric cells or in the gastrointestinal tract and neurotoxins which disrupt the functions of neurons in the human body. WHO [52] states that all herbal materials, preparations, and pretreated and finished products must not have a trace of *Clostridium* in them. According to Ameri et al. [143], *Bacillus cereus* also secretes enterotoxins mainly lecithinase C, phospholipase, protease, and haemolysin which are responsible for diarrhoea-related illnesses. Iran's national standard conditions state that all herbal products must not have more than 100 CFU/g or ml of *Bacillus cereus,* while the European Pharmacopeia [135] has a limit of less than 10^3 CFU/g or ml. *Pseudomonas aeruginosa* is an opportunistic bacterium found in the environment and implicated as the causative agent for several diseases; especially in immune-compromised individuals, it is antibiotic resistant, hence making it a challenge to overcome [49]. The European Pharmacopeia states it must be completely absent from all classes of herbal medicines. Faecal streptococci are pathogens associated with intestinal tract infections; according to Kporou et al. [142], herbal medicines that have colonies of faecal streptococci have likely been in contact with animal or human faeces because like *Escherichia coli* other coliforms are indicators of poor personal and process hygiene making them unsuitable for use. From European Pharmacopeia [135], faecal streptococci must be absent from all herbal medicines. *Micrococcus* species are found in soil, fresh and salt water, and also as normal flora on the human skin and other animals; though not considered to be pathogenic, certain strains of the bacteria can become opportunistic pathogens causing pneumonia, bacteremia, and other diseases [144]. *Listeria monocytogenes* is found in water, soil, decaying plant materials, and in the digestive tract of humans and is known to cause meningitis in newborns and immuno-compromised people.

6 Microbial Density Found in Herbal Medicine

From previous studies (Table 1), several bacterial and fungal species have been recorded in herbs, herbal materials, and finished herbal products which are above the recommended standard by WHO and other pharmacopoeias (Table 2). Some of these microorganisms are normal flora from humans which can be attributed to poor harvesting, processing, and storage methods especially fungal species that are dominant in herbal products in which proper drying and storage methods were not employed. Certain species of bacteria like *Enterobacter* and other coliforms are indicators of faecal contamination either from soil contaminated with human excreta or cross-contamination from humans to the herbs or herbal materials through

Table 1 Microbial counts from previous studies

Bacteria diversity	Fungi diversity	Status (liquid or powdered)	Location/ country of study	References
3.0×10^5	1.1×10^5	Liquid	Davo/cote D'ivoire	[142]
1.2×10^2 (crude plant) 2.0×10^4–6.3×10^6 (finished herbal product)	3×10^3 (crude plant) 1.1×10^2–3.3×10^4 (finished herbal product)	Powder Powder	Syria	[145]
5.2×10^7	2.5×10^4	Powder	Dhaka/Bangladesh	[66]
1.62×10^5	$<10^5$	Powder	India	[139]
5.03 ± 0.03–6.25 ± 0.03	3.02 ± 0.00 to 4.78 ± 0.06	Powder	Ahvaz/Iran	[143]
1.35×10^4–2.53×10^4	–	Powder	Gombe/Nigeria	[146]
1.0×10^2–1.2×10^6	1.0×10^2–6.0×10^4	Powder	Poland	[147]
2.3 (3 ± 0.5) 10.6 (14 ± 1.5)	3.0 (4 ± 1.8) 3.0 (4 ± 2.3)	Powder Liquid	Macapa/Brazil	[140]
$1.0 \pm 0.02 \times 101$–$2.3 \pm 0.30 \times 106$	–	Liquid	Upper west region/ Ghana	[51]
3.1×10^2–2.65×10^3 9×10^1–1.5×10^2	2×10^1–1.9×10^2 1×10^1–1.0×10^2	Liquid Powder	South-west and south-south/ Nigeria	[71]
>104	>103	Liquid	Alexandria/Egypt	[148]
0–25.0×10^8	3.0×106–3.5×108	Liquid	Minna/Nigeria	[46]
1.1×10^1–2.6×10^2	3.2×10^2–4.5×10^5	Powder	Benin City/ Nigeria	[125]
1×10^2	1×10^3	Liquid	Bangladesh	[141]
1.3×10^2–5.6×10^9	2.8×10^2 to 8.6×10^6	Powder	Peshawar/Pakistan	[149]
1.0×10^7–1.8×10^8	–	Powder	Kaduna/Nigeria	[150]
2.6×10^8	1.1×10^4 (yeasts) 2.0×10^4 (Moulds)	Powder	Cairo/Egypt	[151]

vectors. Some examples of bacterial and fungal counts in herbal remedies from previous studies are presented in Table 2.

7 Health Implications of Microbial Contaminants Found in Herbal Remedies

Medicines are expected to reduce the burden of infectious diseases not to contribute to it; hence, it is alarming when medicinal and therapeutic remedies are found to be contaminated with the microorganisms they are expected to annihilate or reduce. This is a public health concern as the majority of the world's population in

Table 2 Acceptable microbial standards for herbal materials, preparations, and finished products

Classes of herbal products	Total viable counts	*Salmonella* counts	*Shigella* counts	*Staphylococcus* counts	Total coliform	Fungi counts	*Escherichia coli*	References
Herbal medicines to which boiling water is added before use	10^7	Absence	Absence	–	10^4	10^4	10	[52]
Herbal materials that have been pretreated (e.g. boiling water is used like herbal teas and infusions) or that are used in topical forms	10^7	Absence	Absence	–	10^4	10^4	10^2	
Other herbal materials for internal use	10^5	Absence	Absence	–	10^3	10^3	10	
Herbal products, for making of infusions and decoctions using boiling water	10^7	Absence	Absent	Absence	Unspecified	10^5	10^3	[136]
Pretreated herbal products containing extracts and/or herbal drugs	10^4	Absent	Absent	Absent	10^2	10^2	Absent	
Herbal products containing extracts and/or herbal products not pretreated	10^5	Absent	Absent	Absent	10^4	10^4	Absent	

Dried or powdered botanicals	10^5	Absent	Absent	Unspecified	10^3	10^3	Absent	[152]
Powdered botanical extracts	10^4	Absent	Absent	Unspecified	Unspecified	10^3	Absent	
Containing botanical ingredient	10^4	Absent	Absent	Unspecified	Unspecified	10^3	Absent	
Herbal materials need boiling water before use	10^7	Absent	Absent	Absent	Absent	10^5	10^3	[135]
Herbal materials (not pretreated) for oral use	10^4	Absent	Absent	Absent	Absent	10^2	Absent	
Powdered preparations for oral use	10^3	Absent	Absent	Unspecified	Unspecified	10^2	Absent	
Liquid preparations for oral use	10^2	Absent	Absent	Unspecified	Unspecified	101	Absent	
Herbal drugs got through hot water extraction process	10^7	Absent	Unspecified	Unspecified	10^4	10^4	10^2	[153]
Herbal drugs got through cold extraction process	10^5	Absent	Unspecified	Unspecified	10^3	10^3	10	
Finished products for oral use	10^4	Absent	Unspecified	Unspecified	10^2	10^2	Absent	

developing countries rely on herbal medicines; it is therefore very important for relevant organisations and regulatory bodies to understand the potential harm any given preparation or product that has not been properly ascertained for its quality can have on the end users. Some microorganisms associated with herbal medicines have been known to cause serious harm to children, the elderly, and immune-compromised people who may or may not be battling with other ailments (Table 3).

Diseases arising from microbial contaminants in herbal remedies are numerous, depending on the class of contaminants. For instance, *Escherichia coli* has been reported to cause gastroenteritis, urinary tract infections, and neonatal meningitis, which mainly depends on the strain [161, 162] *Enterococcus faecalis* tends to cause urinary tract infections, biliary tract infections, ulcers, and endocarditis [146]. *Staphylococcus aureus* can cause food poisoning and gastrointestinal illnesses through the production of toxins. Toxins produced by these bacteria are heat-resistant [51, 146, 161, 162]. *Salmonella* species have been implicated in causing typhoid fever and gastroenteritis. *Salmonella* infections may present symptoms including diarrhoea, stomach cramps, vomiting, and fever [51, 161]. *Bacillus* species are characterised by diarrhoea. *Pseudomonas aeruginosa* has been reported to cause urinary tract infections, respiratory tract infections, bacteremia, dermatitis, bone and joint infections, and soft tissue infections [161, 162]. Izah and Ohimain [2] reported that respiratory tract infections, septicemia, and meningitis are occasionally caused by *Enterobacter* species; bacteremia and endocarditis are caused by some *Bacillus* species. Urinary tract infections are caused by some *Proteus* species. Endocarditis, arthritis, meningitis, intracranial abscesses, pneumonia, and septic arthritis are caused by some species of *Micrococcus*. Aspergillosis, onychomycosis, sarcoidosis, sinusitis, otomycosis, endophthalmitis, meningitis, osteomyelitis, endocarditis, myocarditis, pneumoconiosis, ankylosing spondylitis, hepatosplenic, tracheobronchitis, aspergilloma, allergic bronchopulmonary aspergilloma, rhinitis, and farmers' lung are caused by some species of *Aspergillus*; candidiasis and otomycosis are caused by some species of *Candida*. Mucormycosis is caused by *Mucor* species. Mycotic keratitis and otomycosis are caused by some species of *Penicillium*.

8 Quality Control Strategies

The herbal medicinal market is booming worldwide; with each day when a new product is being included in the market, the concern regarding quality and safety also increases. In this run, the quality standards are often compromised. It is to be remembered that every drug has its effective dose at which it shows its best efficacy; beyond or below, it may be deleterious to the subject consuming it. Thus, it becomes imperative to ascertain that the quality, stability, and toxicity level of herbal medicine is maintained at every level.

Compared to synthetic medicines, there are certain critical aspects about maintaining the quality standard of an herbal formulation starting from the origin of the plant to its clinical application. The chemical composition of the finished product may vary depending on the location of the plant, soil fertility, agricultural

Table 3 Diversity of microbes that have been reported in herbal medicine in different parts of the world

Microbes	Locations	References
Escherichia coli, salmonella species, *Pseudomonas aeruginosa,* and *Staphylococcus aureus*	Macapa, Brazil	[140]
Klebsiella, Pseudomonas, *Candida* species, and *Escherichia coli*	Dar es Salaam, Tanzania	[154]
Bacillus, Aspergillus, and Penicillium species	Southern Nigeria	[71]
Bacillus subtilis, Klebsiella pneumonia, Staphylococcus aureus, Pseudomonas aeruginosa, Enterococcus feacalis, Escherichia coli, Proteus and *Shigella* species, *Aspergillus niger, Aspergillus flavus, Aspergillus fumigatus, Cladosporium cladosporius, Rhizopus arrhizus, Mucor,* and *Alterneria* species	Gombe metropolis, Nigeria	[146]
Staphylococcus aureus, Pseudomonas, Proteus, Streptococcus species, *Aspergillus niger, Aspergillus flavus,* and *Candida* species	Abraka, Delta state, Nigeria	[155]
Escherichia coli, Pseudomonas aeruginosa, Citrobacter divergens, Staphylococcus aureus, Shigella sonnei, Moraxella catarrhalis, Serretia marcenscens, Bacillus, Citrobacter, Enterobacter, Staphylococcus, and *Candida* species	Ghanaian market, Accra	[156]
Staphylococcus aureus, Escherichia coli, Pseudomonas aeuroginosa, Shigella, and *Salmonella* species	Trichy, Tamil Nadu, India	[157]
Escherichia coli, Klebsiella, Bacillus and *Staphylococcus species*	Port Harcourt, Nigeria	[158]
Klebsiella species, Bacillus subtilis, Enterobcter species, Staphylococcus aureus, and *Serretia marcescens*	Ilorin-Kwara State Nigeria	[159]
Escherichia coli, Salmonella species, *Staphylococcus aureus,* and *Pseudomonas aeruginosa*	–	[160]

practice, harvesting season, collection time of the plant parts, post-harvest handling, preservation method of the collected parts, extraction procedure, herbal preparation, and many more aspects [163]. For example, the quality of the herbal product may vary if the leaves are not collected during bloom or the rhizomes are uprooted before the completion of the vegetation period, or the fruits and seeds are collected before ripening or attending maturity. Thus, several intricate aspects are important when it comes to maintaining originality and quality standard. For the production of herbal products at the industrial level, three very vital aspects need to be kept in mind: identification, purity, and composition of active compounds as often they are adulterated and are not at par with the standards laid down for authentic drugs. Often there are chances of wrong identification of plant material, and so voucher specimens should be considered reliable reference sources. Due to the complex nature of herbal formulations, the purity of the product is a serious concern. Indeed, one universal procedure of extraction cannot be considered suitable for every herbal compound, rather it is unique to each of them. However, advanced extraction techniques should be considered. To obtain pure compounds, different chromatographic isolation techniques as mentioned earlier in this chapter are used. Often isolated compounds are adulterated with heavy metals, dust, pesticidal residues, microbial contaminations, insect contaminations, and toxins from fungal infection, which reduces the shelf life and purity of the effective ingredients. Thus, decontamination is an essential step, which can be carried out through conventional methods such as steam heating, γ-irradiation, or using ethylene oxide gas. However, these conventional methods often interfere with the quality of the product and lead to the formation of toxic by-products [164]. To avoid these, new technologies such as decontamination with ultraviolet rays, infrared rays, ozonation, radiofrequency heating, and cold plasma are considered [165]. Ash value assessment also needs to be performed to determine the sulphate, acid-insoluble, and water-soluble ash content in the herbal product to maintain quality [166].

Complex herbal formulations need to be validated with modern chromatographic techniques such as high-performance liquid chromatography, gas chromatography-mass spectrophotometry GC-MS, Fourier-transform infrared spectroscopy (FTIR), ultra-performance liquid chromatography (UPLC), and high-speed counter-current chromatography (HSCCC). These techniques help in determining the consistency of the product. Various other modern techniques such as DNA fingerprinting and immunoassays with monoclonal antibodies are also used to evaluate the purity of the herbal compound [167]. An herbal extract is a chemically complex ingredient, and thus it is a difficult task to validate with a single analytical technique. Thus, a new term called "chromatographic fingerprint" has come into the picture, which is a comprehensive qualitative method recognized by WHO, the US Food and Drug Administration (FDA), and other regulatory bodies of different countries [168]. This technique helps in identifying various biologically active functional groups and compares the chemical composition of different compounds using the total chromatograms obtained from several chromatographic techniques (such as HPLC, TLC, GC-MS, and HS-CCC). The entire chromatograms obtained are coupled to a multivariate analysis that not only helps in determining the identity and authenticity of the herbal preparation but also helps in identifying new biomarkers and active molecules [169].

Pharmacovigilance is also a very important aspect to maintain the quality and safety of herbal drugs as often unregistered herbal formulations are marketed freely, thereby misleading people. WHO has released guidelines in the pharmacovigilance system [170, 171] where it has been addressed that the poor quality and improper use of herbal products lead to the majority of adverse events. Mostly in developing countries, it is often easy to market herbal products without any restraint leading to uncontrolled distribution. At the same time, people have the perception that herbal medicines are not toxic and will not show adverse effects. Unfortunately, this is a serious misconception as often herbal formulations are consumed without proper prescription. Poor-quality herbal medicines can cause severe toxicity, showing deleterious effects and even leading to death if the appropriate dose regime is not followed. Research studies have shown that several unauthorized herbal drugs contained high levels of metals such as mercury, lead, and arsenic. Herbal medicine adulterated with aristolochic acid can lead to serious kidney failure and carcinogenic effects. Betamethasone present in herbal cream often can cause life-threatening anaphylactic reactions [172]. Thus, for monitoring safety, herbal products must include proper labelling specifying the content, instructions for usage instructions, storage conditions, patient information, shelf life, stability, and quality control information.

Recently, the World Health Assembly (WHA) introduced the WHO Traditional Medicine Strategy 2014–2023, where impetus has been given to sharing knowledge, formulating policies, and regulating the quality, safety, and effectiveness of traditional and complementary medicine in the national health systems. It is a universal participatory approach where the government, healthcare practitioners, and consumers will have safe, cost-effective, and quality healthcare service through integrative therapeutic research [173]. Newer and strict guidelines for standardization are of utmost need. The approach has to be multidisciplinary combining pharmacology, biochemistry, analytical chemistry, microbiology, molecular and cell biology, genomics, proteomics, and bioinformatics.

9 Conclusion

As long as there have been people on earth, people have been using herbs. Herbal products and their formulation are as old as man which makes their user unavoidable especially where modern healthcare resources are scarce. These herbal remedies possess some pharmacological properties which can be harnessed to support modern and traditional health care. In recent years, interest in herbal medicine has grown exponentially which has led to an increasing number of unskilled people formulating herbal remedies solely for profit leading to a greater number of contaminants in herbal medicine products as well as many adulterated products in the sector. In the same vein, there has been a parallel growth in technology that may be used for the extraction of active ingredients using internationally standardized methods and best-known practices. Given that herbal remedies represent the first line of therapy for certain diseases for more than 80% of the world's population, prompt public health concerns about the safety of such remedies are justified. Mechanisms need to be put in place to guarantee that they are suitable for human consumption and free from microbial and other contaminants like toxins, microbial contaminants, heavy metals, dust, and pesticide residues. Plants are known to have essential phytochemicals with antimicrobial properties, but several factors must be taken into consideration, including the toxicity of the crude extracts, the extraction and purification processes, the dosage, and contamination in the different formulation and production phases. Additionally, many of the bacteria connected to herbal remedies have been known to seriously affect younger people, the elderly, and immune-compromised individuals. Some of the microbial contaminants have been implicated in several life-threatening illnesses. Therefore, it is necessary to implement more strategic quality control procedures to decrease the variety and density of microorganisms present in the herbal medications that have been created.

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Adoption and Application of Biotechnology in Herbal Medicine Practices

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Contents

Abstract

According to the World Health Organization, 80% of people in underdeveloped nations and 60% of the global population, respectively, rely exclusively on herbal medications for their basic medical requirements. The public, policymakers, and healthcare experts are all becoming more concerned about the safety, standardization, effectiveness, quality, availability, and preservation of herbal products as a consequence of the rising demand for herbal medications. The most significant challenges in studying herbal medicine are incorrect plant identification and speciation, limited yields of chemically synthesized plant bioactive metabolites,

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diversity of traditional techniques, and others. Thus, it is crucial to choose, reproduce, enhance, and study therapeutic plants and products using biotechnological technologies. Difficulties in plant species identification can be resolved with DNA-based technique and sequencing. Under some circumstances, the production of bioactive metabolites is possible, utilizing in vitro tissue culture methods. Pharmaceuticals made from proteins may be created using recombinant DNA methods that can alter metabolic pathways. Drug development from plant-based compounds may use the emerging fields of bioinformatics and genomics. This chapter covers an introduction of herbal medicine, biotechnological methods for producing herbal medicines, biotechnological methods for conserving herbal medicinal plant products, and bioinformatics methods as well as possibilities and problems.

Keywords

Biotechnology · Herbal medicine · Tissue and organ culture · Bioinformatics · Recombinant DNA · Genomics

Abbreviations

AFLP	Amplified restriction fragment length polymorphism
BA	6-Benzylaminopurine
CAM	Complementary and alternative medicine
CAMT	Complementary and alternative medicine therapy
CHM	Chinese herbal medicine
DNA	Deoxyribonucleic acid
FDA	Food and Drug Administration
HM	Herbal medicine
HPLC	High-performance liquid chromatography
KEGG	Kyoto Encyclopedia of Genes and Genomes
LC-MS	Liquid chromatography-mass spectrometry
NBPGR	National Bureau of Plant Genetic Resources
PCR	Polymerase chain reaction
R & D	Research and development
RAPD	Randomly amplified polymorphic DNA
RFLP	Restriction fragment length polymorphism
Spp.	Species
T-DNA	Transfer deoxyribonucleic acid
TDZ	Thidiazuron
WHO	World Health Organization

1 Introduction

Common natural catastrophes, food scarcity, excessive resource consumption, environmental degradation, and other issues are the main issues the world is now dealing with and significantly impede the sustainable growth of human civilization. By

2050, the nearly 6.8 billion people who now live on the planet should have doubled and about 97% of the worldwide rise in population is due to this, which is concentrated largely in emerging nations [1]. So, the problem for the future is ensuring global food security, which will require tripling production of food over the next 50 years to accommodate demand from the world population [2].

Biotechnology application in plant breeding has brought about a positive effect almost entirely on high-demanding crops including rice, wheat, maize, soybean, potatoes, and sunflower across the globe [2]. According to estimates, 70–80 percent of people worldwide depend mostly on conventional medications, many of which are herbal. Traditional medicine includes herbal medicine (HM), often known as botanical medicines or phytomedicines. Herbal medicine (HM) is the use of herbs, herbal materials, herbal preparations, and completed herbal products that include active substances made from plant parts or other plant materials [3–31]. Plant parts used in the product may include seeds, berries, roots, leaves, bark, or flowers [32]. Many of the medications used in traditional medicine were first derived from plants.

Many modern medicines come from plants whose organs or parts contain chemicals that can be used as either therapeutics [33–45] or, in a more cutting-edge concept, precursors for the production of pharmaceuticals. One-fourth or more of prescription medications are thought to include plant extracts, active chemicals derived from or based on plant components, or both, while some of the most effective anticancer medications, like paclitaxel and vinblastine, are only made from plant sources [46–48].

The demand for herbal medicine is not only substantial but also rising globally. Several technological advancements have been made to improve the bioactive chemicals found in medicinal plants. By using methods like in vitro regeneration and genetic alterations, biotechnological instruments are crucial for the genetic improvement and genetic proliferation of medicinal plants. It may also be used to produce secondary metabolites in bioreactors made of plants. Therefore, this chapter discusses herbal medicine overview, biotechnological approaches for herbal medicine production, biotechnology approaches in conservation of herbal medicinal plant products, bioinformatics approaches, and opportunities and challenges.

2 Herbal Medicine Overview

Herbalism is a kind of traditional medicine or folk medicine that relies on using plants and plant extracts to treat ailments. One of the primary forms of life on earth is plants and herbs, which make up the majority of traditional *materia medica* across the globe. There are thought to be 350,000 plant species in existence (including ferns, bryophytes, and seed plants), of which 287,655 have been recognized as of 2004 [49]. Herbs, herbal materials, herbal preparations, and completed herbal products that include plant parts or other materials as active components are referred to as herbal medicine (HM), also known as botanical medicine, phytomedicine, or phytotherapy [50]. In herbal treatment, seeds, berries, roots, leaves, fruits, bark, flowers, or even whole plants are employed as plant parts. Before aspirin was developed from *Spiraea ulmaria*, which was already prescribed for fever and swelling in Egyptian papyri and recommended by the Greek Hippocrates for pain and fever, man was primarily dependent on crude botanical material for medical needs to maintain vitality and cure diseases [51].

Archeological studies have revealed that the use of herbal medicine dates back as far as 60,000 years in Iraq and 8000 years in China, despite written records about medicinal plants dating back at least 5000 years to the Sumerians, who described well-established medicinal uses for plants like laurel, caraway, and thyme [52–54]. Herbal medicine has been contested by practitioners of mainstream medicine since the development of western medicine (or "conventional" medicine) during the last century because of the lack of empirical proof in the context of modern medicine, despite its long history of successful usage. Due to the adverse effects of chemical medications, the absence of effective current treatments for a number of chronic illnesses, microbial resistance, and the unprecedented expenditure in pharmaceutical research and development (R&D), the use of herbs has increased recently [55]. For instance, the US Food and Drug Administration (FDA) has only authorized around 1200 new medications since 1950 [56]. As a consequence, during the last 40 years, both in developing and industrialized nations, the use of herbs and herbal products for health reasons has become more widespread [57]. Additionally, with the aid of contemporary science, technology, and ideas, multinational pharmaceutical companies have started to rediscover herbs as a potential source of fresh drug candidates and have updated their strategies in favor of the development and discovery of drugs from natural sources [58–60].

Nowadays, many "conventional" medical professionals do not hesitate to suggest herbs, herbal products, or complementary and alternative medicine (CAM) therapy to their patients for the successful management of certain disorders [61, 62]. According to a poll conducted in 2007, around 40% of adults and 11% of children used complementary and alternative medicine (CAM). On adult individuals both black- and white-skinned color users made up to 25.5% and 43.1%, respectively of the population [63]. Additionally, those with greater incomes and levels of education utilize CAM and herbal medications more often [64, 65]. In this regard, a 2012 research revealed that the use of complementary and alternative medicine (CAM) substantially linked with tertiary level of education, hoping to increase the usage among young patients with breast cancer disease [66]. The public's excitement for accepting CAM/CAMT is unaffected by the fact that we currently do not completely grasp the precise facts and processes behind most traditional medicines and/or how they prevent illness [67]. CAM and CAMT are widely practiced globally; however, they may all be categorized into one of the two groups: (1) nondrug-based CAM and CAMT and (2) drug-based CAM and CAMT [68].

Previous studies on promoting the use of organic and conventional plant material for advanced food or diet treatment and healthcare practices focused on various scientific findings and discovery of traditional herbal medicine and recent trends in pharmaceutical design, including different complementary and alternative medicines (CAM) which were conducted by Pan et al. [69–71].

3 Concept of Biotechnology

The term biotechnology can be defined as the application of biological components or phenomenon and living organisms and their derivatives to produce or modify products or processes for special utilization. Hence, conventional biotechnology has been in existence right from human history. Most of these traditional biotechnological applications are the brewing of beer, baking breads, breeding animals and food crops, etc. At the advent of molecular biology, a new dimension of biotechnology has emerged, which is commonly known as new or modern biotechnology. Modern biotechnology has brought tremendous breakthrough in global economy and society [72].

Modern biotechnology is the application of plant and animal genetic information for manufacturing of product and services. Genetic engineering alters the working of genes in the same genus or species to produce genetically modified organisms (GMOs). The development of modern biotechnology has been accelerated in the twentieth century, for example, the accomplishment of human genome mapping which identifies about 30,000 genes that encode the human hereditary features – a huge achievement in biology [73].

In the healthcare system, commercialized human insulin was first produced to meet the societal needs. Others are the treatment of genetic diseases and chronic disorders using gene therapy. The production of proteins with pharmaceutical values from cloning of mammals are all very significant exploits of modern biotechnology [72]. The genetic modification of plant and animals by implanting genetic materials referred as crossbreeding has been fruitful in agriculture, for example, crops like corns, oilseeds, etc. have been modified genetically [73]. Another application of biotechnology is the use of microorganisms to prevent, monitor, and remediate polluted environments. Biotechnology also enhances the manufacturing of vaccines, antibiotics, enzymes, etc. and the successful use of in vitro fertilization (IVF) in medicine, among others [74].

4 Biotechnological Approaches for Herbal Medicine Production

Field or wild plants in their natural habitats produce valuable bioactive compounds [75–77]. The availability and accessibility of these naturally derived products, however, limits the potential usage for the benefit of humanity. Utilization of medicinal plants has increased along with the secondary metabolite demand over the past years. Therefore, it is crucial to find new ways to produce high-quality plants and metabolites on a large scale. When it comes to homogeneous, controlled production, biotechnology offers a compelling substitute for whole-plant extraction, particularly when we consider the commercial demand. Additionally, regardless of the seasons and regions, it leads to high production of different herbal formulations [78–81]. By cultivating different plant tissues or whole organisms using in vitro techniques which can genetically alter the cells or organs to produce beneficial secondary metabolites, biotechnology provides an opportunity to profit from them [82]. There are several biotechnological techniques that may be used to increase the synthesis of secondary metabolites from medicinal plants, including embryogenesis, organogenesis, cell line screening, media modification, and elicitation. The sections that follow briefly go over several in vitro cultivation methods that may be utilized to create herbal medicines**.**

4.1 In Vitro Culture Techniques

See Fig. 1.

4.1.1 Micropropagation

The manufacture of high-quality plant-based medications has a lot of promise, thanks to in vitro plant reproduction [83]. With the use of micropropagation techniques, this process can be accomplished. Traditional plant propagation methods have several limitations when compared with micropropagation techniques which provides so many advantages [84]. The increase of plant production rate has been made possible due to micropropagation. Moreover, the techniques produce materials that are free from pathogenic microorganisms. Several medicinal plants can be known to micropropagate [85–90].

According to reports, a variety of conditions may affect how well certain medicinal plants can be propagated in vitro [91–94]. The plant hormones cytokinins with auxins can affect the growth rate of several medicinal plants' shoots. The compound 6-benzylaminopurine (BA) can increase the growth rate of *A. belladonna*'s plant

Fig. 1 In vitro culture techniques

with higher concentration of 1–5 ppm according to Benjamin et al. [95]. Using 1.0–5.0 mg/l of kinetin, Lal and Ahuja [96] discovered fast rate of multiplication of *P. kurroa*. Moreover, the regeneration of several plantlets from male medicinal yam using florescence on 13.94 mM kinetin-enriched media has been shown [97]. Using 2.2 mM of thidiazuron (TDZ) medium can produce higher proliferation of *N. foetida* [98]. In line with the above, the plant hormone cytokinin is very significant for shoot proliferation in a lot of genotypes but with the addition of auxin concentrations can increase the rate of shoot proliferation [99–101].

The use of micropropagation methods for medicinal plants assists breeders in many ways since it makes it possible, as demonstrated in Fig. 2.

The term "micropropagation" refers to the tiny branches or plantlets that are initially generated by this form of plant multiplication. This process offers a quick and dependable way to produce genetically pure plantlets. This is an important aspect of commercial plant propagation made by plant tissue culture, and it has significant implications [102]. The development of vegetation and its multiplication from plant known as micropropagation includes numerous in vitro aseptic culture methods that allow plant components to be artificially cultured on nutritional media [103]. It is designed to support a range of growth patterns, such as plant regeneration, cell multiplication, and organ development. The idea of transforming plant cells to new plants is the foundation of micropropagation [104, 105].

Many plants have extremely poor seed germination rates, produce blooms and seeds only when certain climatic conditions are fulfilled, or take a very lengthy time to mature and multiply. Since this technique enables the conservation of a broad

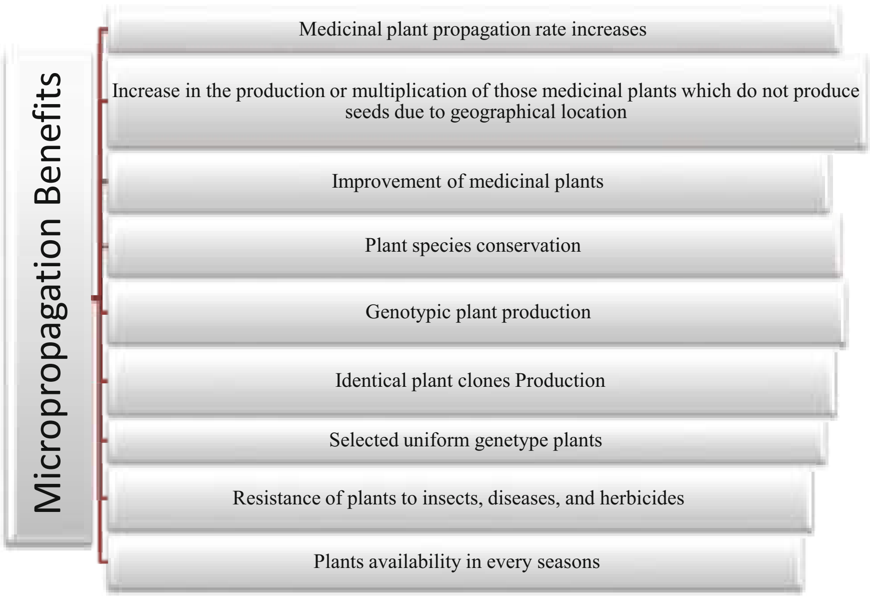

Fig. 2 Micropropagation benefits

genetic background, using seeds as the beginning material should be more appropriate for species that thrive in the wild [106]. Using the least amount of area and time possible, micropropagation provides a constant supply of therapeutic plants [107]. In order to cultivate and investigate the physiological behavior of isolated plant organs, tissues, cells, protoplasts, and even cell organelles under carefully regulated physical and chemical circumstances, micropropagation has now become a well-established technology [108]. Using in vitro techniques to preserve plant material has several benefits over keeping live collections in field condition [106]. Figure 3 illustrates the benefits of micropropagating medicinal plants.

Basic requirements are met by plant genetic resources, which also aid in finding solutions to issues like hunger and poverty. The most significant and necessary fundamental raw materials for crop development programs' present and foreseeable future demands are plant genetic resources. A number of factors, such as habitat fragmentation, overexploitation (overgrazing and excessive harvesting), competition from exotic species (accidental and deliberate introductions), changes in land use (deforestation and land clearance), population growth, and climate change, are all contributing to the genetic erosion that is causing them to disappear. The growth of contemporary commercial agriculture and the substitution of contemporary hybrid plant varieties for traditional farmer plant types during the last 40 years have been the primary causes of genetic loss. The circumstance justifies stepping up efforts to

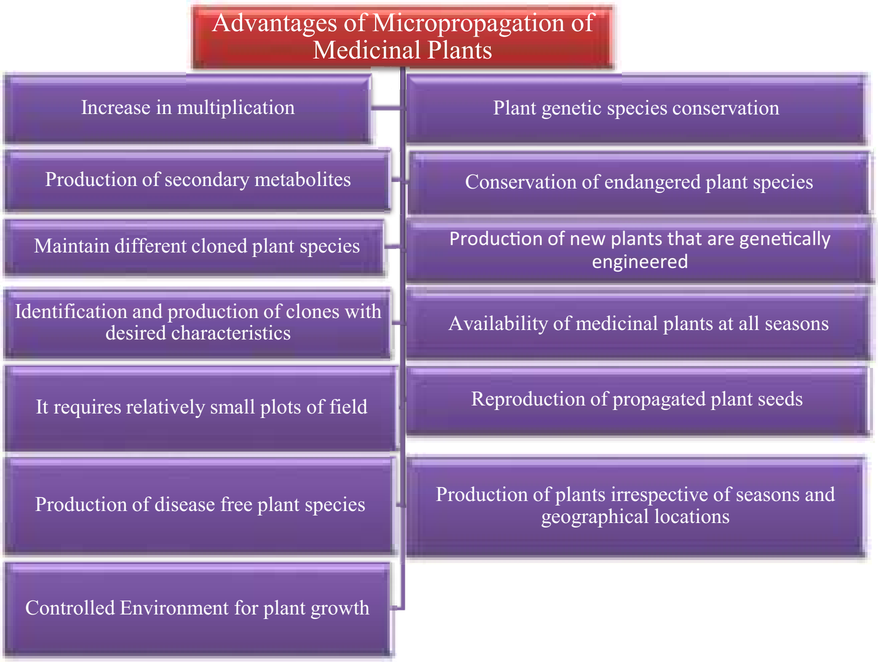

Fig. 3 Advantages of micropropagation of medicinal plants

create strategies for their germplasm preservation. We must protect them due to their critical relevance for the current and the new generations to come [109].

The management and conservation of plant variety have benefited greatly from improvement of biotechnology of plant breeding, particularly the ones relating to genetic engineering and tissue culture techniques. Nowadays, disease-free plant materials, special plants, and genetic variety may be preserved for short, medium, and long terms, thanks to biotechnological approaches employed to conserve endangered, uncommon, decorative, medicinal, and forest species. In an aseptic setting, cell culture methods enable the fast growth of plant parts. In vitro plant culture system has been widely established for regeneration and propagation of more than 1000 distinct vegetative species which include several scarce and vulnerable species, using two alternate morphogenic processes: shoot organogenesis or somatic embryogenesis.

Micropropagation is a technique for multiplying plants by cultivating extremely minute components. The preservation of these priceless genotypes is essential since the widespread harvesting of medicinal plants from their native habitats is depleting plant resources. Such germplasm may be effectively preserved by micropropagation. Another strategy to increase the plant's ability to produce drugs is genetic improvement [110].

4.1.2 Organ Cultures

Depending on the outcomes, the best approach should be chosen. Dedifferentiated (callus) cultures are not preferred for medicinal plant species which are localized in specialized tissues, although organogenic cultures that have been generated would be beneficial. Redifferentiation is often linked to better secondary metabolite production in in vitro settings [111]. This is likely caused by the emergence of complex, metabolically more capable cells and tissues. Even though it is nonmorphogenic, it may exhibit some cell-level differentiation and, as a result of coevolution, mimic the biochemistry of redifferentiated cells [112]. According to findings on *A. annua* and *A. indica*, redifferentiation was necessary for the formation of compounds since the production of artemisinin was extremely low in uniform callosity cultures [113].

In addition, it was shown that organogenesis was a necessary condition for *Ruscus aculeatus* to produce steroidal saponin. Similar findings were reported with the production of *picroside* in *Picrorhiza kurroa*, where it was shown that the metabolite only appeared in the redifferentiated cultures which cannot accrue in uniform callosity culture. Moreover, Berkov et al. [114] showed that tissue differentiation and alkaloid production are tightly connected in *Pancratium maritimum*.

There were attempts for regeneration of a whole plant tissue such as roots or shoots in favorable conditions from explants or through an intermediary callosity stage, due to the observation of higher production of secondary metabolite from well-arranged plant cells. As might be predicted, these regenerative cultures generate secondary metabolite patterns that are comparable to those of the parent plant cultivated in the field, with the extra benefit of increased metabolite synthesis. Another benefit of utilizing structured cultures is that they produce secondary metabolites more steadily when compared with uniform cultured tissues in either suspension or callosity cultures [82].

4.1.3 Callus Cultures

A uniform group of tissues or cells derived from explants implanted in vitro inside of a media with comparatively greater auxin concentrations or a mix of auxin and cytokinin concentrations at equal levels is referred to as a callus culture. Plant parts that are made of beneficial bioactive compounds can be therapeutically utilized after plant extraction process using in vitro techniques. High-performance liquid chromatography and liquid chromatography-mass spectrometry techniques can be employed for identification and quantification of therapeutic compounds from various plant species [113].

There are several biotechnological techniques that may be used to increase the synthesis of secondary metabolites from medicinal plants, including embryogenesis, organogenesis, cell line screening, media modification, and elicitation. The sections that follow briefly go over several in vitro cultivation methods that may be utilized to create herbal medicines.

4.1.4 Suspension Cultures

With the effective development of cell lines capable of synthesizing secondary chemicals in cell suspension cultures in high yields, a breakthrough in cell culture technology was achieved. This strategy has attracted corporate attention during the last several decades. The idea of the synthesis of plant tissue totipotency means that different tissues or cells in a culture media preserve the full hereditary information for producing the variety of metabolites in the entire plant, which is the foundation for the technique of using plant cell suspension cultures for the production of secondary metabolites [115].

By inoculating them into liquid medium, established callus cultures are transformed into cell suspension cultures. After that, the cultures are maintained in glass flasks while being constantly shaken by rotatory shakers; ultimately, it may be moved into a dedicated fermentor. Since oxygen and nutrients are more effectively mixed under shaking conditions, cells in suspension cultures develop significantly more rapidly than those in semisolid medium. The implementation of plant tissue technology for the generation of secondary compounds depends on the productivity of suspension cultures. There are a number of methods that may be utilized to boast the synthesis of bioactive compounds in in vitro cultures, including altering the environmental and media characteristics, choosing high multiplication clone cells, feeding precursors, and elicitation [113].

4.2 Genetic Transformation and Production Technology of Medicinal Plant Products

In plant production of crops, recombinant DNA technology or biogenetics plays very vital roles in their proliferation. The technology can be used to solve several agricultural problems that traditional plant breeding cannot able to solve. There have been improvements recorded by recent research of bioactive compound productions using recombinant DNA technology. At present, using *Agrobacterium*-assisted methods has able to generate about 150 plant species from nothing less than 35 plant families of different types of plants [116–119].

Two plant diseases, namely, hairy root and crown gall tumor are caused by *A. tumefaciens* spp., a gram-negative soil bacteria related to *A. rhizogenes*. While there have been examples of monocotyledonous plants being infected, these species, which are members of the *Rhizobiaceae*, mostly alter or change dicotyledonous plants [120–125]. The genetics involved in the production of the plant hormones (cytokinins and auxins) tumors and opiates are two different types of T-DNA genes.

Several plants especially dicotyledonous have frequently used *A. rhizogenes* for genetic transfer. This gram-negative bacteria promotes infections in plants which results in the multiplication of different branched roots, or "hairy roots," at the site of the plant infection [125]. The study was focused on the use of hereditary plant modification and transformation applying the soil gram-negative bacterial *A. rhizogenes* to enhance the production of the beneficial bioactive compounds, which are synthesized in the mother plant roots.

Various hairy roots that have been converted often show greater production rates and replicate the biochemical mechanisms that exist which makes it active in the regular root products. Many therapeutic plant species have claimed to pass through several genetic changes. According to Naina, Gupta, and Mascarenhas [126], genetically modified *Azadirachta indica* plants were successfully regenerated with the use of *Agrobacterium tumefaciens* strain that included variant of pTi A6 plasmid [126–128]. Using different explant leaves, agrobacterium-assisted modification of *Echinacea purpurea* was shown [129]. A very important technique necessary for making high yield of bioactive compounds is the application of genetic transformation or recombinant DNA due to its efficacy and production of hormone-free growth conditions.

Artemisia annua medicinal plant's root culture process was created via an infectious microorganism *Agrobacterium rhizogenes*, and 4.8 mg/L of artemisinin concentration is ideal for its formation [130]. Using *Agrobacterium rhizogenes*, Giri et al. [131] stimulated the hairy root growth in *Aconitum heterophyllum* [131]. Pradel et al. [132] created a method for growing altered plants from *Digitalis lanata* root explants. They tested several natural *Agrobacterium rhizogenes* strains for their ability to produce secondary products from transgenic plants and hairy roots. The altered hairy roots contained more anthraquinones and flavonoids than the untransformed roots did, according to their findings. It is stated that an effective method for creating plant cultures of the some transgenic plants, namely, *Eschscholzia californica* and *Papaver somniferum*, utilizing *Agrobacterium rhizogenes* has been developed [133]. After being transformed by *Agrobacterium rhizogenes*, the hairy roots of *Atropa belladonna* produced tropane alkaloids, according to Bonhomme et al. [134]. According to Argolo et al. [135], the roots of *Solanum aviculare* that have been altered by *Agrobacterium rhizogenes* control the formation of solasodine. The sesquiterpene artemisinin production of *A. annua*'s converted roots has been shown by Souret et al. [136] to be higher than that of entire plants. Shi and Kintzios have shown how *Agrobacterium rhizogenes* genetically modified *Pueraria phaseoloides*, producing puerarin in the hairy roots [137]. Puerarin concentrations in hairy roots increased to 1.2 mg/g dry weight, which is 1.067 times higher than concentrations in untransformed plant roots. As a result, these altered hairy roots have enormous promise as a reliable supply of secondary metabolites for commerce.

4.3 Bioprocess Production Technology of Secondary Metabolites

Plant cells are said to be particularly sensitive to shear pressures in the literature, calling for the usage of specialized low-shear fermentor, such as air lift fermentor. Such bioreactors are uncommon in industry, because most operations take place in stirred-tank vessels. Since it is the least expensive process unit, such a bioreactor is preferred for plant cell cultivation [138]. Recent laboratory investigations, among others, on the shear sensitivity of plant cells have shown in reality that they are generally fairly shearing stress resistant [139]. This is corroborated by the several large-scale procedures using plant cell cultures, such as the manufacture of shikonin that has been documented. Even plant cells have been grown in 60-meter agitated tanks [140]. Given that technology is practical, what about the economy? Many studies have been published on this [141]). A price of 1500 US dollars/kg was determined using an annual production rate of 3000 kg/year of a chemical generated by a cell culture at a level of 0.3 g/L. Price increases to 430 $/kg with a factor of 10 (3 g/L) increase in production [142]. A fed-batch method of processing was used in both situations. While these costs are exorbitant, other natural products (such as vincristine, vinblastine, and Taxol) are considerably more expensive [143]. Nevertheless, in plant cell cultures, the majority of valuable specialty chemicals are generated at insufficient amounts [144]. So, their output must be raised to enable an industrial process [145].

4.4 Molecular Genotyping Method

While physical traits will continue to be used in plant identification to determine the family, genus, and species, uncertainties at the level of the species and strain may be cleared up by contemporary genotyping techniques. The fundamental premise is that each species and, in certain cases, each creature have a unique DNA sequence that may be used to unmistakably identify it**.** With the development of genome technology, DNA sequencing has become automated and practical on a wide scale [146]. Yet, it is not practical to routinely study whole plant genomes. Targeted, recognized, and sequenced diagnostic genes or fragments may be very specific. In these situations, the polymerase chain reaction (PCR), which enables the targeted amplification of DNA fragments, is often utilized. After the diagnostic segment has been located, it is possible to design and create particular primers to employ in its amplification. In fact, these methods are now often employed to diagnose bacterial, viral, and parasite illnesses in animals [147].

The use of DNA fingerprinting may reveal subtle variations among species. The restriction fragment length polymorphism (RFLP) is a well-known technique for DNA fingerprinting. In describing the plants, genomic or chloroplastic DNA is obtained, digested with certain restriction endonucleases, separated by agarose gel electrophoresis, blotted onto nitrocellulose paper, and probed with radiolabeled particular gene probes. Each distinct species is given a distinctive pattern of horizontal lines. After distinct DNA bands have been removed and sequenced, the organism (or plant) in issue may be identified without a doubt. The amplified fragment length polymorphism (AFLP) analysis is a useful RFLP modification [148].

The PCR used in the randomly amplified polymorphic DNA (RAPD) technique uses a set of primers that are chosen at random. They create DNA fragments that are specific to the target organism by binding to a number of complimentary places along the genome. The identification and speciation of medicinal plants is expected to be made simpler by a thoughtful mix of conventional morphological identification with DNA profiling, as described above. For the detection of plant pathogens and specification, unique monoclonal antibodies may be generated in addition to DNA-based techniques [148].

A feasible approach would be to first determine a plant's medicinal value before developing a DNA-based test for it. If the bioactive plant product is a protein, this technique could specifically target the gene that codes for the protein. The technique may also concentrate on the genes that code for important enzymes in a secondary metabolite's synthesis pathway.

5 Biotechnological Approaches for Conservation of Medicinal Plant Products

Biotechnology is rapidly becoming a crucial technology for sustainable environmental management and preservation [149–151]. Almost every element of human existence has been impacted by biotechnological advancements, including food, fuel, cosmetics, medications, and drinks. Most significantly, biotechnology-based approaches provide a consistent supply of natural products and raw materials for the food, pharmaceutical, and cosmetic sectors [152]. Prescott-Allen and Prescott-Allen proposed a methodical notion of sustainability. They contend that both people and the ecology are interconnected and reliant on one another. So, in a conceptual sense, the link between humans and the ecology around them expresses the core of sustainable development. They went on to say that it is believed that a society is sustainable when both the status of the environment and the state of the population are adequate or improving. They came to the conclusion that the system only becomes better when both the ecological and human situations get better [153].

Various in situ and ex situ approaches (including in vitro methods, botanical gardens, plant banks, GenBank, gene sanctuaries, and seed banks) have been proposed for the conservation of critically endangered plant species in order to provide medicinal plants or medicinal plant-based raw materials in a sustainable manner [154–156]. Maintaining genetic variety is crucial for protecting plant genetic resources. Ex situ and in situ methods are used to save plant genetic material. The in situ technique involves preserving plant species and/or populations in their native environments, where they may develop, thrive, and breed. Ex situ conservation, on the other hand, focuses on preserving plant species' genetic material under controlled circumstances [157, 158]. At nurseries, a large number of plants are produced by the traditional processes of cutting, budding, layering, and/or grafting. Yet, biotechnological techniques like genetic engineering, metabolic engineering, and micropropagation are particularly suitable for species that are challenging to reproduce in vivo [159].

Hence, preserving medicinal plants in situ would not be an efficient and successful technique for their preservation and growth. According to Krishnan et al. [160], judicious use of propagating biotechnology tools in plant conservation programs is a need for success (in the sustainable utilization of therapeutic plants) and serves to supplement the ex situ measures already in place. Since it will be useful in creating management strategies, preserving their genetic variety, and guaranteeing their long-term existence, the comprehensive study of the genetic diversity of uncommon and endangered plant species is crucial [161]. Micropropagation, mycorrhization, genetic modification, and the creation of DNA banks are biotechnological strategies for the long-term supply and preservation of medicinal plants [158, 162]. Using biotechnological methods, any group of plants may be conserved.

5.1 Cryopreservation

Cryopreservation, which involves freezing biological material or storing it in a cryogenic environment at a very low temperature, is one of the key biotechnological methods in the conservation of plant species [163]. When more conventional methods like seed banking and vegetative propagation are ineffective for the particular plant species, cryopreservation is utilized to preserve plant germplasm. Hence, liquid nitrogen(-196 °C) is used to retain plant germplasm generated from in vitro for long-term conservation [164].

The preservation of surviving individuals is regarded to be of the utmost significance for vulnerable and endangered species since they either generate few or no viable seeds or are dormant [165]. Such problematic species need to be preserved by cryo-collections [166]. When necessary, material may be easily removed and reinitiated, reestablishing ideal clonal lines. Cryopreservation provides the effective and safely long preservation of species without compromising viability [167]. Plant materials are preserved for future use, including cells, tissues, gametes, oocytes, organs, DNA samples, etc. [168]. One of the most significant conservation strategies is cryopreservation, which has been used effectively to preserve a variety of medicinal plants. Examples of preserved species are *Atropa belladonna*, *Digitalis lanata*, *Hyoscyamus* sp., and *Rauvolfia serpentine* [163]. Some institutions have already made steps to conserve medicinally significant yet endangered plants. For instance, the Kings Park and Botanic Garden in Australia uses the cryopreservation strategy to conserve over 110 succession of scarce or vulnerable plant species [169].

Similar to this, more than 1200 accessions from 50 distinct plant species are kept at the National Bureau for Plant Genetic Resources (NBPGR, New Delhi, India) [170]. The similar method has been used to preserve the shoot tips of *Picrorhiza kurroa*, an endangered plant with significant medicinal value [168]. *Dioscorea bulbifera* embryogenic cultures were cryo-preserved utilizing the encapsulation-dehydration method. The subculturing after cryopreservation revealed a 53.3% recovery in the development of the embryogenic culture [171]. These investigations unmistakably shed light on the value of cryo-techniques in the preservation of priceless medicinal plants [159]. Cryopreservation approaches have, however, only received a very small amount of study; hence, further work in this field is required to effectively use this technique or strategy for the preservation of medicinal plants.

6 Bioinformatics Approaches

The current pharmaceutical business has also made use of the information from ancient systems of plant-based treatments. So, based on the ethno-medicinal information, there is a huge potential for the identification of novel medications from plants [172, 173]. Almost one-third of medications now on the market are made from plant-based natural ingredients [174]. While research on plant-based cures is behind (particularly when contrasted to the focus in creating synthetic pharmaceuticals for commercial use), they offer a great deal of promise to advance contemporary medical therapies [175]. This could be in part due to the time- and money-consuming nature of traditional plant drug development processes [176]. Yet, it would be beneficial to do further study into the cultivation of therapeutic plants and herbal medicines. The literature and resources in this field are often dispersed, which makes it difficult to quickly take use of the knowledge about medicinal plants and the manufacture of herbal medicines that is already accessible.

The diversity of chemicals may be analyzed computationally using a variety of methods. These methods have been crucial to computer-aided drug design [177]. To keep up with the increasingly demanding drug needs, the area of pharmaceutical discovery and design from therapeutical plants necessitates the implementation of the methodologies to bring about speedier and more effective advancement. Huge amounts of data produced by molecular biology-based procedures may be analyzed and interpreted utilizing a variety of vital tools provided by bioinformatics. These methods have become more important in data analysis and integration to infer information from a whole systems perspective as high-throughput techniques have advanced. A deep examination of metabolomics, genomic, and proteomic data is necessary to improve our knowledge of the biological processes connected to plants. Genetic information and its mechanism identifications can be linked to significant production of biologically active compounds from therapeutical plants which have been facilitated by the use of bioinformatics methods [178].

Coexpression networks may be generated from transcriptome data using bioinformatics techniques, potentially pointing to new genes in closely related plant species. Particularly, the use of comparative genomes offers a foundation for information sharing across various species. For instance, a coexpression network was developed using transcriptome data from barley. Moreover, subnetworks (or "modules") linked with biological processes were developed using the results of coexpression investigations, with a focus on finding modules connected to drought stress and cellulose production. It has been shown that this genome-scale sequence comparison reveals numerous *Triticeae* species-specific genes connected to certain regulatory networks [179].

A useful method for determining putative functional actions of genes is pathway analysis. A framework for pathway study of secondary metabolites from many species is provided by the Kyoto Encyclopedia of Genes and Genomes (KEGG) resource [180]. Bioinformatics plays a crucial function in aiding the metabolic pathways of processes knowledge in light of the many advancements in "omic" technology. Many possibilities exist to research metabolic pathways, thanks to the homogenization of metabolomics and transcriptomic information using knowledge discovery in data (KDD) tools [181]. In order to find potential genetic makeup metabolite involved in the manufacture of alkaloids and terpenoids, Rischer et al. [182] examined a "gene-metabolite coexpression network" of the therapeutic plant *Catharanthus roseus*. The ability to examine metabolic regulation of nonmodel plants with potential medical utility has therefore been shown by integrated gene-metabolite expression analysis. Bioinformatics offers vital tools for processing large amounts of data produced by high-throughput methods. These methods have made it feasible to identify candidate genetic processes that are rapt in the production of secondary metabolites in therapeutic plants, in particular. Bioinformatics techniques may be crucial for combining disparate bits of evidence into coherent hypotheses, creating possible leads for experimental validation [173].

7 Opportunities and Challenges

The use of traditional medications and the significance of its business are both rapidly expanding and pervasive. The practice of herbal medicine, which dates back thousands of years, is now advancing quickly and experiencing something of a revival around the globe, especially in affluent nations. The majority of the components in herbal medicines come from wild plants, and as a result of habitat loss, climate change, and rising demand for medicinal herbs, many species are under threat. Wild resource extraction conducted without discrimination has resulted in genetic diversity loss, declining populations, local extinctions, and habitat devastation [3].

For an optimum production of therapeutic plants, it will provide a workable conserving approach and gets rid of issues with misidentification, genetic and phenotypic variability, extract variability and instability, poisonous components, and contamination that are often present in herbal extracts. Cultivation may also result in homogeneous high-quality output and an optimized yield. But, the planter must choose which specific species to produce in a market that is always changing and fashion-sensitive. Consequently, another significant barrier to the effective commercial production of therapeutic plants is the difficulty in determining which extract will stay marketable [154].

Despite the fact that many of the plants utilized in traditional medicine are grown, the vast bulk of these plants are obtained from the natural resources. Due to poor germination rates, a lack of high-quality planting materials, or certain ecological needs, producers of herbal plants have several challenges. The primary obstacles to growing herbal plants are a lack of understanding of the precise needs for pollination, seed germination, and growth. Low germination rates are typically caused by fungus or mechanical damage, but they may be readily addressed with better seed treatments and by maintaining ideal storage conditions. However, employing controlled settings, such as hydroponic systems, it is simple to produce challenging herbal plants on a commercial scale [97].

The fact that primary metabolites are key component in the traditional medication which is generated relatively in little amount of the plant species presents another significant problem in the development of herbal medicines. This is clearly shown that plant cells mostly generate the bioactive components in modest amounts as secondary metabolites. This results in the destruction of several herbal plants in order to produce a single medicine. Nowadays, however, this wasteful harvesting method may be readily avoided by using contemporary methods and their modifications genetically. The commerce in medicinal plants is growing yet mainly unregulated in the modern world. Many current harvesting methods are not sustainable, endangering both the populations of medicinal plants and their ecosystems as well as the means of subsistence for individuals who participate in their gathering. It is now up to us as individuals, as well as the government, to discover effective, worldwide answers [155].

8 Conclusion

Nowadays, the oldest and most popular kind of medicine in the world is herbal. In both developing and developed countries, a significant section of the population still depends on herbal medicine as their main form of treatment. As a consequence of this rising demand, policymakers, medical experts, as well as the general public have concerns regarding the safety, standardization, effectiveness, quality, availability, and preservation of herbal products. The most significant challenges in studying herbal medicine include incorrect plant identification and speciation, low yields of chemically prepared plant bioactive metabolites, variability of traditional protocols, and others. Due to the effective application of in vitro technique for bioactive compound production in conjunction with other techniques such as in vitro regeneration and hereditary transformation, the use of biotechnological tools is crucial for the multiplication and genetic enhancement of the medicinal plants. An important set of tools for creating effective and focused searches for herbal medicine practice may also be provided by bioinformatics methodologies. Combining bioinformatics techniques may open the door to a new era of herbal medicine use that will greatly advance the field.

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Challenges and Future of Nanotechnology in Global Herbal Medicine Practices

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Abstract

Herbal remedies have been practiced all over the globe for many years, but the prevalence of their use in India is significantly based on nanotechnology; it is necessary to make concessions about the drug distribution techniques observed. Herbal medicines have gained popularity as a therapy option due to their range of applications and lesser likelihood of adverse responses in comparison to traditional pharmaceuticals. This is because herbal medications can be used to treat various conditions. It is essential to develop novel nanodrug delivery to circumvent such low bioavailability, in vivo stability, liquid insolubility, intestinal uptake, and unspecific site of action. The incorporation of nanotechnology as a novel drug delivery system (NDDS) into conventional methods of medical practice has the potential to increase the efficacy of herbal medications in the treatment of chronic illnesses such as cancer and other fatal diseases. In order to effectively produce herbal medicine work based on nanotechnology, it is necessary to make concessions with regard to drug distribution techniques based on nanotechnology. This is essential whenever observed as a component of the global economy. This chapter emphasizes the present issues that herbal medicine practitioners are facing, as well as the ways in which these challenges might be overcome by utilizing new solutions.

Keywords

Nanotechnology · Herbal medicine · NDDS – novel drug delivery System · Chronic diseases · Traditional medicine · Global economy · Development · Healthcare

Abbreviations

AFM	Atomic force microscopy
BBB	Blood-brain barriers
BD	Biodistribution
BET	Brunauer, Emmett, and Teller
CE	Capillary electrophoresis
CPT	Current procedural terminology
CVD	Cardiovascular disorders
DLS	Dynamic light scattering
DNA	Deoxyribonucleic acid
EU	European Union

FDA	Food and Drug Administration
GC-MS	Gas chromatography-mass spectroscopy
GNPs	Nanoparticles made using green synthesis
HPLC	High-performance liquid chromatography
HPTLC	High-performance thin layer chromatography
LC-MS	Liquid chromatography-mass spectroscopy
MAPs	Medicinal and aromatic plants
MNs	Magnetic nanomaterials
MRI	Magnetic resonance imaging
NCs	Nanocarriers
NDDS	Novel drug delivery system
NPs	Nanoparticles
PCS	Photon correlation spectroscopy
PK	Pharmacokinetics
RES	Reticuloendothelial system
SEM	Scanning electron microscope
SLNs	Nanomaterials made up of polymeric material, solid-state nanoparticles of the lipids
TEM	Transmission electron microscope
UPLC	Ultra-performance liquid chromatography
WHO	World Health Organization

1 Introduction

The field of science and engineering known as nanotechnology is concerned with designing, synthesizing, measuring, and implementing devices and materials whose lowest functional organization appears on the nanoscale in any one of their dimensions (a trillionth of a yard) [1, 2]. A vast area of science called nanotechnology has recently made major advancements and is rapidly growing [3–6]. Due to their perceived efficacy compared to contemporary conventional medications and reduced cost, herbal medicines are growing in popularity worldwide. They have considerable potential for providing treatment, sustaining and enhancing health, and avoiding and curing several illnesses [7, 8]. However, many of these physiologically active phytochemicals have some drawbacks, like sluggish reception and distribution rates and typically poor target selectivity. Poor bioavailability results in lower cellular function. Furthermore, substantial quantities of these phytochemical compounds must be consumed to have the desired effects, and some of these compounds are unstable and highly vulnerable to acidic environments [9–12]. It is challenging to apply them therapeutically because of these restrictions.

Herbal medicines numerous diverse ingredients all work to treat diseases concurrently [13]. For phytotherapeutics to increase patient cooperation and eliminate the need for repetitive administration, a systematic approach must introduce the processes gradually. This requires the development of innovative drug delivery systems (NDDSs) made especially for plants to become a scientific fact. Innovative drug distribution techniques will reduce the need for repeated administering, which is required to counteract noncompliance. Among other things, lowering the toxic effects of the medication and raising the pharmacokinetics will also enable the achievement of higher therapeutic benefits [14, 15].

Nanotechnology-based drug carriers have the potential to overcome the many obstacles that plant medicines face, including enhancing their plant chemicals and their biological function and bioavailability. One strategy that has the potential to develop into a cutting-edge new technique that will improve the effectiveness of the usage of phytotherapy in medicinal products is the incorporation of nanoscience into phytochemical components. Developing and implementing a dependable and risk-free pharmaceutical distribution system is a significant objective of this study. Recent developments in nanotechnology have reignited interest in the creation of herbal medications. Some of the techniques proposed as possible means of substance transport include phytosomes, nanomaterials made up of polymeric material, solid-state nanoparticles of the lipids (SLN), the nanostructured transporters of triglycerides, and nanoemulsions. The pharmacokinetics and biochemical effects of herbal medicines are anticipated to improve with the implementation of nanotechnology [7, 11, 16, 17]. This is because a significant number of drugs have had their pharmacokinetic properties altered and improved by the use of nanoparticles. This chapter's primary goal is to briefly overview recent advances in creating medicinal product formulations using nanotechnology that boost the activity of herbal components.

1.1 Background of Nanotechnology

The term "nano" originates from the Greek term for "dwarf." One nanometer (nm), often known as a billionth of a yard, is equivalent to approximately six atoms of carbon or ten water molecules. Compared to the width of a red blood cell, which is only about 7000 nm, a nanometer is more significant than an atom, yet many substances, including specific proteins, can be larger than a nanometer [18]. The philosophical underpinnings of nanotechnology were first introduced by Richard Feynman, a physicist, in his 1959 talk "There's Plenty of Space at the Bottom." In the words of Q. Feynman, who looked into the concept of impacting material at the size of distinct atoms and molecules, the complete *Encyclopedia Britannica* could probably fit on the tip of a pin. Feynman anticipated that shortly, we would be able to observe and manipulate materials at the nanoscale. The term "nanotechnology" was created in 1974 at University of Tokyo by scholar Norio Taniguchi to characterize the ability to develop materials at the nanoscale precisely. During this period, the primary driver of miniaturization was the electronics sector, which tried to create methods for producing smaller (and subsequently faster and more complex) electrical gadgets on silicon chips. IBM utilized another technique known as electron beam lithography in the early 1970s in the USA to produce nanoparticles and devices with dimensions ranging from 40–70 nm.

1.2 History and Development

Natural ingredients, particularly plants, have always been the cornerstone of healing human illnesses. Conventional treatments and medicine still provide the philosophical underpinnings for advancing contemporary medicine [19, 20]. Long before known history, people worldwide used plants for medical purposes, particularly in medieval China, Egypt, America, South Africa, and India. This practice may even be traced back to prehistoric times. The first efforts at extracting and altering plant components date back to the early nineteenth century, thanks to advances in chemical research [19, 21]. For a long time, herbal remedies were neglected to create innovative formulations due to a lack of research backing and processing challenges such as standardization, extraction, and identification of specific medicinal components in complicated polyherbal systems. This was due to the fact that herbal remedies are notoriously difficult to work with. Research on current phytopharmaceuticals, on the other hand, not only meets the scientific standards for herbal treatments in modern medicine but also paves the way for the creation of innovative formulations such as nanoparticles, microemulsion matrix frameworks, solids dispersion, liposomes, and SLNs, among other things. Curcumin can now be used alone or with other chemotherapy medicines like paclitaxel, thanks to creating aqueous nanogels, nanotubes, and nanomicellar systems [22].

1.3 Nanoherbal Medication Delivery Method

Over the past 20 years, much attention has been placed on creating an innovative medication delivery system (NDDS) for herbal medicines. Humans have used plant-based medicines to cure various illnesses for a very long time. However, these phytomedicines have many limitations due to issues with their durability and fat solubility. Different botanical medicines are being created with herbal components to solve these issues. Such herbal medication delivery methods may one day be used to handle problems with plant-based therapies by enhancing their activity. Such delivery mechanisms facilitate the discharge of active substances into the body and are simple to produce [23]. Other studies have examined the various applications of NCs in nanomedicine and the challenges and new opportunities in the field [24]. As a result, to fight the effects of chronic illnesses like asthma, diabetes, cancer, and others, the traditional medical system must adopt nanocarriers as a distinctive nanodrug delivery technology. Some nano-based drug delivery methods currently under investigation include liposomes, dendrimers, polymer nanocarriers, synthetic subatomic particles, nanocapsules, nanospheres, and nanogels [25] (Fig. 1). Some of them have been used in clinical contexts and have received FDA approval, particularly for cancer treatment. Active and passive targeting are the two methods that have been applied to the use of medication distribution devices in cancer therapy. The final goal of these targeted methods is to reduce side effects while boosting drug accumulation in tumor tissue [26].

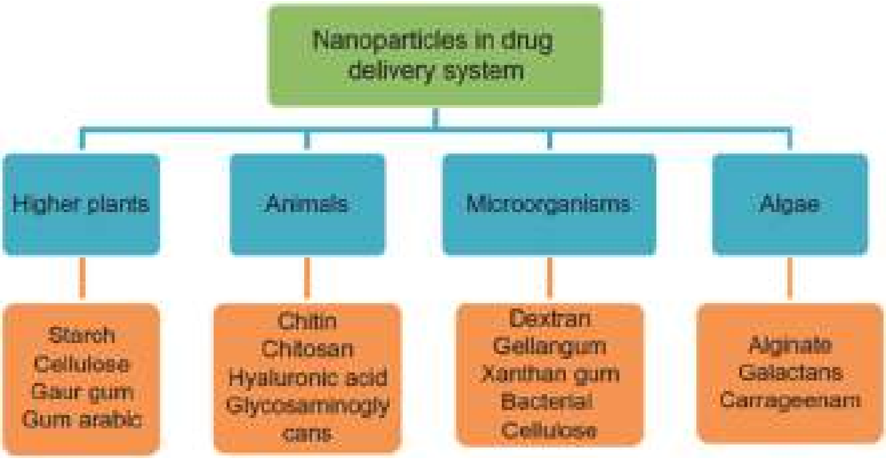

Fig. 1 The natural biopolymers that are utilized in the nanoherbal medication delivery system as well as the sources of these materials

1.4 Nanotechnology Applications in the Field of Medicinal Plants

Nanoparticles (NPs) are adaptable materials that may be employed in a wide variety of contexts; nevertheless, they have shown to be of particular utility in the field of nanomedicine, where they have been utilized in sensitive diagnostic procedures and to repair damage to tissues caused by extremely high temperatures, treatment with radiation enhancement, drug administration, and gene transport [27]. The metallic NPs-herb union may more effectively treat many pathophysiological ailments, such as cardiovascular disorders (CVD). Given the increasing incidence of CVD, research into plants and medicines derived from them with antioxidant, anti-inflammatory, anti-bacterial, and antihypertensive properties is crucial for further evaluating organically generated medicine's compatibility with people [28]. Investigating natural products and their prospective for use with NPs is also critical to grasp better current technical developments and the medicinal advantages of organically derived substances. The application of nanotechnology in the medical industry is gaining more and more favor, particularly for the delivery of medications. Pharmaceutical studies have employed nanomaterials to reduce the harmful adverse effects of medicines for millennia. Unfortunately, not all cancer treatments reach the cancerous site without harming nearby healthy tissue and organs [29]. To successfully treat cancer, nanotechnology might be able to directly deliver therapeutic compounds by beating biological obstacles. The particulars of metallic NPs elements, including nanoparticles of silver (AgNPs), which have produced much study because of their degree of cytotoxicity, offer novel possibilities for cancer treatments [30]. According to the research, AgNPs could be used as a cancer-fighting weapon, a medical breakthrough.

A very safe technique for the development of nanomedicine is the nano-encapsulation of medicinally important drugs derived from plants in NCs, which include liposomes and micelles, which are synthesized or nanoparticles that are solids of lipids (SLNs), dendrimers of lipids or albumin nanocapsules, nanovesicles, nanogels, nanosuspensions, and nanoemulsions [31]. The contemporary nanoencapsulation of pharmaceuticals method reduces drug toxicity by enabling the appropriate quantity of drug compounds to be loaded within the NCs. Numerous MAPs generate essential oils with antibacterial properties that are effective against various pathogens, including bacteria, viruses, and fungi. The two significant factors of critical oil responsiveness are the features of the functional categories present in the oils and the perspective of the essential oil. Because they can promote membrane permeability, disrupt cellular metabolism, and do other things, essential oils made from MAPs are believed to be helpful against a wide range of infections, liberate important intracellular components, and change the enzyme kinetics of pathogens [32]. These variables eventually cause the cell walls of the bacteria to deform. The literature provides details regarding essential oil usage in studies conducted in vitro; however, multiple research investigations utilizing these kinds of oils fell short of presenting precise information regarding their chemical structure and the mechanisms of action, clouding the reproducible nature and competence of their discoveries. With the above prospects, future studies should focus on deciphering extracted oils, chemical constituents and molecular processes. Biopharmaceutical companies are frantic for ecologically friendly alternative drug molecules for treating bacterial and human metabolism disorders. Plant essential oils that have been extracted may be a source of antibacterial agents, according to in vitro studies, and may thus play a vital part in curing many kinds of pathogenic microbes by introducing new medicines.

2 Synthesis of Nanoparticles

2.1 Polymer Nanomaterials

Polymer nanoparticles can be either solid or liquid, ranging from 10–1000 nanometers. Polymer nanoparticles are also known as nanospheres and nanocapsules. They can be produced from prefabricated polymers or straight from monomers through polymerization. Fluid evaporation, salting out, dialysis, rapid growth of extremely critical solutions, and vaporization are all used [33]. When selecting a preparation process, the kind of polymeric material, the scope of usage, the size requirement, and other parameters are all considered. Polymeric nanoparticles had displayed successful therapeutic activity using any of these techniques. Figure 2 depicts a conceptual model of polymeric nanoparticle synthesis.

2.2 Magnetic Nanomaterials

Metal complexes like FePt and CoPt were used in addition to pure metals like CO, Fe, and Ni in the creation of the magnetic nanoparticles [34]. Using nanomaterials with magnetic properties makes producing particles with dimensions of about 3 nm feasible. Nanoparticles are several hundred atoms, enabling us to create media for

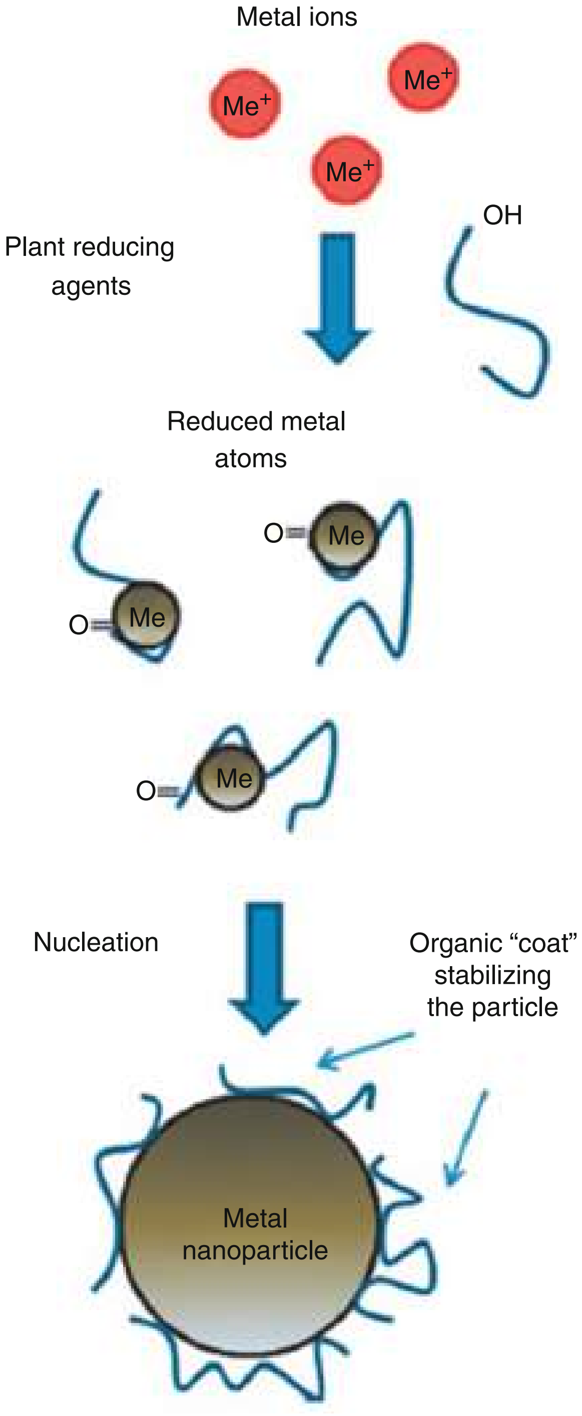

Fig. 2 An illustration of the creation of metal nanoparticles in a plant extract. The reducing metabolites and stabilizing substances bind to the metal ions, reducing them to metal atoms. The metal ion and metabolite complex that forms as a result interacts with other complexes to create a tiny metal nanoparticle. Next, the coarsening process causes distinct tiny particles to expand and merge into larger ones. Until the particles take on a stable shape and size, this process is repeated [39]. (CC BY 4.0)

recording with a recording density of up to 1 Tb/in2 if the particles are properly organized. Coprecipitation, sonochemistry, colloidal methodology, solvothermal, combustion synthesizing, hydrothermal method, microemulsion, and thermal breakdown procedures are a few of the approaches that have been reported [35]. Magnetic nanomaterials (MNs) serve as the primary purpose in the bioseparation technique in which a combination of biomolecules of interest made functioning with particular receptors leads to mixtures that are easily attracted by the strong magnetic field that is produced and generated from the pristine mix. This method is quick and easy compared to conventional techniques like centrifugation and filtration. Furthermore, this technique is used in MRI, hyperthermia, medication administration, and biological detection [36].

2.3 Metallic Nanoparticles

Metal nanoparticles are those measuring sizes (length, breadth, and thickness) ranging from 1–100 nm. Multiple liquid-phase techniques, such as biochemical diminution, solvent gel, and reversed micelle, can create metallic nanoparticles. The metal nanomaterials that are recently discovered are produced by using chemical reduction mechanisms having forms of sphere-like structures and sizes [37, 38]. This diagram depicts the various methods used to make the metallic nanoparticles shown in Fig. 2. Metal nanoparticles are extensively used due to their unique properties, such as handling many low coordination locations and offering unique electronic configurations between molecular and solid states. These variables are used for a variety of purposes, which include the magnetic division of designated cells and other biological components, the therapeutic transportation of drugs, gene sequences, and radioactive substances, radiofrequency methods for tumor hyperthermic catabolism, and contrasting drivers for magnetic resonance imaging (MRI).

3 Characterization of Synthesized Nanoparticles

The scientific investigation of a material's constituents, configurations, and chemical and physical characteristics is known as characterization. Nanoparticles are frequently recognized by their size, form, and charge on the surface using state-of-the-art microscope methods like atomic force, electron transmission, and scanning electron microscopy. The general condition of nanoparticles made of polymers, which can affect their toxicity, may be evaluated using electron microscopy technologies, which are highly beneficial. The physical durability of these nanoparticles, as well as their redispersibility and efficiency in vivo, are all influenced by the surface charge they possess.

3.1 Scanning Electron Microscope (SEM)

Scanning electron microscopy (SEM) enables direct anatomical investigation viewing. We employed electron imaging for morphological and measurement analysis;

however, these images only give limited information on the scale distribution. The approach is primarily and fully reliant on electron imaging. To do an SEM study, nanoparticles must initially be reduced to powdery. Once this has been accomplished, the nanoparticles must be deposited on an organized carrier and blanketed with a reactive metal, such as gold, utilizing a stutter coater. After that, a focused first-rate electron beam is used to conduct an inspection of the design. The surface of the ground is determined using secondary electrons. In order to prevent the polymer from being damaged, the nanocrystals need to be able to tolerate high pressure and electron beams. The effects acquired using dynamic moderate scattering are equivalent to the suggested length that was achieved through SEM [40].

3.2 Transmission Electron Microscope (TEM)

Preparing samples for transmission electron microscopes is complex and time-consuming, mainly because the specimen must be excellent for electron transmission [40]. When applied to support sheets or grids, nanoparticle dispersion is utilized. Nanomaterials are either bonded with a damaging staining agent that includes phosphorous tungstic acidic compounds or its associated products, the acetate material salt of uranium oxide, or plastic embedding so that they may better withstand the device hoover and so that their handling is simplified. This helps the nanomaterials better endure the instrument hoover. The sample can then be heated to the temperatures of liquid nitrogen after first being completely submerged in vitreous ice. The surface characteristics of a model can be reconstructed by passing a fragile specimen through an apparatus that reacts to the sample as it moves through the machine.

3.3 Particle Size Analyzer

The structure and distribution of particle sizes of developed nanoparticles are the most critical elements in characterizing them. Nanoparticles are mainly used in the transport and targeting of medications. It has been found that particle size influences the rate at which medicines are discharged. Because smaller fragments have a greater surface area, most medicines applied to them will be subjected to the nanoparticle surface. Therapy, on its counterpart, progressively spreads inside larger fragments. The drawback of the distribution of nanoparticles is that tiny particles tend to clump together when kept and transported. Consequently, nanoparticles' compact structure and maximum safety are jeopardized [41].

3.4 Dynamic Light Scattering (DLS)

Photon correlation spectroscopy (PCS), also known as dynamic light scattering, is now the technology used most frequently to determine the size of particles. DLS is often utilized to measure the diameters of Brownian nanoparticles in colloidal fluids ranging in size from nano- to submicron structures. The incoming light undergoes a Doppler effect, which causes a change in its wavelength, when a high-energy laser is used to spotlight the suspension of Brownian-moving spherical matter [42].

3.5 Atomic Force Microscopy (AFM)

AFM operates on physically scanning materials up to the submicron level with an atomic scale probe point, and it offers exceptionally high precision in particle size measurement. Samples are usually examined with either interaction or noncontact mode in accordance with their features. A topographical map is produced in contact mode by pushing the probe in opposition to the specimen's surface. The search can float above the conductive surface while the noncontact approach is used. The ability of AFM to picture nonconducting materials without any further processing is its fundamental benefit. This enables it to image biologically complex and synthesized nanoparticles and tiny structures [43]. The most exact appearance of dimension and size dispersion may be obtained using AFM, which doesn't need any form of mathematical processing [44].

3.6 Calculation of Surface Area

Particle surface area is the sum of all particle faces per unit mass. A particle's size is proportional to its inverse squared surface area. The specific surface area of a powder may be calculated using the nitrogen adsorption method. Most frequently, people utilize the Brunauer, Emmett, and Teller (BET) method to calculate the total surface area of a thing [45].

If the particles are spherical and have a limited size distribution, the particular area of the surface gives an average particle dimension in nanometers using the method below: dBET = 6000/ñs

where S denotes the specified area of the surface in m^2/g and T indicates the estimated density in g/cm^3.

In this study, the applications of nanoparticles in the production of herbal formulations and the medicinal uses of a specific drug are discussed to improve bioavailability and therapeutic effectiveness [46].

4 Use of Metabolite-Metal Nanoparticles in Therapeutics

Plant-made NPs have a wide variety of beneficial applications, including medication delivery, medicine, agriculture, biosensors, genetic engineering, the destruction of tumors, anticancer, antidiabetic, and antibacterial effects, and many others [47–56]. Previous research demonstrated that green synthesized silver NPs may be utilized for delivering pharmaceuticals and genetic engineering, as an antioxidant, antifungal, and antibacterial agent, as a sensor for optical signals, and to function as a catalyst in chemical reactions. In addition, it has been found that green synthesized AgNPs have the potential to serve as a substrate in chemical processes [58]. Researchers are working hard in biological systems because there is a growing need to find ways to make NPs that are clean, safe, and good for the environment. NPs additionally have a wide variety of applications in the field of plant research and the agriculture business. For example, the NPs use bioprocessing to turn farming and waste products into power and other valuable things with the help of using products. In 1996, the FDA approved metal-based nanoproducts, like Feridex, a liquid colloidal solution of super-paramagnetic ferrous oxide particles, to be used as a test medium for MRI (magnetic resonance imaging) [59]. Ferumoxytol, which used to be called feraheme, is also an FDA-approved drug that was made to treat iron deficiency anemia in adults. Ferumoxytol is currently being researched as a one-of-a-sort imaging agent for MRI-based cancer studies and different CVDs [59]. Nanoparticles made using green synthesis (GNPs) have secondary molecules. These secondary compounds found in GNPs might enhance these nanoparticles' bioactivity potential and help keep them in balance by acting as stabilizers or capping agents. The antioxidative qualities of silver nanoparticles (AgNPs) produced by employing three distinct polyphenols derived from plants (resveratrol, fisetin, and epigallocatechin-3-gallate) have been strongly linked to the number of secondary metabolites they contain [60]. In contrast to AgNPs made through chemical synthesis, AgNPs made through green synthesis were better at killing harmful bacteria [61]. ZnO NPs synthesized by the green method were more successful at killing bacteria, fungus, and viruses than zinc oxide NPs manufactured through the chemical route. These findings were shown by microbiological experiments that used varied amounts of organic material (Aloe plant) and by chemical means synthesized by nanoparticles made of zinc oxide (ZnO) [62]. Nickel nanoparticles (Ni NPs) were made from a root extract of *Desmodium gangeticum*. The bioactivity of nickel nanoparticles produced using green synthesis was compared to those produced through chemical synthesis. The results indicated that the green synthesized NPs exhibited more excellent antimicrobial and antioxidant effects without damaging the epithelial cell line [63]. Different bio-nanoparticles have been studied to see if they could be used to find and treat diabetes headaches caused by glycation [64]. Also, it's interesting that the anti-diabetic effects of green synthesized ZnO NPs stabilized by secondary metabolites change in streptozotocin-induced diabetic mice depending on their size [57]. Good antioxidant activity was demonstrated by gold nanoparticles (AuNPs) derived using the methanol-based extract of *Azolla microphylla*. In contrast, antibacterial activity was shown by gold nanoparticles (HNPs) derived from *Hypoxis hemerocallidea* versus *Staphylococcus aureus, Escherichia*, *Pseudomonas Aeruginosa*, and *Staphylococcus epidermidis* [65]. However, antibiotic activity was only found in AuNPs made from *Galenia africana* against *Pseudomonas aeruginosa* [66]. Figure 3 shows a summary of the different ways in which nanoparticles can be used.

Some commonly accessible mending products based on NPs have undergone a gradual but consistent development process throughout the past few years. In the year 2006, the European Science and Technology Observatory conducted a global assessment and discovered that over 150 businesses are producing therapeutic goods at the nanoscale level. This information was published in a report titled "Worldwide

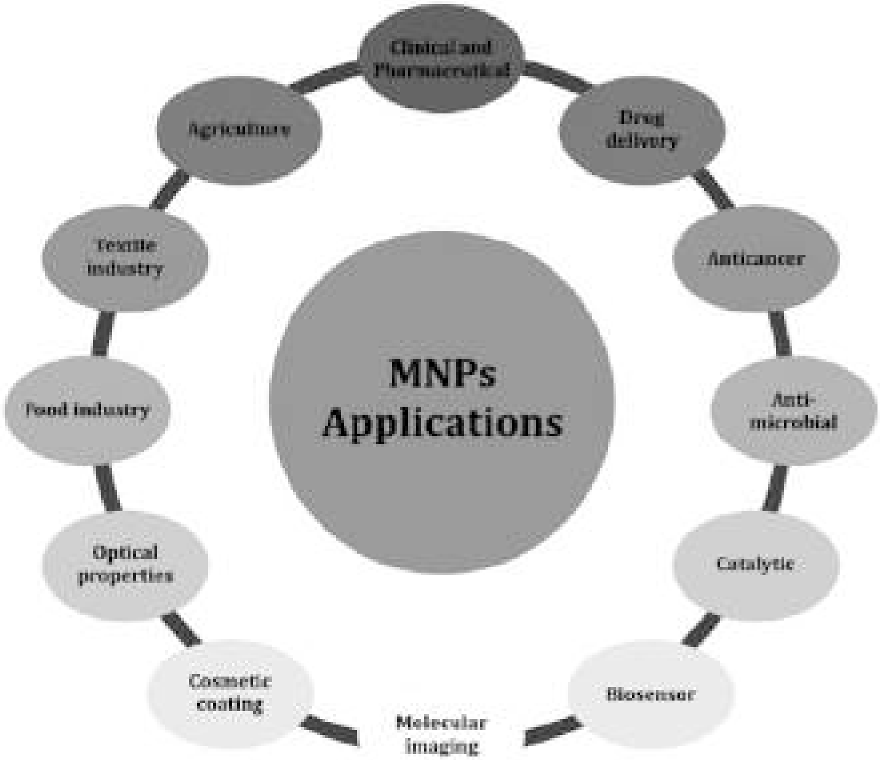

Fig. 3 Applications of metal nanoparticles [67]. (CC BY 4.0)

Review of Nanoscale Manufacturing of Healing Products." From a medical point of view, 24 mending products based on NPs have been approved so far, and they have made around $5.4 billion in sales. The best ways to take these products are in the form of polymeric pellets and liposomal drug conjugates, which account for more than 80% of the whole amount. NPs made from the breakdown products of MAPs have great packages that can help people. For example, the plant *Euphorbia hirta L.* was used to make AuNPs. Leaf extract was made into NPs, and those NPs were able to kill bacteria [68, 69]. The leaf extract of *Mimosa pudica* was used to make AuNPs on breast cancer cell lines. *Azadirachta indica L.* has phenol chemicals. Leaf extract is to blame for stabilizing AuNPs and lowering the amount of Au(III) [70]. Also, AuNPs were made using the dried leaves of *Bauhinia tomentosa*, which also showed anticancer action in a test tube [71]. So, we can say that NPs are becoming more popular to treat cancer in the modern world. In contrast to standard therapies, NPs can improve results for cancer patients and simultaneously reduce overall toxicity. Also, they have the power to get around drug resistance, which leads to better drug buildup inside cells. Nanotechnologies are now being utilized in the production of protein and DNA microarray biochips. These biochips are comparable to molecular diagnostic procedures and are associated with molecular diagnosis. Recent advances in nanotechnology for cancer treatment have opened up a wide variety of innovative administration methods for certain medications.

5 "Nanocarriers": A Demand for Nanodrug Delivery Techniques

Many herbal medication components are destroyed in the highly acidic gastric juice medium before reaching the plasma, while the liver may degrade other parts. This result shows that blood may not benefit from herbal medicines. There will be no way to establish the therapeutic action of the preparation if the elements are not delivered in the best possible quantity to the diseased site at a "minimum effective level." Because of their diminutive size and their capacity to provide a controlled dose of the medication directly to the area of action, they have become more popular; while avoiding all other obstacles like liver metabolism and the stomach's acidic pH, herbal preparations have been designated as suitable drug candidates for delivery via nanodelivery systems. The following herbal preparations have been given this designation [72]:

- Since these are wholesale medications, dose reduction is advised.
- Although it is feasible to generate acetone, chloroform, and methanol-based extracts, they may not be suitable for delivery.
- Advertised treatments for many chronic diseases lack target specificity.
- The patient's reluctance since the current formulations are less effective and require big doses.
- At present, commercialized formulations have been linked to an extensive array of additional side effects. Some of the herbal extract ingredients which are used in medications are given in Table 1.

6 New Methods for Recognizing and Characterizing Nanoscale Medicinal Plants

6.1 High-Performance Liquid Chromatography (HPLC)

The medicine industry and the industry that cleans computers often use HPLC preparation and research. The main bioactive compound of *Adhatoda vasica* is vasicine, which is tested by the HPLC method in two polyherbal drug forms – Shereeshadi Kashaya and Jastyadivati. It is observed that 18.1 mg per 100 g of vasicine in Shereeshadi Kashaya and 0.7 mg per gram of vasicine in Jastyadivati are found in both drug forms, respectively. Using an RP-18 column and an acidic mobile phase, the use of high-performance liquid chromatography (HPLC), a mixture of equal parts of *Terminalia chebula*, *Emblica officinalis*, and *Terminalia belerica* is called Triphala suspension (antioxidant-rich herbal preparation). Now, an excellent method exists to manufacture 100% plant medicines like liquorice, which are pure.

Table 1 Some herbal drug nanoparticle

Formulations	Active ingredients	Biological activity	Method of preparation
Curcuminoids solid lipid nanoparticles	Curcuminoids	Antitumor, antioxidant, antiamylodin, antiplatelet aggregation and anti-inflammatory, antimalarial	Microemulsion technique [22, 73]
Glycyrrhizic acid loaded nanoparticles	Glycyrrhizin acid	Anti-inflammatory anti-hypertensive	Rotary-evaporated film ultrasonication method [22, 74]
Nanoparticles of cuscuta chinensis	Flavonoids and lignans	Hepatoprotective and antioxidant effects	Nanosuspension method [22, 75]
Taxel-loaded nanoparticles	Taxel	Anticancer	Emulsion solvent evaporation method [22, 76]
Artemisinin nanocapsules	Artemisinin	Anticancer	Self-assembly procedure [22, 77]
CPT encapsulated nanoparticles	Camptothecin	Anticancer	Dialysis method [22, 78]
Berberine-loaded nanoparticles	Berberine	Anticancer	Ionic gelation method [22, 79]

6.2 High-Performance Thin Layer Chromatography (HPTLC)

With HPTLC, phytochemical tests and quality analyses are done on pharmaceutical medicines and formulas. Since HPTLC only uses a small amount of cellular parts, it can analyze more than one material simultaneously. HPTLC allows for precisely quantifying *Terminalia chebula* constituents such as rutin, quercetin, and gallic acids. Using the HPTLC method, the biomarkers withaferin A and the sitosterol d-glucoside had been identified in Ashwagandha at the same time. *Syzygium jambolanum's* tannin, ellagic acids, glycoside, and gallic acid were all analyzed using high-performance liquid chromatography (HPTLC) during its manufacturing for stability, longevity, accuracy, and measurement. The HPTLC method is a good way to measure diosgenin because it is quick, accurate, and cheap.

6.3 Ultra-Performance Liquid Chromatography (UPLC)

Ultra-performance liquid chromatography (UPLC) was recently used to compare traditional and powder decoctions to see the chemical responses and chemical instability.

6.4 Liquid Chromatography-Mass Spectroscopy (LC-MS)

LC-MS has become recognized as a viable choice across several drug creation phases. According to an LC-MS analysis of these medications, aminoglycosides have a low affinity for coupling to plasma proteins, a high water-soluble content, and a higher kidney excretion rate. In addition, the utilization of this methodology facilitates the recognition of amino glycosides within samples of plasma via ion chromatography using investigations with LC-MS into the pharmacokinetics of traditional Chinese medicinal medications.

6.5 Gas Chromatography-Mass Spectroscopy (GC-MS)

Using this method, it's possible to identify significant quantities of compounds found within ecological and biological processes. The GC-MS method was utilized to remember and process the chemical components used in manufacturing poly-herbal oil (Megni). Megni includes nine constituents, the majority of which are Myristica fragrances, *Mentha piperita*, *Eucalyptus globulus*, and *Gaultheria procumbens*. The GC-MS method was deployed to identify and process the chemical components. There are currently 35 flexible chemicals that have been separated and characterized.

6.6 Capillary Electrophoresis

The CE method was developed to evaluate a singular herbal remedy's description, sensitivity, and precision. There have been several published CE studies on herbal medications, but alkaloids and flavonoids, two categories of medicinal compounds, have been the subject of significant research. Additionally, the analysis period required by the CE technique was half as fast as that needed for the HPLC method, but it required more than one hundred times the amount of solvent. It has been found that it is necessary to use CE apparatus such as CE-MS, CE-NMR, and CE-diode array detection.

7 Using Nanotechnology to Create a Nanodrug Delivery System

Nanodrug delivery methods are emerging as a newest methodology for lowering the restrictions imposed by traditional drug delivery methods [80]. This is because nanodrugs have the characteristic of being extremely small. There are a variety of causes behind the rise of nanoscale drug delivery techniques including the following:

1. Infected regions can receive the most significant medication due to their exceptional size and high-loading capabilities.
2. It is best to provide the medication to the patient in tiny particles, as this can enhance the overall surface area of the drug.
3. Exhibit enhanced benefits in terms of retention and penetration.

4. Getting to the location of action of the illness does not require any additional ligand component.
5. Cause exceedingly few, if any, unintended consequences.
6. A decrease in the amount of the active ingredient in the pharmaceutical composition.

8 Recent Progress in Designing NDDS and Herbal Drug Manufacturing

Recently, the focus of specialists in the pharmaceutical industry has switched to the creation of the most effective drug delivery technique for herbal medications by using several recent scientific breakthroughs. The plant *Cuscuta chinensis* is used in ancient Chinese medicine to nourish the kidneys and the liver. Because the main components of its compositions are flavonoids and lignans, which tend to be less water soluble, taking it orally may result in inadequate assimilation. Because of this, nanoparticles used for the same purpose were manufactured commercially [22, 81]. Nanomaterials of polylactic acid used in the treatment of a lipophilic cancer-fighting herb (cucurbitacins and curcuminoids) have been the subject of recent research [22, 82] that was conducted using a technique called precipitation [22, 82]. Due to their targeted operation, appropriate bioavailability and efficacy of traditional Chinese medicine's SLNs have likewise made strides in the creation and description during the past few years. Recent studies have been undertaken on an array of nanostructured delivery systems, including SLNs, polymeric micelles, nanoparticles made of polymers, liposomes, nanoemulsions, and others [22, 83, 84]. This research focuses on using these systems to administer chemotherapy medications directly. In addition, the administration of cytotoxic drugs through the oral pathway has been hypothesized to have a significant likelihood of success. Consequently, there has been a focus on advancing oral chemotherapy for malignancy [22, 85, 86].

9 Challenges in Designing NDDS and Herbal Drug Manufacturing

At this time a wide variety of therapeutic choices are accessible to manage this fatal situation. Cancer treatment is by far the most common choice for addressing health problems. This approach may be successful to some degree; however, it does come with its fair share of drawbacks, including poor tissue access, excessive doses, unwanted cytotoxicity, drug resistance, and nonspecific targeting. Therefore, there is an immediate need to create cutting-edge technology that might aid in overcoming obstacles offered by standard cancer therapy [87].

The demand for nanoparticulate medication delivery devices is rising as nanotechnology develops quickly [88, 89]. Modern nanoparticulate dose forms can all improve a drug's bioavailability, such as polymerized nanocapsules, liposomes, nanoemulsions, and solid lipid nanoparticles. Nanoparticulate drug administration often enhances solubility and bioavailability, creates pharmacological characteristics, and disperses tissue macrophages. In contrast, traditional drug delivery lacks a drug release profile and is poorly absorbed, preventing toxicity because of prolonged exposure while hindering physical and chemical degradation [90, 91]. Characterizing the toxicological properties and efficacy of the manufactured nanoparticles is difficult [92–94]. Extensive research is still needed into the connection between the nanoparticles' forms and their functions. Plasma proteins are frequently adsorbable by nanoparticles, which interferes with the body's defense system. They might create free radicals, which might have genotoxic properties. Nanomedicines cost significantly more than conventional ones because characterizing them requires costly machinery. These nanomedicines also have a higher level of safety when compared to traditional medications, but doctors hardly ever suggest raising the level of effectiveness.

The difficulties with nanotechnology encountered during the production of natural medicines. The bioavailability of this nanoformulation has increased because of its bigger surface area and reduced particle size. However, the crucial issue is how safe and efficient it is. Overall, the intriguing promise of nanomedicines to decrease dosage frequency is a limitation of nanomedicine if a formulation enhances the pharmacokinetic characteristics. Instead, it fails to solve safety issues, resulting in toxicity in the body. High amounts of energy are required during the processing of nanoformulations like niosomes, liposomes, and polymer-based nanoparticles to solve stability-related difficulties such as coalescence and creaming [95]. Either a top-down (through pulverization or micronization) or a bottom-up approach is used to create nanoformulations (nucleation). The fundamental difficulty in expanding the formulation is controlling the growth of the particles. As a consequence, a batch-to-batch variance occurs [96].

Liposomes are an example of a nanotherapeutic substance that usually triggers an immune-mediated hypersensitive response when administered to the body. Additionally, they can be found accumulated in organs such as the liver, kidneys, and spleen [97–100]. Thrombocytopenia is a common side effect of liposomes use because they frequently interact with platelets and clotting agents like XII and XI. In order to promote their adhesion and coagulation, they also engage alongside the extracellular matrix, the lipoproteins and platelets [101, 102].

9.1 Challenges Faced to Use Nanotechnology in Herbal Medicine

The following is a list of properties of nanoparticles that are examples of factors that might impact the effective transport of herbal drugs for a particular purpose:

1. The nanoparticles' dimensions in terms of both shape and size.
2. The chemical and physical attributes of the extracts, such as their ability to remain stable and solubility in water.
3. Surface properties of the nanoparticles, including their charges, morphologies, porosity, and permeability, are considered here.

4. The level of biodegradation that has occurred.
5. The nanoparticles' biocompatibility and how they interact, in addition to any other aspects that may be pertinent.
6. Toxicity.
7. The amount of drug packed into nanoparticles and the effectiveness of encapsulation.
8. The release profile of the active component from the product, which includes information such as the component's kind (immediate, controlled, or focused release), discharge type (zero-order or first-order release), and discharge procedure (dissolution, diffusion, matrix system, etc.).
9. Antigenicity and other things.

Medications with poor solubility in water have traditionally been challenging for pharmaceutical specialists to formulate [103]. There is a substantial problem with the inability of therapeutic herbal extracts or their phytoconstituents to dissolve easily in water. Micronization, cosolvent solubilization, surfactant dispersion, and precipitation techniques are some of the standard procedures adopted in the pharmaceutical sector to improve the water solubility of medications that generally have poor water solubility [104]. On the contrary, it was seen that these approaches are ineffective when used in medicines that are not soluble in organic as well as water-based solvents.

9.2 Health Implications of Nanoparticles

It is essential to recognize the risks associated with using nanopharmaceuticals [105], despite the fact that there may be a wide variety of probable applications of nanopharmaceuticals in dispensing medicines for the diagnosis and treatment of various illnesses. There is a possibility that nanomaterials will enter the human body via several routes, including the epidermis, the gastrointestinal tract, and the lungs [106]. Nanoscale particles that enter the body through the lungs have the ability to quickly cross the blood-brain barriers (BBB) and reach crucial organs. This is because of the BBB's porous nature.

9.3 Challenges in Nanoformulation and Characterization

The size of a particle can significantly affect the pharmacokinetics (PK), biodistribution (BD), and safety of pharmaceuticals. Particulate size and its variation are some of nanoparticle-based medications' widely recognized distinguishing features. Small nanoparticles less than 20–30 nm in size are rapidly excreted by the kidneys following administration. In contrast, fragments 200 nm or bigger have a better absorption by the mononuclear phagocyte system, that is further known as the reticuloendothelial system (RES), which consists of living cells in the spleen, bone marrow, and liver [107]. The liver rapidly eliminates the particles within 200–400 nanometers, while nanomaterials within 150–300 nanometers in size are found predominantly in the spleen and liver [108, 109]. Consequently, particle size optimization is a crucial phase in manufacturing an efficacious nanoherbal composition. Nanoparticle behaviors and relationships with proteins and organelles are also greatly affected by surface characteristics [110]. Many surface attributes, including charge, hydrophobicity, and functional structures, can significantly influence nanomaterials' survivability over the long term and the opsonization process [111, 112]. With respect to traditional small molecules, the dimensions, morphologies, and physical and chemical attributes of nanoparticles may affect the pharmacokinetics (PK) and biodistribution of the active ingredients within them. Consequently, developing herbal medicines based on nanoparticles faces challenges with excellent pharmacological resistance. It is feasible to determine the detrimental effects of nanoparticles by quantifying the size, zeta capacity (surface charge), index of polydispersity, and solubility [113]. For instance, inhaling nanostructures smaller than 100 nm may result in oxidative stress and pulmonary inflammation [114]. Unstable nanomaterials possess the potential to form large aggregates that can become confined in the capillary bed of a patient's airways and cause significant problems. There is a correlation between the immunogenicity of nanomaterials and their size, surface properties, hydrophobicity, solubility, and charge. Nanomaterials could develop antigenic properties, become opsonized by plasma protein molecules, and are recognized as foreign components, triggering complement and activating rapid phagocytosis in macrophages and clearance if these characteristics are not effectively managed.

10 Past Studies and Future Prospective

Natural product and medicinal herb research has been conducted worldwide. A great number of research institutions are now working on developing herbal treatments for use in drug delivery systems at both the primary and clinical trial levels. The only requirement is to develop improved methods for administering these drugs to the appropriate sites and throughout the body in amounts that have no impact on the treatment options that are now accessible. It would be ideal to have a drug that not only lessens the severity of side effects like poisoning and allergic reactions but also makes the patient more resilient on the inside. This would be the ideal situation. Future research groups may be captivated by the idea of using botanical nanoparticles to transport anticancer medications, and they may produce noteworthy results.

Through the inclusion of "herbal medicine" in nanocarriers, the possibilities for healing many kinds of chronic illnesses and providing medical advantages will increase. Numerous successful examples supported by concrete evidence point to the right way of nanoresearch. Herbal remedies are exceptionally high in antioxidants and components which can be incorporated into foods for specific purposes [115]. This type of collaborative research between traditional "herbal treatments" alongside more modern techniques of the present medication delivery system, i.e. "nanotechnology," has created the pharmaceutical drugs that will enhance the wellness of people shortly. It is predicted that the advantageous and important applicability of natural items and herbal treatments used with the nanocarrier would raise the relevance of the present-day technique for the delivery of drugs [22].

Herbal remedies which are nanosized particles are able to enhance biological function and resolve problems with herbal medicines alone. The two new challenges in the development of nanotechnology-based drug delivery methods are the potential to obtain systems with multiple functions that satisfy distinct biological and therapeutic requirements and the feasibility of growing procedures that quickly develop novel approaches for therapy to the market. The study of the possible toxicological consequences of nanoparticles, known as nanotoxicology, has recently emerged as a new branch of toxicology. In previous decades, the beneficial effects of medical advances and their impact on patients' quality lifestyles were taken into account. The expenses related to medical treatment must now be considered. The objective of nanotherapeutic gadgets, which have more complex structures and are more expensive than conventional alternatives, is to decrease the total expense of healthcare [116].

Increased nanotherapeutic efficacy, shortened hospital stays, lower personal healthcare expenditures, and the successful management of costly primary maladies are likely contributors to this decline in costs associated with healthcare. The pharma regulation environment, healthcare policy, demographics, and general economic climate significantly impact the market for nanomedicine-related pharmaceuticals. Companies specializing in nanomedicine used specific approaches to surmount the challenges that arise in this fiercely competitive industry. BCC [117] found that the EU healthcare systems will be exposed to an increasing number of novel nanomedicine products due to the increased interest in and investment in nanotherapeutics. Incorporating nanotherapeutic treatments in country medication reimbursement policies will significantly impact the accessibility of these therapies across all healthcare systems. It is important to exercise caution when introducing nanoparticles into an ecosystem vulnerable to their effects, as the surroundings are already laden with nanoparticles of particulate matter. Investigating the targeting efficacy of nanomaterials and meeting global toxicological and biocompatibility standards are two additional new issues. Prospective use of nanopharma may alter human health in unimaginable ways, therefore evaluating their benefits and drawbacks has become essential.

11 Conclusion

The multidisciplinary development of nanotechnology is hastening the creation of the nearly impossibly compact gadget. Herbal extracts for medicinal purposes have already been recognized for having fewer negative impacts and are commercially accepted globally. When it comes to pharmaceuticals, nanotechnology offers a wide range of potential uses because it may enhance drug bioavailability and usage by decreasing drug size. National laws that govern the evaluation of medicinal plants' safety, quality, and effectiveness, alongside WHO attempts to support the development of model standards in these fields, have helped to strengthen recognition of medicinal plants' position in healthcare. Despite several breakthroughs, providing therapeutically effective drugs in this industry continues to face numerous challenges. One of the most significant obstacles in transforming these technologies into therapies is testing modern techniques to control how nanoparticles interact with biological constituents. Using nanoparticles and novel nanomedicines in conjunction with herbal treatments is less widespread since there is still a need for evidence of their safety and hazardous danger. However, if clinical validation information is obtained methodically, it can readily rub shoulders with contemporary medicine and lead to an entirely new segment of therapy in the medical world.

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Eco-metabolomic Studies of Medicinal Plants and Herbal Medicine

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Contents

Abstract

Ecometabolomics is becoming recognized as a valuable approach in ecological research, owing to recent advancements in technology, such as HNMR spectrometers, and bioinformatic progress. These improvements have facilitated the inclusive analysis of the metabolome, encompassing the entirety of metabolites, and their dynamic alterations in response to ecological perturbations. This quickly

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evolving technological platform offers a thorough qualitative and quantitative analysis of a particular metabolite set or the entire metabolite composition of a part. As a result of this technology, drugs are being discovered from natural plant resources, medicinal plant quality is being controlled, and new lead compounds are being discovered in recent years. From the perspective of small-molecule metabolism, metabolomics can be a terrific way to understand and enhance herbal remedies. By using a metabolic approach, development of novel pharmaceuticals, and encouraging the research. The scientific obstacles surrounding herbal remedies can be surmounted. In this chapter, we discussed the popular analytical methods used in metabolomic investigation and highlighted the relevance of eco-metabolomics in addressing several metabolites.

Keywords

Herbal medicine · Eco metabolomics · Mass spectrometry · Medicinal plants · Metabolomics

Abbreviations

CE	Capillary electrophoresis
FDA	Food and Drug Administration
GC-MS	Gas chromatography-mass spectrometry
HPLC	High-performance liquid chromatography
HSQC	Heteronuclear single quantum coherence spectroscopy
IMS	Ion mobility-mass spectrometry
NMR	Nuclear magnetic resonance
WHO	World Health Organization

1 Introduction

Natural habitats and the environment have been the most preferable sources since the beginning of human life. Food, energy, and therapeutics can all be found within local habitats and in nature. The current annual global pharmaceutical industry is worth roughly US$1.1 trillion. A third of these medications (about 35%) are made from natural ingredients, including substances derived from plants (about 25%), microorganisms (about 13%), and mammals (about 3%) [1]. Additionally, natural goods or their derivatives make up about 30% of the top-selling medications worldwide [2]. Furthermore, of the 1562 new medications that the US FDA approved between 1981 and 2014, 141 (9.1%) were botanical medicines comprising plants (defined mixtures), 64 (4%) were natural products that had not been altered, 61 (4%) were artificial medications that had a natural product pharmacophore, and 320 (21%) were natural product derivatives [1, 2].

The expected number of metabolites produced by plants in the plant kingdom ranges from 200,000 to 1,000,000 [3]. Numerous developed to withstand environmental challenges, defend against predatory animals, and function as chemical attractants [4]. Plants with these metabolites have been utilized for generations to treat and prevent illness. The bioactivity of the majority of these products against various malignant tumors and infectious illnesses has been established. Additionally, it is predicted that the global market for these natural goods is worth US$60 billion annually [5]. According to estimates from the WHO, medicinal plants are the main source of medication for about 80% of the world's population. Pharmacologically active molecules, medical agents, and drug research leads have increasingly been derived from plant natural products in recent years [6]. Medicinal plant-based metabolomics research is crucial because plants produce various primary and secondary compounds in large amount, some of which might be useful therapeutically. Approximately 200,000 secondary chemicals derived from plants have been subjected to scientific investigation [7]. For instance, *Arabidopsis thaliana* has produced 5000 secondary metabolites, while 1500 and 2500, respectively, have been derived from microorganisms and mammals. For the investigation of a variety of properties, such as metabolite profiling in plants, various genomics-based "phytochemical arrays" (proteome, transcriptome, genome, and metabolome) have been developed [8]. There have been reports of potential anticancer action in plant secondary metabolites such as camptothecinpaclitaxel (taxol) and podophyllo toxins among others [9]. As a result, alternate methods for finding new chemical entities (NCEs) for medication advancement are being explored, such as medicinal plants or natural products. From *Salix alba*, aspirin which is a semi-derivative chemical, has been utilized for many years [10] as a miracle medication to treat issues like pain. The best techniques for finding/isolating novel compounds, like morphine from opium, etc., are through ethnic applications of plants as traditional remedies (in the form of crude or extract). This was a technique for finding drugs from natural materials at the beginning of the nineteenth century. Because of their rarity in plants and consequently weak therapeutic action, a few plants secondary metabolites (single compounds) are frequently only weakly identified during drug analysis. However, since herbs and their formulations contain a variety of constituents, the biological activities can be generated synergistically. Metabolomics can be a useful tool in this situation to comprehend the phytochemical underpinnings of such medically active phytoconstituents [11]. Through various cutting-edge hyphenated technologies, in the realm of herbal medicine, metabolomic fingerprinting can be very beneficial for medication development, gene-function research, systems biology, and a variety of diagnostic methods. Targeting metabolomics is the study of a cluster of identified metabolites, and it typically combines NMR-MS techniques, which are employed in these kinds of analyses [12]. This aimed metabolomics technique determines the comparative concentration of around 200 predefined metabolites in a trial.

Through GC-MS and LC-MS analysis, the undefined or unknown plant metabolites may be found in a "untargeted" metabolomics investigation. This can be very beneficial for categorizing and identifying metabolites as well as for assessing herbal medicine [13]. Numerous areas of application can benefit from metabolic studies. Metabolite fingerprinting is a versatile analytical technique that encompasses various applications, such as the quantitative and qualitative assessment of specific molecules, the identification of chemical groups, the quantification of all metabolites, and the quick study of metabolites [14]. According to recent research, herbal remedies may be helpful in the treatment of a number of illnesses. The complexity of their chemical makeup now hinders the development of herbal pharmaceuticals, which use bioactive compounds with uncertain modes of action, unknown therapeutic targets, non-specific properties for drug metabolism, etc. To overcome these problems, metabolomics can be a useful method for learning about and bettering the small-molecule metabolism of herbal medicines, also with a metabolic perspective, metabolomics could accelerate the discovery and development of new pharmaceuticals by resolving scientific issues with natural therapies. Recently, there has been a significant utilization of MS-based metabolomics for the purpose of examining the in vitro and in vivo constituents of herbal extracts. In this study, we emphasize the importance of metabolomics and metabolism based on mass spectrometry for addressing the complication of herbal medicines in systems. Research on phytomedicine is now receiving special attention as a result of this investigation. It can be extremely helpful in changing the way that drugs are discovered and developed using natural materials.

2 Eco Metabolomics

The growing use of new technological advancements, such as contemporary GC-MSand HNMR spectrometers coupled with bioinformatic advancements, has raised understanding of Eco metabolomics, a discipline that investigates the metabolome, encompassing the complete assemblage of metabolites, and its dynamic alterations in reaction to shifts in the environment. For metabolomic approaches used in ecological investigations, we utilize nomenclature based on that of [15, 16] but simplified. Metabolomic profiling is the designated study type where the aim is to quantitatively analyze a collection of metabolites belonging to specific metabolic paths or distinct metabolite categories (e.g., polar, semi-polar, and non-polar), with as well as without their identification. The term "partial ecometabolomic studies (PEM)" is used to describe the investigation of an organism's response to ecological changes through the detection of specific metabolites or its group. This is achieved by employing quantitative and qualitative analytical procedures such as NMR or GC-MS. PEM refers to metabolomic profiling investigations conducted in the context of eco-physiological or ecological research endeavors, with the objective of unravelling a particular comprehensive pathway or the interconnections between many pathways. Hence, target studies that use many target techniques, like HPLC, and are concentrated on the analysis of a small number of metabolites are not regarded as ecometabolomic investigations.

Partial ecometabolomic studies are what we propose to name them. It is frequently sufficient to quickly categorize samples in accordance with their origin or their eco-physiological or ecological relevance when it is not necessary or possible to identify every metabolite. Metabolic fingerprinting is the term used for this procedure, which we now propose to refer to as exometabolic fingerprinting in an ecological setting. Through a suitable interpretation, the aforementioned "holistic" approach facilitates the objective investigation and analysis of molecular biochemistry samples, hence enabling the research of plants' responses to alterations in their surroundings [17, 18].

Analytical methods that allow for the measurement and identification of a wide range of metabolites are essential for determining and quantifying the maximum number of metabolites within a particular biological system. The term "metabolomics" refers to a method that reveals as much as possible about the metabolome of the biological system being studied. The term "ecometabolomics" is suggested to be used in reference to the investigation of an organism's comprehensive metabolomic response to alterations in the environment within an ecological framework. In actuality, the latter method ecometabolomics is the most suitable for understanding an organism's full reaction to environmental changes.

3 Metbolomics an Important Tool

Metabolomics is an emerging discipline within post-genomic biological research that is dedicated to investigating the characteristics and functions of tiny compounds, typically weighing less than 1000 Daltons, which are synthesized or utilized within biological systems. Metabolomics provides a comprehensive methodology for comprehending the molecular alterations that transpire across a complete body by examining the impacts of these molecules on biological or natural systems. This discipline gives significant insights into the influence of diverse elements on biological systems and presents a comprehensive viewpoint on the interaction between molecules and biological processes [19, 20]. Because metabolites change more quickly than genetic materials, they are thought to be a better indicator of ongoing physiological events than phenotypes [21]. In essence, the scientific community has used metabolomics frequently and regards it as a method for identifying biomarkers [22, 23], disease diagnosis [24, 25], novel drug discovery [26], drug mechanism study [27, 28], drug safety assessment [29], and toxicology evaluation [29], because it offers extremely dependable and repeatable findings. The metabolome is the entire set of metabolic molecules, varies from the genome, transcriptome, and proteome in some ways. Metabolites are more diverse in terms of their physicochemical nature than DNA or proteins [15, 30, 31]. The present difficulties in the study of plant metabolites are their heterogeneity and complexity considering that only one-fourth of the estimated 200,000 metabolites in plants have been fully characterized [15, 32, 33]. The metabolome reflects the intricate connections that occur among the genome, transcriptome, proteome, interactome, and interaction networks [34]. Consequently, metabolomes not only supply the cell with the metabolic components it needs, but also serve as a diagnostic tool [35]. The metabolome serves as the culmination of interactions between the environment and biological system, whereas the genome specifies indirectly what might occur [36]. Furthermore, the use of metabolomics data analysis does not vary on the presence of organism-specific genome data, unlike proteomics data analysis, genomics, and transcriptomics, which necessitate such information [37]. Regardless of the creature that produces them, this special characteristic comes from the same molecular object as metabolites [38].

4 Mass Spectrometry Techniques Applied to Metabolomics

Despite NMR and MS being the most frequently used analytical methods in metabolomics, MS is usually chosen because of its unparalleled structural specificity, high resolution, and sensitivity. In practical applications, the pre-separation process involves the integration of IMS, Gas Chromatography (GC), Liquid Chromatography (LC), or other methodologies with Mass Spectrometry (MS) [39]. The advancement of MS-imaging (MSI) tools has contributed to an enhanced comprehension of metabolomics by broadening our knowledge of the spatial allocation of metabolites.

4.1 Gas Chromatography-Mass Spectrometry

In the early 1970s, Devaux conducted pioneering research on metabolite profiling, employing GC-MS to investigate various constituents of steroids as well as urine drug metabolites [40]. GC-MS is a cost-effective as well as reproducible analytical technology that offers a high peak capacity for MS analysis. It is particularly well-suited for detecting thermo-stable volatile compounds or their derivatives. However, it necessitates intricate and time-consuming sample preparation, which can potentially alter the structure of the analytes during the process. Despite its shortcomings, GC-MS is nevertheless often employed in metabolomics due to the abundance of commercial and public libraries for chemical identification [41]. Using GC-MS, researchers examined the medicinal benefits of Jieduquyuziyin, a traditional Chinese medicine formulation containing Rehmanniae Radix, Trionycis Carapax, and Artemisiae annuae herba (the dried aboveground portions of *Artemisia annua* L.). Multivariate statistics revealed 11 metabolites that were linked to the metabolism of amino acids and energy, including malic acid, citric acid, hippuric acid, etc. The disruption of these networks may prevent the body's regular physiological processes and hasten the onset of systemic lupus erythematosus. For preferred therapeutic outcome, the concentrations of metabolites seen after administering the Jieduquyuziyin prescription were found to be similar to those observed in the control group. The findings of this analysis provide clarification regarding the therapeutic effects of the Jieduquyuziyin prescription on systemic lupus erythematosus, as observed by the analysis of urinary metabolomics using GC-MS. A significant male infertility condition in traditional Chinese medicine is called KYDS. In this study, a metabolomics approach utilizing GC-MS was created to investigate the modernization of KYDS (Kampo-Yonin-to-Dokudami-San) for the treatment of male infertility. A total of 67 infertile males diagnosed with KYDS and 55 healthy controls were recruited as participants for this research endeavor. A total of 37 potential biomarkers were identified across six metabolic pathways. Among these indications, ten were found to effectively differentiate infertile individuals with KYDS from healthy individuals, hence offering valuable insights into the underlying process [42].

4.2 Liquid Chromatography-Mass Spectrometry

Traditional medicine recognizes several treatments for the dampness-heat jaundice syndrome, including Artemisia capillaris Thunb., Rhei Radix et Rhizoma, and Gardeniae Fructus (the dried mature seeds of *Gardenia jasminoides* Ellis). In a study conducted by Sun [43], the therapeutic mechanism of YCHT was examined. The researcher employed UPLC-MS-based technology to analyze metabolite alterations that occurred during the treatment. As a result, an integrated biological interpretation was developed. Consequently, the association between the syndrome of dampness-heat jaundice and a total of 22 biomarkers and 8 metabolic pathways was established. The development of the biomarker-protein interaction network was also undertaken in order to forecast potential targets. In their study, Zhu [44] employed UPLC-Q-TOF MS to exhaustively elucidate the mechanism as well as targets of Huanglian Jiedu Decoction in the context of cerebral ischemia. This decoction consists of Coptidis Rhizoma, Scutellariae Radix, Phellodendri Amurensis, Gardeniae Fructus, and Cortex. A total of 19 biomarkers associated with ten metabolic pathways were identified in the experimental groups that underwent therapy with Huanglian Jiedu Decoction, in comparison to the control group, as well as the groups with cerebral artery blockage and normal conditions. The study of metabolomics and pathophysiology has established that Huanglian Jiedu Decoction exerts its pharmacological effect on cerebral artery occlusion through the maintenance of cholinergic neuron function, modulation of metabolic stress, and regulation of the glutamate/gamma-aminobutyric acid (GABA)-glutamine cycle. These studies were able to effectively elucidate the targets and mechanisms of drug action using UPLC-MS-based metabolomics.

4.3 Capillary Electrophoresis-Mass Spectrometry

Charged analytes can be quickly and precisely separated via CE using minimal injection volumes. It is a potent analytical technique that, when combined with MS, allows for molecular characterization based on fragmentation and provides (precise) mass information. Highly charged metabolites and nucleotides are among the kinds of metabolites that the platform can easily capture, despite being incredibly sensitive and capable of doing so. The separation efficiency observed in CE is generally superior by a factor of ten compared to that achieved in traditional high-performance liquid chromatography [45]. Whereas the performance of LC is typically constrained, CE is applicable for the split of polar drugs (charged) or metabolites [46]. For the same sample, different settings can be swapped out to expand coverage. The productivity of the separation procedure in CE is completely influenced by the magnitude of the applied electric field, which effectively facilitates heat dissipation. The biggest obstacle to its widespread adoption is its inherent lack of consistency. Though there aren't many instances where CE-MS is used in metabolomics, those who are adept at using it can achieve separation efficiencies that are on level with or even superior to GC-MS and LC-MS results. In the study conducted by Kimura [47], plasma obtained from a prospective cohort of 112 patients was utilized for the purpose of performing a CE-MS metabolic profiling of end-stage kidney disease. There were 218 metabolites found in total, and 16 of them have predictive readings for end-stage kidney disease, which is a condition that is associated with a high risk. In addition, 73 of the metabolites had a substantial relationship to the estimated glomerular filtration rate. The assessment of metabolites can potentially aid in the management of chronic renal disease by assessing the probability of its progression to end-stage kidney disease.

4.4 Ion Mobility-Mass Spectrometry

IMS is an electrophoretic method that operates in a gas phase, enables rapid segregation of ions according to their size, charge state, and structure [48]. The utilization of collisional cross-section (CCS) values, which are a unique molecular physicochemical attribute that represents the dimension and structure of ions, has the ability to facilitate the process of identification of metabolites [49]. Recently, an improved method of metabolite analysis has developed that combines IMS and MS. This technique improves the identification and coverage depth of metabolomics by providing millisecond structural resolution and multi-dimensional in addition to mass information [49]. The utilization of IMS has facilitated a novel approach in the discipline of metabolomics, allowing for the acquisition of multi-dimensional molecular data suitable for both semi-targeted and untargeted research purposes. Lipids constitute the principal constituents of cellular membranes and play a critical role in the metabolic processes essential for sustaining life. The precise determination of their natural functions has been impeded by the prevalence of isomers and the absence of a proficient analytical apparatus. The application of IMS was extensively investigated for the purpose of distinguishing between lipid isomers. The success rate in differentiating the four primary categories of glycero- and phospholipids was found to be 75%. These lipids can be distinguished based on variations in sn, cis/trans configurations, chain length, and the position of double bond. The results of this study showcased the efficacy of IMS as an effective implement for conducting lipidomic investigations [50].

4.5 Mass Spectrometry Imaging

The first step in the investigation of biological tissues using LC-MS-based metabolomics involves the homogenization of samples to extract metabolites [51]. The method employed in this study involved combining the metabolites from all cells within the specimen to evaluate the overall influence on the tissue as a whole. However, it should be noted that this technique does not account for the spatial characteristics of the tissue. The utilization of MSI technology in the context of identifying metabolites in situ holds significant promise as it allows for the visualization and characterization of metabolite distribution and changes. The most advanced imaging method used today is MALDI MSI. The sectioned tissues are subsequently transferred onto a MALDI plate, where they are subjected to matrix spraying, laser rastering, and subsequent collection of their MS spectra, peak intensities can be transformed and then translated into two-dimensional images, allowing for the natural visualization of metabolite changes. Other ion sources besides MALDI can be used for MSI, including secondary IMS, laser ablation electrospray ionization, and desorption electrospray ionization. Medical research, microbiology, and natural product chemistry are just a few of the life sciences fields where MSI methods have created new opportunities [52]. Sarabia [53] developed a high-resolution matrix-assisted laser ionization MSI technology to investigate the impact of salinity stress on the structural distribution of metabolites in the barley seedlings roots. The researchers conducted a comparative analysis of metabolites prior to and during exposure to salt-induced stress. They successfully identified, visualized, and distinguished several chemicals in meticulous detail throughout longitudinal sections of the root. The levels of Phosphatidylcholine (PC [54: n]) exhibited a general decrease in salt-treated roots, indicating a reaction to the stress induced by high salinity. To investigate the spatial dynamics of plant growth and metabolites, researchers employed a novel methodology to grab a deeper understanding of the spatial metabolomics occurring in the maturation zone and elongation region during root development following an exposure to salt-induced stress. This approach facilitated the elucidation of underlying mechanisms involved in plant responses.

5 The Methabolomics Approaches Used in Herbal Medicine Research

The fundamental method and consistent detection for the precise identification of unidentified metabolites is NMR-based metabolomics. But compared to MS, this technology is less sensitive. Currently, MS is the most popular tool in plant metabolomics [55]. The first technique in plant metabolomics discipline is the utilization of gas chromatography coupled with electron impact and time-of-flight mass spectrometry (GC-TOF-MS), which is widely implemented [56]. GC-TOF-MS enables the achievement of high reproducibility in the separation and fragmentation patterns of lipophilic compounds and non-volatile primary metabolism metabolites [57]. CE-MS offers the most optimal means of detecting and separating a variety of polar primary metabolites. Contrary to GC-TOF-MS, it is a technique that is infrequently employed [58]. The most effective approach for the characterization and identification of semi-polar metabolites involves the utilization of LC-MS coupled with atmospheric ionization techniques, such as chemical ionization, atmospheric pressure, and electrospray ionization. In general, LC-MS is commonly employed for the identification of many plant secondary metabolites, including but not limited to alkaloids, saponins, flavonoids, phenolic acids, polyamines, and their derivatives [59, 60]. These metabolites can be immediately analyzed without any derivatization because they are typically extracted with hydroalcoholic solutions [61]. Direct flow injection mass spectrometry (DFI-MS) enables the direct assessment of metabolic signatures without the need for earlier separation. Nevertheless, the utilization of this direct injection technique may give rise to complications pertaining to ion suppression and the development of substantial adducts. Additionally, DFI-MS is unable to distinguish between multiple molecule isomers. Therefore, a separation method must be used in plant metabolomics research using MS-based platforms. In general, vegetation contains a lot of isomeric compounds. Such isomeric substances can be detected using LC in conjunction with MS. Additionally, in addition to allowing, it also offers helpful details about the metabolite's structure that can be collected online [62]. A single crude plant sample can contain hundreds of metabolites when LC and high-resolution MS are combined. The LC-MS technique, known for its rapid advancements in data collecting and processing, is considered the most advantageous approach for metabolite identification. Among the advanced techniques employed in plant metabolomics, LC-NMR and LC-NMR-MS are notable for their sophisticated identification capabilities [63–66].

6 The Analysis of Untargeted Metabolites in Medical Plants Based on LC-MS

Polar sugars and different lipids are among the thermolabile and non-volatile substances that can be analyzed using LC-MS. Due to its ubiquitous adaptability to the majority of compounds, LC has been the most effective method for studying herbal medicines. Capillary LC, monolithic column (single piece column) UPLC, and LC were created to increase segregation effectiveness; this approach offers a high level of precision and provides an untargeted analysis. As a result, it can be used to analyze a variety of plant compounds. Untargeted metabolomics investigates the entire plant metabolome without favoring any particular compound. Untargeted metabolomics is the best method for thorough qualitative and quantitative study of herbal medications.

6.1 The Typical Workflow for Untargeted Metabolomics Using LC-MS Studies

The process of untargeted metabolomics investigations utilizing LC-MS typically involves many fundamental phases. The preliminary stage of the research process involves the preparation of the experimental design, which is subsequently succeeded by the groundwork of the sample and the collecting of data. The efficacy of these steps is contingent upon the specific aim and purpose of the research inquiry [67]. Sample preparation is a very important stage, and it can be affected by a number of sample integrity factors [68]. The choice of extraction technique and solvents employed significantly influence both the quantity and diversity of metabolites obtained. For the extraction of metabolites, a singular solvent solution is typically employed. However, it is also possible to extract a variety of metabolites from the plant sample using a mix of solvents [69, 70]. The advancements in automatic data processing means such as XCMS, Metalign, and MZmine have facilitated the analysis of complex untargeted metabolomics data and the abstraction of crucial information from raw chromatographic data. This approach facilitates and empowers the analysis of vast quantities of untargeted metabolomics data [71, 72]. Once the procedure of data processing has been concluded, it becomes imperative to analyze the acquired data in alignment with the research's purpose and objective, employing the suitable methodology. Identification and measurement of the entire plant metabolome is the primary goal of untargeted metabolomics (both known and unknown metabolites). With untargeted metabolomics, it is difficult to use simple statistical analysis or other traditional statistical techniques. As a result, this technique uses multivariate statistical methods to analyze huge data sets of unidentified metabolites. The application of untargeted metabolomics also enables the identification of synergistic effects that may not be discernible at the individual level. The issue under consideration can be effectively addressed by applying multivariate statistical approaches that are grounded in the concept of projections. In this approach, it is possible to integrate the elements under research with the latent variable. Principal component analysis (PCA) is generally utilized in the examination of exploratory data. In contrast, projection to latent structures discriminant analysis (PLS-DA) and orthogonal projection to latent structures discriminant analysis (OPLS-DA) are widely recognized as highly successful statistical methodologies for extracting concealed information that enables the categorization and characterization of system behavior [73]. The two primary processes for transforming secondary metabolites in plants are esterification and glycosylation. The pharmacological effects of these various secondary compounds are produced. However, it is very tough to perfectly determine the metabolites in untargeted metabolomics. The coverage of metabolites in existing databases such as HMDB, MASSBANK, MS2T, METLIN, Campus Chemical Instrument Centre, and Birmingham Metabolite Library is inadequate due to the extensive diversity of plant metabolites [74, 75]. The fragmentation and mass pattern of plant metabolites can be accurately determined with the help of tandem mass spectrometry, which simplifies structural analysis.

7 The Analysis of Untargeted Metabolites in Medical Plants Based on NMR Analysis

The integration of contemporary multivariate statistical analysis techniques, in conjunction with the latest breakthroughs in analytical chemistry focused on the identification and characterization of tiny molecule compounds using high-field NMR, has facilitated the development of very efficient methodologies for doing extensive analysis of data derived from metabolomics investigations. NMR procedures exhibit notable attributes such as expeditiousness, dependability, and the ability to produce consistent results. One advantage of this method is its universality in detecting all NMR active substances, distinguishing it from mass spectrometry-based techniques. This is attributed to the proportional relationship between signal strength of protons and metabolite quantity, enabling direct comparison of compound concentrations. In the subject of plant metabolomics, the utilization of solid phase extraction in combination with LC and NMR, or HPLC-NMR has quickly increased.

7.1 A Workflow for Untargeted Metabolomics Studies Utilizing NMR

The fundamental procedures involved in untargeted metabolomics investigations utilizing NMR are analogous to those employed in studies utilizing LC-MS. In these studies, the step of sample collection and extraction is essential. Deuterated solvents are typically used as the extraction fluid. For the NMR investigation, combinations of KH_2 PO_4 and D_2O and CD_3OD are employed to extract the full metabolome [69]. The development of data processing technologies like as SIMCA and ACD Labs has facilitated untargeted metabolomics investigations by efficiently analyzing several variables and extracting significant information from NMR spectroscopic data. In the untargeted metabolomics approach, multivariate statistical methods will be employed to analyze a diverse range of unidentified plant metabolites using the data obtained from NMR. With the use of this technology, multivariate statistical methods for LC-MS-based metabolomics can be applied [76]. Several variables, including concentration, pH, ionic condition, temperature, can affect the sensitivity of NMR parameters [77]. Consequently, the detection of metabolites is the most difficult. In order to enhance the discernment of plant metabolites, it is imperative that the NMR spectra be obtained under identical conditions as those employed during their acquisition for the database. Multiple multidimensional nuclear magnetic resonance (NMR) tests must be performed to validate the identified metabolites within a cohort of similar samples, including TOCSY, HMQC, HMBC, HSQC, or a combination of HSQC-TOCSY/HMQC [78]. Derivatizing agents, which are chemically selective probes, can be employed for the purpose of measuring functional groups. These agents have the ability to precisely target aldehydes, ketones, thiols, and carboxylates. The chosen metabolites containing specific functional groups can be identified by strategically including isotopes such as 15N or 13C at specific locations inside the chemically selective probe, utilizing the isotope-editing capability of Nuclear Magnetic Resonance (NMR) [79, 80]. Metabolites fingerprinting has been done to analyze the Indian medicinal plants metabolites which are listed in Table 1.

Table 1 Metabolite fingerprinting of traditional Indian medicines: the table showing Indian medicinal plants which undergo various metabolomics approaches

Indian medicinal plant	Metabolomics analysis approach	Metabolites analyzed	References
Commiphora wightii	GC–MS, NMR, HPLC	n-Methylpyrrolidone; GuggulsteroneZ,E; Isoeugenol; α-tocopherol; β-Caryophyllene; β-Myrcene; δ-Cadiene; D-limonene; Quinic acid; β-Elemene; Guaiacol; α-Caryophyllene; Verticiol; transFarnesol; Prostaglandin F2; myo-inositol; Protocatechuic; Cinnamic acids: Gallic acid.	[81]
Psoralea corylifolia	HPLC/UV–MS GC/MS	Neophytadiene; Psoralene; Daidzein; Stearic acid; Myristic acid; 1,2-Benzenedicarboxylic acid; Phytol; Caryophyllene oxide; 2Furancarboxaldehyde,5-(hydroxymethyl); Linoleic acid; Bakuchiol; Phytol; Palmitic acid; 2,6Dimethoxyphenol; 2,8-Diisopropyl-peri-xanthenoxanthene4,10-quinone; 1-Eicosanol.	[82]
Echinacea species (Echinacea pallida, angustifolia and *purpurea)*	Biplot, HPLC/ESI/MS, GAP	Trideca-2E,7Zdien-10,12-diynoic acid-isobutylamide (DETBA); Undeca-2Z(E),4E(Z)-dien-8,10-diynoic acid-isobutylamide; Undeca-2E (Z)-en-8,10-diynoic acid-isobutylamide; Undeca-2Z-en-8,10-diynoic acid-2-methylbutylamide; Pentadeca-2E,9Zdien-12,14-diynoic acid-isobutylamide; γ-Gurjunene; Germacrene-D; Aromadendren; p-Menth-1-ene-6-ol; Dodeca-2Z,4E-dien-8,10-diynoic acid-isobutylamide.	[83]
Withania somnifera	GC-MS,NMR and HPLC	Palmitic acid; Oleic acid; Linoleic acid; Citric acid; Fructose-5 TMS; Fructose-5 TMS, Fructose-5 TMS, Fumaric acid (L); Glycine (L); Myo-inositol (L); Isoleucine (L); Lactic acid; Malic acid 3 TMS; Lysine; Leucine (L); N-acetyl-glucosamine (L); Phenyl alanine (L); Tartaric acid (L); Benzoic acid (L & R); Butandioic acid (L); Phenyl acetic acid (L & R); β-Sitosterol (L); p-Hydroxy, phenyl acetic acid (R); 3,4,5-Trihydroxy cinnamic acid (R).	[84]
Tinospora cordifolia (Guduchi) in Sanskrit	UPLC-QTOFMS	Mangoflorine; Jatrorrhizine; Columbamine; Berberine; Tinosporoside; Menisperine.	[85]

(continued)

Table 1 (continued)

Indian medicinal plant	Metabolomics analysis approach	Metabolites analyzed	References
Saraca asoca	UPLC-QTOFMS	Prunasin; Ophosphocholine; Delphinidin (antioxidant); (−) Epicatechin; Procyanidin B1.	[86]
Zanthoxylum armatum Commonly known as (Winged Prickly Ash)	UPLC-DAD-ESI-QTOF-MS/MS	Zanthosin; Rubemamin; Horsfieldin; Magnolin or epimagnolin; Isomer of hydroxysanshool; Eudesmin; Armatamide; Fargesin; Xanthoxylin; Kobusin; Sesamin; Dioxamin; Asarinin; Hydroxy-α-sanshool.	[87]
Curcuma longa (turmeric)	LC-ESI-MS	Bisdemethoxycurcumin (curcuminoid); Demethoxycurcumin (beta-diketone); Curcumin; 5-Hydroxy-1,7-bis(4-hydroxyphenyl)-3-heptanone; 11-(4-Hydroxy-3-methoxyphenyl)-7-(4-hydroxy-3,5-dimethoxyphenyl)-4,6-heptadien-3-one; 7-Bis(4-hydroxy-3-methoxyphenyl)-4,6-heptadien-3-one.	[88]
Boerrhaevia diffusa (Red Spiderling)	HS-SPME, headspace solid-phase microextraction	3 Limonene; a-Pinene; 2-Decen-1-ol; 2-Nonen-1-ol; Camphor; Menthol; cis 4-Hexen-1-ol; Geranylacetone; Phellandrene; Isomenthone; Methylpyrrole; β-Cyclocitral; 3-Phenyl-2-(20-pyridyl)-indole; trans 2-Octanal; Vanilin; 4-Oxoisophorone; Resorcinol monoacetate; β-Ionone; Dihydroactinidiolide; Eugenol; Benzothiazol; Benzophenone; Linalyl anthranilate.	[89]
Glycyrrhiza species (*Glycyrrhiza echinate, glabra, inflata* and *uralensis*)	1HNMR, GC-MS and LC-MS, PCA	Rhamnose (glycosides); Isoliquiritigenin; Glycyrrhizin; Liquiritigenin; Hydroxyphenyl acetic acid; Liquiritin; Sucrose; Licochalcone; Isoliquiritin; 4-Licochalcone.	[90]

8 The Advantage of Metabolomics in Herbal Medicine Research

Evaluation of metabolic products is of utmost relevance due to their crucial role as constituents of biochemical pathways, their potential utility as biomarkers for various biological states, and their nutritional significance [55]. Multiple studies have indicated that metabolites have a significant impact on the gene expression regulation and the modulation of an organism's response as a signal to stress and development [91]. Recent progress in characterizing metabolic alterations in unconditional pathways has aided our understanding of how cells prioritize and allocate scarce nutrients. Investigating things at the metabolite level may have an immediate effect on biological performance. In order to determine how effectively herbal medicinal products can treat illness, it can therefore be applied in a promising way in herbal metabolomics. Advantages of this technique are not limited; it is widespread across the world. Some common herbal medicines across the world are listed in Table 2.

9 Promising Ways to Tackle Challenges in Untargeted Metabolomics Studies

Untargeted metabolomics methodologies, such as LC-MS and NMR, offer insights into the structural aspects of metabolites, including stereochemical characteristics. Additionally, these techniques enable rapid, automated, and high-throughput analysis of raw plant extracts, facilitating the quantification of diverse metabolites [92, 102].

NMR	Functions	Overreliance	Challenges	Promising ways to tackle challenges
	NMR provides structural information about the metabolites, but it has very low and poor chemical resolution [103].	The magnetic field's strength affects the NMR signals' sensitivity and intensity.	1. As a result, stronger magnets are needed to enhance the performance of the NMR technique. 2. In the assessment of plant extraction by the utilization of NMR spectroscopy, the substantial resonance overlap may make it difficult to quantify and identify metabolites.	1. The utilization of complementary metal oxide semiconductors in the increased high-throughput NMR spectrometer involves the integration of several interconnected radio-frequency circuits and high-sensitivity micro-coils on a single chip. This integration aims to enhance both sensitivity and

(continued)

				throughput [104]. 2. Numerous NMR investigations using enrichment or fraction step have been successful in overcoming this [105].
LC-MS	Compared to the NMR, the LC-MS platform offers greater sensitivity and selectivity. Additionally, it enables the detection and measurement of metabolites at very low concentrations [106]. Semi-polar phytochemical identification is frequently done using LC-MS. The detection of unidentified spectrum characteristics in the utilization of heteronuclear and multidimensional NMR techniques, such as 31P and 15N, can enhance the efficacy of NMR-based investigations.	When comparing the database, the combination of heteronuclear scalar coupling and chemical shift measurements can provide enough information to identify metabolites.	The proper functional groups can be difficult to identify using chemical shift values.	Hence, more investigation is necessary to confirm the existence of amine functional groups and carboxylic such as through the utilization of pH titrations [107].

10 Conclusion

In light of the strengthening global demand and expanding market for herbal drugs, it is imperative to conduct thorough investigations and get comprehensive insights into the fundamental breakthroughs in this field by leveraging modern research technology. The incorporation of a wide range of analytical instruments, in conjunction with multiple techniques and online bioassays, is helpful for evaluating efficiency. Presently, the drug development process primarily utilizes the reductionist method in order to identify compounds that demonstrate intriguing features. Nevertheless, the successful recognition of active chemicals can be realized by the utilization of the

Table 2 Analytical techniques to identify the components after quantitative analysis by different plant parts or from herbal medicine

Herbal medicine/plant	Plant part	Analytical technique	Components of quantitative analysis	References
Ocimum Sanctum	Entire plant of *Ocimum sanctum*	MS/UPLC-ESI MS	Vanillic acid, Apigenin, Catechin, Sinapinic acid, Quercetin dihydrate. Caffeic acid, Rutin, Ferulic acid, Rosmarinic acid.	[92]
Rabdosiae Rubescentis Herba	*Rabdosiarubescens dried* above ground parts	HPLC-ESIMS/MS	Effusanin, Rosmarininc acid, Quercetin, Rutin, Isorhamnetin, Lasidonin, Chlorogenic acid, Ferulic acid, Salicylic acid, Caffeic acid, Lasiokaurinol, Nodosin	[93]
Decoction Mahuang Fizi	Derived from ephedra Herba, Asari radix et rhizome	HPLC-QQQ MS/MS	Methyl ephedrine, Songrine, Neoline, Talatisamine, Chasmanine, Benzoylmesaconine.	[94]
Injection Shenqi Fuzheng (SQFZ)	Derived from extract of radix astragali and radix codonopsis	UPHPLC-Q-TOF MS	Adenosine, L-phenylalanine, Codnopsine, Tryptophan, Ononin, Fructose, Glucose, Sucrose, Soyasaponin I, Cyclocephaloside II.	[95]
Capsule Wuzhi	Preparation of an herbal extract (ethanolic) of *Schisandra sphenanthera*a deciduous climber (Nan-Wuweizi).	LC-MS/MS	Deoxyshisandrin, Schisandrin, Schisandrol B, Schisantherin A, Schisanhenol.	[96]
Lyophilized injection YiqiFumai (YQFM)	Derived from dried roots and rhizome of ginseng radix. A modern preparation derived from Shengmai powder.	UFLC-IT-TOF-MS	Ginsenosides Re, Ginsenosides Rg1, Ginsenosides Rc, Ginsenosides F2, Ginsenosides R-Rg3, Rh1, Ro, Schizandrol A, B, Gomisin D, Ophiogenin.	[97]

(continued)

Table 2 (continued)

Herbal medicine/plant	Plant part	Analytical technique	Components of quantitative analysis	References
Sanhuang tablet	Prepared from powdered form extract of *Scutellariabaicalensis* Georgi.	UPHLC-Q-TOF-MS	Sennoside, Daidzein, Rhein, Emodin, Coptisine, Palmatine hydrochloride, Berberine, Sucrose, Wogonin, Jatrorrhizine, Eriodictyol, Baicalin, Physcion.	[98]
Fructus Sophorae pill	The substance is obtained from the fully developed seeds of *Sophora japonica* L. and the dehydrated roots of *Sanguisorba officinalis* L.	HPLC-MS	Kaempferol, Baicalein, Quercetin, Cimifugin, Naringin, Hesperidin, Sophoricoside, Genistin, Rutin.	[99]
Capsule Dengzham Shengmai	Derived from entire plant of *Erigeron breviscapus*	HPLC-DAD-ESI-IT MS	Chlorogenic acid, scutellarin, caffeic acid, Schisantherin B, A, E, Ophiopognin D.	[100]
Centipedae Herb	Derived from entire plant of *Centipeda minima* L.	HPLC-Q-TOF MS	Caffeic acid, Chlorogenic acid, Kaempferol-3-o-rutinoside, Bervilin A, 3-Methoxyquercitin, Isochlorogenic acid C.	[101]

metabolomics approach in combination with multivariate data processing techniques. The utilization of NMR and MS methodologies in recent scientific progress has greatly augmented the capability to assess a wide range of chemical compounds. These developments have moreover bolstered the practice of acquiring data for newly discovered compounds by making use of the open access natural products library. Without a doubt, metabolomics is widely recognized as the most efficient method for assessing the effectiveness of herbal medicines. This methodology effectively captures the diverse range of changes exhibited by the key constituent elements that exert significant influence on the fundamental active elements. The metabolomics approach enables rapid analysis of a diverse array of metabolites. As a result, the product of metabolomics in the realm of personalized medicine is experiencing an upward trend. The approach outlined in this study allows for the correlation between the metabolic profile of individual patients and their distinct responses to a particular herbal medicine. This correlation considers factors such as efficacy, clearance, and toxicity. The utilization of metabolomics approach greatly enhances the process of discovering new leads and identifying targets that have not been previously studied. Metabolic tests possess the capacity to provide valuable insights into both the underlying mechanisms of action and the promising adverse effects. The utilization of tandem mass spectrometry enables the examination of metabolite structures, hence enabling accurate purpose of the mass and fragmentation pattern of plant metabolites.

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Physiological Ecology of Medicinal Plants: Implications for Phytochemical Constituents

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Abstract

Herbal plants have been valued for their medicinal, flavoring, and aromatic properties for centuries. The global market for medicinal and aromatic plants is estimated to be $62 billion, with demand expected to reach $5 trillion by 2050. Aromatic plants and herbs have been widely cultivated and used for medical purposes not only for humans but for animals as well. According to the World Health Organization, herbal medicines are characterized as substances or products derived from plants that possess therapeutic or other beneficial effects for human health. These herbal remedies can be sourced from one or more plants and may include both raw and processed ingredients. This chapter describes the curative values, botanical characteristics, and climatic and cultural requirements of medicinal plants. These include psyllium husk, black cumin or black seed, stevia, fennel, bishop's weed/carom seeds, sweet basil, holy basil, linseed, and turmeric, all medical plants that are grown worldwide. Despite having a higher market value than major field crops, medicinal plants are not cultivated on the same scale as significant as other crops. One of the major reasons for the globally low production of medicinal plants is the lack of information regarding their suitable cultivable climate and their poor adaptability to growing environments. It is imperative to give careful consideration to the selection of soil and cropping strategies to achieve satisfactory yields of high-quality products while ensuring the preservation of their safety and nutritional value. Physiological ecology is also discussed in this chapter in relation to medicinal plant therapeutic potential and medicinal qualities. It is still necessary to explore and understand the cultural practices necessary to cultivate medicinal plants successfully. This chapter aims to address these gaps by discussing the input requirements, therapeutic value, and suitable cultivation areas for medicinal plants. This chapter aims to fill these knowledge gaps by disseminating insights into cultivation practices and addressing these knowledge gaps. By addressing these gaps and disseminating insights into cultivation techniques, it hopes to develop sustainable and optimized cultivation techniques for medicinal plants. As a result of such advancements, medicinal plants will be available for future generations and will meet the growing demand for them.

Keywords

Climate change effects · Cultural practices · Ecological interactions · Geographical distribution · Herbal plants · Irrigation and nutrient requirements · Phytochemical profiling · Therapeutic or medicinal properties

1 Introduction

Aromatic plants, spices, and herbs are widely used for medical purposes due to their antimicrobial activities, in addition to their flavor and fragrance qualities [1, 2]. The positive effects of aromatic plants, herbs, and their essential oils in various diseases have been evidenced throughout human history [3, 4]. The utilization of herbal remedies for disease treatment is prevalent in nonindustrialized societies across the globe [5]. The advancement of chemical and phytochemical analysis has resulted in a growing reliance on herbal medicine for addressing human ailments. It has been observed that the characteristics of medicinal and aromatic plants can vary significantly based on soil composition. Therefore, careful consideration must be given to soil selection and cropping strategies to ensure the production of high-quality, valuable products that prioritize safety and nutritional value. Medicinal plants require different soil and cropping strategies to obtain superior yield and quality [6]. While sustainable production of raw materials from these plants can be achieved through growth manipulation.

Medicinal plants have the potential for enhancing the consistency, purity, and quality, as well as the biomass production of raw materials [7]. The inconsistent and low availability of high-quality plant raw materials often hinders the sustainable production of various innovatively developed plant-based products. By cultivating these plants, it becomes possible to improve biomass production, ensuring a sustainable supply without compromising the bioactivities of the medicinal plants through the manipulation of growing conditions. For example, the cultivation of *P. sidoides* under well-watered conditions significantly accelerated the biomass, without significant alteration in the content of active compounds [8].

Despite efforts to improve the cultivation of medicinal plants, comprehensive information on production techniques and product biochemical composition and active ingredients of these plants is still lacking, particularly high-quality research data on appropriate cultivation technologies for geologically diverse agricultural systems and ecological zones. Research on promoting innovative cultivation and application technologies for medicinal plants could help improve their yield and quality. Further, understanding the biochemical composition of plant products will assist in identifying suitable allopathic ingredients for disease treatment. A detailed analysis of the relationship between the plant's physiological processes and the production of phytochemical constituents is described in this chapter. Additionally, this chapter examined how various environmental factors, such as light, temperature, and water availability, affect phytochemical synthesis and accumulation. This knowledge will contribute to developing climate-specific crop production technologies for superior growth, yield, and quality of herbal plants. In this chapter, we aim to (1) cover a comprehensive overview of the origins, geographical distribution, and botanical characteristics of medicinal plants, (2) examine their physiological characteristics, phenological devolvement, and chemical composition, with a particular focus on the production process of medicines and (3) therapeutic use of medicinal plants. By bridging the gap between physiological ecology and phytochemical constituent accumulation in medicinal plants, this chapter provides valuable insights for researchers, practitioners, and stakeholders who are involved in the cultivation, processing, and use of medicinal plants.

2 Psyllium Husk (*Plantago ovata*)

The medicinal plant known as psyllium was brought into use by Indian Muslims, who initially gathered its seeds from wild species [9]. Psyllium is an annual plant of the *Plantago* genus, which has around 200 different species. Scientifically known as *Plantago ovata,* it is the most important and known species with a wide variety of uses. It is also called Isabgol, meaning "horse ear," which describes the shape of the seed [10, 11]. In India, psyllium production is 0.432 million metric tons from an area of 0.448 million hectares [12]. Gujarat and Rajasthan states are the major producing states of psyllium because the climatic conditions of these regions are most suitable for psyllium cultivation. *Plantago* ranks sixth in the economic trade of medicinal plants, and India dominates its world market while the United States is the main importer of its seeds and husks, consuming 8000 tons annually [13, 14]. *P. ovata* is a short-stemmed or stemless annual herb having 45–69 flowers with a height of 30–45 cm while leaves are 7.5–25 cm long [15, 16]. The plant has a tap root system with a group of secondary roots. After 60 days of planting, white-colored flowers appear on the flowering shoots. The psyllium is mature and ready to harvest when flower spikes turn reddish brown and lower leaves get dried up. Psyllium husk is the main product of Isabgol. Through mechanical process, the outer skin of the husk is detached, and the husk recovery from seed is about 25–26% [9].

The psyllium husk is the outermost skin of the seed that contains protein (0.94%), ash (4.07%), and total carbohydrates (84.98%) [17]. Linoleic acid (40.6% of the oil recovered from Plantago seeds) and oleic acid (39.1%) each made up a sizeable component of the oil's makeup. In addition, 6.9% of linolenic acid was also found, though in a lesser amount. It is considered a good source of soluble and insoluble fiber. The diet fibers produce low calorie food and are extracted from plants having pharmaceutical properties [18]. All the properties show the nutritional and medicinal values of *plantago ovata* [19].

2.1 Effects of Abiotic Stresses

Environmental variations limit the growth, physiology, and geographical distribution of plants. Seed germination and growth of *Plantago ovata* are affected severely in alkaline soils and hot climates, for example, air temperature $> 30\ ^\circ\mathrm{C}$ [20]. The high-temperature stress reduces the yield by affecting the photosynthetic rate and plant cell metabolism [21]. Drought stress increases the soil salinity, which inhibits the cell function and pharmaceutical properties of plants [22, 23].

2.2 Cultural Practices and Management

P. ovata generally grows in semi-arid to arid environments under the cold season (October–March) in poor fertile sandy-loam to loamy soils [16, 24]. The optimum temperature ranges between 15 °C and 30 °C, and the ideal soil pH ranges between 4 and 6 for maximum seed germination and plant growth [25]. Before sowing, seeds are treated with fungicide to control root rot disease while nitrogenous and potash fertilizers should be used in a balanced way [26]. The two main insects of *P. ovata* are aphids and white grubs while downy mildew and Plantago wilt are the major diseases [27]. *P. ovata* is a 119- to 130-day-crop, and its harvesting starts when flower spikes turn reddish brown, upper leaves turn yellow, and lower leaves dry up [28]. Psyllium husk has shelf life of 6 months, and the husk recovery from seeds is around 25–26% [9].

2.3 Phytochemicals

The presence of phytochemicals in *Plantago ovata* shows its importance for different industries and in anti-inflammatory activities [29]. Numerous metabolites are found in their seeds and used to treat different diseases (Table 1).

Table 1 Chemical constituents of Plantago ovata seeds

Source	Group	Constituents	References
Seed	Primary and secondary metabolites	Fumaric acid, malic acid, stearic acid, linoleic acid, linoleic acid, kaempferol, glucose, acetic acid, sucrose, fructose, xylose, formic acid, GABA as γ-amino butyric acid, citric acid, glutamine, cystine, valine, glutamic acid, alanine, tyrosine succinic acid, glycine, leucine, and lysine	[30, 31]
Seed coat	Hydrocolloid	Lrhamnose, L-arabinose, galacturonic acid, D-galactose, and D-xylose	[32]

2.4 Geographical Distribution

Psyllium husk originated from the Mediterranean region and is now used in North Africa, Russia, and central and southwest Asia in different herbal medicines [33, 34].

3 Black Cumin or Black Seed (*Nigella sativa*)

Black cumin, also called black seed, roman coriander, or kalonji, is the annual plant and belongs to the ranunculus family (Ranunculaceae) grown for its pungent seeds, which are used as a spice and in herbal medicines. Black cumin sowing and reaping have been mentioned in the Holy Bible, and The Prophet Mohammed (PBUH) declared it as the remedy for all diseases, except death. Black cumin cultivation was reported more than 3000 years ago in the tomb of Egyptian King Tutankhamun [35] and was spread in North Africa, the Middle East, South Asia, Eastern Europe, North America, the Mediterranean, Russia, Turkey, and France, where its seed was used in medicine, bread, and cakes [36]. In traditional foods as well as baked goods, where it is used to enhance flavor and taste, black cumin is used as a food preservative and a spice [37]. Black cumin essential oil contains significant amounts of carbohydrates, proteins, and lipids, which indicate the nutritional value of black seeds [38]. India is the largest grower of black cumin with an area of 6.24 m ha and production of 254 thousand tons, followed by Turkey with an area of 0.81 m ha and production of 689 thousand tons [39].

Black cumin is a tall annual plant, typically ranging from 20 to 60 cm in height. It features an upright growth habit, with a branching stem and a taproot system [40, 41]. The flowers are hermaphrodite and often cross-pollinated that are solitary on the main axis [42]. During the initial stages of blooming, the flower exhibits a pale green coloration in its petaloid sepals. However, upon reaching full bloom, the color transforms into a distinctive pigeon blue shade [43]. On average, approximately 10% of the flowers are infertile. The flowers lack bracts as involucres, and the peduncle is elongated and upright. The sepals of the flower are petal-like and have a wide, ovate shape. They are arranged in a single whorl, typically consisting of 4–6 sepals [44]. Once mature, the fruit, in the form of a capsule composed of 3–7 united follicles, contains numerous minuscule seeds [35].

The seeds of black cumin have a distinct triangular shape, featuring a flat side and a convex side [45]. Black cumin oil contains tocopherol, phenolic compounds, thymoquinone, and antioxidants, and it has a great market for pharmaceutical applications [46]. The essential oil can be used in the cosmetics industry due to the presence of a high amount of thymoquinone in the oil [47].

3.1 Effects of Abiotic Stresses

Sudden changes in the environment during the crop growth period affect the pharmacological and pharmaceutical activities of black cumin [48]. The abiotic stresses like drought and salinity produce free radicals in black cumin, which disrupt the osmotic balance and plant metabolism [49].

3.2 Cultural Practices and Management

Black cumin grows well in cool and dry weather, which favors flowering and seed setting. It is sown on ridges and beds, but a better yield has been obtained from bed sowing with a seed rate of 20 kg ha^{-1} [50, 51]. The seeds are soaked in water before sowing. Cumin is relatively drought tolerant, but the availability of water at the reproductive stage increases grain yield and 120- to 400-mm rainfall is sufficient during the season [52, 53]. The ideal temperature for its cultivation is 12–14 °C, but it can grow well up to 25 °C [35]. Black cumin requires low fertilizer, but moderate fertilization is important to get optimum yield, while its deficiency may lead to yield reduction [54]. The amount of NPK at the rates of 50, 40, and 20 kg/ha, respectively, is recommended for a good yield of black cumin [55]. Full sunshine during vegetative growth increases plant growth and lead development and direct sunlight is required at flowering stage [36]. Weeds have the potential to significantly diminish crop yield, with reductions ranging from 60% to 85%. It is recommended to conduct 3–5 weeding at intervals of 20–25 days in order to ensure a weed-free environment throughout this crucial period [56, 57]. Caterpillars and armyworms are the major insects of black cumin that can be controlled through 5% aldrin solution and 0.05% methyl parathion mixture, respectively [58]. Major diseases of black cumin are fusarium wilt (yield loss up to 72%), blight (yield loss up to 88%), and powdery mildew (yield loss up to 60%), which can be controlled by treating the seeds with fungicides [58].

3.3 Chemical Constituents

Black cumin is used in various traditional medicines due to its certain chemical constituents [59, 60] (Table 2).

Table 2 Chemical constituents of black cumin

Chemical groups	Subgroup	Active constituents
Fixed oil	Saturated fatty acids	Palmitic acid and stearic acid
	Unsaturated fatty acids	Oleic acid, eicodadienoic acid, dihomolinoleic acid, and linoleic acid
Terpenes	Isoquinoline	Nigellicimine and nigellicimine N-oxide
	Aliphatic	Citronellol, carvacrol, limonene, dithymoquinone, 4-terpineol, p-cymene, carvone, α-pinene, thymohydroquinone, thymoquinone, and anethol
Amino acids	Essential amino acids	Tryptophan, phenylalanine, lysine threonine, leucine, valine, isoleucine, and histidine
Phenolics	Acidic phenolics	Syringic acid, p-coumaric acids, vanillic acid, and hydroxybenzoic acid
Metals	Trace elements	Phosphorus, calcium, iron, potassium, and zinc
Saponins	Triterpenes	Steryl glucosides and acetyl-steryl-glucoside
Flavonoids	Flavonoidal glycoside	3-Galactosyl glucoside, Kaempferol 3-glucosyl galactosyl glucoside, and trigillin quercetin-3-glucoside quercetin

3.4 Geographical Distribution

Black cumin has been used for many years to treat many illnesses in both humans and animals [61, 62]. In addition to Western and Southern Europe, it is used in North Africa, the Mediterranean basin, and Western and Southern Asia [63, 64].

4 Stevia (*Stevia rebaudiana*)

Stevia rebaudiana (Bertoni) is a herbaceous perennial plant that belongs to the Asteraceae family. It originated in Paraguay (South America) and is commonly known as honey leaf, sweet leaf, sweet weed, or sweet herb. Stevia was introduced as an alternative sweetener for the first time in Japan when saccharin was banned [65], although the commercial cultivation of stevia was started in Paraguay in 1964 [66]. Stevioside, the major sweetener present in leaf and stem tissues of stevia, was first considered as a sugar substitute in the early 1970s by a Japanese consortium formed to commercialize stevioside and stevia extracts [67]. During World War II, Stevia was used as a sweetener to combat food shortages in Britain. Diterpene glycosides produced by stevia leaves are noncaloric and more sweetener than sucrose [68]. Subsequently, stevia was marketed in Latin America, Canada, China, Japan, Indonesia, and the United States [69]. Stevia is mainly cultivated in Japan, China, Taiwan, Thailand, Korea, Brazil, Malaysia, and Paraguay. China has a 75% share in stevia cultivation, and it is cultivated on an area of 32,000 hectares worldwide. Stevia grows up to a height of 65–80 cm as it is a small perennial shrub [70]. The roots are the only part of plants that do not contain stevioside (sweetness) and are fibrous, filiform, and perennial in nature [71]. Stevia is an insect-pollinated plant [72, 73], and the seed is called achene having feathery pappus. Stevia is very popular in many regions of the world and is expected to become a major source of high-potential sweeteners. The sativoside sugar in leaves has no calories and has a large market [74]. It became economically important owing to its major contribution to the sugar industry, which generously uses its highly sweet leaves to sweeten beverages and make herbal tea as well. This popular plant's leaves have a wonderful sweetness that makes them a great option for people managing their intake of sugar and carbohydrates. The leaves can also be used as a color and flavor enhancer as well as a sweetener in teas and salads [75].

4.1 Effects of Abiotic Stresses

Stevia plants have some resistance against salinity and could control the reactive oxygen species through the production of antioxidants [76]. The abiotic stresses for a short duration increase the biosynthesis of steviol glycosides while the prevalence of these stresses for a long time affects the photosynthetic rate and elongation of shoots. Long-term drought stress decreases the protein and chlorophyll content of leaves due to the increased activity of proteases [77]. High-temperature stress and hot climate result in early flowering of stevia plants, which results in low yield [78].

4.2 Cultural Practices and Management

The ideal climate for Stevia is semi-humid subtropical, ranging from 6 °C to 43 °C, with an average temperature of 23 °C [79]. The synthesis of terpenes in stevia plants is affected by climatic and environmental factors [80]. Stevia yield and quality are affected by temperature and photoperiod. Temperature, length, and intensity of the photoperiod have a significant impact on the growth, quality, and secondary metabolites of stevia plants, and higher yields are recorded in summer crops than in winter crops [81]. When the temperature drops below 20 °C and the day length decreases by 12 hours, vegetative growth significantly reduces. Vegetative propagation with a stem having two leaves is used for multiplication because of low seed germination in stevia. Plant growth and stevioside content in the leaves of the plants grown from stem tips were more uniform than in plants grown from seeds [82]. In Pakistan, the best time of sowing for stevia seedlings is December–January or August–September and the seedlings can be transplanted after 2 months, while the concentration of stevioside is higher in January sowing plants [83]. Planting density and transplanting date are the most important agronomic factors that may affect the quality and quantity of yield. Stevia herb is very sensitive to water stress, thereby frequent irrigation in summer at an interval of 3–5 days is useful to avoid the effects of drought and heat stress; sprinkler irrigation gives better results [84]. Balanced soil nutrition increases the contents of steviol glycosides in stevia tissues, although this crop can be sown on poor quality soil [85, 86] while the deficiency symptoms of N, P, and K appear on stevia leaves. Nutrient requirements of this crop are low to moderate since this crop is adaptable to poor-quality soils in its natural habitat. Stevia has a poor capacity to compete with weeds during the initial growth period, which limits the crop yield more than any other factor. However, using black plastic mulch and increasing plant densities inhibit the weeds population [87]. Because of the presence of the inherent sweetness that acts as an insect repellant, no insect attack has been reported in stevia; therefore, no insecticide has been used in this crop, making this crop organic stevia. Moreover, several diseases such as alterneria leaf spot (*Alternaria alternata*), stem rot (*Sclerotium dephinii* Welch.), root rot (*Sclerotium rolfsii*), powdery mildew (*Erysiphe cichoracearum* DC), damping-off (*Rhizoctonia solani* Kuehn.), *Sclerotinia sclerotoirum,* and *Septoria stevia* (leaf spot) attack this crop. Neem-based pesticides give good results in integrated disease management [88–90]. Stevia leaves, along with stem tips, are harvested 4 months after transplanting, and the subsequent harvesting is taken 3 months after the first harvest. The best time to harvest is mid-September to late September when plants attain 50–70 cm height (before flowering) to get maximum steviol glycoside contents [91]. Three commercial cuttings are taken, leaving only a 10 cm stem after each cutting to get a better subsequent flush [84].

4.3 Chemical Constituents

Table 3 shows the chemical constituents of Stevia leaves.

Table 3 Chemical constituents of Stevia leaves

Plant part	Group	Constituents	References
Stevia leaves	Phytochemicals	Chlorogenic acids, folic acid, austroinullin, stevioside, β-carotene, non-glycosidic diterpenes, steviol, thiamine, riboflavin, glucoside, niacin, anthraquinones, rebaudi, vitamin C, oxides, and flavonoids	[92, 93]
	Complex compounds	Rebaudioside, diterpene glycosides-stevioside, rebaudioside A, dulcoside A, rebaudioside C, and steviolbioside	[94, 95]

4.4 Geographical Distribution

Stevia is mainly grown in the northeast regions of South America [96] and is also used in China, Japan, Korea, and India.

5 Fennel (*Foeniculum vulgare*)

Foeniculum vulgare is an aromatic annual medicinal herb that belongs to the family Apiaceae commonly known as fennel. This herb is native to Southern Europe and the Mediterranean region where it is grown as an important cash crop. India dominates the economic market of fennel and exports it as seed, powder, and volatile oil to the United States, Singapore, the United Kingdom, the United Arab Emirates, Sri Lanka, Malaysia, Saudi Arabia, and Japan [97]. Fennel is an erect, glaucous green plant, having a height of 3.3–12 feet with a smooth, bright green and hollow stem [98]. Fennel is a branched herb that has hairy, feathery, and soft foliage and contains greenish gray, ovate, ribbed, and approximately 6–10-mm-long fruit. Due to its flush foliage and seeds, it is grown in vegetable gardens and harvested for its usage in cooking. The leaves are 40 cm long and 0.5 mm wide and are branched and cut into fine segments. The flowers are present on terminal compound umbels with each umbel containing about 20–50 tiny yellow flowers on the pedicels. The pericarp contains essential oil and the spicy licorice fragrance present in the whole plant.

5.1 Effects of Abiotic Stresses

Drought and high-temperature stresses in arid and semi-arid areas are major issues of yield reduction of different plants [99]. Hot climate and water shortage affect the photosynthetic rate, quality, and quantity of essential oil and decrease the vegetative growth and yield of fennel [100]. High-temperature stress alters physiological processes like photosynthetic and transpiration rates, which negatively affects the growth and development of fennel plants [101].

5.2 Cultural Practices and Management

Fennel likes cool and dry seasons and a temperature of 15–20 °C for optimal growth and production, so high temperature during flowering results in low grain filling. The loam soils and the soils that are rich in organic matter are best while sandy soils are not suitable for its sowing. Waterlogged and salt-affected soils are not suitable for their cultivation, and appropriate drainage is required for proper growth of fennel [102]. Fennel is sown from mid-September to the first week of October and 18 °C is the ideal temperature for germination [103]. Hydro priming of seeds for 8–12 hours before sowing results in better germination and good seedling establishment and seeds are sown at 1.5–2 cm depth. It can tolerate a soil pH of 4.8–8.3 and can be grown as an intercrop with other crops. To cultivate it through transplanting, about 2.5 kg of seed is required for each hectare of nursery while 10 kg/ha is required for direct seeding of the crop. Maximum grain yield was obtained at 20 × 15 spacing [104]. Fennel requires 90 kg N, 40 kg P_2O_5 and 30 kg K_2O and 15 tons of farmyard manure per hectare for good crop yield. If plants seem weak after 3–4 weeks of sowing, 1% urea should be applied exogenously for better crop stand. The application of phosphorus-solubilizing bacteria increases the seed yield as well as essential oil content in fennel seeds [105]. The critical period of weed competition in fennel is the first 50 days [106], and irrigation is applied at 10-day intervals depending on crop requirement [107]. *Leveillula taurica* (powdery mildew), *Ramularia foeniculi* (Ramularia blight), *Cercospora foeniculi* Magn (Cercospora blight), *Ascochyta foeniculum* Mc Alpine (Ascochyta blight), *Rhizoctonia solani* (Rhizoctonia root rot), *Alternaria alternata* (Alternaria blight), *Pythium aphanidermatum* (damping off of seedlings), and *Plasmophara nivea* (Casparg) Schroeter (Downy mildew) are important diseases of fennel [108]. Thrips are the main insect of fennel. Fennel umbels are harvested 4–5 times. Before full ripening, when the umbels change their color from deep green to slightly yellow, they are plucked, and the umbels are dried in shade by avoiding their heaps. Afterward, umbels are threshed to separate the seeds by winnowing [109].

5.3 Chemical Constituents

Some of the essential oil and chemical constituents of Fennels leaves are shown in Table 4.

Table 4 Average essential oil and chemical constituents of Fennels leaves

Plant parts	Constituents	References
Essential oil	A-phellandrene, fenchone, trans-anethole, methyl chavicol (estragole)	[110]
Fennel seeds	Terpenoids, fat, alkaloids, minerals, protein, saponins, carbohydrates, fiber tannins, carotenoids, cardiac glycosides, polyphenols, flavonoids, glycosides, and vitamins like K, E, C, and A	[111]

5.4 Geographical Distribution

Fennel originated from Mediterranean areas and now it is grown and used in Russia, India, China, and Japan [112, 113]. In Europe and Mediterranean areas, it is used as antispasmodic, diuretic, anti-inflammatory, analgesic, secretomotor, secretolytic, galactagogue, eye lotion, and an antioxidant agent [114].

6 Bishop's Weed/Carom Seeds (*Trachyspermum ammi*)

Bishop's weed, belonging to the family Apiaceae, is a herbal medicinal plant and has some secondary metabolites with potential bioactivities. It is used as an anti-inflammatory, antibacterial, antifungal, antiviral, and anticancer agent [115]. Bishop's weed originated from Eastern Mediterranean and spread to India during the Greek conquest of Central Asia [116]. Now, it is grown as an important spice and aromatic herb all over the world, including in the Middle East and Asia [117]. Bishop's weed is a grassy, aromatic annual herbaceous plant with an erect stem that reaches up to a height of 90 cm. The seeds are 2 mm long and 1 mm wide and are grayish brown, ovoid in shape, aromatic in odor, and pungent in taste [118]. Carom seeds have fiber 21.8%, protein 17.1%, and minerals 7.9% (Ca, P, Na, and riboflavin). India is the major producer of Bishop's weed with a production of 28,973 tons from an area of 41,134 ha [119]. The demand and area under cultivation of carom seeds have increased over time due to their usage in herbal medicines and cosmetics [120].

6.1 Effects of Abiotic Stresses

Different biotic and abiotic factors affect the chemical composition of this medicinal plant. Drought stress affects the chlorophyll contents, phenolic compounds, and net carbon dioxide assimilation rate in bishops weed plants [117].The high-temperature stress limits the thymol and some other phytochemicals in the essential oil of carom seeds [121].

6.2 Cultural Practices and Management

Bishop's weed is grown in arid and semi-arid climates of the world and is sown from October to November [122]. The optimum temperature for its growth ranges from 15 °C to 25 °C with 65%–70% relative humidity [118]. Well-drained loam soil is best for this crop as this crop is severely affected by heavy rainfall and waterlogging. It is typically grown in the winter season between October and November, has flowers and fruits from January to April, and is harvested in May and June. A shallow soil cultivation with a fine-seed bed is required for better germination and 10–15 ton/ha well-rotten farmyard manure should be applied before the last plowing. About 3.5–4 kg

Table 5 Chemical found in Carom

Plant part	Constituents	References
Carom seeds	38.6% carbohydrates, 11.9% fiber, 15.4% protein, 18.1% fat saponins, flavone, glycosides, nicotinic acid, tannins and minerals such as iron, calcium, and phosphorous	[124–126]
Essential oil of carom seeds	Styrene, δ-3-carene, β-pinene, terpinene-4-ol, α-phyllanderene, β-phyllanderene, carvacrol, α-pinene, γ-terpinene pcymene, and 35–60% thymol	
Fruit of bishops weed	Carotene, iron, chromium, calcium, iodine, zinc, copper, manganese, cobalt, thiamine, phosphorus, and riboflavin	

per acre of seed is required in drill sowing and 2 kg of seed per acre is required in manual sowing of seed on beds. The plant-to-plant distance is maintained at 6 inches while row to row distance is maintained at 2 feet and the crop is usually cross-pollinated [122]. Three to four irrigations are required for this crop without missing irrigation at flowering and grain filling, and 80 kg nitrogen and 50 kg phosphorous are required for successful crop growth. The Rabi weeds compete with this crop for the first 54 days, and two weeding at an interval of 15 days are performed to reduce the weed population. Aphids, jassids, stem flies, and seed bugs are major insects of this crop while bacterial blight, root rot, and powdery mildew are the major diseases of this crop. The newly developed varieties should be grown and the optimum plant population should be maintained for weed and insect control while some copper-containing fungicides are used to control the disease infestation [123].

6.3 Chemical Constituents

The chemicals present in Carom are presented in Table 5 and includes carotene, thiamine, cobalt, etc.

6.4 Geographical Distribution

Bishop's weed originated from the Middle East. It is grown and used in Egypt, Iran, Iraq, India, Pakistan, and Afghanistan [127].

7 Sweet Basil (*Ocimum basilicum*)

Sweet basil, belonging to the family Lamiaceae, is an aromatic herb that originated in India. This culinary herb, also known as the "king of herbs," is being grown commercially in Iran, Africa, Italy, Belgium, Hungary, France, Spain, Bulgaria, Poland, and the United States [128–130]. Since the last few years, North America has been the largest producer of basal leaves with a 51.23% share in the world market while the Asia Pacific contributes 37.87% and Europe contributes 4.35%. The market value of basal is projected to increase from 57 million USD in 2020 to 62 million USD in 2026. Basil is a fragrant erect-type plant that goes up to a height of 62–90 cm. The stem has strongly scented green or violet leaves while white or pale purple flowers of three blooms appear in the form of branch racemes [131]. The world's leading suppliers of basil leaves in the world market are McCormick, Litehouse, Fresh Origins, Frontier Natural Products, and Herbs Egypt. *O. basilicum* contains linalool (43.78%), eugenol (13.66%), 1, 8-cineole (10.18%), methyl chavicol (81.82%), β-(E)-ocimene (2.93%), and α-(E)-bergamothene (2.45%) [132].

7.1 Effects of Abiotic Stresses

The synergistic effects of drought, salinity, and high-temperature stress limit the photosynthetic pigments and other physiological processes of plants of sweet basil. Drought stress causes alteration in the metabolism of plant leaves and water shortage increases the salinity issues in soil, which leads to oxidative stress, resulting in low yield of sweet basil [133, 134]. The high-temperature stress decreases the chlorophyll contents of leaves and plant produces glycine betaine and proline as secondary osmolytes, which plays a vital role in osmoregulation adjustment of plants [135].

7.2 Cultural Practices and Management

The genus *Ocimum* has more than 150 species, 30 of which are grown in tropical to subtropical areas, and some are grown in temperate areas. Environmental factors such as drought, frost, and cold stresses affect the oil composition while the ideal plant growth takes place from 7 °C to 27 °C. Well-drained loamy fertile soil having a sufficient amount of organic matter is required for this crop, although it can grow in saline soils [136]. For 1-hectare nursery, about 200–300 g of seed is sufficient, and the nursery is sown at the end of February and is transplanted on well-drained raised beds 6 weeks after sowing. Plants are transplanted at 40 × 40 cm spacing. *Ocimum basilicum* requires 12–15 irrigations with the first irrigation at the time of nursery transplantation, and four irrigations are applied in summer. The weeds competition period is for the first 2 months, and to control the weeds, the first weeding is done 1 month after nursey transplanting while the second hoeing is done 4 weeks after the first hoeing [137]. Leaf roller and lacewing are the major insects of *Ocimum basilicum,* whereas powdery mildew (*Oidium* spp.), root-rot (*Rhizoctonia bataticola*), and seedling blight (*Rhizoctonia solani*) are the major diseases of this crop.

7.3 Chemical Constituents

The essential oil of sweet basil contains α-terpineol, α-bergamotene, α-bisabolol, α-cariophylene, α-copaene, α-guaiene, β-elemene, β-cadinene, β-cariophylene, β-farnesene, estragol, ocimene, eucalyptol, linalool acetate, menthol, eugenol, menthone, bornil acetate, germacrene-D, camphor, cubenol, elixen, eugenol, cyclohexanone, cyclohexanol, nerol, and myrcenol [138]. The plant leaves also have some chemical constituents like alkaloids, terpenoids, flavonoids, saponin glycosides, tannins, and ascorbic acid [139]. The chemical analysis of essential oil also revealed the presence of the following constituents [140] (Table 6):

Table 6 Chemicals found in Ocimum

Chemical groups	Constituents
Phenylpropene	Syringing, glycoside esculin, and estragole
Coumarin	P-coumaric acid, aesculetin, and coumarin
Oxygenated sesquiterpenes	Nerolidol, cubenol, dihydroactinidiolide, viridiflorol, isospathulenol, caryophyllene oxide, zpathulenol, muurolol, α-cadinol, alloaromadendrene, α-humulene oxide, β-basibolol, β-eudesmol, and β-basibolol
Flavonoids	Kaempferol, isoquercetrin, quercetin, 3,5-pyridinedicarboxylic acid, galactoside, quercetin-3-O-a-L-rhamnoside, kaempferol-3-O-B-D-glucoside, quercetin-3-O-B-D-glucoside-2″-gallate, quercetin-3-O-diglycoside, kaempferol-3-O-rotinoside, 2,6-dimethyldiethyl ester, quercetin-3-O-B-D-glucoside, and quercetin-3-O-(2”-O-galloyl)-rutinoside
Triterpene	Alphitolic acid, basilol, ocimol, betulin, oleonolic acid, 3-epimaslinic acid, ursolic acid, betulinic acid, pomolic acid, and euscaphic acids
Aliphatic ketones	3-Hydroxy-2-butanone, 6-Methyl-5-heptenone, cis-Jasmone, and β-ionone
Aromatic compounds	Anisaldehyde, benzaldehyde, anethole, ethyl cinnamate, safrole, methyl cinnamate, benzyl alcohol, methoxycinnam aldehyde, p- phenethyl alcohol, cumin aldehyde, estragole, methyl benzoate, methyl eugenol, methyl salicylate, phenyl acetaldehyde, and 4-allylphenol
Aliphatic alcohol	1-penten-3-ol, 1-octen-3-ol, 3-hexanol, 2-pentenol, hexanol, cyclohexanol, and octanol
Sesquiterpene hydrocarbons	Aromadendrene, bicyclogermacrene, valencene, caryophyllene, calamenene, bicycloelemene, cadinene, α-copaene, germacrene-a, α-humulene, α-acoradiene, α-(z)-bergamotene, α-cedrene, α-cubebene, α-guaiene, α-amorphene, α-ylangene, α-bulnesene, δ-cadinene, α-cadinene, α-bisabolene, α-gurjunene, γ-gurjunene, α-zingiberene, β-acoradiene, β-ocimene, β-bourbonene, β-caryophyllene, β-elemene, β-cedrene, β-copaene, β-selinene, β-cubebene, β-guaiene, β-farnesene, germacrene-b, germacrene-d, iso-caryophyllene, isoledene, longifolene, α-bergamotene, β-farnesene, β-ocimene, γ-terpin, γ-muurolene, and γ-cadinene
Monosaccharide	D-xylose, L-arabinose, and L-rhamnose
Monoterpenes	Pinocarvone, ocimene oxide, iso-pinocamphone, fenchone, menthol, eugenol, myrtenol, α-myrcene, hotrienol, α-phellandrene, linalyl acetate, α-pinene, α-fenchyl acetate, α-terpinolene, β-phellandrene, β-pinene, camphene, myrtenal, limonene, myrcene, p-cymene, sabinene, terpinolene, thujene, γ-terpinene, α-citral, α-terpineol borneol, bornyl acetate, camphor, carvacrol, carvone, L-camphor, 1,8-cineole, cis-linalool oxide, lavandulol, cis-rose oxide, citronellol, endo-fenchol, estragol, exo-2-hydroxycineole-acetate, geranial, geraniol, geranyl acetate, iso-neomenthol, trans-pinocamphone, L-carvone, linalool, cis-furanoid, and methyl chavicol

7.4 Geographical Distribution

Sweet basil originated from India and has been used in Unani and Ayurvedic herbal medicines since ancient times [141]. It is used as a culinary herb and is widely cultivated in Iran, Africa, Italy, Belgium, Hungary, France, Spain, Bulgaria, Poland, and the United States [130].

8 Holy Basil (*Ocimum tenuiflorum*)

Holy basil (*Ocimum tenuiflorum*) is an aromatic herb belonging to the family lamiaceae and is known as the Queen of Plants and the Mother Medicine of Nature [142, 143]. Out of 150 genera of *Ocimum,* only holy basil or tulsi (*Ocimum tenuiflorum* L.) and sweet basil (*Ocimum basilicum*) have culinary and religious uses [144]. Holy basil originated in India and is grown throughout Asia, Northern and Eastern Africa, Malaysia, South and North America, Taiwan, Hainan Island, and China [145, 146]. Holy basil is cultivated both for essential oil and medical purposes [140]. In India, about 250–300 tons of oil is extracted annually from an area of 25,000 acres [147]. Holy basil shrubby plant has a hairy stem and green- to purple-colored leaves, and it attains a height of 30–60 cm. The leaves reach a length of 5 cm while the flowers are purple that are arranged in racemes [145]. The leaves of holy basil contain 0.7% volatile oil, which has 71% eugenol and 20% methyl eugenol. Furthermore, the oil contains sesquiterpene hydrocarbon caryophyllene, fatty acids, carvacrol, phenolics, flavonoids, linoleic, sitosterol, mucilage, terpenoids, and polysaccharides [148].

8.1 Effects of Abiotic Stresses

Environmental factors like cold, drought, salinity, and flooding have an adverse impact on the growth and development of holy basil plants. Plants develop physiological and biochemical homeostasis against abiotic stresses. Cold stress prior to flowering has drastic impacts on the eugenol contents of the essential oil of holy basil [149]. Excess rainfall or water logging increases the activity of 1-aminocyclopropane-1-carboxylic acid in submerged roots, which impacts the biochemistry and physiological processes within the plant body of holy basil [150].

8.2 Cultural Practices and Management

The tropical climate, along with long days and high temperature, favors the growth of holy basil, and it can be grown in any soil except salt-affected and water-logged soils [151]. Light spectrum, nutrients, water, and temperature affect the growth,

Table 7 Chemical constituents of holy basil

Plant part	Constituents	References
Seeds	β-Sitosterol, fixed oil, mucilage, and polysaccharides	[152, 153]
Fruit	Carbohydrates, β-carotene, protein (30 kcal), fat, insoluble oxalates, vitamin A, vitamin C, and chlorophyll	[154]
Leaves	Gallic acid, volatile oil, vanillin, carotenoids, derivatives of eugenol, ursolic acid, phenolics, flavonoid, ascorbic acid, riboflavin, vanillic acid, thiamine terpenoids, fatty acid, and rosmarinic acid	[155]

yield, and oil contents of holy basil, although it can tolerate frost and drought up to a certain extent [136]. Holy basil is sown on a well-prepared bed or ridge, and ridge-to-ridge distance is maintained at 2 feet distance. Holy basil is sensitive to water deficiency, and 2–3 irrigations are applied at an interval of 1 week while the remaining irrigations are applied at an interval of 2 weeks. The first 2 weeks are important for weed control, and the first hoeing is done 1 week after transplanting while the second hoeing is done 4 weeks after the first hoeing [137]. Leaf rollers and lacewings are the important insects of holy basil while powdery mildew (*Oidium* spp.), root-rot (*Rhizoctonia bataticola*), and seedling blight (*Rhizoctonia solani*) are important diseases of holy basil.

8.3 Chemical Constituents

The chemical constituents of holy basil is presented in Table 7.

8.4 Geographical Distribution

Holy basil originated in India, and now it can be found across tropical and subtropical areas of Asia, Northern and Eastern Africa, Malaysia, South and North America, Taiwan, Hainan Island, and some regions of China [145, 146, 156].

9 Linseed (*Linum usitatissimum*)

Linseed belongs to the genus Linum and to the family Linaceae. It originated from the Fertile Crescent, a region east of the Mediterranean Sea and toward India [157]. Worldwide, about 3180 thousand tons of linseed are produced from an area of 3270 thousand hectares. Linseed is mainly produced in Russia (23%), China (12.4%), Canada (19.8%), Ukraine (3.2%), the United States (7.5%), India (4.3%), Kazakhstan (19%), Ethiopia (3.0%), the United Kingdom (1.6%), France (1.5%), Sweden (0.6%), Argentina (0.6%), Brazil (0.4%), Belgium (0.3%), and Poland (0.3%) [158]. Linseed attains a height of 30–120 cm having slender, green, and glabrous stems with some secondary branches near the top of the stem. The plant completes its life cycle from 90 to 150 days while the seed matures in 30–60 days after flowering [159]. The seeds of linseed contain 20–24% polyunsaturated fatty acids, 42% oil contents, 5–9% carbohydrates, 15–19% protein contents, 5–9% crude fiber, and 2–4% ash contents [160].

9.1 Effects of Abiotic Stresses

Different abiotic stresses like drought, high temperature, salinity, and flooding affect the productivity of linseed. Drought stress occurs when humidity in the air is low and soil moisture is not enough for good plant growth and development. Drought stress at the vegetative growth stage decreases the photosynthetic assimilation rate and quality of essential oil of holy basil [161]. Heat stress for five or more days at the reproductive stage decreases the pollen viability and seed setting of linseed [162].

9.2 Cultural Practices and Management

This crop grows better in temperate climates having high humidity but moderately warm conditions [163]. A temperature of 10–30 °C, average relative humidity of 60–70%, and rainfall of 150–200 mm throughout the crop growth period are suitable for linseed growth [164]. A well-drained fertile soil is suitable for linseed [165]. Seed is sown manually or with a small seed drill with a seed rate of 19.75 kg per hectare in rain-fed areas and 14.8 kg per hectare in irrigated areas. This crop requires fewer irrigations with the first irrigation 3 weeks after sowing, and the subsequent irrigations are applied at flowering, pollination, and seed setting. Seed canary grass and common wild oat are the major weeds of this crop that compete for resources with major crops. Aphid is the major insect of linseed while rust, crinkle, curly top, fusarium wilt, gray mold, alternaria blight, and powdery mildew are the diseases of this crop [166].

9.3 Chemical Constituents

The chemical constituents of linseed is presented in Table 8.

Table 8 Chemical found in linseed

Plant part	Constituents	References
Linseed oil	Dotriacontane, 2-palmitoylglycerol, palmitoyl chloride, squalene, palmitic acid, oleic acid, tetradecanoic acid, methyl tetradecanoate, isopropyl myristate ethyl myristate, palmitoyl chloride, and methyl palmitate	[167]
Seeds	Protein, omega-3 fatty acids (*alpha*-linolenic acid), lignin, and dietary fiber	[168]
Fruit	Carbohydrates, phenolic compounds, proteins, alcaloids, flavonoids, terpenoids, glycosides, and saponins	[169]

9.4 Geographical Distribution

Linseed is grown in Canada, Ethiopia, India, and China and is exported to different areas of the world [165].

10 Turmeric (*Curcuma longa*)

Turmeric belongs to the genus *Curcuma* and the family Zingiberaceae [170]. It is a perennial plant that originated from Asia, Australia, and South America. It is also known as Indian saffron or golden spice due to the brilliant yellow color of its rhizome, which makes it an important spice. Turmeric is a rhizomatous herbaceous plant that originated in India and is cultivated throughout the world [171]. Out of more than 1.1 million tons of world annual production, India's share is 80% [160, 172, 173]. Over the last decade, the consumption of turmeric has increased due to awareness about its benefits for health as it contains valuable compounds called curcuminoids [174]. As an Asian spice, it is cultivated in India, China, Indonesia, Pakistan, Bangladesh, Malaysia, Haiti, Taiwan, and El Salvador [156]. India is the biggest exporter of fresh rhizomes and powder turmeric throughout the world [175]. Turmeric has yellowish to orange-scented rhizomes and can reach up to a height of 1 m. The leaves are arranged in two alternate rows while the rhizomes are in branching form [176–178].

10.1 Effects of Abiotic Stresses

Environmental factors affect the growth and physiology of turmeric plants and the plants show different responses to survive different kinds of stresses. At the reproductive stage, very low temperatures and water deficit conditions negatively affect the curcumin contents of turmeric [179]. Salt stress during vegetative growth decreases the osmotic potential and photosynthesis of turmeric plants, which decreases the quality and quantity of rhizomes [180].

10.2 Cultural Practices and Management

Turmeric is grown in tropical to subtropical climates [179]. The average temperature for proper growth ranges from 20 °C to 35 °C with an annual rainfall of 1500–2250 mm. Turmeric requires sandy loam to clay loam soil, but humus-rich soil with 4.5 and 7.5 pH is best for its growth. For rhizome sowing, ridges and beds are prepared with 2–3 times plowing followed by planking. The ridge-to-ridge or bed-to-bed distance is maintained at 2 feet while the crop is sown in May by keeping a seed-to-seed distance of 20 cm. Turmeric requires 20–25 irrigations while, in summer, the irrigation is applied at an interval of 7–10 days. Weeds are controlled at 60, 90, and 120 days after sowing [181]. Turmeric is an exhaustive crop and

Table 9 Chemical found in turmeric

Plant parts	Constituents	References
Essential oil	Diterpines, monoterpines (cineole, β-phellandrane, p-cymene, terpinolone), and sesquiterpenes (α-turmerone, β-turmerone, Ar-turmerone)	[185]
Leaves	Isoeugenol, α-terpinolene, isomaltol, beta-pinene, tumerone, γ-terpenine, eucalyptol, caryophyllene, and thymol	[186]
Aqueous extract of turmeric	Phenols, alkaloids, saponins, cardiac glycosides, tannins, and flavonoids	[187]
Rhizome	Saponins, carbohydrates, alkaloids, starch, proteins, tannins, amino acids, glycoside, flavonoids, and steroids	[188]

requires 150 kg of nitrogen, 60 kg of phosphorus, and 108 kg of potassium per hectare to get better rhizome yield [182]. Shoot borer, rhizome scale, lacewing bug, leaf beetles, and nematodes are important insects of turmeric [183] while rhizome rot and root rot are important diseases of rhizomes although leaf blotch and leaf spot also attack turmeric crops [184].

10.3 Chemical Constituents

The chemical constituents of turmeric is presented in Table 9.

10.4 Geographical Distribution

Turmeric is widely cultivated in Pakistan, Bangladesh, India, China, Indonesia, Malaysia, Haiti, Taiwan, the United Kingdom, the United States, and Japan, and is the key component of Ayurveda, Unani medicines, and traditional Chinese medicines [189]. The rhizome powder is commonly used in southern and southeastern tropical regions of Asia [190].

11 Therapeutic Uses of Medicinal Plants

The therapeutic uses of some medicinal plants are presented in Table 10.

12 Conclusion

This chapter describes the medicinal, aromatic, and flavoring properties of herbal plants and highlights the importance of their physiological ecology that helps to understand the therapeutic potential of these plants. The chapter also underscores the

Table 10 Therapeutic uses of some medicinal plants

Medicinal plants	Therapeutic application/diseases to be cured	References
Psyllium husk	Colon cancer, hypertension, ulcerative colitis, diabetes, and weight loss.	[191–193]
Black cumin	It has anticancer, antidiabetic, antioxidant, antimicrobial, and neuroprotective properties and is used to cure the following diseases: nasal ulcers, abscesses, back pain, eczema, orchitis, swollen joints, migraine, chronic headache, dizziness, chest congestion, dysmenorrhea, obesity, diabetes, paralysis, hemiplegia, and gastrointestinal problems such as stomach aches, dyspepsia, diarrhea, flatulence, and dysentery.	[194–198]
Stevia	Control blood glucose, cholesterol level, insulin production, cardiovascular functions, blood pressure, and inhibits the activity of bacteria and fungus.	[199–204]
Fennel	It has antitumor, cytoprotective, chemopreventive, hepatoprotective, estrogenic activities and hypoglycemic, and is used to cure nausea, kidney stones, and obesity menopausal problems and also improves digestion system of human body and prevent the infants from flatulence.	[205–208]
Bishop's weed/ carom seeds	It prevents heart diseases by regulating cholesterol levels and has phenolic compounds that quench the free radicals of oxygen in the human body.	[209–211]
Sweet basil	It has antifungal, antimicrobial, antiseptic, anti-inflammatory, and antiviral properties and helps to reduce migraines, mental fatigue, and insomnia. It is used to cure urinary tract, respiratory system, kidney and cardiovascular diseases, improves memory, and relaxes the neurological system.	[212–218]
Holy basil	It has antioxidative, analgesic, antibacterial, antiviral, antidiabetic, antifungal, anticancer, adaptogenic, cardioprotective, and hepatoprotective properties and is used to cure malignancies, diabetes, heart diseases, stomachaches, headaches, malaria, and inflammation.	[152, 219, 220]
Linseed	It is used to treat arthritis, cancer, and cardiovascular disorders. The alpha-linolenic acid contains omega-3 fatty acid (58%) that enhances the absorption of long-chain polyunsaturated fatty acids and is used in the treatments of heart, arthritis, inflammatory, cancer, and autoimmune diseases.	[221–223]
Turmeric	It has anticancer, antitoxic, antioxidant, anti-inflammatory, and antibacterial properties and is used as a brain and heart tonic to lower cholesterol levels. It is used against many disorders such as diabetic ulcers, anorexia, cough, biliary disorders, jaundice, hepatic infections, malaria, to treat tuberculosis glands in the neck, tumors, cancer, leucorrhea, liver, hemorrhoids, leukoderma, diabetes, and respiratory diseases. It is used to disinfect cuts and burns while its mixer with calcium hydroxide is a good home remedy to treat wounds swelling and to heal sprains. It purifies the blood, maintains the levels of triglycerides and cholesterol, helps to enhance the immune system, relieves joint pain, controls blood sugar, soothes irritated tissue, and cures allergies, and digestive and eye disorders.	[224–232]

importance of understanding the curative properties, botanical characteristics, and climatic requirements of medicinal plants. Cultivating medicinal plants successfully requires a deeper exploration of cultural practices specific to their growth. Several valuable medicinal plants, such as psyllium husk, black cumin or black seed, stevia, fennel, bishop's weed/carom seeds, sweet basil, holy basil, linseed, and turmeric, are grown worldwide and have multibillion-dollar market value. Despite their potential, medicinal plants have not been cultivated on a scale commensurate with their market value, primarily due to limited knowledge about their optimal growing conditions and poor adaptability to various environments. Authors, in this section, have also mentioned the botanical characteristics and climatic requirements of medicinal plants. To ensure successful cultivation and sustainable production, careful consideration must be given in the future to factors like soil selection, climatic conditions, and cropping strategies while preserving the safety and nutritional value of these plants.

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Big Data Application in Herbal Medicine: The Need for a Consolidated Database

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Christiana Eleojo Aruwa and Saheed Sabiu

Contents

Abstract

Advances in technology have created and will continue to create changes in traditional and herbal medicine development and applications. Globally, and on a daily and annual basis, an increasing amount of data is generated from the traditional and orthodox health care systems. While the term 'big data' is loosely used, the phrase seems to have stuck with many but with the possibility of evolving in the near future. 'Big data' connotes huge and complicated data sets that usually lack structure, are multidimensional, and exhibit heterogeneity. With the promise of data-driven knowledge and intelligent decision-making, big data projects are ongoing and massively being explored for candidate therapeutics identification and development using cheminformatics and bioinformatics as well as other advanced data analytic approaches. However, the process of engaging stakeholders to benefit from the information provided by big data remain a major issue, knowing well that such data sets have the capacity to be utilized as backup

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for the support of medicines (herbal or otherwise) regulation. In the last decade, there has been an emergence of new data sets within the traditional medicine sector, with evolution of several institutional, national, and regional databases (DBs), providing new insights on phytochemicals, their efficacies, co-interactions, and mechanisms, and without much consolidation or unification. As such, repetitive data sets among databases have become the norm. This chapter presents how big data and big data analytics are being used and can be further explored in herbal medicine (HM) advancement, while advocating for consolidated HM databases. It also highlights the challenges that could be encountered in attempts to consolidate traditional medicines DBs, how they can be circumvented, and the many potential benefits of consolidated HM DBs. Likewise, other intertwined factors that are integral for advancing HM-based research, and drug discovery and development efforts for enhanced future health applications are discussed.

Keywords

Big data · Data analytics · Database · Database consolidation · Bioinformatics · Computational drug discovery · Herbal medicines · Health care systems

Abbreviations

ABC	American Botanical Council
AMF	Alternative Medicine Foundation
ARS, USDA	Agricultural Research Service (ARS), United States Department of Agriculture
BD	Big data
CAM	Complementary and alternative medicine
CHMD	Chinese herbal medicine database
CMAUP	Collective molecular activities of useful plants
COCONUT	Collection of Open Natural Products
CVDHD	Cardiovascular disease herbal database
DBs	Databases
DNP	Dictionary of Natural Products
EMA	European Medicines Agency
ETM-DB	Ethiopian Traditional Medicine Database
EU	European Union
FAQs	Frequently asked questions
GAD	Genetic Association DB
GPs	Good practices
HAPPI	Human protein-protein interaction
HIT	Herbal Ingredients' Targets
HMA	Heads of Medicines Agencies
HMPs	Herbal medicinal products
HMs	Herbal medicines
HPRD	Human Protein Reference

IMPPAT	Indian Medicinal Plants, Phytochemistry and Therapeutics
InPACdb	Indian Plant Anticancer Compounds Database
ITM	Iranian traditional medicine
KUIHerb	Knowledge Unifying Initiator for Herbal Information
MB	Molecular biology
MeSH	Medical Subject Headings
MINT	Molecular INTeraction
MPD3	Medicinal Plants Database for Drug Designing
NAL	National Agricultural Library
NCCIH	National Center for Complementary and Integrative Health
NDA	Natural Database for Africa
NIH-ODS	National Institutes of Health-Office of Dietary Supplement
NLM	National Library of Medicine
NLP	Natural Language Processing
NMCD	Natural Medicines Comprehensive Database
NPASS	Natural Product Activity and Species Source
OMIM	Online Mendelian Inheritance in Man
p-ANAPL	Pan-African Natural Products Library
PubMed	Public search engine for MEDLINE and NLM databases
RAMP	Reviews on Aromatic and Medicinal Plants
SANCDB	South African Natural Compound Database
SHPIS	Saudi Herbal Plant Information System
T&CM	Traditional and Complementary Medicine
TCM	Traditional Chinese Medicine
TCM-ID	Traditional Chinese Medicine Integrated Medicine
TCMSP	Traditional Chinese Medicine Systems Pharmacology
TradiMed	Traditional Medicine
UNaProd	Universal Natural Products
UniProt	Universal Protein database
Vietherb	Vietnamese Herbal Medicine Database
WHO	World Health Organization
YaTCM	Yet another Traditional Chinese Medicine

1 Introduction

Worldwide, the consciousness of research institutions, governments, regulatory agencies, and other health sector stakeholders has been heightened to possibilities of big data use in checking herbal medicines (HMs) impacts and efficacies. For example, in traditional Chinese medicine (TCM) or Japanese ‘kampo’, a plan has been evolved for data collation from millions of patients taking herbal medication. The prescription of ‘kampo’ medicines depend on herb mechanism and effectiveness, and patients’ experiences. Still, elucidation on mechanisms leading to various adverse reactions requires further scientific studies [1]. Agencies with roles linked to welfare and health care policy making have key functions in verifying herbal medicines effectiveness and side impacts. These roles cannot be extricated from care facilities, clinics, and hospitals cooperation since they provide the basic data from patient records which are uploaded into a big data (BD) server. Usually, data is first fed in as 'anonymous' prior to being statically analysed. The statistics generated then give insight into herb efficacy, best duration of use, frequency, and effective means of taking the medicines to cure or manage a disease with exclusion of adverse effects [2].

In other climes, the European Medicines Agency [1] in conjunction with Heads of Medicines Agencies (HMA) work to identify practical steps and state the regulatory frameworks for big data use to facilitate innovative developments and enhance public health among European Union (EU) countries. Medicines regulators can also use big data insights to measure the benefits and risks attributed to herbal medicines [1]. The foregoing is possible due to the large amount of data streaming on herbal medicine types being deposited in several medicine and health care-based databases. The compendium of data sets available from different platforms gives scientific backing for the effective use of herbal medicines without side effects. Thus, BD application in herbal medicine is also useful for health promotion and disease prevention [2].

Despite the potential benefits of big data application in herbal medicine, the health sector has failed to fully utilize data sets in varied databases to reduce disturbing disease trends and make health care more cost-effective. This is largely due to data inadequacy and/or unavailability, or availability of data in unusable forms. Again, health care and herbal medicine databases make data pooling and linking almost impossible. This has led to the emergence of many several and small institution based databased (DBs). While many health care practitioners take to ordering that several tests for effective treatment, the tests are usually ineffective and quite expensive. As a result, institutions like the University of Florida made available free and virtual public health data that is capable of efficiently analysing health-related information and fast identification in cases of chronic disease tracking. The 'Obamacare' initiative was also based on 'big data' utilization, much like other industry providers, for example, Humedica and Recombinant Data [3].

To date, not much is known about the science behind traditional medicines that make up a relevant aspect of global health care systems [4]. In clinical and everyday practices, the popularity of herbal preparations and medications continues to show a gradual increase, even to the developed Western world. The level of acceptance and impact of HMs is also encouraging. These drive HMs standardization using current pharmacological benchmarks. In the storage of knowledge and science behind trado-medical therapies, there have been the development of several databases (DBs) having varied focal points. The emergence of the traditional medicine DBs fast-tracked pharmaceutically oriented studies on products and metabolites from natural sources [5]. Natural products in HMs have continued to provide new insights and are key in current drug discovery research drives. Considering the information and data explosion especially from the life and molecular research fields, a new age of powerful networks and 'big data' was born. The building of sturdy DBs and networks is needed now more than ever to enhance the sharing of useful and interdisciplinary knowledge finding. As such, within the past decade, efforts have been put into verifying, restoring, digitalizing, and computerizing HMs documents in the form of DBs [6, 7]. However, each DB developer's outlook or focus may differ, but with a basic feature on plant metabolites and/or co-interaction details (plant/herb compound-biological activity-treatment links) and based on thousands of published research reports [8–10]. Many open access, subscription, and online DBs also target giving the user (both non-expert and expert) web-based experiences like compound to target protein interactions and molecular docking analyses [11, 12]. Unfortunately, there are also scenarios of recent DBs that came up sporadically which are no longer accessible, resulting in data loss on natural products of HM origin. In other case, consolidated or aggregator DBs are not enough to reduce redundancy and repetitive data set compilations [13].

Given these challenges with 'big data' use and access in health care systems, and sporadic emergence of single and localized platforms collating and providing information on bioactive herbal medications and phytochemicals efficacies and mechanisms, as well as herbal drugs' adverse effects, it has become important to expatiate on the need for consolidated databases to strengthen the science behind herbal medicines use and better guide future research on trado-medical applications.

2 Big Data Analytics, Platforms, and Herbal Medicine

Generally, data is passed through a cleansing, data joining, extraction of attributes, modelling, and prediction (grouping, categorizing, clustering, and so on), data presentation or visualization, interpretation, and reporting [14]. Analysis of health care-related big data can further be done with open-source platforms (MapReduce, Hadoop, etc.) [15]. For storage of distributed databases, Hadoop can be used. However, when merged with streams of continuously monitored health data, it is required that system adaptability and process optimization be improved using advanced data caching models and in-memory computation [16]. Within the health care industry, big data platforms like MapReduce carry out data processing and computation. Others such as Infogram and FusionCharts are good for data visualization, and Big Data Appliance and Pentaho Data Integration for data management. Large volumes of data sets with different structures have been handled with Oozie, Cassandra, PIG, Mahout, MapReduce, and HBase platforms, all of which are supported within the Hadoop distributed platform [17]. The Apache Spark Streaming, IBM InfoSphere, TIBCO Spotfire, and QlikView are some of the advanced tools used in analysis of big data [18]. Some data visualization packages include 'ggplot2' (open-source program for graphics mapping and creation) and 'igraph' (a library for graphs modelling and network analysis), which also exist within the R statistical programming language. The R software can be linked to many other visualization tools (RCytoscap for Cytoscape, RGraphViz for GraphViz, and RCircos for Circos). Data visualization using Tableau does not require programming skills. Circos is common with genome chromosomes visualization; GraphViz for laying out networks with large number of nodes and edges; and Cytoscape provides rich data presentation styles while saving layout of networks [19].

Furthermore, research has shown that there is a lack of study consensus on internally and externally sourced big data operation in health care systems. Again, big data is mostly analysed using natural language processing (NLP) tools which are based on Hadoop. Big data analytics is also applicable for optimizing clinical operations (decision support and cost reduction) but has a key drawback of its adoption being the lack of proof to back practical health care benefits. This challenge is closely linked to higher number of studies using qualitative strategies (describing possible gains) compared to quantitative approaches. In addition, it had been observed that more research reports were derived from developed than developing countries, hence the need to promote studies on health care BD analytics in developing or third world nations [20].

Data from trado-medical medicine applications require treatment and big data protection because they are vulnerable and sensitive, with immense potential to enhance disease diagnosis and prevention, and medical therapies [21]. Transforming unstructured data to usable forms that improve knowledge with benefits to patients comes with challenges [22]. Health care big data are presented as 'flat' and 'csv' files, tables, and so on, which emanate from internal and external sources [15]. Big data analysis and application in health care covers text mining (mostly unstructured data), machine learning and artificial intelligence (for predicting the unknown from learned known facts), and intelligent agent (for enhanced information flexibility, proactivity, and autonomy) technologies. Thus, big data analytics can be used in predicting outcomes for at risk individuals, identifying patients requiring preventive care, carrying out real-time analysis, provide public health services for at risk populations, conducting toxicity, side effects and trial design assessments, and accelerating acceptance of new therapies to target markets [23]. In order to attain a high level of precision in HM applications, particular accuracy is needed with respect to drug targets, disease markers and mechanism, among others [24].

3 An Overview of Herbal Medicine Databases

Some of the localized or institution-based, national, and international big data-based platforms providing free, limited, or subscribed access to processed and unprocessed information to guide research and decision-making as it relates to herbal medicines are discussed in the proceeding sections. We must, however, note that while each database has its unique qualities, main tables, central components, and structure, none is standardized for universal and unified global research. This may be attributed to none of the DBs allowing certified HM specialists scientists add their insights [25]. As such, the list presented here (Table 1) are in no way exhaustive.

1. *PubMed*: This platform is controlled and updated by the National Library of Medicine (NLM). It majorly targets life sciences, health care professionals, and physicians. It is the single largest and global biomedical database that allows the

Table 1 A non-exhaustive list of databases for natural product and herbal medicine studies

S/N	Database	URL (link)
1	PubMed	https://pubmed.ncbi.nlm.nih.gov/
2	HerbMed	http://www.herbmed.org
3	Duke's Phytochemical and Ethnobotanical Databases	https://phytochem.nal.usda.gov/phytochem/search
4	CAB Database of Plant Science, and Review of Aromatic and Medicinal Plants (RAMP)	http://www.cabi.org/
5	TarNet (bio-target to network construction) platform	http://www.herbbol.org:8001/tarnet
6	Universal Natural Products (UNaProd)	http://jafarilab.com/unaprod/
7	National Center for Complementary and Integrative Health (NCCIH)	https://www.nccih.nih.gov/
8	Natural Medicines Comprehensive Database (NMCD)	http://www.naturaldatabase.com
9	DynaMedex (formerly Micromedex)	www.dynamedex.com
10	GreenMolBD (Bangladesh based)	https://www.greenmolbd.gov.bd/
11	VietHerb	http://vietherb.com.vn
12	Ethiopian Traditional Medicine Database (ETM-DB)	http://biosoft.kaist.ac.kr/etm
13	Phytochemdb	https://phytochemdb.com/
14	Indian Medicinal Plants, Phytochemistry and Therepeutics (IMPPAT)	https://cb.imsc.res.in/imppat
15	MedPServer	http://bif.uohyd.ac.in/medserver
16	Medicinal Plants Database for Drug Designing (MPD3)	https://www.bioinformation.info
17	PharmDB-K	http://pharmdb-k.org http://biomart.i-pharm.or
18	SuperTCM	http://tcm.charite.de/supertcm
19	Agricola	https://www.nal.usda.gov/agricola/
20	AromaDb	http://bioinfo.cimap.res.in/aromadb
21	AfroDb	http://african-compounds.org/about/afrodb/
22	Traditional Chinese Medicine (TCM) Systems Pharmacology Database and Analysis Platform	https://old.tcmsp-e.com/tcmsp.php http://tcmspw.com/tcmsp.php
23	TCM Database@Taiwan	http://tcm.cmu.edu.tw/
24	Traditional Chinese Integrated Medicine (TCM-ID)	http://www.megabionet.org/tcmid
25	Saudi Herbal Plant Information System (SHPIS)	https://www.shpis.com
26	Collection of Open Natural Products (COCONUT)	https://coconut.naturalproducts.net/
27	Encyclopaedia of Traditional Chinese Medicine	https://www.nrc.ac.cn:9090/ETCM/
28	Embase	https://www.embase.com/

(continued)

Table 1 (continued)

S/N	Database	URL (link)
29	Reaxys	https://www.elsevier.com/en-xm/solutions/reaxys http://new.reaxys.com
30	Natural Product Activity and Species Source (NPASS)	http://bidd2.nus.edu.sg/NPASS/
31	South African Natural Compound Database (SANCDB)	https://sancdb.rubi.ru.ac.za/
32	Yet another Traditional Chinese Medicine (YaTCM)	http://cadd.pharmacy.nankai.edu.cn/yatcm/home
33	Indian Plant Anticancer Compounds Database (InPACdb)	http://www.inpacdb.org
34	Foundation for Revitalisation of Local Health Traditions (FRLHT)	https://www.frlht.org/
35	Medline Plus	www.nlm.nih.gov/medlineplus/herbalmedicine.html
36	Botanical Medicine Information Resources – Directory of databases	www.cpmcnet.columbia.edu/dept/rosenthal/Botanicals.html
37	American Herbalists Guild-Health	www.healthy.net/herbalists/
38	KUIHerb: Knowledge Unifying Initiator Herbal Information	http://inf.pharm.su.ac.th/~kuiherb/kui-herb-web/
39	MedHerb	www.medherb.com/
40	Phytotherapies	Phytotherapies.org
41	Herbal Hall	www.herb.com/herbal.htm
42	STITCH database	http://stitch.embl.de/
43	PRELUDE	http://www.africamuseum.be/collections/external/prelude
44	Botanica Ethiopia	https://botanicaethiopia.com/herbs
45	PROTAbase	https://www.prota4u.org/database/
46	PlantZAfrica	http://pza.sanbi.org
47	Francophone	http://www.ulb.ac.be/sciences/bota/pharmel.htm

inclusion of MeSH (Medical Subject Headings) terminologies like 'complementary therapies' and 'alternative medicine', with links to over 20 major headings [26].

2. *Micromedex*: This is database that is subscribed to and covers clinical procedures, dietary supplements, herbal medicines, herbal and dietary supplement co-interactions, and patient education. Subfield under acute care, regulatory compliance, chemical safety, pharmacology, and toxicology are also addressed. Although a subscribed DB, a study showed that health science librarians and educators need to strengthen and reiterate trainings, especially for students, on the proper utilization of drug information DBs like Micromedex [27].

3. *Duke's Phytochemical and Ethnobotanical Database*: On this platform, keyword searches can be performed on biological activity, plants, chemicals, and ethnobotanical fields and information derived may be pooled to other external databases. The platform is run by the Agricultural Research Service (ARS) of the US Department of Agriculture (USDA).
4. *Agricola*: This platform is used to find information on medicinal or herbal plants and is fed and driven by the National Agricultural Library (NAL).
5. *Complementary and Alternative Medicine (CAM)*: This platform is a rich resource for studies in complementary and alternative medicine (CAM). Other databases linked to CAM are the National Center for Complementary and Integrative Health (NCCIH), National Institutes of Health-Office of Dietary Supplement (NIH-ODS), and Natural Medicines Comprehensive Database (NMCD) [28]. The CAM DB is also a subset of PubMed linked to certain subfields within complementary and alternative medicine.
6. *Alternative Medicine – Health & Wellness Resource Centre*: This platform was launched from the Gale Encyclopaedia of Alternative Medicine.
7. *HerbMed or HerbMedPro*: This platform is a study-based resource providing information on the use of herbs as health remedies. It is a compendium of published processed data on toxicity, side effects, and contraindications on certain herbal medicines and provides links to research on medicinal plants, their preparation methods, efficacies, and mechanism of action. It is hosted by the American Botanical Council (ABC) and funded by the Alternative Medicine Foundation (AMF). The ABC provides scientifically backed information to promote effective and safe application of plants and herbal medicines. It also publishes the Expanded German Commission E monographs, HerbClip (reviews/summaries of seminal research), and HerbalGram journal. An article by Pollard [29] in HerbalGram emphasized the need for rational phytotherapy backed by product scientific evidence, especially for widely studied HM products derived for gingko (*Ginkgo biloba*), garlic, turmeric (*Curcuma longa*), and St. John's wort (*Hypericum perforatum*), among others.
8. *Review of Natural Products – Facts and Comparisons*: This provides several data subsets like facts on drug identifiers, off-label drugs, natural products review, and herbal co-interaction.
9. *Natural Medicines*: This database is a compilation done by physicians and pharmacists, and that requires subscription prior to access and use. It provides updated clinical data on herbal and natural medicines, as well as dietary supplements used by the Western nations.
10. *RXList-Alternatives*: This online TRUSTe certified drug index database contains herbal medicine groups from Chinese herbs and homeopathies, and the Western world. It also offers complete monographs and frequently asked questions (FAQs) sections. The major current and formidable competitors with this database include online platforms like webmd.com, singlecare.com, goodrx.com, and drugs.com. the latter online resources having over 30 million views quite early in the year 2023.

11. *Embase*: A pharmaceutical and biomedical database having more than 30 million indexed records, over 8000 peer-reviewed journal articles, and more than 2500 journals not accessible through MEDLINE and other international platforms. A study covering searches through Embase, MEDLINE, and other DBs showed that while the reported prevalence of herbal products and practices use was high, their use remained largely underestimated. As such, best practices covering submission of research reports on HM operational definitions like exclusion and inclusion criteria for treatments, practices or products, context of use, and detailed data on HMs exposure requirements were recommended. Likewise, the description of criteria used in coding the data, analytical strategies used, and publication of study survey questions used are needed [30].
12. *Examine*: This platform provides evidence-based herb/supplement, health, and nutrition health information, with coverage of study deep dives and summaries, trendy health topics, with detailed discussion of findings.
13. *Natural Medicines*: This platform reviews dietary and botanical supplements, and other CAM treatments, as well as provision of evidence tables which assess key research and baseline summaries of herbal medicines efficacies.
14. *Review of Aromatic and Medicinal Plants (RAMP)*: This platform provides reviews on aromatic and medicinal plants (RAMP). RAMP is an abstract database of globally published studies on wild and grown spice and herb species, medicinal plants, and essential oils. Coverage includes the biological activity, botany, cultivation, and use of these plants [31].
15. *VietHerb*: A database of plants and herbs used in Vietnamese traditional medicines and aimed at initiating data communication among other available disease, metabolites, geography, and plants ontologies in order to portray a concise description of each plant species. Information is provided on thousands of species and metabolites from hundreds of locations, and close to 10,000 oriental treatment efficacies. In addition, it provides links between herb morphology and species distribution, herb species and metabolite, and herbs and therapeutic impacts [32].
16. *Collection of Open Natural Products (COCONUT)*: This was believed to be the largest DB on natural metabolites. It is an online DB that is freely accessible and provides data on natural molecules or products in one single data set. Metabolites sparse annotations and structures are available for more than 350,000 non-redundant natural products [13]. It was based on the attempt to create an online resource that regroups identified natural metabolites into a single source location and simplify and facilitate natural products research and allow other downstream processing such as *in silico* screenings and related applications [33].
17. *The Pan-African Natural Products Library (p-ANAPL)*: This DB contains data and information on about 600 natural molecules of African plants' origin, especially those derived from the northern, southern, western, and central African territories. A study that screened NPs from this DB identified the compounds, boldine and ixoratannin A-2, as possible HIV-1 and hepatitis C virus (HCV) inhibitors. As such, these NPs could be incorporated in novel drug

formulation targeting the mitigation of these diseases [34]. The p-ANAPL library is believed to be the largest collection of NPs from African ethnomedicinal plants [35].

18. *Traditional Chinese Medicine (TCM) and linked DBs*: The presence of built-in algorithms and tools in this DB contributes greatly to enhancing its application in HM-related drug discovery, design, and development, and network pharmacology-based research studies. Network pharmacology effectively links HM NPs/compounds to protein or genetic disease targets, as well as underlying disease regulatory pathways, under high-throughput conditions. It is a relatively new and integral approach in drug development and mechanism studies. In addition, network pharmacology allows TCM and other HM-based mixtures or drugs synergy or combinations, target predictions, compatibility, and toxicological analyses. As such, it is a comprehensive, rational, and powerful tool for studies in HM [36]. Databases for HM-network pharmacology studies include the Traditional Chinese Integrated Medicine (TCMID) [37], Traditional Chinese Medicine Systems Pharmacology (TCMSP) [38], ChEMBL (https://www.ebi.ac.uk/chembl/) [39], TCM database@Taiwan [8, 40], TCM-Mesh [41], Herbal Ingredients' Targets (HIT) [42], PubChem [43], and STITCH (to find association among phenotypic, cellular, and molecular data) [44]. Among the databases to study interactions between proteins or gene targets, IntAct (a molecular interaction DB; https://www.ebi.ac.uk/intact/home) [45], human protein-protein interaction DB (HAPPI; http://discovery.informatics.uab.edu/HAPPI) [46], Human Protein Reference (HPRD; https://www.hprd.org/) [47], Molecular INTeraction DB (MINT; https://mint.bio.uniroma2.it/) [48], STRING (for protein-protein interaction networks functional enrichment analysis; https://string-db.org/) [49], and Reactome [50] have been used. On the other hand, Genetic Association DB (GAD; https://geneticassociationdb.nih.gov/) [51] and Online Mendelian Inheritance in Man (OMIM; https://www.omim.org/) [52] were useful to link various gene targets to diseases.
19. *Ethiopian Traditional Medicine database (ETM-DB)*: The search for novel drugs and drug designs has pushed scientist back to nature for answers. The collation of valuable data and information on the resplendent natural resources found globally has led to the development of trado-medical DBs like the ETM-DB. In order to expand on drug discovery strategies based on natural NPs, the ETM-DB was floated as the largest, open access, structured, online database synergizing indigenous HM knowledge from over 80 ethnic groups. Moreover, close to 80% of the population access herbal medicines for their primary health care needs. Within this user-friendly DB web interface, the search menu provides and array of topics or entities (target genes, species phenotypes, compound, prescription, and herb) to choose from. It also allows users to search for relationships between and among entities within it and other natural products DBs like TCM, Natural Database for Africa (NDA), and South African Natural Compound (SANCDB). Such DB consolidation efforts were aimed at fast-tracking the drug development and discovery processes based on NPs of Ethiopian origin. The DB website (http://biosoft.kaist.ac.kr/etm)

provides essential raw and processed data on NP chemistry and composition, including associated human protein or gene targets [53].

20. *Natural Database for Africa (NDA)*: This DB was emanated from an initial effort to consolidate HM and NPs information from East Africa and Ethiopia. Interestingly, the NDA covers data on about 7000 African plants, but majorly of Ethiopian origin, and the rest from African nations along its borders. Information such as plant species local or traditional, and botanical names, and their herbal applications can be found within the NDA. This DB is, however, limited for drug discovery focused studies, because it does not provide additional information such as herbal medicines components and target genes [54].
21. *Universal Natural Product (UNaProd)*: This is DB derived from Iranian traditional medicines (ITMs) and consists of an array of therapeutic HMs. UNaProd was initially built on Makhzan al-Advieh (a medical encyclopaedia covering simple drugs) using manual and text mining and had the widest coverage of ITM monographs. UNaProd is an advancement of this and allows access to close to 3000 ITM-based monographs including mineral, herbal, and animal compounds with over 15 different features. It was created as baseline upon which to build a systems pharmacology platform capable of correlating disease, drug, and target attributes. Hyperlinks exist within UNaProd connecting to Collective Molecular Activities of Useful Plants (CMAUP) (National University of Singapore-based) and other DBs [55]. CMAUP covers over 2500 medicinal plants, as well as data on garden, agricultural, and food plant species [56].
22. *Other herbal metabolites and plants databases*: These include the BDTOI (a bilingual platform of Indonesian origin and catering to Indonesian therapeutic plants with metabolites referenced with KNApSAcK and PubChem); CVDHD (a cardiovascular disease herbal database providing annotated structures with docking score and thousands of targets with optimized 3D structures); ChemieBase (a Thailand database for herbal plants' metabolites); Chinese Herbal Medicine Database (CHMD); Dictionary of Natural Products (DNP) (a sub-database of CHEMnet-BASE that allows concurrent integrated search); HIT (a curated and comprehensive database offering data on herbal metabolites interaction with target proteins); NPASS (database for natural products, their biological targets and bioactivities) [57]; HerDing (recommends herbs to treat infections using information from bioactive molecules and genes through text mining and integration with many other databases); IMPPAT (manually curated database of Indian medicinal plants metabolites) [58]; Reaxys (a commercial, Elsevier-owned database, offering a vast metabolites library from data generated from years of scientific findings); TradiMed (a commercial platform that merges Korean and Chinese traditional medicines knowledge and insights to facilitate modern drug discovery from HMs); and TCM Database@Taiwan and Super Natural II, among a vast array of other databases [5, 59]. Others include the South African Natural Compound (SANCDB) [60], Yet another Traditional Chinese Medicine (YaTCM) databases [61], KUIHerb [62, 63], MedHerb and HerbalHall [64], and InPACdb [65]. Additional examples of African traditional

medicine database include the Botanica Ethiopia, PROTAbase, PlantZAfrica, PRELUDE, and Francophone, among others [60, 66–69].

4 Initiating Databases Consolidation, Challenges, and Benefits

Integration of data or data consolidation (DC) implies collation, combination, and storage of data within a single harmonized one-point access space. Such consolidated platforms may facilitate the derivation of insightful information from inherent raw or processed data by users for enhanced decision-making. Database consolidation also helps in trends discovery and data sources streamlining. Achieving this may typically involve technological layers such as the data source, a data pipeline for its extraction, loading, and transformation, and the data warehouse. A data pipeline may be built through hand coding and automated cloud-based tools. In hand coding, data engineers write scripts to integrate data from already known sources. Hand coding is a laborious task that is best tailored to fewer data sources or when the data store or source is not linked to other tools. On the other and, the use of cloud-based tools to automate data pipelines fast-track the process of data consolidation. Cloud-based platforms tools are periodically evaluated, maintained, and updated by the provider. Some data consolidation practices include checking compatibility of in-source data types and target, maintenance of data and processed output copies for improved traceability, compliance checks, accountability, and surveillance. In addition, standardizing data character set conversions enhances data consolidation for more dependable results after data processing [70].

Common impediments to data consolidation include those attributed to the demerits of hand coding. Here, new levels of code may often be required to be written, managed, and processed with incorporation of new online data sources, thus lengthening the process of consolidation. Again, both potential and real security issues plague DC. Data breaches can occur prior to and after DC. Data can also be deleted, corrupted, or compromised during backup development. The problems with data exfiltration (data copying, transfer, or retrieval without authority to access it) from DB servers or computers also exists. Inconsistency of data types stored in different locations, missing data, and similar or duplicate data may also be found across several sources or databases, making consolidation a herculean task [70].

Despite the aforementioned and highlighted challenges, the benefits of DC cannot be overemphasized. Databases consolidation ensure the decreased complexity, enhanced data infrastructure, and greater control of the data environment. Importantly, DC minimizes perceived number of DB issues, optimizes DB availability and space, performance, and systems security. As essential building blocks, DBs enable the storage of accessible and organized formats for smooth functioning in HM research. Nonetheless, keeping up with the pace of change and the generation of spurious, unverified data sources, the best time is now to welcome a consolidated future for HM research. Adopting DC could greatly reduce cyberattack risk through easily enforceable polices covering fewer DB platforms, scaling back of data access points by reduced systems vulnerabilities, and improved monitoring and reaction to perceived and real threats. Again, more modern consolidated systems tend to have a generally reduced risk of intrusion through regular hard and software updates. The use of consolidated DBs also serve to reduce cost and increase platforms efficiency, compared to many separated DBs with limited data sets and scale. In consolidating DBs into optimized systems, operational cost is reduced due to decreased maintenance and hardware, and limited number of DB licenses required. A modern and standardized DB may also offer better strategic data architecture and access restriction tools such that one can only access data they require for particular tasks. A disjointed data and information infrastructure makes it hard to derive a general overview of the desired data set. Furthermore, DB consolidation ensures data protection and streamlined recovery through use of a robust disaster recovery approach and backup systems [71].

More specifically, other gains from HM DBs consolidation include a greater base on which to build multi-strata policies for HMs use and its integration into health care systems alongside World Health Organization (WHO) guidelines. Database consolidation facilitates the establishment of regulations for HM practices, products quality, and safety controls, and increases awareness on HMs efficacy and safe treatment regimens among products end users and general public. In the same vein, it ensures the sustainability of ethnomedicinal plants through their conservation and cultivation while constituting an integral scientific evidence base on trado-medicals quality, safety, and efficacy [72, 73]. In order to solidify consolidation efforts, more collaborations between countries and international regulatory agencies like the WHO and USDA could contribute to encouraging HMs valorization, cost effectiveness, and availability [74, 75].

5 Emphasizing and Ensuring the Need for HM DBs Consolidation

5.1 Herbal Medicine Products and HM-Based Studies: Case Scenarios and Approaches

This section briefly zooms in on some research efforts on herbal medicinal products (HMPs) and related new findings from HM-based studies. This is in efforts to entrench the need for update of HMs new findings, information, and DB consolidation, and which could be spread across board to analytical techniques and assay methods to bridge widening study gaps. Some research, such as that conducted at the University of Oulu, focused on the toxicological and pharmacokinetic attributes of developed HMPs to strengthen potential market base and give scientific proof to claimed therapeutic value. The application of chemogenomic tools gives key predictions into potential toxic impact as shown for some Ayurvedic and TCM products [76]. Nonetheless, validation of the *in silico* predictions remains necessary through actual clinical and experimental research. Here, 'omic' tools prove to be indispensable to delineate and assess different mechanisms and toxicities [77]. Application of existing and modified *in vitro* interaction, metabolism, and transport assay protocols have also sufficed for HMPs, as they have for orthodox medicines [78–80]. While the multi-component structure of HMs is expected to evince multi-level dynamic and kinetic interactions, the therapeutic responses of herbal products may also be predicted to a limited extent based on their individual components [81].

In other research efforts, underutilized plants are scientifically valorized to enhance their industrial applications. For example, new molecules identified in prickly *Opuntia ficus-indica* like polyhydroxypregnane glycoside, neohancoside C, and isovitexin 7-O-xyloside-2″-O-glucoside from cladodes [82], and pinellic acid from fruit peel and pulp [83] contribute to the reported pharmacological properties (antibacterial and antioxidant) of the extracts. Yet, these molecules or extracts have not been subjected to molecular biological techniques to gain further insights on potential therapeutic functions. Another study on a south American plant species, *Baccharis dracunculifolia* in the family Asteraceae, has shown good anti-inflammatory action for use in the treatment of skin diseases [84–86]. As such, the commercial and economic value of this plant has shown steady increase in past years since it is also a natural source of Brazilian green propolis [87]. The plant's leaf extract has also shown the ability to modulate immune system responses [88, 89] and protect the liver [90], as well as have antimicrobial [91] and analgesic properties [85]. These features were linked to the extract's flavonoid (e.g., isosakuranetin) and phenolic acid (mostly artepillin C) contents [88]. In the root extract, more recent results identified the presence of monoterpenes, triterpenes, and fatty acid derivatives, and a major plant-specific compound reported for the first time as baccharis oxide. Both baccharis oxide and extract showed good topical anti-inflammatory activity, thus supporting its use in skin ailments like dermatitis, psoriasis, and other skin inflammatory diseases. The study, however, noted that peculiar action mechanisms and assessment of toxicity and safety levels of both the new compound and extract required additional exploration [87].

In providing added examples, in case of chronic neurodegenerative disorders, neurological inflammation is key to the disease progression, and the peripheral organs, brain, neurons, and the elderly are mostly adversely affected [92, 93]. The disease also has negative impacts on health care systems, communities, families, and communities [93]. Furthermore, curative or preventative therapy options are not available as a result of dearth of insight on its complicated pathophysiology and causal factors [94–96]. In order to make headway, the multi-target and component system action of HMPs could be practical in addressing the complicated neuroinflammatory mechanisms and pathology. A study analysed the synergistic effect of HM formulas in diminishing neuroinflammation by applying molecular docking, network pharmacology, and experimental analyses. Results showed that combinations of 6-shogaol and baicalein, and 6-shogaol and andrographolide, at 50% inhibitory concentrations (IC) of 3.28 and 2.85 g/mL synergistically slowed lipopolysaccharides-induced nitric oxide (NO) generation in immune cells of the brain. The key genetic pathways put forward for the action of the three identified molecules are the mitogen-activated protein kinase 8 (MAPK8) and 14 (MAPK14) and nitric oxide synthase 3 (NOS3) pathways, using network pharmacology analysis. On the other hand, docking analysis showed significant binding affinities of the three molecules to respective protein targets. Studies like these which utilize an array of *in silico*, *in vitro*, and experimental tools, platforms, and information from HM DBs show a practical strategy in new herbal molecules and product formulations development. At the same time, they give scientific baselines and novel understanding into synergistic combinations against neuroinflammation. Such study designs, workflow, and tools could be extended to other disease research [97].

5.2 Need for Consolidated Databases in Herbal Medicine

Given the sporadic emergence of HM databases, it is also important to discuss the need for their consolidation to strengthen herbal health care access and delivery systems in the face of 'big data' availability. Health care BD drawbacks include large data volume, dimension, and format variability, data value, presence of poor quality or 'noisy' data (trustworthiness or veracity issues) (Fig. 1), strengthening and processing isolated (silo) data sets from individual health care institutions. Other drawbacks include classification and analysis of semi-structured or unstructured data, processing and compiling continuous data streams, that is, data velocity (rate at which data is generated, collated, and analysed), intensive computational needs for analysis of genomic data, and problems with big data visualization. Problems with cybersecurity, data privacy (Fig. 1), ownership, and governance constitute other

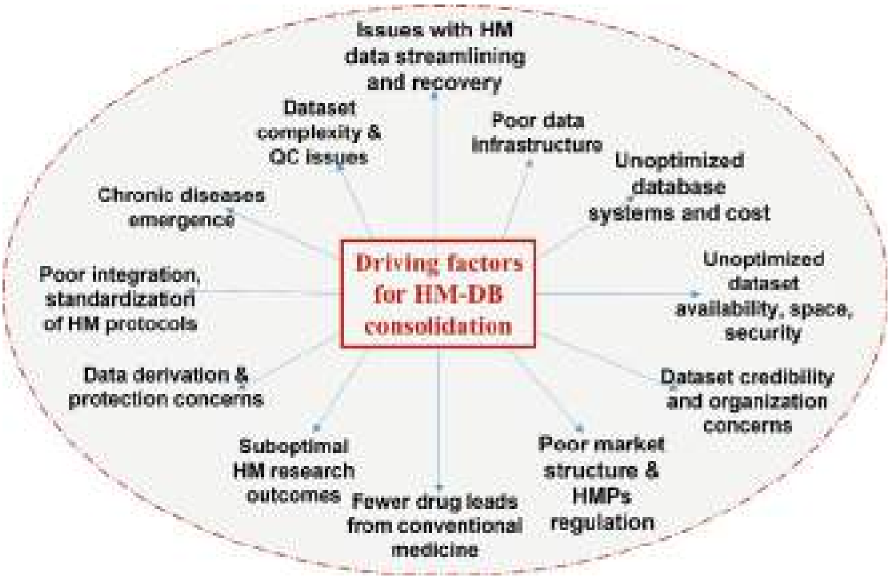

Fig. 1 An overview of factors that drive the need for herbal medicine database consolidation. *QC* quality control, *HM* herbal medicine, *HMPs* herbal medicinal products, *HM-DB* herbal medicine database

valid challenges [13, 50, 98–100]. In line with these challenges, even when a patient's privacy is protected, quite several health care practitioners are unwilling to share data due to market competition. As such, maintaining a balance between patient confidentiality and data integrity (Fig. 1) remains a daunting task. Again, the number of available open access databases or resources, the integration of useful and structured data, and their standardization make up present challenges [101].

Privacy and cybersecurity problems include unauthorized access or hacking, data sets combination without analysis resulting in future breaches in privacy, merging of varied data sets in a way that contravenes privacy laws, and data capturing which is bereft of human activity that raise ethical issues. Usually, outcomes of health care data analysis are traced back to the patient to aid in disease diagnosis, prognosis, and decision-making on most effective therapy regime needed. Thus, herbal medicine data is not fully 'anonymous', and some form of false, complex anonymity protocol is created to ensure some level of 'k-anonymity' which is acceptable on a legal and ethical bases. Yet, if big data subsets are re-associated or merged with other data, 'k-anonymity' reduces greatly with increased risk to data privacy. Hence, adaptable and advanced algorithms are needed which circumvent scenarios where decreased k-anonymity fall below acceptable levels [102].

Furthermore, in many African nations, despite the use of Western medicines, over 60% of Africans still patronize traditional HMs as a complementary medical source to ameliorate disease effects [103, 104]. However, it has been reported that only partial knowledge of herbal medicine is available such that hair-brained herbal practitioners' cash in on the situation by providing contentious medical care with prepared medicines that do not perform as expected and portend several dangers to health of herbal medicine clients [105]. Again, poor communication exists among herbal care practitioners, and this is aggravated by little or documentation of their knowledge leading to underdeveloped knowledge systems or knowledge loss [103]. This herbal medicine knowledge system is inappropriate, inefficient, difficult to transfer and access, and tacit, particularly for durability in knowledge use and preservation [106]. This explains why trado-medical knowledge is lost over time. In certain cases, herbal knowledge misuse, abuse, misappropriation, and misinformation have also been reported [105]. This is a cause for concern since improper documentation of herbal medical practices that involve methods of preparation, ingredients selection, and mode of administration have led to a lack of trust in herbal medicines effectiveness, safety, and quality [107]. Hence, the importance of HMs knowledge preservation cannot be overlooked, because it strategically affects the levels of competitive advantage, sustenance, and growth that HMs have in various institutions, environments, and on persons [108]. As a result, mitigating strategies are required to potentially rectify the drawbacks (Fig. 1) facing HM practices with a view to restoring prevailing circumstances [104].

Considering the above challenges, a recent study proposed an approach for data capture, storage, compilation, and semantic search of African herbal medicines based on a consolidated hybrid (incorporating ontology and machine learning tools) knowledge-based model for enhanced data retrieval and preservation. The strategy allows multi-word searches and functions as a useful resource for extraction of herbal medicines knowledge from a real-world application context [109]. In addition, it is essential that herbal medicines BD should emanate from actual patient data to achieve beneficial health knowledge. Big data application in herbal medicine cannot be fully optimized if it does not translate to enhanced patient health care. Although a challenging process, it is clear that a continuous co-interaction between small (individual) and big (population) data, medical care, and clinical studies cannot be overemphasized. In predictive modelling applications, the combination of varied data types (clinical, -omics, molecular, physiology, and imaging) should also be the rule of thumb [110].

Generally, health care big data challenges are linked with large data volume, data formats, data streaming, reliability and veracity, data visualization, privacy, and cybersecurity, and so on (Fig. 1). In addition, the main problems with big data and their analysis lie particularly in segmented data consolidation and processing, mining of data from distributed and heterogeneous health databases, cross-border health care management (different cultures, languages, and countries), and discrepancies between 'individual' and big 'population' data. Therefore, a continuously streamlined co-interaction between small and big data could be circumvented by through herbal medicine databases consolidation. This would further portray a more global, unified, and synchronized front, provide added scientific backing for herbal medicines-related research, while strengthening the veracity platforms, tools, and databases and minimizing the core concerns with big data use and application. The emergence of consolidated herbal medicine databases will further encourage the exploration of herbs or plant bioactive constituents in herbal drug formulations to better tackle specific infections and comprehend and predict treatment impacts based on metabolite profiles and a scientific perspective. Some thought also needs to put into making consolidated HMs database designs user-friendly with interactive elements that facilitate easy information retrieval [111].

Furthermore, of the over 120 DBs on HM metabolites and products that are cited in literature, less than half can be freely accessed. Yet, major content overlap exists in these DBs, where about 40 of them share data sets on half the number of metabolites with at least one other data set. Some consolidators or aggregator DBs like the SuperNatural II and ZINC catalogue for natural compounds exist. However, they do not suffice to efficiently synchronize all possible and available data and information on known natural metabolites. These aggregators also do not allow submissions of newly discovered compounds. Hence, there is a major need for consolidation of organized and globally recognized DBs for natural compounds linked to HMs, to facilitate the easy addition or submission of novel metabolites, as seen with the UniProt proteins DB [13].

In further analysis to drive home the need for HM DBs consolidation, a transient look at HM-related boards such as traditional and complementary medicine (T&CM) methods, new HM findings, and protocols such as herbal medicinal products (HMPs) development are briefly discussed. Consolidation of these HM-related aspects could further strengthen the quality of consolidated HM DBs. Herbal medicinal products are the derivatives of increasing HM studies which are increasingly published as articles in many Western journals primarily focused on T&CM.

These HM-based articles cover subjects ranging from products interactions and efficacy to safety concerns with review papers never failing to highlight the dearth of HM quality-based research. This concern remains valid since clinical trials aimed at elucidating on any drug's efficacy, toxicity, and safety eventually show the HMP's therapeutic use/application. Again, such trials give scientific backing and proof for the use of HMs and ensure HMP quality control, standardization, authenticity, provided good practices (GPs) in T&CM are applied [112]. Within these practices, care must be taken to scientifically define important controls and evaluated outcomes. This cannot be overemphasized since HMPs are usually a complex mix of herbal extracts. As such, study strategies aiming to expatiate on HMPs mechanisms of action, especially with 'omic' and network pharmacology technologies, are needed. The impacts of these tools in revolutionizing HM-based research through emphasis placed on disease gene or protein targets and their interaction with compounds and compounded mixtures, as they have done with conventional orthodox drugs, are immense [81].

In addition, new approaches have become integral in HM studies because chronic diseases continue to emerge and continue to worsen death and morbidity trends in both the developed and developing climates. Whether communicable or non-communicable, these chronic disorders are associated with many mechanisms, underlying molecular networks and pathways etiological factors [113, 114]. Although new conventional orthodox medicines have been slowly developed within the last decade, there has been little or no improvement in drug therapeutic outcomes in terms of associated drug risks versus benefits analysis. This may be closely linked to the myopic research focus on particular disease to particular drug paradigm, or one compound to target outlook with emphasis on molecular and biological techniques and strategies [115]. Molecular biology (MB) tools are key in elucidating possible medicine targets with selection of potent therapeutic molecules. However, these MB results have not amounted to feasible drug development to cater to clinical realities. Consequently, a much better go-to plan would be to engage a range of targets at a time with several mechanisms, as opposed to one target. Still, a balanced approach has to be employed in line with good practise in T&CM. HMs could be the answer to challenging multiple disease targets at once and aid in reducing the search gap for promising drug leads, assays, and analytical protocols development [112]. The scientific validation of HMPs using advanced analytical tools could further aid in redesigning the HMs market [116–118]. It can also help change the 'untrusting' viewpoints of the Western, developed societies and thier HM regulators. This is important since the Western world mostly believes that HMs research lack quality, are outdated, sporadic, and uncoordinated when compared with conventional medicines [119, 120]. As such, ensuring the proper coordination of the varied aspects of HM-based studies as well as its update and consolidation of its related databases are key to advancing science behind T&CM. Also, as part of HMs DBs consolidation, policies outlining the steps to take in verifying and depositing new NPs or herbal compounds and other novel findings and information from plant species should also be explicitly described at local, national, regional, and international levels.

6 Conclusions

Over time, data generation has been reported to go through a passive data input stage, an active phase where users input data directly via the internet or web, and the automatic phase. While the process of data generation cannot be totally reversed, the discovery of health care big data features, improving therapeutic efficacies, and promoting the benefits of big data in health care systems constitute valid future outlooks. The concept of 'big data' is here to stay and will continue to give new insights to herbal medicine applications. Big data cannot be extricated from herbal and integrated medicine advancements and is key to evolving to evidence-based medicine from experience-based medicine. The introduction of computational, bioinformatics, and algorithm-based platforms has further aided in big data analysis, categorization, and organization [121]. Regarding natural molecules efficiency and assessment, novel advancements in HM safety and mechanism attributes are still needed to ensure gradual attainment of wholesome data quality, consistency, and control [122]. Most importantly, the projected impact of and need for increase in consolidated DBs cannot be overemphasized in ensuring the availability of credible, non-repetitive, and up-to-date data or information on both existing and new HMs and natural metabolites. Overall, the benefits of HM DBs consolidation outweigh the disadvantages. As such, the challenges to herbal medicine databases consolidation could be overcome by collaborative efforts from all relevant industry stakeholder. The near future gains in advancing the course of HM-based studies, especially as regards drug discovery and development to improve health care therapeutic outcomes and applications, should not be undermined.

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Challenges in the Storage of Herbal Medicine Products and Strategies for Sustainable Management

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Priyanka Devi and Prasann Kumar

Contents

Abstract

Medicinal plants, the world's richest source of herbal products, are rapidly disappearing as global demand for these resources surges. This chapter aims to serve as a reliable guide for the conservation and responsible utilization of medicinal plants by examining current and future global trends, developments,

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and preservation strategies. The sustainable use of these resources, incorporating good agricultural practices, conservation methods like in situ and ex situ conservation, and cultivation practices, is deemed essential for their long-term viability. To enhance the yield and potency of therapeutic plants, we propose adopting biotechnological methods, such as tissue culture, micropropagation, synthetic seed technology, and molecular marker-based techniques. Additionally, it is crucial to ensure the safe and effective use of Ayurvedic herbal medicines through proper storage and handling of plants intended for various Ayurvedic dosage forms. This involves establishing sound storage facilities, raw herb packaging, adherence to API format for herb testing, microbial elimination from herbs, sterilization processes, and best practices for spice storage. Regulating the storage, packaging, and processing of raw herbs for various Ayurvedic herbal medicines is pivotal to enhance the evaluation of quality, safety, and effectiveness in the realm of medicinal plants. Standardizing these practices will produce safer, more effective, and higher-quality herbal products, potentially expediting the global adoption of the Indian medical system.

Keywords

Herbal · Agriculture · Potency · Seed · Objectives · Seed · Medicinal · Resources

Abbreviations

BHC	Benzene hexachlorides
CBD	Cannabidiol
CPMs	Chinese patent medicines
DDT	Dichloro-diphenyl-trichloroethane
EPO	European Patent Office
GACP	Good Agricultural and Collection Practices
GMP	Good Manufacturing Practice
GSP	Good Supply Practices
HPTLC	High-Performance Liquid Chromatography
IAMs	Indian Ayurvedic medicines
MHRA	Medicines and Healthcare Products Regulatory Agency
NMPB	National Medicinal Plants Board
OCPs	Organochlorine pesticides
PCNB	Pentachloronitrobenzene
SOPs	Standard operating procedures
TKDL	Traditional Knowledge Digital Library
TLC	Thin-layer chromatography
USP	United States Pharmacopeia
UV	Ultraviolet
WHO	World Health Organization

1 Introduction

Traditional herbal remedies predate all other forms of medical treatment used by humans. Because of this, herbal medicine is an integral part of many different types of conventional medical practice all over the globe [78]. The World Health Organization (WHO) reports that roughly 74% of the 119 plant-derived prescription medications used today have been used for centuries in traditional treatment [7]. Today, 90% of the global population still uses unprocessed preparations and uncooked plants for medical purposes. Concerns about natural medications' quality, effectiveness, and safety have been raised. In light of WHO's efforts to evaluate medical plants' quality, safety, and efficacy, the botanicals used in the different Ayurvedic Dosage Forms must be appropriately stored, packaged, and handled. The therapeutic effectiveness of a given medicinal plant varies as its plant material does in terms of the proportion of active compound, which in turn changes according to the location of collection, the time of collection, the environmental variables, and the surrounding of its growth [15, 26, 27, 33–36]. Insects, bugs, fungi, microorganisms, pesticides, and other animal matter (including animal excreta) should not be present in vegetable medicines used as such or in composition. These could be free of any off-putting odors, colors, sliminess, mold, or other signs of decay and within the legal and recommended limits for lead, arsenic, and heavy metals essential components for standards according to the Ayurvedic Pharmacopoeia of India, including quantitative tests such as total ash, water-soluble ash, acid insoluble ash, alcohol-soluble extractive, ether-soluble extractive, moisture content, volatile oil content, and assay. The quality control measures taken during the production of herbal medicine, as well as the storing, packing, and processing of the raw materials, the shelf life of the finished product, the presence or absence of preservatives, herbicide residue, and other factors, all contribute to the product's overall safety and effectiveness. Because of these factors, it is clear that several in-depth studies are needed to determine whether or not Ayurvedic natural plants are safe and effective before they can be used in treatment. The plants' freshness and viability are directly impacted by their proximity to light, air, and microorganisms. Since the unprocessed plants are oxidized by air and decomposed by light, their security decreases as time passes. The appropriate packing substance for raw plants must be researched and chosen to ensure product compatibility and the preservation of qualities. It is possible to keep raw plant medicines in a way that prevents them from getting contaminated or going bad. The active component in fresh plants can be preserved for an extended time if proper precautions are taken to protect them from the impacts of air, wetness, heat, and light. Changes in color, flavor, taste, and physical structure indicate the degradation of therapeutic components of plants. Proper storing techniques guarantee the safety and effectiveness of completed plant goods by preventing contamination and degradation. Raw plants can be cleansed, rinsed to eliminate organic material and other impurities, sanitized to avoid infection, and dried correctly before storing. Ayurvedic Pharmacopoeia of India guidelines can also be used to evaluate the product's identification, purity, potency, boundaries of heavy metals, microbiological burden, herbicide residue, and aflatoxins [15].

The use of natural remedies has skyrocketed all over the world. The Secretariat of the Convention on Biological Diversity reports that in 2000, the worldwide market for plant remedies generated an estimated \$60 billion [76]. The worldwide market for plant supplements was valued at around US \$83 billion in 2008, with annual development rates between 3% and 12%. Unfortunately, there are still significant dangers involved with natural remedies [19]. Most adverse outcomes can be traced back to subpar manufacturing standards, some of which can be traced back to tainted plant ingredients. Herbal medicine, as the World Health Organization (WHO) describes, is any substance or product produced from vegetation used to treat or prevent human disease. There are three types of herbal medicines. There are three types of herbal products: (1) raw herbal materials, (2) traditional herbal products, and (3) controlled herbal products. Raw herbal materials include powdered or segmented botanicals, while processed herbal materials include tablets, capsules, and decoctions (formulations containing standardized extracts or purified substances).

Nature has been generous to humans since the beginning of time, meeting our requirements regarding food, housing, medication, and other resources [11]. The decline in ecological resilience and the accompanying increase in susceptibility to shocks and disruptions directly result from human activities such as removing wildlife areas and felling trees and vegetation for their wood. The state of human health is inextricably linked to the state of the world in which we find ourselves; for people to be healthy, they require both environmentally conducive settings and a well-functioning healthcare system that offers eco-friendly, bio-friendly, cost-effective, and reasonably secure therapies [57]. Synthetic medicines have vastly aided in treating or avoiding several acute/chronic/life-threatening illnesses, thanks to humanity's desire and contemporary technical and scientific breakthroughs creating success in finding synthetic medicines. However, contemporary society faces grave danger from the harmful adverse effects of overuse and extended use of these medications. Alarming changes in the structure and function of the environment have occurred recently due to increases in pollution, bad habits, tension, and loss of plant variety. Many people in the modern era prefer to take preventative measures and opt for treatments that pose less risk, leading to the popularity of therapeutic plants in conventional medicine. Herbal products are becoming increasingly prevalent worldwide because of the widespread belief that they are safer than conventional pharmaceuticals because they contain fewer toxic ingredients, have fewer side effects, and work better with the body's natural ecosystem than synthetic chemicals [14, 61]. Traditional medical methods often include using healing herbs as an essential component. Herbal remedies have been used and recorded in the medical practices of ancient India, China, Egypt, Greece, and Rome for at least the past 5000 years, according to the earliest documents available. America, the Arabian Peninsula, and Japan are just a few places where traditional plant therapy has been used since primordial times. The Rigveda, the Ayurveda, the Charak Samhita, and the Sushruta Samhita are all transcriptions of ancient Indian conventional medical practices. The indigenous healthcare system also heavily relies on folk (tribe) remedies. In early times, India was recognized as a hub for collecting and preserving valuable medical plants. Many beneficial and pleasant-smelling flora can be found in

India's woods [12]. Studying medical plants, isolating their valuable components, and conducting chemical screening can lead to discovering novel, clinically active medicines. Codified systems like Ayurveda, Siddha, and Unani practices, as well as traditional medicine of India, are believed to include more than 6000 higher plant species [70]. India is poised for a botanical renaissance thanks to its abundance of valuable plants and ability to satisfy the rising global demand for therapeutic plants. Medicinal plants can be crucial to the expansion and improvement of the economy, and not just in the healthcare sector. Due to the biodiversity loss, many important therapeutic plants are in danger of disappearing [4]. India has a significant opportunity to play a leading role in ensuring the continued survival of plant species and the responsible application of those species' genetic resources in the service of human health and economic prosperity.

2 Plants Used for Medical Purposes for Healthcare Systems

Throughout history, plants have consistently served as a valuable resource for the discovery of new therapeutic compounds, and this trend remains robust, as such they have been variously studied for their pharmacological properties. Due to the ubiquitous nature of microbes, plant with antimicrobial potentials are often studied [28–32, 41–43]. While only a fraction, approximately 5–15%, of the total 250,000 higher plant species have been extensively characterized to date; there's a growing anticipation that many plants hold untapped potential for novel bioactive compounds. This vast reservoir of plant-derived resources has only begun to be tapped into [11].

Herbal remedies and other plant-based products are integral to the healthcare industry, especially in remote and less-developed regions. Contemporary medicine has harnessed various medicinal plant-derived compounds such as reserpine, deserpidine, vincristine, morphine, etoposide, teniposide, nabilone, z-guggulsterone, lectinan, artemisinin, and ginkgolides [49, 71]. These natural products often serve as foundational elements for research in the fields of biology and pharmacology, often leading to the development of synthetic medicines. An estimated 75% of globally used medicines find their origins in traditional remedies [71]. In India, nearly 70% of contemporary medications are derived from natural sources, and many synthetic counterparts have evolved from prototype compounds obtained from plants, with these natural resources forming the basis for most of these synthetic equivalents. This percentage is projected to increase in the future [61].

Despite the wealth of plant species, only a small fraction has undergone comprehensive chemical and pharmacological research, approximately 6% of all identified species. The natural product industry is experiencing remarkable growth, driven by rising demand for products related to weight management and skin health [53]. Within the botanical product market, conditions associated with the cardiovascular system claim a 27% market share, respiratory system-related diseases hold 15.3%, intestinal disorders represent 14.5%, hypnotics and sedatives make up 9.6%, while other conditions collectively account for approximately 13% of the market.

Compared to the lengthy and expensive process of discovering, researching, and developing synthetic medicines, natural medicines have demonstrated the potential to require significantly less time and financial investment [72].

The undesired and, at times, the unpredictability of side effects brought on by the use of synthetic drugs is the primary disadvantage associated with this practice. These unwanted effects can potentially be more harmful than the conditions that synthetic drugs are purported to address. Natural medications are generally more secure and have fewer adverse side effects than synthetic ones. Manufactured drugs, even though some of the drugs derived from plants, such as vincristine, also have some potentially life-threatening side effects. Compared to synthetic medicine, herbal medicine is predicated on the premise that it contains natural substances that can support health and alleviate illness. Keeping the body disease-free and healthy has also been shown to be possible through the consumption of plant-based foods and the use of botanical remedies as nutritional supplements [58, 59]. According to the World Health Organization, most people in underdeveloped countries across Asia and Africa rely on botanical medication as their primary healthcare method. In recent years, there has been a significant increase in industrialized countries using conventional or supplementary alternative treatment. Dietary supplements derived from plant materials are currently becoming increasingly popular. Because of the steep rise in the consumption of herbal supplements and medications in industrialized nations, it is anticipated that the market for natural products is expected to see growth [75].

Herbal medicine incorporates a diverse array of pharmacological compounds derived from plants. These compounds have the potential to address a wide spectrum of health conditions and diseases. The Table 1 highlights various pharmacological compounds commonly found in herbal medicine and their applications in treating specific medical conditions. It's important to emphasize that these are just a few examples of plant-derived pharmacological substances used in herbal medicine. Furthermore, it's imperative to underline that the use of these substances, despite their natural origin, should be supervised by healthcare professionals to ensure safety and efficacy. Additionally, it is essential to be aware that the quality and dosage of herbal products can vary significantly, underscoring the need to source these remedies from reputable and recognized brands and suppliers.

3 The Situation Regarding Herbal Medication Worldwide

The global population is increasingly turning to natural remedies. In particular, multiple investigations have shown a generalized high prevalence [68] in individuals with chronic illnesses like kidney disease. However, the potential for damage from herbal medications is a source of worry. Literature examples emphatically demonstrate the link between herb consumption and renal illness. The need to ensure the safety of natural medications has grown in tandem with their popularity.

The World Health Organization (WHO) has recommended the integration of herbal medicines into established state pharmacovigilance systems [44]. To assess

Table 1 Pharmaceutical compounds derived from plants and the diseases they are used to treat

Compound	Derived from	Medical application
Artemisinin	*Artemisia annua* (sweet wormwood)	Malaria treatment
Morphine	*Papaver somniferum* (opium poppy)	Pain management, especially in cancer patients
Vincristine and Vinblastine	*Catharanthus roseus* (Madagascar periwinkle)	Cancer chemotherapy
Salicylates (Aspirin)	*Salix* spp. (willow bark) and other plants	Pain relief, anti-inflammatory, fever reduction
Quinine	*Cinchona* spp. (cinchona bark)	Malaria treatment
Reserpine	*Rauwolfia serpentina* (Indian snakeroot)	High blood pressure management
Etoposide and Teniposide	*Podophyllum peltatum* (Mayapple)	Cancer chemotherapy
Digoxin	*Digitalis* spp. (foxglove)	Treatment of heart conditions like congestive heart failure and atrial fibrillation
Atropine	*Atropa belladonna* (deadly nightshade)	Pupil dilation and treatment of certain heart and nerve-related conditions
Colchicine	*Colchicum autumnale* (autumn crocus)	Gout management and related inflammatory conditions
Ginkgolides	*Ginkgo biloba* (ginkgo tree)	Cognitive enhancement, particularly in age-related cognitive decline
Paclitaxel (Taxol)	*Taxus brevifolia* (Pacific yew)	Cancer chemotherapy
Cannabidiol (CBD)	*Cannabis sativa* (hemp and marijuana)	Pain management, epilepsy, anxiety, and various neurological conditions
Curcumin	*Curcuma longa* (turmeric)	Anti-inflammatory and antioxidant properties, used in health supplements
Ephedrine	*Ephedra* spp. (ephedra)	Used as a decongestant and for weight loss; restricted due to safety concerns

the risks and benefits of phytotherapy, a process known as phytovigilance, or the pharmacovigilance of medicinal plants, is employed. The ultimate goal is to prevent people from experiencing health issues as a result of using herbal remedies. It is of utmost importance to establish reliable data on the safety of natural medications [5]. Phytovigilance becomes a regulatory necessity when it comes to interactions between closely related plants due to the potential risks of toxicity, both acute and chronic, and medication interactions, whether of pharmacokinetic or pharmacodynamic nature.

India, being one of the 12 major biodiversity zones globally, boasts unparalleled biodiversity thanks to the presence of 16 distinct agro-climatic zones and 10 special agricultural zones. India is unique because it harbors not just one but two "biodiversity hotspots" [66]. In India, there are more than 45,000 plant species, including 15,000–18,000 flowering plant species, 23,000 types of fungi, 2500 types of algae, 1600 types of lichens, 1800 types of bryophytes, and 30 million microorganism species [6, 39, 40, 61]. The Indian subcontinent experiences a wide range of temperatures and environmental conditions, from glacial in the Himalayas to equatorial and humid in the south, and from the desert in Rajasthan to the north-eastern states with diverse topography, including grasslands, plateaus, mountains, and lowlands that accompany the mountains. The favorable environmental conditions, including soil quality, temperature, and humidity, contribute to the rich and diverse plant life found in the Indian subcontinent [69].

India's traditional medical practitioners, numbering over one and a half million, extensively utilize medicinal plants for disease prevention, development, and treatment. Additionally, more than 500,000 non-allopathic practitioners receive their education from one of the over 400 medical institutions to meet the healthcare needs of their respective systems [20, 69, 71]. The Indian subcontinent is renowned for its vast wealth of traditional knowledge and diverse vegetation, and it is crucial, both nationally and internationally, to preserve these resources for the benefit of all. In addition to its role in healthcare, the exchange of medicinal plants serves as a vital source of income for economically disadvantaged communities.

4 Global Trends in the Botanical Medicine Market and India's Role in the Industry

The market for botanical medicines has expanded at a phenomenal rate in recent years as a direct result of the widespread interest in alternative and conventional methods of medical treatment. Because of the profitable nature of the global botanical medicine market, the total value of world commerce in herbs is anticipated to reach USD 7 trillion by the year 2050. India only accounts for a relatively small portion of this global commerce [69]. According to the information that was readily accessible, the valuation of the market for botanical medicines in European countries was approximately USD 6 billion in 1991. This figure increased to USD 10 billion in 1996, which was anticipated to surpass USD 20 billion by 2000. In 1996, the United States contributed approximately 4 million USD to the global market for botanical medicines, while the total contribution from all other countries was 5 billion USD. Between May 1996 and May 1998 in the United States, sales of herbal products in conventional marketplaces increased by 101% [23, 40]. According to a publication that was released not too long ago by the WHO on traditional medicine, the total amount of money spent on traditional medicine in Japan in 2006 was just slightly more than one billion dollars, while the total amount of money spent on botanical medicine in Australia in 2004–2005 was 1.86 billion dollars. In China, traditional pharmaceutical preparations account for between 30% and 50% of the overall medication consumption, corresponding to a selling value of 14 billion USD in 2005 (an increase of more than 28.1% from the previous year). Even though expenditures in the United Kingdom and Brazil in 2007 were comparatively lower, totaling only 230 million and 160 million dollars, respectively [54]. Approximately 40% of all medical treatment delivered in China is in the form of traditional medicine. According to estimates made in 2002, Malaysia spent only about USD

300 million on allopathic medicine yearly while spending approximately USD 500 million annually on traditional and supplementary alternative medicine [75]. According to a different report by the WHO titled "Traditional Medicine," the yearly value of the worldwide conventional medicine business at the time was projected to be USD 83.1 billion in 2008. Additionally, this market is experiencing explosive growth. The annual revenue increase rate generated by sales of products relating to conventional medicine ranges between 5% and 18% [54]. Despite the country's enormous availability of natural resources, there is significantly less interest in using botanical medication in India than elsewhere. India and China are the two countries in Asia that have the largest populations and the most natural components taken together. According to Wakdikar [72], China has approximately 4942 species of therapeutic plants out of a total of 26,092 species, whereas India has only 3000 out of 15,000 plant species. However, compared to China and India, the number of plant varieties found in Indonesia, Malaysia, and Nepal is significantly lower – the total number of plants and therapeutic plants found across the globe. In the rapidly expanding international market for natural medication, India holds only a modest portion (1.6%). The percentage of the world's total output from India increased by 4.95% between 1991 and 2002, while the percentage from China increased by 7.38% during the same period [64, 71]. The Ayush Department of Health and Family Welfare of India reports that in 2009, India was the second biggest producer of therapeutic plants globally, behind China [13].

5 Renal-Related Health Consequences of Botanical and Supplement Usage

Traditional medical practices from various cultures worldwide have harnessed the therapeutic properties of various plants for centuries to address a diverse range of illnesses and conditions. While many of these medicinal plants have proven therapeutic value, it's crucial to recognize that certain plants can adversely affect kidney functions within the human body. The severity of these unintended effects can vary from mild to severe. Therefore, individuals and healthcare professionals must be aware of the potential risks involved. Some medicinal plants known to have detrimental side effects on renal functions.

(i) **Aristolochia Species:** Some plants within the Aristolochia genus, commonly known as birthwort or Dutchman's pipe, contain aristolochic acid. Exposure to aristolochic acid has been linked to developing a serious kidney condition called aristolochic acid nephropathy. This condition can lead to long-lasting kidney damage and, in severe cases, renal failure.

(ii) **Ephedra:** Also known as ma-huang, Ephedra has a history of use in traditional Chinese medicine for respiratory conditions and weight reduction. However, it contains ephedrine, associated with adverse effects like high blood pressure, dehydration, and kidney damage. Consumption of products containing ephedra has been linked to kidney-related health problems.

(iii) **Ginger (*Zingiber officinale*):** While ginger is generally considered safe and offers various health benefits, excessive or prolonged use of ginger supplements may negatively affect the kidneys. Those with pre-existing kidney issues should exercise caution.
(iv) ***Glycyrrhiza glabra* (Licorice):** Licorice root contains glycyrrhizin, linked to high blood pressure, low potassium levels, and potential renal impact. Extended or excessive consumption of licorice can lead to kidney problems.
(v) **Horsetail (*Equisetum arvense*):** Traditionally used as a diuretic, excessive horsetail use can lead to potassium deficiency, potentially affecting kidney function.
(vi) **Cascara Sagrada (*Rhamnus purshiana*):** Known for its laxative properties, prolonged or excessive use of cascara sagrada may result in dehydration, electrolyte imbalances, and potential kidney issues.
(vii) **Aloe Vera:** Topical use of aloe vera is generally considered safe. However, ingestion of aloe vera latex, which has laxative properties, can lead to diarrhea and potassium loss, affecting kidney function.

Other specific medicinal plants with adverse effects on the renal functions are summarized in Table 2. Furthermore, it is essential to note that the adverse consequences associated with these medicinal plants typically arise from excessive or prolonged use. When consumed in safe and moderate amounts, many of these plants may not pose a significant risk to renal function. Nevertheless, exercising caution and seeking advice from a qualified healthcare professional is essential before using medicinal plants, especially if you have a history of kidney problems, are pregnant, or are taking medications.

Additionally, many countries have established regulations and restrictions on the use of specific medicinal plants due to their potential for adverse effects. Therefore, always consult with a healthcare practitioner or herbalist before using any medicinal plant for therapeutic purposes to ensure its safety and effectiveness for your specific circumstances.

6 External Challenges in Herbal Medicine: Contaminants, Adulteration, and Misidentification

External quality challenges in herbal medicine are a significant concern due to their potential risks, including contamination, adulteration, and misidentification, which can have adverse consequences for patients.

6.1 Contamination

Heavy metals, herbicides, bacteria, and mycotoxins are some of the contaminants most likely to be discovered in botanical materials and herbal products; however, this list is not exhaustive [80]. Although the issues are everywhere, they appear more

Table 2 Renal-related health consequences of botanical and supplement usage

S. no	Herb	Country	Gender	Age	Reason	Preparation	Renal side effects	Causality	Reference
1.	*Dioscorea quinqueloba*	Korea	Male	51	Cardiovascular disease	Raw extract	Biopsy-proven acute interstitial nephritis	Certain	Sharma et al. [62]
2.	*Dioscorea quinqueloba*	Korea	Male	58	Health supplements and diabetes mellitus	Extract from tubers	Acute kidney injury	Possible	Muthuswamy and Senthamarai [50]
3.	*Glycyrrhiza glabra*	Serbia	Female	39	Sterility	Take liquorice-containing plant supplements daily for 8 weeks (50–100 g)	Acute renal failure	Probable	Payne et al. [52]
4.	*Chenopodium polyspermum*	Anatolia region, Turkey	Male	46	To regulate his blood glucose level	The medicine should be taken orally thrice in the month leading up to admittance, with the final dosage occurring around 10 days beforehand	Chronic renal failure	Possible	Mann et al. [47]
5.	*African mango (Irvingia gabonensis)*	Turkey	Female	42	Slimming purposes	She began taking two capsules of 500-mg African mango daily 3 months prior	Rapid renal progression	Probable	Payne et al. [52]
6.	*Lawsonia inermis*	Myanmar	Male	34	His face was puffy, and his complexion looked pale	consumed roughly 700 mL/day for 3 days' worth of cooked henna plants	Acute kidney injury	Probable	Mann et al. [47]

(continued)

Table 2 (continued)

S. no	Herb	Country	Gender	Age	Reason	Preparation	Renal side effects	Causality	Reference
7.	*Euphorbia paralias*	Tunisia	Male	28	Edema	Consumption of a single dose of a cooked Euphorbia para las shrub 10 days prior to	Acute renal failure	Possible	Mativandlela et al. [48]
8.	*Ayurvedic powder*	India	Male	32	Eczematous skin lesions	Ingestion of ayurvedic contained arsenic for the last 6 months	Acute kidney injury	Possible	Sivakumar and Jayaraman [65]
9.	*Carica papaya*	India	Male	63	Increase the platelet count	Juice extracted from papaya leaves	Acute kidney injury	Possible	Rehman et al. [55]
10.	*Trigonella foenumgraecum*	Iran	Female	61	Urinary tract infection	Taking a fenugreek seed decoction that has been heated every day	Died of renal failure	Certain	Mann et al. [47]

widespread in countries across Asia, most notably China and India, the most populous regions that manufacture and consume the most natural medications. In addition, contaminations are a significant barrier to the commercial distribution of natural medications.

6.1.1 Heavy Elements

Heavy metals are a recurring issue in natural remedies [1, 2, 37], with the most common culprits being mercury, arsenic, and lead. Contaminants like cadmium, copper, and thallium have also been identified [16–18]. These toxic heavy metals are frequently detected in traditional Asian medicines, particularly Chinese patent medicines (CPMs) and Indian Ayurvedic medicines (IAMs). Both are abbreviated as [CPMs] and [IAMs] [51].

Contamination with heavy metals can occur due to several factors:

1. **Environmental Accumulation:** This can result from heavy metals accumulating in the environment, such as contaminated soil or air.
2. **Accidental Pollution:** During the manufacturing process, accidental contamination can occur.
3. **Intentional Inclusion:** Some traditional medical systems intentionally use minerals, including toxic metals, in the preparation of remedies. For instance, certain cosmetic formulations require the use of substances like Zhusha (Cinnabaris) and Xionghuang (Realgar), which contain mercury and arsenic. Realgar is also known as Xionghuang [84].

It's important to note that labeling this intentional inclusion of heavy metals as "contamination" is inaccurate. This practice is based on outdated beliefs and doesn't align with modern scientific research, as there is no evidence to suggest that the benefits of heavy metals outweigh the associated health risks [46, 73]. Stricter quality control measures in industrialized countries reflect the need to prevent such practices and ensure the safety of natural medicines.

6.1.2 Pesticides

Pesticides, which encompass insecticides, fungicides, and herbicides, can have far-reaching implications for the quality and safety of botanical medications. These chemicals and their byproducts can persist in plants, animals, and the environment where they were used, posing a significant contamination risk to herbal products. To address this concern, the World Health Organization (WHO) and other organizations have established guidelines to limit the presence of pesticides in botanical products [76].

Organochlorine pesticides (OCPs), such as benzene hexachlorides (BHC), pentachloronitrobenzene, and dichloro-diphenyl-trichloroethane (DDT), are often identified in natural medications (PCNB). Despite being banned in many countries for approximately three decades due to their detrimental effects on human health, OCPs can persist in the environment and accumulate in the food chain due to their slow degradation. However, they are less common than other types of insecticides, particularly those containing organophosphorus and carbamate compounds, which are frequently detected in natural medications. Some naturally occurring substances have even been found to contain traces of more modern chemicals like fenvalerate and deltamethrin.

Modern agriculture extensively employs pyrethroid herbicides due to their high effectiveness and low toxicity to humans and the environment [9]. Research in China has shed light on the issue of herbicide residues in medicinal materials, affecting both domestic and international sources, including plants like *Panax notoginseng*, *Panax quinquefolium*, and *Panax ginseng*, among others [45]. These studies identified five types of pyrethroid insecticides, with 14.8% of samples exceeding permissible limits, and revealed more significant damage to above-ground parts compared to roots [79]. OCPs, consisting of nine different types, were found to exceed permissible limits in 7.09% of samples, with roots being more affected than above-ground parts [83].

Furthermore, contamination can result from the improper application of pesticides used for soil disinfection (e.g., PCNB) and excessive or untimely application close to harvest. The use of insecticides during storage and fumigants like sulfur dioxide to prevent insect or fungal infestations can also contribute to residue contamination [84]. The environmental consequences of OCPs leaching into the soil further compound these challenges.

6.1.3 Microorganisms as well as Mycotoxins

The risk of microbial contamination in natural medications is a prevalent concern. Botanical compounds have been found to provide a conducive environment for pathogenic microorganisms, including enterobacter, enterococcus, shigella, and streptococcus [8]. Mycotoxins, such as fusarium toxin, ochratoxin, penicillic acid, citreoviridin, and particularly aflatoxins, pose significant health hazards and can be present in considerable quantities.

Studies have indicated that over half of the samples of medicinal plants sourced from a Brazilian market exceeded the microbial contamination limits set by the United States Pharmacopeia (USP). Similar findings from various research have consistently highlighted issues of fungal contamination and mycotoxins in natural medications, not only in India but also in countries like China, South Africa [67], Malaysia [3], and Indonesia.

The risk of microbial contamination can occur at various stages of the manufacturing and distribution processes of herbal medicines. Quality control is particularly affected by storage and manufacturing conditions. High temperatures and elevated moisture levels, prevalent in tropical and subtropical regions, create an environment conducive to fungal growth and mycotoxin production [56].

6.1.4 Other Unidentified Foreign Material

Additional forms of foreign matter that can contribute to the external contamination of botanical medications include ash, adjuvants, and organic solvents [22]. To maintain a high level of quality in the final product, minimizing the impact of these contaminants is crucial.

Adulteration:

1. Can involve replacing authentic plant materials with substandard or false alternatives, further undermining product quality.
2. **Incorporation of Foreign Materials:** The inclusion of foreign substances, such as non-officinal herb parts, sand, or metals, can also occur.

There have been documented instances of pharmaceuticals being discovered in herbal medications, with reports of adulteration rates estimated at 1.24% in Singapore, 7.1% in New York, and 7% in California [81]. More recently, sildenafil, the active ingredient in Viagra, has been found in products labeled as tonics.

Adulteration is often motivated by financial gain, as unscrupulous parties fraudulently seek to increase the weight or abundance of botanical materials. These deceptive practices can occur at various stages of the manufacturing process, from the early production steps to later stages.

It's important to note that while including lawful, manufactured medications within natural remedies is not classified as adulteration, it is illegal to market herbal medications containing undisclosed conventional pharmaceuticals. Adequate supervision and control are essential to prevent the mislabeling and adulteration of herbal products.

Addressing the issue of adulteration requires the implementation of rigorous regulatory measures. For these measures to be effective, they must be enforced and adhered to consistently. Stringent oversight and regulation are essential to maintain the quality and safety of herbal remedies.

6.2 Misidentification

In contrast to deliberate acts like adulteration or displacement, typically arises from unintended errors. It occurs when a supplier or merchant mistakenly confuses one botanical material for another, often due to incorrect labeling and a superficial resemblance between various plant specimens [10].

Several factors can contribute to such misidentification. One primary cause is the often confusing nomenclature of plants. A single plant may have multiple names, including common names, Latin-derived names, terms from original languages, and brand-specific names. Furthermore, different medicinal plants from distinct species, each containing various therapeutic components, may share the same or similar titles. This situation leads to the terms "Mutong" and "Fangji" being applied to multiple plant species originating from various biological families that are not related.

The consequences of misidentification can be severe. For instance, the dangerous condition known as aristolochic acid nephropathy has resulted from the improper identification of "Mutong," wherein Caulis akebiae was mistakenly substituted for Caulis aristolochiae manshuriensis. The complexities of plant identification are exacerbated by factors such as the visual resemblance of plants, ambiguities in historical records, and variations in regional usage patterns. The use of perplexing terminologies and linguistic differences across countries further complicates the matter.

In many cases, the common names of plants do not accurately reflect their scientific classifications. The physical description of a plant or its examination under a microscope may not always reveal the specific components relevant to herbal medicine. As a result, the accurate authentication of botanical materials often necessitates a combination of historical research and the application of contemporary scientific methods.

Efforts are being made by organizations like the World Health Organization (WHO) and others to standardize and harmonize biological nomenclature to mitigate misidentification issues in the herbal and botanical product industry [10].

7 Excellent Facility for the Stockpiling of Plants

The storage facility should be equipped to accommodate plants designated as “Under test,” “Approved,” and “Rejected.” It must also provide the necessary infrastructure for washing, drying, and orderly storage of herbs, along with climate control to regulate temperature and humidity. Compliance with Good Manufacturing Practice (GMP) regulations is essential in the storage area, covering aspects such as plant handling, drying space allocation, and overall facility management.

For storage areas falling under GMP regulations, several requirements must be met. These include adequate air circulation, moisture control, and sufficient space for various types of materials. In line with GMP guidelines (A), the following list identifies the types of storage suitable for various herb-related products:

- Fresh plants
- Dried herbs
- Concentrates and plant material preparations

These measures are essential to ensure that the storage of herbs and related products complies with GMP standards and maintains their quality.

8 Quality Control and Proper Storage of Fresh Medicinal Plants

8.1 Processes Involved in the Correct Storing of Fresh Plants

The process involves several key steps:

(i) **Herb Identification:** Initially, herbs are identified using human senses. This includes assessing their appearance, aroma, and other sensory characteristics to ensure they match the desired specifications.

(ii) **Cleaning of Herbs:** Herbs can include various plant parts like flowers, vegetables, and other components. Cleaning involves washing unprocessed drug materials, such as stems, foliage, roots, leaves, and stalks, with potable water to remove impurities and contaminants.

(iii) **Drying of Unprocessed Plants:** After cleaning, uncooked plants are dried in their basic form. This step helps preserve the herbs and prevents moisture-related issues.

(iv) **Microorganism Elimination:** To ensure the safety of herbal products for internal use, it's crucial to consider bacterial limits and the risk of mold contamination, similar to food products. Many herbal products undergo sterilization, typically using ethylene oxide and specialized equipment. It's important to note that disinfection kills bacteria but not their spores, while sterilization eradicates all forms of bacterial life, including spores.

(v) **Sterility Testing:** To confirm that a product is sterile, it must be entirely free of any living organisms. Sterility testing is performed to ensure that the product meets these criteria.

These steps are vital in the preparation and quality control of herbal products, guaranteeing their safety and effectiveness for consumers.

8.2 Process of Sterilizing Fresh Plants

Raw plants intended for preservation can undergo sterilization using several methods, depending on the specific requirements of the situation. These sterilization techniques include:

(i) **Dry Heat Sterilization:** Dry heat sterilization is the most commonly used method. It involves the use of high temperatures to eliminate microorganisms and other contaminants from raw herbal materials [25].

(ii) **Steam Sterilization:** This method employs steam to achieve sterilization.

(iii) **Radiation Sterilization:** Radiation-based methods like infrared radiation, UV rays, X-rays, gamma rays, alpha radiations, and beta radiation can be used for sterilization.

(iv) **Chemical Sterilization:** Gaseous sterilization, such as using ethylene oxide, is another effective technique to ensure sterilization.

Raw herbal medicines should be stored at a specific temperature range between 80 °C and 250 °C. Protecting them from environmental factors like moisture, freezing temperatures, excessive heat, and light is crucial. These precautions are essential to prevent the deterioration of raw plants during storage.

8.3 Evaluation of High-Grade Unprocessed Plants

Following the guidelines set forth by the WHO, testing of fresh plants is essential and involves a series of crucial assessments:

(i) **Authentication:** Taxonomic research and botanical identification are conducted to verify the plant's species and origin.
(ii) **Contamination from Foreign Matter:** Herbs must be free of contaminants like dirt, stones, pollen, insects, or other foreign materials.
(iii) **Organoleptic Assessment:** This assessment considers the macroscopic appearance of the substance, including its odor, flavor, texture, color, and overall sensory characteristics.
(iv) **Microscopic Investigation:** Microscopic analysis provides valuable data for species identification, including details such as stomata type and artery thickenings.
(v) **Extraction of Volatile Substances:** The extraction of volatile oils, like in species such as mentha and ocimum, through techniques like steam distillation, is essential for quality control assessment.
(vi) **Ash Content:** The ash content in a medicinal plant can be measured as Total Ash, Acid Insoluble Ash, and more. The presence of insoluble acid in water suggests contamination with organic substances.
(vii) **Extractive Values:** Extractive values offer additional insights into the quality of raw plant materials.
(viii) **Chromatographic Profile and Marker Component:** Techniques like Thin Layer Chromatography and High-Performance Liquid Chromatography (HPTLC) are used to determine the chromatographic profile and identify marker components in the substance.
(ix) **Pharmacological Evaluation:** This assessment involves determining the foaming index, hemolytic activity, and bitterness value of the plant material, considering its pharmacological aspects.
(x) **Residues of Toxicological Pesticides:** Pesticides are used to reduce insect, fungi, and mold prevalence in fresh plants. It's essential to analyze for toxicological pesticide residues, especially if the medicinal plants are consumed over an extended period.
(xi) **Microbial Contamination:** Analysis for the presence of bacteria, molds, and aflatoxins is carried out to ensure the plant material's safety and quality.

These assessments collectively ensure the quality, safety, and authenticity of fresh plants used in medicinal applications.

8.4 Testing of Fresh Plants

Testing fresh plants serves various purposes, including morphological examination, microscopic evaluation, identification testing, purity and strength assessment, thin-layer chromatography (TLC), and constituent testing. To maintain the quality, safety, and potency of plant materials, adherence to proper handling, packaging, and storage is crucial.

Ensuring the correct storage of raw botanical substances is paramount, despite the absence of specific recommendations to date. Various environmental factors like oxygen, moisture, sunlight, and pollen can adversely affect plant materials and should be minimized or avoided. The ideal storage area should be dry, free from moisture, humidity, and protected from intrusions by animals, birds, insects, and similar factors.

Furthermore, distinct storage areas are necessary for different classifications of medicinal and aromatic plants, including fresh herbs, dried herbs, extracts, gums, essential oils, and potentially harmful substances, as outlined in the Drug and Cosmetic Act of 1940.

To facilitate efficiency, separate sections should be designated for items “Being Tested,” “Approved,” and “Rejected.” It’s essential to avoid storing plants that appear nearly identical in the same area to prevent potential misidentification.

Each batch of stored plants should bear an identification label, including details such as the test result date, consignment number, arrival date, best-before date, batch number, and storage location.

Authenticated reference samples must be maintained within the facility for reference. Additionally, to ensure proper storage, plants should be kept in containers placed on hardwood or plastic shelves, with each container housing a single plant. The appropriate choice of container material is crucial.

It’s crucial to avoid certain practices, such as storing plants in exposed areas, placing them immediately on surfaces, storing them near similar drugs, using inappropriate container materials, mixing authorized and unverified quantities, and storing the substances for extended periods at abnormal temperature and humidity levels. Proper storage practices are vital to maintain the integrity and quality of plant materials.

9 Strategies for Dealing with Exterior Obstacles

Environmental concerns related to natural medicinal products encompass cultivation, production, and distribution. Strict adherence to “Good Agricultural and Collection Practices” (GACP), often referred to as “Good Practices,” is a critical step in elevating the overall quality of herbal medicines. In 2003, the World Health Organization (WHO) recommended the adoption of GACP for medicinal plants. This standard has been adopted at the national and regional levels in various countries and regions, including the European Union, China, and Japan. In addition to adhering to GACP, the manufacturing of herbal products must follow the Generally Accepted Manufacturing Procedures (GMP) [77].

Furthermore, the distribution of natural remedies should be carried out following Good Supply Practices (GSP). Between 2004 and 2009, Wang et al. [73] certified various national GACP bases in China, which are specific locations for cultivating botanical medicines in compliance with GACP standards. These bases were established across China, totaling 430 municipal GACP sites covering an extensive cultivation area of approximately 11,000 km^2 [84]. The implementation of GACP has notably improved the quality of medicinal ingredients used in Chinese medicine.

However, it's essential to acknowledge that the journey toward this achievement has not been without its challenges.

Sustained issues persist due to the inadequate application of scientifically sound standard operating procedures (SOPs), standardized management practices, well-trained farming personnel, and other relevant experts [82]. The effective implementation of these principles, such as GACP and GMP, is vital for ongoing success. Companies specializing in herbal products are at the forefront of advocating for the high quality of natural remedies. Manufacturers of herbal products must provide assurances of the quality of their products to comply with national regulations and ensure the safety and effectiveness of natural medicinal products.

10 Obstacles to Quality and Regulations

The extensive use of botanical medications for disease prevention, detection, treatment, and management presents significant global challenges in terms of product control and relevant regulations. The quality, safety, and effectiveness of these substances are of paramount concern due to their widespread and increasing utilization in healthcare. To advance conventional medical practices, scientific validation and technological standardization of botanical medicine are indispensable.

The application of products adhering to established quality assurance standards holds the potential to significantly reduce the risks associated with natural medicine. However, herbal medicine regulations and legislation have been implemented in only a few countries, leaving a majority without adequate oversight of botanicals, leading to a lack of guaranteed quality in herbal products despite existing regulations [74, 75].

Recently, several European countries adopted the European Directive on Traditional Herbal Medicinal Products, which aims to regulate and ensure the quality of herbal products. In the United Kingdom, the Medicines and Healthcare Products Regulatory Agency (MHRA), formed in April 2003 through the merger of the Medicines Control Agency and the Medical Devices Agency, is responsible for enforcing this directive, ensuring the quality and safety of all medicines.

Given the absence of such regulations in many countries, particularly in parts of Africa and Asia, the establishment of global regulations is urgently required. While extensive research is conducted on natural products, only a few clinical studies yield satisfactory findings that can be integrated into the conventional medical system. Research into herbal products remains notably deficient, with clinical studies for isolated active phytoconstituents being few, having small sample sizes, and insufficient controls.

The evaluation of botanical remedies' efficacy is inherently challenging, primarily due to the accurate identification of plants and the isolation of their active components. Medicinal plants' characteristics can vary significantly over time, and a single plant may contain hundreds of naturally occurring components depending on factors like harvesting, geographical origin, and environmental conditions. This complex process necessitates prompt action on various fronts, including legislation and supervision for botanical products, treatment practices, education, training, practitioner licensing, research and development, and the allocation of financial and other resources [40, 75].

Several critical obstacles need to be overcome to promote the development of herbal medicine:

1. **Education and Training:** The first challenge is to ensure that both conventional and alternative medicine practitioners possess the necessary knowledge, certifications, and training. Educational programs are required to help these practitioners recognize and appreciate the complementary nature of their respective practices.
2. **Protection of Indigenous Knowledge:** It is imperative to address issues related to the protection of indigenous traditional knowledge and the proper use of this knowledge. Many traditional herbal medicine systems lack robust scientific evidence of their efficacy, and concerns about intellectual property need to be resolved to prevent the theft of indigenous natural resources and traditional knowledge to produce traditional medicines [75].
3. **Sustainable Plant Harvesting:** Sustainable harvesting of medicinal plants is essential to prevent their depletion. Many therapeutic plants are sourced from natural resources, with unsustainable harvesting practices being a primary cause of their destruction. Over 70% of plant gathering is detrimental due to the extraction of parts or the entire plant, leading to plant eradication, which is particularly problematic for species with low populations. To preserve the genetic diversity and complexity of plants used for therapeutic purposes, sustainable harvesting practices are imperative [61].
4. **Regulations and Standards:** Establishing regulations and standards for inexpensive pharmaceuticals and herbal medicines is another hurdle that must be addressed. Inadequate legislation, underdeveloped infrastructure for small-scale businesses, and a lack of stringent quality standards for botanical substances and compositions hinder the standardization of herbal medicine [71].
5. **Good Manufacturing Practices (GMP):** Implementing GMP is crucial to ensure that herbal medicines meet the highest quality standards. Good Manufacturing Practices are designed to guarantee the quality of natural medicines, and their adoption is an immediate necessity.
6. **Clinical Studies:** Rigorous clinical studies demonstrating the safety and efficacy of herbal treatments, particularly when combined with allopathic pharmaceuticals, are essential. These studies should employ randomized, double-blind, placebo-controlled methodologies to meet the high standards expected by the public. Conducting systematic clinical studies will open up new opportunities for fundamental and applied research on botanicals and Ayurvedic treatments in India.
7. **Pharmacopoeias:** The development and expansion of pharmacopoeias is vital. In India, GMP was introduced for Ayurvedic, Siddha, and Unani medicines in 2000. The Indian Pharmacopoeia also included many books on flora and medicinal products. The 2010 edition featured 89 such books, compared to 59 in 2007. Additionally, the Ayurvedic pharmacopoeia documented 258 medications in

2005, and the Indian Herbal pharmacopoeia detailed 52 different medicinal plants, both of which are Indian publications [24, 74].

Addressing these challenges is essential to ensure the growth and acceptance of herbal medicine and to provide safe, effective, and standardized treatments to the public.

11 Initiatives Were Taken in the Field of Natural Medicine

In India, there is an urgent need to expand research in agro-pharmaceutics, plant biotechnology, and the pharmaceutical industry to boost the production of medicinal plants and their active constituents by harnessing modern genetic and biotechnological techniques. Despite numerous institutions and colleges across India, research in this field is notably insufficient compared to countries such as Mainland China, Tokyo, Europe, and Jerusalem. To address this shortfall, it’s imperative to encourage the active involvement of Agricultural, Medical, and Pharmaceutical schools, along with botanical research organizations, in conducting fundamental research.

Another significant challenge hindering the global expansion of Indian traditional medicine is biopiracy. Developed countries tend to dominate the patenting of products, putting less developed nations like India at a disadvantage [39]. Moreover, the degradation of biodiversity, excessive commercialization, and industrialization pose obstacles to the growth of natural medicine. In contemporary times, the reluctance or inability of people to apply traditional knowledge inherited from their ancestors regarding plants, medicines, and remedies contributes to the scarcity of educated individuals in the field of alternative and complementary health [70]. Consequently, the development of Indian traditional medicine requires comprehensive documentation of traditional knowledge, patenting efforts, and the continuation of previous research efforts.

Key priorities for this sector include standardizing drug quality, increasing the supply of medicinal plant materials, conducting further research, developing new treatments, and raising awareness about the effectiveness of traditional systems. During the government’s 10th five-year plan, the AYUSH Department received approximately 158 million US dollars (about 775 billion Indian rupees) in funding. In November 2000, the Government of India established the National Medicinal Plants Board (NMPB) to address issues related to medicinal plants and promote their trade, conservation, and sustainable production. Since then, the NMPB has been diligently pursuing these objectives, developing various programs and initiatives during the 11th 5-year plan (2007–2012) to protect rare, endangered, and threatened plant species while promoting the sustainable management of medicinal plants.

Government initiatives and programs in India allocated USD 65.21 million (INR 320 crores) for addressing various issues related to patents and intellectual property rights [20]. An essential strategy to combat these problems involves providing financial support for research aimed at documenting indigenous knowledge.

Concerns about biopiracy have driven India's efforts to safeguard its traditional knowledge, particularly in light of several instances.

The Traditional Knowledge Digital Library (TKDL) is a pioneering effort in India to tackle these issues. Established in 2001 as a collaborative project between the Council of Scientific and Industrial Research of India and the Department of AYUSH of the Government of India, the TKDL is the first of its kind. It is dedicated to archiving traditional knowledge that is already part of the public domain, particularly information related to Ayurveda, Unani, and Siddha systems of medicine. The TKDL contains approximately 212,000 different medicinal formulations, including 82,900 from Ayurveda, 115,300 from Unani, and 12,950 from Siddha.

By September 2010, the TKDL database reported that the European Patent Office (EPO) had revoked 23 patent applications and set aside three patents. In addition, the EPO withdrew three patents in the same period. Consequently, the number of patent applications for Indian medical practices at the European Patent Office saw a significant reduction of 44%. This suggests that the TKDL has been effective in protecting against biopiracy.

Furthermore, India has implemented a global biopiracy monitoring system within the TKDL to scrutinize patent applications associated with Indian medical practices [21, 63].

12 Challenges Associated with the Medicinal Plants Practice

Working with medicinal plants presents unique challenges that must be understood and addressed to guarantee the safety and efficacy of herbal therapies. Table 3 offers a summary of the most significant issues associated with the use of medicinal plants. To address these challenges, collaboration between researchers, healthcare professionals, traditional practitioners, and government officials will be essential.

Balancing the preservation of traditional knowledge with its integration into modern healthcare systems is crucial. Simultaneously, it is imperative to ensure that medicinal plants are both safe and effective while remaining readily accessible for the benefit of all individuals.

13 The Function and Its Prospects for the Future

The demand and development of natural medications are set to continue, ensuring a prosperous future. This ongoing internationalization of medicinal herbs and ethnobotanical research will lead to the discovery and utilization of novel plant products for various applications, including construction, medicine, and industry [38, 57]. Notably, the expansion of natural medicine holds significant economic potential for India. Consumers must have access to untampered products, highlighting the need for quality and safety assurance in herbal remedies.

The proliferation of botanical medicine has provided an alternative healthcare system, essential for enhancing global healthcare accessibility. The delayed

Table 3 Challenges associated with medicinal plant practice

Challenge	Description
1. Quality control	Ensuring consistent quality of medicinal plants despite variations in climate, soil, and harvesting methods
2. Standardization	Standardizing herbal remedies for consistent dosages of active ingredients, critical for clinical efficacy
3. Regulation and quality assurance	Herbal products often have less stringent regulation, posing risks due to lack of standardization and control
4. Authentication and identification	Correctly identifying and authenticating plant species to prevent misidentification and ineffective treatments
5. Sustainability	Preventing environmental degradation and plant species endangerment due to overharvesting
6. Cultural and ethical considerations	Integrating traditional cultural and ethical aspects into modern healthcare can be challenging
7. Research and evidence	The need for rigorous scientific research to validate the safety and efficacy of medicinal plants
8. Biopiracy	Protecting indigenous knowledge and ensuring fair benefit-sharing to counter unauthorized commercial use
9. Globalization	Challenges arising from global trade, regulation, and standardization discrepancies between countries
10. Education and training	Building a skilled workforce knowledgeable about medicinal plants through training and education
11. Patient safety	Ensuring practitioners are aware of potential interactions between herbal and conventional medications
12. Access and affordability	Balancing the accessibility and affordability of herbal remedies for all socioeconomic groups

awareness of declining medicinal plant supplies is attributed to the increasing global demand for herbal remedies. India's diverse environmental conditions have endowed it with a wealth of therapeutic plants, making it a leader in this field [63]. It's imperative to combine modern knowledge with traditional wisdom.

Culturing profitable medicinal plants in neglected areas can boost national economies, as seen in North Eastern India [60]. The government should promote private cultivation of these plants to ensure a sustainable supply, ultimately benefiting the economy's long-term health and creating employment opportunities in remote regions. Policies and planning strategies must align with the economic forecast to fully unlock this sector's potential [40, 63].

In today's world, harnessing the potential of traditional botanical medicines and integrating them into public healthcare systems is crucial. India's abundant therapeutic plants, in high demand in international markets, should be explored further in areas like Agrotechnology, Natural Products, Quality Assurance, and Control of homoeopathic medicines. Understanding the socioeconomic context and supporting research are essential for the industry's sustainable growth.

Preserving the biodiversity of therapeutic plants and preventing biopiracy are critical factors for the industry's expansion. The time has come to compile and document traditional wisdom about our invaluable plant resources and conduct comprehensive research into their phytochemical, biological, and pharmacological properties to demonstrate their effectiveness.

14 Conclusion

This chapter primarily emphasizes the quality concerns associated with Eastern medicinal medications. Strict adherence to Good Agricultural and Collection Practices (GACP) and Good Manufacturing Practices (GMP) can significantly reduce the risks of contamination, adulteration, and other issues. By incorporating contemporary scientific methods and techniques from the pharmaceutical industry, it becomes possible to produce standardized botanical products that are easily manageable in terms of quality. However, it's important to note that there has been limited success in achieving consistent quality control for natural medications.

To achieve an overall improvement in the quality of these products, it is essential to prioritize two key areas: enhancing regulatory measures and conducting additional methodological studies. Government organizations should play a crucial role in providing active guidance and effective legislation in this regard.

Furthermore, utilizing specific packaging materials tailored to different parts of medicinal herbs allows for controlled storage conditions, including temperature, pressure, air, and light. Proper labeling should be a part of this provision to ensure the preservation of the herbs' active constituents during storage in various facilities. This approach not only safeguards the efficacy, safety, and quality of Ayurvedic medicines but also taps into the substantial economic opportunities presented by the expanding pharmaceutical market in our country.

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Herbal Medicine Formulation, Standardization, and Commercialization Challenges and Sustainable Strategies for Improvement

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Contents

Abstract

Herbal or Phyto medicines have been used since time immemorial as a form of traditional medicine systems and practices like Ayurveda, Siddha, Unani, and homeopathy. This medical system uses herbal medicines, or phytomedicines that are extracted from herbal plants, for the treatment of diverse diseases. People in developing countries such as Asia and Africa rely on herbal medicines for the treatment of various diseases the most because it is believed that they are cheaper, more effective, and risk-free. Herbal medicines are preferred over modern med-

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icines in developing countries. However, herbal medicines suffer from a lack of scientific data on their effectiveness and adverse effects. But for the treatment of some serious disease conditions, phytomedicine formulation must be standardized and developed. Following the COVID-19 pandemic, the demand for herbal medicines has grown and commercialization challenges for herbal medicines are expected to also increase. Nonetheless, efforts should be made to increase the commercialization of herbal medicines by searching for their formulations, development, and methodology so that medicines can be made available for addressing serious diseases. This chapter focuses on some of these challenges and how to improve them.

Keywords

Phytomedicines · Herbal formulation · Standardization · Commercialization · Treatment

Abbreviations

BMS	Bristol Meyers Squibb
CAM	Complementary or alternative medicines
CDDS	Colloidal drug delivery systems
DCGI	The Drug Controller General of India
DSHEA	Dietary Supplement Health and Education Act
FDA	Food drug administration
GAP	Good Agricultural Practice
GCP	Good Clinical Trial Practice
GLP	Good Laboratory Practice
GMP	Good Manufacturing Practice
GSP	Good Sourcing Practice
HDS	Herbal and dietary supplements
HM	Herbal medicine
HMPs	Herbal medicinal products
ICMR	Indian Council of Medical Research
ICSBD	Indonesian Country Study on Biodiversity
NPs	Natural products
TCM	Traditional Chinese medicine
TM	Traditional medicine
WHO	World Health Organization

1 Introduction

Herbal medicine has been regarded as a promising future medicine for the management of healthcare in the twenty-first century due to its increased pharmacological efficacy. The worldwide trend toward herbal medicine, which is known as a “Return to Nature,” has recently changed. Since ancient times, medicinal plants have been used to treat a variety of ailments and diseases, and they are widely regarded as a rich source of therapeutic agents [1].

In industrialized nations, herbal medicine sales have skyrocketed over the past 10 years. Some of the conditions for which herbal medicines are increasingly being used, i.e., the emergence of new, severe diseases, include insomnia, anxiety, obesity, bronchial asthma, congestion, gingivitis, Vincent's disease, eczema, varicosity, and immunodeficiency syndrome. Other conditions include (i) varicosity and immunodeficiency syndrome; (ii) problems for which there is currently no effective treatment and the belief that herbal remedies are safer than pharmaceuticals; (iii) the idea that only good things can come from nature; (iv) the specific spotlight on the climate. In Western nations, movements promote herbal remedies and the belief that natural herbal remedies are superior to synthetic ones. The advancements made in the production and preservation of herbal medicines are another factor. Today, medicinal plants that are rich in desirable active compounds can be introduced to the market thanks to the possibility of cultivating genetically modified varieties [2]. One percent of the estimated 25,000 to 75,000 plant species used in traditional medicine are recognized by scientists and accepted for commercial use [3]. Traditional medicine, as opposed to conventional "Western" drug therapy, is still the only health service that most people in many developing countries can afford. For people who live in these areas, traditional medicine is the first line of defense and basic healthcare [4]. The utilization of plants, portions of plants, and secluded phytochemicals for the counteraction and treatment of different well-being diseases have been practically speaking from days of yore. One hundred twenty-one such active compounds are in use, accounting for approximately 25% of all prescribed medications worldwide [5]. However, conventional medicines have largely influenced the development of the current pharmacovigilance model, its activities, and associated tools. Due to the particular characteristics of herbal medicines as well as the various ways in which herbal medicines are regulated, used, and perceived, applying these methods to monitor the safety of herbal medicines presents additional challenges [6]. At several different levels, people are becoming more aware of how important it is to develop pharmacovigilance procedures for herbal medicines. The current pharmacovigilance model and the tools that go along with it were made for synthetic drugs. Using these methods to monitor the safety of herbal medicines is different from using them to monitor the safety of conventional medicines [7]. Traditional Chinese medicine (TCM) was used in 91.50% of COVID-19 cases in China, showing encouraging results in improving symptom management and reducing the rates of deterioration, mortality, and recurrence, as the coronavirus disease 2019 (COVID-19) pandemic poses a significant threat to global public health. For the treatment of COVID-19 in China, a total of 166 modified herbal formulas containing 179 individual herbal medicines were gathered [8].

Atropine, colchicine, digoxin, taxol, and vincristine are just a few of the isolated plant constituents that have made their way into the prescription drug inventory. Based on the depth that pharmacognosy brought to pharmacy curricula in the past, the significance of plant-derived medicine (phytomedicine) is obvious. However, just being familiar with chemistry and botany does not always translate into a comprehension of clinical pharmacology and therapeutics. More and more clinical courses and continuing education programs are including herbal medicine, and pharmacists are now able to access additional published sources of information [8].

Traditional medicine, as opposed to conventional, “Western” drug therapy, is still the only health service that can be afforded by the majority of people in many developing countries. Traditional medicine serves as a first line and basic health service for people who live in these areas. In industrialized countries, the situation is quite different because modern medicine’s services are readily available. However, traditional medicine has gained popularity in recent years and is now one of many important options for healthcare. Patients may be looking for less invasive alternatives to healthcare in this consumer-aware era. The revival of traditional medicine’s interest and popularity also follows its promising potential for treating certain diseases, such as malaria and chronic eczema [9].

Due to an epidemic like Corona, people had to take care of their traditional herbal remedies at that time. As a consequence of this, people remembered the Ayurvedic, Unani, and Chinese medicine systems, which in turn brought to mind herbs and herbal remedies, so this book chapter aims to achieve the development of herbs, herbal medicines, and herbal preparations with the regulations of herbal medicines in developing countries, its uses and commercialization by the evaluation of herbal preparations.

2 Herbal Medicines: An Overview

The creation of new Pharmaceuticals relies heavily on medicinal plants. The World Health Organization says that plants used in traditional medicine have been the source of nearly 25% of modern medicines [10]. The practice of using a plant’s seeds, berries, roots, leaves, bark, or flowers for therapeutic purposes is known as herbal medicine, botanical medicine, or phytotherapy. How Plant-based medicines are used in different countries in very different ways. Herbal medicine can be considered either a traditional or a complementary treatment depending on where and how it is used. Because the use of herbal medicines is frequently intertwined with local customs and beliefs about the workings of the body and the nature of health and disease, herbal medicines are an important part of the majority of indigenous forms of medicine, which are called ethnomedicine [4]. In many developing nations, herbal medicines are the most common form of treatment for a wide range of conditions. Currently, herbal products can only be sold as food supplements in the United States, and the FDA (Food drug administration) requires FDA approval before a manufacturer or distributor of herbs can make any specific health claims. The Dietary Supplement Health and Education Act (DSHEA) of 1994 regulates herbal medicines as dietary supplements, so they are exempt from the premarketing regulatory clearance that is required for drugs. The Food and Drug Administration of the United States bears the burden of proving that a dietary supplement is unsafe. In contrast, pharmaceutical drugs cannot be approved until their safety and effectiveness have been demonstrated by the manufacturer [11]. It was reported that natural products are the basis for more than 60% of cancer drugs currently on the market or in testing. Presently, around 80% of antimicrobial, immunosuppressive, cardiovascular, and anticancer medications are gotten from plant sources. Among the 177 approved anticancer drugs, more than 70% are based on natural products or mimics. Nearly 121 plant-based prescription medications are in use worldwide, accounting for approximately 25% of all prescription medications. Between 2005 and 2007, 13 natural product-based drugs were approved in the United States, and over 100 natural product-based drugs are currently undergoing clinical trials. Additionally, it was estimated that 11% of the 252 medicines on the WHO essential medicine list are not of plant origin [12]. Products derived from plants became the most popular option for biological and Pharmacological research, as well as the starting point for the creation of synthetic drugs. It is estimated that indigenous medicine was used to make up nearly 75% of all herbal medicines used today [13]. The current pharmacopeia, which is based on the pharmaceutical market, was inspired in large part by herbal medicines. Due to their lack of side effects and healthier alternatives for patients, herbal medicines are preferred over modern ones. The majority of the world's population, 85% used herbal medicines to treat skin conditions like viral, fungal, diabetic, hypersensitive, and other conditions. However, despite their appropriate pharmacological activity, they are less utilized in medical practices for a variety of reasons, including difficulties with solubility, bioavailability, and the need for high doses, among others [14]. Herbs, herbal materials, herbal preparations, and finished herbal products are included in the WHO definition of herbal medicines. Herbs are plant parts like leaves, flowers, fruits, seeds, stems, wood, bark, roots, rhizomes, and other parts that can be whole, broken up, or ground up into a powder. Herbal materials are either whole medicinal plants or crude parts of plants. Fresh juices, gums, herbs, fixed oils, essential oils, resins, and dry herb powders are all included [15]. In Western nations, herbal medicinal products (HMPs) are becoming increasingly popular. In 2005, a survey found that 71% of Canadians were using natural health products, which include vitamins and minerals as well as HMPs. In this study, herbal remedies and algal/fungal products were used by 11% of those surveyed. According to Kennedy (2005), about 19% of adults in the United States were using HMPs in 2002. According to another study, approximately 36% of Norwegian pregnant women use herbs [16].

There are many reasons why people use herbs and other alternative treatments. It is common to cite a "sense of control" or mental comfort for taking action. Regular items are additionally remembered to be more grounded than drugs that are made. Additionally, conventional medicine is not popular. Patients believe that their physicians:

1. do not respect or pay attention to cultural customs or beliefs;
2. are unwilling to discuss alternative therapies or lack knowledge of them; what's more;
3. put a more prominent accentuation on treating the infection or condition than they do on the patient. Traditional medicine is a quick fix that focuses on curing or treating diseases once they've developed.

Patients frequently believe that conventional medicine has failed them due to chronic or incurable diseases like diabetes, arthritis, and depression [17].

3 Status of Phytomedicines in Developing Countries

The herbal medicine market is growing at an ever-increasing rate. The domestic restorative industry in India generates approximately Rs. 2300 crores, contrasted with the Rs. 2 trillion turnovers of the drug business. 14,500 crores, addressing a development pace of 15%. India is the second-largest producer of castor seeds in the world, producing approximately 1,25,000 tons annually [18]. It has been estimated that 70% of all doctors in France and Germany regularly prescribe herbal medicines, indicating that herbal medicine is used extensively in developed countries as well. According to available records, the herbal medicine market in the countries of the European Union in 1991 was approximately \$6 billion (it may now be over \$20 billion), with Germany accounting for \$3 billion, France for \$1.6 billion, and Italy for \$0.6 billion. In 1996, the US market for herbal medicines was about \$4 billion, and it has since doubled. The commodity of natural rough concentrate and the home-grown drug market in India absolute around \$80 million [19]. Almost 80% of individuals in Africa utilize this medication for their essential considerations. It was estimated that between 30 and 50% of all medicines consumed in China were traditional herbal remedies. In nations like Ghana, Mali, Nigeria, and Zambia, most individuals (around 60%) utilize conventional home-grown drugs as a first line of treatment for jungle fever-related high fever. Around 48% of individuals in Australia, 70% in Canada, 80% in Germany, 42% in the US, 39% in Belgium, and 76% in France utilize conventional and correlative medication something like once. In South Africa, London, and San Francisco, approximately 75% of HIV/AIDS patients use complementary and alternative medicine. Their annual consumption continues to rise significantly, reaching over \$3 billion. Additionally, the Internet has entered the market. The majority of consumers believe that herbal remedies are at least as safe and effective as patented drugs [20–26] (Fig. 1).

4 Formulation Challenges for Herbal Medicine Preparation

There are a plethora of potential side effects, reactions, interactions, nephrotoxicity, hepatotoxicity, and carcinogens that can arise from taking herbal medicines. Contaminated, counterfeit, or adulterated herbal medicines of substandard quality can result in potentially fatal side effects [28]. The fact that the majority of herbal medicines can be obtained without a prescription from a variety of retail and other establishments causes additional issues. As a result, herbal medicines face the same challenges as conventional non-prescription medicines in terms of pharmacovigilance. The World Health Organization (WHO) has issued guidelines as one example of the growing awareness of the need to develop pharmacovigilance practices for herbal medicines [29]. Other disadvantages include drug-herb

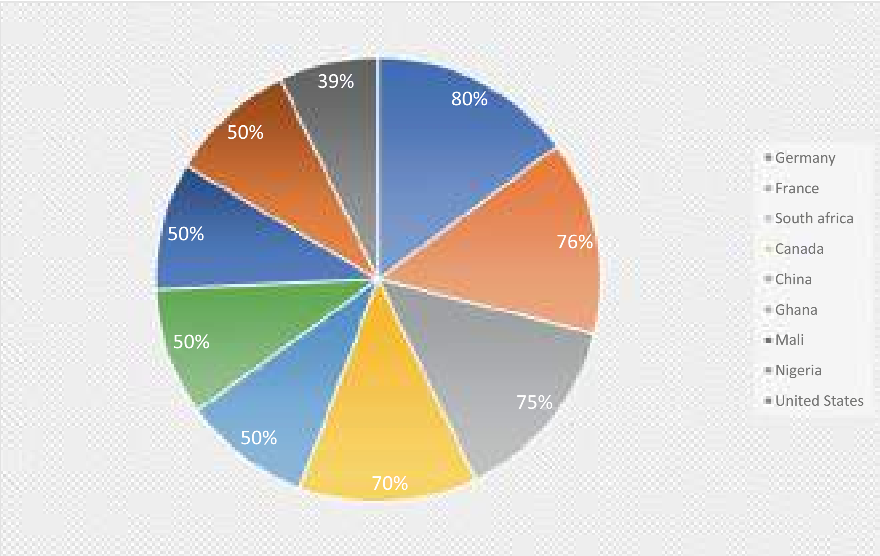

Fig. 1 Consumption of herbal drugs by different countries for healing of disease. (Source: Adapted from Sen and Chakraborty [27])

interactions and quality control. Several herbal products can interact with food, allopathic medicines, or pharmaceuticals, but this information is, unfortunately, lacking for the majority of products. Herbs generally lose their medicinal value after a year of collection; powders made from these herbs last for nearly 6 months, whereas pastes or ointments last for a year [30]. Negative effects of taking herbal medicines include accidentally using the wrong species of plant, adulterating herbal products with other, undeclared medicines, contaminating them with toxic or hazardous substances, overdosing, healthcare providers or consumers misusing herbal medicines, and taking herbal medicines concurrently with other medicines.

The analysis of adverse events associated with the use of herbal medicines is significantly more complex than that of conventional pharmaceuticals, even though the evaluation of the safety of herbal medicines has emerged as a significant concern for consumers, regulatory authorities, and healthcare professionals [31]. There is still a lot of research to be done on the topic of standardizing herbal drugs and formulations in the vast field. The standard books Indian Pharmacopeia, Ayurvedic Pharmacopeia of India, Abundance of India, and Ayurvedic Model all gather monographs that give each of the subtleties to the different tests that should be completed to decide if a rough or planned natural medication follows the guidelines laid. A variety of factors, such as the environment, climate, growth conditions, and storage conditions, influence the potency of a crude drug as well as the formulation prepared using it as a whole, as an extract, or as a constituent isolated. It is essential to standardize not only the primary drug component but also the other included excipients and additives [32]. Numerous pharmaceutical drugs and medicinal herbs are both therapeutic at one dose and toxic at another. The pharmacological or toxicological effects of herbs and drugs may be increased or decreased through interactions. The administration of long-term medications may be complicated by synergistic therapeutic effects. For instance, if taken in conjunction with conventional medications, herbs that are typically used to lower glucose concentrations in diabetic-1 could theoretically lead to hypoglycemia [33].

According to a survey writing study, customary medicine is used by 48.50% of people in Australia, 90% of people in China, 60% of people in Hong Kong, 49% of people in Japan, 60% of people in Nauru, 69% of people in the Republic of Korea, 57.40% of people in the Philippines, 45% of people in Singapore, and half of people in Vietnam [34]. Ayurveda, Kampo, Siddha, traditional Chinese medicine, traditional Korean medicine, and Unani, among other traditional medicine systems, plants are frequently used as ingredients. The discovery of morphine in 1804 by a German pharmacist named Friedrich Serturner marked the beginning of the modern use of drugs.

Around 60% of the medications have been gotten from normal assets since the center of the 1980s. The discovery of chloral hydrate by Justus von Liebig, a professor of chemistry in Giessen (Germany), in 1869 marked the beginning of the era of synthetic drugs [35].

Studies on herbal medicines are now regarded as significant. Typically, herbal medications are administered in the form of pills, decoctions, syrups, tablets, capsules, and other forms. These herbal formulations or raw materials for herbal drugs are physically unstable due to their high moisture content, bacterial and fungal contamination, chemical instability, improper harvesting, storage conditions, and others, Additionally, high moisture content frequently facilitates active component biodegradation. Legitimate drying conditions, considering the deterioration conduct of dynamic constituents, nanoparticle covering of the medication, the utilization of a chelating specialist, definition of emulsion and suspension, and so forth, can diminish these normal reasons for insecurity [36]. When herbal medicine practitioners directly provide products, there is no standardized level of pharmacological knowledge required, and they may not be aware of the knowledge required to safely advise consumers about interactions.

The national systems of the European Union encourage spontaneous reporting of adverse drug reactions. However, this system is known to be prone to underreporting, and the Medicines and Healthcare Products Regulatory Agency in the United Kingdom receives approximately 20,000 yellow card reports annually, but only about 100 herbal reports [37] (Fig. 2, Table 1).

5 Standardization Challenges for Phytomedicines

Nonetheless, normalization can apply to a couple of fixings. Thus, there might in any case be varieties in different fixings, which could affect the item's viability as well as security. Therefore, to ensure the reproducibility of herbal remedy scientific tests,

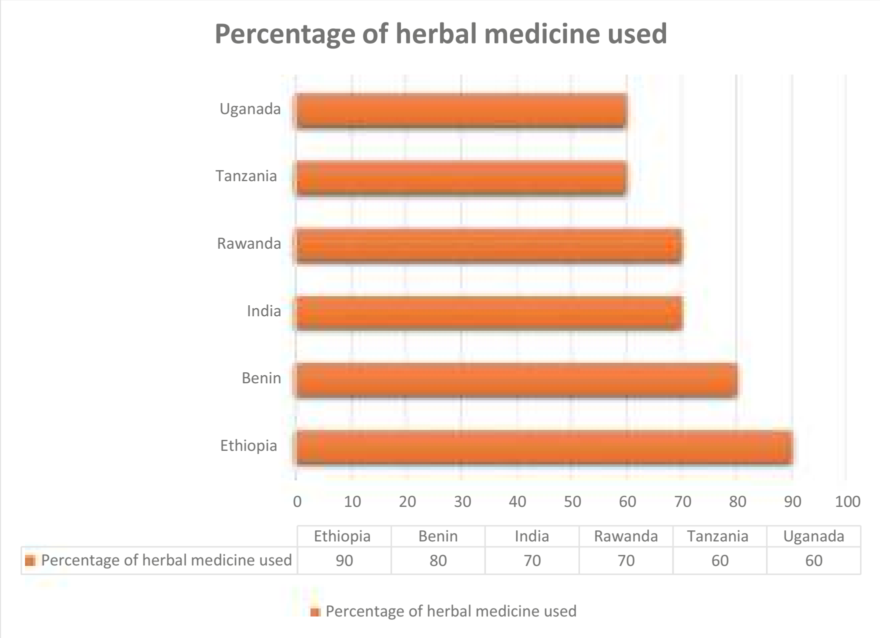

Fig. 2 In some developing nations, herbal medicine is used extensively for primary care. (Source: Author reproduced from WHO data)

Table 1 The Traditional Use of some Herbs as herbal medicine to Treat Common Conditions

S. No.	Name of plants	Common Name	Medicinal uses in some health ailments
1.	*Allium sativum*	Garlic, Lashun	Blood pressure regulation, maintaining the level of cholesterol
2.	*Boerhavia diffusa*	Tarvine, punarnava	Hypertension
3.	*Boswellia serrata*	Indian frankincense	Hypertension
4.	*Butea monosperma*	Palash, dhak	Hypertension
5.	*Cassia Alata*	Candlestick cassia, ringworm cassia	Skin diseases (Pruritis, eczema, ringworm), constipation
6.	*Commiphora mukul*	Guggul	Obesity
7.	*Momordica Charantia*	Karela, bitter gourd	Control sugar level
8.	*Psidium guajava*	Guava, Amrud	Antioxidant, anti-plasmodium, anti-bacterial, anti-inflammatory
9.	*Plumbago zeylanica*	Wild leadwort	Cancer
10.	*Terminalia arjuna*	Arjuna, arjun tree	Cardiovascular diseases

complete product characterization and quality control are required [38]. The dosage formulation, which can influence the bioavailability of the active chemical constituents in the standardized clinical extract, is an important factor to take into account when evaluating herbal supplements and medicines for safety and efficacy [39]. In natural materials, standardization, or the assurance of consistent and dependable ingredients, has been extremely challenging. Plant and herb constituents are extremely complex, and frequently multiple active chemicals with distinct pharmacological and clinical effects are present. This is further complicated by the fact that natural herbs have different chemical compositions and ratios of their constituents depending on many important factors, such as genetic composition, variations in growing conditions, and processing methods (such as harvesting, storage, and manufacturing techniques) [40]. Some herbal medicines are thought to have low associated safety risks, but little is known about how safe they are relative to other medicines. When herbal medicines are used, standard pharmacovigilance tools have additional limitations. Even when a pharmaceutical drug is taken concurrently with an herbal product, adverse effects may still be reported as being caused by the pharmaceutical drug. Inefficient pharmacovigilance procedures always result in a relative underreporting of adverse effects. Numerous authors in Iran, Nigeria, and sub-Saharan Africa have conducted pharmacovigilance studies on herbal medicines [41].

A few of the international authorities and organizations that have begun developing a new strategy to induce and regulate quality control and standardization of botanical medicine include the World Health Organization (WHO), the European Agency for the Evaluation of Medicinal Products, and European Scientific Cooperation of Phytomedicine, the United States Agency for Health Care Policy and Research, the European Pharmacopeia Commission, and the Department of Indian System of Medicine. The Department of Indian System of Medicine is one more. Recent usage of the term “Nutraceutical” refers to foods with health benefits that have been enhanced nutritionally or medicinally. Nutraceuticals include genetically modified or enriched soy products, cereals with added vitamins and minerals, engineered grain, and canola oil that does not contain trans-fatty acids [42]. Numerous nations have added test things of microorganisms, additives, pesticide deposits, weighty metals, and aflatoxin for customary medications and home-grown well-being food sources, and formed their public norms introducing huge cross-country varieties [43].

Standardization and quality control of herbals, as defined by WHO (World health organization) (1996a, b, 1992), is the process of physiochemically evaluating a crude drug. This evaluation covers aspects such as selecting and handling the crude material, evaluating the finished product’s safety, efficacy, and stability, documenting safety and risk based on experience, providing consumer information, and promoting the product [44].

Phytochemical standardization encompasses all available information regarding the chemical components of herbal medicines. As a result, the standardized phytochemical evaluation includes the following:

1. Preliminary testing to determine whether various chemical groups are present.
2. Quantification of the relevant chemical groups, such as total alkaloids, total phenols, total triterpene acids, and total tannins, for instance. Fingerprint profiles are created.
3. Multiple fingerprint profiles based on markers.
4. Determination of significant chemical constituents in numbers [45].

Researchers, manufacturers, and regulatory agencies must use rigorous scientific methods to guarantee the quality and lot-to-lot consistency of traditional herbal products to gain public trust and integrate them into the mainstream of today's healthcare system. The need for standardization and quality control in herbal products can be summed up as follows:

1. Technology and the idea of standardization were very different when traditional medicines were developed.
2. The dynamic process of evolution may have altered the identity of plant material over the past 1000 years.
3. The availability of genuine raw materials has become difficult as a result of commercialization.
4. The effects of time and the environment may have altered the properties of botanicals [46].

The task of quality standardization for herbal medicines (HMs) presents significant obstacles. A major method for quality control of HMs is currently the selection of abundant compounds as markers. However, in many instances, these marker compounds are irrelevant to the bioactivities. As the global demand for herbal medicines (HMs) grows, their quality standardization and control become increasingly important in commercial and clinical settings.

The uniformity of herbal products from batch to batch could be affected by a variety of factors, including variations in growing conditions due to geography and environmental factors. Since phytochemical compounds in HMs are not as refined as in Western drugs, choosing the right marker compounds for quality standardization presents significant difficulties [47]. The most common methods for standardizing herbal medicines are chromatography and spectroscopy, but the herbal system is difficult to analyze due to its complex chemical composition [48]. However, standardization will be more challenging than it would be with synthetic compounds. It is necessary to treat preparations with known and unknown active constituents in different ways. In any case, phytopharmaceuticals must be adequately labeled and characterized for reliable quality and transparency to be achieved. The label should at the very least specify the raw material, extraction method, solvent, ratio of drug to native extract, and quantity of active constituents to facilitate rational phytotherapy [49].

The safety and efficacy of herbal medicines are directly impacted by quality control. When collecting herbal materials, environmental and agricultural practices are important. The WHO has fostered a progression of specialized rules and reports connecting with the security and quality confirmation of restorative plants and natural materials. WHO distributed the ‘Quality Control Strategies for Restorative Plant Materials’, an assortment of suggested test techniques for surveying the personality, immaculateness, and content of therapeutic plant materials to help public research centers participate in drug quality control. The “WHO Guidelines for Assessing the quality of herbal medicines concerning with reference to contaminants and residues” were formulated in 2007 and the “Guidelines on good agricultural and collection practices (GACP) for Medicinal Plants” were published in 2003 [50].

5.1 Current Research on Natural Products

Integrating knowledge of evidence-based herbal medicine into the healthcare curriculum of Western medicine is perhaps the most crucial strategy for the safe use of herbs. The management of herbs may benefit from a few different strategies. They consist of ensuring open communication with patients, educating providers and patients about the potential benefits and risks of herbs, encouraging providers to inquire about their patients’ herb use without being judgmental, and patients should also be cautious when herbal claims are made, and they should only buy herbs from a reputable provider, business, or website [51].

Currently available plant-derived drugs fall into the following main categories: polyketides and other (18%), alkaloids (16%), and terpenes (34%), glycosides (32%), and for instance, digoxin derived from *Digitalis lanata* and used to treat atrial fibrillation, atrial flutter, and heart failure, belongs to the glycoside (cardiac glycoside) class, while artemisinin an antimalarial frontline drug, belongs to the terpene class (sesquiterpene lactone). Lovastatin, preferably isolated and fermented Aspergillus terreus (a hypolipidemic agent used to lower cholesterol), etc. [52].

The following actions are required to set international standards for natural medicines:

Accurate identification is of the utmost importance because medicinal plants that have been incorrectly identified may contain poisonous components. To standardize the manufacturing of Chinese herbal prescriptions, international quality guidelines like Good Sourcing Practice (GSP), Good Agricultural Practice (GAP), Good Laboratory Practice (GLP), and Good Clinical Trial Practice (GCP) are utilized. The processing of herbal products needs to be standardized. Regular items or items got from normal items make up numerous significant medications. From the 1940s to the present, natural products or derivatives make up 48.6% of all cancer drugs registered, and natural products make up 39.1% of all drugs approved by the Food and Drug Administration (FDA). Natural products play a significant role in the process of developing new drugs. More than 200,000 natural metabolites with various bioactive properties demonstrate the significance of natural products in drug discovery. The medicinal properties of between 35,000 and 70,000 plant species have been investigated [53].

6 Strategies for Improvement of Herbal Medicines

The low quality of the drugs is another major problem. Although international standards for drug quality are becoming more stringent, the actual quality of drugs in many nations is not sufficient. Surveys carried out in several developing nations indicate that between 10% and 20% of sampled drugs fail quality control tests. This is partly because there aren't enough resources for drug regulation. It is estimated that only one in three developing nations has fully operational drug regulatory authorities. Another factor is that bad manufacturing practices are not followed, which frequently results in products that are hazardous and even fatal. In the meantime, global quality assurance issues arise from the pharmaceutical trade. Customary medication is additionally impacted by quality issues. In developing nations, traditional medicine provides healthcare for up to 80% of the population. This medication is not only affordable but also widely available and dependable. In developed nations, complementary and alternative medicine has always been viewed as less important than modern Western medicine, but its popularity is currently rising [54].

6.1 Some Strategies for the Advancement of the Natural Medication Industry

By and by, north of 100 million ha of badlands are not just lying inactive, denying the poor valuable open doors for money age yet additionally representing a danger to the environment and climate. These areas can be used to cultivate highly sought-after medicinal herbs. By connecting with bulk buyers, medicinal herb commercial cultivation can be promoted. The following steps are suggested for the successful promotion of the herbal medicine industry in the nation.

1. Establishment of data banks and herbariums, as well as the identification of medicinal plant species.
2. Preservation of restorative spices in their regular living spaces – in situ and in safeguarded regions (ex situ).
3. Choice of prevalent germplasm in light of research facility examination and field studies, for taming and business creation.
4. Normalization of spread and development rehearses for delivering unrivaled quality natural materials, which can be taken on by ranchers.
5. Standardization of the protocols, documentation, and validation of the various medicinal herb cures for various diseases.
6. Intensive research into herbal cures for various diseases, particularly those that are just becoming known.
7. Patenting of traditional methods and plant material. Traditional healers can use facilitation counters to help patent their treatments.
8. The inclusion of herbal medicine as a core course in medical schools that study various medical systems.

9. Using specific prescriptions, social marketing, and widespread awareness to promote the use of various herbal medicines.
10. Assisting traditional healers and Ayurvedic practitioners by sharing knowledge, networking, providing high-quality germplasm, and establishing a connection with farmers for raw material supply.
11. Laying out a conventional linkage between cultivators, mass buyers, and drug enterprises at public and global levels.
12. Changes to government policies to help with herbal medicine research and development and to stop people from using medicinal herbs too much in natural forests [55].

Some of the strategies that are utilized in the process of developing and improving herbal medicines include liposomes, phytosomes, colloidal drug delivery systems (CDDS), super emulsifying drug delivery systems (SEDDS), and other more recent drug delivery system strategies. The development of a novel, safer method for transporting natural active compounds has been significantly aided by CDDS. Numerous natural medications have been delivered through these innovative systems. It was demonstrated that CDDS has a remarkable capacity to increase phytochemical bioavailability and stability. When administered via CDDS, natural active compounds like flavonoids, terpenoids, tannins, and xanthones can cross biological membranes and increase their bioavailability, which results in a better absorption profile. Additionally, CDDS are nontoxic due to their formulation with highly biocompatible raw materials. Since natural excipients have a high-security profile and cause less unfavorable impacts than current drugs, the last medicine is sure to be consented to by patients [56].

The WHO-AFRO's top priority interventions for the first and second decades of African herbal medicine development are as follows:

1. Putting together plans.
2. Upgrading limit.
3. Advancement of research.
4. Promotion of the local production of traditional medicines and medicinal plants.
5. Protection of intellectual property rights and traditional medical knowledge [57].

Three main steps go into making herbal medicine more competitive as part of a management strategy. Strategies like determining the current state of herbal medicine through a literature review (books, journals, and patent documents) and field observation, which were then examined to extract data, are the first steps in the process. Natural medications can be improved from various sources, including HR like workers, volunteers, and ideal interest groups; admittance to brand names, copyrights, and normal assets; Additionally, monetary assets include current cycles like programming frameworks, divisional pecking orders, and representative projects. Physical resources include things like your company's location, facilities, and equipment, such as human resources. Working on Natural Medication's intensity through three primary advances is an administration procedure – the initial step will be figuring out systems like deciding the genuine condition of home-grown medication through a writing survey (books, diaries, patent reports) and field perception, which was then inspected to separate information. Herbal medicines can be improved from a variety of sources, including human resources like employees, volunteers, and target audiences; access to trademarks, copyrights, and natural resources; what's more, current cycles, for example, representative projects, division orders, and programming frameworks are instances of monetary assets. Actual assets, like your organization's area, offices, and hardware, are instances of HR. Each business, association, and individual is impacted by outer powers. It is vital to note and record every one of these variables, whether or not they are related to an open door or danger straightforwardly or in a roundabout way. The term "external factors" typically refers to things over which neither you nor your company has any influence, such as economic trends, as financial trends on a local, national, or international scale; financing, like gifts, the lawmaking body, and different sources; demographics, such as a target audience's age, race, gender, and culture; relations with suppliers and partners; and economic, political, and environmental regulations [58].

Traditional herbal medicine employs the "multi-target/multi-component" strategy because, despite having no analytically defined mechanisms, it has therapeutic effects that have been clinically confirmed thanks to experience and knowledge gained over a long history of treating chronic illnesses. By re-engineering conventional herbal medicine, reverse pharmacology develops novel botanical drugs. Traditional herbal medicine's active ingredients and formulations are studied in reverse pharmacology, and drug candidates or formulations are then developed for preclinical and clinical research [59] (Fig. 3).

7 Prospects of Herbal Medicines in Global Markets

As per WHO (2000, 2002, 2003, and 2005), natural prescriptions and home-grown definitions assume a huge and filling part in worldwide medical services in Asia, Africa, the Americas, Australasia, and Europe. As per WHO gauges (WHO 2000b), 80% of Africans and 70% of Indians utilize natural medication in some structure. "The total of the knowledge, skills, and practices based on the theories, beliefs, and experiences indigenous to different cultures, whether explicable or not, used in the maintenance of health as well as in the prevention, diagnosis, improvement, or treatment of physical and mental illness," according to WHO (2002, 2013) [60]. Amid the COVID-19 crisis, the global market for herbal medicines, which was estimated to be worth US$110.2 billion in 2020, is expected to grow to US $190.4 billion in 2027, growing at a compound annual growth rate of 8.1% from 2020 to 2027. One of the segments examined in the report, herbal pharmaceuticals, is anticipated to reach US$84.1 billion by the conclusion of the analysis period and grow at a CAGR of 8.6%. Growth in the Herbal Functional Foods segment is adjusted to a revised 7.9% CAGR for the next 7 years, taking into account the ongoing recovery from the pandemic [61].

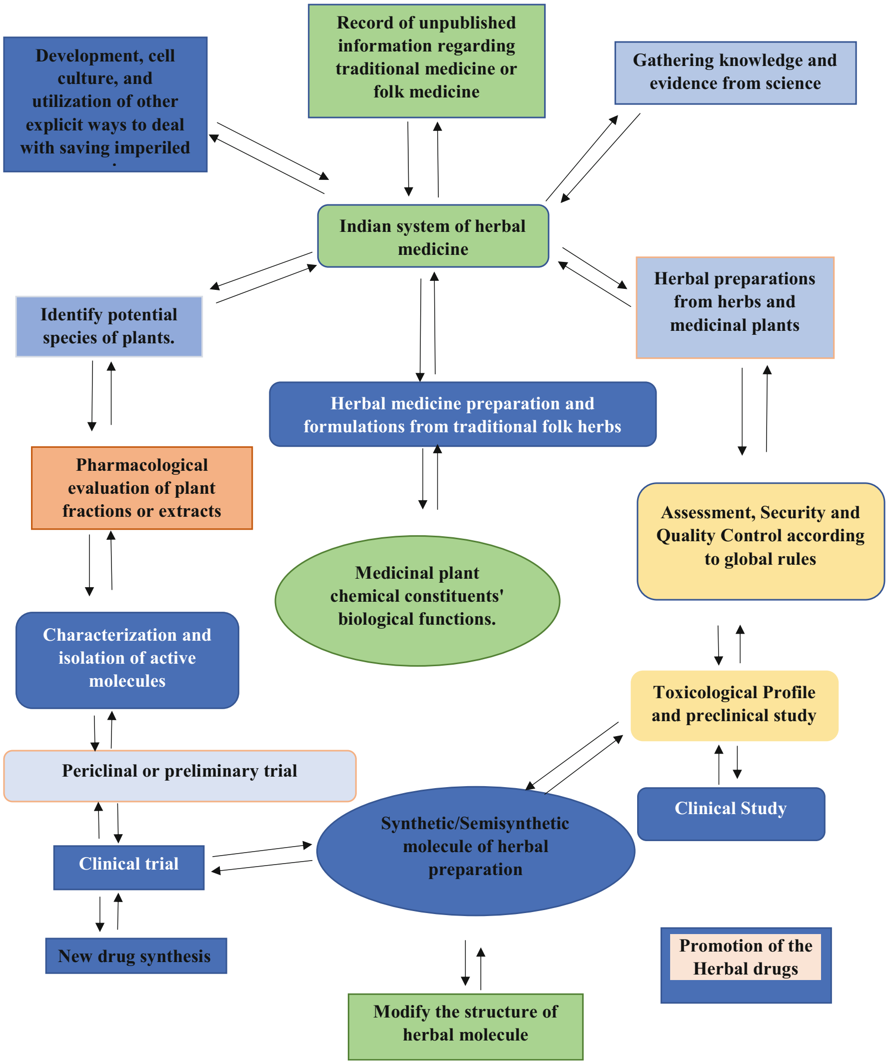

Fig. 3 Strategies for the advancement and development of traditional herbal medicine and its integration into modern medicine. (Source: Adapted from Sen and Chakraborty [27])

Worldwide, the use of herbal medicines has significantly increased. In recent decades, concerted efforts to globalize China's botanical medicines have led to a boom in the herbal industry. Mahuang and ginkgo, two Chinese herbal remedies, have successfully entered the international market, and their use is on the rise all over the world. Herbal and dietary supplements (HDS) make up a $180 billion market in the United States, and 18.9% of people reported using them in 2004. HDS is purchased by American consumers annually for approximately $6 billion. It has been estimated that the global market for herbal medicine, which includes herbal products and raw materials, grows at a rate of 5% to 15% annually. Botanical products account for approximately 30% of international drug sales. It is anticipated that the global herbal drug market will reach USD 5 trillion by 2050 [62].

India is the world's largest exporter, surpassing China, with approximately 13% of all exports. With 50% of exports, the United States is the primary market for Indian medicinal plants. The EXIM study found that 880 species of medicinal plants are traded in the country. There are approximately 42 imported species and 48 exported species among these. Over 8000 species of medicinal plants are grown in India, according to another survey by the Ministry of Environment and Forests of the Government of India. Seventy percent of these species are found in the tropical forests of the Western and Eastern Ghats. In its report for 1997, the Export-Import Bank of India estimates that India's trade in medicinal plants is worth $5.5 billion, an enormous increase [63].

According to WHO, traditional medicine has received renewed global attention and interest over the past decade. Traditional medicine is used for about 40% of all healthcare in China. Seventy-one percent of the population in Chile and 40% in Colombia have used such medication. Ayurveda and medicinal plants are used by 65% of the rural population in India to meet their primary healthcare requirements. Traditional, complementary, and alternative medicines are growing in popularity in developed nations. For instance, 48% of people in Australia, 31% in Belgium, 70% in Canada, 49% in France, and 42% in the United States have taken such medications at least once. Even though herbal products are subject to regulation or registration in nearly 70 nations, only 25 countries reported having a national policy for traditional medicine as of the year 2000. It is common to refer to traditional medicine as "complementary," "alternative," or "non-conventional" medicine in some nations where it has not been integrated into the national healthcare system. Due to the widespread misperception that "natural" means "safe," many consumers use traditional medicine for self-care. It's possible that they don't know about herbal medicines' safe use or potential side effects. Either there is no safety monitoring system at all or herbal medicines are not included in the safety monitoring system that is in place. Cases of improper use of herbal preparations have been reported due to a lack of quality control and consumer use. For instance, in 1996, an herbal remedy that contained the toxic plant *Aristolochia fangchi* rather than *Stephania tetrandra* or Magnolia officinalis led to the kidney failure of more than 50 people in Belgium [64].

Consumers now have much greater access to herbal products from anywhere in the world thanks to the development of the internet and the growing emphasis on a global economy. Also, businesses are using websites to boost sales, and the majority of them are more concerned with making a profit than protecting the public. While many of these websites may assert that their products are pure, standardized, safe, and effective, these claims are impossible to verify. With the growing use of medicinal plants and herbal remedies, public interest in natural therapies has skyrocketed in industrialized nations over the past few decades. Traditional medicines come in many different forms and have developed in response to a wide range of cultural, climatic, geographical, and even philosophical contexts. There are significant obstacles to overcome when evaluating these products and ensuring their safety and effectiveness through registration and regulation [65].

During the Middle Ages, trade along important terrestrial or maritime routes was linked to the first historical trend toward the globalization of herbal and traditional medicines. Global product trade is common today. Traditional Chinese medicine (TCM) and other herbal and traditional medicines are becoming increasingly popular worldwide. When it comes to trying to get into different markets for herbal medicines, businesses face a lot of obstacles. They must adhere to various laws, accept diverse requirements, and accept diverse standards. Pharmaceutical companies must therefore create distinct dossiers for the same products to present them to various national authorities.

To create the ideal regulatory environments for herbal and traditional medicines, international communication between the scientific community and regulators is required [66].

It would appear that one of the most crucial steps in implementing targeted marketing, market segmentation, has been overlooked. It is possible to identify customer attribution and select the appropriate sectors for targeting the market, establishing an appropriate market position, achieving a competitive advantage by creating distinct competitors, and increasing profitability based on effective market segmentation. Additionally, these market agents ought to discover methods for more precisely and effectively identifying customer behavior to identify, formulate, and implement their marketing strategies. The study's objective was to determine the appropriate factors for segmenting the herbal medicine market [67] (Fig. 4).

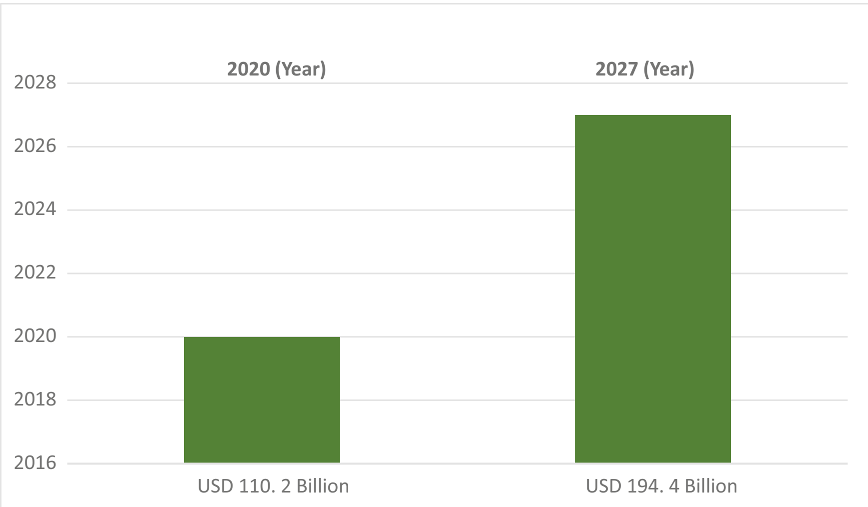

Fig. 4 Estimated growth in herbal medicine industry. (Source: Modified from Research and Markets [68])

8 Herbal Medicine Commercialization Challenges for Global Market

A major obstacle is verifying the herbal materials used and evaluating contradictory toxicological, epidemiological, and other data objectively. These significant issues persist. Pharmacovigilance Analyzing "drug" interactions and comprehending why harmful additives work. Requirements with clinical preliminaries and individuals accessible Normalization, Security and adequacy evaluation, The executives inside risk ranges, correspondence of vulnerability, Pharmacological, Toxicological, and Clinical documentation [69].

It also appears that medicinal herbs and their products have very diverse value chains. Surprisingly, very few studies have examined the value chain of medicinal herbs and herbal products, despite the volume of trade. The World Health Organization (WHO) has estimated that there is a $14 billion annual demand for medicinal plants (2006), and this demand is growing at a rate of 15–25% annually. The World Health Organization predicts that trade will reach up to $5 trillion by 2050 (Fig. 4). The global herbal medicine value chain approach is a useful analytical tool for comprehending the consequences of these shifting internal trade patterns in herbal products and medicinal herbs. Biopiracy, illegal collection, a lengthy gestational period before maturity, unresponsive permit/license systems, dispersed producers, and a lack of linkages among chain actors are the primary sources of risks to the commercialization of medicinal plants through cultivation (domestication).

According to many types of research and collected data by researchers, the commercialization of herbal drugs for the market has many obstacles:

1. Different requirements from the law.
2. Restricted market.
3. Issues with raw material standardization.
4. The limited or absence of regulations [70].

Regulatory and legislative processes for herbal medicines vary from country to country. This is due, in large part, to cultural factors as well as the fact that herbal remedies are rarely studied scientifically. As a result, few herbal preparations have been evaluated for safety and effectiveness. To assist national regulatory authorities, scientific organizations, and manufacturers in this particular field, the WHO has published guidelines that define fundamental criteria for evaluating the quality, safety, and efficacy of herbal medicines [71]. There are frequently several obstacles that must be overcome before traditional or herbal medicines can be regulated in various regions of the world. Challenges that are prevalent in many nations include regulatory status, evaluation of safety and efficacy, quality control, safety monitoring, and inadequate knowledge of traditional, complementary, and herbal medicines within national drug regulatory authorities. Even though the worldwide exchange of restorative plants is developing, the advantages to agricultural countries, especially cultivators and makers, are still low. This is because the poor collectors of medicinal plants from the wild lack organization and networking, as well as intermediaries.

This raises transaction costs, which are limited by the increasing stringency of health and safety regulations in developed nations' main markets [72].

The actual issues begin with the commercialization and industrial production of herbal plants, and bio-entrepreneurial skills take center stage. The bioproduct can be sold if there is already a market for it and there is a shortage of it, or if it is valuable enough or there are enough of it. The three nations that currently account for the majority of the commercialization of bioresources and medicinal plants are Germany, the United States, and Hong Kong. Altogether, 12 countries are answerable for 80% of the exchange that happens around the world. The second issue is industrial production or commercial harvesting. Due to the ongoing development of biotechnology processes and the high cost of chemical synthesis, Paclitaxel or Taxol, a well-known cancer drug that has been produced by Bristol Meyers Squibb (BMS) since 1992 and has recently been produced by other companies after BMS' exclusivity ended, still heavily relies on plant-based production [73]. One more example of how precious our lives are in tropical regions all over the world is the fact that tropical plants have produced more than 50 major drugs. Worldwide rural associations have progressively seen the general target of food security as a connection to the more extensive objective of further developing occupations throughout the last 10 years or something like that. However, as the globalization and commercialization processes progress, so do the number of partners and the information they produce. Lack of data on information sources, ad-hoc access to data, and incompatibility of a great deal of data that is accessible by hundreds of organizations, government agencies, and individuals in a dispersed manner are additional obstacles in information management [74]. The importance of the trade in traditional medicines to consumers' overall well-being and the improvement of healers' and/or traders' livelihoods is growing. However, this trade's unregulated activities jeopardize harvesting's long-term viability and, as a result, the availability of sufficient ethnomedicinal plant resources [75].

A marketing strategy is the foundation for any drug development beyond a certain point. A pharmaceutical company must decide whether to spend millions of dollars on advanced pharmacological studies and eventually clinical trials at a crucial point in the development process. When making such a decision, there are numerous factors to take into account. One of these is whether the medication will be able to generate approximately $500 million in sales within 2–3 years of its release [76].

The creation and implementation of HM regulations are difficult in many nations. Regulations, evaluation of safety and efficacy, quality control, and a lack of understanding of TM/CAM among national drug regulatory authorities are to blame for these difficulties. The WHO TM Strategy was developed with four primary goals to address these issues: forming policies; boosting quality, efficiency, and safety; determining access; and promoting sound use [77]. The use of HM and its legal status differ greatly from country to country. HM in India is governed by the Drugs and Cosmetics Act of 1940 (Amendment), the Department of AYUSH, the Central Council of Indian Medicine Act, and the Research Councils Indian Council of Medical Research (ICMR) and Council of Scientific and Industrial Research (CSIR). The Drug Controller General of India (DCGI) has set guidelines for the use of botanicals in allopathy (Table 2).

Table 2 A list of some Indian medicinal plants that are important to the market [12, 53]

S.No.	Name of Medicinal Plants	Biological Name	Family of the plants
1.	Makoi	*Solanum nigrum*	Solanaceae
2.	Punarnava	*Boerhavia diffusa*	Nyctaginaceae
3	Brahmi	*Bacopa monnieri*	Plantaginaceae
4.	Brahmi booti	*Centella asiatica*	Apiaceae
5.	Tulsi	*Ocimum sanctum*	Lamiaceae
6.	Jatamansi	*Nardostachys grandiflor*	Caprifoliaceae
7.	Shatavari	*Asparagus racemosus*	Asparagaceae
8.	Ashwagandha	*Withania somnifera*	Solanaceae
9.	Giloe	*Tinospora cordifolia*	Menispermacea
10.	Mulethi	*Glycyrrhiza glabra*	Fabaceae
11.	Sarpagandha	*Rauvolfia serpentina*	Apocynaceae

9 Future Prospects of Herbal Medicine

In light of their growing use and rapidly expanding market in both developed and developing nations, policymakers, health professionals, and the general public are increasingly raising concerns about the safety, efficacy, quality, availability, preservation, and future development issues of herbal products. Proof of natural items' security, viability, and quality has likewise expanded popularity among people in general. Due to the high value they bring to healthcare as well as the commercial advantages they provide, extensive research on herbal medicines is required to alleviate these concerns and satisfy public demand. Fortunately, there are already a lot of phytochemical and pharmacological studies on herbal medicines and medicinal plants all over the world. Attempts are being made to isolate and identify the active chemical constituents of these medicines and medicinal plants to support claims that they are safe and effective. More often than not, it has been demonstrated that natural cures don't need logical help because the vast majority of them contain significant substance compounds and carry out the asserted role. In addition, randomized clinical trials have shown that many herbal medicines have strong scientific support.

It is abundantly clear that, regardless of how significant the demands and concerns of policymakers, medical professionals, and the general public may be, they will never stop the growing use of herbal medicines. As a consequence of this, herbal medicine-based Traditional Medicine (TM) practices are still prevalent in developing nations, whereas CAM (complementary, alternative, or CAM) practices are rapidly expanding in developed nations. As specialists produce increasingly more logical proof of their quality, adequacy, and well-being, this overall pattern of expanding and broad utilization of natural meds is probably going to proceed.

However, just like with allopathic medicines, established rules and regulations should be used to control the production, sale, and use of herbal medicines to ensure their quality and safety. In any case, the nature of home-grown items sold isn't ensured, and guidelines and enrollment of home-grown prescriptions are not well-developed in many countries. Thus, endeavors ought to be made to raise public mindfulness about the dangers and advantages of utilizing natural meds and to get home-grown drugs under legitimate control in all nations where they are utilized for clinical and restorative purposes.

When utilized accurately, home-grown prescriptions of "guaranteed quality" make certain to affect clients and lower their gamble. In addition, it is important to keep in mind that herbal products and remedies, just like allopathic medicines, can have negative effects. In addition, they will probably have negative effects. Additionally, the use of adulterated herbal ingredients and formulations that are not appropriate may facilitate the production of low-quality, harmful, or even dangerous herbal medicines. To deliver home-grown medications, it is important to comply with GMP guidelines stringently.

With this caution in mind, it is safe to say that herbal medicines have a bright future and may one day be better alternatives to allopathic medicines made of synthetic chemicals or even replace them.

10 Conclusion

For the treatment of both laid out and arising infections, conventional prescriptions are presently the essential strategy for drug disclosure. Because they could be used as herbal remedies or for the isolation of compounds that could be patentable, researchers in Tanzania and other locations continue to find them appealing. Nevertheless, it is essential to guarantee that the information is utilized for both financial gain and the enhancement of our populace's health. There is a need to establish the expertise necessary for the creation of traditional medicines for cultivation to become feasible and, as a result, contribute to the reduction of poverty. Furthermore, conscious endeavors ought to be made to energize neighborhood modern creation of conventional and natural meds. Through administrative rules for quality control, normalization, and the assembling system, existing information on home-grown drugs should be checked and reported. Chemical consistency throughout all manufacturing processes extraction, standardization, quality control, quality assurance, stability, shelf life, and purity are essential to ensuring that medicines are safe and effective. Because herbal medicines are an essential component of traditional healing systems, it is necessary to preserve, promote, and further develop the knowledge base necessary to guarantee the identity, purity, and quality of botanical ingredients. Pharmacopeias assume an essential part in these cycles. Clinical evidence of the therapeutic effects of herbal products has led to the study of numerous additional herbs for their therapeutic, either curative or preventative, roles. Future research should focus on the isolation and identification of specific active substances from plant extracts because this could reveal compounds with greater therapeutic value. Modern and traditional knowledge can be combined to create novel medications that heal wounds and have fewer undesirable side effects. In created countries, natural-based conventional medication has acquired prevalence as of late and is probably going to fill in ubiquity before long. This method performs better than the allopathic method because it is preventative. Utilizing these plants sustainably, conserving them, and studying reproductive biology, phytochemistry, and pharmacological validation are all necessary because the increased use of herbs has a direct impact on the collection of raw materials. Another crucial requirement for quality control is the standardization of chemical fingerprinting (TLC, HPLC).

Herbal medicine will continue to serve as a broad foundation for drug discovery alongside bioactive lead compounds, active fractions, and active formulations. The strategy by which new single mixtures and dynamic parts can be created as clinical competitors will be given by profoundly effective bioactivity-coordinated fractionation and separation, structure portrayal, simple amalgamation, and unthinking investigations. To confirm and guarantee the safety and efficacy of prescriptions for multiple components and single herbs, clinical trials and dependable pharmacological and toxicological methods and standards are required. Using organic and compound markers, present-day innovation should be utilized to lay out quality controls and normalize natural items. In conclusion, Herbal ought to employ three strategies for discovering and developing novel drugs: formulations, pure active principals, and active fractions that have been tested or improved.

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Research Needs of Medicinal Plants Used in the Management and Treatment of Some Diseases Caused by Microorganisms

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Contents

Abstract

An increasing number of people around the world rely on herbal medicine to manage or treat medical problems caused by microorganisms, either directly or indirectly. Microbial infections may lead to an overabundance or the presence of

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microorganisms in certain parts of the body culminating in diverse medical problems. The use of herbal medicine to cure and manage these microbial diseases requires a certain amount of training (formal or informal) for efficiency and success. Improper composition of herbal mixtures may end up containing microbes like *Staphylococcus* species, *Escherichia coli*, *Pseudomonas aeruginosa*, and others that can exacerbate the initial illness. The training and practice rely heavily on available plant resources of the locale as well as local knowledge and cultural practices and the indigenous people. The bioactive substances found in medicinal plants have been linked to their therapeutic and microbial load reduction or eliminating effects. Depending on their species, regional distribution, and prevailing environmental conditions, different metabolites are known to be present in plants. Most diseases brought on by microorganisms may be grouped into one of four categories: bacteria, parasites, viruses, and fungi. Some of the common diseases that are treatable by plant extracts or herbal medicine include malaria, trichomoniasis, fascioliasis, helminthiasis, ascariasis, trichinellosis, dracunculiasis, gonorrhoea, syphilis, cholera, tuberculosis, pneumonia, food poisoning, tooth decay and gingival inflammation, typhoid fever, salmonellosis, skin infections, blood poisoning, and urinary tract infections (bacteria). Some of the causative agents are resistant to conventional antibiotics from modern medicine like methicillin-resistant *Staphylococcus aureus*, and *Candida albicans*, and vancomycin-resistant *Enterococcus* species. These microorganisms and the disease they cause are easily treated with plant extracts or herbal remedies. However, there is a need for research to focus on the identification, isolation, and purification of bioactive compounds in plant extracts and herbal remedies to encourage sustainable utilization and production in commercial quantities. Furthermore, the procedures involved in the investigation of raw resources, the production of herbal treatments, branding, and marketing, as well as sustainable utilization, need to be standardized and most importantly, require aseptic techniques to minimize microbial contamination of the raw materials and finished products.

Keywords

Microbial infections · Antibiotic-resistant microbes · Herbal medicine · Medicinal plants · Microorganisms · Phytomedical research

Abbreviations

%	Percentage
COVID-19	Corona Virus 2019
DNA	Deoxyribonucleic acid
FDA	Food and Drug Administration
FIV	Feline immunodeficiency virus
HCoV-NL63	Human coronavirus NL63
HIV	Human immunodeficiency virus
HPLC	High-performance liquid chromatography

MRSA	Methicillin-resistant *Staphylococcus aureus*
MTT	3-(4, 5-Dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide
NAFDAC	National Agency for Food and Drug Administration
SARS-CoV	Severe Acute Respiratory Syndrome Coronavirus
SON	Standard Organization of Nigeria
SYBR green	Asymmetric cyanines (fluorophores) group dyes
VRE	Vancomycin-resistant *Enterococcus* sp

1 Introduction

Microbes are everywhere, and many tend to thrive in extreme environments. Different groups of microbes have been reported in many human foods and food sources [1–16] and in different environmental matrices, including water [17–24], sediment [25], and air [26]. Many of these microbes are useful and have been implicated in food safety and quality; whereas some are used industrially in the production of useful products and metabolites. Quite a number of them are pathogenic and have the tendency to cause disease conditions in humans that are exposed to infectious agents. The diseases caused by microbes are often grouped into viruses, bacteria, parasites, and fungi.

In our world today, products of plant origin have become reliable sources for medications used to combat diverse microbial infections and are considered vital in the management of diseases caused by microbes [21]. Medicinal plants have been used for centuries in traditional medicine for the management of various diseases, including those caused by microbes. The efficacy of plant-based medications in treating microbial infections, especially in traditional medicine, is well known and has been reported by researchers from Africa [27–31], Asia [32], South America [33], and North America [34]. Plant-based products have always been sought out across the globe in the maintenance of health and treatment of diseases irrespective of culture and ethnicity thanks to their easy access and low cost [35–38]. In our world today, despite the availability of modern medicine, herbal medicines are still relied on. In different parts of the world, for example, Nigeria, China, Pakistan, and the United States, it is estimated that about 50–70% of their population depend on medicinal plants [39]. Even in an industrialized country like Japan, the demand for plant-based medicines in comparison to modern medicines has increased significantly [39].

Phytochemical substances with antimicrobial properties released during plants' biochemical reactions such as free radicals, flavonoids, reducing compounds, alkaloids, tannins, saponins, anthraquinones, terpenoids, and phlobatinins are now being used for combating various microbial infections [40]. The active compounds present in these plants have antimicrobial properties that can inhibit the growth of microorganisms, including bacteria, viruses, and fungi. The use of medicinal plants as a form of alternative medicine has gained popularity in recent years due to the increasing resistance of microorganisms to conventional antibiotics [41–58] and control of insects [59–64]. In this section, we will discuss medicinal plants that have been traditionally used for the management of diseases caused by bacteria, fungi, and viruses and the scientific evidence supporting their use [27, 28, 32, 34, 65–67]. There are enormous research needs around the use of medicinal plants in the management and treatment of microbial infectious diseases.

This chapter focuses on the plants, plant extracts, and herbal remedies used to address diseases caused by diverse microorganisms, including fungi, bacteria, viruses, and parasites, and the research gaps that are pertinent to the promotion of the practice. The chapter first presents an overview of herbal medicine practices and then proceeds to herbal medicine used in the management of diseases caused by microorganisms from parasitic, fungal, bacterial, and viral standpoints. Then, the chapter focuses on the treatment of antibiotic-resistant microbes and herbal medicine research needs specifically for medicinal plants and herbal remedies used for addressing microorganism-caused diseases.

2 Overview of Herbal Medicine Practices

Traditional medicine is viewed as a combination of different skills, practices, and knowledge based on beliefs and experiences used by different cultures in diagnosing, preventing, and eliminating [68]. Traditional medical knowledge may be passed on from one generation to the next in the form of stories (orally), by spiritual ancestors, and in recent times in writing [69]. Herbal medicine, also called phytomedicine or phytotherapy, is a core part of traditional medicine and involves the use of medicinal plants, their water or solvent extracts, oils, gums, resins, or exudates made from the plant parts, and other natural substances as sources of drugs for the treatment, prevention, cure of diseases, as well as maintenance of health and general well-being [70, 71]. According to the World Health Organization, herbal medicine comprises herbs, herbal preparations, and finished herbal products that contain plant materials like flowers, roots, berries, leaves bark, etc., or their extracts as active ingredients used for therapeutic interventions in humans as well as animals [72]. Throughout history people have used plants and various plant parts like leaves, stems, bark, and roots for healing and curing diseases and health promotion with varying levels of success [73].

Documented records of medicinal plants used in herbal formulations date back over 5000 years to the Sumerians with archaeological records even indicating an earlier use of medicinal plants thousands of years before [70]. Ancient records as well as archaeological data have shown the use of medicinal plants in ancient Egypt, Greece, China, and India [74–77]. In the last century, there have been massive developments and production of chemically synthesized medicines which have changed the healthcare system in most parts of the world, but about 70–80% of the world's population, especially developing countries, still rely on herbal medicine as their primary healthcare option. In Africa, up to 90% and, in India, up to 70% of the population rely on herbal medicine to meet their healthcare needs [73]. The use of herbal medicine is not only limited to developing countries as even industrialized countries have in recent years expanded their use of ethnobotanicals. About 38% of adults and 12% of children were found to be using some form of herbal medicine [78]. The use of herbal medicine is now widely embraced in many developed countries and is now becoming mainstream as these remedies are now available in drug stores, supermarkets, and even food stores [79]. There has been a growing interest in alternative medicine and the use of natural products as therapeutic agents, especially plant products, in the last 20 years [70]. There are many reasons why herbal medicine use has become widespread and now has an increasing appeal to the world's population. In some countries, herbal medicine is more accessible as a large percentage of the world's population does not have access to conventional allopathic medicine [80]. Herbal medicine generally also more closely corresponds to the patient's ideology as folk medicine, and ecological awareness suggests that these products are natural, harmless, and non-toxic compared with the side/adverse effects associated with synthetic medicine [73]. This is not necessarily true as when these medicines are taken with other herbs or over-the-counter medicine which is sometimes the case, they can lose their efficacy and even become harmful [81]. It is generally also more affordable when compared with the high cost of modern drugs. Herbal medicine also satisfies the desire for more personalized healthcare and establishes a movement towards self-medication [82]. Some people also have the erroneous belief that herbal products are superior to manufactured conventional medicine because of the presence of multiple active compounds that can provide effects that may not be achievable by a single compound [73]. They also claim that herbal products are highly effective against their health challenges [79]. Herbal products are generally used for the promotion of health and as therapy for chronic conditions rather than fatal conditions; however, where conventional medicine has failed or is ineffective such as advanced cancer, herbal medicine is employed [73]. Recent technological advancements leading to improvements in the quality, efficacy, and safety of herbal medicines have also contributed to the recent resurgence of public interest in herbal remedies [79]. Because of the greater access to health information, some patients believe that their doctor has not properly identified and diagnosed their problems and thus feel that herbal medicines are the next options to try [79].

Herbals and herbal preparations can be processed and administered in several ways, the most common being in the form of liquids either as teas or diluted plant extracts such as infusions, decoctions, macerations, tinctures, elixirs, extracts, etc. [83]. Herbal products can also be utilized as whole herbs, syrups, oils (essential oils), capsules, tablets, salves, rubes, etc. [73]. Regardless of how herbal products are processed and used, it does provide a significant healthcare service whether or not allopathic medicine is available [84]. The general perception by the population that herbal remedies are safe, harmless, or non-toxic, and devoid of side effects is very wrong as some herbs have been shown to produce a wide array of undesirable side/adverse effects, some even life-threatening under certain circumstances. A study by Ekor et al. [85] revealed that the herbal formula Yoyo 'Cleanser' bitters, a very popular herbal product marketed and sold in Nigeria, was associated with elevated levels of liver enzymes, resulting in hypokalaemia and arrhythmias. Another report by John et al. [86] revealed that some herbal blood tonics sold in Nigeria caused splenic enlargement in 10% of the recipient mice and even a case of lung tumour. Rietjens et al. [87] also reported on several Chinese herbal medicines implicated in cases of poisoning. Herbal products may also be contaminated and adulterated, and because most times there are no standardized set doses, it could lead to herb–drug or herb–herb interactions which could be fatal sometimes [73].

In most countries, herbal medicines are not standardized and do not follow mandatory safety or toxicology testing as they lack the machinery to regulate the manufacturing, packaging, and distribution of these products, leading to substandard products which could be potential hazards to the consumers [82]. Also, the increasingly growing market for herbal products could drive the over-harvesting of these medicinal plants, leading to the destruction of biodiversity and extinction of these plants [88–91]. Medicinal plants and ethnobotanicals have been used throughout history for the promotion of health and the treatment of disease. Although herbal products do not meet the accepted scientific standard for drugs, it is, however, very important to note that most herbal products from history have become the basis for today's modern medicine and contribute largely to the commercial conventional drugs manufactured today [92]. Examples include quinine isolated from the bark of a cinchona tree previously used for shivering but now used for treating malaria [93]. Aspirin, found in the bark of the willow tree, was previously used as a herbal product for fevers, pain, and inflammation in ancient times but is now used for different conditions, including rheumatic fever, heart attacks, blood clots, pain, fever, pericarditis, amongst others [94]. Digitalis from the plant *Digitalis* is a group of herbaceous plants commonly called foxgloves used previously by herbalists but now employed in conventional medicine for the treatment of heart conditions [95]. Opium from the plant *Papaver somniferum is* used as a herbal product for anaesthetics, suppositories, medical poultices, etc., but now conventionally used as morphine for pain and also as a recreational drug (heroin) [96]. Artemisinin from the plant *Artemisia annua* used in Chinese traditional medicine is now used commercially for the treatment of malaria due to *Plasmodium falciparum* for which its founder obtained a Nobel Prize in 2015 [97]. Herbal medicine is the oldest and still the most widely used system of medicine in our world today, and although the mechanisms underlying these plants derived remedies are still not well understood, the successful use of herbal medicine in the treatment and prevention of infectious diseases shows that many plants have the potential to mitigate bacterial fungal viral and even parasitic infections as will be discussed shortly [98].

3 Herbal Medicines Used in the Treatment and Management of Diseases Caused by Microorganisms

Phytotherapy is one if not the oldest system employed for the treatment of infectious diseases and has been employed to treat even contemporary infectious diseases like human immunodeficiency virus (HIV) and cancer with varying levels of success [99]. In recent years, the search and use of drugs obtained from plants and natural sources have been on the rise with scientists combing the earth for different potent

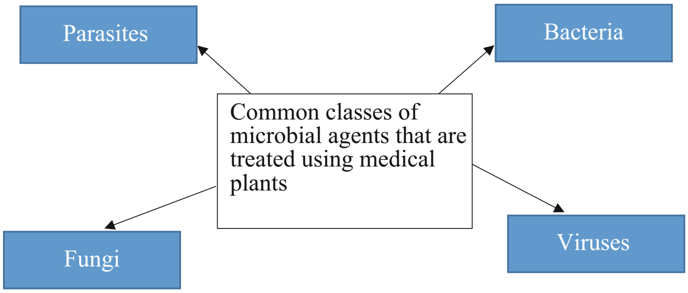

Fig. 1 Classes of infectious agents that are commonly treated using medicinal plants

phytochemicals that can be developed into medicine for the prevention and treatment of various diseases [100]. The use of herbal medicine seems to be universal across cultures, but the types of plants and methods applied may vary. When compared with conventional allopathic medicine, herbal medicine is easily and freely available, especially in West Africa (Nigeria), where 80–90% of the population still rely on herbal medicine as their basic healthcare system [101]. The diseases caused by microbes are often classified into four classes (Fig. 1).

3.1 Herbal Medicine and Medicinal Plants Used for the Treatment of Parasitic Infections

Infections caused by endoparasites like nematodes, protozoa, trematodes, etc., can affect more than 20–30% of the population because they cannot be prevented by vaccination. Most of the parasites have become drug resistant over the years, and the production of new bioactive antiparasitic drugs has not been a priority because these types of infection are mostly found in poor underdeveloped countries. This has resulted in the increased need for plant extracts and herbal formulations which can interfere with membrane structure, signal transduction, and DNA alkylation and intercalation of these parasites. Several plants have shown promising effects towards the management and treatment of diseases caused by parasites (Table 1).

3.2 Herbal Medicine and Medicinal Plants Used for the Treatment of Bacterial Infections

Medicinal plants have been used for centuries as a source of natural cures for various ailments, including infections caused by bacteria [114, 115]. Bacterial infections are a common problem that affects a wide range of individuals, and traditional societies have used plant-based remedies to treat them. The use of medicinal plants in the treatment of bacterial infections has been the subject of many studies, and the aim has been to identify new drugs to combat them, as well as provide new treatment options for patients who are resistant to conventional antibiotics.

Table 1 Medicinal plants and their formulations that are used to control parasitic infections

Plant species (or herbal products)	Parts used	Herbal preparation	Disease used against	Causative agent	Reference
Azadirachta indica	Leaves	Boiled in water to create an infusion which is then taken	Malaria	*Plasmodium falciparum*	[102]
Allium cepa	Bulb	Aqueous extract	Trichomoniasis	*Trichomonas vaginalis*	[103]
Allium sativum	Bulb	Decoction	Fascioliasis	*Fasciola gigantica*	[104]
Carica papaya	Seeds	Aqueous extract		*Ascaris lumbricoides* and *Ascaridia galli*	[105]
Chenopodium ambrosioides	Twigs, leaves, and flowers	Boiled in water to create an infusion which is then taken	Helminthiasis	Intestinal parasites (roundworm, pinworms, *Giardia lamblia*)	[106]
Aristolochia trilobata	Leaves	Boiled in water to create an infusion which is then taken or soaked in alcohol (rum) and taken as shots	Helminthiasis	Intestinal parasites (roundworm, pinworms, *Giardia lamblia*)	[106]
Vermicin (a mixture of *Olea europaea*, *Thymus vulgaris*, *Hydrastis canadensis Berberis aristata*, *Inulae radix*, and *Caryophylli flos*)	Leaves, rhizome, flower buds	Syrup	Helminthiasis	Intestinal parasites (roundworm, pinworms, *Giardia lamblia*)	[98]

Giardinophyt contains *Thymus vulgaris, Chrysanthemum vulgare, Anethum graveolens, Eugenia caryophyllata,* and propolis cera	Leaves, seeds	Ethyl alcohol extract	Helminthiasis	Intestinal parasites (*Giardia*, pinworms, *Ascaris*, etc.)	[98]
Cucurbita pepo and *Punica granatum*	Seeds and peels	Ethanolic and aqueous extract	Ascariasis	*Ascaridia galli*	[107]
Nigella sativa	Seeds	Alcoholic extract	Trichomoniasis	*Trichomonas vaginalis*	[108]
Artemisia vulgaris and *Artemisia absinthium*	Leaves	Methanolic extracts	Trichinellosis	*Trichinella spiralis*	[109]
Vernonia amygdalina	Leaves	Hot water extract	Helminthiasis	Intestinal parasites (*Giardia*, pinworms, *Ascaris*, etc.)	[110]
Jatropha curcas	Roots, leaves	Poultice with salts for roots and decoction for leaves	Dracunculiasis	Guinea worm	[111]
Combretum mucronatum	Leaves	Decoction	Dracunculiasis Helminthiases Enterobiasis	Guinea worm, hookworm, and pinworm	[112]
Triphyophyllum peltatum, Ancistrocladus abbreviatus, and *Ancistrocladus barteri*	Root and stem bark	Dichloromethane extract	Malaria	*Plasmodium berghei*	[113]

Echinacea is an example of a medicinal plant used in the treatment of bacterial, fungal, and viral infections [114–116]. As a herb, Echinacea has been traditionally used to boost the immune system and to treat a variety of infections, including bacterial infections [117]. Studies have shown that the antibacterial properties of Echinacea are due to the presence of active secondary metabolites such as alkamides (from *Echinacea*), glycoproteins, caffeic acid derivatives, and polysaccharides [115]. Extracts of Echinacea obtained from ultrasound-assisted extraction and other classical methods have been reported to effectively inhibit the growth of various bacteria species, including *Staphylococcus aureus*, *Streptococcus pyogenes*, *Listeria monocytogenes, Haemophilus influenza, Pseudomonas aeruginosa,* and *Escherichia coli* [115, 118].

Ginger is another medicinal plant with antimicrobial properties. Its active compounds are shogaol and gingerol, which have been found to have antimicrobial activity. A research study published in the *Journal of Sharif Medical and Dental College*, Lahore, Pakistan, found that gingerol, the active compound in ginger, had strong antimicrobial activity against several types of bacteria, including *Staphylococcus aureus* and *Escherichia coli* [119]. This suggests that ginger may be an effective alternative treatment option for bacterial infections. Another example of a plant with medicinal value used in treating a variety of bacterial infections is turmeric. Turmeric is commonly used in Indian and Middle Eastern cuisine as a spice and colouring agent [120, 121]. Traditionally, it has been used to treat a variety of ailments, including bacterial infections. Curcumin is the active compound found in turmeric and is known to be an antioxidant and possesses antibacterial properties, including antiplatelet, cholesterol-lowering, anti-inflammatory, and antifungal properties [120]. According to research studies, curcumin can effectively inhibit the growth of various species of bacteria, including *Helicobacter pylori* (which is the common cause of peptic ulcers) and *Staphylococcus aureus* [120, 121].

Manuka honey is also another traditionally important plant-based remedy for treating bacterial infections. Manuka honey is obtained from the nectar of the Manuka tree found in New Zealand [122, 123]. Extensive research studies on Manuka honey have revealed that Manuka honey possesses a significantly high antioxidant capacity which contributes to its antibacterial properties [124]. According to Swift et al. [123], high levels of a stable antimicrobial compound called methylglyoxal present in Manuka honey make it potent against a variety of bacterial species, including *Staphylococcus aureus* resistant to Methicillin, commonly known as Methicillin-resistant *Staphylococcus aureus* (MRSA), and *Clostridium difficile*, a nosocomial agent responsible for a significant percentage of hospital-acquired infections [122], by effectively inhibiting their growth.

Oregano is another medicinal plant that contains phytochemical compounds such as thymol and carvacrol, which are effective against bacterial infections [125]. According to Lu et al. [125], oregano oil was effective against multi-drug-resistant clinical isolates, including *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and methicillin-resistant *Staphylococcus aureus*. Thymol and carvacrol, the active compounds in thyme, have been found to have antimicrobial properties.

Medicines of plant origin have been used for centuries to cure bacterial infections and remain the preferred treatment option in many traditional societies. *Echinacea*, turmeric, ginger, and Manuka honey are examples of plant-based medicines with antibacterial properties that are effective in treating bacterial infections. Several plants have shown promising effects towards the management and treatment of diseases caused by bacteria (Table 2).

3.3 Medicinal Plants Used in the Treatment of Fungal Diseases

Fungal infections are a universal problem that affects a wide range of individuals, from healthy persons to those with compromised immune systems. The use of medicinal plants in the treatment of fungal infections has a lengthy history, and many traditional societies rely on them as remedies for treating fungal infections. An example of a medicinal plant used in the treatment of fungal infections is garlic. Traditionally, garlic has been used to treat a variety of fungal infections, such as aspergillosis and candidiasis. Researchers have shown that garlic has antifungal properties and can effectively inhibit the growth of various fungal species [144–146]. According to Dikasso et al. [144], garlic juice was observed to have a more efficient antifungal effect on pathogenic moulds, dermatophytes, and yeasts in comparison to conventional antifungal medications. The major antimicrobial components of garlic are allicin, organosulfur compounds, and diallyl sulphides [146].

Tea tree oil is also a good example of a medicinal plant used in the treatment of fungal infections. It is obtained from the leaves of the *Melaleuca alternifolia* plant commonly found in Australia and has traditionally been used for treating fungal infections [147]. Studies have shown that the active antimicrobial component of tea tree oil Terpinene-4-ol has antibacterial activity against *Propionibacterium acnes* as well as antifungal activity. Terpinene-4-ol was seen to effectively inhibit the growth of various fungal species, including *Candida albicans*, which is a common cause of candidiasis [147]. Another example is the use of neem, a tree native to India. Traditionally, it has been used to treat diverse fungal infections, including athlete's foot and ringworm. According to Singh et al. [148], salicylic acids and phenolic acids such as caffeic, vanillic, o-coumaric, and cinnamic that are extracted by high-performance liquid chromatography (HPLC) showed high antifungal activity in vitro. Neem and its antifungal components have been reported to effectively inhibit the growth of various fungal species, including *Trichophyton mentagrophytes*, which is a common cause of ringworm [149]. Medicinal plants have been relied on for centuries to treat fungal infections, and for many traditional societies, they remain an alternative to conventional cures for treating fungal infections. Garlic, tea tree oil, and neem are examples of medicinal plants with antifungal properties that are effective in the treatment of fungal infections.

Table 2 Medicinal plants and their formulations that are used to control bacterial infections

Plant species (or herbal products)	Parts used	Herbal preparation	Disease used against	Causative agent	Reference
Gloriosa superba	Roots	Root infusions	Gonorrhoea Syphilis	*Neisseria gonorrhoeae*	[126, 127]
Terminalia chebula	Fruit	Methanolic extract	Cholera	*Vibrio cholera*	[128]
Anogeissus leiocarpa	Stem and root barks	Methanolic extracts	Tuberculosis	*Mycobacterium tuberculosis*	[129]
Urtica urens	Whole plant	Aqueous extract	Pneumonia	*Streptococcus pneumoniae*	[130]
Ribes nigrum and *Hippophae rhamnoides*	Fruits	Fruit juices	Tooth decay and gingival inflammation	*Streptococcus mitis*, *Staphylococcus aureus*, and *Pseudomonas aeruginosa*	[131]
Calluna vulgaris L. and *Vaccinium vitis-idaea*	Leaves and flowers	Water, ethanol and ethyl acetate extract	Urinary tract infection	*Escherichia coli*, *Enterococcus faecalis*, and *Proteus vulgaris*	[132]
Punica granatum	Fruit	Methanolic extracts	Ulcer	*Helicobacter pylori*	[133]
Harrisonia perforata	Twigs	Methanolic extract	Plague	*Streptococcus mutans*	[134]
Psidium guajava	Leaves	Essential oil, acetone, methanol, and hexane extracts	Food poisoning	*Bacillus cereus*, *Enterobacter aerogenes*, and *Pseudomonas fluorescens*	[135]
Cannabis sativa	Leaves	Oils and aqueous and alcoholic extracts	Skin infections, blood poisoning, and shock	Methicillin-resistant *Staphylococcus aureus*	[136]
Sideritis italica	Flowers and leaves	Essential oil and acetone extracts	Ulcer	*Helicobacter pylori*	[137]
Barringtonia asiatica	Leaves	The aqueous extract and essential oil	Typhoid fever Salmonellosis	*Salmonella typhi*	[138]
Juglans regia L.	Bark	Ether extract	Oral infections	*Staphylococcus aureus*, *Pseudomonas aeruginosa*, and *Escherichia coli*	[139]

(continued)

Table 2 (continued)

Plant species (or herbal products)	Parts used	Herbal preparation	Disease used against	Causative agent	Reference
Guatteria multivenia	Root	Ethanolic extract	Skin infections, blood poisoning, and shock	Methicillin-resistant *Staphylococcus aureus*	[140]
Acanthus montanus	Root	Aqueous root extract made by hot water maceration of the root powder	Boils (furuncles)	*Staphylococcus aureus*	[141]
Amaranthus spinosus	Leaves	Ethanolic extract	Diarrhoea	*Staphylococcus aureus* and *Escherichia coli*	[142]
Bridelia ferruginea	Bark	Aqueous and ethanolic extracts	Urinary tract infections, diarrhoea, and food poisoning	*S. aureus*, *S. epidermidis*, *E. coli*, and *Proteus vulgaris*	[143]

3.4 Medicinal Plants Used in the Treatment of Viral Diseases

For centuries, medicinal plants have been used as a source of natural remedies for various ailments, including viral infections. In today's world, viral infections have become common among a wide range of individuals, thereby prompting the use of plant-based remedies to treat them. The use of medicinal plants in treating viral infections has been the subject of many studies to identify new drug formulations to combat viral infections, as well as make available new treatment options for patients who are resistant to conventional antiviral drugs [150–154].

Andrographis paniculata is one of the medicinal plants used in the treatment of viral infections. *A. paniculata* is a herbaceous plant traditionally used in Ayurvedic and Chinese medicine to treat diverse viral infections [150]. *A. paniculata* possesses active compounds such as andrographolide and neoandrographolide which have been found to have antiviral properties. Aqueous extracts of *A. paniculata* using 3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide (MTT) method and SYBR green quantitative real-time polymerase chain reaction (PCR) method have been reported to be potent against the dengue-2 virus responsible for dengue fever [150]. Significant antiviral activity of andrographolide and dihydro andrographolide, both secondary metabolites of *A. paniculata,* has also been reported against the highly pathogenic influenza A viruses and avian influenza (H5N1) virus. Research studies have shown that *Andrographis paniculata* can effectively inhibit the replication of various viruses by inhibiting neuraminidase activity, thereby reducing the symptoms experienced during upper respiratory tract infections [150, 151]. Another study carried out by Shi et al. [152] using silver nanoparticles biologically synthesized from herbs, including *A. paniculata*, showed an inhibition of the replication process of the Chikungunya virus, thereby suggesting an alternative cure for Chikungunya virus infection. Nitrobenzoxadiazole-conjugated andrographolide (Andro-NBD), a derivative of andrographolide, has been reported to suppress protease activities of Severe Acute Respiratory Syndrome Coronavirus (SARS-CoV) which suggests a possible alternative for combating COVID-19 infection [152]. Other derivatives of andrographolide with quinoline moiety have been shown to protect against Zika virus infection [153] while others have been reported to prevent HIV infections [154].

Elderberry is another example of a plant-based cure for viral infections [155]. Elderberry has been traditionally used for centuries to treat a variety of upper respiratory tract and viral infections, including influenza and the common cold [155]. Recent studies have shown that extracts of elderberry effectively inhibited the replication of influenza viruses, and feline immunodeficiency virus (FIV). Black elderberry fruit extract has also been reported to exhibit antiviral activity against human coronavirus HCoV-NL63 in vitro [155]. Reductions in the severity of symptoms and duration of infections caused by these viruses were also recorded during the use of elderberry extracts [155]. The active compounds found in elderberry include phenolic acid components and flavonoids [155]. Another example is the use of licorice root, *Glycyrrhiza glabra,* to treat viral infections [156]. Traditionally, licorice root has been used in Ayurvedic and Chinese medicine to treat a variety of viral infections. Fukuchi et al. [157] reported high antiviral activity of alkaline extracts of licorice roots and their possible application as an HIV agent [157]. The researchers also identified significant potency of water extracts of licorice roots rich in flavonoids in the treatment of herpes simplex virus [157]. The active compounds found in licorice root include glycyrrhizin and liquiritin [157]. *Andrographis paniculata*, elderberry, and licorice root are examples of plant-based treatment options which are effective in the treatment of viral infections.

4 Medicinal Plants Used in the Treatment of Antibiotic-Resistant Microbial Infections

Antibiotic-resistant microbes, also known as superbugs [158], have become a growing problem worldwide, and the search for new treatment options for antibiotic-resistant microbial infections has become increasingly important. One potential solution is the use of medicinal plants as a source of natural remedies for these infections [159].

An example of a plant-based medicine used in the treatment and management of antibiotic-resistant bacterial infections is goldenseal (*Hydrastis canadensis*) native to Appalachia [160, 161]. The rhizome and root of goldenseal have been traditionally used to treat a variety of infections, including bacterial infections [160, 161]. Studies have shown that goldenseal contains a strong antioxidant called berberine [160] which has strong antibacterial and antifungal properties, and in synergy can effectively inhibit the proliferation of various antibiotic-resistant bacteria species, including *Pseudomonas aeruginosa*, *Escherichia coli* [162], methicillin-resistant *Staphylococcus aureus* (MRSA), and vancomycin-resistant *Enterococcus* sp. (VRE) [161]. The main bioactive compounds in goldenseal are alkaloids – berberine, canadine, and hydrastine [160, 161].

Although allicin in garlic is traditionally used to combat bacterial infections, it has also been reported to be potent against antibiotic-resistant infections. A study by Saripilli and Pikkala [163] showed that garlic was as potent as ciprofloxacin in combating antibiotic resistance. Its strong antibacterial properties can effectively inhibit the growth of various antibiotic-resistant bacteria species, including *Escherichia coli* and methicillin-resistant *Staphylococcus aureus* (MRSA) [158]. Another example is the use of neem (*Azadirachta indica*), commonly referred to as 'village pharmacy' [148]. *Azadirachta indica* has been traditionally used to treat a variety of bacterial infections, including antibiotic-resistant infections. Studies have shown that neem leaf extract and neem oil have antibacterial properties and can effectively inhibit the growth of various antibiotic-resistant bacteria species, including MRSA and *Candida albicans* [164].

With rising cases of resistance to antibiotics, the need for phytochemical substances with medicinal value cannot be overemphasized. New approaches to combating microbial infection have become essential, and medicinal plants such as garlic, goldenseal, and neem are examples of medicinal plants with antimicrobial properties that are effective against antibiotic-resistant bacterial infections [40].

5 Research Needs in Herbal Medicine and Medicinal Plants Used in the Management and Treatment of Diseases Caused by Microorganisms

The research needs in the field of herbal medicines are very many, but this may be balanced out by the potential benefits derived from these products as evidenced by the use by large populations, especially in underdeveloped countries. The area of research needs includes evaluation of safety and efficacy, quality control, regulatory and safety monitoring, and education and training of health authorities and control agencies [165]. In industrialized and civilized countries, standardization and consistent quality control of herbal medicinal products are considered one of the major bottlenecks affecting the overall acceptance of herbal formulations [166]. Considering the increasing popularity of medicinal plants, there is a need for extensive research on herbal medicine to ascertain their toxic level and overall safety. Some key research areas requiring attention include a thorough analysis of side effects associated with herbal medicines and ways of bypassing their negative interactions with other prescription drugs [166]. It is worth noting that some herbal plant species bear the same name which necessitates a taxonomic and systematic review to avoid their inappropriate usage during the treatment of microbial diseases [35].

Another issue affecting research on herbal medicine is the lack of quality clinical trials, and in some cases, the research is either irreproducible or bereft of methodology or interpretation. Safe dosage standards concerning factors such as body weight and drug interactions are a key aspect of herbal medicine research requiring attention, especially during clinical trials. This could be resolved through aggressive testing to determine the safest and most effective doses. Research is needed to ensure the identification of the exact compounds and bioactive elements in these herbal formulations so that they can be purified to use for specific diseases as against using the whole plant [42, 43]. A single medicinal plant may contain hundreds of bioactive compounds, and when used in combination with other herbs or natural products, the herbal product may contain several hundreds of bioactive molecules which makes it difficult to determine what exact component is responsible for the therapeutic effects seen when used and what components may be responsible for adverse effects. This is well observed and described in conventional medicine where there is only one active ingredient, and the benefits and side effects are well elucidated [73]. Research is also needed on the use of whole herbs or whether the use of extracted bioactive compounds is better. The isolation and identification of each active ingredient from each herb in the product may be very costly, time-consuming, and not cost-effective for the manufacturers of these products, but with new emerging techniques like genomic testing and chemical fingerprinting definitive authentication and standardization of herbal products may be achieved [73]. Standardization in the preparation methods of herbal products is still very lacking and needs more research. There is a lack of adequate or accepted research methodology for the evaluation of herbal medicine. Methods for preparing these medicines need to be accurate and reproducible to ensure the release of the same herbal products irrespective of the brands and manufacturers [72, 167].

Research is needed to ensure the safety of herbal products as there is generally a lack of systematic approaches to assess the safety and effectiveness of herbal products. The general notion that herbal products are safe or non-toxic is very untrue and misleading as several products and medicinal plants have caused toxicity to humans and animals. For example, aristolochic acids found in Asian medicinal plants and occasionally in slimming products have been shown to have genotoxicity, which is hard to repair and persists for years in the DNA [168]. Another herb, *Ephedra sinica,* which is routinely used for respiratory conditions and is now marketed as a weight loss dietary supplement, is now associated with cardiovascular and central nervous system adverse effects [169] and also hepatotoxicity [170]. The evaluation of the safety of these herbal products is also complicated by the different processing techniques, route of administration, dosage, compatibility with other herbs or conventional medicine, and even the geographical origin of the plant as well as environmental factors in the area of harvest [171]. Research must be carried out to properly identify and authenticate the names of medicinal plants to avoid confusion and potential risks. More manufacturers rely on common names and not the universal binomial taxonomic identifier which can lead to toxicity. For example, the herb *Heliotropium europaeum* (heliotrope), which contains hepatotoxic compounds, could be confused with *Valeriana officinalis* (garden heliotrope), known to contain sedative and muscle relaxant properties. National health authorities and control agencies should work collaboratively with botanists, phytochemists, pharmacologists, and other major stakeholders to establish published databases with the taxonomic identifiers of medicinal plants to avoid misidentification and mislabelling which can potentially lead to toxicity [79]. Also, the issue of herb-to-herb interaction and herb-to-drug interactions needs to be analysed and awareness increased to avoid toxic adverse reactions. Concurrent use of herbs with drugs could mimic or magnify or oppose the effects of conventional drugs. For example, bleeding is seen when warfarin is combined with garlic (*Allium sativum*), don quai (*Angelica sinensis*), ginkgo (*Ginkgo biloba*), or danshen (*Salvia miltiorrhiza*), mania is seen in depressed patients who mix antidepressants and *Panax ginseng*, increased risk of hypertension when tricyclic antidepressants are mixed with yohimbine (*Pausinystalia yohimbe*), amongst other adverse effects [172]. Herb-to-herb interactions have also been observed with turnip juice capable of abolishing the geno protective effect of ginseng [173].

There needs to be increased education, training, and expertise within the national health authorities and control agencies like the Food and Drug Administration (FDA), Standard Organization of Nigeria (SON), National Agency for Food and Drug Administration (NAFDAC). These bodies should work collaboratively to ensure that the necessary regulations and legal and ethical mechanisms are established for the promotion and maintenance of policies bordering the safety, efficacy, and authenticity of herbal products [165]. A common regulatory framework should be established across different countries to ensure that the use, efficacy, and safety of herbal products are not influenced by the traditional experiences available in a particular location [174]. More control should be given to these controlling bodies to ensure the safety and efficacy of herbal products before they are released into the market. The manufacturers should be made to follow specific guidelines and good manufacturing practices to ensure the quality of herbal products and also made to obtain licences stating the source of herbs used, potency, other medicinal ingredients and recommended use, and other vital information [175]. There is also a general lack of data on the composition and quality of most herbal products as a result of the lack of policies or governmental requirements. Regulatory policies on herbal products need to be standardized across the globe and the relevant authorities in each country need to be proactive and put appropriate measures in place to ensure public health and that the herbal medicines approved for sale are of good quality [79].

It is imperative to determine that herbal products will be made from plants from a uniform genetic source with records of the exact species and cultivar to ensure manufacturers do not use different species of medicinal plants that could potentially be toxic. Herbs should be grown under specific conditions because the environment (temperature, drought, pollutants, etc.) can affect the phytochemical components of these herbs. Techniques like biochemical profiling should be employed to ensure that a consistent plant is always used to produce herbal medicine [73]. When it comes to herbal medicine there is a general lack of research data. Therefore, another important area of research is the production of scientific evidence concerning health claims by carrying out clinical trials. Questions of efficacy will always remain except adequate amounts of scientific evidence are produced and accumulated from experimental animal models and then human trials [176]. Ethically this would be very hard to accomplish; therefore, there is a need to find and document new predictor biomarkers and detectable signs of early cellular degeneration that relate to health and disease outcomes [73].

In many developing countries, traditional medicine, of which herbal medicine is a core part, is the only system of healthcare available or affordable. Therefore, as the global use of herbal medicine continues to increase and many more herbal products are released into the market, many more public health concerns and research needs will be identified accordingly. Although many herbal medicines have great potential, the lack of substantial scientific evidence of their efficacy is lacking and many of them even remain untested; therefore, more research is needed in this area. As with other conventional medicines, herbal medicines must be covered by a country's regulatory framework to ensure their safety, quality, and efficacy, and the consumers should be provided with science-based information on the dosage, origin of ingredients, antagonist, efficacy, and adverse effects of herbal medicine [73, 79]. Other bottlenecks closely associated with research on herbal medicines include the improper selection of patients' batches, inconsistency in the duration of treatments, inadequate number of patients during clinical trials making it difficult to reach a statistical significance, and inability to set up adequate placebos due to the taste and aroma of herbal formulations [177].

6 Conclusion

Approximately 80% of the world's population currently uses herbal medicine for addressing diverse medical problems, either directly or indirectly. A large proportion of these medical problems may be linked to microbial infection or appearance in diverse parts of the body. Herbal remedies or plant extracts have been widely documented to be effective against these microorganisms' caused diseases but many lack supporting experimental or research data and are considered as claims. There is a need for research validation as well as the enumeration of the mechanism of action for these herbal remedies. Research techniques, tools and approaches in herbal medicine need to adopt contemporary practices. Plant extracts or herbal remedies prepared from plants will continue to be used for addressing viral, fungal, bacterial, and parasitic medical problems in humans. Diverse plants and plant parts (used either alone or in combination) are formulated mostly using local strategies and consumed in varying dosage are adopted to address specific microorganism-caused diseases. Considering that the majority of herbal medicine used for addressing diverse illnesses including those caused by microorganisms is within the Global South where modern or conventional healthcare is rare, it would be hugely beneficial to support the practice with modern research and techniques. There is a need for increased education, training, and expertise within the national and international health authorities and control agencies about the efficacy of herbal remedies and plant extracts in treating and managing microorganism-caused medical issues. Other pertinent research areas needing urgent attention are the discovery, enactment, and purification of the bioactive ingredient to be employed in large-scale production of the ingredients. Together, these practices can help address the lack of substantial evidence to support the efficacy and use of herbal medicine. Herbal medicine used for treating and managing diseases caused by microorganisms needs to be aligned with modern medicine in terms of incorporating clinical trials as well as standardization and quality control during the formulation phase of herbal remedies. Also, training and education can help reduce cases of microbial contamination of plant extracts and herbal remedies used for addressing diseases caused by microorganisms.

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Conservation and Sustainable Uses of Medicinal Plants Phytochemicals

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Abhishek Raj and Manoj Kumar Jhariya

Contents

Abstract

Plants play a key role in humans by providing uncountable ecosystem services and have sophisticated traditional medicinal importance. Food, flavors, clothing, fragrances, and shelters are major benefits of plant ecosystem. Medicinal plants gained wider recognition due to varying herbal products but these are disappearing at alarming rate. Climate change also affects medicinal resources and their conservation is of prime concern today. However, conservation strategies and resource management techniques minimize medicinal plant depletions at greater extent. Good farming practices and sustainable use solutions are

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progress by resource management techniques. Whereas conservation strategies promote both in situ and ex situ methods, which are further geared by sustainable usage of medicinal plant resources. Protecting medicinal plant resources are highly advocated in updated and reframed policy. Also, an awareness of the importance of medicinal plant and its sustainable usage is recommended from local to global platforms. Constructive policies and plans are needed for conserving and sustainable usage of important medicinal plants, which having herbal and ethnobotanical importance. Therefore, healthy medicinal plants maximize species diversity and intensity of many ecosystem services, which maintains soil, food, and climate security at the global scale. Further, a transdisciplinary approach is needed for targeting production, conservation, and sustainable usage of medicinal resources, which strengthen socioecological and economic paradigm. Moreover, scientific research and future recommendation are necessitated for defending medicinal losses as threatened and species rarity in the safeguards of humans which maintains environmental sustainability and ecological stability.

Keywords

Medicinal plants · Ecosystem services · Climate change · Conservation strategies · Ethnobotany · Sustainability

Abbreviations

BDRs	Biological Diversity Rules
BRs	Biosphere reserves
CCRAS	Central Council for Research in Ayurvedic Sciences
CCRH	Central Council for Research in Homoeopathy
CCRS	Central Council for Research in Siddha
CCRUM	Central Council for Research in Unani Medicine
CIMAP	Central Institute of Medicinal and Aromatic Plants
CITES	The Convention on International Trade in Endangered Species of Wild Fauna and Flora
CR	Critically Endangered
DANIDA	Danish International Development Aid
DD	Data Deficient
DST	Department of Science and Technology
EIP	Export Import Policy
EN	Endangered
ES	Ecosystem services
EW	Extinct in the Wild
EX	Extinct
FRA	Forest Dwellers Act
ICAR	Indian Council of Agricultural Research

ICFRE	Indian Council of Forestry Research and Education
IUCN	The International Union for Conservation of Nature
JFM	Joint forest management
JNTBGRI	Jawaharlal Nehru Tropical Botanic Garden and Research Institute
LC	Least concern
LR/cd	Lower Risk-Conservation Dependent
MEA	Millennium Ecosystem Assessment
MPCDAs	Medicinal Plants Conservation and Development Areas
MPs	Medicinal plants
NBPGR	National Bureau of Plant Genetic Research
NPs	National parks
NT	Near threatened
NWAP	National Wildlife Action Plan
O_3	Ozone
Pas	Protected areas
PMC	Primary health care
SDGs	Sustainable development goals
SFDs	State Forest Departments
VU	Vulnerable

1 Introduction

Medicinal plants represent plant system cherish with medicinal properties which needs for biodiversity and human society. These plants consist of important compounds, which are used in the drug industry [1]. Plant parts such as leaves, roots, stem bark, fruits, flower, or even whole plants can be used for medicines [2, 3]. These plant parts contain different active compounds, which induce therapeutic effects directly or indirectly and are used as medicinal agents [4]. The production and storage of active compounds in the plant body have certain physiological effects on different organisms [5]. Raw plant materials are needed by humans for meeting medicinal needs, which further maintains healthy ecosystem [6]. Medicinal plants perform synergistic actions due to presence of certain active compounds that are used for treatment of many diseases [7, 8]. It plays key significant role in people's life by ensuring medicinal security to them. Medicinal plants used in society since time immemorial which maintains wellbeing and minimize illness including epidemics. However, peoples treated themselves by medicinal uses due to their interests and knowledge of medicinal plants [9].

Medicinal plants have been used in healthcare since ancient times in many countries such as India, Egypt, China, and Greece [10]. Plants are used as drugs, aromatic agents, and disinfectant due to the presence of active compounds [11, 12]. Cosmetic and pharmaceutical products used more than one-tenth of the plant species for different medicinal uses. However, medicinal plants population and its distribution are not uniform across the world [13, 14]. Further, demands of medicinal plants increased by 8–15% per year^{-1} in the countries of Asia, North America, and Europe [15]. This is due to increasing importance and utility of medicinal plants throughout the world. But their overuses cause decline in medicinal resources, which affects biodiversity and ecosystem services. Similarly, written documents on traditional knowledge systems are scarce and vanishing day by day due to negligence of importance and services. Overexploitation and unsustainable usage of medicinal plants exert pressure on natural forests, which affects the overall health of the ecosystem and its productivity [16, 17]. Overexploitation and indiscriminate uses of medicinal plants not only affects regeneration, distribution, and diversity but also affects livelihood of forest fringe peoples [18, 19]. However, plants-based herbal medicine gained wide recognition due to fewer side effects compared to allopathic medicines, which satisfies the human population by providing cures for many diseases. Thus, medicinal plants ensure healthy human life with greater biodiversity, which maintains ecosystem health and environmental sustainability. Managing medicinal resources and its conservation through many scientific strategies (in situ and ex situ conservation) manage quality and quantity of medicinal products for people and industrial uses [20]. Medicinal plant can be conserved through in situ and ex situ practices. National parks (NPs), biosphere reserves (BRs), joint forest management (JFM), and some medicinal plant conservation areas are included under in situ conservation. Many herbal and botanical gardens, seed bank, cryopreservation, gene banks, different tissue culture repositories, etc., are included under ex situ conservation techniques [21, 22]. Proper harvesting and sustainable uses of medicinal plant along with its good market structure augment its productivity and medicinal quality. Construction policies and good governance are needed for harvesting, marketing, and conservation of different medicinal resources across the world. Moreover, a scientific research and future recommendation are necessitated for defending medicinal losses as threatened and species rarity in the safeguards of humans.

This chapter discusses medicinal plant diversity, the importance of traditional medicinal plants, and its related ecosystem services. Phytochemicals of medicinal plants and its relevant medicinal properties and antimicrobial affects are also discussed. Threatened medicinal plant resources due to climate change and unsustainable harvesting practices along with their conservation strategies are included in this chapter. Many constraints and challenges of medicinal plant resources are included, which can be overcome through better conservation techniques, constructive policies, and scientific research.

2 Medicinal Plant Diversity: Global Context

Medicinal plant's population, diversity and its distribution are uneven globally [23]. Drugs and health products used approximately one-tenth of the total plant species. China and India stand in topmost position in medicinal plants usage. Approx. 11146.0 and 7500 number of species were used as medicinal plants in

China and India, respectively, which was followed by other countries [23, 24]. Most of these plants are threatened species among the medicinal plants [13]. These threatened species exist due to resource destruction and genetic erosion of certain medicinal plants, which are listed under IUCN status [25]. Approximately 38,660 numbers of medicinal plants exist in Asian continent of which 78.0 number of medicinal plant species grown and commercialized whereas 26.0 species was accounted by China [26, 27]. Therefore, an extraction and cultivation of medicinal plants popularized and becomes integral part of many countries such as India, China, Bangladesh, Pakistan, Indonesia, and Myanmar [28, 29].

Medicinal plant diversity and its importance gained wider recognition globally due to higher medicinal usage which often represented as "Medicinal or Botanical garden" [30]. Globally, 9500 number of species (of total 48,655 number of documented plant species) represented ethno-botanical importance [31, 32]. Similarly, modern system of medicines and indigenous health practices used approx. 7500 number of medicinal plant species [32]. Diversified medicinal resources have greater medicinal significance, which cures many diseases and ailments in the world [33]. For example, *Withania somnifera*, *Phyllanthus emblica*, *Asparagus racemosus,* and *Tinospora cordifolia* have greater potential to deprive the negative effects of COVID-19 and SARS-CoV-2 [34]. So, many plants have been shown to inhibit both gramme-positive and gramme-negative organisms, thereby being an effective source of antimicrobials [17, 35–37]. Similarly, the characteristics and significances of organic farming-based medicinal plants are depicted in Fig. 1 [38–40].

3 Traditional Medicinal Plants: Drugs of Tomorrow

From Asia, traditional medicinal practices have been popularized in the world since time immemorial. *Ayurveda* practices in Himalayas, Kampo practices in the region of Japan, *Jamu* practices in the region of Indonesia, Sowa Rigpa practices in the region of in Bhutan, traditional Chinese medicine practices in the region of China, and Thai medicine practices in the region of in Thailand are critical examples of traditional medicinal practices in the world [41, 42]. However, traditional herbal medicines are also based on practicing different indigenous knowledge systems. These traditional systems guided people in using different medicinal plants resources, which play greater role in managing the lifestyle of human beings [43, 44]. Similarly, use and practice of medicinal plant *Rosmarinus officinalis* (rosemary) can be quantified based on past documentary proof and different statistical and linguistic approaches [45].

Plants usage can be determined by different cultural practices [46]. Medicinal plants are traditionally used in different cultures due to its diversified groups in nature [47, 48]. Diversion and shifting of traditional medicine practices to modern medicine system are an interesting part of research today [49, 50]. Therefore, ethnobotany science gained wider recognition and ensures intrinsic the relationship between plants and local communities [41]. Indigenous plant species play greater role in traditional knowledge protection and biodiversity conservation. Traditional

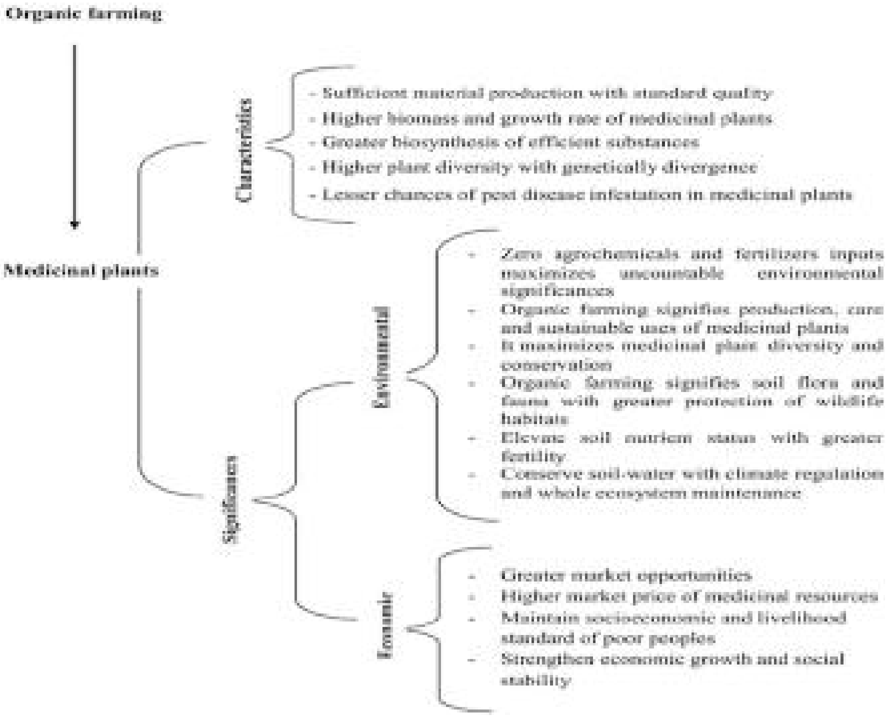

Fig. 1 The characteristics and significances of organic farming-based medicinal plants [38–40]

medicines promote PMC (primary health care)-based system and sustains approximately 70% of global populations [51]. Industrialized countries directly or indirectly depend on different medicinal plants for generating pharmaceutical products [52]. However, both India and China are top listed Asian countries that play greater role in production and management of medicinal plants. Approximately 50 and 45% of both export and earning of medicinal plants are contributed by Asian countries [53].

4 Medicinal Plants for Ecosystem Services

Plants play an important role in delivery of uncountable ecosystem services. However, medicinal plants are listed as provisioning ecosystem services under the United Nations MEA (*Millennium Ecosystem Assessment*) of 2005. Diversified medicinal plants ensure variety of ecosystem services, which maintains ecological stability and environmental services. Medicinal plants cure many diseases and ailments for better lifestyle of humans. Humans, animals, and other living organisms depend on plants for their life sustenance. However, herbal medicines promise good health of ecosystem and maintain health security for greater sustainability. A variety of plants are to treat many different diseases for better human and animal life. These plants not only provide shelters to many organisms but also maintain soil health and ecosystem productivity. Medicinal plants ensure soil fertility by providing nutrient loads in unfertile soils. Soil enrichment, water regulation, health security by disease management, greater plant productivity, and climate change mitigation are key ecosystem services provided by medicinal plants. However, medicinal plants also mitigate C footprints and climate change by capturing atmospheric carbon through carbon sequestration process. Thus, medicinal plants ensure soil, food, climate, and health security for maintaining ecosystem health and environmental sustainability. Similarly, different medicinal pastures can feed many grazing animals; therefore, feed supply is also counted as ecosystem services. Global productions of medicinal plants deliver multifarious ecosystem services for ensuring health and wealth of peoples and regulate climate [54].

5 Medicinal Plant Utility and Sustainability

Overexploitation and unsustainable use of medicinal plants affects biodiversity, ecosystem health, and ecological stability. It not only deprives health and productivity of medicinal plants but also affects people and animals they are depended on them. Destructive or illicit harvesting, slow growth performance, and restricted distribution are major causes of medicinal plants destruction and species disappearance [55]. Deforestation, intensive farming practices, and other anthropogenic activities affects distribution, diversity, and regeneration of medicinal plants. Therefore, sustainable use of medicinal plants, its conservation, and scientific management are necessary for ecosystem health and ecological stability. Sustainable utility of whole plant parts in formulating herbal drugs delivers greater services and advantages to people. Moreover, different parts of the same plants have similar pharmacological activities but have varying sustainable resources. For examples, extractives of roots and leaf-stem from ginseng (*Panax ginseng*) have equal pharmacological activities but leaf-stem of ginseng is a more sustainable resource [56]. Most of the medicinal plants have great taste and strong odor along with their greater medicinal value. These medicinal plants are also a good source of food and peoples used in daily life for their sustenance of life [57]. Sustainable uses of some important medicinal plants are depicted in Table 1 [58–60].

Sustainable harvesting and usage of medicinal plants are gaining wider recognition, which helps in biological resource conservation. Medicinal resource conservation is an urgent concern, which requires many scientific research, policy supports, and academic assistance for enhancing sustainability. Harvest timing, techniques, equipments, and material to be harvested, etc., are important parameters for conservation, management, and sustainable production of diversified medicinal resources. Moreover, good quality medicinal plants and its sustainable collection must be promoted for efficient utilization of diversified medicinal resources. Anthropogenic pressure for illicit harvesting can also be reduced by sustainable collection and optimum utilization of these medicinal resources. Therefore, case studies on conservation and sustainable use of medicinal plants in the world is depicted in Table 2.

Table 1 Sustainable uses of important medicinal plants [58–60]

Medicinal plant details	Distributions in India	Sustainable medicinal uses
Aonla (*Emblica officinalis*), Euphorbiaceae	Region of Maharashtra, Gujarat, Rajasthan, and Uttar Pradesh	Plants used for curing hemorrhagic disorders, anemia, asthmatic problem, and jaundice
Guggal (*Commiphora wightii*),Burseraceae	Distributed throughout Indian continent	It is a gum producing tree of medicinal value used for curing of rheumatoid arthritis
Senna (*Cassia angustifolia*), Caesalpinaceae	Region of Maharashtra, Rajasthan, Gujarat, Tamil Nadu, and New Delhi	Plant parts such as pod and leaves used for treatment of constipation
Safed musli (*Chlorophytum borivillanum*), Liliaceae	Mostly found in Southern region of the Rajasthan, Madhya Pradesh, and Gujarat	Plant characterized by tuberous root systems, which is highly recommended for Ayurvedic medicines for treatment of various disease
Satavari (*Asparagus racemosus*), Liliaceae	Distributed throughout India	Plant characterized by tuberous root systems, which is highly recommended for treatment of hemophilic disorders, diarrhea, swellings, and blood dysentery
Long pepper (*Piper longum*), Piperaceae	Distributed throughout India	Root is highly medicinal and used for treatment of chronic bronchitis, leukoderma, cough, epilepsy, and asthma
Brahmi (*Bacopa monnieri*), Scrophulariaceae	Distributed throughout India	Plant part such as leaves is used for nervous disorder especially as memory enhancer
Ashoka (*Saraca asoca*), Leguminosae	Distributed in the region of Himalayas and Western Peninsula	Bark system is highly medicinal and used for treatment of uterine affections and internal bleeding
Ashwagandha (*Withania somnifera*), Solanaceae	Mostly found in Rajasthan and Madhya Pradesh	Plant root system is used for treatment of rheumatism and weakness from old age
Indian barbery (*Berberis aristata*), Berberidaceae	Distributed in the region of Himalayas, Bihar, and Assam	Whole plants have medicinal properties, which are used for the treatment of skin disease, diarrhea, and jaundice
Isabgol (*Plantago ovata*), Plantaginaceae	Distributed throughout the plain region of Punjab, Sindh, and rarely found in Baluchistan	Seed husk has high medicinal properties used for treatment of dysentery and constipation
Kokum (*Garcinia indica*), Clusiaceae	Mostly found in Goa, Assam, West Bengal, Maharashtra, and Kerala	Mainly ripe fruits are utilized for preparation of Kokum syrup, which helps in eradication of many diseases

(continued)

Table 1 (continued)

Medicinal plant details	Distributions in India	Sustainable medicinal uses
Kalmegh (*Andrographis paniculata*), Acanthaceae	Distributed over Indian continent	Plant parts such as leaves used for treatment of many chronic diseases including malaria, anemia, jaundice, etc.
Acorus calamus L., *Bacopa monnieri* (L.), *Leonotis leonurus* (L.), *Uncaria rhynchophylla*, *Calotropis gigantea* (L.)	Distributed over most part of Asia, Africa, and Latin America in the world	Plant parts helps in eradication of neurological disease such as epilepsy

6 Selected Phytochemicals of Medicinal Plants

Medicinal plants enhance the health security of people and become a greater alternative for modern medical treatments. Plants are synergistically connected with people's life, death, illness, and aging and maintains socioeconomic and cultural relations. Medicinal plants help in diagnosis of many diseases and its treatments. Plants are greater source of safe medicine due to presence of varying phytochemicals [68]. Phytochemicals characterized by various medicinal properties help ecosystem health and functioning. Antimicrobial, anticancerous, and wound healing properties are key characteristics of phytochemicals. It also promotes metabolic activity, helpful in oral health and neuropathy. Therefore, medicinal properties of phytochemicals are depicted in Fig. 2 [69, 70].

Understanding phytochemicals, its identification, and compositions in medicinal plants is an important baseline for many researchers for treatment of disease and infections [71]. Alkaloids, steroids, tannins, cardiac glycosides, anthraquinones, favonoids, reducing sugars, saponins, and phlobatannins are important phytochemicals in medicinal plants [72]. Phytochemical components induce medicinal power in varying traditional plants that significantly promotes different pharmacological action on the human body [73]. Moreover, phytochemicals components are classified into two major groups, viz., primary and secondary, which are based on varying metabolic action in the plants. Amino acids, proteins, sugars, and chlorophyll constitute the primary group whereas saponins, alkaloids, phenolic compounds, flavonoids, and tannins constitute the secondary group of phytochemicals components [74]. Phytochemicals and its significant antimicrobial effects are depicted in Table 3. Similarly, pathogens can be suppressed by various defense mechanisms due to the production of phytochemicals in the plants. These plants are used to treat many immunological, metabolic, and neurological disorders in the people across the world. Practicing tissue culture helps in production of different phytochemicals and bioactive compounds in medicinal plants. Phytochemicals production can also be enhanced by artificial plant culture techniques of medicinal plants [70].

Table 2 Case studies on conservation and sustainable use of medicinal plants in the world

Medicinal plants (MPs)	Methodology and study features	Results and conclusions	References
A total number of 249 medicinal plants (MPs) selected in the Suriname	Study carried for understanding of unsustainable uses and overharvesting of medicinal plants. It includes monitoring and analysis of survivorship of the selected medicinal plants	Not more than half of the medicinal plant was exploited from wild. Plant part, such as leaves, was harvested extensively. Man-made and secondary forests were used for harvesting of medicinal plants. Losses and decline of medicinal resources were not reported in this study	[61]
Study includes ten of endangered and rare medicinal plants (MPs)	Study carried in the region of Kedarnath Wildlife Sanctuary, Central Himalayas, India for estimating population structure, distribution pattern, and their conservation status by analyzing species diversity, density, and occurrences in selected regions	This study resulted in medicinal plants grouped into two levels as healthy distribution with less pressure and limited distribution with high pressure. It provides insights for sustainable usage with better management and conservation strategies of selected medicinal resources	[62]
Study includes medicinal plants (MPs) such as *Amorphophallus, Camellia sinensis* (tea plant) and *Musella lasiocarpa* (Chinese dwarf banana) in selected regions of the China	Chinese Yunnan Province was selected for the study, which dealt for conservation and management of medicinal plants-based agro-biodiversity conservation and their sustainable uses	Study emphasized both in situ and ex situ conservation of medicinal resources, which helps in maintaining agro-biodiversity conservation and relevant ecosystem services for ensuring environmental sustainability	[63]
Study includes medicinal plant such as *Abies guatemalensis* in the Guatemala region of Central America	Study carried out a nation-wide survey with effective methodology on plantation owners, retailers, and urban consumers between 2004 and 2007	Study observed a declining production cost of plantation with increasing importance of legal supplies by involvement of local peoples in conservation and management of available medicinal resources	[64]

(continued)

Table 2 (continued)

Medicinal plants (MPs)	Methodology and study features	Results and conclusions	References
Study includes medicinal plant such as *Rheum tanguticum* in the region of China	Study conducted for analyzing distribution pattern of *Rheum tanguticum* through TCMGIS-II (a GIS-based program), which integrate climatic situation, geographical areas, and soil types of selected sites of China	Create a roadmap for conservation, scientific management, and sustainable usage of medicinal plant	[65]
Study includes medicinal plant such as *Scutellaria baicalensis* in the region of Northeast to Northwest China	Study carried for analyzing structure and genetic diversity of wild and cultivated population of *Scutellaria baicalensis* by involving polymorphic chloroplast fragments	Study focused on conservation-by-cultivation, which is effective for genetic resource protection and its sustainable uses among local communities	[66]
Study includes 27 threatened medicinal plants from 40 European countries	This study has undertaken 51 number of conservation projects, which surveyed 119 and 25 botanical gardens in 29 and 14 European and other countries	Study showed many medicinal plants were not efficiently collected and were outside of the country jurisdiction. However, collection from botanical gardens were slanted toward different ornamental plant species which were a poor reflection of conservation priority	[67]

7 Threatened Medicinal Plant Resources and Species Rarity

India, which recorded 2.4% of global geographical area, encompasses approximately 8% of world biodiversity [84, 85]. Indo-Burma, Sundaland, Western Ghats, and Himalayas are four main biodiversity hotspots of the Indian territory among the total 35 GBH (global biodiversity hotspots) [86]. Phyto-diversity richness is experienced due to the presence of a total of 15 agro-climatic zones, which have diversified ecological habitats. India accounts for approximately 10.45% of the world's floral. Similarly, 35 GBH (global biodiversity hotspots) hold approximately 50% of the global plant diversity that enveloped 15.9% of global land areas that is now reduced to only 2.30% [87]. However, the existing number of hotspots harbor many endemic species, which have been extinct due to natural and anthropogenic activities [88]. In this context, the first publication of "The IUCN Red List of

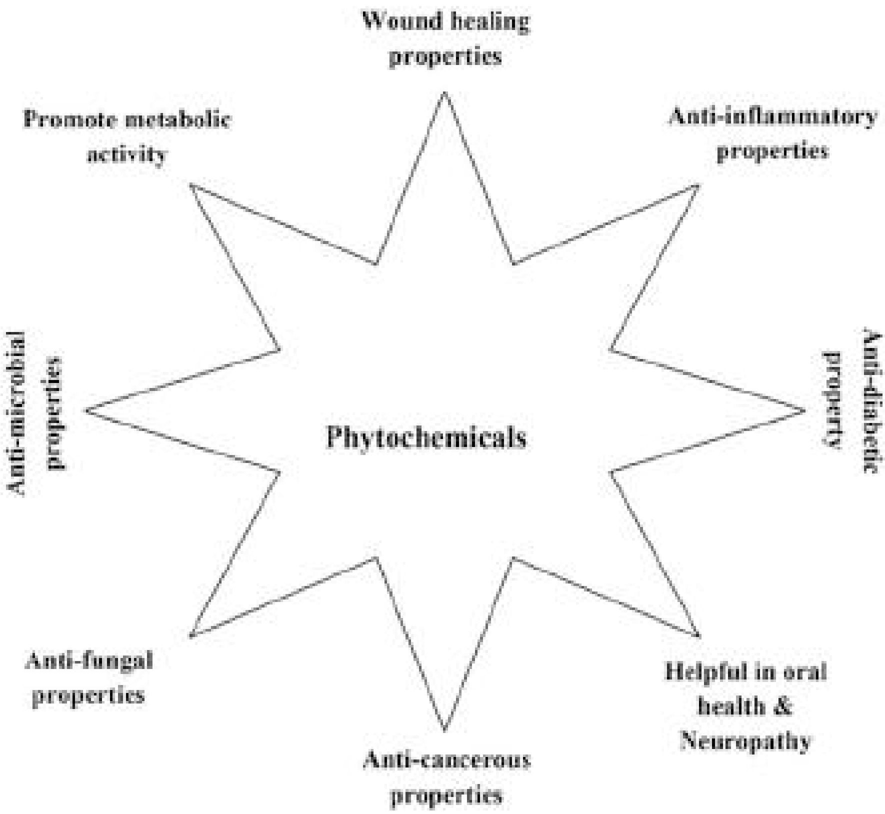

Fig. 2 Medicinal property of phytochemicals [69, 70]

Table 3 Phytochemicals and its significant antimicrobial effects

Source	Observed microbial species	Effects	References
Carrot (*Daucus carota*) seed oil	*Helicobacter pylori*	Antibacterial effects	[75]
Tea (*Camellia sinensis*) oil	Mycoplasmas species		[76]
Essential oils of medicinal plant	*Haemophilus Influenza, Streptococcus pyogenes, Moraxella* catarrhalis, and *Streptococcus pneumoniae*		[77]
Essential oils of medicinal plant	HSV-1 (Herpes simplex virus)	Antiviral effect	[78]
Clove (*Syzygium aromaticum*) and oregano (*Origanum vulgare*) oil	Poliovirus and Adenovirus		[79]
Essential oils of medicinal plant	Coronavirus		[80]
Essential extract of wild stevia (*Stevia rebaudiana*) plant	Fusarium and Aspergillus species	Antifungal effects	[81, 82]
Lycopene chemical	Zearalenone producing species		[83]

Threatened Plants" was reported in the year 1998, which reported 8000 species under threat [89]. Approximately 13.49% of the global vascular plants numbering 40,468 species are reported under varying degrees of threat. A total of 123 and 37 number of plant species are extinct in the wild due to anthropogenic activities, reproduction losses, population and pollinator losses, and habitat destruction [90, 91]. Similarly, approximately 11.53% of vascular plants are red listed under IUCN threatened category [90]. However, 90% of medicinal plants are threatened due to overexploitation and unskilled harvesting techniques [92]. Similarly, 1000 medicinal plant species are reported threatened at varying degrees in different ecosystems of India [93]. However, 602 threatened vascular plants in the Red Data Book has further increased up to 1255 and 2152 in the year 2003 and 2020, respectively [90, 94]. IUCN has categorized medicinal plants into various threatened categories at the world and national levels. IUCN categorized threatened plant species into Extinct (EX), Extinct in the Wild (EW), Critically Endangered (CR), Endangered (EN), Vulnerable (VU), Lower Risk-Conservation Dependent (LR/cd), Near threatened (NT), Least concern (LC), and Data Deficient (DD). Least concern (LC) species recorded maximum as 18,403 (global) and 1545 (national) whereas 7072 and 6063 species are Vulnerable (VU) and Endangered (EN) in the world. Hundred and twenty-three species are Extinct (EX) globally due to overexploitation and its unsustainable usage. Similarly, 2774 (global) and 103 (national) number of plant species are categorized under Data Deficient (DD), respectively. Therefore, plant species threatened at world and national levels as per IUCN estimation is depicted in Fig. 3 [90]. Similarly, IUCN threatened status of medicinal plants in different regions of India is depicted in Table 4 [90, 95, 96].

Rising demand for medicinal plants leads to overharvesting of many medicinal resources from the,wild which later decline existing populations. For examples, indiscriminate extraction of taxol from *Taxus baccata* (Himalayan yew), which helps in ovarian cancer treatment cause decline in population. Similarly, over-exploitation of many other medicinal plants such as *Polygonatum verticillatum, Nardostachys grandiflora, Gloriosa superba, Aconitum heterophyllum, Arnebia benthamii,* and *Megacarpoea polyandra* caused these species to be ranked under rare and endangered categories. Similarly, many medicinal plants are exploited for curing many diseases. For examples, medicinal plants such *Hemidesmus indicus*, *Aegle marmelos*, *Phyllanthus emblica*, and *Gloriosa superba* were used for treatment of 34, 31, 29, and 28 types of diseases [98]. Therefore, overexploitation and indiscriminate uses of medicinal plants not only affects regeneration, distribution, and diversity but also affects livelihood of forest fringe people [18].

8 Medicinal Plant at Risk Under Changing Climate: Implications on Its Phytochemical Constituents

Climate change becomes a major hurdle in the success of SDGs (Sustainable development goals) of the nation [99–101]. Extreme weather due to changing climate has deleterious effects on harvester's and cultivar's potential to cultivate and collect many medicinal resources. This type of experience is observed in

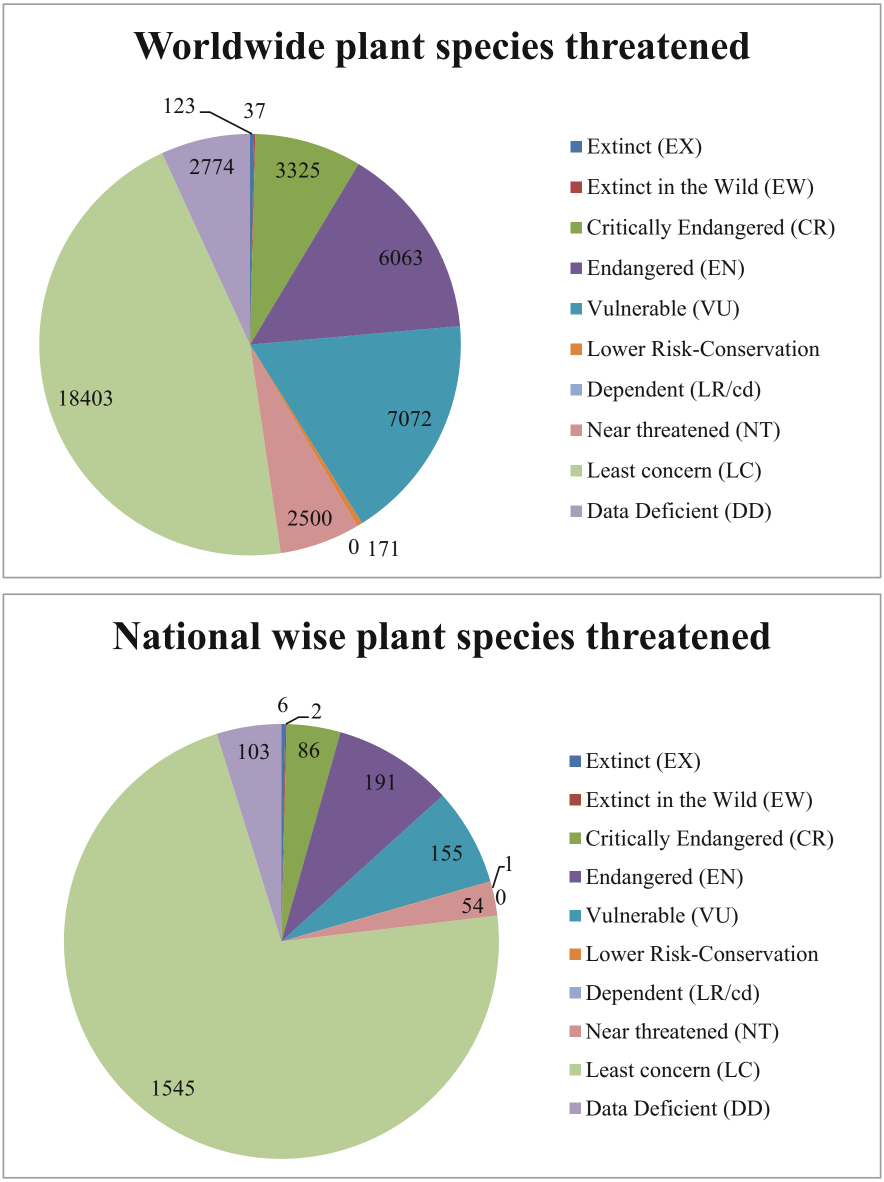

Fig. 3 Plant species threatened at world and national level as per IUCN estimation [90]

Europe, Poland, and Germany and across the entire world [102]. Africa's Sahel region also experienced severe droughts in the twentieth century. Extreme and severe drought badly affected many medicinal plants such as Boswellia spp., *Adansonia digitata* (baobab), *Moringa oleifera* (moringa), *Commiphora africana* (myrrh),

Table 4 IUCN threatened status of medicinal plants in different regions of the world including India [64, 90, 95–97]

IUCN threatened status	Medicinal plants	Regions
Critically endangered (CR)	*Aconitum chasmanthum*	J&K in India
	Aquilaria malaccensis	Tripura, Assam, Arunachal Pradesh, and Meghalaya
	Lilium polyphyllum	J&K, Himachal Pradesh, and Uttarakhand
	Valeriana leschenaultii	Tamil Nadu, Karnataka, and Kerala
Least concern (LC)	*Acorus calamus*	Tripura and Chhattisgarh
	Berberis aristata	West Bengal, Himachal Pradesh, and Sikkim
	Drosera indica	Madhya Pradesh and Karnataka
	Fraxinus floribunda	Sikkim in India
	Gloriosa superba	J&K, West Bengal, Kerala, Uttarakhand, Chhattisgarh, and Maharashtra
	Gmelina arborea	Madhya Pradesh in India
	Juniperus communis	Himachal Pradesh in India
	Toona ciliata	West Bengal in India
	Zeuxine strateumatica	Madhya Pradesh in India
Vulnerable (VU)	*Boswellia ovalifoliolata*	Andhra Pradesh in India
	Diospyros paniculata	Karnataka and Kerala
	Nilgirianthus ciliates	Mostly in Kerala and Tamil Nadu
	Phyllanthus indofischeri	Andhra Pradesh in India
	Santalum album	Mostly in Kerala, Tamil Nadu, and Maharashtra
	Abies guatemalensis Rehder	Mostly in Central America in Mexico, Guatemala, Honduras, and El Salvador
Endangered (EN)	*Cinnamomum wightii*	Mostly in Kerala and Tamil Nadu
	Humboldtia vahliana	Mostly in Kerala and Tamil Nadu
	Cycas circinalis	Orissa, Karnataka, Tamil Nadu, and Kerala
	Piper barberi	Tamil Nadu and Kerala
	Syzygium alternifolium	Andhra Pradesh in India
	Taxus wallichiana	J&K, Himachal Pradesh, Arunachal Pradesh, Meghalaya, and Sikkim
	Diplostephium rhododendroides	Central and South America.

(continued)

Table 4 (continued)

IUCN threatened status	Medicinal plants	Regions
Data deficient (DD)	*Coscinium fenestratum*	Kerala and Karnataka
Near threatened (NT)	*Pterocarpus marsupium*	Orissa, Chhattisgarh, West Bengal, Madhya Pradesh, and Rajasthan
	Pterocarpus santalinus	Tamil Nadu and Andhra Pradesh

Hibiscus sabdariffa (hibiscus), and Aloe spp. Therefore, rising cases of climate change has distressed ecosystem health and harvesting of medicinal plants [103]. *Triticuma estivum* (wheat), *Plantago ovate* (psyllium), *Cuminum cyminum* (cumin), and Indian psyllium crops are damaged by frequent rainstorms and hails due to climate change. Heavy rainfall in monsoon also affects the presence of menthol crystals in *Mentha arvensis* plant of Northern India [104]. Therefore, hailstorms and uncertain rainfall becomes key factors, which severely affects biological, reproductive, and phytochemicals constituents of many medicinal plants. Climate change also affects hurricane seasons, which further influence diversity and production attributes of medicinal plants [105]. Furthermore, elevated CO_2 has influenced the quality and productivity of medicinal plants. Increased value of root numbers and leaf fresh weight in *Mentha piperita* (peppermint) and *Ocimum basilicum* (lemon basil) have been reported due to increased value of CO_2 levels which is reported as 3000 μl CO_2 L^{-1} of air [106]. Similarly, alteration of secondary chemicals productions in many medicinal plants are also due to changing value of ozone (O_3) concentrations in the environment. Elevated O_3 promotes physiological stresses, which further encourage stimulation of different metabolic pathways (for examples, jasmonic acid and salicylic acid pathways) in the plants [107]. Further, elevated temperature may enhance many volatile organic compounds in plants [108].

9 Conservation Strategy

Medicinal plant can be conserved through in situ and ex situ practices. National parks (NPs), biosphere reserves (BRs), joint forest management (JFM), and some mmedicinal plant conservation areas are included under in situ conservation. Many herbal and botanical gardens, seed bank, cryopreservation, gene banks, different tissue culture repositories, etc., are included under ex situ conservation technique. A systematic cultivation of many medicinal plants can be adopted for biodiversity conservation and protection of various threatened species. However, different social, scientific, and political challenges become hurdles behind conservation and management of biodiversity [109]. Assessing the status of threatened plants and accordingly prioritizing the species conservation are prerequisites under multifarious biodiversity and limited resources. In this context, an IUCN (the International

Union for Conservation of Nature)-based “Red List of Threatened Species” has been established, which provides various scientific data on species diversity, distribution, and their extinction risk [110, 111]. IUCN has categorized species into different categories, viz., EX (extinct), EW (extinct in wild), CR (critically endangered), EN (endangered), VU (vulnerable), NT (near threatened), LC (least concern), and DD (data deficient) as per their threatened nature. These categories are based on quantitative criteria, which is a good indicator of different extinction risk [112]. Approximately, 16.5 m ha (5.02%) and 70.8 m ha (21.54%) of the total geographical area is shared by protected areas and forests in India. Similarly, National parks (NPs), wildlife sanctuaries (WS), community reserves, and conservation reserves (CRs) accounted 104, 551, 127, and 88 of the total 870 numbers of protected areas (PAs). Similarly, sacred groves (SGs) ranges from 100,000 to 150,000 in India [113]. However, many organizations have initiated research on conservation and management of medicinal plants. Institutes such as CCRS (Central Council for Research in Siddha), CCRUM (Central Council for Research in Unani Medicine), CCRAS (Central Council for Research in Ayurvedic Sciences), CCRH (Central Council for Research in Homoeopathy), ICFRE (Indian Council of Forestry Research and Education), and CIMAP (Central Institute of Medicinal and Aromatic Plants) have undertaken several researches on medicinal resource conservation. Several institutes of ICAR (Indian Council of Agricultural Research) have also strengthened policy and roadmap for adopting better scientific strategies of medicinal resource conservation. Moreover, many international organization, ayurvedic practitioners, and industries have successfully participated in cultivation and scientific conservation of these medicinal resources across the world [114]. National Gene Banks for Medicinal and Aromatic Plants has also coordinated with four institutes such as CIMAP, RRL as Regional Research Laboratory, ICAR-based NBPGR (National Bureau of Plant Genetic Research), and Jawaharlal Nehru Tropical Botanic Garden and Research Institute (JNTBGRI). This coordination has emphasized on better collection, propagation, and sustainable uses of medicinal plants [9]. Similarly, 1200 taxa herbals were collected in 10 acres of land from and Jawaharlal Nehru Tropical Botanic Garden and Research Institute (JNTBGRI). A total of 3334 and 2476 accession in seed and field gene bank are being maintained at CIMAP [115]. Cryo-gene bank, seed bank, in vitro, and field gene bank are key facilities provided by ICAR-NBPGR for scientific management and conservation of medicinal plants across the country. A total 8071, 1041, and 178 accession in seed gene bank, cryo-gene bank, and in vitro gene bank is being maintained at ICAR-based NBPGR [116, 117].

NMPB has been constituted under the Ministry of AYUSH, which was established by Government of India (GoI) in the year 2000. It provides funds for research and development for medicinal plants. DST (Department of Science and Technology) and DBT also promote many researchers to focus on medicinal plant-based research. *Aconitum ferox*, *Crocus sativus, Commiphora wightii*, *Cassia angustifolia*, *Saraca asoca, Bacopa monnieri*, *Phyllanthus amarus*, *Ocimum sanctum*, *Withania somnifera*, *Tinospora cordifolia, Piper longum, Emblica officinalis, Santalum album*, *Garcinia indica, Aegle marmelos, Swertia chirata*, *Plantago ovate*, *Solanum nigrum*, *Piper longum,* and *Andrographis paniculata* were important medicinal plants that needed conservation priority by NMPB in 2019 [118]. Similarly, 72 MPCDAs (Medicinal Plants Conservation and Development Areas) have been created for in situ conservation of medicinal plants in 13 states which covered 10,935 ha in forest areas [119]. Moreover, 108 MPCAs (Medicinal Plant Conservation Areas) and 18 MPCPs (Medicinal Plants Conservation Parks) have been established under partnership with State Forest Departments (SFDs) and DANIDA (Danish International Development Aid) to give financial supports to them [120]. These conservation strategies have emphasized on medicinal plant conservation and its sustainable uses across the country. DBT has also started a program for conservation of threatened and extinct plants with the help of biotechnological tools and by which approximately 100 medicinal plant species are conserved [121]. A total of 115 species were reintroduced under this project whereas 106 and 76 species were standardized through macro- and micropropagation protocols [121]. Therefore, "Reintroduction" is great strategy or scientific approach for re-establishment of threatened plants into suitable areas. *Syzygium travancorium*, *Calophyllum apetalum*, *Vanda coerulea*, *Ceropegia fantastica*, *Decalepis arayalpathra*, *Mahonia leschenaultii*, *Blepharistemma serratum, Berberis napaulensis*, and *Heracleum candolleanum* are key threatened medicinal plants conserved through successful application of reintroduction [122–125]. A total 156 medicinal plant species reported as highly threatened have been conserved through species-specific recovery programs adopted by DBT in the recent years [126].

10 Constraint and Challenges

A scientific research on medicinal plants gained significant attention from national and international platform. Medicinal plants also received wide recognition due to uncountable ES (ecosystem services), which includes biodiversity conservation, poverty alleviation, climate change mitigation, and health security. However, a deep fragmentation remains occur in data related to institutional arrangements, utilization methods, and medicinal plants production system. This creates hurdles in the development of an inclusive research agenda related to medicinal plant conservation and management in Asia [127]. Rising global population and their dependency on plant resources resulted in pressure on natural forests. This led to overexploitation and unsustainable usage of medicinal plants, which becomes a greater challenge for environmentalist. Overexploitation and unsustainable usage of medicinal resources reduce regeneration, species distribution, productivity, and ecosystem health. This is a key constraint and global challenge faced across the world. Increasing global population cause resource exploitation for meeting daily requirements of food and medicines, which increases the dependency on natural forests. This incident obviously increases erosion of forests and its related products [128]. Thus, it necessitates global challenges for conserving useful medicinal resources. A lack of correct information of demand, supply, and profits of various medicinal resources are other constraints. Lack of good market structure affects revenue generation from many medicinal plant resources, which becomes challenging from marketing and economic perspective. A poor knowledge on traditional medicinal system also affects people's health care system. Unsustainable harvesting of these medicinal plants is further excavated by poor scientific knowledge within the community stakeholders. Similarly, poor scientific knowledge on plant propagation, lack of proper resources, and negligence of fund flow are common constraints that affects production and market economy of medicinal plants. Adopting intensive farming practices utilize excessive agrochemicals and pesticides which affects valuable medicinal plants by affecting health and productivity. Peoples get affected negatively by consuming theses pesticides-based medicinal plants. Agrochemicals-induced medicinal plant productions also affects soil health and productivity, which becomes a global challenge for soil scientists. Higher use of agrochemicals enhances agricultural productivity but affects humans, plants, and environmental health negatively. Higher synthetic inputs not only degrade soil quality but also affect humans and animals due to disturbance in the food web and ecosystems [129, 130]. Further, agrochemical usage in agriculture poisons the food system that affects human, animal, and environmental health. Excessive usage of pesticides and its dependence is detrimental to human and the whole health and sustainability of the ecosystem [131].

Adopting sustainable or ecologically oriented cultivation practices of medicinal plants can overcome all these constraints. Moreover, practicing innovative agriculture ensures sustainable food system and creates resilience in the farming system the world over [132–135]. Thus, an eco-friendly scientific approach is needed for removal of harmful agrochemicals or pesticides residuals that can ensure healthy plants, human, animal, and whole the ecosystem, which is framed in "One Health" concept for sustainable environment. A good agriculture practice with efficient harvesting techniques also enhances medicinal plant productivity. Efficient processing unit, adequate infrastructure, and value addition of various medicinal plant products can ensure sustainability and production of medicinal resources. Poor infrastructure, lack of technological support and expertise, and low economic support also resulted in challenging situations for the global markets. Further, global competition among various pharmaceutical industries leads to overexploitation of medicinal plants. This unsustainable practice leads to scarcity of medicinal resources, which directly or indirectly affects ecosystem productivity and the global health care system. Thus, a perfect examination of such complex issues is required for cultivation, conservation, and sustainable usage of medicinal plants, which will further open the door for global market in the medical industry.

11 Strategic Plan and Policy Framework

Exploiting medicinal resources and its unsustainable uses under limited natural resources caused decline in medicinal plants and poor health care system. Climate change also affects the availability of medicinal plants by depriving its productivity, quality, and health system. Similarly, intensive farming practices utilized excessive agrochemicals and pesticides, which also affect valuable medicinal plants by affecting health and productivity [136–138]. These natural and anthropogenic practices push medicinal plants into threatened at higher risk which needs constructive strategic plan and policy reformation for better conservation and management of medicinal plants. The Government of India (GoI) has developed policy and some regulatory frameworks for medicinal plants conservation and its sustainable uses from both wild and cultivated areas. Indian Forest Act (1927), The Forest Conservation Act (1980), Wildlife Protection Act (1972), National Biodiversity Strategy and Action Plan (NBSAP, 2002), Biological Diversity Rules (BDRs, 2004), Forest Dwellers Act (FRA, 2006), National Wildlife Action Plan (NWAP, 2017–31), Export Import Policy (EIP, 2015–20), etc., have developed effective policy and some regulatory frameworks for medicinal plants conservation and its sustainable uses across the world. Similarly, National Medicinal Plant Board (NMPB) has also initiated many constructive policies, strategic plans, and scientific programs for low cost cultivation practices, proper collection, harvesting, processing, and sustainable uses of raw and value-added products of medicinal plants. Policy governance and institutional support are also needed in scientific research and development in different marketing sectors of medicinal plants. CITES agency has initiated an agreement with government for prohibiting illegal trade of many medicinal plants, which leads to species being threatened. A total of 28,000 and 5000 number of plant and animal species, respectively, have been protected by CITES through adopting better strategic action plan and constructive policy [139].

12 Scientific Research and Future Recommendation

Medicinal plants gained wider recognition due to varying herbal products but these are disappearing at an alarming rate. Scientific research and future recommendation are necessitated for defending medicinal losses as threatened and species rarity in the safeguards of humans. Diversion and shifting of traditional medicine practices to modern medicine system are interesting part of the research today [50, 140]. A degraded land can also be utilized by planting many medicinal and aromatic plants, which helps in eco-restoration and environmental management along with climate change mitigation [141]. Understanding phytochemicals, its identification, and compositions in medicinal plants are important baselines for many researchers for treatment of disease and infections [71]. However, many organizations have initiated research on the conservation of medicinal plants. Scientific research on medicinal plants has gained significant attention from national and international platforms. Policy governance and institutional support are also needed for scientific research and development in different marketing sectors of medicinal plants. A scientific research is needed on mega-network project for both in situ and ex situ conservation of medicinal plants. Similarly, a challenge has been rising due to high cost of botanical garden and gene bank establishment, which may also lead to plant losses under extreme climate, disease infestation, and frequent natural disasters. Therefore, cryopreservation, seed conservation, and in vitro conservation-based practices are prioritized for effective research to claim long-term safe conservation of medicinal plants. Moreover, research and development are required for collection of indigenous and quality plant material for strengthening its higher productivity and sustainable uses. Similarly, a future roadmap should be created for conservation of medicinal plants with relevant information on national gene/seed bank, which can ensure availability of diversified medicinal plants for current and future generation [142].

13 Conclusion

Traditional medicinal practices have been popularized in the world since time immemorial. Medicinal plants cure many diseases and ailments due to the presence of important phytochemicals, which provides better lifestyle for humans. Medicinal plant diversity and its importance gained wider recognition globally due to higher medicinal usage. Diversified medicinal resources have greater medicinal significance, which cures many diseases and ailments in the world. However, overexploitation and unsustainable uses of medicinal plants affects biodiversity, ecosystem health, and ecological stability. It not only deprives health and productivity of medicinal plants but also affects people and animals as they are dependent on them. Climate change also affects availability of medicinal plants by depriving its productivity, quality, and health system. Similarly, intensive farming practices utilize excessive agrochemicals and pesticides, which affect valuable medicinal plants by affecting health and productivity. In this context, maintaining medicinal plant diversity ensures a variety of ecosystem services, which maintains ecological stability and environmental services. Soil enrichment, water regulation, health security by disease management, greater plant productivity, and climate change mitigation are key ecosystem services provided by medicinal plants. Thus, global productions of medicinal plants deliver multifarious ecosystem services for ensuring health and wealth of people and regulate climate. Moreover, good quality medicinal plants and their sustainable collection must be promoted for efficient utilization of diversified medicinal resources. Therefore, herbal medicines promise good health of the ecosystem and provides health security for greater sustainability. Constructive policies and plans are needed for conserving and sustainable usage of important medicinal plants that have herbal and ethanobotanical importance. Moreover, a scientific research and future recommendation are necessitated for defending medicinal losses as threatened and species rarity in the safeguards of humans.

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Threats and Conservation Strategies of Common Edible Vegetables That Possess Pharmacological Potentials in Nigeria

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Abstract

Man uses plants as a source of fuel, food, medicine, shelter, fodder, and other forestry products. Vegetables are any component of a plant that may be eaten whole or in part, raw or cooked and include leaves, stems, roots, flowers, seeds, fruits, bulbs, tubers, and fungus. Nigeria is endowed with an abundance of vegetables that are used both as food and for therapeutic reasons. Huge amounts of vegetables are grown in Nigeria, and astounding numbers are sometimes cited as the projected yearly output. Vegetables are particularly essential for nutrition and health, especially since they include chemicals that control or promote digestion, function as laxatives or diuretics, and contain pectins and phenoic components that help control the intestines' pH levels. Both urban and rural residents' incomes are boosted by vegetables. Native American tribes' documentation of wild food plants helps in planning, identifying dangers, developing conservation strategies, and determining nutrient and pharmacological contents. As a result, this chapter covers an overview of edible vegetables in Nigeria, risks to common edible plants with pharmaceutical potential in Nigeria, and conservation efforts for those common edible vegetables.

Keywords

Vegetables · Pharmacology · Dangers · Nutrition · Plants · Nigeria · Conservation

Abbreviations

CBC	Community-based conservation
FRIN	Forestry Research Institute of Nigeria
IITA	International Institute of Tropical Agriculture
IPGRI	International Plant Genetic Resources Institute
NACGRAB	National Centre for Genetic Resources and Biotechnology
NCF	Nigerian Conservation Foundation

1 Introduction

Plants of every colour and kind have been used as sustenance for both people and animals from the beginning of time. Numerous crops have historically served as sources of food, medicine, and nourishment. Today, both humans and animals regularly consume about 10,000 indigenous food species throughout the globe. Again, during the last five centuries, there has been a rise in population contact as well as worldwide trade, and as a result, around 20% of these crop species are now utilized internationally and form the foundation of most of the world's agricultural production. However, many of the plant species that are grown for food across the globe are either underused, neglected, or both. According to studies, consuming the micronutrients and some phytochemicals found in these plants contributes to better health and lower rates of illness especially the ones associated with microorganisms [1–24]. Furthermore, they can be used in the control of insects such as mosquitoes [25–29]. Indigenous plants provide advantages for human health, in addition to enhancing local communities' access to food. There is widespread food insecurity in Nigeria due to hunger, malnutrition, and a lack of certain micronutrients in the diet, which is the main source of energy and sustenance [30].

Vegetables are any component of a plant that may be eaten whole or in part, raw or cooked, and they can include leaves, stems, roots, flowers, seeds, fruits, bulbs, tubers, and fungus [31]. However, similar vegetables are referred to as leafy vegetables when the component consumed as a vegetable is mostly from the leaves. They may also be referred to as potherbs, salad greens, vegetable greens, or leafy greens. Although they originate from a very diverse range of plants, some of them share a lot in terms of nutrients and preparation techniques with other green vegetables [32]. Due to their contributions to the nutrition security of millions of people, indigenous vegetables are becoming more and more popular [33]. The most affordable and widely accessible sources of vitamins, proteins, minerals, antioxidants, carbs, and amino acids are found in them. They keep the body's alkalinity balanced by serving as buffers for the acidic byproducts of digestion. Constipation is avoided by the fibre content, which supplies the bulk needed for digestion. In all parts of Nigeria, traditional medicine also uses vegetables as a kind of treatment. Their bioactive phytochemical components, which cause distinct physiological changes or activity in the body, are what give them their therapeutic potential. In certain situations, these bioactive phytochemicals have served as helpful building blocks for the creation of medicinal agents and practical pharmaceuticals [34].

The intake of diverse vegetables by a sizable portion of the world's rural population, according to Singh and Arora [35], helps them satisfy a portion of their nutritional needs. Vegetables have been shown to be very important to the food culture of people in Nigeria and all of Africa [36]. Many African households often consume a lot of green vegetables. In total, 115 of the 150 food plants that men regularly eat are native to Africa [37]. Furthermore, eating leafy greens has been linked to a number of health benefits, including the ability to stave against certain age-related degenerative disorders including arteriosclerosis and stroke [38]. According to Kimiywe et al. [39], several green vegetables have been linked to the treatment of various ailments. They give low resource people an alternative form of income in addition to acting as supplemental food and medicine [40]. Malnutrition and hunger have been noted by Obel-Lawson [41] to be threats to millions of people in sub-Saharan Africa, according to Jansen van Rensburg et al. [42].

However, it has been shown that eating green vegetables benefits both rural and urban people' nutrition, health, and economic well-being. Nigeria is renowned around the globe for its rich botanical diversity, which may be utilized in a variety of ways for culinary, medical, therapeutic, and nutritional reasons [43]. Leafy vegetables are a part of this enormous plant variety. Green leafy vegetables have been widely accepted as nutritional components in various regions of Nigeria, where they often make up a significant section of the diet when used to make soups and stews. It is regrettable that several of the traditional green vegetables that have long been utilized as food and medicine are now being gradually neglected. Traditional green vegetables have often been underused due to neglect. Civilization and a lack of knowledge about the nutritional and therapeutic advantages to many populations are factors that contribute to such underutilization [44]. Other concerns include industrialization; lack of suitable propagation practices to promote yield, quality, and quantity; regional conflicts that cause significant rural to urban migration; and the displacement of people [45]. As a result, this chapter covers an overview of edible vegetables in Nigeria, risks to common edible plants with pharmaceutical potential in Nigeria, and conservation efforts for those common edible vegetables.

2 Overview of Edible Vegetables in Nigeria

In sub-Saharan Africa, vegetables are often referred to as green or soft edible fruits that may be eaten fresh as a salad or cooked in a stew [46]. Some of these edible parts may be leafy, such as bitter leaves, water leaves, fluted pumpkins, mint leaves, lemon grass, and *Amaranthus hybridus*. African green vegetables aid in supplying the daily needs for nutrients. Sub-Saharan African rural residents were recognized as having a mandatory requirement to boost their intake of vegetables [47]. As they include both macro- and micronutrients, including dietary fibre; different vitamins like A, C, and K; as well as minerals like iron, potassium, and zinc to support human health development, vegetables play a significant part in the diet and nutrition of humans [48, 49].

Leaf vegetables are inexpensive, easy to prepare, and packed with nutrients, including vitamin C and B-carotene, which are critical for human health. Vegetable intake has been closely linked to stress reduction and a lower risk of developing chronic illnesses [49]. It is crucial to enhance this information since, for example, potassium-rich foods may lower blood pressure [50–52]. According to Gido et al. [53], different families eat a broad variety of green vegetables in varied ways. It may be argued that leafy greens alone have the ability to preserve maximum health. However, by enhancing the diet and diversifying the food options that are easily accessible in the field, traditional vegetables have the potential to help reduce malnutrition among the people in rural regions [54]. According to the RDA, these native vegetables in Nigeria have not been consumed to their full potential in Nigeria [55, 56]. The following list of frequently used edible vegetables is discussed:

2.1 Water Leaf (*Talinum triangulare*)

Talinum triangulare, a short-lived perennial succulent shrub that is susceptible to cold weather, is a member of the Portulacaceae family. The term "bologi" is used locally in Nigeria and is pronounced "gbologi" in the Ukwani dialect of Delta State. Without cultivation, it may grow naturally on its own. The flavour, suppleness, and taste of *Talinum triangulare* are well recognized. It is an excellent source of several crucial minerals including magnesium and potassium as well as antioxidant vitamins [57]. It is also high in fibre, protein, and other vitamins needed for a healthy human diet [58]. According to research by Ikewuchi et al. [57], waterleaf has significant amounts of antioxidant, antibacterial, and anti-inflammatory effects. Managing the metabolic syndrome, which includes diabetes, high cholesterol, and hypertension, is also a possibility [59–62]. Due to its high potassium content (Ikewuchi et al., [57], waterleaf soup is used in Nigeria as a treatment for hypertension [58].

2.2 Fluted Pumpkin (*Telfairia occidentalis*)

Fluted pumpkin is also known as ikongubong (Efik/Ibibio), eweroko (Yoruba), ugu (Igbo), and offi (Ukwani) in Nigeria. *T. occidentalis* is mostly grown in West Africa; it is also known by other regional names in other regions of Africa, including pondokoko or Gonugbe (Sierra Leone) and krobonko (Ghana). This vegetable crop is a perennial in the Cucurbitaceae family that is grown extensively for its tasty and nutrient-rich leaves. It is often planted as an outdoor home food plant. *T. occidentalis* is not an exception to the fact that vegetables are recognized for both their antioxidant and phenolic chemical characteristics [63, 64]. As compared to other regularly consumed green vegetables, fluted pumpkin leaves are exceptionally rich in minerals including calcium, phosphorus, and iron [65]. Although it may not be a trustworthy source of protein, the moderate amount of fibre and high nutritious value support its many uses [66]. The consumption of fluted pumpkin may be able to shield people from the negative effects of oxidative stress on their health [65]. Based on mineral contents, it is hypothesized that pumpkin leaf's very high potassium concentration may help to reduce hypertension [51, 52]. Additionally, foods with high iron contents may be utilized to treat anaemia and may help to increase blood production. Additionally, the presence of zinc increases the antioxidant capacity of the food [67], which may help, for instance, with the control of diabetes [60, 68].

2.3 Bitter Leaf (*Vernonia amygdalina*)

Nearly everywhere in the world grows bitter leaf plants for home consumption, particularly in the tropical parts of Africa like Nigeria. It is a natural-regenerating shrub or small tree [69]. They thrive in environments that are fully humid and sunny. It typically grows in the wild or is farmed near lakes, rivers, wooded areas, and open grasslands [70]. Etidot (Efik/Ibibio), onugbu (Igbo and Ukwani), and Ewuro (Yoruba) are some of the native names used in Nigeria. In Nigeria, the renowned bitter-leaf soup contains spices made from *V. amygdalina* leaves [71]. Extracts include a variety of phenolic and phytochemical components. According to a recent study, the nutritional qualities are equivalent to those of mint leaf [72], which is what follows in the text. In comparison to all other vegetables, it has reportedly been shown to be extremely high in folic acids and vitamin C [47]. Traditional herbal therapy employs *V. amygdalina*. According to Uusiku et al. [47], one of this vegetable's many traditional applications is to treat malnutrition, which may be brought on by a lack of certain micronutrients. It is also said to decrease cholesterol [73], cure infections [71], and be helpful in the treatment of diabetes [74, 68].

2.4 Mint Leaf (*Ocimum gratissimum*)

Local names for the smell leaf, also known as mint leaf (*Ocimum gratissimum*), in Nigeria include efinrin nla (Yoruba), alulu-nta (Ukwani), and dadoya (Hausa). It is grown in residential vegetable gardens and has the ability to recover as well as grow wildly. The high antioxidant content of mint leaves helps to flavour food, reduce salt consumption, and give taste to dishes. Mint and bitter leaf are equivalent in terms of the close makeup of their nutritional contents [75]. It is a spice that is often used in African cuisines as well as in tea. Because of the menthol and mint essential oil in mint leaves, aromatherapy has long been connected with them [76]. Asthma, a cough, and a cold have all been treated with it [77]. It is used to treat diabetes [78] and has been suggested as a possible anticancer [79].

2.5 Green Amaranth (*Amaranthus hybridus*)

Local names for *Amaranthus hybridus*, also known as green tete or morogo, include tete (Yoruba), alaiyaho (Hausa), inine (Igbo), and shorokotom yokotor (Ukwani). It may be farmed but also grows naturally in the farms. Amaranthus is very nutrient-dense; both the leaves and the grain amaranth are used as food for both humans and animals. The amaranth was found to be the healthiest in terms of fat content while being equivalent to bitter leaf, mint leaf, and waterleaf in terms of fibres in a research that examined the proximate composition of 10 plants in Southern Nigeria [80]. Due to its iron and antioxidant characteristics, *A. hybridus* has a significant potential to ameliorate micronutrient deficiency [81]. This green vegetable grows organically in the Ndokwa communities' fields in Delta State, Nigeria, making it a free source of food rich in micronutrients. It is important to highlight that zinc affects fertility and that it may help control diabetes [82].

2.6 *Moringa oleifera*

The *Moringa oleifera* plant, often known as the miracle tree, horseradish, or drumstick tree, is grown for both medicinal and culinary uses [83]. In Igbo and Ukwani, respectively, it is known as okwe or okwe-oyibo in Nigeria. It thrives as a tree in both rural and populated areas, although in the latter it is all but gone. *M. oleifera* is recognized as a highly beneficial dietary source to maintain excellent health as well as a conventional treatment for a variety of illnesses, according to Vergara-Jimenez et al. [84]. The leaves of the plant may be used raw, cooked, or powdered while retaining their nutritional value. *Moringa oleifera* leaves provide more vitamin C than lemon and orange, according to Mbailao et al. [85]. Vergara-Jimenez et al. [84] found that these leaves contain a large number of bioactive compounds, including antioxidants. A variety of disorders, such as high cholesterol, hypertension, liver problems, stress, infection, and diabetes, have been studied in relation to the use of moringa oleifera leaf as an alternative therapeutic agent [83, 86–88]. It's noteworthy to note that it has just recently begun to be used to treat a variety of medical conditions [89].

2.7 Lemon Grass (*Cymbopogon citratus*)

In the family Poaceae, commonly referred to as non-native plants, lemon grass is a native species. It is used to prepare chicken dishes and is known as atta-okuku and atta-oyibo in the Ukwani community in Delta State, Nigeria. It grows naturally but is also farmed in other nations, for example, as a cash crop. Lemon grass has equivalent nutritional contents as chives and pumpkin leaves. Particularly, it is said that lemon grass contains around 14 times as much manganese as chives [50]. In fact, as herbal infusion, lemon grass is utilized in tea [90, 50]. In traditional medicine, lemon grass has been used for a very long time to cure a variety of ailments, including diabetes and high cholesterol [91, 92]. Recently, the therapeutic value of lemongrass was evaluated [93].

2.8 *Piper guineense*

Nigeria and other countries in West Africa place a high value on the vegetable species *Piper guineense*, also known as West African black pepper or Ashanti pepper. Its aromatic flavour and medicinal properties have made it a popular spice and condiment in traditional cuisine. Ogunwande et al. [94] found that *P. guineense* extracts had antimicrobial and anti-inflammatory properties. Adedire et al. [95] found that it had promising results as a natural insecticide and larvicide.

2.9 *Gongronema latifolium*

Gongronema latifolium, or Utazi as it is known in Nigeria, is a leafy green vegetable that has both culinary and medicinal applications. *G. latifolium* leaves are frequently used as a seasoning in classic dishes like soups and stews. Because of their antioxidant and anti-inflammatory properties, they may be useful in the treatment of diseases brought on by oxidative stress. Animal research has also shown that *G. latifolium* extracts have antidiabetic effects [96].

2.10 *Murraya koenigii*

Curry leaf, or *Murraya koenigii*, is a common ingredient in Indian cooking thanks to its pungent aroma and flavour. *M. koenigii* has important medicinal properties in addition to its culinary uses. Alkaloids, flavonoids, and phenols are just some of the bioactive compounds found in abundance in the leaves; these compounds have antioxidant, antimicrobial, and anti-inflammatory properties [96]. Because of these characteristics, *M. koenigii* shows promise as a treatment for a wide range of conditions, including diabetes, inflammation, and gastrointestinal issues.

2.11 *Colocasia esculenta*

Starchy root vegetable *Colocasia esculenta*, also called taro or cocoyam, is commonly grown in tropical areas. Many people rely on it as a main source of nutrition because of the high amount of carbohydrates it contains. *C. esculenta* is also well-known for its therapeutic benefits [97]. Its antimicrobial and anti-inflammatory properties have shown promise in wound healing research [98]. It has been found that *C. esculenta* leaves have diuretic and antihypertensive properties, so they are used in traditional medicine as well [99].

2.12 *Amaranthus hybridus*

Slender amaranth, or *Amaranthus hybridus*, is a leafy vegetable that is popular in many parts of the world. Vitamins, minerals, and anti-oxidants abound in it, making it an excellent source of nourishment. Several health advantages have been linked to *Amaranthus hybridus* consumption. Due to its high bioactive compound content [100], it has been shown to have the potential to lower blood pressure and improve cardiovascular health. *Amaranthus hybridus* has also shown promise in warding off chronic diseases thanks to its extracts' anti-inflammatory and anticancer properties [101].

2.13 *Solanum macrocarpon*

The African aubergine, or *Solanum macrocarpon*, as it is more commonly known, is a staple of traditional African cuisine. Because of its high nutrient content and possible health benefits, it is widely consumed. *Solanum macrocarpon* fruits have anti-inflammatory and free radical-scavenging properties because they are high in antioxidants, especially phenolic compounds. *Solanum macrocarpon* is an important component of traditional medicine due to its antimicrobial and antidiabetic properties [102].

2.14 *Pterocarpus mildbraedii*

Pterocarpus mildbraedii, or African padauk, is a tree species with multiple traditional medical applications. Bioactive compounds with anti-inflammatory, antimicrobial, and antioxidant properties have been found in *Pterocarpus mildbraedii* bark, leaves, and roots [103]. These characteristics suggest that it may be useful in the search for natural therapeutic agents to combat inflammatory and infectious diseases. Wound healing properties have also been observed in experimental studies using *Pterocarpus mildbraedii* extracts [104].

2.15 *Basella alba*

Basella alba is a leafy green that is popular in many Asian and African cuisines. It is also known as Malabar spinach and vine spinach. It's a great source of nutrients like B vitamins, minerals, and fibre. The antioxidant and anti-inflammatory properties of *Basella alba* have been studied and reported to have promising medical applications. Traditional medicine has long made use of *Basella alba* for its beneficial effects, particularly on digestion and the treatment of gastrointestinal disorders [105].

2.16 *Heinsia crinita*

Vegetables native to Central and West Africa include the species *Heinsia crinita*, also known as false sesame or Ntonga. *Heinsia crinita* seeds are loaded with healthy nutrients like protein, fat, and minerals [106, 107]. They have traditional medicinal and culinary uses. Antioxidant and anti-inflammatory properties may be present in the seeds [108]. The antimicrobial properties shown by *Heinsia crinita* extracts also make them a candidate for use as a natural preservative.

2.17 *Trigonella foenum-graecum*

Fenugreek, or *Trigonella foenum-graecum*, is a herbaceous plant that has both culinary and medicinal applications [109]. Bioactive compounds like flavonoids and saponins abound in *Trigonella foenum-graecum* seed. Historically, they have been used to aid digestion and reduce inflammation and blood sugar levels [110]. Trigonella foenum-graecum extracts have been studied and reported to have potential cholesterol-lowering effects [111]. In addition, studies on the effects of fenugreek seeds on breastfeeding mothers have been encouraging [112].

2.18 *Taraxacum officinale*

The herbaceous plant known as dandelion, or *Taraxacum officinale*, can be found in many different regions around the world. It has long been used for its medicinal properties. The antioxidant, anti-inflammatory, and hepatoprotective properties of *Taraxacum officinale* can be attributed to the plant's high concentration of bioactive compounds, such as sesquiterpene lactones, phenolic acids, and flavonoids [113]. In addition, diuretic and antimicrobial activities have been observed in tests of *Taraxacum officinale* extracts [114]. Because of these qualities, it has long been used in alternative medicine to treat conditions like liver disease, UTIs, and gastrointestinal distress.

2.19 *Jatropha tanjorensis* Ellis

The plant species *Jatropha tanjorensis* Ellis, also known as bellyache bush, is native to various parts of Africa. For centuries, its healing properties have led to its use in alternative medicine. Anti-inflammatory and analgesic bioactive compounds have been found in *Jatropha tanjorensis* Ellis [115] in both the leaves and the roots. In addition, Ukuuiwe et al. [116] found that *Jatropha tanjorensis* Ellis extracts may have antifungal and antimicrobial activities. Based on these characteristics, it may be useful in relieving pain, reducing inflammation, and fighting off infections.

2.20 *Crassocephalum rubens*

Crassocephalum rubens, also known as red flower rag leaf, is a popular leafy vegetable across many countries in Africa. It's useful as both food and medicine. Vitamins, minerals, and phytochemicals like flavonoids and phenolic compounds can all be found in *Crassocephalum rubens* leaves. They are able to combat free radicals and reduce inflammation [117]. Donga and Chanda [118] found that *Crassocephalum rubens* extracts had a hepatoprotective effect and an anti-diabetic effect in animal studies. These results point to its possible use in the treatment of metabolic disorders and the avoidance of chronic diseases.

2.21 *Solanum americanum*

American nightshade, or *Solanum americanum*, is a plant species found all over the world. It is also known as glossy nightshade. It has been utilized for its therapeutic qualities in conventional medicine. Bioactive compounds with antioxidant, anti-inflammatory, and anticancer activities have been found in *Solanum americanum* leaves and fruits [119]. Kumar et al. [120] review the research on its antimicrobial and wound-healing properties. *Solanum americanum* extracts have also been used traditionally for the management of diabetes, and research has shown that they have antidiabetic effects [121].

2.22 *Celosia argentea*

Cockscomb, also called silver cock's comb, is the common name for the ornamental and edible plant species *Celosia argentea*. It has long been valued for both its edible and medicinal qualities. The antioxidant and anti-inflammatory properties of *Celosia argentea* can be attributed to the presence of bioactive compounds like flavonoids and phenolic acids in the plant's leaves and flowers. Antimicrobial activity against a wide range of pathogens has been suggested for the plant extracts [122]. *Celosia argentea* has also been used to treat gastrointestinal disorders, skin conditions, and cardiovascular diseases in traditional medicine.

2.23 *Solanum nigrum*

Black nightshade, or *Solanum nigrum*, is a wildly distributed plant species that can be eaten and used medicinally. Bioactive compounds with antioxidant, anti-inflammatory, and antimicrobial properties have been found in *Solanum nigrum*'s leaves, fruits, and roots [123]. In addition to its traditional uses for treating diabetes, skin disorders, and gastrointestinal issues, *Solanum nigrum* extracts have shown promising anticancer effects [124]. It should be noted, however, that some *Solanum nigrum* varieties are toxic, so use caution if you plan to eat them.

Some other vegetable species with ethno-medicinal properties are indicated in Table 1 below.

3 Threats of Common Edible Vegetables That Possess Pharmacological Potentials in Nigeria

The accessibility of common edible vegetables in Nigeria that have the capacity to be used in medicine is being put in jeopardy by a variety of variables which encompass everything from the effects of global warming to the effects of social and economic variables. All of these variables have been contributing to a restricted access of these plants as well as their costly nature, both of which may result in major consequences

Table 1 Common edible vegetables that possess pharmacological potentials in Nigeria

S/N	Scientific name	Family	Type of vegetable	References
1	*Talinum triangulare*	Portulacaceae	Leafy	[125]
2	*Cymbopogon citratus*	Poaceae	Grass	[126]
3	*Moringa oleifera*	Moringaceae	Leafy	[127]
4	*Amaranthus viridis*	Amaranthaceae	Leafy	[128]
5	*Ocimum gratissimum*	Lamiaceae	Leafy	[128]
6	*Vernonia amygdalina*	Asteraceae	Leafy	[129]
7	*Telfairia occidentalis*	Cucurbitaceae	Leafy	[129, 125]
8	*Gnetum africanum*	Gnetaceae	Leafy	[130]
9	*Piper guineense*	Piperaceae	Leafy	[126]
10	*Gongronema latifolium*	Asclepiadaceae	Leafy	[131]
11	*Murraya koenigii*	Rutaceae	Curry tree	[130]
12	*Colocasia esculenta*	Araceae	Leafy	[126]
13	*Amaranthus hybridus*	Amaranthaceae	Leafy	[127]
14	*Solanum macrocarpon*	Solanaceae	Fleshy fruit	[132]
15	*Pterocarpus mildbraedii*	Fabaceae	Legumes	[128]
16	*Basella alba*	Basellaceae	Leafy	[127]
17	*Heinsia crinita*	Rubiaceae	Leafy	[126]
18	*Trigonella foenum-graecum*	Fabaceae	Leafy	[130]
19	*Taraxacum officinale*	Asteraceae	Sunflower	[126]
20	*Jatropha tanjorensis ellis*	Euphorbiaceae	Leafy	[131]
21	*Telfairia occidentalis*	Cucurbitaceae	Leafy	[130]
22	*Crassocephalum rubens*	Asteraceae	Leafy	[127]
23	*Solanum americanum*	Solanaceae	Flower	[128]
24	*Celosia argentea*	Amaranthaceae	Leafy	[131]
25	*Solanum nigrum*	Solanaceae	Flower	[130]
26	*Solanum scabrum*	Solanaceae	Flower	[133]
27	*Sesamum radiatum*	Pedaliaceae	Leafy	[133]
28	*Senecio biafrae*	Solanaceae	Leafy	[126]
29	*Myrianthus arboreus*	Urticaceae	Leafy	[126]
30	*Manihot esculenta*	Euphorbiaceae	Root	[134]
31	*Launaea taraxacifolia*	Asteraceae	Lettuce	[135]
32	*Cucurbita pepo*	Cucurbitaceae	Flower	[126]
33	*Crassocephalum crepidioides*	Asteraceae	Leafy	[131]
34	*Corchorus olitorius*	Tiliaceae	Leafy	[127]
35	*Cnidoscolus aconitifolius*	Euphorbiaceae	Leafy	[135]
36	*Ceiba pentandra*	Malvaceae	Leafy	[135]
37	*Bidens pilosa*	Asteraceae	Flower	[133]
38	*Abelmoschus esculentus*	Malvaceae	Flower	[135]
39	*Amaranthus dubius*	Amaranthaceae	Leafy	[128]
40	*Adenia cissampeloides*	Passifloraceae	Leafy	[135]
41	*Gongronema latifolium*	Asclepiadaceae	Leafy	[126]

(continued)

Table 1 (continued)

S/N	Scientific name	Family	Type of vegetable	References
42	*Ipomoea aquatica*	Convolvulaceae	Leafy	[127]
43	*Ipomoea batatas*	Convolvulaceae	Root	[127]
44	*Newbouldia laevis*	Bignoniaceae	Leafy	[131]
45	*Pterocarpus soyauxii*	Papilionaceae	Leafy	[135]
46	*Pterocarpus santalinoides*	Papilionaceae	Leafy	[133]
47	*Vitex doniana*	Verbenaceae	Flower	[130]
48	*Zanthoxylum zanthoxyloides*	Rutaceae	Leafy	[127]
49	*Gnetum buchholzianum*	Gnetaceae	Leafy	[131]
50	*Ocimum basilicum*	Lamiaceae	Leafy	[131]
51	*Uapaca heudelotii*	Euphorbiaceae	Leafy	[130]
52	*Solanum gilo*	Solanaceae	Fleshy fruit	[128]
53	*Zheneria cordifolia*	Cucurbitaceae	Leafy	[126]
54	*Albizia zygia*	Fabaceae	Leafy	[127]
55	*Ficus glumosa*	Moraceae	Leafy	[127]
56	*Erythrina senegalensis*	Fabaceae-papilionaceae	Shrub	[133]

for the general well-being of the general public. In the following subsection, discussions on the challenges that posed threats to the accessibility of common edible vegetables in Nigeria that have the capacity to be used in pharmaceutical uses are made (Fig. 1).

3.1 Global Warming

The availability of widely consumed vegetables in Nigeria has been significantly impacted by global warming. Diminished harvest rates due to temporal shifts, altered precipitation sequences, and severe weather patterns have impacted both the quantity and quality of these plants [136].

Changing patterns of precipitation have been identified to be among global warming's greatest influences the accessibility of common edible vegetables in Nigeria. Desertification and floods, caused by shifts in the frequency and intensity of precipitation, are capable of having a catastrophic impact on harvests. The elevated temperatures along with inadequate precipitation, for instance, have been illustrated to reduce the output of bitter leaf, a popular vegetable in Nigeria [137]. Shifts in precipitation and temperature additionally influence the harvest of African aubergine, a different widely consumed vegetable [138]. These shifts have the potential to lessen supply and raise the prices of these veggies.

The temperature climbs due to global warming also have an effect on the production of crops, alongside shifts in precipitation patterns. Diminished development, delayed blossoming, and fewer harvests are all possible outcomes associated

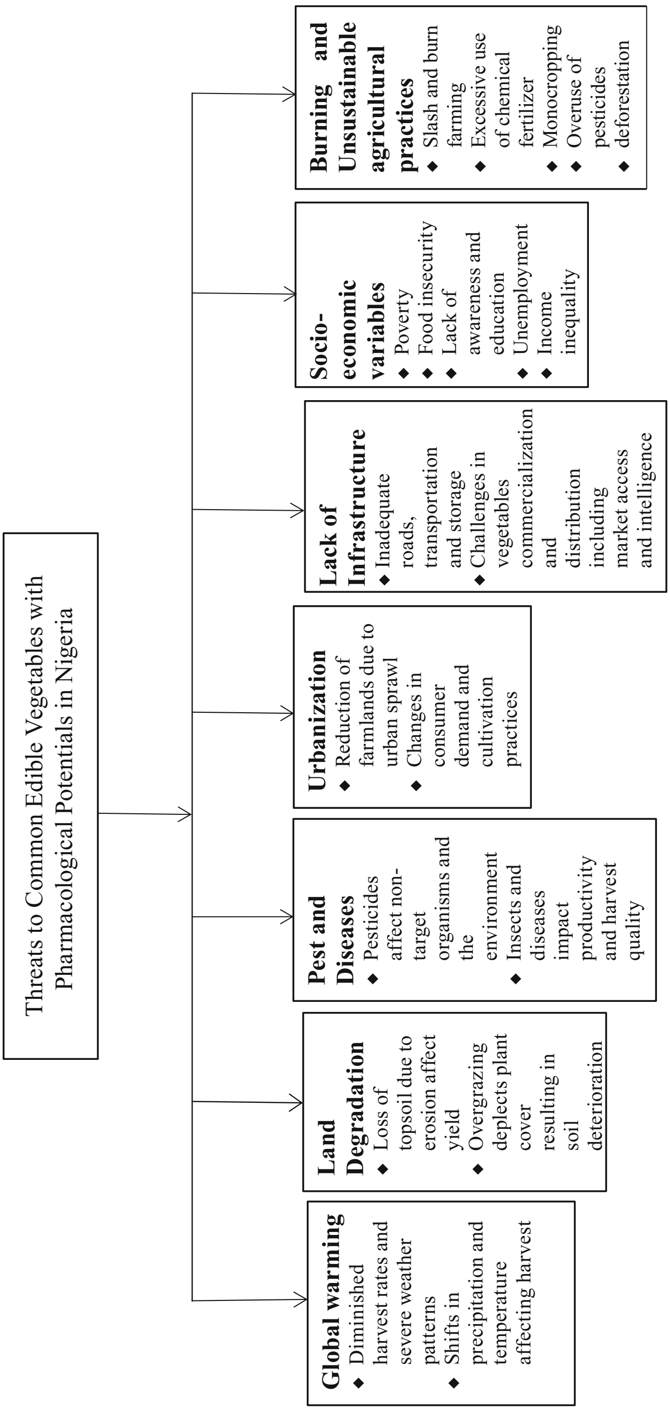

Fig. 1 Threats of Common Edible Vegetables of Pharmacological Potentias in Nigeria

with temperature fatigue, which can be brought on by rising temperatures. Okra, amaranth, and jute mallow are all widely consumed veggies in Nigeria, but research shows that rising temperatures may diminish the yields they produce [139].

Inundations and windstorms are just two examples of catastrophic weather conditions that can significantly reduce harvests [140]. Agricultural products can be harmed by inundation, resulting in a decreased yield, and by violent gales, that has the potential to spoil or otherwise harm agricultural produce. For instance, Babatunde et al. [141] reported that widespread inundation in Nigeria in 2012 wrecked farming operations, which led to shortfalls of pumpkins, waterleaf, and leafy greens.

The soil's state is also impacted by global warming. Destitution of the soil as a consequence of fluctuating temperatures, altered precipitation patterns, or additional grave climate-related circumstances possess the potential to cause a detrimental effect on the production of crops. As an example, a decrease in okra output has proved to be associated to the deterioration of the soil in recent years. Additional grave climate-related circumstances possess the potential to cause a detrimental effect on the production of crops [144]. As an example, a decrease in okra output has proved association to the deterioration of the soil in recent years. Okra is a popular Nigerian vegetable [143], driving up the price of already-expensive vegetables.

3.2 Land Degradation

Deterioration of land is a major ecological challenge that has an immense effect on the accessibility of staple foods like veggies in Nigeria [142]. The clearing of forests, overgrazing, and bad farming techniques are all examples of the degradation of the environment that humans have caused [143–149, 25]. The accessibility and value of commonly consumed veggies in Nigeria may be negatively impacted by these actions due to eroding soil, poor soil fertility, and decreased productivity of crops [150].

When the topsoil is lost due to erosion of the soil, soil that is less productive that is capable of supporting good crop development is left behind. Some of the frequently consumed veggies in the country have had their yields decreased due to the loss of soil [143] which causes the loss of vital nutrients for plants like nitrogen and phosphorus.

The accessible nature of staple fruits and vegetables in Nigeria can be negatively affected by overgrazing. When an excessive number of livestock nibble an area, excessive grazing causes plant life to be depleted and the ground's condition to deteriorate [146]. The depletion of plant cover due to excessive grazing is a risk factor for soil deterioration and decreased fertility. Fruits and vegetables like leafy greens and pumpkins are affected because they need a lot of cover from plants and soil that is fertile [141].

The application of synthetic pesticides and fertilizers are two examples of unsuitable farming techniques that have contributed to land deterioration and reduce the supply of staple vegetables in Nigeria. The acidification of the soil and the accumulation of salt caused by too much application of inorganic fertilizers may diminish fertility in the soil and impact the development of crops [136]. Soil quality and quantity of agricultural produce are likewise adversely influenced by the elimination of insects that are beneficial and microbes caused by the application of pesticides.

3.3 Pest and Diseases

Due to the rise in pests, the use of pesticides including weedicides, insecticides, fungicides, acaricides, etc. are on the rise. Many of these pesticides have the ability to affect non-target organisms and cause an alteration in the environmental matrices [151–162]. Significant financial consequences diminished availability of food and raised cost of food may culminate from the negative effects of pests and diseases on the productivity of crops [163]. Insects and disorders have a significant effect on agricultural production in Nigeria, thereby affecting the accessibility of common vegetables.

The decline in agricultural productivity is a major problem caused by pests and diseases. Tomatoes, the okra plant, and peppers are especially vulnerable to harm from pests like the aphid, mites, and thrips in Nigeria [164]. Insect pests like the aforementioned feed on plants by extracting the juices out of the stems, foliage, and fresh fruit. This kind of feeding behaviour can cause the plant to develop growth defects, lose its vegetative cover, and produce lesser quantity fruits. Vegetable plants are susceptible to harm from a variety of diseases, including bacterial wilt, powdery mould, and blight [165], and can as well affect the quantity and quality of harvests.

The cost of groceries for customers may rise if pests and diseases force agriculturalists to spend more money on the production process. In order to forestall and control pests and diseases, agriculturalists can find themselves spending money on chemicals such as pesticides and fungicides and other prevention techniques. Farmers on small farms might have no accessibility to cost-effective pesticides, making these steps prohibitive. When deployed incorrectly or in excessive amounts, pesticides along with other forms of control measures can be harmful to the well-being of humans and the environment [166].

Pests and diseases also pose a threat to the integrity of our food supply. *Salmonella* and *E. coli* can also pose a threat to the quality and health of fruits and vegetables. Therefore, pest and disease control is fundamental to guaranteeing the freshness, quality, and hygiene of food for shoppers.

3.4 Urbanization

The accessible nature of commonly consumed vegetables in Nigeria has been significantly impacted by urbanization [167]. Vegetable consumer demand has shifted as cities expand, resulting in modifications to the cultivation and distribution process.

The reduction of farmland due to urbanization is a major factor in reducing access to fresh produce. There is less agricultural land readily accessible for farming because of urban sprawl converting farmland into housing and business districts [168].

As a result of this shortage of farmland, vegetable production has fallen in certain regions. For instance, accelerated urbanization has reduced land for farming and lowered vegetable production in the city of Lagos, the nation's most significant urban area [169], thereby reducing the accessibility of fruits and vegetables.

The changing consumer needs for particular vegetable forms is yet a further consequence of urbanization on vegetable being accessible. Exotic and highly valuable vegetables like lettuce, cucumbers, and broccoli are in greater popular demand as urban areas expand. Variations in the cultivation and distribution of vegetables have resulted from the changes in appetite, with producers and marketers making adjustments to satisfy the increasing need for exceptionally valuable vegetables at the expense of more common varieties [170].

Vegetable commercialization and distribution have also been impacted by urbanization [167]. As cities expand, the travel distance between agricultural operations and marketplaces grows, increasing the difficulty of transporting vegetables and fruits in the absence of suitable logistics and storage infrastructure. As a result, deterioration and transit damage have increased losses following harvest. Furthermore, rural farmers have faced growing rivalry as a result of the increased shipment of vegetables to meet the frequent use in urban areas [169].

3.5 Lack of Infrastructure

Common edible vegetables in Nigeria are significantly harder to come by due to the absence of facilities. Transmission and sales of vegetables have been hampered by the country's inadequate facilities, which include roads, conveyance, and warehouses, leading to a shortage of these foods. Significant losses following the harvest are one way in which poor infrastructure reduces the accessibility of vegetables. Because of poor infrastructure, a large portion of Nigeria's vegetable produce is lost after being harvested. The majority of farmers do not have the ability to access adequate facilities for storage, which results in wasted food and crops that never make it to the marketplace. The damage is compounded because vegetables and fruits cannot easily be transported from remote regions to urban centres because of the poor condition of highways and transportation infrastructure [171].

Moreover, because of these challenges, farmers are losing money on their crops and fewer vegetables are available on the marketplace as a whole. Vegetable storage has become problematic due to a scarcity of cold storage spaces, forcing farmers to offload their harvest as soon as possible at a loss [172].

Additionally, the low output of Nigerian vegetable farms can be attributed to the country's inadequate infrastructure. Low yields are a result of the fact that most vegetable farms lack accessibility to essential commodities like water and fertilizer. Vegetable farming in the country hinges on precipitation, which makes it challenging for producing fruits and vegetables throughout the entire year due to a shortage of infrastructure for irrigation. Because of this, vegetable accessibility in the marketplace is now seasonal in nature resulting in higher prices outside of peak growing months [96].

3.6 Socioeconomic Variables

Common edible vegetables in Nigeria are significantly affected by socioeconomic factors. Vegetable availability and demand are affected by socioeconomic variables such as financial status, schooling, and accessibility to marketplaces [163, 173].

The financial standing of household members is a socioeconomic variable that has a substantial effect on the accessibility of vegetables [145]. Because of price, economically disadvantaged Nigerian families have restricted accessibility to many food items, as well as vegetables. Nutritional deficiencies along with additional health issues arise because people in such families eat a narrow diet that is constantly nutrient-wise inadequate. However, because they are able to spend to buy more produce, households with higher incomes have a wider selection of foods available to them [174].

Vegetable accessibility in Nigeria is also affected by schooling level of a family. Vegetables are an essential part of a diet that is nutritious, but they aren't always seen as such. Those with a higher level of education are a greater probability to understand the importance of eating vegetables on a regular schedule. Farmers are encouraged by the rising interest for vegetables to produce greater quantities of these foods for sale [175].

In addition, the accessibility of marketplaces is a crucial socioeconomic component influencing the accessibility of vegetables in Nigeria. Agriculturalists who have convenient accessing in the marketplaces are capable to market their goods to a larger customer base, driving up both requests and sales of vegetables. However, vegetable cultivation and accessibility fall when producers are incapacitated to maximize earnings from marketing their goods [176] due to a lack of marketplace access [177].

The absence of a proper distribution system in farming communities where the majority of vegetables originate is also a factor. Vegetables being available in urban centres, in which they are in tremendous demand, tends to decrease because of the difficulty of transporting them there due to bad road network and coveyance facilities. The farmers' inability to market their crops usually caused charges to drop and output to decrease for farmers growing vegetables in remote locations [178].

3.7 Bush Burning and Unsustainable Agricultural Practices

The intentional burning of plant material in undeveloped land, known as "bush burning," is a common agricultural practice in Nigeria for a number of motives, such as clearing land for cultivation, controlling insects, as well as encouraging regeneration [163, 166]. As a result of the depletion of natural environments, fragile ecosystems are disrupted [179, 180], which has devastating effects on Nigerian vegetables. Vegetable populations are dwindling as a direct result of habitat loss, which is also responsible for the extinction of some plant species [181, 182]. Destroying this cover through bushfires increases the risk of deterioration and decreases soil fertility [181]. This has a chilling effect on the development and output of Nigerian vegetables [168]. Vegetables in Nigeria come in a wide variety of species, and many of them have special health benefits. Thus, the loss of biodiversity is exacerbated by the practice of bush burning as it kills off plants and reduces genetic diversity. Also, vegetables' ability to withstand and adapt to environmental changes is diminished as a result of this loss [183, 184].

The threats to Nigerian vegetables are also exacerbated by ineffective agricultural practices, such as the wasteful use of agrochemicals, poor land management, and excessive exploitation [184]. Excessive tillage, poor soil nutrient administration, and forest loss are all examples of poor land management practices that contribute to soil degradation. Consequently, vegetable growth and yield are negatively impacted because of the soil's diminished organic matter, fertility, and structure [185, 186]. Water depletion and pollution are also exacerbated by unsustainable irrigation practices like wasteful use and inadequate management [187]. As such, vegetable production is hindered by the lack of water, while polluted water supplies endanger both vegetable crops and human health. Furthermore, pest management practices that encourage the excessive and careless use of synthetic pesticides have negative outcomes for people and the planet. Thus, vegetable growth and nutrient content are negatively affected by the pesticide contamination of soil, water, and air [188].

4 Conservation Strategies of Common Edible Vegetables That Possess Pharmacological Potentials in Nigeria

It is essential to preserve common edible vegetables that have the potential to be used in pharmaceutical applications in order to guarantee their continued use in Nigeria. The preservation of these vegetables' genetic diversity, the prevention of their extinction, and the guarantee that generations to come will have access to them are the primary goals of conservation efforts. In this segment, we will talk about a few of the conservation strategies that can be used in Nigeria to protect common edible vegetables that also have pharmacological potential as indicated in the chart below (Fig. 2).

4.1 Seed Banks

The preservation of common edible vegetables in Nigeria relies heavily on the use of seed banks as an important tool. In accordance with Kehinde et al. [189], seed storage facilities contribute to the preservation of the unique genetic makeup found in plant species, which in turn helps to guarantee the continued existence of different kinds of plants. This is crucial as it facilitates plants to become accustomed to evolving conditions in their surroundings as well as create novel characteristics that include durability against illnesses and pests. The diversity of genes additionally provides opportunities for crops to adjust to shifting circumstances in the

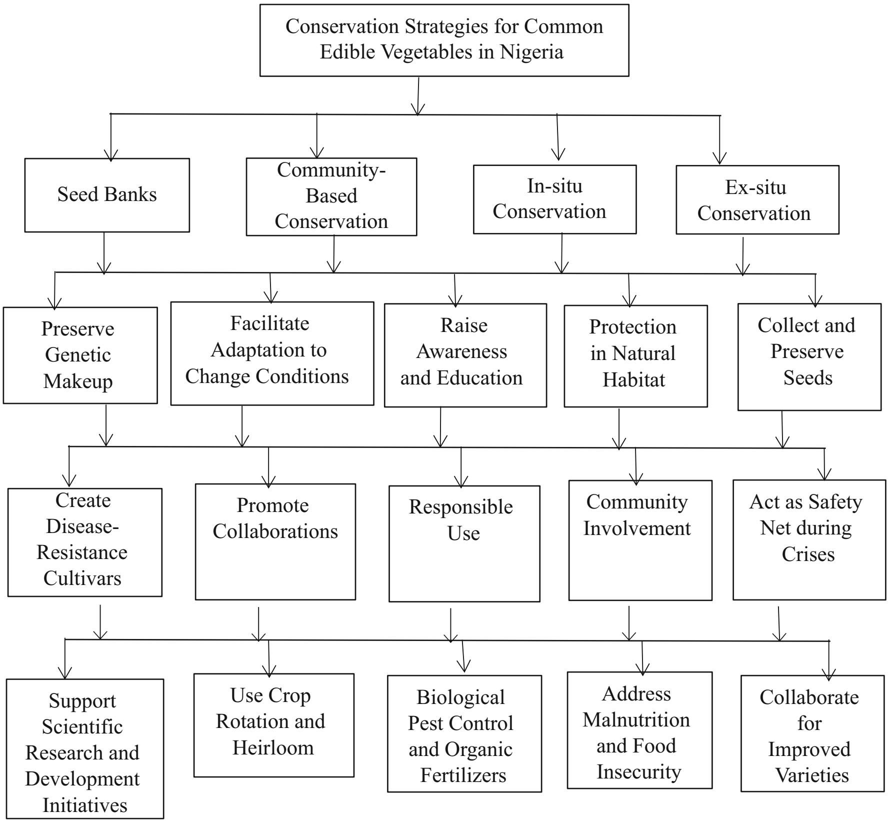

Fig. 2 Conservation strategies for common edible vegetables in Nigeria Conservation strategies of common edible vegetables that possess pharmacological potentials in Nigeria. (Note: The organizational chart provides a hierarchical representation of the conservation strategies, with the main categories branching out into their respective subcategories.)

environment. A seed bank for many different kinds of plants, as well as common edible vegetables, was set up in Nigeria in 2004 by the National Centre for Genetic Resources and Biotechnology (NACGRAB). This bank is known as the National Gene Bank.

Scientists, cultivators, and agriculturalists are all helped greatly from seed banks because they offer a dependable source of gene materials from plants to draw from. This constitutes the single most significant benefit of seed banks. For instance, scientists are able to investigate the genetic makeup of plants, ecological function, and biological processes of various kinds of plants by using seeds that have been stored in seed banks. Geneticists are able to employ seeds to come up with novel cultivars of common edible vegetables that can be immune to pests and diseases, have increased productivity, and are additionally healthy nutrients. These new variations can be produced using seeds. Thus, agriculturalists are able to use seeds to cultivate agricultural products that have improved outputs and are well-suited to the unique circumstances of their particular region.

In order to stop the extinction of vegetative species, seed banks are also an extremely important component. According to Kehinde et al. [189], many plant species including common edible vegetables in Nigeria are on the verge of extinction due to the depletion of their natural habitat, the effects of global warming, as well as additional events carried out by humans. Seed banks play an important role in the conservation of those kinds of plants by preserving the seeds of these plants and making sure that future generations of people will have access to them. For instance, if a kind of plant becomes vanished in its native environment, the seeds of that plant will be reintroduced into its natural environment so that it can continue to exist.

In addition to this, seed banks encourage international collaboration for the purpose of preserving the world's variety of vegetation. According to Doku et al. [190], seed banks are able to swap seeds of many different kinds of plants with one another, which makes it easier for nations to share their expertise and assets. The National Gene Bank at NACGRAB in Nigeria has worked with other internationally recognized organizations, including the International Plant Genetic Resources Institute (IPGRI), to pass on and safeguard the genetic materials of plants. This collaboration has taken place in recent years. This contributes to ensuring that the unique genetic makeup of commonly consumed vegetables is preserved across the world.

4.2 Community-Based Conservation

Okali et al. [191] conducted research into the use of Community-Based Conservation (CBC) in Nigeria to safeguard and oversee a wide variety of flora species, including those that are commonly used in cooking. Because of their often-valuable understanding as well as practices that were handed down across several generations, communities in the rural areas must be involved in initiatives to conserve flora and fauna resources [192–197].

Accordingly, the UNEP [198] argued that CBC serves as a venue for raising awareness and educating people about the value of preserving commonly consumed vegetables. Common edible vegetables, their dietary advantages, and the dangers they encounter are all topics that are discussed and taught in grassroots programmes. By teaching people why and how these vegetables need to be preserved, communities can learn to feel more invested in the conservation effort.

Additionally, Pretty et al. [199] in their study pointed out that CBC encourages the responsible use of commonly consumed vegetables. In many cases, these vegetables are primarily consumed by the local population. CBC guarantees that individual and environmental needs are met through the long-term utilization of these vegetables by engaging them in conservation efforts.

Lastly, CBC aids in generating income and reducing impoverished circumstances in local economies, as stated by the UNEP [198]. The possibilities for seeds and fruit collection and sale can result from efforts to preserve commonly consumed vegetables. Sustainable agricultural practices that raise harvest rates and boost food security can also be fostered by CBC [200]. CBC improves the quality of life for people in rural areas by engaging them in the preservation of widely consumed forest resources [201, 202].

4.3 In Situ Conservation

The term "in situ conservation" refers to the practice of protecting wildlife and their habitats without removing them from their original settings. The conservation of widely available vegetables in Nigeria benefits greatly from this method.

In situ conservation has proved useful in protecting many different kinds of flora species in Nigeria, including some that are commonly eaten, as stated by Onuminya et al. [203]. The authors emphasized that this method includes the conservation and administration of wild plant communities. This method ensures that common edible vegetables will continue to be available for future generations of people by protecting their native environment and genetic variation.

Common edible vegetables are managed and conserved with the help of community members in in situ conservation. Omokanye et al. [204] reported that some local communities in the country have participated in in situ conservation of commonly consumed vegetables. The authors pointed out that these local communities are able to conserve these plants in their communities by keeping these fruits and vegetables in their native environments by employing indigenous expertise and practices.

In situ conservation also encourages the long-term utilization of commonly consumed vegetables. Protecting the areas where these vegetables naturally grow helps to ensure that they will always be around for human consumption. To ensure food security in Nigeria, especially in local communities where these vegetables are a vital source of nutrition, Akpabio and Udo [205] found that the long-term utilization of common edible vegetables is crucial.

In addition, in situ conservation helps to keep cultural artefacts and biological diversity around for future generations [206]. The common edible vegetables studied by Adedayo et al. [207] were found to have major societal and customary worth in Nigeria, in addition to their nutritional benefits. The environment in which they occur and the genetic variation of these vegetables are better protected through in situ conservation efforts.

4.4 Ex Situ Conservation

In addition to in situ conservation, common edible vegetables in Nigeria are also protected through ex situ conservation. Adewale and Ojo [208] describe this strategy as collecting and preserving seeds, plant tissues, and other materials in seed storage facilities, plant museums, and other ecological preservation centres other than of their native environment.

The preservation of commonly consumed vegetables in Nigeria is greatly influenced by ex situ conservation. For starters, it's a safety net in times of crisis, like when extreme weather, maladies, or global warming strike. Avoiding the extinction of key floral species that contribute to our nutritional and food security is made possible through ex situ conservation [209].

Also, ex situ conservation can help revive deteriorating ecosystems. Restoring degraded ecosystems can boost crop yields of staple food crops and increase the variety of plants. Degenerated regions can be revitalized by planting indigenous plant species that thrive in the area's climate and soil conditions with the aid of ex situ conservation [210].

Moreso, germplasm from ex situ conservation efforts can be used in development initiatives. Improved crop varieties with increased yields, resistance to disease and insect pests, and adaptability to shifting climates can't be developed without access to a wide range of genetic resources. Plant breeding programmes can access useful germplasm from ex situ conservation centres, leading to the creation of novel cultivars of crops with enhanced properties [211].

Furthermore, scientific inquiry can benefit from ex situ conservation. Research, including the creation of novel pharmaceuticals, biofuels, as well as other materials, can benefit from the safeguarding of plant remains in ex situ conservation centres. It is possible to learn more about the biology, environment, and genetic variation of commonly eaten vegetables by studying them in ex situ conservation centres [212].

The National Centre for Genetic Resources and Biotechnology (NACGRAB), the Forestry Research Institute of Nigeria (FRIN), and the International Institute of Tropical Agriculture (IITA) are just some of the organizations working on ex situ conservation projects in Nigeria. Common edible vegetables and other plant species are being preserved by these organizations for the benefit of future generations of people [212].

4.5 Promotion of Traditional Farming Practices

Studies have shown that agroforestry systems increase biological diversity and boost the fertility of the soil. In addition to providing cover for crops, preventing the erosion of soil, and encouraging the integration of locally adopted species of naturally occurring plants, planting trees and shrubs alongside crops has many other benefits. Rotations of crops are useful in reviving the fertility of the soil and decreasing diseases transmitted by soil. In addition to aiding in the preservation of the genetic material of plants, this system encourages the use of heirloom cultivars for farming. Biological pest control and fertilizers made from organic materials have been shown to improve agricultural sustainability and mitigate the harmful effects of chemical compounds used as fertilizers on plant life [212, 213].

One group that encourages farmers to stick to tried-and-true methods is the Nigerian Conservation Foundation (NCF). The organization claims that as an outcome of their efforts, rural communities have adopted environmentally friendly agriculture practices like integrating agriculture and forestry, and crop rotation, which have boosted the yields of crops and helped preserve biodiversity.

Consequently, since traditional farming methods encourage environmentally friendly farming and aid in ensuring the preservation of genetic material from plants, they may possess an enormous effect on the preservation of commonly consumed vegetables in Nigeria.

4.6 Plant Breeding and Selection

The preservation of widely consumed vegetables in Nigeria can be greatly aided by plant breeding and selection. These methods focus on creating and selecting novel varieties for agriculture that are efficient, resilient to pests and diseases, and adaptable to a range of environmental conditions.

By encouraging the cultivation of a wide range of cultivars of crops that are suited to the regional climate, plant breeding and selection can aid in the preservation of genetic material from plants. Common edible vegetable yields can be increased and the adverse consequences of global warming mitigated with the assistance of these varieties of crops. Nwalieji et al. [213], for instance, observed that plant breeding to create new cowpea versions increased both the quantity and the quality of the crop in the country.

Plant breeding and selection can also aid in combating malnutrition and extreme food insecurity by creating more nutrient-dense cultivars of crops. Examples of biofortified crops that may assist with addressing deficiencies in micronutrients in at-risk populations include vitamin A cassava, iron pearl millet, and zinc maize [214].

Many organizations and campaigns in Nigeria are working to preserve staple vegetables through breeding and selection programmes. Improved varieties of cassava, yam, cowpea, maize, and soybean have all been created by the International Institute of Tropical Agriculture (IITA), one such organization. Farmers in Nigeria have embraced these varieties of crops in large numbers, which is good news for the future of Nigeria's plant genetic assets as well as food supply.

5 Conclusion

Indigenous vegetables are vital to the Nigerian people's nutritional, physical, and financial well-being. Vitamins, minerals, antioxidants, and other essential nutrients abound in them, making them an important factor in maintaining good health and reducing the incidence of disease. These vegetables are not only an essential source of earnings for many communities, but they also have traditional medicinal uses.

However, there are a number of challenges to preserving and ensuring access to commonly consumed vegetables in Nigeria. The affordability and accessibility of these plants are threatened by factors like climate change, socioeconomic parameters, industrialization, and insufficient propagation practices. Heightened malnutrition and hunger are just two of the ways in which this situation threatens the health of the population as a whole. These problems necessitate the implementation of conservation strategies. The continued existence and variation in genes of common edible vegetables rely on seed banks. They provide a pool of genetic material from which new and better cultivars can be bred and developed. International cooperation and the exchange of genetic resources are two additional benefits brought about by seed banks. In both in situ and community-based conservation strategies, locals are encouraged to take part in the protection and long-term maintenance of staple foods like vegetables. These methods improve the standard of living in rural areas by increasing knowledge, encouraging responsible use, creating economic opportunities, and fostering social cohesion. These methods aid in preserving the native environment and the genetic diversity of these vegetables by engaging local communities in conservation efforts.

In a nutshell, protecting Nigeria's staple food crops is important for a number of reasons, including maintaining food security, improving people's diets and overall health, and maintaining the country's rich cultural heritage. Important steps towards protecting these precious resources for future generations include establishing seed banks, engaging in community-based conservation, and engaging in in situ conservation.

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Sustainable Supply Chain Management in the Herbal Medicine Industry

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Abstract

Supply chain management (SCM) is a strategic approach that orchestrates the movement of goods, products, and services from their origin to where they are utilized or consumed as a linchpin of organizational success, customer satisfaction, and overall business performance. In the herbal medicine industry (HMI) sector, SCM refers to the process of managing the whole range of value chain activities connected to the sourcing, processing, distribution, and use of plants for medicinal purposes. Sustainable supply chain management (SSCM) is the concept of striving to harmonize economic performance with environmental stewardship and sociocultural equity. The chapter proposed a sustainable supply chain system for the HMI that includes: (1) sustainable cultivation and harvesting of herbal resources, (2) sustainable manufacturing of herbal medicine products, (3) the green distribution process for herbal medicine products, (4) eco-friendly wholesalers, retailers, and vendors of herbal medicine products, (5) sustainable herbal medicine utilization, and (6) herbal medicine waste recovery. There is also a need for a supply chain monitoring system for HMI to ensure quality control and improve relationships and experience among and between the different HMI value chain players. In addition to environmental stewardship, social responsibility, economic viability, regulatory compliance, risk management, and innovation and learning. By integrating the principles of SSCM into the HMI, this chapter contributes to the ongoing discourse on sustainability in healthcare and provides insights for practitioners and policymakers within the industry. It is hoped that the chapter will stimulate further research in this area and inform the development of strategies and policies that promote the sustainable growth of the HMI.

Keywords

Supply chain management · Herbal medicine · Value chain · Healthcare · Sustainability

Abbreviations

BOP	Base-of-the-pyramid
CO2	Carbon dioxide
COVID-19	Coronavirus disease 2019
GCLSCN	Green closed-loop supply chain network
HMI	Herbal medicine industry
LSCM	Logistics and supply chain management

PAT	Process analytical technology
PI	Physical internet
PPE	Personal protective equipment
Pt catalyst	Platinum catalyst
QbD	Quality by design and control
SARS-CoV-2	Severe acute respiratory syndrome *coronavirus 2*
SCM	Supply chain management
SSCM	Sustainable supply chain management

1 Introduction

In the contemporary era of globalization, the intricacies of business operations have escalated exponentially, necessitating the evolution of robust and efficient supply chain management (SCM) systems. SCM is a strategic approach that orchestrates the movement of goods and products from their origin to where they are utilized or consumed. This concept, first introduced by Hugos [1], has become a linchpin of organizational success, influencing cost efficiency, customer satisfaction, and overall business performance. However, as the business landscape has evolved, so too have the expectations and responsibilities of organizations. The traditional focus on economic efficiency and profitability has been supplanted by a more comprehensive perspective that incorporates environmental and social sustainability. This paradigm shift is manifested in the burgeoning interest in sustainable supply chain management (SSCM), a concept that strives to harmonize economic performance with environmental stewardship and social equity. This approach, as outlined by Seuring and Müller [2], represents a significant departure from traditional SCM practices, reflecting a broader societal shift toward sustainability and corporate social responsibility.

The advent of digital technologies and data analytics has further transformed the landscape of supply chain management. Recent literature has underscored the increasing role of these tools in shaping SSCM practices. Studies such as those by Ada et al. [3], and Utami, Anna, and Hidayati (2022) highlight the potential of tools such as big data analytics, Industry 4.0 technologies, and value chain analysis in driving SSCM practices. These technologies offer unprecedented opportunities for organizations to monitor and manage their supply chains in real time, enabling them to respond more effectively to changes in demand, supply, and market conditions. They also provide a means for organizations to track and reduce their environmental impact, contributing to the broader goals of sustainable development.

The herbal medicine industry (HMI) is a significant yet growing sector within the broader healthcare industry and is not immune to these trends. Herbal, botanical, or phytomedicine refers to the use of plants (either wholly or in their parts) for medicinal purposes. This practice has been a cornerstone of healthcare for centuries across various cultures and is now gaining recognition in the modern healthcare system due to its natural origins, perceived safety, and potential health benefits [4, 5, 6, 7, 8]. The World Health Organization [9] has highlighted the importance of herbal medicine in global health, noting its role in primary healthcare and its potential to contribute to the achievement of the United Nations Sustainable Development Goals and targets. However, the HMI faces unique challenges in terms of supply chain management. Given the nature of its raw materials, the complexity of production processes, and the stringent regulatory requirements, managing supply chains in the HMI requires a nuanced and sophisticated approach. The global nature of these supply chains, which often involve sourcing ingredients from diverse locations and exporting products worldwide, adds further complexity to this task. Moreover, the industry must contend with the challenges of sustainability, plant systematics, and urbanization and include the need to conserve biodiversity, reduce environmental impact, and ensure the fair and equitable distribution of resources [10–28].

This chapter aims to evaluate SSCM in the HMI by exploring the application and value of the concepts to the herbal medicine value chain to highlight the key aspects, players, and subdivisions. The chapter also examines the development of the HMI by discussing the status, trends, issues, and potential and actual benefits of SSCM. It will also critically analyze the supply chain systems within the HMI by evaluating how SCM systems can be incorporated into herbal medicine resource sourcing, production, processing, sale, use, and waste management as well as identify potential and actual supply chain issues within the HMI and how these issues may impact the system. The chapter will contribute to ascertaining the benefits of sustainable supply chain development and management practices within the herbal medicine sector and concludes by recommending sustainable supply chain systems and management practices that can enhance the efficiency and sustainability of the HMI. The chapter is organized into six main sections. The first section provides a theoretical background on SCM and SSCM, followed by an overview of the HMI in the second section. The third section delves into the specifics of the supply chain in the HMI, while the fourth section discusses the challenges and issues faced by the industry. The fifth section presents the benefits of SSCM in the herbal medicine sector, and the final section offers recommendations for improving supply chain sustainability in the industry. The chapter concludes with a summary of the main findings and implications for future research and practice. Also, by integrating the principles of SSCM into the HMI, this chapter contributes to the ongoing discourse on sustainability in healthcare and provides insights for practitioners and policymakers in the industry. It is hoped that this chapter will stimulate further research in this area and inform the development of strategies and policies that promote the sustainable growth of the HMI.

2 Concept of Supply Chain Management

Supply Chain Management (SCM) is a complex network of interconnected businesses, each playing a crucial role in delivering products and services to the end customer. This chapter explores the roles and networks of key players in SCM, drawing insights from various scholarly articles. The essence of SCM, as described by Hossain et al. [29], is brought about by the optimal management of the interconnectedness of activities involved in the provision of products and services to meet the needs of the end customers and business owners. This network includes various key players, each with a specific role, contributing to the overall performance of the supply chain.

One of the key players in SCM is the supplier, whose role is to provide raw materials for the production process. The study by Hossain et al. [29] emphasizes the importance of strategic supplier partnerships in improving supply chain performance. These partnerships are crucial in ensuring a steady supply of high-quality raw materials, thereby enhancing the efficiency of the production process. Manufacturers, another key player in SCM, are responsible for transforming raw materials into finished goods. The efficiency of the manufacturing process significantly influences the overall performance of the supply chain. The study by Hossain et al. [29] further highlights the importance of effective management of the manufacturing process in improving supply chain performance. Distributors and retailers are also key players in SCM. Distributors ensure that goods reach retailers, who then sell these goods to the end customers. The study by Kumar and Ostor [30] highlights the importance of effective distribution and retailing strategies in ensuring that goods reach the end customer in a timely and efficient manner. The study by Huang et al. [31] adds another layer of complexity to SCM by exploring the influence of horizontal collaborations on vertical collaborations. The authors argue that these collaborations can significantly influence the dynamics of supply chain networks, thereby affecting the overall performance of the supply chain. The concept of supply chain finance introduces financial institutions as another key player in SCM. These institutions provide comprehensive financial services to enterprises in the supply chain, thereby enhancing the financial stability and efficiency of the supply chain. Therefore, SCM involves a complex network of key players, each with a specific role and responsibilities. The effective management of these roles and the relationships between these players is crucial in enhancing the overall performance of the supply chain. Future research should focus on exploring strategies for enhancing the efficiency of these roles and relationships, thereby improving supply chain performance.

3 Key Supply Chain Responsibilities and Roles

3.1 Suppliers in the Supply Chain: The Starting Point and Beyond

Suppliers, the initial entities in the supply chain, are responsible for providing the raw materials or components that are transformed into finished goods by manufacturers. Their role is critical, as their performance directly impacts the efficiency and effectiveness of downstream operations. The primary responsibility of suppliers is to ensure the availability and quality of the raw materials or components they provide. They must manage their operations to meet the demands of their customers in terms of volume, delivery schedules, and quality specifications. This requires effective production planning, quality control, and logistics management [32]. In addition to providing raw materials or vital production components, suppliers often play a role in product innovation. They also contribute to the development of new products and the improvement of existing ones by providing innovative materials or components, or by collaborating with their customers in the design and development process [32]. Suppliers are also critical in the management of supply chain risks. They can cause disruptions in the supply chain due to factors such as production failures, quality issues, or delivery delays. Therefore, they need to implement effective risk management practices to prevent or mitigate these disruptions [33].

In the context of sustainable supply chain management, suppliers are increasingly expected to comply with environmental and social standards. This can involve implementing green practices in their operations, such as reducing waste and emissions and ensuring fair labor practices [34]. Moreover, suppliers can play multiple roles in the supply chain, not just as providers of raw materials or components, but also as employees, entrepreneurs, and even customers, particularly in the base-of-the-pyramid (BOP) segment. This perspective expands the traditional understanding of the BOP segment as merely a customer segment, highlighting the potential for the BOP to contribute to the supply chain in various ways [34]. The choice of particular suppliers within a given supply chain is complex. A timely, and continuous, task that requires efficiency to ensure it is cost-effective to the business practice and ultimately suppliers lead to improved quality production. Optimal supplier choice affects not only product quality but also the formulation of its price and quality [35]. In recent years, the concept of a sustainable closed-loop supply chain has emerged and incorporates the impacts of backup suppliers and lateral transshipment/resupply on the supply chain design. This approach aims to decrease the shortage that may occur during the transmission of produced goods in the network, thereby enhancing the resilience of the supply chain [36].

In summary, suppliers play a fundamental role in the supply chain, with their responsibilities extending beyond the provision of raw materials or components to include aspects such as product innovation, risk management, and sustainability. A clear understanding of the roles of suppliers in the supply chain is crucial for managing and optimizing the supply chain.

3.2 Manufacturers in the Supply Chain: An Integral Role

The supply chain is considered a network of interconnected links within a business, industry, or sector that ultimately determines the production and provision of products and services required by end customers. It is a complex system. At the heart of this system are manufacturers, who play a pivotal role in transforming raw materials into finished goods. This chapter will delve into the multifaceted role of manufacturers in the supply chain, drawing upon recent academic literature to provide a comprehensive understanding of their responsibilities and their impact on the overall supply chain. The primary role of manufacturers is the production of goods. They are the entities that take raw materials and, through a series of processes, transform them into finished products [37]. This could involve anything from assembling parts to creating a car to baking ingredients to produce bread. This transformation process is the core function of manufacturers and the reason for their existence in the supply chain. However, the role of manufacturers extends far beyond the production process. They are also responsible for ensuring that the products they produce meet certain quality standards. This involves testing products, inspecting materials, and monitoring production processes to ensure that the final product is free from defects and meets the specifications set by the company or industry standards [37]. This quality control function is crucial as it directly impacts the reputation of the manufacturer and the satisfaction of the end customer. In addition to production and quality control, manufacturers also play a crucial role in managing production costs. They need to balance the cost of raw materials, labor, and overheads with the need to produce goods that are competitively priced in the market. This often involves finding ways to improve efficiency and reduce waste in the production process [38]. The ability to manage costs effectively is a key determinant of a manufacturer's profitability and competitiveness.

Moreover, manufacturers are responsible for managing the inventory of raw materials as well as the work-in-progress, and finished goods, which involves forecasting demand, scheduling production, and coordinating with suppliers and distributors [39]. Effective inventory management is essential to prevent stockouts or overstocking, both of which can have significant financial implications. Furthermore, manufacturers often work closely with suppliers to ensure a steady supply of raw materials. This can involve negotiating contracts, managing orders, and resolving any issues that arise [40]. The relationship between manufacturers and suppliers is a critical aspect of the supply chain, as any disruptions in the supply of raw materials can have a significant impact on the production process. Importantly, in many industries, manufacturers are also responsible for product innovation. This can involve researching and developing new products, improving existing products, or finding new ways to produce goods more efficiently [38]. Innovation is a key driver of competitive advantage and is increasingly important in today's rapidly changing business environment. Manufacturers play a critical and multifaceted role in the supply chain. Their responsibilities extend beyond the production process to include quality control, cost management, inventory management, supplier relationships, and innovation. Understanding the role of manufacturers in the supply chain is crucial for managing and optimizing the supply chain.

3.3 Distributors in the Supply Chain: The Bridge to the Market

Distributors, often referred to as the bridge between manufacturers and consumers, play a crucial role in the supply chain. They are responsible for the storage, transportation, and delivery of goods from manufacturers to retailers or directly to consumers. Their performance and efficiency significantly impact the overall effectiveness of the supply chain. Distributors are primarily responsible for the distribution of goods. They manage the logistics of getting products from manufacturers to the market, ensuring that goods are delivered in the right quantities, time, and condition [41]. This involves tasks such as inventory management, order processing, transportation management, and customer service. In addition to their logistics role, distributors often provide value-added services such as product customization, marketing support, and after-sales service. They can also play a role in gathering market information and providing feedback to manufacturers, helping them to better understand and respond to market trends and customer needs [42].

Distributors also play a critical role in managing supply chain risks, particularly those related to logistics and transportation. They need to plan and implement contingency measures to deal with potential disruptions such as transportation delays, inventory shortages, or changes in demand [43]. In the context of the COVID-19 pandemic, the role of distributors in the healthcare supply chain has been highlighted. They have been instrumental in ensuring the timely delivery of essential medical supplies, demonstrating the importance of resilience and adaptability in the face of unexpected disruptions [44]. Moreover, with the increasing emphasis on sustainability in supply chain management, distributors are expected to adopt green practices in their operations. This can involve measures such as optimizing transportation routes to reduce fuel consumption, using environmentally friendly packaging materials, and implementing waste reduction and recycling programs [43]. Distributors play a vital role in the supply chain, serving as the bridge between manufacturers and the market. Their responsibilities extend beyond logistics to include value-added services, risk management, and sustainability. Understanding the role of distributors in the supply chain is crucial for managing and optimizing the supply chain.

3.4 Retailers in the Supply Chain: The Final Link

Retailers, often the final link in the supply chain, play a critical role in delivering goods to the end consumer. They are responsible for the sale of goods, often in small quantities, to the consumer. Their performance and efficiency significantly impact the overall effectiveness of the supply chain. Retailers are primarily responsible for the sale of goods. They manage the interface with the end consumer, ensuring that goods are available in the right quantities, time, and condition. This involves tasks such as inventory management, order processing, customer service, and marketing [45]. In addition to their sales role, retailers often provide value-added services such as product customization, after-sales service, and gathering market information. They also provide feedback to manufacturers, helping them to better understand and respond to market trends and customer needs [46]. Retailers also play a critical role in managing supply chain risks, particularly those related to inventory management and demand forecasting. They need to plan and implement measures to deal with potential disruptions such as demand fluctuations, inventory shortages, or changes in consumer preferences [46].

In the context of the COVID-19 pandemic, the role of retailers in the healthcare supply chain has been highlighted. They have been instrumental in ensuring the timely delivery of essential medical supplies, demonstrating the importance of resilience and adaptability in the face of unexpected disruptions [45]. Moreover, with the increasing emphasis on sustainability in supply chain management, retailers are expected to adopt green practices in their operations. This can involve measures such as reducing waste, promoting recycling, and selling environmentally friendly products [45].

Furthermore, retailers are increasingly involved in the recycling process, contributing to the development of a closed-loop supply chain. This involves collecting used products from consumers and returning them to the supply chain, either by selling them as second-hand goods or by sending them back to manufacturers for remanufacturing [46]. Retailers play a vital role in the supply chain, serving as the final link between manufacturers and consumers. Their responsibilities extend beyond sales to include value-added services, risk management, and sustainability. Understanding the role of retailers in the supply chain is crucial for managing and optimizing the supply chain.

3.5 Customers in the Supply Chain: The Ultimate Recipient

Customers, often the ultimate recipients in the supply chain, play a critical role in shaping the supply chain's operations. They are the end-users of the goods and services produced and distributed by the supply chain. Their preferences, behaviors, and feedback significantly impact the overall effectiveness of the supply chain. Customers are primarily responsible for the consumption of goods and services and are referred to as the destination of the supply chain as their demand drives the production, distribution, and sales activities of the supply chain. This involves aspects such as purchasing behavior, product preferences, and feedback on product quality and service [41]. In addition to their consumption role, customers often provide valuable feedback and market information. They can play a role in shaping product development, marketing strategies, and service improvements by providing feedback on their needs, preferences, and experiences [46]. Customers also play a critical role in managing supply chain risks, particularly those related to demand fluctuations and changes in consumer preferences. Their purchasing behavior can be influenced by various factors, including price, product quality, brand reputation, and external factors such as economic conditions and social trends [46].

In the context of the COVID-19 pandemic, the role of customers in the supply chain has been highlighted. Their changing needs and behaviors, driven by factors such as lockdown measures and health concerns, have led to significant shifts in supply chain operations, including increased demand for online shopping and home delivery services [47]. Moreover, with the increasing emphasis on sustainability in supply chain management, customers are playing a growing role in driving sustainable practices. Their demand for environmentally friendly products and their willingness to support companies with sustainable practices are influencing companies to adopt green practices in their operations [48]. Customers play a vital role in the supply chain, serving as the ultimate recipient of the goods and services produced by the supply chain. Their responsibilities extend beyond consumption to include providing feedback, managing risk, and driving sustainability. Understanding the role of customers in the supply chain is crucial for managing and optimizing the supply chain.

4 Herbal Medicine Industry Development: Status and Trends

The HMI represents a significant segment of the global healthcare framework. It involves the use of whole plants, plant parts, or extracts that may be consumed wholly, used to prepare a concussion or infusion, or applied to the ailing body parts to treat or prevent diseases and enhance overall health and wellbeing. In recent years, the industry has witnessed a resurgence, fueled by an increasing consumer preference for natural and organic remedies, a growing awareness of the potential side effects of synthetic drugs, and a renewed interest in traditional and alternative medicine systems [49]. The importance of the HMI is underscored by its vast potential in addressing a multitude of health conditions, its role in healthcare systems, especially in low- and middle-income countries, and its contribution to economic development [49]. The industry is not only a source of healthcare but also a provider of livelihoods and a contributor to biodiversity conservation [49]. Consumer satisfaction is a critical aspect of the industry, influenced by factors such as perceived safety, service quality, convenience, and pricing [50]. As the industry grows, so does the need for quality control and standardization of herbal extracts or products to ensure their safety, quality, and efficacy [49].

4.1 Brief Historical Context

The use of plants for healing purposes is as old as human history and origin and not only predates modern medicine but is considered the foundation of the practice. The medicinal value of these plants is linked to their phytochemical constituents. Many old and new conventional medicine drugs are from plant sources which is a testament to the enduring value of medicinal plants. Aspirin, for instance, was developed from compounds found in the bark and leaves of the willow tree. Similarly, the heart medicine digoxin was derived from foxglove. Quinine, which is used to treat malaria, is derived from the bark of the *Cinchona* tree species, while morphine is derived from the opium poppy plant (*Papaver somniferum*).

Herbal medicine has a rich history that spans different cultures and traditions. In ancient Egypt, one of the earliest civilizations to have a structured medical system, herbs were used extensively for their medicinal properties. Papyrus scrolls from as far back as 1500 BC provide evidence of an organized medical system, where physicians used herbs like garlic, juniper, cannabis, aloe, and mandrake to treat ailments. These ancient Egyptian practices influenced Greek and Roman medicine and had a significant impact on the later development of Western medicine. In the East (Orient), traditional medicine systems like Ayurveda in India and Traditional Chinese Medicine have a long history of using herbs for treatment. These systems have a holistic approach, viewing the body as an interconnected whole and using herbs to restore balance. For instance, Ayurveda, which dates back over 3000 years, uses a wide variety of plants like turmeric, neem, and ashwagandha for their therapeutic properties [51]. In Traditional Chinese Medicine, which has a history of over 2000 years, herbs are a fundamental element. The Shen Nong Ben Cao Jing, a classic Chinese text on materia medica compiled around 200–300 AD, records the use of 365 medicinal plants. Today, herbs like ginseng, goji berry, and astragalus are widely used in Chinese medicine [51].

In more recent times, the use of herbal medicine has seen a resurgence. This is driven by factors such as the rising cost of prescription medications, side effects of synthetic drugs, and a growing interest in natural remedies. Today, herbal medicine is not only used for healthcare but is also a significant contributor to economic development and biodiversity conservation [52]. Despite the long history and widespread use, the HMI faces challenges such as standardization, quality control, and safety regulations. However, advancements in technology, such as the development of electrospinning technology for incorporating herbal medicine into nanofibers, offer new possibilities for the industry [53]. The history of herbal medicine is a testament to the enduring value of nature's pharmacy. As we move forward, it is crucial to integrate this traditional wisdom with modern scientific methods to ensure the safety, efficacy, and sustainability of herbal medicine.

4.2 Current State of the Herbal Medicine Industry

4.2.1 Market Size and Key Players

The HMI, a significant segment of the global healthcare sector, has been enjoying substantial growth over the past few years that is likely driven by a confluence of factors, including an increasing consumer preference for natural and organic remedies, a growing awareness of the potential side effects of synthetic drugs, and a renewed interest in traditional and alternative medicine systems.

The market size of the HMI is challenging to estimate due to its diverse nature, encompassing everything from small-scale local businesses to multinational corporations. However, various market research reports suggest that the industry is worth billions of dollars globally, with a steady annual growth rate. This growth is expected to continue as more consumers turn to herbal remedies for their health needs. Key players in the industry range from traditional herbal medicine practitioners who have been practicing for generations to modern companies that combine traditional knowledge with scientific research to produce and distribute herbal products on a large scale. These companies operate in various segments of the industry, including cultivation, extraction, processing, and distribution of herbal medicines. For instance, in India, *Taverniera cuneifolia*, a potent substitute for Licorice, is used extensively for its medicinal properties. The plant has expectorant, anti-ulcer, anti-inflammatory, blood purifier, wound healing, etc. properties. Recent research has focused on identifying specific functional genes involved in the biosynthesis of essential secondary metabolites in this plant, highlighting the role of scientific research in advancing the HMI [54]. However, it's important to note that the HMI is not without its challenges. Issues such as quality control, standardization of products, and regulatory hurdles pose significant challenges to the growth and development of the industry. It is crucial to understand these challenges and work toward solutions that ensure the safety, efficacy, and sustainability of herbal medicine.

4.2.2 Consumer Trends and Demands

Consumer trends in the HMI are shaped by a variety of factors, including health and wellness trends, cultural and traditional beliefs, and the perceived benefits and risks of herbal versus synthetic medicines.

1. **Preference for Natural Products**: There is a growing consumer preference for natural and organic remedies [55, 56]. This trend is driven by increasing awareness of the potential side effects of synthetic drugs and a desire for holistic and preventive healthcare approaches.
2. **Personalized Herbal Treatments**: Consumers are increasingly interested in personalized herbal treatments that are tailored to their specific health needs and conditions. This trend is facilitated by advances in technology and research, which allow for more precise customization of herbal remedies.
3. **Sustainability and Ethical Sourcing**: Herbal medicine consumers are becoming more conscious of the environmental and social impacts of the herbal products they consume. There is a growing demand for herbal products that are sustainably sourced and produced ethically.
4. **Quality and Safety**: As the HMI grows, so does consumer demand for high-quality and safe products. Consumers are looking for products that are tested for safety and efficacy, and that meet regulatory standards.
5. **Education and Information**: Consumers are seeking more information about herbal medicines, including their uses, benefits, and potential risks. There is a demand for reliable and accessible sources of information, as well as education about how to use herbal medicines safely and effectively [57].
6. **Brand Awareness and Preference**: In some markets, such as India, there is a growing awareness and preference for specific brands of herbal personal care products [58].

These trends reflect a broader shift toward more natural, personalized, and conscious consumption in the healthcare sector. They present both opportunities and challenges for the HMI and will likely continue to shape the industry in the coming years.

4.2.3 Regulatory Environment

The regulatory environment for the HMI is complex and varies by country and region. In many places, herbal medicines are regulated as dietary supplements rather than drugs and are not subjected to the same rigorous testing and quality control standards as conventional pharmaceuticals. However, there is a growing push for stricter regulations to ensure the safety and efficacy of herbal products, as well as to combat the issue of counterfeit or adulterated products in the market [59]. Standardization of herbal products is a critical aspect of these regulations. For instance, herbal antihypertensive formulations are being standardized according to Ayurvedic Pharmacopeia and WHO guidelines to provide safe and effective medicines to patients [60]. Supporting local pharmaceutical manufacturing, including the HMI, is also seen as a significant step toward reducing the cost of medical bills [61]. Furthermore, understanding the regulatory networks of herbal medicine biosynthesis, such as ginsenoside biosynthesis in ginseng, is crucial for maintaining the quality and efficacy of herbal products [62].

4.3 Key Developments in the Herbal Medicine Industry

The HMI has witnessed several significant developments in recent years, particularly in the areas of disease treatment, consumption trends, and marketing strategies. This review highlights some of these key developments based on recent scientific literature.

4.3.1 Therapeutic Applications of Herbal Medicine

Recent research has underscored the potential of herbal medicine in treating complex diseases. For instance, a study by Liu et al. [53] highlighted the role of Chinese herbal medicine and its active compounds in treating diabetic and kidney diseases. The study suggested that these herbal medicines could ameliorate diabetic kidney injury by regulating autophagy, a process that maintains cellular homeostasis and energy balance [53]. Similarly, a study by Li et al. [43] discussed the potential of herbal medicine in treating atherosclerosis, a major cause of cardiovascular disease. The researchers proposed that herbal medicines could help restore mitochondrial function and intestinal microbiota disorders, offering new insights into the prevention and treatment of atherosclerosis [43].

4.3.2 Consumption Trends in Herbal Medicine

A survey on the consumption of herbal medicine in Korea revealed interesting trends in the industry. The study by Park et al. [63] found that the consumption of herbal medicine and the size of the HMI has continued to expand. This growth was attributed to the large-scale branding of Korean Medicine institutions and the expansion of health insurance coverage [63].

4.3.3 Marketing Strategies in the Herbal Medicine Industry

The development and marketing of new herbal medicine products have also seen significant advancements. A study by Ganeshfit Company [64] discussed the development of a marketing strategy for a new anti-inflammatory herbal medicine. The researchers emphasized that the development of new herbal medicine products requires a comprehensive marketing strategy, including market segmentation, targeting, positioning, and the marketing mix (Ganeshfit Company, [64]). In summary, these developments highlight the diverse applications of herbal medicine in treating various diseases, the growing acceptance of herbal medicine in healthcare systems, and the importance of strategic marketing in the HMI. As the industry continues to evolve, it will be crucial to stay abreast of these developments to leverage opportunities and address challenges effectively.

5 Potential and Actual Supply Chain Issues Within the Herbal Medicine Industry

Some issues in the herbal medicine supply chain include the following.

5.1 Distribution and Quality Control

The distribution process is a critical component of integrated product SCM in the HMI. This process involves the processing, storage, and distribution of high-quality herbal products, making it a crucial aspect of the industry. However, the distribution of counterfeit or substandard products by unauthorized or unregistered distributors poses a significant challenge. These fake products not only undermine the integrity of the HMI but also pose serious health risks to consumers [65]. In addition to distribution, quality control is another significant issue in the HMI. The quality of herbal products is influenced by various factors, including the quality of raw materials, manufacturing processes, and storage conditions. For instance, the formulation of herbal tea infusions with botanical extracts presents several challenges, such as sourcing ingredients in appropriate quantity and quality, as well as developing quality control methods and addressing processing constraints like hygroscopicity, homogeneity of distribution, solubility, packaging machinability, and dispersibility [66]. Furthermore, the development of herbal cosmeceuticals in developing countries faces several difficulties. These include the lack of trained personnel with entrepreneurial skills, low levels of public-private partnerships, regulatory barriers, unavailability of accurate methods for the characterization/authentication of raw materials, unsustainable collection practices of medicinal plants, quality control issues, insufficient manufacturing capacity, and lack of financing [67]. These issues highlight the need for robust quality control and distribution systems in the HMI. Addressing these issues requires a comprehensive approach that includes strengthening regulatory frameworks, enhancing public-private partnerships, investing in research and development, and promoting sustainable practices.

5.2 Supply Chain Design and Sustainability

In the design of supply chains, different goals are considered. Sustainability in supply chains plays an important role in goal setting to reduce economic, social, and environmental risks. Therefore, it promotes optimization of and stability within the supply chain network. In addition, a lesser-known aspect of sustainability, the political aspect, has been added to the previous aspects [68]. The design of a sustainable supply chain involves the consideration of various factors, including economic, social, environmental, and political aspects. For instance, in the case of the supply chain for COVID-19 drugs, a mathematical model was developed to optimize the stability of the supply chain network. This model considers multiple functional objectives including economic, political, and environmental goals for a metaheuristic algorithm designed to find an optimal configuration [68]. Whereas the green closed-loop supply chain network (GCLSCN) mainly considers trade-offs between environmental and economic aspects. This model was used during the COVID-19 pandemic to address hygiene costs like disinfection and sanitizer costs, as well as personal protective equipment (PPE) costs, cost of COVID-19 tests, and education, medicines, and vaccination costs [47]. In the context of the HMI, the design of a sustainable supply chain is crucial. The supply chain should not only ensure the efficient distribution of herbal products but also promote sustainable practices, such as the sustainable harvesting of medicinal plants. A study on the dual trade of *Nardostachy jatamansi* and *Fritillaria cirrhosa* and Tibetan people's harvesting behavior revealed that the sustainability of the former species was largely dependent on the latter one (Jingjing [69]).

5.3 Physical Internet and Supply Chain Management

The Physical Internet (PI) is a recent innovation in logistics and supply chain management (LSCM) that aims to improve the economic, environmental, and social ways that physical objects are designed, supplied, and used. The PI is described as "a global logistics system based on the interconnection of logistics networks by a standardized set of collaboration protocols, modular containers, and smart interfaces for increased efficiency and sustainability" [70]. Considering that everything in the physical world, in the future would be replicated in the digital space, PI would play a greater role [71]. In the context of the HMI, the concept of PI can potentially revolutionize supply chain management. By leveraging the principles of PI, the industry can enhance the efficiency and sustainability of its supply chains. This can involve the use of standardized protocols and smart interfaces to facilitate the global movement and distribution of herbal products. Furthermore, the use of digital technologies can enable the replication of physical supply chain processes in the digital space, thereby enhancing visibility, traceability, and control.

6 Sustainable Supply Chain in Herbal Medicine Industry

The incorporation of a formal supply chain system into the HMI has the potential to greatly improve the value and benefits of the sector. The proposed HMI supply chain (Fig. 1) will have the following six core parts:

- Sustainable Cultivation and Harvesting of Herbal Resources
- Sustainable Manufacturing of Herbal Medicine Products
- Green Distribution Process for Herbal Medicine Products
- Eco-friendly Wholesalers, Retailers, and Vendors of Herbal Medicine Products
- Sustainable Herbal Medicine Utilization
- Herbal Medicine Waste Recovery

6.1 Sustainable Cultivation and Harvesting of Herbal Resources

The proposed first component of this sustainable herbal medicine supply chain will involve the implementation of sustainable agricultural practices to ensure the long-term availability of medicinal plants. Sustainable cultivation and harvesting of herbal resources are crucial to the preservation of these resources and the environment. It involves practices such as in situ and ex situ conservation, cultivation practices, and good agricultural practices [72]. In situ conservation refers to the conservation of species in their natural habitats, while ex situ conservation involves the conservation of species outside their natural habitats, for example, in botanical gardens or seed banks [72]. Cultivation practices refer to the growing of medicinal plants in a controlled environment to reduce the pressure on wild populations. Good

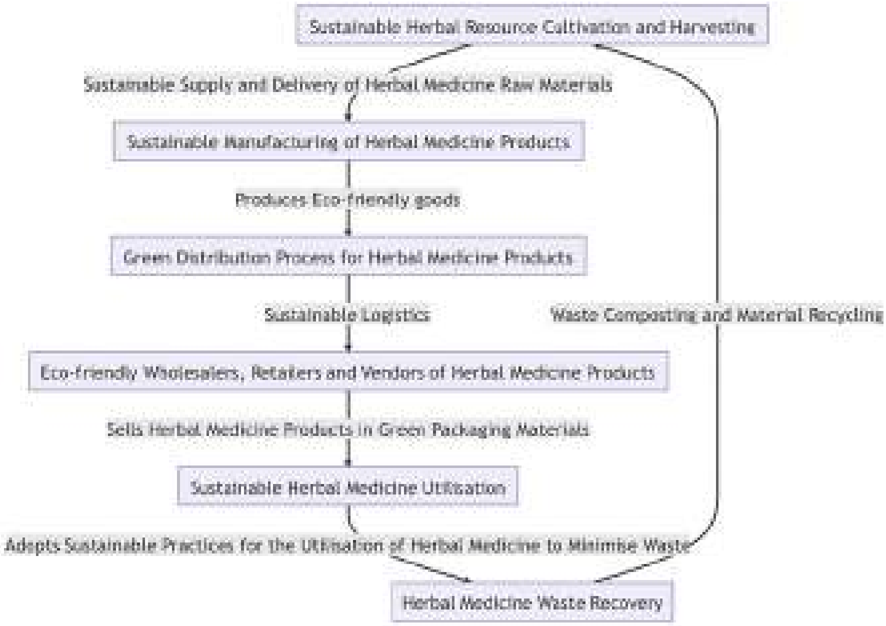

Fig. 1 Proposed herbal medicine sustainable supply chain

agricultural practices involve the application of methods and techniques that minimize environmental impacts, maintain the health and welfare of workers, and ensure the quality and safety of herbal products [72].

Moreover, sustainable cultivation and harvesting also involve the application of biotechnical approaches such as tissue culture, micropropagation, synthetic seed technology, and molecular marker-based approaches. These techniques can improve the yield and modify the potency of medicinal plants, thereby enhancing their therapeutic effectiveness [72]. In the context of herbal medicine, sustainable cultivation, and harvesting are particularly important. For example, Barbados aloe (*Aloe vera* (L.) Burm. F.), a plant traditionally used for healing in natural medicine, is now attracting great interest in the global market due to its bioactive chemicals. The most important factors affecting aloe production include propagation techniques, sustainable agronomic practices, and efficient post-harvesting and processing systems [73]. Sustainable cultivation and harvesting of herbal resources are critical components of a sustainable herbal medicine supply chain. They ensure the long-term availability of medicinal plants and contribute to the preservation of the environment.

6.2 Sustainable Manufacturing of Herbal Medicine Products

The second component of the proposed sustainable herbal medicine supply chain will involve the application of sustainable practices in the manufacturing process to ensure the production of eco-friendly goods. Sustainable manufacturing of herbal medicine products is a critical aspect of the supply chain. It involves the use of environmentally friendly methods and technologies to transform raw herbal materials into finished products. This process should minimize waste, reduce energy consumption, and limit the use of non-renewable resources [74].

In the context of herbal medicine, sustainable manufacturing can involve the use of innovative process design methods like Quality by Design (QbD) and quality control in the form of Process Analytical Technology (PAT). These methods can help to ensure the quality and safety of herbal products [74]. Moreover, sustainable manufacturing can also involve the use of standardized laboratory equipment combined with physicochemical predictive process modeling. These techniques can improve the efficiency of the manufacturing process and reduce the cost of production [74]. For example, in the production of *Indigo naturalis*, a traditional Chinese medicine used for various ailments, sustainable manufacturing practices are crucial. The manufacturing process involves the use of environmentally friendly methods to transform the raw Indigo naturalis into a finished product. This process minimizes waste and reduces energy consumption (Yang [75]). Sustainable manufacturing of herbal medicine products is a critical component of a sustainable herbal medicine supply chain. It ensures the production of eco-friendly goods and contributes to the preservation of the environment.

6.3 Green Distribution Process for Herbal Medicine Products

The proposed third component of the sustainable herbal medicine supply chain is a critical aspect that requires strategic planning and implementation. This involves the sustainable logistics of distributing herbal medicine products using low-carbon emission processes. To enhance the sustainability of the herbal medicine supply chain, it is proposed that companies should adopt green distribution strategies. This could involve the use of vehicles that emit fewer greenhouse gases or the optimization of delivery routes to reduce fuel consumption and emissions. This not only contributes to environmental sustainability but also can lead to cost savings in the long run [76].

In the context of the COVID-19 pandemic, the importance of green distribution has been highlighted. Companies should consider using digital platforms to facilitate the distribution of herbal medicine products. This can reduce the need for physical transportation, thus reducing carbon emissions [77]. Moreover, companies should also consider collaborating with local suppliers and distributors. This can reduce the distance that products need to be transported, thus reducing carbon emissions. In addition, companies should also consider using environmentally friendly packaging materials to further reduce the environmental impact [78]. Finally, the adoption of green distribution strategies is a critical component of a sustainable supply chain for herbal medicine products. It not only contributes to environmental sustainability but also can lead to cost savings and enhanced customer satisfaction.

6.4 Eco-friendly Wholesalers, Retailers, and Vendors of Herbal Medicine Products

The fourth proposed component of the sustainable herbal medicine supply chain will involve selling herbal medicine products in green packaging materials that minimize loss and environmental damage.

Eco-friendly wholesalers, retailers, and vendors play a crucial role in the sustainable supply chain of herbal medicine. They are the link between the manufacturers and the consumers, and their practices significantly impact the overall sustainability of the supply chain. One of the key practices is the use of green packaging materials. Traditional packaging materials often lead to significant environmental pollution due to their non-biodegradable nature. Green packaging, on the other hand, is made from renewable, biodegradable materials, which significantly reduces the environmental impact [79]. Moreover, eco-friendly wholesalers, retailers, and vendors can also contribute to sustainability by reducing waste. This can be achieved by implementing practices such as just-in-time inventory management, which ensures that only the necessary number of products is in stock, thereby reducing waste from expired or unsold products [80].

Furthermore, these stakeholders can also promote sustainability by educating consumers about the importance of sustainable practices in the use of herbal medicine. This can include information about the sustainable sourcing of the products, the environmental benefits of using green packaging, and the importance of proper disposal of used products [80]. Eco-friendly wholesalers, retailers, and vendors are a critical component of a sustainable herbal medicine supply chain. Their practices significantly impact the overall sustainability of the supply chain, and they play a crucial role in promoting sustainability to consumers.

6.5 Sustainable Herbal Medicine Utilization

The fifth component of the proposed sustainable herbal medicine supply chain will involve adopting sustainable practices for the utilization of herbal medicine to minimize waste and ensure the efficient use of resources. Sustainable utilization of herbal medicine is a critical aspect of the supply chain. It involves the efficient use of medicinal plants to ensure that the benefits are maximized while the negative impacts on the environment and biodiversity are minimized [81]. One of the key aspects of sustainable utilization is the efficient use of medicinal plants. This involves using all parts of the plant effectively and reducing waste. For instance, the Garo tribe in Bangladesh, which is heavily dependent on natural resources, has demonstrated how to effectively use various parts of plants for food, energy, timber, and healthcare. Their practices reflect a deep understanding of the plants and their uses, which contributes to the sustainable utilization of these resources [81]. Another important aspect is the use of traditional knowledge in the utilization of herbal medicine. Traditional knowledge, which has been accumulated over generations, often contains valuable information about the sustainable use of medicinal plants. For example, the Garo tribe's practices are based on their indigenous knowledge, which has been passed down through generations. This knowledge can be an important tool for promoting the sustainable utilization of herbal medicine [81]. Furthermore, sustainable utilization also involves the responsible use of herbal medicine. This includes following recommended dosages and avoiding overuse, which can lead to the depletion of medicinal plant resources. It also involves promoting the use of sustainable and eco-friendly practices among consumers [82]. Sustainable herbal medicine utilization is a critical component of a sustainable herbal medicine supply chain. It involves the efficient and responsible use of medicinal plants, the application of traditional knowledge, and the promotion of sustainable practices among consumers.

6.6 Herbal Medicine Waste Recovery

The final component of the proposed sustainable herbal medicine supply chain involves the management of waste produced by the HMI. One of the ways this can be achieved is through the process of pyrolysis. Pyrolysis is a promising method to dispose of biomass and organic waste such as herbal medicine waste. A study by Lee et al. [83] applied an activated carbon-supported Pt catalyst and carbon dioxide (CO_2) to the pyrolysis of real herbal medicine waste. The study found that the generation of benzyl species, which are harmful to the environment and human health, was suppressed. Another study by Mi et al. [84] also used pyrolysis to create activated carbon from Chinese herbal medicine waste. The study found that herbal medicine waste is a good precursor material for preparing activated carbon. In addition to pyrolysis, the recovery of herbal medicine waste can also be achieved through the inhibition of harmful chemicals. A study by Oh et al. [85] found that several single herbal medicines have the potential to inhibit the activation of harmful chemicals in an in vitro AKI-mimicked condition. Lastly, a study by Jin et al. [77] found that the potential mechanisms involved in the treatment of COVID-19-related pulmonary fibrosis and SARS-CoV-2 infection involve multiple components, targets, and pathways, indicating the potential of herbal medicine waste in the treatment of diseases. In summary, the recovery of herbal medicine waste is a crucial component of a sustainable herbal medicine supply chain. It involves the management of waste through methods such as pyrolysis and the inhibition of harmful chemicals. It also involves the potential use of herbal medicine waste in the treatment of diseases.

7 Herbal Medicine Supply Chain Monitoring System

There is a need for a supply chain monitoring system for HMI to ensure QbD and improve relationships and experience among and between the different HMI value chain players (Fig. 2).

1. **Sustainable Management and Conservation of Herbal Resources**: This is the first and foundational step in the supply chain. It involves the creation of policies and strategies that promote the sustainable use and conservation of medicinal plant resources. These policies should be based on scientific research and best practices in the field. They should address issues such as overharvesting, habitat conservation, and the promotion of biodiversity. Training programs should be

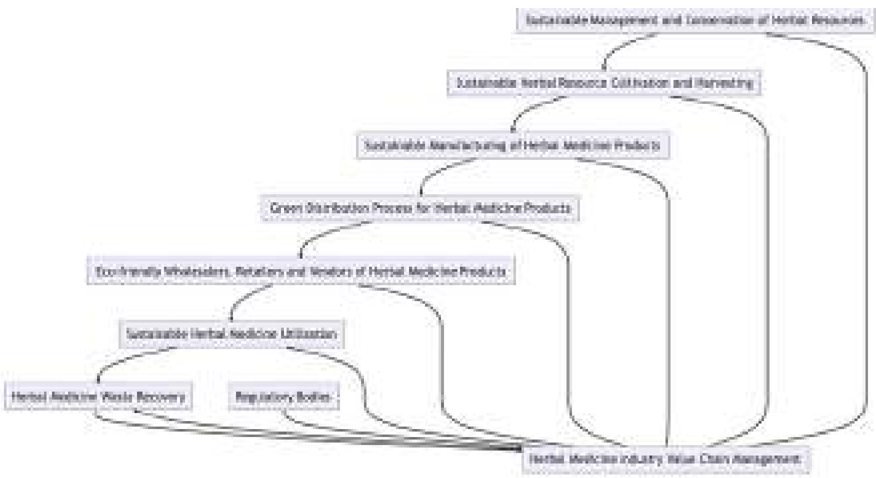

Fig. 2 Proposed herbal medicine industry supply chain monitoring system

developed to educate stakeholders, including farmers and manufacturers, about these policies and how to implement them. Research initiatives should be undertaken to continually improve these policies and practices. Regulatory bodies should regularly review these policies, training materials, and research findings to ensure they are effective and up-to-date.

2. **Sustainable Herbal Resource Cultivation and Harvesting**: This stage involves the cultivation and harvesting of medicinal plants in a sustainable manner. This includes practices such as crop rotation, organic farming, and the use of sustainable irrigation systems. Farmers should be trained in these practices and provided with the necessary resources and support. Regulatory bodies should conduct regular field inspections to ensure these practices are being followed. They should also review the training provided to farmers to ensure it is effective and up to date.
3. **Sustainable Manufacturing of Herbal Medicine Products**: At this stage, the harvested plants are processed into herbal medicine products. The manufacturing process should be designed to minimize waste and reduce environmental impact. This could involve the use of energy-efficient machinery, the recycling of waste materials, and the minimization of water use. Regulatory bodies should conduct regular facility inspections to ensure these practices are being followed. They should also review the manufacturing processes to ensure they meet environmental standards.
4. **Green Distribution Process for Herbal Medicine Products**: This stage involves the distribution of the manufactured products. The distribution process should be designed to minimize carbon emissions and reduce environmental impact. This could involve the use of fuel-efficient vehicles, the optimization of delivery routes, and the use of renewable energy sources. Regulatory bodies should review logistics processes and transportation methods regularly to ensure they are sustainable.
5. **Eco-friendly Wholesalers, Retailers, and Vendors of Herbal Medicine Products**: At this stage, the products are sold to consumers. The packaging used should be eco-friendly and designed to minimize loss and environmental damage. This could involve the use of biodegradable or recyclable materials, the minimization of packaging, and the promotion of reusable containers. Regulatory bodies should conduct regular inspections of packaging materials and review the practices of wholesalers, retailers, and vendors to ensure they are sustainable.
6. **Sustainable Herbal Medicine Utilization**: This stage involves the use of herbal medicine products by consumers. Consumers should be educated about the sustainable use of these products to minimize waste. This could involve the provision of information about dosage, storage, and disposal. Regulatory bodies should conduct regular surveys or inspections to assess consumer knowledge and behavior.
7. **Herbal Medicine Waste Recovery**: The final stage involves the management and disposal of waste produced from herbal medicine. Waste management practices should be environmentally friendly and comply with waste management regulations. This could involve the composting of organic waste, the recycling of packaging materials, and the safe disposal of hazardous waste. Regulatory bodies should conduct regular inspections of waste management practices to ensure they are sustainable.

8. **Herbal Medicine Industry Value Chain Management**: This is the overarching management system that oversees the entire value chain. It ensures the implementation of sustainable practices at each stage and compliance with all relevant laws, regulations, and standards. This system should be transparent and accountable, with regular audits and reviews conducted by regulatory bodies.
9. **Regulatory Bodies**: These organizations play a crucial role in ensuring the sustainability of the supply chain. They have the authority to enforce compliance with laws, regulations, and standards, and to penalize non-compliance. They also have a responsibility to provide guidance and support to stakeholders, promote best practices, and facilitate continuous improvement in the industry.

By understanding each stage of the value chain and the role of regulatory bodies, businesses can better manage their operations to create value in an environmentally friendly manner. This not only benefits the environment but also enhances the business's reputation and competitiveness in the market. Regular monitoring and evaluation by regulatory bodies will ensure that the plan is being followed and that any necessary adjustments are made promptly.

8 Benefits of Sustainable Supply Chain Development and Management Within the Herbal Medicine Sector

There are numerous benefits to developing and managing a sustainable supply chain within the herbal medicine sector. These benefits span across environmental, social, and economic dimensions, contributing to the overall sustainability of the industry. Here are some key benefits:

1. **Environmental Stewardship**: A sustainable supply chain in the herbal medicine sector emphasizes the responsible use and conservation of natural resources. This involves practices such as organic farming, crop rotation, and sustainable irrigation systems that are designed to minimize environmental impact. An example is organic farming, which avoids the use of synthetic agrochemicals that can harm soil health and biodiversity. Crop rotation can help to maintain soil fertility and prevent pest outbreaks, reducing the need for chemical inputs. Sustainable irrigation systems can conserve water and reduce energy use. These practices not only help to ensure a steady supply of medicinal plants for future generations but also contribute to broader environmental goals such as climate change mitigation and biodiversity conservation.
2. **Social Responsibility**: A sustainable supply chain also promotes social equity and community well-being. This can be achieved through fair trade practices that ensure farmers and workers receive a fair price for their products and labor. This can help to improve living standards and reduce poverty in communities where medicinal plants are grown and harvested. Additionally, the promotion of traditional knowledge and practices in herbal medicine can contribute to cultural preservation and community empowerment. For instance, indigenous

communities that have been custodians of medicinal plant knowledge for generations can be recognized and compensated for their contributions, helping to maintain their cultural heritage and livelihoods.

3. **Economic Viability**: From a business perspective, a sustainable supply chain can enhance profitability and competitiveness. Efficient use of resources and waste reduction can lower operational costs. For instance, organic farming practices can improve soil health and productivity, reducing the need for costly chemical inputs. Waste reduction strategies such as composting and recycling can turn waste into valuable resources, further reducing costs. Moreover, the production of high-quality, sustainable herbal medicine products can attract a growing market of consumers who value sustainability, enhancing sales and customer loyalty. Companies that demonstrate a strong commitment to sustainability can also enhance their brand reputation, gaining a competitive edge in the market.
4. **Regulatory Compliance**: As governments and international bodies impose stricter environmental and social regulations, having a sustainable supply chain can help companies in the herbal medicine sector to comply with these regulations. This can prevent legal issues and financial penalties that can arise from non-compliance. Moreover, companies that proactively adopt sustainable practices can influence regulatory frameworks and position themselves favorably for potential future regulations.
5. **Risk Management**: A sustainable supply chain can help to mitigate a variety of risks. Environmental risks such as resource depletion and climate change can disrupt supply chains, but these can be mitigated through sustainable resource management and climate adaptation strategies. Social risks such as labor disputes and community conflicts can be mitigated through fair trade practices and community engagement. Financial and reputational risks due to regulatory penalties or consumer backlash against unsustainable practices can be mitigated through proactive compliance and sustainability communication strategies.
6. **Innovation and Learning**: The pursuit of sustainability can stimulate innovation and learning within the organization. It will promote the development of new products, processes, and business models that are more sustainable and competitive. For instance, research and development efforts can lead to the discovery of new medicinal plants or the development of more efficient processing techniques. Learning and capacity-building initiatives can enhance the skills and knowledge of employees, enabling them to implement sustainable practices more effectively. This culture of innovation and learning can enhance the organization's adaptability and resilience in a rapidly changing world.

9 Conclusion

The development and management of a sustainable supply chain in the herbal medicine sector can yield substantial environmental, social, and economic benefits. It can contribute to the sustainability of the sector and the broader society, while also enhancing business performance and resilience. Therefore, it is a strategic imperative for any company in this sector. SCM developments in the HMI will highlight and enhance the diverse applications and intrinsic value of herbal medicine in treating various diseases and contribute to growing the acceptance and integration of herbal medicine into formal healthcare systems as well as the strategic marketing of herbal medicine products.

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Consumer Perception and Demand for Sustainable Herbal Medicine Products and Market

63

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Abstract

The healthcare modality that capitalizes on the therapeutic and spiritual attributes of plants and is deeply rooted in indigenous people's traditions, local knowledge, and practices as well as plant resources in the locale is considered herbal medicine. This system of healthcare has been the cornerstone of various cultures globally for centuries but is currently experiencing a renaissance due to the need to incorporate sustainability considerations in plant resource management, phytochemical extraction, herbal product design and development, production processes, marketing, packaging, distribution, utilization, and waste management. Using an online survey, this chapter attempts to comprehend and discuss consumer perception and demand for sustainable herbal medicine products (HMPs) by assessing the level of consumer awareness and knowledge about sustainable HMPs as well as factors influencing consumer demand and perceptions about the efficacy and safety of HMPs. Such information will provide invaluable insights to HMP manufacturers, marketers, policymakers, and other stakeholders in the herbal medicine sector and effectively aid the incorporation of sustainability in the production, demand, supply, and consumption of HMPs. Findings suggest that personal networks and recommendations (63.2%) are the most effective channels for spreading awareness about herbal medicine, while public advertisements play a lesser role. A greater number of people continue to use and accept HMPs (74.8%) mostly due to their perceived efficacy (72.5%) and safety. Precisely, 45.1% of respondents expressed that herbal medicines perform "a bit better" than conventional medicine. Interestingly, price appears to be less significant in the decision-making process (26.5%). A substantial number of respondents (83.3%) affirmed their preference for the heightened presence of herbal medicine products in the market. The most common issue reported by respondents (65.2%) was a lack of sufficient information about HMPs. Positive consumers' perception of HMPs is mostly linked to the concept of "nature and naturalness" as well as "safety and healthiness" and "efficacy and effectiveness," while some things that respondents would like to see improved include "dosage, prescription, packaging, labelling, regulation, authentication, research, development, and safety concerns, accessibility, availability, taste and aesthetics, and support from government and non-governmental organizations." In conclusion, there is a need for more research to address pertinent sustainability issues in herbal medicine practices and HMPs that are linked to consumer perception.

Keywords

Consumer preference · Consumer demand · Herbal medicine products · Market dynamics · Supply chain management · Consumer perception · Sustainability

Abbreviations

%	Percent or percentage
COVID-19	Corona virus disease of 2019
DNA	Deoxyribonucleic acid

HCl	Hydrochloride
HMPs	Herbal medicine products
IR	Infrared
ITS2	Internal transcribed spacer 2
LCMS	Liquid chromatography-mass spectroscopy
matK	Megakaryocyte-associated tyrosine kinase
MODS	Multi-organ dysfunction syndrome
n	Number of respondents
NGOs	Nongovernmental organizations
PHWE	Pressurized hot water extraction
psbA	Photosystem II protein D1
rbcL	Ribulose bisphosphate carboxylase
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
TEOS	Tetraethyl orthosilicate
THM	Traditional herbal medicine

1 Introduction

Herbal medicine, a healthcare modality that capitalizes on the therapeutic and spiritual attributes of plants, has been a cornerstone of various cultures globally for centuries ([1, 2]). The practice is deeply rooted in indigenous people's tradition, local knowledge, and practices as well as plant resources in the locale and has been experiencing a renaissance in recent years. Interestingly, this resurgence is largely driven by escalating consumer demand for herbal medicine products, particularly those that are sustainably sourced and produced ([3]). In addition, herbal medicine has found relevance in the treatment and management of new and existing human disease conditions including Lassa fever and COVID-19 [4]. This paradigm shift in consumer perception is catalyzing a transformation in market dynamics, necessitating novel businesses in the herbal medicine sector to recalibrate their strategies in alignment with evolving market trends and consumer preferences.

In the realm of herbal medicine, sustainability has emerged as a pivotal concept in the contemporary discourse surrounding plant resource management, extraction, product design and development, production processes, marketing, packaging, distribution, utilization, and waste management ([5, 6, 7]). This new direction requires a multidimensional approach to forge, encompassing environmental, social, and economic facets. From an environmental perspective, sustainable herbal medicine resources and products are sourced, produced, and consumed in a manner that minimizes their environmental legacy through techniques like organic farming, eschewing the use of synthetic pesticides and fertilizers, and sustainable harvesting and conservation to ensure that plant populations are not compromised ([8, 9]). On the social front, sustainable herbal medicine products foster fair and equitable relationships among all stakeholders. This includes fair trade practices, ensuring that farmers and workers receive fair remuneration for their labor and that working conditions are safe and humane. It also involves acknowledging the cultural significance of medicinal plants and respecting indigenous knowledge and rights ([10]). From an economic perspective, sustainable HMPs confer benefits to all stakeholders in the supply chain. This includes not only profits for manufacturers and retailers but also minimizing healthcare expenditures and ensuring economic stability and resilience for farming communities ([11]).

The demand for sustainable HMPs represents a significant intersection of health and sustainability trends. Modern consumers are increasingly seeking out products that not only bolster their health but also align with their values and concerns about the environment and societal well-being. In response to this demand, herbal medicine product manufacturers are implementing more sustainable practices in their production processes and offering products that are certified as organic, fair trade, or sustainably harvested. However, despite the burgeoning interest and demand for sustainable HMPs, there exists a notable gap in comprehensive research on consumer perception and demand for herbal products. Moving forward, this lacuna is likely to have significant implications for manufacturers, marketers, and policymakers involved in the production and promotion of HMPs ([12, 13]). Hence, understanding consumer perception is pivotal for manufacturers and marketers, as it is the consumers who ultimately determine the success or failure of their herbal products in the market as well as overall marketing strategies. Consumer perceptions, preferences, and purchasing behaviors in relation to herbal products shape the market dynamics. Moreover, the factors influencing consumers' decisions to purchase sustainable products are multifaceted, encompassing disease conditions, health awareness, risk perception, and perceptions of a company's environmental and social practices.

For policymakers, understanding consumer perception and demand for sustainable herbal medicine products is crucial for developing effective policies and regulations. Policymakers may use relevant instruments and mechanisms to promote the sustainable production and consumption of herbal medicine products and to avoid greenwashing and whitewashing of consumers, their products, and the environment. They can implement policies and regulations that encourage manufacturers to adopt sustainable practices and ensure the quality and safety of these products. They can also promote consumer education and awareness about the benefits of sustainable herbal medicine products).

This chapter delves into and attempts to comprehend consumer perception and demand for sustainable herbal medicine products. The objectives include:

1. Investigating the level of consumer awareness and knowledge about sustainable herbal medicine products.
2. Examining the factors influencing consumer demand for sustainable herbal medicine products.
3. Assessing consumer perceptions toward the efficacy and safety of these products.
4. Identifying the barriers and challenges consumers face in accessing these products.
5. Providing recommendations for enhancing the sustainability and consumer acceptance of herbal medicine products in Nigeria.

Attempting an understanding of consumer perception and demand for sustainable herbal medicine products is a complex yet crucial endeavor, but the findings presented in this chapter will provide invaluable insights to herbal medicine product manufacturers, marketers, policymakers, and other stakeholders in the herbal medicine sector in order to aid them in effectively promoting the sustainable production and consumption of herbal medicine products. Also, this chapter includes a review of the multifaceted world of herbal medicine products, examining their classifications, applications, production processes, and the research methodologies employed in pertinent studies.

2 Importance of Herbal Medicine Products and Herbal Medicine Practices

2.1 Herbal Medicine Products

Herbal medicine products, both branded and unbranded, have been a cornerstone of healthcare systems worldwide for centuries. Since herbal medicine began to attract big pharmaceutical companies, branded herbal products are commonly found in the Global North whereas unbranded or poorly packaged ones are found in the Global South ([14, 15]). Nonetheless, derived from plants, these herbal products serve therapeutic purposes and often act as alternatives to conventional pharmaceuticals. Herbal medicine products exhibit a diverse range of forms, usage, and branding. They can be presented in various forms such as tablets, capsules, powders, extracts, or teas, among others, and can be put together by the patient or their loved ones or purchased from the market for treating and managing diverse ailments [16, 17]. Each form has its unique advantages and is chosen based on the desired route of administration, convenience, and patient preference. Their applications are equally varied, addressing a wide spectrum of health conditions ranging from minor ailments like colds and coughs to chronic diseases such as diabetes and hypertension. Branded products, marketed with specific health claims, are subject to regulatory standards, ensuring their safety, efficacy, and quality [18]. Conversely, unbranded products are often traditional remedies passed down through generations or self-prepared mixtures, whose efficacy and safety are usually based on anecdotal evidence and personal experiences.

Maramba-Lazarte [19] highlights the pivotal role of traditional herbal medicine in healthcare, particularly in rural areas where access to conventional medicine may be limited. The study employed a qualitative approach, using interviews and focus group discussions to investigate healthcare system challenges, notably the high cost of branded and generic drugs. The findings suggest that research into traditional medicinal plants, including herbal medicine, could provide a cost-effective solution to these challenges. However, the study also emphasizes the need for quality and standardization of herbal medicine, preclinical pharmacological assessments, and clinical efficacy and safety assessments to ensure that these traditional remedies are safe and effective. The study's limitation lies in its focus on the Philippine context, which may not be representative of other countries with different healthcare systems and cultural practices. Sriharini and Syafira [20] explore the use of herbal medicine in the village of Rejowinangun. The villagers have capitalized on their inherited knowledge of herbal medicine, integrating it into a tourism-based model. The study reveals a transition from traditional usage to a tourist village concept, increasing the economic value of herbal medicines in modern society. This innovative approach not only preserves the traditional knowledge of herbal medicine but also provides a sustainable source of income for the community. However, the study's findings may not apply to other Indonesian communities or communities lacking the potential to develop a tourism-based model. In the Global North, Welz et al. [21] delve into the motivations behind the use of herbal medicine. They identify three influencing factors:

1. **Push factors**, which include dissatisfaction with conventional medicine
2. **Pull factors**, which include an attraction to herbal medicine
3. **Traditional factors** include long-standing family traditions and the cultural importance of herbal medicine

The study by Welz et al. [21] provides valuable insights into the complex interplay of factors influencing the use of herbal medicine, which could be useful for healthcare providers and policymakers in designing strategies to promote the safe and effective use of herbal medicine. However, the study does not explore the potential influence of other factors, such as economic factors, on herbal medicine usage, and its findings may not be applicable outside the German context. The study by Shah and Pokhrel [22] focused on the microbial quality and antibacterial activity of different herbal medicines. The study demonstrates the in vitro antibacterial activity of herbal medicines, suggesting their potential use in treating bacterial infections. This finding underscores the therapeutic potential of herbal medicines, which could be harnessed to address the growing problem of antibiotic resistance. However, the study's focus on microbial quality does not explore other aspects of herbal medicine usage in the Nepalese context, such as cultural beliefs, traditional practices, or economic factors.

Furthermore, the use of herbal medicine varies significantly across different cultural contexts. Factors such as community traditions, dissatisfaction with conventional medicine, and the need for culturally appropriate healthcare education influence the use of herbal medicine. However, more research is needed to explore the use of herbal medicine in other cultural contexts and to address the limitations identified in these studies. As the demand for herbal medicine continues to grow, ensuring their safety, efficacy, and quality will remain a critical area of focus in healthcare.

Nevertheless, despite the widespread use of herbal medicine, several studies have identified the issue of adulteration in herbal medicine. The adulteration of herbal medicine products is a significant concern in the field of herbal medicine. This issue is not only a matter of consumer deception but also a significant health risk, as adulterants can have harmful effects. Various studies have been conducted to detect and understand the extent of this problem, employing different methodologies and focusing on various aspects of adulteration. A study by reveals that some herbal medicines were adulterated with acetaminophen, a chemical drug. The study utilizes liquid chromatography-mass spectroscopy (LCMS) to identify the fragmentation patterns of herbal medicine. This finding underscores the need for stringent quality control and regulation in the herbal medicine industry, highlighting the potential risks associated with unregulated or poorly regulated products. However, the study's limitation is that it only tested for acetaminophen, leaving the possibility of other adulterants undetected.

In a similar vein, a study by Andawiyah et al. [23] developed an optical sensor-based test strip for detecting sibutramine HCl (hydrochloride) adulteration in traditional herbal products. Sibutramine HCl is an anti-obesity agent that has been withdrawn from the Indonesian market due to its adverse effects such as palpitations, insomnia, and cardiovascular disease. The researchers used a cellulosic paper immobilized with Dragendorff's reagent and tetraethyl orthosilicate (TEOS) as the precursor using the sol-gel method. The presence of sibutramine HCl changed the color of the test strip from yellow to orange-red, indicating adulteration. This method provided a low-cost, simple, and efficient way to detect sibutramine HCl in herbal products. However, the study only tested for sibutramine HCl, leaving the possibility of other adulterants undetected. In another study, Teshager et al. [24] investigated the association between traditional herbal medicine (THM) use and the risk of multi-organ dysfunction syndrome (MODS) in a pediatric intensive care unit in Ethiopia. The study found that patients who used THM had a threefold increased risk of MODS, suggesting the potential harm of adulterated or inappropriate use of herbal medicines. However, the study did not specifically identify the adulterants in the herbal medicines used by the patients. A different approach was taken by Balaji and Parani [25], who used DNA barcoding to authenticate single-drug herbal powders collected from markets in Tamil Nadu, India. They found that 46% of the tested samples were adulterated with taxonomically unrelated plant species. This study underscores the prevalence of adulteration in herbal medicine products and the need for effective detection methods. However, the study did not identify the specific adulterants in the herbal powders. Therefore, the adulteration of herbal medicine products is a widespread and complex issue that poses significant health risks. Various methodologies, from optical sensor-based test strips to DNA barcoding, have been employed to detect adulteration. However, these studies are limited in their ability to detect a wide range of potential adulterants, highlighting the need for comprehensive testing methods. Furthermore, more research is needed to understand the health impacts of adulterated herbal medicine products and to develop effective strategies for prevention and regulation. It is therefore paramount for this study to examine the production processes involved in the production of herbal medicine.

2.2 Production of Herbal Medicine

The production of herbal medicine products involves several steps, including cultivation, harvesting, extraction, formulation, and packaging. In a study by Sixt and Strube [26], a modern process engineering concept is provided for the extraction of traditional herbal medicines from plants. The study critically assesses traditional solvent-based percolation and compares it to pressurized hot water extraction (PHWE) as a potential alternative. The study employs a systematic process design for the extraction of hawthorn (*Crataegus*) and incorporates natural batch variability into the physicochemical process modelling concept. The findings reveal that PHWE is the best choice from both technical and economic perspectives, suggesting a potential shift in extraction methodologies for herbal medicine production. However, the study is limited by its focus on hawthorn, and the results may not apply to other herbal medicines.

In a different approach to waste management in herbal medicine production, Soetrisnanto et al. [27] discuss the use of phytoremediation, an environmentally friendly method, to reduce contaminants and nutrients in herbal medicine waste. The study posits that this waste, due to its high nutrient content, could potentially be used as a growth medium for the algae, *Spirulina*. The study employs a two-stage process, with the effluent of the first phytoremediation transferred to the second stage for the cultivation of *Spirulina*. This innovative approach to waste management not only addresses environmental concerns but also proposes a potential avenue for resource optimization. However, the study is limited by its focus on *Spirulina*, and the results may not apply to other algae or plant species.

Lastly, in a study by Liu et al. [28], the shotgun metabarcoding method is used for the molecular identification of the biological ingredients in Qingguo Wan, a herbal medicine used for treating pharyngitis. The study employs the Illumina NovaSeq platform for targeting the ITS2, psbA-trnH, matK, and rbcL sequences to survey the species composition of laboratory-made and commercial samples. The findings demonstrate that the shotgun metabarcoding method is a powerful tool for the molecular identification of the biological ingredients in herbal medicine products, suggesting a potential shift in quality control methodologies for herbal medicine production. However, the study is limited by its focus on Qingguo Wan, and the results may not apply to other herbal medicines.

In summary, the reviewed studies underscore the importance and potential of herbal medicine products in healthcare. However, they also highlight the challenges in the field, such as the high cost of branded drugs, adulteration of products, and the need for more research and stringent regulation and quality control of herbal medicines. One of the limitations in the field of herbal medicine research is the lack of innovation due to stringent regulation and approval processes. Additionally, the natural variability of herbal ingredients can pose challenges in standardizing product quality. Furthermore, while modern methods like shotgun metabarcoding provide precise identification, they may not be accessible to all researchers due to their cost and technical requirements. Future research should focus on developing innovative, cost-effective, and reliable methods for the production and quality control of these products. It is therefore paramount for this study to examine the quality measures of herbal medicine.

2.3 Quality and Reliability of Herbal Medicine Products

Herbal medicine products (HMPs) have been used for centuries for disease prevention, treatment, and management as well as for general healthcare and maintenance.

However, the quality and reliability of these products are of paramount importance to ensure their efficacy and safety. This literature review critically analyses four recent studies that focus on the quality control and reliability of HMPs.

Chen et al. [29] highlighted the importance of modern techniques in the quality control of herbs. They argue that traditional analysis methods are time-consuming and laborious, necessitating the need for faster, cheaper, and more environment-friendly techniques. Infrared (IR) and Raman spectroscopy techniques are proposed as potential solutions for quality control and safety inspection. These techniques offer the advantage of being nondestructive, rapid, and capable of providing detailed information about the chemical composition of herbs. However, the practical application of these techniques on a large scale remains a challenge due to the complexity of herbal products and the need for specialized equipment and expertise.

Furthermore, in the traditional Chinese medicine context, Xiao et al. [30] provide a comprehensive review of the traditional uses, phytochemistry, and pharmacology of *Euodiae Fructus* in their study. The authors emphasize the importance of quality control in ensuring the safety and efficacy of this traditional Chinese medicine. They highlight the various pharmacological activities of *Euodiae Fructus*, such as its anti-inflammatory, antibacterial, and anticancer properties. However, the study does not provide a detailed methodology for quality control, which could limit its practical application. This omission underscores the need for more research into practical, effective quality control methods for traditional Chinese medicine.

Building on the theme of quality control, present a compelling study wherein they highlighted the need for reliable detection methods for aflatoxin, a potent carcinogen produced by certain molds. They underscore the importance of quality control in ensuring the safety of herbal products. The study provides a comprehensive review of recent developments in screening assays for aflatoxin qualification or quantitation, including lateral flow immunoassays, enzyme-linked immunosorbent assays, aptamer-based lateral flow assays, and cytometric bead arrays. However, the global occurrence of aflatoxin contamination and the complexity of testing for it in herbal products present significant challenges. This study underscores the need for more research into effective, practical methods for detecting aflatoxin in herbal products.

Finally, Noviana et al. [31] contribute to the discussion on standardization and quality control of herbal medicines in their study. The authors argue that fingerprint analysis allows for the evaluation of the whole chemical profile of herbal drugs, thereby ensuring their quality and reliability. They present a comprehensive review of various fingerprinting techniques and their application in the standardization and quality control of herbal medicines. However, the application of these techniques requires advanced analytical tools and expertise, which may not be readily available in all settings. This limitation underscores the need for more research into cost-effective, user-friendly techniques for the quality control of herbal products.

A critical analysis of the above literature reveals that quality and reliability are indeed critical aspects of HMPs. The reviewed studies highlight the importance of modern analytical techniques in ensuring the quality and safety of these products. However, the practical application of these techniques on a large scale remains a challenge due to the complexity of herbal products and the need for specialized equipment, expertise, and increased cost. This further justifies the need for understanding consumer perception and demand for sustainable herbal medicine products and market, as this will provide an understanding of market demand and dynamics and can promote the need for future research focusing on developing cost-effective and user-friendly techniques for quality control of herbal products. Despite the limitations of the current studies, they provide valuable insights into the current state of quality control and reliability in the field of herbal medicine products and pave the way for future research in this area.

2.4 Global Demand for Herbal Medicine

The escalating global demand for herbal medicine products (HMPs) is a phenomenon that has been propelled by a confluence of factors. These include heightened awareness of the potential benefits of herbal medicine, the prevalence of chronic diseases, and the pursuit of natural alternatives to conventional medicine. This trend is particularly pronounced in the traditional medicinal systems of Asia and India, which are at the vanguard of this surge in demand.

In the study by Li et al. (2021), the authors employed a systematic review methodology to explore the potential of traditional Asian herbal medicine in the treatment and prevention of COVID-19. They discovered that several plant bioactive compounds exhibited potent antiviral activity against SARS-CoV-2, suggesting a promising avenue for future research. This study underscores the potential of herbal medicine in addressing global health crises, such as the COVID-19 pandemic. The authors argue that the antiviral properties of certain plant bioactive compounds could be harnessed to develop effective treatments for COVID-19. This revelation is particularly significant in the current global health landscape, where the need for effective treatments for emerging diseases is more pressing than ever.

In a similar vein, the study by Kumar et al. (2020) illuminates the diverse health practices of traditional Indian medicine, including the use of herbal medicine for the treatment of diabetes. The authors conducted an extensive review of the literature and found that herbal medicines, particularly those used in Ayurveda, have shown promising results in managing diabetes and its complications. This study provides evidence of the effectiveness of herbal medicine in managing chronic diseases, such as diabetes. The authors argue that the use of herbal medicine in traditional Indian medicinal systems could be leveraged to meet the growing global demand for natural and effective treatments for chronic diseases. This is a significant finding, considering the global burden of chronic diseases and the need for effective management strategies.

However, the landscape of herbal medicine is not static, with technology playing a pivotal role in its transformation. The study by Banerjee and Mitra (2011) discusses the increasing demand for herbal medicine, attributing this to an herbal "renaissance." The authors found that scientific research and modern technology have played a crucial role in the industrial growth of herbal medicine. They highlighted the importance of technological advancements in improving the quality, safety, and efficacy of herbal medicines. This study underscores the role of technology in meeting the growing global demand for high-quality herbal medicine. The authors argue that technological advancements, coupled with scientific research, could be harnessed to improve the quality, safety, and efficacy of herbal medicines, thereby meeting the growing global demand for high-quality herbal medicines.

This increasing demand for herbal medicine has led to the expansion of partnerships between farmers and the herbal medicine industry, as discussed in the research by Mulyono Daru [32]. The author found that this partnership was crucial for the development of the economic community, leading to increased income for farmers and improved availability of herbal medicines for consumers. This study underscores the role of partnerships in meeting the growing global demand for herbal medicine. The authors argue that partnerships between farmers and the herbal medicine industry could be leveraged to increase the production of herbal medicines, thereby meeting the growing global demand for these products.

On the other hand, the article by Sollini et al. (2020) discusses the use of neural networks for market segmentation in the herbal medicine industry. The authors found that a fit strategy for each market segment should be formulated and executed to add value to both customers and the market. This study underscores the role of market segmentation in meeting the growing global demand for herbal medicine. The authors argue that market segmentation strategies, formulated and executed using neural networks, could be used to add value to both customers and the market, thereby meeting the growing global demand for herbal medicine. While these developments are promising, they also highlight several challenges. The lack of standardization in these processes can lead to significant variations in the quality and reliability of HMPs. Moreover, while the economic benefits of high-quality and reliable HMPs are evident, the cost implications of achieving these standards can be substantial. This necessitates rigorous quality assurance and validation processes to ensure the safety and efficacy of these products.

In summary, the global demand for herbal medicine is on the rise, and understanding consumer perception toward these products is crucial for the sustainable growth of the herbal medicine market. Despite the challenges, high-quality and reliable HMPs can provide substantial economic benefits by making healthcare more accessible and affordable, thereby justifying the need for further research in this area.

3 Assessment of Consumer Perception and Demand for Sustainable Herbal Medicine Products and Market Using Online Survey Instrument

3.1 Philosophical Framework, Approach, and Design

The philosophy adopted to assess consumer perception and demand for sustainable herbal medicine products and markets using an online survey instrument is pragmatism, which combines the positivist and interpretivist approaches to solve practical problems and improve human conditions as presented in Feilzer (2009). Pragmatism was chosen because it allows for the practical application of knowledge in a specific context (Sauders et al., 2016). The research problem required the investigation of consumer perception and demand for sustainable HMPs, which involved both objective and subjective aspects. The study required a flexible method to obtain answers to the key assessment questions and neither positivism nor interpretivism could successfully bridge this gap.

The assessment approach chosen for this assessment focused on addressing practical problems and generating solutions that can be directly implemented. This approach aligned well with the research objectives, which aimed to explore consumer perception and demand for sustainable herbal medicine products. It allowed the evaluation of a subject with an incomplete set of available data, making it the most suitable approach for a pragmatic philosophical stance (Williamson, 2016).

The applied research adopted a mixed-methods approach that combined the strengths of both qualitative and quantitative research methods to provide a more comprehensive understanding of the research problem. It involved using both inductive and deductive approaches to data collection and analysis, allowing exploration of different aspects of the research question and enhancing the validity and credibility of the findings (Creswell, 2013).

The assessment strategy adopted was that of a mixed-methods approach, using both qualitative and quantitative research methods to collect data (Creswell and Plano, 2017). Surveys were chosen as they were a cost-effective strategy to collect a large amount of data quickly and easily, while also allowing for the standardization of questions and responses, making the data easier to analyze (Nardi, 2016).

The design adopted for this work was a cross-sectional online-based survey, a type of observational study that involves the analysis of data collected from a population, or a representative subset, at a specific point in time. Cross-sectional surveys are particularly suited to this study as they allow for the collection of data on multiple variables simultaneously, providing a comprehensive snapshot of the current state of consumer perceptions and demand for sustainable HMPs [33].

3.2 Data Collection Participants Population and Sample Selection

Data collection was a pivotal aspect as it enabled the quality and reliability of the assessment data obtained (Saunders et al., 2016). In this assessment, a mixed-method approach was employed using both closed and open-ended questionnaires to evaluate consumer perception and demand for sustainable herbal medicine products. The data collection procedure involved distributing the questionnaire to the selected sample. This is done either in person or online, depending on the preferences and accessibility of the participants. Participants are provided with information about the assessment and their rights as participants and are required to provide informed consent before participating. The questionnaire was self-administered, allowing participants to complete it at their own pace and in their own time. Assistance was provided if needed, and confidentiality is maintained throughout the process [33].

The target population for this assessment primarily includes consumers of herbal medicine products. Given the broad nature of this population, a convenience sampling technique is employed. This non-probability sampling technique involves selecting participants based on their availability and willingness to participate in the assessment. While this method may not provide a perfectly representative sample of the entire population, it is practical and efficient for studies where the population is diverse and hard to reach (Etikan et al., 2016).

The sample size is determined using the Krejcie and Morgan table [34] for determining sample size, which provides a recommended sample size based on the population size. This ensures that the sample is large enough to provide reliable results but not so large as to be impractical or unnecessarily costly [34].

3.2.1 Quantitative Data Collection

The primary method of quantitative data collection in this study was using closed-ended questionnaires. The questionnaires were structured with a series of statements, and participants were asked to indicate their level of agreement or disagreement on a Likert scale. This allowed for the collection of quantitative data that could be easily analyzed using statistical methods.

The use of closed-ended questionnaires allowed for the collection of a large amount of data in a relatively short amount of time, making it a cost-effective and efficient method of data collection. However, one potential limitation of this method is the possibility of response bias, where participants may not answer truthfully or may not fully understand the questions being asked, leading to inaccurate or incomplete data.

3.2.2 Qualitative Data Collection

In addition to closed-ended questionnaires, open-ended questionnaires were used to collect qualitative data. The open-ended questions were designed to allow participants to express their views and experiences in their own words. This provided rich, detailed data that could provide insights into the reasons behind the attitudes and perceptions identified in the closed-ended questions. The use of open-ended questionnaires allowed for the collection of qualitative data that could provide a deeper understanding of the research problem. However, one potential limitation of this method is that it can be time-consuming to analyze due to the need to code and categorize the responses.

3.2.3 Data Collection Procedure

The data collection procedure involved distributing the questionnaire to the selected sample. This was done either in person or online, depending on the preferences and accessibility of the participants. Participants were provided with information about the study and their rights as participants and were required to provide informed consent before participating. The questionnaire was self-administered, allowing participants to complete it at their own pace and in their own time. Assistance was provided if needed, and confidentiality was maintained throughout the process.

3.2.4 Data Analysis

The data analysis for this study involves both quantitative and qualitative methods. Quantitative data from the customer surveys, collected via Google Forms, was analyzed using Microsoft Excel. This statistical tool facilitated the computation of descriptive statistics such as frequencies, means, and standard deviations [35]. Utilizing Microsoft Excel for data analysis enhanced the accuracy of the results and minimized the potential for human error during the analysis process.

In addition to the quantitative data, qualitative data were analyzed using a thematic coding approach, which involved identifying and studying recurring patterns in the data (Braun & Clarke, 2006). This six-step process, outlined below, allowed for a comprehensive understanding of the data:

1. **Familiarization with the Data**: The initial step involved thoroughly reading the data multiple times to gain a comprehensive understanding of the information provided by the participants. This step also involved identifying initial patterns and themes within the data.
2. **Generation of Initial Codes**: After understanding the data, areas of interest were identified, and initial codes were generated. These codes represented key aspects of the data that were relevant to the research question.
3. **Search for Themes**: The next step involved identifying common themes across the data and linking them to the relevant codes.
4. **Review of Themes**: The identified themes were then reviewed to ensure they were comprehensive and relevant to the research objectives.
5. **Definition and Naming of Themes**: Each theme was then defined and named in a way that accurately represented the data and was relevant to the research objectives.
6. **Production of the Report**: Finally, a detailed report was produced that described how the themes related to the research objectives. This report was used to draw conclusions and make recommendations based on the findings of the study.

3.3 Validity and Reliability of Data

Ensuring validity and reliability in research is crucial for producing accurate, credible, and trustworthy results. Validity refers to the extent to which a study measures what it intends to measure, while reliability refers to the consistency and stability of the results over time (Saunders et al., 2016). In this assessment, validity was ensured by carefully designing the questionnaires to accurately measure consumer perception and demand for sustainable HMPs. The reliability of the data was ensured by using a consistent method of data collection and analysis.

3.4 Ethical Consideration

Ethical considerations are fundamental principles that guide the conduct of research, ensuring it is carried out responsibly and with integrity [36]. When conducting research involving human participants, it is crucial to consider these ethical principles. These principles safeguard the rights and welfare of the participants and ensure that they are not subjected to harm. Adherence to ethical principles also upholds the integrity of the research and fosters trust in the research findings. This study adhered to the following ethical considerations:

1. **Voluntary Participation**: The principle of voluntary participation was strictly observed throughout this research. Participants were invited to participate in the study and were informed that their involvement was entirely voluntary. They were also made aware that they could withdraw from the study at any time without any repercussions, aligning with the recommendations of Bryman and Bell [36].
2. **Informed Consent**: Consistent with the guidelines outlined by Mbuagbaw et al. (2013), informed consent was obtained from all participants. Before their participation, participants were provided with comprehensive information about the study's purpose, nature, potential risks, and benefits. They were then asked to sign an informed consent form, indicating their voluntary agreement to participate in the study.
3. **Confidentiality and Anonymity**: The confidentiality and anonymity of participants were maintained throughout the study. Personal information collected during the study was securely stored and kept confidential. Participants were assured that their responses would remain anonymous and that their participation would not result in any adverse consequences. This approach aligns with the recommendations of Lwin et al. (2017).
4. **Fair Treatment**: Participants were treated fairly and respectfully throughout the study. The selection of participants was based on established criteria, ensuring there was no discrimination or bias in the data analysis. Participants were not exposed to any harm or discomfort during the assessment, and they were free to withdraw from the assessment without penalty, as emphasized by Ahn et al. (2018).
5. **Data Protection**: The protection of data was a key ethical consideration in this study. All data collected during the assessment was securely stored and kept confidential, accessible only to authorized personnel. The data was stored on a password-protected computer and was used solely for this assessment. This approach aligns with the recommendations of Saunders et al. (2016).

3.5 Survey Respondents

There was a total of 120 respondents, and the response rate of 96% ($n = 115$) was deemed adequate and representative of the sample (Table 1).

3.6 Objective One: To Investigate the Level of Consumer Awareness and Knowledge About Sustainable Herbal Medicine Products

To achieve the first objective, respondents were asked three questions as presented in Sects. 3.6.1, 3.6.2, and 3.6.3.

3.6.1 Awareness of Herbal Medicine

A significant majority of respondents, 63.2% (n = 72), reported that they heard about herbal medicine through family and friends (Fig. 1). The finding supports the report of Abdelmola et al. [37] that the views of family and friends influence patients' views of HMP. In turn, this will contribute toward traditional knowledge and practices surrounding herbal medicine as posited by Vujicic and Cohall [38] wherein they stated that herbal medicine knowledge is mostly acquired from family members and loved ones. The trend observed in the present study indicates a strong influence of personal networks in the dissemination of information about herbal medicine. The second most common source of information was recommendations, with 20.2% of respondents stating that someone recommended herbal medicine to them. This suggests that word-of-mouth plays a crucial role in spreading awareness about herbal medicine. Public advertisements were the third most common source, with 11.4% of respondents learning about herbal medicine this way. This shows that despite the prevalence of modern advertising methods, they are less influential in this context compared to personal networks and recommendations. Finally, a small portion of respondents, 5.3%, mentioned other sources of information. This category could include a variety of sources such as online research, health professionals, books, etc. Therefore, personal networks and recommendations appear to be the most effective

Table 1 Respondents of the online survey on consumer perception and demand for sustainable herbal medicine products and market

Respondents	Number	Percentage
Total targeted customers	120	100
Total questionnaires filled by customers	115	96

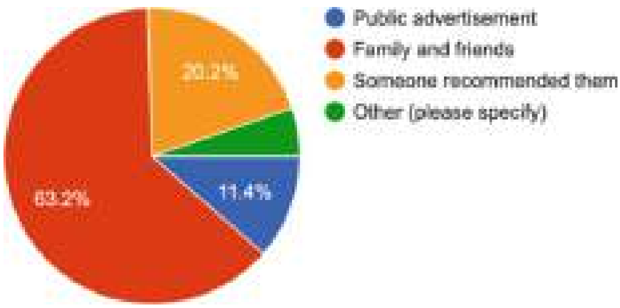


Fig. 1 Respondent's awareness of herbal medicine

channels for spreading awareness about herbal medicine, while public advertisements play a lesser role. Further research could explore the “others” category in more detail to identify any other significant sources of information.

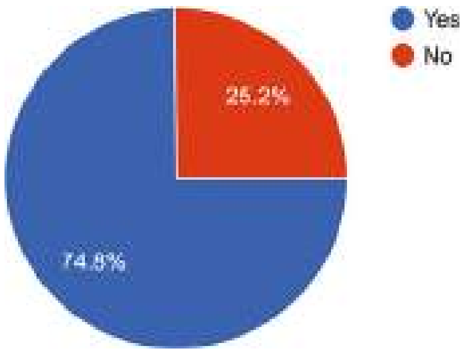
In support of the findings reported by Ahmed et al. (2022) and Hasen et al. (2021), the results of the current survey indicate a high level of awareness about herbal medicine among respondents. A significant majority (63.2%) reported learning about herbal medicine through family and friends, while 20.2% heard about it through recommendations. This suggests that information about herbal medicine is widely disseminated through personal networks.

3.6.2 Usage of Herbal Medicine

Ekor [39] suggests around 80% of the world’s population relies on HMPs and practices for their healthcare. In the current survey, the respondents were probed about their usage of herbal medicine with the question as presented in Fig. 2. A significant majority of respondents (74.8%) reported that either they or their loved ones have used herbal medicine. This indicates a high level of engagement with herbal medicine within the respondent’s personal or familial context. According to Niemeyer et al. [40], engagement with herbal medicine for healthcare is a nonlinear process that has both individual and global effects beyond symptom reduction. The current survey also revealed that 25.2% of respondents stated that neither they nor their loved ones have used herbal medicine. This group represents a quarter of the respondents, suggesting that while herbal medicine usage is prevalent, it is not universal among the survey participants. It therefore means most respondents or their loved ones have experience with using herbal medicine, demonstrating its widespread usage. However, a significant minority have not engaged with herbal medicine, indicating potential areas for growth and outreach.

The high level of engagement (74.8%) with herbal medicines reporting here will likely have a correlative effect on healthcare as well as in the use and conservation of plant resources [41–45]. In essence, a greater number of people and their loved ones use to have personal or familial experience with herbal medicine, which could

Fig. 2 Usage of herbal medicine among respondents



contribute to their knowledge about these products as well as a desire for an improvement in the sector through incorporating sustainability.

3.6.3 Purchase or Recommend the Use of Herbal Medicine

The respondents were further probed on their use of herbal medicine and their responses are presented in Fig. 3. The results for the future intent to purchase or recommend herbal medicine indicate a significant majority of respondents (80.9%), indicating a positive response. This suggests a strong inclination toward the use of herbal medicine, either for personal use or as a recommendation to others. The use is mostly for the promotion of health as well as a therapy for chronic health issues as opposed to life-threatening conditions, especially for conditions that conventional medicine has proven ineffective to address [46, 47]. However, it is important to note that 19.1% of respondents stated that they would not buy or recommend herbal medicine. This group represents nearly one-fifth of the respondents, indicating a degree of resistance or hesitation toward herbal medicine. Therefore, while most respondents expressed a willingness to buy or recommend herbal medicine, a significant minority expressed the opposite. This suggests that while herbal medicine has a strong acceptance, there is still a segment of the population that remains unconvinced, indicating potential areas for education and outreach. In this regard, Soltani et al. (2022) noted that interest and desire to use HMPs are linked to their availability, safety perception, and lack of healthcare options.

Therefore, the future intent to purchase or recommend herbal medicine is also high among respondents, with 80.9% (n = 93) indicating a positive response. This suggests a strong inclination toward the use of herbal medicine and could reflect a level of knowledge and trust in these products.

3.7 Objective Two: To Examine the Factors Influencing Consumer Demand for Sustainable Herbal Medicine Products

To achieve objective two, respondents were asked three questions as outlined in Sects. 3.7.1, 3.7.2, and 3.7.3. Factors considered in this objective included efficacy,

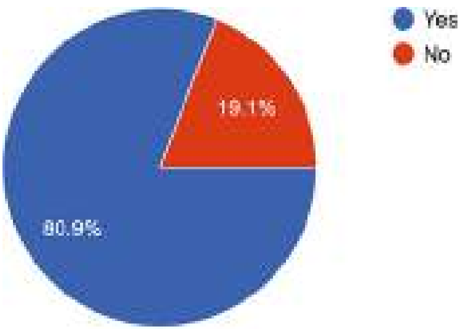


Fig. 3 The likelihood that respondents will buy or use herbal medicine

availability, safety, price, packaging, and overall availability of these products and findings may be summarized as follows.

1. **Efficacy**: A significant 72.5% of respondents indicated that the perceived effectiveness of herbal medicines was a primary factor influencing their purchasing decisions. This suggests that consumers place a high value on the health benefits of these products.
2. **Availability**: Approximately 28.4% of respondents purchase herbal medicines because they are easy to find, indicating the importance of widespread accessibility and distribution. Additionally, 32.5% of respondents stated that availability greatly influences their use of herbal medicine products, further emphasizing the role of accessibility in consumer demand.
3. **Safety**: Around 29.4% of respondents purchase herbal medicines due to their belief in their safety, reflecting a level of trust in these products and possibly a perception of herbal medicines being a safer alternative to conventional medicines.
4. **Price**: Only 26.5% of respondents base their purchasing decisions on price, suggesting a willingness to pay a premium for products perceived as effective and safe.
5. **Packaging**: Packaging significantly influences 37.2% (n = 42) of respondents, while 26.5% (n = 30) state it sometimes does. This indicates that packaging can play a role in shaping consumer perceptions and potentially their purchasing decisions.

3.7.1 What Makes You Buy or Use Herbal Medicine

Respondents were asked what makes them buy and use herbal medicine and their responses are presented in Fig. 4. The results indicate factors influencing their decision to purchase and use herbal medicines and provide insights into consumer behavior in this market. Efficacy emerged as the most significant factor, with 72.5% of respondents indicating that they purchase herbal medicines due to their perceived effectiveness. This suggests a high value placed on the health benefits of these

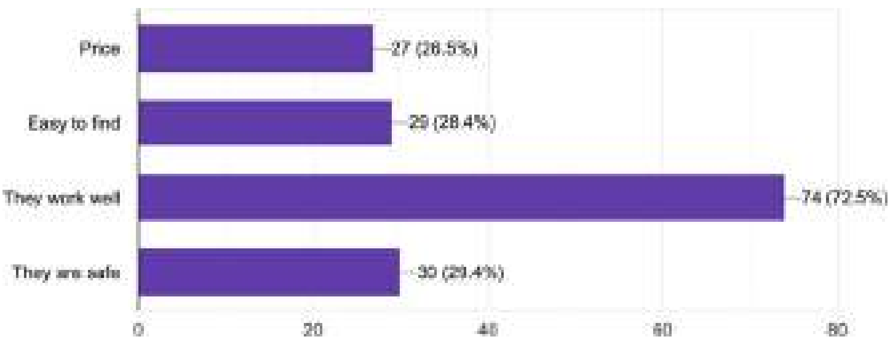


Fig. 4 Reason for buying or using herbal medicines

products by consumers. Nonetheless, Wachtel-Galor and Benzie [47] opined that research is needed to ensure the quality, molecular effects, safety, and clinical efficacy of HMPs to the perceived public view of the efficacy of the products. In the same, Bent [48] opined that HMPs will not become an important alternative to modern medical therapies unless important regulatory changes are made. The survey also revealed that the availability of herbal products also influences purchasing decisions. Approximately 28.4% of respondents purchase herbal medicines because they are easy to find, indicating the importance of widespread accessibility and distribution. Safety is another key consideration, with 29.4% of respondents purchasing herbal medicines due to their belief in their safety. This reflects a level of trust in these products and could be interpreted as a perception of herbal medicines being a safer alternative to conventional medicines. Interestingly, price appears to be less significant in the decision-making process. Only 26.5% of respondents base their purchasing decisions on price, suggesting a willingness to pay a premium for products perceived as effective and safe.

This study, therefore, noted the perceived efficacy, availability, and safety of herbal medicines are key drivers of purchase decisions. While price is a consideration, it is less influential than the other factors. These insights can inform marketing strategies in this sector.

3.7.2 Impact of Packaging on the Perception of Herbal Medicine

Furthermore, respondents were asked about the impact of packaging on the perception of herbal medicine and their response is response in Fig. 5. Recently, Aziato and Odai [49] and Vaezi et al. [50] reported that bad packaging alongside inadequate or unclear dosage information and labelling are some major barriers militating the use of herbal medicines. Here, respondents were queried about the impact of packaging on their perception of herbal medicine products as shown in Fig. 5. The findings reveal that packaging plays a varying degree of role in shaping consumer perceptions. A significant proportion of respondents, 37.2%, indicated that packaging greatly influences their perception of herbal medicine products. This suggests that for a considerable segment of consumers, packaging is a critical factor in shaping their views and potentially their purchasing decisions. Approximately 26.5% of

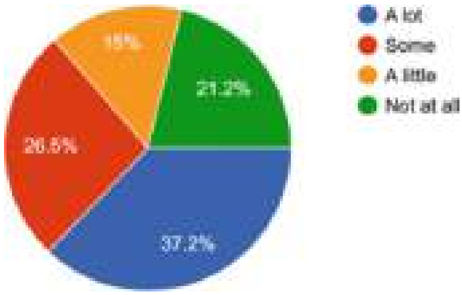


Fig. 5 Impact of herbal medicine packaging on consumer perception

respondents stated that packaging sometimes affects their perception. This indicates that while packaging may not be the primary factor influencing their views, it does have an impact under certain circumstances or for specific products. A smaller percentage, 15%, reported that packaging has a minimal effect on their perception of herbal medicine products. For these consumers, other factors may be more influential in shaping their views. Finally, 21% of respondents stated that packaging does not affect their perception at all. This group appears to be indifferent to packaging and likely bases their perceptions and decisions on other factors. The observed patterns may be linked to the views of Zhou et al. [51] that packaging and labelling help in distinguishing adulterated from unadulterated HMPs as well as quality control and positive consumer perception. In summary, while packaging significantly influences a substantial segment of consumers, its impact varies, with a noteworthy percentage of consumers being unaffected by it. These insights can guide businesses in the herbal medicine sector in their packaging and marketing strategies.

3.7.3 Availability of Herbal Medicine and Consumer Perception

Respondents were asked about the impact of availability on their use of herbal medicine products and the responses as presented in Fig. 6. The findings reveal that availability plays a varying degree of role in influencing consumers' usage of these products. A significant proportion of respondents, 32.5%, indicated that availability greatly influences their use of herbal medicine products. This suggests that for a considerable segment of consumers, the ease of access or procurement of these products is a critical factor in their usage decisions. It is in line with the findings of Mbali et al. [52] that the availability and accessibility HMPs contribute to their use and acceptance, especially in low-resource communities. Approximately 25.4% of respondents stated that availability sometimes affects their use of herbal medicine products. This indicates that while availability may not be the primary factor influencing their usage, it does have an impact under certain circumstances or for specific products. A smaller percentage, 16.7%, reported that availability has a minimal effect on their use of herbal medicine products. For these consumers, other factors may be more influential in shaping their usage

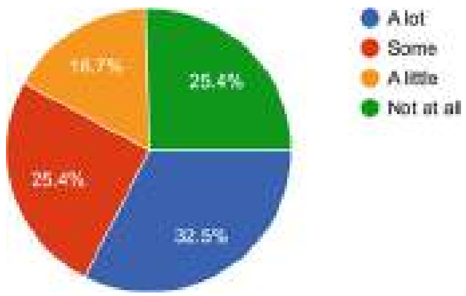


Fig. 6 Impact of herbal medicine availability and consumer perception

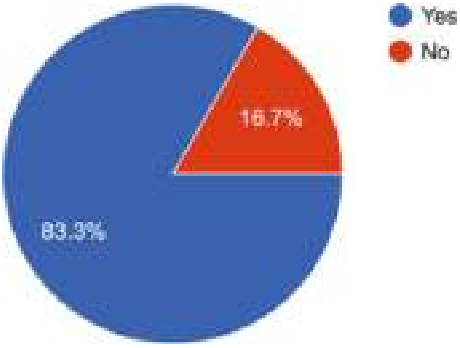
decisions. Finally, 25.4% of respondents stated that availability does not affect their use of herbal medicine products at all. This group appears to be indifferent to the availability of products and likely bases their usage decisions on other factors. In summary, while availability significantly influences a substantial segment of consumers in their usage of herbal medicine products, its impact varies, with a noteworthy percentage of consumers being unaffected by it. These insights can guide businesses in the herbal medicine sector in their distribution and marketing strategies.

3.8 Objective Three: To Assess Consumers' Perception of the Efficacy and Safety of Sustainable Herbal Medicine Products

To achieve this research's third objective, the study examined the following questions outlined in Sects. 3.8.1, 3.8.2, 3.8.3, 3.8.4, and 3.8.5. The third objective was set to assess the perception of consumers toward the efficacy and safety of sustainable herbal medicine products. The study revealed the following perceptions:

1. **Perception of Efficacy**: In the study, a substantial majority perceived herbal medicines as being equally or more effective than conventional medicines. Specifically, 45.1% believed herbal medicines were "a bit better," and 22.1% considered them "much better." In contrast, 18.6% viewed herbal and conventional medicines as equally effective, while a minor yet significant portion viewed herbal medicines as less effective. This indicates a prevalent belief in the high efficacy of herbal medicines, albeit with some reservations.
2. **Perception of Safety**: In assessing the perceived safety of herbal medicines compared to conventional ones, most respondents deemed herbal medicines to be safer. Approximately 27.2% agreed that herbal medicines are "a lot safer," and another 25.4% considered them "a bit safer." However, a noteworthy proportion of respondents expressed safety concerns or perceived equal safety with conventional medicines, highlighting a need for further safety-focused consumer education and possibly enhanced safety procedures for herbal medicines.
3. **Perception of Sustainability and Environmental Impact**: The survey results suggested that most respondents view herbal medicines as environmentally friendly. Specifically, 46.5% believed they are "a lot" environmentally friendly, and 36.8% believed they are "somewhat" environmentally friendly. Nevertheless, a small yet significant proportion of respondents did not share this view, indicating a potential area for increased communication about the environmental sustainability of herbal medicine production.
4. **Perception of General Health Impact**: When asked about their belief in the overall positive health impact of herbal medicines, 80% of the respondents agreed to some degree, indicating a generally favorable perception. However, a smaller yet significant portion expressed reservations or disbelief, demonstrating an area for potential improvement in product communication and education.

Fig. 7 Desire to see an increase in herbal medicine products in stores



3.8.1 Would You Prefer to See an Increase in the Availability of Herbal Medicine Within Stores?

The response to this question is presented in Fig. 7. This quantitative inquiry aimed to ascertain whether a significant demand exists and to further identify potential market dynamics. Of the respondents, a substantial 83.3% affirmed their preference for the heightened presence of herbal medicine products in the market. This denotes a clear majority favoring the increased accessibility of such merchandise, potentially elucidating a promising avenue for businesses considering the proliferation of their herbal medicine offerings. In contrast, a minority contingent of 16.7% ($n = 19$) expressed a lack of interest in seeing an augmentation in the availability of herbal medicines within retail venues. This dissenting perspective is essential, as it embodies a subset of the market that might not be amenable to the enhanced prevalence of these products. In summation, the data collected presents a convincing inclination toward the increased availability of herbal medicines among most participants, providing valuable insight for potential market expansion strategies. Nevertheless, the dissenting minority underscores the significance of market diversification and hints at the necessity for a more comprehensive understanding of varied consumer preferences.

3.8.2 Do You Believe That Herbal Medicines Work Better, Worse, or the Same as Conventional Medicines?

Respondents were asked about the efficacy of herbal medicine in comparison with orthodox medicine with the question presented in Fig. 8. Most respondents indicated a belief in the superior or equal effectiveness of herbal medicines relative to conventional ones. Precisely, 45.1% of respondents expressed that herbal medicines perform “a bit better,” while 22.1% suggested they work “much better.” There also exists a significant portion of the market, 18.6%, that perceives herbal and conventional medicines to have equivalent effectiveness. This data identifies a group that does not distinguish between the two types of medicine in terms of their therapeutic impact. Conversely, a minority of respondents held a less favorable view of herbal medicines. About 8% indicated that herbal medicines perform “a bit worse,” and a

Fig. 8 Respondents believe herbal medicines work better than regular medicines

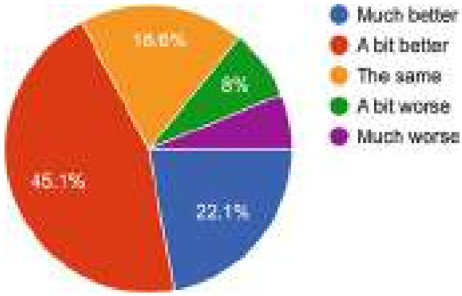
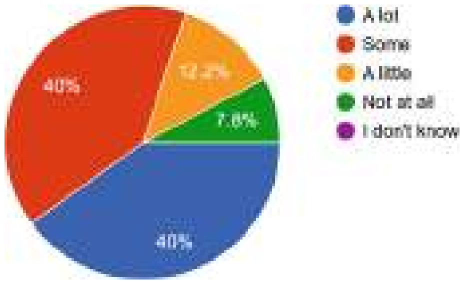


Fig. 9 Respondent's perception of the benefits of herbal medicines

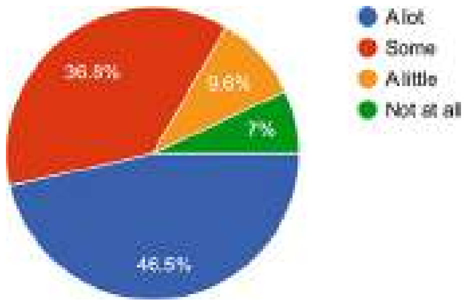


marginal 6.2% believed they work "much worse" than conventional medicines. In summary, the findings of this survey reveal that most participants perceive herbal medicines as being equally or more effective than their conventional counterparts. This indicates a promising potential for expansion within the herbal medicine market. However, attention must be given to the minority perception of lesser effectiveness, which underscores a need for further consumer education or potential product enhancement strategies.

3.8.3 Do You Believe That Herbal Medicines Are Good (Safe and Work Well) for Healthcare and Management?

Furthermore, respondents were asked "Do you believe that herbal medicines are good (safe and work well) for health care and management?" as shown in Fig. 9. A significant proportion of the respondents, constituting 40%, firmly agreed that herbal medicines are significantly good for healthcare and management. Another equally substantial group, again 40%, agreed to some extent, demonstrating a moderate level of confidence in the effectiveness and safety of herbal medicines. However, a noticeable percentage of participants held more reservations. Precisely, 12.2% of respondents believed only "a little" in the effectiveness and safety of herbal medicines for healthcare and management. Lastly, a minority group of respondents, 7.8%, disagreed entirely with the notion that herbal medicines are good for healthcare and management, indicating no belief in the safety or effectiveness of these products. Therefore, the data reveal a substantial belief in the safety and

Fig. 10 Perception of the environmental benefits of herbal medicines



effectiveness of herbal medicines among most surveyed individuals, providing a promising outlook for the herbal medicine market. However, a small yet significant proportion of respondents hold reservations or outright disbelieve the value of herbal medicines in healthcare and management. These perceptions are crucial to acknowledge for future market strategies, particularly those involving consumer education and communication.

3.8.4 Do You Think Herbal Medicines Are Environmentally Friendly?

In addition, respondents were asked "Do you think herbal medicines are environmentally friendly?" and the result is presented in Fig. 10. A significant majority of the respondents hold a strong belief in the environmental friendliness of herbal medicines. Specifically, 46.5% indicated that herbal medicines are "a lot" environmentally friendly. Furthermore, a substantial 36.8% hold the belief to "some" extent, showing a moderate level of agreement with the environmental friendliness of these products. Nevertheless, there is a minority of respondents who express less certainty. About 9.6% believe that herbal medicines are "a little" environmentally friendly, indicating a lower level of agreement. Moreover, a smaller group, 7%, disagreed entirely with the proposition that herbal medicines are environmentally friendly. In summary, the findings reveal that a substantial majority of respondents perceive herbal medicines as environmentally friendly to varying degrees. This indicates a promising alignment with growing global trends toward environmentally friendly products. However, there remains a smaller segment of respondents who either express reservations or disagree entirely, suggesting the need for further education and communication about the environmental impact of herbal medicines.

3.8.5 Do You Think Herbal Medicines Are Safer Than Regular Medicine?

Furthermore, respondents were asked: "Do you think herbal medicines are safer than regular medicine?" as shown in Fig. 11. A clear majority of the respondents demonstrated a perception of herbal medicines as being safer than conventional medicines. Specifically, 27.2% of respondents agreed that herbal medicines are "a lot safer," and another 25.4% considered them "a bit safer." However, a significant proportion of respondents, 21.9%, believed that herbal medicines are "a bit less safe"

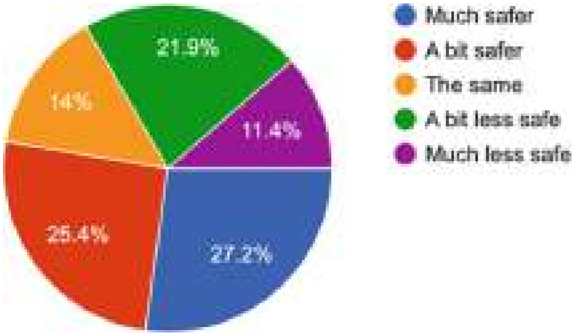


Fig. 11 Comparative safety perception of herbal versus regular medicines

than conventional medicines, indicating a more skeptical view. Additionally, 14% of respondents perceived the safety of herbal and conventional medicines as being the same, suggesting neutrality regarding the relative safety of these two categories of medicines. Lastly, a minority of respondents, 11.4%, believed that herbal medicines are "much less safe" than conventional medicines. In summary, the data reveals a notable majority of respondents perceive herbal medicines as safer than conventional medicines, which could be promising for market trends. However, a significant minority expressed safety concerns or held a perception of equal safety, highlighting the need for further consumer education and potential investigation into safety procedures for herbal medicines.

3.9 Objective Four: To Identify the Barriers and Challenges Consumers Face in Accessing Sustainable Herbal Medicine Products

To access the barriers and challenges consumers face in accessing sustainable medicine. This aspect of the assessment examined the perception of consumers regarding the regulation of herbal medicine. The study set to identify the barriers and challenges consumers face in accessing sustainable herbal medicine products. The survey results provide valuable insights into the challenges and barriers that consumers face when accessing herbal medicine products. They include:

1. **Regulation Perception**: A significant majority of respondents (86.7%) do not believe that herbal medicine is well-regulated. This perception could deter consumers from accessing these products due to concerns about safety, efficacy, and quality control.
2. **Information Gap**: The most common challenge faced when trying to find herbal medicine was a lack of sufficient information, as reported by 65.2% of respondents. This information gap could be related to understanding what herbal medicine products are available, how to use them, where to find them, and their benefits and potential side effects.

3. **Availability**: The second most common challenge was the difficulty in finding herbal medicine, as reported by 45.7% of respondents. This could be due to limited availability in local stores, lack of online options, or geographical constraints.
4. **Cost**: Although only a small proportion of respondents (3.3%) identified cost as a barrier, for those individuals, the price of herbal medicine products could be prohibitive.
5. **Frequency of Difficulty**: When asked about the frequency of difficulty in finding herbal medicine, 48.2% of respondents reported sometimes having trouble, and 10.7% reported often having trouble. This indicates that for a significant proportion of respondents, accessing herbal medicine is not always straightforward.

3.9.1 Do You Think Herbal Medicine Is Well-Regulated?

Response to the current regulation of herbal medicine by respondents is presented in Fig. 12. Most respondents, 86.7%, expressed the belief that herbal medicine is not well-regulated. This indicates a significant concern about the oversight and standards applied to herbal medicine in the current system. On the other hand, a much smaller proportion of respondents, 13.3%, believe that herbal medicine is well-regulated. This group represents a minority view that the existing regulations are sufficient. These results highlight a clear call for increased scrutiny and potentially more stringent regulation in the field of herbal medicine. Further research could explore the specific areas where respondents feel regulation is lacking and propose targeted solutions to address these concerns.

3.9.2 Have You Ever Had Trouble Finding Herbal Medicine?

Respondents were asked about their experiences with finding herbal medicine and their responses are presented in Fig. 13. The responses indicate a mixed experience among the respondents. A significant proportion, 48.2%, reported that they sometimes encounter difficulties when trying to find herbal medicine. This suggests that while they do not consistently struggle to find these products, there are instances where it becomes a challenge. A smaller group, 10.7%, reported often having trouble

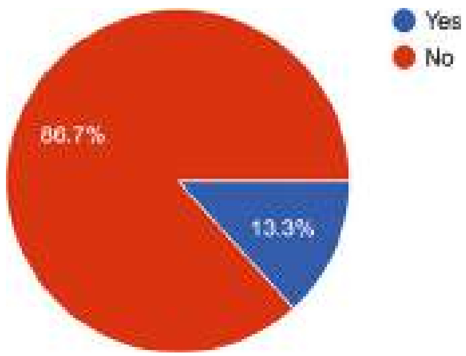


Fig. 12 Respondents believe herbal medicine is not well-regulated

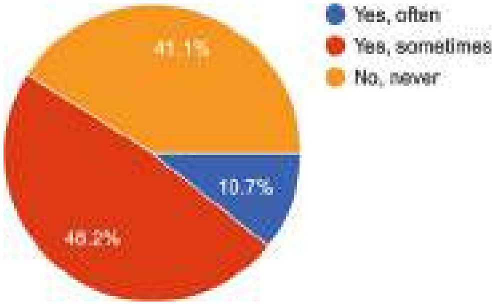


Fig. 13 Most respondents reported having had challenges with finding herbal medicine

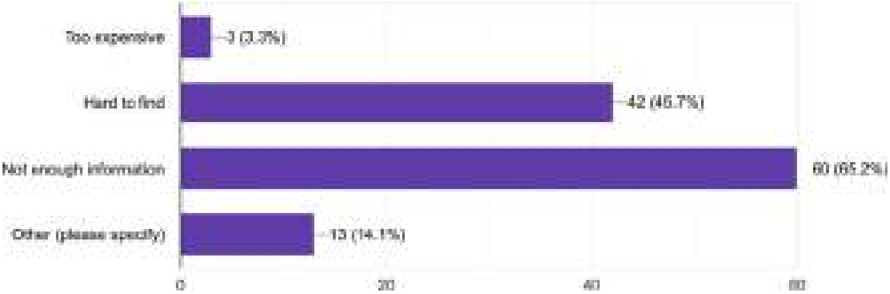


Fig. 14 Challenges associated with finding herbal medicines

finding herbal medicine. This indicates a more consistent struggle, suggesting that for these individuals, the availability or accessibility of herbal medicine is a recurring issue. On the other hand, 41.1% of respondents reported that they do not have trouble finding herbal medicine. This suggests that a substantial proportion of the surveyed population does not encounter significant barriers when seeking out these products. These findings highlight the variability in experiences when it comes to finding herbal medicine. They suggest the need for further investigation into the factors that contribute to these difficulties, particularly for those who often struggle to find these products. This could include factors such as geographical location, availability in local stores, or online access.

3.9.3 If You Had Trouble Finding Herbal Medicine, What Was the Problem?

Respondents were asked to identify the challenges they faced when trying to find herbal medicine and their responses are presented in Fig. 14. The most common issue, reported by 65.2% of respondents, was a lack of sufficient information. This suggests that there is a significant knowledge gap when it comes to understanding, locating, and using herbal medicine. This could be due to a lack of accessible, reliable, and comprehensive resources about herbal medicine. The second most common problem, reported by 45.7% of respondents, was that herbal medicine was hard to find. This could indicate issues with the availability or distribution of herbal medicine in certain areas or markets. A smaller proportion of respondents, 14.1%, reported other unspecified problems. Further research could be conducted to identify and understand these miscellaneous challenges. Finally, a very small group of respondents, 3.3%, reported that the cost of herbal medicine was prohibitive. While this was the least common issue, it is still a significant barrier for those individuals affected. These findings suggest that there are several key areas to address to improve access to and understanding of herbal medicine. These include enhancing the availability of information, improving distribution and availability, understanding, and addressing other miscellaneous challenges, and considering cost factors.

3.10 Open-Ended Questions on Consumer Perception of Herbal Medicine

3.10.1 What Do You Associate with Herbal Medicine?

Participants were asked, "What do you associate with herbal medicine? (i.e., when you think of herbal medicine what comes to mind)." The data was analyzed using thematic analysis and generated the following themes with the frequency of respondents as presented in Table 2. This result provides an overview of the themes identified in people's perceptions and associations with herbal medicines, their frequency, the respondents who mentioned each theme, and a brief description of each theme. The concept of "nature and naturalness" is the most frequently mentioned theme, followed by "safety and healthiness," and "concerns about quality and regulation." "Traditional medicine," "efficacy and effectiveness," and "concerns about dosage and prescription" also appear significantly in responses. The themes of "associations with illness and treatment," "taste and use," and "accessibility and affordability" were mentioned less frequently.

3.10.2 What Would You Like to See Improved Regarding Herbal Medicines?

Respondents were asked: "What would you like to see improved regarding herbal medicines?" and their responses are presented in Table 3. The result provides an overview of the themes identified in perceptions of improvement needs in herbal medicines, their frequency, the respondents who mentioned each theme, and a brief description of each theme. "Information and awareness" is the most frequently mentioned theme, followed by "dosage and prescription" and "packaging and labelling." "Regulation and authentication," "research and development," and "safety concerns" also showed up significantly in the responses. Themes of "accessibility and availability," "taste and aesthetics," and "support from government and NGOs" were mentioned less frequently. Some of these themes align with the earlier report of Fienzouli and Gori (2007) about the safety evaluation and verification of the pharmaceutical efficacy mechanisms of herbal medicines. This is also supported by the report of Bent [48] about the standardization of herbal medicine products and practices, increased regulations, and more funding for research into the products and practices.

Table 2 Perception and association with herbal medicines

Theme	Frequency	Description
Nature and naturalness	27	Associations of herbal medicine with natural products, herbs, plant-based substances, and organic qualities
Safety and healthiness	14	Perception of herbal medicines as safe, less harmful, pure, and offering health benefits with fewer side effects compared to synthetic drugs
Concerns about quality and regulation	16	Concerns about poor regulation, quality, preparation hygiene, and possible adulteration. These concerns often link to perceived risks
Traditional medicine	9	Viewing herbal medicine as traditional, local, or complementary to orthodox medicine, often connected with native doctors/herbalists
Efficacy and effectiveness	11	Beliefs about the effectiveness, potency, and immediacy of herbal medicines in treating illnesses
Concerns about dosage and prescription	8	Worries about proper dosage instructions, lack of research-based data on dosage, and possible health risks due to incorrect dosing
Associations with illness and treatment	8	Associations of herbal medicine with treatment of specific conditions, such as pain, fractures, fertility, and childbirth issues
Taste and use	5	Associations with the taste of herbal medicines, often described as bitter, and their use in food and drinks
Accessibility and affordability	2	Perceptions of herbal medicines as more accessible and affordable compared to orthodox medicine

4 Conclusion

While the cross-sectional design adopted for the assessment reported in this chapter enables the provision of a comprehensive snapshot of consumer perceptions and demand at a specific point in time, it does not allow for the examination of changes over time. Moreover, future research could employ a longitudinal design to track changes in perceptions and demand over time. Additionally, while the use of both closed and open-ended questions in the online assessment tool enabled a comprehensive view of consumer perceptions and demands, the responses are still subject to participant interpretation and bias. Future research could employ more objective measures, such as behavioral observations or physiological measures, to complement the self-report data as suggested in Bryman [33]. Nonetheless, this chapter noted that a significant minority expressed a lack of awareness, usage, or intent to purchase/recommend herbal medicine, which suggests that while there is a high level of awareness and engagement with herbal medicine, there is still a segment of the population that may lack knowledge about these products. Nevertheless, there is a high level of consumer awareness and

Table 3 Perception of improvement needs in herbal medicines

Theme	Frequency	Description
Regulation and authentication	11	Calls for better regulation, authentication, and production standards for herbal medicines
Information and awareness	20	A desire for more enlightenment, information, and awareness about herbal medicines, including details about content, usage, and potential risks
Packaging and labelling	16	Need for improvements in packaging, hygiene, and clarity in labelling with ingredients
Dosage and prescription	17	Emphasizing the importance of clear dosage instructions and prescriptions to avoid potential issues such as overdosing
Research and development	13	Calls for increased research on the safety, efficacy, and contraindications of herbal medicines
Accessibility and availability	7	Suggestions for better access to, and visibility of, herbal medicines
Taste and aesthetics	3	A desire for improvements in the taste and presentation of herbal medicines
Safety concerns	12	Expressions of concern regarding the safety of herbal medicines, including possible side effects and growing conditions
Support from the government and NGOs	2	Suggestions for increased governmental and nongovernmental support and involvement in promoting and regulating herbal medicine

knowledge about herbal medicine products even though there is still a segment of the population that may require further education and outreach. Importantly, the perceived efficacy, availability, safety, and packaging of herbal medicines are key drivers of consumer demand. While price is also key a consideration, it is less influential compared to the other factors considered in this assessment. These insights can inform strategies for businesses in the herbal medicine sector, particularly those focusing on sustainable products. Also, the majority of consumer perception of the efficacy and safety of sustainable herbal medicines is largely positive, suggesting promising prospects for this market. However, there exist some concerns related to safety, efficacy, and environmental impact among a minor yet significant proportion of consumers, highlighting the need for effective communication strategies and potential product enhancements. In conclusion, the key barriers to accessing sustainable herbal medicine products appear to be concerns about regulation, a lack of information, issues with availability, and for some, cost. To improve access to these products, efforts could be made to enhance regulatory transparency, provide more comprehensive and accessible information, improve distribution and availability, and consider affordability. Further research could also be conducted to understand the specific “other” challenges faced by some respondents.

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Indigenous Knowledge and Phytochemistry: Deciphering the Healing Power of Herbal Medicine

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Abstract

Indigenous phytochemistry (i.e., the understanding of the chemical constituents of plants by people of a certain locale) is key to the use of plants in herbal medicine. The chemical composition, beneficial interactions, and therapeutic applications of medicinal plants are discussed, and the importance of indigenous knowledge systems and cultural wisdom is emphasized in this chapter. It delves into the cultural ideas and perceptions that contribute to the success of traditional healing practices, as well as the rituals that accompany them. A more holistic and culturally acceptable approach to healthcare can be achieved by learning about the cultural importance of medicinal plants. Ethical considerations and working together with indigenous communities are also highlighted. Thus, as partners in research, they must treat each other with respect and work together to safeguard intellectual property and distribute profits fairly. Moreover, herbal medicine can be more welcoming as well as culturally sensitive if indigenous knowledge is given due respect and indigenous populations are included in the research process since indigenous phytochemistry has the potential to revolutionize our understanding of herbal medicine's healing properties. Researchers can tap into the potential of indigenous knowledge systems by combining cultural understanding, ecological expertise, and cutting-edge science to create effective, culturally grounded herbal treatments that also support long-term health.

Keywords

Indigenous knowledge · Phytochemistry · Traditional healing · Herbal medicine · Modern science

Abbreviations

CE	Capillary electrophoresis
FTIR	Fourier transform infrared spectroscopy
GC	Gas chromatography
GC-MS	Gas chromatography-mass spectrometry
HPLC	High-performance liquid chromatography
HRMS	High-resolution mass spectrometry
ICP-MS	Inductively coupled plasma mass spectrometry
LC	Liquid chromatography
LC-MS	Liquid chromatography-mass spectrometry
MALDI	Matrix-assisted laser desorption/ionization
MRI	Nuclear magnetic resonance imaging

MSI	Mass spectrometry imaging
NMR	Nuclear magnetic resonance
SFE	Supercritical fluid extraction
SPME	Solid-phase microextraction
TCM	Traditional Chinese medicine
TLC	Thin-layer chromatography
UV-Vis	Ultraviolet-visible spectroscopy
WHO	World Health Organization
XRD	X-ray diffraction

1 Introduction

Indigenous phytochemistry (i.e., the understanding of the chemical constituents of plants by people of a certain locale) is a branch of ethnopharmacology, which is the scientific investigation of the pharmacological effects of natural products used by different cultures. Consequently, culture, customary expertise, and the curative properties of medicinal plants are deeply intertwined in the indigenous herbal medicine practiced by indigenous populations for generations [11, 12, 52, 130, 135, 238, 278]. Traditional indigenous practices and beliefs have been continuously passed down through the generations in indigenous communities and make up the bulk of indigenous knowledge systems [92, 132]. The practice of traditional medicine is an integral part of these systems of expertise because it is grounded on cultural understanding and experiential learning about the therapeutic effects of medicinal plants [67, 136].

Research into medicinal plants from an ethnobotanical perspective is crucial to preserve traditional healing practices [81, 82]. Researching the historical and cultural importance of medicinal plants, these studies integrate anthropology, botany, and chemistry [256, 285]. Ethnobotanical research guarantees the preservation of traditional knowledge and validates the value of traditional medicine by recording it using rigorous scientific approaches [266].

Understanding the chemical components and bioactive chemicals present in medicinal plants requires the use of phytochemical analysis [25, 153–157]. Chromatography, mass spectrometry, and nuclear magnetic resonance spectroscopy are just a few examples of modern analytical techniques that can be used to isolate and characterize individual chemicals [171]. The complex blend of active chemicals responsible for the therapeutic benefits of traditional herbal treatments can be uncovered through phytochemical analysis [287, 21]. This research lends credence to the age-old practice of using plants as medicine and may lead to the development of novel therapeutics.

The notion of synergy motivates the use of many plant constituents in complicated formulations common in traditional herbal therapy. The therapeutic efficacy is increased when many drugs interact synergistically [166]. This indicates that traditional formulations that combine several plant ingredients can increase bioavailability and pharmacokinetics. A greater appreciation for herbal medicine's curative potential can be gained by investigating the synergistic interactions seen in traditional formulations.

Traditional herbal medicine relies heavily on cultural knowledge to be effective. Traditional practitioners have extensive expertise with various plant species, medicinal preparations, and dosages [91]. This accumulated knowledge from previous generations guarantees that medicinal herbs are used effectively within their designated cultural frameworks. Therefore, realizing herbal medicine's full therapeutic potential requires appreciating the reciprocal relationship between cultural awareness and native phytochemistry. Phytochemical examination reveals the chemical elements of therapeutic plants, while ethnobotanical investigations chronicle and authenticate indigenous understanding [278].

In summary, this chapter will deal with the chemical composition, beneficial interactions, and therapeutic applications of medicinal plants by indigenous people. The importance of indigenous knowledge systems and cultural wisdom will also be emphasized in this chapter. It will also delve into the cultural ideas and perceptions that contribute to the success of traditional healing practices, as well as the rituals that accompany them. The need for a more holistic and culturally acceptable approach to healthcare by learning about the cultural importance of medicinal plants will also be highlighted. Ethical considerations and collaboration with indigenous communities will also be highlighted as partners in research. Hence, they need to treat each other with respect and work together to safeguard intellectual property and distribute profits fairly.

2 Integrating Cultural Wisdom and Modern Science

Culture and science can be integrated into today's fast-changing environment. The wisdom of cultures is regarded as the total of humankind's collected knowledge, ideals, and practices across time and societies [191]. It is strongly ingrained in global cultures and traditions. On the contrary, the scope of our modern-day science extends to the exhaustive investigation and clarification of the environment via tested theories and methods [101]. Despite their differences, indigenous knowledge and modern cutting-edge research can work together to address complex global concerns and promote comprehensive health [223].

Insights into sustainable practices, ecological responsibility, and the advancement of general healthcare can be gained through cultural wisdom gleaned via indigenous knowledge systems [141]. It includes in-depth knowledge of ecological connections, traditional medical practices, and the links between people and the natural world. Historical environmental knowledge and indigenous practices have supported communities for generations. Thus, incorporating cultural understanding into scientific study will help us to tap into these resources. This synthesis will allow for the protection of historical artifacts while also yielding fresh approaches to pressing problems of the present.

With its focus on empirical data and rigorous techniques, modern science provides systematic ways to make sense of the world. It has helped propel the fields of medicine, technology, and the hard sciences forward by leaps and bounds. Traditional practices can be made more effective and secure by incorporating current scientific approaches with cultural wisdom. Both indigenous physicians and the

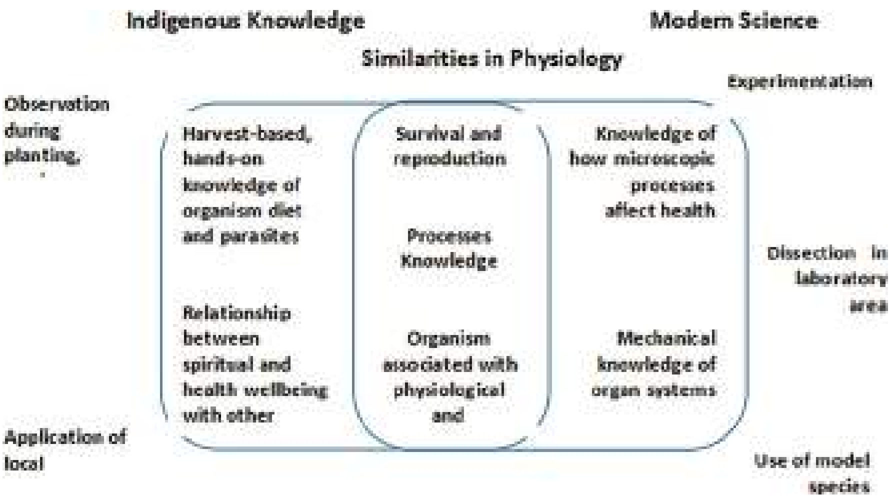

Fig. 1 Indigenous knowledge and modern science share complementary conceptual underpinnings in physiological research

scientific community can gain from cross-cultural partnerships that are made possible by this integration [16].

Moreover, healthcare is one field where traditional knowledge and cutting-edge research both have a place. Holistic methods of wellness and rejuvenation are common in traditional medical treatments because they are grounded in cultural wisdom. Physical attributes, psychological, and spiritual wellness are all seen as interdependent in indigenous healing practices [99]. Thus, care that is both culturally competent and patient-centered can be achieved when conventional medicine is combined with alternative therapies [87].

Also, indigenous knowledge systems are better able to survive and thrive when current scientific information is integrated with cultural knowledge [70, 141]. Sustainability in managing resources, safeguarding biodiversity, and adapting to climate change are all areas in which indigenous cultures excel [27]. Therefore, improved long-term and robust strategies for environmental protection will be possible once we recognize and incorporate this information. Traditional knowledge is preserved and passed on from generation to generation by the incorporation of cultural wisdom into educational and research practices [284] (Fig. 1).

3 Exploration of Bioactive Compounds to Unveil Therapeutic Secrets

The medicinal potential of bioactive chemicals isolated from plants, animals, and microbes has been known for quite some time. The use of these chemicals to create new medications and therapies to treat disease and enhance human health has great potential. According to Newman and Cragg [208], bioactive chemicals are "chemical molecules detected in an environment that exhibit a pharmacological effect on living organisms." Antioxidant, antibacterial, anti-inflammatory, anticancer, and neuroprotective are only some of the activities they display.

Discovering bioactive chemicals entails isolating and studying molecules from nature that may have useful medical properties. Medicinal plants, marine species, and microbial extracts are just some of the natural sources that can be screened for bioactive chemicals [106]. The discovery of new bioactive chemicals has been greatly aided by the development of cutting-edge technologies and methods [142, 262].

The medicinal potential of bioactive chemicals discovered through exploration is illustrated by several examples. The cancer therapy paradigm was shifted with the introduction of paclitaxel, which was isolated from the Pacific yew tree (*Taxus brevifolia*) [212]. Isolated from the *Artemisia annua* plant, artemisinin is useful in treating malaria [172]. In each of these cases, bioactive chemicals are used to their fullest potential, illustrating their promise in resolving serious health issues.

Bioactive chemicals are also excellent starting points for developing new medicines. The efficacy, preference, and pharmacokinetic features of these molecules can be improved through structural alterations and optimization once they have been identified [199]. Researchers can design and produce more effective and safer medications that target specific diseases by tapping into the curative value of bioactive molecules.

Therefore, this chapter focuses on cultural knowledge and indigenous phytochemistry with a special focus on the significance of indigenous knowledge in phytochemistry, the chemical composition of medicinal plants in indigenous knowledge systems, traditional healing practices and their phytochemical foundations, an intertwining of culture and phytochemistry, indigenous wisdom as a treasure trove of phytochemical discoveries and biodiversity conservation, challenges and ethical considerations in indigenous knowledge research, approaches that respects and safeguard indigenous knowledge, synergies between indigenous knowledge and modern phytochemistry in herbal medicine, and the future directions in advancing indigenous phytochemistry research.

4 The Significance of Indigenous Knowledge in Phytochemistry

The science of phytochemistry benefits greatly from indigenous knowledge, which originates in the traditional understandings and practices of indigenous societies. Traditional ecological knowledge and the in-depth insight indigenous peoples hold are invaluable resources for the field of phytochemistry, which studies chemical compounds originating from plants. The section delves into how indigenous knowledge has helped uncover and make use of bioactive substances found in therapeutic plants through the field of phytochemistry.

Native peoples have relied on the natural world for centuries, both as a source of food and medical care [133, 206, 207]. A wide range of plant species, their medicinal properties, and techniques of preparation are all well known to these people [91]. This historical understanding is essential for locating and employing medicinal plants in modern phytochemical studies.

The ability to recognize plants with therapeutic potential is one of the indigenous knowledge's greatest contributions to phytochemistry. Through millennia of observation and testing, indigenous cultures have amassed knowledge about the healing capabilities of numerous plant species [92]. Researchers can benefit greatly from their knowledge of medicinal plants and their indications [117, 120, 260, 261, 312] as it helps them zero in on plants with the most potential for further phytochemical investigation.

4.1 Indigenous Knowledge Systems as a Holistic Approach to Healing

Indigenous knowledge systems provide a comprehensive approach to healing that spans one's physical, psychological, and spiritual health [177]. These systems are profoundly anchored in the cultural legacy and wisdom of Indigenous groups, building on centuries of customary procedures and views.

Indigenous knowledge systems describe health as a situation of harmonious integration between people, groups, and the natural environment. The healing techniques linked with these traditions of knowledge frequently comprise a blend of herbal remedies, rituals, faith, and civic engagement. These practices reflect the interdependence of all elements of life and the necessity to address imbalances at numerous levels for comprehensive healing [110].

For instance, in many Indigenous societies, the usage of herbal remedies plays a fundamental part in healing. These plants are considered to offer great healing powers that help promote equilibrium and enhance well-being. The understanding of these plants and their applications is passed down through generations, indicating an intimate relationship between Indigenous communities and their natural environs [65, 78, 134].

Spirituality is another key component of Indigenous healing traditions. Indigenous cultures generally emphasize the necessity of connecting with the spiritual realm and having a peaceful relationship with ancestors, supernatural beings, and the higher powers. Spiritual rituals and rites are undertaken to correct spiritual deficiencies, maintain peace, and facilitate recovery [20].

Thus, the holistic approach of Indigenous knowledge systems spans beyond the individual to cover the well-being of the community as a whole. All members of the community are considered essential to the healing process. Social bonds, group rituals, and the participation of elders and traditional healers are commonly emphasized in indigenous healing practices [110].

When used as a whole, Indigenous knowledge systems have a profound effect on health that goes far beyond the scope of any single person. Indigenous communities benefit from these structures because they foster a sense of pride, belonging, and cultural continuity [110]. The gap between Western medicine and Indigenous healing practices can be closed by greater recognition and incorporation of Indigenous knowledge into mainstream healthcare systems, hence promoting cultural respect and better health outcomes for Indigenous communities [20].

4.2 Cultural Diversity and Traditional Healing Practices

Traditions, beliefs, and practices that have been passed down from one generation to the next are all part of cultural variety [110]. Traditional medicine is an important part of many different cultures. Different civilizations' historic approaches to health and wellness provide valuable insights into the spiritual dimensions of life. Traditional healing practices have developed over the ages and throughout cultures [86]. Herbal medication, spiritual rituals, energy healing, and mind–body therapies are all included in the broad category of traditional healing.

Traditional medical practices are a reflection of a culture's worldview and medical understanding of illness and health. Traditional healing, for instance, has strong ties to both the natural and spiritual realms in Indigenous cultures (Table 1). Traditional healing for them includes the use of herbs, rituals, and shamanism [110]. These methods promote total health by emphasizing the interdependence of the individual, the group, and the environment.

Traditional Chinese medicine (TCM) has also been practiced for thousands of years and is an integral part of Chinese culture [111]. TCM uses a wide range of techniques, including acupuncture, herbal medicine, and qi gong, to help the body return to a state of health and harmony. The principles of Yin and Yang, Qi (energy), and the dynamic relationship between the various body parts form the foundation of these practices [175].

Because they provide complementary and alternative methods to Western medicine, traditional healing modalities can better meet the requirements of a wide range of people and help to create a more equitable healthcare system [84]. They tend to take into account not only the physical but also the mental and spiritual well-being of their patients. There is growing consensus that underserved groups can benefit from traditional healing practices. Access to care, patient happiness, and patient agency in the healing process are all bolstered by culturally competent practices, according to the literature [71].

For the sake of future generations and the preservation of knowledge, it is essential to protect cultural variety and traditional healing practices. Fostering cultural pride, promoting international collaboration, and contributing to a more inclusive and fair society can all result from encouraging discourse, knowledge, and respect for varied healing traditions.

4.3 Preservation and Transmission of Indigenous Knowledge

Native American, Inuit, and other Indigenous peoples have accumulated and passed on a wealth of expertise through the ages [210]. Indigenous peoples' understanding

Table 1 Some cultures and their healing practices

Cultural diversity	Traditional healing practices	References
Ayurveda	Holistic system of medicine originating in India, focusing on balance and natural remedies	[224, 253]
Traditional Chinese medicine (TCM)	Ancient system of medicine in China was based on balancing Yin and Yang and using herbs, acupuncture, and other modalities	[188, 189]
Indigenous healing practices	Diverse healing practices from indigenous cultures worldwide, often incorporating rituals, plant medicines, and energy work	[73, 203, 6]
Shamanism	Spiritual healing practices involve communication with the spirit world, often through shamans or medicine people	[137, 305]
African traditional medicine	Traditional healing practices are rooted in African cultures, utilizing medicinal plants, divination, and spiritual beliefs	[8, 76, 217]
Unani medicine	Traditional healing system originating from Greece and popular in the Middle East, incorporating natural medicines and humoral theory	[113]
Native American healing practices	Healing practices of Indigenous peoples of North America involve ceremonies, herbal remedies, and sacred rituals	[237]
Australian Aboriginal healing	Traditional healing practices of Aboriginal Australians, including bush medicine, storytelling, and connection to the land	[299]
Tibetan medicine	Traditional healing system of Tibet, integrating herbal medicine, acupuncture, and spiritual practices	[218]
Folk medicine	Traditional healing practices are passed down through generations in various cultures, using local remedies and rituals	[184]
Traditional Mexican healing	Indigenous healing practices of Mexico, combine ancient Aztec and Mayan traditions with Catholicism	[68, 108]
Traditional Korean medicine	The holistic medical system in Korea incorporates acupuncture, herbal medicine, and lifestyle practices	[311]
Amazonian Shamanic healing	Healing practices of Amazonian indigenous tribes, utilizing sacred plants like ayahuasca and traditional ceremonies	[249, 138]
Hmong traditional healing	Healing practices of the Hmong people involve herbal remedies, spiritual rituals, and ancestral beliefs	[245, 310]
Islamic traditional healing	Healing practices are influenced by Islamic teachings, combining faith, prayers, and natural remedies	[102, 242]
Hawaiian traditional healing	Healing practices of Native Hawaiians, known as Lāʼau Lapaʼau, use plants, prayers, and spiritual connection	[149]
Siberian Shamanism	Traditional healing practices of Siberian shamans involve spiritual journeys, rituals, and herbal remedies	[77, 297]
Caribbean traditional healing	Healing practices of Caribbean indigenous cultures, incorporating plant medicine, rituals, and ancestral wisdom	[48]

(continued)

Table 1 (continued)

Cultural diversity	Traditional healing practices	References
Brazilian traditional healing	Traditional healing practices in Brazil, blend indigenous, African, and European influences with plant medicine	[37]
Maori healing	Traditional healing practices of the Maori people of New Zealand, known as Rongoā, use plants and spiritual connection	[186]
Romani (Gypsy) traditional healing	Healing practices of the Romani people, including herbal remedies, divination, and cultural ceremonies	[36]

of the world is intrinsically linked to their homelands, their traditions, and their own identities. There can be no sustainable development, cultural preservation, or community resilience without the transmission of Indigenous knowledge.

It is crucial to preserve Indigenous knowledge to ensure the survival of Indigenous cultures. Indigenous recuperation, ecological understanding, cultural practices, and sustainable resource management are just a few of the many areas that might benefit from Indigenous knowledge, which is grounded in the knowledge and expertise of Indigenous populations. Keeping Indigenous knowledge safe ensures that future generations can learn from the past and continue to feel a deep connection to their heritage.

Many methods are being used in the fight to keep Indigenous knowledge alive. Traditional methods, oral histories, and ecological knowledge can all benefit from careful documentation. Indigenous communities, academics, and researchers can work together to accomplish this [210]. Indigenous knowledge is best protected through documentation, which also serves as a significant resource for learning, teaching, and cultural renewal.

Reviving Indigenous cultural practices inside communities is also crucial to ensuring their survival. Facilitating the transfer of wisdom from older to younger generations is essential to this end. When it comes to preserving and spreading Indigenous knowledge, community-based activities like cultural camps, seminars, and mentorship programs play a crucial role [27]. Communities can benefit from these endeavors because they provide venues where people can discuss and learn about their history and traditions from those who have lived with them the longest.

However, there are considerable obstacles to conserving and passing on Indigenous knowledge. Indigenous knowledge systems have been eroded and lost due to colonization, globalization, and cultural assimilation. The transmission of Indigenous knowledge from one generation to the next has been hampered by historical trauma, forced relocation, and discriminatory laws, according to research by Nicholas and Markey [210]. The problems of conserving and conveying Indigenous knowledge are exacerbated by the decline of languages, the spread of urbanization, and the dominance of dominant cultures.

Together, respect and acknowledgment of Indigenous rights are essential to overcoming these obstacles. Indigenous peoples should have a say in how their knowledge is used and disseminated [27] and be included in the process of decision-making that impacts their knowledge systems. Supporting the safeguarding and propagation of Indigenous knowledge through initiatives like developing curriculum, language rejuvenation, and the integration of Indigenous perspectives into mainstream education can be achieved through the involvement of the indigenous commnities in its planning and implementation [143].

4.4 Contributions of Indigenous Knowledge to Phytochemistry Research

Indigenous knowledge has enhanced phytochemistry studies on medicinal plants and their uses (Table 2). Indigenous peoples know the qualities, uses, and preparation of local flora. This understanding helped find bioactive chemicals, produce novel medications, and advance phytochemical studies. Indigenous knowledge in phytochemistry research and its impact on science are examined in this chapter.

Indigenous knowledge systems have long recognized plant medicine and developed complex strategies for its use. Indigenous knowledge holders – traditional healers – have studied therapeutic plants for ages. This expertise covers medicinal plant identification, components, preparation, and treatment [278].

Native American wisdom is also a valuable resource for information about when and how to harvest plants for maximum bioavailability of their medicinal chemicals and which parts of the plant to use [278].

Artemisinin, a powerful antimalarial chemical, was discovered with the help of Indigenous knowledge and has since become a cornerstone of phytochemistry study. Traditional Chinese medicine practitioners recommended *Artemisia annua* (sweet wormwood) for treating fever and malaria symptoms [314]. Artemisinin, a potent antimalarial medication, was isolated and characterized as a result of research motivated by this ancient knowledge. Millions of lives were spared all around the world thanks to this breakthrough in malaria treatment.

Traditional healing practices and indigenous knowledge have both played important roles in the discovery of new bioactive substances with therapeutic potential. For instance, the Mapuche people of Chile were responsible for the discovery of antibacterial chemicals in the tree *Quillaja saponaria* thanks to their indigenous knowledge. The antibacterial, anticancer, and immunomodulatory effects of these saponins have been the subject of substantial research. Other medicinal plants like *Echinacea purpurea* and *Uncaria tomentosa* (cat's claw) have been discovered thanks to indigenous people's awareness of their bioactive chemicals and possible health benefits [278].

Sustainable practices have been furthered by combining Indigenous wisdom with advances in phytochemistry research. To ensure the continued availability of medicinal herbs, indigenous cultures practice environmentally responsible harvesting methods. Scientists can learn from Indigenous knowledge holders about these eco-friendly methods, and incorporate them into the creation of environmentally and ethically responsible strategies for plant-based therapy [314].

Table 2 Contribution of Indigenous knowledge to phytochemistry research

Contributions	Description	References
Traditional plant selection criteria	Indigenous knowledge provides valuable insights into the selection of plants for phytochemical research	[1]
Ethnobotanical surveys	Indigenous communities' knowledge of medicinal plants guides ethnobotanical surveys, identifying potential sources of bioactive compounds	[195]
Traditional extraction techniques	Indigenous knowledge contributes to the development of traditional extraction methods to isolate bioactive compounds effectively	[295]
Identification of medicinal plants	Indigenous knowledge aids in the accurate identification of medicinal plants, ensuring the use of the correct species for phytochemical analysis	[85]
Traditional knowledge of plant properties	Indigenous knowledge provides information on the medicinal properties of plants, guiding phytochemical studies toward specific therapeutic areas	[257]
Traditional methods of plant preparation	Indigenous knowledge informs traditional methods of plant preparation, influencing the extraction techniques used in phytochemical analysis	[72, 139]
Sustainable harvesting practices	Indigenous knowledge promotes sustainable harvesting practices, ensuring the conservation of medicinal plants for future research and use	[221, 259]
Traditional knowledge of plant conservation	Indigenous knowledge contributes to plant conservation efforts, preserving biodiversity for future phytochemical research and medicinal use	[204]
Traditional pharmacopoeias and formulations	Indigenous pharmacopeias and traditional formulations serve as valuable resources for phytochemical research and drug development	[222]

5 Chemical Composition of Medicinal Plants in Indigenous Knowledge Systems

Traditional Indigenous knowledge and modern scientific inquiry come together in the study of the chemical makeup of medicinal plants. This section identifies the bioactive compounds accountable for the plant's therapeutic properties, thereby lending credence to traditional remedies and opening the door to the development of novel pharmaceuticals.

5.1 Methods and Techniques Used for Phytochemical Analysis

Analysis of medicinal plants via the lens of their chemical components is called phytochemical analysis. The medicinal potential of plants can be better understood through the use of various approaches and techniques for isolating and characterizing bioactive substances (Table 3). Extraction, fractionation, and component identification are common steps in phytochemical investigation.

Table 3 Some methods and techniques used for phytochemical analysis

Method/technique	Description	References
Liquid chromatography-mass spectrometry (LC-MS)	Separation and identification of compounds in complex mixtures	[183]
Gas chromatography-mass spectrometry (GC-MS)	Separation and identification of volatile and semivolatile compounds	[145]
High-performance liquid chromatography (HPLC)	Separation and quantification of individual compounds	[219]
Thin-layer chromatography (TLC)	Separation and identification of compounds using a thin layer of adsorbent	[147]
Nuclear magnetic resonance (NMR)	Structural elucidation and identification of compounds	[19]
Fourier transform infrared spectroscopy (FTIR)	Identification of functional groups and structural analysis	[182]
Ultraviolet-visible spectroscopy (UV-Vis)	Quantification and analysis of compounds based on the absorption of light	[151]
Mass spectrometry imaging (MSI)	Spatial distribution and identification of compounds in tissues	[246]
Capillary electrophoresis (CE)	Separation of charged compounds based on electrophoretic mobility	[263]
Solid-phase microextraction	Extraction and pre-concentration of volatile compounds	[22]
High-resolution mass spectrometry	Accurate mass determination and identification of compounds	[205]
Supercritical fluid extraction	Extraction of compounds using supercritical fluids	[9]
Matrix-assisted laser desorption/ionization	Analysis of large biomolecules and polymers	[58]
Electrochemical analysis	Determination of redox properties and electroactive compounds	[115]
Nuclear magnetic resonance imaging	Imaging and localization of compounds in intact samples	[79]
X-ray diffraction	Structural analysis and identification of crystalline compounds	[275]
Gas chromatography	Separation and analysis of volatile and semivolatile compounds	[160]
Liquid chromatography	Separation and analysis of compounds in liquid samples	[140]
Inductively coupled plasma mass spectrometry	Quantification of trace elements and metals	[159]

Extracting bioactive chemicals from plant materials using appropriate solvents is the first step in phytochemical analysis. Methods such as maceration, Soxhlet extraction, and supercritical fluid extraction (SFE) are frequently used for extraction. Each approach has benefits and drawbacks; which one is chosen depends on the substances of interest and the specifics of the plant being studied [53].

The physical and chemical features of materials can be used in chromatography to separate and purify them [304]. Thus, the phytochemical analysis relies heavily on the identification and characterization of bioactive chemicals. Nuclear magnetic resonance (NMR) and mass spectrometry (MS) spectroscopy are two common spectroscopic methods used to identify compounds. To learn more about the compounds' structures, NMR is used in conjunction with MS [53].

5.2 Medicinal Plants and Their Bioactive Constituents

Bioactive chemicals are the chemical components of medicinal plants that are responsible for the plants' observed biological activity and therapeutic benefits [80, 118, 119, 126, 127] A wide variety of chemical structures, such as alkaloids, flavonoids, terpenoids, and phenolic chemicals, can fall within this category. Antioxidant, antibacterial, anti-inflammatory, and anticancer activity are only some of the many biological effects attributed to bioactive substances [63].

One of the most widely recognized classes of bioactive molecules extracted from plants used for medicine is alkaloids (Table 4). Their pharmacological effects, such as analgesic, antibacterial, and anticancer properties, are sometimes rather strong. Morphine, extracted from the opium poppy (*Papaver somniferum*), and quinine, extracted from the bark of the Cinchona tree (*Cinchona spp*.) [63], are both examples of alkaloids.

Antioxidant and anti-inflammatory characteristics are also possessed by flavonoids, another major class of bioactive chemicals. Fruits, veggies, and medicinal plants all contain these chemicals. Onions (*Allium cepa*) and chamomile (*Matricaria chamomilla*) both contain flavonoids such as quercetin and apigenin, respectively [63].

Aromatic plants typically contain terpenoids, such as essential oils. They have properties that make them effective against microbes, fungi, and insects. Menthol, found in peppermint (*Mentha piperita*), and limonene, found in lemon (*Citrus limon*), are both terpenoids [63].

Understanding the processes that govern the action and therapeutic applications of these medicinal plants requires the isolation and characterization of their bioactive components [116, 121–125]. As either lead molecules or sources of inspiration, they provide the backbone of the drug discovery process [63].

5.3 The Importance of Traditional Knowledge for Locating Medicinal Plants

Medicinal plant identification relies heavily on Indigenous knowledge, which draws from Indigenous communities' centuries-long store of tacit knowledge and experience. Traditional physicians and wisdom keepers have an in-depth familiarity with the native flora and the medicinal benefits of individual plants. This information includes the recognition, preparation, and usage of medicinal plants for diverse ailments, and it is typically transmitted orally. Researchers can take advantage of this wealth of knowledge and focus their hunt for potential medicinal plants by consulting Indigenous knowledge systems.

Table 4 Medicinal plants and their bioactive constituents

Medicinal plant	Bioactive constituents	References
Curcuma longa (Turmeric)	Curcumin, turmerones, curcuminoids	[105]
Panax ginseng (Ginseng)	Ginsenosides, polysaccharides, polyacetylenes	[158]
Hypericum perforatum (St. John's Wort)	Hypericin, hyperforin, flavonoids	[306]
Vinca rosea (periwinkle)	Vinblastine, vincristine, alkaloids	[190]
Camellia sinensis (green tea)	Epigallocatechin gallate (EGCG), catechins	[54]
Ginkgo biloba (ginkgo)	Flavonoids, terpenoids, ginkgolides	[178]
Allium sativum (garlic)	Allicin, sulfur compounds, flavonoids	[3]
Aloe vera	Anthraquinones, polysaccharides, flavonoids	[10]
Salvia officinalis (sage)	Rosmarinic acid, flavonoids, triterpenoids	[97]
Curcuma zedoaria (zedoary)	Curzerene, curcumenol, zedoarondiol	[7]
Silybum marianum (milk thistle)	Silymarin, silybin, flavonolignans	[2]
Echinacea purpurea (purple coneflower)	Alkamides, polysaccharides, flavonoids	[40]
Zingiber officinale (ginger)	Gingerol, shogaol, zingerone	[173]
Centella asiatica (gotu kola)	Asiaticoside, madecassoside, triterpenoids	[236]
Rosmarinus officinalis (rosemary)	Rosmarinic acid, carnosic acid, flavonoids	[293]
Mentha piperita (peppermint)	Menthol, menthone, rosmarinic acid	[194]
Camellia japonica (camellia)	Saponins, tannins, flavonoids	[227]
Cinnamomum verum (cinnamon)	Cinnamaldehyde, cinnamic acid, procyanidins	[31]
Withania somnifera (ashwagandha lehya)	Withanolides, alkaloids, steroids	[29]
Melissa officinalis (lemon balm)	Rosmarinic acid, flavonoids, terpenes	[181]
Matricaria chamomilla (chamomile)	Apigenin, chamazulene, bisabolol	[276]
Glycyrrhiza glabra (licorice)	Glycyrrhizin, flavonoids, coumarins	[239]
Berberis vulgaris (barberry)	Berberine, isoquinoline alkaloids, flavonoids	[114]
Coptis chinensis (goldthread)	Berberine, coptisine, palmatine	[56]
Curcuma aromatica (wild turmeric)	Curcumin, turmerones, curcuminoids	[98]
Urtica dioica (nettle)	Flavonoids, phenolic acids, lectins	[30]
Punica granatum (pomegranate)	Punicalagin, ellagic acid, anthocyanins	[143]

Based on their experiences in the natural world and their knowledge of the relationships between plants, ecosystems, and human health, indigenous knowledge holders have developed complex systems for identifying and selecting medicinal plants [278]. They are experts on the local flora and fauna and can identify plants by sight and sound alone. Traditional medicine identifies certain characteristics in plants as indicators of their therapeutic efficacy. For instance, the anti-inflammatory properties of *Curcuma longa* (turmeric) are indicated by its bright yellow color, and the wound-healing properties of *Calotropis procera* (milkweed) are indicated by the presence of milky latex [278].

Indigenous knowledge also includes an appreciation of how plants influence their surrounding ecosystems and other forms of life. Some plants are known to attract certain insects, animals, or soil types, and this information is used by traditional healers. These connections are extremely helpful for determining whether or not a plant has any therapeutic value. For instance, if certain species of butterflies or bees prefer the nectar of a specific flower, this may indicate that the plant has therapeutic properties [42]. Indigenous knowledge also includes an appreciation of how plants influence their surrounding ecosystems and other forms of life. Traditional practitioners of medicine take note of the connections between plants and their ecosystems, such as the presence or absence of particular insects, animals, or types of soil. Insights into a plant's possible medicinal capabilities can be gleaned from these correlations. If butterflies or bees go toward a particular flower for its nectar, that could indicate the plant has therapeutic properties [42].

The effectiveness and safety of medicinal plants are evaluated by Indigenous knowledge holders based on their own experiences and the knowledge of their communities. This understanding is gleaned from experience gained via the frequent usage of particular plants for medicinal reasons and the subsequent observation of their effects. To ensure the reliability of their selection criteria, indigenous people frequently use prescribed procedures and traditional practices to verify the efficacy of medicinal plants [278].

5.4 Phytochemical Diversity and Classical Methods of Plant Selection

In Indigenous knowledge systems, the rationale for choosing plants is generally based on accumulated observational and empirical information. Plants with potential therapeutic characteristics can be found using these parameters as a starting point. When choosing plants, indigenous peoples think about more than just their looks; they also examine the plants' natural setting, traditional usage, and cultural importance. Because of this selection, medicinal plants have a wide variety of phytochemical substances.

The interdependence of flora, fauna, and human well-being is a central tenet of indigenous knowledge systems. Conventional plant selection criteria think about where plants naturally grow and how they are dispersed [42]. Indigenous peoples have long known that plants' bioactive substances reflect their ecological traits and how they have adapted to their environments. Indigenous cultures make use of the unique chemical profiles of plants that have adapted to different conditions by selecting plants from places like high-altitude regions and coastal areas.

Indigenous knowledge systems also consider a plant's morphology when deciding which plants to use as medicines. Certain characteristics of plants have been linked by traditional healers' observation to beneficial effects. As stated by

Bussmann et al. [42], visual and sensory indications of medicinal potential include color, shape, texture, and scent. There is some evidence that antioxidant selection favors plants with red pigments, while antimicrobial selection favors those with strong smells.

Cultural value and Indigenous people's accumulated knowledge also factor into the traditional plant selection criteria. Medicinal plants with deep cultural roots tend to be given more attention. Plant selection criteria are influenced by indigenous groups' respect for and sharing of their predecessors' knowledge and experiences [278]. Some cultures place a high value on plants because of their significance in rituals and folklore.

Indigenous knowledge systems employ a wide variety of selection factors for choosing medicinal plants, which increases their phytochemical diversity. Plants contain a broad variety of bioactive substances, such as alkaloids, flavonoids, terpenoids, and phenolic compounds, which are together referred to as phytochemical diversity. The medicinal benefits of plants are based in part on the distinct chemical compounds that are produced by each species.

Indigenous tribes increase the number of plants with useful phytochemical compounds by using traditional plant selection criteria to include them in their pharmacopeias. The chance of finding new bioactive chemicals with pharmaceutical applications is boosted by this phytochemical variety. Thus, indigenous knowledge systems aid in the search for therapeutic plants by providing useful information about how to choose plants with a wide range of chemical constituents.

6 Traditional Healing Practices and Their Phytochemical Foundations

The use of medicinal plants is common in traditional healing practices since they are based on accumulated cultural knowledge. Various phytochemical substances found in these plants give them their medicinal effects. Hence, there is a need for insight into the efficacy and potential of these ancient medicinal practices to understand their phytochemical roots.

6.1 Traditional Healing Systems and Their Philosophies

Different cultures all over the world have evolved and passed down their own unique sets of beliefs and practices when it comes to traditional healing. The values and beliefs of the people from which these systems sprang are the bedrock upon which they stand.

The idea that everything in life is interrelated is central to many different types of alternative medicine. Traditional healers understand the linked nature of a person's bodily, mental, cognitive, and spiritual health. To them, health is achieved when all of these components are in tune with one another. For instance, the Yin and Yang theory is central to traditional Chinese medicine (TCM), which stresses the

Table 5 Some traditional healing systems and their philosophies

Traditional healing systems	Philosophies	References
Ayurveda	Holistic approach to balance body and mind	[282]
Traditional Chinese medicine	Balance of Yin and Yang, Qi flow	[188]
Unani medicine	Balancing the four humors, vital force	[13]
Indigenous medicine	Harmony with nature, interconnectedness	[1]
Siddha medicine	Balance of elements and energy	[104]
Shamanism	Interaction with spirits and energies	[164]
Traditional African medicine	Ancestral spirits, energy balance	[300]
Native American medicine	Connection to nature, spirit guides	[274]
Traditional Korean medicine	Balance of Yin and Yang, Qi flow	[60]
Tibetan medicine	Balance of three humors, mind-body–spirit unity	[38]
Kampo medicine	Balance of Qi and body fluids	[269]
Australian Aboriginal medicine	Connection to land, dreamtime	[251]
Hmong traditional medicine	Balance of elements and energies	[281]
Native Hawaiian medicine	Connection to land and ancestors	[41]
Amazonian traditional medicine	Connection to the rainforest, plant spirits	[43]
Islamic medicine	Balance of bodily humors and spiritual well-being	[234]
Traditional Mexican medicine	Connection to Aztec and Mayan traditions	[44]
Maori Rongoā medicine	Connection to ancestors, spiritual healing	[301]

importance of maintaining a healthy equilibrium between seemingly opposed energies (Table 5).

Traditional medical tenets also place a premium on tailoring treatments to the specific needs of each patient. Because of this, traditional healers customize their care for each patient based on their distinctive characteristics and the nature of their illness or injury. This individualized treatment plan considers the whole person, not just their physical ailments. Traditional Native American medicine, for example, seeks to balance both the individual and the group [110].

Preventative care and general wellness upkeep are fundamental tenets of both modern and traditional medical philosophies. The systems in question place a premium on taking preventative actions for one's health. Traditional healing systems typically include practices such as recommending good lifestyle choices, nutritional guidelines, and stress reduction strategies. The goal of Ayurveda, India's traditional healing practice, is to avoid sickness by promoting health through a balanced way of living [95].

Often at the center of the ideologies of traditional healing methods are spirituality and the acknowledgment of a higher power Spiritual rituals, prayer, and communion with a higher power are integral parts of many time-honored approaches to healthcare. According to proponents of alternative medicine, one's emotional and physical health are inextricably linked. To re-establish the balance between the spiritual and material worlds, for instance, ceremonies and rituals are frequently incorporated into Indigenous healing practices [110].

The core tenets of conventional medical philosophies stress the significance of considering the whole person when treating illness or injury. Traditional healers understand the interconnected nature of a person's bodily, emotional, mental, and spiritual selves. Traditional healing systems seek to establish balance and harmony by attending to these facets as a whole, so bolstering the well-being of individuals and communities. Thus, underlying traditional treatment systems prioritize respect for the whole person, attention to detail, a focus on prevention, and a spiritual dimension. These beliefs underpin comprehensive strategies for health and well-being, which take into account the connections between a person's mental, emotional, and spiritual states.

6.2 Traditional Diagnosis and Treatment Methods

An essential aspect of any traditional healing system is the use of tried-and-true methods of diagnosis and treatment that have been passed down through the generations. These strategies are part of a more all-encompassing approach to healthcare that takes into account the patient as a whole, not just their physical symptoms. To promote holistic and individualized care, this chapter delves into the history of conventional approaches to diagnosis and therapy.

Observations, interviews, and tests are all components of the conventional diagnostic process. To diagnose a patient, traditional healers must rely on the five senses and their intuition. They learn about a person's health by observing outward manifestations of it, such as skin tone, facial expressions, and bodily movements [308]. Additional information on manifestations, history of illness, way of life, and emotional well-being can be gleaned from chats and interviews with the individual and their family. Traditional healers can establish a holistic picture of an individual's health and detect imbalances or disharmony by taking this all-encompassing approach.

Energy flow and balance are fundamental to both assessment and treatment in traditional healing modalities (Table 6). To heal from disease, according to traditional healers, one must first restore energy balance. For instance, in traditional Chinese medicine (TCM), abnormalities in the body's energy can be determined through pulse diagnosis and tongue examination [169]. Feeling the pulse at several locations on the wrist can reveal information about the health of various organs by picking up on variations in the pulse's quality, rhythm, and strength. Examination of the tongue can reveal information about a person's general health and the state of several organs. These diagnostic tools aid in making treatment decisions that bring about equilibrium and health.

Herbal therapy, acupuncture, massage, spiritual rituals, nutritional advice, and lifestyle changes are just a few of the many modalities that make up traditional treatment methods. In many alternative medical practices, plants and plant extracts

Table 6 Some traditional diagnosis and treatment methods

Traditional method	Description	References
Ayurveda pulse diagnosis	Assessment of health through pulse examination	[96]
Traditional Chinese medicine (TCM) diagnosis	Pattern differentiation based on Yin-Yang and five-element theory	[176]
Unani humoral diagnosis	Diagnosis based on the balance of four humors (blood, phlegm, yellow bile, black bile)	[94]
Indigenous Shamanic diagnosis	Communication with spirits and ancestors for diagnosis and treatment	[35]
Native American medicine diagnosis	Connection to nature, visions, and dreams for diagnosis	[185]
Tibetan medicine pulse diagnosis	Examination of the pulse for diagnosing imbalances and diseases	[211]
African divination methods	Interpretation of signs, symbols, and rituals for diagnosis and treatment	[215]
Kampo diagnosis	Assessment based on the theory of Qi and pattern identification	[213]
Native Hawaiian Ho′oponopono	Emotional and spiritual healing through reconciliation and forgiveness	[146]
Mexican Curanderismo diagnosis	Observation, palpation, and use of prayer for diagnosis and treatment	[174]
Korean Sasang constitutional medicine diagnosis	Assessment based on individual constitutional types and imbalances	[214]
Australian Aboriginal dreamtime healing	Connection to land, dreaming, and ancestral guidance for diagnosis and treatment	[198]
Hmong spirituality and rituals	Communication with spirits and use of rituals for diagnosis and healing	[47]
Native American sweat lodge	Purification and spiritual healing through sweat baths	[90]
Maori Rongoā traditional diagnosis	Observation, inquiry, and spiritual connection for diagnosis and treatment	[187]
Amazonian Shamanic diagnosis	Interaction with spirits and use of medicinal plants for diagnosis and treatment	[45]
Islamic Tibb diagnosis	Assessment based on the balance of body, mind, and soul	[112]
Traditional Mexican Sobadores	Manual manipulation, massage, and herbal remedies for diagnosis and treatment	[232]

are used as therapeutic tools. To treat their patients, traditional healers use their extensive knowledge of medicinal plants and their therapeutic characteristics to create unique herbal treatments [278]. Acupuncture is a key component of TCM, and it involves inserting very thin needles into the body at strategic spots along energy meridians to balance the flow of chi and induce healing. Ayurvedic massage and other forms of indigenous bodywork are two types of massage that focus on re-establishing equilibrium and improving blood flow.

Traditional healing practices also make use of spiritual ceremonies and rituals. Invoking the spirit world, making contact with ancestors or deities, and praying to them are all part of these healing rites. Prayer, chanting, smudging, and the usage of sacred artifacts are all examples of spiritual healing rituals [110]. These methods take into account the fact that one's physical, mental, emotional, and spiritual health are all intertwined and work to revitalize each of these areas.

A personalized and patient-centered focus is common in conventional medical practice. When treating a patient, traditional healers consider the person's history, lifestyle choices, and cultural context. Individualized meal plans, healing exercises and stretches, and tips for improving one's quality of life are all components of effective treatment regimens [308]. Such individualized attention promotes patient agency and active participation in their recovery by ensuring that treatment is tailored to their unique medical needs.

6.3 Phytochemical Basis of Traditional Healing Practices

Cultures all across the world have relied on time-tested methods of healing for hundreds of years. The use of medicinal plants and their derivatives is a common component of these practices. The presence of these plants' phytochemicals has been linked to the success of traditional healing methods. To put it simply, phytochemicals are substances found in plants that have biological activity and may contribute to their therapeutic effects. The traditional usage of plants for healing can be supported and expanded upon by learning more about the phytochemical underpinnings of these practices.

Traditional therapeutic practices such as India's Ayurvedic medicine are one example. The efficacy of the many different medicinal plants used in Ayurvedic medicine is often credited to the plants' phytochemical components. Curcumin, the active ingredient in turmeric (*Curcuma longa*), has been the subject of much research due to its anti-inflammatory, antioxidant, and anticancer characteristics [224]. Similarly, it has been established that the ginsenosides in *Panax ginseng* have adaptogenic, immunomodulatory, and neuroprotective properties [290].

Similarly, medicinal plants have a significant role in traditional Chinese medicine (TCM). In TCM, internal harmony and equilibrium are paramount. Phytochemicals are a key component of many TCM medicinal plants' therapeutic properties. Artemisinin, isolated from the herb *Artemisia annua,* is a powerful antimalarial drug (Leung and Wong 2021). Anti-inflammatory, antiviral, and anticancer properties are among the many pharmacological activities indicated by the bioactive chemicals found in TCM plants.

Medicinal herbs are used in indigenous healing practices throughout the world. These methods frequently necessitate an in-depth familiarity with the indigenous flora and its therapeutic characteristics. Native American societies in North America have long relied on the immune-boosting and antiviral capabilities of herbs like

Echinacea purpurea and *Sambucus nigra* [298]. The medicinal effects of these plants can be attributed in part to the presence of phytochemicals such polysaccharides, flavonoids, and phenolic compounds.

Traditional therapeutic modalities draw on the phytochemical base of a wide range of plant species. The therapeutic efficacy of herbs is typically multiplied in traditional herbal formulations because of the synergistic effect of combining numerous plant constituents. Plant mixtures with synergistic phytochemical profiles are more effective than single plant preparations [254]. Traditional remedies such as Triphala in Ayurveda are illustrative of this synergistic effect as they combine the fruits of three plants (*Emblica officinalis, Terminalia chebula*, and *Terminalia bellerica*). Antioxidant, anti-inflammatory, and cancer-preventing effects have all been observed in studies using Triphala.

Bioactive phytochemicals have been found in traditional therapeutic plants, according to recent scientific studies. These investigations into phytochemicals' pharmacokinetics and pharmacodynamics have shed light on their possible medicinal effects. The identification and quantification of medicinal plants' phytochemical ingredients have been aided by developments in phytochemical analysis techniques including liquid chromatography-mass spectrometry (LC-MS) and nuclear magnetic resonance (NMR).

6.4 Traditional Healing Rituals and Their Influence on Phytochemistry

Throughout history, traditional healing rituals have played a crucial role in the health and well-being of people from a wide range of cultural backgrounds. The phytochemistry of medicinal plants is profoundly affected by the rituals in which they are used [126–129]. A vast variety of rituals, from spiritual rites to the preparation and administration of herbal treatments, make up traditional healing practices. Traditional practices that incorporate an understanding of the phytochemical components of medicinal plants have helped to ensure their survival.

The Ayahuasca ceremony is a traditional healing ritual performed by natives of the Amazon rainforest. The *Banisteriopsis caapi* vine, together with other plants, is used to create a psychedelic drink known as ayahuasca. Participants in the ceremony are led by the shaman through a process of personal transformation that is thought to promote emotional, mental, and spiritual well-being. Phytochemical analyses have revealed that Ayahuasca plants like *Psychotria viridis* and *Diplopterys cabrerana* contain chemicals including N, N-dimethyltryptamine (DMT), and harmine. The psychedelic effects of Ayahuasca have been linked to the presence of these bioactive ingredients in the plants [243].

Acupuncture is largely accepted as a legitimate medical modality within TCM. The goal of acupuncture is to restore the natural flow of life force energy (Qi) through the body by inserting very fine needles into precise sites along the body's meridians. In addition to acupuncture, several herbal formulae are also employed to treat a wide range of conditions. Gui Zhi Tang, a traditional Chinese medicine formulation, has been scientifically proven to include bioactive chemicals including cinnamaldehyde thanks to the inclusion of the herb *Cinnamomum cassia*. The historic usage of the formula for pain treatment is supported by the chemicals' anti-inflammatory and analgesic effects [55].

Medicinal plant phytochemistry is heavily influenced by traditional indigenous healing practices. Smudging is a ritual in which sacred plants are burned to clear the air and encourage healing; it is widespread in Native American cultures. Sacred plants include white sage (*Salvia apiana*) and cedar (*Thuja* species). The aromatic qualities and possible antibacterial benefits of these herbs come from volatile components like terpenes and phenolic chemicals. Cineole and thujone, among other chemicals, have been isolated from these plants through phytochemical study [226]. The presence of these bioactive substances lends credence to the age-old practice of smudging as a means of purification and spiritual upliftment.

Traditional healing practices like these illustrate the relationship between phytochemistry and medicinal plants. Knowledge of medicinal plant identification, preparation, and application has been preserved and passed down through traditional ceremonies. Scientific support for their traditional use is found through the phytochemical study of these plants, which also confirms the presence of bioactive chemicals responsible for their medicinal qualities.

7 Traditional Herbal Remedies: An Intertwining of Culture and Phytochemistry

This section analyses how traditional herbal remedies are used, which reflects the diverse cultural heritage of people all over the world. Their effectiveness is frequently credited to the intricate interactions between phytochemical components and the human body.

7.1 Traditional Herbal Formulations and Their Preparation

For centuries, people have been making and using herbal remedies that adhere to precise protocols that have been passed down through oral history and cultural norms (Sharma et al. 2020). The specific cultural environment and prevailing body of traditional knowledge determine the range of possible approaches (Table 7).

Herbal formulations are commonly made by methods including decoction, powdering, and oil extraction in Ayurveda, India's traditional medical system (Sharma et al. 2020). To make a decoction, plants are combined and boiled in water to release their medicinal properties. Herbs can be powdered and utilized internally or topically after being ground into a fine powder. The therapeutic properties of herbs are extracted by oil extraction techniques like taila and sneha kalpana by boiling the herbs in a specific oil or ghee [279].

Decoction, soaking, and steaming are also common methods of preparation for herbal formulations in TCM [247]. To make a decoction, plants are boiled together in

Table 7 Some traditional Indian herbal formulations and their preparation

Traditional herbal formulation	Preparations	References
Triphala	Decoction, powder, capsules	[250]
Chyawanprash	Jam-like confection	[264]
Ashwagandhadi lehya	Herbal jam	[235]
Maha Sudarshan Churna	Herbal powder	[51]
Hingvadi Churna	Herbal powder	[23]
Rasayana Churna	Herbal powder	[267]
Bhringarajasava	Fermented herbal tonic	[220]
Panchagavya Ghrita	Medicated ghee	[264]
Chandanasava	Fermented herbal tonic	[296]
Brahmi Ghrita	Medicated ghee	[148]
Shilajit Rasayan Vati	Mineral and herbal tablets	[303]
Chandraprabha Vati	Herbal tablets	[264]
Mahanarayan oil	Herbal oil	[50]
Mahamanjisthadi Kwath	Herbal decoction	[270]

water, usually in predetermined amounts and times, to draw out their medicinal properties. Herbs can be soaked to extract their healing properties by submerging them in a liquid like wine or vinegar for an allotted amount of time. Herbs' medicinal capabilities and flavor profiles can be altered by the steaming process.

The World Health Organization [307] notes that in traditional African medicine, herbal formulations frequently involve infusion, maceration, or fermentation. Herbs' medicinal properties can be extracted by infusion by steeping them in hot water. Maceration is the process of soaking plants in a liquid (often water or alcohol) for a set amount of time to extract active chemicals that have therapeutic effects. Fermentation is the process of transforming and enhancing the therapeutic properties of herbs by allowing them to undergo microbial fermentation in a liquid.

It is worth noting that the ingredients and methods used to prepare traditional herbal remedies have significant cultural and historical roots. The recipes and methods used to make them are usually unique to a given region or community and have been passed down through the generations. It is vital to respect and conserve the old methods and knowledge systems while there is a growing interest in modernizing the creation of traditional herbal compositions. These preparations are effective and holistic because of the cultural background and traditional wisdom that went into them.

7.2 Combining Multiple Plant Ingredients for Synergistic Effects

Herbal formulations often contain several different plants, each used for different purposes [11, 12, 130]. This strategy stems from the idea that complementary interactions between different plant components can boost treatment outcomes.

Each plant was chosen for its intended medical characteristics and how well it would work with the others.

Ayurveda, TCM, and traditional African medicine are just a few examples of traditional medical systems that have long understood the benefits of mixing herbs for synergistic effects. A person's distinctive make-up and the root causes of imbalance or sickness are central to the holistic therapeutic emphases of these systems.

When many herbs are combined into one concoction, practitioners of Ayurveda refer to it as a "herbal formula" or "polyherbal preparation." These preparations are developed to treat particular ailments by combining herbs with synergistic qualities. Classical Ayurvedic formulations like "Triphala" blend three fruits for a variety of health benefits [230]: *Terminalia chebula*, *Terminalia bellerica*, and *Emblica officinalis*.

Herbal formulae in traditional Chinese medicine (TCM) are generally composed of many herbs that function synergistically to reestablish harmony and improve health. When putting together a prescription, it is important to think about how each herb will work with the others. To nourish Yin, tonify the kidneys, and promote the body's vitality, for example, the TCM prescription "Liu Wei Di Huang Wan" comprises six herbs [268].

Multiple plant elements are often combined in traditional African medicine to increase their medicinal benefits. In West Africa, a popular herbal remedy called "Agbo" combines the healing powers of many plants. These preparations are often formulated for individual patients and are effective in treating a wide variety of diseases and disorders [76].

Multiple plant components are included because of the potential for synergistic interactions between their bioactive constituents. Alkaloids, flavonoids, terpenoids, and phenolic compounds are only a few examples of phytochemicals that can be found in various plants and combined to create a more powerful medicinal impact [209].

The potential advantages of mixing plant substances have also been recognized by scientific inquiry. Antioxidant, anti-inflammatory, antibacterial, and anticancer effects of specific herbal combinations have been demonstrated to be significantly higher than those of single plant extracts. Multiple mechanisms, such as the modification of signaling pathways, enhanced bioavailability, and enhanced cellular absorption of active substances, have been proposed to underlie these synergistic effects [309] (Table 8).

7.3 Indigenous Practices of Herbal Medicine Safety and Quality Control

Traditional herbal therapy has a long and substantial history of use in healthcare systems around the world. To guarantee the efficacy and safety of herbal treatments, these procedures frequently incorporate safety safeguards and quality control processes. For example, many indigenous peoples have their own methods for

Table 8 Combination of multiple plant ingredients for synergistic effects

Indigenous medicine system	Herbal formulation	Plant ingredients	Synergistic effects	References
Ayurveda	Triphala	*Terminalia chebula, Terminalia bellerica, Emblica officinalis*	Promotes digestion, detoxification, and overall well-being	[18]
Traditional Chinese medicine (TCM)	Liu Wei Di Huang Wan	*Rehmannia glutinosa, Dioscorea opposita, Cornus officinalis, Alisma orientale, Poria cocos*	Nourishes yin, tonifies kidneys, and supports vitality	[57]
Indigenous medicine (West Africa)	Agbo	Various combinations of medicinal plants	Used for a wide range of ailments	[76]
Native American	Four Thieves Vinegar	Garlic, rosemary, sage, thyme, lavender, and other aromatic herbs	Antimicrobial, antiviral, and anti-inflammatory properties	[74]
Maori medicine (New Zealand)	Kawakawa balm	*Macropiper excelsum* (Kawakawa), combined with carrier oils or beeswax	Soothes skin irritations, eases muscle and joint pain	[103]
Mayan medicine (Maya, Mexico)	Xunan Kab	*Artemisia vulgaris, Simmondsia chinensis, Haematoxylum brasiletto, Calophyllum brasiliense*	Treats menstrual disorders, strengthens reproductive health	[233]
Aboriginal medicine (Indigenous peoples of North America)	Smudging	Sage, sweetgrass, cedar	Cleansing, purifying, and spiritual rituals	[252]
Traditional Hawaiian medicine (Lomilomi)	La'au Lapa'au	Noni, Kukui, Awapuhi, Olena	Healing various ailments and injuries	[165]
Inuit medicine (Inuit, Arctic)	Aakulujjusi	Labrador tea, crowberry, Arctic willow	Promotes physical and mental well-being	[17]
Traditional Amazonian medicine (Shamanism)	Ayahuasca	*Banisteriopsis caapi, Psychotria viridis*	Spiritual and healing ceremonies	[144]

Table 9 Indigenous practices for herbal medicine safety and quality control

Indigenous practice	Description	References
Ethnobotanical knowledge	In-depth understanding of traditional plant uses and safety precautions based on indigenous knowledge systems	[196]
Gathering and harvesting practices	Sustainable collection of medicinal plants, considering specific seasons, plant parts, and growth stages	[49]
Preparation methods	Traditional techniques such as decoction, infusion, maceration, or fermentation, ensuring optimal extraction of active constituents	[179]
Dosage and administration	Customary guidelines for dosage, frequency, and duration of herbal treatments, considering individual characteristics and specific conditions	[14]
Adverse event monitoring	Recognition and documentation of any adverse effects or interactions through continuous observation and communication within the community	[271]
Quality assessment	Traditional methods of assessing herbal quality, such as organoleptic evaluation, sensory perception, and specific tests for authenticity and purity	[289]
Conservation strategies	Indigenous practices for sustainable use, conservation, and cultivation of medicinal plants, including sacred groves and community-managed gardens	[168]
Traditional practitioner training	Apprenticeship and mentorship, passing down knowledge, skills, and ethical practices to future generations of healers	[34]
Collaboration with modern science	Integration of traditional knowledge with scientific research for safety evaluation, efficacy studies, and development of evidence-based practices	[107]

determining whether or not herbal remedies are safe to use. Observation, experience, and following established procedures are common components of such methods (Table 9). For instance, rongo is a practice used by the Maori people of New Zealand that entails the precise identification, preparation, and administration of plants in accordance with cultural protocols and conventional wisdom [180]. Many different indigenous cultures around the world, like the Quechua of the Andes, the Aboriginal of Australia, and the Native American tribes of North America, share similar rituals and beliefs.

Additionally, indigenous herbal medicine systems are dependent on traditional quality control practices. Traditional indigenous healers typically have years of education and apprenticeship under their belts before they are allowed to practice independently. To select the right species and use plants obtained at the right time and under the best conditions, these practices are guided by specialized rituals, cultural norms, and traditional ecological knowledge [278]. The purity and effectiveness of herbal medicines are maintained in part by indigenous tribes' emphasis on the need of precise plant identification and harvesting procedures.

The importance of indigenous knowledge systems in ensuring the quality and safety of herbal medicines has been increasingly acknowledged in recent years. The goal of cooperative projects between indigenous people and contemporary scientific organizations is to lessen the chasm between the two. This partnership utilizes cutting-edge analytical methods for quality control without compromising indigenous people's cultural norms, ethics, or IP rights.

To identify and quantify bioactive chemicals in medicinal plants, analytical techniques like high-performance liquid chromatography (HPLC), gas chromatography-mass spectrometry (GC-MS), and nuclear magnetic resonance (NMR) spectroscopy are used [170]. These methods are helpful for determining the chemical make-up of herbal medicines, which is essential for quality assurance, standardization, and the elimination of any potentially dangerous chemicals or adulterants.

Guidelines and regulatory frameworks that incorporate traditional knowledge into herbal medicine safety and quality control are now under development. The World Health Organization (WHO) has produced guidelines to help member nations set up regulatory frameworks for traditional medicine items including herbal medicines that guarantee their safety, efficacy, and quality [33].

The intellectual property rights of indigenous tribes must be recognized and respected before their traditional knowledge can be used in scientific study or for commercial purposes. A framework for guaranteeing the fair and equitable sharing of advantages generated from traditional knowledge and genetic resources is provided by the Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Advantages Arising from Their Utilisation [109].

8 Indigenous Wisdom as a Treasure Trove of Phytochemical Discoveries and Biodiversity Conservation

Indigenous peoples' relationships with their ecosystems, particularly the many plant species therein, form the foundation of their knowledge systems. Insights into the healing powers of plants and the need to preserve biodiversity can be found in this knowledge, which is a veritable treasure trove of phytochemical findings (Table 10).

Traditional plant identification and therapeutic use have long been depended upon by indigenous populations. Intricate knowledge of the medicinal powers of plants has been acquired by indigenous healers via meticulously observing, exploring, and transferring of expertise from one era to the next. Specific plant components, methods of preparation, and dosages for various conditions are commonly included in this body of knowledge [34]. Many phytochemicals with important medical significance have been discovered thanks to indigenous knowledge.

Indigenous Amazonians, for instance, know everything about the plants in their rainforest that may treat various ailments. Their knowledge helped determine that the Cinchona tree (*Cinchona* species), namely its bark, contains the antimalarial chem-

Table 10 Indigenous wisdom in phytochemical discoveries and biodiversity conservation

Indigenous wisdom	Description and examples	References
Traditional ecological knowledge (TEK)	Indigenous communities possess extensive knowledge about their ecosystems and the interconnections between species. They have developed sustainable practices for managing biodiversity and conserving medicinal plants. For instance, the Indigenous people of the Amazon rainforest employ agroforestry techniques, such as cultivating medicinal plants alongside food crops, to maintain biodiversity	[27, 229]
Sacred sites and rituals	Indigenous cultures often recognize certain areas as sacred sites, considering them crucial for the protection and preservation of biodiversity and medicinal plants. These sites are protected through cultural practices, rituals, and taboos that restrict access and promote sustainable use. The Maasai people of East Africa, for example, have sacred groves where medicinal plants are conserved and only used in specific rituals	[65, 66, 131, 286, 288]
Traditional resource management systems	Indigenous communities have intricate resource management systems that promote the sustainable use and conservation of medicinal plants. These systems include rules, regulations, and community-led governance mechanisms to ensure the responsible harvesting and cultivation of medicinal plants. The Quechua people in the Andes practice "ayllu," a collective farming system that includes the cultivation and conservation of medicinal plants	[135, 283, 294]
Inter-generational knowledge transmission	Indigenous knowledge about biodiversity and medicinal plants is passed down through generations via oral traditions, storytelling, and apprenticeship. Elders play a crucial role in transferring knowledge to younger community members, ensuring the continuity of traditional practices and conservation efforts. The Indigenous communities of Australia have a strong oral tradition, where elders teach the younger generation about the identification, preparation, and sustainable harvesting of medicinal plants.	[61, 152]
Traditional conservation practices	Indigenous communities have developed unique conservation practices that are deeply rooted in their cultural beliefs and spiritual connections to nature. These practices often involve seasonal harvesting, rotational cultivation, and the protection of specific areas to allow natural regeneration. The Kuna people of Panama, for instance, have established marine protected areas to safeguard marine biodiversity and medicinal plant resources.	[201, 216]

ical quinine [258]. Similarly, the antimalarial effects of plants like *Cryptolepis sanguinolenta* have been uncovered by the traditional healing practices of the Maasai culture in East Africa [69].

When it comes to protecting biodiversity, indigenous knowledge is just as important. Native peoples have a deep emotional investment in their homelands and have created environmentally friendly methods of living that help protect endangered plant species and entire ecosystems. Sustainable ways of gathering medicinal plants, guarding sacred sites, and maintaining traditional landscapes are all within their sphere of expertise [231].

Native crops such as quinoa (*Chenopodium quinoa*), amaranth (*Amaranthus* species), and many medicinal plants have been cultivated and preserved by the Quechua people of the Andean region in South America for centuries [241]. Because of their dedication to sustainable farming methods, the region's rich plant life has been preserved for future generations.

When it comes to the use of medicinal plants and the protection of biodiversity, indigenous knowledge systems provide a more comprehensive perspective. They shed light on the need for harmony and balance in the natural world by showing how people and plants are interdependent. Traditional knowledge from local communities can help scientists learn more about the potential medicinal uses of plants and develop more effective conservation strategies [89].

In summary, traditional knowledge is an important tool for exploring new phytochemicals and protecting existing species. Sustainable practices and in-depth knowledge of medicinal plants among indigenous tribe's aid in the discovery of important bioactive components and the protection of endangered plant species. Sustainable use of plant resources and protection of cultural and ecological variety depends on the recognition and respect of indigenous knowledge systems.

9 Challenges and Ethical Considerations in Indigenous Knowledge Research

To work in a way that is both respectful and equitable with indigenous populations, researchers examining indigenous knowledge must first face several particular obstacles and ethical considerations (Table 11). The complicated background of colonization, power disparities, and the desire to protect cultural rights and patents all contribute to these difficulties. Research that upholds indigenous rights, creates fruitful relationships, and encourages the preservation and revival of indigenous knowledge systems requires an appreciation for and response to these obstacles.

Written authorization and community engagement is a major obstacle to advancing indigenous knowledge research. Involving indigenous people early on in the research process is crucial for obtaining their express, informed, and free permission and ensuring their active participation. Trust and collaborative partnerships based on mutual respect and collaboration are essential for researchers. To ensure the research is in line with the community's interests, values, and cultural norms, this procedure

Table 11 Challenges and ethical considerations in indigenous knowledge research

Challenges	Ethical considerations	Authors
Knowledge ownership	Recognizing and respecting Indigenous knowledge holders' rights to control and own their knowledge	[26, 273]
Cultural appropriation	Avoiding the appropriation or exploitation of Indigenous knowledge and ensuring appropriate attribution	[62, 228]
Informed consent	Obtaining free, prior, and informed consent from Indigenous communities, individuals, and knowledge holders	[24, 197]
Power imbalances	Acknowledging and addressing power imbalances between researchers and Indigenous communities during the research process	[100, 244]
Intellectual property rights	Protecting Indigenous intellectual property rights and ensuring fair and equitable benefit-sharing	[32, 202]
Community engagement	Engaging with Indigenous communities in a meaningful and collaborative manner throughout the research process	[88, 292]
Confidentiality and privacy	Respecting the confidentiality and privacy of Indigenous knowledge holders and community-specific information	[150, 161]
Data sovereignty	Recognizing Indigenous communities' rights to control and manage their own data, including Indigenous data sovereignty	[192, 200]
Research methodologies	Adopting culturally appropriate research methodologies and frameworks that align with Indigenous worldviews and knowledge systems	[15, 273]
Preservation of cultural integrity	Preserving and promoting the cultural integrity, values, and practices associated with Indigenous knowledge	[273, 46]

may incorporate community standards, research agreements, and continuing participation [4, 273].

Another difficulty could be the theft or misuse of traditional information. Researchers face a balancing act between doing what's best for humanity and respecting the rights of indigenous peoples to their intellectual property and cultural traditions. In order to act ethically, it is important to gain permission before using indigenous knowledge, give credit where credit is due, and divide any rewards fairly among those who contributed to the discovery. The Nagoya Protocol and similar initiatives provide best practices for the equitable and fair distribution of benefits accruing from the use of traditional knowledge.

Indigenous knowledge studies must also take into account cultural appropriateness and cultural security. Researchers have a responsibility to be sensitive to local customs, refrain from plagiarizing someone else's work, and not offend anyone throughout the course of their investigation. Reflexive practices are important because they force the researcher to reflect on his or her own positionality, biases, and the power dynamics at play in the research. Respect for native cultures, languages, and methods of knowing is essential. This includes acknowledging the variety among indigenous groups.

Furthermore, great care must be taken in the reporting and sharing of study results. Indigenous peoples should have a say in the dissemination of their knowledge, and the results of research should be presented in a respectful way that allows for community input. Feedback loops and community involvement in knowledge translation activities are examples of open and transparent communication that can help ensure research benefits the community and is in line with their self-determined goals [28, 93].

Ethical indigenous knowledge research also places a premium on creating local capacity and sharing existing expertise. Learning from and teaching indigenous peoples, as well as helping to preserve and revitalize indigenous knowledge systems, should be high priorities for researchers. Sharing research results, facilitating community-led research projects, and enabling indigenous people to document, administer, and apply their own knowledge are all essential steps in this direction [46].

10 Collaborative Approaches That Respect and Safeguard Indigenous Knowledge

To ensure that research processes and findings are congruent with the values, needs, and ambitions of Indigenous people, collaborative approaches are vital for honoring and safeguarding Indigenous knowledge. Honoring Indigenous knowledge systems, fostering self-determination, and supporting the preservation and revitalization of traditional practices can be accomplished when researchers engage in meaningful relationships and employ inclusive techniques. These cooperative methods originate from the values of equality, fairness, and consensus (Table 12).

One important method that includes Indigenous populations in the study process is called participatory research. It goes beyond simple consultation to include Indigenous knowledge bearers at all stages of the research process, from conception to execution. This strategy places Indigenous peoples at the center of the research process, giving them credit for their knowledge and insights [273, 313].

Collaborative methods also heavily rely on indigenous research frameworks. Indigenous values, philosophies, and practices inform these models. They incorporate cultural practices and storytelling and place an emphasis on holistic approaches that respect community values. Researchers are guided in performing research that is in accordance with Indigenous protocols, honors cultural practices, and represents Indigenous ways of knowing by using Indigenous research methodology like Two-Eyed Seeing, yarning circles, and Two-Spirit research procedures. Researchers show they value Indigenous knowledge by using these frameworks [163, 162] and so producing results that are culturally suitable and relevant.

Knowledge sharing and creation are at the heart of collaborative methods. Researchers in these collaborations deliberately seek out opportunities for conver-

Table 12 Types of collaborative approaches that respects and safeguard indigenous knowledge

Collaborative approaches	Description and examples	References
Participatory research	Adopting participatory research methodologies that actively involve Indigenous communities in the research process. This includes collaborative planning, decision-making, and co-design of research objectives, methods, and outcomes	[163, 273]
Indigenous research frameworks	Incorporating Indigenous research frameworks and methodologies that align with Indigenous worldviews, protocols, and knowledge systems. This ensures that research is conducted in a manner that respects and reflects Indigenous values and perspectives	[273]
Knowledge co-creation and sharing	Fostering knowledge co-creation and sharing by acknowledging and valuing the expertise and perspectives of Indigenous knowledge holders. This involves creating spaces for dialogue, active listening, and reciprocal learning, where both Indigenous and scientific knowledge systems can be mutually respected and integrated	[26, 46]
Community engagement and capacity building	Engaging with Indigenous communities in a respectful and ongoing manner, building trusting relationships, and promoting capacity building within the community. This includes providing opportunities for skill development, training, and knowledge exchange, empowering communities to conduct their research and advocate for their rights	[193]
Intellectual property rights and benefit sharing	Recognizing and respecting Indigenous intellectual property rights, including the right to control, own, and benefit from their knowledge. Researchers should ensure that any commercial or intellectual property outcomes resulting from the research are shared equitably and in consultation with the community	[225]
Ethical protocols and informed consent	Following rigorous ethical protocols and obtaining free, prior, and informed consent from Indigenous communities and individuals. Researchers should be aware of and adhere to relevant guidelines and protocols, respecting privacy, confidentiality, and cultural protocols regarding the use and dissemination of Indigenous knowledge	[273]
Long-term collaboration and reciprocity	Building long-term collaborative partnerships with Indigenous communities based on trust, reciprocity, and shared decision-making. Researchers should prioritize sustained engagement beyond the research project, supporting community-led initiatives, and contributing to the long-term well-being and self-determination of Indigenous communities	[26]

sation, attentive listening, and mutual education with Indigenous knowledge keepers. This method recognizes the complementary capabilities and insights of Indigenous and scientific knowledge systems. Researchers and Indigenous communities can better respect community procedures and preferences during the dissemination of research findings by encouraging knowledge co-creation and sharing [26, 46].

Collaborative strategies must always include community involvement and capacity building. Researchers maintain mutually beneficial connections with Indigenous people based on mutual respect and a commitment to capacity building. This includes facilitating research, rights advocacy, and decision-making by communities through providing opportunities for skill development, training, and knowledge exchange. Through community involvement, research is done with Indigenous communities rather than on them, and results are tailored to their needs and goals [273].

In the context of collaborative research, individual patent rights and benefit sharing are crucial factors to address. Researchers have an obligation to honor the rights of Indigenous people to the fruits of their intellectual labor. This involves making sure the community has a voice in and benefits from any financial or intellectual property decisions made as a consequence of the study. Financial compensation, capacity-building initiatives, and community-driven projects that improve the well-being and self-determination of Indigenous communities are all examples of the types of benefits that can be shared under benefit-sharing agreements [225].

Collaborative research with Indigenous populations must adhere strictly to ethical protocols, including informed consent. In order to conduct their studies ethically, researchers must first have the free, prior, and informed consent of Indigenous groups and individuals. This requires open discussion of the research's goals, methods, outcomes, and drawbacks. Researchers also have a responsibility to adhere to cultural norms and practices around the sharing and use of Indigenous knowledge. Respect for the rights, well-being, and self-determination of Indigenous communities is essential to ethical research [163].

Collaboration strategies rely heavily on long-term cooperation and mutual benefit. It is critical to establish mutually beneficial partnerships with Indigenous communities that can last for decades. These collaborations go beyond the scope of a single study and instead contribute to the long-term health and autonomy of Indigenous communities through sustained participation and backing for community-led initiatives. Researchers show their dedication to Indigenous knowledge, community goals, and producing research with lasting good consequences by working together over the long term [59].

Researchers may safeguard Indigenous knowledge as an essential element of human heritage by adopting these collaborative methods. These methods respect the rights and ambitions of Indigenous people while also fostering mutually beneficial partnerships, bolstering cultural resilience, and adding to the co-creation of knowledge.

11 Synergies Between Indigenous Knowledge and Modern Phytochemistry in Herbal Medicine

When it comes to the study and development of herbal medicine, indigenous knowledge and modern phytochemistry are two separate but complementary fields. What we mean when we talk about "indigenous knowledge" is the wisdom that has been passed down through Indigenous cultures from one generation to the next. Meaningful use of plants for therapeutic reasons, including their recognition, gathering, processing, and administration. In contrast, contemporary phytochemistry entails the systematic investigation of plant chemicals to determine their chemical make-up and their medicinal applications [64].

Indigenous wisdom and modern phytochemistry complement one another in several ways that advance our knowledge of and access to herbal medicine. For example, native American lore is a treasure trove of information on medicinal plants, including their identities, traits, and historical applications. This information can help direct current phytochemical studies by pinpointing plants with the most potential. The chemical constituents present in these plants can be validated and their potential therapeutic activities determined by modern phytochemistry [76]. For instance, the discovery of quinine, an effective antimalarial compound, was made possible by indigenous people's familiarity with the tree's bark used in treating malaria (*Cinchona* species).

Comparison of Modern Therapeutic Methods to Traditional Healing Methods: Traditional indigenous healing focuses on the whole person, taking into account the body, mind, and spirit. This comprehensive method is consistent with current ideas in integrative medicine, which aim to combine conventional and scientific methods. Researchers can pinpoint the bioactive compounds responsible for the therapeutic effects seen in traditional healing practices by combining Indigenous knowledge with modern phytochemistry [92]. For instance, artemisinin, a powerful antimalarial compound, was initially isolated from *Artemisia annua* due to its historical use in malaria treatment [291].

Indigenous peoples' knowledge of the interdependence of plants, their environments, and the healing properties they contain is invaluable for the conservation and sustainable use of medicinal plants. Insights like these help keep medicinal plants around for future generations by facilitating their responsible collection, cultivation, and preservation. Sustainable harvesting practices and conservation methods can be developed with the help of modern phytochemistry by identifying plant chemicals and understanding their ecological roles [255]. The identification of plant compounds with potential therapeutic applications is just one example of the positive impact of collaboration between Indigenous communities and scientists in the Amazon rainforest, which has also led to the promotion of sustainable use and conservation of the region's biodiversity [167].

The identification of promising lead chemicals for drug research is aided by indigenous peoples' knowledge. The bioactive components in traditional medicines that have been used for generations often have therapeutic effects. Insight into their methods of action, safety profiles, and prospective applications in modern medicine can be gained through the use of isolation and analysis techniques made possible by modern phytochemistry [83]. By pooling resources, this collaborative strategy can speed up the drug development process by drawing on Indigenous tribes' rich store of knowledge. For instance, indigenous peoples' use of the Pacific yew tree (*Taxus brevifolia*) resulted in the creation of paclitaxel, an effective cancer medicine [248].

Integrating traditional Indigenous knowledge with contemporary phytochemistry honors and celebrates Indigenous communities' rich cultural histories and valuable contributions. It encourages Indigenous communities' cultural maintenance, knowledge transmission, and self-determination through the participation of Indigenous knowledge holders in research and the acknowledgment of their expertise. By working together, we can guarantee Indigenous voices are heard and respected, which is essential to building cultural strength and autonomy [5]. Additionally, it strengthens the recognition and protection of Indigenous intellectual property rights, which helps with the fair distribution of profits made from selling herbal medicine items [75].

Indigenous wisdom and contemporary phytochemistry complement one another, creating a robust foundation for the development of herbal medicine. Researchers can learn more about medicinal plants' therapeutic potential by combining traditional wisdom with scientific rigor. Together, academics and Indigenous communities can ensure that Indigenous knowledge is respected, safeguarded, and used ethically, all while gaining from the experience.

12 Future Directions in Advancing Indigenous Phytochemistry Research

The integration of traditional and Western medicine, as well as the establishment of ethical research collaborations, are all necessary steps toward advancing indigenous phytochemistry research. Researchers may help preserve and advance indigenous cultures by embracing these future directions, which will also advance the scientific community's knowledge of the tremendous potential of molecules originating from plants. The combination of these two approaches has the potential to enhance healthcare, discover novel medications, and ensure the long-term survival of medicinal plants utilized by indigenous peoples. Indigenous phytochemistry research has the potential to make a significant effect in the future if it moves in a few key directions.

First, scientists and indigenous populations need to work together in an ethical manner to advance indigenous phytochemistry research [277]. Community involvement, informed consent, and fair distribution of benefits are all crucial to the success of such partnerships. To a large extent, phytochemical studies can benefit from indigenous peoples' extensive familiarity with local flora and traditional therapeutic practices [240]. Researchers can learn more and ensure that indigenous community's benefit from their studies if they treat indigenous people as equal collaborators [302].

Prioritizing the preservation and protection of indigenous knowledge is crucial to the success of indigenous phytochemistry studies. In order to preserve their cultural and medicinal expertise, traditional knowledge holders should be recognized as the principal custodians [272]. To avoid exploiting or appropriating holy practices, researchers should consult with elders and healers as they navigate the complexity of indigenous knowledge systems.

Third, using current analytical techniques Indigenous phytochemistry research stands to benefit much from the development of modern analytical methods. The identification and characterization of bioactive chemicals from traditional medicinal plants can be sped up by integrating technologies like liquid chromatography-mass spectrometry (LC-MS) and nuclear magnetic resonance (NMR) spectroscopy [39]. These methods can also be used to standardize herbal compositions, boosting confidence in their reliability [280] and usefulness.

The future of indigenous phytochemistry research should center on bridging Western and non-Western approaches to healthcare. Therapeutic benefits from combining plant chemicals with conventional medications may be further investigated through collaborative research. Integrating these two fields can also help researchers find ways to prove the value of traditional medicine through scientific inquiry.

13 Conclusion

As a result of the extensive historical observations, cultural transmission practices, and profound understanding of the medicinal properties of plants among the local population, collaboration between traditional expertise and indigenous knowledge of phytochemistry holds significant promise in advancing our understanding of the therapeutic effects of herbs. Traditional healing practices contribute to the foundation of Indigenous wisdom, which in turn provides valuable insights into plant medicinal properties. Also, combining rigorous scientific analysis of plant chemicals with insights from cultural traditions will improve researchers' ability to effectively identify, validate, and harness plant medicinal properties. It is important that herbal therapy research is based on solid evidence and takes into account cultural context and societal significance through collaborative efforts.

Furthermore, the occurrence of a synergistic relationship between traditional knowledge and indigenous phytochemistry helps to promote the preservation and sustainable use of medicinal plants. Thus, increasing public awareness and understanding of the ecological significance of medicinal plants plays an important role in the conservation of natural resources for future generations through the advancement of increased awareness, sustainable harvesting techniques, promotion of conservation effort, and avoidance of wasteful practices. Furthermore, the integration of traditional medicine and modern science will significantly advance drug research and discovery. This convergence provides an opportunity to create novel pharmaceuticals based on the invaluable wisdom of ancient Indigenous knowledge.

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The Nexus of Business, Sustainability, and Herbal Medicine

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Abstract

The fields of business, sustainability, and herbal medicine are linked and continue to interact for the actual and potential benefits of all stakeholders. This chapter provides a comprehensive understanding of the interplay between herbal medicine, business, and sustainability by offering invaluable insights for scholars, practitioners, health policymakers, and industry and other stakeholders. The herbal medicine industry is at the crossroads of tradition and modernity. Therefore, an examination of the economic perspectives of herbal medicine from a sustainability viewpoint as well as the present evolving roles of herbal medicine in today's global market is critical. The current global landscape of Integrative and Complementary Medicine is driving the rise of a form of conventional medicine practice that is considered Traditional, Complementary, Alternative, and Integrative Medicine that encompasses a diverse range of medical and healthcare interventions, practices, products, and disciplines and has heightened the need for sustainability within the business of herbal medicine. For instance, bringing herbal cosmeceuticals to the global market as natural cosmetics presents unique challenges, especially in but also beyond developing nations. Manufacturers face hurdles in developing eco- and user-friendly cum value-added herbal products with sustainable attributes. Moreover, establishing new herbal products from indigenous medicinal practices requires bolstering Private-Public Partnerships in collaboration with universities, research institutes, private industries, and other key medical stakeholders. This chapter discusses the challenges, solutions, and business perspectives of key interactions that can improve the nexus of herbal medicine, business, and sustainability like eco-friendly packing, regenerative sourcing, community engagement, research on sustainable practices, transparent communication, consumer and medical stakeholder feedbacks, education and training, policy and partnerships, and the adoption of innovation. Finally, the chapter highlights the interconnectedness and feedback loop of the nexus as well as major stakeholder groups as well as their influence and interests.

Keywords

Herbal medicine industry · Sustainable Development · Global business · Healthcare systems · Medical stakeholders

Abbreviations

%	Percent or Percentage
AfCFTA	African Continental Free Trade Area
ALS	Amyotrophic Lateral Sclerosis
AYUSH	Ayurveda, Yoga and Naturopathy, Unani, Siddha, and Homoeopathy
ANSM	Agence nationale de sécurité du médicament et des produits de santé
BRIC nations	Brazil, Russia, India, and China
CAM	Complementary and Alternative Medicine
CFTA	Continental Free Trade Area
CHM	Chinese herbal medicines
CMD	Chinese medicinal drugs
CO2	Carbon dioxide
COVID-19	Corona virus 2019
CRC	Colorectal cancer
E-commerce	Electronic (internet-based) commerce
EMA	European Medicinal Agency
EU	European Union
FDR	Fagopyri Dibotryis Rhizoma
GDP	Gross Domestic Product
GVC	global value chains
HMI	Herbal Medicine Industry
HMP	Herbal medicinal products
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals
ICM	Integrative and Complementary Medicine
LEKMA	Ledzokuku-Krowor Municipal Assembly
LNG	Liquefied natural gas
MHRA	Medicines and Healthcare products Regulatory Agency
NAFDAC	National Agency for Food and Drug Administration and Control
NGO	Nongovernmental Organization
NMPA	National Medical Products Administration
PESTEL	Political, Economic, Social, Technological, Legal, and Environmental factors
PPP	Private-Public Partnerships
Q&A	Question and answer
R & D	Research and Development

SWOT	Strengths, weaknesses, opportunities, and threats
TCAIM	Traditional, Complementary, Alternative, and Integrative Medicine
TCIM	Traditional, complementary, and integrative medicine
U.S. or U.S.A	United States of America
UK	United Kingdom
VRIO	value, rarity, imitability, and organization
WHO	World Health Organization

1 Introduction

The importance of herbal medicine, in today's world cannot be overstated. It not only plays a role in promoting good health but also has a strong presence in the business sector [1–7]. When one delves into studies conducted on the herbal practice, products, and the dynamic world of international business, an interesting overlap becomes apparent. This yin-yan loop is a form of common or socio-economic and ecological collectivism that is currently without an institution-based paradigm [8–10]. The ancient practices of herbal medicine seamlessly merge with business methodologies creating a harmonious synergy. Adding to this blend is the concept of sustainability, which acts as a binding force fostering respect and reliance, between traditional and modern approaches. Together these three elements – herbal medicine, business, and sustainability – form a network of interconnectedness and shared influence.

Furthermore, herbal medicine has been a cornerstone of holistic health and wellness since its inception in ancient civilizations [11–22]. Its guiding principles, which are based on the harmonious coexistence of humans and nature, place a strong emphasis on the responsible use of natural resources. As a result, medicinal plants must be grown, harvested, and used in a way that preserves their potency and ensures their availability for years to come [23–25]. Herbal medicine has experienced global popularity, which is evidence of both its ongoing relevance and the growing appreciation of its capacity to address contemporary health issues.

Conversely, the realm of business, especially in the digital age, has witnessed transformative changes. The rise of e-commerce platforms, such as Alibaba and Zongteng, has revolutionized the way businesses operate, reaching global audiences and tapping into new markets [26–28]. These platforms leverage advanced technologies such as blockchain, to enhance supply chain efficiency, ensuring transparency and reliability in transactions. The integration of technology into business models has not only streamlined operations but has also opened avenues for sustainable business practices [29]. For instance, the establishment of overseas warehouses by cross-border e-commerce companies exemplifies supply chain innovations that prioritize localization, reducing carbon footprints, and promoting sustainable logistics.

Nevertheless, in the intricate landscape of today's global discourse, the prominence of sustainability cannot be emphasized enough [2]. This principle, much like the revered practices of herbal medicine, not only shapes our understanding of ecological balance but also carves a niche in the realm of modern business methodologies. As one navigates through the extensive research on global adversities, notably the profound impacts of climate change and the alarming erosion of biodiversity, a compelling narrative unfolds. This narrative accentuates the pressing demand for initiatives that are not only in harmony with nature but also economically pragmatic and socially just. Furthermore, venturing deeper into the vast expanse of the global economic tableau, particularly when reflecting upon the tumultuous waves created by unforeseen calamities such as the COVID-19 pandemic, the essence of sustainable business frameworks becomes even more palpable. E-commerce platforms, sensing the evolving pulse of consumer dynamics, have elegantly shifted their compass. Their strategies now resonate with a broader ethos, one that transcends mere fiscal achievements and embraces the ideals of environmental guardianship and societal enrichment.

The chapter aims to elucidate the nexus of business, sustainability, and herbal medicine from a multifaceted standpoint that includes an examination of the economic perspectives of herbal medicine, shedding light on its historical significance from a sustainability viewpoint and the present evolving role in today's global market. The chapter also provides a critical dissection of the current business focus of herbal medicine products and practices, offering insights into market trends, consumer preferences, and commercial strategies. Furthermore, the incorporation of sustainability into herbal medicine practices and products is interrogated. This objective seeks to determine the conceptualization of sustainability within the herbal medicine industry, probing its current manifestations and suggesting avenues for enhancement. The subsequent objective is to forecast the realm of sustainable herbal medicine business, presenting a prospective outlook that envisions a harmonious blend of economic prosperity, environmental stewardship, and holistic health. Additionally, the work ascertains the specific business needs of herbal medicine, pinpointing pivotal elements that can fortify the industry and ensure its continued growth and relevance. Finally, the prerequisites for sustainable herbal medicine products and practices are demarcated, outlining foundational requirements and best practices that can steer the domain toward a sustainable trajectory. In synthesizing these objectives, this chapter seeks to provide a comprehensive understanding of the interplay between herbal medicine, business, and sustainability, offering valuable insights for scholars, practitioners, policymakers, and industry and medical stakeholders.

2 Economic Perspective of Herbal Medicine

In the evolving landscape of healthcare and commerce, the economic implications of herbal medicine have garnered significant attention. Recent studies have sought to elucidate the potential economic benefits and challenges associated with the integration of herbal medicine into modern healthcare systems. Gautam et al. [30] highlighted the socio-economic importance of wild Ayurvedic medicinal plants in the Western Himalayas, emphasizing the potential economic upliftment of local communities through large-scale cultivation. This perspective is supported by Sharma et al. (2019), who noted the increasing global demand for Ayurvedic products and the potential for local communities to tap into this lucrative market. However, critics like Patel et al. (2020) argue that while there is economic potential, overharvesting and lack of sustainable practices could lead to the depletion of valuable resources and long-term economic losses. Aside from that, the review by Geng et al. [31] on Fagopyri Dibotryis Rhizoma (FDR) underscores its therapeutic and economic benefits, particularly its antitumor pharmacological activity. Zhang et al. (2021) concurred, noting the rising demand for natural antitumor agents in the global market. However, Wang et al. [32] caution that while FDR shows promise, over-reliance on a single herbal product could lead to market saturation and reduced economic returns. Nevertheless, Mehboob's [33] study on the use of Complementary and Alternative Medicine (CAM) in healthcare emphasizes the economic significance of herbal medicine as a subset of CAM. This observation is echoed by Lee et al. (2019), who noted the increasing global expenditure on CAM products. Contrarily, Thompson et al. (2021) argue that while CAM, including herbal medicine, is popular, it often leads to out-of-pocket expenses, potentially straining individual finances. Dewi et al.'s [34] research on the indirect costs faced by chronic kidney failure patients underscores the economic considerations of incorporating herbal treatments. This is in line with the findings of Kumar et al. [28], who noted the potential cost-saving benefits of herbal treatments in chronic diseases. However, critics like Singh et al. (2022) point out that while initial costs might be lower, the long-term efficacy and potential side effects of herbal treatments could lead to increased economic burdens in the future.

2.1 Economic Implications of the Herbal Medicine Industry

The economic implications of the herbal medicine industry are vast and multifaceted. With a significant portion of the global population relying on herbal remedies, the market potential is enormous. However, this burgeoning demand also brings forth challenges related to sustainable cultivation, fair trade practices, and equitable distribution of economic benefits, especially in regions rich in biodiversity but poor economically.

In regions like the Swat Valley in northwest Pakistan, many practice subsistence farming, supplementing their income by collecting and selling wild harvested plants for use in herbal medicine. The cultivation of medicinal and aromatic plants in such areas not only contributes to economic development but also plays a pivotal role in biodiversity conservation [35].

Note that regions like Aritar in East Sikkim (India) are rich in ethnomedicinal knowledge, with diverse tribal communities relying on traditional medicines for their health needs. However, the economic potential of this knowledge remains largely untapped due to various challenges, including lack of awareness, standardization, and market access. However, the economic benefits derived from the herbal medicine industry are not always equitably distributed. In many biodiverse regions, local communities may not receive their fair share of the profits. This disparity is often exacerbated by a lack of regulatory frameworks and the dominance of middlemen in the supply chain. Ethical trade practices are essential to ensure that the communities responsible for cultivating and harvesting these medicinal plants benefit economically [36–38]. Nevertheless, the shift toward environmentally friendly agricultural practices, such as the cultivation of floating rice in Vietnam, represents a sustainable approach to farming. Floating rice, a traditional method of rice cultivation, offers environmental, economic, and social benefits. However, promoting such practices in international markets requires understanding consumer preferences. For instance, UK consumers have shown a willingness to pay premiums for ethical attributes like fair trade but may not necessarily value other environmental attributes like CO_2 emission reduction [39]. Moreover, the commercial success of herbal medicinal products (HMP) hinges on their compliance with regulatory requirements pertaining to quality, safety, and efficacy. However, the standards and regulations governing HMPs vary across countries, posing challenges for manufacturers aiming to introduce standardized herbal products in the global market. A collaborative effort between regulatory bodies and international organizations like the World Health Organization (WHO) is imperative to establish harmonized regulations for HMPs.

The works of Malathi et al. (2019) supported the earlier work of Chawla et al. [40] on diabetes, a prevalent lifestyle disorder, that has seen a surge in the use of herbal remedies for its management and noted that the transition from traditional herbal treatments to evidence-based herbal drug development is fraught with challenges, especially in terms of standardization. The lack of universally accepted technical standards, coupled with the absence of comprehensive regulatory guidelines, hampers the development and commercialization of effective herbal treatments for diabetes [40]. In addition, the works of Sahoo and Manchikanti [41] further support that of Chawla et al. [40] in their study on the Indian herbal drug industry, a significant player in the global herbal medicine market, which faces unique challenges in the manufacturing and commercialization of herbal medicines. Divergent regulatory requirements, especially in terms of quality control and safety standards, impede the industry's growth. Moreover, the weak implementation of existing regulations further exacerbates the challenges faced by manufacturers [41]. Furthermore, the production of herbal cosmeceuticals, especially in developing countries, faces numerous challenges. While these products offer both cosmetic and therapeutic benefits, their development requires a nuanced understanding of indigenous medicinal knowledge. The commercialization of such products is often hindered by factors such as lack of awareness, standardization issues, and limited market access. Collaborative efforts between research institutions, indigenous practitioners, and the private sector can help overcome these challenges [42].

2.2 Current Business Focus of Herbal Medicine Products and Practices

The global resurgence of herbal medicine, deeply rooted in ancient traditions, has evolved into a significant business sector in contemporary times. One of the key objectives set for this chapter is to critically analyze the current business focus of herbal medicine products and practices, emphasizing the diverse types of herbal medicine practices and their respective business orientations. Pan et al. [43] underscores the global re-emergence of herbal medicines, noting that approximately 75% of the world's population relies on herbs for primary healthcare. This "herbal renaissance" signifies not only a return to ancient practices but also a testament to the efficacy of herbal remedies. The study further emphasizes the dual-edged sword of demand, where the increasing global appetite for herbs has led to the potential overexploitation of more than 53,000 species. However, after this study, several studies have attempted to classify herbal medicine practices into different categories some of which have been identified and discussed in the next section.

3 Types of Herbal Medicine Practices

3.1 Traditional Herbal Medicine Practices: Global Perspectives

Traditional herbal medicines, which form a major branch of global healthcare systems, have been practiced within various cultural contexts and are often utilized based on different traditions and beliefs. These medicines are most times passed down through generations and constitute the backbone of many communities' healthcare systems. They offer not only therapeutic benefits but also an opportunity to connect with various ancestral wisdom and cultural identity.

Several studies have highlighted the use of traditional herbal medicine around the world. Particularly, Kunwar et al. [44] conducted a study in Nepal, a country that is very rich in biodiversity as well as several ethnic groups. The researchers highlighted the role of medicinal herbs in traditional treatments and therapies. These practices, deeply embedded in the cultural fabric, have been sustained over centuries, reflecting the intricate relationship between the people and their natural environment. In addition, Devkota et al. [45] emphasized the significance of medicinal plants in traditional medicine systems across Nepal. The country, with its diverse ecosystems, hosts a plethora of medicinal plants used in various indigenous therapeutic practices. However, these practices and the associated knowledge are not always well-documented, leading to the potential loss of invaluable information. The study underscores the need for detailed documentation, conservation efforts, and research to strengthen these traditional systems [16, 17, 46].

Another dimension to consider is the socio-cultural context surrounding traditional practices. A study by Thapa et al. [47] delves into the traditional menstrual practices in far-western Nepal, revealing the deep-rooted beliefs and cultural norms that influence such practices. While this study primarily focuses on menstrual practices, it sheds light on the broader context of how traditions, beliefs, and societal norms shape health-related behaviors in the community.

Furthermore, Sriharini and Syafira [48] discuss the concept of community empowerment by leveraging local potential with cultural value. In the village of Rejowinangun, herbal medicine, a hereditary heritage, is optimized for marketing in the concept of a tourist village, thereby increasing its economic and cultural value in modern society. In a similar vein, China has been at the forefront of traditional herbal medicine research, with a significant contribution to global research productivity in this domain. A study by Musa et al. [49] provides an overview of traditional herbal medicine research, emphasizing the annual increase in research after 1990. The study also highlights that the topmost-ranked authors and funding agencies were predominantly from Asia, especially China [49]. Traditional Chinese medicine has demonstrated potential in addressing various health conditions. For instance, a study by Xu et al. [50] emphasizes the role of traditional Chinese herbal medicine in ameliorating hyperuricemia. The research underscores the potential of certain natural compounds in treating hyperuricemia based on experimental studies [50]. In addition, East Asia has been known for the use of herbal medicine as well, as traditional East Asian herbal medicine offers potential treatments for various conditions, including neurodegenerative diseases like Amyotrophic Lateral Sclerosis (ALS). A scoping review by Suh et al. [51] delves into the potential of East Asian herbal medicine in treating ALS, emphasizing the need for large-scale and rigorously designed clinical studies [51].

3.2 Integrative and Complementary Medicine

Modern healthcare systems across the globe are increasingly recognizing the value of traditional herbal practices, leading to their integration into contemporary medical frameworks. This integration, often termed Integrative and Complementary Medicine (ICM), seeks to combine the best of both traditional and modern therapeutic approaches to provide holistic patient care. The current global landscape of ICM is driving the rise of a form of practice that is considered Traditional, Complementary, Alternative, and Integrative Medicine (TCAIM). TCAIM encompasses a diverse range of medical and healthcare interventions, practices, products, or disciplines that are not typically considered part of conventional medicine. Despite the variability in safety and effectiveness evidence profiles inherent in TCAIM, its use is highly prevalent among patients worldwide. A bibliometric analysis by Ng [52] provides insights into the characteristics of research published in TCAIM journals, emphasizing the upward trend in the volume of publications since the 1940s, with a significant increase observed between the mid-2000s and mid-2010s [52]. In addition to the works of Ng [52], Kenu et al. [53] conducted a study in Ghana, they discovered that approximately 70% of the population relies on traditional, complementary, and integrative medicine (TCIM) practices for primary healthcare needs. Recognizing its significance, the Ministry of Health integrated TCIM into the mainstream healthcare delivery system in 2012. The study conducted at the

LEKMA hospital in Ghana further revealed that the general belief in the potency, perceived effectiveness, location, and availability of TCIM services are key determinants of the high preference for its usage [53].

The findings of Ng [52] justify the works of Nugraha et al. [54] who noted that the recent COVID-19 pandemic has seen a surge in interest in herbal medicine as potential complementary treatments. Nugraha et al. [54] discuss various herbal agents, including *Echinacea, Cinchona*, and *Curcuma*, considered for the treatment of COVID-19. The study provides insights into the pros and cons of utilizing herbal medicine during global health crises [54]. The integration of traditional and complementary medicine into mainstream healthcare necessitates effective interprofessional communication. Hunter et al. [55] emphasize the importance of respectful, culturally competent interprofessional communication and collaboration that mutually supports patients' care needs. The study underscores the need for educational initiatives that promote inclusive, culturally competent, interprofessional communication and collaboration between conventional and TCAIM healthcare practitioners [55].

3.3 Commercial Herbal Medicine Production

The globalization of herbal medicine has ushered in a new era of large-scale commercial production of herbal products. This transformation encompasses the cultivation, processing, and distribution of herbal medicines, catering to an ever-expanding global market. The commercialization of herbal products has been driven by increasing consumer demand for natural remedies, the perceived health benefits of herbal medicines, and the integration of traditional herbal practices into modern healthcare systems.

Bringing herbal cosmeceuticals to the global market as natural cosmetics presents unique challenges, especially in developing nations. Manufacturers face hurdles in developing eco-friendly and user-friendly value-added cosmeceutical products with medicinal attributes. In many developing countries, establishing new cosmeceuticals from indigenous medicine requires fostering Private-Public Partnerships (PPP) in collaboration with universities, research institutes, and private industries. However, challenges such as knowledge-sharing barriers, lack of trained personnel with entrepreneurial skills, regulatory barriers, unsustainable collection practices of medicinal plants, and quality control issues persist [42]. Furthermore, the growing demand for agricultural commodities, including medicinal herbs, necessitates an optimization of crop distribution. A study by Davis et al. (2017) highlighted that the current global distribution of crops neither maximizes production nor minimizes water use. By reshaping the global distribution of crops based on total water consumption, it's possible to achieve greater food production while reducing water use, without compromising crop diversity or requiring massive investments in modern technology. Aside from that, the application of bacteriophages, or phages, in food production and processing is an emerging trend. Phages are viruses that infect bacteria and have been explored for their potential to improve food safety.

The use of phage biocontrol in food production offers a novel approach to combat bacterial contaminants, ensuring the safety and quality of food products, including herbal preparations [56–58].

4 Business Focus in Herbal Medicine

4.1 Herbal Medicine Market Dynamics and Globalization

The globalization of herbal medicine has led to a myriad of opportunities and challenges. As traditional remedies find audiences in distant parts of the world, the business landscape of herbal medicine is undergoing significant transformation with a mix of both positive and negative consequences. The global market potential of herbal medicine has seen a significant shift in product development and marketing strategies. In countries like Indonesia, companies are at the forefront of developing herbal medicinal products that promise safer long-term effects. The introduction of new herbal products necessitates the formulation of effective marketing strategies, which often involve both internal and external analyses. This includes understanding competitors, gauging consumer considerations, and leveraging frameworks like PESTEL analysis, Porter's Five Forces analysis (i.e., threats from new entrants, threats from new substitutes, bargaining power of suppliers, bargaining power of buyers, and influence of competitive rivalry), and SWOT analysis. Kunwar et al. [44] emphasize the challenges presented by uncontrolled exploitation, especially in regions like Nepal where many medicinal plants face potential extinction. The unregulated harvesting of medicinal plants can lead to a decline in biodiversity and disrupt the ecological balance. This underscores the need for sustainable harvesting practices and conservation efforts to ensure the long-term viability of these valuable resources [44]. In addition, the last few decades have seen unprecedented economic volatility, altering the political dynamics and socio-cultural landscape globally. New markets, such as the BRIC nations (Brazil, Russia, India, and China), have emerged as strong economic powers. As multinational firms expand into these markets, they face cultural challenges. Understanding these cultural elements is critical for the success of global firms, especially in the herbal medicine sector, where cultural beliefs and practices play a significant role [59].

4.2 Therapeutic Benefits and Modern Drug Discovery: "-omics" Approaches in Herbal Medicine Research

The nexus between traditional herbal medicine and modern drug discovery is a burgeoning area of interest, offering a plethora of opportunities for the pharmaceutical industry. This intersection of age-old wisdom and cutting-edge science is paving the way for the development of innovative therapeutic solutions. Furthermore, Tran et al. [60] have delved deep into the realm of bioactive compounds present in antidiabetic plants. Their research emphasizes the potential of certain bioactive drugs isolated from these plants, which have shown more efficacy than conventional oral hypoglycemic agents. The study underscores the importance of understanding the intricate molecular structures and mechanisms of these compounds, which can lead to the development of more effective and safer therapeutic agents for diabetes management. In addition, natural products, derived from plants, animals, or microorganisms, have been recognized for their myriad health benefits. In developing countries, a staggering 80% of the population relies on traditional or folk medicines, primarily plant-based, for disease prevention and treatment. These traditional medicines, often perceived as more affordable and having fewer adverse effects than modern drugs, have piqued the interest of the pharmaceutical industry. The diverse biological properties of plant-derived secondary metabolites, such as their anti-inflammatory, antidiabetic, and antioxidant activities, present a goldmine of opportunities for drug discovery.

Aside from that, recent research has highlighted the role of gut dysbiosis in the pathogenesis of colorectal cancer (CRC). Herbal medicine, with its rich repository of natural bioactive products, has shown promise in improving microbiota composition, repairing the intestinal barrier, and reducing inflammation. These therapeutic effects can potentially prevent the onset and progression of CRC. The study underscores the importance of understanding the intricate interplay between gut microbiota and herbal compounds, which can lead to the development of effective preventive strategies against CRC [61, 62]. The advent of omics approaches, encompassing transcriptomics, proteomics, genomics, metabolomics, and microbiomics, has revolutionized herbal medicine research and studies. These approaches provide a comprehensive understanding of the molecular mechanisms and interaction networks underlying the therapeutic effects of herbal compounds. For instance, omics-based studies have shed light on the effects and mechanisms of Chinese herbal medicines in gastrointestinal cancers, offering novel insights into potential therapeutic strategies.

4.3 Some Challenges in Herbal Medicine Production

Herbal medicine, an ancient practice that harnesses the therapeutic properties of plants, has witnessed a resurgence in recent years. As the global demand for natural remedies grows, the herbal medicine industry has responded by introducing a plethora of products. The major herbal medicine products are, however, still facing various issues including the following (Fig. 1):

4.3.1 Standardization of Traditional Herbal Formulations and Microbial Quality

Traditional herbal formulations have been an integral part of various cultures for centuries, offering therapeutic solutions to a myriad of health issues. However, as these formulations gain traction in the global market, concerns regarding their microbial quality have emerged. In Divo City, Côte d'Ivoire, traditional herbal formulations are a staple in the local healthcare system. A study by Elisée et al.

Fig. 1 Some challenges in herbal medicine production

(2022) sought to assess the microbial quality of 30 liquid herbal remedies commonly consumed in the region. The findings were alarming, revealing significant levels of microbial contamination in these products. The contamination ranged from yeasts and molds to coliforms and fecal streptococci. Such high levels of contamination underscore the potential health risks associated with the consumption of these remedies.

Furthermore, Traditional Chinese Medicine (TCM), a principal form of herbal medicine, is composed of treatments using multiple herbs with complex chemical profiles. Due to a lack of understanding of its modality and a lack of standardization, there are significant challenges associated with regulating TCM's safety for practice and understanding its mechanisms of efficacy [63]. In addition, despite recent research by Kexiang et al. (2022) highlights the antitumor effects of Chinese herbal medicine compounds, making the therapeutic potential of herbal formulations not limited to general health, the microbial quality of these formulations remains a concern. Ensuring the safety and efficacy of these products is crucial, especially when they are used to treat life-threatening conditions like cancer.

4.3.2 Quality Control in Traditional Chinese Medicine

Ensuring reproducible quality, efficacy, and safety from herbal drugs remains a challenging task. Herbal medicines typically consist of many constituents whose presence and quantity may vary among different sources of materials. For instance, Traditional Chinese Medicine (TCM) is another domain where herbal formulations play a pivotal role. However, concerns about the safety and quality of Chinese medicinal drugs (CMD) have been raised. A study conducted at the TCM hospital Bad Kötzting implemented a quality control program to ensure the safe administration of TCM drugs. The study found that several samples showed contamination, emphasizing the need for rigorous quality control measures in the production and distribution of these drugs [64]. However, fingerprint analysis has emerged as a very useful technique to assess the quality of herbal drug materials and formulations for establishing standardized herbal products [65].

4.3.3 Regulatory Challenges and Standardization

One of the critical challenges in the business of herbal medicine is the lack of standardization and regulatory oversight. While herbal medicines have been used for millennia, their integration into modern healthcare requires rigorous testing, standardization, and quality control to ensure safety and efficacy.

4.3.4 Safety Concerns and Toxicity

The safety of herbal medicines, especially those containing minerals like Cinnabar, has been a topic of concern. However, modern scientific evaluations have shown that when properly prepared, such medicines can meet safety standards [66].

5 Global Perspective on Herbal Drug Development

Despite the significant increase in global interest in the investigation and development of new botanical products, only a few have been approved till now. Natural product medication development has significant technical and financial hurdles. Regulatory agencies like EMA (Europe), ICH and WHO (International/Global), AYUSH and DCGI (India), ANSM (France), MHRA (UK), FDA (U.S.A), NMPA (China), NAFDAC (Nigeria), and others work to bring these herbal drugs under a regulatory pipeline to ensure their safety and efficacy [11, 67, 68].

5.1 Major Herbal Medicine Products in the Global Market

Herbal medicine, rooted in ancient traditions, has experienced a renaissance in the modern era. As consumers globally become more health-conscious and inclined toward natural remedies, the global market has seen a surge in the demand for herbal products especially in Africa and Asia.

5.1.1 Traditional Herbal Formulations in Africa

In regions such as Divo City in Côte d'Ivoire, traditional herbal formulations are not just a part of the cultural heritage but are also widely consumed for their perceived health benefits. However, a study by Elisée et al. (2022) highlighted concerns regarding the microbial quality of these products. The research assessed the microbial contamination of 30 liquid herbal remedies and found significant contamination levels, emphasizing the urgent need for stringent quality control measures to ensure the safety of these products for consumers.

In addition, obesity, a global health concern, has seen a rise in the use of herbal products as potential remedies. In Egypt, regime herbal teas in the form of fine powder are commonly used as phytotherapy for weight management. A study aimed at evaluating the quality control of two commercial herbal tea products available in the Egyptian market revealed that these products matched the identity and purity of genuine monoherbs, suggesting their potential for further clinical trials [69].

In bustling cities like Kampala, Uganda, there has been a noticeable proliferation in the trade of herbal medicines, especially those targeting common ailments like diarrhea and cough. A study conducted by Walusansa et al. [70] highlighted the widespread consumption of traditional herbal formulations in the city. However, concerns regarding the microbial quality of these products have been raised. The study assessed the microbial contamination of 30 liquid herbal remedies and found significant contamination levels. This underscores the pressing need for stringent quality control measures to ensure the safety of these products for consumers. The findings also emphasize the importance of establishing and adhering to standardized production and storage practices to maintain the efficacy and safety of herbal medicines [70–75].

5.1.2 Chinese Herbal Medicines in the European Union

Chinese herbal medicines (CHMs) have gained significant traction in international markets, especially in the European Union (EU). However, the quality control of CHMs remains a challenge for their acceptance in the EU. A comprehensive review by Mei Wang et al. compared the quality requirements for CHMs in the EU and China. The study highlighted the key factors of quality control, including regulations, extracts, and products. The review also discussed the legal status of traditional Chinese medicine granules in the EU, providing insights into the challenges and opportunities for CHMs in the European market [32].

5.2 Innovative Herbal Products and Marketing Strategies

The global market potential of herbal medicine has led to the emergence of innovative herbal products. Ganeshfit Company, for instance, has delved into the development of anti-inflammatory herbal medicinal products. Recognizing the need for a robust marketing strategy for new products, the company conducted extensive research, integrating internal and external analyses, to formulate potential marketing strategies. The study underscored the importance of understanding market dynamics, consumer preferences, and the competitive landscape to ensure the successful launch and acceptance of new herbal products in the market. The global herbal medicine market is vast and diverse, offering a plethora of products catering to various health needs. As the demand for these products grows, ensuring their quality, safety, and efficacy becomes paramount. Rigorous research, stringent quality control measures, and effective marketing strategies are essential to navigate the challenges and harness the opportunities in this burgeoning industry.

5.3 Herbal Medicine Products Market Strategy in the Global Market

The global market for herbal medicine products has witnessed significant growth over the past few decades. As consumers become more health-conscious and seek natural alternatives to conventional medicine, the demand for herbal products has surged and companies have utilized various strategies to survive in the market.

In Indonesia, for instance, the global health awareness trend has led to the rise of new herbal medicinal product innovations. The Ganeshfit Company's Research and Development Team, for instance, has been at the forefront of developing anti-inflammatory herbal medicinal products. These products are touted to have safer long-term effects compared to their synthetic counterparts. To successfully introduce these new products to the market, the company has employed a comprehensive marketing strategy. This strategy encompasses both internal and external analyses, including competitor analysis, consumer considerations, and market dynamics. The company's approach underscores the importance of a well-researched and tailored marketing strategy in ensuring the successful launch and acceptance of new herbal products in the market [76]. The agricultural sector, particularly the micro-small-scale segment in the Sukoharjo Regency (Indonesia), dominates all other economic activities in the community. Herbal medicine stalls have been identified as a priority for development within this sector. The focus has been on product development strategies that align with market demand and offer competitive advantages. The strategy formulation process involves a combination of external analyses, such as PESTEL analysis and Porter's Five Forces analysis, and internal analyses, including SWOT and VRIO (i.e., a four-question framework focusing on value, rarity, imitability, and organization) analyses [77]. However, the pharmaceutical market, which encompasses herbal medicine products, operates predominantly under monopolistic competition. This market structure allows for the extensive use of marketing tools, innovation, research, branding, and more. Certain segments of the pharmaceutical market, however, function within different market structures, such as monopoly and oligopoly. The global pharmaceutical market's growth forecasts up to 2027 indicate varying market development factors across different periods. Notably, Indian, and Chinese companies have shown high activity in developing the global pharmaceutical market, emphasizing the importance of strategic approaches, business model adaptation, and significant R&D investments [78].

5.4 External Factors Influencing Herbal Medicine Products in the Global Market

The global market for herbal medicine products has witnessed a significant surge in demand over the past few decades. As consumers become more health-conscious and seek natural alternatives to conventional medicine, the herbal medicine industry has capitalized on this trend. However, several external factors greatly impact the potential distribution of these medicines, and they include (Fig. 2):

5.4.1 Global Trade Dynamics

The dynamics of global trade have a profound impact on the herbal medicine market. For instance, the trade war between the USA and China, two of the world's largest economies, has implications for the global distribution chain, international trade, and

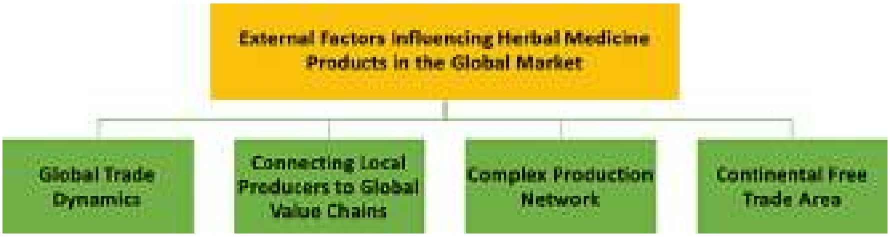

Fig. 2 External factors influencing herbal products in the global market

the economy at large. Such geopolitical events can influence the flow of herbal products between countries, affecting availability, pricing, and consumer choices [79].

5.4.2 Connecting Local Producers to Global Value Chains

The integration of local producers in developing countries into global value chains (GVCs) has been a focal point of discussion. While GVCs allow firms in developing countries to participate in international trade, the benefits of such participation are not automatic. Policies that build productive capacity, ensure inclusive growth, and upgrade capabilities are essential. The report by Bamber et al. [80] highlights the need for data and analysis to inform trade-related policy implications for developing countries, providing a foundation for discussions on domestic policies and actions required to promote beneficial participation in value chains.

5.4.3 An Increasingly Complex Production Network

Synonymous with the evolution of liquefied natural gas (LNG) in today's global market, the production network for herbal products is increasingly becoming more complex. While seemingly unrelated, the global market dynamics of (LNG) offer insights into the potential evolution of the herbal medicine market. Just as LNG has transitioned from a simple floating pipeline model to a more complex production network, the herbal medicine market is also transforming. The increasing demand for herbal products worldwide necessitates the development of robust supply chains, innovative distribution strategies, and adherence to international quality standards [81].

5.4.4 Continental Free Trade Area

The establishment of Continental Free Trade Areas (CFTAs) like the African Continental Free Trade Area (AfCFTA) creates enormous opportunities for the Herbal Medicine Industry (HMI) to tap into. For instance, the AfCFTA agreement, encompassing 55 nations and a combined GDP of $3.4 trillion, can potentially double intra-regional trade. For the herbal medicine sector, this means increased market access, collaborative research opportunities, and the potential for standardized production practices across member countries. The AfCFTA could also facilitate the sharing of traditional herbal knowledge, leading to the development of innovative products that cater to a global audience [82].

6 The Nexus of Business, Sustainability, and Herbal Medicine

The HMI stands at the crossroads of tradition and modernity. As the demand for natural remedies grows, so does the imperative need for a sustainable business approach. It is known that the fields of business, sustainability, and herbal medicine are linked and continue to interloop and interplay for the benefit of medical stakeholders. However, various issues and challenges continually impact and affect the nexus of herbal medicine, business, and sustainability, including some like the challenge of microbial contamination, toxicity, and standardization of herbal medicine products, which is tough to surmount. Aside from that, the economic implications of the HMI are profound, with vast opportunities for growth and innovation. There are still several challenges related to regulatory compliance, standardization, and sustainable production. To solve these challenges and bridge the gap between these fields it is important to incorporate the field of sustainability and business with herbal medicine as demonstrated in the proposed incorporation cycle in Fig. 3.

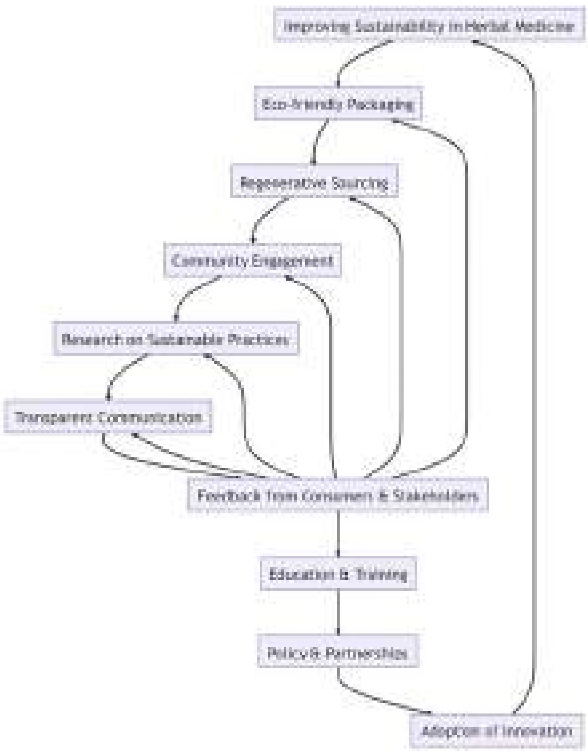

Fig. 3 Interactions within the nexus of herbal medicine, business, and sustainability

6.1 Eco-friendly Packaging

In improving sustainability and encouraging business growth the first step will be to provide eco-friendly packaging. This is because of the awareness of the majority of the population globally on the environmental issues with the indiscriminate use of plastics.

The global waste crisis is one of the most pressing environmental issues of our time. As urbanization and consumerism rise, so does the production of waste, particularly from packaging materials. A significant portion of this waste is non-biodegradable, meaning it doesn't break down naturally over time. Instead, it persists in the environment for hundreds, if not thousands, of years.

Traditional packaging solutions, often made from plastics and other synthetic materials, are a primary contributor to this crisis. These materials, once discarded, find their way into various ecosystems. Landfills overflow with waste, much of which seeps into the ground, contaminating soil and groundwater. Oceans are littered with plastic debris, leading to the formation of vast "plastic islands" that disrupt marine life. Wildlife, both marine and terrestrial, often mistake these materials for food, leading to ingestion, entanglement, and, in many cases, death. Furthermore, the production of traditional packaging materials is resource-intensive, often requiring significant amounts of water, energy, and raw materials. The carbon footprint associated with the production, transportation, and disposal of these materials further exacerbates global warming.

Eco-friendly packaging offers a beacon of hope in addressing the global waste crisis. At its core, eco-friendly packaging prioritizes the environment, both in terms of the materials used and the processes involved in its production, and typically involves:

- **Biodegradable Materials**: These are materials that can be broken down by natural processes, turning into water, carbon dioxide, and biomass. They decompose within a reasonably short time, leaving no trace behind. Examples include certain types of plant-based plastics and paper.
- **Recyclable Materials**: Packaging made from these materials can be processed and transformed into new products. This cycle can be repeated multiple times, reducing the need for raw materials and energy. Common recyclable materials include glass, aluminum, and certain types of plastics.
- **Reusable Materials**: Some packaging solutions are designed for multiple uses. Instead of being discarded after a single use, they can be cleaned and reused, significantly reducing waste. Examples include cloth bags, glass containers, and metal straws.

By adopting eco-friendly packaging, businesses can significantly reduce their environmental footprint. The reduced reliance on raw materials, coupled with decreased waste production, ensures that the environment is preserved for future generations. This does not only improve the environment (sustainability) but also plays a huge role in the business perspective as explained below.

In today's world, where information is at one's fingertips, consumers are more informed and conscious about their choices. Environmental concerns rank high on their list of priorities. Consumers are increasingly seeking out brands that reflect their values, and eco-friendly packaging is a clear indicator of a brand's commitment to sustainability. This may end up in one or more of the following:

- **Brand Differentiation**: In a saturated market, eco-friendly packaging offers brands a unique selling proposition. It sets them apart from competitors and appeals to a growing demographic of environmentally conscious consumers.
- **Increased Sales and Brand Loyalty**: Consumers are willing to pay a premium for products that are sustainably packaged. They are also more likely to remain loyal to brands that demonstrate a genuine commitment to the environment.
- **Cost Savings**: While the initial investment in eco-friendly packaging might be higher, the long-term benefits are manifold. Reduced waste management costs, decreased reliance on expensive raw materials, and potential tax benefits associated with sustainable practices can lead to significant cost savings.
- **Risk Mitigation**: With governments around the world implementing stricter environmental regulations, businesses that proactively adopt sustainable practices are better positioned to navigate these changes. They face fewer regulatory hurdles and are less likely to incur penalties.
- In summary, eco-friendly packaging is not just an environmental imperative but also a smart business strategy. It offers a win-win solution, benefiting both the planet and the bottom line.

Furthermore, after incorporating eco-friendly packaging into the herbal medicine industry, the authors of this chapter recommend the next step to include regenerating sourcing as this would further strengthen the corporate social responsibility of herbal medicine business owners as well as improve their brand value proposition which directly impacts the business and solves the challenges presented below.

6.2 Regenerative Sourcing

The global demand for herbal medicines has seen a meteoric rise in recent decades. This surge in demand, coupled with often short-sighted agricultural practices, has put immense pressure on the natural reserves of medicinal plants. Overharvesting, or the act of extracting more from the land than it can naturally replenish, has become a widespread issue. This not only depletes the immediate stock of valuable medicinal plants but also disrupts the natural balance of ecosystems.

Unsustainable agricultural practices further compound this problem. The excessive use of chemical fertilizers and pesticides, monocropping, and deforestation for agricultural expansion degrade the soil, reduce its fertility, and threaten local biodiversity [83]. The loss of biodiversity is particularly concerning as it reduces the resilience of ecosystems, making them more susceptible to diseases, pests, and climate change impacts.

Furthermore, the depletion of key medicinal plants jeopardizes the long-term availability of essential ingredients for the herbal medicine industry. This poses a significant threat to the health and well-being of countless individuals who rely on these medicines. Regenerative sourcing emerges as a beacon of hope in this challenging scenario. It goes beyond mere sustainability, aiming not just to maintain but to enhance and rejuvenate the health of ecosystems through one or more of the following:

- **Crop Rotation**: This age-old practice involves growing different types of crops in succession on the same land. Each crop type has different nutrient requirements and pest resistance. By rotating them, the soil's nutrient balance is maintained, and the build-up of pests and diseases is prevented.
- **Organic Farming**: Eschewing the use of synthetic fertilizers, pesticides, and genetically modified organisms, organic farming relies on natural processes and materials to cultivate crops. This ensures that the soil remains rich in organic matter and beneficial microorganisms, promoting long-term soil health.
- **Agroforestry**: This practice integrates trees, shrubs, and crops on the same plot of land. Trees and shrubs provide shade, reduce soil erosion, and enhance soil fertility through nitrogen fixation. They also act as windbreaks and improve water retention, creating a microclimate conducive to the growth of crops.

Together, these practices ensure that the land remains fertile and productive. They also enhance local biodiversity, ensuring that ecosystems are resilient and can support the sustainable harvesting of medicinal plants for generations to come. Hence, adopting regenerative sourcing practices offers a plethora of benefits from a business standpoint that include:

- **Long-term Supplier Relationships**: Consistent quality is paramount in the herbal medicine industry. By engaging in regenerative sourcing, businesses can foster long-term relationships with suppliers who adhere to these practices. This ensures a steady supply of high-quality ingredients.
- **Reduced Supply Chain Risks**: With the increasing global awareness of sustainability issues, there's a growing risk of resource scarcity. Regenerative sourcing mitigates this risk by ensuring the continuous availability of key ingredients.
- **Ethical Branding**: Modern consumers are well informed and value driven. They prefer brands that align with their ethical and environmental beliefs. By marketing products as ethically and sustainably sourced, brands can appeal to this growing demographic, enhancing their market share and brand loyalty.
- **Cost Efficiency**: While the initial transition to regenerative practices might require investment, the long-term benefits, such as reduced dependency on chemical inputs and enhanced soil fertility, can lead to cost savings.

Therefore, regenerative sourcing is not just an environmental and ethical imperative but also a strategic business decision. It ensures the long-term viability of the HMI while safeguarding the planet's precious biodiversity. Furthermore, after regenerating sources and ensuring the products are sourced through highly sustainable methods, the authors of this chapter recommend the next step to include community engagement as most of these herbal medicine as discussed in the previous sections hold some ties to certain cultural beliefs which have greatly been exploited. Community engagement will bring about more trust in the herbal medicine businesses.

6.3 Community Engagement

The rich tapestry of traditional knowledge, especially in indigenous regions, has been the backbone of herbal medicine for centuries. However, with the commercialization of herbal products and the race to source rare and potent ingredients, there's been a growing trend of exploitative sourcing. This often involves bypassing local communities, disregarding their rights, and harvesting resources without proper compensation or acknowledgment.

Such practices can lead to several adverse outcomes including:

- **Conflicts**: Disregarding the rights and wishes of local communities can lead to conflicts. These can range from disputes over land rights to disagreements about compensation and benefits.
- **Loss of Traditional Knowledge**: Exploitative sourcing often overlooks the wealth of knowledge that local communities possess. This knowledge passed down through generations can offer insights into the medicinal properties of plants, sustainable harvesting techniques, and more. Ignoring this can lead to the loss of invaluable information.
- **Socio-economic Disparities**: When businesses profit from resources without adequately compensating or involving local communities, it can exacerbate socio-economic disparities. While businesses thrive, the communities that have been stewards of these resources for generations may continue to face poverty and marginalization.

Community engagement emerges as a holistic solution to these challenges. It emphasizes the importance of building relationships, fostering trust, and collaborating with local communities through:

- **Respecting Traditional Knowledge**: Local communities, especially indigenous ones, have a deep understanding of their environment. By respecting and incorporating this knowledge, businesses can ensure that they are sourcing resources most effectively and sustainably.
- **Collaborative Partnerships**: Instead of a top-down approach, businesses can form collaborative partnerships with communities. This involves involving them in decision-making processes, from identifying resources to determining harvesting techniques and compensation structures.

- **Sustainable and Ethical Harvesting**: With the involvement of local communities, harvesting can be done in a manner that ensures the long-term viability of resources. This includes adhering to traditional harvesting calendars, using non-destructive methods, and ensuring that a portion of the benefits goes back to the community.

Engaging with communities is not just a moral obligation but also a strategic business decision with multiple benefits:

- **Positive Brand Image**: In an age of conscious consumerism, businesses that engage ethically with communities are viewed favorably. This can lead to increased brand loyalty, positive word-of-mouth, and a competitive edge in the market.
- **Unique Collaborations**: Trust-building at the grassroots level can pave the way for unique collaborations. This could involve joint research projects, knowledge-sharing initiatives, or access to rare resources that are not available through conventional channels.
- **Long-term Sustainability**: By making local community stakeholders in the business, companies ensure the long-term sustainability of their sourcing practices. Communities that benefit from the business are more likely to support and collaborate with it, ensuring a steady supply of resources.
- **Risk Mitigation**: Engaging with communities can help businesses pre-empt and address potential conflicts, reducing operational risks and ensuring smoother supply chains.

Therefore, community engagement offers a win-win approach. It ensures that businesses operate ethically and sustainably while also unlocking a plethora of opportunities for growth, collaboration, and innovation. Furthermore, it is recommended that after the process of community engagement, HMI practitioners delve into more research on sustainability as information obtained from community engagements can help improve sustainability practices.

6.4 Research on Sustainable Practices

Sustainability is not a static concept. As our understanding of the environment, ecosystems, and human impact evolves, so do the benchmarks for what is considered sustainable. The herbal medicine industry, like all sectors, faces the challenge of keeping pace with these changing dynamics through:

- **Evolving Standards**: What was considered sustainable a decade ago might not meet today's standards. As global awareness about environmental issues grows, industries are held to higher standards of sustainability.
- **Complex Ecosystem Interactions**: The interactions between various components of an ecosystem are intricate. A change in one component can have ripple

effects, making it crucial to have a holistic understanding before implementing any intervention.

- **Rapid Technological Advancements**: The pace of technological change is rapid. New technologies can offer more sustainable alternatives, but they also come with their own set of challenges and environmental impacts.

In the same way, research emerges as the cornerstone to address these challenges. It provides the knowledge and tools to navigate the complex landscape of sustainability such as:

- **Ecological Impact Studies**: Understanding the impact of farming practices, harvesting techniques, and production processes on the environment is crucial. Research can provide insights into the carbon footprint, water usage, soil health, and biodiversity impact of various practices.
- **Waste Reduction Innovations**: Waste, especially non-biodegradable waste, is a significant environmental concern. Research can explore innovative ways to reduce waste from developing biodegradable packaging to finding uses for production by-products.
- **Sustainable Farming Techniques**: Research can lead to the development of farming techniques that are both productive and sustainable. This includes exploring alternative pest control methods, soil conservation techniques, and sustainable irrigation practices.

From a business standpoint, investing in research is not just an environmental imperative but also a strategic business decision:

- **Innovation and Differentiation**: Research can lead to breakthrough innovations that set a brand apart. This could be a new sustainable farming technique, a novel product derived from waste, or a patented production process.
- **Brand Leadership**: Brands that are at the forefront of sustainability research are seen as industry leaders. This leadership position can translate into a strong brand image, customer trust, and a loyal customer base.
- **Economic Benefits**: Sustainable practices, derived from research, can lead to cost savings in the long run. For instance, reducing waste can lower disposal costs, and sustainable farming can reduce the dependency on expensive chemical inputs.
- **Premium Pricing**: Products that are backed by solid research and are verifiably sustainable can command premium prices in the market. Consumers are willing to pay a premium for products that align with their values.

Research on sustainable practices is a multifaceted endeavor that offers environmental, social, and economic benefits. For the herbal medicine industry, it ensures long-term viability, brand leadership, and a competitive edge in a market that is increasingly valuing sustainability. However, despite the need for research on sustainability, it is paramount that organizations encourage transparent communication based on research generated to inform all respective medical stakeholders.

6.5 Transparent Communication

The major challenge associated with non-transparent communication in the nexus of herbal medicine, business, and sustainability is greenwashing. Greenwashing is the practice of making misleading or unsubstantiated claims about the environmental benefits of a product, service, or company. In the Herbal Medicine Industry, this could manifest as exaggerated claims about organic sourcing, ethical practices, or environmental impact. The challenge here is significant because consumers are becoming increasingly savvy and skeptical. They demand authentic information and can easily spot inconsistencies or false claims, thanks to the plethora of information available online.

The solution includes:

- **Open and Honest Communication**: Transparent communication goes beyond merely sharing information; it's about creating an open dialogue with medical stakeholders, particularly consumers. This involves:
- **Full Disclosure**: Clearly stating the origins of raw materials, the methods of production, and the environmental impact of these processes.
- **Accountability**: Owning up to areas where the company may fall short of sustainability goals and outlining improvement plans.
- **Third-Party Verification**: Utilizing external audits or certifications to validate claims can add an extra layer of credibility.
- **Regular Updates**: Consistently update stakeholders on progress made toward sustainability goals, setbacks faced, and lessons learned.
- **Interactive Platforms**: Using social media, blogs, or dedicated sections on the company website where consumers can ask questions and receive prompt, honest answers.

In today's digital age, consumers are not just passive recipients of information; they are active participants in brand narratives. Transparency in communication can yield several business benefits:

- **Brand Loyalty**: Consumers are more likely to stick with brands that they trust. Transparency fosters this trust by showing that a company has nothing to hide.
- **Positive Reviews**: Satisfied, informed customers are more likely to leave positive reviews, which can significantly influence potential buyers.
- **Word-of-Mouth Marketing**: Transparency can turn consumers into brand advocates. People are more likely to recommend a brand that they feel is honest and aligned with their values.
- **Competitive Advantage**: In a market saturated with claims of 'natural' and 'sustainable,' genuine transparency can set a brand apart.
- **Risk Mitigation**: Being upfront about challenges and shortcomings can help mitigate the impact of any negative publicity and demonstrate a commitment to continuous improvement.

- **Stronger and Resilient Herbal Brands**: By embracing transparent communication, companies in the HMI can not only meet the growing consumer demand for sustainability but also build stronger, more resilient brands. In addition to this, seeking feedback can greatly improve sustainability practices.

6.6 Feedback from Consumers and Stakeholders

The challenge within this aspect of the nexus is linked to the imperfect nature of diverse sustainability models. Sustainability is a complex, multifaceted issue that requires ongoing efforts for improvement. No model or initiative is perfect out of the gate. The challenge lies in identifying the gaps, inefficiencies, or unintended consequences that may arise during the implementation of sustainability practices. This is particularly relevant in the Herbal Medicine Industry, where the stakes are high due to the involvement of natural resources, community welfare, and consumer health.

Actively seeking feedback from consumers and other medical stakeholders is crucial for continuous improvement. This can be achieved through:

- **Surveys and Questionnaires**: These can be sent to consumers post-purchase or be made available on the company's website.
- **Focus Groups**: Inviting a diverse set of stakeholders to discuss various aspects of the company's sustainability efforts can provide in-depth insights.
- **Public Forums**: Hosting webinars, Q&A sessions, or community meetings where stakeholders can voice their opinions and ask questions.
- **Social Media Engagement**: Utilizing platforms like Twitter, Instagram, and Facebook to gather feedback in real time.
- **Third-Party Audits**: External evaluations can offer an unbiased view of the effectiveness of sustainability initiatives.

To ensure there is a win-win business scenario, there needs to be an incorporation of feedback-driven sustainability approaches that offer several advantages including:

- **Product Improvement**: Consumer feedback can highlight areas where the product can be made more sustainable, whether it's the packaging, the sourcing, or the production process.
- **Process Efficiency**: Stakeholder feedback can help identify inefficiencies in the supply chain or production methods, leading to cost savings and a reduced environmental footprint.
- **Consumer Satisfaction**: Meeting or exceeding consumer expectations in terms of sustainability can lead to higher satisfaction rates, repeat purchases, and brand loyalty.
- **Regulatory Compliance**: Feedback from regulatory bodies can help the company stay ahead of legal requirements, reducing the risk of fines or sanctions.
- **Innovation**: Fresh perspectives from stakeholders can inspire new ways to approach sustainability, leading to innovative solutions that benefit both the company and the environment.

By incorporating feedback into their sustainability models, companies in the HMI can create a cycle of continuous improvement. This not only enhances their sustainability efforts but also builds a stronger, more competitive, and more resilient business. For these improvements to be implemented, education and training is highly paramount.

6.7 Education and Training

The challenge within this aspect of the nexus is linked to knowledge gaps, especially from a sustainability viewpoint. The adoption of sustainable practices is often hindered by a lack of knowledge and awareness. This is especially true in the Herbal Medicine Industry, where the supply chain involves various stakeholders, from farmers and suppliers to manufacturers and retailers. Without proper education and training, these stakeholders may inadvertently engage in practices that are detrimental to sustainability goals.

A potentially viable solution would be comprehensive education and training programs. Education and training are essential tools for bridging these knowledge gaps. The scope of these programs can be diverse, including but not limited to:

- **Workshops for Farmers**: Training on sustainable farming techniques, soil conservation, and responsible use of water resources.
- **Supplier Seminars**: Educating suppliers on the importance of ethical sourcing and providing them with the tools to achieve it.
- **Employee Training**: Regular training sessions for staff on waste management, energy conservation, and other sustainability practices relevant to their roles.
- **Stakeholder Webinars**: Online seminars that educate stakeholders on the company's sustainability initiatives and how they can contribute.
- **Public Awareness Campaigns**: Educational materials and campaigns aimed at consumers to inform them about the sustainability features of the products they are buying.

From the business perspective, there are numerous innumerable multifaceted benefits of education and training. Investing in education and training offers several business advantages:

- **Increased Efficiency**: An educated workforce is more likely to identify and implement efficient, sustainable practices, reducing waste and lowering costs.
- **Innovation**: Training can spark new ideas and approaches to sustainability, leading to innovative solutions that could give the company a competitive edge.
- **Regulatory Compliance**: Educated staff and suppliers are less likely to violate environmental laws and regulations, reducing the risk of legal repercussions.
- **Brand Image**: A company that invests in education is seen as responsible and committed to sustainability, enhancing its reputation among consumers and stakeholders alike.

- **Employee Satisfaction**: Employees take pride in working for a company that is committed to sustainability, leading to higher job satisfaction and lower turnover rates.

By prioritizing education and training, companies in the HMI can ensure that all stakeholders are aligned in their sustainability efforts. This not only accelerates the adoption of sustainable practices but also fosters a culture of continuous learning and improvement, benefiting both the environment and the business.

6.8 Policies and Partnerships

The challenge within this aspect of the nexus is linked to the limits of individual efforts. While individual companies may make significant strides in sustainability, the scale of environmental challenges often requires collective action. In the Herbal Medicine Industry, this is particularly relevant given the complex supply chains, diverse stakeholders, and global reach of many companies. Individual efforts, no matter how well intentioned, may fall short of creating the systemic change needed for true sustainability.

A potentially viable solution would be strategic collaborations and policy influence. To overcome this challenge, companies can engage in two main avenues:

- **Policy Advocacy**: Collaborating with governmental bodies to influence policies that promote sustainability. This could involve lobbying for stricter regulations on pesticide use, advocating for fair trade, or pushing for tax incentives for sustainable practices.
- **Partnerships**: Forming alliances with NGOs, academic institutions, and even competitors to amplify the impact of sustainability initiatives. These partnerships can focus on research, community outreach, or joint ventures that promote sustainability.
- **Certification Programs**: Partnering with organizations that offer sustainability certifications can add credibility to a company's efforts and attract a more conscious consumer base.
- **Public-Private Partnerships**: Engaging in collaborations that bring together the public and private sectors can leverage the strengths of both to achieve more significant results.

From the business perspective, there are multiple avenues to reap benefits from partnerships and policies. Engaging in policy advocacy and partnerships offers several advantages:

- **Financial Support**: Collaborations can often unlock access to grants, subsidies, or low-interest loans aimed at promoting sustainability, making it more financially feasible to undertake significant initiatives.

- **Policy Influence**: Being part of a larger coalition gives companies a stronger voice in policy discussions, allowing them to steer public policy in a direction that aligns with their sustainability goals.
- **Brand Enhancement**: Partnerships with reputable organizations can significantly boost a brand's image, positioning it as a leader in sustainability and attracting a more conscious consumer base.
- **Knowledge Sharing**: Collaborations often involve an exchange of expertise and resources, leading to more effective and innovative sustainability solutions.
- **Risk Mitigation**: Aligning with industry standards and governmental policies reduces the risk of non-compliance and the associated legal repercussions.

By actively engaging in policy advocacy and forming strategic partnerships, companies in the HMI can significantly amplify the impact of their sustainability efforts. This not only benefits the environment but also creates a more resilient and competitive business landscape.

6.9 Adoption of Innovation

The challenge within this aspect of the nexus is linked to the double-edged sword nature of technological innovations and evolution. The rapid pace of technological advancements offers a wealth of opportunities for enhancing sustainability. However, it also presents challenges in terms of adaptability, cost, and effective implementation. In the Herbal Medicine Industry, where traditional practices often coexist with modern methods, the integration of new technologies can be particularly complex.

A potentially viable solution would be the strategic adoption of innovative technologies. To harness the power of innovation for sustainability, companies can focus on:

- **Research and Development (R &D)**: Investing in R&D to explore new technologies that can make farming, production, and distribution more sustainable.
- **Pilot Programs**: Before full-scale implementation, running pilot programs can help assess the effectiveness and ROI of new technologies.
- **Technology Partnerships**: Collaborating with tech companies or start-ups that specialize in sustainable solutions can accelerate the adoption of new technologies.
- **Employee Training**: Ensuring that staff are well equipped to operate new technologies can maximize their effectiveness and sustainability impact.
- **Consumer Education**: Educating consumers about the benefits of new technologies, such as blockchain for traceability, can drive demand and acceptance.

From a business perspective, the strategic adoption of innovative technologies offers multiple advantages:

- **Operational Efficiency**: Technologies like the adoption of innovation can optimize various aspects of the business, from supply chain logistics to inventory management, leading to reduced waste and lower costs.

- **Cost Savings**: While the initial investment may be high, the long-term savings from reduced resource consumption can be significant. For example, adopting renewable energy sources can lower utility bills.
- **Competitive Edge**: Being an early adopter of sustainable technologies can position a company as a leader in the industry, attracting both consumer attention and potential partnerships.
- **Environmental Impact**: Technologies focused on waste reduction, energy efficiency, or sustainable sourcing have a direct positive impact on the environment.
- **Regulatory Advantage**: Staying ahead of the curve in terms of sustainable technologies can also mean staying ahead of regulatory changes, reducing the risk of non-compliance.

By being open to change and willing to invest in innovative technologies, companies in the HMI can significantly enhance their sustainability efforts. This not only benefits the environment but also translates into operational efficiencies, cost savings, and a stronger competitive position.

7 The Interconnectedness and Feedback Loop: A Holistic Approach to Sustainability

The model's strength lies in its design as a feedback loop, which underscores the interconnectedness of its various components. This loop is not just a theoretical construct but a practical tool for continuous improvement and alignment with broader sustainability goals.

7.1 Feedback from Consumers and Stakeholders

This is the cornerstone of the loop. Consumer and stakeholder feedback informs every other component, acting as both a catalyst and a checkpoint. For example, if consumers express concerns about packaging waste, this feedback can trigger changes in transparent communication to address these concerns. Similarly, this feedback can influence policy and partnerships, ensuring that any lobbying or advocacy efforts align with public sentiment.

7.2 Feedback from Education and Training to Policy and Partnerships

Education and training serve as the foundation for informed policy decisions. A well-educated workforce and stakeholder community are more likely to support and contribute to effective policies. This educated base can also be a powerful ally in advocacy efforts, lending credibility and weight to Policy and partnerships.

7.3 Feedback from Policy and Partnerships to Adoption of Innovation

Effective policies create a fertile ground for innovation. For instance, a policy that offers tax incentives for renewable energy can encourage companies to invest in new, sustainable technologies. These policies can also facilitate partnerships with tech companies, expediting the Adoption of Innovation.

7.4 Feedback from Adoption of Innovation to Improving Sustainability

The technologies and practices adopted feed directly into improving sustainability across the board. Whether it's a new waste reduction technology or a renewable energy source, these innovations contribute to the overall sustainability goals of the company.

7.5 Completing the Loop

The loop comes full circle with these innovations feeding back into the system, leading to further improvements. This could manifest as updates to existing policies, new training programs, or even shifts in consumer behavior due to more effective Transparent Communication. In essence, the feedback loop creates a self-sustaining cycle of continuous improvement. Each component not only contributes to sustainability but also informs and enhances the other components, creating a holistic, adaptable, and effective sustainability model with consideration of the business practices in the herbal medicine industry. However, to encourage implementation, this chapter examined the key stakeholders their roles and their impacts on this implementation.

8 Stakeholder Groups in the Herbal Medicine Industry

This chapter further delves into each stakeholder group to understand their roles, interests, and potential impact on sustainability initiatives in the HMI.

8.1 High Influence, High Interest

This quadrant is crucial for immediate action and long-term strategy. These stakeholders have both the interest and the power to drive or hinder sustainability initiatives. Regular engagement, transparent communication, and active involvement in decision-making processes are essential for maintaining a positive relationship with this group (Fig. 4).

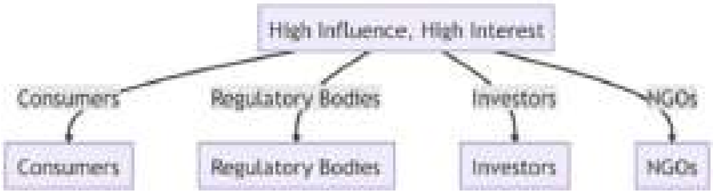

Fig. 4 High influence, high interest decision-making processes for maintaining a positive relationship within the nexus of business, herbal medicine, and sustainability

Consumers

I. **Role**: End-users of herbal medicine products.
II. **Interest**: Concerned about the quality, safety, and environmental impact of the products they consume.
III. **Impact**: Their purchasing decisions can significantly influence a company's sustainability practices. Brands that fail to meet consumer expectations risk losing market share.

Regulatory Bodies

I. **Role**: Organizations that set and enforce standards and regulations.
II. **Interest**: Ensuring that the industry operates within legal and ethical boundaries, including sustainability.
III. **Impact**: Can impose fines, sanctions, or even shut down operations for non-compliance, making their influence substantial.

Investors

I. **Role**: Individuals or organizations that provide capital.
II. **Interest**: Maximizing ROI while aligning investments with ethical and sustainable practices.
III. **Impact**: Can provide the necessary funding for sustainability initiatives or withdraw support if a company fails to meet sustainability criteria.

Non-governmental Organizations (NGOs)

I. **Role**: Non-profit organizations focused on various causes, including environmental sustainability.
II. **Interest**: Ensuring that the industry adopts sustainable and ethical practices.
III. **Impact**: Can mobilize public opinion and drive campaigns that force companies to adopt sustainable practices.

8.2 Low Influence, High Interest

While these stakeholders may not have the direct power to enforce changes, their high level of interest makes them valuable allies. They can offer insights, data, and public support, which can indirectly influence more powerful stakeholders. Periodic updates and collaborative projects can help keep this group engaged. The low

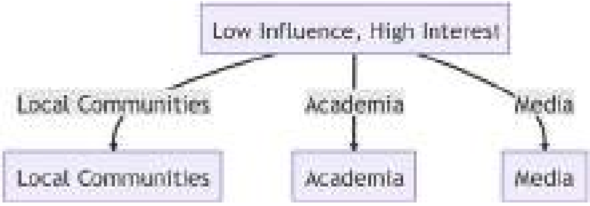

Fig. 5 Low influence, high interest decision-making processes for maintaining a positive relationship within the nexus of business, herbal medicine, and sustainability

influence, high interest decision-making processes for maintaining a positive relationship within the nexus of business, herbal medicine, and sustainability are presented in Fig. 5.

Local Communities

I. **Role**: Communities living in areas where herbal plants are sourced.
II. **Interest**: Sustainable and ethical sourcing, community development.
III. **Impact**: While they may lack formal influence, their cooperation is crucial for sustainable sourcing and long-term business operations.

Academia

I. **Role**: Educational institutions and researchers.
II. **Interest**: Researching and promoting sustainable practices.
III. **Impact**: Can provide valuable insights and data but generally lack the power to enforce change.

Media

I. **Role**: Channels of mass communication.
II. **Interest**: Reporting on industry trends, including sustainability efforts.
III. **Impact**: Can shape public opinion but cannot enforce regulatory changes.

8.3 High Influence, Low Interest

This group has the power to make or break sustainability initiatives but may not have a vested interest in them. The key is to make them see the value – whether it's financial, reputational, or regulatory – in participating in sustainability efforts. Tailored communication that speaks to their specific interests can be effective here. The high influence, low interest decision-making processes for maintaining a positive relationship within the nexus of business, herbal medicine, and sustainability are presented in Fig. 6.

Suppliers

I. **Role**: Provide raw materials for herbal medicine products.
II. **Interest**: Business continuity and profitability.

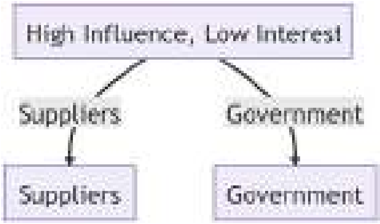

Fig. 6 High influence, low interest decision-making processes for maintaining a positive relationship within the nexus of business, herbal medicine, and sustainability

III. **Impact**: Have the power to choose sustainable sourcing methods, although they may not have a vested interest in sustainability.

Government

I. **Role**: State and federal governing bodies.
II. **Interest**: Economic growth, public safety.
III. **Impact**: Can enact laws that mandate sustainable practices but may prioritize other interests like economic development.

8.4 Low Influence, Low Interest

These stakeholders may seem less relevant due to their low influence and interest, but they can't be ignored. Changes in public opinion can gradually move them to a quadrant of higher interest or influence. Occasional updates and public awareness campaigns can be beneficial for this group. The low influence, low interest decision-making processes for maintaining a positive relationship within the nexus of business, herbal medicine, and sustainability are presented in Fig. 7.

Competitors

I. **Role**: Other companies in the herbal medicine industry.
II. **Interest**: Market share and profitability.
III. **Impact**: May adopt similar sustainability practices to stay competitive but have limited influence on other companies.

General Public

I. **Role**: The broader population is not directly involved in the industry.
II. **Interest**: General well-being and environmental sustainability.
III. **Impact**: Limited direct influence but can sway public opinion, which indirectly affects the industry.

Understanding the unique roles and interests of each stakeholder group can help tailor sustainability initiatives and communication strategies, ensuring a more effective and comprehensive approach.

It is important to note that by understanding the dynamics of each quadrant, herbal medicine businesses can tailor their sustainability strategies and

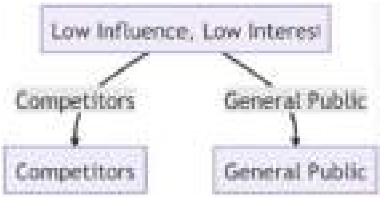

Fig. 7 Low influence, low interest decision-making processes for maintaining a positive relationship within the nexus of business, herbal medicine, and sustainability

communication plans to effectively engage each stakeholder group. This nuanced approach will likely yield better results in incorporating sustainability into the HMI with the business ethos in mind.

9 Conclusion

Sustainability in the HMI is not a one-off effort but a continuous process that involves multiple stakeholders, from farmers and local communities to policymakers within the healthcare sector and consumers. The comprehensive model presented in this chapter offers a multifaceted approach to sustainability, emphasizing the need for a holistic strategy that addresses various challenges and opportunities. By understanding and implementing each component of the model proposed in this chapter, the HMI can make significant strides toward a more sustainable future. While the economic potential of herbal medicine is evident, it is essential to approach the topic with a balanced perspective. The integration of herbal medicine into modern healthcare and commerce presents both opportunities and challenges. Sustainable practices, evidence-based approaches, and a focus on long-term benefits will be crucial in harnessing the full economic potential of herbal medicine. In addition, the herbal medicine industry, with its vast economic potential, presents both opportunities and challenges. Sustainable cultivation, fair trade practices, and equitable economic distribution are crucial to harnessing this potential. As the global demand for herbal remedies continues to grow, it is imperative to address these challenges to ensure the long-term viability and sustainability of the industry.

It is notable that traditional herbal medicine practices is deeply rooted in history and culture, but remains dynamic and continue to evolve in response to changing societal needs and global influences. As these practices navigate the challenges of modernity, it is crucial to ensure their preservation, documentation, and sustainable utilization, ensuring that future generations can benefit from this rich heritage. Nonetheless, the integration of traditional herbal practices into modern healthcare systems signifies a paradigm shift toward holistic patient care and a more robust healthcare system. While the potential benefits of Integrative and Complementary Medicine are evident, it is crucial to approach its implementation with a balance of respect for tradition, scientific rigor, and interprofessional collaboration. The commercial production of herbal products is a multifaceted endeavor, influenced by global trends, technological advancements, and consumer preferences. As the demand for herbal products continues to grow, it is imperative to address the challenges faced by the industry, ensuring sustainable practices, product quality, and safety. The globalization of herbal medicine presents a complex interplay of opportunities and challenges. While it offers a broader market reach and potential for growth, it also brings to the forefront issues related to sustainability, cultural sensitivity, and market dynamics. As the herbal medicine industry continues to evolve, businesses must adapt and innovate to navigate the global landscape effectively. The confluence of traditional herbal medicine and modern drug discovery is reshaping the therapeutic landscape. As research continues to unravel the mysteries of herbal compounds, there is immense potential for the development of novel drugs that are both effective and safe. The business focus in this domain promises lucrative opportunities, driven by scientific innovation and consumer demand for natural remedies. At the moment, the global demand for traditional herbal formulations is undeniable and will clearly grow in the future. However, as these new and old herbal products find their way into global markets, ensuring their microbial quality, safety, standardization, for standardization, quality control, innovative distribution strategies sustainability and alignment with global value chains and pricing regimes will be becomes paramount. Rigorous quality control measures, standardization, and continuous research are essential to guarantee the safety and efficacy of these age-old remedies. In the same way, the vast and diverse global herbal medicine market will continually need to evolve. As the demand for these products grows, understanding their distribution channels, market dynamics, and economic implications becomes paramount. Strategic approaches, informed by rigorous research and market analyses, are essential for navigating the challenges and harnessing the opportunities in this burgeoning industry. The nexus of business, herbal medicine, and sustainability has an immense growth potential but is faced with numerous pressing needs constantly emerging from their interaction. Hence, the need for the HMI to remain adaptive to global trade dynamics and geopolitical events to ensure sustained growth and consumer trust.

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Healing Trails: Integrating Medicinal Plant Walks into Recreational Development

66

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Abstract

The potential for nature-based healing pathways to promote physical, mental, and emotional repair has received a lot of attention. Tourists have a closer connection to nature and the restorative advantages of medicinal flora by adding medicinal plant walks into recreational development. This chapter discusses the development of healing trails as a revolutionary ecotherapy technique and continues to study its many features. Incorporating cultural history and traditional medical knowledge along these pathways strengthens communities' relationships with their natural environments. Healing trials that combine indigenous medicinal approaches not only help to preserve historical knowledge but also promote responsible plant usage and conservation. As more urban green spaces and parks are added, the positive impact on the well-being of city dwellers becomes more obvious. Furthermore, research into the combining of medicinal plant walks with other healing techniques provides a promising path toward a holistic approach to health. However, keeping historical relics secure, welcoming all people, and adhering to ethical foraging techniques provide challenges and concerns along these roads. Implementing high-quality certification programs is critical for tackling these difficulties. This protects the integrity of healing paths, prioritizes trail user safety, and defends fragile habitats. The chapter concludes with a suggestion to use universal design principles to construct healing paths that are accessible to persons with impairments and from varied cultural backgrounds.

Keywords

Healing Trails · Medicinal plant walks · Recreational development · Indigenous knowledge · Sustainability

Acronyms

AHG	American Herbalists Guild
COPD	Chronic Obstructive Pulmonary Disease
FSC	Forest Stewardship Council
ICOMOS	International Council on Monuments and Sites
NAMA	National Ayurvedic Medical Association
TCM	Traditional Chinese Medicine

1 Introduction

Since ancient times, people have understood that spending time in nature can have positive effects on their health, and this understanding has only grown in recent decades. Healing trails or medicinal plant walks have been practiced for millennia as part of several traditional therapeutic modalities, thus the idea is not new. Medicinal plant walks have been an integral part of many healthcare systems [1–5], including those that place a high value on local flora (such as indigenous healing practices) and those with complex materia medica (such as Traditional Chinese Medicine and Ayurveda) [6–9]. Practitioners can draw on a wealth of traditional knowledge by merging these tried-and-true methods into modern alternative medicine, all the while taking into account the latest in scientific findings and safety concerns.

Healing Trails are an innovative and comprehensive way to improving human health and fostering environmental protection by incorporating medicinal plant walks into recreational projects [10]. Physical exertion, mental de-stressing, and the opportunity to learn about the medicinal plants that grow wild on Healing Trails make for a potent therapeutic experience. Consequently, a growing number of people in today's culture are leaving the bustle of city life to find peace and rejuvenation in the great outdoors. Ecotherapy, often known as green therapy or nature-based healing, is a relatively new field that investigates the psychological and physiological advantages of relaxing in natural settings. Interest in incorporating environment-based experiences into numerous areas of life, particularly recreational development, has been prompted by the expanding body of studies supporting the good beneficial effects of nature on human health.

The mental and physiological advantages offered by spending time in nature have been well-documented. For instance, studies by Astell-Burt et al. [11], Jacob et al. [12, 13], and Udoakpan et al. [14] indicated that time spent in green spaces including parks, forests, and gardens alleviated symptoms of anxiety, depression, and stress-related disorders. Trees, water, and vegetation have been shown to have a healing impact on the mind, boosting one's disposition and ability to concentrate [15].

The biophilia theory, put out by E. O. Wilson, states that people have an inbuilt connection to nature and that our well-being is profoundly influenced by how we engage with the environment [16]. This idea of healing in nature is consistent with this notion. As cities expand into more and more green areas, residents are looking for ways to reduce the stress of city life by spending time in nature.

Changing our perspectives on public spaces may be as simple as incorporating nature-based therapy into recreational development. Recreational areas can be made into safe havens of good health and recuperation by including natural components like parks, landscaped areas, and healing paths with medicinal plant walks [12, 17]. This new emphasis places greater weight on the design of spaces that encourage interactions between people and the natural world, acknowledging the therapeutic potential of nature.

Also, with the growing awareness of the value of green spaces in boosting the general quality of life as urbanization continues to increase, people have become more alienated from natural areas. In response to the stresses of city life, healing trails have emerged as a way to help people relax, de-stress, and recharge in natural settings without having to go too far from their homes [18]. Thus, the popularity of healing trails has increased and more people now look for natural and holistic ways to improve their health. This network of paths provides an excellent opportunity to learn about the intricate web of connections between human well-being and the natural world.

Therefore, within the context of conventional medicine, historical preservation, and city landscaping, this chapter delves into the many facets of healing trails. We explore into the healing trails' potential to alleviate stress, anxiety, and despair while encouraging a profound bond with the natural world. We also discuss the value of indigenous healing methods, long-term economic growth, and the improvement of certain health ailments thanks to these natural havens. Inclusion, responsible foraging, and proper plant identification are all stressed. The need for quality standards and certification systems to guarantee healing trails' veracity and visitors' safety is also addressed. It also highlights the rich potential of healing trails as transforming spaces that nourish well-being through the embrace of nature's healing embrace and highlights the importance of evidence-based design, cultural integration, and accessibility.

2 Recreational Development and the Roles of Medicinal Plants

This section delves into the concept of recreational development, components, and the significance of Healing Trails in protecting indigenous medicinal plant knowledge. This facet of Healing Trails emphasizes the significance of recognizing and adapting indigenous healing practices into contemporary recreational development. Different indigenous groups have amassed extensive knowledge of the local ecosystems over time, including the medicinal benefits of local flora. By incorporating this understanding into recovery Trails, we can encourage the preservation of cultural history while simultaneously providing tourists with a one-of-a-kind and holistic approach to recovery.

2.1 Concept of Recreational Development

Recreational development is an essential part of city and neighborhood planning which includes the design, construction, and maintenance of facilities and programs for amusement, relaxation, and exercise [19]. Its value resides in its capacity to promote community prosperity, harmony, and long-term growth [20, 21].

The establishment of public areas like parks, playgrounds, and sports fields is an essential part of recreational development. All sorts of activities, from sports and physical exercise to relaxing and socializing, rely on these places as their primary locations. Particularly parks have been acknowledged for their beneficial effects on physical health, stress reduction, and providing a respite from city life [22]. In addition, recreational places that are planned correctly can motivate people to be more physically active, which aids in the fight against obesity and cardiovascular disease [23].

Cultural and entertainment amenities also fall under the umbrella of recreational development. The cultural fabric of a community benefits greatly from the establishment of performance venues, educational institutions, and social hubs such as museums, art galleries, and community centers. These places help people feel like they have a place in the community and give them something to be proud of [24].

Recreational planning in the modern era relies heavily on sustainable development principles. Studies have found that spending time in natural settings, especially green spaces, can improve one's mental health and well-being by providing a welcome break from the stresses of city life [25]. Sustainable features, such as energy-efficient lighting, water conservation measures, and the use of recycled materials, contribute to environmental stewardship and improve the ecological balance of communities when included into recreational areas.

Community involvement and input are essential to the success of any recreational development project. When planning and developing parks and other recreational areas, it is essential to get feedback from local residents, stakeholders, and user groups. Involving the locals in the planning process helps make sure the ultimate result is suitable for the people who will be using it. The incorporation of recreational development into urban and community planning is essential to the promotion of health, social integration, and long-term prosperity. Communities may improve their residents' physical and mental well-being as well as cultural and environmental sustainability by investing in the development of a wide range of recreational places and services. Recreational infrastructure helps build strong, diverse communities via participation and cooperation.

2.2 Concept and Components of Healing Trails

Healing trails, also known as therapeutic or nature trails, are a type of outdoor recreational development with the goal of fostering psychological, physiological, and spiritual health in its users [26, 27]. These trails are an upgrade from the standard hiking route since they include amenities designed to stimulate the senses, calm the mind, and bring you closer to nature [28, 29]. To promote healing, stress reduction, and general well-being, healing paths are constructed with careful planning and design.

The idea behind healing trails stems from research showing the health benefits of spending time in nature. The notion was inspired by the Japanese tradition of forest healing, also known as "Shinrin-yoku" or "forest bathing." This method entails spending time in nature and interacting with it consciously in order to improve one's health and well-being [26].

Consequently, the integration of nature-based healing practices into the planning, design, and management of public areas, healing trails serve a novel and game-changing concept in recreational development. The desire for environments that promote psychological, emotional, and physical health has led to the development of healing trails in response to the rising popularity of nature-based healing. Careful planning has gone into including picturesque spots and areas rich in biodiversity along these paths, giving hikers opportunities for more than just passive enjoyment. The addition of therapeutic plant walks to these pathways gives visitors a new option to connect with nature and learn about the plants' healing capabilities.

The components of healing trails as indicated in Table 1 include the following:

(i) **Design for Accessibility and Inclusivity:** Accessibility and inclusivity are key considerations in the construction of healing trails. This is to ensure that people of all ages and abilities may take part in the therapeutic experience. Paths are made easier to navigate by including features like handrails, ramps, and mild inclines [36].

(ii) **Natural Landscape and Biodiversity:** Healing paths are often situated in scenic, natural settings rich in flora and animals. These settings offer a plethora

Table 1 Concept and components of healing trails

Components	Description	References
Design for accessibility	Incorporates features for people with diverse physical abilities, ensuring inclusivity and access	[30]
Natural landscapes and biodiversity	Located in natural environments with diverse flora and fauna, offering sensory experiences	[31]
Rest areas and meditation points	Designated spaces for visitors to pause, practice mindfulness, and immerse in the surroundings	[32]
Interpretive signage and education	Provides information about nature, ecosystems, and health benefits, enhancing connection	[33]
Sensory elements	Incorporates features like wind chimes, fragrant plants, and textures to engage visitors' senses	[34]
Nature art and installations	Introduces artistic elements, sculptures, or installations that enhance aesthetic and engagement	[35]
Guided experiences	Facilitated walks led by trained guides, encouraging mindfulness and deeper nature connection	[32]
Ecotherapy and mental well-being	Supports mental health through nature immersion, reducing stress, anxiety, and promoting well-being	[33]
Physical activity and exercise	Encourages gentle physical activity such as walking, contributing to cardiovascular health	[23]

of sensory input, from sight to smell to touch to sound. Birdsong, running water, and native flora all help create a peaceful and rejuvenating environment [31].

(iii) **Rest and Meditation Points:** Along the route, there are a number of spots set aside as rest spaces or meditation spots. These areas encourage guests to take a moment for reflection, practice meditation, or simply unwind in the midst of breathtaking scenery.

(iv) **Interpretive Signage and Education:** Many healing trails have educational and interpretive signage to inform visitors about the ecosystems they're visiting and the possible health benefits of spending time outdoors. This improves the experience as a whole by encouraging a more intimate bond with one's environment [32].

(v) **Sensory Elements:** Healing pathways sometimes include wind chimes, scented plants, and textured surfaces to stimulate the senses of visitors and promote a meditative state of mind. These factors aid in relieving tension and encouraging calm [34].

(vi) **Nature Arts and Installations:** Healing pathways can incorporate artistic installations, sculptures, and other forms of creative expression into nature. These additions improve the therapeutic atmosphere by providing visual stimulation and room for individual expression [29].

(vii) **Guided Experiences:** Mindful Nature Interaction is facilitated on some healing trails through the use of guided experiences conducted by qualified facilitators. Deep breathing, exploring your senses, and thinking about the world around you are all possible on these walks [32].

(viii) **Ecotherapy and Mental Well-Being:** Healing trails are intended to promote psychological wellness as a form of ecotherapy. Spending time in natural settings has been shown to alleviate stress, depression, and anxiety. People in need of peace and restoration can find it on a healing trail [33].

(ix) **Physical Activity:** Walking or light trekking on a healing trail is a great way to get some exercise and improve your cardiovascular health, muscle strength, and general fitness. Different levels of difficulty on the paths could be an intentional design choice.

(x) **Social Interaction and Community Engagement:** Opportunities for social contact and community involvement can be found in healing paths. Group walks, outdoor workshops, and events allow people who are interested in the same things to meet each other [23, 37].

2.3 Historical Perspectives of Healing Trails

The word "healing trails" itself may not have been regularly used in all periods and civilizations, but the idea of healing pathways has historical roots. The basic idea of using natural environments for therapeutic purposes has been around for a long time, even if the language and specific practices have changed through time. Thus, integrating indigenous medicinal plant walks into the Healing Trails concept draws on a long heritage of people using the outdoors to improve their health and well-being. Cultures all around the world have historically depended on their understanding of herbal remedies to treat a wide range of illnesses and boost general health. Humanity's continuing connection with nature healing is illuminated via the past accounts of historical significance and traditional medical value linked with Healing Trails.

The Egyptians, the Greeks, and the Chinese all left behind written documents about the medical uses of plants. Ancient Egyptians widely used herbal medicine for the prevention and treatment of illness [38]. Traditional Chinese medicine, which dates back thousands of years, takes a similar holistic approach by focusing on restoring the body's natural energy balance through the use of herbs [39]. Hippocrates, the ancient Greek physician frequently called the "father of medicine," promoted the therapeutic application of medicinal plants as a means of treating illness and promoting wellness [40, 41].

Medicinal plant knowledge is an important part of indigenous tribes' history and identity, and it has been handed down orally from generation to generation. For example, for generations, Native American communities have relied on plant-based medicines thanks to a thorough grasp of the plants' therapeutic characteristics [42]. Like their Asian and South American counterparts, indigenous physicians in African cultures have long used their familiarity with native plants to aid in physical recovery and mental reorientation [43–45].

The following eras illustrate the development of healing trails:

(i) **Ancient Cultures and Nature Retreats:** Recognizing the restorative and revitalizing advantages of time spent in natural environments, humans have been taking use of them since ancient times. For example, the ancient Greeks built special hospitals called "Asclepia" in honor of the deity Asclepius, where people may go to be healed on all levels of being [46]. The tranquilly of these havens was aided by the presence of things like beautiful gardens and calming water features.

(ii) **Japanese Tradition of Forest Bathing:** The Japanese practice of "forest bathing" is emblematic of the country's long-held belief in the curative powers of nature. Since the 1980s, Japan has actively pushed "Shinrin-yoku," also known as "forest bathing," in which people spend time in a forested setting. Ancient Shinto and Buddhist teachings that honor the spiritual link between humans and nature are the inspiration for this practice [26].

(iii) **Nineteenth-Century Nature Cure Movement:** The "nature treatment" movement of the nineteenth century gained traction in both the Old World and the New World [47]. The proponents of nature cure held that exposure to natural elements, such as the outdoors, might have beneficial health effects. By encouraging exercise, fresh air, and connection with nature, healing trails and sanatoriums aim to facilitate complete health and wellness [48].

(iv) **Growth of National Parks and Recreation Areas:** Many countries began creating national parks and other protected areas in the late-nineteenth and early twentieth centuries. The purpose of these preserved green areas was to offer people a place to unwind, recharge, and enjoy nature. In order to promote the health of both visitors' bodies and minds, trails were built inside these locations [49].

(v) **Modern Ecotherapy and Mindfulness Practices:** In recent decades, the idea of healing trails within the context of ecotherapy and mindfulness practices has received increased interest. Spending time in natural environments has been studied by many including Zhu et al. [50], Bratman et al. [51], and Pretty et al. [52] in the mental health and research fields for its potential to alleviate stress, anxiety, and depression. Individuals can take time for reflection and contemplation on trails created with an emphasis on mindfulness and sensory sensations.

(vi) **Contemporary Wellness and Green Infrastructure:** As wellness and environmentally responsible city planning have gained prominence in recent years, so too have healing trails become a standard feature of today's green infrastructure. In an effort to improve residents' health and well-being, as well as foster a stronger sense of community, several cities are including nature paths and green spaces into their urban development. Even in highly crowded places, these pathways provide city inhabitants with access to nature and its restorative benefits [53].

2.4 Indigenous Medicinal Practices in Healing Trails

Indigenous healing practices are a big part of what makes Healing Trails what they are. They give these recreational areas a sense of cultural history, indigenous wisdom, and an in-depth knowledge of how medicinal plants heal. Indigenous people all over the world have used their close relationship with nature to find and use plants to treat different illnesses and improve their general health for hundreds of years. Integrating native healing practices into Healing Trails not merely shows respect for the knowledge of these communities, but it also gives tourists an unforgettable experience that helps them understand how nature and health are connected.

Indigenous people in places like South America have a deep understanding of medicinal plants that have been passed down orally and is used by shamans and other traditional physicians [54, 55]. For example, the Amazon jungle is a storehouse of biodiversity that is home to a wide range of plants that have been used by native tribes for healing [56]. By adding these herbal remedy walks to Healing Trails, visitors can learn from the deep knowledge of indigenous people. They can learn about their traditional ways of healing and what these plants mean to them culturally.

Native American tribes in North America have used plants as medicine and for spiritual reasons for a long time. For example, the Cherokee people have a long history of using different plants to treat sicknesses and keep themselves healthy [57]. By adding traditional Cherokee herbal remedy walks to Healing Trails, the cultural history and indigenous understanding of these native communities are kept alive and shared with a wider audience.

Indigenous healing practices in Healing Trails also show how important it is to harvest and save resources in a way that doesn't hurt the environment [58, 59]. Voeks and Leony [60] observed that many indigenous groups have come up with ways to harvest plants in a way that protects medicinal plant species and keeps these

Table 2 Some indigenous medicinal practices and healing trail components

Indigenous medicinal practices	Healing trail components	References
Amazon rainforest medicinal plants	Medicinal plant walks, traditional healing	[61]
Cherokee herbal tradition	Medicinal plant walks, cultural heritage	[57]
Maori Rongoā (New Zealand)	Medicinal plant walks, traditional knowledge	[62]
Ayurvedic medicine (India)	Medicinal plant walks, holistic healing	[63, 64]
Traditional Chinese medicine	Medicinal plant walks, energy balance	[65]
Inuit healing traditions (Arctic)	Medicinal plant walks, spiritual practices	[66]
Sami traditional medicine (Scandinavia)	Medicinal plant walks, reindeer herding	[67, 68]
Navajo healing practices (USA)	Medicinal plant walks, sacred landscapes	[69]
Hawaiian Lāʻau Lapaʻau	Medicinal plant walks, cultural heritage	[70]
Ainu traditional medicine (Japan)	Medicinal plant walks, cultural revival	[71]

resources around for a long time. Healing Trails can help protect medicinal plants and protect culture and biodiversity by working with indigenous groups and learning from how they do things.

Thus, healing trails celebrate the cultural history and indigenous knowledge of communities by incorporating their healing practices. This gives visitors a deep experience of healing and connection with nature. Healing Trails become places of cultural appreciation, natural conservation, and whole-person well-being when they include medicinal plant walks based on indigenous knowledge (Table 2).

2.5 The Importance of Healing Trails in Safekeeping Historical Relics

Healing trails are meant to help people feel better about themselves as a whole and get closer to nature. These places also have important historical relics, artifacts, or ancient sites that need to be protected (Table 3). To safeguard the rich tapestry of human culture and history and to give visitors a valuable and educational experience, it is important to keep these historical artifacts safe in Healing Trails.

Along Healing Trails, you might find old buildings, artifacts made by native people, or traces of past cultures. These items tell us a lot about how our ancestors lived, what they believed in, and what kind of societies they lived in [82]. Healing Trails are becoming more popular as places to go for fun. This means that they are getting more and more users, so it is important to take steps to protect these relics from being accidentally damaged or worn down.

By adding historical artifacts to Healing Trails, there is a chance to teach about and protect cultural history through sustainable tourism. Signs and interpretive installations can be put in a way that tells visitors about the relics' historical importance and helps them understand their cultural worth and significance [83]. Also, barring direct entry to delicate regions can keep tourists from accidentally hurting the artifacts while still letting them understand their historical significance.

Table 3 Examples of historical sites and relics that benefit from healing trails

Historical site or relic	Country	References
Angkor Wat	Cambodia	[72]
Petra	Jordan	[15]
Acropolis of Athens	Greece	[57]
Machu Picchu	Peru	[73]
Chaco Culture National Historical Park	USA	[74]
The Great Wall of China	China	[39]
Ayutthaya Historical Park	Thailand	[15]
Stonehenge	United Kingdom	[75]
Teotihuacan	Mexico	[76]
Great Zimbabwe	Zimbabwe	[77]
Palmyra	Syria	[78]
Borobudur	Indonesia	[79]
Ephesus historical sanctuary	Turkey	[80]
Stone circles of Senegambia and the Gambia	Senegal and the Gambia	[81]

For the safekeeping of historical artifacts, it is important to work with community members, archaeologists, and antique experts. Engaging these groups can help create management plans that meet the needs of both leisure and preservation [84]. Regular monitoring and care, based on the advice of experts, can make sure that the relics stay in good shape and can be seen by future generations.

The Australia International Council on Monuments and Sites (ICOMOS) and para llevar a cabo Estudios report of 1999 says that the growth of Healing Trails can be guided by ethical principles like those in the ICOMOS International Cultural Tourism Charter [85]. These principles can help protect the integrity of historical artifacts. Healing Trails can help promote cultural respect, ecological responsibility, and sustainable tourism by finding a compromise between leisurely pursuits and cultural preservation. In summary, keeping historical artifacts safe in Healing Trails is an important part of preserving cultural history and being a good tourist.

2.6 Benefits of Cultural Integration of Healing Trails and Recreational Development

Healing Trails' cultural incorporation into recreational planning and development provides numerous, substantial benefits. Healing Trails are more than just trails; they develop into places for cultural festivities, learning, and conservation efforts as they incorporate indigenous knowledge and practices. Some major advantages are as follows:

(a) **Preservation of Cultural Heritage**

Traditional indigenous knowledge, customs, and practices can be preserved by the incorporation of cultural aspects into Healing Trails. Healing Trails ensure that cultural traditions are passed on from one generation to the next by serving as a showcase for such elements [72].

(b) **Improved Service for Tourists**

Incorporating local culture enhances the atmosphere and makes guests feel like they have stepped back in time. Recreating in a place where you have a deeper understanding of its cultural value through the study of its herbal remedies, traditional healing practices, and historical artifacts is a rewarding experience [57].

(c) **Participatory Democracy and Local Autonomy**

Planning and developing Healing Trails in collaboration with community members and indigenous groups gives them a voice in the preservation of their culture. Environmentally friendly and responsible tourist practices are encouraged, and a sense of ownership, as well as satisfaction, is fostered [15].

(d) **Environmental Protection**

The preservation of natural resources and ecosystems is a central theme in many culturally integrated societies. Healing Trails promote environmental stewardship and appropriate use of land, leading to long-term conservation efforts [86] by emphasizing the cultural relevance of the terrain and its biodiversity.

(e) **Education and Awareness**

Visitors to healing trails that make use of cultural features are usually able to learn about other people's customs, history, and healing methods. As such, integrating healing trails with cultural integration encourages people to learn about other people's cultures and appreciate them [87]. In a nutshell, there are several benefits for incorporating Healing Trails' cultural aspects into recreational design and development. Together, people, nature, and cultural diversity are celebrated in these holistic and meaningful spaces created through the integration of paths that serve several purposes, such as the preservation of cultural heritage, the promotion of environmental protection, and the enrichment of visitor experiences.

3 Healing Trails and Mental Health

Integrating therapeutic experiences in nature into Healing Trails has shown promise as a method of improving people's emotional and psychological well-being. Healing Trails are frequently built using biophilic design concepts, which involve the intentional use of natural materials and designs to foster feelings of well-being and oneness with the outdoors. The incorporation of the biophilic design helps to alleviate stress and boost cognitive performance [88]. Also, rhythms in nature, such as the trickle of water or the rustle of leaves, can have a relaxing effect on people, and Healing Trails often lead their visitors through a variety of natural settings. These ambient noises have been shown to reduce stress and promote calmness [89]. Other benefits include the following:

(a) **Reducing Stress and Improving Relaxation**

People can get away from the hustle and bustle of everyday life and recuperate their lives on a Healing Trail. Because of the increased stress levels caused by urbanization and frequent exposure to technology, therapies rooted in nature are gaining popularity. Studies have shown that visiting a Healing Trail might help you unwind and forget about your worries. The stress hormone cortisol was significantly reduced after exposure to natural habitats like those found on Healing Trails, according to research by Hartig et al. [15]. We can find the tranquilly and peace that in nature make it the perfect place to forget about the past and focus on the here and now.

(b) **Enhanced Happiness and General Emotional Health**

Participation in Healing Trails has been shown to improve mental and emotional health. Endorphins are neurotransmitters that are released in response to physical activity, and these trails are a great place to get some exercise and feel good about life. The peace and splendor of nature have the power to inspire awe and astonishment and boost the spirit. Joye et al. (2020) and Hartig et al. [15] found that spending time in natural settings boosted good emotions and dampened negative ones. These results demonstrate the therapeutic value of Healing Trails for fostering emotional health and optimism.

(c) **Improved Brain Power**

Healing Trails emphasizes the ways in which contact with nature can improve not just physical health but also mental acuity. Directed attention fatigue, brought on by the constant demand placed on one's brain in a city, can be alleviated through time spent in a more natural setting. Kaplan's [90] Attention Restoration Theory (ART) suggests that spending time in natural settings has positive effects on brain function, such as restoring focus and sparking inspiration. Healing Trails provide a respite from the stresses of daily life, restoring one's ability to think clearly and concentrate.

(d) **Decreased Worry and Depression**

Anxiety and despair can be greatly alleviated with the help of Healing Trails. Healing Trails' peaceful setting and proximity to nature are ideal for relieving stress and emotional distress. Participants in the study by Bratman et al. [91] who went on a 90-min walk in the woods reported significantly reduced ruminating, an important risk factor for depression. Individuals with mental health issues can find comfort and emotional relief through the healing power of nature through the Healing Trails' immersive experience.

(e) **Enhanced Resilience and Coping**

The time spent in nature healing trails strengthens emotional fortitude and enables more effective coping. Having access to natural areas that provide solace and support can help make life's hardships more manageable. Inspiring hope and a sense of rebirth in the face of adversity, nature's dynamic and adaptable qualities serve as a potent reminder of life's cyclical nature [92].

(f) **Social Connection and Support**

Nature healing trails makes it easier to connect with others, which is important for our mental health. Outdoor activities are a great way to spend quality time with loved ones and build memories that will last a lifetime. Nature-based interactions can boost the sense of connectivity between people, which is an effective defense against anxiety, depression, and stressful situations [93].

(g) **Mindfulness and Presence**

Being outside is a great way to practice mindfulness and live in the here and now. Individuals can take a mental vacation from thinking about their past or their future by taking in the sights, sounds, and fragrances of their natural surroundings as they move along the healing trails [92]. Research has found a correlation between mindfulness and improved mental and emotional health as these methods can help people feel less anxious and more at ease [94].

3.1 The Calming or Mood-Lifting Effects of Healing Trails

Medicinal plants have been used into Healing Trails and ecotherapy practices for their relaxing and mood-lifting benefits. These plants (Table 4) have natural therapeutic characteristics that help reduce stress, ease anxiety, and boost mood. Specific medicinal herbs that have been shown to have a sedative or mood-enhancing effect include as follows:

One of the most well-known plants for its sedative effects is lavender (*Lavandula* spp.). The calming and comforting odor of this plant has been studied [110]. Visitors to Healing Trails will find a relaxing atmosphere thanks to the abundance of lavender plants.

The relaxing and tranquil qualities of chamomile (*Matricaria chamomilla*) make it a popular herbal remedy. Anxiety symptoms are alleviated and sleep quality is enhanced, according to research [97]. A visit to Healing Trails is sure to calm and relax you thanks to the calming chamomile aroma that permeates the area.

Table 4 Calming and mood-lifting effects of some medicinal plants

Plant	Calming or mood-lifting effects	References
Lavender (*Lavandula* spp.)	Anxiety reduction, relaxation, improved sleep quality	[95, 96]
Chamomile (*Matricaria chamomilla*)	Anxiety reduction, improved sleep quality	[97]
St. John's Wort (*Hypericum perforatum*)	Potential mood improvement	[98]
Passionflower (*Passiflora incarnata*)	Anxiety reduction, relaxation	[99]
Lemon Balm (*Melissa officinalis*)	Uplifting, potential mood enhancement	[100]
Rose (*Rosa* spp.)	Stress reduction, mood improvement	[101]
Peppermint plant	Uplifting, increased alertness, mental clarity	[102]
Bergamot (*Citrus bergamia*)	Anxiety reduction, relaxation	[103]
Valerian (*Valeriana officinalis*)	Relaxation, potential anxiety reduction	[98]
Frankincense (*Boswellia* spp.)	Stress reduction, potential mood enhancement	[104]
Vetiver (*Vetiveria zizanioides*)	Calming, relaxation	[105]
Clary Sage (*Salvia sclarea*)	Potential mood enhancement, relaxation	[106]
Sandalwood (*Santalum* spp.)	Relaxation, potential anxiety reduction	[107]
Ylang Ylang (*Cananga odorata*)	Uplifting, relaxation	[101]
Patchouli (*Pogostemon cablin*)	Calming, relaxation	[101]
Geranium (*Pelargonium* spp.)	Uplifting, potential mood enhancement	[108]
Cedarwood (*Cedrus* spp.)	Relaxation, potential anxiety reduction	[109]
Marjoram (*Origanum majorana*)	Calming, relaxation	[101]
Neroli (*Citrus aurantium*)	Uplifting, potential anxiety reduction	[101]

Melissa officinalis (Lemon Balm) is a plant that has been used for centuries for its calming effects. Historically, it has been employed in the treatment of depression and anxiety [100]. Adding lemon balm to Healing Trails has been shown to increase feelings of happiness and calm.

Rose (*Rosa* spp.) fragrance has been shown to reduce stress and boost mood. Hongratanaworakit [101] found that inhaling rose essential oil reduced anxiety and increased calmness. The healing and restorative atmosphere of Healing Trails is enhanced by the roses that bloom there.

Citrus bergamia (Bergamot) essential oil has been shown to reduce anxiety and boost mood. Alkofahi and Atta [103] found that bergamot aromatherapy reduced both heart rate and blood pressure, producing a calming effect. Visitors to Healing Trails with bergamot should expect to feel revitalized and refreshed.

Cananga odorata (Ylang Ylang) is a flowering plant whose essential oil has been used for centuries for its calming and sedative properties. Psychological stress responses can be mitigated through the use of ylang ylang aromatherapy [101]. Ylang ylang is a pleasant addition to Healing Trails, helping visitors unwind and regain equilibrium.

Vetiver (*Vetiveria zizanioides*) has been used for centuries for its relaxing and grounding effects. The aroma has been shown to have calming effects on the mind and body [105]. Adding vetiver to Healing Trails can help people feel more grounded and relaxed.

Salvia sclarea, or clary sage, is also on the list because of its calming and mood-boosting effects. The stress hormone cortisol is lowered by inhaling clary sage essential oil [106]. Clary sage, which is used in Healing Trails, has been shown to improve mental health.

Frankincense (*Boswellia* spp.) also has a calming and inspiring properties. The scent of this plant has been shown to calm anxious feelings and boost mood [104]. When visiting Healing Trails, you may find the presence of frankincense to be enlightening to your soul.

Patchouli (*Pogostemon cablin*) is also considered beneficial because of its sedative and mood-balancing properties. It has been found to have calming effects simply through its scent [101]. Adding patchouli to Healing Trails is a great way to calm your mind and restore your equilibrium.

4 Urban Green Spaces and Healing Trails

The decline in urban greenery has contributed to a host of health problems experienced by city dwellers. However, a promising option to improve city dwellers' well-being is to incorporate Healing Trails, replete with medicinal plant walks, into urban green spaces. When it comes to counteracting the drawbacks of city life, urban green spaces are indispensable. As urban areas continue to grow, people's opportunities to connect with the environment are dwindling. Daily inhabitants can get away from the hustle and bustle of daily life and enjoy some time in nature by visiting one of the city's many green parks. Stress is reduced, air quality is enhanced, and community spirit is strengthened in these settings [111]. But having parks and other green areas may not be enough to improve people's quality of life in cities. The impact on public health is heavily dependent on the planning and function of these areas.

The goal of creating Healing Trails is to use the therapeutic effects of nature on the body, mind, and spirit (Fig. 1). The incorporation of Healing Trails into urban greenery provides inhabitants with a rare opportunity to reap the therapeutic advantages of nature in a curated and organized setting. There are usually designated spots along these routes where you can find medicinal herbs. Physical exercise, time spent in nature, and interaction with medicinal plants all work together to provide an integrative experience that benefits the health of city inhabitants.

4.1 Physical Health Benefits

Healing Trails make it simple for city inhabitants to fit exercise into their schedules by connecting parks and other green areas. There are likely medicinal plants that aid with respiratory health all along these pathways. Plants like eucalyptus have been

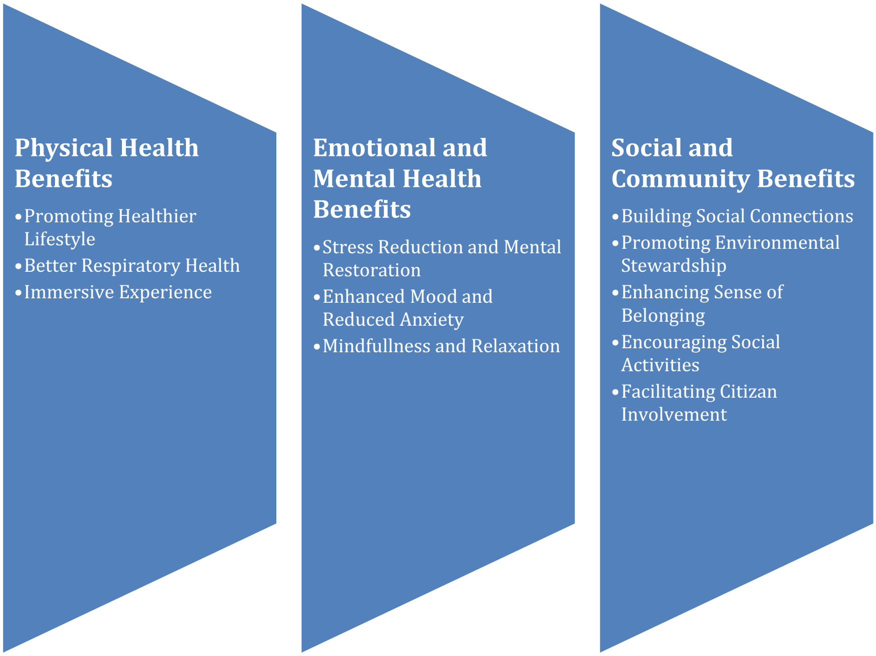

Fig. 1 Benefits of Healing Trails in urban green spaces

shown to improve lung function and reduce respiratory distress [112]. As a result, Healing Trails provide a potent tool for improving physical health and mitigating the negative consequences of city life. These include the following:

(a) **Promoting Healthier Lifestyles**

Walking, jogging, and hiking are just some of the physical activities that may be enjoyed in urban green spaces with healing trails. These routines not only improve cardiovascular health, but they also help prevent serious conditions like diabetes, heart disease, and obesity [113]. Urban green spaces provide easy access to exercise areas in visually pleasing surroundings, which is especially important for the health and happiness of city people. Healing trails motivate people to get out and about in these green areas, turning exercise into something people look forward to doing on a regular basis.

(b) **Immersive Experience**

For those who need a break from the city's typical concrete surrounds, healing pathways in urban green spaces provide an immersive experience in nature. Research has found that spending time in nature can improve health in several ways, including lowering blood pressure, decreasing cortisol levels, and boosting the immune system [114]. Humans have what is called a biophilic link or an inbuilt preference for natural settings that has been shown to improve health outcomes. Nature has a stress-relieving and mood-boosting effect because of its relaxing and rejuvenating qualities. The incorporation of healing paths into urban green spaces allows city inhabitants to find peace and quiet away from the chaos of the city.

(c) **Better Respiratory Health**

People who live in cities can benefit from medicinal plant walks since they are exposed to phytochemicals generated by different plants. For example, studies have linked eucalyptus with reduced respiratory symptoms [112]. Air pollution is a common problem in urban areas and can have harmful effects on people's respiratory systems. The negative effects of urban air pollution on the respiratory system can be mitigated by the existence of healing trails including specialized medicinal plants, such as eucalyptus. During medicinal plant walks, inhaling the plants' natural aromas may help clear airways, lessen respiratory irritation, and improve lung function.

4.2 Emotional and Mental Health Benefits

Research has linked time spent in green spaces and interacting with nature to lower stress levels, higher levels of happiness, and better memory and reasoning skills [91]. In contrast to the hectic pace of the city, Healing Trails offer a calm and peaceful setting ideal for introspection and rejuvenation. The therapeutic value of these treks is enhanced by the inclusion of interactions with medicinal plants. Lavender and chamomile, for example, are commonly used for their sedative and anti-anxiety properties [97]. The therapeutic plant aromas combined with the tranquil setting produce a potent environment for relieving stress and restoring emotional balance. Other emotional and mental health benefits include the following:

(a) **Stress Reduction and Mental Restoration**

Opportunities for relaxation and psychological recovery are made possible by the combination of urban green spaces with healing pathways and medicinal plant walks. Spending time in nature has been demonstrated to boost concentration and memory [94]. Access to green spaces provides a natural refuge for mental renewal, especially in metropolitan environments where the flow of life is often fast and stress levels are high. Healing trails help people relax, boost their mood, and regain concentration by combining time in nature with vigorous exercise.

(b) **Enhanced Mood and Reduced Anxiety**

Spending time in green places and interacting with nature has been shown to have a good effect on mood and alleviate anxiety and depression symptoms. Happiness and life satisfaction have both been shown to increase with time spent in nature [91]. Some of the medicinal plants found along healing paths have been shown to have mood-boosting effects, adding another layer to the overall experience. Lavender, for instance, has been linked to calmness and relaxation, while lemon balm is renowned for its energizing properties. Visitors can enjoy psychological wellness through their connection to nature and these plants in particular at urban green spaces that feature medicinal plant walks.

(c) **Mindfulness and Relaxation**

Healing pathways in urban green spaces provide a peaceful backdrop for mindfulness and relaxation exercises. It has been shown that interacting with medicinal herbs while meditation increases feelings of peace and relaxation [115]. Being in a beautiful natural setting can help with mindfulness practices, which focus on the present moment. Participants in medicinal plant walks often report feeling more relaxed and at peace after experiencing the sights, sounds, and scents of nature. This focused state helps you unwind, lowers your tension, and boosts your mood.

4.3 Social and Community Benefits

Urban green spaces linked with healing trails offer significant social and communal benefits in addition to physical and mental health benefits. These efforts may help city inhabitants feel more connected to one another, expand their social circles, and take better care of their local environments.

(a) **Building Social Connections**

Urban parks with healing trails can be community hubs where people can meet and get to know one another. Group walks and walks in search of medicinal plants are just two examples of the kinds of outdoor activities that bring people of all ages and walks of life together. This is because people are more likely to talk to one another and share ideas after participating in group activities [116]. Urbanites who take advantage of healing trails together strengthen their social networks through a shared experience of nature's restorative power.

(b) **Promoting Environmental Stewardship**

Incorporating medicinal plant walks into urban green spaces helps teach the community the value of safeguarding and preserving indigenous plants and ecosystems, which in turn promotes environmental stewardship. The medicinal value, ecological importance, and traditional uses of the plants along healing paths are often explained in interpretive materials and educational signage [117]. As city inhabitants gain knowledge of the plant's medicinal and cultural significance, they have a more meaningful relationship with nature and become more aware of the importance of preserving it. This increased environmental stewardship has the potential to go beyond parks and plazas to inspire sustainable lifestyle choices and citywide conservation drives.

(c) **Enhancing Sense of Belonging**

Urban parks with healing trails provide a shared space where urban people of all backgrounds and socioeconomic statuses can come together and strengthen their sense of community. By providing a place where everyone feels welcome, these places help residents develop a common sense of self [118]. The existence of healing trails increases people's awareness of the value of parks in urban areas and the therapeutic effects of spending time in nature. When people feel a connection to a therapeutic space, they become invested in keeping it healthy for the sake of the whole community.

(d) **Encouraging Social Activities**

Healing paths in urban green spaces make it possible for people to gather for a variety of group pursuits. Group activities, workshops, and educational programs about medicinal plant walks and nature-based therapy can be organized by collaborating community organizations, schools, and local businesses [119]. These events encourage participation from all individuals and bring together groups with varying backgrounds for dialogue and education. When people come together, the healing pathways blossom into bustling centers of community life that enhance the cityscape.

(e) **Facilitating Citizen Involvement**

Neighborhoods can take an active role in the development and upkeep of their urban green spaces by incorporating healing pathways and medicinal plant walks. Residents have the opportunity to contribute their ideas and experience to the planning, design, and implementation stages of these projects [120]. When people see how their efforts improve the lives of themselves and their neighbors, they feel a greater sense of pride in the green places in their community.

5 Complementary Medicine and Healing Trails

Complementary medicine, also known as holistic healthcare, is an umbrella term for a wide variety of non-Western approaches to health and wellness. Complementary medicine, which includes everything from acupuncture and herbal medicine to Ayurveda and Traditional Chinese Medicine, is an approach to health and wellness that takes into account the whole person rather than just treating symptoms [121, 122]. Complementary medicine relies heavily on the therapeutic use of medicinal plants, which highlights the powerful healing capabilities present in the natural world [123–128].

The value of healing trails along medicinal plants as part of complementary medicine practices has been increasingly acknowledged in recent years [129–132]. To learn about the identification, characteristics, and traditional usage of many medicinal plants, practitioners and participants go on medicinal plant walks, which are guided trips in natural environments [133]. Participants can interact with the plants up close and personal, smelling, touching, and even tasting them on these walks.

When complementary medicine, like healing trails, is used with conventional treatment, a synergistic effect is created that benefits both the patient and the healthcare provider [134]. These integrative therapeutic approaches provide a more thorough understanding of plant-based treatments and their uses by incorporating knowledge of herbal remedies from many cultural and historical viewpoints.

There is a rising curiosity in investigating natural and alternative methods of health and well-being as the globe struggles under an increasing load of serious illnesses and mental health difficulties. To achieve a more compassionate and inclusive healthcare paradigm, complementary medicine's emphasis on holistic care and individualized therapies is a good fit [122]. Practitioners and patients alike can benefit from a deeper understanding of nature, the healing power of plants, and self-care by participating in medicinal plant walks as part of the range of complementary medicine practices.

Besides helping healthcare professionals, healing trails encourage patient agency in the healing process. Insights on the safe and efficient use of natural treatments for minor medical conditions and preventive care are gained through medicinal plant walks [121]. By giving people more control over their own lives, we may increase their affinity with the natural world and inspire them to take action to protect medicinal plants and the habitats in which they thrive.

5.1 Benefits of Integrating Healing Trails in Complementary Medicine

(a) **Diversity of Healing Traditions**

When we talk about complementary medicine, we're talking about a vast range of healing modalities from all around the world. Practitioners and patients alike can benefit from medicinal plant walks of healing trails by learning about the wide range of plant usage in different healing modalities, such as TCM, Ayurveda, indigenous medicine, and European herbalism [122]. This sharing of ideas across cultures deepens our respect for traditional wisdom and creates a diverse tapestry of therapeutic approaches.

(b) **Personalized and Holistic Healing**

Healing that is both individualized and holistic can be achieved via the use of healing trails or medicinal plant walks and other forms of integrative medicine. Plants and cures are chosen with careful consideration of the individual's needs and physical make-up [135]. Practitioners gain further insight into their patients' emotional and mental health through the experiential side of medicinal plant walks, in which they may see how patients connect with plants and nature. This all-encompassing assessment helps create a more thorough, patient-specific therapy strategy.

(c) **Self-Care and Empowerment**

Medicinal plant walks encourage people to engage in self-care activities that improve their health and well-being. It has been shown that education on medicinal plants and their therapeutic capabilities can increase people's comfort and success with utilizing such treatments for common ailments and as a kind of preventative medicine [121]. This sense of control helps people feel more attuned to the natural world and more in charge of their own recovery.

(d) **Awareness and Conservation of the Environment**

The use of healing trails or medicinal plant walks as part of integrative healing raise awareness of and appreciation for the natural world. When people learn about medicinal plants and their use in healing, they gain a greater awareness of the interdependence of all living things in the natural world [136]. Increased environmental awareness promotes eco-friendly herb collection and responsible use, ensuring medicinal plants will be around for future generations to benefit from (Fig. 2).

5.2 Challenges and Considerations for Healing Trails and Complimentary Medicine

(a) **Ethical Harvesting and Sustainability:**

The use of medicinal plant trail walks for integrated treatment poses ethical harvesting and sustainability issues. Over harvesting and the extinction of some plant species may result from the rising demand for therapeutic herbs [137]. To guarantee the ethical use of plants, practitioners and participants must prioritize ethical practices such as wild-crafting rules and supporting cultivation efforts. The Forest Stewardship Council (FSC), for instance, publishes recommendations for responsible plant resource management and advocates for sustainable wild harvesting [138].

(b) **Safety and Knowledge of Practitioners:**

Practitioners using an integrative approach to therapy should be well-versed in the qualities and interactions of the plants they are working with to ensure the safety of their patients. When combining plant-based cures with other therapies or drugs, safety must be the top priority [139]. Integrative healing trail experiences can only be beneficial if practitioners have the knowledge and skills necessary to ensure the safety of their patients. Educators and Ayurvedic doctors have groups like the American Herbalists Guild (AHG) and the National Ayurvedic Medical Association (NAMA) to thank for their efforts to improve the field.

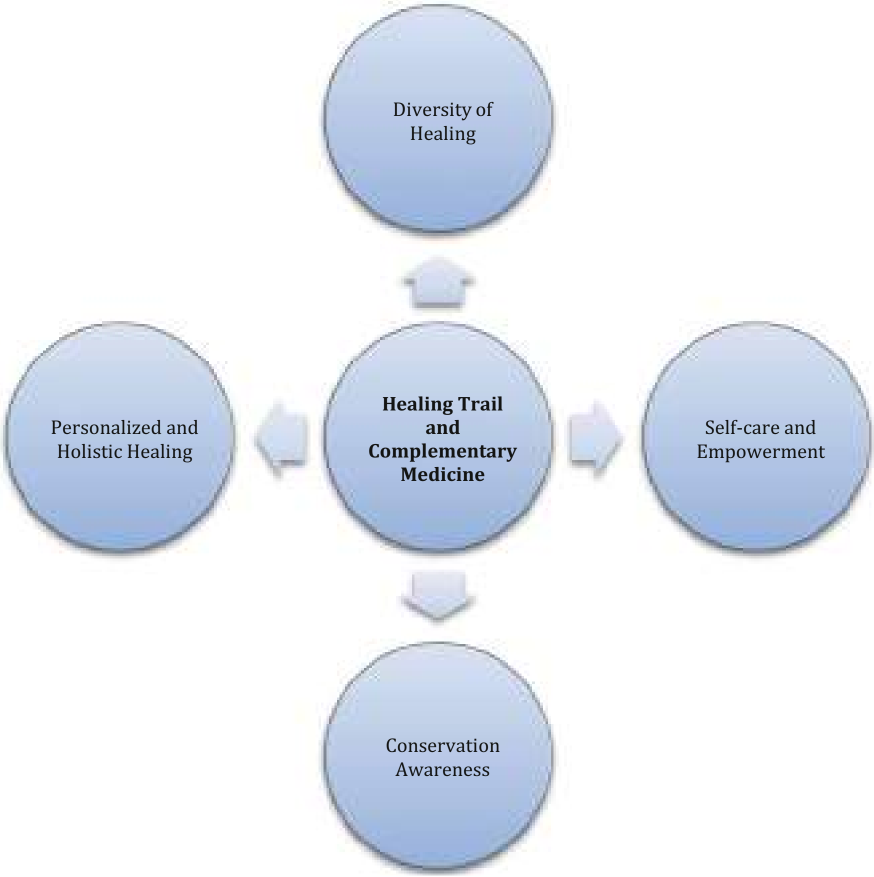

Fig. 2 Benefits of integrating healing trails in complementary medicine

(c) **Cultural Appropriation and Respect for Indigenous Knowledge:**

Given that medicinal plant trail walks rely on the expertise of indigenous and local people, it is important to recognize and address the potential for cultural appropriation [140]. Traditional healing knowledge should be used with an appreciation for and understanding of its historical context on the part of all practitioners and participants. Respect for and appreciation of indigenous elders' and healers' cultural practices and points of view can be achieved through collaboration and partnership [141]. In addition, the preservation and regeneration of traditional healing systems can benefit from appropriate recompense for the use of traditional knowledge and active efforts to promote Indigenous-led projects.

(d) **Evidence-Based Practices and Integration with Conventional Medicine:**

The integration of medicinal plant walks with complementary medicine practices and conventional medicine should be evidence-based, although traditional healing traditions provide vital insights into the usage of medicinal plants trail walks. Validating the safety and efficacy of plant-based medicines and informing their integration with traditional medical treatments requires research and clinical investigations [135]. To provide the most thorough and coordinated care for their patients, practitioners should work together with traditional medical specialists. Better integrated therapeutic practices can result from increased transparency and knowledge-sharing among various healthcare methods.

(e) **Individual Variability and Personalization:**

Because people have different reactions to plant-based treatments, an individualized strategy is necessary for effective integrative healing with medicinal plant trail walks. Individual differences in heredity, health problems, and allergies must be taken into account when prescribing herbal remedies [8] because not all plants are appropriate for all people. For optimal results from the use of medicinal plants, practitioners should conduct in-depth evaluations and consultations with patients.

6 Ecotourism and Sustainable Economic Development in Healing Trails

Ecotourism is described as "responsible travel to natural places that safeguard ecosystems and improves the well-being of local communities" [142]. When incorporated into ecotourism programs, healing trails and medicinal plant walks provide visitors with unique and enlightening experiences while also creating economic advantages for local communities. This section examines the possible benefits of healing trails and medicinal plant walks in ecotourism, focusing on their role in bolstering local economies and promoting long-term, sustainable growth.

6.1 The Impact on Neighborhood Economies

The creation of tourism earnings is one of the key economic benefits of healing trails and medicinal plant walks based on ecotourism. The opportunity to learn about plants' healing capabilities, forge meaningful connections with the natural world, and fully submerge oneself in restorative settings are some of the reasons why tourists flock to these special events [143]. Communities in proximity to healing trails might benefit from the resulting increase in tourism by providing services such as lodging, dining, and sightseeing to the growing number of visitors [44, 144–150]. Local economies benefit from these services because they help firms expand and new jobs are created.

(a) **Participation in Community Events and Cross-Cultural Dialogue**

Travelers and locals alike can learn from one another through ecotourism's use of healing trails and plant-based remedy hikes. By consulting with local guides and healers, tourists can acquire insight into the country's rich cultural history and traditional medical practices [151]. These kinds of exchanges improve everyone's capacity to appreciate and understand one another, which in turn boosts the ecotourism experience as a whole and makes visitors more likely to extend their stay and spend money in the local economy. The local community can also reap financial rewards from ecotourism by providing services such as guiding therapeutic plant walks, selling handicrafts, and imparting traditional knowledge.

(b) **Environmental Stewardship and Conservation Efforts**

Healing trails and therapeutic plant walks that are part of ecotourism help spread awareness about the importance of protecting natural resources. Communities have an interest in preserving ecological resources because efforts like these depend on the availability of natural habitats and therapeutic plants [152]. For ecotourism to thrive along the healing route into the future, residents must take an active role in conservation efforts and prudent land use [153]. This dedication to environmental conservation helps to preserve biodiversity and natural habitats, which in turn helps to secure the future prosperity of ecotourism.

(c) **Specialized Tourism and Market Diversification**

Visitors interested in alternative medicine and folklore are the target market for healing trails and medicinal plant walks, both of which fall under the umbrella of niche tourism. By highlighting these one-of-a-kind activities, towns may increase the variety of their tourism offers and appeal to a wider spectrum of visitors [154]. It is especially important for less popular tourist locations to diversify their visitor base. Communities can strengthen their economies and become less dependent on a select few sectors if they focus on satisfying the needs of specific subsets of the population.

(d) **The Renewal of Traditional Wisdom**

By combining ecotourism with the study of medicinal plants, traditional knowledge can be revitalized and preserved [155]. Tourists can learn from the experience and knowledge of local physicians and traditional healthcare providers by joining one of these walks [156]. Reviving cultural practices and instilling a sense of pride in the local population are two outcomes of recognizing and promoting indigenous knowledge systems. Additional financial opportunities may arise as a result of cooperation and partnerships with institutions of higher learning and research organizations made possible by such recognition.

6.2 Challenges and Considerations

While there is potential for substantial financial gain from ecotourism that centers on healing trails and medicinal plant walks, there are a number of obstacles that must be overcome to achieve long-term financial health. These include the following:

(a) **Responsible Tourism Management**

Ecotourism must be managed and regulated properly to prevent overcrowding and reduce the strain on natural and cultural resources [157]. To prevent damage to healing pathways and the ecosystems that surround them, sustainable tourism practices including visitor carrying capacity and trash management should be put into place.

(b) **Community Empowerment and Benefit Sharing**

Benefits and responsibilities from ecotourism projects should be shared with local populations, according to a report by Jacob and Ogogo [158] and Ukpong and Jacob [159, 160]. Economic advantages from ecotourism should be allocated equally among community members through established benefit-sharing systems to reduce wealth gaps and increase community cohesion.

(c) **Cultural Sensitivity and Respect**

Activities related to tourism should be carried out with sensitivity to and respect for local culture [154]. Positive ties between locals and tourists can be maintained by urging visitors to observe local laws and customs.

(d) **Environmental Conservation and Biodiversity Protection**

It is imperative that conservation efforts be prioritized in order to protect the natural ecosystems and medicinal plants that form the basis of healing pathways [152]. Sustainable ecotourism relies on the active participation of local populations in ecological restoration and biodiversity protection activities.

7 Healing Trails for Specific Health Conditions

There is growing evidence that suggests these methods can help with the treatment of a variety of health problems, including those connected to the respiratory system, the skin, and stress. Therapeutic advantages from nature can be maximized by creating specialized healing trails and therapeutic plant walks for those with various diseases. Some of the ailments include the following:

7.1 Respiratory Health Challenges

(a) **Asthma**

People who suffer from asthma should seek out healing paths in areas with low concentrations of air pollution, allergens, and irritants. The clean air found in forest and coastal locations is especially helpful for people with breathing problems [161]. Shinrin-yoku, or forest bathing, has been shown to help people with asthma by lowering their stress and inflammatory levels and enhancing their lung function [162]. Certain trees, such as conifers, have been proven to have therapeutic potential due to the essential oils they exude [163] for lowering bronchial inflammation and alleviating respiratory symptoms.

(b) **Chronic Obstructive Pulmonary Disease (COPD)**

Healing trails for people with COPD typically include lung- and heart-health boosting exercises. Moderate activity, including walking and climbing, can boost respiratory muscle strength and general fitness, and trails with varying terrains provide those possibilities [164]. Patients with chronic obstructive pulmonary disease (COPD) who spend time in natural settings may have less anxiety and depression, which could improve their ability to take care of their condition [165].

7.2 Stress-Relieving Hiking Trails

(a) **General Stress Reduction**

Healing trails for stress reduction aim to create an atmosphere of calm and serenity to aid in the process of unwinding and improving one's state of mind. Ecotherapy, often known as green therapy, is the practice of spending time in natural environments with the intention of improving one's mental and emotional health [166]. Stress can be reduced and mental stability can be fostered by spending time in natural environments like parks and gardens [167].

(b) **Post-Traumatic Stress Disorder (PTSD)**

Healing paths for people with PTSD typically feature serene and secure settings. Trauma survivors can benefit from spending time in natural settings that have been stripped of triggering elements [168]. Healing trails can integrate mindfulness-based practices like meditation and guided nature walks to help people with PTSD deal with anxiety and unwanted thoughts [169].

7.3 Evidence Supporting the Effectiveness

(a) **Physiological Benefits**

Numerous studies have demonstrated the physiological advantages of time spent outdoors. It has been shown that spending time in nature can help reduce stress and improve physiological responses [170, 171] by lowering cortisol levels, slowing the sympathetic nervous system, and speeding up the parasympathetic nervous system. Researchers have found that people whose lungs aren't functioning properly benefit from spending time in forest surroundings [172].

(b) **Psychological Well-Being**

Healing paths and walks among medicinal plants have been shown to improve mental health in numerous studies. Berto [94], Capaldi et al. [173], and Roe and Aspinall [119] all found that people who engaged in nature-based therapy reported fewer anxious and depressed sensations, greater enjoyment and contentment, and enhanced cognitive function. Time spent in nature has also been linked to improved stress resistance and coping strategies [118, 166].

(c) **Immune System Enhancement**

To boost the immune system, which can be especially helpful for people with chronic health disorders, nature-based therapies have shown promise [174]. It has been observed that being exposed to phytoncides, natural substances emitted by trees, increases natural killer cell activity and the synthesis of anti-cancer proteins [175]. Improved immune response and decreased inflammation have also been linked to mindful meditation in healing trails [176].

(d) **Medicinal Plant Walks' Potential Health Benefits**

Different plant chemicals have shown promise in the management of health disorders, therefore medicinal plant walks can have a definite therapeutic effect. For instance, the chamomile plant (*Matricaria chamomilla*) has been shown to provide sedative effects and alleviate tension [97]. Anxiety and insomnia can be alleviated with the use of lavender (*Lavandula angustifolia*), according to research [177]. The anti-inflammatory and wound-healing characteristics of aloe vera (*Aloe barbadensis miller*) make it useful for treating skin diseases [178]. Due to its antibacterial and anti-inflammatory qualities, eucalyptus (*Eucalyptus globulus*) is beneficial for the respiratory system [179].

8 Visitors' Accessibility and Inclusivity in Healing Trails

Accessibility and inclusion for people with disabilities and people of different cultural backgrounds are of the utmost importance if nature-based therapies are to reach their full potential. Healing trails and therapeutic herb walks can be inclusive settings that give people agency regardless of their limitations, sensory preferences, or cultural background if they remove obstacles to entry. Embracing diversity and fostering fairness in nature-based healing experiences are discussed in this section, along with the significance of making healing trails accessible and inclusive.

(a) **Physical Accessibility**

Creating healing trails and medicinal plant walks accessible to those with disabilities and limited mobility begins with addressing the issue of physical access. The use of wheelchairs and other mobility aids is made easier on trails that have been built with accessible walkways, ramps, and railings [180]. Visitors can better plan their time on the path with the use of clear signs, maps, and information regarding the trail's accessible features [181]. Moreover, trail planners should think about where to put benches so that people of all abilities can stop and take in the scenery.

(b) **Sensory Accessibility**

Constructing sensory-friendly healing trails can greatly improve the quality of nature-based activities for those with sensory disorders. Those who have trouble seeing print may benefit from audio or tactile interpretations [182]. Those with sensory processing impairments can participate in meaningful outdoor activities thanks to sensory gardens that include a variety of scented plants and textures [183]. It is possible to create a pleasant and stress-free environment for everyone by strategically placing calm areas along the trail.

(c) **Cultural Inclusion**

To ensure that people from all walks of life may benefit from healing trail walks, it is important to encourage cultural inclusion. Cultural exchange and acknowledgement are encouraged by involving local communities in the trail's planning and development, as well as by incorporating their traditional knowledge and practices into the trail's interpretation [184]. Because of the diversity of trail visitors, it is important to provide information and interpretation in multiple languages. Trail guides and staff that have received empathy and cultural awareness training can better meet the needs and preferences of all tourists [185].

8.1 Barriers to Accessibility and Participation in Healing Trail Walks

The Barriers to accessibility and participation of individuals in healing trails (Fig. 3) are discussed below.

(a) **Physical Barriers**

(i) **Inadequate Infrastructure:** Without proper infrastructure like accessible walkways, ramps, and handrails, people with mobility issues or physical disabilities face substantial challenges when trying to use healing trails [186]. Wheelchair users and others who rely on mobility aids may have difficulty navigating the trail due to its uneven surface, steep inclines, and lack of resting spots.

(ii) **Limited Restroom Facilities:** The lack of wheelchair-accessible restrooms along the trail is a major barrier for people with mobility impairments. People who are visiting must have easy access to restrooms in case they need to take frequent breaks or have special needs.

(b) **Sensory Barriers**

(i) **Lack of Interpretive Materials:** One barrier to full participation in healing trails for people with visual impairments is the absence of accessible interpretive materials in formats such as braille or audio [187]. Individuals with visual impairments may be limited in their ability to participate due to a lack of descriptive audio or tactile components.

(ii) **Noisy or Overstimulating Environments:** Individuals with sensory processing issues or neurodevelopmental conditions may have difficulty on healing trails that offer noisy or overstimulating situations. Discomfort

• Physical Barriers	◆ Inadequate Infrastructure ◆ Limited Restroom Facilities
• Sensory Barriers	◆ Lack of Interpretive Materials ◆ Noisy or Overstimulating Environment
• Cultural Barriers	◆ Lack of Representation ◆ Communication Issues
• Financial Barriers	◆ High Entry Fees ◆ Lack of Funding for Accessibility
• Knowledge and Awareness Barriers	◆ Limited Information ◆ Lack of Education and Training
• Social Barriers	◆ Stigma and Misconceptions ◆ Lack of Social Support
• Transport and Location Barriers	◆ Ease of accessing ◆ Distance

Fig. 3 Barriers to accessibility and participation in healing trail walks

caused by excessive crowding or noise may discourage people from taking part.

(c) **Cultural Barriers**

(i) **Lack of Representation:**

To begin, healing trails may not connect with people of different cultural communities if they don't include cultural representation or information from multiple backgrounds [188]. People of different cultural origins may feel excluded and like the healing process doesn't apply to them if their culture isn't well represented.

(ii) **Communication Issues:** Some people may feel left out if healing paths only provide information in one language. Making the experience accessible to persons of different language origins requires the use of multilingual signs and interpretative materials [189].

(d) **Financial Barrier**

(i) **High Entry Fees**: People with fewer financial resources may be deterred from visiting healing trails with high entry costs [190]. Access to nature-based therapeutic experiences might be hindered by barriers such as the price of admission, transportation costs, and other related fees. All people, regardless of their financial situation, should be able to make use of healing paths and medicinal plant hikes. Providing free or discounted entry, or community-based programs, can widen access to nature-based therapeutic experiences and eliminate cost obstacles [190]. Through partnerships with neighborhood institutions like schools, community centers, and social service organizations, we may reach out to disadvantaged neighborhoods and remove financial barriers to participation.

(ii) **Lack of Funding for Accessibility:** Inadequate funding for accessible healing pathways can impede attempts to eliminate physical and sensory barriers. Ramps, accessible walkways, and alternate format interpretative materials may not be possible due to financial limits.

(e) **Knowledge and Awareness Barriers:**

(i) **Limited Information:** Planning a trip to a healing trail can be difficult for some because of a lack of knowledge regarding the path's accessibility features [181]. It can be discouraging to try out a new path if you have doubts about whether or not it will accommodate your particular demands.

(ii) **Lack of Education and Training:** There may not be enough knowledge and training for trail guides and personnel on how to accommodate people with impairments or from different cultural backgrounds [191]. Barriers to participation may arise unintentionally due to a lack of understanding of accessible communication and cultural sensitivity.

(f) **Social Barriers**

(i) **Stigma and Misconceptions:** Exclusion and prejudice can result from social stigma and false beliefs about someone based on their disability or cultural background [192]. People's participation in healing trails could be hindered by preconceived notions and biases.

(ii) **Lack of Social Support:** Due to a lack of social support, it may be difficult for people with disabilities or from marginalized populations to participate in healing trails on their own [193]. Isolation and lack of engagement may result from a lack of support from friends and family.

(g) **Transportation and Location Barriers**

Accessibility to healing trails and medicinal plant hikes should be planned with transportation and location in mind. Those without access to a car will have an easier time visiting trails that are close to major transportation nodes or have convenient parking options [194]. The inclusion of people with mobility problems might be bolstered by advertising shuttle services and other accessible transportation options.

8.2 Benefits of Accessibility and Inclusivity of Healing Trails

(a) **Enhanced Well-Being and Health Outcomes**

Expanding access to nature-based therapies by creating universally accessible healing pathways and medicinal plant walks. Researchers have found that spending time in nature has positive effects on health and happiness [118, 166]. More people will benefit from these outcomes and be able to contribute to better public health if obstacles are removed and everyone has the same opportunity to participate.

(b) **Social Cohesion and Community Engagement**

Healing trails that are easily accessible to all people bring people from different walks of life together, which is good for promoting social cohesion and community involvement [195]. A sense of community is fostered through outdoor activities that bring people of different backgrounds together, according to research [196]. Having more people to talk to helps build a community where everyone feels welcome.

(c) **Empowerment and Sense of Agency**

Accessible healing trails walk give people with disabilities the freedom to discover the outdoors on their own terms while still receiving the benefits of nature-based healing [197]. This boosts their sense of autonomy and allows them to take control of their own health and well-being. Fox and Magnus [198] asserted that when people feel that they have some control over their own lives, they tend to have a more favorable view of themselves. Furthermore, fostering diversity and inclusion in outdoor activities strengthens the conviction that nature is a place for all people.

(d) **Cultural Preservation and Promotion**

Healing paths that incorporate traditional knowledge and practices from a variety of cultures help to maintain and promote that heritage [199]. Visitors can gain a deeper understanding of and appreciation for alternative approaches to health and wellness by participating in events that highlight cultural diversity [200]. By serving as meeting places for people from different cultures, healing paths help revitalize and preserve indigenous ways of life.

9 Plant Identification and Ethical Foraging in Healing Trails

Healing trails and medicinal plant walks rely heavily on participants' ability to identify plants and to forage in a responsible manner. Through these activities, people are able to forge meaningful connections with the natural world, gain insight into the healing potential of plants, and experience these treatments themselves. To guarantee the safety and sustainability of medicinal plant use, however, it is essential to conduct plant identification and foraging with integrity and reverence for the environment. This section reinforces ethical foraging techniques to protect nature's healing treasures and offers tips for identifying plants along healing pathways.

9.1 The Value of Proper Plant Labelling and Identification in Healing Trails

In order to improve the healing trail experience for visitors and encourage the responsible and safe use of medicinal plants, accurate plant labeling and identification are of the utmost importance. Proper plant labeling educates and encourages visitors to actively participate in their own health by providing them with information about the plants they are considering using for therapeutic purposes. Below are some of the importance of identifying and labeling plants:

(a) **Enhancing Educational Opportunities**

The labeling of plants along healing paths is a great way to teach tourists about the medicinal properties of the area's flora. Educating tourists about the healing power of plants through their common and scientific names, as well as their traditional and therapeutic uses, is important [201]. The educational materials help people feel more connected to the natural world and develop an appreciation for the variety of life they'll encounter on their hike.

(b) **Ensuring Safe and Responsible Use**

The incorrect usage of hazardous or harmful plants can be avoided with proper plant identification. Visitors can reliably identify and avoid potentially toxic species with the help of clear and trustworthy plant labels [135]. By knowing this, you can avoid any potential health problems associated with misusing therapeutic plants.

(c) **Increasing the Depth of the Healing Process**

A more holistic and transforming healing experience is provided for guests by correctly identified medicinal plants. The healing benefits of the plants can really sink in as people go along the trail and experience them with their labels [202]. Accurate information fosters faith in the restorative power of nature, elevating the quality of one's experience.

(d) **Facilitating Individualized Instruction**

The ability to study and explore on one's own is enhanced by healing paths that feature clearly labeled therapeutic plants. With the information provided throughout the path, visitors can tailor their experience based on their own needs or areas of interest [203]. Visitors are empowered to take charge of their health and recovery by providing them with the means to do so.

(e) **Encouraging Respect for Nature**

Labelling plants helps spread an attitude of gratitude toward nature and the resources it provides. Visitors who are educated about the plants' historical and cultural significance are more likely to value the environment for what it contributes to human health and happiness [204]. This respect calls for eco-friendly, low-impact behavior on the healing trail.

9.2 Plant Identification Techniques in Healing Trails

The correct identification of plants is essential if tourists are to benefit from their curative effects. In healing trails, which place an emphasis on learning about and connecting with nature, visitors can learn more about local plant life by using a variety of plant identification methods. Some of these methods include the following:

(a) **Field Guides and Reference Books**

The plants used in healing trails can be identified with the use of field guides and other reference materials. Numerous plant species have been profiled in these books, along with their habitats, features, and medicinal uses [205]. Improve your plant identification skills and your appreciation for the local ecosystem by consulting a regional flora or medicinal plants field guide.

(b) **Herbaria and Botanic Gardens**

Visitors to botanical gardens and herbaria can examine the unique characteristics of plants in close proximity. Learning by doing and side-by-side species comparisons are only two of the many benefits of visiting a botanical garden [206]. Herbaria are great for research and fact-checking since they store preserved plant specimens.

(c) **Professional Guided Walks**

Visitors can gain first-hand experience with plant identification on guided walks conducted by botanists, herbalists, or expert naturalists [207]. These professionals can impart their extensive understanding of medicinal plants, highlight key characteristics, and provide direction for the responsible and secure identification of plants. Visitors can learn more and feel more connected to the natural world on walks led by experts.

(d) **Digital and Mobile Platforms and Applications**

Thanks to digital resources and application development, plant identification has become more convenient for the general public. Numerous smart phone programs employ neural networks for image recognition to determine the species of a plant from a photograph the user has taken [208]. Accessing information about a plant's medical characteristics and historic applications is a common feature of these apps.

(e) **Training Sessions on Plant Recognition**

Visitors can engage in hands-on learning during plant identification seminars hosted by healing path managers and local experts. Activities such as collecting botanical specimens, consulting field guides, and honing identification abilities are often incorporated into workshops [167]. Participating in such programs gives guests more experience in recognizing therapeutic plants on their own.

(f) **Citizen Science Projects, Number**

Visitors are encouraged to participate in plant identification attempts through citizen science activities. Scientists and conservationists can learn a lot about the distribution and presence of plants along healing paths if they get the public involved in data gathering [209]. Both the scientific community and the healing trail experience benefit from these kinds of activities.

(g) **Working with People Who Already Have the Knowledge**

Cooperation with local knowledge holders is crucial for healing paths situated in areas inhabited by indigenous communities. Native peoples have been keeping track of the medicinal and ecological value of local plants for decades [210]. Engaging with indigenous knowledge in a respectful manner promotes cross-cultural understanding and deepens insight into the practice of plant identification.

9.3 Ethical Foraging Practices in Healing Trails

For sustainable usage of medicinal plants and to protect ecological balance, ethical foraging methods are crucial along healing pathways. Responsible foraging methods are essential in ensuring that medicinal plants will be available for future generations as healing trails continue to attract people looking to experience the healing effects of nature. Ethical foraging guidelines and principles that visitors should follow when interacting with therapeutic plants include the following:

(i) **Sustainability and Biodiversity Conservation**

The sustainability of medicinal plant populations is given top priority in ethical foraging techniques. It's important for foragers to respect the plants' normal growth cycles and refrain from taking too much from them [211]. Visitors can help preserve plant diversity and maintain the long-term availability of these resources by taking only what is necessary and leaving a great deal of plants alone.

(ii) **Harvesting with Care**

Ethical foragers take great care to protect both the plant and its natural habitat when they gather their supplies. To protect the plant and its environment, they are careful not to pull it out by its roots [212]. Selective harvesting promotes plant health and growth by allowing plants to self-renew and spread.

(iii) **Respect for Cultural and Indigenous Knowledge**

Paths to health tend to pass through areas where there is a wealth of traditional and local knowledge about the therapeutic properties of plants. Foragers who operate ethically take into account the customs and expertise of the local population [210]. The people do not use any flora considered sacred or culturally significant without first obtaining permission from the local indigenous population.

(iv) **Avoiding Endangered Species**

Plants that are endangered or protected by conservation regulations should never be taken from the wild [147, 213]. It is important for foragers to be informed of the conservation status of indigenous plants and to avoid taking any endangered or threatened varieties.

(v) **Controlling Invasive Species**

Ethical foraging guidelines may inadvertently promote the collection of noxious weeds. Harvesting invasive species for medical use is one way to limit their spread, which can be damaging to native ecosystems [214]. To avoid unintentionally aiding the spread of invasive species, foragers should use prudence.

(vi) **Plant Propagation and Care**

When it comes to medicinal plants, ethical foraging also includes growing them from seed or cuttings [215]. If more people grew their own medicine, it would put less of a strain on wild species while still providing a steady supply.

(vii) **Adopting the "No Trace" Principle**

Ethical foragers must adhere strictly to the "Leave No Trace" code of conduct. It is recommended that foragers leave the place as they found it and make as little of an influence as possible [216]. The natural integrity of the healing trail will be preserved if visitors refrain from trampling plants and harming wildlife habitats.

10 Healing Trail Design and Landscaping

Healing paths can have positive effects on visitors' bodies, minds, and spirits depending on how they've been planned and landscaped. Creating a deep connection with nature, relaxation, and healing, a well-designed healing trail provides a harmonic and visually beautiful setting. In order to maximize the therapeutic effects of healing trails, this chapter delves into a number of design considerations, including trail layout, landscaping options, and the strategic use of medicinal plants.

10.1 Healing Trail Design

The effectiveness and pleasantness of a healing trail depend heavily on its design. Accessibility, aesthetics, and the natural flow of the landscape are all factors that should be considered when designing a trail. Trail design factors to think about include:

(a) **Availability and Acceptance**

To ensure that individuals of all abilities can benefit from nature's healing powers, it is crucial to develop a healing trail that is accessible to all. Wheelchair users and anyone with mobility issues should be given special consideration in the trail's layout. Paths should be large and flat or have a gentle incline so that people may easily navigate them [217].

(b) **Circular Versus Linear Paths**

The layout of healing trails can take the form of either a loop or a straight path. Hikers may get that same sense of accomplishment without having to retrace their steps on loop trails. In order to provide visitors with a wide range of experiences, many trails are now being designed as loops [218]. Linear trails, on the other hand, provide a continuous route across the natural landscape to designated destinations like lookout sites or unique geological formations.

(c) **Integration of Rest Areas**

The walk features a number of carefully located benches where hikers may sit back, relax, and take in the scenery. Seating, picnic tables, and naturally looking canopies are all appropriate amenities for rest stops [219]. Healing is aided by the opportunity for introspection and contemplation provided by such settings.

(d) **Directional and Interpretive Signage**

The educational value of the healing trail can be greatly improved by installing clear and informative interpretive signage at key locations along the path. Explanations of the area's geology, wildlife, and human history can all be included on signs. Trail users benefit from navigational signs because they make it easier for them to get around and learn more about the trail's layout and major features [220].

10.2 Landscaping Choices for Healing Trails

The mood and feel of a healing trail greatly depends on the landscaping that is used. The healing potential of the trail is boosted by the careful design of its surroundings:

(a) **Organic Components**

The design of the healing route is based on preserving and highlighting natural features, such as trees, bushes, and rock formations that are indigenous to the area. Genuineness and a sense of oneness with nature are provided by materials taken from nature [221]. Also crucial to local ecosystems and biodiversity, these components are often overlooked.

(b) **Water Attractions**

Small streams, ponds, or waterfalls might be incorporated into the healing route to provide a meditative and relaxing atmosphere. The soothing sound of water flowing has been shown to have a calming impact on visitors (Berman et al. 2020). Wildlife can be attracted to water features, adding to the whole natural experience.

(c) **Landscaping with Native Plants**

Landscaping the pathway with native plants helps sustain biodiversity in the area and makes the path blend in with its surroundings. There is less need for upkeep and supplies when using native plants because they are already accustomed to the weather and soil [167].

(d) **Changes with the Seasons**

The dynamic beauty of nature is accentuated by the healing trail's design, which takes into account seasonal changes in plant life and natural events. A greater understanding for nature's cyclical processes can be developed through witnessing the yearly transformation of the landscape as leaves turn, flowers blossom, and animals migrates.

10.3 Strategic Planting of Medicinal Plants

The therapeutic value of an area can be amplified via the strategic placement of medicinal plants along healing routes. The following factors should be taken into account when deciding where to strategically position these plants:

(a) **Naturally Occurring Medicinal Plants**

The design of a healing trail should prioritize the use of native and indigenous medicinal plant species whenever possible. Plants that are indigenous to an area have a greater chance of surviving and thriving there [167]. According to Giampiccoli and Kalis [222], indigenous plants and animals have cultural relevance because they introduce tourists to local traditions and knowledge.

(b) **Gardens for Medicinal and Educational Plants**

Visitors can learn more about the medical benefits of plants by visiting gardens or educational spaces dedicated to them [205]. A plant's historical applications, therapeutic properties, and any necessary precautions can all be communicated through informative signage.

(c) **Classification Based on Medicinal Values**

By grouping medicinal plants together according to their healing capabilities, the trail becomes a resource for exploring a variety of health-related topics [205]. The educational value of the trail is increased, and visitors are encouraged to focus on specific health concerns, when plants with similar healing benefits are grouped together.

(d) **Creation of Sensory Gardens**

Healing paths can include sensory gardens, which are landscaped with a wide range of plants chosen for their ability to evoke a range of sensory experiences. Aromatic plants like lavender and rosemary added to a space can create a peaceful and energizing atmosphere [223].

11 Healing Trail Certification and Quality Standards

Authenticity, safety, and efficacy of healing trails and medicinal plant walks must be guaranteed by standardized processes and quality assurance as their popularity grows. Connecting with nature, experiencing its therapeutic advantages, and learning about medicinal plants are all possible through healing paths. It is critical to create certification systems and quality criteria for healing trail offers to ensure the continued credibility of these services and the safety of trail users. Focusing on authenticity, sustainability, safety, and educational benefits, this chapter investigates the potential implementation of such certification schemes and quality criteria for healing trails with medicinal plant walks. Furthermore, the identification, isolation, and purification of the active compounds that makes the plants possesses specific therapeutic potentials is also key in the certification and quality standard analysis [224–235].

11.1 Ensuring Credibility and Ethical Practices

(a) **Professional Guidance**

Expert review should be part of a certification process for healing trails [222] to verify that trails are run ethically, that proper criteria are followed for medicinal plant use, and that the cultural relevance of indigenous medicinal knowledge is honored. The fields of ethnobotany, traditional medicine, and ecotourism all have experts who can contribute to the development of these guidelines.

(b) **Indigenous Community Participation**

It is crucial to include local communities in the certification process in areas where there is indigenous knowledge of medicinal plants. Traditional knowledge is better protected, respectful customs are maintained, and tourists have a more meaningful and meaningful encounter with indigenous peoples when they are involved [236].

(c) **Validating Medicinal Plant Data**

Healing trails may be required by certification programs to include trustworthy information about the therapeutic plants along the trail. It is important that visitors receive accurate educational content, so it is important to verify plant identification, traditional applications, and safety precautions.

(d) **Plant Source Documentation**

In order to meet certification standards, it may be necessary to provide proof of where the medicinal plants were harvested along the path. This paperwork guarantees that plants were harvested in a way that didn't impact natural populations [237].

11.2 Sustainability and Risk Management in Healing Trails

(a) **Assessing Risks and Taking Precautions**

Certification standards for healing trails should center on safety first and foremost. Trail designers and guides can benefit from a thorough understanding of potential dangers thanks to a hazard assessment, which may include the identification of plants that could be toxic or hazardous. Clear warnings and signage are examples of mitigation actions that can be used to protect tourists [238].

(b) **Methods of Sustainable Harvesting**

Sustainable methods of medicinal plant harvesting can be encouraged through certification schemes. Selective harvesting and culture are examples of responsible harvesting methods that can be used on trails without severely harming wild populations [239].

(c) **Assessment of Environmental Impact**

An environmental impact assessment may be required for certification if it is deemed necessary to protect the natural habitat of the healing trail. This report takes into account the trail's possible effects on the surrounding ecology and suggests ways to lessen those problems.

(d) **First Aid and Emergency Preparedness**

Healing trails that want to be certified may be required to have proper first aid equipment and emergency procedures in place. Visitor security can be improved by providing first aid training for guides and establishing secure lines of communication with emergency services.

11.3 Instructional Material and Interpretation in Healing Trails

(a) **Professional Tour Guides**

Accredited guides with knowledge of botany, ethnobotany, or traditional medicine may be required by certification programs for healing trails. It is imperative that people have access to reliable information on medicinal plants, their applications, and their cultural value, and guides play a pivotal role in this regard [211].

(b) **Cultural Relevance Interpretation**

Cultural interpretation of medicinal plants is a top priority along healing routes that pass through areas with significant indigenous history and tradition. This method promotes appreciation for indigenous wisdom and facilitates communication between tourists and locals [236].

(c) **Signage and Resources for Interpretation**

Visitors' knowledge of medicinal plants and their therapeutic properties is enhanced by the healing trail's well-designed interpretive signage and educational materials. The reliability and precision of these resources can be specified by means of certification standards.

(d) **Instructional Seminars and Courses**

Healing trails can be encouraged to host workshops and programs on medicinal plants and traditional healing modalities if they are certified to do so. Visitors can walk away from a workshop with practical skills they can use immediately [219].

12 Conclusion

The concept of healing trails and incorporating medicinal plant walks into recreational development has developed as a strong and transforming approach to holistic well-being. Throughout this extensive debate, we have covered the different features of healing trails, from their historical emergence to their incorporation into urban green spaces and parks. We have delved into the cultural legacy and traditional medicinal practices that infuse these routes with authenticity and meaning. Healing paths offer an unmatched opportunity for individuals to bond with nature, experience the therapeutic advantages of natural habitats, and receive insights into the traditional knowledge of medicinal plants. The merging of cultural heritage and traditional medicine not only encourages respect for varied traditions but also enhances the relationships between individuals and their natural surroundings.

As we explore further the area of ecotherapy and complementary medicine, the potential of healing trails to promote mental health, reduce stress, and enhance overall well-being becomes clear. Specific plant genera with calming or mood-lifting effects further show the deep impact of nature on human emotions and mental states. However, as healing paths gain popularity, it becomes vital to address the concerns of accessibility, inclusion, and ethical harvesting. By creating certification procedures and quality standards, we can ensure that healing paths preserve their originality, emphasize safety, and safeguard the ecosystem for future generations.

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Harmonizing Tradition and Technology: The Synergy of Artificial Intelligence in Traditional Medicine

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Abstract

Herbal medicine, deeply rooted in tradition, emphasizes locally sourced, renewable resources and is crucial in preserving cultural heritage. Technology has expanded its reach globally and encouraged inclusivity. Phytochemistry enhances the effectiveness of herbal treatments by identifying bioactive compounds, understanding their mechanisms, and ensuring product quality. The convergence of technology and tradition has far-reaching implications, promoting innovation, preserving heritage, and affecting various aspects of life. Artificial intelligence (AI) supplements conventional medicine without replacing it, enhancing accuracy, accessibility, and results. AI supports individualized therapy recommendations, predictive analytics, research, and diagnosis accuracy. It aids in herbal medicine development, extends healthcare access, and empowers patients. However, integrating AI into traditional medicine poses challenges such as data privacy, ethical dilemmas, and cultural value preservation. Overcoming generational technology gaps is vital. Successful integration addresses issues like data quality, regulatory compliance, resistance to adoption, risk of overreliance, cost, education, training, liability, and patient trust. This chapter advocates for AI integration into traditional medicine, which offers numerous benefits, including enhanced diagnosis, personalized treatment, research support, and patient empowerment. Overcoming the challenges is essential to fully leveraging this synergy, ensuring the advancement of healthcare while preserving cultural traditions and values.

Keywords

Herbal medicine · Machine learning · Phytochemistry · Medicinal plants · Healthcare · Traditional medicine

Abbreviations

AI	Artificial intelligence
TCM	Traditional Chinese medicine

1 Introduction

In many parts of the world, people have long relied on treatments and therapies that have their origins in traditional medicine. Traditional medicine, which is defined by methods that have been passed down through generations and refined over ages, has shown consistent effectiveness across a wide range of health concerns [9, 52]. Presently, there is a growing need to improve traditional medicine accuracy, broaden its availability, and streamline its incorporation into cutting-edge healthcare infrastructure. To meet this challenge, the incorporation of artificial intelligence (AI) stands out as a promising path since it can help bridge the gap between traditional medicine and cutting-edge medical advances. Combining AI with conventional medicine is a giant step forward in improving patient care [20, 49].

Healthcare is no exception to the impressive advancements being made in a variety of fields by AI, a branch of computer science that aims to build intelligent robots that are capable of carrying out activities that traditionally require human intelligence [5]. Ayurveda, herbal medicine, acupuncture, and traditional Chinese medicine (TCM), all of which have their own distinct beliefs and practices, are just a few of the therapeutic modalities that fall under the umbrella of traditional medicine [28, 50–52]. The addition of AI to these established methods has the potential to improve patient care by increasing the precision of diagnoses, the effectiveness of therapies, and, eventually, their overall health. Traditional medicine has played a significant role in delivering healthcare treatments that have been firmly ingrained in the history and cultures of many different countries for ages. This broad field includes a variety of therapeutic modalities, including manual therapies, spiritual practices, herbal cures, and energy-based therapies. These techniques have been developed as a result of careful observation, empirical information that has been passed down through the years, and practical experience [19, 47].

TCM, a long-standing tradition dating back more than two millennia, serves as one example. TCM includes a variety of therapeutic methods, such as acupuncture, herbal medicine, nutritional advice, and mind-body practices like Tai Chi and Qigong [80]. This all-encompassing approach has helped treat a variety of health problems and promote general well-being. Similar to this, restoring harmony and balance between the body, mind, and spirit is a major focus of the traditional Indian medicinal system known as Ayurveda [23–29, 52]. To improve health and avoid disease, this holistic approach makes use of nutritional advice, yoga, meditation, and herbal medicines. These venerable belief systems have a profound grasp of the complicated relationships that exist between the human body and the natural environment, highlighting how crucial it is to achieve and maintain equilibrium to maintain good health. Due to the potential role of AI in traditional medical procedures and beliefs, it is essential to consider how AI may improve the precision and efficiency of conventional treatments, modernize their delivery, and solve enduring problems [60, 81].

The ethical issues and potential difficulties associated with the integration of AI into conventional healthcare systems are equally important. The importance of conserving traditional medicine through culture and historical diversity while utilizing AI could ensure that these age-old therapeutic techniques continue to make a substantial contribution to global healthcare in the twenty-first century and beyond. AI and conventional medicine working together may have the potential to improve healthcare outcomes, increase access to high-quality care, and combine the best aspects of both worlds: conventional medicine and cutting-edge technology. This chapter comprehensively explores the revolutionary effects of AI on conventional medical procedures by emphasizing the complex interactions between tried-and-true treatment modalities and AI-driven innovations. Importantly, this is because AI plays a complementary function in conventional medicine, enhancing rather than replacing the skills of traditional practitioners [18, 63, 84].

2 Exploring the Significance of Herbal Medicine in Traditional Practices and the Role of Phytochemistry in Efficacy

Herbal medicine, phytotherapy, or plant medicine is a crucial component of traditional medicine throughout the world [2, 3, 34, 43]. It uses a variety of plant-derived substances, including those found in the roots, leaves, stems, flowers, and fruits, to prevent, identify, and cure a wide range of medical diseases [32, 40–42, 72, 73, 83]. Herbal medicine is still a key part of conventional medicine since it provides important knowledge about holistic approaches to health, the preservation of cultural heritage, and natural cures. As it completes and expands the variety of therapeutic options accessible to people and communities around the world, its enduring relevance is obvious [81, 84]. Table 1 lists the historical importance, cultural relevance, holistic approach, sustainable and local resources, complementary care, preservation of cultural legacy, and accessibility of herbal medicine in traditional practices [8, 30, 44, 50–52].

Therefore, understanding the crucial role of phytochemistry in the effectiveness of herbal medicine is vital for comprehending how medicinal plants function and their potential health benefits [15–17, 31, 33, 53–56]. Phytochemistry is a subfield of chemistry that focuses on investigating the chemical constituents within plants, detailing their structure, composition, and functions. The significant contributions

Table 1 Role of herbal medicine in traditional practices

Aspect	Roles
Historical significance	Herbal medicine, practiced for centuries, holds historical importance. Ancient societies worldwide, including China, India, and indigenous communities, have extensive knowledge of herbal remedies
Cultural relevance	Herbal medicine is deeply ingrained in cultural practices, beliefs, and traditions. Herbal medicine practice is a vital part of cultural heritage, expressing community identity, often passed down through generations
Holistic approach	Traditional healing systems, like herbal medicine, take a holistic approach. They address physical symptoms and the interplay of mind, body, and spirit, promoting overall well-being
Sustainable and local resources	Traditional medicine, such as herbal remedies, values local, sustainable resources. This aligns with modern environmental goals, supporting self-sufficiency and minimizing environmental impact
Complementary care	Herbal medicine complements modern healthcare, providing additional treatment options for a wide range of diseases. Patients can choose therapies that align with their cultural backgrounds
Cultural heritage preservation	Herbal medicine plays a vital role in preserving cultural traditions and passing regional knowledge to future generations. Efforts like digitization and collaboration with indigenous groups are crucial
Global accessibility	Advancements in technology have enabled the worldwide dissemination of knowledge about herbal medicine. This fosters inclusivity, making herbal remedies more accessible and bridging cultural divides

of phytochemistry to the efficacy of herbal therapy include the identification of bioactive compounds, understanding mechanisms of action, quality control, and standardization, optimizing extraction techniques, safety and toxicology, compound synergy, bioavailability and pharmacokinetics, drug-plant interactions, ethnopharmacology, and evidence-based medicine [69, 78] (Table 2).

Therefore, phytochemistry is integral to herbal medicine research and practice. It enhances the understanding of how medicinal plants work, supports quality control, ensures safety, and promotes the integration of herbal medicine into evidence-based healthcare. The knowledge of the chemical composition of medicinal plants is central to harnessing their full therapeutic potential [8, 9, 24, 26, 27, 29].

Table 2 Significant contributions of phytochemistry to the efficacy of herbal therapy

Aspect	Significant contributions of phytochemistry to the efficacy of herbal therapy
Identification of bioactive compounds	Phytochemistry is crucial for identifying and characterizing bioactive compounds in medicinal plants, including alkaloids, flavonoids, terpenes, glycosides, tannins, saponins, phenolic compounds, etc. essential for understanding therapeutic properties
Understanding mechanisms of action	Analyzing phytochemical composition helps researchers understand how medicinal plants exert therapeutic effects. This understanding is vital for defining unique mechanisms of action
Quality control and standardization	Phytochemical analysis ensures the quality and standardization of herbal products, maintaining consistent levels of active compounds for predictable and reproducible effects
Optimizing extraction techniques	Knowledge of phytochemistry guides the selection of appropriate extraction methods to efficiently isolate bioactive compounds, maximizing yields and therapeutic potential
Safety and toxicology	Phytochemistry identifies potentially toxic compounds in medicinal plants, essential for safety assessments and establishing proper dosage guidelines to prevent adverse effects
Compound synergy	Medicinal plants contain multiple bioactive compounds that work synergistically for therapeutic effects. A deeper understanding of phytochemistry helps identify potential synergies and optimize formulations
Bioavailability and pharmacokinetics	Phytochemistry contributes to knowledge about how the body absorbs, distributes, metabolizes, and eliminates bioactive substances from medicinal plants, aiding in developing high bioavailability formulations
Drug-plant interactions	Some bioactive compounds in medicinal plants may interact with pharmaceutical medications. Phytochemical research helps identify potential interactions for safe and effective care
Ethnopharmacology	Phytochemistry aligns with ethnopharmacological research, combining traditional and indigenous knowledge of plant-based medicine with a focus on cultural and phytochemical insights
Evidence-based medicine	Phytochemistry contributes to the evidence base for herbal therapy. Scientific studies examining the phytochemical profile of medicinal plants provide essential data supporting their use in healthcare

3 The Nexus of Tradition and Technology

The convergence of technological advancement and cultural customs has become a subject of intense interest and a crucial aspect of contemporary life. Progress and innovation are often linked with technological advancement; tradition signifies the timeless principles and customs that connect us to our cultural origins. These divergent forces create a dynamic space where the past and future intersect, fostering innovation, the preservation of traditions, and exact change.

The rapid evolution of technology has brought profound changes in how we work, communicate, and live. It has enabled us to transcend geographical boundaries, disseminate knowledge at unprecedented rates, and enhance the quality of life through medical breakthroughs and sustainable inventions. Humanity has propelled itself into a technology-driven era; therefore, concerning tradition, striving to strike a balance between the efficiencies offered by technology and the depth of experience offered by tradition [75].

One domain where this synergy is particularly evident is in the realm of culture and the arts. Modern technology has brought new life into time-honored art forms such as music, dance, and storytelling [21]. Streaming services have made it possible for people worldwide to access traditional melodies once confined to isolated communities. Artists blend traditional techniques with modern digital tools to create compelling experiences across various mediums [57]. The advancements in virtual reality and augmented reality have opened up new avenues for cultural preservation, allowing individuals to explore heritage sites and artifacts from the comfort of their homes.

Education is another significant arena where this convergence plays a pivotal role. Online learning platforms have made the dissemination of knowledge, both ancient and modern, a global phenomenon, reaching the farthest corners of the world [77]. Students now have access to the teachings of ancient philosophers alongside the latest scientific discoveries. Traditional classroom learning coexists with innovative educational approaches enabled by technological progress.

The corporate world, too, has embraced this nexus by fostering a corporate culture that respects tradition while embracing innovation, creating a win-win situation for all. Many businesses recognize the importance of diversity and inclusivity, which often involves respecting employees' cultural backgrounds and traditions [68]. Technology is harnessed to create flexible work environments, enabling individuals to uphold religious customs or cultural practices while maintaining productivity.

In healthcare, the nexus has a significant impact as well. Traditional and modern forms of medicine are integrated to provide holistic healthcare solutions [46]. Cutting-edge research and diagnostic methods are used to analyze and enhance traditional remedies [1]. This harmonious blend of antiquity and modernity offers a more comprehensive approach to healthcare, acknowledging the significance of religious and cultural observances in overall well-being.

Nevertheless, the nexus is not without challenges. The rapid pace of technological advancement can render traditional practices obsolete, and generational disparities in

Table 3 Overview of the convergence of technology and tradition

Aspect	The convergence of technology and tradition
Convergence significance	The fusion of technology and tradition is of intense interest and a critical aspect of modern life. It bridges progress and innovation with timeless cultural principles, fostering innovation, tradition preservation, and transformative change
Technological impact	Rapid technological evolution transforms work, communication, and daily life, breaking geographical barriers, disseminating knowledge, enhancing healthcare, and igniting sustainability
Cultural and artistic revival	Technology revitalizes traditional cultural forms such as music, dance, and storytelling, granting global access to heritage, and merging conventional techniques with digital tools for multimedia experiences
Educational transformation	Online platforms democratize knowledge, offering ancient wisdom alongside cutting-edge discoveries, while blending traditional and innovative educational methods
Corporate embrace of tradition	Businesses promote diversity and inclusivity, respecting employees' cultural backgrounds. Technology facilitates flexible work settings that honor cultural practices
Holistic healthcare integration	Traditional and modern medicine harmonize, employing advanced research and diagnostics to enhance traditional remedies, and acknowledge cultural and religious significance in healthcare
Challenges and preservation	Rapid technological advancement risks erasing traditions, and generational gaps in tech adoption may undermine cultural knowledge. Navigating the intersection requires preserving wisdom while harnessing technology's power

technology adoption can lead to divisions within societies, potentially eroding cultural knowledge and values [6]. It is crucial to navigate this intersection thoughtfully, preserving traditional wisdom while harnessing the power of technology.

Table 3 shows the overview of the convergence of technology and tradition. The intersection of tradition and technology is a dynamic space where the past and future coexist harmoniously. It serves as a bridge between generations, a driving force in preserving cultural traditions, and a wellspring of boundless creative potential. This harmony underscores our capacity to adapt and evolve, maintaining a connection to our heritage while embracing the opportunities of the digital age. To navigate this intricate crossroads successfully, we must continue to find ways to honor tradition, harness technology for the greater good, and ensure that progress does not come at the expense of our cultural and historical identity.

4 The Complementary Role of Artificial Intelligence in Traditional Medicine Practices

Traditional medicine practices are increasingly recognizing the benefits of integrating AI into their operations within the ever-evolving healthcare industry. This transition does not involve replacing traditional practitioners with new ones, rather it's about enhancing the precision, accessibility, and outcomes of traditional therapeutic systems by complementing the expertise of traditional practitioners [4, 67]. In traditional medicine, AI could play a complementary role, bolstering the capabilities of traditional healers in various ways. This comprehensive analysis highlights the significant and symbiotic impact of AI on advancing traditional medicine practices (Table 4). It involves envisioning a promising future where the preservation of cultural heritage and the integration of cutting-edge technology coexist harmoniously, advancing healthcare and enhancing patient well-being through the collaboration of artificial intelligence and traditional healing systems [5, 12, 59].

Table 4 Benefits of artificial intelligence in traditional medical practices presented in a tabular form

AI application	Benefits
Enhanced diagnosis accuracy	AI could improve the precision of traditional diagnostic methods by analyzing patient records and medical images, enabling early disease detection
Personalized therapy recommendations	AI can use patient-specific data to create tailored treatment plans based on medical history, genetics, and lifestyle, resulting in more effective and precise therapies
Predictive analytics	Due to the predictive capabilities of AI, it can help clinicians foresee patient outcomes and treatment responses through predictive analytics, allowing for timely adjustments and improved patient care
Medical research support	Due to AI's speed, it can process data rapidly, thereby aiding in medical research by identifying patterns, exploring new treatment options, and enhancing traditional herbal medicines
Continuous patient monitoring and follow-up	AI-powered systems can continually monitor patient health, issuing alerts for deviations, enabling prompt interventions, and improving patient outcomes
Expanded reach in underserved areas	AI can act as decision support for traditional practitioners, extending healthcare services to remote areas and optimizing resources, particularly in regions with limited healthcare access
Herbal medicine development	AI can analyze extensive data on plant characteristics and bioactive compounds, enhancing traditional herbal medicine by determining the most effective treatments
Facilitating telemedicine	AI can enable remote consultations, simplifying access to healthcare for patients in rural or underserved areas and facilitating follow-up care
Streamlined administrative tasks	AI can automate administrative duties, such as appointment scheduling and patient information management, improving efficiency, and allowing practitioners to focus on patient care
Reducing human errors	AI systems are less error-prone, resulting in more reliable diagnoses and treatments, ultimately enhancing patient safety and reducing incorrect diagnoses
Empowering patients	AI can provide patients with access to health data and educational resources, enabling informed decision-making and fostering better collaboration with healthcare providers
Continuous learning and adaptation	AI can continually learn and adapt to evolving medical knowledge, ensuring traditional practices stay current with the latest advancements in healthcare

5 Utilizing Artificial Intelligence for Enhanced Diagnostics

The field of medical diagnosis may undergone a profound transformation with the advent of AI. AI systems, powered by intricate algorithms and machine learning, can analyze extensive medical databases, encompassing patient records and medical imagery, with unparalleled precision and speed [62, 70]. This innovation tends to bring forth several significant advantages as shown in Table 5.

Traditional medicine encompasses a rich reservoir of knowledge and practices passed down through generations. It often encompasses herbal remedies, dietary modifications, and holistic approaches to healing. Traditional medicine presents several advantages when harmonized with AI-enhanced diagnostics (Table 6).

The integration of AI diagnostics with conventional medical practices holds immense promise. However, this is not without its challenges. Rapid technological advancements have the potential to render longstanding practices obsolete, and the rate at which different generations embrace new technologies may vary. Successfully

Table 5 Benefits of artificial intelligence for enhanced diagnostics

Aspect	Benefits of artificial intelligence for enhanced diagnostics
Improved accuracy	AI can assist in precise and timely disease diagnosis by analyzing patient records and medical images and identifying subtle patterns and anomalies for early detection
Personalized diagnosis	AI can consider individual patient data, including genetics and medical history, to provide tailored diagnoses, enhancing accuracy, and ensuring suitable treatment strategies
Predictive analytics	Due to the predictive capabilities of AI, it can enable healthcare professionals to anticipate patient outcomes and treatment responses by analyzing extensive datasets, aiding informed decision-making

Table 6 Synergizing traditional medicine with AI-enhanced diagnostics: Advantages and opportunities

Aspect	Traditional medicine with AI-enhanced diagnostics; synergy perspective
Holistic healthcare solutions	Traditional medicine focuses on holistic patient care, considering environmental, emotional, and physiological factors. AI-enhanced diagnostics enable tailored treatment plans based on individual medical histories and lifestyles
Evidence-based traditional medicine	AI can play a pivotal role in scientifically validating traditional treatments by analyzing data on plant properties, bioactive substances, and their interactions with the human body. This enhances the credibility of traditional medicine
Remote healthcare services	AI-facilitated telemedicine can allow traditional medicine practitioners to consult with patients regardless of their location, particularly benefiting those in underserved or remote areas

navigating this convergence necessitates the thoughtful preservation of traditional wisdom while harnessing the power of technology for the healthcare sector for the greater good [48, 58].

6 The Ongoing Relevance of Traditional Medicine in the Modern Era

Traditional medicine has been a long-standing and crucial part of healthcare globally. It is rooted in centuries-old practices and frequently draws on cultural knowledge. It coexists and occasionally combines with traditional Western medicine in the modern period, presenting potential and difficulties for the field. Modern society still finds value in traditional medicine because of its cultural relevance, all-natural treatments, and sustainable practices [23, 25, 30]. As such, about 80% of the world's population still seeks treatment from traditional medicine practitioners [35–39]. Many of these people reside in rural areas in many developing nations. To guarantee the continued relevance and contributions of traditional medicine to the health and well-being of the world, its scientific confirmation and knowledge preservation are crucial. Traditional medical knowledge is well-positioned to make a lasting impact in today's healthcare landscape, which is increasingly focused on individualization and comprehensiveness.

6.1 The Significance of Traditional Medicine

Traditional medicine is deeply rooted in cultural identity and legacy, serving as a source of pride and a means of preserving ancestral values and wisdom passed down through generations. It often adopts a holistic approach, recognizing the interconnectedness of the mind, body, and spirit, thereby complementing Western medicine by addressing not only physical symptoms but also the mental, emotional, and spiritual well-being of patients. Additionally, traditional medical systems prioritize preventive care and advocate for a balanced lifestyle, making them particularly relevant in tackling modern health challenges like stress, obesity, and chronic diseases. Moreover, traditional medicine frequently relies on local and sustainable resources, aligning with contemporary sustainability goals and reducing its environmental impact while promoting self-sufficiency. Figure 1 summarizes the significance of traditional medicine.

6.2 Opportunities and Challenges of Traditional Medicine in the Modern World

Traditional medicine faces the pressing challenge of scientific validation to ensure the safety and effectiveness of its therapies, leading to an increasing collaboration between modern researchers and traditional healers [20, 45]. The regulation of

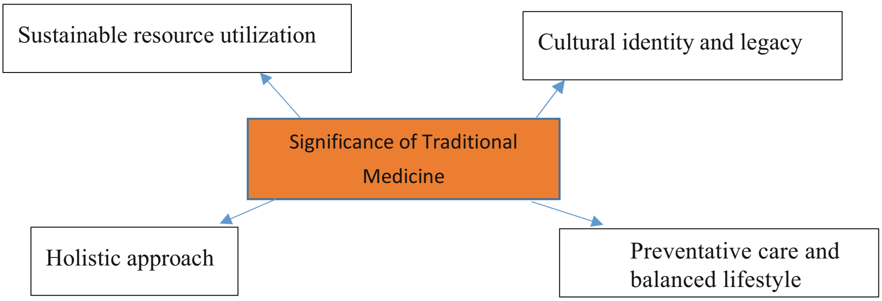

Fig. 1 The significance of traditional medicine

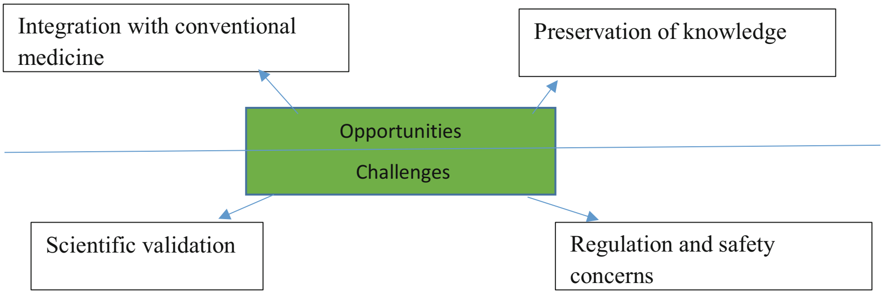

Fig. 2 Opportunities and challenges of traditional medicine in the modern world

traditional medical practices can be intricate, particularly when they involve potentially harmful or ethically questionable activities, with a strong focus on ensuring patient safety, prompting various nations to enact regulations for traditional medicine. Integrative medicine, which combines traditional and conventional approaches, is gaining prominence in modern healthcare, offering patients a more comprehensive and personalized approach to their well-being [13]. The preservation of traditional medical knowledge is of paramount importance, with ongoing initiatives utilizing digitization, training programs, and partnerships with indigenous communities to document and transmit this invaluable wisdom to future generations [22, 79]. Figure 2 summarizes the opportunities and challenges of traditional medicine in the modern world.

6.3 Case Studies in Modern Traditional Medicine

In India, Ayurveda, a traditional system of medicine, has evolved to offer contemporary services like personalized treatments, wellness retreats, and herbal products, earning international recognition for its holistic approach to health [14]. Similarly, TCM has become an integral part of healthcare in China and worldwide, combining acupuncture, herbal remedies, and lifestyle guidance, providing treatment options for various conditions, often used alongside Western medicine. Indigenous healing practices are also making strides, with indigenous communities striving to maintain their traditions while collaborating with modern healthcare providers [61]. These initiatives emphasize the cultural significance of these practices and their role in promoting community health.

7 Navigating the Ethical Challenges of AI Integration in Traditional Medical Practices

AI can potentially revolutionize the healthcare sector, including traditional medical practices. However, this transformation may bring many challenges and ethical considerations that must be carefully addressed. There are some of the essential concerns that need to be resolved when incorporating AI into traditional healing systems. These challenges and ethical dilemmas are pivotal for successfully integrating AI into traditional medicine and are summarized in Table 7.

Therefore, balancing the potential benefits of AI in traditional medicine with these challenges and ethical considerations is paramount. An ethical framework that prioritizes patient well-being, cultural sensitivity, and accountability is essential for the harmonious integration of AI into traditional healing systems. Addressing these issues ensures that AI positively contributes to preserving and advancing traditional medicine while upholding the highest standards of healthcare ethics [62, 67].

8 The Future of Traditional Medicine with Artificial Intelligence

The fusion of AI and traditional medical practices holds immense potential for advancing healthcare in the future. Traditional healing methods are on the brink of a profound transformation driven by the rapid progress of AI technologies. This section explores the potential impact, advantages, and challenges of integrating AI into traditional medicine, envisioning a future where time-honored wisdom harmoniously blends with cutting-edge technology [59, 66].

The promises of artificial intelligence in traditional medicine are as follows: enhanced diagnosis and treatment, accessibility and cultural preservation resource optimization, and environmental stewardship. AI algorithms can empower traditional practitioners by providing accurate diagnoses and personalized treatment plans, seamlessly integrating traditional knowledge with data-driven insights. This may result in a more comprehensive understanding of a patient's condition. The incorporation of AI will equip practitioners to make better-informed decisions, ultimately improving patient outcomes. AI-driven digital platforms and telemedicine solutions will enable the preservation of traditional healing methods, making them accessible to patients worldwide. Indigenous communities can connect with modern healthcare, preserving their rich traditions while extending medical assistance to

Table 7 Challenges and ethical considerations of AI integration in traditional medical practices

Challenges and ethical considerations	Ethical consideration
Data privacy and security: The challenge	It will be good to protect patients' right to confidentiality with strict data protection protocols and informed consent requirements
Cultural context and sensitivity: The challenge	AI developers should strive to incorporate cultural awareness in algorithms, with medical professionals offering context for culturally diverse patients
Excellence and personal responsibility: Problem	It will be good to incorporate accountability criteria that are vital to ensure responsibility for AI development and its outcomes in healthcare
Informed consent: The obstacle	It will be good for end users in the health and medical sector to ensure that patients fully understand AI use in treatment and its implications
Bias and fairness: The challenge	It will be good to do vigilant monitoring and audits when required to identify and rectify biases, ensuring equal care for all demographic groups
Compliance with regulations: Challenge	It will be good to develop and update regulations covering AI in traditional medicine, aligning with legal standards and the unique nature of traditional healing
Professional training: Challenge	It will be good for traditional medicine practitioners to have access to educational programs to effectively use AI, in other words to showcase their commitment to patient care
Transparency and explainability: Obstacle	It will be good to prioritize transparency and explainability in AI algorithms, allowing practitioners to comprehend and validate AI's decisions
Patient autonomy: Challenge	It will be good to ensure patients retain the final say in treatment choices, using AI as an informative tool rather than a decision-maker
Long-term impact: Challenge	It will be good to continuously evaluate and research AI to understand the long-term effects of AI on traditional medicine, enabling timely adjustments and improvements

remote regions. AI may facilitate the sustainable utilization of local resources in traditional medicine, aligning with contemporary environmental sustainability goals. This optimization ensures self-sufficiency and reduces the environmental impact of traditional practices [12, 63, 66].

Challenges and ethical considerations in the future of AI include data privacy and security, cultural context and sensitivity, accountability and quality assurance, bias and fairness, and informed consent. The application of AI requires the responsible handling of sensitive patient data, necessitating robust data protection measures and obtaining informed consent from patients. Upholding ethical standards is essential to safeguard patient privacy. Incorporating cultural context and awareness into AI algorithms is imperative to ensure respect for diverse cultural beliefs and practices. Medical professionals need to deliver care that is culturally sensitive, including the interpretation of AI-driven results [18, 48]. Ensuring AI algorithms' reliability and high quality prevents erroneous diagnoses and treatment recommendations.

Accountability standards need to be established to hold those involved in AI development and usage responsible for the outcomes of AI-assisted treatments. Patients should be made responsible for fully understanding the role of AI in their treatment and the potential consequences of providing informed consent. Medical practitioners may play a pivotal role in educating and informing patients about the integration of AI. AI systems may inherit biases from training data, potentially leading to unequal treatment and outcomes for different demographic groups. Ethical considerations necessitate continuous monitoring and correcting biases to ensure equitable and just care for all patients [4, 5, 70].

Despite the role of AI in more accurate diagnosis and treatment alongside preserving culturally significant procedures, it comes with ethical considerations, including accountability, cultural sensitivity, and data security. Therefore striking a balance between tradition and technology can shape a future where AI and traditional medicine collaborate to provide comprehensive and culturally sensitive healthcare solutions. Attempting to address these challenges, the incorporation of AI into traditional medicine can positively impact the well-being of individuals and communities, resulting in a harmonious fusion of traditional healing practices with state-of-the-art technological advancements [65, 81].

9 Preserving Cultural Heritage and Healing Wisdom: The Role of Herbal and Traditional Medicine

Herbal and traditional medicine, which are firmly entrenched in centuries-old customs and cultural knowledge, serve as both a vital steward of our cultural legacy and a priceless resource for complementary medical procedures. In-depth research on the value of traditional and herbal medicine in conserving cultural legacies and the priceless medical information passed down through the centuries is provided in this section. Table 4 provides a summary of the historical background and therapeutic knowledge of herbal and conventional medicine. Cultural identity is deeply linked with its use of herbal and traditional medicine, which reflects the values, beliefs, and knowledge that have been passed down through the years [20, 84]. These habits may serve as a source of cultural pride and a way to keep information about earlier behaviors and traditions alive. A holistic approach emphasizing the interdependence of the patient mind, body, and spirit is also frequently embraced by herbal and conventional therapies. According to cultural standards, this holistic approach takes into account the physical, mental, emotional, and spiritual wellness of the patient. Traditional medicine usually uses locally sourced, environmentally friendly resources, minimizing its detrimental environmental impact and encouraging self-sufficiency. Modern sustainability objectives are consistent with the emphasis on natural remedies [9, 23, 25–27, 29, 50, 51] (Table 8).

The function of herbal and traditional medicine in modern health is summarized in Table 9. Modern healthcare is complemented by herbal and traditional medicine in contemporary health by giving patients more options for treating a range of ailments.

Table 8 Cultural heritage and healing wisdom

Aspect	Cultural heritage and healing wisdom
Cultural significance	Herbal and traditional medicines are deeply entwined with cultural identity
	Reflect cultural values, beliefs, and ancestral knowledge
	Source of cultural pride and preservation of ancient customs
Healing from a holistic perspective	It embraces a holistic approach to health and well-being
	It emphasizes the interconnectedness of mind, body, and spirit
	It considers mental, emotional, and spiritual aspects alongside physical health
Sustainable healing and environmental responsibility	It relies on local, sustainable resources
	It reduces the negative impact on the environment
	It aligns with contemporary sustainability goals

Table 9 The role of herbal and traditional medicine in contemporary health

Aspect	Key points
Complementary care	Herbal and traditional medicine enhance modern healthcare
	It offers a wide range of additional treatment options
	The patients can choose therapies that align with cultural heritage and beliefs
Preservation of cultural traditions	It may play a vital role in preserving cultural practices and local knowledge
	It may ensure traditions remain relevant and continue for future generations
	Preservation efforts include digitization, educational programs, and collaboration
Global access and inclusivity	Technological advancements enable the global dissemination of traditional medicine
	It will enable greater accessibility to bridge cultural divides, making healing more inclusive

They provide a variety of therapeutic methods, enabling patients to select therapies that are compatible with their cultural background and worldview. To ensure that these traditions remain relevant, traditional medicine is essential for maintaining cultural practices and passing on regional expertise to future generations. The preservation depends on several initiatives, such as the digitalization of documents, the execution of educational initiatives, and joint initiatives with indigenous groups. By allowing the global distribution of traditional medicine knowledge via digital channels, technological improvements can promote inclusion on a global level. Traditional and herbal medicine are becoming more widely available, bridging cultural gaps and providing healing options to a global audience [8, 23, 29, 30, 50, 51, 71].

There are opportunities and challenges for herbal and traditional medicine in modern health (Fig. 1). Traditional medicine needs additional research to verify the efficacy and safety of its treatments because it lacks scientific confirmation. Collaborations between modern academics and conventional practitioners can close this gap. To ensure patient safety and the upkeep of moral norms, traditional and conventional therapies must be regulated. Nations' newly passed laws attempt to strike a balance between traditional values and the demands of the present. Integrative medicine mixes traditional and modern practices to offer complete healthcare solutions. Patients benefit from customized therapy solutions that take into account their cultural and medical needs. To successfully incorporate contemporary knowledge into their techniques, traditional practitioners might need further training. Their capacity to bridge the gap between conventional and modern medicine is improved by access to educational opportunities [7, 76].

Therefore, herbal medicine and traditional practices serve as essential custodians of cultural heritage, preserving healing wisdom and holistic approaches to well-being. As these practices evolve and integrate with modern healthcare, they offer diverse treatment alternatives while respecting cultural traditions. Collaboration between traditional and modern medicine is crucial for preserving and advancing healthcare practices that honor our heritage and contribute to the well-being of individuals and communities. With its millennia-old history, traditional medicine continues to enrich our understanding of health and healing [64, 74, 82].

10 Benefits and Challenges of Artificial Intelligence in Traditional Medicine Practices

Integrating AI in traditional medicine practices offers a range of potential benefits, which can significantly enhance the healthcare field. The key benefits are summarized in Table 10.

The integration of AI into traditional medicine practices presents several challenges that must be carefully addressed. These challenges encompass technical, ethical, and practical concerns. Table 11 summarizes the challenges of AI in herbal medicine practices.

Therefore, since AI offers numerous benefits in traditional medicine, addressing these challenges is crucial to ensure its successful integration while preserving the principles and ethics of traditional healthcare practices. Artificial intelligence has enormous and varied potential applications in traditional medical procedures. AI has the potential to completely transform conventional medicine by enhancing diagnosis and treatment, personalizing care, and overcoming resource limitations. However, it is crucial to go slowly, address moral issues, and make sure that AI augments rather than replaces the knowledge of practitioners of conventional medicine. When used appropriately, AI can be a useful tool for maintaining and improving the efficacy of conventional medical procedures [10, 11].

Table 10 Benefits of artificial intelligence in traditional medicine practices

Benefit	Benefits of artificial intelligence in traditional medicine practices
Enhanced diagnosis and disease detection	AI can analyze extensive data for accurate early disease detection, improving treatment success rates
Improved treatment recommendations	AI can assist in personalized treatment plans based on a patient's medical history, genetics, and lifestyle
Predictive analytics	AI can be used to predict patient outcomes, helping practitioners anticipate complications and adjust treatment plans
Data analysis and research	AI can aid in processing vast medical data, identifying trends, and enhancing herbal remedies
Patient monitoring and follow-up	AI-driven systems can continuously monitor patients, providing timely interventions and improving outcomes
Resource optimization	AI can extend healthcare services to underserved areas by providing decision support and remote consultations
Herbal medicine development	AI may identify effective herbal remedies through data analysis, promoting evidence-based herbal medicine
Telemedicine	AI facilitates remote consultations, benefiting patients in rural areas and supporting follow-up care
Efficiency and productivity	AI automates administrative tasks, streamlining workflow and allowing practitioners to focus on patient care
Reduced human error	AI systems reduce errors in procedures and diagnoses, enhancing reliability and patient safety
Patient empowerment	AI provides access to health data and educational resources, empowering patients to make informed decisions
Continuous learning and adaptation	AI continuously learns and adapts to evolving medical knowledge, keeping traditional medicine practices up-to-date

11 Conclusion

A turning point in healthcare will be reached when AI is integrated into conventional medical procedures. This will allow for the integration of cutting-edge technology with the wisdom of long-standing traditions. This paradigm shift could be made possible by the harmonic fusion of technology and tradition, forging a connection between development and cultural heritage. Beyond improving healthcare, this union may protect cultural traditions and knowledge, ensuring its applicability to coming generations. Traditional medicine has a long cultural and historical history and is firmly rooted in communities all around the world. It not only provides a wide range of culturally rich treatment alternatives, but it also enhances contemporary healthcare by adopting a holistic therapeutic philosophy and utilizing locally sourced, sustainable materials. The preservation of cultural heritage and the global transmission of traditional knowledge are essential components of this fusion of technology and tradition. Phytochemistry, which clarifies bioactive components and their modes of action and ensures quality control and standardization, is a major factor in the effectiveness of herbal therapy. It aids in safety and toxicological

Table 11 Challenges of artificial intelligence in traditional medicine practices

Challenge	Challenges of artificial intelligence in traditional medicine practices
Data privacy and security	Traditional medicine practices handle sensitive patient data. AI systems must ensure robust data protection and encryption to safeguard patient privacy. Compliance with data protection regulations may be crucial
Ethical dilemmas	AI in healthcare may raise ethical issues, such as the potential for bias in algorithms, data ownership, informed consent, and the ethical use of patient data. Traditional medicine must navigate these ethical commemorates
Integration with traditional practices	AI may need to complement, not replace, traditional medicine. Finding the right balance between AI and the human touch is a challenge
Quality of data	The effectiveness of AI may rely on the quality of the data it uses for training and analysis. Inaccurate or biased data can lead to incorrect diagnoses and recommendations. Traditional medicine needs to ensure data quality
Regulatory compliance	Healthcare regulations and standards are stringent. Therefore, AI systems in traditional medicine need to adhere to these standards, which can be complex and vary from region to region. Compliance is a critical challenge
Resistance to adoption	Traditional medicine practitioners may resist adopting AI, fearing a loss of professional autonomy or skills
Risk of overreliance	An overreliance on AI for diagnosis and treatment decisions can be risky. There is a risk of overlooking critical clinical judgment. Therefore, striking the right balance between AI and human expertise is challenging
Cost and infrastructure	Implementing AI systems can be expensive, and not all traditional medicine practices may have the financial resources to invest in the necessary infrastructure and training. Cost-effectiveness is often a concern
Training and education	Traditional medicine practitioners need training to use AI tools effectively. Bridging the knowledge gap and ensuring practitioners are proficient in AI technologies is a significant challenge
Liability and malpractice	Determining liability in case of AI-related errors or malpractice can be complex. Establishing clear legal frameworks and accountability standards for AI use in traditional medicine is challenging
Patient trust and communication	Patients need to trust and understand the AI systems used in their healthcare. Ensuring effective communication and transparency about AI's role in their treatment can be challenging

evaluations, improves extraction methods, identifies chemical synergies, and provides information on pharmacokinetics, bioavailability, and potential drug-plant interactions. Research on phytochemicals supports the use of herbal remedies in healthcare and supports evidence-based medicine. AI has a variety of benefits over conventional medical procedures. The benefits includes diagnostic precision, personalized treatment suggestions, predictive analytics, supporting medical research, ongoing patient monitoring, increasing access to healthcare in underserved areas, developments in herbal medicine, facilitation of telemedicine, streamlining of administrative tasks, reduction of errors, patient empowerment, and encouraging continuous learning and adaptation. AI has opened up new doors for creativity and precision in conventional medicine, but it does not replace them. However, the integration of AI with conventional medicine faces significant obstacles and moral quandaries. Data security and privacy issues, ethical dilemmas, and regulatory compliance are all crucial factors. While guaranteeing the highest data integrity standards and legal adherence in integrating AI with conventional treatments, it is imperative to uphold the holistic and patient-centered ethos of traditional medicine. To successfully integrate AI into traditional medicine, additional challenges must be overcome, including those related to adoption resistance, overreliance on AI, infrastructure and cost concerns, training and education requirements, liability and malpractice worries, patient trust, and effective communication. Therefore, there is significant promise for improving healthcare while simultaneously maintaining cultural legacy and expertise, thanks to the synergy between AI and traditional medicine. To fully take advantage of these advantages, it is essential to overcome the related difficulties and moral issues. Finding a careful balance between tradition and technology, traditional medicine may survive and develop in the contemporary world while still providing the best possible patient care. This is a remarkable and groundbreaking method of providing healthcare in the twenty-first century: the seamless integration of conventional medicine and AI.

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Antimicrobial Resistance and the Role of Herbal Medicine: Challenges, Opportunities, and Future Prospects

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Abstract

The use of herbal medicine for antimicrobial treatment has a rich history that spans cultures and civilizations worldwide. This chapter focuses on the various aspects of herbal medicine, including its extensive classification, antimicrobial properties, factors influencing its efficacy, benefits, challenges, and future prospects. The antimicrobial properties of herbal medicine are remarkable, stemming from phytochemicals like alkaloids, flavonoids, terpenes, and polyphenols. These compounds interfere with microorganisms, creating broad-spectrum activity. Additionally, the synergistic effects of combining medicinal plants enhance their efficacy, thereby potentially reducing the development of resistance. However, herbal medicine's effectiveness is influenced by many factors, including plant species, plant parts used, geographical origin, and mode of administration. Quality control, scientific validation, and standardization are essential to overcome the challenges associated with herbal remedies. Furthermore, the cultural integration and sustainable practices surrounding herbal medicine are vital for its continued success. The prospects of herbal medicine in antimicrobial treatment look promising. Herbal medicine offers alternatives to combat antibiotic resistance, developing new drugs, personalized treatments, and sustainable healthcare practices. Synergistic approaches and global collaboration are expected to expand herbal medicine's role in modern healthcare. Therefore, herbal medicine remains a valuable and versatile tool for antimicrobial treatment. As we navigate the challenges and harness the potential of herbal remedies, they can continue to play a significant role in the treatment of microbial infections, paying homage to the wisdom of our ancestors while addressing the healthcare challenges of our time.

Keywords

Microorganisms · Mode of action · Medicinal plants · Herbal medicine · Human health · Traditional medicine

1 Introduction

A wide range of microorganisms, including bacteria, fungi, and viruses, can cause diseases [36, 39, 51]. Infections caused by bacteria, which are single-celled germs, are quite prevalent. These infections have the potential to cause many different human diseases. Accurate diagnosis and treatment are crucial for the efficient management and cure of bacterial infections, which can vary greatly in severity. Tuberculosis is a typical example of an infectious disease caused by the bacteria *Mycobacterium tuberculosis* with high severity. Infected people can transmit it to others through coughing or sneezing. *Streptococci* species are another type of bacterium that can cause a wide range of illnesses, including the common cold, ear infections, and pneumonia. Infections from Streptococci species can range from mild to severe. *Salmonella* species is another leading bacteria that could cause diseases in humans. *Salmonella* has been widely isolated in food [37] and are characterized by symptoms such as nausea, vomiting, and diarrhea. *Borrelia burgdorferi*, which causes Lyme disease, spreads through the bite of an infected tick and is another bacterium that causes diseases. Lyme disease is characterized by symptoms such as a rash and discomfort in the joints. Cholera, caused by the bacterium *Vibrio cholerae*, is characterized by severe diarrhea. One common route of exposure is through the ingestion of contaminated food, water, or drink. Bacterial meningitis can be caused by various bacteria, including *Neisseria meningitidis* and *Streptococcus pneumoniae*, but there are many other possible culprits as well. Syphilis, caused by the bacterium *Treponema pallidum*, is a chronic infection that worsens with time and can lead to organ failure if not treated. Anthrax, which is caused by the bacterium *Bacillus anthracis*, can show up in a few distinct ways: cutaneously, pulmonarily, or gastrointestinally.

Infections caused by the many species of fungi can show up anywhere in the body in a wide variety of forms [26]. The severity of a fungal infection varies from case to case. Antifungal agents are commonly used as standard treatment, although the specific medication used to treat an infection depends on the kind of fungus involved and its location. For efficient management of fungal infections, a precise diagnosis and a clearly defined treatment plan are essential. For instance, infections of the skin (cutaneous candidiasis), genitalia (vaginal yeast infections), and mouth (oral thrush) are all caused by the same fungus, *Candida*. The type of treatments required for the causative agent for each of these sites differs. Dermatophytes, a group of fungi that causes ringworm, can result in round, itchy rashes when it affects the skin (*Tinea corporis*), scalp (*Tinea capitis*), nails (*Tinea unguium*), and other body regions. Thus, the type of treatment plans also differs. Infections caused by *Aspergillus* fungus are common in immunocompromised people and tend to manifest in the respiratory system. Its extreme form is known as invasive aspergillosis. According to Vuong et al. [62], aspergillosis encompasses a range of infections caused by fungi within the *Aspergillus* genus. Prominent species contributing to these infections include *Aspergillus niger*, *Aspergillus fumigatus*, *Aspergillus terreus*, and *Aspergillus flavus*. The authors further stated that clinical manifestations of aspergillosis are often linked to the state of the immune system, with invasive forms primarily affecting individuals with compromised immune function. Notable examples of invasive diseases include rhinosinusitis and pulmonary aspergillosis. On the other hand, patients with preexisting lung conditions are more prone to developing slower-developing forms of aspergillosis, such as chronic pulmonary aspergillosis, aspergilloma, and allergic bronchopulmonary aspergillosis [62].

The fungus *Histoplasma capsulatum* is responsible for this infection, which spreads through the inhalation of spores found in bird or bat poop. Pneumocystis Pneumonia, caused by the fungus *Pneumocystis jirovecii*, is a serious lung infection that usually affects people with compromised immune systems, such as those with human immunodeficiency virus/acquired immunodeficiency syndrome (HIV/AIDS). Meningitis can be caused by the fungus *Cryptococcus neoformans* or *Cryptococcus gattii*, and it primarily affects people with weakened immune systems. According to Marco et al. [48], cryptococcosis, an opportunistic infection, leads to over 100,000 human immunodeficiency virus (HIV)-related deaths annually. The authors further reported that HIV is a common cause of this infection, it can also affect individuals with various other medical conditions such as persons undergoing immunosuppressive therapy, experiencing organ failure syndrome, recipients of organ transplants, individuals with inherent immunological disorders, common variable immunodeficiency syndrome, or hematological diseases.

Infections caused by viruses are common and can have far-reaching effects on the human body [36, 39]. The intensity, mode of transmission, and symptoms of these viral diseases are all different. Viral infections can be prevented and treated with vaccination, antiviral medication, and supportive care. The influenza virus is an infectious respiratory illness that causes fever, coughing, sore throats, and muscle aches. Acquired immunodeficiency syndromes (AIDS) is a disease that can develop if HIV is allowed to continue weakening the immune system. It can have far-reaching effects on one's health. Numerous viruses, including rhinoviruses, can be the cause of the common cold symptoms, which include a cough, a slight fever, a runny or stuffy nose, and mucus production. The new coronavirus severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) is what caused the 2019 global pandemic, COVID-19, which primarily affects the respiratory system [55]. The severity of the respiratory distress or pneumonia-like symptoms varies. Hepatitis A, B, C, etc. are all capable of causing liver inflammation, with symptoms ranging from those of the common flu to permanent liver damage. Herpes simplex virus (HSV) can cause painful blisters or sores on the lips or vaginal area, and it can also cause genital herpes [60]. The varicella-zoster virus, which causes chickenpox, is an infectious disease that most frequently affects children and results with an itchy rash and fever. The measles virus causes a high body temperature, a characteristic red rash, and potentially fatal consequences such as pneumonia and encephalitis [52]. The rubella virus causes a minor rash and fever, but its potential to cause birth defects is serious, especially in pregnant women [7]. High fever, bleeding, and organ failure are hallmarks of the Ebola virus disease, which almost usually results in death. Zika virus infection is worrying because of its link to birth abnormalities, even though it may only cause minor symptoms in adults [63]. Rabies is a fatal virus that is spread through animal bites and can have devastating effects on the central nervous system if treatment is delayed [21]. Variola virus (which caused smallpox) has been eradicated thanks to the widespread immunization campaigns, the sickness it produced was extremely dangerous and even lethal [64]. The human body is home to numerous helpful bacteria, and it is important to remember that not all microbes are hazardous.

When bacteria and other microbes develop the ability to resist the drugs intended to combat them, a worldwide health crisis is created [11, 12, 25, 44]. There are far-reaching ramifications to this issue, as well as several serious difficulties. Antibiotic usage in healthcare and agriculture, better diagnostics, increased public knowledge, and research into new antibiotics and alternative treatments are all necessary to combat antibiotic resistance. Actions including reducing inappropriate antibiotic use, increasing vaccination, and creating responsible antibiotic stewardship are necessary to address this critical global health concern. Due to the problem of antimicrobial resistance, plants have emerged as the most promising sources for novel antimicrobials [22, 45–47] and anti-pesticidal agents [23, 27, 33–35, 57, 58, 65]. Throughout human history, countless plants have been harvested from the wild and used for their medical value [6, 33–35]. Phytotherapy, or herbal medicine, has long been an integral part of conventional medical practices across cultural contexts [1, 2, 38]. Herbal medicine is unique because of the strong antibacterial qualities it contains [41]. Herbal remedies have been used for centuries to combat infections caused by bacteria, fungi, viruses, and other microorganisms. This chapter explores the antimicrobial activities of herbal medicine, shedding light on their challenges, opportunities, and future prospects.

2 The Looming Threat of Antimicrobial Resistance: A Global Health Crisis and Multifaceted Challenge

Antimicrobial resistance, commonly called AMR, presents a mounting challenge that jeopardizes our ability to combat bacterial, fungal, and viral infections effectively. This predicament has evolved into a global health crisis, stemming from the capacity of microorganisms such as bacteria, fungi, and viruses to develop resistance to drugs designed to eradicate them. Table 1 summarizes vital information about antibiotic resistance.

2.1 The Complex Nature of Antimicrobial Resistance

AMR encompasses a broad spectrum, encompassing resistance to antibiotics, antifungals, and antiviral medications, with antibiotics being most closely associated with this issue. Resistance can emerge through various mechanisms, including

Table 1 Overview of antibiotic resistance

Aspect	Note about AMR
Complex nature of AMR	AMR includes resistance to antibiotics, antifungals, and antiviral drugs, with antibiotics being most closely associated. Resistance can develop through genetic mutations, gene transfer, and misuse of drugs, leading to a reduction in the efficiency in the use for the treatment of diseases
Public health implications	AMR could lead to prolonged, severe infections, and increased healthcare costs
Global impact of AMR	AMR affects both developed and developing nations, crossing international borders. The economic burden includes rising healthcare costs, reduced productivity, and expenses for developing new antimicrobials for effective use in the treatment of diseases caused by microbial agents
Possible contributing factors to AMR	Inadequate infection control, poor sanitation, and antibiotic use in animal agriculture are contributing to AMR
Strategies for addressing AMR	Combatting AMR requires research into new antibiotics and alternative treatments, educating the local patent dealer and the society as a whole on how this products are utilized
International initiatives	Global organizations and national action plans, such as WHO and the global action plan on AMR, focus on surveillance, responsible use, and research into alternative treatments to address AMR and bringing awareness of the importance and aftermath of this product through media aid
The role of research	Research is essential to understand AMR better and develop innovative solutions. Advancements in diagnostics, therapies, and vaccines are crucial to addressing to challenges posed by antimicrobial resistance
A global effort to address AMR	Addressing AMR requires a global effort involving medical experts, researchers, policymakers, and the public. Public awareness, responsible antibiotic use, and international collaboration are essential to ensure the continued effectiveness of antimicrobial drugs in treating diseases

genetic mutations, horizontal gene transfer, and the inappropriate or excessive utilization of antimicrobial drugs. Infections become increasingly challenging to treat when they involve resilient microorganisms capable of withstanding one or more drugs.

2.2 Implications for Public Health

AMR leads to protracted and severe infections, amplified mortality rates, and escalated healthcare costs. This could be due to diminished availability of effective antibiotics compounds, and the difficulty in treating common diseases caused by microbial agents. Surgical procedures, cancer treatments, and organ transplants become fraught with more significant risks due to the indispensability of antibiotics in averting postoperative infections.

2.3 Global Impact

AMR transcends national boundaries, affecting both developed and developing nations. Its transnational nature enables resistant microorganisms to traverse international borders and disseminate globally. This issue confers a substantial economic burden through elevated healthcare costs, reduced labor productivity, and the outlays associated with developing novel treatments. According to Prestinaci et al. [53], AMR is a grave global public health concern, particularly concerning antibacterial resistance. The authors also reported that the first WHO Global report on AMR, published in 2014, emphasized the extensive impact of antibacterial resistance, its economic and health consequences, and risk factors, and highlighted antibiotic-resistant pathogens, including *Staphylococcus aureus*, *Klebsiella pneumoniae*, non-typhoidal *Salmonella*, and *Mycobacterium* tuberculosis.

2.4 Contributing Factors to Antimicrobial Resistance

The burgeoning issue of antibiotic resistance is aggravated by the appropriate use of antibiotics in clinical settings and their unwarranted agricultural deployment, especially in poultry science and animal husbandry. Substandard infection prevention and control measures in healthcare settings especially in developing countries facilitate the dispersion of resistant microorganisms. Furthermore, inadequate sanitation and hygiene in communities augment the proliferation of antibiotic-resistant microbes, particularly bacteria.

2.5 Strategies for Addressing Antimicrobial Resistance

Effectively combating antibiotic resistance mandates the research and development of novel antibiotics and alternative treatment modalities. AMR can be partly mitigated through judicious antibiotic administration, advancing precise diagnostic techniques, and establishing robust surveillance systems. Vaccination initiatives, aimed at diminishing the impact of infectious diseases, indirectly curtail the demand for antibiotics. Encouraging prudent antibiotic stewardship practices in healthcare settings diminishes superfluous antibiotic use. According to Uchil et al. [59], the strategies to combat this problem involve rational drug use, regulation of over-the-counter availability, enhanced hygiene, infection control, and the development of innovative drugs and vaccines, necessitating a multidisciplinary, collaborative, and regulatory approach.

2.6 International Initiatives

To combat AMR, global organizations and initiatives, such as the WHO and other agencies saddled action plans on AMR, need to encourage research and development in the field. Furthermore, setting up action plans at the national and even state levels focusing mainly on surveillance, responsible antibiotic use, and exploring alternative treatments to combat this issue could yield the desired result.

2.7 The Significance of Research in the Field

Research is imperative for a deeper comprehension of AMR and the formulation of innovative strategies to combat it. Breakthroughs in diagnostics, therapeutic interventions, and vaccines are pivotal in tackling the AMR problem. According to Masterton [49], surveillance studies play a crucial role in detecting emerging pathogens, identifying trends in pathogen incidence, and understanding antibiotic resistance, contributing significantly to the fight against AMR. The author further stated that monitoring pathogen susceptibility and antimicrobial use aids in developing effective strategies to combat AMR and informs clinical decision-making.

2.8 The Menace of Antibiotic Resistance: Impact and Implications on Global Health

Antibiotic resistance threatens the historical value of these medical marvels, with multidrug resistance emerging due to overuse and limited development of new antibiotics [3]. Antibiotic resistance poses a multifaceted threat to our ability to combat bacterial infections, with far-reaching implications. Comprehensive efforts, spanning healthcare, agriculture, and the environment, are needed to combat resistance, and promising alternatives like probiotics, antibodies, and vaccinations offer potential preventive and supplementary treatments in this fight [3]. Table 2 summarizes the menace of antibiotic resistance. In light of these consequences, it is essential to prioritize efforts to combat antibiotic resistance, a critical issue that imperils global health and warrants a concerted, worldwide response.

Therefore, AMR is a complex and pressing challenge requiring a unified global endeavor. The involvement of medical experts, researchers, policymakers, and the general populace is essential to ensure the continued effectiveness of antibiotics and other antimicrobial agents for treating infectious agents.

3 Herbal Medicine: An Overview of Practices and Philosophies

Herbal medicine, or phytotherapy or botanical medicine uses plants and plant-derived compounds for their therapeutic properties. It encompasses a wide range of herbal remedies and approaches to healthcare [4, 5, 20, 40, 42]. Table 3 summarizes the diverse world of herbal medicine classification. Furthermore, herbal medicine is a diverse field encompassing various traditions, plants, preparations,

Table 2 Menace of antibiotic resistance

Consequence	Description
Reduced treatment options	Antibiotic resistance diminishes the efficacy of medications, reducing the available treatments for bacterial infections and potentially turning curable diseases into life-threatening ones
Increased mortality	Antibiotic-resistant microbes particularly bacteria could lead to higher mortality rates in common diseases due to delayed or ineffective treatments, posing risks of severe disease conditions and fatalities
Prolonged illness	Treating antibiotic-resistant infections often demands more intensive interventions, resulting in extended diseases, higher medical costs, and longer hospital stays
Cross-resistance	Bacteria can develop resistance to single drugs and entire classes of antibiotics, thereby intensifying the shortage of effective treatment options
Financial strain	Antibiotic resistance places a significant financial burden on healthcare systems, contributing to increased costs due to extended treatments and possibly prolonged hospitalizations
Public health hazards	Uncontrolled transmission of antibiotic-resistant microbes particularly bacteria in communities and healthcare settings could pose a severe public health risk with rapid dissemination and challenging containment
Impact on surgical procedures and cancer treatment	Antibiotics are crucial for various surgical and cancer treatments, and antibiotic resistance could pose risks to these critical medical interventions
Agricultural use	The use of antibiotics in agriculture to promote livestock growth and disease prevention fosters the development of antibiotic-resistant microorganisms, particularly bacteria, thereby jeopardizing food safety and human health
Global transmission	Antibiotic-resistant microbes particularly bacteria disregard national borders and can spread worldwide, necessitating international collaboration to address the challenge

and philosophies. This section also offers a holistic and natural approach to health and wellness, often complementing conventional medical practices. The choice of herbal remedies and approaches depends on individual health needs, cultural contexts, and personal preferences [13–19, 43].

4 Antimicrobial Properties of Herbal Medicine

Herbal medicine, an ancient practice, has been employed for centuries to address various health conditions. Many herbs possess antimicrobial properties, making them valuable allies against bacterial [28–30], fungal, and viral infections [41]. While herbal medicine can serve as an alternative or complement to conventional antibiotics and antifungal drugs, it's essential to understand its applications in combating infections. This section discusses the various aspects of antimicrobial activities of herbal medicine.

Table 3 Exploring the diverse world of herbal medicine

Herbal medicine categories	Description
Traditional herbal medicine	Ayurvedic medicine (originated from India): Utilizing medicinal plants to enhance overall human health Traditional Chinese medicine: Combining herbal remedies with acupuncture, cupping, and therapies to enhance good health
Western herbalism	Native American herbalism: Indigenous Americans use native medicinal plants for therapeutic purposes Herbalism in Europe: Draws from herbal practices rooted in Greece, Rome, etc. Physiomedicalism: This blends traditional herbal therapy with early modern European medical practices
Modern herbal treatment	Modern Western herbalism: A holistic approach of integrating traditional knowledge and modern research Clinical herbalism: Collaborative patient-specific treatment plans using medicinal plants. Though not encouraged in clinical practice Homeopathy: The use of highly diluted plant substances to enhance natural healing Phytopharmacology: Scientific study of plant compounds or phytochemicals for the development of therapeutic agents
Herbal remedies by plant part	Leaves and flowers, roots and rhizomes, stem bark, seed, and nuts, etc. have shown that they can be used to inhibit the growth of microorganisms
Herbal preparations	Tinctures: Medicinal plant extracts preserved with alcohol Teas: Brewed from dried medicinal plants Infusions and decoctions: Herbal teas are prepared through infusion in boiling water Powders: A ground medicinal plant used in capsules or topically Salves and ointments: Topical herbal applications mixed with bases like beeswax Essential oils: Concentrated plant extracts for topical use and aromatherapy
Effects and qualities of herbal products	Antioxidants: Plants high in antioxidants, e.g., green tea, offer protection against oxidative stress Anti-inflammatory: Turmeric, ginger, etc. possess anti-inflammatory properties Antimicrobial: Herbs like garlic, paw paw, cashew, and lime have antimicrobial properties
Methods for herbal medicine practice	Herbal detoxification: Medicinal plant supporting natural detox processes Herbal tonics: Used for long-term health and vitality Herbal first aid: Herbs like aloe Vera, rosemary etc. for minor injuries and wounds
Herbal medicine systems and philosophies	Holistic herbalism: Focuses on addressing underlying causes of diseases Bioregional herbalism: Emphasizes local medicinal plant use Eclectic herbalism: Draws from diverse herbal traditions and systems

4.1 Types of Antimicrobial Herbal Medicine and Medicinal Plants

A diverse array of antimicrobial herbal medicines and medicinal plants offers unique natural remedies for addressing microbial infections [10, 24, 31, 32]. This section shows some prominent categories of antimicrobial herbal medicine and their associated medicinal plants:

4.1.1 Antimicrobial Medicinal Plants

The microbes under consideration here are viruses, bacteria, fungi, and parasites. Authors have variously reported that plant possess antimicrobial activities including *Anacardium occidentale* [28], *Carica papaya L* [29], *Vernonia amygdalina L.*, and *Ocimum gratissimum L* [30], *Myristica fragrans* [31], *Buchholzia coriacea* [32], *Tetrapleura tetraptera* [10], *Cymbopogon citratus*, *Zingiber officinale* [24], *Medicago sativa, Pimenta dioica, Aloe barbadensis, Aloe vera, Malus sylvestris, Withania somniferum, Euphorbia tirucalli, Momordica charantia, Berberis vulgaris, Ocimum basilicum, Laurus nobilis, Piper betel, Piper nigrum, Vaccinium spp., Arctium lappa, Carum carvi, Rhamnus purshiana, Anacardium pulsatilla, Matricaria chamomilla, Larrea tridentate, Capsicum annuum, Syzygium aromaticum, Erythroxylum coca, Coriandrum sativum, Vaccinium spp., Anethum graveolens, Echinaceae angustifolia, Eucalyptus globulus, Vicia faba, Garcinia hanburyi, Allium sativum, Panax notoginseng, Hydrastis canadensis, Citrus paradise, Camellia sinensis, Peganum harmala, Cannabis sativa, Lawsonia inermis, Humulus lupulus, Armoracia rusticana, Hyssopus officinalis, Rabdosia trichocarpa, Lantana camara, Lawsonia, Santolina chamaecyparissus, Millettia thonningii, Melissa officinalis, Aloysia triphylla, Calendula officinalis, Glycyrrhiza glabra, Thevetia peruviana, Myristica fragrans, Prosopis juliflora, Arnica Montana,* and *Quercus rubra*, among others [8].

Furthermore, Cowan [8] in a review study reported that *Aegle marmelos*, *Taraxacum officinale*, *Coriandrum sativum*, *Panax notoginseng*, *Armoracia rusticana*, *Lantana camara*, *Humulus lupulus*, *Santolina chamaecyparissus*, *Myristica fragrans*, *Prosopis juliflora*, *Arnica Montana*, *Laurus nobilis*, *Piper nigrum*, *Arctium lappa*, *Carum carvi*, *Rhamnus purshiana*, *Citrus paradisi*, *Camellia sinensis*, and *Peganum harmala* among others possess antifungi properties. Cowan [8] in a review study reported that *Arctium lappa, Carum carvi, Rhamnus purshiana, Matricaria chamomilla*, *Eucalyptus globulus*, *Cannabis sativa*, *Hyssopus officinalis*, and *Melissa officinalis* possess antiviral properties. Also, Cowan [8] in a review study reported that *Matricaria chamomilla*, *Hydrastis canadensis*, *Camellia sinensis*, *Lantana camara*, *Millettia thonningii*, *Aloysia triphylla*, *Thevetia peruviana*, *Myristica fragrans*, *Prosopis juliflora*, and *Arnica Montana* among others possess antiparasitic properties. Studies have attributed the antimicrobial potential of these plants to their phytochemical constituents [33–35].

4.1.2 Broad-Spectrum Antimicrobial Agents

Some medicinal plants have demonstrated potentials toward broad-spectrum antimicrobial agents. This is because of the ability to inhibit both gram-positive and negative microbes. Some of these plants include *Anacardium occidentale* [28], *Carica papaya* L [29], *Vernonia amygdalina L.* and *Ocimum gratissimum L* [30], *Myristica fragrans* [31], *Buchholzia coriacea* [32], *Tetrapleura tetraptera* [10], *Cymbopogon citratus*, and *Zingiber officinale* [24]. These antimicrobial medicinal plants exemplify the vast spectrum of natural treatments effective against microbes. When using medicinal plant extracts or parts for therapeutic purposes, seeking guidance on the correct dosage and approach from a healthcare professional or herbalist through consultation is crucial.

4.2 Mechanisms of Action

Antimicrobial agents, antifungals, and antiviral medications deploy various mechanisms to combat microbial infections effectively. These mechanisms are indispensable for identifying and eradicating harmful microbes, particularly bacteria. The diverse mechanisms of action are employed by antimicrobials, especially bacteria.

4.2.1 Protein Synthesis Inhibition

Antibiotics such as tetracyclines and macrolides focus on bacterial ribosomes responsible for protein synthesis. These antibiotics impede bacterial protein production by interfering with this process, inhibiting growth and replication. Also, antibiotics like macrolides, tetracyclines, aminoglycosides, and oxazolidinones target bacterial ribosomal subunits, inhibiting protein production. Amikacin, for example, permanently binds to 16S rRNA and the RNA-binding S12 protein, altering the ribosome's structure and disrupting mRNA codon reading, thereby obstructing tRNA anticodon engagement [61].

4.2.2 Nucleic Acid Synthesis Inhibition

Antimicrobials, including fluoroquinolones, block bacterial DNA replication and transcription, a process known as nucleic acid synthesis inhibition. This disruption hampers bacterial proliferation, leading to cell death. Also, antibiotics like ciprofloxacin target the enzyme DNA gyrase, essential for bacterial DNA synthesis, replication, repair, and transcription. By blocking topoisomerase IV and type II topoisomerase, these antibiotics prevent cell division [61].

4.2.3 Cell Wall Biosynthesis

Bifunctional enzymes involved in bacterial cell wall construction are targeted by bactericidal antibiotics such as penicillin, cephalosporins, and vancomycin [61]. They inhibit enzyme processes by binding to peptide substrates of the peptidoglycan layer [61]. For instance, vancomycin interferes with gram-positive bacteria's cell wall synthesis by binding to D-Ala-D-Ala moieties and preventing the growth of N-acetylglucosamine/N-acetylmuramic acid peptide strands that make up the cell wall's backbone [61].

4.2.4 Destruction of Bacterial Cell Wall

Many antibiotics, like penicillins and cephalosporins, target bacteria by disrupting cell wall synthesis, a vital process in bacterial growth. The absence of cell walls compromises structural integrity, rendering bacteria vulnerable to damage and eventual cell death. Antibiotics like polymyxins modify the structural makeup of lipopolysaccharides in the bacterial outer membrane. Polymyxin B, for example, attaches to the lipid component of lipopolysaccharides, causing structural changes that lead to osmotic imbalance and bacterial death. It also disrupts the integrity of the cytoplasmic membrane, leading to molecular leakage and inhibiting cellular respiration [61].

4.2.5 Interference with Metabolic Pathways

Certain antimicrobials, like sulfonamides, can mimic essential metabolic substrates of microorganisms. This interference disrupts microbial metabolism, hindering the production of critical metabolites and impeding their growth.

4.2.6 Targeting Specific Enzymes

Antifungal drugs, such as azoles, target specific enzymes essential for fungal cell membrane synthesis [56]. Inhibiting these enzymes disrupts fungal cell membranes.

4.2.7 Inhibition of Viral Replication

Antiviral drugs interfere at various stages of the viral replication cycle. Some hinder viral DNA or RNA synthesis, while others prevent viral entry into host cells [50]. Nucleoside analogs, for instance, induce chain termination during viral replication.

4.2.8 Enhancement of the Immune Response

Many antiviral medications, especially those used for retroviruses like HIV, enhance the immune response [9]. This may involve boosting immune cell activity, such as CD4+ T cells, to control viral replication [9].

4.2.9 Interference with Parasitic Metabolism

Antiparasitic drugs disrupt various metabolic pathways specific to parasites, often targeting unique enzymes. For example, antimalarial drugs inhibit heme-detoxification in *Plasmodium* species [54].

Furthermore, the choice of antimicrobial treatment depends on the specific microbial pathogen, its susceptibility to various drugs, and the infection site. A thorough understanding of these mechanisms of action is crucial for devising effective treatment strategies and preventing microbial resistance.

4.3 Broad-Spectrum Antimicrobial Efficacy

Broad-spectrum antimicrobial efficacy is of paramount importance in microbiology and medicine as it signifies the remarkable ability of specific antimicrobial agents, including antibiotics and medicinal plants, to effectively combat a wide range of infectious agents, such as bacteria, fungi, and viruses. This broad-spectrum capability is invaluable in the management of infectious diseases, providing an efficient means to address various illnesses caused by diverse agents. The essential aspects of broad-spectrum antimicrobial efficacy are outlined in Table 4, making it a cornerstone of medical science for combatting a wide array of pathogens responsible for bacterial, fungal, and viral infections.

4.4 Herbal Medicine Formulations

Herbal medicine formulations encompass a wide range of preparations that utilize medicinal plants for health improvement, treatment of various ailments, and addressing medical issues. These formulations have a long history of use across diverse cultures, with their continued popularity driven by the perception of their natural and holistic qualities. Table 5 summarizes the various herbal medicine formulations and uses [4, 15–17, 19]. Herbal medicine formulations offer diverse

Table 4 Aspects of broad-spectrum antimicrobial efficacy

Specific	Activities
Versatility in pathogen eradication	Broad-spectrum antimicrobial agents effectively combat a wide range of microorganisms, and they could be effective in treating patients with mixed infections
Bacterial infections	Broad-spectrum antibiotics address diverse bacteria, including gram-positive and gram-negative types, targeting pathogens such as *Staphylococcus aureus, Escherichia coli*, and *Streptococcus pneumoniae*. Ciprofloxacin and amoxicillin-clavulanate are notable examples
Fungal infections	Broad-spectrum antifungal agents tackle various fungal species, such as yeasts (e.g., *Candida*) and molds (e.g., *Aspergillus*). Common examples of antifungal agents are fluconazole and voriconazole
Viral infections	Broad-spectrum antiviral drugs are vital for treating viral infections, especially when dealing with novel or poorly understood viruses
Complex infections	Complex infections, involving various pathogens (bacteria, fungi, and viruses), benefit from broad-spectrum antimicrobials that address multiple causal agents concurrently
Risks of broad-spectrum usage	Misuse or overuse of broad-spectrum antimicrobials can lead to issues, including antimicrobial resistance, disruption of the body's normal microbiota, and other potential adverse effects
Combination therapy	Combining complementary broad-spectrum antimicrobials can enhance treatment efficacy in complex infections or resistance concerns
Diagnostic dilemmas	The early use of broad-spectrum antimicrobials can pose diagnostic challenges, hindering the identification of the specific pathogen and complicating the selection of targeted therapies
Future of antimicrobial therapy	Broad-spectrum antimicrobials remain crucial in the ongoing battle against infectious diseases, adapting to evolving pathogens and new threats

Table 5 Herbal medicine formulations and their common uses

Formulation	Description	Common uses
Tinctures	Concentrated liquid herbal extracts are preserved with alcohol	Precise dosing, extended shelf life
Teas (infusions/ decoctions)	Herbal teas are produced by steeping dried/fresh medicinal plants in hot water, either as infusions or decoctions	Gentle and calming effects, easy to prepare
Powders	Fine powders are made by grinding dried medicinal plant material, suitable for direct ingestion or mixing into foods	Convenient and versatile for various uses
Ointments and salves	Topical remedies are made by combining medicinal plants in oils (e.g., olive, palm kernel oil, or coconut) with beeswax	Treating skin conditions, cuts, burns, etc.
Capsules	Encased dried or powdered medicinal plant in gelatin or vegetarian capsules, convenient for controlled dosages	Tasteless and easy herbal consumption
Essential oils	Highly concentrated plant extracts capture aromatic and therapeutic qualities for topical or inhalation	Aromatherapy, various therapeutic applications
Syrups	Herbal infusions combined with sweeteners such as honey for flavor enhancement	Palatable herbal remedies, soothing for coughs
Poultices	Herbal pastes are produced by mashing fresh or dried medicinal plants for topical application to localized concerns	Treating muscle discomfort, insect stings, etc.
Compresses	Cloths soaked in herbal infusions or decoctions are applied to affected areas for pain relief or skin issues	Reducing inflammation and alleviating pain
Herbal baths	Herbal additions to bathwater for skin absorption of herbal benefits are believed to aid various conditions	Relaxation, muscle relief, and stress reduction

options, allowing individuals to choose the most suitable approach to address their health needs. However, it’s crucial to exercise caution, particularly when combining herbal medicines with conventional medications or during pregnancy. Consulting a qualified herbalist or healthcare practitioner is advisable to ensure the safe and effective use of herbal medicine formulations [13, 15].

4.5 The Synergistic Antimicrobial Effects of Herbal Medicine and Medicinal Plants

The synergistic antimicrobial effects of herbal medicine and medicinal plants offer a promising approach to treating various infectious diseases [11, 12]. Combining medicinal plants with complementary mechanisms of action, broad-spectrum activity, and increased efficacy can achieve a holistic treatment approach, reducing the risk of resistance development. Table 6 summarizes the synergistic antimicrobial effects of herbal medicine and medicinal plants.

Table 6 Synergistic antimicrobial effects of herbal medicine and medicinal plants

Point	Summary
Enhanced efficacy	Combining medicinal plants enhances antimicrobial effectiveness through synergistic effects
Broad-spectrum action	Certain medicinal plant combinations combat various infectious agents simultaneously, offering a holistic approach
Preventing antibiotic resistance	Herbal synergies can hinder pathogen resistance compared to single antimicrobial agents
Complementary mechanisms	Combining medicinal plants provides multiple modes of action, making it harder for pathogens to develop resistance
Improved tolerance	Herbal mixtures reduce adverse effects associated with high doses of a single medicinal plant, thereby enhancing patient tolerance
Enhanced bioavailability	Some medicinal plants boost the body's ability to absorb antimicrobial compounds effectively when used together
Personalized therapies	Herbal practitioners tailor treatments to individual health needs using synergistic medicinal plant combinations
Traditional wisdom	Traditional medicine systems like Ayurveda and TCM offer valuable knowledge for modern herbal therapy
Use with caution	Careful medicinal plant selection, proper dosages, and guidance from experienced herbalists are crucial for safe treatment
Validation and research	More research is required to bridge traditional and evidence-based medicine to assess herbal combinations' effectiveness

4.6 Safety and Side Effects

Herbal remedies are generally safe at low concentrations for plants that are not toxic to humans. However, caution is advised, this is because some individuals may exhibit sensitivity or allergies to certain medicinal plants. Pregnant or breastfeeding women require special attention when using herbal remedies.

4.7 Precautions

During the use of herbal remedies and medicinal plants, it is crucial to prioritize safety for the patient's well-being and to ensure the treatment's effectiveness. Table 7 summarizes essential safety measures when using herbal medicine and medicinal plants.

4.8 Future Prospects of Herbal Medicine in Antimicrobial Treatment

The future of herbal medicine in antimicrobial treatment is promising and holds the potential to address significant contemporary healthcare challenges. Herbal therapy offers a sustainable, holistic, and potentially more effective approach to combating infectious diseases than synthetic pharmaceuticals. This is particularly relevant as society becomes increasingly aware of the limitations and drawbacks of synthetic drugs. Table 8 summarizes the future prospects of herbal medicine in antimicrobial treatment.

Table 7 Essential safety measures when using herbal medicine and medicinal plants

Safety measure	Summary
Consult a qualified healthcare provider	Seek guidance from a healthcare provider or herbalist before herbal treatment, especially with health issues, pregnancy, or medication use
Adhere to recommended dosages	Follow recommended dosages and preparation methods to avoid adverse effects
Recognize allergies and sensitivities	Be aware of allergies and sensitivities; conduct patch tests when applying herbal remedies to the skin
Quality and source of herbal remedies	Use herbal medicine or medicinal plants from reputable suppliers to ensure quality and authenticity
Educate yourself	It's good to learn about potential side effects, interactions, and contraindications to make informed decisions
Avoid self-diagnosis	It's good to complement rather than replace conventional treatments for serious medical conditions
Start with lower dosages	It is advisable to begin with lower dosages when trying new herbal remedies, monitor effects, and adjust as needed
Maintain records	It is advisable to keep a record of herbal medicine uses including dosages and their effects
Medicinal plant-drug interactions	It is advisable to be mindful of interactions with pharmaceuticals, as medicinal plants may affect medication efficacy or cause reactions
Consider age and health status	It is advisable to tailor herbal therapies based on age and health status; some medicinal plants may not be suitable for certain age groups
Check expiration dates and store properly	It is advisable to verify herbal remedies' expiration dates and store them in a cool, dry place to maintain potency and avoid spoilage
Herbal combinations	Always ensure compatibility and prevent adverse interactions when using combinations of herbs
Seek professional guidance	Always consult with qualified herbalists or healthcare providers for any doubts or concerns about herbal therapy

4.9 Ethical and Sustainable Practices of Herbal Medicine in Antimicrobial Treatment

During the use of herbal medicine and medicinal plants for antimicrobial treatment, ethical and sustainable practices are essential. These practices could ensure responsible and environmentally friendly use of medicinal plants while respecting cultural and ethical considerations. Table 9 summarizes the aspects of ethical and sustainable practices in herbal medicine for antimicrobial treatment.

5 Factors Affecting the Antimicrobial Properties of Herbal Medicine

The effectiveness of herbal medicine in combatting microbial illnesses is influenced by many factors. To successfully apply herbal medicine in healthcare, it is crucial to understand these elements. The primary factors that contribute to the antibacterial

Table 8 Future prospects of herbal medicine in antimicrobial treatment

Prospect	Prospects
Combating antibiotic resistance	Herbal medicine may provide alternative treatments with diverse bioactive compounds, reducing the risk of contributing to antibiotic resistance
Synergistic methods	Research into synergistic medicinal plant combinations can lead to more effective antimicrobial medications by understanding how different herbs interact
Scientific validation and standardization	Extensive scientific research and standardization of herbal medicine could enhance the credibility and reliability of herbal treatments, validating their safety and effectiveness
Identification and isolation of bioactive compounds	The isolation of specific bioactive compounds from medicinal plants can create innovative medications, enhancing the therapeutic effects of herbal therapy
Personalized medicine	The integration of herbal medicine into modern healthcare could allow personalized treatment plans based on an individual's health profile, optimizing herbal antibacterial treatments
Combination therapies	Herbal medicine could complement traditional antimicrobial therapies, thereby potentially reducing synthetic drug dosages, and minimizing adverse effects and resistance development
Green and sustainable healthcare	Herbal medicine aligns with environmentally friendly healthcare as it involves sustainable cultivation and gathering of medicinal plants
Cultural integration	The integration of herbal medicine with modern practices could bridge traditional wisdom and contemporary science, thereby enhancing patient outcomes through collaboration
Educational initiatives	Increasing education and awareness about evidence-based herbal medicine use could empower healthcare providers and the public to make informed choices
Global collaboration	Collaborative efforts between countries and cultures could facilitate the exchange of knowledge about herbal remedies, thereby enhancing the understanding of antimicrobial properties
Regulatory frameworks	The development of effective regulatory frameworks for herbal medicine products could ensure their safety and efficacy while considering their unique characteristics in antimicrobial treatment

effects of herbal medicine are summarized in Table 10. The various factors, from plant characteristics to processing and application methods, influence the antimicrobial properties of herbal medicine. During herbal medicine use in the treatment of microbial infections, it is essential to consider these elements to ensure the proper production and administration of herbal preparations.

Table 9 Aspects of ethical and sustainable practices in herbal medicine for antimicrobial treatment

Practice	Aspects of ethical and sustainable practices in herbal medicine
Ethical sourcing	Medicinal plants should be sourced from ethical suppliers who follow fair trade practices and ensure proper compensation for local communities involved in cultivation and harvesting. Ethical sourcing could promote social justice and community well-being
Biodiversity conservation	Herbal practitioners and growers should prioritize the conservation of plant biodiversity by using sustainable cultivation and harvesting methods, preventing the overexploitation of plant species
Sustainable cultivation and harvesting	Medicinal plants should be grown and harvested using sustainable practices to prevent habitat destruction and depletion of plant populations
Traditional knowledge respect	Ethical herbal medicine should acknowledge and respect the traditional knowledge of indigenous communities and traditional healers. This could be achieved through fair collaboration, consultation with these communities, and respecting their intellectual property rights
Environmental impact consideration	Herbal medicine production should consider the environmental impact of cultivation and processing
Labeling and certification	Herbal medicine products and medicinal plants should be accurately labeled, indicating the botanical name, part of the plant used, and cultivation or harvesting methods
No overharvesting of endangered species	Ethical practices should ensure that endangered or threatened plant species are not overharvested, and alternatives are sought to prevent their extinction
Waste reduction	Sustainable herbal medicine production minimizes waste and utilizes plant parts efficiently to reduce environmental impact
Research on cultivation techniques	Ethical herbal medicine involves research and development to improve cultivation techniques and ensure the long-term availability of medicinal plants
Fair compensation for herbal practitioners	Practitioners of herbal medicine should be made to receive fair compensation for their expertise and efforts, reflecting the value of their knowledge and skills
Traditional healers collaboration	Collaboration between traditional healers and modern healthcare practitioners respects and integrates the wisdom and experience of both to enhance patient outcomes
Education and public awareness	Raising public awareness about evidence-based use of herbal medicine is essential for promoting ethical and sustainable practices. Education could help consumers make informed choices
Regulatory compliance	Compliance with local and international regulations and standards could ensure that herbal medicines meet safety and quality requirements
Scientific validation	All herbal remedies need to undergo scientific validation to assess their effectiveness and safety, thereby bridging the gap between traditional wisdom and evidence-based medicine

Table 10 Factors affecting the antimicrobial properties of herbal medicine

Factor	Effects
Antimicrobial compound levels	Levels of antimicrobial compounds tend to vary among plant species and varieties, affecting their efficacy against specific microbes
Plant part used	The portion of the plant used (leaves, roots, bark, etc.) influences the type and quantity of antimicrobial chemicals present
Geographical origin	Environmental factors like soil, climate, and altitude could impact the chemical makeup of plants, resulting in varying antimicrobial capabilities
Timing of harvest	The plant's developmental stage during harvesting can affect the concentration of active chemicals
Processing and preparation	The processing and preparation methods, such as drying or extraction, can alter the antimicrobial properties of medicinal plants
Storage conditions	Storage conditions, including light, heat, and moisture exposure, can affect herbal products' shelf life and efficacy
Plant maturity	The age of the plant influences the presence and concentration of antimicrobial compounds, thereby impacting on its efficacy
Chemical composition	The specific phytochemicals in medicinal plants, including alkaloids, flavonoids, terpenes, polyphenols etc., determine their antimicrobial qualities
Synergistic effects	Combinations of medicinal plants can create synergistic effects, thereby enhancing antibacterial activity
Dosage and concentration	The appropriate dosage and concentration of herbal remedies affect their efficacy against microbes
Administration method	The manner of administering herbal medication, such as inhalation, topical application, or ingestion, could impact effectiveness of the remedies
Microorganism strain susceptibility	The specific microbial strain and its susceptibility to herbal compounds can influence the effectiveness of herbal therapies
Host factors	Factors related to the host, including age, overall health, and immune function, can affect the efficacy of herbal medicines
Quality control and standardization	Strict quality control methods are required to ensure the consistency and quality of herbal products, thereby influencing their antimicrobial properties
Scientific validation	The extent of scientific research and validation of antimicrobial agents from herbal medicine or medicinal plants can determine its acceptance in modern medicine
Cultural and traditional knowledge	Traditional wisdom and knowledge of herbal use for specific infections play a vital role in herbal medicine's efficacy

6 Benefits of Herbal Medicine in Antimicrobial Treatment

Herbal medicine has maintained its popularity and is increasingly gaining attention in contemporary medical practice due to its numerous benefits in antimicrobial therapy. The benefits of herbal medicine in antimicrobial treatment are summarized in Table 11. While herbal medicine offers numerous benefits in antimicrobial

Table 11 Benefits of herbal medicine in antimicrobial treatment

Benefit	Description
Broad antimicrobial activity	Herbal medicines have a wide spectrum of antimicrobial actions, effectively targeting bacteria, fungi, viruses, and parasites, making them versatile for treating various infectious diseases
Reduced antibiotic resistance	Herbal treatments, rich in diverse phytochemicals, challenge microorganisms in developing resistance, addressing the concern of drug-resistant strains in conventional medicine
Fewer adverse effects	Herbal medicines are generally better tolerated by the body, leading to fewer side effects, suitable for extended use, and ideal for individuals sensitive to prescription drug side effects
Enhanced immune response	Certain medicinal plants such as *Echinacea purpurea*, *Lycium barbarum*, *Ganoderma lucidum*, *Agaricus blazei*, *Clerodendrum splendens*, *Achyranthes bidentata*, *Morinda citrifolia*, *Dimocarpus longan*, *Abrus cantoniensis*, *Rhizoma gastrodiae*, *Momordica charantia*, *Gynostemma pentaphyllum*, *Astragalus membranaceus* etc. possess immunomodulatory properties, strengthening the immune system's ability to combat diseases
Anti-inflammatory effects	Many herbal remedies inherently have natural anti-inflammatory properties, aiding infection management and reducing associated inflammation and discomfort
Synergistic effects	Traditional medical practices, like Ayurveda and TCM, use herbal combinations that create synergistic effects, thereby enhancing infectious disease treatment
Gentle healing approach	Herbal medicine is often considered a gentler approach, working with the body's natural processes, supporting and encouraging them rather than imposing forceful treatments
Cultural and traditional wisdom	Herbal remedies are deeply rooted in cultural and traditional practices, passed down through generations, making them widely accessible
Cultural inclusivity	Herbal medicine is practiced globally and can be culturally inclusive, offering alternative treatment options that respect diverse communities and belief systems

treatment, it's vital to know that it is not a panacea. The effectiveness of herbal remedies varies, and scientific validation is often necessary to confirm their safety and efficacy [4, 13, 16, 17, 19, 43].

7 Challenges of Herbal Medicine in Antimicrobial Treatment

Despite its potential in antimicrobial treatment, herbal medicine has several significant problems that must be carefully addressed before it can be used safely and effectively to treat microbial illnesses. The disadvantages of herbal medicine as an antimicrobial treatment are summarized in Table 12. The use of herbal medicine to treat infectious diseases shows promise, but there are still many obstacles that must be overcome.

Table 12 Challenges in herbal medicine for antimicrobial treatment

Challenge	Description and effects
Lack of standardization	Herbal remedies often lack standardized formulations, making ensuring consistent quality and efficacy difficult. Variations in active substance concentrations can impact treatment outcomes
Quality control	Ensuring the quality of herbal products is challenging due to susceptibility to contamination, adulteration, and mishandling during manufacturing and distribution, compromising safety and effectiveness
Regulatory concerns	Regulatory frameworks for herbal medicine vary between jurisdictions, potentially subjecting herbal products to less rigorous testing and oversight than pharmaceuticals, raising safety and efficacy concerns
Insufficient scientific validation	Although there is growing evidence supporting herbal remedies' antimicrobial properties, rigorous scientific investigations are needed to establish their efficacy and safety conclusively through robust clinical trials
Medicinal plant-drug interactions	Herbal medicines can interact with pharmaceutical drugs, thereby potentially altering the effectiveness of both treatments. Vigilance is necessary to prevent adverse effects and ensure compatibility with conventional therapies
Lack of dosage guidelines	Determining optimal herbal remedy dosages is challenging due to the absence of standardized recommendations. Reliance on traditional knowledge and expert guidance may not always be readily available
Variability in active compounds	Herbal treatment composition varies based on plant species, geographical origin, harvesting conditions, and processing methods, resulting in inconsistent treatment efficacy among patients
Safety concerns	Herbal medicines are generally considered safer than many synthetic drugs; some medicinal plants can induce adverse effects, allergic reactions, or side effects when misused or in high doses, necessitating safety measures
Mislabeling and misrepresentation	The herbal product market faces issues of mislabeling and misrepresentation, with products often fraudulently claimed to treat conditions or inaccurately labeled, leading to consumer confusion and potential harm
Limited access to qualified practitioners	Limited access to qualified herbalists and healthcare providers with herbal medicine expertise in certain regions may result in improper or unsafe herbal remedy use
Resistance to scientific integration	Resistance to integrating herbal medicine into modern healthcare by some traditional practitioners and advocates can hinder collaboration and the development of evidence-based practices
Cultural and ethical considerations	Use of specific herbs for antimicrobial treatment may raise cultural and ethical questions, particularly concerning sustainable sourcing of rare or endangered plant species, necessitating responsible practices

8 Conclusion

There is a wealth of traditional knowledge and growing scientific backing for the antimicrobial remedies found in herbal therapy. Herbal medicines' antibacterial effects show that nature can provide effective treatments for many different kinds of infections. Herbal medicine is unique among other methods of treatment because of the wide variety of herbal traditions, plant sources, and preparation methods. The classification, characteristics, benefits, and potential of herbal medicine in antimicrobial therapy all indicate a promising and fruitful future. The diverse range of approaches, attitudes, and practices that constitute herbal medicine are all reflected in its comprehensive taxonomy. Therapies can range from the ancient knowledge of Ayurveda and TCM to more modern practices like clinical herbalism and phytopharmacology, allowing individuals to choose treatments depending on their cultural background and personal health needs. Bioactive substances found in plants have been hypothesized to explain herbal medicine's antimicrobial effects. Herbal medicine's effectiveness can be affected by many variables, including the type of plant used, how it was prepared, and how it was given to the patient. The overall efficacy of herbal therapy in treating microbial infections can be improved by understanding and optimizing these parameters. Due to benefits of using herbal medicine, the holistic approach to advocate for it is in line with the current movement toward environmentally friendly and sustainable healthcare practices, which seek to meet the requirements of people from all walks of life and all cultures. There are, however, a number of obstacles that need to be overcome before herbal therapy may realize its full potential. Herbal treatments can only be trusted if they have undergone rigorous standardization, quality control, and scientific validation. The abundance of traditional medicinal knowledge should be recognized and incorporated into modern healthcare systems, and regulatory frameworks should be established to oversee herbal goods. Sustainable practices are also essential to defending the ecosystem and keeping the wide variety of medicinal plants available for future generations.

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